ACC FOR MINIMAL LOG DISCREPANCIES IN DIMENSION THREE, III

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Abstract. We prove Shokurov's ACC conjecture for minimal log discrepancies in dimension three: the minimal log discrepancies at closed points of log canonical threefold pairs with boundary coefficients in a fixed DCC set satisfy the ascending chain condition.

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1. Introduction

We work over the field of complex numbers $\mathbb{C}$.

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Shokurov asked whether the minimal log discrepancies of log terminal singularities of a fixed dimension satisfy the ACC [Sho88, Problem 5(c)]. For pairs (X, B) whose coefficients belong to a DCC set Γ (we write $B \in \Gamma$), the question becomes the following conjecture [Sho96, Conjecture 4.2] (see also [Sho04, p. 316]).

Conjecture 1.1 (cf. [Sho88, Problem 5]). *Let d be a positive integer and let $\Gamma \subset [0,1]$ be a DCC set. Then the set of minimal log discrepancies $\mathrm{mld}(X \ni x, B)$, where (X, B) is an lc pair of dimension d, $x \in X$ is a closed point, and $B \in \Gamma$, satisfies the ACC.*

Shokurov proved that Conjecture 1.1 and the lower semicontinuity conjecture of Ambro [Amb99, Conjecture 2.4] imply the termination of flips [Sho04, Theorem, p. 317]. The implication only uses Conjecture 1.1 for finite sets Γ [Nak16a, Section 1]. The termination of flips in dimension at most d also follows from the termination of $\mathbb{Q}$-factorial terminal flips in dimension d and the ACC for the minimal log discrepancies

$$\mathrm{mld}(X, B) := \inf\{a(E, X, B) \mid E \text{ is an exceptional prime divisor over } X\}$$

of exceptionally non-canonical pairs (X, B) of dimension d whose coefficients belong to a finite set [HL25b, Theorem 1.1], and in dimension 4 the second condition alone implies the termination of flips [HL25b, Theorem 1.2]. Here (X, B) is exceptionally non-canonical if $\mathrm{mld}(X, B) < 1$ and exactly one exceptional prime divisor E over X satisfies $a(E, X, B) \leq 1$ [HL25b, Definition 3.8(1)], and for $d = 3$ the ACC for $\mathrm{mld}(X, B)$ of exceptionally non-canonical pairs (X, B) holds when the coefficients belong to a DCC set [HL25b, Theorem 1.4].

Before this paper, Conjecture 1.1 was known in the following cases, where a condition on the value $\mathrm{mld}(X \ni x, B)$ means that the ACC holds for the subset of values satisfying the condition:

(1) When $d = 2$ [Ale93, Sho94, HL24].
(2) When X is toric and B is torus-invariant [Amb06] (see also [Bor97]).
(3) When $(X \ni x, B)$ is exceptional [HLS26].
(4) When $(X \ni x, B)$ is klt and the local volume $\widehat{\mathrm{vol}}(X \ni x, B)$ of $(X \ni x, B)$ satisfies $\widehat{\mathrm{vol}}(X \ni x, B) \geq \epsilon$ for a fixed positive real number ϵ [HLQ24, Theorem 1.3].
(5) When $B = 0$, $\mathrm{mld}(X \ni x) < \epsilon_d$ for a positive real number ϵ_d depending only on d, and either X is $\mathbb{Q}$-factorial klt and $x \in X$ has regularity one, that is, the largest dimension of the dual complexes of the lc pairs (X, B') near x is 1 [Mor23a, Section 2.5 and Theorem 3], or $x \in X$ is a quotient singularity [Mor23b].
(6) When $d \leq 3$ and, for a fixed positive real number ϵ and a fixed positive integer N,
$$\mathrm{mld}(X \ni x, B) \leq 1 \quad \text{and} \quad \#\{E \mid \mathrm{center}_X E = \{x\},\ a(E, X, B) < 1 + \epsilon\} \leq N,$$
where E runs through the exceptional prime divisors over X [Sho96].
(7) When $d = 3$ and $x \in X$ is smooth [Kaw26]. The case $\mathrm{mld}(X \ni x, B) \in (1, 3]$ was proved earlier [Kaw15].
(8) When $d = 3$ and $x \in X$ is a terminal quotient singularity of index at most a fixed r [Kaw26].
(9) When $d = 3$ and X is fixed [Kaw24].
(10) When $d = 3$, (X, B) is canonical, and Γ is finite [Nak16a].
(11) When $d = 3$, X is terminal near x, and $\mathrm{mld}(X \ni x, B) \geq 1$ [HLL25].
(12) When $d = 3$ and the following hold [HL25b, Theorem 6.8] (see also [HL25a, Theorems 1.2 and 1.3]):
  (i) X is $\mathbb{Q}$-factorial.
  (ii) $(X \ni x, B)$ is exceptionally non-canonical, that is, (X, B) is exceptionally non-canonical near x and $\mathrm{mld}(X \ni x, B) = \mathrm{mld}(X, B)$ [HL25b, Definition 3.8(2)].
  (iii) $x \in X$ is an isolated non-canonical singularity whose index 1 cover is terminal and not smooth.
(13) When $d = 3$ and either $\mathrm{mld}(X \ni x, B) \geq 1$ or $\Gamma = \{1\} \cup \{1 - \frac{1}{n} \mid n \in \mathbb{N}^+\}$ [Liu26a, Theorems 1.2 and 1.3] (see Theorem 1.2).

(14) When $d = 3$ and Γ is finite [Liu26b, Theorem 1.3] (see Theorem 1.3).

The main results of the first two papers [Liu26a, Liu26b] of the series are the following.

Theorem 1.2 ([Liu26a, Theorems 1.2, 1.3 and 1.5]). *The following statements hold.*

(1) Let $\Gamma \subset [0,1]$ be a DCC set. Then the set of minimal log discrepancies $\mathrm{mld}(X \ni x, B)$, where (X, B) is an lc pair of dimension 3, $x \in X$ is a closed point, $B \in \Gamma$, and $\mathrm{mld}(X \ni x, B) \geq 1$, satisfies the ACC.

(2) Conjecture 1.1 holds for $d = 3$ and $\Gamma = \Gamma_{\mathrm{std}}$, where

$$\Gamma_{\mathrm{std}} := \{1\} \cup \left\{1 - \frac{1}{n} \;\middle|\; n \in \mathbb{N}^+\right\}.$$

(3) Let $\Gamma \subset [0,1]$ be a DCC set. Then the set of positive numbers $\mathrm{mld}((X \ni x, B)/G)$ (see Definition 2.2), where $(X \ni x)$ is a smooth threefold germ, G is a finite abelian group acting faithfully on X and fixing x, and $B = \sum_{j=1}^{n} c_j \operatorname{div}(h_j)$ with $c_j \in \Gamma$ and non-zero regular G-semi-invariant functions h_j near x such that (X, B) is klt near x, satisfies the ACC.

Theorem 1.3 ([Liu26b, Theorem 1.3]). *Let $\Gamma \subset [0,1]$ be a finite set of real numbers. Then the set of minimal log discrepancies $\mathrm{mld}(X \ni x, B)$, where (X, B) is an lc pair of dimension 3, $x \in X$ is a closed point, and $B \in \Gamma$, satisfies the ACC.*

Throughout the paper, results of [Liu26a] and [Liu26b] are cited with the numbering of the versions on the author's homepage.

Other results fix part of the data. For a fixed positive integer r, the set of minimal log discrepancies $\mathrm{mld}(X \ni x, \mathfrak{a})$, where $(X, \mathfrak{a})$ is an lc pair of dimension d with rK_X Cartier and $\mathfrak{a}$ is an $\mathbb{R}$-ideal with exponents in a fixed finite set, is discrete in $\mathbb{R}$ [Nak16a, Corollary 1.3]. In dimension 3, fixing the minimal log discrepancy instead bounds the index: for a positive real number a and a finite set $\Gamma \subset [0,1] \cap \mathbb{Q}$, there exists a positive integer r depending only on a and Γ such that $r(K_X + B)$ is Cartier near x for any threefold germ $(X \ni x, B)$ with $\mathrm{mld}(X \ni x, B) = a$ and $B \in \Gamma$ [Liu26c, Theorems 1.2 and 1.3].

For a fixed pair (X, B), the set of $\mathrm{mld}(X \ni x, B, \mathfrak{a})$, where $x \in X$ is a closed point and $\mathfrak{a}$ is an $\mathbb{R}$-ideal with exponents in a fixed finite set, is finite [Kaw14, Theorem 1.2]. For a fixed klt germ $(X \ni x)$, the ACC for $\mathrm{mld}(X \ni x, \mathfrak{a})$, where the exponents of $\mathfrak{a}$ belong to a DCC set, follows [MN18, Theorem 1.5] from the following boundedness [MN18, Conjecture 1.1]: for any finite set $I \subset \mathbb{R}_{\geq 0}$ there exists a positive integer ℓ depending only on $(X \ni x)$ and I such that, for any $\mathbb{R}$-ideal $\mathfrak{a}$ with exponents in I, some prime divisor E over X computing $\mathrm{mld}(X \ni x, \mathfrak{a})$ (when $\mathrm{mld}(X \ni x, \mathfrak{a}) = -\infty$, this means $\operatorname{center}_X E = \{x\}$ and $a(E, X, 0, \mathfrak{a}) < 0$ [MN18, Section 2]) satisfies $a(E, X, 0) \leq \ell$. The ACC for $\mathrm{mld}(X \ni x, \mathfrak{a})$ with the exponents of $\mathfrak{a}$ in a DCC set is equivalent to the ACC for a-lc thresholds on $(X \ni x)$ [Kaw21, Conjecture 4.2(ii) and Theorem 4.6].

Let $\Gamma \subset [0,1]$ be a DCC set. In each of the following cases, there exists a real number ℓ depending only on the listed data such that some prime divisor E over X satisfies $\operatorname{center}_X E = \{x\}$, $a(E, X, B) = \mathrm{mld}(X \ni x, B)$, and $a(E, X, 0) \leq \ell$:

(1) $(X \ni x, B)$ is an lc surface germ with $B \in \Gamma$, with ℓ depending on Γ [HL23, Theorem 1.2].
(2) X is a fixed klt threefold, $B \in \Gamma$ is $\mathbb{R}$-Cartier, and (X, B) is lc near x, with ℓ depending on Γ and X [Kaw24, Theorem 1.4].
(3) X is a terminal threefold, $B \in \Gamma$, and $\mathrm{mld}(X \ni x, B) \geq 1$, with ℓ depending on Γ [HLL25, Theorem 1.2].
(4) $(X \ni x, B)$ is a klt germ of dimension d, X is $\mathbb{Q}$-Gorenstein, $B \in \Gamma$, and $\widehat{\mathrm{vol}}(X \ni x, B) \geq \epsilon$ for a positive real number ϵ, with ℓ depending on d, ϵ, and Γ [HLQ24, Theorem 1.6].

For varying germs, in each dimension, the ACC for a-lc thresholds [BS10, Definition 1.4 and Conjecture 1.7] (see also [Liu18, Definition 1.2 and Conjecture 1.4]) for all $0 \leq a < 1$ is equivalent

to the statement that, for any DCC set $\Gamma \subset [0,1)$, the set of $\mathrm{mld}(X \ni x, B)$ with $B \in \Gamma$ has no accumulation point from below in $[0,1)$ [Liu18, Theorem 1.6].

The global minimal log discrepancy $\mathrm{mld}(X,B)$ is a different invariant. For a DCC set $\Gamma \subset [0,1]$, there exists a positive real number ϵ depending only on Γ such that

$$\{\mathrm{mld}(X,B) \mid (X,B) \text{ is a pair of dimension } 3,\ B \in \Gamma\} \cap [1-\epsilon, +\infty)$$

satisfies the ACC [HLL25, Theorem 1.1]. Without boundary, there exists a positive real number ϵ_0 such that, for any threefold X, $\mathrm{mld}(X) < 1$ implies $\mathrm{mld}(X) \leq 1-\epsilon_0$ [Jia21, Theorem 1.3]. The optimal value is $\epsilon_0 = \frac{1}{13}$, and $\mathrm{mld}(X) = \frac{12}{13}$ for the quotient $X = \mathbb{A}^3/\mu_{13}$ with weights $(3,4,5)$ [LX21, Theorem 1.4 and Example 1.5]. For threefolds X, a terminal point of index $r \geq 2$ has minimal log discrepancy $1+\frac{1}{r}$ [Kaw93, Theorem], so the largest accumulation point of the set of $\mathrm{mld}(X)$ is 1, the second largest is $\frac{5}{6}$ [LL22, Theorem 1.3], and the set of $\mathrm{mld}(X) \in [\frac{5}{6}, +\infty)$ satisfies the ACC [LL22, Theorem 1.5]. For a DCC set $\Gamma \subset [0,1]$, the ACC for $\mathrm{mld}(X,B)$ with $B \in \Gamma$ holds for klt threefold pairs (X,B) with

$$\#\{a(E,X,B) \in [0,1] \mid E \text{ is an exceptional prime divisor over } X\}$$

bounded from above [HL25b, Theorem 1.6]. The ACC also holds for the minimum of $a(E,X,B)$ over all prime divisors E over X, the components of B included, where (X,B) is a $\mathbb{Q}$-factorial lc threefold pair with $B \in \Gamma$ and X carries a very ample divisor H with H^3 bounded from above [CL26, Theorem 1.3].

For $d = 3$, after Theorems 1.2 and 1.3, Conjecture 1.1 was still open for the values $\mathrm{mld}(X \ni x, B) \in (0,1)$ when Γ is infinite and $\Gamma \not\subset \Gamma_{\mathrm{std}}$. This paper proves this case (Theorem 1.4). For $d \geq 4$, Conjecture 1.1 remains open in general.

This paper is the third and last paper of the series after [Liu26a] and [Liu26b]. Our first main result is the following theorem.

Theorem 1.4. *Let $\Gamma \subset [0,1]$ be a DCC set. Then the set of minimal log discrepancies $\mathrm{mld}(X \ni x, B)$, where (X,B) is an lc pair of dimension 3, $x \in X$ is a closed point, and $B \in \Gamma$, satisfies the ACC.*

By [BS10, Corollary 1.10(i)], Theorem 1.4 implies the ACC for a-lc thresholds in dimension 3 [BS10, Conjecture 1.7].

Our second main result is the following theorem.

Theorem 1.5. *Let a be a positive rational number and $\Gamma \subset [0,1] \cap \mathbb{Q}$ a finite set. Then there exists a positive real number $\eta(a,\Gamma)$ with the following property. Let X be a $\mathbb{Q}$-factorial klt threefold, $x \in X$ a closed point with $\mathrm{mld}(X \ni x) \geq a$, and $D \geq 0$ a $\mathbb{Q}$-divisor on X whose coefficients belong to Γ such that (X,D) is lc near x. Then*

$$\tau_a(X,D;x) := \sup\{t \in [0,1] \mid \mathrm{mld}(X \ni x, tD) \geq a\} \notin (1-\eta(a,\Gamma), 1).$$

Theorem 1.4 follows from Theorems 1.5 and 1.3, together with [Liu26a, Lemma 6.2 and Proposition 6.3] and [HLS26, Theorem 5.6] (see Section 20).

Sketch of the proof. Let $\Gamma \subset [0,1]$ be a DCC set, and let (X_i, B_i) be lc threefold pairs with $B_i \in \Gamma$ and closed points $x_i \in X_i$ such that $a_i := \mathrm{mld}(X_i \ni x_i, B_i)$ is strictly increasing. We may assume that (X_i, B_i) is a $\mathbb{Q}$-factorial klt pair, that $B_i = \sum_{j=1}^r b_{ij} D_{ij}$ for distinct prime divisors D_{ij}, where b_{ij} is increasing in i with limit $b_j > 0$, and that $(X_i, \overline{B}_i)$ is lc for $\overline{B}_i := \sum_j b_j D_{ij}$ [Liu26a, Lemma 6.2 and Proposition 6.3]. Let $a := \lim_i a_i$. For any $t \in (0,1)$ and $i \gg 0$, $t\overline{B}_i \leq B_i$, so $\mathrm{mld}(X_i \ni x_i, t\overline{B}_i) \geq a_i$. The coefficients of $t\overline{B}_i$ belong to the finite set $\{tb_1, \dots, tb_r\}$, so $\{\mathrm{mld}(X_i \ni x_i, t\overline{B}_i)\}_i$ satisfies the ACC (Theorem 1.3), which together with $a_i \to a$ implies $\mathrm{mld}(X_i \ni x_i, t\overline{B}_i) \geq a$ for $i \gg 0$ (depending on t). In particular, $\mathrm{mld}(X_i \ni x_i) \geq a$ for these i. Since $B_i \leq \overline{B}_i$, $\mathrm{mld}(X_i \ni x_i, \overline{B}_i) \leq a_i < a$. Thus

$$\tau_a(X_i, \overline{B}_i; x_i) < 1 \quad \text{and} \quad \lim_{i \to +\infty} \tau_a(X_i, \overline{B}_i; x_i) = 1.$$

Since $\tau_a(X, D; x)$ is the minimum of 1 and the a-lc threshold of $(X \ni x, 0)$ with respect to D [BS10, Definition 1.4] and [Liu18, Definition 1.2], Theorem 1.5 is implied by the ACC for a-lc thresholds [BS10, Conjecture 1.7] and [Liu18, Conjecture 1.4].

In Section 20, the level a and the coefficients b_j, which may be irrational, are replaced by rational numbers. Let $\delta_0 > 0$ satisfy $\mathrm{mld}(X_i \ni x_i, \overline{B}_i) \le a - \delta_0$ for any i (Theorem 1.3). In the smallest affine subspace of $\mathbb{R}^r$ defined over $\mathbb{Q}$ containing $(b_1, \dots, b_r)$, there exists a neighborhood $\mathcal{U}$ of $(b_1, \dots, b_r)$ such that $(X_i, \sum_j u_j D_{ij})$ is lc for any $(u_1, \dots, u_r) \in \mathcal{U}$ and any i [HLS26, Theorem 5.6]. For a small real number $\epsilon > 0$ and a rational vector $(b'_1, \dots, b'_r) \in \mathcal{U} \cap (0, 1]^r$ close enough to $(b_1, \dots, b_r)$,

$$(1-\epsilon)a(E, X_i, t\overline{B}_i) \le a\Big(E, X_i, t\sum_j b'_j D_{ij}\Big) \le (1+\epsilon)a(E, X_i, t\overline{B}_i)$$

for any $t \in [0, 1]$ and any prime divisor E over X_i with center x_i, so that, for a rational number $\alpha \in ((1+\epsilon)(a-\delta_0), (1-\epsilon)a)$,

$$\tau_a(X_i, \overline{B}_i; x_i) \le \tau_\alpha\Big(X_i, \sum_j b'_j D_{ij}; x_i\Big) < 1.$$

Since $\tau_a(X_i, \overline{B}_i; x_i) \to 1$, this contradicts Theorem 1.5 for $(\alpha, \{b'_1, \dots, b'_r\})$.

We prove Theorem 1.5 by contradiction in Section 19. Let $(W \ni o)$ be a normal threefold germ, G a finite group acting faithfully on W and fixing o, m a positive integer, and h a mark, that is, a non-zero G-semi-invariant regular function near o (Definition 2.4), such that $(W, \frac{1}{m}\mathrm{div}(h))$ is lc near o. For $t \in [0, 1]$, the radial function (Definition 4.1) is

$$\begin{aligned} q(t) &:= \mathrm{mld}\Big(\Big(W \ni o, \frac{t}{m}\mathrm{div}(h)\Big)/G\Big) \\ &= \inf\Big\{\frac{a(F, W, 0) - \frac{t}{m}\mathrm{mult}_F h}{e_F} \;\Big|\; F \text{ is a prime divisor over } W \text{ with center } o\Big\}, \end{aligned}$$

where e_F is the order of the inertia group of F in G (Definition 2.2). The value $q(t)$ is also the minimal log discrepancy of the quotient of $(W, \frac{t}{m}\mathrm{div}(h))$ by G at the image of o (Lemma 2.15). A prime divisor attaining the infimum for $t = 1$ is a computing lift of $q(1)$ (Definition 2.4).

Let a be a positive rational number, $\Gamma \subset [0, 1] \cap \mathbb{Q}$ a finite set, and m a positive integer with $m\Gamma \subset \mathbb{Z}$. If Theorem 1.5 fails for (a, Γ), then there exist $\delta \in (0, a)$ and a radial failure sequence $(W_i \ni o_i, G_i, h_i, \tau_i)$ with constants (m, a, δ) (Construction-Theorem 4.7(C)). Here $(W_i \ni o_i)$ is an isolated Gorenstein terminal hypersurface threefold germ whose quadratic rank, the rank of the quadratic part of a local equation (Definition 2.2), is 1, 2, or 3, and the radial functions q_i of $(W_i \ni o_i, G_i, h_i, m)$ satisfy

$$b_i := q_i(1) \le a - \delta, \qquad q_i(\tau_i) \ge a, \qquad \text{and} \qquad \lim_{i\to+\infty} \tau_i = 1.$$

More precisely, $\tau_i = \tau_a(X_i, D_i; x_i)$ for a new sequence (X_i, x_i, D_i) of counterexamples to Theorem 1.5, and $(W_i \ni o_i)$ is the germ of an equivariant Gorenstein terminal crepant higher model of a finite abelian Galois cover V_i of X_i near x_i such that the pullbacks of K_{X_i} and mD_i to V_i are Cartier, at the center, a closed point, of a prime divisor F'_i over V_i lying over a prime divisor over X_i that computes $\mathrm{mld}(X_i \ni x_i, D_i)$. The group G_i is the inertia group of F'_i in the Galois group of $V_i \to X_i$, hence cyclic, $\mathrm{div}(h_i)$ is the pullback of mD_i, and $q_i(t) \ge \mathrm{mld}(X_i \ni x_i, tD_i)$ for $t \in [0, 1]$, with equality for $t = 1$ (Construction-Theorem 4.7).

Each term of the infimum defining q is a decreasing affine function of t, so a computing lift F of $q(1)$ satisfies $q(1) \le q(t) \le q(1) + (1-t)\,\mathrm{mult}_F h/(me_F)$ for any $t \in [0, 1]$. Along the radial failure sequence, any computing lift F of b_i thus satisfies

$$\frac{a(F, W_i, 0)}{e_F} \ge \frac{\mathrm{mult}_F h_i}{me_F} \ge \frac{q_i(\tau_i) - b_i}{1 - \tau_i} \ge \frac{\delta}{1 - \tau_i} \to +\infty$$

(Lemma 4.3). Hence computing lifts F_i of b_i with $a(F_i, W_i, 0)/e_{F_i}$ bounded, for infinitely many i, lead to a contradiction. So does a real number $\theta > 1$ such that $(W_i, \frac{\theta}{m}\operatorname{div}(h_i))$ is lc near o_i for infinitely many i, since then $\operatorname{mult}_F h_i/(me_F) \leq b_i/(\theta - 1)$ for any computing lift F of b_i.

After passing to subsequences, the proof in Section 19 establishes the following in turn.

(1) $b_i > 0$, and in quadratic ranks 1 and 2 moreover $\lfloor \frac{1}{m}\operatorname{div}(h_i) \rfloor = 0$ and $(W_i, \frac{1}{m}\operatorname{div}(h_i))$ is klt near o_i (Theorem 9.1).
(2) The quadratic rank is 1 or 2.
(3) $|G_i| \leq M$ for a positive integer M (Theorem 12.1).
(4) By (3), after passing to a subsequence, $G_i = G$. With the linear action of G on coordinates $x_1, \dots, x_4$ and the characters of the equations and marks fixed, the local equations $f_i \in \mathbb{C}[[x_1, \dots, x_4]]$ and the marks h_i have a compatible generic limit (f_*, h_*) with coefficients in an algebraically closed field $\Bbbk \supset \mathbb{C}$ (Definition 2.5), the prepared limit of Construction-Theorem 13.9. Let
$$W_* := (f_* = 0) \subset \mathcal{X}_* := \operatorname{Spec} \Bbbk[[x_1, \dots, x_4]].$$
Then W_* is normal, canonical, and cDV, and $(W_*, \frac{1}{m}\operatorname{div}(h_*))$ is lc with positive minimal log discrepancy at the closed point.
(5) $(W_*, \frac{1}{m}\operatorname{div}(h_*))$ is not klt at the generic point of some 1-dimensional component of $\operatorname{Sing}(W_*)$ (Theorem 14.1).
(6) No component of $\frac{1}{m}\operatorname{div}(h_*)$ has coefficient 1 (Theorem 16.7).
(7) The non-klt locus $\operatorname{Nklt}(W_*, \frac{1}{m}\operatorname{div}(h_*))$ is a regular G-invariant curve $C \subset \operatorname{Sing}(W_*)$. It is the minimal lc center (Lemma 13.5), which is not the closed point by (4) and not a surface by (6), and it is the 1-dimensional component of $\operatorname{Sing}(W_*)$ of (5).
(8) The quadratic rank is not 2 (Theorem 18.3).
(9) In quadratic rank 1, the generic transverse type of W_* along C (Definition 15.1) is A_n for some $n \geq 1$ (Theorem 17.2), which is impossible (Theorem 18.3).

For (2), in quadratic rank 3, $q_i \leq 2$ (Lemma 4.2(3)) and $q_i(\tau_i) - b_i \geq \delta$ yield $t_i \in [\tau_i, 1)$ strictly increasing to 1 with $q_i(t_i)$ strictly increasing (Lemma 4.5). Since the labels $\frac{t_i}{m}$ of the boundaries $\frac{t_i}{m}\operatorname{div}(h_i)$ form a DCC set, the strictly increasing quotient values $q_i(t_i)$ (Definition 2.2) contradict the ACC for quotient values on smooth germs and on germs of quadratic rank at least 3 with labels in a DCC set (Theorem 10.1). For klt pairs, [Liu26b, Theorem 7.1] proves the ACC for finite sets of labels, and for DCC sets of labels except on isolated germs of quadratic rank 3, which are formally isomorphic to the germs $W_N = (xy + z^2 + w^N = 0)$ with $N \geq 3$ (Definition 2.9) and reduce to diagonal actions on them of groups of bounded order [Liu26b, Proposition 7.31 and Theorem 7.2(1)]. For bounded N, the ACC holds on finitely many fixed quotients (Theorem 2.22(1)).

The new case is that of labelled boundaries $\sum_j c_{ij}\operatorname{div}(h_{ij})$ on W_{N_i} with $N_i \to +\infty$ and labels c_{ij} increasing to limits κ_j. The case of finitely many labels, where $c_{ij} = \kappa_j$ for $i \gg 0$, is treated in [Liu26b, Theorem 7.2(2)]. Prime divisors F over W_{N_i} with $a(F, W_{N_i}, 0)/e_F$ bounded are transported to $W_0 = (xy + z^2 = 0)$ (Proposition 3.14(2)), and the marks are pulled back to the smooth double cover of W_0, where a strict increase of the quotient values can only come from the deficits $\kappa_j - c_{ij}$ multiplied by diverging orders of the marks along the computing lifts (Proposition 10.5). If the deck involution acts non-trivially on the lc center of the limit boundary, charts étale-locally isomorphic to one fixed germ exclude this (Theorem 10.7). Otherwise the rescalings $(x, y, z) \to (w^k x, w^k y, w^k z)$ or the rigidity of critical pairs below exclude it (Theorems 10.8 and 10.9).

For (3), for any $\epsilon > 0$ there exist $N_\epsilon \in \mathbb{N}^+$ and $\eta \in (0, 1)$, depending only on m and ϵ, such that
$$q(t) - q(1) < \epsilon \quad \text{for any } t \in (1 - \eta, 1]$$
whenever G is cyclic with $|G| \geq N_\epsilon$, $(W \ni o)$ is an isolated Gorenstein terminal hypersurface threefold germ of quadratic rank 1 or 2, h is a mark on $(W \ni o)$, and $(W, \frac{1}{m}\operatorname{div}(h))$ is lc near

o (Theorems 12.2, 12.3, and 12.4). With $\epsilon = \delta/2$, the bound contradicts $q_i(\tau_i) - b_i \geq \delta$. If the bound fails along a sequence $(W_i \ni o_i, G_i, h_i)$ with $|G_i| \to +\infty$, then for some $t_i \to 1$, $\alpha_i := q_i(t_i)$ is strictly increasing with a limit α_∞ such that $\alpha_\infty \geq q_i(1) + \epsilon/2$ for any i (Lemma 4.5), and, after passing to a subsequence, for a primitive circle $\mathbf{T}$ (Definition 2.9), $r_i := |G_i \cap \mathbf{T}| \to +\infty$ and $[G_i : G_i \cap \mathbf{T}]$ is bounded.

Along a circle $\mathbf{T}$, the radial functions are compared with the minimal log discrepancies β_i^+, β_i^- of two surface quotients by $\mathbf{T}$, of signs $+$ and $-$, which belong to a fixed finite set (see Set-up 11.1 and Proposition 11.2). Let

$$\beta_i := \min\{\beta_i^+, \beta_i^-\}.$$

Gauss extensions (Definition 2.3) of bounded computing divisors on fixed quotients of $\mathbb{A}^3$ bound $q_i(t) - \beta_i$, for $i \gg 0$, from above by a multiple, independent of i and of $t \in [0,1]$, of $\frac{1}{r_i} + 1 - t$ (Proposition 11.4), so $\beta_i \geq \alpha_\infty$ after passing to a subsequence. Lower bounds come from sections with coefficient near 1 (Lemma 11.5) and from a dichotomy: a compatible generic limit of plane pairs computing β_i is either klt, which yields a margin $\theta > 1$ for $(W_i, \frac{1}{m}\operatorname{div}(h_i))$, or has a unique smooth floor, a curve with coefficient 1. In the second case, a plane coordinate restricting to a parameter of the floor yields a $\mathbf{T}$-invariant polynomial U, semi-invariant with a character of order n_U, such that

$$\Big(W_i, \frac{1}{m}\operatorname{div}(h_i) + n_U\beta_i \operatorname{div}(U)\Big) \text{ is lc},$$

whence $q_i(1) \geq \beta_i$ (Proposition 11.7, [Liu26b, Lemma 5.8]). In the first case,

$$q_i(t_i) - q_i(1) \leq (1 - t_i)q_i(1)/(\theta - 1) \to 0$$

as in the paragraph containing Lemma 4.3, while

$$q_i(t_i) - q_i(1) \geq \alpha_i - \alpha_\infty + \frac{\epsilon}{2} \to \frac{\epsilon}{2}.$$

In the second case, $q_i(1) \geq \beta_i \geq \alpha_\infty$ contradicts $q_i(1) \leq \alpha_\infty - \epsilon/2$. In [Liu26b] such bounds along a circle are proved for klt labelled boundaries with labels in a DCC set on the split sources of [Liu26b, Set-up B.17] ([Liu26b, Lemmas B.20(3), B.22 and B.23], used for [Liu26b, Theorem 8.13]), and under [Liu26b, Set-up 11.1] ([Liu26b, Propositions E.3 and E.4]), while here one mark is compared at the labels $\frac{t}{m}$ and $\frac{1}{m}$ and $(W_i, \frac{1}{m}\operatorname{div}(h_i))$ is only lc.

For (4), the positivity of $\operatorname{mld}(W_i \ni o_i, \frac{1}{m}\operatorname{div}(h_i))$ does not pass to the limit by itself, but a uniform margin does. There exists $\eta_m > 0$, depending only on m, such that $(W, \frac{1}{m}\operatorname{div}(h), \mathfrak{m}_o^{\eta_m})$ is lc near o whenever $(W \ni o)$ is a smooth threefold germ or a normal hypersurface threefold germ, $\mathfrak{m}_o$ is the ideal of o, h is a non-zero regular function, and $\operatorname{mld}(W \ni o, \frac{1}{m}\operatorname{div}(h)) > 0$ (Theorem 5.1(2)). Inversion of adjunction [EMY03, Corollary 1.8] turns the margin into an lc property on the fixed smooth ambient germ, which passes to compatible generic limits (Theorem 2.18(1)). Thus any prime divisor P over $\mathcal{X}_*$ with center the closed point satisfies

$$a\big(P, \mathcal{X}_*, (f_*)^1(h_*)^{1/m}\big) \geq \eta_m \operatorname{mult}_P \mathfrak{n}_* \geq \eta_m,$$

where $\mathfrak{n}_*$ is the maximal ideal (Lemma 13.4), and adjunction from $\mathcal{X}_*$ to W_* (Lemma D.9(1)) yields the positivity in (4).

For (5), suppose that $(W_*, \frac{1}{m}\operatorname{div}(h_*))$ is klt at the generic points of the 1-dimensional components of $\operatorname{Sing}(W_*)$. Then each f_i, after an equivariant change of coordinates and multiplication by a unit, agrees with a polynomial reference of bounded degree modulo a high power of the multiplier ideal $\mathcal{J}(f_i^\gamma h_i^{\gamma\mu})$, for fixed rational numbers $\gamma \in (0,1)$ and $\mu > \frac{1}{m}$, so that bounded computing lifts on the references return to W_i (see Lemma 6.1 and Theorems 8.2, 14.2, 14.4, and 14.5).

For (6), (8), and (9), let Jac_W be the ideal generated by the partial derivatives of a local equation of W (Definition 5.2). If F is a computing lift of $q(1)$, then some computing lift P of $q(1)$ satisfies $a(P, W, 0) \leq L$ for a real number L depending only on m, $|G|$, and $\operatorname{mult}_F \operatorname{Jac}_W$ (Theorem 15.8(1)). In (6), the Jacobian orders of computing lifts are bounded when the minimal lc center is not contained in $\operatorname{Sing}(W_*)$ (Proposition 15.5, Theorem 15.8(2)). Otherwise the

minimal lc center is a regular curve in $\mathrm{Sing}(W_*)$, and on the smooth uniformizing cover of the generic transverse Du Val surface along the curve, the lc property leaves one or two coefficient-one branches, which are excluded (Theorems 16.5 and 16.6). In (8) and (9), the Jacobian orders of the computing lifts diverge (Theorem 15.8(1)), and the computing lifts concentrate along C: there exist G-eigenparameters (Definition 2.3) $(s_i, u_{i1}, u_{i2}, u_{i3})$ of the ambient germs with limits (s, u_1, u_2, u_3) such that $C = (u_1 = u_2 = u_3 = 0)$ and

$$\sup_i \mathrm{mult}_{F_i} s_i < +\infty \quad \text{and} \quad \min_{1\le j\le 3} \mathrm{mult}_{F_i} u_{ij} \to +\infty$$

for computing lifts F_i of b_i (Theorem 16.1(2)). Thus the computing lifts only see the generic transverse surface of W_* along C, a Du Val germ over an algebraic closure of $\Bbbk((s))$, whose types D and E (Theorem 17.2) and A (Theorem 18.3) are treated separately.

In the types A, there exist G-eigenparameters (x, y, z, s) with $C = (x = y = z = 0)$ in which f_* is a unit times $x^2 + g(y, z, s)$, where the quadratic part of g in y, z is $s^\nu y^2$, $s^{\nu_1} y^2 + s^{\nu_2} z^2$, or $s^\nu yz$ (Lemma 17.3). After the extension $s = u^2$, the coefficients s^ν, s^{ν_1}, and s^{ν_2} are squares, so that, for instance, $x = u^\nu x'$ turns $x^2 + s^\nu y^2$ into $u^{2\nu}(x'^2 + y^2)$, where $x'^2 + y^2$ has quadratic rank 2. The extension $s = u^2$ and fixed weighted point charts along C return the computing lifts to isolated Gorenstein terminal germs $(Z_i \ni 0)$ of quadratic rank at least 2, with groups of order $2|G_i|$, integers $N' \ge 0$ and $\kappa \ge 1$ independent of i, and marks l_i on Z_i such that the pullback of h_i to Z_i is $u^{N'} l_i$ and b_i is the quotient value of

$$\Big(Z_i \ni 0, \frac{1}{m}\,\mathrm{div}(l_i) + \beta\,\mathrm{div}(u)\Big), \qquad \beta := \frac{N'}{m} - \kappa < 1,$$

and any prime divisor P over Z_i with center 0 restricts to a prime divisor F over W_i with ramification index ρ (Definition 2.2) and

$$e_P = \rho e_F \quad \text{and} \quad \rho a(F, W_i, 0) = a(P, Z_i, 0) + \kappa\,\mathrm{mult}_P u$$

(Theorem 18.1). Propositions 17.5, 17.7, and 17.13 are similar returns in types D and E. For infinitely many i, the selection theorem for such signed boundaries, whose coefficient β along $\mathrm{div}(u)$ may be negative (Theorem 18.2), yields computing lifts P_i with $a(P_i, Z_i, 0) \le C_0$ and $\mathrm{mult}_{P_i} u \le C_0$ for a real number C_0, whose restrictions F_i compute b_i with

$$a(F_i, W_i, 0)/e_{F_i} \le (1 + \kappa)C_0,$$

a contradiction. In quadratic rank 2, the theorem is used with $\beta = 0$ for split forms $xy + g_i$ of the equations with limit $xy + z^d\vartheta(z, s)$.

Theorems 10.9, 17.6, 17.11, and 18.2 and Definition-Lemma 7.13 depend on the rigidity of weighted critical pairs (Section 7). Let w_1, w_2 be coprime positive integers, $Q_j \in \Bbbk[p, v]$ finitely many polynomials whose top parts T_j, the parts of maximal (w_1, w_2)-weighted degree, have degrees d_j, and $\kappa_j > 0$ real numbers with

$$\sum_j \kappa_j d_j = w_1 + w_2,$$

so that $(\mathbb{A}^2, \sum_j \kappa_j\,\mathrm{div}(T_j))$ is a critical pair (Definition 7.1). If $(\mathbb{A}^2, \sum_j \kappa_j\,\mathrm{div}(Q_j))$ is lc near 0, not klt at 0, and any component through 0 has coefficient less than 1, then there exists a unique automorphism (p', v') of $\mathbb{A}^2$ fixing 0 and inducing the identity on the graded ring of the weighted-degree filtration such that $Q_j(p, v) = T_j(p', v')$ for any j (Proposition 7.4), and similarly on $(xy + z^d = 0)$ and on the Du Val singularities of types D and E (Propositions 7.6 and 7.7). The proof uses the connectedness principle on the weighted projective closure of $\mathbb{A}^2$ (see Lemma 7.3).

Structure of the paper. In Section 2 we recall preliminaries, in particular labelled boundaries and quotient values, compatible generic limits, circles and floors, and the results of [Liu26a] and [Liu26b] that are cited several times in this paper. In Section 3 we state and prove the versions of results of [Liu26a] and [Liu26b] that this paper uses and that those papers do not contain. In Section 4 we define radial functions, and we reduce a failure of Theorem 1.5 to a radial failure sequence on hypersurface germs with actions of cyclic groups. In Section 5 we prove uniform margins, construct ambient lifts, bound Jacobian orders on split germs, choose computing lifts on surface quotients with lc marks, resolve marked hypersurfaces of bounded degree with a finite table of log discrepancies and multiplicities, and compare germs near products with products. In Section 6 we spread multiplier ideal memberships from compatible generic limits to infinitely many members of the sequence, and we treat bounded contact, concentration and degenerations on a fixed ambient space, and bounded marked orders off the singular locus. In Section 7 we prove the rigidity of weighted critical pairs and a signed adjunction, and we prove a lemma on critical stretching. In Section 8 we compare radial functions with fixed reference germs. In Section 9 we treat radial failure sequences with zero values, with boundaries having coefficient-one components, or with non-klt pairs. In Section 10 we prove the ACC for quotient values on smooth threefold germs and on hypersurface germs of quadratic rank at least 3 with labelled boundaries whose labels belong to a DCC set. In Section 11 we fix a circle and compare radial functions for lc marks with fiber references along it. In Section 12 we prove that the orders of the groups of a radial failure sequence of quadratic rank 1 or 2 are bounded. In Section 13 we construct prepared limits. In Section 14 we exclude limits that are klt along the singular locus. In Section 15 we choose bounded computing lifts when the Jacobian orders are bounded. In Section 16 we exclude limits with a coefficient-one component. In Section 17 we prove that the generic transverse type along the non-klt curve of a limit is A_n for some $n \geq 1$. In Section 18 we exclude quadratic rank 2 and generic transverse type A_n. In Section 19 we prove Theorem 1.5, and in Section 20 we prove Theorem 1.4. Appendices A–G contain the long proofs for Sections 9, 10 and 12–18.

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2. Preliminaries

We adopt the standard notation and definitions for the MMP from [Sho92, KM98, BCHM10] and use them freely.

2.1. Pairs and quotients.

Definition 2.1. A *pair* $(X/Z \ni z, B)$ consists of a contraction $\pi : X \to Z$ between normal quasi-projective varieties, a point $z \in Z$, and an $\mathbb{R}$-divisor $B \geq 0$ on X such that $K_X + B$ is $\mathbb{R}$-Cartier over a neighborhood of z. If π is the identity morphism, then we may drop Z and say that $(X \ni z, B)$ is a pair. If $(X/Z \ni z, B)$ is a pair for any $z \in Z$, then we say that $(X/Z, B)$ is a pair and also say that (X, B) is a pair. If we do not require that $B \geq 0$, then we say that (X, B) is a *sub-pair*. Let $h : X' \to X$ be any log resolution of (X, B) and E a prime divisor on X', and write $K_{X'} + B' := h^*(K_X + B)$. Then we define

$$a(E, X, B) := 1 - \operatorname{mult}_E B'$$

as the *log discrepancy* of E with respect to (X, B). For any divisorial valuation $v = c \operatorname{ord}_E$ for some real number $c > 0$, we denote by $a(v, X, B) := ca(E, X, B)$.

For any (sub-)pair (X, B), we say that (X, B) is *(sub-)lc* if $a(E, X, B) \geq 0$ for any prime divisor E over X, and *(sub-)klt* if $a(E, X, B) > 0$ for any prime divisor E over X. An *lc place*

of (X, B) is a prime divisor E over X such that $a(E, X, B) = 0$. An *lc center* of (X, B) is the center of an lc place of (X, B) on X.

Definition 2.2. We use the $\mathbb{R}$-ideals $\mathfrak{a} = \prod_j \mathfrak{a}_j^{r_j}$, the log discrepancies $a(E, X, B, \mathfrak{a})$ and the minimal log discrepancies $\mathrm{mld}(X \ni \eta, B, \mathfrak{a})$ at points $\eta \in X$ of [Liu26a, Definition 2.2], the convention $B \in \Gamma$ of [Liu26a, Definition 2.3], the set Γ_+ of finite sums of elements of a set $\Gamma \subset \mathbb{R}$ of [Liu26a, Definition 2.9], which satisfies the DCC if $\Gamma \subset [0, 1]$ does [Liu26a, Lemma 2.10(1)], the lc thresholds $\mathrm{lct}(X, \Delta; M)$ and $\mathrm{lct}(X, \Delta; \mathfrak{a})$ and their local versions $\mathrm{lct}(X \ni x, \Delta; M)$ and $\mathrm{lct}(X \ni x, \Delta; \mathfrak{a})$ of [Liu26b, Definition 2.11], with Δ omitted if $\Delta = 0$, and the pairs on $\operatorname{Spec} \Bbbk[[x_1, \dots, x_n]]$ for a field $\Bbbk \supset \mathbb{C}$ and their lc and klt properties in the sense of [dFEM11, Section 2]. For a finite group G acting faithfully on a normal quasi-projective variety X and a G-invariant $\mathbb{R}$-divisor B with $K_X + B$ $\mathbb{R}$-Cartier, $(X, B)/G = (X/G, B_{X/G})$ is the *crepant quotient*, where $B_{X/G}$ is the $\mathbb{R}$-divisor on X/G with $\pi^*(K_{X/G} + B_{X/G}) = K_X + B$ for the quotient morphism $\pi : X \to X/G$, and for a prime divisor F over X, $\mathrm{Stab}_G(F)$ and $I_G(F)$ are the *decomposition group* and the *inertia group* of F, and $e_F := |I_G(F)|$ [Liu26a, Definition 2.5]. For a dominant morphism $\phi : V \to W$ of normal integral schemes, a prime divisor E over V and a prime divisor F over W such that $\mathrm{mult}_E(\phi^*\zeta) = \rho \,\mathrm{mult}_F \zeta$ for a positive integer ρ and any non-zero rational function ζ on W, we call F the *restriction* of E and ρ the *ramification index* of E over F. For a quotient morphism ϕ, F is the prime divisor under E of [Liu26a, Definition 2.7]. For a point $x \in X$ fixed by G, not necessarily closed, and an $\mathbb{R}$-ideal $\mathfrak{a}$ with G-invariant factors,

$$\mathrm{mld}((X \ni x, B, \mathfrak{a})/G) := \inf_F \frac{a(F, X, B, \mathfrak{a})}{e_F},$$

where F runs over the prime divisors over X with $\mathrm{center}_X F = \overline{\{x\}}$ [Liu26b, Definition 2.4]. As [Liu26b, Definition 2.7] does for labelled boundaries, we call it the *quotient value* of $(X \ni x, B, \mathfrak{a})$ and G. For a point $x \in X$ not necessarily fixed by G, the infimum in the display defines $\mathrm{mld}((X \ni x, B, \mathfrak{a})/G)$, and we call it the *quotient value at the image of* x. For a closed point x fixed by G, the infimum of $a(F, X, B, \mathfrak{a})/e_F$ over the formal prime divisors F of [Liu26a, Definition 3.2] with center the closed point of the completion defines the quotient value on the completion at x over a field $\Bbbk \supset \mathbb{C}$. For a threefold germ $(W \ni o)$ as in [Liu26a, Definition-Lemma 11.1], we use the *quadratic rank* of $(W \ni o)$, the rank of the quadratic part f_2 of f for a presentation $\widehat{\mathcal{O}}_{W,o} \cong \mathbb{C}[[x_1, \dots, x_4]]/(f)$ with $f \in (x_1, \dots, x_4)^2$, and, if the quadratic rank is 1, the *residual cubic* of $(W \ni o)$, the restriction of the cubic part of f to the kernel of f_2 [Liu26a, Definition-Lemma 11.1].

Definition 2.3. We use the *Gauss extension* v_c of a valuation v with value c on an added variable [Liu26a, Definition-Proposition 2.11], the semi-invariant functions and their characters χ_h [Liu26a, Definition 2.14], the torus $\mathbb{T}^d$, the lattice N_H of a subgroup $H \subset \mathbb{T}^d$, the cyclic groups $\mu_r(a_1, \dots, a_d)$, the 1-dimensional subtori $\mathbf{T}_{(w)}$ and the eigenparameters at a fixed point [Liu26a, Definition 2.15], and the w-weight wt_w, the lowest w-weight part in_w and the lattice Newton minima $\psi_{\min}(H, J, c)$ [Liu26a, Definition 2.16]. As in [Liu26b, Definition 2.6], we identify $\mathbb{T}^d = \mathbb{R}^d/\mathbb{Z}^d$ with $(S^1)^d \subset (\mathbb{C}^*)^d$ through

$$\rho(u) \to (e^{2\pi\sqrt{-1}u_1}, \dots, e^{2\pi\sqrt{-1}u_d}),$$

where $\rho : \mathbb{R}^d \to \mathbb{T}^d$ is the quotient map, so that a finite diagonal group $G \subset (\mathbb{C}^*)^d$ is regarded as a subgroup of $\mathbb{T}^d$, and we also write $\mathbf{T}_{(w)}$ for $\rho(\mathbb{R}w)$.

2.2. **Marks, limits and circles.**

Definition 2.4 (cf. [Liu26b, Definition 2.7])**.** Let W be a normal variety or a formal germ, that is, the spectrum of a complete normal local ring containing $\mathbb{C}$, $o \in W$ a closed point, and G a finite group acting faithfully on W and fixing o. A *mark* on $(W \ni o)$ is a non-zero regular G-semi-invariant function h on a G-invariant open neighborhood of o, or a non-zero G-semi-invariant element of $\widehat{\mathcal{O}}_{W,o}$ when W is a formal germ (Definition 2.2). It may be

a unit, and $\mathrm{div}(h)$ need not be reduced. A *labelled boundary* on $(W \ni o)$ is an $\mathbb{R}$-divisor $D = \sum_{j=1}^{n} c_j \,\mathrm{div}(h_j)$, where the h_j are marks and the *labels* c_j are non-negative real numbers. For a labelled boundary D, or more generally a G-invariant $\mathbb{R}$-divisor D near o, with $K_W + D$ $\mathbb{R}$-Cartier near o, the number $\mathrm{mld}((W \ni o, D)/G)$ is the quotient value of $(W \ni o, D)$ and G (Definition 2.2), which for a formal germ W is the infimum of $a(F, W, D)/e_F$ over the prime divisors F with center o on normal schemes proper and birational over W, with log discrepancies and inertia groups as in [Liu26a, Definition 3.2], and a prime divisor F over W with center o and $a(F, W, D)/e_F = \mathrm{mld}((W \ni o, D)/G)$ is a *computing lift*.

Definition 2.5 (cf. [Liu26b, Construction-Proposition 4.1(a)])**.** Let $H \subset GL_n(\mathbb{C})$ be a finite subgroup acting linearly on $\mathbb{A}^n$ and N a positive integer. For $i \geq 1$ let $(f_{i,1}, \dots, f_{i,N})$ be N-tuples of non-zero H-semi-invariant regular functions on $\mathbb{A}^n$, or more generally of non-zero H-semi-invariant elements of $\mathbb{C}[[x_1, \dots, x_n]]$, whose characters depend only on j and whose orders at 0 are bounded. For a positive integer d, the *d-jet* of such a tuple is the tuple of the parts of degree less than d, and the d-jets of the tuples $(f_{i,j})_j$ belong to the finite-dimensional affine space $\mathcal{J}_d$ of N-tuples $(p_1, \dots, p_N)$ of polynomials in $x_1, \dots, x_n$ with complex coefficients and of degree less than d such that $g^* p_j = \chi_j(g) p_j$ for any $g \in H$ and any j, where χ_j is the character of the $f_{i,j}$. We write $\mathrm{jet}_d(\cdot)$ for the d-jet of a tuple. For an N-tuple of H-semi-invariant power series over a field $\Bbbk \supset \mathbb{C}$ whose j-th entry has the character of the $f_{i,j}$ for any j, $\mathrm{jet}_d(\cdot)$ is a $\Bbbk$-point of $\mathcal{J}_d$. Let $\mathcal{I} \subset \mathbb{N}^+$ be an infinite subset and $\Bbbk \supset \mathbb{C}$ an algebraically closed field. An N-tuple $(f_{*,j})_j$ of non-zero H-semi-invariant power series $f_{*,j} \in \Bbbk[[x_1, \dots, x_n]]$ with the characters of the $f_{i,j}$ is a *compatible generic limit* of the tuples $(f_{i,j})_j$ along $\mathcal{I}$, with *coefficient field* $\Bbbk$, if for any positive integer d the d-jet of $(f_{*,j})_j$ maps to the generic point of an irreducible closed subset $Z_d \subset \mathcal{J}_d$, and any non-empty open subset of Z_d contains the d-jets of $(f_{i,j})_j$ for all but finitely many $i \in \mathcal{I}$.

Lemma 2.6. *Let H, N and the tuples $(f_{i,j})_j$ be as in Definition 2.5. Then the following hold.*

1. *There exist an infinite subset $\mathcal{I} \subset \mathbb{N}^+$ and a compatible generic limit of the tuples $(f_{i,j})_j$ along $\mathcal{I}$* [Liu26b, Construction-Proposition 4.1(a)].
2. *Let $(f_{*,j})_j$ be a compatible generic limit of the tuples $(f_{i,j})_j$ along an infinite subset $\mathcal{I} \subset \mathbb{N}^+$, with coefficient field $\Bbbk$. Then for any infinite subset $\mathcal{I}' \subset \mathcal{I}$ and any subset $J \subset \{1, \dots, N\}$, $(f_{*,j})_{j \in J}$ is a compatible generic limit of the tuples $(f_{i,j})_{j \in J}$ along $\mathcal{I}'$, with coefficient field $\Bbbk$.*
3. *Notation as in (2). Let $(g_{i,1}, \dots, g_{i,N'})$ for $i \geq 1$ be tuples as in Definition 2.5, with N' in place of N. Then there exist an infinite subset $\mathcal{I}' \subset \mathcal{I}$, a compatible generic limit $((f'_{*,j})_j, (g_{*,k})_k)$ of the tuples $((f_{i,j})_j, (g_{i,k})_k)$ along $\mathcal{I}'$, with coefficient field $\Bbbk'$, and an embedding $\Bbbk \to \Bbbk'$ over $\mathbb{C}$ under which $f_{*,j}$ maps to $f'_{*,j}$ for any j.*
4. *Notation as in (2). Let $g_{*,1}, \dots, g_{*,N'} \in \Bbbk[[x_1, \dots, x_n]]$ be non-zero H-semi-invariant power series. Then there exist an infinite subset $\mathcal{I}' \subset \mathcal{I}$ and, for $i \in \mathcal{I}'$, non-zero H-semi-invariant polynomials $g_{i,1}, \dots, g_{i,N'}$ with the characters of $g_{*,1}, \dots, g_{*,N'}$ such that $((f_{*,j})_j, (g_{*,k})_k)$ is a compatible generic limit of the tuples $((f_{i,j})_j, (g_{i,k})_k)$ along $\mathcal{I}'$, with coefficient field $\Bbbk$. In particular, any polynomial identity with complex coefficients among finitely many coefficients of the $f_{*,j}$ and the $g_{*,k}$ holds for the corresponding coefficients of the $f_{i,j}$ and the $g_{i,k}$ for all but finitely many $i \in \mathcal{I}'$.*

Proof. (1) If all the $f_{i,j}$ are regular functions on $\mathbb{A}^n$, then (1) is [Liu26b, Construction-Proposition 4.1(a)]. In general, since H acts linearly on $\mathbb{A}^n$, the part $\widetilde{f}_{i,j}$ of $f_{i,j}$ of degree less than $n_0 + i$, where n_0 is a positive integer with $n_0 > \mathrm{mult}_0 f_{i,j}$ for any i and j, is a non-zero H-semi-invariant polynomial with the character of $f_{i,j}$. For any positive integer d,

$$\mathrm{jet}_d((\widetilde{f}_{i,j})_j) = \mathrm{jet}_d((f_{i,j})_j) \quad \text{for any } i \geq d.$$

Hence any compatible generic limit of the tuples $(\widetilde{f}_{i,j})_j$ along an infinite subset $\mathcal{I} \subset \mathbb{N}^+$, which exists by the polynomial case, is a compatible generic limit of the tuples $(f_{i,j})_j$ along $\mathcal{I}$.

(2) Passing to $\mathcal{I}' \subset \mathcal{I}$ preserves Definition 2.5, and, for $Z_d \subset \mathcal{J}_d$ as in Definition 2.5, the projection q to the entries in J maps $\mathrm{jet}_d((f_{*,j})_j)$ to the generic point of $\overline{q(Z_d)}$, and the preimage under $q|_{Z_d}$ of a non-empty open subset of $\overline{q(Z_d)}$ is non-empty, so (2) holds with $\overline{q(Z_d)}$ in place of Z_d.

(3) Let $\mathcal{J}'_d$ be the affine space of d-jets of the tuples $(g_{i,k})_k$, so that $\mathcal{J}_d \times \mathcal{J}'_d$ is the affine space of d-jets of the tuples $((f_{i,j})_j, (g_{i,k})_k)$. Indexing $\mathcal{I}$ increasingly by $\mathbb{N}^+$, (1) for the tuples $((f_{i,j})_j, (g_{i,k})_k)$ with $i \in \mathcal{I}$ implies that there exist an infinite subset $\mathcal{I}' \subset \mathcal{I}$ and a compatible generic limit $((f'_{*,j})_j, (g_{*,k})_k)$ of the tuples $((f_{i,j})_j, (g_{i,k})_k)$ along $\mathcal{I}'$, with coefficient field $\Bbbk'$. Let $Z_d \subset \mathcal{J}_d$ and $Z'_d \subset \mathcal{J}_d \times \mathcal{J}'_d$ be the irreducible closed subsets of Definition 2.5 for $(f_{*,j})_j$ and for $((f'_{*,j})_j, (g_{*,k})_k)$ respectively, let η_d be the generic point of Z_d, and let $Y_d \subset \mathcal{J}_d$ be the closure of the image of Z'_d.

***Step 1**: $Y_d = Z_d$, so $\mathrm{jet}_d((f_{*,j})_j)$ and $\mathrm{jet}_d((f'_{*,j})_j)$ both map to η_d.* By (2) and its proof, $(f'_{*,j})_j$ is a compatible generic limit of the tuples $(f_{i,j})_j$ along $\mathcal{I}'$ with Y_d in place of Z_d, and $(f_{*,j})_j$ is a compatible generic limit of the tuples $(f_{i,j})_j$ along $\mathcal{I}'$ with Z_d. Taking the open subsets Y_d and Z_d in Definition 2.5, we have

$$\mathrm{jet}_d((f_{i,j})_j) \in Y_d \cap Z_d \quad \text{for all but finitely many } i \in \mathcal{I}'.$$

If $Y_d \backslash Z_d \neq \emptyset$, then $Y_d \backslash Z_d$ is a non-empty open subset of Y_d that contains $\mathrm{jet}_d((f_{i,j})_j)$ for only finitely many $i \in \mathcal{I}'$, which contradicts Definition 2.5 since $\mathcal{I}'$ is infinite. If $Z_d \backslash Y_d \neq \emptyset$, then $Z_d \backslash Y_d$ is a non-empty open subset of Z_d that contains $\mathrm{jet}_d((f_{i,j})_j)$ for only finitely many $i \in \mathcal{I}'$, which again contradicts Definition 2.5. Thus $Y_d = Z_d$.

***Step 2**: the embedding.* The coordinates of $\mathcal{J}_d$ restrict to generators of $\mathbb{C}(Z_d)$ over $\mathbb{C}$. By **Step 1**, sending each coordinate of $\mathcal{J}_d$ to the corresponding coefficient of $\mathrm{jet}_d((f_{*,j})_j)$, resp. of $\mathrm{jet}_d((f'_{*,j})_j)$, defines an embedding $\sigma_d : \mathbb{C}(Z_d) \to \Bbbk$, resp. $\sigma'_d : \mathbb{C}(Z_d) \to \Bbbk'$, over $\mathbb{C}$. The truncation $\mathcal{J}_{d+1} \to \mathcal{J}_d$ maps η_{d+1} to η_d, hence induces an embedding $\mathbb{C}(Z_d) \to \mathbb{C}(Z_{d+1})$, and σ_{d+1} and σ'_{d+1} restrict to σ_d and σ'_d. Let $\mathbb{K} := \varinjlim_d \mathbb{C}(Z_d)$ with the induced embeddings $\sigma : \mathbb{K} \to \Bbbk$ and $\sigma' : \mathbb{K} \to \Bbbk'$. Since $\Bbbk \otimes_{\mathbb{K}} \Bbbk' \neq 0$, it has a prime ideal $\mathfrak{p}$. Let $\Bbbk''$ be an algebraic closure of the residue field of $\mathfrak{p}$, and let $\iota : \Bbbk \to \Bbbk''$ and $\iota' : \Bbbk' \to \Bbbk''$ be the induced embeddings, so that $\iota \circ \sigma = \iota' \circ \sigma'$. Since, for any d and any coordinate ξ of $\mathcal{J}_d$ with image $\bar{\xi} \in \mathbb{K}$, the coefficients of $\mathrm{jet}_d((f_{*,j})_j)$ and of $\mathrm{jet}_d((f'_{*,j})_j)$ at ξ are $\sigma(\bar{\xi})$ and $\sigma'(\bar{\xi})$,

$$\iota(f_{*,j}) = \iota'(f'_{*,j}) \quad \text{for any } j.$$

For any d, the d-jet of $((\iota'(f'_{*,j}))_j, (\iota'(g_{*,k}))_k)$ is the image under ι' of the d-jet of $((f'_{*,j})_j, (g_{*,k})_k)$, so both map to the generic point of Z'_d, and $((\iota'(f'_{*,j}))_j, (\iota'(g_{*,k}))_k)$ is a compatible generic limit of the tuples $((f_{i,j})_j, (g_{i,k})_k)$ along $\mathcal{I}'$ with coefficient field $\Bbbk''$. Hence the field $\Bbbk''$, the compatible generic limit $((\iota'(f'_{*,j}))_j, (\iota'(g_{*,k}))_k)$ and the embedding ι satisfy (3).

(4) For a positive integer d, let $\mathcal{J}'_d$ be the affine space of N'-tuples of polynomials of degree less than d whose k-th entry is 0 or H-semi-invariant with the character of $g_{*,k}$. Since H acts linearly on $\mathbb{A}^n$, $\gamma_d := \mathrm{jet}_d(((f_{*,j})_j, (g_{*,k})_k))$ is a $\Bbbk$-point of $\mathcal{J}_d \times \mathcal{J}'_d$. Let $Z_d \subset \mathcal{J}_d$ be as in Definition 2.5, let $Z'_d \subset \mathcal{J}_d \times \mathcal{J}'_d$ be the closure of the image of γ_d, which is irreducible with generic point the image of γ_d, and for $d \leq d'$ let

$$\rho_d : \mathcal{J}_d \times \mathcal{J}'_d \to \mathcal{J}_d \quad \text{and} \quad \pi_{d',d} : \mathcal{J}_{d'} \times \mathcal{J}'_{d'} \to \mathcal{J}_d \times \mathcal{J}'_d$$

be the projection and the truncation. Since $\rho_d(\gamma_d) = \mathrm{jet}_d((f_{*,j})_j)$ and $\pi_{d',d}(\gamma_{d'}) = \gamma_d$, the morphisms ρ_d and $\pi_{d',d}$ restrict to dominant morphisms $Z'_d \to Z_d$ and $Z'_{d'} \to Z'_d$.

We choose inductively integers $i_1 < i_2 < \cdots$ in $\mathcal{I}$ and closed points $y_m \in Z'_m$ with

$$\rho_m(y_m) = \mathrm{jet}_m((f_{i_m,j})_j).$$

Let $\mathbb{K}_0 := \mathbb{Q}$ and

$$\mathbb{K}_m := \mathbb{K}_{m-1}(\text{coordinates of } y_m) \subset \mathbb{C}.$$

The set Φ_m of pairs (d,ϕ), where d is a positive integer and ϕ is a polynomial with coefficients in $\mathbb{K}_m$ in the coordinates of $\mathcal{J}_d \times \mathcal{J}'_d$ not vanishing identically on Z'_d, is countable. Fix an enumeration $(d_{m,r},\phi_{m,r})_{r\geq 1}$ of Φ_m. At step m, let $\mathcal{A}_m$ be the finite set of pairs (m',r) with $m'<m$, $r\leq m$, and $d_{m',r}\leq m$, and let

$$\mathcal{V}_m := Z'_m \setminus \bigcup_{(m',r)\in\mathcal{A}_m} \big(\phi_{m',r}\circ \pi_{m,d_{m',r}} = 0\big).$$

Since Z'_m is irreducible and each $\pi_{m,d_{m',r}}|_{Z'_m}$ is dominant onto $Z'_{d_{m',r}}$, $\mathcal{V}_m$ is a non-empty open subset of Z'_m, so by Chevalley's theorem $\rho_m(\mathcal{V}_m)$ contains a non-empty open subset $U_m\subset Z_m$. By Definition 2.5 there exists $i_m\in\mathcal{I}$ with $i_m>i_{m-1}$, where $i_0:=0$, and $\mathrm{jet}_m((f_{i_m,j})_j)\in U_m$. Take a closed point $y_m\in\mathcal{V}_m$ with $\rho_m(y_m)=\mathrm{jet}_m((f_{i_m,j})_j)$, and let $(g_{i_m,k})_k$ be the $\mathcal{J}'_m$-component of y_m. Then

$$\mathrm{jet}_d(((f_{i_m,j})_j,(g_{i_m,k})_k)) = \pi_{m,d}(y_m)\in Z'_d \quad \text{for any } d\leq m.$$

We show that for any positive integer d and any non-empty open subset $\mathcal{V}\subset Z'_d$, $\pi_{m,d}(y_m)\in\mathcal{V}$ for all but finitely many $m\geq d$. Otherwise $\pi_{m,d}(y_m)\in T:=Z'_d\setminus\mathcal{V}$ for m in an infinite set $\{m_1<m_2<\cdots\}$ of integers $\geq d$. Take a polynomial ψ that vanishes on T but not on Z'_d, let $\mathcal{P}$ be the finite-dimensional space of complex polynomials of degree at most $\deg\psi$ in the coordinates of $\mathcal{J}_d\times\mathcal{J}'_d$, and let $\mathcal{P}_a\subset\mathcal{P}$ be the subspace of those vanishing at $\pi_{m_b,d}(y_{m_b})$ for any $b\leq a$. The descending chain $(\mathcal{P}_a)_a$ stabilizes at some a_0, and $\psi\in\mathcal{P}_{a_0}$. Since $\mathcal{P}_{a_0}$ is the solution space of a linear system with coefficients in $\mathbb{K}_{m_{a_0}}$, it has a basis of polynomials with coefficients in $\mathbb{K}_{m_{a_0}}$, and one of them, ϕ, does not vanish identically on Z'_d. Then $(d,\phi)=(d_{m_{a_0},r},\phi_{m_{a_0},r})$ for some r, so $(m_{a_0},r)\in\mathcal{A}_{m_b}$ and $\phi(\pi_{m_b,d}(y_{m_b}))\neq 0$ for any $m_b>\max\{m_{a_0},r\}$, while $\phi\in\mathcal{P}_a$ for any a vanishes at $\pi_{m_b,d}(y_{m_b})$, a contradiction.

Let d_0 be an integer with $d_0>\mathrm{mult}_0\, g_{*,k}$ for any k. The open subset of Z'_{d_0} where all the $\mathcal{J}'_{d_0}$-entries are non-zero contains the image of γ_{d_0}, so $g_{i_m,k}\neq 0$ and $\mathrm{mult}_0\, g_{i_m,k}<d_0$ for any k and all but finitely many m. Let $\mathcal{I}'$ be the set of these i_m. Then $((f_{*,j})_j,(g_{*,k})_k)$ is a compatible generic limit of the tuples $((f_{i,j})_j,(g_{i,k})_k)$ along $\mathcal{I}'$, with coefficient field $\Bbbk$ and with Z'_d in place of Z_d. Finally, a polynomial Θ with complex coefficients in the coordinates of $\mathcal{J}_d\times\mathcal{J}'_d$ with $\Theta(\gamma_d)=0$ vanishes at the generic point of Z'_d, hence on Z'_d, which contains $\mathrm{jet}_d(((f_{i_m,j})_j,(g_{i_m,k})_k))$ for any $m\geq d$. □

Lemma 2.7. *Notation as in Definition 2.5, with a compatible generic limit $(f_{*,j})_j$ of the tuples $(f_{i,j})_j$ along $\mathcal{I}$ and coefficient field $\Bbbk$. Let d be a positive integer, $m\in\mathbb{N}$, $\lambda\in\Bbbk^m$, and $\Phi_1,\dots,\Phi_r,\Psi$ polynomials with complex coefficients in the coordinates of $\mathcal{J}_d$ and m further variables, where $\Bbbk^0$ and $\mathbb{C}^0$ are points. Assume that*

$$\Phi_k(\mathrm{jet}_d((f_{*,j})_j),\lambda)=0 \text{ for any } k \quad \text{and} \quad \Psi(\mathrm{jet}_d((f_{*,j})_j),\lambda)\neq 0.$$

Then for all but finitely many $i\in\mathcal{I}$ there exists $\lambda_i\in\mathbb{C}^m$ with

$$\Phi_k(\mathrm{jet}_d((f_{i,j})_j),\lambda_i)=0 \text{ for any } k \quad \text{and} \quad \Psi(\mathrm{jet}_d((f_{i,j})_j),\lambda_i)\neq 0.$$

Proof. Let $Z_d\subset\mathcal{J}_d$ be as in Definition 2.5, and let $Z'\subset\mathcal{J}_d\times\mathbb{A}^m$ be the closure of the image of the $\Bbbk$-point $(\mathrm{jet}_d((f_{*,j})_j),\lambda)$. Then Z' is irreducible with generic point η' the image of the $\Bbbk$-point, and the projection $Z'\to\mathcal{J}_d$ maps η' to the generic point of Z_d, so it induces a dominant morphism $Z'\to Z_d$. Since the Φ_k vanish at η', they vanish on Z'. Since Ψ does not vanish at η',

$$Z'_0 := Z'\cap(\Psi\neq 0)$$

is a dense open subset of Z'. By Chevalley's theorem, the image of Z'_0 in Z_d is constructible and dense, so it contains a non-empty open subset $U\subset Z_d$. By Definition 2.5, $\mathrm{jet}_d((f_{i,j})_j)\in U$ for all but finitely many $i\in\mathcal{I}$. For such i, the fiber of $Z'_0\to Z_d$ over the closed point $\mathrm{jet}_d((f_{i,j})_j)$ is non-empty, so it contains a closed point $(\mathrm{jet}_d((f_{i,j})_j),\lambda_i)$ with $\lambda_i\in\mathbb{C}^m$. The lemma follows. □

Lemma 2.8. *Notation as in Definition 2.5, with a compatible generic limit $(f_{*,j})_j$ of the tuples $(f_{i,j})_j$ along $\mathcal{I}$ and coefficient field $\Bbbk$. Let d be a positive integer and $\Sigma \subset \mathcal{J}_d$ a constructible subset. Then the following are equivalent.*

(1) $\mathrm{jet}_d((f_{*,j})_j) \in \Sigma$.
(2) $\mathrm{jet}_d((f_{i,j})_j) \in \Sigma$ *for infinitely many* $i \in \mathcal{I}$.
(3) $\mathrm{jet}_d((f_{i,j})_j) \in \Sigma$ *for all but finitely many* $i \in \mathcal{I}$.

Proof. Let $Z_d \subset \mathcal{J}_d$ be as in Definition 2.5. Since Z_d is irreducible and $\Sigma \cap Z_d$ is constructible, $\Sigma \cap Z_d$ contains the generic point of Z_d, to which $\mathrm{jet}_d((f_{*,j})_j)$ maps, if and only if it contains a non-empty open subset of Z_d. Thus (1) implies (3) by Definition 2.5, and (3) implies (2) since $\mathcal{I}$ is infinite. If (1) fails, then the closure of $\Sigma \cap Z_d$ is a proper closed subset of Z_d, so by Definition 2.5 its complement in Z_d, which is disjoint from Σ, contains $\mathrm{jet}_d((f_{i,j})_j)$ for all but finitely many $i \in \mathcal{I}$, and (2) fails. □

Definition 2.9. For a pair (X, B) with B $\mathbb{R}$-Cartier and a closed subset $V \subset X$, a *margin* of (X, B) near V is a real number $\theta > 1$ such that $(X, \theta B)$ is lc near V [Liu26b, Definition 5.2]. For a primitive vector w, a finite diagonal group G, a positive integer e_0 and a labelled boundary D as in [Liu26b, Set-up 5.1], we use the *primitive circle* $\mathbf{T} = \mathbf{T}_{(w)}$, the group $R = G \cap \mathbf{T}$ with $r = |R|$, which satisfies $[G : R] \leq e_0$ by [Liu26b, Set-up 5.1(3)], the $\mathbf{T}$-weights, the *principal blocks*, the *mass* and the *adjusted mass* of D along $\mathbf{T}$ of [Liu26b, Set-up 5.1]. For positive real numbers c_j and a compatible generic limit $(f_{*,j})_j$ over $\Bbbk$ of tuples of functions on $\mathbb{A}^2$, the *floors* of the pair

$$\Big(\operatorname{Spec} \Bbbk[[x_1, x_2]], \sum_j c_j \operatorname{div}(f_{*,j})\Big)$$

are the curves of [Liu26b, Definition-Proposition 5.7]. An *augmentation* is an $\mathbb{R}$-divisor $\lambda \operatorname{div}(U)$ as in [Liu26b, Lemma 5.8]. We write $W_N := (xy + z^2 + t^N = 0) \subset \mathbb{A}^4$ for an integer $N \geq 2$ as in [Liu26b, Theorem 7.2] and $W_0 := (xy + z^2 = 0) \subset \mathbb{A}^4$ as in [Liu26b, Proposition 7.10], and we use the types cD and cE of [Liu26b, Definition 9.1] and the classes of residual cubics of [Liu26b, Definition-Lemma 6.4].

2.3. Quotients, completions and polynomial models.

Theorem 2.10 ([Liu26a, Theorem 8.3])**.** *Let $n \geq 2$ and $1 \leq c < n$ be integers, G a finite abelian group acting diagonally on $\mathbb{A}^n$, $f_1, \dots, f_c$ G-semi-invariant polynomials vanishing at 0 with common zero scheme X, $h_1, \dots, h_N$ G-semi-invariant polynomials with non-zero restrictions to X, and $b_1, \dots, b_N$ positive real numbers. Let*

$$B := \sum_j b_j \operatorname{div}(h_j|_X) \quad \text{and} \quad B_{\mathbb{A}^n} := \sum_i \operatorname{div}(f_i) + \sum_j b_j \operatorname{div}(h_j).$$

Assume that X is, near 0, a normal integral complete intersection of dimension $n - c$ with klt singularities, that G acts faithfully on X near 0, and that (X, B) is klt near 0. Then, for any subgroup $H \subset G$,

$$\operatorname{mld}((X \ni 0, B)/H) = \operatorname{mld}((\mathbb{A}^n \ni 0, B_{\mathbb{A}^n})/H) > 0.$$

In particular, any prime divisor over $\mathbb{A}^n/H$ with center the image of 0 has log discrepancy at least $\operatorname{mld}((X \ni 0, B)/H)$ with respect to $(\mathbb{A}^n, B_{\mathbb{A}^n})/H$.

Construction-Theorem 2.11 ([Liu26a, Construction-Theorem 7.2])**.** Let H be a finite abelian group and $n \in \{3, 4\}$, and assume the following.

(i) $(X \ni x)$ is a smooth threefold germ or an isolated Gorenstein terminal threefold germ, and H acts faithfully on X and fixes x.
(ii) X has an H-equivariant closed embedding into a smooth germ $(Y \ni x)$ of dimension n, with $Y = X$ if $n = 3$ and $X = (f = 0)$ for a regular H-semi-invariant function f if $n = 4$, and $x_1, \dots, x_n$ are H-semi-invariant regular parameters of Y at x.

(iii) $h_1, \dots, h_N$ are regular H-semi-invariant functions on Y, non-zero on X, $\beta_1, \dots, \beta_N$ are positive real numbers, and (X, B) is klt near x for $B := \sum_k \beta_k \operatorname{div}(h_k|_X)$ ($N = 0$ is allowed, in which case $B = 0$).
(iv) $\mathcal{P}$ is a finite set of prime divisors over X with center x.

Let $\widehat{h}_k$, and $\widehat{f}$ if $n = 4$, be the expansions of h_k and f in $\mathfrak{R} := \mathbb{C}[[x_1, \dots, x_n]]$ with the induced diagonal action of H, and for a power series g and a positive integer L, let $g_{<L}$ be the sum of the terms of g of degree less than L. Then there exist a faithful diagonal action of H on $\mathbb{A}^n$ in which the coordinates have the characters of the x_i, a germ $X_0 = \mathbb{A}^3$ at 0 if $n = 3$, or $X_0 = (f_0 = 0) \subset \mathbb{A}^4$ at 0 for a semi-invariant polynomial f_0 with the character of f such that X_0 is an isolated Gorenstein terminal germ if $n = 4$, and semi-invariant polynomials $h_{1,0}, \dots, h_{N,0}$ with the characters of $h_1, \dots, h_N$, non-zero on X_0, such that the following hold for

$$B_0 := \sum_k \beta_k \operatorname{div}(h_{k,0}|_{X_0}).$$

(1) (X_0, B_0) is klt near 0, H acts faithfully on X_0, and

$$\operatorname{mld}((X_0 \ni 0, B_0)/H') = \operatorname{mld}((X \ni x, B)/H')$$

for any subgroup $H' \subset H$.
(2) There exists an H-equivariant isomorphism between the completions of the local rings of X at x and of X_0 at 0, under which each $P \in \mathcal{P}$ corresponds to a prime divisor P_0 over X_0 with center 0 having isomorphic residue field and equal decomposition group and inertia group, with

$$a(P_0, X_0, 0) = a(P, X, 0) \quad \text{and} \quad \operatorname{mult}_{P_0}(h_{k,0}|_{X_0}) = \operatorname{mult}_P(h_k|_X) \text{ for any } k.$$

(3) For any H-equivariant automorphism ψ of $\mathfrak{R}$ and any H-invariant unit v of $\mathfrak{R}$, there exists a positive integer L_1' such that for any integer $L' \geq L_1'$ there exists a positive integer L_1 such that for any integer $L \geq L_1$, the polynomials $h_{k,0} := \psi(\widehat{h}_k)_{<L'}$ for any k and, if $n = 4$, $f_0 := (v\psi(\widehat{f}))_{<L}$ have the properties of $h_{k,0}$, f_0, and X_0 listed above and satisfy (1) and (2).

Lemma 2.12 ([Liu26a, Lemma 3.3]). *Let $\Bbbk \supset \mathbb{C}$ be a field, $(X \ni x)$ a germ of a normal variety over $\Bbbk$, and B an $\mathbb{R}$-divisor on X such that $K_X + B$ is $\mathbb{R}$-Cartier near x. Let H be a finite group acting faithfully on X, fixing x and preserving B. Let $\widehat{X} := \operatorname{Spec} \widehat{\mathcal{O}}_{X,x}$ with closed point $\widehat{x}$ and the induced action of H, as in* [Liu26a, Definition 3.2]*, and let $\widehat{B}$ be the pullback of B to $\widehat{X}$. Then $K_{\widehat{X}} + \widehat{B}$ is the pullback of $K_X + B$ and hence $\mathbb{R}$-Cartier, and the following hold.*

(1) The completion of the local ring of X/H at the image of x is $\widehat{\mathcal{O}}^H_{X,x}$.
(2) We have

$$\operatorname{mld}(\widehat{X} \ni \widehat{x}, \widehat{B}) = \operatorname{mld}(X \ni x, B) \quad \text{and} \quad \operatorname{mld}((\widehat{X} \ni \widehat{x}, \widehat{B})/H) = \operatorname{mld}((X \ni x, B)/H).$$

(3) Any formal prime divisor over $\widehat{X}$ with center $\widehat{x}$ restricts to a prime divisor over X with center x which has isomorphic residue field, equal log discrepancy with respect to $\widehat{B}$ and B, and equal decomposition group and inertia group, and any prime divisor over X with center x arises in this way.

Lemma 2.13 ([Liu26a, Lemma 9.3]). *Let $\psi : X' \to X$ be a morphism between normal varieties, $x' \in X'$ a closed point, $x := \psi(x')$, and B an $\mathbb{R}$-divisor on X with $K_X + B$ $\mathbb{R}$-Cartier. Assume that ψ is étale at x' and $\psi^{-1}(x) = \{x'\}$. Then for any prime divisor E over X with center x there exists a unique prime divisor E' over X' with center x' such that, for some projective birational morphisms $X_1' \to X'$ and $X_1 \to X$ from normal varieties on which E' and E are divisors and some morphism $\psi_1 : X_1' \to X_1$ over ψ, the morphism ψ_1 maps E' onto E with ramification index*

1, *and any prime divisor over* X' *with center* x' *arises in this way. Moreover,* E *and* E' *have isomorphic residue fields and*

$$a(E', X', \psi^* B) = a(E, X, B),$$

and if a finite group G *acts on* X *and* X'*, fixing* x *and* x'*, and* ψ *is* G*-equivariant, then*

$$\mathrm{Stab}_G(E') = \mathrm{Stab}_G(E) \quad \textit{and} \quad I_G(E') = I_G(E).$$

Construction-Theorem 2.14 ([Liu26a, Construction-Theorem 5.1])**.** Let $\alpha_1, \dots, \alpha_N$ be positive real numbers such that

$$1 \in \mathcal{S} := \Big\{ \sum_{j=1}^{N} \alpha_j c_j \;\Big|\; c_j \in \mathbb{N} \Big\},$$

and let

$$\Lambda_{\mathcal{S}} := \{1 - \sigma \mid \sigma \in \mathcal{S},\ 0 < \sigma \le 1\}, \quad \eta_{\mathcal{S}} := \min\{\sigma \in \mathcal{S} \mid \sigma > 1\} - 1,$$
$$R_{\mathcal{S}} := \lfloor 1/\eta_{\mathcal{S}} \rfloor, \quad \text{and} \quad M_{\mathcal{S}} := \mathrm{lcm}(1, \dots, R_{\mathcal{S}}).$$

Then $\Lambda_{\mathcal{S}} \subset [0, 1)$ is finite and $0 < \eta_{\mathcal{S}} \le 1$. Let (X, B) be a threefold pair, $x \in X$ a closed point, and H a finite group, and assume the following.

(i) (X, B) is klt near x.
(ii) $U \ni x$ is an open neighborhood on which H acts faithfully, fixing x and preserving B.
(iii) $a(E, X, B) \in \mathcal{S}$ for any prime divisor E over U.

Then, after shrinking to an H-invariant neighborhood of x, there exist an H-equivariant projective birational morphism $g : Y \to X$ and an $\mathbb{R}$-divisor $B_Y \ge 0$ on Y with the following properties.

(1) $K_Y + B_Y = g^*(K_X + B)$, the g-exceptional prime divisors are exactly the exceptional prime divisors E over X with $a(E, X, B) \le 1$, and $\mathrm{mult}_E B_Y = 1 - a(E, X, B)$ for them.
(2) $B_Y \in \Lambda_{\mathcal{S}}$, the pair (Y, B_Y) is terminal, and all log discrepancies of (Y, B_Y) belong to $\mathcal{S}$.
(3) Any H-invariant Weil divisor on Y is $\mathbb{Q}$-Cartier. Each reduced H-orbit sum of components of B_Y is $\mathbb{Q}$-Cartier, and B_Y is a non-negative real linear combination of such orbit sums.
(4) K_Y is $\mathbb{Q}$-Cartier, Y is terminal, and for any closed point $y \in Y$ we have

$$\mathrm{mld}(Y \ni y) \ge \mathrm{mld}(Y \ni y, B_Y) \ge 1 + \eta_{\mathcal{S}},$$

and the Cartier index of K_Y at y is $\le R_{\mathcal{S}}$. In particular $M_{\mathcal{S}} K_Y$ is Cartier.

Lemma 2.15 ([Liu26a, Lemma 2.6])**.** *Let* $\Bbbk \supset \mathbb{C}$ *be a field,* X *a normal quasi-projective variety over* $\Bbbk$*,* G *a finite group acting faithfully on* X *by* $\Bbbk$*-automorphisms, and* B *a* G*-invariant* $\mathbb{R}$*-divisor on* X *such that* $K_X + B$ *is* $\mathbb{R}$*-Cartier. Let* $\pi : X \to X/G$ *be the quotient morphism and* F *a prime divisor over* X*. Then there exist a unique prime divisor* E *over* X/G *and a unique positive integer* e *such that, for any* G*-equivariant projective birational morphism* $X' \to X$ *from a normal variety* X' *with a* G*-action on which* F *is a divisor (such* X' *exist), the induced finite morphism* $X' \to X'/G$ *maps* F *onto* E *with ramification index* e*. Moreover, the following hold.*

(0) $K_X + B = \pi^*(K_{X/G} + B_{X/G})$*,* $K_{X/G} + B_{X/G}$ *is* $\mathbb{R}$*-Cartier, and* $B_{X/G} \ge 0$ *if* $B \ge 0$*. In particular* $(X, B)/G$ *is a sub-pair, and it is a pair if* $B \ge 0$*.*
(1) $e = e_F$*, the group* $I_G(F)$ *is cyclic, and* $I_G(F)$ *acts faithfully on* $\mathfrak{m}_F/\mathfrak{m}_F^2$*, where* $\mathfrak{m}_F$ *is the maximal ideal of the valuation ring of* F*.*
(2) We have

$$a(F, X, B) = e_F \cdot a(E, X/G, B_{X/G}) \quad \textit{and} \quad \mathrm{center}_{X/G} E = \pi(\mathrm{center}_X F),$$

and

$$a(F, X, B, \pi^* \mathfrak{b}) = e_F \cdot a(E, X/G, B_{X/G}, \mathfrak{b})$$

for any $\mathbb{R}$*-ideal* $\mathfrak{b} = \prod_j \mathfrak{b}_j^{r_j}$ *on* X/G*, where* $\pi^* \mathfrak{b} := \prod_j (\mathfrak{b}_j \mathcal{O}_X)^{r_j}$*.*

(3) *For any prime divisor E' over X/G, there exist a G-equivariant projective birational morphism $X' \to X$ from a normal variety X' with a G-action and a prime divisor F' on X' such that $X' \to X'/G$ maps F' onto E'.*

(4) *If P is a prime divisor on X with $\operatorname{mult}_P B = b_P$ and ramification index e_P over $\pi(P)$, then*
$$\operatorname{mult}_{\pi(P)} B_{X/G} = 1 - \frac{1 - b_P}{e_P}.$$

(5) *If $G' \subset G$ is a normal subgroup, then G/G' acts faithfully on X/G', $B_{X/G'}$ is G/G'-invariant, and $(X, B)/G = ((X, B)/G')/(G/G')$, where X/G is identified with $(X/G')/(G/G')$.*

In particular, for a closed point $x \in X$ fixed by G,
$$\operatorname{mld}((X \ni x, B)/G) = \operatorname{mld}(X/G \ni \pi(x), B_{X/G}).$$

Theorem 2.16 ([Nak16b, Theorem 1.2(ii) and the paragraph after it])**.** *Let X be a $\mathbb{Q}$-Gorenstein normal variety which has a crepant resolution in the category of Deligne–Mumford stacks, T a variety, $\Delta \geq 0$ an $\mathbb{R}$-Cartier $\mathbb{R}$-divisor on X, and $x \in X$ a closed point. For an $\mathbb{R}$-ideal sheaf $\mathfrak{A}$ on $X \times T$, the function on the set of closed points of T given by $y \to \operatorname{mld}(X \ni x, \Delta, \mathfrak{A}_y)$, where $\mathfrak{A}_y$ is the restriction of $\mathfrak{A}$ to $X \times \{y\}$, is lower semi-continuous. The conclusion holds in particular if $X = X'/G$, where X' is a smooth variety and G a finite group acting on X' freely in codimension 1.*

2.4. **Adjunction and compatible generic limits.**

Theorem 2.17 ([Liu26b, Theorem 3.3])**.** *Let n, c be integers with $c \geq 1$ and $n - c \geq 2$, let G be a finite abelian group acting diagonally on $\mathbb{A}^n$, and let $f_1, \dots, f_c$ be G-semi-invariant polynomials vanishing at 0 whose common zero scheme X is, near 0, a normal integral complete intersection of dimension $n - c$ with klt singularities, on which G acts faithfully. Let $\mathfrak{a} = \prod_{j=1}^N \mathfrak{a}_j^{r_j}$ be an $\mathbb{R}$-ideal on $\mathbb{A}^n$ with $r_j > 0$ and each $\mathfrak{a}_j$ generated by G-semi-invariant polynomials, such that $\mathfrak{a}|_X$ is non-zero near 0. Let $H \subset G$ be a subgroup.*

(1) *If $(X, \mathfrak{a}|_X)$ is lc near 0, then*
$$\operatorname{mld}((X \ni 0, \mathfrak{a}|_X)/H) = \operatorname{mld}\Big(\Big(\mathbb{A}^n \ni 0, \sum_{i=1}^{c} \operatorname{div}(f_i), \mathfrak{a}\Big)/H\Big).$$

(2) *Let f_{c+1} be a G-semi-invariant polynomial such that $S := X \cap (f_{c+1} = 0)$ is, near 0, a normal integral complete intersection of dimension $n - c - 1$ in $\mathbb{A}^n$ with klt singularities, on which H acts faithfully, and such that $\mathfrak{a}|_S$ is non-zero near 0. If $(S, \mathfrak{a}|_S)$ is lc near 0, then*
$$\operatorname{mld}((S \ni 0, \mathfrak{a}|_S)/H) = \operatorname{mld}((X \ni 0, S, \mathfrak{a}|_X)/H) = \operatorname{mld}\Big(\Big(\mathbb{A}^n \ni 0, \sum_{i=1}^{c+1} \operatorname{div}(f_i), \mathfrak{a}\Big)/H\Big),$$
where S has coefficient one in the middle term.

(3) *Let ξ be the generic point of an H-invariant integral curve through 0 contained in X. If $(X, \mathfrak{a}|_X)$ is lc near ξ, then the equality of (1) holds with 0 replaced by ξ. If $n - c \geq 3$, $\overline{\{\xi\}} \subset S$, and $(S, \mathfrak{a}|_S)$ is lc near ξ, then the equalities of (2) hold with 0 replaced by ξ. Here the minimal log discrepancies are taken at the images of ξ in the quotients.*

Theorem 2.18 ([Liu26b, Theorem 4.2])**.** *Let H, N and the tuples $(f_{i,j})_j$ of regular functions on $\mathbb{A}^n$ be as in Definition 2.5, let $(f_{*,j})_j$ be a compatible generic limit of the tuples $(f_{i,j})_j$ along an infinite subset $\mathcal{I} \subset \mathbb{N}^+$, with coefficient field $\Bbbk$, and let $\mathcal{A}_\Bbbk := \operatorname{Spec} \Bbbk[[x_1, \dots, x_n]]$ with the induced action of H. Let $c_1, \dots, c_N$ be positive real numbers and $\mathfrak{c} = \prod_k \mathfrak{c}_k^{r_k}$ an $\mathbb{R}$-ideal on $\mathbb{A}^n$ with $r_k > 0$ and H-invariant ideals $\mathfrak{c}_k$ such that $(\mathbb{A}^n, \mathfrak{c})$ is klt near 0, write $\mathfrak{c}$ also for the pullback of $\mathfrak{c}$ to $\mathcal{A}_\Bbbk$, and let $C_i := \sum_j c_j \operatorname{div}(f_{i,j})$ and $C_* := \sum_j c_j \operatorname{div}(f_{*,j})$. Then there exists an infinite subset $\mathcal{I}' \subset \mathcal{I}$ such that the following hold for $i \in \mathcal{I}'$.*

(1) *The lc thresholds of C_i with respect to $(\mathbb{A}^n, \mathfrak{c})$ near 0 converge to the lc threshold of C_* with respect to $(\mathcal{A}_{\Bbbk}, \mathfrak{c})$. For any real number $\theta > 0$, $(\mathcal{A}_{\Bbbk}, \theta C_*, \mathfrak{c})$ is lc near 0 if and only if $(\mathbb{A}^n, \theta C_i, \mathfrak{c})$ is lc near 0 for all i. If $(\mathcal{A}_{\Bbbk}, \theta C_*, \mathfrak{c})$ is klt near 0, then there exists $\theta' > \theta$ such that $(\mathbb{A}^n, \theta' C_i, \mathfrak{c})$ is lc near 0 for all i.*

(2) *If $n = 2$, H is diagonal when $\mathfrak{c} \neq \mathcal{O}$, and $(\mathcal{A}_{\Bbbk}, C_*, \mathfrak{c})$ is lc near 0, then*

$$\operatorname{mld}((\mathbb{A}^2 \ni 0, C_i, \mathfrak{c})/H) = \operatorname{mld}((\mathcal{A}_{\Bbbk} \ni 0, C_*, \mathfrak{c})/H)$$

for any i.

(3) *For any sufficiently large integer d there exist a non-empty open subset $Z_d^\circ \subset Z_d$ containing the d-jets of $(f_{i,j})_j$ for all but finitely many $i \in \mathcal{I}'$, and an H-equivariant log resolution near 0 of the family of pairs defined by the d-jets in Z_d°, such that the following are constant along Z_d°:*

   (a) *The dual complex.*

   (b) *The discrepancies.*

   (c) *The multiplicities along the exceptional divisors of the divisor of each entry of the d-jets and of each ideal $\mathfrak{c}_k$.*

   (d) *The decomposition and inertia groups.*

   Moreover, for the points z of Z_d° with values in algebraically closed fields $k' \supset \mathbb{C}$, the irreducible components of the inverse image of 0, of the exceptional locus, and of the strict transforms of the zero loci of the entries of z and of the $\mathfrak{c}_k$ in the fiber over z of the log resolution can be indexed by one finite set, independent of z, so that the sets of indices of the components with non-empty intersection, the indices of the components mapping to 0, the log discrepancies of the components over $\mathbb{A}^n_{k'}$, and the multiplicities along the components of the divisor of each entry of z and of each $\mathfrak{c}_k$ do not depend on z.

Lemma 2.19 ([Liu26b, Lemma 4.14]). *Let n and N be positive integers and $\Lambda \subset \mathbb{R}_{>0}$ a DCC set. Then there exists a positive integer L, depending only on n, N, and Λ, such that the following holds. Let $\Bbbk \supset \mathbb{C}$ be an algebraically closed field, and $f_1, \dots, f_N$ and $f'_1, \dots, f'_N$ non-zero elements of $\Bbbk[[x_1, \dots, x_n]]$ with $f_j - f'_j \in (x_1, \dots, x_n)^L$, and $c_1, \dots, c_N \in \Lambda$. If $(\operatorname{Spec} \Bbbk[[x_1, \dots, x_n]], \prod_j (f_j)^{c_j})$ is lc, then so is $(\operatorname{Spec} \Bbbk[[x_1, \dots, x_n]], \prod_j (f'_j)^{c_j})$.*

Lemma 2.20 ([Liu26b, Lemma 4.8]). *Fix positive integers N and M. Let $(Y \ni y)$ be a normal germ with $\dim Y \geq 1$, B an $\mathbb{R}$-divisor on Y with $N(K_Y + B)$ Cartier near y, G a finite group acting faithfully on Y, fixing y and preserving B, and $(Z \ni z, \Delta) := (Y \ni y, B)/G$. Assume that $|I_G(F)| \leq M$ for any prime divisor F over Y with center y. Then*

$$a(E, Z, \Delta) \in (N \operatorname{lcm}(1, 2, \dots, M))^{-1}\mathbb{Z}$$

for any prime divisor E over Z with center z. If (Y, B) is lc near y, then $\operatorname{mld}((Y \ni y, B)/G)$ is attained and belongs to $(N \operatorname{lcm}(1, 2, \dots, M))^{-1}\mathbb{N}$.

2.5. **Surfaces, fixed threefolds and thresholds.**

Theorem 2.21 ([HL24, Theorem 1.2], [HL23, Theorem 1.2]). *Let $\Gamma \subset [0, 1]$ be a DCC set and $\Bbbk$ an algebraically closed field containing $\mathbb{C}$. Then the following hold.*

(1) *The set of $\operatorname{mld}(X \ni x, B)$, where (X, B) is an lc pair over $\Bbbk$ with $\dim X = 2$, $x \in X$ is a closed point and $B \in \Gamma$, satisfies the ACC.*

(2) *There exists a positive integer L depending only on Γ such that any lc surface germ $(X \ni x, B)$ over $\Bbbk$ with $B \in \Gamma$ has a prime divisor E over X with $\operatorname{center}_X E = \{x\}$, $a(E, X, B) = \operatorname{mld}(X \ni x, B)$, and $a(E, X, 0) \leq L$.*

Both [HL24] and [HL23] work over an algebraically closed field of arbitrary characteristic [HL24, Section 1], [HL23, Section 1].

Theorem 2.22 ([Kaw24, Theorems 1.1, 1.3 and 1.4]). *Let X be a normal threefold and $\Gamma \subset \mathbb{R}_{>0}$ a DCC set. Then the following hold.*

(1) *The set of* $\mathrm{mld}(X \ni \eta, B)$*, where* $\eta \in X$ *is a scheme-theoretic point and* (X, B) *is a pair with* $B \in \Gamma$*, satisfies the ACC.*
(2) *There exists a positive integer* L *depending only on* X *and* Γ *such that*

$$\mathrm{mld}(X \ni \eta, B, \mathfrak{a}) = \mathrm{mld}(X \ni \eta, B, \mathfrak{b})$$

for any scheme-theoretic point $\eta \in X$*, any pair* (X, B) *with* $B \in \Gamma$*, and any two* $\mathbb{R}$*-ideals* $\mathfrak{a} = \prod_j \mathfrak{a}_j^{r_j}$ *and* $\mathfrak{b} = \prod_j \mathfrak{b}_j^{r_j}$ *with* $r_j \in \Gamma$ *and*

$$\mathfrak{a}_j \mathcal{O}_{X,\eta} + \mathfrak{m}_\eta^L = \mathfrak{b}_j \mathcal{O}_{X,\eta} + \mathfrak{m}_\eta^L$$

for any j*.*
(3) *If moreover* X *is klt, then there exists a positive integer* L *depending only on* X *and* Γ *such that for any scheme-theoretic point* $\eta \in X$ *and any* $\mathbb{R}$*-Cartier* $\mathbb{R}$*-divisor* $B \geq 0$ *on* X *with* $B \in \Gamma$ *such that* (X, B) *is lc near* η*, there exists a prime divisor* E *over* X *which computes* $\mathrm{mld}(X \ni \eta, B)$ *and satisfies* $a(E, X, 0) \leq L$*.*

Theorem 2.23 ([HMX14, Theorem 1.1], [dFEM11, Propositions 2.11 and 2.19])**.** *The following hold.*

(1) *Let* n *be a positive integer, and* $\Gamma \subset [0, 1]$ *and* $\Gamma' \subset \mathbb{R}_{>0}$ *DCC sets. Then the set of log canonical thresholds* $\mathrm{lct}(X, \Delta; D)$*, where* (X, Δ) *is an lc pair of dimension* n *with* $\Delta \in \Gamma$ *and* D *is an* $\mathbb{R}$*-Cartier* $\mathbb{R}$*-divisor with* $D \in \Gamma'$*, satisfies the ACC.*
(2) *Let* $(X \ni x)$ *be a klt germ of dimension* n*,* $\epsilon > 0$*, and* $\mathfrak{a} = \prod_{i=1}^N \mathfrak{a}_i^{r_i}$ *and* $\mathfrak{b} = \prod_{i=1}^N \mathfrak{b}_i^{r_i}$ *two* $\mathbb{R}$*-ideals on* X*, or on* $\operatorname{Spec} \widehat{\mathcal{O}}_{X,x}$*, with* $r_i > 0$ *such that* $\mathrm{lct}(X \ni x; \mathfrak{a})$ *is finite. Let* L *be a positive integer with* $L \geq n/(\epsilon \min_i r_i)$*, and assume that* $\mathfrak{a}_i + \mathfrak{m}_x^L = \mathfrak{b}_i + \mathfrak{m}_x^L$ *for all* i*. Then*

$$|\mathrm{lct}(X \ni x; \mathfrak{a}) - \mathrm{lct}(X \ni x; \mathfrak{b})| \leq \epsilon.$$

3. Complements to the first two papers

In this section, we prove versions of results of [Liu26a] and [Liu26b] in the form used in later sections, which those papers do not contain.

3.1. Tangent data.

Lemma 3.1 (cf. [Liu26b, Theorem 6.2(2) and Lemma C.2])**.** *Let* $(W \ni o)$ *be an isolated cDV threefold germ singular at* o*, embedded as* $W = (f = 0)$ *in a smooth fourfold germ* $(\mathcal{X} \ni o)$*, and let* $\mathfrak{m}_o$ *be the maximal ideal of* $\mathcal{O}_{W,o}$*. Then the following hold.*

(1) *Assume that* $(W \ni o)$ *has quadratic rank* ≥ 2*. Then any irreducible component of the exceptional locus of the blow-up of* W *at* o *defines a prime divisor* E *over* W *with*

$$\mathrm{mult}_E \mathfrak{m}_o = 1 \quad \text{and} \quad a(E, W, 0) = 2.$$

If moreover $(W \ni o)$ *has quadratic rank* ≥ 3*, then the exceptional locus is irreducible.*
(2) *Assume that* $(W \ni o)$ *has quadratic rank* 1*. Then there exist regular parameters* x, y, z, t *of* $\mathcal{X}$ *at* o *in which the quadratic part of* f *is* cx^2 *for some* $c \in \mathbb{C}^*$*. For any such parameters, let* $g_3(y, z, t)$ *be the cubic part of* $f(0, y, z, t)$*, where* f *is expanded in* $\mathbb{C}[[x, y, z, t]]$*. Let* $w := (3, 2, 2, 2)$*, and let* wt_w *and* in_w *denote the* w*-weight and the lowest* w*-weight part of a power series in* x, y, z, t [Liu26a, Definition 2.16]*. Then the following hold.*
   (a) $g_3 \neq 0$*, and* g_3 *is the residual cubic of* $(W \ni o)$ [Liu26a, Definition-Lemma 11.1].
   (b) $\mathrm{in}_w(f) = cx^2 + g_3$*.*
   (c) *The strict transform of* W *in the weighted blow-up of* $\mathcal{X}$ *with weights* w *in* x, y, z, t *intersects the exceptional divisor in a single prime divisor* E_0*, and*

$$a(E_0, W, 0) = 3 \quad \text{and} \quad \mathrm{mult}_{E_0} \mathfrak{m}_o = 2.$$

If moreover a finite abelian group G *acts on* $\mathcal{X}$*, fixing* o*, and* f *is* G*-semi-invariant, then* x, y, z, t *can be taken* G*-semi-invariant.*

Proof. Let $\mathfrak{n}$ be the maximal ideal of $\mathcal{O}_{\mathcal{X},o}$. Since W is singular at o, $f \in \mathfrak{n}^2$. The quadratic part f_2 of f is the class of f in

$$\mathfrak{n}^2/\mathfrak{n}^3 = \mathrm{Sym}^2(\mathfrak{n}/\mathfrak{n}^2).$$

By [Liu26a, Definition-Lemma 11.1], the quadratic rank of $(W \ni o)$ is the rank of f_2. Since $(W \ni o)$ is an isolated cDV singularity, W is normal near o.

Proof of (1). Let $\pi : \mathcal{X}' \to \mathcal{X}$ be the blow-up of o, let F be the exceptional prime divisor of π, let $W' := \pi_*^{-1}W$, and let $\pi_W : W' \to W$ be the induced morphism, which is the blow-up of W at o. Let x_1, x_2, x_3, x_4 be regular parameters of $\mathcal{X}$ at o, so that $F \cong \mathbb{P}^3$ with homogeneous coordinates x_1, x_2, x_3, x_4, and regard f_2 as a quadratic form in x_1, x_2, x_3, x_4. On the chart of $\mathcal{X}'$ on which $\mathfrak{n}\mathcal{O}_{\mathcal{X}'} = (x_1)$, since $f - f_2 \in \mathfrak{n}^3$, we have

$$\pi^* f = x_1^2\Big(f_2\big(1, \tfrac{x_2}{x_1}, \tfrac{x_3}{x_1}, \tfrac{x_4}{x_1}\big) + x_1 r\Big)$$

for a regular function r, and similarly on the other three charts. Thus

$$\pi^* W = W' + 2F,$$

and the scheme-theoretic intersection $W' \cap F$ is the quadric

$$Q := (f_2 = 0) \subset \mathbb{P}^3.$$

In particular, the exceptional locus of π_W is Q.

Since f_2 has rank ≥ 2 (resp. ≥ 3), the quadric Q is reduced (resp. irreducible). This proves the last sentence of (1).

Possibly replacing x_1, x_2, x_3, x_4 with general linear combinations of them, we may assume that no irreducible component of Q is contained in the hyperplane $(x_1 = 0) \subset \mathbb{P}^3$. Let Q_1 be an irreducible component of Q with generic point η. Since $W' \cap F = Q$ is reduced and $F = (x_1 = 0)$ near η, W' is smooth at η with uniformizer x_1, so Q_1 defines a prime divisor E over W, and $\mathrm{mult}_E \mathfrak{m}_o = \mathrm{mult}_E(x_1|_W) = 1$ since x_j/x_1 is regular at η for any j.

Since $K_{\mathcal{X}'} = \pi^* K_{\mathcal{X}} + 3F$ and W, W' are Cartier divisors on the smooth varieties $\mathcal{X}$, $\mathcal{X}'$, adjunction implies

$$K_{W'} = (K_{\mathcal{X}'} + W')|_{W'} = \big(\pi^*(K_{\mathcal{X}} + W) + F\big)|_{W'} = \pi_W^* K_W + F|_{W'}.$$

Near η, $F|_{W'} = \mathrm{div}(x_1) = E$. Thus $a(E, W, 0) = 1 + 1 = 2$. This proves (1).

Proof of (2). Since the rank of f_2 is 1, $f_2 = c\overline{x}^2$ for some $c \in \mathbb{C}^*$ and some non-zero $\overline{x} \in \mathfrak{n}/\mathfrak{n}^2$. Lifting $\overline{x}$ to $x \in \mathfrak{n}$ and extending x to regular parameters x, y, z, t of $\mathcal{X}$ at o proves the first sentence of (2).

Let x, y, z, t be regular parameters of $\mathcal{X}$ at o in which $f_2 = cx^2$, so that $\widehat{\mathcal{O}}_{\mathcal{X},o} = \mathbb{C}[[x, y, z, t]]$. We write

$$f = f(0, y, z, t) + x\alpha(y, z, t) + x^2\beta(x, y, z, t) \tag{3.1}$$

with $\alpha \in \mathbb{C}[[y, z, t]]$ and $\beta \in \mathbb{C}[[x, y, z, t]]$. Since $f \in \mathfrak{n}^2$ and $f_2 = cx^2$,

$$f(0, y, z, t) \in (y, z, t)^3, \quad \alpha \in (y, z, t)^2, \quad \text{and} \quad \beta(0, 0, 0, 0) = c.$$

Thus $\mathrm{wt}_w(x\alpha) \geq 3 + 4 = 7$, $\mathrm{wt}_w(x^2(\beta - c)) \geq 6 + 2 = 8$, and $\mathrm{wt}_w(f(0, y, z, t) - g_3) \geq 8$, while $cx^2 + g_3$ is non-zero and homogeneous of w-weight 6. Hence

$$\mathrm{in}_w(f) = cx^2 + g_3.$$

This proves (2)(b). Since $\widehat{\mathcal{O}}_{W,o} = \mathbb{C}[[x, y, z, t]]/(f)$, the kernel of $f_2 = cx^2$ is $(x = 0)$, and the restriction of the cubic part of f to $(x = 0)$ is g_3, the cubic form g_3 is the residual cubic of $(W \ni o)$ [Liu26a, Definition-Lemma 11.1]. In particular, whether $g_3 = 0$ does not depend on the choice of x, y, z, t.

We show that $g_3 \neq 0$. Since $(W \ni o)$ is cDV, there exists $\ell \in \mathfrak{m}_o$ such that $(S \ni o)$, where $S := (\ell = 0) \subset W$, is a Du Val singularity. Let $\widetilde{\ell} \in \mathfrak{n}$ be a lift of ℓ. Since S has embedding dimension 3 at o and $f \in \mathfrak{n}^2$, $\widetilde{\ell} \notin \mathfrak{n}^2$. Since $(S \ni o)$ has multiplicity 2, the image of f in the regular

local ring $\widehat{\mathcal{O}}_{\mathcal{X},o}/(\widetilde{\ell})$ does not belong to the cube of the maximal ideal, so the restriction of $f_2 = cx^2$ to the hyperplane $(\overline{\ell} = 0)$ of the tangent space, where $\overline{\ell} \in \mathfrak{n}/\mathfrak{n}^2$ is the class of $\widetilde{\ell}$, is non-zero. Thus $\overline{x}$ and $\overline{\ell}$ are linearly independent. Since whether $g_3 = 0$ does not depend on the choice of x, y, z, t, we may assume that $t = \widetilde{\ell}$. Let $f_S := f(x, y, z, 0)$, so that $\widehat{\mathcal{O}}_{S,o} = \mathbb{C}[[x, y, z]]/(f_S)$. By (3.1), the quadratic part of f_S is cx^2, and the cubic part of $f_S(0, y, z)$ is $g_3(y, z, 0)$.

The quadratic rank of $(S \ni o)$, that is, the rank of the quadratic part cx^2 of f_S, is 1, and it does not depend on the presentation of $\widehat{\mathcal{O}}_{S,o}$ [Liu26a, Definition-Lemma 11.1], which covers presentations in any number $d \geq 3$ of variables, here $d = 3$. Since the normal forms $x'^2+y'^2+z'^{k+1}$ of the Du Val singularities of type A_k have quadratic rank ≥ 2, $(S \ni o)$ is of type D or E, that is, $\widehat{\mathcal{O}}_{S,o} \cong \mathbb{C}[[x', y', z']]/(\phi)$, where ϕ is one of

$$x'^2 + y'^2 z' + z'^{k-1} \ (k \geq 4), \quad x'^2 + y'^3 + z'^4, \quad x'^2 + y'^3 + y' z'^3, \quad \text{and} \quad x'^2 + y'^3 + z'^5.$$

In the presentation $\widehat{\mathcal{O}}_{S,o} \cong \mathbb{C}[[x', y', z']]/(\phi)$, the kernel of the quadratic part x'^2 of ϕ is $(x' = 0)$, so the residual cubic ϕ_3 of $(S \ni o)$ is the cubic part of $\phi(0, y', z')$:

$$\phi_3 = y'^2 z' + z'^3 \text{ if } \phi = x'^2 + y'^2 z' + z'^3, \quad \phi_3 = y'^2 z' \text{ if } \phi = x'^2 + y'^2 z' + z'^{k-1} \ (k \geq 5),$$

and $\phi_3 = y'^3$ in the three other cases. Thus $\phi_3 \neq 0$. In the presentation $\widehat{\mathcal{O}}_{S,o} = \mathbb{C}[[x, y, z]]/(f_S)$, the kernel of cx^2 is $(x = 0)$, so the residual cubic of $(S \ni o)$ is $g_3(y, z, 0)$. By [Liu26a, Definition-Lemma 11.1], $g_3(y, z, 0) = \lambda\phi_3\big(A(y, z)\big)$ for some $A \in GL_2(\mathbb{C})$ and $\lambda \in \mathbb{C}^*$, so $g_3(y, z, 0) \neq 0$. Thus $g_3 \neq 0$. This proves (2)(a).

We prove (2)(c). Since $g_3 \neq 0$ is homogeneous of degree 3, $cx^2 + g_3$ is irreducible in $\mathbb{C}[x, y, z, t]$: a factorization into two polynomials of degree 1 in x would make g_3 a constant multiple of the square of a polynomial in y, z, t, which is impossible since $g_3 \neq 0$ is homogeneous of odd degree 3, and a factor of degree 0 in x divides c. Moreover, x does not divide $cx^2 + g_3$ since $g_3 \neq 0$, and none of y, z, t divides $cx^2 + g_3$ since $c \neq 0$. Since W is normal near o and w is primitive, by [Liu26b, Lemma 4.12], the strict transform of W in the weighted blow-up of $\mathcal{X}$ with weights w in x, y, z, t intersects the exceptional divisor in a single prime divisor E_0,

$$a(E_0, W, 0) = 3 + 2 + 2 + 2 - \mathrm{wt}_w(f) = 9 - 6 = 3,$$

and for any $\Phi \in \mathcal{O}_{\mathcal{X},o}$,

$$\mathrm{mult}_{E_0}(\Phi|_W) \geq \mathrm{wt}_w(\Phi),$$

with equality if $cx^2 + g_3$ does not divide $\mathrm{in}_w(\Phi)$. Since any element of $\mathfrak{m}_o$ is the restriction of some $\Phi \in \mathfrak{n}$, and $\mathrm{wt}_w(\Phi) \geq 2$ for any $\Phi \in \mathfrak{n}$, we have $\mathrm{mult}_{E_0} \mathfrak{m}_o \geq 2$. Since $\mathrm{in}_w(y) = y$ is not divisible by $cx^2 + g_3$, $\mathrm{mult}_{E_0}(y|_W) = 2$. Thus $\mathrm{mult}_{E_0} \mathfrak{m}_o = 2$. This proves (2)(c).

Finally, if G is as in (2), then $\mathbb{C}\overline{x} \subset \mathfrak{n}/\mathfrak{n}^2$ is G-stable since $f_2 = c\overline{x}^2$ is G-semi-invariant, so, G being finite abelian, a G-eigenbasis $\overline{x}, \overline{y}, \overline{z}, \overline{t}$ of $\mathfrak{n}/\mathfrak{n}^2$ lifts to G-semi-invariant regular parameters x, y, z, t with $f_2 = cx^2$. The lemma follows. □

3.2. **Truncation of boundaries and contact determinacy.**

Lemma 3.2 (cf. [Liu26a, Lemma 7.1 and Construction-Theorem 7.2])**.**

(1) Let H be a finite abelian group acting faithfully and diagonally on $\mathbb{A}^n$, where $n \in \{3, 4\}$. Let $V_0 := \mathbb{A}^3$ if $n = 3$, and let $V_0 := (f_0 = 0) \subset \mathbb{A}^4$ if $n = 4$, where f_0 is an H-semi-invariant polynomial such that $(V_0 \ni 0)$ is an isolated Gorenstein terminal germ. Let $\widehat{V}_0$ be the completion of V_0 at 0, with closed point $\widehat{0}$. Let $g_1, \dots, g_N \in \mathbb{C}[[x_1, \dots, x_n]]$ be H-semi-invariant elements with non-zero restrictions to $\widehat{V}_0$, let $\beta_1, \dots, \beta_N$ be positive real numbers, and let

$$B := \sum_{k=1}^{N} \beta_k \,\mathrm{div}(g_k) \quad \text{on } \widehat{V}_0.$$

For $h \in \mathbb{C}[[x_1, \dots, x_n]]$ and a formal prime divisor P over $\widehat{V}_0$, we write $\mathrm{mult}_P h$ for the order along P of $h|_{\widehat{V}_0}$, with $\mathrm{mult}_P h := +\infty$ if $h|_{\widehat{V}_0} = 0$. Assume that H acts faithfully

on V_0 and $(\widehat{V}_0, B)$ is klt. Then for any real number γ there exists an integer L_γ satisfying the following. Let $L \geq L_\gamma$ be an integer, let $g'_1, \dots, g'_N$ be H-semi-invariant polynomials such that

$$\chi_{g'_k} = \chi_{g_k} \quad \text{and} \quad g'_k - g_k \in (x_1, \dots, x_n)^L \quad \text{for any } k,$$

and let $B' := \sum_{k=1}^N \beta_k \operatorname{div}(g'_k)$ on $\widehat{V}_0$. Then the following hold.

(a) $g'_k|_{V_0} \neq 0$ for any k, and $(V_0, \sum_{k=1}^N \beta_k \operatorname{div}(g'_k|_{V_0}))$ is klt near 0.

(b) For any subgroup $H' \subset H$ and any formal prime divisor P over $\widehat{V}_0$ with center $\widehat{0}$ such that $\operatorname{mult}_P g'_k \neq \operatorname{mult}_P g_k$ for some k, we have

$$\frac{a(P, \widehat{V}_0, B)}{|I_{H'}(P)|} > \gamma \quad \text{and} \quad \frac{a(P, \widehat{V}_0, B')}{|I_{H'}(P)|} > \gamma.$$

(c) Assume that $\gamma > \operatorname{mld}(\widehat{V}_0 \ni \widehat{0}, B)$. Then for any subgroup $H' \subset H$,

$$\operatorname{mld}((\widehat{V}_0 \ni \widehat{0}, B)/H') = \operatorname{mld}((\widehat{V}_0 \ni \widehat{0}, B')/H').$$

Moreover, let P be a formal prime divisor over $\widehat{V}_0$ with center $\widehat{0}$ such that

$$\frac{a(P, \widehat{V}_0, B)}{|I_{H'}(P)|} = \operatorname{mld}((\widehat{V}_0 \ni \widehat{0}, B)/H') \quad \text{or} \quad \frac{a(P, \widehat{V}_0, B')}{|I_{H'}(P)|} = \operatorname{mld}((\widehat{V}_0 \ni \widehat{0}, B')/H').$$

Then $\operatorname{mult}_P g'_k = \operatorname{mult}_P g_k$ for any k, and

$$\frac{a(P, \widehat{V}_0, B)}{|I_{H'}(P)|} = \operatorname{mld}((\widehat{V}_0 \ni \widehat{0}, B)/H') \quad \text{and} \quad \frac{a(P, \widehat{V}_0, B')}{|I_{H'}(P)|} = \operatorname{mld}((\widehat{V}_0 \ni \widehat{0}, B')/H').$$

(2) Let n be a positive integer, let $\Bbbk \supset \mathbb{C}$ be an algebraically closed field, let G be a finite abelian group acting diagonally on $\Bbbk[[x_1, \dots, x_n]]$, and let $\mathfrak{m}$ be the maximal ideal of $\Bbbk[[x_1, \dots, x_n]]$. Let $F \in \Bbbk[[x_1, \dots, x_n]]$ be a G-semi-invariant element, and let c be a positive integer such that

$$\mathfrak{m}^c \subset (F) + \mathfrak{m} \cdot (\partial_1 F, \dots, \partial_n F).$$

Then for any integer $L > 2c$ and any G-semi-invariant element F_0 such that $\chi_{F_0} = \chi_F$ and $F_0 - F \in \mathfrak{m}^L$, there exist a G-equivariant automorphism φ of $\Bbbk[[x_1, \dots, x_n]]$ and a G-invariant unit v of $\Bbbk[[x_1, \dots, x_n]]$ such that

$$v\varphi(F) = F_0, \qquad \varphi(x_i) - x_i \in \mathfrak{m}^{L-c+1} \text{ for any } i, \quad \text{and} \quad v - 1 \in \mathfrak{m}^{L-c}.$$

(3) Let n, $\Bbbk$, G, and $\mathfrak{m}$ be as in (2). Let $F_0 \in \Bbbk[[x_1, \dots, x_n]]$ be a G-semi-invariant element, let I be a G-stable ideal of $\Bbbk[[x_1, \dots, x_n]]$, and let c be a positive integer such that

$$\mathfrak{m}^c \subset (F_0) + \mathfrak{m} \cdot (\partial_1 F_0, \dots, \partial_n F_0) + I. \tag{3.2}$$

Then for any integer $L > 2c$ and any G-semi-invariant element F such that $\chi_F = \chi_{F_0}$ and $F - F_0 \in \mathfrak{m}^L$, there exist a G-equivariant automorphism φ of $\Bbbk[[x_1, \dots, x_n]]$ and a G-invariant unit v of $\Bbbk[[x_1, \dots, x_n]]$ such that

$$v\varphi(F) - F_0 \in \varphi(I), \qquad \varphi(x_i) - x_i \in \mathfrak{m}^{L-c+1} \text{ for any } i, \quad \text{and} \quad v - 1 \in \mathfrak{m}^{L-c}. \tag{3.3}$$

The previous sentence still holds if (3.2) is replaced by

$$\mathfrak{m}^c \subset (F) + \mathfrak{m} \cdot (\partial_1 F, \dots, \partial_n F) + I. \tag{3.4}$$

Proof. (1) Let $\widehat{\mathcal{O}} := \widehat{\mathcal{O}}_{V_0,0}$, so that $\widehat{V}_0 = \operatorname{Spec} \widehat{\mathcal{O}}$ and

$$\widehat{\mathcal{O}} = \mathbb{C}[[x_1, x_2, x_3]] \text{ if } n = 3, \qquad \widehat{\mathcal{O}} = \mathbb{C}[[x_1, \dots, x_4]]/(f_0) \text{ if } n = 4.$$

Let $\mathfrak{m}_{\widehat{0}}$ be the maximal ideal of $\widehat{\mathcal{O}}$. If the center of a formal prime divisor P over $\widehat{V}_0$ is $\widehat{0}$, then $\operatorname{mult}_P x_i \geq 1$ for any i, hence

$$\operatorname{mult}_P h \geq e \quad \text{for any integer } e \geq 0 \text{ and any } h \in (x_1, \dots, x_n)^e. \tag{3.5}$$

Let $J_0 := \{k \mid g_k(0) = 0\}$. For $k \notin J_0$ and any $L \geq 1$, both g_k and g'_k are units, and $\operatorname{mult}_P g_k = \operatorname{mult}_P g'_k = 0$ for any P. If $J_0 = \emptyset$, then $B = B' = 0$, $(V_0 \ni 0)$ is klt, and no formal prime divisor P satisfies $\operatorname{mult}_P g'_k \neq \operatorname{mult}_P g_k$ for some k, so $L_\gamma := 1$ satisfies (a)–(c). From now on we assume that $J_0 \neq \emptyset$. Let

$$\beta_{\min} := \min_{k \in J_0} \beta_k \quad \text{and} \quad \mathfrak{a} := \prod_{k \in J_0} (g_k \widehat{\mathcal{O}})^{\beta_k},$$

an $\mathbb{R}$-ideal on $\widehat{V}_0$ with $\operatorname{mult}_P B = \operatorname{mult}_P \mathfrak{a}$ for any P.

We let P_* be a formal prime divisor over $\widehat{V}_0$ with center $\widehat{0}$ such that $a(P_*, \widehat{V}_0, B) < \gamma$ if $\gamma > \operatorname{mld}(\widehat{V}_0 \ni \widehat{0}, B)$. Since $g_k|_{\widehat{V}_0} \neq 0$, $\operatorname{mult}_{P_*} g_k < +\infty$ for any k, and $\operatorname{mult}_{P_*} \mathfrak{a} \geq \beta_{\min} > 0$. Thus

$$\operatorname{lct}(\widehat{V}_0 \ni \widehat{0}; \mathfrak{a}) \leq \frac{a(P_*, \widehat{V}_0, 0)}{\operatorname{mult}_{P_*} \mathfrak{a}} < +\infty.$$

Since $(\widehat{V}_0, B)$ is klt, $\operatorname{lct}(\widehat{V}_0 \ni \widehat{0}; \mathfrak{a}) > 1$. We let δ be a positive real number such that

$$\operatorname{lct}(\widehat{V}_0 \ni \widehat{0}; \mathfrak{a}) > 1 + 2\delta,$$

and we let L_γ be an integer such that

$$L_\gamma \geq \frac{3}{\delta \beta_{\min}}, \qquad L_\gamma \geq \frac{|\gamma| \cdot |H|}{\delta \beta_{\min}}, \quad \text{and} \quad L_\gamma > \operatorname{mult}_{P_*} g_k \text{ for any } k.$$

Let L and $g'_1, \dots, g'_N$ be as in (1), and let

$$\mathfrak{b} := \prod_{k \in J_0} (g'_k \widehat{\mathcal{O}})^{\beta_k}.$$

(a) By (3.5), $\operatorname{mult}_{P_*}(g'_k - g_k) \geq L > \operatorname{mult}_{P_*} g_k$, hence

$$\operatorname{mult}_{P_*} g'_k = \operatorname{mult}_{P_*} g_k < +\infty \quad \text{for any } k. \tag{3.6}$$

Thus the image of g'_k in $\widehat{\mathcal{O}}$ is non-zero, and since $\mathcal{O}_{V_0,0} \subset \widehat{\mathcal{O}}$, $g'_k|_{V_0} \neq 0$. Since

$$g'_k - g_k \in (x_1, \dots, x_n)^L, \quad \text{we have} \quad g_k \widehat{\mathcal{O}} + \mathfrak{m}_{\widehat{0}}^L = g'_k \widehat{\mathcal{O}} + \mathfrak{m}_{\widehat{0}}^L \quad \text{for any } k \in J_0.$$

Since $(V_0 \ni 0)$ is a klt germ of dimension 3 and $L \geq 3/(\delta \beta_{\min})$, Theorem 2.23(2) on $\operatorname{Spec} \widehat{\mathcal{O}}$ implies that

$$|\operatorname{lct}(\widehat{V}_0 \ni \widehat{0}; \mathfrak{a}) - \operatorname{lct}(\widehat{V}_0 \ni \widehat{0}; \mathfrak{b})| \leq \delta.$$

Hence

$$\operatorname{lct}(\widehat{V}_0 \ni \widehat{0}; \mathfrak{a}) > 1 + \delta \quad \text{and} \quad \operatorname{lct}(\widehat{V}_0 \ni \widehat{0}; \mathfrak{b}) > 1 + \delta. \tag{3.7}$$

By [dFEM11, Proposition 2.11], the log canonical threshold at 0 of the $\mathbb{R}$-ideal $\prod_{k \in J_0} (g'_k|_{V_0})^{\beta_k}$ on V_0 equals $\operatorname{lct}(\widehat{V}_0 \ni \widehat{0}; \mathfrak{b}) > 1$. Since g'_k is a unit for any $k \notin J_0$, $(V_0, \sum_{k=1}^N \beta_k \operatorname{div}(g'_k|_{V_0}))$ is klt near 0.

(b) Let P be a formal prime divisor over $\widehat{V}_0$ with center $\widehat{0}$. Since V_0 is klt, $a(P, \widehat{V}_0, 0) > 0$. By the definition of the log canonical threshold,

$$a(P, \widehat{V}_0, 0) \geq \operatorname{lct}(\widehat{V}_0 \ni \widehat{0}; \mathfrak{a}) \operatorname{mult}_P \mathfrak{a} \quad \text{and} \quad a(P, \widehat{V}_0, 0) \geq \operatorname{lct}(\widehat{V}_0 \ni \widehat{0}; \mathfrak{b}) \operatorname{mult}_P \mathfrak{b}.$$

By (3.7), we obtain

$$a(P, \widehat{V}_0, 0) > (1 + \delta) \operatorname{mult}_P \mathfrak{a} \quad \text{and} \quad a(P, \widehat{V}_0, 0) > (1 + \delta) \operatorname{mult}_P \mathfrak{b}.$$

Since $a(P, \widehat{V}_0, B) = a(P, \widehat{V}_0, 0) - \operatorname{mult}_P \mathfrak{a}$ and $a(P, \widehat{V}_0, B') = a(P, \widehat{V}_0, 0) - \operatorname{mult}_P \mathfrak{b}$, we have

$$a(P, \widehat{V}_0, B) > \delta \operatorname{mult}_P \mathfrak{a} \quad \text{and} \quad a(P, \widehat{V}_0, B') > \delta \operatorname{mult}_P \mathfrak{b}. \tag{3.8}$$

Assume that $\mathrm{mult}_P g'_k \neq \mathrm{mult}_P g_k$ for some k. Then $k \in J_0$. Since $\mathrm{mult}_P(g'_k - g_k) \geq L$ by (3.5) and mult_P is a valuation, $\mathrm{mult}_P g_k \geq L$ and $\mathrm{mult}_P g'_k \geq L$. Thus $\mathrm{mult}_P \mathfrak{a} \geq \beta_k L \geq \beta_{\min} L$ and $\mathrm{mult}_P \mathfrak{b} \geq \beta_{\min} L$. By (3.8) and $L \geq |\gamma| \cdot |H|/(\delta\beta_{\min})$,

$$a(P, \widehat{V}_0, B) > |\gamma| \cdot |H| \quad \text{and} \quad a(P, \widehat{V}_0, B') > |\gamma| \cdot |H|.$$

Since $1 \leq |I_{H'}(P)| \leq |H|$ for any subgroup $H' \subset H$, (b) follows.

(c) Fix a subgroup $H' \subset H$. Since the g_k and the g'_k are H-semi-invariant, H preserves B and B'. For a formal prime divisor P over $\widehat{V}_0$ with center $\widehat{0}$, let

$$s(P) := \frac{a(P, \widehat{V}_0, B)}{|I_{H'}(P)|} \quad \text{and} \quad s'(P) := \frac{a(P, \widehat{V}_0, B')}{|I_{H'}(P)|}.$$

By Definition 2.2, $\mathrm{mld}((\widehat{V}_0 \ni \widehat{0}, B)/H') = \inf_P s(P)$ and $\mathrm{mld}((\widehat{V}_0 \ni \widehat{0}, B')/H') = \inf_P s'(P)$, where P runs over the formal prime divisors over $\widehat{V}_0$ with center $\widehat{0}$. Let $\mathcal{U}$ be the set of formal prime divisors P over $\widehat{V}_0$ with center $\widehat{0}$ such that $\mathrm{mult}_P g'_k = \mathrm{mult}_P g_k$ for any k. The numbers $a(P, \widehat{V}_0, 0)$ and $|I_{H'}(P)|$ depend only on $\widehat{V}_0$, H', and P, so

$$s(P) = s'(P) \quad \text{for any } P \in \mathcal{U}.$$

By (b), $s(P) > \gamma$ and $s'(P) > \gamma$ for any $P \notin \mathcal{U}$. By (3.6), $P_* \in \mathcal{U}$, and since $\gamma > \mathrm{mld}(\widehat{V}_0 \ni \widehat{0}, B)$,

$$s'(P_*) = s(P_*) \leq a(P_*, \widehat{V}_0, B) < \gamma.$$

Since $\inf_P s(P) \leq s(P_*) < \gamma$ and $\inf_P s'(P) \leq s'(P_*) < \gamma$, we have $\inf_P s(P) = \inf_{P \in \mathcal{U}} s(P)$ and $\inf_P s'(P) = \inf_{P \in \mathcal{U}} s'(P)$, and hence

$$\mathrm{mld}((\widehat{V}_0 \ni \widehat{0}, B)/H') = \inf_{P \in \mathcal{U}} s(P) = \inf_{P \in \mathcal{U}} s'(P) = \mathrm{mld}((\widehat{V}_0 \ni \widehat{0}, B')/H').$$

If $s(P) = \inf_{P'} s(P')$, then $s(P) < \gamma$, so $P \in \mathcal{U}$ and $s'(P) = s(P) = \mathrm{mld}((\widehat{V}_0 \ni \widehat{0}, B')/H')$. Similarly, if $s'(P) = \inf_{P'} s'(P')$, then $P \in \mathcal{U}$ and $s(P) = s'(P) = \mathrm{mld}((\widehat{V}_0 \ni \widehat{0}, B)/H')$. This proves (c).

(2) and (3). Let $\mathfrak{R} := \Bbbk[[x_1, \ldots, x_n]]$ and $\chi_i := \chi_{x_i}$, so that $g(x_i) = \chi_i(g) x_i$ for any $g \in G$ and any i. For $h \in \mathfrak{R}$, let

$$\mathcal{T}(h) := (h) + \mathfrak{m} \cdot (\partial_1 h, \ldots, \partial_n h).$$

We prove (3) under (3.2) and under (3.4), for any integer $L > 2c$. Since $\varphi(0) = 0$ for any automorphism φ, (2) is the case $I = 0$ of (3) under (3.4).

(i) For a character χ of G and $h \in \mathfrak{R}$, let

$$\pi_\chi(h) := \frac{1}{|G|} \sum_{g \in G} \chi(g)^{-1} g(h),$$

the sum of the terms of h whose monomials have character χ. Thus π_χ fixes the semi-invariants of character χ, preserves $\mathfrak{m}^e$ for any integer $e \geq 0$ and the G-stable ideals, and $\pi_\chi(sh) = s\pi_{\chi\chi_s^{-1}}(h)$ for semi-invariant s. Moreover, $\partial_i h$ is semi-invariant with character $\chi_h \chi_i^{-1}$ for semi-invariant h.

(ii) For $y_1, \ldots, y_n \in \mathfrak{m}$, substituting y_i for x_i defines a $\Bbbk$-algebra endomorphism σ of $\mathfrak{R}$ with $\sigma(h) = h(y_1, \ldots, y_n)$. For $y'_1, \ldots, y'_n \in \mathfrak{m}$, writing $\mathfrak{d} := (y_1 - y'_1, \ldots, y_n - y'_n)$, we have

$$\prod_i y_i^{\alpha_i} - \prod_i y_i'^{\alpha_i} \in \mathfrak{d} \cdot \mathfrak{m}^{|\alpha|-1} \quad \text{for any monomial } x^\alpha \neq 1,$$

and since ideals of $\mathfrak{R}$ are closed in the $\mathfrak{m}$-adic topology, it follows that

$$\begin{aligned} &h(y) - h(y') \in \mathfrak{d} \quad \text{for any } h \in \mathfrak{R}, \\ &h(y) - h(y') \in \mathfrak{d} \cdot \mathfrak{m}^{d-1} \quad \text{for any integer } d \geq 1 \text{ and any } h \in \mathfrak{m}^d. \end{aligned} \tag{3.9}$$

If $y_i - x_i \in \mathfrak{m}^2$ for any i, then σ is an automorphism by the formal inverse function theorem. If y_i is semi-invariant with character χ_i for any i, then σ is G-equivariant, since $\sigma \circ g$ and $g \circ \sigma$ send x_i to $\chi_i(g)y_i$ for any $g \in G$. For $b_1, \dots, b_n \in \mathfrak{m}$ and $h \in \mathfrak{R}$,

$$h(x_1 - b_1, \dots, x_n - b_n) - h + \sum_{i=1}^{n} b_i \partial_i h \in (b_1, \dots, b_n)^2. \tag{3.10}$$

(iii) Assume that $\mathfrak{m}^c \subset \mathcal{T}(F_0) + I$, and let ψ be a G-equivariant automorphism of $\mathfrak{R}$ such that $\psi(x_i) - x_i \in \mathfrak{m}^{c+1}$ for any i. Then $\psi(I)$ is G-stable, since $g(\psi(I)) = \psi(g(I)) = \psi(I)$ for any $g \in G$. By (3.9), $h - \psi(h) \in \mathfrak{m}^{c+1}$ for any $h \in I$, so $\mathfrak{m}^c \subset \mathcal{T}(F_0) + \psi(I) + \mathfrak{m}^{c+1}$, and Nakayama's lemma for the finitely generated $\mathfrak{R}$-module $(\mathfrak{m}^c + \mathcal{T}(F_0) + \psi(I))/(\mathcal{T}(F_0) + \psi(I))$ implies $\mathfrak{m}^c \subset \mathcal{T}(F_0) + \psi(I)$.

First we reduce (3) under (3.4) to (3) under (3.2). Assume that $\mathfrak{m}^c \subset \mathcal{T}(F) + I$. Since $F - F_0 \in \mathfrak{m}^L$, $\mathfrak{m} \cdot \partial_i(F - F_0) \subset \mathfrak{m}^L$ for any i, so $\mathcal{T}(F) \subset \mathcal{T}(F_0) + \mathfrak{m}^L$. Since $L \geq c + 1$, $\mathfrak{m}^c \subset \mathcal{T}(F_0) + I + \mathfrak{m}^{c+1}$, and Nakayama's lemma implies $\mathfrak{m}^c \subset \mathcal{T}(F_0) + I$. Thus we may assume that $\mathfrak{m}^c \subset \mathcal{T}(F_0) + I$.

For any integer $j \geq 0$, let $N_j := 2c + 2^j(L - 2c)$. Then

$$N_0 = L, \qquad N_{j+1} = 2N_j - 2c, \qquad N_j \geq L \geq 2c + 1, \quad \text{and} \quad \lim_{j \to +\infty} N_j = +\infty,$$

where the last two relations use $L > 2c$. Let $\tau_{-1} := \mathrm{id}_{\mathfrak{R}}$ and $v_{-1} := 1$. We construct inductively, for $j \geq 0$, a G-invariant element $a_j \in \mathfrak{m}^{N_j - c}$ and a G-equivariant automorphism φ_j of $\mathfrak{R}$ given by substitution, and we let

$$\tau_j := \varphi_j \circ \tau_{j-1}, \quad v_j := (1 - a_j)\varphi_j(v_{j-1}), \quad I_j := \tau_{j-1}(I), \quad \text{and} \quad F^{(j)} := v_{j-1}\tau_{j-1}(F).$$

Then $F^{(0)} = F$, $I_0 = I$, and $F^{(j+1)} = (1 - a_j)\varphi_j(F^{(j)})$. We choose a_j and φ_j so that the following hold for any $j \geq 0$.

(A) τ_{j-1} is a G-equivariant automorphism of $\mathfrak{R}$ with $\tau_{j-1}(x_i) - x_i \in \mathfrak{m}^{L-c+1}$ for any i, and v_{j-1} is a G-invariant element with $v_{j-1} - 1 \in \mathfrak{m}^{L-c}$.

(B) $F^{(j)} = F_0 + \kappa_j + \rho_j$ for some $\kappa_j \in I_j$ and $\rho_j \in \mathfrak{m}^{N_j}$, both semi-invariant with character χ_{F_0}.

For $j = 0$, (A) holds, and (B) holds with $\kappa_0 := 0$ and $\rho_0 := F - F_0$.

Assume that (A) and (B) hold for j. Since $L - c + 1 \geq c + 1$, (iii) with $\psi = \tau_{j-1}$ implies that I_j is G-stable and $\mathfrak{m}^c \subset \mathcal{T}(F_0) + I_j$. Multiplying by $\mathfrak{m}^{N_j - c}$, we have

$$\rho_j \in \mathfrak{m}^{N_j} \subset \mathfrak{m}^{N_j - c}F_0 + \sum_{i=1}^{n} \mathfrak{m}^{N_j - c + 1}\partial_i F_0 + \mathfrak{m}^{N_j - c} I_j.$$

We write

$$\rho_j = a_j F_0 + \sum_{i=1}^{n} b_{ij} \partial_i F_0 + z_j, \qquad a_j \in \mathfrak{m}^{N_j - c}, \quad b_{ij} \in \mathfrak{m}^{N_j - c + 1}, \quad \text{and} \quad z_j \in \mathfrak{m}^{N_j - c} I_j.$$

Since $\pi_{\chi_{F_0}}(\rho_j) = \rho_j$ and $\mathfrak{m}^{N_j - c} I_j$ is G-stable, applying $\pi_{\chi_{F_0}}$ and (i), we may replace a_j, b_{ij}, and z_j by $\pi_{\mathbf{1}}(a_j)$, $\pi_{\chi_i}(b_{ij})$, and $\pi_{\chi_{F_0}}(z_j)$ respectively, where $\mathbf{1}$ is the trivial character. Thus we may assume that a_j is G-invariant, b_{ij} is semi-invariant with character χ_i for any i, and z_j is semi-invariant with character χ_{F_0}. Let φ_j be the substitution of $x_i - b_{ij}$ for x_i. Since $N_j - c + 1 \geq 2$, (ii) implies that φ_j is a G-equivariant automorphism of $\mathfrak{R}$.

By (3.10) and (3.9), the elements

$$Q_j := \varphi_j(F_0) - F_0 + \sum_{i=1}^{n} b_{ij} \partial_i F_0, \qquad \eta_j := \varphi_j(\rho_j) - \rho_j, \quad \text{and} \quad \zeta_j := \varphi_j(z_j) - z_j$$

satisfy

$$\begin{aligned} Q_j &\in (b_{1j},\dots,b_{nj})^2 \subset \mathfrak{m}^{2N_j-2c+2},\\ \eta_j &\in \mathfrak{m}^{(N_j-c+1)+(N_j-1)} = \mathfrak{m}^{2N_j-c}, \quad \text{and}\\ \zeta_j &\in \mathfrak{m}^{(N_j-c+1)+(N_j-c-1)} = \mathfrak{m}^{2N_j-2c}. \end{aligned}$$

Since $F^{(j)} = F_0 + \kappa_j + \rho_j$ and $\rho_j - \sum_i b_{ij}\partial_i F_0 = a_j F_0 + z_j$, we obtain

$$\varphi_j(F^{(j)}) = (1+a_j)F_0 + \varphi_j(\kappa_j + z_j) + Q_j + \eta_j - \zeta_j.$$

Thus

$$F^{(j+1)} = (1-a_j)\varphi_j(F^{(j)}) = F_0 + \kappa_{j+1} + \rho_{j+1},$$

where

$$\kappa_{j+1} := (1-a_j)\varphi_j(\kappa_j + z_j) \quad \text{and} \quad \rho_{j+1} := -a_j^2 F_0 + (1-a_j)(Q_j + \eta_j - \zeta_j).$$

Since $\kappa_j + z_j \in I_j$, $\kappa_{j+1} \in \varphi_j(I_j) = I_{j+1}$. Since $a_j^2 \in \mathfrak{m}^{2N_j-2c}$, $\rho_{j+1} \in \mathfrak{m}^{2N_j-2c} = \mathfrak{m}^{N_{j+1}}$. Since $1-a_j$ is G-invariant, φ_j is G-equivariant, and $\kappa_j + z_j$ and $F^{(j+1)} - F_0$ are semi-invariant with character χ_{F_0}, so are κ_{j+1} and ρ_{j+1}. This proves (B) for $j+1$. By (3.9) with $d=1$,

$$\begin{aligned} &\tau_j(x_i) - \tau_{j-1}(x_i) = \varphi_j(\tau_{j-1}(x_i)) - \tau_{j-1}(x_i) \in \mathfrak{m}^{N_j-c+1} \quad \text{for any } i,\\ &v_j - v_{j-1} = \big(\varphi_j(v_{j-1}) - v_{j-1}\big) - a_j\varphi_j(v_{j-1}) \in \mathfrak{m}^{N_j-c}. \end{aligned} \tag{3.11}$$

Since $N_j \geq L$, (3.11) and (A) for j imply (A) for $j+1$, where v_j is G-invariant because $1-a_j$ is G-invariant and φ_j is G-equivariant. By induction on j, a_j and φ_j exist and (A) and (B) hold for any $j \geq 0$.

Since $\lim_j N_j = +\infty$ and $\mathfrak{R}$ is complete, (3.11) implies that the limits $y_i := \lim_j \tau_j(x_i)$ and $v := \lim_j v_j$ exist, and, since any power of $\mathfrak{m}$ is closed in $\mathfrak{R}$,

$$y_i - x_i \in \mathfrak{m}^{L-c+1}, \qquad y_i - \tau_j(x_i) \in \mathfrak{m}^{N_{j+1}-c+1}, \quad \text{and} \quad v - 1 \in \mathfrak{m}^{L-c}$$

for any i and j. By (i), the semi-invariant elements of a given character form a closed subspace of $\mathfrak{R}$, so y_i is semi-invariant with character χ_i and v is G-invariant. Since $L - c \geq 1$, v is a unit. Let φ be the substitution of y_i for x_i. Since $L-c+1 \geq 2$, (ii) implies that φ is a G-equivariant automorphism of $\mathfrak{R}$ with $\varphi(x_i) - x_i \in \mathfrak{m}^{L-c+1}$ for any i. Since τ_j is the substitution of $\tau_j(x_i)$ for x_i, (3.9) implies

$$\tau_j(h) - \varphi(h) \in \mathfrak{m}^{N_{j+1}-c+1} \quad \text{for any } h \in \mathfrak{R},$$

hence $\kappa_{j+1} \in \tau_j(I) \subset \varphi(I) + \mathfrak{m}^{N_{j+1}-c+1}$, and $v_j\tau_j(F)$ converges to $v\varphi(F)$. Since

$$v_j\tau_j(F) = F^{(j+1)} = F_0 + \kappa_{j+1} + \rho_{j+1} \quad \text{and} \quad \rho_{j+1} \in \mathfrak{m}^{N_{j+1}},$$

we have $v\varphi(F) - F_0 \in \varphi(I) + \mathfrak{m}^e$ for any integer $e \geq 1$. By the Krull intersection theorem, $\bigcap_{e\geq 1}(\varphi(I) + \mathfrak{m}^e) = \varphi(I)$, hence

$$v\varphi(F) - F_0 \in \varphi(I).$$

This proves (3) for any integer $L > 2c$, and hence (2) and (3). □

3.3. **Surfaces, constant extensions and products.**

Lemma 3.3 (cf. [Liu26a, Theorem 16.13(1)]). *Let $\Bbbk \supset \mathbb{C}$ be an algebraically closed field, $\Theta \subset (0,1]$ a non-empty DCC set, $c_0 \geq 0$ a real number and*

$$\lambda := 2 + \frac{1+c_0}{\delta(\Theta)}$$

(see [Liu26a, Definition 2.9]). Let $(T \ni p)$ be a smooth surface germ over $\Bbbk$, D a non-zero reduced curve through p, B_- an $\mathbb{R}$-divisor supported on D with coefficients in $[-c_0, 0]$, and $B_+ := \sum_{j=1}^N \beta_j \operatorname{div}(g_j)$ with $\beta_j \in \Theta$ and non-zero regular functions g_j on a neighborhood of p. Assume that

$$a_0 := \operatorname{mld}(T \ni p, B_- + B_+) \geq 0.$$

Let E be a prime divisor over T with center p and $a(E,T,B_-+B_+)=a_0$. Let $E_1,\dots,E_n=E$ be the exceptional divisors of the blow-ups of closed points extracting E, the center of the i-th blow-up being a point of E_{i-1} for $i\geq 2$, and let E' be the first of $E_1,\dots,E_n$ with $a(E',T,B_-+B_+)=a_0$. Then

$$a(E',T,0)\leq \lambda \operatorname{mult}_{E'} D\leq \lambda \operatorname{mult}_E D.$$

Proof. Let $B:=B_-+B_+$ and $\delta:=\delta(\Theta)$. Since Θ is a non-empty DCC subset of $(0,1]$, $\delta>0$ [Liu26a, Lemma 2.10(2)]. Let $T_0:=T$ and $P_0:=p$. For $1\leq i\leq n$, let $T_i\to T_{i-1}$ be the blow-up of the closed point P_{i-1} with exceptional curve E_i, where $P_i\in E_i$ is the center of E_{i+1} on T_i for $1\leq i<n$. For $0\leq i\leq i'\leq n$, let $h_{i',i}:T_{i'}\to T_i$ be the composite of the blow-ups $T_{i''}\to T_{i''-1}$ with $i<i''\leq i'$, and for $i\geq 1$ denote $(h_{i',i})_*^{-1}E_i$ again by E_i. Let $i_0\in\{1,\dots,n\}$ be the index with $E'=E_{i_0}$, and for $1\leq i\leq n$ let

$$A_i:=a(E_i,T,0),\qquad a_i:=a(E_i,T,B),\quad\text{and}\quad d_i:=\operatorname{mult}_{E_i} D.$$

Since the center of E_i on T is p, we have $a_i\geq a_0\geq 0$ for any i, and by the choice of E',

$$a_i>a_0\quad\text{for any } 1\leq i<i_0. \tag{3.12}$$

For each j, let $G_{j,0}:=\operatorname{div}(g_j)$, so that $G_{j,0}\geq 0$ near p, and let $G_{j,i}:=(h_{i,0})_*^{-1}G_{j,0}$. Let

$$B_{+,i}:=\sum_j \beta_j G_{j,i}\quad\text{and}\quad m_i:=\operatorname{mult}_{P_{i-1}} B_{+,i-1}\quad\text{for } 1\leq i\leq n.$$

Step 1. In this step we compare the numbers m_i. Since $\operatorname{mult}_{P_{i-1}} G_{j,i-1}\in\mathbb{N}$ and $\beta_j\in\Theta$, m_i is a finite sum of elements of Θ. Since $\delta(\Theta)\leq s-1$ for any finite sum $s>1$ of elements of Θ [Liu26a, Definition 2.9],

$$m_i>1\implies m_i\geq 1+\delta. \tag{3.13}$$

For $1\leq i<n$, since $B_{+,i}\cdot E_i=m_i$, we obtain

$$m_{i+1}\leq (B_{+,i}\cdot E_i)_{P_i}\leq B_{+,i}\cdot E_i=m_i. \tag{3.14}$$

Step 2. In this step we write the recurrences for A_i and d_i. For $1\leq i\leq n$ let

$$J_i:=\{i'<i\mid P_{i-1}\in E_{i'}\}.$$

For $i\geq 2$ we have $i-1\in J_i$ and $\#J_i\leq 2$, because the exceptional curves of $T_{i-1}\to T$ form a simple normal crossing divisor. Pulling back K_T and D to T_i, we obtain

$$A_i=2+\sum_{i'\in J_i}(A_{i'}-1)\quad\text{and}\quad d_i=\operatorname{mult}_{P_{i-1}} D_{i-1}+\sum_{i'\in J_i} d_{i'},$$

where $D_{i-1}:=(h_{i-1,0})_*^{-1}D$. Thus $A_1=2$ and $d_1=\operatorname{mult}_p D\geq 1$. Since $i-1\in J_i$ for $i\geq 2$, the recurrence for d_i gives $d_i\geq d_{i-1}$ for $2\leq i\leq n$, so

$$d_{i_0}\leq d_n=\operatorname{mult}_E D. \tag{3.15}$$

For $i\geq 2$ we call P_{i-1} *free* if $J_i=\{i-1\}$, and *satellite* if $J_i=\{i-1,i'\}$ with $i'<i-1$. Since $d_{i'}\geq 1$ for any i',

$$\begin{aligned}&\frac{A_i}{d_i}\leq\frac{A_{i-1}}{d_{i-1}}+1 && \text{if } P_{i-1}\text{ is free, and}\\ &\frac{A_i}{d_i}\leq\max\left\{\frac{A_{i-1}}{d_{i-1}},\frac{A_{i'}}{d_{i'}}\right\} && \text{if } P_{i-1}\text{ is satellite with } J_i=\{i-1,i'\}.\end{aligned} \tag{3.16}$$

Fix $1\leq i\leq n$, and for $1\leq i'\leq i$ let $\rho_{i,i'}$ be the order along E_i of the pullback to T_i of the ideal of $P_{i'-1}$ on $T_{i'-1}$. Since the center of E_i on $T_{i'-1}$ is $P_{i'-1}$, $\rho_{i,i'}$ is a positive integer. The ideal of

$P_{i'-1}$ generates $\mathcal{O}_{T_{i'}}(-E_{i'})$, $K_{T_{i'}}$ is the pullback of $K_{T_{i'-1}}$ plus $E_{i'}$, and the pullback of $G_{j,i'-1}$ to $T_{i'}$ is $G_{j,i'} + (\operatorname{mult}_{P_{i'-1}} G_{j,i'-1})E_{i'}$. Thus

$$A_i = 1 + \sum_{i'=1}^{i} \rho_{i,i'} \quad \text{and} \quad \operatorname{mult}_{E_i} B_+ = \sum_{i'=1}^{i} \rho_{i,i'} m_{i'}. \tag{3.17}$$

Step 3. In this step we prove: if $m_1, \dots, m_i > 1$, then $A_i \le (\lambda - 1)d_i$. Since B_- is supported on D with coefficients in $[-c_0, 0]$, $\operatorname{mult}_{E_i} B_- \ge -c_0 d_i$. By (3.13) and (3.17),

$$0 \le a_i = A_i - \operatorname{mult}_{E_i} B_- - \operatorname{mult}_{E_i} B_+ \le 1 + \sum_{i'=1}^{i} \rho_{i,i'} + c_0 d_i - (1+\delta) \sum_{i'=1}^{i} \rho_{i,i'} = 1 + c_0 d_i - \delta \sum_{i'=1}^{i} \rho_{i,i'}.$$

Since $d_i \ge 1$, we obtain

$$A_i = 1 + \sum_{i'=1}^{i} \rho_{i,i'} \le 1 + \frac{1 + c_0 d_i}{\delta} \le \Big(1 + \frac{1 + c_0}{\delta}\Big) d_i = (\lambda - 1) d_i.$$

Step 4. In this step we prove: if $1 \le i < i_0$ and $m_i \le 1$, then P_i is satellite. Suppose that P_i is free, that is, E_i is the only exceptional curve of $T_i \to T$ through P_i. Let $B_{-,i} := (h_{i,0})_*^{-1} B_-$. Then $B_{-,i} \le 0$ and, near P_i,

$$K_{T_i} + (1 - a_i)E_i + B_{-,i} + B_{+,i}$$

is the pullback of $K_T + B$. By (3.14), $(B_{+,i} \cdot E_i)_{P_i} \le m_i \le 1$. For any real number $0 < \epsilon < 1$, the pair $(E_i, (1-\epsilon)B_{+,i}|_{E_i})$ is klt near P_i, so $(T_i, E_i + (1-\epsilon)B_{+,i})$ is plt near P_i by inversion of adjunction. Letting ϵ tend to 0, we have $a(F, T_i, E_i + B_{+,i}) \ge 0$ for any prime divisor F over T_i with center P_i. The center of E_{i_0} on T_i is P_i, so $\mu := \operatorname{mult}_{E_{i_0}} E_i$ is a positive integer, and

$$a_0 = a_{i_0} = a(E_{i_0}, T_i, E_i + B_{+,i}) + a_i \mu - \operatorname{mult}_{E_{i_0}} B_{-,i} \ge a_i \mu \ge a_i,$$

where we use $a_i \ge 0$. This contradicts (3.12).

Step 5. In this step we prove $A_i \le \lambda d_i$ for $1 \le i \le i_0$, by induction on i. If $m_1, \dots, m_i > 1$, this follows from **Step 3**. Otherwise let $i_1 \le i$ be the least index with $m_{i_1} \le 1$. By **Step 3**, $A_{i'} \le (\lambda - 1)d_{i'}$ for any $i' < i_1$.

- If $i = i_1 = 1$, then $A_1 = 2 < \lambda d_1$.
- If $i = i_1 \ge 2$, then $\max J_i < i_1$, so $A_i/d_i \le \lambda$ by (3.16).
- If $i > i_1$, then $m_{i-1} \le m_{i_1} \le 1$ by (3.14) and $i - 1 < i_0$, so P_{i-1} is satellite by **Step 4**, and $A_i/d_i \le \lambda$ by (3.16) and the induction hypothesis.

Taking $i = i_0$ and using (3.15), we obtain $a(E', T, 0) = A_{i_0} \le \lambda d_{i_0} \le \lambda \operatorname{mult}_E D$. □

In Lemma 3.4 and its proof, for a field k of characteristic zero and a birational morphism $Y \to S$ of regular schemes of finite type over k or over $\operatorname{Spec} k[[x_1, x_2]]$, the relative canonical divisor $K_{Y/S}$ in the definition of log discrepancies is the divisor of the zeroth Fitting ideal of $\Omega_{Y/S}$ [dFEM11, Remark A.12].

Lemma 3.4 (cf. [Kaw17, Theorem 1] and [Kaw21, Lemma 6.7 and Proposition 6.9])**.** *Let k be a field of characteristic zero, $(S \ni P)$ the germ at a k-point of a regular k-variety of dimension 2, or $S = \operatorname{Spec} k[[x_1, x_2]]$ and P the closed point of S, and $\mathfrak{a}$ an $\mathbb{R}$-ideal on S. Assume that $(S, \mathfrak{a})$ is lc near P and $\operatorname{mld}(S \ni P, \mathfrak{a}) = 0$. Then there exist an integer $r \ge 1$ and closed points $P_0 = P, P_1, \dots, P_{r-1}$ such that, for $S_0 := S$ and the blow-up $S_{j+1} \to S_j$ of P_j with exceptional curve E_{j+1}, the following hold.*

(1) $a(E_j, S, \mathfrak{a}) \in (0, 1)$ for $1 \le j < r$, and $a(E_r, S, \mathfrak{a}) = 0$. For $1 \le j < r$, P_j is the unique closed point of E_j which is the center on S_j of a prime divisor over S with log discrepancy zero with respect to $(S, \mathfrak{a})$, and P_j is a k-point.

(2) *E_r is the exceptional divisor of the weighted blow-up of S with coprime weights in some regular parameters of $\mathcal{O}_{S,P}$.*
(3) *Let γ be an automorphism of S, not necessarily over k, fixing P and with*

$$\operatorname{mult}_F(\gamma^*\mathfrak{a}) = \operatorname{mult}_F \mathfrak{a} \quad \text{for any prime divisor } F \text{ over } S.$$

Then γ lifts to an automorphism of S_j fixing P_j for any $j < r$, and its lift to S_r preserves E_r.

Proof. For $S = \operatorname{Spec} k[[x_1, x_2]]$ we use blow-ups, multiplicities and log discrepancies as in [dFEM11, Section 2], and in **Step 2** we use [Kaw17, Proposition 6] and [Kaw07, Theorem] also for the regular surfaces over $\operatorname{Spec} \overline{k}[[x_1, x_2]]$ obtained by blowing up closed points, and the termination of the blow-ups of the centers of F_0 also over $\operatorname{Spec} k[[x_1, x_2]]$, which holds for any prime divisor over a regular local scheme of dimension 2 with center the closed point [Abh56, Proposition 3]. Write $\mathfrak{a} = \prod_{i=1}^m \mathfrak{a}_i^{\alpha_i}$ with non-zero ideals $\mathfrak{a}_i$ and real numbers $\alpha_i > 0$.

Step 1. In this step we write the crepant pullback to a tower. Assume that $S_0 = S, \dots, S_j$ are obtained by blowing up k-points $P_0, \dots, P_{j-1}$, with exceptional curves $E_1, \dots, E_j$ (for $j' \le j$ we write $E_{j'}$ also for $(h_{j,j'})_*^{-1} E_{j'}$, where $h_{j,j'} : S_j \to S_{j'}$ is the composite of the blow-ups), and that $a(E_{j'}, S, \mathfrak{a}) \in (0,1)$ for $1 \le j' \le j$. For any i, let $M_{i,j} := \sum_{j'=1}^{j} (\operatorname{mult}_{E_{j'}} \mathfrak{a}_i) E_{j'}$, and let $\mathfrak{b}_{i,j}$ be the ideal with

$$\mathfrak{a}_i \mathcal{O}_{S_j} = \mathfrak{b}_{i,j} \cdot \mathcal{O}_{S_j}(-M_{i,j}).$$

Then $\mathfrak{b}_{i,j} \not\subset \mathcal{O}_{S_j}(-E_{j'})$ for any $j' \le j$. Let

$$\Delta_j := \sum_{j'=1}^{j} \big(1 - a(E_{j'}, S, \mathfrak{a})\big) E_{j'} \quad \text{and} \quad \mathfrak{b}_j := \prod_{i=1}^{m} \mathfrak{b}_{i,j}^{\alpha_i}.$$

Since $K_{S_j/S} = \sum_{j'} (a(E_{j'}, S, 0) - 1) E_{j'}$, we have $\Delta_j = \sum_i \alpha_i M_{i,j} - K_{S_j/S}$, so

$$a(F, S, \mathfrak{a}) = a(F, S_j, \Delta_j, \mathfrak{b}_j) \quad \text{for any prime divisor } F \text{ over } S_j. \tag{3.18}$$

In particular $(S_j, \Delta_j, \mathfrak{b}_j)$ is lc over a neighborhood of P, and $\Delta_j \in (0,1)$.

Step 2. In this step we construct the points P_j and prove (1). Since $(S, \mathfrak{a})$ is lc near P, $\operatorname{mld}(S \ni P, \mathfrak{a}) = 0$ is the minimum of $a(\cdot, S, \mathfrak{a})$ over the exceptional prime divisors with center P on a log resolution of $(S, \mathfrak{a}\mathfrak{m}_P)$ (for $S = \operatorname{Spec} k[[x_1, x_2]]$, log resolutions exist by [dFEM11, Theorems 2.3 and 2.4]), so there exists a prime divisor F_0 over S with center P and $a(F_0, S, \mathfrak{a}) = 0$. We prove by induction on $j \ge 0$: if S_j is as in **Step 1** and the center of F_0 on S_j is a k-point P_j, with $P_j \in E_j$ if $j \ge 1$, then for the blow-up $S_{j+1} \to S_j$ of P_j:

(a) $a(E_{j+1}, S, \mathfrak{a}) \in [0,1)$.
(b) If $a(E_{j+1}, S, \mathfrak{a}) > 0$, then the center P_{j+1} of F_0 on S_{j+1} is a k-point of E_{j+1}, and it is the unique closed point of E_{j+1} which is the center of a prime divisor F over S with $a(F, S, \mathfrak{a}) = 0$.

For (a): by (3.18),

$$a(E_{j+1}, S, \mathfrak{a}) = 2 - \operatorname{mult}_{P_j} \Delta_j - \operatorname{mult}_{P_j} \mathfrak{b}_j \ge 0.$$

The prime divisor F_0 has center P_j and $a(F_0, S_j, \Delta_j, \mathfrak{b}_j) = 0$, so $\operatorname{mld}(S_j \ni P_j, \Delta_j, \mathfrak{b}_j) < 1$. Since $\Delta_j \ge 0$, after the base change to $\overline{k}$, which preserves log discrepancies and multiplicities [Kaw21, Remark 2.6] (P_j is a k-point), [Kaw17, Proposition 6] implies $\operatorname{mult}_{P_j} \Delta_j + \operatorname{mult}_{P_j} \mathfrak{b}_j > 1$, so $a(E_{j+1}, S, \mathfrak{a}) < 1$.

For (b): let $c := 1 - a(E_{j+1}, S, \mathfrak{a}) \in (0,1)$ and $E := E_{j+1}$. Since $a(F_0, S, \mathfrak{a}) = 0 < a(E, S, \mathfrak{a})$, $F_0 \ne E$, so the center of F_0 on S_{j+1} is a closed point of E. Let $Q \in E$ be a closed point which is the center of a prime divisor F over S with $a(F, S, \mathfrak{a}) = 0$, and let $\Delta' := \Delta_{j+1} - cE$. Then $\Delta' \ge 0$, $E \not\subset \operatorname{Supp} \Delta'$, and since $\operatorname{mult}_F E > 0$,

$$a(F, S_{j+1}, E + \Delta', \mathfrak{b}_{j+1}) = a(F, S, \mathfrak{a}) - (1 - c) \operatorname{mult}_F E < 0.$$

Thus $(S_{j+1}, E+\Delta', \mathfrak{b}_{j+1})$ is not lc near Q, and by inversion of adjunction [Kaw07, Theorem] after the base change to $\overline{k}$, which preserves log discrepancies [Kaw21, Remark 2.6], $(E, \Delta'|_E, \mathfrak{b}_{j+1}\mathcal{O}_E)$ is not lc at Q. Since P_j is a k-point,

$$E \cong \mathbb{P}^1_k, \quad E^2 = -1, \quad \text{and} \quad K_{S_{j+1}/S} \cdot E = E^2 = -1.$$

On the regular curve E we have $\mathfrak{b}_{i,j+1}\mathcal{O}_E = \mathcal{O}_E(-Z_i)$ for a divisor $Z_i \geq 0$. Let

$$\Delta_E := \Delta'|_E + \sum_{i=1}^{m} \alpha_i Z_i.$$

Then $\operatorname{mult}_Q \Delta_E > 1$. The ideal $\mathfrak{a}_i$ has a generator whose pullback restricts to a non-zero section of $\mathcal{O}_{S_{j+1}}(-M_{i,j+1})|_E$ vanishing on Z_i, so $\deg Z_i \leq -M_{i,j+1} \cdot E$. With

$$\Delta' = \sum_i \alpha_i M_{i,j+1} - K_{S_{j+1}/S} - cE$$

from **Step 1**,

$$\sum_{Q'} [k(Q') : k] \operatorname{mult}_{Q'} \Delta_E = \deg \Delta_E \leq \Delta' \cdot E - \sum_{i=1}^{m} \alpha_i M_{i,j+1} \cdot E = -K_{S_{j+1}/S} \cdot E - cE^2 = 1 + c < 2,$$

where Q' runs over the closed points of E. Since any such Q satisfies $\operatorname{mult}_Q \Delta_E > 1$, there exists exactly one such Q, and $[k(Q) : k] = 1$. The center of F_0 is such a point, which proves (b).

By (a) and (b), starting from $P_0 = P$, the centers P_j of F_0 are k-points as long as $a(E_j, S, \mathfrak{a}) > 0$. After finitely many blow-ups of the center, F_0 becomes the exceptional curve, and $a(F_0, S, \mathfrak{a}) = 0$. Thus there exists a least $r \geq 1$ with $a(E_r, S, \mathfrak{a}) = 0$, and (1) follows from (a) and (b). By the uniqueness in (b), the points P_j and the integer r do not depend on F_0.

Step 3. In this step we reduce (2) to an $\mathfrak{m}_P$-primary $\mathbb{R}$-ideal on $\mathbb{A}^2_k$. Let N be a positive integer with $N \operatorname{mult}_{E_j} \mathfrak{m}_P \geq \operatorname{mult}_{E_j} \mathfrak{a}_i$ for any i and any $j \leq r$, and let

$$\mathfrak{a}' := \prod_i (\mathfrak{a}_i + \mathfrak{m}_P^N)^{\alpha_i}.$$

Then $a(E_j, S, \mathfrak{a}') = a(E_j, S, \mathfrak{a})$ for $j \leq r$, and $a(F, S, \mathfrak{a}') \geq a(F, S, \mathfrak{a})$ for any prime divisor F over S. Thus $(S, \mathfrak{a}')$ is lc near P, $\operatorname{mld}(S \ni P, \mathfrak{a}') = 0$, and any prime divisor with log discrepancy zero with respect to $(S, \mathfrak{a}')$ has log discrepancy zero with respect to $(S, \mathfrak{a})$. Since E_r has center P_j on S_j, the uniqueness in (1) for $\mathfrak{a}'$ implies that the tower of (1) for $\mathfrak{a}'$ is $P_0, \dots, P_{r-1}$, $E_1, \dots, E_r$. Thus we may replace $\mathfrak{a}$ by $\mathfrak{a}'$. Fix regular parameters x_1, x_2 of $\mathcal{O}_{S,P}$. Each $\mathfrak{a}_i + \mathfrak{m}_P^N$ is generated by an ideal $\mathfrak{a}_{i,0} \subset k[x_1, x_2]$. Let $\mathfrak{a}_0 := \prod_i \mathfrak{a}_{i,0}^{\alpha_i}$ on the germ at 0 of $\mathbb{A}^2_k$ with coordinates x_1, x_2. By [Kaw21, Remark 2.6 and Lemma 2.7], the blow-ups of closed points over P and the prime divisors over S with center P correspond to the blow-ups of closed points over 0 and the prime divisors over $\mathbb{A}^2_k$ with center 0, and corresponding prime divisors F and F' satisfy $a(F, S, \mathfrak{a}') = a(F', \mathbb{A}^2_k, \mathfrak{a}_0)$. Thus we may assume that S is the germ of $\mathbb{A}^2_k$ at $P = 0$. If $r = 1$, then E_1 is the weighted blow-up with weights $(1, 1)$ in x_1, x_2, so we may assume $r \geq 2$.

Step 4. In this step we find weights over $\overline{k}$. Let $\overline{S} := S \times_k \overline{k}$, $\overline{P}$ the point over P, and $\overline{E}_j := E_j \times_k \overline{k}$. Since $P_0, \dots, P_{r-1}$ are k-points, $\overline{E}_1, \dots, \overline{E}_r$ are the exceptional curves of the blow-ups of the $\overline{k}$-points $P_j \times_k \overline{k}$, and the valuation $v := \operatorname{mult}_{\overline{E}_r}$ restricts to $\operatorname{mult}_{E_r}$ on $k(S)$. By [Kaw21, Remark 2.6], the log discrepancies and the minimal log discrepancy at P do not change under the base change to $\overline{k}$, so $(\overline{S}, \mathfrak{a}\mathcal{O}_{\overline{S}})$ is lc near $\overline{P}$ and $\overline{E}_r$ computes $\operatorname{mld}(\overline{S} \ni \overline{P}, \mathfrak{a}\mathcal{O}_{\overline{S}}) = 0$. By [Kaw17, Theorem 1(i)], there exist regular parameters $\overline{y}_1, \overline{y}_2$ of $\mathcal{O}_{\overline{S},\overline{P}}$ and coprime positive integers $w_1 \geq w_2$ such that v is the monomial valuation with weights (w_1, w_2) in $\overline{y}_1, \overline{y}_2$ on $\widehat{\mathcal{O}} := \overline{k}[[\overline{y}_1, \overline{y}_2]]$. Since $r \geq 2$, $\overline{E}_r$ is not the blow-up of $\overline{P}$, so $w_1 > w_2$. The graded ring of v is the polynomial ring $\overline{k}[U_1, U_2]$, where U_i is the initial form of $\overline{y}_i$, of degree w_i. We write $\operatorname{in}(f)$ for the initial form of $f \in \widehat{\mathcal{O}} \setminus \{0\}$.

Step 5. In this step we descend the parameters to k and prove (2). The set of $\ell \in \overline{k}x_1 + \overline{k}x_2$ with $v(\ell) > w_2$ is a $\overline{k}$-linear subspace, which is proper since $v(\mathfrak{m}_{\overline{P}}) = w_2$ and $v(\mathfrak{m}_{\overline{P}}^2) \geq 2w_2$. As $kx_1 + kx_2$ spans $\overline{k}x_1 + \overline{k}x_2$, there exists $y_2 \in kx_1 + kx_2$ with $v(y_2) = w_2$, and we choose $y \in \{x_1, x_2\}$ such that y, y_2 are regular parameters of $\mathcal{O}_{S,P}$. Since $w_1 > w_2$, U_2 is the only monomial of degree w_2, so $\mathrm{in}(y_2) = c_2U_2$ with $c_2 \in \overline{k}^\times$.

Let $\mathfrak{J} := \{f \in \widehat{\mathcal{O}} \mid v(f) \geq w_1\}$. Modulo $\mathfrak{J}$, any element of $\widehat{\mathcal{O}}$ is congruent to a polynomial in $\overline{y}_2$ of degree less than w_1/w_2, and

$$y_2 \equiv c_2\overline{y}_2 + (\text{terms of degree} \geq 2 \text{ in } \overline{y}_2).$$

Thus $\overline{y}_2$ is congruent to a polynomial in y_2, and there exists $\phi \in \overline{k}[T]$ with

$$\phi(0) = 0, \quad \deg\phi < w_1/w_2, \quad \text{and} \quad y - \phi(y_2) \in \mathfrak{J}.$$

Such ϕ is unique: for a non-zero $\psi \in \overline{k}[T]$ with $\deg\psi < w_1/w_2$ and least exponent e_ψ, $\mathrm{in}(\psi(y_2))$ is a non-zero multiple of $U_2^{e_\psi}$, so $v(\psi(y_2)) = e_\psi w_2 < w_1$. Any $\sigma \in \mathrm{Gal}(\overline{k}/k)$ acts on $\widehat{\mathcal{O}} = \widehat{\mathcal{O}}_{S,P}\widehat{\otimes}_k\overline{k}$ through $\overline{k}$, fixes y and y_2, and satisfies $v \circ \sigma = v$, since σ preserves $\overline{E}_r = E_r \times_k \overline{k}$. Thus the coefficientwise conjugate ϕ^σ satisfies $y - \phi^\sigma(y_2) \in \mathfrak{J}$, so $\phi^\sigma = \phi$ and $\phi \in k[T]$.

Let $y_1 := y - \phi(y_2) \in \mathcal{O}_{S,P}$. Since $\phi(y_2) \in ky_2 + \mathfrak{m}_P^2$, y_1, y_2 are regular parameters of $\mathcal{O}_{S,P}$. Write $y_1 = \overline{y}_1 u + \psi_1(\overline{y}_2)$ with $u \in \widehat{\mathcal{O}}$ and $\psi_1 \in \overline{y}_2\overline{k}[[\overline{y}_2]]$. Since $v(\overline{y}_1 u) \geq w_1$ and $v(y_1) \geq w_1$, we have $v(\psi_1(\overline{y}_2)) \geq w_1 > w_2$, so $\psi_1(\overline{y}_2) \in \mathfrak{m}_{\overline{P}}^2$. As $y_1 \notin \mathfrak{m}_{\overline{P}}^2$, $u(0) \neq 0$. Thus $\mathrm{in}(y_2) = c_2U_2$, and $\mathrm{in}(y_1) = u(0)U_1$ if $w_2 \geq 2$, while $\mathrm{in}(y_1) = u(0)U_1 + \beta U_2^{w_1}$ for some $\beta \in \overline{k}$ if $w_2 = 1$. The graded $\overline{k}$-algebra endomorphism of $\overline{k}[U_1, U_2]$ sending U_1, U_2 to $\mathrm{in}(y_1), \mathrm{in}(y_2)$ is an automorphism. Let $f \in \widehat{\mathcal{O}}$ be non-zero, and write

$$f = \sum a_{e_1e_2} y_1^{e_1} y_2^{e_2} \quad \text{and} \quad n_f := \min\{e_1w_1 + e_2w_2 \mid a_{e_1e_2} \neq 0\}.$$

The terms of weight greater than n_f belong to the closed ideal $\{v \geq n_f + 1\} \supset \mathfrak{m}_{\overline{P}}^{n_f+1}$, and the part of weight n_f has the initial form

$$\sum_{e_1w_1+e_2w_2=n_f} a_{e_1e_2}\,\mathrm{in}(y_1)^{e_1}\,\mathrm{in}(y_2)^{e_2} \neq 0.$$

Thus $v(f) = n_f$, that is, v is the monomial valuation with weights (w_1, w_2) in y_1, y_2. Restricting to $k(S)$, E_r is the exceptional divisor of the weighted blow-up of S with weights (w_1, w_2) in the regular parameters y_1, y_2 of $\mathcal{O}_{S,P}$. For the germ S before the reduction of **Step 3**, y_1, y_2 are polynomials over k in the regular parameters x_1, x_2 of $\mathcal{O}_{S,P}$, which proves (2).

Step 6. In this step we prove (3). Let γ be as in (3). Since $K_{Y/S}$ only depends on the morphism $Y \to S$ and not on the k-structure of S, and $\mathrm{mult}_F(\gamma^*\mathfrak{a}) = \mathrm{mult}_F\,\mathfrak{a}$, $a(F, S, \mathfrak{a})$ is preserved under γ for any prime divisor F over S. Thus γ permutes the prime divisors over S with log discrepancy zero. Assume that γ lifts to an automorphism of S_j fixing P_j for some $j < r$. Then γ lifts to S_{j+1} and preserves E_{j+1}. If $j + 1 < r$, the closed point $\gamma(P_{j+1}) \in E_{j+1}$ is the center of a prime divisor with log discrepancy zero, so $\gamma(P_{j+1}) = P_{j+1}$ by (1). As γ fixes $P_0 = P$, by induction on j, γ lifts to an automorphism of S_j fixing P_j for any $j < r$, and its lift to S_r preserves E_r. □

Lemma 3.5 (cf. [Liu26b, Lemma 3.22])**.** *Let (X, B) be a threefold sub-pair, $C \subset X$ an integral curve with generic point η, and H a finite group acting faithfully on X, preserving B and fixing C pointwise. Let t be an H-invariant rational function on X which is regular at η and non-constant on C, $U \subset X$ the open subset on which t is regular, $k := \mathbb{C}(t)$, X_k the generic fiber of $t : U \to \mathbb{A}^1$, $c \in X_k$ the closed point given by η, and B_k the restriction of B. Let Ω be an algebraic closure of k, c' a closed point of $X_k \otimes_k \Omega$ over c, X_Ω the connected component containing c', B_Ω the restriction of $B_k \otimes_k \Omega$, and $\varpi : X_\Omega \to X$ the induced morphism. Then the following hold.*

(0) $\mathcal{O}_{X_k,c} = \mathcal{O}_{X,\eta}$, so that the prime divisors over X with center C are exactly the prime divisors over X_k with center c, and $a(F, X, B) = a(F, X_k, B_k)$ for any such prime divisor

F. The group H, acting on $X_k \otimes_k \Omega$ through the first factor, preserves X_Ω, acts faithfully on X_Ω and fixes c', and ϖ is H-equivariant.

(1) *For any prime divisor F over X_k with center c, there exists a prime divisor F' over X_Ω with center c' such that $\operatorname{mult}_{F'}(\varpi^*\zeta) = \operatorname{mult}_F \zeta$ for any non-zero rational function ζ on X.*

(2) *For any prime divisor F'' over X_Ω with center c', there exists a prime divisor F over X_k with center c such that*

$$\operatorname{mult}_F \zeta = \operatorname{mult}_{F''}(\varpi^*\zeta) \quad \text{for any non-zero rational function } \zeta \text{ on } X,$$
$$a(F, X_k, B_k) = a(F'', X_\Omega, B_\Omega), \quad \text{and} \quad I_H(F) = I_H(F'').$$

In particular, if F'' computes $\operatorname{mld}((X_\Omega \ni c', B_\Omega)/H)$, then F computes $\operatorname{mld}((X_k \ni c, B_k)/H) = \operatorname{mld}(X/H \ni \overline{\eta}, B_{X/H})$, where $\overline{\eta}$ is the image of η, and F is a prime divisor over X with center C.

(3) *If (X, B) is sub-klt near C, then (X_Ω, B_Ω) is sub-klt near c'.*

Proof. (0). Since $t|_C$ is non-constant, $g(t) \notin \mathfrak{m}_{X,\eta}$ for any non-zero $g \in \mathbb{C}[t]$, so any non-zero element of k is a unit of $\mathcal{O}_{X,\eta}$. Thus $\mathcal{O}_{X_k,c} = \mathcal{O}_{X,\eta}$, and $\mathbb{C}(X) = k(X_k)$. A valuation of $\mathbb{C}(X)$ with center η on X is non-negative on $\mathcal{O}_{X,\eta} \supset k$, hence trivial on k, and the transcendence degree over $\mathbb{C}$ of the residue field of such a valuation is one more than that over k. Thus the prime divisors over X with center C are exactly the prime divisors over X_k with center c. For any higher model $Y \to U$ with generic fiber Y_k over $\mathbb{A}^1$, Y_k is smooth over k in codimension 1, and the wedge product with dt is an isomorphism from $\omega_{Y_k/k}$ onto the restriction of ω_Y there. Thus K_{Y_k} is the restriction of K_Y, and likewise K_{X_k} is the restriction of K_X. Thus $a(F, X, B) = a(F, X_k, B_k)$ for any prime divisor F over X with center C.

The group H acts on U over $\mathbb{A}^1$, hence on X_k over k and on $X_k \otimes_k \Omega$ through the first factor, and ϖ is H-equivariant on X_Ω. Since H fixes C pointwise, H restricts to the identity of C, hence acts trivially on the generic point η of C, on c, whose residue field is that of η, and on $c \times_k \Omega$. Thus H fixes c' and preserves X_Ω, and H acts faithfully on X_Ω, since $g \circ \varpi = \varpi$ for $g \in H$ acting trivially on X_Ω, ϖ is dominant and H acts faithfully on X. Since $\operatorname{char} k = 0$ and X_k is normal, the connected components of $X_k \otimes_k k'$ are normal integral schemes for any algebraic extension k' of k.

We use the following fact. Let Y_0 be a normal variety over k, P a prime divisor on Y_0 with uniformizer τ of $\mathcal{O}_{Y_0,P}$, and k' an algebraic extension of k. Then $\mathcal{O}_{Y_0,P} \otimes_k k'$ is integral over $\mathcal{O}_{Y_0,P}$, the primes of $\mathcal{O}_{Y_0,P} \otimes_k k'$ containing τ correspond to the primes of $\kappa(P) \otimes_k k'$, a reduced ring of dimension 0, and the localization of $\mathcal{O}_{Y_0,P} \otimes_k k'$ at any such prime is a discrete valuation ring with uniformizer τ (†). This holds because $\mathcal{O}_{Y_0,P} \otimes_k k'$ is finite and étale over $\mathcal{O}_{Y_0,P}$ for k' finite over k, and the general case is the direct limit over the finite subextensions.

(1). Let $Y \to X_k$ be a higher model on which F is a prime divisor. By (†), the generic point of any irreducible component F' of $F \otimes_k \Omega \subset Y \otimes_k \Omega$ has a discrete valuation ring with uniformizer a uniformizer of $\mathcal{O}_{Y,F}$, so $\operatorname{mult}_{F'} \zeta = \operatorname{mult}_F \zeta$ for any $\zeta \in k(X_k)^\times$. The morphism $Y \otimes_k \Omega \to X_k \otimes_k \Omega$ is projective and birational on any connected component, and maps $F \otimes_k \Omega$ onto $c \times_k \Omega$, a finite set of closed points. Thus some component F' of $F \otimes_k \Omega$ maps to c', and F' is a prime divisor over X_Ω with center c' and $\operatorname{mult}_{F'}(\varpi^*\zeta) = \operatorname{mult}_F \zeta$.

(2). Let $Y'' \to X_\Omega$ be a higher model on which F'' is a prime divisor. Since Y'' and F'' are of finite type over Ω and Ω is the union of the finite extensions of k in it [Gro66, Théorème 8.8.2 and Théorème 8.10.5], there exist a finite extension $k' \subset \Omega$ of k, a higher model $Y_{k'} \to X_{k'}$ with $Y_{k'}$ normal, where $X_{k'}$ is the connected component of $X_k \otimes_k k'$ containing the image of c', and a prime divisor $F_{k'}$ on $Y_{k'}$ such that $Y'' = Y_{k'} \otimes_{k'} \Omega$ over X_Ω and F'' is a component of $F_{k'} \otimes_{k'} \Omega$. By (†), $\operatorname{mult}_{F''}$ extends $\operatorname{mult}_{F_{k'}}$ with ramification index 1.

The morphism $X_{k'} \to X_k$ is finite and étale, and $k'(X_{k'})$ is a finite extension of $k(X_k)$. The restriction of $\operatorname{mult}_{F_{k'}}$ to $k(X_k)$ is a discrete valuation v, and the residue field of $\operatorname{mult}_{F_{k'}}$ is a finite extension of that of v. Thus the residue field of v has transcendence degree $1 = \dim X_k - 1$ over

k, and $v = e\,\mathrm{mult}_F$ for a positive integer e and a prime divisor F over X_k. Let $Y \to X_k$ be a higher model on which F is a prime divisor, and let Y' be the connected component of $Y \otimes_k k'$ over $X_{k'}$. Then Y' is normal, finite and étale over Y, with function field $k'(X_{k'})$, so Y' is the normalization of Y in $k'(X_{k'})$. The valuations of $k'(X_{k'})$ extending mult_F are the local rings of Y' at the generic points of the components of $F \otimes_k k'$ in Y', which have ramification index 1 over mult_F by (†). As $\mathrm{mult}_{F_{k'}}$ is one of them, $e = 1$, and we may assume that $Y_{k'} = Y'$ and that $F_{k'}$ is a component of $F \otimes_k k'$. By (†), F'' is then a component of $F_{k'} \otimes_{k'} \Omega$. Since the center c' of F'' maps to c, the center of F on X_k is c. Thus $\mathrm{mult}_F \zeta = \mathrm{mult}_{F_{k'}} \zeta = \mathrm{mult}_{F''}(\varpi^*\zeta)$ for any $\zeta \in k(X_k)^\times$.

For the log discrepancies, let f denote the morphism $Y \to X_k$. The morphisms $Y' \to Y$ and $Y' \otimes_{k'} \Omega \to Y'$ are étale and flat respectively, and the differentials commute with base change, so the pullbacks of the $\mathbb{R}$-divisor $K_Y - f^*(K_{X_k} + B_k)$ are the corresponding $\mathbb{R}$-divisors on Y' and on $Y' \otimes_{k'} \Omega$. As $F_{k'}$ and F'' are components of $F \otimes_k k'$ and $F_{k'} \otimes_{k'} \Omega$ with ramification index 1, we obtain

$$a(F, X_k, B_k) = a(F_{k'}, X_{k'}, B_{k'}) = a(F'', X_\Omega, B_\Omega),$$

where $B_{k'}$ is the pullback of B_k.

For the inertia groups, let $g \in I_H(F'')$. Then g preserves the restriction mult_F of $\mathrm{mult}_{F''}$ and acts trivially on the residue field of mult_F, which embeds in that of $\mathrm{mult}_{F''}$. Thus $I_H(F'') \subset I_H(F)$. Conversely, let R be the valuation ring of mult_F in $k(X_k)$, τ a uniformizer of R, and $R_\Omega := R \otimes_k \Omega$. The valuation ring of $\mathrm{mult}_{F''}$ contains the image of R_Ω and $\mathrm{mult}_{F''}(\tau) = 1$, so it dominates the localization of R_Ω at a prime $\mathfrak{q} \ni \tau$. By (†), $(R_\Omega)_\mathfrak{q}$ is a discrete valuation ring with fraction field equal to that of the valuation ring of $\mathrm{mult}_{F''}$, so the valuation ring of $\mathrm{mult}_{F''}$ is $(R_\Omega)_\mathfrak{q}$. Let $g \in I_H(F)$. Then g preserves R, acts Ω-linearly on R_Ω and trivially on $R_\Omega/\tau R_\Omega = \kappa(F) \otimes_k \Omega$, so g fixes $\mathfrak{q}$, preserves $(R_\Omega)_\mathfrak{q}$, and acts trivially on the residue field of $(R_\Omega)_\mathfrak{q}$, a localization of $\kappa(F) \otimes_k \Omega$. Since g also preserves X_Ω, $g \in I_H(F'')$. Thus $I_H(F) = I_H(F'')$.

Assume that F'' computes $\mathrm{mld}((X_\Omega \ni c', B_\Omega)/H)$. By [Liu26b, Lemma 3.22(2)(3)],

$$\mathrm{mld}((X_\Omega \ni c', B_\Omega)/H) = \mathrm{mld}((X_k \ni c, B_k)/H) = \mathrm{mld}(X/H \ni \overline{\eta}, B_{X/H}).$$

Since

$$\frac{a(F, X_k, B_k)}{e_F} = \frac{a(F'', X_\Omega, B_\Omega)}{e_{F''}},$$

F computes $\mathrm{mld}((X_k \ni c, B_k)/H)$, and F is a prime divisor over X with center C by (0).

(3). Let $f : Y \to V$ be a log resolution of $(V, B|_V)$ for an open neighborhood $V \subset U$ of η on which (X, B) is sub-klt. Its base change to X_Ω is a log resolution of (X_Ω, B_Ω) near c': as $\mathrm{char}\, k = 0$, the base change of each component of the exceptional locus of f and of $f_*^{-1}B$ is reduced and smooth, and the coefficient of any component of the base change of a prime divisor P on Y in the base change of $K_Y - f^*(K_V + B|_V)$ is $\mathrm{mult}_P(K_Y - f^*(K_V + B|_V))$, as for the log discrepancies in (2). Thus (X_Ω, B_Ω) is sub-klt near c'. □

Lemma 3.6 (cf. [Liu26a, Lemma 8.1], [Liu26b, Lemmas 3.1 and 3.2])**.** *Let $\Bbbk \supset \mathbb{C}$ be an algebraically closed field, X a normal quasi-projective variety over $\Bbbk$ with $\dim X > 0$, $x \in X$ a closed point, G a finite group acting faithfully on X by $\Bbbk$-automorphisms and fixing x, and $\mathfrak{a} = \prod_j \mathfrak{a}_j^{r_j}$ an $\mathbb{R}$-ideal on X with G-invariant factors $\mathfrak{a}_j$. Let B be an $\mathbb{R}$-divisor on X preserved by G such that $K_X + B$ is $\mathbb{R}$-Cartier. Let $\chi_1, \dots, \chi_n$ be characters of G, let G act on $X \times \mathbb{A}^n$, with coordinates $t_1, \dots, t_n$ on $\mathbb{A}^n$, by the given action on X and by $g^*t_i = \chi_i(g)t_i$, let $\varpi : X \times \mathbb{A}^n \to X$ be the projection, $B^{\mathrm{c}} := \varpi^*B + \sum_{i=1}^n \mathrm{div}(t_i)$,*

$$\mu := \mathrm{mld}((X \ni x, B, \mathfrak{a})/G) \quad \textit{and} \quad \mu^{\mathrm{c}} := \mathrm{mld}((X \times \mathbb{A}^n \ni (x, 0), B^{\mathrm{c}}, \varpi^*\mathfrak{a})/G).$$

Then the following hold.

(1) For any prime divisor F over $X \times \mathbb{A}^n$ with center $(x, 0)$, there exist a positive integer ϱ and a prime divisor E over X with center x such that $\mathrm{mult}_F(\varpi^*\zeta) = \varrho\,\mathrm{mult}_E \zeta$ for any

non-zero rational function ζ on X, $I_G(F) \subset I_G(E)$, and

$$a(F, X \times \mathbb{A}^n, B^{\mathrm{c}}, \varpi^* \mathfrak{a}) \geq \varrho a(E, X, B, \mathfrak{a}).$$

(2) *For any prime divisor E over X with center x, there exists a prime divisor F over $X \times \mathbb{A}^n$ with center $(x, 0)$ such that $I_G(F) = I_G(E)$ and*

$$a(F, X \times \mathbb{A}^n, B^{\mathrm{c}}, \varpi^* \mathfrak{a}) = a(E, X, B, \mathfrak{a}).$$

In particular, $\mu^{\mathrm{c}} \leq \mu$.

(3) *If $\mu \geq 0$ or $\mu^{\mathrm{c}} \geq 0$, then $\mu^{\mathrm{c}} = \mu$.*

(4) *Assume in addition that G is abelian and acts diagonally on $\mathbb{A}^n$ with coordinates $x_1, \ldots, x_n$ by $g^* x_i = \chi_i(g) x_i$, and that $X \subset \mathbb{A}^n$ is a G-invariant closed subvariety with $x = 0$ on which G acts faithfully near 0. Let G act on $\mathbb{A}^n \times \mathbb{A}^n$ by the action on the first factor and by $g^* t_i = \chi_i(g) t_i$. Then, near $(0, 0)$, for any $g \in G \backslash \{1\}$,*

$$\dim \mathrm{Fix}_{\mathbb{A}^n \times \mathbb{A}^n}(g) \leq 2n - 2 \quad \text{and} \quad \dim \mathrm{Fix}_{X \times \mathbb{A}^n}(g) \leq \dim X + n - 2.$$

In particular, the following hold.

(a) *G acts on $\mathbb{A}^n \times \mathbb{A}^n$ without pseudoreflections.*

(b) *The quotient morphisms of $\mathbb{A}^n \times \mathbb{A}^n$ and of $X \times \mathbb{A}^n$ are étale in codimension 1.*

(c) *Near the image of $(0, 0)$, the image in $(\mathbb{A}^n \times \mathbb{A}^n)/G$ of the locus where $\mathbb{A}^n \times \mathbb{A}^n \to (\mathbb{A}^n \times \mathbb{A}^n)/G$ is not étale intersects $(X \times \mathbb{A}^n)/G$ in a subset of dimension at most $\dim X + n - 2$.*

(5) *The statements (1)–(3) hold as well with X replaced by $\operatorname{Spec} \widehat{\mathcal{O}}_{X,x}$, x by the closed point $\widehat{x}$ of $\operatorname{Spec} \widehat{\mathcal{O}}_{X,x}$, $X \times \mathbb{A}^n$ by $\operatorname{Spec} \widehat{\mathcal{O}}_{X,x}[[t_1, \ldots, t_n]]$, the completion of $X \times \mathbb{A}^n$ at $(x, 0)$, $(x, 0)$ by its closed point $(\widehat{x}, 0)$, B by a G-invariant $\mathbb{R}$-divisor B on $\operatorname{Spec} \widehat{\mathcal{O}}_{X,x}$ with $K_{\operatorname{Spec} \widehat{\mathcal{O}}_{X,x}} + B$ $\mathbb{R}$-Cartier, and $\mathfrak{a}$ by an $\mathbb{R}$-ideal on $\operatorname{Spec} \widehat{\mathcal{O}}_{X,x}$ with G-invariant factors.*

Proof. (1). Let v be the valuation of F. Since the center of F is $(x, 0)$, v is positive on the non-zero functions on X vanishing at x, so the restriction of v to $\Bbbk(X)$ is non-trivial, and the value group of $v|_{\Bbbk(X)}$ is $\varrho\mathbb{Z}$ for a positive integer ϱ. Let $w := v|_{\Bbbk(X)}/\varrho$. By [Bou72, Chapter VI, §10, no. 3, Corollary 1 to Theorem 1] for the extension $\Bbbk(X \times \mathbb{A}^n)/\Bbbk(X)$, the residue field of w has transcendence degree at least $(\dim X + n - 1) - n = \dim X - 1$ over $\Bbbk$. By [Bou72, Chapter VI, §10, no. 3, Corollary 1 to Theorem 1] for $\Bbbk(X)/\Bbbk$, it has transcendence degree at most $\dim X - 1$. Thus $w = \mathrm{mult}_E$ for a prime divisor E over X, and the center of E is x, as $w \geq 0$ on $\mathcal{O}_{X,x}$ and $w > 0$ on $\mathfrak{m}_x$. Since G preserves $\Bbbk(X) \subset \Bbbk(X \times \mathbb{A}^n)$, any $g \in I_G(F)$ preserves w and acts trivially on the residue field of w, which embeds in that of v. Thus $I_G(F) \subset I_G(E)$.

Let $f : Y \to X$ be a higher model with Y normal on which E is a prime divisor, and let $K_Y + B_Y := f^*(K_X + B)$, so that $\mathrm{mult}_E B_Y = 1 - a(E, X, B)$. Possibly shrinking Y to an open subset meeting E, we may assume that Y and E are smooth. The restriction of v to $\Bbbk(Y)$ is $\varrho\,\mathrm{mult}_E$, and $v(t_i) > 0$ for any i, so the center of F on $Y \times \mathbb{A}^n$ is the generic point ξ of $E \times \{0\}$. Let τ be a local equation of E at the generic point of E. Then $\tau, t_1, \ldots, t_n$ are regular parameters at ξ, and $v(\tau) = \varrho$. Since $(Y \times \mathbb{A}^n, E \times \mathbb{A}^n + \sum_i \mathrm{div}(t_i))$ is log smooth near ξ, it is lc there, so

$$a(F, Y \times \mathbb{A}^n, 0) \geq v(\tau) + \sum_{i=1}^{n} v(t_i).$$

The pullback of $K_{X \times \mathbb{A}^n} + B^{\mathrm{c}}$ to $Y \times \mathbb{A}^n$ is

$$K_{Y \times \mathbb{A}^n} + \mathrm{pr}_Y^* B_Y + \sum_i \mathrm{div}(t_i),$$

and near ξ the only component of $\mathrm{pr}_Y^* B_Y$ is $E \times \mathbb{A}^n$. Thus

$$a(F, X \times \mathbb{A}^n, B^{\mathrm{c}}) = a(F, Y \times \mathbb{A}^n, 0) - (1 - a(E, X, B))v(\tau) - \sum_{i=1}^{n} v(t_i)$$
$$\geq a(E, X, B)v(\tau) = \varrho a(E, X, B).$$

Since

$$\mathrm{mult}_F(\mathfrak{a}_j \mathcal{O}_{X \times \mathbb{A}^n}) = \varrho \, \mathrm{mult}_E \, \mathfrak{a}_j \quad \text{for any } j,$$

we obtain

$$a(F, X \times \mathbb{A}^n, B^{\mathrm{c}}, \varpi^* \mathfrak{a}) \geq \varrho a(E, X, B, \mathfrak{a}).$$

(2). Let $I := I_G(E)$. By Lemma 2.15(1), I is cyclic and acts faithfully on the line $\mathfrak{m}_E/\mathfrak{m}_E^2$ through a character χ_E, so χ_E generates the character group of I, and for any i there exists $q_i \in \{1, \dots, e_E\}$ with $\chi_i|_I = \chi_E^{q_i}$. Let $\tau \in \Bbbk(X)$ with $\mathrm{mult}_E \tau = 1$. For $0 \leq i \leq n$, let $V_i := X \times \mathbb{A}^i$ with coordinates $t_1, \dots, t_i$, let

$$B_i := \mathrm{pr}_X^* B + \sum_{i' \leq i} \mathrm{div}(t_{i'}),$$

let $v_0 := \mathrm{mult}_E$ and, for $i \geq 1$, let v_i be the Gauss extension of v_{i-1} to $\Bbbk(V_{i-1})(t_i)$ with t_i-value q_i (Definition 2.3). We prove by induction on i the following statement (S_i):

(a) $v_i = \mathrm{mult}_{F_i}$ for a prime divisor F_i over V_i with center $(x, 0)$, where $F_0 := E$.
(b) The valuation of F_i restricts to mult_E on $\Bbbk(X)$.
(c) $I_G(F_i) = I$.
(d) The function τ is a uniformizer of mult_{F_i} on which I acts through χ_E modulo $\mathfrak{m}_{F_i}^2$.
(e) $a(F_i, V_i, B_i) = a(E, X, B)$.

Since $F_0 = E$, $V_0 = X$, and $B_0 = B$, (S_0) holds. Let $i \geq 1$ and suppose that (S_{i-1}) holds. Since $I_G(F_{i-1}) = I$ and I acts on $\mathfrak{m}_{F_{i-1}}/\mathfrak{m}_{F_{i-1}}^2$ through χ_E, [Liu26b, Lemma 3.1], with the integer q_i, implies that $v_i = \mathrm{mult}_{F_i}$ for a prime divisor F_i over V_i with center $(x, 0)$, $I_G(F_i) = I$, and $a(F_i, V_i, B_i) = a(F_{i-1}, V_{i-1}, B_{i-1})$. The valuation v_i restricts to v_{i-1} with value group $\mathbb{Z}$, so τ is a uniformizer of mult_{F_i}. Since $\mathfrak{m}_{F_{i-1}}^2 \subset \mathfrak{m}_{F_i}^2$, I acts on the class of τ in $\mathfrak{m}_{F_i}/\mathfrak{m}_{F_i}^2$ through χ_E. This proves (S_i). Let $F := F_n$. Since $V_n = X \times \mathbb{A}^n$, $B_n = B^{\mathrm{c}}$ and mult_F restricts to mult_E, for any j we have

$$\mathrm{mult}_F(\mathfrak{a}_j \mathcal{O}_{X \times \mathbb{A}^n}) = \mathrm{mult}_E \, \mathfrak{a}_j, \quad \text{so} \quad a(F, X \times \mathbb{A}^n, B^{\mathrm{c}}, \varpi^* \mathfrak{a}) = a(E, X, B, \mathfrak{a}).$$

Thus

$$\mu^{\mathrm{c}} \leq \frac{a(F, X \times \mathbb{A}^n, B^{\mathrm{c}}, \varpi^* \mathfrak{a})}{e_F} = \frac{a(E, X, B, \mathfrak{a})}{e_E},$$

and taking the infimum over E, $\mu^{\mathrm{c}} \leq \mu$.

(3). By (2), $\mu^{\mathrm{c}} \leq \mu$, and if $\mu^{\mathrm{c}} \geq 0$ then $\mu \geq 0$. Assume that $\mu \geq 0$, and let F be a prime divisor over $X \times \mathbb{A}^n$ with center $(x, 0)$. Let ϱ and E be as in (1). Since e_F divides e_E and $a(E, X, B, \mathfrak{a}) \geq e_E \mu \geq 0$,

$$\frac{a(F, X \times \mathbb{A}^n, B^{\mathrm{c}}, \varpi^* \mathfrak{a})}{e_F} \geq \varrho \frac{e_E}{e_F} \cdot \frac{a(E, X, B, \mathfrak{a})}{e_E} \geq \frac{a(E, X, B, \mathfrak{a})}{e_E} \geq \mu.$$

Thus $\mu^{\mathrm{c}} \geq \mu$, and $\mu^{\mathrm{c}} = \mu$.

(4). Let $g \in G \backslash \{1\}$. Since G acts faithfully on X near 0, g acts non-trivially on X, hence on $\mathbb{A}^n$, so $\chi_{i_g}(g) \neq 1$ for some i_g. Then

$$\mathrm{Fix}_{\mathbb{A}^n \times \mathbb{A}^n}(g) = \{x_i = t_i = 0 \text{ for any } i \text{ with } \chi_i(g) \neq 1\} \subset (x_{i_g} = t_{i_g} = 0),$$

so $\dim \mathrm{Fix}_{\mathbb{A}^n \times \mathbb{A}^n}(g) \leq 2n - 2$. Moreover $\mathrm{Fix}_{X \times \mathbb{A}^n}(g) = \mathrm{Fix}_X(g) \times \mathrm{Fix}_{\mathbb{A}^n}(g)$, where $\mathrm{Fix}_X(g) \subsetneq X$ near 0 as X is irreducible and g acts non-trivially near 0, and $\mathrm{Fix}_{\mathbb{A}^n}(g) \subset (t_{i_g} = 0)$. Thus $\dim \mathrm{Fix}_{X \times \mathbb{A}^n}(g) \leq (\dim X - 1) + (n - 1)$.

In particular no $g \neq 1$ fixes a hyperplane of $\mathbb{A}^n \times \mathbb{A}^n$, so G has no pseudoreflections, which proves (a). The quotient morphism $q : \mathbb{A}^n \times \mathbb{A}^n \to (\mathbb{A}^n \times \mathbb{A}^n)/G$ is étale off $N := \bigcup_{g \neq 1} \mathrm{Fix}_{\mathbb{A}^n \times \mathbb{A}^n}(g)$, as

the inertia groups of the points off N are trivial [Gro71, Exposé V, Corollaire 2.3], and likewise the quotient morphism of $X\times\mathbb{A}^n$ is étale off $N\cap(X\times\mathbb{A}^n)=\bigcup_{g\neq1}\mathrm{Fix}_{X\times\mathbb{A}^n}(g)$. By the dimension bounds, q and the quotient morphism of $X\times\mathbb{A}^n$ are étale in codimension 1, which proves (b). Since $X\times\mathbb{A}^n$ is G-invariant and q is finite, $(X\times\mathbb{A}^n)/G=q(X\times\mathbb{A}^n)$ and

$$q(N)\cap q(X\times\mathbb{A}^n)=q\Big(\bigcup_{g\neq1}\mathrm{Fix}_{X\times\mathbb{A}^n}(g)\Big).$$

Thus $\dim\big(q(N)\cap q(X\times\mathbb{A}^n)\big)\leq\dim X+n-2$ near the image of $(0,0)$, which proves (c). This proves (4).

(5). The proofs of (1)–(3) use a normal higher model on which a given prime divisor is a divisor, the inequality $a(F,Y\times\mathbb{A}^n,0)\geq v(\tau)+\sum_i v(t_i)$ at a point ξ with regular parameters $\tau,t_1,\dots,t_n$ [dFEM11, Lemma A.11(iii)], orders of finitely many functions, the transcendence degree count in (1), Lemma 2.15(1) and [Liu26b, Lemma 3.1]. Over $\operatorname{Spec}\widehat{\mathcal{O}}_{X,x}$ and $\operatorname{Spec}\widehat{\mathcal{O}}_{X,x}[t_1,\dots,t_n]$, a prime divisor is, as in [Liu26a, Definition 3.2], a prime divisor on a normal scheme proper and birational over it, and such a scheme is excellent with open regular locus [Gro65, Scholie 7.8.3 (ii)–(iv)], so the shrinking in (1) is possible. The other inputs hold over $\widehat{\mathcal{O}}_{X,x}$ and $\widehat{\mathcal{O}}_{X,x}[t_1,\dots,t_n]$. In (1), $\operatorname{trdeg}_{\Bbbk}\kappa(F)=\dim X+n-1$ by [Mat89, Theorem 15.6], so Abhyankar's inequality [Abh56, Theorem 1(1)] and [HS06, Theorem 9.3.2] make w a formal prime divisor. Lemma 2.15(1) is transferred by Lemma 2.12(3), and [Liu26b, Lemma 3.1] holds there, as its proof (which constructs the prime divisor and computes its inertia group and its log discrepancies) is a computation at the generic point of the prime divisor on a model regular there, with the relative canonical divisor of [dFEM11, Section 2]. Thus (1)–(3) hold over the local ring of $\operatorname{Spec}\widehat{\mathcal{O}}_{X,x}[t_1,\dots,t_n]$ at $(\widehat{x},0)$. Finally, we show that the prime divisors with center the closed point over $\operatorname{Spec}\widehat{\mathcal{O}}_{X,x}[[t_1,\dots,t_n]]$ and over the local ring R of $\operatorname{Spec}\widehat{\mathcal{O}}_{X,x}[t_1,\dots,t_n]$ at $(\widehat{x},0)$ correspond, with isomorphic residue fields, equal inertia groups and equal log discrepancies with respect to B^{c} (cf. [Liu26a, Lemma 3.3(3)]). Let $\mathfrak{m}_R$ be the maximal ideal of R. As $\widehat{\mathcal{O}}_{X,x}$ is complete, the completion of R is $\widehat{R}:=\widehat{\mathcal{O}}_{X,x}[[t_1,\dots,t_n]]$, and $R\to\widehat{R}$ is faithfully flat and regular, as R is excellent [Mat89, Theorem 8.14], [Gro65, Scholie 7.8.3 (ii)–(v)]. Since $\widehat{R}/\mathfrak{m}_R^N\widehat{R}=R/\mathfrak{m}_R^N$ for any $N\geq1$, the maps $J\to J\cap R$ and $I\to I\widehat{R}$ are mutually inverse bijections between the ideals of $\widehat{R}$ containing $\mathfrak{m}_R^N\widehat{R}$ and the ideals of R containing $\mathfrak{m}_R^N$.

Let P be a prime divisor over $\operatorname{Spec}\widehat{R}$ with center the closed point, $v:=\mathrm{mult}_P$, and, for $m\geq0$, $\mathfrak{c}_m:=\{u\in\widehat{R}\mid v(u)\geq m\}$ and $\mathfrak{b}_m:=\mathfrak{c}_m\cap R$. As $v\geq1$ on $\mathfrak{m}_R\widehat{R}$, we have $\mathfrak{m}_R^m\widehat{R}\subset\mathfrak{c}_m$, so

$$\mathfrak{c}_m=\mathfrak{b}_m\widehat{R}\quad\text{and}\quad\mathfrak{c}_m/\mathfrak{c}_{m+1}=\mathfrak{b}_m/\mathfrak{b}_{m+1}.$$

Thus v is determined by its restriction v_0 to the fraction field of R, v_0 has value group $\mathbb{Z}$, and the residue field of v_0 is that of v, since the residue of u_1/u_2 for $u_1,u_2\in\mathfrak{c}_m\backslash\mathfrak{c}_{m+1}$ depends only on the classes of u_1,u_2 in $\mathfrak{c}_m/\mathfrak{c}_{m+1}$. The residue field of v_0 has transcendence degree $\dim R-1$ over $\Bbbk$ [Mat89, Theorem 15.6], so $v_0=\mathrm{mult}_E$ for a prime divisor E over $\operatorname{Spec}R$ with center $\mathfrak{m}_R$ [HS06, Theorem 9.3.2].

Conversely, let E be a prime divisor over $\operatorname{Spec}R$ with center $\mathfrak{m}_R$, $Y\to\operatorname{Spec}R$ a resolution on which E is a prime divisor [dFEM11, Theorems 2.3 and 2.4], $W:=Y\times_{\operatorname{Spec}R}\operatorname{Spec}\widehat{R}$, and $h:W\to Y$ the projection. Then h is flat and regular, W is regular, and $W\to\operatorname{Spec}\widehat{R}$ is proper and birational, as $\widehat{R}$ is a domain. As $\mathfrak{m}_R\mathcal{O}_E=0$, the morphism $h^{-1}(E)=E\times_{\operatorname{Spec}R}\operatorname{Spec}\widehat{R}\to E$ is an isomorphism, so the local ring of W at the generic point of $h^{-1}(E)$ is flat over the discrete valuation ring $\mathcal{O}_{Y,E}$ with closed fiber a field. Hence $P:=h^*E$ is a prime divisor over $\operatorname{Spec}\widehat{R}$ with center the closed point, and mult_P restricts to mult_E.

The group G acts compatibly on R and $\widehat{R}$, and $g\in G$ preserves v if and only if it preserves $\mathfrak{c}_m$ for any m, if and only if it preserves $\mathfrak{b}_m$ for any m, if and only if it preserves v_0. The identification of residue fields is G-equivariant, so P and E have equal inertia groups. The special differentials of [dFEM11, Appendix A] of $\widehat{R}$ are those of R tensored with $\widehat{R}$, since each module is generated

by the special differentials of $\widehat{\mathcal{O}}_{X,x}$ and $dt_1,\dots,dt_n$, so $K_W = h^*K_Y$ and $K_{\operatorname{Spec}\widehat{R}}$ is the pullback of $K_{\operatorname{Spec}R}$. Writing $K_Y + B_Y$ for the pullback of $K_{\operatorname{Spec}R} + B^{\mathrm{c}}$ to Y, the pullback of $K_{\operatorname{Spec}\widehat{R}} + B^{\mathrm{c}}$ to W is $K_W + h^*B_Y$, and $h^*E = P$, so $a(P, \operatorname{Spec}\widehat{R}, B^{\mathrm{c}}) = a(E, \operatorname{Spec}R, B^{\mathrm{c}})$. □

3.4. **Normal forms and rays near circles.**

Lemma 3.7 (cf. [Liu26a, Lemmas 11.3, 12.4 and 12.12], [Liu26b, Definition-Lemma 11.5(1)])**.** *Let $(W \ni o)$ be an isolated Gorenstein terminal hypersurface threefold germ of quadratic rank 1 whose residual cubic is the cube of a linear form, G a finite abelian group acting faithfully on W and fixing o, and h a mark on $(W \ni o)$. Then there exist a smooth fourfold germ $(\mathcal{X} \ni o)$ with a faithful G-action, a G-equivariant closed embedding $W \subset \mathcal{X}$, and G-semi-invariant elements x, y, z, t of the maximal ideal of $\widehat{\mathcal{O}}_{\mathcal{X},o}$ with $\widehat{\mathcal{O}}_{\mathcal{X},o} = \mathbb{C}[[x,y,z,t]]$, on which G acts diagonally, satisfying the following.*

(1) We have

$$\widehat{\mathcal{O}}_{W,o} = \mathbb{C}[[x,y,z,t]]/(x^2 + y^3 + y\alpha(z,t) + \beta(z,t)),$$

where $\alpha \in (z,t)^3$ and $\beta \in (z,t)^4$ are power series in z,t, each zero or G-semi-invariant, with characters χ_y^2 and χ_y^3 respectively, χ_y being the character of y.

(2) Let α_3 be the homogeneous part of degree 3 of α, and β_j the homogeneous part of degree j of β for $j \in \{4,5\}$. Then exactly one of the following holds.

(E_6) $\beta_4 \neq 0$.

(E_7) $\beta_4 = 0$ and $\alpha_3 \neq 0$.

(E_8) $\beta_4 = \alpha_3 = 0$ and $\beta_5 \neq 0$.

(3) The image of h in $\widehat{\mathcal{O}}_{W,o}$ is the class of a unique element $\varphi(y,z,t) + x\xi(y,z,t)$ with $\varphi, \xi \in \mathbb{C}[[y,z,t]]$, and $\varphi + x\xi$ is G-semi-invariant with character χ_h.

Proof. By [Liu26a, Lemma 11.3(1)], there exist $(\mathcal{X} \ni o)$, $W \subset \mathcal{X}$, and x, y_1, y_2, y_3 as in the lemma, with y_1, y_2, y_3 in place of y, z, t, such that

$$\widehat{\mathcal{O}}_{W,o} = \mathbb{C}[[x, y_1, y_2, y_3]]/(x^2 + g)$$

for a G-semi-invariant $g \in (y_1, y_2, y_3)^3 \subset \mathbb{C}[[y_1, y_2, y_3]]$ whose cubic part g_3 is the residual cubic. By assumption, $g_3 = l^3$ for a non-zero linear form l in y_1, y_2, y_3.

We show that l is G-semi-invariant. By [Liu26a, Lemma 11.3(2)], the set of linear factors of g_3, counted with multiplicities, is G-stable. Any linear factor of $g_3 = l^3$ is a non-zero multiple of l, so for any $\gamma \in G$, $\gamma^* l = c_\gamma l$ for some $c_\gamma \in \mathbb{C}^*$. Thus l is G-semi-invariant. Since G acts diagonally on y_1, y_2, y_3, any y_j occurring in l with non-zero coefficient has the character of l. Possibly reordering y_1, y_2, y_3, we may assume that y_1 occurs in l with non-zero coefficient. Let

$$y := l, \quad z := y_2, \quad \text{and} \quad t := y_3.$$

Then x, y, z, t are G-semi-invariant, G acts diagonally on them, $\mathbb{C}[[x, y_1, y_2, y_3]] = \mathbb{C}[[x,y,z,t]]$, and the cubic part of g as a power series in y, z, t is y^3.

By [Liu26a, Lemma 12.4], there exist a G-semi-invariant power series $y' \in \mathbb{C}[[y,z,t]]$ with character χ_y and $y' - y \in (y,z,t)^2$, and power series $\alpha, \beta \in \mathbb{C}[[z,t]]$ with $\alpha \in (z,t)^3$ and $\beta \in (z,t)^4$, each zero or G-semi-invariant, with characters χ_y^2 and χ_y^3 respectively, such that

$$g = y'^3 + \alpha(z,t)y' + \beta(z,t).$$

Since $y' - y \in (y,z,t)^2$, the elements x, y', z, t generate the maximal ideal of $\mathbb{C}[[x,y,z,t]]$, and $\mathbb{C}[[x,y,z,t]] = \mathbb{C}[[x,y',z,t]]$. Replacing y with y', we obtain (1). By [Liu26a, Lemma 12.12], α_3, β_4, and β_5 are not all zero, which implies (2).

Let $f := x^2 + y^3 + y\alpha + \beta$. Since $f(x,0,0,0) = x^2$, by the Weierstrass division theorem [GLS07, Theorem I.1.8], any $P \in \mathbb{C}[[x,y,z,t]]$ is uniquely of the form

$$P = Qf + \varphi + x\xi, \quad Q \in \mathbb{C}[[x,y,z,t]], \ \varphi, \xi \in \mathbb{C}[[y,z,t]].$$

Thus the image of h is the class of a unique $\varphi + x\xi$ with $\varphi, \xi \in \mathbb{C}[[y,z,t]]$. The surjection $\mathbb{C}[[x,y,z,t]] \to \widehat{\mathcal{O}}_{W,o}$ is G-equivariant, and G acts diagonally on x,y,z,t. Hence for any $\gamma \in G$,

$$\gamma^*(\varphi + x\xi) = \gamma^*\varphi + \chi_x(\gamma)x\gamma^*\xi, \quad \gamma^*\varphi, \gamma^*\xi \in \mathbb{C}[[y,z,t]],$$

where χ_x is the character of x, and the class of $\gamma^*(\varphi + x\xi)$ is the image of $\gamma^* h = \chi_h(\gamma)h$. By uniqueness,

$$\gamma^*(\varphi + x\xi) = \chi_h(\gamma)(\varphi + x\xi),$$

which implies (3). □

Definition 3.8. Let $(x_1, x_2, x_3, x_4) := (x, y, z, t)$ be the coordinates of $\mathbb{A}^4$, $\omega = (\omega_1, \ldots, \omega_4)$ a primitive vector in $(\mathbb{N}^+)^4$, and $i \in \{1, 2, 3, 4\}$.

(1) The *x_i-chart* of the ω-weighted blow-up of $\mathbb{A}^4$ is $\mathbb{A}^4_{(i)}/\mu_{\omega_i}$, where $\mathbb{A}^4_{(i)}$ is the affine space with the coordinates ρ and X_j for $j \neq i$, the morphism to $\mathbb{A}^4$ is given by

$$x_i = \rho^{\omega_i}, \quad x_j = \rho^{\omega_j} X_j \ (j \neq i),$$

and $\zeta \in \mu_{\omega_i}$ acts by $\rho \to \zeta\rho$ and $X_j \to \zeta^{-\omega_j} X_j$ for $j \neq i$. We call ρ the *chart parameter*, and write X, Y, Z, T for X_1, X_2, X_3, X_4.

(2) Let $W = (f = 0) \subset \mathbb{A}^4$ be a hypersurface through 0, and

$$\widetilde{f} := \rho^{-\operatorname{wt}_\omega(f)} f(\rho^{\omega_1} X_1, \ldots, \rho^{\omega_4} X_4) \quad \text{with } X_i := 1.$$

Then $\widetilde{f}$ is a polynomial in ρ and X_j, $j \neq i$, with

$$\widetilde{f}|_{\rho=0} = \operatorname{in}_\omega(f)|_{X_i=1} \neq 0,$$

and $(\widetilde{f} = 0)$ is the closure in $\mathbb{A}^4_{(i)}$ of the preimage of $W \backslash \{0\}$, that is, the strict transform of W. The *central surface* of W on the x_i-chart is the surface $(\widetilde{f} = \rho = 0)$ in $\mathbb{A}^4_{(i)}$, that is, the surface $(\operatorname{in}_\omega(f)|_{X_i=1} = 0)$ in $\mathbb{A}^3$ with the coordinates X_j, $j \neq i$.

(3) Let $w \in \mathbb{Z}^4$ be a primitive vector, G a finite group acting diagonally on x, y, z, t, by $x_j \to \gamma_j x_j$ for $\gamma = (\gamma_1, \ldots, \gamma_4) \in G$, and preserving W, $R := G \cap \mathbf{T}_{(w)}$, $r := |R|$, and L an integer with $\gcd(L, r) = 1$ and $\omega \equiv Lw \pmod{r}$. Let

$$\widetilde{G}_i := \{(\rho, X_j)_{j\neq i} \to (\lambda\rho, \gamma_j \lambda^{-\omega_j} X_j)_{j\neq i} \mid \gamma \in G,\ \lambda \in \mathbb{C}^*,\ \lambda^{\omega_i} = \gamma_i\}$$

and

$$\mu_r^{\mathrm{rad}} := \{(\rho, X_j)_{j\neq i} \to (\lambda\rho, X_j)_{j\neq i} \mid \lambda \in \mu_r\},$$

groups of automorphisms of $\mathbb{A}^4_{(i)}$. The quotient of $\mathbb{A}^4_{(i)}$ by μ_r^{rad} is the affine space $\mathbb{A}^4_{\mathrm{rad}}$ with the coordinates X_j, $j \neq i$, and $\varsigma := \rho^r$. Let $\overline{G}_i$ be the group of automorphisms of $\mathbb{A}^4_{\mathrm{rad}}$ induced by $\widetilde{G}_i$. We call $\mathbb{A}^4_{\mathrm{rad}}/\overline{G}_i$ the *quotient chart* and $\overline{G}_i$ the *group of the quotient chart*.

Lemma 3.9. *Notation as in Definition 3.8(3). Then $\widetilde{G}_i$ is a diagonal group of order $\omega_i|G|$ containing μ_{ω_i} and μ_r^{rad}, and the quotient of the x_i-chart by G is $\mathbb{A}^4_{(i)}/\widetilde{G}_i$. The kernel of $\widetilde{G}_i \to \overline{G}_i$ is μ_r^{rad}, so the quotient chart is the quotient of the x_i-chart by G, $|\overline{G}_i| = \omega_i[G : R]$, and μ_{ω_i} maps injectively into $\overline{G}_i$.*

Proof. For each $\gamma \in G$ exactly ω_i values of λ satisfy $\lambda^{\omega_i} = \gamma_i$, and distinct pairs (λ, γ) give distinct automorphisms, so $|\widetilde{G}_i| = \omega_i|G|$. The pairs with $\gamma = 1$ form μ_{ω_i}, and $\widetilde{G}_i/\mu_{\omega_i}$ acts on the x_i-chart as G, so the quotient of the chart by G is $\mathbb{A}^4_{(i)}/\widetilde{G}_i$.

Let L' be an integer with $LL' \equiv 1 \pmod{r}$. Any $\gamma \in R$ is $(\zeta^{w_1}, \ldots, \zeta^{w_4})$ for some $\zeta \in \mu_r$, and $\lambda := \zeta^{L'}$ satisfies $\lambda^{\omega_j} = \zeta^{w_j}$ for any j, since $L'\omega_j \equiv w_j \pmod{r}$, so (λ, γ) acts by $\rho \to \lambda\rho$ and $X_j \to X_j$. As ζ runs over μ_r so does λ, hence $\mu_r^{\mathrm{rad}} \subset \widetilde{G}_i$. An element (λ, γ) acts trivially on $\mathbb{A}^4_{\mathrm{rad}}$ if and only if $\lambda^r = 1$ and $\gamma_j = \lambda^{\omega_j}$ for $j \neq i$, that is, if and only if $(\lambda, \gamma) \in \mu_r^{\mathrm{rad}}$. Thus

$$|\overline{G}_i| = \omega_i|G|/r = \omega_i[G : R].$$

Since ω is primitive, the identity is the only element of μ_{ω_i} fixing all X_j, so $\mu_{\omega_i} \cap \mu_r^{\mathrm{rad}} = \{1\}$. The lemma follows. □

Lemma 3.10. *For $e \in \mathbb{N}^4$ let $\mathbf{x}^e := x^{e_1}y^{e_2}z^{e_3}t^{e_4}$ and $|e| := e_1+e_2+e_3+e_4$. Let $w \in \mathbb{Z}^4$ be a primitive vector, $M_w := \max(|w_1|,|w_2|,|w_3|,|w_4|)$, $\omega_0 \in \mathbb{Z}^4$, $r \geq 1$, and L integers, and $\Lambda \geq 1$ and $\sigma_+ > 0$ real numbers. Let $\omega := r\omega_0 + Lw$, and assume that $\omega \in [r/\Lambda, \Lambda r]^4$ and $0 < L/r \leq \sigma_+$. Let κ be an integer, $h \neq 0$ a polynomial in x,y,z,t whose monomials have $\mathbf{T}_{(w)}$-weights in $\kappa + r\mathbb{Z}$, with a monomial of $\mathbf{T}_{(w)}$-weight κ, d_h the least degree of a monomial of h of $\mathbf{T}_{(w)}$-weight κ, and*

$$j := \min\{\langle \omega_0, e\rangle \mid \mathbf{x}^e \text{ is a monomial of } h,\ \langle w, e\rangle = \kappa\}.$$

Assume that $r > |\kappa| + \Lambda^2 M_w d_h$. Then

$$|j| \leq \Lambda d_h + \sigma_+|\kappa| \quad \text{and} \quad \mathrm{wt}_\omega(h) = rj + L\kappa,$$

and $\mathrm{in}_\omega(h)$ is the sum of the terms of h whose monomials $\mathbf{x}^e$ satisfy $\langle w, e\rangle = \kappa$ and $\langle \omega_0, e\rangle = j$.

Proof. For any $e \in \mathbb{N}^4$,

$$\langle \omega, e\rangle = r\langle \omega_0, e\rangle + L\langle w, e\rangle \quad \text{and} \quad r|e|/\Lambda \leq \langle \omega, e\rangle \leq \Lambda r|e|.$$

Let $\mathbf{x}^{e_h}$ be a monomial of h with $\langle w, e_h\rangle = \kappa$ and $|e_h| = d_h$. Then

$$rj + L\kappa \leq \langle \omega, e_h\rangle \leq \Lambda r d_h,$$

so $j \leq \Lambda d_h + \sigma_+|\kappa|$. For any monomial $\mathbf{x}^e$ of h with $\langle w, e\rangle = \kappa$,

$$r\langle \omega_0, e\rangle = \langle \omega, e\rangle - L\kappa \geq -r\sigma_+|\kappa|,$$

so $j \geq -\sigma_+|\kappa|$. For any monomial $\mathbf{x}^e$ of h with $\langle w, e\rangle = \kappa + nr$ for an integer $n \neq 0$, we have $M_w|e| \geq |\langle w, e\rangle| \geq r - |\kappa|$, so

$$\langle \omega, e\rangle \geq \frac{r(r-|\kappa|)}{\Lambda M_w} > \Lambda r d_h \geq \langle \omega, e_h\rangle.$$

Thus the least ω-weight of a monomial of h is attained only at monomials $\mathbf{x}^e$ with $\langle w, e\rangle = \kappa$, at which $\langle \omega, e\rangle = r\langle \omega_0, e\rangle + L\kappa$. The lemma follows. □

Lemma 3.11 (cf. [Liu26b, Lemmas 4.9 and 4.12])**.** *Let $v > 0$ and c_1 be coprime integers with $c_1 \leq 0$ or $c_1 > 3v$, and $(v, c_1) \neq (1, 4)$. Let $w := (3v, c_1, 6v - 2c_1, 2v)$ and $\mathbf{T} := \mathbf{T}_{(w)}$.*

(1) If $(v, c_1) \neq (1, 5)$, then there exist integers s_0, y_0 with $c_1 s_0 - v y_0 = \mathrm{sgn}(c_1 - 3v)$ and a non-empty open interval $I \subset (0, +\infty)$ such that

$$s_0 + v\sigma > 0, \quad 0 < y_0 + c_1\sigma < 3(s_0 + v\sigma), \quad \text{and} \quad 5(s_0 + v\sigma) - (y_0 + c_1\sigma) < 1$$

for any $\sigma \in I$. If $(v, c_1) = (1, 5)$, then $s_0 = 0$, $y_0 = -1$, and $I = (\frac{1}{5}, \frac{1}{2})$ satisfy $c_1 s_0 - v y_0 = 1$, $s_0 + v\sigma > 0$, $0 < y_0 + c_1\sigma < 3(s_0 + v\sigma)$, and $5(s_0 + v\sigma) - (y_0 + c_1\sigma) = 1$ for any $\sigma \in I$.

(2) Fix s_0, y_0, I as in (1). For integers $r \geq 1$ and L with $\gcd(L, r) = 1$ and $L/r \in I$, let $S_r := rs_0 + vL$ and $Y_r := ry_0 + c_1 L$. Then $c_1 S_r - v Y_r = \mathrm{sgn}(c_1 - 3v)r$, $\gcd(S_r, Y_r) = 1$, the vector

$$\omega := (3S_r, Y_r, 6S_r - 2Y_r, 2S_r)$$

is primitive and positive with $\omega \equiv Lw \pmod r$, and $0 < 5S_r - Y_r \leq r$, with equality exactly when $(v, c_1) = (1, 5)$. For any compact subinterval $I' \subset I$, there exists a real number $\Lambda \geq 1$, depending only on v, c_1, s_0, y_0, I', such that $\omega \in [r/\Lambda, \Lambda r]^4$ whenever $L/r \in I'$.

(3) Let $I' \subset I$ be a compact subinterval, $g \in (y, z, t)^3 \subset \mathbb{C}[y, z, t]$ a polynomial with cubic part $y^2 z + t^3$ such that all monomials of $x^2 + g$ have $\mathbf{T}$-weight in $6v + r\mathbb{Z}$, and $W := (x^2 + g = 0) \subset \mathbb{A}^4$. If $L/r \in I'$ and r is large in terms of v, c_1, s_0, y_0, I', then $\mathrm{in}_\omega(x^2 + g) = x^2 + y^2 z + t^3$, the strict transform of W in the ω-weighted blow-up of $\mathbb{A}^4$ intersects the exceptional divisor in a single prime divisor E_ω, and $a(E_\omega, W, 0) = 5S_r - Y_r$. Let κ be an integer, $h \neq 0$ a polynomial of x-degree at most 1 whose monomials have $\mathbf{T}$-weights

in $\kappa + r\mathbb{Z}$, with a monomial of $\mathbf{T}$-weight κ, and d_h the least degree of a monomial of h of $\mathbf{T}$-weight κ. If $L/r \in I'$ and r is large in terms of $v, c_1, s_0, y_0, I', \kappa, d_h$, then $\mathrm{wt}_\omega(h)$ and $\mathrm{mult}_{E_\omega}(h|_W)$ are both equal to $rj + L\kappa$ for an integer j bounded in terms of $v, c_1, s_0, y_0, I', \kappa, d_h$.

(4) In the situation of (3), let G be a finite group acting diagonally on x, y, z, t and preserving W, with $R := G \cap \mathbf{T}$ of order r, and let $d := 3S_r - Y_r$. Then the central surfaces of W on the x-, y- and t-charts of the ω-weighted blow-up are smooth, and that on the z-chart has only an A_2 point. Let (X, Y, T, ς) be the coordinates of $\mathbb{A}^4_{\mathrm{rad}}$ for the z-chart (Definition 3.8(3)), in which the image of the A_2 point is the origin. Then the image of the subgroup $\mu_d \subset \mu_{2d} = \mu_{\omega_3}$ in the group $\overline{G}_3$ of the quotient chart acts on $\mathbb{A}^4_{\mathrm{rad}}$ as $\mu_d(-3, -3, -2, |c_1 - 3v|)$ and has index $2[G : R]$ in $\overline{G}_3$. If moreover $r > 2\Lambda(3|c_1 - 3v| + 6)$, where Λ is as in (2) for I', then the weights of $\mu_d(-3, -3, -2, |c_1 - 3v|)$ exclude from the equation F of the strict transform of W in the coordinates (X, Y, T, ς) a linear term in ς, any quadratic term involving ς, and any cubic term in T, ς other than T^3.

Proof. (1) Let $\ell(\eta_1, \eta_2) := c_1\eta_1 - v\eta_2$, and let

$$\Sigma := \{(\eta_1, \eta_2) \in \mathbb{R}^2 \mid \eta_1 > 0,\ 0 < \eta_2 < 3\eta_1,\ 5\eta_1 - \eta_2 < 1\}$$

be the open triangle with vertices $(0, 0)$, $(\frac{1}{5}, 0)$, and $(\frac{1}{2}, \frac{3}{2})$. Since $\gcd(v, c_1) = 1$, there exist integers s_0, y_0 with $c_1 s_0 - v y_0 = \mathrm{sgn}(c_1 - 3v)$, and then

$$(\ell = \mathrm{sgn}(c_1 - 3v)) = \{(s_0 + v\sigma, y_0 + c_1\sigma) \mid \sigma \in \mathbb{R}\}.$$

The three inequalities in (1) say that $(s_0 + v\sigma, y_0 + c_1\sigma) \in \Sigma$. Thus, if $(\ell = \mathrm{sgn}(c_1 - 3v)) \cap \Sigma \neq \emptyset$, then the set I_0 of such σ is a non-empty open interval. Replacing (s_0, y_0) with $(s_0 - nv, y_0 - nc_1)$ replaces I_0 with $I_0 + n$, and for an integer $n \gg 0$ we let $I := I_0 + n \subset (0, +\infty)$. It remains to show that $(\ell = \mathrm{sgn}(c_1 - 3v)) \cap \Sigma \neq \emptyset$ if $(v, c_1) \neq (1, 5)$, and to check the case $(v, c_1) = (1, 5)$.

Suppose that $c_1 \leq 0$. On $(\ell = -1)$ we have $\eta_2 = (c_1\eta_1 + 1)/v$, so $\eta_2 > 0$ if and only if $c_1\eta_1 > -1$, $\eta_2 < 3\eta_1$ if and only if $(3v - c_1)\eta_1 > 1$, and $5\eta_1 - \eta_2 < 1$ if and only if $(5v - c_1)\eta_1 < v + 1$. Hence $(\ell = -1) \cap \Sigma$ is the set of points of $(\ell = -1)$ with

$$\frac{1}{3v - c_1} < \eta_1 < \min\Big(\frac{1}{|c_1|}, \frac{v + 1}{5v - c_1}\Big),$$

where $\frac{1}{|c_1|}$ is omitted if $c_1 = 0$. The lower bound is less than both upper bounds, since $3v - c_1 > |c_1|$ and

$$(v + 1)(3v - c_1) - (5v - c_1) = v(3v - c_1 - 2) > 0.$$

Suppose that $c_1 > 3v$. The values of ℓ at the vertices of Σ are 0, $\frac{c_1}{5}$, and $\frac{c_1 - 3v}{2}$, and $\ell(\Sigma)$ is the open interval between the least and the largest of them. If $c_1 \leq 5$, then $3v < 5$, so $v = 1$ and $c_1 \in \{4, 5\}$. Since $(v, c_1) \neq (1, 4)$, if moreover $(v, c_1) \neq (1, 5)$, then $c_1 \geq 6$, so $\frac{c_1}{5} > 1$ and $1 \in \ell(\Sigma)$. If $(v, c_1) = (1, 5)$, then $c_1 s_0 - v y_0 = 1$ for $(s_0, y_0) = (0, -1)$, and for any $\sigma \in (\frac{1}{5}, \frac{1}{2})$,

$$s_0 + v\sigma = \sigma > 0, \quad 0 < 5\sigma - 1 < 3\sigma, \quad \text{and} \quad 5\sigma - (5\sigma - 1) = 1.$$

(2) Let $\omega_0 := (3s_0, y_0, 6s_0 - 2y_0, 2s_0)$. Then $\omega = r\omega_0 + Lw$, so $\omega \equiv Lw \pmod r$, and

$$c_1 S_r - v Y_r = r(c_1 s_0 - v y_0) = \mathrm{sgn}(c_1 - 3v) r.$$

If a prime p divides S_r and Y_r, then $p \mid r$, so $p \nmid L$, and $vL \equiv S_r \equiv 0$ and $c_1 L \equiv Y_r \equiv 0 \pmod p$ imply $p \mid \gcd(v, c_1) = 1$, a contradiction. Thus $\gcd(S_r, Y_r) = 1$, and ω is primitive since the greatest common divisor of the entries of ω divides $\gcd(3S_r, Y_r, 2S_r) = 1$. For $\sigma := L/r$ we have $(S_r, Y_r) = r(s_0 + v\sigma, y_0 + c_1\sigma)$. By (1), $S_r > 0$ and $0 < Y_r < 3S_r$, so ω is positive, and

$$0 < 5S_r - Y_r = r\big(5(s_0 + v\sigma) - (y_0 + c_1\sigma)\big) \leq r,$$

with equality exactly when $(v, c_1) = (1, 5)$. The entries of

$$\omega/r = (3, 0, 6, 2)(s_0 + v\sigma) + (0, 1, -2, 0)(y_0 + c_1\sigma)$$

are positive continuous functions of $\sigma \in I$. Hence for any compact subinterval $I' \subset I$ there exists $\Lambda \geq 1$, depending only on v, c_1, s_0, y_0, I', with $\omega/r \in [1/\Lambda, \Lambda]^4$ for any $\sigma \in I'$, which proves the last sentence of (2).

(3) Let $f := x^2 + g$, $\sigma_+ := \max I'$, Λ the constant of (2) for I', and

$$M_w := \max(3v, |c_1|, |6v - 2c_1|).$$

Since the cubic part $y^2z + t^3$ of g is reduced, W is normal near 0.

We compute $\langle \omega_0, e\rangle$ for the monomials $\mathbf{x}^e$ of f with $\langle w, e\rangle = 6v$. For $\mathbf{x}^e = x^2$, $\langle \omega_0, e\rangle = 6s_0$. Let $\mathbf{x}^e = y^{e_2}z^{e_3}t^{e_4}$ be a monomial of g with $\langle w, e\rangle = 6v$, and $m_e := e_2 - 2e_3$. Then $|e| \geq 3$ and

$$\begin{aligned} c_1 m_e &= 2v(3 - 3e_3 - e_4), \quad \text{so} \\ \langle \omega_0, e\rangle &= y_0 m_e + 2s_0(3e_3 + e_4) = 6s_0 - \frac{\operatorname{sgn}(c_1 - 3v)}{v} m_e. \end{aligned} \tag{3.19}$$

We show that $\operatorname{sgn}(c_1 - 3v)m_e \leq 0$, with equality exactly for y^2z and t^3. Suppose that $c_1 \leq 0$. If $e_3 = 0$, then $m_e = e_2 \geq 0$, and $m_e = 0$ forces $e_4 = 3$ by (3.19). If $e_3 \geq 1$ and $c_1 < 0$, then $-c_1 m_e = 2v(3e_3 + e_4 - 3) \geq 0$, with equality exactly when $(e_3, e_4) = (1, 0)$, that is, $\mathbf{x}^e = y^2z$. If $e_3 \geq 1$ and $c_1 = 0$, then (3.19) implies $(e_3, e_4) = (1, 0)$, so $e_2 \geq 2$ by $|e| \geq 3$, and $m_e = e_2 - 2 \geq 0$, with equality exactly for y^2z. Suppose that $c_1 > 3v$, so $c_1 > 0$. If $e_3 \geq 1$, then $-c_1 m_e = 2v(3e_3 + e_4 - 3) \geq 0$, with equality exactly for y^2z. If $e_3 = 0$ and $e_2 \geq 1$, then (3.19) reads $c_1 e_2 = 2v(3 - e_4)$, so $3 - e_4 > \frac{3}{2}e_2$, while $3 - e_4 \leq e_2$ by $|e| \geq 3$, a contradiction. If $e_3 = e_2 = 0$, then $\mathbf{x}^e = t^3$.

Thus, if $r > 6v + 2\Lambda^2 M_w$, then Lemma 3.10, for $h = f$, $\kappa = 6v$ and $d_h = 2$, implies

$$\operatorname{in}_\omega(f) = x^2 + y^2z + t^3 \quad \text{and} \quad \operatorname{wt}_\omega(f) = 6rs_0 + 6vL = 6S_r.$$

The polynomial $x^2 + y^2z + t^3$ is irreducible, since $y^2z + t^3$ is not a square, and is not divisible by any coordinate. By [Liu26b, Lemma 4.12], the strict transform of W intersects the exceptional divisor in a single prime divisor E_ω, and

$$a(E_\omega, W, 0) = 3S_r + Y_r + (6S_r - 2Y_r) + 2S_r - 6S_r = 5S_r - Y_r.$$

If moreover $r > |\kappa| + \Lambda^2 M_w d_h$, then by Lemma 3.10, $\operatorname{wt}_\omega(h) = rj + L\kappa$ with $|j| \leq \Lambda d_h + \sigma_+|\kappa|$, and $\operatorname{in}_\omega(h)$ is a non-zero polynomial of x-degree at most 1. Thus $x^2 + y^2z + t^3$ does not divide $\operatorname{in}_\omega(h)$, and $\operatorname{mult}_{E_\omega}(h|_W) = rj + L\kappa$ by [Liu26b, Lemma 4.12].

(4) By (3) and Definition 3.8(2), the central surfaces of the x-, y-, t- and z-charts are

$$(1 + Y^2Z + T^3 = 0), \quad (X^2 + Z + T^3 = 0),$$
$$(X^2 + Y^2Z + 1 = 0), \quad \text{and} \quad (X^2 + Y^2 + T^3 = 0).$$

By the Jacobian criterion, the first three surfaces are smooth, and the fourth is singular only at $X = Y = T = 0$, an A_2 point.

Let $c := |c_1 - 3v| > 0$. By (2), $\gcd(L, r) = 1$ and $\omega \equiv Lw \pmod r$, so the quotient chart of Definition 3.8(3) is defined for the z-chart. Since $\omega_3 = 6S_r - 2Y_r = 2d$, Lemma 3.9 gives $|\overline{G}_3| = 2d[G : R]$ and an injection of $\mu_{2d} = \mu_{\omega_3}$ into $\overline{G}_3$, so the image of $\mu_d \subset \mu_{2d}$ in $\overline{G}_3$ has index $2[G : R]$.

For any monomial $\mathbf{x}^e$ of f, $\langle w, e\rangle \in 6v + r\mathbb{Z}$, so

$$\langle \omega, e\rangle - 6S_r = r(\langle \omega_0, e\rangle - 6s_0) + L(\langle w, e\rangle - 6v) \in r\mathbb{Z},$$

and $\langle \omega, e\rangle - 6S_r \geq 0$ by (3). Thus on $\mathbb{A}^4_{(3)}$, $\widetilde{f}$ is a polynomial F in X, Y, T and $\varsigma = \rho^r$, which is the equation of the strict transform of W in the coordinates (X, Y, T, ς), and

$$F(X, Y, T, 0) = X^2 + Y^2 + T^3.$$

The element $\zeta \in \mu_{2d}$ acts by $\rho \to \zeta\rho$, $X \to \zeta^{-3S_r}X$, $Y \to \zeta^{-Y_r}Y$, and $T \to \zeta^{-2S_r}T$, so $\varsigma \to \zeta^r\varsigma$ and $F \to \zeta^{-6S_r}F$. Since $d = 3S_r - Y_r$ and $r = \operatorname{sgn}(c_1 - 3v)(c_1S_r - vY_r)$, we have

$$Y_r \equiv 3S_r \quad \text{and} \quad r \equiv \operatorname{sgn}(c_1 - 3v)(c_1 - 3v)S_r = cS_r \pmod d.$$

Thus μ_d acts on (X,Y,T,ς) with weights $S_r(-3,-3,-2,c)$ modulo d, and F has weight $-6S_r$. Since $\gcd(S_r,d)=\gcd(S_r,Y_r)=1$, $\zeta\to\zeta^{S_r}$ is an automorphism of μ_d, so μ_d acts as $\mu_d(-3,-3,-2,c)$, and F is semi-invariant of weight -6.

The monomials ς, $X\varsigma$, $Y\varsigma$, $T\varsigma$, ς^2, $T^2\varsigma$, $T\varsigma^2$, and ς^3 have weights c, $c-3$, $c-3$, $c-2$, $2c$, $c-4$, $2c-2$, and $3c$ respectively, which differ from the weight -6 of F by $c+6$, $c+3$, $c+3$, $c+4$, $2c+6$, $c+2$, $2c+4$, and $3c+6$, integers in $[1,3c+6]$. By (2), $d=\omega_3/2\geq r/(2\Lambda)$. Hence if $r>2\Lambda(3c+6)$, then $d>3c+6$, no weight in the list is congruent to -6 modulo d, and no monomial in the list occurs in F. □

Lemma 3.12 (cf. [Liu26b, Lemmas 4.9 and 4.12])**.** *Let c_2,k be coprime positive integers with $(c_2,k)\neq(1,1)$, $c_3:=-k$, $w:=(2c_2+c_3,2c_2,2c_3,c_2+c_3)$, and $\mathbf{T}:=\mathbf{T}_{(w)}$.*

(1) There exist integers u_0,v_0 with $ku_0+c_2v_0=1$ and a non-empty open interval $I\subset(0,+\infty)$ such that $u_0+c_2\sigma>0$ and $v_0-k\sigma>0$ for any $\sigma\in I$. If $(c_2,k)\neq(2,1)$, then moreover $(u_0+c_2\sigma)+2(v_0-k\sigma)<1$ on I, and if $(c_2,k)=(2,1)$, then $(u_0+c_2\sigma)+2(v_0-k\sigma)=1$.

(2) Fix u_0,v_0,I as in (1). For integers $r\geq1$ and L with $\gcd(L,r)=1$ and $L/r\in I$, the integers $N_1:=ru_0+c_2L$ and $N_2:=rv_0-kL$ are positive and coprime, $kN_1+c_2N_2=r$, the vector

$$\omega:=(2N_1+N_2,2N_1,2N_2,N_1+N_2)$$

is primitive and positive with $\omega\equiv Lw\pmod r$, and $0<N_1+2N_2\leq r$, with equality exactly when $(c_2,k)=(2,1)$. For any compact subinterval $I'\subset I$, there exists a real number $\Lambda\geq1$, depending only on c_2,k,u_0,v_0,I', such that $\omega\in[r/\Lambda,\Lambda r]^4$ whenever $L/r\in I'$.

(3) The monomials x^2, y^2z, and yt^2 have ω-weight $4N_1+2N_2$. Let $I'\subset I$ be a compact subinterval, $\psi\in(z,t)^4\subset\mathbb{C}[z,t]$ a polynomial such that all monomials of $x^2+y^2z+yt^2+\psi$ have $\mathbf{T}$-weight in $4c_2-2k+r\mathbb{Z}$, and $W:=(x^2+y^2z+yt^2+\psi=0)\subset\mathbb{A}^4$. If $L/r\in I'$ and r is large in terms of c_2,k,u_0,v_0,I', then $\mathrm{in}_\omega(x^2+y^2z+yt^2+\psi)=x^2+y^2z+yt^2$, the strict transform of W in the ω-weighted blow-up of $\mathbb{A}^4$ intersects the exceptional divisor in a single prime divisor E_ω, and $a(E_\omega,W,0)=N_1+2N_2$. Let κ be an integer, $h\neq0$ a polynomial of x-degree at most 1 whose monomials have $\mathbf{T}$-weights in $\kappa+r\mathbb{Z}$, with a monomial of $\mathbf{T}$-weight κ, and d_h the least degree of a monomial of h of $\mathbf{T}$-weight κ. If $L/r\in I'$ and r is large in terms of $c_2,k,u_0,v_0,I',\kappa,d_h$, then $\mathrm{mult}_{E_\omega}(h|_W)=rj+L\kappa$ for an integer j bounded in terms of $c_2,k,u_0,v_0,I',\kappa,d_h$.

(4) In the situation of (3), let G be a finite group acting diagonally on x,y,z,t and preserving W, with $R:=G\cap\mathbf{T}$ of order r. Then the central surfaces of W on the x-, y- and t-charts of the ω-weighted blow-up are smooth, and that on the z-chart has only an A_3 point. Let (X,Y,T,ς) be the coordinates of $\mathbb{A}^4_{\mathrm{rad}}$ for the z-chart (Definition 3.8(3)), in which the image of the A_3 point is the origin. Then the image of the subgroup $\mu_{N_2}\subset\mu_{2N_2}=\mu_{\omega_3}$ in the group $\overline{G}_3$ of the quotient chart acts on $\mathbb{A}^4_{\mathrm{rad}}$ as $\mu_{N_2}(-2,-2,-1,k)$ and has index $2[G:R]$ in $\overline{G}_3$. If moreover $r>2\Lambda(2k+4)$, where Λ is as in (2) for I', then the weights of $\mu_{N_2}(-2,-2,-1,k)$ exclude from the equation F of the strict transform of W in the coordinates (X,Y,T,ς) a linear term in ς and any quadratic term involving ς.

Proof. (1) Let $\ell(\eta_1,\eta_2):=k\eta_1+c_2\eta_2$, and let

$$\Sigma:=\{(\eta_1,\eta_2)\in\mathbb{R}^2\mid\eta_1>0,\ \eta_2>0,\ \eta_1+2\eta_2<1\}$$

be the open triangle with vertices $(0,0)$, $(1,0)$, and $(0,\frac12)$. Since $\gcd(c_2,k)=1$, there exist integers u_0,v_0 with $ku_0+c_2v_0=1$, and then

$$(\ell=1)=\{(u_0+c_2\sigma,v_0-k\sigma)\mid\sigma\in\mathbb{R}\}.$$

Replacing (u_0,v_0) with (u_0-nc_2,v_0+nk) for an integer n replaces the parameter σ of a point of $(\ell=1)$ with $\sigma+n$. Thus it suffices to find a non-empty open interval I_0 of values of σ with the properties of (1): replacing (u_0,v_0) with (u_0-nc_2,v_0+nk) for an integer $n\gg0$ replaces I_0 with $I:=I_0+n\subset(0,+\infty)$.

Suppose that $(c_2,k)\neq(2,1)$. The values of ℓ at the vertices of Σ are 0, k, and $\frac{c_2}{2}$, and $\ell(\Sigma)$ is the open interval between the least and the largest of them. Since $(c_2,k)\notin\{(1,1),(2,1)\}$, either $k\geq 2$ or $c_2\geq 3$, so $\max(k,\frac{c_2}{2})>1$ and $1\in\ell(\Sigma)$. We let I_0 be the non-empty open interval of the σ with $(u_0+c_2\sigma, v_0-k\sigma)\in\Sigma$. Suppose that $(c_2,k)=(2,1)$. Then $(u_0+2\sigma)+2(v_0-\sigma)=u_0+2v_0=1$ for any σ, and we let

$$I_0:=\{\sigma\mid u_0+2\sigma>0,\ v_0-\sigma>0\}=\Big(-\frac{u_0}{2},v_0\Big),$$

which is non-empty since $v_0+\frac{u_0}{2}=\frac{1}{2}$.

(2) Let $\omega_0:=(2u_0+v_0,2u_0,2v_0,u_0+v_0)$. Then $\omega=r\omega_0+Lw$, so $\omega\equiv Lw \pmod r$, and

$$kN_1+c_2N_2=r(ku_0+c_2v_0)=r.$$

If a prime p divides N_1 and N_2, then $p\mid r$, so $p\nmid L$, and $c_2L\equiv N_1\equiv 0$ and $kL\equiv -N_2\equiv 0 \pmod p$ imply $p\mid\gcd(c_2,k)=1$, a contradiction. Thus $\gcd(N_1,N_2)=1$, and ω is primitive since the greatest common divisor of the entries of ω divides

$$\gcd(2N_1+N_2-(N_1+N_2),N_1+N_2)=\gcd(N_1,N_2)=1.$$

For $\sigma:=L/r$ we have $(N_1,N_2)=r(u_0+c_2\sigma,v_0-k\sigma)$. By (1), $N_1,N_2>0$, so ω is positive, and

$$0<N_1+2N_2=r\big((u_0+c_2\sigma)+2(v_0-k\sigma)\big)\leq r,$$

with equality exactly when $(c_2,k)=(2,1)$. The entries of

$$\omega/r=(2,2,0,1)(u_0+c_2\sigma)+(1,0,2,1)(v_0-k\sigma)$$

are positive continuous functions of $\sigma\in I$. Hence for any compact subinterval $I'\subset I$ there exists $\Lambda\geq 1$, depending only on c_2,k,u_0,v_0,I', with $\omega/r\in[1/\Lambda,\Lambda]^4$ for any $\sigma\in I'$, which proves the last sentence of (2).

(3) The ω-weights of x^2, y^2z, and yt^2 are $2(2N_1+N_2)$, $4N_1+2N_2$, and $2N_1+2(N_1+N_2)$, all equal to $4N_1+2N_2$. Let $f:=x^2+y^2z+yt^2+\psi$, $\sigma_+:=\max I'$, Λ the constant of (2) for I', and $M_w:=\max(|2c_2-k|,2c_2,2k,|c_2-k|)$. The cubic part $y(yz+t^2)$ of $y^2z+yt^2+\psi$ has no multiple factor, so $y^2z+yt^2+\psi$ is reduced at 0. Thus $\operatorname{Sing}W=\{0\}\times\operatorname{Sing}(y^2z+yt^2+\psi=0)$ and $\dim\operatorname{Sing}W\leq 1$ near 0, so the hypersurface W is normal near 0.

The monomials x^2, y^2z, and yt^2 have $\mathbf{T}$-weight $4c_2-2k$ and $\langle\omega_0,e\rangle=4u_0+2v_0$. Let $\mathbf{x}^e=z^{e_3}t^{e_4}$ be a monomial of ψ with $\mathbf{T}$-weight $-2ke_3+(c_2-k)e_4=4c_2-2k$, so $e_3+e_4\geq 4$ and

$$c_2(e_4-4)=k(2e_3+e_4-2).$$

Since $\gcd(c_2,k)=1$, there exists an integer n with $e_4-4=kn$ and $2e_3+e_4-2=c_2n$. If $n<0$, then $e_3+e_4\leq 2e_3+e_4<2$, and if $n=0$, then $e_3=-1$. Both are impossible. Thus $n\geq 1$, and

$$\langle\omega_0,e\rangle-(4u_0+2v_0)=u_0(e_4-4)+v_0(2e_3+e_4-2)=n(ku_0+c_2v_0)=n\geq 1.$$

Thus, if $r>|4c_2-2k|+2\Lambda^2M_w$, then Lemma 3.10, for $h=f$, $\kappa=4c_2-2k$ and $d_h=2$, implies

$$\operatorname{in}_\omega(f)=x^2+y^2z+yt^2\quad\text{and}\quad\operatorname{wt}_\omega(f)=r(4u_0+2v_0)+L(4c_2-2k)=4N_1+2N_2.$$

The polynomial $x^2+y(yz+t^2)$ is irreducible, since $y(yz+t^2)$ is not a square, and is not divisible by any coordinate. By [Liu26b, Lemma 4.12], the strict transform of W intersects the exceptional divisor in a single prime divisor E_ω, and

$$a(E_\omega,W,0)=(2N_1+N_2)+2N_1+2N_2+(N_1+N_2)-(4N_1+2N_2)=N_1+2N_2.$$

If moreover $r>|\kappa|+\Lambda^2M_wd_h$, then by Lemma 3.10, $\operatorname{wt}_\omega(h)=rj+L\kappa$ with $|j|\leq\Lambda d_h+\sigma_+|\kappa|$, and $\operatorname{in}_\omega(h)$ is a non-zero polynomial of x-degree at most 1. Thus $x^2+y^2z+yt^2$ does not divide $\operatorname{in}_\omega(h)$, and $\operatorname{mult}_{E_\omega}(h|_W)=rj+L\kappa$ by [Liu26b, Lemma 4.12].

(4) By (3) and Definition 3.8(2), the central surfaces of the x-, y-, t- and z-charts are

$$(1+Y^2Z+YT^2=0),\quad(X^2+Z+T^2=0),$$
$$(X^2+Y^2Z+Y=0),\quad\text{and}\quad(X^2+Y^2+YT^2=0).$$

The partial derivatives of the first equation in Y, Z, T are $2YZ + T^2, Y^2, 2YT$, which vanish only if $Y = T = 0$, where the equation reads $1 = 0$. The second equation has derivative 1 in Z. The partial derivatives of the third equation in X, Y, Z are $2X, 2YZ + 1, Y^2$, which do not vanish simultaneously. The partial derivatives of the fourth equation in X, Y, T are $2X, 2Y + T^2, 2YT$, which vanish only if $X = Y = T = 0$, and

$$X^2 + Y^2 + YT^2 = X^2 + \Big(Y + \frac{T^2}{2}\Big)^2 - \frac{T^4}{4},$$

so the fourth surface has only an A_3 point.

By (2), $\gcd(L, r) = 1$ and $\omega \equiv Lw \pmod r$, so the quotient chart of Definition 3.8(3) is defined for the z-chart. Since $\omega_3 = 2N_2$, Lemma 3.9 gives $|\overline{G}_3| = 2N_2[G : R]$ and an injection of $\mu_{2N_2} = \mu_{\omega_3}$ into $\overline{G}_3$, so the image of $\mu_{N_2} \subset \mu_{2N_2}$ in $\overline{G}_3$ has index $2[G : R]$.

For any monomial $\mathbf{x}^e$ of f, $\langle w, e\rangle \in 4c_2 - 2k + r\mathbb{Z}$, so

$$\langle \omega, e\rangle - (4N_1 + 2N_2) = r\big(\langle \omega_0, e\rangle - (4u_0 + 2v_0)\big) + L\big(\langle w, e\rangle - (4c_2 - 2k)\big) \in r\mathbb{Z},$$

and $\langle \omega, e\rangle - (4N_1 + 2N_2) \geq 0$ by (3). Thus on $\mathbb{A}^4_{(3)}$, $\widetilde{f}$ is a polynomial F in X, Y, T and $\varsigma = \rho^r$, which is the equation of the strict transform of W in the coordinates (X, Y, T, ς), and $F(X, Y, T, 0) = X^2 + Y^2 + YT^2$.

The element $\zeta \in \mu_{2N_2}$ acts by $\rho \to \zeta\rho$, $X \to \zeta^{-(2N_1+N_2)}X$, $Y \to \zeta^{-2N_1}Y$, and $T \to \zeta^{-(N_1+N_2)}T$, so $\varsigma \to \zeta^r\varsigma$ and $F \to \zeta^{-2(2N_1+N_2)}F$. Since $r = kN_1 + c_2N_2 \equiv kN_1 \pmod{N_2}$, the group μ_{N_2} acts on (X, Y, T, ς) with weights $N_1(-2, -2, -1, k)$ modulo N_2, and F has weight $-4N_1$. Since $\gcd(N_1, N_2) = 1$, $\zeta \to \zeta^{N_1}$ is an automorphism of μ_{N_2}, so μ_{N_2} acts as $\mu_{N_2}(-2, -2, -1, k)$, and F is semi-invariant of weight -4.

Since F is semi-invariant of weight -4, a monomial of weight a occurs in F only if $N_2 \mid a + 4$. The monomials ς, $X\varsigma$, $Y\varsigma$, $T\varsigma$, and ς^2 have weights k, $k-2$, $k-2$, $k-1$, and $2k$ respectively, so their weights a satisfy $a+4 \in \{k+4, k+2, k+3, 2k+4\} \subset [1, 2k+4]$. By (2), $N_2 = \omega_3/2 \geq r/(2\Lambda)$. Hence if $r > 2\Lambda(2k + 4)$, then $N_2 > 2k + 4$, so N_2 divides none of $k + 4$, $k + 2$, $k + 3$, and $2k + 4$, and no monomial in the list occurs in F. □

3.5. **Smooth surface quotients with a negative curve.** For a smooth germ with a finite diagonal group action, [Liu26a, Lemma 3.4] computes the quotient value as a lattice minimum when the boundary is supported on the coordinate divisors with coefficients less than 1.

Proposition 3.13 (cf. [Liu26a, Lemma 3.4 and Theorem 4.5] and [LX23, Theorem 1.2])**.** *Let $\Bbbk \supset \mathbb{C}$ be an algebraically closed field, $(S \ni p)$ a smooth surface germ over $\Bbbk$, D a smooth curve through p, and*

$$B = bD + B_+, \quad B_+ = \sum_{j=1}^{n_+} c_j H_j,$$

where $b < 0$, the $H_j \neq D$ are distinct reduced prime curves through p, and $0 < c_j < 1$ for any j. Let G be a finite group acting faithfully on $(S \ni p)$ and preserving D and B, and let $\mu := \mathrm{mld}((S \ni p, B)/G)$ be the minimal log discrepancy of the crepant quotient $(S, B)/G$ at the image of p. Assume that (S, B) is sub-klt near p and

$$(B_+ \cdot D)_p < 2.$$

Then $\mu > 0$, there exists a prime divisor E over S with center p and $a(E, S, B) = |I_G(E)|\mu$, and the following hold.

(A) Let E be such a prime divisor, $I := I_G(E)$, and $e := |I|$. Then $\mu = \mathrm{mld}((S \ni p, B)/I)$, and there exist I-eigenparameters ζ_1, ζ_2 of $\mathcal{O}_{S,p}$ with $D = (\zeta_1 = 0)$ in which E is the exceptional divisor of the weighted blow-up with coprime positive weights (u, v). Moreover, let $N_I \subset \mathbb{R}^2$ be the lattice of the image of I in $\mathbb{T}^2$ under the characters of ζ_1, ζ_2. For any I-orbit $\mathcal{H}$ of the components of B_+, let $h_{\mathcal{H}}$ be an I-semi-invariant equation of $\sum_{H_j \in \mathcal{H}} H_j$ near p, let $d_{\mathcal{H}}$ be the coefficient in B_+ of the components in $\mathcal{H}$, and let $J(h_{\mathcal{H}}) \subset \Bbbk[\zeta_1, \zeta_2]$ be

the ideal generated by the monomials occurring in the expansion of $h_{\mathcal{H}}$ *in* $\widehat{\mathcal{O}}_{S,p} = \Bbbk[[\zeta_1, \zeta_2]]$. *Then*

$$\mu = \min_{w \in N_I \cap \mathbb{R}^2_{>0}} \Big((1-b)w_1 + w_2 - \sum_{\mathcal{H}} d_{\mathcal{H}} \operatorname{wt}_w(J(h_{\mathcal{H}})) \Big),$$

attained at $w = (u, v)/e$, *where* $\mathcal{H}$ *runs over the* I*-orbits of the components of* B_+.

(B) *Let* $\Lambda \subset (-\infty, 0)$ *and* $\Gamma \subset [0, 1]$ *be DCC sets. Then the set of the numbers* μ *above with* $b \in \Lambda$ *and* $c_j \in \Gamma$ *for any* j, *where* $\Bbbk$, $(S \ni p)$, D, B, *and* G *vary, satisfies the ACC.*

Proof. The argument of **Steps 3 and 4** below follows [Kaw17, Proof of Theorem 1]. For a subgroup $G' \subset G$ and a prime divisor E' over S with center p, $I_{G'}(E') = G' \cap I_G(E')$, and by Definition 2.2

$$\operatorname{mld}((S \ni p, B)/G') = \inf \left\{ \frac{a(E', S, B)}{|I_{G'}(E')|} \;\middle|\; E' \text{ is a prime divisor over } S \text{ with center } p \right\}. \tag{3.20}$$

Since (S, B) is sub-klt near p, $a(E', S, B) > 0$ for any such E'.

Step 1. In this step we show that $\mu > 0$ and that the infimum (3.20) for $G' = G$ is attained, prove $\mu = \operatorname{mld}((S \ni p, B)/I)$, and compute the characters of I.

Let $\overline{p} \in S/G$ be the image of p and $(S/G, B_{S/G})$ the crepant quotient, so that by Lemma 2.15,

$$\mu = \operatorname{mld}(S/G \ni \overline{p}, B_{S/G}).$$

Since (S, B) is sub-klt near p, $(S/G, B_{S/G})$ is sub-klt near $\overline{p}$ by Lemma 2.15(2)(3). Possibly shrinking S to a G-invariant neighborhood of p, we may assume that $S/G \backslash \{\overline{p}\}$ is smooth and that $\operatorname{Supp} B_{S/G} \backslash \{\overline{p}\}$ is a disjoint union of smooth curves. Let $\rho : X \to S/G$ be a log resolution of $(S/G, B_{S/G})$ which is an isomorphism over $S/G \backslash \{\overline{p}\}$ and such that $\Sigma_0 := \rho^{-1}(\overline{p})$ is a divisor. Since $(S/G, B_{S/G})$ is sub-klt near $\overline{p}$, μ is the minimum of the log discrepancies $a(P, S/G, B_{S/G}) > 0$ of the finitely many components P of Σ_0. In particular $\mu > 0$ and μ is attained. Let P be a component of Σ_0 with $a(P, S/G, B_{S/G}) = \mu$. By Lemma 2.15(1)–(3), there exists a prime divisor E over S whose valuation restricts to $|I_G(E)|$ times that of P. Since p is the only point of S over $\overline{p}$, E has center p and

$$a(E, S, B) = |I_G(E)|\mu. \tag{3.21}$$

From now until the end of **Step 5**, let E, I, and e be as in (A). For any prime divisor E' over S with center p, since $|I_I(E')| \le |I_G(E')|$ and $a(E', S, B) > 0$,

$$\frac{a(E', S, B)}{|I_I(E')|} \ge \frac{a(E', S, B)}{|I_G(E')|} \ge \mu,$$

and $a(E, S, B)/|I_I(E)| = a(E, S, B)/e = \mu$ since $I_I(E) = I$. By (3.20) for $G' = I$,

$$\mu = \operatorname{mld}((S \ni p, B)/I), \tag{3.22}$$

that is, $a(E', S, B) \ge |I_I(E')|\mu$ for any prime divisor E' over S with center p.

Let $\mathcal{O}_E$ be the valuation ring of E, $\mathfrak{m}_E$ its maximal ideal, $\kappa(E) = \mathcal{O}_E/\mathfrak{m}_E$, and ϖ a generator of $\mathfrak{m}_E$. Since I maps E to itself and acts trivially on $\kappa(E)$, I acts $\kappa(E)$-linearly on the 1-dimensional $\kappa(E)$-vector space $\mathfrak{m}_E/\mathfrak{m}_E^2$, by a character $\eta : I \to \kappa(E)^*$. Since any value of η is an e-th root of unity and $\Bbbk$ is algebraically closed, $\eta(I) \subset \Bbbk^*$. By Lemma 2.15(1), η is faithful. Hence η is an isomorphism from I onto the group of e-th roots of unity in $\Bbbk$. Let $\phi \in \mathcal{O}_{S,p}$ be a non-zero I-semi-invariant function. For $g \in I$, $\phi\varpi^{-\operatorname{mult}_E \phi} \in \mathcal{O}_E^*$ and g acts trivially on $\kappa(E)$, so

$$\chi_\phi(g)\phi = g(\phi) \equiv \eta(g)^{\operatorname{mult}_E \phi}\phi \mod \mathfrak{m}_E^{\operatorname{mult}_E \phi + 1}.$$

Since $\chi_\phi(g), \eta(g) \in \Bbbk$,

$$\chi_\phi = \eta^{\operatorname{mult}_E \phi}. \tag{3.23}$$

Step 2. In this step we choose the eigenparameters ζ_1, ζ_2.

Let $R := \mathcal{O}_{S,p}$, $\mathfrak{m} \subset R$ the maximal ideal, and $\mathfrak{p} \subset R$ the ideal of D. Since D is I-invariant, I acts on the 1-dimensional $\Bbbk$-vector space $\mathfrak{p}/\mathfrak{m}\mathfrak{p}$ by a character α. Let t be a generator of $\mathfrak{p}$ and

$$\zeta_1 := \frac{1}{e}\sum_{g\in I} \alpha(g)^{-1} g(t).$$

Then $g(\zeta_1) = \alpha(g)\zeta_1$ for any $g \in I$, and $\zeta_1 \equiv t \mod \mathfrak{m}\mathfrak{p}$, so ζ_1 generates $\mathfrak{p}$ by Nakayama's lemma. Since D is smooth at p, $\zeta_1 \in \mathfrak{m}\backslash\mathfrak{m}^2$.

Let $\mathcal{P}$ be the set of $s \in \mathfrak{m}$ with non-zero image in $\mathfrak{m}/(\mathfrak{m}^2 + \zeta_1 R)$, that is, the set of $s \in R$ such that (ζ_1, s) is a regular system of parameters of R. For $s \in \mathcal{P}$, the pair $(S, D + (s = 0))$ is log smooth near p, so $a(E, S, D + (s = 0)) \geq 0$, that is,

$$\operatorname{mult}_E \zeta_1 + \operatorname{mult}_E s \leq a(E, S, 0).$$

Thus there exists $s \in \mathcal{P}$ with $\operatorname{mult}_E s$ maximal. The group I acts on the 1-dimensional $\Bbbk$-vector space $\mathfrak{m}/(\mathfrak{m}^2 + \zeta_1 R)$ by a character β. Let

$$\zeta_2 := \frac{1}{e}\sum_{g\in I} \beta(g)^{-1} g(s).$$

Then ζ_2 is I-semi-invariant with character β, and $\zeta_2 - s \in \mathfrak{m}^2 + \zeta_1 R$, so $\zeta_2 \in \mathcal{P}$. Since any $g \in I$ maps E to itself, $\operatorname{mult}_E g(s) = \operatorname{mult}_E s$, so $\operatorname{mult}_E \zeta_2 \geq \operatorname{mult}_E s$ and

$$\operatorname{mult}_E \zeta_2 = \max\{\operatorname{mult}_E s \mid s \in \mathcal{P}\}. \tag{3.24}$$

Thus ζ_1, ζ_2 are I-eigenparameters of R with $D = (\zeta_1 = 0)$. Since p is the center of E, $\operatorname{mult}_E \zeta_1$ and $\operatorname{mult}_E \zeta_2$ are positive integers. Let

$$n := \gcd(\operatorname{mult}_E \zeta_1, \operatorname{mult}_E \zeta_2), \quad u := \frac{\operatorname{mult}_E \zeta_1}{n}, \quad \text{and} \quad v := \frac{\operatorname{mult}_E \zeta_2}{n}.$$

Then u, v are coprime positive integers, and by (3.23),

$$\chi_{\zeta_1} = \eta^{nu} \quad \text{and} \quad \chi_{\zeta_2} = \eta^{nv}. \tag{3.25}$$

We identify $\widehat{R} = \Bbbk[[\zeta_1, \zeta_2]]$. For coprime positive integers u', v', let F' be the exceptional divisor of the weighted blow-up of p with weights (u', v') in (ζ_1, ζ_2). Then $a(F', S, 0) = u' + v'$,

$$\operatorname{mult}_{F'} \phi = \operatorname{wt}_{(u',v')}(\phi) = \min\{u'i_1 + v'i_2 \mid \text{the coefficient of } \zeta_1^{i_1}\zeta_2^{i_2} \text{ in } \phi \text{ is non-zero}\}$$

for $\phi \in R\backslash\{0\}$, and $\kappa(F') = \Bbbk(\zeta_2^{u'}\zeta_1^{-v'})$. Since $g(\zeta_i) = \chi_{\zeta_i}(g)\zeta_i$ for $g \in I$ and $i \in \{1, 2\}$, I maps F' to itself, and $g \in I$ acts on $\kappa(F')$ by multiplying $\zeta_2^{u'}\zeta_1^{-v'}$ by $\chi_{\zeta_2}(g)^{u'}\chi_{\zeta_1}(g)^{-v'}$. Thus

$$I_I(F') = \{g \in I \mid \chi_{\zeta_2}(g)^{u'} = \chi_{\zeta_1}(g)^{v'}\}. \tag{3.26}$$

Step 3. In this step we compare E with the weighted blow-up F of weights (u, v).

Let $\pi : Y \to S$ be the weighted blow-up of p with weights (u, v) in (ζ_1, ζ_2) and F the exceptional divisor. By (3.25), $\chi_{\zeta_2}^u = \eta^{nuv} = \chi_{\zeta_1}^v$, so $I_I(F) = I$ by (3.26). By (3.22),

$$a(E, S, B) = e\mu \leq a(F, S, B). \tag{3.27}$$

Assume that $E \neq F$. Let $D_Y := \pi_*^{-1}D$ and $C_Y := \pi_*^{-1}(\zeta_2 = 0)$. Then $D_Y \cap C_Y = \emptyset$. Since $E \neq F$ and $\pi(\operatorname{center}_Y E) = p$, the center $Q := \operatorname{center}_Y E$ is a closed point of F. Since $\pi^*D = D_Y + uF$ and $\pi^*(\zeta_2 = 0) = C_Y + vF$ as $\mathbb{Q}$-Cartier divisors,

$$nu = u \operatorname{mult}_E F + \operatorname{mult}_E D_Y \quad \text{and} \quad nv = v \operatorname{mult}_E F + \operatorname{mult}_E C_Y.$$

As $D_Y \cap C_Y = \emptyset$, $\operatorname{mult}_E D_Y = 0$ or $\operatorname{mult}_E C_Y = 0$. In either case $\operatorname{mult}_E F = n$, and then the other one is also 0. Thus

$$Q \notin D_Y \cup C_Y \quad \text{and} \quad \operatorname{mult}_E F = n.$$

Let q_1, q_2 be integers with $uq_1 + vq_2 = 1$, and let

$$z := \zeta_1^{q_1}\zeta_2^{q_2} \quad \text{and} \quad \tau := \zeta_2^u\zeta_1^{-v}, \quad \text{so that} \quad \zeta_1 = z^u\tau^{-q_2} \quad \text{and} \quad \zeta_2 = z^v\tau^{q_1}.$$

Since (ζ_1,ζ_2) is étale at p, $Y\backslash(D_Y\cup C_Y)$ is, over a neighborhood of p, the base change of the open subset $\mathbb{A}^1_z\times\mathbb{G}_{m,\tau}$ of the weighted blow-up of $\mathbb{A}^2$. Thus Y is smooth at Q, $\xi_0:=\tau(Q)\in\Bbbk^*$, $F=(z=0)$ near Q, and $\widehat{\mathcal{O}}_{Y,Q}=\Bbbk[[z,\tau-\xi_0]]$. Since

$$\zeta_1^{i_1}\zeta_2^{i_2}=z^{ui_1+vi_2}\tau^{q_1i_2-q_2i_1}$$

for any $i_1,i_2\in\mathbb{N}$, any $\phi\in R\backslash\{0\}$ satisfies

$$\phi=z^{\operatorname{mult}_F\phi}\big(\operatorname{in}_{(u,v)}(\phi)(\tau^{-q_2},\tau^{q_1})+z\psi_\phi\big)\quad\text{in }\widehat{\mathcal{O}}_{Y,Q}\tag{3.28}$$

for some $\psi_\phi\in\widehat{\mathcal{O}}_{Y,Q}$, where $\operatorname{in}_{(u,v)}(\phi)$ is the lowest (u,v)-weight part of ϕ.

Let $B_{+,Y}:=\pi_*^{-1}B_+$. Since $Q\notin D_Y$, near Q we have

$$K_Y+(1-a(F,S,B))F+B_{+,Y}=\pi^*(K_S+B).$$

Since $\operatorname{mult}_E F=n\geq 1$ and $a(F,S,B)>0$, (3.27) implies

$$a(E,Y,F+B_{+,Y})=a(E,S,B)-na(F,S,B)\leq(1-n)a(F,S,B)\leq 0.\tag{3.29}$$

Step 4. In this step we show that $E=F$.

We continue to assume that $E\neq F$ and first show that

$$(B_{+,Y}\cdot F)_Q\geq 1.\tag{3.30}$$

Suppose that $(B_{+,Y}\cdot F)_Q<1$. Since Y is smooth at Q, F is smooth near Q, and $F\not\subset\operatorname{Supp}B_{+,Y}$, we have $\operatorname{mult}_Q(B_{+,Y}|_F)=(B_{+,Y}\cdot F)_Q<1$, so $(Y,F+B_{+,Y})$ is plt near Q by inversion of adjunction. Since E is exceptional over Y with center Q, $a(E,Y,F+B_{+,Y})>0$, contradicting (3.29). This proves (3.30).

For any j, let $f_j\in R$ be a local equation of H_j. The polynomial $\operatorname{in}_{(u,v)}(f_j)$ is (u,v)-weighted homogeneous of weight $\operatorname{mult}_F f_j$, and, as $\gcd(u,v)=1$, the quotient of two monomials $\zeta_1^{i_1}\zeta_2^{i_2}$ and $\zeta_1^{i_1'}\zeta_2^{i_2'}$ with $ui_1+vi_2=ui_1'+vi_2'$ is an integer power of $\zeta_2^u\zeta_1^{-v}$. Factoring a polynomial in one variable over $\Bbbk$, we obtain

$$\operatorname{in}_{(u,v)}(f_j)=\gamma_j\zeta_1^{m_j}\zeta_2^{m_j'}\prod_{\xi\in\Bbbk^*}(\zeta_2^u-\xi\zeta_1^v)^{m_{j,\xi}}$$

with $\gamma_j\in\Bbbk^*$ and $m_j,m_j',m_{j,\xi}\in\mathbb{N}$, only finitely many $m_{j,\xi}$ non-zero. Hence

$$\operatorname{mult}_F f_j=um_j+vm_j'+uv\sum_{\xi\in\Bbbk^*}m_{j,\xi}\geq uvm_{j,\xi_0}.$$

Since substituting $\zeta_1=\tau^{-q_2}$ and $\zeta_2=\tau^{q_1}$ in $\zeta_2^u-\xi\zeta_1^v$ yields $\tau^{-q_2v}(\tau-\xi)$, and since τ is a unit of $\widehat{\mathcal{O}}_{Y,Q}$, (3.28) implies that $\pi_*^{-1}H_j$ is defined near Q by the sum of $z\psi_{f_j}$ and a unit of $\widehat{\mathcal{O}}_{Y,Q}$ times $\prod_\xi(\tau-\xi)^{m_{j,\xi}}$. Thus

$$(\pi_*^{-1}H_j\cdot F)_Q=\dim_\Bbbk\Bbbk[[z,\tau-\xi_0]]/\Big(z,\prod_\xi(\tau-\xi)^{m_{j,\xi}}\Big)=m_{j,\xi_0}\leq\frac{\operatorname{mult}_F f_j}{uv}.$$

Let $r_j:=(H_j\cdot D)_p=\dim_\Bbbk R/(\zeta_1,f_j)$. Since $H_j\neq D$, the image of f_j in $\widehat{R}/(\zeta_1)=\Bbbk[[\zeta_2]]$ is a unit times $\zeta_2^{r_j}$, so the coefficient of $\zeta_2^{r_j}$ in f_j is non-zero, and $\operatorname{mult}_F f_j\leq vr_j$. By (3.30),

$$1\leq(B_{+,Y}\cdot F)_Q=\sum_j c_j(\pi_*^{-1}H_j\cdot F)_Q\leq\sum_j c_j\frac{vr_j}{uv}=\frac{(B_+\cdot D)_p}{u}<\frac{2}{u}.$$

Thus $u=1$. Taking $q_1=1$ and $q_2=0$, we have $z=\zeta_1$ and $\zeta_2=\zeta_1^v\tau$. Let

$$s':=\zeta_2-\xi_0\zeta_1^v=\zeta_1^v(\tau-\xi_0)\in R.$$

Since $s'-\zeta_2\in\zeta_1R$, $s'\in\mathcal{P}$. Since $\operatorname{mult}_E\zeta_1=n$ and $\tau-\xi_0$ vanishes at $Q=\operatorname{center}_Y E$,

$$\operatorname{mult}_E s'=nv+\operatorname{mult}_E(\tau-\xi_0)>nv=\operatorname{mult}_E\zeta_2,$$

contradicting (3.24). Thus $E = F$. In particular $(\mathrm{mult}_E\,\zeta_1, \mathrm{mult}_E\,\zeta_2) = (u, v)$ and $n = 1$. Together with (3.22), this proves (A) up to the lattice formula.

Step 5. In this step we prove the lattice formula of (A).

The group I preserves D and B, hence B_+, so I permutes the components H_j and preserves their coefficients. For any I-orbit $\mathcal{H}$ of the components of B_+, the ideal of $\sum_{H_j\in\mathcal{H}} H_j$ is principal and I-stable, so the projection of **Step 2** applied to any generator yields an I-semi-invariant generator $h_{\mathcal{H}}$. Thus near p,

$$B_+ = \sum_{\mathcal{H}} d_{\mathcal{H}}\,\mathrm{div}(h_{\mathcal{H}}).$$

For $w \in \mathbb{R}^2_{>0}$ let

$$\Psi(w) := (1-b)w_1 + w_2 - \sum_{\mathcal{H}} d_{\mathcal{H}}\,\mathrm{wt}_w(J(h_{\mathcal{H}})).$$

Then $\Psi(tw) = t\Psi(w)$ for any real number $t > 0$. For coprime positive integers u', v' and F' as in **Step 2**, by **Step 2** and [Liu26a, Definition 2.16],

$$\mathrm{mult}_{F'} h_{\mathcal{H}} = \mathrm{wt}_{(u',v')}(h_{\mathcal{H}}) = \mathrm{wt}_{(u',v')}(J(h_{\mathcal{H}})),$$

so

$$a(F', S, B) = u' + v' - bu' - \sum_{\mathcal{H}} d_{\mathcal{H}}\,\mathrm{wt}_{(u',v')}(J(h_{\mathcal{H}})) = \Psi((u', v')). \tag{3.31}$$

By Definition 2.3, N_I is the set of $w \in \mathbb{R}^2$ for which there exists $g \in I$ with

$$\chi_{\zeta_1}(g) = \exp(2\pi\sqrt{-1}w_1) \quad \text{and} \quad \chi_{\zeta_2}(g) = \exp(2\pi\sqrt{-1}w_2).$$

By **Step 4** and (3.25), $\chi_{\zeta_1} = \eta^u$ and $\chi_{\zeta_2} = \eta^v$. By **Step 1**, there exists $g_0 \in I$ with $\eta(g_0) = \exp(2\pi\sqrt{-1}/e)$, so $(u, v)/e \in N_I$. Since $E = F$, (3.31) implies

$$\Psi\Big(\frac{(u,v)}{e}\Big) = \frac{a(E,S,B)}{e} = \mu.$$

Let $w \in N_I \cap \mathbb{R}^2_{>0}$. Since $N_I \subset \frac{1}{e}\mathbb{Z}^2$, we may write $w = t(u', v')$ with coprime positive integers u', v' and a rational number $t > 0$. Let F' be as in **Step 2**, $e' := |I_I(F')|$, and q'_1, q'_2 integers with $u'q'_1 + v'q'_2 = 1$. For $g \in I_I(F')$, let $\omega(g) := \chi_{\zeta_1}(g)^{q'_1}\chi_{\zeta_2}(g)^{q'_2}$. By (3.26), $\chi_{\zeta_1}(g) = \omega(g)^{u'}$ and $\chi_{\zeta_2}(g) = \omega(g)^{v'}$. Thus $\omega : I_I(F') \to \Bbbk^*$ is a homomorphism, injective because I acts faithfully on $(S \ni p)$ and $g \in I$ with $\chi_{\zeta_1}(g) = \chi_{\zeta_2}(g) = 1$ acts trivially on $\widehat{R}$. Hence $\omega(I_I(F'))$ is the group of e'-th roots of unity. Let $g \in I$ with $\chi_{\zeta_i}(g) = \exp(2\pi\sqrt{-1}w_i)$ for $i \in \{1, 2\}$. Then

$$\chi_{\zeta_2}(g)^{u'} = \exp(2\pi\sqrt{-1}tu'v') = \chi_{\zeta_1}(g)^{v'},$$

so $g \in I_I(F')$ by (3.26), and

$$\omega(g) = \exp(2\pi\sqrt{-1}t(u'q'_1 + v'q'_2)) = \exp(2\pi\sqrt{-1}t).$$

Thus $\exp(2\pi\sqrt{-1}t)^{e'} = 1$, so $te' \in \mathbb{N}^+$ and $t \geq 1/e'$. By (3.31) and (3.22),

$$\Psi(w) = ta(F', S, B) \geq te'\mu \geq \mu.$$

Hence $\mu = \min_{w\in N_I\cap\mathbb{R}^2_{>0}} \Psi(w)$, attained at $w = (u, v)/e$. This proves (A).

Step 6. In this step we prove (B).

We may assume that $\Lambda \neq \emptyset$. Let $\lambda := -\min\Lambda > 0$. If $\Gamma\backslash\{0\} \neq \emptyset$, let $\gamma := \min(\Gamma\backslash\{0\}) > 0$ and $n_0 := \lceil 2/\gamma \rceil$. Otherwise, let $n_0 := 0$. Let μ be a number of the set in (B), with data $\Bbbk$, $(S \ni p)$, D, $B = bD + \sum_{j=1}^{n_+} c_j H_j$, and G. Let E, I, ζ_1, ζ_2, N_I, $h_{\mathcal{H}}$, and $d_{\mathcal{H}}$ be as in (A), and let $\mathcal{H}_1, \ldots, \mathcal{H}_m$ be the I-orbits of the components of B_+. Since $(H_j \cdot D)_p \geq 1$ for any j,

$$\gamma n_+ \leq \sum_{j=1}^{n_+} c_j (H_j \cdot D)_p = (B_+ \cdot D)_p < 2$$

if $\Gamma\backslash\{0\} \neq \emptyset$, and $n_+ = 0$ otherwise. Thus $m \le n_+ \le n_0$. Let $\overline{I} \subset \mathbb{T}^2$ be the image of I under the characters of ζ_1, ζ_2, so that $N_{\overline{I}} = N_I$. Let

$$\mathbf{J} = (\mathbf{J}_1, \dots, \mathbf{J}_{n_0+2}) := \big((\zeta_1), (\zeta_2), J(h_{\mathcal{H}_1}), \dots, J(h_{\mathcal{H}_m}), (1), \dots, (1)\big)$$

and

$$\mathbf{c} = (\mathbf{c}_1, \dots, \mathbf{c}_{n_0+2}) := \Big(\frac{b+\lambda}{\lambda+1}, \frac{\lambda}{\lambda+1}, \frac{d_{\mathcal{H}_1}}{\lambda+1}, \dots, \frac{d_{\mathcal{H}_m}}{\lambda+1}, 0, \dots, 0\Big) \in \mathbb{R}^{n_0+2}.$$

Since $-\lambda \le b < 0$ and $d_{\mathcal{H}} \in \Gamma$, any entry of $\mathbf{c}$ is in

$$\Lambda' := \Big\{0, \frac{\lambda}{\lambda+1}\Big\} \cup \frac{\Lambda+\lambda}{\lambda+1} \cup \frac{\Gamma}{\lambda+1} \subset [0,1),$$

which is a DCC set depending only on Λ and Γ. By [Liu26a, Definition 2.16], for $w \in \mathbb{R}^2_{>0}$,

$$\psi_{\mathbf{J},\mathbf{c}}(w) = w_1 + w_2 - \sum_{i=1}^{n_0+2} \mathbf{c}_i \operatorname{wt}_w(\mathbf{J}_i) \quad \text{and} \quad \psi_{\min}(\overline{I}, \mathbf{J}, \mathbf{c}) = \inf\{\psi_{\mathbf{J},\mathbf{c}}(w) \mid w \in N_{\overline{I}} \cap \mathbb{R}^2_{>0}\}.$$

Since $\operatorname{wt}_w((\zeta_1)) = w_1$, $\operatorname{wt}_w((\zeta_2)) = w_2$, and $\operatorname{wt}_w((1)) = 0$,

$$(\lambda+1)\psi_{\mathbf{J},\mathbf{c}}(w) = (\lambda+1)(w_1+w_2) - (b+\lambda)w_1 - \lambda w_2 - \sum_{\mathcal{H}} d_{\mathcal{H}} \operatorname{wt}_w(J(h_{\mathcal{H}})) = \Psi(w),$$

with Ψ as in **Step 5**. By (A),

$$\mu = (\lambda+1)\psi_{\min}(\overline{I}, \mathbf{J}, \mathbf{c}).$$

Each $\mathbf{J}_i$ is generated by monomials, and $\psi_{\min}(\overline{I}, \mathbf{J}, \mathbf{c})$ depends only on $N_{\overline{I}}$, on the sets of exponents of the monomials in the $\mathbf{J}_i$, and on $\mathbf{c}$. The characters of I take values in the group of roots of unity, which does not depend on $\Bbbk$. Thus

$$\psi_{\min}(\overline{I}, \mathbf{J}, \mathbf{c}) = \psi_{\min}(\overline{I}, \mathbf{J}', \mathbf{c}),$$

where $\mathbf{J}' = (\mathbf{J}'_1, \dots, \mathbf{J}'_{n_0+2})$ and $\mathbf{J}'_i \subset \mathbb{C}[x_1, x_2]$ is the ideal generated by the monomials $x_1^{\nu_1}x_2^{\nu_2}$ with $\zeta_1^{\nu_1}\zeta_2^{\nu_2} \in \mathbf{J}_i$. By [Liu26a, Theorem 4.5] for $d = 2$, $s = n_0 + 2$, and the DCC set Λ', the numbers $\psi_{\min}(\overline{I}, \mathbf{J}, \mathbf{c})$ arising in this way, where $\Bbbk$ also varies, satisfy the ACC. Since $\lambda + 1$ depends only on Λ, the numbers μ satisfy the ACC. This proves (B). □

3.6. **The germs W_N and W_0.** Let $\widetilde{W}_0 := \mathbb{A}^3$ with coordinates p, v, t, let $\widetilde{W}_0 \to W_0$ be the double cover $(p, v, t) \to (p^2, -v^2, pv, t)$, and let $\iota : (p, v, t) \to (-p, -v, t)$ be its deck involution. To compare quotient values on W_N and on W_0, we need the following proposition.

Proposition 3.14 (cf. [Liu26b, Proposition 7.10])**.** *Let M be a positive integer, $\Gamma \subset [0,1]$ a DCC set, and $\gamma := \min(\Gamma\backslash\{0\})$ if $\Gamma \not\subset \{0\}$ and $\gamma := 1$ otherwise. Let $N \ge 2$ be an integer and G a finite diagonal abelian group of order at most M preserving the equation $xy + z^2 + t^N$ of W_N up to a character and acting faithfully on W_N, with $g(x, y, z, t) = (\chi_x(g)x, \chi_y(g)y, \chi_z(g)z, \chi_t(g)t)$ for $g \in G$, and let G act on $W_0 = (xy + z^2 = 0)$ by the restriction of the diagonal action of G on $\mathbb{A}^4$, which preserves W_0 since $\chi_x\chi_y = \chi_z^2$. Let $D_N = \sum_j c_j \operatorname{div}(h_j)$ be a labelled boundary on $(W_N \ni 0)$ with labels in Γ, each h_j a G-semi-invariant polynomial without monomials divisible by xy, such that (W_N, D_N) is klt near 0. Let*

$$D_0 := \sum_j c_j \operatorname{div}(h_j|_{W_0}), \quad a_N := \operatorname{mld}((W_N \ni 0, D_N)/G), \quad \text{and} \quad b_0 := \operatorname{mld}((W_0 \ni 0, D_0)/G),$$

where $h_j|_{W_0} \neq 0$ by [Liu26b, Lemma A.1(1)], and let $\varphi : W_N \dashrightarrow W_0$ be the G-equivariant birational map $(x, y, z, t) \to (x, y + t^N/x, z, t)$, whose inverse is $(x, y, z, t) \to (x, y - t^N/x, z, t)$. Then the following hold.

(1) *Let n_z and n_t be the orders of the subgroups of G acting only on z and only on t, respectively. Then the ramification divisors of $W_N \to W_N/G$ and of $W_0 \to W_0/G$ are both $(n_z-1)\operatorname{div}(z)+(n_t-1)\operatorname{div}(t)$. The quotients W_N/G and W_0/G are $\mathbb{Q}$-Gorenstein klt threefolds, the boundaries of $(W_N,0)/G$ and $(W_0,0)/G$ are $\mathbb{Q}$-Cartier, and the Cartier indices of $K_{W_N/G}$ and $K_{W_0/G}$ at the image of 0 both equal the order of $\chi_z^{n_z}\chi_t^{n_t}$. Moreover, the functions $z^{M!}$, $t^{M!}$, $h_j^{M!}$, and $(h_j|_{W_0})^{M!}$ are G-invariant. Write $(W_N,D_N)/G=(W_N/G,B_N)$, $(W_0,D_0)/G=(W_0/G,B_0)$, and*

$$\mathfrak{a}_N := (z^{M!})^{(n_z-1)/M!}(t^{M!})^{(n_t-1)/M!}\prod_j (h_j^{M!})^{c_j/M!},$$

$$\mathfrak{a}_0 := (z^{M!})^{(n_z-1)/M!}(t^{M!})^{(n_t-1)/M!}\prod_j \big((h_j|_{W_0})^{M!}\big)^{c_j/M!}.$$

Then $a(E,W_N/G,B_N)=a(E,W_N/G,\mathfrak{a}_N)$ for any prime divisor E over W_N/G, and $a(E,W_0/G,B_0)=a(E,W_0/G,\mathfrak{a}_0)$ for any prime divisor E over W_0/G.

(2) (a) *For any real number $\lambda>0$ there exists $N_\lambda=N_\lambda(M,\Gamma,\lambda)$ such that the following holds for $N\geq N_\lambda$. If F is a prime divisor over W_N with center 0 and $a(F,W_N,0)/e_F\leq\lambda$, then the birational transform F_0 of F under φ is a prime divisor over W_0 with center 0. Conversely, if F_0 is a prime divisor over W_0 with center 0 and $a(F_0,W_0,0)/e_{F_0}\leq\lambda$, then the birational transform F of F_0 under φ^{-1} is a prime divisor over W_N with center 0. In either case $\operatorname{Stab}_G(F)=\operatorname{Stab}_G(F_0)$, $I_G(F)=I_G(F_0)$, φ^* induces a G-equivariant isomorphism from the residue field of F_0 to that of F, $a(F,W_N,0)=a(F_0,W_0,0)$, and $\operatorname{mult}_F(h_j|_{W_N})=\operatorname{mult}_{F_0}(h_j|_{W_0})$ for any j with $c_j>0$, so that $a(F,W_N,D_N)=a(F_0,W_0,D_0)$.*

(b) *There exist $\lambda_0=\lambda_0(M,\Gamma)$ and $N_{\mathrm{ref}}=N_{\mathrm{ref}}(M,\Gamma)$ such that the following holds for $N\geq N_{\mathrm{ref}}$. The value b_0 has a computing lift F_0 over W_0 with $a(F_0,W_0,0)/e_{F_0}\leq\lambda_0$, the birational transform F of F_0 under φ^{-1} is a prime divisor over W_N with center 0 and $a(F,W_N,D_N)/e_F=b_0$, and $0<a_N\leq b_0\leq 2$. If moreover $a_N=b_0$, then F is a computing lift of a_N.*

(c) *If $N\geq N_{\mathrm{ref}}$, then any prime divisor F' over W_N with center 0 and $a(F',W_N,D_N)/e_{F'}<b_0$ satisfies $a(F',W_N,0)/e_{F'}\geq N/[M(2+1/\gamma)]$.*

(3) *If θ is a margin of (W_0,D_0) near 0, then $a_N=b_0$ for $N\geq N'_\theta(M,\Gamma,\theta)$. Moreover, for fixed M, Γ, and θ, the set of the numbers a_N above for which θ is a margin of (W_0,D_0) near 0, where $N\geq 2$, G, and D_N vary, satisfies the ACC.*

Proof. We let $\pi_N : W_N \to W_N/G$ and $\pi_0 : W_0 \to W_0/G$ be the quotient morphisms. By Definition 2.2,

$$a_N=\inf_F \frac{a(F,W_N,D_N)}{e_F} \quad\text{and}\quad b_0=\inf_{F_0}\frac{a(F_0,W_0,D_0)}{e_{F_0}}, \tag{3.32}$$

where F (resp. F_0) runs over the prime divisors over W_N (resp. W_0) with center 0. Since G fixes 0, we have $\pi_N^{-1}(\pi_N(0))=\{0\}$ and $\pi_0^{-1}(\pi_0(0))=\{0\}$. For any prime divisor F over W_N or W_0, we have $e_F\leq|G|\leq M$, and $c_j\geq\gamma$ for any j with $c_j>0$.

Step 1. Let $W\in\{W_N,W_0\}$. By adjunction, the rational 3-form

$$\Omega := \frac{dx\wedge dz\wedge dt}{x}$$

generates ω_W, so

$$a(F,W,0)=1+\operatorname{mult}_F\Omega \tag{3.33}$$

for any prime divisor F over W. We show that W is klt and that

$$\operatorname{mult}_F z\leq a(F,W,0) \tag{3.34}$$

for any prime divisor F over W with center 0.

Since the partial derivatives $y, x, 2z, Nt^{N-1}$ of $xy + z^2 + t^N$ vanish only at 0, W_N is smooth outside 0. The scheme $W_N \cap (z = 0) = (xy + t^N = 0) \subset \mathbb{A}^3_{x,y,t}$ is a normal surface with a Du Val singularity at 0. By inversion of adjunction, $(W_N, \operatorname{div}(z))$ is plt near 0. Thus W_N is klt, and $(W_N, \operatorname{div}(z))$ is lc near 0.

The double cover $\widetilde{W}_0 \to W_0$ is the quotient morphism by ι. Since ι fixes only the curve $(p = v = 0)$, the morphism $\widetilde{W}_0 \to W_0$ is étale in codimension 1, and the pullback of $K_{W_0} + \operatorname{div}(z)$ is

$$K_{\widetilde{W}_0} + \operatorname{div}(p) + \operatorname{div}(v).$$

Since $\widetilde{W}_0$ is smooth and $\operatorname{div}(p) + \operatorname{div}(v)$ is snc, W_0 is klt and $(W_0, \operatorname{div}(z))$ is lc. Thus $a(F, W, \operatorname{div}(z)) = a(F, W, 0) - \operatorname{mult}_F z \geq 0$ for $W \in \{W_N, W_0\}$, which is (3.34).

Step 2. We prove (1). Since G preserves $xy + z^2 + t^N$ up to a character,

$$\chi_x \chi_y = \chi_z^2 = \chi_t^N. \tag{3.35}$$

Let

$$G_z := \{g \in G \mid \chi_x(g) = \chi_y(g) = \chi_t(g) = 1\} \quad \text{and} \quad G_t := \{g \in G \mid \chi_x(g) = \chi_y(g) = \chi_z(g) = 1\},$$

so that $n_z = |G_z|$ and $n_t = |G_t|$. Let $W \in \{W_N, W_0\}$ and let $\pi \in \{\pi_N, \pi_0\}$ be the corresponding quotient morphism.

Let P be a prime divisor on W and $g \in G \backslash \{1\}$ an element fixing P pointwise, and let V_g be the set of $u \in \{x, y, z, t\}$ with $\chi_u(g) \neq 1$. Then

$$P \subset W \cap \bigcap_{u \in V_g} (u = 0).$$

Assume that $\#V_g \geq 2$. Then $P \subset W \cap (u = u' = 0)$ for two distinct $u, u' \in V_g$. Since $\dim P = 2$, the plane $(u = u' = 0) \subset \mathbb{A}^4$ is contained in W and equals P, so $\#V_g = 2$. The restriction of $xy + z^2 + t^N$ to any of the six coordinate planes is one of $z^2 + t^N$, t^N, z^2, and xy, none of which is 0, so $W = W_0$. The restriction of $xy + z^2$ vanishes only on the planes $(x = z = 0)$ and $(y = z = 0)$, so $V_g = \{x, z\}$ or $V_g = \{y, z\}$. In either case $\chi_t(g) = 1$, so $\chi_x(g)\chi_y(g) = \chi_z(g)^2 = 1$ by (3.35). Since one of $\chi_x(g)$ and $\chi_y(g)$ is 1, both are 1, which contradicts $\#(V_g \cap \{x, y\}) = 1$. Thus $\#V_g = 1$. By (3.35), $V_g \neq \{x\}$ and $V_g \neq \{y\}$. Hence $g \in G_z \cup G_t$, and P is a component of $\operatorname{div}(z)$ or of $\operatorname{div}(t)$.

Conversely, G_z fixes $\operatorname{div}(z)$ pointwise and G_t fixes $\operatorname{div}(t)$ pointwise. Since

$$W \cap (z = t = 0) = (xy = 0) \subset \mathbb{A}^2_{x,y}$$

is a curve, no element of $G_t \backslash \{1\}$ fixes a component of $\operatorname{div}(z)$ pointwise, and no element of $G_z \backslash \{1\}$ fixes $\operatorname{div}(t)$ pointwise. Thus the inertia group of any component of $\operatorname{div}(z)$ is G_z, that of $\operatorname{div}(t)$ is G_t, and any other prime divisor on W has trivial inertia group. The schemes $W_N \cap (z = 0)$, $W_0 \cap (z = 0) = (xy = 0) \subset \mathbb{A}^3_{x,y,t}$, and $W \cap (t = 0) = (xy + z^2 = 0) \subset \mathbb{A}^3_{x,y,z}$ are reduced, so $\operatorname{div}(z)$ and $\operatorname{div}(t)$ are reduced divisors on W. By the Hurwitz formula,

$$K_W = \pi^* K_{W/G} + R(\pi), \qquad R(\pi) = (n_z - 1)\operatorname{div}(z) + (n_t - 1)\operatorname{div}(t). \tag{3.36}$$

Let $\psi := \chi_z^{n_z} \chi_t^{n_t}$ and let n be the order of ψ. Since $g^* z = \chi_z(g) z$, $g^* t = \chi_t(g) t$ and $g^* \Omega = \chi_z(g)\chi_t(g)\Omega$ for $g \in G$, the rational section

$$\sigma := z^{n(n_z - 1)} t^{n(n_t - 1)} \Omega^{\otimes n}$$

of $\omega_W^{\otimes n}$ is G-invariant, and $\operatorname{div}(\sigma) = nR(\pi)$ because Ω generates ω_W. Thus $\sigma = \pi^* \eta$ for a rational section η of $\mathcal{O}_{W/G}(nK_{W/G})$, and by (3.36),

$$\pi^* \operatorname{div}(\eta) = \operatorname{div}(\sigma) - nR(\pi) = 0.$$

Since π^* is injective on $\mathbb{R}$-divisors, $\operatorname{div}(\eta) = 0$, so $nK_{W/G}$ is Cartier. Conversely, let n' be a positive integer such that $n'K_{W/G}$ is Cartier near $\pi(0)$, and let η' be a local generator of $\mathcal{O}_{W/G}(n'K_{W/G})$ at $\pi(0)$. By (3.36), $\operatorname{div}(\pi^*\eta') = n'R(\pi)$ near 0, so

$$\pi^*\eta' = w \cdot z^{n'(n_z-1)}t^{n'(n_t-1)}\Omega^{\otimes n'}$$

for a function w which is invertible near 0. Since $\pi^*\eta'$ is G-invariant, $g^*w = \psi(g)^{-n'}w$ for $g \in G$. Since g fixes 0 and $w(0) \neq 0$, $w(0) = (g^*w)(0) = \psi(g)^{-n'}w(0)$, so $\psi^{n'} = 1$. Thus the Cartier index of $K_{W/G}$ at $\pi(0)$ is n, and $K_{W/G}$ is $\mathbb{Q}$-Cartier.

Since $K_W - R(\pi) = \pi^*K_{W/G}$ is $\mathbb{Q}$-Cartier, Lemma 2.15(2) for the G-invariant divisor $-R(\pi)$ implies that for any prime divisor E over W/G and any prime divisor F over W whose valuation restricts to e_F times that of E,

$$a(E, W/G, 0) = \frac{a(F, W, 0) + \operatorname{mult}_F R(\pi)}{e_F} \geq \frac{a(F, W, 0)}{e_F} > 0, \tag{3.37}$$

where we use $R(\pi) \geq 0$ and **Step 1**. By Lemma 2.15(3), any E has such an F, so W/G is klt.

Since $|G| \leq M$, the order of any character of G divides $M!$. Since h_j is G-semi-invariant, the functions $z^{M!}$, $t^{M!}$, and $h_j^{M!}$ are G-invariant, that is, regular functions on W_N/G, and

$$\pi_N^* \operatorname{div}(z^{M!}) = M! \operatorname{div}(z), \quad \pi_N^* \operatorname{div}(t^{M!}) = M! \operatorname{div}(t), \quad \text{and} \quad \pi_N^* \operatorname{div}(h_j^{M!}) = M! \operatorname{div}(h_j).$$

By Lemma 2.15(0) and (3.36), $\pi_N^* B_N = K_{W_N} + D_N - \pi_N^* K_{W_N/G} = R(\pi_N) + D_N$. Since π_N^* is injective on $\mathbb{R}$-divisors,

$$B_N = \frac{n_z - 1}{M!} \operatorname{div}(z^{M!}) + \frac{n_t - 1}{M!} \operatorname{div}(t^{M!}) + \sum_j \frac{c_j}{M!} \operatorname{div}(h_j^{M!}).$$

Thus $\operatorname{mult}_E B_N = \operatorname{mult}_E \mathfrak{a}_N$ for any prime divisor E over W_N/G, and

$$a(E, W_N/G, B_N) = a(E, W_N/G, \mathfrak{a}_N).$$

Since the polynomial h_j is G-semi-invariant and G preserves W_0, $h_j|_{W_0}$ is G-semi-invariant, so $(h_j|_{W_0})^{M!}$ is G-invariant and D_0 is G-invariant. On W_0, (3.36) for π_0 and the G-invariant functions $z^{M!}$, $t^{M!}$, and $(h_j|_{W_0})^{M!}$ imply

$$B_0 = \frac{n_z - 1}{M!} \operatorname{div}(z^{M!}) + \frac{n_t - 1}{M!} \operatorname{div}(t^{M!}) + \sum_j \frac{c_j}{M!} \operatorname{div}\big((h_j|_{W_0})^{M!}\big),$$

so $a(E, W_0/G, B_0) = a(E, W_0/G, \mathfrak{a}_0)$ for any prime divisor E over W_0/G. Setting $D_N = 0$ and $D_0 = 0$ in the formulas for B_N and B_0 shows that the boundaries of $(W_N, 0)/G$ and $(W_0, 0)/G$ are $\mathbb{Q}$-linear combinations of principal divisors, hence $\mathbb{Q}$-Cartier. This proves (1).

Step 3. We compare prime divisors over W_N and over W_0. The map φ is G-equivariant since $\chi_t^N/\chi_x = \chi_y$ by (3.35), and it restricts to a G-equivariant isomorphism $W_N \cap (x \neq 0) \to W_0 \cap (x \neq 0)$. We have

$$\varphi^*x = x, \quad \varphi^*z = z, \quad \varphi^*t = t, \quad \varphi^*y = y + \frac{t^N}{x} = -\frac{z^2}{x}, \quad \text{and} \quad \varphi^*\Omega = \Omega.$$

Let F be a prime divisor over W_N and F_0 the birational transform of F under φ, or let F_0 be a prime divisor over W_0 and F the birational transform of F_0 under φ^{-1}. In either case, $\operatorname{mult}_{F_0} u = \operatorname{mult}_F \varphi^* u$ for any rational function u on W_0. Let

$$\alpha := \operatorname{mult}_F x, \quad \beta := \operatorname{mult}_F y, \quad \zeta := \operatorname{mult}_F z, \quad \tau := \operatorname{mult}_F t, \quad \text{and}$$

$$\beta_0 := \operatorname{mult}_{F_0} y = \operatorname{mult}_F \frac{-z^2}{x}.$$

We claim the following. Assume that F has center 0 on W_N or F_0 has center 0 on W_0, and that

$$N\tau > 2\zeta. \tag{3.38}$$

Then F is a prime divisor over W_N with center 0, F_0 is a prime divisor over W_0 with center 0, and

$$\operatorname{mult}_F\bigl(\varphi^*(h|_{W_0}) - h|_{W_N}\bigr) > N\tau - 2\zeta \tag{3.39}$$

for any polynomial h without monomials divisible by xy.

We prove the claim. We have $\alpha, \zeta, \tau \geq 1$ and $\alpha + \beta_0 = 2\zeta$. By (3.38),

$$\operatorname{mult}_F \frac{t^N}{x} = N\tau - \alpha = N\tau - 2\zeta + \beta_0 > \beta_0 = \operatorname{mult}_F \varphi^* y.$$

Since $y = \varphi^* y - t^N/x$ on W_N, we obtain $\beta = \beta_0$. Since F has center 0 on W_N or F_0 has center 0 on W_0, we have $\beta > 0$ or $\beta_0 > 0$, so $\beta = \beta_0 > 0$. The integers $\alpha, \beta, \zeta, \tau$ are the orders along F of the coordinate functions on W_N, and $\alpha, \beta_0, \zeta, \tau$ are the orders along F_0 of the coordinate functions on W_0. Thus F is a prime divisor over W_N with center 0, and F_0 is a prime divisor over W_0 with center 0. Moreover $\alpha = 2\zeta - \beta_0 < 2\zeta$, so $\operatorname{mult}_F(t^N/x) > N\tau - 2\zeta > 0$. Write

$$h = h^{(0)}(x, z, t) + \sum_{i \geq 1} y^i h^{(i)}(z, t)$$

with polynomials $h^{(0)}$ and $h^{(i)}$. Then

$$\varphi^*(h|_{W_0}) - h|_{W_N} = \sum_{i \geq 1} \sum_{k=1}^{i} \binom{i}{k} y^{i-k} \Bigl(\frac{t^N}{x}\Bigr)^k h^{(i)}(z, t),$$

and since $\operatorname{mult}_F y \geq 0$ and $\operatorname{mult}_F h^{(i)}(z, t) \geq 0$, each term has order at least

$$k \operatorname{mult}_F(t^N/x) > N\tau - 2\zeta.$$

This proves the claim.

Now assume that F has center 0 on W_N and

$$(2 + 1/\gamma) a(F, W_N, 0) < N. \tag{3.40}$$

By (3.34) and $\tau \geq 1$,

$$N\tau - 2\zeta \geq N - 2a(F, W_N, 0) > \frac{a(F, W_N, 0)}{\gamma} \geq 0,$$

so the claim holds. By (3.33) and $\varphi^*\Omega = \Omega$, $a(F_0, W_0, 0) = a(F, W_N, 0)$. Since (W_N, D_N) is lc near 0, $c_j \operatorname{mult}_F(h_j|_{W_N}) \leq a(F, W_N, 0)$ for any j. Thus, for any j with $c_j > 0$,

$$\operatorname{mult}_F(h_j|_{W_N}) \leq \frac{a(F, W_N, 0)}{\gamma} < N\tau - 2\zeta,$$

and (3.39) implies $\operatorname{mult}_{F_0}(h_j|_{W_0}) = \operatorname{mult}_F(h_j|_{W_N})$. If instead (W_0, D_0) is lc near 0, F_0 has center 0 on W_0, and $(2 + 1/\gamma) a(F_0, W_0, 0) < N$, then the claim holds by (3.34) on W_0, and (3.33), the inequalities $c_j \operatorname{mult}_{F_0}(h_j|_{W_0}) \leq a(F_0, W_0, 0)$, and (3.39) imply $a(F, W_N, 0) = a(F_0, W_0, 0)$ and $\operatorname{mult}_F(h_j|_{W_N}) = \operatorname{mult}_{F_0}(h_j|_{W_0})$ for any j with $c_j > 0$. In either case

$$a(F, W_N, D_N) = a(F, W_N, 0) - \sum_{c_j > 0} c_j \operatorname{mult}_F(h_j|_{W_N}) = a(F_0, W_0, D_0).$$

Since φ is G-equivariant, $\operatorname{mult}_{F_0} \circ g^* = \operatorname{mult}_F \circ g^* \circ \varphi^*$ for $g \in G$, so g maps F_0 to itself if and only if g maps F to itself. Thus $\operatorname{Stab}_G(F) = \operatorname{Stab}_G(F_0)$, and φ^* induces a G-equivariant isomorphism from the residue field of F_0 to that of F. Thus $I_G(F) = I_G(F_0)$, and $e_F = e_{F_0}$.

By [Liu26b, Proposition 7.10(1)], there exists $N_0 = N_0(\Gamma)$ such that (W_0, D_0) is lc near 0 for $N \geq N_0$. For $\lambda > 0$ let

$$N_\lambda := \max\bigl\{N_0, \lfloor (2 + 1/\gamma) M\lambda \rfloor + 1\bigr\}.$$

If $a(F, W_N, 0)/e_F \leq \lambda$, then $a(F, W_N, 0) \leq M\lambda$, so (3.40) holds for $N \geq N_\lambda$. If $N \geq N_\lambda$ and $a(F_0, W_0, 0)/e_{F_0} \leq \lambda$, then

$$(2 + 1/\gamma) a(F_0, W_0, 0) \leq (2 + 1/\gamma) M\lambda < N,$$

and (W_0, D_0) is lc near 0 since $N \geq N_0$. This proves (2)(a).

Step 4. We prove (2)(b) and (2)(c). Since the set Γ_+ (Definition 2.2) is a DCC set,

$$\Gamma_M := \left\{1 - \frac{1-c}{e} \;\middle|\; c \in \Gamma_+,\ e \in \{1, \dots, M\}\right\} \backslash \{0\}$$

is a DCC subset of $\mathbb{R}_{>0}$. For any prime divisor P on W_0,

$$\operatorname{mult}_P D_0 = \sum_j c_j \operatorname{mult}_P(h_j|_{W_0}) \in \Gamma_+,$$

and $e_P \leq M$, so $B_0 \in \Gamma_M$ by Lemma 2.15(4). By (1), B_0 is an $\mathbb{R}$-linear combination of principal divisors, hence $\mathbb{R}$-Cartier.

Since any element of G has order at most M, G is a subgroup of the finite group of diagonal matrices whose entries are $M!$-th roots of unity. Thus G ranges over finitely many groups, and W_0/G over finitely many klt threefolds (see (1)). Let $\lambda_0 = \lambda_0(M, \Gamma)$ be the largest of the integers given by Theorem 2.22(3) for the DCC set Γ_M and the finitely many threefolds $X = W_0/G$, and let $N_{\mathrm{ref}} := N_{\lambda_0}$.

Assume that $N \geq N_{\mathrm{ref}}$. Then (W_0, D_0) is lc near 0, so $(W_0/G, B_0)$ is lc near $\pi_0(0)$ by Lemma 2.15(2)(3). By Theorem 2.22(3), there exists a prime divisor E over W_0/G which computes $\operatorname{mld}(W_0/G \ni \pi_0(0), B_0) = b_0$ and satisfies $a(E, W_0/G, 0) \leq \lambda_0$. By Lemma 2.15(3), there exists a prime divisor F_0 over W_0 whose valuation restricts to e_{F_0} times that of E. Then F_0 has center 0, $a(F_0, W_0, D_0)/e_{F_0} = b_0$, and by (3.37),

$$\frac{a(F_0, W_0, 0)}{e_{F_0}} \leq \frac{a(F_0, W_0, 0) + \operatorname{mult}_{F_0} R(\pi_0)}{e_{F_0}} = a(E, W_0/G, 0) \leq \lambda_0.$$

By (2)(a) with $\lambda = \lambda_0$, the birational transform F of F_0 under φ^{-1} is a prime divisor over W_N with center 0 and $a(F, W_N, D_N)/e_F = b_0$. Thus $a_N \leq b_0$ by (3.32).

Let E_1 be the exceptional divisor of the blow-up of W_0 at 0. Then $a(E_1, W_0, 0) = 2$, so by (3.32), $b_0 \leq a(E_1, W_0, D_0) \leq 2$ since $D_0 \geq 0$. Since (W_N, D_N) is klt near 0, $(W_N/G, B_N)$ is klt near $\pi_N(0)$ by Lemma 2.15(2)(3), so $a_N > 0$. Thus $0 < a_N \leq b_0 \leq 2$.

Let F' be a prime divisor over W_N with center 0 and $a(F', W_N, D_N)/e_{F'} < b_0$, and suppose that $a(F', W_N, 0)/e_{F'} < N/[M(2 + 1/\gamma)]$. Then $(2 + 1/\gamma)a(F', W_N, 0) < N$, so by **Step 3** the birational transform F'_0 of F' under φ is a prime divisor over W_0 with center 0 and

$$\frac{a(F'_0, W_0, D_0)}{e_{F'_0}} = \frac{a(F', W_N, D_N)}{e_{F'}} < b_0,$$

which contradicts (3.32). Thus $a(F', W_N, 0)/e_{F'} \geq N/[M(2+1/\gamma)]$. If $a_N = b_0$, then F computes a_N. This proves (2).

Step 5. Assume that θ is a margin of (W_0, D_0) near 0. Let $\Lambda := \{1\} \cup \theta(\Gamma \backslash \{0\})$, a DCC subset of $\mathbb{R}_{>0}$, and let $L = L(\Gamma, \theta)$ be the largest of the integers given by Lemma 2.19 for $n = 4$, Λ, and at most $2/\gamma + 1$ functions. We show that $(W_N, \theta D_N)$ is lc near 0 for $N \geq L$. For any j with $c_j = 0$ or $h_j(0) \neq 0$, we have $c_j \operatorname{div}(h_j) = 0$ near 0 on W_N and $c_j \operatorname{div}(h_j|_{W_0}) = 0$ near 0 on W_0. Possibly reordering indices, we may assume that $h_1, \dots, h_m$ are the h_j with $c_j > 0$ and $h_j(0) = 0$. Since $(W_N, \sum_{j=1}^m c_j \operatorname{div}(h_j))$ is klt near 0 and $c_j \geq \gamma$ for $j \leq m$, we have $m < 2/\gamma$ by [Liu26b, Lemma A.1(3)].

Let $\mathcal{R} := \mathbb{C}[[x, y, z, t]]$, $\widehat{W}_N := \operatorname{Spec} \mathcal{R}/(xy + z^2 + t^N)$, and $\widehat{W}_0 := \operatorname{Spec} \mathcal{R}/(xy + z^2)$, which are normal since W_N and W_0 are normal. For $j \leq m$, the function $h_j|_{W_N}$ is non-zero since $\operatorname{div}(h_j)$ occurs in D_N with coefficient $c_j > 0$, and $h_j|_{W_0}$ is non-zero by [Liu26b, Lemma A.1(1)], so they are non-zero in the completions $\mathcal{R}/(xy + z^2 + t^N)$ and $\mathcal{R}/(xy + z^2)$ of the local rings of W_N and W_0 at 0. Equivalently, $h_j \notin (xy + z^2 + t^N)$ and $h_j \notin (xy + z^2)$ in $\mathcal{R}$. Since $(W_0, \theta D_0)$ is lc near

0, the pair $(\widehat{W}_0, \prod_{j=1}^m (h_j)^{\theta c_j})$ is lc by [dFEM11, Proposition 2.11], so

$$\Big(\operatorname{Spec} \mathcal{R}, (xy+z^2)\cdot \prod_{j=1}^m (h_j)^{\theta c_j}\Big)$$

is lc by [Liu26b, Lemma D.14]. Assume that $N \geq L$. Then

$$(xy+z^2+t^N)-(xy+z^2) = t^N \in (x,y,z,t)^L.$$

By Lemma 2.19 applied to the $m+1$ functions $xy+z^2, h_1, \dots, h_m$ and $xy+z^2+t^N, h_1, \dots, h_m$ with exponents $1, \theta c_1, \dots, \theta c_m \in \Lambda$, the pair

$$\Big(\operatorname{Spec} \mathcal{R}, (xy+z^2+t^N)\cdot \prod_{j=1}^m (h_j)^{\theta c_j}\Big)$$

is lc. By [Liu26b, Lemma D.14], $(\widehat{W}_N, \prod_{j=1}^m (h_j)^{\theta c_j})$ is lc, and by [dFEM11, Proposition 2.11], $(W_N, \theta D_N)$ is lc near 0.

Step 6. We prove (3). Let

$$\lambda := \frac{2\theta}{\theta-1} \quad \text{and} \quad N'_\theta = N'_\theta(M,\Gamma,\theta) := \max\{L, N_{\mathrm{ref}}, N_\lambda\}.$$

Assume that $N \geq N'_\theta$. By (2)(b), $a_N \leq b_0 \leq 2$. Since $(W_N/G, B_N)$ is klt near $\pi_N(0)$, there exists a prime divisor E' over W_N/G which computes $a_N = \operatorname{mld}(W_N/G \ni \pi_N(0), B_N)$ [Liu26a, Lemma 3.5]. By Lemma 2.15(3), there exists a prime divisor F' over W_N whose valuation restricts to $e_{F'}$ times that of E'. Then F' has center 0 and $a(F', W_N, D_N) = e_{F'} a_N$. By **Step 5**, $(W_N, \theta D_N)$ is lc near 0, so

$$\theta \sum_j c_j \operatorname{mult}_{F'}(h_j|_{W_N}) \leq a(F', W_N, 0) = a(F', W_N, D_N) + \sum_j c_j \operatorname{mult}_{F'}(h_j|_{W_N}).$$

Thus $(1-1/\theta)a(F', W_N, 0) \leq a(F', W_N, D_N)$, and

$$\frac{a(F', W_N, 0)}{e_{F'}} \leq \frac{\theta}{\theta-1} a_N \leq \lambda.$$

By (2)(a), the birational transform F'_0 of F' under φ is a prime divisor over W_0 with center 0 and $a(F'_0, W_0, D_0)/e_{F'_0} = a_N$, so $b_0 \leq a_N$ by (3.32). Hence $a_N = b_0$.

Suppose that the numbers a_N in the second sentence of (3) contain a strictly increasing sequence

$$a^{(1)} < a^{(2)} < \cdots,$$

where $a^{(i)} = \operatorname{mld}((W_{N_i} \ni 0, D^{(i)})/G_i)$ is the value of the data $(N_i, G_i, D^{(i)})$. If $\{N_i\}_i$ is bounded, then, possibly passing to a subsequence, $N_i = N$ for any i, and $\{a^{(i)}\}_i$ is a strictly increasing sequence of positive values $\operatorname{mld}((W_N \ni 0, D^{(i)})/G_i)$, which contradicts [Liu26b, Theorem 7.8]. Otherwise, possibly passing to a subsequence, $N_i \geq N'_\theta$ for any i and, since G_i ranges over finitely many groups (**Step 4**), $G_i = G$ for any i. Let $D_0^{(i)}$ be the restriction of $D^{(i)}$ to W_0 as in the statement, and let $(W_0, D_0^{(i)})/G = (W_0/G, B_0^{(i)})$. Then

$$a^{(i)} = \operatorname{mld}(W_0/G \ni \pi_0(0), B_0^{(i)})$$

by the first part of (3) and Lemma 2.15, and $B_0^{(i)} \in \Gamma_M$ by **Step 4**. This contradicts Theorem 2.22(1) for the normal threefold W_0/G and the DCC set Γ_M. This proves (3). □

3.7. **Specialization of prime divisors.**

Lemma 3.15 (cf. [Kaw21, Remark 3.8(i)]). *Let H, N and the tuples $(f_{i,j})_j$ of regular functions on $\mathbb{A}^n$ be as in Definition 2.5, let $(f_{*,j})_j$ be a compatible generic limit of the tuples $(f_{i,j})_j$ along an infinite subset $\mathcal{I} \subset \mathbb{N}^+$, with coefficient field $\Bbbk$, and let $\mathcal{A}_\Bbbk := \operatorname{Spec} \Bbbk[[x_1, \dots, x_n]]$. Let P be a formal prime divisor over $\mathcal{A}_\Bbbk$ with center the closed point. Then for all but finitely many $i \in \mathcal{I}$ there exists a prime divisor P_i over $\mathbb{A}^n$ with center 0 such that $a(P_i, \mathbb{A}^n, 0) = a(P, \mathcal{A}_\Bbbk, 0)$ and $\operatorname{mult}_{P_i} f_{i,j} = \operatorname{mult}_P f_{*,j}$ for any j.*

Proof. We let $\mathfrak{m} := (x_1, \dots, x_n)$. For any $j \in \{1, \dots, N\}$ let $\mu_j := \operatorname{mult}_P f_{*,j}$. Let d be a positive integer with $d > \mu_j$ for any j, and for any j let $q_j \in \Bbbk[x_1, \dots, x_n]$ be the part of degree less than d of $f_{*,j}$, so that $(q_j)_j$ is the d-jet of $(f_{*,j})_j$.

Step 1. *Reduction to $\mathbb{A}^n_\Bbbk$.* Since the center of P is the closed point of $\mathcal{A}_\Bbbk$, we have $\operatorname{mult}_P \mathfrak{m} \geq 1$. Since $f_{*,j} - q_j \in \mathfrak{m}^d \Bbbk[[x_1, \dots, x_n]]$, we have $\operatorname{mult}_P(f_{*,j} - q_j) \geq d > \mu_j$, so

$$\operatorname{mult}_P q_j = \mu_j \quad \text{for any } j \in \{1, \dots, N\}. \tag{3.41}$$

By Lemma 2.12(3) for $(X \ni x) = (\mathbb{A}^n_\Bbbk \ni 0)$, $B = 0$ and the trivial group, P restricts to a prime divisor $P_\Bbbk$ over $\mathbb{A}^n_\Bbbk$ with center 0, so that $\operatorname{mult}_{P_\Bbbk} g = \operatorname{mult}_P g$ for any $g \in \Bbbk[x_1, \dots, x_n]$, and

$$a(P_\Bbbk, \mathbb{A}^n_\Bbbk, 0) = a(P, \mathcal{A}_\Bbbk, 0).$$

By (3.41), $\operatorname{mult}_{P_\Bbbk} q_j = \mu_j$ for any j.

Step 2. *An affine chart of $P_\Bbbk$.* Let $\pi_0 : Y_0 \to \mathbb{A}^n_\Bbbk$ be a birational morphism from a normal variety Y_0 such that $P_\Bbbk$ is a prime divisor on Y_0. Since Y_0 is normal, Y_0 is smooth at the generic point of $P_\Bbbk$. Thus there exist an affine open subset $Y_1 \subset Y_0$ and $u \in \mathcal{O}(Y_1)$ such that Y_1 is smooth, $P_\Bbbk \cap Y_1$ is non-empty and smooth, and $P_\Bbbk \cap Y_1 = (u = 0)$ as schemes. We fix a closed point $o \in P_\Bbbk \cap Y_1$ and a closed embedding $Y_1 \subset \mathbb{A}^m_\Bbbk$, with coordinates $y_1, \dots, y_m$, such that $y_m|_{Y_1} = u$.

By the Jacobian criterion, there exist polynomials $\varphi_1, \dots, \varphi_{m-n} \in \Bbbk[y_1, \dots, y_m]$ which generate the ideal of Y_1 near o and whose differentials at o are linearly independent. Since $(u = 0) = P_\Bbbk \cap Y_1$ is smooth of dimension $n-1$ at o, $dy_m(o)$ is not a linear combination of $d\varphi_1(o), \dots, d\varphi_{m-n}(o)$, so the Jacobian matrix of $(\varphi_1, \dots, \varphi_{m-n}, y_m)$ at o has a non-zero $(m-n+1)$-minor. Expanding it along the row of y_m and possibly reordering $y_1, \dots, y_{m-1}$, we may assume that

$$\delta := \det \frac{\partial(\varphi_1, \dots, \varphi_{m-n})}{\partial(y_1, \dots, y_{m-n})}$$

does not vanish at o. Thus there exists a polynomial $\psi \in \Bbbk[y_1, \dots, y_m]$, divisible by δ and not vanishing at o, such that $Y := Y_1 \cap (\psi \neq 0)$ satisfies

$$Y = \operatorname{Spec} R, \quad R := \Bbbk[y_1, \dots, y_m]_\psi / (\varphi_1, \dots, \varphi_{m-n}).$$

We let $\pi := \pi_0|_Y$, and for any $k \in \{1, \dots, n\}$ we let $\gamma_k \in \Bbbk[y_1, \dots, y_m]_\psi$ be an element whose image in R is $\pi^* x_k$. Since δ is invertible in R, Ω_Y is free with basis $dy_{m-n+1}, \dots, dy_m$, and eliminating $dy_1, \dots, dy_{m-n}$ by $d\varphi_1 = \cdots = d\varphi_{m-n} = 0$ implies

$$\begin{aligned} &\pi^*(dx_1 \wedge \cdots \wedge dx_n) = J dy_{m-n+1} \wedge \cdots \wedge dy_m, \\ &J := \delta^{-1} \det \frac{\partial(\varphi_1, \dots, \varphi_{m-n}, \gamma_1, \dots, \gamma_n)}{\partial(y_1, \dots, y_m)} \in R. \end{aligned} \tag{3.42}$$

Thus $K_Y - \pi^* K_{\mathbb{A}^n_\Bbbk} = \operatorname{div}(J)$, and

$$a(P_\Bbbk, \mathbb{A}^n_\Bbbk, 0) = 1 + \mu_0, \quad \text{where } \mu_0 := \operatorname{mult}_{P_\Bbbk} J.$$

The local ring $\mathcal{O}_{Y, P_\Bbbk}$ of Y at the generic point of $P_\Bbbk$ is a discrete valuation ring with uniformizer u. Since $\operatorname{mult}_{P_\Bbbk} \mathfrak{m} \geq 1$, $\operatorname{mult}_{P_\Bbbk} q_j = \mu_j$ and $\operatorname{mult}_{P_\Bbbk} J = \mu_0$, there exist elements $r_1, \dots, r_n$ and units $w_0, \dots, w_N$ of $\mathcal{O}_{Y, P_\Bbbk}$ such that

$$\pi^* x_k = u r_k, \quad \pi^* q_j = u^{\mu_j} w_j, \quad \text{and} \quad J = u^{\mu_0} w_0 \tag{3.43}$$

for any $k \in \{1, \dots, n\}$ and $j \in \{1, \dots, N\}$. Possibly replacing ψ with the product of ψ with a polynomial not vanishing identically on $P_{\Bbbk} \cap Y$, we may assume that $r_k \in R$ for any k and $w_j, w_j^{-1} \in R$ for any $j \in \{0, \dots, N\}$. Since π is birational, there exist polynomials $\theta \neq 0$ and $\alpha_0, \dots, \alpha_m$ in $\Bbbk[x_1, \dots, x_n]$ such that

$$\pi^*\theta \cdot \psi^{-1} = \pi^*\alpha_0 \quad \text{and} \quad \pi^*\theta \cdot y_t = \pi^*\alpha_t \quad \text{in } R \text{ for any } t \in \{1, \dots, m\}. \tag{3.44}$$

Step 3. *Spreading out.* We write any element of $\Bbbk[y_1, \dots, y_m]_\psi$ as g/ψ^s with $g \in \Bbbk[y_1, \dots, y_m]$ and $s \in \mathbb{N}$. Let $\Lambda \subset \Bbbk$ be the $\mathbb{C}$-subalgebra generated by the following finitely many elements of $\Bbbk$:

(1) The coefficients of $\varphi_1, \dots, \varphi_{m-n}$, ψ, ψ/δ, $q_1, \dots, q_N$, θ, and $\alpha_0, \dots, \alpha_m$, and the inverse of a non-zero coefficient of θ.
(2) The coefficients of the numerators of $\gamma_1, \dots, \gamma_n$ and of fixed elements of $\Bbbk[y_1, \dots, y_m]_\psi$ representing $r_1, \dots, r_n$ and w_j, w_j^{-1} for $j \in \{0, \dots, N\}$.
(3) For any one of the identities (3.43), (3.44), and $w_j w_j^{-1} = 1$, the coefficients of the numerators of elements $c_1, \dots, c_{m-n} \in \Bbbk[y_1, \dots, y_m]_\psi$ such that $\sum_l c_l \varphi_l$ is the left-hand side minus the right-hand side of the identity, computed in $\Bbbk[y_1, \dots, y_m]_\psi$ with the representatives of (2).

Then Λ is a finitely generated $\mathbb{C}$-algebra and a domain. We let

$$R_\Lambda := \Lambda[y_1, \dots, y_m]_\psi/(\varphi_1, \dots, \varphi_{m-n}),$$

so that $R = R_\Lambda \otimes_\Lambda \Bbbk$. Since $\delta^{-1} = (\psi/\delta) \cdot \psi^{-1}$, the formula for J in (3.42) defines an element $J \in R_\Lambda$, and $u = y_m$, r_k, $w_j^{\pm 1}$, and J are elements of R_Λ satisfying (3.43), (3.44), and $w_j w_j^{-1} = 1$ in R_Λ.

Let $\rho : \operatorname{Spec} \Lambda \to \mathcal{J}_d$ be the morphism given by the coefficients of $q_1, \dots, q_N$. The composition of $\operatorname{Spec} \Bbbk \to \operatorname{Spec} \Lambda$ with ρ is the d-jet of $(f_{*,j})_j$, and $\operatorname{Spec} \Bbbk$ maps to the generic point of $\operatorname{Spec} \Lambda$ as $\Lambda \subset \Bbbk$. By Definition 2.5, ρ maps the generic point of $\operatorname{Spec} \Lambda$ to the generic point of Z_d, so $\rho(\operatorname{Spec} \Lambda) \subset Z_d$. Since

$$(R_\Lambda/(u)) \otimes_\Lambda \Bbbk = R/(u) \neq 0,$$

the image of $\operatorname{Spec} R_\Lambda/(u) \to \operatorname{Spec} \Lambda$ contains the generic point of $\operatorname{Spec} \Lambda$. By Chevalley's theorem, the image of $\operatorname{Spec} R_\Lambda/(u) \to \operatorname{Spec} \Lambda$ contains a non-empty open subset $U \subset \operatorname{Spec} \Lambda$, and $\rho(U)$ contains a non-empty open subset $Z_d^\circ \subset Z_d$. By Definition 2.5, there exists a finite subset $\mathcal{I}_0 \subset \mathcal{I}$ such that the d-jet z_i of $(f_{i,j})_j$ belongs to Z_d° for any $i \in \mathcal{I} \backslash \mathcal{I}_0$.

Step 4. *The prime divisor P_i.* We fix $i \in \mathcal{I} \backslash \mathcal{I}_0$. Since $\rho^{-1}(z_i) \cap U$ is a non-empty locally closed subset of $\operatorname{Spec} \Lambda$, it contains a closed point of $\operatorname{Spec} \Lambda$, that is, there exists a $\mathbb{C}$-algebra homomorphism $\beta : \Lambda \to \mathbb{C}$ such that $\ker \beta \in U$ and $\rho(\ker \beta) = z_i$. For an element e of Λ, of R_Λ or of a polynomial ring over Λ, we let e_β be the image of e under β. We let

$$R_\beta := R_\Lambda \otimes_{\Lambda,\beta} \mathbb{C} = \mathbb{C}[y_1, \dots, y_m]_{\psi_\beta}/(\varphi_{1,\beta}, \dots, \varphi_{m-n,\beta}), \quad Y_\beta := \operatorname{Spec} R_\beta,$$

and let $\pi_\beta : Y_\beta \to \mathbb{A}^n$ be the morphism with $\pi_\beta^* x_k = \gamma_{k,\beta}$ for any k. Since $\rho(\ker \beta) = z_i$, $q_{j,\beta}$ is the part of degree less than d of $f_{i,j}$ for any j. The identities (3.43) and (3.44) hold in R_β with the subscript β added to π, u, r_k, q_j, w_j, J, θ, α_t, and ψ, the elements $w_{0,\beta}, \dots, w_{N,\beta}$ are units of R_β, $\theta_\beta \neq 0$, and δ_β divides ψ_β.

(a) Since Y_β is cut out by $m - n$ equations in an open subset of $\mathbb{A}^m$, $\dim Y' \geq n$ for any irreducible component Y' of Y_β. Since δ_β is invertible on Y_β, the Zariski tangent space of Y_β at any closed point is contained in the kernel of the differentials of $\varphi_{1,\beta}, \dots, \varphi_{m-n,\beta}$, of dimension n. Thus Y_β is smooth of pure dimension n, ω_{Y_β} is generated by $dy_{m-n+1} \wedge \cdots \wedge dy_m$, and (3.42) with the subscript β implies

$$\pi_\beta^*(dx_1 \wedge \cdots \wedge dx_n) = J_\beta dy_{m-n+1} \wedge \cdots \wedge dy_m.$$

(b) Since $\ker\beta \in U$, we have

$$R_\beta/(u_\beta) = (R_\Lambda/(u)) \otimes_{\Lambda,\beta} \mathbb{C} \neq 0.$$

The minor of the Jacobian matrix of $(\varphi_{1,\beta}, \dots, \varphi_{m-n,\beta}, y_m)$ in the columns $1, \dots, m-n, m$ is δ_β, which vanishes nowhere on Y_β. As in (a), $(u_\beta = 0) \subset Y_\beta$ is non-empty and smooth of pure dimension $n-1$, and $du_\beta \neq 0$ in the cotangent space of Y_β at any closed point. Since Y_β is smooth, u_β vanishes identically on no irreducible component of Y_β.

(c) Let Y' be an irreducible component of Y_β. Since $J_\beta = u_\beta^{\mu_0} w_{0,\beta}$ with $w_{0,\beta}$ a unit, (b) implies that J_β vanishes identically on no Y', so by (a), π_β is étale at some closed point of Y', and $\pi_\beta(Y')$ is dense in $\mathbb{A}^n$. Since $\theta_\beta \neq 0$, $\pi_\beta^*\theta_\beta$ vanishes identically on no Y'. By (3.44) with the subscript β, the homomorphism

$$\mathbb{C}[x_1, \dots, x_n]_{\theta_\beta} \to (R_\beta)_{\pi_\beta^*\theta_\beta}$$

is surjective, and it is injective since R_β is reduced and any Y' dominates $\mathbb{A}^n$. Thus π_β induces an isomorphism

$$Y_\beta \cap (\pi_\beta^*\theta_\beta \neq 0) \cong (\theta_\beta \neq 0) \subset \mathbb{A}^n,$$

so the open subset $Y_\beta \cap (\pi_\beta^*\theta_\beta \neq 0)$ is irreducible, and $Y' \cap (\pi_\beta^*\theta_\beta \neq 0) \neq \emptyset$ for any Y'. Since the irreducible components of the smooth scheme Y_β are disjoint, Y_β is irreducible and π_β is birational.

(d) Let Q be an irreducible component of $(u_\beta = 0)$. By (a), (b), and (c), Q is a prime divisor on the smooth variety Y_β, u_β is a uniformizer of the local ring of Y_β at the generic point of Q, and $\pi_\beta : Y_\beta \to \mathbb{A}^n$ is a birational morphism. Thus Q is a prime divisor over $\mathbb{A}^n$, which we denote by P_i. Since $\pi_\beta^* x_k = u_\beta r_{k,\beta}$ with $r_{k,\beta} \in R_\beta$ for any k, we have $\operatorname{mult}_{P_i} \mathfrak{m} \geq 1$, so the center $\overline{\pi_\beta(Q)}$ of P_i on $\mathbb{A}^n$ is 0. By (3.43) with the subscript β, (a) and **Steps 1–2**,

$$\operatorname{mult}_{P_i} q_{j,\beta} = \mu_j \quad \text{and} \quad a(P_i, \mathbb{A}^n, 0) = 1 + \operatorname{mult}_Q J_\beta = 1 + \mu_0 = a(P, \mathcal{A}_\Bbbk, 0).$$

Since $f_{i,j} - q_{j,\beta} \in \mathfrak{m}^d$ and $\operatorname{mult}_{P_i} \mathfrak{m} \geq 1$, we have

$$\operatorname{mult}_{P_i}(f_{i,j} - q_{j,\beta}) \geq d > \mu_j, \quad \text{so} \quad \operatorname{mult}_{P_i} f_{i,j} = \mu_j = \operatorname{mult}_P f_{*,j}$$

for any j. The lemma follows. □

4. Radial functions and the reduction to cyclic quotients

In this section, a failure of the gap of Theorem 1.5 is reduced to a radial failure sequence (see Definition 4.1 and Construction-Theorem 4.7).

4.1. **Radial functions.**

Definition 4.1. Let $(W \ni o)$ be a normal threefold germ, G a finite group acting faithfully on W and fixing o, h a mark on $(W \ni o)$, and m a positive integer such that $(W, \frac{1}{m}\operatorname{div}(h))$ is lc near o. The *radial function* of $(W \ni o, G, h, m)$ is

$$q(t) := \operatorname{mld}\Big(\Big(W \ni o, \frac{t}{m}\operatorname{div}(h)\Big)/G\Big), \qquad t \in [0,1].$$

For a positive integer m and real numbers $a > \delta > 0$, a *radial failure sequence* with constants (m, a, δ) is a sequence of data $(W_i \ni o_i, G_i, h_i, \tau_i)$ such that the following hold for any i.

(1) $(W_i \ni o_i)$ is an isolated Gorenstein terminal hypersurface threefold germ, and G_i is a finite cyclic group acting faithfully on W_i and fixing o_i.
(2) h_i is a mark on $(W_i \ni o_i)$ with $h_i(o_i) = 0$ such that $(W_i, \frac{1}{m}\operatorname{div}(h_i))$ is lc near o_i.
(3) $\tau_i \in [0,1)$, $\lim_{i\to+\infty} \tau_i = 1$, and the radial function q_i of $(W_i \ni o_i, G_i, h_i, m)$ satisfies $0 \leq q_i(1) \leq a - \delta$ and $q_i(\tau_i) \geq a$.

We write $b_i := q_i(1)$.

Lemma 4.2. *Notation as in Definition 4.1, with radial function q of $(W \ni o, G, h, m)$. Then the following hold.*

(1) *q is non-negative, decreasing, concave and continuous on $[0,1]$, and $q(t)$ has a computing lift for any $t \in [0,1]$. If K_W is Cartier near o, then $q(1) \in (m|G|)^{-1}\mathbb{N}$.*

(2) *For any computing lift F of $q(1)$,*

$$\frac{\operatorname{mult}_F h}{me_F} = \frac{a(F,W,0)}{e_F} - q(1) \quad \text{and} \quad q(1) \le q(t) \le q(1) + (1-t)\frac{\operatorname{mult}_F h}{me_F} \quad \text{for any } t \in [0,1].$$

(3) *If $(W \ni o)$ is a hypersurface threefold germ, then $q \le 3$, and if moreover W is singular at o, then $q \le 2$.*

Proof. (1) Let $\pi : W \to W/G$ be the quotient morphism and $z := \pi(o)$. Since G fixes o, $\pi^{-1}(z) = \{o\}$. Since $K_W + \frac{1}{m}\operatorname{div}(h)$ is $\mathbb{R}$-Cartier near o and $\operatorname{div}(h)$ is Cartier, K_W is $\mathbb{R}$-Cartier near o. For any $t \in [0,1]$, let

$$(W/G, \Delta_t) := \Big(W, \frac{t}{m}\operatorname{div}(h)\Big)/G.$$

Since h is regular and $t \le 1$, $(W, \frac{t}{m}\operatorname{div}(h))$ is lc near o. Since π is finite and $\pi^{-1}(z) = \{o\}$, Lemma 2.15(2)(3) implies that $(W/G, \Delta_t)$ is lc near z, and that $q(t) \ge 0$.

For any prime divisor P on W with ramification index e_P over $\pi(P)$, Lemma 2.15(4) implies

$$\operatorname{mult}_{\pi(P)} \Delta_t = 1 - \frac{1}{e_P} + t\frac{\operatorname{mult}_P h}{me_P}.$$

Since any prime divisor on W/G is of the form $\pi(P)$, we have

$$\Delta_t = (1-t)\Delta_0 + t\Delta_1 \quad \text{and} \quad \Delta_1 - \Delta_0 \ge 0,$$

and $\Delta_1 - \Delta_0 = (K_{W/G} + \Delta_1) - (K_{W/G} + \Delta_0)$ is $\mathbb{R}$-Cartier by Lemma 2.15(0). Let $\mathfrak{m}_z$ be the maximal ideal of $\mathcal{O}_{W/G,z}$, and $g : Y \to W/G$ a log resolution of $(W/G, \Delta_0 + \Delta_1)$ such that $\mathfrak{m}_z\mathcal{O}_Y$ is invertible and $g^{-1}(z) \cup \operatorname{Exc}(g) \cup \operatorname{Supp} g_*^{-1}(\Delta_0 + \Delta_1)$ is snc. Let $E_1, \dots, E_n$ be the irreducible components of $g^{-1}(z)$. For any $t \in [0,1]$, let $K_Y + B_{Y,t} := g^*(K_{W/G} + \Delta_t)$. Then $\operatorname{Supp} \Delta_t \subset \operatorname{Supp}(\Delta_0 + \Delta_1)$, and, since $(W/G, \Delta_t)$ is lc near z, any prime divisor on Y meeting $g^{-1}(z)$ has coefficient at most 1 in $B_{Y,t}$. By the last sentence of Lemma 2.15 and [Liu26a, Lemma 3.5],

$$\begin{aligned} q(t) &= \operatorname{mld}(W/G \ni z, \Delta_t) = \min_{1 \le j \le n} a(E_j, W/G, \Delta_t) \\ &= \min_{1 \le j \le n} \big(a(E_j, W/G, \Delta_0) - t \operatorname{mult}_{E_j} g^*(\Delta_1 - \Delta_0)\big). \end{aligned}$$

Since $\operatorname{mult}_{E_j} g^*(\Delta_1 - \Delta_0) \ge 0$ for any j, q is the minimum of n decreasing affine functions on $[0,1]$. Thus q is decreasing, concave and continuous on $[0,1]$.

Let $t \in [0,1]$. Since $q(t)$ is the minimum of the numbers $a(E_j, W/G, \Delta_t)$, we have

$$q(t) = a(E_j, W/G, \Delta_t)$$

for some j. By Lemma 2.15(3), there exists a prime divisor F over W whose valuation restricts to a positive integral multiple of the valuation of E_j. By Lemma 2.15(1)(2), $\pi(\operatorname{center}_W F) = \operatorname{center}_{W/G} E_j = \{z\}$, so $\operatorname{center}_W F = \{o\}$, and

$$\frac{a(F, W, \frac{t}{m}\operatorname{div}(h))}{e_F} = a(E_j, W/G, \Delta_t) = q(t).$$

Thus F is a computing lift of $q(t)$.

Assume that K_W is Cartier near o. For any prime divisor F over W with $\operatorname{center}_W F = \{o\}$, $a(F,W,0)$ and $\operatorname{mult}_F h$ are integers and $e_F = |I_G(F)|$ divides $|G|$, so

$$\frac{a(F, W, \frac{1}{m}\operatorname{div}(h))}{e_F} = \frac{ma(F,W,0) - \operatorname{mult}_F h}{me_F} \in \frac{1}{m|G|}\mathbb{Z},$$

and $a(F, W, \frac{1}{m}\operatorname{div}(h))/e_F \ge 0$ since $(W, \frac{1}{m}\operatorname{div}(h))$ is lc near o. Taking F to be a computing lift of $q(1)$, we obtain $q(1) \in (m|G|)^{-1}\mathbb{N}$.

(2) Let F be a computing lift of $q(1)$. Then $\operatorname{center}_W F = \{o\}$ and

$$q(1) = \frac{a(F, W, \frac{1}{m}\operatorname{div}(h))}{e_F} = \frac{a(F, W, 0)}{e_F} - \frac{\operatorname{mult}_F h}{me_F},$$

which is the equality in (2). For any $t \in [0,1]$, $q(1) \leq q(t)$ by (1), and, by the definition of $q(t)$ as an infimum (Definition 2.2),

$$q(t) \leq \frac{a(F, W, \frac{t}{m}\operatorname{div}(h))}{e_F} = \frac{a(F, W, 0)}{e_F} - t\frac{\operatorname{mult}_F h}{me_F} = q(1) + (1-t)\frac{\operatorname{mult}_F h}{me_F}.$$

(3) Since $(W, \frac{1}{m}\operatorname{div}(h))$ is lc near o, W is lc near o, so $a(F, W, 0) \geq 0$ for any prime divisor F over W with center o. Since q is decreasing and $e_F \geq 1$, for any $t \in [0,1]$,

$$q(t) \leq q(0) \leq \operatorname{mld}(W \ni o).$$

If W is smooth at o, then the exceptional divisor E_1 of the blow-up of W at o satisfies

$$a(E_1, W, 0) = 3.$$

If W is singular at o, then $W = (f = 0)$ for a regular function f on a smooth fourfold germ $(\mathcal{X} \ni o)$ with $\operatorname{mult}_o f \geq 2$, and the exceptional divisor E_1' of the blow-up of $\mathcal{X}$ at o satisfies $a(E_1', \mathcal{X}, W) = 4 - \operatorname{mult}_o f \leq 2$. Since W is normal, $\operatorname{mld}(W \ni o) = \operatorname{mld}(\mathcal{X} \ni o, W) \leq 2$ by [EMY03, Theorem 1.6]. The lemma follows. □

Lemma 4.3. *Notation as in Definition 4.1.*

(1) If some computing lift F of $q(1)$ satisfies $a(F, W, 0)/e_F \leq \lambda$ for a real number λ, then $q(1) \leq q(t) \leq q(1) + \lambda(1-t)$ for any $t \in [0,1]$.

(2) Let $(W_i \ni o_i, G_i, h_i, \tau_i)$ be a radial failure sequence with constants (m, a, δ). Then any computing lift F of $q_i(1)$ satisfies

$$\frac{a(F, W_i, 0)}{e_F} \geq \frac{\operatorname{mult}_F h_i}{me_F} \geq \frac{\delta}{1-\tau_i}.$$

In particular, for any real number λ, there exist only finitely many i for which $q_i(1)$ has a computing lift F_i with $a(F_i, W_i, 0)/e_{F_i} \leq \lambda$.

Proof. (1) Since $q(1) \geq 0$ by Lemma 4.2(1), Lemma 4.2(2) implies

$$\frac{\operatorname{mult}_F h}{me_F} = \frac{a(F, W, 0)}{e_F} - q(1) \leq \lambda,$$

and (1) follows from the inequalities of Lemma 4.2(2).

(2) Let F be a computing lift of $q_i(1)$. By Definition 4.1(3) and Lemma 4.2(2) with $t = \tau_i$,

$$a \leq q_i(\tau_i) \leq q_i(1) + (1-\tau_i)\frac{\operatorname{mult}_F h_i}{me_F} \leq a - \delta + (1-\tau_i)\frac{\operatorname{mult}_F h_i}{me_F}.$$

Since $\tau_i < 1$, we have $\operatorname{mult}_F h_i/(me_F) \geq \delta/(1-\tau_i)$. Since $q_i(1) \geq 0$ by Lemma 4.2(1), Lemma 4.2(2) implies

$$\frac{a(F, W_i, 0)}{e_F} = q_i(1) + \frac{\operatorname{mult}_F h_i}{me_F} \geq \frac{\operatorname{mult}_F h_i}{me_F}.$$

If $a(F_i, W_i, 0)/e_{F_i} \leq \lambda$ for a real number λ, infinitely many i, and computing lifts F_i of $q_i(1)$, then $\delta/(1-\tau_i) \leq \lambda$ for infinitely many i, which contradicts $\lim_{i\to+\infty} \tau_i = 1$. □

Lemma 4.4. *Notation as in Definition 4.1. Assume that K_W is Cartier near o, and that there exist real numbers $\lambda > 0$ and $t_0 < 1$ with $q(t) \leq q(1) + \lambda(1-t)$ for any $t \in [t_0, 1]$. Then $q(1)$ has a computing lift F with*

$$\frac{\operatorname{mult}_F h}{me_F} \leq \lambda \quad \text{and} \quad \frac{a(F, W, 0)}{e_F} \leq q(1) + \lambda.$$

Proof. Let $t \in [\max\{t_0, 0\}, 1)$ be a real number such that $\lambda(1-t) < 1/(m|G|)$. By Lemma 4.2(1), $q(t)$ has a computing lift F, that is, $\operatorname{center}_W F = \{o\}$ and

$$\frac{a(F, W, \frac{t}{m}\operatorname{div}(h))}{e_F} = q(t).$$

Let

$$b_F := \frac{a(F, W, \frac{1}{m}\operatorname{div}(h))}{e_F} = \frac{a(F, W, 0) - \operatorname{mult}_F h/m}{e_F}.$$

Since K_W is Cartier near o, $a(F, W, 0)$ and $\operatorname{mult}_F h$ are integers and e_F divides $|G|$, so

$$b_F \in (m|G|)^{-1}\mathbb{Z}.$$

By Lemma 4.2(1), $q(1) \in (m|G|)^{-1}\mathbb{Z}$. Since $\operatorname{center}_W F = \{o\}$, $q(1) \leq b_F$. Since $\operatorname{mult}_F h \geq 0$ and $t \leq 1$,

$$b_F \leq \frac{a(F, W, \frac{t}{m}\operatorname{div}(h))}{e_F} = q(t).$$

Thus

$$0 \leq b_F - q(1) \leq q(t) - q(1) \leq \lambda(1-t) < \frac{1}{m|G|},$$

so $b_F = q(1)$, and F is a computing lift of $q(1)$. Then

$$q(1) + (1-t)\frac{\operatorname{mult}_F h}{me_F} = b_F + (1-t)\frac{\operatorname{mult}_F h}{me_F} = q(t) \leq q(1) + \lambda(1-t).$$

Since $t < 1$, $\operatorname{mult}_F h/(me_F) \leq \lambda$. By Lemma 4.2(2),

$$\frac{a(F, W, 0)}{e_F} = q(1) + \frac{\operatorname{mult}_F h}{me_F} \leq q(1) + \lambda.$$

The lemma follows. □

4.2. **Separated sequences and admission.**

Lemma 4.5. *Let m be a positive integer and ϵ a positive real number. For any $i \in \mathbb{N}^+$, let $(W_i \ni o_i, G_i, h_i, m)$ be data as in Definition 4.1 with radial function q_i, and let $s_i \in [0, 1)$. Assume the following.*

(i) The sequence $(q_i(0))_{i\in\mathbb{N}^+}$ is bounded from above.
(ii) $\lim_{i\to+\infty} s_i = 1$.
(iii) $q_i(s_i) - q_i(1) \geq \epsilon$ for any i.

Then, possibly passing to a subsequence, there exist real numbers $t_i \in [s_i, 1)$ satisfying the following.

(1) The sequence $(t_i)_{i\in\mathbb{N}^+}$ is strictly increasing and $\lim_{i\to+\infty} t_i = 1$.
(2) The sequence $(\alpha_i)_{i\in\mathbb{N}^+}$, where $\alpha_i := q_i(t_i)$, is strictly increasing and

$$\alpha_\infty := \lim_{i\to+\infty} \alpha_i > 0.$$

(3) For any i,

$$q_i(1) \leq \alpha_\infty - \frac{\epsilon}{2}.$$

Proof. Since $(W_i, \frac{1}{m}\operatorname{div}(h_i))$ is lc near o_i, $q_i(1) \geq 0$. By Lemma 4.2(1), q_i is decreasing, so

$$\epsilon \leq q_i(1) + \epsilon \leq q_i(s_i) \leq q_i(0) \quad \text{for any } i.$$

By (i), possibly passing to a subsequence, we may assume that $\beta := \lim_{i\to+\infty} q_i(s_i)$ exists. Then $\beta \geq \epsilon > 0$. Possibly removing finitely many indices, we may assume that $q_i(s_i) < \beta + \frac{\epsilon}{2}$ for any i. By (iii),

$$q_i(1) \leq q_i(s_i) - \epsilon < \beta - \frac{\epsilon}{2} \quad \text{for any } i. \tag{4.1}$$

Let $(\lambda_k)_{k\in\mathbb{N}^+}$ be a strictly increasing sequence of real numbers with

$$\beta - \frac{\epsilon}{2} < \lambda_1 \quad \text{and} \quad \lim_{k\to+\infty} \lambda_k = \beta.$$

Then $\lambda_k < \beta$ for any k. We choose indices $i_1 < i_2 < \cdots$ and real numbers $t_{i_k} \in (s_{i_k}, 1)$ recursively. Let $k \in \mathbb{N}^+$, and assume that i_{k-1} and $t_{i_{k-1}}$ are defined, where we set $i_0 := 0$ and $t_{i_0} := 0$. Since

$$\lim_{i\to+\infty} q_i(s_i) = \beta > \lambda_k \quad \text{and} \quad \lim_{i\to+\infty} s_i = 1 > t_{i_{k-1}},$$

there exists an index $i_k > i_{k-1}$ with

$$q_{i_k}(s_{i_k}) > \lambda_k \quad \text{and} \quad s_{i_k} > t_{i_{k-1}}.$$

By (4.1),

$$q_{i_k}(1) < \beta - \frac{\epsilon}{2} < \lambda_k < q_{i_k}(s_{i_k}).$$

Since q_{i_k} is continuous on $[0,1]$ (Lemma 4.2(1)), by the intermediate value theorem there exists $t_{i_k} \in (s_{i_k}, 1)$ with $q_{i_k}(t_{i_k}) = \lambda_k$.

We replace the sequence by the subsequence indexed by $(i_k)_{k\in\mathbb{N}^+}$. For any $k \geq 2$,

$$t_{i_k} > s_{i_k} > t_{i_{k-1}},$$

so $(t_{i_k})_k$ is strictly increasing, and $\lim_k t_{i_k} = 1$ since $s_{i_k} < t_{i_k} < 1$ and $\lim_k s_{i_k} = 1$. This is (1). Since $q_{i_k}(t_{i_k}) = \lambda_k$, the values $\alpha_{i_k} = \lambda_k$ strictly increase to β, so $\alpha_\infty = \beta > 0$, which is (2). By (4.1),

$$q_{i_k}(1) < \beta - \frac{\epsilon}{2} = \alpha_\infty - \frac{\epsilon}{2} \quad \text{for any } k,$$

which is (3). The lemma follows. □

Lemma 4.6. *Let $n \in \{3,4\}$, let $(W \ni o)$ be a smooth threefold germ if $n = 3$ and an isolated Gorenstein terminal threefold germ if $n = 4$, and let G be a finite abelian group acting faithfully on W and fixing o. Assume the following.*

- *(i) There exists a G-equivariant closed embedding of W into a smooth germ $(Y \ni o)$ of dimension n, with $Y = W$ if $n = 3$ and $W = (f = 0)$ for a regular G-semi-invariant function f on Y if $n = 4$.*
- *(ii) $x_1, \dots, x_n$ are G-eigenparameters of Y at o, so that*

$$\widehat{\mathcal{O}}_{Y,o} = \mathbb{C}[[x_1, \dots, x_n]] =: \mathfrak{R}.$$

- *(iii) h is a regular G-semi-invariant function on Y such that $h|_W$ is a mark on $(W \ni o)$.*
- *(iv) m is a positive integer such that $(W, \frac{1}{m}\operatorname{div}(h))$ is lc near o.*
- *(v) $t \in [0,1)$ is a real number.*
- *(vi) $\mathcal{P}$ is a finite set of prime divisors over W with center o.*
- *(vii) ψ is a G-equivariant automorphism of $\mathfrak{R}$.*
- *(viii) u and v are G-invariant units of $\mathfrak{R}$.*

Here and below, $\operatorname{div}(h)$ and $\operatorname{mult}_P h$ denote $\operatorname{div}(h|_W)$ and $\operatorname{mult}_P(h|_W)$, and similarly for a polynomial restricted to a hypersurface in $\mathbb{A}^4$. Let $\widehat{h}$, and $\widehat{f}$ if $n = 4$, be the images of h and f in $\mathfrak{R}$. We let G act diagonally on $\mathbb{A}^n$ with coordinates $x_1, \dots, x_n$, the coordinate x_i having the character of the eigenparameter x_i, so that $\mathbb{C}[x_1, \dots, x_n] \subset \mathfrak{R}$ G-equivariantly. For any $g \in \mathfrak{R}$ and any positive integer L, let $g_{<L}$ be the sum of the terms of g of degree less than L, and let

$$h' := (u\psi(\widehat{h}))_{<L}, \qquad W' := \begin{cases} \mathbb{A}^3 & \text{if } n = 3, \\ (f' = 0) \subset \mathbb{A}^4 \text{ with } f' := (v\psi(\widehat{f}))_{<L} & \text{if } n = 4. \end{cases}$$

Then for any sufficiently large integer L, h' is a G-semi-invariant polynomial with the character of h, G acts faithfully on W', $(W' \ni 0)$ is an isolated Gorenstein terminal threefold germ if $n = 4$, and there exists a G-equivariant isomorphism between the completions of the local rings of W at o and of W' at 0 such that the following hold.

(1) *For any subgroup* $G' \subset G$,

$$\operatorname{mld}\Big(\Big(W' \ni 0, \frac{t}{m}\operatorname{div}(h')\Big)/G'\Big) = \operatorname{mld}\Big(\Big(W \ni o, \frac{t}{m}\operatorname{div}(h)\Big)/G'\Big).$$

(2) *Under the isomorphism, any* $P \in \mathcal{P}$ *corresponds to a prime divisor* P' *over* W' *with center* 0 *satisfying the following: the residue fields of* P' *and* P *are isomorphic, the decomposition groups and the inertia groups of* P' *and* P *in* G *are equal, and*

$$a(P', W', 0) = a(P, W, 0), \qquad \operatorname{mult}_{P'} h' = \operatorname{mult}_P h.$$

(3) $(W', \frac{1}{m}\operatorname{div}(h'))$ *is lc near* 0.

Proof. Let $\mathfrak{m}$ be the maximal ideal of $\mathfrak{R}$. Since W is klt and $(W, \frac{1}{m}\operatorname{div}(h))$ is lc near o, the pairs $(W, \frac{1}{2m}\operatorname{div}(h))$ and $(W, \frac{t}{m}\operatorname{div}(h))$ are klt near o.

We use [Liu26b, Construction-Theorem 4.13] for the single function h with coefficient $\frac{1}{2m}$, the coefficient vectors $(\frac{1}{2m})$ and $(\frac{t}{m})$, the prime divisors in $\mathcal{P}$, the automorphism ψ, and the units v and u. Thus there exists a positive integer N_1 such that for any integer $N \geq N_1$ there exists an integer L_1 with the following property: for any integer $L \geq L_1$ and any G-semi-invariant polynomials h_0 and, if $n = 4$, f_0, with the characters of h and f and with

$$h_0 \equiv u\psi(\widehat{h}) \pmod{\mathfrak{m}^N}, \qquad f_0 \equiv v\psi(\widehat{f}) \pmod{\mathfrak{m}^L},$$

the germ $W_0 := \mathbb{A}^3$ if $n = 3$, or $W_0 := (f_0 = 0) \subset \mathbb{A}^4$ if $n = 4$, with the action of G on $\mathbb{A}^n$ fixed in the statement, satisfies the following [Liu26b, Construction-Theorem 4.13(1)–(3)]: G acts faithfully on W_0, $(W_0, \frac{1}{2m}\operatorname{div}(h_0))$ is klt near 0, the equality in (1) holds for any subgroup $G' \subset G$ with (W', h') replaced by (W_0, h_0), there exists a G-equivariant isomorphism between the completions of the local rings of W at o and of W_0 at 0 satisfying (2) with (W', h') replaced by (W_0, h_0), and $(W_0 \ni 0)$ is an isolated Gorenstein terminal threefold germ if $n = 4$.

Let L_{cut} be the integer of Lemma 2.19 for the dimension n, the number $n-2$ of factors (the factor from h if $n = 3$, and the factors from f and h if $n = 4$), and the DCC set $\{1, \frac{1}{m}\}$. Let $N := \max\{N_1, L_{\mathrm{cut}}\}$, let L_1 be the corresponding integer, and let L be an integer with $L \geq \max\{L_1, N\}$. Since ψ is G-equivariant and u and v are G-invariant, $u\psi(\widehat{h})$ and $v\psi(\widehat{f})$ are G-semi-invariant with the characters of h and f. Since G acts diagonally on $x_1, \dots, x_n$, any monomial is G-semi-invariant, so h' and f' are G-semi-invariant polynomials with the characters of h and f. Moreover,

$$h' - u\psi(\widehat{h}) \in \mathfrak{m}^L \subset \mathfrak{m}^N \quad \text{and} \quad f' - v\psi(\widehat{f}) \in \mathfrak{m}^L.$$

Hence $h_0 := h'$ and $f_0 := f'$ are allowed above, and $W_0 = W'$. Thus the conclusions of the previous paragraph hold for $(W_0, h_0) = (W', h')$. Hence (1) and (2) hold, G acts faithfully on W', and $(W' \ni 0)$ is an isolated Gorenstein terminal threefold germ if $n = 4$. Since $(W', \frac{1}{2m}\operatorname{div}(h'))$ is klt near 0, $h'|_{W'} \neq 0$.

It remains to prove (3). If $n = 3$, then $Y = W$ and $W' = \mathbb{A}^3$, and we omit the factors $(\widehat{f})$, $(v\psi(\widehat{f}))$, and (f'), the boundary components W and W', and the uses of [Kaw07, Theorem] below. If $n = 4$, then $f' \neq 0$ since $(W' \ni 0)$ is a threefold germ. By (iv), since W is a normal Cartier divisor on the smooth germ Y, [Kaw07, Theorem] implies that $(Y, W + \frac{1}{m}\operatorname{div}_Y(h))$ is lc near o. By [dFEM11, Proposition 2.11], $(\operatorname{Spec}\mathfrak{R}, (\widehat{f})(\widehat{h})^{\frac{1}{m}})$ is lc. Since ψ is an automorphism of $\mathfrak{R}$ and u and v are units,

$$\Big(\operatorname{Spec}\mathfrak{R}, \big(v\psi(\widehat{f})\big)\big(u\psi(\widehat{h})\big)^{\frac{1}{m}}\Big)$$

is lc. Since $L \geq L_{\mathrm{cut}}$, $f' \neq 0$ and $h' \neq 0$, Lemma 2.19 implies that $(\operatorname{Spec}\mathfrak{R}, (f')(h')^{\frac{1}{m}})$ is lc. By [dFEM11, Proposition 2.11], $(\mathbb{A}^n, W' + \frac{1}{m}\operatorname{div}_{\mathbb{A}^n}(h'))$ is lc near 0. Since W' is a normal Cartier divisor on $\mathbb{A}^n$, by [Kaw07, Theorem], $(W', \frac{1}{m}\operatorname{div}(h'))$ is lc near 0. This is (3). □

4.3. **The reduction to radial failure sequences.** If Theorem 1.5 fails, then there exists a radial failure sequence satisfying Construction-Theorem 4.7(C).

Construction-Theorem 4.7. Let $\Gamma \subset [0,1] \cap \mathbb{Q}$ be a finite set and m a positive integer with $m\Gamma \subset \mathbb{Z}$.

(A) Assume the following.
  (i) X is a $\mathbb{Q}$-factorial klt threefold.
  (ii) $x \in X$ is a closed point.
  (iii) $D \geq 0$ is a $\mathbb{Q}$-divisor on X with $D \in \Gamma$.
  (iv) (X, D) is lc near x.

  Let $\mu(t) := \mathrm{mld}(X \ni x, tD)$ for $t \in [0,1]$. Then, possibly shrinking X to an affine neighborhood of x, there exist a finite quasi-étale abelian cover $\rho : (V \ni o) \to (X \ni x)$ with Galois group G and $\rho^{-1}(x) = \{o\}$, and a G-equivariant projective birational morphism $g : Y \to V$, satisfying the following for $D_V := \rho^* D$ and $L := g^* D_V$.
  (a) K_V and mD_V are Cartier, $K_Y = g^* K_V$, and Y is Gorenstein terminal.
  (b) The g-exceptional prime divisors are exactly the exceptional prime divisors E over V with $a(E, V, 0) = 1$.
  (c) Any G-invariant Weil divisor on Y is $\mathbb{Q}$-Cartier.
  (d) $L \geq 0$, mL is Cartier, (Y, L) is lc, and $K_Y + tL = (\rho \circ g)^*(K_X + tD)$ for any $t \in [0,1]$.
  (e) For any $t \in [0,1]$, $\mu(t)$ is the infimum of $a(F, Y, tL)/e_F$ over the prime divisors F over Y with center contained in $(\rho \circ g)^{-1}(x)$.

(B) In (A), assume the following.
  (i) E is a prime divisor over X with $\mathrm{center}_X E = \{x\}$.
  (ii) F is a prime divisor over V whose valuation restricts to a positive integral multiple of the valuation of E.

  Let $J := I_G(F)$, let $Z := \mathrm{center}_Y F$, and let ξ_Z be the generic point of Z. Then J is cyclic and fixes Z pointwise, $a(F, Y, 0) = |J| \cdot a(E, X, 0)$, and $\mathrm{mult}_F L = |J| \, \mathrm{mult}_E D$. Moreover, for any $t \in [0,1]$, $q_Z(t) := \mathrm{mld}((Y \ni \xi_Z, tL)/J)$ satisfies $\mu(t) \leq q_Z(t) \leq a(E, X, tD)$, and $q_Z(t) = \mu(t)$ if E computes $\mu(t)$. If Z is a closed point y, then there exists a mark h on $(W \ni y) := (Y \ni y)$ with the action of J such that $\mathrm{div}(h) = mL$ near y, and for any such h, q_Z is the radial function of $(W \ni y, J, h, m)$.

(C) Assume the following.
  (i) a is a positive rational number.
  (ii) Theorem 1.5 fails for (a, Γ).

  Then there exist $\delta \in (0, a)$ and, for any $i \in \mathbb{N}^+$, data (X_i, x_i, D_i) satisfying (i)–(iv) of (A) with $\mathrm{mld}(X_i \ni x_i) \geq a$ and $\tau_i := \tau_a(X_i, D_i; x_i) \in (1 - \frac{1}{i}, 1)$, a prime divisor E_i over X_i computing $\mathrm{mld}(X_i \ni x_i, D_i)$, and, for $\rho_i : V_i \to X_i$ and $g_i : Y_i \to V_i$ as in (A) for (X_i, x_i, D_i), a prime divisor F_i over V_i as in (B)(ii) for E_i, with the following property. After passing to a subsequence, $\mathrm{center}_{Y_i} F_i$ is a closed point y_i for any i, and $(W_i \ni o_i, G_i, h_i, \tau_i)$ is a radial failure sequence with constants (m, a, δ), where $(W_i \ni o_i) := (Y_i \ni y_i)$, $G_i := I_{\mathrm{Gal}(V_i/X_i)}(F_i)$, and h_i is a mark on $(W_i \ni o_i)$ as in (B). Moreover, for any i, $(W_i \ni o_i)$ is singular of quadratic rank 1, 2, or 3, $q_i(1) = \mathrm{mld}(X_i \ni x_i, D_i)$, and $q_i \geq \mu_i$ on $[0,1]$ for $\mu_i(t) := \mathrm{mld}(X_i \ni x_i, tD_i)$.

Proof of (A) and (B) of Construction-Theorem 4.7. We prove (A). Possibly shrinking X to an affine neighborhood of x whenever needed, we may assume that (X, D) is lc. Since X is $\mathbb{Q}$-factorial and $m\Gamma \subset \mathbb{Z}$, K_X and mD are $\mathbb{Q}$-Cartier Weil divisors. By [Zhu24, Lemma-Definition 2.15], there exists a finite quasi-étale abelian Galois cover $\rho : V \to X$ with Galois group G such that $\rho^{-1}(x)$ is a single point o and $\rho^* K_X$ and $\rho^*(mD)$ are Cartier. Since ρ is quasi-étale, $K_V = \rho^* K_X$, so K_V and mD_V are Cartier. The group G acts faithfully on V and fixes o, and $X = V/G$.

Since ρ is quasi-étale, any prime divisor P on V has ramification index 1 over $\rho(P)$, and $\mathrm{mult}_P(tD_V) = t \, \mathrm{mult}_{\rho(P)} D$. By Lemma 2.15(0)(4), $(V, tD_V)/G = (X, tD)$ for any $t \in [0,1]$. Thus, by Lemma 2.15(1)(2), for any prime divisor T over V, the valuation of T restricts to

$|I_G(T)|$ times the valuation of a prime divisor E_T over X, and, for any $t \in [0,1]$,

$$a(T,V,tD_V) = |I_G(T)| \cdot a(E_T,X,tD) \quad \text{and} \quad \operatorname{center}_X E_T = \rho(\operatorname{center}_V T). \tag{4.2}$$

Since X is klt and (X,D) is lc, (4.2) with $t=0$ and $t=1$ implies that V is klt and (V,D_V) is lc. Since K_V is Cartier and V is klt, $a(E,V,0) \in \mathbb{N}^+$ for any exceptional prime divisor E over V, and $a(P,V,0)=1$ for any prime divisor P on V.

We use Construction-Theorem 2.14 with $N=1$, $\alpha_1=1$, $(X,B)=(V,0)$, $x=o$, and $H=G$. Then

$$\mathcal{S}=\mathbb{N}, \quad \Lambda_{\mathcal{S}}=\{0\}, \quad \eta_{\mathcal{S}}=1, \quad \text{and} \quad R_{\mathcal{S}}=M_{\mathcal{S}}=1.$$

There exist a G-equivariant projective birational morphism $g: Y \to V$ and an $\mathbb{R}$-divisor $B_Y \geq 0$ on Y satisfying (1)–(4) of Construction-Theorem 2.14. Thus, by Construction-Theorem 2.14(1), the g-exceptional prime divisors are exactly the exceptional prime divisors E over V with $a(E,V,0)=1$, and $\operatorname{mult}_E B_Y = 1-1=0$ for them. Since $K_V + g_*B_Y = g_*(K_Y+B_Y) = K_V$, we have $g_*B_Y=0$. Therefore $B_Y=0$ and

$$K_Y = g^*K_V.$$

In particular, K_Y is Cartier. By Construction-Theorem 2.14(4), Y is terminal, so Y is Gorenstein terminal. By Construction-Theorem 2.14(3), any G-invariant Weil divisor on Y is $\mathbb{Q}$-Cartier.

Let $L := g^*D_V$. Since $mD_V \geq 0$ is a Cartier divisor, $mL = g^*(mD_V) \geq 0$ is a Cartier divisor. Since $K_V + tD_V = \rho^*(K_X+tD)$,

$$K_Y + tL = g^*(K_V+tD_V) = (\rho\circ g)^*(K_X+tD) \qquad \text{for any } t\in[0,1].$$

Thus $a(T,Y,tL) = a(T,V,tD_V)$ for any prime divisor T over V and any $t\in[0,1]$, and (Y,L) is lc since (V,D_V) is lc.

Let $t\in[0,1]$, and let F be a prime divisor over Y. Since g is proper, $\operatorname{center}_V F = g(\operatorname{center}_Y F)$. Since $\rho^{-1}(x)=\{o\}$, by (4.2), we have

$$\operatorname{center}_Y F \subset (\rho\circ g)^{-1}(x) \iff \operatorname{center}_V F = \{o\} \iff \operatorname{center}_X E_F = \{x\}.$$

For such F, (4.2) implies

$$\frac{a(F,Y,tL)}{e_F} = a(E_F,X,tD) \geq \mu(t).$$

Conversely, let E be a prime divisor over X with $\operatorname{center}_X E=\{x\}$. By Lemma 2.15(3), there exists a prime divisor F over V whose valuation restricts to a positive integral multiple of the valuation of E. By the uniqueness in Lemma 2.15, $E_F=E$, so $\operatorname{center}_Y F \subset (\rho\circ g)^{-1}(x)$ and $a(F,Y,tL)/e_F = a(E,X,tD)$. Since $\mu(t)$ is the infimum of $a(E,X,tD)$ over the prime divisors E over X with $\operatorname{center}_X E=\{x\}$, (A) follows.

We prove (B). Let E, F, J, and Z be as in (B). By the uniqueness in Lemma 2.15, $E_F=E$. By (4.2), $\rho(\operatorname{center}_V F) = \operatorname{center}_X E = \{x\}$, so $\operatorname{center}_V F=\{o\}$ and $Z\subset g^{-1}(o)$. By Lemma 2.15(1), J is cyclic. Let $\sigma\in J$. Since σ preserves the valuation of F, $\sigma(Z)=Z$. By the definition of the inertia group [Liu26a, Definition 2.5], σ acts trivially on the residue field of the valuation ring of F, which contains $\mathbb{C}(Z)$. Since Z is integral, $\sigma|_Z = \mathrm{id}_Z$. Thus J fixes Z pointwise.

Since $I_G(F)=J$ and $E_F=E$, (4.2) with $T=F$ and $a(F,Y,tL)=a(F,V,tD_V)$ imply

$$\begin{aligned} a(F,Y,0) - t\operatorname{mult}_F L &= a(F,Y,tL) = |J|\cdot a(E,X,tD) \\ &= |J|\cdot\big(a(E,X,0) - t\operatorname{mult}_E D\big) \qquad \text{for any } t\in[0,1]. \end{aligned}$$

Comparing the values at $t=0$ and the coefficients of t, we obtain $a(F,Y,0) = |J|\cdot a(E,X,0)$ and $\operatorname{mult}_F L = |J|\operatorname{mult}_E D$.

Let $t\in[0,1]$ and let T be a prime divisor over Y with $\operatorname{center}_Y T = Z$. Then

$$\operatorname{center}_V T = g(Z) = \{o\},$$

so $\operatorname{center}_X E_T = \{x\}$ by (4.2). By the definition of the inertia group [Liu26a, Definition 2.5], $I_J(T) = J \cap I_G(T)$, so $[I_G(T) : I_J(T)]$ is a positive integer. Since $a(E_T, X, tD) \geq \mu(t) \geq 0$, (4.2) and $a(T, Y, tL) = a(T, V, tD_V)$ imply

$$\frac{a(T, Y, tL)}{|I_J(T)|} = [I_G(T) : I_J(T)] \cdot a(E_T, X, tD) \geq \mu(t).$$

For $T = F$ the left-hand side equals $a(E, X, tD)$, since $I_J(F) = J = I_G(F)$ and $E_F = E$. Since $q_Z(t)$ is the infimum of $a(T, Y, tL)/|I_J(T)|$ over these T (Definition 2.2),

$$\mu(t) \leq q_Z(t) \leq a(E, X, tD),$$

and $q_Z(t) = \mu(t)$ if E computes $\mu(t)$.

Assume that $Z = \{y\}$ for a closed point y. Since $L = (\rho \circ g)^* D$ is G-invariant, mL is Cartier, and J is a finite abelian group fixing y, there exist a J-invariant affine open neighborhood W of y in Y and a regular J-semi-invariant function h on W with $\operatorname{div}(h) = mL_W$, where $L_W := L|_W$. Thus h is a mark on $(W \ni y)$. For any mark h' on $(W \ni y)$ with $\operatorname{div}(h') = mL$ near y, the group J acts faithfully on W and fixes y, (W, L_W) is lc, and the radial function of $(W \ni y, J, h', m)$ is $t \to \operatorname{mld}((W \ni y, tL_W)/J) = q_Z(t)$, since W is an open neighborhood of y in Y (Definitions 2.2 and 4.1). This proves (B). □

Proof of (C) of Construction-Theorem 4.7. We fix a, Γ, and m.

Step 1. We construct the pairs (X_i, D_i), the constant δ and the prime divisors E_i.

Since Theorem 1.5 fails for (a, Γ), for any $i \in \mathbb{N}^+$ there exist a $\mathbb{Q}$-factorial klt threefold X_i, a closed point $x_i \in X_i$ with $\operatorname{mld}(X_i \ni x_i) \geq a$, and a $\mathbb{Q}$-divisor $D_i \geq 0$ on X_i with $D_i \in \Gamma$, such that (X_i, D_i) is lc near x_i and

$$\tau_i := \tau_a(X_i, D_i; x_i) \in \left(1 - \frac{1}{i}, 1\right).$$

Possibly shrinking X_i to a neighborhood of x_i, we may assume that (X_i, D_i) is lc. For $t \in [0, 1]$ let

$$\mu_i(t) := \operatorname{mld}(X_i \ni x_i, tD_i).$$

Since $\mu_i \geq 0$ is the infimum of the decreasing affine functions

$$t \to a(E, X_i, tD_i) = a(E, X_i, 0) - t \operatorname{mult}_E D_i$$

over the prime divisors E over X_i with $\operatorname{center}_{X_i} E = \{x_i\}$, it is decreasing and continuous on $(0, 1)$. Since moreover $\mu_i(0) = \operatorname{mld}(X_i \ni x_i) \geq a$ and $\tau_i \in (0, 1)$, we have

$$\mu_i(\tau_i) = a \quad \text{and} \quad \mu_i(1) < a.$$

The numbers $\mu_i(1)$ are minimal log discrepancies at closed points of lc threefold pairs whose coefficients belong to the finite set $\Gamma \subset [0, 1]$. By Theorem 1.3, $\{\mu_i(1) \mid i \in \mathbb{N}^+\}$ satisfies the ACC, and since $\mu_i(1) < a$ for any i, there exists $\delta \in (0, a)$ with $\mu_i(1) \leq a - \delta$ for any $i \in \mathbb{N}^+$.

By [Liu26a, Lemma 3.5], there exists a prime divisor E_i over X_i with $\operatorname{center}_{X_i} E_i = \{x_i\}$ and $a(E_i, X_i, D_i) = \mu_i(1)$. Then

$$a = \mu_i(\tau_i) \leq a(E_i, X_i, \tau_i D_i) = \mu_i(1) + (1 - \tau_i) \operatorname{mult}_{E_i} D_i \leq a - \delta + (1 - \tau_i) \operatorname{mult}_{E_i} D_i,$$

so

$$\operatorname{mult}_{E_i} D_i \geq \frac{\delta}{1 - \tau_i}. \tag{4.3}$$

Step 2. We apply (A) and (B), and exclude divisors as centers.

Let $\rho_i : V_i \to X_i$, $g_i : Y_i \to V_i$, and L_i be as in (A) for (X_i, x_i, D_i), and let $\operatorname{Gal}(V_i/X_i)$ be the Galois group of ρ_i, which acts on V_i and, since g_i is equivariant, on Y_i. By Lemma 2.15(3), there exists a prime divisor F_i over V_i whose valuation restricts to a positive integral multiple of the valuation of E_i. Let

$$G_i := I_{\operatorname{Gal}(V_i/X_i)}(F_i),$$

which is the group J of (B), and let $Z_i := \mathrm{center}_{Y_i} F_i$. By (B), G_i is cyclic and fixes Z_i pointwise.

Suppose that $\dim Z_i = 2$. Then F_i is a prime divisor on Y_i. Since (Y_i, L_i) is lc, (B) implies

$$|G_i| \operatorname{mult}_{E_i} D_i = \operatorname{mult}_{F_i} L_i \le 1,$$

so $\delta/(1-\tau_i) \le 1$ by (4.3). Since $\lim_{i\to+\infty} \tau_i = 1$, $\dim Z_i = 2$ only for finitely many i. Possibly removing finitely many indices, we may assume that $\dim Z_i \le 1$ for any i.

Step 3. We define the local functions q_i and choose parameters s_i.

For any i, let ξ_i be the generic point of Z_i, and let $q_i(t) := \mathrm{mld}((Y_i \ni \xi_i, tL_i)/G_i)$ for $t \in [0,1]$, the function q_{Z_i} of (B). If $Z_i = \{y_i\}$ is a closed point, let $(W_i \ni o_i) := (Y_i \ni y_i)$ with the action of G_i, and let h_i be a mark on $(W_i \ni o_i)$ with $\mathrm{div}(h_i) = mL_i$ near y_i, which exists by (B). Then q_i is the radial function of $(W_i \ni o_i, G_i, h_i, m)$ by (B).

By (B), for any $t \in [0,1]$,

$$\mu_i(t) \le q_i(t) \le a(E_i, X_i, tD_i) = \mu_i(1) + (1-t)\operatorname{mult}_{E_i} D_i, \quad \text{in particular} \quad q_i(1) = \mu_i(1). \tag{4.4}$$

Moreover, $q_i \ge 0$ is the infimum of the decreasing affine functions $(a(T, Y_i, 0) - t \operatorname{mult}_T L_i)/|I_{G_i}(T)|$ of t, where T runs over the prime divisors over Y_i with $\mathrm{center}_{Y_i} T = Z_i$, so q_i is decreasing and concave on $[0,1]$, and q_i is continuous on $(0,1)$.

Let

$$\sigma_i := \sup\{t \in [0,1] \mid q_i(t) \ge a\}.$$

By (4.4), $q_i(\tau_i) \ge \mu_i(\tau_i) = a$, so $\sigma_i \ge \tau_i$. By (4.4) and $\mu_i(1) \le a - \delta$, $q_i(t) < a$ for any $t < 1$ close to 1, so $\sigma_i < 1$. Since q_i is decreasing and continuous at $\sigma_i \in (0,1)$,

$$q_i(\sigma_i) = a \quad \text{and} \quad q_i(t) < a \text{ for any } t \in (\sigma_i, 1].$$

Finally, since Y_i is terminal and (Y_i, L_i) is lc, (Y_i, sL_i) is klt for any $s \in [0,1)$.

Claim 4.8. *Let $I \subset \mathbb{N}^+$ be an infinite set. Then there exist an infinite subset $I' \subset I$ and real numbers $s_i \in (\sigma_i, 1)$ for $i \in I'$ such that the sequences $(s_i)_{i\in I'}$ and $(q_i(s_i))_{i\in I'}$ are strictly increasing and $q_i(s_i) > a/2$ for any $i \in I'$.*

Proof. Since $\tau_i \le \sigma_i < 1$ and $\lim_{i\to+\infty} \tau_i = 1$, there exist indices $i_1 < i_2 < \cdots$ in I such that $\sigma_{i_1} < \sigma_{i_2} < \cdots$. Let $I' := \{i_k \mid k \in \mathbb{N}^+\}$. We choose s_{i_k} by induction on k. Since q_{i_k} is continuous at $\sigma_{i_k} \in (0,1)$ and $q_{i_k}(\sigma_{i_k}) = a$, and since $q_{i_{k-1}}(s_{i_{k-1}}) < a$ for $k \ge 2$ by $s_{i_{k-1}} > \sigma_{i_{k-1}}$, there exists $s_{i_k} \in (\sigma_{i_k}, \sigma_{i_{k+1}})$ such that $q_{i_k}(s_{i_k}) > a/2$ and, if $k \ge 2$, $q_{i_k}(s_{i_k}) > q_{i_{k-1}}(s_{i_{k-1}})$. Since $s_{i_k} < \sigma_{i_{k+1}} < s_{i_{k+1}}$ for any k, the claim follows. □

Step 4. We exclude curves as centers.

Suppose that the set $I := \{i \in \mathbb{N}^+ \mid \dim Z_i = 1\}$ is infinite. By Claim 4.8, there exist an infinite subset $I' \subset I$ and real numbers $s_i \in (\sigma_i, 1)$, $i \in I'$, such that $(s_i)_{i\in I'}$ and $(q_i(s_i))_{i\in I'}$ are strictly increasing and $q_i(s_i) > a/2$ for any $i \in I'$. Let Ω be an algebraic closure of the field $\mathbb{C}(v)$ of rational functions in one variable v, and let

$$\Gamma' := \left\{ 1 - \frac{1 - s_j c}{n} \;\middle|\; j \in I',\ c \in \frac{1}{m}\mathbb{Z} \cap [0,1],\ n \in \mathbb{N}^+ \right\}.$$

Since the set

$$\left\{ s_j c \;\middle|\; j \in I',\ c \in \frac{1}{m}\mathbb{Z} \cap [0,1] \right\} \subset [0,1)$$

satisfies the DCC, so does Γ'.

Fix $i \in I'$. Since Y_i is terminal, Y_i is smooth near ξ_i. Let v_i be the pullback to Y_i of a rational function on Y_i/G_i which is regular at the image of ξ_i and non-constant on the image of Z_i. Then v_i is G_i-invariant, regular at ξ_i and non-constant on Z_i. Let $k_i := \mathbb{C}(v_i)$. Since v_i is transcendental over $\mathbb{C}$, there exists a $\mathbb{C}$-isomorphism $\phi_i : k_i \to \mathbb{C}(v)$ with $\phi_i(v_i) = v$, and via ϕ_i, Ω is an algebraic closure of k_i. Let Y_{k_i}, ζ_i, B_{k_i}, ζ_i', $Y_{i,\Omega}$, and $B_{i,\Omega}$ be the objects X_k, c, B_k, c', X_Ω, and B_Ω of Lemma 3.5 for the threefold sub-pair $(Y_i, s_i L_i)$, the curve Z_i, the group G_i, the function v_i, and the field Ω. By Lemma 3.5(0), G_i, acting on $Y_{k_i} \otimes_{k_i} \Omega$ through the first

factor, acts faithfully on $Y_{i,\Omega}$ and fixes ζ_i'. Let $\pi_i : Y_{i,\Omega} \to Y_{i,\Omega}/G_i$ be the quotient morphism, $\overline{\zeta}_i := \pi_i(\zeta_i')$, and define $\overline{B}_i$ by

$$K_{Y_{i,\Omega}} + B_{i,\Omega} = \pi_i^*(K_{Y_{i,\Omega}/G_i} + \overline{B}_i).$$

By Lemma 3.5(0), the prime divisors T over Y_i with $\operatorname{center}_{Y_i} T = Z_i$ are exactly the prime divisors over Y_{k_i} with center ζ_i, with $a(T, Y_i, s_iL_i) = a(T, Y_{k_i}, B_{k_i})$, and the inertia group $I_{G_i}(T)$ of T as a prime divisor over Y_i equals that of T as a prime divisor over Y_{k_i}. Thus

$$q_i(s_i) = \operatorname{mld}((Y_{k_i} \ni \zeta_i, B_{k_i})/G_i),$$

and $a(T, Y_{k_i}, B_{k_i}) > 0$ for any such T since (Y_i, s_iL_i) is klt. Since $\mathcal{O}_{Y_{k_i},\zeta_i} = \mathcal{O}_{Y_i,\xi_i}$ is regular, $Y_{i,\Omega}$ is smooth near ζ_i'. Since mL_i is Cartier and (Y_i, L_i) is lc, $L_i \in \frac{1}{m}\mathbb{Z} \cap [0,1]$, so

$$B_{i,\Omega} \in s_i\Big(\frac{1}{m}\mathbb{Z} \cap [0,1]\Big) \subset [0, s_i].$$

By Lemma 3.5(2), any prime divisor over $Y_{i,\Omega}$ with center ζ_i' has positive log discrepancy with respect to $B_{i,\Omega}$. Since $s_i < 1$, $(Y_{i,\Omega}, B_{i,\Omega})$ is klt near ζ_i', so $(Y_{i,\Omega}/G_i, \overline{B}_i)$ is klt near $\overline{\zeta}_i$. By Lemma 2.15(4), a prime divisor P_1 on $Y_{i,\Omega}$ with ramification index n over $\pi_i(P_1)$ satisfies

$$\operatorname{mult}_{\pi_i(P_1)} \overline{B}_i = 1 - \frac{1 - \operatorname{mult}_{P_1} B_{i,\Omega}}{n},$$

so the coefficients of $\overline{B}_i$ near $\overline{\zeta}_i$ belong to Γ'.

By Theorem 2.21(2) with $\Bbbk = \Omega$, there exists a prime divisor P over $Y_{i,\Omega}/G_i$ with center $\overline{\zeta}_i$ and

$$a(P, Y_{i,\Omega}/G_i, \overline{B}_i) = \operatorname{mld}(Y_{i,\Omega}/G_i \ni \overline{\zeta}_i, \overline{B}_i).$$

By Lemma 2.15(3), there exists a prime divisor F'' over $Y_{i,\Omega}$ whose valuation restricts to a positive integral multiple of the valuation of P. Since $\pi_i^{-1}(\overline{\zeta}_i) = \{\zeta_i'\}$, F'' has center ζ_i', and

$$\frac{a(F'', Y_{i,\Omega}, B_{i,\Omega})}{|I_{G_i}(F'')|} = a(P, Y_{i,\Omega}/G_i, \overline{B}_i) = \operatorname{mld}(Y_{i,\Omega}/G_i \ni \overline{\zeta}_i, \overline{B}_i) = \operatorname{mld}((Y_{i,\Omega} \ni \zeta_i', B_{i,\Omega})/G_i).$$

Thus F'' computes $\operatorname{mld}((Y_{i,\Omega} \ni \zeta_i', B_{i,\Omega})/G_i)$. By Lemma 3.5(2), the prime divisor F' over Y_{k_i} defined by F'' computes $\operatorname{mld}((Y_{k_i} \ni \zeta_i, B_{k_i})/G_i)$, with $a(F', Y_{k_i}, B_{k_i}) = a(F'', Y_{i,\Omega}, B_{i,\Omega})$ and $I_{G_i}(F') = I_{G_i}(F'')$. Therefore

$$q_i(s_i) = \operatorname{mld}(Y_{i,\Omega}/G_i \ni \overline{\zeta}_i, \overline{B}_i) \quad \text{for any } i \in I'.$$

The pairs $(Y_{i,\Omega}/G_i, \overline{B}_i)$, $i \in I'$, are klt surface pairs over the algebraically closed field $\Omega \supset \mathbb{C}$ near $\overline{\zeta}_i$, with coefficients in the DCC set Γ'. By Theorem 2.21(1), $\{q_i(s_i) \mid i \in I'\}$ satisfies the ACC, which contradicts that $(q_i(s_i))_{i \in I'}$ is strictly increasing. Hence $\dim Z_i = 1$ only for finitely many i. Possibly removing finitely many indices, we may assume that $Z_i = \{y_i\}$ is a closed point for any i.

Step 5. We exclude smooth germs and germs of quadratic rank 4, and obtain the radial failure sequence.

Let $(W_i \ni o_i)$, h_i and the radial function q_i be as in **Step 3**. The group $G_i \subset \operatorname{Gal}(V_i/X_i)$ is cyclic, acts faithfully on W_i, and fixes o_i. Near o_i we have $\frac{1}{m}\operatorname{div}(h_i) = L_i$, so $(W_i, \frac{1}{m}\operatorname{div}(h_i))$ is lc near o_i. If h_i were a unit at o_i, then $\operatorname{div}(h_i) = 0$ near o_i and q_i would be constant, which contradicts $q_i(\tau_i) \geq a > \mu_i(1) = q_i(1)$ (see (4.4)). Thus $h_i(o_i) = 0$. Since Y_i is Gorenstein terminal, $(W_i \ni o_i)$ is either smooth or an isolated cDV point. If $(W_i \ni o_i)$ is singular, it is a hypersurface double point of quadratic rank 1, 2, 3, or 4.

Suppose that the set $I := \{i \in \mathbb{N}^+ \mid W_i \text{ is smooth at } o_i\}$ is infinite. By Claim 4.8, there exist an infinite subset $I' \subset I$ and real numbers $s_i \in (\sigma_i, 1)$, $i \in I'$, such that $(s_i)_{i \in I'}$ and $(q_i(s_i))_{i \in I'}$ are

strictly increasing and $q_i(s_i) > a/2$ for any $i \in I'$. For $i \in I'$, since $(W_i, \frac{s_i}{m}\operatorname{div}(h_i)) = (Y_i, s_iL_i)$ is klt near o_i by **Step 3** and the labels $\frac{s_i}{m}$ belong to the DCC set $\{\frac{s_j}{m} \mid j \in I'\}$, the positive numbers

$$q_i(s_i) = \operatorname{mld}\Big(\Big(W_i \ni o_i, \frac{s_i}{m}\operatorname{div}(h_i)\Big)/G_i\Big), \quad i \in I',$$

satisfy the ACC by [Liu26a, Theorem 1.5], which contradicts that $(q_i(s_i))_{i\in I'}$ is strictly increasing. Hence W_i is smooth at o_i only for finitely many i. Let I be the set of $i \in \mathbb{N}^+$ such that $(W_i \ni o_i)$ is singular of quadratic rank 4, and suppose that I is infinite. By Claim 4.8, there exist an infinite subset $I' \subset I$ and real numbers $s_i \in (\sigma_i, 1)$, $i \in I'$, such that $(s_i)_{i\in I'}$ and $(q_i(s_i))_{i\in I'}$ are strictly increasing and $q_i(s_i) > a/2$ for any $i \in I'$. For $i \in I'$, since $(W_i, \frac{s_i}{m}\operatorname{div}(h_i)) = (Y_i, s_iL_i)$ is klt near o_i by **Step 3** and the labels $\frac{s_i}{m}$ belong to the DCC set $\{\frac{s_j}{m} \mid j \in I'\}$, the second sentence of [Liu26b, Theorem 7.1] implies that the positive numbers $q_i(s_i)$, $i \in I'$, satisfy the ACC, which contradicts that $(q_i(s_i))_{i\in I'}$ is strictly increasing. Hence $(W_i \ni o_i)$ has quadratic rank 4 only for finitely many i. Possibly removing finitely many indices, we may assume that $(W_i \ni o_i)$ is an isolated Gorenstein terminal hypersurface threefold germ, singular of quadratic rank 1, 2, or 3, for any i.

By (4.4) and **Step 1**, $0 \le q_i(1) = \mu_i(1) \le a - \delta$ and $q_i(\tau_i) \ge \mu_i(\tau_i) = a$. Since moreover $\lim_{i\to+\infty} \tau_i = 1$, $(W_i \ni o_i, G_i, h_i, \tau_i)$ is a radial failure sequence with constants (m, a, δ), and the last sentence of (C) holds by the previous paragraph and (4.4). This proves (C). □

5. Margins, ambient lifts and products

In this section we prove a uniform margin by a fixed power of the maximal ideal, construct lifts of prime divisors over a hypersurface to the ambient space and Hensel transport of prime divisors between nearby hypersurfaces, bound Jacobian orders on split germs, choose computing lifts on surface quotients with lc marks, resolve marked hypersurfaces of bounded degree with a finite table of log discrepancies and multiplicities, and compare germs close to products with product germs.

5.1. **Uniform margins and ambient lifts.**

Theorem 5.1. *The following hold.*

(1) Let $d \ge 2$ be an integer and $I \subset \mathbb{R}_{>0}$ a finite set. Then there exists a real number $\eta = \eta(d, I) > 0$ such that for any smooth germ $(X \ni 0)$ of dimension d, algebraic or $X = \operatorname{Spec}\mathbb{C}[[x_1, \dots, x_d]]$, and any $\mathbb{R}$-ideal $\mathfrak{a}$ on X with exponents in I and $\operatorname{mld}(X \ni 0, \mathfrak{a}) > 0$, the pair

$$(X, \mathfrak{a} \cdot \mathfrak{m}_0^{\eta}) \text{ is lc near } 0,$$

where $\mathfrak{m}_0 \subset \mathcal{O}_X$ is the ideal of 0.

(2) Let m be a positive integer. Then there exists a real number $\eta_m > 0$ such that for any germ $(W \ni o)$ which is either smooth of dimension 3 or a normal hypersurface threefold germ in a smooth fourfold germ, and any non-zero regular function h on W such that $\operatorname{mld}(W \ni o, \frac{1}{m}\operatorname{div}(h)) > 0$, the pair

$$\Big(W, \frac{1}{m}\operatorname{div}(h), \mathfrak{m}_o^{\eta_m}\Big) \text{ is lc near } o,$$

where $\mathfrak{m}_o \subset \mathcal{O}_W$ is the ideal of o. In particular,

$$a\Big(F, W, \frac{1}{m}\operatorname{div}(h)\Big) \ge \eta_m \operatorname{mult}_F \mathfrak{m}_o$$

for any prime divisor F over W with center o.

(3) In (2), let G be a finite group acting faithfully on W and fixing o such that h is a mark, and let

$$b := \operatorname{mld}\Big(\Big(W \ni o, \frac{1}{m}\operatorname{div}(h)\Big)\Big/G\Big).$$

Then for any real number $L > 0$, any prime divisor F over W with center o and $a(F, W, \frac{1}{m}\operatorname{div}(h))/e_F \leq L$ satisfies

$$\frac{\operatorname{mult}_F \mathfrak{m}_o}{e_F} \leq \frac{L}{\eta_m}.$$

In particular, any computing lift F of b satisfies $\operatorname{mult}_F \mathfrak{m}_o/e_F \leq b/\eta_m \leq 3/\eta_m$, and if $|G| \leq M$ for a positive integer M, then $\operatorname{mult}_F \mathfrak{m}_o \leq 3M/\eta_m$.

Proof. **Step 1.** In this step we prove (1) for $X = \operatorname{Spec} \mathbb{C}[[x_1, \dots, x_d]]$.

Let $R := \mathbb{C}[[x_1, \dots, x_d]]$, so that $X = \operatorname{Spec} R$, and let $\mathfrak{n} := (x_1, \dots, x_d) \subset R$. Since any open subset of X containing the closed point 0 is X, "lc near 0" means lc on X. Suppose that (1) does not hold for X. Then there exist $\mathbb{R}$-ideals $\mathfrak{a}_i$ on X, $i \in \mathbb{N}^+$, with exponents in I, such that $\operatorname{mld}(X \ni 0, \mathfrak{a}_i) > 0$ and

$$(X, \mathfrak{a}_i \mathfrak{n}^{1/i}) \text{ is not lc.} \tag{5.1}$$

Below, passing to a subsequence means replacing $\mathbb{N}^+$ with an infinite subset and keeping the indices. Write $I = \{\lambda_1, \dots, \lambda_n\}$. Since

$$\mathfrak{c}^\lambda \mathfrak{c}'^\lambda = (\mathfrak{c}\mathfrak{c}')^\lambda$$

for ideals $\mathfrak{c}, \mathfrak{c}'$ and $\lambda > 0$, we may write

$$\mathfrak{a}_i = \prod_{j=1}^n \mathfrak{a}_{i,j}^{\lambda_j}$$

with non-zero ideals $\mathfrak{a}_{i,j} \subset R$. Possibly passing to a subsequence, for any j, either $\mathfrak{a}_{i,j} = R$ for any i, or $\mathfrak{a}_{i,j} \neq R$ for any i. Removing the indices j of the first kind and renumbering, we may assume that $\mathfrak{a}_{i,j} \subsetneq R$ for any i, j. If $n = 0$, then $\mathfrak{a}_i = R$ and $(X, \mathfrak{n}^{1/i})$ is lc since $1/i < d = \operatorname{lct}(X; \mathfrak{n})$, contradicting (5.1). Thus $n \geq 1$. Since $\operatorname{mld}(X \ni 0, \mathfrak{a}_i) > 0$, the pair $(X, \mathfrak{a}_i)$ is lc, that is, $\operatorname{lct}(X; \mathfrak{a}_i) \geq 1$.

Possibly passing to a subsequence, let $(\mathfrak{a}_{*,1}, \dots, \mathfrak{a}_{*,n}, \mathfrak{n}_*)$ be a generic limit of the sequence $(\mathfrak{a}_{i,1}, \dots, \mathfrak{a}_{i,n}, \mathfrak{n})_i$ [dFEM11, Section 3] (cf. [MN18, Proposition 3.1]). Here $\mathfrak{a}_{*,j}$ and $\mathfrak{n}_*$ are ideals of $R_* := \mathbb{K}[[x_1, \dots, x_d]]$ for a field $\mathbb{K} \supset \mathbb{C}$, and $\mathfrak{n}_* = (x_1, \dots, x_d)$ is the maximal ideal of R_*. Let $X_* := \operatorname{Spec} R_*$ and $\mathfrak{a}_* := \prod_{j=1}^n \mathfrak{a}_{*,j}^{\lambda_j}$. We use the following two properties of the generic limit [dFEM11, Section 3 and Corollary 3.4].

(a) For any real numbers $\mu_1, \dots, \mu_n > 0$ and $\mu \geq 0$,

$$\operatorname{lct}\left(X_*; \prod_{j=1}^n \mathfrak{a}_{*,j}^{\mu_j} \cdot \mathfrak{n}_*^\mu\right) \text{ is a limit point of the sequence } \left(\operatorname{lct}\left(X; \prod_{j=1}^n \mathfrak{a}_{i,j}^{\mu_j} \cdot \mathfrak{n}^\mu\right)\right)_i.$$

(b) For any positive integer N, let $\mathcal{H}_N$ be the scheme of finite type over $\mathbb{C}$ parametrizing n-tuples of ideals of $\mathbb{C}[x_1, \dots, x_d]/(x_1, \dots, x_d)^N = R/\mathfrak{n}^N$, and let $z_{i,N} \in \mathcal{H}_N$ be the closed point given by the images of $\mathfrak{a}_{i,1}, \dots, \mathfrak{a}_{i,n}$ in $R/\mathfrak{n}^N$. Then there exist an irreducible closed subset $Z_N \subset \mathcal{H}_N$ and a $\mathbb{C}$-embedding $\mathbb{C}(Z_N) \hookrightarrow \mathbb{K}$ such that any non-empty open subset of Z_N contains $z_{i,N}$ for infinitely many i, and the images of $\mathfrak{a}_{*,1}, \dots, \mathfrak{a}_{*,n}$ in $R_*/\mathfrak{n}_*^N$ are the base change to $\mathbb{K}$ of the n-tuple of ideals given by the generic point of Z_N.

We claim that

$$\operatorname{mld}(X_* \ni 0, \mathfrak{a}_*) > 0. \tag{5.2}$$

By (a) with $\mu_j = \lambda_j$ and $\mu = 0$, $\operatorname{lct}(X_*; \mathfrak{a}_*) \geq 1$. Thus $\mathfrak{a}_{*,j} \neq 0$ for any j, the pair $(X_*, \mathfrak{a}_*)$ is lc, and $\operatorname{mld}(X_* \ni 0, \mathfrak{a}_*) \geq 0$. Suppose that $\operatorname{mld}(X_* \ni 0, \mathfrak{a}_*) = 0$. Then a log resolution of $(X_*, \mathfrak{a}_*\mathfrak{n}_*)$

[dFEM11, Theorems 2.3–2.4] carries a prime divisor E with center the closed point 0 of X_* and $a(E, X_*, \mathfrak{a}_*) = 0$. Since X_* is smooth, $a(E, X_*, 0) > 0$, so

$$\operatorname{mult}_E \mathfrak{a}_* := \sum_{j=1}^{n} \lambda_j \operatorname{mult}_E \mathfrak{a}_{*,j} = a(E, X_*, 0) > 0.$$

Let N be a positive integer such that $N \operatorname{mult}_E \mathfrak{n}_* > \operatorname{mult}_E \mathfrak{a}_{*,j}$ for any j, and let $\mathfrak{b}_{i,j} := \mathfrak{a}_{i,j} + \mathfrak{n}^N$, $\mathfrak{b}_{*,j} := \mathfrak{a}_{*,j} + \mathfrak{n}_*^N$, $\mathfrak{b}_i := \prod_{j=1}^n \mathfrak{b}_{i,j}^{\lambda_j}$, and $\mathfrak{b}_* := \prod_{j=1}^n \mathfrak{b}_{*,j}^{\lambda_j}$. Then

$$\operatorname{mult}_E \mathfrak{b}_{*,j} = \operatorname{mult}_E \mathfrak{a}_{*,j} \quad \text{for any } j,$$

so $a(E, X_*, \mathfrak{b}_*) = 0$ and $\operatorname{mult}_E \mathfrak{b}_* > 0$. Since $(X_*, \mathfrak{a}_*)$ is lc and $\mathfrak{a}_{*,j} \subset \mathfrak{b}_{*,j}$, the pair $(X_*, \mathfrak{b}_*)$ is lc. Thus

$$\operatorname{lct}(X_*; \mathfrak{b}_*) = 1.$$

Next we show that $\operatorname{lct}(X; \mathfrak{b}_i) = 1$ for infinitely many i. Let $\mathcal{B}_1, \dots, \mathcal{B}_n$ be the ideals of $\mathbb{C}(Z_N)[x_1, \dots, x_d]$ containing $(x_1, \dots, x_d)^N$ given by the generic point of Z_N, and for a closed point $z \in Z_N$, let $\mathcal{B}_{z,1}, \dots, \mathcal{B}_{z,n}$ be the ideals of $\mathbb{C}[x_1, \dots, x_d]$ containing $(x_1, \dots, x_d)^N$ given by z. Since $\mathfrak{n}^N \subset \mathfrak{b}_{i,j}$, the ideal $\mathfrak{b}_{i,j}$ is the completion of $\mathcal{B}_{z_{i,N},j}$, and by (b), $\mathfrak{b}_{*,j}$ is the completion of $\mathcal{B}_j \otimes_{\mathbb{C}(Z_N)} \mathbb{K}$. Thus by [dFEM11, Proposition 2.11 and Lemma 2.14],

$$\operatorname{lct}\left(\mathbb{A}^d_{\mathbb{C}(Z_N)}; \prod_{j=1}^n \mathcal{B}_j^{\lambda_j}\right) = \operatorname{lct}(X_*; \mathfrak{b}_*) = 1 \quad \text{and} \quad \operatorname{lct}\left(\mathbb{A}^d; \prod_{j=1}^n \mathcal{B}_{z_{i,N},j}^{\lambda_j}\right) = \operatorname{lct}(X; \mathfrak{b}_i).$$

A log resolution $V \to \mathbb{A}^d_{\mathbb{C}(Z_N)}$ of $(\mathbb{A}^d_{\mathbb{C}(Z_N)}, \prod_j \mathcal{B}_j)$ spreads out over a non-empty open subset $U \subset Z_N$ to a morphism whose fiber over any closed point $z \in U$ is a log resolution of $(\mathbb{A}^d, \prod_j \mathcal{B}_{z,j})$, such that for any prime divisor E' on V and any component E'_z of the fiber over z of the closure of E',

$$a(E'_z, \mathbb{A}^d, 0) = a(E', \mathbb{A}^d_{\mathbb{C}(Z_N)}, 0) \quad \text{and} \quad \operatorname{mult}_{E'_z} \mathcal{B}_{z,j} = \operatorname{mult}_{E'} \mathcal{B}_j \text{ for any } j.$$

Thus $\operatorname{lct}(\mathbb{A}^d; \prod_j \mathcal{B}_{z,j}^{\lambda_j}) = 1$ for any closed point $z \in U$. By (b), $z_{i,N} \in U$ for infinitely many i, and for such i, $\operatorname{lct}(X; \mathfrak{b}_i) = 1$.

Fix such an i. Since $\operatorname{lct}(X; \mathfrak{b}_i) = 1$, there exists a prime divisor E_i over X with $\operatorname{mult}_{E_i} \mathfrak{b}_i > 0$ and $a(E_i, X, \mathfrak{b}_i) = 0$. Since $\mathfrak{n}^N \subset \mathfrak{b}_{i,j}$ for any j, the ideals $\mathfrak{b}_{i,j}$ are trivial outside 0, so $\operatorname{center}_X E_i = \{0\}$. Since $\mathfrak{a}_{i,j} \subset \mathfrak{b}_{i,j}$ for any j,

$$a(E_i, X, \mathfrak{a}_i) \le a(E_i, X, \mathfrak{b}_i) = 0,$$

contradicting $\operatorname{mld}(X \ni 0, \mathfrak{a}_i) > 0$. This proves (5.2).

Let $g : Y \to X_*$ be a log resolution of $(X_*, \mathfrak{a}_*\mathfrak{n}_*)$, and let

$$\eta_* := \min\left\{ \frac{a(E', X_*, \mathfrak{a}_*)}{\operatorname{mult}_{E'} \mathfrak{n}_*} \;\middle|\; E' \text{ is a prime divisor on } Y \text{ with } \operatorname{mult}_{E'} \mathfrak{n}_* > 0 \right\}$$

(cf. [MN18, Proposition 2.1]). Any prime divisor E' on Y with $\operatorname{mult}_{E'} \mathfrak{n}_* > 0$ has center 0, so $a(E', X_*, \mathfrak{a}_*) > 0$ by (5.2). Thus $\eta_* > 0$. For any prime divisor E' on Y,

$$a(E', X_*, \mathfrak{a}_*\mathfrak{n}_*^{\eta_*}) = a(E', X_*, \mathfrak{a}_*) - \eta_* \operatorname{mult}_{E'} \mathfrak{n}_* \ge 0 :$$

if $\operatorname{mult}_{E'} \mathfrak{n}_* > 0$, by the definition of η_*, and otherwise since $(X_*, \mathfrak{a}_*)$ is lc. Equality holds for a prime divisor E'_0 on Y attaining the minimum, and

$$\operatorname{mult}_{E'_0}(\mathfrak{a}_*\mathfrak{n}_*^{\eta_*}) \ge \eta_* \operatorname{mult}_{E'_0} \mathfrak{n}_* > 0.$$

Since g is a log resolution of $(X_*, \mathfrak{a}_*\mathfrak{n}_*)$,

$$\operatorname{lct}(X_*; \mathfrak{a}_*\mathfrak{n}_*^{\eta_*}) = 1.$$

We show that

$$(X, \mathfrak{a}_i\mathfrak{n}^{\eta_*}) \text{ is lc} \tag{5.3}$$

for infinitely many i. Suppose not. Then $c_i := \mathrm{lct}(X; \mathfrak{a}_i\mathfrak{n}^{\eta_*}) < 1$ for any $i \gg 0$, and 1 is a limit point of $(c_i)_i$ by (a) with $\mu_j = \lambda_j$ and $\mu = \eta_*$. Possibly passing to a subsequence, we may assume that $c_i < 1$ for any i and $\lim_{i\to+\infty} c_i = 1$. For any i, let N_i be an integer such that

$$N_i \geq \frac{2d}{(1-c_i)\min\{\lambda_1,\dots,\lambda_n,\eta_*\}}, \quad \text{and let} \quad c_i' := \mathrm{lct}\Big(X; \prod_{j=1}^n (\mathfrak{a}_{i,j} + \mathfrak{n}^{N_i})^{\lambda_j} \cdot \mathfrak{n}^{\eta_*}\Big).$$

By Theorem 2.23(2) with $\epsilon = (1-c_i)/2$, $|c_i - c_i'| \leq (1-c_i)/2$, so

$$\frac{3c_i - 1}{2} \leq c_i' \leq \frac{1+c_i}{2} < 1.$$

Thus $c_i' < 1$ for any i and $\lim_{i\to+\infty} c_i' = 1$. Possibly passing to a subsequence, $(c_i')_i$ is strictly increasing. Since $\mathfrak{n}^{N_i} \subset \mathfrak{a}_{i,j} + \mathfrak{n}^{N_i}$, the ideal $\mathfrak{a}_{i,j} + \mathfrak{n}^{N_i}$ is the completion of an ideal $\mathfrak{a}'_{i,j}$ of $\mathbb{C}[x_1,\dots,x_d]$ containing $(x_1,\dots,x_d)^{N_i}$, and by [dFEM11, Proposition 2.11],

$$c_i' = \mathrm{lct}\Big(\mathbb{A}^d \ni 0; \prod_{j=1}^n (\mathfrak{a}'_{i,j})^{\lambda_j} \cdot (x_1,\dots,x_d)^{\eta_*}\Big).$$

Thus $(c_i')_i$ is a strictly increasing sequence of lc thresholds on the fixed germ $(\mathbb{A}^d \ni 0)$ of $\mathbb{R}$-ideals with exponents in the finite set $I \cup \{\eta_*\}$, contradicting [dFEM11, Theorem 1.1]. This proves (5.3).

Choose i with (5.3) and $1/i < \eta_*$. Since decreasing the exponent of $\mathfrak{n}$ preserves lc, $(X, \mathfrak{a}_i\mathfrak{n}^{1/i})$ is lc, contradicting (5.1). Thus (1) holds for X.

Step 2. In this step we prove (1) for algebraic germs, with the constant $\eta(d, I)$ of **Step 1**.

Let $(X \ni 0)$ be a smooth algebraic germ of dimension d and $\mathfrak{a}$ an $\mathbb{R}$-ideal on X with exponents in I and $\mathrm{mld}(X \ni 0, \mathfrak{a}) > 0$. Let

$$\widehat{X} := \mathrm{Spec}\,\widehat{\mathcal{O}}_{X,0} \cong \mathrm{Spec}\,\mathbb{C}[[x_1,\dots,x_d]].$$

By [dFEM11, Proposition 2.11, Lemma 2.21, Proposition A.14 and Remark A.15],

$$\mathrm{mld}(\widehat{X} \ni 0, \mathfrak{a}\mathcal{O}_{\widehat{X}}) = \mathrm{mld}(X \ni 0, \mathfrak{a}) > 0,$$

and, for $\eta = \eta(d, I)$, the pair $(X, \mathfrak{a} \cdot \mathfrak{m}_0^\eta)$ is lc near 0 if and only if

$$\big(\widehat{X}, (\mathfrak{a} \cdot \mathfrak{m}_0^\eta)\mathcal{O}_{\widehat{X}}\big) \text{ is lc.}$$

Thus **Step 1** implies (1) for $(X \ni 0)$.

Step 3. In this step we prove (2). Let

$$\eta_m := \min\Big\{\eta\Big(3, \Big\{\frac{1}{m}\Big\}\Big),\ \eta\Big(4, \Big\{1, \frac{1}{m}\Big\}\Big)\Big\}$$

with the constants of (1), and let $(W \ni o)$ and h be as in (2). Since decreasing the exponent of $\mathfrak{m}_o$ preserves lc, it suffices to show that $(W, \frac{1}{m}\mathrm{div}(h), \mathfrak{m}_o^{\eta'})$ is lc near o for $\eta' = \eta(3, \{\frac{1}{m}\})$ if $(W \ni o)$ is smooth of dimension 3, and for $\eta' = \eta(4, \{1, \frac{1}{m}\})$ if W is a normal hypersurface in a smooth fourfold germ.

If $(W \ni o)$ is smooth of dimension 3, then

$$\mathrm{mld}(W \ni o, (h)^{1/m}) = \mathrm{mld}\Big(W \ni o, \frac{1}{m}\mathrm{div}(h)\Big) > 0,$$

so (1) with $d = 3$ and $I = \{\frac{1}{m}\}$ implies that $(W, (h)^{1/m}\mathfrak{m}_o^{\eta'})$ is lc near o.

Assume that $W = (f = 0)$ is a normal hypersurface in a smooth fourfold germ $(\mathcal{X} \ni o)$, and let $\mathfrak{m}_{\mathcal{X},o} \subset \mathcal{O}_{\mathcal{X}}$ be the ideal of o, so $\mathfrak{m}_{\mathcal{X},o}\mathcal{O}_W = \mathfrak{m}_o$. Possibly shrinking $\mathcal{X}$, let $\widetilde{h}$ be a regular

function on $\mathcal{X}$ with $\widetilde{h}|_W = h$. Since $h \neq 0$, $\widetilde{h} \notin (f)$. Since $\mathcal{X}$ is smooth and W is normal, by precise inversion of adjunction [EMY03, Theorem 1.6],

$$\operatorname{mld}\big(\mathcal{X} \ni o, (f)\cdot(\widetilde{h})^{1/m}\big) = \operatorname{mld}\Big(W \ni o, \frac{1}{m}\operatorname{div}(h)\Big) > 0. \tag{5.4}$$

By (1) with $d = 4$ and $I = \{1, \frac{1}{m}\}$,

$$\big(\mathcal{X}, (f)\cdot(\widetilde{h})^{1/m}\cdot \mathfrak{m}_{\mathcal{X},o}^{\eta'}\big) \text{ is lc near } o.$$

By [EMY03, Theorem 1.6] again, now with the closed subscheme $\{o\}$ of $\mathcal{X}$ added with coefficient η',

$$\operatorname{mld}\Big(W \ni o, \frac{1}{m}\operatorname{div}(h), \mathfrak{m}_o^{\eta'}\Big) = \operatorname{mld}\big(\mathcal{X} \ni o, (f)\cdot(\widetilde{h})^{1/m}\cdot\mathfrak{m}_{\mathcal{X},o}^{\eta'}\big) \geq 0.$$

Thus $(W, \frac{1}{m}\operatorname{div}(h), \mathfrak{m}_o^{\eta'})$ is lc near o.

The inequality in (2) holds since for any prime divisor F over W with center o,

$$a\Big(F, W, \frac{1}{m}\operatorname{div}(h)\Big) - \eta_m \operatorname{mult}_F \mathfrak{m}_o = a\Big(F, W, \frac{1}{m}\operatorname{div}(h), \mathfrak{m}_o^{\eta_m}\Big) \geq 0.$$

Step 4. In this step we prove (3).

Let F be a prime divisor over W with center o and $a(F, W, \frac{1}{m}\operatorname{div}(h))/e_F \leq L$. By (2),

$$\eta_m \frac{\operatorname{mult}_F \mathfrak{m}_o}{e_F} \leq \frac{a(F, W, \frac{1}{m}\operatorname{div}(h))}{e_F} \leq L,$$

which proves the first sentence of (3). A computing lift F of b satisfies $a(F, W, \frac{1}{m}\operatorname{div}(h))/e_F = b$ (Definition 2.4), so the case $L = b$ implies $\operatorname{mult}_F \mathfrak{m}_o/e_F \leq b/\eta_m$.

By Lemma 4.2(3) for the radial function q of $(W \ni o, G, h, m)$, $b = q(1) \leq 3$. Finally, if $|G| \leq M$, then $e_F = |I_G(F)| \leq M$, so any computing lift F of b satisfies $\operatorname{mult}_F \mathfrak{m}_o \leq 3e_F/\eta_m \leq 3M/\eta_m$. The theorem follows. □

Definition 5.2. Let $(\mathcal{X} \ni o)$ be a smooth germ of dimension n, either algebraic over $\mathbb{C}$ or $\mathcal{X} = \operatorname{Spec} \Bbbk[[x_1, \dots, x_n]]$ for a field $\Bbbk \supset \mathbb{C}$, with regular parameters $x_1, \dots, x_n$ at o, and let $W = (f = 0) \subset \mathcal{X}$ be an integral hypersurface. The *Jacobian ideal* $\operatorname{Jac}_W \subset \mathcal{O}_W$ is the ideal generated by

$$\frac{\partial f}{\partial x_1}\Big|_W, \dots, \frac{\partial f}{\partial x_n}\Big|_W.$$

By the chain rule, Jac_W does not depend on the choice of f and $x_1, \dots, x_n$. For a prime divisor F over W (a formal prime divisor if $\mathcal{X} = \operatorname{Spec} \Bbbk[[x_1, \dots, x_n]]$, see Definition 2.2), the *Jacobian order* of F is $\operatorname{mult}_F \operatorname{Jac}_W$.

Lemma 5.3. *Let Z be a normal variety over an algebraically closed field $\Bbbk \supset \mathbb{C}$, or the spectrum of a complete normal local domain containing $\mathbb{C}$ with residue field $\mathbb{C}$, in which case let $\Bbbk := \mathbb{C}$. Let E be a prime divisor over Z, with center the closed point in the second case, $\mathcal{O}_E$ the valuation ring of E, $\mathfrak{m}_E$ its maximal ideal, $\widehat{\mathcal{O}}_E$ the completion of $\mathcal{O}_E$, $\widehat{\mathfrak{m}}_E$ the maximal ideal of $\widehat{\mathcal{O}}_E$, and $\kappa(E)$ the residue field. The valuation mult_E extends continuously to the fraction field of $\widehat{\mathcal{O}}_E$. Then the following hold.*

(1) Let π be a uniformizer of $\widehat{\mathcal{O}}_E$, let $\zeta_1, \dots, \zeta_{\dim Z-1} \in \widehat{\mathcal{O}}_E$ be elements whose residues form a transcendence basis of $\kappa(E)$ over $\Bbbk$, and let ω be a generator of ω_Z near the generic point of $\operatorname{center}_Z E$. Then the $\widehat{\mathfrak{m}}_E$-adic completion of $\Omega_{\widehat{\mathcal{O}}_E/\Bbbk}$ is free with basis $d\pi, d\zeta_1, \dots, d\zeta_{\dim Z-1}$, and

$$\omega = \pi^{a(E,Z,0)-1} d\pi \wedge d\zeta_1 \wedge \cdots \wedge d\zeta_{\dim Z-1} \quad \text{up to a unit of } \widehat{\mathcal{O}}_E. \tag{5.5}$$

(2) *Let $\widehat{\mathcal{O}}$ be a complete discrete valuation ring containing $\mathbb{C}$, with valuation v and maximal ideal $\widehat{\mathfrak{m}}$, let $x_0, y_1, y_2, y_3 \in \widehat{\mathfrak{m}}$, $b \in \widehat{\mathcal{O}}$, and $\varphi \in \mathbb{C}[[x, y_1, y_2, y_3]]$, and write $\varphi(X) := \varphi(X, y_1, y_2, y_3)$ and $\varphi_x := \partial\varphi/\partial x$. Then for any $\psi \in \mathbb{C}[[x, y_1, y_2, y_3]]$ and $X_1, X_2 \in \widehat{\mathfrak{m}}$,*

$$\psi(X_1) - \psi(X_2) - (X_1 - X_2)\psi_x(X_2) \in (X_1 - X_2)^2\widehat{\mathcal{O}}. \tag{5.6}$$

If $v(\varphi(x_0) - b) > 2v(\varphi_x(x_0))$, then there exists a unique $X \in \widehat{\mathfrak{m}}$ with $\varphi(X) = b$ and $v(X - x_0) > v(\varphi_x(x_0))$, X satisfies

$$v(X - x_0) = v(\varphi(x_0) - b) - v(\varphi_x(x_0)), \tag{5.7}$$

and X belongs to any closed subfield of the fraction field of $\widehat{\mathcal{O}}$ containing $\mathbb{C}$, x_0, y_1, y_2, y_3, and b.

(3) *Assume that a finite group G acts faithfully on Z, and let $I := I_G(E)$. Then I acts on the 1-dimensional $\kappa(E)$-vector space $\mathfrak{m}_E/\mathfrak{m}_E^2$ by an injective character $\lambda : I \to \Bbbk^*$. In particular, I is cyclic of order e_E, and any character of I is λ^k for an integer k unique modulo e_E. Moreover, there exists a uniformizer $\pi \in \mathcal{O}_E$ with $\sigma(\pi) = \lambda(\sigma)\pi$ for any $\sigma \in I$.*

(4) *Let $\Bbbk(Z)$ be the function field of Z (the fraction field of the complete local domain in the second case), s an indeterminate, k a positive integer, π a uniformizer of $\mathcal{O}_E$, $v := (\mathrm{mult}_E)_k$ the Gauss extension of mult_E to $\Bbbk(Z)(s)$ with value k on s (Definition 2.3), and $w := s/\pi^k$. Then the residue $\overline{w}$ of w is transcendental over $\kappa(E)$, the residue field of v is $\kappa(E)(\overline{w})$, and, for ω and $\zeta_1, \dots, \zeta_{\dim Z-1}$ as in (1),*

$$\omega \wedge ds = \pi^{a(E,Z,0)-1+k} d\pi \wedge d\zeta_1 \wedge \cdots \wedge d\zeta_{\dim Z-1} \wedge dw \tag{5.8}$$

up to a unit of the completion of the valuation ring of v. If moreover G, λ, and π are as in (3), and G acts on $\Bbbk(Z)(s)$ extending the action on $\Bbbk(Z)$ with $\sigma(s) = \chi(\sigma)s$ for a character χ of G, then $v \circ \sigma = v$ for any $\sigma \in \mathrm{Stab}_G(E)$, and if $\chi|_I = \lambda^k$, then I acts trivially on $\kappa(E)(\overline{w})$.

(5) *Let R be $\mathcal{O}_{\mathbb{A}^4,0}$ or $\mathbb{C}[[x_1, \dots, x_4]]$, and let $\iota : R \to \widehat{\mathcal{O}}$ be a homomorphism to a complete discrete valuation ring $\widehat{\mathcal{O}} \supset \mathbb{C}$ with valuation v, sending $x_1, \dots, x_4$ into the maximal ideal, such that the fraction field of $\iota(R)$ is dense in the fraction field of $\widehat{\mathcal{O}}$ and the residue field of $\widehat{\mathcal{O}}$ has transcendence degree r over $\mathbb{C}$. Then $\dim R/\operatorname{Ker}\iota \geq r + 1$. If equality holds, then $v \circ \iota = \mathrm{mult}_P$ for a prime divisor P over $\operatorname{Spec} R/\operatorname{Ker}\iota$ with center the closed point.*

Proof. For (1), the $\widehat{\mathfrak{m}}_E$-adic completion of $\Omega_{\widehat{\mathcal{O}}_E/\Bbbk}$ depends only on the rings $\mathcal{O}_E/\mathfrak{m}_E^m$, so it is the completion of the module of special differentials of $\mathcal{O}_E$, a finite free module [dFEM11, Lemma A.1 and Proposition A.8]. Here (5.5) is the local form at the generic point of E of $K_Y = \rho^* K_Z + (a(E, Z, 0) - 1)E$ for a birational model $\rho : Y \to Z$ on which E is a divisor, with $K_{Y/Z}$ defined through special differentials as in [dFEM11, Proposition A.8 and Lemma A.11] in the second case. All differentials below are elements of the completions of the modules $\Omega_{\widehat{\mathcal{O}}_E/\Bbbk}$, in which d commutes with convergent power series.

For (2), (5.6) holds by Taylor expansion in the first variable. If $v(\varphi(x_0) - b) > 2v(\varphi_x(x_0))$, then Newton's method gives the unique X and (5.7): the iterates

$$X_0 := x_0 \quad \text{and} \quad X_{m+1} := X_m - \frac{\varphi(X_m) - b}{\varphi_x(X_m)}$$

converge to X. Since φ and φ_x are power series with coefficients in $\mathbb{C}$, by induction on m, each X_m, hence X, belongs to any closed subfield of the fraction field of $\widehat{\mathcal{O}}$ containing $\mathbb{C}$, x_0, y_1, y_2, y_3, and b.

For (3), the group I preserves $\mathcal{O}_E$ and acts trivially on $\kappa(E)$, so I acts on the 1-dimensional $\kappa(E)$-vector space $\mathfrak{m}_E/\mathfrak{m}_E^2$ by a character λ, whose values are roots of unity. Let $\sigma \in \operatorname{Ker}\lambda$ and $\delta := \sigma - \mathrm{id}_{\mathcal{O}_E}$. Then $\delta(\mathfrak{m}_E^i) \subset \mathfrak{m}_E^{i+1}$ for any $i \geq 0$. Suppose that $\delta \neq 0$, and let $r \geq 1$ be the

largest integer with $\delta(\mathfrak{m}_E^i) \subset \mathfrak{m}_E^{i+r}$ for any $i \geq 0$, which exists since $\bigcap_i \mathfrak{m}_E^i = 0$. Let N_σ be the order of σ in G. Then

$$0 = \sigma^{N_\sigma} - \mathrm{id}_{\mathcal{O}_E} = N_\sigma \delta + \sum_{i'=2}^{N_\sigma} \binom{N_\sigma}{i'} \delta^{i'},$$

and, for $i' \geq 2$,

$$\delta^{i'}(\mathfrak{m}_E^i) \subset \mathfrak{m}_E^{i+i'r} \subset \mathfrak{m}_E^{i+r+1},$$

so $\delta(\mathfrak{m}_E^i) \subset \mathfrak{m}_E^{i+r+1}$ for any i, a contradiction. Thus σ acts trivially on $\mathcal{O}_E$, hence on the function field of Z, and $\sigma = \mathrm{id}_Z$ since G acts faithfully. Thus λ is injective, so I is cyclic of order e_E, and any character of I is λ^k for an integer k unique modulo e_E. For a uniformizer $\pi_0 \in \mathcal{O}_E$, the element

$$\pi := e_E^{-1} \sum_{\sigma \in I} \lambda(\sigma)^{-1} \sigma(\pi_0)$$

is congruent to π_0 modulo $\mathfrak{m}_E^2$, hence a uniformizer, and satisfies $\sigma(\pi) = \lambda(\sigma)\pi$ for any $\sigma \in I$.

For (4), by definition,

$$v\Big(\sum_{i \geq 0} b_i s^i\Big) = \min_i (\mathrm{mult}_E\, b_i + ik) \quad \text{for } b_i \in \Bbbk(Z),$$

so v has value group $\mathbb{Z}$. Since $v(\sum_i c_i w^i) = \min_i \mathrm{mult}_E\, c_i$ for $c_i \in \Bbbk(Z)$, the residue $\overline{w}$ is transcendental over $\kappa(E)$, and the residue field of v is $\kappa(E)(\overline{w})$. Since $s = \pi^k w$, the term $k\pi^{k-1} w d\pi$ of ds vanishes after wedging with $d\pi$, so (5.5) implies (5.8). In the equivariant case, any $\sigma \in \mathrm{Stab}_G(E)$ satisfies $v \circ \sigma = v$, since $\mathrm{mult}_E\, \sigma(b_i) = \mathrm{mult}_E\, b_i$. If $\chi|_I = \lambda^k$, then any $\sigma \in I$ satisfies $\sigma(w) = \chi(\sigma)\lambda(\sigma)^{-k} w = w$ and acts trivially on $\kappa(E)$, so I acts trivially on $\kappa(E)(\overline{w})$.

For (5), since the fraction field of $\iota(R)$ is dense in the fraction field of $\widehat{\mathcal{O}}$, the valuation $v \circ \iota$ of the fraction field of $R/\operatorname{Ker}\iota$ has value group $\mathbb{Z}$ and the residue field of $\widehat{\mathcal{O}}$ as residue field, and it is positive on the maximal ideal of $R/\operatorname{Ker}\iota$ since ι sends $x_1, \dots, x_4$ into the maximal ideal. So $\dim R/\operatorname{Ker}\iota \geq r+1$ [Abh56, Theorem 1(1)]. If equality holds, then the valuation ring of $v \circ \iota$ is essentially of finite type over $R/\operatorname{Ker}\iota$ ([HS06, Theorem 9.3.2] if R is formal), so $v \circ \iota = \mathrm{mult}_P$ for a prime divisor P over $\operatorname{Spec} R/\operatorname{Ker}\iota$ with center the closed point. □

Lemma 5.4. *The following hold.*

(1) Let $\Bbbk \supset \mathbb{C}$ be an algebraically closed field, V a smooth variety over $\Bbbk$, g a regular function on V such that $T := (g = 0)$ is a normal hypersurface, G a finite group acting on V, preserving T and acting faithfully on T, F a prime divisor over T, and $H_1, \dots, H_n$ regular functions on V with non-zero restrictions to T. Then there exists a residue class k_0 modulo e_F such that for any sufficiently large integer $k \equiv k_0 \pmod{e_F}$, there exists a prime divisor $\widehat{F}_k$ over V satisfying

$$\mathrm{center}_V \widehat{F}_k = \mathrm{center}_T F, \quad \mathrm{mult}_{\widehat{F}_k} g = k, \quad a(\widehat{F}_k, V, 0) = a(F, T, 0) + k,$$
$$I_G(\widehat{F}_k) = I_G(F), \quad \text{and} \quad \mathrm{mult}_{\widehat{F}_k} H_i = \mathrm{mult}_F(H_i|_T) \text{ for any } i.$$

If moreover $V = \mathbb{A}^4_{\Bbbk}$ with coordinates x, y, z, t, $g = xy + g_0(z,t)$, and G acts diagonally, then k_0 can be taken such that for any sufficiently large integer $k \equiv k_0 \pmod{e_F}$, the prime divisor $\widehat{F}_k$ can be taken with $\mathrm{mult}_{\widehat{F}_k} \varphi = \mathrm{mult}_F(\varphi|_T)$ for any polynomial $\varphi \in \Bbbk[x, z, t]$.

(2) Let $\mathcal{X}$ be $\mathbb{A}^4$ or $\operatorname{Spec} \mathbb{C}[[x_1, \dots, x_4]]$, with coordinates $x_1, \dots, x_4$ and closed point 0, let f be an element of $\mathcal{O}_{\mathcal{X},0}$ such that $W := (f = 0)$ is normal near 0, let F be a prime divisor over W with center 0, $j := \mathrm{mult}_F \operatorname{Jac}_W$, let $x \in \{x_1, \dots, x_4\}$ with $\mathrm{mult}_F(\partial f/\partial x|_W) = j$, and let $H_1, \dots, H_n \in \mathcal{O}_{\mathcal{X},0}$ with non-zero restrictions to W. Then for any integer k satisfying

$$k > 2j \quad \text{and} \quad k - j > \max_i \mathrm{mult}_F(H_i|_W),$$

there exists a prime divisor P over $\mathcal{X}$ with center 0 satisfying

$$\operatorname{mult}_P f = k, \quad \operatorname{mult}_P(\partial f/\partial x) = j, \quad a(P,\mathcal{X},0) = a(F,W,0)+k,$$
$$\text{and} \quad \operatorname{mult}_P H_i = \operatorname{mult}_F(H_i|_W) \text{ for any } i.$$

If moreover a finite abelian group G acts diagonally on $x_1,\dots,x_4$, preserving W and acting faithfully on W, then there exists a residue class k_0 modulo e_F such that for any such k with $k \equiv k_0 \pmod{e_F}$, the prime divisor P can be taken with $I_G(P) = I_G(F)$.

(3) *Let G be a finite abelian group acting diagonally on $\mathbb{C}[[x_1,\dots,x_4]]$, and let $f, f' \in \mathbb{C}[[x_1,\dots,x_4]]$ be G-semi-invariant with a common character such that $W := (f=0)$ and $W' := (f'=0)$ are G-equivariantly isomorphic to the completions of normal algebraic hypersurface threefold germs at closed points. Let F be a prime divisor over W with center the closed point, $j := \operatorname{mult}_F \operatorname{Jac}_W$, $x \in \{x_1,\dots,x_4\}$ with $\operatorname{mult}_F(\partial f/\partial x|_W) = j$, and $c := \operatorname{mult}_F((f'-f)|_W)$. Write y_1,y_2,y_3 for the other three coordinates and identify the coordinates with their images in the completion $\widehat{\mathcal{O}}_F$ of the valuation ring of F. Assume that*

$$c > 2j \quad \text{and} \quad \operatorname{mult}_F\Big(\frac{\partial(f'-f)}{\partial x}\Big|_W\Big) > j.$$

Then the following hold.

(a) *There exists a unique $x' \in \widehat{\mathcal{O}}_F$ with $f'(x',y_1,y_2,y_3) = 0$ and $\operatorname{mult}_F(x'-x) > j$. Moreover, $\operatorname{mult}_F(x'-x) = c-j$.*

(b) *The homomorphism $\mathbb{C}[[x_1,\dots,x_4]] \to \widehat{\mathcal{O}}_F$ sending x to x' and y_i to y_i for $i \in \{1,2,3\}$ has kernel (f') and defines a prime divisor F' over W' with center the closed point. Moreover, $\widehat{\mathcal{O}}_{F'} = \widehat{\mathcal{O}}_F$, $\kappa(F') = \kappa(F)$, $\operatorname{Stab}_G(F') = \operatorname{Stab}_G(F)$, $I_G(F') = I_G(F)$, $a(F',W',0) = a(F,W,0)$, and x is the unique $X \in \widehat{\mathcal{O}}_F$ with $f(X,y_1,y_2,y_3) = 0$ and $\operatorname{mult}_F(X-x') > j$. The assignment $F \to F'$ is G-equivariant.*

(c) *If $\operatorname{mult}_F(\partial(f'-f)/\partial x_i|_W) > j$ for any $i \in \{1,\dots,4\}$, then $\operatorname{mult}_{F'} \operatorname{Jac}_{W'} = j$.*

(d) *If $H_1,\dots,H_n \in \mathbb{C}[[x_1,\dots,x_4]]$ have non-zero restrictions to W and $c > j + \max_i \operatorname{mult}_F(H_i|_W)$, then $\operatorname{mult}_{F'}(H_i|_{W'}) = \operatorname{mult}_F(H_i|_W)$ for any i.*

Proof. We use Lemma 5.3. All differentials are elements of the completed modules of differentials of Lemma 5.3(1).

Step 1: proof of (1). There exists an ideal $\mathfrak{b} \subset \mathcal{O}_T$ such that F is a divisor on the normalization of the blow-up of T along $\mathfrak{b}$. Let $\mathfrak{a}_V \subset \mathcal{O}_V$ be the preimage of the G-stable ideal $\prod_{\sigma\in G} \sigma^*\mathfrak{b}$, let $\phi : Y \to V$ be a G-equivariant log resolution of (V,T) which principalizes $\mathfrak{a}_V$, and let $T_Y := \phi_*^{-1}T$. Since T_Y is smooth and $(\prod_\sigma \sigma^*\mathfrak{b})\mathcal{O}_{T_Y}$ is invertible, $\mathfrak{b}\mathcal{O}_{T_Y}$ is invertible, so $T_Y \to T$ factors through the normalization of the blow-up along $\mathfrak{b}$, and F is a divisor on T_Y.

Let ξ be the generic point of $F \subset T_Y \subset Y$. Then $\mathcal{O}_{Y,\xi}$ is a regular local ring of dimension 2 with residue field $\kappa(F)$, and $\mathcal{O}_{T_Y,\xi} = \mathcal{O}_F$. Since $\operatorname{Exc}(\phi) \cup T_Y$ is an snc divisor, at most one exceptional prime divisor E_1 of ϕ contains ξ. Let $I := I_G(F)$, and let λ be the character of Lemma 5.3(3) for $E = F$. Any $\sigma \in I$ fixes ξ and preserves T_Y and E_1. Let $q \in \mathcal{O}_{Y,\xi}$ be a local equation of T_Y. Let $\pi \in \mathcal{O}_{Y,\xi}$ be a local equation of E_1 if E_1 exists, and an element with $(\pi,q) = \mathfrak{m}_\xi$ otherwise. Then $\pi|_{T_Y}$ is a uniformizer of $\mathcal{O}_F$. The group I acts on the 1-dimensional $\kappa(F)$-vector spaces

$$(q)/\mathfrak{m}_\xi(q) \quad \text{and} \quad \mathfrak{m}_\xi/(\mathfrak{m}_\xi^2+(q)) \cong \mathfrak{m}_F/\mathfrak{m}_F^2$$

by characters ψ and λ. Possibly replacing q with $e_F^{-1}\sum_{\sigma\in I}\psi(\sigma)^{-1}\sigma(q)$ and π with $e_F^{-1}\sum_{\sigma\in I}\lambda(\sigma)^{-1}\sigma(\pi)$, where the new q and π are still local equations of T_Y and of E_1 (if E_1 exists) since any $\sigma \in I$ preserves T_Y and E_1, we may assume by Nakayama's lemma that

$$\sigma(q) = \psi(\sigma)q \quad \text{and} \quad \sigma(\pi) = \lambda(\sigma)\pi \quad \text{for any } \sigma \in I.$$

Let $\zeta_1,\dots,\zeta_{\dim V-2} \in \mathcal{O}_{Y,\xi}$ be elements whose residues form a transcendence basis of $\kappa(F)$ over $\Bbbk$, and let ω_V^0 be a generator of ω_V near the generic point of $\phi(\overline{\{\xi\}}) = \operatorname{center}_T F$. Let $\alpha := \operatorname{mult}_{E_1} g$

and $\gamma := a(E_1, V, 0) - 1$ if E_1 exists, and $\alpha = \gamma = 0$ otherwise. Since $\operatorname{div}(\phi^* g) = T_Y + \alpha E_1$ and $K_Y = \phi^* K_V + \gamma E_1$ near ξ, up to units of $\mathcal{O}_{Y,\xi}$,

$$\phi^* g = \pi^\alpha q \quad \text{and} \quad \phi^* \omega_V^0 = \pi^\gamma d\pi \wedge dq \wedge d\zeta_1 \wedge \cdots \wedge d\zeta_{\dim V - 2}. \tag{5.9}$$

By adjunction, ω_T is generated by $\operatorname{Res}_T(\omega_V^0/g)$, and by (5.9), up to a unit,

$$\phi^* \operatorname{Res}_T(\omega_V^0/g) = \pi^{\gamma-\alpha} d\pi \wedge d\zeta_1 \wedge \cdots \wedge d\zeta_{\dim V - 2} \quad \text{on } T_Y \text{ near } \xi.$$

Thus (5.5) implies

$$a(F, T, 0) = 1 + \gamma - \alpha.$$

By Lemma 5.3(3), $\psi = \lambda^{N_0}$ for an integer N_0. Let $k_0 := \alpha + N_0$. Let $k \equiv k_0 \pmod{e_F}$ be an integer with $N := k - \alpha > \max_i \operatorname{mult}_F(H_i|_T)$. Then $N \geq 1$ and $N \equiv N_0 \pmod{e_F}$. Let v be the monomial valuation on $\mathcal{O}_{Y,\xi}$ with $v(\pi) = 1$ and $v(q) = N$, that is,

$$v(\varphi) := \max\{N' \in \mathbb{Z} \mid \varphi \in (\pi^{i_1} q^{i_2} \mid i_1 + N i_2 \geq N')\}.$$

Then v has value group $\mathbb{Z}$ and residue field $\kappa(F)(\overline{q'})$, where $q' := q/\pi^N$ and $\overline{q'}$ is transcendental over $\kappa(F)$. Thus $v = \operatorname{mult}_{\widehat{F}_k}$ for a prime divisor $\widehat{F}_k$ over Y, hence over V. Since $v \geq 0$ on $\mathcal{O}_{Y,\xi}$ and $v > 0$ on $\mathfrak{m}_\xi$, $\operatorname{center}_Y \widehat{F}_k = \overline{\{\xi\}}$, so $\operatorname{center}_V \widehat{F}_k = \operatorname{center}_T F$. By (5.9), $\operatorname{mult}_{\widehat{F}_k} g = \alpha + N = k$. For any i, $\phi^* H_i - u_i \pi^{\operatorname{mult}_F(H_i|_T)} \in (q)$ for a unit $u_i \in \mathcal{O}_{Y,\xi}$, and $v(q) = N > \operatorname{mult}_F(H_i|_T)$, so $\operatorname{mult}_{\widehat{F}_k} H_i = \operatorname{mult}_F(H_i|_T)$. Since $dq = \pi^N dq' + N\pi^{N-1} q' d\pi$, (5.9) implies that, up to a unit,

$$\phi^* \omega_V^0 = \pi^{\gamma+N} d\pi \wedge dq' \wedge d\zeta_1 \wedge \cdots \wedge d\zeta_{\dim V - 2},$$

and the residues of $\zeta_1, \ldots, \zeta_{\dim V - 2}, q'$ form a transcendence basis of $\kappa(\widehat{F}_k)$. By (5.5),

$$a(\widehat{F}_k, V, 0) = 1 + \gamma + N = a(F, T, 0) + \alpha + N = a(F, T, 0) + k.$$

For $\sigma \in I$, σ multiplies each monomial $\pi^{i_1} q^{i_2}$ by a root of unity, so $v \circ \sigma = v$. Moreover, $\sigma(q') = \psi(\sigma)\lambda(\sigma)^{-N} q' = q'$ and σ acts trivially on $\kappa(F)$, so σ acts trivially on $\kappa(\widehat{F}_k)$. Thus $I \subset I_G(\widehat{F}_k)$. Conversely, any $\sigma \in I_G(\widehat{F}_k)$ preserves v, hence preserves $\operatorname{center}_Y \widehat{F}_k = F$ and T_Y, so $\sigma \in \operatorname{Stab}_G(F)$. Since σ acts trivially on $\kappa(F) \subset \kappa(\widehat{F}_k)$, $\sigma \in I$. Thus $I_G(\widehat{F}_k) = I_G(F)$.

Now assume that $V = \mathbb{A}^4_{\Bbbk}$, $g = xy + g_0(z,t)$, and G acts diagonally. Since the ideal (g) is G-stable and the units of $\Bbbk[x,y,z,t]$ are constants, g is G-semi-invariant with a character χ_g. Since T is normal, $g_0 \neq 0$, so $x|_T \neq 0$ and $y|_T = -g_0/x$. Let $p : \mathbb{A}^4_{\Bbbk} \to \mathbb{A}^3_{\Bbbk}$ be the projection $(x,y,z,t) \to (x,z,t)$. Then the morphisms

$$p|_T : T \to \mathbb{A}^3_{\Bbbk} \quad \text{and} \quad (p, g) : \mathbb{A}^4_{\Bbbk} \to \mathbb{A}^3_{\Bbbk} \times \mathbb{A}^1_{\Bbbk}$$

are birational.

Let λ be the character and π the uniformizer of Lemma 5.3(3) for $E = F$, and let k_0 be an integer with $\chi_g|_I = \lambda^{k_0}$. Let $k \equiv k_0 \pmod{e_F}$ be an integer with $k > \operatorname{mult}_F x + \max_i \operatorname{mult}_F(H_i|_T)$, and let $v := (\operatorname{mult}_F)_k$ be the Gauss extension of mult_F with value k on $s := g$ (Definition 2.3), where the function field of T is identified through $p|_T$ with the rational functions in x, z, t, and the function field of V is identified through (p, g) with the rational functions in x, z, t, g. By Lemma 5.3(4), $v = \operatorname{mult}_{\widehat{F}_k}$ for a prime divisor $\widehat{F}_k$ over V, and $v(\varphi) = \operatorname{mult}_F(\varphi|_T)$ for any polynomial $\varphi \in \Bbbk[x,z,t]$. For $Q \in \Bbbk[x,y,z,t]$, by Taylor expansion in $y = y|_T + g/x$, $Q = \sum_{i' \geq 0} b_{i'} g^{i'}$ with the rational functions $b_{i'}$ in x, z, t given by

$$b_{i'} = \frac{(\partial^{i'} Q/\partial y^{i'})|_T}{i'! x^{i'}}, \quad b_0 = Q|_T.$$

Since $(\partial^{i'} Q/\partial y^{i'})|_T$ is regular on T, for $i' \geq 1$,

$$\operatorname{mult}_F b_{i'} + i'k \geq i'(k - \operatorname{mult}_F x) > \max_i \operatorname{mult}_F(H_i|_T) \geq 0.$$

Thus $v(Q) \geq 0$, $v(Q) > 0$ if and only if $\mathrm{mult}_F(Q|_T) > 0$, and $v(H_i) = \mathrm{mult}_F(H_i|_T)$ for any i. The first two properties imply $\mathrm{center}_V \widehat{F}_k = \mathrm{center}_T F$, and $v(g) = k$ by the definition of v. Since $\partial g/\partial y = x$, ω_T is generated by

$$\omega_T^0 := \mathrm{Res}_T(dx \wedge dy \wedge dz \wedge dt/g) = dx \wedge dz \wedge dt/x,$$

and

$$dg \wedge \omega_T^0 = -dx \wedge dy \wedge dz \wedge dt.$$

By (5.8) with $\omega = \omega_T^0$ and $s = g$, and (5.5), $a(\widehat{F}_k, V, 0) = a(F, T, 0) + k$. Since G acts diagonally, G preserves $\Bbbk[x, z, t]$ and $\sigma(g) = \chi_g(\sigma)g$. By Lemma 5.3(4), I preserves v and acts trivially on $\kappa(\widehat{F}_k) = \kappa(F)(\overline{w})$, so $I \subset I_G(\widehat{F}_k)$. Conversely, any $\sigma \in I_G(\widehat{F}_k)$ preserves v and $\Bbbk[x, z, t]$, so $\sigma \in \mathrm{Stab}_G(F)$, and σ acts trivially on $\kappa(F) \subset \kappa(\widehat{F}_k)$, so $\sigma \in I$. This proves (1).

Step 2: proof of (2). Possibly reordering coordinates, we may assume that $x = x_1$. Let $y_i := x_{i+1}$ for $i \in \{1, 2, 3\}$, and $\mathcal{O}_W := \mathcal{O}_{\mathcal{X},0}/(f)$, which is a normal local domain. Write $f(X) := f(X, y_1, y_2, y_3)$.

If G is given, G acts on the 1-dimensional $\mathbb{C}$-vector space $(f)/\mathfrak{m}_0(f)$ by a character χ, where $\mathfrak{m}_0$ is the maximal ideal of $\mathcal{O}_{\mathcal{X},0}$, and the element

$$f_1 := |G|^{-1} \sum_{\sigma \in G} \chi(\sigma)^{-1} \sigma(f)$$

is G-semi-invariant with $f_1 = uf$ for a unit u by Nakayama's lemma. Replacing f with f_1 changes neither W nor Jac_W, and $\partial f_1/\partial x|_W = u\partial f/\partial x|_W$. For a prime divisor P over $\mathcal{X}$ with center 0 and $\mathrm{mult}_P f_1 = k > j$, we have $\mathrm{mult}_P f = k$, and $\mathrm{mult}_P(\partial f_1/\partial x) = j$ implies $\mathrm{mult}_P(\partial f/\partial x) = j$ since

$$\partial f_1/\partial x - u\partial f/\partial x = f\partial u/\partial x.$$

Thus we may assume that f is G-semi-invariant with character χ when G is given.

Let $f_x := \partial f/\partial x$. Let $v := (\mathrm{mult}_F)_k$ be the Gauss extension of mult_F with value k on an indeterminate s over $\mathrm{Frac}(\mathcal{O}_W)$ (Definition 2.3), $\widehat{L}$ the completion of $(\mathrm{Frac}(\mathcal{O}_W)(s), v)$, and $\widehat{\mathcal{O}}_v$ the valuation ring of $\widehat{L}$, with maximal ideal $\widehat{\mathfrak{m}}_v$. We write v also for the valuation of $\widehat{L}$. Identify the coordinates with their images in $\widehat{\mathfrak{m}}_F$, viewed in $\widehat{\mathcal{O}}_v$ through $\widehat{\mathcal{O}}_F \subset \widehat{\mathcal{O}}_v$. Then $f(x) = 0$, $v(f_x(x)) = j$, and $v(f(x) - s) = k > 2j$. By Lemma 5.3(2), there exists a unique $\widetilde{x} \in \widehat{\mathcal{O}}_v$ with

$$f(\widetilde{x}) = s \quad \text{and} \quad v(\widetilde{x} - x) > j,$$

and $v(\widetilde{x} - x) = k - j$. By (5.6) and $k - j > j$, $k - j > \mathrm{mult}_F(H_i|_W)$,

$$v(f_x(\widetilde{x})) = j \quad \text{and} \quad v(H_i(\widetilde{x}, y_1, y_2, y_3)) = \mathrm{mult}_F(H_i|_W) \text{ for any } i. \tag{5.10}$$

Let ι be the homomorphism from $\mathcal{O}_{\mathcal{X},0}$ to $\widehat{\mathcal{O}}_v$ sending φ to $\varphi(\widetilde{x}, y_1, y_2, y_3)$, and $L' \subset \widehat{L}$ the closure of the fraction field of $\iota(\mathcal{O}_{\mathcal{X},0})$. Then $\widetilde{x}, y_1, y_2, y_3 \in L'$, $s = \iota(f) \in L'$, and

$$v(f(\widetilde{x}) - 0) = k > 2j = 2v(f_x(\widetilde{x})).$$

By Lemma 5.3(2), the unique root X of $f(X) = 0$ with $v(X - \widetilde{x}) > j$ belongs to L'. Since $v(x - \widetilde{x}) = k - j > j$, $X = x$. Thus L' contains the image of $\mathcal{O}_W$ and s, hence $\mathrm{Frac}(\mathcal{O}_W)(s)$, which is dense in $\widehat{L}$. So $L' = \widehat{L}$. By Lemma 5.3(4), the residue field $\kappa(F)(\overline{w})$ of $\widehat{L}$ has transcendence degree 3 over $\mathbb{C}$. Since $\widetilde{x}, y_1, y_2, y_3 \in \widehat{\mathfrak{m}}_v$, Lemma 5.3(5) for ι and $r = 3$ gives $\dim \mathcal{O}_{\mathcal{X},0}/\operatorname{Ker}\iota \geq 4$, so $\operatorname{Ker}\iota = 0$, and $v \circ \iota = \mathrm{mult}_P$ for a prime divisor P over $\mathcal{X}$ with center 0, with completion $\widehat{L}$ and residue field $\kappa(P) = \kappa(F)(\overline{w})$. We have $\mathrm{mult}_P f = v(s) = k$, and (5.10) implies $\mathrm{mult}_P f_x = j$ and $\mathrm{mult}_P H_i = \mathrm{mult}_F(H_i|_W)$ for any i.

Let

$$\omega_{\mathcal{X}}^0 := dx \wedge dy_1 \wedge dy_2 \wedge dy_3 \quad \text{and} \quad \omega_W^0 := \mathrm{Res}_W(\omega_{\mathcal{X}}^0/f) = dy_1 \wedge dy_2 \wedge dy_3/f_x|_W,$$

which generate $\omega_{\mathcal{X}}$ and ω_W. Differentiating $f(\widetilde{x}) = s$, we obtain

$$f_x(\widetilde{x})d\widetilde{x} + \sum_i (\partial f/\partial y_i)(\widetilde{x})dy_i = ds,$$

so

$$\iota(\omega^0_{\mathcal{X}}) = d\widetilde{x} \wedge dy_1 \wedge dy_2 \wedge dy_3 = \pm\frac{f_x(x)}{f_x(\widetilde{x})}\omega^0_W \wedge ds.$$

By (5.10), $f_x(x)/f_x(\widetilde{x})$ is a unit of $\widehat{\mathcal{O}}_v$. By (5.8) for $E = F$ and (5.5) for $E = P$,

$$a(P, \mathcal{X}, 0) = a(F, W, 0) + k.$$

Assume that G is given, so f is G-semi-invariant with character χ. Let λ be the character and π the uniformizer of Lemma 5.3(3) for $E = F$, let k_0 be an integer with $\chi|_{I_G(F)} = \lambda^{k_0}$, and assume that $k \equiv k_0 \pmod{e_F}$. Let G act on $\mathrm{Frac}(\mathcal{O}_W)(s)$ by $\sigma(s) = \chi(\sigma)s$. For $\sigma \in G$, write $\sigma(x_i) = \chi_{x_i}(\sigma)x_i$. By Lemma 5.3(4), any $\sigma \in \mathrm{Stab}_G(F)$ preserves v and extends to $\widehat{L}$. Since

$$f(\chi_{x_1}(\sigma)X_1, \dots, \chi_{x_4}(\sigma)X_4) = \chi(\sigma)f(X_1, \dots, X_4),$$

the element $\chi_x(\sigma)^{-1}\sigma(\widetilde{x})$ is a root X of $f(X) = s$ with $v(X - x) > j$, so $\sigma(\widetilde{x}) = \chi_x(\sigma)\widetilde{x}$ by the uniqueness in Lemma 5.3(2). Thus $\iota \circ \sigma = \sigma \circ \iota$ and $\mathrm{mult}_P \circ \sigma = \mathrm{mult}_P$. Conversely, any $\sigma \in \mathrm{Stab}_G(P)$ extends to the completion $\widehat{L}$ of P, and the uniqueness of the root x of $f(X) = 0$ with $v(X - \widetilde{x}) > j$ implies $\sigma(x) = \chi_x(\sigma)x$. So σ preserves the image of $\mathcal{O}_W$ and the restriction mult_F of v, that is, $\sigma \in \mathrm{Stab}_G(F)$. Thus $\mathrm{Stab}_G(P) = \mathrm{Stab}_G(F)$. For $\sigma \in \mathrm{Stab}_G(F)$, the extension of σ to $\widehat{L}$ through v and the extension of σ to $\widehat{L}$ through mult_P coincide on the fraction field of $\iota(\mathcal{O}_{\mathcal{X},0})$, which is dense in $\widehat{L}$, hence define one action on $\kappa(P) = \kappa(F)(\overline{w})$. By Lemma 5.3(4), $I_G(F)$ acts trivially on $\kappa(F)(\overline{w})$, so $I_G(F) \subset I_G(P)$. Conversely, any element of $I_G(P)$ acts trivially on $\kappa(F) \subset \kappa(P)$, so $I_G(P) \subset I_G(F)$. This proves (2).

Step 3: proof of (3). Let $v := \mathrm{mult}_F$ on $\widehat{\mathcal{O}}_F$, and write $\varphi(X) := \varphi(X, y_1, y_2, y_3)$ for $\varphi \in \mathbb{C}[[x_1, \dots, x_4]]$ and $X \in \widehat{\mathfrak{m}}_F$. Then

$$f(x) = 0, \quad v(f_x(x)) = j, \quad v(f'(x)) = v((f' - f)(x)) = c, \quad \text{and} \quad v(f'_x(x)) = j,$$

where the last equality holds since $v((f' - f)_x(x)) > j$. By Lemma 5.3(2) and $c > 2j$, there exists a unique $x' \in \widehat{\mathfrak{m}}_F$ with $f'(x') = 0$ and $v(x' - x) > j$, and $v(x' - x) = c - j$ by (5.7). Any root of $f'(X) = 0$ in $\widehat{\mathcal{O}}_F$ with $v(X - x) > j$ belongs to $\widehat{\mathfrak{m}}_F$, so this proves (a). By (5.6) and $c - j > j$, for any $\psi \in \mathbb{C}[[x_1, \dots, x_4]]$,

$$v(\psi(x') - \psi(x)) \geq c - j > j, \tag{5.11}$$

so $v(f_x(x')) = v(f'_x(x')) = j$, and, since $2(c - j) > c$, $f(x') = (x' - x)f_x(x)$ up to a term of value at least $2(c - j)$ implies $v(f(x')) = c$.

Let ι' be the homomorphism from $\mathbb{C}[[x_1, \dots, x_4]]$ to $\widehat{\mathcal{O}}_F$ sending x to x' and y_i to y_i, and L'' the closure of the fraction field of $\iota'(\mathbb{C}[[x_1, \dots, x_4]])$ in $\mathrm{Frac}(\widehat{\mathcal{O}}_F)$. Then $\iota'(f') = 0$. Since $v(f(x')) = c > 2j = 2v(f_x(x'))$, by Lemma 5.3(2), x is the unique root X of $f(X) = 0$ with $v(X - x') > j$, and $x \in L''$. Thus L'' contains the image of $\mathbb{C}[[x_1, \dots, x_4]]/(f)$, whose fraction field is dense in $\mathrm{Frac}(\widehat{\mathcal{O}}_F)$, so $L'' = \mathrm{Frac}(\widehat{\mathcal{O}}_F)$. Since $x', y_1, y_2, y_3 \in \widehat{\mathfrak{m}}_F$ and $\kappa(F)$ has transcendence degree 2 over $\mathbb{C}$, Lemma 5.3(5) for ι' and $r = 2$ gives $\dim \mathbb{C}[[x_1, \dots, x_4]]/\operatorname{Ker}\iota' \geq 3$. Since $(f') \subset \operatorname{Ker}\iota'$ and $\mathbb{C}[[x_1, \dots, x_4]]/(f')$ is a normal domain of dimension 3, $\operatorname{Ker}\iota' = (f')$, equality holds, and $v \circ \iota' = \mathrm{mult}_{F'}$ for a prime divisor F' over W' with center the closed point, $\widehat{\mathcal{O}}_{F'} = \widehat{\mathcal{O}}_F$ and $\kappa(F') = \kappa(F)$.

The forms $\omega^0_W := dy_1 \wedge dy_2 \wedge dy_3/f_x|_W$ and $\omega^0_{W'} := dy_1 \wedge dy_2 \wedge dy_3/f'_x|_{W'}$ generate ω_W and $\omega_{W'}$, and their images in the completed module of differentials of $\widehat{\mathcal{O}}_F$ differ by the factor $f_x(x)/f'_x(x')$, which is a unit since $v(f_x(x)) = v(f'_x(x')) = j$. By (5.5), $a(F', W', 0) = a(F, W, 0)$.

Let $\sigma \in G$, and write $\sigma(x) = \chi_x(\sigma)x$. Since σ preserves W and $f' - f$ is G-semi-invariant, $\sigma(F)$ satisfies the assumptions of (3) with $\mathrm{mult}_{\sigma(F)} \mathrm{Jac}_W = j$ and $\mathrm{mult}_{\sigma(F)}((f' - f)|_W) = c$. Since x is an eigenparameter and f, f' are G-semi-invariant with a common character, the uniqueness in (a)

implies that the isomorphism $\widehat{\mathcal{O}}_F \to \widehat{\mathcal{O}}_{\sigma(F)}$ induced by σ intertwines the homomorphisms ι' for F and for $\sigma(F)$ with the action of σ on $\mathbb{C}[[x_1,\dots,x_4]]$. Thus $\sigma(F)' = \sigma(F')$, and $\sigma \in \mathrm{Stab}_G(F)$ implies $\sigma \in \mathrm{Stab}_G(F')$. Conversely, F is defined from F' by the unique root x of $f(X) = 0$ with $v(X - x') > j$, so the uniqueness of x as such a root shows that $\sigma \in \mathrm{Stab}_G(F')$ implies $\sigma \in \mathrm{Stab}_G(F)$. For $\sigma \in \mathrm{Stab}_G(F) = \mathrm{Stab}_G(F')$, the uniqueness in (a) implies $\sigma(x') = \chi_x(\sigma)x'$. Thus $\iota' \circ \sigma = \sigma \circ \iota'$, so the actions of σ on $\mathrm{Frac}(\widehat{\mathcal{O}}_F)$ as the completion of F and of F' coincide on the fraction field of $\iota'(\mathbb{C}[[x_1,\dots,x_4]])$, which is dense in $\mathrm{Frac}(\widehat{\mathcal{O}}_F)$, hence on $\mathrm{Frac}(\widehat{\mathcal{O}}_F)$ and on $\kappa(F) = \kappa(F')$. Thus $I_G(F') = I_G(F)$. This proves (b).

For (c), let $i \in \{1,\dots,4\}$ and $\partial_i := \partial/\partial x_i$. Since $v(\partial_i f(x)) \geq j$, (5.11) implies $v(\partial_i f(x')) \geq j$, and the assumption with (5.11) implies $v(\partial_i(f' - f)(x')) > j$. Thus $v(\partial_i f'(x')) \geq j$ for any i, and $v(f'_x(x')) = j$, so $\mathrm{mult}_{F'} \mathrm{Jac}_{W'} = j$. For (d), (5.6) implies

$$v(H_i(x') - H_i(x)) \geq c - j > \mathrm{mult}_F(H_i|_W) = v(H_i(x)),$$

so $\mathrm{mult}_{F'}(H_i|_{W'}) = v(H_i(x')) = \mathrm{mult}_F(H_i|_W)$. □

Lemma 5.5. *Let $\mathcal{X}$ be $\mathbb{A}^4$ or $\mathrm{Spec}\,\mathbb{C}[[x_1,\dots,x_4]]$, with closed point 0, let $f, h \in \mathcal{O}_{\mathcal{X},0}$ with $W := (f = 0)$ normal near 0 and $h|_W \neq 0$, let F be a prime divisor over W with center 0 and $j := \mathrm{mult}_F \mathrm{Jac}_W$, and let $\theta \in (0,1)$ and $\mu > 0$ be real numbers. Then the following hold.*

(1) Let $g \in \mathcal{O}_{\mathcal{X},0}$ with $g|_W \neq 0$, and let ρ be a positive integer or $+\infty$. Assume that any prime divisor P over $\mathcal{X}$ with center 0 and $\mathrm{mult}_P g < \rho$ satisfies

$$\mathrm{mult}_P g + a(P,\mathcal{X},0) > \theta\,\mathrm{mult}_P f + \theta\mu\,\mathrm{mult}_P h,$$

and that $\mathrm{mult}_F(g|_W) < \rho$. Then

$$(\theta\mu - 1 + \theta)\,\mathrm{mult}_F(h|_W) < (2-\theta)\,\mathrm{mult}_F(g|_W) + a(F,W,0) + 2(1-\theta)j + 1 - \theta.$$

(2) If the multiplier ideal $\mathcal{J} := \mathcal{J}(f^\theta h^{\theta\mu})$ [Laz04, Definition 9.2.3 and Generalization 9.2.8], [JM12, Section 1.3] satisfies $\mathcal{J}\mathcal{O}_W \neq 0$ near 0, then

$$(\theta\mu - 1 + \theta)\,\mathrm{mult}_F(h|_W) < (2-\theta)\,\mathrm{mult}_F(\mathcal{J}\mathcal{O}_W) + a(F,W,0) + 2(1-\theta)j + 1 - \theta.$$

Proof. Let x be one of the coordinates $x_1,\dots,x_4$ of $\mathcal{X}$ with $\mathrm{mult}_F(\partial f/\partial x|_W) = j$, and

$$k := \mathrm{mult}_F(h|_W) + \mathrm{mult}_F(g|_W) + 2j + 1.$$

Then $k > 2j$ and $k - j > \max\{\mathrm{mult}_F(h|_W), \mathrm{mult}_F(g|_W)\}$. By Lemma 5.4(2) with $H_1 = h$ and $H_2 = g$, there exists a prime divisor P over $\mathcal{X}$ with center 0 satisfying

$$\mathrm{mult}_P f = k, \quad \mathrm{mult}_P h = \mathrm{mult}_F(h|_W), \quad \mathrm{mult}_P g = \mathrm{mult}_F(g|_W), \quad \text{and} \quad a(P,\mathcal{X},0) = a(F,W,0)+k.$$

Since $\mathrm{mult}_P g = \mathrm{mult}_F(g|_W) < \rho$, the assumption on g gives

$$\mathrm{mult}_F(g|_W) + a(F,W,0) + k > \theta k + \theta\mu\,\mathrm{mult}_F(h|_W).$$

Substituting the value of k, we obtain

$$\theta\mu\,\mathrm{mult}_F(h|_W) < (2-\theta)\,\mathrm{mult}_F(g|_W) + a(F,W,0) + (1-\theta)\,\mathrm{mult}_F(h|_W) + 2(1-\theta)j + 1 - \theta,$$

which is (1).

We prove (2). Let $g \in \mathcal{J}$, and let P be a prime divisor over $\mathcal{X}$ with center 0. If $\mathcal{X} = \mathbb{A}^4$, possibly shrinking $\mathcal{X}$ to a neighborhood of 0, we may assume that f, h, g are regular on $\mathcal{X}$. Let $\varphi : Y \to \mathcal{X}$ be a log resolution of $(\mathcal{X}, \mathrm{div}(fh))$ on which P is a divisor. By [Laz04, Definition 9.2.1, Theorem 9.2.18 and Remark 9.2.20] if $\mathcal{X} = \mathbb{A}^4$, and by [JM12, Section 1.3] if $\mathcal{X}$ is formal,

$$\mathcal{J} = \varphi_*\mathcal{O}_Y(K_{Y/\mathcal{X}} - \lfloor \theta\varphi^*\,\mathrm{div}(f) + \theta\mu\varphi^*\,\mathrm{div}(h)\rfloor),$$

so

$$\begin{aligned}\mathrm{mult}_P g &\geq \lfloor \theta\,\mathrm{mult}_P f + \theta\mu\,\mathrm{mult}_P h\rfloor - (a(P,\mathcal{X},0) - 1)\\ &> \theta\,\mathrm{mult}_P f + \theta\mu\,\mathrm{mult}_P h - a(P,\mathcal{X},0).\end{aligned}$$

Thus g satisfies the assumption of (1) with $\rho = +\infty$. If $\mathcal{J}\mathcal{O}_W \neq 0$ near 0, then, since $\mathcal{J}$ is finitely generated, some $g_0 \in \mathcal{J}$ satisfies $\operatorname{mult}_F(g_0|_W) = \operatorname{mult}_F(\mathcal{J}\mathcal{O}_W)$ and $g_0|_W \neq 0$, and (1) for $g = g_0$ and $\rho = +\infty$ gives (2). □

5.2. **Split germs and surface choices.** We bound the Jacobian order of a prime divisor over a split hypersurface germ $(xy + g(z,t) = 0)$ by the log discrepancy (Lemma 5.6), and choose computing lifts with bounded invariants on surface quotients (Lemma 5.7).

Lemma 5.6. *Let $(W \ni o)$ be a normal hypersurface threefold germ, and let $k \geq 2$ be an integer. Assume that*

$$\widehat{\mathcal{O}}_{W,o} \cong \mathbb{C}[[x,y,z,t]]/(xy + g(z,t))$$

for some $g \in (z,t)^k \backslash (z,t)^{k+1}$. For a prime divisor F over W with $\operatorname{center}_W F = \{o\}$ and for $\phi \in \widehat{\mathcal{O}}_{W,o}$, let $\operatorname{mult}_F \phi$ be the value at ϕ of the continuous extension of mult_F to $\widehat{\mathcal{O}}_{W,o}$. Then, after a linear change of the coordinates z,t depending only on g, any prime divisor F over W with $\operatorname{center}_W F = \{o\}$ satisfies

$$\operatorname{mult}_F x + \operatorname{mult}_F y = \operatorname{mult}_F g \leq k\big(a(F,W,0) - \operatorname{mult}_F t\big) \tag{5.12}$$

and

$$\operatorname{mult}_F \operatorname{Jac}_W \leq \min\{\operatorname{mult}_F x, \operatorname{mult}_F y\} \leq \frac{k}{2}\big(a(F,W,0) - \operatorname{mult}_F t\big) \leq \frac{k}{2}\big(a(F,W,0) - 1\big). \tag{5.13}$$

Proof. Let $f := xy + g$, so that $\widehat{\mathcal{O}}_{W,o} = \mathbb{C}[[x,y,z,t]]/(f)$. By Lemma 2.12(3), F corresponds to a formal prime divisor over $\operatorname{Spec} \widehat{\mathcal{O}}_{W,o}$ with center the closed point, with isomorphic residue field $\kappa(F)$ and equal log discrepancy $a(F,W,0)$. We denote the formal prime divisor again by F. The valuation mult_F of the formal prime divisor restricts to mult_F on $\mathcal{O}_{W,o}$ and satisfies $\operatorname{mult}_F(\mathfrak{m}_o^N \widehat{\mathcal{O}}_{W,o}) \geq N$ for any $N \in \mathbb{N}$, so on $\widehat{\mathcal{O}}_{W,o}$ it is the continuous extension of the statement, and it is finite on non-zero elements.

Step 1. Let $x_1, \dots, x_4$ be regular parameters at o of a smooth fourfold germ $(\mathcal{X} \ni o)$ containing W. Since $f \in (x,y,z,t)^2$, lifting their images in $\widehat{\mathcal{O}}_{W,o}$ to $\mathbb{C}[[x,y,z,t]]$ defines an isomorphism

$$\mathbb{C}[[x_1, \dots, x_4]] \to \mathbb{C}[[x,y,z,t]]$$

compatible with the surjections onto $\widehat{\mathcal{O}}_{W,o}$ and mapping a local equation of W to a unit multiple of f. Hence, by Definition 5.2,

$$\operatorname{Jac}_W \widehat{\mathcal{O}}_{W,o} = (\partial_x f, \partial_y f, \partial_z f, \partial_t f) = (x, y, \partial_z g, \partial_t g),$$

so

$$\operatorname{mult}_F \operatorname{Jac}_W = \min\{\operatorname{mult}_F x, \operatorname{mult}_F y, \operatorname{mult}_F \partial_z g, \operatorname{mult}_F \partial_t g\} \leq \min\{\operatorname{mult}_F x, \operatorname{mult}_F y\}.$$

Step 2. We show that for any $\phi \in \mathbb{C}[[z,t]]$ with $\partial_z \phi \neq 0$ in $\widehat{\mathcal{O}}_{W,o}$, we have $\phi \neq 0$ in $\widehat{\mathcal{O}}_{W,o}$ and

$$\operatorname{mult}_F \phi \leq \operatorname{mult}_F \partial_z \phi + a(F,W,0) - \operatorname{mult}_F t. \tag{5.14}$$

Let $\widehat{\mathcal{O}}_F$ be the completion of the valuation ring of mult_F in the fraction field of $\widehat{\mathcal{O}}_{W,o}$, let π be a uniformizer of $\widehat{\mathcal{O}}_F$, and let $\zeta_1, \zeta_2 \in \widehat{\mathcal{O}}_F$ be elements whose residues form a transcendence basis of $\kappa(F)$ over $\mathbb{C}$. The completion $\widehat{\Omega}_F$ of $\Omega_{\widehat{\mathcal{O}}_F/\mathbb{C}}$ with respect to the maximal ideal of $\widehat{\mathcal{O}}_F$ is free with basis $d\pi, d\zeta_1, d\zeta_2$ (see Lemma 5.3(1) for $Z = \operatorname{Spec} \widehat{\mathcal{O}}_{W,o}$), and for a non-zero $\eta \in \bigwedge^3 \widehat{\Omega}_F[\pi^{-1}]$ we write $\operatorname{mult}_F \eta$ for mult_F of the coefficient of η with respect to $d\pi \wedge d\zeta_1 \wedge d\zeta_2$. Since d commutes with convergent power series, $d\phi = \partial_z \phi dz + \partial_t \phi dt$ in $\widehat{\Omega}_F$ for any $\phi \in \mathbb{C}[[z,t]]$. The form

$$\omega := \frac{dx \wedge dz \wedge dt}{x} = \frac{dx \wedge dz \wedge dt}{\partial_y f}$$

generates the canonical module of $\widehat{\mathcal{O}}_{W,o}$, so by Lemma 5.3(1) for $Z = \operatorname{Spec} \widehat{\mathcal{O}}_{W,o}$, up to a unit of $\widehat{\mathcal{O}}_F$ we have

$$\omega = \pi^{a(F,W,0)-1} d\pi \wedge d\zeta_1 \wedge d\zeta_2,$$

that is, $\operatorname{mult}_F \omega = a(F,W,0) - 1$.

Let $\phi \in \mathbb{C}[[z,t]]$ with $\partial_z \phi \neq 0$ in $\widehat{\mathcal{O}}_{W,o}$. Since $d\phi = \partial_z \phi dz + \partial_t \phi dt$ and $dt \wedge dt = 0$,

$$\frac{dx}{x} \wedge d\phi \wedge dt = \partial_z \phi \cdot \omega, \tag{5.15}$$

whose right-hand side is non-zero and satisfies

$$\operatorname{mult}_F(\partial_z \phi \cdot \omega) = \operatorname{mult}_F \partial_z \phi + a(F,W,0) - 1.$$

Thus $d\phi \neq 0$, so $\phi \neq 0$ in $\widehat{\mathcal{O}}_{W,o}$. In $\widehat{\mathcal{O}}_F$ write

$$x = \pi^{\operatorname{mult}_F x} u_x, \quad \phi = \pi^{\operatorname{mult}_F \phi} u_\phi, \quad \text{and} \quad t = \pi^{\operatorname{mult}_F t} u_t$$

with units $u_x, u_\phi, u_t \in \widehat{\mathcal{O}}_F$. Then

$$\frac{dx}{x} = \operatorname{mult}_F x \cdot \frac{d\pi}{\pi} + \frac{du_x}{u_x}, \qquad d\phi = \pi^{\operatorname{mult}_F \phi}\Big(\operatorname{mult}_F \phi \cdot u_\phi \frac{d\pi}{\pi} + du_\phi \Big), \quad \text{and}$$
$$dt = \pi^{\operatorname{mult}_F t}\Big(\operatorname{mult}_F t \cdot u_t \frac{d\pi}{\pi} + du_t \Big),$$

where $du_x/u_x, du_\phi, du_t \in \widehat{\Omega}_F$. In the expansion of the left-hand side of (5.15), the terms with two factors $d\pi/\pi$ vanish, the term with no factor $d\pi/\pi$ is $\pi^{\operatorname{mult}_F \phi + \operatorname{mult}_F t}$ times an element of $\bigwedge^3 \widehat{\Omega}_F$, and the terms with exactly one factor $d\pi/\pi$ are $\pi^{\operatorname{mult}_F \phi + \operatorname{mult}_F t - 1}$ times elements of $\bigwedge^3 \widehat{\Omega}_F$. Hence

$$\operatorname{mult}_F \partial_z \phi + a(F,W,0) - 1 = \operatorname{mult}_F \Big(\frac{dx}{x} \wedge d\phi \wedge dt \Big) \geq \operatorname{mult}_F \phi + \operatorname{mult}_F t - 1,$$

which is (5.14).

Step 3. Let $g_k \neq 0$ be the homogeneous part of degree k of g, and let $(z_0, t_0) \in \mathbb{C}^2$ with $g_k(z_0, t_0) \neq 0$. Possibly applying a linear change of the coordinates z, t that maps (z_0, t_0) to $(1, 0)$, which depends only on g, we may assume that the coefficient $g_k(1,0)$ of z^k in g is non-zero. Then $\partial_z^k g - k! g_k(1,0) \in (z,t)$, so $\partial_z^k g$ is a unit and $\operatorname{mult}_F \partial_z^k g = 0$. By **Step 2** and descending induction on j, $\partial_z^j g \neq 0$ in $\widehat{\mathcal{O}}_{W,o}$ for any $j \in \{0, \dots, k\}$, and (5.14) for $\phi = \partial_z^j g$ with $j \in \{0, \dots, k-1\}$ implies

$$\operatorname{mult}_F g = \sum_{j=0}^{k-1} \big(\operatorname{mult}_F \partial_z^j g - \operatorname{mult}_F \partial_z^{j+1} g \big) + \operatorname{mult}_F \partial_z^k g \leq k\big(a(F,W,0) - \operatorname{mult}_F t \big).$$

Since $xy = -g$ in $\widehat{\mathcal{O}}_{W,o}$, we have $\operatorname{mult}_F x + \operatorname{mult}_F y = \operatorname{mult}_F g$, which implies (5.12). Since

$$\min\{\operatorname{mult}_F x, \operatorname{mult}_F y\} \leq \frac{1}{2}\big(\operatorname{mult}_F x + \operatorname{mult}_F y \big)$$

and $\operatorname{mult}_F t \geq 1$ as t vanishes at o, (5.12) and **Step 1** imply (5.13). The lemma follows. □

Lemma 5.7. *Let m be a positive integer. Then there exists a positive real number C_m satisfying the following. Let $\Bbbk \supset \mathbb{C}$ be an algebraically closed field, and let $(S \ni o)$ be a Du Val surface germ over $\mathbb{C}$, a smooth surface germ over $\Bbbk$, or $S = \operatorname{Spec} \Bbbk[[x_1, x_2]]$ with closed point o. Let G be a finite group acting faithfully on S, by $\mathbb{C}$-automorphisms if S is singular and by $\Bbbk$-automorphisms otherwise, and fixing o, and let h be a mark on $(S \ni o)$ (Definition 2.4). Assume that $(S, \frac{1}{m}\operatorname{div}(h))$ is lc near o. Then $\operatorname{mld}((S \ni o, \frac{1}{m}\operatorname{div}(h))/G)$ has a computing lift F (Definition 2.4), which is a formal prime divisor over S if $S = \operatorname{Spec} \Bbbk[[x_1, x_2]]$, with*

$$\frac{a(F,S,0)}{e_F} \leq C_m, \qquad \frac{\operatorname{mult}_F \operatorname{Jac}_S}{e_F} \leq C_m, \qquad \text{and} \qquad \frac{\operatorname{mult}_F h}{e_F} \leq m C_m,$$

where Jac_S *is the Jacobian ideal of Definition 5.2 if* S *is singular, and* $\mathrm{Jac}_S = \mathcal{O}_S$ *if* S *is smooth.*

Proof. We first fix C_m. Let

$$\Gamma_m := \Big\{1 - \frac{1 - j/m}{r} \;\Big|\; j \in \{0, 1, \dots, m\},\ r \in \mathbb{N}^+\Big\} \subset [0, 1].$$

Then Γ_m is a DCC set. Let L_m be the positive integer L of Theorem 2.21(2) for $\Gamma = \Gamma_m$. Since L_m depends only on Γ_m, it serves for every algebraically closed field $\Bbbk \supset \mathbb{C}$. Let $(S_1 \ni 0), \dots, (S_5 \ni 0)$ be the Du Val germs in $(\mathbb{A}^3 \ni 0)$ of types A_1, D_4, E_6, E_7, E_8 defined by

$$x^2 + y^2 + z^2, \qquad x^2 + y^2 z + z^3, \qquad x^2 + y^3 + z^4, \qquad x^2 + y^3 + yz^3, \qquad \text{and} \qquad x^2 + y^3 + z^5$$

respectively. Since each S_i is klt and $\mathrm{Jac}_{S_i} \neq 0$, there exists a real number $\epsilon > 0$ such that $(S_i, \mathrm{Jac}_{S_i}^{\epsilon})$ is lc near 0 for any $i \in \{1, \dots, 5\}$. Let

$$C_m := \max\Big\{(2m + 4)L_m + 1, \frac{L_m}{\epsilon}\Big\}.$$

Then C_m depends only on m, and $C_m \geq 2mL_m + 1 \geq 2L_m$.

Now let $\Bbbk$, $(S \ni o)$, G, and h be as in the lemma, and let $D := \frac{1}{m}\operatorname{div}(h)$. If $S = \operatorname{Spec} \Bbbk[[x_1, x_2]]$, the lemma is proved in **Step 6**. Until **Step 6**, $(S \ni o)$ is an algebraic germ over $\Bbbk$, with $\Bbbk$ replaced by $\mathbb{C}$ if S is singular. Possibly replacing S with a G-invariant open neighborhood of o, we may assume that (S, D) is lc. Let $\pi : S \to Q := S/G$ be the quotient morphism, $\overline{o} := \pi(o)$, and Δ the boundary on Q with $K_S + D = \pi^*(K_Q + \Delta)$ (Definition 2.2).

For a prime divisor F over S with $\operatorname{center}_S F = \{o\}$, we also write F for the corresponding formal prime divisor over $\operatorname{Spec} \widehat{\mathcal{O}}_{S,o}$ (Lemma 2.12(3)). If S is singular, then an equation of S and regular parameters of the smooth germ of Definition 5.2 give a presentation $\widehat{\mathcal{O}}_{S,o} \cong \mathbb{C}[[x_1, x_2, x_3]]/(f)$, so $\mathrm{Jac}_S \widehat{\mathcal{O}}_{S,o}$ is the Jacobian ideal of Definition 5.2 for (x_1, x_2, x_3, f), hence for any presentation $\widehat{\mathcal{O}}_{S,o} \cong \mathbb{C}[[x_1, x_2, x_3]]/(f)$.

By Definition 2.2, we have, for any prime divisor F over S with $\operatorname{center}_S F = \{o\}$,

$$a(F, S, D) \geq e_F \operatorname{mld}((S \ni o, D)/G), \tag{5.16}$$

with equality if and only if F is a computing lift (Definition 2.4). Since (S, D) is lc, any prime divisor F over S satisfies

$$\operatorname{mult}_F h = m\big(a(F, S, 0) - a(F, S, D)\big) \leq m a(F, S, 0). \tag{5.17}$$

Step 1. By Lemma 2.15(0), $K_S + D = \pi^*(K_Q + \Delta)$ and $K_Q + \Delta$ is $\mathbb{R}$-Cartier, so (Q, Δ) is lc since π is finite and (S, D) is lc. For any prime divisor P on S, $\operatorname{mult}_P h \in \{0, 1, \dots, m\}$ since h is regular and (S, D) is lc. Thus by Lemma 2.15(4), the coefficient of Δ along $\pi(P)$ is

$$1 - \frac{1 - \operatorname{mult}_P h/m}{r} \in \Gamma_m,$$

where r is the ramification index of π along P. By Theorem 2.21(2), there exists a prime divisor E over Q with $\operatorname{center}_Q E = \{\overline{o}\}$ such that

$$a(E, Q, \Delta) = \operatorname{mld}(Q \ni \overline{o}, \Delta) \qquad \text{and} \qquad a(E, Q, 0) \leq L_m.$$

By Lemma 2.15(3), there exists a prime divisor F_0 over S whose restriction to Q is E. Then $\operatorname{center}_S F_0 = \{o\}$, since $\pi(\operatorname{center}_S F_0) = \operatorname{center}_Q E = \{\overline{o}\}$ by Lemma 2.15(2) and $\pi^{-1}(\overline{o}) = \{o\}$. By Lemma 2.15(0)(2)(4) with $B = 0$, we have $K_S = \pi^*(K_Q + \Delta_0)$ for the boundary $\Delta_0 \geq 0$ on Q with coefficients $1 - 1/r$, and $a(F_0, S, 0) = e_{F_0} a(E, Q, \Delta_0) \leq e_{F_0} a(E, Q, 0)$. Therefore, by Lemma 2.15(2) and the equality $\operatorname{mld}((S \ni o, D)/G) = \operatorname{mld}(Q \ni \overline{o}, \Delta)$ of Lemma 2.15,

$$a(F_0, S, D) = e_{F_0} \operatorname{mld}((S \ni o, D)/G) \qquad \text{and} \qquad a(F_0, S, 0) \leq L_m e_{F_0}, \tag{5.18}$$

so F_0 is a computing lift. If S is smooth, then $\operatorname{mult}_{F_0} \mathrm{Jac}_S = 0$, and F_0 satisfies the lemma by (5.17), (5.18), and $L_m \leq C_m$.

From now on, assume that S is singular. Let $F_\sharp$ be an exceptional prime divisor of the minimal resolution of S. Then $a(F_\sharp, S, 0) = 1$ and $\operatorname{center}_S F_\sharp = \{o\}$. Since (S, D) is lc and h is regular, (5.16) and (5.18) imply

$$0 \le \operatorname{mld}((S \ni o, D)/G) \le \frac{a(F_\sharp, S, D)}{e_{F_\sharp}} \le a(F_\sharp, S, 0) = 1. \tag{5.19}$$

Step 2. Assume that S is of type A_1, D_4, E_6, E_7, or E_8, and let S_i be the germ among $S_1, \dots, S_5$ whose type is the type of S. By the Du Val classification, there exists an isomorphism $\widehat{\mathcal{O}}_{S,o} \cong \widehat{\mathcal{O}}_{S_i,0}$, which carries $\operatorname{Jac}_S \widehat{\mathcal{O}}_{S,o}$ to $\operatorname{Jac}_{S_i} \widehat{\mathcal{O}}_{S_i,0}$ since the Jacobian ideal does not depend on the presentation. By Lemma 2.12(3) on S and on S_i, any prime divisor F over S with $\operatorname{center}_S F = \{o\}$ corresponds to a prime divisor F_i over S_i with $\operatorname{center}_{S_i} F_i = \{0\}$,

$$a(F_i, S_i, 0) = a(F, S, 0), \qquad \text{and} \qquad \operatorname{mult}_{F_i} \operatorname{Jac}_{S_i} = \operatorname{mult}_F \operatorname{Jac}_S .$$

Since $(S_i, \operatorname{Jac}_{S_i}^{\epsilon})$ is lc near 0, we obtain

$$\epsilon \operatorname{mult}_F \operatorname{Jac}_S \le a(F, S, 0).$$

For $F = F_0$, (5.18) implies

$$\frac{a(F_0, S, 0)}{e_{F_0}} \le L_m \le C_m \qquad \text{and} \qquad \frac{\operatorname{mult}_{F_0} \operatorname{Jac}_S}{e_{F_0}} \le \frac{L_m}{\epsilon} \le C_m.$$

Step 3. Let $N \ge 3$ be an integer, R a $\mathbb{C}$-algebra isomorphic to $\widehat{\mathcal{O}}_{T,t}$ for a normal surface germ $(T \ni t)$ and to $\mathbb{C}[[x, y, z]]/(xy - z^N)$, $S_R := \operatorname{Spec} R$ with closed point o_R, and $x, y, z \in R$ any elements inducing an isomorphism $R \cong \mathbb{C}[[x, y, z]]/(xy - z^N)$. Let F be a formal prime divisor over S_R with center o_R. Then $\operatorname{mult}_F x$, $\operatorname{mult}_F y$, and $\operatorname{mult}_F z$ are positive integers with $\operatorname{mult}_F x + \operatorname{mult}_F y = N \operatorname{mult}_F z$. Since $\operatorname{Jac}_{S_R} = (y, x, Nz^{N-1})R$ and

$$\min\{\operatorname{mult}_F x, \operatorname{mult}_F y\} \le \frac{N}{2} \operatorname{mult}_F z \le (N-1) \operatorname{mult}_F z,$$

we have

$$\operatorname{mult}_F \operatorname{Jac}_{S_R} = \min\{\operatorname{mult}_F x, \operatorname{mult}_F y\}. \tag{5.20}$$

The residue form $\omega := dx \wedge dz/x$ generates the canonical module of R, so by Lemma 5.3(1) for $Z = S_R$, the order of ω along F is $a(F, S_R, 0) - 1$. At a general closed point of F on a higher model of S_R, let η be a local equation of F, and write $x = \eta^{\operatorname{mult}_F x} u_x$ and $z = \eta^{\operatorname{mult}_F z} u_z$ with units u_x, u_z. Then

$$\omega = \Big(\operatorname{mult}_F x \frac{d\eta}{\eta} + \frac{du_x}{u_x} \Big) \wedge \eta^{\operatorname{mult}_F z - 1} \big(\operatorname{mult}_F z u_z d\eta + \eta du_z \big)$$

has order at least $\operatorname{mult}_F z - 1$ along F, so

$$a(F, S_R, 0) \ge \operatorname{mult}_F z. \tag{5.21}$$

Claim 5.8. *With m and L_m as in the beginning of the proof, let N, R, S_R, o_R, and $(T \ni t)$ be as at the beginning of **Step 3**, G_R a finite group acting faithfully on R by $\mathbb{C}$-algebra automorphisms, $h_R \in R \setminus \{0\}$ a G_R-semi-invariant element, and $\alpha \in [0, 1]$ a real number, and assume that R is G_R-equivariantly isomorphic to $\widehat{\mathcal{O}}_{T,t}$ for a faithful action of G_R on $(T \ni t)$ fixing t. Let $D_R := \frac{1}{m} \operatorname{div}(h_R)$, and let $e_F := |I_{G_R}(F)|$ for any formal prime divisor F over S_R with center o_R. Assume that the following hold.*

(1) Any formal prime divisor F over S_R with center o_R satisfies $a(F, S_R, D_R) \ge \alpha e_F$.

(2) There exists a formal prime divisor F_0 over S_R with center o_R such that

$$a(F_0, S_R, D_R) = \alpha e_{F_0} \quad \text{and} \quad a(F_0, S_R, 0) \le L_m e_{F_0}.$$

Then there exists a formal prime divisor F' over S_R with center o_R such that

$$a(F', S_R, D_R) = \alpha e_{F'}, \qquad a(F', S_R, 0) \le 2L_m e_{F'}, \qquad \text{and} \qquad \operatorname{mult}_{F'} \operatorname{Jac}_{S_R} \le (2mL_m + 1) e_{F'}.$$

Proof. Let F_0 be as in (2) and $e := e_{F_0}$. By Lemma 2.15(1) for the action on T and Lemma 2.12(3), $I_{G_R}(F_0)$ is cyclic of order e and acts faithfully on $\mathfrak{m}_{F_0}/\mathfrak{m}_{F_0}^2$ through a character $\lambda : I_{G_R}(F_0) \to \mathbb{C}^*$, where $\mathfrak{m}_{F_0}$ is the maximal ideal of the valuation ring of F_0. Let ϖ be a generator of $\mathfrak{m}_{F_0}$. Since $I_{G_R}(F_0)$ acts trivially on the residue field of F_0, any $g \in I_{G_R}(F_0)$ satisfies $g \cdot \varpi \in \lambda(g)\varpi + \mathfrak{m}_{F_0}^2$ and $g \cdot u - u \in \mathfrak{m}_{F_0}$ for any unit u of the valuation ring. Writing $\phi = \varpi^{\mathrm{mult}_{F_0}\phi} u$ with a unit u, we obtain

$$g \cdot \phi = \lambda(g)^{\mathrm{mult}_{F_0}\phi}\phi \qquad \text{for any } g \in I_{G_R}(F_0) \text{ and } \phi \in R\backslash\{0\} \text{ with } g \cdot \phi \in \mathbb{C}^*\phi. \tag{5.22}$$

Similarly, let $g \in I_{G_R}(F_0)$, and let ω' be a generator of the canonical module of R with $g\cdot\omega' \in \mathbb{C}^*\omega'$. Let $\widehat{\mathcal{O}}_{F_0}$ be the completion of the valuation ring of F_0, $\widehat{\mathfrak{m}}_{F_0}$ its maximal ideal, $\widehat{\Omega}_{F_0}$ the $\widehat{\mathfrak{m}}_{F_0}$-adic completion of $\Omega_{\widehat{\mathcal{O}}_{F_0}/\mathbb{C}}$, and $\varpi' \in \widehat{\mathcal{O}}_{F_0}$ an element whose residue is transcendental over $\mathbb{C}$. By Lemma 5.3(1) for $Z = S_R$, $\widehat{\Omega}_{F_0}$ is free with basis $d\varpi, d\varpi'$, and $\omega' = \varpi^{a(F_0,S_R,0)-1}ud\varpi \wedge d\varpi'$ with a unit u of $\widehat{\mathcal{O}}_{F_0}$. Since $g \cdot \varpi' - \varpi' \in \widehat{\mathfrak{m}}_{F_0}$, we have

$$g \cdot \omega' - \lambda(g)^{a(F_0,S_R,0)}\omega' \in \varpi^{a(F_0,S_R,0)} \bigwedge^2 \widehat{\Omega}_{F_0},$$

so

$$g \cdot \omega' = \lambda(g)^{a(F_0,S_R,0)}\omega'. \tag{5.23}$$

Let $\mathfrak{m}_R$ be the maximal ideal of R. Since $R \cong \mathbb{C}[[x,y,z]]/(xy-z^N)$ with $N \geq 3$, $\dim \mathfrak{m}_R/\mathfrak{m}_R^2 = 3$ and the kernel of the multiplication map

$$\mathrm{Sym}^2(\mathfrak{m}_R/\mathfrak{m}_R^2) \to \mathfrak{m}_R^2/\mathfrak{m}_R^3$$

is spanned by $\xi_1\xi_2$ for linearly independent $\xi_1, \xi_2 \in \mathfrak{m}_R/\mathfrak{m}_R^2$, and the pair of lines $\{\mathbb{C}\xi_1, \mathbb{C}\xi_2\}$ is determined by R. Hence $I_{G_R}(F_0)$ permutes $\{\mathbb{C}\xi_1, \mathbb{C}\xi_2\}$. Let $G_1 \subset I_{G_R}(F_0)$ be the stabilizer of $\mathbb{C}\xi_1$, so $[I_{G_R}(F_0) : G_1] \leq 2$. Let $x \in \mathfrak{m}_R$ be a G_1-eigenvector with class in $\mathbb{C}^*\xi_1$. If $G_1 = I_{G_R}(F_0)$, let $y \in \mathfrak{m}_R$ be an $I_{G_R}(F_0)$-eigenvector with class in $\mathbb{C}^*\xi_2$. Otherwise, let g_0 be a generator of $I_{G_R}(F_0)$ and $y := g_0 \cdot x$. Let $z \in \mathfrak{m}_R$ be an $I_{G_R}(F_0)$-eigenvector whose class spans an $I_{G_R}(F_0)$-stable complement of $\mathbb{C}\xi_1 + \mathbb{C}\xi_2$. Since $I_{G_R}(F_0)$ is abelian, any $g \in I_{G_R}(F_0)$ acts by

$$\begin{pmatrix} x \\ y \end{pmatrix} \to \Lambda_g \begin{pmatrix} x \\ y \end{pmatrix}, \qquad z \to \chi_z(g)z$$

with $\Lambda_g \in GL_2(\mathbb{C})$ diagonal if $g \in G_1$ and anti-diagonal otherwise. Let any $g \in I_{G_R}(F_0)$ act on $\mathbb{C}[[x,y,z]]$ by $(x,y)^T \to \Lambda_g(x,y)^T$ and $z \to \chi_z(g)z$. Then the surjection $\mathbb{C}[[x,y,z]] \to R$ is equivariant, and the kernel $\mathfrak{K}$ of the surjection satisfies $\dim \mathfrak{K}/(x,y,z)\mathfrak{K} = 1$, so $\mathfrak{K}$ is generated by a semi-invariant element f with character χ_f. The quadratic part of f is a non-zero multiple of xy, so after scaling, $f - xy \in (x,y,z)^3$.

Since $\partial_x f, \partial_y f, z$ form a regular system of parameters of $\mathbb{C}[[x,y,z]]$, there exist $\varphi_1, \varphi_2 \in z\mathbb{C}[[z]]$ with

$$(\partial_x f, \partial_y f) = (x - \varphi_1(z), y - \varphi_2(z)).$$

The ideal $(\partial_x f, \partial_y f)$ is $I_{G_R}(F_0)$-stable since $g \cdot f = \chi_f(g)f$ and the action is linear on $\mathbb{C}x + \mathbb{C}y$ and on $\mathbb{C}z$. Let $x_1 := x - \varphi_1(z)$, $y_1 := y - \varphi_2(z)$, and $\boldsymbol{x} := (x_1, y_1)^T$. For $g \in I_{G_R}(F_0)$, the entries of $g \cdot \boldsymbol{x} - \Lambda_g\boldsymbol{x}$ belong to $\mathbb{C}[[z]] \cap (x_1, y_1) = 0$, so $g \cdot \boldsymbol{x} = \Lambda_g\boldsymbol{x}$. In the coordinates x_1, y_1, z, we have $\partial_{x_1} f, \partial_{y_1} f \in (x_1, y_1)$, so Taylor's formula with integral remainder in $\boldsymbol{x}$, applied to each monomial, implies

$$f = \varphi_0(z) + \boldsymbol{x}^T\mathcal{H}\boldsymbol{x}, \qquad \mathcal{H} := \int_0^1 (1-s)\,\mathrm{Hess}_{\boldsymbol{x}}\, f(s\boldsymbol{x}, z)ds,$$

where $\varphi_0(z) := f|_{x_1=y_1=0}$ and $\mathcal{H}$ is a symmetric 2×2 matrix over $\mathbb{C}[[x_1, y_1, z]]$ with constant term

$$\mathcal{H}_0 = \begin{pmatrix} 0 & 1/2 \\ 1/2 & 0 \end{pmatrix}.$$

Since $g \cdot f = \chi_f(g) f$ and $g \cdot \boldsymbol{x} = \Lambda_g \boldsymbol{x}$, applying g to the entries of $\mathcal{H}$ yields

$$g \cdot \mathcal{H} = \chi_f(g) \Lambda_g^{-T} \mathcal{H} \Lambda_g^{-1}, \qquad \text{in particular} \qquad \mathcal{H}_0 = \chi_f(g) \Lambda_g^{-T} \mathcal{H}_0 \Lambda_g^{-1}.$$

Let $\Psi := (\mathcal{H}_0^{-1}\mathcal{H})^{1/2}$ be defined by the binomial series in $\mathcal{H}_0^{-1}\mathcal{H} - 1$. Since $\mathcal{H}_0$ and $\mathcal{H}$ are symmetric, $\mathcal{H}_0 \Psi = \Psi^T \mathcal{H}_0$, hence $\Psi^T \mathcal{H}_0 \Psi = \mathcal{H}_0 \Psi^2 = \mathcal{H}$. Since

$$g \cdot (\mathcal{H}_0^{-1}\mathcal{H}) = \Lambda_g \mathcal{H}_0^{-1} \mathcal{H} \Lambda_g^{-1},$$

we have $g \cdot \Psi = \Lambda_g \Psi \Lambda_g^{-1}$. Thus $(x_2, y_2)^T := \Psi \boldsymbol{x}$ satisfies

$$f = x_2 y_2 + \varphi_0(z) \qquad \text{and} \qquad g \cdot (x_2, y_2)^T = \Lambda_g (x_2, y_2)^T.$$

Moreover, $\dim_{\mathbb{C}} R/\operatorname{Jac}_{S_R} R$ is $N-1$ by the presentation $xy - z^N$ and is $\operatorname{mult}_0 \varphi_0 - 1$ by the presentation $x_2 y_2 + \varphi_0(z)$, so $\operatorname{mult}_0 \varphi_0 = N$. Comparing the components of $g \cdot f = \chi_f(g) f$ in $\mathbb{C}[[z]]$ and in (x_2, y_2), we obtain $\varphi_0(\chi_z(g) z) = \chi_f(g) \varphi_0(z)$. Write $\varphi_0 = -\beta_0^N z^N u_0(z)$ with $\beta_0 \in \mathbb{C}^*$ and $u_0(0) = 1$. Then $\chi_f = \chi_z^N$ and u_0 is $I_{G_R}(F_0)$-invariant, hence so is the square root $u_0^{1/2}$ of u_0 with constant term 1. Since

$$f = u_0 \cdot \big((u_0^{-1/2} x_2)(u_0^{-1/2} y_2) - (\beta_0 z)^N \big),$$

possibly replacing x, y, z with $u_0^{-1/2} x_2$, $u_0^{-1/2} y_2$, and $\beta_0 z$ respectively, we may assume that

$$R = \mathbb{C}[[x, y, z]]/(xy - z^N), \qquad g \cdot z = \chi_z(g) z, \qquad \text{and} \qquad g \cdot (x, y)^T = \Lambda_g (x, y)^T \tag{5.24}$$

for any $g \in I_{G_R}(F_0)$, with Λ_g diagonal if and only if $g \in G_1$. In the presentation (5.24), $\omega = dx \wedge dz/x$ is an eigenvector of any $g \in I_{G_R}(F_0)$, and x, y, z are eigenvectors of any $g \in G_1$.

By (2), $a(F_0, S_R, D_R) = \alpha e \geq 0$, so

$$\operatorname{mult}_{F_0} h_R \leq m a(F_0, S_R, 0) \leq m L_m e. \tag{5.25}$$

Case 1. $\min\{\operatorname{mult}_{F_0} x, \operatorname{mult}_{F_0} y\} \leq \operatorname{mult}_{F_0} h_R$. Let $F' := F_0$. By (5.20) and (5.25),

$$\operatorname{mult}_{F'} \operatorname{Jac}_{S_R} \leq \operatorname{mult}_{F_0} h_R \leq m L_m e,$$

and F' satisfies the claim by (2).

Case 2. $\operatorname{mult}_{F_0} x > \operatorname{mult}_{F_0} h_R$ and $\operatorname{mult}_{F_0} y > \operatorname{mult}_{F_0} h_R$. Write $h_R = \psi_0(z) + x\psi_1 + y\psi_2$ with $\psi_0 \in \mathbb{C}[[z]]$ and $\psi_1, \psi_2 \in R$. Since

$$\operatorname{mult}_{F_0}(x\psi_1 + y\psi_2) \geq \min\{\operatorname{mult}_{F_0} x, \operatorname{mult}_{F_0} y\} > \operatorname{mult}_{F_0} h_R,$$

we have $\psi_0 \neq 0$ and $\operatorname{mult}_{F_0} \psi_0(z) = \operatorname{mult}_{F_0} h_R$. Thus $\psi_0 = z^i w(z)$ with $i \in \mathbb{N}$ and $w(0) \neq 0$, and

$$\operatorname{mult}_{F_0} h_R = i \operatorname{mult}_{F_0} z. \tag{5.26}$$

Let c be the smallest integer such that $c > \operatorname{mult}_{F_0} h_R$ and $c \equiv \operatorname{mult}_{F_0} x \pmod{|G_1|}$. Since $\operatorname{mult}_{F_0} x$ satisfies both conditions and $\operatorname{mult}_{F_0} y = N \operatorname{mult}_{F_0} z - \operatorname{mult}_{F_0} x$,

$$\begin{gathered} \operatorname{mult}_{F_0} h_R < c \leq \min\{\operatorname{mult}_{F_0} x,\ \operatorname{mult}_{F_0} h_R + |G_1|\} \quad \text{and} \\ N \operatorname{mult}_{F_0} z - c \geq \operatorname{mult}_{F_0} y > \operatorname{mult}_{F_0} h_R. \end{gathered} \tag{5.27}$$

Let $\delta := \gcd(c, \operatorname{mult}_{F_0} z)$. The monomials $x^{j_1} z^{j_2}$ with $(j_1, j_2) \in \mathbb{Z}^2$ are the characters of the torus of the toric surface $(xy = z^N)$, and $(c/\delta, \operatorname{mult}_{F_0} z/\delta)$ is a primitive vector of the dual lattice. Let F' be the corresponding toric formal prime divisor over S_R. The valuation v of F' is the monomial valuation with

$$v(x) = \frac{c}{\delta}, \qquad v(z) = \frac{\operatorname{mult}_{F_0} z}{\delta}, \qquad \text{and} \qquad v(y) = \frac{N \operatorname{mult}_{F_0} z - c}{\delta}.$$

By (5.27), $v(x), v(y), v(z) > 0$, so the center of F' is o_R, and the residue field of F' is $\mathbb{C}(x^{\operatorname{mult}_{F_0} z/\delta} z^{-c/\delta})$. Let s_1, s_2 be a basis of the character lattice with $v(s_1) = 1$ and $v(s_2) = 0$.

Then x and z are Laurent monomials in s_1, s_2 with an invertible exponent matrix over $\mathbb{Z}$, so the form

$$\frac{dx}{x} \wedge \frac{dz}{z} = \pm \frac{ds_1}{s_1} \wedge \frac{ds_2}{s_2}$$

has a pole of order 1 along F'. Since $\omega = z \cdot \frac{dx}{x} \wedge \frac{dz}{z}$, the order of ω along F' is $v(z) - 1$, so $a(F', S_R, 0) = v(z)$ by Lemma 5.3(1) for $Z = S_R$. By (5.26) and (5.27), $v(\psi_0(z)) = \operatorname{mult}_{F_0} h_R/\delta$, while $v(x\psi_1) \geq c/\delta$ and $v(y\psi_2) \geq (N \operatorname{mult}_{F_0} z - c)/\delta$ are larger. Together with (5.20), we obtain

$$a(F', S_R, 0) = \frac{\operatorname{mult}_{F_0} z}{\delta}, \qquad \operatorname{mult}_{F'} h_R = \frac{\operatorname{mult}_{F_0} h_R}{\delta}, \qquad \text{and} \qquad \operatorname{mult}_{F'} \operatorname{Jac}_{S_R} \leq \frac{c}{\delta}. \tag{5.28}$$

We show that $G_1 \subset I_{G_R}(F')$. Any $g \in G_1$ multiplies x, y, z, hence any monomial $x^{j_1} z^{j_2}$ and $y^{j_1} z^{j_2}$, by a constant, so g preserves v. By (5.22),

$$g \cdot \left(x^{\operatorname{mult}_{F_0} z/\delta} z^{-c/\delta}\right) = \lambda(g)^{\frac{\operatorname{mult}_{F_0} z}{\delta}(\operatorname{mult}_{F_0} x - c)} x^{\operatorname{mult}_{F_0} z/\delta} z^{-c/\delta} = x^{\operatorname{mult}_{F_0} z/\delta} z^{-c/\delta},$$

since $|G_1|$ divides $\operatorname{mult}_{F_0} x - c$ and $\lambda(g)^{|G_1|} = 1$. Thus g acts trivially on the residue field of F', so $G_1 \subset I_{G_R}(F')$ and $e_{F'} \geq |G_1|$.

By (5.28) and (1),

$$\frac{\operatorname{mult}_{F_0} z - \operatorname{mult}_{F_0} h_R/m}{\delta} = a(F', S_R, D_R) \geq \alpha e_{F'} \geq 0,$$

so, as $\delta \geq 1$ and $e_{F'} \geq |G_1|$,

$$\frac{a(F', S_R, D_R)}{e_{F'}} \leq \frac{\operatorname{mult}_{F_0} z - \operatorname{mult}_{F_0} h_R/m}{|G_1|}. \tag{5.29}$$

Assume that $G_1 = I_{G_R}(F_0)$, so $|G_1| = e$. By (5.29), (5.21) and (2),

$$\frac{a(F', S_R, D_R)}{e_{F'}} \leq \frac{a(F_0, S_R, 0) - \operatorname{mult}_{F_0} h_R/m}{e} = \frac{a(F_0, S_R, D_R)}{e} = \alpha,$$

so $a(F', S_R, D_R) = \alpha e_{F'}$ by (1). By (5.28), (5.21), (5.27), and (5.25),

$$\frac{a(F', S_R, 0)}{e_{F'}} \leq \frac{\operatorname{mult}_{F_0} z}{e} \leq \frac{a(F_0, S_R, 0)}{e} \leq L_m \quad \text{and}$$
$$\frac{\operatorname{mult}_{F'} \operatorname{Jac}_{S_R}}{e_{F'}} \leq \frac{c}{e} \leq \frac{\operatorname{mult}_{F_0} h_R + e}{e} \leq mL_m + 1.$$

Assume that $G_1 \neq I_{G_R}(F_0)$, so $|G_1| = e/2$. Then $g_0 \notin G_1$, so $g_0 \cdot x \in \mathbb{C}^* y$ by (5.24). Since $\frac{dx}{x} + \frac{dy}{y} = N\frac{dz}{z}$ on S_R,

$$g_0 \cdot \omega = \chi_z(g_0) \frac{dy}{y} \wedge dz = -\chi_z(g_0) \frac{dx}{x} \wedge dz = -\chi_z(g_0)\omega.$$

By (5.22) for z and (5.23) for ω, $\lambda(g_0)^{a(F_0, S_R, 0)} = -\lambda(g_0)^{\operatorname{mult}_{F_0} z}$. Since $\lambda(g_0)$ is a primitive e-th root of unity, $a(F_0, S_R, 0) - \operatorname{mult}_{F_0} z \equiv e/2 \pmod{e}$, so by (5.21),

$$a(F_0, S_R, 0) - \operatorname{mult}_{F_0} z \geq \frac{e}{2}.$$

Hence by (2),

$$\alpha = \frac{a(F_0, S_R, 0) - \operatorname{mult}_{F_0} h_R/m}{e} \geq \frac{\operatorname{mult}_{F_0} z - \operatorname{mult}_{F_0} h_R/m}{e} + \frac{1}{2}.$$

Since $\alpha \leq 1$, $(\operatorname{mult}_{F_0} z - \operatorname{mult}_{F_0} h_R/m)/e \leq \frac{1}{2}$, and (5.29) implies

$$\frac{a(F', S_R, D_R)}{e_{F'}} \leq \frac{2(\operatorname{mult}_{F_0} z - \operatorname{mult}_{F_0} h_R/m)}{e} \leq \frac{\operatorname{mult}_{F_0} z - \operatorname{mult}_{F_0} h_R/m}{e} + \frac{1}{2} \leq \alpha.$$

Thus $a(F', S_R, D_R) = \alpha e_{F'}$ by (1). By (2), (5.28), (5.21), (5.27), and (5.25),

$$\frac{a(F', S_R, 0)}{e_{F'}} \le \frac{2\operatorname{mult}_{F_0} z}{e} \le 2L_m, \qquad \frac{\operatorname{mult}_{F'} \operatorname{Jac}_{S_R}}{e_{F'}} \le \frac{2c}{e} \le \frac{2\operatorname{mult}_{F_0} h_R + e}{e} \le 2mL_m + 1.$$

The claim follows. □

Step 4. Assume that S is of type A_{N-1} with $N \ge 3$. Then $\widehat{\mathcal{O}}_{S,o} \cong \mathbb{C}[[x, y, z]]/(xy - z^N)$, and G acts faithfully on $\widehat{\mathcal{O}}_{S,o}$. By Lemma 2.12(3), any formal prime divisor over $\operatorname{Spec} \widehat{\mathcal{O}}_{S,o}$ whose center is the closed point corresponds to a prime divisor over S with center o. Thus by (5.16), (5.18), and (5.19), the data $R := \widehat{\mathcal{O}}_{S,o}$, $G_R := G$, $(T \ni t) := (S \ni o)$, $h_R := h$, $\alpha := \operatorname{mld}((S \ni o, D)/G)$, and F_0 satisfy the hypotheses of Claim 5.8. Let F' be the prime divisor over S with center o corresponding to the formal prime divisor of Claim 5.8. By (5.16), F' is a computing lift, and

$$\frac{a(F', S, 0)}{e_{F'}} \le 2L_m \le C_m \qquad \text{and} \qquad \frac{\operatorname{mult}_{F'} \operatorname{Jac}_S}{e_{F'}} \le 2mL_m + 1 \le C_m.$$

Step 5. Assume that S is of type D_n with $n \ge 5$, and let $k := n - 2 \ge 3$. By the Du Val classification, $(S \ni o) \cong (\mathbb{C}^2/\operatorname{BD}_{4k} \ni 0)$ as analytic germs, where $\operatorname{BD}_{4k} \subset SL_2(\mathbb{C})$ is the binary dihedral group of order $4k$. Let $q : (\mathbb{C}^2 \ni 0) \to (S \ni o)$ be the quotient morphism by BD_{4k} composed with the isomorphism $(\mathbb{C}^2/\operatorname{BD}_{4k} \ni 0) \cong (S \ni o)$. For an automorphism g of the germ $(S \ni o)$, a lift of g is a holomorphic map germ $\tau : (\mathbb{C}^2 \ni 0) \to (\mathbb{C}^2 \ni 0)$ with $q \circ \tau = g \circ q$. Let $V \subset \mathbb{C}^2$ be an open ball centered at 0 on which q is defined. Since $\operatorname{BD}_{4k} \subset SL_2(\mathbb{C})$ acts freely on $V\backslash\{0\}$ and $V\backslash\{0\}$ is simply connected, the restriction

$$q|_{V\backslash\{0\}} : V\backslash\{0\} \to q(V)\backslash\{o\}$$

is a universal covering with deck transformation group BD_{4k}. Hence the lifts of id_S are exactly the elements of BD_{4k} [Pri67, Proposition 1]. Let g be an automorphism of the germ $(S \ni o)$, and $V' \subset V$ an open ball centered at 0 with $g(q(V')) \subset q(V)$. Since $q^{-1}(o) = \{0\}$, the map $g \circ q|_{V'\backslash\{0\}}$ lifts through $q|_{V\backslash\{0\}}$ to a holomorphic map $V'\backslash\{0\} \to V$, which is bounded and hence extends to a lift τ_g of g with $\tau_g(0) = 0$. As $\tau_{g^{-1}} \circ \tau_g$ and $\tau_g \circ \tau_{g^{-1}}$ are lifts of id_S, they belong to BD_{4k}, so τ_g is a biholomorphic automorphism of $(\mathbb{C}^2 \ni 0)$. For $b \in \operatorname{BD}_{4k}$, we have

$$q \circ \tau_g \circ b \circ \tau_g^{-1} = g \circ q \circ \tau_g^{-1} = q,$$

so τ_g normalizes BD_{4k}.

Let $\overline{G}$ be the group, under composition, of the lifts of the elements of G. Thus the lifts of any $g \in G$ form the coset $\tau_g \operatorname{BD}_{4k}$, so $\overline{G}$ is finite and $|\overline{G}| = 4k|G|$. For $\tau \in \overline{G}$, let $d\tau_0 \in GL_2(\mathbb{C})$ be the differential of τ at 0, and let

$$\Phi := \frac{1}{|\overline{G}|} \sum_{\sigma \in \overline{G}} (d\sigma_0)^{-1} \circ \sigma.$$

Then $d\Phi_0 = \mathrm{id}_{\mathbb{C}^2}$, so Φ is a biholomorphic automorphism of $(\mathbb{C}^2 \ni 0)$. For $\tau \in \overline{G}$, substituting $\sigma' := \sigma \circ \tau$, with $(d\sigma_0)^{-1} = d\tau_0 \circ (d\sigma'_0)^{-1}$, we obtain $\Phi \circ \tau = d\tau_0 \circ \Phi$. Since the elements of BD_{4k} are linear, Φ commutes with BD_{4k}, so $q \circ \Phi^{-1} : (\mathbb{C}^2 \ni 0) \to (S \ni o)$ is again the quotient morphism by BD_{4k} composed with an isomorphism $(\mathbb{C}^2/\operatorname{BD}_{4k} \ni 0) \cong (S \ni o)$, and the lifts of $g \in G$ with respect to $q \circ \Phi^{-1}$ are the maps $\Phi \circ \tau \circ \Phi^{-1} = d\tau_0$ for the lifts τ of g with respect to q. Possibly replacing q with $q \circ \Phi^{-1}$, we may assume that $\overline{G} \subset GL_2(\mathbb{C})$.

The rotation subgroup $\mu_{2k} \subset \operatorname{BD}_{4k}$ is characteristic: any element of $\operatorname{BD}_{4k} \backslash \mu_{2k}$ has order 4, so μ_{2k} is the unique cyclic subgroup of order $2k > 4$. Hence the lifts normalize μ_{2k} and induce automorphisms of $\widetilde{S} := \mathbb{C}^2/\mu_{2k}$, a germ of type A_{2k-1} at the image $\widetilde{o}$ of 0. Let $\widetilde{G}$ be the group of the automorphisms of $(\widetilde{S} \ni \widetilde{o})$ induced by the lifts. Since an automorphism of $(\mathbb{C}^2 \ni 0)$ inducing $\mathrm{id}_{\widetilde{S}}$ is a deck transformation of

$$\mathbb{C}^2\backslash\{0\} \to \widetilde{S}\backslash\{\widetilde{o}\},$$

that is, an element of μ_{2k}, the kernel of $\overline{G} \to \widetilde{G}$ is μ_{2k}, and there exists an exact sequence

$$1 \to \mathrm{BD}_{4k}/\mu_{2k} \cong \mathbb{Z}/2 \to \widetilde{G} \to G \to 1.$$

The quotient morphism $p : \widetilde{S} \to S$ by $\mathbb{Z}/2 = \mathrm{BD}_{4k}/\mu_{2k}$ is $\widetilde{G}$-equivariant and étale over $S\backslash\{o\}$. Since $\overline{G} \subset GL_2(\mathbb{C})$, the group $\widetilde{G} = \overline{G}/\mu_{2k}$ acts algebraically on the affine variety $\widetilde{S} = \mathbb{C}^2/\mu_{2k}$, the quotient $\widetilde{S}/(\mathbb{Z}/2)$ is $\mathbb{C}^2/\mathrm{BD}_{4k}$, and q induces a G-equivariant isomorphism between $\widehat{\mathcal{O}}_{S,o}$ and the completion of the local ring of $\mathbb{C}^2/\mathrm{BD}_{4k}$ at the image of 0, where $G = \overline{G}/\mathrm{BD}_{4k}$ acts on $\mathbb{C}^2/\mathrm{BD}_{4k}$.

Let $\widetilde{R} := \widehat{\mathcal{O}}_{\widetilde{S},\widetilde{o}}$. Then $\widetilde{G}$ acts faithfully on $\widetilde{R}$, and $\widetilde{R}^{\mathbb{Z}/2} = \widehat{\mathcal{O}}_{S,o}$ by Lemma 2.12(1) and the isomorphism induced by q. By [Rei02, Section 1.1] and [Rei97, Exercises 9 and 10 to Chapter 4], there exist $\widetilde{x}, \widetilde{y}, \widetilde{z} \in \widetilde{R}$ with

$$\widetilde{R} = \mathbb{C}[[\widetilde{x}, \widetilde{y}, \widetilde{z}]]/(\widetilde{x}\widetilde{y} - \widetilde{z}^{2k})$$

such that the generator of $\mathbb{Z}/2$ maps $(\widetilde{x}, \widetilde{y}, \widetilde{z})$ to $(\widetilde{y}, \widetilde{x}, -\widetilde{z})$ and

$$\begin{gathered}\widehat{\mathcal{O}}_{S,o} = \mathbb{C}[[y_1, y_2, y_3]]/\big(y_1^2 - y_3y_2^2 + 4y_3^{n-1}\big),\\ y_1 = \widetilde{z}(\widetilde{x} - \widetilde{y}), \quad y_2 = \widetilde{x} + \widetilde{y}, \quad \text{and} \quad y_3 = \widetilde{z}^2.\end{gathered} \tag{5.30}$$

For log discrepancies over $\operatorname{Spec}\widetilde{R}$ we write $a(\cdot, \widetilde{S}, \cdot)$. Let $\widetilde{h} := p^*h$ and $\widetilde{D} := \frac{1}{m}\operatorname{div}(\widetilde{h}) = p^*D$. The element $\widetilde{h}$ is $\widetilde{G}$-semi-invariant. Let $\widetilde{F}$ be a formal prime divisor over $\operatorname{Spec}\widetilde{R}$ whose center is the closed point, F the restriction of $\widetilde{F}$ to $\operatorname{Spec}\widehat{\mathcal{O}}_{S,o}$, E the restriction of F to Q, $\rho := |I_{\mathbb{Z}/2}(\widetilde{F})|$, and $e_{\widetilde{F}} := |I_{\widetilde{G}}(\widetilde{F})|$. Then for any $\phi \in \widehat{\mathcal{O}}_{S,o}\backslash\{0\}$,

$$\begin{gathered}\operatorname{mult}_{\widetilde{F}}(p^*\phi) = \rho \operatorname{mult}_F \phi, \qquad a(\widetilde{F}, \widetilde{S}, \widetilde{D}) = \rho a(F, S, D),\\ a(\widetilde{F}, \widetilde{S}, 0) = \rho a(F, S, 0), \qquad \text{and} \qquad e_{\widetilde{F}} = \rho e_F.\end{gathered} \tag{5.31}$$

The first and the third equalities follow from Lemma 2.15(1)(2) for the action of $\mathbb{Z}/2$ on $\widetilde{S}$ with $B = 0$ and Lemma 2.12(3), since $K_{\widetilde{S}} = p^*K_S$ as p is étale over $S\backslash\{o\}$. The second follows from the first for $\phi = h$ and the third, as $\widetilde{D} = \frac{1}{m}\operatorname{div}(p^*h)$. For the last one, F is the restriction of $\widetilde{F}$, with ramification index ρ, and E is the restriction of F, with ramification index e_F (Definition 2.2). Since $\widetilde{R}^{\widetilde{G}} = \widehat{\mathcal{O}}_{Q,\bar{o}}$, Lemma 2.15(1)(5) for the actions of $\mathbb{Z}/2 \subset \widetilde{G}$ on $\widetilde{S}$ and Lemma 2.12(3) imply $e_{\widetilde{F}} = \rho e_F$.

By (5.31), $a(\widetilde{F}, \widetilde{S}, \widetilde{D})/e_{\widetilde{F}} = a(F, S, D)/e_F$ and $a(\widetilde{F}, \widetilde{S}, 0)/e_{\widetilde{F}} = a(F, S, 0)/e_F$. Thus by (5.16), (5.18), and (5.19), the data

$$R := \widetilde{R}, \quad G_R := \widetilde{G}, \quad (T \ni t) := (\widetilde{S} \ni \widetilde{o}), \quad h_R := \widetilde{h}, \quad \text{and} \quad \alpha := \operatorname{mld}((S \ni o, D)/G),$$

and any formal prime divisor $\widetilde{F}_0$ over $\operatorname{Spec}\widetilde{R}$ whose restriction to $\operatorname{Spec}\widehat{\mathcal{O}}_{S,o}$ is F_0 satisfy the hypotheses of Claim 5.8 with $N = 2k$. Let $\widetilde{F}'$ be the formal prime divisor of Claim 5.8, and F' the restriction of $\widetilde{F}'$ to $\operatorname{Spec}\widehat{\mathcal{O}}_{S,o}$, which corresponds to a prime divisor over S with center o. By (5.31) and (5.16), F' is a computing lift and

$$\frac{a(F', S, 0)}{e_{F'}} = \frac{a(\widetilde{F}', \widetilde{S}, 0)}{e_{\widetilde{F}'}} \le 2L_m \le C_m.$$

It remains to bound $\operatorname{mult}_{F'}\operatorname{Jac}_S$. By (5.30), $\operatorname{Jac}_S \widehat{\mathcal{O}}_{S,o}$ contains $2y_1$ and $-2y_3y_2$, whose pullbacks are $2\widetilde{z}(\widetilde{x} - \widetilde{y})$ and $-2\widetilde{z}^2(\widetilde{x} + \widetilde{y})$. Since $(\widetilde{x} - \widetilde{y}, \widetilde{x} + \widetilde{y}) = (\widetilde{x}, \widetilde{y})$, (5.20) gives

$$\min\{\operatorname{mult}_{\widetilde{F}'}(\widetilde{x} - \widetilde{y}), \operatorname{mult}_{\widetilde{F}'}(\widetilde{x} + \widetilde{y})\} = \operatorname{mult}_{\widetilde{F}'}\operatorname{Jac}_{\widetilde{S}}.$$

Thus by (5.21) and Claim 5.8,

$$\begin{aligned}\operatorname{mult}_{\widetilde{F}'}\big(p^*\operatorname{Jac}_S\big) &\le \operatorname{mult}_{\widetilde{F}'}\operatorname{Jac}_{\widetilde{S}} + 2\operatorname{mult}_{\widetilde{F}'}\widetilde{z}\\ &\le \operatorname{mult}_{\widetilde{F}'}\operatorname{Jac}_{\widetilde{S}} + 2a(\widetilde{F}', \widetilde{S}, 0) \le \big((2m+4)L_m + 1\big)e_{\widetilde{F}'}.\end{aligned}$$

By (5.31),

$$\operatorname{mult}_{\widetilde{F}'}(p^*\operatorname{Jac}_S) = \rho \operatorname{mult}_{F'}\operatorname{Jac}_S \qquad \text{and} \qquad e_{\widetilde{F}'} = \rho e_{F'},$$

so

$$\frac{\operatorname{mult}_{F'}\operatorname{Jac}_S}{e_{F'}} \le (2m+4)L_m + 1 \le C_m.$$

Step 6. Assume that $S = \operatorname{Spec}\Bbbk[[x_1,x_2]]$, let $\mathfrak{m}$ be the maximal ideal of $\Bbbk[[x_1,x_2]]$, and let $\alpha := \operatorname{mld}((S \ni o, D)/G)$. Let $\sigma_0 : \mathfrak{m}/\mathfrak{m}^2 \to \mathfrak{m}$ be a $\Bbbk$-linear section of the quotient map, and

$$\sigma := \frac{1}{|G|}\sum_{g\in G} g\circ\sigma_0\circ g^{-1}.$$

Then σ is a G-equivariant section of $\mathfrak{m} \to \mathfrak{m}/\mathfrak{m}^2$, so for a basis v_1, v_2 of $\mathfrak{m}/\mathfrak{m}^2$, the elements $u_1 := \sigma(v_1)$ and $u_2 := \sigma(v_2)$ are regular parameters with $\Bbbk[[x_1,x_2]] = \Bbbk[[u_1,u_2]]$, and G acts on $\Bbbk u_1 + \Bbbk u_2$ by linear maps. Since a $\Bbbk$-automorphism of $\Bbbk[[u_1,u_2]]$ is continuous for the $\mathfrak{m}$-adic topology, it is determined by the images of u_1 and u_2. Hence G preserves $\Bbbk[u_1,u_2]$ and acts faithfully on $\mathbb{A}^2_\Bbbk := \operatorname{Spec}\Bbbk[u_1,u_2]$ fixing 0, and $S = \operatorname{Spec}\widehat{\mathcal{O}}_{\mathbb{A}^2_\Bbbk,0}$ G-equivariantly. Let χ be the character of h, so that $g\cdot h = \chi(g)h$ for any $g \in G$. For an integer $d > \operatorname{mult}_o h$, let $h_d \in \Bbbk[u_1,u_2]$ be the sum of the homogeneous parts of h of degree less than d in u_1, u_2, and $D_d := \frac{1}{m}\operatorname{div}(h_d)$. Since G preserves degrees, $g\cdot h_d = \chi(g)h_d$ for any $g \in G$, so h_d is a mark on $(\mathbb{A}^2_\Bbbk \ni 0)$, and $h - h_d \in \mathfrak{m}^d$. For any formal prime divisor F over S with center o,

$$a(F,S,D) = a(F,S,0) - \frac{\operatorname{mult}_F h}{m}, \qquad a(F,S,D_d) = a(F,S,0) - \frac{\operatorname{mult}_F h_d}{m}, \quad \text{and} \tag{5.32}$$
$$\operatorname{mult}_F(h-h_d) \ge d\operatorname{mult}_F\mathfrak{m} \ge d,$$

so $\operatorname{mult}_F h = \operatorname{mult}_F h_d$ if $\operatorname{mult}_F h < d$ or $\operatorname{mult}_F h_d < d$.

If $h \notin \mathfrak{m}$, then $h_d(0) = h(0) \neq 0$ for any $d \ge 1$, so $D_d = 0$ near 0, and we let $d_1 := 1$. Otherwise, let $c := \operatorname{lct}(S;(h))$ and $c_d := \operatorname{lct}(\mathbb{A}^2_\Bbbk \ni 0;(h_d))$. Since (S,D) is lc, $c \ge \frac{1}{m}$. By [dFEM11, Proposition 2.11], $c_d = \operatorname{lct}(S;(h_d))$, and since $(h)+\mathfrak{m}^d = (h_d)+\mathfrak{m}^d$, [dFEM11, Proposition 2.19(ii)] with $n=2$ gives $|c-c_d| \le \frac{2}{d}$. If $c_d < c$ for infinitely many d, then, as $\lim_d c_d = c$, the set $\{c_d \mid c_d < c\}$ contains a strictly increasing infinite sequence of log canonical thresholds on the fixed smooth germ $(\mathbb{A}^2_\Bbbk \ni 0)$, which contradicts [dFEM11, Theorem 1.1]. Hence there exists an integer d_1 such that $c_d \ge c \ge \frac{1}{m}$ for any $d \ge d_1$. In both cases, $(\mathbb{A}^2_\Bbbk, D_d)$ is lc near 0 for any $d \ge d_1$.

Let $d \ge \max\{d_1, \operatorname{mult}_o h + 1, mL_m|G|+1\}$ be an integer, and $U \subset \mathbb{A}^2_\Bbbk$ a G-invariant open neighborhood of 0 with $(U, D_d|_U)$ lc. The argument of **Step 1** for the smooth germ $(U \ni 0)$ over $\Bbbk$, the group G, and the mark $h_d|_U$ gives a prime divisor F_d over U with center 0 such that

$$a(F_d,U,D_d) = e_{F_d}\operatorname{mld}((U\ni 0, D_d)/G) \qquad \text{and} \qquad a(F_d,U,0) \le L_m e_{F_d}.$$

By Lemma 2.12(2)(3) for $X = U$, $H = G$, and $B \in \{D_d, 0\}$, we also write F_d for the corresponding formal prime divisor over S with center o. Then the inertia groups of F_d over S and over U are equal, and

$$a(F_d,S,D_d) = a(F_d,U,D_d), \qquad a(F_d,S,0) = a(F_d,U,0), \quad \text{and}$$
$$\alpha_d := \operatorname{mld}((S\ni o, D_d)/G) = \operatorname{mld}((U\ni 0, D_d)/G).$$

Since $(U, D_d|_U)$ is lc, (5.32) gives

$$\operatorname{mult}_{F_d} h_d \le m a(F_d,S,0) \le mL_m e_{F_d} \le mL_m|G| < d,$$

so $\operatorname{mult}_{F_d} h = \operatorname{mult}_{F_d} h_d$ and $a(F_d,S,D) = a(F_d,S,D_d) = \alpha_d e_{F_d}$. Hence $\alpha \le \alpha_d$. On the other hand, by (5.32),

$$\alpha_d \le \beta_d := \inf\left\{\frac{a(F,S,D)}{e_F} \;\middle|\; F \text{ a formal prime divisor over } S \text{ with center } o \text{ and } \operatorname{mult}_F h < d\right\}.$$

The sequence $(\beta_d)_d$ is decreasing, $\beta_d \geq \alpha$, and any F satisfies $\operatorname{mult}_F h < d$ for $d \gg 0$, so $\lim_d \beta_d = \alpha$, and thus $\lim_d \alpha_d = \alpha$. Since $a(F_d, S, 0) = a(F_d, U, 0)$ is a positive integer,

$$\alpha_d = \frac{a(F_d, S, 0) - \operatorname{mult}_{F_d} h_d/m}{e_{F_d}} \in \left\{ \frac{j_1 - j_2/m}{j_3} \;\middle|\; j_1, j_2, j_3 \in \mathbb{Z},\ 1 \leq j_1 \leq L_m|G|,\ 0 \leq j_2 \leq mL_m|G|,\ 1 \leq j_3 \leq |G| \right\},$$

a finite set independent of d. Hence $\alpha_d = \alpha$ for any $d \gg 0$. Fix such a d. Then $F := F_d$ is a computing lift of α with

$$\frac{a(F, S, 0)}{e_F} \leq L_m \leq C_m, \qquad \operatorname{mult}_F \operatorname{Jac}_S = \operatorname{mult}_F \mathcal{O}_S = 0, \quad \text{and}$$
$$\frac{\operatorname{mult}_F h}{e_F} = \frac{\operatorname{mult}_F h_d}{e_F} \leq mL_m \leq mC_m.$$

By the Du Val classification, a Du Val germ is of type A_{N-1} with $N \geq 2$, D_n with $n \geq 4$, or E_6, E_7, E_8, so **Steps 1, 2, 4, and 5** treat all algebraic germs $(S \ni o)$ of the lemma. In each of **Steps 1, 2, 4, and 5**, we found a computing lift F of $\operatorname{mld}((S \ni o, D)/G)$ with $a(F, S, 0)/e_F \leq C_m$ and $\operatorname{mult}_F \operatorname{Jac}_S / e_F \leq C_m$, and $\operatorname{mult}_F h/e_F \leq ma(F, S, 0)/e_F \leq mC_m$ by (5.17). Together with **Step 6**, the lemma follows. □

5.3. **Products and bounded degree.**

Theorem 5.9. *Let L and n be positive integers. Then there exists a finite set $\Lambda \subset \mathbb{Z}^{n+1}$, depending only on L and n, satisfying the following. Let $G \subset (\mathbb{C}^*)^4$ be a finite subgroup acting diagonally on $\mathbb{A}^4$, f a G-semi-invariant polynomial of degree at most L with $f(0) = 0$, and W a G-invariant normal open neighborhood of 0 in $(f = 0) \subset \mathbb{A}^4$. Let $h_1, \dots, h_n$ be G-semi-invariant polynomials of degree at most L with $h_j|_W \neq 0$, let $c_1, \dots, c_n$ be real numbers, and let $B := \sum_{j=1}^n c_j \operatorname{div}(h_j|_W)$. Then there exist a G-equivariant projective birational morphism $g : Y \to W$ from a smooth variety Y and a G-invariant reduced simple normal crossing divisor T_Y on Y such that, with $K_Y + C_Y := g^*(K_W + B)$, the following hold.*

(1) $g^{-1}(0)$ is a divisor and $\operatorname{Supp} C_Y \cup g^{-1}(0) \subset T_Y$.
(2) For any closed point $y \in Y$, the stabilizer G_y preserves every local irreducible branch of T_Y through y.
(3) For any prime divisor E on Y contained in $\operatorname{Supp} C_Y \cup g^{-1}(0)$,

$$(1 - a(E, W, 0), \operatorname{mult}_E(h_1|_W), \dots, \operatorname{mult}_E(h_n|_W)) \in \Lambda.$$

Proof. We divide the proof into four steps.

Step 1. In this step we reduce to an irreducible f and show that G is contained in one of finitely many diagonalizable groups $\mathcal{H}$.

Let $d := 4$, and let $x_1, \dots, x_d$ be the coordinates of $\mathbb{A}^d$. For $\gamma \in (\mathbb{C}^*)^d$ and $\alpha \in \mathbb{N}^d$ we have $\gamma^* x^\alpha = \gamma^\alpha x^\alpha$, where $\gamma^\alpha := \prod_{k=1}^d \gamma_k^{\alpha_k}$. Since W is a normal variety, $W \subset (f_1 = 0)$ for an irreducible factor f_1 of f. Since W is reduced and contains the generic point of $(f_1 = 0)$, $f_1^2 \nmid f$. Since $\mathcal{O}_{W,w}$ is a domain for any $w \in W$, no other irreducible factor of f vanishes at a point of W. Thus W is an open subset of $(f_1 = 0)$. For any $\gamma \in G$, $\gamma^* f_1$ is an irreducible factor of $\gamma^* f = \chi_f(\gamma) f$ vanishing on $\gamma^{-1}(W) = W$, so $\gamma^* f_1 \in \mathbb{C}^* f_1$. Possibly replacing f with f_1, we may assume that f is irreducible. Then f is still G-semi-invariant of degree at most L with $f(0) = 0$.

The exponent set of a polynomial is the set of $\alpha \in \mathbb{N}^d$ such that x^α has non-zero coefficient in it. Let $\mathfrak{M} = (\mathfrak{M}_j)_{0 \leq j \leq n}$ be a tuple of non-empty subsets of $\{\alpha \in \mathbb{N}^d \mid \alpha_1 + \cdots + \alpha_d \leq L\}$. There exist finitely many such tuples. Let $\mathcal{P} := \prod_{j=0}^n (\mathbb{C}^*)^{\mathfrak{M}_j}$ with coordinates $y_{j,\alpha}$, and on $\mathbb{A}^d \times \mathcal{P}$ let

$$\mathbf{f} := \sum_{\alpha \in \mathfrak{M}_0} y_{0,\alpha} x^\alpha, \qquad \mathbf{h}_j := \sum_{\alpha \in \mathfrak{M}_j} y_{j,\alpha} x^\alpha \quad (1 \leq j \leq n).$$

The tuples $(f, h_1, \dots, h_n)$ with exponent sets $\mathfrak{M}$ are the restrictions

$$(\mathbf{f}|_{\mathbb{A}^d\times\{t\}}, \mathbf{h}_1|_{\mathbb{A}^d\times\{t\}}, \dots, \mathbf{h}_n|_{\mathbb{A}^d\times\{t\}})$$

for $t \in \mathcal{P}$. Let

$$\mathcal{H} := \{\gamma \in (\mathbb{C}^*)^d \mid \gamma^\alpha = \gamma^{\alpha'} \text{ for any } 0 \le j \le n \text{ and any } \alpha, \alpha' \in \mathfrak{M}_j\},$$

a diagonalizable algebraic group acting on $\mathbb{A}^d \times \mathcal{P}$ through the first factor. The ideals $(\mathbf{f})$, $(\mathbf{h}_j)$, and $J_0 := (x_1, \dots, x_d)$ are $\mathcal{H}$-invariant. If the exponent sets of $(f, h_1, \dots, h_n)$ form $\mathfrak{M}$, then $G \subset \mathcal{H}$: for $\gamma \in G$ and $\alpha \in \mathfrak{M}_j$, comparing the coefficients of x^α in $\gamma^* h_j = \chi_{h_j}(\gamma) h_j$ implies $\gamma^\alpha = \chi_{h_j}(\gamma)$, and similarly for f.

Step 2. In this step we construct equivariant embedded resolutions over finitely many pieces of $\mathcal{P}$, and define Λ.

Let $\mathcal{P}_0 \subset \mathcal{P}$ be a smooth irreducible locally closed subvariety, and on $\mathbb{A}^d \times \mathcal{P}_0$ let

$$\mathcal{I} := J_0 \cdot (\mathbf{f}) \cdot \prod_{j=1}^{n} (\mathbf{h}_j),$$

a non-zero $\mathcal{H}$-invariant ideal. By [Kol07, Theorem 3.35], there exists a composite $\mu : V' \to \mathbb{A}^d \times \mathcal{P}_0$ of blow-ups of smooth centers, $\mathcal{H}$-equivariant since the principalization of [Kol07, Theorem 3.35] commutes with smooth morphisms [Kol07, Theorem 3.35(4)] and the action and the projection from $\mathcal{H} \times \mathbb{A}^d \times \mathcal{P}_0$ to $\mathbb{A}^d \times \mathcal{P}_0$ are smooth, such that μ is an isomorphism over the complement of the cosupport of $\mathcal{I}$, $\mathcal{I}\mathcal{O}_{V'}$ is invertible and $\mu^{-1}(\operatorname{cosupp}\mathcal{I})$ is an snc divisor. We omit the blow-ups of centers of codimension 1, which are isomorphisms. Then μ is an isomorphism over the complement of the union $\mathcal{Z}$ of the images of the centers. Let $D_1, \dots, D_N$ be the irreducible components of $\mu^{-1}(\operatorname{cosupp}\mathcal{I})$. Then $\operatorname{Exc}(\mu) \subset \bigcup_i D_i$. Since $(\mathbf{f})$ and $(\mathbf{h}_j)$ are principal, $J_0\mathcal{O}_{V'}$ is invertible as well.

Applying generic smoothness to V', to the irreducible components of all intersections of the D_i and to the irreducible components of the centers of μ, and removing the closures of the images of those which do not dominate $\mathcal{P}_0$, we obtain a dense open subset $\mathcal{P}_1 \subset \mathcal{P}_0$ over which all of them are smooth. Let $t \in \mathcal{P}_1$, let $\mu_t : V'_t \to \mathbb{A}^d$ be the fiber of μ over t, and let $D_{i,t} := D_i \cap V'_t$. Then V'_t and $D_{i,t}$ are smooth, $\sum_i D_{i,t}$ is snc, and μ_t is an isomorphism over $\mathbb{A}^d \backslash Z_t$, where $Z_t := \mathcal{Z} \cap (\mathbb{A}^d \times \{t\})$ satisfies $\dim Z_t \le d-2$, since each center is of codimension at least 2 and smooth over $\mathcal{P}_1$. Over $\mathcal{P}_1$, let κ_i be the multiplicity along D_i of the divisor of zeros R_μ of $\mu^*(dx_1 \wedge \cdots \wedge dx_d)$, regarded as a section of $\omega_{V'/\mathcal{P}_1}$. Then $\operatorname{Supp} R_\mu \subset \operatorname{Exc}(\mu)$. Restricting to V'_t, we obtain

$$K_{V'_t} = \mu_t^* K_{\mathbb{A}^d} + \sum_i \kappa_i D_{i,t}, \qquad \mu_t^* \operatorname{div}(\mathbf{h}_j|_{\mathbb{A}^d\times\{t\}}) = \sum_i \operatorname{mult}_{D_i}(\mu^*\mathbf{h}_j) D_{i,t},$$

and similarly for $\mathbf{f}$ and J_0. Let

$$k^{(i)} := \big(\operatorname{mult}_{D_i}(\mu^*\mathbf{f}) - \kappa_i,\ \operatorname{mult}_{D_i}(\mu^*\mathbf{h}_1), \dots, \operatorname{mult}_{D_i}(\mu^*\mathbf{h}_n)\big) \in \mathbb{Z}^{n+1}.$$

By Noetherian induction on $\mathcal{P}$, repeating the choice of μ and $\mathcal{P}_1$ for finitely many smooth irreducible locally closed subvarieties covering $\mathcal{P}_0 \backslash \mathcal{P}_1$, we obtain finitely many subsets $\mathcal{P}_1^{(1)}, \dots, \mathcal{P}_1^{(N')}$ covering $\mathcal{P}$, the ν-th one given with its morphism $\mu^{(\nu)}$, components $D_i^{(\nu)}$, multiplicities $\kappa_i^{(\nu)}$ and vectors $k^{(\nu,i)}$. Let Λ_1 be the finite set of the vectors $k^{(\nu,i)}$ for all tuples $\mathfrak{M}$, all ν and all i, and let

$$\Lambda := \Lambda_1 \cup \{k + k' - (1, 0, \dots, 0) \mid k, k' \in \Lambda_1\} \cup \{k + k' + k'' - (2, 0, \dots, 0) \mid k, k', k'' \in \Lambda_1\}.$$

Then Λ is finite and depends only on L and n.

Step 3. In this step we construct a G-equivariant resolution $g_0 : Y_0 \to W$ and compute the coefficients on it.

Let $(f, h_1, \dots, h_n)$ be a datum as in the theorem, with exponent sets $\mathfrak{M}$, and let t be the corresponding point of $\mathcal{P}$. Then $t \in \mathcal{P}_1^{(\nu)}$ for some ν, and we write μ, D_i, κ_i, $k^{(i)}$ for $\mu^{(\nu)}$, $D_i^{(\nu)}$, $\kappa_i^{(\nu)}$, $k^{(\nu,i)}$. Since $G \subset \mathcal{H}$ and $\mathcal{H}$ acts trivially on $\mathcal{P}$, G acts on V_t' and μ_t is G-equivariant. Since V_t' is smooth, the connected components of V_t' are irreducible. Let V_t be the one containing $\mu_t^{-1}(\mathbb{A}^d \backslash Z_t) \cong \mathbb{A}^d \backslash Z_t$. Then V_t is G-invariant and $\mu_t|_{V_t} : V_t \to \mathbb{A}^d$ is projective and birational.

Since $\dim Z_t \le 2$, μ_t is an isomorphism over the generic point of the irreducible hypersurface $(f = 0)$. Let $\widetilde{W}$ be the closure of $\mu_t^{-1}((f = 0) \backslash Z_t)$ in V_t. Then $\widetilde{W}$ is a prime divisor contained in $\operatorname{Supp} \mu_t^* \operatorname{div}(f) \subset \bigcup_i D_{i,t}$, hence a connected component of $D_{i_0,t}$ for some i_0. In particular $\widetilde{W}$ is smooth and does not intersect the other components of $D_{i_0,t}$. Since f is irreducible (**Step 1**), μ_t is an isomorphism at the generic point of $\widetilde{W}$, and h_j does not vanish on the dense open subset W of $(f = 0)$,

$$\operatorname{mult}_{D_{i_0}}(\mu^* \mathbf{f}) = 1, \qquad \kappa_{i_0} = 0, \quad \text{and} \quad \operatorname{mult}_{D_{i_0}}(\mu^* \mathbf{h}_j) = 0 \quad (1 \le j \le n).$$

Let $Y_0 := \widetilde{W} \cap \mu_t^{-1}(W)$, $g_0 := \mu_t|_{Y_0} : Y_0 \to W$, and $T_0 := (\bigcup_{i \ne i_0} D_{i,t}) \cap Y_0$. Then g_0 is a G-equivariant projective birational morphism, and T_0 is a G-invariant reduced snc divisor on Y_0, since $\widetilde{W}$ is a component of the snc divisor $\sum_i D_{i,t}$, and $T_0 = \overline{(\bigcup_i D_{i,t}) \backslash \widetilde{W}} \cap Y_0$, where $\widetilde{W}$, being the closure of the G-invariant set $\mu_t^{-1}((f = 0) \backslash Z_t)$, and

$$\bigcup_i D_{i,t} = \mu_t^{-1}(\operatorname{cosupp} \mathcal{I}) \cap V_t'$$

are G-invariant.

Any component P of T_0 is a component of $D_{i,t} \cap Y_0$ for exactly one $i =: i(P)$: otherwise P would be contained in $D_{i,t} \cap D_{i',t} \cap Y_0$ for some $i \ne i'$, while

$$\dim(D_{i,t} \cap D_{i',t} \cap Y_0) \le 1$$

since $\sum_i D_{i,t}$ is snc. Since $\sum_i D_{i,t}$ is snc, $\operatorname{mult}_P(D_{i(P),t}|_{Y_0}) = 1$. We have $K_W = (K_{\mathbb{A}^4} + \operatorname{div}(f))|_W$, and near Y_0

$$K_{V_t} + Y_0 = \mu_t^*(K_{\mathbb{A}^4} + \operatorname{div}(f)) + \sum_{i \ne i_0} \big(\kappa_i - \operatorname{mult}_{D_i}(\mu^* \mathbf{f})\big) D_{i,t},$$

so by adjunction

$$K_{Y_0} = g_0^* K_W + \sum_{i \ne i_0} (\kappa_i - \operatorname{mult}_{D_i}(\mu^* \mathbf{f})) D_{i,t}|_{Y_0}.$$

Together with **Step 2**, for any component P of T_0,

$$\big(1 - a(P, W, 0), \operatorname{mult}_P(h_1|_W), \dots, \operatorname{mult}_P(h_n|_W)\big) = k^{(i(P))} \in \Lambda_1. \tag{5.33}$$

Since $\mathbb{A}^d$ is smooth, $\operatorname{Exc}(\mu_t)$ is a divisor, and $\operatorname{Exc}(\mu_t) \subset \operatorname{Exc}(\mu) \subset \bigcup_i D_i$. Since $\widetilde{W} \not\subset \operatorname{Exc}(\mu_t)$ and the components of $D_{i_0,t}$ other than $\widetilde{W}$ do not meet $\widetilde{W}$, $\operatorname{Exc}(\mu_t) \cap Y_0 \subset T_0$. Thus the prime divisors P on Y_0 which are not contained in T_0 are not g_0-exceptional and satisfy $\operatorname{mult}_P(h_j|_W) = 0$. Since the coefficient of a prime divisor P in C_{Y_0}, defined by $K_{Y_0} + C_{Y_0} = g_0^*(K_W + B)$, is

$$1 - a(P, W, 0) + \sum_{j=1}^n c_j \operatorname{mult}_P(h_j|_W),$$

we have $\operatorname{Supp} C_{Y_0} \subset T_0$. Finally, $J_0 \mathcal{O}_{Y_0}$ is invertible and non-zero, so $g_0^{-1}(0)$ is the reduced divisor $\operatorname{Supp} \sum_i \operatorname{mult}_{D_i}(\mu^* J_0) D_{i,t}|_{Y_0} \subset T_0$.

Step 4. In this step we separate the branches of T_0 and conclude the proof.

Since T_0 is snc on the threefold Y_0, at most three components of T_0 contain a point, and the set S_0 of points contained in three components is finite. Let $g_1 : Y_1 \to Y_0$ be the blow-up of S_0. For components $P \ne P'$ of T_0, the curve $P \cap P'$ is smooth. Two such curves intersect only at points contained in three components, and in the local model where three coordinate planes contain such a point, the blow-up separates the three coordinate axes. Thus the strict

transforms in Y_1 of the curves $P \cap P'$ are smooth and pairwise disjoint. Let $S_1 \subset Y_1$ be their union, let $g_2 : Y \to Y_1$ be the blow-up of S_1, let $g := g_0 \circ g_1 \circ g_2 : Y \to W$, and let T_Y be the reduced preimage of T_0 in Y. The centers S_0 and S_1 are G-invariant strata of the snc divisors T_0 and $(g_1^{-1}(T_0))_{\mathrm{red}}$, so g is a G-equivariant projective birational morphism, Y is smooth, T_Y is a G-invariant reduced snc divisor, and $g^{-1}(0)$ is a divisor. Since any prime divisor on Y which is not contained in T_Y is not g-exceptional and is not contained in $\operatorname{Supp} g^* \operatorname{div}(h_j|_W)$ for any j, and since $g_0^{-1}(0) \subset T_0$, we have $\operatorname{Supp} C_Y \cup g^{-1}(0) \subset T_Y$.

The components of T_Y are the prime divisors $(g_1 \circ g_2)_*^{-1} P$ for the components P of T_0 (type 1), the exceptional divisors of g_2 (type 2), and the prime divisors $(g_2)_*^{-1} E'$ for the exceptional divisors E' of g_1 (type 3). Two distinct components of one type are disjoint: for type 1, the prime divisors $(g_1)_*^{-1} P$ and $(g_1)_*^{-1} P'$ intersect exactly along the strict transform of $P \cap P'$, which g_2 separates. For types 2 and 3, the components of S_1 and the points of S_0 are pairwise disjoint. Since G preserves the types, for any closed point $y \in Y$, G_y maps any component E of T_Y through y to a component through y of the type of E, that is, to E. Thus G_y preserves every local irreducible branch of T_Y through y.

Let E be the exceptional divisor over a point contained in the components P_1, P_2, P_3 of T_0 (resp. over the strict transform of a curve $P_1 \cap P_2$, whose generic point is contained only in $(g_1)_*^{-1} P_1$ and $(g_1)_*^{-1} P_2$ among the components of $g_1^{-1}(T_0)$). Since the blow-up of a smooth center of codimension 3 (resp. 2) has discrepancy 2 (resp. 1) and the pullbacks of P_1, P_2, P_3 (resp. P_1, P_2) have multiplicity 1 along E,

$$a(E, W, 0) = \sum_{k=1}^{3} a(P_k, W, 0) \quad \Big(\text{resp. } \sum_{k=1}^{2} a(P_k, W, 0)\Big),$$

$$\operatorname{mult}_E(h_j|_W) = \sum_{k=1}^{3} \operatorname{mult}_{P_k}(h_j|_W) \quad \Big(\text{resp. } \sum_{k=1}^{2} \operatorname{mult}_{P_k}(h_j|_W)\Big).$$

By (5.33), the vector $(1 - a(E, W, 0), \operatorname{mult}_E(h_1|_W), \dots, \operatorname{mult}_E(h_n|_W))$ is

$$k^{(i(P_1))} + k^{(i(P_2))} + k^{(i(P_3))} - (2, 0, \dots, 0) \quad \big(\text{resp. } k^{(i(P_1))} + k^{(i(P_2))} - (1, 0, \dots, 0)\big),$$

which belongs to Λ. Together with (5.33) for the components of type 1, we obtain

$$(1 - a(E, W, 0), \operatorname{mult}_E(h_1|_W), \dots, \operatorname{mult}_E(h_n|_W)) \in \Lambda$$

for any component E of T_Y. Together with the inclusion $\operatorname{Supp} C_Y \cup g^{-1}(0) \subset T_Y$ and the preservation of the local branches of T_Y by the stabilizers G_y, the theorem follows. □

Theorem 5.10. *Let $f_0(x_1, x_2, x_3)$ be a polynomial with $f_0(0) = 0$ such that $S := (f_0 = 0) \subset \mathbb{A}^3$ is a normal surface with at most Du Val singularities, and m, M, and L positive integers. Then there exists a positive integer N, depending only on f_0, m, M, L, satisfying the following. Let $(\mathcal{X} \ni 0)$ be a smooth fourfold germ with regular parameters x_1, x_2, x_3, s at 0, $\mathfrak{n}$ the maximal ideal of $\mathcal{O}_{\mathcal{X},0}$, and $W_0 := (f_0(x_1, x_2, x_3) = 0) \subset \mathcal{X}$. Let G be a finite abelian group of order at most M acting on $(\mathcal{X} \ni 0)$, with x_1, x_2, x_3, s and $f_0(x_1, x_2, x_3)$ G-semi-invariant, and $f, h \in \mathcal{O}_{\mathcal{X},0}$, f with the character of f_0 and h G-semi-invariant. Assume the following.*

(i) $W := (f = 0)$ is normal near 0.
(ii) G acts faithfully on W and on W_0.
(iii) $f - f_0 \in (s)$.
(iv) $h - h_0 \in (s)$ for a polynomial $h_0(x_1, x_2, x_3)$ of degree at most L with $h_0|_S \neq 0$ and $(S, \frac{1}{m} \operatorname{div}(h_0|_S))$ lc near 0.

Let $\beta < 1$ be a real number,

$$B := \frac{1}{m} \operatorname{div}(h|_W) + \beta \operatorname{div}(s|_W), \quad \text{and} \quad B_0 := \frac{1}{m} \operatorname{div}(h_0|_{W_0}) + \beta \operatorname{div}(s|_{W_0}).$$

Then:

(1) We have

$$\operatorname{mld}((W \ni 0, B)/G) \geq \operatorname{mld}((W_0 \ni 0, B_0)/G) \geq \frac{1-\beta}{|G|} > 0.$$

(2) If $f - f_0 \in \mathfrak{n}^N$ *and* $h - h_0 \in \mathfrak{n}^N$, *then* $\operatorname{mld}((W \ni 0, B)/G) = \operatorname{mld}((W_0 \ni 0, B_0)/G)$, *and there exists a prime divisor* F *over* W *with center* 0 *and*

$$a(F, W, B)/e_F = \operatorname{mld}((W \ni 0, B)/G)$$

such that $a(F, W, 0) \leq N$, $\operatorname{mult}_F h \leq N$, $\operatorname{mult}_F \operatorname{Jac}_W \leq N$, *and* $\operatorname{mult}_F s \leq N$.

Proof. Let $\widehat{\mathcal{O}} := \mathbb{C}[[x_1, x_2, x_3, s]]$ be the completion of $\mathcal{O}_{\mathcal{X},0}$. It is also the completion of $\mathcal{O}_{\mathbb{A}^4,0}$, where G acts on $\mathbb{A}^4$ diagonally through the characters of x_1, x_2, x_3, s. By Lemma 2.12(2)(3), quotient values at 0, and prime divisors with center 0 together with their inertia groups, log discrepancies and the orders along them of elements of the completed local ring, depend only on the completion at 0. Thus $\operatorname{mld}((W_0 \ni 0, B_0)/G)$ is computed on $S \times \mathbb{A}^1_s \subset \mathbb{A}^4$, which we again denote by W_0, and we regard f and h as G-semi-invariant elements of $\widehat{\mathcal{O}}$. Since $f - f_0 \in (s)$ and $h - h_0 \in (s)$,

$$W \cap (s = 0) = S \times \{0\} \quad \text{and} \quad h|_{S\times\{0\}} = h_0|_S \neq 0. \tag{5.34}$$

We prove (1). Since $(S, \frac{1}{m}\operatorname{div}(h_0|_S))$ is lc near 0 and $(\mathbb{A}^1_s, \operatorname{div}(s))$ is lc, the pair

$$\Big(W_0, \frac{1}{m}\operatorname{div}(h_0|_{W_0}) + \operatorname{div}(s|_{W_0})\Big)$$

is lc near 0. Thus, for any prime divisor P_0 over W_0 with center 0, since $\operatorname{mult}_{P_0} s \geq 1$,

$$a(P_0, W_0, B_0) = a\Big(P_0, W_0, \frac{1}{m}\operatorname{div}(h_0|_{W_0}) + \operatorname{div}(s|_{W_0})\Big) + (1-\beta)\operatorname{mult}_{P_0} s \geq 1 - \beta > 0. \tag{5.35}$$

Since $e_{P_0} \leq |G|$ (Definition 2.2), (5.35) implies $\operatorname{mld}((W_0 \ni 0, B_0)/G) \geq \frac{1-\beta}{|G|} > 0$.

Fix a prime divisor F over W with center 0. Let $I := I_G(F)$, $e := e_F = |I|$,

$$B^+ := \frac{1}{m}\operatorname{div}(h|_W) + \operatorname{div}(s|_W), \quad a_F := a(F, W, B^+), \quad \text{and} \quad k := \operatorname{mult}_F s.$$

Since F has center 0, $k \geq 1$, and

$$a(F, W, B) = a_F + (1-\beta)k. \tag{5.36}$$

Since F is arbitrary, the first inequality of (1) follows from

$$\operatorname{mld}((W_0 \ni 0, B_0)/G) \leq \frac{a_F + (1-\beta)k}{e}. \tag{5.37}$$

For a prime divisor E over a normal variety with a faithful action of a finite group H, let $\mathcal{O}_E$ be the valuation ring of E, $\mathfrak{m}_E \subset \mathcal{O}_E$ the maximal ideal, and $\pi_E \in \mathfrak{m}_E$ a uniformizer. By Lemma 2.15(1), $I_H(E)$ is cyclic and acts on $\mathfrak{m}_E/\mathfrak{m}_E^2$ through an injective character

$$\lambda_E : I_H(E) \to \mathbb{C}^*,$$

that is, $g^*\pi_E/\pi_E \equiv \lambda_E(g) \bmod \mathfrak{m}_E$ for any $g \in I_H(E)$. Thus, writing $g^*\varphi = \chi_\varphi(g)\varphi$ for an H-semi-invariant rational function φ, for any $g \in I_H(E)$

$$\chi_\varphi(g) = \lambda_E(g)^{\operatorname{mult}_E \varphi}. \tag{5.38}$$

Since G acts faithfully on W, $\lambda := \lambda_F$ is injective and I is cyclic of order e. Let

$$\zeta_e := \exp(2\pi\sqrt{-1}/e),$$

and let $g_F \in I$ be the element with $\lambda(g_F) = \zeta_e$. By (5.38) with $\varphi = s$,

$$\chi_s(g_F) = \zeta_e^k. \tag{5.39}$$

Let $k_+ \in \{1, \dots, e\}$ with $k_+ \equiv k \pmod e$. Since $k \geq 1$, $k_+ \leq k$.

Let n be an integer with $n \geq 2$ and $n > a_F$, and let I act on $\mathbb{A}^n$, with coordinates $u_1, \dots, u_n$, by $g'^* u_i = \lambda(g') u_i$ for any $g' \in I$ and any i. For any $g' \in I \backslash \{1\}$, $\lambda(g') \neq 1$, so

$$\operatorname{Fix}(g') \subset (u_1 = \cdots = u_n = 0).$$

Thus

$$\operatorname{codim}_{\mathbb{A}^4 \times \mathbb{A}^n} \operatorname{Fix}(g') \geq n \geq 2 \quad \text{and} \quad \operatorname{codim}_{S \times \mathbb{A}^n} (\operatorname{Fix}(g') \cap (S \times \mathbb{A}^n)) \geq 2, \tag{5.40}$$

where $S \times \mathbb{A}^n = (f_0 = s = 0) \subset \mathbb{A}^4 \times \mathbb{A}^n$.

Let F^{c} be the prime divisor over $W \times \mathbb{A}^n$ whose valuation is the order of vanishing at the maximal ideal of the regular local ring $\mathcal{O}_F[u_1, \dots, u_n]_{(\pi_F, u_1, \dots, u_n)}$. If $Y_F \to W$ is a projective birational morphism from a normal variety on which F is a prime divisor, F^{c} is the exceptional divisor of the blow-up of $Y_F \times \mathbb{A}^n$ at the generic point of $F \times \{0\}$, a smooth point of $Y_F \times \mathbb{A}^n$ of codimension $n+1$. Near the generic point of $F \times \{0\}$ the crepant pullback of $\varpi^* B^+$ to $Y_F \times \mathbb{A}^n$, where $\varpi : W \times \mathbb{A}^n \to W$ is the projection, has the single component $F \times \mathbb{A}^n$, with coefficient $1 - a_F$. Thus

$$a(F^{\mathrm{c}}, W \times \mathbb{A}^n, \varpi^* B^+) = n + 1 - (1 - a_F) = a_F + n, \quad \operatorname{mult}_{F^{\mathrm{c}}} u_i = 1, \tag{5.41}$$

and $\operatorname{mult}_{F^{\mathrm{c}}} \varphi = \operatorname{mult}_F \varphi$ for any non-zero rational function φ on W. The group I preserves $\mathcal{O}_F$ and the ideal $(\pi_F, u_1, \dots, u_n)$, hence the valuation of F^{c}. The residue field of F^{c} is generated over $\mathcal{O}_F / \mathfrak{m}_F$ by the classes of u_i / π_F, and for any $g' \in I$ the class of $g'^*(u_i/\pi_F) = \lambda(g') u_i / g'^* \pi_F$ is the class of u_i / π_F. Thus $I_I(F^{\mathrm{c}}) = I$.

Possibly replacing $\mathcal{X}$ with a G-invariant affine open neighborhood of 0, we may assume that $V := \mathcal{X} \times \mathbb{A}^n$ is a smooth variety and $W \times \mathbb{A}^n = (f = 0) \subset V$ is a normal hypersurface on which I acts faithfully. The restrictions of $s, h, u_1, \dots, u_n$ to $W \times \mathbb{A}^n$ are non-zero by (5.34). By Lemma 5.4(1), with $T := W \times \mathbb{A}^n$, the group I, the prime divisor F^{c} and the functions $s, h, u_1, \dots, u_n$, there exists a prime divisor $\widehat{F}$ over V with center $(0,0)$ such that $I_I(\widehat{F}) = I$,

$$\operatorname{mult}_{\widehat{F}} s = k, \quad \operatorname{mult}_{\widehat{F}} h = \operatorname{mult}_F h, \quad \text{and} \quad a(\widehat{F}, V, 0) = a(F^{\mathrm{c}}, W \times \mathbb{A}^n, 0) + \operatorname{mult}_{\widehat{F}} f.$$

Thus, by (5.41),

$$a\Big(\widehat{F}, V, \operatorname{div}(f) + \operatorname{div}(s) + \frac{1}{m} \operatorname{div}(h)\Big) = a(F^{\mathrm{c}}, W \times \mathbb{A}^n, \varpi^* B^+) = a_F + n. \tag{5.42}$$

Let r be an integer with $r \geq \max\{\deg f_0, L\}$ and $r > \max\{\operatorname{mult}_{\widehat{F}} f, \operatorname{mult}_F h\}$, and let $f_{\leq r}$ and $h_{\leq r}$ be the sums of the monomials of degree at most r in the power series f and h. Since G acts diagonally, $f_{\leq r}$ and $h_{\leq r}$ are G-semi-invariant polynomials with the characters of f and h. Since $f - f_0, h - h_0 \in (s)$ and $\deg f_0, \deg h_0 \leq r$, we have $f_{\leq r} - f_0, h_{\leq r} - h_0 \in (s)$. Since $\widehat{F}$ has center 0, $\operatorname{mult}_{\widehat{F}}(f - f_{\leq r}) \geq r + 1$ and $\operatorname{mult}_{\widehat{F}}(h - h_{\leq r}) \geq r + 1$, so $\operatorname{mult}_{\widehat{F}} f_{\leq r} = \operatorname{mult}_{\widehat{F}} f$ and $\operatorname{mult}_{\widehat{F}} h_{\leq r} = \operatorname{mult}_{\widehat{F}} h$. Regarding $\widehat{F}$ as a prime divisor over $\mathbb{A}^4 \times \mathbb{A}^n$, whose inertia group in I, log discrepancy and orders do not change (Lemma 2.12(3) for $\mathcal{X} \times \mathbb{A}^n$ at $(0,0)$ and for $\mathbb{A}^4 \times \mathbb{A}^n$ at 0, whose completions are both $\widehat{\mathcal{O}}[[u_1, \dots, u_n]]$), (5.42) implies

$$\operatorname{mld}\Big(\Big(\mathbb{A}^4 \times \mathbb{A}^n \ni 0, \operatorname{div}(f_{\leq r}) + \operatorname{div}(s) + \frac{1}{m} \operatorname{div}(h_{\leq r})\Big) \Big/ I\Big) \leq \frac{a_F + n}{e}. \tag{5.43}$$

The ideal $(f_{\leq r}, s) = (f_0, s)$ of $\mathbb{C}[x_1, x_2, x_3, s, u_1, \dots, u_n]$ defines $S \times \mathbb{A}^n$, and $f_{\leq r}, s$ is a regular sequence of I-semi-invariant polynomials vanishing at 0. By (5.40), I contains no pseudo-reflection, the quotient morphism $\mathbb{A}^4 \times \mathbb{A}^n \to (\mathbb{A}^4 \times \mathbb{A}^n)/I$ is étale in codimension 1, and the image of the locus where $\mathbb{A}^4 \times \mathbb{A}^n \to (\mathbb{A}^4 \times \mathbb{A}^n)/I$ is not étale intersects $(S \times \mathbb{A}^n)/I$ in codimension at least 2. Since S is a normal surface with at most Du Val singularities, $S \times \mathbb{A}^n$ is normal and klt, and since $S \times \mathbb{A}^n \to (S \times \mathbb{A}^n)/I$ is étale in codimension 1 by (5.40), $(S \times \mathbb{A}^n)/I$ is normal and klt. Thus $(S \times \mathbb{A}^n)/I$ and (5.40) give the hypotheses of [NS26, Section 3] and the klt hypothesis of [NS26, Theorem 6.4] for the linear action of I on $\mathbb{A}^4 \times \mathbb{A}^n$ and the regular sequence $f_{\leq r}, s$. Since $e = |I|$, the function $h_{\leq r}^e$ is I-invariant, so $(h_{\leq r}^e)$ is an ideal of the coordinate ring of $(\mathbb{A}^4 \times \mathbb{A}^n)/I$, as [NS26, Theorem 6.4] requires, and its restriction to $S \times \mathbb{A}^n$ is $(h_0^e) \neq 0$ by (5.34).

For the exponent $\frac{1}{me}$, [NS26, Theorem 6.4] states that the minimal log discrepancy at the image of 0 of $(\mathbb{A}^4 \times \mathbb{A}^n)/I$ with the images of $(f_{\le r} = 0)$ and $(s = 0)$ and the ideal $(h^e_{\le r})^{\frac{1}{me}}$ equals that of $(S \times \mathbb{A}^n)/I$ with the restriction of $(h^e_{\le r})^{\frac{1}{me}}$. Since the quotient morphisms of $\mathbb{A}^4 \times \mathbb{A}^n$ and of $S \times \mathbb{A}^n$ by I are étale in codimension 1 and $(h^e_{\le r})^{\frac{1}{me}}$ pulls back to $\frac{1}{m}\operatorname{div}(h_{\le r})$, the minimal log discrepancy on $(\mathbb{A}^4 \times \mathbb{A}^n)/I$ is the left-hand side and that on $(S \times \mathbb{A}^n)/I$ is the right-hand side of

$$\begin{aligned} &\operatorname{mld}\Big(\Big(\mathbb{A}^4 \times \mathbb{A}^n \ni 0, \operatorname{div}(f_{\le r}) + \operatorname{div}(s) + \frac{1}{m}\operatorname{div}(h_{\le r})\Big)\Big/I\Big) \\ &= \operatorname{mld}\Big(\Big(S \times \mathbb{A}^n \ni 0, \frac{1}{m}\operatorname{div}(h_0|_{S\times\mathbb{A}^n})\Big)\Big/I\Big). \end{aligned} \tag{5.44}$$

Let $\Delta := \frac{1}{m}\operatorname{div}(h_0|_{S\times\mathbb{A}^n})$, and let E be a prime divisor over $S \times \mathbb{A}^n$ with center 0 computing the right-hand side of (5.44), which exists by Lemma 2.20, since $(S \times \mathbb{A}^n, \Delta)$ is lc near 0 and $m(K_{S\times\mathbb{A}^n} + \Delta)$ is Cartier. Let

$$a_E := a(E, S \times \mathbb{A}^n, \Delta) \quad \text{and} \quad w_i := \operatorname{mult}_E u_i \ge 1 \quad \text{for } 1 \le i \le n.$$

Since

$$\Big(S \times \mathbb{A}^n, \Delta + \sum_{i=1}^n \operatorname{div}(u_i)\Big)$$

is lc near 0, $a_E \ge \sum_{i=1}^n w_i$. Thus, by (5.43) and (5.44),

$$\sum_{i=1}^n \frac{w_i}{|I_I(E)|} \le \frac{a_E}{|I_I(E)|} \le \frac{a_F + n}{e}. \tag{5.45}$$

The group I acts faithfully on $S \times \mathbb{A}^n$ since λ is injective. Since λ_E is injective, let g_E be the generator of the cyclic group $I_I(E)$ with $\lambda_E(g_E) = \exp(2\pi\sqrt{-1}/|I_I(E)|)$. By (5.38) with $\varphi = u_i$,

$$\lambda(g_E) = \exp\big(2\pi\sqrt{-1}w_i/|I_I(E)|\big) \quad \text{for any } i.$$

Write $\lambda(g_E) = \zeta_e^p$ with $p \in \{1, \dots, e\}$. Then $w_i/|I_I(E)| \in p/e + \mathbb{Z}$, so $w_i/|I_I(E)| \ge p/e$ for any i. If $g_E \ne g_F$, then $p \ge 2$ since λ is injective, and (5.45) implies $2n/e \le (a_F + n)/e$, which contradicts $n > a_F$. Thus $g_E = g_F$, so

$$I_I(E) = \langle g_F \rangle = I, \quad |I_I(E)| = e, \quad \text{and} \quad \lambda_E(g_F) = \zeta_e,$$

and, since $\sum_{i=1}^n w_i \ge n$, (5.45) implies

$$a_E - \sum_{i=1}^n w_i \le a_F. \tag{5.46}$$

Let P be the prime divisor over $(S \times \mathbb{A}^n) \times \mathbb{A}^1_s = W_0 \times \mathbb{A}^n$ whose valuation is the monomial valuation of the regular local ring $\mathcal{O}_E[s]_{(\pi_E, s)}$ of dimension 2 with value 1 at π_E and value k_+ at s. If $Y_E \to S \times \mathbb{A}^n$ is a projective birational morphism from a normal variety on which E is a prime divisor, P is the exceptional divisor of the weighted blow-up of $Y_E \times \mathbb{A}^1_s$ at the generic point of $E \times \{0\}$ with weights 1 and k_+, and the center of P on $W_0 \times \mathbb{A}^n$ is 0. Let $\varpi_0 : W_0 \times \mathbb{A}^n \to W_0$ be the projection. Near the generic point of $E \times \{0\}$, the crepant pullback of $\varpi_0^* B_0$ to $Y_E \times \mathbb{A}^1_s$ has exactly two components: $E \times \mathbb{A}^1_s$, with coefficient $1 - a_E$, and $Y_E \times \{0\}$, with coefficient β. Since $u_i \pi_E^{-w_i} \in \mathcal{O}_E^*$,

$$a(P, W_0 \times \mathbb{A}^n, \varpi_0^* B_0) = a_E + (1 - \beta)k_+ \quad \text{and} \quad \operatorname{mult}_P u_i = w_i \quad \text{for } 1 \le i \le n. \tag{5.47}$$

The group $I = I_I(E)$ preserves $\mathcal{O}_E$, the ideal (π_E) and the ideal (s), hence the valuation of P. The residue field of P is generated over $\mathcal{O}_E/\mathfrak{m}_E$, on which I acts trivially, by the class $\overline{y}$ of $s/\pi_E^{k_+}$, and by (5.39) and $\lambda_E(g_F) = \zeta_e$,

$$g_F^* \overline{y} = \chi_s(g_F)\zeta_e^{-k_+}\overline{y} = \zeta_e^{k-k_+}\overline{y} = \overline{y}.$$

Thus $I_I(P) = \langle g_F \rangle = I$.

By Lemma 3.6(1), with $X := W_0$, $x := 0$, the boundary B_0, the group I, which acts faithfully on W_0, the unit ideal, and the characters $\chi_1 = \cdots = \chi_n = \lambda$, there exist a positive integer ϱ and a prime divisor P_0 over W_0 with center 0 such that $I_I(P) \subset I_I(P_0)$ and

$$\begin{aligned}\varrho a(P_0, W_0, B_0) &\le a\Big(P, W_0 \times \mathbb{A}^n, \varpi_0^* B_0 + \sum_{i=1}^n \operatorname{div}(u_i)\Big) \\ &= a_E + (1-\beta)k_+ - \sum_{i=1}^n w_i \le a_F + (1-\beta)k_+,\end{aligned}$$

where the equality follows from (5.47) and the last inequality from (5.46). Since

$$I = I_I(P) \subset I_I(P_0) \subset I_G(P_0),$$

we have $e_{P_0} \ge e$. Since $a(P_0, W_0, B_0) > 0$ by (5.35), $\varrho \ge 1$, $k_+ \le k$, and $1 - \beta > 0$,

$$\operatorname{mld}((W_0 \ni 0, B_0)/G) \le \frac{a(P_0, W_0, B_0)}{e_{P_0}} \le \frac{\varrho a(P_0, W_0, B_0)}{e} \le \frac{a_F + (1-\beta)k_+}{e} \le \frac{a_F + (1-\beta)k}{e}.$$

Thus (5.37) holds, and (1) is proved. The proof of (2) continues below.

Proof of Theorem 5.10, continued. We now choose the integer N and prove (2). The variety $W_0 = S \times \mathbb{A}^1_s$ is klt near 0 and Jac_{W_0} is a non-zero ideal, so there exists a real number $\kappa \in (0, 1]$, depending only on f_0, such that

$$(W_0, \operatorname{Jac}_{W_0}^\kappa) \text{ is lc near } 0.$$

Let $L' := \max\{L, \deg f_0\}$. Let $\Lambda \subset \mathbb{Z}^3$ be the finite set of Theorem 5.9 for the integers L' and 2, and let $c_\Lambda \ge 0$ be the largest absolute value of an entry of a vector in Λ ($c_\Lambda := 0$ if $\Lambda = \emptyset$). We let

$$N_1 := 3M(1 + c_\Lambda), \qquad N_2 := \lceil N_1/\kappa \rceil, \quad \text{and} \quad N := N_1 + 2N_2 + 3. \tag{5.48}$$

Then N depends only on f_0, M, and L.

Since G acts faithfully on W_0, the characters of x_1, x_2, x_3, s embed G into $(\mathbb{C}^*)^4$. Let

$$\Delta_0 := \frac{1}{m}\operatorname{div}(h_0|_{W_0}) + \operatorname{div}(s|_{W_0}), \quad \text{so that} \quad B_0 = \Delta_0 - (1-\beta)\operatorname{div}(s|_{W_0}).$$

A resolution of W_0 with a finite table. Since x_1, x_2, x_3, s are G-semi-invariant and $h - h_0 \in (s)$, the polynomial h_0 is G-semi-invariant with the character of h. By Theorem 5.9 for W_0, G, $h_1 := h_0$, $h_2 := s$, $c_1 := \frac{\sqrt{2}}{2m}$, and $c_2 := \frac{\sqrt{2}}{2}$, there exist a G-equivariant projective birational morphism $g : Y \to W_0$ from a smooth variety Y and a G-invariant reduced simple normal crossing divisor T_Y on Y satisfying the following.

(1) $g^{-1}(0)$ is a divisor and $T_Y \supset \operatorname{Supp} C_Y \cup g^{-1}(0)$, where $K_Y + C_Y := g^*(K_{W_0} + \frac{\sqrt{2}}{2}\Delta_0)$.
(2) For any closed point $y \in Y$, the stabilizer G_y preserves any local branch of T_Y through y.
(3) For any prime divisor E on Y contained in $\operatorname{Supp} C_Y \cup g^{-1}(0)$, we have

$$(1 - a(E, W_0, 0), \operatorname{mult}_E h_0, \operatorname{mult}_E s) \in \Lambda.$$

Let E be a prime divisor on Y. Since W_0 is a hypersurface, K_{W_0} is Cartier, so $a(E, W_0, 0) \in \mathbb{Z}$, and

$$\operatorname{mult}_E C_Y = \big(1 - a(E, W_0, 0)\big) + \frac{\sqrt{2}}{2m}\big(\operatorname{mult}_E h_0 + m \operatorname{mult}_E s\big).$$

Since $\sqrt{2} \notin \mathbb{Q}$ and $\operatorname{mult}_E h_0, \operatorname{mult}_E s \ge 0$, $\operatorname{mult}_E C_Y = 0$ if and only if $a(E, W_0, 0) = 1$ and $\operatorname{mult}_E h_0 = \operatorname{mult}_E s = 0$. Thus the supports of $K_Y - g^* K_{W_0}$, $g^* \operatorname{div}(h_0|_{W_0})$, and $g^* \operatorname{div}(s|_{W_0})$ are contained in $\operatorname{Supp} C_Y \subset T_Y$, and for any prime divisor $E \subset T_Y$,

$$r_E := \big(1 - a(E, W_0, 0), \operatorname{mult}_E h_0, \operatorname{mult}_E s\big) \in \Lambda \cup \{0\}. \tag{5.49}$$

We let B_Y be the $\mathbb{R}$-divisor with $K_Y + B_Y = g^*(K_{W_0} + B_0)$. Then $\operatorname{Supp} B_Y \subset T_Y$, and for any prime divisor $E \subset T_Y$ with $0 \in g(E)$ and $r_E = (r_0, r_1, r_2)$,

$$\operatorname{mult}_E B_Y = r_0 + \frac{r_1}{m} + \beta r_2 = 1 - a(E, W_0, \Delta_0) - (1-\beta)\operatorname{mult}_E s \le 1,$$

where the inequality holds since (W_0, Δ_0) is lc near 0 and $\beta < 1$.

Local charts. Fix a closed point $y \in g^{-1}(0)$. Since G is abelian, G_y is abelian. Since G_y preserves any local branch of T_Y through y, there exist G_y-eigenparameters y_1, y_2, y_3 of Y at y (Definition 2.3) such that any local branch of T_Y through y is $E_i := (y_i = 0)$ near y for some i. Let χ_i be the character of y_i, that is, $\sigma^* y_i = \chi_i(\sigma) y_i$ for any $\sigma \in G_y$. If E_i is a branch of T_Y, let $r_{E_i} = (r_{i,0}, r_{i,1}, r_{i,2})$ be the vector (5.49) of the component of T_Y containing E_i. Otherwise, let $(r_{i,0}, r_{i,1}, r_{i,2}) := 0$. Let

$$d_i := r_{i,0} + \frac{r_{i,1}}{m} + \beta r_{i,2} \le 1 \quad \text{and} \quad \psi_y(u) := \sum_{i=1}^{3} (1 - d_i) u_i \ \text{ for } u \in \mathbb{R}^3.$$

Then $\psi_y \ge 0$ on $\mathbb{R}^3_{\ge 0}$. Near y we have

$$K_Y - g^* K_{W_0} = -\sum_{i=1}^{3} r_{i,0} E_i, \quad g^* \operatorname{div}(h_0|_{W_0}) = \sum_{i=1}^{3} r_{i,1} E_i, \quad \text{and} \quad g^* \operatorname{div}(s|_{W_0}) = \sum_{i=1}^{3} r_{i,2} E_i.$$

Let P be a prime divisor over Y whose center on Y contains y, and let $w = (w_1, w_2, w_3)$ with $w_i := \operatorname{mult}_P y_i$. Then

$$\begin{aligned} a(P, W_0, 0) &= a(P, Y, 0) - \sum_{i=1}^{3} r_{i,0} w_i, \\ \operatorname{mult}_P h_0 &= \sum_{i=1}^{3} r_{i,1} w_i, \quad \text{and} \quad \operatorname{mult}_P s = \sum_{i=1}^{3} r_{i,2} w_i. \end{aligned} \tag{5.50}$$

Moreover, $a(P, W_0, B_0) = a(P, Y, B_Y) = a(P, Y, 0) - \sum_i d_i w_i$. Since $(Y, E_1 + E_2 + E_3)$ is lc near y, $a(P, Y, 0) \ge w_1 + w_2 + w_3$, with equality if mult_P is the monomial valuation in y_1, y_2, y_3 with weights w. Hence

$$a(P, W_0, B_0) \ge \psi_y(w), \quad \text{with equality if } \operatorname{mult}_P \text{ is monomial in } y_1, y_2, y_3. \tag{5.51}$$

For $\alpha \in \mathbb{Z}^3$ write $y^\alpha := \prod_i y_i^{\alpha_i}$ and $\chi^\alpha := \prod_i \chi_i^{\alpha_i}$. Let

$$\mathcal{K}_y := \{\alpha \in \mathbb{Z}^3 \mid \chi^\alpha = 1\}.$$

Since G acts faithfully on W_0 and g is G-equivariant and birational, G acts faithfully on Y, and since $\widehat{\mathcal{O}}_{Y,y} = \mathbb{C}[[y_1, y_2, y_3]]$, the homomorphism

$$G_y \to (\mathbb{C}^*)^3, \quad \sigma \to (\chi_i(\sigma))_i,$$

is injective. Thus $\alpha \to \chi^\alpha$ maps $\mathbb{Z}^3$ onto the character group $\widehat{G}_y$ of G_y, with kernel $\mathcal{K}_y$. Let $N_{G_y} \subset \mathbb{R}^3$ be the lattice of $G_y \subset (\mathbb{C}^*)^3$ (Definition 2.3), that is,

$$N_{G_y} = \{u \in \mathbb{R}^3 \mid \langle \alpha, u \rangle \in \mathbb{Z} \text{ for any } \alpha \in \mathcal{K}_y\}.$$

Since $\chi_i^{|G_y|} = 1$, we have

$$|G_y| \mathbb{Z}^3 \subset \mathcal{K}_y \subset \mathbb{Z}^3, \quad \text{hence} \quad \mathbb{Z}^3 \subset N_{G_y} \subset \frac{1}{|G_y|} \mathbb{Z}^3.$$

Let

$$\mathcal{U}_y := \{u \in N_{G_y} \cap [0,1]^3 \mid u_i > 0 \text{ for some } i \text{ with } E_i \subset g^{-1}(0) \text{ near } y\}.$$

Lower bound. Let P be a prime divisor over W_0 with center 0, and $e_P := |I_G(P)|$ (Definition 2.2). Then $\operatorname{center}_Y P \subset g^{-1}(0)$. Let y be a closed point of $\operatorname{center}_Y P$ and $w := (\operatorname{mult}_P y_i)_i$. The group

$I_G(P)$ acts trivially on the residue field of P, so it fixes $\mathrm{center}_Y P$ pointwise. Thus $I_G(P) \subset G_y$, so $I_{G_y}(P) = I_G(P)$, and P has ramification index $|I_{G_y}(P)| = e_P$ over the image of P on Y/G_y (Lemma 2.15(1) for the action of G_y on Y). For $\alpha \in \mathcal{K}_y$, the rational function y^α is G_y-invariant, hence the pullback of a rational function on Y/G_y, so $\langle \alpha, w\rangle = \mathrm{mult}_P(y^\alpha) \in e_P\mathbb{Z}$, that is,

$$u := \frac{w}{e_P} \in N_{G_y} \cap \mathbb{R}^3_{\geq 0}.$$

Since $\mathrm{center}_Y P$ is irreducible and contained in $g^{-1}(0) \subset T_Y$, there exists i with $\mathrm{center}_Y P \subset E_i \subset g^{-1}(0)$ near y, and then $u_i > 0$. Let $u'_i := u_i - \lceil u_i \rceil + 1$ if $u_i > 0$ and $u'_i := 0$ if $u_i = 0$. Then $u - u' \in \mathbb{Z}^3_{\geq 0} \subset N_{G_y}$, so $u' \in \mathcal{U}_y$ and $u' \leq u$ componentwise. By (5.51) and $1 - d_i \geq 0$,

$$\frac{a(P, W_0, B_0)}{e_P} \geq \psi_y(u) \geq \psi_y(u'). \tag{5.52}$$

Toric competitors. Conversely, let $y \in g^{-1}(0)$ be a closed point and $u \in \mathcal{U}_y$. Let $w \in \mathbb{Z}^3_{\geq 0}$ be the primitive vector with $u \in \mathbb{R}_{>0}w$, and write $u = tw$ with $t > 0$. Let v_w be the monomial valuation in y_1, y_2, y_3 with weights w, that is,

$$v_w(\phi) = \min\{\langle \alpha, w\rangle \mid a_\alpha \neq 0\} \quad \text{for } \phi = \sum_\alpha a_\alpha y^\alpha \in \mathbb{C}[[y_1, y_2, y_3]],$$

and let Q be the prime divisor over Y with $\mathrm{mult}_Q = v_w$. Then $\mathrm{center}_Y Q$ is the closure of the component through y of $\bigcap_{w_i>0} E_i$. Since $u_i > 0$ for some i with $E_i \subset g^{-1}(0)$ near y, $\mathrm{center}_Y Q \subset g^{-1}(0)$, so Q has center 0 on W_0. By (5.51),

$$a(Q, W_0, B_0) = \psi_y(w).$$

Let e_w be the positive integer with

$$\{\langle \alpha, w\rangle \mid \alpha \in \mathcal{K}_y\} = e_w\mathbb{Z}.$$

It exists since $|G_y|\mathbb{Z}^3 \subset \mathcal{K}_y$ and $w \neq 0$. Since $u \in N_{G_y}$, $te_w \in \mathbb{N}^+$, so $t \geq \frac{1}{e_w}$. Let

$$H_w := \{\sigma \in G_y \mid \text{there exists } \zeta \in \mathbb{C}^* \text{ with } \chi_i(\sigma) = \zeta^{w_i} \text{ for } i = 1, 2, 3\}.$$

For $\sigma \in H_w$, σ^* multiplies any monomial y^α by $\zeta^{\langle \alpha, w\rangle}$. Hence for any ϕ with $v_w(\phi) = \delta$, $v_w(\sigma^*\phi - \zeta^\delta\phi) > \delta$. Thus σ preserves v_w and acts trivially on the residue field of Q, and $H_w \subset I_G(Q)$. Let

$$w^\perp := \{\alpha \in \mathbb{Z}^3 \mid \langle \alpha, w\rangle = 0\}.$$

Since w is primitive, $\alpha \to \langle \alpha, w\rangle$ maps $\mathbb{Z}^3$ onto $\mathbb{Z}$ with kernel $w^\perp$. For $\sigma \in G_y$, the homomorphism $\alpha \to \chi^\alpha(\sigma)$ factors through $\alpha \to \langle \alpha, w\rangle$ if and only if it is trivial on $w^\perp$. Thus H_w is the annihilator in G_y of the image of $w^\perp$ in $\widehat{G}_y \cong \mathbb{Z}^3/\mathcal{K}_y$, and

$$|H_w| = [\mathbb{Z}^3 : w^\perp + \mathcal{K}_y] = [\mathbb{Z} : e_w\mathbb{Z}] = e_w.$$

Therefore $e_Q \geq e_w$,

$$\frac{a(Q, W_0, B_0)}{e_Q} \leq \frac{\psi_y(w)}{e_w} \leq t\psi_y(w) = \psi_y(u), \tag{5.53}$$

and

$$w_i = \frac{u_i}{t} \leq e_w = |H_w| \leq |G_y| \leq M \quad \text{for } i = 1, 2, 3.$$

A bounded computing divisor. By (5.52) and (5.53),

$$\mathrm{mld}((W_0 \ni 0, B_0)/G) = \min\{\psi_y(u) \mid y \in g^{-1}(0) \text{ a closed point}, \ u \in \mathcal{U}_y\}.$$

The minimum exists: $\mathcal{U}_y \subset \frac{1}{M!}\mathbb{Z}^3 \cap [0,1]^3$, and ψ_y is determined by $(r_{i,0}, r_{i,1}, r_{i,2})_{i=1,2,3} \in (\Lambda \cup \{0\})^3$, so the right-hand side is a minimum over a finite set of pairs (rows, vector). Let (y, u) attain the minimum, and let Q and w be as in (5.53). Then

$$\frac{a(Q, W_0, B_0)}{e_Q} \le \psi_y(u) = \operatorname{mld}((W_0 \ni 0, B_0)/G) \le \frac{a(Q, W_0, B_0)}{e_Q},$$

so Q computes $\operatorname{mld}((W_0 \ni 0, B_0)/G)$. Since $\operatorname{mult}_Q y_i = w_i \le M$, $a(Q, Y, 0) = w_1 + w_2 + w_3$, and $|r_{i,k}| \le c_\Lambda$ for $k = 0, 1, 2$, by (5.50) and (5.48) we have

$$a(Q, W_0, 0) = \sum_{i=1}^{3} (1 - r_{i,0}) w_i \le N_1, \qquad \operatorname{mult}_Q h_0 \le N_1, \quad \text{and} \quad \operatorname{mult}_Q s \le N_1. \tag{5.54}$$

Since $(W_0, \operatorname{Jac}_{W_0}^{\kappa})$ is lc near 0, $\kappa \operatorname{mult}_Q \operatorname{Jac}_{W_0} \le a(Q, W_0, 0)$, so

$$j := \operatorname{mult}_Q \operatorname{Jac}_{W_0} \le \frac{N_1}{\kappa} \le N_2.$$

Transport to W. Assume now that $f - f_0 \in \mathfrak{n}^N$ and $h - h_0 \in \mathfrak{n}^N$. Since $\partial f_0/\partial s = 0$, there exists $x \in \{x_1, x_2, x_3\}$ with $\operatorname{mult}_Q(\partial f_0/\partial x|_{W_0}) = j$ (Definition 5.2). Since Q has center 0, $\operatorname{mult}_Q \mathfrak{n} \ge 1$. By (5.48) and (5.54),

$$\operatorname{mult}_Q\big((f - f_0)|_{W_0}\big) \ge N > \max\{2j,\ j + \operatorname{mult}_Q h_0,\ j + \operatorname{mult}_Q s\},$$

and the restrictions to W_0 of the four partial derivatives of $f - f_0$ have order at least $N - 1 > j$ along Q. Both W_0 and W are normal algebraic hypersurface germs at 0, and f has the character of f_0. By Lemma 5.4(3) with f_0, f, Q, and (h_0, s) in place of f, f', F, and (H_1, H_2), and by Lemma 2.12(3), there exists a prime divisor F over W with center 0 such that

$$I_G(F) = I_G(Q), \quad a(F, W, 0) = a(Q, W_0, 0), \quad \operatorname{mult}_F \operatorname{Jac}_W = j,$$
$$\operatorname{mult}_F h_0 = \operatorname{mult}_Q h_0, \quad \text{and} \quad \operatorname{mult}_F s = \operatorname{mult}_Q s.$$

Since $h - h_0 \in \mathfrak{n}^N$, $\operatorname{mult}_F(h - h_0) \ge N > N_1 \ge \operatorname{mult}_F h_0$, hence $\operatorname{mult}_F h = \operatorname{mult}_Q h_0$. Thus $e_F = e_Q$ and

$$\begin{aligned} a(F, W, B) &= a(F, W, 0) - \frac{1}{m} \operatorname{mult}_F h - \beta \operatorname{mult}_F s \\ &= a(Q, W_0, 0) - \frac{1}{m} \operatorname{mult}_Q h_0 - \beta \operatorname{mult}_Q s = a(Q, W_0, B_0). \end{aligned}$$

By the first inequality of (1),

$$\begin{aligned} \operatorname{mld}((W \ni 0, B)/G) \le \frac{a(F, W, B)}{e_F} &= \frac{a(Q, W_0, B_0)}{e_Q} \\ &= \operatorname{mld}((W_0 \ni 0, B_0)/G) \le \operatorname{mld}((W \ni 0, B)/G). \end{aligned}$$

Hence $\operatorname{mld}((W \ni 0, B)/G) = \operatorname{mld}((W_0 \ni 0, B_0)/G) = a(F, W, B)/e_F$. By (5.54) and (5.48),

$$a(F, W, 0) \le N_1 < N, \quad \operatorname{mult}_F h \le N_1 < N,$$
$$\operatorname{mult}_F \operatorname{Jac}_W = j \le N_2 < N, \quad \text{and} \quad \operatorname{mult}_F s \le N_1 < N.$$

This proves (2). □

6. Multiplier ideals, bounded contact and degenerations

6.1. Multiplier ideals and bounded contact.

Lemma 6.1. *Let n and N be positive integers, let $\mathbf{f}_i = (f_{i,1}, \dots, f_{i,N})$, $i \in \mathbb{N}^+$, be N-tuples of non-zero elements of $\mathbb{C}[[x_1, \dots, x_n]]$, and let $\mathbf{f}_* = (f_{*,1}, \dots, f_{*,N})$ be a compatible generic limit of the $\mathbf{f}_i$ along an infinite subset $\mathcal{I} \subset \mathbb{N}^+$, with coefficient field $\Bbbk$ (Definition 2.5 with H trivial). Let $\mathfrak{m}$ be the maximal ideal of $\mathbb{C}[[x_1, \dots, x_n]]$, let $X_* := \operatorname{Spec} \Bbbk[[x_1, \dots, x_n]]$, and let $\mathfrak{n}_*$ be the maximal ideal of $\Bbbk[[x_1, \dots, x_n]]$. For an $\mathbb{R}$-ideal $\mathfrak{a}$ on a smooth variety over a field containing $\mathbb{C}$, or on*

the spectrum of a formal power series ring over such a field, $\mathcal{J}(\mathfrak{a})$ denotes the multiplier ideal of $\mathfrak{a}$ [Laz04, Definition 9.2.3 and Generalization 9.2.8], [JM12, Section 1.3]. *For a polynomial Φ with complex coefficients in the variables $y_{k,\alpha}$ ($1 \le k \le N$, $\alpha \in \mathbb{N}^n$), $u_1, \dots, u_q$ ($q \ge 0$), and $x_1, \dots, x_n$, an N-tuple $\mathbf{g} = (g_1, \dots, g_N)$ of power series in $x_1, \dots, x_n$, and $\beta \in \mathbb{C}^q$ or $\beta \in \Bbbk^q$, let $\Phi(\mathbf{g}, \beta)$ be the power series obtained by substituting $\partial^\alpha g_k$ for $y_{k,\alpha}$ and β for $(u_1, \dots, u_q)$.*

(1) Let $\gamma_1, \dots, \gamma_N, \gamma'_1, \dots, \gamma'_N$ be rational numbers with $0 < \gamma_k < \gamma'_k$ for any k, let L be a positive integer, let $\Phi_1, \dots, \Phi_m$ be polynomials with complex coefficients in $y_{k,\alpha}$, $u_1, \dots, u_q$, and $x_1, \dots, x_n$, and let $\beta \in \Bbbk^q$. Assume that

$$\Phi_j(\mathbf{f}_*, \beta) \in \mathcal{J}\Big(\prod_k (f_{*,k})^{\gamma'_k}\Big) + \mathfrak{n}_*^L$$

for any j. Then for all but finitely many $i \in \mathcal{I}$, there exists $\beta_i \in \mathbb{C}^q$ such that the elements $z_{j,i} := \Phi_j(\mathbf{f}_i, \beta_i)$ satisfy (2) below. If moreover for each i there exist a smooth algebraic germ $(Y_i \ni y_i)$ of dimension n and an isomorphism $\widehat{\mathcal{O}}_{Y_i,y_i} \cong \mathbb{C}[[x_1, \dots, x_n]]$ under which each $f_{i,k}$ is a unit times the image of a regular function on Y_i, then

$$z_{j,i} \in \mathcal{J}\Big(\prod_k (f_{i,k})^{\gamma_k}\Big) + \mathfrak{m}^L$$

for any j, where the multiplier ideal is taken on $\operatorname{Spec} \mathbb{C}[[x_1, \dots, x_n]]$. If moreover $\Psi_1, \dots, \Psi_{m'}$ are polynomials with complex coefficients in $y_{k,\alpha}$, $u_1, \dots, u_q$, and $x_1, \dots, x_n$, and c and P are positive integers with $L \ge c+1$ and

$$\mathfrak{n}_*^c \subset \big(\Psi_1(\mathbf{f}_*, \beta), \dots, \Psi_{m'}(\mathbf{f}_*, \beta)\big) + \big(\Phi_1(\mathbf{f}_*, \beta), \dots, \Phi_m(\mathbf{f}_*, \beta)\big)^P,$$

then the β_i can be taken such that moreover

$$\mathfrak{m}^c \subset \big(\Psi_1(\mathbf{f}_i, \beta_i), \dots, \Psi_{m'}(\mathbf{f}_i, \beta_i)\big) + (z_{1,i}, \dots, z_{m,i})^P,$$

so that, if for each i there exist a smooth algebraic germ $(Y_i \ni y_i)$ of dimension n and an isomorphism $\widehat{\mathcal{O}}_{Y_i,y_i} \cong \mathbb{C}[[x_1, \dots, x_n]]$ under which each $f_{i,k}$ is a unit times the image of a regular function on Y_i, then

$$\mathfrak{m}^c \subset \big(\Psi_1(\mathbf{f}_i, \beta_i), \dots, \Psi_{m'}(\mathbf{f}_i, \beta_i)\big) + \mathcal{J}\Big(\prod_k (f_{i,k})^{\gamma_k}\Big)^P.$$

Finally, given an integer d' and a finite set $\mathcal{Q}$ of polynomials with complex coefficients in the coefficients of the d'-jets of N-tuples of power series and in $u_1, \dots, u_q$, the β_i can be taken such that for any $Q \in \mathcal{Q}$, $Q(\operatorname{jet}_{d'}(\mathbf{f}_i), \beta_i) = 0$ if and only if $Q(\operatorname{jet}_{d'}(\mathbf{f}_*), \beta) = 0$.

(2) In (1), for any i and j there, any prime divisor F over $\operatorname{Spec} \mathbb{C}[[x_1, \dots, x_n]]$ with center the closed point and $\operatorname{mult}_F(\mathfrak{m}^L) > \operatorname{mult}_F z_{j,i}$ satisfies

$$\operatorname{mult}_F z_{j,i} + a(F, \operatorname{Spec} \mathbb{C}[[x_1, \dots, x_n]], 0) > \sum_k \gamma_k \operatorname{mult}_F f_{i,k}.$$

If the $f_{i,k}$ are regular functions near $0 \in \mathbb{A}^n$, then any prime divisor F over $\mathbb{A}^n$ with center 0 and $\operatorname{mult}_F((x_1, \dots, x_n)^L) > \operatorname{mult}_F z_{j,i}$ satisfies

$$\operatorname{mult}_F z_{j,i} + a(F, \mathbb{A}^n, 0) > \sum_k \gamma_k \operatorname{mult}_F f_{i,k}.$$

(3) Assume that $n = 4$. Let $f_*, h_* \in \Bbbk[[x_1, \dots, x_4]]$ be power series such that the hypersurface

$$W_* := (f_* = 0) \subset X_*$$

is normal and $h_*|_{W_*} \ne 0$, let $\mu > 0$ and $\theta_2 \in (0,1)$ be real numbers, and let $V \subset W_*$ be a closed subset, containing the closed point if W_* is not regular, such that $(W_*, \mu \operatorname{div}(h_*|_{W_*}))$ is lc on $W_* \backslash V$. Then $\mathcal{J}(f_*^{\theta_2} h_*^{\theta_2 \mu})$ is the unit ideal near any point of $W_* \backslash V$.

(4) *Let $\lambda \in \Bbbk^{N'}$, let ℓ be a positive integer, let $\mathfrak{b} \subset \Bbbk[x_1, \dots, x_n]$ be an ideal containing $(x_1, \dots, x_n)^\ell$ and generated by polynomials whose coefficients are the values at λ of polynomials with complex coefficients, let $\phi_1, \dots, \phi_r \in \Bbbk[x_1, \dots, x_n]$ be polynomials whose coefficients are such values, and let $\gamma > 0$ be a rational number such that $\phi_j \in \mathcal{J}(\mathfrak{b}^\gamma)$ near $0 \in \mathbb{A}^n_\Bbbk$ for any j. Then there exists a polynomial Θ with complex coefficients in N' variables with $\Theta(\lambda) \neq 0$ such that, for any closed point λ' of the Zariski closure $T \subset \mathbb{A}^{N'}$ of λ with $\Theta(\lambda') \neq 0$, the specializations $\mathfrak{b}_{\lambda'}$ and $\phi_{j,\lambda'}$ obtained by evaluating the given polynomials at λ' satisfy, for any j,*

$$\phi_{j,\lambda'} \in \mathcal{J}(\mathfrak{b}_{\lambda'}^\gamma) \quad \text{near } 0 \in \mathbb{A}^n.$$

Proof. (1) and (2). We let $a_1, \dots, a_N$ and M be positive integers such that $\gamma'_k = a_k/M$ for any k, let t be a rational number such that

$$\max_k \frac{\gamma_k}{a_k} < t < \frac{1}{M},$$

let $\epsilon := \frac{1}{M} - t$, and let ℓ be a positive integer such that $\lfloor \ell\epsilon \rfloor - n + 1 \geq L$. Let

$$\mathfrak{a}_* := \Big(\prod_k f_{*,k}^{a_k}\Big) \subset \Bbbk[[x_1, \dots, x_n]], \quad \mathfrak{a}_i := \Big(\prod_k f_{i,k}^{a_k}\Big) \subset \mathbb{C}[[x_1, \dots, x_n]],$$

$$\text{and} \quad N_0 := \max\Big\{0, \Big\lfloor \frac{\ell}{M} \Big\rfloor - n + 1\Big\}.$$

Since $\epsilon < 1/M$,

$$N_0 \geq \lfloor \ell\epsilon \rfloor - n + 1 \geq L.$$

Since multiplier ideals are monotone in the ideal and in the exponents, and $\mathcal{J}(\mathfrak{n}_*^{\ell/M}) = \mathfrak{n}_*^{N_0}$ and $\mathcal{J}((x_1, \dots, x_n)^{\ell/M}) = (x_1, \dots, x_n)^{N_0}$ near $0 \in \mathbb{A}^n_\Bbbk$ [Laz04, Proposition 9.2.32(iii) and Example 9.2.14], we have

$$\mathcal{J}\Big(\prod_k (f_{*,k})^{\gamma'_k}\Big) = \mathcal{J}(\mathfrak{a}_*^{1/M}) \subset \mathcal{J}\big((\mathfrak{a}_* + \mathfrak{n}_*^\ell)^{1/M}\big),$$

and the hypothesis gives

$$\Phi_j(\mathbf{f}_*, \beta) \in \mathcal{J}\big((\mathfrak{a}_* + \mathfrak{n}_*^\ell)^{1/M}\big) + \mathfrak{n}_*^L \quad \text{and} \quad \mathfrak{n}_*^{N_0} \subset \mathcal{J}\big((\mathfrak{a}_* + \mathfrak{n}_*^\ell)^{1/M}\big). \tag{6.1}$$

We let D_0 be the largest $|\alpha|$ for which a variable $y_{k,\alpha}$ occurs in some Φ_j or Ψ_j ($D_0 := 0$ if no such variable occurs), let $c := 0$ if no Ψ_j are given, and let $d := \max\{\ell + N_0 + c + D_0, d'\}$ ($d' := 1$ if no $\mathcal{Q}$ is given). For a power series g, let $g^{<d}$ be the sum of the terms of g of degree less than d, and let $\mathbf{g}^{<d} := (g_1^{<d}, \dots, g_N^{<d})$, so that $\mathrm{jet}_d(\mathbf{g})$ is the tuple of coefficients of $\mathbf{g}^{<d}$. Since

$$\partial^\alpha g - \partial^\alpha(g^{<d}) \in (x_1, \dots, x_n)^{d-|\alpha|},$$

for $\mathbf{g} = \mathbf{f}_*$ or $\mathbf{g} = \mathbf{f}_i$ and for any β' we have

$$\begin{gathered}\Phi_j(\mathbf{g}^{<d}, \beta') - \Phi_j(\mathbf{g}, \beta'),\ \Psi_j(\mathbf{g}^{<d}, \beta') - \Psi_j(\mathbf{g}, \beta') \in (x_1, \dots, x_n)^{N_0+c+1}, \\ \prod_k (g_k^{<d})^{a_k} - \prod_k g_k^{a_k} \in (x_1, \dots, x_n)^\ell.\end{gathered} \tag{6.2}$$

Let

$$\mathfrak{b} := \Big(\prod_k (f_{*,k}^{<d})^{a_k}\Big) + (x_1, \dots, x_n)^\ell \subset \Bbbk[x_1, \dots, x_n], \quad \phi_j := \Phi_j(\mathbf{f}_*^{<d}, \beta), \quad \text{and} \quad \psi_j := \Psi_j(\mathbf{f}_*^{<d}, \beta).$$

By (6.2) for the first relation, and by (6.1) and (6.2) for the second,

$$\mathfrak{b}\Bbbk[[x_1, \dots, x_n]] = \mathfrak{a}_* + \mathfrak{n}_*^\ell \quad \text{and} \quad \phi_j \in \mathcal{J}\big((\mathfrak{a}_* + \mathfrak{n}_*^\ell)^{1/M}\big) + \mathfrak{n}_*^L.$$

A log resolution near the closed point of a smooth algebraic germ pulls back to a log resolution over the completion of the germ, and the relative canonical divisor pulls back to the relative canonical divisor [dFEM11, Proposition A.14], so by flat base change multiplier ideals commute with the

completions $\mathcal{O}_{\mathbb{A}^n_{\Bbbk},0} \to \Bbbk[[x_1,\dots,x_n]]$, $\mathcal{O}_{\mathbb{A}^n,0} \to \mathbb{C}[[x_1,\dots,x_n]]$, and $\mathcal{O}_{Y_i,y_i} \to \mathbb{C}[[x_1,\dots,x_n]]$ for $(Y_i \ni y_i)$ as in (1). Moreover, the completions are faithfully flat. Thus

$$\phi_j \in \mathcal{J}(\mathfrak{b}^{1/M}) + (x_1,\dots,x_n)^L \quad \text{near } 0 \in \mathbb{A}^n_{\Bbbk}.$$

Since $N_0 \geq L$ and

$$\mathcal{J}(\mathfrak{b}^{1/M}) \supset \mathcal{J}\big((x_1,\dots,x_n)^{\ell/M}\big) = (x_1,\dots,x_n)^{N_0} \quad \text{near } 0,$$

cutting the Taylor expansion at 0 of the summand in $(x_1,\dots,x_n)^L$ at degree N_0, we may write

$$\phi_j = \phi'_j + \phi''_j,$$

where $\phi'_j, \phi''_j \in \Bbbk[x_1,\dots,x_n]$, $\phi'_j \in \mathcal{J}(\mathfrak{b}^{1/M})$ near 0, and every monomial of ϕ''_j has degree at least L and less than N_0. For the moreover statement of (1), the hypothesis modulo $\mathfrak{n}_*^{c+1}$ and (6.2) imply that for any monomial x^ν of degree c there exist polynomials $\omega_{\nu,j}, \omega'_{\nu,\rho} \in \Bbbk[x_1,\dots,x_n]$ with

$$x^\nu - \sum_{j=1}^{m'} \omega_{\nu,j}\psi_j - \sum_{\rho\in\mathbb{N}^m,|\rho|=P} \omega'_{\nu,\rho} \prod_{j=1}^{m} \phi_j^{\rho_j} \in (x_1,\dots,x_n)^{c+1}. \tag{6.3}$$

We let $\lambda \in \Bbbk^{N'}$ be the tuple of the coefficients of $\mathbf{f}_*^{<d}$, the entries of β, and the coefficients of the ϕ''_j, $\omega_{\nu,j}$, and $\omega'_{\nu,\rho}$. The generators $\prod_k (f_{*,k}^{<d})^{a_k}$ and x^ν $(|\nu| = \ell)$ of $\mathfrak{b}$ and the polynomials $\phi'_j = \Phi_j(\mathbf{f}_*^{<d},\beta) - \phi''_j$ have coefficients which are values at λ of polynomials with complex coefficients. Let Θ be as in (4) for $\mathfrak{b}$, $\phi'_1,\dots,\phi'_m$, and $\gamma = 1/M$, and let $T \subset \mathbb{A}^{N'}$ be the Zariski closure of λ. For $Q \in \mathcal{Q}$, since $d \geq d'$, substituting for the coefficients of the d'-jets and for $u_1,\dots,u_q$ the corresponding coordinates of $\mathbb{A}^{N'}$ (those of the terms of $\mathbf{f}_*^{<d}$ of degree less than d', and those of β) gives a polynomial $\widetilde{Q}$ with complex coefficients in N' variables with

$$\widetilde{Q}(\lambda) = Q(\mathrm{jet}_{d'}(\mathbf{f}_*),\beta).$$

Let

$$\Theta' := \Theta \cdot \prod_{Q\in\mathcal{Q},\ \widetilde{Q}(\lambda)\neq 0} \widetilde{Q},$$

so that $\Theta'(\lambda) \neq 0$. The projection of $\mathbb{A}^{N'}$ to the coordinates corresponding to $\mathbf{f}_*^{<d}$ maps λ to $\mathrm{jet}_d(\mathbf{f}_*)$, which maps to the generic point of Z_d (Definition 2.5), so it restricts to a dominant morphism $\tau : T \to Z_d$. By Chevalley's theorem, $\tau(T \cap (\Theta' \neq 0))$ contains a non-empty open subset Z_d° of Z_d. By Definition 2.5, $\mathrm{jet}_d(\mathbf{f}_i) \in Z_d^\circ$ for all but finitely many $i \in \mathcal{I}$. For such i, the scheme

$$T_i := \tau^{-1}(\mathrm{jet}_d(\mathbf{f}_i)) \cap (\Theta' \neq 0)$$

is non-empty and of finite type over $\mathbb{C}$. We let $\xi \in \mathbb{C}^{N'}$ be a closed point of T_i, and let $\beta_i \in \mathbb{C}^q$ be the coordinates of ξ corresponding to β. We write $\mathfrak{b}_\xi$, $\phi_{j,\xi}$, $\phi'_{j,\xi}$, $\phi''_{j,\xi}$, $\psi_{j,\xi}$, $\omega_{\nu,j,\xi}$, and $\omega'_{\nu,\rho,\xi}$ for the polynomials obtained by evaluating at ξ. Then $\phi_{j,\xi} = \Phi_j(\mathbf{f}_i^{<d},\beta_i)$, $\psi_{j,\xi} = \Psi_j(\mathbf{f}_i^{<d},\beta_i)$, and $\phi''_{j,\xi} \in (x_1,\dots,x_n)^L$, so by (4) and (6.2),

$$\begin{gathered} \phi'_{j,\xi} \in \mathcal{J}(\mathfrak{b}_\xi^{1/M}) \text{ near } 0 \in \mathbb{A}^n, \quad \mathfrak{b}_\xi\mathbb{C}[[x_1,\dots,x_n]] = \mathfrak{a}_i + \mathfrak{m}^\ell, \\ z_{j,i} - \phi_{j,\xi} \in \mathfrak{m}^{N_0+c+1}, \quad \text{and} \quad z_{j,i} - \phi'_{j,\xi} \in \mathfrak{m}^L. \end{gathered} \tag{6.4}$$

For $Q \in \mathcal{Q}$ with $Q(\mathrm{jet}_{d'}(\mathbf{f}_*),\beta) = 0$, $\widetilde{Q}$ vanishes at λ, hence on T, so

$$Q(\mathrm{jet}_{d'}(\mathbf{f}_i),\beta_i) = \widetilde{Q}(\xi) = 0.$$

For the other $Q \in \mathcal{Q}$, $\widetilde{Q}$ divides Θ', so

$$Q(\mathrm{jet}_{d'}(\mathbf{f}_i),\beta_i) = \widetilde{Q}(\xi) \neq 0.$$

Assume that $(Y_i \ni y_i)$ is as in (1). We let $\widetilde{f}_{i,k}$ be regular functions on Y_i whose images in $\mathbb{C}[[x_1,\dots,x_n]]$ are unit multiples of the $f_{i,k}$, let $\mathfrak{m}_{y_i}$ be the maximal ideal of $\mathcal{O}_{Y_i,y_i}$, and let

$$\widetilde{\mathfrak{a}}_i := \Big(\prod_k \widetilde{f}_{i,k}^{a_k}\Big) \subset \mathcal{O}_{Y_i,y_i}.$$

Then $\mathfrak{a}_i$ and $\mathfrak{a}_i + \mathfrak{m}^\ell$ are the completions of $\widetilde{\mathfrak{a}}_i$ and $\widetilde{\mathfrak{a}}_i + \mathfrak{m}_{y_i}^\ell$. By [Mus02, Proposition 2.1] on Y_i, as $t + \epsilon = 1/M$,

$$\mathcal{J}\big((\widetilde{\mathfrak{a}}_i + \mathfrak{m}_{y_i}^\ell)^{t+\epsilon}\big) \subset \mathcal{J}(\widetilde{\mathfrak{a}}_i^t) + \mathfrak{m}_{y_i}^{\lfloor \ell\epsilon \rfloor - n + 1}.$$

By (6.4),

$$z_{j,i} \in \mathcal{J}\big((\mathfrak{a}_i + \mathfrak{m}^\ell)^{1/M}\big) + \mathfrak{m}^L.$$

Since $ta_k > \gamma_k$ for any k and $\lfloor \ell\epsilon \rfloor - n + 1 \ge L$, we have

$$\mathcal{J}(\mathfrak{a}_i^t) \subset \mathcal{J}\Big(\prod_k (f_{i,k})^{\gamma_k}\Big) \quad \text{and} \quad z_{j,i} \in \mathcal{J}\Big(\prod_k (f_{i,k})^{\gamma_k}\Big) + \mathfrak{m}^L.$$

For the moreover statement of (1), the coefficients of the monomials of degree at most c of the left-hand side of (6.3) are values at λ of polynomials with complex coefficients which vanish at λ, hence on T, so (6.3) holds with $\psi_{j,\xi}$, $\phi_{j,\xi}$, $\omega_{\nu,j,\xi}$, and $\omega'_{\nu,\rho,\xi}$ in place of ψ_j, ϕ_j, $\omega_{\nu,j}$, and $\omega'_{\nu,\rho}$. By (6.2), $\psi_{j,\xi} - \Psi_j(\mathbf{f}_i, \beta_i)$ and $\phi_{j,\xi} - z_{j,i}$ belong to $\mathfrak{m}^{c+1}$. Thus

$$\mathfrak{m}^c \subset \big(\Psi_1(\mathbf{f}_i,\beta_i),\dots,\Psi_{m'}(\mathbf{f}_i,\beta_i)\big) + (z_{1,i},\dots,z_{m,i})^P + \mathfrak{m}\cdot\mathfrak{m}^c,$$

and by Nakayama's lemma,

$$\mathfrak{m}^c \subset \big(\Psi_1(\mathbf{f}_i,\beta_i),\dots,\Psi_{m'}(\mathbf{f}_i,\beta_i)\big) + (z_{1,i},\dots,z_{m,i})^P.$$

If the $f_{i,k}$ come from $(Y_i \ni y_i)$, let $\mathfrak{c}_i := \mathcal{J}(\prod_k (f_{i,k})^{\gamma_k})$. Then

$$(z_{1,i},\dots,z_{m,i})^P \subset (\mathfrak{c}_i + \mathfrak{m}^L)^P \subset \mathfrak{c}_i^P + \mathfrak{m}^L,$$

and $L \ge c+1$, so by Nakayama's lemma,

$$\mathfrak{m}^c \subset \big(\Psi_1(\mathbf{f}_i,\beta_i),\dots,\Psi_{m'}(\mathbf{f}_i,\beta_i)\big) + \mathfrak{c}_i^P.$$

We show that the $z_{j,i}$ satisfy (2) for any $i \in \mathcal{I}$ with $\operatorname{jet}_d(\mathbf{f}_i) \in Z_d^\circ$, with the corresponding closed point ξ, and for any j. Let $X := \operatorname{Spec}\mathbb{C}[[x_1,\dots,x_n]]$, or $X := \mathbb{A}^n$ if the $f_{i,k}$ are regular functions near $0 \in \mathbb{A}^n$, and let F be a prime divisor over X with center the closed point 0 of X and $\operatorname{mult}_F((x_1,\dots,x_n)^L) > \operatorname{mult}_F z_{j,i}$. We let d_2 be an integer with $d_2 \ge d$ and $d_2 > \max_k \operatorname{mult}_F f_{i,k}$, and let

$$\mathfrak{a}_i^{(d_2)} := \Big(\prod_k (f_{i,k}^{<d_2})^{a_k}\Big) \subset \mathbb{C}[x_1,\dots,x_n].$$

Since $d_2 \ge d \ge \ell$, $\mathfrak{a}_i^{(d_2)} + (x_1,\dots,x_n)^\ell = \mathfrak{b}_\xi$. By (6.4), [Mus02, Proposition 2.1] on $\mathbb{A}^n$, $t+\epsilon = 1/M$, and $\lfloor \ell\epsilon \rfloor - n + 1 \ge L$,

$$\phi'_{j,\xi} \in \mathcal{J}\big((\mathfrak{a}_i^{(d_2)} + (x_1,\dots,x_n)^\ell)^{t+\epsilon}\big) \subset \mathcal{J}\big((\mathfrak{a}_i^{(d_2)})^t\big) + (x_1,\dots,x_n)^L \quad \text{near } 0 \in \mathbb{A}^n.$$

By (6.4), $z_{j,i} - \phi'_{j,\xi} \in (x_1,\dots,x_n)^L\mathcal{O}_X$ at 0. Thus $z_{j,i} = w' + w''$ for some

$$w' \in \mathcal{J}((\mathfrak{a}_i^{(d_2)}\mathcal{O}_X)^t) \quad \text{and} \quad w'' \in (x_1,\dots,x_n)^L\mathcal{O}_X \quad \text{at } 0.$$

Since $\operatorname{mult}_F w'' \ge \operatorname{mult}_F((x_1,\dots,x_n)^L) > \operatorname{mult}_F z_{j,i}$, we have

$$\operatorname{mult}_F w' = \operatorname{mult}_F z_{j,i}.$$

Since $f_{i,k} - f_{i,k}^{<d_2} \in (x_1,\dots,x_n)^{d_2}\mathcal{O}_X$ at 0 and $\operatorname{mult}_F((x_1,\dots,x_n)^{d_2}) \ge d_2 > \operatorname{mult}_F f_{i,k}$, we have $\operatorname{mult}_F f_{i,k}^{<d_2} = \operatorname{mult}_F f_{i,k}$, so $\operatorname{mult}_F \mathfrak{a}_i^{(d_2)} = \sum_k a_k \operatorname{mult}_F f_{i,k}$. Let $\pi_1 : Y_1 \to X$ be a log resolution of $\mathfrak{a}_i^{(d_2)}\mathcal{O}_X$ such that F is a prime divisor on Y_1 [dFEM11, Theorems 2.3 and 2.4]. Since

$\operatorname{center}_X F = \{0\}$, computing $\mathcal{J}((\mathfrak{a}_i^{(d_2)}\mathcal{O}_X)^t)$ on Y_1 [Laz04, Theorem 9.2.18], [JM12, Section 1.3], we have

$$\operatorname{mult}_F w' + \operatorname{mult}_F K_{Y_1/X} \geq \left\lfloor t \sum_k a_k \operatorname{mult}_F f_{i,k} \right\rfloor > \sum_k \gamma_k \operatorname{mult}_F f_{i,k} - 1,$$

since $ta_k > \gamma_k$ for any k. Since $a(F, X, 0) = \operatorname{mult}_F K_{Y_1/X} + 1$, the $z_{j,i}$ satisfy (2).

(3) Let $\eta \in W_* \backslash V$, and let

$$\Delta_* := \operatorname{div}(f_*) + \mu \operatorname{div}(h_*) \quad \text{on } X_*.$$

Since W_* is normal and $\dim W_* = 3$, $\dim \operatorname{Sing}(W_*) \leq 1$. If V contains the closed point, then $\dim \overline{\{\eta\}} \geq 1$, so no point of $\operatorname{Sing}(W_*)$ other than η is a generization of η, since any generization of η other than η has closure of dimension at least $2 > \dim \operatorname{Sing}(W_*)$, that is, $\operatorname{Spec} \mathcal{O}_{W_*,\eta}$ is regular outside the closed point. If V does not contain the closed point, then W_* is regular, so $\operatorname{Spec} \mathcal{O}_{W_*,\eta}$ is regular. Since $\mathcal{O}_{X_*,\eta}$ is excellent, $\operatorname{Spec} \widehat{\mathcal{O}}_{W_*,\eta}$ is normal and regular outside the closed point.

We let $n_\eta := \dim \mathcal{O}_{X_*,\eta}$. By the Cohen structure theorem,

$$\widehat{\mathcal{O}}_{X_*,\eta} \cong \kappa(\eta)[[y_1, \dots, y_{n_\eta}]].$$

We let $\overline{\kappa}$ be an algebraic closure of $\kappa(\eta)$, let $R := \overline{\kappa}[[y_1, \dots, y_{n_\eta}]]$ with maximal ideal $\mathfrak{m}_R$, let $\partial_k := \partial/\partial y_k$ on $\kappa(\eta)[[y_1, \dots, y_{n_\eta}]]$, which extends to R, and denote the images of f_* and h_* in R again by f_* and h_*. By the Jacobian criterion [Mat89, p. 239 and Theorem 30.8], the ideal

$$\mathfrak{j}_\eta := (f_*, \partial_1 f_*, \dots, \partial_{n_\eta} f_*) \subset \kappa(\eta)[[y_1, \dots, y_{n_\eta}]]$$

contains a power of the maximal ideal, and $\mathfrak{j}_\eta$ is the unit ideal if $\widehat{\mathcal{O}}_{W_*,\eta}$ is regular. If W_* is regular or $n_\eta \leq 2$, then $\widehat{\mathcal{O}}_{W_*,\eta}$ is regular (normal of dimension at most 1 in the second case), so $\mathfrak{j}_\eta$ is the unit ideal. Let $c'' \geq 0$ be an integer such that $\mathfrak{j}_\eta$ contains the c''-th power of the maximal ideal of $\kappa(\eta)[[y_1, \dots, y_{n_\eta}]]$ ($c'' = 0$ if $\mathfrak{j}_\eta$ is the unit ideal). Let $c' := c'' + 1$. Then

$$\mathfrak{m}_R^{c'} \subset \mathfrak{m}_R \cdot \mathfrak{j}_\eta R,$$

so by the definition of $\mathfrak{j}_\eta$,

$$\mathfrak{m}_R^{c'} \subset (f_*) + \mathfrak{m}_R \cdot (\partial_1 f_*, \dots, \partial_{n_\eta} f_*). \tag{6.5}$$

Since $\mathfrak{j}_\eta R$ contains a power of $\mathfrak{m}_R$, by the Jacobian criterion [Mat89, p. 239 and Theorem 30.8] the hypersurface $W_R := \operatorname{Spec} R/(f_*)$ is regular outside the closed point, and regular if $c'' = 0$. Since $c'' = 0$ if $n_\eta \leq 2$, W_R is regular in codimension 1, hence normal by Serre's criterion. Since $\operatorname{Spec} R \to \operatorname{Spec} \mathcal{O}_{X_*,\eta}$ is a regular morphism of regular schemes (the completion of the excellent ring $\mathcal{O}_{X_*,\eta}$ followed by $\kappa(\eta)[[y_1, \dots, y_{n_\eta}]] \to R$, which is flat with closed fiber $\overline{\kappa}$ separable over $\kappa(\eta)$, hence regular by [Mat89, Theorem 28.10] and André's theorem [Mat89, p. 260]), log resolutions and relative canonical divisors pull back along it [JM12, Lemma 1.2], [dFEM11, Remark 2.10], so lc properties and multiplier ideals are compatible with $\mathcal{O}_{X_*,\eta} \to R$. Let $\pi' : X' \to \operatorname{Spec} \mathcal{O}_{X_*,\eta}$ be an embedded log resolution of $W_* + \operatorname{div}(h_*)$ [dFEM11, Theorems 2.3 and 2.4], let $W' := \pi_*'^{-1} W_*$ and $\sigma := \pi'|_{W'}$, and let $\pi_R' : X_R' \to \operatorname{Spec} R$, W_R', and σ_R be their base changes along $\operatorname{Spec} R \to \operatorname{Spec} \mathcal{O}_{X_*,\eta}$. Then π_R' is an embedded log resolution of $W_R + \operatorname{div}(h_*)$ with $W_R' = \pi_{R*}'^{-1} W_R$, and by adjunction

$$K_{W'/W_*} = (K_{X'/X_*} + W' - \pi'^* W_*)|_{W'} \quad \text{and} \quad K_{W_R'/W_R} = (K_{X_R'/\operatorname{Spec} R} + W_R' - \pi_R'^* W_R)|_{W_R'}.$$

Hence $K_{W_R'/W_R} - \mu\sigma_R^* \operatorname{div}(h_*|_{W_R})$ is the pullback of $K_{W'/W_*} - \mu\sigma^* \operatorname{div}(h_*|_{W_*})$ along the regular morphism $W_R' \to W'$, which pulls back reduced divisors to reduced divisors, and the coefficients of $K_{W'/W_*} - \mu\sigma^* \operatorname{div}(h_*|_{W_*})$ are at least -1 since $(W_*, \mu \operatorname{div}(h_*|_{W_*}))$ is lc near η. Thus $(W_R, \mu \operatorname{div}(h_*|_{W_R}))$ is lc, and since the non-lc locus of (X_*, Δ_*) is closed, (X_*, Δ_*) is lc near η provided that $(\operatorname{Spec} R, \Delta_*)$ is lc near the closed point.

Let F_0 be the sum of the terms of f_* of degree at most $2c'$, a polynomial with $F_0 - f_* \in \mathfrak{m}_R^{2c'+1}$. By (6.5) and Lemma 3.2(2) (G trivial, $c = c'$, $L = 2c'+1$), there exist an automorphism φ of R and a unit $v \in R$ such that $v\varphi(f_*) = F_0$. We let $h' := \varphi(h_*)$ and $S := (F_0 = 0) \subset \mathbb{A}_{\overline{\kappa}}^{n_\eta}$. Then $\operatorname{Spec}\varphi$ is an isomorphism from $(\operatorname{Spec} R, \operatorname{div}(F_0) + \mu \operatorname{div}(h'))$ to $(\operatorname{Spec} R, \Delta_*)$. In particular, the completion $\widehat{S} := \operatorname{Spec} R/(F_0)$ of S at 0 satisfies $\widehat{S} \cong W_R$, so S is normal near 0, and $(\widehat{S}, \mu \operatorname{div}(h'|_{\widehat{S}}))$ is lc. It suffices to show that $(\operatorname{Spec} R, \operatorname{div}(F_0) + \mu \operatorname{div}(h'))$ is lc near the closed point. Finitely many polynomials over $\overline{\kappa}$ are defined over an algebraically closed subfield of $\overline{\kappa}$ of finite transcendence degree over $\mathbb{Q}$, which embeds into $\mathbb{C}$. Since lc properties are invariant under extension of the coefficient field [dFEM11, Lemma 2.14], we may use [EMY03, Corollary 1.8] for pairs on $\mathbb{A}_{\overline{\kappa}}^{n_\eta}$ defined by polynomials.

If $h' \notin \mathfrak{m}_R$, then $\operatorname{div}(h') = 0$ near the closed point, $(S \ni 0)$ is lc (completion at 0 preserves lc properties and lc thresholds, Lemma 2.12(2), [dFEM11, Proposition 2.11]), and by [EMY03, Corollary 1.8], $(\mathbb{A}_{\overline{\kappa}}^{n_\eta}, S)$ is lc near 0. Thus $(\operatorname{Spec} R, \operatorname{div}(F_0))$ is lc.

Assume that $h' \in \mathfrak{m}_R$. Then $\widehat{S}$ is klt: a prime divisor E over $\widehat{S}$ with center the closed point satisfies

$$a(E, \widehat{S}, 0) = a\big(E, \widehat{S}, \mu \operatorname{div}(h'|_{\widehat{S}})\big) + \mu \operatorname{mult}_E h' > 0,$$

and a prime divisor over $\widehat{S}$ with any other center satisfies $a(E, \widehat{S}, 0) \geq 1$, since $\widehat{S}$ is regular outside the closed point. Let $\mu' \in (0, \mu)$. Since $h' \notin (F_0)$, by Krull's intersection theorem and [dFEM11, Proposition 2.19(ii)] on $\widehat{S}$, for any $e \gg 0$, the e-jet $h'^{<e}$ of h' satisfies $h'^{<e}|_{\widehat{S}} \neq 0$ and $\operatorname{lct}(\widehat{S}; h'^{<e}|_{\widehat{S}}) \geq \mu'$. For such e, $(S, \mu' \operatorname{div}(h'^{<e}|_S))$ is lc near 0, so by [EMY03, Corollary 1.8],

$$\big(\mathbb{A}_{\overline{\kappa}}^{n_\eta}, S + \mu' \operatorname{div}(h'^{<e})\big)$$

is lc near 0. Thus the $\mathbb{R}$-ideal $(F_0)(h'^{<e})^{\mu'}$ on $\operatorname{Spec} R$ has lc threshold at least 1 at the closed point for any $e \gg 0$, and by [dFEM11, Proposition 2.19(ii)] on $\operatorname{Spec} R$, so does $(F_0)(h')^{\mu'}$. Thus

$$\big(\operatorname{Spec} R, \operatorname{div}(F_0) + \mu' \operatorname{div}(h')\big)$$

is lc near the closed point for any $\mu' \in (0, \mu)$. Since log discrepancies are affine functions of μ', $(\operatorname{Spec} R, \operatorname{div}(F_0) + \mu \operatorname{div}(h'))$ is lc near the closed point.

Thus (X_*, Δ_*) is lc near η. For any prime divisor E over X_*,

$$a(E, X_*, \theta_2\Delta_*) = \theta_2 a(E, X_*, \Delta_*) + (1 - \theta_2) a(E, X_*, 0),$$

and $a(E, X_*, 0) \geq 1$ since X_* is regular. Thus $(X_*, \theta_2\Delta_*)$ is klt near η. Let $\pi_2 : Y_2 \to X_*$ be a log resolution of $f_* h_*$. Then $K_{Y_2/X_*} - \lfloor \theta_2 \pi_2^* \Delta_* \rfloor \geq 0$ over a neighborhood of η, so $\mathcal{J}(f_*^{\theta_2} h_*^{\theta_2\mu})$ is the unit ideal near η.

(4) Let $\pi : Y_3 \to \mathbb{A}_{\Bbbk}^n$ be a log resolution of $\mathfrak{b} \cdot \prod_{\phi_j \neq 0} \phi_j$. Then $\phi_j \in \mathcal{J}(\mathfrak{b}^\gamma)$ near 0 means that

$$\operatorname{mult}_E \phi_j + \operatorname{mult}_E K_{Y_3/\mathbb{A}_{\Bbbk}^n} - \lfloor \gamma \operatorname{mult}_E \mathfrak{b} \rfloor \geq 0 \tag{6.6}$$

for any prime divisor E on Y_3 with $0 \in \pi(E)$. Let $\mathbb{C}[T] \subset \Bbbk$ be the $\mathbb{C}$-subalgebra generated by the entries of λ, the coordinate ring of T. We let $R_T \subset \Bbbk$ be a finitely generated $\mathbb{C}[T]$-subalgebra over which π and the prime divisors on Y_3 contained in

$$D := \operatorname{Exc}(\pi) \cup \operatorname{Supp} \pi^{-1}\Big(\mathfrak{b} \cdot \prod_{\phi_j \neq 0} \phi_j\Big)$$

are defined. For a closed point $\xi \in \operatorname{Spec} R_T$ we write $\mathfrak{b}_\xi$, $\phi_{j,\xi}$, $\pi_\xi : Y_{3,\xi} \to \mathbb{A}^n$, and E_ξ for the specializations at ξ. By generic smoothness and generic flatness, possibly replacing R_T with $R_T[1/a]$ for a non-zero $a \in R_T$, we may assume that for any closed point $\xi \in \operatorname{Spec} R_T$:

- π_ξ is a log resolution of $\mathfrak{b}_\xi \cdot \prod_{\phi_j \neq 0} \phi_{j,\xi}$.
- $E \to E_\xi$ is a bijection from the prime divisors on Y_3 contained in D to those contained in

$$D_\xi := \operatorname{Exc}(\pi_\xi) \cup \operatorname{Supp} \pi_\xi^{-1}\Big(\mathfrak{b}_\xi \cdot \prod_{\phi_j \neq 0} \phi_{j,\xi}\Big)$$

which preserves the multiplicities of $\mathfrak{b}$, of the ϕ_j and of the relative canonical divisor, and $0 \in \pi_\xi(E_\xi)$ if and only if $0 \in \pi(E)$.

Thus (6.6) holds at ξ (along the prime divisors on $Y_{3,\xi}$ not contained in D_ξ its three terms vanish), so $\phi_{j,\xi} \in \mathcal{J}(\mathfrak{b}_\xi^\gamma)$ near $0 \in \mathbb{A}^n$. The inclusion $\mathbb{C}[T] \subset R_T$ defines a dominant morphism $\operatorname{Spec} R_T \to T$, so by Chevalley's theorem its image contains a non-empty open subset of T, hence contains $T \cap (\Theta \neq 0)$ for a polynomial Θ with complex coefficients in N' variables with $\Theta(\lambda) \neq 0$. For a closed point $\lambda' \in T$ with $\Theta(\lambda') \neq 0$, the fiber of $\operatorname{Spec} R_T \to T$ over λ' is a non-empty scheme of finite type over $\mathbb{C}$, and a closed point ξ of it satisfies $\mathfrak{b}_\xi = \mathfrak{b}_{\lambda'}$ and $\phi_{j,\xi} = \phi_{j,\lambda'}$. The lemma follows. □

Proposition 6.2. *Let N be a positive integer, let $\kappa_1, \dots, \kappa_N$ be positive real numbers, let $R_0 > 0$ and $C_0 \geq 0$ be real numbers, and let $r \in \{z_1, z_1 z_2\} \subset \mathbb{C}[z_1, z_2, z_3]$. For any $i \in \mathbb{N}^+$, let $g_{i,1}, \dots, g_{i,N} \in \mathbb{C}[z_1, z_2, z_3]$ be non-zero polynomials, and let F_i be a prime divisor over $\mathbb{A}^3$ with center 0. Let $(g_{*,1}, \dots, g_{*,N})$ be a compatible generic limit of the tuples $(g_{i,1}, \dots, g_{i,N})$ along an infinite subset $\mathcal{I} \subset \mathbb{N}^+$, with coefficient field $\Bbbk$ (Definition 2.5 with H trivial). Assume that the following hold.*

(1) $\operatorname{mult}_{F_i} r \leq R_0$ *for any i.*
(2) $a(F_i, \mathbb{A}^3, 0) - \sum_k \kappa_k \operatorname{mult}_{F_i} g_{i,k} \leq C_0$ *for any i.*
(3) The pair

$$\left(\operatorname{Spec} \Bbbk[[z_1, z_2, z_3]][r^{-1}], \prod_k (g_{*,k})^{\kappa_k} \right)$$

is klt.

Then there exists a real number T such that $\operatorname{mult}_{F_i} g_{i,k} \leq T$ for any k and for all but finitely many $i \in \mathcal{I}$.

Proof of Proposition 6.2. Let $X_* := \operatorname{Spec} \Bbbk[[z_1, z_2, z_3]]$, and let $\mathfrak{m}$ be the maximal ideal of $\mathcal{O}_{\mathbb{A}^3,0}$. Let $\pi : Y \to X_*$ be a log resolution of $\prod_k g_{*,k}$ [dFEM11, Theorems 2.3 and 2.4]. By (3),

$$a(E, X_*, 0) - \sum_k \kappa_k \operatorname{mult}_E g_{*,k} > 0$$

for any prime divisor E on Y with $\pi(E) \not\subset (r = 0)$. Since $\operatorname{Supp}(K_{Y/X_*} + \sum_k \pi^* \operatorname{div}(g_{*,k}))$ contains only finitely many such E, and for the others $a(E, X_*, 0) = 1$ and $\operatorname{mult}_E g_{*,k} = 0$, there exist rational numbers $\mu_k > \kappa_k$ such that

$$a(E, X_*, 0) - \sum_k \mu_k \operatorname{mult}_E g_{*,k} > 0$$

for any such E. Thus for any such E,

$$\operatorname{mult}_E \left(K_{Y/X_*} - \left\lfloor \sum_k \mu_k \pi^* \operatorname{div}(g_{*,k}) \right\rfloor \right) \geq 0,$$

and the multiplier ideal $\mathcal{J}_* := \mathcal{J}(\prod_k (g_{*,k})^{\mu_k})$ satisfies

$$\mathcal{J}_* \Bbbk[[z_1, z_2, z_3]][r^{-1}] = \Bbbk[[z_1, z_2, z_3]][r^{-1}].$$

Since $\Bbbk[[z_1, z_2, z_3]]/\mathcal{J}_*$ is a finitely generated module whose localization at r is zero, there exists $m_0 \in \mathbb{N}$ such that

$$r^{m_0} \in \mathcal{J}_*.$$

We let $\gamma_1, \dots, \gamma_N$ be rational numbers with $\kappa_k < \gamma_k < \mu_k$ for any k, and let L be an integer with $L > m_0 R_0$. By Lemma 6.1(1) ($n = 3$, $(x_1, x_2, x_3) = (z_1, z_2, z_3)$, $q = 0$, $m = 1$, $\Phi_1 = r^{m_0}$, $\gamma'_k = \mu_k$), for all but finitely many $i \in \mathcal{I}$, Lemma 6.1(2) holds with r^{m_0} in place of $z_{1,i}$. For such i, since $\operatorname{center}_{\mathbb{A}^3} F_i = \{0\}$, we have $\operatorname{mult}_{F_i} \mathfrak{m} \geq 1$, so by (1),

$$\operatorname{mult}_{F_i}(\mathfrak{m}^L) \geq L > m_0 R_0 \geq \operatorname{mult}_{F_i} r^{m_0}.$$

By Lemma 6.1(2),

$$m_0 \operatorname{mult}_{F_i} r + a(F_i, \mathbb{A}^3, 0) > \sum_k \gamma_k \operatorname{mult}_{F_i} g_{i,k}.$$

By (1) and (2),

$$\sum_k (\gamma_k - \kappa_k) \operatorname{mult}_{F_i} g_{i,k} < C_0 + m_0 \operatorname{mult}_{F_i} r \leq C_0 + m_0 R_0.$$

Let $\sigma := \min_k(\gamma_k - \kappa_k) > 0$. Since $\operatorname{mult}_{F_i} g_{i,k} \geq 0$ for any k, we have

$$\operatorname{mult}_{F_i} g_{i,k} < T := \frac{C_0 + m_0 R_0}{\sigma}$$

for any k. The proposition follows. □

6.2. **Concentration and degenerations.** For a power series $\phi \in \Bbbk[[z_1, z_2, z_3]]$ and a positive integer L, we let $\phi^{<L}$ be the sum of the terms of ϕ of degree less than L. For a tuple $g = (g_k)_k$ of power series, $g^{<L} := (g_k^{<L})_k$ is the L-jet of g in the sense of Definition 2.5. For a pair (X, B) and an $\mathbb{R}$-ideal $\mathfrak{a}$ on X, $\operatorname{Nklt}(X, B, \mathfrak{a})$ denotes the union of the centers on X of the prime divisors E over X with $a(E, X, B, \mathfrak{a}) \leq 0$. We omit B or $\mathfrak{a}$ when it is trivial.

Proposition 6.3. *Let $r \in \{z_1, z_1 z_2\}$, let N be a positive integer, let $\kappa_1, \dots, \kappa_N$ be positive real numbers, and let $R_0 > 0$ and $C_0 \geq 0$ be real numbers. For any integer $i \geq 1$, let $g_{i,1}, \dots, g_{i,N}$ be non-zero polynomials in z_1, z_2, z_3, and let F_i be a prime divisor over $\mathbb{A}^3$ with center 0. Let $(g_{*,k})_k$ be a compatible generic limit of the tuples $(g_{i,k})_k$ along an infinite subset $\mathcal{I} \subset \mathbb{N}^+$, with coefficient field $\Bbbk$ (Definition 2.5, with H trivial), and let $S := \Bbbk[[z_1, z_2, z_3]]$. Let Z be the reduced closure in $\operatorname{Spec} S$ of $\operatorname{Nklt}(\operatorname{Spec} S[r^{-1}], \prod_k (g_{*,k})^{\kappa_k})$, and let $I_Z \subset S$ be the ideal of Z. Assume that the following hold.*

(i) The orders of the $g_{i,k}$ at 0 are bounded.

(ii) For any i,

$$\operatorname{mult}_{F_i} r \leq R_0 \quad \text{and} \quad a(F_i, \mathbb{A}^3, 0) - \sum_{k=1}^{N} \kappa_k \operatorname{mult}_{F_i} g_{i,k} \leq C_0.$$

(iii) $\lim_{i \to +\infty} \max_k \operatorname{mult}_{F_i} g_{i,k} = +\infty$.

(iv) $(\operatorname{Spec} S[r^{-1}], \prod_k (g_{*,k})^{\kappa_k})$ is lc.

Then the following hold.

(A) Let T, L, d, q, and n' be positive integers with $L \geq T$, let $n'' \in \mathbb{N}$, let $\beta \in \Bbbk^q$, let $\Phi_1, \dots, \Phi_{n'}$ be complex polynomials in $y_1, \dots, y_q, z_1, z_2, z_3$ such that $\Phi_1(\beta, z), \dots, \Phi_{n'}(\beta, z)$ are the parts of degree less than L of generators of I_Z, and let $\Psi_1, \dots, \Psi_{n''}$ be polynomial functions on $\mathcal{J}_d \times \mathbb{A}^q$ vanishing at $((g_{*,k}^{<d})_k, \beta)$, where $\mathcal{J}_d$ is the affine space of d-jets in Definition 2.5. Then for all but finitely many $i \in \mathcal{I}$, there exists a vector $\beta_i \in \mathbb{C}^q$ such that

$$\Psi_j((g_{i,k}^{<d})_k, \beta_i) = 0 \text{ for any } 1 \leq j \leq n'' \quad \text{and} \quad \operatorname{mult}_{F_i} \Phi_j(\beta_i, z) \geq T \text{ for any } 1 \leq j \leq n'.$$

In particular, $Z \neq \emptyset$.

(B) Assume that Z has no irreducible component of dimension 2. Let $\mathcal{I}' \subset \mathcal{I}$ be an infinite subset and $\{1, \dots, N\} = \mathcal{H} \sqcup \mathcal{B}$ a partition. Assume that the following hold.

(1) For any $j \in \{1, 2\}$ with $z_j \mid r$, the integer $t_j := \operatorname{mult}_{F_i} z_j$ does not depend on $i \in \mathcal{I}'$.

(2) For any $k \in \mathcal{H}$,

$$\lim_{i \in \mathcal{I}', i \to +\infty} \operatorname{mult}_{F_i} g_{i,k} = +\infty.$$

(3) $\sup_{i \in \mathcal{I}'} \operatorname{mult}_{F_i} g_{i,k} < +\infty$ for any $k \in \mathcal{B}$.

Then there exists an irreducible component C of Z with $\dim C = 1$ satisfying the following.

(B1) $C \subset V(g_{*,k})$ for any $k \in \mathcal{H}$, and $C \not\subset V(g_{*,k})$ for any $k \in \mathcal{B}$.

(B2) The normalization of S/I_C, where I_C is the ideal of C, is isomorphic to $\Bbbk[[u]]$, and for any j with $z_j \mid r$, the order n_j of $z_j|_C$ in u is a positive divisor of t_j. In particular,

$$\sum_{z_j|r} n_j \le \sum_{z_j|r} t_j = \operatorname{mult}_{F_i} r \le R_0 \quad \text{for any } i \in \mathcal{I}'.$$

Proof. We let $X_* := \operatorname{Spec} S$, $X_*^\circ := \operatorname{Spec} S[r^{-1}]$, $\mathfrak{n}_*$ the maximal ideal of S, $\mathfrak{n}$ the ideal of 0 in $\Bbbk[z_1, z_2, z_3]$, and $\mathfrak{m}$ the ideal of 0 in $\mathbb{C}[z_1, z_2, z_3]$. Log discrepancies on X_* are taken in the sense of [dFEM11, Section 2], and multiplier ideals on X_* in the sense of [JM12, Section 1.3], where they are computed on any log resolution.

Step 1. We show that there exist positive integers $Q, N_1, N_2, p_1, \dots, p_N$ such that $p_k/Q > \kappa_k$ for any k and

$$r^{N_2} I_Z^{N_1} \subset \mathcal{J}_* := \mathcal{J}\Big(\big(\prod_{k=1}^N g_{*,k}^{p_k} \big)^{1/Q} \Big). \tag{6.7}$$

By [dFEM11, Theorems 2.3 and 2.4], there exists a log resolution $h : Y \to X_*$ of $(X_*, \operatorname{div}(r \prod_k g_{*,k}))$. Let $\mathcal{E}$ be the finite set of prime divisors on Y which are exceptional over X_* or are components of $h^* \operatorname{div}(r \prod_k g_{*,k})$, and let $\mathcal{E}^\circ := \{E \in \mathcal{E} \mid \operatorname{mult}_E r = 0\}$. For $\gamma = (\gamma_k)_k \in \mathbb{R}_{>0}^N$ and $E \in \mathcal{E}$, let

$$a_\gamma(E) := a(E, X_*, 0) - \sum_{k=1}^N \gamma_k \operatorname{mult}_E g_{*,k}.$$

Since h is a log resolution, for any $\gamma \in \mathbb{R}_{>0}^N$, $(X_*^\circ, \prod_k (g_{*,k})^{\gamma_k})$ is lc if and only if $a_\gamma(E) \ge 0$ for any $E \in \mathcal{E}^\circ$, and

$$\operatorname{Nklt}\Big(X_*^\circ, \prod_k (g_{*,k})^{\gamma_k} \Big) = X_*^\circ \cap \bigcup_{E \in \mathcal{E}^\circ,\ a_\gamma(E) \le 0} h(E). \tag{6.8}$$

Since $\mathcal{E}^\circ$ is finite and $a_\kappa(E) \ge 0$ for any $E \in \mathcal{E}^\circ$, there exist positive integers $Q, p_1, \dots, p_N$ such that $\gamma' := (p_k/Q)_k$ satisfies $p_k/Q > \kappa_k$ for any k and $a_{\gamma'}(E) > 0$ for any $E \in \mathcal{E}^\circ$ with $a_\kappa(E) > 0$. If $E \in \mathcal{E}^\circ$ and $a_\kappa(E) = 0$, then $\operatorname{mult}_E g_{*,k} > 0$ for some k because $a(E, X_*, 0) > 0$, so $a_{\gamma'}(E) < 0$. By (6.8),

$$\operatorname{Nklt}\Big(X_*^\circ, \prod_k (g_{*,k})^{p_k/Q} \Big) = \operatorname{Nklt}\Big(X_*^\circ, \prod_k (g_{*,k})^{\kappa_k} \Big) = Z \cap X_*^\circ.$$

By the resolution formula for multiplier ideals [Laz04, Definition 9.2.3], the left hand side is $V(\mathcal{J}_*) \cap X_*^\circ$. Let

$$\mathcal{J}_* : r^\infty := \mathcal{J}_* S[r^{-1}] \cap S = \{\phi \in S \mid r^{N'} \phi \in \mathcal{J}_* \text{ for some integer } N' \ge 0\}.$$

Then $V(\mathcal{J}_* : r^\infty)$ is the closure of $V(\mathcal{J}_*) \cap X_*^\circ$, that is, Z. Thus $I_Z = \sqrt{\mathcal{J}_* : r^\infty}$. Since S is Noetherian, the increasing chain of ideals $\mathcal{J}_* : r^{N'}$, $N' \ge 0$, stabilizes at some N_2, so $\mathcal{J}_* : r^\infty = \mathcal{J}_* : r^{N_2}$, and $I_Z^{N_1} \subset \mathcal{J}_* : r^\infty$ for some N_1. This proves (6.7).

We fix a rational number λ with $\max_k \kappa_k/p_k < \lambda < 1/Q$, and let

$$\epsilon := \frac{1}{Q} - \lambda > 0 \quad \text{and} \quad \sigma := \min_k (\lambda p_k - \kappa_k) > 0.$$

Step 2. We prove (A). Let $L, T, d, q, \beta, \Phi_1, \dots, \Phi_{n'}, \Psi_1, \dots, \Psi_{n''}$ be as in (A), and let $\varphi_1, \dots, \varphi_{n'}$ be generators of I_Z with $\varphi_j^{<L} = \Phi_j(\beta, z)$ for any j. We fix integers

$$N_3 > N_2 R_0 + N_1 T, \qquad N_5 \ge 1 \text{ with } \lfloor N_5 \epsilon \rfloor - 2 \ge N_3, \quad \text{and} \quad N_4 \ge \max\{L, N_5\},$$

and let

$$\mathfrak{b} := \Big(\prod_{k=1}^N (g_{*,k}^{<N_5})^{p_k} \Big) + \mathfrak{n}^{N_5} \subset \Bbbk[z_1, z_2, z_3] \quad \text{and} \quad P_j := r^{N_2} (\varphi_j^{<N_4})^{N_1} \text{ for } 1 \le j \le n'.$$

Since

$$\prod_k g_{*,k}^{p_k} - \prod_k (g_{*,k}^{<N_5})^{p_k} \in \mathfrak{n}_*^{N_5},$$

the ideal $\mathfrak{b}S$ contains $\prod_k g_{*,k}^{p_k}$. Multiplier ideals commute with the completion S of $\Bbbk[z_1, z_2, z_3]_{\mathfrak{n}}$ by [dFEM11, Proposition A.14] and flat base change, so

$$\mathcal{J}(\mathfrak{b}^{1/Q})S = \mathcal{J}((\mathfrak{b}S)^{1/Q}) \supset \mathcal{J}_*.$$

Since $\mathfrak{n}^{N_5} \subset \mathfrak{b}$, [Laz04, Example 9.2.14] implies

$$\mathcal{J}(\mathfrak{b}^{1/Q}) \supset \mathcal{J}(\mathfrak{n}^{N_5/Q}) = \mathfrak{n}^{\max\{0, \lfloor N_5/Q \rfloor - 2\}} \supset \mathfrak{n}^{N_4}.$$

By (6.7), $r^{N_2}\varphi_j^{N_1} \in \mathcal{J}_*$, and $P_j - r^{N_2}\varphi_j^{N_1} \in \mathfrak{n}_*^{N_4}$. Thus

$$P_j \in \mathcal{J}(\mathfrak{b}^{1/Q})S \cap \Bbbk[z_1, z_2, z_3] = \mathcal{J}(\mathfrak{b}^{1/Q}),$$

where the equality holds because $\mathfrak{n}^{N_4} \subset \mathcal{J}(\mathfrak{b}^{1/Q})$ and

$$\Bbbk[z_1, z_2, z_3]/\mathfrak{n}^{N_4} \cong S/\mathfrak{n}_*^{N_4}.$$

Let $h' : Y' \to \mathbb{A}^3_{\Bbbk}$ be a log resolution of $\mathfrak{b}$, let D' be the divisor on Y' with $\mathfrak{b}\mathcal{O}_{Y'} = \mathcal{O}_{Y'}(-D')$, and let $\mathcal{E}'$ be the finite set of prime divisors contained in $\operatorname{Exc}(h') \cup \operatorname{Supp} D'$. By the resolution formula [Laz04, Definition 9.2.3], a polynomial $P \in \Bbbk[z_1, z_2, z_3]$ belongs to $\mathcal{J}(\mathfrak{b}^{1/Q})$ if and only if

$$\operatorname{mult}_E P \geq \left\lfloor \frac{1}{Q} \operatorname{mult}_E D' \right\rfloor - \operatorname{mult}_E K_{Y'/\mathbb{A}^3_{\Bbbk}} \quad \text{for any } E \in \mathcal{E}'. \tag{6.9}$$

Let $d_1 := \max\{d, N_5\}$. By Definition 2.5, $(g_{*,k}^{<d_1})_k \in \mathcal{J}_{d_1}(\Bbbk)$ maps to the generic point of the irreducible closed subset $Z_{d_1} \subset \mathcal{J}_{d_1}$, which defines an injection $\mathbb{C}[Z_{d_1}] \hookrightarrow \Bbbk$. Let $\Lambda \subset \Bbbk$ be a finitely generated $\mathbb{C}$-subalgebra containing the image of $\mathbb{C}[Z_{d_1}]$, the entries of β and the coefficients of $\varphi_1^{<N_4}, \dots, \varphi_{n'}^{<N_4}$, over which h', D' and the members of $\mathcal{E}'$ are defined. Possibly replacing Λ with a localization at one non-zero element, by generic smoothness we may assume that, for any closed point $\tau \in \operatorname{Spec} \Lambda$, the fiber h'_τ of h' over τ is a log resolution of the fiber $\mathfrak{b}_\tau \subset \mathbb{C}[z_1, z_2, z_3]$ of $\mathfrak{b}$, the fibers of the members of $\mathcal{E}'$ are the prime divisors contained in $\operatorname{Exc}(h'_\tau) \cup \operatorname{Supp} D'_\tau$, and the multiplicities appearing in (6.9) for $P = P_j$ do not change under passing to the fibers. Since (6.9) holds for $P = P_j$, the fiber $P_{j,\tau}$ of P_j over τ belongs to $\mathcal{J}(\mathfrak{b}_\tau^{1/Q})$. Moreover, $\Psi_j((g_{*,k}^{<d})_k, \beta) = 0$ is an identity in Λ, so it holds on the fiber over τ.

Since $\mathbb{C}[Z_{d_1}] \to \Lambda$ is injective, the morphism $\operatorname{Spec} \Lambda \to Z_{d_1}$ is dominant, so by Chevalley's theorem the image of $\operatorname{Spec} \Lambda$ contains a non-empty open subset $U \subset Z_{d_1}$. By Definition 2.5, $(g_{i,k}^{<d_1})_k \in U$ for all but finitely many $i \in \mathcal{I}$. For such i, we fix a closed point $\tau_i \in \operatorname{Spec} \Lambda$ over $(g_{i,k}^{<d_1})_k$, let $\beta_i \in \mathbb{C}^q$ be the fiber of β over τ_i, and let $\varphi_{j,i} \in \mathbb{C}[z_1, z_2, z_3]$ be the fiber of $\varphi_j^{<N_4}$ over τ_i. The coordinates of $\mathcal{J}_{d_1}$ take at τ_i the values of $(g_{i,k}^{<d_1})_k$, and $(\varphi_j^{<N_4})^{<L} = \Phi_j(\beta, z)$ as $N_4 \geq L$. Thus:

- For any j, $\Psi_j((g_{i,k}^{<d})_k, \beta_i) = 0$.
- The fiber of $\mathfrak{b}$ over τ_i is

$$\mathfrak{b}_{\tau_i} = \Big(\prod_k (g_{i,k}^{<N_5})^{p_k}\Big) + \mathfrak{m}^{N_5} = \Big(\prod_k g_{i,k}^{p_k}\Big) + \mathfrak{m}^{N_5}.$$

- For any j,

$$r^{N_2}\varphi_{j,i}^{N_1} = P_{j,\tau_i} \in \mathcal{J}(\mathfrak{b}_{\tau_i}^{1/Q}) \quad \text{and} \quad \varphi_{j,i} - \Phi_j(\beta_i, z) \in \mathfrak{m}^L.$$

Since $1/Q = \lambda + \epsilon$, [Mus02, Proposition 2.1] implies

$$\mathcal{J}(\mathfrak{b}_{\tau_i}^{1/Q}) \subset \mathcal{J}\Big(\big(\prod_k g_{i,k}^{p_k}\big)^{\lambda}\Big) + \mathfrak{m}^{\lfloor N_5 \epsilon \rfloor - 2}.$$

Thus $r^{N_2}\varphi_{j,i}^{N_1} = \psi_{j,i} + \psi'_{j,i}$ for some $\psi_{j,i} \in \mathcal{J}((\prod_k g_{i,k}^{p_k})^\lambda)$ and $\psi'_{j,i} \in \mathfrak{m}^{N_3}$. By [Laz04, Definition 9.2.3 and Theorem 9.2.18] and (ii),

$$\begin{aligned}\operatorname{mult}_{F_i}\psi_{j,i} &> \lambda\sum_k p_k \operatorname{mult}_{F_i} g_{i,k} - a(F_i,\mathbb{A}^3,0)\\ &\geq \sigma\sum_k \operatorname{mult}_{F_i} g_{i,k} + \Big(\sum_k \kappa_k \operatorname{mult}_{F_i} g_{i,k} - a(F_i,\mathbb{A}^3,0)\Big)\\ &\geq \sigma\sum_k \operatorname{mult}_{F_i} g_{i,k} - C_0.\end{aligned}$$

By (iii), possibly discarding finitely many i, we may assume that

$$\sigma\sum_k \operatorname{mult}_{F_i} g_{i,k} - C_0 > N_2R_0 + N_1T.$$

Suppose that $\operatorname{mult}_{F_i}\Phi_j(\beta_i,z) < T$ for some j. Since F_i has center 0, $\operatorname{mult}_{F_i}\mathfrak{m}^L \geq L \geq T$, so $\operatorname{mult}_{F_i}\varphi_{j,i} = \operatorname{mult}_{F_i}\Phi_j(\beta_i,z) < T$. By (ii),

$$\operatorname{mult}_{F_i}(r^{N_2}\varphi_{j,i}^{N_1}) < N_2R_0 + N_1T < N_3 \leq \operatorname{mult}_{F_i}\psi'_{j,i},$$

so

$$\operatorname{mult}_{F_i}\psi_{j,i} = \operatorname{mult}_{F_i}(r^{N_2}\varphi_{j,i}^{N_1}) < N_2R_0 + N_1T,$$

which contradicts the bound $\operatorname{mult}_{F_i}\psi_{j,i} > N_2R_0 + N_1T$ above. This proves the existence of the β_i in (A). If $Z = \emptyset$, then $I_Z = S$ is generated by 1, and the existence of β_i with $L = T = d = q = n' = 1$, $\beta = 0$, $\Phi_1 = 1$, and $n'' = 0$ implies $0 = \operatorname{mult}_{F_i} 1 \geq 1$, a contradiction. Thus $Z \neq \emptyset$.

Step 3. We assume the hypotheses of (B) and decompose Z into irreducible curves. Since $g_{*,k} \neq 0$ for any k, $(X_*^\circ, \prod_k (g_{*,k})^{\kappa_k})$ is klt at the generic point of X_*, so $\dim Z \leq 2$. Any irreducible component of Z is the closure of a subset of X_*°, hence is not the closed point of X_*, which is contained in $V(r)$. By (A), $Z \neq \emptyset$. Since Z has no irreducible component of dimension 2 by assumption, $Z = C_1 \cup \cdots \cup C_m$ for some positive integer m and irreducible curves $C_1, \ldots, C_m$. Let $\mathfrak{p}_\mu \subset S$ be the prime ideal of C_μ. Then

$$\prod_{\mu=1}^m \mathfrak{p}_\mu \subset \bigcap_{\mu=1}^m \mathfrak{p}_\mu = I_Z. \tag{6.10}$$

Fix μ. The ring $S/\mathfrak{p}_\mu$ is a complete local domain with $\dim S/\mathfrak{p}_\mu = 1$, so the normalization $\widetilde{S}_\mu$ of $S/\mathfrak{p}_\mu$ is finite over $S/\mathfrak{p}_\mu$ since complete Noetherian local rings are excellent [Mat89, p. 260]. Hence $\widetilde{S}_\mu$ is a complete local domain. Thus $\widetilde{S}_\mu$ is a normal local ring of dimension 1 whose residue field is a finite extension of $\Bbbk$, hence equal to $\Bbbk$, so $\widetilde{S}_\mu \cong \Bbbk[[u]]$ by the Cohen structure theorem. For any j with $z_j \mid r$, $z_j \notin \mathfrak{p}_\mu$ because $C_\mu \not\subset V(r)$, so $z_j|_{C_\mu} \in u^{n_{\mu,j}}\Bbbk[[u]]^\times$ for a unique positive integer $n_{\mu,j}$. Since $S/\mathfrak{p}_\mu$ is a local domain of dimension 1, for any $\phi \in S\backslash\mathfrak{p}_\mu$, the ideal $\mathfrak{p}_\mu + (\phi)$ is S or $\mathfrak{n}_*$-primary.

Suppose that (B) does not hold. Since by the previous paragraph and (ii), C_μ satisfies (B2) as soon as $n_{\mu,j} \mid t_j$ for any j with $z_j \mid r$, for any μ one of the following cases occurs, and we fix the indicated data.

(a) $g_{*,k} \notin \mathfrak{p}_\mu$ for some $k \in \mathcal{H}$. We fix such k and a positive integer b_μ with $z_1^{b_\mu} \in \mathfrak{p}_\mu + (g_{*,k})$, and write $z_1^{b_\mu} - w_\mu g_{*,k} \in \mathfrak{p}_\mu$ with $w_\mu \in S$.

(b) $g_{*,k} \in \mathfrak{p}_\mu$ for some $k \in \mathcal{B}$. We fix such k.

(c) $n_{\mu,j} \nmid t_j$ for some j with $z_j \mid r$. We fix such j, let $n := n_{\mu,j}$, and write $z_j|_{C_\mu} = u^n v$ with $v \in \Bbbk[[u]]^\times$. Since $\Bbbk$ is algebraically closed of characteristic 0, $v = v'^n$ for some $v' \in \Bbbk[[u]]^\times$, so $u' := uv'$ satisfies $z_j|_{C_\mu} = u'^n$. As $\Bbbk[[u]]$ is contained in the fraction field of $S/\mathfrak{p}_\mu$, there exist $\xi_\mu, \eta_\mu \in S$ with $\eta_\mu \notin \mathfrak{p}_\mu$ such that u' is the image of ξ_μ divided by

the image of η_μ. Then $z_j\eta_\mu^n - \xi_\mu^n \in \mathfrak{p}_\mu$. Since $\mathfrak{p}_\mu + (\eta_\mu)$ is S or $\mathfrak{n}_*$-primary, there exist a positive integer b_μ and $\zeta_\mu \in S$ with $z_j^{b_\mu} - \eta_\mu\zeta_\mu \in \mathfrak{p}_\mu$.

Step 4. We derive a contradiction, which proves (B). We fix an integer T greater than

- $b_\mu t_1$ for any μ in case (a) (as $z_1 \mid r$, $t_1 = \operatorname{mult}_{F_i} z_1$ is defined),
- $\sup_{i\in\mathcal{I}'} \operatorname{mult}_{F_i} g_{i,k}$ for any μ in case (b), with k as in (b), and
- $(1 + n_{\mu,j}b_\mu)t_j$ for any μ in case (c), with j as in (c),

and an integer $L > mT$. Let $\varphi_1, \dots, \varphi_{n'}$ be generators of I_Z, and for any μ let $\omega_{\mu,1}, \dots, \omega_{\mu,n'_\mu}$ be generators of $\mathfrak{p}_\mu$. We fix expressions of the following elements as S-linear combinations of the φ_j, respectively of the $\omega_{\mu,j}$:

(I0) $\prod_\mu \omega_{\mu,j_\mu} \in I_Z$ for any $(j_1, \dots, j_m)$ (see (6.10)).
(I1) $z_1^{b_\mu} - w_\mu g_{*,k} \in \mathfrak{p}_\mu$ for μ in case (a).
(I2) $g_{*,k} \in \mathfrak{p}_\mu$ for μ in case (b).
(I3) $z_j\eta_\mu^n - \xi_\mu^n \in \mathfrak{p}_\mu$ and $z_j^{b_\mu} - \eta_\mu\zeta_\mu \in \mathfrak{p}_\mu$ for μ in case (c).

Let $\Sigma \subset S$ be the finite set consisting of the φ_j, the $\omega_{\mu,j}$, the w_μ, ξ_μ, η_μ, ζ_μ, and the coefficients of the expressions, let $\beta \in \Bbbk^q$ be the vector of the coefficients of $\phi^{<L}$ for $\phi \in \Sigma$, and for any $\phi \in \Sigma$ let $\Phi_\phi \in \mathbb{C}[y_1, \dots, y_q, z_1, z_2, z_3]$ be the polynomial, linear in $y_1, \dots, y_q$, with $\Phi_\phi(\beta, z) = \phi^{<L}$. Each expression is an identity $\Theta(z, (\phi)_{\phi\in\Sigma}, (g_{*,k})_k) = 0$ for a polynomial Θ with complex coefficients. Since $\phi - \phi^{<L} \in \mathfrak{n}_*^L$ for any ϕ, the coefficients of the part of degree less than L of $\Theta(z, (\Phi_\phi(\beta, z))_\phi, (g_{*,k}^{<L})_k)$ vanish. The coefficients are polynomial functions on $\mathcal{J}_L \times \mathbb{A}^q$ vanishing at $((g_{*,k}^{<L})_k, \beta)$. Let $\Psi_1, \dots, \Psi_{n''}$ be all such coefficients for all the expressions.

Since $\mathcal{I}'$ is infinite, (A) with $mT+1$ in place of T, $d = L$, and $\Phi_j := \Phi_{\varphi_j}$ implies that there exist $i \in \mathcal{I}'$ and $\beta_i \in \mathbb{C}^q$ satisfying the conclusion of (A) and such that $\operatorname{mult}_{F_i} g_{i,k} > b_\mu t_1$ for any μ in case (a), with k as in (a). For $\phi \in \Sigma$ let $\phi[i] := \Phi_\phi(\beta_i, z) \in \mathbb{C}[z_1, z_2, z_3]$, and let $\mathfrak{p}_\mu[i] \subset \mathbb{C}[z_1, z_2, z_3]$ be the ideal generated by $\omega_{\mu,1}[i], \dots, \omega_{\mu,n'_\mu}[i]$. The relations $\Psi_j((g_{i,k}^{<L})_k, \beta_i) = 0$ mean that each identity $\Theta = 0$ above implies

$$\Theta(z, (\phi[i])_\phi, (g_{i,k}^{<L})_k) \in \mathfrak{m}^L.$$

Moreover, $\operatorname{mult}_{F_i} \varphi_j[i] \ge mT + 1$ for any j, and $\operatorname{mult}_{F_i} \mathfrak{m}^L \ge L > mT$ since F_i has center 0.

For any μ, let j_μ be an index with

$$\operatorname{mult}_{F_i} \omega_{\mu,j_\mu}[i] = \operatorname{mult}_{F_i} \mathfrak{p}_\mu[i].$$

By (I0),

$$\prod_\mu \omega_{\mu,j_\mu}[i] \in (\varphi_1[i], \dots, \varphi_{n'}[i]) + \mathfrak{m}^L,$$

so

$$\sum_{\mu=1}^m \operatorname{mult}_{F_i} \mathfrak{p}_\mu[i] = \operatorname{mult}_{F_i} \prod_\mu \omega_{\mu,j_\mu}[i] \ge mT + 1.$$

Thus there exists μ with $\operatorname{mult}_{F_i} \mathfrak{p}_\mu[i] > T$, and then $\operatorname{mult}_{F_i}(\mathfrak{p}_\mu[i] + \mathfrak{m}^L) > T$. Since $g_{i,k} - g_{i,k}^{<L} \in \mathfrak{m}^L$, we have $\operatorname{mult}_{F_i} g_{i,k}^{<L} \ge \min\{\operatorname{mult}_{F_i} g_{i,k}, L\}$ and $\operatorname{mult}_{F_i} g_{i,k} \ge \min\{\operatorname{mult}_{F_i} g_{i,k}^{<L}, L\}$.

- In case (a), (I1) implies

  $$z_1^{b_\mu} - w_\mu[i] g_{i,k}^{<L} \in \mathfrak{p}_\mu[i] + \mathfrak{m}^L.$$

  Since

  $$\operatorname{mult}_{F_i} g_{i,k}^{<L} \ge \min\{\operatorname{mult}_{F_i} g_{i,k}, L\} > b_\mu t_1$$

  and $T > b_\mu t_1$, we get $b_\mu t_1 = \operatorname{mult}_{F_i} z_1^{b_\mu} > b_\mu t_1$, a contradiction.
- In case (b), (I2) implies $g_{i,k}^{<L} \in \mathfrak{p}_\mu[i] + \mathfrak{m}^L$, so $\operatorname{mult}_{F_i} g_{i,k} \ge \min\{T + 1, L\} > T$, which contradicts the choice of T.

- In case (c), (I3) implies $\operatorname{mult}_{F_i}(z_j^{b_\mu} - \eta_\mu[i]\zeta_\mu[i]) > T$ and $\operatorname{mult}_{F_i}(z_j\eta_\mu[i]^n - \xi_\mu[i]^n) > T$. Since $\operatorname{mult}_{F_i} z_j^{b_\mu} = b_\mu t_j < T$, we have $\operatorname{mult}_{F_i}(\eta_\mu[i]\zeta_\mu[i]) = b_\mu t_j$, so $\operatorname{mult}_{F_i}\eta_\mu[i] \le b_\mu t_j$. Then
$$\operatorname{mult}_{F_i}(z_j\eta_\mu[i]^n) = t_j + n\operatorname{mult}_{F_i}\eta_\mu[i] \le (1+nb_\mu)t_j < T,$$
so $n\operatorname{mult}_{F_i}\xi_\mu[i] = t_j + n\operatorname{mult}_{F_i}\eta_\mu[i]$, that is, $t_j = n(\operatorname{mult}_{F_i}\xi_\mu[i] - \operatorname{mult}_{F_i}\eta_\mu[i])$. Thus $n \mid t_j$, a contradiction.

Therefore some C_μ satisfies (B1) and $n_{\mu,j} \mid t_j$ for any j with $z_j \mid r$. Let $C := C_\mu$. By **Step 3**, C satisfies (B2). The proposition follows. □

Lemma 6.4. *Let n and N be positive integers, let G be a finite diagonal group acting on $\mathbb{A}^n$, let $w \in (\mathbb{N}^+)^n$, let $f_1, \dots, f_N$ be non-zero G-semi-invariant regular functions on a neighborhood of $0 \in \mathbb{A}^n$, and let $c_1, \dots, c_N$ be positive real numbers. Let*
$$B := \sum_{k=1}^N c_k \operatorname{div}(f_k) \quad \text{and} \quad B_w := \sum_{k=1}^N c_k \operatorname{div}(\operatorname{in}_w f_k).$$
Then the following hold.

(1) We have
$$\operatorname{mld}((\mathbb{A}^n \ni 0, B)/G) \ge \operatorname{mld}((\mathbb{A}^n \ni 0, B_w)/G).$$
(2) Assume moreover that $n = 3$ and the following hold.
  (i) For any k, $\operatorname{in}_w f_k$ is a reduced polynomial vanishing at 0, and for any $k \ne k'$, $\operatorname{in}_w f_k$ and $\operatorname{in}_w f_{k'}$ are coprime.
  (ii) $c_k \le 1$ for any k.
  (iii) For any $w' \in \mathbb{R}^3_{>0}$, the pair
$$\Big((\mathbb{C}^*)^3, \sum_k c_k \operatorname{div}(\operatorname{in}_{w'}\operatorname{in}_w f_k)\Big)$$
  is lc.
  (iv) $(\mathbb{A}^3, B_w)$ is lc near 0.
  Then
$$\operatorname{mld}((\mathbb{A}^3 \ni 0, B_w)/G) = \psi_{\min}(G, J, c),$$
  where $J := (J_1, \dots, J_N)$, $J_k \subset \mathbb{C}[x_1, x_2, x_3]$ is the ideal generated by the monomials occurring in $\operatorname{in}_w f_k$, and $c := (c_1, \dots, c_N)$ (Definition 2.3).
(3) (1) holds as well for non-zero G-semi-invariant elements $f_1, \dots, f_N$ of $\mathbb{C}[[x_1, \dots, x_n]]$, where the minimal log discrepancy on the left is taken on $\operatorname{Spec}\mathbb{C}[[x_1, \dots, x_n]]$ at its closed point (Definition 2.2).

Proof. Let $x_1, \dots, x_n$ be the coordinates of $\mathbb{A}^n$ on which G acts diagonally, $q := |G|$, the quotient morphism
$$\pi : \mathbb{A}^n \to X := \mathbb{A}^n/G,$$
$o := \pi(0)$, and $B_{\mathrm{br}} \ge 0$ the $\mathbb{Q}$-divisor on X with
$$K_{\mathbb{A}^n} = \pi^*(K_X + B_{\mathrm{br}}).$$
The components of B_{br} are images of hyperplanes fixed pointwise by non-trivial elements of G, that is, of coordinate hyperplanes $(x_j = 0)$. Since x_j^q is a G-invariant function whose divisor on X is supported on $\pi((x_j = 0))$, B_{br} is $\mathbb{Q}$-Cartier. For a non-zero G-semi-invariant regular function f near 0, f^q is G-invariant, so $f^q = \pi^* f_X$ for a regular function f_X near o, and $\pi^*(\frac{1}{q}\operatorname{div}(f_X)) = \operatorname{div}(f)$. Thus, for positive real numbers c'_k and non-zero G-semi-invariant regular functions f'_k near 0, the $\mathbb{R}$-divisors $D := \sum_k c'_k \operatorname{div}(f'_k)$ and $D_X := \sum_k \frac{c'_k}{q}\operatorname{div}(f'_{k,X})$ satisfy $K_{\mathbb{A}^n} + D = \pi^*(K_X + B_{\mathrm{br}} + D_X)$, and
$$\operatorname{mld}((\mathbb{A}^n \ni 0, D)/G) = \operatorname{mld}(X \ni o, B_{\mathrm{br}} + D_X) \tag{6.11}$$

(Definition 2.2 and [Liu26a, Lemma 2.6]).

(1) Write $f_k = h_k/u_k$ with $h_k, u_k \in \mathbb{C}[x_1, \dots, x_n]$ and $u_k(0) \neq 0$, and let $u'_k := \prod_{g\in G} g^* u_k$. Then u'_k is G-invariant with $u'_k(0) = u_k(0)^q \neq 0$, and $f_k u'_k = h_k \prod_{g\in G\setminus\{1\}} g^* u_k$ is a G-semi-invariant polynomial. Near 0, $\operatorname{div}(f_k u'_k) = \operatorname{div}(f_k)$, and $\operatorname{in}_w(f_k u'_k) = u'_k(0) \operatorname{in}_w f_k$ because in_w is multiplicative and $\operatorname{in}_w u'_k = u'_k(0)$. Since both sides of the inequality in (1) depend only on a neighborhood of 0, possibly replacing f_k with $f_k u'_k$, we may assume that each f_k is a polynomial.

Let t be the coordinate of $\mathbb{A}^1$. For any k, let

$$f_{k,t} := t^{-\operatorname{wt}_w(f_k)} f_k(t^{w_1} x_1, \dots, t^{w_n} x_n).$$

Since any monomial $x_1^{\alpha_1} \cdots x_n^{\alpha_n}$ occurring in f_k satisfies $\sum_j w_j \alpha_j \geq \operatorname{wt}_w(f_k)$, we have

$$f_{k,t} \in \mathbb{C}[x_1, \dots, x_n, t] \quad \text{and} \quad f_{k,0} = \operatorname{in}_w f_k.$$

Since G is diagonal, any monomial occurring in f_k has the character of f_k, so for any $t_0 \in \mathbb{C}$, f_{k,t_0} is a non-zero G-semi-invariant polynomial with the character of f_k. Let $B_{t_0} := \sum_k c_k \operatorname{div}(f_{k,t_0})$, so $B_0 = B_w$. For $t_0 \neq 0$, the automorphism ϕ_{t_0} of $\mathbb{A}^n$ given by $\phi_{t_0}^* x_j = t_0^{w_j} x_j$ commutes with G and fixes 0, so ϕ_{t_0} descends to an automorphism ϕ_{X,t_0} of X fixing o. Since B_{br} is determined by π and $\pi \circ \phi_{t_0} = \phi_{X,t_0} \circ \pi$, we have $\phi_{X,t_0}^* B_{\mathrm{br}} = B_{\mathrm{br}}$. As $f_{k,t_0} = t_0^{-\operatorname{wt}_w(f_k)} \phi_{t_0}^* f_k$, (6.11) implies

$$\operatorname{mld}((\mathbb{A}^n \ni 0, B_{t_0})/G) = \operatorname{mld}((\mathbb{A}^n \ni 0, B)/G) \quad \text{for any } t_0 \in \mathbb{C}^*. \tag{6.12}$$

The polynomials $f_{k,t}^q$ belong to $\mathbb{C}[x_1, \dots, x_n, t]^G = \mathbb{C}[X \times \mathbb{A}^1]$. Let $\mathcal{I}_k \subset \mathcal{O}_{X\times\mathbb{A}^1}$ be the ideal sheaf generated by $f_{k,t}^q$, and let $\mathfrak{A} := \prod_k \mathcal{I}_k^{c_k/q}$. For any $t_0 \in \mathbb{C}$, the restriction $\mathfrak{A}_{t_0}$ of $\mathfrak{A}$ to $X \times \{t_0\}$ is $\prod_k (f_{k,t_0,X})^{c_k/q}$, so by (6.11),

$$\operatorname{mld}(X \ni o, B_{\mathrm{br}}, \mathfrak{A}_{t_0}) = \operatorname{mld}((\mathbb{A}^n \ni 0, B_{t_0})/G),$$

where both sides may be $-\infty$.

Let $G_0 \subset G$ be the subgroup generated by the pseudo-reflections in G. For any j, let $G^{(j)} \subset G$ be the subgroup of elements acting trivially on $x_{j'}$ for any $j' \neq j$, and $r_j := |G^{(j)}|$. Since G is diagonal, $G_0 = G^{(1)} \times \cdots \times G^{(n)}$, so $\mathbb{A}^n/G_0 \cong \mathbb{A}^n$ with coordinates $x_1^{r_1}, \dots, x_n^{r_n}$, on which G/G_0 acts diagonally. If $g = \operatorname{diag}(\zeta_1, \dots, \zeta_n) \in G$ acts on $\mathbb{A}^n/G_0$ as a pseudo-reflection, say $\zeta_j^{r_j} = 1$ for any $j \neq j_0$, then $\operatorname{diag}(1, \dots, \zeta_j, \dots, 1) \in G_0$ for any $j \neq j_0$, so $\operatorname{diag}(1, \dots, \zeta_{j_0}, \dots, 1) \in G$ and $g \in G_0$. Thus G/G_0 contains no pseudo-reflection of $\mathbb{A}^n/G_0$. As G/G_0 acts diagonally, the fixed locus of any non-trivial element of G/G_0 is a coordinate subspace of codimension ≥ 2, so G/G_0 acts on $\mathbb{A}^n/G_0$ freely in codimension 1. Thus $X = (\mathbb{A}^n/G_0)/(G/G_0)$ with $\mathbb{A}^n/G_0 \cong \mathbb{A}^n$ and G/G_0 acting freely in codimension 1, so by Theorem 2.16 the function $t_0 \to \operatorname{mld}(X \ni o, B_{\mathrm{br}}, \mathfrak{A}_{t_0})$ on $\mathbb{C}$ is lower semi-continuous.

Suppose that $\operatorname{mld}((\mathbb{A}^n \ni 0, B_w)/G) > \operatorname{mld}((\mathbb{A}^n \ni 0, B)/G)$, and fix a real number α_0 strictly between them. By lower semi-continuity, the set U_0 of $t_0 \in \mathbb{C}$ with $\operatorname{mld}(X \ni o, B_{\mathrm{br}}, \mathfrak{A}_{t_0}) > \alpha_0$ is open, and $0 \in U_0$ since $B_0 = B_w$. Thus U_0 contains some $t_0 \in \mathbb{C}^*$, which contradicts (6.12). This proves (1).

(2) By [Liu26a, Definition 2.16],

$$\operatorname{wt}_v(J_k) = \operatorname{wt}_v(\operatorname{in}_w f_k) \quad \text{for any } v \in \mathbb{R}^3_{>0} \text{ and any } k,$$

so [Liu26a, Proposition 8.11(3)] for the polynomials $\operatorname{in}_w f_1, \dots, \operatorname{in}_w f_N$ with coefficients $c_1, \dots, c_N$ implies

$$\operatorname{mld}((\mathbb{A}^3 \ni 0, B_w)/G) = \min_{v\in N_G \cap \mathbb{R}^3_{>0}} \psi_{J,c}(v) = \psi_{\min}(G, J, c),$$

where, by [Liu26a, Definition 2.16],

$$\psi_{J,c}(v) := v_1 + v_2 + v_3 - \sum_k c_k \operatorname{wt}_v(J_k), \quad \text{so that} \quad \psi_{\min}(G, J, c) = \inf\{\psi_{J,c}(v) \mid v \in N_G \cap \mathbb{R}^3_{>0}\}.$$

(3) Let $\widehat{\mathbb{A}} := \operatorname{Spec} \mathbb{C}[[x_1, \dots, x_n]]$, let E be a formal prime divisor over $\widehat{\mathbb{A}}$ with center the closed point, and fix an integer L greater than $\operatorname{mult}_E f_k$ and than the degrees of the monomials occurring

in $\mathrm{in}_w f_k$ for any k. Since G is diagonal, the sum $f_k^{<L}$ of the terms of f_k of degree less than L is a non-zero G-semi-invariant polynomial with $\mathrm{in}_w f_k^{<L} = \mathrm{in}_w f_k$, and $\mathrm{mult}_E f_k^{<L} = \mathrm{mult}_E f_k$ because $\mathrm{mult}_E(f_k - f_k^{<L}) \geq L$. Let $B^{(L)} := \sum_k c_k \operatorname{div}(f_k^{<L})$, and let E_0 be the prime divisor over $\mathbb{A}^n$ with center 0 to which E restricts. Then Lemma 2.12(3) for $(\mathbb{A}^n \ni 0, B^{(L)})$ and $H = G$, and (1) for the polynomials $f_k^{<L}$ imply

$$\frac{a(E, \widehat{\mathbb{A}}, B)}{e_E} = \frac{a(E_0, \mathbb{A}^n, B^{(L)})}{e_{E_0}} \geq \mathrm{mld}((\mathbb{A}^n \ni 0, B^{(L)})/G) \geq \mathrm{mld}((\mathbb{A}^n \ni 0, B_w)/G).$$

Taking the infimum over E proves (3). The lemma follows. □

6.3. **Bounded marked orders.**

Theorem 6.5. *Let m and R be positive integers and $B_0 \geq 0$ a real number. For any integer $i \geq 1$, let f_i and h_i be polynomial functions on $\mathbb{A}^4$ such that $W_i := (f_i = 0) \subset \mathbb{A}^4$ is a hypersurface threefold normal near 0 and $h_i|_{W_i} \neq 0$, and for a prime divisor F over W_i write $\mathrm{mult}_F h_i := \mathrm{mult}_F(h_i|_{W_i})$. Let (f_*, h_*) be a compatible generic limit of the tuples (f_i, h_i) along an infinite subset $\mathcal{I} \subset \mathbb{N}^+$, with coefficient field $\Bbbk$ (Definition 2.5, with H trivial). Assume that*

(i) $W_* := \operatorname{Spec} \Bbbk[[x_1, \dots, x_4]]/(f_*)$ is normal,
(ii) $h_*|_{W_*} \neq 0$, and
(iii) $(W_*, \frac{1}{m} \operatorname{div}(h_*|_{W_*}))$ is klt on $W_* \setminus \operatorname{Sing}(W_*)$.

Then there exists a real number T such that for all but finitely many $i \in \mathcal{I}$, any prime divisor F over W_i with center 0 satisfying

$$a(F, W_i, 0) - \frac{1}{m} \mathrm{mult}_F h_i \leq B_0 \quad \text{and} \quad \mathrm{mult}_F \operatorname{Jac}_{W_i} \leq R$$

satisfies

$$\mathrm{mult}_F h_i \leq T \quad \text{and} \quad a(F, W_i, 0) \leq B_0 + \frac{T}{m}.$$

Proof. We prove the theorem in three steps. Let $\mathfrak{m}$ be the maximal ideal of $\mathcal{O}_{\mathbb{A}^4,0}$, let $\mathfrak{n}_*$ be the maximal ideal of $\Bbbk[[x_1, \dots, x_4]]$, and let

$$\mathcal{X}_* := \operatorname{Spec} \Bbbk[[x_1, \dots, x_4]].$$

Step 1. Let $g : Y \to W_*$ be a log resolution of $(W_*, \operatorname{div}(h_*|_{W_*}))$ [dFEM11, Theorems 2.3 and 2.4], and let $\mathcal{E}$ be the finite set of prime divisors E on Y contained in $\operatorname{Exc}(g) \cup \operatorname{Supp} g_*^{-1} \operatorname{div}(h_*|_{W_*})$ with $g(E) \not\subset \operatorname{Sing}(W_*)$. By (iii),

$$a(E, W_*, 0) - \frac{1}{m} \mathrm{mult}_E h_* > 0 \quad \text{for any } E \in \mathcal{E}.$$

Since $\mathcal{E}$ is finite, there exists a rational number $\mu > \frac{1}{m}$ such that $a(E, W_*, 0) - \mu \,\mathrm{mult}_E h_* > 0$ for any $E \in \mathcal{E}$. Thus $(W_*, \mu \operatorname{div}(h_*|_{W_*}))$ is klt, in particular lc, on $W_* \setminus \operatorname{Sing}(W_*)$. Since $\mu > \frac{1}{m}$, there exist rational numbers $0 < \theta_1 < \theta_2 < 1$ such that

$$c_0 := \theta_1 \mu - \frac{1}{m} - (1 - \theta_1) > 0. \tag{6.13}$$

Let

$$\mathcal{J}_* := \mathcal{J}\big((f_*)^{\theta_2} (h_*)^{\theta_2 \mu}\big).$$

By Lemma 6.1(3), with $V = \operatorname{Sing}(W_*)$, $\mathcal{J}_*$ is the unit ideal near any point of $W_* \setminus \operatorname{Sing}(W_*)$. Thus

$$\operatorname{Supp}\big(\mathcal{O}_{\mathcal{X}_*}/(\mathcal{J}_* + (f_*))\big) \subset \operatorname{Sing}(W_*).$$

By the Jacobian criterion [Mat89, Theorem 30.4(ii)],

$$\operatorname{Sing}(W_*) \subset V\Big(f_*, \frac{\partial f_*}{\partial x_1}, \dots, \frac{\partial f_*}{\partial x_4}\Big).$$

Hence each $\partial f_*/\partial x_j$ belongs to the radical of $\mathcal{J}_* + (f_*)$. Since $\Bbbk[[x_1, \dots, x_4]]$ is Noetherian, there exist a positive integer N and $b_1^*, \dots, b_4^* \in \Bbbk[[x_1, \dots, x_4]]$ such that

$$\left(\frac{\partial f_*}{\partial x_j}\right)^N - f_* b_j^* \in \mathcal{J}_* \quad \text{for any } j \in \{1, \dots, 4\}. \tag{6.14}$$

Step 2. We spread (6.14) to the W_i. Let L be an integer with $L > NR$. For $j \in \{1, \dots, 4\}$, let

$$\Phi_j := y_{1,e_j}^N - y_{1,0} \sum_{\alpha \in \mathbb{N}^4,\ |\alpha| < L} u_{j,\alpha} x^\alpha,$$

where $e_j \in \mathbb{N}^4$ is the j-th unit vector, $|\alpha|$ is the sum of the entries of α, and the $u_{j,\alpha}$ ($1 \le j \le 4$, $|\alpha| < L$) are the parameter variables $u_1, \dots, u_q$ of Lemma 6.1, indexed by (j, α). Let $\beta := (\beta_{j,\alpha})_{j,\alpha}$, where $\beta_{j,\alpha}$ is the coefficient of x^α in b_j^*. Then

$$\Phi_j((f_*, h_*), \beta) = \left(\frac{\partial f_*}{\partial x_j}\right)^N - f_* \sum_{|\alpha| < L} \beta_{j,\alpha} x^\alpha,$$

and it differs from the element in (6.14) by an element of $f_* \mathfrak{n}_*^L$. Thus

$$\Phi_j((f_*, h_*), \beta) \in \mathcal{J}_* + \mathfrak{n}_*^L \quad \text{for any } j \in \{1, \dots, 4\}.$$

Let $\mathcal{I}_0$ be the set of indices $i \in \mathcal{I}$ for which there exists $\beta_i = (\beta_{i,j,\alpha})_{j,\alpha}$ with complex entries such that

$$z_{j,i} := \left(\frac{\partial f_i}{\partial x_j}\right)^N - f_i b_{j,i} \in \mathcal{J}\big((f_i)^{\theta_1}(h_i)^{\theta_1 \mu}\big) + \mathfrak{m}^L \quad \text{for any } j \in \{1, \dots, 4\}, \tag{6.15}$$

where the multiplier ideal is taken on $\operatorname{Spec} \mathbb{C}[[x_1, \dots, x_4]]$, $\mathfrak{m}^L$ stands for $\mathfrak{m}^L \mathbb{C}[[x_1, \dots, x_4]]$, and

$$b_{j,i} := \sum_{|\alpha| < L} \beta_{i,j,\alpha} x^\alpha,$$

that is, $z_{j,i} = \Phi_j((f_i, h_i), \beta_i)$. By Lemma 6.1(1), with the pairs (f_i, h_i) and (f_*, h_*) in place of $\mathbf{f}_i$ and $\mathbf{f}_*$, $(\gamma_1, \gamma_2) = (\theta_1, \theta_1\mu)$, $(\gamma_1', \gamma_2') = (\theta_2, \theta_2\mu)$, and $(Y_i \ni y_i) = (\mathbb{A}^4 \ni 0)$, $\mathcal{I} \setminus \mathcal{I}_0$ is finite. For $i \in \mathcal{I}_0$ we fix such β_i.

Step 3. Let

$$T := \frac{B_0 + NR + (1 - \theta_1)((N+2)R + 1)}{c_0},$$

which depends only on m, B_0, R, μ, θ_1, and N. Let $i \in \mathcal{I}_0$, and let F be a prime divisor over W_i with center 0 such that

$$a(F, W_i, 0) - \frac{1}{m} \operatorname{mult}_F h_i \le B_0 \quad \text{and} \quad \operatorname{mult}_F \operatorname{Jac}_{W_i} \le R.$$

By Definition 5.2, there exists $j \in \{1, \dots, 4\}$ such that

$$\operatorname{mult}_F \left(\frac{\partial f_i}{\partial x_j}\Big|_{W_i}\right) = \operatorname{mult}_F \operatorname{Jac}_{W_i}.$$

Let $k := \operatorname{mult}_F h_i + (N+2)R + 1$. Then

$$k > 2R \ge 2 \operatorname{mult}_F \operatorname{Jac}_{W_i} \quad \text{and} \quad k - \operatorname{mult}_F \operatorname{Jac}_{W_i} \ge k - R > \operatorname{mult}_F h_i,$$

so Lemma 5.4(2) with $\mathcal{X} = \mathbb{A}^4$, $x = x_j$, and $H_1 = h_i$ gives a prime divisor P over $\mathbb{A}^4$ with center 0 such that

$$\begin{gathered} \operatorname{mult}_P f_i = k, \quad \operatorname{mult}_P h_i = \operatorname{mult}_F h_i, \quad \operatorname{mult}_P \frac{\partial f_i}{\partial x_j} = \operatorname{mult}_F \operatorname{Jac}_{W_i}, \\ \text{and} \quad a(P, \mathbb{A}^4, 0) = a(F, W_i, 0) + k. \end{gathered} \tag{6.16}$$

Since $b_{j,i}$ is a polynomial and $\operatorname{center}_{\mathbb{A}^4} P = \{0\}$,

$$\operatorname{mult}_P(f_i b_{j,i}) \geq \operatorname{mult}_P f_i = k > NR \geq N \operatorname{mult}_F \operatorname{Jac}_{W_i} = \operatorname{mult}_P \Big(\frac{\partial f_i}{\partial x_j}\Big)^N.$$

Thus

$$\operatorname{mult}_P z_{j,i} = N \operatorname{mult}_F \operatorname{Jac}_{W_i} \leq NR < L \leq \operatorname{mult}_P(\mathfrak{m}^L).$$

By (6.15) and the resolution formula for $\mathcal{J}((f_i)^{\theta_1}(h_i)^{\theta_1\mu})$ on a log resolution on which P is a prime divisor, as at the end of the proof of Lemma 6.1(2),

$$\operatorname{mult}_P z_{j,i} + a(P, \mathbb{A}^4, 0) > \theta_1 \operatorname{mult}_P f_i + \theta_1\mu \operatorname{mult}_P h_i,$$

and by (6.16),

$$N \operatorname{mult}_F \operatorname{Jac}_{W_i} + a(F, W_i, 0) + k > \theta_1 k + \theta_1\mu \operatorname{mult}_F h_i.$$

Since $a(F, W_i, 0) \leq B_0 + \frac{1}{m} \operatorname{mult}_F h_i$ and $\operatorname{mult}_F \operatorname{Jac}_{W_i} \leq R$,

$$\Big(\theta_1\mu - \frac{1}{m}\Big) \operatorname{mult}_F h_i < B_0 + NR + (1-\theta_1)k = B_0 + NR + (1-\theta_1)\big(\operatorname{mult}_F h_i + (N+2)R + 1\big).$$

By (6.13),

$$c_0 \operatorname{mult}_F h_i < B_0 + NR + (1-\theta_1)((N+2)R+1) = c_0 T,$$

so $\operatorname{mult}_F h_i < T$ and

$$a(F, W_i, 0) \leq B_0 + \frac{1}{m} \operatorname{mult}_F h_i \leq B_0 + \frac{T}{m}.$$

Since $\mathcal{I}\backslash\mathcal{I}_0$ is finite, the theorem follows. □

7. Rigidity of weighted critical pairs and signed adjunction

In this section, we show that if a pair on $\mathbb{A}^2$, or on a surface in $\mathbb{A}^3$ whose equation has as top part one of the Du Val normal forms of Lemma 7.3, is lc but not klt at a closed point o, no component of the boundary through o has coefficient one, and the top parts of the equations of the surface and of the boundary form a critical pair, then after a triangular change of coordinates the equation of the surface becomes its top part and the equations of the boundary and their top parts have equal restrictions to the surface. We also prove an adjunction on the A-cone for signed boundaries, that is, boundaries whose coefficient along one coordinate hyperplane is a real number $\beta < 1$, possibly negative, and whose other components are effective, and a critical stretching lemma.

7.1. Critical pairs and $\mathbb{P}^1$-links.

Definition 7.1. Let $\Bbbk$ be an algebraically closed field containing $\mathbb{C}$, let $w = (w_1, \dots, w_n)$ be a vector of positive integers, and let V be either $\mathbb{A}^n_\Bbbk$, or, when $n = 3$, a normal surface $(F = 0) \subset \mathbb{A}^3_\Bbbk$ for a w-homogeneous polynomial F. Let $\deg_w V := 0$ in the first case and $\deg_w V := \operatorname{wt}_w(F)$ in the second case. Let $P_1, \dots, P_N$ be w-homogeneous polynomials with $P_j|_V \neq 0$ for any j, let $d_j := \operatorname{wt}_w(P_j)$, and let $\kappa_1, \dots, \kappa_N$ be positive real numbers. The pair $(V, \sum_j \kappa_j \operatorname{div}(P_j|_V))$ is called a *critical pair* for w if

$$\sum_{j=1}^N \kappa_j d_j = \sum_{i=1}^n w_i - \deg_w V.$$

For a non-zero polynomial Q, the w-homogeneous part of Q of maximal w-weight is called the *top part* of Q.

A *dlt model* of a pair (X, B) is a $\mathbb{Q}$-factorial dlt pair (X', B') with a projective birational morphism $\sigma : X' \to X$ as in [FS23, Theorem 2.9]: any exceptional prime divisor of σ has log discrepancy at most 0 with respect to (X, B), and $B' = \sigma_*^{-1} B^{\leq 1} + \operatorname{Exc}(\sigma)$, where $B^{\leq 1}$ is B with all coefficients larger than 1 replaced by 1. A dlt model exists without any assumption on the singularities of (X, B).

Theorem 7.2 ([FS23, Theorem 1.1 and the remark after it]). *Let (X, B) be a pair, $f : X \to Z$ a projective morphism to a variety with $K_X + B \sim_{\mathbb{Q},f} 0$, and $z \in Z$ a point such that $f^{-1}(z)$ is connected and $f^{-1}(z) \cap \mathrm{Nklt}(X, B)$ is disconnected. Then $f^{-1}(z) \cap \mathrm{Nklt}(X, B)$ has exactly two connected components, and (X, B) is lc near $f^{-1}(z)$. Moreover, any dlt model $(X', B') \to Z$ of (X, B) is plt near the fiber over z.*

Both [FS23] and [Mor24] work over an arbitrary algebraically closed field of characteristic zero [FS23, Section 2.1], [Mor24, Section 2], so Theorem 7.2 and the results of [Mor24] used below hold over $\Bbbk$.

Lemma 7.3. *Let $\Bbbk$ be an algebraically closed field containing $\mathbb{C}$, and let V, V_0, and w be as in one of the following cases.*

(a) $V = V_0 = \mathbb{A}^2_\Bbbk$, and $w = (w_1, w_2)$ for coprime positive integers w_1, w_2.

(b) $w = (w_1, w_2, w_3)$ is a vector of positive integers with $\gcd(w_1, w_2, w_3) = 1$, F is a polynomial in the coordinates x, y, z of $\mathbb{A}^3_\Bbbk$ whose top part f_0 is one of the following normal forms of Du Val singularities of types A_{d-1}, D_k, E_6, E_7, and E_8:

$$xy + z^d \ (d \geq 2), \quad x^2 + zy^2 + z^{k-1} \ (k \geq 4), \quad x^2 + y^3 + z^4, \quad x^2 + y^3 + yz^3, \quad \text{and} \quad x^2 + y^3 + z^5,$$

$V := (F = 0)$, and $V_0 := (f_0 = 0)$.

Let $Q_1, \dots, Q_N$ be polynomials whose top parts $P_1, \dots, P_N$ satisfy $P_j|_{V_0} \neq 0$ for any j, let $\kappa_1, \dots, \kappa_N$ be positive real numbers such that $(V_0, \sum_j \kappa_j \operatorname{div}(P_j|_{V_0}))$ is a critical pair for w, and let $B := \sum_j \kappa_j \operatorname{div}(Q_j|_V)$. Let $X \subset \mathbb{P}(1, w)$ be the closure of V, the first homogeneous coordinate having weight 1, let $\Delta_\infty := X \backslash V$, and let $\overline{B}$ be the closure of B in X. Then V is normal with at worst Du Val singularities, X is normal and klt, and

$$K_X + \overline{B} + \Delta_\infty \sim_\mathbb{R} 0.$$

If $(V_0, \sum_j \kappa_j \operatorname{div}(P_j|_{V_0}))$ is lc and has no coefficient-one component, then $(X, \overline{B} + \Delta_\infty)$ is plt near Δ_∞.

Now let $o \in V$ be a closed point. Assume that (V, B) is lc near o but not klt at o, and that $\mathrm{mult}_C B < 1$ for any prime divisor C on V with $o \in C$. Then the following hold.

(1) The pair $(X, \overline{B} + \Delta_\infty)$ is lc and plt near Δ_∞, and

$$\mathrm{Nklt}(X, \overline{B} + \Delta_\infty) = \Delta_\infty \sqcup \{o\}.$$

(2) The pair (V, B) has a unique lc place over o.

(3) There exists a polynomial automorphism φ of $\mathbb{A}^2_\Bbbk$, resp. $\mathbb{A}^3_\Bbbk$, whose coordinate functions vanish at o and which induces the identity on the graded ring of the w-weighted degree filtration, such that $Q_j = P_j \circ \varphi$ for any j in case (a), and

$$F = f_0 \circ \varphi \quad \text{and} \quad Q_j|_V = (P_j \circ \varphi)|_V$$

for any j in case (b). Here φ induces the identity on the graded ring of the w-weighted degree filtration if the top part of $g \circ \varphi$ is equal to the top part of g for any non-zero polynomial g.

Proof. Let $n := 2$ in case (a) and $n := 3$ in case (b), and let $x_1, \dots, x_n$ be the coordinates of $\mathbb{A}^n_\Bbbk$, with $(x_1, x_2, x_3) = (x, y, z)$ in case (b). Let $d_0 := \deg_w V_0$, that is, $d_0 = 0$ in case (a) and $d_0 = \mathrm{wt}_w(f_0)$ in case (b), and let $d_j := \mathrm{wt}_w(P_j)$ for $1 \leq j \leq N$. The critical equality reads

$$\sum_{j=1}^{N} \kappa_j d_j = \sum_{i=1}^{n} w_i - d_0. \tag{7.1}$$

For a polynomial g whose top part has w-weight m, let

$$g^{\mathrm{h}}(x_0, x_1, \dots, x_n) := x_0^m g(x_0^{-w_1} x_1, \dots, x_0^{-w_n} x_n).$$

Then g^{h} is a $(1,w)$-homogeneous polynomial and $g^{\mathrm{h}}(0,x_1,\dots,x_n)$ is the top part of g. In case (b), f_0 is irreducible since V_0 has an isolated singularity at 0. If $F=g_1g_2$, then f_0 is the product of the top parts of g_1 and g_2, so the top part of g_1 or of g_2 is a constant, hence so is g_1 or g_2, as w is positive. Thus F is irreducible, and so is F^{h}. Thus $X=(F^{\mathrm{h}}=0)\subset\mathbb{P}(1,w)$ in case (b), $X=\mathbb{P}(1,w_1,w_2)$ in case (a), and $\Delta_\infty=X\cap(x_0=0)$.

We divide the proof into several steps.

Step 1. In this step we show that V is normal with at worst Du Val singularities, X is normal, $\mathbb{Q}$-factorial, and klt, and $K_X+\overline{B}+\Delta_\infty\sim_{\mathbb{R}}0$.

In case (a), V is smooth. In case (b), let

$$V^{\mathrm{h}}:=(F^{\mathrm{h}}=0)\subset\mathbb{A}^4_{\Bbbk}$$

with coordinates x_0,x_1,x_2,x_3, and let $\mathbb{G}_m$ act on $\mathbb{A}^4_{\Bbbk}$ by

$$\tau\cdot(x_0,x_1,x_2,x_3)=(\tau x_0,\tau^{w_1}x_1,\tau^{w_2}x_2,\tau^{w_3}x_3).$$

Then V^{h} is $\mathbb{G}_m$-invariant, $V^{\mathrm{h}}\cap(x_0=0)=\{0\}\times V_0$, and the morphism $V^{\mathrm{h}}\cap(x_0\neq0)\to V\times\mathbb{G}_m$ sending (x_0,x_1,x_2,x_3) to $((x_0^{-w_1}x_1,x_0^{-w_2}x_2,x_0^{-w_3}x_3),x_0)$ is an isomorphism. Since f_0 is a normal form of a Du Val singularity, V_0 is normal with a Du Val singularity at 0 and smooth elsewhere. Since the Cartier divisor $\{0\}\times V_0$ of the hypersurface V^{h} is normal, V^{h} is normal near $\{0\}\times V_0$, and by inversion of adjunction, $(V^{\mathrm{h}},\{0\}\times V_0)$ is plt near $\{0\}\times V_0$. Thus V^{h} is normal and klt near $\{0\}\times V_0$. For any point $v\in V^{\mathrm{h}}$, $\tau\cdot v$ tends to 0 as τ tends to 0, so V^{h} is normal and klt at $\tau\cdot v$ for some $\tau\in\mathbb{G}_m$, hence at v since τ acts on V^{h} by an automorphism. Thus V^{h}, hence $V\times\mathbb{G}_m$, hence V, is normal and klt. Since V is a hypersurface, K_V is Cartier, so V is canonical, that is, V has at worst Du Val singularities.

Fix $i\in\{1,\dots,n\}$. The open subset $X\cap(x_i\neq0)$ is the quotient of

$$\widetilde{X}_i:=\{(y_0,y_1,\dots,y_n)\in\mathbb{A}^{n+1}_{\Bbbk}\mid y_i=1,\ F^{\mathrm{h}}(y_0,y_1,\dots,y_n)=0\}$$

(the equation $F^{\mathrm{h}}=0$ is omitted in case (a)) by μ_{w_i} acting by

$$\zeta\cdot(y_0,y_1,\dots,y_n)=(\zeta y_0,\zeta^{w_1}y_1,\dots,\zeta^{w_n}y_n)$$

(for $\mathbb{P}(1,w)$ this is [Ian00, 5.3]). Let $q_i:\widetilde{X}_i\to X\cap(x_i\neq0)$ be the quotient morphism, and let

$$\widetilde{\Delta}_i:=q_i^{-1}(\Delta_\infty)=\widetilde{X}_i\cap(y_0=0).$$

The morphism

$$\eta_i:\mathbb{G}_m\times\widetilde{\Delta}_i\to V_0\cap(x_i\neq0)$$

sending $(\tau,(0,y_1,\dots,y_n))$ to $(\tau^{w_1}y_1,\dots,\tau^{w_n}y_n)$ is the quotient by the free action of μ_{w_i} given by

$$\zeta\cdot(\tau,y)=(\zeta^{-1}\tau,\zeta\cdot y),$$

hence finite étale. In case (a), $\widetilde{X}_i\cong\mathbb{A}^2_{\Bbbk}$ and $\widetilde{\Delta}_i\cong\mathbb{A}^1_{\Bbbk}$. In case (b), since 0 is the only singular point of V_0 and

$$\sum_{j=1}^{3}w_jx_j\frac{\partial f_0}{\partial x_j}=d_0f_0,$$

at any point of V_0 with $x_i=1$ some $\partial f_0/\partial x_j$ with $j\neq i$ is non-zero. Since

$$\frac{\partial F^{\mathrm{h}}}{\partial y_j}(0,y_1,\dots,y_n)=\frac{\partial f_0}{\partial x_j}(y_1,\dots,y_n)\quad\text{for } j\geq1,$$

the variety $\widetilde{X}_i$ is smooth along $\widetilde{\Delta}_i$ and $\widetilde{\Delta}_i$ is a smooth curve. Thus in cases (a) and (b), $\widetilde{X}_i$ is smooth near $\widetilde{\Delta}_i$ and $\widetilde{\Delta}_i=(y_0=0)$ is a smooth Cartier curve on $\widetilde{X}_i$. Since $\gcd(w_1,\dots,w_n)=1$, a non-trivial $\zeta\in\mu_{w_i}$ fixes only points with $y_0=0$ and $y_j=0$ for some j with $\zeta^{w_j}\neq1$. Since $f_0|_{(x_j=0)}\neq0$ for any j (check the five equations), V_0 is irreducible, and η_i is finite étale, no component of $\widetilde{\Delta}_i$ is contained in $(y_j=0)$. Thus q_i is étale in codimension 1, $q_i^*\Delta_\infty=\widetilde{\Delta}_i$,

$K_{\widetilde{X}_i} = q_i^* K_X$, and X has only cyclic quotient singularities near Δ_∞. Since V_0 is irreducible, Δ_∞ is a prime divisor. Since V is a klt surface, hence $\mathbb{Q}$-factorial, X is normal, $\mathbb{Q}$-factorial, and klt.

For $g \in \mathcal{O}(V)\backslash\{0\}$, let $\deg_w g$ be the minimum of the w-weights of the top parts of the polynomials representing g. Let $\widetilde{g}$ be a representative of g whose top part $\widetilde{g}_{\mathrm{top}}$ has w-weight $\deg_w g$. In case (b), $f_0 \nmid \widetilde{g}_{\mathrm{top}}$, since otherwise subtracting a multiple of F from $\widetilde{g}$ lowers the weight of the top part. On $\widetilde{X}_i\backslash\widetilde{\Delta}_i$ we have $x_j = y_0^{-w_j} y_j$, so

$$q_i^* g = y_0^{-\deg_w g}\widetilde{g}^{\mathrm{h}}(y_0, y_1, \dots, y_n) \quad \text{and} \quad \widetilde{g}^{\mathrm{h}}(0, y_1, \dots, y_n) = \widetilde{g}_{\mathrm{top}}(y_1, \dots, y_n).$$

Since

$$\eta_i^* \widetilde{g}_{\mathrm{top}} = \tau^{\deg_w g} \cdot \widetilde{g}_{\mathrm{top}}|_{\widetilde{\Delta}_i}$$

and $\widetilde{g}_{\mathrm{top}}|_{V_0} \neq 0$, $\widetilde{g}_{\mathrm{top}}$ does not vanish identically on any component of $\widetilde{\Delta}_i$. Since q_i is étale over the generic point of Δ_∞,

$$\mathrm{mult}_{\Delta_\infty} \mathrm{div}_X(g) = -\deg_w g. \tag{7.2}$$

For any j, let $\overline{B}_j$ be the closure of $\mathrm{div}(Q_j|_V)$ in X. Since $P_j|_{V_0} \neq 0$, we have $\deg_w(Q_j|_V) = d_j$. Thus by (7.2),

$$\mathrm{div}_X(Q_j) = \overline{B}_j - d_j\Delta_\infty, \quad \text{and} \quad q_i^*\overline{B}_j = \mathrm{div}\left(Q_j^{\mathrm{h}}(y_0, y_1, \dots, y_n)\right) \text{ near } \widetilde{\Delta}_i. \tag{7.3}$$

Let Ω be the residue of $dx_1 \wedge dx_2 \wedge dx_3/F$ along V in case (b), and $\Omega := dx_1 \wedge dx_2$ in case (a). Since V is a hypersurface, Ω is a nowhere vanishing section of ω_V, so $\mathrm{div}_X(\Omega)$ is supported on Δ_∞. Let $\widetilde{\Omega}_i$ be the residue of $dy_0 \wedge \bigwedge_{j\neq i,0} dy_j/F^{\mathrm{h}}$ along $\widetilde{X}_i$ in case (b), and $\widetilde{\Omega}_i := dy_0 \wedge dy_{3-i}$ in case (a). Then $\widetilde{\Omega}_i$ is nowhere vanishing near $\widetilde{\Delta}_i$. Since $x_i = y_0^{-w_i}$, $x_j = y_0^{-w_j} y_j$ for $j \neq i$, and $F(x_1, \dots, x_n) = y_0^{-d_0} F^{\mathrm{h}}(y_0, \dots, y_n)$, we have

$$dx_1 \wedge \cdots \wedge dx_n = \pm w_i y_0^{-1-\sum_{j=1}^n w_j} dy_0 \wedge \bigwedge_{j\neq i,0} dy_j \quad \text{and} \quad q_i^*\Omega = \pm w_i y_0^{d_0-1-\sum_{j=1}^n w_j}\widetilde{\Omega}_i.$$

Since q_i is étale over the generic point of Δ_∞,

$$K_X \sim \mathrm{div}_X(\Omega) = \Big(d_0 - 1 - \sum_{j=1}^n w_j\Big)\Delta_\infty. \tag{7.4}$$

By (7.3), $\overline{B}_j \sim d_j\Delta_\infty$, so by (7.1) and (7.4),

$$K_X + \overline{B} + \Delta_\infty \sim_{\mathbb{R}} \Big(d_0 - 1 - \sum_{j=1}^n w_j + \sum_{j=1}^N \kappa_j d_j + 1\Big)\Delta_\infty = 0.$$

Step 2. In this step we assume that $(V_0, \sum_j \kappa_j \mathrm{div}(P_j|_{V_0}))$ is lc and has no coefficient-one component, and show that $(X, \overline{B} + \Delta_\infty)$ is plt near Δ_∞.

Fix $i \in \{1, \dots, n\}$. Since $\eta_i^* P_j = \tau^{d_j} \cdot P_j|_{\widetilde{\Delta}_i}$ and η_i is étale, for any closed point $p \in \widetilde{\Delta}_i$,

$$\mathrm{mult}_p \sum_j \kappa_j \mathrm{div}(P_j|_{\widetilde{\Delta}_i}) = \mathrm{mult}_{\overline{\eta_i(\mathbb{G}_m \times \{p\})}} \sum_j \kappa_j \mathrm{div}(P_j|_{V_0}) < 1.$$

Thus the pair

$$\Big(\widetilde{\Delta}_i, \sum_j \kappa_j \mathrm{div}(P_j|_{\widetilde{\Delta}_i})\Big)$$

is klt. By (7.3) and $Q_j^{\mathrm{h}}(0, y_1, \dots, y_n) = P_j(y_1, \dots, y_n)$, the restriction of $q_i^*\overline{B}$ to $\widetilde{\Delta}_i$ is $\sum_j \kappa_j \mathrm{div}(P_j|_{\widetilde{\Delta}_i})$. By inversion of adjunction, $(\widetilde{X}_i, \widetilde{\Delta}_i + q_i^*\overline{B})$ is plt near $\widetilde{\Delta}_i$. Since q_i is finite and étale in codimension 1,

$$K_{\widetilde{X}_i} + \widetilde{\Delta}_i + q_i^*\overline{B} = q_i^*(K_X + \Delta_\infty + \overline{B}),$$

so $(X, \overline{B}+\Delta_\infty)$ is plt near $\Delta_\infty \cap (x_i \neq 0)$. Since the open subsets $(x_i \neq 0)$, $1 \leq i \leq n$, cover Δ_∞, $(X, \overline{B}+\Delta_\infty)$ is plt near Δ_∞.

Step 3. From now on, we assume that (V, B) is lc near o but not klt at o, and that $\operatorname{mult}_C B < 1$ for any prime divisor C on V with $o \in C$. In this step we replace the labels κ_j by rational numbers.

Let $\pi_W : W \to X$ be a log resolution of $(X, \sum_j \overline{B}_j + \Delta_\infty)$. For $c = (c_1, \dots, c_N) \in \mathbb{R}^N_{>0}$, let $\overline{B}^c := \sum_j c_j \overline{B}_j$, and for a prime divisor S over X, let

$$a_S(c) := a(S, X, \overline{B}^c + \Delta_\infty).$$

Since X is $\mathbb{Q}$-factorial, a_S is the restriction of an affine function with rational coefficients. Let $\kappa := (\kappa_1, \dots, \kappa_N)$, so $\overline{B} = \overline{B}^\kappa$, and let $L \subset \mathbb{R}^N$ be the intersection of all affine subspaces containing κ which are zero sets of affine functions with rational coefficients. Then any affine function with rational coefficients vanishing at κ vanishes on L, and $L \cap \mathbb{Q}^N$ is dense in L. Let Λ be the finite set of the functions $c_1, \dots, c_N$ and a_S for the prime divisors S on W. We let $\lambda = (\lambda_1, \dots, \lambda_N) \in L \cap \mathbb{Q}^N$ be a point close to κ such that

$$\operatorname{sgn} \alpha(\lambda) = \operatorname{sgn} \alpha(\kappa) \quad \text{for any } \alpha \in \Lambda.$$

Then $\lambda_j > 0$ for any j, and since (7.1) is an affine equation in κ with integer coefficients and $\lambda \in L$,

$$\sum_{j=1}^N \lambda_j d_j = \sum_{j=1}^n w_j - d_0.$$

By (7.3) and (7.4),

$$K_X + \overline{B}^\lambda + \Delta_\infty \sim_{\mathbb{Q}} 0.$$

Moreover, the following hold.

(i) For any prime divisor S over X with $a_S(\kappa) = 0$, we have $a_S(\lambda) = 0$.

(ii) $(X, \overline{B}^\lambda + \Delta_\infty)$ is lc near o but not klt at o, any component of $\overline{B}^\lambda$ through o has coefficient less than 1, and there exists a prime divisor S_o on W with $\pi_W(S_o) = \{o\}$ and $a_{S_o}(\kappa) = a_{S_o}(\lambda) = 0$.

Since a_S is affine with rational coefficients, $a_S(\kappa) = 0$ implies $a_S|_L = 0$, so (i) holds. Since $(X, \overline{B}^c + \Delta_\infty)$ is lc near o if and only if $a_S(c) \geq 0$ for any prime divisor S on W with $o \in \pi_W(S)$, the pair $(X, \overline{B}^\lambda + \Delta_\infty)$ is lc near o. Since (V, B) is not klt at o, there exists a prime divisor S_o on W with $o \in \pi_W(S_o)$ and $a_{S_o}(\kappa) = 0$. If $\pi_W(S_o)$ were a curve, then it would be a component of B through o with coefficient 1. Thus $\pi_W(S_o) = \{o\}$, and $a_{S_o}(\lambda) = 0$ by (i). For a component C of $\overline{B}$ through o, the coefficient of C in $\overline{B}^c$ is $1 - a_C(c)$, and $a_C(\kappa) > 0$, so $a_C(\lambda) > 0$. This proves (ii).

Step 4. In this step we prove (1) and (2).

Since V is normal, a curve through o contained in $\operatorname{Nklt}(X, \overline{B}^\lambda + \Delta_\infty)$ is a component of $\overline{B}^\lambda$ through o with coefficient at least 1. By **Step 3**(ii), no such curve exists. Thus o is an isolated point of $\operatorname{Nklt}(X, \overline{B}^\lambda + \Delta_\infty)$. Since

$$o \notin \Delta_\infty \subset \operatorname{Nklt}(X, \overline{B}^\lambda + \Delta_\infty),$$

the non-klt locus of $(X, \overline{B}^\lambda + \Delta_\infty)$ is disconnected. By Theorem 7.2 for $X \to \operatorname{Spec} \Bbbk$, $(X, \overline{B}^\lambda + \Delta_\infty)$ is lc and

$$\operatorname{Nklt}(X, \overline{B}^\lambda + \Delta_\infty) = \Gamma \sqcup \{o\}$$

for a connected closed subset $\Gamma \supset \Delta_\infty$ of X.

Let $\pi : Y \to X$ be a dlt model of $(X, \overline{B}^{\lambda} + \Delta_\infty)$ [FS23, Theorem 2.9], let $B_Y := \pi_*^{-1}\overline{B}^{\lambda}$, and let $\Delta_Y := \pi_*^{-1}\Delta_\infty$. Since $(X, \overline{B}^{\lambda} + \Delta_\infty)$ is lc, any π-exceptional prime divisor has log discrepancy 0, so the boundary of the dlt model is $B_Y + \Delta_Y + \mathrm{Exc}(\pi)$ and

$$K_Y + B_Y + \Delta_Y + \mathrm{Exc}(\pi) = \pi^*(K_X + \overline{B}^{\lambda} + \Delta_\infty).$$

Let $\mathrm{Nklt}_Y := \mathrm{Nklt}(Y, B_Y + \Delta_Y + \mathrm{Exc}(\pi))$. By Theorem 7.2, $(Y, B_Y + \Delta_Y + \mathrm{Exc}(\pi))$ is plt. Thus $\mathrm{Nklt}_Y = \mathrm{Supp}\lfloor B_Y + \Delta_Y + \mathrm{Exc}(\pi)\rfloor$, and the components of $\lfloor B_Y + \Delta_Y + \mathrm{Exc}(\pi)\rfloor$ are pairwise disjoint. By the connectedness lemma, $\pi^{-1}(x) \cap \mathrm{Nklt}_Y$ is connected for any $x \in X$. Since $\pi(\mathrm{Nklt}_Y) = \Gamma \sqcup \{o\}$ and π is proper, $\pi^{-1}(\Gamma) \cap \mathrm{Nklt}_Y$ is connected, and $\pi^{-1}(o) \cap \mathrm{Nklt}_Y$ is non-empty and connected. Since each component of $\lfloor B_Y + \Delta_Y + \mathrm{Exc}(\pi)\rfloor$ is connected and its image is contained in $\Gamma \sqcup \{o\}$, each of $\pi^{-1}(\Gamma) \cap \mathrm{Nklt}_Y$ and $\pi^{-1}(o) \cap \mathrm{Nklt}_Y$ is a union of such components, so

$$\pi^{-1}(\Gamma) \cap \mathrm{Nklt}_Y = \Delta_Y \quad \text{and} \quad \pi^{-1}(o) \cap \mathrm{Nklt}_Y = E$$

for a π-exceptional prime divisor E over o. Thus $\Gamma = \Delta_\infty$, $\mathrm{Exc}(\pi) = E$, $\lfloor B_Y + \Delta_Y + E\rfloor = \Delta_Y + E$, and $\lfloor B_Y\rfloor = 0$. Since $(Y, B_Y + \Delta_Y + E)$ is plt, any prime divisor exceptional over Y has positive log discrepancy with respect to $(Y, B_Y + \Delta_Y + E)$. Thus

$$\text{the lc places of } (X, \overline{B}^{\lambda} + \Delta_\infty) \text{ are exactly } \Delta_\infty \text{ and } E. \tag{7.5}$$

We return to κ. If $a_S(\kappa) < 0$ for a prime divisor S on W, then $a_S(\lambda) < 0$, which is impossible. Thus $(X, \overline{B} + \Delta_\infty)$ is lc. By **Step 3**(i) and (7.5), any lc place of $(X, \overline{B} + \Delta_\infty)$ is Δ_∞ or E. Since S_o is an lc place of $(X, \overline{B} + \Delta_\infty)$ over o, $S_o = E$. Thus the lc places of $(X, \overline{B} + \Delta_\infty)$ are exactly Δ_∞ and E. Since $(X, \overline{B} + \Delta_\infty)$ is lc, $\mathrm{Nklt}(X, \overline{B} + \Delta_\infty)$ is the union of the centers of the lc places, that is,

$$\mathrm{Nklt}(X, \overline{B} + \Delta_\infty) = \Delta_\infty \sqcup \{o\}.$$

Since E is the only exceptional lc place and $\mathrm{center}_X E = \{o\}$ is disjoint from Δ_∞, $(X, \overline{B} + \Delta_\infty)$ is plt near Δ_∞. This proves (1). Since $(X, \overline{B} + \Delta_\infty) = (V, B)$ near o, E is the unique lc place of (V, B) over o. This proves (2). Moreover, $K_Y + B_Y + \Delta_Y + E \sim_{\mathbb{Q}} 0$ by **Step 3**.

Step 5. In this step we show that Y maps by a sequence of divisorial contractions to a standard $\mathbb{P}^1$-link whose two sections are the images of Δ_Y and E. A *standard* $\mathbb{P}^1$*-link* is defined in [FS23, Definition 2.21]. For pairs we use [Mor24, Definition 2.19].

For m divisible by $w_1 \cdots w_n$, x_0^m is a section of the ample line bundle $\mathcal{O}_X(m)$ with zero divisor $m\Delta_\infty$, so Δ_∞ is ample. Since $\mathrm{Exc}(\pi) = E$ and $o \notin \Delta_\infty$, we have $\Delta_Y = \pi^*\Delta_\infty$ and $\Delta_Y \cap E = \emptyset$. Thus Δ_Y is nef, $\Delta_Y^2 = \Delta_\infty^2 > 0$, $\Delta_Y \cdot C > 0$ for any curve $C \neq E$ on Y, and $E^2 < 0$. Since $(Y, B_Y + \Delta_Y + E)$ is plt and $\lfloor B_Y\rfloor = 0$, the pair (Y, B_Y) is $\mathbb{Q}$-factorial klt. Since Δ_Y is nef, $K_Y + B_Y \sim_{\mathbb{Q}} -\Delta_Y - E$, and $(K_Y + B_Y) \cdot \Delta_Y = -\Delta_Y^2 < 0$, $K_Y + B_Y$ is not pseudo-effective. We run a $(K_Y + B_Y)$-MMP, which ends with a Mori fiber space:

$$Y = Y_0 \xrightarrow{g_0} Y_1 \xrightarrow{g_1} \cdots \xrightarrow{g_{k_0-1}} Y_{k_0} =: Y' \xrightarrow{g} T.$$

Since $\dim Y = 2$, each g_k is a divisorial contraction of a curve C_k. For any k, let B_k, Δ_k, and E_k be the images of B_Y, Δ_Y, and E on Y_k. We show by induction on k that the following hold:

- We have
$$K_{Y_0} + B_0 + \Delta_0 + E_0 = (g_{k-1} \circ \cdots \circ g_0)^*(K_{Y_k} + B_k + \Delta_k + E_k),$$
and $(Y_k, B_k + \Delta_k + E_k)$ is plt with $\lfloor B_k + \Delta_k + E_k\rfloor = \Delta_k + E_k$.
- Δ_k is nef, $\Delta_k^2 > 0$, $\Delta_k \cdot C > 0$ for any curve $C \neq E_k$ on Y_k, $\Delta_k \cap E_k = \emptyset$, and $E_k^2 < 0$.

The case $k = 0$ is proved above. Assume that the items above hold for some $k < k_0$. Since

$$(K_{Y_k} + B_k) \cdot E_k = -(\Delta_k + E_k) \cdot E_k = -E_k^2 > 0,$$

we have $C_k \neq E_k$, and since $C_k^2 < 0 < \Delta_k^2$, we have $C_k \neq \Delta_k$. Thus $\Delta_k \cdot C_k > 0$ and $\mathrm{mult}_{C_k} B_k < 1$. Since $K_{Y_k} + B_k + \Delta_k + E_k \equiv 0$, the $\mathbb{Q}$-divisor

$$g_k^*(K_{Y_{k+1}} + B_{k+1} + \Delta_{k+1} + E_{k+1}) - (K_{Y_k} + B_k + \Delta_k + E_k)$$

is a multiple of C_k with zero intersection with C_k, so it is zero since $C_k^2 < 0$. Thus the pair

$$(Y_{k+1}, B_{k+1} + \Delta_{k+1} + E_{k+1})$$

is plt: a prime divisor exceptional over Y_{k+1} is either exceptional over Y_k or C_k, whose log discrepancy is

$$1 - \mathrm{mult}_{C_k} B_k > 0.$$

If $E_k \cdot C_k > 0$, then both Δ_{k+1} and E_{k+1} contain the point $g_k(C_k)$, which is impossible since the components of the coefficient-one part of a plt surface pair are disjoint. Thus $E_k \cdot C_k = 0$, so $g_k^* E_{k+1} = E_k$, $E_{k+1}^2 = E_k^2 < 0$, and $\Delta_{k+1} \cap E_{k+1} = \emptyset$. Moreover,

$$g_k^* \Delta_{k+1} = \Delta_k + u C_k, \quad u := -\frac{\Delta_k \cdot C_k}{C_k^2} > 0.$$

Thus $\Delta_{k+1}^2 = \Delta_k^2 + u \Delta_k \cdot C_k > 0$, and for any curve C' on Y_{k+1} and $C := (g_k)_*^{-1} C'$,

$$\Delta_{k+1} \cdot C' = \Delta_k \cdot C + u C_k \cdot C \geq \Delta_k \cdot C,$$

which is positive unless $C = E_k$. This finishes the induction.

Let $\Delta' := \Delta_{k_0}$, $E' := E_{k_0}$, $B' := B_{k_0}$, and $h := g_{k_0-1} \circ \cdots \circ g_0 : Y \to Y'$. If $\dim T = 0$, then $\rho(Y') = 1$, which contradicts $E'^2 < 0$. Thus T is a curve. Since $\rho(Y'/T) = 1$, any vertical curve on Y' is numerically a positive multiple of a fiber of g, so it has self-intersection 0 and positive intersection with any horizontal curve. Since $\Delta'^2 > 0$, Δ' is horizontal over T, and since $\Delta' \cdot E' = 0$, E' is horizontal over T. By the induction, $(Y', B' + \Delta' + E')$ is plt, $\lfloor B' + \Delta' + E' \rfloor = \Delta' + E'$, and $K_{Y'} + B' + \Delta' + E' \sim_{\mathbb{Q}} 0$. Since moreover g is a Mori fiber space and Δ', E' are disjoint horizontal $\mathbb{Q}$-Cartier curves, $g : (Y', B' + \Delta' + E') \to T$ is a standard $\mathbb{P}^1$-link with sections Δ' and E' [Mor24, Lemma 2.21]. Let F_T be a general fiber of g. Then $F_T \cong \mathbb{P}^1$ is contained in the non-singular locus of Y', $K_{Y'} \cdot F_T = -2$, $\Delta' \cdot F_T \geq 1$, and $E' \cdot F_T \geq 1$. Since $(K_{Y'} + B' + \Delta' + E') \cdot F_T = 0$, we have $B' \cdot F_T \leq 0$, so any component of B' is vertical over T.

Step 6. In this step we construct a non-trivial $\mathbb{G}_m$-action on X which fixes Δ_∞ pointwise, fixes o, and preserves any component of $\overline{B}$.

By [Mor24, Theorem 1.4], there exist a $\mathbb{Q}$-divisor D_T on T and an isomorphism

$$Y' \cong \mathbb{P}_T(\mathcal{O}_T \oplus \mathcal{O}_T(D_T))$$

over T under which Δ' and E' are the sections corresponding to the 2 summands of the split rank 2 $\mathbb{Q}$-vector bundle $\mathcal{O}_T \oplus \mathcal{O}_T(D_T)$. Here, with

$$\mathcal{R}_{\geq 0} := \bigoplus_{m \geq 0} \mathcal{O}_T(\lfloor m D_T \rfloor), \quad \mathcal{R}_{\leq 0} := \bigoplus_{m \leq 0} \mathcal{O}_T(\lfloor m D_T \rfloor), \quad \text{and} \quad \mathcal{R} := \bigoplus_{m \in \mathbb{Z}} \mathcal{O}_T(\lfloor m D_T \rfloor),$$

$\mathbb{P}_T(\mathcal{O}_T \oplus \mathcal{O}_T(D_T))$ is the T-scheme obtained by gluing $Y'_+ := \mathrm{Spec}_T \mathcal{R}_{\geq 0}$ and $Y'_- := \mathrm{Spec}_T \mathcal{R}_{\leq 0}$ along $\mathrm{Spec}_T \mathcal{R}$, and the sections corresponding to the 2 summands are the zero loci in Y'_+ and in Y'_- of the ideals generated by the summands of degree $m \neq 0$. Letting $\tau \in \mathbb{G}_m$ act on the summand of degree m by multiplication by τ^m defines a $\mathbb{G}_m$-action on Y' over T. It is non-trivial since $\mathcal{O}_T(\lfloor D_T \rfloor) \neq 0$, and it fixes Δ' and E' pointwise since it acts trivially on the summand $\mathcal{O}_T$ of degree 0, which is the structure sheaf of either section.

Let $\mathbb{G}_m$ act non-trivially on Y' over T, fixing Δ' and E' pointwise. Since $\mathbb{G}_m$ is connected and acts over T, it preserves any vertical curve on Y'. Since any component of B' is vertical over T, $\mathbb{G}_m$ acts on the pair $(Y', B' + \Delta' + E')$ over T. For any h-exceptional prime divisor C, by **Step 5**,

$$a(C, Y', B' + \Delta' + E') = 1 - \mathrm{mult}_C B_Y \in (0, 1].$$

We show that for any h-exceptional prime divisor C, any $\tau \in \mathbb{G}_m$, and any non-zero rational function g on Y',

$$\operatorname{mult}_C \operatorname{div}(\tau \cdot g) = \operatorname{mult}_C \operatorname{div}(g). \tag{7.6}$$

Since any component of $B' + \Delta' + E'$ is $\mathbb{G}_m$-invariant, there exists a $\mathbb{G}_m$-equivariant log resolution $\psi_0 : Z_0 \to Y'$ of $(Y', B' + \Delta' + E')$. For resolutions instead of log resolutions, this is [Kol07, Proposition 3.9.1]. By [Kol07, Theorems 3.35(4) and 3.36(4)], $(Y', B'+\Delta'+E')$ has a log resolution $\psi_0 : Z_0 \to Y'$ which commutes with smooth morphisms: for any smooth morphism $p : Y'' \to Y'$, the base change $Z_0 \times_{Y'} Y'' \to Y''$ is the log resolution of $(Y'', p^{-1} \operatorname{Supp}(B' + \Delta' + E'))$ given by [Kol07, Theorems 3.35 and 3.36]. Let $\alpha : \mathbb{G}_m \times Y' \to Y'$ be the action and $q : \mathbb{G}_m \times Y' \to Y'$ the projection. Then α and q are smooth, and

$$\alpha^{-1} \operatorname{Supp}(B' + \Delta' + E') = q^{-1} \operatorname{Supp}(B' + \Delta' + E')$$

since any component of $B' + \Delta' + E'$ is $\mathbb{G}_m$-invariant. Thus the base changes of ψ_0 by α and by q both equal the log resolution of $(\mathbb{G}_m \times Y', q^{-1} \operatorname{Supp}(B'+\Delta'+E'))$ given by [Kol07, Theorems 3.35 and 3.36], and we obtain an isomorphism

$$\mathbb{G}_m \times Z_0 \cong Z_0 \times_{Y',\alpha} (\mathbb{G}_m \times Y')$$

over $\mathbb{G}_m \times Y'$. Composing it with the projection to Z_0, we obtain a morphism $\widetilde{\alpha} : \mathbb{G}_m \times Z_0 \to Z_0$ with

$$\psi_0 \circ \widetilde{\alpha} = \alpha \circ (\mathrm{id}_{\mathbb{G}_m} \times \psi_0).$$

Let $U \subset Y'$ be a dense open subset over which ψ_0 is an isomorphism, let $S := \mathbb{G}_m \times \mathbb{G}_m \times Z_0$, and let $\beta_1, \beta_2 : S \to Z_0$ be the morphisms

$$\beta_1(\tau, \tau', z) := \widetilde{\alpha}(\tau\tau', z) \quad \text{and} \quad \beta_2(\tau, \tau', z) := \widetilde{\alpha}(\tau, \widetilde{\alpha}(\tau', z)).$$

Since α is an action, $\psi_0 \circ \beta_1 = \psi_0 \circ \beta_2$, so $\beta_1 = \beta_2$ on $(\psi_0 \circ \beta_1)^{-1}(U)$. The open subset $(\psi_0 \circ \beta_1)^{-1}(U)$ is non-empty since $\psi_0 \circ \beta_1$ is surjective, hence dense since S is irreducible. Since S is reduced and Z_0 is separated, $\beta_1 = \beta_2$. Similarly, $\widetilde{\alpha}|_{\{1\}\times Z_0} = \mathrm{id}_{Z_0}$. Thus $\widetilde{\alpha}$ is a $\mathbb{G}_m$-action on Z_0, and ψ_0 is $\mathbb{G}_m$-equivariant.

Fix an h-exceptional prime divisor C. Since Z_0 is a smooth surface, there exist $r \in \mathbb{N}$ and a sequence of blow-ups

$$Z_r \to Z_{r-1} \to \cdots \to Z_0$$

such that C is a divisor on Z_r and, for $0 \le i < r$, the center of C on Z_i is a closed point z_i and $Z_{i+1} \to Z_i$ is the blow-up of z_i, with exceptional curve R_{i+1}. Then $C = R_r$ if $r > 0$. For $0 \le i \le r$, let $\psi_i : Z_i \to Y'$ be the induced morphism, let $\Delta_{Z_i} := \psi_{i*}^{-1}\Delta'$ and $E_{Z_i} := \psi_{i*}^{-1}E'$, let D_{Z_i} be the $\mathbb{R}$-divisor with

$$K_{Z_i} + D_{Z_i} = \psi_i^*(K_{Y'} + B' + \Delta' + E') \quad \text{and} \quad \psi_{i*}D_{Z_i} = B' + \Delta' + E',$$

let $\mathcal{P}_i$ be the set of prime divisors contained in $\operatorname{Supp} D_{Z_i} \cup \operatorname{Exc}(\psi_i)$, and for $R \in \mathcal{P}_i$ let

$$a(R) := 1 - \operatorname{mult}_R D_{Z_i} = a(R, Y', B' + \Delta' + E').$$

Then $\operatorname{Supp} D_{Z_i} \cup \operatorname{Exc}(\psi_i)$ is an snc divisor, so any point of Z_i is contained in at most 2 elements of $\mathcal{P}_i$. Since $(Y', B' + \Delta' + E')$ is plt and $\lfloor B' + \Delta' + E' \rfloor = \Delta' + E'$, $a(\Delta_{Z_i}) = a(E_{Z_i}) = 0$, and $a(R) > 0$ for any other $R \in \mathcal{P}_i$. For $0 \le i < r$, by the formula for the log discrepancy of the blow-up of a point on a smooth surface,

$$a(R_{i+1}) = 2 - \#\{R \in \mathcal{P}_i \mid z_i \in R\} + \sum_{R \in \mathcal{P}_i, z_i \in R} a(R). \tag{7.7}$$

For $1 \le i < r$, we have $z_i \in R_i$, so $a(R_{i+1}) \ge a(R_i)$ by (7.7). Thus for $0 \le i < r$,

$$a(R_{i+1}) \le a(R_r) = a(C, Y', B' + \Delta' + E') \le 1.$$

We show by induction on i that the $\mathbb{G}_m$-action on Y' lifts to Z_i with ψ_i equivariant. The case $i = 0$ holds by the choice of ψ_0. Assume that the action lifts to Z_i for some $i < r$. Since

$$\operatorname{Supp} D_{Z_i} \cup \operatorname{Exc}(\psi_i) = \operatorname{Exc}(\psi_i) \cup \operatorname{Supp} \psi_{i*}^{-1}(B' + \Delta' + E')$$

is invariant, the connected group $\mathbb{G}_m$ permutes the finitely many elements of $\mathcal{P}_i$, hence preserves each of them. The set of points contained in 2 elements of $\mathcal{P}_i$ is finite and invariant, and the orbit of a point is irreducible, so $\mathbb{G}_m$ fixes any such point. Since the action on Δ' and E' is trivial and $\Delta_{Z_i} \to \Delta'$ and $E_{Z_i} \to E'$ are birational, the action on Δ_{Z_i} and E_{Z_i} is trivial. Since $a(R_{i+1}) \leq 1$, by (7.7) either z_i is contained in 2 elements of $\mathcal{P}_i$, or z_i is contained in exactly 1 element R of $\mathcal{P}_i$ and $a(R) = 0$, so $R \in \{\Delta_{Z_i}, E_{Z_i}\}$. Thus z_i is fixed, the ideal sheaf of z_i is invariant, and the action lifts to the blow-up Z_{i+1} of z_i. This finishes the induction. Since $C \in \mathcal{P}_r$ is preserved by the action on Z_r and $\operatorname{div}(\tau \cdot g)$ is the image of $\operatorname{div}(g)$ under the automorphism of Z_r given by τ, we obtain (7.6).

We lift the action to Y. Let A be an h-ample Cartier divisor on Y. Since Y' is $\mathbb{Q}$-factorial, possibly replacing A with a multiple, we may assume that h_*A is Cartier. Then $D_A := A - h^*h_*A$ is h-exceptional and h-ample, so $D_A \leq 0$ by the negativity lemma. If the coefficient of D_A along an h-exceptional curve C were 0, then $D_A \cdot C \leq 0$, since the other coefficients are non-positive and distinct curves have non-negative intersection, which contradicts h-ampleness. Possibly replacing A with a multiple, we may write $D_A = -\sum_{C \subset \operatorname{Exc}(h)} n_C C$ with $n_C \in \mathbb{N}^+$ and D_A Cartier. Since Y' is normal, for any open subset $U \subset Y'$ and any $m \geq 0$,

$$h_*\mathcal{O}_Y(mD_A)(U) = \{g \in \mathcal{O}_{Y'}(U) \mid \operatorname{mult}_C \operatorname{div}(g) \geq mn_C \text{ for any } h\text{-exceptional } C \text{ with } h(C) \in U\},$$

which is $\mathbb{G}_m$-invariant by (7.6). Since D_A is h-ample,

$$Y \cong \operatorname{Proj}_{Y'} \bigoplus_{m \geq 0} h_*\mathcal{O}_Y(mD_A)$$

over Y', so the $\mathbb{G}_m$-action on Y' lifts to a $\mathbb{G}_m$-action on Y such that h is equivariant. The lifted action preserves $\Delta_Y = h_*^{-1}\Delta'$ and $E = h_*^{-1}E'$, since Δ' and E' are invariant. Since the action on Δ' is trivial, the action on Δ_Y is trivial over the dense open subset of Δ' over which $h|_{\Delta_Y} : \Delta_Y \to \Delta'$ is an isomorphism, hence trivial on Δ_Y. Similarly, the action on E is trivial. Any component of B_Y is either h-exceptional, hence invariant by (7.6), or $h_*^{-1}C$ for a vertical curve C on Y', hence invariant.

We descend the action to X. Let m_0 be a positive integer such that $m_0\Delta_\infty$ is Cartier. Since $\Delta_Y = \pi^*\Delta_\infty$ is invariant and $\pi_*\mathcal{O}_Y = \mathcal{O}_X$, $\mathbb{G}_m$ acts on

$$\bigoplus_{m \geq 0} H^0(Y, \mathcal{O}_Y(mm_0\Delta_Y)) = \bigoplus_{m \geq 0} H^0(X, \mathcal{O}_X(mm_0\Delta_\infty)),$$

whose Proj is X since Δ_∞ is ample. Thus $\mathbb{G}_m$ acts on X and π is equivariant. Since the action on Y is trivial on Δ_Y and on E, the action on X fixes $\Delta_\infty = \pi(\Delta_Y)$ pointwise and fixes $o = \pi(E)$. Since $\operatorname{Supp} \overline{B} = \operatorname{Supp} \overline{B}^\lambda$, the action preserves any component of $\overline{B}$. Since π and h are birational and equivariant and the action on Y' is non-trivial, the action on X is non-trivial.

Step 7. In this step we prove (3).

Fix a non-trivial $\mathbb{G}_m$-action on X as in **Step 6**. It preserves $V = X \backslash \Delta_\infty$, and we let $\mathbb{G}_m$ act on the coordinate ring $\mathcal{O}(V)$ by $(\tau \cdot g)(v) := g(\tau^{-1} \cdot v)$. For any integer m, let

$$\begin{aligned}\mathcal{O}(V)_{\leq m} &:= \{g \in \mathcal{O}(V) \backslash \{0\} \mid \deg_w g \leq m\} \cup \{0\} \\ &= \{g \in \mathcal{O}(V) \backslash \{0\} \mid \operatorname{mult}_{\Delta_\infty} \operatorname{div}_X(g) \geq -m\} \cup \{0\},\end{aligned}$$

where the equality follows from (7.2). Thus $\mathcal{O}(V)_{\leq m}$ is $\mathbb{G}_m$-invariant, $\dim \mathcal{O}(V)_{\leq m} < +\infty$, $\mathcal{O}(V)_{\leq 0} = \Bbbk$, and $\mathcal{O}(V)_{\leq -1} = 0$. Taking top parts of representatives of minimal weight gives a well-defined map from $\mathcal{O}(V)_{\leq m}/\mathcal{O}(V)_{\leq m-1}$ to $(\Bbbk[x_1, \ldots, x_n]/(f_0))_m$, resp. $\Bbbk[x_1, x_2]_m$, injective

since in case (b) such top parts are not divisible by f_0 (**Step 1**) and surjective since any w-homogeneous polynomial of weight m restricts to an element of $\mathcal{O}(V)_{\le m}$. Thus the graded ring

$$\bigoplus_{m\ge 0} \mathcal{O}(V)_{\le m}/\mathcal{O}(V)_{\le m-1}$$

is isomorphic to $\Bbbk[x_1,\dots,x_n]/(f_0)$ in case (b), resp. $\Bbbk[x_1,x_2]$ in case (a), which is a domain since f_0 is irreducible. In particular, $\deg_w(gg') = \deg_w g + \deg_w g'$ for any non-zero $g, g' \in \mathcal{O}(V)$, so the units of $\mathcal{O}(V)$ are the elements of $\Bbbk^*$.

Since X is smooth at the generic point of Δ_∞, the local ring $\mathcal{O}_{X,\Delta_\infty}$ is a discrete valuation ring. Let t be a uniformizer and $\mathfrak{m}$ the maximal ideal. Since $\mathbb{G}_m$ fixes Δ_∞ pointwise, it acts trivially on $\mathcal{O}_{X,\Delta_\infty}/\mathfrak{m}$, and acts on the 1-dimensional $\mathcal{O}_{X,\Delta_\infty}/\mathfrak{m}$-vector space $\mathfrak{m}/\mathfrak{m}^2$ by $\tau\cdot t \in \tau^{-e}t + \mathfrak{m}^2$ for an integer e. For $g \in \mathcal{O}(V)$ with $\deg_w g = m$, by (7.2) we may write $g = t^{-m}g_1$ with $g_1 \in \mathcal{O}^*_{X,\Delta_\infty}$. Since $\tau\cdot g_1 \in g_1 + \mathfrak{m}$, the element $\tau\cdot g - \tau^{em}g$ has pole order less than m along Δ_∞. Thus

$$\tau\cdot g - \tau^{em} g \in \mathcal{O}(V)_{\le m-1} \quad \text{for any } g \in \mathcal{O}(V)_{\le m}. \tag{7.8}$$

If $e = 0$, then $\mathbb{G}_m$ acts unipotently on each $\mathcal{O}(V)_{\le m}$, hence trivially since $\mathbb{G}_m$ acts diagonalizably, so $\mathbb{G}_m$ acts trivially on V and on X, a contradiction. Thus $e \ne 0$. For $m \ge 0$, let $\mathcal{O}(V)^{(m)}$ be the set of elements $g \in \mathcal{O}(V)_{\le m}$ with $\tau\cdot g = \tau^{em}g$ for any τ. Since $\mathbb{G}_m$ acts diagonalizably on $\mathcal{O}(V)_{\le m}$, by (7.8) and induction on m, the characters of $\mathcal{O}(V)_{\le m}$ are $\tau^{em'}$ for $0 \le m' \le m$, and the projection

$$\mathcal{O}(V)^{(m)} \to \mathcal{O}(V)_{\le m}/\mathcal{O}(V)_{\le m-1}$$

is an isomorphism. In particular,

$$\mathcal{O}(V)^{(m)} \cap \mathcal{O}(V)_{\le m-1} = 0. \tag{7.9}$$

For $1 \le i \le n$, let $u_i \in \mathcal{O}(V)^{(w_i)}$ be the element whose class in $\mathcal{O}(V)_{\le w_i}/\mathcal{O}(V)_{\le w_i-1}$ is the class of x_i. Since o is fixed, $u_i(o) = (\tau\cdot u_i)(o) = \tau^{ew_i}u_i(o)$ for any τ, so $u_i(o) = 0$. Since $u_i - x_i \in \mathcal{O}(V)_{\le w_i-1}$, there exists a polynomial φ_i representing u_i such that the top part of $\varphi_i - x_i$ has w-weight at most $w_i - 1$. Any monomial of w-weight less than w_i involves only the variables x_j with $w_j < w_i$. Thus, ordering the variables by increasing weight, $\varphi := (\varphi_1,\dots,\varphi_n)$ is a triangular polynomial automorphism of $\mathbb{A}^n_\Bbbk$ with $\varphi_i - x_i$ a polynomial in the variables of weight less than w_i. In particular, φ induces the identity on the graded ring of the w-weighted degree filtration, and the coordinate functions φ_i vanish at o.

In case (b), since f_0 is w-homogeneous of weight d_0 and $u_i \in \mathcal{O}(V)^{(w_i)}$,

$$(f_0\circ\varphi)|_V = f_0(u_1,u_2,u_3) \in \mathcal{O}(V)^{(d_0)},$$

and since $u_i - x_i \in \mathcal{O}(V)_{\le w_i-1}$, the class of $f_0(u_1,u_2,u_3)$ in $\mathcal{O}(V)_{\le d_0}/\mathcal{O}(V)_{\le d_0-1}$ is the class of f_0, which is 0. By (7.9), $(f_0\circ\varphi)|_V = 0$, so $f_0\circ\varphi = F\cdot g_2$ for a polynomial g_2. Since both $f_0\circ\varphi$ and F have top part f_0, the top part of g_2 is 1, so $g_2 = 1$ and $F = f_0\circ\varphi$.

Fix j. Since $\mathbb{G}_m$ preserves any component of $\overline{B}$, it preserves $\operatorname{div}(Q_j|_V)$. Since the units of $\mathcal{O}(V)$ are the elements of $\Bbbk^*$, $\tau\cdot Q_j|_V = \chi(\tau)Q_j|_V$ for a character χ of $\mathbb{G}_m$. Since $\deg_w(Q_j|_V) = d_j$, $\chi(\tau) = \tau^{ed_j}$ by (7.8), so $Q_j|_V \in \mathcal{O}(V)^{(d_j)}$. Since P_j is w-homogeneous of weight d_j and $u_i \in \mathcal{O}(V)^{(w_i)}$, we have

$$(P_j\circ\varphi)|_V = P_j(u_1,\dots,u_n) \in \mathcal{O}(V)^{(d_j)},$$

and since $u_i - x_i \in \mathcal{O}(V)_{\le w_i-1}$ and P_j is the top part of Q_j, the classes of $(P_j\circ\varphi)|_V$ and of $Q_j|_V$ in $\mathcal{O}(V)_{\le d_j}/\mathcal{O}(V)_{\le d_j-1}$ are equal to $P_j|_{V_0}$. By (7.9), $Q_j|_V = (P_j\circ\varphi)|_V$. In case (a), $\mathcal{O}(V) = \Bbbk[x_1,x_2]$, so $Q_j = P_j\circ\varphi$. This proves (3). □

Proposition 7.4. *Let $\Bbbk \supset \mathbb{C}$ be an algebraically closed field, let w_1 and w_2 be coprime positive integers, and let the coordinates p and v of $\mathbb{A}^2_\Bbbk$ have the weights w_1 and w_2. The* weighted degree *of a non-zero polynomial f is*

$$\max\{iw_1 + kw_2 \mid p^iv^k \text{ is a monomial of } f\}.$$

Let $Q_1, \dots, Q_n \in \Bbbk[p, v]$ be non-zero polynomials of weighted degrees $d_1, \dots, d_n$, let P_j be the part of Q_j of weighted degree d_j for any $j \in \{1, \dots, n\}$, and let $\kappa_1, \dots, \kappa_n$ be positive real numbers such that

$$\sum_{j=1}^{n} \kappa_j d_j = w_1 + w_2.$$

Let $B := \sum_{j=1}^{n} \kappa_j \operatorname{div}(Q_j)$ and $B_0 := \sum_{j=1}^{n} \kappa_j \operatorname{div}(P_j)$. Assume that $(\mathbb{A}^2_{\Bbbk}, B)$ is lc near 0 but not klt at 0, and that

$$\operatorname{mult}_C B < 1 \text{ for any prime divisor } C \text{ on } \mathbb{A}^2_{\Bbbk} \text{ with } 0 \in C.$$

Then the following hold.

(1) *There exists a unique pair of polynomials $p', v' \in \Bbbk[p, v]$ with $p'(0) = v'(0) = 0$ such that the morphism $(p', v') : \mathbb{A}^2_{\Bbbk} \to \mathbb{A}^2_{\Bbbk}$ is an automorphism inducing the identity on the graded ring of the weighted-degree filtration of $\Bbbk[p, v]$ and*

$$Q_j(p, v) = P_j(p', v') \quad \text{for any } j \in \{1, \dots, n\}.$$

If $w_1 > w_2$, then $v' = v$ and $p' = p + \sigma(v)$ for a polynomial $\sigma \in \Bbbk[v]$ whose monomials v^e satisfy $0 < ew_2 < w_1$. If $w_1 < w_2$, then $p' = p$ and $v' = v + \sigma(p)$ for a polynomial $\sigma \in \Bbbk[p]$ whose monomials p^e satisfy $0 < ew_1 < w_2$. If $w_1 = w_2 = 1$, then $(p', v') = (p, v)$. The pairs $(\mathbb{A}^2_{\Bbbk}, B)$ and $(\mathbb{A}^2_{\Bbbk}, B_0)$ are lc, any component of B_0 has coefficient less than one,

$$\operatorname{Nklt}(\mathbb{A}^2_{\Bbbk}, B) = \{0\},$$

and the exceptional divisor of the (w_1, w_2)-weighted blow-up of $\mathbb{A}^2_{\Bbbk}$ in the coordinates p', v' is the unique lc place of $(\mathbb{A}^2_{\Bbbk}, B)$ over 0. If $\Bbbk_0 \subset \Bbbk$ is a subfield containing the coefficients of $Q_1, \dots, Q_n$, then $p', v' \in \Bbbk_0[p, v]$. If g is a ring automorphism of $\Bbbk[p, v]$ with $g(\Bbbk) = \Bbbk$, $g(p) = \alpha p$, and $g(v) = \beta v$ for some $\alpha, \beta \in \Bbbk^*$, and if $g(Q_j) = \chi_j Q_j$ and $g(P_j) = \chi_j P_j$ for some $\chi_j \in \Bbbk^*$ for any j, then

$$g(p') = \alpha p' \quad \text{and} \quad g(v') = \beta v'.$$

(2) *Let $\overline{L}$ be an algebraic closure of $L := \Bbbk((t))$. Assume that the hypotheses before (1) hold with $\overline{L}$ in place of $\Bbbk$, that $Q_1, \dots, Q_n \in \Bbbk[[t]][p, v]$, so that $P_1, \dots, P_n \in \Bbbk[[t]][p, v]$, and that*

$$\operatorname{mld}\Big(\operatorname{Spec} \Bbbk[[p, v, t]] \ni 0, \sum_{j=1}^{n} \kappa_j \operatorname{div}(P_j)\Big) > 0.$$

Then the polynomials $p', v' \in \overline{L}[p, v]$ given by (1) belong to $\Bbbk[[t]][p, v]$. Let G be a group acting on $\Bbbk[[t]][p, v]$ by ring automorphisms such that for any $g \in G$, we have $g(\Bbbk) = \Bbbk$,

$$g(p) = \alpha_g p, \quad g(v) = \beta_g v, \quad \text{and} \quad g\Big(\sum_{e \geq 0} c_e t^e\Big) = \sum_{e \geq 0} g(c_e) \gamma_g^e t^e \text{ for any } c_e \in \Bbbk,$$

and $g(Q_j) = \chi_j(g) Q_j$ and $g(P_j) = \chi_j(g) P_j$ for any j, where $\alpha_g, \beta_g, \gamma_g, \chi_j(g) \in \Bbbk^*$. We let G act on $\Bbbk[[x, y, z, t]]$ by applying $g|_{\Bbbk}$ to the coefficients of power series and by

$$g(x) = \alpha_g^2 x, \quad g(y) = \beta_g^2 y, \quad g(z) = \alpha_g \beta_g z, \quad \text{and} \quad g(t) = \gamma_g t.$$

Assume that $w_1 > w_2$ and that G contains the $\Bbbk$-linear automorphism ι with $\iota(p) = -p$, $\iota(v) = -v$, and $\iota(t) = t$. Then $p' = p + v\phi(-v^2, t)$ for a polynomial $\phi(y, t) \in \Bbbk[[t]][y]$, and the elements

$$x' := x + 2z\phi(y, t) - y\phi(y, t)^2, \quad y' := y, \quad \text{and} \quad z' := z - y\phi(y, t)$$

of $\Bbbk[[t]][x, y, z]$ satisfy

$$x'y' + z'^2 = xy + z^2.$$

Under the substitution $(x, y, z) = (p^2, -v^2, pv)$, the triple (x', y', z') becomes $(p'^2, -v'^2, p'v')$. The $\Bbbk$-algebra endomorphism of $\Bbbk[[x, y, z, t]]$ sending x, y, z, t to x', y', z', t is an automorphism commuting with the action of G. For any integer $N \geq 2$, it induces

an automorphism of the completed local ring $\Bbbk[[x,y,z,t]]/(xy+z^2+t^N)$ *of* $W_N := (xy+z^2+t^N=0) \subset \mathbb{A}^4_{\Bbbk}$ *at* 0, *and if* $(\alpha_g\beta_g)^2 = \gamma_g^N$ *for any* $g \in G$, *then* G *acts on* $\Bbbk[[x,y,z,t]]/(xy+z^2+t^N)$ *and the induced automorphism is* G*-equivariant.*

Proof. **Step 1.** We prove the existence and the form of p', v' in (1).

Since $P_j \neq 0$ is the top part of Q_j for any j and $\sum_j \kappa_j d_j = w_1 + w_2$, the pair $(\mathbb{A}^2_{\Bbbk}, B_0)$ is a critical pair for (w_1, w_2), and the hypotheses of Lemma 7.3 hold in case (a) with $o = 0$. By Lemma 7.3(3), there exist $p', v' \in \Bbbk[p, v]$ with $p'(0) = v'(0) = 0$ such that (p', v') is an automorphism of $\mathbb{A}^2_{\Bbbk}$ inducing the identity on the graded ring of the weighted-degree filtration and $Q_j(p, v) = P_j(p', v')$ for any j.

For any $d \in \mathbb{N}$, let $\mathcal{F}_d \subset \Bbbk[p, v]$ be the span of the monomials of weight at most d. Since $\mathcal{F}_d\mathcal{F}_{d'} \subset \mathcal{F}_{d+d'}$, an automorphism (p', v') of $\mathbb{A}^2_{\Bbbk}$ induces the identity on the graded ring if and only if

$$p' - p \in \mathcal{F}_{w_1-1} \quad \text{and} \quad v' - v \in \mathcal{F}_{w_2-1}. \tag{7.10}$$

Assume that $w_1 > w_2$. Since any non-constant monomial has weight at least w_2, $\mathcal{F}_{w_2-1} = \Bbbk$, so $v' - v$ is a constant, which is 0 as $v'(0) = v(0) = 0$. Any monomial $p^i v^k$ of weight less than w_1 satisfies $i = 0$ and $kw_2 < w_1$. Thus $p' = p + \sigma(v)$ for a polynomial $\sigma \in \Bbbk[v]$ whose monomials v^e satisfy $ew_2 < w_1$, and $e > 0$ since $\sigma(0) = p'(0) = 0$. The case $w_1 < w_2$ is symmetric, and if $w_1 = w_2 = 1$, then $\mathcal{F}_0 = \Bbbk$ and $(p', v') = (p, v)$. Conversely, for any $\delta \in \Bbbk[v]$ whose monomials v^e satisfy $0 < ew_2 < w_1$, the pair $(p + \delta(v), v)$ is an automorphism of $\mathbb{A}^2_{\Bbbk}$ with inverse $(p - \delta(v), v)$, fixes 0, and satisfies (7.10).

Step 2. We prove the conclusions of (1) on singularities.

Let $X := \mathbb{P}(1, w_1, w_2)$, which contains $\mathbb{A}^2_{\Bbbk}$ as the open subset where the homogeneous coordinate of weight 1 is non-zero, so that X is the closure of $\mathbb{A}^2_{\Bbbk}$ in case (a) of Lemma 7.3. Let $\Delta_\infty := X \backslash \mathbb{A}^2_{\Bbbk}$, and let $\overline{B}$ be the closure of B in X. By Lemma 7.3(1), $(X, \overline{B} + \Delta_\infty)$ is lc and

$$\operatorname{Nklt}(X, \overline{B} + \Delta_\infty) = \Delta_\infty \sqcup \{0\}.$$

Restricting to $\mathbb{A}^2_{\Bbbk} = X \backslash \Delta_\infty$, $(\mathbb{A}^2_{\Bbbk}, B)$ is lc and $\operatorname{Nklt}(\mathbb{A}^2_{\Bbbk}, B) = \{0\}$. Since $\operatorname{div}(Q_j) = \operatorname{div}(P_j(p', v'))$ for any j, the automorphism (p', v') of $\mathbb{A}^2_{\Bbbk}$ pulls B_0 back to B and fixes 0. Hence $(\mathbb{A}^2_{\Bbbk}, B_0)$ is lc and $\operatorname{Nklt}(\mathbb{A}^2_{\Bbbk}, B_0) = \{0\}$. In particular, any component of B_0 has coefficient less than one, since a component with coefficient at least one is contained in $\operatorname{Nklt}(\mathbb{A}^2_{\Bbbk}, B_0)$.

Let E be the exceptional divisor of the (w_1, w_2)-weighted blow-up of $\mathbb{A}^2_{\Bbbk}$ in the coordinates p', v'. Since $\gcd(w_1, w_2) = 1$ and $Q_j = P_j(p', v')$ is weighted homogeneous of degree d_j in p', v', E has log discrepancy $w_1 + w_2$ with respect to $\mathbb{A}^2_{\Bbbk}$ and $\operatorname{mult}_E \operatorname{div}(Q_j) = d_j$ for any j [JM12, Propositions 3.1 and 5.1], so $a(E, \mathbb{A}^2_{\Bbbk}, B) = w_1 + w_2 - \sum_j \kappa_j d_j = 0$. Since $\operatorname{center}_{\mathbb{A}^2_{\Bbbk}} E = \{0\}$, E is an lc place of $(\mathbb{A}^2_{\Bbbk}, B)$ over 0. By Lemma 7.3(2), $(\mathbb{A}^2_{\Bbbk}, B)$ has a unique lc place over 0, so it is E.

Step 3. We prove the uniqueness in (1).

Assume that $w_1 > w_2$, and let p'', v'' be another pair of polynomials satisfying the conditions of (1). By **Step 1**, $p'' = p + \sigma_2(v)$ and $v'' = v$ for a polynomial $\sigma_2 \in \Bbbk[v]$. Let $\delta := \sigma_2 - \sigma$. Since $P_j(p + \sigma(v), v) = Q_j = P_j(p + \sigma_2(v), v)$, substituting $p - \sigma(v)$ for p, we obtain

$$P_j(p, v) = P_j(p + \delta(v), v) \quad \text{for any } j \in \{1, \dots, n\}.$$

Suppose that $\delta \neq 0$. If P_j has degree $i \geq 1$ in p and $c \in \Bbbk[v] \backslash \{0\}$ is the coefficient of p^i, then the coefficient of p^{i-1} in $P_j(p + \delta(v), v) - P_j(p, v)$ is $ic\delta \neq 0$, a contradiction. Thus $P_j \in \Bbbk[v]$, and since P_j is weighted homogeneous of degree d_j, $P_j = c_j v^{d_j/w_2}$ for some $c_j \in \Bbbk^*$. Hence

$$\operatorname{mult}_{(v=0)} B_0 = \sum_{j=1}^{n} \frac{\kappa_j d_j}{w_2} = \frac{w_1 + w_2}{w_2} > 1,$$

which contradicts **Step 2**. Thus $\delta = 0$. The case $w_1 < w_2$ is symmetric, and if $w_1 = w_2 = 1$, the uniqueness follows from **Step 1**.

Step 4. We prove the descent and the equivariance in (1).

We may assume that $w_1 > w_2$. The case $w_1 < w_2$ is symmetric, and the case $w_1 = w_2 = 1$ follows from **Step 1**. Let $\mathcal{E} := \{e \in \mathbb{N}^+ \mid ew_2 < w_1\}$, let $(s_e)_{e\in\mathcal{E}}$ be the coordinates of $\mathbb{A}^{\mathcal{E}}_{\Bbbk_0}$, and let $\mathcal{S} \subset \mathbb{A}^{\mathcal{E}}_{\Bbbk_0}$ be the closed subscheme defined by the coefficients of the polynomials

$$Q_j(p,v) - P_j\Big(p + \sum_{e\in\mathcal{E}} s_e v^e, v\Big), \quad j \in \{1,\dots,n\},$$

in p, v. Since P_j is a part of Q_j, $Q_j, P_j \in \Bbbk_0[p,v]$, and the defining equations are polynomials in $(s_e)_{e\in\mathcal{E}}$ with coefficients in $\Bbbk_0$. By the last sentence of **Step 1**, any point of $\mathcal{S}(\Bbbk)$ gives a pair satisfying the conditions of (1), so by **Steps 1 and 3**, $\mathcal{S}(\Bbbk)$ consists of exactly one point s^0, the coefficient vector of σ. Since $\mathcal{S}$ is defined over $\Bbbk_0$ and $\operatorname{char}\Bbbk_0 = 0$, $s^0 \in \mathcal{S}(\Bbbk_0)$, so $p', v' \in \Bbbk_0[p,v]$.

Let g, α, β, and χ_j be as in (1). For any $f \in \Bbbk[p,v]$, let f^g be the polynomial obtained by applying $g|_{\Bbbk}$ to the coefficients of f, so that $g(f) = f^g(\alpha p, \beta v)$. Since $g(P_j) = \chi_j P_j$, we have $P_j^g(\alpha p, \beta v) = \chi_j P_j(p,v)$. Applying g to $Q_j = P_j(p + \sigma(v), v)$, we obtain

$$\chi_j Q_j = g(Q_j) = P_j^g\big(\alpha p + \sigma^g(\beta v), \beta v\big) = \chi_j P_j\big(p + \alpha^{-1}\sigma^g(\beta v), v\big).$$

The monomials v^e of the polynomial $\alpha^{-1}\sigma^g(\beta v)$ satisfy $e \in \mathcal{E}$, so by **Step 1** the pair

$$\big(p + \alpha^{-1}\sigma^g(\beta v), v\big)$$

satisfies the conditions of (1). By **Step 3**, $\alpha^{-1}\sigma^g(\beta v) = \sigma(v)$, that is,

$$g(p') = \alpha p + \sigma^g(\beta v) = \alpha p' \quad \text{and} \quad g(v') = \beta v'.$$

Step 5. We prove that $p', v' \in \Bbbk[[t]][p,v]$ in (2).

Since $\overline{L} \supset \Bbbk \supset \mathbb{C}$, we may use (1) over $\overline{L}$, and by **Step 4** with $\Bbbk_0 = L$, $p', v' \in L[p,v]$. If $w_1 = w_2 = 1$, there is nothing to prove, and the case $w_1 < w_2$ is symmetric to the case $w_1 > w_2$. Assume that $w_1 > w_2$, and write $\sigma = \sum_{e\in\mathcal{E}} s_e v^e$ with $s_e \in L$. Suppose that $s_e \notin \Bbbk[[t]]$ for some e. For $\xi \in L^*$, let $\operatorname{mult}_{(t=0)}\xi$ be the coefficient of the prime divisor $(t=0)$ in $\operatorname{div}(\xi)$ on $\operatorname{Spec}\Bbbk[[t]]$. Let

$$m := \max\{-\operatorname{mult}_{(t=0)} s_e \mid e \in \mathcal{E},\ s_e \neq 0\} \geq 1,$$

and write

$$\sigma = t^{-m}\big(\psi(v) + t\sigma_1\big), \quad \psi \in \Bbbk[v]\backslash\{0\}, \quad \text{and} \quad \sigma_1 \in \Bbbk[[t]][v].$$

Since the monomials v^e of σ satisfy $0 < ew_2 < w_1$, we have $\psi(0) = 0$ and $w_2 \deg\psi < w_1$.

Write

$$P_j = \sum_{(i,k)\in\mathcal{M}_j} c_{j,i,k} p^i v^k,$$

where $\mathcal{M}_j$ is a finite set of pairs $(i,k) \in \mathbb{N}^2$ with $iw_1 + kw_2 = d_j$ and $c_{j,i,k} \in \Bbbk[[t]]\backslash\{0\}$, let $\overline{c}_{j,i,k} \in \Bbbk^*$ be the coefficient of $t^{\operatorname{mult}_{(t=0)} c_{j,i,k}}$ in $c_{j,i,k}$, and let

$$n_j := \min_{(i,k)\in\mathcal{M}_j}\big(\operatorname{mult}_{(t=0)} c_{j,i,k} - mi\big).$$

We show that $n_j \geq 0$. Let $\mathcal{M}'_j \subset \mathcal{M}_j$ be the subset where the minimum is attained. Setting $p = 0$ in $Q_j = P_j(p + \sigma, v)$, we have

$$Q_j(0,v) = \sum_{(i,k)\in\mathcal{M}_j} c_{j,i,k}\sigma^i v^k, \quad c_{j,i,k}\sigma^i v^k \in t^{\operatorname{mult}_{(t=0)} c_{j,i,k} - mi}\big(\overline{c}_{j,i,k}\psi^i v^k + t\Bbbk[[t]][v]\big).$$

Thus the coefficient of t^{n_j} in $Q_j(0,v)$ is $\sum_{(i,k)\in\mathcal{M}'_j} \overline{c}_{j,i,k}\psi^i v^k$. Let $(i_0,k_0) \in \mathcal{M}'_j$ be the pair with the least i. For any $(i,k) \in \mathcal{M}'_j$, we have $(i - i_0)w_1 = (k_0 - k)w_2$ and $\gcd(w_1,w_2) = 1$, so

$i - i_0 = bw_2$ and $k_0 - k = bw_1$ for some $b \in \mathbb{N}$, and distinct pairs have distinct b. Hence, with $\rho := \psi^{w_2} v^{-w_1} \in \Bbbk(v)$,

$$\sum_{(i,k)\in\mathcal{M}'_j} \overline{c}_{j,i,k}\psi^i v^k = \psi^{i_0} v^{k_0} \sum_{(i,k)\in\mathcal{M}'_j} \overline{c}_{j,i,k}\rho^{(i-i_0)/w_2},$$

and the last sum is a non-zero polynomial in ρ with coefficients in $\Bbbk$. Since $\psi^{w_2} = v^{w_1}\rho$ is non-zero and $\deg \psi^{w_2} = w_2 \deg \psi < w_1$, $\rho \notin \Bbbk[v]$, so $\rho \notin \Bbbk$. Since $\Bbbk$ is algebraically closed, ρ is transcendental over $\Bbbk$, and the coefficient of t^{n_j} in $Q_j(0,v)$ is non-zero. Since $Q_j(0,v) \in \Bbbk[[t]][v]$, $n_j \geq 0$.

Let q be a positive integer with $qw_1 > m$, and let F be the exceptional divisor of the weighted blow-up of $\operatorname{Spec} \Bbbk[[p,v,t]]$ at 0 with weights $(qw_1 - m, qw_2, 1)$ on (p,v,t). Since the weight of t is 1, F has log discrepancy $q(w_1+w_2)+1-m$ with respect to $\operatorname{Spec} \Bbbk[[p,v,t]]$, and for any $f \in \Bbbk[[p,v,t]]$, $\operatorname{mult}_F \operatorname{div}(f)$ is the least weight of the monomials of f [JM12, Propositions 3.1 and 5.1]. For any $(i,k) \in \mathcal{M}_j$, the least weight of the monomials of $c_{j,i,k}p^i v^k$ is

$$\operatorname{mult}_{(t=0)} c_{j,i,k} + i(qw_1 - m) + kqw_2 = qd_j + \operatorname{mult}_{(t=0)} c_{j,i,k} - mi,$$

and for distinct (i,k) the monomials of $c_{j,i,k}p^i v^k$ are distinct. Thus $\operatorname{mult}_F \operatorname{div}(P_j) = qd_j + n_j$, and since $n_j \geq 0$ for any j, the log discrepancy of F with respect to $(\operatorname{Spec} \Bbbk[[p,v,t]], \sum_j \kappa_j \operatorname{div}(P_j))$ is

$$q(w_1+w_2)+1-m-\sum_{j=1}^{n}\kappa_j(qd_j+n_j) = 1-m-\sum_{j=1}^{n}\kappa_j n_j \leq 1-m \leq 0.$$

Since $\operatorname{center}_{\operatorname{Spec}\Bbbk[[p,v,t]]} F = \{0\}$, this contradicts the positivity of the minimal log discrepancy at 0. Hence $s_e \in \Bbbk[[t]]$ for any e, and $p', v' \in \Bbbk[[t]][p,v]$.

Step 6. We prove the conclusions of (2) on ι and W_N.

Let $g \in G$. The formula for g on $\Bbbk[[t]]$ extends to an automorphism of L, with $e \in \mathbb{Z}$, and then to an automorphism of $\overline{L}$. With $g(p) = \alpha_g p$ and $g(v) = \beta_g v$, it acts on $\overline{L}[p,v]$ as in (1) with $\overline{L}$ in place of $\Bbbk$, so by (1),

$$g(p') = \alpha_g p'. \tag{7.11}$$

For $g = \iota$, (7.11) reads $-p + \sigma(-v) = -p - \sigma(v)$. Thus σ is odd in v, and since $\sigma \in \Bbbk[[t]][v]$ by **Step 5**, $\sigma = v\phi(-v^2, t)$ for a polynomial $\phi(y,t) \in \Bbbk[[t]][y]$. We obtain

$$x'y' + z'^2 = \big(x + 2z\phi - y\phi^2\big)y + \big(z - y\phi\big)^2 = xy + z^2,$$

where $\phi = \phi(y,t)$. Under $(x,y,z) = (p^2, -v^2, pv)$, we have $\phi(y,t) = \phi(-v^2,t)$, $p' = p + v\phi$, and

$$x' = p^2 + 2pv\phi + v^2\phi^2 = p'^2, \quad y' = -v^2 = -v'^2, \quad \text{and} \quad z' = pv + v^2\phi = p'v'.$$

The elements $2z\phi(y,t) - y\phi(y,t)^2$ and $y\phi(y,t)$ belong to the maximal ideal of $\Bbbk[[x,y,z,t]]$, so $x,y,z,t \to x',y',z',t$ defines a $\Bbbk$-algebra endomorphism of $\Bbbk[[x,y,z,t]]$. Since $y' = y$, the endomorphism

$$(x,y,z,t) \to \big(x - 2z\phi(y,t) - y\phi(y,t)^2, y, z + y\phi(y,t), t\big),$$

obtained by replacing ϕ with $-\phi$, is inverse to it, so it is an automorphism.

Let $g \in G$. Comparing $(gh)(p)$, $(gh)(v)$, and $(gh)(t)$ with $g(h(p))$, $g(h(v))$, and $g(h(t))$, we have $\alpha_{gh} = g(\alpha_h)\alpha_g$, $\beta_{gh} = g(\beta_h)\beta_g$, and $\gamma_{gh} = g(\gamma_h)\gamma_g$ for any $h \in G$, so the formulas in (2) define an action of G on $\Bbbk[[x,y,z,t]]$. Let g act on $\Bbbk[[t]][y]$ by applying $g|_{\Bbbk}$ to the coefficients and by $g(t) = \gamma_g t$ and $g(y) = \beta_g^2 y$. The homomorphism $\Bbbk[[t]][y] \to \Bbbk[[t]][v]$ with $y \to -v^2$ is injective and commutes with g. Since (7.11) reads $\beta_g v \cdot g(\phi(-v^2,t)) = \alpha_g v\phi(-v^2,t)$, we have

$$g(\phi(y,t)) = \alpha_g\beta_g^{-1}\phi(y,t) \quad \text{in } \Bbbk[[t]][y].$$

Hence

$$g(x') = \alpha_g^2 x + 2\alpha_g\beta_g z \cdot \alpha_g\beta_g^{-1}\phi - \beta_g^2 y \cdot \alpha_g^2\beta_g^{-2}\phi^2 = \alpha_g^2 x',$$
$$g(y') = \beta_g^2 y', \quad \text{and}$$
$$g(z') = \alpha_g\beta_g z - \beta_g^2 y \cdot \alpha_g\beta_g^{-1}\phi = \alpha_g\beta_g z'.$$

Thus g acts on x', y', z', t as on x, y, z, t, and the automorphism sending x, y, z, t to x', y', z', t commutes with g. Since $x'y' + z'^2 + t^N = xy + z^2 + t^N$, the automorphism preserves the ideal $(xy + z^2 + t^N)$ and induces an automorphism of $\Bbbk[[x, y, z, t]]/(xy + z^2 + t^N)$. If $(\alpha_g\beta_g)^2 = \gamma_g^N$, then

$$g(xy + z^2 + t^N) = (\alpha_g\beta_g)^2(xy + z^2 + t^N),$$

so g preserves the ideal $(xy + z^2 + t^N)$, and the induced automorphism commutes with g. The proposition follows. □

Lemma 7.5. *Let $\Bbbk \supset \mathbb{C}$ be an algebraically closed field.*

(1) Let $B = \sum_j b_j C_j$ be an $\mathbb{R}$-divisor on $\mathbb{P}^2_\Bbbk$ with distinct integral curves C_j, positive real numbers b_j, and $\deg B = \sum_j b_j \deg C_j \le 2$, and let q be a closed point with $b_j < 1$ for every $C_j \ni q$. Then $(\mathbb{P}^2, B)$ is klt near q, unless $\deg B = 2$ and every component of B is a line through q, in which case $(\mathbb{P}^2, B)$ is lc near q, the exceptional divisor of the blow-up of q is the unique lc place of $(\mathbb{P}^2, B)$ with center q, and $\operatorname{Nklt}(\mathbb{P}^2, B) = \{q\}$ near q.

(2) Let $d_1, \dots, d_N$ be non-negative integers, $\kappa_1, \dots, \kappa_N$ positive real numbers with $\sum_j \kappa_j d_j \le 1$, and $h_j \in \Bbbk[x, y, z]/(xy + z^2)$ non-zero elements whose representatives without monomials divisible by xy have degree at most d_j. On the double cover $\mathbb{A}^2 \to (xy + z^2 = 0) \subset \mathbb{A}^3_\Bbbk$, $(x, y, z) = (p^2, -v^2, pv)$, let $B := \sum_j \kappa_j \operatorname{div}(h_j(p^2, -v^2, pv))$, and assume that every component of B through 0 has coefficient less than one. If $(\mathbb{A}^2, B)$ is not klt at 0, then $\sum_j \kappa_j d_j = 1$, every h_j is homogeneous of degree d_j, $(\mathbb{A}^2, B)$ is lc, $B = \sum_k \beta_k L_k$ for distinct lines L_k through 0 and positive real numbers β_k with $\sum_k \beta_k = 2$, and the exceptional divisor of the blow-up of 0 is the unique lc place of $(\mathbb{A}^2, B)$ over 0.

Proof. By the Lefschetz principle, the results of [EM01] and [EMY03] used in **Steps 1 and 2** hold over $\Bbbk$.

We prove (1). Let F be any prime divisor over $\mathbb{P}^2$ with $\operatorname{center}_{\mathbb{P}^2} F = q$, and let

$$\alpha := a(F, \mathbb{P}^2, 0) > 0.$$

Step 1. For any integral curve $C \subset \mathbb{P}^2$ of degree $d \ge 2$, we have

$$\operatorname{mult}_F C < \frac{d}{2}\alpha. \tag{7.12}$$

If $d = 2$, then C is a smooth conic, so $(\mathbb{P}^2, C)$ is plt. Since F is exceptional over $\mathbb{P}^2$, $\alpha - \operatorname{mult}_F C = a(F, \mathbb{P}^2, C) > 0$, which is (7.12).

Assume that $d \ge 3$. We show that

$$\operatorname{lct}(\mathbb{P}^2; C) > \frac{2}{d}.$$

Then $(\mathbb{P}^2, \operatorname{lct}(\mathbb{P}^2; C)C)$ is lc, so $\operatorname{mult}_F C \le \alpha/\operatorname{lct}(\mathbb{P}^2; C) < \frac{d}{2}\alpha$.

Let $Z \subset \mathbb{A}^3$ be the affine cone over C, defined by an irreducible homogeneous polynomial f of degree d. The projection $\mathbb{A}^3\backslash\{0\} \to \mathbb{P}^2$ is a Zariski locally trivial $\mathbb{G}_m$-bundle and $Z\backslash\{0\}$ is the preimage of C, so locally over $\mathbb{P}^2$ the pair $(\mathbb{A}^3\backslash\{0\}, Z\backslash\{0\})$ is the product of $(\mathbb{P}^2, C)$ with $\mathbb{G}_m$. Hence, as in [EM01, Remark 1.3],

$$\operatorname{lct}(\mathbb{P}^2; C) = \operatorname{lct}(\mathbb{A}^3\backslash\{0\}; Z\backslash\{0\}) \ge \operatorname{lct}(\mathbb{A}^3; Z). \tag{7.13}$$

Let $r := \dim \operatorname{Sing} Z$.

- If C is smooth, then $\operatorname{Sing} Z = \{0\}$ and $r = 0$. By [EM01, Theorem 1.1], $\operatorname{lct}(\mathbb{A}^3; Z) \geq \min\{\frac{3}{d}, 1\} > \frac{2}{d}$.
- If C is singular, then $\operatorname{Sing} Z$ is the cone over the finite non-empty set $\operatorname{Sing} C$ and $r = 1$. By [EM01, Theorem 1.1], $\operatorname{lct}(\mathbb{A}^3; Z) \geq \min\{\frac{2}{d}, 1\} = \frac{2}{d}$.

By (7.13), $\operatorname{lct}(\mathbb{P}^2; C) > \frac{2}{d}$ in the first case and $\operatorname{lct}(\mathbb{P}^2; C) \geq \frac{2}{d}$ in the second case. Suppose that $\operatorname{lct}(\mathbb{P}^2; C) = \frac{2}{d}$ in the second case. Then by (7.13),

$$\frac{2}{d} = \operatorname{lct}(\mathbb{P}^2; C) \geq \operatorname{lct}(\mathbb{A}^3; Z) \geq \frac{2}{d},$$

so $\operatorname{lct}(\mathbb{A}^3; Z) = \frac{2}{d} = \frac{3-r}{d}$. Since $d \geq 3 = 3 - r + 1$, by [EM01, Theorem 1.2], after a linear change of coordinates (x_1, x_2, x_3) on $\mathbb{A}^3$ we have $Z = T \times \mathbb{A}^1_{x_3}$ for a curve $T \subset \mathbb{A}^2_{x_1,x_2}$. Thus $f \in \Bbbk[x_1, x_2]$ is an irreducible homogeneous polynomial in two variables, so f is linear, which contradicts $d \geq 3$. Hence $\operatorname{lct}(\mathbb{P}^2; C) > \frac{2}{d}$.

Step 2. For any line $L \ni q$ and any integral curve $C \neq L$ of degree $d \geq 1$, we have

$$0 < \operatorname{mult}_F L < \alpha \quad \text{and} \quad \operatorname{mult}_F C \leq d(\alpha - \operatorname{mult}_F L). \tag{7.14}$$

Since $q \in L$, $\operatorname{mult}_F L > 0$. Since $(\mathbb{P}^2, L)$ is plt and F is exceptional over $\mathbb{P}^2$,

$$\alpha - \operatorname{mult}_F L = a(F, \mathbb{P}^2, L) > 0.$$

By Bézout's theorem, $(C \cdot L)_x \leq d$ for any $x \in L$, so $(L, \frac{1}{d}C|_L)$ is lc, hence $(\mathbb{P}^2, L + \frac{1}{d}C)$ is lc near L [EMY03, Corollary 1.8]. Since $q \in L$,

$$0 \leq a\Big(F, \mathbb{P}^2, L + \frac{1}{d}C\Big) = \alpha - \operatorname{mult}_F L - \frac{1}{d}\operatorname{mult}_F C.$$

Step 3. We show that $a(F, \mathbb{P}^2, B) \geq 0$, and that $a(F, \mathbb{P}^2, B) = 0$ only if $\deg B = 2$ and every component of B is a line through q.

Let $J_q := \{j \mid q \in C_j\}$ and $B_q := \sum_{j \in J_q} b_j C_j$. Then

$$\operatorname{mult}_F B = \operatorname{mult}_F B_q, \quad \deg B_q \leq \deg B \leq 2, \quad \text{and} \quad b_j < 1 \text{ for any } j \in J_q.$$

If $J_q = \emptyset$, then $a(F, \mathbb{P}^2, B) = \alpha > 0$. Assume that $J_q \neq \emptyset$.

Case 1. $\operatorname{mult}_F C_j < \frac{\alpha}{2}$ for any $j \in J_q$ with C_j a line. By (7.12), $\operatorname{mult}_F C_j < \frac{\deg C_j}{2}\alpha$ for any $j \in J_q$. Since $b_j > 0$ and $J_q \neq \emptyset$,

$$\operatorname{mult}_F B_q = \sum_{j \in J_q} b_j \operatorname{mult}_F C_j < \frac{\deg B_q}{2}\alpha \leq \alpha,$$

so $a(F, \mathbb{P}^2, B) > 0$.

Case 2. $\operatorname{mult}_F C_j \geq \frac{\alpha}{2}$ for some $j \in J_q$ with C_j a line. Possibly reordering indices, we may assume that $1 \in J_q$, C_1 is a line, and $\operatorname{mult}_F C_1 \geq \frac{\alpha}{2}$. Let $\rho := \operatorname{mult}_F C_1/\alpha$. By the assumption of **Case 2** and (7.14),

$$\frac{1}{2} \leq \rho < 1,$$

and for any $j \in J_q \backslash \{1\}$, (7.14) with $L = C_1$ and $C = C_j$ implies

$$\operatorname{mult}_F C_j \leq \deg C_j (1 - \rho)\alpha.$$

Since $\sum_{j \in J_q \backslash \{1\}} b_j \deg C_j = \deg B_q - b_1$, we obtain

$$\frac{\operatorname{mult}_F B_q}{\alpha} \leq b_1 \rho + (\deg B_q - b_1)(1 - \rho) =: \phi(\rho).$$

The function ϕ is affine in ρ, with $\phi(\frac{1}{2}) = \frac{\deg B_q}{2}$ and $\phi(1) = b_1 < 1$.

- If $\deg B_q < 2$, then $\phi(\frac{1}{2}) < 1$ and $\phi(1) < 1$, so $\phi(\rho) < 1$.

- If $\deg B_q = 2$, then

$$\phi(\rho) = 1 - 2(1-b_1)\Big(\rho - \frac{1}{2}\Big) \le 1,$$

with equality if and only if $\rho = \frac{1}{2}$.

Thus

$$a(F, \mathbb{P}^2, B) = \alpha - \operatorname{mult}_F B_q \ge (1-\phi(\rho))\alpha \ge 0.$$

If $a(F, \mathbb{P}^2, B) = 0$, then $\deg B_q = 2$, $\rho = \frac{1}{2}$, and, as $b_j > 0$, $\operatorname{mult}_F C_j = \frac{\deg C_j}{2}\alpha$ for any $j \in J_q \backslash \{1\}$. By (7.12), every C_j with $j \in J_q$ is a line through q. Since $\deg B_q = 2 \ge \deg B$ and $b_j > 0$ for any j, $B = B_q$. This proves **Step 3**.

Step 4. We conclude the proof of (1).

Let $f : Y' \to \mathbb{P}^2$ be a log resolution of $(\mathbb{P}^2, B)$ and let F' be a prime divisor on Y' with $q \in f(F')$. If $f(F') = \{q\}$, then $a(F', \mathbb{P}^2, B) \ge 0$ by **Step 3**, with equality only in the exceptional case of (1). Otherwise $F' = f_*^{-1}C$ for a curve $C \ni q$ and $a(F', \mathbb{P}^2, B) = 1 - \operatorname{mult}_C B > 0$, since $b_j < 1$ for every $C_j \ni q$. Since $\operatorname{Nklt}(\mathbb{P}^2, B)$ is the union of the closed sets $f(F')$ over the prime divisors F' on Y' with $a(F', \mathbb{P}^2, B) \le 0$, $(\mathbb{P}^2, B)$ is klt near q outside the exceptional case of (1), and $(\mathbb{P}^2, B)$ is lc near q in the exceptional case.

Assume that $\deg B = 2$ and every C_j is a line through q, so $\sum_j b_j = 2$ and $b_j < 1$ for any j. Let $g : Y \to \mathbb{P}^2$ be the blow-up of q, E the exceptional divisor, and $C'_j := g_*^{-1}C_j$. Since $a(E, \mathbb{P}^2, 0) = 2$ and $\operatorname{mult}_E B = \sum_j b_j = 2$,

$$K_Y + E + \sum_j b_j C'_j = g^*(K_{\mathbb{P}^2} + B).$$

Distinct lines through q intersect only at q and have distinct tangent directions at q, so the curves C'_j are pairwise disjoint and intersect E transversally. Thus $(Y, E + \sum_j b_j C'_j)$ is log smooth, with coefficient 1 along E and $b_j < 1$ along C'_j, so $(Y, E + \sum_j b_j C'_j)$ is plt. Any prime divisor $F \ne E$ over $\mathbb{P}^2$ with $\operatorname{center}_{\mathbb{P}^2} F = q$ satisfies $\operatorname{center}_Y F \subset E$, hence

$$a(F, \mathbb{P}^2, B) = a\Big(F, Y, E + \sum_j b_j C'_j\Big) > 0.$$

Since $a(E, \mathbb{P}^2, B) = 0$, E is the unique lc place of $(\mathbb{P}^2, B)$ with center q, and $q \in \operatorname{Nklt}(\mathbb{P}^2, B)$. On $\mathbb{P}^2 \backslash \{q\}$, B is a sum of pairwise disjoint smooth curves $C_j \backslash \{q\}$ with coefficients $b_j < 1$, so $\operatorname{Nklt}(\mathbb{P}^2, B) = \{q\}$. This proves (1).

We prove (2). Let $\psi^* : \Bbbk[x, y, z] \to \Bbbk[p, v]$ be the homomorphism with

$$(\psi^* x, \psi^* y, \psi^* z) = (p^2, -v^2, pv).$$

Since $\psi^*(xy + z^2) = 0$, ψ^* induces the double cover $\psi : \mathbb{A}^2 \to (xy + z^2 = 0)$. For any j, let $\widetilde{h}_j \in \Bbbk[x, y, z]$ be the representative of h_j without monomials divisible by xy, and let $h'_j := \psi^* \widetilde{h}_j = h_j(p^2, -v^2, pv)$. For non-negative integers i_1, i_2, i_3 with $i_1 i_2 = 0$,

$$\psi^*(x^{i_1} y^{i_2} z^{i_3}) = (-1)^{i_2} p^{2i_1 + i_3} v^{2i_2 + i_3},$$

which has degree $2(i_1 + i_2 + i_3)$, and (i_1, i_2, i_3) is determined by the exponents $(2i_1 + i_3, 2i_2 + i_3)$ since $i_3 = \min\{2i_1 + i_3, 2i_2 + i_3\}$. Thus ψ^* maps distinct monomials not divisible by xy to distinct monomials, so $\widetilde{h}_j$ exists and is unique: replacing xy with $-z^2$ lowers the power of x and keeps the degree, and the difference of two such representatives of h_j is in $(xy + z^2) \subset \ker \psi^*$, hence is 0. For any integer $e \ge 0$, the homogeneous part of degree $2e$ of h'_j is thus the image of the homogeneous part of degree e of $\widetilde{h}_j$. Since $\widetilde{h}_j \ne 0$, we obtain

$$h'_j \ne 0, \quad \deg h'_j = 2 \deg \widetilde{h}_j \le 2d_j,$$

and $\widetilde{h}_j$ is homogeneous if and only if h'_j is homogeneous.

Let $[p : v : s]$ be homogeneous coordinates on $\mathbb{P}^2_{\Bbbk} \supset \mathbb{A}^2 = (s \neq 0)$ and let $\overline{B}$ be the closure of B in $\mathbb{P}^2$, that is,

$$\overline{B} = \sum_j \kappa_j \operatorname{div}\left(s^{\deg h'_j} h'_j(p/s, v/s)\right).$$

Then

$$\deg \overline{B} = \sum_j \kappa_j \deg h'_j \leq 2 \sum_j \kappa_j d_j \leq 2,$$

and the components of $\overline{B}$ through the origin $0 = [0 : 0 : 1]$ are the closures $\overline{D}$ of the components D of B through 0, and the coefficient of $\overline{D}$ in $\overline{B}$ equals the coefficient of D in B, which is less than one.

Assume that $(\mathbb{A}^2, B)$ is not klt at 0. Since $\overline{B}|_{\mathbb{A}^2} = B$, $(\mathbb{P}^2, \overline{B})$ is not klt near 0. We apply (1) with q the origin: $\deg \overline{B} = 2$, every component of $\overline{B}$ is a line through 0, $(\mathbb{P}^2, \overline{B})$ is lc near 0, and the exceptional divisor of the blow-up of 0 is the unique lc place of $(\mathbb{P}^2, \overline{B})$ with center 0. Thus

$$2 = \sum_j \kappa_j \deg h'_j \leq 2 \sum_j \kappa_j d_j \leq 2,$$

so $\sum_j \kappa_j d_j = 1$, and since $\kappa_j > 0$ and $\deg h'_j \leq 2d_j$ for any j, $\deg h'_j = 2d_j$ for any j. Since $\kappa_j > 0$, any irreducible factor of h'_j defines a component of B, hence a line through 0, so it is a linear form in p, v. Thus h'_j is homogeneous of degree $2d_j$ (a non-zero constant if $d_j = 0$), so $\widetilde{h}_j$ is homogeneous of degree d_j. We write $B = \sum_k \beta_k L_k$ with distinct lines L_k through 0 and $\beta_k > 0$. Then $\sum_k \beta_k = \deg \overline{B} = 2$, and $\beta_k < 1$ for any k by assumption. The pair $(\mathbb{A}^2, B)$ is lc near 0 by (1), and klt on $\mathbb{A}^2 \backslash \{0\}$ since the curves $L_k \backslash \{0\}$ are smooth, pairwise disjoint, with coefficients $\beta_k < 1$. Hence $(\mathbb{A}^2, B)$ is lc. A prime divisor over $\mathbb{A}^2$ with center 0 is a prime divisor over $\mathbb{P}^2$ with center 0, so the uniqueness of the lc place follows from (1). The lemma follows. □

7.2. **Rigidity.**

Proposition 7.6. *Let $\Bbbk \supset \mathbb{C}$ be an algebraically closed field, $d \geq 2$ an integer, and $w_1 \geq w_2$ coprime positive integers, and let x, y, z have the weights dw_1, dw_2, $w_1 + w_2$. Let $P \in \Bbbk[z]$ be a monic polynomial of degree d with $P(0) = 0$, and let*

$$f := xy + P(z), \quad V := (f = 0) \subset \mathbb{A}^3_{\Bbbk}, \quad \text{and} \quad o := 0 \in V.$$

Let $Q_1, \ldots, Q_N$ be non-zero elements of $\Bbbk[V]$. For any j, we denote again by Q_j the unique polynomial in $\Bbbk[x, y, z]$ representing Q_j with no monomial divisible by xy, we let D_j be the weighted degree of Q_j, and we let T_j be the part of Q_j of weighted degree D_j. Let $\kappa_1, \ldots, \kappa_N$ be positive real numbers such that

$$\sum_{j=1}^{N} \kappa_j D_j = w_1 + w_2,$$

and let $B := \sum_{j=1}^{N} \kappa_j \operatorname{div}(Q_j)$. Assume that (V, B) is lc near o but not klt at o, and that any component of B through o has coefficient less than one. Then the following hold.

(1) The following hold.

(a) $P = z^d$.

(b) There exists a unique polynomial

$$U = \sum_{k \geq 1,\ kdw_2 < w_1 + w_2} u_k y^k \in \Bbbk[y]$$

such that $Q_j = T_j(x', y', z')$ in $\Bbbk[V]$ for any j, where

$$x' := x - \frac{(z + U)^d - z^d}{y}, \quad y' := y, \quad \text{and} \quad z' := z + U.$$

If $w_1 = w_2 = 1$, then the sum is empty and $U = 0$.

(c) *The $\Bbbk$-algebra endomorphism Φ of $\Bbbk[x,y,z]$ with $(x,y,z)\to(x',y',z')$ of (b) is an automorphism with $\Phi(xy+z^d)=xy+z^d$ and Jacobian determinant one, the polynomials x',y',z' vanish at o, and Φ induces the identity on the graded ring of the weighted degree filtration of $\Bbbk[x,y,z]$.*

(d) *The automorphism Φ of (c) descends to any field of definition: if $\Bbbk_0\subset\Bbbk$ is a subfield with $Q_1,\dots,Q_N\in\Bbbk_0[x,y,z]$, then $U\in\Bbbk_0[y]$. Moreover, Φ is equivariant in the following sense. Let G be a group acting on the ring $\Bbbk[x,y,z]$ by automorphisms preserving $\Bbbk$, the ideal (f), and the weighted grading, such that $\eta(z)\in\Bbbk^*z$ and $\{\eta(x),\eta(y)\}\subset\Bbbk^*x\cup\Bbbk^*y$ for any $\eta\in G$, and assume that for any j there exists a map $\chi_j:G\to\Bbbk^*$ with*

$$\eta(Q_j)=\chi_j(\eta)Q_j \text{ in } \Bbbk[V] \quad \text{and} \quad \eta(T_j)=\chi_j(\eta)T_j \text{ in } \Bbbk[x,y,z] \quad \text{for any } \eta\in G.$$

Then $\eta\circ\Phi=\Phi\circ\eta$ for any $\eta\in G$, and if $\eta(x)\in\Bbbk^*y$ for some $\eta\in G$, then $w_1=w_2=1$.

(2) *Let $P\in\Bbbk[[t]][z]$ and $Q_1,\dots,Q_N\in\Bbbk[[t]][x,y,z]$, where no monomial of any Q_j is divisible by xy and t has weight 0, and assume that the hypotheses above hold with $\Bbbk$ replaced by an algebraic closure $\overline{\Bbbk((t))}$ of $\Bbbk((t))$. Then $P=z^d$, and the polynomial U of (1)(b) over $\overline{\Bbbk((t))}$ belongs to $\Bbbk((t))[y]$.*

(3) *Let R be a commutative $\Bbbk$-algebra, $U\in yR[y]$, and $g\in R[z]$. Then the R-algebra endomorphism*

$$\Phi_g:(x,y,z)\to\Big(x-\frac{g(z+U(y))-g(z)}{y},\ y,\ z+U(y)\Big)$$

of $R[x,y,z]$ is an automorphism with inverse

$$\Phi_g^{-1}:(x,y,z)\to\Big(x-\frac{g(z-U(y))-g(z)}{y},\ y,\ z-U(y)\Big).$$

Moreover, $\Phi_g(xy+g)=xy+g$, the Jacobian determinant of Φ_g is one, and $\eta\circ\Phi_g=\Phi_g\circ\eta$ for any $\eta\in G$, where G is any group acting on R and diagonally on x,y,z such that for any $\eta\in G$, writing $\eta(x)=\alpha x$, $\eta(y)=\beta y$, $\eta(z)=\gamma z$ with $\alpha,\beta,\gamma\in R^\times$, we have $\eta(g)=\alpha\beta g$ and $\eta(U)=\gamma U$.

Proof. **Step 1.** We prove (3). Since $U\in yR[y]$ and $g(z+U)-g(z)\in U\cdot R[y,z]$, the polynomial $g(z+U)-g(z)$ is divisible by y in $R[y,z]$, and the quotient is unique because y is a non-zerodivisor. Thus Φ_g is well defined. Let Φ_g^{-1} be the R-algebra endomorphism in (3). Since $\Phi_g^{-1}(z+U)=z$ and

$$\Phi_g^{-1}\big(\Phi_g(x)\big)=x-\frac{g(z-U)-g(z)}{y}-\frac{g(z)-g(z-U)}{y}=x,$$

we have $\Phi_g^{-1}\circ\Phi_g=\mathrm{id}_{R[x,y,z]}$, and exchanging U and $-U$ we have $\Phi_g\circ\Phi_g^{-1}=\mathrm{id}_{R[x,y,z]}$. Next,

$$\Phi_g(xy+g)=xy-\big(g(z+U)-g(z)\big)+g(z+U)=xy+g.$$

In the order y,z,x of the coordinates, $\Phi_g(y)$ depends only on y, $\Phi_g(z)-z$ depends only on y, and $\Phi_g(x)-x$ depends only on y,z, so the Jacobian matrix of Φ_g is triangular with diagonal entries one, and the Jacobian determinant is one.

Let $\eta\in G$, and write $\eta(x)=\alpha x$, $\eta(y)=\beta y$, $\eta(z)=\gamma z$ with $\alpha,\beta,\gamma\in R^\times$, $\eta(g)=\alpha\beta g$, and $\eta(U)=\gamma U$. The ring homomorphisms $\eta\circ\Phi_g$ and $\Phi_g\circ\eta$ coincide on R and on z, since $\eta(z+U)=\gamma(z+U)$, so they coincide on $R[z]$. Evaluating them at g, we obtain

$$\eta(g(z+U))=\Phi_g(\alpha\beta g)=\alpha\beta g(z+U).$$

Thus

$$\beta y\cdot\eta\Big(\frac{g(z+U)-g(z)}{y}\Big)=\eta\big(g(z+U)-g(z)\big)=\alpha\beta\big(g(z+U)-g(z)\big),$$

so $\eta(\Phi_g(x))=\alpha\Phi_g(x)=\Phi_g(\eta(x))$. Since also $\eta(\Phi_g(y))=\Phi_g(\eta(y))$, $\eta(\Phi_g(z))=\Phi_g(\eta(z))$, and both $\eta\circ\Phi_g$ and $\Phi_g\circ\eta$ restrict to η on R, we have $\eta\circ\Phi_g=\Phi_g\circ\eta$. This proves (3).

Step 2. We prove $P = z^d$. Let $V_0 := (xy + z^d = 0) \subset \mathbb{A}^3_{\Bbbk}$. For the lexicographic order with $x > y > z$, the leading monomial of f and of $xy + z^d$ is xy. Replacing xy with $-P(z)$ reduces any polynomial modulo f to one with no monomial divisible by xy, and a non-zero multiple of f has leading monomial divisible by xy. Thus the monomials not divisible by xy form a basis of $\Bbbk[V]$, and likewise of $\Bbbk[V_0]$. In particular the representatives Q_j are unique, and $T_j|_{V_0} \neq 0$ for any j.

Let $e := \gcd(d, w_1 + w_2)$. Since $\gcd(dw_1, dw_2) = d$, the vector

$$w := \frac{1}{e}(dw_1, dw_2, w_1 + w_2)$$

has entries with greatest common divisor one. For the w-weights, the top part of f is $xy + z^d$, of w-degree $d(w_1 + w_2)/e$, the top part of Q_j is T_j, of w-degree D_j/e, and

$$\sum_{j=1}^{N} \kappa_j \frac{D_j}{e} = \frac{w_1 + w_2}{e} = \frac{dw_1 + dw_2 + (w_1 + w_2)}{e} - \frac{d(w_1 + w_2)}{e}.$$

Thus $(V_0, \sum_j \kappa_j \operatorname{div}(T_j))$ is a critical pair (Definition 7.1), and Lemma 7.3(3), in case (b) with $F = f$ and $f_0 = xy + z^d$, implies that there exists a polynomial automorphism $\varphi = (\varphi_1, \varphi_2, \varphi_3)$ of $\mathbb{A}^3_{\Bbbk}$ with $\varphi_1(o) = \varphi_2(o) = \varphi_3(o) = 0$ and

$$f = \varphi_1\varphi_2 + \varphi_3^d.$$

Write $P = z^k P_1$ with $1 \leq k \leq d$ and $P_1(0) \neq 0$. Then $(V \ni o)$ is an A_{k-1} singularity (a smooth point if $k = 1$), while φ maps $(V \ni o)$ isomorphically onto the A_{d-1} singularity $(V_0 \ni 0)$, so $k = d$ and $P = z^d$. From now on $V = V_0$ and $f = xy + z^d$.

Step 3. We construct U and prove uniqueness. Let $\psi : \mathbb{A}^2_{\Bbbk} \to V$ be the morphism with

$$\psi^* x = p^d, \quad \psi^* y = -v^d, \quad \text{and} \quad \psi^* z = pv,$$

and let p, v have the weights w_1, w_2. A monomial $x^a y^b z^c$ with $ab = 0$ maps to $\pm p^{da+c} v^{db+c}$, of weighted degree $(da+c)w_1 + (db+c)w_2 = \deg(x^a y^b z^c)$, and (a, b, c) is recovered from $(da+c, db+c)$ because $c = \min\{da + c, db + c\}$. Since the monomials not divisible by xy form a basis of $\Bbbk[V]$ (**Step 2**), $\psi^* : \Bbbk[V] \to \Bbbk[p, v]$ is injective, and $\psi^* Q_j$ has weighted degree D_j and top part $\psi^* T_j \neq 0$. The image of ψ^* is the ring of invariants of the group μ_d acting by $(p, v) \to (\zeta p, \zeta^{-1} v)$, since the invariant monomials $p^a v^b$, $a \equiv b \pmod d$, are products of p^d, v^d, pv, so ψ is the quotient by μ_d. The group μ_d acts freely on $\mathbb{A}^2 \backslash \{0\}$, so ψ is étale over $V \backslash \{o\}$ and $\psi^{-1}(o) = \{0\}$.

Let $B_{\mathbb{A}^2} := \sum_j \kappa_j \operatorname{div}(\psi^* Q_j)$. Since ψ is étale in codimension 1, $K_{\mathbb{A}^2} + B_{\mathbb{A}^2} = \psi^*(K_V + B)$, and any prime divisor C on $\mathbb{A}^2$ through 0 satisfies $\operatorname{mult}_C B_{\mathbb{A}^2} = \operatorname{mult}_{\psi(C)} B < 1$. Hence $(\mathbb{A}^2, B_{\mathbb{A}^2})$ is lc near 0 but not klt at 0. Since $\sum_j \kappa_j D_j = w_1 + w_2$, Proposition 7.4(1) implies that there exists a unique polynomial coordinate change $(p, v) \to (p', v')$ fixing 0 and inducing the identity on the graded ring of the weighted degree filtration such that

$$\psi^* Q_j(p, v) = \psi^* T_j(p', v') \quad \text{for any } j.$$

By Proposition 7.4(1), $p' = p + \sigma(v)$ and $v' = v$ for a polynomial σ whose exponents m satisfy $0 < mw_2 < w_1$ ($\sigma = 0$ if $w_1 = w_2 = 1$). The automorphism $(p, v) \to (\zeta p, \zeta^{-1} v)$ is $\Bbbk$-linear, diagonal, and fixes $\psi^* Q_j$ and $\psi^* T_j$, so by Proposition 7.4(1) it commutes with $(p, v) \to (p', v')$, that is,

$$\zeta p + \sigma(\zeta^{-1} v) = \zeta\big(p + \sigma(v)\big) \quad \text{for any } \zeta \in \mu_d.$$

Thus any exponent m of σ satisfies $m \equiv -1 \pmod d$, and writing $m = dk - 1$, the condition $0 < mw_2 < w_1$ is equivalent to $k \geq 1$ and $kdw_2 < w_1 + w_2$. Hence there exists a unique $U = \sum_{k \geq 1,\ kdw_2 < w_1 + w_2} u_k y^k \in \Bbbk[y]$ with

$$U(-v^d) = v\sigma(v).$$

Then $\psi^*y' = -v^d$, $\psi^*z' = pv + v\sigma(v) = p'v$, and

$$\psi^*x' = p^d - \frac{(p'v)^d - (pv)^d}{-v^d} = p'^d.$$

Thus $\psi^*\big(T_j(x', y', z')\big) = \psi^*T_j(p', v) = \psi^*Q_j$, and since ψ^* is injective, $Q_j = T_j(x', y', z')$ in $\Bbbk[V]$ for any j.

Let $U_1 = \sum_{k\geq 1,\ kdw_2<w_1+w_2} u_{1,k}y^k$ be a polynomial, let

$$x_1 := x - \frac{(z+U_1)^d - z^d}{y}, \quad y_1 := y, \quad \text{and} \quad z_1 := z + U_1,$$

and assume that $Q_j = T_j(x_1, y_1, z_1)$ in $\Bbbk[V]$ for any j. Let $\sigma_1(v) := U_1(-v^d)/v$. Any exponent m of σ_1 satisfies $0 < mw_2 < w_1$, so $(p, v) \to (p + \sigma_1(v), v)$ fixes 0 and induces the identity on the graded ring, and the computation above shows that $\psi^*Q_j = \psi^*T_j(p + \sigma_1(v), v)$ for any j. By the uniqueness in Proposition 7.4(1), $\sigma_1 = \sigma$, so $U_1 = U$.

Step 4. We prove the rest of (1). The endomorphism Φ is Φ_{z^d} of (3) for $R = \Bbbk$, so by (3) it is an automorphism with $\Phi(xy + z^d) = xy + z^d$ and Jacobian determinant one. If $U = 0$, then $\Phi = \mathrm{id}_{\Bbbk[x,y,z]}$. Assume that $U \neq 0$, and let deg denote the weighted degree. Then $y \mid U$ and $\deg U < w_1 + w_2$, and

$$x' - x = -\frac{U}{y}\sum_{i=1}^{d}\binom{d}{i}z^{d-i}U^{i-1}.$$

Since $d \geq 2$, any summand vanishes at o, and

$$\deg\Big(\frac{U}{y}z^{d-i}U^{i-1}\Big) = (d-i)(w_1+w_2) + i\deg U - dw_2 < dw_1 = \deg x.$$

Together with $z' - z = U$ and $\deg U < \deg z$, this shows that x', y', z' vanish at o and that Φ induces the identity on the graded ring.

Let $\Bbbk_0 \subset \Bbbk$ be a subfield with $Q_1, \dots, Q_N \in \Bbbk_0[x, y, z]$. Then $T_j \in \Bbbk_0[x, y, z]$, and

$$\psi^*Q_j,\ \psi^*T_j \in \Bbbk_0[p, v] \quad \text{for any } j,$$

so by Proposition 7.4(1), $\sigma \in \Bbbk_0[v]$, hence $U \in \Bbbk_0[y]$.

Let $\eta \in G$. If $\eta(x) \in \Bbbk^*y$ or $\eta(y) \in \Bbbk^*x$, then since η preserves the weighted grading, $dw_1 = dw_2$, so $w_1 = w_2 = 1$, $\Phi = \mathrm{id}_{\Bbbk[x,y,z]}$, and $\eta \circ \Phi = \Phi \circ \eta$. Otherwise write $\eta(x) = \alpha x$, $\eta(y) = \beta y$, $\eta(z) = \gamma z$ with $\alpha, \beta, \gamma \in \Bbbk^*$. Since $\eta(xy + z^d) = \alpha\beta xy + \gamma^d z^d \in (xy + z^d)$, we have $\alpha\beta = \gamma^d$. For a polynomial H over $\Bbbk$, let H^η be the polynomial obtained by applying η to the coefficients of H. Let

$$\widetilde{U}(y) := \gamma^{-1}U^\eta(\beta y),$$

whose monomials are those of U, and let

$$\widetilde{x} := x - \frac{(z+\widetilde{U})^d - z^d}{y}, \quad \widetilde{y} := y, \quad \text{and} \quad \widetilde{z} := z + \widetilde{U}.$$

Then $\eta(y') = \beta\widetilde{y}$, $\eta(z') = \gamma z + U^\eta(\beta y) = \gamma\widetilde{z}$, and

$$\eta(x') = \alpha x - \frac{\gamma^d\big((z+\widetilde{U})^d - z^d\big)}{\beta y} = \alpha\widetilde{x}.$$

Since no monomial of T_j is divisible by xy, the equality $\eta(T_j) = \chi_j(\eta)T_j$ is the polynomial identity $T_j^\eta(\alpha x, \beta y, \gamma z) = \chi_j(\eta)T_j(x, y, z)$. Applying η to $Q_j = T_j(x', y', z')$ in $\Bbbk[V]$, we obtain

$$\chi_j(\eta)Q_j = T_j^\eta(\alpha\widetilde{x}, \beta\widetilde{y}, \gamma\widetilde{z}) = \chi_j(\eta)T_j(\widetilde{x}, \widetilde{y}, \widetilde{z}) \quad \text{in } \Bbbk[V].$$

By the uniqueness in **Step 3**, $\widetilde{U} = U$, that is, $\eta(U) = \gamma U$. Since $\eta(z^d) = \gamma^d z^d = \alpha\beta z^d$, (3) for $R = \Bbbk$ and $g = z^d$ implies that $\eta \circ \Phi = \Phi \circ \eta$. This proves (1).

Step 5. We prove (2). Since $\overline{\Bbbk((t))} \supset \Bbbk \supset \mathbb{C}$, (1) over $\overline{\Bbbk((t))}$ implies that $P = z^d$ and $U \in \overline{\Bbbk((t))}[y]$, and the descent in (1)(d) with $\Bbbk_0 = \Bbbk((t))$ implies that $U \in \Bbbk((t))[y]$. □

Proposition 7.7. *Let $\Bbbk \supset \mathbb{C}$ be an algebraically closed field, and let $f_0 \in \Bbbk[x,y,z]$ be a polynomial and $w = (w_x, w_y, w_z)$ a weight vector as in one of the following cases.*

(D_k) $f_0 = x^2 + zy^2 + z^{k-1}$ and $w = (k-1, k-2, 2)$ for an integer $k \geq 4$.
(E_6) $f_0 = x^2 + y^3 + z^4$ and $w = (6,4,3)$.
(E_7) $f_0 = x^2 + y^3 + yz^3$ and $w = (9,6,4)$.
(E_8) $f_0 = x^2 + y^3 + z^5$ and $w = (15,10,6)$.

The w-weight of a monomial $x^{a_1}y^{a_2}z^{a_3}$ is $a_1w_x + a_2w_y + a_3w_z$. For a non-zero polynomial $q \in \Bbbk[x,y,z]$, let $\deg_w q$ be the largest w-weight of a monomial of q, and let $\deg_w 0 := -\infty$. Let $F = f_0 + F'$ for a polynomial $F' \in \Bbbk[x,y,z]$ with $\deg_w F' < \deg_w f_0$, and let $V := (F = 0) \subset \mathbb{A}^3_{\Bbbk}$. Then V is normal and has at worst Du Val singularities.

Now let $Q_1, \dots, Q_N \in \Bbbk[x,y,z]$ be non-zero polynomials of x-degree at most 1, for any j let $d_j := \deg_w Q_j$ and let P_j be the top part of Q_j, let $\kappa_1, \dots, \kappa_N$ be positive real numbers such that

$$\sum_{j=1}^{N} \kappa_j d_j = 1,$$

let $B := \sum_{j=1}^N \kappa_j \operatorname{div}(Q_j|_V)$, and let o be a closed point of V. Assume that (V,B) is lc near o but not klt at o, and that any component of B through o has coefficient less than one. For a polynomial automorphism φ of $\mathbb{A}^3_{\Bbbk}$ with coordinate functions x', y', z', let φ^* be the $\Bbbk$-algebra automorphism of $\Bbbk[x,y,z]$ with $\varphi^*(q) = q(x',y',z')$. Then the following hold.

(1) There exists a unique polynomial automorphism φ of $\mathbb{A}^3_{\Bbbk}$ whose coordinate functions are

$$x' = x + g_1(y,z), \quad y' = y + g_2(z), \quad \text{and} \quad z' = z + c$$

for some $g_1 \in \Bbbk[y,z]$ with $\deg_w g_1 < w_x$, $g_2 \in \Bbbk[z]$ with $\deg_w g_2 < w_y$, and $c \in \Bbbk$, such that

$$F = f_0(x', y', z').$$

Moreover, $Q_j = P_j(x',y',z')$ for any j, $\varphi(o) = (0,0,0)$, and the Jacobian determinant of φ is one.

(2) For any subring $R_0 \subset \Bbbk$ containing $\mathbb{Q}$ and the coefficients of F, we have $g_1 \in R_0[y,z]$, $g_2 \in R_0[z]$, and $c \in R_0$.

(3) Let Γ be a group acting on $\Bbbk[x,y,z]$ by ring automorphisms such that $\gamma(\Bbbk) = \Bbbk$, $\gamma(x) \in \Bbbk^* x$, $\gamma(y) \in \Bbbk^* y$, and $\gamma(z) \in \Bbbk^* z$ for any $\gamma \in \Gamma$. Assume that for any $\gamma \in \Gamma$ there exists $a_\gamma \in \Bbbk^*$ such that $\gamma(F) = a_\gamma F$ and $\gamma(f_0) = a_\gamma f_0$. Then $\gamma \circ \varphi^* = \varphi^* \circ \gamma$ for any $\gamma \in \Gamma$.

(4) The pair (V,B) is lc, $\operatorname{Nklt}(V,B) = \{o\}$, and $\operatorname{Sing} V = \{o\}$.

(5) Assume that $\Bbbk$ is an algebraic closure of $\Bbbk_1((u))$ for an algebraically closed field $\Bbbk_1 \supset \mathbb{C}$, and that $F \in R[x,y,z]$ for $R := \Bbbk_1[[u]]$. Then the coordinate functions of φ and of φ^{-1} belong to $R[x,y,z]$. If moreover $o = (0,0,0)$, then the coordinate functions of φ and of φ^{-1} belong to the ideal $(x,y,z)R[x,y,z]$, and φ^* extends, by $\varphi^*(u) = u$, to an automorphism of $\Bbbk_1[[x,y,z,u]]$ with Jacobian determinant one.

Proof. In each of the four cases,

$$\gcd(w_x, w_y, w_z) = 1, \quad 2w_x = \deg_w f_0, \quad w_x > w_y \geq w_z, \quad \text{and} \quad w_x + w_y + w_z - \deg_w f_0 = 1. \tag{7.15}$$

Let $V_0 := (f_0 = 0) \subset \mathbb{A}^3_{\Bbbk}$. Since $\deg_w F' < \deg_w f_0$, f_0 is the top part of F. The polynomial z is w-homogeneous of w-weight w_z and $z|_{V_0} \neq 0$, and $(V_0, \frac{1}{w_z}\operatorname{div}(z))$ is a critical pair (Definition 7.1) by (7.15). Thus, by Lemma 7.3 in case (b), applied to the single polynomial z with coefficient $\frac{1}{w_z}$, V is normal and has at worst Du Val singularities.

We prove the existence part of (1). Since f_0 is monic of degree 2 in x, any non-zero multiple of f_0 has x-degree at least 2. Since $P_j \neq 0$ has x-degree at most 1, $P_j|_{V_0} \neq 0$ for any j. Since $\sum_j \kappa_j d_j = 1 = w_x + w_y + w_z - \deg_w f_0$, the pair $(V_0, \sum_j \kappa_j \operatorname{div}(P_j))$ is a critical pair. By Lemma 7.3(3) in case (b), there exists a polynomial automorphism φ of $\mathbb{A}^3_{\Bbbk}$ inducing the identity on the graded ring of the w-weighted degree filtration, whose coordinate functions x', y', z' vanish at o, such that

$$F = f_0(x', y', z') \quad \text{and} \quad Q_j - P_j(x', y', z') \in (F) \text{ for any } j.$$

Since φ induces the identity on the graded ring,

$$\deg_w(x' - x) < w_x, \quad \deg_w(y' - y) < w_y, \quad \text{and} \quad \deg_w(z' - z) < w_z.$$

By (7.15), w_z is the smallest of the three weights, so a monomial of w-weight less than w_x does not contain x, a monomial of w-weight less than w_y is a power of z, and a monomial of w-weight less than w_z is a constant. Thus $x' = x + g_1(y, z)$, $y' = y + g_2(z)$, and $z' = z + c$ with g_1, g_2, c as in (1) (in case (D_4), $w_y = w_z = 2$ and g_2 is a constant). The Jacobian matrix $\partial(x', y', z')/\partial(x, y, z)$ is upper triangular with diagonal entries 1, so the Jacobian determinant of φ is one. Since x', y', z' vanish at o, $\varphi(o) = (0, 0, 0)$.

By (7.15), any monomial of x-degree at least 2 has w-weight at least $2w_x = \deg_w f_0 > \deg_w F'$, so F' has x-degree at most 1 and F is monic of degree 2 in x over $\Bbbk[y, z]$. Thus any non-zero element of (F) has x-degree at least 2. Since $x' - x, y', z' \in \Bbbk[y, z]$ and P_j has x-degree at most 1, $P_j(x', y', z')$ has x-degree at most 1, and so does Q_j. Since $Q_j - P_j(x', y', z') \in (F)$, we have $Q_j = P_j(x', y', z')$ for any j.

We prove the uniqueness part of (1) and (2). Let φ be any polynomial automorphism with coordinate functions x', y', z' of the form in (1) such that $F = f_0(x', y', z')$, and let R_0 be a subring of $\Bbbk$ containing $\mathbb{Q}$ and the coefficients of F. We show that g_1, g_2, and c are determined by F and have coefficients in R_0. For $R_0 = \Bbbk$ this is the uniqueness. Since $f_0 - x^2 = f_0(0, y, z)$,

$$F = (x + g_1(y, z))^2 + f_0(0, y + g_2(z), z + c).$$

As a polynomial in x over $\Bbbk[y, z]$, the coefficient of x in F is $2g_1$. Thus g_1 is determined by F, $g_1 \in R_0[y, z]$, and

$$F_1 := F - (x + g_1)^2 = f_0(0, y + g_2, z + c) \in R_0[y, z].$$

We regard F_1 as a polynomial in y over $\Bbbk[z]$.

- In case (D_k), $F_1 = (z+c)(y+g_2)^2 + (z+c)^{k-1}$. The coefficient of y^2 in F_1 is $z+c$, and the coefficient of y is $2(z+c)g_2$. Thus $c \in R_0$ is determined by F. Since $z+c \in R_0[z]$ is monic and $2(z+c)g_2 \in R_0[z]$, long division in $R_0[z]$ shows that $2g_2 = 2(z+c)g_2/(z+c) \in R_0[z]$, and g_2 is determined by F.
- In cases (E_6), (E_7), and (E_8), the coefficient of y^2 in F_1 is $3g_2$. Thus $g_2 \in R_0[z]$ is determined by F. Moreover, $F_1 - (y + g_2)^3 \in R_0[y, z]$ is equal to

  $$(z + c)^4, \quad (y + g_2)(z + c)^3, \quad \text{and} \quad (z + c)^5$$

  respectively, whose coefficients of z^3, yz^2, and z^4 are $4c$, $3c$, and $5c$ respectively. Thus $c \in R_0$ is determined by F.

This proves the uniqueness part of (1), and (2).

We prove (3). Let $\gamma \in \Gamma$, and let $\varphi^*_\gamma := \gamma \circ \varphi^* \circ \gamma^{-1}$. Since $\gamma(\Bbbk) = \Bbbk$, φ^*_γ is a $\Bbbk$-algebra automorphism of $\Bbbk[x, y, z]$. Write $\gamma(x) = \alpha_x x$, $\gamma(y) = \alpha_y y$, and $\gamma(z) = \alpha_z z$ with $\alpha_x, \alpha_y, \alpha_z \in \Bbbk^*$. Then $\gamma^{-1}(x) = \gamma^{-1}(\alpha_x^{-1})x$, so

$$\varphi^*_\gamma(x) = \gamma\big(\gamma^{-1}(\alpha_x^{-1})(x + g_1(y, z))\big) = x + \alpha_x^{-1}\gamma(g_1(y, z)),$$

and similarly $\varphi^*_\gamma(y) = y + \alpha_y^{-1}\gamma(g_2(z))$ and $\varphi^*_\gamma(z) = z + \alpha_z^{-1}\gamma(c)$. Since γ maps any monomial to a $\Bbbk^*$-multiple of itself, $\alpha_x^{-1}\gamma(g_1(y, z)) \in \Bbbk[y, z]$ has $\deg_w < w_x$ and $\alpha_y^{-1}\gamma(g_2(z)) \in \Bbbk[z]$

has $\deg_w < w_y$, so the coordinate functions $\varphi_\gamma^*(x), \varphi_\gamma^*(y), \varphi_\gamma^*(z)$ are of the form in (1). Since $\gamma^{-1}(f_0) = \gamma^{-1}(a_\gamma^{-1})f_0$ and $\gamma(F) = a_\gamma F$,

$$\varphi_\gamma^*(f_0) = \gamma\big(\gamma^{-1}(a_\gamma^{-1})F\big) = a_\gamma^{-1}a_\gamma F = F.$$

By the uniqueness part of (1), $\varphi_\gamma^* = \varphi^*$, that is, $\gamma \circ \varphi^* = \varphi^* \circ \gamma$.

We prove (4). Let $X \subset \mathbb{P}(1, w)$ be the closure of V, let $\Delta_\infty := X \backslash V$, and let $\overline{B}$ be the closure of B. By Lemma 7.3(1), $(X, \overline{B} + \Delta_\infty)$ is lc and $\operatorname{Nklt}(X, \overline{B} + \Delta_\infty) = \Delta_\infty \sqcup \{o\}$. Since $V = X \backslash \Delta_\infty$ is open and $\overline{B}|_V = B$, (V, B) is lc and $\operatorname{Nklt}(V, B) = \{o\}$. We have $\partial f_0/\partial x = 2x$, and $(\partial f_0/\partial y, \partial f_0/\partial z)$ is

$$(2yz,\ y^2 + (k-1)z^{k-2}), \quad (3y^2,\ 4z^3), \quad (3y^2 + z^3,\ 3yz^2), \quad \text{and} \quad (3y^2,\ 5z^4)$$

in cases (D_k), (E_6), (E_7), and (E_8) respectively. In each case the three partial derivatives vanish simultaneously only at $(0, 0, 0)$. Thus $\operatorname{Sing} V_0 = \{(0, 0, 0)\}$. Since $F = f_0(x', y', z')$, φ restricts to an isomorphism $V \to V_0$, so $\operatorname{Sing} V = \varphi^{-1}(0, 0, 0) = \{o\}$ by (1).

We prove (5). Since $R \supset \mathbb{Q}$, (2) with $R_0 = R$ implies $g_1 \in R[y, z]$, $g_2 \in R[z]$, and $c \in R$. The coordinate functions of φ^{-1} are

$$x - g_1\big(y - g_2(z - c), z - c\big), \quad y - g_2(z - c), \quad \text{and} \quad z - c,$$

which belong to $R[x, y, z]$. Assume that $o = (0, 0, 0)$. Since $\varphi(o) = (0, 0, 0)$ by (1), the constants $g_1(0, 0)$, $g_2(0)$, and c vanish, so x', y', z' and the coordinate functions of φ^{-1} belong to $(x, y, z)R[x, y, z]$. Thus φ^* and $(\varphi^*)^{-1}$ restrict to R-algebra automorphisms of $R[x, y, z]$ which map the ideal (x, y, z, u) into itself, so they extend to mutually inverse automorphisms of the (x, y, z, u)-adic completion $\Bbbk_1[[x, y, z, u]]$ fixing u. The Jacobian matrix $\partial(x', y', z', u)/\partial(x, y, z, u)$ has last row $(0, 0, 0, 1)$ and upper left block $\partial(x', y', z')/\partial(x, y, z)$, whose determinant is one by (1). Thus the Jacobian determinant is one. □

7.3. **Maximal weights and signed adjunction.**

Construction-Proposition 7.8. Let $\Bbbk \supset \mathbb{C}$ be an algebraically closed field, let $n \geq 2$ and $N \geq 1$ be integers, let $\mathbb{A}^n_\Bbbk$ have coordinates $x_0 := u, x_1, \dots, x_{n-1}$ and completion $\widehat{\mathcal{X}} := \operatorname{Spec} \Bbbk[[x_0, \dots, x_{n-1}]]$ at 0, and let G be a finite diagonal group acting faithfully on $\mathbb{A}^n_\Bbbk$. For $0 \leq \mu \leq n - 1$, let $H_\mu \subset G$ be the subgroup acting trivially on x_ν for any $\nu \neq \mu$, $d_\mu := |H_\mu|$ and $y_\mu := x_\mu^{d_\mu}$. Let $H \subset G$ be the subgroup generated by $H_0, \dots, H_{n-1}$, $G' := G/H$, $\varrho : \mathbb{A}^n_\Bbbk \to \mathbb{A}^n_\Bbbk$ the morphism given by $y_0, \dots, y_{n-1}$, and $\widehat{\mathcal{X}}_y := \operatorname{Spec} \Bbbk[[y_0, \dots, y_{n-1}]]$. Let β and $c_1, \dots, c_N$ be real numbers, let $Q_1, \dots, Q_N \in \Bbbk[[x_0, \dots, x_{n-1}]]$ be non-zero G-semi-invariant elements, let $\beta' := 1 - \frac{1-\beta}{d_0}$, and let

$$D := \beta \operatorname{div}(u) + \sum_{j=1}^{N} c_j \operatorname{div}(Q_j).$$

Then the following hold.

(1) $H = H_0 \times \cdots \times H_{n-1}$ is the subgroup generated by the pseudoreflections in G, ϱ is the quotient by H, and G' acts faithfully and diagonally in the coordinates $y_0, \dots, y_{n-1}$ and contains no pseudoreflection.

(2) For any non-zero G-semi-invariant $f \in \Bbbk[[x_0, \dots, x_{n-1}]]$, there exists a unique $\overline{f}$ in $\Bbbk[[y_0, \dots, y_{n-1}]]$ with $\varrho^*\overline{f} = f^{|H|}$, the element $\overline{f}$ is non-zero and G'-semi-invariant, and $\operatorname{mult}_0 \overline{f} \leq |H| \operatorname{mult}_0 f$.

(3) For any f in (2) and any positive integer m, the terms of $\overline{f}$ of degree less than m are polynomials in the coefficients of the terms of f of degree less than $m \max_\mu d_\mu$.

(4) For any f in (2) with $f \in \Bbbk[[u]][x_1, \dots, x_{n-1}]$, we have $\overline{f} \in \Bbbk[[y_0]][y_1, \dots, y_{n-1}]$.

(5) For the $\mathbb{R}$-divisor

$$D_y := \beta' \operatorname{div}(y_0) + \sum_{\mu=1}^{n-1} \Big(1 - \frac{1}{d_\mu}\Big) \operatorname{div}(y_\mu) + \sum_{j=1}^{N} \frac{c_j}{|H|} \operatorname{div}(\overline{Q}_j),$$

we have

$$\varrho^*(K_{\widehat{\mathcal{X}}_y} + D_y) = K_{\widehat{\mathcal{X}}} + D,$$

and the quotient value of $(\widehat{\mathcal{X}} \ni 0, D)$ and G equals that of $(\widehat{\mathcal{X}}_y \ni 0, D_y)$ and G'.

(6) For any prime divisor F over $\widehat{\mathcal{X}}$ with center the closed point, mapping under ϱ onto the prime divisor F' over $\widehat{\mathcal{X}}_y$ with ramification index r_F, and any $\varphi \in \Bbbk[[y_0, \dots, y_{n-1}]]$, we have

$$\frac{a(F, \widehat{\mathcal{X}}, D)}{e_F} = \frac{a(F', \widehat{\mathcal{X}}_y, D_y)}{e_{F'}},$$

$$a(F, \widehat{\mathcal{X}}, 0) = r_F a\Big(F', \widehat{\mathcal{X}}_y, \sum_{\mu=0}^{n-1} \Big(1 - \frac{1}{d_\mu}\Big) \operatorname{div}(y_\mu)\Big),$$

$$\text{and} \quad \operatorname{mult}_F(\varrho^*\varphi) = r_F \operatorname{mult}_{F'} \varphi,$$

where $e_{F'}$ is the order of the inertia group of F' in G'.

(7) Any prime divisor over $\widehat{\mathcal{X}}_y$ with center the closed point is the image of a prime divisor F as in (6).

Proof. (1) A diagonal element fixing a hyperplane pointwise has at most one eigenvalue different from 1, so the pseudoreflections in G are the non-trivial elements of the H_μ, and $H = \prod_\mu H_\mu$. The group H_μ is a finite subgroup of $\Bbbk^*$ acting on x_μ, hence acts through all d_μ-th roots of unity, so the H-invariant power series are $\Bbbk[[y_0, \dots, y_{n-1}]]$ and ϱ is the quotient by H. Let $g \in G$ act trivially on y_ν for any $\nu \neq \mu$. For $\nu \neq \mu$, the eigenvalue of g on x_ν is a d_ν-th root of unity, hence equals that of some $h_\nu \in H_\nu$, and $g \prod_{\nu \neq \mu} h_\nu^{-1} \in H_\mu$. Thus $g \in H$, which proves that G' acts faithfully and contains no pseudoreflection.

(2)–(4) The character of f restricted to H has order dividing $|H|$, so $f^{|H|}$ is H-invariant and $\overline{f}$ exists by (1), and $\overline{f}$ is unique since ϱ^* is injective. The element $\overline{f}$ is G'-semi-invariant with the $|H|$-th power of the character of f. The monomial $y_0^{b_0} \cdots y_{n-1}^{b_{n-1}}$ pulls back to a monomial of degree $\sum_\mu d_\mu b_\mu$, and

$$\sum_\mu b_\mu \leq \sum_\mu d_\mu b_\mu \leq \max_\mu d_\mu \sum_\mu b_\mu,$$

which implies the inequality in (2), and implies (3) since the coefficients of the terms of $f^{|H|}$ of degree less than $m \max_\mu d_\mu$ are polynomials in the coefficients of the terms of f of degree less than $m \max_\mu d_\mu$. An H-invariant element of $\Bbbk[[u]][x_1, \dots, x_{n-1}]$ has exponents of x_μ divisible by d_μ, hence belongs to $\Bbbk[[y_0]][y_1, \dots, y_{n-1}]$, which proves (4).

(5)–(7) Let

$$B_y := \sum_{\mu=0}^{n-1} \Big(1 - \frac{1}{d_\mu}\Big) \operatorname{div}(y_\mu).$$

By (1), ϱ is the quotient by H, and it is ramified exactly along the hyperplanes $(x_\mu = 0)$ with $d_\mu \geq 2$, with index d_μ. By Lemma 2.15(4) for H acting on $\mathbb{A}^n_\Bbbk$ with boundary 0, we have $(\mathbb{A}^n_\Bbbk, 0)/H = (\mathbb{A}^n_\Bbbk, B_y)$, where $\mathbb{A}^n_\Bbbk/H$ is identified with $\mathbb{A}^n_\Bbbk$ by ϱ, with coordinates $y_0, \dots, y_{n-1}$. Hence

$$K_{\widehat{\mathcal{X}}} = \varrho^*(K_{\widehat{\mathcal{X}}_y} + B_y) \quad \text{and} \quad \varrho^* \operatorname{div}(\overline{Q}_j) = |H| \operatorname{div}(Q_j).$$

Since $d_0\beta' - (d_0 - 1) = \beta$, we obtain

$$\varrho^*(K_{\widehat{\mathcal{X}}_y} + D_y) = K_{\widehat{\mathcal{X}}} + D \quad \text{and} \quad \varrho^*(D_y - B_y) = D.$$

We first work on $\mathbb{A}^n_{\Bbbk}$. Let F be a prime divisor over $\mathbb{A}^n_{\Bbbk}$ with center 0. By Lemma 2.15 for H, including (1) and (2), F maps onto a prime divisor F' over $\mathbb{A}^n_{\Bbbk}/H$ with ramification index $r_F = |I_H(F)|$, center $\varrho(0) = 0$, and $a(F, \mathbb{A}^n_{\Bbbk}, 0) = r_F a(F', \mathbb{A}^n_{\Bbbk}/H, B_y)$. By Lemma 2.15(3), any prime divisor over $\mathbb{A}^n_{\Bbbk}/H$ with center 0 is such an F', since a prime divisor over $\mathbb{A}^n_{\Bbbk}$ mapping onto it has center in $\varrho^{-1}(0) = \{0\}$. By Lemma 2.15(5), $\mathbb{A}^n_{\Bbbk}/G = (\mathbb{A}^n_{\Bbbk}/H)/G'$, and F' and F map onto a common prime divisor F'' over $\mathbb{A}^n_{\Bbbk}/G$ with ramification indices $e_{F'}$ and e_F by Lemma 2.15(1) for G' and for G. Since ramification indices multiply under composition, $e_F = r_F e_{F'}$.

By Lemma 2.12(1)(3) for $\mathbb{A}^n_{\Bbbk}$ with G and H and boundary 0, and for $\mathbb{A}^n_{\Bbbk}/H$ with G' and boundary B_y, the completion of $\mathbb{A}^n_{\Bbbk}/H$ at 0 is $\widehat{\mathcal{X}}_y$, and the prime divisors over $\widehat{\mathcal{X}}$ and $\widehat{\mathcal{X}}_y$ with center the closed point correspond to the prime divisors over $\mathbb{A}^n_{\Bbbk}$ and $\mathbb{A}^n_{\Bbbk}/H$ with center 0, with equal log discrepancies, equal inertia groups, and compatibly with images under ϱ and ramification indices. Thus the second equality in (6), the equality $e_F = r_F e_{F'}$, and (7) hold over $\widehat{\mathcal{X}}$ and $\widehat{\mathcal{X}}_y$, and the third equality in (6) holds for polynomials φ. For a non-zero power series φ and a positive integer T, let $\varphi^{<T}$ be the sum of the terms of φ of total degree less than T. Since $\operatorname{mult}_{F'} y_\mu \geq 1$ and $\operatorname{mult}_F x_\mu \geq 1$ for any μ, we have $\operatorname{mult}_{F'}(\varphi - \varphi^{<T}) \geq T$ and $\operatorname{mult}_F \varrho^*(\varphi - \varphi^{<T}) \geq T$. Taking $T > r_F \operatorname{mult}_{F'} \varphi$, we get $\operatorname{mult}_{F'} \varphi^{<T} = \operatorname{mult}_{F'} \varphi$ and $\operatorname{mult}_F \varrho^* \varphi^{<T} = r_F \operatorname{mult}_{F'} \varphi < T$, so $\operatorname{mult}_F(\varrho^*\varphi) = r_F \operatorname{mult}_{F'} \varphi$. Applying the third equality in (6) to y_0 and the $\overline{Q}_j$ and using $\varrho^*(D_y - B_y) = D$, we obtain

$$a(F, \widehat{\mathcal{X}}, D) = a(F, \widehat{\mathcal{X}}, 0) - \operatorname{mult}_F D = r_F\big(a(F', \widehat{\mathcal{X}}_y, B_y) - \operatorname{mult}_{F'}(D_y - B_y)\big) = r_F a(F', \widehat{\mathcal{X}}_y, D_y),$$

and dividing by $e_F = r_F e_{F'}$ gives the first equality in (6). Taking the infimum over F in (6), which by (7) gives the infimum over F', proves the statement on quotient values in (5). □

Let $\Bbbk \supset \mathbb{C}$ be an algebraically closed field, let $n \geq 2$, let $\mathbb{A}^n_{\Bbbk}$ have coordinates $u, x_1, \dots, x_{n-1}$, with $x_0 := u$, and completion $\widehat{\mathcal{X}} := \operatorname{Spec} \Bbbk[[u, x_1, \dots, x_{n-1}]]$ at 0, and let $G \subset GL_n(\mathbb{C})$ be a finite group of diagonal matrices acting on $\mathbb{A}^n_{\Bbbk}$. For $g \in G$ with eigenvalues $\exp(2\pi\sqrt{-1}\alpha_\mu)$ on x_μ, $0 \leq \alpha_\mu < 1$, a *g-twisted arc* at the image of 0 over a field $\mathbb{K} \supset \Bbbk$ is a tuple $\gamma = (x_\mu(t))_{0 \leq \mu \leq n-1}$ with

$$x_\mu(t) = t^{\alpha_\mu} \sum_{r \geq 0} X_{\mu,r} t^r, \quad X_{\mu,r} \in \mathbb{K}, \quad \text{and} \quad X_{\mu,0} = 0 \text{ if } \alpha_\mu = 0$$

[NS26, Proposition 3.10]. The g-twisted arcs at the image of 0 form the infinite affine space $\mathcal{A}_g$ with coordinates $X_{\mu,r}$. A *cylinder* is a subset $C = \pi_L^{-1}(S)$, where π_L is the projection to the affine space $\mathcal{A}_{g,L}$ of all tuples $(X_{\mu,r})_{0 \leq \mu \leq n-1, r < L}$, of dimension nL, in which $\pi_L(\mathcal{A}_g)$ is the linear subspace defined by $X_{\mu,0} = 0$ for the μ with $\alpha_\mu = 0$, and $S \subset \pi_L(\mathcal{A}_g)$ is constructible, and $\operatorname{codim} C := nL - \dim S$, with $\operatorname{codim} \emptyset = +\infty$. Since $g^{|G|} = 1$, we have $\alpha_\mu \in \frac{1}{|G|}\mathbb{Z}$, and since $X_{\mu,0} = 0$ if $\alpha_\mu = 0$,

$$x_\mu(t) \in t^{1/|G|}\mathbb{K}[[t^{1/|G|}]] \quad \text{for any } 0 \leq \mu \leq n-1 \text{ and any } \gamma \in \mathcal{A}_g. \tag{7.16}$$

For a non-zero G-semi-invariant $f \in \Bbbk[[u, x_1, \dots, x_{n-1}]]$, the *order* $\operatorname{mult}_\gamma f$ of f along γ is the smallest exponent of t in $f(\gamma)$, which belongs to $(\theta + \mathbb{N}) \cup \{+\infty\}$ for some $\theta \in [0, 1)$ determined by g and the character of f. We call $\operatorname{mult}_\gamma u$ the *u-order* of γ. For a positive integer T, we let $f^{<T}$ be the sum of the terms of f of total degree less than T, which is G-semi-invariant and non-zero for $T > \operatorname{mult}_0 f$. By (7.16),

$$\operatorname{mult}_\gamma(f - f^{<T}) \geq \frac{T}{|G|} \quad \text{and} \quad \operatorname{mult}_E(f - f^{<T}) \geq T \tag{7.17}$$

for any $\gamma \in \mathcal{A}_g$ and any prime divisor E over $\widehat{\mathcal{X}}$ with center the closed point, the second because $\operatorname{mult}_E x_\mu \geq 1$ for any μ. By Lemma 2.12(3), we regard a prime divisor E over $\mathbb{A}^n_{\Bbbk}$ with center 0 as a prime divisor over $\widehat{\mathcal{X}}$ with center the closed point (unique by [dFEM11, Lemma 2.21 and Remark A.15]), which defines $\operatorname{mult}_E f$ for $f \in \Bbbk[[u, x_1, \dots, x_{n-1}]]$.

Lemma 7.9. *Let $\Bbbk \supset \mathbb{C}$ be an algebraically closed field, let $n \geq 2$, let $\mathbb{A}^n_\Bbbk$ have coordinates $x_0 := u, x_1, \dots, x_{n-1}$, and let $G \subset GL_n(\mathbb{C})$ be a finite group of diagonal matrices acting on $\mathbb{A}^n_\Bbbk$ and containing no pseudoreflection, with the spaces $\mathcal{A}_g$ of g-twisted arcs at the image of 0, their cylinders, the codimensions of cylinders, and the orders mult_γ along twisted arcs γ as defined above. For $g \in G$ with eigenvalues $\exp(2\pi\sqrt{-1}\alpha_\mu)$ on x_μ, where $0 \leq \alpha_\mu < 1$, let $\mathrm{age}(g) := \alpha_0 + \cdots + \alpha_{n-1}$. Then the following hold.*

(1) For any prime divisor E over $\mathbb{A}^n_\Bbbk$ with center 0, there exist $g \in G$ and an irreducible cylinder $C_E \subset \mathcal{A}_g$ such that

$$\operatorname{codim} C_E + \mathrm{age}(g) = \frac{a(E, \mathbb{A}^n_\Bbbk, 0)}{e_E},$$

and the generic arc γ of C_E satisfies $\mathrm{mult}_\gamma f = \mathrm{mult}_E f / e_E$ for any non-zero G-semi-invariant $f \in \Bbbk[[u, x_1, \dots, x_{n-1}]]$.

(2) For any $g \in G$ and any irreducible cylinder $C' \subset \mathcal{A}_g$, there exist a prime divisor E over $\mathbb{A}^n_\Bbbk$ with center 0 and a positive integer q such that

$$\operatorname{codim} C' + \mathrm{age}(g) \geq q\frac{a(E, \mathbb{A}^n_\Bbbk, 0)}{e_E},$$

and the generic arc γ of C' satisfies $\mathrm{mult}_\gamma u = q\,\mathrm{mult}_E u / e_E$ and $\mathrm{mult}_E f \geq e_E \kappa / q$ for any non-zero G-semi-invariant $f \in \Bbbk[[u, x_1, \dots, x_{n-1}]]$ and any real number $\kappa \leq \mathrm{mult}_\gamma f$.

Proof. Let $\varpi : \mathbb{A}^n_\Bbbk \to X := \mathbb{A}^n_\Bbbk / G$ be the quotient and $x := \varpi(0)$. Since G contains no pseudoreflection, ϖ is étale in codimension 1, and for a prime divisor E over $\mathbb{A}^n_\Bbbk$ with center 0 and with image E_X over X,

$$a(E, \mathbb{A}^n_\Bbbk, 0) = e_E a(E_X, X, 0) \quad \text{and} \quad \mathrm{mult}_E(\varpi^*\varphi) = e_E\,\mathrm{mult}_{E_X}\varphi \text{ for any } \varphi \in \mathcal{O}_{X,x} \tag{7.18}$$

(Lemma 2.15(1)(2)(4)). Any prime divisor over X with center x is such an E_X (Lemma 2.15(3)).

The field $\Bbbk$ is uncountable, algebraically closed and of characteristic zero, so [EM09], [NS26], and [dFEI08] hold over $\Bbbk$. Let m be a positive integer such that mK_X is Cartier and $\mathfrak{n}_{m,X}$ the ideal of the m-th Nash subscheme of X [EM09, Section 7]. For an arc γ' of X through x and $\varphi \in \mathcal{O}_{X,x}$, let $\mathrm{mult}_{\gamma'}\varphi$ be the smallest exponent of t in $\varphi(\gamma')$, and let $\mathrm{mult}_{\gamma'}\mathfrak{n}_{m,X}$ be the minimum of $\mathrm{mult}_{\gamma'}\varphi$ over $\varphi \in \mathfrak{n}_{m,X}$. Any $g \in G$ defines a morphism ψ_g from $\mathcal{A}_g$ to the arcs of X through x such that $\mathrm{mult}_{\psi_g(\gamma)}\varphi = \mathrm{mult}_\gamma(\varpi^*\varphi)$ for $\varphi \in \mathcal{O}_{X,x}$ [NS26, Proposition 3.10]. Let $Z \subsetneq X$ be the minimal closed subset outside which ϖ is étale. Since G is abelian, an arc of X through x that does not factor through Z belongs to $\psi_g(\mathcal{A}_g)$ for some g [NS26, Proposition 3.10(3)]. Let $C \subset \mathcal{A}_g$ be an irreducible cylinder on which $\mathrm{mult}_{\psi_g(\gamma)}\mathfrak{n}_{m,X}$ is a constant ℓ and the other contact orders of [NS26, Lemma 5.2] are constant. Then $\psi_g(C) = \psi_g(G \cdot C)$ is a cylinder and

$$\operatorname{codim}\psi_g(C) - \frac{\ell}{m} = \operatorname{codim} C + \mathrm{age}(g), \tag{7.19}$$

by [NS26, Lemma 5.2, Propositions 2.15 and 2.16] for the G-saturation $G \cdot C$, which is G-invariant with $\operatorname{codim} G \cdot C = \operatorname{codim} C$. In (7.19) the Jacobian and character terms of [NS26, Lemma 5.2] vanish, since X is the quotient of $\mathbb{A}^n_\Bbbk$ itself.

Let C_X be an irreducible cylinder of arcs of X through x on which $\mathrm{mult}_{\gamma'}\mathfrak{n}_{m,X}$ is a constant ℓ and along whose arcs the Jacobian ideal $\mathrm{Jac}_X := \mathrm{Fitt}^n(\Omega_X)$ of X [NS26, Definition 2.1(1)] has finite order, and let γ_{C_X} be its generic arc. Then C_X is fat by [dFEI08, Corollary 3.7]. Hence there exist a prime divisor E_X over X with center x and a positive integer q such that $\mathrm{mult}_{\gamma_{C_X}}\varphi = q\,\mathrm{mult}_{E_X}\varphi$ for any $\varphi \in \mathcal{O}_{X,x}$ [dFEI08, Proposition 3.10, Proposition 2.12, Remark 2.11]. Let $h : X' \to X$ be a resolution factoring through the Nash blow-up of X on which E_X is a divisor, and let $\widehat{K}_{X'/X}$ be the relative Mather canonical divisor of h [dFEI08, Definition 1.6]. Then $\operatorname{codim} C_X \geq q(\mathrm{mult}_{E_X}\widehat{K}_{X'/X} + 1)$ [dFEI08, Theorem 3.9]. Comparing the

images of $(h^*\Omega^n_X)^{\otimes m}$ in $(\Omega^n_{X'})^{\otimes m}$ [dFEI08, Proposition 1.7] and of $(\Omega^n_X)^{\otimes m}$ in $\mathcal{O}_X(mK_X)$ [EM09, Section 7] gives

$$\mathfrak{n}_{m,X}\mathcal{O}_{X'} = \mathcal{O}_{X'}(-m\widehat{K}_{X'/X} + mK_{X'/X}).$$

Since

$$\ell = \operatorname{mult}_{\gamma_{C_X}} \mathfrak{n}_{m,X} = q \operatorname{mult}_{E_X} \mathfrak{n}_{m,X},$$

we have $q \operatorname{mult}_{E_X} \widehat{K}_{X'/X} = q \operatorname{mult}_{E_X} K_{X'/X} + \ell/m$, so $\operatorname{codim} C_X - \ell/m \geq qa(E_X, X, 0)$. For any E_X, the cylinder C_{E_X} of arcs whose lifts to a log resolution of $(X, \operatorname{Jac}_X \cdot \mathfrak{n}_{m,X})$ on which E_X is a divisor have order exactly 1 along E_X and 0 along the other exceptional divisors attains equality with $q = 1$ [dFEI08, Theorem 3.9], since C_{E_X} is a dense open subset of the maximal divisorial set of E_X with multiplicity 1 and $\mathfrak{n}_{m,X}$ and Jac_X have the constant orders $\ell = \operatorname{mult}_{E_X} \mathfrak{n}_{m,X}$ and $\operatorname{mult}_{E_X} \operatorname{Jac}_X$ along its arcs.

Given E, take C_{E_X}. Since C_{E_X} is fat, its generic arc does not factor through Z [dFEI08, Definition 2.2], so there exists $g \in G$ with $\psi_g(\mathcal{A}_g)$ containing the generic arc of C_{E_X} (after a finite extension of the field of definition of the generic arc). Take an irreducible component C_E of $\psi_g^{-1}(C_{E_X})$ dominating C_{E_X}, and replace C_E by its dense open subset on which the other contact orders of [NS26, Lemma 5.2] are constant, which are finite at the generic arc by [NS26, Lemma 5.1(2)]. Given C', replace C' by its dense open subset on which $\operatorname{mult}_{\psi_g(\gamma)} \mathfrak{n}_{m,X}$ and the other contact orders of [NS26, Lemma 5.2], among them that of Jac_X, are constant, which are finite at the generic arc by [NS26, Lemma 5.1(2)], take $C_X := \psi_g(C')$, let E_X and q be the prime divisor and the integer given by the previous paragraph for C_X, and let E be a prime divisor over $\mathbb{A}^n_{\Bbbk}$ with image E_X. By (7.18) and (7.19), the codimension statements of (1) and (2) hold, and $\operatorname{mult}_\gamma f = \operatorname{mult}_E f/e_E$ in (1), resp. $\operatorname{mult}_\gamma f = q \operatorname{mult}_E f/e_E$ in (2), for any semi-invariant polynomial f, since $f^{|G|} = \varpi^*\varphi$ for some $\varphi \in \mathcal{O}_{X,x}$.

For a semi-invariant power series f in (1), let $T > |G| \operatorname{mult}_E f$. Then $\operatorname{mult}_E f^{<T} = \operatorname{mult}_E f$ and $\operatorname{mult}_\gamma f^{<T} = \operatorname{mult}_E f/e_E < T/|G|$ by (7.17), so $\operatorname{mult}_\gamma f = \operatorname{mult}_E f/e_E$. For f and κ in (2), let $T > |G|\kappa$. By (7.17), $\operatorname{mult}_\gamma f^{<T} \geq \min\{\kappa, T/|G|\} = \kappa$, so $\operatorname{mult}_E f^{<T} \geq e_E\kappa/q$, and $\operatorname{mult}_E(f - f^{<T}) \geq T > e_E\kappa/q$, so $\operatorname{mult}_E f \geq e_E\kappa/q$. □

Lemma 7.10. *Let $\Bbbk \supset \mathbb{C}$ be an algebraically closed field, let $n \geq 2$ and $N \geq 1$ be integers, let $\mathbb{A}^n_{\Bbbk}$ have coordinates $(u, x_1, \dots, x_{n-1})$ and completion $\widehat{\mathcal{X}} := \operatorname{Spec} \Bbbk[[u, x_1, \dots, x_{n-1}]]$ at 0, let $G \subset GL_n(\mathbb{C})$ be a finite group of diagonal matrices acting on $\mathbb{A}^n_{\Bbbk}$ and on $\mathbb{A}^n_{\mathbb{C}}$, let β be a real number, and let $c_1, \dots, c_N \geq 0$ be real numbers.*

(1) Let $Q_1, \dots, Q_N \in \Bbbk[[u, x_1, \dots, x_{n-1}]]$ be non-zero G-semi-invariant elements, and let

$$a_Q := \operatorname{mld}\Big(\Big(\widehat{\mathcal{X}} \ni 0, \beta \operatorname{div}(u) + \sum_{j=1}^N c_j \operatorname{div}(Q_j)\Big)/G\Big)$$

be the quotient value (Definition 2.2). Assume that G contains no pseudoreflection, and let I be a positive integer divisible by $|G|$. For $g \in G$ with eigenvalues $\exp(2\pi\sqrt{-1}\alpha_0)$ on u and $\exp(2\pi\sqrt{-1}\alpha_\mu)$ on x_μ for $1 \leq \mu \leq n-1$, where $0 \leq \alpha_0, \dots, \alpha_{n-1} < 1$, let $\operatorname{age}(g) := \alpha_0 + \cdots + \alpha_{n-1}$, and for $k = (k_0, \dots, k_N) \in \mathbb{N}^{N+1}$ let $C(g,k) \subset \mathcal{A}_g$ be the cylinder

$$C(g,k) := \{\gamma \in \mathcal{A}_g \mid \operatorname{mult}_\gamma u = \alpha_0 + k_0 \text{ and } \operatorname{mult}_\gamma Q_j^I \geq k_j \text{ for } 1 \leq j \leq N\},$$

with $\mathcal{A}_g$ and $\operatorname{mult}_\gamma$ as defined before Lemma 7.9. Then

$$a_Q = \inf_{g \in G, k \in \mathbb{N}^{N+1}} \Big(\operatorname{codim} C(g,k) + \operatorname{age}(g) - \beta(\alpha_0 + k_0) - \sum_{j=1}^N \frac{c_j}{I} k_j\Big).$$

(2) Let $\mathcal{I} \subset \mathbb{N}^+$ be an infinite subset. For $i \in \mathcal{I}$ let $(Q_{i,1}, \dots, Q_{i,N})$ be a tuple of non-zero G-semi-invariant elements of $\mathbb{C}[[u, x_1, \dots, x_{n-1}]]$, and assume that the tuples $(Q_{i,j})_j$ have a compatible generic limit $(Q_{*,1}, \dots, Q_{*,N})$ along $\mathcal{I}$ with coefficient field $\Bbbk$ (Definition 2.5).

For $i \in \mathcal{I}$ let F_i be a prime divisor over $\mathbb{A}^n_{\mathbb{C}}$ with center 0, and assume that $\operatorname{mult}_{F_i} u$, $\operatorname{mult}_{F_i} Q_{i,j}$ for $1 \le j \le N$ (Lemma 2.12(3)) and $a(F_i, \mathbb{A}^n_{\mathbb{C}}, 0)$ are bounded from above independently of i. Then

$$\operatorname{mld}\Big(\Big(\widehat{\mathcal{X}} \ni 0, \beta \operatorname{div}(u) + \sum_{j=1}^{N} c_j \operatorname{div}(Q_{*,j})\Big)/G\Big) \le \liminf_{i \in \mathcal{I}} \frac{a(F_i, \mathbb{A}^n_{\mathbb{C}}, 0) - \beta \operatorname{mult}_{F_i} u - \sum_{j=1}^{N} c_j \operatorname{mult}_{F_i} Q_{i,j}}{e_{F_i}}.$$

Proof. Let

$$D := \beta \operatorname{div}(u) + \sum_j c_j \operatorname{div}(Q_j). \tag{7.20}$$

By Lemma 2.12(3), a prime divisor E over $\widehat{\mathcal{X}}$ with center the closed point corresponds to a prime divisor over $\mathbb{A}^n_{\Bbbk}$ with center 0, with $a(E, \widehat{\mathcal{X}}, 0)$ equal to $a(E, \mathbb{A}^n_{\Bbbk}, 0)$ and with inertia group in G equal to that of E over $\widehat{\mathcal{X}}$.

Proof of (1). For a tuple $Q = (Q_j)_j$ we write $C_Q(g,k)$ for $C(g,k)$ and

$$\Phi_Q(g,k) := \operatorname{codim} C_Q(g,k) + \operatorname{age}(g) - \beta(\alpha_0 + k_0) - \sum_{j=1}^{N} \frac{c_j}{I} k_j.$$

Since $\operatorname{mult}_\gamma Q_j^I \ge k_j$ if and only if $\operatorname{mult}_\gamma Q_j \ge k_j/I$, (7.17) implies

$$C_Q(g,k) = C_{Q^{<T}}(g,k) \quad \text{if } T > \max_j \max\Big\{\frac{|G|}{I} k_j, \operatorname{mult}_0 Q_j\Big\}, \quad \text{where } Q^{<T} := (Q_j^{<T})_j. \tag{7.21}$$

In particular $C_Q(g,k)$ is a cylinder, defined by polynomial conditions on finitely many coordinates $X_{\mu,r}$.

Step 1: $a_Q \ge \inf_{g,k} \Phi_Q(g,k)$. Let E be a prime divisor over $\widehat{\mathcal{X}}$ with center the closed point, and let $g \in G$ and $C_E \subset \mathcal{A}_g$ be the element and the irreducible cylinder given by Lemma 7.9(1) for E. Let $k_0 := \operatorname{mult}_E u/e_E - \alpha_0$ and $k_j := I \operatorname{mult}_E Q_j/e_E$ for $1 \le j \le N$. Here $k_0, \dots, k_N$ are non-negative integers, since $\operatorname{mult}_\gamma u \in \alpha_0 + \mathbb{N}$ and $\operatorname{mult}_\gamma Q_j^I \in \mathbb{N}$ for the generic arc γ of C_E. Thus $\gamma \in C_Q(g,k)$, so $C_Q(g,k)$ contains a non-empty open subset of C_E, and $\operatorname{codim} C_Q(g,k) \le \operatorname{codim} C_E$. By Lemma 7.9(1),

$$\Phi_Q(g,k) \le \frac{a(E, \widehat{\mathcal{X}}, D)}{e_E}.$$

Taking the infimum over E proves the inequality.

Step 2: $a_Q \le \Phi_Q(g,k)$ *for any* (g,k). If $C_Q(g,k) = \emptyset$, then $\Phi_Q(g,k) = +\infty$. If $a_Q < 0$, then $a_Q = -\infty$, since $n \ge 2$ and $e_E \le |G|$ for any E. Hence we may assume that $a_Q \ge 0$ and $C_Q(g,k) \ne \emptyset$. Let C' be an irreducible component of $C_Q(g,k)$ with $\operatorname{codim} C' = \operatorname{codim} C_Q(g,k)$, which exists by (7.21), and let E and q be the prime divisor and the integer given by Lemma 7.9(2) for C'. Since $\operatorname{mult}_\gamma u = \alpha_0 + k_0$ and $\operatorname{mult}_\gamma Q_j \ge k_j/I$ for the generic arc γ of C', Lemma 7.9(2) implies

$$\operatorname{mult}_E u = \frac{e_E}{q}(\alpha_0 + k_0) \quad \text{and} \quad \operatorname{mult}_E Q_j \ge \frac{e_E}{q} \cdot \frac{k_j}{I}.$$

Since $c_j \ge 0$ for any j, we obtain

$$q \cdot \frac{a(E, \widehat{\mathcal{X}}, D)}{e_E} \le \operatorname{codim} C' + \operatorname{age}(g) - \beta(\alpha_0 + k_0) - \sum_j \frac{c_j}{I} k_j = \Phi_Q(g,k).$$

Since $a(E, \widehat{\mathcal{X}}, D)/e_E \ge a_Q \ge 0$ and $q \ge 1$, we obtain $a_Q \le \Phi_Q(g,k)$. **Steps 1 and 2** prove (1).

Proof of (2) when G contains no pseudoreflection. Let $I := |G|$. Let s be the right-hand side of (2). The non-negative integers $\mathrm{mult}_{F_i} u$, $\mathrm{mult}_{F_i} Q_{i,j}$, $a(F_i, \mathbb{A}^n_{\mathbb{C}}, 0)$, and e_{F_i} take finitely many values, hence so does the quotient under $\liminf$ in (2), and there exists an infinite subset $\mathcal{I}' \subset \mathcal{I}$ such that for any $i \in \mathcal{I}'$,

$$\mathrm{mult}_{F_i} u = \lambda_0, \quad \mathrm{mult}_{F_i} Q_{i,j} = \lambda_j \ (1 \le j \le N), \quad a(F_i, \mathbb{A}^n_{\mathbb{C}}, 0) = \lambda', \quad e_{F_i} = e,$$
$$\text{and} \quad s = \frac{\lambda' - \beta\lambda_0 - \sum_j c_j\lambda_j}{e}.$$

By Lemma 2.6(2), $(Q_{*,j})_j$ is a compatible generic limit along $\mathcal{I}'$. For $i \in \mathcal{I}'$, let $g_i \in G$ and $C_{F_i} \subset \mathcal{A}_{g_i}$ be the element and the irreducible cylinder given by Lemma 7.9(1) over $\mathbb{C}$ for F_i. Possibly replacing $\mathcal{I}'$ with an infinite subset, we may assume that $g_i = g$ for any $i \in \mathcal{I}'$. Let $k_0 := \lambda_0/e - \alpha_0$ and $k_j := I\lambda_j/e$ for $1 \le j \le N$, and $Q_i := (Q_{i,j})_j$. As in **Step 1**, Lemma 7.9(1) implies that the generic arc of C_{F_i} belongs to $C_{Q_i}(g,k)$, so

$$\operatorname{codim} C_{Q_i}(g,k) \le \operatorname{codim} C_{F_i} = \frac{\lambda'}{e} - \operatorname{age}(g) =: \delta \quad \text{for any } i \in \mathcal{I}'. \tag{7.22}$$

Let T be an integer with $T > \frac{|G|}{I}\max_j k_j$ and $T > \mathrm{mult}_0 Q_{i,j}$ for any $i \in \mathcal{I}'$ and any j, and let $L > \max\{k_0, \max_j k_j/I\}$ be an integer. For $\vartheta = (\vartheta_j)_j$ in the space $\mathcal{J}_T$ of T-jets of Definition 2.5, defined over $\mathbb{C}$, the conditions defining the cylinder $C_\vartheta(g,k)$ of the tuple of polynomials ϑ involve only the coordinates $X_{\mu,r}$ with $r < L$ and are polynomial in the $X_{\mu,r}$ and in the coefficients of ϑ, together with the open condition $X_{0,k_0} \ne 0$. Thus they define a constructible subset $\mathcal{S} \subset \mathcal{J}_T \times \mathcal{A}_{g,L}$ whose fiber $\mathcal{S}_\vartheta$ over a point ϑ is $\pi_L(C_\vartheta(g,k))$. By Chevalley's theorem and the upper semicontinuity of the dimension of fibers, the set

$$\mathcal{W} := \{\vartheta \in \mathcal{J}_T \mid \dim \mathcal{S}_\vartheta \ge \dim \mathcal{A}_{g,L} - \delta\}$$

is constructible. By (7.21) and (7.22), $\mathcal{W}$ contains the T-jet of Q_i for any $i \in \mathcal{I}'$. Let η be the generic point of Z_T. If $\eta \notin \mathcal{W}$, then $\mathcal{W} \cap Z_T$ is contained in a proper closed subset $Z' \subsetneq Z_T$, and the non-empty open subset $Z_T \backslash Z'$ contains the T-jets of Q_i for all but finitely many $i \in \mathcal{I}'$, a contradiction. Thus $\eta \in \mathcal{W}$. The T-jet ϑ_* of $Q_* := (Q_{*,j})_j$ is a $\Bbbk$-point over η, so

$$\dim \mathcal{S}_{\vartheta_*} = \dim \mathcal{S}_\eta \ge \dim \mathcal{A}_{g,L} - \delta,$$

and each entry of ϑ_* is non-zero, since ϑ_* maps to the generic point of Z_T and the T-jets of the Q_i in Z_T have non-zero entries. Thus by (7.21) over $\Bbbk$,

$$\operatorname{codim} C_{Q_*}(g,k) = \operatorname{codim} C_{\vartheta_*}(g,k) \le \delta.$$

Since $\alpha_0 + k_0 = \lambda_0/e$ and $k_j/I = \lambda_j/e$, (1) over $\Bbbk$ implies

$$\operatorname{mld}\Big(\Big(\widehat{\mathcal{X}} \ni 0, \beta \operatorname{div}(u) + \sum_j c_j \operatorname{div}(Q_{*,j})\Big)/G\Big) \le \Phi_{Q_*}(g,k) \le \delta + \operatorname{age}(g) - \frac{\beta\lambda_0 + \sum_j c_j\lambda_j}{e} = s.$$

Proof of (2) in general. Let H, d_μ, y_μ, ϱ, G', and $\overline{f}$ be as in Construction-Proposition 7.8 for $\Bbbk$ and for $\mathbb{C}$, and for $i \in \mathcal{I}$ let F_i' be the image of F_i over $\mathbb{A}^n_{\mathbb{C}}/H$, where $\mathbb{A}^n_{\mathbb{C}}/H$ is identified with $\mathbb{A}^n_{\mathbb{C}}$ by ϱ, with ramification index $r_i := r_{F_i} \le |H|$. By Lemma 2.12(3), we regard F_i and F_i' as prime divisors over the completions. By Construction-Proposition 7.8(5) and (6), the quotient under $\liminf$ in (2) for F_i equals the corresponding quotient for F_i', with the group G', the coefficient β' of $\operatorname{div}(y_0)$, the functions $\overline{Q}_{i,j}$ with coefficients $c_j/|H|$, and the functions y_μ, $\mu \ge 1$, with coefficients $1 - \frac{1}{d_\mu} \ge 0$. The left-hand side of (2) equals the quotient value with G' of

$$\beta' \operatorname{div}(y_0) + \sum_{\mu=1}^{n-1}\Big(1 - \frac{1}{d_\mu}\Big)\operatorname{div}(y_\mu) + \sum_j \frac{c_j}{|H|}\operatorname{div}(\overline{Q}_{*,j}).$$

The hypotheses of (2) hold for G', $(\overline{Q}_{i,j})_j$, and F_i', as follows. Since $(\mathbb{A}^n_{\mathbb{C}}, \sum_\mu \operatorname{div}(x_\mu))$ is lc, $\operatorname{mult}_{F_i} x_\mu \le a(F_i, \mathbb{A}^n_{\mathbb{C}}, 0)$, so by Construction-Proposition 7.8(6),

$$\operatorname{mult}_{F_i'} y_\mu = \frac{d_\mu \operatorname{mult}_{F_i} x_\mu}{r_i}, \quad \operatorname{mult}_{F_i'} \overline{Q}_{i,j} = \frac{|H| \operatorname{mult}_{F_i} Q_{i,j}}{r_i},$$
$$\text{and} \quad a(F_i', \mathbb{A}^n_{\mathbb{C}}, 0) = \frac{a(F_i, \mathbb{A}^n_{\mathbb{C}}, 0)}{r_i} + \sum_\mu \Big(1 - \frac{1}{d_\mu}\Big) \operatorname{mult}_{F_i'} y_\mu$$

are bounded independently of i. For a positive integer d, let $d' := d \max_\mu d_\mu$. By Construction-Proposition 7.8(2) and (3), the d-jets of the tuples $(\overline{Q}_{i,j})_j$ are the images of the d'-jets of the tuples $(Q_{i,j})_j$ under a morphism of affine spaces defined over $\mathbb{C}$, and the orders at 0 stay bounded. The closure Z_d' of the image of $Z_{d'}$ is irreducible, the d-jet of $(\overline{Q}_{*,j})_j$ maps to the generic point of Z_d', and the preimage in $Z_{d'}$ of a non-empty open subset of Z_d' is a non-empty open subset of $Z_{d'}$. Thus $(\overline{Q}_{*,j})_j$ is a compatible generic limit of the tuples $(\overline{Q}_{i,j})_j$ along $\mathcal{I}$, and $(\overline{Q}_{*,1}, \dots, \overline{Q}_{*,N}, y_1, \dots, y_{n-1})$ is a compatible generic limit of the tuples $(\overline{Q}_{i,1}, \dots, \overline{Q}_{i,N}, y_1, \dots, y_{n-1})$ along $\mathcal{I}$. Since G' contains no pseudoreflection (Construction-Proposition 7.8(1)), the case of (2) proved above implies (2). □

Lemma 7.11. *Let $\Bbbk \supset \mathbb{C}$ be an algebraically closed field, let $n \ge 2$ and $N \ge 1$ be integers, let $\mathbb{A}^n_\Bbbk$ have coordinates $(u, x_1, \dots, x_{n-1})$ and completion $\widehat{\mathcal{X}} := \operatorname{Spec} \Bbbk[[u, x_1, \dots, x_{n-1}]]$ at 0, let G be a finite diagonal group acting faithfully on $\mathbb{A}^n_\Bbbk$, let β be a real number, let $Q_1, \dots, Q_N$ be non-zero G-semi-invariant elements of $\Bbbk[[u]][x_1, \dots, x_{n-1}]$, and let $c_1, \dots, c_N \ge 0$ be real numbers. Let $w_1, \dots, w_{n-1}$ be positive integers, let the weight of $u^{\nu_0} x_1^{\nu_1} \cdots x_{n-1}^{\nu_{n-1}}$ be $\sum_{\mu=1}^{n-1} w_\mu \nu_\mu$, and let P_j be the sum of the terms of Q_j of maximal weight for $1 \le j \le N$. Let*

$$a_Q := \operatorname{mld}\Big(\Big(\widehat{\mathcal{X}} \ni 0, \beta \operatorname{div}(u) + \sum_{j=1}^N c_j \operatorname{div}(Q_j)\Big)/G\Big)$$
$$\text{and} \quad a_P := \operatorname{mld}\Big(\Big(\widehat{\mathcal{X}} \ni 0, \beta \operatorname{div}(u) + \sum_{j=1}^N c_j \operatorname{div}(P_j)\Big)/G\Big)$$

be the quotient values (Definition 2.2). Then any prime divisor E over $\widehat{\mathcal{X}}$ with center the closed point satisfies

$$\frac{a(E, \widehat{\mathcal{X}}, 0) - \beta \operatorname{mult}_E u - \sum_j c_j \operatorname{mult}_E Q_j}{e_E} \ge a_P.$$

Equivalently, $a_P \le a_Q$, possibly $a_P = -\infty$.

Proof. By the definition of a_Q, the left-hand side is $\ge a_Q$, so we only need to prove $a_P \le a_Q$. Each P_j is non-zero and G-semi-invariant, since G acts diagonally and the weight depends only on the monomial.

***Step 1**: reduction to G without pseudoreflections.* Let H, d_μ, y_μ, ϱ, G', and $\overline{f}$ be as in Construction-Proposition 7.8. By Construction-Proposition 7.8(4), $\overline{Q}_j, \overline{P}_j \in \Bbbk[[y_0]][y_1, \dots, y_{n-1}]$. Let y_0 have weight 0 and y_μ weight $d_\mu w_\mu$ for $\mu \ge 1$, so that ϱ^* preserves weights. Since $\Bbbk[[u]][x_1, \dots, x_{n-1}]$ is a domain graded by the weight, the part of maximal weight of $Q_j^{|H|}$ is $P_j^{|H|}$, so the part of maximal weight of $\overline{Q}_j$ is $\overline{P}_j$. By Construction-Proposition 7.8(5), a_Q and a_P are the quotient values with G' of

$$\beta' \operatorname{div}(y_0) + \sum_{\mu=1}^{n-1} \Big(1 - \frac{1}{d_\mu}\Big) \operatorname{div}(y_\mu) + \sum_j \frac{c_j}{|H|} \operatorname{div}(\overline{Q}_j)$$
$$\text{and} \quad \beta' \operatorname{div}(y_0) + \sum_{\mu=1}^{n-1} \Big(1 - \frac{1}{d_\mu}\Big) \operatorname{div}(y_\mu) + \sum_j \frac{c_j}{|H|} \operatorname{div}(\overline{P}_j).$$

Since G' contains no pseudoreflection, replacing G, β, $(Q_j)_j$, $(P_j)_j$, $(c_j)_j$, the coordinates $(u, x_1, \dots, x_{n-1})$, the weights $(w_\mu)_\mu$, and N with G', β', $(\overline{Q}_j)_j \cup (y_\mu)_{\mu\ge1}$, $(\overline{P}_j)_j \cup (y_\mu)_{\mu\ge1}$, $(c_j/|H|)_j \cup (1-\frac{1}{d_\mu})_{\mu\ge1}$, $(y_0, \dots, y_{n-1})$, $(d_\mu w_\mu)_\mu$, and $N+n-1$, we may assume that G contains no pseudoreflection.

***Step 2**: cylinders.* Let $I := |G|$, $x_0 := u$, and $P := (P_j)_j$. For $g \in G$ and $k \in \mathbb{N}^{N+1}$, let $C_Q(g,k)$ and $C_P(g,k)$ be the cylinders $C(g,k)$ of Lemma 7.10(1) for $(Q_j)_j$ and for $(P_j)_j$, with $\mathcal{A}_g$, $\mathcal{A}_{g,L}$ and mult_γ as defined before Lemma 7.9. By (7.16), $\mathrm{mult}_\gamma x_\mu \ge \frac{1}{|G|}$ for any μ and any $\gamma \in \mathcal{A}_g$. By Lemma 7.10(1), a_Q and a_P are the infima over (g,k) of

$$\operatorname{codim} C_Q(g,k) + \operatorname{age}(g) - \beta(\alpha_0+k_0) - \sum_j \frac{c_j}{I}k_j$$

$$\text{and} \quad \operatorname{codim} C_P(g,k) + \operatorname{age}(g) - \beta(\alpha_0+k_0) - \sum_j \frac{c_j}{I}k_j.$$

Thus it suffices to show that for any $g \in G$ and any $k \in \mathbb{N}^{N+1}$,

$$\operatorname{codim} C_P(g,k) \le \operatorname{codim} C_Q(g,k). \tag{7.23}$$

Fix (g,k), and let $T > |G| \max_j k_j/I$ be an integer such that the terms of P_j of u-degree less than T are non-zero for any j. Let $Q_j^{u<T}$ and $P_j^{u<T}$ be the sums of the terms of Q_j and P_j of u-degree less than T, which are polynomials. Since $\mathrm{mult}_\gamma(Q_j - Q_j^{u<T}) \ge T/|G| > k_j/I$ for any $\gamma \in \mathcal{A}_g$, and since $\mathrm{mult}_\gamma Q_j^I \ge k_j$ if and only if $\mathrm{mult}_\gamma Q_j \ge k_j/I$, we have $C_Q(g,k) = C_{Q^{u<T}}(g,k)$, and likewise $C_P(g,k) = C_{P^{u<T}}(g,k)$. The weight does not depend on the u-degree, so $P_j^{u<T}$ is the part of maximal weight of $Q_j^{u<T}$. Possibly replacing Q_j and P_j with $Q_j^{u<T}$ and $P_j^{u<T}$, we may assume that Q_j and P_j are polynomials.

***Step 3**: jet ideals.* Let $L > \max\{k_0, \max_j k_j/I\}$ be an integer and

$$R := \Bbbk[X_{\mu,r}, Y \mid 0 \le \mu \le n-1,\ 0 \le r < L].$$

For a semi-invariant polynomial f and $\gamma \in \mathcal{A}_g$, write $f(\gamma) = \sum_\tau \varphi_{f,\tau} t^\tau$, where τ runs through $\theta + \mathbb{N}$ for some $\theta \in [0,1)$. For $\tau < L$, the coefficient $\varphi_{f,\tau}$ is a polynomial in the $X_{\mu,r}$ with $r < L$. Let $J_Q \subset R$ be the ideal generated by

(1) $X_{\mu,0}$ for any μ with $\alpha_\mu = 0$,
(2) $X_{0,r}$ for $r < k_0$, and $X_{0,k_0}Y - 1$,
(3) $\varphi_{Q_j,\tau}$ for $1 \le j \le N$ and $\tau < k_j/I$,

and let $J_P \subset R$ be the ideal generated by the generators of the first two kinds and by $\varphi_{P_j,\tau}$ for $1 \le j \le N$ and $\tau < k_j/I$. The projection forgetting Y maps $\operatorname{Spec} R/J_Q$ isomorphically onto the image of $C_Q(g,k)$ in $\mathcal{A}_{g,L}$, with inverse $Y = X_{0,k_0}^{-1}$, so $\operatorname{codim} C_Q(g,k) = \dim \mathcal{A}_{g,L} - \dim R/J_Q$, where $\dim R/R = -\infty$. Likewise, $\operatorname{codim} C_P(g,k) = \dim \mathcal{A}_{g,L} - \dim R/J_P$.

Let λ be the weight on R with $\lambda(X_{0,r}) = \lambda(Y) = 0$ and $\lambda(X_{\mu,r}) = w_\mu$ for $\mu \ge 1$, which is non-negative and integral. Substituting γ into a monomial of weight W, each coefficient $\varphi_{f,\tau}$ is λ-homogeneous of degree W. Let W_j be the maximal weight of the terms of Q_j. Then $\varphi_{P_j,\tau}$ is λ-homogeneous of degree W_j, and $\varphi_{Q_j-P_j,\tau}$ has λ-degree less than W_j, so $\varphi_{P_j,\tau}$ is zero or the part of maximal λ-degree of $\varphi_{Q_j,\tau}$. The generators of the first two kinds are λ-homogeneous, $X_{0,k_0}Y - 1$ of degree 0. Let $\operatorname{top}_\lambda(J_Q)$ be the ideal generated by the parts of maximal λ-degree of the non-zero elements of J_Q. Thus

$$J_P \subset \operatorname{top}_\lambda(J_Q). \tag{7.24}$$

***Step 4**: dimensions.* We show $\dim R/\operatorname{top}_\lambda(J_Q) \ge \dim R/J_Q$. We may assume $J_Q \ne R$. Let $\widetilde{J} \subset R[s]$ be the ideal generated by $s^{\deg_\lambda f} f(s^{-\lambda}\xi)$ for $f \in J_Q$, where ξ denotes the variables of R and $\deg_\lambda f$ the maximal λ-degree of the terms of f. By [Eis95, Theorem 15.17], $R[s]/\widetilde{J}$ is flat over $\Bbbk[s]$, the fiber over $s=0$ is $\operatorname{Spec} R/\operatorname{top}_\lambda(J_Q)$ and $\operatorname{Spec} R[s,s^{-1}]/\widetilde{J} \cong \operatorname{Spec} R/J_Q \times \mathbb{G}_m$. The

action $c\cdot(\xi,s)=(c^\lambda\xi,cs)$ of $\mathbb{G}_m$ preserves $V(\widetilde{J})$, since the generators of $\widetilde{J}$ are homogeneous for the action. Let Z be an irreducible component of $V(\widetilde{J})$ with $\dim Z=\dim R/J_Q+1$ dominating $\mathbb{A}^1_s$. Since $\mathbb{G}_m$ is connected, Z is $\mathbb{G}_m$-stable, and since $\lambda\geq 0$, for a point $(\xi,1)\in Z$ the limit of $(c^\lambda\xi,c)$ as $c\to 0$ exists and belongs to $Z\cap(s=0)$. Thus $Z\cap(s=0)$ is non-empty and defined by one equation not vanishing on Z, so $\dim Z\cap(s=0)=\dim Z-1$ by Krull's principal ideal theorem. Hence $\dim R/\operatorname{top}_\lambda(J_Q)\geq\dim R/J_Q$.

By (7.24),

$$\dim R/J_P\geq\dim R/\operatorname{top}_\lambda(J_Q)\geq\dim R/J_Q,$$

which implies (7.23). Thus $a_P\leq a_Q$, and the lemma follows. □

Proposition 7.12. *Let $\Bbbk\supset\mathbb{C}$ be an algebraically closed field, $d\geq 2$ and $r_1,r_2,c\geq 1$ integers with $r_1+r_2=dc$. On $\widehat{\mathcal{X}}:=\operatorname{Spec}\Bbbk[[x,y,z,u]]$ with weights $(r_1,r_2,c,0)$ on (x,y,z,u) let $f_0:=xy+z^d$ and $W_0:=(f_0=0)$. Let G be a finite abelian group acting on $\widehat{\mathcal{X}}$ monomially on the unordered pair $\{x,y\}$ and by characters on z and u, and commuting with the grading. Assume that f_0 is G-semi-invariant and that G acts faithfully on W_0. Let β be a real number, $b_1,\dots,b_N>0$, and $P_1,\dots,P_N\in\Bbbk[[u]][x,y,z]$ non-zero G-semi-invariant elements with non-zero restrictions to W_0, each homogeneous of some weight $\operatorname{wt}P_j$ for the weights $(r_1,r_2,c,0)$, and let*

$$B_0:=\beta\operatorname{div}(u|_{W_0})+\sum_j b_j\operatorname{div}(P_j|_{W_0})$$

and $a_0:=\operatorname{mld}((W_0\ni 0,B_0)/G)$. Assume that a_0 is finite and non-negative. Then

$$a_0=\operatorname{mld}\Big(\Big(\widehat{\mathcal{X}}\ni 0,\operatorname{div}(f_0)+\beta\operatorname{div}(u)+\sum_{j=1}^N b_j\operatorname{div}(P_j)\Big)/G\Big).$$

Proof. Let $a_{\widehat{\mathcal{X}}}$ be the right-hand side of the equality in the proposition, and let W_0^{alg} be the hypersurface $(f_0=0)$ in $\mathbb{A}^4_\Bbbk$. By Lemma 2.12(3) for $X=\mathbb{A}^4_\Bbbk$ and for $X=W_0^{\mathrm{alg}}$, with $H=G$ and boundary 0, restriction maps the formal prime divisors over $\widehat{\mathcal{X}}$ (resp. W_0) with center the closed point 0 onto the prime divisors over $\mathbb{A}^4_\Bbbk$ (resp. W_0^{alg}) with center 0, preserving log discrepancies and inertia groups. For such prime divisors E over $\widehat{\mathcal{X}}$ and F over W_0 let

$$\sigma(E):=\frac{a(E,\widehat{\mathcal{X}},0)-\operatorname{mult}_E f_0-\beta\operatorname{mult}_E u-\sum_{j=1}^N b_j\operatorname{mult}_E P_j}{e_E},$$

$$\sigma_0(F):=\frac{a(F,W_0,0)-\beta\operatorname{mult}_F u-\sum_{j=1}^N b_j\operatorname{mult}_F P_j}{e_F},$$

where mult_F is taken of the restrictions to W_0. By Definition 2.2,

$$a_{\widehat{\mathcal{X}}}=\inf_E\sigma(E)\quad\text{and}\quad a_0=\inf_F\sigma_0(F).\tag{7.25}$$

We first show $a_{\widehat{\mathcal{X}}}\leq a_0$. Let F be a prime divisor over W_0 with center 0, let M be an integer with $M>\operatorname{mult}_F P_j$ for $1\leq j\leq N$, and for $h\in\Bbbk[[u]][x,y,z]$ let $h^{<M}$ be the sum of the terms of h of degree less than M in u. Since u vanishes at 0, $\operatorname{mult}_F u\geq 1$. Since $P_j-P_j^{<M}\in u^M\Bbbk[[u]][x,y,z]$, we have $\operatorname{mult}_F(P_j-P_j^{<M})\geq M>\operatorname{mult}_F P_j$, so $\operatorname{mult}_F P_j^{<M}=\operatorname{mult}_F P_j$. By Lemma 5.4(1) with $V=\mathbb{A}^4_\Bbbk$, $T=W_0^{\mathrm{alg}}$, $g=f_0$ and the functions $u,P_1^{<M},\dots,P_N^{<M}$, there exist a positive integer ℓ and a prime divisor $\widehat{F}$ over $\mathbb{A}^4_\Bbbk$ with center 0 such that $I_G(\widehat{F})=I_G(F)$ and

$$\operatorname{mult}_{\widehat{F}} f_0=\ell,\quad a(\widehat{F},\mathbb{A}^4_\Bbbk,0)=a(F,W_0,0)+\ell,$$

$$\operatorname{mult}_{\widehat{F}} u=\operatorname{mult}_F u,\quad\text{and}\quad\operatorname{mult}_{\widehat{F}} P_j^{<M}=\operatorname{mult}_F P_j\text{ for }1\leq j\leq N.$$

Thus

$$\operatorname{mult}_{\widehat{F}}(P_j-P_j^{<M})\geq M\operatorname{mult}_{\widehat{F}} u>\operatorname{mult}_{\widehat{F}} P_j^{<M},$$

so $\operatorname{mult}_{\widehat{F}} P_j = \operatorname{mult}_F P_j$ for any j. Since $e_{\widehat{F}} = e_F$, we obtain

$$\sigma(\widehat{F}) = \frac{a(F, W_0, 0) + \ell - \ell - \beta \operatorname{mult}_F u - \sum_{j=1}^N b_j \operatorname{mult}_F P_j}{e_F} = \sigma_0(F). \tag{7.26}$$

Since F is arbitrary, (7.25) implies

$$a_{\widehat{\mathcal{X}}} \le a_0. \tag{7.27}$$

By (7.25) and (7.27), the proposition is equivalent to $\sigma(E) \ge a_0$ for any prime divisor E over $\widehat{\mathcal{X}}$ with center 0. From now until (7.37) we assume moreover

(†) G acts on $\widehat{\mathcal{X}}$ without pseudoreflections, and the image in $\widehat{\mathcal{X}}/G$ of the locus where $\widehat{\mathcal{X}} \to \widehat{\mathcal{X}}/G$ is not étale intersects W_0/G in a subset of codimension at least 2.

Fix $g \in G$. Either $g^*x \in \Bbbk x$ and $g^*y \in \Bbbk y$, or $g^*x \in \Bbbk y$ and $g^*y \in \Bbbk x$. In the second case we say that g exchanges x and y. If g does not exchange x and y, let $(x_1, x_2) := (x, y)$. If g exchanges x and y, write $g^*x = \omega_x y$ and $g^*y = \omega_y x$ with $\omega_x, \omega_y \in \Bbbk^*$. Then $r_1 = r_2$ since g commutes with the grading, and $(g^*)^2 = \omega_x\omega_y$ on $\Bbbk x + \Bbbk y$. Let $\zeta \in \Bbbk$ with $\zeta^2 = \omega_x\omega_y$, and let $U, V \in \Bbbk x + \Bbbk y$ with $g^*U = \zeta U$ and $g^*V = -\zeta V$. Let $(x_1, x_2) := (U, V)$. Since $g^*(xy) = \zeta^2 xy$, $g^*(UV) = -\zeta^2 UV$ and xy is a quadratic form of rank 2, we have $xy = \theta_1 U^2 + \theta_2 V^2$ for some $\theta_1, \theta_2 \in \Bbbk^*$, so

$$f_0 = \theta_1 U^2 + \theta_2 V^2 + z^d, \tag{7.28}$$

and U, V are homogeneous of weight $r_1 = r_2$. Whether or not g exchanges x and y, g acts diagonally on the coordinates x_1, x_2, z, u of weights $r_1, r_2, c, 0$. For $\xi \in \{x_1, x_2, z, u\}$ let s_ξ be the integer in $\{0, \dots, |G| - 1\}$ with $g^*\xi = \exp(2\pi\sqrt{-1}s_\xi/|G|)\xi$, and let $\operatorname{age}(g) := \sum_\xi s_\xi/|G|$.

Let $\mathcal{A}_g$ be the set of tuples $\gamma = (\gamma_\xi)_\xi \in \Bbbk[[t]]^4$, indexed by $\xi \in \{x_1, x_2, z, u\}$, with $\gamma_\xi(0) = 0$ for any ξ with $s_\xi = 0$. We regard γ as the g-twisted arc $\xi = t^{s_\xi/|G|}\gamma_\xi(t)$ of $\widehat{\mathcal{X}}$ at 0. For a g-semi-invariant $h \in \Bbbk[[u]][x, y, z]$ let $w_g(h) \in [0, 1)$ be the fractional part of $\sum_\xi i_\xi s_\xi/|G|$ for one, hence any, monomial $\prod_\xi \xi^{i_\xi}$ of h, and

$$h^g(t, \gamma) := t^{-w_g(h)} h\big((t^{s_\xi/|G|}\gamma_\xi(t))_\xi\big) \in \Bbbk[[t]],$$
$$\operatorname{mult}_\gamma h := w_g(h) + \operatorname{mult}_0 h^g(t, \gamma),$$

where for $\varphi \in \Bbbk[[t]]$ we write $\operatorname{mult}_0 \varphi$ for the order of φ at $t = 0$ ($\operatorname{mult}_0 0 := +\infty$). Regard f_0^g as a polynomial in $\gamma_{x_1}, \gamma_{x_2}, \gamma_z$ with coefficients in $\Bbbk[t]$. The arcs of W_0 in $\mathcal{A}_g$ form

$$\mathcal{W}_g := \{\gamma \in \mathcal{A}_g \mid f_0^g(t, \gamma) = 0\},$$

and for $\gamma \in \mathcal{A}_g$ let

$$m(\gamma) := \min_{\xi \in \{x_1, x_2, z\}} \operatorname{mult}_0(\partial f_0^g/\partial\gamma_\xi)(t, \gamma) \in \mathbb{N} \cup \{+\infty\}.$$

For an integer $k \ge 0$ let π_k be the reduction of the four series modulo t^{k+1}, with values in $J_k := \Bbbk^{4(k+1)}$. A cylinder of $\mathcal{A}_g$ is a subset $\pi_k^{-1}(\Sigma)$ with $\Sigma \subset J_k$ constructible, and a cylinder of $\mathcal{W}_g$ is a subset $\pi_k^{-1}(\Sigma) \cap \mathcal{W}_g$. For $k \gg 0$, the codimension of a cylinder $\mathcal{C}$ of $\mathcal{A}_g$ is $\operatorname{codim}\mathcal{C} := 4(k+1) - \dim \pi_k(\mathcal{C})$, and that of a cylinder $\mathcal{D}$ of $\mathcal{W}_g$ is $\operatorname{codim}\mathcal{D} := 3(k+1) - \dim \pi_k(\mathcal{D})$. For an irreducible cylinder $\mathcal{C}$ and a g-semi-invariant $h \in \Bbbk[[u]][x, y, z]$ let

$$\operatorname{mult}_\mathcal{C} h := \min_{\gamma \in \mathcal{C}} \operatorname{mult}_\gamma h,$$

which is attained on a dense open subset of $\mathcal{C}$.

We use the change of variables for twisted arcs behind [NS26, Theorem 5.3 and Remark 5.4] in the following form.

(a) For any prime divisor E over $\widehat{\mathcal{X}}$ with center 0 there exist $g_E \in G$ and an irreducible cylinder $\mathcal{C}_E \subset \mathcal{A}_{g_E}$ such that $\operatorname{mult}_{\mathcal{C}_E} h = \operatorname{mult}_E h/e_E$ for any non-zero G-semi-invariant $h \in \Bbbk[[u]][x,y,z]$ and

$$\operatorname{codim} \mathcal{C}_E + \operatorname{age}(g_E) = a(E, \widehat{\mathcal{X}}, 0)/e_E.$$

(b) For any irreducible cylinder $\mathcal{D}$ of $\mathcal{W}_g$ on which $m(\gamma)$ equals a constant $m_{\mathcal{D}} < +\infty$, there exist a prime divisor F over W_0 with center 0 and a positive integer q such that $\operatorname{mult}_{\mathcal{D}} h = q \operatorname{mult}_F h/e_F$ for any G-semi-invariant $h \in \Bbbk[[u]][x,y,z]$ with $h|_{W_0} \neq 0$, and

$$\operatorname{codim} \mathcal{D} + \operatorname{age}(g) - w_g(f_0) - m_{\mathcal{D}} \geq q \frac{a(F, W_0, 0)}{e_F}.$$

We prove (b). Let $\varpi_0 : W_0^{\mathrm{alg}} \to X_0 := W_0^{\mathrm{alg}}/G$ be the quotient morphism. The $\mathbb{G}_m$-action on $\mathbb{A}^4_{\Bbbk}$ of weights $(r_1, r_2, c, 1)$ on (x, y, z, u) commutes with G, preserves W_0^{alg} and contracts $\mathbb{A}^4_{\Bbbk}$ to 0. Each closed subset in (†) is stable under it, so its codimension in $\mathbb{A}^4_{\Bbbk}$, resp. in X_0, equals its codimension at 0, resp. at $\varpi_0(0)$. Thus G acting on $\mathbb{A}^4_{\Bbbk}$ and W_0^{alg} satisfy the assumptions of [NS26, Section 3], and ϖ_0 is étale in codimension 1. Since $W_0^{\mathrm{alg}} = (xy + z^d = 0) \times \mathbb{A}^1_u$ has canonical singularities, X_0 is klt, in particular normal with IK_{X_0} Cartier for some positive integer I. As f_0 is a polynomial, $\mathcal{W}_g$ is the space of g-twisted arcs of W_0^{alg} at 0. Let ψ_g be the map $\overline{\mu}_{g\infty}$ of [NS26, Proposition 3.10], sending $\gamma \in \mathcal{W}_g$ to the arc

$$\varphi \to \varphi\big((t^{s_\xi/|G|}\gamma_\xi(t))_\xi\big)$$

of X_0. Possibly replacing $\mathcal{D}$ with a dense open subset, which changes neither $\operatorname{codim} \mathcal{D}$ nor any $\operatorname{mult}_{\mathcal{D}} h$, we may assume that the orders along the arcs of $\mathcal{D}$ of the ideals defining the cylinders of [NS26, Lemma 5.2], with the ideal $\mathfrak{a}$ there equal to $\mathcal{O}_{X_0}$, are constant and finite [NS26, Lemma 5.1]. Let $\ell_{\mathcal{D}}$ be the order along $\psi_g(\gamma)$, $\gamma \in \mathcal{D}$, of the I-th Nash ideal $\mathfrak{n}_{I,X_0}$ [EM09, Section 7]. Then [NS26, Lemma 5.2], which holds over any algebraically closed field of characteristic zero [NS26, Notation] and whose Jacobian term is $m_{\mathcal{D}}$ and character term is $w_g(f_0)$, applied through its proof, that is, through [NS26, Propositions 2.15 and 2.16], to the G-saturation $G \cdot \mathcal{D}$, a G-invariant subcylinder of the cylinder of [NS26, Lemma 5.2] given by the constant orders along $\mathcal{D}$ with $\operatorname{codim} G \cdot \mathcal{D} = \operatorname{codim} \mathcal{D}$ and, G being abelian, $\psi_g(G \cdot \mathcal{D}) = \psi_g(\mathcal{D})$, gives

$$\operatorname{codim} \psi_g(\mathcal{D}) - \ell_{\mathcal{D}}/I = \operatorname{codim} \mathcal{D} + \operatorname{age}(g) - w_g(f_0) - m_{\mathcal{D}}.$$

By [dFEI08, Proposition 3.10, Proposition 2.12, Remark 2.11, and Theorem 3.9] and [dFEI08, Proposition 1.7] for the irreducible cylinder $\psi_g(\mathcal{D})$ of arcs of X_0 at $\varpi_0(0)$, which is fat by [dFEI08, Corollary 3.7] since the Jacobian ideal Jac_{X_0} [NS26, Definition 2.1(1)], whose pullback is one of the ideals of [NS26, Lemma 5.2], has finite order along its arcs, there exist a prime divisor F_{X_0} over X_0 with center $\varpi_0(0)$ and a positive integer q such that

$$\operatorname{codim} \psi_g(\mathcal{D}) - \ell_{\mathcal{D}}/I \geq q a(F_{X_0}, X_0, 0)$$

and the order of any $\varphi \in \mathcal{O}_{X_0, \varpi_0(0)}$ along a general arc of $\psi_g(\mathcal{D})$ is $q \operatorname{mult}_{F_{X_0}} \varphi$.

By Lemma 2.15(3) let F be a prime divisor over W_0^{alg} mapped onto F_{X_0}. Its center is 0 since $\varpi_0^{-1}(\varpi_0(0)) = \{0\}$. By Lemma 2.15(1)(2), with $B = 0$ and $B_{X_0} = 0$ by Lemma 2.15(4) as ϖ_0 is étale in codimension 1, we have $\operatorname{mult}_F(\varpi_0^* \varphi) = e_F \operatorname{mult}_{F_{X_0}} \varphi$ and $a(F, W_0^{\mathrm{alg}}, 0) = e_F a(F_{X_0}, X_0, 0)$. By Lemma 2.12(3) we regard F as a formal prime divisor over W_0, with log discrepancy, inertia group and orders of polynomials equal to those of F over W_0^{alg}. Thus

$$\operatorname{codim} \mathcal{D} + \operatorname{age}(g) - w_g(f_0) - m_{\mathcal{D}} = \operatorname{codim} \psi_g(\mathcal{D}) - \frac{\ell_{\mathcal{D}}}{I} \geq q a(F_{X_0}, X_0, 0) = q \frac{a(F, W_0, 0)}{e_F},$$

which is the inequality in (b). For a G-semi-invariant polynomial h with $h|_{W_0} \neq 0$, $h^{|G|}|_{W_0^{\mathrm{alg}}} = \varpi_0^*\varphi$ for a non-zero φ, and $|G| \operatorname{mult}_\gamma h = \operatorname{mult}_\gamma h^{|G|}$ is the order of φ along $\psi_g(\gamma)$, so

$$\operatorname{mult}_{\mathcal{D}} h = q \operatorname{mult}_{F_{X_0}} \varphi/|G| = q \operatorname{mult}_F h/e_F.$$

For a G-semi-invariant $h \in \Bbbk[[u]][x,y,z]$ with $h|_{W_0} \neq 0$, take an integer M with $M \operatorname{mult}_F u > \operatorname{mult}_F h$. Since u^M divides $h - h^{<M}$ in $\Bbbk[[u]][x,y,z]$, we have $\operatorname{mult}_F h^{<M} = \operatorname{mult}_F h$ and

$$\operatorname{mult}_{\mathcal{D}}(h - h^{<M}) \geq M \operatorname{mult}_{\mathcal{D}} u = qM \operatorname{mult}_F u/e_F > \operatorname{mult}_{\mathcal{D}} h^{<M},$$

so $\operatorname{mult}_{\mathcal{D}} h = \operatorname{mult}_{\mathcal{D}} h^{<M} = q \operatorname{mult}_F h/e_F$.

Here (a) is Lemma 7.9(1) for the action of G on $\mathbb{A}^4_{\Bbbk}$, which contains no pseudoreflection by (†), together with Lemma 2.12(3). Indeed G is abelian. If no element of G exchanges x and y, then G is diagonal in (x,y,z,u). If an element $g' \in G$ exchanges x and y, the elements not exchanging them act on $\Bbbk x + \Bbbk y$ by scalars, G is diagonal in the coordinates (U,V,z,u) of g', and for each $g \in G$ the coordinates (x_1,x_2,z,u) above differ from (U,V,z,u) by a linear change, which preserves $\mathcal{A}_g$, its cylinders, their codimensions and $\operatorname{mult}_\gamma$.

Let $\gamma \in \mathcal{A}_g$ with $\gamma_u \neq 0$. Then $\rho := \operatorname{mult}_\gamma u = s_u/|G| + \operatorname{mult}_0 \gamma_u > 0$, since $\gamma_u(0) = 0$ if $s_u = 0$. For a g-semi-invariant $h \in \Bbbk[[u]][x,y,z]$ we have $\operatorname{mult}_\gamma(h - h^{<M}) \geq M\rho$, so $\operatorname{mult}_\gamma h = \operatorname{mult}_\gamma h^{<M}$ for any integer M with $M\rho > \operatorname{mult}_\gamma h^{<M}$. Likewise, $\operatorname{mult}_F h = \operatorname{mult}_F h^{<M}$ for any prime divisor F over $\widehat{\mathcal{X}}$ or W_0 with center 0 and $M \operatorname{mult}_F u > \operatorname{mult}_F h^{<M}$. Thus on the arcs with $\operatorname{mult}_\gamma u = \rho$, a condition $\operatorname{mult}_\gamma P_j \geq \eta$ depends only on $P_j^{<M}$ for $M\rho > \eta$, and only on $\pi_k(\gamma)$ for $k \gg 0$. On each such stratum only finitely many coefficients of $P_1, \dots, P_N$ in u enter. For an irreducible cylinder $\mathcal{C}$ of $\mathcal{A}_g$ and an irreducible cylinder $\mathcal{D}$ as in (b) let

$$\sigma(\mathcal{C}) := \operatorname{codim} \mathcal{C} + \operatorname{age}(g) - \operatorname{mult}_{\mathcal{C}} f_0 - \beta \operatorname{mult}_{\mathcal{C}} u - \sum_{j=1}^N b_j \operatorname{mult}_{\mathcal{C}} P_j$$

and

$$\sigma_0(\mathcal{D}) := \operatorname{codim} \mathcal{D} + \operatorname{age}(g) - w_g(f_0) - m_{\mathcal{D}} - \beta \operatorname{mult}_{\mathcal{D}} u - \sum_{j=1}^N b_j \operatorname{mult}_{\mathcal{D}} P_j.$$

By (a), for any prime divisor E over $\widehat{\mathcal{X}}$ with center 0,

$$\sigma(\mathcal{C}_E) = \sigma(E). \tag{7.29}$$

By (b), with F and q as there, $\sigma_0(\mathcal{D}) \geq q\sigma_0(F)$. Since $\sigma_0(F) \geq a_0 \geq 0$ by (7.25) and $q \geq 1$,

$$\sigma_0(\mathcal{D}) \geq a_0 \quad \text{for any irreducible cylinder } \mathcal{D} \text{ as in (b).} \tag{7.30}$$

Fix a rational number $\rho > 0$ with $\rho - s_u/|G| \in \mathbb{N}$, an integer $\eta_0 \geq 0$ and rational numbers $\eta_1, \dots, \eta_N \geq 0$, write $\eta := (\eta_0, \dots, \eta_N)$, and let

$$\mathcal{K}_g(\rho,\eta) := \{\gamma \in \mathcal{A}_g \mid \operatorname{mult}_\gamma u = \rho,\ \operatorname{mult}_0 f_0^g(t,\gamma) \geq \eta_0, \\ \operatorname{mult}_\gamma P_j \geq \eta_j \text{ for } 1 \leq j \leq N\},$$

a cylinder of $\mathcal{A}_g$. For an irreducible component $\mathcal{C}$ of $\mathcal{K}_g(\rho,\eta)$ let

$$\sigma_\eta(\mathcal{C}) := \operatorname{codim} \mathcal{C} + \operatorname{age}(g) - w_g(f_0) - \eta_0 - \beta\rho - \sum_{j=1}^N b_j \eta_j.$$

We show

$$a_0 \leq \sigma_\eta(\mathcal{C}) \quad \text{if } m(\gamma) < +\infty \text{ for some } \gamma \in \mathcal{C} \cap \mathcal{W}_g. \tag{7.31}$$

Let μ be the least value of $m(\gamma)$ for $\gamma \in \mathcal{C} \cap \mathcal{W}_g$, and let

$$\mathcal{C}^\circ := \{\gamma \in \mathcal{C} \mid m(\gamma) \leq \mu\}.$$

Then $\mathcal{C}^\circ$ is a non-empty open subset of the irreducible cylinder $\mathcal{C}$, so $\operatorname{codim}\mathcal{C}^\circ = \operatorname{codim}\mathcal{C}$, and $m(\gamma) = \mu$ for any arc γ of $\mathcal{D} := \mathcal{C}^\circ \cap \mathcal{W}_g$ by the minimality of μ. Since $\operatorname{mult}_0 f_0^g(t,\gamma) \geq \eta_0$ for any $\gamma \in \mathcal{C}$, [NS26, Lemma 2.17] (compare [EM04, Theorem 2.2]) implies

$$\operatorname{codim}\mathcal{D} \leq \operatorname{codim}\mathcal{C} + \mu - \eta_0. \tag{7.32}$$

Let $\mathcal{D}_1$ be an irreducible component of $\mathcal{D}$ with $\operatorname{codim}\mathcal{D}_1 = \operatorname{codim}\mathcal{D}$. Then $m(\gamma) = \mu$ for any $\gamma \in \mathcal{D}_1$, $\operatorname{mult}_{\mathcal{D}_1} u = \rho$ and $\operatorname{mult}_{\mathcal{D}_1} P_j \geq \eta_j$ for any j. Since $b_j > 0$ for any j, (7.30) and (7.32) imply

$$a_0 \leq \sigma_0(\mathcal{D}_1) \leq \operatorname{codim}\mathcal{D} + \operatorname{age}(g) - w_g(f_0) - \mu - \beta\rho - \sum_{j=1}^{N} b_j\eta_j \leq \sigma_\eta(\mathcal{C}),$$

which proves (7.31).

It remains to show that the hypothesis of (7.31) holds for any irreducible component $\mathcal{C}$ of $\mathcal{K}_g(\rho,\eta)$, that is,

$$m(\gamma) < +\infty \text{ for some } \gamma \in \mathcal{C} \cap \mathcal{W}_g. \tag{7.33}$$

We contract only $\gamma_{x_1}, \gamma_{x_2}, \gamma_z$, adapting the contraction used for [EMY03, Lemma 3.2]. Let $k \gg 0$, so that $\mathcal{K}_g(\rho,\eta) = \pi_k^{-1}(\mathcal{K}_k)$ with $\mathcal{K}_k := \pi_k(\mathcal{K}_g(\rho,\eta))$, $\mathcal{C} = \pi_k^{-1}(\pi_k(\mathcal{C}))$, and $\pi_k(\mathcal{C})$ is an irreducible component of $\mathcal{K}_k$. Let $J_k^\circ \subset J_k$ be the open subset where the coefficient of $t^{\rho - s_u/|G|}$ in γ_u is non-zero. The remaining conditions defining $\mathcal{K}_k$ are closed, so $\mathcal{K}_k$ and $\pi_k(\mathcal{C})$ are closed in J_k°. Let $\Bbbk^*$ act on $\mathcal{A}_g$ and on J_k by

$$\lambda \cdot \gamma := (\lambda^{r_1}\gamma_{x_1}, \lambda^{r_2}\gamma_{x_2}, \lambda^c\gamma_z, \gamma_u).$$

Since f_0 and P_j are homogeneous of weights dc and $\operatorname{wt} P_j$, u has weight 0, and U, V have weight $r_1 = r_2$ when g exchanges x and y, we have

$$f_0^g(t, \lambda\cdot\gamma) = \lambda^{dc} f_0^g(t,\gamma) \quad \text{and} \quad P_j^g(t,\lambda\cdot\gamma) = \lambda^{\operatorname{wt} P_j} P_j^g(t,\gamma) \quad \text{for } \lambda \in \Bbbk^*.$$

The action also preserves the conditions $\gamma_\xi(0) = 0$ and fixes γ_u, so $\mathcal{K}_k$ is $\Bbbk^*$-stable. Since $\Bbbk^*$ is connected, it preserves each irreducible component of $\mathcal{K}_k$, in particular $\pi_k(\mathcal{C})$. Let $\gamma \in \mathcal{C}$ and $\gamma^0 := (0,0,0,\gamma_u)$. Since $r_1, r_2, c > 0$, the morphism $\Bbbk^* \to \pi_k(\mathcal{C})$, $\lambda \to \lambda\cdot\pi_k(\gamma)$, extends to $\mathbb{A}^1 \to J_k$ with $0 \to \pi_k(\gamma^0)$. Thus $\pi_k(\gamma^0)$ is in the closure of $\pi_k(\mathcal{C})$ and in J_k°, hence

$$\pi_k(\gamma^0) \in \pi_k(\mathcal{C}). \tag{7.34}$$

We now construct, for any integer $n \gg 0$, an arc $\gamma^n \in \mathcal{W}_g$ with $m(\gamma^n) < +\infty$, $\gamma_u^n = \gamma_u$, and $\operatorname{mult}_0 \gamma_\xi^n > k$ for $\xi \in \{x_1, x_2, z\}$. Then $\pi_k(\gamma^n) = \pi_k(\gamma^0)$, so $\gamma^n \in \mathcal{C}$ by (7.34), which proves (7.33).

Assume that g does not exchange x and y, so that $(x_1, x_2) = (x, y)$. Then

$$f_0^g(t,\gamma) = t^{(s_x+s_y)/|G| - w_g(f_0)}\gamma_x\gamma_y + t^{ds_z/|G| - w_g(f_0)}\gamma_z^d.$$

Since $g^* f_0 \in \Bbbk^* f_0$ and g acts diagonally, $g^*(xy)/(xy) = g^*(z^d)/z^d$, so $q_0 := (ds_z - s_x - s_y)/|G| \in \mathbb{Z}$. Let

$$\gamma_x^n := t^n, \quad \gamma_y^n := -t^{(d-1)n+q_0}, \quad \gamma_z^n := t^n, \quad \text{and} \quad \gamma_u^n := \gamma_u. \tag{7.35}$$

Then the orders n and $(d-1)n + q_0$ exceed k for $n \gg 0$,

$$f_0^g(t,\gamma^n) = -t^{ds_z/|G| - w_g(f_0) + dn} + t^{ds_z/|G| - w_g(f_0) + dn} = 0,$$
$$\text{and} \quad (\partial f_0^g/\partial\gamma_y)(t,\gamma^n) = t^{(s_x+s_y)/|G| - w_g(f_0) + n} \neq 0,$$

so $m(\gamma^n) < +\infty$.

Assume that g exchanges x and y, so that $(x_1, x_2) = (U, V)$. By (7.28),

$$f_0^g(t,\gamma) = \theta_1 t^{2s_U/|G| - w_g(f_0)}\gamma_U^2 + \theta_2 t^{2s_V/|G| - w_g(f_0)}\gamma_V^2 + t^{ds_z/|G| - w_g(f_0)}\gamma_z^d.$$

Since $g^* f_0 \in \Bbbk^* f_0$, (7.28) implies $g^*(U^2)/U^2 = g^*(V^2)/V^2 = g^*(z^d)/z^d$, so $q_U := (ds_z - 2s_U)/|G|$ and $q_V := (ds_z - 2s_V)/|G|$ are integers. Since the eigenvalues of g^* on U and V are ζ and $-\zeta$, we have $|s_U - s_V| = |G|/2$, so $q_U - q_V = 2(s_V - s_U)/|G| = \pm 1$. If d is odd, let $S := U$ and consider

only n with $dn+q_U$ even. If d is even, let $S\in\{U,V\}$ with q_S even. Let θ_S be the coefficient of S^2 in (7.28), $\kappa\in\Bbbk$ with $\theta_S\kappa^2=-1$, and let S' be the element of $\{U,V\}$ other than S. Let

$$\gamma_z^n:=t^n,\quad \gamma_S^n:=\kappa t^{(dn+q_S)/2},\quad \gamma_{S'}^n:=0,\quad\text{and}\quad \gamma_u^n:=\gamma_u. \tag{7.36}$$

Then the orders n and $(dn+q_S)/2$ exceed k for $n\gg 0$,

$$f_0^g(t,\gamma^n)=\theta_S\kappa^2 t^{ds_z/|G|-w_g(f_0)+dn}+t^{ds_z/|G|-w_g(f_0)+dn}=0,$$
$$\text{and}\quad (\partial f_0^g/\partial\gamma_S)(t,\gamma^n)=2\theta_S\kappa t^{2s_S/|G|-w_g(f_0)+(dn+q_S)/2}\neq 0,$$

so $m(\gamma^n)<+\infty$.

By (7.31) and (7.33), under (†),

$$a_0\le\sigma_\eta(\mathcal{C})\quad\text{for any } g,\rho,\eta \text{ and any irreducible component } \mathcal{C}\text{ of }\mathcal{K}_g(\rho,\eta). \tag{7.37}$$

The proof continues below.

Proof of Proposition 7.12, continued. We first show that (†) implies

$$\sigma(E)\ge a_0\quad\text{for any prime divisor } E \text{ over } \widehat{\mathcal{X}} \text{ with center } 0. \tag{7.38}$$

Let E be such a prime divisor, g_E and $\mathcal{C}_E$ as in (a), $\rho:=\operatorname{mult}_E u/e_E$, $\eta_0:=\operatorname{mult}_E f_0/e_E-w_{g_E}(f_0)$, $\eta_j:=\operatorname{mult}_E P_j/e_E$ for $1\le j\le N$, and $\eta:=(\eta_0,\eta_1,\dots,\eta_N)$ (by (a), $\rho-s_u/|G|\in\mathbb{N}$ and $\eta_0\in\mathbb{N}$). The arcs $\gamma\in\mathcal{C}_E$ with $\operatorname{mult}_\gamma u=\rho$ form a dense open subset of $\mathcal{C}_E$ contained in $\mathcal{K}_{g_E}(\rho,\eta)$, hence in an irreducible component $\mathcal{C}$ of $\mathcal{K}_{g_E}(\rho,\eta)$ with $\operatorname{codim}\mathcal{C}\le\operatorname{codim}\mathcal{C}_E$. By (7.37), (a) and (7.29),

$$a_0\le\sigma_\eta(\mathcal{C})\le\operatorname{codim}\mathcal{C}_E+\operatorname{age}(g_E)-\operatorname{mult}_{\mathcal{C}_E}f_0-\beta\operatorname{mult}_{\mathcal{C}_E}u-\sum_{j=1}^N b_j\operatorname{mult}_{\mathcal{C}_E}P_j=\sigma(E).$$

We remove the assumption (†) by adjoining coefficient-one eigenparameters. If G preserves the lines $\Bbbk x$ and $\Bbbk y$, let $(\tau_1,\tau_2,\tau_3,\tau_4):=(x,y,z,u)$. Otherwise some $g'\in G$ exchanges $\Bbbk x$ and $\Bbbk y$. Let U,V be as in (7.28) for g'. Since G is abelian and the eigenvalues of g' on U,V differ, G acts diagonally on U,V. Let $(\tau_1,\tau_2,\tau_3,\tau_4):=(U,V,z,u)$. In both cases G acts diagonally on $(\tau_1,\dots,\tau_4)$, the grading is preserved, and τ_1,τ_2,τ_3 have positive weights.

Let $v_1,\dots,v_4$ be further coordinates of weight 1 with $g^*v_i=\chi_{\tau_i}(g)v_i$ for any $g\in G$ and any i, where $g^*\tau_i=\chi_{\tau_i}(g)\tau_i$. Let

$$\widehat{\mathcal{X}}^{\mathrm{c}}:=\operatorname{Spec}\Bbbk[[x,y,z,u,v_1,\dots,v_4]]\quad\text{and}\quad W_0^{\mathrm{c}}:=(f_0=0)\subset\widehat{\mathcal{X}}^{\mathrm{c}},$$

and let

$$D_0:=\operatorname{div}(f_0)+\beta\operatorname{div}(u)+\sum_{j=1}^N b_j\operatorname{div}(P_j)\text{ on }\widehat{\mathcal{X}},$$

$$D_0^{\mathrm{c}}:=D_0+\sum_{i=1}^4\operatorname{div}(v_i),\quad\text{and}\quad B_0^{\mathrm{c}}:=B_0+\sum_{i=1}^4\operatorname{div}(v_i|_{W_0^{\mathrm{c}}}).$$

The following hold.

(i) By Lemma 3.6(3)(5), with $X=W_0^{\mathrm{alg}}$, whose completion at 0 is W_0, the closed point $x=0$, $B=B_0$, $n=4$, and the characters $\chi_{\tau_1},\dots,\chi_{\tau_4}$, since $a_0\ge 0$, we have $\operatorname{mld}((W_0^{\mathrm{c}}\ni 0,B_0^{\mathrm{c}})/G)=a_0$.

(ii) By Lemma 3.6(4), with $X=W_0^{\mathrm{alg}}\subset\mathbb{A}^4_\Bbbk$ in the coordinates $(\tau_1,\dots,\tau_4)$ and $n=4$, G acts on $\mathbb{A}^4_\Bbbk\times\mathbb{A}^4_\Bbbk$ without pseudoreflections, and near the image of 0, the image of the locus where the quotient map

$$\mathbb{A}^4_\Bbbk\times\mathbb{A}^4_\Bbbk\to(\mathbb{A}^4_\Bbbk\times\mathbb{A}^4_\Bbbk)/G$$

is not étale intersects $(W_0^{\mathrm{alg}}\times\mathbb{A}^4_\Bbbk)/G$ in a subset of codimension ≥ 2. Thus $(\widehat{\mathcal{X}}^{\mathrm{c}},W_0^{\mathrm{c}})$ in place of $(\widehat{\mathcal{X}},W_0)$ satisfies (†).

(iii) W_0^{c}/G is klt, since W_0 is the product of the A_{d-1} surface singularity $(xy+z^d=0)$ with a line and W_0^{c} is the product of W_0 with $\mathbb{A}^4$.

(iv) The $\Bbbk^*$-action of weights $(r_1, r_2, c, 0, 1, 1, 1, 1)$ on the arc components γ_ξ, where ξ runs over the eight coordinates $x_1, x_2, z, u, v_1, \dots, v_4$, fixes γ_u and preserves the conditions on $\operatorname{mult}_\gamma v_1, \dots, \operatorname{mult}_\gamma v_4$, since $v_1, \dots, v_4$ enter $\mathcal{K}_g(\rho, \eta)$ as further P_j with coefficient one and are homogeneous of weight 1. The arcs γ^n of (7.35) and (7.36), completed by $\gamma^n_{v_i} := 0$, satisfy $\gamma^n \in \mathcal{W}_g$ and have k-jet equal to that of γ^0. Since f_0 does not involve $v_1, \dots, v_4$, $m(\gamma^n) < +\infty$.

By (i)–(iv), replacing

$$(\widehat{\mathcal{X}}, W_0, D_0, B_0) \quad \text{with} \quad (\widehat{\mathcal{X}}^{\mathrm{c}}, W_0^{\mathrm{c}}, D_0^{\mathrm{c}}, B_0^{\mathrm{c}})$$

and the four coordinates x_1, x_2, z, u with the eight coordinates $x_1, x_2, z, u, v_1, \dots, v_4$, the argument from (†) to (7.38) implies

$$\frac{a(E^{\mathrm{c}}, \widehat{\mathcal{X}}^{\mathrm{c}}, D_0^{\mathrm{c}})}{e_{E^{\mathrm{c}}}} \geq a_0 \quad \text{for any prime divisor } E^{\mathrm{c}} \text{ over } \widehat{\mathcal{X}}^{\mathrm{c}} \text{ with center 0.}$$

Here $\mathcal{A}_g \subset \Bbbk[[t]]^8$, $J_k = \Bbbk^{8(k+1)}$, the codimensions of cylinders of $\mathcal{A}_g$ and of $\mathcal{W}_g$ are

$$8(k+1) - \dim \pi_k(\cdot) \quad \text{and} \quad 7(k+1) - \dim \pi_k(\cdot),$$

$v_1, \dots, v_4$ enter as further P_j with coefficient one, $\gamma^0 := (0, 0, 0, \gamma_u, 0, 0, 0, 0)$, (a) holds by Lemma 7.9(1) since G acts diagonally and without pseudoreflections by (ii), and (b) uses (†) for W_0^{c}, which holds by (ii), and the klt property of W_0^{c}/G, which holds by (iii). Let E be a prime divisor over $\widehat{\mathcal{X}}$ with center 0. By Lemma 3.6(2)(5), with $X = \mathbb{A}^4_{\Bbbk}$, whose completion at 0 is $\widehat{\mathcal{X}}$, the closed point $x = 0$, $B = D_0$, $n = 4$, and the characters $\chi_{\tau_1}, \dots, \chi_{\tau_4}$, there exists a prime divisor E^{c} over $\widehat{\mathcal{X}}^{\mathrm{c}}$ with center 0 such that $I_G(E^{\mathrm{c}}) = I_G(E)$ and

$$a(E^{\mathrm{c}}, \widehat{\mathcal{X}}^{\mathrm{c}}, D_0^{\mathrm{c}}) = a(E, \widehat{\mathcal{X}}, D_0).$$

Thus

$$\sigma(E) = \frac{a(E, \widehat{\mathcal{X}}, D_0)}{e_E} = \frac{a(E^{\mathrm{c}}, \widehat{\mathcal{X}}^{\mathrm{c}}, D_0^{\mathrm{c}})}{e_{E^{\mathrm{c}}}} \geq a_0.$$

Hence (7.38) holds without (†). Together with (7.27) and (7.25), this proves the proposition. □

7.4. **Critical stretching.**

Definition-Lemma 7.13. Let G be a finite abelian group, m a positive integer, $\beta \in \frac{1}{m}\mathbb{Z}$, and $\bar{b}$, η, R' positive real numbers. For any $i \in \mathbb{N}^+$, let

- $(V_i \ni o_i)$ be a smooth threefold germ with a faithful action of G and regular parameters s_i, p_i, q_i at o_i which are G-eigenfunctions whose characters χ_s, χ_p, χ_q do not depend on i,
- h_i a G-semi-invariant mark on $(V_i \ni o_i)$ (Definition 2.4) whose character does not depend on i,
- the $\mathbb{Q}$-divisor

$$B_i := \frac{1}{m}\operatorname{div}(h_i) + \beta \operatorname{div}(s_i),$$

and
- F_i a prime divisor over V_i with center o_i.

Let $\Bbbk \supset \mathbb{C}$ be an algebraically closed field, $h_* \in \Bbbk[[s, p, q]]$ a compatible generic limit of $(h_i)_i$ along an infinite subset of $\mathbb{N}^+$ (Definition 2.5), where $\widehat{\mathcal{O}}_{V_i, o_i}$ is identified with $\mathbb{C}[[s, p, q]]$ via (s_i, p_i, q_i), $\widehat{\mathcal{X}} := \operatorname{Spec} \Bbbk[[s, p, q]]$, and $C_* := (p = q = 0) \subset \widehat{\mathcal{X}}$. Let $\widehat{X} := \operatorname{Spec} \Bbbk((s))[[p, q]]$, with closed point x, be the completion of the local ring of $\widehat{\mathcal{X}}$ at the generic point of C_* (Cohen structure theorem), on which G acts semilinearly. Assume that

(i) (V_i, B_i) is sub-klt near o_i and $b_i := \operatorname{mld}((V_i \ni o_i, B_i)/G) \in (0, \bar{b}]$ for any i,

(ii) $a(T, V_i, B_i) \geq \eta \operatorname{mult}_T s_i$ for any i and any prime divisor T over V_i with center o_i,

(iii) $a(F_i, V_i, B_i) = b_i e_{F_i}$ and $\operatorname{mult}_{F_i} s_i \le R'$ for any i, and

$$\lim_{i\to+\infty} \min\{\operatorname{mult}_{F_i} p_i, \operatorname{mult}_{F_i} q_i\} = +\infty,$$

(iv) the pair

$$\Big(\widehat{\mathcal{X}}\backslash(s=0), \frac{1}{m}\operatorname{div}(h_*)\Big)$$

is lc near the generic point of C_* and not klt at the generic point of C_*, and any prime divisor on $\widehat{\mathcal{X}}$ containing C_* has coefficient less than one in $\frac{1}{m}\operatorname{div}(h_*)$.

Then the following hold.

(1) There exists a G-invariant prime divisor E over $\widehat{X}$ with center x such that

$$a\Big(E, \widehat{X}, \frac{1}{m}\operatorname{div}(h_*)\Big) = 0,$$

and $a(E', \widehat{X}, \frac{1}{m}\operatorname{div}(h_*)) > 0$ for any prime divisor $E' \neq E$ exceptional for the sequence of blow-ups that starts from $\widehat{X}$, blows up the center of E at each step, and ends when E is the exceptional curve of the last blow-up. Possibly exchanging p and q, there exist coprime integers $w_1 \ge w_2 \ge 1$ and $\sigma \in \Bbbk((s))[q]$ with $\sigma(s,0) = 0$ and $\deg_q \sigma < w_1/w_2$ satisfying the following.

- E is the exceptional divisor of the weighted blow-up of $\widehat{X}$ with weights (w_1, w_2) in $(p - \sigma(s,q), q)$.
- The automorphism $(p,q) \to (p + \sigma(s,q), q)$ of $\Bbbk((s))[[p,q]]$ commutes with the action of G.
- Let $d := m(w_1 + w_2)$. Then $h_*(s, p + \sigma(s,q), q)$ has no monomial of (w_1, w_2)-weight less than d, and the part Φ of $h_*(s, p + \sigma(s,q), q)$ of (w_1, w_2)-weight d is a non-zero polynomial such that $(\mathbb{A}^2_{\Bbbk((s))}, \frac{1}{m}\operatorname{div}(\Phi))$ is an lc critical pair for (w_1, w_2) (Definition 7.1).

(2) Assume that $\frac{1}{m}\operatorname{div}(\Phi)$ has no component of coefficient one on $\mathbb{A}^2_{\Bbbk((s))}$. Then there exist a real number L and an infinite subset $\mathcal{I} \subset \mathbb{N}^+$ such that for any $i \in \mathcal{I}$, there exists a prime divisor E_i over V_i with center o_i satisfying

$$a(E_i, V_i, B_i) = b_i e_{E_i} \quad \text{and} \quad a(E_i, V_i, 0) \le L.$$

Proof. Any statement about the indices i is made after passing to an infinite subset of indices, which keeps (i)–(iv) and suffices for (2).

Step 1. In this step we construct E.

By (iv), $(\widehat{X}, \frac{1}{m}\operatorname{div}(h_*))$ is lc and not klt at x, and any prime divisor on $\widehat{X}$ through x has coefficient less than one in $\frac{1}{m}\operatorname{div}(h_*)$: log discrepancies of prime divisors over $\widehat{X}$ are computed on log resolutions, which commute with completion. Thus some prime divisor E° over $\widehat{X}$ with center x satisfies $a(E^\circ, \widehat{X}, \frac{1}{m}\operatorname{div}(h_*)) = 0$. For $N \in \mathbb{N}^+$, let $h_*^{<N} \in \Bbbk[[s]][p,q]$ be the sum of the terms of h_* of degree less than N in (p,q), and let

$$\mathfrak{a}_N := \big((h_*^{<N}) + (p,q)^N\big)^{1/m},$$

a G-invariant $\mathbb{R}$-ideal on $\widehat{X}$. Since $h_* - h_*^{<N} \in (p,q)^N$, we have $(h_*) \subset (h_*^{<N}) + (p,q)^N$, so $(\widehat{X}, \mathfrak{a}_N)$ is lc and

$$a(F, \widehat{X}, \mathfrak{a}_N) \ge a\Big(F, \widehat{X}, \frac{1}{m}\operatorname{div}(h_*)\Big) \quad \text{for any prime divisor } F \text{ over } \widehat{X}, \tag{7.39}$$

with equality when $N \operatorname{mult}_F(p,q) > \operatorname{mult}_F h_*$. Fix N with $N \operatorname{mult}_{E^\circ}(p,q) > \operatorname{mult}_{E^\circ} h_*$. Then $\operatorname{mld}(\widehat{X} \ni x, \mathfrak{a}_N) = 0$, and Lemma 3.4, with $k = \Bbbk((s))$ and $S = \widehat{X}$, yields closed points

$x^{(0)} = x, x^{(1)}, \dots, x^{(r-1)}$ and exceptional curves $E_1, \dots, E_r$. Let $j \le r$ be the smallest index with $a(E_j, \widehat{X}, \frac{1}{m}\operatorname{div}(h_*)) = 0$. It exists since, by (7.39) and lc,

$$0 \le a\Big(E_r, \widehat{X}, \frac{1}{m}\operatorname{div}(h_*)\Big) \le a(E_r, \widehat{X}, \mathfrak{a}_N) = 0.$$

Let $E := E_j$. By lc and the minimality of j, $a(E_{j'}, \widehat{X}, \frac{1}{m}\operatorname{div}(h_*)) > 0$ for $1 \le j' < j$.

Fix $N' \ge N$ with $N' \operatorname{mult}_{E_{j'}}(p,q) > \operatorname{mult}_{E_{j'}} h_*$ for $1 \le j' \le j$. By (7.39),

$$a(E_{j'}, \widehat{X}, \mathfrak{a}_{N'}) = a\Big(E_{j'}, \widehat{X}, \frac{1}{m}\operatorname{div}(h_*)\Big) \quad \text{for } 1 \le j' \le j.$$

In particular $\operatorname{mld}(\widehat{X} \ni x, \mathfrak{a}_{N'}) = 0$. Let $x'^{(0)} = x, x'^{(1)}, \dots, x'^{(r'-1)}$ and $E'_1, \dots, E'_{r'}$ be the points and curves of Lemma 3.4 for $\mathfrak{a}_{N'}$. Since $x'^{(0)} = x^{(0)} = x$, we have $E'_1 = E_1$. If $E'_{j'} = E_{j'}$ for some $j' < j$, then $a(E'_{j'}, \widehat{X}, \mathfrak{a}_{N'}) > 0$, so $j' < r'$, and $x'^{(j')}$ is the unique closed point of $E_{j'}$ which is the center of a prime divisor of log discrepancy zero with respect to $(\widehat{X}, \mathfrak{a}_{N'})$ (Lemma 3.4(1)). Since E_j is such a prime divisor with center $x^{(j')}$, we have $x'^{(j')} = x^{(j')}$ and $E'_{j'+1} = E_{j'+1}$. By induction $E'_j = E_j$, and $a(E_j, \widehat{X}, \mathfrak{a}_{N'}) = 0$ forces $r' = j$ by Lemma 3.4(1). By Lemma 3.4(2)(3), E is the exceptional divisor of the weighted blow-up of $\widehat{X}$ with coprime weights $w_1 \ge w_2 \ge 1$ in some regular parameters y_1, y_2 of $\Bbbk((s))[[p,q]]$, and E is invariant under any automorphism of $\widehat{X}$ fixing x and preserving $\mathfrak{a}_{N'}$, in particular under G. Since $x^{(j')}$ is the center of E on the surface $S_{j'}$ of Lemma 3.4 for $0 \le j' < j$, with $S_0 = \widehat{X}$, the sequence of blow-ups in (1) consists of the blow-ups of $x^{(0)}, \dots, x^{(j-1)}$, with exceptional curves $E_1, \dots, E_j$, so E satisfies the first sentence of (1).

Step 2. In this step we construct σ and Φ.

Let v be the valuation mult_E on $\Bbbk((s))[[p,q]]$. It is the monomial valuation with $v(y_1) = w_1$ and $v(y_2) = w_2$, and the graded ring of v is $\Bbbk((s))[Y_1, Y_2]$ with Y_1, Y_2 the initial forms of y_1, y_2. Since $(p,q) = (y_1, y_2)$, $\min\{v(p), v(q)\} = w_2$. Possibly exchanging p and q, we may assume that $v(q) = w_2$. If $w_1 = w_2$, then $w_1 = w_2 = 1$, v is the order along (p,q), and we let $\sigma := 0$.

Assume that $w_1 > w_2$. The only monomial of weight w_2 is Y_2, so the initial form of q is κY_2 for some $\kappa \in \Bbbk((s))^\times$. If $f \in (p,q)$ and $v(f) < w_1$, then the initial form of f is homogeneous of weight $v(f) < w_1$, hence equal to $\kappa' Y_2^n$ with $nw_2 = v(f)$, $n \ge 1$, and $\kappa' \in \Bbbk((s))$, so $v(f - \kappa'\kappa^{-n} q^n) > v(f)$. Starting from $f = p$ and repeating, we obtain $\sigma \in \Bbbk((s))[q]$ with $\sigma(s,0) = 0$, $\deg_q \sigma < w_1/w_2$, and $v(p - \sigma(s,q)) \ge w_1$. Such a σ is unique: for another one σ', $v(\sigma - \sigma') \ge w_1$, whereas a non-zero element of $\Bbbk((s))[q]$ of degree less than w_1/w_2 has value less than w_1.

Let $p_1 := p - \sigma(s,q)$. Modulo $(y_1, y_2)^2$, write $q \equiv \kappa y_2 + \kappa_1 y_1$ and $p_1 \equiv \alpha y_1 + \gamma y_2$. Since $v(p_1) \ge w_1 > w_2$ and the monomials of $(y_1, y_2)^2$ have weight greater than w_2, we have $\gamma = 0$. Since (p_1, q) are regular parameters, $\alpha \ne 0$. Hence the initial form of p_1 is $\alpha Y_1 + \kappa_2 Y_2^{w_1/w_2}$ for some $\kappa_2 \in \Bbbk((s))$ ($\kappa_2 = 0$ unless $w_2 = 1$), and $v(p_1) = w_1$. The initial forms of p_1 and q generate $\Bbbk((s))[Y_1, Y_2]$, so the initial forms of the monomials $p_1^{a_1} q^{a_2}$ are linearly independent. Thus for any $f = \sum f_{a_1,a_2} p_1^{a_1} q^{a_2} \in \Bbbk((s))[[p_1, q]]$,

$$v(f) = \min\{a_1 w_1 + a_2 w_2 \mid f_{a_1,a_2} \ne 0\},$$

that is, v is the monomial valuation with weights (w_1, w_2) in (p_1, q), and E is the exceptional divisor of the weighted blow-up in (p_1, q).

For $g \in G$, g fixes E, so $v(g \cdot p_1) = w_1$. Since $g \cdot p = \chi_p(g) p$, $g \cdot q = \chi_q(g) q$, and g acts on $\Bbbk((s))$ by $s \to \chi_s(g) s$, the polynomial $\chi_p(g)^{-1} \sigma(\chi_s(g) s, \chi_q(g) q)$ satisfies the conditions defining σ, so

$$\sigma(\chi_s(g) s, \chi_q(g) q) = \chi_p(g) \sigma(s,q) \quad \text{for any } g \in G, \tag{7.40}$$

that is, $(p,q) \to (p + \sigma(s,q), q)$ commutes with G.

Let

$$h_*^\sigma(s,p,q) := h_*(s, p + \sigma(s,q), q) \in \Bbbk((s))[[p,q]],$$

so that $h_* = h_*^\sigma(s, p_1, q)$. Since $a(E, \widehat{X}, 0) = w_1 + w_2$ and $a(E, \widehat{X}, \frac{1}{m}\operatorname{div}(h_*)) = 0$, we have $v(h_*) = d$. Thus h_*^σ has no monomial of weight less than d, and the part Φ of h_*^σ of weight d is a non-zero (w_1, w_2)-homogeneous polynomial in $\Bbbk((s))[p, q]$. Since $\frac{1}{m} \cdot d = w_1 + w_2$, $(\mathbb{A}^2_{\Bbbk((s))}, \frac{1}{m}\operatorname{div}(\Phi))$ is a critical pair for (w_1, w_2). We show in **Step 6** that $(\mathbb{A}^2_{\Bbbk((s))}, \frac{1}{m}\operatorname{div}(\Phi))$ is lc.

Step 3. In this step we construct the charts used below.

Fix i. Possibly shrinking V_i to a G-invariant neighborhood of o_i, we may assume that $(s_i, p_i, q_i) : V_i \to \mathbb{A}^3$ is étale. Let $k \in |G|\mathbb{N}$ and $c \in \mathbb{Z}$, let $\lambda \in \mathbb{C}[s]$ with $\lambda(0) = 0$, and let $\mu \in \mathbb{C}[s, s^{-1}][q]$ such that $\mu(s, s^{kw_2}q) \in \mathbb{C}[s, q]$ vanishes at 0, and such that for any $g \in G$,

$$\lambda(\chi_s(g)s) = \chi_q(g)\lambda(s) \quad \text{and} \quad \mu(\chi_s(g)s, \chi_q(g)q) = \chi_p(g)\mu(s, q).$$

Let $\phi : \mathbb{A}^3 \to \mathbb{A}^3$ be the morphism

$$(s, p', q') \to \big(s,\ s^{kw_1}p' - \mu(s, s^{kw_2}q'),\ s^{kw_2}q' - \lambda(s)\big). \tag{7.41}$$

Then $\phi(0) = 0$, ϕ is an isomorphism over $(s \neq 0)$, the Jacobian determinant of ϕ is $s^{k(w_1+w_2)}$, and ϕ is G-equivariant when G acts on p', q' by χ_p, χ_q, since $\chi_s^k = 1$. Let $(Y \ni y)$ be the germ at $(o_i, 0)$ of the fiber product of (s_i, p_i, q_i) and ϕ, and $\pi : Y \to V_i$ the first projection. Then Y is smooth with G-eigen regular parameters s, p', q' at y, π is G-equivariant and birational, $\pi(y) = o_i$, and

$$K_Y = \pi^* K_{V_i} + k(w_1 + w_2)\operatorname{div}(s).$$

We call $(Y \ni y)$ the *chart of depth k with translations (λ, μ) and offset c*. It is *h_i-divisible* if s^{kd+c} divides $\pi^* h_i$ in $\widehat{\mathcal{O}}_{Y,y} = \mathbb{C}[[s, p', q']]$. In this case let $Q := s^{-kd-c}\pi^* h_i$. Then

$$K_Y + B_Y = \pi^*(K_{V_i} + B_i), \quad \text{where } B_Y := \frac{1}{m}\operatorname{div}(Q) + \Big(\beta + \frac{c}{m}\Big)\operatorname{div}(s). \tag{7.42}$$

Any prime divisor T over Y with center y is a prime divisor over V_i with center o_i. Since π is G-equivariant and birational, e_T computed over Y equals e_T computed over V_i, and $\pi^* s_i = s$. By (7.42) and $\pi^* s_i = s$,

$$a(T, Y, B_Y) = a(T, V_i, B_i) \quad \text{and} \quad \operatorname{mult}_T s = \operatorname{mult}_T s_i,$$

so $\operatorname{mld}((Y \ni y, B_Y)/G) \geq b_i$, and

$$a(T, V_i, 0) = a(T, Y, 0) + k(w_1 + w_2)\operatorname{mult}_T s. \tag{7.43}$$

Conversely, $\mathcal{O}_{Y,y}$ is a localization of $\mathcal{O}_{V_i,o_i}[p', q']$ at a maximal ideal containing p', q' and the maximal ideal of o_i, so a prime divisor F over V_i with center o_i and $\operatorname{mult}_F p', \operatorname{mult}_F q' > 0$ has center y on Y. We then say that the chart *captures* F. If the chart is h_i-divisible and captures F_i, then by (iii) and (7.42),

$$\operatorname{mld}((Y \ni y, B_Y)/G) = b_i,$$

F_i computes it, and by (ii), $a(T, Y, B_Y) \geq \eta \operatorname{mult}_T s$ for any prime divisor T over Y with center y.

Step 4. In this step we pass prime divisors from limits to the members. Let $\widehat{\mathcal{X}}^\circ := \widehat{\mathcal{X}} \backslash (s = 0)$.

Claim 7.14. *For any i, let $(Y_i \ni y_i)$ be an h_i-divisible chart of offset c. Let Q_i be the function Q of **Step 3** for $(Y_i \ni y_i)$, and $Q_* \in \Bbbk[[s, p, q]]$ a compatible generic limit of $(Q_i)_i$. Let T be a prime divisor over $\widehat{\mathcal{X}}$ with center 0. Then for infinitely many i, there exists a prime divisor T_i over Y_i with center y_i such that*

$$a(T_i, Y_i, 0) = a(T, \widehat{\mathcal{X}}, 0), \quad \operatorname{mult}_{T_i} s = \operatorname{mult}_T s, \quad \operatorname{mult}_{T_i} Q_i = \operatorname{mult}_T Q_*, \quad \text{and} \quad e_{T_i} = e_T.$$

In particular

$$a\Big(T, \widehat{\mathcal{X}}, \frac{1}{m}\operatorname{div}(Q_*) + \Big(\beta + \frac{c}{m}\Big)\operatorname{div}(s)\Big) \geq \eta \operatorname{mult}_T s, \quad \text{and}$$
$$a\Big(T, \widehat{\mathcal{X}}, \frac{1}{m}\operatorname{div}(Q_*) + \Big(\beta + \frac{c}{m}\Big)\operatorname{div}(s)\Big) \geq b e_T \quad \text{if } b_i = b \text{ for any } i.$$

Proof. We identify $\widehat{\mathcal{O}}_{Y_i,y_i}$ with $\mathbb{C}[[s,p,q]]$ via the regular parameters s,p',q' of **Step 3**. By Lemma 2.13 for the morphism $(s,p',q') : Y_i \to \mathbb{A}^3$, étale at y_i (after shrinking Y_i, y_i is the only point over 0), prime divisors over Y_i with center y_i correspond to prime divisors over $\mathbb{A}^3$ with center 0, preserving log discrepancies, orders, decomposition groups, and inertia groups. Let M be an integer greater than $\operatorname{mult}_T Q_*$, and let $Q_i^{<M}$ and $Q_*^{<M}$ be the sums of the terms of Q_i and Q_* of degree less than M. Then $Q_*^{<M}$ is a compatible generic limit of $(Q_i^{<M})_i$, $\operatorname{mult}_T Q_*^{<M} = \operatorname{mult}_T Q_*$, and $\operatorname{mult}_{T'} Q_i = \operatorname{mult}_{T'} Q_i^{<M}$ for any prime divisor T' over $\mathbb{A}^3$ with center 0 and $\operatorname{mult}_{T'} Q_i^{<M} < M$, since $\operatorname{mult}_{T'}(s,p,q) \geq 1$. By [Kaw21, Remark 3.8(i)] for the polynomials $Q_i^{<M}$ and s, T descends to a prime divisor T_Λ over $\mathbb{A}^3 \times \Lambda$, where Λ is a variety whose function field, in $\Bbbk$, contains a field of definition of T, with a dominant morphism to the set Z_M of Definition 2.5 for the tuples $(Q_i^{<M}, s)$, and for infinitely many i the fiber T_i of T_Λ over a point of Λ over $\operatorname{jet}_M(Q_i^{<M}, s)$ satisfies $a(T_i, \mathbb{A}^3, 0) = a(T, \widehat{\mathcal{X}}, 0)$, $\operatorname{mult}_{T_i} Q_i^{<M} = \operatorname{mult}_T Q_*^{<M}$, and $\operatorname{mult}_{T_i} s = \operatorname{mult}_T s$. We also write T_i for the corresponding prime divisor over Y_i with center y_i. Since G acts on $\mathbb{A}^3$ and commutes with the projection to Λ, gT_Λ is the descent of gT for any $g \in G$. Possibly shrinking Λ to a non-empty open subset, the fibers of T_Λ are irreducible with center the origin, and for any $g \in G$ the conditions $gT_\Lambda = T_\Lambda$ and that g acts trivially on T_Λ hold at the generic point of Λ if and only if they hold on any fiber, so $e_{T_i} = e_T$. Since $\operatorname{mult}_{T_i} Q_i^{<M} = \operatorname{mult}_T Q_* < M$, $\operatorname{mult}_{T_i} Q_i = \operatorname{mult}_T Q_*$. The inequalities in the display of the claim follow from (ii), from $\operatorname{mld}((Y_i \ni y_i, B_{Y_i})/G) \geq b_i$ (**Step 3**), and from Definition 2.2. □

Claim 7.15. *Let $f \in \Bbbk[[s,p,q]]$ be non-zero and $\beta' \in \mathbb{R}$, and assume that*

$$a\Big(T, \widehat{\mathcal{X}}, \frac{1}{m}\operatorname{div}(f) + \beta' \operatorname{div}(s)\Big) \geq 0$$

for any prime divisor T over $\widehat{\mathcal{X}}$ with center 0. Then $(\widehat{\mathcal{X}}^\circ, \frac{1}{m}\operatorname{div}(f))$ is lc.

Proof. If a prime divisor D over $\widehat{\mathcal{X}}$ whose center is not contained in $(s = 0)$ had $a(D, \widehat{\mathcal{X}}, \frac{1}{m}\operatorname{div}(f)) < 0$, then, since the center of D contains 0 and $\operatorname{mult}_D s = 0$, on a log resolution of $(\widehat{\mathcal{X}}, \operatorname{div}(sf))$ on which D is a divisor, the weighted blow-ups with weights $(n,1,1)$, $n \gg 0$, at a closed point of D over 0 in snc coordinates (z_1,z_2,z_3) with $D = (z_1 = 0)$ would yield prime divisors over $\widehat{\mathcal{X}}$ with center 0 and arbitrarily negative log discrepancy with respect to $\frac{1}{m}\operatorname{div}(f) + \beta' \operatorname{div}(s)$, contrary to the assumption. □

Step 5. In this step we reduce to the case $\sigma = 0$ and $\Phi \in \Bbbk[[s]][p,q] \backslash s\Bbbk[[s]][p,q]$.

Let $c_\sigma \in \mathbb{N}$ with $s^{c_\sigma}\sigma \in \Bbbk[[s]][q]$. For a monomial $p^{a_1}q^{a_2}$ of h_*, any monomial $p^{a_1'}q^{a_2'}$ of $(p + \sigma(s,q))^{a_1}q^{a_2}$ has a coefficient of s-order at least $-c_\sigma(a_1 - a_1')$, and $a_2' \geq a_2 + a_1 - a_1'$ since $\sigma(s,0) = 0$. Thus the coefficient of a monomial of weight w in h_*^σ is a finite sum of s-order at least $-c_\sigma w/w_2$. Let c be the minimum of the s-orders of the coefficients of Φ, and fix a positive integer k_0 divisible by $|G|$ with

$$k_0 w_2 - c_\sigma \geq 1, \quad k_0 d + c \geq 0, \quad \text{and} \quad k_0 \geq \frac{c_\sigma(d+1)}{w_2} + c;$$

since $k_0 - c_\sigma/w_2 > 0$, the last inequality implies $k_0(w-d) - c_\sigma w/w_2 - c \geq 0$ for any integer $w \geq d+1$.

For $n \in \mathbb{N}^+$ let $Z_n \subset \mathcal{J}_n$ be the irreducible closed subset of Definition 2.5 with generic point the n-jet $\operatorname{jet}_n(h_*)$. By **Step 2**, σ is determined by E, and by **Step 1**, E is invariant under any automorphism of $\widehat{X}$ fixing x and preserving $\mathfrak{a}_{N'}$. Thus any automorphism of $\Bbbk$ over the algebraic closure $\overline{\Bbbk_h}$ in $\Bbbk$ of the field $\Bbbk_h$ generated over $\mathbb{C}$ by the coefficients of h_*, acting on coefficients, fixes σ, so $\sigma \in \overline{\Bbbk_h}((s))[q]$, and finitely many coefficients of σ are algebraic over the field generated over $\mathbb{C}$ by the coefficients of $\operatorname{jet}_{n'}(h_*)$ for $n' \gg 0$. Since $k_0 w_2 - c_\sigma \geq 1$, for $n \in \mathbb{N}^+$ there exist integers $n' \geq n$ and $N_n > 0$ such that

$$\operatorname{jet}_n\big(s^{-k_0 d - c} h_*^\sigma(s, s^{k_0 w_1}p, s^{k_0 w_2}q)\big) = \tau_n\big(\operatorname{jet}_{n'}(h_*), \xi_n\big)$$

for a polynomial map τ_n with complex coefficients, where $\xi_n \in \Bbbk^{N_n w_1}$ is the vector of the coefficients of $s^{c_\sigma}\sigma$ of s-degree less than N_n and q-degree less than w_1, and

$$\operatorname{jet}_n\left(s^{-k_0 d-c} h_i(s, s^{k_0 w_1} p + \sigma'(s, s^{k_0 w_2} q), s^{k_0 w_2} q)\right) = \tau_n\left(\operatorname{jet}_{n'}(h_i), \xi'\right)$$

for any $\sigma' \in \mathbb{C}[s, s^{-1}][q]$ with $s^{c_\sigma}\sigma' \in \mathbb{C}[s][q]$, $\sigma'(s,0) = 0$, and $\deg_q \sigma' < w_1/w_2$, and the vector ξ' of the coefficients of $s^{c_\sigma}\sigma'$ of s-degree less than N_n and q-degree less than w_1. We choose n' and N_n increasing in n and such that ξ_n is algebraic over the field generated over $\mathbb{C}$ by the coefficients of $\operatorname{jet}_{n'}(h_*)$. Let $W_n \subset Z_{n'} \times \mathbb{A}^{N_n w_1}$ be the closure of the point $(\operatorname{jet}_{n'}(h_*), \xi_n)$. Then $W_n \to Z_{n'}$ is dominant and generically finite, and the projection maps W_{n+1} into W_n.

By Lemma 2.7, with the generators of the ideal of W_n as identities, for any n and all but finitely many i there exists $\xi_{n,i} \in \mathbb{C}^{N_n w_1}$ with $(\operatorname{jet}_{n'}(h_i), \xi_{n,i}) \in W_n$. Choose $n(i) \in \mathbb{N}^+$ with $\lim_i n(i) = +\infty$ such that $\xi_{n(i),i}$ exists for all but finitely many i, let $\sigma_i \in \mathbb{C}[s, s^{-1}][q]$ be the polynomial such that the coefficients of $s^{c_\sigma}\sigma_i$ of s-degree less than $N_{n(i)}$ and q-degree less than w_1 are $\xi_{n(i),i}$ and the others vanish, and for $n \le n(i)$ let $\omega_{n,i} \in W_n$ be the image of the point of $W_{n(i)}$ with last coordinates $\xi_{n(i),i}$. Then $s^{c_\sigma}\sigma_i \in \mathbb{C}[s][q]$, and since the conditions $\sigma(s,0) = 0$, $\deg_q \sigma < w_1/w_2$, and (7.40) say that the coefficients of $s^{c_\sigma}\sigma$ outside a set of positions vanish, $\sigma_i(s,0) = 0$, $\deg_q \sigma_i < w_1/w_2$, and σ_i satisfies (7.40).

Consider the chart of depth k_0 with translations $(0, -\sigma_i)$ and offset c, that is,

$$p_i = s^{k_0 w_1} p' + \sigma_i(s, s^{k_0 w_2} q'), \quad q_i = s^{k_0 w_2} q'.$$

The chart is h_i-divisible if and only if for any monomial of weight w, the coefficient of the monomial in $h_i(s, p + \sigma_i(s,q), q)$ has s-order at least $k_0(d-w)+c$. For $w \ge d+1$ it holds by the choice of k_0. For the finitely many monomials of weight $w \le d$ it is the vanishing of finitely many s-jets of coefficients, which holds for h_*^σ since h_*^σ has no monomial of weight less than d and c is the minimal s-order of Φ. The conditions for $w \le d$ are polynomial identities in $(\operatorname{jet}_{n'}(h_*), \xi_n)$ for some n, so they hold on W_n, hence at $\omega_{n,i}$, and the chart is h_i-divisible for all but finitely many i. Since $\operatorname{mult}_{F_i} s \le R'$ and $\operatorname{mult}_{F_i} \sigma_i(s, q_i) \ge \operatorname{mult}_{F_i} q_i - c_\sigma R'$,

$$\operatorname{mult}_{F_i} q' \ge \operatorname{mult}_{F_i} q_i - k_0 w_2 R' \quad \text{and} \quad \operatorname{mult}_{F_i} p' \ge \min\{\operatorname{mult}_{F_i} p_i, \operatorname{mult}_{F_i} q_i - c_\sigma R'\} - k_0 w_1 R',$$

which tend to $+\infty$. In particular the chart captures F_i for $i \gg 0$. Let h_i' be the function Q of **Step 3** for the chart of depth k_0 with translations $(0, -\sigma_i)$ and offset c, and

$$h_*' := s^{-k_0 d - c} h_*^\sigma(s, s^{k_0 w_1} p, s^{k_0 w_2} q) \in \Bbbk[[s, p, q]],$$

whose part of weight w is $s^{k_0(w-d)-c}$ times the part of weight w of h_*^σ. Thus h_*' has no monomial of weight less than d, and the part of h_*' of weight d is $s^{-c}\Phi \in \Bbbk[[s]][p,q] \setminus s\Bbbk[[s]][p,q]$, and $\operatorname{div}(s^{-c}\Phi) = \operatorname{div}(\Phi)$ on $\mathbb{A}^2_{\Bbbk((s))}$ since s is a unit there. We show that h_*' is a compatible generic limit of $(h_i')_i$. Fix n. Let $W' \subset W_n$ be a closed subset containing $\omega_{n,i}$ for infinitely many i, and W'' an irreducible component of W' containing $\omega_{n,i}$ for infinitely many i. By Definition 2.5, a proper closed subset of $Z_{n'}$ contains $\operatorname{jet}_{n'}(h_i)$ for only finitely many i, so the image of W'' is dense in $Z_{n'}$, $\dim W'' \ge \dim Z_{n'} = \dim W_n$, and $W' = W'' = W_n$. Thus a proper closed subset of W_n contains $\omega_{n,i}$ for only finitely many i. Since

$$\operatorname{jet}_n(h_*') = \tau_n(\operatorname{jet}_{n'}(h_*), \xi_n)$$

is the generic point of the closure Z_n' of $\tau_n(W_n)$ and $\operatorname{jet}_n(h_i') = \tau_n(\omega_{n,i})$ for all but finitely many i, any non-empty open subset of Z_n' contains $\operatorname{jet}_n(h_i')$ for all but finitely many i. As $s^{-c}\Phi$ has a monomial $p^{a_1}q^{a_2}$ whose coefficient has non-zero constant term, the coefficient of $p^{a_1}q^{a_2}$ in h_i' has non-zero constant term for all but finitely many i, so $\operatorname{mult}_0 h_i' \le a_1 + a_2$.

Possibly replacing $(V_i \ni o_i)$, (s_i, p_i, q_i), h_i, β, h_*, and Φ with the chart, (s, p', q'), h_i', $\beta + \frac{c}{m}$, h_*', and $s^{-c}\Phi$, we may assume that $\sigma = 0$ and $\Phi \in \Bbbk[[s]][p,q] \setminus s\Bbbk[[s]][p,q]$. By **Step 3**, (i), (ii), and (iii) still hold, and by (7.43) it suffices to prove (2) for prime divisors E_i with $\operatorname{mult}_{E_i} s$ bounded independently of i. Below we only use that h_* has no monomial of weight less than d and that Φ is the part of h_* of weight d.

Step 6. In this step we prove that $(\mathbb{A}^2_{\overline{\Bbbk((s))}}, \frac{1}{m}\operatorname{div}(\Phi))$ is lc, which finishes the proof of (1).

Let $k \in |G|\mathbb{N}^+$. Consider the chart of depth k with translations $(0,0)$ and offset 0, that is, $p_i = s^{kw_1}p'$ and $q_i = s^{kw_2}q'$. It is h_i-divisible for all but finitely many i: the coefficient of a monomial of weight $w < d$ of h_i must be divisible by $s^{k(d-w)}$, a polynomial identity which holds for h_*, so for h_i by Lemma 2.7. It captures F_i for $i \gg 0$ by (iii). The functions $Q = s^{-kd}h_i(s, s^{kw_1}p, s^{kw_2}q)$ of **Step 3** for the charts of depth k with translations $(0,0)$ and offset 0 have, by the argument of **Step 5** with $\sigma = 0$, $c = 0$, and k in place of k_0, the compatible generic limit $h_*^{(k)} := s^{-kd}h_*(s, s^{kw_1}p, s^{kw_2}q)$, and since the part of weight $w > d$ is multiplied by $s^{k(w-d)}$,

$$h_*^{(k)} - \Phi \in s^k\Bbbk[[s,p,q]]. \tag{7.44}$$

Let T be a prime divisor over $\widehat{\mathcal{X}}$ with center 0. Then $\operatorname{mult}_T s \geq 1$. Taking $k > \operatorname{mult}_T \Phi$, (7.44) implies $\operatorname{mult}_T h_*^{(k)} = \operatorname{mult}_T \Phi$, and Claim 7.14 implies

$$a\Big(T, \widehat{\mathcal{X}}, \frac{1}{m}\operatorname{div}(\Phi) + \beta \operatorname{div}(s)\Big) \geq \eta \operatorname{mult}_T s > 0. \tag{7.45}$$

By Claim 7.15, $(\widehat{\mathcal{X}}^\circ, \frac{1}{m}\operatorname{div}(\Phi))$ is lc. The completions at the generic point of C_* and at the origin of $\mathbb{A}^2_{\Bbbk((s))}$ are both $(\widehat{X}, \frac{1}{m}\operatorname{div}(\Phi))$, so $(\mathbb{A}^2_{\Bbbk((s))}, \frac{1}{m}\operatorname{div}(\Phi))$ is lc near the origin. The non-lc locus of $(\mathbb{A}^2_{\Bbbk((s))}, \frac{1}{m}\operatorname{div}(\Phi))$ is closed and invariant under the $\mathbb{G}_m$-action of weights (w_1, w_2) on (p,q), and the closure of any orbit contains the origin, so the pair is lc. A log resolution over $\Bbbk((s))$ stays a log resolution over $\overline{\Bbbk((s))}$, which proves (1).

Step 7. From now on we assume that $\frac{1}{m}\operatorname{div}(\Phi)$ has no component of coefficient one on $\mathbb{A}^2_{\overline{\Bbbk((s))}}$. In this step we construct charts of maximal depth.

For any prime divisor T over V_i with center o_i, $a(T, V_i, B_i) \in \frac{1}{m}\mathbb{Z}$ and e_T divides $|G|$, so $b_i \in \frac{1}{m|G|}\mathbb{Z} \cap (0, \bar{b}]$, a finite set. Passing to an infinite subset of indices, $b_i = b$ for any i.

For $k \in |G|\mathbb{N}^+$, the chart of depth k of **Step 6** is h_i-divisible and captures F_i for $i \gg 0$. Thus for $i \gg 0$ the largest $k_i^0 \in |G|\mathbb{N}^+$ with $k_i^0 \leq i$ such that the chart of depth k_i^0 of **Step 6** is h_i-divisible and captures F_i is defined, and $\lim_i k_i^0 = +\infty$. Fix positive integers n_i with $\lim_i n_i = +\infty$ and $\lim_i n_i/k_i^0 = 0$. We say that $k \in |G|\mathbb{N}^+$ with $k \geq k_i^0$ is *admissible* for i if there exist $\lambda \in s^{n_i}\mathbb{C}[s]$ and $\mu \in s^{n_i}\mathbb{C}[s][q]$ with $\deg_q \mu < w_1/w_2$, satisfying the character conditions of **Step 3**, such that the chart of depth k with translations (λ, μ) and offset 0 is h_i-divisible and captures F_i. The depth k_i^0 is admissible with $\lambda = \mu = 0$.

For such a chart, $x_1 := p_i + \mu(s, q_i + \lambda(s))$, $x_2 := q_i + \lambda(s)$, and s are regular parameters at o_i, and $x_1 = s^{kw_1}p'$, $x_2 = s^{kw_2}q'$ by (7.41). Since $(V_i, \operatorname{div}(x_1x_2s))$ is lc and the chart captures F_i,

$$a(F_i, V_i, 0) \geq \operatorname{mult}_{F_i} x_1 + \operatorname{mult}_{F_i} x_2 + \operatorname{mult}_{F_i} s \geq \big(k(w_1+w_2)+1\big)\operatorname{mult}_{F_i} s + 2 > k.$$

Hence the set of admissible depths for i is finite. Let k_i be the largest one, (λ_i, μ_i) translations for k_i, $(Y_i \ni y_i)$ the chart, and Q_i the function Q of **Step 3** for it. By **Step 3**,

$$\operatorname{mld}\Big(\Big(Y_i \ni y_i, \frac{1}{m}\operatorname{div}(Q_i) + \beta\operatorname{div}(s)\Big)\Big/G\Big) = b,$$

F_i computes it, and (ii) holds on $(Y_i \ni y_i)$.

Step 8. In this step we prove that Q_i and h_* have the common top part Φ in the limit, and we compare the quotient values.

Let $\widetilde{h}_i(s,p,q) := h_i(s, p - \mu_i(s,q), q - \lambda_i(s))$. Then

$$Q_i = s^{-k_id}\widetilde{h}_i(s, s^{k_iw_1}p, s^{k_iw_2}q) \quad \text{and} \quad \widetilde{h}_i - h_i \in s^{n_i}\mathbb{C}[[s,p,q]].$$

The orders of Q_i at 0 are bounded: since $\Phi \notin s\Bbbk[[s]][p,q]$, h_* has a monomial $p^{a_1}q^{a_2}$ of weight d whose coefficient has non-zero constant term. By Lemma 2.7 so does h_i for all but finitely many i, hence so does $\widetilde{h}_i$ since $\widetilde{h}_i - h_i \in (s^{n_i})$, and the coefficient of a monomial of weight d in

Q_i equals that in $\widetilde{h}_i$, so $\operatorname{mult}_0 Q_i \le a_1 + a_2 \le d/w_2$. By Lemma 2.6(3), passing to an infinite subset of indices, there exist an algebraically closed field $\Bbbk'$, an embedding $\Bbbk \to \Bbbk'$ over $\mathbb{C}$, and a compatible generic limit of $(h_i, Q_i)_i$ with coefficient field $\Bbbk'$ whose first entry is the image of h_* and whose second entry we denote by Q_*. Replacing $\Bbbk$, h_*, and Φ with $\Bbbk'$ and the images of h_* and Φ, the conclusions of (1), the assumption of (2), and the reduction of **Step 5** still hold, since log resolutions and irreducible factorizations over an algebraically closed field stay so over a larger one. Thus $Q_* \in \Bbbk[[s,p,q]]$. Since $\lim_i n_i = +\infty$, any jet of $\widetilde{h}_i$ coincides with that of h_i for $i \gg 0$. The coefficient of a monomial of weight w in Q_i is $s^{k_i(w-d)}$ times that in $\widetilde{h}_i$, and $\lim_i k_i = +\infty$, so

$$\begin{gathered} Q_* \in \Bbbk[[s]][p,q] \text{ has no monomial of weight greater than } d, \\ \text{and the part of weight } d \text{ of } Q_* \text{ is } \Phi. \end{gathered} \tag{7.46}$$

In particular $Q_* \ne 0$ and Φ is the top part of Q_*. By Claim 7.14, for any prime divisor T over $\widehat{\mathcal{X}}$ with center 0,

$$a\Big(T, \widehat{\mathcal{X}}, \frac{1}{m}\operatorname{div}(Q_*) + \beta \operatorname{div}(s)\Big) \ge \eta \operatorname{mult}_T s > 0. \tag{7.47}$$

Let a_Φ and a_Q be the quotient values with G at the closed point of $\widehat{\mathcal{X}}$ of $\frac{1}{m}\operatorname{div}(\Phi) + \beta \operatorname{div}(s)$ and of $\frac{1}{m}\operatorname{div}(Q_*) + \beta \operatorname{div}(s)$ (Definition 2.2). For a prime divisor T over $\widehat{\mathcal{X}}$ with center 0, take $k > \operatorname{mult}_T \Phi$ as in **Step 6**. Claim 7.14 for the charts of depth k implies

$$a\Big(T, \widehat{\mathcal{X}}, \frac{1}{m}\operatorname{div}(\Phi) + \beta \operatorname{div}(s)\Big) \ge b e_T.$$

Thus $a_\Phi \ge b > 0$, and $a_\Phi < +\infty$. By Lemma 7.11, with $u = s$, $(x_1, x_2) = (p, q)$ of weights (w_1, w_2), $N = 1$, $Q_1 = Q_*$, whose P_1 is the top part Φ of Q_* by (7.46), and $c_1 = \frac{1}{m}$,

$$b \le a_\Phi \le a_Q. \tag{7.48}$$

We show that any component of $\frac{1}{m}\operatorname{div}(Q_*)$ on $\mathbb{A}^2_{\Bbbk((s))}$ has coefficient less than one. Let ψ be an irreducible factor of Q_* of multiplicity at least m. Since ψ is not constant and $w_1, w_2 > 0$, the top part ψ^+ of ψ is not constant, and since the top part of a product is the product of the top parts, $(\psi^+)^m$ divides Φ. Hence $\frac{1}{m}\operatorname{div}(\Phi)$ has a component of coefficient at least one, which is one by (1), a contradiction. By (7.47) and Claim 7.15, $(\widehat{\mathcal{X}}^\circ, \frac{1}{m}\operatorname{div}(Q_*))$ is lc.

Step 9. In this step we prove that, passing to an infinite subset of indices, $\operatorname{mult}_{F_i} Q_i$ is bounded.

Suppose not. Passing to an infinite subset of indices, $\lim_i \operatorname{mult}_{F_i} Q_i = +\infty$. By **Step 7** and (iii),

$$a(F_i, Y_i, 0) - \frac{1}{m}\operatorname{mult}_{F_i} Q_i = b e_{F_i} + \beta \operatorname{mult}_{F_i} s \le \bar{b}|G| + |\beta| R' \quad \text{and} \quad \operatorname{mult}_{F_i} s \le R'. \tag{7.49}$$

By Lemma 2.13 for the morphism $(s,p,q) : Y_i \to \mathbb{A}^3$, étale at y_i (after shrinking Y_i, y_i is the only point over 0), prime divisors over Y_i with center y_i correspond to prime divisors over $\mathbb{A}^3$ with center 0, preserving log discrepancies and orders. Replacing Q_i with the sum of the terms of Q_i of total degree at most $i + \operatorname{mult}_{F_i} Q_i$ changes neither $\operatorname{mult}_{F_i} Q_i$ nor the limit Q_*, and the new Q_i are polynomials. By **Step 8**, the orders of the Q_i at 0 are bounded, and the truncation keeps them since $i + \operatorname{mult}_{F_i} Q_i \ge \operatorname{mult}_0 Q_i$. Since $\lim_i \operatorname{mult}_{F_i} Q_i = +\infty$ and conditions (1) and (2) of Proposition 6.2, with $(z_1, z_2, z_3) = (s, p, q)$, $r = z_1$, $N = 1$, $g_{i,1} = Q_i$, $\kappa_1 = \frac{1}{m}$, $R_0 = R'$, and $C_0 = \bar{b}|G| + |\beta|R'$, hold by (7.49), condition (3) of Proposition 6.2 fails, that is, $(\widehat{\mathcal{X}}^\circ, \frac{1}{m}\operatorname{div}(Q_*))$ is not klt.

Let z be a non-klt point of $(\widehat{\mathcal{X}}^\circ, \frac{1}{m}\operatorname{div}(Q_*))$. Since the pair is lc with coefficients less than one, z is a closed point of $\widehat{\mathcal{X}}^\circ$. For the prime ideal $\mathfrak{p}_z$ of z, $\Bbbk[[s,p,q]]/\mathfrak{p}_z$ is finite over $\Bbbk[[s]]$,

so z corresponds to a point (p_0, q_0) of $\mathbb{A}^2(\overline{\Bbbk((s))})$ with p_0, q_0 of positive s-order, at which $(\mathbb{A}^2_{\overline{\Bbbk((s))}}, \frac{1}{m}\operatorname{div}(Q_*))$ is lc and not klt, since

$$\widehat{\mathcal{O}}_{\widehat{\mathcal{X}},z} \cong \widehat{\mathcal{O}}_{\mathbb{A}^2_{\overline{\Bbbk((s))}},z'}$$

for the closed point z' of $\mathbb{A}^2_{\overline{\Bbbk((s))}}$ corresponding to z. The polynomial $Q_*(s, p + p_0, q + q_0)$ also has top part Φ, and any component through 0 of the divisor it defines has coefficient less than one. By Proposition 7.4(1), with $n = 1$, $\kappa_1 = \frac{1}{m}$, and $d_1 = d$, (p_0, q_0) is the only non-klt point of $(\mathbb{A}^2_{\overline{\Bbbk((s))}}, \frac{1}{m}\operatorname{div}(Q_*))$, and

$$Q_*(s, p + p_0, q + q_0) = \Phi(s, p + U(s, q), q) \tag{7.50}$$

for some $U \in \overline{\Bbbk((s))}[q]$ whose exponents j satisfy $0 < jw_2 < w_1$ ($U = 0$ if $w_1 = w_2 = 1$). By the uniqueness of (p_0, q_0), (p_0, q_0) is invariant under G and under $\operatorname{Gal}(\overline{\Bbbk((s))}/\Bbbk((s)))$, so $p_0, q_0 \in s\Bbbk[[s]]$, and by Proposition 7.4(1), $U \in \Bbbk((s))[q]$ and the coordinate change

$$(p, q) \to (p + p_0 - U(s, q - q_0), q)$$

commutes with G. Let Z be the closure in $\widehat{\mathcal{X}}$ of

$$\operatorname{Nklt}\Big(\widehat{\mathcal{X}}^\circ, \frac{1}{m}\operatorname{div}(Q_*)\Big).$$

Then $I_Z = (p - p_0(s), q - q_0(s))$.

We bound the poles of U. Let $c_U := \min\{c' \in \mathbb{N} \mid s^{c'}U \in \Bbbk[[s]][q]\}$, and write $\Phi = \sum \delta_{a_1,a_2} p^{a_1} q^{a_2}$ with $a_1w_1 + a_2w_2 = d$. We claim that $\delta_{a_1,a_2} \in s^{c_U a_1}\Bbbk[[s]]$ for any (a_1, a_2). The claim is trivial if $c_U = 0$. If $c_U > 0$, let $\overline{U} \in \Bbbk[q]$ be the reduction of $s^{c_U}U$ modulo s. Then $\overline{U} \neq 0$ and $\deg \overline{U} < w_1/w_2$. Suppose that the claim fails, so that

$$t := \max\{t' \in \mathbb{Z} \mid \delta_{a_1,a_2} \in s^{c_U a_1 + t'}\Bbbk[[s]] \text{ for any } (a_1, a_2)\} < 0.$$

Let $\overline{\delta}_{a_1,a_2} \in \Bbbk$ be the coefficient of $s^{c_U a_1 + t}$ in δ_{a_1,a_2}. Not all $\overline{\delta}_{a_1,a_2}$ are zero. Setting $p = 0$ in (7.50), $\Phi(s, U(s, q), q) \in \Bbbk[[s]][q]$, so the coefficient of s^t vanishes:

$$\sum_{(a_1,a_2)} \overline{\delta}_{a_1,a_2} \overline{U}^{a_1} q^{a_2} = 0.$$

Let a_1^0 be the smallest a_1 with $\overline{\delta}_{a_1,a_2} \neq 0$ and $a_2^0 := (d - a_1^0 w_1)/w_2$. Since w_1, w_2 are coprime, the pairs (a_1, a_2) with $\overline{\delta}_{a_1,a_2} \neq 0$ are among the $(a_1^0 + jw_2, a_2^0 - jw_1)$, $j \in \mathbb{N}$, so the left side is $\overline{U}^{a_1^0} q^{a_2^0}$ times a non-zero polynomial in $\rho := \overline{U}^{w_2} q^{-w_1}$. Since $w_2 \deg \overline{U} < w_1$, ρ is not constant, hence transcendental over $\Bbbk$, a contradiction.

Let $\widetilde{p} := s^{c_U}(p - p_0 + U(s, q - q_0))$ and $\widetilde{q} := q - q_0$. By (7.50),

$$Q_* = \sum \delta_{a_1,a_2} s^{-c_U a_1} \widetilde{p}^{a_1} \widetilde{q}^{a_2},$$

and by the claim $\delta_{a_1,a_2} s^{-c_U a_1} \in \Bbbk[[s]]$ for any (a_1, a_2). Let $v_0 \geq 0$ be the minimum of their s-orders. Substituting $p = p_0 + s^{-c_U}\widetilde{p} - U(s, \widetilde{q})$ and $q = q_0 + \widetilde{q}$ writes any element of $\Bbbk[[s, p, q]]$ as a series in $\widetilde{p}, \widetilde{q}$ whose coefficient of $\widetilde{p}^{a_1}\widetilde{q}^{a_2}$ has s-order at least $-(a_1 + a_2)c_U$. Thus for any integer $n > c_U + 1$ the monomial valuation ν_n with weights $(1, nw_1, nw_2)$ in $(s, \widetilde{p}, \widetilde{q})$ is defined on $\Bbbk[[s, p, q]]$ and positive on (s, p, q), and, since s, $s^{-c_U}\widetilde{p}$, and $s^{-c_U}\widetilde{q}$ are regular parameters at the origin of the open subset of the blow-up of $(s^{c_U}, q - q_0)$ on which $s^{-c_U}\widetilde{q}$ is regular, and ν_n is the monomial valuation with weights $(1, nw_1 - c_U, nw_2 - c_U)$ in them, $\nu_n = \iota_n \operatorname{mult}_{T_n}$ for a prime divisor T_n over $\widehat{\mathcal{X}}$ with center 0 and some $\iota_n \in \mathbb{N}^+$. Since

$$ds \wedge dp \wedge dq = s^{-c_U} ds \wedge d\widetilde{p} \wedge d\widetilde{q}$$

and $\nu_n(Q_*) = nd + v_0$,

$$\iota_n a\Big(T_n, \widehat{\mathcal{X}}, \frac{1}{m}\operatorname{div}(Q_*) + \beta \operatorname{div}(s)\Big) = 1 + n(w_1 + w_2) - c_U - \frac{nd + v_0}{m} - \beta = 1 - c_U - \beta - \frac{v_0}{m}. \tag{7.51}$$

By (7.51) and (7.47) for $T = T_n$, $1 - c_U - \beta - \frac{v_0}{m} > 0$. Since $v_0 \geq 0$ and $c_U \in \mathbb{N}$,

$$c_U \leq C_U := \max\{0, \lceil -\beta \rceil\}.$$

Fix $k' \in |G|\mathbb{N}^+$ with $k' \geq C_U + 2$, let $C_1 := k'(w_1 + w_2) + C_U + 3$, and fix an integer J with

$$J > \max\{k'd + C_U, C_1 R'\}.$$

Let $p_0^{<J}, q_0^{<J} \in s\Bbbk[s]$ and $\widetilde{U} \in \Bbbk[s, s^{-1}][q]$ be the truncations of p_0, q_0, U in s-degree less than J. Consider the substitution

$$q = q_0^{<J} + s^{k'w_2} q'', \quad p = p_0^{<J} + s^{k'w_1} p'' - \widetilde{U}(s, s^{k'w_2} q''). \tag{7.52}$$

Since $k'w_2 - C_U \geq 2$, it maps s, p, q into $s\Bbbk[[s, p'', q'']]$. By (7.50), Q_* becomes

$$\Phi(s, s^{k'w_1} p'' + \epsilon_1, s^{k'w_2} q'' + \epsilon_2) \quad \text{with } \epsilon_1, \epsilon_2 \in s^{J - C_U}\Bbbk[[s, q'']],$$

and $\Phi(s, s^{k'w_1} p'', s^{k'w_2} q'') = s^{k'd}\Phi(s, p'', q'')$. Since $J - C_U > k'd$, $s^{k'd}$ divides the image of Q_*, a condition on the J-jet of Q_*. We use the following data in Proposition 6.3(A).

- $(z_1, z_2, z_3) = (s, p, q)$, $r = z_1$, $N = 1$, $\kappa_1 = \frac{1}{m}$, $g_{i,1} = Q_i$, $R_0 = R'$, and $C_0 = \bar{b}|G| + |\beta|R'$, as for Proposition 6.2 above. Hypotheses (i)–(iv) of Proposition 6.3 hold by **Step 8**, (7.49), $\lim_i \operatorname{mult}_{F_i} Q_i = +\infty$, and the lc property at the end of **Step 8**.
- J in place of d and L, and $T = \lfloor C_1 R' \rfloor + 1$.
- The vector $\zeta \in \Bbbk^\ell$ of the coefficients of $p_0^{<J}$, $q_0^{<J}$, and $s^{C_U}\widetilde{U}$ in place of $\beta \in \Bbbk^q$.
- Ξ_1, Ξ_2 in place of $\Phi_1, \dots, \Phi_{n'}$, where Ξ_1, Ξ_2 are the polynomials with $\Xi_1(\zeta, z) = p - p_0^{<J}$ and $\Xi_2(\zeta, z) = q - q_0^{<J}$.
- $\Psi_1, \dots, \Psi_{n''}$ the polynomial functions on $\mathcal{J}_J \times \mathbb{A}^\ell$ expressing the character identities and the divisibility by $s^{k'd}$ of the image of $Q_*^{<J}$ under (7.52).

By Proposition 6.3(A), for all but finitely many i there exist $p_{0,i}, q_{0,i} \in s\mathbb{C}[s]$ and $\widetilde{U}_i \in \mathbb{C}[s, s^{-1}][q]$ with $s^{C_U}\widetilde{U}_i \in \mathbb{C}[s][q]$ and exponents j in q with $0 < jw_2 < w_1$, satisfying the character identities and the divisibility of the image of $Q_i^{<J}$ under (7.52) with $(p_{0,i}, q_{0,i}, \widetilde{U}_i)$ in place of $(p_0^{<J}, q_0^{<J}, \widetilde{U})$, such that

$$\operatorname{mult}_{F_i}(p - p_{0,i}) > C_1 R' \quad \text{and} \quad \operatorname{mult}_{F_i}(q - q_{0,i}) > C_1 R'. \tag{7.53}$$

The terms of Q_i of degree at least J map into $s^J\mathbb{C}[[s, p'', q'']]$, so $s^{k'd}$ divides the image of Q_i. By (7.53) and $\operatorname{mult}_{F_i} s \leq R'$, we have $\operatorname{mult}_{F_i} q'' > 0$ and

$$\operatorname{mult}_{F_i} p'' \geq \min\{\operatorname{mult}_{F_i}(p - p_{0,i}), \operatorname{mult}_{F_i}(q - q_{0,i}) - C_U R'\} - k'w_1 R' > 0.$$

Composing (7.52) for Q_i with the chart of **Step 7**, we obtain

$$q_i = s^{(k_i + k')w_2} q'' - \lambda_i'(s), \quad p_i = s^{(k_i + k')w_1} p'' - \mu_i'\big(s, s^{(k_i + k')w_2} q''\big),$$

where $\lambda_i' := \lambda_i - s^{k_i w_2} q_{0,i}$ and

$$\mu_i'(s, q) := \mu_i\big(s, q + s^{k_i w_2} q_{0,i}\big) + s^{k_i w_1}\widetilde{U}_i\big(s, s^{-k_i w_2} q\big) - s^{k_i w_1} p_{0,i}.$$

The coefficient of q^j in the middle term has s-order at least $k_i(w_1 - jw_2) - C_U \geq k_i - C_U$, and the other new terms have s-order at least $k_i w_2 + 1$. Since $k_i \geq k_i^0$ and $\lim_i n_i / k_i^0 = 0$, for $i \gg 0$ we have $n_i \leq k_i - C_U$ and $n_i \leq k_i w_2 + 1$, so $\lambda_i' \in s^{n_i}\mathbb{C}[s]$ and $\mu_i' \in s^{n_i}\mathbb{C}[s][q]$, with $\deg_q \mu_i' < w_1/w_2$ and the character conditions of **Step 3**. Since h_i maps to $s^{k_i d}$ times the image of Q_i, which is divisible by $s^{k'd}$, the composite chart is h_i-divisible, and it captures F_i since $\operatorname{mult}_{F_i} p'' > 0$ and $\operatorname{mult}_{F_i} q'' > 0$. Thus $k_i + k'$ is admissible for i, which contradicts the maximality of k_i.

Step 10. In this step we finish the proof of (2).

By **Step 9**, (iii), and (7.49), $\operatorname{mult}_{F_i} s$, $\operatorname{mult}_{F_i} Q_i$, and $a(F_i, Y_i, 0)$ are bounded. By Lemma 7.10(2), with $u = s$, $(x_1, x_2) = (p, q)$, $N = 1$, and $c_1 = \frac{1}{m}$,

$$a_Q \leq \liminf_i \frac{a(F_i, Y_i, 0) - \beta \operatorname{mult}_{F_i} s - \frac{1}{m}\operatorname{mult}_{F_i} Q_i}{e_{F_i}} = b.$$

By (7.48), $a_\Phi = a_Q = b$. For any prime divisor T over $\widehat{\mathcal{X}}$ with center 0,

$$\frac{1}{e_T} a\Big(T, \widehat{\mathcal{X}}, \frac{1}{m}\operatorname{div}(\Phi) + \beta \operatorname{div}(s)\Big) \in \frac{1}{m|G|}\mathbb{Z},$$

a discrete set, so some T_0 attains $a_\Phi = b$. Fix $k \in |G|\mathbb{N}^+$ with $k > \operatorname{mult}_{T_0}\Phi$. By (7.44) and Claim 7.14 for the charts of depth k of **Step 6**, for infinitely many i there exists a prime divisor E_i over the chart of depth k with center the origin, the T_i of Claim 7.14 for $T = T_0$. Since $b_i = b$ (**Step 7**), by Claim 7.14 and (7.42), $e_{E_i} = e_{T_0}$ and

$$a(E_i, V_i, B_i) = a\Big(T_0, \widehat{\mathcal{X}}, \frac{1}{m}\operatorname{div}(\Phi) + \beta \operatorname{div}(s)\Big) = be_{T_0} = b_i e_{E_i},$$

and by (7.43),

$$a(E_i, V_i, 0) = a(T_0, \widehat{\mathcal{X}}, 0) + k(w_1 + w_2)\operatorname{mult}_{T_0} s \quad \text{and} \quad \operatorname{mult}_{E_i} s = \operatorname{mult}_{T_0} s.$$

The right sides are independent of i, so (2) follows by **Step 5**. □

8. Comparison with fixed reference germs

8.1. **Fixed references.**

Lemma 8.1. *Let G be a finite abelian group acting diagonally on $\mathbb{A}^4$ with coordinates $x_1, \dots, x_4$, and let $\widehat{\mathcal{O}} := \mathbb{C}[[x_1, \dots, x_4]]$ be the completion of the local ring of $\mathbb{A}^4$ at 0, with maximal ideal $\mathfrak{n}$ and the induced action of G.*

(1) Let $f_0 \in \mathfrak{n}$ be a G-semi-invariant element such that $\widehat{W_0} := \operatorname{Spec}\widehat{\mathcal{O}}/(f_0)$ is G-equivariantly isomorphic to the completion of a normal algebraic hypersurface threefold germ at a closed point. Let $f' \in \mathfrak{n}$ be a G-semi-invariant element with the character of f_0 such that $W' := \operatorname{Spec}\widehat{\mathcal{O}}/(f')$ is G-equivariantly isomorphic to the completion of a normal algebraic hypersurface threefold germ at a closed point. Let $I \subset \widehat{\mathcal{O}}$ be a G-stable ideal and P' a positive integer with $f' - f_0 \in I^{P'}$. Let (W_1, f_1) be one of $(\widehat{W_0}, f_0)$ and (W', f'), and let (W_2, f_2) be the other one. Let F be a prime divisor over W_1 with center the closed point, let $j := \operatorname{mult}_F \operatorname{Jac}_{W_1}$ and $\operatorname{mult}_F I := \operatorname{mult}_F(I \cdot \mathcal{O}_{W_1})$, and let x_i be a coordinate with $\operatorname{mult}_F(\partial_i f_1|_{W_1}) = j$. Assume that $0 < \operatorname{mult}_F I < +\infty$ and

$$P' \operatorname{mult}_F I > 2j + 2\operatorname{mult}_F I.$$

Then Lemma 5.4(3) associates with F and x_i a prime divisor F' over W_2 with center the closed point, and the following hold.

(i) $a(F', W_2, 0) = a(F, W_1, 0)$, $\operatorname{mult}_{F'} \operatorname{Jac}_{W_2} = j$, and $\operatorname{mult}_{F'} I = \operatorname{mult}_F I$.

(ii) $\kappa(F') \cong \kappa(F)$, $\operatorname{Stab}_G(F') = \operatorname{Stab}_G(F)$, and $I_G(F') = I_G(F)$.

(iii) $\operatorname{mult}_{F'}(h|_{W_2}) = \operatorname{mult}_F(h|_{W_1})$ for any $h \in \widehat{\mathcal{O}}$ with $\operatorname{mult}_F(h|_{W_1}) < P' \operatorname{mult}_F I - j$, in particular for any $h \in \widehat{\mathcal{O}}$ with $I = (h^R)$ for a positive integer R.

(iv) For any G-semi-invariant h as in (iii), any subgroup $H \subset G$ and any real number $\beta \geq 0$,

$$\frac{a(F', W_2, \beta \operatorname{div}(h|_{W_2}))}{|I_H(F')|} = \frac{a(F, W_1, \beta \operatorname{div}(h|_{W_1}))}{|I_H(F)|}.$$

(v) F' and x_i satisfy the assumptions of (1) with (W_1, f_1) and (W_2, f_2) exchanged, and the prime divisor over W_1 associated with F' and x_i by Lemma 5.4(3) is F.

(2) Let $f_0 \in \mathfrak{n}\backslash\{0\}$ be a G-semi-invariant element, let $(Y, y) \to (\mathbb{A}^4, 0)$ be a G-equivariant morphism from a smooth variety Y with a G-action, étale at a G-fixed point y over 0, for instance a G-stable open neighborhood of 0 in $\mathbb{A}^4$, through which we identify $\widehat{\mathcal{O}}_{Y,y}$ with $\widehat{\mathcal{O}}$, and let f and h be non-zero G-semi-invariant regular functions on Y, f with the character of f_0. Let $\mu > 0$ and $0 < \theta_1 < \theta_2 < 1$ be rational numbers, Q a positive integer

with $Q\mu \in \mathbb{Z}$, $P' \geq 2$ an integer with $P'(\theta_2 - \theta_1) \geq 10Q$, and c a positive integer. Let $\mathcal{J}$ be the stalk at y of the multiplier ideal $\mathcal{J}(Y, f^{\theta_2} h^{\theta_2 \mu})$. Assume that $\mathcal{J} \neq \mathcal{O}_{Y,y}$, and that

$$\mathfrak{n}^c \subset (f_0) + \mathfrak{n} \cdot (\partial_1 f_0, \dots, \partial_4 f_0) + (\mathcal{J}\widehat{\mathcal{O}})^{P'} \quad \text{and} \quad f - f_0 \in \mathfrak{n}^{2c+4}.$$

Then there exist a G-equivariant automorphism φ of $\widehat{\mathcal{O}}$ and a G-invariant unit $v \in \widehat{\mathcal{O}}$ with

$$v\varphi(f) - f_0 \in \varphi(\mathcal{J}\widehat{\mathcal{O}})^{P'} \quad \text{and} \quad \varphi(\mathcal{J}\widehat{\mathcal{O}}) \subset \mathcal{J}(\operatorname{Spec} \widehat{\mathcal{O}}, f_0^{\theta_1} \varphi(h)^{\theta_1 \mu}).$$

If moreover $(f = 0)$ is normal near y and $h|_{(f=0)} \neq 0$ near y, then $\varphi(\mathcal{J}\widehat{\mathcal{O}}) \not\subset (f_0)$.

Proof. **(1).** Since $I^{P'}$ is generated by products of P' elements of I, the Leibniz rule implies

$$\partial_k(f_2 - f_1) \in I^{P'-1} \quad \text{for any } k.$$

Since $f_2 - f_1 \in I^{P'}$ and $\operatorname{mult}_F I > 0$,

$$\begin{aligned} c_F := \operatorname{mult}_F((f_2 - f_1)|_{W_1}) &\geq P' \operatorname{mult}_F I > 2j + 2 \operatorname{mult}_F I \quad \text{and} \\ \operatorname{mult}_F(\partial_k(f_2 - f_1)|_{W_1}) &\geq (P' - 1) \operatorname{mult}_F I > j \end{aligned} \tag{8.1}$$

for any $k \in \{1, \dots, 4\}$. By (8.1) and Lemma 5.4(3), f_2 has a unique root x_i' in the completion $\widehat{\mathcal{O}}_F$ of the valuation ring of F in the direction of x_i with $\operatorname{mult}_F(x_i' - x_i) \geq c_F - j$, the prime divisor F' over W_2 defined by x_i' has the value group of F, $\kappa(F') \cong \kappa(F)$, $\operatorname{Stab}_G(F') = \operatorname{Stab}_G(F)$, and $I_G(F') = I_G(F)$, and

$$a(F', W_2, 0) = a(F, W_1, 0), \qquad \operatorname{mult}_{F'} \operatorname{Jac}_{W_2} = j.$$

Moreover $\operatorname{mult}_{F'}(h|_{W_2}) = \operatorname{mult}_F(h|_{W_1})$ for any $h \in \widehat{\mathcal{O}}$ with $j + \operatorname{mult}_F(h|_{W_1}) < c_F$. Since $c_F \geq P' \operatorname{mult}_F I$, this proves (ii) and the first two equalities of (i), and proves (iii) for any h with $\operatorname{mult}_F(h|_{W_1}) < P' \operatorname{mult}_F I - j$. If $I = (h^R)$ for a positive integer R, then

$$\operatorname{mult}_F(h|_{W_1}) = \frac{\operatorname{mult}_F I}{R} \leq \operatorname{mult}_F I < j + 2 \operatorname{mult}_F I < P' \operatorname{mult}_F I - j,$$

which completes the proof of (iii).

Let $z_1, \dots, z_n$ be generators of I with $\operatorname{mult}_F(z_1|_{W_1}) = \operatorname{mult}_F I$. Since $\operatorname{mult}_F I < P' \operatorname{mult}_F I - j$, (iii) implies $\operatorname{mult}_{F'}(z_1|_{W_2}) = \operatorname{mult}_F I$. Let $k \in \{2, \dots, n\}$. If $\operatorname{mult}_F(z_k|_{W_1}) < P' \operatorname{mult}_F I - j$, then $\operatorname{mult}_{F'}(z_k|_{W_2}) = \operatorname{mult}_F(z_k|_{W_1}) \geq \operatorname{mult}_F I$ by (iii). Otherwise $\operatorname{mult}_F((z_1 + z_k)|_{W_1}) = \operatorname{mult}_F I$, so $\operatorname{mult}_{F'}((z_1 + z_k)|_{W_2}) = \operatorname{mult}_F I$ by (iii), and again $\operatorname{mult}_{F'}(z_k|_{W_2}) \geq \operatorname{mult}_F I$. Thus

$$\operatorname{mult}_{F'} I = \operatorname{mult}_F I,$$

which completes the proof of (i).

By Lemma 5.4(3)(d) for the function $\partial_i f_1$, since $c_F > 2j$,

$$\operatorname{mult}_{F'}(\partial_i f_1|_{W_2}) = j.$$

Since $\partial_i(f_2 - f_1) \in I^{P'-1}$ and $\operatorname{mult}_{F'} I = \operatorname{mult}_F I$, (8.1) implies $\operatorname{mult}_{F'}(\partial_i(f_2 - f_1)|_{W_2}) \geq (P' - 1) \operatorname{mult}_F I > j$. Thus $\operatorname{mult}_{F'}(\partial_i f_2|_{W_2}) = j$.

Since $I_H(F) = I_G(F) \cap H$ and $I_H(F') = I_G(F') \cap H$, (iv) follows from (i), (ii) and (iii). By (i) and the previous paragraph, $\operatorname{mult}_{F'} I = \operatorname{mult}_F I \in (0, +\infty)$, $P' \operatorname{mult}_{F'} I > 2j + 2 \operatorname{mult}_{F'} I$, and

$$\operatorname{mult}_{F'} \operatorname{Jac}_{W_2} = \operatorname{mult}_{F'}(\partial_i f_2|_{W_2}) = j,$$

so F' and x_i satisfy the assumptions of (1) with (W_1, f_1) and (W_2, f_2) exchanged. Since the correspondence of Lemma 5.4(3) is invertible, the prime divisor over W_1 associated with F' and x_i is F. This proves (v).

(2). ***Step 1***. Since f and h are G-semi-invariant, G preserves $\operatorname{div}(f)$ and $\operatorname{div}(h)$, so $\mathcal{J}$ and $(\mathcal{J}\widehat{\mathcal{O}})^{P'}$ are G-stable. By Lemma 3.2(3) for $\Bbbk = \mathbb{C}$, $n = 4$, $F_0 = f_0$, $F = f$, $I = (\mathcal{J}\widehat{\mathcal{O}})^{P'}$, and $L = 2c + 4$, there exist a G-equivariant automorphism φ of $\widehat{\mathcal{O}}$ and a G-invariant unit v with

$$v\varphi(f) - f_0 \in \varphi(\mathcal{J}\widehat{\mathcal{O}})^{P'}. \tag{8.2}$$

Step 2. We prove

$$\varphi(\mathcal{J}\widehat{\mathcal{O}}) \subset \mathcal{J}(\operatorname{Spec}\widehat{\mathcal{O}}, f_0^{\theta_1}\varphi(h)^{\theta_1\mu}).$$

Let $S := \mathcal{O}_{Y,y}$, with maximal ideal $\mathfrak{m}_S$, so that $\widehat{\mathcal{O}}$ is the completion of S and $\mathfrak{m}_S\widehat{\mathcal{O}} = \mathfrak{n}$. For an ideal $\mathfrak{b} \subset S$ and a rational number $\gamma > 0$, $\mathcal{J}(S, \mathfrak{b}^\gamma)$ denotes the stalk at y of the multiplier ideal of $\mathfrak{b}^\gamma$ on a neighborhood of y in Y on which generators of $\mathfrak{b}$ are regular. Let

$$\psi := f^Q h^{Q\mu} \in S \quad \text{and} \quad \gamma := \frac{\theta_1 + \theta_2}{2Q}.$$

Since

$$\frac{\theta_2}{Q}\operatorname{div}(\psi) = \theta_2 \operatorname{div}(f) + \theta_2\mu \operatorname{div}(h),$$

we have $\mathcal{J} = \mathcal{J}(S, (\psi)^{\theta_2/Q})$. Moreover

$$P'\Big(\frac{\theta_2}{Q} - \gamma\Big) = \frac{P'(\theta_2 - \theta_1)}{2Q} \geq 5.$$

Let $u := \varphi^{-1}(v)$, a unit, and $\psi' := u^{-Q}\varphi^{-1}(f_0)^Q h^{Q\mu} \in \widehat{\mathcal{O}}$, which is non-zero since $f_0 \neq 0$. By (8.2), $uf - \varphi^{-1}(f_0) \in (\mathcal{J}\widehat{\mathcal{O}})^{P'}$, and since $(uf)^Q - \varphi^{-1}(f_0)^Q$ is divisible by $uf - \varphi^{-1}(f_0)$,

$$\psi' - \psi = u^{-Q}h^{Q\mu}\big(\varphi^{-1}(f_0)^Q - (uf)^Q\big) \in (\mathcal{J}\widehat{\mathcal{O}})^{P'}.$$

Let $\xi_1, \dots, \xi_n \in S$ be generators of $\mathcal{J}^{P'}$, and write $\psi' - \psi = \sum_{k=1}^n \rho_k\xi_k$ with $\rho_k \in \widehat{\mathcal{O}}$. For any positive integer i choose $\rho_{k,i} \in S$ with $\rho_{k,i} - \rho_k \in \mathfrak{n}^i$, and let

$$\psi_i := \psi + \sum_{k=1}^n \rho_{k,i}\xi_k \in S.$$

Then $\psi_i - \psi \in \mathcal{J}^{P'}$ and $\psi_i - \psi' \in \mathfrak{n}^i$. Since $\psi' \neq 0$, $\psi_i \neq 0$ for any $i \gg 0$. For any such i, since $(\psi) + \mathcal{J}^{P'} = (\psi_i) + \mathcal{J}^{P'}$, monotonicity of multiplier ideals and the summation theorem [Mus02, Theorem 0.1] imply

$$\mathcal{J} = \mathcal{J}(S, (\psi)^{\theta_2/Q}) \subset \mathcal{J}\big(S, ((\psi_i) + \mathcal{J}^{P'})^{\theta_2/Q}\big) \subset \sum_{\alpha_1+\alpha_2=\theta_2/Q,\ \alpha_1,\alpha_2\geq 0} \mathcal{J}(S, (\psi_i)^{\alpha_1}) \cdot \mathcal{J}(S, (\mathcal{J}^{P'})^{\alpha_2}).$$

If $\alpha_1 \geq \gamma$, the corresponding summand is contained in $\mathcal{J}(S, (\psi_i)^\gamma)$. If $\alpha_1 < \gamma$, then

$$P'\alpha_2 > P'\Big(\frac{\theta_2}{Q} - \gamma\Big) \geq 5,$$

so by Skoda's theorem [Laz04, Theorem 9.6.21(i)] in dimension 4 the summand is contained in

$$\mathcal{J}(S, \mathcal{J}^{P'\alpha_2}) \subset \mathcal{J}(S, \mathcal{J}^5) = \mathcal{J}^2 \cdot \mathcal{J}(S, \mathcal{J}^3) \subset \mathcal{J}^2.$$

Thus $\mathcal{J} \subset \mathcal{J}(S, (\psi_i)^\gamma) + \mathcal{J}^2$. Since $\mathcal{J} \subset \mathfrak{m}_S$, Nakayama's lemma implies

$$\mathcal{J} \subset \mathcal{J}(S, (\psi_i)^\gamma) \quad \text{for any } i \gg 0. \tag{8.3}$$

Let $z \in \mathcal{J}$ be non-zero, and let E be a prime divisor over $\operatorname{Spec}\widehat{\mathcal{O}}$ with center the closed point. By [dFEM11, Lemma 2.21, Proposition A.14], E restricts to a prime divisor E_Y over Y with center y and $a(E_Y, Y, 0) = a(E, \operatorname{Spec}\widehat{\mathcal{O}}, 0)$. For $i \gg 0$ with $i \operatorname{mult}_E \mathfrak{n} > \operatorname{mult}_E \psi'$, we have $\operatorname{mult}_E \psi_i = \operatorname{mult}_E \psi'$, so (8.3) and [Laz04, Definition 9.2.3, Theorem 9.2.18] imply

$$a(E, \operatorname{Spec}\widehat{\mathcal{O}}, 0) + \operatorname{mult}_E z > \gamma \operatorname{mult}_E \psi'. \tag{8.4}$$

We claim that any prime divisor D over $\operatorname{Spec}\widehat{\mathcal{O}}$ satisfies

$$a(D, \operatorname{Spec}\widehat{\mathcal{O}}, 0) + \operatorname{mult}_D z \geq \gamma \operatorname{mult}_D \psi'. \tag{8.5}$$

Suppose not. By [dFEM11, Theorems 2.3–2.4] there exists a log resolution $g_X : X \to \operatorname{Spec}\widehat{\mathcal{O}}$ of $\psi' z \cdot \mathfrak{n}$ on which D is a prime divisor. Since $\operatorname{center}_{\operatorname{Spec}\widehat{\mathcal{O}}} D$ contains the closed point 0, D

intersects $g_X^{-1}(0)$. Let q be a closed point of $D \cap g_X^{-1}(0)$, and let $t_1, \dots, t_4$ be regular parameters at q such that t_1 defines D and the components through q of the simple normal crossing divisor

$$\operatorname{Supp} g_X^* \operatorname{div}(\psi' z) \cup g_X^{-1}(0) \cup \operatorname{Exc}(g_X)$$

are defined by some of the t_k. For a positive integer N, let E_N be the prime divisor over $\operatorname{Spec} \widehat{\mathcal{O}}$ whose valuation is the monomial valuation at q with weights $(N, 1, 1, 1)$ on $(t_1, \dots, t_4)$. Then $\operatorname{center}_{\operatorname{Spec} \widehat{\mathcal{O}}} E_N$ is the closed point. The discrepancy formula on X implies

$$a(E_N, \operatorname{Spec} \widehat{\mathcal{O}}, 0) + \operatorname{mult}_{E_N} z - \gamma \operatorname{mult}_{E_N} \psi' = N\big(a(D, \operatorname{Spec} \widehat{\mathcal{O}}, 0) + \operatorname{mult}_D z - \gamma \operatorname{mult}_D \psi'\big) + c_q,$$

where c_q does not depend on N. For $N \gg 0$ the right hand side is negative, which contradicts (8.4). This proves (8.5).

Let D be a prime divisor over $\operatorname{Spec} \widehat{\mathcal{O}}$. We show

$$a(D, \operatorname{Spec} \widehat{\mathcal{O}}, 0) + \operatorname{mult}_D z > \frac{\theta_1}{Q} \operatorname{mult}_D \psi'. \tag{8.6}$$

If $\operatorname{mult}_D \psi' > 0$, this follows from (8.5) and $\gamma > \frac{\theta_1}{Q}$. If $\operatorname{mult}_D \psi' = 0$, it follows from $a(D, \operatorname{Spec} \widehat{\mathcal{O}}, 0) \geq 1$. The automorphism φ maps the prime divisors D over $\operatorname{Spec} \widehat{\mathcal{O}}$ bijectively to the prime divisors D^φ over $\operatorname{Spec} \widehat{\mathcal{O}}$ with $\operatorname{mult}_{D^\varphi} \varphi(z') = \operatorname{mult}_D z'$ for any $z' \in \widehat{\mathcal{O}}$ and

$$a(D^\varphi, \operatorname{Spec} \widehat{\mathcal{O}}, 0) = a(D, \operatorname{Spec} \widehat{\mathcal{O}}, 0),$$

and

$$\frac{\theta_1}{Q} \operatorname{div}(\varphi(\psi')) = \frac{\theta_1}{Q} \operatorname{div}\left(v^{-Q} f_0^Q \varphi(h)^{Q\mu}\right) = \theta_1 \operatorname{div}(f_0) + \theta_1 \mu \operatorname{div}(\varphi(h)).$$

Hence, for each prime divisor D on a log resolution of $f_0 \varphi(h)$ [dFEM11, Theorems 2.3–2.4], (8.6) for the prime divisor D' with $D'^\varphi = D$ implies

$$a(D, \operatorname{Spec} \widehat{\mathcal{O}}, 0) + \operatorname{mult}_D \varphi(z) > \theta_1 \operatorname{mult}_D f_0 + \theta_1 \mu \operatorname{mult}_D \varphi(h),$$

and since $a(D, \operatorname{Spec} \widehat{\mathcal{O}}, 0)$ and $\operatorname{mult}_D \varphi(z)$ are integers,

$$\operatorname{mult}_D \varphi(z) \geq \lfloor \theta_1 \operatorname{mult}_D f_0 + \theta_1 \mu \operatorname{mult}_D \varphi(h) \rfloor - a(D, \operatorname{Spec} \widehat{\mathcal{O}}, 0) + 1,$$

so

$$\varphi(z) \in \mathcal{J}(\operatorname{Spec} \widehat{\mathcal{O}}, f_0^{\theta_1} \varphi(h)^{\theta_1 \mu}) \quad \text{for any } z \in \mathcal{J}$$

by the log resolution formula of [JM12, Section 1.3]. Since $\varphi(\mathcal{J})$ generates the ideal $\varphi(\mathcal{J}\widehat{\mathcal{O}})$, **Step 2** follows.

Step 3. Assume that $(f = 0)$ is normal near y and $h|_{(f=0)} \neq 0$ near y. Since $f - f_0 \in \mathfrak{n}^{2c+4}$ and $f_0 \in \mathfrak{n}$, $f(y) = 0$. Since $(f = 0)$ is normal near y, f is a prime element of S. At the generic point of the component of $(f = 0)$ through y, h is a unit and $\theta_2 < 1$, so the multiplier ideal $\mathcal{J}(Y, f^{\theta_2} h^{\theta_2 \mu})$ is trivial there. Thus $\mathcal{J} \not\subset fS$. Since $\widehat{\mathcal{O}}$ is faithfully flat over S, $\mathcal{J}\widehat{\mathcal{O}} \not\subset f\widehat{\mathcal{O}}$, so

$$\varphi(\mathcal{J}\widehat{\mathcal{O}}) \not\subset \varphi(f)\widehat{\mathcal{O}} = v\varphi(f)\widehat{\mathcal{O}}. \tag{8.7}$$

Suppose that $\varphi(\mathcal{J}\widehat{\mathcal{O}}) \subset (f_0)$. Then (8.2) implies $v\varphi(f) - f_0 \in (f_0^{P'})$, so $v\varphi(f) = f_0(1 + f_0^{P'-1} u')$ for some $u' \in \widehat{\mathcal{O}}$. Since $P' \geq 2$ and $f_0 \in \mathfrak{n}$, $1 + f_0^{P'-1} u'$ is a unit, so $(f_0) = v\varphi(f)\widehat{\mathcal{O}}$, and the assumption $\varphi(\mathcal{J}\widehat{\mathcal{O}}) \subset (f_0)$ contradicts (8.7). The lemma follows. □

8.2. **Exact multiplier contact.**

Theorem 8.2. *Let m and M be positive integers, $\lambda, B_0 > 0$, $\mu > \frac{1}{m}$ rational, and $0 < \theta_1 < \theta_2 < 1$ rational with $\theta_1\mu - \frac{1}{m} - (1-\theta_1)(1+\frac{2\lambda}{m}) > 0$. Then there exist an integer $P' \geq 2$ and a real number $C > 0$, depending only on $m, M, \lambda, B_0, \mu, \theta_1, \theta_2$, such that the following holds for any integer $c \geq 1$. Let $\widehat{\mathcal{O}}$ be the completion of the local ring of $\mathbb{A}^4$ at 0, with maximal ideal $\mathfrak{n}$, let G be a finite abelian group of order at most M acting diagonally on $\mathbb{A}^4$, let $f, f_0, h \in \widehat{\mathcal{O}}$ be G-semi-invariant, f with the character of f_0, and let $W := (f = 0)$, $W_0 := (f_0 = 0)$, and*

$$\mathcal{J} := \mathcal{J}(\operatorname{Spec}\widehat{\mathcal{O}}, f^{\theta_2} h^{\theta_2\mu}).$$

Assume that the following hold.

- *(i) W and W_0 are G-equivariantly isomorphic to the completions at closed points of normal hypersurface threefold germs, G acts faithfully on W and on W_0, and $h|_W$ is non-zero and a non-unit.*
- *(ii) After a G-equivariant automorphism of $\widehat{\mathcal{O}}$ and multiplication of f by a G-invariant unit, f and h become the images in $\widehat{\mathcal{O}}$ of regular functions on a G-equivariant étale neighborhood (Y, y) of $0 \in \mathbb{A}^4$, that is, a smooth variety Y with a G-action and a G-equivariant morphism $Y \to \mathbb{A}^4$ which is étale at a G-fixed point y over 0, under the induced isomorphism $\widehat{\mathcal{O}}_{Y,y} \cong \widehat{\mathcal{O}}$. For instance, f and h may be regular functions near $0 \in \mathbb{A}^4$.*
- *(iii) $(W, \frac{1}{m}\operatorname{div}(h|_W))$ is lc, and $b := \operatorname{mld}((W \ni 0, \frac{1}{m}\operatorname{div}(h|_W))/G) \geq 0$.*
- *(iv) Some prime divisor over W with center 0 has log discrepancy at most B_0.*
- *(v) $\operatorname{mult}_F \operatorname{Jac}_V \leq \lambda a(F, V, 0)$ for $V = W$ and for $V = W_0$, and any prime divisor F over V with center 0.*

Then the following hold.

- *(1) If $\mathcal{J} = \widehat{\mathcal{O}}$, then b has a computing lift E with $a(E, W, 0)/e_E \leq C$.*
- *(2) Assume that $\mathcal{J} \neq \widehat{\mathcal{O}}$, $f - f_0 \in \mathfrak{n}^{2c+4}$, and*

$$\mathfrak{n}^c \subset (f_0) + \mathfrak{n} \cdot (\partial_1 f_0, \dots, \partial_4 f_0) + \mathcal{J}^{P'}.$$

Then there exist a G-equivariant automorphism φ of $\widehat{\mathcal{O}}$ and a G-invariant unit v with $v\varphi(f) - f_0 \in \varphi(\mathcal{J})^{P'}$, the restriction $h_0 := \varphi(h)|_{W_0}$ is non-zero and a non-unit, and the following hold.

- *(a) For any subgroup $H \subset G$ and any $t \in [0, 1]$,*

$$\operatorname{mld}((W \ni 0, \tfrac{t}{m}\operatorname{div}(h|_W))/H) = \operatorname{mld}((W_0 \ni 0, \tfrac{t}{m}\operatorname{div}(h_0))/H).$$

- *(b) If $\operatorname{mld}((W_0 \ni 0, \frac{1}{m}\operatorname{div}(h_0))/G)$ has a computing lift E_0 with $a(E_0, W_0, 0)/e_{E_0} \leq L$ for a real number L, then b has a computing lift E with $a(E, W, 0)/e_E \leq L$.*

Proof. We let

$$\gamma := \theta_1\mu - \frac{1}{m} - (1-\theta_1)\Big(1 + \frac{2\lambda}{m}\Big),$$

so $\gamma > 0$ by assumption, and we let Q be the smallest positive integer with $Q\mu \in \mathbb{Z}$. We let

$$C_1 := \big(1 + 2\lambda(1-\theta_1)\big)MB_0 + 1 - \theta_1, \qquad C_2 := \frac{C_1 + 2}{\gamma} + 1,$$

$$C_3 := \lambda\Big(\frac{C_2}{m} + MB_0\Big) + 1, \qquad C := \frac{C_1}{\gamma m} + MB_0 + 1,$$

and we let P' be the smallest integer with $P' \geq 2$, $P'(\theta_2 - \theta_1) \geq 10Q$, and $P' > 2C_2 + 2C_3 + 4$.

Now let c, G, f, f_0, h be as in the theorem, let $x_1, \dots, x_4$ be coordinates of $\mathbb{A}^4$ on which G acts diagonally, and let $\partial_i := \partial/\partial x_i$.

Let $g \in \mathfrak{n}$ with $V := (g = 0)$ normal. For a prime divisor F over V, an element $z \in \widehat{\mathcal{O}}$ and an ideal $\mathcal{I} \subset \widehat{\mathcal{O}}$, we write $\operatorname{mult}_F z := \operatorname{mult}_F(z|_V)$ and $\operatorname{mult}_F \mathcal{I} := \min_{z \in \mathcal{I}} \operatorname{mult}_F z$.

Let α be an automorphism of $\widehat{\mathcal{O}}$ and u a unit of $\widehat{\mathcal{O}}$. By the chain rule,

$$\alpha\big((g)+\mathfrak{n}\cdot(\partial_1 g,\dots,\partial_4 g)\big)=(u\alpha(g))+\mathfrak{n}\cdot\big(\partial_1(u\alpha(g)),\dots,\partial_4(u\alpha(g))\big), \tag{8.8}$$

and the isomorphism

$$\widehat{\mathcal{O}}/(g)\to\widehat{\mathcal{O}}/(u\alpha(g))$$

induced by α maps $\mathrm{Jac}_{(g=0)}$ onto $\mathrm{Jac}_{(u\alpha(g)=0)}$. Since α maps log resolutions to log resolutions,

$$\alpha\big(\mathcal{J}(\operatorname{Spec}\widehat{\mathcal{O}},g_1^{\beta_1}g_2^{\beta_2})\big)=\mathcal{J}\big(\operatorname{Spec}\widehat{\mathcal{O}},(u\alpha(g_1))^{\beta_1}\alpha(g_2)^{\beta_2}\big) \tag{8.9}$$

for any non-zero $g_1,g_2\in\widehat{\mathcal{O}}$ and any real numbers $\beta_1,\beta_2\ge 0$. In particular $\mathcal{J}$ is G-stable, since f and h are G-semi-invariant.

By (iv) there exists a prime divisor F_1 over W with center 0 and $a(F_1,W,0)\le B_0$. For any subgroup $H\subset G$ and any $t\in[0,1]$, since $(W,\frac{1}{m}\operatorname{div}(h|_W))$ is lc and $t\le 1$, we have

$$0\le a(F_1,W,\tfrac{t}{m}\operatorname{div}(h|_W))\le a(F_1,W,0),$$

and $|I_H(F_1)|\ge 1$, so

$$\frac{a(F_1,W,\frac{t}{m}\operatorname{div}(h|_W))}{|I_H(F_1)|}\le a(F_1,W,0)\le B_0. \tag{8.10}$$

In particular, $b\le B_0$ by Definition 2.2.

Claim 8.3. *Let $g\in\mathfrak{n}$ with $V:=(g=0)$ normal, let $h_1\in\widehat{\mathcal{O}}$ with $h_1|_V\ne 0$, and let $\mathcal{I}$ be an ideal of $\widehat{\mathcal{O}}$ with*

$$\mathcal{I}\not\subset(g)\quad\text{and}\quad\mathcal{I}\subset\mathcal{J}(\operatorname{Spec}\widehat{\mathcal{O}},g^{\theta_1}h_1^{\theta_1\mu}).$$

Let F be a prime divisor over V with center 0, and let $\nu:=\operatorname{mult}_F\mathcal{I}$ and $j:=\operatorname{mult}_F\operatorname{Jac}_V$. Assume that $j\le\lambda a(F,V,0)$, and that $a(F,V,\frac{t}{m}\operatorname{div}(h_1|_V))\le eB_0$ for some integer $e\in[1,M]$ and some $t\in[0,1]$. Then the following hold.

(1) If $\nu=0$, then $a(F,V,0)<C$.

(2) If $\nu\ge 1$, then $\operatorname{mult}_F h_1<C_2\nu$ and $j<C_3\nu$.

Proof of the claim. Since $\mathcal{I}\not\subset(g)$ and V is integral, $\nu<+\infty$. Pick $z\in\mathcal{I}$ with $\operatorname{mult}_F z=\nu$, and pick $x\in\{x_1,\dots,x_4\}$ with $\operatorname{mult}_F(\partial g/\partial x)=j$. Let

$$k:=\operatorname{mult}_F h_1+\nu+2j+1.$$

Since $k>2j$ and $k-j>\max\{\operatorname{mult}_F h_1,\nu\}$, Lemma 5.4(2), with $\mathcal{X}=\operatorname{Spec}\widehat{\mathcal{O}}$, $(f,W)=(g,V)$, and $(H_1,H_2)=(h_1,z)$, implies that there exists a prime divisor $\widehat{F}$ over $\operatorname{Spec}\widehat{\mathcal{O}}$ with center 0 such that

$$\operatorname{mult}_{\widehat{F}}g=k,\quad\operatorname{mult}_{\widehat{F}}h_1=\operatorname{mult}_F h_1,\quad\operatorname{mult}_{\widehat{F}}z=\nu,\quad\text{and}\quad a(\widehat{F},\operatorname{Spec}\widehat{\mathcal{O}},0)=a(F,V,0)+k.$$

Since $z\in\mathcal{I}$, the hypothesis on $\mathcal{I}$ and the definition of multiplier ideals imply

$$a(\widehat{F},\operatorname{Spec}\widehat{\mathcal{O}},0)+\operatorname{mult}_{\widehat{F}}z>\theta_1\operatorname{mult}_{\widehat{F}}g+\theta_1\mu\operatorname{mult}_{\widehat{F}}h_1,$$

that is, $a(F,V,0)+(1-\theta_1)k+\nu>\theta_1\mu\operatorname{mult}_F h_1$. Substituting k, we obtain

$$\big(\theta_1\mu-(1-\theta_1)\big)\operatorname{mult}_F h_1<a(F,V,0)+(2-\theta_1)\nu+(1-\theta_1)(2j+1).$$

Since $a(F,V,0)-\frac{t}{m}\operatorname{mult}_F h_1\le eB_0$, we have

$$a(F,V,0)\le\frac{1}{m}\operatorname{mult}_F h_1+MB_0,\quad\text{so}\quad j\le\lambda\Big(\frac{1}{m}\operatorname{mult}_F h_1+MB_0\Big). \tag{8.11}$$

Substituting (8.11) into the previous inequality, we obtain

$$\gamma\operatorname{mult}_F h_1<C_1+(2-\theta_1)\nu\le C_1+2\nu.$$

If $\nu=0$, then $\operatorname{mult}_F h_1<C_1/\gamma$, so $a(F,V,0)<\frac{C_1}{\gamma m}+MB_0<C$ by (8.11). If $\nu\ge 1$, then

$$\operatorname{mult}_F h_1<\frac{(C_1+2)\nu}{\gamma}<C_2\nu\quad\text{and}\quad j<\lambda\Big(\frac{C_2\nu}{m}+MB_0\Big)\le\lambda\Big(\frac{C_2}{m}+MB_0\Big)\nu<C_3\nu$$

by (8.11). □

Proof of (1). Assume that $\mathcal{J} = \widehat{\mathcal{O}}$. By (ii), there exist a G-stable hypersurface $Z \subset Y$ and an $\mathbb{R}$-divisor D_Z on Z such that $(W, \frac{1}{m}\operatorname{div}(h|_W))$ is G-equivariantly isomorphic to the completion at y of (Z, D_Z). Thus by Lemma 2.12(1)–(3) and Lemma 2.15(0)–(3), $(Z, D_Z)/G$ is lc, b is the minimal log discrepancy of $(Z, D_Z)/G$ at the image of y, and a prime divisor on a log resolution of $(Z, D_Z)/G$ attaining $b \geq 0$ is the image of a prime divisor over W with center 0, which is a computing lift E of b. Since $\theta_1 < \theta_2$,

$$\widehat{\mathcal{O}} = \mathcal{J} \subset \mathcal{J}(\operatorname{Spec}\widehat{\mathcal{O}}, f^{\theta_1} h^{\theta_1 \mu}).$$

Moreover $a(E, W, \frac{1}{m}\operatorname{div}(h|_W)) = be_E \leq B_0 e_E$ and $e_E \leq M$. By the claim (1), with f, h, $\widehat{\mathcal{O}}$, E, e_E, and 1 in place of g, h_1, $\mathcal{I}$, F, e, and t,

$$\frac{a(E, W, 0)}{e_E} \leq a(E, W, 0) < C.$$

This proves (1).

Proof of (2). Assume the hypotheses of (2).

Step 1. By (ii), there exist a G-equivariant automorphism ψ of $\widehat{\mathcal{O}}$ and a G-invariant unit w such that $\overline{f} := w\psi(f)$ and $\overline{h} := \psi(h)$ are the images in $\widehat{\mathcal{O}}$ of regular functions on a G-equivariant étale neighborhood (Y, y) of $0 \in \mathbb{A}^4$ as in (ii), also denoted by $\overline{f}$ and $\overline{h}$. Possibly replacing Y with its connected component containing y, we may assume that the regular functions $\overline{f}$ and $\overline{h}$ on Y are non-zero and G-semi-invariant, since $\mathcal{O}_{Y,y} \to \widehat{\mathcal{O}}$ is injective and G-equivariant. We let $\overline{f}_0 := w\psi(f_0)$, and we let $\overline{\mathcal{J}}$ be the stalk at y of $\mathcal{J}(Y, \overline{f}^{\theta_2}\overline{h}^{\theta_2\mu})$. The base change along $\operatorname{Spec}\widehat{\mathcal{O}} \to Y$ of a log resolution of $(Y, \overline{f}\overline{h})$ near y is a log resolution of $(\operatorname{Spec}\widehat{\mathcal{O}}, \overline{f}\overline{h})$ with the base-changed relative canonical divisor [dFEM11, Proposition A.14], and push-forward commutes with flat base change, so (8.9) gives

$$\psi(\mathcal{J}) = \mathcal{J}(\operatorname{Spec}\widehat{\mathcal{O}}, \overline{f}^{\theta_2}\overline{h}^{\theta_2\mu}) = \overline{\mathcal{J}}\widehat{\mathcal{O}}.$$

The data (Y, y), $\overline{f}$, $\overline{h}$, $\overline{f}_0$ satisfy the hypotheses of Lemma 8.1(2):

- $\overline{f}_0 \in \mathfrak{n}\backslash\{0\}$, since W_0 is a threefold germ at 0 and $w\psi$ maps $\mathfrak{n}$ onto $\mathfrak{n}$.
- $\overline{f}$, $\overline{h}$, $\overline{f}_0$ are G-semi-invariant, and $\overline{f}$, $\overline{f}_0$ have the character of f, since ψ is G-equivariant and w is G-invariant.
- $\overline{\mathcal{J}} \neq \mathcal{O}_{Y,y}$, since $\overline{\mathcal{J}}\widehat{\mathcal{O}} = \psi(\mathcal{J})$ and $\mathcal{J} \neq \widehat{\mathcal{O}}$.
- $\overline{f} - \overline{f}_0 = w\psi(f - f_0) \in \mathfrak{n}^{2c+4}$, since $\psi(\mathfrak{n}) = \mathfrak{n}$.
- By applying ψ to the inclusion assumed in (2) and by (8.8) with $(\alpha, u, g) = (\psi, w, f_0)$,

$$\mathfrak{n}^c \subset (\overline{f}_0) + \mathfrak{n} \cdot (\partial_1\overline{f}_0, \dots, \partial_4\overline{f}_0) + (\overline{\mathcal{J}}\widehat{\mathcal{O}})^{P'}.$$

- $P' \geq 2$ and $P'(\theta_2 - \theta_1) \geq 10Q$ with $Q\mu \in \mathbb{Z}$.
- $(\overline{f} = 0) \subset Y$ is normal near y, since the completion at y of $(\overline{f} = 0)$ is $\operatorname{Spec}\widehat{\mathcal{O}}/(\psi(f)) \cong W$, which is normal. Moreover, $\overline{h}|_{(\overline{f}=0)} \neq 0$ near y, since $h \notin (f)$.

Thus Lemma 8.1(2) implies that there exist a G-equivariant automorphism $\overline{\varphi}$ of $\widehat{\mathcal{O}}$ and a G-invariant unit $\overline{v}$ such that

$$\overline{v\varphi}(\overline{f}) - \overline{f}_0 \in \overline{\varphi}(\overline{\mathcal{J}}\widehat{\mathcal{O}})^{P'}, \quad \overline{\varphi}(\overline{\mathcal{J}}\widehat{\mathcal{O}}) \subset \mathcal{J}(\operatorname{Spec}\widehat{\mathcal{O}}, \overline{f}_0^{\theta_1}\overline{\varphi}(\overline{h})^{\theta_1\mu}), \quad \text{and} \quad \overline{\varphi}(\overline{\mathcal{J}}\widehat{\mathcal{O}}) \not\subset (\overline{f}_0).$$

We let

$$\varphi := \psi^{-1} \circ \overline{\varphi} \circ \psi \quad \text{and} \quad v := \psi^{-1}\big(\overline{v\varphi}(w)w^{-1}\big).$$

Then φ is a G-equivariant automorphism of $\widehat{\mathcal{O}}$ and v is a G-invariant unit. Since

$$\overline{\varphi}(\overline{\mathcal{J}}\widehat{\mathcal{O}}) = \psi(\varphi(\mathcal{J})), \qquad \psi^{-1}(\overline{v\varphi}(\overline{f})) = \psi^{-1}(w)v\varphi(f),$$
$$\psi^{-1}(\overline{f}_0) = \psi^{-1}(w)f_0, \quad \text{and} \quad \psi^{-1}(\overline{\varphi}(\overline{h})) = \varphi(h),$$

applying ψ^{-1} and (8.9), we obtain

$$v\varphi(f) - f_0 \in \varphi(\mathcal{J})^{P'}, \tag{8.12}$$

$$\varphi(\mathcal{J}) \subset \mathcal{J}(\operatorname{Spec}\widehat{\mathcal{O}}, f_0^{\theta_1}\varphi(h)^{\theta_1\mu}), \tag{8.13}$$

$$\varphi(\mathcal{J}) \not\subset (f_0). \tag{8.14}$$

By (8.12), φ and v satisfy the first conclusion of (2). We let

$$f' := v\varphi(f), \qquad W' := (f' = 0), \qquad \text{and} \qquad \mathcal{I} := \varphi(\mathcal{J}).$$

Since $\mathcal{J}$ is a G-stable proper ideal and φ is G-equivariant, $\mathcal{I}$ is a G-stable ideal contained in $\mathfrak{n}$.

If $\mathcal{I} \subset (f')$, then $f' - f_0 \in \mathcal{I}^{P'} \subset f'^{P'}\widehat{\mathcal{O}}$, so $f_0 \in f'(1 + f'^{P'-1}\widehat{\mathcal{O}})$. Since $P' \geq 2$ and $f' \in \mathfrak{n}$, we get $(f_0) = (f')$ and $\mathcal{I} \subset (f_0)$, contradicting (8.14). Thus

$$\mathcal{I} \not\subset (f'). \tag{8.15}$$

We show that $h_0 \neq 0$. Suppose that $\varphi(h) \in (f_0)$. Since W_0 is normal, f_0 is a prime element of $\widehat{\mathcal{O}}$, and $D_0 := (f_0 = 0)$ is a prime divisor on $\operatorname{Spec}\widehat{\mathcal{O}}$ with $a(D_0, \operatorname{Spec}\widehat{\mathcal{O}}, 0) = 1$ and $\operatorname{mult}_{D_0}\varphi(h) \geq 1$. Since $\gamma > 0$, $\theta_1\mu > (1-\theta_1)(1+\frac{2\lambda}{m}) \geq 1 - \theta_1$, so

$$\theta_1 \operatorname{mult}_{D_0} f_0 + \theta_1\mu \operatorname{mult}_{D_0}\varphi(h) \geq \theta_1 + \theta_1\mu > 1.$$

By the definition of multiplier ideals, any z in the multiplier ideal on the right-hand side of (8.13) satisfies $1 + \operatorname{mult}_{D_0} z > \theta_1 + \theta_1\mu > 1$, so $z \in (f_0)$. By (8.13), $\mathcal{I} \subset (f_0)$, contradicting (8.14). Thus $h_0 \neq 0$. Since $h|_W$ is a non-unit, $h \in \mathfrak{n}$, and $\varphi(\mathfrak{n}) = \mathfrak{n}$. Thus h_0 is a non-unit.

Since $\varphi(f) = v^{-1}f'$, φ induces a G-equivariant isomorphism $\widehat{\mathcal{O}}/(f) \to \widehat{\mathcal{O}}/(f')$, which maps $h|_W$ to $\varphi(h)|_{W'}$, and maps Jac_W onto $\operatorname{Jac}_{W'}$ by the chain rule. For a prime divisor F over W with center 0, let F^φ be the prime divisor over W' with $\operatorname{mult}_{F^\varphi}\varphi(z) = \operatorname{mult}_F z$ for any $z \in \widehat{\mathcal{O}}$. Then $F \to F^\varphi$ is a bijection from the prime divisors over W with center 0 to those over W' with center 0. It preserves the log discrepancy, the Jacobian order, the residue field, and the decomposition and inertia groups, and $\operatorname{mult}_{F^\varphi}\varphi(h) = \operatorname{mult}_F h$. Hence W' is normal, $(W', \frac{1}{m}\operatorname{div}(\varphi(h)|_{W'}))$ is lc, (v) holds for $V = W'$, and it suffices to prove (a) and (b) for W' and $\varphi(h)|_{W'}$ instead of W and $h|_W$.

Step 2. Let $V \in \{W', W_0\}$ with equation $g \in \{f', f_0\}$, and let V' be the other one of W', W_0. Let $H \subset G$ be a subgroup, $t \in [0,1]$, and F a prime divisor over V with center 0 such that

$$a(F, V, \tfrac{t}{m}\operatorname{div}(\varphi(h)|_V)) \leq B_0|I_H(F)|. \tag{8.16}$$

The hypotheses of the claim hold for $h_1 = \varphi(h)$ and $e = |I_H(F)| \leq M$. First, $\varphi(h)|_V \neq 0$, since $h|_W \neq 0$ and $h_0 \neq 0$. Second, $\mathcal{I} \not\subset (g)$ by (8.14) and (8.15). Third,

$$\mathcal{I} \subset \mathcal{J}(\operatorname{Spec}\widehat{\mathcal{O}}, g^{\theta_1}\varphi(h)^{\theta_1\mu}).$$

For $g = f_0$ this is (8.13). For $g = f'$, since $\theta_1 < \theta_2$,

$$\mathcal{I} = \mathcal{J}(\operatorname{Spec}\widehat{\mathcal{O}}, f'^{\theta_2}\varphi(h)^{\theta_2\mu}) \subset \mathcal{J}(\operatorname{Spec}\widehat{\mathcal{O}}, f'^{\theta_1}\varphi(h)^{\theta_1\mu})$$

by (8.9) with $(\alpha, u) = (\varphi, v)$. Finally, the inequality in (v) holds for V, by (v) if $V = W_0$ and by **Step 1** if $V = W'$. Let $\nu := \operatorname{mult}_F\mathcal{I}$ and $j := \operatorname{mult}_F \operatorname{Jac}_V$. Since $\mathcal{I} \subset \mathfrak{n}$ and $\operatorname{center} F = 0$, $\nu \geq 1$. By the claim (2), $\operatorname{mult}_F\varphi(h) < C_2\nu$ and $j < C_3\nu$. Since $P' > 2C_2 + 2C_3 + 4$, we have

$$P'\nu > 2j + 2\nu \quad \text{and} \quad P'\nu - j > \operatorname{mult}_F\varphi(h).$$

Moreover f' is G-semi-invariant with the character of f_0, W' is G-equivariantly isomorphic through φ to W, by (i) W and W_0 are G-equivariantly isomorphic to the completions of normal algebraic germs, $\mathcal{I}$ is G-stable, and $f' - f_0 \in \mathcal{I}^{P'}$ by (8.12). Fix a coordinate x_i with $\operatorname{mult}_F \partial_i g = j$. Thus by Lemma 8.1(1), F and x_i give a prime divisor F' over V' with center 0, whose log discrepancy,

Jacobian order, order of $\mathcal{I}$, residue field, decomposition group, and inertia group equal those of F, such that $\operatorname{mult}_{F'}\varphi(h) = \operatorname{mult}_F \varphi(h)$ and

$$\frac{a(F', V', \frac{t'}{m}\operatorname{div}(\varphi(h)|_{V'}))}{|I_{H'}(F')|} = \frac{a(F, V, \frac{t'}{m}\operatorname{div}(\varphi(h)|_V))}{|I_{H'}(F)|} \tag{8.17}$$

for any subgroup $H' \subset G$ and any $t' \in [0,1]$. In particular F' satisfies (8.16) over V'.

Step 3. We prove (a). Fix a subgroup $H \subset G$ and $t \in [0,1]$. Let $\mathcal{S}'$, resp. $\mathcal{S}_0$, be the set of prime divisors F over W', resp. over W_0, with center 0 satisfying (8.16). By **Step 2** for $V = W'$ and for $V = W_0$, each element of $\mathcal{S}'$ corresponds to an element of $\mathcal{S}_0$, and each element of $\mathcal{S}_0$ to an element of $\mathcal{S}'$, with (8.17). By (8.10) and **Step 1**, $F_1^{\varphi} \in \mathcal{S}'$, so $\mathcal{S}' \neq \emptyset$ and $\mathcal{S}_0 \neq \emptyset$. For a prime divisor F over W', resp. over W_0, with center 0, we let

$$\rho'(F) := \frac{a(F, W', \frac{t}{m}\operatorname{div}(\varphi(h)|_{W'}))}{|I_H(F)|}, \quad \text{resp.} \quad \rho_0(F) := \frac{a(F, W_0, \frac{t}{m}\operatorname{div}(h_0))}{|I_H(F)|}.$$

Then $\rho' > B_0$ outside $\mathcal{S}'$ and $\rho' \leq B_0$ on $\mathcal{S}'$, and $\rho_0 > B_0$ outside $\mathcal{S}_0$ and $\rho_0 \leq B_0$ on $\mathcal{S}_0$. Thus by Definition 2.2 for the subgroup H and by (8.17), $\inf_{\mathcal{S}'} \rho' = \inf_{\mathcal{S}_0} \rho_0$, and

$$\begin{aligned}\operatorname{mld}((W' \ni 0, \tfrac{t}{m}\operatorname{div}(\varphi(h)|_{W'}))/H) &= \inf_{F \in \mathcal{S}'} \rho'(F) \\ &= \operatorname{mld}((W_0 \ni 0, \tfrac{t}{m}\operatorname{div}(h_0))/H).\end{aligned}$$

By **Step 1**, the left-hand side is $\operatorname{mld}((W \ni 0, \frac{t}{m}\operatorname{div}(h|_W))/H)$. This proves (a).

Step 4. We prove (b). Let E_0 be a computing lift of $\operatorname{mld}((W_0 \ni 0, \frac{1}{m}\operatorname{div}(h_0))/G)$ with $a(E_0, W_0, 0)/e_{E_0} \leq L$. By (a) with $H = G$ and $t = 1$, $a(E_0, W_0, \frac{1}{m}\operatorname{div}(h_0))/e_{E_0} = b \leq B_0$, so E_0 satisfies (8.16) over W_0 with $H = G$ and $t = 1$. By **Step 2** and **Step 1**, there exists a prime divisor E over W with center 0 such that

$$a(E, W, 0) = a(E_0, W_0, 0), \quad e_E = e_{E_0}, \quad \text{and} \quad \operatorname{mult}_E h = \operatorname{mult}_{E_0} \varphi(h),$$

so $a(E, W, \frac{1}{m}\operatorname{div}(h|_W))/e_E = b$ and $a(E, W, 0)/e_E \leq L$. Thus E is a computing lift of b. This proves (b). □

8.3. Jacobian orders of germs with Du Val sections of types D_n and E.

Theorem 8.4. *Let $n \geq 4$ be an integer, let $S_n := (x^2 + y^2 z + z^{n-1} = 0) \subset \mathbb{A}^3$, and let $\kappa_n > 0$ be a real number such that $(S_n, \operatorname{Jac}_{S_n}^{\kappa_n})$ is lc near 0. Let $(V \ni 0)$ be a normal hypersurface threefold germ in a smooth fourfold germ $(\mathcal{X} \ni 0)$, and let ℓ_V be a regular parameter of $\mathcal{X}$ at 0 such that $(V \cap (\ell_V = 0) \ni 0)$ is a Du Val surface germ of type D_n. Then any prime divisor F over V with center 0 satisfies*

$$\kappa_n \operatorname{mult}_F \operatorname{Jac}_V \leq a(F, V, 0) - \operatorname{mult}_F \ell_V.$$

Remark 8.5. Since $(S_n \ni 0)$ is klt, any sufficiently small rational number $\kappa_n > 0$ satisfies the hypothesis of Theorem 8.4.

Proof of Theorem 8.4. Let f_V be an equation of V in $\mathcal{X}$, and let (w_1, w_2, w_3, ℓ_V) be a regular system of parameters of $\mathcal{X}$ at 0. Let J_{rel} be the ideal of $\mathcal{O}_V$ generated by $\partial f_V/\partial w_1$, $\partial f_V/\partial w_2$, and $\partial f_V/\partial w_3$, the partial derivatives being taken with respect to (w_1, w_2, w_3, ℓ_V). Then $J_{\mathrm{rel}} \subset \operatorname{Jac}_V$, so

$$\operatorname{mult}_F \operatorname{Jac}_V \leq \operatorname{mult}_F J_{\mathrm{rel}} \quad \text{for any prime divisor } F \text{ over } V. \tag{8.18}$$

Let $S := V \cap (\ell_V = 0)$, a hypersurface in the smooth threefold germ $(\ell_V = 0)$ with regular parameters $w_1|_{(\ell_V=0)}, w_2|_{(\ell_V=0)}, w_3|_{(\ell_V=0)}$. Since $f_V|_{(\ell_V=0)}$ is an equation of S in $(\ell_V = 0)$ and, for $i \in \{1,2,3\}$, $\partial/\partial w_i$ commutes with restriction to $(\ell_V = 0)$, we have

$$J_{\mathrm{rel}}|_S = \operatorname{Jac}_S .$$

Since $(S \ni 0)$ and $(S_n \ni 0)$ are both Du Val of type D_n, their completions are isomorphic, and, since $x^2 + y^2 z + z^{n-1}$ and the equation of S belong to the squares of the maximal ideals, the isomorphism lifts to an isomorphism of the completed ambient threefold germs, which maps the ideal of S_n onto the ideal of S and hence carries the equation of S_n to a unit multiple of the equation of S. Thus it carries Jac_{S_n} to Jac_S (Definition 5.2). Since $(S_n, \mathrm{Jac}_{S_n}^{\kappa_n})$ is lc near 0, Lemma 2.12(3) with $B = 0$, used for S_n and for S, implies, since any formal prime divisor $\widehat{P}$ with restriction P satisfies $\mathrm{mult}_{\widehat{P}} \widehat{\mathfrak{a}} = \mathrm{mult}_P \mathfrak{a}$ for any ideal $\mathfrak{a}$ with extension $\widehat{\mathfrak{a}}$ to the completion, that $\kappa_n \mathrm{mult}_E \mathrm{Jac}_S \le a(E, S, 0)$ for any prime divisor E over S with center 0, that is,

$$\mathrm{mld}(S \ni 0, \mathrm{Jac}_S^{\kappa_n}) \ge 0.$$

Since $J_{\mathrm{rel}}|_S = \mathrm{Jac}_S$ is primary to the maximal ideal of $\mathcal{O}_{S,0}$, as S has an isolated singularity at 0, $S \not\subset V(J_{\mathrm{rel}})$. Since V is a normal local complete intersection and S is a normal Cartier divisor on V, [EM04, Theorem 1.1] with $D = S$, $Y = \kappa_n V(J_{\mathrm{rel}})$, and the closed subset $\{0\}$ implies

$$\mathrm{mld}(V \ni 0, S, J_{\mathrm{rel}}^{\kappa_n}) = \mathrm{mld}(S \ni 0, \mathrm{Jac}_S^{\kappa_n}) \ge 0.$$

Since $S = \mathrm{div}(\ell_V)$, any prime divisor F over V with center 0 satisfies

$$a(F, V, 0) - \mathrm{mult}_F \ell_V - \kappa_n \mathrm{mult}_F J_{\mathrm{rel}} \ge 0,$$

and the theorem follows from (8.18). □

Theorem 8.6. *Let $(V \ni 0)$ be a normal hypersurface threefold germ in a smooth fourfold germ $(\mathcal{X} \ni 0)$, of quadratic rank 1, whose general hyperplane section through 0 is a Du Val surface germ of type E_6, E_7, or E_8. Then*

$$\mathrm{mult}_F \mathrm{Jac}_V \le 15 a(F, V, 0)$$

for any prime divisor F over V with center 0.

Proof. Let $y_1, \dots, y_4$ be regular parameters of $\mathcal{X}$ at 0, and let f_V be a local equation of V. We first show that

(8.19) $\quad (V, \frac{1}{15}\mathrm{div}(g))$ is lc near 0 for any regular function $g \in \mathfrak{m}_{V,0} \backslash \mathfrak{m}_{V,0}^2$.

Let $g \in \mathfrak{m}_{V,0} \backslash \mathfrak{m}_{V,0}^2$. Since $f_V \in \mathfrak{m}_{\mathcal{X},0}^2$, we have

$$\mathfrak{m}_{V,0}/\mathfrak{m}_{V,0}^2 = \mathfrak{m}_{\mathcal{X},0}/\mathfrak{m}_{\mathcal{X},0}^2.$$

Let $H := (\ell = 0)$ for a general linear combination ℓ of $y_1, \dots, y_4$, and let $S := V \cap H$. Then S is a Du Val surface of type E_6, E_7, or E_8, and $g_S := g|_S \in \mathfrak{m}_{S,0} \backslash \mathfrak{m}_{S,0}^2$, since the class of g in $\mathfrak{m}_{\mathcal{X},0}/\mathfrak{m}_{\mathcal{X},0}^2$ is not a multiple of the class of ℓ.

Let $\mathcal{B} \subset SL_2(\mathbb{C})$ be the binary tetrahedral, octahedral, or icosahedral group if S is of type E_6, E_7, or E_8 respectively, acting linearly on $\mathbb{A}^2$ with coordinates z_1, z_2, let $\pi : \mathbb{A}^2 \to \overline{S} := \mathbb{A}^2/\mathcal{B}$ be the quotient morphism, and let $\overline{0} := \pi(0)$. By [Roa04, Table 1], $\mathbb{C}[z_1, z_2]^{\mathcal{B}}$ is generated by homogeneous invariants $\gamma_1, \gamma_2, \gamma_3$ of degrees

$$(d_1, d_2, d_3) = (6, 8, 12),\ (8, 12, 18),\ \text{or } (12, 20, 30)$$

respectively, and the kernel of the homomorphism

$$\mathbb{C}[\Gamma_1, \Gamma_2, \Gamma_3] \to \mathbb{C}[z_1, z_2]^{\mathcal{B}}, \quad \Gamma_i \to \gamma_i,$$

is generated by one polynomial $\Phi \in (\Gamma_1, \Gamma_2, \Gamma_3)^2$. By Lemma 2.12(1),

$$\widehat{\mathcal{O}}_{\overline{S},\overline{0}} = \mathbb{C}[[z_1, z_2]]^{\mathcal{B}}.$$

Since S and $\overline{S}$ are Du Val surfaces, both of type E_6, both of type E_7, or both of type E_8, we fix an isomorphism

$$\widehat{\mathcal{O}}_{S,0} \cong \mathbb{C}[[z_1, z_2]]^{\mathcal{B}}$$

and identify $\widehat{S} := \operatorname{Spec} \widehat{\mathcal{O}}_{S,0}$ with $\operatorname{Spec} \widehat{\mathcal{O}}_{\overline{S},\overline{0}}$. Let $\mathfrak{m} = \mathfrak{m}_{S,0}\widehat{\mathcal{O}}_{S,0}$ be the maximal ideal of

$$\mathbb{C}[[z_1, z_2]]^{\mathcal{B}} \cong \mathbb{C}[[\Gamma_1, \Gamma_2, \Gamma_3]]/(\Phi), \quad \Gamma_i \to \gamma_i,$$

and let $\widehat{g} \in \mathbb{C}[[z_1, z_2]]^{\mathcal{B}}$ be the image of g_S.

We show that $\mu := \operatorname{mult}_0 \widehat{g} \leq 30$. Write

$$\widehat{g} = \sum_{i=1}^{3} c_i\gamma_i + \Psi(\gamma_1, \gamma_2, \gamma_3), \qquad \text{with } c_i \in \mathbb{C} \text{ and } \Psi \in (\Gamma_1, \Gamma_2, \Gamma_3)^2 \cdot \mathbb{C}[[\Gamma_1, \Gamma_2, \Gamma_3]].$$

Since (Φ) is contained in $(\Gamma_1, \Gamma_2, \Gamma_3)^2$, the classes of $\gamma_1, \gamma_2, \gamma_3$ form a basis of $\mathfrak{m}/\mathfrak{m}^2$, so $c_i \neq 0$ for some i, as $g_S \notin \mathfrak{m}_{S,0}^2$. Fix i with $c_i \neq 0$. Let Ψ_i be the sum of the terms of Ψ of weighted degree d_i for the weights (d_1, d_2, d_3). Since $d_1 < d_2 < d_3$, the homogeneous component of $\widehat{g}$ of degree d_i is $c_i\gamma_i + \Psi_i(\gamma_1, \gamma_2, \gamma_3)$. It is non-zero: otherwise $c_i\Gamma_i + \Psi_i$ belongs to (Φ), which is contained in $(\Gamma_1, \Gamma_2, \Gamma_3)^2$, contradicting $c_i \neq 0$ and $\Psi_i \in (\Gamma_1, \Gamma_2, \Gamma_3)^2$. Thus

$$\mu \leq d_i \leq 30.$$

Let $\widehat{g}_\mu$ be the homogeneous component of $\widehat{g}$ of degree μ. Since $\mathcal{B}$ acts linearly, $\widehat{g}_\mu$ is $\mathcal{B}$-invariant. Any linear factor of $\widehat{g}_\mu$ has multiplicity at most $\frac{\mu}{2}$ in $\widehat{g}_\mu$. Otherwise exactly one linear factor λ has multiplicity $> \frac{\mu}{2}$, so $\mathcal{B}$ preserves the line $(\lambda = 0) \subset \mathbb{C}^2$. By Maschke's theorem $\mathcal{B}$ also preserves a complementary line, so $\mathcal{B}$ is abelian, which is not the case.

For any integer $N \geq \mu$, let $\widehat{g}_{\leq N}$ be the sum of the homogeneous components of $\widehat{g}$ of degree at most N. Then $\widehat{g}_{\leq N} \in \mathbb{C}[z_1, z_2]^{\mathcal{B}}$ and $\operatorname{mult}_0 \widehat{g}_{\leq N} = \mu$. Since any linear factor of $\widehat{g}_\mu$ has multiplicity at most $\frac{\mu}{2}$, $(\mathbb{A}^2, \frac{2}{\mu}\operatorname{div}(\widehat{g}_{\leq N}))$ is lc near 0 by inversion of adjunction on the blow-up of 0. Since $\mathcal{B} \subset SL_2(\mathbb{C})$ contains no pseudo-reflection, π is quasi-étale, and $\widehat{g}_{\leq N} = \pi^*\overline{g}_{\leq N}$ for a regular function $\overline{g}_{\leq N}$ on $\overline{S}$. Thus $(\overline{S}, \frac{2}{\mu}\operatorname{div}(\overline{g}_{\leq N}))$ is lc near $\overline{0}$ for any integer $N \geq \mu$.

Let F be a prime divisor over S with center 0. By Lemma 2.12(3) for S, F is the restriction of a formal prime divisor $\widehat{F}$ over $\widehat{S}$ with center the closed point, and

$$a(F, S, \tfrac{2}{\mu}\operatorname{div}(g_S)) = a(\widehat{F}, \widehat{S}, \tfrac{2}{\mu}\operatorname{div}(\widehat{g})).$$

By Lemma 2.12(3) for $\overline{S}$, $\widehat{F}$ restricts to a prime divisor $\overline{F}$ over $\overline{S}$ with center $\overline{0}$, and

$$a(\overline{F}, \overline{S}, \tfrac{2}{\mu}\operatorname{div}(\overline{g}_{\leq N})) = a(\widehat{F}, \widehat{S}, \tfrac{2}{\mu}\operatorname{div}(\widehat{g}_{\leq N}))$$

for any $N \geq \mu$. A homogeneous $\mathcal{B}$-invariant polynomial of degree e is a weighted homogeneous polynomial in $\gamma_1, \gamma_2, \gamma_3$ of weighted degree e, hence belongs to $\mathfrak{m}^{\lceil e/30 \rceil}$. Thus $\widehat{g} - \widehat{g}_{\leq N} \in \mathfrak{m}^{\lceil (N+1)/30 \rceil}$. Since

$$\operatorname{mult}_{\widehat{F}}(\widehat{g} - \widehat{g}_{\leq N}) \geq \lceil \tfrac{N+1}{30} \rceil \operatorname{mult}_{\widehat{F}} \mathfrak{m} \to +\infty,$$

$\operatorname{mult}_{\widehat{F}} \widehat{g}_{\leq N} = \operatorname{mult}_{\widehat{F}} \widehat{g}$ for any $N \gg 0$, and

$$a(F, S, \tfrac{2}{\mu}\operatorname{div}(g_S)) = a(\overline{F}, \overline{S}, \tfrac{2}{\mu}\operatorname{div}(\overline{g}_{\leq N})) \geq 0.$$

Hence $\operatorname{mld}(S \ni 0, \frac{2}{\mu}\operatorname{div}(g_S)) \geq 0$, so $(S, \frac{2}{\mu}\operatorname{div}(g_S))$ is lc near 0. Since $\frac{1}{15} \leq \frac{2}{\mu}$, $(S, \frac{1}{15}\operatorname{div}(g_S))$ is lc near 0.

Since S is klt, $(S, (\frac{1}{15} - \epsilon)\operatorname{div}(g_S))$ is klt near 0 for any rational number $0 < \epsilon < \frac{1}{15}$, so $(V, S + (\frac{1}{15} - \epsilon)\operatorname{div}(g))$ is plt near 0 by inversion of adjunction. Letting $\epsilon \to 0$, $(V, S + \frac{1}{15}\operatorname{div}(g))$ is lc near 0, which implies (8.19).

Possibly changing $y_1, \ldots, y_4$ linearly, we may assume that $f_V - y_1^2 \in \mathfrak{m}_{\mathcal{X},0}^3$. Then

$$\partial f_V/\partial y_1 - 2y_1 \in \mathfrak{m}_{\mathcal{X},0}^2,$$

so

$$g_1 := (\partial f_V/\partial y_1)|_V \in \mathfrak{m}_{V,0} \backslash \mathfrak{m}_{V,0}^2.$$

Since Jac_V is generated by the restrictions of $\partial f_V/\partial y_1, \ldots, \partial f_V/\partial y_4$ to V, (8.19) implies

$$\operatorname{mult}_F \operatorname{Jac}_V \leq \operatorname{mult}_F g_1 \leq 15a(F, V, 0)$$

for any prime divisor F over V with center 0. □

9. Zero values, coefficient-one components and non-klt pairs

In this section we show that the radial failure sequences $(W_i \ni o_i, G_i, h_i, \tau_i)$ of Construction-Theorem 4.7(C) have, after passing to a subsequence, positive values at 1, and that, after passing to a further subsequence, in quadratic ranks 1 and 2 the pairs defined by their marks satisfy $\lfloor \frac{1}{m}\operatorname{div}(h_i)\rfloor = 0$ near o_i and are klt.

Theorem 9.1. *Let $(W_i \ni o_i, G_i, h_i, \tau_i)$ be a radial failure sequence with constants (m, a, δ) given by Construction-Theorem 4.7(C). Then, after passing to a subsequence, $b_i > 0$ for any i. If moreover $(W_i \ni o_i)$ has quadratic rank 1 or 2 for any i, then, after passing to a further subsequence,*

$$\left\lfloor \frac{1}{m}\operatorname{div}(h_i)\right\rfloor = 0 \text{ near } o_i \quad \text{and} \quad \left(W_i, \frac{1}{m}\operatorname{div}(h_i)\right) \text{ is klt near } o_i$$

for any i.

Theorems 9.2–9.4 are proved in Appendix A.

Theorem 9.2. *Let m be a positive integer. Then there exists a real number L_m such that the following holds. Let $(W \ni o)$ be a smooth threefold germ or an isolated Gorenstein terminal hypersurface threefold germ, G a finite abelian group acting faithfully on W and fixing o, and h a mark on $(W \ni o)$ such that $(W, \frac{1}{m}\operatorname{div}(h))$ is lc near o. Let q be the radial function of $(W \ni o, G, h, m)$, and assume that $q(1) = 0$. Then $q(1)$ has a computing lift F with*

$$\frac{a(F, W, 0)}{e_F} \le L_m.$$

Theorem 9.3. *Let m be a positive integer. Then there exists a real number $L(m)$ such that the following holds. Let $(W \ni o)$ be an isolated Gorenstein terminal hypersurface threefold germ of quadratic rank 1 or 2, G a finite cyclic group acting faithfully on W and fixing o, h a mark on $(W \ni o)$, $D := \frac{1}{m}\operatorname{div}(h)$, and q the radial function of $(W \ni o, G, h, m)$. Assume that (W, D) is lc near o, $\lfloor D\rfloor \ne 0$ at o, and $\lfloor D\rfloor$ is $\mathbb{Q}$-Cartier. Then $q(1)$ has a computing lift F with*

$$\frac{a(F, W, 0)}{e_F} \le L(m).$$

Theorem 9.4. *Let m be a positive integer. Then there exists a real number $L(m)$ such that the following holds. Let $(W \ni o)$ be a smooth threefold germ or an isolated Gorenstein terminal hypersurface threefold germ, G a finite group acting faithfully on W and fixing o, h a mark on $(W \ni o)$, $D := \frac{1}{m}\operatorname{div}(h)$, and q the radial function of $(W \ni o, G, h, m)$. Assume that*

(i) (W, D) is lc near o,
(ii) $\lfloor D\rfloor = 0$ near o,
(iii) (W, D) is not klt near o, and
(iv) $q(1) > 0$.

Then $q(1)$ has a computing lift F with

$$\frac{a(F, W, 0)}{e_F} \le L(m).$$

Proof of Theorem 9.1. Let (X_i, x_i, D_i) be the data of Construction-Theorem 4.7(C) from which the sequence is obtained. For any i, let $\rho_i : V_i \to X_i$ and $g_i : Y_i \to V_i$ be the cover and the morphism of Construction-Theorem 4.7(A) for (X_i, x_i, D_i), and let $D_{V_i} := \rho_i^* D_i$. Then $(W_i \ni o_i) = (Y_i \ni y_i)$ for a closed point $y_i \in Y_i$, and $\operatorname{div}(h_i) = m g_i^* D_{V_i}$ near o_i by Construction-Theorem 4.7(B). Let $\mathcal{G}_i$ be the Galois group of ρ_i (the group G of Construction-Theorem 4.7(A)),

so that $G_i \subset \mathcal{G}_i$. Since $D_{V_i} = \rho_i^* D_i$ is $\mathcal{G}_i$-invariant and g_i is $\mathcal{G}_i$-equivariant, $\lfloor g_i^* D_{V_i} \rfloor$ is a $\mathcal{G}_i$-invariant Weil divisor on Y_i, hence $\mathbb{Q}$-Cartier by Construction-Theorem 4.7(A)(3). Thus

$$\left\lfloor \frac{1}{m} \operatorname{div}(h_i) \right\rfloor \text{ is } \mathbb{Q}\text{-Cartier near } o_i \text{ for any } i. \tag{9.1}$$

We show that each of the following cases holds for only finitely many indices i.

(a) $b_i = 0$.
(b) $(W_i \ni o_i)$ has quadratic rank 1 or 2, and $\lfloor \frac{1}{m} \operatorname{div}(h_i) \rfloor \neq 0$ at o_i.
(c) $b_i > 0$, $\lfloor \frac{1}{m} \operatorname{div}(h_i) \rfloor = 0$ near o_i, and $(W_i, \frac{1}{m} \operatorname{div}(h_i))$ is not klt near o_i.

Since $\lim_{i \to +\infty} \tau_i = 1$, Lemma 4.3(2) implies that, for any real number λ,

$$\#\left\{ i \;\middle|\; b_i \text{ has a computing lift } F_i \text{ with } \frac{a(F_i, W_i, 0)}{e_{F_i}} \leq \lambda \right\} < +\infty. \tag{9.2}$$

For any i in case (a), by Theorem 9.2, b_i has a computing lift F_i with $a(F_i, W_i, 0)/e_{F_i} \leq L_m$. For any i in case (c), by Theorem 9.4, b_i has a computing lift F_i with $a(F_i, W_i, 0)/e_{F_i}$ at most the constant $L(m)$ of Theorem 9.4. By (9.2), only finitely many indices i are in case (a) or (c). Passing to a subsequence, we may assume that no index is in case (a) or (c). In particular, $b_i > 0$ for any i.

Assume that $(W_i \ni o_i)$ has quadratic rank 1 or 2 for any i, and let λ be the constant $L(m)$ of Theorem 9.3. Let i be an index in case (b). Then $(W_i \ni o_i)$ has quadratic rank 1 or 2, and $\lfloor \frac{1}{m} \operatorname{div}(h_i) \rfloor$ is $\mathbb{Q}$-Cartier near o_i by (9.1). Thus, by Theorem 9.3, b_i has a computing lift F_i with $a(F_i, W_i, 0)/e_{F_i} \leq \lambda$. By (9.2), only finitely many indices i are in case (b). Passing to a further subsequence, we may assume that no index is in case (b). Then $\lfloor \frac{1}{m} \operatorname{div}(h_i) \rfloor = 0$ near o_i for any i, and since no index is in case (c) and $b_i > 0$, $(W_i, \frac{1}{m} \operatorname{div}(h_i))$ is klt near o_i for any i. The theorem follows. □

10. ACC for germs of quadratic rank at least three

In this section we prove the following theorem.

Theorem 10.1. *Let $\Gamma \subset [0,1]$ be a DCC set. Then the set of values*

$$\operatorname{mld}((W \ni o, D)/G)$$

satisfies the ACC, where $(W \ni o)$ is a smooth threefold germ or a hypersurface threefold germ of quadratic rank at least 3, G is a finite group acting faithfully on W and fixing o, and $D = \sum_{j=1}^n c_j \operatorname{div}(h_j)$ is a labelled boundary with $c_j \in \Gamma$ such that (W, D) is lc near o.

The proof of Theorem 10.1 is given at the end of this section.

10.1. **The set-up and floors.** The proofs of Theorem 10.4 and Proposition 10.5 below are given in Appendix B.

Lemma 10.2. *Let $\Bbbk \supset \mathbb{C}$ be a field, and let $\pi^\sharp : \Bbbk[[x,y,z,t]] \to \Bbbk[[p,v,t]]$ be the homomorphism with $(x,y,z,t) \to (p^2, -v^2, pv, t)$.*

(1) $\pi^\sharp$ maps distinct monomials not divisible by xy to distinct monomials up to sign, and a monomial of total degree d in x,y,z to a monomial of total degree $2d$ in p,v.
(2) Let $w = (w_1, w_2, w_3) \in \mathbb{R}^3_{>0}$ be a weight vector on (p,v,t), and let $w^\flat$ be the weight vector on (x,y,z,t) given by

$$w^\flat := (2w_1, 2w_2, w_1 + w_2, w_3).$$

Then

$$\operatorname{wt}_w(\pi^\sharp h) = \operatorname{wt}_{w^\flat}(h) \quad \text{and} \quad \operatorname{in}_w(\pi^\sharp h) = \pi^\sharp(\operatorname{in}_{w^\flat} h)$$

for any non-zero $h \in \Bbbk[[x,y,z,t]]$ without monomials divisible by xy.

Proof. (1) A monomial not divisible by xy is $x^a z^c t^e$ or $y^b z^c t^e$ with $a, b, c, e \geq 0$, and

$$\pi^\sharp(x^a z^c t^e) = p^{2a+c} v^c t^e, \qquad \pi^\sharp(y^b z^c t^e) = (-1)^b p^c v^{2b+c} t^e.$$

The exponent of p in the first image is at least, resp. in the second image at most, the exponent of v, with equality if and only if $a = 0$, resp. $b = 0$, that is, exactly when the monomial is $z^c t^e$. Since the exponents of the image determine (a, c, e), resp. (b, c, e), distinct monomials not divisible by xy have distinct images up to sign. The images have total degree $2(a + c)$, resp. $2(b + c)$, in p, v.

(2) The w-weight of $p^{2a+c} v^c t^e$ is $(2a + c)w_1 + cw_2 + ew_3 = 2aw_1 + c(w_1 + w_2) + ew_3$, the $w^\flat$-weight of $x^a z^c t^e$, and similarly the w-weight of $p^c v^{2b+c} t^e$ is the $w^\flat$-weight of $y^b z^c t^e$. By (1), the monomials of $\pi^\sharp h$ are, up to sign, the images of the monomials of h, without cancellation, and the image of a monomial μ of h has w-weight equal to the $w^\flat$-weight of μ. Hence, for $\lambda := \mathrm{wt}_{w^\flat}(h)$, we have $\mathrm{wt}_w(\pi^\sharp h) = \lambda$, and the terms of $\pi^\sharp h$ of w-weight λ are the images of the terms of h of $w^\flat$-weight λ, which is the second equality. □

Set-up 10.3. Let Γ, M, $(N_i)_{i\geq 1}$, G, n, $(D_i)_{i\geq 1}$, and $(h_{*,j})_j$ be as follows.

(1) $\Gamma \subset [0, 1]$ is a DCC set and M is a positive integer.
(2) $N_i \geq 2$ is an integer for any $i \geq 1$, and $\lim_i N_i = +\infty$.
(3) G is a finite diagonal abelian group of order at most M, independent of i, acting on $\mathbb{A}^4$ by $g(x, y, z, t) = (\chi_x(g)x, \chi_y(g)y, \chi_z(g)z, \chi_t(g)t)$ for $g \in G$, such that for any i, G preserves the equation of $W_{N_i} = (xy + z^2 + t^{N_i} = 0) \subset \mathbb{A}^4$ up to a character and acts faithfully on W_{N_i}.
(4) For any i, $D_i = \sum_{j=1}^n c_{ij} \operatorname{div}(h_{ij})$ is a labelled boundary on $(W_{N_i} \ni 0)$ such that (W_{N_i}, D_i) is klt near 0, where n does not depend on i, $c_{ij} \in \Gamma\backslash\{0\}$, and the marks h_{ij} are polynomials without monomials divisible by xy whose characters do not depend on i, hence G-semi-invariant polynomials, since G is diagonal and a polynomial without monomials divisible by xy is determined by its restriction to W_{N_i}.
(5) For any j, the sequence $(c_{ij})_{i\geq 1}$ is increasing with limit $\kappa_j > 0$.
(6) The orders at 0 of the h_{ij} are bounded.
(7) $(h_{*,j})_j$ is a compatible generic limit of the tuples $(h_{ij})_j$ along $\mathbb{N}^+$, with coefficient field $\Bbbk$, such that, for $H_{ij} := h_{ij}(p^2, -v^2, pv, t)$ and $H^*_j := h_{*,j}(p^2, -v^2, pv, t)$ with variables p, v, t, the tuple $(H^*_j)_j$ is a compatible generic limit of the tuples $(H_{ij})_j$ along $\mathbb{N}^+$.

Let $a_i := \mathrm{mld}((W_{N_i} \ni 0, D_i)/G)$. Since the exceptional divisor E of the blow-up of W_{N_i} at 0 satisfies $a(E, W_{N_i}, 0) = 2$ [Liu26b, Lemma 4.12] and (W_{N_i}, D_i) is klt near 0, we have $0 < a_i \leq 2$ for any i.

Let $\pi_0 : \widetilde{W}_0 = \mathbb{A}^3 \to W_0 = (xy + z^2 = 0)$ be the double cover of Subsection 3.6, given by $(x, y, z, t) = (p^2, -v^2, pv, t)$, and $\iota : (p, v, t) \to (-p, -v, t)$ its deck involution. For any i let $D_{0,i} := \sum_j c_{ij} \operatorname{div}(h_{ij}|_{W_0})$. Let

$$\widetilde{G} := \{(g, \alpha) \mid g \in G,\ \alpha \in \mathbb{C},\ \alpha^2 = \chi_x(g)\}$$

act on $\widetilde{W}_0$ by

$$(p, v, t) \to (\alpha p, \chi_z(g)\alpha^{-1} v, \chi_t(g)t).$$

Let $W_* := \operatorname{Spec} \Bbbk[[x, y, z, t]]/(xy+z^2)$ and $\widetilde{W}_* := \operatorname{Spec} \Bbbk[[p, v, t]]$, and write π_0 also for the induced double cover $\widetilde{W}_* \to W_*$. By (4), $h_{*,j}$ has no monomials divisible by xy, since for any d the d-jets without such monomials form a closed linear subspace of the space $\mathcal{J}_d$ of Definition 2.5 containing the d-jets of the tuples $(h_{ij})_j$, hence containing Z_d. For any j let d_j be the least total degree in x, y, z of a non-zero term of $h_{*,j}$. By Lemma 10.2(1), $2d_j$ is the order of H^*_j along $(p = v = 0)$. Let

$$D_* := \sum_{j=1}^n \kappa_j \operatorname{div}(h_{*,j}|_{W_*}) \quad \text{and} \quad B^* := \sum_{j=1}^n \kappa_j \operatorname{div}(H^*_j) \quad \text{on } \widetilde{W}_*.$$

A *floor* of B^* is a $\widetilde{G}$-stable prime divisor F through 0 with coefficient one in B^*. For a floor F we write $H_j^* = F^{e_j} R_j^*$ for any j, where F also denotes a $\widetilde{G}$-semi-invariant equation of F and F does not divide R_j^*, so that $\sum_j e_j \kappa_j = 1$, and we let $B_F := \sum_j \kappa_j \operatorname{div}(R_j^*|_F)$ on F. A floor F is *deck-odd*, resp. *deck-even*, if F has an equation which is anti-invariant, resp. invariant, under ι. Since $(1,-1) \in \widetilde{G}$ acts as ι, any floor is deck-odd or deck-even.

Theorem 10.4. *In Set-up 10.3, assume that B^* is lc and $\lfloor B^* \rfloor \neq 0$. Then $(a_i)_{i\geq 1}$ is not strictly increasing.*

Proposition 10.5. *In Set-up 10.3, assume that $(a_i)_{i\geq 1}$ is strictly increasing. Then, after passing to a subsequence, the following hold.*

(1) B^* is lc and not klt, $\lfloor B^* \rfloor = 0$, the unique lc center of B^* through 0 is a smooth $\widetilde{G}$-stable curve C, and
$$0 < \lim_i a_i \leq \operatorname{mld}(\widetilde{W}_* \ni 0, B^*) \leq 1.$$
(2) If ι acts trivially on C, then $C = (p = v = 0)$. Otherwise, after a linear change of coordinates $(p, v) \to (P, Q)$ of determinant one such that P and Q are $\widetilde{G}$-semi-invariant,
$$C = (Q - Pf(P^2) = t - g(P^2) = 0)$$
for power series f and g in one variable over $\Bbbk$ with $f(0) = g(0) = 0$.

(3) Let J_0, resp. J_1, be the set of the indices j such that H_j^* vanishes, resp. does not vanish, identically on C. Then there exists a real number $\theta > 1$ such that
$$\Big(W_{N_i}, \sum_{j\in J_0} \kappa_j \operatorname{div}(h_{ij}) + \theta \sum_{j\in J_1} c_{ij} \operatorname{div}(h_{ij})\Big)$$
is lc near 0 for infinitely many i. For any such i, let F_i be a computing lift of a_i, and let
$$w_{ij} := \frac{\operatorname{mult}_{F_i} h_{ij}}{e_{F_i}} \text{ for any } j, \qquad R_i := \sum_{j\in J_0} (\kappa_j - c_{ij}) w_{ij}.$$
Then, after passing to a further subsequence of such i, the numbers w_{ij} for $j \in J_1$ and the number $a_i - R_i + \sum_{j\in J_1} c_{ij} w_{ij}$ do not depend on i, and $(R_i)_i$ is strictly increasing with a positive limit.

10.2. **Limits without floor.**

Set-up 10.6. Notation and conditions as in Set-up 10.3. Assume moreover that B^* is lc, $\lfloor B^* \rfloor = 0$, $\operatorname{mld}(\widetilde{W}_* \ni 0, B^*) > 0$, and B^* has exactly one lc center C of dimension 1 through 0, which is a smooth curve.

By Proposition 10.5(1)(2), if $(a_i)_{i\geq 1}$ is strictly increasing, then, after passing to a subsequence, the data satisfy Set-up 10.6, and either ι acts non-trivially on C or $C = (p = v = 0)$. We prove Theorem 10.7 in Subsection B.3, and Theorems 10.8 and 10.9 in Subsection B.4.

Theorem 10.7. *Notation and conditions as in Set-up 10.6. Assume that ι acts non-trivially on C. Then $(a_i)_{i\geq 1}$ is not strictly increasing.*

Theorem 10.8. *Notation and conditions as in Set-up 10.6. Assume that $C = (p = v = 0)$ and*
$$\sum_j \kappa_j d_j = 1.$$
Then $(a_i)_{i\geq 1}$ is not strictly increasing.

Theorem 10.9. *Notation and conditions as in Set-up 10.6. Assume that $C = (p = v = 0)$ and*
$$\sum_j \kappa_j d_j < 1.$$
Then $(a_i)_{i\geq 1}$ is not strictly increasing.

10.3. **Bounded groups and the proof of the ACC.**

Theorem 10.10. *Let M be a positive integer and $\Gamma \subset [0,1]$ a DCC set. Then the set of* $\operatorname{mld}((W_N \ni 0, D)/G)$ *satisfies the ACC, where N, G, and D satisfy the following.*

(i) $N \geq 2$ *is an integer.*
(ii) G *is a finite diagonal abelian group of order at most M which preserves the equation of W_N up to a character and acts faithfully on W_N.*
(iii) $D = \sum_j c_j \operatorname{div}(h_j)$ *is a labelled boundary on $(W_N \ni 0)$ with $c_j \in \Gamma\backslash\{0\}$, and (W_N, D) is klt near 0.*

Proof. Possibly replacing Γ with $\Gamma \cup \{1\}$, we may assume that $\Gamma\backslash\{0\} \neq \emptyset$. Let

$$\gamma := \min(\Gamma\backslash\{0\}) > 0.$$

Suppose that there exist data N_i, G_i, and $D_i = \sum_{j=1}^{n_i} c_{ij} \operatorname{div}(h_{ij})$ as in the theorem, $i \in \mathbb{N}^+$, such that

$$a_i := \operatorname{mld}((W_{N_i} \ni 0, D_i)/G_i)$$

is strictly increasing.

We first normalize the marks. Fix i and j. There exist polynomials ϕ and ψ in x, y, z, t with $\psi(0) \neq 0$ and $h_{ij} = \phi/\psi$ near 0. The function $\prod_{\sigma \in G_i} \sigma^*\psi$ is a G_i-invariant unit near 0, and

$$h_{ij} \prod_{\sigma \in G_i} \sigma^*\psi = \phi \prod_{\sigma \in G_i \backslash \{\mathrm{id}_{\mathbb{A}^4}\}} \sigma^*\psi$$

is the restriction to W_{N_i} of a polynomial. Using $xy = -z^2 - t^{N_i}$ on W_{N_i}, there exists a unique polynomial Φ_{ij} without monomials divisible by xy whose restriction to W_{N_i} is $h_{ij} \prod_{\sigma \in G_i} \sigma^*\psi$. Since G_i acts diagonally, for any $\sigma \in G_i$ the polynomial $\sigma^*\Phi_{ij}$ has no monomials divisible by xy and restricts on W_{N_i} to

$$\chi_{h_{ij}}(\sigma) h_{ij} \prod_{\sigma' \in G_i} \sigma'^*\psi,$$

so $\sigma^*\Phi_{ij} = \chi_{h_{ij}}(\sigma)\Phi_{ij}$ by uniqueness. Hence Φ_{ij} is G_i-semi-invariant with the character of h_{ij}. Replacing each h_{ij} with Φ_{ij} and then deleting the marks h_{ij} with $h_{ij}(0) \neq 0$ changes neither D_i near 0 nor a_i. Thus we may assume that each h_{ij} is a polynomial without monomials divisible by xy and $h_{ij}(0) = 0$. Let $\operatorname{mult}_0 h_{ij} \geq 1$ be the order of h_{ij} at $0 \in \mathbb{A}^4$.

Next we bound the number and the orders of the marks. Consider the blow-up of $0 \in W_{N_i}$, and let E_1 be the exceptional prime divisor. Since 0 is a double point of the hypersurface $W_{N_i} \subset \mathbb{A}^4$, $a(E_1, W_{N_i}, 0) = 2$ and $\operatorname{mult}_{E_1} h_{ij} \geq \operatorname{mult}_0 h_{ij} \geq 1$ for any j. Since (W_{N_i}, D_i) is klt near 0 and $c_{ij} \geq \gamma$ for any j,

$$\gamma \max\Big\{n_i, \max_j \operatorname{mult}_0 h_{ij}\Big\} \leq \sum_{j=1}^{n_i} c_{ij} \operatorname{mult}_{E_1} h_{ij} < a(E_1, W_{N_i}, 0) = 2.$$

Thus

$$n_i < \frac{2}{\gamma} \quad \text{and} \quad \operatorname{mult}_0 h_{ij} < \frac{2}{\gamma} \quad \text{for any } i \text{ and } j. \tag{10.1}$$

The elements of G_i are diagonal matrices whose entries are roots of unity of order at most M, so only finitely many groups occur as G_i. Possibly passing to a subsequence, we may assume that $G_i = G$ and $n_i = n$ do not depend on i.

Case 1: the N_i are bounded. Possibly passing to a subsequence, we may assume that $N_i = N$ does not depend on i. Let $\pi : W_N \to X := W_N/G$ be the quotient morphism, and Δ_i the $\mathbb{R}$-divisor with $K_{W_N} + D_i = \pi^*(K_X + \Delta_i)$. By Lemma 2.15(0), (X, Δ_i) is a pair, and

$a_i = \mathrm{mld}(X \ni \pi(0), \Delta_i)$ by Lemma 2.15(2)(3). Let P be a prime divisor on W_N, and e_P the ramification index of π along P. Then $e_P \le |G| \le M$ and, by Lemma 2.15(4),

$$\mathrm{mult}_{\pi(P)} \Delta_i = 1 - \frac{1 - \mathrm{mult}_P D_i}{e_P} \quad \text{and}$$

$$\mathrm{mult}_P D_i = \sum_{j=1}^{n} c_{ij} \,\mathrm{mult}_P h_{ij} \in \Lambda := \Big\{ \sum_{j=1}^{n} c'_j m_j \;\Big|\; c'_j \in \Gamma,\ m_j \in \mathbb{N} \Big\}.$$

Since Γ and $\mathbb{N}$ are DCC subsets of $\mathbb{R}_{\ge 0}$, Λ is a DCC set. Thus the non-zero coefficients of all Δ_i belong to the DCC set

$$\Big\{ 1 - \frac{1-\lambda}{e} \;\Big|\; \lambda \in \Lambda,\ e \in \{1, \dots, M\} \Big\} \cap \mathbb{R}_{>0}.$$

By Theorem 2.22(1) for the normal threefold X, the set $\{a_i\}_i$ satisfies the ACC, a contradiction.

Case 2: the N_i are unbounded. Possibly passing to a subsequence, we may assume that $\lim_i N_i = +\infty$ and that the characters of the h_{ij} do not depend on i, since G has finitely many characters. Since Γ is a DCC set, possibly passing to a subsequence, we may assume that c_{ij} is increasing in i for any j. Let $\kappa_j := \lim_i c_{ij} \in [\gamma, 1]$.

Let $H_{ij} := h_{ij}(p^2, -v^2, pv, t)$ as in Set-up 10.3. By Lemma 10.2(1), $\mathrm{mult}_0 H_{ij} \le 2\,\mathrm{mult}_0 h_{ij}$, so by (10.1) the orders at 0 of the h_{ij} and of the H_{ij} are bounded. By Lemma 2.6(1), possibly passing to a subsequence, there exists a compatible generic limit $(h_{*,j})_j$ of the tuples $(h_{ij})_j$, with coefficient field $\Bbbk$.

Let d be a positive integer, and let $Z_d \subset \mathcal{J}_d$ be the irreducible closed subset of Definition 2.5 for $(h_{*,j})_j$. Let $\Sigma_d \subset \mathcal{J}_d$ be the linear subspace of tuples without monomials divisible by xy. It is closed and contains the d-jets of all the tuples $(h_{ij})_j$, so $Z_d \backslash \Sigma_d$ is an open subset of Z_d containing none of these d-jets, hence empty (Definition 2.5). Thus $Z_d \subset \Sigma_d$, and $\mathrm{jet}_d((h_{*,j})_j) \in \Sigma_d$. Since d is arbitrary, $h_{*,j}$ has no monomials divisible by xy, so

$$H^*_j := h_{*,j}(p^2, -v^2, pv, t) \neq 0$$

by Lemma 10.2(1), and H^*_j is $\widetilde{G}$-semi-invariant with the character of H_{ij}, since $h_{*,j}$ is G-semi-invariant with the character of h_{ij} (Definition 2.5) and the substitution $(x, y, z, t) = (p^2, -v^2, pv, t)$ is equivariant for $\widetilde{G} \to G$.

Let $\mathcal{J}'_d$ be the affine space of d-jets of the tuples $(H_{ij})_j$. Since the substitution does not lower degrees, it induces a linear map $\rho_d : \mathcal{J}_d \to \mathcal{J}'_d$ sending $\mathrm{jet}_d((h_{ij})_j)$ to $\mathrm{jet}_d((H_{ij})_j)$ and $\mathrm{jet}_d((h_{*,j})_j)$ to $\mathrm{jet}_d((H^*_j)_j)$. The closure Z'_d of $\rho_d(Z_d)$ is irreducible, and $\mathrm{jet}_d((H^*_j)_j)$ maps to the generic point of Z'_d. For a non-empty open subset $U' \subset Z'_d$, $Z_d \cap \rho_d^{-1}(U')$ is a non-empty open subset of Z_d, so it contains $\mathrm{jet}_d((h_{ij})_j)$ for all but finitely many i, and their images $\mathrm{jet}_d((H_{ij})_j)$ belong to U'. Since d is arbitrary, $(H^*_j)_j$ is a compatible generic limit of the tuples $(H_{ij})_j$ along $\mathbb{N}^+$, with coefficient field $\Bbbk$. Since the orders at 0 of the h_{ij} are bounded by (10.1), the data satisfy Set-up 10.3, and the a_i are strictly increasing.

By Proposition 10.5(1), possibly passing to a subsequence, B^* is lc, $\lfloor B^* \rfloor = 0$,

$$\mathrm{mld}(\widetilde{W}_* \ni 0, B^*) \ge \lim_i a_i > 0,$$

and the unique lc center of B^* through 0 is a smooth curve C, so the data satisfy Set-up 10.6. If ι acts non-trivially on C, then Theorem 10.7 contradicts the strict increase of the a_i. Thus ι acts trivially on C, and $C = (p = v = 0)$ by Proposition 10.5(2).

For any j, let $\mathrm{mult}_C H^*_j$ be the order of H^*_j in the ideal (p, v). By Lemma 10.2(1),

$$\mathrm{mult}_C H^*_j = 2d_j.$$

Consider the blow-up of C, and let E_C be the exceptional prime divisor. Since B^* is lc,

$$0 \le a(E_C, \widetilde{W}_*, B^*) = 2 - \sum_j \kappa_j \,\mathrm{mult}_C H^*_j = 2 - 2\sum_j \kappa_j d_j,$$

so $\sum_j \kappa_j d_j \le 1$. If $\sum_j \kappa_j d_j = 1$, then Theorem 10.8 contradicts the strict increase of the a_i, and if $\sum_j \kappa_j d_j < 1$, then Theorem 10.9 does. □

Proof of Theorem 10.1. **Step 1.** We prove the theorem under the additional assumption that (W, D) is klt near o. Suppose that there exist data $(W_i \ni o_i)$, G_i, and D_i as in the theorem, $i \in \mathbb{N}^+$, such that (W_i, D_i) is klt near o_i for any i and

$$a_i := \mathrm{mld}((W_i \ni o_i, D_i)/G_i)$$

is strictly increasing. Then $a_i > 0$ since (W_i, D_i) is klt near o_i. By the second sentence of [Liu26b, Theorem 7.1], only finitely many of the germs $(W_i \ni o_i)$ are smooth, of quadratic rank 4, or non-isolated singularities of quadratic rank 3. Possibly passing to a subsequence, we may assume that each $(W_i \ni o_i)$ is an isolated singularity of quadratic rank 3, hence an isolated cDV singularity, which is Gorenstein and terminal.

By Definition 2.2, for any subgroup $H \subset G_i$,

$$\mathrm{mld}((W_i \ni o_i, D_i)/H) = \inf_F \frac{a(F, W_i, D_i)}{|I_H(F)|}, \tag{10.2}$$

where F runs over the prime divisors over W_i with center o_i. By [Liu26a, Lemma 3.5] and Lemma 2.15(2)(3), the infimum in (10.2) for $H = G_i$ is attained by a prime divisor F_i, that is, $a_i = a(F_i, W_i, D_i)/e_{F_i}$. Let $I_i := I_{G_i}(F_i)$, which is cyclic by Lemma 2.15(1). For any F, $I_{I_i}(F) = I_i \cap I_{G_i}(F)$, so $|I_{I_i}(F)| \le |I_{G_i}(F)|$. Since $a(F, W_i, D_i) > 0$, the term of (10.2) at F for $H = I_i$ is at least the term at F for $H = G_i$, with equality at $F = F_i$ because $I_{I_i}(F_i) = I_i$. Thus

$$\mathrm{mld}((W_i \ni o_i, D_i)/I_i) = a_i.$$

Write $D_i = \sum_j c_{ij} \operatorname{div}(h_{ij})$. Deleting the terms with $c_{ij} = 0$ or $h_{ij}(o_i) \ne 0$ changes neither D_i near o_i nor any quotient value. Thus we may assume that $c_{ij} \in \Gamma \setminus \{0\}$ and $h_{ij}(o_i) = 0$ for any j. By [Liu26b, Proposition 7.31] for I_i, the marks h_{ij} and the vector $(c_{ij})_j$, there exist an integer $N_i \ge 3$, a faithful diagonal action of I_i on W_{N_i} preserving the equation of W_{N_i} up to a character, and I_i-semi-invariant polynomials h'_{ij} with the characters of the h_{ij}, such that (W_{N_i}, D'_i) is klt near 0 for $D'_i := \sum_j c_{ij} \operatorname{div}(h'_{ij})$ and $(W_{N_i}, (h'_{ij})_j)$ is a numerical admission of $(W_i, (h_{ij})_j)$ in the sense of [Liu26b, Construction-Theorem 4.13]. By [Liu26b, Construction-Theorem 4.13(1)] for the subgroup I_i,

$$\mathrm{mld}((W_{N_i} \ni 0, D'_i)/I_i) = \mathrm{mld}((W_i \ni o_i, D_i)/I_i) = a_i.$$

By [Liu26b, Theorem 7.2(1)], possibly passing to a subsequence, there exist a positive integer M and cyclic subgroups $I'_i \subset I_i$ of order at most M such that

$$a_i = \mathrm{mld}((W_{N_i} \ni 0, D'_i)/I'_i) \quad \text{for any } i.$$

By Theorem 10.10, the set $\{a_i\}_i$ satisfies the ACC, a contradiction. This proves the theorem when (W, D) is klt near o.

Step 2. Suppose that there exist data $(W_i \ni o_i)$, G_i, and $D_i = \sum_j c_{ij} \operatorname{div}(h_{ij})$ as in the theorem, $i \in \mathbb{N}^+$, such that

$$a_i := \mathrm{mld}((W_i \ni o_i, D_i)/G_i)$$

is strictly increasing. Possibly removing the first term, we may assume that $a_i > 0$ for any i. Let $\pi_i : W_i \to W_i/G_i$ be the quotient morphism and $x_i := \pi_i(o_i)$. For $u \in [0, 1]$, let $\Delta_{i,u}$ be the $\mathbb{R}$-divisor with

$$K_{W_i} + (1-u)D_i = \pi_i^*(K_{W_i/G_i} + \Delta_{i,u}),$$

and for $u \in [0, 1)$ let

$$f_i(u) := \mathrm{mld}((W_i \ni o_i, (1-u)D_i)/G_i) = \mathrm{mld}(W_i/G_i \ni x_i, \Delta_{i,u}).$$

Since $K_{W_i} + (1-u)D_i = (1-u)(K_{W_i} + D_i) + uK_{W_i}$,

$$\Delta_{i,u} = (1-u)\Delta_{i,0} + u\Delta_{i,1},$$

and $\Delta_{i,0}, \Delta_{i,1} \ge 0$ by Lemma 2.15(0).

We show that $f_i(u) \geq a_i$ for $u \in [0,1)$ and $\lim_{u\to 0^+} f_i(u) = a_i$. For $u \in [0,1)$, $(1-u)D_i \leq D_i$, so $(W_i, (1-u)D_i)$ is lc near o_i and $f_i(u) \geq f_i(0) = a_i$. Since $(W_i/G_i, \Delta_{i,0})$ is lc near x_i by Lemma 2.15(2)(3), there exists a prime divisor E_i over W_i/G_i with center x_i and $a(E_i, W_i/G_i, \Delta_{i,0}) = a_i$ [Liu26a, Lemma 3.5]. Since $a_i \leq f_i(u) \leq (1-u)a_i + ua(E_i, W_i/G_i, \Delta_{i,1})$, we have $\lim_{u\to 0^+} f_i(u) = a_i$.

Since $f_i(0) = a_i < a_{i+1}$, we may choose inductively $u_i \in (0,1)$ with $u_i < u_{i-1}$ for $i \geq 2$ and $f_i(u_i) < a_{i+1}$. Then

$$f_i(u_i) < a_{i+1} \leq f_{i+1}(u_{i+1}),$$

so $f_i(u_i)$ is strictly increasing and positive. If W_i is singular at o_i, then W_i is klt near o_i by [Liu26b, Lemma 3.16]. Thus, in any case, for any prime divisor P over W_i whose center contains o_i,

$$a(P, W_i, (1-u_i)D_i) = u_i a(P, W_i, 0) + (1-u_i)a(P, W_i, D_i) > 0.$$

Thus $(W_i, (1-u_i)D_i)$ is klt near o_i. The labels $(1-u_i)c_{ij}$ of $(1-u_i)D_i$ belong to

$$\Gamma' := \{(1-u_i)c \mid i \in \mathbb{N}^+,\ c \in \Gamma\} \subset [0,1],$$

which is the set of products of elements of the DCC subsets $\{1-u_i\}_i$ and Γ of $\mathbb{R}_{\geq 0}$, hence a DCC set. By **Step 1** for Γ', the set $\{f_i(u_i)\}_i$ satisfies the ACC, a contradiction. □

11. Bounds along a circle for unbounded groups

11.1. Fiber references and the radial gap.

Set-up 11.1. Let $\mathbf{T}$, V_0, G, e_0, h^0, κ, and c be as follows.

(1) $\mathbf{T} = \mathbf{T}_{(w)}$ is a primitive circle whose weights $w_1, \dots, w_4$ on the coordinates $z_1, \dots, z_4$ of $\mathbb{A}^4$ satisfy $\min_j w_j < 0 < \max_j w_j$.
(2) $V_0 = (f^0 = 0) \subset \mathbb{A}^4$ is a normal klt hypersurface through 0 not contained in any coordinate hyperplane, and f^0 is $\mathbf{T}$-homogeneous.
(3) G is a finite abelian group acting diagonally on $\mathbb{A}^4$ and faithfully on V_0, and e_0 is a positive integer with $[G : R] \leq e_0$, where $R := G \cap \mathbf{T}$.
(4) h^0 is a $\mathbf{T}$-homogeneous G-semi-invariant polynomial of $\mathbf{T}$-weight κ with $h^0|_{V_0} \neq 0$.
(5) $c > 0$ is a real number.

Here $s \in \mathbf{T}$ acts by $s \cdot (z_1, \dots, z_4) = (s^{w_1} z_1, \dots, s^{w_4} z_4)$, a rational function ϕ on V_0 has $\mathbf{T}$-weight k if $\phi(s \cdot p) = s^k \phi(p)$ for any $s \in \mathbf{T}$, and $\mathbb{C}[V_0]_k$ denotes the space of regular functions on V_0 of $\mathbf{T}$-weight k. Let $r := |R|$, $H := G/R$, $Z_0 := V_0 /\!/ \mathbf{T} = \operatorname{Spec} \mathbb{C}[V_0]_0$, and $Q_{\mathbf{T}} := e_0! \cdot \operatorname{lcm}\{|w_j| : w_j \neq 0\}$, let $o \in Z_0$ and $\overline{o} \in Z_0/H$ be the images of 0, let

$$\Omega_{V_0} := \operatorname{Res}_{V_0} \frac{dz_1 \wedge \cdots \wedge dz_4}{f^0},$$

which generates ω_{V_0} and has $\mathbf{T}$-weight $\sum_j w_j - \operatorname{wt}_w(f^0)$, and let

$$\delta_{\mathbf{T}} := c\kappa - \Big(\sum_j w_j - \operatorname{wt}_w(f^0)\Big)$$

be the adjusted mass of [Liu26b, Set-up 5.1]. Let

$$Y^+ := \operatorname{Proj} \bigoplus_{k\geq 0} \mathbb{C}[V_0]_k \quad \text{and} \quad Y^- := \operatorname{Proj} \bigoplus_{k\geq 0} \mathbb{C}[V_0]_{-k}$$

be the graded Proj quotients of V_0 by $\mathbf{T}$ of sign $+$ and $-$. They are projective over Z_0.

For a coordinate z_j with $w_j > 0$, resp. $w_j < 0$, let $\mathbf{T}_j := \{s \in \mathbf{T} \mid s^{w_j} = 1\}$ be the stabilizer of the hyperplane $(z_j = 1) \subset \mathbb{A}^4$, and let $S_j := V_0 \cap (z_j = 1)$. The *root section* at z_j is the open subset $D_+(z_j) \cong S_j/\mathbf{T}_j$ of Y^+, resp. Y^-, and the *ambient* of the root section is $(z_j = 1)/\mathbf{T}_j$ (see Proposition 11.2(1)). Let G_j be the stabilizer of $(z_j = 1)$ in the group of automorphisms of $\mathbb{A}^4$ generated by G and $\mathbf{T}$. The *root-section pair* at z_j is $(S_j, c \operatorname{div}(h^0|_{S_j}))$ with the action of G_j, and the *ambient pair* at z_j is $((z_j = 1), S_j + c \operatorname{div}(h^0|_{(z_j=1)}))$ with the action of G_j. Let B^+, resp. B^-,

be the $\mathbb{R}$-divisor on Y^+, resp. Y^-, whose restriction to the root section at any z_j with $w_j > 0$, resp. $w_j < 0$, is the boundary of the crepant quotient $(S_j, c\,\mathrm{div}(h^0|_{S_j}))/\mathbf{T}_j$ (Lemma 2.15(0)). By Proposition 11.2(1) and (2), B^+ and B^- are well defined. The group H acts on $Y^\pm$ and preserves $B^\pm$ (see Proposition 11.2(2)). The *fiber reference* of sign $+$, resp. $-$, is the crepant quotient $(Y^+, B^+)/H$, resp. $(Y^-, B^-)/H$, near the fiber of $Y^+/H \to Z_0/H$, resp. $Y^-/H \to Z_0/H$, over $\overline{o}$. Let β^+, resp. β^-, be the infimum of the log discrepancies of the fiber reference of sign $+$, resp. $-$, over the prime divisors with center in that fiber, and let $\beta_{\mathbf{T}} := \beta^+$ if $\delta_{\mathbf{T}} \le 0$ and $\beta_{\mathbf{T}} := \beta^-$ if $\delta_{\mathbf{T}} > 0$.

We call a triple $(W \ni 0, G, h)$ a *marked germ* over (V_0, h^0) if

$$(W \ni 0) = (f = 0) \subset (\mathbb{A}^4 \ni 0)$$

for a G-semi-invariant polynomial f, G acts faithfully on W, h is a G-semi-invariant polynomial, and $f - f^0$ and $h - h^0$ are sums of $\mathbf{T}$-homogeneous polynomials whose $\mathbf{T}$-weights are congruent modulo r to, and different from, $\mathrm{wt}_w(f^0)$, resp. κ.

For $V_0 = (xy + z^a t^b = 0)$ and the sign used in $\beta_{\mathbf{T}}$, [Liu26b, Lemma 8.12(3)] expresses the log discrepancies of $(Y^\pm, B^\pm)$ through the base Z_0, and gaps between β^+ and β^- appear in [Liu26b, Lemma B.20(3)] for the circles $\mathbf{T}_{(-A,-A,-C,D)}$ of [Liu26b, Set-up B.17] and in [Liu26b, Propositions E.3 and E.4] under [Liu26b, Set-up 11.1]. The following proposition proves, for any V_0 as in Set-up 11.1, the log discrepancy formulas for the signs $+$ and $-$, and, if $(V_0, c\,\mathrm{div}(h^0))$ is lc near 0, that the fiber references are lc near their fibers and the gap between β^+ and β^-.

Proposition 11.2 (cf. [Liu26b, Propositions E.3 and E.4])**.** *In Set-up 11.1, the following hold.*

(1) The root sections at the coordinates z_j with $w_j > 0$ cover Y^+, and those at the coordinates z_j with $w_j < 0$ cover Y^-.

(2) The group H acts on $Y^\pm$, preserves $B^\pm$, and acts faithfully on each root section, and the boundaries of the crepant quotients $(S_j, c\,\mathrm{div}(h^0|_{S_j}))/\mathbf{T}_j$ coincide on the overlaps of the root sections. Moreover, for any j with $w_j \neq 0$, the morphism

$$\rho_j : \mathbf{T} \times S_j \to V_0 \cap (z_j \neq 0), \qquad \rho_j(s, p) := s \cdot p,$$

is the quotient by a free action of $\mathbf{T}_j$, S_j is normal and klt, each connected component of S_j contains a point with non-zero coordinates, $|G_j| = |w_j| \cdot |H|$, the stabilizer in G_j of each connected component of S_j acts faithfully on it, and over $D_+(z_j)/H$ the fiber reference of the sign of w_j is the crepant quotient of the root-section pair at z_j by G_j.

(3) If $(V_0, c\,\mathrm{div}(h^0))$ is lc near 0, then the fiber references of sign $+$ and $-$ are lc near their fibers.

(4) Let ξ be a rational function on V_0 of $\mathbf{T}$-weight 1. Then the rational functions on V_0 are the rational functions in ξ over the function field of Z_0, and ξ is transcendental over it. Write $z_j = \xi^{w_j}\psi_j$ with rational functions ψ_j on Z_0. Let P be a prime divisor over Z_0/H, let ν be an extension of mult_P to a $\mathbb{Q}$-valued valuation on the rational functions on Z_0, for $\vartheta \in \mathbb{R}$ let ν_ϑ be the Gauss extension of ν with value ϑ on ξ (Definition 2.3), and let

$$\vartheta_- := \max_{w_j > 0}\Big(-\frac{\nu(\psi_j)}{w_j}\Big) \quad \text{and} \quad \vartheta_+ := \min_{w_j < 0}\frac{\nu(\psi_j)}{|w_j|}.$$

Then the following hold.

(a) There exists a real number $\varphi(\nu)$ such that $a(P, (Y^+, B^+)/H) = \varphi(\nu) - \delta_{\mathbf{T}}\vartheta_-$ and $a(P, (Y^-, B^-)/H) = \varphi(\nu) - \delta_{\mathbf{T}}\vartheta_+$.

(b) If $\mathrm{center}_{Z_0/H} P = \overline{o}$, then $\nu(z_j) > 0$ for any j with $w_j = 0$, and $\nu_\vartheta(z_j) > 0$ for any j if and only if $\vartheta_- < \vartheta < \vartheta_+$.

(c) If $\mathrm{center}_{Z_0/H} P = \overline{o}$, then

$$\vartheta_+ - \vartheta_- \ge \frac{1}{Q_{\mathbf{T}}}.$$

(5) Assume that $(V_0, c\,\mathrm{div}(h^0))$ *is lc near* 0*. Then*

$$\beta^- \geq \beta^+ + \frac{|\delta_{\mathbf{T}}|}{Q_{\mathbf{T}}} \text{ if } \delta_{\mathbf{T}} < 0, \qquad \beta^+ \geq \beta^- + \frac{\delta_{\mathbf{T}}}{Q_{\mathbf{T}}} \text{ if } \delta_{\mathbf{T}} > 0, \quad \text{and} \quad \beta^+ = \beta^- \text{ if } \delta_{\mathbf{T}} = 0.$$

If moreover $c = \frac{1}{m}$ *for a positive integer* m*, then* β^+ *and* β^- *belong to a finite subset of* $[0, 2]$ *depending only on* m*,* e_0*, and* w*.*

Proof. In this proof, for a prime divisor F over V_0 and a rational number $\lambda > 0$, let

$$a_{V_0}(\lambda\,\mathrm{mult}_F) := \lambda a(F, V_0, c\,\mathrm{div}(h^0)),$$

and for a prime divisor D over Y^+, resp. Y^-, let $a_+(\lambda\,\mathrm{mult}_D) := \lambda a(D, Y^+, B^+)$, resp. $a_-(\lambda\,\mathrm{mult}_D) := \lambda a(D, Y^-, B^-)$. Write $z_k(g \cdot p) = \chi_{z_k}(g) z_k(p)$ for $g \in G$. For j with $w_j \neq 0$ let $B_j := c\,\mathrm{div}(h^0|_{S_j})$.

Step 1: (1) and (2). For j with $w_j \neq 0$, the morphism

$$\rho_j : \mathbf{T} \times S_j \to V_0 \cap (z_j \neq 0), \qquad \rho_j(s, p) := s \cdot p,$$

is surjective: for $p' \in V_0 \cap (z_j \neq 0)$ and $s \in \mathbf{T}$ with $s^{w_j} = z_j(p')$, we have $s^{-1} \cdot p' \in S_j$. The fibers of ρ_j are the orbits of the free action $s' \cdot (s, p) = (ss'^{-1}, s' \cdot p)$ of $\mathbf{T}_j$ on $\mathbf{T} \times S_j$. Thus ρ_j is the quotient by the free action of $\mathbf{T}_j$, hence étale. In particular, S_j is normal with klt singularities, $\mathbf{T}_j$ permutes the connected components of S_j transitively since V_0 is irreducible, and ρ_j^* induces an isomorphism from the ring of $\mathbf{T}$-invariant regular functions on $V_0 \cap (z_j \neq 0)$ onto the ring of $\mathbf{T}_j$-invariant regular functions on S_j, and similarly for rational functions. When Lemma 2.15 is used for S_j, we tacitly replace S_j by one connected component and the group by the stabilizer of that component. Since V_0 is irreducible and not contained in any coordinate hyperplane (Set-up 11.1(2)), it contains a point with non-zero coordinates. As ρ_j is surjective and $\mathbf{T}_j$ permutes the connected components of S_j transitively, each connected component of S_j contains a point with non-zero coordinates.

The element z_j is homogeneous of degree $|w_j|$ in the graded ring defining Y^+ if $w_j > 0$, resp. Y^- if $w_j < 0$, and the degree 0 part of the localization at z_j is

$$\mathbb{C}[V_0][z_j^{-1}]^{\mathbf{T}} = \{x/z_j^N \mid N \in \mathbb{N},\ x \in \mathbb{C}[V_0]_{Nw_j}\}.$$

By the previous paragraph,

$$D_+(z_j) = \mathrm{Spec}\,\mathbb{C}[V_0][z_j^{-1}]^{\mathbf{T}} \cong S_j/\mathbf{T}_j.$$

On the overlap of two root sections, both crepant quotient boundaries have coefficient

$$1 - (1 - b_P)/e_P$$

along a prime divisor P by Lemma 2.15(4), where b_P is the multiplicity of $c\,\mathrm{div}(h^0)$ along the preimage of P in V_0 and e_P is the order of the stabilizer in $\mathbf{T}$ of a general point of the preimage (the ramification index of $S_j \to S_j/\mathbf{T}_j$ along a prime divisor is $|\mathrm{Stab}_{\mathbf{T}_j}(q)|$ for a general point q of that divisor, and $\mathrm{Stab}_{\mathbf{T}}(q) \subset \mathbf{T}_j$). Thus $B^{\pm}$ is well defined.

Let j, k be indices with $w_j > 0 > w_k$, and let $u_{jk} := z_j^{|w_k|} z_k^{w_j} \in \mathbb{C}[V_0]_0$, which is non-zero since V_0 is not contained in a coordinate hyperplane. For $N \in \mathbb{N}$, $x \in \mathbb{C}[V_0]_{Nw_j}$, and $x' \in \mathbb{C}[V_0]_{Nw_k}$,

$$\frac{x}{z_j^N} = \frac{x z_j^{N(|w_k|-1)} z_k^{Nw_j}}{u_{jk}^N} \quad \text{and} \quad \frac{x'}{z_k^N} = \frac{x' z_k^{N(w_j-1)} z_j^{N|w_k|}}{u_{jk}^N},$$

with numerators in $\mathbb{C}[V_0]_0$. Thus $D_+(z_j) \to Z_0$ and $D_+(z_k) \to Z_0$ are isomorphisms over $(u_{jk} \neq 0) \subset Z_0$, so $Y^{\pm} \to Z_0$ is birational. The pull-backs of the rational functions on Z_0 are exactly the $\mathbf{T}$-invariant rational functions on V_0, and their pull-backs by ρ_j are exactly the $\mathbf{T}_j$-invariant rational functions on S_j. Since $\dim S_j = \dim V_0 - 1 = 2$, we have $\dim Z_0 = 2$, and the $\mathbf{T}$-invariant ψ_j in (4) are rational functions on Z_0.

(1) Any monomial of positive $\mathbf{T}$-weight is divisible by some z_j with $w_j > 0$, so the ideal generated by $\{z_j \mid w_j > 0\}$ has the irrelevant ideal as radical, and the root sections at the z_j with $w_j > 0$ cover Y^+. Replacing w with $-w$ gives the statement for Y^-.

The group G commutes with $\mathbf{T}$ and preserves V_0, hence each $\mathbb{C}[V_0]_k$, and $R \subset \mathbf{T}$ acts on $\mathbb{C}[V_0]_k$ by scalars. Thus H acts on $Y^\pm$, preserves each root section, and preserves $B^\pm$ since h^0 is G-semi-invariant. Let $g \in G$ act trivially on the root section at z_j, and let $p \in V_0$ be a point with non-zero coordinates. The fibers of $V_0 \cap (z_j \neq 0) \to S_j/\mathbf{T}_j$ are $\mathbf{T}$-orbits, so $g \cdot p = s \cdot p$ for some $s \in \mathbf{T}$, that is, $\chi_{z_k}(g) = s^{w_k}$ for any k. Thus $g \cdot q = s \cdot q$ for any $q \in \mathbb{A}^4$, and $g \in R$. Hence H acts faithfully on each root section.

The group G_j of Set-up 11.1 consists of the diagonal automorphisms gs of $\mathbb{A}^4$ with $g \in G$, $s \in \mathbf{T}$, and $\chi_{z_j}(g)s^{w_j} = 1$. The surjection $G_j \to G\mathbf{T}/\mathbf{T} \cong H$ has kernel $G_j \cap \mathbf{T} = \mathbf{T}_j$, so

$$|G_j| = |w_j| \cdot |H| \le e_0 \max_k |w_k|. \tag{11.1}$$

An element of G_j acting trivially on a connected component of S_j fixes a point with non-zero coordinates, hence is the identity. Thus the stabilizer in G_j of each connected component of S_j acts faithfully on it. An element $gs \in G_j$ acts on $\mathbf{T}$-invariant functions as g. By Lemma 2.15(5) for the normal subgroup $\mathbf{T}_j \subset G_j$, the fiber reference of the sign of w_j over $D_+(z_j)/H$ is

$$(S_j, B_j)/G_j. \tag{11.2}$$

This proves (2).

Step 2: Gauss extensions and log discrepancies. A function ξ as in (4) exists: since w is primitive, $\sum_j d_j w_j = 1$ for some integers d_j, and $\prod_j z_j^{d_j}$ is a rational function on V_0 of $\mathbf{T}$-weight 1 by Set-up 11.1(2). Any element of $\mathbb{C}[V_0]_k$ is ξ^k times a $\mathbf{T}$-invariant rational function on V_0, which is the pull-back of a rational function on Z_0, and $\mathbb{C}[V_0] = \bigoplus_k \mathbb{C}[V_0]_k$. Let $\iota : V_0 \dashrightarrow Z_0 \times \mathbb{A}^1_\xi$ be the rational map given by the quotient morphism $V_0 \to Z_0$ and ξ. Thus any rational function on V_0 is pulled back by ι, so ι is birational onto $\overline{\iota(V_0)}$, which is $Z_0 \times \mathbb{A}^1_\xi$ since $\dim V_0 = \dim Z_0 + 1$. Thus the rational functions on V_0 are the rational functions in ξ over the function field of Z_0, and ξ is transcendental over it. For a non-zero rational 2-form ω on Z_0, $\Omega_{V_0} = \gamma \frac{d\xi}{\xi} \wedge \omega$ for some rational function γ on V_0. Since $\frac{d\xi}{\xi}$ and ω are $\mathbf{T}$-invariant and Ω_{V_0} has $\mathbf{T}$-weight $\sum_j w_j - \mathrm{wt}_w(f^0)$, $\gamma\xi^{-(\sum_j w_j - \mathrm{wt}_w(f^0))}$ is the pull-back of a rational function on Z_0. Thus

$$h^0 = \xi^\kappa h_Z \quad \text{and} \quad \Omega_{V_0} = \xi^{\sum_j w_j - \mathrm{wt}_w(f^0)} \frac{d\xi}{\xi} \wedge \omega_0$$

for some rational function h_Z on Z_0 and some rational 2-form ω_0 on Z_0.

Let D be a prime divisor over Z_0, $e \in \mathbb{N}^+$, and $v := \frac{1}{e}\,\mathrm{mult}_D$. Let

$$\vartheta_-(v) := \max_{w_j > 0} \Big(-\frac{v(\psi_j)}{w_j} \Big) \quad \text{and} \quad \vartheta_+(v) := \min_{w_j < 0} \frac{v(\psi_j)}{|w_j|},$$

let v_ϑ be the Gauss extensions (Definition 2.3) of v with value ϑ on ξ, and let

$$\varphi(v) := \frac{1}{e}(1 + \mathrm{mult}_D\,\omega_0 - c\,\mathrm{mult}_D\,h_Z),$$

where $\mathrm{mult}_D\,\omega_0$ is the order of ω_0 along D. Then $\vartheta_\pm(v) \in \mathbb{Q}$, and we have $\vartheta_\pm(v) = \frac{1}{e}\vartheta_\pm(\mathrm{mult}_D)$, $\varphi(v) = \frac{1}{e}\varphi(\mathrm{mult}_D)$, and $v_{\vartheta/e} = \frac{1}{e}(\mathrm{mult}_D)_\vartheta$ for $\vartheta \in \mathbb{R}$. We prove the following.

(i) $\vartheta_-(v) \le \vartheta_+(v)$. For $\vartheta \in [\vartheta_-(v), \vartheta_+(v)]$, v_ϑ is non-negative on $\mathbb{C}[V_0]$. If the center of D on Z_0 is o, then $\vartheta_-(v) < \vartheta_+(v)$, $v(z_k) > 0$ for any k with $w_k = 0$, and $v_\vartheta(z_k) > 0$ for any k if and only if $\vartheta_-(v) < \vartheta < \vartheta_+(v)$.

(ii) For $\vartheta \in [\vartheta_-(v), \vartheta_+(v)] \cap \mathbb{Q}$, v_ϑ is a positive rational multiple of mult_F for a prime divisor F over V_0, and

$$a_{V_0}(v_\vartheta) = \varphi(v) - \delta_{\mathbf{T}}\vartheta.$$

(iii) $a_+(v) = a_{V_0}(v_{\vartheta_-(v)})$ and $a_-(v) = a_{V_0}(v_{\vartheta_+(v)})$.

(i) For $w_j > 0 > w_k$, $u_{jk} = \psi_j^{|w_k|}\psi_k^{w_j}$, and u_{jk} and the z_k with $w_k = 0$ are regular functions on Z_0 vanishing at o. Thus $|w_k|v(\psi_j) + w_j v(\psi_k) \geq 0$ and $v(z_k) \geq 0$ for $w_k = 0$, with strict inequalities if the center of D is o. This proves the inequalities between $\vartheta_-(v)$ and $\vartheta_+(v)$. Since

$$v_\vartheta(z_k) = v(\psi_k) + w_k\vartheta \quad \text{for any } k,$$

$v_\vartheta(z_k) \geq 0$ for any k if $\vartheta \in [\vartheta_-(v), \vartheta_+(v)]$, and, when the center of D is o, $v_\vartheta(z_k) > 0$ for any k exactly when $\vartheta_-(v) < \vartheta < \vartheta_+(v)$. As the z_k generate $\mathbb{C}[V_0]$, (i) follows.

(ii) By homogeneity we may assume $e = 1$. Let Z' be a normal variety birational to Z_0 on which D is a prime divisor, and let y_1, y_2 be rational functions on Z_0 with $\operatorname{mult}_D y_1 = 1$ and $y_2|_D$ non-constant. Then $\omega_0 = \tau\frac{dy_1}{y_1} \wedge dy_2$ for some rational function τ on Z_0 with $\operatorname{mult}_D \tau = 1 + \operatorname{mult}_D \omega_0$, and

$$\Omega_{V_0} = \pm\xi^{\sum_j w_j - \operatorname{wt}_w(f^0)}\tau\frac{dy_1}{y_1} \wedge \frac{d\xi}{\xi} \wedge dy_2.$$

Write $\vartheta = n_1/n_2$ with coprime integers n_1 and $n_2 > 0$. Let $\xi_1 := \xi$ if $n_1 \geq 0$ and $\xi_1 := \xi^{-1}$ if $n_1 < 0$. Since V_0 is birational to $Z_0 \times \mathbb{A}^1_\xi$, it is birational to $Z' \times \mathbb{A}^1_{\xi_1}$, on which (y_1, ξ_1) is a regular system of parameters at the generic point of $D \times \{0\}$. Expanding $\sum_k \alpha_k \xi^k$, for rational functions α_k on Z_0, in y_1 and ξ_1 shows that v_ϑ is the monomial valuation with $v_\vartheta(y_1) = 1$ and $v_\vartheta(\xi_1) = |\vartheta|$. Thus $n_2 v_\vartheta = \operatorname{mult}_F$ for the exceptional divisor F of the weighted blow-up of (y_1, ξ_1) with weights $(n_2, |n_1|)$ near the generic point of $D \times \{0\}$ (for $n_1 = 0$, $F = D \times \mathbb{A}^1_{\xi_1}$). The form

$$\frac{dy_1}{y_1} \wedge \frac{d\xi}{\xi} \wedge dy_2 = \pm\frac{dy_1}{y_1} \wedge \frac{d\xi_1}{\xi_1} \wedge dy_2$$

has order -1 along F, and F has a center on V_0 by (i), near which Ω_{V_0} generates ω_{V_0}. Hence

$$a(F, V_0, 0) = 1 + \operatorname{mult}_F \Omega_{V_0} = \operatorname{mult}_F \left(\xi^{\sum_j w_j - \operatorname{wt}_w(f^0)}\tau\right),$$

and, by the definition of $\delta_{\mathbf{T}}$,

$$\begin{aligned} a_{V_0}(v_\vartheta) &= v_\vartheta\left(\xi^{\sum_j w_j - \operatorname{wt}_w(f^0)}\tau\right) - c v_\vartheta(\xi^\kappa h_Z) \\ &= \Big(\sum_j w_j - \operatorname{wt}_w(f^0)\Big)\vartheta + 1 + \operatorname{mult}_D \omega_0 - c\kappa\vartheta - c\operatorname{mult}_D h_Z = \varphi(v) - \delta_{\mathbf{T}}\vartheta. \end{aligned}$$

(iii) By homogeneity we may assume $e = 1$. Replacing w with $-w$ and ξ with ξ^{-1} interchanges Y^+ and Y^-, does not change the ψ_j, h_Z, and $\varphi(v)$, replaces $\delta_{\mathbf{T}}$ and $\vartheta_-(v)$ with $-\delta_{\mathbf{T}}$ and $-\vartheta_+(v)$, and turns the Gauss extension with value ϑ on ξ into the one with value $-\vartheta$ on ξ^{-1}, so it suffices to prove the first equality. Let j be an index with $w_j > 0$ and $\vartheta_-(v) = -v(\psi_j)/w_j$, and write $\vartheta_- := \vartheta_-(v)$. Then $v_{\vartheta_-}(z_j) = 0$. For $x \in \mathbb{C}[V_0]_{Nw_j}$, x/z_j^N is a rational function on Z_0, so

$$v(x/z_j^N) = v_{\vartheta_-}(x) - N v_{\vartheta_-}(z_j) = v_{\vartheta_-}(x) \geq 0$$

by (i). Thus the center of D on Y^+ is in the root section $D_+(z_j) \cong S_j/\mathbf{T}_j$. By Lemma 2.15(1)–(3), there exist a prime divisor E over S_j and $e' \in \mathbb{N}^+$ such that $\operatorname{mult}_E \alpha = e' \operatorname{mult}_D \alpha$ for any rational function α on Z_0, viewed on S_j by pull-back, and

$$a(E, S_j, B_j) = e' a(D, Y^+, B^+).$$

Let s be the coordinate of $\mathbf{T} \cong \mathbb{G}_m$. Since ξ has $\mathbf{T}$-weight 1, $\rho_j^*\xi = s\zeta$ for some rational function ζ on S_j, and since $\rho_j^* z_j = s^{w_j}$, $\rho_j^*\psi_j = \zeta^{-w_j}$. Thus

$$\operatorname{mult}_E \zeta = -\frac{e' \operatorname{mult}_D \psi_j}{w_j} = e'\vartheta_-.$$

Let $\widetilde{E} := \mathbf{T} \times E$, a prime divisor over $\mathbf{T} \times S_j$ with

$$a(\widetilde{E}, \mathbf{T} \times S_j, \operatorname{pr}_2^* B_j) = a(E, S_j, B_j).$$

For rational functions α_k on Z_0, $\rho_j^*(\sum_k \alpha_k \xi^k) = \sum_k \alpha_k \zeta^k s^k$, so

$$\operatorname{mult}_{\widetilde{E}}\Big(\rho_j^* \sum_k \alpha_k \xi^k\Big) = \min_k \big(e' \operatorname{mult}_D \alpha_k + k e' \vartheta_-\big) = e' v_{\vartheta_-}\Big(\sum_k \alpha_k \xi^k\Big).$$

Any element of the inertia group of $\widetilde{E}$ in $\mathbf{T}_j$ fixes the center of $\widetilde{E}$ pointwise, hence is trivial as $\mathbf{T}_j$ acts freely. Since $\rho_j^* h^0 = s^\kappa h^0|_{S_j}$, Lemma 2.15(1), (2), and (4) imply that $e' v_{\vartheta_-} = \operatorname{mult}_F$ for a prime divisor F over V_0 with

$$a(F, V_0, c\operatorname{div}(h^0)) = a(\widetilde{E}, \mathbf{T} \times S_j, \operatorname{pr}_2^* B_j).$$

Therefore

$$a_+(v) = \frac{a(E, S_j, B_j)}{e'} = \frac{a(F, V_0, c\operatorname{div}(h^0))}{e'} = a_{V_0}(v_{\vartheta_-}).$$

By (ii) and (iii), for any prime divisor D over Z_0 and $e \in \mathbb{N}^+$,

$$\begin{aligned} a_+(\tfrac{1}{e}\operatorname{mult}_D) &= \varphi(\tfrac{1}{e}\operatorname{mult}_D) - \delta_{\mathbf{T}} \vartheta_-(\tfrac{1}{e}\operatorname{mult}_D), \\ a_-(\tfrac{1}{e}\operatorname{mult}_D) &= \varphi(\tfrac{1}{e}\operatorname{mult}_D) - \delta_{\mathbf{T}} \vartheta_+(\tfrac{1}{e}\operatorname{mult}_D). \end{aligned} \tag{11.3}$$

Step 3: (4). The extension ν is a discrete valuation on the rational functions on Z_0 whose residue field is finite over that of mult_P, so $\nu = \frac{1}{e}\operatorname{mult}_D$ for a prime divisor D over Z_0 and some $e \in \mathbb{N}^+$. By (2) and Lemma 2.15(1) and (2) for $(Y^\pm, B^\pm)$ and H, we have $e = |I_H(D)| \le |H| \le e_0$ and $a(D, Y^\pm, B^\pm) = e \cdot a(P, (Y^\pm, B^\pm)/H)$. Hence $a(P, (Y^\pm, B^\pm)/H) = a_\pm(\frac{1}{e}\operatorname{mult}_D)$, and (11.3) proves (4)(a) with $\varphi(\nu) := \varphi(\frac{1}{e}\operatorname{mult}_D)$.

Assume that $\operatorname{center}_{Z_0/H} P = \overline{o}$. Since 0 is fixed by G, the center of D on Z_0 is o, and (i) implies (4)(b). For $w_j > 0 > w_k$, $\nu(\psi_j)$ and $\nu(\psi_k)$ belong to $\frac{1}{e}\mathbb{Z}$ and $e \mid e_0!$, so

$$-\frac{\nu(\psi_j)}{w_j} \in \frac{1}{e_0! w_j}\mathbb{Z} \quad \text{and} \quad \frac{\nu(\psi_k)}{|w_k|} \in \frac{1}{e_0! |w_k|}\mathbb{Z}.$$

Thus $\vartheta_+ - \vartheta_- \in \frac{1}{Q_{\mathbf{T}}}\mathbb{Z}$. Since $\vartheta_+ - \vartheta_- > 0$ by (i), $\vartheta_+ - \vartheta_- \ge \frac{1}{Q_{\mathbf{T}}}$, which proves (4)(c).

Step 4: (3). Assume that $(V_0, c\operatorname{div}(h^0))$ is lc near 0. By Lemma 2.15(2) and (3), and by the symmetry in the proof of (iii), it suffices to show that (Y^+, B^+) is lc near the fiber over o. Suppose that (Y^+, B^+) is not lc near a closed point y of the fiber over o. Then there exists a prime divisor D over Y^+ with center y and $a(D, Y^+, B^+) < 0$ (repeatedly blow up the point over y on the strict transform of a component of coefficient > 1 of the crepant pull-back of B^+ to a log resolution). The center of D on Z_0 is o. Let $v := \operatorname{mult}_D$. By (i) and (11.3), $\vartheta_-(v) < \vartheta_+(v)$ and $\varphi(v) - \delta_{\mathbf{T}}\vartheta_-(v) = a_+(v) < 0$. Choose $\vartheta \in (\vartheta_-(v), \vartheta_+(v)) \cap \mathbb{Q}$ with $\varphi(v) - \delta_{\mathbf{T}}\vartheta < 0$. By (i) and (ii), v_ϑ is a positive multiple of mult_F for a prime divisor F over V_0 with center 0 and

$$a(F, V_0, c\operatorname{div}(h^0)) < 0,$$

which contradicts the assumption that $(V_0, c\operatorname{div}(h^0))$ is lc near 0.

Step 5: (5). Assume that $(V_0, c\operatorname{div}(h^0))$ is lc near 0. Write $(Y^\pm, B^\pm)/H = (Y^\pm/H, B_{Y^\pm/H})$. Since $Y^\pm/H \to Z_0/H$ is proper and birational, a prime divisor over $Y^\pm/H$ has center in the fiber over $\overline{o}$ if and only if the center on Z_0/H is $\overline{o}$. Thus β^+ and β^- are both infima over the set of prime divisors P over Z_0/H with center $\overline{o}$, and by (3) both are non-negative. For such P and an extension ν of mult_P as in (4), (4) implies

$$a(P, (Y^-, B^-)/H) - a(P, (Y^+, B^+)/H) = -\delta_{\mathbf{T}}(\vartheta_+ - \vartheta_-) \quad \text{and} \quad \vartheta_+ - \vartheta_- \ge \frac{1}{Q_{\mathbf{T}}}.$$

Taking the infimum over P proves the three relations between β^+ and β^-.

Let $\overline{y}$ be a closed point of the fiber over $\overline{o}$. If $\overline{y}$ is a smooth point of $Y^\pm/H$, then the blow-up of $\overline{y}$ has log discrepancy at most 2, and otherwise the exceptional curves of the minimal resolution of $\overline{y}$ have log discrepancies at most 1, since $B_{Y^\pm/H} \ge 0$. Thus $\beta^\pm \le 2$.

Assume that $c = \frac{1}{m}$, and let $M := e_0 \max_k |w_k|$. By (1), $\overline{y}$ belongs to $D_+(z_j)/H$ for some j with $\pm w_j > 0$. Let $p \in S_j$ be a point over $\overline{y}$ and $G_{j,p}$ the stabilizer of p in G_j. By (11.2), near $\overline{y}$ the fiber reference is $(S_j, B_j)/G_j$. For a prime divisor F over S_j with center p, $I_{G_j}(F) = I_{G_{j,p}}(F)$ by Lemma 11.3(4). Thus, by Lemma 2.15(2) and (3) and after translating F by G_j, the log discrepancies of the prime divisors over $(Y^{\pm}/H, B_{Y^{\pm}/H})$ with center $\overline{y}$ are the numbers $a(F, S_j, B_j)/|I_{G_{j,p}}(F)|$, which are the log discrepancies of the prime divisors over $(S_j \ni p, B_j)/G_{j,p}$ with center the image of p. Since $m(K_{S_j} + B_j) = mK_{S_j} + \operatorname{div}(h^0|_{S_j})$ is Cartier and $|I_{G_{j,p}}(F)| \le |G_j| \le M$ by (11.1), Lemma 2.20, with m in place of N, shows that the numbers $a(F, S_j, B_j)/|I_{G_{j,p}}(F)|$ belong to $(m \operatorname{lcm}(1, 2, \dots, M))^{-1}\mathbb{Z}$. A prime divisor P on $Y^{\pm}/H$ contained in the fiber has log discrepancy

$$1 - \operatorname{mult}_P B_{Y^{\pm}/H} = \frac{1-b}{e} \in (m \operatorname{lcm}(1, 2, \dots, M))^{-1}\mathbb{Z}$$

by Lemma 2.15(4) for $S_j \to S_j/G_j$, where $b \in \frac{1}{m}\mathbb{Z}$ is the multiplicity of B_j along a prime divisor of S_j over P and $e \le |G_j| \le M$ is the ramification index. Any prime divisor over $Y^{\pm}/H$ with center in the fiber has a closed point as center or is a prime divisor on $Y^{\pm}/H$ contained in the fiber. Hence $\beta^{\pm}$ belongs to the finite set $[0, 2] \cap (m \operatorname{lcm}(1, 2, \dots, M))^{-1}\mathbb{Z}$, which depends only on m, e_0, and w. □

Lemma 11.3. *Let G be a finite abelian group acting diagonally on $\mathbb{A}^n$, $\mathfrak{a}$ an $\mathbb{R}$-ideal on $\mathbb{A}^n$ with G-semi-invariant generators, and $f_1, \dots, f_N$ G-semi-invariant polynomials with common zero scheme X such that no non-identity element of G acts trivially on an irreducible component of X. Let $\eta \in X$ be a closed point, or the generic point of an integral curve $C \subset X$ not necessarily through 0 or G-invariant, such that near η, X is a normal complete intersection with klt singularities, $\mathfrak{a}|_X$ is non-zero, and $(X, \mathfrak{a}|_X)$ is lc. Let $G_\eta \subset G$ be the stabilizer of η if η is a closed point, and the subgroup of G fixing C pointwise if η is the generic point of C. Then the following hold.*

1. *G_η fixes a closed point $x_0 \in \overline{\{\eta\}}$ near which X is a normal complete intersection with klt singularities and $(X, \mathfrak{a}|_X)$ is lc, and G_η acts diagonally in the coordinates of $\mathbb{A}^n$ translated to x_0.*
2. *At the image of η, the quotient values (Definition 2.2) satisfy*
$$\operatorname{mld}((X \ni \eta, \mathfrak{a}|_X)/G) = \operatorname{mld}((X \ni \eta, \mathfrak{a}|_X)/G_\eta).$$
3. *At the image of η, the quotient values satisfy*
$$\operatorname{mld}\Big(\Big(\mathbb{A}^n \ni \eta, \sum_{i=1}^N \operatorname{div}(f_i), \mathfrak{a}\Big)/G\Big) = \operatorname{mld}\Big(\Big(\mathbb{A}^n \ni \eta, \sum_{i=1}^N \operatorname{div}(f_i), \mathfrak{a}\Big)/G_\eta\Big).$$
4. *$I_G(F) = I_{G_\eta}(F)$ for any prime divisor F over X, resp. $\mathbb{A}^n$, with $\operatorname{center}_X F = \overline{\{\eta\}}$, resp. $\operatorname{center}_{\mathbb{A}^n} F = \overline{\{\eta\}}$.*
5. *If $n - N \ge 2$ and G_η acts faithfully on X near x_0, then, in the coordinates of $\mathbb{A}^n$ translated to x_0, the polynomials $f_1, \dots, f_N$, the group G_η, and $\mathfrak{a}$ satisfy the hypotheses of Theorem 2.17 with N in place of c and x_0 in place of 0, and, if η is the generic point of C, then C is a G_η-invariant integral curve through x_0.*

Proof. (4) Let F be a prime divisor over X, resp. $\mathbb{A}^n$, with

$$\operatorname{center}_X F = \overline{\{\eta\}}, \quad \text{resp.} \quad \operatorname{center}_{\mathbb{A}^n} F = \overline{\{\eta\}}.$$

Any $g \in I_G(F)$ preserves the valuation ring of F and acts trivially on the residue field of F. Thus g preserves $\overline{\{\eta\}}$ and acts trivially on the residue field of η, which is a subfield of the residue field of F. If η is a closed point, then g fixes η. If η is the generic point of C, then g acts trivially on the rational functions on C, so g fixes C pointwise. Hence $g \in G_\eta$, and

$$I_G(F) = I_{G_\eta}(F). \tag{11.4}$$

This proves (4).

(1) If η is a closed point, let $x_0 := \eta$. If η is the generic point of C, let $U \ni \eta$ be the intersection of the open subset of X on which X is a normal complete intersection with klt singularities and the open subset on which $(X, \mathfrak{a}|_X)$ is lc, and let x_0 be a closed point of $C \cap U$, which is fixed by G_η since G_η fixes C pointwise. Let $y_1, \dots, y_n$ be the coordinates of $\mathbb{A}^n$ and write $y_k(g \cdot p) = \chi_{y_k}(g) y_k(p)$ for $g \in G$. For $g \in G_\eta$ we have $\chi_{y_k}(g) y_k(x_0) = y_k(x_0)$, hence

$$(y_k - y_k(x_0))(g \cdot p) = \chi_{y_k}(g)\big(y_k(p) - y_k(x_0)\big) \quad \text{for any } k.$$

This proves (1).

(5) In the coordinates $y_k - y_k(x_0)$, the f_i and the generators of $\mathfrak{a}$ are G_η-semi-invariant polynomials, the f_i vanish at x_0, X is a normal integral complete intersection with klt singularities near x_0, $(X, \mathfrak{a}|_X)$ is lc near x_0, and, if η is the generic point of C, then C is a G_η-invariant integral curve through x_0. Thus, if $n - N \geq 2$ and G_η acts faithfully on X near x_0, the hypotheses of Theorem 2.17, with N in place of c, G_η in place of G, and x_0 in place of 0, hold. This proves (5).

(2) The conditions on X and $(X, \mathfrak{a}|_X)$ near η hold near $g\eta$ for any $g \in G$, since G preserves X and $\mathfrak{a}$. Possibly shrinking X to a G-stable open neighborhood of $\{g\eta \mid g \in G\}$, we may assume that X is normal. Then G still acts faithfully on X, since no non-identity element of G acts trivially on an irreducible component of X. Let $M := |G|$ and $\mathfrak{a} = \prod_j \mathfrak{a}_j^{r_j}$. The M-th powers of the G-semi-invariant generators of $\mathfrak{a}_j$ are G-invariant and generate an ideal $\mathfrak{b}_j$ on $\mathbb{A}^n/G$, and $\mathfrak{b}_j|_{X/G}$ satisfies

$$\operatorname{mult}_F(\mathfrak{b}_j \mathcal{O}_X) = M \operatorname{mult}_F(\mathfrak{a}_j \mathcal{O}_X) \quad \text{for any prime divisor } F \text{ over } X$$

and, for (3),

$$\operatorname{mult}_F(\mathfrak{b}_j \mathcal{O}_{\mathbb{A}^n}) = M \operatorname{mult}_F \mathfrak{a}_j \quad \text{for any prime divisor } F \text{ over } \mathbb{A}^n.$$

We use the $\mathbb{R}$-ideal $\prod_j (\mathfrak{b}_j|_{X/G})^{r_j/M}$ in Lemma 2.15(2). Let $\pi : X \to X/G$ be the quotient morphism. Since π is finite and

$$\pi^{-1}(\pi(\overline{\{\eta\}})) = \bigcup_{g \in G} g\overline{\{\eta\}},$$

Lemma 2.15(2) and (3) imply that, for $(X/G, B_{X/G}) := (X, 0)/G$,

$$\operatorname{mld}\Big(X/G \ni \pi(\eta), B_{X/G}, \prod_j (\mathfrak{b}_j|_{X/G})^{r_j/M}\Big)$$

is the infimum of $a(F, X, \mathfrak{a}|_X)/|I_G(F)|$ over the prime divisors F over X with $\operatorname{center}_X F = g\overline{\{\eta\}}$ for some $g \in G$. Since $a(gF, X, \mathfrak{a}|_X) = a(F, X, \mathfrak{a}|_X)$ and $I_G(gF) = g I_G(F) g^{-1} = I_G(F)$, we may take $\operatorname{center}_X F = \overline{\{\eta\}}$ in the infimum, which is the left side of (2) by Definition 2.2. Since G_η fixes η, the right side of (2) is the infimum of $a(F, X, \mathfrak{a}|_X)/|I_{G_\eta}(F)|$ over the prime divisors F over X with $\operatorname{center}_X F = \overline{\{\eta\}}$. Thus (2) follows from (11.4).

(3) Since no non-identity element of G acts trivially on X, G acts faithfully on $\mathbb{A}^n$, and $\sum_{i=1}^N \operatorname{div}(f_i)$ is G-invariant. Possibly replacing $(X, \mathfrak{a}|_X)$ with $(\mathbb{A}^n, \sum_{i=1}^N \operatorname{div}(f_i), \mathfrak{a})$, the argument of (2) implies (3). □

11.2. **Upper and lower bounds.**

Proposition 11.4 (cf. [Liu26b, Proposition 5.4]). *Let $\mathbf{T} = \mathbf{T}_{(w)}$, e_0, m, and κ be fixed. For any i let $(V_{0,i}, G_i, h_i^0)$ be data as in Set-up 11.1 with these $\mathbf{T}$, e_0, and κ, with $c = \frac{1}{m}$ and with $r_i := |G_i \cap \mathbf{T}| \to +\infty$. Let $(W_i \ni 0, G_i, h_i)$ be marked germs over $(V_{0,i}, h_i^0)$ such that W_i is normal and klt near 0 and $(W_i, \frac{1}{m}\operatorname{div}(h_i))$ is lc near 0, let q_i be the radial function of $(W_i \ni 0, G_i, h_i, m)$, and let β_i be the number $\beta_{\mathbf{T}}$ of Set-up 11.1 for $(V_{0,i}, G_i, h_i^0)$. Then there exist real numbers N_0*

and N_1 such that for any $i \gg 0$ there exists a prime divisor F_i over $\mathbb{A}^4$ with center 0 satisfying

$$\frac{a(F_i, \mathbb{A}^4, W_i + \frac{1}{m}\operatorname{div}(h_i))}{e_{F_i}} \le \beta_i + \frac{N_0}{r_i} \quad \textit{and} \quad \frac{\operatorname{mult}_{F_i} h_i}{e_{F_i}} \le N_1$$

for the action of G_i on $\mathbb{A}^4$. Hence $q_i(t) \le \beta_i + N_0/r_i + \frac{1}{m}N_1(1-t)$ for any $i \gg 0$ and any $t \in [0,1]$.

Proof. Let $c := \frac{1}{m}$, let $w_1, \dots, w_4$ be the weights of $\mathbf{T}$ on the coordinates $z_1, \dots, z_4$ of $\mathbb{A}^4$, and let $w_{\max} := \max_k |w_k|$. For any i let $R_i := G_i \cap \mathbf{T}$, let f_i^0 and f_i be the equations of $V_{0,i}$ and W_i in $\mathbb{A}^4$, and let δ_i, $Y_i^{\pm}$, $Z_{0,i}$, β_i^+, and β_i^- be the number $\delta_{\mathbf{T}}$, the graded quotients $Y^{\pm}$, the quotient Z_0, and the numbers β^+ and β^- of Set-up 11.1 for $(V_{0,i}, G_i, h_i^0)$.

Step 1. In this step we show that $(V_{0,i}, c\operatorname{div}(h_i^0))$ is lc near 0 for any $i \gg 0$.

Since $V_{0,i}$ is normal and klt, $(\mathbb{A}^4, V_{0,i})$ is lc near 0 [Kaw07, Theorem]. The exceptional divisor of the blow-up of $0 \in \mathbb{A}^4$ has log discrepancy $4 - \operatorname{mult}_0 f_i^0$ with respect to $(\mathbb{A}^4, V_{0,i})$, so $\operatorname{mult}_0 f_i^0 \le 4$. Since f_i^0 is $\mathbf{T}$-homogeneous and has a monomial of degree $\operatorname{mult}_0 f_i^0$, and since $\delta_i = c\kappa - \sum_k w_k + \operatorname{wt}_w(f_i^0)$, we obtain

$$|\operatorname{wt}_w(f_i^0)| \le 4w_{\max} \quad \text{and} \quad |\delta_i| \le c|\kappa| + 8w_{\max}. \tag{11.5}$$

Since W_i is normal and $(W_i, c\operatorname{div}(h_i))$ is lc near 0, the pair $(\mathbb{A}^4, W_i + c\operatorname{div}(h_i))$ is lc near 0 [Kaw07, Theorem]. By the definition of marked germs, the monomials of $f_i - f_i^0$, resp. $h_i - h_i^0$, have $\mathbf{T}$-weights $\operatorname{wt}_w(f_i^0) + nr_i$, resp. $\kappa + nr_i$, with $n \ne 0$, hence degrees at least $(r_i - 4w_{\max})/w_{\max}$, resp. $(r_i - |\kappa|)/w_{\max}$, by (11.5) (cf. [Liu26b, Lemma 5.3]), which tend to $+\infty$ as $r_i \to +\infty$ since κ is fixed. Let L be the integer of Lemma 2.19 for 4 variables, 2 functions, and $\Lambda = \{1, c\}$. For any $i \gg 0$ we have $f_i - f_i^0, h_i - h_i^0 \in (z_1, \dots, z_4)^L$, so by Lemma 2.19 (applied to the images of f_i, h_i, f_i^0, h_i^0 in $\mathbb{C}[[z_1, \dots, z_4]]$, see [dFEM11, Proposition 2.11(i)]) the pair $(\mathbb{A}^4, V_{0,i} + c\operatorname{div}(h_i^0))$ is lc near 0, and by [Kaw07, Theorem] again, $(V_{0,i}, c\operatorname{div}(h_i^0))$ is lc near 0. Thus the assumption of Proposition 11.2(3)(5) holds for $(V_{0,i}, G_i, h_i^0)$. From now on we only consider such i.

Step 2. In this step we fix notation for the root sections and prove (11.6) below.

Fix i and an index j with $w_j \ne 0$. Let $\mathbb{A}_j^3 := (z_j = 1) \subset \mathbb{A}^4$, with coordinates $\overline{z}_k := z_k|_{\mathbb{A}_j^3}$ for $k \ne j$. We write $\overline{f}_i^0, \overline{h}_i^0, \overline{f}_{i,n}, \overline{h}_{i,n}$ for the restrictions to $\mathbb{A}_j^3$. We write an element $\gamma \in G_i$ as $(\gamma_1, \dots, \gamma_4)$, acting on $\mathbb{A}^4$ by $z_k \to \gamma_k z_k$. The abelian group

$$\widetilde{G}_{i,j} := \{(\gamma, s) \in G_i \times \mathbb{C}^* \mid s^{w_j} = \gamma_j\}$$

acts on $\mathbb{A}_j^3 \times \mathbb{G}_m$, with coordinates $\overline{z}_k$ $(k \ne j)$ and ξ, by $\overline{z}_k \to \gamma_k s^{-w_k}\overline{z}_k$ and $\xi \to s\xi$. The morphism

$$\widetilde{\rho}_j : \mathbb{A}_j^3 \times \mathbb{G}_m \to \mathbb{A}^4 \backslash (z_j = 0), \qquad z_j = \xi^{w_j} \quad \text{and} \quad z_k = \xi^{w_k}\overline{z}_k \ (k \ne j),$$

is the quotient by the free action of the subgroup $\{(1, s) \mid s^{w_j} = 1\}$, and $\widetilde{\rho}_j$ is equivariant when (γ, s) acts on $\mathbb{A}^4$ by γ. Let $S_{i,j}$ and $G_{i,j}$ be the S_j and G_j of Set-up 11.1 for $(V_{0,i}, G_i, h_i^0)$, so that $S_{i,j} = (\overline{f}_i^0 = 0) \subset \mathbb{A}_j^3$, and $G_{i,j}$ is the image of $\widetilde{G}_{i,j}$ in $\operatorname{Aut}(\mathbb{A}_j^3)$, as (γ, s) acts on $\mathbb{A}_j^3$ as the element $s^{-1}\gamma$ of the group generated by G_i and $\mathbf{T}$. By Proposition 11.2(2), $S_{i,j}$ is normal and klt, the stabilizer in $G_{i,j}$ of each connected component of $S_{i,j}$ acts faithfully on it, over $D_+(z_j)/(G_i/R_i)$ the fiber reference of the sign of w_j is the crepant quotient of the root-section pair $(S_{i,j}, c\operatorname{div}(\overline{h}_i^0))$ by $G_{i,j}$, and

$$|G_{i,j}| = |w_j|[G_i : R_i] \le e_0 w_{\max},$$

so $G_{i,j}$ is one of the finitely many diagonal subgroups of $GL_3(\mathbb{C})$ of order at most $e_0 w_{\max}$.

Let

$$\Phi_{i,j} := S_{i,j} \cap \{\overline{z}_k = 0 \text{ for any } k \ne j \text{ with } w_k w_j \le 0\}.$$

Any non-constant $\mathbf{T}$-invariant monomial $z_1^{\alpha_1}\cdots z_4^{\alpha_4}$ has $\alpha_k>0$ for some $k\neq j$ with $w_kw_j\leq 0$, and for any such k the monomial $z_k^{|w_j|}z_j^{|w_k|}$ is $\mathbf{T}$-invariant and restricts to $\overline{z}_k^{|w_j|}$ on $\mathbb{A}^3_j$. Hence $\Phi_{i,j}$ is the preimage in $S_{i,j}$ of the fiber of $Y_i^{\pm}\to Z_{0,i}$ over the image of 0, where $\pm$ is the sign of w_j.

By Proposition 11.2(3), the fiber reference is lc near the fiber. Since lc is preserved by crepant pull-back under the finite morphism $S_{i,j}\to S_{i,j}/G_{i,j}$, the root-section pair $(S_{i,j},c\operatorname{div}(\overline{h}_i^0))$ is lc near $\Phi_{i,j}$.

Let $\eta\in\Phi_{i,j}$ be a closed point or the generic point of a curve. We show the equality

$$\operatorname{mld}((S_{i,j}\ni\eta,c\operatorname{div}(\overline{h}_i^0))/G_{i,j})=\operatorname{mld}((\mathbb{A}^3_j\ni\eta,S_{i,j}+c\operatorname{div}(\overline{h}_i^0))/G_{i,j}). \tag{11.6}$$

Let $G'_\eta\subset G_{i,j}$ be the group of Lemma 11.3 for $X=S_{i,j}$, the zero scheme of $\overline{f}_i^0$ on $\mathbb{A}^3_j$, and for $\mathfrak{a}=(\overline{h}_i^0)^c$, and let x_0 be the point of Lemma 11.3(1). The assumptions of the lemma hold by the previous paragraphs and since $\overline{h}_i^0|_{S_{i,j}}\neq 0$, and G'_η acts faithfully on $S_{i,j}$ near x_0, since G'_η fixes x_0 and hence is contained in the stabilizer of the connected component of $S_{i,j}$ through x_0, which acts faithfully on that component. By Lemma 11.3(2)(3), (11.6) is equivalent to

$$\operatorname{mld}((S_{i,j}\ni\eta,c\operatorname{div}(\overline{h}_i^0))/G'_\eta)=\operatorname{mld}((\mathbb{A}^3_j\ni\eta,S_{i,j}+c\operatorname{div}(\overline{h}_i^0))/G'_\eta). \tag{11.7}$$

By Lemma 11.3(1)(5), Theorem 2.17(1)(3) at x_0 for $\overline{f}_i^0$, G'_η, and $\mathfrak{a}=(\overline{h}_i^0)^c$ implies (11.7).

Step 3. In this step we choose a prime divisor computing β_i on a root-section ambient, with bounded log discrepancy.

Replacing w with $-w$ does not change the subgroup $\mathbf{T}$, replaces κ, $\operatorname{wt}_w(f_i^0)$, and δ_i with their negatives, and interchanges Y_i^+ and Y_i^-, hence β_i^+ and β_i^-. When $\delta_i>0$, it preserves β_i. Treating separately the indices i with $\delta_i\leq 0$ and those with $\delta_i>0$, possibly replacing w with $-w$, we may assume that $\delta_i\leq 0$ for any i, so that $\beta_i=\beta_i^+$.

For any of the finitely many diagonal subgroups $G'\subset GL_3(\mathbb{C})$ of order at most $e_0w_{\max}$, let $L_{G'}$ be the integer of Theorem 2.22(3) for $X=\mathbb{A}^3/G'$ and the DCC set $\frac{1}{m\cdot|G'|!}\mathbb{N}^+$, and let

$$N_2:=\max_{G'}|G'|L_{G'}\quad\text{and}\quad N_3:=\max_{G'}|G'|.$$

Fix i. For any j with $w_j>0$ and any prime divisor P' over $S_{i,j}$, the number

$$m\cdot a(P',S_{i,j},c\operatorname{div}(\overline{h}_i^0))$$

is an integer, since $S_{i,j}$ is a hypersurface and $mc=1$. Since $|I_{G_{i,j}}(P')|\leq N_3$ and the root-section pair is lc near $\Phi_{i,j}$, the left-hand side of (11.6) belongs to $\frac{1}{m\cdot N_3!}\mathbb{N}$ for any η as in **Step 2**. The centers of prime divisors over $Y_i^+/(G_i/R_i)$ with center in the fiber are closed points and curves of the fiber, and by Proposition 11.2(1) and **Step 2** they are images of closed points and generic points of curves in $\Phi_{i,j}$ for indices j with $w_j>0$. Hence $\beta_i=\beta_i^+$ is the minimum of the left-hand side of (11.6) over all such j and η. We fix j with $w_j>0$ and η such that the left-hand side of (11.6) equals β_i, and let

$$D_i:=S_{i,j}+c\operatorname{div}(\overline{h}_i^0)\quad\text{on }\mathbb{A}^3_j.$$

By (11.6), $\beta_i=\operatorname{mld}((\mathbb{A}^3_j\ni\eta,D_i)/G_{i,j})$.

Let $\pi:\mathbb{A}^3_j\to X:=\mathbb{A}^3_j/G_{i,j}$ be the quotient morphism, $\overline{\eta}:=\pi(\eta)$, and let B and B_0 be the $\mathbb{R}$-divisors on X such that

$$K_{\mathbb{A}^3_j}+D_i=\pi^*(K_X+B)\quad\text{and}\quad K_{\mathbb{A}^3_j}=\pi^*(K_X+B_0).$$

Then $\beta_i=\operatorname{mld}(X\ni\overline{\eta},B)$ and $B_0\geq 0$. Since $D_i\in c\mathbb{N}$, Lemma 2.15(4) implies $B\in\frac{1}{m\cdot|G_{i,j}|!}\mathbb{N}^+$. The threefold X is klt and $\mathbb{Q}$-factorial, and (X,B) is lc near $\overline{\eta}$, since $(\mathbb{A}^3_j,D_i)$ is lc near η by [Kaw07, Theorem]. By Theorem 2.22(3) there exists a prime divisor E over X computing

$\mathrm{mld}(X \ni \overline{\eta}, B)$ such that $a(E, X, 0) \le L_{G_{i,j}}$. Let E' be a prime divisor over $\mathbb{A}^3_j$ over E with center $\overline{\{\eta\}}$, and let $e_{E'} := |I_{G_{i,j}}(E')|$. By Lemma 2.15(2),

$$a(E', \mathbb{A}^3_j, D_i) = e_{E'}\beta_i \quad \text{and} \quad a(E', \mathbb{A}^3_j, 0) = e_{E'}a(E, X, B_0) \le e_{E'}a(E, X, 0) \le N_2.$$

Since $\beta_i \ge 0$, we have

$$\mathrm{mult}_{E'} \overline{f}_i^0 + c\,\mathrm{mult}_{E'} \overline{h}_i^0 = a(E', \mathbb{A}^3_j, 0) - e_{E'}\beta_i \le N_2.$$

Together with $\eta \in \Phi_{i,j}$, we obtain

$$\begin{gathered} \mathrm{mult}_{E'} \overline{f}_i^0 \le N_2, \qquad \mathrm{mult}_{E'} \overline{h}_i^0 \le mN_2, \\ \text{and} \quad \mathrm{mult}_{E'} \overline{z}_k \ge 1 \text{ for any } k \ne j \text{ with } w_k \le 0. \end{gathered} \tag{11.8}$$

Step 4. In this step we construct F_i and prove the first statement.

Let $\varrho := \xi^{r_i}$. The subgroup

$$\widetilde{R}_i := \{(s^{w_1}, \dots, s^{w_4}, s) \mid s^{r_i} = 1\}$$

of $\widetilde{G}_{i,j}$ acts trivially on $\mathbb{A}^3_j$ and by $\xi \to s\xi$, so $(\overline{z}, \xi) \to (\overline{z}, \varrho)$ is the quotient of $\mathbb{A}^3_j \times \mathbb{G}_m$ by $\widetilde{R}_i$. If $(\gamma, s) \in \widetilde{G}_{i,j}$ acts trivially on $\mathbb{A}^3_j$, then $\gamma_k = s^{w_k}$ for any k, so $\gamma \in R_i$. Since $\mathbf{T}$ is primitive, $(\gamma, s) \in \widetilde{R}_i$. Hence $\widetilde{G}_{i,j}/\widetilde{R}_i \to G_{i,j}$ is an isomorphism, and $G_{i,j}$ acts on $\mathbb{A}^3_j \times \mathbb{A}^1_\varrho$, faithfully on $\mathbb{A}^3_j$, and on ϱ by the character $(\gamma, s) \to s^{r_i}$. Moreover, $\widetilde{\rho}_j$ is the quotient by the subgroup $\{(1, s) \mid s^{w_j} = 1\}$ of $\widetilde{G}_{i,j}$, with quotient group G_i, and $(\overline{z}, \xi) \to (\overline{z}, \varrho)$ is the quotient by $\widetilde{R}_i$, so they induce isomorphisms

$$(\mathbb{A}^4 \backslash (z_j = 0))/G_i \cong (\mathbb{A}^3_j \times \mathbb{G}_m)/\widetilde{G}_{i,j} \cong (\mathbb{A}^3_j \times (\mathbb{A}^1_\varrho \backslash \{0\}))/G_{i,j}.$$

In particular, $\mathbb{A}^4/G_i$ and $(\mathbb{A}^3_j \times \mathbb{A}^1_\varrho)/G_{i,j}$ are birational.

By [Liu26b, Lemma 3.1] for the group G'_η of **Step 2** acting on $\mathbb{A}^3_j$ with coordinates translated to η if η is a closed point, there exists a residue class modulo $e_{E'}$ such that for any positive integer ϑ in the residue class, the Gauss extension $(\mathrm{mult}_{E'})_\vartheta$ of $\mathrm{mult}_{E'}$ with value ϑ on ϱ (Definition 2.3) is $\mathrm{mult}_{F'}$ for a prime divisor F' over $\mathbb{A}^3_j \times \mathbb{A}^1_\varrho$ with center $\overline{\{\eta\}} \times \{0\}$, satisfying

$$I_{G_{i,j}}(F') = I_{G_{i,j}}(E') \quad \text{and} \quad a(F', \mathbb{A}^3_j \times \mathbb{A}^1_\varrho, \mathrm{pr}^* D) = a(E', \mathbb{A}^3_j, D) + \vartheta$$

for any $\mathbb{R}$-divisor D on $\mathbb{A}^3_j$, where pr is the projection to $\mathbb{A}^3_j$. Here $I_{G_{i,j}}(F')$ and $I_{G_{i,j}}(E')$ fix the centers of F' and E' pointwise, so they are contained in G'_η and are the inertia groups of F' and E' for G'_η. We show that such a residue class and such prime divisors F' also exist if η is the generic point of a curve C (cf. [Liu26b, Lemma 3.1]).

Let η be the generic point of a curve C, let χ be the character of $G_{i,j}$ with $g^*\varrho = \chi(g)\varrho$, and let $I := I_{G_{i,j}}(E')$. By Lemma 2.15(1), I is cyclic and acts on $\mathfrak{m}_{E'}/\mathfrak{m}^2_{E'}$ by a faithful character λ through the pull-back of functions, so $\chi|_I = \lambda^{\nu_0}$ for an integer ν_0. We take the residue class of ν_0 modulo $e_{E'}$. Let $\vartheta \equiv \nu_0 \pmod{e_{E'}}$ be a positive integer, D an $\mathbb{R}$-divisor on $\mathbb{A}^3_j$, $Y \to \mathbb{A}^3_j$ a higher model on which E' is a divisor, $K_Y + D_Y$ the pull-back of $K_{\mathbb{A}^3_j} + D$, τ a rational function on $\mathbb{A}^3_j$ with $\mathrm{mult}_{E'} \tau = 1$, and $U \subset Y$ a smooth affine open subset with a nowhere vanishing top form σ such that $E' \cap U \ne \emptyset$, $\mathrm{div}(\tau|_U) = E' \cap U$, and $D_Y|_U = (1 - a(E', \mathbb{A}^3_j, D))(E' \cap U)$. The morphism

$$h : U \times \mathbb{A}^1_{\varrho'} \to U \times \mathbb{A}^1_\varrho, \qquad (y, \varrho') \to (y, \tau(y)^\vartheta \varrho'),$$

is birational, and $F' := (E' \cap U) \times \mathbb{A}^1_{\varrho'}$ is a prime divisor over $\mathbb{A}^3_j \times \mathbb{A}^1_\varrho$ with center $C \times \{0\}$. For rational functions g_n on $\mathbb{A}^3_j$, write $g_n = \epsilon_n \tau^{\mathrm{mult}_{E'} g_n}$ with ϵ_n invertible at the generic point of E'. Then

$$h^* \sum_n g_n \varrho^n = \sum_n \epsilon_n \tau^{\mathrm{mult}_{E'} g_n + n\vartheta} \varrho'^n,$$

and the terms of least order in τ restrict to a non-zero function on F', so $\mathrm{mult}_{F'} = (\mathrm{mult}_{E'})_\vartheta$. In particular, F' does not depend on D. Since

$$h^*(\sigma \wedge d\varrho) = \tau^\vartheta \sigma \wedge d\varrho' \quad \text{and} \quad h^* \operatorname{pr}^*(E' \cap U) = F',$$

we have

$$K_{U\times\mathbb{A}^1_{\varrho'}} + (1 - a(E', \mathbb{A}^3_j, D) - \vartheta)F' = h^*(K_{U\times\mathbb{A}^1_\varrho} + \operatorname{pr}^* D_Y|_U),$$

so

$$a(F', \mathbb{A}^3_j \times \mathbb{A}^1_\varrho, \operatorname{pr}^* D) = a(E', \mathbb{A}^3_j, D) + \vartheta.$$

Any $g \in I$ preserves $\mathrm{mult}_{E'}$ and multiplies ϱ by $\chi(g)$, hence preserves $\mathrm{mult}_{F'}$. The residue field of F' is generated over the residue field of E' by the residue of $\varrho' = \varrho/\tau^\vartheta$. Since $g^*\tau/\tau$ has residue $\lambda(g)$, the residue of $g^*\varrho' = \chi(g)(\tau/g^*\tau)^\vartheta \varrho'$ is $\lambda(g)^{\nu_0-\vartheta} = 1$ times that of ϱ'. Thus $I \subset I_{G_{i,j}}(F')$. Conversely, any element of $I_{G_{i,j}}(F')$ preserves the restriction $\mathrm{mult}_{E'}$ of $\mathrm{mult}_{F'}$ to the rational functions on $\mathbb{A}^3_j$ and acts trivially on the residue field of E', a subfield of that of F', so it belongs to I. Hence $I_{G_{i,j}}(F') = I_{G_{i,j}}(E')$.

We fix such a $\vartheta = \vartheta_i$ with

$$mN_2 < \vartheta_i \le mN_2 + N_3,$$

and we assume that $r_i > w_{\max}(2mN_2 + N_3 + 4) + |\kappa|$, which holds for any $i \gg 0$.

By the definition of marked germs, $f_i = \sum_{n\in\mathbb{Z}} f_{i,n}$ and $h_i = \sum_{n\in\mathbb{Z}} h_{i,n}$, where $f_{i,n}$ and $h_{i,n}$ are $\mathbf{T}$-homogeneous of weights $\mathrm{wt}_w(f_i^0) + nr_i$ and $\kappa + nr_i$, $f_{i,0} = f_i^0$, and $h_{i,0} = h_i^0$. Then

$$\widetilde{\rho}_j^* f_i = \xi^{\mathrm{wt}_w(f_i^0)} \widehat{f}_i, \quad \widehat{f}_i := \sum_n \varrho^n \overline{f}_{i,n}, \qquad \widetilde{\rho}_j^* h_i = \xi^\kappa \widehat{h}_i, \quad \text{and} \quad \widehat{h}_i := \sum_n \varrho^n \overline{h}_{i,n}.$$

We show

$$\mathrm{mult}_{F'} \widehat{f}_i = \mathrm{mult}_{E'} \overline{f}_i^0 \quad \text{and} \quad \mathrm{mult}_{F'} \widehat{h}_i = \mathrm{mult}_{E'} \overline{h}_i^0. \tag{11.9}$$

If $n \ge 1$, then $\mathrm{mult}_{F'}(\varrho^n \overline{f}_{i,n}) \ge \vartheta_i > mN_2$ and $\mathrm{mult}_{F'}(\varrho^n \overline{h}_{i,n}) \ge \vartheta_i > mN_2$, since $\overline{f}_{i,n}$ and $\overline{h}_{i,n}$ are polynomials and the center of E' is contained in $\mathbb{A}^3_j$. If $n \le -1$, a monomial $z_1^{\alpha_1} \cdots z_4^{\alpha_4}$ of $f_{i,n}$ satisfies

$$\sum_{w_k<0} \alpha_k |w_k| = \sum_{w_k>0} \alpha_k w_k - \mathrm{wt}_w(f_i^0) + |n| r_i \ge |n| r_i - 4w_{\max}$$

by (11.5), so $\sum_{w_k<0} \alpha_k \ge |n| r_i / w_{\max} - 4$. Since $w_j > 0$, (11.8) implies

$$\mathrm{mult}_{F'}(\varrho^n \overline{f}_{i,n}) \ge |n| \Big(\frac{r_i}{w_{\max}} - \vartheta_i \Big) - 4 \ge \frac{r_i}{w_{\max}} - \vartheta_i - 4 > mN_2.$$

If $n \le -1$, a monomial $z_1^{\alpha_1} \cdots z_4^{\alpha_4}$ of $h_{i,n}$ satisfies

$$\sum_{w_k<0} \alpha_k |w_k| \ge |n| r_i - |\kappa|,$$

so

$$\mathrm{mult}_{F'}(\varrho^n \overline{h}_{i,n}) \ge |n| \Big(\frac{r_i}{w_{\max}} - \vartheta_i \Big) - \frac{|\kappa|}{w_{\max}} \ge \frac{r_i}{w_{\max}} - \vartheta_i - \frac{|\kappa|}{w_{\max}} > mN_2.$$

Since $\mathrm{mult}_{E'} \overline{f}_i^0 \le mN_2$ and $\mathrm{mult}_{E'} \overline{h}_i^0 \le mN_2$ by (11.8), the minima defining $\mathrm{mult}_{F'} \widehat{f}_i$ and $\mathrm{mult}_{F'} \widehat{h}_i$ are attained at $n = 0$, which proves (11.9).

We pull back rational functions on $\mathbb{A}^4$ and on $\mathbb{A}^3_j \times \mathbb{A}^1_\varrho$ to $\mathbb{A}^3_j \times \mathbb{G}_m$ by $\widetilde{\rho}_j$ and by $\varrho = \xi^{r_i}$. Let v be a valuation on the rational functions on $\mathbb{A}^3_j \times \mathbb{G}_m$ extending $\frac{1}{e_{E'}} \mathrm{mult}_{F'}$, so that $v(\xi) = \frac{\vartheta_i}{r_i e_{E'}}$ and $v(g) = \frac{1}{e_{E'}} \mathrm{mult}_{E'} g$ for any rational function g on $\mathbb{A}^3_j$, and let F_i be the prime divisor over $\mathbb{A}^4$ whose valuation is a positive multiple of the restriction of v to the rational functions on $\mathbb{A}^4$. Since $|I_{G_{i,j}}(F')| = e_{E'}$, the image $\overline{F}$ of F' over $(\mathbb{A}^3_j \times \mathbb{A}^1_\varrho)/G_{i,j}$ satisfies $\mathrm{mult}_{\overline{F}} = \frac{1}{e_{E'}} \mathrm{mult}_{F'}$ on $G_{i,j}$-invariant rational functions. Through the birational map above, $\overline{F}$ is also the image of

F_i over $\mathbb{A}^4/G_i$, so $\frac{1}{e_{F_i}}\operatorname{mult}_{F_i}$ and v coincide on G_i-invariant rational functions. Since they are proportional on the rational functions on $\mathbb{A}^4$,

$$\frac{\operatorname{mult}_{F_i} g}{e_{F_i}} = v(g) \quad \text{for any rational function } g \text{ on } \mathbb{A}^4.$$

We have

$$v(z_j) = \frac{w_j\vartheta_i}{r_i e_{E'}} > 0 \quad \text{and} \quad v(z_k) = \frac{1}{e_{E'}}\Big(\frac{w_k\vartheta_i}{r_i} + \operatorname{mult}_{E'} \overline{z}_k\Big) \quad \text{for any } k \neq j.$$

For $k \neq j$, $v(z_k) > 0$ if $w_k > 0$, and also if $w_k \leq 0$ by (11.8) and $r_i > w_{\max}\vartheta_i$. Thus F_i has center 0 on $\mathbb{A}^4$.

Let $d\overline{z}$ be the wedge product of the $d\overline{z}_k$ $(k \neq j)$ in increasing order. Then

$$\widetilde{\rho}_j^*(dz_1 \wedge \cdots \wedge dz_4) = \pm w_j \xi^{\sum_k w_k - 1} d\xi \wedge d\overline{z}.$$

Since $d\xi/\xi = d\varrho/(r_i\varrho)$ and $\sum_k w_k - \operatorname{wt}_w(f_i^0) - c\kappa = -\delta_i$, where $m\delta_i = \kappa - m(\sum_k w_k - \operatorname{wt}_w(f_i^0))$ is an integer, we have

$$\widetilde{\rho}_j^*\frac{(dz_1 \wedge \cdots \wedge dz_4)^{\otimes mr_i}}{f_i^{mr_i} h_i^{r_i}} = \Big(\pm\frac{w_j}{r_i}\Big)^{mr_i} \frac{(d\varrho \wedge d\overline{z})^{\otimes mr_i}}{\varrho^{mr_i + m\delta_i} \widehat{f}_i^{mr_i} \widehat{h}_i^{r_i}}. \tag{11.10}$$

Let

$$\Delta_i := \operatorname{div}(\widehat{f}_i) + c\operatorname{div}(\widehat{h}_i) + \Big(1 + \frac{\delta_i}{r_i}\Big)\operatorname{div}(\varrho) \quad \text{on } \mathbb{A}^3_j \times \mathbb{A}^1_\varrho.$$

The left-hand side and the right-hand side of (11.10) are $\widetilde{G}_{i,j}$-semi-invariant rational sections of $\omega^{\otimes mr_i}$ with divisors $-mr_i(W_i + c\operatorname{div}(h_i))$ on $\mathbb{A}^4$ and $-mr_i\Delta_i$ on $\mathbb{A}^3_j \times \mathbb{A}^1_\varrho$, so a power of them descends to $\mathbb{A}^4/G_i$ and to $(\mathbb{A}^3_j \times \mathbb{A}^1_\varrho)/G_{i,j}$, and the crepant quotients $(\mathbb{A}^4, W_i + c\operatorname{div}(h_i))/G_i$ and $(\mathbb{A}^3_j \times \mathbb{A}^1_\varrho, \Delta_i)/G_{i,j}$ are crepant birational through the birational map above. Since F_i and F' have the image $\overline{F}$ over $\mathbb{A}^4/G_i$ and over $(\mathbb{A}^3_j \times \mathbb{A}^1_\varrho)/G_{i,j}$, Lemma 2.15(2) for the quotients by G_i and by $G_{i,j}$ implies

$$\frac{a(F_i, \mathbb{A}^4, W_i + c\operatorname{div}(h_i))}{e_{F_i}} = \frac{a(F', \mathbb{A}^3_j \times \mathbb{A}^1_\varrho, \Delta_i)}{e_{E'}}.$$

By (11.9), $\Delta_i - \operatorname{pr}^* D_i$ has multiplicity $(1 + \frac{\delta_i}{r_i})\vartheta_i$ along F', so

$$a(F', \mathbb{A}^3_j \times \mathbb{A}^1_\varrho, \Delta_i) = a(E', \mathbb{A}^3_j, D_i) + \vartheta_i - \Big(1 + \frac{\delta_i}{r_i}\Big)\vartheta_i = e_{E'}\beta_i - \frac{\delta_i\vartheta_i}{r_i}.$$

Since $\delta_i \leq 0$ and $e_{E'} \geq 1$, (11.5) implies

$$\frac{a(F_i, \mathbb{A}^4, W_i + c\operatorname{div}(h_i))}{e_{F_i}} \leq \beta_i + \frac{|\delta_i|\vartheta_i}{r_i} \leq \beta_i + \frac{N_0}{r_i}, \qquad N_0 := (c|\kappa| + 8w_{\max})(mN_2 + N_3).$$

By (11.9), (11.8) and $\vartheta_i < r_i$,

$$\frac{\operatorname{mult}_{F_i} h_i}{e_{F_i}} = v(\xi^\kappa \widehat{h}_i) = \frac{1}{e_{E'}}\Big(\frac{\kappa\vartheta_i}{r_i} + \operatorname{mult}_{E'} \overline{h}_i^0\Big) \leq N_1, \qquad N_1 := |\kappa| + mN_2.$$

The numbers N_0 and N_1 depend only on w, e_0, m, and κ. This proves the first statement.

Step 5. In this step we prove the last statement.

Let $t \in [0,1]$. Since $(W_i, c\operatorname{div}(h_i))$ is lc near 0, so is $(W_i, tc\operatorname{div}(h_i))$, and $h_i|_{W_i} \neq 0$. Theorem 2.17(1) for f_i, G_i, and $\mathfrak{a} = (h_i)^{tc}$ ($\mathfrak{a} = \mathcal{O}_{\mathbb{A}^4}$ if $t = 0$) implies

$$q_i(t) = \operatorname{mld}\big((\mathbb{A}^4 \ni 0, W_i + tc\operatorname{div}(h_i))/G_i\big).$$

Since F_i has center 0, the first statement implies

$$q_i(t) \le \frac{a(F_i, \mathbb{A}^4, W_i + tc\,\mathrm{div}(h_i))}{e_{F_i}} = \frac{a(F_i, \mathbb{A}^4, W_i + c\,\mathrm{div}(h_i))}{e_{F_i}} + (1-t)c\frac{\mathrm{mult}_{F_i} h_i}{e_{F_i}}$$
$$\le \beta_i + \frac{N_0}{r_i} + \frac{1}{m}N_1(1-t).$$

□

Lemma 11.5 (cf. [Liu26b, Lemma 5.5]). *Let X be $\mathbb{A}^3$ or a normal klt hypersurface in $\mathbb{A}^4$ through 0, with a diagonal action of a finite abelian group H which is faithful on X, let u be a semi-invariant coordinate, let $S_0 := X \cap (u = 0)$, and let $\mathfrak{a}$ be an $\mathbb{R}$-ideal with H-semi-invariant generators such that $\mathfrak{a}|_{S_0} \neq 0$. Let $\eta \in S_0$ be a closed point or the generic point of a curve in S_0. Assume that*

(1) S_0 is normal and klt near η,
(2) no non-identity element of H acts trivially on an irreducible component of S_0, and
(3) $(S_0, \mathfrak{a}|_{S_0})$ is lc near η.

Let $\beta_0 := \mathrm{mld}((S_0 \ni \eta, \mathfrak{a}|_{S_0})/H)$, taken at the image of η. Then for any real number c_S and any prime divisor P over X with center $\overline{\{\eta\}}$,

$$\frac{a(P, X, c_S S_0, \mathfrak{a})}{e_P} \ge \beta_0 + (1-c_S)\frac{\mathrm{mult}_P u}{e_P},$$

where $e_P := |I_H(P)|$.

Proof. Since $S_0 = \mathrm{div}_X(u)$ and K_X is Cartier, for any real number c_S we have

$$a(P, X, c_S S_0, \mathfrak{a}) = a(P, X, 0, \mathfrak{a}) - c_S\,\mathrm{mult}_P u = a(P, X, S_0, \mathfrak{a}) + (1-c_S)\,\mathrm{mult}_P u. \tag{11.11}$$

Thus we only need to prove $a(P, X, S_0, \mathfrak{a}) \ge e_P\beta_0$.

Let $H_\eta \subset H$ be the group of Lemma 11.3 for the zero scheme S_0 of u (and of the equation of X if $X \neq \mathbb{A}^3$) and for $\mathfrak{a}$, and let x_0 be the point of Lemma 11.3(1). The assumptions of the lemma hold near η by assumption. Any element of $I_H(P)$ acts trivially on the residue field of P, hence fixes $\overline{\{\eta\}}$ pointwise. Thus $I_H(P) \subset H_\eta$, so $I_{H_\eta}(P) = I_H(P)$ and

$$\frac{a(P, X, S_0, \mathfrak{a})}{e_P} \ge \mathrm{mld}((X \ni \eta, S_0, \mathfrak{a})/H_\eta). \tag{11.12}$$

Since S_0 is normal near x_0, it has a unique irreducible component through x_0, which H_η preserves, so by assumption (2), H_η acts faithfully on S_0 near x_0. If $X = \mathbb{A}^3$, by Lemma 11.3(1)(5), Theorem 2.17(1)(3) at x_0 for u, H_η, and $\mathfrak{a}$ implies

$$\mathrm{mld}((S_0 \ni \eta, \mathfrak{a}|_{S_0})/H_\eta) = \mathrm{mld}((X \ni \eta, S_0, \mathfrak{a})/H_\eta). \tag{11.13}$$

If $X = (f = 0) \subset \mathbb{A}^4$, Theorem 2.17(2)(3) at x_0 for f, the additional equation u, H_η, and $\mathfrak{a}$ implies (11.13). By Lemma 11.3(2) for S_0,

$$\mathrm{mld}((S_0 \ni \eta, \mathfrak{a}|_{S_0})/H_\eta) = \mathrm{mld}((S_0 \ni \eta, \mathfrak{a}|_{S_0})/H) = \beta_0.$$

By (11.12) and (11.13), $a(P, X, S_0, \mathfrak{a}) \ge e_P\beta_0$, and by (11.11) the lemma follows. □

11.3. **Homogeneous adjunction.**

Lemma 11.6 (cf. [Liu26b, Proposition 11.12]). *Notation and conditions as in Set-up 11.1. Let U be a non-zero regular $\mathbf{T}$-invariant G-semi-invariant function on V_0, regarded as a regular function on Z_0 and on $Y^\pm$, and let $\lambda \ge 0$ be a real number. Assume that one of the following holds.*

(1) $\delta_{\mathbf{T}} \le 0$, and $(Y^+, B^+ + \lambda\,\mathrm{div}(U))/H$ is lc near the fiber of $Y^+/H \to Z_0/H$ over $\overline{o}$.
(2) $\delta_{\mathbf{T}} \ge 0$, and $(Y^-, B^- + \lambda\,\mathrm{div}(U))/H$ is lc near the fiber of $Y^-/H \to Z_0/H$ over $\overline{o}$.

Then $(V_0, c\,\mathrm{div}(h^0) + \lambda\,\mathrm{div}(U))$ is lc near 0.

The following proof is similar to [Liu26b, Proof of Proposition 11.12, Step 5].

Proof of Lemma 11.6. We let

$$B_{V_0} := c\,\mathrm{div}(h^0) + \lambda\,\mathrm{div}(U),$$

let $\pi : V_0 \to Z_0$ be the quotient morphism, and let Ω_{V_0} be the generator of ω_{V_0} in Set-up 11.1, so that $K_{V_0} = \mathrm{div}(\Omega_{V_0}) = 0$. Thus for any prime divisor P over V_0,

$$a(P, V_0, B_{V_0}) = 1 + \mathrm{mult}_P\,\Omega_{V_0} - c\,\mathrm{mult}_P\,h^0 - \lambda\,\mathrm{mult}_P\,U. \tag{11.14}$$

Replacing w with $-w$ replaces $\delta_{\mathbf{T}}$ with $-\delta_{\mathbf{T}}$ and interchanges (Y^+, B^+) and (Y^-, B^-), hence turns case (2) into case (1). Thus we may assume (1).

Step 1. It suffices to show that $a(P, V_0, B_{V_0}) \geq 0$ for any prime divisor P over V_0 whose center $g(P)$ on V_0 contains 0, where $g : V' \to V_0$ is a projective birational morphism from a normal variety on which P is a divisor. We fix such P and g in **Steps 2–4**.

Step 2. In this step we choose an index j_0 and pass to the section $(z_{j_0} = 1)$.

Since $z_j|_{V_0}$ is a non-zero regular function (Set-up 11.1(2)), $\mathrm{mult}_P\,z_j$ is a non-negative integer for any j. Let j_0 be an index with $w_{j_0} > 0$ such that

$$\frac{\mathrm{mult}_P\,z_{j_0}}{w_{j_0}} = \min\left\{\frac{\mathrm{mult}_P\,z_j}{w_j}\;\middle|\; w_j > 0\right\}. \tag{11.15}$$

Let $S := S_{j_0} = V_0 \cap (z_{j_0} = 1)$ be as in Set-up 11.1, and let

$$\phi : S \times \mathbb{G}_m \to V_0 \cap (z_{j_0} \neq 0), \quad (z, \xi) \to \xi \cdot z,$$

which is the composition of the isomorphism $S \times \mathbb{G}_m \to \mathbf{T} \times S$, $(z, \xi) \to (\xi, z)$, and the morphism ρ_{j_0} of Proposition 11.2(2), where $\mathbf{T}$ is identified with $\mathbb{G}_m$. Here and below we regard a function on S as a function on $S \times \mathbb{G}_m$ via the first projection, so that $\phi^* z_j = \xi^{w_j} z_j|_S$ for any j. By Proposition 11.2(2), ϕ is the quotient by a free action of $\mathbf{T}_{j_0} = \mu_{|w_{j_0}|}$ on $S \times \mathbb{G}_m$, so ϕ is étale, and S is normal.

Let

$$\Omega_S := \mathrm{Res}_S \frac{\bigwedge_{j \neq j_0} dz_j}{f^0|_{z_{j_0}=1}},$$

where the wedge product is taken in increasing order of j. Since S is a normal hypersurface in $(z_{j_0} = 1) \cong \mathbb{A}^3$, $K_S = \mathrm{div}(\Omega_S) = 0$. The morphism $\widetilde{\phi} : (z_{j_0} = 1) \times \mathbb{G}_m \to \mathbb{A}^4$, $(z, \xi) \to \xi \cdot z$, satisfies $\widetilde{\phi}^* dz_{j_0} = w_{j_0} \xi^{w_{j_0}} \frac{d\xi}{\xi}$, and $\widetilde{\phi}^* dz_j - \xi^{w_j} dz_j$ is a multiple of $d\xi$ for any $j \neq j_0$. Thus

$$\widetilde{\phi}^*(dz_1 \wedge dz_2 \wedge dz_3 \wedge dz_4) = \pm w_{j_0} \xi^{\sum_j w_j} \frac{d\xi}{\xi} \wedge \bigwedge_{j \neq j_0} dz_j \quad \text{and} \quad \widetilde{\phi}^* f^0 = \xi^{\mathrm{wt}_w(f^0)} f^0|_{z_{j_0}=1}.$$

Taking residues along $\widetilde{\phi}^{-1}(V_0) = S \times \mathbb{G}_m$, we obtain

$$\phi^* \Omega_{V_0} = \pm w_{j_0} \xi^{\sum_j w_j - \mathrm{wt}_w(f^0)} \frac{d\xi}{\xi} \wedge \Omega_S, \qquad \phi^* h^0 = \xi^\kappa h^0|_S, \quad \text{and} \quad \phi^* U = U|_S. \tag{11.16}$$

Since $h^0|_{V_0} \neq 0$, $U \neq 0$, and ϕ maps any connected component of $S \times \mathbb{G}_m$ onto $V_0 \cap (z_{j_0} \neq 0)$, the functions $h^0|_S$ and $U|_S$ are non-zero on any connected component of S. We let

$$B_S := B_{V_0}|_S = c\,\mathrm{div}(h^0|_S) + \lambda\,\mathrm{div}(U|_S).$$

Let $B_{Y^+} := B^+ + \lambda\,\mathrm{div}(U)$. Since $Y^+ \to Z_0$ is projective, by assumption (1) the crepant quotient of (Y^+, B_{Y^+}) by H is lc over an open subset $(Z_0/H)^\circ \subset Z_0/H$ containing $\bar{o}$. Let $Z_0^\circ \subset Z_0$ be the preimage of $(Z_0/H)^\circ$. Then Z_0° is an open neighborhood of o. Since lc is preserved by finite crepant pullback, (Y^+, B_{Y^+}) is lc over Z_0°. By the definition of B^+, the pullback of (Y^+, B_{Y^+}) under the composition $S \to S/\mathbf{T}_{j_0} \subset Y^+$ is (S, B_S), and the composition $S \to Y^+ \to Z_0$ is $\pi|_S$. Hence

$$(S, B_S) \text{ is lc on } (\pi|_S)^{-1}(Z_0^\circ). \tag{11.17}$$

Step 3. In this step we lift P to $S \times \mathbb{G}_m$ and express its log discrepancy there.

Let V'' be the normalization of V' in the ring of rational functions on $S \times \mathbb{G}_m$. The group $\mu_{|w_{j_0}|}$ acts on V'' with quotient V'. Let P' be a prime divisor on V'' whose image in V' is P. Let S_1 be the connected component of S such that $\operatorname{mult}_{P'}$ is a valuation of the field of rational functions on $S_1 \times \mathbb{G}_m$. Let V_1'' be the connected component of V'' containing P', and let $\Gamma \subset \mu_{|w_{j_0}|}$ be the stabilizer of V_1''. By Lemma 2.15(1) and (2) for V_1'' with the action of Γ,

$$\operatorname{mult}_{P'} \phi^*\psi = e_{P'} \operatorname{mult}_P \psi \quad \text{and} \quad 1 + \operatorname{mult}_{P'} \phi^*\omega = e_{P'}(1 + \operatorname{mult}_P \omega)$$

for any non-zero rational function ψ and any non-zero rational differential form ω of top degree on V_0, where $e_{P'} := |I_{\mu_{|w_{j_0}|}}(P')|$. Let

$$\rho := \operatorname{mult}_{P'} \xi.$$

By (11.14), (11.16), and the definition of $\delta_{\mathbf{T}}$ in Set-up 11.1, we have

$$\begin{aligned} e_{P'} a(P, V_0, B_{V_0}) = 1 &+ \operatorname{mult}_{P'} \Big(\frac{d\xi}{\xi} \wedge \Omega_S\Big) \\ &- c \operatorname{mult}_{P'}(h^0|_{S_1}) - \lambda \operatorname{mult}_{P'}(U|_{S_1}) - \delta_{\mathbf{T}} \rho. \end{aligned} \tag{11.18}$$

Since $\phi^* z_j = \xi^{w_j} z_j|_{S_1}$ on $S_1 \times \mathbb{G}_m$, we have

$$\operatorname{mult}_{P'}(z_j|_{S_1}) = e_{P'} \operatorname{mult}_P z_j - w_j \rho$$

for any j. Since $z_{j_0}|_{S_1} = 1$, $\rho = e_{P'} \operatorname{mult}_P z_{j_0}/w_{j_0} \geq 0$. By (11.15),

$$\begin{aligned} &\operatorname{mult}_{P'}(z_j|_{S_1}) = e_{P'} w_j \Big(\frac{\operatorname{mult}_P z_j}{w_j} - \frac{\operatorname{mult}_P z_{j_0}}{w_{j_0}}\Big) \geq 0 && \text{if } w_j > 0, \\ &\operatorname{mult}_{P'}(z_j|_{S_1}) = e_{P'} \operatorname{mult}_P z_j + |w_j|\rho \geq 0 && \text{if } w_j \leq 0. \end{aligned}$$

Since the functions $z_j|_{S_1}$ generate $\mathbb{C}[S_1]$,

$$\operatorname{mult}_{P'} \psi \geq 0 \text{ for any non-zero } \psi \in \mathbb{C}[S_1]. \tag{11.19}$$

Step 4. In this step we compare P' with a product pair on $S_1 \times \mathbb{A}^1$ and prove the lemma.

By $\rho \geq 0$ and (11.19), $\operatorname{mult}_{P'} \geq 0$ on $\mathbb{C}[S_1][\xi]$, so P' is a prime divisor over $S_1 \times \mathbb{A}^1$ whose center on $S_1 \times \mathbb{A}^1$ exists. Since $S_1 \times \mathbb{A}^1$ is a normal hypersurface in $(z_{j_0} = 1) \times \mathbb{A}^1$ whose residue form is $\pm d\xi \wedge \Omega_S$, we have $K_{S_1 \times \mathbb{A}^1} = \operatorname{div}(d\xi \wedge \Omega_S) = 0$. As

$$\operatorname{mult}_{P'} \Big(\frac{d\xi}{\xi} \wedge \Omega_S\Big) = \operatorname{mult}_{P'}(d\xi \wedge \Omega_S) - \rho \quad \text{and} \quad \rho = \operatorname{mult}_{P'}(S_1 \times \{0\}),$$

(11.18) becomes

$$e_{P'} a(P, V_0, B_{V_0}) = a\big(P', S_1 \times \mathbb{A}^1, B_{S_1} \times \mathbb{A}^1 + S_1 \times \{0\}\big) - \delta_{\mathbf{T}} \rho,$$

where $B_{S_1} := B_S|_{S_1}$.

By (11.17), $(S \times \mathbb{A}^1, B_S \times \mathbb{A}^1 + S \times \{0\})$ is lc on $(\pi|_S)^{-1}(Z_0^\circ) \times \mathbb{A}^1$.

Let η' be the generic point of the center of P' on $S_1 \times \mathbb{A}^1$, and let η_Z be the image of η' under the composition $S_1 \times \mathbb{A}^1 \to S_1 \to Z_0$ of the first projection and $\pi|_{S_1}$. For any $\psi \in \mathbb{C}[Z_0]$, $\pi^*\psi$ is $\mathbf{T}$-invariant, so $\phi^*\pi^*\psi = (\pi^*\psi)|_{S_1}$ on $S_1 \times \mathbb{G}_m$ and

$$\operatorname{mult}_{P'}((\pi^*\psi)|_{S_1}) = e_{P'} \operatorname{mult}_P \pi^*\psi.$$

Thus ψ vanishes at η_Z if and only if $\operatorname{mult}_P \pi^*\psi > 0$, if and only if ψ vanishes on $\pi(g(P))$. Hence η_Z is the generic point of $\overline{\pi(g(P))}$. Since $0 \in g(P)$, $o \in \overline{\pi(g(P))}$, so $\eta_Z \in Z_0^\circ$. Thus $\eta' \in (\pi|_S)^{-1}(Z_0^\circ) \times \mathbb{A}^1$, and

$$a\big(P', S_1 \times \mathbb{A}^1, B_{S_1} \times \mathbb{A}^1 + S_1 \times \{0\}\big) \geq 0.$$

Since $\rho \geq 0$ and $\delta_{\mathbf{T}} \leq 0$, we have $a(P, V_0, B_{V_0}) \geq 0$. This proves the lemma. □

11.4. **The reference dichotomy.**

Proposition 11.7 (cf. [Liu26b, Proposition 11.12]). *Let $\mathbf{T} = \mathbf{T}_{(w)}$ be a primitive circle with weights $w = (w_x, w_y, w_z, w_t)$ on the coordinates $(x, y, z, t) = (z_1, \dots, z_4)$ of $\mathbb{A}^4$, let e_0 be a positive integer, $c > 0$ a real number, and κ an integer. For any $i \in \mathbb{N}^+$, let $(V_{0,i}, G_i, h_i^0)$ be data as in Set-up 11.1 for $\mathbf{T}$, e_0, and c, where h_i^0 has $\mathbf{T}$-weight κ, let f_i^0 be the $\mathbf{T}$-homogeneous equation of $V_{0,i}$, let β_i be the number $\beta_{\mathbf{T}}$ of Set-up 11.1 for $(V_{0,i}, G_i, h_i^0)$, and let $(W_i \ni 0, G_i, h_i)$ be a marked germ over $(V_{0,i}, h_i^0)$ such that W_i is normal near 0 and $(W_i, c\,\mathrm{div}(h_i))$ is lc near 0. Assume that the following hold.*

(i) $r_i := |G_i \cap \mathbf{T}| \to +\infty$.
(ii) The polynomials f_i^0 have bounded degree.
(iii) $\beta_i > 0$ for any i.
(iv) Each f_i^0 belongs to one of the following classes.
  (a) $xy + z^{a'}t^{b'}$ with integers $a', b' \geq 0$ and $a' + b' \geq 2$.
  (b) $xy + g(z,t)$ with a polynomial $g \neq 0$, $w_x < 0$, and $w_y, w_z, w_t > 0$.
  (c) $x^2 + y^2 z + z^{a'}t^{b'}$ with integers $a', b' \geq 0$, $a' + b' \geq 3$, and $b' \geq 3$ if $a' = 0$, or $x^2 + y^2 z + yt^{b'}$ with an integer $b' \geq 2$.
  (d) $x^2 + y^3 + z^{a'}t^{b'}$ with integers $a', b' \geq 1$ and $a' + b' \in \{4, 5\}$, or $x^2 + y^3 + yz^{a'}t^{b'}$ with integers $a', b' \geq 1$ and $a' + b' = 3$, and $w_z < 0 < w_t$.

Then, after passing to a subsequence, there exist a finite subgroup $\overline{G} \subset GL_2(\mathbb{C})$ acting linearly on $\mathbb{A}^2_{\zeta_1,\zeta_2}$, positive real numbers $c_1, \dots, c_N$, and N-tuples $(f_{i,j})_j$ of non-zero $\overline{G}$-semi-invariant polynomials as in Definition 2.5 with a compatible generic limit $(f_{*,j})_j$ along $\mathbb{N}^+$, with coefficient field $\Bbbk$, such that

$$\beta_i = \mathrm{mld}\big((\mathbb{A}^2 \ni 0, C_i)/\overline{G}\big) \quad \text{for any } i, \quad \text{where } C_i := \sum_{j=1}^{N} c_j\,\mathrm{div}(f_{i,j}),$$

and such that, for $C_* := \sum_{j=1}^N c_j\,\mathrm{div}(f_{*,j})$ on $\mathcal{A}_\Bbbk := \mathrm{Spec}\,\Bbbk[[\zeta_1, \zeta_2]]$ with the induced action of $\overline{G}$, the following hold.

(1) If $(\mathcal{A}_\Bbbk, C_*)$ is klt, then there exists a real number $\theta > 1$ such that $(V_{0,i}, \theta c\,\mathrm{div}(h_i^0))$ and $(W_i, \theta c\,\mathrm{div}(h_i))$ are lc near 0 for any $i \gg 0$.
(2) If $(\mathcal{A}_\Bbbk, C_*)$ is not klt, then it has a unique smooth floor F, and for any i

$$\beta_i = \beta := \frac{1 - \mu_F}{n_F},$$

where $\mu_F := \mathrm{mult}_0((C_* - F)|_F)$ and n_F is the order of the image of $\overline{G}$ in $GL(T_0F)$. Moreover, there exists a $\mathbf{T}$-invariant polynomial U on $\mathbb{A}^4$ with $U(0) = 0$, independent of i and G_i-semi-invariant with a character of order n_U for any i, such that for $\lambda := n_U\beta$ and any $i \gg 0$, the pair $(W_i, c\,\mathrm{div}(h_i) + \lambda\,\mathrm{div}(U))$ is lc near 0, and

$$\frac{a(E, W_i, c\,\mathrm{div}(h_i))}{e_E} \geq \beta$$

for any prime divisor E over W_i with center 0, where $e_E := |I_{G_i}(E)|$.

Proof. Let $H_i := G_i/(G_i \cap \mathbf{T})$ and $Z_{0,i} := V_{0,i}/\!/\mathbf{T}$, let $\bar{o}_i \in Z_{0,i}/H_i$ be the image of 0, and let Y_i^+ and Y_i^- be the graded Proj quotients of $V_{0,i}$ of Set-up 11.1. For a real number $\theta > 0$, a sign $\pm$, and data (V, G, h) for which Set-up 11.1 defines the fiber reference of sign $\pm$, the *fiber reference at θc of sign $\pm$ of* (V, G, h) is the fiber reference of sign $\pm$ of (V, G, h) with c replaced by θc. Without mention of the data, it refers to $(V_{0,i}, G_i, h_i^0)$. The fiber reference at θc of sign $\pm$ of $(V_{0,i}, G_i, h_i^0)$ is a pair on $Y_i^\pm/H_i$, which is proper over $Z_{0,i}/H_i$, and birational since $\min_u w_u < 0 < \max_u w_u$.

Step 1: fixed data. Let $D_4 \subset GL_4(\mathbb{C})$ be the diagonal torus. Since $[G_i : G_i \cap \mathbf{T}] \leq e_0$, we have $g^{e_0!} \in \mathbf{T}$ for any $g \in G_i$, so the image of G_i in the torus $D_4/\mathbf{T}$ is contained in the finite subgroup

of elements of order dividing $e_0!$. Possibly passing to a subsequence, we may assume that the group $G_i\mathbf{T} \subset D_4$ does not depend on i, that one class among (a)–(d) contains all f_i^0, and that the exponents a', b' do not depend on i. For the last condition we use that $\deg f_i^0$ is bounded. In particular $H_i \cong G_i\mathbf{T}/\mathbf{T}$ does not depend on i.

We have $\mathrm{wt}_w(f_i^0) = w_x + w_y$ in the classes (a) and (b), and $\mathrm{wt}_w(f_i^0) = 2w_x$ in the classes (c) and (d). Hence, for a real number $\theta > 0$, the adjusted mass of Set-up 11.1 with c replaced by θc,

$$\delta_\theta := \theta c\kappa - \big(w_x + w_y + w_z + w_t - \mathrm{wt}_w(f_i^0)\big),$$

does not depend on i, and $\delta_1 = \delta_{\mathbf{T}}$. Let $s := +1$ if $\delta_1 < 0$, or if $\delta_1 = 0$ and $\kappa \le 0$, and let $s := -1$ otherwise. We also write s for the sign $+$, resp. $-$. Since $\delta_\theta = \delta_1 + (\theta - 1)c\kappa$, there exists a real number $\theta_0 > 1$ such that

$$s\delta_\theta \le 0 \quad \text{for any } \theta \in [1, \theta_0]. \tag{11.20}$$

Step 2: the data $(V_{0,i}, h_i^0)$ and marked germs. Write $W_i = (f_i = 0)$ as in Set-up 11.1. We show that, for any real number $\theta' > 0$ and any $i \gg 0$ depending on θ',

$$(V_{0,i}, \theta' c\,\mathrm{div}(h_i^0)) \text{ is lc near } 0 \iff (W_i, \theta' c\,\mathrm{div}(h_i)) \text{ is lc near } 0. \tag{11.21}$$

The hypersurfaces $V_{0,i}$ and W_i are normal and not contained in the zero loci of h_i^0, resp. h_i, where $h_i|_{W_i} \ne 0$ since $(W_i, c\,\mathrm{div}(h_i))$ is a pair. By [Kaw07, Theorem], the left, resp. right, side of (11.21) is equivalent to lc near 0 of

$$(\mathbb{A}^4, \mathrm{div}(f_i^0) + \theta' c\,\mathrm{div}(h_i^0)), \quad \text{resp.} \quad (\mathbb{A}^4, \mathrm{div}(f_i) + \theta' c\,\mathrm{div}(h_i)).$$

Hence, passing to the completion at 0, the left, resp. right, side of (11.21) is equivalent to lc of

$$(\mathrm{Spec}\,\mathbb{C}[[z_1, \dots, z_4]], (f_i^0)^1 (h_i^0)^{\theta' c}), \quad \text{resp.} \quad (\mathrm{Spec}\,\mathbb{C}[[z_1, \dots, z_4]], (f_i)^1 (h_i)^{\theta' c}).$$

By Set-up 11.1, $f_i - f_i^0$ and $h_i - h_i^0$ are sums of $\mathbf{T}$-homogeneous polynomials whose $\mathbf{T}$-weights are congruent modulo r_i to, and different from, $\mathrm{wt}_w(f_i^0)$, resp. κ. A monomial of $\mathbf{T}$-weight k' has degree $\ge |k'| / \max_{u \in \{x,y,z,t\}} |w_u|$. Hence, for any positive integer L,

$$f_i - f_i^0,\ h_i - h_i^0 \in (z_1, \dots, z_4)^L \quad \text{if } r_i \ge L \max_u |w_u| + \max\{|\,\mathrm{wt}_w(f_i^0)|, |\kappa|\}.$$

Let L be the integer of Lemma 2.19 for $n = 4$, two functions, and the DCC set $\{1, \theta' c\}$. Since $r_i \to +\infty$, Lemma 2.19, applied to (f_i^0, h_i^0) and (f_i, h_i) in both orders, implies (11.21).

Since $(W_i, c\,\mathrm{div}(h_i))$ is lc near 0, (11.21) with $\theta' = 1$ shows that, possibly removing finitely many indices, $(V_{0,i}, c\,\mathrm{div}(h_i^0))$ is lc near 0 for any i. Then $(V_{0,i}, \theta c\,\mathrm{div}(h_i^0))$ is lc near 0 for any $\theta \in (0, 1]$, so Proposition 11.2 holds for the data $(V_{0,i}, G_i, h_i^0)$ with c replaced by θc. By Proposition 11.2(5), $\beta^+ = \beta^-$ if $\delta_1 = 0$. Hence, for any i,

$$\beta_i = \beta_i^s, \tag{11.22}$$

where β_i^+ and β_i^- are the numbers β^+ and β^- of Set-up 11.1 for $(V_{0,i}, G_i, h_i^0)$.

Step 3: plane pairs. For a normal variety Z, two pairs, possibly with ideals, admitting proper birational morphisms to Z are *crepant over* Z if any prime divisor over Z has equal log discrepancies with respect to them. We show that, after passing to a subsequence, there exist

(i) a finite subgroup $\overline{G} \subset GL_2(\mathbb{C})$ and, for any i, a finite morphism $\mathbb{A}^2_{\zeta_1,\zeta_2} \to Z_{0,i}/H_i$ which is the quotient by $\overline{G}$ and maps 0 to $\overline{o}_i$,

(ii) for any i, a non-zero $\overline{G}$-semi-invariant polynomial ϕ_i on $\mathbb{A}^2$,

(iii) $\overline{G}$-semi-invariant polynomials $\omega_1, \dots, \omega_{n_1}$ on $\mathbb{A}^2$, independent of i, and affine functions $\varrho_1, \dots, \varrho_{n_1}$ of θ which are non-negative on $[1, \theta_0]$, and

(iv) a real number $\sigma_0 \ge 0$, a positive integer N_0, and, for any i, an ideal $\mathfrak{a}_i \subset \mathbb{C}[\zeta_1, \zeta_2]$ generated by $\overline{G}$-invariant polynomials $\gamma_{i,1}, \dots, \gamma_{i,n_2}$ of degree at most N_0, such that either $\mathfrak{a}_i = (1)$ for any i, or $(\zeta_1, \zeta_2)^{N_0} \subset \mathfrak{a}_i$ for any i,

such that for any $\theta \in [1,\theta_0]$ and any i, the fiber reference at θc of sign s is crepant over $Z_{0,i}/H_i$ to the quotient by $\overline{G}$ of

$$(\mathbb{A}^2, \Delta_i(\theta), \mathfrak{a}_i^{\sigma(\theta)}), \quad \text{where } \Delta_i(\theta) := \theta c \operatorname{div}(\phi_i) + \sum_{j=1}^{n_1} \varrho_j(\theta) \operatorname{div}(\omega_j) \text{ and } \sigma(\theta) := \sigma_0 |\delta_\theta|. \tag{11.23}$$

For a pair with an ideal generated by finitely many invariant polynomials $\gamma_1, \dots, \gamma_m$, the quotient is taken with the ideal generated by $\gamma_1, \dots, \gamma_m$, regarded as functions on the quotient. Its pullback is the given ideal, so log discrepancies over the quotient are those of Lemma 2.15(2). Since 0 is the only point of $\mathbb{A}^2$ over $\overline{o}_i$, (11.23), (11.22), and the behavior of log discrepancies under the finite quotient $\mathbb{A}^2 \to \mathbb{A}^2/\overline{G}$ imply, for any $\theta \in [1,\theta_0]$ and any i,

$$\begin{aligned} &\text{the fiber reference at } \theta c \text{ of sign } s \text{ is lc near the fiber over } \overline{o}_i \\ \iff\ &(\mathbb{A}^2, \Delta_i(\theta), \mathfrak{a}_i^{\sigma(\theta)}) \text{ is lc near } 0, \\ &\beta_i = \operatorname{mld}\big((\mathbb{A}^2 \ni 0, \Delta_i(1), \mathfrak{a}_i^{\sigma(1)})/\overline{G}\big). \end{aligned} \tag{11.24}$$

Step 3a: a coordinate of unique sign. In this step, assume that there exist a sign $s_v \in \{+1,-1\}$ and a coordinate $v \in \{x,y,z,t\}$ such that $s_v w_v > 0$ and $s_v w_u \le 0$ for any coordinate $u \ne v$. Let

$$\mathcal{U} := \{u \in \{x,y,z,t\} \mid s_v w_u < 0\},$$

which is non-empty since $\min_u w_u < 0 < \max_u w_u$. Let $S_i := V_{0,i} \cap (v=1)$, and let $G_S \subset G_i\mathbf{T}$ be the stabilizer of the hyperplane $(v=1)$, so that S_i and G_S are the set S_j and the group G_j of Set-up 11.1 for $z_j = v$. The group G_S is finite and, by **Step 1**, does not depend on i. Let

$$N_S := |w_v| \cdot |G_S| \cdot \operatorname{lcm}\{|w_u| \mid u \in \mathcal{U}\},$$

and let $\mathfrak{a}_{S,i} \subset \mathbb{C}[S_i]$ be the ideal generated by $u|_{S_i}^{N_S/|w_u|}$ for $u \in \mathcal{U}$. Since G_S is diagonal and $|G_S|$ divides $N_S/|w_u|$, the generators $u|_{S_i}^{N_S/|w_u|}$ are G_S-invariant. We show that for any $\theta \in [1,\theta_0]$ and any i, the fiber reference at θc of sign s is crepant over $Z_{0,i}/H_i$ to the quotient by G_S of

$$\big(S_i, \theta c \operatorname{div}(h_i^0|_{S_i}), \mathfrak{a}_{S,i}^{\sigma_S(\theta)}\big), \quad \text{where } \sigma_S(\theta) := 0 \text{ if } s = s_v, \text{ and } \sigma_S(\theta) := \frac{|\delta_\theta|}{N_S} \text{ if } s = -s_v. \tag{11.25}$$

By Proposition 11.2(1), $Y_i^{s_v}$ is covered by the root section at v, so $Y_i^{s_v} = S_i/\mathbf{T}_v$ is affine, where $\mathbf{T}_v \subset \mathbf{T}$ is the stabilizer of $(v=1)$, the group $\mathbf{T}_j$ of Set-up 11.1 for $z_j = v$. The morphism $Y_i^{s_v} \to Z_{0,i}$ is projective and birational, hence finite and birational, so it is an isomorphism since $Z_{0,i}$ is normal. Thus $Z_{0,i}/H_i = S_i/G_S$, since $G_S \cap \mathbf{T} = \mathbf{T}_v$ and $G_S\mathbf{T} = G_i\mathbf{T}$, so that $G_S/\mathbf{T}_v \cong H_i$, and by Proposition 11.2(2) the fiber reference at θc of sign s_v is the quotient by G_S of $(S_i, \theta c \operatorname{div}(h_i^0|_{S_i}))$. This proves (11.25) if $s = s_v$.

Assume that $s = -s_v$. For $u \in \mathcal{U}$, the polynomial $v^{|w_u|}u^{|w_v|}$ has $\mathbf{T}$-weight $|w_u|w_v + |w_v|w_u = 0$, so it is a regular function on $Z_{0,i}$ with $(v^{|w_u|}u^{|w_v|})|_{S_i} = u|_{S_i}^{|w_v|}$. Hence $\mathfrak{a}_{S,i}$ is generated by the functions $(v^{|w_u|}u^{|w_v|})^{N_S/(|w_u||w_v|)}$, $u \in \mathcal{U}$, on $Z_{0,i}/H_i$. Let P be a prime divisor over $Z_{0,i}/H_i$. Let ξ be a rational function on $V_{0,i}$ of $\mathbf{T}$-weight 1, and write $z_j = \xi^{w_j}\psi_j$ with rational functions ψ_j on $Z_{0,i}$ (Proposition 11.2(4)). We write ψ_u for the function ψ_j with $z_j = u$. Let ν be an extension of mult_P to a $\mathbb{Q}$-valued valuation on the rational functions on $Z_{0,i}$, and let

$$\vartheta_- := \max_{w_j>0}\Big(-\frac{\nu(\psi_j)}{w_j}\Big) \quad \text{and} \quad \vartheta_+ := \min_{w_j<0}\frac{\nu(\psi_j)}{|w_j|}.$$

Since v is the only coordinate whose weight has sign s_v, the definition of $\vartheta_\pm$ implies

$$\vartheta_+ - \vartheta_- = \min_{u\in\mathcal{U}}\Big(\frac{\nu(\psi_u)}{|w_u|} + \frac{\nu(\psi_v)}{|w_v|}\Big) = \min_{u\in\mathcal{U}}\frac{\nu(v^{|w_u|}u^{|w_v|})}{|w_u||w_v|} = \frac{\nu(\mathfrak{a}_{S,i})}{N_S}.$$

For $\theta > 0$, let $a^{\pm}(P,\theta)$ be the log discrepancy of P with respect to the fiber reference at θc of sign $\pm$. By Proposition 11.2(4)(a), for any $\theta > 0$,

$$a^{+}(P,\theta) - a^{-}(P,\theta) = \delta_\theta(\vartheta_+ - \vartheta_-) = \frac{\delta_\theta}{N_S}\nu(\mathfrak{a}_{S,i}).$$

Since $s = -s_v$, (11.20) implies

$$a^{s}(P,\theta) = a^{s_v}(P,\theta) - \frac{|\delta_\theta|}{N_S}\nu(\mathfrak{a}_{S,i}) \quad \text{for any } \theta \in [1,\theta_0].$$

As $a^{s_v}(P,\theta)$ is the log discrepancy of P with respect to the quotient by G_S of $(S_i, \theta c\,\mathrm{div}(h_i^0|_{S_i}))$, this proves (11.25).

Step 3b: the classes (b) and (d), and the equations $x^2+y^2z+t^{b'}$. In this step, assume that f_i^0 belongs to the class (b) or (d), or that $f_i^0 = x^2+y^2z+t^{b'}$ for an integer $b' \geq 2$. The last case contains the class (c) with $a' = 0$. We choose v and s_v as follows, so that the assumption of **Step 3**a holds.

- Class (b): $v := x$ and $s_v := -1$. Then $\mathcal{U} = \{y,z,t\}$, and $S_i = (y = -g_i(z,t)) \cong \mathbb{A}^2_{z,t}$, where g_i is the polynomial g of f_i^0.
- $f_i^0 = x^2+y^2z+t^{b'}$: then $2w_x = 2w_y + w_z = b'w_t$. If $w_t = 0$, then $w_x = 0$ and $w_z = -2w_y \neq 0$, and we let $v := y$ and $s_v :=$ the sign of w_y, so that $\mathcal{U} = \{z\}$ and $S_i = (x^2+z+t^{b'} = 0) \cong \mathbb{A}^2_{x,t}$. If $w_t \neq 0$, let s' be the sign of w_t, so that $s'w_x > 0$. If $s'w_y < 0$, then $s'w_z = 2s'(w_x - w_y) > 0$, and we let $v := y$ and $s_v := -s'$, so that $\mathcal{U} = \{x,z,t\}$ and $S_i \cong \mathbb{A}^2_{x,t}$. If $s'w_y \geq 0$, then $s'w_z < 0$ since some weight has sign $-s'$, $s'w_y = s'(w_x - w_z/2) > 0$, and we let $v := z$ and $s_v := -s'$, so that $\mathcal{U} = \{x,y,t\}$ and $S_i = (x^2+y^2+t^{b'} = 0)$.
- Class (d): $2w_x = 3w_y$. If $w_x \geq 0$, let $v := z$ and $s_v := -1$, so that $\mathcal{U} = \{t\}$ if $w_x = 0$, $\mathcal{U} = \{x,y,t\}$ if $w_x > 0$, and S_i is $(x^2+y^3+t^{b'} = 0)$, resp. $(x^2+y^3+yt^{b'} = 0)$. If $w_x < 0$, let $v := t$ and $s_v := +1$, so that $\mathcal{U} = \{x,y,z\}$ and S_i is $(x^2+y^3+z^{a'} = 0)$, resp. $(x^2+y^3+yz^{a'} = 0)$.

If $S_i \cong \mathbb{A}^2$ by the coordinates named above, we let (ζ_1,ζ_2) be the coordinates named above and $\overline{G} \subset GL_2(\mathbb{C})$ the image of G_S, which is diagonal, and we let $\varpi_S := \mathrm{id}_{\mathbb{A}^2}$.

In the remaining cases $S := S_i$ does not depend on i. If $v = z$, resp. $v = t$, we write x,y,t, resp. x,y,z, also for their restrictions to S. Since $a', b' \geq 1$, we have $b' \leq 4$, resp. $a' \leq 4$, in the first form and $b' \leq 2$, resp. $a' \leq 2$, in the second form of (d). If $S = (x^2+y^3+t = 0)$, resp. $S = (x^2+y^3+z = 0)$, then $S \cong \mathbb{A}^2_{x,y}$, we let $(\zeta_1,\zeta_2) := (x,y)$, $\varpi_S := \mathrm{id}_{\mathbb{A}^2}$, and $\Gamma := \{1\}$, and we let $\overline{G} \subset GL_2(\mathbb{C})$ be the image of G_S, which is diagonal. Otherwise S is $(x^2+y^2+t^{b'} = 0)$ of type $A_{b'-1}$, or $(x^2+y^3+t^{b'} = 0)$, resp. $(x^2+y^3+z^{a'} = 0)$, of type A_2, D_4, E_6 for $b' = 2,3,4$, resp. $a' = 2,3,4$, or $(x^2+y^3+yt^{b'} = 0)$, resp. $(x^2+y^3+yz^{a'} = 0)$, of type A_1, D_4 for $b' = 1,2$, resp. $a' = 1,2$. In the case $x^2+y^3+yt = 0$, resp. $x^2+y^3+yz = 0$, we replace t by $t+y^2$, resp. z by $z+y^2$, which is G_S-semi-invariant since x^2+y^3+yt, resp. x^2+y^3+yz, is G_S-semi-invariant, so that $\chi_y^3 = \chi_y\chi_t$, resp. $\chi_y^3 = \chi_y\chi_z$. Then S is a Du Val surface defined by a weighted homogeneous equation in coordinates on which G_S acts diagonally. In each case, a linear change of the coordinates of equal weight carries the equation of S to one of the quasi-homogeneous equations of [DBW18, Sections 5.1–5.2], so $S \cong \mathbb{A}^2/\Gamma$ for a finite subgroup $\Gamma \subset SL_2(\mathbb{C})$, the quotient map $\varpi_S : \mathbb{A}^2_{\zeta_1,\zeta_2} \to S$ is quasi-étale, and the coordinates of S pull back to homogeneous polynomials [DBW18, Section 5.1]. In particular, the scalar action of $\mathbb{G}_m$ on $\mathbb{A}^2$ descends to a $\mathbb{G}_m$-action on S which is diagonal in the coordinates of S.

Since G_S acts diagonally, it fixes $0 \in S$. Since $\Gamma \subset SL_2(\mathbb{C})$ is finite, it acts freely on $\mathbb{A}^2\backslash\{0\}$, so ϖ_S restricts to a Galois covering $\mathbb{A}^2\backslash\{0\} \to S\backslash\{0\}$ with group Γ, which is a universal covering as $\mathbb{A}^2\backslash\{0\}$ is simply connected. Hence for any $g \in G_S$ there exists a biholomorphism $\widetilde{g}$ of $\mathbb{A}^2\backslash\{0\}$ with $\varpi_S \circ \widetilde{g} = g \circ \varpi_S$, which extends over 0 by Hartogs' theorem, and the lifts $\widetilde{g}$ of all $g \in G_S$ form

a finite group $\overline{G}$ with $\Gamma \subset \overline{G}$ and $\overline{G}/\Gamma \cong G_S$. Any $\widetilde{g} \in \overline{G}$ is linear: $\widetilde{g}$ lifts a diagonal automorphism of S, which commutes with the $\mathbb{G}_m$-action, so for any $\varsigma \in \mathbb{G}_m$ the automorphism $\widetilde{g} \circ \varsigma \circ \widetilde{g}^{-1} \circ \varsigma^{-1}$ lifts the identity and belongs to Γ. As $\mathbb{G}_m$ is connected and Γ is finite, $\widetilde{g} \circ \varsigma \circ \widetilde{g}^{-1} \circ \varsigma^{-1}$ is the identity for any $\varsigma \in \mathbb{G}_m$, so $\widetilde{g}$ commutes with the scalar multiplications, and the Taylor series of $\widetilde{g}$ at 0 has only terms of degree 1. Thus $\overline{G} \subset GL_2(\mathbb{C})$, and $\overline{G}$ does not depend on i.

In all cases $\mathbb{A}^2 \to S_i/G_S = Z_{0,i}/H_i$ is the quotient by $\overline{G}$. Let $\phi_i := \varpi_S^*(h_i^0|_{S_i})$. Since ϖ_S is quasi-étale, the quotient by G_S of the pair (11.25) is the quotient by $\overline{G}$ of

$$\big(\mathbb{A}^2, \theta c \operatorname{div}(\phi_i), (\varpi_S^{-1}\mathfrak{a}_{S,i} \cdot \mathcal{O}_{\mathbb{A}^2})^{\sigma_S(\theta)}\big).$$

Since h_i^0 is $G_i\mathbf{T}$-semi-invariant and $\mathbf{T}$-homogeneous with $h_i^0|_{V_{0,i}} \neq 0$, and $\mathbf{T} \cdot S_i = V_{0,i} \cap (v \neq 0)$ is dense in $V_{0,i}$, ϕ_i is a non-zero $\overline{G}$-semi-invariant polynomial. If $|\mathcal{U}| \geq 2$, then in each case above the common zero locus of the generators of $\mathfrak{a}_{S,i}$ is the origin of S_i, since $\mathcal{U}$ contains the coordinates of S_i named above. We let $n_1 := 0$, let $\gamma_{i,1}, \dots, \gamma_{i,n_2}$ be the pullbacks $\varpi_S^*(u|_{S_i}^{N_S/|w_u|})$, $u \in \mathcal{U}$, which are $\overline{G}$-invariant, and let $\sigma_0 := 0$ if $s = s_v$ and $\sigma_0 := 1/N_S$ if $s = -s_v$. In class (d) and in the case $f_i^0 = x^2 + y^2 z + t^{b'}$, S_i and hence $\mathfrak{a}_i$ do not depend on i. In the class (b), $\mathfrak{a}_i$ contains z^{N_S/w_z} and t^{N_S/w_t}, and the generators $\gamma_{i,j}$ have degree at most $\max\{N_S/w_z, N_S/w_t, (N_S/w_y)\deg g_i\}$, which is bounded. Thus (iv) holds for some N_0. If $\mathcal{U} = \{u\}$, then $\mathfrak{a}_{S,i} = (u|_{S_i}^{N_S/|w_u|})$, and

$$\big(S_i, \theta c \operatorname{div}(h_i^0|_{S_i}), \mathfrak{a}_{S,i}^{\sigma_S(\theta)}\big) \quad \text{and} \quad \Big(S_i, \theta c \operatorname{div}(h_i^0|_{S_i}) + \frac{N_S \sigma_S(\theta)}{|w_u|} \operatorname{div}(u|_{S_i})\Big)$$

have equal log discrepancies for any prime divisor over S_i. Here S_i does not depend on i, since $|\mathcal{U}| = 1$ occurs only for $w_t = 0$ and in class (d) with $w_x = 0$, so ω_1 below does not depend on i. We let $\mathfrak{a}_i := (1)$, $\sigma_0 := 0$, $n_1 := 1$, $\omega_1 := \varpi_S^*(u|_{S_i})$, and $\varrho_1(\theta) := 0$ if $s = s_v$ and $\varrho_1(\theta) := -s\delta_\theta/|w_u|$ if $s = -s_v$. By (11.20), $\varrho_1(\theta) = N_S\sigma_S(\theta)/|w_u| \geq 0$ for $\theta \in [1, \theta_0]$. By (11.25), the data $\overline{G}$, ϕ_i, ω_j, ϱ_j, σ_0, N_0, and $\mathfrak{a}_i$ satisfy (11.23).

Step 3c: binomial references. In this step we assume the following.

(1) $V' = (x'y' + z'^{a''}t'^{b''} = 0) \subset \mathbb{A}^4$ with integers $a'', b'' \geq 0$ and $a'' + b'' \geq 2$.
(2) $\mathbf{T}' = \mathbf{T}_{(w')}$ is a primitive circle such that V' is $\mathbf{T}'$-invariant and both w' and $-w'$ have a negative entry, so that $\mathbb{R}w'$ intersects the cone
$$\{\mathbf{n} \in \mathbb{R}^4_{\geq 0} \mid n_{x'} + n_{y'} = a''n_{z'} + b''n_{t'}\}$$
of the affine toric variety V' only at 0.
(3) For any i, G_i' is a finite abelian group acting faithfully on V', monomially on $\{x', y'\}$ and diagonally on z', t', and commuting with $\mathbf{T}'$, such that $G_i'\mathbf{T}'$ does not depend on i.
(4) For any i, h_i' is a $\mathbf{T}'$-homogeneous G_i'-semi-invariant polynomial of $\mathbf{T}'$-weight κ' with $h_i'|_{V'} \neq 0$.
(5) $\delta_\theta' := \theta c\kappa' - w_{z'}' - w_{t'}'$ satisfies $s\delta_\theta' \leq 0$ for $\theta \in [1, \theta_0]$.
(6) $H_i' := G_i'/(G_i' \cap \mathbf{T}')$ and $Z_0' := V'//\mathbf{T}'$.
(7) Y'^+ and Y'^- are the graded Proj quotients of V', and the fiber references at θc of (V', G_i', h_i') are defined as in Set-up 11.1, with the group G_i' in place of G.
(8) The fiber reference at c of sign s of (V', G_i', h_i') is lc near the fiber over the image of 0.

We show that the fiber reference at θc of sign s of (V', G_i', h_i') is crepant over Z_0'/H_i' to the quotient by a finite group $\overline{G} \subset GL_2(\mathbb{C})$ of a pair (11.23) with $|\delta_\theta'|$ in place of $|\delta_\theta|$ and with data as in (i)–(iv), where $Z_{0,i}/H_i$ and $\overline{o}_i$ are replaced by Z_0'/H_i' and the image of 0 in Z_0'/H_i'. The class (a) is the case $V' = V_{0,i}$, $\mathbf{T}' = \mathbf{T}$, $G_i' = G_i$, $h_i' = h_i^0$, and $\kappa' = \kappa$, for which $\delta_\theta' = \delta_\theta$ and the fiber reference at c of sign s is lc near the fiber over $\overline{o}_i$ by Proposition 11.2(3).

Let I and N_I be the H_i'-invariant monomial ideal on Z_0' and the positive integer of [Liu26b, Lemma 8.12(3)] for negative adjusted mass if $s = +1$, resp. positive adjusted mass if $s = -1$. The ideal I and the integer N_I depend only on a'', b'', w', and s. By [Liu26b, Lemma 8.12(3)], the fiber reference at θc of sign s of (V', G_i', h_i') is crepant over Z_0'/H_i' to the quotient by H_i' of

$(Z'_0, B'_{0,i}(\theta), I^{|\delta'_\theta|/N_I})$, where $B'_{0,i}(\theta)$ is the push-forward to Z'_0 of the boundary of the pair on Y'^s. For $\delta'_\theta = 0$ the ideal does not occur. If the cone of the affine toric surface Z'_0 contained a line, then there would be non-zero vectors $\mathbf{n}, \mathbf{n}'$ of the cone of V' with $\mathbf{n} + \mathbf{n}' \in \mathbb{R}w'$, contradicting the choice of w'. Hence the cone of Z'_0 is strictly convex.

Let $\mathbf{v}_1, \mathbf{v}_2$ be the primitive generators of the rays of the cone of Z'_0 in the cocharacter lattice N_Z of the torus of Z'_0. The toric morphism $\pi : \mathbb{A}^2_{\zeta_1,\zeta_2} \to Z'_0$ given by $\mathbb{Z}\mathbf{v}_1 + \mathbb{Z}\mathbf{v}_2 \subset N_Z$ is the quotient by the finite diagonal group $\Xi' \cong N_Z/(\mathbb{Z}\mathbf{v}_1 + \mathbb{Z}\mathbf{v}_2)$, and it is étale in codimension 1 since $\mathbf{v}_1$ and $\mathbf{v}_2$ are primitive. Since G'_i acts monomially on $\{x', y'\}$ and diagonally on z', t' and commutes with $\mathbf{T}'$, the group H'_i acts on Z'_0 by toric automorphisms, that is, by compositions of translations by points of the torus with lattice automorphisms permuting $\mathbf{v}_1$ and $\mathbf{v}_2$. Any such automorphism lifts to a linear automorphism of $\mathbb{A}^2$, diagonal or interchanging the axes $(\zeta_1 = 0)$ and $(\zeta_2 = 0)$, and the lifts of the elements of H'_i form a finite group $\overline{G} \subset GL_2(\mathbb{C})$ with $\Xi' \subset \overline{G}$. The group $\overline{G}$ does not depend on i since $G'_i\mathbf{T}'$ does not. Thus $\mathbb{A}^2 \to Z'_0/H'_i$ is the quotient by $\overline{G}$.

Since w' is primitive, $\mathbf{T}'$ acts freely on the torus of V', so over the torus of Z'_0 the boundary of Y'^s is θc times the divisor descended from $\operatorname{div}(h'_i)$. Hence

$$\pi^* B'_{0,i}(\theta) = \theta c \operatorname{div}(\phi_i) + \varrho_{i,1}(\theta) \operatorname{div}(\zeta_1) + \varrho_{i,2}(\theta) \operatorname{div}(\zeta_2),$$

where ϕ_i is a polynomial divisible by neither ζ_1 nor ζ_2. The polynomial ϕ_i is $\overline{G}$-semi-invariant because $\operatorname{div}(\phi_i)$ is $\overline{G}$-invariant. Since π is étale in codimension 1, for $j \in \{1, 2\}$, $\varrho_{i,j}(\theta)$ is the coefficient of the prime divisor P_j of Y'^s mapping onto $\pi(\zeta_j = 0)$. Computing it on a root section of Y'^s containing the generic point of P_j, the quotient of a slice $V' \cap (u = 1)$ by its stabilizer in $\mathbf{T}'$ for a coordinate u with $sw'_u > 0$, we obtain

$$\varrho_{i,j}(\theta) = 1 - \frac{1}{q_j} + \theta c \frac{m_{i,j}}{q_j},$$

where q_j, which does not depend on i, is the ramification index along P_j of the quotient map from the slice to the root section, and $m_{i,j} \in \mathbb{N}$ is the order of vanishing of the restriction of h'_i to the slice along the preimage of P_j. The fiber reference at c of sign s of (V', G'_i, h'_i) is lc near the fiber over the image of 0, so $\varrho_{i,j}(1) \le 1$ and $m_{i,j} \le 1/c$. Possibly passing to a subsequence, $m_{i,j}$ does not depend on i, and we write $\varrho_j := \varrho_{i,j}$. If $\overline{G}$ is diagonal, let $n_1 := 2$ and $\omega_j := \zeta_j$ for $j \in \{1, 2\}$. Otherwise $\overline{G}$ interchanges the axes, $\varrho_1 = \varrho_2$, and we let $n_1 := 1$ and $\omega_1 := \zeta_1\zeta_2$.

The ideal $\mathfrak{b} := \pi^{-1} I \cdot \mathcal{O}_{\mathbb{A}^2}$ is a $\overline{G}$-invariant monomial ideal, equal to (1) or primary at 0, and independent of i. If $\mathfrak{b} = (1)$, let $\mathfrak{a}_i := (1)$ and $\sigma_0 := 0$. Otherwise let $\mathfrak{a}_i$ be the ideal generated by the polynomials $\prod_{g \in \overline{G}} g^*\gamma$ for $\gamma \in \mathfrak{b}$. The ideal $\mathfrak{a}_i$ is generated by finitely many $\overline{G}$-invariant polynomials and does not depend on i. For a divisorial valuation ν over $\mathbb{A}^2$ and $\gamma \in \mathfrak{b}$, we have

$$\nu(\prod_g g^*\gamma) = \sum_g \nu(g^*\gamma) \ge \sum_g \nu(g^*\mathfrak{b}) = |\overline{G}|\nu(\mathfrak{b}),$$

with equality for a general $\gamma \in \mathfrak{b}$, as the valuations $\nu \circ g^*$ are finitely many. Hence $\nu(\mathfrak{a}_i) = |\overline{G}|\nu(\mathfrak{b})$. For $p \in \mathbb{A}^2 \backslash \{0\}$, a general $\gamma \in \mathfrak{b}$ does not vanish at the finitely many points $g(p)$, so $\mathfrak{a}_i$ is primary at 0 and (iv) holds for some N_0. Let $\sigma_0 := 1/(N_I|\overline{G}|)$. Since π is étale in codimension 1, the quotient by H'_i of $(Z'_0, B'_{0,i}(\theta), I^{|\delta'_\theta|/N_I})$ is the quotient by $\overline{G}$ of

$$(\mathbb{A}^2, \Delta_i(\theta), \mathfrak{a}_i^{\sigma_0|\delta'_\theta|})$$

with $\Delta_i(\theta)$ as in (11.23). This proves the claim of this step, and (11.23) in the class (a).

Step 3d: the class (c) by double covers. It remains to treat $f_i^0 = x^2 + y^2 z + z^{a'} t^{b'}$ with $a' \ge 1$, and $f_i^0 = x^2 + y^2 z + y t^{b'}$. In both cases below, G'_i denotes the group of automorphisms of V' lifting elements of G_i, $h'_i := \varpi'^* h_i^0$, κ' is the $\mathbf{T}'$-weight of h'_i, and δ'_θ is the adjusted mass of (V', h'_i) at θc.

The case $f_i^0 = x^2 + y^2z + z^{a'}t^{b'}$. Let $V' \subset \mathbb{A}^4_{x',y',z',t'}$, a morphism $\varpi' : V' \to \mathbb{A}^4$, and an involution ι of V' be

$$V' = (x'y' + z'^{2a'-2}t'^{b'} = 0), \quad \varpi' = \Big(\frac{z'(x'+y')}{2}, \frac{x'-y'}{2\sqrt{-1}}, z'^2, t'\Big), \quad \text{and} \quad \iota = (-y', -x', -z', t').$$

Then $f_i^0 \circ \varpi'$ is z'^2 times the equation of V', and the fixed locus of ι on V' is contained in $(z' = x' + y' = 0)$. Since $2w_x = 2w_y + w_z$, w_z is even. Let $\mathbf{T}'$ be the primitive circle with weights $(w_y, w_y, w_z/2, w_t)$. Then ϖ' is equivariant for an isomorphism $\mathbf{T}' \to \mathbf{T}$, and the pullback of a $\mathbf{T}$-homogeneous function of weight k' has $\mathbf{T}'$-weight k'. For $g \in G_i$ with characters χ_u, choose $\chi_{z'}$ with $\chi_{z'}^2 = \chi_z$. As $\chi_x^2 = \chi_y^2\chi_z$, we have $\chi_x/\chi_{z'} = \pm\chi_y$, and g lifts to

$$(x', y', z', t') \to (\chi_y x', \chi_y y', \chi_{z'} z', \chi_t t'), \quad \text{resp.} \quad (x', y', z', t') \to (-\chi_y y', -\chi_y x', \chi_{z'} z', \chi_t t'),$$

for the sign $+$, resp. $-$. The group G_i' acts monomially on $\{x', y'\}$ and diagonally on z', t'. We have $\kappa' = \kappa$, and a computation of the weights shows that $\delta'_\theta = \delta_\theta$.

The case $f_i^0 = x^2 + y^2z + yt^{b'}$. Let

$$V' = (x'^2 + y'^2z' + t'^{b'} = 0), \quad \varpi' = (y'x', y'^2, z', t'), \quad \text{and} \quad \iota = (-x', -y', z', t').$$

Then $f_i^0 \circ \varpi'$ is y'^2 times the equation of V', and the fixed locus of ι on V' is contained in

$$(x' = y' = t' = 0).$$

Let $\mathbf{T}'$ be the primitive circle with weights proportional to $(2w_x - w_y, w_y, 2w_z, 2w_t)$. Since w is primitive, $\gcd(2w_x - w_y, w_y, 2w_z, 2w_t) = \gcd(2w_x, w_y, 2w_z, 2w_t)$ is 1 if w_y is odd and 2 if w_y is even. Hence ϖ' is equivariant for a homomorphism $\mathbf{T}' \to \mathbf{T}$ of degree 2, resp. 1, if w_y is odd, resp. even, and the pullback of a $\mathbf{T}$-homogeneous function of weight k' has $\mathbf{T}'$-weight $2k'$, resp. k'. For $g \in G_i$ with characters χ_u, choose $\chi_{y'}$ with $\chi_{y'}^2 = \chi_y$. Then g lifts to $(\chi_x\chi_{y'}^{-1}x', \chi_{y'}y', \chi_z z', \chi_t t')$. The group G_i' acts diagonally. We have $\kappa' = 2\kappa$, resp. $\kappa' = \kappa$, and a computation of the weights shows that $\delta'_\theta = 2\delta_\theta$, resp. $\delta'_\theta = \delta_\theta$.

In both cases, a direct computation shows that $\mathbb{C}[V']^\iota$ is generated by the pullbacks of x, y, z, t, so $\varpi' : V' \to V_{0,i}$ is the quotient by ι, and ϖ' is quasi-étale by the description of the fixed locus of ι. Since $\dim \operatorname{Sing} V' \le 1$, V' is normal, and V' is klt since $V_{0,i}$ is klt and ϖ' is quasi-étale. The group G_i' is a finite abelian group containing ι with $G_i'/\langle\iota\rangle \cong G_i$, and $G_i'\mathbf{T}'$ does not depend on i.

Define H_i', Z_0', $Y'^{\pm}$, and the fiber references of (V', G_i', h_i') as in **Step 3**c. Since ϖ' is the quotient by ι and the $\mathbf{T}$-weights are multiplied by the degree of $\mathbf{T}' \to \mathbf{T}$, we have $Z_0'/\langle\iota\rangle = Z_{0,i}$ and $Y'^{\pm}/\langle\iota\rangle = Y_i^{\pm}$, so $Z_0'/H_i' = Z_{0,i}/H_i$ and $Y'^{\pm}/H_i' = Y_i^{\pm}/H_i$. For a coordinate u with $\pm w_u > 0$, let $Q_u := V_{0,i} \cap (u = 1)$, the set S_j of Set-up 11.1 for $z_j = u$, and $Q_u' := \varpi'^{-1}(Q_u)$. The fixed locus V'^ι of ι is $\mathbf{T}'$-invariant with $\dim V'^\iota \le 1$, so $\dim(V'^\iota \cap Q_u') \le 0$, and $(Q_u', \theta c \operatorname{div}(h_i'))$ is the crepant pullback of $(Q_u, \theta c \operatorname{div}(h_i^0))$ by a finite quasi-étale morphism. The quotients of the slices Q_u' by their stabilizers in $\mathbf{T}'$ cover $Y'^{\pm}$, and by Lemma 2.15(4) the crepant quotient boundary of a slice pair has, along a prime divisor P of $Y'^{\pm}$, the coefficient $1 - (1 - \theta c m_P)/q_P$, where m_P is the order of h_i' along the $\mathbf{T}'$-invariant prime divisor $\widetilde{P}$ of V' over P and q_P is the order of the generic stabilizer of $\widetilde{P}$ in $\mathbf{T}'$. The numbers m_P and q_P do not depend on the $\mathbf{T}'$-homogeneous function cutting the slice, so on such a quotient the boundary of the fiber reference is the crepant quotient boundary of the slice pair. Therefore, for any $\theta > 0$, the fiber reference at θc of sign $\pm$ of $(V_{0,i}, G_i, h_i^0)$ is crepant over $Z_{0,i}/H_i$ to the fiber reference at θc of sign $\pm$ of (V', G_i', h_i'). As $(\delta'_\theta, \kappa')$ is (δ_θ, κ) times the degree of $\mathbf{T}' \to \mathbf{T}$, the sign defined in **Step 1** from (δ'_1, κ') is s, and $s\delta'_\theta \le 0$ for any $\theta \in [1, \theta_0]$ by (11.20). Since the fiber references at c of sign s of $(V_{0,i}, G_i, h_i^0)$ and of (V', G_i', h_i') are crepant over $Z_{0,i}/H_i$, Proposition 11.2(3) for $(V_{0,i}, G_i, h_i^0)$ shows that the fiber reference at c of sign s of (V', G_i', h_i') is lc near the fiber over $\overline{o}_i$.

In the first case, $(V', \mathbf{T}', G_i', h_i')$ satisfies the assumptions of **Step 3**c with $a'' = 2a' - 2$ and $b'' = b'$: $a'' + b'' \ge 2$ since $a' \ge 1$ and $a' + b' \ge 3$, and w' and $-w'$ have a negative entry since w and $-w$ do. In the second case, V' is defined by $x'^2 + y'^2z' + t'^{b'}$ with $b' \ge 2$, the data (V', G_i', h_i')

satisfy Set-up 11.1 with e_0 replaced by $2e_0$, and $(V', c\,\mathrm{div}(h_i'))$ is lc near 0 since ϖ' is quasi-étale. Hence **Steps 3**a and 3b hold for (V', G_i', h_i'). In both cases we obtain (11.23) for (V', G_i', h_i') with

$$\sigma(\theta) = \sigma_0|\delta_\theta'|,$$

hence (11.23) for $(V_{0,i}, G_i, h_i^0)$, with σ_0 times the degree of $\mathbf{T}' \to \mathbf{T}$ in place of σ_0.

Step 3e: bounded marks. By **Steps 3**a–3d, (11.23) holds in all classes. By Proposition 11.2(3) and (11.24), $(\mathbb{A}^2, \Delta_i(1), \mathfrak{a}_i^{\sigma(1)})$ is lc near 0, so $(\mathbb{A}^2, c\,\mathrm{div}(\phi_i))$ is lc near 0, whence $c\,\mathrm{mult}_0\,\phi_i \le 2$. Since $\overline{G}$ has finitely many characters, possibly passing to a subsequence, the character of ϕ_i does not depend on i.

Step 4: general members and the compatible generic limit. If $\overline{G}$ is abelian and not diagonal in (ζ_1, ζ_2), we change the linear coordinates (ζ_1, ζ_2) so that $\overline{G}$ is diagonal. For any i, let $\Delta_{0,i}$ be the reduced divisor on $\mathbb{A}^2$ whose support is the union of the zero loci of $\phi_i, \omega_1, \dots, \omega_{n_1}$ and, if $\overline{G}$ is abelian, of $\zeta_1\zeta_2$. Then $\Delta_{0,i}$ is $\overline{G}$-invariant. Since δ_θ is an affine function of θ which does not change sign on $[1, \theta_0]$ by (11.20), σ is affine on $[1, \theta_0]$. Let n_3 be an integer with $n_3 > \max\{\sigma(1), \sigma(\theta_0)\}$. If $\mathfrak{a}_i \ne (1)$, then, as $\mathfrak{a}_i \supset (\zeta_1, \zeta_2)^{N_0}$ is generated by the $\overline{G}$-invariant polynomials $\gamma_{i,1}, \dots, \gamma_{i,n_2}$, by [Liu26b, Lemma E.8] there exist non-zero $\overline{G}$-invariant polynomials $\eta_{i,1}, \dots, \eta_{i,n_3} \in \mathfrak{a}_i$ with $\mathrm{mult}_0\,\eta_{i,j} \le N_0$ such that for any $\overline{G}$-invariant $\mathbb{R}$-divisor $\Delta \ge 0$ with $\mathrm{Supp}\,\Delta \subset \Delta_{0,i}$ and any $\sigma' \in [0, n_3)$,

$$\begin{aligned}(\mathbb{A}^2, \Delta, \mathfrak{a}_i^{\sigma'}) \text{ is lc near } 0 \iff \Big(\mathbb{A}^2, \Delta + \frac{\sigma'}{n_3}\sum_{j=1}^{n_3}\mathrm{div}(\eta_{i,j})\Big) \text{ is lc near } 0, \text{ and in this case}\\ \mathrm{mld}\big((\mathbb{A}^2 \ni 0, \Delta, \mathfrak{a}_i^{\sigma'})/\overline{G}\big) = \mathrm{mld}\Big(\Big(\mathbb{A}^2 \ni 0, \Delta + \frac{\sigma'}{n_3}\sum_{j=1}^{n_3}\mathrm{div}(\eta_{i,j})\Big)/\overline{G}\Big).\end{aligned} \tag{11.26}$$

If $\mathfrak{a}_i = (1)$, then $\sigma_0 = 0$, and we omit the $\eta_{i,j}$ below.

Consider the tuples

$$(\phi_i, \omega_1, \dots, \omega_{n_1}, \eta_{i,1}, \dots, \eta_{i,n_3}, \zeta_1, \zeta_2), \tag{11.27}$$

where ζ_1, ζ_2 occur only if $\overline{G}$ is abelian. By **Step 3**e, the invariance of the $\eta_{i,j}$ and the choice of the coordinates, the entries are non-zero $\overline{G}$-semi-invariant polynomials whose characters do not depend on i, and their multiplicities at 0 are bounded. By Lemma 2.6(1), after passing to a subsequence, re-indexed by $\mathbb{N}^+$, the tuples (11.27) have a compatible generic limit

$$(\phi_*, \omega_1, \dots, \omega_{n_1}, \eta_{*,1}, \dots, \eta_{*,n_3}, \zeta_1, \zeta_2)$$

with coefficient field $\Bbbk$. An entry ω independent of i is the limit of the constant sequence ω, since the d-jets of ω form one point of $\mathcal{J}_d$. By Lemma 2.6(2), the corresponding entries of the limit of the tuples (11.27) form a compatible generic limit of any sub-tuples, also along any infinite subset of indices, so we may pass to further subsequences below. For $\theta \in [1, \theta_0]$, let $C_i(\theta)$ on $\mathbb{A}^2$ and $C_*(\theta)$ on $\mathcal{A}_\Bbbk$ be

$$\begin{aligned}C_i(\theta) &:= \Delta_i(\theta) + \frac{\sigma(\theta)}{n_3}\sum_{j=1}^{n_3}\mathrm{div}(\eta_{i,j}),\\ C_*(\theta) &:= \theta c\,\mathrm{div}(\phi_*) + \sum_{j=1}^{n_1}\varrho_j(\theta)\,\mathrm{div}(\omega_j) + \frac{\sigma(\theta)}{n_3}\sum_{j=1}^{n_3}\mathrm{div}(\eta_{*,j}),\end{aligned}$$

so that $C_i(\theta)$ and $C_*(\theta)$ are the boundaries C_i and C_* of Theorem 2.18 for the sub-tuples of the entries with positive coefficient. By (11.24) and (11.26), which apply as $\mathrm{Supp}\,\Delta_i(\theta) \subset \Delta_{0,i}$ and $\sigma(\theta) < n_3$, we have, for any $\theta \in [1, \theta_0]$ and any i,

$$\begin{aligned}&\text{the fiber reference at } \theta c \text{ of sign } s \text{ is lc near the fiber over } \overline{o}_i\\ &\iff (\mathbb{A}^2, C_i(\theta)) \text{ is lc near } 0.\end{aligned} \tag{11.28}$$

By **Step 3**e, $(\mathbb{A}^2, \Delta_i(1), \mathfrak{a}_i^{\sigma(1)})$ is lc near 0, so (11.24) and (11.26) imply

$$\beta_i = \mathrm{mld}\big((\mathbb{A}^2 \ni 0, C_i(1))/\overline{G}\big) \quad \text{for any } i. \tag{11.29}$$

Let $c_1, \dots, c_N$ be the positive coefficients of $C_i(1)$ and $(f_{i,j})_j$ the corresponding sub-tuples of (11.27). Then $\overline{G}$, the tuples $(f_{i,j})_j$ and their compatible generic limit satisfy the part of Proposition 11.7 before (1), with $C_i = C_i(1)$ and $C_* = C_*(1)$.

Step 5: a klt limit. Assume that $(\mathcal{A}_\Bbbk, C_*)$ is klt. Let

$$\Phi_* := \phi_* \omega_1 \cdots \omega_{n_1} \eta_{*,1} \cdots \eta_{*,n_3},$$

and let $\tau : \mathcal{Y} \to \mathcal{A}_\Bbbk$ be a log resolution of $(\mathcal{A}_\Bbbk, \mathrm{div}(\Phi_*))$. The log discrepancies with respect to $(\mathcal{A}_\Bbbk, C_*(\theta))$ of the finitely many τ-exceptional prime divisors and components of $\tau_*^{-1}\,\mathrm{div}(\Phi_*)$ are affine functions of θ, and they are positive for $\theta = 1$. Hence there exists $\theta \in (1, \theta_0]$ such that they are positive, so $(\mathcal{A}_\Bbbk, C_*(\theta))$ is klt. We fix such a θ. Since the coefficients of $C_*(\theta)$ need not be proportional to those of C_*, we use the second sentence of Theorem 2.18(1), applied to the sub-tuples of the entries with positive coefficient in $C_i(\theta)$, the coefficients of $C_i(\theta)$, $\mathfrak{c} = \mathcal{O}$, and 1 in place of the number θ there: possibly passing to a subsequence, $(\mathbb{A}^2, C_i(\theta))$ is lc near 0 for any i. By (11.28), the fiber reference at θc of sign s is lc near the fiber over $\overline{o}_i$. Since $s\delta_\theta \le 0$ by (11.20), Lemma 11.6, applied to the data of Set-up 11.1 with c replaced by θc, $U = 1$, and $\lambda = 0$, implies that $(V_{0,i}, \theta c\,\mathrm{div}(h_i^0))$ is lc near 0. By (11.21) with $\theta' = \theta$, $(W_i, \theta c\,\mathrm{div}(h_i))$ is lc near 0 for any $i \gg 0$. This proves (1).

It remains to prove (2), which we do in the continuation of the proof below.

Proof of Proposition 11.7, continued. It remains to treat the case where $(\mathcal{A}_\Bbbk, C_*)$ is not klt. We keep the notation of (11.23), (11.26), and (11.27), and write $\Delta_i := \Delta_i(1)$ and $\sigma := \sigma(1)$. The quotient map $\mathbb{A}^2 \to Z_{0,i}/H_i$ of (11.23) factors through a finite morphism $\varpi : \mathbb{A}^2 \to Z_{0,i}$: in class (a) it is π, in classes (b) and (d) and for $f_i^0 = x^2 + y^2 z + t^{b'}$ it is ϖ_S followed by $S_i \to S_i/\mathbf{T}_v = Z_{0,i}$, and in the remaining cases of class (c) it is the map to Z_0' followed by $Z_0' \to Z_0'/\langle\iota\rangle = Z_{0,i}$. In each case $\overline{G}$ consists of lifts through ϖ of the automorphisms of $Z_{0,i}$ induced by G_i, so ϖ is the quotient by the kernel Ξ of $\overline{G} \to \mathrm{Aut}(Z_{0,i})$, and the images of $\overline{G}$ and G_i in $\mathrm{Aut}(Z_{0,i})$ are equal. In classes (a), (c), and (d), $V_{0,i} = V_0$ and ϖ do not depend on i.

In class (b), $S_i \cong \mathbb{A}^2_{z,t}$, so Ξ acts on $\mathbb{A}^2_{z,t}$ as $\mathbf{T}_x = \mathbb{Z}/|w_x|\mathbb{Z}$, that is, on z and t with weights w_z and w_t modulo $|w_x|$.

The unique floor. Since $(\mathbb{A}^2, C_i)$ is lc near 0 for any i by Proposition 11.2(3) and (11.28) with $\theta = 1$, Theorem 2.18(1), applied with $\theta = 1$ and $\mathfrak{c} = \mathcal{O}_{\mathbb{A}^2}$, shows that $(\mathcal{A}_\Bbbk, C_*)$ is lc near 0. By (11.29) and Theorem 2.18(2), possibly passing to a subsequence, we have

$$\beta_i = \mathrm{mld}((\mathbb{A}^2 \ni 0, C_i)/\overline{G}) = \mathrm{mld}((\mathcal{A}_\Bbbk \ni 0, C_*)/\overline{G})$$

for any i. Since $\beta_i > 0$ and $(\mathcal{A}_\Bbbk, C_*)$ is lc but not klt, by [Liu26b, Definition-Proposition 5.7], $(\mathcal{A}_\Bbbk, C_*)$ has a unique floor F, F is a smooth component of C_* with coefficient one, and

$$\beta_i = \beta := \frac{1 - \mu_F}{n_F}$$

for any i, where $\mu_F := \mathrm{mult}_0((C_* - F)|_F)$ and n_F is the order of the image of $\overline{G}$ in $GL(T_0F)$. Since $\beta > 0$, $\mu_F < 1$.

Since C_* is $\overline{G}$-invariant and F is the unique floor, $\overline{G}$ preserves F. Since $\overline{G}$ preserves the line T_0F and, by Maschke's theorem, a complement of it, $\overline{G}$ is diagonalizable, hence abelian. Since $\overline{G}$ is abelian, ζ_1 and ζ_2 are $\overline{G}$-semi-invariant and are entries of (11.27). In class (b), $(\zeta_1, \zeta_2) = (z, t)$. Choose $j_0 \in \{1, 2\}$ with $\zeta_{j_0}|_{T_0F} \neq 0$, let $\zeta' := \zeta_{j_0}$, and let χ be the character of ζ'. Since T_0F is $\overline{G}$-invariant and $\zeta'|_{T_0F}$ is a basis of T_0^*F, $\overline{G}$ acts on T_0^*F through χ, so

$$n_F = \#\chi(\overline{G}).$$

Since F is smooth, $\zeta'|_F$ is a parameter of F. Since $\zeta' \in \{\zeta_1, \zeta_2\}$, the polynomial U constructed below is defined over $\mathbb{C}$, and no identification of $\Bbbk$ with $\mathbb{C}$ is used.

Since $\operatorname{mult}_0(\operatorname{div}(\zeta')|_F) = 1$,

$$\operatorname{mult}_0\left((C_* - F + (1-\mu_F)\operatorname{div}(\zeta'))|_F\right) = \mu_F + (1-\mu_F) = 1,$$

so by [Liu26b, Lemma 2.14],

$$(\mathcal{A}_{\Bbbk}, C_* + (1-\mu_F)\operatorname{div}(\zeta')) \text{ is lc.}$$

Since ζ' is an entry of (11.27) independent of i, $((f_{*,j})_j, \zeta')$ is a compatible generic limit of the sub-tuples $((f_{i,j})_j, \zeta')$. By Theorem 2.18(1), applied to the tuples $((f_{i,j})_j, \zeta')$ with the coefficients of C_i on the entries $f_{i,j}$ and the coefficient $1-\mu_F$ on ζ', $\theta = 1$, and $\mathfrak{c} = \mathcal{O}_{\mathbb{A}^2}$, possibly passing to a subsequence, $(\mathbb{A}^2, C_i + (1-\mu_F)\operatorname{div}(\zeta'))$ is lc near 0 for any i. For any prime divisor E over $\mathbb{A}^2$, since $\eta_{i,j} \in \mathfrak{a}_i$ [Liu26b, Lemma E.8], we have $\operatorname{mult}_E \eta_{i,j} \geq \operatorname{mult}_E \mathfrak{a}_i$, so

$$a(E, \mathbb{A}^2, \Delta_i + (1-\mu_F)\operatorname{div}(\zeta'), \mathfrak{a}_i^{\sigma}) \geq a(E, \mathbb{A}^2, C_i + (1-\mu_F)\operatorname{div}(\zeta')).$$

Therefore,

$$(\mathbb{A}^2, \Delta_i + (1-\mu_F)\operatorname{div}(\zeta'), \mathfrak{a}_i^{\sigma}) \text{ is lc near 0 for any } i. \tag{11.30}$$

The augmenting function. Let n_0 be the order of $\chi|_{\Xi}$. Then $n_0 \mid n_F$ and ζ'^{n_0} is Ξ-invariant. Since ϖ is the quotient by Ξ, there exists a unique $U_{0,i} \in \mathbb{C}[Z_{0,i}] = \mathbb{C}[V_{0,i}]^{\mathbf{T}}$ such that

$$\varpi^* U_{0,i} = \zeta'^{n_0}.$$

Since $\zeta'(0) = 0$ and ϖ maps 0 to the image of $0 \in V_{0,i}$, $U_{0,i}$ vanishes at 0. Let $g \in G_i$, and let $\widetilde{g} \in \overline{G}$ be an element whose image in $\operatorname{Aut}(Z_{0,i})$ is equal to the image of g, which exists because the images of $\overline{G}$ and G_i in $\operatorname{Aut}(Z_{0,i})$ are equal. Since $\varpi \circ \widetilde{g} = g \circ \varpi$,

$$\varpi^*(g^* U_{0,i}) = \widetilde{g}^*(\zeta'^{n_0}) = \chi(\widetilde{g})^{n_0} \zeta'^{n_0},$$

so $g^* U_{0,i} = \chi(\widetilde{g})^{n_0} U_{0,i}$. Since χ^{n_0} is trivial on Ξ and the images of $\overline{G}$ and G_i in $\operatorname{Aut}(Z_{0,i})$ are equal, $U_{0,i}$ is G_i-semi-invariant, and the image of the character of $U_{0,i}$ on G_i is the image of χ^{n_0} on $\overline{G}$. Thus the order of the character of $U_{0,i}$ on G_i is

$$n_U := \#\chi^{n_0}(\overline{G}) = \frac{n_F}{n_0}$$

for any i. Let

$$\lambda := \frac{1-\mu_F}{n_0}, \quad \text{so that} \quad \frac{\lambda}{n_U} = \frac{1-\mu_F}{n_F} = \beta \quad \text{and} \quad \lambda = n_U \beta.$$

We write $U_{0,i}$ as the restriction of a polynomial U on $\mathbb{A}^4$ independent of i, so that

$$U(0) = U_{0,i}(0) = 0.$$

In class (b), $\zeta' \in \{z, t\}$. Let $w_U \in \{w_z, w_t\}$ be the weight of ζ', so $w_U > 0$ and

$$n_0 = |w_x| / \gcd(w_U, |w_x|).$$

Let

$$U := \zeta'^{n_0} x^{w_U / \gcd(w_U, |w_x|)}.$$

Then

$$\operatorname{wt}_w(U) = n_0 w_U - \frac{|w_x| w_U}{\gcd(w_U, |w_x|)} = 0,$$

so $U|_{V_{0,i}}$ is a regular function on $Z_{0,i}$. Since $U|_{(x=1)} = \zeta'^{n_0}$, we have $\varpi^*(U|_{V_{0,i}}) = \zeta'^{n_0} = \varpi^* U_{0,i}$, hence $U|_{V_{0,i}} = U_{0,i}$ as ϖ is dominant. The monomial U is G_i-semi-invariant for any i because G_i acts diagonally on $\mathbb{A}^4$, and the character of U is that of $U|_{V_{0,i}} = U_{0,i}$.

In classes (a), (c), and (d), $V_{0,i} = V_0$ and ϖ do not depend on i, so $U_0 := U_{0,i}$ does not depend on i. Let U be the unique polynomial with $U|_{V_0} = U_0$ such that no monomial of U is divisible by xy in class (a), resp. such that $\deg_x U \leq 1$ in classes (c) and (d). It exists and is

unique by division by $f^0 = xy + z^{a'}t^{b'}$ with respect to the monomial xy in class (a), resp. by division by f^0, which is monic of degree 2 in x, in classes (c) and (d). The groups $\mathbf{T}$ and G_i act diagonally on monomials, and f^0 is $\mathbf{T}$-homogeneous and G_i-semi-invariant because V_0 is $\mathbf{T}$-stable and G_i-stable. For $g \in \mathbf{T}$ or $g \in G_i$, the polynomial g^*U thus again has no monomial divisible by xy, resp. satisfies $\deg_x(g^*U) \le 1$, and restricts to g^*U_0, so by uniqueness U is $\mathbf{T}$-invariant and G_i-semi-invariant with the character of U_0.

The augmented pairs. Fix i. Since $U|_{V_{0,i}}$ is the pullback of $U_{0,i}$ and $\varpi^*U_{0,i} = \zeta'^{n_0}$,

$$\lambda \operatorname{div}(\varpi^*U_{0,i}) = \lambda n_0 \operatorname{div}(\zeta') = (1-\mu_F)\operatorname{div}(\zeta').$$

For any prime divisor P over $\mathbb{A}^2/\overline{G}$, adding $\lambda\operatorname{div}(U)$ to the fiber reference defining β_i and adding $\lambda\operatorname{div}(\varpi^*U_{0,i})$ to $(\mathbb{A}^2, \Delta_i, \mathfrak{a}_i^\sigma)$ decrease each of the log discrepancies of P in (11.23) with $\theta = 1$ by $\frac{\lambda}{n_U}\operatorname{mult}_P \operatorname{div}(U_{0,i}^{n_U})$, where $U_{0,i}^{n_U}$ is a regular function on $\mathbb{A}^2/\overline{G} = Z_{0,i}/H_i$. By (11.30), the crepant quotient by $\overline{G}$ of the pair in (11.30) is lc near $\overline{o}_i$. Since the fiber reference is proper over $\mathbb{A}^2/\overline{G}$ near $\overline{o}_i$, any prime divisor whose center on the fiber reference intersects the fiber has center on $\mathbb{A}^2/\overline{G}$ containing $\overline{o}_i$. Hence the fiber reference defining β_i with the boundary $\lambda\operatorname{div}(U)$ added is lc near the fiber over $\overline{o}_i$. The fiber reference in (11.22) has sign s, and $s\delta_{\mathbf{T}} = s\delta_1 \le 0$ by (11.20), so by Lemma 11.6,

$$(V_{0,i}, c\operatorname{div}(h_i^0) + \lambda\operatorname{div}(U)) \text{ is lc near } 0.$$

Since $V_{0,i}$ is normal and $U|_{V_{0,i}} \neq 0$ and $h_i^0|_{V_{0,i}} \neq 0$, by [Kaw07, Theorem],

$$(\mathbb{A}^4, V_{0,i} + c\operatorname{div}(h_i^0) + \lambda\operatorname{div}(U)) \text{ is lc near } 0.$$

Let L be the integer of Lemma 2.19 for $n = 4$, three functions, and $\Lambda = \{1, c, \lambda\}$. Any monomial m of $f_i - f_i^0$ or of $h_i - h_i^0$ satisfies

$$r_i - \max\left(|\mathrm{wt}_w(f_i^0)|, |\kappa|\right) \le |\mathrm{wt}_w(m)| \le \deg(m)\max_j |w_j|,$$

so, since $r_i \to +\infty$ and the degrees of the f_i^0 are bounded, $f_i - f_i^0,\ h_i - h_i^0 \in (x,y,z,t)^L$ for any $i \gg 0$. By Lemma 2.19, applied to (f_i^0, h_i^0, U) and (f_i, h_i, U) with exponents $(1, c, \lambda)$,

$$(\mathbb{A}^4, W_i + c\operatorname{div}(h_i) + \lambda\operatorname{div}(U)) \text{ is lc near } 0$$

for any $i \gg 0$. In particular, W_i is not a component of $\operatorname{div}(U)$, as otherwise the coefficient of W_i in $W_i + \lambda\operatorname{div}(U)$ would be at least $1 + \lambda > 1$. Since W_i is normal near 0, by [Kaw07, Theorem],

$$(W_i, c\operatorname{div}(h_i) + \lambda\operatorname{div}(U)) \text{ is lc near } 0$$

for any $i \gg 0$.

Finally, fix $i \gg 0$. The group G_i acts faithfully on W_i, fixes 0, and preserves $c\operatorname{div}(h_i)$, and U is a G_i-semi-invariant regular function vanishing at 0 whose character has order n_U. By [Liu26b, Lemma 5.8], any prime divisor E over W_i with center 0 satisfies

$$\frac{a(E, W_i, c\operatorname{div}(h_i))}{e_E} \ge \frac{\lambda}{n_U} = \beta,$$

where $e_E = |I_{G_i}(E)|$. This proves (2), and the proposition follows. □

12. Bounding the group orders

In this section we bound the orders of the groups along a radial failure sequence whose germs have quadratic rank 1 or 2.

Theorem 12.1. *Let $(W_i \ni o_i, G_i, h_i, \tau_i)$ be a radial failure sequence with constants (m, a, δ) (see Definition 4.1). Assume that $(W_i \ni o_i)$ has quadratic rank 1 or 2 for any i. Then $|G_i|$ is bounded.*

The proofs of Theorems 12.2, 12.3, and 12.4 are given in Sections C.1, C.2, and C.3, respectively. Quadratic ranks and residual cubics are as in Definition 2.2, and marks are as in Definition 2.4.

Theorem 12.2. *Let m be a positive integer and $\epsilon > 0$ a real number. Then there exist a positive integer N and a real number $\eta \in (0,1)$ satisfying the following. Let $(W \ni o)$ be an isolated Gorenstein terminal hypersurface threefold germ of quadratic rank 2, G a finite abelian group of order at least N acting faithfully on W and fixing o, and h a mark on $(W \ni o)$ such that $(W, \frac{1}{m}\operatorname{div}(h))$ is lc near o. Then the radial function q of $(W \ni o, G, h, m)$ satisfies*

$$q(t) - q(1) < \epsilon \quad \text{for any } t \in (1-\eta, 1].$$

Theorem 12.3. *Let m be a positive integer and $\epsilon > 0$ a real number. Then there exist a positive integer N and a real number $\eta \in (0,1)$ satisfying the following. Let $(W \ni o)$ be an isolated Gorenstein terminal hypersurface threefold germ of quadratic rank 1 whose residual cubic is not the cube of a linear form, G a finite cyclic group of order at least N acting faithfully on W and fixing o, and h a mark on $(W \ni o)$ such that $(W, \frac{1}{m}\operatorname{div}(h))$ is lc near o. Then the radial function q of $(W \ni o, G, h, m)$ satisfies*

$$q(t) - q(1) < \epsilon \quad \text{for any } t \in (1-\eta, 1].$$

Theorem 12.4. *Let m be a positive integer and $\epsilon > 0$ a real number. Then there exist a positive integer N and a real number $\eta \in (0,1)$ satisfying the following. Let $(W \ni o)$ be an isolated Gorenstein terminal hypersurface threefold germ of quadratic rank 1 whose residual cubic is the cube of a linear form, G a finite abelian group of order at least N acting faithfully on W and fixing o, and h a mark on $(W \ni o)$ such that $(W, \frac{1}{m}\operatorname{div}(h))$ is lc near o. Then the radial function q of $(W \ni o, G, h, m)$ satisfies*

$$q(t) - q(1) < \epsilon \quad \text{for any } t \in (1-\eta, 1].$$

Proof of Theorem 12.1. Let (N_1, η_1), (N_2, η_2), and (N_3, η_3) be the pairs (N, η) of Theorems 12.2, 12.3, and 12.4, respectively, for the integer m and $\epsilon := \frac{\delta}{2}$. Let

$$N := \max\{N_1, N_2, N_3\} \quad \text{and} \quad \eta := \min\{\eta_1, \eta_2, \eta_3\}.$$

Since $\tau_i < 1$ for any i and $\lim_{i\to+\infty} \tau_i = 1$, possibly removing finitely many terms, we may assume that $\tau_i \in (1-\eta, 1)$ for any i.

Fix i, and suppose that $|G_i| \geq N$. Since $(W_i \ni o_i)$ has quadratic rank 1 or 2, exactly one of the following holds.

(1) $(W_i \ni o_i)$ has quadratic rank 2.
(2) $(W_i \ni o_i)$ has quadratic rank 1, and the residual cubic of $(W_i \ni o_i)$ is not the cube of a linear form.
(3) $(W_i \ni o_i)$ has quadratic rank 1, and the residual cubic of $(W_i \ni o_i)$ is the cube of a linear form.

In case (k), since $|G_i| \geq N \geq N_k$ and $\tau_i \in (1-\eta, 1) \subset (1-\eta_k, 1]$, Definition 4.1(1)(2) and Theorem 12.2, 12.3, or 12.4, respectively, imply

$$q_i(\tau_i) - q_i(1) < \frac{\delta}{2}.$$

On the other hand, by Definition 4.1(3), $q_i(\tau_i) \geq a$ and $q_i(1) \leq a - \delta$, so

$$q_i(\tau_i) - q_i(1) \geq a - (a - \delta) = \delta > \frac{\delta}{2},$$

a contradiction. Thus $|G_i| < N$ for all but finitely many i, and the theorem follows. □

13. Bounded groups and prepared limits

In Section 19 we reach radial failure sequences with $|G_i| \leq M$, $b_i > 0$, and $(W_i, \frac{1}{m}\operatorname{div}(h_i))$ klt near o_i. For them we take compatible generic limits of the equations and marks in a fixed smooth fourfold germ, the prepared limits of Construction-Theorem 13.9, and show that the ambient pairs of the limits are lc with positive minimal log discrepancy at the closed point.

Except for Lemma 13.4 and Construction-Theorem 13.9, the results of this section are proved in Appendix D.

For a hypersurface threefold germ $(W \ni o)$ with a faithful action of a finite group G fixing o, a *completed equivariant presentation* of W consists of a smooth fourfold germ $(\mathcal{X} \ni o)$ with an action of G containing W equivariantly, elements $x_1, \dots, x_4$ of the maximal ideal of $\widehat{\mathcal{O}}_{\mathcal{X},o}$ spanning a G-stable subspace with $\widehat{\mathcal{O}}_{\mathcal{X},o} = \mathbb{C}[[x_1, \dots, x_4]]$, and a generator f of the ideal of W in $\widehat{\mathcal{O}}_{\mathcal{X},o}$. We denote such a presentation by $f(x_1, \dots, x_4)$. A *completed presentation* of W is a completed equivariant presentation for the trivial group.

Theorem 13.1. *Let m and M be positive integers. Then there exist an integer $N(m, M) \geq 3$ and a real number $L(m) > 0$ satisfying the following. Let $(W \ni o)$ be an isolated Gorenstein terminal hypersurface threefold germ, G a finite cyclic group acting faithfully on W and fixing o, h a mark on $(W \ni o)$, and q the radial function of $(W \ni o, G, h, m)$. Assume that the following hold.*

(i) $|G| \leq M$, and $(W, \frac{1}{m}\operatorname{div}(h))$ is lc near o.

(ii) W has a completed equivariant presentation $xy + g(z,t)$ with $\operatorname{mult}_0 g \geq N(m, M)$, in which G acts diagonally on x, y, z, t.

Then the following hold.

(1) $q(1)$ has a computing lift F with

$$\frac{a(F, W, 0)}{e_F} \leq L(m) \quad \text{and} \quad \frac{\operatorname{mult}_F h}{m e_F} \leq L(m).$$

(2) Assume moreover that $xy + g$ is a polynomial and that h is the restriction to W of a polynomial $\widetilde{h} \in \mathbb{C}[x, y, z, t]$ with no monomial divisible by xy. Then $h_0 := \widetilde{h}(0, 0, z, t) \neq 0$, $(\mathbb{A}^2, \frac{1}{m}\operatorname{div}(h_0))$ is lc near 0, and

$$q(1) = \operatorname{mld}\Big(\Big(\mathbb{A}^2 \ni 0, \frac{1}{m}\operatorname{div}(h_0)\Big)/\overline{G}\Big),$$

where $\overline{G}$ is the image of G in $GL_2(\mathbb{C})$ acting on the (z,t)-plane.

Theorem 13.2. *Let m and M be positive integers. Then there exist an integer $N'(m, M) \geq 3$ and a real number $L(m, M) > 0$ satisfying the following. Let $(W \ni o)$ be an isolated Gorenstein terminal hypersurface threefold germ of quadratic rank 2, G a finite cyclic group acting faithfully on W and fixing o, h a mark on $(W \ni o)$, and q the radial function of $(W \ni o, G, h, m)$. Assume that $|G| \leq M$, that $(W, \frac{1}{m}\operatorname{div}(h))$ is lc near o, and that W has a completed equivariant presentation $xy + g(z,t)$ with $\operatorname{mult}_0 g \geq N'(m, M)$, in which G may exchange the factors x and y. Then $q(1)$ has a computing lift F with $a(F, W, 0)/e_F \leq L(m, M)$.*

Theorem 13.3. *(1) Let m, M, d be positive integers. Then there exists a real number $L(m, M, d) > 0$ satisfying the following. Let $(W \ni 0) = (f = 0) \subset \mathbb{A}^4$ be an isolated Gorenstein terminal threefold germ with $\deg f \leq d$, G a finite diagonal abelian group of order at most M preserving f up to a character and acting faithfully on W, h a* polynomial mark *on $(W \ni 0)$, that is, a mark which is the restriction to W of a G-semi-invariant polynomial, and q the radial function of $(W \ni 0, G, h, m)$. Assume that $(W, \frac{1}{m}\operatorname{div}(h))$ is lc near 0 and $b := q(1) > 0$. Then some computing lift F of b satisfies $a(F, W, 0)/e_F \leq L(m, M, d)$.*

(2) Let m, M, c be positive integers. Then there exists a real number $L > 0$, depending only on m, M, c, satisfying the following. Let $(W \ni 0)$ be an isolated Gorenstein terminal hypersurface threefold germ, G a finite abelian group of order at most M acting faithfully on W and fixing 0, h a mark on $(W \ni 0)$, and q the radial function of $(W \ni 0, G, h, m)$. Assume that $(W, \frac{1}{m}\operatorname{div}(h))$ is klt near 0, and that W has a completed equivariant presentation $f \in \mathbb{C}[[x_1, \dots, x_4]]$ with

$$\mathfrak{n}^c \subset (f) + \mathfrak{n} \cdot (\partial_1 f, \dots, \partial_4 f), \quad \text{where } \mathfrak{n} = (x_1, \dots, x_4).$$

Then some computing lift F of $b := q(1)$ satisfies $a(F, W, 0)/e_F \le L$.

(3) *Let m and M be positive integers. For any i, let $(W_i \ni 0) = (f_i = 0)$, with $f_i \in \mathbb{C}[[x_1, \dots, x_4]]$, be an isolated Gorenstein terminal hypersurface threefold germ, G_i a finite abelian group of order at most M acting faithfully and diagonally, h_i a mark on $(W_i \ni 0)$ with $(W_i, \frac{1}{m}\operatorname{div}(h_i))$ klt near 0, and $b_i := q_i(1)$ for the radial function q_i of $(W_i \ni 0, G_i, h_i, m)$. Assume that*

$$\lim_{i\to+\infty} \min_F \frac{a(F, W_i, 0)}{e_F} = +\infty,$$

the minimum over the computing lifts F of b_i. Let (f_*, h_*) be a compatible generic limit of the (f_i, h_i) with coefficient field $\Bbbk$ along an infinite set of indices on which $G = G_i$, the action of G and the characters of f_i and h_i are independent of i, and let $W_* := \operatorname{Spec} \Bbbk[[x_1, \dots, x_4]]/(f_*)$. Then $\dim \operatorname{Sing}(W_*) \ge 1$.

Lemma 13.4. *Let $H \subset GL_4(\mathbb{C})$ be a finite subgroup, m a positive integer, η_m the constant of Theorem 5.1(2), and $\mathcal{X} := \operatorname{Spec} \mathbb{C}[[x_1, \dots, x_4]]$ with maximal ideal $\mathfrak{n}$. For any i, let $f_i, h_i \in \mathbb{C}[[x_1, \dots, x_4]]$ be non-zero H-semi-invariant elements whose characters are independent of i, with $\operatorname{mult}_0 f_i$ and $\operatorname{mult}_0 h_i$ bounded independently of i, such that $W_i := (f_i = 0) \subset \mathcal{X}$ is regular outside the closed point 0 of $\mathcal{X}$, $h_i|_{W_i} \ne 0$, and $(W_i, \frac{1}{m}\operatorname{div}(h_i|_{W_i}))$ is klt. Then the following hold.*

(1) *$(\mathcal{X}, (f_i)^1 (h_i)^{1/m} \mathfrak{n}^{\eta_m})$ is lc for any i.*

(2) *Let (f_*, h_*) be a compatible generic limit of the (f_i, h_i) with coefficient field $\Bbbk$, $\mathcal{X}_* := \operatorname{Spec} \Bbbk[[x_1, \dots, x_4]]$, and $\mathfrak{n}_*$ the maximal ideal of $\Bbbk[[x_1, \dots, x_4]]$. Then $(\mathcal{X}_*, (f_*)^1 (h_*)^{1/m} \mathfrak{n}_*^{\eta_m})$ is lc, and any prime divisor P over $\mathcal{X}_*$ with center the closed point satisfies*

$$a\big(P, \mathcal{X}_*, (f_*)^1 (h_*)^{1/m}\big) \ge \eta_m \operatorname{mult}_P \mathfrak{n}_* \ge \eta_m.$$

In particular, $(\mathcal{X}_*, (f_*)^1 (h_*)^{1/m})$ is lc and $\operatorname{mld}(\mathcal{X}_* \ni 0, (f_*)^1 (h_*)^{1/m}) \ge \eta_m > 0$.

Proof. Since $\mathcal{X}$ is the spectrum of a local ring, the center of any prime divisor over $\mathcal{X}$ contains the closed point 0. Weighted blow-ups of closed points over 0 on a log resolution show that for an $\mathbb{R}$-ideal $\mathfrak{a}$ on $\mathcal{X}$, the pair $(\mathcal{X}, \mathfrak{a})$ is lc once $a(E, \mathcal{X}, \mathfrak{a}) \ge 0$ for any prime divisor E over $\mathcal{X}$ with $\operatorname{center}_{\mathcal{X}} E = 0$.

(1) Fix i and a prime divisor P over $\mathcal{X}$ with center the closed point. Since W_i is regular outside the closed point, the Jacobian criterion gives a positive integer c' with

$$\mathfrak{n}^{c'} \subset (f_i, \partial_1 f_i, \dots, \partial_4 f_i),$$

so $c := c' + 1$ satisfies

$$\mathfrak{n}^c \subset (f_i) + \mathfrak{n} \cdot (\partial_1 f_i, \dots, \partial_4 f_i).$$

Let $n_1 > 2c$ be an integer and $f_i^{<n_1}$ the sum of the terms of f_i of degree less than n_1. Lemma 3.2(2), with the trivial group, $F = f_i$ and $F_0 = f_i^{<n_1}$, gives an automorphism φ of $\mathbb{C}[[x_1, \dots, x_4]]$ and a unit v with $v\varphi(f_i) = f_i^{<n_1}$. Hence the completion at 0 of $W_i^{\mathrm{alg}} := (f_i^{<n_1} = 0) \subset \mathbb{A}^4$ is isomorphic to W_i via φ, so W_i^{alg} is normal near 0 and the pair

$$\Big(\widehat{W}_i^{\mathrm{alg}}, \frac{1}{m}\operatorname{div}(\varphi(h_i))\Big)$$

is klt, where $\widehat{W}_i^{\mathrm{alg}}$ is that completion. Since the log canonical threshold of $\operatorname{div}(\varphi(h_i))$ on $\widehat{W}_i^{\mathrm{alg}}$ is greater than $\frac{1}{m}$, [dFEM11, Propositions 2.11(i) and 2.19(ii)] give an integer n_0 such that for any $n_2 \ge n_0$ the pair

$$\Big(W_i^{\mathrm{alg}}, \frac{1}{m}\operatorname{div}\big(\varphi(h_i)^{<n_2}|_{W_i^{\mathrm{alg}}}\big)\Big)$$

is klt near 0, where $\varphi(h_i)^{<n_2}$ is the sum of the terms of $\varphi(h_i)$ of degree less than n_2. Let P' be the prime divisor over $\mathcal{X}$ corresponding to P under the automorphism of $\mathcal{X}$ induced by φ, so

that $\operatorname{mult}_{P'}\varphi(g) = \operatorname{mult}_P g$ for any g. Fix $n_2 \geq n_0$ with $n_2 > \operatorname{mult}_P h_i$, and let $h_i' := \varphi(h_i)^{<n_2}$. Since $(\varphi(f_i)) = (f_i^{<n_1})$, $\varphi(\mathfrak{n}) = \mathfrak{n}$ and $\varphi(h_i) - h_i' \in \mathfrak{n}^{n_2}$, we have $\operatorname{mult}_{P'} h_i' = \operatorname{mult}_P h_i$ and

$$a\big(P, \mathcal{X}, (f_i)^1 (h_i)^{1/m} \mathfrak{n}^{\eta_m}\big) = a\big(P', \mathcal{X}, (f_i^{<n_1})^1 (h_i')^{1/m} \mathfrak{n}^{\eta_m}\big). \tag{13.1}$$

Let $D_i := \frac{1}{m} \operatorname{div}(h_i'|_{W_i^{\mathrm{alg}}})$. Since $n_2 \geq n_0$, the pair $(W_i^{\mathrm{alg}}, D_i)$ is klt near 0, so its minimal log discrepancy at 0 is positive, and by Theorem 5.1(2), possibly shrinking $\mathbb{A}^4$ to a neighborhood of 0, we may assume that

$$\big(W_i^{\mathrm{alg}}, D_i, \mathfrak{m}_0^{\eta_m}\big)$$

is lc, where $\mathfrak{m}_0$ is the ideal of 0. Since W_i^{alg} is a normal divisor on the smooth $\mathbb{A}^4$ and $W_i^{\mathrm{alg}} \not\subset V(h_i') \cup \{0\}$, inversion of adjunction [EMY03, Corollary 1.8] implies that the pair

$$\big(\mathbb{A}^4, (f_i^{<n_1})^1 (h_i')^{1/m} \mathfrak{m}_0^{\eta_m}\big)$$

is lc near W_i^{alg}. Since the log discrepancy of a prime divisor over $\mathbb{A}^4$ with center 0 equals that of the corresponding prime divisor over the completion [Kaw21, Remark 2.6 and Lemma 2.7(ii)], the right-hand side of (13.1) is non-negative. By the criterion above, $(\mathcal{X}, (f_i)^1 (h_i)^{1/m} \mathfrak{n}^{\eta_m})$ is lc.

(2) Let E_0 be the exceptional divisor of the blow-up of the closed point of $\mathcal{X}$. Since $a(E_0, \mathcal{X}, 0) = 4$, $\operatorname{mult}_{E_0} f_i \geq 1$ and $\operatorname{mult}_{E_0} \mathfrak{n} = 1$, (1) implies $\eta_m \leq 3$, so $(\mathbb{A}^4, (x_1, \dots, x_4)^{\eta_m})$ is klt near 0, and the ideal $(x_1, \dots, x_4)$ is H-invariant. Fix i, let c be as in (1) for f_i, and let $k > \max\{2c, \operatorname{mult}_0 f_i, \operatorname{mult}_0 h_i\}$ be an integer. Since H acts linearly, $f_i^{<k}$ and $h_i^{<k}$ are non-zero H-semi-invariant polynomials with the characters and the orders at 0 of f_i and h_i. As in (1), Lemma 3.2(2) gives an automorphism φ of $\mathbb{C}[[x_1, \dots, x_4]]$ with $(\varphi(f_i)) = (f_i^{<k})$ and $\varphi(x_j) - x_j \in \mathfrak{n}^{k-c+1}$ for any j, hence $\varphi(h_i) - h_i^{<k} \in \mathfrak{n}^{k-c+1}$. Thus $W_i' := (f_i^{<k} = 0) \subset \mathcal{X}$ is isomorphic to W_i via φ, and for k large, depending on i, [dFEM11, Propositions 2.11(i) and 2.19] show that $h_i^{<k}|_{W_i'} \neq 0$ and that $(W_i', \frac{1}{m} \operatorname{div}(h_i^{<k}|_{W_i'}))$ is klt. For such k the tuple $(f_i^{<k}, h_i^{<k})$ satisfies the assumptions of the lemma, and the proof of (1) for it, with φ the identity and with n_1 and n_2 greater than the degrees of $f_i^{<k}$ and $h_i^{<k}$, shows that

$$\big(\mathbb{A}^4, (f_i^{<k})^1 (h_i^{<k})^{1/m} \mathfrak{m}_0^{\eta_m}\big) \text{ is lc near } 0.$$

Choose such integers $k = k_i$ with $\lim_{i \to +\infty} k_i = +\infty$. For any d,

$$\operatorname{jet}_d(f_i^{<k_i}, h_i^{<k_i}) = \operatorname{jet}_d(f_i, h_i) \quad \text{whenever } k_i \geq d,$$

that is, for all but finitely many i, so (f_*, h_*) is also a compatible generic limit of the tuples $(f_i^{<k_i}, h_i^{<k_i})$ with coefficient field $\Bbbk$. By Theorem 2.18(1) for the tuples $(f_i^{<k_i}, h_i^{<k_i})$, with $\mathfrak{c} = (x_1, \dots, x_4)^{\eta_m}$, the pair $(\mathcal{X}_*, (f_*)^1 (h_*)^{1/m} \mathfrak{n}_*^{\eta_m})$ is lc. Thus any prime divisor P over $\mathcal{X}_*$ with center the closed point satisfies

$$\begin{aligned} a\big(P, \mathcal{X}_*, (f_*)^1 (h_*)^{1/m}\big) &= a\big(P, \mathcal{X}_*, (f_*)^1 (h_*)^{1/m} \mathfrak{n}_*^{\eta_m}\big) + \eta_m \operatorname{mult}_P \mathfrak{n}_* \\ &\geq \eta_m \operatorname{mult}_P \mathfrak{n}_* \geq \eta_m. \end{aligned}$$

Removing the factor $\mathfrak{n}_*^{\eta_m}$ does not decrease log discrepancies, so $(\mathcal{X}_*, (f_*)^1 (h_*)^{1/m})$ is lc. The lemma follows. □

Lemma 13.5. *Let $\Bbbk$ be an algebraically closed field of characteristic zero, R a complete normal local $\Bbbk$-algebra with residue field $\Bbbk$ which is a quotient of $\Bbbk[[x_1, \dots, x_n]]$ for some n, $X := \operatorname{Spec} R$ with closed point o, and $B \geq 0$ a $\mathbb{Q}$-divisor on X such that $K_X + B$ is $\mathbb{Q}$-Cartier. Assume that (X, B) is lc but not klt. Then (X, B) has finitely many lc centers and a unique minimal lc center C, which is contained in any lc center of (X, B), is normal, and is invariant under any finite group of automorphisms of X preserving B. If moreover $\dim X = 3$ and $\operatorname{mld}(X \ni o, B) > 0$, then $\dim C \in \{1, 2\}$, and $C \cong \operatorname{Spec} \Bbbk[[s]]$ if $\dim C = 1$.*

Let moreover m be a positive integer, G a finite group acting diagonally on $\Bbbk[[x_1, \dots, x_4]]$, and f_*, h_* G-semi-invariant elements of $\Bbbk[[x_1, \dots, x_4]]$, with f_* a non-zero element of the maximal ideal, such that $W_* := \operatorname{Spec} \Bbbk[[x_1, \dots, x_4]]/(f_*)$ is normal and klt and $h_*|_{W_*} \neq 0$.

Let $D := \frac{1}{m}\operatorname{div}(h_*|_{W_*})$, *assume that* (W_*, D) *is lc but not klt and* $\operatorname{mld}(W_* \ni 0, D) > 0$, *and let* C *be the minimal lc center of* (W_*, D). *Then exactly one of the following holds.*

(1) $\dim C = 2$, C *is a component of* D *with coefficient one,* C *is the only lc center of* (W_*, D), *and* (W_*, D) *is plt.*

(2) $\dim C = 1$, C *is regular, and there exist* G*-eigenparameters* (s, u_1, u_2, u_3) *of* $\Bbbk[[x_1, \dots, x_4]]$ *with* $C = (u_1 = u_2 = u_3 = 0)$ *such that*

$$h_* \in (u_1, u_2, u_3), \quad \text{and} \quad f_* \in (u_1, u_2, u_3)^2 \quad \text{if } C \subset \operatorname{Sing}(W_*).$$

Construction-Theorem 13.6. Let $(W_i \ni o_i, G_i, h_i, \tau_i)$ be a radial failure sequence with constants (m, a, δ) and M a positive integer. Assume that for any i, $|G_i| \le M$, $b_i > 0$, $(W_i, \frac{1}{m}\operatorname{div}(h_i))$ is klt near o_i, and $(W_i \ni o_i)$ is of type cE. Then, after passing to a subsequence and to G_i-equivariant isomorphisms of the completed local rings, the group $G = G_i$, the diagonal representation of G on $\mathbb{C}[[x, y, z, t]]$ and the characters of the equations and marks are independent of i, and the equation $f_i \in \mathbb{C}[[x, y, z, t]]$ of W_i and the expansion h_i of the mark in a completed equivariant presentation of W_i with coordinates x, y, z, t satisfy

$$f_i = x^2 + y^3 + y\alpha_i(z, t) + \beta_i(z, t) \quad \text{and} \quad h_i = \varphi_i(y, z, t) + x\xi_i(y, z, t)$$

with one row (E_6), (E_7) or (E_8) of Lemma 3.7 independent of i. Moreover, there exists a compatible generic limit (f_*, h_*) of the (f_i, h_i) with coefficient field $\Bbbk$ satisfying the following, where $\mathcal{X}_* := \operatorname{Spec}\Bbbk[[x, y, z, t]]$ and $W_* := \operatorname{Spec}\Bbbk[[x, y, z, t]]/(f_*)$.

(1) W_* is normal, Gorenstein, canonical and cDV, and there exists $c' \in \{0, 1, \dots, 5\}$ such that for $\ell := t - c'z$, the surface $(\ell = 0)$ on W_* and on W_i for infinitely many i is Du Val of type E_6, E_7 or E_8 according to the row.

(2) The singular locus $\operatorname{Sing}(W_*)$ satisfies the following, where ℓ is as in (1).
   (a) $\operatorname{Sing}(W_*)$ with the reduced structure is finite over $\operatorname{Spec}\Bbbk[[\ell]]$.
   (b) If $\Gamma_1, \dots, \Gamma_k$ are the 1-dimensional irreducible components of $\operatorname{Sing}(W_*)$ and r_j is the degree of the normalization of Γ_j over $\operatorname{Spec}\Bbbk[[\ell]]$, then $\sum_{j=1}^{k} r_j \le 8$.

(3) $(W_*, \frac{1}{m}\operatorname{div}(h_*))$ is lc but not klt, $\operatorname{mld}(W_* \ni 0, \frac{1}{m}\operatorname{div}(h_*)) \ge a$, and any prime divisor P over $\mathcal{X}_*$ with center the closed point satisfies

$$\frac{a(P, \mathcal{X}_*, 0) - \operatorname{mult}_P f_* - \frac{1}{m}\operatorname{mult}_P h_*}{e_P} \ge a.$$

We call such (f_*, h_*) a *prepared limit of type cE.*

Construction-Theorem 13.7. Let $(W_i \ni o_i, G_i, h_i, \tau_i)$ be a radial failure sequence with constants (m, a, δ) and M a positive integer. Assume that for any i, $|G_i| \le M$, $b_i > 0$, $(W_i, \frac{1}{m}\operatorname{div}(h_i))$ is klt near o_i, and $(W_i \ni o_i)$ has quadratic rank 1 and is of type cD. For any i let F_i be a computing lift of b_i. Then, after passing to a subsequence and to G_i-equivariant isomorphisms of the completed local rings, the group $G = G_i$, the diagonal representation of G on $\mathbb{C}[[x, y, z, t]]$ and the characters of the equations and marks are independent of i, and the equation $f_i \in \mathbb{C}[[x, y, z, t]]$ of W_i and the expansion h_i of the mark in a completed equivariant presentation of W_i with coordinates x, y, z, t satisfy

$$f_i = x^2 + g_i(y, z, t) \quad \text{and} \quad h_i = \alpha_i(y, z, t) + x\xi_i(y, z, t)$$

with $\alpha_i, \xi_i \in \mathbb{C}[[y, z, t]]$. Moreover, there exists a compatible generic limit (f_*, h_*) of the (f_i, h_i) with coefficient field $\Bbbk$ satisfying the following, where $W_* := \operatorname{Spec}\Bbbk[[x, y, z, t]]/(f_*)$ and η_m is the constant of Theorem 5.1(2).

(1) W_* is normal, not isolated, Gorenstein, canonical and cDV.

(2) There exist an integer $n \ge 4$ and linearly independent linear forms ℓ_1, ℓ_2, ℓ_3 in y, z, t with coefficients in $\mathbb{C}$ such that for any $k \in \{1, 2, 3\}$ and any i, the surfaces $(\ell_k = 0)$ on W_* and on W_i are Du Val of type D_n.

(3) There exists $j \in \{1,2,3\}$ such that for any i,
$$\operatorname{mult}_{F_i} \ell_j = \operatorname{mult}_{F_i} \mathfrak{m}_{o_i} \leq \frac{3M}{\eta_m},$$
and there exists a regular function $\ell_{j,i}$ on W_i with $\operatorname{mult}_{F_i} \ell_{j,i} = \operatorname{mult}_{F_i} \ell_j$ such that $(\ell_{j,i} = 0)$ is a Du Val surface of type D_n.
(4) $(W_*, \frac{1}{m}\operatorname{div}(h_*))$ is lc but not klt, and $\operatorname{mld}(W_* \ni 0, \frac{1}{m}\operatorname{div}(h_*)) \geq a$.
(5) The minimal lc center C of $(W_*, \frac{1}{m}\operatorname{div}(h_*))$ (Lemma 13.5) is either a component of $\frac{1}{m}\operatorname{div}(h_*)$ with coefficient one, in which case $(W_*, \frac{1}{m}\operatorname{div}(h_*))$ is plt, or a regular curve, in which case ℓ_1, ℓ_2, ℓ_3 in (2) can be taken so that $\ell_k|_C$ is a regular parameter of C for any $k \in \{1,2,3\}$.
(6) If the residual cubic of the W_i is the union of a double line and a different line, then the coordinates x, y, z, t can be taken so that
$$f_i = x^2 + zy^2 + \gamma_i(t)y + \psi_i(z,t) \quad \text{and} \quad f_* = x^2 + zy^2 + \gamma_*(t)y + \psi_*(z,t),$$
with $\operatorname{mult}_0 \gamma_i, \operatorname{mult}_0 \gamma_* \geq 3$ and $\operatorname{mult}_0 \psi_i, \operatorname{mult}_0 \psi_* \geq 4$.

We call such (f_*, h_*) a *prepared limit of type cD*.

Theorem 13.8. *Let $(W_i \ni o_i, G_i, h_i, \tau_i)$ be a radial failure sequence with constants (m, a, δ) and M a positive integer. Assume that for any i, $|G_i| \leq M$, $b_i > 0$, and $(W_i, \frac{1}{m}\operatorname{div}(h_i))$ is klt near o_i, and that either all $(W_i \ni o_i)$ have quadratic rank 2, or all have quadratic rank 1 with residual cubic neither zero nor the cube of a linear form. For a compatible generic limit with coefficient field $\Bbbk$, let $\mathcal{X}_* := \operatorname{Spec} \Bbbk[[x_1, \dots, x_4]]$.*

(1) Pass to a subsequence on which $G = G_i$, the linear action of G on the coordinates $(x,y,z,t) = (x_1, \dots, x_4)$ and the characters are independent of i, and let (f_*, h_*) be a compatible generic limit with coefficient field $\Bbbk$ of the equations f_i and of G-semi-invariant lifts $h_i \in \mathbb{C}[[x_1, \dots, x_4]]$ of the marks, formed in completed equivariant presentations $x^2 + g_i(y,z,t)$ in quadratic rank 1 and $xy + g_i(z,t)$ in quadratic rank 2. If $W_* := \operatorname{Spec} \Bbbk[[x_1, \dots, x_4]]/(f_*)$ is not normal, then there exist an infinite set I of indices and a real number L such that for any $i \in I$, some computing lift F_i of b_i satisfies $a(F_i, W_i, 0)/e_{F_i} \leq L$.
(2) Assume that
$$\lim_{i \to +\infty} \min_F \frac{a(F, W_i, 0)}{e_F} = +\infty,$$
the minimum over the computing lifts F of b_i. Then the equations and marks have a compatible generic limit (f_*, h_*), formed as in (1), satisfying the following.
(a) W_* is normal, not isolated, Gorenstein and canonical.
(b) Both $(\mathcal{X}_*, (f_*)^1 (h_*)^{1/m})$ and $(W_*, \frac{1}{m}\operatorname{div}(h_*))$ are lc with positive minimal log discrepancy at the closed point.

Construction-Theorem 13.9. Let $(W_i \ni o_i, G_i, h_i, \tau_i)$ be a radial failure sequence with constants (m, a, δ) and M a positive integer. Assume that for any i, $|G_i| \leq M$, $b_i > 0$, $(W_i, \frac{1}{m}\operatorname{div}(h_i))$ is klt near o_i, and $(W_i \ni o_i)$ has quadratic rank 1 or 2. Then, after passing to a subsequence, the following are independent of i:

(i) The quadratic rank of $(W_i \ni o_i)$ and, in quadratic rank 1, whether its residual cubic is the cube of a linear form.
(ii) The group $G = G_i$ and its linear action on the coordinates $(x,y,z,t) = (x_1, \dots, x_4)$, which is diagonal except that in quadratic rank 2 it may exchange x and y.
(iii) The characters of the equations f_i and of the marks h_i.

Moreover the following hold, where $\mathcal{X}_* := \operatorname{Spec} \Bbbk[[x_1, \dots, x_4]]$.

(1) If the quadratic rank is 2, then there exists an integer $k_1 \geq 3$ such that $\operatorname{mult}_0 g_i \leq k_1$ for any i and any completed equivariant presentation $xy + g_i(z,t)$ of W_i.

(2) The equations f_i and the marks h_i, in the completed equivariant presentations of Construction-Theorems 13.7 and 13.6 in quadratic rank 1 and in completed equivariant presentations $xy + g_i(z,t)$ in quadratic rank 2, have a compatible generic limit (f_*, h_*) with coefficient field $\Bbbk$, which we call a *prepared limit*, and $W_* := \operatorname{Spec} \Bbbk[[x_1, \dots, x_4]]/(f_*)$ satisfies the following.
   (a) In quadratic rank 2, $f_* = xy + g_*(z,t)$ with $3 \le \operatorname{mult}_0 g_* \le k_1$, W_* is normal, Gorenstein, canonical and cDV, and $(W_*, \frac{1}{m}\operatorname{div}(h_*))$ is lc with positive minimal log discrepancy at the closed point.
   (b) In quadratic rank 1, (f_*, h_*) is a prepared limit of type cD of Construction-Theorem 13.7 if the residual cubic is not the cube of a linear form, and a prepared limit of type cE of Construction-Theorem 13.6 if it is.

(3) $(\mathcal{X}_*, (f_*)^1 (h_*)^{1/m})$ is lc, and $\operatorname{mld}(\mathcal{X}_* \ni 0, (f_*)^1 (h_*)^{1/m}) > 0$.

(4) Let (n_i) be positive integers with $\lim_{i\to+\infty} n_i = +\infty$, and assume that the quadratic rank is 1.
   (a) For any i there exist a G-equivariant smooth algebraic fourfold germ $(\mathcal{X}_i \ni o_i)$ containing W_i and G-semi-invariant regular parameters of $\mathcal{X}_i$ at o_i in which the expansion $\widetilde{f}_i$ of an equation of W_i multiplied by a G-invariant regular unit and the expansion $\widetilde{h}_i$ of a regular function on $\mathcal{X}_i$ restricting to the mark satisfy
   $$\widetilde{f}_i - f_i \in \mathfrak{n}^{n_i} \quad \text{and} \quad \widetilde{h}_i - h_i \in \mathfrak{n}^{n_i},$$
   where $\mathfrak{n}$ is the maximal ideal of $\mathbb{C}[[x_1, \dots, x_4]]$.
   (b) The prepared limit (f_*, h_*) of (2) is also a compatible generic limit of the $(\widetilde{f}_i, \widetilde{h}_i)$.

Proof. Since $|G_i| \le M$, there exist only finitely many possibilities for the data (i)–(iii), so we may fix them after passing to a subsequence. By Lemma 4.3(2), any computing lift F of b_i satisfies

$$\frac{a(F, W_i, 0)}{e_F} \ge \frac{\delta}{1-\tau_i}, \quad \text{and} \quad \lim_{i\to+\infty} \frac{\delta}{1-\tau_i} = +\infty. \tag{13.2}$$

(1) Let $N'(m,M) \ge 3$ and $L(m,M) > 0$ be the constants of Theorem 13.2 for m and M, and $k_1 := \max\{N'(m,M) - 1, 3\}$. If infinitely many W_i had completed equivariant presentations $xy + g_i(z,t)$ with $\operatorname{mult}_0 g_i \ge N'(m,M)$, then by Theorem 13.2, for any such i some computing lift F_i of b_i would satisfy $a(F_i, W_i, 0)/e_{F_i} \le L(m,M)$, contradicting (13.2). Removing finitely many indices, we obtain (1).

(2) In quadratic rank 1, $(W_i \ni o_i)$ is of type cD if the residual cubic is not the cube of a linear form and of type cE if it is [Liu26a, Lemma 11.3(3)], and (2) is Construction-Theorem 13.7, resp. Construction-Theorem 13.6.

Assume that the quadratic rank is two. By (13.2) and Theorem 13.8(2), the equations and marks have a compatible generic limit (f_*, h_*) with coefficient field $\Bbbk$, formed in completed equivariant presentations $f_i = xy + g_i(z,t)$, such that W_* is normal, Gorenstein and canonical, and $(W_*, \frac{1}{m}\operatorname{div}(h_*))$ is lc with positive minimal log discrepancy at the closed point. Since quadratic rank 2 forces $g_i \in (z,t)^3$, (1) implies

$$3 \le \operatorname{mult}_0 g_i \le k_1 \quad \text{for any } i.$$

For any integer $d > k_1$, the d-jets $(p_1, p_2) \in \mathcal{J}_d$ with p_1 of the form $xy + g(z,t)$, where

$$g \in (z,t)^3 \backslash (z,t)^{k_1+1},$$

form a constructible subset of $\mathcal{J}_d$ containing $\operatorname{jet}_d(f_i, h_i)$ for any i, so it contains $\operatorname{jet}_d(f_*, h_*)$ by Lemma 2.8. Thus $f_* = xy + g_*(z,t)$ with

$$3 \le n := \operatorname{mult}_0 g_* \le k_1.$$

For a general $\lambda \in \Bbbk$, $g_*(z, \lambda z) = z^n v(z)$ for a unit v of $\Bbbk[[z]]$, so the surface $(t = \lambda z)$ on W_* is $(xy + z^n v(z) = 0) \subset \operatorname{Spec} \Bbbk[[x,y,z]]$, a Du Val surface of type A_{n-1}. Hence W_* is cDV, which proves (2).

(3) By Definition 4.1(1), the completion $(f_i = 0) \subset \operatorname{Spec} \mathbb{C}[[x_1, \dots, x_4]]$ of W_i in the completed equivariant presentation of (2) is regular outside the closed point, and the induced pair is klt since $(W_i, \frac{1}{m} \operatorname{div}(h_i))$ is klt near o_i [dFEM11, Proposition 2.11]. Thus Lemma 13.4(2), for H the image of G in $GL_4(\mathbb{C})$, implies (3).

(4) The action of G on $x_1, \dots, x_4$ is diagonal in quadratic rank 1 by (ii). For any i, approximating the equivariant isomorphism of completed local rings modulo $\mathfrak{n}^{n_i}$ by G-semi-invariant regular functions on the smooth fourfold germ $(\mathcal{X}_i \ni o_i)$ of the completed equivariant presentation of W_i in (2), we obtain G-semi-invariant regular parameters of $\mathcal{X}_i$ at o_i in which the expansion $\widetilde{f}_i$ of the equation of W_i multiplied by an invariant regular unit, and the expansion $\widetilde{h}_i$ of a lift of the mark plus a regular multiple of the equation, satisfy

$$\widetilde{f}_i - f_i \in \mathfrak{n}^{n_i} \quad \text{and} \quad \widetilde{h}_i - h_i \in \mathfrak{n}^{n_i}.$$

For any d, the d-jets of $(\widetilde{f}_i, \widetilde{h}_i)$ and of (f_i, h_i) coincide whenever $n_i \geq d$, that is, for all but finitely many i. Hence (f_*, h_*) is also a compatible generic limit of the $(\widetilde{f}_i, \widetilde{h}_i)$. □

Set-up 13.10. Let $(W_i \ni o_i, G_i, h_i, \tau_i)$, (m, a, δ), M, G, and (f_*, h_*) be as follows.

(1) $(W_i \ni o_i, G_i, h_i, \tau_i)$ is a radial failure sequence with constants (m, a, δ), and M is a positive integer such that for any i, $|G_i| \leq M$, $b_i > 0$, $(W_i, \frac{1}{m} \operatorname{div}(h_i))$ is klt near o_i, and $(W_i \ni o_i)$ has quadratic rank 1 or 2.

(2) The data of Construction-Theorem 13.9(i)–(iii), in particular the group $G = G_i$ and its action on the coordinates $(x, y, z, t) = (x_1, \dots, x_4)$, do not depend on i, and (f_*, h_*) is a prepared limit of Construction-Theorem 13.9(2), with $W_* := \operatorname{Spec} \Bbbk[[x_1, \dots, x_4]]/(f_*)$.

14. Limits that are klt along the singular locus

In this section we prove Theorem 14.1 (see the sketch of the proof in Section 1).

Theorem 14.1. *Notation and conditions as in Set-up 13.10. Then $(W_*, \frac{1}{m} \operatorname{div}(h_*))$ is not klt at the generic point of some 1-dimensional component of $\operatorname{Sing}(W_*)$.*

Theorem 14.2. *Let $k_1 \geq 3$, M, and m be positive integers. For any i let $(W_i \ni o_i)$ be an isolated Gorenstein terminal hypersurface threefold germ of quadratic rank 2, let G_i be a finite cyclic group of order at most M acting faithfully on W_i and fixing o_i, and let h_i be a mark with $(W_i, \frac{1}{m} \operatorname{div}(h_i))$ klt near o_i. Let q_i be the radial function of $(W_i \ni o_i, G_i, h_i, m)$ and $b_i := q_i(1)$. After passing to a subsequence, fix $G = G_i$, the action of G on the ambient coordinates, which permutes the lines $\mathbb{C}x$ and $\mathbb{C}y$ and is linear on the (z, t)-plane, and the characters of the equations and marks, and let (f_*, h_*), $f_* = xy + g_*(z, t)$, be a compatible generic limit (Definition 2.5), with coefficient field $\Bbbk$, of completed equivariant presentations $xy + g_i(z, t)$ of the $(W_i \ni o_i)$ with $\operatorname{mult}_0 g_i \in [3, k_1]$ and of the h_i, and let $W_* := (f_* = 0) \subset \mathcal{X}_* := \operatorname{Spec} \Bbbk[[x, y, z, t]]$. Assume that*

(1) W_* is normal and canonical,

(2) $(W_*, \frac{1}{m} \operatorname{div}(h_*))$ is lc, and

(3) $\operatorname{Nklt}(W_*, \frac{1}{m} \operatorname{div}(h_*))$ and $\operatorname{Sing}(W_*)$ have no common point other than 0.

Then there exist a real number L and an infinite set of indices i for which some computing lift P_i of b_i satisfies $a(P_i, W_i, 0)/e_{P_i} \leq L$.

Set-up 14.3. Let m, M, G, $(W_i \ni o_i)$, h_i, $(\mathcal{X}_i \ni o_i)$, f_i, and (f_*, h_*) be as follows.

(1) m and M are positive integers, and G is a finite abelian group of order at most M.

(2) For any i, $(W_i \ni o_i)$ is an isolated Gorenstein terminal hypersurface threefold germ with a faithful action of G fixing o_i, and h_i is a mark on $(W_i \ni o_i)$ which is not a unit.

(3) The following hold.

  (a) For any i, $(\mathcal{X}_i \ni o_i)$ is a G-equivariant smooth algebraic fourfold germ containing W_i, with G-semi-invariant regular parameters $x_1, \dots, x_4$ at o_i.

  (b) For any i, f_i is a G-invariant unit times the expansion in $x_1, \dots, x_4$ of an equation of W_i.

(c) For any i, h_i also denotes the expansion in $x_1, \dots, x_4$ of a regular function on $\mathcal{X}_i$ restricting to the mark.
(d) The characters of $x_1, \dots, x_4$, of f_i, and of h_i do not depend on i.

(4) (f_*, h_*) is a compatible generic limit of the (f_i, h_i) (Definition 2.5) with coefficient field $\Bbbk$, and $W_* := \operatorname{Spec} \Bbbk[[x_1, \dots, x_4]]/(f_*)$.

If $(W_i, \frac{1}{m}\operatorname{div}(h_i))$ is lc near o_i for any i, let q_i be the radial function of $(W_i \ni o_i, G, h_i, m)$ and $b_i := q_i(1)$.

Theorem 14.4. *Notation and conditions as in Set-up 14.3. Assume that for any i, $(W_i \ni o_i)$ is of type cE, $(W_i, \frac{1}{m}\operatorname{div}(h_i))$ is lc near o_i, and $b_i > 0$, that W_* is normal, and that $(W_*, \frac{1}{m}\operatorname{div}(h_*))$ is lc, and klt at the generic point of any 1-dimensional component of $\operatorname{Sing}(W_*)$. Then there exist a real number L and an infinite set of indices i for which some computing lift F_i of b_i satisfies $a(F_i, W_i, 0)/e_{F_i} \le L$.*

Theorem 14.5. *Notation and conditions as in Set-up 14.3, and let $n \ge 4$ be an integer. Assume that for any i, $(W_i \ni o_i)$ has quadratic rank 1 and its residual cubic is not the cube of a linear form, $(W_i, \frac{1}{m}\operatorname{div}(h_i))$ is klt near o_i, and ℓ_i is a linear form in $x_1, \dots, x_4$ such that $(W_i \cap (\ell_i = 0) \ni o_i)$ is a Du Val germ of type D_n. Assume moreover that W_* is normal, and that $(W_*, \frac{1}{m}\operatorname{div}(h_*))$ is lc, and klt at the generic point of any 1-dimensional component of $\operatorname{Sing}(W_*)$. Then there exist a real number L and an infinite set of indices i for which some computing lift F_i of b_i satisfies $a(F_i, W_i, 0)/e_{F_i} \le L$.*

We prove Theorem 14.2 in Subsection D.2, and Theorems 14.4 and 14.5 in Subsection D.3.

Proof of Theorem 14.1. Let $(W_i \ni o_i, G_i, h_i, \tau_i)$, (m, a, δ), M, G, and (f_*, h_*) be the data of Set-up 13.10. By Lemma 4.3(2), there does not exist an infinite subsequence with computing lifts F_i of b_i such that $a(F_i, W_i, 0)/e_{F_i}$ is bounded.

By Construction-Theorem 13.9(2)(a), Construction-Theorem 13.7(4), and Construction-Theorem 13.6(3), $(W_*, \frac{1}{m}\operatorname{div}(h_*))$ is lc. Suppose that $(W_*, \frac{1}{m}\operatorname{div}(h_*))$ is klt at the generic point of any 1-dimensional component of $\operatorname{Sing}(W_*)$. Since W_* is normal, $\dim \operatorname{Sing}(W_*) \le 1$, so the points of $\operatorname{Sing}(W_*)$ other than 0 are the generic points of the 1-dimensional components of $\operatorname{Sing}(W_*)$, and

$$\operatorname{Nklt}(W_*, \tfrac{1}{m}\operatorname{div}(h_*)) \cap \operatorname{Sing}(W_*) \subset \{0\}.$$

We obtain an infinite subfamily with computing lifts F_i of b_i such that $a(F_i, W_i, 0)/e_{F_i}$ is bounded, a contradiction, in each of the following three cases of Construction-Theorem 13.9(2).

If the quadratic rank of the W_i is two, then G_i is cyclic by Definition 4.1(1), and by Construction-Theorem 13.9(1)(2)(a) there exists an integer $k_1 \ge 3$ with $\operatorname{mult}_0 g_i \in [3, k_1]$ for completed equivariant presentations $xy + g_i(z,t)$ of the W_i, $f_* = xy + g_*(z,t)$, and W_* is normal and canonical. Theorem 14.2 implies the desired subfamily.

If the quadratic rank of the W_i is one and the residual cubic is not the cube of a linear form, then (f_*, h_*) is a prepared limit of type cD by Construction-Theorem 13.9(2)(b), so by Construction-Theorem 13.7(1)(2), W_* is normal and canonical, and for an integer $n \ge 4$ a linear form ℓ_1 cuts out a Du Val section of type D_n of any W_i. Let $(\widetilde{f}_i, \widetilde{h}_i)$ be the expansions of Construction-Theorem 13.9(4) for a sequence $n_i \to +\infty$. Their compatible generic limit is (f_*, h_*), so the conditions of Set-up 14.3 hold for the $(\widetilde{f}_i, \widetilde{h}_i)$ and the parameters of Construction-Theorem 13.9(4), and the assumptions on W_* and $(W_*, \frac{1}{m}\operatorname{div}(h_*))$ hold. Since a surface germ of type D_n has Tjurina number n, it is contact $2n$-determined [GP19, Theorem 4.8], and since $\widetilde{f}_i - f_i \in \mathfrak{n}^{n_i}$, ℓ_1 cuts out a D_n section of $(\widetilde{f}_i = 0)$ for any $i \gg 0$. Thus Theorem 14.5 with $\ell_i := \ell_1$ implies the desired subfamily.

If the quadratic rank of the W_i is one and the residual cubic is the cube of a linear form, then the W_i are of type cE and (f_*, h_*) is a prepared limit of type cE by Construction-Theorem 13.9(2)(b), so W_* is normal and canonical by Construction-Theorem 13.6(1). Theorem 14.4, applied to the expansions $(\widetilde{f}_i, \widetilde{h}_i)$ as in the previous case, implies the desired subfamily. □

15. Bounded Jacobian orders

In this section we prove Theorem 15.8: a computing lift with bounded Jacobian order yields a computing lift of bounded log discrepancy, and compatible limits with no lc center in $\mathrm{Sing}(W_*)$ have bounded Jacobian orders (see the sketch of the proof in Section 1). The proofs of the results of this section other than Theorem 15.8 are given in Appendix E.

Definition 15.1. Let $\Bbbk \supset \mathbb{C}$ be an algebraically closed field, $\mathcal{X}_* := \operatorname{Spec} \Bbbk[[x_1, \dots, x_4]]$, and $C \subset W_*$ a regular curve through the closed point of a normal threefold germ

$$W_* = (f_* = 0) \subset \mathcal{X}_*.$$

Let (s, u_1, u_2, u_3) be regular parameters of $\mathcal{X}_*$ with $C = (u_1 = u_2 = u_3 = 0)$, and $\overline{\Bbbk((s))}$ an algebraic closure of $\Bbbk((s))$. The *generic transverse surface* of W_* along C is the surface germ

$$\operatorname{Spec} \overline{\Bbbk((s))}[[u_1, u_2, u_3]]/(f_*),$$

the base change to $\overline{\Bbbk((s))}$ of the completion $\Bbbk((s))[[u_1, u_2, u_3]]/(f_*)$ of the local ring of W_* at the generic point of C. If it is Du Val, then its Du Val type (A_n, D_n, or E_n) is the *generic transverse type* of W_* along C.

The generic transverse surface of W_* along C is the completion of a strict henselization of the local ring of W_* at the generic point of C [Sta26, Tag 06LJ], so up to isomorphism it depends only on W_* and C. If W_* is canonical, then restricting a resolution of W_* [Tem18, Theorem 1.1.9] over the generic point of C, completing, and extending the residue field to $\overline{\Bbbk((s))}$ [Kaw21, Remark 2.6] keeps all discrepancies non-negative, so the generic transverse surface is canonical, hence Du Val [Kol13, Theorem 2.29(2) and Example 3.26]. It is singular if $C \subset \mathrm{Sing}(W_*)$.

Theorem 15.2. *Let m, M, N, N_0 be positive integers. Then there exists a real number $L > 0$ depending only on m, M, N, and N_0 satisfying the following. Let $(T \ni o)$ be a smooth surface germ over an algebraically closed field $\Bbbk \supset \mathbb{C}$ with a faithful action of a finite abelian group G of order at most M fixing o, h a non-zero regular G-semi-invariant function, and $D_T \geq 0$ a G-invariant integral Cartier divisor on T. Assume that $(T, \frac{1}{m}\operatorname{div}(h) - D_T)$ is sub-lc near o, that*

$$\mu := \operatorname{mld}\Big(\Big(T \ni o, \frac{1}{m}\operatorname{div}(h) - D_T\Big)/G\Big) > 0,$$

and that some computing lift F of μ satisfies $\operatorname{mult}_F D_T \leq N$ and $\operatorname{mult}_F \mathfrak{m}_o \leq N_0$. Then some computing lift P of μ satisfies

$$a(P, T, 0) \leq L \quad \text{and} \quad \operatorname{mult}_P D_T \leq L.$$

Proposition 15.3. *Let m, M, N be positive integers and η_m the constant of Theorem 5.1(2). Then there exists a real number $L > 0$ depending only on m, M, and N satisfying the following. Let $(W \ni o)$ be a normal canonical hypersurface threefold germ in a smooth fourfold germ, G a finite group of order at most M acting faithfully on the embedded germ and fixing o, h a mark on $(W \ni o)$, and F a prime divisor over W. Assume that $(W, \frac{1}{m}\operatorname{div}(h))$ is lc near o, that*

$$b := \operatorname{mld}\Big(\Big(W \ni o, \frac{1}{m}\operatorname{div}(h)\Big)/G\Big) > 0,$$

and that F is a computing lift of b with $\operatorname{mult}_F \operatorname{Jac}_W \leq N$. Then there exist a G-equivariant proper birational morphism $g : Y \to W$ and a smooth G-invariant open subset $U \subset Y$ containing the generic points of the orbit of $Z := \operatorname{center}_Y F$ with the following properties.

(1) $K_{U/W} := K_U - g^* K_W$ is an integral Cartier divisor on U with

$$K_{U/W} \geq 0, \quad g(\operatorname{Supp} K_{U/W}) \subset \{o\}, \quad \text{and} \quad \operatorname{mult}_F K_{U/W} \leq N.$$

(2) For any prime divisor P over Y with center Z, we have

$$a(P, W, 0) = a(P, U, 0) + \operatorname{mult}_P K_{U/W}.$$

(3) If $\dim Z \geq 1$, *then some computing lift* P *of* b *satisfies*

$$\frac{a(P,W,0)}{e_P} \leq L \quad \textit{and} \quad \frac{\operatorname{mult}_P h}{m e_P} \leq L.$$

(4) If Z *is a closed point, then* $Z \in U$, *the coefficients of* $\frac{1}{m}\operatorname{div}(g^*h) - K_{U/W}$ *near* Z *belong to* $\frac{1}{m}\mathbb{Z} \cap [-N, 1]$, *and* $\operatorname{mult}_F \mathfrak{m}_{U,Z} \leq 3M/\eta_m$.

Set-up 15.4. Let m, M, B_0, G, $(W_i \ni o_i)$, h_i, and (f_*, h_*) be as follows.

(1) m and M are positive integers and $B_0 > 0$ is a real number.
(2) $\mathcal{X} := \operatorname{Spec} \mathbb{C}[[x_1, \dots, x_4]]$, and G is a finite abelian group of order at most M acting diagonally on $\mathcal{X}$.
(3) $(W_i \ni o_i) = (f_i = 0) \subset \mathcal{X}$, $i \geq 1$, are isolated Gorenstein terminal threefold germs, where o_i is the closed point of $\mathcal{X}$, the f_i are G-semi-invariant with a character independent of i, and G acts faithfully on each W_i.
(4) The h_i are G-semi-invariant elements of $\mathbb{C}[[x_1, \dots, x_4]]$ with a character independent of i, non-zero on W_i, such that $(W_i, \frac{1}{m}\operatorname{div}(h_i))$ is klt near o_i and $0 < b_i \leq B_0$, where $b_i := \operatorname{mld}((W_i \ni o_i, \frac{1}{m}\operatorname{div}(h_i))/G)$.
(5) (f_*, h_*) is a compatible generic limit of the (f_i, h_i) along $\mathbb{N}^+$, with coefficient field $\Bbbk$ (Definition 2.5), $\mathcal{X}_* := \operatorname{Spec} \Bbbk[[x_1, \dots, x_4]]$, and $W_* := (f_* = 0) \subset \mathcal{X}_*$.
(6) W_* is normal and canonical.
(7) $(W_*, \frac{1}{m}\operatorname{div}(h_*))$ is lc and not klt, with positive minimal log discrepancy at the closed point.

Proposition 15.5. *Notation and conditions as in Set-up 15.4. Then the following hold.*

(1) If no lc center of $(W_*, \frac{1}{m}\operatorname{div}(h_*))$ *is contained in* $\operatorname{Sing}(W_*)$, *then there exists a rational number* $\epsilon > 0$ *such that*

$$\left(W_i, \frac{1}{m}\operatorname{div}(h_i), \operatorname{Jac}^{\epsilon}_{W_i}\right) \textit{ is lc near } o_i$$

for infinitely many i, *and for any such* i, *any computing lift* F *of* b_i *satisfies*

$$\operatorname{mult}_F \operatorname{Jac}_{W_i} \leq \left\lceil \frac{MB_0}{\epsilon} \right\rceil.$$

(2) If $(W_*, \frac{1}{m}\operatorname{div}(h_*))$ *has a coefficient-one component through the closed point, then one of the following holds.*
  (i) The minimal lc center of $(W_*, \frac{1}{m}\operatorname{div}(h_*))$ *is not contained in* $\operatorname{Sing}(W_*)$, *so that no lc center of* $(W_*, \frac{1}{m}\operatorname{div}(h_*))$ *is contained in* $\operatorname{Sing}(W_*)$.
  (ii) The minimal lc center of $(W_*, \frac{1}{m}\operatorname{div}(h_*))$ *is a regular* G*-invariant curve* $C \subset \operatorname{Sing}(W_*)$ *contained in each coefficient-one component through the closed point,* $(W_*, \frac{1}{m}\operatorname{div}(h_*))$ *is not plt at the generic point of* C, *and if the generic transverse surface of* W_* *along* C *is Du Val, then the generic transverse type of* W_* *along* C *is* A_n *or* D_n.

Proposition 15.6. *Let* m, M, N, N_0 *be positive integers and* $N_1 := 2(N + 2NN_0)$. *Then there exists a real number* $L > 0$ *depending only on* m, M, N, *and* N_0 *satisfying the following. Let* $(W \ni o)$ *be a normal Gorenstein canonical threefold germ with a faithful action of a finite abelian group* G *of order at most* M *fixing* o, h *a mark on* $(W \ni o)$, *and* F *a prime divisor over* W. *Let* $g_0 : Y_0 \to W$ *be a* G*-equivariant proper birational morphism such that* $z_0 := \operatorname{center}_{Y_0} F$ *is a closed point. Let* $U_0 \subset Y_0$ *be a smooth* G*-invariant open neighborhood of the orbit of* z_0. *Assume the following.*

(i) $(W, \frac{1}{m}\operatorname{div}(h))$ *is klt near* o, $b := \operatorname{mld}((W \ni o, \frac{1}{m}\operatorname{div}(h))/G) > 0$, *and* F *is a computing lift of* b.
(ii) $K_{U_0/W} := K_{U_0} - g_0^* K_W$ *is an integral divisor with* $K_{U_0/W} \geq 0$ *and* $g_0(\operatorname{Supp} K_{U_0/W}) \subset \{o\}$.
(iii) $\operatorname{mult}_F K_{U_0/W} \leq N$ *and* $\operatorname{mult}_F \mathfrak{m}_o \leq N_0$.

Then either some computing lift P of b satisfies $a(P,W,0) \le L$, or there exist a G-equivariant proper birational morphism $g : Y \to W$ and a smooth G-invariant open neighborhood $U \subset Y$ of the orbit of the closed point $z := \operatorname{center}_Y F$ with the following properties.

(1) Near z, $K_{U/W} := K_U - g^*K_W = \alpha S$ for a smooth prime divisor S with $g(S) = o$ and an integer $1 \le \alpha \le N_1$. Moreover,

$$\alpha \operatorname{mult}_F S \le N_1 \quad \text{and} \quad \operatorname{mult}_F \mathfrak{m}_{U,z} \le N_0.$$

(2) There exist regular G_z-eigenparameters s, x_1, x_2 at z, where G_z is the stabilizer of z, with $S = (s = 0)$, and $g^*h = s^n h_U$ for a regular G_z-semi-invariant function h_U not divisible by s and an integer $0 \le n \le m(\alpha+1) - 1$. Let

$$B_U := \frac{1}{m}\operatorname{div}(g^*h) - \alpha S = \Big(\frac{n}{m} - \alpha\Big)S + \frac{1}{m}\operatorname{div}(h_U) \quad \text{near } z,$$

where $\frac{n}{m} - \alpha \in \frac{1}{m}\mathbb{Z} \cap [-N_1, 1)$.

(3) For any prime divisor P over U with center z, we have $I_{G_z}(P) = I_G(P)$,

$$a(P,W,0) = a(P,U,0) + \alpha \operatorname{mult}_P s \le (1+N_1)a(P,U,0),$$
$$\operatorname{mult}_P(g^*h) = n \operatorname{mult}_P s + \operatorname{mult}_P h_U,$$

and $a(P,W,\frac{1}{m}\operatorname{div}(h)) = a(P,U,B_U)$.

(4) $\operatorname{mld}((U \ni z, B_U)/G_z) = b$, and any computing lift of $\operatorname{mld}((U \ni z, B_U)/G_z)$ is a computing lift of b.

(5) If $\eta > 0$ and

$$a\Big(P, W, \frac{1}{m}\operatorname{div}(h)\Big) \ge \eta \operatorname{mult}_P \mathfrak{m}_o$$

for any prime divisor P over W with center o, then $a(P,U,B_U) \ge \eta \operatorname{mult}_P s$ for any prime divisor P over U with center z.

Theorem 15.7. *Let m, M, R' be positive integers and $A_0 \ge 0$, $B_0 > 0$, $\eta > 0$ real numbers. Then there exists a real number $L > 0$ depending only on m, M, R', A_0, B_0, and η satisfying the following. Let $(V \ni o)$ be a smooth threefold germ with a faithful action of a finite abelian group G of order at most M and eigenparameters s, p, q, let h be a mark on $(V \ni o)$, $S := (s = 0)$, and $\beta \in \frac{1}{m}\mathbb{Z} \cap [-A_0, 1)$. Assume the following.*

(i) $(V, \frac{1}{m}\operatorname{div}(h) + \beta S)$ is sub-klt and

$$b := \operatorname{mld}\Big(\Big(V \ni o, \frac{1}{m}\operatorname{div}(h) + \beta S\Big)/G\Big) \in (0, B_0].$$

(ii) $a(P,V,\frac{1}{m}\operatorname{div}(h) + \beta S) \ge \eta \operatorname{mult}_P s$ for any prime divisor P over V with center o.

(iii) Some computing lift F of b satisfies $\operatorname{mult}_F s \le R'$.

Then some computing lift P of b satisfies

$$a(P,V,0) \le L.$$

Theorem 15.8. *(1) Let m, M, N be positive integers. Then there exists a real number $L > 0$ depending only on m, M, and N satisfying the following. Let $(W \ni o)$ be a normal Gorenstein canonical hypersurface threefold germ in a smooth fourfold germ, G a finite abelian group of order at most M acting faithfully on the embedded germ and fixing o, and h a mark on $(W \ni o)$. Let q be the radial function of $(W \ni o, G, h, m)$ (Definition 4.1). Assume that $(W, \frac{1}{m}\operatorname{div}(h))$ is klt near o, that $b := q(1) > 0$, and that some computing lift F of b satisfies $\operatorname{mult}_F \operatorname{Jac}_W \le N$. Then some computing lift P of b satisfies*

$$a(P,W,0) \le L \quad \text{and} \quad \operatorname{mult}_P h \le mL,$$

and

$$b \le q(t) \le b + L(1-t) \quad \text{for any } t \in [0,1].$$

(2) *Notation and conditions as in Set-up 15.4. Assume that each* $(W_i \ni o_i) \subset \mathcal{X}$ *is the completion of an algebraic germ* $(W_i' \ni o_i')$ *in a completed equivariant presentation for* G*, that* h_i *is the expansion of a mark* h_i' *on* $(W_i' \ni o_i')$*, and that no lc center of* $(W_*, \frac{1}{m}\operatorname{div}(h_*))$ *is contained in* $\operatorname{Sing}(W_*)$*, and let* q_i *be the radial function of* $(W_i \ni o_i, G, h_i, m)$*. Then there exists a real number* $L > 0$ *such that, for infinitely many* i*, some computing lift* P_i *of* b_i *satisfies*

$$a(P_i, W_i, 0) \le L \quad \text{and} \quad \operatorname{mult}_{P_i} h_i \le mL,$$

and

$$b_i \le q_i(t) \le b_i + L(1-t) \quad \text{for any } t \in [0,1].$$

Proof. We prove (1). Let

$$N_0 := \left\lceil \frac{3M}{\eta_m} \right\rceil \quad \text{and} \quad N_1 := 2(N + 2NN_0).$$

Let L_1 be the constant L of Proposition 15.3 for (m, M, N), L_2 the constant L of Proposition 15.6 for (m, M, N, N_0), and L_3 the constant L of Theorem 15.7 for

$$(m, M, R', A_0, B_0, \eta) = (m, M, N_1, N_1, 3, \eta_m),$$

and let

$$L := \max(ML_1, L_2, (1+N_1)L_3).$$

Let $(W \ni o)$, G, h, q, b, and F be as in (1). We first show that some computing lift P of b satisfies

$$a(P, W, 0) \le L. \tag{15.1}$$

By Theorem 5.1(2),

$$a\Big(Q, W, \frac{1}{m}\operatorname{div}(h)\Big) \ge \eta_m \operatorname{mult}_Q \mathfrak{m}_o \quad \text{for any prime divisor } Q \text{ over } W \text{ with center } o. \tag{15.2}$$

By Theorem 5.1(3), $b \le 3$ and

$$\operatorname{mult}_F \mathfrak{m}_o \le \frac{3M}{\eta_m} \le N_0.$$

By Proposition 15.3, there exist a G-equivariant proper birational morphism $g_0 : Y_0 \to W$ and a smooth G-invariant open subset $U_0 \subset Y_0$ containing the generic points of the orbit of

$$Z_0 := \operatorname{center}_{Y_0} F$$

with properties (1)–(4) of Proposition 15.3. If $\dim Z_0 \ge 1$, then Proposition 15.3(3) gives a computing lift P of b with

$$a(P, W, 0) \le e_P L_1 \le ML_1 \le L,$$

since $e_P \le |G| \le M$.

Assume that $z_0 := Z_0$ is a closed point. By Proposition 15.3(4), $z_0 \in U_0$, so the G-invariant open subset U_0 is a neighborhood of the orbit of z_0. By Proposition 15.3(1) and $\operatorname{mult}_F \mathfrak{m}_o \le N_0$, the morphism g_0 and the open subset U_0 satisfy the assumptions of Proposition 15.6 for (m, M, N, N_0). If some computing lift P of b satisfies $a(P, W, 0) \le L_2$, then (15.1) holds. Otherwise let $g : Y \to W$, U, z, G_z, α, S, s, x_1, x_2, h_U, n, and B_U be the objects of Proposition 15.6(1)–(2).

The hypotheses of Theorem 15.7 hold for $(U \ni z)$, G_z, s, x_1, x_2, h_U, $\frac{n}{m} - \alpha$, and

$$(R', A_0, B_0, \eta) = (N_1, N_1, 3, \eta_m),$$

since

- near z, $K_U + B_U = g^*(K_W + \frac{1}{m}\operatorname{div}(h))$ by Proposition 15.6(1)(2), so (U, B_U) is sub-klt near z,
- by (15.2) and Proposition 15.6(5) with $\eta = \eta_m$, $a(Q, U, B_U) \ge \eta_m \operatorname{mult}_Q s$ for any prime divisor Q over U with center z, and $\operatorname{mult}_F s = \operatorname{mult}_F S \le N_1$ by Proposition 15.6(1), and
- by Proposition 15.6(3)(4), $I_{G_z}(F) = I_G(F)$ and $a(F, U, B_U) = a(F, W, \frac{1}{m}\operatorname{div}(h)) = e_F b$, so F is a computing lift of $\operatorname{mld}((U \ni z, B_U)/G_z) = b \in (0, 3]$.

By Theorem 15.7, some computing lift P of $\mathrm{mld}((U \ni z, B_U)/G_z)$ satisfies $a(P,U,0) \le L_3$. By Proposition 15.6(4), P is a computing lift of b. Since the center of P on U is z, Proposition 15.6(3) implies

$$a(P,W,0) \le (1+N_1)a(P,U,0) \le (1+N_1)L_3 \le L.$$

Thus (15.1) holds in any case.

Let P be a computing lift of b satisfying (15.1). Since P computes b,

$$a(P,W,0) - \frac{1}{m}\operatorname{mult}_P h = a\Big(P, W, \frac{1}{m}\operatorname{div}(h)\Big) = e_P b > 0,$$

so $\operatorname{mult}_P h < ma(P,W,0) \le mL$. Since $e_P \ge 1$, we have $a(P,W,0)/e_P \le L$, and Lemma 4.3(1) with $\lambda = L$ implies $b \le q(t) \le b + L(1-t)$ for any $t \in [0,1]$. This proves (1).

We prove (2). By Proposition 15.5(1), there exist a rational number $\epsilon > 0$ and an infinite subset $\mathcal{I} \subset \mathbb{N}^+$ such that for any $i \in \mathcal{I}$, any computing lift F of b_i satisfies

$$\operatorname{mult}_F \operatorname{Jac}_{W_i} \le N := \Big\lceil \frac{MB_0}{\epsilon} \Big\rceil.$$

Let L be the constant of (1) for (m, M, N), and let $i \in \mathcal{I}$. By Lemma 2.12(2)(3) and Definition 5.2, q_i, b_i, and the computing lifts of b_i with their log discrepancies, inertia groups, orders of h_i, and Jacobian orders are those of $(W_i' \ni o_i')$, G, and h_i', and by Set-up 15.4(2)–(4) and [dFEM11, Proposition 2.11(i)], the hypotheses of (1) other than the existence of F hold for $(W_i' \ni o_i')$, G, and h_i'. By Lemma 4.2(1), $b_i = q_i(1)$ has a computing lift F_i, and $\operatorname{mult}_{F_i} \operatorname{Jac}_{W_i} \le N$ since $i \in \mathcal{I}$. By (1), some computing lift P_i of b_i satisfies $a(P_i, W_i, 0) \le L$ and $\operatorname{mult}_{P_i} h_i \le mL$, and $b_i \le q_i(t) \le b_i + L(1-t)$ for any $t \in [0,1]$. This proves (2). □

16. Limits with a coefficient-one component

In this section we prove Theorem 16.7.

16.1. **Concentration.**

Theorem 16.1. *Notation and conditions as in Set-up 15.4. Assume that the minimal lc center of $(W_*, \frac{1}{m}\operatorname{div}(h_*))$ is a regular curve $C \subset \operatorname{Sing}(W_*)$. Let Σ be the union of the irreducible components of $\operatorname{Sing}(W_*)$ other than C, and $\mathfrak{b}_* \subset \mathcal{O}_{\mathcal{X}_*}$ the ideal of Σ, with $\mathfrak{b}_* = \mathcal{O}_{\mathcal{X}_*}$ if $\Sigma = \emptyset$. Then there exist a rational number $\epsilon > 0$, a positive integer n, and G-semi-invariant generators $g_1, \dots, g_{N_2}$ of $\mathfrak{b}_*$ such that the following hold for any G-eigenparameters (s, u_1, u_2, u_3) of $\mathcal{X}_*$ with $C = (u_1 = u_2 = u_3 = 0)$.*

(1) For any positive integer k, after passing to a subsequence, there exist G-eigenparameters $(s_i, u_{i1}, u_{i2}, u_{i3})$ of $\mathcal{X}$ and G-semi-invariant elements $g_{i,1}, \dots, g_{i,N_2} \in \mathcal{O}_{\mathcal{X}}$ satisfying the following, where $\mathfrak{b}_i := (g_{i,1}, \dots, g_{i,N_2})$. The tuples

$$(f_i, h_i, s_i, u_{i1}, u_{i2}, u_{i3}, g_{i,1}, \dots, g_{i,N_2})$$

have a compatible generic limit

$$(f_*, h_*, s_*', u_{*,1}', u_{*,2}', u_{*,3}', g_{*,1}', \dots, g_{*,N_2}')$$

with coefficient field $\Bbbk$ such that

$$s_* - s,\ u_{*,j}' - u_j,\ g_{*,j'}' - g_{j'} \in (x_1, \dots, x_4)^k$$

for any $j \in \{1,2,3\}$ and any $j' \in \{1, \dots, N_2\}$. Moreover, for any i in the subsequence the following hold.

(a) The triple

$$(W_i, \tfrac{1}{m}\operatorname{div}(h_i), (\mathfrak{b}_i \mathcal{O}_{W_i})^\epsilon \mathfrak{m}_{o_i}^\epsilon)$$

is lc near o_i.

(b) Any prime divisor F over W_i with center o_i and $a(F, W_i, \frac{1}{m}\operatorname{div}(h_i)) \le e_F B_0$ satisfies $\operatorname{mult}_F \mathfrak{m}_{o_i} \le \lceil MB_0/\epsilon \rceil$. If moreover $\operatorname{mult}_F((u_{i1}, u_{i2}, u_{i3})\mathcal{O}_{W_i}) \le k$, then

$$\operatorname{mult}_F \operatorname{Jac}_{W_i} \le n(k + \lceil MB_0/\epsilon \rceil).$$

(2) For any i let F_i be a computing lift of b_i, and assume that

$$\lim_{i\to+\infty} \operatorname{mult}_{F_i} \operatorname{Jac}_{W_i} = +\infty.$$

Then, after passing to a subsequence, there exist positive integers k_i with $\lim_{i\to+\infty} k_i = +\infty$ and, for any i, G-eigenparameters $(s_i, u_{i1}, u_{i2}, u_{i3})$ of $\mathcal{X}$ and G-semi-invariant elements $g_{i,1}, \dots, g_{i,N_2} \in \mathcal{O}_{\mathcal{X}}$ satisfying (1)(a) and (1)(b) with $k = k_i$ and $\mathfrak{b}_i := (g_{i,1}, \dots, g_{i,N_2})$, such that

$$\operatorname{mult}_{F_i} s_i \le \lceil MB_0/\epsilon \rceil \quad \text{and} \quad \lim_{i\to+\infty} \min_j \operatorname{mult}_{F_i} u_{ij} = +\infty.$$

If moreover

$$\lim_{i\to+\infty} \min\{\operatorname{mult}_F \operatorname{Jac}_{W_i} \mid F \text{ is a computing lift of } b_i\} = +\infty,$$

then the subsequence and the data can be taken so that in addition $\operatorname{mult}_F s_i \le \lceil MB_0/\epsilon \rceil$ for any computing lift F of b_i and

$$\lim_{i\to+\infty} \min\{\operatorname{mult}_F u_{ij} \mid 1 \le j \le 3,\ F \text{ is a computing lift of } b_i\} = +\infty.$$

In each conclusion above, the subsequence and the data can moreover be taken so that $s_i, u_{i1}, u_{i2}, u_{i3}$ have the characters of s, u_1, u_2, u_3, respectively, and the tuples (f_i, h_i) written in $(s_i, u_{i1}, u_{i2}, u_{i3})$ have a compatible generic limit with coefficient field $\Bbbk$, which is (f_*, h_*) written in (s, u_1, u_2, u_3), where φ written in regular parameters $y = (y_1, \dots, y_4)$ is the power series φ^y in $t_1, \dots, t_4$ with $\varphi = \varphi^y(y_1, \dots, y_4)$, and G acts on $t_1, \dots, t_4$ by the characters of s, u_1, u_2, u_3.

Proof. Let $\mathfrak{n}_*$ be the maximal ideal of $\mathcal{O}_{\mathcal{X}_*}$, $D_* := \frac{1}{m}\operatorname{div}(h_*|_{W_*})$, and, for a prime divisor E over $\mathcal{X}_*$,

$$a(E) := a(E, \mathcal{X}_*, (f_*)^1 (h_*)^{1/m}).$$

By Set-up 15.4(2)–(5) and Lemma 13.4(2), $(\mathcal{X}_*, (f_*)^1 (h_*)^{1/m})$ is lc, and

$$a(E) \ge \eta_m \operatorname{mult}_E \mathfrak{n}_* > 0 \tag{16.1}$$

for any prime divisor E over $\mathcal{X}_*$ with center the closed point, where η_m is the constant of Theorem 5.1(2). We divide the proof into several steps.

Step 1. In this step, let $\Bbbk_0$ be an algebraically closed field of characteristic zero, $(X \ni o)$ a klt germ of a variety over $\Bbbk_0$, $\widehat{\mathcal{O}}_{X,o}$ the completion of $\mathcal{O}_{X,o}$ with maximal ideal $\mathfrak{m}_o$, and N_1 a positive integer. For a non-zero ideal $\mathfrak{a} \subset \widehat{\mathcal{O}}_{X,o}$ we write $\operatorname{lct}(\mathfrak{a})$ for the lc threshold of $\mathfrak{a}$ with respect to $\operatorname{Spec} \widehat{\mathcal{O}}_{X,o}$. We show that there exists a positive integer N_0 satisfying the following: for any non-zero ideals $\mathfrak{a}, \mathfrak{a}' \subset \widehat{\mathcal{O}}_{X,o}$ with

$$\mathfrak{a} + \mathfrak{m}_o^{N_0} = \mathfrak{a}' + \mathfrak{m}_o^{N_0} \quad \text{and} \quad \operatorname{lct}(\mathfrak{a}) \ge \frac{1}{N_1}, \tag{16.2}$$

we have $\operatorname{lct}(\mathfrak{a}') \ge \frac{1}{N_1}$.

If an ideal $\mathfrak{d} \subset \widehat{\mathcal{O}}_{X,o}$ contains a power of $\mathfrak{m}_o$, then $\mathfrak{d} = \mathfrak{d}_0 \widehat{\mathcal{O}}_{X,o}$ for an ideal $\mathfrak{d}_0 \subset \mathcal{O}_{X,o}$ with $\operatorname{lct}(\mathfrak{d}_0) = \operatorname{lct}(\mathfrak{d})$ [dFEM11, Proposition 2.11]. The lc thresholds of non-zero ideals of $\mathcal{O}_{X,o}$ satisfy the ACC [dFEM11, Theorem 1.1], so there exists a real number $\delta_1 > 0$ such that none of them is contained in $(\frac{1}{N_1} - \delta_1, \frac{1}{N_1})$. Let $N_0 := \lceil 2 \dim X/\delta_1 \rceil$, and let $\mathfrak{a}, \mathfrak{a}' \subset \widehat{\mathcal{O}}_{X,o}$ be non-zero ideals satisfying (16.2). We may assume $\mathfrak{a}, \mathfrak{a}' \subset \mathfrak{m}_o$. For any integer $N \ge N_0$,

$$(\mathfrak{a}' + \mathfrak{m}_o^N) + \mathfrak{m}_o^{N_0} = \mathfrak{a} + \mathfrak{m}_o^{N_0} = (\mathfrak{a} + \mathfrak{m}_o^N) + \mathfrak{m}_o^{N_0},$$

so by Theorem 2.23(2) with $\epsilon = \delta_1/2$ and $L = N_0$, which holds over $\Bbbk_0$ since [dFEM11] works over any algebraically closed field of characteristic zero,

$$\operatorname{lct}(\mathfrak{a}' + \mathfrak{m}_o^N) \ge \operatorname{lct}(\mathfrak{a} + \mathfrak{m}_o^N) - \frac{\delta_1}{2} \ge \operatorname{lct}(\mathfrak{a}) - \frac{\delta_1}{2} > \frac{1}{N_1} - \delta_1.$$

As $\mathfrak{a}' + \mathfrak{m}_o^N$ contains $\mathfrak{m}_o^N$, its lc threshold is the lc threshold of an ideal of $\mathcal{O}_{X,o}$, so it is not in $(\frac{1}{N_1} - \delta_1, \frac{1}{N_1})$ by the choice of δ_1, hence $\operatorname{lct}(\mathfrak{a}' + \mathfrak{m}_o^N) \geq \frac{1}{N_1}$. By Theorem 2.23(2) with $\epsilon = \dim X/N$ and $L = N$,

$$\operatorname{lct}(\mathfrak{a}') = \lim_{N \to +\infty} \operatorname{lct}(\mathfrak{a}' + \mathfrak{m}_o^N) \geq \frac{1}{N_1}.$$

Let moreover n_0 be a positive integer, $\mathcal{Y} := \operatorname{Spec} \Bbbk_0[[y_1, \dots, y_{n_0}]]$ with maximal ideal $\mathfrak{m}$, $p \in \mathfrak{m}$ a non-zero element such that $W := \operatorname{Spec} \Bbbk_0[[y_1, \dots, y_{n_0}]]/(p)$ is normal, klt and regular outside the closed point, and $\mathfrak{a} = \prod_j \mathfrak{a}_j^{\gamma_j}$ with finitely many ideals $\mathfrak{a}_j \not\subset (p)$ and rational numbers $\gamma_j > 0$. We show that

$$(\mathcal{Y}, (p)^1\mathfrak{a}) \text{ is lc if and only if } (W, \mathfrak{a}\mathcal{O}_W) \text{ is lc.} \tag{16.3}$$

Since W is regular outside the closed point, the Tjurina number

$$\tau(p) := \dim_{\Bbbk_0} \Bbbk_0[[y_1, \dots, y_{n_0}]] \Big/ \Big(p, \frac{\partial p}{\partial y_1}, \dots, \frac{\partial p}{\partial y_{n_0}}\Big)$$

is finite, and p is contact $(2\tau(p) - \operatorname{mult}_0 p + 2)$-determined [GP19, Theorem 4.8]. Thus the polynomial p_0 of degree at most $2\tau(p) - \operatorname{mult}_0 p + 2$ with $p - p_0 \in \mathfrak{m}^{2\tau(p) - \operatorname{mult}_0 p + 3}$ satisfies $(\varphi(p)) = (p_0)$ for an automorphism φ of $\Bbbk_0[[y_1, \dots, y_{n_0}]]$ over $\Bbbk_0$. Let $W_0 := (p_0 = 0) \subset \mathbb{A}^{n_0}_{\Bbbk_0}$ and $\widehat{W_0}$ its completion at 0. Then φ maps $(\mathcal{Y}, (p)^1\mathfrak{a})$ and $(W, \mathfrak{a}\mathcal{O}_W)$ isomorphically to $(\mathcal{Y}, (p_0)^1\varphi(\mathfrak{a}))$ and $(\widehat{W_0}, \varphi(\mathfrak{a})\mathcal{O}_{\widehat{W_0}})$, so W_0 is klt near 0 [dFEM11, Proposition 2.11]. Let N_1 be a positive integer with $N_1\gamma_j \in \mathbb{Z}$ for any j. Let N' be an integer not less than the integers N_0 of the first part of **Step 1** for N_1 and for $(\mathbb{A}^{n_0}_{\Bbbk_0} \ni 0)$ and $(W_0 \ni 0)$. Let $\mathfrak{a}'_j \subset \Bbbk_0[y_1, \dots, y_{n_0}]$ be ideals with

$$\mathfrak{a}'_j \Bbbk_0[[y_1, \dots, y_{n_0}]] + \mathfrak{m}^{N'} = \varphi(\mathfrak{a}_j) + \mathfrak{m}^{N'},$$

and $\mathfrak{a}' := \prod_j (\mathfrak{a}'_j)^{\gamma_j}$. Since $\varphi(\mathfrak{a}_j) \not\subset (p_0)$ and $\bigcap_{N'}((p_0) + \mathfrak{m}^{N'}) = (p_0)$, we have $\mathfrak{a}'_j \not\subset (p_0)$ for N' large. By the first part of **Step 1** for N_1 on $(\mathbb{A}^{n_0}_{\Bbbk_0} \ni 0)$ and on $(W_0 \ni 0)$, $(\mathcal{Y}, (p_0)^1\varphi(\mathfrak{a}))$ is lc if and only if $(\mathcal{Y}, (p_0)^1\mathfrak{a}')$ is lc, and $(\widehat{W_0}, \varphi(\mathfrak{a})\mathcal{O}_{\widehat{W_0}})$ is lc if and only if $(\widehat{W_0}, \mathfrak{a}'\mathcal{O}_{\widehat{W_0}})$ is lc. By [dFEM11, Proposition 2.11], $(\mathcal{Y}, (p_0)^1\mathfrak{a}')$ and $(\widehat{W_0}, \mathfrak{a}'\mathcal{O}_{\widehat{W_0}})$ are lc if and only if $(\mathbb{A}^{n_0}_{\Bbbk_0}, (p_0)^1\mathfrak{a}')$ and $(W_0, \mathfrak{a}'\mathcal{O}_{W_0})$ are lc near 0. The coefficients of p_0 and of finitely many generators of the $\mathfrak{a}'_j$ are contained in an algebraically closed subfield $\Bbbk_1 \subset \Bbbk_0$ of finite transcendence degree over $\mathbb{Q}$, the field $\Bbbk_1$ embeds into $\mathbb{C}$, and normality and lc are preserved under extension of the coefficient field [dFEM11, Lemma 2.14(ii)]. Thus inversion of adjunction [EMY03, Corollary 1.8], on an open neighbourhood of 0 in $\mathbb{A}^{n_0}$ on which W_0 is normal, shows that $(\mathbb{A}^{n_0}_{\Bbbk_0}, (p_0)^1\mathfrak{a}')$ is lc near 0 if and only if $(W_0, \mathfrak{a}'\mathcal{O}_{W_0})$ is lc near 0, which proves (16.3).

Step 2. In this step we assume that $\Sigma \neq \emptyset$, and we show that, for any prime divisor E over $\mathcal{X}_*$ whose center is an irreducible component of Σ,

$$a(E) > 0. \tag{16.4}$$

Since W_* is normal, $\dim \operatorname{Sing}(W_*) \leq 1$. The closed point is contained in C, so any irreducible component of Σ is a curve. Let Σ_1 be an irreducible component of Σ and ξ the generic point of Σ_1. The local ring $\mathcal{O}_{\mathcal{X}_*,\xi}$ is regular of dimension 3, so

$$\widehat{\mathcal{O}}_{\mathcal{X}_*,\xi} \cong \kappa(\xi)[[y_1, y_2, y_3]],$$

where $\kappa(\xi)$ is the residue field of ξ. Let $\overline{\kappa(\xi)}$ be an algebraic closure of $\kappa(\xi)$,

$$R_\xi := \overline{\kappa(\xi)}[[y_1, y_2, y_3]],$$

$\mathfrak{m}_\xi$ the maximal ideal of R_ξ, f_ξ and h_ξ the images of f_* and h_* in R_ξ, $T := \operatorname{Spec} R_\xi/(f_\xi)$, and $D_T := \frac{1}{m} \operatorname{div}(h_\xi|_T)$. The homomorphism $\mathcal{O}_{\mathcal{X}_*,\xi} \to R_\xi$, the completion followed by the extension of the coefficient field, is flat with regular fibers, so T is normal and regular at the points over the regular points of W_* [Mat89, Theorem 23.7 and Corollary to Theorem 23.9]. Since $T \to W_*$ is the base change of the regular homomorphism $\mathcal{O}_{\mathcal{X}_*,\xi} \to R_\xi$, log resolutions on $\mathcal{X}_*$ and on

W_* pull back to log resolutions on $\operatorname{Spec} R_\xi$ and on T (cf. [dFEM11, Remark 2.10]), and any prime divisor on the pulled-back resolution has, in the log pullback, the coefficient of the prime divisor whose preimage contains it: on $\mathcal{X}_*$ by [JM12, Lemma 1.2] and [Kaw15, Corollary 2.3], and on W_*, which is a Cartier divisor on the regular scheme $\mathcal{X}_*$ and hence Gorenstein, because the relative canonical divisors are determined by the dualizing sheaves, which commute with flat base change (cf. [dFEM11, Proposition 2.11 and Lemma 2.14]). Since W_* is normal and canonical, $\xi \in \operatorname{Sing}(W_*)$ and $\dim \operatorname{Sing}(W_*) \le 1$, T is a normal canonical surface germ with an isolated singularity at the closed point, and $h_\xi|_T \ne 0$ as $h_*|_{W_*} \ne 0$. Since Σ is reduced and ξ is not contained in any irreducible component of Σ other than Σ_1, $\mathfrak{b}_*\mathcal{O}_{\mathcal{X}_*,\xi}$ is the maximal ideal of $\mathcal{O}_{\mathcal{X}_*,\xi}$, hence

$$\mathfrak{b}_* R_\xi = \mathfrak{m}_\xi.$$

We first show that, for any prime divisor P over T with center the closed point,

$$(T, D_T) \text{ is lc, and } a(P, T, D_T) > 0. \tag{16.5}$$

By Lemma 13.5, C is contained in any lc center of (W_*, D_*). Since C and Σ_1 are distinct irreducible curves, $C \not\subset \Sigma_1$, so Σ_1 is not an lc center of (W_*, D_*). Let $\mu_W : Y_W \to W_*$ be a log resolution of (W_*, D_*) [dFEM11, Theorems 2.3–2.4]. No stratum of the coefficient-one part of the log pullback of (W_*, D_*) to Y_W has image Σ_1. By the base change property above, the base change of μ_W to T is a log resolution of (T, D_T), and any prime divisor on it has the coefficient of the prime divisor P' of Y_W whose preimage contains it. Any stratum of the coefficient-one part of the log pullback of (T, D_T) over the closed point of T is contained in the base change of a stratum on Y_W with image Σ_1. Thus there does not exist such a stratum over the closed point of T, which implies (16.5).

Since (T, D_T) is lc and $a(P, T, D_T) > 0$ for any prime divisor P over T with center the closed point, $(T, D_T, \mathfrak{m}_\xi^\rho)$ is lc for some rational number $\rho > 0$. Since T is canonical and regular outside the closed point and $h_\xi|_T \ne 0$, the second part of **Step 1** shows that $(\operatorname{Spec} R_\xi, (f_\xi)^1 (h_\xi)^{1/m} \mathfrak{m}_\xi^\rho)$ is lc, hence

$$a(P, \operatorname{Spec} R_\xi, (f_\xi)^1 (h_\xi)^{1/m}) \ge \rho \operatorname{mult}_P \mathfrak{m}_\xi > 0$$

for any prime divisor P over $\operatorname{Spec} R_\xi$ with center the closed point.

Finally let E be a prime divisor over $\mathcal{X}_*$ with center Σ_1, and $\mu_\mathcal{X} : Y_\mathcal{X} \to \mathcal{X}_*$ a log resolution of $(\mathcal{X}_*, (f_* h_*))$ on which E is a divisor [dFEM11, Theorems 2.3–2.4]. By the base change property above, the base change of $\mu_\mathcal{X}$ to $\operatorname{Spec} R_\xi$ is a log resolution of $(\operatorname{Spec} R_\xi, (f_\xi h_\xi))$, and any prime divisor P on it has, in the log pullback of $(\operatorname{Spec} R_\xi, (f_\xi)^1 (h_\xi)^{1/m})$, the coefficient of the prime divisor on $Y_\mathcal{X}$ whose preimage contains P. Moreover, the base change of E is a non-empty union of prime divisors P over $\operatorname{Spec} R_\xi$ with center the closed point and $a(P, \operatorname{Spec} R_\xi, (f_\xi)^1 (h_\xi)^{1/m}) = a(E)$. Thus $a(E) > 0$. As Σ_1 is arbitrary, (16.4) holds.

Step 3. In this step we construct ϵ, n, and $g_1, \dots, g_{N_2}$, which only depend on (f_*, h_*). Let $\mathfrak{c}_* \subset \mathcal{O}_{\mathcal{X}_*}$ be the ideal of C. From now on (s, u_1, u_2, u_3) denotes any G-eigenparameters of $\mathcal{X}_*$ with $C = (u_1 = u_2 = u_3 = 0)$. Since $(\mathcal{X}_*, (f_*)^1 (h_*)^{1/m})$ is lc and the zero locus of $\mathfrak{b}_* \mathfrak{n}_*$ is Σ union the closed point, (16.1) and (16.4) give a rational number $\epsilon > 0$ such that

$$(\mathcal{X}_*, (f_*)^1 (h_*)^{1/m} (\mathfrak{b}_* \mathfrak{n}_*)^\epsilon) \text{ is lc.} \tag{16.6}$$

Let

$$J_* := \left(f_*, \frac{\partial f_*}{\partial x_1}, \dots, \frac{\partial f_*}{\partial x_4} \right) \subset \mathcal{O}_{\mathcal{X}_*}.$$

By the Jacobian criterion, the zero locus of J_* is $\operatorname{Sing}(W_*) = C \cup \Sigma$, so

$$\sqrt{J_*} = \mathfrak{c}_* \cap \mathfrak{b}_* \supset \mathfrak{c}_* \mathfrak{b}_*.$$

Thus for some positive integer n,

$$(\mathfrak{c}_* \mathfrak{b}_*)^n \subset J_*. \tag{16.7}$$

Since f_* and h_* are G-semi-invariant, G preserves (W_*, D_*), so C is G-invariant by Lemma 13.5, and G preserves $\mathrm{Sing}(W_*)$, hence Σ. Thus $\mathfrak{b}_*$ is G-invariant, and applying the projections

$$\mathrm{pr}_\chi(\psi) := \frac{1}{|G|}\sum_{\sigma\in G}\chi(\sigma)^{-1}\sigma(\psi), \quad \psi \in \mathcal{O}_{\mathcal{X}_*},$$

for the characters χ of G to finitely many generators of $\mathfrak{b}_*$ gives G-semi-invariant generators $g_1, \dots, g_{N_2}$ of $\mathfrak{b}_*$ ($N_2 = 1$ and $g_1 = 1$ if $\Sigma = \emptyset$). Moreover $\mathfrak{c}_* = (u_1, u_2, u_3)$.

Step 4. In this step we fix a positive integer k and construct the data of (1). Let

$$r := \left\lceil \frac{MB_0}{\epsilon} \right\rceil,$$

N_1 a positive integer with $N_1/m, N_1\epsilon \in \mathbb{Z}$, N_0 the integer of the first part of **Step 1** for $(\mathbb{A}^4_{\mathbb{C}} \ni 0)$ and N_1, and

$$N := \max\{N_0, n(k+r)+1, \mathrm{mult}_0\, g_1 + 1, \dots, \mathrm{mult}_0\, g_{N_2} + 1\}.$$

By (16.7), for any monomial β of degree n in the products $u_j g_{j'}$, $1 \le j \le 3$, $1 \le j' \le N_2$, there exist $w_{\beta,0}, \dots, w_{\beta,4} \in \mathcal{O}_{\mathcal{X}_*}$ with

$$\beta = w_{\beta,0} f_* + \sum_{j''=1}^{4} w_{\beta,j''}\frac{\partial f_*}{\partial x_{j''}}. \tag{16.8}$$

Applying pr_χ for the character χ of β to both sides, we may assume that any $w_{\beta,j''}$ is G-semi-invariant, and replacing the $w_{\beta,j''}$ contained in $\mathfrak{n}_*^N$ by 0, we may assume that (16.8) holds modulo $\mathfrak{n}_*^N$ and that any non-zero $w_{\beta,j''}$ is not contained in $\mathfrak{n}_*^N$.

Let $\lambda \in \Bbbk^{N_3}$ be the vector of the coefficients, in degrees less than N, of s, the u_j, the $g_{j'}$, and the non-zero $w_{\beta,j''}$, where for s, u_j, $g_{j'}$, and $w_{\beta,j''}$ we only take the monomials with the character of s, u_j, $g_{j'}$, and $w_{\beta,j''}$, respectively. Let $\mathcal{J}_{N+1}$ be the affine space of $(N+1)$-jets of the pairs (f_i, h_i) (Definition 2.5). For a point Q of $\mathcal{J}_{N+1} \times \mathbb{A}^{N_3}$ over a field, let f^Q, h^Q be the polynomials of degree at most N given by the first factor, and $s^Q, u_j^Q, g_{j'}^Q, w_{\beta,j''}^Q$ the polynomials of degree less than N whose coefficients at the monomials taken for λ are the corresponding coordinates of the second factor and whose other coefficients are 0. Let β^Q be the monomial β evaluated at the $u_j^Q, g_{j'}^Q$, and

$$\mathfrak{a}^Q := (f^Q)^{N_1}(h^Q)^{N_1/m}\big((g_1^Q, \dots, g_{N_2}^Q)(x_1, \dots, x_4)\big)^{N_1\epsilon} + (x_1, \dots, x_4)^N.$$

Consider the following conditions on Q.

(C1) For any β,

$$\beta^Q - w_{\beta,0}^Q f^Q - \sum_{j''=1}^{4} w_{\beta,j''}^Q \frac{\partial f^Q}{\partial x_{j''}} \in (x_1, \dots, x_4)^N.$$

(C2) The linear parts of s^Q, u_1^Q, u_2^Q, u_3^Q are linearly independent. For any j', the coefficient of $g_{j'}^Q$ at one fixed monomial of degree less than N at which the coefficient of $g_{j'}$ is non-zero is non-zero. For any non-zero $w_{\beta,j''}$, the coefficient of $w_{\beta,j''}^Q$ at one fixed monomial of degree less than N at which the coefficient of $w_{\beta,j''}$ is non-zero is non-zero.

(C3) $\mathrm{lct}_0(\mathfrak{a}^Q) \ge \frac{1}{N_1}$.

Let $P_* := (\mathrm{jet}_{N+1}(f_*, h_*), \lambda)$. Then P_* satisfies (C1) by (16.8), and (C2) as (s, u_1, u_2, u_3) are regular parameters of $\mathcal{O}_{\mathcal{X}_*}$ and $\mathrm{mult}_0\, g_{j'} < N$ for any j'. Moreover,

$$\mathfrak{a}^{P_*}\mathcal{O}_{\mathcal{X}_*} = \mathfrak{a}_* + \mathfrak{n}_*^N \quad \text{for} \quad \mathfrak{a}_* := (f_*)^{N_1}(h_*)^{N_1/m}(\mathfrak{b}_*\mathfrak{n}_*)^{N_1\epsilon},$$

and by (16.6)

$$\mathrm{lct}(\mathfrak{a}_* + \mathfrak{n}_*^N) \ge \mathrm{lct}(\mathfrak{a}_*) \ge \frac{1}{N_1},$$

so P_* satisfies (C3) [dFEM11, Proposition 2.11].

Let $V \subset \mathcal{J}_{N+1} \times \mathbb{A}^{N_3}$ be the Zariski closure over $\mathbb{C}$ of the image of P_*. Then V is an irreducible variety with generic point the image of P_*. Condition (C1) is given by polynomials with integer coefficients vanishing at P_*, so it holds on V, and (C2) holds on a non-empty open subset of V. We show that condition (C3) also holds at any $\mathbb{C}$-point of a non-empty open subset of V. The point P_* induces an embedding of the function field $\mathbb{C}(V)$ into $\Bbbk$. Let $\mathfrak{a}^{\mathbb{C}(V)}$ be the ideal of $\mathbb{C}(V)[x_1, \dots, x_4]$ defined by the generic point of V. Then

$$\operatorname{lct}_0(\mathfrak{a}^{P_*}) = \operatorname{lct}_0(\mathfrak{a}^{\mathbb{C}(V)})$$

[dFEM11, Lemma 2.14]. By generic smoothness, a log resolution of $(\mathbb{A}^4, \mathfrak{a}^{\mathbb{C}(V)})$, defined over the function field of a variety V_1 finite and dominant over V, spreads out over a non-empty open subset of V_1 to log resolutions of the $(\mathbb{A}^4, \mathfrak{a}^Q)$ with equal discrepancies and equal multiplicities, so there exists a non-empty open subset $V^\circ \subset V$ with

$$\operatorname{lct}_0(\mathfrak{a}^Q) = \operatorname{lct}_0(\mathfrak{a}^{\mathbb{C}(V)}) \geq \frac{1}{N_1}$$

for any $\mathbb{C}$-point $Q \in V^\circ$. Thus there exists a polynomial Ψ with complex coefficients such that $\Psi(P_*) \neq 0$ and any $\mathbb{C}$-point Q of V with $\Psi(Q) \neq 0$ satisfies (C1)–(C3).

Next we realize $s, u_j, g_{j'}$ on the members. Let $\mathcal{I}$ be the infinite set along which (f_*, h_*) is a compatible generic limit of the (f_i, h_i). The polynomials s^{P_*}, $u_j^{P_*}$, $g_{j'}^{P_*}$, and $w_{\beta,j''}^{P_*}$ for the non-zero $w_{\beta,j''}$ are non-zero by (C2) and G-semi-invariant with the characters of s, u_j, $g_{j'}$, and $w_{\beta,j''}$, and

$$s^{P_*} - s,\ u_j^{P_*} - u_j,\ g_{j'}^{P_*} - g_{j'} \in \mathfrak{n}_*^N.$$

By Lemma 2.6(4), there exist an infinite subset $\mathcal{I}' \subset \mathcal{I}$ and, for $i \in \mathcal{I}'$, non-zero G-semi-invariant polynomials s_i, u_{ij}, $g_{i,j'}$, and $w_{i,\beta,j''}$ with the characters of s, u_j, $g_{j'}$, and $w_{\beta,j''}$ such that (f_*, h_*) together with the $s^{P_*}, u_j^{P_*}, g_{j'}^{P_*}, w_{\beta,j''}^{P_*}$ is a compatible generic limit of the joint tuples of the $f_i, h_i, s_i, u_{ij}, g_{i,j'}, w_{i,\beta,j''}$ along $\mathcal{I}'$, with coefficient field $\Bbbk$.

For $i \in \mathcal{I}'$ let $Q_i \in \mathcal{J}_{N+1} \times \mathbb{A}^{N_3}$ be the point given by $\operatorname{jet}_{N+1}(f_i, h_i)$ and by the coefficients of $s_i, u_{ij}, g_{i,j'}, w_{i,\beta,j''}$ at the monomials taken for λ. As $s_i, u_{ij}, g_{i,j'}, w_{i,\beta,j''}$ have the characters of $s, u_j, g_{j'}, w_{\beta,j''}$,

$$s_i - s^{Q_i},\ u_{ij} - u_j^{Q_i},\ g_{i,j'} - g_{j'}^{Q_i},\ w_{i,\beta,j''} - w_{\beta,j''}^{Q_i} \in \mathfrak{n}^N, \tag{16.9}$$

where $\mathfrak{n}$ is the maximal ideal of $\mathcal{O}_\mathcal{X}$. The linear projection π from the space of $(N+1)$-jets of the joint tuples to $\mathcal{J}_{N+1} \times \mathbb{A}^{N_3}$ sending the $(N+1)$-jet of the i-th tuple to Q_i sends the $(N+1)$-jet of the limit to P_*, so it maps the irreducible closed subset $\mathcal{Z}'_{N+1}$ (Z_{N+1} in Definition 2.5) for the joint limit into V, and $\mathcal{Z}'_{N+1} \cap \pi^{-1}(\Psi \neq 0)$ is a non-empty open subset of $\mathcal{Z}'_{N+1}$. Removing finitely many indices from $\mathcal{I}'$, we get $Q_i \in V$ and $\Psi(Q_i) \neq 0$, so Q_i satisfies (C1)–(C3), for any $i \in \mathcal{I}'$. Let

$$\mathfrak{b}_i := (g_{i,1}, \dots, g_{i,N_2}).$$

By (C2) and (16.9), $(s_i, u_{i1}, u_{i2}, u_{i3})$ are G-eigenparameters of $\mathcal{X}$. By Lemma 2.6(2), dropping the entries $w_{i,\beta,j''}$, the tuple

$$(f_*, h_*, s^{P_*}, u_1^{P_*}, u_2^{P_*}, u_3^{P_*}, g_1^{P_*}, \dots, g_{N_2}^{P_*})$$

is a compatible generic limit of the tuples

$$(f_i, h_i, s_i, u_{i1}, u_{i2}, u_{i3}, g_{i,1}, \dots, g_{i,N_2})$$

along $\mathcal{I}'$, with coefficient field $\Bbbk$. As $N > k$, this proves the first part of (1).

Step 5. In this step we prove (1)(a) and (1)(b) for $i \in \mathcal{I}'$. Let

$$\mathfrak{a}_i := (f_i)^{N_1} (h_i)^{N_1/m} (\mathfrak{b}_i \mathfrak{n})^{N_1 \epsilon} \subset \mathcal{O}_\mathcal{X}.$$

Since $f^{Q_i} - f_i, h^{Q_i} - h_i \in \mathfrak{n}^{N+1}$ and $g_{j'}^{Q_i} - g_{i,j'} \in \mathfrak{n}^N$ by (16.9),

$$\mathfrak{a}^{Q_i} \mathcal{O}_\mathcal{X} = \mathfrak{a}_i + \mathfrak{n}^N,$$

so $\operatorname{lct}(\mathfrak{a}_i+\mathfrak{n}^N)\geq\frac{1}{N_1}$ by (C3) and [dFEM11, Proposition 2.11]. As $N\geq N_0$, the first part of **Step 1** for $(\mathbb{A}^4_{\mathbb{C}}\ni 0)$ and N_1 shows that $\operatorname{lct}(\mathfrak{a}_i)\geq\frac{1}{N_1}$, that is,

$$(\mathcal{X},(f_i)^1(h_i)^{1/m}(\mathfrak{b}_i\mathfrak{n})^\epsilon)\text{ is lc.}\tag{16.10}$$

In particular $\mathfrak{b}_i\not\subset(f_i)$, as otherwise the prime divisor W_i of $\mathcal{X}$ would have log discrepancy at most $-\epsilon$ with respect to the pair in (16.10). Since W_i is klt and regular outside o_i, and $h_i\notin(f_i)$ and $\mathfrak{b}_i\mathfrak{n}\not\subset(f_i)$ as (f_i) is prime, the second part of **Step 1** shows that

$$(W_i,\tfrac{1}{m}\operatorname{div}(h_i),(\mathfrak{b}_i\mathcal{O}_{W_i})^\epsilon\mathfrak{m}_{o_i}^\epsilon)\text{ is lc near }o_i.$$

This proves (1)(a).

Let F be a prime divisor over W_i with center o_i and $a(F,W_i,\frac{1}{m}\operatorname{div}(h_i))\leq e_FB_0$. Since $e_F\leq|G|\leq M$, (1)(a) implies

$$\epsilon(\operatorname{mult}_F\mathfrak{b}_i+\operatorname{mult}_F\mathfrak{m}_{o_i})\leq a(F,W_i,\tfrac{1}{m}\operatorname{div}(h_i))\leq MB_0,$$

hence

$$\operatorname{mult}_F\mathfrak{b}_i+\operatorname{mult}_F\mathfrak{m}_{o_i}\leq r.\tag{16.11}$$

In particular $\operatorname{mult}_F\mathfrak{m}_{o_i}\leq r$. Let $\mathfrak{c}_i:=(u_{i1},u_{i2},u_{i3})\mathcal{O}_{W_i}$. The ideal $(\mathfrak{c}_i\mathfrak{b}_i\mathcal{O}_{W_i})^n$ is generated by the restrictions of the β_i, where β_i is the monomial β evaluated at the $u_{ij},g_{i,j'}$, and $\beta_i-\beta^{Q_i}\in\mathfrak{n}^N$ by (16.9). Since $f^{Q_i}-f_i\in\mathfrak{n}^{N+1}$, we have

$$\frac{\partial f^{Q_i}}{\partial x_{j''}}-\frac{\partial f_i}{\partial x_{j''}}\in\mathfrak{n}^N,$$

so (C1) implies

$$(\mathfrak{c}_i\mathfrak{b}_i\mathcal{O}_{W_i})^n\subset\operatorname{Jac}_{W_i}+\mathfrak{m}_{o_i}^N.\tag{16.12}$$

Assume that $\operatorname{mult}_F\mathfrak{c}_i\leq k$. By (16.12),

$$\min\{\operatorname{mult}_F\operatorname{Jac}_{W_i},N\operatorname{mult}_F\mathfrak{m}_{o_i}\}\leq n(\operatorname{mult}_F\mathfrak{c}_i+\operatorname{mult}_F\mathfrak{b}_i),$$

which is at most $n(k+r)<N\leq N\operatorname{mult}_F\mathfrak{m}_{o_i}$ by (16.11) and $\operatorname{mult}_F\mathfrak{m}_{o_i}\geq1$, so

$$\operatorname{mult}_F\operatorname{Jac}_{W_i}\leq n(k+r).$$

This proves (1)(b).

Step 6. In this step we prove (2). For any integer $k\geq r$, let $\mathcal{I}_k$ be the set $\mathcal{I}'$ and $(s_i^{(k)},u_{ij}^{(k)},g_{i,j'}^{(k)})_{i\in\mathcal{I}_k}$ the data constructed in **Step 4** for k, and let $s^{(k)},u_j^{(k)}$ be the polynomials $s^{P_*},u_j^{P_*}$ of **Step 4** for k. Since $N\geq n(k+r)+1\geq k+1$,

$$s^{(k)}-s,\ u_j^{(k)}-u_j\in\mathfrak{n}_*^{k+1}.$$

For G-eigenparameters $y=(y_1,\dots,y_4)$ of $\mathcal{X}$ or of $\mathcal{X}_*$ with the characters of s,u_1,u_2,u_3 and $\psi\in\mathcal{O}_{\mathcal{X}}$ or $\psi\in\mathcal{O}_{\mathcal{X}_*}$, we write ψ^y for ψ written in y as in (2). Then ψ^y is G-semi-invariant with the character of ψ if ψ is G-semi-invariant, and $\operatorname{mult}_0\psi^y=\operatorname{mult}_0\psi$. For $d\geq1$, the d-jet of ψ^y is a polynomial with rational coefficients in the coefficients of the $(d+1)$-jets of ψ and y, divided by a power of the determinant $\det(y)$ of the linear parts of y. So there exists a morphism ω_d, defined over $\mathbb{Q}$ on the open subset $(\det\neq0)$ of the space of $(d+1)$-jets of the tuples (ψ_1,ψ_2,y), with

$$\omega_d(\operatorname{jet}_{d+1}(\psi_1,\psi_2,y))=\operatorname{jet}_d(\psi_1^y,\psi_2^y).$$

Let

$$y_*:=(s,u_1,u_2,u_3),\quad y_*^{(k)}:=(s^{(k)},u_1^{(k)},u_2^{(k)},u_3^{(k)}),\quad\text{and}\quad y_i^{(k)}:=(s_i^{(k)},u_{i1}^{(k)},u_{i2}^{(k)},u_{i3}^{(k)}),$$

and for $d\geq1$ let $\mathcal{Z}_d$ be the Zariski closure over $\mathbb{C}$ of the point $\operatorname{jet}_d(f_*^{y_*},h_*^{y_*})$, an irreducible variety with generic point $\operatorname{jet}_d(f_*^{y_*},h_*^{y_*})$. Let $1\leq d\leq k$. Since $y_*^{(k)}-y_*\in\mathfrak{n}_*^{k+1}$, we have

$$f_*^{y_*^{(k)}}-f_*^{y_*},\ h_*^{y_*^{(k)}}-h_*^{y_*}\in(t_1,\dots,t_4)^{k+1},$$

so ω_d maps $\mathrm{jet}_{d+1}(f_*, h_*, y_*^{(k)})$ to the generic point of $\mathcal{Z}_d$. Let $\mathcal{Z}_{d+1}^{(k)}$ be the irreducible closed subset Z_{d+1} of Definition 2.5 for the compatible generic limit $(f_*, h_*, y_*^{(k)})$ of the tuples $(f_i, h_i, y_i^{(k)})$ along $\mathcal{I}_k$ (**Step 4** and Lemma 2.6(2)). Then ω_d maps $\mathcal{Z}_{d+1}^{(k)} \cap (\det \neq 0)$ into $\mathcal{Z}_d$ and its generic point to the generic point of $\mathcal{Z}_d$, so the preimage of any non-empty open subset $U \subset \mathcal{Z}_d$ is a non-empty open subset of $\mathcal{Z}_{d+1}^{(k)}$, and hence

$$\mathrm{jet}_d\left(f_i^{y_i^{(k)}}, h_i^{y_i^{(k)}}\right) \in U \quad \text{for all but finitely many } i \in \mathcal{I}_k. \tag{16.13}$$

For integers $d \geq 1$ and $c \geq 0$ let $\mathcal{P}_{d,c}$ be the $\mathbb{C}$-vector space of the restrictions to $\mathcal{Z}_d$ of the polynomials of degree at most c with complex coefficients on the space of d-jets, which is a finite-dimensional vector space. We choose indices $i_r < i_{r+1} < \cdots$ with $i_k \in \mathcal{I}_k$ inductively, and write $z_{d,k} := \mathrm{jet}_d(f_{i_k}^y, h_{i_k}^y)$ for $y = y_{i_k}^{(k)}$. Let $k \geq r$ and assume that we have $i_{k'}$ for $r \leq k' < k$. We choose $i_k \in \mathcal{I}_k$, with $i_k > i_{k-1}$ if $k > r$, satisfying the following.

- (O) $\mathrm{mult}_{F_{i_k}} \mathrm{Jac}_{W_{i_k}} > n(k+r)$. Under the additional assumption in the second part of (2), $\mathrm{mult}_F \mathrm{Jac}_{W_{i_k}} > n(k+r)$ for any computing lift F of b_{i_k}.
- (J) For any integers $1 \leq d < k$ and $0 \leq c < k$, $z_{d,k} \in \mathcal{Z}_d$, and for any subset $S \subset \{k' \mid \max\{d,c\} < k' < k\}$ with $\#S < \dim \mathcal{P}_{d,c}$, $z_{d,k}$ is not contained in the common zero locus in $\mathcal{Z}_d$ of

$$L_S := \{\phi \in \mathcal{P}_{d,c} \mid \phi(z_{d,k'}) = 0 \text{ for any } k' \in S\}.$$

Here $z_{d,k'} \in \mathcal{Z}_d$ for $k' \in S$ by (J) for k', and $\dim L_S \geq \dim \mathcal{P}_{d,c} - \#S > 0$, so the common zero locus of L_S is a proper closed subset of $\mathcal{Z}_d$. Thus (J) consists of finitely many conditions, each of the form $z_{d,k} \in U$ for a non-empty open subset $U \subset \mathcal{Z}_d$ with $d < k$, and by (16.13) all but finitely many $i \in \mathcal{I}_k$ satisfy (J). By the assumptions of (2), all but finitely many i satisfy (O). Hence i_k exists.

Fix $d \geq 1$ and $c \geq 0$. By induction on q, for any $k_1 < \cdots < k_q$ with $k_1 > \max\{d,c\}$ and $q \leq \dim \mathcal{P}_{d,c}$, the evaluations at $z_{d,k_1}, \ldots, z_{d,k_q}$ are linearly independent on $\mathcal{P}_{d,c}$: by (J) for k_q and $S = \{k_1, \ldots, k_{q-1}\}$, some $\phi \in L_S$ satisfies $\phi(z_{d,k_q}) \neq 0$. Hence any non-zero $\phi \in \mathcal{P}_{d,c}$ vanishes at $z_{d,k}$ for fewer than $\dim \mathcal{P}_{d,c}$ integers $k > \max\{d,c\}$. Any proper closed subset of $\mathcal{Z}_d$ is contained in the zero locus of a non-zero element of $\mathcal{P}_{d,c}$ for some c, so it contains $z_{d,k}$ for only finitely many k. As $z_{d,k} \in \mathcal{Z}_d$ for any $k > d$, any non-empty open subset of $\mathcal{Z}_d$ contains $z_{d,k}$ for all but finitely many k. Therefore, along the subsequence $(i_k)_{k \geq r}$, the tuples (f_{i_k}, h_{i_k}) written in $y_{i_k}^{(k)}$ have the compatible generic limit $(f_*^{y_*}, h_*^{y_*})$ with coefficient field $\Bbbk$. Since $s_{i_k}^{(k)}, u_{i_k j}^{(k)}$ have the characters of s, u_j by **Step 4**, this is the last sentence of (2).

We pass to the subsequence $(i_k)_{k \geq r}$, take $k_{i_k} := k$ and the data $(s_{i_k}^{(k)}, u_{i_k j}^{(k)}, g_{i_k, j'}^{(k)})$ at the index i_k, and drop the superscripts. Let F be a computing lift of b_{i_k} with $\mathrm{mult}_F \mathrm{Jac}_{W_{i_k}} > n(k+r)$. Then F has center o_{i_k} and

$$a(F, W_{i_k}, \tfrac{1}{m} \mathrm{div}(h_{i_k})) = e_F b_{i_k} \leq e_F B_0,$$

so $\mathrm{mult}_F \mathfrak{m}_{o_{i_k}} \leq r$ and $\min_j \mathrm{mult}_F u_{i_k j} > k$ by (1)(b). Since $(s_{i_k}, u_{i_k 1}, u_{i_k 2}, u_{i_k 3})$ are regular parameters of $\mathcal{O}_{\mathcal{X}}$, their restrictions to W_{i_k} generate $\mathfrak{m}_{o_{i_k}}$, so

$$\min\{\mathrm{mult}_F s_{i_k}, \min_j \mathrm{mult}_F u_{i_k j}\} = \mathrm{mult}_F \mathfrak{m}_{o_{i_k}} \leq r \leq k,$$

hence $\mathrm{mult}_F s_{i_k} \leq r$. By (O), F_{i_k} is such an F, which proves the first part of (2), and, under the additional assumption in the second part of (2), so is any computing lift of b_{i_k}, which proves (2). □

16.2. **One or two branches.**

Set-up 16.2. Let m, a, δ, M, $(W_i \ni o_i, G_i, h_i, \tau_i)$, G, $\Bbbk$, (f_*, h_*), and W_* be as follows.

(1) M is a positive integer, and $(W_i \ni o_i, G_i, h_i, \tau_i)$ is a radial failure sequence with constants (m, a, δ) (Definition 4.1) such that for any i, $|G_i| \leq M$, $b_i > 0$, $(W_i, \frac{1}{m}\operatorname{div}(h_i))$ is klt near o_i, and $(W_i \ni o_i)$ has quadratic rank 1 or 2.
(2) $G = G_i$ and (f_*, h_*), with coefficient field $\Bbbk$, are the group and the prepared limit of Construction-Theorem 13.9, after passing to the subsequence there, and

$$W_* := (f_* = 0) \subset \mathcal{X}_* := \operatorname{Spec} \Bbbk[[x_1, \dots, x_4]].$$

(3) W_* is normal, canonical and cDV.
(4) $(W_*, \frac{1}{m}\operatorname{div}(h_*))$ is lc and not klt, has positive minimal log discrepancy at the closed point, and has a coefficient-one component through the closed point.

Definition 16.3. Notation and conditions as in Set-up 16.2. Assume that the minimal lc center C of $(W_*, \frac{1}{m}\operatorname{div}(h_*))$ is a regular curve with $C \subset \operatorname{Sing}(W_*)$, let (s, u_1, u_2, u_3) be regular parameters of $\mathcal{X}_*$ with $C = (u_1 = u_2 = u_3 = 0)$, and let T be the generic transverse surface of W_* along C (Definition 15.1). Since W_* is canonical (Set-up 16.2(3)), restricting a resolution of W_* [Tem18, Theorem 1.1.9] over the generic point of C, completing, and extending the residue field to $\overline{\Bbbk((s))}$ [Kaw21, Remark 2.6] keeps all discrepancies non-negative, so T is canonical, hence a Du Val germ over $\overline{\Bbbk((s))}$ [Kol13, Theorem 2.29(2) and Example 3.26], and T is singular since $C \subset \operatorname{Sing}(W_*)$. Let

$$D_T := \frac{1}{m}\operatorname{div}(h_*|_T),$$

and let $\pi : T' \to T$ be the smooth uniformizing cover of T: T' is a smooth surface germ, and π is finite and unramified in codimension 1. A *coefficient-one branch* of $(W_*, \frac{1}{m}\operatorname{div}(h_*))$ along C is a prime divisor on T' with coefficient one in $\pi^* D_T$.

Proposition 16.4. *Let m, M, n be positive integers, and let $f_0, h_0 \in \mathbb{C}[x, y, z]$ be one of the following.*

Row A. $n \geq 2$, $f_0 = xy - z^n$, and $h_0 = z^m$.
Row D. $n \geq 4$, m is even, $f_0 = x^2 + y^2 z + z^{n-1}$, and $h_0 = z^{m/2}$.

Let $J := 1 + 4M(n + m + 2)$. Let $(\mathcal{X} \ni 0)$ be a smooth fourfold germ with regular parameters x, y, z, s at 0, and let G be a finite group of order at most M acting faithfully on $(\mathcal{X} \ni 0)$ such that x, y, z, s are G-semi-invariant. Let f, h be G-semi-invariant regular functions near 0 such that

$$f - f_0 \in (x, y, z, s)^J \quad \text{and} \quad h - h_0 \in (x, y, z, s)^J.$$

Let R be the order of the character χ_s of s. Assume that $W := (f = 0)$ is normal and integral near 0. Then the following hold.

(1) The pair

$$(W, \frac{1}{m}\operatorname{div}(h|_W) + \operatorname{div}(s|_W))$$

is lc near 0.

(2) $\operatorname{mld}((W \ni 0, \frac{1}{m}\operatorname{div}(h|_W))/G) = \frac{1}{R}$, and there exist a weighted blow-up $\phi : \mathcal{X}' \to \mathcal{X}$ with respect to x, y, z, s and a prime divisor F over W with center 0, an irreducible component of the intersection of $\phi_*^{-1} W$ with the exceptional divisor of ϕ, such that

$$\frac{a(F, W, \frac{1}{m}\operatorname{div}(h|_W))}{e_F} = \frac{1}{R}, \qquad \frac{a(F, W, 0)}{e_F} \leq 2 + \frac{1}{R}, \quad \text{and} \quad \frac{\operatorname{mult}_F h}{m e_F} \leq 2.$$

Theorem 16.5. *There does not exist a radial failure sequence satisfying Set-up 16.2 such that the minimal lc center of $(W_*, \frac{1}{m}\operatorname{div}(h_*))$ is a regular curve $C \subset \operatorname{Sing}(W_*)$ and $(W_*, \frac{1}{m}\operatorname{div}(h_*))$ has exactly one coefficient-one branch along C (Definition 16.3).*

Theorem 16.6. *There does not exist a radial failure sequence satisfying Set-up 16.2 such that the minimal lc center of $(W_*, \frac{1}{m}\operatorname{div}(h_*))$ is a regular curve $C \subset \operatorname{Sing}(W_*)$ and $(W_*, \frac{1}{m}\operatorname{div}(h_*))$ has exactly two coefficient-one branches along C (Definition 16.3).*

We prove Proposition 16.4 and Theorems 16.5 and 16.6 in Subsection F.1.

Theorem 16.7. *There does not exist a radial failure sequence satisfying Set-up 16.2.*

Proof. Suppose that a radial failure sequence $(W_i \ni o_i, G_i, h_i, \tau_i)$ satisfies Set-up 16.2, and let

$$D_* := \frac{1}{m}\operatorname{div}(h_*|_{W_*}).$$

By Set-up 16.2(2), $G = G_i$ for any i, and G acts linearly on the ambient coordinates $x_1, \dots, x_4$. By Definition 4.1(1), G is cyclic, so possibly replacing $x_1, \dots, x_4$ with linear combinations independent of i, we may assume that G acts diagonally. Then the data satisfy Set-up 15.4 with $B_0 = a$.

Let C be the minimal lc center of (W_*, D_*) (Lemma 13.5). By Set-up 16.2(4), D_* has a coefficient-one component through the closed point, so by Proposition 15.5(2), either no lc center of (W_*, D_*) is contained in $\operatorname{Sing}(W_*)$, or C is a regular curve with $C \subset \operatorname{Sing}(W_*)$ and C is contained in any coefficient-one component of D_* through the closed point. In the first case, by Theorem 15.8(2), there exists a real number L such that for infinitely many i some computing lift P_i of b_i satisfies $a(P_i, W_i, 0) \le L$, so $a(P_i, W_i, 0)/e_{P_i} \le L$, which contradicts Lemma 4.3(2).

In the second case, let T, D_T, and $\pi : T' \to T$ be as in Definition 16.3, and let o' be the closed point of T'. A log resolution of (W_*, D_*) induces a log resolution of (T, D_T) whose discrepancies are discrepancies of (W_*, D_*) [dFEM11, Proposition 2.11, Remark 2.13, Lemma 2.14 and Appendix A], so (T, D_T) is lc. Since π is unramified in codimension 1, $K_{T'} + \pi^* D_T = \pi^*(K_T + D_T)$, so the pair $(T', \pi^* D_T)$ is lc. In particular, any coefficient of $\pi^* D_T$ is ≤ 1.

We show that there exists a coefficient-one branch. Let S be a coefficient-one component of D_* through the closed point. Then $C \subset S$, so $S_T := S|_T$ is a non-zero divisor on T with positive integer coefficients and $D_T \ge S_T$. Since π is unramified in codimension 1, $\pi^* S_T$ is a non-zero divisor with positive integer coefficients, and $\pi^* D_T \ge \pi^* S_T$. Since any coefficient of $\pi^* D_T$ is ≤ 1, any prime component of $\pi^* S_T$ has coefficient one in $\pi^* D_T$, hence is a coefficient-one branch.

We show that the number of coefficient-one branches is ≤ 2. Write

$$\pi^* D_T = \sum_{j=1}^{N} c_j P_j,$$

where $P_1, \dots, P_N$ are distinct prime divisors on T' and $0 < c_j \le 1$ for any j. Let $\mu_j := \operatorname{mult}_{o'} P_j$, so that $\mu_j \ge 1$. Let E_1 be the exceptional prime divisor of the blow-up of o'. Then $a(E_1, T', 0) = 2$ and $\operatorname{mult}_{E_1} \pi^* D_T = \sum_{j=1}^{N} c_j \mu_j$. Since $(T', \pi^* D_T)$ is lc,

$$\sum_{j=1}^{N} c_j \mu_j \le 2.$$

Any j with $c_j = 1$ contributes $\mu_j \ge 1$ to the left-hand side, so

$$\#\{j \mid c_j = 1\} \le 2.$$

Therefore (W_*, D_*) has exactly one or exactly two coefficient-one branches along C. Since C is a regular curve with $C \subset \operatorname{Sing}(W_*)$, exactly one branch contradicts Theorem 16.5, and exactly two branches contradict Theorem 16.6. □

17. Transverse types D and E

In this section we exclude the generic transverse types D and E of W_* along C in Set-up 17.1 (Definition 15.1). The type A is treated in Section 18.

17.1. **The set-up, the splitting, and type cE.** The proofs of Theorem 17.4, Proposition 17.5, Theorem 17.6, and Proposition 17.7 are given in Appendix F.

Set-up 17.1. Let m, M, a, δ, $(W_i \ni o_i, G_i, h_i, \tau_i)$, G, $\Bbbk$, (f_*, h_*), $W_* \subset \mathcal{X}_*$, C, and (s, u_1, u_2, u_3) be as follows.

(1) m and M are positive integers, and $a > \delta > 0$ are real numbers.
(2) $(W_i \ni o_i, G_i, h_i, \tau_i)$ is a radial failure sequence with constants (m, a, δ) (Definition 4.1) such that for any i, $|G_i| \le M$, $b_i > 0$, $(W_i, \frac{1}{m}\operatorname{div}(h_i))$ is klt near o_i, and $(W_i \ni o_i)$ has quadratic rank 1.
(3) (f_*, h_*) is the prepared limit of type cD or cE of the sequence in (2) given by Construction-Theorem 13.9, after passing to a subsequence, with the group $G = G_i$ and the coefficient field $\Bbbk$, and
$$W_* := (f_* = 0) \subset \mathcal{X}_* := \operatorname{Spec} \Bbbk[[x_1, \dots, x_4]].$$
(4) $(W_*, \frac{1}{m}\operatorname{div}(h_*))$ has no coefficient-one component and has positive minimal log discrepancy at the closed point.
(5) $\operatorname{Nklt}(W_*, \frac{1}{m}\operatorname{div}(h_*))$ is a regular G-invariant curve $C \subset \operatorname{Sing}(W_*)$.
(6) (s, u_1, u_2, u_3) are G-eigenparameters of $\mathcal{X}_*$ such that
$$C = (u_1 = u_2 = u_3 = 0), \quad h_* \in (u_1, u_2, u_3) \text{ modulo } f_*, \quad \text{and} \quad f_* \in (u_1, u_2, u_3)^2.$$

In Set-up 17.1, by Construction-Theorems 13.9, 13.7 and 13.6, W_* is normal canonical cDV, f_* and h_* are G-semi-invariant (Definition 2.5), $(W_*, \frac{1}{m}\operatorname{div}(h_*))$ is lc, and the ambient pair $(\mathcal{X}_*, (f_*)^1(h_*)^{1/m})$ is lc and has positive minimal log discrepancy at the closed point. Since G is cyclic and acts diagonally and $(W_*, \frac{1}{m}\operatorname{div}(h_*))$ has no coefficient-one component, Lemma 13.5 shows that the minimal lc center C' of $(W_*, \frac{1}{m}\operatorname{div}(h_*))$ is a regular curve. As $C' \subset C$, we have $C' = C$, and eigenparameters as in Set-up 17.1(6) exist by Lemma 13.5(2).

Theorem 17.2. *In Set-up 17.1, the generic transverse type of W_* along C is A_n for some $n \ge 1$.*

Since W_* is canonical, restricting a resolution of W_* [Tem18, Theorem 1.1.9] over the generic point of C, completing, and extending the residue field to $\overline{\Bbbk((s))}$ [Kaw21, Remark 2.6] keeps all discrepancies non-negative, so the generic transverse surface of W_* along C is canonical, hence Du Val [Kol13, Theorem 2.29(2) and Example 3.26], and it is singular as $C \subset \operatorname{Sing}(W_*)$. Thus the generic transverse type of W_* along C is one of A_n $(n \ge 1)$, D_k $(k \ge 4)$, E_6, E_7, and E_8, and the types D and E are excluded as follows.

(1) If the prepared limit is of type cD, the type E does not occur (see the end of this section), and the types D_k are excluded by Theorem 17.11.
(2) If the prepared limit is of type cE, the types E_7 and E_8 are excluded by Theorem 17.4. The type E_6 is excluded by Theorem 17.6, applied to the charts of Proposition 17.5. The types D_k with $k \ge 5$ are excluded by Theorem 17.12, which starts from the charts of Proposition 17.7. The type D_4 is excluded by Theorem 17.14.

Lemma 17.3. *In Set-up 17.1, there exist G-eigenparameters (x, y, z, s) of $\mathcal{X}_*$, with the parameter s of Set-up 17.1(6), such that $C = (x = y = z = 0)$ and*
$$f_* = v \cdot \big(x^2 + g(y, z, s)\big)$$
for an invariant unit v and some $g \in (y, z)^2 \subset \Bbbk[[y, z, s]]$, where the character of x^2 is that of f_*, and the (y, z)-quadratic part Q of g has coefficients in $s\Bbbk[[s]]$. Moreover, exactly one of the following holds.

(1) The generic transverse type along C is A_n with $n \ge 2$, and $Q = s^\nu y^2$ for an integer $\nu \ge 1$.
(2) The generic transverse type along C is A_1, and $Q = s^{\nu_1} y^2 + s^{\nu_2} z^2$ or $Q = s^\nu yz$ for integers $\nu_1, \nu_2, \nu \ge 1$.
(3) The generic transverse type along C is of type D or E, and $g \in (y, z)^3$.

Proof. Let $f_*^{(2)}$ be the quadratic part of f_* at the closed point.

***Step 1**: $f_*^{(2)}$ has rank* 1. The quadratic parts of the equations f_i of the W_i have rank 1. The 3-jets of the tuples whose first entry has quadratic part of rank at most 1 form a closed subset $\mathcal{R} \subset \mathcal{J}_3$. If the irreducible closed subset $Z_3 \subset \mathcal{J}_3$ of Definition 2.5 were not contained in $\mathcal{R}$, then $Z_3 \backslash \mathcal{R}$ would be a non-empty open subset of Z_3 containing no 3-jet of the (f_i, h_i), which contradicts Definition 2.5. Thus $Z_3 \subset \mathcal{R}$, and since $\operatorname{jet}_3(f_*, h_*)$ maps to the generic point of Z_3, $\operatorname{rank} f_*^{(2)} \le 1$. Since W_* is cDV, $\operatorname{mult}_0 f_* = 2$, so $f_*^{(2)} \neq 0$.

***Step 2**: an eigenparameter for $f_*^{(2)}$.* Since $f_* \in (u_1, u_2, u_3)^2$, $f_*^{(2)}$ is a quadratic form in u_1, u_2, u_3. By **Step 1**, $f_*^{(2)} = \lambda \ell^2$ for some $\lambda \in \Bbbk^*$ and a linear form ℓ in u_1, u_2, u_3. For any $\gamma \in G$ we have $\gamma^* f_*^{(2)} = \chi_{f_*}(\gamma) f_*^{(2)}$, so $\gamma^* \ell \in \Bbbk \ell$, that is, ℓ is semi-invariant. Since G acts diagonally on u_1, u_2, u_3, every u_j occurring in ℓ has the character χ_ℓ. Possibly reordering indices and replacing u_1 with ℓ, we may assume that $f_*^{(2)} = \lambda u_1^2$. The ideal (u_1, u_2, u_3) does not change, and $\chi_{u_1}^2 = \chi_{f_*}$.

***Step 3**: splitting off x.* Since $f_* \in (u_1, u_2, u_3)^2$ and $f_*^{(2)} = \lambda u_1^2$, we have

$$\frac{\partial f_*}{\partial u_1} = 2\lambda u_1 + (\text{terms of order at least } 2) \in (u_1, u_2, u_3).$$

By the formal implicit function theorem there exists a unique $\phi \in \Bbbk[[u_2, u_3, s]]$ with $\phi(0) = 0$ and $\frac{\partial f_*}{\partial u_1}(\phi, u_2, u_3, s) = 0$. As $\frac{\partial f_*}{\partial u_1} \in (u_1, u_2, u_3)$, $u_1 = 0$ solves the equation $\frac{\partial f_*}{\partial u_1} = 0$ when $u_2 = u_3 = 0$, so $\phi(0, 0, s) = 0$ by uniqueness, that is, $\phi \in (u_2, u_3)$. For any $\gamma \in G$,

$$\gamma^* \frac{\partial f_*}{\partial u_1} = \chi_{f_*}(\gamma) \chi_{u_1}(\gamma)^{-1} \frac{\partial f_*}{\partial u_1},$$

so $\chi_{u_1}(\gamma)^{-1} \gamma^* \phi$ is also a solution. By uniqueness, ϕ is semi-invariant and $\chi_\phi = \chi_{u_1}$. Let $\widetilde{x} := u_1 - \phi$. The Taylor expansion of f_* in $\widetilde{x}$ over $\Bbbk[[u_2, u_3, s]]$ has no term of degree 1 in $\widetilde{x}$ by the choice of ϕ, so

$$f_* = f_*(\phi, u_2, u_3, s) + \widetilde{x}^2 w$$

for some $w \in \Bbbk[[\widetilde{x}, u_2, u_3, s]]$ with $w(0) = \lambda$. Since $\widetilde{x}^2$ and f_* both have the character $\chi_{u_1}^2 = \chi_{f_*}$, w is invariant, and so is the square root $(w/\lambda)^{1/2}$ with constant term 1. Let

$$x := \widetilde{x}(w/\lambda)^{1/2}, \quad y := u_2, \quad z := u_3, \quad v := \lambda, \quad \text{and} \quad g := \lambda^{-1} f_*(\phi, y, z, s).$$

Then $f_* = v \cdot (x^2 + g)$, $\chi_x = \chi_{u_1}$, so $\chi_{x^2} = \chi_{f_*}$, and the ideals (x, y, z) and (u_1, u_2, u_3) are equal, so $C = (x = y = z = 0)$. Since $f_* \in (u_1, u_2, u_3)^2$ and $\phi \in (y, z)$, we have $g \in (y, z)^2$.

***Step 4**: the coefficients of Q.* Write

$$Q = a_{yy} y^2 + 2 a_{yz} yz + a_{zz} z^2, \qquad a_{yy}, a_{yz}, a_{zz} \in \Bbbk[[s]].$$

In the parameters (x, y, z, s), $f_*^{(2)} = v \cdot (x^2 + Q(y, z, 0))$. Since $\operatorname{rank} f_*^{(2)} = 1$, $Q(y, z, 0) = 0$, that is, $a_{yy}, a_{yz}, a_{zz} \in s\Bbbk[[s]]$. Since $\chi_g = \chi_{f_*}$, each of $a_{yy} y^2$, $a_{yz} yz$, and $a_{zz} z^2$ is zero or semi-invariant with the character χ_{f_*}.

***Step 5**: the transverse type.* The completion of the local ring of $\mathcal{X}_*$ at the generic point of C is $\Bbbk((s))[[x, y, z]]$, so the generic transverse surface of W_* along C is the germ at 0 of $(x^2 + g(y, z, s) = 0) \subset \operatorname{Spec} \overline{\Bbbk((s))}[[x, y, z]]$, with the coefficients of g in $\Bbbk[[s]]$. The quadratic part of $x^2 + g$ is $x^2 + Q$, of rank $1 + \operatorname{rank} Q$, where $\operatorname{rank} Q$ is the rank of Q as a quadratic form over $\overline{\Bbbk((s))}$. By the normal forms of Du Val singularities [Kol13, Example 3.26], the quadratic part of a Du Val germ has rank 3 for A_1, rank 2 for A_n with $n \ge 2$, and rank 1 for the types D and E. Thus

$$\operatorname{rank} Q = \begin{cases} 2 & \text{for } A_1, \\ 1 & \text{for } A_n,\ n \ge 2, \\ 0 & \text{for } D \text{ and } E. \end{cases}$$

For the types D and E, $Q = 0$, so $g \in (y,z)^3$, which is (3).

For the types A_n, $Q \neq 0$. Let $\nu \geq 1$ be the least s-order of a_{yy}, a_{yz}, and a_{zz}. The changes of coordinates below are linear in (y,z) with coefficients in $\Bbbk[[s]]$ and invertible, map y and z to eigenparameters with the characters χ_y and χ_z, respectively, and preserve s, the ideal (y,z), and the conditions $g \in (y,z)^2$ and $Q \in s\Bbbk[[s]][y,z]$.

Suppose first that a_{yy} or a_{zz} has s-order ν. Possibly interchanging y and z, a_{yy} has s-order ν. Then $a_{yz}/a_{yy} \in \Bbbk[[s]]$, and $(a_{yz}/a_{yy})z$ has the character χ_y by **Step 4**. Replacing y with $y + (a_{yz}/a_{yy})z$, we obtain

$$Q = a_{yy}y^2 + a'_{zz}z^2, \qquad a'_{zz} := a_{zz} - a_{yz}^2/a_{yy} \in s\Bbbk[[s]],$$

where $a'_{zz} \in s\Bbbk[[s]]$ since $a_{yz}^2/a_{yy} \in s^\nu\Bbbk[[s]]$. The leading term of a_{yy} is a scalar multiple of s^ν and has the character of a_{yy}, so a_{yy}/s^ν is an invariant unit. Replacing y with $(a_{yy}/s^\nu)^{1/2}y$, where the square root is the product of a square root of the constant term and the square root with constant term 1, we obtain $a_{yy} = s^\nu$. If $\operatorname{rank} Q = 1$, then $a'_{zz} = 0$, which is (1). If $\operatorname{rank} Q = 2$, then $a'_{zz} \neq 0$. Let $\nu_2 \geq 1$ be the s-order of a'_{zz}. The leading term of a'_{zz} is a scalar multiple of s^{ν_2} and has the character of a'_{zz}, so a'_{zz}/s^{ν_2} is an invariant unit, and replacing z with $(a'_{zz}/s^{\nu_2})^{1/2}z$, with the square root taken as for a_{yy}/s^ν, we obtain $Q = s^\nu y^2 + s^{\nu_2}z^2$, which is the first alternative of (2) with $\nu_1 := \nu$.

Suppose next that only a_{yz} has s-order ν. Let

$$a' := a_{yy}/s^\nu, \quad b' := a_{yz}/s^\nu, \quad \text{and} \quad c' := a_{zz}/s^\nu.$$

Then $a', c' \in s\Bbbk[[s]]$ and b' is a unit. Since $\chi_{b'yz} = \chi_{s^{-\nu}Q}$ by **Step 4** and $b'(0) \neq 0$, b' is invariant and $\chi_{s^{-\nu}Q} = \chi_y\chi_z$. Hence $\chi_{a'} = \chi_z\chi_y^{-1}$, $\chi_{c'} = \chi_y\chi_z^{-1}$, and $a'c'$ is invariant. The invariant series $b'^2 - a'c'$ is a unit with constant term $b'(0)^2$, so

$$e := b' + (b'^2 - a'c')^{1/2},$$

with the square root whose constant term is $b'(0)$, is an invariant unit with $e^2 - 2b'e + a'c' = 0$. Let $\mu_1 := c'/e$ and $\mu_2 := a'/e$, both in $s\Bbbk[[s]]$. Then

$$e(y + \mu_1 z)(z + \mu_2 y) = a'y^2 + e(1 + \mu_1\mu_2)yz + c'z^2 = a'y^2 + 2b'yz + c'z^2 = s^{-\nu}Q,$$

where $e(1 + \mu_1\mu_2) = e + a'c'/e = 2b'$. The functions $y + \mu_1 z$ and $e(z + \mu_2 y)$ are semi-invariant with the characters χ_y and χ_z, and the determinant $e(1 - \mu_1\mu_2)$ of the change is a unit. Replacing y with $y + \mu_1 z$ and z with $e(z + \mu_2 y)$, we obtain $Q = s^\nu yz$. Since $\operatorname{rank} Q = 2$, the type is A_1, which is the second alternative of (2).

The three cases are distinguished by the generic transverse type, so exactly one of (1), (2) and (3) holds. The lemma follows. □

Theorem 17.4. *There does not exist a radial failure sequence satisfying Set-up 17.1 whose prepared limit is of type cE and has generic transverse type E_7 or E_8 along C.*

Proposition 17.5. *In Set-up 17.1, assume that the prepared limit is of type cE and that the generic transverse type along C is E_6. Then there exist formal G-eigenparameters (x,y,z,s) with $C = (x = y = z = 0)$ in which*

$$f_* = x^2 + y^3 + s^\nu z^4 + yz^3\phi(z,s) + z^5\psi(z,s)$$

with $0 \leq \nu \leq 5$ and power series ϕ, ψ. Let $\mathcal{E} \subset \mathbb{N}^4$ be the set of the exponents α of the non-zero monomials $x^{\alpha_1}y^{\alpha_2}z^{\alpha_3}s^{\alpha_4}$ of the representative of h_* modulo f_* of x-degree at most 1, and let $w(\alpha) := 6\alpha_1 + 4\alpha_2 + 3\alpha_3$. Then $w(\alpha) \geq m$ for any $\alpha \in \mathcal{E}$, with equality for some α. Let $d_\nu := 1$ if ν is even and $d_\nu := 2$ if ν is odd, let $\nu' := d_\nu\nu/2$, and let

$$c := \min\{\nu'(3\alpha_1 + 2\alpha_2 + \alpha_3) + d_\nu\alpha_4 \mid \alpha \in \mathcal{E},\ w(\alpha) = m\}.$$

Then

$$\lceil d_\nu\nu m/6 \rceil \leq c < d_\nu m.$$

There exist an integer $r \geq 1$ and an infinite subfamily with the following properties, where f_i and h_i are written in G-eigenparameters of the ambient germ of W_i, again denoted (x, y, z, s), which have the characters of the parameters (x, y, z, s) of $\mathcal{X}_*$ above, u is a parameter with $u^{d_\nu} = s$, Y_i is the germ at the origin p_i of the hypersurface defined by

$$u^{-(12r+6\nu')} f_i\big(u^{6r+3\nu'}x', u^{4r+2\nu'}y', u^{3r+\nu'}z', u^{d_\nu}\big),$$

$$\Gamma_i := \{(\gamma, \xi) \in G_i \times \mathbb{C}^* \mid \xi^{d_\nu} = \chi_s(\gamma)\},$$

an abelian group of order $d_\nu|G_i| \leq 2M$ acting faithfully and diagonally on (x', y', z', u), and

$$l_i := h_i/u^{mr+c}, \quad S_i := (u = 0), \quad \beta := c/m - (d_\nu - 1),$$

$$\kappa := r + d_\nu - 1, \quad \text{and} \quad B_i := \tfrac{1}{m}\operatorname{div}(l_i) + \beta S_i.$$

(1) *$(Y_i \ni p_i)$ is an isolated Gorenstein terminal hypersurface germ, and the equations of the Y_i have a compatible generic limit $x'^2 + y'^3 + z'^4$ plus terms of transverse $(6, 4, 3)$-weight greater than 12, with coefficients in $\Bbbk'[[u]]$ for an algebraically closed field $\Bbbk' \supset \Bbbk$, where for $w = (w_x, w_y, w_z)$ the transverse w-weight of $x'^{j_1}y'^{j_2}z'^{j_3}u^{j_4}$ is $w_xj_1 + w_yj_2 + w_zj_3$.*

(2) *l_i is regular and does not contain S_i, $\operatorname{mld}((Y_i \ni p_i, B_i)/\Gamma_i) = b_i$, every prime divisor over Y_i whose valuation extends that of a computing lift of b_i has center p_i, and every computing lift of $\operatorname{mld}((Y_i \ni p_i, B_i)/\Gamma_i)$ restricts to a computing lift of b_i.*

(3) *Every prime divisor P over Y_i with center p_i restricts to a prime divisor F over W_i with center o_i, and, with ρ the ramification index of P over F and e_P, e_F the orders of the inertia groups in Γ_i, G_i, respectively,*

$$e_P = \rho e_F, \quad \rho a(F, W_i, 0) = a(P, Y_i, 0) + \kappa \operatorname{mult}_P u, \quad \text{and}$$

$$\rho \operatorname{mult}_F h_i = \operatorname{mult}_P l_i + (mr + c) \operatorname{mult}_P u.$$

(4) *There exists a positive integer R' such that $\operatorname{mult}_P u \leq R'$ for any i and any computing lift P of $\operatorname{mld}((Y_i \ni p_i, B_i)/\Gamma_i)$, and $\min\{\operatorname{mult}_P x', \operatorname{mult}_P y', \operatorname{mult}_P z'\} \to +\infty$ for any such P.*

(5) *The pairs formed by the equations of the Y_i and the representatives of the l_i modulo the equations of the Y_i of x'-degree at most 1 have a compatible generic limit (f'_*, l_*) with coefficients in $\Bbbk''[[u]]$ for an algebraically closed field $\Bbbk'' \supset \Bbbk'$, where f'_* is $x'^2 + y'^3 + z'^4$ plus terms of transverse $(6, 4, 3)$-weight greater than 12 as in (1). The lowest transverse $(6, 4, 3)$-weight part $\operatorname{in}_w(l_*)$ has weight m, is non-zero and has x'-degree at most 1, and the pair*

$$\big(x'^2 + y'^3 + z'^4 = 0, \tfrac{1}{m}\operatorname{div}(\operatorname{in}_w(l_*))\big)$$

over an algebraic closure of $\Bbbk''((u))$ is lc and has no coefficient-one component.

(6) *In (1)–(5), the parameters (x, y, z, s) can be taken to be regular functions near o_i on the ambient germ $(\mathcal{X}_i \ni o_i)$ of W_i, and f_i and h_i to be regular G-semi-invariant functions near o_i on $\mathcal{X}_i$, where f_i generates the ideal of W_i and h_i restricts to the mark on W_i. This can be done so that there exist a smooth fourfold germ $(\mathcal{Y}_i \ni p_i)$ with an action of Γ_i fixing p_i, on which x', y', z', u are regular Γ_i-eigenparameters at p_i, and a morphism $\widetilde{\varpi}_i : \mathcal{Y}_i \to \mathcal{X}_i$, equivariant for the projection $\Gamma_i \to G_i$, with*

$$x \circ \widetilde{\varpi}_i = u^{6r+3\nu'}x', \quad y \circ \widetilde{\varpi}_i = u^{4r+2\nu'}y', \quad z \circ \widetilde{\varpi}_i = u^{3r+\nu'}z', \quad \text{and} \quad s \circ \widetilde{\varpi}_i = u^{d_\nu},$$

such that $f'_i := u^{-(12r+6\nu')}(f_i \circ \widetilde{\varpi}_i)$ and $l'_i := u^{-(mr+c)}(h_i \circ \widetilde{\varpi}_i)$ are regular Γ_i-semi-invariant functions near p_i, $(Y_i \ni p_i)$ is the germ at p_i of the normal algebraic hypersurface $(f'_i = 0) \subset \mathcal{Y}_i$, whose completion at p_i is $\operatorname{Spec}\mathbb{C}[[x', y', z', u]]/(f'_i)$, $l_i = l'_i|_{Y_i}$, and the expansions in x', y', z', u of the pairs (f'_i, l'_i) have the compatible generic limit (f'_*, l_*) of (5).

Theorem 17.6. *Let f_0 be one of the weighted homogeneous polynomials of Proposition 7.7, or $x^2 + y^3 + z^3$, with the weights $w = (w_x, w_y, w_z)$ and the degree d given there ($w = (3, 2, 2)$ and $d = 6$ for $x^2 + y^3 + z^3$). Let m, M, and R' be positive integers, $L_0 > 0$ a real number, $\beta \in \frac{1}{m}\mathbb{Z}$,*

and G a finite abelian group of order at most M acting faithfully and diagonally on (x,y,z,u). For any i let $W_i=(f_i=0)$ be a G-invariant normal klt Gorenstein hypersurface threefold germ at 0 in a smooth fourfold germ with regular G-eigenparameters (x,y,z,u), where f_i and h_i are regular functions near 0 on the smooth fourfold germ and the restriction of h_i to W_i is a mark, and let

$$B_i := \frac{1}{m}\operatorname{div}(h_i)+\beta\operatorname{div}(u).$$

Assume that f_0 and the f_i are semi-invariant with one character, that the h_i are semi-invariant with one character, and that the following hold.

(1) $0<\operatorname{mld}((W_i\ni 0,B_i)/G)\le L_0$.

(2) There exist computing lifts F_i of $\operatorname{mld}((W_i\ni 0,B_i)/G)$ with $\operatorname{mult}_{F_i}u\le R'$ and

$$\min\{\operatorname{mult}_{F_i}x,\operatorname{mult}_{F_i}y,\operatorname{mult}_{F_i}z\}\to+\infty.$$

(3) The (f_i,h_i) have a compatible generic limit (f_*,h_*) with coefficient field $\Bbbk$ and coefficients in $\Bbbk[[u]]$ such that f_*-f_0 consists of terms of transverse w-weight greater than d, the lowest transverse w-weight part $\operatorname{in}_w(h_*)$ of h_* has weight m, is non-zero and has x-degree at most 1, and $(f_0=0,\frac{1}{m}\operatorname{div}(\operatorname{in}_w(h_*)))$ over an algebraic closure of $\Bbbk((u))$ is lc and has no coefficient-one component.

Then on an infinite subfamily some computing lift E_i of $\operatorname{mld}((W_i\ni 0,B_i)/G)$ satisfies $a(E_i,W_i,0)\le L$ for a constant L independent of i. Moreover, by Proposition 17.5 and Lemma 4.3(2), there does not exist a radial failure sequence satisfying Set-up 17.1 whose prepared limit is of type cE and has generic transverse type E_6 along C.

Proposition 17.7. *In Set-up 17.1, assume that the prepared limit is of type cE and that the generic transverse type along C is D_k with $k\ge 5$. Then there exist formal G-eigenparameters (x,y,z,s) with $C=(x=y=z=0)$, an integer $\nu\ge 1$ and $R\in(y,z)^4$ such that a unit multiple of f_* is*

$$x^2+y^3-3s^{2\nu}yz^2+2s^{3\nu}z^3+R,$$

and there exist integers $r\ge 1$ and $N\ge 0$ and an infinite subfamily with the following properties.

(1) A fixed chart

$$x=s^{3r+2\nu}X,\quad y=s^{2r+\nu}\big(\tfrac{2}{3}V+s^{\nu}Z\big),\quad \text{and}\quad z=s^{2r}\big(-\tfrac{1}{3}V+s^{\nu}Z\big),$$

in G-eigenparameters of the ambient germs of the W_i, again denoted (x,y,z,s), which have the characters of the parameters (x,y,z,s) above and in which the (f_i,h_i) have the compatible generic limit (f_*,h_*), defines normal Gorenstein canonical germs $(Y_i\ni 0)$ with the action of G_i and a fixed Du Val section of type D_n for an integer $n\ge 4$ independent of i, such that $S_i:=(s=0)$ is a reduced integral Cartier divisor, $h_i=s^Nl_i$ with l_i regular and not containing S_i, $\beta:=N/m-r\in\frac{1}{m}\mathbb{Z}$ satisfies $-r\le\beta<1$, and (Y_i,B_i) is sub-klt for $B_i:=\frac{1}{m}\operatorname{div}(l_i)+\beta S_i$. Moreover $\operatorname{mld}((Y_i\ni 0,B_i)/G_i)=b_i$, every computing lift of b_i has center 0 on Y_i, every computing lift of $\operatorname{mld}((Y_i\ni 0,B_i)/G_i)$ is a computing lift of b_i, every prime divisor F over Y_i with center 0 has equal inertia groups over Y_i and over W_i and satisfies

$$a(F,W_i,0)=a(F,Y_i,0)+r\operatorname{mult}_F s,\qquad \operatorname{mult}_F h_i=\operatorname{mult}_F l_i+N\operatorname{mult}_F s,$$

$$a\big(F,W_i,\tfrac{t}{m}\operatorname{div}(h_i)\big)=a\big(F,Y_i,tB_i+(t-1)rS_i\big)\quad \text{for } 0\le t\le 1,$$

and every computing lift F of b_i satisfies $\operatorname{mult}_F s/e_F\le 3/\eta_m$, where η_m is the constant of Theorem 5.1(2).

(2) The pairs formed by the equations of the Y_i and the l_i have a compatible generic limit (g_*,l_*) such that, in G-eigenparameters (X,V,Z,s) preserving s, $g_*=X^2+3V^2Z+\Psi(Z,s)$ with $\Psi\in Z^{k-1}\Bbbk[[Z,s]]$ of order $k-1$ in Z over $\Bbbk((s))$, and, over $\overline{\Bbbk((s))}$,

$$\Big((g_*=0),\frac{1}{m}\operatorname{div}(l_*)\Big)\cong\Big(T,\frac{1}{m}\operatorname{div}(h_*|_T)\Big)$$

for the generic transverse surface T of W_* along C (Definition 15.1). Every computing lift F of b_i satisfies $\min\{\operatorname{mult}_F X, \operatorname{mult}_F V, \operatorname{mult}_F Z\} \to +\infty$.

17.2. **Types** D**.** The results of this subsection are proved in Appendix F.

Proposition 17.8. *In Set-up 17.1, assume that the prepared limit is of type cD, that*

$$f_* = x^2 + zy^2 + \gamma(s)y + \phi(z,s)$$

in G-eigenparameters (x,y,z,s) with $\operatorname{mult}_0\gamma \geq 3$ *and* $\operatorname{mult}_0\phi \geq 4$*, and that the generic transverse type along C is D_k. Then C is the s-axis, $\gamma = 0$, and, after substituting $\mu(s)^{-1/2}y$ for y and $\mu(s)z$ for z, where μ is a G-invariant unit of $\Bbbk[[s]]$,*

$$\phi = z^{k-1}\bigl(s^\nu + z\phi_1(z,s)\bigr)$$

with $\nu \geq 0$ and $\phi_1 \in \Bbbk[[z,s]]$. Let $d := \lfloor \nu/2 \rfloor$ and $\nu_0 := \nu - 2d$. For any integer $N \geq 2k+8$ there exist an integer $r \geq N$, an infinite subfamily, and for each i in it a G-semi-invariant lift of the mark h_i to the ambient fourfold, again denoted h_i, with the following properties, where the chart below is taken in G-eigenparameters of the ambient germ of W_i, again denoted (x,y,z,s), which have the characters of the parameters (x,y,z,s) above after the substitution and in which the f_i and the expansions of the marks have the compatible generic limit (f_*, h_*), and $(Y_i \ni 0)$ is the germ at the origin of the point chart given by

$$x = s^{(k-1)r+d}x', \quad y = s^{(k-2)r+d}y', \quad \text{and} \quad z = s^{2r}z',$$

and

$$l_i := s^{-mr}h_i\bigl(s^{(k-1)r+d}x', s^{(k-2)r+d}y', s^{2r}z', s\bigr).$$

(1) *$(Y_i \ni 0)$ is an isolated Gorenstein terminal germ on which the induced action of G is faithful, defined in the coordinates (x',y',z',s) by $g_i = 0$ for*

$$g_i := x'^2 + z'y'^2 + s^{\nu_0}z'^{k-1} + R_i, \quad R_i \in (x',y',z',s)^N.$$

The function l_i is regular, $\operatorname{mult}_{S_i} l_i = c$ *for the prime divisor $S_i := (s=0) \subset Y_i$ and an integer c with $0 \leq c < m$ independent of i, and $\frac{1}{m}\operatorname{div}(l_i)$ is the crepant transform of $\frac{1}{m}\operatorname{div}(h_i)$.*

(2) *Any computing lift of b_i has center the origin of Y_i, and*

$$\operatorname{mld}\Bigl(\Bigl(Y_i \ni 0, \frac{1}{m}\operatorname{div}(l_i)\Bigr)/G\Bigr) = b_i.$$

Any prime divisor E over Y_i with center the origin is a prime divisor over W_i with center o_i, the inertia groups of E in G over Y_i and over W_i are equal, and

$$a(E,W_i,0) = a(E,Y_i,0) + r\operatorname{mult}_E s, \quad \operatorname{mult}_E h_i = \operatorname{mult}_E l_i + mr\operatorname{mult}_E s.$$

(3) *Any computing lift E of b_i satisfies* $\operatorname{mult}_E s/e_E \leq 3/\eta_m$*, and*

$$\min_E \operatorname{mult}_E \operatorname{Jac}_{Y_i} \to +\infty \quad \text{as } i \to +\infty,$$

the minimum over the computing lifts E of b_i.

(4) *The pairs (g_i, l_i) have a compatible generic limit (g_*, l_*) with*

$$g_* = x'^2 + z'y'^2 + s^{\nu_0}z'^{k-1} + z'^k\psi(z',s)$$

for a power series $\psi \in \Bbbk[[z',s]]$. Let $\operatorname{in}_w(h_*)$ be the lowest-weight part, for the weights $w = (k-1, k-2, 2)$ on (x,y,z), of the representative of h_* modulo f_* of x-degree at most 1. Then $\operatorname{in}_w(h_*)$ has weight m, and the lowest-weight part, for the weights $(k-1,k-2,2)$ on (x',y',z'), of the representative of l_* modulo g_* of x'-degree at most 1 has weight m and is equal to $\operatorname{in}_w(h_*)(s^dx', s^dy', z')$. Any computing lift E of b_i satisfies $\min\{\operatorname{mult}_E x', \operatorname{mult}_E y', \operatorname{mult}_E z'\} \to +\infty$.

Theorem 17.9. *In Set-up 17.1, assume that the prepared limit is of type cD, with the presentation of f_* and the generic transverse type D_k of Proposition 17.8, and let ν be the integer given there and $\mathrm{in}_w(h_*)$ the lowest-weight part, for the weights $w = (k-1, k-2, 2)$ on (x, y, z), of the representative of h_* modulo f_* of x-degree at most 1. Let*

$$S^0 := (x^2 + zy^2 + s^\nu z^{k-1} = 0)$$

over an algebraic closure of $\Bbbk((s))$, and assume that $(S^0, \frac{1}{m}\operatorname{div}(\mathrm{in}_w(h_*)))$ has a coefficient-one component. Then the following hold.

(1) m is even, and there exists a non-zero $\lambda \in \Bbbk[[s]]$ such that one of the following holds.
  (a) $\mathrm{in}_w(h_*) = \lambda(s) z^{m/2}$.
  (b) $k = 4$, ν is even, and

$$\mathrm{in}_w(h_*) = \lambda(s)(y \pm \sqrt{-1} s^{\nu/2} z)^{m/2}.$$

(2) There exists an integer $N \geq 2k+8$ such that, for the integers r and c and the infinite subfamily given by Proposition 17.8 for this N, the following holds for infinitely many i in the subfamily: $b_i = (1 - c/m)/R$ for the order R of the character of s, and some computing lift F_i of b_i satisfies

$$a(F_i, W_i, 0)/e_{F_i} \leq 3 + 3r/\eta_m.$$

By (2) and Lemma 4.3(2), no radial failure sequence satisfies Set-up 17.1 together with the assumptions above.

Theorem 17.10. *There does not exist a radial failure sequence satisfying Set-up 17.1 whose prepared limit is of type cD, with the presentation of f_* of Proposition 17.8 and generic transverse type D_k with $k \geq 4$ along C.*

Theorem 17.11. *There does not exist a radial failure sequence satisfying Set-up 17.1 whose prepared limit is of type cD, with generic transverse type D_k ($k \geq 4$) along C and residual cubic of W_* reduced or a double line and a distinct line.*

Theorem 17.12. *There does not exist a radial failure sequence satisfying Set-up 17.1 whose prepared limit is of type cE and has generic transverse type D_k with $k \geq 5$ along C.*

Proposition 17.13. *In Set-up 17.1, assume that the prepared limit is of type cE and that the generic transverse type along C is D_4. Then there exist formal G-eigenparameters (x, y, z, s) with $C = (x = y = z = 0)$ in which $f_* = x^2 + y^3 + \alpha(s) y z^2 + \gamma(s) z^3 + R$ with $R \in (y, z)^4 \Bbbk[[y, z, s]]$, $\alpha(0) = \gamma(0) = 0$, and $4\alpha^3 + 27\gamma^2 \neq 0$. Let $v := 12 \min(\mathrm{mult}_0\, \alpha/2, \mathrm{mult}_0\, \gamma/3)$. There exist integers $r \geq 1$, $N \geq 0$, and $j_0 \in \{0, 1\}$ with $\kappa := r - v/2 + 11 > 0$, and on an infinite subfamily, with f_i and h_i written in G-eigenparameters of the ambient germ of W_i, again denoted (x, y, z, s), whose restrictions to W_i are regular functions on the algebraic germ $(W_i \ni o_i)$, which have the characters of the parameters (x, y, z, s) above and in which the (f_i, h_i) have the compatible generic limit (f_*, h_*), after the extensions $s = u^{12}$ with the abelian groups*

$$\Gamma_i := \{(g, \zeta) \in G_i \times \mathbb{C}^* \mid \zeta^{12} = \chi_s(g)\}$$

of order $12|G_i|$, point charts $(Z_i \ni 0)$ given by $x = u^{3r+3v/2} x'$, $y = u^{2r+v} y'$, $z = u^{2r} z'$, followed, if $j_0 = 1$, by a Γ_i-equivariant coordinate change which fixes x' and u and replaces y', z' by elements of $(y', z')\mathbb{C}[y', z', u]$, each Z_i being the completion at 0 of a normal algebraic germ $(Z_i^{\mathrm{alg}} \ni 0)$ with function field $\mathbb{C}(W_i)(u)$ on which the coordinates of Z_i are regular functions, where Γ_i acts on Z_i^{alg} fixing 0, inducing the action on Z_i, and Z_i^{alg} has a morphism to W_i, equivariant for the projection $\Gamma_i \to G_i$, which induces on completions the map given by the chart, with the following properties, where $h_i = u^N l_i$ with l_i regular and not containing $S_i := (u = 0) \subset Z_i$, and $\beta := N/m - \kappa$. If $j_0 = 0$, then $(Z_i \ni 0)$ is an isolated Gorenstein terminal germ, and if $j_0 = 1$, then $(Z_i \ni 0)$ is a normal Gorenstein hypersurface germ which is smooth near 0 outside S_i.

(1) $\beta \in \frac{1}{m}\mathbb{Z} \cap [-10, 1)$, *every computing lift of* b_i *has center the origin of* Z_i,

$$\operatorname{mld}\Big(\Big(Z_i \ni 0, \frac{1}{m}\operatorname{div}(l_i) + \beta S_i\Big)/\Gamma_i\Big) = b_i,$$

and every computing lift of the left-hand side restricts to a computing lift of b_i.

(2) *Every prime divisor* P *over* Z_i *with center the origin restricts to a prime divisor* F *over* W_i *with center* o_i, *and, with* ρ *the ramification index of* P *over* F, $e_P = \rho e_F$, $\rho a(F, W_i, 0) = a(P, Z_i, 0) + \kappa \operatorname{mult}_P u$, $\rho \operatorname{mult}_F h_i = \operatorname{mult}_P l_i + N \operatorname{mult}_P u$, *and* $\rho \operatorname{mult}_F \mathfrak{m}_{o_i} = 12 \operatorname{mult}_P u$.

(3) *Every computing lift* P *of the quotient value in (1) satisfies* $\operatorname{mult}_P u/e_P \leq 3/(12\eta_m)$.

(4) *For any integer* J, *after passing to a further infinite subfamily, every computing lift of* b_i *has order greater than* J *along each of* x', y', z' *and, if* $j_0 = 1$, *along each of the three transverse coordinates after the coordinate change.*

(5) $\min_P a(P, Z_i, 0)/e_P \to +\infty$ *and* $\min_P \operatorname{mult}_P \operatorname{Jac}_{Z_i} \to +\infty$, *the minima over the computing lifts* P *of the quotient value in (1).*

(6) *The equations of the point charts have a compatible generic limit which, in* Γ_i*-eigenparameters preserving* u, *is* $x'^2 + y'^3 + \alpha'(u)y'z'^2 + \gamma'(u)z'^3 + u^L R'$ *with* $\alpha'(u) := u^{-2v}\alpha(u^{12})$ *and* $\gamma'(u) := u^{-3v}\gamma(u^{12})$ *in* $\Bbbk[[u]]$, $L > 0$, $R' \in (y', z')^4\Bbbk[[y', z', u]]$, *and* $\alpha'(0), \gamma'(0)$ *not both zero. The binary cubic* $y'^3 + \alpha'(0)y'z'^2 + \gamma'(0)z'^3$ *has three distinct roots if* $j_0 = 0$ *and one double root if* $j_0 = 1$, *and if* $j_0 = 1$, *then the equations of the* Z_i *have a compatible generic limit which, in* Γ_i*-eigenparameters preserving* u, *is* $x'^2 + z'y'^2 + \phi(z', u)$ *with* $\phi \in z'^3\Bbbk[[z', u]]$, $\phi(z', 0) = 0$ *and non-zero coefficient of* z'^3. *Moreover, the* l_i *have representatives of* x'*-degree at most* 1 *which, together with the equations of the* Z_i, *have a compatible generic limit* $(f_{Z,*}, l_*)$ *with* $f_{Z,*}$ *of the form given above for the equations of the* Z_i *(the first form if* $j_0 = 0$ *and the second if* $j_0 = 1$*), and over an algebraic closure of* $\Bbbk((u))$ *the charts and, if* $j_0 = 1$, *a coordinate change over* $\Bbbk[[u]]$ *which fixes* x' *and* u *and replaces* y', z' *by elements of* $(y', z')\Bbbk[[y', z', u]]$ *induce an automorphism of the power series ring in the three transverse coordinates fixing* 0 *which carries* f_* *to a unit times a power of* u *times* $f_{Z,*}$ *and* h_* *to* $u^N l_*$ *modulo* f_*.

Theorem 17.14. *There does not exist a radial failure sequence satisfying Set-up 17.1 whose prepared limit is of type cE and has generic transverse type* D_4 *along* C.

Proof of Theorem 17.2. We use the generic transverse surface and the generic transverse type of W_* along C (Definition 15.1). Since W_* is canonical (Construction-Theorems 13.9, 13.7 and 13.6), restricting a resolution of W_* [Tem18, Theorem 1.1.9] over the generic point of C, completing, and extending the residue field to an algebraic closure $\overline{\Bbbk((s))}$ of $\Bbbk((s))$ [Kaw21, Remark 2.6] keeps all discrepancies non-negative, so the generic transverse surface is canonical, hence Du Val [Kol13, Theorem 2.29(2) and Example 3.26], and it is singular since $C \subset \operatorname{Sing}(W_*)$. Thus the generic transverse type is A_n with $n \geq 1$, D_k with $k \geq 4$, E_6, E_7, or E_8. We use the following consequence of the classification of Du Val germs [Kol13, Example 3.26]: a Du Val germ $(x^2 + g'(y', z') = 0)$ with $g' \in (y', z')^3$ is of type D if and only if the cubic part of g' is neither 0 nor the cube of a linear form, and it is of type E if and only if the cubic part of g' is the cube of a non-zero linear form.

Type cE. By Theorem 17.4, the type is not E_7 or E_8. By the last sentence of Theorem 17.6, the type is not E_6. By Theorems 17.12 and 17.14, the type is not D_k for any $k \geq 4$. Thus the type is A_n for some $n \geq 1$.

Type cD. Assume that the type is D_k or E_j. By Lemma 17.3(3), there exist G-eigenparameters (x, y, z, s) with $C = (x = y = z = 0)$ in which

$$f_* = v \cdot (x^2 + g(y, z, s)), \quad g \in (y, z)^3,$$

for an invariant unit v. Let $g_3(y, z; s)$ be the (y, z)-cubic part of g, a binary cubic form in y, z with coefficients in $\Bbbk[[s]]$. Since $g \in (y, z)^3$, the quadratic part of f_* at the closed point is $v(0)x^2$,

so W_* has quadratic rank 1, and the residual cubic of W_* is

$$\Phi := v(0)g_3(y,z;0).$$

The residual cubic is well defined up to a linear change of variables and a non-zero scalar.

We first show that Φ is neither 0 nor the cube of a linear form. By Construction-Theorem 13.7, in the eigenparameters (x,y,z,t) given there, $f_i = x^2+g_i(y,z,t)$, so the limit is $f_* = x^2+g_*(y,z,t)$ for some $g_* \in \Bbbk[[y,z,t]]$, and there exists a linear form ℓ_1 in y,z,t such that $W_* \cap (\ell_1 = 0)$ is a Du Val germ of type D_n for some $n \geq 4$. Since W_* has quadratic rank 1, $g_* \in (y,z,t)^3$, and the residual cubic of W_* is the cubic part Φ_* of g_*. Choose linear forms y', z' in y,z,t such that (ℓ_1, y', z') is a basis of the linear forms in y,z,t. Then $W_* \cap (\ell_1 = 0)$ is defined by $x^2 + g_*|_{(\ell_1=0)}$ with $g_*|_{(\ell_1=0)} \in (y',z')^3$ and cubic part $\Phi_*|_{(\ell_1=0)}$. Since $W_* \cap (\ell_1 = 0)$ is of type D_n, $\Phi_*|_{(\ell_1=0)}$ is neither 0 nor the cube of a linear form. Thus Φ_* is neither 0 nor the cube of a linear form, and since Φ and Φ_* differ by a linear change of variables and a non-zero scalar, neither is Φ.

Suppose that the type is E_j. Over the algebraic closure of $\Bbbk((s))$ fixed above, the generic transverse surface is defined by $x^2 + g(y,z,s)$ with $g \in (y,z)^3$, so $g_3(y,z;s) = \ell^3$ for a non-zero linear form ℓ in y,z with coefficients in the algebraic closure of $\Bbbk((s))$. Write a binary cubic form as $a_0y^3 + 3a_1y^2z + 3a_2yz^2 + a_3z^3$. Over the algebraic closure of $\Bbbk((s))$ and over $\Bbbk$, the binary cubic forms which are 0 or the cube of a linear form form the closed cone defined by

$$\operatorname{rank}\begin{pmatrix} a_0 & a_1 & a_2 \\ a_1 & a_2 & a_3 \end{pmatrix} \leq 1,$$

that is, by the vanishing of the three 2×2 minors, which are polynomials with rational coefficients. The coefficients of $g_3(y,z;s)$ are elements of $\Bbbk[[s]]$ satisfying the three equations, hence so do the values at $s=0$ of the coefficients. Thus $g_3(y,z;0)$, and hence Φ, is 0 or the cube of a linear form, a contradiction.

Suppose that the type is D_k with $k \geq 4$. Since Φ is a non-zero cubic form in y,z and $\Bbbk$ is algebraically closed, Φ is a product of three linear forms in y,z. Since Φ is not the cube of a linear form, either the three factors are pairwise non-proportional or Φ is the product of the square of a linear form and a non-proportional linear form. Thus the residual cubic of W_* is reduced or a double line and a distinct line, and Theorem 17.11 contradicts the existence of the radial failure sequence of Set-up 17.1.

Thus the generic transverse type is A_n for some $n \geq 1$ also when the prepared limit is of type cD. The theorem follows. □

18. Quadratic rank two and transverse type A

Theorems 18.1 and 18.2 are proved in Appendix G.

Theorem 18.1. *In Set-up 17.1, assume that the generic transverse type along C is A_n for some $n \geq 1$. Then there exist integers $\kappa \geq 1$ and $N' \geq 0$ and an infinite subfamily with degree 2 extensions $s = u^2$ of the function fields of the W_i and fixed point charts (one chart if $n = 1$, and the composite of two charts if $n \geq 2$), giving isolated Gorenstein terminal hypersurface germs $(Z_i \ni 0)$ of quadratic rank at least 2 with faithful actions of finite abelian groups Γ_i of order $2|G_i|$, such that the following hold. Here, for i in the subfamily, s on W_i denotes a G-semi-invariant function $s_i \in \mathfrak{m}_{o_i}\backslash\mathfrak{m}_{o_i}^2$ with the character of s such that the tuples (f_i, h_i, s_i) have the compatible generic limit (f_*, h_*, s), and the fixed point charts are the Γ_i-equivariant chart (G.3), followed, if $n \geq 2$, by a change of coordinates and a second chart. The function s_i and the charts are constructed in Appendix G.1.*

(1) $h_i = u^{N'}l_i$ for a mark l_i on $(Z_i \ni 0)$ with $u \nmid l_i$, the divisor $U_i := \operatorname{div}(u)$ on Z_i is reduced with one or two components,

$$\beta := \frac{N'}{m} - \kappa \in \frac{1}{m}\mathbb{Z} \cap [-\kappa, 1),$$

any component of $D_i' := \frac{1}{m}\operatorname{div}(l_i) + \beta U_i$ *has coefficient less than* 1, (Z_i, D_i') *is sub-klt near* 0, *and*

$$\operatorname{mld}((Z_i \ni 0, D_i')/\Gamma_i) = b_i.$$

(2) *Any prime divisor* P *over* Z_i *with center* 0 *restricts to a prime divisor* F *over* W_i *with center* o_i, *and*

$$e_P = \rho e_F, \quad \rho a(F, W_i, 0) = a(P, Z_i, 0) + \kappa \operatorname{mult}_P u,$$
$$\text{and} \quad \rho \operatorname{mult}_F h_i = \operatorname{mult}_P l_i + N' \operatorname{mult}_P u,$$

where ρ *is the ramification index of* P *over* F, *and* e_P *and* e_F *are the orders of the inertia groups of* P *in* Γ_i *and of* F *in* G_i.

(3) *Any prime divisor over* Z_i *restricting to a computing lift of* b_i *has center* 0 *on* Z_i, *and any computing lift* P *of* $\operatorname{mld}((Z_i \ni 0, D_i')/\Gamma_i)$ *restricts to a computing lift of* b_i *and satisfies*

$$\frac{\operatorname{mult}_P u}{e_P} \le \frac{3}{2\eta_m} \quad \text{and} \quad \operatorname{mult}_P u \le \frac{3M}{\eta_m},$$

where η_m *is the constant of Theorem 5.1(2).*

(4) *The equations of the* Z_i *and the* l_i *have a compatible generic limit* (f_*', l_*), *the extension* $s = u^2$ *and the fixed point charts above, taken over* $\Bbbk$, *transform* (f_*, h_*) *into* $(f_*', u^{N'} l_*)$, *and over* $(u \ne 0)$ *the composite of the charts with* $Z_i \to W_i$ *is finite étale followed by invertible coordinate changes. In particular, over* $(u \ne 0)$ *the pair* $((f_*' = 0), \frac{1}{m}\operatorname{div}(l_*))$ *is étale locally* $(W_*, \frac{1}{m}\operatorname{div}(h_*))$, *and the inverse image of* C *in* $(f_*' = 0)$ *is a regular curve with parameter* u *along which the chart coordinates other than* u *vanish.*

Theorem 18.2. *Let* m, M, *and* R' *be positive integers,* $B_0 > 0$ *a real number,* $\beta \in \frac{1}{m}\mathbb{Z}$, *and* G *a finite abelian group of order at most* M. *For any* $i \in \mathbb{N}^+$ *let* $(W_i \ni o_i)$ *be an isolated Gorenstein terminal or smooth threefold germ with pointed étale coordinates* (x, y, z, u) *at* o_i *in which* $W_i = (xy + g_i(z, u) = 0)$, *and let* h_i *be a regular function on* W_i, *non-zero on* W_i. *Here pointed étale coordinates* (x, y, z, u) *at* o_i *in which* $W_i = (xy + g_i(z, u) = 0)$ *mean regular functions* z, u *on* W_i, *a smooth surface germ* $(V_i \ni 0)$ *with parameters also denoted by* z, u, *a regular function* g_i *on* V_i, *and an action of* G *on* $\mathbb{A}^2_{x,y} \times V_i$ *as in (1), such that* $(xy + g_i = 0) \subset \mathbb{A}^2_{x,y} \times V_i$ *and* $(W_i \ni o_i)$ *have a common* G*-equivariant pointed étale neighborhood on which the pullback of* z *(resp.* u*) from* W_i *equals the pullback of* z *(resp.* u*) from* $\mathbb{A}^2_{x,y} \times V_i$ *(*[Liu26b, Proposition 3.15]*). In (4),* $xy + g_i$ *and* h_i *denote* $xy + \widehat{g}_i(z, u)$, *where* $\widehat{g}_i$ *is the expansion of* g_i, *and a* G*-semi-invariant representative of* h_i *under the induced isomorphism*

$$\widehat{\mathcal{O}}_{W_i, o_i} \cong \mathbb{C}[[x, y, z, u]]/(xy + \widehat{g}_i).$$

Let

$$D_i := \frac{1}{m}\operatorname{div}(h_i) + \beta \operatorname{div}(u) \quad \text{and} \quad b_i := \operatorname{mld}((W_i \ni o_i, D_i)/G).$$

A computing lift of b_i *is a prime divisor* P *over* W_i *with center* o_i *and* $a(P, W_i, D_i)/e_P = b_i$. *Assume that the following hold.*

(1) G *acts faithfully on* W_i, *monomially on* $\{x, y\}$ *and diagonally on* z, u, *and* $xy + g_i$ *and* h_i *are* G*-semi-invariant with characters independent of* i.

(2) (W_i, D_i) *is sub-klt near* o_i *and* $b_i \in (0, B_0]$.

(3) *Some computing lift* F_i *of* b_i *satisfies* $\operatorname{mult}_{F_i} u \le R'$.

(4) *The tuples* $(xy + g_i, h_i)$ *have a compatible generic limit* $(xy + z^d \vartheta(z, u), h_*)$ *with coefficient field* $\Bbbk$, *where* $d \ge 2$ *and* $\vartheta(0, u) \ne 0$. *Let*

$$W_* := (xy + z^d \vartheta(z, u) = 0) \subset \operatorname{Spec} \Bbbk[[x, y, z, u]] \quad \text{and} \quad C := (x = y = z = 0).$$

On the open subset $(u \ne 0)$ *of* W_*, *the pair* $(W_*, \frac{1}{m}\operatorname{div}(h_*))$ *is lc without coefficient-one component, and*

$$\operatorname{Nklt}\left(W_*, \frac{1}{m}\operatorname{div}(h_*)\right) \cap (u \ne 0) = C \cap (u \ne 0).$$

Then there exist a real number C_0 and an infinite subset $\mathcal{I} \subset \mathbb{N}^+$ such that for any $i \in \mathcal{I}$, some computing lift P_i of b_i satisfies

$$a(P_i, W_i, 0) \le C_0, \qquad \operatorname{mult}_{P_i} u \le C_0, \quad \text{and} \quad \operatorname{mult}_{P_i} h_i \le C_0.$$

Theorem 18.3. *Let m and M be positive integers and $a > \delta > 0$ real numbers, and let*

$$(W_i \ni o_i, G_i, h_i, \tau_i)$$

be a radial failure sequence with constants (m, a, δ) such that for any i, $|G_i| \le M$, $b_i > 0$, $(W_i, \frac{1}{m}\operatorname{div}(h_i))$ is klt near o_i, and $(W_i \ni o_i)$ has quadratic rank 1 or 2. Let G and (f_*, h_*) be the group and the prepared limit of the sequence given by Construction-Theorem 13.9, and $W_* := (f_* = 0)$. Assume that $(W_*, \frac{1}{m}\operatorname{div}(h_*))$ is lc without coefficient-one component and with positive minimal log discrepancy at the closed point, and that $\operatorname{Nklt}(W_*, \frac{1}{m}\operatorname{div}(h_*))$ is a regular G-invariant curve $C \subset \operatorname{Sing}(W_*)$. Then the quadratic rank of $(W_i \ni o_i)$ is not two, and if it is one, then the generic transverse type along C is not A_n for any $n \ge 1$.

Proof. Possibly passing to a subsequence, we may assume that the quadratic rank of $(W_i \ni o_i)$ does not depend on i. Let C' be the minimal lc center of $(W_*, \frac{1}{m}\operatorname{div}(h_*))$ (Lemma 13.5). Then $C' \subset C$, and $\dim C' \in \{1, 2\}$ since the minimal log discrepancy at the closed point is positive. Thus $\dim C' = \dim C = 1$, and $C' = C$ as C is irreducible.

Step 1. *The quadratic rank is not two.* Suppose that the quadratic rank is two. For any i let

$$N_i := \min_F \operatorname{mult}_F \operatorname{Jac}_{W_i},$$

the minimum over the computing lifts F of b_i. We show that $N_i \to +\infty$. Otherwise there exist a positive integer N and an infinite set $\mathcal{I}$ of indices with $N_i \le N$ for any $i \in \mathcal{I}$. For $i \in \mathcal{I}$, Theorem 15.8(1), applied to the embedded germs $(W_i \ni o_i) \subset (\mathcal{X}_i \ni o_i)$, where $(\mathcal{X}_i \ni o_i)$ is the smooth fourfold germ of the completed equivariant presentation $xy + g_i(z,t)$ of W_i in Construction-Theorem 13.9(2), on which G_i acts faithfully as it does on W_i, gives a computing lift P_i of b_i with $a(P_i, W_i, 0) \le L(m, M, N)$. Since $e_{P_i} \ge 1$, this contradicts Lemma 4.3(2).

Since G is abelian, after a linear change of the coordinates x, y of the presentation

$$f_* = xy + g_*$$

of Construction-Theorem 13.9(2)(a), which G may exchange, G acts diagonally on the ambient coordinates. Then, by Definition 4.1, the hypotheses of the theorem, and Construction-Theorem 13.9(2)(a), the conditions of Set-up 15.4 hold with $B_0 = a$, and the minimal lc center C is a regular curve contained in $\operatorname{Sing}(W_*)$. Let $\epsilon > 0$ be the rational number given by Theorem 16.1, $R := \lceil Ma/\epsilon \rceil$, and $k \ge \max\{R, 2\}$ an integer. By Lemma 13.5(2) there exist G-eigenparameters (s', v_1, v_2, v_3) of $\mathcal{X}_*$ with $C = (v_1 = v_2 = v_3 = 0)$. Theorem 16.1(1), for (s', v_1, v_2, v_3) in place of (s, u_1, u_2, u_3) and for k, gives, possibly passing to a subsequence, G-eigenparameters $(s_i, v_{i1}, v_{i2}, v_{i3})$ of $\mathcal{X}$ such that the tuples (f_i, h_i, s_i) have a compatible generic limit (f_*, h_*, s) with coefficient field $\Bbbk$ (Lemma 2.6(2)) and $s - s' \in \mathfrak{n}_*^k \subset \mathfrak{n}_*^2$, where $\mathfrak{n}_*$ is the maximal ideal of $\mathcal{O}_{\mathcal{X}_*}$. Thus $s|_C$ is a uniformizer of $\widehat{\mathcal{O}}_C$. Moreover, after dropping finitely many i,

$$\operatorname{mult}_F s_i \le R \quad \text{for any } i \text{ and any computing lift } F \text{ of } b_i : \tag{18.1}$$

as $N_i \to +\infty$, Theorem 16.1(1)(b) with $B_0 = a$ shows that for $i \gg 0$ any computing lift F of b_i satisfies

$$\operatorname{mult}_F((v_{i1}, v_{i2}, v_{i3})\mathcal{O}_{W_i}) > k \ge R \ge \operatorname{mult}_F \mathfrak{m}_{o_i},$$

so $\operatorname{mult}_F s_i = \operatorname{mult}_F \mathfrak{m}_{o_i} \le R$.

We now write W_i in split form, keeping s_i. Let x, y be the coordinates of the presentation $f_* = xy + g_*$ of Construction-Theorem 13.9(2)(a), before the linear change above. Since $\partial f_*/\partial x = y$ and $\partial f_*/\partial y = x$, we have $C \subset \operatorname{Sing}(W_*) \subset (x = y = 0)$. Let $I_C \subset \mathcal{O}_{\mathcal{X}_*}$ be the ideal of C. Then $x, y \in I_C$, and I_C is G-stable. By Maschke's theorem and projection to a character, there exists a G-semi-invariant $z \in I_C$ whose class in $\mathfrak{n}_*/\mathfrak{n}_*^2$ is not in the span of the classes of

x and y. Since C is regular and $s|_C$ is a uniformizer, (x, y, z, s) are regular parameters of $\mathcal{X}_*$. By Lemma 2.6(4), after passing to an infinite subset, and truncating s and the s_i in degrees greater than R, we take G-semi-invariant regular functions z_i and s_i on the ambient germs, namely G-semi-invariant regular functions on the $\mathcal{X}_i$ congruent, modulo the n_i-th power of the maximal ideal of $\widehat{\mathcal{O}}_{\mathcal{X}_i,o_i}$ for integers $n_i > R$ with $n_i \to +\infty$, to the polynomials so obtained in the coordinates of the presentations, such that the tuples (f_i, h_i, s_i, z_i) have the compatible generic limit (f_*, h_*, s, z). The truncation and the approximation by regular functions keep (18.1): each changes s_i by an element of $\mathfrak{m}_{o_i}^{R+1}$, whose order along any computing lift F of b_i is at least $R + 1 > \mathrm{mult}_F\, s_i$. As $R \geq 1$, the truncation keeps the class of s in $\mathfrak{n}_*/\mathfrak{n}_*^2$, so $s|_C$ is still a uniformizer and (x, y, z, s) are still regular parameters.

Since the quadratic part xy of f_* is non-degenerate on $(z = s = 0)$, the quadratic part of f_i is non-degenerate on $(z_i = s_i = 0)$ for all but finitely many i. By [Liu26b, Proposition 3.15], there exist G-equivariant morphisms, étale at the marked points, which keep z_i and s_i and yield pointed étale coordinates (x, y, z_i, s_i) of W_i at o_i in which $W_i = (xy + g_i'(z_i, s_i) = 0)$ and G acts monomially on $\{x, y\}$ and diagonally on z_i, s_i. If the split quadratic form is $x^2 + y^2$, we replace x, y by $x + \sqrt{-1}y, x - \sqrt{-1}y$. The splitting is given by the implicit function theorem, so the jet of fixed order of g_i' is determined by a jet of fixed order of (f_i, z_i, s_i). Hence, possibly passing to a subsequence, the tuples $(xy + g_i', h_i)$ have a compatible generic limit $(xy + g_*'(z, s), h_*)$, where $xy + g_*'$ is the splitting of f_* with z and s kept, up to an invariant unit. In the coordinates (x, y, z, s) of the limit, $C \subset \mathrm{Sing}(W_*) \subset (x = y = 0)$ and $C \subset (z = 0)$. Since $(x = y = z = 0)$ is a regular curve, $C = (x = y = z = 0)$. As g_*' and $\partial g_*'/\partial z$ vanish along C, $g_*' \in (z^2)$, and $g_* \neq 0$ since W_* is normal. Therefore

$$g_*' = z^d \vartheta(z, s), \qquad d \geq 2, \quad \text{and} \quad \vartheta(0, s) \neq 0.$$

By Theorem 18.2 for (x, y, z_i, s_i) in place of (x, y, z, u), the group G, $\beta = 0$, $B_0 = a$, and $R' = R$, whose hypotheses hold by the splitting, Definition 4.1, the hypotheses of the theorem, (18.1), and the form of g_*', there exist a real number C_0 and, for infinitely many i, computing lifts P_i of b_i with $a(P_i, W_i, 0) \leq C_0$. As $e_{P_i} \geq 1$, this contradicts Lemma 4.3(2).

Step 2. *In quadratic rank* 1*, the generic transverse type along C is not A_n.* Suppose that the quadratic rank is one and that the generic transverse type along C is A_n for some $n \geq 1$. By the hypotheses, Construction-Theorem 13.9(2)(b), and Lemma 13.5(2) for the minimal lc center C, the conditions of Set-up 17.1 hold. Let κ, N', β, and, for i in the infinite subfamily, Z_i, Γ_i, l_i, U_i, and D_i' be given by Theorem 18.1. Since $|\Gamma_i| = 2|G_i| \leq 2M$, possibly passing to a subsequence, we may assume that the group $\Gamma := \Gamma_i$, the action of Γ on the coordinates of the charts, and the characters of the equations of the Z_i and of the l_i do not depend on i.

By Theorem 18.1(4) and the hypotheses on $(W_*, \frac{1}{m}\,\mathrm{div}(h_*))$, on $(u \neq 0)$ the pair $(W_*', \frac{1}{m}\,\mathrm{div}(l_*))$, where (f_*', l_*) is the compatible generic limit there and $W_*' := (f_*' = 0)$, is lc without coefficient-one component with non-klt locus C°, the inverse image of C. Thus the generic transverse type of W_*' along C° is A_n.

Hence W_*' is singular along C°. Let I_{C° be the ideal of C°. As the chart coordinates other than u vanish along the regular curve C° with parameter u (Theorem 18.1(4)), they generate I_{C°, so $f_*' \in I_{C^\circ}^2$ and the quadratic part Q_0 of f_*' at 0 is a Γ-semi-invariant quadratic form in the chart coordinates other than u. It has rank at least 2, since rank at most 1 is a closed condition on 3-jets and the Z_i have quadratic rank at least 2 (Definition 2.5). Hence there exists a Γ-semi-invariant $\mathbb{C}$-linear combination z of the chart coordinates other than u such that Q_0 is non-degenerate on $(z = u = 0)$. If $\mathrm{rank}\, Q_0 = 2$, we take z non-zero on the Γ-stable kernel of Q_0. If $\mathrm{rank}\, Q_0 = 3$, the dual form of Q_0 is Γ-semi-invariant and non-degenerate on a space of odd dimension, hence non-zero on some eigenspace, and we take z in that eigenspace with non-zero value. Complex coefficients suffice, as a non-zero polynomial with coefficients in $\Bbbk$ does not vanish at all complex points. As in **Step 1**, the quadratic part of the equation of Z_i at 0 is non-degenerate on $(z = u = 0)$ for all but finitely many i.

By [Liu26b, Proposition 3.15] there exist Γ-equivariant morphisms, étale at the marked points, which keep z and u and yield pointed étale coordinates (x, y, z, u) of Z_i at 0 in which $Z_i = (xy + g_i(z, u) = 0)$ and Γ acts monomially on $\{x, y\}$ and diagonally on z, u. As in **Step 1**, the jet of fixed order of g_i is determined by a jet of fixed order of the equation of Z_i, so, possibly passing to a subsequence, the tuples $(xy+g_i, l_i)$ have a compatible generic limit $(xy+g_*(z, u), l_*)$. In the coordinates (x, y, z, u) of the limit, $C^\circ \subset \mathrm{Sing}(W'_*) \subset (x = y = 0)$ and $C^\circ \subset (z = 0)$. Since $(x = y = z = 0)$ is a regular curve, $C^\circ = (x = y = z = 0)$. Since the generic transverse type along C° is A_n, the order in z of g_* over an algebraic closure of $\Bbbk((u))$ is $n + 1$, that is,

$$g_* = z^{n+1}\vartheta(z, u), \qquad \vartheta(0, u) \neq 0.$$

By Theorem 18.2 for $(Z_i \ni 0)$, l_i, β, the group Γ of order at most $2M$, $d = n + 1$, $B_0 = a$, and $R' := \lceil 3M/\eta_m \rceil$, whose hypotheses hold by the splitting, the choice of the subsequence, Theorem 18.1(1)(3), $b_i \leq a - \delta$, and the description of $(xy + g_*, l_*)$ above, there exist a real number C_0 and, for infinitely many i, computing lifts P_i of $\mathrm{mld}((Z_i \ni 0, D'_i)/\Gamma)$ over Z_i with

$$a(P_i, Z_i, 0) \leq C_0 \quad \text{and} \quad \mathrm{mult}_{P_i} u \leq C_0.$$

By Theorem 18.1(3), the restriction F_i of P_i is a computing lift of b_i. Let ρ_i be the ramification index of P_i over F_i. By Theorem 18.1(2),

$$\frac{a(F_i, W_i, 0)}{e_{F_i}} = \frac{\rho_i a(F_i, W_i, 0)}{e_{P_i}} = \frac{a(P_i, Z_i, 0) + \kappa \,\mathrm{mult}_{P_i} u}{e_{P_i}} \leq (1 + \kappa)C_0.$$

As this holds for infinitely many i, it contradicts Lemma 4.3(2). The theorem follows. □

19. Proof of the threshold gap

Proof of Theorem 1.5. Let a be a positive rational number and $\Gamma \subset [0, 1] \cap \mathbb{Q}$ a finite set. Assume that there does not exist a positive real number $\eta(a, \Gamma)$ with the property of Theorem 1.5. We derive a contradiction in several steps.

Step 1. In this step we construct a radial failure sequence. Let m be a positive integer with $m\Gamma \subset \mathbb{Z}$. By Construction-Theorem 4.7(C), there exist a real number $\delta \in (0, a)$ and a radial failure sequence $(W_i \ni o_i, G_i, h_i, \tau_i)$ with constants (m, a, δ) such that $(W_i \ni o_i)$ is singular of quadratic rank 1, 2, or 3 for any i. Any subsequence of $(W_i \ni o_i, G_i, h_i, \tau_i)$ is again a radial failure sequence with constants (m, a, δ) given by Construction-Theorem 4.7(C). In the following steps we pass to subsequences without changing the notation.

Step 2. By Theorem 9.1, after passing to a subsequence, $b_i > 0$ for any i.

Step 3. In this step we show that, after passing to a subsequence, $(W_i \ni o_i)$ has quadratic rank 1 or 2 and $(W_i, \frac{1}{m}\mathrm{div}(h_i))$ is klt near o_i for any i.

Assume that $(W_i \ni o_i)$ has quadratic rank 3 for infinitely many i. After passing to a subsequence, $(W_i \ni o_i)$ has quadratic rank 3 for any i. By Lemma 4.2(3), $q_i(0) \leq 2$ for any i.

By Definition 4.1(3), $q_i(\tau_i) - q_i(1) \geq \delta$ for any i. By Lemma 4.5 with $s_i = \tau_i$ and $\epsilon = \delta$, after passing to a subsequence, there exist $t_i \in [\tau_i, 1)$, strictly increasing with limit 1, such that $\alpha_i := q_i(t_i)$ is strictly increasing with limit $\alpha_\infty > 0$. By Lemma 4.2(1), q_i is decreasing on $[0, 1]$, so $\alpha_i \geq b_i > 0$ by **Step 2**. Since $(t_i)_i$ is strictly increasing, the set $\{t_i/m\}_i$ is DCC. By Theorem 10.1, the set

$$\{\alpha_i\}_i = \left\{\mathrm{mld}\Big(\Big(W_i \ni o_i, \frac{t_i}{m}\mathrm{div}(h_i)\Big)/G_i\Big)\right\}_i$$

satisfies the ACC, which contradicts the fact that $(\alpha_i)_i$ is strictly increasing.

Thus, after passing to a subsequence, $(W_i \ni o_i)$ has quadratic rank 1 or 2 for any i. By the second sentence of Theorem 9.1, after passing to a further subsequence, $(W_i, \frac{1}{m}\mathrm{div}(h_i))$ is klt near o_i for any i.

Step 4. By Theorem 12.1, after passing to a subsequence, there exists a positive integer M such that $|G_i| \leq M$ for any i.

Step 5. In this step we take the prepared limit. After passing to a subsequence, there exists $r \in \{1,2\}$ such that $(W_i \ni o_i)$ has quadratic rank r for any i. By **Steps 2–4** and Construction-Theorem 13.9, after passing to a subsequence, let $G = G_i$ and (f_*, h_*), with coefficient field $\Bbbk$, be the group and the prepared limit given there, let

$$W_* := (f_* = 0) \subset \mathcal{X}_* := \operatorname{Spec} \Bbbk[[x_1, \dots, x_4]],$$

and denote by 0 the closed point. By Construction-Theorems 13.9(2)(3), 13.7(1)(4) and 13.6(1)(3), W_* is normal, canonical and cDV, the ambient pair $(\mathcal{X}_*, (f_*)^1 (h_*)^{1/m})$ is lc and has positive minimal log discrepancy at 0, and

$$(W_*, \tfrac{1}{m} \operatorname{div}(h_*)) \text{ is lc} \quad \text{and} \quad \operatorname{mld}(W_* \ni 0, \tfrac{1}{m} \operatorname{div}(h_*)) > 0. \tag{19.1}$$

In particular, W_* is klt and integral, and f_* generates a prime ideal of $\Bbbk[[x_1, \dots, x_4]]$.

Step 6. In this step we show that $h_*|_{W_*} \neq 0$ and that G preserves W_* and $\operatorname{div}(h_*|_{W_*})$.

If $h_* \in (f_*)$, then the $\mathbb{Q}$-divisor $\operatorname{div}(f_*) + \frac{1}{m} \operatorname{div}(h_*)$ on $\mathcal{X}_*$ has coefficient at least $1 + \frac{1}{m} > 1$ along the prime divisor W_*, which contradicts the lc property of the ambient pair in **Step 5**. Thus $h_*|_{W_*} \neq 0$.

By Definition 2.5, f_* and h_* are G-semi-invariant, so G preserves W_* and $\operatorname{div}(h_*|_{W_*})$.

Step 7. In this step we show that $(W_*, \frac{1}{m} \operatorname{div}(h_*))$ has no coefficient-one component, and that $\operatorname{Nklt}(W_*, \frac{1}{m} \operatorname{div}(h_*))$ is a regular G-invariant curve $C \subset \operatorname{Sing}(W_*)$.

By **Steps 1–5** and Theorem 14.1, there exists a 1-dimensional component C' of $\operatorname{Sing}(W_*)$ such that $(W_*, \frac{1}{m} \operatorname{div}(h_*))$ is not klt at the generic point of C'. In particular, by (19.1), $(W_*, \frac{1}{m} \operatorname{div}(h_*))$ is lc and not klt.

Assume that $(W_*, \frac{1}{m} \operatorname{div}(h_*))$ has a coefficient-one component. Since W_* is the spectrum of a local ring, the component contains 0. By **Steps 1–6** and the first paragraph of this step, the conditions of Set-up 16.2 hold, which contradicts Theorem 16.7. Thus $(W_*, \frac{1}{m} \operatorname{div}(h_*))$ has no coefficient-one component.

By Lemma 13.5, (19.1) and **Step 6**, $(W_*, \frac{1}{m} \operatorname{div}(h_*))$ has a unique minimal lc center C, which is contained in any lc center, is G-invariant, and satisfies $\dim C \in \{1,2\}$, and C is regular if $\dim C = 1$. Since W_* is normal and $(W_*, \frac{1}{m} \operatorname{div}(h_*))$ has no coefficient-one component, $\dim V \neq 2$ for any lc center V of $(W_*, \frac{1}{m} \operatorname{div}(h_*))$. Thus $\dim C = 1$, C is regular, and any lc center V contains C and satisfies $\dim V = 1$, so $V = C$. Thus C is the only lc center, and

$$\operatorname{Nklt}\left(W_*, \tfrac{1}{m} \operatorname{div}(h_*)\right) = C.$$

Since $(W_*, \frac{1}{m} \operatorname{div}(h_*))$ is not klt at the generic point of C', $C' \subset C$. Since C' and C are irreducible curves, $C' = C$, so $C \subset \operatorname{Sing}(W_*)$.

Step 8. In this step we derive the contradiction. By **Steps 1–7** and Theorem 18.3, $r \neq 2$, that is, $r = 1$.

Since $r = 1$, G is a finite cyclic group acting diagonally on $\Bbbk[[x_1, \dots, x_4]]$ by Definition 4.1(1) and Construction-Theorem 13.9(ii). By **Step 6**, f_* and h_* are G-semi-invariant and $h_*|_{W_*} \neq 0$. By **Step 5**, W_* is normal and klt. By (19.1) and **Step 7**, $(W_*, \frac{1}{m} \operatorname{div}(h_*))$ is lc and not klt, and $\operatorname{mld}(W_* \ni 0, \frac{1}{m} \operatorname{div}(h_*)) > 0$. Since $\dim C = 1$, by Lemma 13.5(2), there exist G-eigenparameters (s, u_1, u_2, u_3) of $\Bbbk[[x_1, \dots, x_4]]$ such that

$$C = (u_1 = u_2 = u_3 = 0), \quad h_* \in (u_1, u_2, u_3) + (f_*), \quad \text{and} \quad f_* \in (u_1, u_2, u_3)^2,$$

where $f_* \in (u_1, u_2, u_3)^2$ holds since $C \subset \operatorname{Sing}(W_*)$, and f_* and h_* are G-semi-invariant. By **Steps 1–7**, $r = 1$ and the eigenparameters above, the conditions of Set-up 17.1 hold.

By Theorem 17.2, the generic transverse type of W_* along C is A_n for some $n \geq 1$. By Theorem 18.3, the generic transverse type of W_* along C is not A_n for any $n \geq 1$, a contradiction.

Thus there exists a positive real number $\eta(a, \Gamma)$ with the property of Theorem 1.5. Since a and Γ are arbitrary, the theorem follows. □

20. Proof of the main theorem

Proof of Theorem 1.4. Let X be a normal $\mathbb{Q}$-factorial variety and $x \in X$ a closed point. Since log discrepancies are affine in the boundary,

$$\operatorname{mld}(X \ni x, B) \geq \operatorname{mld}(X \ni x, B') \quad \text{for any } \mathbb{R}\text{-divisors } 0 \leq B \leq B'. \tag{20.1}$$

Let a be a positive real number and $D \geq 0$ an $\mathbb{R}$-divisor on X, and assume that

$$\operatorname{mld}(X \ni x) := \operatorname{mld}(X \ni x, 0) \geq a.$$

We use $\tau_a(X, D; x)$ of Theorem 1.5 also when a is real and D is an $\mathbb{R}$-divisor. The set in the definition of $\tau_a(X, D; x)$ contains 0 since $\operatorname{mld}(X \ni x) \geq a$. Then

$$\tau_a(X, D; x) < 1 \quad \text{if and only if} \quad \operatorname{mld}(X \ni x, D) < a. \tag{20.2}$$

Step 1. In this step we prove the following version of Theorem 1.5 for real levels and finite sets of real coefficients.

Claim 20.1. *Let a be a positive real number and $\Gamma_0 \subset [0, 1]$ a finite set. Then there exists a positive real number η satisfying the following. Let X be a $\mathbb{Q}$-factorial klt threefold, $x \in X$ a closed point with $\operatorname{mld}(X \ni x) \geq a$, and $D \geq 0$ an $\mathbb{R}$-divisor on X whose coefficients belong to Γ_0 such that (X, D) is lc near x. Then*

$$\tau_a(X, D; x) \notin (1 - \eta, 1).$$

Proof. Suppose that the claim fails for some a and Γ_0. Then, for any $k \in \mathbb{N}^+$, there exist a $\mathbb{Q}$-factorial klt threefold X_k, a closed point $x_k \in X_k$ with $\operatorname{mld}(X_k \ni x_k) \geq a$, and an $\mathbb{R}$-divisor $D_k \geq 0$ on X_k whose coefficients belong to Γ_0 such that (X_k, D_k) is lc near x_k and

$$1 - \frac{1}{k} < \tau_k := \tau_a(X_k, D_k; x_k) < 1.$$

Possibly shrinking X_k to a neighborhood of x_k, we may assume that (X_k, D_k) is lc.

Let $s_1, \dots, s_n$ be the positive elements of Γ_0, and let $\mathbf{s} := (s_1, \dots, s_n)$. For any k and any $j \in \{1, \dots, n\}$, let D_{kj} be the sum of the prime components of D_k with coefficient s_j. Then $D_{k1}, \dots, D_{kn}$ are reduced divisors without common components and

$$D_k = \sum_{j=1}^{n} s_j D_{kj}.$$

If $n = 0$, then $D_k = 0$ and $\tau_k = 1$, a contradiction. Thus $n \geq 1$.

By (20.2), $\operatorname{mld}(X_k \ni x_k, D_k) < a$ for any k. Since, by Theorem 1.3 for the finite set Γ_0, the set $\{\operatorname{mld}(X_k \ni x_k, D_k) \mid k \in \mathbb{N}^+\}$ satisfies the ACC, there exists a positive real number δ such that

$$\operatorname{mld}(X_k \ni x_k, D_k) \leq a - \delta \quad \text{for any } k. \tag{20.3}$$

Since (X_k, D_k) is lc, $\operatorname{mld}(X_k \ni x_k, D_k) \geq 0$, so $\delta \leq a$. Let $\epsilon := \frac{\delta}{4a} \in (0, 1)$. Since

$$(1 + \epsilon)(a - \delta) < (1 - \epsilon)a,$$

we may fix a rational number α such that

$$(1 + \epsilon)(a - \delta) < \alpha < (1 - \epsilon)a. \tag{20.4}$$

In particular, $\alpha > 0$.

Let $V \subset \mathbb{R}^n$ be the smallest affine subspace defined over $\mathbb{Q}$ containing $\mathbf{s}$. Since X_k is $\mathbb{Q}$-factorial, $K_{X_k} + \sum_j u_j D_{kj}$ is $\mathbb{R}$-Cartier for any $u \in \mathbb{R}^n$. By [HLS26, Theorem 5.6], there exists an open subset $U \subset V$ containing $\mathbf{s}$ and independent of k such that

$$\Big(X_k, \sum_{j=1}^{n} u_j D_{kj}\Big) \text{ is lc for any } u = (u_1, \dots, u_n) \in U \text{ and any } k.$$

For any j with $s_j = 1$, the hyperplane $\{u \in \mathbb{R}^n \mid u_j = 1\}$ is defined over $\mathbb{Q}$ and contains $\mathbf{s}$, so it contains V. Thus $u_j = 1$ for any $u \in V$ and any j with $s_j = 1$, and $s_j \in (0,1)$ for the other j, so there exists a positive real number ρ such that

$$\{u \in V \mid \|u - \mathbf{s}\| \le \rho\} \subset U \cap (0,1]^n,$$

where $\|\cdot\|$ is the Euclidean norm on $\mathbb{R}^n$. Since V is defined over $\mathbb{Q}$, $V \cap \mathbb{Q}^n$ is dense in V. We let $\mathbf{s}' = (s'_1, \dots, s'_n) \in V \cap \mathbb{Q}^n$ be a vector with $\|\mathbf{s}' - \mathbf{s}\| \le \epsilon\rho$, and let

$$D'_k := \sum_{j=1}^{n} s'_j D_{kj}.$$

Since $\|\mathbf{s}' - \mathbf{s}\| \le \epsilon\rho < \rho$, $\mathbf{s}' \in U \cap (0,1]^n$, so (X_k, D'_k) is lc, and D'_k is a $\mathbb{Q}$-divisor whose coefficients belong to $\{s'_1, \dots, s'_n\} \subset (0,1] \cap \mathbb{Q}$.

Let E be a prime divisor over X_k. We show that

$$(1-\epsilon)a(E, X_k, tD_k) \le a(E, X_k, tD'_k) \le (1+\epsilon)a(E, X_k, tD_k) \quad \text{for any } t \in [0,1]. \tag{20.5}$$

If $\mathbf{s}' = \mathbf{s}$, then $D'_k = D_k$ and there is nothing to prove. Otherwise, let $w := (\mathbf{s}' - \mathbf{s})/\|\mathbf{s}' - \mathbf{s}\|$ and $\mu := \sum_j w_j \operatorname{mult}_E D_{kj}$, where $\operatorname{mult}_E D_{kj}$ is the coefficient of E in the pullback of D_{kj}. Since log discrepancies are affine in the boundary,

$$a\Big(E, X_k, \sum_{j=1}^{n}(s_j + \lambda w_j)D_{kj}\Big) = a(E, X_k, D_k) - \lambda\mu \quad \text{for any } \lambda \in \mathbb{R}. \tag{20.6}$$

Since $\mathbf{s}, \mathbf{s}' \in V$, we have

$$\mathbf{s} \pm \rho w \in \{u \in V \mid \|u - \mathbf{s}\| \le \rho\} \subset U,$$

so the left-hand side of (20.6) is non-negative for $\lambda = \pm\rho$. Thus $|\mu| \le a(E, X_k, D_k)/\rho$. Taking

$$\lambda = \|\mathbf{s}' - \mathbf{s}\| \le \epsilon\rho,$$

we obtain

$$|a(E, X_k, D'_k) - a(E, X_k, D_k)| = \|\mathbf{s}' - \mathbf{s}\| \cdot |\mu| \le \epsilon a(E, X_k, D_k).$$

Since X_k is klt, $a(E, X_k, 0) > 0$. Since log discrepancies are affine in the boundary,

$$\begin{aligned} |a(E, X_k, tD'_k) - a(E, X_k, tD_k)| &= t|a(E, X_k, D'_k) - a(E, X_k, D_k)| \\ &\le t\epsilon a(E, X_k, D_k) \le \epsilon a(E, X_k, tD_k) \end{aligned}$$

for any $t \in [0,1]$, which is (20.5). Taking the infimum over the prime divisors E over X_k with $\operatorname{center}_{X_k} E = \{x_k\}$, we obtain

$$(1-\epsilon)\operatorname{mld}(X_k \ni x_k, tD_k) \le \operatorname{mld}(X_k \ni x_k, tD'_k) \le (1+\epsilon)\operatorname{mld}(X_k \ni x_k, tD_k) \tag{20.7}$$

for any $t \in [0,1]$ and any k.

By Theorem 1.5 for the level α and the finite set $\{s'_1, \dots, s'_n\}$, since $\operatorname{mld}(X_k \ni x_k) \ge a > \alpha$,

$$\sigma_k := \tau_\alpha(X_k, D'_k; x_k) \notin \big(1 - \eta(\alpha, \{s'_1, \dots, s'_n\}), 1\big) \quad \text{for any } k. \tag{20.8}$$

By (20.7) with $t = 1$, (20.3), and (20.4),

$$\operatorname{mld}(X_k \ni x_k, D'_k) \le (1+\epsilon)(a - \delta) < \alpha,$$

so $\sigma_k < 1$ by (20.2). Since τ_k is a supremum, there exists $t_k \in [0,1]$ such that

$$\operatorname{mld}(X_k \ni x_k, t_k D_k) \ge a \quad \text{and} \quad t_k > \tau_k - \frac{1}{k}.$$

By (20.7) and (20.4),

$$\operatorname{mld}(X_k \ni x_k, t_k D'_k) \ge (1-\epsilon)a > \alpha,$$

so $\sigma_k \ge t_k$. Thus

$$1 - \frac{2}{k} < \tau_k - \frac{1}{k} < t_k \le \sigma_k < 1 \quad \text{for any } k,$$

which contradicts (20.8) for any $k > 2/\eta(\alpha, \{s'_1, \dots, s'_n\})$. The claim follows. □

Step 2. By [Liu26a, Lemma 6.2], we only need to prove Theorem 1.4 for $\mathbb{Q}$-factorial klt pairs. Suppose to the contrary that there exist a DCC set $\Gamma \subset [0,1]$ and, for any $i \in \mathbb{N}^+$, a $\mathbb{Q}$-factorial klt threefold pair (X_i, B_i) and a closed point $x_i \in X_i$ such that $B_i \in \Gamma$ and

$$a_i := \mathrm{mld}(X_i \ni x_i, B_i)$$

is strictly increasing in i.

By [Liu26a, Proposition 6.3], possibly shrinking X_i to a neighborhood of x_i and passing to a subsequence, we may assume that there exist a non-negative integer r and expressions

$$B_i = \sum_{j=1}^{r} b_{ij} D_{ij}$$

satisfying the following.

(1) For any i, $D_{i1}, \dots, D_{ir}$ are distinct prime divisors on X_i through x_i.
(2) For any j, the sequence $(b_{ij})_{i \in \mathbb{N}^+}$ is increasing with positive limit $b_j := \lim_{i \to +\infty} b_{ij}$.
(3) For any i, the pair $(X_i, \overline{B}_i)$ is lc, where $\overline{B}_i := \sum_{j=1}^{r} b_j D_{ij}$.

If $r = 0$, then $B_i = 0$ for any i, and the strictly increasing sequence $(a_i)_{i \in \mathbb{N}^+}$ contradicts Theorem 1.3 for the finite set $\{0\}$. Thus $r \geq 1$. Since $\Gamma \subset [0,1]$, $b_j \in (0,1]$ for any j. Since $b_{ij} \leq b_j$ for any i and j,

$$B_i \leq \overline{B}_i \quad \text{for any } i. \tag{20.9}$$

Step 3. For any $t \in [0,1]$, the pair $(X_i, t\overline{B}_i)$ is lc since $0 \leq t\overline{B}_i \leq \overline{B}_i$, and the coefficients of $t\overline{B}_i$ belong to the finite set $\{tb_1, \dots, tb_r\}$. Thus, by Theorem 1.3,

$$\{\mathrm{mld}(X_i \ni x_i, t\overline{B}_i) \mid i \in \mathbb{N}^+\} \text{ satisfies the ACC for any } t \in [0,1]. \tag{20.10}$$

Fix $t \in (0,1)$. Since $tb_j < b_j = \lim_{i \to +\infty} b_{ij}$ for any j, we have $t\overline{B}_i \leq B_i$ for any $i \gg 0$. Thus, by (20.1),

$$\mathrm{mld}(X_i \ni x_i, t\overline{B}_i) \geq a_i \quad \text{for any } i \gg 0, \tag{20.11}$$

where the bound on i may depend on t. By (20.10) with $t = \frac{1}{2}$, the numbers $\mathrm{mld}(X_i \ni x_i, \frac{1}{2}\overline{B}_i)$ are bounded from above, so by (20.11) with $t = \frac{1}{2}$ the strictly increasing sequence $(a_i)_{i \in \mathbb{N}^+}$ is bounded from above. Thus $a := \lim_{i \to +\infty} a_i$ is a real number, $a_i < a$ for any i, and $a > a_1 > 0$.

We show that

$$\mathrm{mld}(X_i \ni x_i, t\overline{B}_i) \geq a \quad \text{for any } t \in (0,1) \text{ and any } i \gg 0, \tag{20.12}$$

where the bound on i may depend on t. Fix $t \in (0,1)$, and suppose that (20.12) fails for t. Then, by (20.11), there exists an infinite set $I \subset \mathbb{N}^+$ such that

$$a_i \leq \mathrm{mld}(X_i \ni x_i, t\overline{B}_i) < a \quad \text{for any } i \in I.$$

Since $\lim_{i \to +\infty} a_i = a$, the set $\{\mathrm{mld}(X_i \ni x_i, t\overline{B}_i) \mid i \in I\}$ contains a strictly increasing sequence, which contradicts (20.10).

By (20.12) with $t = \frac{1}{2}$ and (20.1), $\mathrm{mld}(X_i \ni x_i) \geq \mathrm{mld}(X_i \ni x_i, \frac{1}{2}\overline{B}_i) \geq a$ for any $i \gg 0$. Possibly passing to a subsequence, we may assume that $\mathrm{mld}(X_i \ni x_i) \geq a$ for any i. Let

$$\tau_i := \tau_a(X_i, \overline{B}_i; x_i).$$

By (20.9) and (20.1), $\mathrm{mld}(X_i \ni x_i, \overline{B}_i) \leq a_i < a$, so $\tau_i < 1$ by (20.2). For any $t \in (0,1)$, (20.12) implies that $\tau_i \geq t$ for any $i \gg 0$. Thus

$$\tau_i < 1 \text{ for any } i \quad \text{and} \quad \lim_{i \to +\infty} \tau_i = 1.$$

By Claim 20.1 for a and $\Gamma_0 := \{b_1, \dots, b_r\}$, there exists a positive real number η such that $\tau_i \notin (1-\eta, 1)$ for any i, a contradiction. Theorem 1.4 follows. □

Appendix A. Proofs for zero values, coefficient-one components and non-klt pairs

Appendix A contains the proofs of Theorems 9.2, 9.3 and 9.4 of Section 9.

A.1. **Zero values.** In this subsection we prove Theorem 9.2. We need the following lemma.

Lemma A.1. *(1) Let $d \geq 2$ be an integer and $\Gamma \subset \mathbb{R}_{>0}$ a finite set. Then there exists a positive real number $L(d,\Gamma)$ satisfying the following. For any $\mathbb{R}$-ideal $\mathfrak{a}$ on the smooth germ $(\mathbb{A}^d \ni 0)$ with exponents in Γ and*

$$\operatorname{mld}(\mathbb{A}^d \ni 0, \mathfrak{a}) = 0,$$

there exists a prime divisor E over $\mathbb{A}^d$ with center 0 such that

$$a(E, \mathbb{A}^d, \mathfrak{a}) = 0 \quad \text{and} \quad a(E, \mathbb{A}^d, 0) \leq L(d,\Gamma).$$

(2) Let m be a positive integer. Let $f \neq 0$ be a polynomial on $\mathbb{A}^4$ vanishing at 0 and h a polynomial on $\mathbb{A}^4$, let $W := (f = 0) \subset \mathbb{A}^4$, and assume that W is normal, integral and klt near 0, $h|_W \neq 0$, and $\operatorname{mld}(W \ni 0, \frac{1}{m}\operatorname{div}(h|_W)) = 0$. Then there exists a prime divisor E over $\mathbb{A}^4$ with center 0 such that

$$a(E, \mathbb{A}^4, 0) = \operatorname{mult}_E f + \frac{1}{m}\operatorname{mult}_E h \leq L\big(4, \{1, \tfrac{1}{m}\}\big),$$

where $L(4, \{1, \frac{1}{m}\})$ is the constant of (1).

Proof. (1) Let $\mathfrak{m}_0$ be the maximal ideal of $0 \in \mathbb{A}^d$. Write $\Gamma = \{\gamma_1, \dots, \gamma_N\}$ with $\gamma_1, \dots, \gamma_N$ pairwise distinct. Multiplying the factors with equal exponents, we write any $\mathbb{R}$-ideal on $(\mathbb{A}^d \ni 0)$ with exponents in Γ as $\prod_{j=1}^N \mathfrak{a}_j^{\gamma_j}$ with non-zero ideals $\mathfrak{a}_1, \dots, \mathfrak{a}_N$, some of which may be the unit ideal.

Suppose that $L(d,\Gamma)$ does not exist. Then for any $i \in \mathbb{N}^+$ there exists an $\mathbb{R}$-ideal

$$\mathfrak{a}_i = \prod_{j=1}^N \mathfrak{a}_{i,j}^{\gamma_j}$$

on $(\mathbb{A}^d \ni 0)$ with $\operatorname{mld}(\mathbb{A}^d \ni 0, \mathfrak{a}_i) = 0$ such that $a(E, \mathbb{A}^d, 0) > i$ for any prime divisor E over $\mathbb{A}^d$ with center 0 and $a(E, \mathbb{A}^d, \mathfrak{a}_i) = 0$.

For any $i \in \mathbb{N}^+$, since $d \geq 2$ and $\operatorname{mld}(\mathbb{A}^d \ni 0, \mathfrak{a}_i) = 0$, there exists a prime divisor E_i over $\mathbb{A}^d$ with center 0 and $a(E_i, \mathbb{A}^d, \mathfrak{a}_i) = 0$, and

$$\operatorname{lct}(\mathbb{A}^d \ni 0; \mathfrak{a}_i) = 1 \quad \text{for any } i \in \mathbb{N}^+. \tag{A.1}$$

By [MN18, Section 3], there exist an algebraically closed field $\Bbbk \supset \mathbb{C}$ and a generic limit $\widetilde{\mathfrak{a}}_1, \dots, \widetilde{\mathfrak{a}}_{N+1}$ on $\widetilde{X} := \operatorname{Spec} \Bbbk[[x_1, \dots, x_d]]$ of the $N+1$ sequences of ideals

$$(\mathfrak{a}_{i,1})_{i\in\mathbb{N}^+}, \ \dots, \ (\mathfrak{a}_{i,N})_{i\in\mathbb{N}^+}, \ (\mathfrak{m}_0)_{i\in\mathbb{N}^+}$$

on the fixed germ $(\mathbb{A}^d \ni 0)$. Let $\widetilde{x}$ be the closed point of $\widetilde{X}$ and $\mathfrak{m}_{\widetilde{x}}$ the maximal ideal of $\widetilde{x}$. The limit of the constant sequence $(\mathfrak{m}_0)_i$ is $\widetilde{\mathfrak{a}}_{N+1} = \mathfrak{m}_{\widetilde{x}}$ [MN18, Section 3]. Let

$$\widetilde{\mathfrak{a}} := \prod_{j=1}^N \widetilde{\mathfrak{a}}_j^{\gamma_j}.$$

Where [MN18, Proposition 3.1(iii) or (iv)] is used below for $\widetilde{\mathfrak{a}}$ and the $\mathfrak{a}_i$, it is used for the $N+1$ sequences above with the exponents $\gamma_1, \dots, \gamma_N, 0$, which is allowed since [MN18, Proposition 3.1(iii) and (iv)] come from [dFEM11, Proposition 3.3 and Corollary 3.4], where the exponents are non-negative.

If $\widetilde{\mathfrak{a}}_j = 0$ for some j, then by [MN18, Proposition 3.1(i)], for an integer $r > d/\gamma_j$ there exist infinitely many i with $\mathfrak{a}_{i,j} \subset \mathfrak{m}_0^r$. For the exceptional prime divisor E_0 of the blow-up of $0 \in \mathbb{A}^d$, any such i satisfies

$$a(E_0, \mathbb{A}^d, \mathfrak{a}_i) \leq d - \gamma_j r < 0,$$

which contradicts $\mathrm{mld}(\mathbb{A}^d \ni 0, \mathfrak{a}_i) = 0$. Thus $\widetilde{\mathfrak{a}}_j \neq 0$ for any j. By [MN18, Proposition 3.1(iii)], $\mathrm{lct}(\widetilde{X} \ni \widetilde{x}; \widetilde{\mathfrak{a}})$ is a limit point of the sequence $(\mathrm{lct}(\mathbb{A}^d \ni 0; \mathfrak{a}_i))_i$, so (A.1) implies

$$\mathrm{lct}(\widetilde{X} \ni \widetilde{x}; \widetilde{\mathfrak{a}}) = 1.$$

In particular $(\widetilde{X}, \widetilde{\mathfrak{a}})$ is lc, and

$$\beta := \mathrm{mld}(\widetilde{X} \ni \widetilde{x}, \widetilde{\mathfrak{a}}) \geq 0.$$

Let $\mu : Y \to \widetilde{X}$ be a log resolution of

$$\Big(\widetilde{X}, \mathfrak{m}_{\widetilde{x}} \prod_{j=1}^{N} \widetilde{\mathfrak{a}}_j\Big)$$

[dFEM11, Theorems 2.3 and 2.4]. Since $\mathfrak{m}_{\widetilde{x}}\mathcal{O}_Y$ is invertible, the set $\mathcal{E}$ of prime divisors on Y with center $\widetilde{x}$ is non-empty, and a prime divisor E on Y satisfies $\mathrm{mult}_E \mathfrak{m}_{\widetilde{x}} > 0$ if and only if $E \in \mathcal{E}$.

We show that $\beta = 0$. Suppose that $\beta > 0$, and let

$$\epsilon := \min_{E \in \mathcal{E}} \frac{a(E, \widetilde{X}, \widetilde{\mathfrak{a}})}{\mathrm{mult}_E \mathfrak{m}_{\widetilde{x}}}.$$

Then

$$\epsilon \geq \frac{\beta}{\max_{E \in \mathcal{E}} \mathrm{mult}_E \mathfrak{m}_{\widetilde{x}}} > 0.$$

Let

$$\widetilde{\mathfrak{c}} := \widetilde{\mathfrak{a}} \cdot \mathfrak{m}_{\widetilde{x}}^{\epsilon}.$$

For any prime divisor E on Y, we have $a(E, \widetilde{X}, \widetilde{\mathfrak{c}}) \geq 0$: if $E \in \mathcal{E}$, this follows from the choice of ϵ, and if $E \notin \mathcal{E}$, then $\mathrm{mult}_E \mathfrak{m}_{\widetilde{x}} = 0$ and $(\widetilde{X}, \widetilde{\mathfrak{a}})$ is lc. Equality holds for some $E_* \in \mathcal{E}$. Since μ is a log resolution of $(\widetilde{X}, \widetilde{\mathfrak{c}})$, the pair $(\widetilde{X}, \widetilde{\mathfrak{c}})$ is lc and

$$\mathrm{mld}(\widetilde{X} \ni \widetilde{x}, \widetilde{\mathfrak{c}}) = 0.$$

Since $\widetilde{X}$ is regular, $\mathrm{lct}(\widetilde{X} \ni \widetilde{x}; \widetilde{\mathfrak{c}}) = 1$. On the other hand, for any i, the prime divisor E_i satisfies

$$a(E_i, \mathbb{A}^d, \mathfrak{a}_i \cdot \mathfrak{m}_0^{\epsilon}) = -\epsilon \,\mathrm{mult}_{E_i} \mathfrak{m}_0 < 0,$$

so

$$\theta_i := \mathrm{lct}(\mathbb{A}^d \ni 0; \mathfrak{a}_i \cdot \mathfrak{m}_0^{\epsilon}) < 1.$$

By [MN18, Proposition 3.1(iii)] for the $N+1$ sequences above and the positive exponents $\gamma_1, \ldots, \gamma_N, \epsilon$, the number 1 is a limit point of the sequence $(\theta_i)_i$ of real numbers less than 1, so $(\theta_i)_i$ has a strictly increasing subsequence. Each θ_i is the lc threshold on the fixed germ $(\mathbb{A}^d \ni 0)$ of the $\mathbb{R}$-ideal

$$\prod_{j=1}^{N} \mathfrak{a}_{i,j}^{\gamma_j} \cdot \mathfrak{m}_0^{\epsilon}$$

with exponents in the finite set $\Gamma \cup \{\epsilon\}$, so a strictly increasing subsequence of $(\theta_i)_i$ contradicts [dFEM11, Theorem 1.1]. Thus $\beta = 0$.

Since $\beta = 0$, there exists a prime divisor F over $\widetilde{X}$ with center $\widetilde{x}$ and $a(F, \widetilde{X}, \widetilde{\mathfrak{a}}) = 0$. Thus F computes $\mathrm{lct}(\widetilde{X} \ni \widetilde{x}; \widetilde{\mathfrak{a}}) = 1$. By [MN18, Proposition 3.1(iv)] for F, the existence of Λ and F_i below holds with the power $\mathfrak{m}_0^k$ for any sufficiently large integer k. Fix such a positive integer k with

$$k \,\mathrm{mult}_F \mathfrak{m}_{\widetilde{x}} > \mathrm{mult}_F \widetilde{\mathfrak{a}}_j \quad \text{for any } j \in \{1, \ldots, N\}.$$

Then

$$\operatorname{mult}_F(\widetilde{\mathfrak{a}}_j + \mathfrak{m}_{\widetilde{x}}^k) = \operatorname{mult}_F \widetilde{\mathfrak{a}}_j \quad \text{for any } j.$$

By [MN18, Proposition 3.1(iv)], there exist an infinite set $\Lambda \subset \mathbb{N}^+$ and, for any $i \in \Lambda$, a prime divisor F_i over $\mathbb{A}^d$ with center 0 such that

$$a(F_i, \mathbb{A}^d, 0) = a(F, \widetilde{X}, 0) \quad \text{and} \quad \operatorname{mult}_{F_i}(\mathfrak{a}_{i,j} + \mathfrak{m}_0^k) = \operatorname{mult}_F(\widetilde{\mathfrak{a}}_j + \mathfrak{m}_{\widetilde{x}}^k) = \operatorname{mult}_F \widetilde{\mathfrak{a}}_j$$

for any j. Let $i \in \Lambda$. Since $\operatorname{mld}(\mathbb{A}^d \ni 0, \mathfrak{a}_i) = 0$,

$$0 \le a(F_i, \mathbb{A}^d, \mathfrak{a}_i) \le a(F_i, \mathbb{A}^d, 0) - \sum_{j=1}^N \gamma_j \operatorname{mult}_{F_i}(\mathfrak{a}_{i,j} + \mathfrak{m}_0^k) = a(F, \widetilde{X}, \widetilde{\mathfrak{a}}) = 0.$$

Thus $a(F_i, \mathbb{A}^d, \mathfrak{a}_i) = 0$ and $a(F_i, \mathbb{A}^d, 0) = a(F, \widetilde{X}, 0)$ for any $i \in \Lambda$, which contradicts the choice of $\mathfrak{a}_i$ for $i \in \Lambda$ with $i > a(F, \widetilde{X}, 0)$. Thus $L(d, \Gamma)$ exists.

(2) Let $D := \frac{1}{m} \operatorname{div}(h|_W)$, and for any $t \in [0, 1]$ let

$$B_t := \operatorname{div}(f) + \frac{t}{m} \operatorname{div}(h) \quad \text{on } \mathbb{A}^4.$$

Since $\dim W = 3$ and $\operatorname{mld}(W \ni 0, D) = 0$, (W, D) is lc near 0. Since W is klt, (W, tD) is klt near 0 for any $t \in [0, 1)$. By Theorem 2.10 with the trivial group, $n = 4$, $c = 1$, $N = 1$, $f_1 = f$, $h_1 = h$, and $b_1 = \frac{t}{m}$, for any $t \in (0, 1)$,

$$\operatorname{mld}(W \ni 0, tD) = \operatorname{mld}(\mathbb{A}^4 \ni 0, B_t) > 0. \tag{A.2}$$

For any prime divisor E over $\mathbb{A}^4$ with center 0, $a(E, \mathbb{A}^4, B_t)$ is an affine function of t, positive on $(0, 1)$ by (A.2), so $a(E, \mathbb{A}^4, B_1) \ge 0$. Thus $\operatorname{mld}(\mathbb{A}^4 \ni 0, B_1) \ge 0$. Let P_0 be a prime divisor over W with center 0 and $a(P_0, W, D) = 0$. For any $t \in (0, 1)$, since $\operatorname{mult}_E h \ge 0$ for any prime divisor E over $\mathbb{A}^4$, by (A.2), and since $a(P_0, W, D) = 0$,

$$\operatorname{mld}(\mathbb{A}^4 \ni 0, B_1) \le \operatorname{mld}(\mathbb{A}^4 \ni 0, B_t) = \operatorname{mld}(W \ni 0, tD) \le a(P_0, W, tD) = (1 - t)a(P_0, W, 0).$$

Letting t tend to 1, we obtain $\operatorname{mld}(\mathbb{A}^4 \ni 0, B_1) = 0$.

Let $\mathfrak{a} := (f) \cdot (h)^{1/m}$, an $\mathbb{R}$-ideal on $(\mathbb{A}^4 \ni 0)$ with exponents in $\{1, \frac{1}{m}\}$, where (f) and (h) are non-zero since $f \ne 0$ and $h|_W \ne 0$. Since $a(E, \mathbb{A}^4, \mathfrak{a}) = a(E, \mathbb{A}^4, B_1)$ for any prime divisor E over $\mathbb{A}^4$, we have $\operatorname{mld}(\mathbb{A}^4 \ni 0, \mathfrak{a}) = 0$. By (1), there exists a prime divisor E over $\mathbb{A}^4$ with center 0 such that $a(E, \mathbb{A}^4, \mathfrak{a}) = 0$ and $a(E, \mathbb{A}^4, 0) \le L(4, \{1, \frac{1}{m}\})$, that is,

$$a(E, \mathbb{A}^4, 0) = \operatorname{mult}_E f + \frac{1}{m} \operatorname{mult}_E h \le L\big(4, \{1, \tfrac{1}{m}\}\big).$$

□

Proof of Theorem 9.2. Let

$$L_m := \max\Big\{1, L\big(3, \{\tfrac{1}{m}\}\big), L\big(4, \{1, \tfrac{1}{m}\}\big)\Big\},$$

where $L(d, \Gamma)$ is the constant of Lemma A.1(1). Let $D := \frac{1}{m} \operatorname{div}(h)$.

Step 1. By Lemma 4.2(1), $q(1)$ has a computing lift P_0. Since $q(1) = 0$, $a(P_0, W, D) = 0$, and $\operatorname{mld}(W \ni o, D) = 0$ since (W, D) is lc near o. Since W is klt, (W, tD) is klt near o for any $t \in [0, 1)$.

Step 2. In this step we assume in addition that $o = 0$ and that one of the following holds.

(a) $W = \mathbb{A}^3$, G acts diagonally on $\mathbb{A}^3$, and h is a G-semi-invariant polynomial.
(b) $W = (f = 0) \subset \mathbb{A}^4$ for a G-semi-invariant polynomial f, G acts diagonally on $\mathbb{A}^4$, and h is the restriction to W of a G-semi-invariant polynomial on $\mathbb{A}^4$, also denoted by h.

We show that

$$q(t) \le L_m(1-t) \quad \text{for any } t \in [0,1]. \tag{A.3}$$

Assume (a). By **Step 1**, $\mathrm{mld}(\mathbb{A}^3 \ni 0, (h)^{1/m}) = \mathrm{mld}(W \ni 0, D) = 0$. By Lemma A.1(1) with $d = 3$ and $\Gamma = \{\frac{1}{m}\}$, there exists a prime divisor E over $\mathbb{A}^3$ with center 0 such that

$$a(E, \mathbb{A}^3, 0) = \frac{1}{m} \operatorname{mult}_E h \le L\big(3, \{\tfrac{1}{m}\}\big) \le L_m.$$

By Definition 2.2, for any $t \in [0,1]$,

$$q(t) \le \frac{a(E, \mathbb{A}^3, 0) - \frac{t}{m} \operatorname{mult}_E h}{e_E} = (1-t)\frac{\operatorname{mult}_E h}{m e_E} \le L_m(1-t).$$

Assume (b). By Lemma A.1(2), there exists a prime divisor E over $\mathbb{A}^4$ with center 0 such that

$$a(E, \mathbb{A}^4, 0) = \operatorname{mult}_E f + \frac{1}{m} \operatorname{mult}_E h \le L\big(4, \{1, \tfrac{1}{m}\}\big).$$

Since $\operatorname{mult}_E f \ge 0$, we have $\frac{1}{m} \operatorname{mult}_E h \le a(E, \mathbb{A}^4, 0) \le L(4, \{1, \frac{1}{m}\})$. Let $e_E := |I_G(E)|$ for the action of G on $\mathbb{A}^4$. For any $t \in (0,1)$, let $B_t := \operatorname{div}(f) + \frac{t}{m} \operatorname{div}(h)$, a G-invariant divisor on $\mathbb{A}^4$. By **Step 1**, (W, tD) is klt near 0. By Theorem 2.10 with $n = 4$, $c = 1$, $N = 1$, $f_1 = f$, $h_1 = h$, $b_1 = \frac{t}{m}$, and $H = G$, and by Definition 2.2 on $\mathbb{A}^4$,

$$\begin{aligned} q(t) = \mathrm{mld}((\mathbb{A}^4 \ni 0, B_t)/G) &\le \frac{a(E, \mathbb{A}^4, 0) - \operatorname{mult}_E f - \frac{t}{m} \operatorname{mult}_E h}{e_E} \\ &= (1-t)\frac{\operatorname{mult}_E h}{m e_E} \le L\big(4, \{1, \tfrac{1}{m}\}\big)(1-t) \le L_m(1-t). \end{aligned}$$

In case (b), (A.3) holds for $t = 1$ since $q(1) = 0$, and for $t = 0$ by the case $t \in (0,1)$ and the continuity of q on $[0,1]$ (Lemma 4.2(1)). Thus (A.3) holds.

Step 3. In this step we show that $q(t) \le L_m(1-t)$ for any $t \in [0,1)$. Together with $q(1) = 0$, this proves $q(t) \le L_m(1-t)$ for any $t \in [0,1]$.

If W is smooth, then by [Liu26a, Lemma 5.4] there exist G-eigenparameters x_1, x_2, x_3 of W at o, and we let $n := 3$ and $Y := W$. Otherwise $(W \ni o)$ is an isolated Gorenstein terminal hypersurface threefold germ, and by [Liu26a, Lemma 5.4] there exist a G-equivariant closed embedding of W into a smooth fourfold germ $(Y \ni o)$ with $W = (f = 0)$ for a regular G-semi-invariant function f on Y, and G-eigenparameters $x_1, \dots, x_4$ of Y at o, and we let $n := 4$. By the last sentence of [Liu26a, Lemma 5.4], h lifts to a regular G-semi-invariant function on Y near o with character χ_h. Thus for $n \in \{3,4\}$, h is the restriction to W of a regular G-semi-invariant function on Y.

Fix $t \in [0,1)$, and let h', W' (and f' if $n = 4$) and the G-equivariant isomorphism between the completions be given by Lemma 4.6 for $\mathcal{P} = \{P_0\}$ (P_0 as in **Step 1**), $\psi = \mathrm{id}_{\mathbb{C}[[x_1,\dots,x_n]]}$, $u = v = 1$, and a sufficiently large L. If $n = 4$, then f' is G-semi-invariant, as a truncation of the G-semi-invariant power series $\widehat{f}$ in the G-eigenparameters $x_1, \dots, x_4$. By Lemma 4.6(2), P_0 corresponds to a prime divisor P_0' over W' with center 0, $I_G(P_0') = I_G(P_0)$, $a(P_0', W', 0) = a(P_0, W, 0)$, and $\operatorname{mult}_{P_0'} h' = \operatorname{mult}_{P_0} h < +\infty$. In particular $h'|_{W'} \ne 0$, so $h'|_{W'}$ is a mark on $(W' \ni 0)$. By Lemma 4.6(3), $(W', \frac{1}{m} \operatorname{div}(h'))$ is lc near 0. Let q' be the radial function of $(W' \ni 0, G, h'|_{W'}, m)$. Then

- $q'(t) = q(t)$ by Lemma 4.6(1) with $G' = G$,
- $q'(1) \ge 0$ by Lemma 4.2(1), and
- by Definition 2.2,

$$q'(1) \le \frac{a(P_0', W', 0) - \frac{1}{m} \operatorname{mult}_{P_0'} h'}{e_{P_0'}} = \frac{a(P_0, W, 0) - \frac{1}{m} \operatorname{mult}_{P_0} h}{e_{P_0}} = q(1) = 0.$$

Thus $q'(1) = 0$, and $(W' \ni 0, G, h'|_{W'})$ satisfies the hypotheses of Theorem 9.2 and (a) or (b) of **Step 2**. By **Step 2**,

$$q(t) = q'(t) \le L_m(1-t).$$

Step 4. In this step we find F. Since W is smooth or Gorenstein, K_W is Cartier near o. By **Step 3**, $q(t) \le q(1) + L_m(1-t)$ for any $t \in [0,1]$, and $L_m \ge 1 > 0$. By Lemma 4.4 with $\lambda = L_m$ and $t_0 = 0$, $q(1)$ has a computing lift F with

$$\frac{\operatorname{mult}_F h}{m e_F} \le L_m \quad \text{and} \quad \frac{a(F,W,0)}{e_F} \le q(1) + L_m = L_m.$$

The theorem follows. □

A.2. **Coefficient-one components.** In this subsection we prove Theorem 9.3 of Section 9. We first state and prove Lemma A.2.

Lemma A.2. *Let $(W \ni o)$ be an isolated Gorenstein terminal hypersurface threefold germ of quadratic rank 1 or 2, $\mathfrak{m}_o$ the maximal ideal of $\mathcal{O}_{W,o}$, G a finite abelian group acting faithfully on W and fixing o, m a positive integer, and h a mark. Let $D := \frac{1}{m}\operatorname{div}(h)$ and $b := \operatorname{mld}((W \ni o, D)/G)$. Assume that (W, D) is lc near o, $\lfloor D \rfloor \ne 0$ at o, and $\lfloor D \rfloor$ is $\mathbb{Q}$-Cartier. Then $\lfloor D \rfloor = \operatorname{div}(u)$ near o for a G-semi-invariant function u, and $h = u^m h_1$ for a mark h_1 coprime to u, that is, such that $\operatorname{div}(h_1)$ and $\operatorname{div}(u)$ have no common components.*

If $u \in \mathfrak{m}_o^2$, then $(W \ni o)$ has quadratic rank 2, h_1 is a unit, and $b = 0$ is computed by a prime divisor E over W with center o and $a(E,W,0) = 2$.

If $u \notin \mathfrak{m}_o^2$, let $S := (u = 0) \subset W$ and G_S the image of G in $\operatorname{Aut}(S \ni o)$. Then there exist a smooth fourfold germ $(\mathcal{X} \ni o)$ with a faithful action of G, a G-equivariant closed embedding $W \subset \mathcal{X}$ with $W = (f = 0)$ for a G-semi-invariant regular function f on $\mathcal{X}$, and G-semi-invariant regular parameters $x, y, z, \widetilde{u}$ of $\mathcal{X}$ at o with $\widetilde{u}|_W = u$, such that x, y, z are invariant under $\operatorname{Ker}(G \to G_S)$ and the character of $\widetilde{u}$ is faithful on $\operatorname{Ker}(G \to G_S)$. Let $V := (\widetilde{u} = 0) \subset \mathcal{X}$ and $f_S := f|_V$. Then

$$b = \operatorname{mld}\Big(\Big(V \ni o, \operatorname{div}(f_S) + \frac{1}{m}\operatorname{div}(\widetilde{h}_1|_V)\Big)/G_S\Big)$$

for any G-semi-invariant regular function $\widetilde{h}_1$ on $\mathcal{X}$ with $\widetilde{h}_1|_W = h_1$. Moreover, one of the following holds.

1. *h_1 is a unit, and $b = 0$ is computed by a prime divisor E over W with center o and $a(E,W,0) \le 7$.*
2. *S is a Du Val surface germ, and $b = \operatorname{mld}((S \ni o, \frac{1}{m}\operatorname{div}(h_1|_S))/G_S)$.*
3. *S has a normal crossing point at o, and $(W \ni o)$ has quadratic rank 2.*
4. *f_S is a unit multiple of $x'^2 + y'^2 z'$ in $\widehat{\mathcal{O}}_{V,o}$ for some G_S-semi-invariant regular parameters x', y', z' of $\widehat{\mathcal{O}}_{V,o}$,*

$$b = \frac{1}{N}\Big(1 - \frac{2c}{m}\Big),$$

where c is the order in z' of the image of $h_1|_S$ in $\widehat{\mathcal{O}}_{S,o}/(x', y') = \mathbb{C}[[z']]$ and N is the order of the character $\chi_{x'}/\chi_{y'}$ of G_S, and b is computed by a prime divisor F over W with center o and $a(F,W,0)/e_F \le 2m + 6$.

Proof. Since (W, D) is lc near o, $b \ge 0$, and by Definition 2.2 any prime divisor P over W with center o and $a(P,W,D) = 0$ computes $b = 0$. By Lemma 2.12(2)(3), for $(W \ni o)$ and for any germ $(Y \ni y)$ below, the quotient values, the prime divisors over Y with center y, their log discrepancies, and their inertia groups do not change when Y is replaced by $\operatorname{Spec}\widehat{\mathcal{O}}_{Y,y}$. We therefore take coordinates in completed local rings, and regard a formal prime divisor over $\operatorname{Spec}\widehat{\mathcal{O}}_{W,o}$ with center the closed point as a prime divisor over W with center o. We divide the proof into five steps.

Step 1. In this step we construct u and h_1. By [Kaw88, Lemma 5.1], $\lfloor D \rfloor$ is Cartier near o, and projecting a generator u_0 of the ideal of $\lfloor D \rfloor$ in $\mathcal{O}_{W,o}$ to the isotypic component of the character of G on $(u_0)/\mathfrak{m}_o(u_0)$, we obtain a G-semi-invariant function u with $\lfloor D \rfloor = \operatorname{div}(u)$ near o. Since (W, D) is lc, $\lfloor D \rfloor$ is reduced, so $h_1 := h/u^m$ is a mark and $\operatorname{div}(h_1)$ and $\operatorname{div}(u)$ have no common components.

Step 2. In this step we treat the case $u \in \mathfrak{m}_o^2$, and prove that $u \notin \mathfrak{m}_o^2$ if the quadratic rank is one. Since $(W \ni o)$ is Gorenstein terminal, it is an isolated cDV germ.

Assume that the quadratic rank is one, and let E_0 be the prime divisor of Lemma 3.1(2), so $a(E_0, W, 0) = 3$ and $\operatorname{mult}_{E_0} \mathfrak{m}_o = 2$. If $u \in \mathfrak{m}_o^2$, then $\operatorname{mult}_{E_0} u \geq 4$, and since $D \geq \operatorname{div}(u)$,

$$0 \leq a(E_0, W, D) \leq a(E_0, W, 0) - \operatorname{mult}_{E_0} u \leq 3 - 4,$$

a contradiction. Thus $u \notin \mathfrak{m}_o^2$.

Assume that the quadratic rank is two and $u \in \mathfrak{m}_o^2$. Let E be a prime divisor of Lemma 3.1(1), so $a(E, W, 0) = 2$ and $\operatorname{mult}_E \mathfrak{m}_o = 1$. Since $u \in \mathfrak{m}_o^2$, $\operatorname{mult}_E u \geq 2$, so

$$0 \leq a(E, W, D) = 2 - \operatorname{mult}_E u - \frac{1}{m} \operatorname{mult}_E h_1 \leq -\frac{1}{m} \operatorname{mult}_E h_1.$$

Thus $\operatorname{mult}_E h_1 = 0$ and $a(E, W, D) = 0$. Since $\operatorname{mult}_E \mathfrak{m}_o = 1$, $h_1 \notin \mathfrak{m}_o$, so h_1 is a unit, and E computes $b = 0$.

Step 3. In this step we assume that $u \notin \mathfrak{m}_o^2$ and prove the formula for b. By [Liu26a, Lemma 5.4], there exist a smooth fourfold germ $(\mathcal{X} \ni o)$ with a faithful action of G, a G-equivariant closed embedding $W \subset \mathcal{X}$ with $W = (f = 0)$ for a G-semi-invariant regular function f on $\mathcal{X}$, and G-semi-invariant regular parameters of $\mathcal{X}$ at o, and any G-semi-invariant regular function ϕ on W is $\phi'|_W$ for a G-semi-invariant regular function ϕ' on $\mathcal{X}$ with $\chi_{\phi'} = \chi_\phi$. Let $\widetilde{u}$ be such a lift of u. We write u also for $\widetilde{u}$. Since W is singular at o, $f \in \mathfrak{m}_{\mathcal{X},o}^2$, so

$$\mathfrak{m}_{\mathcal{X},o}/\mathfrak{m}_{\mathcal{X},o}^2 = \mathfrak{m}_o/\mathfrak{m}_o^2.$$

Since $u \notin \mathfrak{m}_o^2$, replacing one of the G-semi-invariant regular parameters of $\mathcal{X}$ at o given above by $\widetilde{u}$, we obtain G-semi-invariant regular parameters $x, y, z, \widetilde{u}$ of $\mathcal{X}$ at o. We use $\widehat{\mathcal{O}}_{\mathcal{X},o} = \mathbb{C}[[x, y, z, u]]$, on which G acts through $\chi_x, \chi_y, \chi_z, \chi_u$. Then $V = (\widetilde{u} = 0) \subset \mathcal{X}$,

$$\widehat{\mathcal{O}}_{V,o} = \mathbb{C}[[x, y, z]] \subset \mathbb{C}[[x, y, z, u]],$$

and $f_S = f(x, y, z, 0)$ and $f - f_S \in (u)$ in $\mathbb{C}[[x, y, z, u]]$. Since W is integral and singular at o, $f \notin (u)$, so $f_S \neq 0$.

An element of G acts trivially on S if and only if it acts trivially on $\widehat{\mathcal{O}}_{S,o} = \mathbb{C}[[x, y, z]]/(f_S)$, that is, on x, y, z, since $f_S \in (x, y, z)^2$. Let $G_1 := \operatorname{Ker}(G \to G_S)$. Thus χ_x, χ_y, χ_z are trivial on G_1, and G_S acts faithfully on V. Since G acts faithfully on W, the identity is the only element of G acting trivially on x, y, z, u, so χ_u is faithful on G_1. Let

$$r := |G_1| \quad \text{and} \quad v := u^r.$$

Then

$$\mathbb{C}[[x, y, z, u]]^{G_1} = \mathbb{C}[[x, y, z, v]],$$

on which G_S acts diagonally.

Let $\widetilde{h}_1$ be a G-semi-invariant regular function on $\mathcal{X}$ with $\widetilde{h}_1|_W = h_1$, and let $h_V := \widetilde{h}_1|_V$. Since h_1 is coprime to u, $h_V|_S = h_1|_S \neq 0$. The power series f_S and h_V in x, y, z are non-zero and invariant under G_1, so χ_f and $\chi_{\widetilde{h}_1}$ are trivial on G_1, that is, f and $\widetilde{h}_1$ are invariant under G_1.

Let $\mathcal{X}' := \mathcal{X}/G_1$ with the induced action of G_S, let $\pi : W \to W' := W/G_1 \subset \mathcal{X}'$ be the quotient map, and let $o' := \pi(o)$. By Lemma 2.12(1), $\widehat{\mathcal{O}}_{\mathcal{X}',o'} = \mathbb{C}[[x, y, z, v]]$, so $\mathcal{X}'$ is smooth at o' with G_S-semi-invariant regular parameters x, y, z, v. The functions f and $\widetilde{h}_1$ descend to regular functions f', h' on $\mathcal{X}'$, with $f'(x, y, z, 0) = f_S$ and $h'(x, y, z, 0) = h_V$ in $\mathbb{C}[[x, y, z, v]]$, and

$W' = (f' = 0) \subset \mathcal{X}'$, since taking invariants under G_1 is exact and f is invariant. Since π is ramified along $S = \operatorname{div}(u)$ with index r, the Hurwitz formula shows

$$K_W + D = K_W + \operatorname{div}(u) + \frac{1}{m}\operatorname{div}(h_1) = \pi^*(K_{W'} + D'), \qquad D' := \operatorname{div}(v) + \frac{1}{m}\operatorname{div}(h'|_{W'}).$$

Thus $(W, D)/G$ and $(W', D')/G_S$ are equal crepant quotients on $W/G = W'/G_S$, and

$$b = \operatorname{mld}((W' \ni o', D')/G_S).$$

Moreover, W' is klt as W is, and (W', D') is lc near o' as (W, D) is lc near o.

Shrinking $\mathcal{X}'$, we embed it, near o', G_S-equivariantly as a smooth complete intersection $(\Phi_1 = \cdots = \Phi_M = 0)$ in $\mathbb{A}^{4+M}$ with a diagonal action of G_S, in which x, y, z, v are coordinates, and let $\overline{f}, \overline{h}$ be G_S-semi-invariant polynomials restricting to f', h'. Let

$$\overline{B} := \sum_{i=1}^{M} \operatorname{div}(\Phi_i) + \operatorname{div}(\overline{f}) + \operatorname{div}(v) + \frac{1}{m}\operatorname{div}(\overline{h}) \quad \text{on } \mathbb{A}^{4+M}.$$

Near 0, $W' = (\Phi_1 = \cdots = \Phi_M = \overline{f} = 0)$, and $(v) \cdot (\overline{h})^{1/m}$ restricts to the non-zero $\mathbb{R}$-ideal on W' defining D'. By Theorem 2.17(1) with $n = 4 + M$, $c = M + 1$, $X = W'$, $\mathfrak{a} = (v) \cdot (\overline{h})^{1/m}$, and $H = G_S$,

$$b = \operatorname{mld}((\mathbb{A}^{4+M} \ni 0, \overline{B})/G_S).$$

Since $b \geq 0$, $(\mathbb{A}^{4+M}, \overline{B})$ is lc near 0. By adjunction to the smooth complete intersection $(\Phi_1 = \cdots = \Phi_M = v = 0) \cong V$, on which $\overline{f}$ and $\overline{h}$ restrict to f_S and h_V,

$$\Big(V, S + \frac{1}{m}\operatorname{div}(h_V)\Big) \text{ is lc near } o, \tag{A.4}$$

where $S = \operatorname{div}(f_S)$ on V. By Theorem 2.17(1) with

$$n = 4+M, \quad c = M+1, \quad X = V \cong (\Phi_1 = \cdots = \Phi_M = v = 0), \quad \mathfrak{a} = (\overline{f}) \cdot (\overline{h})^{1/m}, \quad \text{and} \quad H = G_S,$$

the value $\operatorname{mld}((\mathbb{A}^{4+M} \ni 0, \overline{B})/G_S)$ is equal to the quotient value on V, that is,

$$b = \operatorname{mld}\Big(\Big(V \ni o, S + \frac{1}{m}\operatorname{div}(h_V)\Big)/G_S\Big). \tag{A.5}$$

Step 4. In this step we determine $(S \ni o)$, and prove (1), (2), (3) and the normal form in (4).

For $w = (w_x, w_y, w_z) \in \mathbb{R}^3_{\geq 0}$ let $|w| := w_x + w_y + w_z$. For a primitive $w \in (\mathbb{N}^+)^3$, let E_w be the exceptional prime divisor of the weighted blow-up of V with weights w in x, y, z. Then $a(E_w, V, 0) = |w|$ [CLS11, Lemma 11.4.10] and $\operatorname{mult}_{E_w} \phi = \operatorname{wt}_w(\phi)$ for any $\phi \in \mathbb{C}[[x, y, z]]$. By (A.4), for any such w,

$$\operatorname{wt}_w(f_S) + \frac{1}{m}\operatorname{wt}_w(h_V) \leq |w|. \tag{A.6}$$

For a primitive $w \in (\mathbb{N}^+)^4$ of weights in x, y, z, u with $\operatorname{in}_w f$ reduced, any irreducible component of $(\operatorname{in}_w f = 0)$ in the exceptional divisor of the weighted blow-up of $\mathcal{X}$ is a prime divisor E over W with center o. Assume that one of the following holds.

(i) Two weights of w are 1, or a coordinate of weight 1 does not divide $\operatorname{in}_w f$.
(ii) E is contained in no coordinate hyperplane.

Then

$$\begin{aligned} a(E, W, 0) &= w_x + w_y + w_z + w_u - \operatorname{wt}_w(f), \\ \operatorname{mult}_E \phi &= \operatorname{wt}_w(\phi), \end{aligned} \tag{A.7}$$

where the second equality holds for any $\phi \in \mathbb{C}[[x, y, z, u]]$ such that the irreducible factor of $\operatorname{in}_w f$ defining E does not divide $\operatorname{in}_w \phi$.

We prove (A.7). Since E is defined by an irreducible factor of $\operatorname{in}_w f$, which is divisible by at most one coordinate, there exists a coordinate x_k among x, y, z, u not vanishing at the generic point of E, of weight 1 in case (i). We use the x_k-chart of Definition 3.8(1)(2), which works

equally for power series in x, y, z, u, writing ϖ for the chart parameter: it has the coordinates ϖ and X_j for $j \neq k$, $x_k = \varpi^{w_k}$, $x_j = \varpi^{w_j} X_j$ for $j \neq k$, and the group μ_{w_k}. In case (i), $w_k = 1$, so the chart is the cover itself. In case (ii), $X_j \neq 0$ for $j \neq k$ at a general closed point of the preimage of E in the cover, so the stabilizer in μ_{w_k} of such a point is $\bigcap_j \mu_{w_j} = 1$ as w is primitive, and the quotient map of the chart is étale over the generic point of E. For $\phi \in \mathbb{C}[[x, y, z, u]]$ write $\phi = \varpi^{\mathrm{wt}_w(\phi)} \widetilde{\phi}$, so $\widetilde{\phi}|_{\varpi=0} = \mathrm{in}_w(\phi)|_{X_k=1}$. Let E' be a prime divisor over E on the strict transform $(\widetilde{f} = 0)$ of W in the cover. Since $\mathrm{in}_w f$ is reduced, $\widetilde{f}|_{\varpi=0}$ has multiplicity 1 along E', so ϖ is a uniformizer of the local ring of $(\widetilde{f} = 0)$ at the generic point of E', which is a discrete valuation ring, and $\mathrm{mult}_E \phi = \mathrm{wt}_w(\phi) + \mathrm{mult}_{E'} \widetilde{\phi}$. If the irreducible factor of $\mathrm{in}_w f$ defining E does not divide $\mathrm{in}_w \phi$, then $\widetilde{\phi}$ does not vanish identically on E', so $\mathrm{mult}_E \phi = \mathrm{wt}_w(\phi)$. Finally, $dx \wedge dy \wedge dz \wedge du$ pulls back to

$$\pm w_k \varpi^{w_x + w_y + w_z + w_u - 1} d\varpi \wedge \bigwedge_{j \neq k} dX_j,$$

so comparing the residue forms on W and on $(\widetilde{f} = 0)$, the coefficient of E in the relative canonical divisor is $w_x + w_y + w_z + w_u - 1 - \mathrm{wt}_w(f)$, that is, $a(E, W, 0) = w_x + w_y + w_z + w_u - \mathrm{wt}_w(f)$.

Below we replace x, y, z by other G_S-semi-invariant regular parameters of $\mathbb{C}[[x, y, z]]$, each regarded as an element of $\mathbb{C}[[x, y, z, u]]$ through $\mathbb{C}[[x, y, z]] \subset \mathbb{C}[[x, y, z, u]]$. The new x, y, z are G-semi-invariant and invariant under G_1, and $f - f_S \in (u)$ still holds. Since Weierstrass preparation is unique and a G_S-semi-invariant unit is G_S-invariant (evaluate it at o), the coordinate changes below are G_S-equivariant: the new coordinates are G_S-semi-invariant, and the units that occur and their roots with fixed constant terms are G_S-invariant.

Claim. Let $w' \in (\mathbb{N}^+)^3$ be primitive with $\mathrm{wt}_{w'}(f_S) = |w'|$ and $\mathrm{in}_{w'} f_S$ reduced, such that two weights of w' are 1 or a coordinate of weight 1 does not divide $\mathrm{in}_{w'} f_S$. Then h_1 is a unit and $b = 0$ is computed by a prime divisor E over W with center o and $a(E, W, 0) = |w'| + 1$.

Proof of the Claim. By (A.6), $\mathrm{wt}_{w'}(h_V) = 0$, so $h_V(o) \neq 0$ and h_1 is a unit. Let

$$w := (w', |w'| + 1).$$

Since $f - f_S \in (u)$,

$$\mathrm{in}_w f = \mathrm{in}_{w'} f_S \in \mathbb{C}[x, y, z],$$

which is reduced, and $\mathrm{wt}_w(f) = |w'|$. Let E be a prime divisor as in (A.7). Then

$$a(E, W, 0) = \mathrm{mult}_E u = |w'| + 1, \qquad a(E, W, D) = a(E, W, 0) - \mathrm{mult}_E u - \frac{1}{m} \mathrm{mult}_E h_1 = 0,$$

and E computes $b = 0$.

Let $f_{S,2}$ and $f_{S,3}$ be the quadratic and cubic parts of f_S. Since $f_{S,2}$ is the restriction to V of the quadratic part of f and $(W \ni o)$ has quadratic rank at most 2, $\mathrm{rank}\, f_{S,2} \leq 2$.

$f_{S,2} = 0$. Then $\mathrm{wt}_{(1,1,1)}(f_S) \geq 3$, so $\mathrm{wt}_{(1,1,1)}(f_S) = 3$ by (A.6). Let $\beta : \widetilde{V} \to V$ be the blow-up of V at o with exceptional divisor $E_1 \cong \mathbb{P}^2$. Since $a(E_1, V, S) = 3 - 3 = 0$, the crepant pullback of (V, S) is $(\widetilde{V}, \beta_*^{-1} S + E_1)$. By (A.4), (V, S) is lc near o, so by adjunction $(E_1, \beta_*^{-1} S|_{E_1}) = (\mathbb{P}^2, (f_{S,3} = 0))$ is lc, and $f_{S,3}$ is reduced. By the Claim with $w' = (1, 1, 1)$, (1) holds with $a(E, W, 0) = 4$.

$f_{S,2}$ has rank 1. Then $f_{S,2}$ is a non-zero multiple of x_0^2 for a linear form x_0, unique up to scalar and hence G_S-semi-invariant. We take $x := x_0$, and y, z to be G_S-semi-invariant linear forms with x, y, z linearly independent. By the Weierstrass preparation theorem in x and completion of the square, after replacing x with $x - \phi(y, z)$ with $\phi \in (y, z)^2$, f_S is a unit multiple of

$$x^2 + g(y, z), \qquad g \in (y, z)^3.$$

Let g_k be the homogeneous part of degree k of g. We distinguish the binary cubic forms g_3.

Assume that $g_3 = 0$. Then $\mathrm{wt}_{(2,1,1)}(f_S) \geq 4 = |(2, 1, 1)|$. If $g_4 = 0$, then

$$\mathrm{wt}_{(5,2,2)}(f_S) \geq 10 > 9 = |(5, 2, 2)|,$$

contradicting (A.6). Thus $g_4 \neq 0$, and $\mathrm{in}_{(2,1,1)} f_S$ is a non-zero multiple of $x^2 + g_4$, which is reduced. By the Claim with $w' = (2,1,1)$, (1) holds with $a(E,W,0) = 5$.

Assume that g_3 is a non-zero multiple of ℓ^3 for a linear form ℓ in y, z. Since ℓ is unique up to scalar, it is G_S-semi-invariant. We take $y := \ell$, and take z to be a G_S-semi-invariant linear form in y, z independent of y. By the Weierstrass preparation theorem in y and the shift of y by an element of $(z)^2$ removing the term in y^2, g is a unit multiple of

$$y^3 + \eta(z)y + \nu(z), \qquad \eta \in (z^3),\ \nu \in (z^4),$$

since g_3 is a multiple of y^3. Let η_k and ν_k be the coefficients of z^k in η and ν. By the classification of Du Val singularities, S is of type E_6 if $\nu_4 \neq 0$, of type E_7 if $\nu_4 = 0 \neq \eta_3$, and of type E_8 if $\nu_4 = \eta_3 = 0 \neq \nu_5$. If $\nu_4 = \eta_3 = \nu_5 = 0$, then $\mathrm{wt}_{(3,2,1)}(f_S) = 6 = |(3,2,1)|$, and $\mathrm{in}_{(3,2,1)} f_S$ is a non-zero multiple of $x^2 + \lambda(y^3 + \eta_4 y z^4 + \nu_6 z^6)$ for some $\lambda \neq 0$. The polynomial $x^2 + \lambda(y^3 + \eta_4 y z^4 + \nu_6 z^6)$ is irreducible, since $y^3 + \eta_4 y z^4 + \nu_6 z^6$ is not a square, and z does not divide it. By the Claim with $w' = (3,2,1)$, (1) holds with $a(E,W,0) = 7$.

Assume that g_3 has three distinct linear factors. Then S is Du Val of type D_4.

Assume that $g_3 = \ell_1^2 \ell_2$ for non-proportional linear forms ℓ_1, ℓ_2 in y, z. The lowest degree part of a product of power series is the product of their lowest degree parts, and the lowest degree part of an irreducible element of $\mathbb{C}[[y,z]]$ is a power of a linear form. Thus g has exactly one irreducible factor g' whose linear part is a non-zero multiple of ℓ_2, and g' has multiplicity 1 in g. The ideal (g') is G_S-stable, so it is generated by a G_S-semi-invariant z with linear part ℓ_2, as in **Step 1**. Since ℓ_1 is unique up to scalar, it is G_S-semi-invariant, and we take $y := \ell_1$. Write $g = z\kappa_1(y,z)$, where the quadratic part of κ_1 is a non-zero multiple of y^2. By the Weierstrass preparation theorem in y and completion of the square, after replacing y with $y - \phi(z)$ with $\phi \in (z)^2$, κ_1 is a unit multiple of $y^2 + \kappa(z)$ with $\kappa \in (z^3)$. If $\kappa \neq 0$, then κ is the product of z^{n-2} and an invariant unit for an integer $n \geq 5$, and absorbing roots of invariant units into z and then into x, we may assume that f_S is a unit multiple of

$$x^2 + y^2 z + z^{n-1},$$

so S is Du Val of type D_n. If $\kappa = 0$, absorbing a root of an invariant unit into x, we may assume that f_S is a unit multiple of $x^2 + y^2 z$, which is the normal form in (4).

$f_{S,2}$ has rank 2. Then $(W \ni o)$ has quadratic rank 2. By the splitting lemma [GLS07, Theorem I.2.47], $(S \ni o) \cong (x^2 + y^2 + \zeta(z) = 0)$ for a power series $\zeta \in (z^3)$. If $\zeta \neq 0$, then ζ is the product of z^{n+1} and a unit for an integer $n \geq 2$, and S is Du Val of type A_n. If $\zeta = 0$, then S has a normal crossing point at o, and (3) holds.

Du Val sections. Assume that S is Du Val. By (A.4) and adjunction to S, $(S, \frac{1}{m}\mathrm{div}(h_1|_S))$ is lc near o. Near 0, $S \cong (\Phi_1 = \cdots = \Phi_M = v = \overline{f} = 0)$ in $\mathbb{A}^{4+M}$, and $\overline{h}$ restricts to $h_1|_S$. By Theorem 2.17(1) with $n = 4 + M$, $c = M + 2$, $X = S$, $\mathfrak{a} = (\overline{h})^{1/m}$, and $H = G_S$, and by **Step 3**,

$$\mathrm{mld}\Big(\Big(S \ni o, \frac{1}{m}\mathrm{div}(h_1|_S)\Big)/G_S\Big) = \mathrm{mld}((\mathbb{A}^{4+M} \ni 0, \overline{B})/G_S) = b,$$

so (2) holds.

Step 5. In this step we prove the rest of (4). Let x, y, z be as in the normal form (so x, y, z are the x', y', z' of (4)). Then f_S is the product of $x^2 + y^2 z$ and an invariant unit. In particular $x^2 + y^2 z$ is G_S-semi-invariant, so $\chi_x^2 = \chi_y^2 \chi_z$. Let

$$\rho := \chi_x/\chi_y, \qquad \text{so that} \qquad \chi_z = \rho^2,$$

and let N be the order of ρ on G_S. Let $\alpha(z) := h_V(0,0,z)$, so that $h_V - \alpha \in (x,y)$ and α is the image of $h_1|_S$ in $\widehat{\mathcal{O}}_{S,o}/(x,y) = \mathbb{C}[[z]]$.

First we prove $\alpha \neq 0$. If $\alpha = 0$, then $h_V \in (x,y)$, and for $w = (m+2, m+1, 1)$,

$$\mathrm{wt}_w(f_S) + \frac{1}{m}\mathrm{wt}_w(h_V) \geq 2m + 3 + \frac{m+1}{m} > 2m + 4 = |w|,$$

contradicting (A.6). Thus $\alpha \neq 0$, and $\alpha \in (z^c)\backslash(z^{c+1})$ for an integer $c \geq 0$. Next we prove $2c \leq m$. For $w = (j+1, j, 2)$ with an integer $j > 2c$, $\mathrm{wt}_w(f_S) = 2j+2$, and $\mathrm{wt}_w(h_V) = 2c$, since $\mathrm{wt}_w(h_V - \alpha) \geq j$. Thus (A.6) shows $2j + 2 + \frac{2c}{m} \leq 2j+3$, that is, $2c \leq m$.

Next we prove $b \geq (1 - \frac{2c}{m})/N$. Fix an integer $j > 2c$, let $w_0 := (j+1, j, 2)$, and let $\alpha_c \neq 0$ be the coefficient of z^c in α. Let P be a prime divisor over V with center o. For an integer k, let $h_V^{\leq k}$ be the sum of the terms of h_V of degree at most k, which is a G_S-semi-invariant polynomial with $\chi_{h_V^{\leq k}} = \chi_{h_V}$. Since $\mathrm{mult}_P(h_V - h_V^{\leq k}) \geq k+1$, we may fix $k \geq c$ with $\mathrm{mult}_P\, h_V^{\leq k} = \mathrm{mult}_P\, h_V$. The completions of V at o and of $\mathbb{A}^3$ at 0 are $\mathbb{C}[[x,y,z]]$ with the diagonal action of G_S through χ_x, χ_y, χ_z. Thus by Lemma 2.12(3), P is also a prime divisor over $\mathbb{A}^3$ with center 0, $I_{G_S}(P)$ is the inertia group of P over V and over $\mathbb{A}^3$, and

$$a\Big(P, V, S + \frac{1}{m}\operatorname{div}(h_V)\Big) = a(P, \mathbb{A}^3, B_k), \qquad B_k := \operatorname{div}(x^2 + y^2 z) + \frac{1}{m}\operatorname{div}(h_V^{\leq k}).$$

Since $\mathrm{in}_{w_0}(x^2+y^2z) = x^2 + y^2 z$ and $\mathrm{in}_{w_0}(h_V^{\leq k}) = \alpha_c z^c$, Lemma 6.4(1) shows

$$\mathrm{mld}((\mathbb{A}^3 \ni 0, B_k)/G_S) \geq \mathrm{mld}((\mathbb{A}^3 \ni 0, B_{w_0})/G_S), \quad \text{where} \quad B_{w_0} := \operatorname{div}(x^2+y^2z) + \frac{c}{m}\operatorname{div}(z),$$

so

$$\frac{a(P, \mathbb{A}^3, B_k)}{|I_{G_S}(P)|} \geq \mathrm{mld}((\mathbb{A}^3 \ni 0, B_{w_0})/G_S).$$

By (A.5) and Definition 2.2, taking the infimum over P implies

$$b \geq \mathrm{mld}((\mathbb{A}^3 \ni 0, B_{w_0})/G_S).$$

For $w \in \mathbb{R}^3_{\geq 0}$, let $\theta := w_x - w_y$ and

$$\psi(w) := |w| - \mathrm{wt}_w(x^2+y^2z) - \frac{c}{m}w_z = \Big(\frac{1}{2} - \frac{c}{m}\Big)w_z + \Big|\theta - \frac{w_z}{2}\Big|.$$

Since $2c \leq m$, $\psi \geq 0$ on $\mathbb{R}^3_{\geq 0}$, so by [Liu26a, Proposition 8.11(2)(3)], $(\mathbb{A}^3, B_{w_0})$ is lc near 0, and

$$\mathrm{mld}((\mathbb{A}^3 \ni 0, B_{w_0})/G_S) = \min_{w \in N_{G_S} \cap \mathbb{R}^3_{>0}} \psi(w)$$

(Definition 2.3). For $w \in N_{G_S}$, $(\exp(2\pi i w_x), \exp(2\pi i w_y), \exp(2\pi i w_z))$ is the image of some $\gamma \in G_S$, so $\exp(2\pi i\theta) = \rho(\gamma)$ and $\exp(2\pi i(w_z - 2\theta)) = \chi_z(\gamma)\rho(\gamma)^{-2} = 1$, so

$$\theta \in \frac{1}{N}\mathbb{Z} \quad \text{and} \quad w_z - 2\theta \in \mathbb{Z}.$$

Let $w \in N_{G_S} \cap \mathbb{R}^3_{>0}$.

- If $\theta > 0$, then $\theta \geq \frac{1}{N}$. If $w_z \leq 2\theta$, then $\psi(w) = \theta - \frac{c}{m}w_z \geq (1 - \frac{2c}{m})\theta$. If $w_z \geq 2\theta$, then $\psi(w) = (1 - \frac{c}{m})w_z - \theta \geq (1 - \frac{2c}{m})\theta$. Thus $\psi(w) \geq (1 - \frac{2c}{m})\theta \geq (1 - \frac{2c}{m})/N$.
- If $\theta < 0$, then
$$\psi(w) = \Big(1 - \frac{c}{m}\Big)w_z - \theta \geq -\theta \geq \frac{1}{N} \geq \Big(1 - \frac{2c}{m}\Big)\frac{1}{N}.$$
- If $\theta = 0$, then $w_z \in \mathbb{N}^+$, so $\psi(w) = (1 - \frac{c}{m})w_z \geq 1 - \frac{2c}{m} \geq (1 - \frac{2c}{m})/N$.

Thus $b \geq (1 - \frac{2c}{m})/N$.

Finally we construct F. Regard ρ as a character of G through the surjection $G \to G_S$. Then $\rho(G)$ is the group of N-th roots of unity. Let $\sigma \in G$ with $\rho(\sigma) = \exp(2\pi i/N)$. Let $\omega_y \in \mathbb{Q}$ with $\exp(2\pi i\omega_y) = \chi_y(\sigma)$ and $\frac{2c}{N} < \omega_y \leq \frac{2c}{N} + 1$, let

$$\omega_x := \omega_y + \frac{1}{N}, \qquad \omega_z := \frac{2}{N},$$

and let $\omega_u \in \mathbb{Q}$ with $\exp(2\pi i\omega_u) = \chi_u(\sigma)$ and $2\omega_x < \omega_u \leq 2\omega_x + 1$. Then the vector

$$\omega := (\omega_x, \omega_y, \omega_z, \omega_u)$$

satisfies $\omega \in \mathbb{Q}^4_{>0}$ and

$$(\exp(2\pi i\omega_x), \exp(2\pi i\omega_y), \exp(2\pi i\omega_z), \exp(2\pi i\omega_u)) = (\chi_x(\sigma), \chi_y(\sigma), \chi_z(\sigma), \chi_u(\sigma)).$$

Let $d := \min\{d' \in \mathbb{N}^+ \mid d'\omega \in \mathbb{Z}^4\}$, let s be the greatest common divisor of the entries of $d\omega$, and let $w := \frac{d}{s}\omega$, which is primitive. Since

$$\mathrm{wt}_\omega(x^2) = \mathrm{wt}_\omega(y^2z) = 2\omega_x, \qquad \mathrm{wt}_\omega(f - f_S) \geq \omega_u > 2\omega_x, \quad \text{and} \quad \mathrm{wt}_\omega(\alpha_c z^c) = \frac{2c}{N},$$

and the other terms of $\widetilde{h}_1$ have ω-weight at least $\min\{\omega_y, \omega_u, \frac{2c+2}{N}\} > \frac{2c}{N}$, we have

$$\mathrm{in}_w f = \lambda(x^2 + y^2z) \ \text{ for some } \lambda \neq 0, \qquad \mathrm{in}_w \widetilde{h}_1 = \alpha_c z^c.$$

Since $x^2 + y^2z$ is irreducible and divisible by no coordinate, the prime divisor F over W with center o given by the weighted blow-up with weights w is contained in no coordinate hyperplane, so (A.7) holds for F, and

$$a(F, W, 0) = \frac{d}{s}\Big(\omega_u + \frac{1}{N}\Big), \qquad \mathrm{mult}_F u = \frac{d}{s}\omega_u, \quad \text{and} \quad \mathrm{mult}_F h_1 = \frac{d}{s} \cdot \frac{2c}{N}.$$

Thus

$$a(F, W, D) = \frac{d}{s}\Big(1 - \frac{2c}{m}\Big)\frac{1}{N}.$$

The action of σ on x, y, z, u is that of $(\xi^{w_x}, \xi^{w_y}, \xi^{w_z}, \xi^{w_u})$ with $\xi = \exp(2\pi is/d)$. The one-parameter subgroup $\xi \to (\xi^{w_x}, \xi^{w_y}, \xi^{w_z}, \xi^{w_u})$ of the diagonal torus acts trivially on the exceptional divisor of the weighted blow-up with weights w, so $\sigma \in I_G(F)$. Since G acts faithfully on x, y, z, u, the order of σ is d, so $d \leq e_F$. Since $1 - \frac{2c}{m} \geq 0$,

$$b \leq \frac{a(F, W, D)}{e_F} \leq \frac{1}{s}\Big(1 - \frac{2c}{m}\Big)\frac{1}{N} \leq \Big(1 - \frac{2c}{m}\Big)\frac{1}{N} \leq b.$$

Thus $b = (1 - \frac{2c}{m})/N$ and F computes b. Moreover, since $N \geq 1$ and $2c \leq m$,

$$\frac{a(F, W, 0)}{e_F} \leq \omega_u + \frac{1}{N} \leq 2\omega_y + \frac{3}{N} + 1 \leq \frac{4c+3}{N} + 3 \leq 2m + 6.$$

The lemma follows. □

Proof of Theorem 9.3. Let C_m be the constant of Lemma 5.7, and let

$$L(m) := \max\{7, 2m + 6, (m + 2)C_m + 1\}.$$

Let $b := q(1) = \mathrm{mld}((W \ni o, D)/G)$, which is non-negative since (W, D) is lc near o. We divide the proof into six steps.

Step 1. In this step we reduce the theorem to the cases (2) and (3) of Lemma A.2.

It suffices to find a computing lift F of b with

$$\frac{a(F, W, 0)}{e_F} \leq L(m). \tag{A.8}$$

By Lemma A.2, $\lfloor D \rfloor = \mathrm{div}(u)$ near o for a G-semi-invariant function u, and $h = u^m h_1$ for a mark h_1 coprime to u. Thus

$$D = \mathrm{div}(u) + \frac{1}{m}\mathrm{div}(h_1).$$

If $u \in \mathfrak{m}_o^2$, Lemma A.2 implies that $b = 0$ is computed by a prime divisor E over W with center o and $a(E, W, 0) = 2$, so E satisfies (A.8) since $2 \leq L(m)$.

Assume that $u \notin \mathfrak{m}_o^2$. Let $S := (u = 0) \subset W$ and G_S the image of G in $\mathrm{Aut}(S \ni o)$. By Lemma A.2, there exist a smooth fourfold germ $(\mathcal{X} \ni o)$ with a faithful action of G, a G-equivariant closed embedding $W \subset \mathcal{X}$ with $W = (f = 0)$ for a G-semi-invariant regular function f on $\mathcal{X}$, and G-semi-invariant regular parameters $x, y, z, \widetilde{u}$ of $\mathcal{X}$ at o with $\widetilde{u}|_W = u$, such that

x, y, z are invariant under $\ker(G \to G_S)$ and the character of $\widetilde{u}$ is faithful on $\ker(G \to G_S)$. Let $V := (\widetilde{u} = 0) \subset \mathcal{X}$ and $f_S := f|_V$. We write u also for $\widetilde{u}$. Let

$$r := |\ker(G \to G_S)|,$$

and fix a G-semi-invariant function $\widetilde{h}_1$ on $\mathcal{X}$ with $\widetilde{h}_1|_W = h_1$ (lift h_1 and take the component of character χ_{h_1}). By Lemma A.2,

$$b = \mathrm{mld}\Big(\Big(V \ni o, \mathrm{div}(f_S) + \frac{1}{m}\mathrm{div}(h_V)\Big)/G_S\Big), \qquad h_V := \widetilde{h}_1|_V. \tag{A.9}$$

In the case (1) of Lemma A.2, $b = 0$ is computed by a prime divisor E over W with center o and $a(E, W, 0) \le 7 \le L(m)$. In the case (4), b is computed by a prime divisor F over W with center o and $a(F, W, 0)/e_F \le 2m + 6 \le L(m)$. In the cases (1) and (4), (A.8) holds.

Step 2. In this step we assume that S is a Du Val surface germ, take a computing lift P over S satisfying (A.10), and extend it to $S \times \mathbb{A}^1_u$.

Let $B_S := \frac{1}{m}\mathrm{div}(h_1|_S)$. By Lemma A.2(2),

$$b = \mathrm{mld}((S \ni o, B_S)/G_S).$$

Since $a(P', S, B_S) \ge e_{P'} b \ge 0$ for any prime divisor P' over S with center o, the pair (S, B_S) is lc near o. By Lemma 5.7 for $(S \ni o)$, G_S and the mark $h_1|_S$, there exists a prime divisor P over S with center o such that, for $e_P := |I_{G_S}(P)|$, $n_J := \mathrm{mult}_P \mathrm{Jac}_S$, and $n_h := \mathrm{mult}_P(h_1|_S)$,

$$a(P, S, B_S) = e_P b, \qquad a(P, S, 0) \le e_P C_m, \qquad n_J \le e_P C_m, \quad \text{and} \quad n_h \le m e_P C_m. \tag{A.10}$$

By Definition 5.2, possibly reordering x, y, z, we may assume that

$$\mathrm{mult}_P\Big(\frac{\partial f_S}{\partial x}\Big|_S\Big) = n_J.$$

Let $G_P \subset G$ be the preimage of $I_{G_S}(P)$, a subgroup of order re_P of the cyclic group G, hence cyclic. Let λ_P be the faithful character of $I_{G_S}(P)$ on the conormal line of P (Lemma 2.15(1) and [Liu26a, Lemma 9.2]): for $\overline{g} \in I_{G_S}(P)$ and a rational function π on S with $\mathrm{mult}_P \pi = 1$, the residue of $\overline{g}^*\pi/\pi$ in the residue field $\kappa(P)$ is $\lambda_P(\overline{g})$. We view λ_P as a character of G_P through $G_P \to I_{G_S}(P)$ and construct a faithful character χ of G_P with $\chi^r = \lambda_P$. Let g_0 be a generator of G_P and ζ a primitive (re_P)-th root of unity, and write $\lambda_P(g_0) = \zeta^{rn_\lambda}$ with $\gcd(n_\lambda, e_P) = 1$. By the Chinese remainder theorem there exists an integer $n_\chi \equiv n_\lambda \bmod e_P$ coprime to re_P, and $\chi(g_0) := \zeta^{n_\chi}$ defines a faithful character χ of G_P with $\chi^r = \lambda_P$. Write $\chi_u|_{G_P} = \chi^{n_u}$ with $n_u \in \mathbb{Z}$. The group $\ker(G \to G_S) \subset G_P$ is generated by $g_0^{e_P}$, $\chi(g_0^{e_P})$ is a primitive r-th root of unity, and χ_u is faithful on $\ker(G \to G_S)$. Thus $\gcd(n_u, r) = 1$. Fix an integer c_u with

$$c_u \equiv n_u \bmod re_P \qquad \text{and} \qquad r\max\{2n_J, n_J + n_h\} < c_u \le r\max\{2n_J, n_J + n_h\} + re_P. \tag{A.11}$$

Then $\gcd(c_u, r) = 1$.

Let v_0 be the Gauss extension of $r\,\mathrm{mult}_P$ with $v_0(u) = c_u$ (Definition 2.3):

$$v_0\Big(\sum_k \phi_k u^k\Big) = \min_k \big(r\,\mathrm{mult}_P \phi_k + kc_u\big) \qquad \text{for rational functions } \phi_k \text{ on } S.$$

Since $\gcd(r, c_u) = 1$, the value group of v_0 is $\mathbb{Z}$. The center of v_0 on $S \times \mathbb{A}^1_u$ is $(o, 0)$, and the residue field of v_0 is $\kappa(P)(\xi)$, where ξ is the residue of $u^r\pi^{-c_u}$ and is transcendental over $\kappa(P)$ [Bou72, Chapter VI, §10, no. 1]. Thus $v_0 = \mathrm{mult}_{F_0}$ for a prime divisor F_0 over $S \times \mathbb{A}^1_u$ with center $(o, 0)$, and

$$a(F_0, S \times \mathbb{A}^1_u, 0) = ra(P, S, 0) + c_u. \tag{A.12}$$

To prove (A.12), let $S' \to S$ be a projective birational morphism from a normal surface on which P is a divisor. Near the generic point of $P \times \{0\}$, the threefold $S' \times \mathbb{A}^1_u$ is smooth with local parameters π, u of $P \times \{0\}$, and v_0 is the monomial valuation with $v_0(\pi) = r$ and $v_0(u) = c_u$. So $a(F_0, S' \times \mathbb{A}^1_u, 0) = r + c_u$. Since, near the generic point of $P \times \mathbb{A}^1_u$, the pullback of $K_{S \times \mathbb{A}^1_u}$ is $K_{S' \times \mathbb{A}^1_u} - (a(P, S, 0) - 1)(P \times \mathbb{A}^1_u)$ and v_0 takes the value r on $P \times \mathbb{A}^1_u$, (A.12) follows.

The group G acts on $S \times \mathbb{A}^1_u$ through G_S on S and through χ_u on $\mathbb{A}^1_u$, faithfully since χ_u is faithful on $\ker(G \to G_S)$. We claim that

$$I_G(F_0) = G_P. \tag{A.13}$$

Let $g \in G_P$, with image $\overline{g} \in I_{G_S}(P)$. Then $\overline{g}$ preserves mult_P and $g^*u = \chi_u(g)u$, so $v_0 \circ g^* = v_0$. The element g acts trivially on $\kappa(P)$, and

$$g^*\left(u^r\pi^{-c_u}\right) = \chi_u(g)^r\left(\frac{\overline{g}^*\pi}{\pi}\right)^{-c_u} u^r\pi^{-c_u},$$

whose residue is $\lambda_P(\overline{g})^{n_u - c_u}\xi = \xi$, since $\chi_u(g)^r = \chi(g)^{rn_u} = \lambda_P(\overline{g})^{n_u}$ and $e_P \mid c_u - n_u$. Hence $G_P \subset I_G(F_0)$. Conversely, if $g \in I_G(F_0)$, then $\overline{g}$ preserves the restriction $r\operatorname{mult}_P$ of v_0 to rational functions on S and acts trivially on $\kappa(P) \subset \kappa(F_0)$, so $\overline{g} \in I_{G_S}(P)$ and $g \in G_P$.

Step 3. In this step we transport F_0 to a prime divisor F over W and prove (A.8) in the case (2) of Lemma A.2.

In $\widehat{\mathcal{O}}_{\mathcal{X},o} = \mathbb{C}[[x,y,z,u]]$, on which G acts diagonally, the completion of W at o is $(f = 0)$, and, identifying $\widehat{\mathcal{O}}_{\mathcal{X},o}$ with the completion of $V \times \mathbb{A}^1_u$ at $(o,0)$ through x,y,z,u, the completion of $S \times \mathbb{A}^1_u$ at $(o,0)$ is $(f_S = 0)$, where $f_S = f(x,y,z,0)$. Both f and f_S are G-semi-invariant with character χ_f, and

$$f - f_S \in (u).$$

By Lemma 2.12(3), F_0 is the restriction of a formal prime divisor over $(f_S = 0)$ with center the closed point, which we also denote by F_0. For $\phi \in \mathbb{C}[[x,y,z,u]]$, we write

$$\operatorname{mult}_{F_0}\phi := \operatorname{mult}_{F_0}\left(\phi|_{(f_S=0)}\right).$$

The ideal $\operatorname{Jac}_{S\times\mathbb{A}^1_u}$ is generated by the restrictions of $\partial f_S/\partial x$, $\partial f_S/\partial y$, $\partial f_S/\partial z$, which are pullbacks of functions on S, and v_0 restricts to $r\operatorname{mult}_P$ on rational functions on S. Thus the choice of x in **Step 2** gives

$$\operatorname{mult}_{F_0}\operatorname{Jac}_{S\times\mathbb{A}^1_u} = rn_J = \operatorname{mult}_{F_0}\left(\frac{\partial f_S}{\partial x}\right).$$

Since $\widetilde{h}_1 - h_V \in (u)$ and h_V restricts to the pullback of $h_1|_S$, and since $c_u > rn_h$ by (A.11), we have $\operatorname{mult}_{F_0}\widetilde{h}_1 = rn_h$. Since $f - f_S$ and $\partial(f - f_S)/\partial x$ belong to (u) and $\operatorname{mult}_{F_0} u = c_u$, (A.11) gives the assumptions of Lemma 5.4(3) and (3)(d) with f_S, f, F_0, x, and $\widetilde{h}_1$ in place of f, f', F, x, and H_1:

$$\operatorname{mult}_{F_0}(f - f_S) \geq c_u > 2rn_J, \qquad \operatorname{mult}_{F_0}\left(\frac{\partial(f - f_S)}{\partial x}\right) \geq c_u > rn_J,$$
$$\text{and} \quad \operatorname{mult}_{F_0}(f - f_S) \geq c_u > r(n_J + n_h).$$

By Lemma 5.4(3)(b)(d) and Lemma 2.12(3), there exists a prime divisor F over W with center o such that $\operatorname{mult}_F h_1 = rn_h$, $\operatorname{mult}_F u = c_u$ (the homomorphism of Lemma 5.4(3)(b) fixes u), $I_G(F) = I_G(F_0)$, and $a(F,W,0) = a(F_0, S \times \mathbb{A}^1_u, 0)$. Thus, by (A.13) and (A.12),

$$I_G(F) = G_P, \qquad \text{so} \qquad e_F = re_P, \tag{A.14}$$

and

$$a(F,W,0) = ra(P,S,0) + c_u.$$

Therefore, by (A.10) and (A.14),

$$\begin{aligned} a(F,W,D) &= a(F,W,0) - \operatorname{mult}_F u - \frac{1}{m}\operatorname{mult}_F h_1 = r\left(a(P,S,0) - \frac{n_h}{m}\right) \\ &= ra(P,S,B_S) = re_P b = e_F b, \end{aligned}$$

so F is a computing lift of b. By (A.10), (A.11) and $m \geq 1$,

$$\frac{a(F,W,0)}{e_F} = \frac{a(P,S,0)}{e_P} + \frac{c_u}{re_P} \leq C_m + \frac{\max\{2n_J, n_J + n_h\}}{e_P} + 1 \leq C_m + (m+1)C_m + 1 \leq L(m),$$

which is (A.8).

Step 4. In this step we assume that S has a normal crossing point at o, and we bring the equation of W into a normal form.

By Lemma 2.12(2)(3), in **Steps 4–6** we may work in $\mathbb{C}[[x,y,z,u]]$, and we replace x,y,z by other parameters, keeping u. The new parameters are combinations of G-semi-invariant functions with characters χ_x,χ_y,χ_z, which are trivial on $\ker(G\to G_S)$. So the new x,y,z are still invariant under $\ker(G\to G_S)$.

Since S has a normal crossing point at o, the quadratic part q_S of f_S has rank 2. The kernel ℓ of q_S is a G-stable line in the tangent space of V at o, so the annihilator of ℓ in the span of x,y,z is spanned by two G-semi-invariant linear forms, and it has a G-semi-invariant complement. Possibly replacing x,y,z with G-semi-invariant linear combinations of x,y,z, we may assume that $q_S\in\mathbb{C}[x,y]$ is non-degenerate.

Then the Hessian of f in (x,y) at o is non-degenerate. By the formal implicit function theorem, $\partial f/\partial x=\partial f/\partial y=0$ has a unique solution

$$x=\sigma_x(z,u),\qquad y=\sigma_y(z,u)\qquad\text{with}\qquad \sigma_x(0,0)=\sigma_y(0,0)=0,$$

and by uniqueness σ_x and σ_y are G-semi-invariant with characters χ_x and χ_y. Possibly replacing x,y with $x-\sigma_x, y-\sigma_y$, we may assume that

$$f-\varphi\in(x,y)^2,\qquad \varphi(z,u):=f(0,0,z,u).$$

Write $f-\varphi=\gamma_{20}x^2+\gamma_{11}xy+\gamma_{02}y^2$. Possibly replacing $\gamma_{20},\gamma_{11},\gamma_{02}$ with their components of characters $\chi_f\chi_x^{-2}$, $\chi_f\chi_x^{-1}\chi_y^{-1}$, $\chi_f\chi_y^{-2}$, we may assume that $\gamma_{20},\gamma_{11},\gamma_{02}$ are G-semi-invariant. The form $\gamma_{20}(0)x^2+\gamma_{11}(0)xy+\gamma_{02}(0)y^2=q_S$ is non-degenerate and semi-invariant with character χ_f. Thus one of the following holds.

(1) $\chi_x^2=\chi_y^2=\chi_f$. Possibly replacing x,y with linear combinations of x,y when $\chi_x=\chi_y$, and rescaling, we may assume that $\gamma_{20}(0)=\gamma_{02}(0)=1$ and $\gamma_{11}(0)=0$. Then γ_{20} and $\gamma_{02}-\gamma_{11}^2/(4\gamma_{20})$ are G-invariant with constant term 1, so their square roots with constant term 1 are G-invariant, and

$$f-\varphi=x'^2+y'^2,\qquad x':=\sqrt{\gamma_{20}}x+\frac{\gamma_{11}y}{2\sqrt{\gamma_{20}}},\quad\text{and}\quad y':=\sqrt{\gamma_{02}-\frac{\gamma_{11}^2}{4\gamma_{20}}}y,$$

with x',y' semi-invariant of characters χ_x,χ_y. Since $\chi_x/\chi_y\in\{\pm1\}$, any $g\in G$ multiplies both $x'+\sqrt{-1}y'$ and $x'-\sqrt{-1}y'$ by $\chi_x(g)$, or exchanges them up to the factor $\chi_x(g)$.

(2) $\chi_x\chi_y=\chi_f$ and $\chi_x^2\neq\chi_f\neq\chi_y^2$. Then $\gamma_{20}(0)=\gamma_{02}(0)=0\neq\gamma_{11}(0)$, and γ_{11} and $\gamma_{20}\gamma_{02}$ are G-invariant. Let

$$\mu:=\frac{1}{2}\Big(\gamma_{11}+\sqrt{\gamma_{11}^2-4\gamma_{20}\gamma_{02}}\Big),$$

with the square root of constant term $\gamma_{11}(0)$. Then μ is a G-invariant unit, $\mu+\gamma_{20}\gamma_{02}/\mu=\gamma_{11}$, and

$$f-\varphi=\mu x'y',\qquad x':=x+\frac{\gamma_{02}y}{\mu},\quad\text{and}\quad y':=y+\frac{\gamma_{20}x}{\mu},$$

with x',y' semi-invariant of characters χ_x,χ_y.

Possibly replacing x,y with $x'+\sqrt{-1}y', x'-\sqrt{-1}y'$ in the case (1), and with x',y' in the case (2), we may assume that

$$f=\mu xy+\varphi(z,u)\tag{A.15}$$

for a G-invariant unit μ ($\mu=1$ in the case (1)), that z,u are G-semi-invariant, and that any $g\in G$ either multiplies x and y by constants or maps x and y to constant multiples of y and x. If $\varphi(z,0)\neq0$, then $S=(\mu|_V xy+\varphi(z,0)=0)$ is smooth or has an isolated singularity at o, which is not a normal crossing point. Thus $\varphi\in u\mathbb{C}[[z,u]]$. Moreover $\varphi\neq0$, since $\widehat{\mathcal{O}}_{W,o}$ is a domain. Let

- $G_0 \subset G$ be the subgroup of elements multiplying x and y by constants, $e_0 := |G_0|$, χ_x, χ_y the characters of G_0 on the new x, y, and n_z the order of $\chi_z|_{G_0}$,
- $\alpha(z) := \widetilde{h}_1(0,0,z,0)$, the restriction of h_V to the curve $(x = y = 0) \subset V$, and $c \in \mathbb{N} \cup \{+\infty\}$ the order of α at 0.

Then $h_V - \alpha(z) \in (x, y)$ and $\widetilde{h}_1 - \alpha(z) \in (x, y, u)$. We have $[G : G_0] \le 2$, since G_0 is the kernel of the action of G on the set $\{(x = 0), (y = 0)\}$, and G_0 acts diagonally on x, y, z, u.

Step 5. In this step we show that $c \le m$ and

$$b \ge \frac{1 - \frac{c}{m}}{n_z}. \tag{A.16}$$

For a prime divisor E over V with center o, with mult_E extended to $\widehat{\mathcal{O}}_{V,o}$, let $e_E := |I_{G_S}(E)|$ and

$$\eta_E := a(E, V, 0) - \mathrm{mult}_E x - \mathrm{mult}_E y - \mathrm{mult}_E z,$$

which is non-negative since $(V, \mathrm{div}(xyz))$ is lc. Here, and in the proof of (A.18) below, the lc and inversion of adjunction statements are applied, for the fixed E, after replacing x, y, z with regular functions on V congruent to them modulo a high power of $\mathfrak{m}_o$, which changes neither $\mathrm{mult}_E x, \mathrm{mult}_E y, \mathrm{mult}_E z$, $a(E, V, 0)$ nor c (Lemma 2.12(3)). By (A.15), $\mathrm{div}(f_S) = \mathrm{div}(xy)$ near o, so

$$a\Big(E, V, \mathrm{div}(f_S) + \frac{1}{m}\,\mathrm{div}(h_V)\Big) = \mathrm{mult}_E z + \eta_E - \frac{1}{m}\,\mathrm{mult}_E h_V, \tag{A.17}$$

and by (A.9), b is the infimum of (A.17) divided by e_E over all such E.

Assume that $c \ge m + 1$, and let E be the exceptional prime divisor of the weighted blow-up of $\mathrm{Spec}\,\mathbb{C}[[x, y, z]]$ with weights $(m + 1, m + 1, 1)$ in x, y, z. Then $\eta_E = 0$, and $\mathrm{mult}_E h_V \ge m + 1$ since $h_V - \alpha(z) \in (x, y)$. Thus

$$\mathrm{mult}_E z + \eta_E - \frac{1}{m}\,\mathrm{mult}_E h_V \le 1 - \frac{m+1}{m} < 0,$$

contradicting $b \ge 0$. Thus $c \le m$, and $\alpha(z) = z^c$ times a unit.

We claim that, for any prime divisor E over V with center o,

$$\mathrm{mult}_E z + \eta_E - \frac{1}{m}\,\mathrm{mult}_E h_V \ge \Big(1 - \frac{c}{m}\Big)(\mathrm{mult}_E z + \eta_E). \tag{A.18}$$

If $c = 0$, then h_V is a unit and (A.18) is an equality. If $c \ge 1$, it suffices to show

$$\mathrm{mult}_E h_V \le c(\mathrm{mult}_E z + \eta_E) = c\big(a(E, V, 0) - \mathrm{mult}_E x - \mathrm{mult}_E y\big),$$

which holds if the pair

$$\Big(V, \mathrm{div}(x) + \mathrm{div}(y) + \frac{1}{c}\,\mathrm{div}(h_V)\Big)$$

is lc near o. Let $\tau \in (0, 1)$. Since $\alpha(z) = z^c$ times a unit, $((x = y = 0), \frac{1-\tau}{c}\,\mathrm{div}(\alpha))$ is klt near o. By inversion of adjunction on the smooth surface $(x = 0)$, the pair

$$\Big((x = 0), \mathrm{div}(y) + \frac{1 - \tau}{c}\,\mathrm{div}(h_V)\Big),$$

with the divisors restricted to $(x = 0)$, is plt near o, so the pair

$$\Big((x = 0), (1 - \tau)\,\mathrm{div}(y) + \frac{1 - \tau}{c}\,\mathrm{div}(h_V)\Big)$$

is klt near o. By inversion of adjunction on V, the pair

$$\Big(V, \mathrm{div}(x) + (1 - \tau)\,\mathrm{div}(y) + \frac{1 - \tau}{c}\,\mathrm{div}(h_V)\Big)$$

is plt near o. Letting $\tau \to 0$ proves the claim.

Let $I_E := I_{G_S}(E)$ and λ_E the faithful character of I_E on the conormal line of E (Lemma 2.15(1)). By [Liu26a, Lemma 9.2(1)], $\chi_\phi|_{I_E} = \lambda_E^{\mathrm{mult}_E \phi}$ for any G_S-semi-invariant ϕ, and by [Liu26a,

Lemma 9.2(2)], I_E acts on $dx \wedge dy \wedge dz$ through $\lambda_E^{a(E,V,0)}$. In the case (1) of **Step 4**, this holds with the G-semi-invariant parameters x', y', z of that case in place of x, y, z, since

$$dx \wedge dy \wedge dz = -2\sqrt{-1}dx' \wedge dy' \wedge dz.$$

Let $\overline{G}_0 \subset G_S$ be the image of G_0. The order of $\chi_z|_{\overline{G}_0}$ is n_z, since $\ker(G \to G_S)$ fixes z.

- If $I_E \subset \overline{G}_0$, then $\lambda_E^{n_z \operatorname{mult}_E z} = \chi_z^{n_z}|_{I_E} = 1$, so $e_E \mid n_z \operatorname{mult}_E z$, and $\operatorname{mult}_E z \geq 1$ implies $\operatorname{mult}_E z/e_E \geq 1/n_z$.
- If $I_E \not\subset \overline{G}_0$, then $I_E \cap \overline{G}_0$ has index 2 in I_E, so λ_E has order $\frac{e_E}{2}$ on $I_E \cap \overline{G}_0$, where $\lambda_E^{n_z \operatorname{mult}_E z} = \chi_z^{n_z} = 1$. Thus $\frac{e_E}{2} \mid n_z \operatorname{mult}_E z$, so $\operatorname{mult}_E z/e_E \geq 1/(2n_z)$. The rational 3-form $\omega := \frac{dx}{x} \wedge \frac{dy}{y} \wedge \frac{dz}{z}$ satisfies $g^*\omega = \omega$ for $g \in \overline{G}_0$, since g^* multiplies x, y, z by constants, and $g^*\omega = -\omega$ for $g \notin \overline{G}_0$, since then $g^*\frac{dx}{x} = \frac{dy}{y}$, $g^*\frac{dy}{y} = \frac{dx}{x}$ and $g^*\frac{dz}{z} = \frac{dz}{z}$. On the other hand, since xy and z are semi-invariant, I_E acts on $\omega = \frac{dx \wedge dy \wedge dz}{xy \cdot z}$ through

$$\lambda_E^{a(E,V,0) - \operatorname{mult}_E x - \operatorname{mult}_E y - \operatorname{mult}_E z} = \lambda_E^{\eta_E}.$$

  Thus $\lambda_E^{\eta_E}$ has order 2, that is, $\eta_E \equiv \frac{e_E}{2} \bmod e_E$, and $\eta_E \geq 0$ implies $\eta_E/e_E \geq \frac{1}{2}$.

Whether or not $I_E \subset \overline{G}_0$, since $c \leq m$, (A.18) implies

$$\frac{1}{e_E}\Big(\operatorname{mult}_E z + \eta_E - \frac{1}{m}\operatorname{mult}_E h_V\Big) \geq \Big(1 - \frac{c}{m}\Big)\min\Big\{\frac{1}{n_z}, \frac{1}{2n_z} + \frac{1}{2}\Big\} = \frac{1 - \frac{c}{m}}{n_z}.$$

This proves (A.16).

Step 6. In this step we construct a computing lift F of b with $a(F, W, 0)/e_F \leq 2m + 3$.

The group G_0 is cyclic of order e_0. Let $w_z := e_0/n_z$, and let χ_1 be a generator of the characters of G_0. Then $\chi_z|_{G_0} = \chi_1^{kw_z}$ with $\gcd(k, n_z) = 1$. By the Chinese remainder theorem there exists $k' \equiv k \bmod n_z$ coprime to e_0, and $\chi_0 := \chi_1^{k'}$ is a faithful character of G_0 with $\chi_z|_{G_0} = \chi_0^{w_z}$. Fix integers w_u, w_x with $\chi_u|_{G_0} = \chi_0^{w_u}$, $\chi_x|_{G_0} = \chi_0^{w_x}$ and

$$2cw_z + e_0 < w_u \leq 2cw_z + 2e_0, \qquad cw_z < w_x \leq cw_z + e_0.$$

Let

$$N_\varphi := \min\{iw_z + jw_u \mid z^i u^j \text{ occurs in } \varphi\},$$

so $N_\varphi \geq w_u$ since $\varphi \in u\mathbb{C}[[z, u]]$, and let $w_y := N_\varphi - w_x$. Then $w_y \geq w_u - w_x > cw_z$. Since μ is invariant and f is semi-invariant, any monomial $z^i u^j$ occurring in φ satisfies

$$\chi_x\chi_y = \chi_f = \chi_z^i\chi_u^j = \chi_0^{iw_z + jw_u} \quad \text{on } G_0.$$

Thus $\chi_y|_{G_0} = \chi_0^{w_y}$. So any monomial M in x, y, z, u is G_0-semi-invariant with character $\chi_0^{\operatorname{wt}_w(M)}$, where $w := (w_x, w_y, w_z, w_u)$.

Let $n_w := \gcd(w_x, w_y, w_z, w_u)$. Since both xy and φ have weight N_φ,

$$\operatorname{in}_w f = \mu(0)xy + \operatorname{in}_w \varphi, \qquad \operatorname{in}_w \varphi \in \mathbb{C}[z, u]\backslash\{0\},$$

which is irreducible, being of degree 1 in x with coprime coefficients, and $(\operatorname{in}_w f = 0)$ is smooth in the torus, since $\partial(\operatorname{in}_w f)/\partial x = \mu(0)y$. Thus the weighted blow-up of $\operatorname{Spec}\mathbb{C}[[x, y, z, u]]$ with weights w/n_w is smooth at the generic point of the intersection of the exceptional divisor with the strict transform of W, the intersection is irreducible and transversal there, and the intersection is a prime divisor F over W with center o. By adjunction,

$$a(F, W, 0) = \frac{w_x + w_y + w_z + w_u - N_\varphi}{n_w} = \frac{w_z + w_u}{n_w},$$

and $\operatorname{mult}_F(\phi|_W) = \operatorname{wt}_w(\phi)/n_w$ for any ϕ such that $\operatorname{in}_w f$ does not divide $\operatorname{in}_w \phi$. In particular $\operatorname{mult}_F u = w_u/n_w$. Since $\widetilde{h}_1 - \alpha(z) \in (x, y, u)$ and $\min\{w_x, w_y, w_u\} > cw_z$, $\operatorname{in}_w \widetilde{h}_1$ is a non-zero

multiple of z^c, so $\operatorname{mult}_F h_1 = cw_z/n_w$. Thus

$$a(F,W,D) = \frac{1}{n_w}\Big(w_z + w_u - w_u - \frac{c}{m}w_z\Big) = \Big(1-\frac{c}{m}\Big)\frac{w_z}{n_w}.$$

We show $G_0 \subset I_G(F)$. Any $g \in G_0$ multiplies any monomial M by $\chi_0(g)^{\mathrm{wt}_w(M)}$, so g^* preserves wt_w, hence F, and g fixes the ratio of any two monomials of equal w-weight. Since $\kappa(F)$ is the function field of $(\mathrm{in}_w f = 0)$ in the weighted projective space, it is generated by residues of such ratios, so g acts trivially on $\kappa(F)$. Thus $e_0 \mid e_F$.

Since $1 - \frac{c}{m} \geq 0$ and $n_w e_F \geq e_0 = n_z w_z$,

$$\frac{a(F,W,D)}{e_F} \leq \Big(1-\frac{c}{m}\Big)\frac{w_z}{e_0} = \frac{1-\frac{c}{m}}{n_z} \leq b$$

by (A.16). Since $b \leq a(F,W,D)/e_F$, F is a computing lift of $b = (1-\frac{c}{m})/n_z$. Finally, since $c \leq m$ and $n_z \geq 1$,

$$\frac{a(F,W,0)}{e_F} \leq \frac{w_z + w_u}{e_0} \leq \frac{(2c+1)w_z + 2e_0}{e_0} = \frac{2c+1}{n_z} + 2 \leq 2m + 3 \leq L(m),$$

which is (A.8). This completes the proof in the case (3), and the theorem follows. □

A.3. **Non-klt pairs.** In this subsection we prove Theorem 9.4 of Section 9.

Lemma A.3. *Let m be a positive integer and $s \in (0,1)$ a rational number. Let $(W \ni o)$ be a Gorenstein terminal threefold germ, G a finite group acting faithfully on W and fixing o, and h a mark. Let $D := \frac{1}{m}\operatorname{div}(h)$, and assume that (W,D) is lc near o. Then, after shrinking W to a G-invariant neighborhood of o, there exists a G-equivariant projective birational morphism $g : Y \to W$ with the following properties.*

(1) (a) Y is terminal.
(b) Any G-invariant Weil divisor on Y is $\mathbb{Q}$-Cartier.
(c) The g-exceptional prime divisors are exactly the exceptional prime divisors E over W with $a(E,W,sD) \leq 1$. Any such E satisfies

$$2 \leq a(E,W,0) \leq \Big\lfloor \frac{1}{1-s} \Big\rfloor \quad \text{and} \quad 0 \leq a(E,W,D) < 1.$$

(2) The $\mathbb{Q}$-divisor D_Y on Y with $K_Y + D_Y = g^*(K_W + D)$ satisfies

$$D_Y \geq 0 \quad \text{and} \quad D_Y \in \Big\{0, \frac{1}{m}, \dots, 1\Big\},$$

and (Y, D_Y) is lc.

Proof. Write $s = \frac{c}{v}$ with coprime positive integers c and v. Since h is G-semi-invariant, D is G-invariant. Possibly shrinking W to a G-invariant neighborhood of o, we may assume that (W,D) is lc. Since $mD = \operatorname{div}(h)$ is Cartier, for any prime divisor E over W we have

$$a(E,W,sD) = (1-s)a(E,W,0) + sa(E,W,D) = a(E,W,0) - \frac{c}{mv}\operatorname{mult}_E h. \tag{A.19}$$

Since W is terminal and (W,D) is lc, $a(E,W,0) \geq 1$ and $a(E,W,D) \geq 0$, so the first equality of (A.19) implies $a(E,W,sD) \geq 1 - s > 0$. Since K_W is Cartier, $a(E,W,0) \in \mathbb{Z}$, and $\operatorname{mult}_E h \in \mathbb{N}$. Thus (W,sD) is klt, and by the second equality of (A.19),

$$a(E,W,sD) \in \mathcal{S} := \frac{1}{mv}\mathbb{N}$$

for any prime divisor E over W. In the notation of Construction-Theorem 2.14 with $N = 1$ and $\alpha_1 = \frac{1}{mv}$, we have $1 \in \mathcal{S}$. By Construction-Theorem 2.14(1)(3)(4) with

$$(X, B, x, H) = (W, sD, o, G),$$

after shrinking W to a G-invariant neighborhood of o, there exists a G-equivariant projective birational morphism $g : Y \to W$ satisfying (1)(a), (1)(b), and the first sentence of (1)(c).

(1). It remains to bound $a(E,W,0)$ and $a(E,W,D)$ for a g-exceptional prime divisor E. Since W is Gorenstein terminal and E is exceptional over W, $a(E,W,0)$ is an integer >1, so $a(E,W,0)\geq 2$. Since $a(E,W,D)\geq 0$, the first equality of (A.19) implies

$$1\geq a(E,W,sD)\geq (1-s)a(E,W,0),$$

so $a(E,W,0)\leq \lfloor \frac{1}{1-s}\rfloor$. Since $a(E,W,0)\geq 2$, the first equality of (A.19) implies

$$sa(E,W,D)=a(E,W,sD)-(1-s)a(E,W,0)\leq 1-2(1-s)=2s-1<s,$$

so $a(E,W,D)<1$.

(2). Since G acts on W by automorphisms and h is G-semi-invariant, $a(E,W,0)$ and $\operatorname{mult}_E h$ are constant on any G-orbit of prime divisors E over W. Let $E_1,\dots,E_n$ be the reduced G-orbit sums of the g-exceptional prime divisors, and for any j let

$$a_j:=a(E,W,0)\quad\text{and}\quad \beta_j:=1-a(E,W,D)$$

for any prime divisor $E\leq E_j$. Choosing K_Y with $g_*K_Y=K_W$, we have

$$K_Y=g^*K_W+\sum_{j=1}^{n}(a_j-1)E_j\quad\text{and}\quad D_Y=g_*^{-1}D+\sum_{j=1}^{n}\beta_jE_j. \tag{A.20}$$

Since (W,D) is lc and $mD=\operatorname{div}(h)$, $g_*^{-1}D\in\{0,\frac{1}{m},\dots,1\}$. Since

$$a(E,W,D)=a(E,W,0)-\frac{1}{m}\operatorname{mult}_E h\in\frac{1}{m}\mathbb{Z}$$

and $0\leq a(E,W,D)<1$ by (1)(c), we have $\beta_j\in\{\frac{1}{m},\dots,1\}$ for any j. Thus $D_Y\geq 0$ and $D_Y\in\{0,\frac{1}{m},\dots,1\}$. Since $a(F,Y,D_Y)=a(F,W,D)\geq 0$ for any prime divisor F over Y, (Y,D_Y) is lc. □

Proposition A.4. *Let m be a positive integer. Then there exists an integer $N_m\geq 2$ satisfying the following. Let $(W\ni o)$ be a smooth threefold germ or an isolated Gorenstein terminal hypersurface threefold germ, G a finite group acting faithfully on W and fixing o, and h a mark. Let $D:=\frac{1}{m}\operatorname{div}(h)$ and $b:=\operatorname{mld}((W\ni o,D)/G)$. Assume the following.*

(i) (W,D) is lc near o, and $\lfloor D\rfloor=0$ near o.
(ii) (W,D) is not klt near o.
(iii) $b>0$.

Then the following hold.

(1) $0<b\leq\operatorname{mld}(W\ni o,D)$.
(2) Near o the non-klt locus $\operatorname{Nklt}(W,D)$ is a non-empty union of curves through o, and for any irreducible component C_0 of $\operatorname{Nklt}(W,D)$ through o there exists a prime divisor E over W with $\operatorname{center}_W E=C_0$ and

$$a(E,W,D)=0\quad\text{and}\quad 2\leq a(E,W,0)\leq N_m.$$

(3) Let $s:=1-\frac{1}{2N_m}$, $g:Y\to W$ a G-equivariant projective birational morphism satisfying (1) and (2) of Lemma A.3, D_Y the $\mathbb{Q}$-divisor on Y with $K_Y+D_Y=g^*(K_W+D)$, and $R:=\lfloor D_Y\rfloor$. Then R is $\mathbb{Q}$-Cartier, $g(\operatorname{Supp}R)$ is the non-klt locus of (W,D) near o, and no component of R maps to o.

Proof. Let $\Gamma:=\{0,\frac{1}{m},\dots,1\}$, let L_Γ be the positive integer of Theorem 2.21(2) for the finite set Γ, and let $N_m:=\max\{2,L_\Gamma\}$. We divide the proof into three steps.

Step 1. In this step we show that $0<b\leq\operatorname{mld}(W\ni o,D)$ and that near o, $\operatorname{Nklt}(W,D)$ is a non-empty union of curves through o.

By Definition 2.2, any prime divisor F over W with $\operatorname{center}_W F=\{o\}$ satisfies

$$a(F,W,D)\geq e_F b\geq b>0. \tag{A.21}$$

Thus $0<b\leq\operatorname{mld}(W\ni o,D)$.

Since (W, D) is lc near o, near o the set $\mathrm{Nklt}(W, D)$ is closed and each irreducible component of $\mathrm{Nklt}(W, D)$ is $\mathrm{center}_W E$ for a prime divisor E over W with $a(E, W, D) = 0$ (consider a log resolution of (W, D)). Since D is G-invariant, so is $\mathrm{Nklt}(W, D)$. Possibly shrinking W to a G-invariant neighborhood of o, we may assume that (W, D) is lc, $\lfloor D \rfloor = 0$, W is smooth outside o, and any irreducible component of $\mathrm{Nklt}(W, D)$ contains o.

Let V be an irreducible component of $\mathrm{Nklt}(W, D)$, and E a prime divisor over W with $\mathrm{center}_W E = V$ and $a(E, W, D) = 0$. By (A.21), $V \neq \{o\}$. If V were a prime divisor on W, then $E = V$ and $a(E, W, D) = 1 - \mathrm{mult}_V D > 0$ since $\lfloor D \rfloor = 0$. Thus V is a curve through o. Since (W, D) is not klt near o, $o \in \mathrm{Nklt}(W, D)$. Thus $\mathrm{Nklt}(W, D)$ is a non-empty union of curves through o.

Step 2. In this step we construct, for an irreducible component C_0 of $\mathrm{Nklt}(W, D)$, a prime divisor E over W with

$$\mathrm{center}_W E = C_0, \quad a(E, W, D) = 0, \quad \text{and} \quad 2 \leq a(E, W, 0) \leq N_m.$$

Let η be the generic point of C_0. Since W is smooth outside o, $\mathcal{O}_{W,\eta}$ is a regular local ring of dimension 2. Let u be a rational function on W which is regular at η and non-constant on C_0. Let $k := \mathbb{C}(u)$, W_k the generic fiber of u, $c \in W_k$ the closed point given by η, D_k the restriction of D, Ω an algebraic closure of k, c' a closed point of $W_k \otimes_k \Omega$ over c, W_Ω the connected component containing c', D_Ω the restriction of $D_k \otimes_k \Omega$, and $\varpi : W_\Omega \to W$ the induced morphism. By Lemma 3.5(0) for the curve C_0, the trivial group, and $t = u$, used for (W, D) and for $(W, 0)$, $\mathcal{O}_{W_k,c} = \mathcal{O}_{W,\eta}$, and the prime divisors over W with center C_0 are exactly the prime divisors over W_k with center c, with $a(E, W, D) = a(E, W_k, D_k)$ and $a(E, W, 0) = a(E, W_k, 0)$ for any such prime divisor E.

Since k has characteristic zero and $\mathcal{O}_{W_k,c}$ is regular, W_k is smooth over k at c, so W_Ω is a smooth surface over Ω near c'. Since k has characteristic zero, the base change to W_Ω of the restriction to W_k of a log resolution of (W, D) is a log resolution of (W_Ω, D_Ω), and the coefficients of the crepant pullbacks do not change. Since (W, D) is lc, (W_Ω, D_Ω) is lc. Since $mD_\Omega = \mathrm{div}(\varpi^* h)$ and (W_Ω, D_Ω) is lc, $D_\Omega \in \Gamma$.

By **Step 1**, there exists a prime divisor E_0 over W with $\mathrm{center}_W E_0 = C_0$ and $a(E_0, W, D) = 0$, so E_0 is a prime divisor over W_k with center c and $a(E_0, W_k, D_k) = 0$. By Lemma 3.5(1), there exists a prime divisor F' over W_Ω with center c' and $\mathrm{mult}_{F'}(\varpi^* \zeta) = \mathrm{mult}_{E_0} \zeta$ for any non-zero rational function ζ on W. By Lemma 3.5(2), the restriction of the valuation of F' is the valuation of a prime divisor over W_k with center c whose multiplicity along any non-zero rational function ζ on W is $\mathrm{mult}_{F'}(\varpi^* \zeta) = \mathrm{mult}_{E_0} \zeta$, so it is E_0, and $a(F', W_\Omega, D_\Omega) = a(E_0, W_k, D_k) = 0$. Since (W_Ω, D_Ω) is lc, $\mathrm{mld}(W_\Omega \ni c', D_\Omega) = 0$.

Since Ω is an algebraically closed field containing $\mathbb{C}$, by Theorem 2.21(2) there exists a prime divisor F'' over W_Ω with center c', $a(F'', W_\Omega, D_\Omega) = 0$, and $a(F'', W_\Omega, 0) \leq L_\Gamma \leq N_m$. By Lemma 3.5(2) and (0), the valuation of F'' restricts to the valuation of a prime divisor E over W with $\mathrm{center}_W E = C_0$ and $a(E, W, D) = a(F'', W_\Omega, D_\Omega) = 0$. By Lemma 3.5(2) for $(W, 0)$, $a(E, W, 0) = a(F'', W_\Omega, 0) \leq N_m$. Since C_0 is a curve, E is exceptional over W, and since W is Gorenstein terminal, $a(E, W, 0)$ is an integer > 1. Thus $2 \leq a(E, W, 0) \leq N_m$.

Step 3. In this step we prove (3).

Possibly shrinking W to a G-invariant neighborhood W° of o and Y to $g^{-1}(W^\circ)$, we may assume that the conclusions of **Step 1** hold. Since g is G-equivariant and D is G-invariant, D_Y and R are G-invariant, so R is a G-invariant Weil divisor on Y, which is $\mathbb{Q}$-Cartier by Lemma A.3(1)(b).

Let P be a component of R, so $\mathrm{mult}_P D_Y = 1$. If P were not g-exceptional, then

$$\mathrm{mult}_P D_Y = \mathrm{mult}_{g(P)} D < 1$$

since $\lfloor D \rfloor = 0$. Thus P is g-exceptional and $a(P, W, D) = 1 - \mathrm{mult}_P D_Y = 0$, so

$$g(P) = \mathrm{center}_W P \subset \mathrm{Nklt}(W, D),$$

and $g(P) \neq \{o\}$ by (A.21). Thus $g(\operatorname{Supp} R) \subset \operatorname{Nklt}(W, D)$, and no component of R maps to o.

Conversely, let C_0 be an irreducible component of $\operatorname{Nklt}(W, D)$, and E the prime divisor of **Step 2**. Since $a(E, W, D) = 0$, $a(E, W, 0) \leq N_m$ and $s = 1 - \frac{1}{2N_m}$,

$$a(E, W, sD) = (1 - s)a(E, W, 0) + sa(E, W, D) = \frac{a(E, W, 0)}{2N_m} \leq \frac{1}{2}.$$

Since E is exceptional over W, Lemma A.3(1)(c) implies that E is a g-exceptional prime divisor on Y, and $\operatorname{mult}_E D_Y = 1 - a(E, W, D) = 1$. Thus $E \leq R$ and $C_0 = g(E) \subset g(\operatorname{Supp} R)$. Since C_0 is an arbitrary irreducible component of $\operatorname{Nklt}(W, D)$, $\operatorname{Nklt}(W, D) \subset g(\operatorname{Supp} R)$. Thus

$$g(\operatorname{Supp} R) = \operatorname{Nklt}(W, D). \quad \square$$

Proposition A.5. *Let m be a positive integer, and let N_m be the integer of Proposition A.4. Let $(W \ni o)$ be a smooth threefold germ or an isolated Gorenstein terminal hypersurface threefold germ, G a finite group acting faithfully on W and fixing o, and h a mark on $(W \ni o)$, and let $D := \frac{1}{m} \operatorname{div}(h)$. Assume the following.*

(i) (W, D) is lc near o, and $\lfloor D \rfloor = 0$ near o.
(ii) (W, D) is not klt near o.
(iii) $b := \operatorname{mld}((W \ni o, D)/G) > 0$.

Then, after shrinking W to a G-invariant neighborhood of o, there exists a G-equivariant projective birational morphism $\varphi : Y' \to W$ from a klt variety Y' with the following property, where $D_{Y'}$ is the $\mathbb{Q}$-divisor with $K_{Y'} + D_{Y'} = \varphi^*(K_W + D)$. Let F be a computing lift of b. Then there exist a Gorenstein terminal threefold X with a generically finite morphism $\gamma : X \to Y'$ onto an open neighborhood of a closed point of $\operatorname{center}_{Y'} F$ such that $K_X = \gamma^* K_{Y'}$, a finite cyclic group G' acting faithfully on X, and a prime divisor P over X whose valuation restricts to a positive integral multiple of the valuation of F. Let D_X be the $\mathbb{Q}$-divisor with $K_X + D_X = \gamma^*(K_{Y'} + D_{Y'})$, $\Sigma := \operatorname{center}_X P$, and $e'_Q := |I_{G'}(Q)|$ for any prime divisor Q over X. Then the following hold.

(a) (X, D_X) is lc, mD_X is Cartier, $\lfloor D_X \rfloor$ is $\mathbb{Q}$-Cartier, $\Sigma \subset \operatorname{Supp} \lfloor D_X \rfloor$, Σ is a closed point or a curve mapped to o, and G' fixes Σ pointwise.
(b) For any closed point $x \in \Sigma$, $(X \ni x)$ is smooth or an isolated hypersurface singularity, and $mD_X = \operatorname{div}(h_X)$ near x for a mark h_X on $(X \ni x)$ with respect to G'.
(c) We have

$$\inf \left\{ \frac{a(Q, X, D_X)}{e'_Q} \;\middle|\; Q \text{ is a prime divisor over } X \text{ with } \operatorname{center}_X Q = \Sigma \right\} = b,$$

and P attains the infimum.
(d) For any prime divisor Q over X with $\operatorname{center}_X Q = \Sigma$, the valuation of Q restricts to a positive integral multiple of the valuation of a prime divisor F_Q over W with $\operatorname{center}_W F_Q = \{o\}$, and

$$\frac{a(Q, X, D_X)}{e'_Q} \geq \frac{a(F_Q, W, D)}{e_{F_Q}} \quad \text{and} \quad \frac{a(F_Q, W, 0)}{e_{F_Q}} \leq (1 + m(2N_m - 1)) \frac{a(Q, X, 0)}{e'_Q}.$$

If $a(Q, X, D_X)/e'_Q = b$, then F_Q is a computing lift of b.

Proof. We divide the proof into six steps.

Step 1. In this step we construct a terminal model $g : Y \to W$ and show that the non-klt locus of the crepant pullback of (W, D) is the support of the coefficient-one part.

Since h is a regular function on a G-invariant open neighborhood of o (Definition 2.4), possibly shrinking W to a G-invariant affine neighborhood of o, we may assume that W is affine and $h \in \mathcal{O}(W)$. Let $s := 1 - \frac{1}{2N_m}$. By Lemma A.3, possibly shrinking W to a G-invariant neighborhood of o, there exists a G-equivariant projective birational morphism $g : Y \to W$ satisfying (1) and

(2) of Lemma A.3. Let D_Y be the $\mathbb{Q}$-divisor with $K_Y + D_Y = g^*(K_W + D)$, and let $R := \lfloor D_Y \rfloor$. Since $\lfloor \frac{1}{1-s} \rfloor = 2N_m$, any g-exceptional prime divisor E satisfies

$$2 \le a(E, W, 0) \le 2N_m \quad \text{and} \quad \frac{1}{m} \le \operatorname{mult}_E D_Y = 1 - a(E, W, D) \le 1, \tag{A.22}$$

where $\operatorname{mult}_E D_Y \ge \frac{1}{m}$ because $a(E, W, D) < 1$ and mD_Y has integral coefficients.

By Proposition A.4, R is $\mathbb{Q}$-Cartier, $\operatorname{mld}(W \ni o, D) \ge b > 0$, no component of R maps to o, and near o we have $g(\operatorname{Supp} R) = \operatorname{Nklt}(W, D)$, which is a non-empty union of curves through o. Possibly shrinking W to a G-invariant neighborhood of o, we may assume that $g(\operatorname{Supp} R) = \operatorname{Nklt}(W, D)$ and that any irreducible component of $\operatorname{Nklt}(W, D)$ is a curve containing o. Since $\operatorname{mld}(W \ni o, D) > 0$,

$$a(F', W, D) > 0 \quad \text{for any prime divisor } F' \text{ over } W \text{ with } \operatorname{center}_W F' = \{o\}. \tag{A.23}$$

We show that, possibly shrinking W to a G-invariant neighborhood of o,

$$\operatorname{Nklt}(Y, D_Y) = \operatorname{Supp} R. \tag{A.24}$$

Since (Y, D_Y) is lc, $\operatorname{Nklt}(Y, D_Y)$ is the union of the lc centers of (Y, D_Y), which are finitely many (consider a log resolution of (Y, D_Y)), and the lc centers of dimension 2 are the components of R. Let V' be an lc center, the center on Y of a prime divisor E' over Y with $a(E', Y, D_Y) = 0$. Then $a(E', W, D) = 0$, so $g(V') = \operatorname{center}_W E' \subset \operatorname{Nklt}(W, D)$. Since G permutes the lc centers, possibly removing from W the union of the sets $g(V')$ not containing o, we may assume that $o \in g(V')$ for any lc center V'. By (A.23), $g(V') \ne \{o\}$, so $g(V')$ is an irreducible component C_0 of $\operatorname{Nklt}(W, D)$.

Suppose that some lc center V' with $g(V') = C_0$ is not contained in $\operatorname{Supp} R$. Then V' is a curve mapping onto C_0. Let $W^\circ \subset W$ be an open subset with $W^\circ \cap C_0 \ne \emptyset$, and let u be a regular function on W° whose restriction to $W^\circ \cap C_0$ is not constant. Let W_η and Y_η be the generic fibers of $u : W^\circ \to \mathbb{A}^1$ and of $u \circ g : g^{-1}(W^\circ) \to \mathbb{A}^1$, that is, their fibers over the generic point η of $\mathbb{A}^1$, which are surfaces, and let $g_\eta : Y_\eta \to W_\eta$ be the projective birational morphism induced by g. Then $c := C_0 \cap W_\eta$ is a closed point of W_η with residue field $\kappa(c)$, and $y_0 := V' \cap Y_\eta$ is a closed point of Y_η with $g_\eta(y_0) = c$. Since Y is terminal, $\operatorname{Sing} Y$ is a finite set of closed points, which map to closed points of $\mathbb{A}^1$. Thus Y_η is a regular surface. Let $D_{Y_\eta} := D_Y|_{Y_\eta}$ and $D_{W_\eta} := D|_{W_\eta}$. Then $K_{Y_\eta} + D_{Y_\eta} = g_\eta^*(K_{W_\eta} + D_{W_\eta})$, so $-(K_{Y_\eta} + D_{Y_\eta})$ is g_η-nef, and it is g_η-big because g_η is birational. Any closed point y of Y_η is $C_y \cap Y_\eta$ for the closure C_y of y in $g^{-1}(W^\circ)$, an integral curve on which $u \circ g$ is not constant. By Lemma 3.5(0) for the sub-pair $(g^{-1}(W^\circ), D_Y)$, the curve C_y, the trivial group, and the function $u \circ g$ in place of t, the prime divisors over Y_η with center y are exactly the prime divisors over $g^{-1}(W^\circ)$ with center C_y, and their log discrepancies with respect to (Y_η, D_{Y_η}) and (Y, D_Y) are equal. The prime divisors on Y_η are the generic fibers $Q_1 \cap Y_\eta$ of the prime divisors Q_1 on $g^{-1}(W^\circ)$ dominating $\mathbb{A}^1$, and $\operatorname{mult}_{Q_1 \cap Y_\eta} D_{Y_\eta} = \operatorname{mult}_{Q_1} D_Y$. Hence (Y_η, D_{Y_η}) is lc and

$$\operatorname{Nklt}(Y_\eta, D_{Y_\eta}) = \operatorname{Nklt}(Y, D_Y) \cap Y_\eta.$$

By [Tan18, Theorem 5.2], the fiber Φ of $\operatorname{Nklt}(Y_\eta, D_{Y_\eta}) \to W_\eta$ over c is empty or geometrically connected over $\kappa(c)$. In particular, the topological space Φ is connected.

Since $g(\operatorname{Supp} R) \supset C_0$ and no component of R maps to o, there exists a component R_1 of R with $g(R_1) = C_0$. Then $R_1 \cap Y_\eta$ is a non-empty closed subset of Φ not containing y_0. On the other hand, an lc center V'' of (Y, D_Y) containing y_0 contains V'. If $V'' \ne V'$, then $\dim V'' = 2$, so V'' is a component of R, which does not contain y_0. Since the lc centers are finitely many and closed, y_0 has an open neighborhood in Y_η whose intersection with $\operatorname{Nklt}(Y_\eta, D_{Y_\eta})$ is $\{y_0\}$. Thus $\{y_0\}$ is open and closed in Φ and $\Phi \ne \{y_0\}$, which contradicts the connectedness of Φ. Therefore any lc center of (Y, D_Y) is contained in $\operatorname{Supp} R$, and (A.24) follows.

Let ϵ be a rational number with $0 < \epsilon < 1$. Then $D_Y - \epsilon R \ge 0$, and for any prime divisor E' over Y,

$$a(E', Y, D_Y - \epsilon R) = a(E', Y, D_Y) + \epsilon \operatorname{mult}_{E'} R.$$

If $a(E', Y, D_Y) = 0$, then $\operatorname{center}_Y E' \subset \operatorname{Supp} R$ by (A.24), so $\operatorname{mult}_{E'} R > 0$. Hence

$$(Y, D_Y - \epsilon R) \text{ is klt.} \tag{A.25}$$

Step 2. In this step we run an MMP on Y/G against $\overline{R}$, construct φ, and obtain (A.31).

Let $\pi_Y : Y \to \overline{Y} := Y/G$ and $\pi_W : W \to \overline{W} := W/G$ be the quotient morphisms, and let $\overline{g} : \overline{Y} \to \overline{W}$ be the projective birational morphism induced by g. The variety $\overline{W}$ is affine. Since any G-invariant Weil divisor on Y is $\mathbb{Q}$-Cartier by Lemma A.3(1)(b), $\overline{Y}$ is $\mathbb{Q}$-factorial.

By Lemma 2.15(0), there exist $\mathbb{Q}$-divisors $D_{\overline{Y}} \geq 0$ on $\overline{Y}$ and $D_{\overline{W}} \geq 0$ on $\overline{W}$ with

$$K_Y + D_Y = \pi_Y^*(K_{\overline{Y}} + D_{\overline{Y}}) \quad \text{and} \quad K_W + D = \pi_W^*(K_{\overline{W}} + D_{\overline{W}}).$$

Since $\pi_W \circ g = \overline{g} \circ \pi_Y$ and π_Y^* is injective,

$$K_{\overline{Y}} + D_{\overline{Y}} = \overline{g}^*(K_{\overline{W}} + D_{\overline{W}}). \tag{A.26}$$

Since R is reduced and G-invariant, $R = \pi_Y^* \overline{R}$ for the $\mathbb{Q}$-divisor

$$\overline{R} := \sum_{\overline{P}} \frac{1}{e_{\overline{P}}} \overline{P},$$

where $\overline{P}$ runs over the components of $\pi_Y(\operatorname{Supp} R)$ and $e_{\overline{P}}$ is the ramification index of π_Y over $\overline{P}$. By Lemma 2.15(4), $\operatorname{mult}_{\overline{P}} D_{\overline{Y}} = 1$ for any such $\overline{P}$, and the quotient of $(Y, D_Y - \epsilon R)$ by G is $(\overline{Y}, D_{\overline{Y}} - \epsilon \overline{R})$, where $D_{\overline{Y}} - \epsilon \overline{R} \geq 0$. By (A.25) and Lemma 2.15(2)(3), $(\overline{Y}, D_{\overline{Y}} - \epsilon \overline{R})$ is klt, and by (A.26),

$$K_{\overline{Y}} + D_{\overline{Y}} - \epsilon \overline{R} \sim_{\mathbb{Q}, \overline{W}} -\epsilon \overline{R}.$$

Since $\overline{g}$ is birational, any $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor on $\overline{Y}$ is $\overline{g}$-big. Let $\overline{A} \geq 0$ be a general $\mathbb{Q}$-divisor that is $\mathbb{Q}$-linearly equivalent to a sufficiently large multiple of a $\overline{g}$-very ample divisor and has coefficients small enough that $(\overline{Y}, D_{\overline{Y}} - \epsilon \overline{R} + \overline{A})$ is klt. Then $K_{\overline{Y}} + D_{\overline{Y}} - \epsilon \overline{R} + \overline{A}$ is $\overline{g}$-nef. By [BCHM10, Theorem 1.2, Corollary 1.4.2], we may run a $(K_{\overline{Y}} + D_{\overline{Y}} - \epsilon \overline{R})$-MMP over $\overline{W}$ with scaling of $\overline{A}$, and it terminates with a log terminal model $\overline{\varphi} : \overline{Y'} \to \overline{W}$. The MMP consists of divisorial contractions and flips, and $\overline{Y'}$ is $\mathbb{Q}$-factorial. For any variety $\overline{Y}_i$ of the MMP, let $f_i : \overline{Y} \dashrightarrow \overline{Y}_i$ be the induced birational map, let $D_i := f_{i*} D_{\overline{Y}}$ and $\overline{R}_i := f_{i*} \overline{R}$, and let $\overline{g}_i : \overline{Y}_i \to \overline{W}$ be the induced morphism. By (A.26), $K_{\overline{Y}_i} + D_i \sim_{\mathbb{Q}, \overline{W}} 0$, so any curve ξ contracted by a step of the MMP starting from $\overline{Y}_i$ satisfies

$$\overline{R}_i \cdot \xi > 0. \tag{A.27}$$

Let $f : \overline{Y} \dashrightarrow \overline{Y'}$ be the induced birational map, and let $D_{\overline{Y'}} := f_* D_{\overline{Y}}$ and $\overline{R}_{\overline{Y'}} := f_* \overline{R}$. Then

$$K_{\overline{Y'}} + D_{\overline{Y'}} = \overline{\varphi}^*(K_{\overline{W}} + D_{\overline{W}}),$$

the pair $(\overline{Y'}, D_{\overline{Y'}} - \epsilon \overline{R}_{\overline{Y'}})$ is klt, and

$$-\overline{R}_{\overline{Y'}} \text{ is } \overline{\varphi}\text{-nef.} \tag{A.28}$$

We show that

$$\overline{\varphi}(\operatorname{Supp} \overline{R}_{\overline{Y'}}) = \overline{g}(\operatorname{Supp} \overline{R}). \tag{A.29}$$

The following argument follows [Kol14, Proof of Lemma 7(2)]. A flip $\overline{Y}_i \dashrightarrow \overline{Y}_{i+1}$ is an isomorphism in codimension 1 over $\overline{W}$, and the image in $\overline{W}$ of a prime divisor is the closure of the image of the generic point, so $\overline{g}_{i+1}(\operatorname{Supp} \overline{R}_{i+1}) = \overline{g}_i(\operatorname{Supp} \overline{R}_i)$. Let $q_i : \overline{Y}_i \to \overline{Y}_{i+1}$ be a divisorial contraction with exceptional prime divisor E_i. If $E_i \not\subset \operatorname{Supp} \overline{R}_i$, then $\operatorname{Supp} \overline{R}_{i+1} = q_i(\operatorname{Supp} \overline{R}_i)$. Assume that $E_i \subset \operatorname{Supp} \overline{R}_i$. It suffices to show $q_i(E_i) \subset \operatorname{Supp} \overline{R}_{i+1}$. Suppose that there exists a closed point $y \in q_i(E_i) \setminus \operatorname{Supp} \overline{R}_{i+1}$, and let $\xi \subset q_i^{-1}(y)$ be a curve. Since the components of $\overline{R}_i$ other than E_i map into $\operatorname{Supp} \overline{R}_{i+1}$, they are disjoint from ξ, so

$$\overline{R}_i \cdot \xi = \operatorname{mult}_{E_i} \overline{R}_i \cdot (E_i \cdot \xi) < 0,$$

because E_i is the exceptional divisor of the contraction of an extremal ray containing the class of ξ. This contradicts (A.27), and (A.29) follows.

Next we show that

$$\overline{\varphi}^{-1}(\overline{\varphi}(\operatorname{Supp}\overline{R}_{\overline{Y'}})) = \operatorname{Supp}\overline{R}_{\overline{Y'}}. \tag{A.30}$$

Let $\overline{w} \in \overline{\varphi}(\operatorname{Supp}\overline{R}_{\overline{Y'}})$ be a closed point. Since $\overline{W}$ is normal, $\overline{\varphi}^{-1}(\overline{w})$ is connected. Suppose that $\overline{\varphi}^{-1}(\overline{w}) \not\subset \operatorname{Supp}\overline{R}_{\overline{Y'}}$. By connectedness, there exists an irreducible component Γ of $\overline{\varphi}^{-1}(\overline{w})$ with $\Gamma \not\subset \operatorname{Supp}\overline{R}_{\overline{Y'}}$ and $\Gamma \cap \operatorname{Supp}\overline{R}_{\overline{Y'}} \neq \emptyset$. In particular, $\dim\Gamma \geq 1$. Let $\xi \subset \Gamma$ be a curve through a point of $\Gamma \cap \operatorname{Supp}\overline{R}_{\overline{Y'}}$ with $\xi \not\subset \operatorname{Supp}\overline{R}_{\overline{Y'}}$. Then $\overline{R}_{\overline{Y'}} \cdot \xi > 0$, which contradicts (A.28).

Let Y' be the normalization of the irreducible component of $\overline{Y'} \times_{\overline{W}} W$ dominating W, and let $\pi_{Y'} : Y' \to \overline{Y'}$ and $\varphi : Y' \to W$ be the induced morphisms, so that $\pi_W \circ \varphi = \overline{\varphi} \circ \pi_{Y'}$. The component is unique and G-invariant, since $\overline{Y'} \times_{\overline{W}} W \to W$ is an isomorphism over the preimage in W of the open subset of $\overline{W}$ over which $\overline{\varphi}$ is an isomorphism. Thus G acts on Y', φ is G-equivariant, birational and projective, and $\pi_{Y'}$ is finite. The morphism $\pi_{Y'}$ is the quotient morphism by G, because the induced morphism $Y'/G \to \overline{Y'}$ is finite and birational and $\overline{Y'}$ is normal.

Let $P_{Y'}$ be a prime divisor on Y'. Since the MMP contracts divisors and extracts none, $\pi_{Y'}(P_{Y'}) = f_*\overline{P}$ for a prime divisor $\overline{P}$ on $\overline{Y}$. Let $\overline{Y'}^\circ \subset \overline{Y'}$ be the largest open subset over which f^{-1} is an isomorphism. It contains the generic point of $\pi_{Y'}(P_{Y'})$, since f extracts no divisor. Let $\widetilde{f} : Y \dashrightarrow Y'$ be the birational map over W, so that $\pi_{Y'} \circ \widetilde{f} = f \circ \pi_Y$. Over $\overline{Y'}^\circ$ the map $\widetilde{f}$ is an isomorphism: the closure of the graph of $\widetilde{f}$ in

$$\pi_Y^{-1}(f^{-1}(\overline{Y'}^\circ)) \times_{\overline{Y'}^\circ} \pi_{Y'}^{-1}(\overline{Y'}^\circ)$$

is finite and birational over both factors, which are normal. Hence $P_{Y'}$ is a prime divisor on Y with $\pi_Y(P_{Y'}) = \overline{P}$, and

$$\operatorname{mult}_{P_{Y'}} D_{Y'} = 1 - a(P_{Y'}, W, D) = \operatorname{mult}_{P_{Y'}} D_Y,$$

and the ramification indices of $\pi_{Y'}$ and π_Y along $P_{Y'}$ are equal. Thus $D_{Y'} \in \{0, \frac{1}{m}, \dots, 1\}$, $R_{Y'} := \lfloor D_{Y'} \rfloor$ is the sum of the components of R that are divisors on Y', and $\pi_{Y'}^*\overline{R}_{\overline{Y'}} = R_{Y'}$. Since $(Y', D_{Y'})$ is crepant to (W, D), it is lc. Any G-invariant Weil divisor on Y' is the pullback by $\pi_{Y'}$ of a $\mathbb{Q}$-divisor on the $\mathbb{Q}$-factorial variety $\overline{Y'}$, hence it is $\mathbb{Q}$-Cartier. In particular, $R_{Y'}$, $D_{Y'}$, and $K_{Y'} = (K_{Y'} + D_{Y'}) - D_{Y'}$ are $\mathbb{Q}$-Cartier. Since

$$K_{Y'} + D_{Y'} - \epsilon R_{Y'} = \varphi^*\pi_W^*(K_{\overline{W}} + D_{\overline{W}}) - \epsilon\pi_{Y'}^*\overline{R}_{\overline{Y'}} = \pi_{Y'}^*(K_{\overline{Y'}} + D_{\overline{Y'}} - \epsilon\overline{R}_{\overline{Y'}}),$$

the quotient of $(Y', D_{Y'} - \epsilon R_{Y'})$ by G is the klt pair $(\overline{Y'}, D_{\overline{Y'}} - \epsilon\overline{R}_{\overline{Y'}})$, so $(Y', D_{Y'} - \epsilon R_{Y'})$ is klt by Lemma 2.15(2). As $D_{Y'} - \epsilon R_{Y'} \geq 0$ is $\mathbb{Q}$-Cartier, Y' is klt.

It remains to show $\varphi(\operatorname{Supp} R_{Y'}) = \operatorname{Nklt}(W, D)$ and $\varphi^{-1}(\varphi(\operatorname{Supp} R_{Y'})) = \operatorname{Supp} R_{Y'}$. The sets $\varphi(\operatorname{Supp} R_{Y'})$ and $\operatorname{Nklt}(W, D) = g(\operatorname{Supp} R)$ are G-invariant and closed. Since

$$\pi_{Y'}(\operatorname{Supp} R_{Y'}) = \operatorname{Supp}\overline{R}_{\overline{Y'}},$$

(A.29) gives

$$\pi_W(\varphi(\operatorname{Supp} R_{Y'})) = \overline{\varphi}(\operatorname{Supp}\overline{R}_{\overline{Y'}}) = \overline{g}(\operatorname{Supp}\overline{R}) = \pi_W(g(\operatorname{Supp} R)).$$

Since a G-invariant subset T_0 of W satisfies $T_0 = \pi_W^{-1}(\pi_W(T_0))$, we obtain

$$\varphi(\operatorname{Supp} R_{Y'}) = \operatorname{Nklt}(W, D).$$

If $z' \in Y'$ satisfies $\varphi(z') \in \varphi(\operatorname{Supp} R_{Y'})$, then

$$\overline{\varphi}(\pi_{Y'}(z')) = \pi_W(\varphi(z')) \in \overline{\varphi}(\operatorname{Supp}\overline{R}_{\overline{Y'}}),$$

so $\pi_{Y'}(z') \in \operatorname{Supp} \overline{R}_{\overline{Y'}}$ by (A.30), and $z' \in \pi_{Y'}^{-1}(\operatorname{Supp} \overline{R}_{\overline{Y'}}) = \operatorname{Supp} R_{Y'}$. Altogether,

$$\begin{gathered} Y' \text{ is klt, and any } G\text{-invariant Weil divisor on } Y' \text{ is } \mathbb{Q}\text{-Cartier,} \\ (Y', D_{Y'}) \text{ is lc, } D_{Y'} \in \{0, \tfrac{1}{m}, \dots, 1\}, \text{ and } R_{Y'} \text{ is } \mathbb{Q}\text{-Cartier,} \\ \varphi(\operatorname{Supp} R_{Y'}) = \operatorname{Nklt}(W, D) \quad \text{and} \quad \varphi^{-1}(\varphi(\operatorname{Supp} R_{Y'})) = \operatorname{Supp} R_{Y'}. \end{gathered} \tag{A.31}$$

Step 3. In this step we obtain (A.32).

By **Step 2**, the φ-exceptional prime divisors are g-exceptional prime divisors on Y. Since K_W is Cartier, choosing $K_{Y'}$ with $\varphi_* K_{Y'} = K_W$, we have

$$K_{Y'} - \varphi^* K_W = \sum_E (a(E, W, 0) - 1)E,$$

where E runs over the φ-exceptional prime divisors. By (A.22) and $\operatorname{mult}_E D_{Y'} = \operatorname{mult}_E D_Y$,

$$0 \le K_{Y'} - \varphi^* K_W \le (2N_m - 1) \sum_E E \le m(2N_m - 1) D_{Y'},$$

and all divisors here are G-invariant, hence $\mathbb{Q}$-Cartier by (A.31). Let F be a prime divisor over W with $\operatorname{center}_W F = \{o\}$. Since $(Y', D_{Y'})$ is lc, $\operatorname{mult}_F D_{Y'} = a(F, Y', 0) - a(F, Y', D_{Y'}) \le a(F, Y', 0)$. Thus

$$\begin{aligned} 0 \le a(F, W, 0) - a(F, Y', 0) &= \operatorname{mult}_F(K_{Y'} - \varphi^* K_W) \\ &\le m(2N_m - 1) \operatorname{mult}_F D_{Y'} \le m(2N_m - 1) a(F, Y', 0), \end{aligned}$$

so

$$a(F, Y', 0) \le a(F, W, 0) \le (1 + m(2N_m - 1)) a(F, Y', 0) \tag{A.32}$$

for any prime divisor F over W with $\operatorname{center}_W F = \{o\}$.

Step 4. In this step we fix a computing lift F of b and construct a Gorenstein canonical cover V of a neighborhood of a closed point of $\operatorname{center}_{Y'} F$ on which the pullback of $R_{Y'}$ is Cartier.

By Definition 2.4, $\operatorname{center}_W F = \{o\}$ and $a(F, W, D) = be_F$. Since $o \in \operatorname{Nklt}(W, D)$, (A.31) implies $\operatorname{center}_{Y'} F \subset \varphi^{-1}(o) \subset \operatorname{Supp} R_{Y'}$. Let $G_F := I_G(F)$. By Lemma 2.15(1), G_F acts faithfully and linearly on the 1-dimensional vector space $\mathfrak{m}_F/\mathfrak{m}_F^2$ over the residue field $\kappa(F)$ of the valuation ring of F, so G_F is cyclic. By Definition 2.2, G_F stabilizes F and acts trivially on $\kappa(F)$. Since the function field of $\operatorname{center}_{Y'} F$ embeds G_F-equivariantly into $\kappa(F)$, G_F fixes $\operatorname{center}_{Y'} F$ pointwise. Let $z \in \operatorname{center}_{Y'} F$ be a closed point, and U a G_F-invariant affine neighborhood of z in Y'. Let $\pi_U : U \to U/G_F$ be the quotient morphism.

Let r be the Cartier index of K_U at z. By [Liu26a, Construction-Theorem 5.3(1)] for the klt germ $(U \ni z)$ with zero boundary and the cyclic group G_F, there exists a finite morphism $p : V_0 \to U$ of degree r with the following properties: p is Galois with group μ_r, $p^{-1}(z)$ is one point v_0, the ramification index of p along any prime divisor of V_0 is 1, V_0 is klt, and $K_{V_0} = p^* K_U$ is Cartier. By [Liu26a, Construction-Theorem 5.3(2)], there exists a finite abelian group G_0 of order $r|G_F|$ acting faithfully on V_0 and fixing v_0, with an exact sequence $1 \to \mu_r \to G_0 \to G_F \to 1$ compatible with p and with $V_0/G_0 = U/G_F$. Thus p is equivariant with respect to $G_0 \to G_F$, so $\pi_U \circ p$ is G_0-invariant.

Let $R_0 := p^*(R_{Y'}|_U)$. Since p has ramification index 1 along any prime divisor, R_0 is a reduced divisor with $\operatorname{mult}_{P_0} R_0 = \operatorname{mult}_{p(P_0)} R_{Y'}$ for any prime divisor P_0 on V_0, and R_0 is $\mathbb{Q}$-Cartier. For any $\sigma \in G_0$, the G_0-invariance of $\pi_U \circ p$ implies $\pi_U(p(P_0)) = \pi_U(p(\sigma(P_0)))$, so the prime divisors $p(P_0)$ and $p(\sigma(P_0))$ are conjugate under G_F. As $R_{Y'}$ is G-invariant, R_0 is G_0-invariant. Let n be the Cartier index of R_0 at v_0. Since G_0 is abelian and fixes v_0, possibly shrinking U to a G_F-invariant affine neighborhood of z and replacing V_0 with its preimage, we may assume that $nR_0 = \operatorname{div}(\beta)$ for a G_0-semi-invariant regular function β on V_0 with character χ, that is, $\sigma^* \beta = \chi(\sigma)\beta$ for any $\sigma \in G_0$.

Let $p_1 : V \to V_0$ be the index 1 cover of R_0 given by β [Zhu24, Lemma-Definition 2.15], that is, V is the normalization of $\operatorname{Spec}_{V_0} \mathcal{O}_{V_0}[\alpha]/(\alpha^n - \beta)$. Then $\deg p_1 = n$, the ramification index of p_1 along any prime divisor of V is 1, $p_1^{-1}(v_0)$ is one point v, and $p_1^* R_0$ is Cartier. For $\sigma \in G_0$ and $\zeta \in \mathbb{C}$ with $\zeta^n = \chi(\sigma)$, σ^* on $\mathcal{O}_{V_0}$ and $\alpha \to \zeta\alpha$ define an automorphism of $\mathcal{O}_{V_0}[\alpha]/(\alpha^n - \beta)$ over σ, since $(\zeta\alpha)^n = \chi(\sigma)\beta = \sigma^*\beta$, hence an automorphism $\tau_{\sigma,\zeta}$ of V. Let G_1 be the group of the automorphisms $\tau_{\sigma,\zeta}$. We have the exact sequence

$$1 \to \mu_n \to G_1 \to G_0 \to 1,$$

and G_1 is abelian, since $\tau_{\sigma,\zeta}\tau_{\sigma',\zeta'}$ and $\tau_{\sigma',\zeta'}\tau_{\sigma,\zeta}$ restrict to $\sigma^*\sigma'^* = \sigma'^*\sigma^*$ on $\mathcal{O}_{V_0}$ and send α to $\zeta\zeta'\alpha$. The group G_1 acts faithfully on V and fixes v, the unique point over the G_0-fixed point v_0. Since μ_n acts faithfully on V over V_0 and $\deg p_1 = n$, we have $V/\mu_n = V_0$, so Lemma 2.15(5) gives

$$V/G_1 = V_0/G_0 = U/G_F. \tag{A.33}$$

Let $\rho := p \circ p_1 : V \to U$. Then ρ is finite of degree nr, $\rho^{-1}(z) = \{v\}$, and the ramification index of ρ along any prime divisor of V is 1. Let $G_2 := \operatorname{Ker}(G_1 \to G_0 \to G_F)$. As $G_1 \to G_F$ is surjective, $|G_2| = |G_1|/|G_F| = nr = \deg \rho$. The group G_2 maps into $\mu_r = \operatorname{Ker}(G_0 \to G_F)$, which acts on V_0 over U, so ρ is G_2-invariant. Since $\deg \rho = |G_2|$, the induced morphism $V/G_2 \to U$ is finite and birational between normal varieties, hence an isomorphism, and ρ is the quotient morphism of V by G_2. Let

$$D_V := \rho^*(D_{Y'}|_U) \quad \text{and} \quad R_V := \rho^*(R_{Y'}|_U).$$

By (A.33), $\pi_U \circ \rho$ is the quotient morphism $V \to V/G_1 = U/G_F$, so the argument for R_0 shows that D_V and R_V are G_1-invariant. By Lemma 2.15(0)(4),

$$K_V = \rho^* K_U \quad \text{and} \quad K_V + D_V = \rho^*\big((K_{Y'} + D_{Y'})|_U\big).$$

By Lemma 2.15(1)(2), for any prime divisor Q over V there exist a unique prime divisor F_Q over U and a unique positive integer $r_Q = |I_{G_2}(Q)|$ such that the valuation of Q restricts to r_Q times the valuation of F_Q, and, as $(Y', D_{Y'})$ is crepant to (W, D),

$$a(Q, V, 0) = r_Q a(F_Q, Y', 0) \quad \text{and} \quad a(Q, V, D_V) = r_Q a(F_Q, W, D). \tag{A.34}$$

The divisor $K_V = p_1^* K_{V_0}$ is Cartier, and $R_V = p_1^* R_0$ is a reduced Cartier divisor. Since $m(K_{Y'} + D_{Y'}) = \varphi^*(mK_W + \operatorname{div}(h))$ is Cartier, so is

$$mD_V = \rho^*\big(m(K_{Y'} + D_{Y'})|_U\big) - mK_V.$$

By (A.34), V is klt because Y' is klt, and (V, D_V) is lc because (W, D) is lc. As K_V is Cartier, $a(Q, V, 0)$ is a positive integer for any prime divisor Q over V, so V is Gorenstein canonical. Finally $D_V \geq R_V$, since $D_{Y'} \geq R_{Y'}$.

Step 5. In this step we construct X, G', and P, and prove (a) and (b).

By Construction-Theorem 2.14 with $\mathcal{S} = \mathbb{N}$ for the pair $(V, 0)$, the point v and the group G_1, for which $\Lambda_{\mathcal{S}} = \{0\}$ and $R_{\mathcal{S}} = 1$, possibly replacing V with a G_1-invariant neighborhood of v and U with the open subset V/G_2 of U, there exists a G_1-equivariant projective birational morphism $\psi : X \to V$ with the following properties. The boundary of Construction-Theorem 2.14(1)(2) belongs to $\Lambda_{\mathcal{S}} = \{0\}$, so $K_X = \psi^* K_V$. Any G_1-invariant Weil divisor on X is $\mathbb{Q}$-Cartier. The variety X is terminal and the Cartier index of K_X is at most $R_{\mathcal{S}} = 1$. Hence X is a Gorenstein terminal threefold. Let $\gamma : X \to Y'$ be the composite of $\rho \circ \psi$ and the open immersion $U \subset Y'$. Since ψ is birational and ρ is finite and surjective, γ is generically finite and $\gamma(X) = U$, and $K_X = \psi^*\rho^* K_U = \gamma^* K_{Y'}$.

Let $D_X := \psi^* D_V$ and $R_X := \psi^* R_V$. Then $K_X + D_X = \psi^*(K_V + D_V) = \gamma^*(K_{Y'} + D_{Y'})$, (X, D_X) is lc, mD_X is Cartier, and D_X is G_1-invariant. Moreover, R_X is a Cartier divisor, hence has integral coefficients, $D_X \geq R_X \geq 0$, and $\psi^{-1}(\operatorname{Supp} R_V) = \operatorname{Supp} R_X$. Since (X, D_X) is lc, any

prime divisor P_X on X with $\mathrm{mult}_{P_X} R_X \neq 0$ satisfies $1 \leq \mathrm{mult}_{P_X} R_X \leq \mathrm{mult}_{P_X} D_X \leq 1$. Hence R_X is reduced and

$$\psi^{-1}(\mathrm{Supp}\, R_V) = \mathrm{Supp}\, R_X \subset \mathrm{Supp}\lfloor D_X \rfloor. \tag{A.35}$$

Since D_X is G_1-invariant, $\lfloor D_X \rfloor$ is G_1-invariant, hence $\mathbb{Q}$-Cartier.

Since $z \in \mathrm{center}_{Y'} F$, F is a prime divisor over V/G_2. By Lemma 2.15(3), there exists a prime divisor P over V whose valuation restricts to a positive integral multiple of the valuation of F, so $F_P = F$. Since ψ is projective and birational, P is a prime divisor over X. Let $\Sigma := \mathrm{center}_X P$. The valuation ring of P dominates the local ring of X at the generic point of Σ, hence the local rings of U and of W at the images of that point. Thus $\rho(\psi(\Sigma)) \subset \mathrm{center}_U F \subset \mathrm{Supp}\, R_{Y'}$ and Σ maps to $\mathrm{center}_W F = \{o\}$. Hence $\psi(\Sigma) \subset \rho^{-1}(\mathrm{Supp}\, R_{Y'}) = \mathrm{Supp}\, R_V$, and (A.35) implies $\Sigma \subset \mathrm{Supp}\lfloor D_X \rfloor$. By (A.34) and $F_P = F$,

$$a(P, X, D_X) = a(P, V, D_V) = r_P a(F, W, D) = r_P e_F b > 0.$$

A prime divisor on X contained in $\mathrm{Supp}\lfloor D_X \rfloor$ has log discrepancy 0 with respect to (X, D_X), so Σ is not a divisor, and Σ is a closed point or a curve.

Let $G' := I_{G_1}(P)$. By Lemma 2.15(1), G' is cyclic, and G' acts faithfully on X as a subgroup of G_1. As for G_F in **Step 4**, G' fixes Σ pointwise. This proves (a).

Let $x \in \Sigma$ be a closed point. Since X is a Gorenstein terminal threefold, $(X \ni x)$ is smooth or an isolated cDV hypersurface singularity. Here mD_X is G'-invariant and Cartier. Since G' is cyclic and fixes x, $mD_X = \mathrm{div}(h_X)$ on a G'-invariant open neighborhood of x for a non-zero regular G'-semi-invariant function h_X. The function h_X is a mark on $(X \ni x)$ with respect to G' (Definition 2.4). This proves (b).

Step 6. In this step we compare inertia groups and prove (c) and (d).

Let Q be a prime divisor over X with $\mathrm{center}_X Q = \Sigma$, and let r_Q and F_Q be as in (A.34). As Σ maps to o, the valuation ring of Q, hence that of F_Q, dominates $\mathcal{O}_{W,o}$, so $\mathrm{center}_W F_Q = \{o\}$. By (A.33), the quotient morphism $V \to V/G_1 = U/G_F$ is the composite of $\rho : V \to V/G_2 = U$ and π_U. The ramification index of Q over F_Q is r_Q, and that of F_Q over its restriction to U/G_F is $|I_{G_F}(F_Q)|$ by Lemma 2.15(1). Hence the ramification index of Q over its restriction to V/G_1 is $r_Q|I_{G_F}(F_Q)|$, and Lemma 2.15(1) for G_1 gives $|I_{G_1}(Q)| = r_Q|I_{G_F}(F_Q)|$. Since $I_{G'}(Q) \subset I_{G_1}(Q)$ and $I_{G_F}(F_Q) \subset I_G(F_Q)$,

$$e'_Q \leq |I_{G_1}(Q)| = r_Q|I_{G_F}(F_Q)| \leq r_Q e_{F_Q}. \tag{A.36}$$

Let $\lambda_Q := r_Q e_{F_Q}/e'_Q$, so that $\lambda_Q \geq 1$. By $K_X = \psi^* K_V$, $K_X + D_X = \psi^*(K_V + D_V)$ and (A.34),

$$\frac{a(Q, X, D_X)}{e'_Q} = \lambda_Q \frac{a(F_Q, W, D)}{e_{F_Q}} \quad \text{and} \quad \frac{a(F_Q, Y', 0)}{e_{F_Q}} = \frac{a(Q, X, 0)}{\lambda_Q e'_Q}. \tag{A.37}$$

Since $a(F_Q, W, D) \geq 0$ and $\lambda_Q \geq 1$, the first equality and Definition 2.2 imply

$$\frac{a(Q, X, D_X)}{e'_Q} \geq \frac{a(F_Q, W, D)}{e_{F_Q}} \geq b.$$

Since $\mathrm{center}_W F_Q = \{o\}$, (A.32) and the second equality in (A.37) imply

$$\frac{a(F_Q, W, 0)}{e_{F_Q}} \leq (1 + m(2N_m - 1)) \frac{a(F_Q, Y', 0)}{e_{F_Q}} \leq (1 + m(2N_m - 1)) \frac{a(Q, X, 0)}{e'_Q}.$$

If $a(Q, X, D_X)/e'_Q = b$, then $b = \lambda_Q a(F_Q, W, D)/e_{F_Q} \geq \lambda_Q b$. Since $b > 0$, we have $\lambda_Q = 1$ and $a(F_Q, W, D)/e_{F_Q} = b$, so F_Q is a computing lift of b. This proves (d).

For $Q = P$, we have $F_P = F$, $I_{G_F}(F) = G_F$, and $I_{G'}(P) = G' = I_{G_1}(P)$. Thus (A.36) implies $e'_P = r_P|G_F| = r_P e_F$, so $\lambda_P = 1$, and by (A.37),

$$\frac{a(P, X, D_X)}{e'_P} = \frac{a(F, W, D)}{e_F} = b.$$

Together with (d), this proves (c). □

Proof of Theorem 9.4. Let T be a Gorenstein terminal threefold, let I be a finite cyclic group acting faithfully on T, let $\Sigma \subset T$ be a closed point or an irreducible curve fixed pointwise by I, and let h_T be an I-semi-invariant regular function on a neighborhood of Σ. Let $B_T := \frac{1}{m}\operatorname{div}(h_T)$, and for a prime divisor Q over T let e_Q be the order of the inertia group of Q in I. We assume the following.

(1) (T, B_T) is lc near Σ, $\lfloor B_T \rfloor$ is $\mathbb{Q}$-Cartier, and $\Sigma \subset \operatorname{Supp}\lfloor B_T \rfloor$.
(2) If $\Sigma = \{o_T\}$ is a closed point, then $(T \ni o_T)$ is smooth or an isolated hypersurface singularity, and h_T is a mark on $(T \ni o_T)$ with respect to I.
(3) The number

$$b_T := \inf\left\{ \frac{a(Q, T, B_T)}{e_Q} \;\middle|\; Q \text{ is a prime divisor over } T \text{ with } \operatorname{center}_T Q = \Sigma \right\} \tag{A.38}$$

is positive.

We say that a prime divisor Q over T attains b_T if $\operatorname{center}_T Q = \Sigma$ and $a(Q, T, B_T) = e_Q b_T$. If $\Sigma = \{o_T\}$, then $b_T = \operatorname{mld}((T \ni o_T, B_T)/I)$ by Definition 2.2, and a prime divisor attaining b_T is a computing lift of b_T (Definition 2.4).

Step 1. In this step we assume that Σ is a curve, and we show that b_T is attained by a prime divisor Q_1 over T with

$$\frac{a(Q_1, T, 0)}{e_{Q_1}} \le m + 1. \tag{A.39}$$

Since T is a terminal threefold, the singular locus of T is finite. Possibly replacing T with an I-invariant open neighborhood of a general closed point x of Σ, we may assume that T and Σ are smooth. The exceptional prime divisor E_1 of the blow-up of T along Σ satisfies $\operatorname{center}_T E_1 = \Sigma$, $a(E_1, T, 0) = 2$, and

$$\operatorname{mult}_{E_1}\lfloor B_T \rfloor = \operatorname{mult}_{\Sigma}\lfloor B_T \rfloor.$$

Since $b_T > 0$, we have $a(E_1, T, B_T) > 0$, so $\operatorname{mult}_{\Sigma}\lfloor B_T \rfloor = 1$. Projecting a generator of the ideal of $\lfloor B_T \rfloor$, and an element of the ideal of Σ which together with it generates the ideal of Σ near x, to character spaces of I, and possibly shrinking T, we obtain I-semi-invariant regular functions u, v on T with $\Sigma = (u = v = 0)$ and $\lfloor B_T \rfloor = \operatorname{div}(u)$. Let χ_u and χ_v be the characters of u and v. As I fixes Σ pointwise, the average over I of a regular function whose restriction to Σ is a local parameter at x is an I-invariant regular function w on T such that u, v, w are regular parameters of T at x.

The homomorphism $(\chi_u, \chi_v) : I \to (\mathbb{C}^*)^2$ is injective, since an element $\sigma \in I$ with

$$\chi_u(\sigma) = \chi_v(\sigma) = 1$$

fixes x and acts trivially on $\mathfrak{m}_x/\mathfrak{m}_x^2$, where $\mathfrak{m}_x$ is the maximal ideal of $\mathcal{O}_{T,x}$, the classes of u, v, w forming a basis, so, being of finite order, σ acts trivially on $\mathcal{O}_{T,x}$, hence on T, and $\sigma = 1$ as I acts faithfully on T.

Since $\lfloor B_T \rfloor = \operatorname{div}(u)$ and the coefficients of B_T are at most 1, $(u = 0)$ has coefficient 1 in B_T, so $h_T = u^m h_1$ for an I-semi-invariant regular function h_1 on T not divisible by u, and $B_T = \operatorname{div}(u) + \frac{1}{m}\operatorname{div}(h_1)$. Since $\Sigma = (v|_{(u=0)} = 0)$ is a smooth curve on the smooth surface $(u = 0)$, possibly shrinking T,

$$h_1 \equiv v^j h_2 \pmod{u} \tag{A.40}$$

for the integer $j := \operatorname{mult}_{\Sigma}(h_1|_{(u=0)}) \ge 0$ and a unit h_2 on T. We have $j < m$: let Q_0 be the exceptional prime divisor dominating Σ of the weighted blow-up of T along Σ with weights $(m, 1)$ on u, v. Then $a(Q_0, T, 0) = m + 1$, $\operatorname{mult}_{Q_0} u = m$, and $\operatorname{mult}_{Q_0} h_1 \ge \min\{j, m\}$ by (A.40). If $j \ge m$, then $a(Q_0, T, B_T) \le m + 1 - m - 1 = 0$, which contradicts $b_T > 0$.

Let r be the order of χ_v, and let $k_1 := |I|/r$. Since I is cyclic, $\chi_v = \rho^{k_1}$ for a faithful character ρ of I. Write $\chi_u = \rho^c$ with $c \in \mathbb{Z}$. The kernel of χ_v is the subgroup of I of order k_1, on which χ_u is faithful because (χ_u, χ_v) is injective. Hence $\gcd(c, k_1) = 1$. Choose an integer N with

$$N \equiv c \pmod{|I|} \qquad \text{and} \qquad k_1 j < N \le k_1 j + |I|.$$

Since k_1 divides $|I|$, $\gcd(N, k_1) = \gcd(c, k_1) = 1$. Let Q_1 be the exceptional prime divisor dominating Σ of the weighted blow-up of T along Σ with weights (N, k_1) on u, v. Then

$$a(Q_1, T, 0) = N + k_1, \qquad \operatorname{mult}_{Q_1} u = N, \quad \text{and} \quad \operatorname{mult}_{Q_1} h_1 = k_1 j,$$

where the last equality follows from (A.40) and $N > k_1 j$. As u and v are I-semi-invariant, I preserves $\operatorname{mult}_{Q_1}$. Since $\gcd(N, k_1) = 1$, the restriction ϑ of the rational function u^{k_1}/v^N to Q_1 is a coordinate on the fibers of $Q_1 \to \Sigma$, and any $\sigma \in I$ multiplies ϑ by

$$\chi_u(\sigma)^{k_1} \chi_v(\sigma)^{-N} = \rho(\sigma)^{k_1(c-N)} = 1.$$

As I also fixes Σ pointwise, I acts trivially on Q_1. Hence $e_{Q_1} = |I|$, and

$$\frac{a(Q_1, T, B_T)}{e_{Q_1}} = \frac{N + k_1 - N - k_1 j/m}{|I|} = \frac{1}{r}\Big(1 - \frac{j}{m}\Big).$$

Let Q be a prime divisor over T with $\operatorname{center}_T Q = \Sigma$. Since $(T, \operatorname{div}(u) + \operatorname{div}(v))$ is lc near Σ, $a(Q, T, \operatorname{div}(u)) \ge \operatorname{mult}_Q v$. If $j > 0$, then $(T, \operatorname{div}(u) + \frac{1}{j}\operatorname{div}(h_1))$ is lc near Σ by inversion of adjunction, since by (A.40) the restriction of $\frac{1}{j}\operatorname{div}(h_1)$ to $(u = 0)$ is $\operatorname{div}(v|_{(u=0)})$. Thus $\operatorname{mult}_Q h_1 \le j a(Q, T, \operatorname{div}(u))$, and

$$a(Q, T, B_T) = a(Q, T, \operatorname{div}(u)) - \frac{\operatorname{mult}_Q h_1}{m} \ge \Big(1 - \frac{j}{m}\Big) a(Q, T, \operatorname{div}(u)) \ge \Big(1 - \frac{j}{m}\Big) \operatorname{mult}_Q v.$$

If $j = 0$, then h_1 does not vanish on Σ, so $\operatorname{mult}_Q h_1 = 0$ and

$$a(Q, T, B_T) = a(Q, T, \operatorname{div}(u)) \ge \operatorname{mult}_Q v.$$

By Lemma 2.15(1) and [Liu26a, Lemma 9.2(1)], the restriction of χ_v to the inertia group of Q is $\lambda_Q^{\operatorname{mult}_Q v}$ for a faithful character λ_Q of the inertia group of Q, which has order e_Q. Since $\chi_v^r = 1$ and $\operatorname{mult}_Q v > 0$, e_Q divides $r \operatorname{mult}_Q v$, so $\operatorname{mult}_Q v \ge e_Q/r$. Thus, for any such Q,

$$\frac{a(Q, T, B_T)}{e_Q} \ge \frac{1}{r}\Big(1 - \frac{j}{m}\Big).$$

Therefore

$$b_T = \frac{1}{r}\Big(1 - \frac{j}{m}\Big) = \frac{a(Q_1, T, B_T)}{e_{Q_1}},$$

so Q_1 attains b_T, and since $j \le m - 1$,

$$\frac{a(Q_1, T, 0)}{e_{Q_1}} = \frac{N + k_1}{|I|} \le \frac{k_1 j + |I| + k_1}{|I|} = \frac{j+1}{r} + 1 \le m + 1.$$

Step 2. In this step and in **Steps 3–6** we assume that $\Sigma = \{o_T\}$ is a closed point and that T is smooth at o_T or $(T \ni o_T)$ is a singularity of quadratic rank at least 3.

Let $R := \lfloor B_T \rfloor$. By Kawamata's lemma [Kaw88, Lemma 5.1], R is Cartier near o_T. Projecting a generator of the ideal of R to a character space of I, we obtain an I-semi-invariant regular function u with $R = \operatorname{div}(u)$ near o_T. Since R is reduced and the coefficients of B_T are at most 1,

$$h_T = u^m h_1, \qquad B_T = R + \frac{1}{m}\operatorname{div}(h_1), \tag{A.41}$$

where h_1 is an I-semi-invariant regular function and no component of R is a component of $\operatorname{div}(h_1)$. Let $\mathfrak{m}_{o_T}$ be the maximal ideal of $\mathcal{O}_{T, o_T}$. If T is singular at o_T, then by Lemma 3.1(1) the blow-up of o_T has exactly one exceptional prime divisor E_1, and $\operatorname{mult}_{E_1} \mathfrak{m}_{o_T} = 1$, $a(E_1, T, 0) = 2$. If $u \in \mathfrak{m}_{o_T}^2$, then $a(E_1, T, B_T) \le 2 - 2 = 0$, which contradicts $b_T > 0$. Hence $u \notin \mathfrak{m}_{o_T}^2$ if T is

singular at o_T. If T is smooth at o_T, then the exceptional prime divisor E_1 of the blow-up of o_T satisfies $a(E_1, T, 0) = 3$ and $\mathrm{mult}_{E_1} u = \mathrm{mult}_{o_T} u$, so $u \notin \mathfrak{m}_{o_T}^3$.

We embed the henselization of T at o_T as a hypersurface $(f = 0)$ in the henselization $(Y \ni o_T)$ of a smooth fourfold Y° with an I-action at an I-fixed closed point, with I-semi-invariant regular parameters x, y, z, u, where f is I-semi-invariant and $u|_T$ is the function u above, in the following way.

(1) If T is singular at o_T, then $\dim \mathfrak{m}_{o_T}/\mathfrak{m}_{o_T}^2 = 4$ and $u \notin \mathfrak{m}_{o_T}^2$. We lift to I-semi-invariant functions x, y, z an eigenbasis of a complement of the class of u in $\mathfrak{m}_{o_T}/\mathfrak{m}_{o_T}^2$, and x, y, z, u define an I-equivariant morphism from a neighborhood of o_T in T to $Y^\circ := \mathbb{A}^4$, on which I acts diagonally by the characters of x, y, z, u, unramified at o_T, so that the henselization of T at o_T is a hypersurface in the henselization of $\mathbb{A}^4$ at 0. The quadratic part of f has rank ≥ 3.

(2) If T is smooth at o_T and $u \in \mathfrak{m}_{o_T}^2$, then we choose I-semi-invariant regular parameters x, y, z of T at o_T, let $Y^\circ := T \times \mathbb{A}^1$, where I acts on the coordinate u' of $\mathbb{A}^1$ by χ_u, let Y be the henselization of Y° at $(o_T, 0)$, and embed T as the graph $(f = 0)$, $f := u' - u$, where u is the function u pulled back from T. We then rename u' as u.

(3) If T is smooth at o_T and $u \notin \mathfrak{m}_{o_T}^2$, then we choose I-semi-invariant y, z such that y, z, u are regular parameters of T at o_T, let $Y^\circ := T \times \mathbb{A}^1$ with coordinate x on $\mathbb{A}^1$ and the trivial I-action on x, let Y be the henselization of Y° at $(o_T, 0)$, and let $f := x$.

Let $V := (u = 0) \subset Y$, $f_0 := f|_V$, and

$$S := T \cap V = (f_0 = 0) \subset V.$$

Then S is the reduced surface R, possibly reducible. In case (3), $f_0 = x$ and S is smooth. In cases (1) and (2), $f_0 \in (x, y, z)^2$ has non-zero quadratic part: in case (1) because restricting a quadratic form of rank at least 3 to a hyperplane leaves rank at least 1, and in case (2) because $u \in \mathfrak{m}_{o_T}^2 \setminus \mathfrak{m}_{o_T}^3$. Let $\widetilde{h}_1$ be an I-semi-invariant function on Y with $\widetilde{h}_1|_T = h_1$, let $h_V := \widetilde{h}_1|_V$ and $h_S := h_1|_S = h_V|_S$. By (A.41), h_S is non-zero on each component of S.

By Lemmas 2.13 and 2.12(3), prime divisors over T with center o_T, their log discrepancies, the orders of functions along them, and their inertia groups do not change when T is replaced by its henselization or its completion at o_T. The finitely many elements $x, y, z, u, f, \widetilde{h}_1$ of $\mathcal{O}_{Y,o_T}$ are regular functions on an I-equivariant pointed étale neighborhood $(Y_1 \ni y_1)$ of $(Y^\circ \ni o_T)$, whose existence we do not prove: $\mathcal{O}_{Y,o_T}$ is the colimit of the local rings of the pointed étale neighborhoods of $(Y^\circ \ni o_T)$ [Sta26, Tag 04GV], and the fiber product over Y° of the I-translates of one such neighborhood carries an I-action. After taking the connected component of Y_1 through y_1 and removing the other points of Y_1 over o_T, y_1 is the only point of Y_1 over o_T. Along $Y_1 \to Y^\circ$, prime divisors with center o_T, their log discrepancies, the orders of functions along them, and their inertia groups do not change (Lemma 2.13).

Let I_S be the subgroup of I acting trivially on S, and $J := I/I_S$. We write 0 for $o_T \in V$, $o_T \in S$, and for the image of o_T in Y/I_S and in T/I_S. The restrictions of y and z to S are non-zero, and so is that of x in cases (1) and (2), since $f_0 \in (x, y, z)^2$ there. As x, y, z are I-semi-invariant and I_S acts trivially on S, χ_x, χ_y, χ_z are trivial on I_S (in case (3), χ_x is trivial). An element of I_S with trivial χ_u acts trivially on x, y, z, u, hence on T, so $\chi_u|_{I_S}$ is faithful. Since I_S acts trivially on V and $f_0 \neq 0$, $h_V \neq 0$, the characters of f and $\widetilde{h}_1$ are trivial on I_S. Let $v := u^{|I_S|}$. Then Y/I_S is smooth with regular parameters x, y, z, v, on which J acts diagonally, and

$$f = f'(x, y, z, v), \qquad \widetilde{h}_1 = h'(x, y, z, v)$$

with $f'(x, y, z, 0) = f_0$ and $h'(x, y, z, 0) = h_V$. The quotient $T/I_S = (f' = 0)$ is klt, J acts faithfully on T/I_S and on V, since J acts faithfully on S, and the quotient map $\pi : T \to T/I_S$ is ramified in codimension 1 exactly along S, with ramification index $|I_S|$. Since S has coefficient 1

in B_T, Lemma 2.15(0)(4) and (A.41) imply

$$K_T + B_T = \pi^*(K_{T/I_S} + B_{T/I_S}), \qquad B_{T/I_S} = \operatorname{div}(v|_{T/I_S}) + \frac{1}{m}\operatorname{div}(h'|_{T/I_S}),$$

so $b_T = \operatorname{mld}((T/I_S \ni 0, B_{T/I_S})/J)$ by Lemma 2.15, namely its part (5) and its formula for quotient values at closed points, and $(T/I_S, B_{T/I_S})$ is lc near 0. Since f and $\widetilde{h}_1$ are I_S-invariant, f' and h' are regular functions on Y_1/I_S, and x, y, z, v are J-semi-invariant regular parameters of Y_1/I_S at the image of y_1, also denoted by 0. We take a J-invariant affine neighborhood U of 0 in Y_1/I_S and J-semi-invariant regular functions $\xi_1, \dots, \xi_M$ on U vanishing at 0 such that $x, y, z, v, \xi_1, \dots, \xi_M$ embed U into $\mathbb{A}^{4+M}$, on which J acts diagonally. We take J-semi-invariant polynomials $g_1, \dots, g_M$ in the ideal of U with $\det\big((\partial g_i/\partial \xi_k)(0)\big) \neq 0$, so that $U = (g_1 = \cdots = g_M = 0)$ near 0, and J-semi-invariant polynomials $\overline{f}$ and $\overline{h}$ with $\overline{f}|_U = f'$ and $\overline{h}|_U = h'$. By Theorem 2.17(1) for $(g_1, \dots, g_M, \overline{f})$ with $\mathfrak{a} = (v) \cdot (\overline{h})^{1/m}$,

$$b_T = \operatorname{mld}\Big(\Big(\mathbb{A}^{4+M} \ni 0, \sum_{i=1}^{M} \operatorname{div}(g_i) + \operatorname{div}(\overline{f}) + \operatorname{div}(v) + \frac{1}{m}\operatorname{div}(\overline{h})\Big)/J\Big).$$

Since $b_T > 0$, the pair on the right hand side is lc near 0, so $(V, \operatorname{div}(f_0) + \frac{1}{m}\operatorname{div}(h_V))$ is lc near 0 by adjunction to $V \cong (g_1 = \cdots = g_M = v = 0)$, where $V \to (v = 0) \subset Y/I_S$ is an isomorphism, both having regular parameters x, y, z. By Theorem 2.17(1) for $(g_1, \dots, g_M, v)$ with $\mathfrak{a} = (\overline{f}) \cdot (\overline{h})^{1/m}$, the right hand side is the quotient value on V, that is,

$$b_T = \operatorname{mld}\Big(\Big(V \ni 0, \operatorname{div}(f_0) + \frac{1}{m}\operatorname{div}(h_V)\Big)/J\Big). \tag{A.42}$$

Step 3. By (A.42), since h_V is regular, any prime divisor E over V with center 0 satisfies

$$a(E, V, \operatorname{div}(f_0)) \geq a\Big(E, V, \operatorname{div}(f_0) + \frac{1}{m}\operatorname{div}(h_V)\Big) \geq e_E b_T > 0.$$

For regular parameters x, y, z of V and a primitive $w \in (\mathbb{N}^+)^3$, the exceptional prime divisor E_w of the weighted blow-up with weights w satisfies

$$a(E_w, V, \operatorname{div}(f_0)) = w_x + w_y + w_z - \operatorname{wt}_w(f_0).$$

Hence

$$\operatorname{wt}_w(f_0) < w_x + w_y + w_z \quad \text{for any regular parameters } x, y, z \text{ of } V \text{ and any primitive } w \in (\mathbb{N}^+)^3. \tag{A.43}$$

Assume that $f_0 \in (x, y, z)^2$.

If the quadratic part of f_0 has rank 3, then S is an A_1 singularity by the Morse lemma.

If the quadratic part of f_0 has rank 2, then the Morse lemma with parameters z, u, applied I-equivariantly to f starting from I-semi-invariant coordinates in which the quadratic part of f_0 is a sum of two squares of coordinates or a product of two coordinates, yields new functions on Y, still denoted by x, y, z, with z I-semi-invariant, such that x, y, z, u are regular parameters of Y,

$$f = xy + \gamma(z, u), \qquad f_0 = xy + \gamma(z, 0), \tag{A.44}$$

and such that for any $\sigma \in I$, $(\sigma^* x, \sigma^* y)$ is $(\alpha x, \beta y)$ or $(\alpha y, \beta x)$ for some $\alpha, \beta \in \mathbb{C}^*$. The second possibility occurs if I exchanges the linear factors of the quadratic part of f_0. If $\gamma(z, 0) \neq 0$, then $\operatorname{mult}_0 \gamma(z, 0) \geq 3$, since $f_0 \in (x, y, z)^2$ and the quadratic part of f_0 has rank 2, and S is a Du Val singularity of type $A_{\operatorname{mult}_0 \gamma(z,0)-1}$. If $\gamma(z, 0) = 0$, then $S = (xy = 0)$ in V.

If the quadratic part of f_0 has rank 1, then in new J-semi-invariant regular parameters of V, still denoted by x, y, z, completing the square yields $f_0 = x^2 + g(y, z)$ with $g \in (y, z)^3$. Let g_3 be the cubic part of g.

(1) If $g_3 = 0$, then $w = (2, 1, 1)$ violates (A.43). So $g_3 \neq 0$.

(2) If $g_3 = \ell^3$ for a linear form ℓ, then ℓ is J-semi-invariant, being unique up to a constant. Taking $y := \ell$, Weierstrass preparation in y, removing the y^2-term and replacing x with $e^{-1/2}x$ yield

$$f_0 = e\big(x^2 + y^3 + \beta_1(z)y + \beta_0(z)\big), \qquad \operatorname{mult}_0 \beta_1 \geq 3, \quad \text{and} \quad \operatorname{mult}_0 \beta_0 \geq 4$$

for a unit e, which changes neither S nor $\operatorname{wt}_w(f_0)$. Let $\beta_{1,3}$, $\beta_{0,4}$, $\beta_{0,5}$ be the coefficients of z^3 in β_1 and of z^4, z^5 in β_0. If $\beta_{0,4} \neq 0$, then S is of type E_6. If $\beta_{0,4} = 0 \neq \beta_{1,3}$, then S is of type E_7. If $\beta_{0,4} = \beta_{1,3} = 0 \neq \beta_{0,5}$, then S is of type E_8. If $\beta_{0,4} = \beta_{1,3} = \beta_{0,5} = 0$, then $w = (3, 2, 1)$ violates (A.43).

(3) If g_3 has 3 distinct linear factors, then S is of type D_4.

(4) If $g_3 = \ell_1^2\ell_2$ for non-proportional linear forms ℓ_1, ℓ_2, then ℓ_1 and ℓ_2 are J-semi-invariant. The curve germ $(g = 0)$ in $(x = 0)$ has a unique smooth branch C tangent to $(\ell_2 = 0)$, and C is J-invariant. Let z be a J-semi-invariant equation of C, so that $g = zg_1$, where the quadratic part of g_1 is a non-zero multiple of y^2, $y := \ell_1$. Solving $\partial g_1/\partial y = 0$ for y and taking a square root of a unit, we may change y so that $g_1 = y^2 + \gamma_1(z)$ with $\operatorname{mult}_0 \gamma_1 \geq 3$. If $\gamma_1 \neq 0$, then $z\gamma_1(z)$ is z^k times a unit for some $k \geq 4$, and replacing z with z times a k-th root of the unit $z^{-k} \cdot z\gamma_1(z)$ and y with y times an inverse $2k$-th root of $z^{-k} \cdot z\gamma_1(z)$, we obtain

$$f_0 = x^2 + y^2 z + z^k \quad (k \geq 4) \qquad \text{or} \qquad f_0 = x^2 + y^2 z, \tag{A.45}$$

where x, y, z are J-semi-invariant regular parameters of V. In the first case S is of type D_{k+1}.

Each coordinate change in this step solves an equation whose derivative at o_T is invertible, or takes a root of a unit with a given value at o_T. By Hensel's lemma the solution exists in the henselian local ring and is unique, so each $\sigma \in I$ carries it to the solution of the transformed equation. Thus I acts on x, y, z as described in the case of rank 2, and in the case of rank 1 the uniqueness shows that the new parameters of V are J-semi-invariant functions of the old ones, hence I-semi-invariant functions on Y, which together with u form regular parameters of Y. Thus S is smooth or Du Val, or $\gamma(z, 0) = 0$ in (A.44), or $f_0 = x^2 + y^2 z$ in (A.45).

Step 4. In this step we assume that S is smooth or Du Val, and we show that b_T is attained by a prime divisor F over T with

$$\frac{a(F, T, 0)}{e_F} \leq (m + 2)C_m + 1, \tag{A.46}$$

where C_m is the constant of Lemma 5.7.

In this step x, y, z, u are the I-semi-invariant regular parameters of **Step 2**, not those of **Step 3**. Since S is smooth or Du Val, $(S, \frac{1}{m}\operatorname{div}(h_S))$ is lc near 0 by adjunction, and Theorem 2.17(1) for $(g_1, \dots, g_M, v, \overline{f})$ with $\mathfrak{a} = (\overline{h})^{1/m}$ implies, since b_T is the quotient value on $\mathbb{A}^{4+M}$ in **Step 2**, that

$$b_T = \operatorname{mld}\Big(\Big(S \ni 0, \frac{1}{m}\operatorname{div}(h_S)\Big)/J\Big). \tag{A.47}$$

Since h_1 is I-semi-invariant, I_S acts trivially on S, and $h_S \neq 0$ on S, the function h_S is a mark on $(S \ni 0)$ with respect to J (Definition 2.4). By (A.47) and Lemma 5.7, b_T has a computing lift P over $(S \ni 0)$ with respect to J and $\frac{1}{m}\operatorname{div}(h_S)$. Let $e := |I_J(P)|$, $n := \operatorname{mult}_P h_S$, and $\mu := \operatorname{mult}_P \operatorname{Jac}_S$. Then

$$\frac{a(P, S, 0)}{e} \leq C_m, \qquad \frac{\mu}{e} \leq C_m, \qquad \frac{n}{e} \leq mC_m, \quad \text{and} \quad \frac{a(P, S, 0) - n/m}{e} = b_T. \tag{A.48}$$

The ideal Jac_S is generated by the restrictions to S of the partial derivatives of f_0. Possibly reordering x, y, z in cases (1) and (2) of **Step 2**, we may assume that $\operatorname{mult}_P(\partial_x f_0|_S) = \mu$. In case (3), $\partial_x f_0 = 1$ and $\mu = 0$.

Let I' be the preimage of $I_J(P)$ in I, so that $|I'| = |I_S|e$. Let λ be the character of I', through $I_J(P)$, on $\mathfrak{m}_P/\mathfrak{m}_P^2$, where $\mathfrak{m}_P$ is the maximal ideal of the valuation ring of P. By Lemma 2.15(1), λ is faithful on $I_J(P)$, so the kernel of λ is I_S. Since I' is cyclic, $\lambda = \rho^{|I_S|}$ for a faithful character ρ of I'. Write $\chi_u|_{I'} = \rho^c$ with $c \in \mathbb{Z}$. Since $\chi_u|_{I_S}$ is faithful and I_S is the subgroup of I' of order $|I_S|$, $\gcd(c, |I_S|) = 1$. Choose an integer N with

$$\text{(A.49)}\quad N \equiv c \pmod{|I_S|e} \qquad \text{and} \qquad |I_S|\max\{2\mu, \mu+n\} < N \le |I_S|\max\{2\mu, \mu+n\} + |I_S|e.$$

Then $\gcd(N, |I_S|) = 1$. Let $\widetilde{P}$ be the prime divisor over $S \times \mathbb{A}^1_u$ such that $\operatorname{mult}_{\widetilde{P}}$ is the Gauss extension of $|I_S|\operatorname{mult}_P$ with value N on u (Definition 2.3). If $S' \to S$ is a projective birational morphism from a smooth surface on which P is a divisor and ϖ is a local equation of P on S', then near the generic point of $P \times \{0\}$ the valuation $\operatorname{mult}_{\widetilde{P}}$ is the monomial valuation in ϖ and u with the coprime weights $|I_S|$ and N. Then $\operatorname{center}_{S\times\mathbb{A}^1} \widetilde{P} = \{(0,0)\}$, $a(\widetilde{P}, S \times \mathbb{A}^1, 0) = |I_S|a(P, S, 0) + N$, and the residue field of $\widetilde{P}$ is $\kappa(P)(\vartheta)$, where ϑ is the class of $u^{|I_S|}/\varpi^N$. Let I act on $S \times \mathbb{A}^1_u$ through J on S and by χ_u on u. An element $\sigma \in I'$ preserves $\operatorname{mult}_{\widetilde{P}}$, acts trivially on $\kappa(P)$, and multiplies ϑ by

$$\chi_u(\sigma)^{|I_S|}\lambda(\sigma)^{-N} = \rho(\sigma)^{|I_S|(c-N)} = 1.$$

Conversely, let σ be an element of the inertia group of $\widetilde{P}$. Then σ preserves the restriction $|I_S|\operatorname{mult}_P$ of $\operatorname{mult}_{\widetilde{P}}$ to rational functions on S and acts trivially on $\kappa(P)$, so the image of σ in J belongs to $I_J(P)$. Hence the inertia group of $\widetilde{P}$ in I is I'.

We now compare $\widetilde{P}$ with a prime divisor over T. In the completion $\mathbb{C}[[x, y, z, u]]$ of Y at o_T, on which I acts diagonally, regarding f_0 and h_V as series in x, y, z, the completion of $S \times \mathbb{A}^1_u$ at $(0,0)$ is $(f_0 = 0)$ and the completion of T at o_T is $(f = 0)$. Since $f_0 = f|_V$, we have $f - f_0 \in (u)$ and $\chi_{f_0} = \chi_f$. By Lemma 2.12(3) we regard $\widetilde{P}$ as a formal prime divisor over $(f_0 = 0)$ with center the closed point. The Jacobian ideal Jac of $(f_0 = 0)$ is generated by $\partial_x f_0, \partial_y f_0, \partial_z f_0$, so for the coordinate x with $\operatorname{mult}_P(\partial_x f_0|_S) = \mu$,

$$\operatorname{mult}_{\widetilde{P}} \operatorname{Jac} = |I_S|\mu = \operatorname{mult}_{\widetilde{P}}(\partial_x f_0).$$

Since $f - f_0 \in (u)$ and $\widetilde{h}_1 - h_V \in (u)$, by (A.49) we have

$$\operatorname{mult}_{\widetilde{P}}\left((f - f_0)|_{S\times\mathbb{A}^1}\right) \ge N > 2|I_S|\mu, \qquad \operatorname{mult}_{\widetilde{P}}\left(\partial_x(f - f_0)|_{S\times\mathbb{A}^1}\right) \ge N > |I_S|\mu,$$

and, as $N > |I_S|n = \operatorname{mult}_{\widetilde{P}} h_V$, the restriction of $\widetilde{h}_1$ to $S \times \mathbb{A}^1$ has value $|I_S|n$. Thus

$$\operatorname{mult}_{\widetilde{P}}\left((f - f_0)|_{S\times\mathbb{A}^1}\right) \ge N > 2|I_S|\mu, \qquad \operatorname{mult}_{\widetilde{P}}\left(\partial_x(f - f_0)|_{S\times\mathbb{A}^1}\right) > |I_S|\mu,$$
$$\text{and} \qquad N > |I_S|\mu + |I_S|n,$$

which give the assumption of Lemma 5.4(3) and the condition of Lemma 5.4(3)(d) with $j = |I_S|\mu$. By Lemma 5.4(3) for the group I, with $f_0, f, \widetilde{P}, \widetilde{h}_1$ in place of f, f', F, H_1, and by Lemma 2.12(3), there exists a prime divisor F over T with $\operatorname{center}_T F = \{o_T\}$ whose inertia group in I is the inertia group I' of $\widetilde{P}$, so $e_F = |I_S|e$, and

$$a(F, T, 0) = a(\widetilde{P}, S \times \mathbb{A}^1, 0) = |I_S|a(P, S, 0) + N, \quad \operatorname{mult}_F h_1 = |I_S|n, \quad \text{and} \quad \operatorname{mult}_F u = N,$$

where $\operatorname{mult}_F u = N$ holds since the homomorphism of Lemma 5.4(3)(b) fixes the coordinate u. By (A.41) and (A.48),

$$\frac{a(F, T, B_T)}{e_F} = \frac{|I_S|a(P, S, 0) + N - N - |I_S|n/m}{|I_S|e} = \frac{a(P, S, 0) - n/m}{e} = b_T,$$

so F attains b_T. By (A.49) and (A.48), and since $\max\{2, m+1\} = m+1$,

$$\frac{a(F, T, 0)}{e_F} = \frac{a(P, S, 0)}{e} + \frac{N}{|I_S|e} \le C_m + \frac{\max\{2\mu, \mu+n\}}{e} + 1 \le C_m + (m+1)C_m + 1 = (m+2)C_m + 1.$$

This proves (A.46). The proof of Theorem 9.4 continues below.

Proof of Theorem 9.4, continued. We keep the germ $(T \ni o_T)$ with the cyclic group I and the mark $h_T = u^m h_1$, the smooth fourfold germ $(Y \ni o_T)$ with the regular parameters x, y, z, u, the equation f, $V = (u = 0)$, $f_0 = f|_V$, $S = (f_0 = 0) \subset V$, the kernel I_S, $J = I/I_S$, the factorization (A.41), and $\widetilde{h}_1$, h_V, h_S introduced above, for which (A.42) holds. It remains to treat the case $\gamma(z, 0) = 0$ in (A.44) and the case $f_0 = x^2 + y^2 z$ in (A.45).

For a prime divisor P over V with center 0, let $e_P := |I_J(P)|$.

Step 5. In this step we treat the normal crossing case: $f = xy + \gamma(z, u)$ with $\gamma(z, 0) = 0$, so $f_0 = xy$. By the choice of x, y following (A.44), the subgroup $J_0 \subset J$ of the elements preserving $(x = 0)$ and $(y = 0)$ satisfies $[J : J_0] \le 2$. Let r be the order of $\chi_z|_{J_0}$. For any J-semi-invariant function h'_V on V with $h'_V|_S = h_S$, the quotient $q_V := (h_V - h'_V)/f_0$ is a J-semi-invariant function on V, and replacing $\widetilde{h}_1$ with $\widetilde{h}_1 - f q_V$, where q_V is regarded as a function of x, y, z on Y, keeps $\widetilde{h}_1|_T = h_1$ and replaces h_V with h'_V. Since any function on V is congruent modulo xy to a unique function of the form $\varphi(z) + x\psi_1(x, z) + y\psi_2(y, z)$, which by uniqueness is J-semi-invariant if the given function is, and (A.42) holds for any I-semi-invariant lift $\widetilde{h}_1$ of h_1, we may take

$$h_V = \varphi(z) + x\psi_1(x, z) + y\psi_2(y, z).$$

Let $\ell := (x = y = 0) \subset V$. If $\varphi = 0$, then h_V vanishes along ℓ, and the exceptional prime divisor E_ℓ of the blow-up of ℓ satisfies

$$a\Big(E_\ell, V, \operatorname{div}(xy) + \frac{1}{m}\operatorname{div}(h_V)\Big) \le 2 - 2 - \frac{1}{m} < 0,$$

which contradicts lc near 0. Thus $\varphi = z^{i_0}\varphi_1(z)$ for an integer $i_0 \ge 0$ and a function φ_1 with $\varphi_1(0) \ne 0$. For an integer $n > i_0$, the monomial valuation with weights $(n, n, 1)$ on (x, y, z) has log discrepancy $2n + 1$, value $2n$ on xy and value i_0 on h_V, so lc near 0 implies $1 - \frac{i_0}{m} \ge 0$. Let

$$\alpha := 1 - \frac{i_0}{m} \in [0, 1].$$

We prove the lower bound $b_T \ge \alpha/r$. Let P be a prime divisor over V with center 0, and let

$$\theta_P := a(P, V, 0) - \operatorname{mult}_P x - \operatorname{mult}_P y - \operatorname{mult}_P z = a(P, V, \operatorname{div}(xyz)) \ge 0.$$

We claim that

$$a\Big(P, V, \operatorname{div}(xy) + \frac{1}{m}\operatorname{div}(h_V)\Big) \ge \alpha\big(\operatorname{mult}_P z + \theta_P\big). \tag{A.50}$$

If $i_0 = 0$, then $\operatorname{mult}_P h_V = 0$ and (A.50) is an equality. Assume that $i_0 > 0$. The restriction of $(x = 0) + (y = 0) + \frac{1}{i_0}\operatorname{div}(h_V)$ to the surface $(x = 0) \subset V$, with coordinates y, z, is

$$(y = 0) + \frac{1}{i_0}\operatorname{div}\big(\varphi(z) + y\psi_2(y, z)\big),$$

whose restriction to the curve ℓ, with coordinate z, is $\frac{1}{i_0}\operatorname{div}(\varphi)$, the point 0 with coefficient 1 near 0. By inversion of adjunction [KM98, Theorem 5.50], used for $\ell \subset (x = 0)$ and then for $(x = 0) \subset V$, each time with the coefficients of the components other than the divisor adjoined to multiplied by $1 - s$ for any $s \in (0, 1)$, the pair

$$\Big(V, \operatorname{div}(xy) + \frac{1}{i_0}\operatorname{div}(h_V)\Big)$$

is lc near 0. Thus $\operatorname{mult}_P h_V \le i_0\big(a(P, V, 0) - \operatorname{mult}_P x - \operatorname{mult}_P y\big)$, and

$$a\Big(P, V, \operatorname{div}(xy) + \frac{1}{m}\operatorname{div}(h_V)\Big) \ge \Big(1 - \frac{i_0}{m}\Big)\big(a(P, V, 0) - \operatorname{mult}_P x - \operatorname{mult}_P y\big) = \alpha\big(\operatorname{mult}_P z + \theta_P\big).$$

By Lemma 2.15(1), $I_J(P)$ is cyclic and its character λ on $\mathfrak{m}_P/\mathfrak{m}_P^2$ is faithful, where $\mathfrak{m}_P$ is the maximal ideal of the valuation ring of P. By [Liu26a, Lemma 9.2(1)], $\chi_g|_{I_J(P)} = \lambda^{\operatorname{mult}_P g}$ for any J-semi-invariant function g, and by [Liu26a, Lemma 9.2(2)] for $V = (v = 0) \subset Y/I_S$ with the

J-eigenparameters x', y', z, v, where $\mathbb{C}x' + \mathbb{C}y' = \mathbb{C}x + \mathbb{C}y$ (the cyclic group J preserves $\mathbb{C}x + \mathbb{C}y$), so that $dx' \wedge dy' \wedge dz \in \mathbb{C}^* dx \wedge dy \wedge dz$, the character of $dx \wedge dy \wedge dz$ on $I_J(P)$ is $\lambda^{a(P,V,0)}$.

- If $I_J(P) \subset J_0$, then χ_z^r is trivial on $I_J(P)$, so $\lambda^{r \operatorname{mult}_P z} = 1$ and e_P divides $r \operatorname{mult}_P z$. Since $\operatorname{mult}_P z > 0$, $\operatorname{mult}_P z / e_P \geq 1/r$.
- If $I_J(P) \not\subset J_0$, let σ be a generator of $I_J(P)$. Then $\sigma \notin J_0$ and $\sigma^2 \in J_0$, so $\lambda(\sigma)^{2r \operatorname{mult}_P z} = \chi_z(\sigma^2)^r = 1$ and $\operatorname{mult}_P z / e_P \geq 1/(2r)$. The function xyz is J-semi-invariant, and the form
$$\omega := \frac{dx}{x} \wedge \frac{dy}{y} \wedge \frac{dz}{z} = \frac{dx \wedge dy \wedge dz}{xyz}$$
satisfies $\sigma^* \omega = -\omega$, since σ exchanges $\frac{dx}{x}$ and $\frac{dy}{y}$ and fixes $\frac{dz}{z}$. By [Liu26a, Lemma 9.2(1)(2)], with (2) for $V \subset Y/I_S$ and x', y', z, v, the character of ω on $I_J(P)$ is $\lambda^{a(P,V,0) - \operatorname{mult}_P(xyz)} = \lambda^{\theta_P}$. Thus $\lambda(\sigma)^{\theta_P} = -1$. Since $\lambda(\sigma)$ is a primitive e_P-th root of unity,
$$\theta_P \equiv \frac{e_P}{2} \pmod{e_P}.$$
Since $\theta_P \geq 0$, $\theta_P / e_P \geq \frac{1}{2}$. Thus $(\operatorname{mult}_P z + \theta_P)/e_P \geq \frac{1}{2r} + \frac{1}{2} \geq \frac{1}{r}$.

In both cases $(\operatorname{mult}_P z + \theta_P)/e_P \geq 1/r$. Since $\alpha \geq 0$, (A.50) and (A.42) imply

$$b_T \geq \frac{\alpha}{r}. \tag{A.51}$$

We construct a prime divisor E over T attaining (A.51). Let $I_0 \subset I$ be the preimage of J_0 and $e_0 := |I_0|$. The group I_0 is cyclic and acts diagonally on x, y, z, u. Since I_S acts trivially on z, $\chi_z|_{I_0}$ has order r. Since I_0 is cyclic, $\chi_z|_{I_0} = \rho^{e_0/r}$ for a faithful character ρ of I_0. Let $w_z := e_0/r$, and choose positive integers w_u and w_x with $\chi_u|_{I_0} = \rho^{w_u}$, $\chi_x|_{I_0} = \rho^{w_x}$ and
$$2i_0 w_z + e_0 < w_u \leq 2i_0 w_z + 2e_0, \qquad i_0 w_z < w_x \leq i_0 w_z + e_0.$$
Since T is normal, $\gamma \neq 0$. Let $w_y := \operatorname{wt}_{(w_z, w_u)}(\gamma) - w_x$. Since u divides γ, $\operatorname{wt}_{(w_z,w_u)}(\gamma) \geq w_u$, so
$$w_y > 2i_0 w_z + e_0 - i_0 w_z - e_0 = i_0 w_z \geq 0.$$
Since f is I_0-semi-invariant, any monomial $z^j u^k$ of γ satisfies $\rho^{j w_z + k w_u} = \chi_f|_{I_0}$, so
$$\chi_f|_{I_0} = \rho^{\operatorname{wt}_{(w_z,w_u)}(\gamma)}.$$
Since $\chi_x \chi_y = \chi_f$, we obtain $\chi_y|_{I_0} = \rho^{w_y}$.

Let $w := (w_x, w_y, w_z, w_u)$, let N' be the greatest common divisor of the entries of w, and let $w' := w/N'$. Then
$$\operatorname{in}_{w'}(f) = xy + \operatorname{in}_{(w_z,w_u)}(\gamma),$$
which is irreducible because $\operatorname{in}_{(w_z,w_u)}(\gamma)$ is a non-zero polynomial in z, u. Since $h_V|_S = h_S = h_1|_S$ and $S = T \cap (u = 0)$, there exists a function H_1 on Y with $h_1 = (h_V + uH_1)|_T$, where h_V is regarded as a function of x, y, z on Y. Any monomial of $h_V + uH_1$ other than those of φ is divisible by x, y, or u, so it has w-weight at least $\min\{w_x, w_y, w_u\} > i_0 w_z$, and
$$\operatorname{in}_{w'}(h_V + uH_1) = \varphi_1(0) z^{i_0},$$
which is not divisible by $\operatorname{in}_{w'}(f)$. By [Liu26b, Lemma 4.12], the weighted blow-up with weights w' defines a prime divisor E over T with center o_T and
$$a(E, T, 0) = \frac{w_x + w_y + w_z + w_u - \operatorname{wt}_{(w_z,w_u)}(\gamma)}{N'} = \frac{w_z + w_u}{N'},$$
$$\operatorname{mult}_E u = \frac{w_u}{N'} \quad \text{and} \quad \operatorname{mult}_E h_1 = \frac{i_0 w_z}{N'}.$$
Thus
$$a(E, T, B_T) = \frac{w_z + w_u - w_u - i_0 w_z/m}{N'} = \frac{\alpha w_z}{N'}.$$

Let g_0 be a generator of I_0 and $\zeta := \rho(g_0)$. Then g_0 multiplies x, y, z, u by $\zeta^{w_x}, \zeta^{w_y}, \zeta^{w_z}, \zeta^{w_u}$, hence multiplies any w'-homogeneous polynomial of w'-weight k by $\zeta^{N'k}$. Thus g_0 acts trivially on the exceptional divisor $\operatorname{Proj}\mathbb{C}[x, y, z, u]$, graded by w', of the weighted blow-up of Y, which contains the center of E, so $I_0 \subset I_I(E)$ and $e_E \geq e_0$. Since $w_z = e_0/r$, $i_0 \leq m$ and $r \geq 1$,

$$\frac{a(E, T, B_T)}{e_E} \leq \frac{\alpha w_z}{e_0} = \frac{\alpha}{r}, \qquad \frac{a(E, T, 0)}{e_E} \leq \frac{w_z + w_u}{e_0} \leq \frac{1 + 2i_0}{r} + 2 \leq 2m + 3.$$

By (A.51), $b_T = \alpha/r$, and E attains b_T with

$$\frac{a(E, T, 0)}{e_E} \leq 2m + 3. \tag{A.52}$$

Step 6. In this step we treat the pinch point case: $f_0 = x^2 + y^2 z$, any monomial of $f - f_0$ is divisible by u, J acts diagonally on x, y, z and I acts diagonally on x, y, z, u. By the replacement of $\widetilde{h}_1$ with $\widetilde{h}_1 - f q_V$ in **Step 5**, we may take

$$h_V = \varphi(z) + y\psi_1(y, z) + x\psi_2(y, z).$$

Let $\ell := (x = y = 0) \subset V$. Since $f_0 \in (x, y)^2$, the exceptional prime divisor E_ℓ of the blow-up of ℓ satisfies $a(E_\ell, V, \operatorname{div}(f_0)) = 0$. If $\varphi = 0$, then h_V vanishes along ℓ and

$$a\Big(E_\ell, V, \operatorname{div}(f_0) + \frac{1}{m}\operatorname{div}(h_V)\Big) < 0,$$

which contradicts lc near 0. Thus $\varphi = z^{i_0}\varphi_1(z)$ for an integer $i_0 \geq 0$ and a function φ_1 with $\varphi_1(0) \neq 0$. For an integer $n > 2i_0$, the monomial valuation with weights $(n + 1, n, 2)$ on (x, y, z) has log discrepancy $2n + 3$, value $2n + 2$ on f_0 and value $2i_0$ on h_V, so lc near 0 implies $2i_0 \leq m$. Let

$$\delta := \frac{\chi_x}{\chi_y} \quad \text{and} \quad \alpha := 1 - \frac{2i_0}{m} \in [0, 1],$$

and let k_δ be the order of δ. Since f_0 is semi-invariant, $\chi_x^2 = \chi_y^2\chi_z$, that is, $\chi_z = \delta^2$. Since I_S acts trivially on x and y, δ has order k_δ both as a character of J and as a character of I.

We prove that, if $i_0 > 0$, then

$$\Big(V, \operatorname{div}(f_0) + \frac{1}{2i_0}\operatorname{div}(h_V)\Big) \text{ is lc near } 0. \tag{A.53}$$

Let $\pi_\ell : V' \to V$ be the blow-up of ℓ, with exceptional divisor E_ℓ. Since $\operatorname{mult}_{E_\ell} f_0 = 2$ and $\operatorname{mult}_{E_\ell} h_V = 0$,

$$K_{V'} + E_\ell + \Delta_{V'} = \pi_\ell^*\Big(K_V + \operatorname{div}(f_0) + \frac{1}{2i_0}\operatorname{div}(h_V)\Big), \qquad \Delta_{V'} := \pi_{\ell*}^{-1}\operatorname{div}(f_0) + \frac{1}{2i_0}\pi_{\ell*}^{-1}\operatorname{div}(h_V).$$

The fiber $\pi_\ell^{-1}(0)$ is $E_\ell \cap (z = 0)$.

- In the chart $x = x'y$, with coordinates x', y, z and $E_\ell = (y = 0)$, we have $\pi_\ell^* f_0 = y^2(x'^2 + z)$ and $\pi_\ell^* h_V|_{E_\ell} = \varphi(z)$. Thus near $\pi_\ell^{-1}(0)$,

  $$\Delta_{V'}|_{E_\ell} = (x'^2 + z = 0) + \frac{1}{2}(z = 0)$$

  on E_ℓ, with coordinates x', z.
- In the chart $y = y'x$, with coordinates x, y', z and $E_\ell = (x = 0)$, we have $\pi_\ell^* f_0 = x^2(1 + y'^2 z)$, so $\pi_{\ell*}^{-1}\operatorname{div}(f_0) = (1 + y'^2 z = 0)$ is disjoint from $(z = 0)$, and $\Delta_{V'}|_{E_\ell} = \frac{1}{2}(z = 0)$ near $\pi_\ell^{-1}(0)$.

Hence $(E_\ell, \Delta_{V'}|_{E_\ell})$ is lc near $\pi_\ell^{-1}(0)$ away from the point $p_0 := (x' = z = 0)$ of the first chart. At p_0 the curves $(x'^2 + z = 0)$ and $(z = 0)$ are smooth with contact order 2, so $(E_\ell, (x'^2 + z = 0) + \frac{1}{2}(z = 0))$ is lc at p_0, as two blow-ups show. Thus $(E_\ell, \Delta_{V'}|_{E_\ell})$ is lc near $\pi_\ell^{-1}(0)$. By inversion of adjunction in the plt form for $(V', E_\ell + (1 - s)\Delta_{V'})$, for any $s \in (0, 1)$, $(V', E_\ell + \Delta_{V'})$ is lc near $\pi_\ell^{-1}(0)$. Since π_ℓ is proper, (A.53) follows.

Let P be a prime divisor over V with center 0. If $i_0 > 0$, (A.53) implies

$$\operatorname{mult}_P h_V \le 2i_0 a(P, V, \operatorname{div}(f_0)).$$

If $i_0 = 0$, then $\operatorname{mult}_P h_V = 0$. In both cases

$$a\Big(P, V, \operatorname{div}(f_0) + \frac{1}{m}\operatorname{div}(h_V)\Big) \ge \alpha a(P, V, \operatorname{div}(f_0)). \tag{A.54}$$

We bound the quotient value of the pair $(V, \operatorname{div}(f_0))$. By Lemma 6.4(2) with the weight vector $(2,1,2)$, for which $\operatorname{in}_{(2,1,2)}(f_0) = f_0$,

$$\operatorname{mld}((V \ni 0, \operatorname{div}(f_0))/J) = \min\{\psi(\beta) \mid \beta \in N_J \cap \mathbb{R}^3_{>0}\},$$
$$\psi(\beta) := \beta_x + \beta_y + \beta_z - \min\{2\beta_x, 2\beta_y + \beta_z\},$$

where N_J is the lattice of J in the coordinates x, y, z (Definition 2.3). For $\beta \in N_J$, δ takes the value $e^{2\pi i(\beta_x - \beta_y)}$ at the diagonal element $(e^{2\pi i\beta_x}, e^{2\pi i\beta_y}, e^{2\pi i\beta_z})$ of J, so $\beta_x - \beta_y \in \frac{1}{k_\delta}\mathbb{Z}$, and $\chi_z = \delta^2$ implies $\beta_z - 2(\beta_x - \beta_y) \in \mathbb{Z}$. Moreover

$$\psi(\beta) = \max\{\beta_x - \beta_y,\ \beta_z - (\beta_x - \beta_y)\}.$$

Let $\beta \in N_J \cap \mathbb{R}^3_{>0}$.

- If $\beta_x > \beta_y$, then $\psi(\beta) \ge \beta_x - \beta_y \ge \frac{1}{k_\delta}$.
- If $\beta_x < \beta_y$, then $\psi(\beta) \ge \beta_z + (\beta_y - \beta_x) > \frac{1}{k_\delta}$.
- If $\beta_x = \beta_y$, then β_z is a positive integer and $\psi(\beta) = \beta_z \ge 1 \ge \frac{1}{k_\delta}$.

Thus $a(P, V, \operatorname{div}(f_0))/e_P \ge 1/k_\delta$ for any prime divisor P over V with center 0. Since $\alpha \ge 0$, (A.54) and (A.42) imply

$$b_T \ge \frac{\alpha}{k_\delta}. \tag{A.55}$$

We construct a prime divisor E over T attaining (A.55). Let $\sigma \in I$ be an element with $\delta(\sigma) = e^{2\pi i/k_\delta}$. Choose rational numbers w_x, w_y, w_z, w_u with

$$\chi_x(\sigma) = e^{2\pi i w_x}, \quad \chi_y(\sigma) = e^{2\pi i w_y}, \quad \chi_z(\sigma) = e^{2\pi i w_z}, \quad \text{and} \quad \chi_u(\sigma) = e^{2\pi i w_u}$$

and

$$\frac{2i_0}{k_\delta} < w_y \le \frac{2i_0}{k_\delta} + 1, \qquad w_x = w_y + \frac{1}{k_\delta}, \qquad w_z = \frac{2}{k_\delta}, \quad \text{and} \quad 2w_x < w_u \le 2w_x + 1.$$

They exist because $\chi_x(\sigma) = \chi_y(\sigma)\delta(\sigma)$ and $\chi_z(\sigma) = \delta(\sigma)^2$. Let $w := (w_x, w_y, w_z, w_u)$. Then x^2 and $y^2 z$ have w-weight $2w_x$, and any monomial of $f - f_0$ has w-weight at least $w_u > 2w_x$, so

$$\operatorname{in}_w(f) = x^2 + y^2 z,$$

which is irreducible. As in **Step 5**, there exists a function H_1 on Y with

$$h_1 = (h_V + uH_1)|_T.$$

Any monomial of $h_V + uH_1$ other than those of φ has w-weight at least

$$\min\{w_y, w_x, w_u\} = w_y > \frac{2i_0}{k_\delta} = \operatorname{wt}_w(z^{i_0}),$$

so

$$\operatorname{in}_w(h_V + uH_1) = \varphi_1(0) z^{i_0},$$

which is not divisible by $x^2 + y^2 z$. Let N be the least positive integer with $Nw \in \mathbb{Z}^4$, N' the greatest common divisor of the entries of Nw, and $w' := Nw/N'$. By [Liu26b, Lemma 4.12], the

weighted blow-up with weights w' defines a prime divisor E over T with center o_T and

$$a(E,T,0)=\frac{N}{N'}(w_x+w_y+w_z+w_u-2w_x)=\frac{N}{N'}\Big(w_u+\frac{1}{k_\delta}\Big),$$
$$\operatorname{mult}_E u=\frac{N}{N'}w_u \quad\text{and}\quad \operatorname{mult}_E h_1=\frac{N}{N'}\cdot\frac{2i_0}{k_\delta}.$$

Thus

$$a(E,T,B_T)=\frac{N}{N'}\Big(\frac{1}{k_\delta}-\frac{2i_0}{mk_\delta}\Big)=\frac{N}{N'}\cdot\frac{\alpha}{k_\delta}.$$

The element σ multiplies the j-th coordinate by $e^{2\pi i w_j}=e^{2\pi i N' w'_j/N}$, hence multiplies any w'-homogeneous polynomial of w'-weight k by $e^{2\pi i N'k/N}$. Thus σ acts trivially on the exceptional divisor $\operatorname{Proj}\mathbb{C}[x,y,z,u]$, graded by w', of the weighted blow-up of Y, which contains the center of E, so $\sigma\in I_I(E)$. Since σ^k fixes x,y,z,u if and only if $kw\in\mathbb{Z}^4$, and I acts faithfully on T, the order of σ is N. Thus $e_E\ge N$, and since $2i_0\le m$ and $k_\delta\ge 1$,

$$\frac{a(E,T,B_T)}{e_E}\le\frac{\alpha}{k_\delta},\qquad \frac{a(E,T,0)}{e_E}\le w_u+\frac{1}{k_\delta}\le 2w_y+\frac{3}{k_\delta}+1\le\frac{4i_0+3}{k_\delta}+3\le 2m+6.$$

By (A.55), $b_T=\alpha/k_\delta$, and E attains b_T with

$$\frac{a(E,T,0)}{e_E}\le 2m+6. \tag{A.56}$$

Step 7. Let $L_2(m)$ be the constant of Theorem 9.3, C_m the constant of Lemma 5.7, N_m the integer of Proposition A.4, and

$$L(m):=(1+m(2N_m-1))\max\{L_2(m),2m+6,(m+2)C_m+1\}.$$

Let q be the radial function of $(W\ni o,G,h,m)$. By Lemma 4.2(1), $b:=q(1)$ has a computing lift F_0. For F_0, Proposition A.5 provides a Gorenstein terminal threefold X, a finite cyclic group G' acting faithfully on X, the crepant pull-back D_X and the center Σ with the properties (a)–(d) there. We show that there exists a prime divisor Q over X with $\operatorname{center}_X Q=\Sigma$ and

$$\frac{a(Q,X,D_X)}{e'_Q}=b,\qquad \frac{a(Q,X,0)}{e'_Q}\le\frac{L(m)}{1+m(2N_m-1)}. \tag{A.57}$$

Assume first that $\Sigma=\{o_X\}$ for a closed point o_X. By Proposition A.5(a)–(c), the quadruple

$$(T,I,\Sigma,h_T):=(X,G',\{o_X\},h_X)$$

satisfies (1)–(3) with $b_T=b$.

- If $(X\ni o_X)$ is smooth or of quadratic rank at least 3, then (A.46), (A.52), and (A.56) give a computing lift Q of b with $a(Q,X,0)/e'_Q\le\max\{(m+2)C_m+1,2m+6\}$.
- If $(X\ni o_X)$ has quadratic rank 1 or 2, Theorem 9.3 implies that b has a computing lift Q with $a(Q,X,0)/e'_Q\le L_2(m)$.

In each case $\operatorname{center}_X Q=\{o_X\}$ and (A.57) holds.

Assume next that Σ is a curve. Let $x\in\Sigma$ be a closed point and X_x a G'-invariant open neighborhood of x on which $mD_X=\operatorname{div}(h_X)$ for the mark h_X of Proposition A.5(b). The prime divisors over X_x with center $\Sigma\cap X_x$ are the prime divisors over X with center Σ, so by Proposition A.5(a)–(c), $(T,I,\Sigma,h_T):=(X_x,G',\Sigma\cap X_x,h_X)$ satisfies (1) and (3) with $b_T=b$. Thus by (A.39) there exists a prime divisor Q over X with $\operatorname{center}_X Q=\Sigma$, $a(Q,X,D_X)/e'_Q=b$, and $a(Q,X,0)/e'_Q\le m+1$. Since $m+1\le 2m+6$, (A.57) holds.

Let Q satisfy (A.57). By Proposition A.5(d), the valuation of Q restricts to a positive multiple of the valuation of a prime divisor $F:=F_Q$ over W with $\operatorname{center}_W F=\{o\}$, F is a computing lift of b, and

$$\frac{a(F,W,0)}{e_F}\le(1+m(2N_m-1))\frac{a(Q,X,0)}{e'_Q}\le L(m).$$

The theorem follows. □

Appendix B. Proofs for germs of quadratic rank at least three

Here we prove the results of Section 10.

B.1. **Floors.** This subsection proves Theorem 10.4. For $\beta \in \mathbb{Q}^d$ we let $\mathbf{e}(\beta)$ be the diagonal matrix with entries $\exp(2\pi\sqrt{-1}\beta_1), \dots, \exp(2\pi\sqrt{-1}\beta_d)$. Thus the lattice of a finite diagonal group H is $N_H = \{\beta \in \mathbb{Q}^d \mid \mathbf{e}(\beta) \in H\}$, and $\mathbb{Z}^d \subset N_H$ (Definition 2.3).

Lemma B.1. *Let $N \geq 2$ be an integer, let G be a finite diagonal abelian group acting faithfully on W_N and preserving the equation of W_N up to a character, and let $\widetilde{W}_0$ be as in Set-up 10.3. Let $\phi(x,y,z)$ be a polynomial semi-invariant with the character χ_t such that $\phi(0) = 0$, and let $T := t + \phi$. Let*

$$\widetilde{G} := \{(g,\alpha) \mid g \in G,\ \alpha \in \mathbb{C},\ \alpha^2 = \chi_x(g)\},$$

which acts on (p,v,T) by

$$(p,v,T) \to (\alpha p, \chi_z(g)\alpha^{-1}v, \chi_t(g)T).$$

Let $w = (\omega_1,\omega_2,\omega_3)$ be a primitive vector with positive entries in the lattice $N_{\widetilde{G}}$ of $\widetilde{G}$ in the coordinates (p,v,T) of $\widetilde{W}_0$, and let

$$b := (2\omega_1, 2\omega_2, \omega_1+\omega_2, \omega_3) \quad \textit{and} \quad \omega_{\min} := \min\{2\omega_1, 2\omega_2, \omega_1+\omega_2, \omega_3\}.$$

Let $h_1, \dots, h_n$ be polynomials in x,y,z,t such that the polynomials

$$H_j := h_j\big(p^2, -v^2, pv, T - \phi(p^2, -v^2, pv)\big)$$

are non-zero. Assume that

$$N\omega_{\min} > \max\{2(\omega_1+\omega_2), \mathrm{wt}_w(H_1), \dots, \mathrm{wt}_w(H_n)\}.$$

Then the weighted blow-up of $\mathbb{A}^4$ in the coordinates (x,y,z,T) with weights the primitive vector $\overline{b}$ of $\mathbb{Z}^4 \cap \mathbb{Q}_{>0}b$ defines a prime divisor F over W_N with center 0 such that

$$\frac{a(F,W_N,0)}{e_F} = \omega_1+\omega_2+\omega_3 \quad \textit{and} \quad \frac{\mathrm{mult}_F\, h_j}{e_F} = \mathrm{wt}_w(H_j) \textit{ for any } j.$$

Proof. Since ϕ is semi-invariant with the character χ_t, the group G acts on (x,y,z,T) by the characters $\chi_x, \chi_y, \chi_z, \chi_t$, and $\widetilde{G}$ acts on (p,v,T) as in the statement. Since G preserves the equation of W_N up to a character, $\chi_x\chi_y = \chi_z^2$.

We first show that b is primitive in the lattice N_G of G in the coordinates (x,y,z,T). Let $\Phi : \mathbb{Q}^3 \to \mathbb{Q}^4$ be the linear map $\Phi(\beta) := (2\beta_1, 2\beta_2, \beta_1+\beta_2, \beta_3)$. If $(g,\alpha) \in \widetilde{G}$ acts on (p,v,T) by $\mathbf{e}(\beta)$, then $\chi_x(g) = \alpha^2$ and $\chi_y(g) = \chi_z(g)^2\alpha^{-2}$, so g acts on (x,y,z,T) by $\mathbf{e}(\Phi(\beta))$. Conversely, let $b' \in N_G$ with $b'_1 + b'_2 = 2b'_3$, let $g := \mathbf{e}(b') \in G$ and let $\alpha := \exp(\pi\sqrt{-1}b'_1)$. Then $\alpha^2 = \chi_x(g)$ and

$$\chi_z(g)\alpha^{-1} = \exp(\pi\sqrt{-1}(2b'_3 - b'_1)) = \exp(\pi\sqrt{-1}b'_2),$$

so $(g,\alpha) \in \widetilde{G}$ acts on (p,v,T) by $\mathbf{e}(\frac{1}{2}b'_1, \frac{1}{2}b'_2, b'_4)$. Hence Φ restricts to a bijection

$$\Phi : N_{\widetilde{G}} \to N_G \cap \{b' \in \mathbb{Q}^4 \mid b'_1 + b'_2 = 2b'_3\}. \tag{B.1}$$

In particular $b = \Phi(w) \in N_G$. If $b/i \in N_G$ for an integer $i \geq 2$, then $b/i = \Phi(w/i)$ belongs to the right-hand side of (B.1), so $w/i \in N_{\widetilde{G}}$, which contradicts the primitivity of w. Thus b is primitive in N_G.

Since N_G is a lattice containing $\mathbb{Z}^4$ and b is primitive in N_G, we have $N_G \cap \mathbb{Q}b = \mathbb{Z}b$. Thus $\overline{b} = rb$ for a positive integer r. Since $\mathbb{Z}^4 \cap \mathbb{Q}b = \mathbb{Z}\overline{b}$, for any $i \in \mathbb{Z}$ the element $\mathbf{e}(ib) \in G$ is the identity if and only if $r \mid i$.

Let $f_N := xy + z^2 + (T - \phi(x,y,z))^N$, so that $W_N = (f_N = 0) \subset \mathbb{A}^4$ in the coordinates (x,y,z,T). Since $\phi(0) = 0$, any monomial of $T - \phi(x,y,z)$ has b-weight at least $\omega_{\min}$, so

$$\mathrm{wt}_b\left((T-\phi)^N\right) \geq N\omega_{\min} > 2(\omega_1+\omega_2) = \mathrm{wt}_b(xy+z^2).$$

Thus the lowest $\bar{b}$-weight part of f_N is $xy + z^2$, and $(xy + z^2 = 0) \subset \mathbb{A}^4$ is integral and not contained in any coordinate hyperplane. By [Liu26b, Lemma 4.12], the weighted blow-up of $\mathbb{A}^4$ with weights $\bar{b}$ defines a prime divisor F over W_N with center 0 such that

$$a(F, W_N, 0) = \sum_{i=1}^{4} \bar{b}_i - \mathrm{wt}_{\bar{b}}(f_N) = r(\omega_1+\omega_2+\omega_3), \tag{B.2}$$

and such that $\mathrm{mult}_F \eta \geq \mathrm{wt}_{\bar{b}}(\eta)$ for any polynomial η in x,y,z,T, with equality if $xy+z^2$ does not divide the lowest $\bar{b}$-weight part of η. Moreover, the intersection of the strict transform of W_N with the exceptional divisor $\mathbb{P}(\bar{b})$ of the weighted blow-up of $\mathbb{A}^4$ is the zero locus of the lowest $\bar{b}$-weight part $xy + z^2$ of f_N, which is irreducible of dimension 2, so F is the hypersurface $(xy + z^2 = 0)$ in $\mathbb{P}(\bar{b})$.

Fix j and let $g_j(x,y,z,T) := h_j(x,y,z,T - \phi(x,y,z))$, so that $h_j|_{W_N} = g_j|_{W_N}$. Replacing xy with $-z^2$ repeatedly, we write $g_j = g_j^0 + (xy + z^2)Q_j$, where Q_j is a polynomial and g_j^0 is a linear combination of monomials $x^{i_1}z^{i_3}T^{i_4}$ and $y^{i_2}z^{i_3}T^{i_4}$ with $i_1 \geq 0$ and $i_2 \geq 1$. Since $xy + z^2 = -(T-\phi)^N$ on W_N,

$$h_j|_{W_N} = \left(g_j^0 - (T-\phi)^N Q_j\right)|_{W_N}. \tag{B.3}$$

The substitution $(x,y,z,T) \to (p^2, -v^2, pv, T)$ sends $xy+z^2$ to 0 and g_j to H_j, hence sends g_j^0 to H_j. It sends $x^{i_1}z^{i_3}T^{i_4}$ to $p^{2i_1+i_3}v^{i_3}T^{i_4}$ and $y^{i_2}z^{i_3}T^{i_4}$ to $(-1)^{i_2}p^{i_3}v^{2i_2+i_3}T^{i_4}$. Distinct monomials of g_j^0 go to distinct monomials, since the exponent of p is at least that of v in the first case and smaller in the second case, and the b-weight of a monomial equals the w-weight of the image. Hence $\mathrm{wt}_b(g_j^0) = \mathrm{wt}_w(H_j)$, and the lowest b-weight part of g_j^0 is not divisible by $xy+z^2$, because the substitution sends it to the lowest w-weight part of $H_j \neq 0$. Thus $\mathrm{mult}_F g_j^0 = r\,\mathrm{wt}_w(H_j)$. Since Q_j has non-negative weight,

$$\mathrm{mult}_F\left((T-\phi)^N Q_j\right) \geq \mathrm{wt}_{\bar{b}}\left((T-\phi)^N\right) \geq rN\omega_{\min} > r\,\mathrm{wt}_w(H_j).$$

By (B.3), $\mathrm{mult}_F h_j = r\,\mathrm{wt}_w(H_j)$.

By (B.2), it remains to show that $e_F = r$. Since G acts diagonally on (x,y,z,T), any element of G maps F to itself. Since F is the hypersurface $(xy+z^2=0)$ in $\mathbb{P}(\bar{b})$, the rational functions on F are the quotients ψ_1/ψ_2 of homogeneous elements $\psi_1, \psi_2 \neq 0$ with $\deg\psi_1 = \deg\psi_2$ of the domain $R := \mathbb{C}[x,y,z,T]/(xy+z^2)$ graded by $\bar{b}$. The substitution $(x,y,z,T) \to (p^2, -v^2, pv, T)$ embeds R into $\mathbb{C}[p,v,T]$ with image spanned by the monomials $p^{i_1}v^{i_2}T^{i_3}$ with i_1+i_2 even, and the degree of such a monomial is $rw \cdot (i_1,i_2,i_3)$. Thus the rational functions on F are generated over $\mathbb{C}$ by the Laurent monomials $p^{i_1}v^{i_2}T^{i_3}$ with (i_1,i_2,i_3) in

$$M_w := \{m \in M_0 \mid w \cdot m = 0\}, \quad \text{where} \quad M_0 := \{m \in \mathbb{Z}^3 \mid m_1 + m_2 \in 2\mathbb{Z}\}.$$

Let $g \in G$ and let $(g,\alpha) \in \widetilde{G}$ act on (p,v,T) by $\mathbf{e}(\beta)$. Then g multiplies $p^{i_1}v^{i_2}T^{i_3}$, $(i_1,i_2,i_3) \in M_0$, by $\exp(2\pi\sqrt{-1}\beta\cdot(i_1,i_2,i_3))$, which does not depend on the choice of α, since (g,α) and $(g,-\alpha)$ differ by $\iota = (1,-1)$, and ι multiplies $p^{i_1}v^{i_2}T^{i_3}$ by $(-1)^{i_1+i_2} = 1$. Since g fixes F pointwise if and only if it fixes $p^{i_1}v^{i_2}T^{i_3}$ for any $(i_1,i_2,i_3) \in M_w$, $g \in I_G(F)$ if and only if $\beta \cdot m \in \mathbb{Z}$ for any $m \in M_w$. Let $M_0^\vee := \mathbb{Z}^3 + \mathbb{Z}(\frac{1}{2},\frac{1}{2},0)$, the dual lattice of M_0. Since $M_w = M_0 \cap w^\perp$ is saturated in M_0, the restriction of β to M_w is integral if and only if there exists $\beta_0 \in M_0^\vee$ with $(\beta - \beta_0)\cdot M_w = 0$. Since M_w spans $w^\perp$, we obtain that $\beta \cdot M_w \subset \mathbb{Z}$ if and only if $\beta \in M_0^\vee + \mathbb{Q}w$. Since $\Phi(\frac{1}{2},\frac{1}{2},0) = (1,1,1,0)$, we have $\Phi(M_0^\vee) \subset \mathbb{Z}^4$. If $\beta = \beta_0 + sw$ with $\beta_0 \in M_0^\vee$ and $s \in \mathbb{Q}$, then

$$\mathbf{e}(M_0^\vee) = \{\mathrm{id}_{\widetilde{W}_0}, \iota\} \subset \widetilde{G}$$

implies $sw \in N_{\widetilde{G}} \cap \mathbb{Q}w = \mathbb{Z}w$, and g acts on (x,y,z,T) by $\mathbf{e}(\Phi(\beta_0)+sb) = \mathbf{e}(sb)$. Conversely, for $i \in \mathbb{Z}$, $\mathbf{e}(ib)$ is the image of $\mathbf{e}(iw) \in \widetilde{G}$, so $\mathbf{e}(ib) \in I_G(F)$. Thus $I_G(F) = \{\mathbf{e}(ib) \mid i \in \mathbb{Z}\}$, which has order r. Hence $e_F = r$, and the lemma follows from (B.2) and $\operatorname{mult}_F h_j = r \operatorname{wt}_w(H_j)$. □

Proposition B.2. *Let $(\alpha_j, \beta_j, \delta_j)$, $1 \le j \le n$, be triples of non-negative integers, none of which is $(0,0,0)$. Then there exist positive integers J_0 and Λ depending only on the triples and, for any positive integer M, a positive integer $J \ge J_0$ depending only on M and the triples, satisfying the following.*

Let $N \ge 2$ be an integer, let G be a finite diagonal abelian group acting faithfully on W_N and preserving the equation of W_N up to a character, and let $\widetilde{W}_0$ be as in Set-up 10.3. Let $\phi(x,y,z)$ be a polynomial semi-invariant with the character χ_t such that $\phi(0) = 0$, let $T := t + \phi$, so that $t = T - \phi(p^2, -v^2, pv)$ on $\widetilde{W}_0$, and let $\mathfrak{m} := (p, v, T) \subset \mathbb{C}[[p, v, T]]$. Let

$$\widetilde{G} := \{(g, \alpha) \mid g \in G,\ \alpha \in \mathbb{C},\ \alpha^2 = \chi_x(g)\},$$

which acts on (p, v, T) by

$$(p, v, T) \to (\alpha p, \chi_z(g)\alpha^{-1} v, \chi_t(g) T).$$

Let (Z_0, Z_1, Z_2) be one of the flags (v, p, T), (v, T, p), and (T, v, p), and let $\nu := |\chi_{Z_2}(\widetilde{G})|$, where χ_{Z_2} is the character of $\widetilde{G}$ on Z_2. Let $R_1, \dots, R_n$ be polynomials in p, v, T, let $\widehat{H}_j := Z_0^{\alpha_j} R_j$, and assume that

$$R_j|_{(Z_0=0)} = Z_1^{\beta_j} S_j(Z_1, Z_2) \quad \text{and} \quad S_j(0, Z_2) = Z_2^{\delta_j} u_j(Z_2)$$

for polynomials S_j and u_j with $u_j(0) \ne 0$. Let $c_1, \dots, c_n$ be positive real numbers such that

$$\mu_0 := \sum_{j=1}^n c_j \alpha_j \le 1, \quad \mu_1 := \sum_{j=1}^n c_j \beta_j \le 1 \quad \text{and} \quad \mu_2 := \sum_{j=1}^n c_j \delta_j \le 1.$$

Let $h_1, \dots, h_n$ be polynomials in x, y, z, t not vanishing identically on W_N, let

$$H_j := h_j\big(p^2, -v^2, pv, T - \phi(p^2, -v^2, pv)\big),$$

and let $D := \sum_{j=1}^n c_j \operatorname{div}(h_j)$ on W_N. Let $L_p := \frac{1}{2}\operatorname{div}(x)$, $L_v := \frac{1}{2}\operatorname{div}(y)$, and $L_T := \operatorname{div}(T)$ on W_N. Then the following hold.

(1) If $N \ge J_0$ and $H_j - \widehat{H}_j \in \mathfrak{m}^{J_0}$ for any j, then the pair

$$\Big(W_N, D + \sum_{i=0}^{2} (1 - \mu_i) L_{Z_i}\Big)$$

is lc near 0, and $a(F, W_N, D)/e_F \ge (1 - \mu_2)/\nu$ for any prime divisor F over W_N with center 0.

(2) If $|G| \le M$, $N \ge J$ and $H_j - \widehat{H}_j \in \mathfrak{m}^J$ for any j, then there exists a prime divisor F over W_N with center 0 such that

$$\frac{a(F, W_N, D)}{e_F} \le \frac{1 - \mu_2}{\nu} + \Lambda\big((1 - \mu_0) + (1 - \mu_1)\big).$$

Proof. For $\tau = (\tau_1, \dots, \tau_n) \in \mathbb{R}^n_{\ge 0}$ let $\mu_0(\tau) := \sum_j \tau_j \alpha_j$, $\mu_1(\tau) := \sum_j \tau_j \beta_j$, and $\mu_2(\tau) := \sum_j \tau_j \delta_j$, and let

$$\Pi := \{\tau \in \mathbb{R}^n_{\ge 0} \mid \mu_0(\tau) \le 1,\ \mu_1(\tau) \le 1,\ \mu_2(\tau) \le 1\}.$$

For any j, $\max\{\alpha_j, \beta_j, \delta_j\} \ge 1$, so $\tau_j \le 1$ for any $\tau \in \Pi$. Thus Π is a bounded polytope defined by rational inequalities, and the set $V(\Pi)$ of vertices of Π is a finite subset of $\mathbb{Q}^n$. Let $\mathcal{R}$ be the set of positive numbers among 1, τ_j, $1 - \mu_i(\tau)$, and $\frac{1}{2}(1 - \mu_i(\tau))$ for $\tau \in V(\Pi)$, $1 \le j \le n$, and $i \in \{0, 1, 2\}$. Then $\mathcal{R} \subset \mathbb{Q}_{>0}$ is finite and depends only on the triples.

Write $S_j = Z_2^{\delta_j} u_j + Z_1 V_j$ and $R_j = Z_1^{\beta_j} S_j + Z_0 V_j'$ with polynomials V_j in Z_1, Z_2 and V_j' in p, v, T. Then

$$\widehat{H}_j = Z_0^{\alpha_j} Z_1^{\beta_j} Z_2^{\delta_j} u_j(Z_2) + Z_0^{\alpha_j} Z_1^{\beta_j + 1} V_j + Z_0^{\alpha_j + 1} V_j'. \tag{B.4}$$

In particular the monomial $Z_0^{\alpha_j}Z_1^{\beta_j}Z_2^{\delta_j}$ has coefficient $u_j(0)\neq 0$ in $\widehat{H}_j$.

Step 1. In this step we show that, for any $\tau\in\Pi$, the pair

$$\Big(\widetilde{W}_0,\ \widehat{B}_\tau:=\sum_{j=1}^{n}\tau_j\operatorname{div}(\widehat{H}_j)+\sum_{i=0}^{2}(1-\mu_i(\tau))\operatorname{div}(Z_i)\Big)$$

is lc near 0. We have $\widehat{B}_\tau\geq 0$ since $\tau\in\Pi$. Since $u_j(0)\neq 0$, Z_1 does not divide S_j, and hence Z_0 does not divide R_j. Thus the coefficient of $(Z_0=0)$ in $\widehat{B}_\tau$ is $\mu_0(\tau)+1-\mu_0(\tau)=1$, and

$$\widehat{B}_\tau=(Z_0=0)+\sum_j\tau_j\operatorname{div}(R_j)+(1-\mu_1(\tau))\operatorname{div}(Z_1)+(1-\mu_2(\tau))\operatorname{div}(Z_2),$$

where $(Z_0=0)$ is not a component of $\widehat{B}_\tau-(Z_0=0)$. The coefficient of $(Z_1=0)$ in the restriction $(\widehat{B}_\tau-(Z_0=0))|_{(Z_0=0)}$ to the plane $(Z_0=0)$ with coordinates Z_1,Z_2 is $\mu_1(\tau)+1-\mu_1(\tau)=1$, so

$$(\widehat{B}_\tau-(Z_0=0))|_{(Z_0=0)}=(Z_1=0)+\sum_j\tau_j\operatorname{div}(S_j)+(1-\mu_2(\tau))\operatorname{div}(Z_2).$$

Since $u_j(Z_2)$ is a unit near 0, the restriction of

$$\sum_j\tau_j\operatorname{div}(S_j)+(1-\mu_2(\tau))\operatorname{div}(Z_2)$$

to the line $(Z_0=Z_1=0)$ is, near 0, the point 0 with coefficient $\mu_2(\tau)+1-\mu_2(\tau)=1$, which is lc. By inversion of adjunction [Kaw07, Theorem], $(\widetilde{W}_0,\widehat{B}_\tau)$ is lc near 0.

Step 2. In this step we prove the first conclusion of (1), with

$$J_0:=\max\Big\{J_3',\ J_4',\ 1+\max_j(\alpha_j+\beta_j+\delta_j)\Big\},$$

where J_3', resp. J_4', is the largest of the integers of Lemma 2.19 for 3, resp. 4, variables, at most $n+3$, resp. $n+4$, functions, and the finite set $\mathcal{R}$. By [Kaw21, Remark 2.6], lc near 0 may be tested on completions, so J_3' and J_4' serve for polynomial pairs near 0. Assume that $N\geq J_0$ and $H_j-\widehat{H}_j\in\mathfrak{m}^{J_0}$ for any j. Since $\alpha_j+\beta_j+\delta_j<J_0$, (B.4) implies $\widehat{H}_j\notin\mathfrak{m}^{J_0}$, so $H_j\neq 0$. Let $\tau\in V(\Pi)$. By **Step 1** and Lemma 2.19, the pair

$$\Big(\widetilde{W}_0,\ B_\tau:=\sum_{j=1}^{n}\tau_j\operatorname{div}(H_j)+\sum_{i=0}^{2}(1-\mu_i(\tau))\operatorname{div}(Z_i)\Big)$$

is lc near 0.

In the coordinates (x,y,z,T) of $\mathbb{A}^4$ let $f_0:=xy+z^2$ and $f_N:=xy+z^2+(T-\phi(x,y,z))^N$, so that $W_0=(f_0=0)$ and $W_N=(f_N=0)$. We regard h_j as the polynomial $h_j(x,y,z,T-\phi(x,y,z))$, and we let $L_{\mathbb{A}^4,p}:=\frac{1}{2}\operatorname{div}(x)$, $L_{\mathbb{A}^4,v}:=\frac{1}{2}\operatorname{div}(y)$, and $L_{\mathbb{A}^4,T}:=\operatorname{div}(T)$ on $\mathbb{A}^4$. In the coordinates (p,v,T) the double cover π_0 of Set-up 10.3 is $(p,v,T)\to(p^2,-v^2,pv,T)$. The map π_0 is the quotient by ι and is étale outside the line $(p=v=0)$, so $K_{\widetilde{W}_0}=\pi_0^*K_{W_0}$. Moreover

$$\pi_0^*(L_{\mathbb{A}^4,Z}|_{W_0})=\operatorname{div}(Z)\text{ for }Z\in\{p,v,T\}\quad\text{and}\quad\pi_0^*\operatorname{div}(h_j|_{W_0})=\operatorname{div}(H_j).$$

Thus $B_\tau=\pi_0^*\Delta_\tau$ for

$$\Delta_\tau:=\sum_j\tau_j\operatorname{div}(h_j|_{W_0})+\sum_{i=0}^{2}(1-\mu_i(\tau))L_{\mathbb{A}^4,Z_i}|_{W_0},$$

and (W_0,Δ_τ) is lc near 0. Since $H_j\neq 0$, f_0 does not divide h_j. Since W_0 is normal, inversion of adjunction [Liu26b, Lemma D.14] implies that

$$\Big(\mathbb{A}^4,\ \operatorname{div}(f_0)+\sum_j\tau_j\operatorname{div}(h_j)+\sum_{i=0}^{2}(1-\mu_i(\tau))L_{\mathbb{A}^4,Z_i}\Big)$$

is lc near 0. Since

$$f_N - f_0 = (T - \phi(x,y,z))^N \in (x,y,z,T)^N \subset (x,y,z,T)^{J_4'}$$

and the positive coefficients belong to $\mathcal{R}$, Lemma 2.19 implies that

$$\Big(\mathbb{A}^4,\ \operatorname{div}(f_N) + \sum_j \tau_j \operatorname{div}(h_j) + \sum_{i=0}^{2}(1-\mu_i(\tau))L_{\mathbb{A}^4,Z_i}\Big)$$

is lc near 0. Since W_N is normal and $h_j|_{W_N} \neq 0$, by adjunction [Liu26b, Lemma D.14], the pair

$$\Big(W_N,\ D_\tau := \sum_j \tau_j \operatorname{div}(h_j) + \sum_{i=0}^{2}(1-\mu_i(\tau))L_{Z_i}\Big)$$

is lc near 0.

Since $c := (c_1,\dots,c_n) \in \Pi$, there exist $\tau^{(1)},\dots,\tau^{(m)} \in V(\Pi)$ and $\rho_1,\dots,\rho_m \geq 0$ with $\sum_k \rho_k = 1$ and $c = \sum_k \rho_k \tau^{(k)}$. Since $\tau \to D_\tau$ is affine, $D_c = \sum_k \rho_k D_{\tau^{(k)}}$, so

$$a(F, W_N, D_c) = \sum_k \rho_k a(F, W_N, D_{\tau^{(k)}}) \geq 0$$

for any prime divisor F over a common neighborhood of 0 on which each $(W_N, D_{\tau^{(k)}})$ is lc. Since $D_c = D + \sum_{i=0}^{2}(1-\mu_i)L_{Z_i}$, the first conclusion of (1) holds.

Step 3. In this step we prove the second conclusion of (1). Let F be a prime divisor over W_N with center 0. By Lemma 2.15(1), e_F is the ramification index of F over W_N/G, so $\operatorname{mult}_F \psi \in e_F\mathbb{Z}$ for any G-invariant rational function ψ on W_N.

By the choice of the flag, $Z_2 \in \{p, T\}$. If $Z_2 = T$, then $\chi_T(\widetilde{G}) = \chi_t(G)$, so $\nu = |\chi_t(G)|$. Thus T^ν is a G-invariant regular function vanishing at 0, and $\operatorname{mult}_F L_T = \operatorname{mult}_F T \geq e_F/\nu$. If $Z_2 = p$, then $\chi_p(\widetilde{G})$ is the set of the α with $(g,\alpha) \in \widetilde{G}$, and $\alpha \to \alpha^2$ maps it onto $\chi_x(G)$ with kernel $\{\pm 1\}$, since $(1,-1) \in \widetilde{G}$. Thus $\nu = 2|\chi_x(G)|$, $x^{\nu/2}$ is G-invariant, and $\operatorname{mult}_F L_p = \frac{1}{2}\operatorname{mult}_F x \geq e_F/\nu$. By the first conclusion of (1),

$$a(F, W_N, D) \geq \sum_{i=0}^{2}(1-\mu_i)\operatorname{mult}_F L_{Z_i} \geq (1-\mu_2)\frac{e_F}{\nu}.$$

Step 4. In this step we prove (2), with

$$\begin{aligned} m_0 &:= \max\{1, \alpha_j, \beta_j, \delta_j \mid 1 \leq j \leq n\},\\ \Lambda &:= (m_0+1)^2, \quad \text{and}\\ J &:= \max\{J_0,\ 4M(m_0+1)^3 + 1\}. \end{aligned}$$

Assume that $|G| \leq M$, $N \geq J$, and $H_j - \widehat{H}_j \in \mathfrak{m}^J$ for any j. Since $|\widetilde{G}| = 2|G| \leq 2M$, we have $\xi_k \geq \frac{1}{2M}$ for any $\xi \in N_{\widetilde{G}}$ and any k with $\xi_k > 0$. For $w \in \mathbb{Q}^3$ in the coordinates (p,v,T) and $Z \in \{p,v,T\}$, let $w(Z)$ be the coordinate of w at Z.

The group $\chi_{Z_2}(\widetilde{G})$ is cyclic of order ν, so there exists $\gamma \in \widetilde{G}$ acting on Z_2 by $\exp(2\pi\sqrt{-1}/\nu)$. Adding integers to the phases of γ on Z_1 and Z_0, we obtain $w \in N_{\widetilde{G}}$ with $\mathbf{e}(w) = \gamma$ and

$$\begin{aligned} w(Z_2) &= \frac{1}{\nu},\\ m_0 w(Z_2) < w(Z_1) &\leq m_0 w(Z_2) + 1,\\ m_0\big(w(Z_1) + w(Z_2)\big) < w(Z_0) &\leq m_0\big(w(Z_1) + w(Z_2)\big) + 1. \end{aligned} \tag{B.5}$$

Thus w has positive entries, and $w(Z_2) \leq 1$, $w(Z_1) \leq m_0 + 1$, and $w(Z_0) \leq (m_0+1)^2$. If $w/i \in N_{\widetilde{G}}$ for an integer $i \geq 2$, then $\mathbf{e}(w/i)$ acts on Z_2 by $\exp(2\pi\sqrt{-1}/(i\nu))$, of order $i\nu > \nu$, while $\chi_{Z_2}(\widetilde{G})$ has order ν. Thus w is primitive in $N_{\widetilde{G}}$.

We compute $\mathrm{wt}_w(\widehat{H}_j)$ by (B.4). The monomials of $Z_2^{\delta_j}u_j(Z_2)$ other than $Z_2^{\delta_j}$ have w-weight larger than $\delta_j w(Z_2)$. Since $w(Z_1) > m_0 w(Z_2) \geq \delta_j w(Z_2)$, any monomial of $Z_0^{\alpha_j}Z_1^{\beta_j+1}V_j$ has w-weight larger than $\alpha_j w(Z_0) + \beta_j w(Z_1) + \delta_j w(Z_2)$. Since

$$w(Z_0) > m_0\big(w(Z_1) + w(Z_2)\big) \geq \beta_j w(Z_1) + \delta_j w(Z_2),$$

any monomial of $Z_0^{\alpha_j+1}V_j'$ has w-weight larger than $\alpha_j w(Z_0) + \beta_j w(Z_1) + \delta_j w(Z_2)$. Hence

$$\mathrm{wt}_w(\widehat{H}_j) = \alpha_j w(Z_0) + \beta_j w(Z_1) + \delta_j w(Z_2) \leq m_0\big((m_0+1)^2 + (m_0+1) + 1\big) < (m_0+1)^3. \tag{B.6}$$

Any monomial of the polynomial $H_j - \widehat{H}_j \in \mathfrak{m}^J$ has w-weight at least $\frac{J}{2M} > 2(m_0+1)^3$, so $\mathrm{wt}_w(H_j) = \mathrm{wt}_w(\widehat{H}_j)$. Write $w = (\omega_1, \omega_2, \omega_3)$ in the coordinates (p, v, T) and let

$$\omega_{\min} := \min\{2\omega_1, 2\omega_2, \omega_1+\omega_2, \omega_3\}.$$

Then $\omega_{\min} \geq \frac{1}{2M}$, and, since $m_0 \geq 1$,

$$2(\omega_1+\omega_2) \leq 2\big((m_0+1)^2 + (m_0+1)\big) < 2(m_0+1)^3.$$

Thus

$$N\omega_{\min} \geq \frac{J}{2M} > 2(m_0+1)^3 > \max\{2(\omega_1+\omega_2), \mathrm{wt}_w(H_1), \dots, \mathrm{wt}_w(H_n)\}.$$

By Lemma B.1 and (B.6), there exists a prime divisor F over W_N with center 0 such that

$$\frac{a(F, W_N, D)}{e_F} = \sum_{i=0}^{2} w(Z_i) - \sum_{j=1}^{n} c_j \,\mathrm{wt}_w(\widehat{H}_j) = (1-\mu_0)w(Z_0) + (1-\mu_1)w(Z_1) + \frac{1-\mu_2}{\nu}.$$

Since $1-\mu_0, 1-\mu_1 \geq 0$ and $w(Z_0), w(Z_1) \leq \Lambda$, (2) holds. □

Lemma B.3. *In Set-up 10.3, let $N \geq 2$ be an integer such that G preserves $xy + z^2 + t^N$ up to a character, let G_B be the image of G in $\mathrm{Aut}(\mathbb{A}^2_{x,t})$ under the projection $(x,y,z,t) \to (x,t)$, let $q : W_N \to \mathbb{A}^2_{x,t}$ be the morphism $(x,y,z,t) \to (x,t)$, and let E be a prime divisor over W_N with center 0. Then the following hold.*

(1) There exist a prime divisor E' over $\mathbb{A}^2_{x,t}$ with center 0 and a positive integer r_E such that $\mathrm{mult}_E(q^*\phi) = r_E \,\mathrm{mult}_{E'}\phi$ for any rational function ϕ on $\mathbb{A}^2_{x,t}$.

(2) Let $e'_{E'}$ be the order of the inertia group of E' in G_B. Then

$$\frac{r_E e'_{E'}}{e_E} \in \mathbb{N}^+.$$

Proof. (1) Let ν be the valuation $\phi \to \mathrm{mult}_E(q^*\phi)$ on the rational functions ϕ on $\mathbb{A}^2_{x,t}$. Since the center of E is 0, $\nu \geq 0$ on $\mathbb{C}[x,t]$ and $\nu(x) \geq 1$, so the value group of ν is $r_E\mathbb{Z}$ for a positive integer r_E. The residue field of mult_E has transcendence degree 2 over $\mathbb{C}$. Since q is dominant of relative dimension 1, Abhyankar's inequality [Bou72, Chapter VI, §10, no. 3, Corollary 1 to Theorem 1] for mult_E and ν implies that the residue field of ν has transcendence degree at least 1 over $\mathbb{C}$. By Abhyankar's inequality for the non-trivial valuation ν on the surface $\mathbb{A}^2_{x,t}$, the residue field of ν has transcendence degree at most 1 over $\mathbb{C}$. Thus ν is a divisorial valuation: $\nu = r_E \,\mathrm{mult}_{E'}$ for a prime divisor E' over $\mathbb{A}^2_{x,t}$. Since $\mathrm{mult}_{E'} \geq 0$ on $\mathbb{C}[x,t]$, $\mathrm{mult}_{E'} x > 0$ and $\mathrm{mult}_{E'} t > 0$, the center of E' is 0.

(2) Let $\psi_G : W_N \to W_N/G$ and $\psi_B : \mathbb{A}^2_{x,t} \to \mathbb{A}^2_{x,t}/G_B$ be the quotient morphisms. Since q is equivariant with respect to $G \to G_B$, there exists a morphism $q_G : W_N/G \to \mathbb{A}^2_{x,t}/G_B$ with $q_G \circ \psi_G = \psi_B \circ q$. By Lemma 2.15(1), there exist a prime divisor E_G over W_N/G and a prime divisor E'_B over $\mathbb{A}^2_{x,t}/G_B$ such that $\mathrm{mult}_E(\psi_G^*\phi) = e_E \,\mathrm{mult}_{E_G}\phi$ for any rational function ϕ on W_N/G and $\mathrm{mult}_{E'}(\psi_B^*\phi) = e'_{E'} \,\mathrm{mult}_{E'_B}\phi$ for any rational function ϕ on $\mathbb{A}^2_{x,t}/G_B$. For any rational function ϕ on $\mathbb{A}^2_{x,t}/G_B$, by (1),

$$e_E \,\mathrm{mult}_{E_G}(q_G^*\phi) = \mathrm{mult}_E(\psi_G^* q_G^* \phi) = \mathrm{mult}_E(q^*\psi_B^*\phi) = r_E e'_{E'} \,\mathrm{mult}_{E'_B}\phi,$$

so

$$\operatorname{mult}_{E_G}(q_G^*\phi) = \frac{r_E e'_{E'}}{e_E} \operatorname{mult}_{E'_B} \phi.$$

Since $\operatorname{mult}_{E_G}$ is $\mathbb{Z}$-valued and $\operatorname{mult}_{E'_B}$ takes the value 1, $r_E e'_{E'}/e_E$ is a positive integer. The lemma follows. □

Lemma B.4. *In Set-up 10.3, the following hold.*

(1) (a) Let $\rho \in \mathbb{C}[x,t]$ be semi-invariant with character χ_z/χ_x. Then the automorphism

$$\sigma_\rho : (x,y,z,t) \to (x, y+2z\rho - x\rho^2, z - x\rho, t)$$

of $\mathbb{A}^4$ satisfies $\sigma_\rho^*(xy+z^2) = xy+z^2$ and $\sigma_\rho^* t = t$, commutes with G, and is induced through $(x,y,z) = (p^2, -v^2, pv)$ by the automorphism $(p,v,t) \to (p, v - p\rho(p^2,t), t)$ of $\widetilde{W}_0$.

(b) Any linear automorphism λ of $\mathbb{A}^2_{p,v}$ which either has determinant one and commutes with $\widetilde{G}$ or interchanges p and v induces through $(x,y,z) = (p^2,-v^2,pv)$ a linear automorphism $\widehat{\lambda}$ of $\mathbb{A}^4$ with $\widehat{\lambda}^* t = t$ and $\widehat{\lambda}^*(xy+z^2) = xy+z^2$, and $\widehat{\lambda}^{-1} G \widehat{\lambda}$ is a diagonal group.

(c) For any integer $N \geq 2$ and any $\eta \in \mathbb{C}[x,y,z,t]$, there exist unique polynomials $\overline{\eta}_N$ and $\overline{\eta}_0$ without monomials divisible by xy such that $\overline{\eta}_N - \eta \in (xy+z^2+t^N)$ and $\overline{\eta}_0 - \eta \in (xy+z^2)$, and

$$\overline{\eta}_N - \overline{\eta}_0 \in t^N \mathbb{C}[x,y,z,t].$$

(2) Let $N \geq 2$ be an integer such that G preserves $xy+z^2+t^N$ up to a character, let $\widehat{W} := \operatorname{Spec} \mathbb{C}[[x,y,z,t]]/(xy+z^2+t^N)$, whose closed point we also denote by 0, let $h_1, \dots, h_n \in \mathbb{C}[[x,y,z,t]]$ be semi-invariant, and let $c_1, \dots, c_n$ be positive real numbers. Assume that $(\widehat{W}, \sum_j c_j \operatorname{div}(h_j))$ is klt. Then for any positive integer J there exist semi-invariant polynomials $k_1, \dots, k_n$ without monomials divisible by xy satisfying the following.

(i) For any j, $\chi_{k_j} = \chi_{h_j}$ and $k_j - h_j \in (x,y,z,t)^J + (xy+z^2+t^N)$, and $k_j - h_j \in (x,y,z,t)^J$ if h_j has no monomials divisible by xy.

(ii) $(W_N, \sum_j c_j \operatorname{div}(k_j))$ is klt near 0, and

$$\operatorname{mld}\Big(\Big(W_N \ni 0, \sum_j c_j \operatorname{div}(k_j)\Big)/G\Big) = \operatorname{mld}\Big(\Big(\widehat{W} \ni 0, \sum_j c_j \operatorname{div}(h_j)\Big)/G\Big).$$

Proof. (1)(a) Since G preserves $xy+z^2+t^{N_i}$ up to a character, $\chi_x \chi_y = \chi_z^2$. Hence the functions

$$x, \quad y + 2z\rho - x\rho^2, \quad z - x\rho, \quad \text{and} \quad t$$

are semi-invariant with characters χ_x, χ_y, χ_z, χ_t, that is, σ_ρ commutes with G. Direct expansion shows that

$$x(y + 2z\rho - x\rho^2) + (z - x\rho)^2 = xy + z^2.$$

Since $\sigma_\rho^* x = x$ and $\sigma_\rho^* t = t$, $\sigma_\rho^* \rho = \rho$, and direct substitution shows that

$$\sigma_\rho \circ \sigma_{-\rho} = \sigma_{-\rho} \circ \sigma_\rho = \mathrm{id}_{\mathbb{A}^4}.$$

With $(x,y,z) = (p^2, -v^2, pv)$ and $\rho = \rho(p^2,t)$,

$$\big(p^2, -(v-p\rho)^2, p(v-p\rho)\big) = (x, y+2z\rho - x\rho^2, z - x\rho),$$

so σ_ρ is induced by $(p,v,t) \to (p, v - p\rho(p^2,t), t)$.

(b) Write

$$\lambda(p,v) = (\lambda_{11}p + \lambda_{12}v, \lambda_{21}p + \lambda_{22}v).$$

Expanding $(\lambda_{11}p+\lambda_{12}v)^2$, $-(\lambda_{21}p+\lambda_{22}v)^2$, and $(\lambda_{11}p+\lambda_{12}v)(\lambda_{21}p+\lambda_{22}v)$ in $x=p^2$, $y=-v^2$, $z=pv$, we obtain a linear automorphism $\widehat{\lambda}$ of $\mathbb{A}^4$ with $\widehat{\lambda}^*t=t$ and $\widehat{\lambda}^*(xy+z^2)=(\det\lambda)^2(xy+z^2)$, which is $xy+z^2$ as $(\det\lambda)^2=1$. An element $(g,\alpha)\in\widetilde{G}$ multiplies $(p^2,-v^2,pv)$ by

$$\big(\alpha^2,\chi_z(g)^2\alpha^{-2},\chi_z(g)\big)=\big(\chi_x(g),\chi_y(g),\chi_z(g)\big),$$

that is, (g,α) acts on (x,y,z) as g does. Since $\widetilde{G}\to G$ is surjective, $\widehat{\lambda}$ commutes with G if λ commutes with $\widetilde{G}$. If λ interchanges p and v, then $\widehat{\lambda}(x,y,z,t)=(-y,-x,z,t)$, and for any $g\in G$, $\widehat{\lambda}^{-1}g\widehat{\lambda}$ multiplies (x,y,z,t) by $(\chi_y(g),\chi_x(g),\chi_z(g),\chi_t(g))$, so $\widehat{\lambda}^{-1}G\widehat{\lambda}$ is a diagonal group.

(c) For a monomial $\mu=x^{m_1}y^{m_2}z^{m_3}t^{m_4}$ and $n_0:=\min\{m_1,m_2\}$, let

$$\overline{\mu}_N:=x^{m_1-n_0}y^{m_2-n_0}z^{m_3}t^{m_4}(-z^2-t^N)^{n_0},\quad \overline{\mu}_0:=x^{m_1-n_0}y^{m_2-n_0}z^{m_3}t^{m_4}(-z^2)^{n_0}. \tag{B.7}$$

Then $\overline{\mu}_N-\mu\in(xy+z^2+t^N)$, $\overline{\mu}_0-\mu\in(xy+z^2)$, neither $\overline{\mu}_N$ nor $\overline{\mu}_0$ has a monomial divisible by xy, and $\overline{\mu}_N-\overline{\mu}_0\in t^N\mathbb{C}[x,y,z,t]$. Moreover, any monomial of $\overline{\mu}_N$ has degree at least $m_1+m_2+m_3+m_4$, and if $\chi_x\chi_y=\chi_z^2=\chi_t^N$, then $\overline{\mu}_N$ is semi-invariant with the character of μ. Thus, for $\eta=\sum_\mu a_\mu\mu$ with $a_\mu\in\mathbb{C}$, the polynomials $\overline{\eta}_N:=\sum_\mu a_\mu\overline{\mu}_N$ and $\overline{\eta}_0:=\sum_\mu a_\mu\overline{\mu}_0$ have no monomials divisible by xy, $\overline{\eta}_N-\eta\in(xy+z^2+t^N)$, $\overline{\eta}_0-\eta\in(xy+z^2)$, and $\overline{\eta}_N-\overline{\eta}_0\in t^N\mathbb{C}[x,y,z,t]$. The lexicographically largest monomial (for $x>y>z>t$) of a non-zero element of $(xy+z^2+t^N)$ or of $(xy+z^2)$ is divisible by xy. Hence $\overline{\eta}_N$ and $\overline{\eta}_0$ are unique.

(2) Let $f_N:=xy+z^2+t^N$, $\widehat{R}:=\mathbb{C}[[x,y,z,t]]$, and $\mathfrak{n}:=(x,y,z,t)\widehat{R}$. Since G preserves f_N up to a character, $\chi_x\chi_y=\chi_z^2=\chi_t^N$. For $q=\sum_\mu a_\mu\mu\in\widehat{R}$ with $a_\mu\in\mathbb{C}$, let $\overline{q}:=\sum_\mu a_\mu\overline{\mu}_N$ with $\overline{\mu}_N$ as in (B.7). Since $\overline{\mu}_N-\mu$ is the product of f_N and a polynomial whose monomials have degree at least $\deg\mu-2$, $\overline{q}$ is a well-defined element of $\widehat{R}$ without monomials divisible by xy, $\overline{q}-q\in(f_N)$, $\overline{q}\in\mathfrak{n}^m$ if $q\in\mathfrak{n}^m$, and $\overline{q}$ is semi-invariant with the character of q if q is semi-invariant.

For any j and any positive integer m, let $k_{j,m}$ be the part of degree less than m of $\overline{h}_j$. For any j and m, $k_{j,m}$ is a semi-invariant polynomial with character χ_{h_j} without monomials divisible by xy, and

$$k_{j,m}-\overline{h}_j\in\mathfrak{n}^m. \tag{B.8}$$

Let $D:=\sum_j c_j\operatorname{div}(h_j)$ on $\widehat{W}$ and $D_m:=\sum_j c_j\operatorname{div}(k_{j,m})$ on W_N.

We show that (W_N,D_m) is klt near 0 for $m\gg0$. Since $(\widehat{W},D)$ is klt, there exists a real number $\theta>1$ such that $(\widehat{W},\theta D)$ is lc. Since $\widehat{R}/(f_N)$ is a normal domain and $\operatorname{div}(\overline{h}_j)=\operatorname{div}(h_j)$ on $\widehat{W}$, the pair

$$\Big(\operatorname{Spec}\widehat{R},\operatorname{div}(f_N)+\theta\sum_j c_j\operatorname{div}(\overline{h}_j)\Big)$$

is lc [Liu26b, Lemma D.14]. By Lemma 2.19 for the finite set $\{1\}\cup\{\theta c_j\}_j$ of exponents, there exists a positive integer N_1' such that if $m\geq N_1'$, then by (B.8), the pair

$$\Big(\operatorname{Spec}\widehat{R},\operatorname{div}(f_N)+\theta\sum_j c_j\operatorname{div}(k_{j,m})\Big)$$

is lc, so $(\widehat{W},\theta D_m)$ is lc [Liu26b, Lemma D.14]. Since $(\widehat{W},0)$ is klt and $(\widehat{W},\theta D_m)$ is lc with $\theta>1$, $(\widehat{W},D_m)$ is klt. By [Kaw21, Remark 2.6], (W_N,D_m) is klt near 0 for $m\geq N_1'$.

We show that $\operatorname{mld}((W_N\ni0,D_m)/G)=\operatorname{mld}((\widehat{W}\ni0,D)/G)$ for $m\gg0$. By Lemma 2.12(3), with $B=0$ and $H=G$, prime divisors over W_N with center 0 and prime divisors over $\widehat{W}$ with center the closed point correspond to each other, and corresponding divisors have equal log discrepancies and equal orders e_E.

Let $Y:=W_N/G$, let $\psi_Y:W_N\to Y$ be the quotient morphism, and for $m\geq N_1'$ let B_m be the $\mathbb{R}$-divisor on Y with $K_{W_N}+D_m=\psi_Y^*(K_Y+B_m)$. By Proposition 3.14(1), Y is a klt threefold

and the branch divisor B_{br} of ψ_Y, with $K_{W_N} = \psi_Y^*(K_Y + B_{\mathrm{br}})$, is $\mathbb{Q}$-Cartier. Since

$$B_m = B_{\mathrm{br}} + \sum_j \frac{c_j}{|G|} \operatorname{div}\left(k_{j,m}^{|G|}\right),$$

where $k_{j,m}^{|G|}$ is a G-invariant function, B_m is $\mathbb{R}$-Cartier. For a prime divisor S_1 of W_N with ramification index r over $\psi_Y(S_1)$, the coefficient of B_m along $\psi_Y(S_1)$ is $1-(1-\operatorname{mult}_{S_1} D_m)/r$ (Lemma 2.15(4)), where $r \le |G|$ and $\operatorname{mult}_{S_1} D_m \in \{\sum_j c_j n_j \mid n_j \in \mathbb{N}\}$. Thus the coefficients of the B_m belong to the DCC set

$$\left\{1 - \frac{1-\sum_j c_j n_j}{r} \;\middle|\; n_j \in \mathbb{N},\ 1 \le r \le |G|\right\} \cap \mathbb{R}_{>0},$$

which does not depend on m. By Theorem 2.22(3), there exists a positive integer L_Y independent of m such that for any $m \ge N_1'$ there exists a prime divisor E_m over Y which computes $\operatorname{mld}(Y \ni \psi_Y(0), B_m)$ and satisfies $a(E_m, Y, 0) \le L_Y$. Let E_m' be a prime divisor over W_N with restriction E_m, which exists by Lemma 2.15(3). Then the center of E_m' is 0, and since $B_{\mathrm{br}} \ge 0$, by Lemma 2.15(2),

$$a(E_m', W_N, 0) = e_{E_m'} a(E_m, Y, B_{\mathrm{br}}) \le |G| L_Y.$$

Since (W_N, D_m) is klt, for any j,

$$c_j \operatorname{mult}_{E_m'} k_{j,m} < a(E_m', W_N, 0) \le |G| L_Y.$$

Let $N_2' := \lfloor |G| L_Y / \min_j c_j \rfloor + 1$. If $m \ge \max\{N_1', N_2'\}$, then by (B.8),

$$\operatorname{mult}_{E_m'}(k_{j,m} - \overline{h}_j) \ge m > \operatorname{mult}_{E_m'} k_{j,m},$$

so $\operatorname{mult}_{E_m'} \overline{h}_j = \operatorname{mult}_{E_m'} k_{j,m}$ for any j. Thus, by the correspondence of Lemma 2.12(3) above, $a(E_m', \widehat{W}, D) = a(E_m', W_N, D_m)$. Since E_m computes $\operatorname{mld}(Y \ni \psi_Y(0), B_m)$, Lemma 2.15(2) and the equality of quotient values at the end of Lemma 2.15 give

$$\frac{a(E_m', W_N, D_m)}{e_{E_m'}} = a(E_m, Y, B_m) = \operatorname{mld}(Y \ni \psi_Y(0), B_m) = \operatorname{mld}\big((W_N \ni 0, D_m)/G\big).$$

Hence

$$\operatorname{mld}\big((\widehat{W} \ni 0, D)/G\big) \le \frac{a(E_m', \widehat{W}, D)}{e_{E_m'}} = \operatorname{mld}\big((W_N \ni 0, D_m)/G\big).$$

Conversely, let $\widehat{B}$ be the $\mathbb{R}$-divisor on $\widehat{W}/G$ with $K_{\widehat{W}} + D$ the pullback of $K_{\widehat{W}/G} + \widehat{B}$. Since $(\widehat{W}, D)$ is klt, a log resolution of $(\widehat{W}/G, \widehat{B})$ [dFEM11, Theorems 2.3 and 2.4] on which the preimage of the closed point is a divisor gives a prime divisor E_* over $\widehat{W}$ with center the closed point and $a(E_*, \widehat{W}, D)/e_{E_*} = \operatorname{mld}((\widehat{W} \ni 0, D)/G)$. If $m > \max_j \operatorname{mult}_{E_*} \overline{h}_j$, then by (B.8), $\operatorname{mult}_{E_*} k_{j,m} = \operatorname{mult}_{E_*} \overline{h}_j$ for any j, so

$$\begin{aligned}\operatorname{mld}\big((W_N \ni 0, D_m)/G\big) &\le \frac{a(E_*, \widehat{W}, D_m)}{e_{E_*}} = \frac{a(E_*, \widehat{W}, D)}{e_{E_*}} \\ &= \operatorname{mld}\big((\widehat{W} \ni 0, D)/G\big).\end{aligned}$$

Let $m := \max\{J, N_1', N_2', \max_j \operatorname{mult}_{E_*} \overline{h}_j + 1\}$ and $k_j := k_{j,m}$. Then (ii) holds. By (B.8) and $\overline{h}_j - h_j \in (f_N)$, $k_j - h_j \in \mathfrak{n}^J + (f_N)$. If moreover h_j has no monomials divisible by xy, then $\overline{h}_j = h_j$ by (B.7), so $k_j - h_j \in \mathfrak{n}^J$. This proves (i), and the lemma follows. □

Theorem B.5 is the case of a smooth deck-odd floor with klt adjunction in the proof of Theorem 10.4.

Theorem B.5. *In Set-up 10.3, assume that B^* has a floor F which is smooth and deck-odd, and that the pair (F, B_F) is klt. Then $(a_i)_{i\ge 1}$ is not strictly increasing.*

Proof. Let

$$\delta := x\partial_z - 2z\partial_y,$$

a vector field on $\mathbb{A}^4$. Since $\delta(xy+z^2) = \delta(t) = 0$, δ is tangent to W_N for any integer $N \geq 2$ and to W_0, so δ acts on rational functions on W_N and on W_0. Let $e_{\max} := \max_j e_j$ and $\kappa_{\min} := \min_j \kappa_j$.

Step 1. In this step we bound log discrepancies on W_N from below by log discrepancies on $\mathbb{A}^2_{x,t}$. Let $N \geq 2$ be an integer such that G preserves $xy + z^2 + t^N$ up to a character, and let E be a prime divisor over W_N with center 0. By Lemma B.3(1), there exist a positive integer r_E and a prime divisor E' over $\mathbb{A}^2_{x,t}$ with center 0 such that mult_E of the pullback of any rational function ϕ on $\mathbb{A}^2_{x,t}$ under the projection $W_N \to \mathbb{A}^2_{x,t}$ is $r_E \operatorname{mult}_{E'} \phi$. We show that for any non-zero rational functions $h_1, \dots, h_n$ on W_N with $\delta^{e_j} h_j \neq 0$ for any j,

$$a(E, W_N, 0) - \sum_j \kappa_j \operatorname{mult}_E h_j \geq r_E a(E', \mathbb{A}^2_{x,t}, 0) - \sum_j \kappa_j \operatorname{mult}_E(\delta^{e_j} h_j). \tag{B.9}$$

Since $y = -(z^2 + t^N)/x$ on W_N, the projection $(x, y, z, t) \to (x, z, t)$ is a birational map $W_N \dashrightarrow \mathbb{A}^3_{x,z,t}$. Let ∂'_z be the vector field $\partial/\partial z$ on $\mathbb{A}^3_{x,z,t}$. Through the birational map $W_N \dashrightarrow \mathbb{A}^3_{x,z,t}$, ∂'_z acts on rational functions on W_N and kills the pullbacks of rational functions on $\mathbb{A}^2_{x,t}$. Since δ and $x\partial'_z$ both send x, z, t to $0, x, 0$, we have $\delta = x\partial'_z$ on rational functions on W_N.

Let $T \to \mathbb{A}^2_{x,t}$ be a smooth higher model on which E' is a divisor, and let u, w be local coordinates of T at a general point of E' with $E' = (u = 0)$ near that point. Then

$$dx \wedge dt = \psi du \wedge dw \quad \text{with} \quad \operatorname{mult}_{E'} \psi = a(E', \mathbb{A}^2_{x,t}, 0) - 1.$$

For a non-zero rational 3-form ω on W_N, let $\operatorname{mult}_E \omega := \operatorname{mult}_E(\omega/\omega')$, where ω' generates $\Omega^3_{W'}$ at the generic point of E on a smooth higher model $W' \to W_N$ on which E is a divisor. The Poincaré residue

$$\Omega := \frac{dx \wedge dz \wedge dt}{x} = \frac{\psi}{x} du \wedge dz \wedge dw$$

of $dx \wedge dy \wedge dz \wedge dt/(xy + z^2 + t^N)$ generates ω_{W_N} near 0, so $a(E, W_N, 0) = \operatorname{mult}_E \Omega + 1$. Let

$$\beta_E := \operatorname{mult}_E(du \wedge dz \wedge dw) + 1 - r_E.$$

Since $\operatorname{mult}_E \psi = r_E \operatorname{mult}_{E'} \psi$ and $\operatorname{mult}_E x = r_E \operatorname{mult}_{E'} x$, we obtain

$$a(E, W_N, 0) = \beta_E + r_E a(E', \mathbb{A}^2_{x,t}, 0) - r_E \operatorname{mult}_{E'} x. \tag{B.10}$$

Let f be a non-zero rational function on W_N with $\partial'_z f \neq 0$. We show that

$$\operatorname{mult}_E f - \operatorname{mult}_E(\partial'_z f) \leq \beta_E. \tag{B.11}$$

Let τ be a local equation of E at a general point of E on W', and write $f = \tau^m f_0$ and $u = \tau^{r_E} u_0$ with $m := \operatorname{mult}_E f$ and units f_0, u_0 at the generic point of E, where $\operatorname{mult}_E u = r_E$. Moreover, w is regular at the generic point of E since $\operatorname{mult}_E w = r_E \operatorname{mult}_{E'} w \geq 0$. As $d\tau \wedge d\tau = 0$, the 2-form $df \wedge du$ is τ^{m+r_E-1} times a regular 2-form near a general point of E, so

$$\operatorname{mult}_E(df \wedge du \wedge dw) \geq m + r_E - 1.$$

Since u, w are pullbacks of rational functions on $\mathbb{A}^2_{x,t}$, $\partial'_z u = \partial'_z w = 0$, and we have

$$df \wedge du \wedge dw = \partial'_z f dz \wedge du \wedge dw,$$

whose order along E is $\operatorname{mult}_E(\partial'_z f) + \beta_E + r_E - 1$. Thus (B.11) holds.

Let h be a non-zero rational function on W_N and e a non-negative integer with $\delta^e h \neq 0$. Then $\delta^k h \neq 0$ and $\partial'_z(\delta^k h) = x^{-1}\delta^{k+1} h \neq 0$ for any $0 \leq k < e$, so $\delta = x\partial'_z$ and (B.11) imply

$$\operatorname{mult}_E(\delta^{k+1} h) \geq \operatorname{mult}_E(\delta^k h) - \beta_E + r_E \operatorname{mult}_{E'} x,$$

and hence

$$\operatorname{mult}_E h \leq \operatorname{mult}_E(\delta^e h) + e\big(\beta_E - r_E \operatorname{mult}_{E'} x\big).$$

We apply the last inequality to $h = h_j$ and $e = e_j$, multiply by κ_j and add up over j. Since $\sum_j e_j\kappa_j = 1$,

$$\sum_j \kappa_j \operatorname{mult}_E h_j \le \sum_j \kappa_j \operatorname{mult}_E(\delta^{e_j} h_j) + \beta_E - r_E \operatorname{mult}_{E'} x.$$

Together with (B.10), we obtain (B.9).

Step 2. Assume, to the contrary, that $(a_i)_{i\ge1}$ is strictly increasing. In this step we fix the constants of the proof and bring the marks into a shape adapted to F.

Let $C_0 := (p = v = 0) \subset \widetilde{W}_*$, the fixed locus of ι. Since $(1,-1) \in \widetilde{G}$ acts as ι, the $\widetilde{G}$-semi-invariant equation F of Set-up 10.3 is an eigenvector of ι, and as F is deck-odd, F is anti-invariant under ι (two equations of F differ by a unit, which is not anti-invariant). Then F vanishes on C_0, that is, $C_0 \subset F$. Since (F, B_F) is klt, there exists a real number $\epsilon \in (0,1)$ such that

$$\Big(F, \sum_j (\kappa_j + 3\epsilon)\operatorname{div}(R_j^*|_F) + \epsilon\Big(\sum_j e_j\Big)C_0\Big) \quad \text{is lc.} \tag{B.12}$$

Let Φ be the set of numbers $(m - 1 + \sum_j n_j\kappa_j)/m$ in $(0,1]$ with $m \in \{1,\dots,M\}$ and $n_j \in \mathbb{N}$. Since $\kappa_j > 0$ for any j, Φ is finite. Let $\mathcal{G}$ be the set of diagonal groups G' of order at most M preserving $xy+z^2$ up to a character and acting faithfully on W_0. The set $\mathcal{G}$ is finite, since such G' consist of diagonal matrices whose entries are $M!$-th roots of unity. For $G' \in \mathcal{G}$, $W_0/G' = \widetilde{W}_0/\widetilde{G}'$ for the group $\widetilde{G}'$ attached to G' as in Set-up 10.3, so W_0/G' is a $\mathbb{Q}$-factorial klt threefold. We fix the following constants, in the order listed.

(1) $\Lambda := 2/\epsilon$ and $\Lambda' := 2\Lambda + 1$.
(2) A positive integer L such that for any $G' \in \mathcal{G}$ and any $\mathbb{R}$-divisor B on $X' := W_0/G'$ with $B \in \Phi$ and (X', B) lc near the image $\overline{0}$ of 0, some prime divisor P over X' computes $\operatorname{mld}(X' \ni \overline{0}, B)$ and satisfies $a(P, X', 0) \le L$ (Theorem 2.22(3) and the finiteness of $\mathcal{G}$).
(3) An integer $J_1 > M(e_{\max}\Lambda' + \Lambda)$.
(4) An integer J satisfying the following.
   (a) $J \ge 2J_1$.
   (b) $J > 2ML/\kappa_{\min}$.
   (c) $J - e_{\max}$ is at least the integer of Lemma 2.19 for power series in 2 variables, $n+1$ functions and the exponents $\kappa_j + 3\epsilon$ and $\epsilon\sum_j e_j$, and J is at least the integer of Lemma 2.19 for power series in 3 variables, $3n$ functions and the exponents κ_j and ϵ.

Since $\lim_i N_i = +\infty$, possibly removing finitely many i, we may assume the following for any i.

(N1) $N_i \ge J$.
(N2) $N_i > 2M\Lambda$.
(N3) $N_i \ge N_L(M, \Gamma, L)$ of Proposition 3.14(2)(a).
(N4) N_i is at least the integer of Lemma 2.19 for power series in 4 variables, $3n+1$ functions and the exponents 1, κ_j, and ϵ.

By [Kaw21, Remark 2.6], lc near 0 may be tested on completions, so the integers of Lemma 2.19 above serve for polynomial pairs near 0.

Let $\mathcal{P}$ be the variety of the automorphisms γ of $\mathbb{A}^3_{p,v,t}$ obtained as a linear automorphism α of $\mathbb{A}^2_{p,v}$ followed, in the coordinates after α, by the shear

$$(p, v, t) \to \big(p, v - p\rho(p^2, t), t\big),$$

where α either has determinant one and commutes with $\widetilde{G}$ or interchanges p and v, and ρ is a polynomial of degree less than J semi-invariant with the character χ_z/χ_x of the diagonal group $\widehat{\alpha}^{-1}G\widehat{\alpha}$, for the linear automorphism $\widehat{\alpha}$ of $\mathbb{A}^4$ induced by α (Lemma B.4(1)(b)). By Lemma B.4(1)(a) and (b), such a γ induces an automorphism of $\mathbb{A}^4$ preserving $xy+z^2$ and t, also denoted by γ. Let $Z_J \subset \mathcal{J}_J$ be the irreducible closed subset of Definition 2.5 for $d = J$ and the compatible generic limit $(h_{*,j})_j$. For $\zeta = (\zeta_j)_j \in Z_J$ and $\gamma \in \mathcal{P}$, let $H_j^{\zeta,\gamma}$ be the pullback of

$\gamma^*\zeta_j$ to $\widetilde{W}_0$ modulo $(p,v,t)^J$, and $v^{e_j}R_j^{\zeta,\gamma}$ the sum of the terms of $H_j^{\zeta,\gamma}$ of v-degree at least e_j. Let $\mathcal{Z} \subset Z_J \times \mathcal{P}$ be the set of (ζ,γ) satisfying the following.

(i) $H_j^{\zeta,\gamma} = v^{e_j}R_j^{\zeta,\gamma}$ modulo $(p,v,t)^J$ for any j.

(ii) The pair

$$\Big(\mathbb{A}^2_{p,t}, \sum_j (\kappa_j + 3\epsilon)\operatorname{div}(R_j^{\zeta,\gamma}(p,0,t)) + \epsilon\Big(\sum_j e_j\Big)\operatorname{div}(p)\Big)$$

is lc near 0.

Condition (i) is closed. By the choice of J and Lemma 2.19, condition (ii) depends only on the classes of the $R_j^{\zeta,\gamma}(p,0,t)$ modulo $(p,t)^{J-e_{\max}}$, and the set of tuples of polynomials of degree less than $J-e_{\max}$ for which the pair in (ii) is lc near 0 is constructible (take log resolutions over a stratification of the parameter space), so (ii) is constructible. Thus the image of $\mathcal{Z}$ in Z_J is constructible.

We show that the image of $\mathcal{Z}$ contains the J-jet ζ_* of $(h_{*,j})_j$, which maps to the generic point of Z_J. Since F is anti-invariant under ι and smooth, the linear part ℓ_F of F is a non-zero linear form in p,v, and ℓ_F is a $\widetilde{G}$-eigenvector. Let χ_p and χ_v be the characters of $\widetilde{G}$ on p and v. If $\chi_p \neq \chi_v$, then ℓ_F is a multiple of p or of v. If $\chi_p = \chi_v$, then any linear automorphism of (p,v) of determinant one commutes with $\widetilde{G}$. Thus there exists a linear automorphism α_* of (p,v), either the interchange of p and v or one of determinant one commuting with $\widetilde{G}$, after which ℓ_F is a multiple of v. Possibly replacing (p,v) with $\alpha_*^*(p,v)$, we may assume that ℓ_F is a multiple of v. By the Weierstrass preparation theorem and the anti-invariance under ι, $F = U\cdot(v - p\rho_*(p^2,t))$ for a unit U and a power series ρ_* semi-invariant with the character $\chi_v/\chi_p = \chi_z/\chi_x$ (characters of $\widehat{\alpha}_*^{-1}G\widehat{\alpha}_*$). Let $\gamma_* \in \mathcal{P}$ be the composite of α_* with the shear by the truncation of ρ_* below degree J. Since $H_j^* = F^{e_j}R_j^*$, the pair (ζ_*,γ_*) satisfies (i), and $R_j^{\zeta_*,\gamma_*}(p,0,t)$ is congruent modulo $(p,t)^{J-e_{\max}}$ to $U^{e_j}R_j^*$ restricted to F, written in the coordinates p,t of F. On F the divisor of p is C_0. Hence (B.12) and Lemma 2.19 imply (ii).

Therefore the image of $\mathcal{Z}$ contains a non-empty open subset of Z_J, which contains the J-jets of $(h_{ij})_j$ for infinitely many i by Definition 2.5. Passing to a subsequence, for any i there exists $\gamma_i \in \mathcal{P}$ such that the J-jet of $(h_{ij})_j$ and γ_i satisfy (i) and (ii).

The automorphism γ_i of $\mathbb{A}^4$ preserves W_{N_i} and W_0, and conjugates G to a diagonal group (Lemma B.4(1)(a) and (b)). We replace G with $\gamma_i^{-1}G\gamma_i$ and h_{ij} with the representative of $\gamma_i^*h_{ij}$ on W_{N_i} without monomials divisible by xy. The values a_i do not change. There exist only finitely many diagonal groups of order at most M, each with finitely many characters, so possibly passing to a subsequence, the new data satisfy Set-up 10.3, with the old c_{ij} and κ_j.

By Lemma B.4(1)(c), the new H_{ij} differs from the pullback of $\gamma_i^*h_{ij}$ (before the replacement) by an element of $t^{N_i}\mathbb{C}[p,v,t] \subset (p,v,t)^J$. Let $v^{e_j}R_{ij}$ be the sum of the terms of H_{ij} of v-degree at least e_j. By (i),

$$H_{ij} - v^{e_j}R_{ij} \in (p,v,t)^J, \quad \text{and} \quad H_{ij} - v^{e_j}R_{ij} \text{ has } v\text{-degree less than } e_j. \tag{B.13}$$

Since $N_i \geq J$, the polynomial $R_{ij}(p,0,t)$ is congruent to $R_j^{\zeta,\gamma_i}(p,0,t)$ modulo $(p,t)^{J-e_{\max}}$ for the J-jet ζ of $(h_{ij})_j$ before the replacement. Thus (ii) implies

$$\Big(\mathbb{A}^2_{p,t}, \sum_j (\kappa_j + 3\epsilon)\operatorname{div}(R_{ij}(p,0,t)) + \epsilon\Big(\sum_j e_j\Big)\operatorname{div}(p)\Big) \quad \text{is lc near 0.} \tag{B.14}$$

In particular, $R_{ij}(p,0,t) \neq 0$ for any j.

The polynomial H_{ij} is ι-invariant and $\widetilde{G}$-semi-invariant with the character of h_{ij}, and ι and $\widetilde{G}$ act diagonally on monomials in p,v,t. Hence $v^{e_j}R_{ij}$ is ι-invariant and $\widetilde{G}$-semi-invariant with the character of h_{ij}. Thus R_{ij} is $\widetilde{G}$-semi-invariant and $\iota^*R_{ij} = (-1)^{e_j}R_{ij}$, and both $v^{e_j}R_{ij}$ and R_{ij}^2 are pullbacks of G-semi-invariant regular functions on W_0. Let $\widehat{h}_{ij}$ be the regular function

on W_0 with pullback $v^{e_j}R_{ij}$, and r_{ij} the polynomial without monomials divisible by xy whose restriction to W_0 has pullback R_{ij}^2. Let

$$g_{ij} := \delta^{e_j} h_{ij} \in \mathbb{C}[x,y,z,t] \quad \text{and} \quad f_{ij}(x,t) := g_{ij}(x,0,0,t).$$

The derivations δ of $\mathbb{C}[x,y,z,t]$ and $p\partial_v$ of $\mathbb{C}[p,v,t]$ are compatible with

$$(x,y,z,t) = (p^2, -v^2, pv, t),$$

since both send x,y,z,t to $0,-2z,x,0$. As $(p\partial_v)^{e_j} = p^{e_j}\partial_v^{e_j}$, (B.13) implies

$$g_{ij}(p^2,-v^2,pv,t) = p^{e_j}\partial_v^{e_j} H_{ij} \quad \text{and} \quad f_{ij}(p^2,t) = e_j! p^{e_j} R_{ij}(p,0,t). \tag{B.15}$$

In particular, $f_{ij} \neq 0$, and g_{ij} does not vanish identically on W_0.

Step 3. In this step we show that for any i,

$$\Big(W_{N_i}, \sum_j \kappa_j \operatorname{div}(h_{ij}) + \epsilon \sum_j \operatorname{div}(g_{ij}) + \epsilon \sum_j \operatorname{div}(r_{ij})\Big) \quad \text{is lc near } 0, \tag{B.16}$$

and deduce that for any prime divisor E over W_{N_i} with center 0 and $a(E, W_{N_i}, D_i) \leq 2e_E$,

$$\operatorname{mult}_E g_{ij} \leq \Lambda e_E \quad \text{and} \quad \operatorname{mult}_E r_{ij} \leq \Lambda e_E \quad \text{for any } j. \tag{B.17}$$

By (B.14) and (B.15), v divides none of R_{ij} and $g_{ij}(p^2,-v^2,pv,t)$, and the restriction of

$$\sum_j \kappa_j \operatorname{div}(R_{ij}) + \epsilon \sum_j \operatorname{div}\left(g_{ij}(p^2,-v^2,pv,t)\right) + \epsilon \sum_j \operatorname{div}(R_{ij}^2)$$

to $(v=0) \cong \mathbb{A}^2_{p,t}$ is

$$\sum_j (\kappa_j + 3\epsilon) \operatorname{div}(R_{ij}(p,0,t)) + \epsilon\Big(\sum_j e_j\Big) \operatorname{div}(p).$$

By (B.14) and inversion of adjunction [Kaw07, Theorem],

$$\Big(\widetilde{W}_0, \operatorname{div}(v) + \sum_j \kappa_j \operatorname{div}(R_{ij}) + \epsilon \sum_j \operatorname{div}\left(g_{ij}(p^2,-v^2,pv,t)\right) + \epsilon \sum_j \operatorname{div}(R_{ij}^2)\Big) \quad \text{is lc near } 0. \tag{B.18}$$

Since $\sum_j e_j\kappa_j = 1$,

$$\operatorname{div}(v) + \sum_j \kappa_j \operatorname{div}(R_{ij}) = \sum_j \kappa_j \operatorname{div}(v^{e_j} R_{ij}).$$

By (B.13), the choice of J and Lemma 2.19, we may replace each $v^{e_j}R_{ij}$ with H_{ij}: the pair

$$\Big(\widetilde{W}_0, \sum_j \kappa_j \operatorname{div}(H_{ij}) + \epsilon \sum_j \operatorname{div}\left(g_{ij}(p^2,-v^2,pv,t)\right) + \epsilon \sum_j \operatorname{div}(R_{ij}^2)\Big)$$

is lc near 0. By (B.15), the boundary of the last pair is the pullback of

$$D_i^{\mathrm{aux}} := \sum_j \kappa_j \operatorname{div}(h_{ij}|_{W_0}) + \epsilon \sum_j \operatorname{div}(g_{ij}|_{W_0}) + \epsilon \sum_j \operatorname{div}(r_{ij}|_{W_0}).$$

Since π_0 is étale in codimension 1, $(W_0, D_i^{\mathrm{aux}})$ is lc near 0. By [dFEM11, Proposition 2.11] and [Liu26b, Lemma D.14], the pair on $\mathbb{A}^4$ with boundary

$$\operatorname{div}(xy+z^2) + \sum_j \kappa_j \operatorname{div}(h_{ij}) + \epsilon \sum_j \operatorname{div}(g_{ij}) + \epsilon \sum_j \operatorname{div}(r_{ij})$$

is lc near 0. Since $xy + z^2 + t^{N_i} \equiv xy + z^2$ modulo $(x,y,z,t)^{N_i}$, the choice of N_i and Lemma 2.19 imply that the pair on $\mathbb{A}^4$ with boundary

$$\operatorname{div}(xy+z^2+t^{N_i}) + \sum_j \kappa_j \operatorname{div}(h_{ij}) + \epsilon \sum_j \operatorname{div}(g_{ij}) + \epsilon \sum_j \operatorname{div}(r_{ij})$$

is lc near 0. In particular, $xy + z^2 + t^{N_i}$ divides none of h_{ij}, g_{ij}, and r_{ij}, and by [dFEM11, Proposition 2.11] and [Liu26b, Lemma D.14], (B.16) holds.

Let E be a prime divisor over W_{N_i} with center 0 and $a(E, W_{N_i}, D_i) \le 2e_E$. Since $c_{ij} \le \kappa_j$ and $\operatorname{mult}_E h_{ij} \ge 0$, (B.16) implies

$$\epsilon \sum_j \big(\operatorname{mult}_E g_{ij} + \operatorname{mult}_E r_{ij} \big) \le a\Big(E, W_{N_i}, \sum_j \kappa_j \operatorname{div}(h_{ij})\Big) \le a(E, W_{N_i}, D_i) \le 2e_E.$$

Since all the orders on the left are non-negative and $\Lambda = 2/\epsilon$, (B.17) follows.

Step 4. In this step we compare the values a_i with quotient values on $\mathbb{A}^2_{x,t}$. Let

$$\Delta_i := \sum_j \kappa_j \operatorname{div}(f_{ij}) \text{ on } \mathbb{A}^2_{x,t}, \quad \mu_i := \operatorname{mld}((\mathbb{A}^2_{x,t} \ni 0, \Delta_i)/G_B),$$

where G_B is the image of G in the automorphism group of $\mathbb{A}^2_{x,t}$. Since $\chi_x \chi_y = \chi_z^2$, the derivation δ is semi-invariant with the character χ_x/χ_z, so the g_{ij} and the f_{ij} are semi-invariant and Δ_i is G_B-invariant. Let $\pi_B : \mathbb{A}^2_{p,t} \to \mathbb{A}^2_{x,t}$ be the double cover $x = p^2$. By (B.15) and $\sum_j e_j \kappa_j = 1$, we have

$$\pi_B^* \Delta_i = \operatorname{div}(p) + \sum_j \kappa_j \operatorname{div}(R_{ij}(p, 0, t)),$$

and $R(\pi_B) = \operatorname{div}(p)$. Thus

$$K_{\mathbb{A}^2_{p,t}} + \sum_j \kappa_j \operatorname{div}(R_{ij}(p, 0, t)) = \pi_B^*(K_{\mathbb{A}^2_{x,t}} + \Delta_i). \tag{B.19}$$

By (B.14), the left side of (B.19) is the log canonical divisor of a klt pair near 0, as the coefficients of $\sum_j \kappa_j \operatorname{div}(R_{ij}(p, 0, t))$ may be multiplied by $1 + 3\epsilon/\max_j \kappa_j$ keeping lc. Hence $(\mathbb{A}^2_{x,t}, \Delta_i)$ is klt near 0.

We show that for any prime divisor E over W_{N_i} with center 0 such that $\operatorname{mult}_E g_{ij} = \operatorname{mult}_E f_{ij}$ for any j,

$$a(E, W_{N_i}, D_i) \ge e_E \mu_i. \tag{B.20}$$

Let r_E and E' be as in **Step 1**, and $e'_{E'}$ the order of the inertia group of E' in G_B. By **Step 3**, the $g_{ij} = \delta^{e_j} h_{ij}$ do not vanish identically on W_{N_i}. Since $c_{ij} \le \kappa_j$, (B.9) and the equalities

$$\operatorname{mult}_E g_{ij} = \operatorname{mult}_E f_{ij} = r_E \operatorname{mult}_{E'} f_{ij}$$

imply

$$\begin{aligned} a(E, W_{N_i}, D_i) &\ge a\Big(E, W_{N_i}, \sum_j \kappa_j \operatorname{div}(h_{ij})\Big) \\ &\ge r_E \Big(a(E', \mathbb{A}^2_{x,t}, 0) - \sum_j \kappa_j \operatorname{mult}_{E'} f_{ij} \Big) = r_E a(E', \mathbb{A}^2_{x,t}, \Delta_i). \end{aligned}$$

By Lemma B.3(2), $r_E e'_{E'}/e_E$ is a positive integer, and $a(E', \mathbb{A}^2_{x,t}, \Delta_i) \ge e'_{E'} \mu_i > 0$ by Definition 2.2. Hence

$$a(E, W_{N_i}, D_i) \ge e_E \cdot \frac{r_E e'_{E'}}{e_E} \mu_i \ge e_E \mu_i.$$

Step 5. In this step we show that

$$a_i \le \mu_i + \frac{L}{\kappa_{\min}} \sigma_i, \quad \text{where} \quad \sigma_i := \sum_j (\kappa_j - c_{ij}) \ge 0. \tag{B.21}$$

Let $\widehat{D}_i := \sum_j \kappa_j \operatorname{div}(\widehat{h}_{ij})$ on W_0 and $\lambda_i := \operatorname{mld}((W_0 \ni 0, \widehat{D}_i)/G)$. Let $X := W_0/G = \widetilde{W}_0/\widetilde{G}$, and let $\overline{0} \in X$ be the image of 0. Let $\Delta_X \ge 0$ be the branch divisor, so that K_{W_0} is the pullback

of $K_X + \Delta_X$. Let B_i be the $\mathbb{R}$-divisor on a neighborhood of $\overline{0}$ in X such that $K_{W_0} + \widehat{D}_i$ is the pullback of $K_X + B_i$ near 0, so that $\lambda_i = \mathrm{mld}(X \ni \overline{0}, B_i)$. The pullback of $\widehat{D}_i$ to $\widetilde{W}_0$ is

$$\sum_j \kappa_j \operatorname{div}(v^{e_j} R_{ij}) = \operatorname{div}(v) + \sum_j \kappa_j \operatorname{div}(R_{ij}),$$

which is at most the boundary in (B.18). Since π_0 is étale in codimension 1, $(W_0, \widehat{D}_i)$ is lc near 0 and $\lambda_i \geq 0$.

We show that $\lambda_i \leq \mu_i$. Let $S \subset X$ be the image of $(v = 0)$. As $\operatorname{div}(v)$ has coefficient one, S has coefficient one in B_i. The image of $\widetilde{G}$ in the automorphism group of $\mathbb{A}^2_{p,t} = (v = 0)$ contains $\iota|_{(v=0)} : (p, t) \to (-p, t)$, and the quotient by $\iota|_{(v=0)}$ is π_B, so the normalization of S is $S^\nu = \mathbb{A}^2_{x,t}/G_B$. Let B_{S^ν} be the different of $B_i - S$ on S^ν. The pullback of $K_X + B_i$ to $\widetilde{W}_0$ near 0 is

$$K_{\widetilde{W}_0} + \operatorname{div}(v) + \sum_j \kappa_j \operatorname{div}(R_{ij}).$$

The different is defined by the Poincaré residue [Kol13, Definition 4.2]. For a positive integer q as in [Kol13, Definition 4.2(4)] and a local section s of $\omega_X^{[q]}(qB_i)$, the residue along $(v = 0)$ of the pullback of s is a non-zero constant times the pullback of the residue of s along S. Thus the pullback of $K_{S^\nu} + B_{S^\nu}$ to $\mathbb{A}^2_{p,t}$ is

$$K_{\mathbb{A}^2_{p,t}} + \sum_j \kappa_j \operatorname{div}(R_{ij}(p, 0, t)).$$

By (B.19), (S^ν, B_{S^ν}) is the quotient of $(\mathbb{A}^2_{x,t}, \Delta_i)$ by G_B, so $\mathrm{mld}(S^\nu \ni \overline{0}, B_{S^\nu}) = \mu_i$. By adjunction [Kol+92, Theorem 17.2], $\lambda_i \leq \mathrm{mld}(S^\nu \ni \overline{0}, B_{S^\nu}) = \mu_i$.

A prime divisor of X over which $W_0 \to X$ has ramification index $m \leq M$ has coefficient $(m - 1 + \sum_j n_j \kappa_j)/m$ in B_i, where $n_j \in \mathbb{N}$ is the order of $\widehat{h}_{ij}$ along a prime divisor of W_0 over it, and $\sum_j n_j \kappa_j \leq 1$ since $(W_0, \widehat{D}_i)$ is lc. Thus $B_i \in \Phi$, and by the choice of L there exists a prime divisor P over X which computes λ_i and satisfies $a(P, X, 0) \leq L$. Let $\widehat{E}$ be a prime divisor over W_0 whose restriction to X is P, so that $\operatorname{mult}_{\widehat{E}}$ of the pullback of any rational function ϕ on X is $e_{\widehat{E}} \operatorname{mult}_P \phi$. Then

$$a(\widehat{E}, W_0, \widehat{D}_i) = e_{\widehat{E}} \lambda_i \quad \text{and} \quad a(\widehat{E}, W_0, 0) = e_{\widehat{E}} a(P, X, \Delta_X) \leq e_{\widehat{E}} a(P, X, 0) \leq e_{\widehat{E}} L.$$

Since $\lambda_i \geq 0$, for any j,

$$\kappa_{\min} \operatorname{mult}_{\widehat{E}} \widehat{h}_{ij} \leq \sum_k \kappa_k \operatorname{mult}_{\widehat{E}} \widehat{h}_{ik} = a(\widehat{E}, W_0, 0) - e_{\widehat{E}} \lambda_i \leq e_{\widehat{E}} L.$$

Since π_0 is finite and surjective, there exists a prime divisor $\widetilde{E}$ over $\widetilde{W}_0$ whose restriction to W_0 is $\widehat{E}$. Let $m_{\widetilde{E}}$ be the ramification index of $\widetilde{E}$ over $\widehat{E}$, so that $\operatorname{mult}_{\widetilde{E}}$ of the pullback of any rational function ϕ on W_0 is $m_{\widetilde{E}} \operatorname{mult}_{\widehat{E}} \phi$. Let $\xi := \operatorname{mult}_{\widetilde{E}} / m_{\widetilde{E}}$. As $\widehat{E}$ has center 0,

$$\xi(p) = \frac{1}{2} \operatorname{mult}_{\widehat{E}} x \geq \frac{1}{2}, \quad \xi(v) = \frac{1}{2} \operatorname{mult}_{\widehat{E}} y \geq \frac{1}{2}, \quad \text{and} \quad \xi(t) \geq 1,$$

so $\xi \geq J/2$ on $(p, v, t)^J$. On the other hand,

$$\xi(v^{e_j} R_{ij}) = \operatorname{mult}_{\widehat{E}} \widehat{h}_{ij} \leq \frac{e_{\widehat{E}} L}{\kappa_{\min}} \leq \frac{ML}{\kappa_{\min}} < \frac{J}{2}.$$

By (B.13), $\xi(H_{ij}) = \xi(v^{e_j} R_{ij})$, that is,

$$\operatorname{mult}_{\widehat{E}}(h_{ij}|_{W_0}) = \operatorname{mult}_{\widehat{E}} \widehat{h}_{ij} \quad \text{for any } j.$$

Since $a(\widehat{E}, W_0, 0)/e_{\widehat{E}} \leq L$ and $N_i \geq N_L(M, \Gamma, L)$, Proposition 3.14(2)(a) implies that the birational transform E of $\widehat{E}$ under the inverse $(x, y, z, t) \to (x, y - t^{N_i}/x, z, t)$ of the birational

map φ of Proposition 3.14 for $N = N_i$ is a prime divisor over W_{N_i} with center 0, $e_E = e_{\widehat{E}}$, $a(E, W_{N_i}, 0) = a(\widehat{E}, W_0, 0)$, and $\operatorname{mult}_E h_{ij} = \operatorname{mult}_{\widehat{E}}(h_{ij}|_{W_0})$ for any j. Therefore

$$\begin{aligned} e_E a_i \le a(E, W_{N_i}, D_i) &= a(\widehat{E}, W_0, 0) - \sum_j c_{ij} \operatorname{mult}_{\widehat{E}} \widehat{h}_{ij} \\ &= e_{\widehat{E}} \lambda_i + \sum_j (\kappa_j - c_{ij}) \operatorname{mult}_{\widehat{E}} \widehat{h}_{ij} \\ &\le e_E \Big(\lambda_i + \frac{L}{\kappa_{\min}} \sigma_i \Big). \end{aligned}$$

Since $\lambda_i \le \mu_i$, (B.21) follows.

Step 6. In this step we show that, possibly removing finitely many i,

$$\operatorname{mult}_{E_i} y > \Lambda' e_{E_i} \quad \text{for any } i \text{ and any computing lift } E_i \text{ of } a_i. \tag{B.22}$$

By (B.13),

$$H_{ij}^2 - v^{2e_j} R_{ij}^2 = (H_{ij} - v^{e_j} R_{ij})(H_{ij} + v^{e_j} R_{ij}) \in (p, v, t)^J,$$

and it is ι-invariant. An ι-invariant monomial $p^{m_1} v^{m_2} t^{m_3}$ has $m_1 + m_2$ even, so it is the pullback of a product of $(m_1 + m_2)/2 + m_3 \ge (m_1 + m_2 + m_3)/2$ elements of $\{x, -y, z, t\}$. Since $J \ge 2J_1$, the function $h_{ij}^2 - (-y)^{e_j} r_{ij}$ on W_0, whose pullback is $H_{ij}^2 - v^{2e_j} R_{ij}^2$, is the restriction of a polynomial $\eta_{ij} \in (x, y, z, t)^{J_1}$. Since W_0 is the prime divisor $(xy + z^2 = 0)$ of $\mathbb{A}^4$,

$$h_{ij}^2 - (-y)^{e_j} r_{ij} - \eta_{ij} = (xy + z^2) Q_{ij}$$

for a polynomial Q_{ij}. As $xy + z^2 = -t^{N_i}$ on W_{N_i} and $N_i \ge J \ge J_1$, for any prime divisor E over W_{N_i} with center 0,

$$\operatorname{mult}_E \left(h_{ij}^2 - (-y)^{e_j} r_{ij} \right) = \operatorname{mult}_E \left(\eta_{ij} - t^{N_i} Q_{ij} \right) \ge J_1 > M(e_{\max} \Lambda' + \Lambda), \tag{B.23}$$

where the order of 0 is $+\infty$.

Let E_i be a computing lift of a_i with $\operatorname{mult}_{E_i} y \le \Lambda' e_{E_i}$. Since $a_i \le 2$, (B.17) holds for $E = E_i$, so

$$\operatorname{mult}_{E_i} \left((-y)^{e_j} r_{ij} \right) \le e_j \Lambda' e_{E_i} + \Lambda e_{E_i} \le (e_{\max} \Lambda' + \Lambda) e_{E_i}.$$

As $e_{E_i} \le |G| \le M$, (B.23) implies

$$2 \operatorname{mult}_{E_i} h_{ij} = \operatorname{mult}_{E_i} \left((-y)^{e_j} r_{ij} \right) \le (e_{\max} \Lambda' + \Lambda) e_{E_i} \quad \text{for any } j.$$

Moreover, since $c_{ij} \le 1$,

$$\frac{a(E_i, W_{N_i}, 0)}{e_{E_i}} = a_i + \sum_j c_{ij} \frac{\operatorname{mult}_{E_i} h_{ij}}{e_{E_i}} \le 2 + \frac{n}{2} (e_{\max} \Lambda' + \Lambda).$$

Since W_{N_i} is Gorenstein, $a(E_i, W_{N_i}, 0)$ is an integer, so $a(E_i, W_{N_i}, 0)/e_{E_i}$ and the $\operatorname{mult}_{E_i} h_{ij}/e_{E_i}$ belong to the finite set

$$\frac{1}{M!} \mathbb{N} \cap \Big[0, 2 + \frac{n}{2} (e_{\max} \Lambda' + \Lambda) \Big].$$

Assume that such E_i exist for infinitely many i. Passing to a subsequence, we may assume that $a(E_i, W_{N_i}, 0)/e_{E_i} = m_0$ and $\operatorname{mult}_{E_i} h_{ij}/e_{E_i} = m_j$ do not depend on i. Then

$$a_i = m_0 - \sum_j c_{ij} m_j$$

is decreasing in i, since $m_j \ge 0$ and the c_{ij} are increasing in i. This contradicts the strict increase of the a_i, and (B.22) follows.

Step 7. In this step we show $a_i \ge \mu_i$ and conclude. Let E_i be a computing lift of a_i. Then $\operatorname{mult}_{E_i} z > \Lambda e_{E_i}$: otherwise $\operatorname{mult}_{E_i}(z^2) \le 2\Lambda e_{E_i}$, whereas (B.22) and $N_i > 2M\Lambda$ imply

$$\operatorname{mult}_{E_i}(xy) \ge \operatorname{mult}_{E_i} y > (2\Lambda + 1) e_{E_i} > 2\Lambda e_{E_i} \quad \text{and} \quad \operatorname{mult}_{E_i}(t^{N_i}) \ge N_i > 2M\Lambda \ge 2\Lambda e_{E_i},$$

so z^2 would be the unique term of $xy+z^2+t^{N_i}=0$ of minimal order along E_i, which is impossible.

Each monomial of $g_{ij}-f_{ij}(x,t)$ is divisible by y or z, so

$$\operatorname{mult}_{E_i}(g_{ij}-f_{ij})\geq\min\{\operatorname{mult}_{E_i}y,\operatorname{mult}_{E_i}z\}>\Lambda e_{E_i}.$$

By (B.17), $\operatorname{mult}_{E_i}g_{ij}\leq\Lambda e_{E_i}$, hence $\operatorname{mult}_{E_i}g_{ij}=\operatorname{mult}_{E_i}f_{ij}$ for any j. By (B.20),

$$a_i=\frac{a(E_i,W_{N_i},D_i)}{e_{E_i}}\geq\mu_i. \tag{B.24}$$

We show that the μ_i satisfy the ACC. Let B_i' be the $\mathbb{R}$-divisor on a neighborhood of $\overline{0}$ in $\mathbb{A}^2_{x,t}/G_B$ such that $K_{\mathbb{A}^2_{x,t}}+\Delta_i$ is the pullback of $K_{\mathbb{A}^2_{x,t}/G_B}+B_i'$ near 0, so that

$$\mu_i=\operatorname{mld}(\mathbb{A}^2_{x,t}/G_B\ni\overline{0},B_i').$$

Since $(\mathbb{A}^2_{x,t},\Delta_i)$ is klt near 0, the computation of the coefficients in **Step 5** shows $B_i'\in\Phi$. As Φ is finite and $(\mathbb{A}^2_{x,t}/G_B,B_i')$ is klt near $\overline{0}$, the μ_i satisfy the ACC by Theorem 2.21(1).

Passing to a subsequence, we may assume that the μ_i are decreasing with limit μ_*. By (B.24), $a_i\geq\mu_i\geq\mu_*$ for any i. Since $c_{ij}\to\kappa_j$, we have $\sigma_i\to0$, so (B.21) implies $\lim_i a_i\leq\mu_*$. Thus $a_1\geq\mu_*\geq\lim_i a_i$, which contradicts the strict increase of the a_i. The theorem follows. □

Theorem B.6. *In Set-up 10.3, assume that B^* has a floor F which is smooth and deck-even, and that (F,B_F) is klt. Then $(a_i)_{i\geq1}$ is not strictly increasing.*

Proof of Theorem B.6. Suppose that $(a_i)_{i\geq1}$ is strictly increasing. We divide the proof into eight steps.

Step 1. In this step we write F as a graph and fix the constants.

The involution ι is the action of $(1,-1)\in\widetilde{G}$, and π_0 is the quotient by ι and is étale outside the t-axis. Thus $\widetilde{W}_0/\widetilde{G}=W_0/G$, and for any G-invariant $\mathbb{R}$-divisor D on W_0 we have $(\widetilde{W}_0,\pi_0^*D)/\widetilde{G}=(W_0,D)/G=(W_0/G,D_G)$, where $K_{W_0}+D$ is the pullback of $K_{W_0/G}+D_G$. Hence $(\widetilde{W}_0,\pi_0^*D)$ is lc (resp. klt) near 0 if and only if (W_0,D) is, and

$$\operatorname{mld}((\widetilde{W}_0\ni0,\pi_0^*D)/\widetilde{G})=\operatorname{mld}((W_0\ni0,D)/G). \tag{B.25}$$

Since F is deck-even and $\widetilde{G}$-stable, F has an equation which is ι-invariant and $\widetilde{G}$-semi-invariant. We denote it also by F. The linear part of F is ι-invariant, hence a multiple of t, and it is non-zero as F is smooth. By the Weierstrass preparation theorem, $F=U\cdot(t+\phi_*)$ for a unit U and a unique $\phi_*\in\Bbbk[[p,v]]$ with $\phi_*(0)=0$. By the uniqueness, ϕ_* is ι-invariant and semi-invariant with character χ_t, so $\phi_*=\phi_*^0(p^2,-v^2,pv)$ for a series $\phi_*^0(x,y,z)$ semi-invariant with character χ_t. Possibly replacing R_j^* with $U^{e_j}R_j^*$, we may assume that $F=T_*:=t+\phi_*$. We expand $R_j^*=\sum_{k\geq0}\varrho_{jk}(p,v)T_*^k$ in the coordinates (p,v,T_*), so that $H_j^*=T_*^{e_j}R_j^*$ and $R_j^*|_{T_*=0}=\varrho_{j0}\neq0$.

Let $e_{\max}:=\max_j e_j$. Then $e_{\max}\geq1$ as $\sum_j e_j\kappa_j=1$. We fix the following constants.

(i) Since (F,B_F) is klt, we may fix a real number $\eta\in(0,1)$ such that

$$\Big(\operatorname{Spec}\Bbbk[[p,v]],\ \sum_j(\kappa_j+2\eta)\operatorname{div}(\varrho_{j0})+2\eta\Big(\sum_j e_j\Big)\operatorname{div}(p)\Big)\text{ is lc.} \tag{B.26}$$

(ii) Let $\gamma_2:=2/\eta$ and $\gamma_1:=\gamma_2+1$.

(iii) Let $\mathcal{C}$ be the finite set consisting of 1 and the numbers $(m-1+b)/m$ with $m\in\{1,\dots,M\}$ and $b\leq1$ a finite sum of elements of $\{\kappa_1,\dots,\kappa_n\}$. Since W_0/G is a klt threefold (Proposition 3.14(1)) and is $\mathbb{Q}$-factorial, being the quotient $\widetilde{W}_0/\widetilde{G}$ of $\widetilde{W}_0\cong\mathbb{A}^3$, by Theorem 2.22(3) there exists a real number $\gamma_3>0$ satisfying the following: for any $\mathbb{R}$-divisor B on W_0/G with coefficients in $\mathcal{C}$ such that $(W_0/G,B)$ is lc near the image $\overline{0}$ of 0, there exists a prime divisor $\overline{P}$ over W_0/G with center $\overline{0}$ such that

$$a(\overline{P},W_0/G,B)=\operatorname{mld}(W_0/G\ni\overline{0},B)\quad\text{and}\quad a(\overline{P},W_0/G,0)\leq\gamma_3.$$

(iv) Let J' be an integer with

$$J' > \max\{M(e_{\max}\gamma_1 + \gamma_2), M\gamma_3/\min_j \kappa_j\}.$$

(v) Let Λ_0 be the finite set consisting of 1, η, $2\eta\sum_j e_j$, and the numbers κ_j, $\kappa_j + 2\eta$. Let L be an integer such that the conclusion of Lemma 2.19 holds with L for power series in 2, 3, or 4 variables, at most $3n+1$ functions and exponents in Λ_0.

(vi) Let $J := \max\{2J', L + e_{\max}\}$.

Step 2. In this step we approximate T_* and the factorizations $H_j^* = T_*^{e_j} R_j^*$ on the members. We show that, possibly discarding finitely many i, for any i there exist a polynomial $\phi_i(x,y,z)$ of degree less than J, semi-invariant with character χ_t, with $\phi_i(0) = 0$, and polynomials $R'_{ij}(p,v,T)$ of degree less than J such that for any j

$$H_{ij}\big(p,v,T - \phi_i(p^2,-v^2,pv)\big) - T^{e_j}R'_{ij} \in (p,v,T)^J, \tag{B.27}$$

and

$$\Big(\mathbb{A}^2_{p,v},\ \sum_j(\kappa_j + 2\eta)\operatorname{div}(R'_{ij}(p,v,0)) + 2\eta\Big(\sum_j e_j\Big)\operatorname{div}(p)\Big) \text{ is lc near } 0. \tag{B.28}$$

Let $\overline{\phi}$ be the sum of the terms of ϕ_*^0 of degree less than J, and $\overline{R}_j(p,v,T)$ the sum of the terms of $\sum_k \varrho_{jk}(p,v)T^k$ of degree less than J. Since $\phi_* - \overline{\phi}(p^2,-v^2,pv) \in (p,v)^J$, the tuple $(h_{*,j})_j$ with $\overline{\phi}$ and $(\overline{R}_j)_j$ satisfies (B.27) over $\Bbbk$, and since $\overline{R}_j(p,v,0) - \varrho_{j0} \in (p,v)^J$ and $J \geq L$, it satisfies (B.28) over $\Bbbk$ by (B.26) and Lemma 2.19.

Let $Z_J \subset \mathcal{J}_J$ be the irreducible closed subset of Definition 2.5 for the tuples $(h_{ij})_j$. Let $\mathcal{P}$ be the affine space of the coefficients of the data $(\phi,(R'_j)_j)$, where $\phi(x,y,z)$ and $R'_j(p,v,T)$ are polynomials of degree less than J and ϕ is semi-invariant with character χ_t with $\phi(0) = 0$, and $\mathcal{Z} \subset Z_J \times \mathcal{P}$ the closed subscheme defined by (B.27), a system of polynomial equations in the coefficients of the data and of the J-jets. The $\Bbbk$-point $((h_{*,j})_j, \overline{\phi}, (\overline{R}_j)_j)$ of $\mathcal{Z}$ maps to a point $\zeta \in \mathcal{Z}$ over the generic point of Z_J. Let $\mathcal{Z}'$ be the closure of ζ. The pair of (B.28) over ζ becomes lc after the extension of the base field to $\Bbbk$, hence it is lc, and a log resolution of it spreads out to log resolutions of the pairs of (B.28) over a non-empty open subset $\mathcal{U} \subset \mathcal{Z}'$, on which each prime divisor has the log discrepancy of the corresponding prime divisor over ζ. By Chevalley's theorem, the image of $\mathcal{U}$ in Z_J is constructible and contains the generic point of Z_J, so it contains a non-empty open subset Z° of Z_J. For all but finitely many i, the J-jet of $(h_{ij})_j$ belongs to Z°, the fiber of $\mathcal{U}$ over it is a non-empty scheme of finite type over $\mathbb{C}$, and a $\mathbb{C}$-point of the fiber provides ϕ_i and $(R'_{ij})_j$.

Step 3. In this step we define the functions T_i, r_{ij}, g_{ij}, and $\widehat{g}_{ij}$ on the members. Possibly discarding finitely many i, we may assume that ϕ_i and $(R'_{ij})_j$ of **Step 2** exist for any i and that

$$N_i \geq \max\{J + e_{\max}, N_{\gamma_3}\},$$

where N_{γ_3} is the integer $N_\lambda(M,\Gamma,\lambda)$ of Proposition 3.14(2)(a) for $\lambda = \gamma_3$. Fix i, and let $T_i := t + \phi_i(x,y,z)$. Then T_i is semi-invariant with character χ_t, (x,y,z,T_i) are coordinates on $\mathbb{A}^4$, (p,v,T_i) are coordinates on $\widetilde{W}_0$, ι fixes T_i, and the ideals (p,v,T_i) and (p,v,t) coincide. Division by $T_i^{e_j}$ in $\mathbb{C}[p,v][T_i]$ writes

$$H_{ij} = T_i^{e_j}R_{ij} + Q_{ij}, \quad \deg_{T_i} Q_{ij} < e_j. \tag{B.29}$$

By the uniqueness of division, R_{ij} and Q_{ij} are ι-invariant and semi-invariant. The monomials of Q_{ij} and of $T_i^{e_j}(R_{ij} - R'_{ij})$ are monomials of $H_{ij} - T_i^{e_j}R'_{ij}$, so (B.27) implies $Q_{ij} \in (p,v,T_i)^J$ and $R_{ij} - R'_{ij} \in (p,v,T_i)^{J-e_j}$. An ι-invariant monomial $p^{k_1}v^{k_2}T_i^{k_3}$ has k_1+k_2 even and is, up to sign, a monomial of degree $(k_1+k_2)/2 + k_3$ in $p^2, -v^2, pv, T_i$. As $J \geq 2J'$, there exist semi-invariant polynomials $r_{ij}(x,y,z,T)$ and $q_{ij}(x,y,z,T) \in (x,y,z,T)^{J'}$ with $R_{ij} = r_{ij}(p^2,-v^2,pv,T_i)$ and $Q_{ij} = q_{ij}(p^2,-v^2,pv,T_i)$. We regard r_{ij} and q_{ij} as polynomials in x,y,z,t through $T = T_i$. Then

$h_{ij} - T_i^{e_j} r_{ij} - q_{ij}$ vanishes on W_0, so it belongs to $(xy + z^2)$. As $xy + z^2 = -t^{N_i}$ on W_{N_i} and $N_i \geq J'$, on both W_0 and W_{N_i}

$$h_{ij} - T_i^{e_j} r_{ij} \in (x, y, z, t)^{J'}. \tag{B.30}$$

Let $\delta_i := x\partial_t - N_i t^{N_i-1}\partial_y$, where ∂_t, ∂_y are the partial derivatives on $\mathbb{C}[x, y, z, t]$. Since $\delta_i(xy + z^2 + t^{N_i}) = 0$ and $\chi_x\chi_y = \chi_z^2 = \chi_t^{N_i}$, δ_i is a G-semi-invariant derivation preserving the ideal of W_{N_i}. Let

$$g_{ij} := \delta_i^{e_j} h_{ij} \quad \text{and} \quad \widehat{g}_{ij} := x^{e_j}\partial_t^{e_j} h_{ij},$$

both semi-invariant polynomials. In the expansion of $\delta_i^{e_j}$ into words in $x\partial_t$ and $-N_i t^{N_i-1}\partial_y$, any word containing $-N_i t^{N_i-1}\partial_y$ maps h_{ij} into $(t^{N_i-e_j})$: after the factor t^{N_i-1} appears, at most $e_j - 1$ derivations follow, each lowering the exponent of t by at most one. Thus $g_{ij} - \widehat{g}_{ij} \in (t^{N_i-e_j})$. Let $\widehat{G}_{ij}$ be the pullback of $\widehat{g}_{ij}$ to $\widetilde{W}_0$. As ϕ_i does not depend on t, the partial derivative in t for the coordinates (p, v, t) is ∂_{T_i} for (p, v, T_i), and by (B.29) and $\deg_{T_i} Q_{ij} < e_j$, $\widehat{G}_{ij} = p^{2e_j}\partial_{T_i}^{e_j}(T_i^{e_j} R_{ij})$. Hence

$$\widehat{G}_{ij}|_{T_i=0} = e_j! p^{2e_j} R_{ij}|_{T_i=0}, \tag{B.31}$$

and $\widehat{G}_{ij} - e_j! p^{2e_j} R_{ij}$ is T_i times an ι-invariant polynomial. Thus $\widehat{g}_{ij} - e_j! x^{e_j} r_{ij} \in (T_i, xy + z^2)$, and on W_{N_i}

$$g_{ij} - e_j! x^{e_j} r_{ij} \in (T_i, t^{N_i-e_j}). \tag{B.32}$$

Step 4. In this step we prove that

$$\Big(W_{N_i},\ \sum_j \kappa_j \operatorname{div}(h_{ij}) + \eta \sum_j \operatorname{div}(g_{ij}) + \eta \sum_j \operatorname{div}(r_{ij})\Big) \text{ is lc near } 0, \tag{B.33}$$

and that $(W_0, \widehat{D}_i)$ is lc near 0, where

$$\widehat{D}_i := \sum_j \kappa_j \operatorname{div}(T_i^{e_j} r_{ij}|_{W_0}).$$

Since $R_{ij}|_{T_i=0} - R'_{ij}(p, v, 0) \in (p, v)^{J-e_{\max}}$ and $J - e_{\max} \geq L$, (B.28) implies by Lemma 2.19 (a pair on $\mathbb{A}^n$ is lc near 0 if and only if its pullback to $\operatorname{Spec}\mathbb{C}[[x_1, \ldots, x_n]]$ is lc, by Lemma 2.12(2)) that

$$\Big(\mathbb{A}^2_{p,v},\ \sum_j (\kappa_j + 2\eta) \operatorname{div}(R_{ij}|_{T_i=0}) + 2\eta\Big(\sum_j e_j\Big) \operatorname{div}(p)\Big) \text{ is lc near } 0; \tag{B.34}$$

in particular $R_{ij}|_{T_i=0} \neq 0$. Let

$$\widehat{B}_i := \pi_0^* \widehat{D}_i = \sum_j \kappa_j \operatorname{div}(T_i^{e_j} R_{ij}).$$

As $\sum_j e_j\kappa_j = 1$,

$$\widehat{\Delta}_i := \widehat{B}_i + \eta \sum_j \operatorname{div}(\widehat{G}_{ij}) + \eta \sum_j \operatorname{div}(R_{ij}) = \operatorname{div}(T_i) + \sum_j (\kappa_j + \eta) \operatorname{div}(R_{ij}) + \eta \sum_j \operatorname{div}(\widehat{G}_{ij}),$$

and by (B.31) no R_{ij} or $\widehat{G}_{ij}$ is divisible by T_i, and the restriction of the last two sums to $(T_i = 0) \cong \mathbb{A}^2_{p,v}$ is the boundary of (B.34). By inversion of adjunction [Kaw07, Theorem], the pair $(\widetilde{W}_0, \widehat{\Delta}_i)$ is lc near 0. In particular $(\widetilde{W}_0, \widehat{B}_i)$ is lc near 0, and so is $(W_0, \widehat{D}_i)$ by (B.25). Let G_{ij} be the pullback of g_{ij} to $\widetilde{W}_0$, and let

$$\Delta_i := \sum_j \kappa_j \operatorname{div}(h_{ij}) + \eta \sum_j \operatorname{div}(g_{ij}) + \eta \sum_j \operatorname{div}(r_{ij})$$

on $\mathbb{A}^4$, so that

$$\pi_0^*(\Delta_i|_{W_0}) = \sum_j \kappa_j \operatorname{div}(H_{ij}) + \eta \sum_j \operatorname{div}(G_{ij}) + \eta \sum_j \operatorname{div}(R_{ij}).$$

We have $H_{ij} - T_i^{e_j} R_{ij} = Q_{ij} \in (p, v, T_i)^J$ and

$$G_{ij} - \widehat{G}_{ij} \in (t^{N_i - e_j}) \subset (p, v, T_i)^{N_i - e_j}.$$

As $J \geq L$ and $N_i - e_{\max} \geq L$, Lemma 2.19 on the completion of $\widetilde{W}_0$ at 0 implies that $(\widetilde{W}_0, \pi_0^*(\Delta_i|_{W_0}))$ is lc near 0. By (B.25), the pair $(W_0, \Delta_i|_{W_0})$ is lc near 0. By [dFEM11, Proposition 2.11] and [Liu26b, Lemma D.14], $(\mathbb{A}^4, \operatorname{div}(xy + z^2) + \Delta_i)$ is lc near 0. As $(xy + z^2 + t^{N_i}) - (xy + z^2) \in (x, y, z, t)^{N_i}$ and $N_i \geq L$, Lemma 2.19 implies that $(\mathbb{A}^4, \operatorname{div}(xy + z^2 + t^{N_i}) + \Delta_i)$ is lc near 0. In particular none of h_{ij}, g_{ij}, r_{ij} vanishes on W_{N_i}, and [dFEM11, Proposition 2.11] and [Liu26b, Lemma D.14] imply (B.33).

Let $E = E_i$ be a prime divisor over W_{N_i} with center 0 and $a(E, W_{N_i}, D_i) = a_i e_E$, where $e_E = |I_G(E)|$. Since $c_{ij} \leq \kappa_j$ and $a_i \leq 2$, (B.33) implies

$$\eta \sum_j (\operatorname{mult}_E g_{ij} + \operatorname{mult}_E r_{ij}) \leq a(E, W_{N_i}, D_i) - \sum_j (\kappa_j - c_{ij}) \operatorname{mult}_E h_{ij} \leq 2e_E,$$

hence

$$\operatorname{mult}_E g_{ij} \leq \gamma_2 e_E \quad \text{and} \quad \operatorname{mult}_E r_{ij} \leq \gamma_2 e_E \quad \text{for any } j. \tag{B.35}$$

Step 5. In this step we prove a lower bound for $a(E, W_{N_i}, D_i)$, with E as in **Step 4**. The projection $(x, y, z, t) \to (x, z, t)$ maps W_{N_i} birationally onto $\mathbb{A}^3_{x,z,t}$, as $y = -(z^2 + t^{N_i})/x$ on W_{N_i}. Since the center of E is 0, $\operatorname{mult}_E x, \operatorname{mult}_E z > 0$. The restriction of mult_E to the rational functions of x, z is non-trivial and its residue field has transcendence degree at least $2 - 1 = 1$, so by Abhyankar's inequality there exist a positive integer e_0 and a prime divisor E_0 over $\mathbb{A}^2_{x,z}$ with center 0 such that $\operatorname{mult}_E f(x, z) = e_0 \operatorname{mult}_{E_0} f$ for any rational function f on $\mathbb{A}^2_{x,z}$. Let $Y_0 \to \mathbb{A}^2_{x,z}$ be a projective birational morphism from a smooth surface on which E_0 is a prime divisor, and let u, s be rational functions on $\mathbb{A}^2_{x,z}$ which are local parameters of Y_0 at the generic point of E_0 with $E_0 = (u = 0)$ there. For a rational 3-form ω on W_{N_i}, $\operatorname{mult}_E \omega$ denotes the order of ω along E on a smooth variety on which E is a divisor. Let

$$a_{\text{prod}}(E) := 1 + \operatorname{mult}_E(du \wedge dt \wedge ds) \quad \text{and} \quad D_{\text{rel}} := a_{\text{prod}}(E) - e_0.$$

The form $dx \wedge dz \wedge dt/x$ generates the canonical sheaf of W_{N_i} near 0, and

$$\operatorname{mult}_{E_0}(dx \wedge dz) = a(E_0, \mathbb{A}^2_{x,z}, 0) - 1,$$

as $\operatorname{mult}_{E_0}(du \wedge ds) = 0$. Hence

$$a(E, W_{N_i}, 0) = D_{\text{rel}} + e_0 a(E_0, \mathbb{A}^2_{x,z}, 0) - e_0 \operatorname{mult}_{E_0} x. \tag{B.36}$$

Let $\partial_i := x^{-1}\delta_i$, the rational vector field on W_{N_i} which is ∂_t in the coordinates (x, z, t) of $\mathbb{A}^3_{x,z,t}$. It kills x and z, hence the rational functions u and s of x, z, and $\partial_i t = 1$. Let f be a rational function on W_{N_i} with $\partial_i f \neq 0$. Then $df \wedge du \wedge ds = \partial_i f dt \wedge du \wedge ds$. Let τ be a local equation of E at the generic point of E on a smooth variety, and write $f = \tau^{\operatorname{mult}_E f} f_1$ and $u = \tau^{e_0} u_1$ with units f_1, u_1. Then $df \wedge du$ is $\tau^{\operatorname{mult}_E f + e_0 - 1}$ times a regular 2-form, and s is regular at the generic point of E as $\operatorname{mult}_E s = e_0 \operatorname{mult}_{E_0} s \geq 0$. Thus $\operatorname{mult}_E(\partial_i f) + a_{\text{prod}}(E) - 1 \geq \operatorname{mult}_E f + e_0 - 1$, that is,

$$\operatorname{mult}_E f - \operatorname{mult}_E(\partial_i f) \leq D_{\text{rel}}.$$

As $\delta_i = x\partial_i$ and $\operatorname{mult}_E x = e_0 \operatorname{mult}_{E_0} x$, $\operatorname{mult}_E f \leq \operatorname{mult}_E(\delta_i f) + D_{\text{rel}} - e_0 \operatorname{mult}_{E_0} x$ whenever $\delta_i f \neq 0$. Since g_{ij} does not vanish on W_{N_i} (**Step 4**), neither does $\delta_i^k h_{ij}$ for $k \leq e_j$, and e_j applications of the last inequality imply

$$\operatorname{mult}_E h_{ij} \leq \operatorname{mult}_E g_{ij} + e_j(D_{\text{rel}} - e_0 \operatorname{mult}_{E_0} x).$$

By (B.36), $\sum_j e_j\kappa_j = 1$ and $c_{ij} \le \kappa_j$,

$$a(E, W_{N_i}, D_i) \ge a\Big(E, W_{N_i}, \sum_j \kappa_j \operatorname{div}(h_{ij})\Big) \ge e_0 a(E_0, \mathbb{A}^2_{x,z}, 0) - \sum_j \kappa_j \operatorname{mult}_E g_{ij}. \tag{B.37}$$

Step 6. In this step we show that, possibly discarding finitely many i, the prime divisor E_i of **Step 4** satisfies $\operatorname{mult}_{E_i} T_i > \gamma_1 e_{E_i}$. Suppose that $\operatorname{mult}_{E_i} T_i \le \gamma_1 e_{E_i}$ for infinitely many i. Possibly passing to a subsequence, we may assume that $\operatorname{mult}_{E_i} T_i \le \gamma_1 e_{E_i}$ for any i. Any coordinate has positive order along E_i and $e_{E_i} \le M$, so by (B.30)

$$\operatorname{mult}_{E_i}(h_{ij} - T_i^{e_j} r_{ij}) \ge J' > M(e_{\max}\gamma_1 + \gamma_2) \ge (e_j\gamma_1 + \gamma_2)e_{E_i},$$

while by (B.35)

$$\operatorname{mult}_{E_i}(T_i^{e_j} r_{ij}) \le (e_j\gamma_1 + \gamma_2)e_{E_i}.$$

Thus

$$\operatorname{mult}_{E_i} h_{ij} = \operatorname{mult}_{E_i}(T_i^{e_j} r_{ij}) \le (e_{\max}\gamma_1 + \gamma_2)e_{E_i},$$

and

$$a(E_i, W_{N_i}, 0) = a_i e_{E_i} + \sum_j c_{ij} \operatorname{mult}_{E_i} h_{ij} \le \big(2 + n(e_{\max}\gamma_1 + \gamma_2)\big)e_{E_i}.$$

Since W_{N_i} is Gorenstein, $a(E_i, W_{N_i}, 0) \in \mathbb{Z}$, and $e_{E_i} \in \{1, \dots, M\}$. Hence the vectors $(a(E_i, W_{N_i}, 0)/e_{E_i}, (\operatorname{mult}_{E_i} h_{ij}/e_{E_i})_j)$ belong to a finite set. Possibly passing to a subsequence, the vector is a constant $(\lambda_0, (\lambda_j)_j)$ with $\lambda_j \ge 0$, and $a_i = \lambda_0 - \sum_j c_{ij}\lambda_j$ is decreasing as the c_{ij} are increasing, a contradiction.

Step 7. In this step we prove $a_i \ge \alpha_i$, where α_i is the quotient value of a klt pair on a fixed surface. Let E be as in **Step 4**. Then $\operatorname{mult}_E T_i > \gamma_1 e_E$ by **Step 6**. By (B.32), $g_{ij} - e_j! x^{e_j} r_{ij}$ has order along E at least $\min\{\operatorname{mult}_E T_i, N_i - e_j\} > \gamma_2 e_E$, as $N_i - e_{\max} \ge J' > M\gamma_2$. By (B.35),

$$\operatorname{mult}_E g_{ij} = e_j \operatorname{mult}_E x + \operatorname{mult}_E r_{ij} \quad \text{for any } j. \tag{B.38}$$

As $e_j > 0$ for some j, $\operatorname{mult}_E x \le \gamma_2 e_E$.

Let $y' := -z^2/x$, so that $y' = y + t^{N_i}/x$ on W_{N_i}, and let $S := (xy' + z^2 = 0) \subset \mathbb{A}^3_{x,y',z}$, a normal surface with origin 0_S which the projection $(x, y', z) \to (x, z)$ maps birationally onto $\mathbb{A}^2_{x,z}$. Since $\operatorname{mult}_E y > 0$ and $\operatorname{mult}_E(t^{N_i}/x) \ge N_i - \gamma_2 e_E > \gamma_2 e_E$, the prime divisor E_0 of **Step 5** is a prime divisor over S with center 0_S. Since $\operatorname{mult}_E T_i > \gamma_1 e_E > \operatorname{mult}_E r_{ij}$, $\operatorname{mult}_E r_{ij} = \operatorname{mult}_E(r_{ij}(x, y, z, 0))$, where $r_{ij}(x, y, z, 0)$ is the polynomial $r_{ij}(x, y, z, T)$ at $T = 0$. Replacing y with y' changes $r_{ij}(x, y, z, 0)$ by t^{N_i}/x times a polynomial in x, y, y', z, whose order along E exceeds $\gamma_2 e_E$. Thus, with the semi-invariant regular function $f_{ij} := r_{ij}(x, y', z, 0)$ on S,

$$\operatorname{mult}_E r_{ij} = \operatorname{mult}_E f_{ij} = e_0 \operatorname{mult}_{E_0} f_{ij}.$$

The canonical sheaf of S is generated near 0_S by $dx \wedge dz/x$, so

$$a(E_0, S, 0) = a(E_0, \mathbb{A}^2_{x,z}, 0) - \operatorname{mult}_{E_0} x.$$

Let $B_{S,i} := \sum_j \kappa_j \operatorname{div}(f_{ij})$. Substituting (B.38) into (B.37) and using $\sum_j e_j\kappa_j = 1$, we obtain

$$a(E, W_{N_i}, D_i) \ge e_0 a(E_0, S, B_{S,i}). \tag{B.39}$$

Let G_S be the image of G in $\operatorname{Aut}(S)$, acting on x, y', z by χ_x, χ_y, χ_z, let $\pi_S : S \to S/G_S$ be the quotient, and let $B_{S/G_S,i}$ be the $\mathbb{R}$-divisor with $K_S + B_{S,i} = \pi_S^*(K_{S/G_S} + B_{S/G_S,i})$. Let

$$\alpha_i := \operatorname{mld}(S/G_S \ni \pi_S(0_S), B_{S/G_S,i}) = \operatorname{mld}((S \ni 0_S, B_{S,i})/G_S).$$

The double cover $\mathbb{A}^2_{p,v} \to S$, $(x, y', z) = (p^2, -v^2, pv)$, is étale outside 0 and pulls f_{ij} back to $R_{ij}|_{T_i=0}$, hence $B_{S,i}$ back to $B_{pv,i} := \sum_j \kappa_j \operatorname{div}(R_{ij}|_{T_i=0})$. For any prime divisor V over $\mathbb{A}^2_{p,v}$ whose center contains 0,

$$a(V, \mathbb{A}^2_{p,v}, B_{pv,i}) \ge a(V, \mathbb{A}^2_{p,v}, \Delta) + 2\eta \operatorname{mult}_V \sum_j \operatorname{div}(R_{ij}|_{T_i=0})$$

for the boundary Δ of (B.34), which is positive if $\operatorname{mult}_V \sum_j \operatorname{div}(R_{ij}|_{T_i=0}) > 0$, and otherwise $a(V, \mathbb{A}^2_{p,v}, B_{pv,i}) = a(V, \mathbb{A}^2_{p,v}, 0) > 0$. Hence $(\mathbb{A}^2_{p,v}, B_{pv,i})$ is klt near 0. Thus $(S, B_{S,i})$ is klt near 0_S and $\alpha_i > 0$.

Let $e_{E_0} := |I_{G_S}(E_0)|$, and let $\overline{E}_0$ and $\overline{E}$ be the images of E_0 and E on S/G_S and W_{N_i}/G. The map $(x, y, z, t) \to (x, y', z)$ is a G-equivariant dominant rational map $W_{N_i} \dashrightarrow S$, under which the pullback of any rational function f on S has order $e_0 \operatorname{mult}_{E_0} f$ along E, and it induces a rational map $W_{N_i}/G \dashrightarrow S/G_S$. For a rational function $\overline{f}$ on S/G_S, the order along E of the pullback of $\overline{f}$ to W_{N_i} is $e_0 e_{E_0} \operatorname{mult}_{\overline{E}_0} \overline{f}$, computed through S, and e_E times the order along $\overline{E}$ of the pullback of $\overline{f}$ to W_{N_i}/G, computed through W_{N_i}/G. Since $\operatorname{mult}_{\overline{E}}$ is $\mathbb{Z}$-valued and $\operatorname{mult}_{\overline{E}_0}$ takes the value 1, $e_0 e_{E_0}/e_E$ is a positive integer, and dividing (B.39) by e_E yields

$$a_i = \frac{a(E, W_{N_i}, D_i)}{e_E} \geq \frac{e_0 e_{E_0}}{e_E} \cdot \frac{a(E_0, S, B_{S,i})}{e_{E_0}} \geq \alpha_i. \tag{B.40}$$

Step 8. In this step we bound a_i from above and conclude. Let $\beta_i := \operatorname{mld}((W_0 \ni 0, \widehat{D}_i)/G)$, with $\widehat{D}_i$ as in **Step 4**.

First we show $\beta_i \leq \alpha_i$. Let $\widetilde{G}_{pv} \subset \mathrm{GL}_2(\mathbb{C})$ be the image of $\widetilde{G}$ acting on (p, v), and G_T the kernel of $\widetilde{G} \to \widetilde{G}_{pv}$, which acts faithfully on T_i through the n_T-th roots of unity, $n_T := |G_T|$. Since R_{ij} is semi-invariant and not divisible by T_i, and G_T fixes p and v, the character of R_{ij} is trivial on G_T, so $R_{ij} = R^\flat_{ij}(p, v, T_i^{n_T})$ for a polynomial $R^\flat_{ij}$. Let $\psi : \widetilde{W}_0 \to \mathbb{A}^3_{p,v,T'}$, $T' = T_i^{n_T}$, be the quotient by G_T. As

$$K_{\widetilde{W}_0} = \psi^* K_{\mathbb{A}^3_{p,v,T'}} + (n_T - 1) \operatorname{div}(T_i) \quad \text{and} \quad \psi^* \operatorname{div}(T') = n_T \operatorname{div}(T_i),$$

the branch divisor of ψ is absorbed by the coefficient-one component $\operatorname{div}(T_i)$ of $\widehat{B}_i$:

$$K_{\widetilde{W}_0} + \widehat{B}_i = \psi^*(K_{\mathbb{A}^3_{p,v,T'}} + B'_i), \quad \text{where} \quad B'_i := \operatorname{div}(T') + \sum_j \kappa_j \operatorname{div}(R^\flat_{ij}).$$

By (B.25) and the last equality, on $W_0/G = \mathbb{A}^3_{p,v,T'}/\widetilde{G}_{pv}$ we have

$$(W_0, \widehat{D}_i)/G = (\widetilde{W}_0, \widehat{B}_i)/\widetilde{G} = (\mathbb{A}^3_{p,v,T'}, B'_i)/\widetilde{G}_{pv},$$

so $\beta_i = \operatorname{mld}((\mathbb{A}^3_{p,v,T'} \ni 0, B'_i)/\widetilde{G}_{pv})$. Since $\mathbb{A}^2_{p,v} \to S$ is the quotient by $-1 \in \widetilde{G}_{pv}$, étale outside 0, and $\widetilde{G}_{pv}/\{\pm 1\} = G_S$, we have $(S, B_{S,i})/G_S = (\mathbb{A}^2_{p,v}, B_{pv,i})/\widetilde{G}_{pv}$ on S/G_S, and

$$\alpha_i = \operatorname{mld}\big((\mathbb{A}^2_{p,v} \ni 0, B_{pv,i})/\widetilde{G}_{pv}\big).$$

Let P be a prime divisor over $\mathbb{A}^2_{p,v}$ with center 0 computing α_i, and $e_P := |I_{\widetilde{G}_{pv}}(P)|$. As G_T fixes T', $\widetilde{G}$ acts on T' through a character $\chi_{T'}$ of $\widetilde{G}_{pv}$. Let $I := I_{\widetilde{G}_{pv}}(P)$, let $\mathfrak{m}_P$ be the maximal ideal of the valuation ring of mult_P, let λ be the character of I on $\mathfrak{m}_P/\mathfrak{m}_P^2$, and let q be an integer with $\chi_{T'}|_I = \lambda^q$. Let ℓ be an integer with $\ell > \max_j \operatorname{mult}_P(R_{ij}|_{T_i=0})$ and $\ell \equiv q \pmod{e_P}$. By [Liu26b, Lemma 3.1], there exists a Gauss extension P' of P over $\mathbb{A}^3_{p,v,T'}$ such that $\operatorname{mult}_{P'} T' = \ell$, $I_{\widetilde{G}_{pv}}(P') = I$, and

$$a(P', \mathbb{A}^3_{p,v,T'}, \operatorname{div}(T')) = a(P, \mathbb{A}^2_{p,v}, 0).$$

As $\operatorname{mult}_{P'}(\sum_k a_k(p, v) T'^k) = \min_k(\operatorname{mult}_P a_k + k\ell)$ and $R^\flat_{ij}|_{T'=0} = R_{ij}|_{T_i=0}$, we have

$$\operatorname{mult}_{P'} R^\flat_{ij} = \operatorname{mult}_P(R_{ij}|_{T_i=0}).$$

Thus

$$\beta_i \leq \frac{a(P', \mathbb{A}^3_{p,v,T'}, B'_i)}{e_P} = \frac{a(P, \mathbb{A}^2_{p,v}, B_{pv,i})}{e_P} = \alpha_i.$$

Next we bound a_i by β_i. By **Step 4**, $(W_0, \widehat{D}_i)$ is lc near 0, so any prime divisor on W_0 through 0 has coefficient $b \leq 1$ in $\widehat{D}_i$, a finite sum of elements of $\{\kappa_1, \dots, \kappa_n\}$, and the image of the

prime divisor on W_0/G has coefficient $(m-1+b)/m$ with $m \le |G| \le M$. Thus the sum of the components through $\overline{0}$ of the boundary of the quotient pair of $(W_0, \widehat{D}_i)$, which is $\mathbb{R}$-Cartier as $W_0/G = \widetilde{W}_0/\widetilde{G}$ is $\mathbb{Q}$-factorial and whose minimal log discrepancy at $\overline{0}$ is β_i, has coefficients in $\mathcal{C}$, and by the choice of γ_3 there exists a prime divisor $\overline{P}$ over W_0/G with center $\overline{0}$ computing β_i with $a(\overline{P}, W_0/G, 0) \le \gamma_3$. Let P_0 be a prime divisor over W_0 whose image on W_0/G is $\overline{P}$. Then $a(P_0, W_0, \widehat{D}_i) = \beta_i e_{P_0} \ge 0$, and $a(P_0, W_0, 0)/e_{P_0} \le a(\overline{P}, W_0/G, 0) \le \gamma_3$ since the boundary Δ_G of the quotient pair $(W_0/G, \Delta_G)$ of $(W_0, 0)$ satisfies $\Delta_G \ge 0$. Let $w_{ij} := \operatorname{mult}_{P_0}(T_i^{e_j} r_{ij})/e_{P_0}$. Then

$$\min_k \kappa_k \cdot w_{ij} \le \sum_k \kappa_k w_{ik} = \frac{a(P_0, W_0, 0) - a(P_0, W_0, \widehat{D}_i)}{e_{P_0}} \le \gamma_3,$$

so $w_{ij} e_{P_0} \le M\gamma_3 / \min_k \kappa_k < J'$. By (B.30) on W_0, $\operatorname{mult}_{P_0}(h_{ij}|_{W_0}) = w_{ij} e_{P_0}$. Then

$$\frac{a(P_0, W_0, D_{0,i})}{e_{P_0}} = \frac{a(P_0, W_0, \widehat{D}_i)}{e_{P_0}} + \sum_j (\kappa_j - c_{ij}) w_{ij} = \beta_i + \sum_j (\kappa_j - c_{ij}) w_{ij}.$$

As $N_i \ge N_{\gamma_3}$, by Proposition 3.14(2)(a) the birational transform P_{N_i} of P_0 under the inverse of the birational map $\varphi : W_{N_i} \dashrightarrow W_0$ of Proposition 3.14 is a prime divisor over W_{N_i} with center 0, $e_{P_{N_i}} = e_{P_0}$, and $a(P_{N_i}, W_{N_i}, D_i) = a(P_0, W_0, D_{0,i})$. Hence, by $\beta_i \le \alpha_i$,

$$a_i \le \beta_i + \sum_j (\kappa_j - c_{ij}) w_{ij} \le \alpha_i + \frac{\gamma_3}{\min_k \kappa_k} \sum_j (\kappa_j - c_{ij}). \tag{B.41}$$

Finally, by [Liu26b, Lemma 3.2(2)] with $k = 1$ and the trivial character, as $\alpha_i > 0$,

$$\alpha_i = \operatorname{mld}\big((S \times \mathbb{A}^1_w \ni (0_S, 0), B_{S,i} \times \mathbb{A}^1_w + \operatorname{div}(w))/G_S\big).$$

The right hand side is the minimal log discrepancy at the image of $(0_S, 0)$ of a pair on the fixed threefold $(S \times \mathbb{A}^1_w)/G_S$, and, after replacing the boundary by the sum of the components of the boundary through the image of $(0_S, 0)$, which is $\mathbb{R}$-Cartier as $(S \times \mathbb{A}^1_w)/G_S = \mathbb{A}^3_{p,v,w}/\widetilde{G}_{pv}$ is $\mathbb{Q}$-factorial, the coefficients of the boundary belong to $\mathcal{C}$ as $(S, B_{S,i})$ is klt near 0_S. By Theorem 2.22(1), the set $\{\alpha_i\}_i$ satisfies the ACC. Possibly passing to a subsequence, α_i is decreasing with limit α_*. By (B.40), (B.41) and $\lim_i c_{ij} = \kappa_j$, we have $a_i \ge \alpha_*$ for any i and $\lim_i a_i = \alpha_*$, which is impossible as $(a_i)_{i \ge 1}$ is strictly increasing. □

Lemma B.7. *In Set-up 10.3, assume that there exists a real number $\epsilon > 0$ such that $a_i \ge \epsilon$ for any i. Then $a(P, \widetilde{W}_*, B^*) \ge \epsilon$ for any prime divisor P over $\widetilde{W}_*$ with center the closed point.*

Proof. Let $\lambda := a(P, \widetilde{W}_*, 0)$ and $n_j := \operatorname{mult}_P H_j^*$. By Set-up 10.3, $(H_j^*)_j$ is a compatible generic limit of the tuples $(H_{ij})_j$ on $\widetilde{W}_0 = \mathbb{A}^3$ with the group $\widetilde{G}$. By Lemma 3.15, for all but finitely many i there exists a prime divisor V_i over $\widetilde{W}_0$ with center 0 such that $a(V_i, \widetilde{W}_0, 0) = \lambda$ and $\operatorname{mult}_{V_i} H_{ij} = n_j$ for any j. For such i, there exist a prime divisor E_i over W_0 with center 0 and an integer $r_i \in \{1, 2\}$ such that $\operatorname{mult}_{V_i} \widetilde{f} = r_i \operatorname{mult}_{E_i} f$ for any rational function f on W_0 with pullback $\widetilde{f}$ to $\widetilde{W}_0$. Since the double cover π_0 is étale outside the t-axis,

$$a(V_i, \widetilde{W}_0, 0) = r_i a(E_i, W_0, 0) \quad \text{and} \quad \operatorname{mult}_{V_i} H_{ij} = r_i \operatorname{mult}_{E_i}(h_{ij}|_{W_0}).$$

In particular $a(E_i, W_0, 0)/e_{E_i} \le \lambda$. For $i \gg 0$ we have $N_i \ge N_\lambda(M, \Gamma, \lambda)$, and by Proposition 3.14(2)(a) the birational transform E_i' of E_i under the inverse of the birational map $\varphi : W_{N_i} \dashrightarrow W_0$ of Proposition 3.14 is a prime divisor over W_{N_i} with center 0, $e_{E_i'} = e_{E_i}$, and $a(E_i', W_{N_i}, D_i) = a(E_i, W_0, D_{0,i})$. Hence

$$\epsilon \le a_i \le \frac{a(E_i', W_{N_i}, D_i)}{e_{E_i'}} = \frac{a(E_i, W_0, D_{0,i})}{e_{E_i}} = \frac{\lambda - \sum_j c_{ij} n_j}{r_i e_{E_i}}.$$

As $r_i e_{E_i} \geq 1$ and $\epsilon > 0$, $\lambda - \sum_j c_{ij} n_j \geq \epsilon$. Since $\lim_i c_{ij} = \kappa_j$, we obtain

$$a(P, \widetilde{W}_*, B^*) = \lambda - \sum_j \kappa_j n_j \geq \epsilon.$$

The lemma follows. □

Theorem B.8. *In Set-up 10.3, assume that B^* has a smooth floor F and that the pair (F, B_F) is lc but not klt. Then $(a_i)_{i\geq 1}$ is not strictly increasing.*

Proof of Theorem B.8. Suppose that $(a_i)_{i\geq 1}$ is strictly increasing, and let $\epsilon := a_1 > 0$, so that $a_i \geq \epsilon$ for any i. If H_j^* is a unit for some j, then $h_{*,j}(0) \neq 0$, and by [Liu26b, Construction-Proposition 4.1] $h_{ij}(0) \neq 0$ for all but finitely many i. Possibly removing such j, we may assume that no H_j^* is a unit. We divide the proof into four steps.

Step 1. In this step we show that

$$a(P, F, B_F) \geq \epsilon \quad \text{for any prime divisor } P \text{ over } F \text{ with center the closed point.} \tag{B.42}$$

We denote also by F a $\widetilde{G}$-semi-invariant equation of the floor, and let $u_1, u_2 \in \Bbbk[[p, v, t]]$ such that (F, u_1, u_2) is a regular system of parameters, so that $F \cong \operatorname{Spec} \Bbbk[[u_1, u_2]]$. Let P be as in (B.42) and ℓ an integer with $\ell > \max_j \operatorname{mult}_P(R_j^*|_F)$. The Gauss extension

$$\sum_k a_k(u_1, u_2) F^k \to \min_k (\operatorname{mult}_P a_k + k\ell)$$

is $\operatorname{mult}_{P'}$ for a prime divisor P' over $\widetilde{W}_*$ with center the closed point, with $a(P', \widetilde{W}_*, 0) = a(P, F, 0) + \ell$ by [Liu26b, Lemma 3.1] for the trivial group, applied on $\mathbb{A}^3_{\Bbbk}$ with coordinates u_1, u_2, F and passed to $\widetilde{W}_* = \operatorname{Spec} \Bbbk[[u_1, u_2, F]]$ by Lemma 2.12(3), and with $\operatorname{mult}_{P'} R_j^* = \operatorname{mult}_P(R_j^*|_F)$. As $H_j^* = F^{e_j} R_j^*$ and $\sum_j e_j \kappa_j = 1$, $a(P', \widetilde{W}_*, B^*) = a(P, F, B_F)$, and Lemma B.7 implies (B.42).

Step 2. In this step we find formal coordinates (Z_0, Z_1, Z_2) adapted to F and to the non-klt locus of (F, B_F). Any prime divisor exceptional over the regular surface F has center the closed point, so by (B.42), as (F, B_F) is lc but not klt, there exists a prime divisor C on F through 0 with coefficient one in B_F. The blow-up of the closed point of F has log discrepancy $2 - \operatorname{mult}_0 B_F \geq \epsilon$ with respect to (F, B_F), so $\operatorname{mult}_0 B_F < 2$, C is smooth and is the unique prime divisor on F with coefficient one through 0. As $\widetilde{G}$ preserves F and B_F, C is $\widetilde{G}$-stable.

We use four changes of coordinates of $\widetilde{W}_*$. The first is the interchange of p and v. The second is a linear automorphism of (p, v) of determinant one commuting with $\widetilde{G}$. The third is a shear $(p, v, t) \to (p, v - p\rho(p^2, t), t)$ with ρ semi-invariant with character χ_z/χ_x. The fourth is a graph coordinate $T := t + \phi(p^2, -v^2, pv)$ with $\phi(x, y, z)$ semi-invariant with character χ_t and $\phi(0) = 0$. For polynomial data, the first three are induced by automorphisms of $\mathbb{A}^4$ which fix t and $xy + z^2$ and conjugate G to a diagonal group (Lemma B.4(1)(a) and (b)). The interchange exchanges the roles of x and y, and below x denotes the coordinate after the interchange. If $\chi_p \neq \chi_v$, where $\chi_p(g, \alpha) := \alpha$ and $\chi_v(g, \alpha) := \chi_z(g)\alpha^{-1}$ are the characters of p and v, a $\widetilde{G}$-eigenvector ℓ_0 in the (p, v)-plane is a multiple of p or of v. If $\chi_p = \chi_v$, $\widetilde{G}$ acts on the plane by scalars. Thus after the interchange or a linear automorphism of (p, v) of determinant one commuting with $\widetilde{G}$, ℓ_0 is a multiple of v.

Suppose that F is deck-odd. Then F has an ι-anti-invariant $\widetilde{G}$-semi-invariant equation g_F, and the linear part of g_F is a non-zero $\widetilde{G}$-eigenvector in the (p, v)-plane. We may assume that it is a multiple of v. By the Weierstrass preparation theorem and the anti-invariance, g_F is a unit times $v - p\rho_*(p^2, t)$ with ρ_* semi-invariant with character $\chi_v/\chi_p = \chi_z/\chi_x$. After the shear by ρ_*, $F = (v = 0)$ with coordinates (p, t) on F, on which ι acts by $(p, t) \to (-p, t)$. A $\widetilde{G}$-semi-invariant equation of C on F is an eigenvector of ι. If it is anti-invariant, it is p times a unit, and $(Z_0, Z_1, Z_2) := (v, p, t)$, with $\phi := 0$. If it is invariant, the linear part is a non-zero

multiple of t, and the equation is a unit times $t+\psi_*(p^2)$ with ψ_* semi-invariant with character χ_t, and $(Z_0,Z_1,Z_2):=(v,T,p)$ with $T:=t+\phi(x)$, $\phi:=\psi_*$.

Suppose that F is deck-even. Then the linear part of an ι-invariant semi-invariant equation of F is a non-zero multiple of t, and by the Weierstrass preparation theorem and the uniqueness, the equation is a unit times $T:=t+\phi_*^0(p^2,-v^2,pv)$ with $\phi_*^0(x,y,z)$ semi-invariant with character χ_t. On $F=(T=0)$ with coordinates (p,v), ι acts by -1, so the linear part of an equation of C is anti-invariant, and the equation is anti-invariant. We may assume that the linear part of the equation of C is a multiple of v. Then C is defined on F by $v-p\rho_*(p^2)$ with ρ_* semi-invariant with character χ_z/χ_x. The interchange, the linear automorphism and the shear by $\rho_*(x)$ fix t and transform ϕ_*^0 into another series in x,y,z semi-invariant with character χ_t. After them, $(Z_0,Z_1,Z_2):=(T,v,p)$.

In each case, $F=(Z_0=0)$ and $C=(Z_0=Z_1=0)$, and we write

$$H_j^*=Z_0^{e_j}R_j^*,\quad R_j^*|_{Z_0=0}=Z_1^{\beta_j}S_j^*,\quad\text{and}\quad S_j^*|_{Z_0=Z_1=0}=Z_2^{\delta_j}u_j^* \tag{B.43}$$

with $\beta_j,\delta_j\in\mathbb{N}$, S_j^* not divisible by Z_1, and units u_j^*. Here Z_0 and Z_1 are equations of F and C up to units U_0 and U_1, and the powers of U_0 and U_1 are absorbed into R_j^* and S_j^*, $R_j^*|_{Z_0=0}\neq 0$ as F does not divide R_j^*, and $S_j^*|_{Z_0=Z_1=0}\neq 0$ as Z_1 does not divide S_j^*. Since no H_j^* is a unit, $e_j+\beta_j+\delta_j>0$. The coefficient of C in B_F is $\sum_j\beta_j\kappa_j=1$. Let $\mu_{2,*}:=\sum_j\delta_j\kappa_j$. For an integer $k>\max_j\delta_j$, the monomial valuation on F with weights k and 1 on Z_1 and Z_2 is a prime divisor over F with center the closed point, along which $R_j^*|_F$ has order $k\beta_j+\delta_j$, and the log discrepancy of the prime divisor with respect to (F,B_F) is $k+1-k-\mu_{2,*}=1-\mu_{2,*}$. By (B.42), $\mu_{2,*}\le 1-\epsilon<1$.

Step 3. In this step we realize the coordinates and the factorizations of **Step 2** on the members up to order J, where J is the integer of Proposition B.2 for M and the triples (e_j,β_j,δ_j). We show that for all but finitely many i there exist

- an automorphism σ_i of $\mathbb{A}^4$ which is the composite of the interchange (if it is used in **Step 2**), a linear automorphism of determinant one commuting with $\widetilde{G}$, and a shear σ_{ρ_i} of Lemma B.4(1)(a) with a polynomial $\rho_i\in\mathbb{C}[x,t]$ semi-invariant with character χ_z/χ_x, where $\rho_i\in\mathbb{C}[x]$ if F is deck-even,
- a polynomial $\phi_i(x,y,z)$ semi-invariant with character χ_t with $\phi_i(0)=0$, where $\phi_i=0$ for $(Z_0,Z_1,Z_2)=(v,p,t)$ and $\phi_i\in\mathbb{C}[x]$ for $(Z_0,Z_1,Z_2)=(v,T,p)$, and
- polynomials $u_{ij}(Z_2)$, $V_{ij}(Z_1,Z_2)$, and $V'_{ij}(Z_0,Z_1,Z_2)$ with $u_{ij}(0)\neq 0$,

such that, with $T_i:=t+\phi_i$, the coordinates (Z_0,Z_1,Z_2) taken from (p,v,T_i) as in **Step 2**, and

$$S_{ij}:=Z_2^{\delta_j}u_{ij}+Z_1V_{ij},\quad R_{ij}:=Z_1^{\beta_j}S_{ij}+Z_0V'_{ij},\quad\text{and}\quad \widehat{H}_{ij}:=Z_0^{e_j}R_{ij},$$

each $\widehat{H}_{ij}$ is $\widetilde{G}$-semi-invariant and ι-invariant, and

$$(\sigma_i^*h_{ij})\big(p^2,-v^2,pv,T_i-\phi_i(p^2,-v^2,pv)\big)-\widehat{H}_{ij}\in(p,v,T_i)^J\quad\text{for any } j. \tag{B.44}$$

Truncating in degree less than J the series ρ_*, ϕ_*^0, or ψ_* of **Step 2**, the series u_j^*, $(S_j^*-Z_2^{\delta_j}u_j^*)/Z_1$, and $(R_j^*-Z_1^{\beta_j}S_j^*)/Z_0$, we obtain such data over $\Bbbk$ for $(h_{*,j})_j$: truncation keeps the characters and the ι-parities, and changes the coordinates and the functions in (B.43) by elements of $(p,v,t)^J$. The conditions on the data are polynomial equations in the coefficients of the data and of the J-jets of the tuples (the condition $\det=1$ and (B.44)), linear conditions (characters and ι-parities), and the open conditions $u_j(0)\neq 0$. Let $Z_J\subset\mathcal{J}_J$ be the irreducible closed subset of Definition 2.5 for the tuples $(h_{ij})_j$. Let $\mathcal{P}$ be the affine space of the coefficients of the data, and $\mathcal{Z}\subset Z_J\times\mathcal{P}$ the locally closed subscheme defined by the conditions. Then $\mathcal{Z}$ has a point over the generic point of Z_J. By Chevalley's theorem, the image of $\mathcal{Z}$ in Z_J contains a non-empty open subset of Z_J, which contains the J-jets of $(h_{ij})_j$ for all but finitely many i, and a $\mathbb{C}$-point of the fiber of $\mathcal{Z}$ over such a jet provides the data.

For such i, σ_i is an automorphism of W_{N_i} which conjugates G to a diagonal group, so replacing h_{ij} with $\sigma_i^* h_{ij}$ and G with the conjugate group keeps a_i, the klt property, and the labels. Since the interchange is used for all i or for none, σ_{ρ_i} commutes with G (Lemma B.4(1)(a)), and the linear automorphism of (p, v) commutes with $\widetilde{G}$, the conjugate group, the characters and $\widetilde{G}$ do not depend on i. Replacing $\sigma_i^* h_{ij}$ with the representative on W_{N_i} without monomials divisible by xy changes the left hand side of (B.44) by the pullback of an element of $t^{N_i}\mathbb{C}[x, y, z, t]$ (Lemma B.4(1)(c)), which belongs to $(p, v, T_i)^{N_i}$. Possibly discarding finitely many i, we may assume that $N_i \geq J$, so (B.44) holds for the representative.

Step 4. In this step we conclude. Let

$$\mu_{0,i} := \sum_j e_j c_{ij}, \quad \mu_{1,i} := \sum_j \beta_j c_{ij}, \quad \text{and} \quad \mu_{2,i} := \sum_j \delta_j c_{ij}.$$

As the c_{ij} are increasing with limits κ_j, the three sequences are increasing, $\mu_{0,i}, \mu_{1,i} \leq 1$, $\lim_i \mu_{0,i} = \lim_i \mu_{1,i} = 1$, and $\lim_i \mu_{2,i} = \mu_{2,*} < 1$. By **Step 3**, the data $D = D_i$, $\phi = \phi_i$, $(\alpha_j, \beta_j, \delta_j) = (e_j, \beta_j, \delta_j)$, and $\widehat{H}_j = \widehat{H}_{ij}$ satisfy the hypotheses of Proposition B.2 with $N_i \geq J$. Let $\nu := |\chi_{Z_2}(\widetilde{G})|$, and let Λ be the integer of Proposition B.2 for the triples (e_j, β_j, δ_j). Thus $\nu = |\chi_t(G)|$ if $Z_2 = t$, and $\nu = 2|\chi_x(G)|$ if $Z_2 = p$, with x the coordinate after the interchange of **Step 2**. Since any prime divisor over W_{N_i} with center 0 satisfies the lower bound of Proposition B.2, and one of them satisfies the upper bound,

$$\frac{1 - \mu_{2,i}}{\nu} \leq a_i \leq \frac{1 - \mu_{2,i}}{\nu} + \Lambda\big((1 - \mu_{0,i}) + (1 - \mu_{1,i})\big).$$

Let $\tau_* := (1 - \mu_{2,*})/\nu$. Then $a_i \geq (1 - \mu_{2,i})/\nu \geq \tau_*$ for any i and $\lim_i a_i = \tau_*$, which is impossible as $(a_i)_{i \geq 1}$ is strictly increasing. □

Theorem B.9. *In Set-up 10.3, assume that B^* has a floor F which is a singular Du Val surface and that (F, B_F) is klt. Then $(a_i)_{i \geq 1}$ is not strictly increasing.*

The proof of Theorem B.9 resolves the floor by the following tower of weighted blow-ups of W_{N_i}.

Lemma B.10. *In Set-up 10.3, let i be an index and k_0 a positive integer with $N_i \geq 4k_0 + 2$. Let $(x_0, y_0, z_0) := (x, y, z)$. For $1 \leq k \leq k_0$ let φ_k be the weighted blow-up with weights $(2, 2, 2, 1)$ on $(x_{k-1}, y_{k-1}, z_{k-1}, t)$ at the origin of the t-chart of φ_{k-1} (of W_{N_i} if $k = 1$), let E_k be the exceptional divisor of φ_k, and let $(x_k, y_k, z_k) := (x_{k-1}, y_{k-1}, z_{k-1})/t^2$. For $1 \leq k \leq k_0$ let f_k be the composite of $\varphi_1, \ldots, \varphi_k$, and let $f := f_{k_0} : Y \to W_{N_i}$. Then the following hold.*

(1) The t-chart of φ_k is $W_{N_i - 4k}$ in the coordinates (x_k, y_k, z_k, t), on which G acts faithfully and diagonally with the characters

$$(\chi_x \chi_t^{-2k}, \chi_y \chi_t^{-2k}, \chi_z \chi_t^{-2k}, \chi_t).$$

(2) Let $N_{i,k} := N_i - 4(k - 1)$. The x-chart of φ_k is the quotient of $\mathbb{A}^3_{s,Z,T}$ by the involution $(s, Z, T) \to (-s, Z, -T)$, through

$$x_{k-1} = s^2, \quad z_{k-1} = s^2 Z, \quad t = sT, \quad \text{and} \quad y_{k-1} = -s^2 Z^2 - s^{N_{i,k}-2} T^{N_{i,k}}.$$

The action of G on it lifts to the faithful diagonal action on $\mathbb{A}^3_{s,Z,T}$ of the group

$$G_k := \{(g, \alpha) \mid g \in G,\ \alpha^2 = \chi_x(g)\chi_t(g)^{-2(k-1)}\}$$

of order $2|G|$ by

$$(s, Z, T) \to \big(\alpha s, \chi_z(g)\chi_t(g)^{-2(k-1)}\alpha^{-2} Z, \chi_t(g)\alpha^{-1} T\big),$$

in which the involution is $(1, -1)$. The action of G_k does not depend on i. The y-chart is obtained by exchanging x and y, and the blow-ups $\varphi_{k+1}, \ldots, \varphi_{k_0}$ do not change the x- and y-charts of φ_k.

(3) *The fiber* $f^{-1}(0)$ *is covered by* $(s=0)$ *in the* x- *and* y-*charts of* φ_1, *by* $(sT=0)$ *in the* x- *and* y-*charts of* φ_k *for* $2 \le k \le k_0$, *and by the origin of the* t-*chart of* φ_{k_0}.

(4) $a(E_k, W_{N_i}, 0) = 2k+1$. *For a mark* h *without monomials divisible by* xy, $\operatorname{mult}_{E_k} h$ *is the weighted order* $\operatorname{wt}_k(h)$ *for the weights* $(2k, 2k, 2k, 1)$ *on* (x, y, z, t), *and* $(f_k)_*^{-1}\operatorname{div}(h)$ *is* $\operatorname{div}(h^{(k)})$ *on the* t-*chart of* φ_k *with*

$$h^{(k)} := t^{-\operatorname{wt}_k(h)} h(t^{2k}x_k, t^{2k}y_k, t^{2k}z_k, t).$$

The coefficient of E_k *in the crepant pullback of* D_i *is*

$$\lambda_{ik} := \sum_j c_{ij} \operatorname{wt}_k(h_{ij}) - 2k.$$

On the t-*chart of* φ_k *the crepant pullback of* D_i *is*

$$\sum_j c_{ij} \operatorname{div}(h_{ij}^{(k)}) + \lambda_{ik} \operatorname{div}(t).$$

On the cover $\mathbb{A}^3_{s,Z,T}$ *of the* x-*chart of* φ_k, *the pullback of the crepant pullback of* D_i *is*

$$\sum_j c_{ij} \operatorname{div}(h'_{ij}) + \lambda_{ik} \operatorname{div}(s) + \lambda_{i,k-1} \operatorname{div}(T),$$

without the last term if $k=1$, *where* h'_{ij} *is the pullback of* h_{ij} *divided by*

$$s^{\operatorname{wt}_k(h_{ij})} T^{\operatorname{wt}_{k-1}(h_{ij})} \quad \text{if } k \ge 2, \qquad s^{\operatorname{wt}_1(h_{ij})} \quad \text{if } k = 1.$$

(5) *Let* V *be a prime divisor over* W_{N_i} *with center* 0. *Then the generic point of the center of* V *on* Y *belongs to the* x- *or* y-*chart of some* φ_k, *or the center of* V *on* Y *is the origin of the* t-*chart of* φ_{k_0}. *In the first case let* $U := \mathbb{A}^3_{s,Z,T}$, *let* G_U *be* G_k *(resp.* G_k *with* x *and* y *exchanged), and let* Δ_U *be the boundary in (4). In the second case let* $U := W_{N_i - 4k_0}$, $G_U := G$, *and*

$$\Delta_U := \sum_j c_{ij} \operatorname{div}(h_{ij}^{(k_0)}) + \lambda_{ik_0} \operatorname{div}(t).$$

We say that a prime divisor V' *over* U *is* above *a prime divisor* V *over* W_{N_i} *if the valuation of* V' *restricted to the pullbacks to* U *of the rational functions on* W_{N_i} *is a positive multiple of the valuation of* V. *Then any prime divisor* V' *over* U *above* V *satisfies*

$$\frac{a(V, W_{N_i}, D_i)}{e_V} = \frac{a(V', U, \Delta_U)}{e_{V'}},$$

where e_V *and* $e_{V'}$ *are the orders of the inertia groups of* V *in* G *and of* V' *in* G_U. *Conversely, any prime divisor over* U *with center in the preimage of* $f^{-1}(0)$ *is above a prime divisor over* W_{N_i} *with center* 0.

Proof. For $1 \le k \le k_0$ we have $N_{i,k} = N_i - 4(k-1) \ge 4(k_0 - k) + 6 \ge 6$. Assume by induction on k that the t-chart of φ_{k-1} is $W_{N_{i,k}} = (x_{k-1}y_{k-1} + z_{k-1}^2 + t^{N_{i,k}} = 0)$ in $(x_{k-1}, y_{k-1}, z_{k-1}, t)$, on which G acts faithfully and diagonally. For $k=1$, $W_{N_{i,1}} = W_{N_i}$. Since the equation of $W_{N_{i,k}}$ is semi-invariant, the characters of the monomials $x_{k-1}y_{k-1}$, z_{k-1}^2, and $t^{N_{i,k}}$ coincide:

$$\chi_{x_{k-1}}\chi_{y_{k-1}} = \chi_{z_{k-1}}^2 = \chi_t^{N_{i,k}}. \tag{B.45}$$

(1) The t-chart of the weighted blow-up of $\mathbb{A}^4$ with weights $(2,2,2,1)$ is $\mathbb{A}^4$ with coordinates (x_k, y_k, z_k, t), and

$$x_{k-1}y_{k-1} + z_{k-1}^2 + t^{N_{i,k}} = t^4(x_k y_k + z_k^2 + t^{N_{i,k}-4}).$$

Since $N_{i,k} - 4 = N_i - 4k$, the t-chart of φ_k is $W_{N_i - 4k}$. Since $x_k = x/t^{2k}$, $y_k = y/t^{2k}$ and $z_k = z/t^{2k}$, the characters are as in (1), and G acts faithfully since $\varphi_1 \circ \cdots \circ \varphi_k$ is G-equivariant and birational.

(2) The x-chart of the weighted blow-up of $\mathbb{A}^4$ is $\mathbb{A}^4_{s,Y',Z,T}/\{\pm1\}$, through

$$x_{k-1} = s^2, \quad y_{k-1} = s^2Y', \quad z_{k-1} = s^2Z, \quad \text{and} \quad t = sT,$$

where -1 acts by $(s, Y', Z, T) \to (-s, Y', Z, -T)$. There

$$x_{k-1}y_{k-1} + z_{k-1}^2 + t^{N_{i,k}} = s^4(Y' + Z^2 + s^{N_{i,k}-4}T^{N_{i,k}}),$$

so the strict transform of $W_{N_{i,k}}$ is the graph of $Y' = -Z^2 - s^{N_{i,k}-4}T^{N_{i,k}}$ over $\mathbb{A}^3_{s,Z,T}$, and

$$y_{k-1} = s^2Y' = -s^2Z^2 - s^{N_{i,k}-2}T^{N_{i,k}}.$$

For $(g, \alpha) \in G_k$, the action in (2) multiplies s^2, s^2Z, and sT by $\chi_{x_{k-1}}(g)$, $\chi_{z_{k-1}}(g)$, and $\chi_t(g)$, and multiplies s^2Z^2 and $s^{N_{i,k}-2}T^{N_{i,k}}$ by $\chi_{z_{k-1}}(g)^2\chi_{x_{k-1}}(g)^{-1}$ and $\chi_t(g)^{N_{i,k}}\chi_{x_{k-1}}(g)^{-1}$, both equal to $\chi_{y_{k-1}}(g)$ by (B.45). Thus G_k lifts the action of G. The homomorphism $G_k \to G$, $(g, \alpha) \to g$, is surjective with kernel $\{(1, \pm1)\}$, so $|G_k| = 2|G|$ and $\mathbb{A}^3_{s,Z,T}/G_k$ is the quotient of the x-chart by G. If (g, α) acts trivially, then $\alpha = 1$ and g fixes x_{k-1}, z_{k-1}, and t, which generate the function field of $W_{N_{i,k}}$, so $g = 1$. The characters defining G_k and the action of G_k only involve χ_x, χ_z, χ_t, and k. The y-chart is obtained by exchanging x and y. The origin of the t-chart of φ_k satisfies $x_k = y_k = 0$, so it is not contained in the x- or y-chart of φ_k. As $\varphi_{k+1}, \dots, \varphi_{k_0}$ are isomorphisms over the complement of the preimage of the origin of the t-chart of φ_k, they do not change the x- and y-charts of φ_k.

(3) Let o_k be the origin of the t-chart of φ_k. We show by induction on k that $f_k^{-1}(0)$ intersects the t-chart of φ_k in $(t = 0)$, and that $f_k^{-1}(0)$ is covered as in (3) with k in place of k_0. A point of the t-chart of φ_k maps to 0 if and only if $t = 0$, since $x = t^{2k}x_k$, $y = t^{2k}y_k$ and $z = t^{2k}z_k$. On $(t = 0)$ we have $x_ky_k + z_k^2 = 0$, so a point of $(t = 0)$ other than o_k has $x_k \neq 0$ or $y_k \neq 0$ and is contained in the x- or y-chart of φ_k. For $k = 1$, $f_1^{-1}(0)$ is $(s = 0)$ in the x- and y-charts, since $x = s^2$ and $t = sT$ there. For $k \geq 2$, $f_k^{-1}(0)$ is the union of $f_{k-1}^{-1}(0)\backslash\{o_{k-1}\}$, covered by induction by the charts of $\varphi_1, \dots, \varphi_{k-1}$ in (3), and of the preimage under φ_k of $(t = 0)$ in the t-chart of φ_{k-1}, which is $(sT = 0)$ in the x- and y-charts of φ_k and $(t = 0)$ in the t-chart of φ_k.

(4) By adjunction, the form $\Omega := x^{-1}dx \wedge dz \wedge dt$ generates $\omega_{W_{N_i}}$ near 0. On the t-chart of φ_k,

$$\Omega = t^{2k}x_k^{-1}dx_k \wedge dz_k \wedge dt,$$

and $x_k^{-1}dx_k \wedge dz_k \wedge dt$ generates $\omega_{W_{N_i-4k}}$. On the t-chart of φ_k, $(t = 0)$ is the integral surface $(x_ky_k + z_k^2 = 0)$, so $\operatorname{div}(t) = E_k$ and $a(E_k, W_{N_i}, 0) = 2k + 1$. Let $w := \operatorname{wt}_k(h)$ and let h_w be the sum of the terms of h of weight w. Then

$$h(t^{2k}x_k, t^{2k}y_k, t^{2k}z_k, t) = t^w\big(h_w(x_k, y_k, z_k, 1) + tr\big)$$

with r regular, and $h_w(x_k, y_k, z_k, 1)$ is a non-zero polynomial without monomials divisible by x_ky_k. The monomials not divisible by xy form a basis of $\mathbb{C}[x, y, z]/(xy + z^2)$, so $h_w(x_k, y_k, z_k, 1)$ does not vanish on $(t = 0)$. Hence $\operatorname{mult}_{E_k} h = w$ and $(f_k)_*^{-1}\operatorname{div}(h)$ is $\operatorname{div}(h^{(k)})$ on the t-chart of φ_k. The coefficient of E_k in the crepant pullback of D_i is

$$1 - a(E_k, W_{N_i}, D_i) = 1 - (2k + 1) + \sum_j c_{ij} \operatorname{mult}_{E_k} h_{ij} = \lambda_{ik}.$$

On the t-chart of φ_k the only exceptional divisor of f_k is $E_k = \operatorname{div}(t)$, which implies the formula there. The x-chart of φ_k maps to the t-chart of φ_{k-1}, on which $(t = 0)$ is E_{k-1} if $k \geq 2$, while for $k = 1$ the x-chart of φ_1 maps to W_{N_i}. Since $t = sT$, the divisors $\operatorname{div}(s)$ and $\operatorname{div}(T)$ on $\mathbb{A}^3_{s,Z,T}$ map onto E_k and onto $(\varphi_k)_*^{-1}(t = 0)$, where $(t = 0)$ is taken on the t-chart of φ_{k-1}, and $E_1, \dots, E_{k-2}$ do not intersect the x-chart of φ_k. The cover $\mathbb{A}^3_{s,Z,T} \to \mathbb{A}^3_{s,Z,T}/\{\pm1\}$ is étale in codimension 1, as the involution fixes only $(s = T = 0)$. Hence the pullbacks of E_k and E_{k-1} are $\operatorname{div}(s)$ and $\operatorname{div}(T)$. The pullback of h_{ij} vanishes to order $\operatorname{wt}_k(h_{ij})$ along $\operatorname{div}(s)$ and, if $k \geq 2$, to order $\operatorname{wt}_{k-1}(h_{ij})$ along $\operatorname{div}(T)$. So h'_{ij} is regular, and the pullback of the crepant pullback of D_i is as in (4).

(5) Since f maps the center of V on Y to 0, the first sentence of (5) follows from (3). By (1) and (2), U/G_U is an open subset of Y/G, so $U/G_U \to W_{N_i}/G$ is birational. Since V is the restriction of V', the prime divisors over W_{N_i}/G induced by V and V' coincide. As $K_U + \Delta_U$ is the pullback of $K_{W_{N_i}} + D_i$ by (4), the equality follows from the Hurwitz formula. Conversely, a prime divisor over U with center in the preimage of $f^{-1}(0)$ restricts to a prime divisor over W_{N_i} whose center is contained in $f(f^{-1}(0)) = \{0\}$. □

Proof of Theorem B.9. Suppose that $(a_i)_{i\geq 1}$ is strictly increasing. We derive a contradiction.

As F and H_j^* are semi-invariant, $R_j^* = H_j^*/F^{e_j}$ is semi-invariant for any j. Since $\sum_j e_j\kappa_j = 1$,

$$B^* = \operatorname{div}(F) + \sum_j \kappa_j \operatorname{div}(R_j^*).$$

As F is normal and (F, B_F) is klt, $(\widetilde{W}_*, B^*)$ is plt by inversion of adjunction.

Let $\operatorname{wt}_{(k,k,1)}$ be the weighted order for the weights $(k,k,1)$ on (p,v,t). The substitution $(x,y,z) \to (p^2, -v^2, pv)$ sends a monomial $x^{n_1}y^{n_2}z^{n_3}t^{n_4}$ with $n_1n_2 = 0$ to $\pm p^{2n_1+n_3}v^{2n_2+n_3}t^{n_4}$. By Lemma 10.2(1), distinct such monomials go to distinct monomials, and both have weight $2k(n_1+n_2+n_3)+n_4$. Hence

$$\operatorname{wt}_{(k,k,1)}\big(h(p^2,-v^2,pv,t)\big) = \operatorname{wt}_k(h) \tag{B.46}$$

for any h without monomials divisible by xy. We use the d-jets and the closed subsets Z_d of Definition 2.5. The d-jets of all but finitely many members belong to Z_d.

Step 1. In this step we pass lc and klt from limits to members. Let $N_i' \geq 2$ be integers with $\lim_i N_i' = +\infty$ such that G preserves the equation of $W_{N_i'}$ up to a character independent of i, let $(f_{ij})_{1\leq j\leq n'}$ be tuples of G-semi-invariant polynomials with a compatible generic limit $(f_{*,j})_j$ such that $(f_{*,j}(p^2,-v^2,pv,t))_j$ is a compatible generic limit of the tuples $(f_{ij}(p^2,-v^2,pv,t))_j$, and let $b_1,\dots,b_{n'}$ be non-negative real numbers. We show:

(A) If

$$\Big(\widetilde{W}_*, \sum_j b_j \operatorname{div}\big(f_{*,j}(p^2,-v^2,pv,t)\big)\Big)$$

is lc (resp. klt) near 0, then for $i \gg 0$ the pairs $(W_{N_i'}, \sum_j b_j \operatorname{div}(f_{ij}|_{W_{N_i'}}))$ and $(W_0, \sum_j b_j \operatorname{div}(f_{ij}|_{W_0}))$ are lc (resp. klt) near 0.

The double cover $\pi_0 : \widetilde{W}_* \to W_*$ is the quotient by ι, which fixes only the t-axis, so it is étale in codimension 1 and $(W_*, \sum_j b_j \operatorname{div}(f_{*,j}|_{W_*}))$ is lc (resp. klt). Assume first that it is lc. Let $\mathcal{A}_{\Bbbk} := \operatorname{Spec} \Bbbk[[x,y,z,t]]$. By [Liu26b, Lemma D.14] for $W_* \subset \mathcal{A}_{\Bbbk}$, the pair

$$\Big(\mathcal{A}_{\Bbbk}, \operatorname{div}(xy+z^2) + \sum_j b_j \operatorname{div}(f_{*,j})\Big)$$

is lc. For $N_i' > d$ the d-jets of

$$xy + z^2 + t^{N_i'} \quad \text{and} \quad xy + z^2$$

coincide, so the tuples $(xy+z^2+t^{N_i'}, (f_{ij})_j)$ have the compatible generic limit $(xy+z^2, (f_{*,j})_j)$. Dropping the entries with $b_j = 0$ (Lemma 2.6(2)), by Theorem 2.18(1) with the coefficients $(1, b_1, \dots, b_{n'})$, the pairs

$$\Big(\mathbb{A}^4, \operatorname{div}(xy+z^2+t^{N_i'}) + \sum_j b_j \operatorname{div}(f_{ij})\Big) \quad \text{and} \quad \Big(\mathbb{A}^4, \operatorname{div}(xy+z^2) + \sum_j b_j \operatorname{div}(f_{ij})\Big)$$

are lc near 0 for $i \gg 0$: otherwise they fail to be lc for i in an infinite set $\mathcal{I}_1$, and Theorem 2.18(1) along $\mathcal{I}_1$, along which the limit is still a compatible generic limit by Lemma 2.6(2), gives $i \in \mathcal{I}_1$ for which they are lc. Then [Liu26b, Lemma D.14] for the completions at 0 of $W_{N_i'} \subset \mathbb{A}^4$ and

$W_0 \subset \mathbb{A}^4$, with [Kaw21, Remark 2.6] for the passage to completions, implies the lc case of (A). In the klt case,

$$\Big(\widetilde{W}_*, \theta \sum_j b_j \operatorname{div}\big(f_{*,j}(p^2, -v^2, pv, t)\big)\Big)$$

is lc near 0 for some $\theta > 1$, so by the lc case $(W_{N_i'}, \theta D_i')$, where $D_i' := \sum_j b_j \operatorname{div}(f_{ij}|_{W_{N_i'}})$, and $(W_0, \theta \sum_j b_j \operatorname{div}(f_{ij}|_{W_0}))$ are lc near 0 for $i \gg 0$. As $W_{N_i'}$ and W_0 are klt, $(W_{N_i'}, D_i')$ and $(W_0, \sum_j b_j \operatorname{div}(f_{ij}|_{W_0}))$ are klt near 0. This proves (A).

By (A) for $(h_{ij})_j$ and the coefficients $(\kappa_j)_j$ (the character of $xy + z^2 + t^{N_i}$ is $\chi_x\chi_y$), the pair

$$\Big(W_{N_i}, D_i^\kappa\Big), \qquad D_i^\kappa := \sum_j \kappa_j \operatorname{div}(h_{ij}), \tag{B.47}$$

is lc near 0 for $i \gg 0$.

Step 2. In this step we blow up the points o_k of the t-axis defined below until the strict transform of F is smooth at o_k or does not pass through o_k. Let o_0 be the closed point of $\widetilde{W}_*^{(0)} := \widetilde{W}_*$. Recursively, as long as the strict transform $F^{(k)}$ of F on $\widetilde{W}_*^{(k)}$ passes through o_k and is singular there, let $\widetilde{W}_*^{(k+1)} \to \widetilde{W}_*^{(k)}$ be the blow-up of o_k followed by the completion at the origin o_{k+1} of the t-chart of the blow-up, so that $\widetilde{W}_*^{(k+1)} = \operatorname{Spec} \Bbbk[[p_{k+1}, v_{k+1}, t]]$ with $p_k = tp_{k+1}$ and $v_k = tv_{k+1}$, where $(p_0, v_0) := (p, v)$. Since $\widetilde{G}$ acts diagonally, it fixes each o_k. Let B_k be the $\mathbb{R}$-divisor on $\widetilde{W}_*^{(k)}$ such that $K_{\widetilde{W}_*^{(k)}} + F^{(k)} + B_k$ is the pullback of $K_{\widetilde{W}_*} + B^*$, so $B_0 = \sum_j \kappa_j \operatorname{div}(R_j^*)$.

We show by induction on k that $B_k \geq 0$, that $(\widetilde{W}_*^{(k)}, F^{(k)} + B_k)$ is plt, and that $\operatorname{mult}_{o_k} F^{(k)} = 2$ if $F^{(k)}$ is singular at o_k. For $k = 0$, $B_0 \geq 0$ and $(\widetilde{W}_*, F + B_0) = (\widetilde{W}_*, B^*)$ is plt. Assume them for k, and that $F^{(k)}$ is singular at o_k. The exceptional divisor E_{o_k} of the blow-up of o_k satisfies

$$0 < a(E_{o_k}, \widetilde{W}_*^{(k)}, F^{(k)} + B_k) = 3 - \operatorname{mult}_{o_k} F^{(k)} - \operatorname{mult}_{o_k} B_k$$

as the pair is plt. Since $\operatorname{mult}_{o_k} F^{(k)} \geq 2$ and $B_k \geq 0$, we have $\operatorname{mult}_{o_k} F^{(k)} = 2$ and $0 \leq \operatorname{mult}_{o_k} B_k < 1$. On $\widetilde{W}_*^{(k+1)}$ the exceptional divisor is $\operatorname{div}(t)$, with coefficient

$$\operatorname{mult}_{o_k}(F^{(k)} + B_k) - 2 = \operatorname{mult}_{o_k} B_k \in [0, 1)$$

in B_{k+1}, and the other components of B_{k+1} are strict transforms of components of B_k. Thus $B_{k+1} \geq 0$, and $(\widetilde{W}_*^{(k+1)}, F^{(k+1)} + B_{k+1})$ is plt since it is crepant to $(\widetilde{W}_*, B^*)$ and $F^{(k+1)}$ is the only component of $F^{(k+1)} + B_{k+1}$ with coefficient one.

Since $\operatorname{mult}_{o_k} F^{(k)} = 2$ for each blown-up point o_k,

$$F^{(k)} = t^{-2k} F(t^k p_k, t^k v_k, t), \tag{B.48}$$

and t does not divide $F^{(k)}$. Hence $\operatorname{wt}_{(k,k,1)}(F) = 2k$. If the process did not stop, then (B.48) would be regular for any k. A monomial $p^{n_1} v^{n_2} t^{n_3}$ of F becomes $t^{n_3 + k(n_1+n_2-2)} p_k^{n_1} v_k^{n_2}$, and distinct monomials stay distinct, so $n_1 + n_2 \geq 2$ for any monomial of F, that is, $F \in (p, v)^2$. Then F would be singular along the t-axis, which contradicts that a Du Val surface has an isolated singularity. Hence the process stops after finitely many blow-ups, say k_0 of them. We have $k_0 \geq 1$ since F is singular at o_0. Then $F^{(k_0)}$ either does not pass through o_{k_0} or is smooth at o_{k_0}.

For $1 \leq k \leq k_0$ let $w_{jk} := \operatorname{wt}_{(k,k,1)}(H_j^*)$, which equals $\operatorname{wt}_k(h_{*,j})$ by (B.46), and let λ_{*k} be the coefficient of $\operatorname{div}(t)$ in B_k. The composite $\widetilde{W}_*^{(k)} \to \widetilde{W}_*$ is $(p, v, t) = (t^k p_k, t^k v_k, t)$, with Jacobian determinant t^{2k}, and H_j^* pulls back to $t^{w_{jk}}$ times a function not divisible by t. Since

$$w_{jk} = 2ke_j + \operatorname{wt}_{(k,k,1)}(R_j^*) \quad \text{and} \quad \sum_j e_j \kappa_j = 1,$$

$$\lambda_{*k} = \sum_j \kappa_j w_{jk} - 2k = \sum_j \kappa_j \operatorname{wt}_{(k,k,1)}(R_j^*) \in [0,1). \tag{B.49}$$

Let

$$R_j^{(k)} := t^{-\operatorname{wt}_{(k,k,1)}(R_j^*)} R_j^*(t^k p_k, t^k v_k, t).$$

Then

$$B_k = \sum_j \kappa_j \operatorname{div}(R_j^{(k)}) + \lambda_{*k} \operatorname{div}(t) \quad \text{near } o_k.$$

By adjunction, if $F^{(k_0)}$ passes through o_{k_0}, then

$$\Big(F^{(k_0)}, \sum_j \kappa_j \operatorname{div}(R_j^{(k_0)}|_{F^{(k_0)}}) + \lambda_{*k_0} \operatorname{div}(t|_{F^{(k_0)}})\Big) \text{ is klt;} \tag{B.50}$$

otherwise $(\widetilde{W}_*^{(k_0)}, B_{k_0})$ is klt near o_{k_0}.

Step 3. In this step we construct the tower of Lemma B.10 on W_{N_i} and the last t-chart of the tower, and we prove a statement (C) on the fixed quotients $\mathbb{A}^3/G_{k,\star}$. Fix $d > \max_{j,k} w_{jk}$. For $1 \le k \le k_0$, the condition $\operatorname{wt}_k \ge w_{jk}$ on the j-th entry of d-jets is closed and holds at the generic point of Z_d, so it holds on Z_d, which contains the d-jets of all but finitely many members. The condition that the terms of weight w_{jk} do not vanish is open and holds at the generic point, hence on a non-empty open subset of Z_d. Hence for $i \gg 0$,

$$\operatorname{wt}_k(h_{ij}) = w_{jk} \quad \text{for any } j \text{ and } 1 \le k \le k_0, \qquad N_i \ge 4k_0 + 2.$$

For such i we use the tower of Lemma B.10 with k_0 of **Step 2**. By Lemma B.10(4),

$$\lambda_{ik} = \sum_j c_{ij} w_{jk} - 2k, \tag{B.51}$$

which is increasing in i with limit λ_{*k} by (B.49), since c_{ij} is increasing with limit κ_j and $w_{jk} \ge 0$.

Let $n_i := N_i - 4k_0$. The last t-chart is W_{n_i} in $(x_{k_0}, y_{k_0}, z_{k_0}, t)$, with the marks $h_{ij}^{(k_0)}$ of Lemma B.10(4), which have no monomials divisible by $x_{k_0} y_{k_0}$. Let

$$D_i^{(k_0)} := \sum_j c_{ij} \operatorname{div}(h_{ij}^{(k_0)}) + \lambda_{ik_0} \operatorname{div}(t) \quad \text{on } W_{n_i},$$

the boundary Δ_U of Lemma B.10(5) in the second case. Let

$$h_{*,j}^{(k_0)} := t^{-w_{jk_0}} h_{*,j}(t^{2k_0} x_{k_0}, t^{2k_0} y_{k_0}, t^{2k_0} z_{k_0}, t),$$

and let $H_{ij}^{(k_0)}$ and $H_{*,j}^{(k_0)}$ be the pullbacks of $h_{ij}^{(k_0)}$ and $h_{*,j}^{(k_0)}$ under

$$(x_{k_0}, y_{k_0}, z_{k_0}) = (p_{k_0}^2, -v_{k_0}^2, p_{k_0} v_{k_0}).$$

Then

$$H_{*,j}^{(k_0)} = t^{-w_{jk_0}} H_j^*(t^{k_0} p_{k_0}, t^{k_0} v_{k_0}, t) = (F^{(k_0)})^{e_j} R_j^{(k_0)}. \tag{B.52}$$

The group of Set-up 10.3 for the characters of Lemma B.10(1) acts on (p_{k_0}, v_{k_0}, t) as $\widetilde{G}$ does through $(g, \alpha) \to (g, \alpha \chi_t(g)^{-k_0})$, so it preserves $F^{(k_0)}$, and the deck involution is

$$(p_{k_0}, v_{k_0}, t) \to (-p_{k_0}, -v_{k_0}, t).$$

We show:

(B) $(h_{*,j}^{(k_0)})_j$ is a compatible generic limit of the tuples $(h_{ij}^{(k_0)})_j$ with coefficient field $\Bbbk$, and $(H_{*,j}^{(k_0)})_j$ is a compatible generic limit of the tuples $(H_{ij}^{(k_0)})_j$. Moreover, $((h_{*,j}^{(k_0)})_j, t)$ and $((H_{*,j}^{(k_0)})_j, t)$ are compatible generic limits of the tuples $((h_{ij}^{(k_0)})_j, t)$ and $((H_{ij}^{(k_0)})_j, t)$, respectively.

Let $w := \max_j w_{jk_0}$ and $d' \in \mathbb{N}^+$. The map

$$h \to t^{-w_{jk_0}} h(t^{2k_0}x_{k_0}, t^{2k_0}y_{k_0}, t^{2k_0}z_{k_0}, t)$$

sends a monomial of degree $\geq d'+w$ to a monomial of degree $\geq d'$, so the d'-jet of $(h_{ij}^{(k_0)})_j$ is the image of the $(d'+w)$-jet of $(h_{ij})_j$ under a linear map ℓ, independent of i, from the linear subspace of $(d'+w)$-jets whose j-th entry has $\mathrm{wt}_{k_0} \geq w_{jk_0}$ for any j, which contains $Z_{d'+w}$ since the closed condition $\mathrm{wt}_{k_0} \geq w_{jk_0}$ on the j-th entry holds at the generic point of $Z_{d'+w}$. Let $Z'_{d'}$ be the closure of $\ell(Z_{d'+w})$. The generic point of $Z'_{d'}$ is the image of the generic point of $Z_{d'+w}$, that is, the d'-jet of $(h_{*,j}^{(k_0)})_j$. For a non-empty open subset Z° of $Z'_{d'}$, $\ell^{-1}(Z^\circ) \cap Z_{d'+w}$ is a non-empty open subset of $Z_{d'+w}$, so Z° contains the d'-jets for all but finitely many i. By the definition of $H_{ij}^{(k_0)}$ and (B.52),

$$H_{ij}^{(k_0)} = t^{-w_{jk_0}} H_{ij}(t^{k_0}p_{k_0}, t^{k_0}v_{k_0}, t), \qquad H_{*,j}^{(k_0)} = t^{-w_{jk_0}} H_j^*(t^{k_0}p_{k_0}, t^{k_0}v_{k_0}, t),$$

and the map

$$H \to t^{-w_{jk_0}} H(t^{k_0}p_{k_0}, t^{k_0}v_{k_0}, t)$$

also sends a monomial of degree $\geq d'+w$ to a monomial of degree $\geq d'$. As $(H_j^*)_j$ is a compatible generic limit of the tuples $(H_{ij})_j$ (Set-up 10.3), the argument above with $(H_{ij})_j$, $(H_j^*)_j$, and $\mathrm{wt}_{(k_0,k_0,1)}$ in place of $(h_{ij})_j$, $(h_{*,j})_j$, and wt_{k_0} proves the claim for $(H_{ij}^{(k_0)})_j$. Adding the fixed function t does not affect the compatible generic limit. This proves (B). By (B), we may use (A) on the last chart (the character of $x_{k_0}y_{k_0} + z_{k_0}^2 + t^{n_i}$ is $\chi_x\chi_y\chi_t^{-4k_0}$), with $(x_{k_0}, y_{k_0}, z_{k_0}, t)$, (p_{k_0}, v_{k_0}, t), $\widetilde{W}_*^{(k_0)}$, and W_{n_i} in place of (x,y,z,t), (p,v,t), $\widetilde{W}_*$, and $W_{N'_i}$.

For $1 \leq k \leq k_0$ and $\star \in \{x, y\}$, let $G_{k,\star}$ be the group G_k of Lemma B.10(2) for $\star = x$ and G_k with x and y exchanged for $\star = y$, let $X_{k,\star} := \mathbb{A}^3/G_{k,\star}$, which does not depend on i, let $\Sigma_k \subset \mathbb{A}^3$ be $(s=0)$ for $k=1$ and $(sT=0)$ for $k \geq 2$, and let $\Delta_{ik,\star}$ be the boundary on $\mathbb{A}^3$ in Lemma B.10(4) for the $\star$-chart of φ_k. We use the following.

(C) Let $\Gamma' \subset [0,1]$ be a DCC set. Then there exist an ACC set $\mathcal{A}$ and a real number L, depending only on $X_{k,\star}$ and Γ', satisfying the following. Let $\Delta = \sum_j b_j \operatorname{div}(u_j)$ be a boundary on a $G_{k,\star}$-stable neighborhood of Σ_k in $\mathbb{A}^3$ with $b_j \in \Gamma'$ and $G_{k,\star}$-semi-invariant regular functions u_j, such that $(\mathbb{A}^3, \Delta)$ is lc near Σ_k. Let μ be the infimum of $a(E', \mathbb{A}^3, \Delta)/e_{E'}$ over prime divisors E' over $\mathbb{A}^3$ with center in Σ_k, where $e_{E'}$ is the order of the inertia group of E' in $G_{k,\star}$. Then $\mu \in \mathcal{A}$, and $\mu = a(E', \mathbb{A}^3, \Delta)/e_{E'}$ for such a prime divisor E' with $a(E', \mathbb{A}^3, 0) \leq e_{E'}L$.

Let $X := X_{k,\star}$, let $\rho : \mathbb{A}^3 \to X$ be the quotient map, $\Sigma_X := \rho(\Sigma_k)$, and let Δ_X be the $\mathbb{R}$-divisor with $K_{\mathbb{A}^3} + \Delta = \rho^*(K_X + \Delta_X)$. A prime divisor P on $\mathbb{A}^3$ with ramification index $e_P \leq |G_{k,\star}| = 2|G| \leq 2M$ and coefficient b_P in Δ has image with coefficient $1-(1-b_P)/e_P$ in Δ_X. Here $b_P \leq 1$ as the pair is lc, and b_P is a sum of elements of Γ' with at most $1/\min(\Gamma' \setminus \{0\})$ non-zero terms. Hence the coefficients of Δ_X belong to the DCC set

$$\Gamma'' := \left\{ 1 - \frac{1-b}{e} \;\middle|\; 1 \leq e \leq 2M,\ b \in [0,1] \text{ a finite sum of elements of } \Gamma' \right\}.$$

Replacing the components of Δ_X by their closures in X does not change the log discrepancies of prime divisors with center in Σ_X. By the Hurwitz formula, μ is the infimum of $a(\overline{E}, X, \Delta_X)$ over prime divisors $\overline{E}$ over X with center in Σ_X.

Let $g : X' \to X$ be a log resolution such that $g^{-1}(\Sigma_X)$ is a divisor and the union Λ of $g^{-1}(\Sigma_X)$, $\operatorname{Exc}(g)$, and $\operatorname{Supp} g_*^{-1}\Delta_X$ is snc, and let $K_{X'} + \Delta' = g^*(K_X + \Delta_X)$. Over a neighborhood of Σ_X the coefficients d_P of Δ' satisfy $d_P \leq 1$. Let $\overline{E}$ be a prime divisor over X with center in Σ_X, let $P_1, \dots, P_m$ be the components of Λ containing the center c of $\overline{E}$ on X', and let Q be the component of $P_1 \cap \cdots \cap P_m$ containing c. Since $c \subset g^{-1}(\Sigma_X)$, we have $m \geq 1$ and $Q \subset g^{-1}(\Sigma_X)$.

Since $(X', \sum_q P_q)$ is lc and $\operatorname{mult}_{\overline{E}} P_q \geq 1$,

$$a(\overline{E}, X, \Delta_X) = a\Big(\overline{E}, X', \sum_q P_q\Big) + \sum_q (1 - d_{P_q}) \operatorname{mult}_{\overline{E}} P_q \geq \sum_q (1 - d_{P_q}) = a(E_Q, X, \Delta_X),$$

where E_Q is the exceptional divisor of the blow-up of Q if $m \geq 2$, and $E_Q := P_1$ if $m = 1$. As Λ has finitely many strata, $\mu = a(E_Q, X, \Delta_X)$ for some Q. Let ξ be the generic point of $g(Q) \subset \Sigma_X$. Any prime divisor with center $\overline{\{\xi\}}$ has center in Σ_X, so $\mu = \operatorname{mld}(X \ni \xi, \Delta_X)$. By Theorem 2.22(1), μ belongs to an ACC set $\mathcal{A}$ depending only on X and Γ''. By Theorem 2.22(3), whose bound L depends only on X and Γ'', there exists a prime divisor $\overline{E}$ over X with center $\overline{\{\xi\}}$ such that $a(\overline{E}, X, \Delta_X) = \mu$ and $a(\overline{E}, X, 0) \leq L$. Let E' be a prime divisor over $\mathbb{A}^3$ above $\overline{E}$, and $\Delta_{X,0}$ the branch divisor with $K_{\mathbb{A}^3} = \rho^*(K_X + \Delta_{X,0})$. By the Hurwitz formula and $\Delta_{X,0} \geq 0$,

$$\frac{a(E', \mathbb{A}^3, 0)}{e_{E'}} = a(\overline{E}, X, \Delta_{X,0}) \leq a(\overline{E}, X, 0) \leq L.$$

This proves (C).

Step 4. In this step we treat the case when $R^*_{j_1}$ is not a unit for some j_1. Then $\operatorname{wt}_{(k,k,1)}(R^*_{j_1}) \geq 1$, so $\lambda_{*k} \geq \kappa_{j_1} > 0$ for $1 \leq k \leq k_0$ by (B.49), and $0 < \lambda_{ik} < 1$ for $i \gg 0$ by (B.51). Thus for $i \gg 0$ the boundaries Δ_U of Lemma B.10(5) satisfy $\Delta_U \geq 0$, and since (W_{N_i}, D_i) is klt near 0, the pairs (U, Δ_U) of Lemma B.10(5) are klt near the preimage of $f^{-1}(0)$. The set

$$\Gamma_1 := \Gamma \cup \{\lambda_{ik} \mid i \gg 0,\ 1 \leq k \leq k_0\} \subset [0, 1]$$

is DCC, since each sequence $(\lambda_{ik})_i$ is increasing. Let $\mu_{ik,\star}$ be the infimum of $a(E', \mathbb{A}^3, \Delta_{ik,\star})/e_{E'}$ over prime divisors E' over $\mathbb{A}^3$ with center in Σ_k, and let

$$\beta_i := \operatorname{mld}\big((W_{n_i} \ni 0, D_i^{(k_0)})/G\big).$$

By Lemma B.10(5),

$$a_i = \min\big(\{\mu_{ik,\star} \mid 1 \leq k \leq k_0,\ \star \in \{x, y\}\} \cup \{\beta_i\}\big).$$

If $a_i = \mu_{ik,\star}$ for fixed $(k, \star)$ and infinitely many i, then the strictly increasing sequence of a_i for such i is contained in the ACC set $\mathcal{A}$ of (C) for $\Gamma' = \Gamma_1$, a contradiction. Hence $a_i = \beta_i$ for $i \gg 0$, and β_i is strictly increasing.

Assume that $F^{(k_0)}$ passes through o_{k_0}. By (B), (B.52), and (B.50), the data W_{n_i}, G, and $D_i^{(k_0)}$, with the marks $h_{ij}^{(k_0)}$ and t and the labels c_{ij} and λ_{ik_0} in $\Gamma_1 \backslash \{0\}$, satisfy Set-up 10.3: the labels are increasing with positive limits κ_j and λ_{*k_0}, the pairs are klt near 0, and the limit boundary is $F^{(k_0)} + B_{k_0}$. Here $F^{(k_0)}$ is a floor, since it has coefficient $\sum_j e_j \kappa_j = 1$ and divides neither t nor any $R_j^{(k_0)}$. It is smooth, and the adjunction pair in (B.50) is klt. By Theorem B.5 or Theorem B.6, according to the deck parity of $F^{(k_0)}$, the sequence of the β_i for $i \gg 0$ is not strictly increasing, a contradiction.

Assume that $F^{(k_0)}$ does not pass through o_{k_0}. Then $(\widetilde{W}_*^{(k_0)}, B_{k_0})$ is klt near o_{k_0}, so there exists $\theta > 1$ such that $(\widetilde{W}_*^{(k_0)}, \theta B_{k_0})$ is lc near o_{k_0}. By (A) for the tuples $((h_{ij}^{(k_0)})_j, t)$ with the coefficients $((\theta\kappa_j)_j, \theta\lambda_{*k_0})$, and since $c_{ij} \leq \kappa_j$ and $\lambda_{ik_0} \leq \lambda_{*k_0}$, the pair

$$\Big(W_0, \theta \sum_j c_{ij} \operatorname{div}(h_{ij}^{(k_0)}|_{W_0}) + \theta\lambda_{ik_0} \operatorname{div}(t|_{W_0})\Big)$$

is lc near 0 for $i \gg 0$. By Proposition 3.14(3) for M, Γ_1, and θ, the values β_i satisfy the ACC, a contradiction.

Step 5. In this step we assume that all R_j^* are units and compare a_i with the boundaries without the exceptional terms. By (B.49), $\lambda_{*k} = 0$ and $w_{jk} = 2ke_j$ for any j and k. Let

$$\sigma_i := \sum_j e_j c_{ij} \leq \sum_j e_j \kappa_j = 1, \qquad \delta_{ik} := 2k(1 - \sigma_i) \geq 0, \quad \text{and} \quad \delta_{i0} := 0.$$

Then $\lim_i \sigma_i = 1$, and $\lambda_{ik} = -\delta_{ik}$ by (B.51). Let

$$\overline{\Delta}_{ik,\star} := \sum_j c_{ij} \operatorname{div}(h'_{ij}), \qquad \overline{D}_i := \sum_j c_{ij} \operatorname{div}(h_{ij}^{(k_0)}),$$

where h'_{ij} is the pullback of h_{ij} to the cover $\mathbb{A}^3_{s,Z,T}$ of the $\star$-chart of φ_k divided by

$$s^{\operatorname{wt}_k(h_{ij})} T^{\operatorname{wt}_{k-1}(h_{ij})} \quad \text{if } k \geq 2, \qquad s^{\operatorname{wt}_1(h_{ij})} \quad \text{if } k = 1.$$

By Lemma B.10(4),

$$\Delta_{ik,\star} = \overline{\Delta}_{ik,\star} - \delta_{ik} \operatorname{div}(s) - \delta_{i,k-1} \operatorname{div}(T), \qquad D_i^{(k_0)} = \overline{D}_i - \delta_{ik_0} \operatorname{div}(t). \tag{B.53}$$

By Lemma B.10(4), the coefficient of E_k in the crepant pullback of D_i^κ is

$$1 - a(E_k, W_{N_i}, D_i^\kappa) = \sum_j \kappa_j w_{jk} - 2k = \lambda_{*k} = 0.$$

Hence the pullbacks of the crepant pullback of D_i^κ to the covers of the charts are

$$\sum_j \kappa_j \operatorname{div}(h'_{ij}) \geq \overline{\Delta}_{ik,\star} \quad \text{and} \quad \overline{D}_i^\kappa := \sum_j \kappa_j \operatorname{div}(h_{ij}^{(k_0)}) \geq \overline{D}_i.$$

By (B.47), for $i \gg 0$ the pairs $(\mathbb{A}^3, \sum_j \kappa_j \operatorname{div}(h'_{ij}))$ and $(W_{n_i}, \overline{D}_i^\kappa)$ are lc near Σ_k and near 0, hence so are $(\mathbb{A}^3, \overline{\Delta}_{ik,\star})$ and $(W_{n_i}, \overline{D}_i)$.

In the case of this step we define $\mu_{ik,\star}$ and β_i with $\overline{\Delta}_{ik,\star}$ and $\overline{D}_i$ in place of the boundaries of **Step 4**: let $\mu_{ik,\star}$ be the infimum of $a(E', \mathbb{A}^3, \overline{\Delta}_{ik,\star})/e_{E'}$ over prime divisors E' over $\mathbb{A}^3$ with center in Σ_k, and let

$$\beta_i := \operatorname{mld}\big((W_{n_i} \ni 0, \overline{D}_i)/G\big), \qquad \mu_i := \min\big(\{\mu_{ik,\star} \mid 1 \leq k \leq k_0,\ \star \in \{x, y\}\} \cup \{\beta_i\}\big).$$

Let E be a prime divisor over W_{N_i} with center 0. By Lemma B.10(5), either the generic point of the center of E on Y belongs to the $\star$-chart of φ_k for some $1 \leq k \leq k_0$ and $\star \in \{x, y\}$, and then we let $U := \mathbb{A}^3$ be the cover of the $\star$-chart of φ_k, or the center of E on Y is the origin of the t-chart of φ_{k_0}, and then we let $U := W_{n_i}$. Let E' be a prime divisor over U above E (Lemma B.10(5)). By Lemma B.10(5) and (B.53), if $U = \mathbb{A}^3$ is the cover of the $\star$-chart of φ_k, then

$$\frac{a(E, W_{N_i}, D_i)}{e_E} = \frac{a(E', \mathbb{A}^3, \overline{\Delta}_{ik,\star})}{e_{E'}} + \frac{\delta_{ik} \operatorname{mult}_{E'} s + \delta_{i,k-1} \operatorname{mult}_{E'} T}{e_{E'}}, \tag{B.54}$$

and if $U = W_{n_i}$, then

$$\frac{a(E, W_{N_i}, D_i)}{e_E} = \frac{a(E', W_{n_i}, \overline{D}_i)}{e_{E'}} + \frac{\delta_{ik_0} \operatorname{mult}_{E'} t}{e_{E'}}. \tag{B.55}$$

The last terms are non-negative, so

$$a_i \geq \mu_i. \tag{B.56}$$

By (C) for $\Gamma' = \Gamma$, there exist L, independent of i, and a prime divisor E' over $\mathbb{A}^3$ with center in Σ_k such that

$$\mu_{ik,\star} = \frac{a(E', \mathbb{A}^3, \overline{\Delta}_{ik,\star})}{e_{E'}} \quad \text{and} \quad a(E', \mathbb{A}^3, 0) \leq e_{E'} L.$$

As there exist only $2k_0$ pairs $(k, \star)$, we take L independent of $(k, \star)$. Since $(\mathbb{A}^3, \operatorname{div}(sT))$ is lc, $\operatorname{mult}_{E'} s + \operatorname{mult}_{E'} T \leq a(E', \mathbb{A}^3, 0)$, so $\operatorname{mult}_{E'} s \leq e_{E'} L$ and $\operatorname{mult}_{E'} T \leq e_{E'} L$. Let E be the prime divisor over W_{N_i} with center 0 below E' (Lemma B.10(5)). Since $\delta_{ik}, \delta_{i,k-1} \leq 2k_0(1 - \sigma_i)$, (B.54) implies

$$a_i \leq \frac{a(E, W_{N_i}, D_i)}{e_E} \leq \mu_{ik,\star} + 4k_0 L(1 - \sigma_i). \tag{B.57}$$

Step 6. In this step we prove that $(W_{n_i}, \overline{D}_i)$ is klt near 0 for $i \gg 0$ and bound a_i in terms of μ_i. First we prove the klt property. If $\sigma_i = 1$, then $c_{ij} = \kappa_j$ whenever $e_j \geq 1$, so $\delta_{ik} = 0$ for any k, and $(W_{n_i}, \overline{D}_i) = (W_{n_i}, D_i^{(k_0)})$ is crepant to (W_{N_i}, D_i) with $D_i^{(k_0)} \geq 0$, hence klt near 0. Assume $\sigma_i < 1$. Then there exists j_0 with $e_{j_0} \geq 1$ and $c_{ij_0} < \kappa_{j_0}$. For any j_0 with $e_{j_0} \geq 1$, by (B.52) the limit boundary $\sum_{j \neq j_0} \kappa_j \operatorname{div}(H_{*,j}^{(k_0)})$ is $(1 - e_{j_0}\kappa_{j_0}) \operatorname{div}(F^{(k_0)})$ near o_{k_0}, which is klt since $F^{(k_0)}$ is smooth or does not pass through o_{k_0} and $0 \leq 1 - e_{j_0}\kappa_{j_0} < 1$. By (A), and by (B) for the subtuple without the j_0-th entry (Lemma 2.6(2)), for $i \gg 0$ and any such j_0, the pair $(W_{n_i}, \overline{D}_i^{-j_0})$ with

$$\overline{D}_i^{-j_0} := \sum_{j \neq j_0} \kappa_j \operatorname{div}(h_{ij}^{(k_0)})$$

is klt near 0. Let $\nu := c_{ij_0}/\kappa_{j_0} \in (0,1)$. Then $\overline{D}_i \leq \nu \overline{D}_i^{\kappa} + (1-\nu)\overline{D}_i^{-j_0}$, so for any prime divisor E' over a neighborhood of 0 on which $(W_{n_i}, \overline{D}_i^{\kappa})$ is lc and $(W_{n_i}, \overline{D}_i^{-j_0})$ is klt,

$$a(E', W_{n_i}, \overline{D}_i) \geq \nu a(E', W_{n_i}, \overline{D}_i^{\kappa}) + (1-\nu) a(E', W_{n_i}, \overline{D}_i^{-j_0}) > 0.$$

Next, let $\eta := 1$ if $F^{(k_0)}$ does not pass through o_{k_0}, and otherwise let $\eta > 0$ with $\eta \operatorname{mult}_{o_{k_0}}(t|_{F^{(k_0)}}) < 1$. Here $t|_{F^{(k_0)}} \neq 0$ as $F^{(k_0)}$ does not divide t. By (B.52), the limit boundary of the tuples $((h_{ij}^{(k_0)})_j, t)$ with the coefficients $((\kappa_j)_j, \eta)$ is $\eta \operatorname{div}(t)$, resp. $F^{(k_0)} + \eta \operatorname{div}(t)$, near o_{k_0}. It is lc: in the second case $(F^{(k_0)}, \eta \operatorname{div}(t|_{F^{(k_0)}}))$ is klt since $F^{(k_0)}$ is smooth, and we use inversion of adjunction. By (A) and (B),

$$\big(W_{n_i}, \overline{D}_i^{\kappa} + \eta \operatorname{div}(t)\big) \text{ is lc near 0 for } i \gg 0. \tag{B.58}$$

The exceptional divisor of the blow-up of $0 \in W_{n_i}$ has log discrepancy 2 over W_{n_i}, and $\overline{D}_i \geq 0$, so $\beta_i \leq 2$. Since $(W_{n_i}, \overline{D}_i)$ is klt, there exists a prime divisor E' over W_{n_i} with center 0 and $a(E', W_{n_i}, \overline{D}_i) = e_{E'}\beta_i$. By (B.58) and $c_{ij} \leq \kappa_j$,

$$0 \leq a\big(E', W_{n_i}, \overline{D}_i^{\kappa} + \eta \operatorname{div}(t)\big) \leq a(E', W_{n_i}, \overline{D}_i) - \eta \operatorname{mult}_{E'} t,$$

so $\operatorname{mult}_{E'} t \leq e_{E'}\beta_i/\eta \leq 2e_{E'}/\eta$. For the prime divisor E over W_{N_i} with center 0 below E', (B.55) and $\delta_{ik_0} = 2k_0(1-\sigma_i)$ imply

$$a_i \leq \frac{a(E, W_{N_i}, D_i)}{e_E} \leq \beta_i + \frac{4k_0}{\eta}(1-\sigma_i). \tag{B.59}$$

With $c_0 := 4k_0 \max\{L, 1/\eta\}$, (B.56), (B.57), and (B.59) imply

$$\mu_i \leq a_i \leq \mu_i + c_0(1-\sigma_i) \quad \text{for } i \gg 0. \tag{B.60}$$

Step 7. In this step we show that μ_i has no strictly increasing subsequence. By (C) for $\Gamma' = \Gamma$, each $\mu_{ik,\star}$ belongs to a fixed ACC set, so $(\mu_{ik,\star})_i$ has no strictly increasing subsequence. Since μ_i is at each i one of the $2k_0 + 1$ values $\mu_{ik,\star}$ and β_i, it suffices to show that $(\beta_i)_i$ has no strictly increasing subsequence.

If $F^{(k_0)}$ does not pass through o_{k_0}, then each $H_{*,j}^{(k_0)}$ is a unit by (B.52), so $h_{*,j}^{(k_0)}(0) \neq 0$. The condition $h(0) \neq 0$ on 1-jets is open and holds at the generic point, so by (B), $h_{ij}^{(k_0)}(0) \neq 0$ for $i \gg 0$. Then $\beta_i = \operatorname{mld}((W_{n_i} \ni 0, 0)/G)$, and by Proposition 3.14(3) with the empty boundary, $\beta_i = \operatorname{mld}((W_0 \ni 0, 0)/G)$ for $i \gg 0$.

Assume that $F^{(k_0)}$ passes through o_{k_0}, and suppose that β_i is strictly increasing along an infinite set I' of indices. By (B) and Lemma 2.6(2), $(h_{*,j}^{(k_0)})_j$ and $(H_{*,j}^{(k_0)})_j$ are compatible generic limits of the tuples $(h_{ij}^{(k_0)})_j$ and $(H_{ij}^{(k_0)})_j$ along I'. Since all $R_j^{(k_0)}$ are units, (B.52) implies

$$\sum_j \kappa_j \operatorname{div}(H_{*,j}^{(k_0)}) = \operatorname{div}(F^{(k_0)}) \quad \text{near } o_{k_0},$$

and $F^{(k_0)}$ is smooth at o_{k_0} by **Step 2**, so $F^{(k_0)}$ is a smooth floor whose adjunction $(F^{(k_0)}, 0)$ is klt. By **Step 6**, $(W_{n_i}, \overline{D}_i)$ is klt near 0 for $i \gg 0$, and the labels c_{ij} are increasing with limits κ_j. Hence the data W_{n_i}, G, and $\overline{D}_i$ for $i \in I'$ satisfy Set-up 10.3 and the hypothesis of Theorem B.5 or Theorem B.6, according to the deck parity of $F^{(k_0)}$. Thus $(\beta_i)_{i\in I'}$ is not strictly increasing, a contradiction.

Step 8. In this step we conclude. By **Step 7**, possibly passing to an infinite subset of indices, we may assume that μ_i is decreasing. Let $\mu_* := \lim_i \mu_i$, which exists as $\mu_i \geq 0$. By (B.60) and $\lim_i \sigma_i = 1$,

$$a_i \geq \mu_i \geq \mu_* \qquad \text{and} \qquad \limsup_i a_i \leq \mu_*.$$

Since a_i is strictly increasing, $a_i < \lim_i a_i \leq \mu_*$ for any i, a contradiction. Together with **Step 4**, this proves the theorem. □

Theorem B.11 is the case of normal floors in the proof of Theorem 10.4.

Theorem B.11. *In Set-up 10.3, assume that B^* is lc and has a normal floor F. Then $(a_i)_{i\geq 1}$ is not strictly increasing.*

Proof. Write 0 also for the closed point of $\widetilde{W}_*$, and, as in Set-up 10.3, let F also denote an equation of F. Since $H_j^* = F^{e_j} R_j^*$ for any j and $\sum_j e_j \kappa_j = 1$,

$$B^* = F + \sum_j \kappa_j \operatorname{div}(R_j^*).$$

Since $\widetilde{W}_*$ is regular and F is a normal Cartier prime divisor on $\widetilde{W}_*$ which divides no R_j^*,

$$(K_{\widetilde{W}_*} + B^*)|_F = K_F + B_F.$$

By [Kaw07, Theorem], (F, B_F) is lc. Since $\operatorname{mult}_F B^* = 1$, $(\widetilde{W}_*, B^*)$ is not klt near 0.

Assume, to the contrary, that $(a_i)_{i\geq 1}$ is strictly increasing.

Step 1. In this step we bound the log discrepancies of $(\widetilde{W}_*, B^*)$ over 0 from below.

Since $(a_i)_{i\geq 1}$ is strictly increasing and $a_1 > 0$ (Set-up 10.3), $a_i \geq a_1$ for any i. By Lemma B.7 with $\epsilon = a_1$,

$$a(P, \widetilde{W}_*, B^*) \geq a_1 \quad \text{for any prime divisor } P \text{ over } \widetilde{W}_* \text{ with } \operatorname{center}_{\widetilde{W}_*} P = \{0\}. \tag{B.61}$$

In particular, $\{0\}$ is not an lc center of $(\widetilde{W}_*, B^*)$.

Step 2. In this step we show that F is smooth or Du Val, and we reduce to the case when F is singular and (F, B_F) is not klt.

Since $B^* \geq F$, the pair $(\widetilde{W}_*, F)$ is lc, and by (B.61),

$$a(P, \widetilde{W}_*, F) \geq a(P, \widetilde{W}_*, B^*) \geq a_1$$

for any prime divisor P over $\widetilde{W}_*$ with $\operatorname{center}_{\widetilde{W}_*} P = \{0\}$. Since F is a normal surface and 0 is the only closed point of $\widetilde{W}_*$, $\operatorname{Sing} F \subset \{0\}$. Thus $F\backslash\{0\}$ is a smooth divisor on the regular scheme $\widetilde{W}_*\backslash\{0\}$, and $a(P, \widetilde{W}_*, F) > 0$ for any exceptional prime divisor P over $\widetilde{W}_*$ with $\operatorname{center}_{\widetilde{W}_*} P \neq \{0\}$. Hence $(\widetilde{W}_*, F)$ is plt, so F is klt by [KM98, Theorem 5.50(1)], and since K_F is Cartier, F is smooth or Du Val.

By Set-up 10.3, F is deck-odd or deck-even. If F is smooth and (F, B_F) is klt, then Theorem B.5 or Theorem B.6 contradicts the strict increase of $(a_i)_{i\geq 1}$. If F is smooth and (F, B_F) is not klt, then Theorem B.8 does. Thus F is a singular Du Val surface, and by Theorem B.9, (F, B_F) is not klt. Since F is klt, $B_F \neq 0$, so there exists an index j_0 with

$$R_{j_0}^*(0) = 0.$$

Step 3. In this step we construct the tower of point blow-ups of $\widetilde{W}_*$ along the t-axis and compute the exceptional coefficients.

Let $S_0 := \widetilde{W}_*$, $(p_0, v_0) := (p, v)$, $o_0 := 0$, and $F^{(0)} := F$. For $k \geq 1$, assume that S_{k-1} with regular parameters p_{k-1}, v_{k-1}, t at the closed point $o_{k-1} := (p_{k-1} = v_{k-1} = t = 0)$ and a prime divisor $F^{(k-1)}$ on S_{k-1} are defined. If $o_{k-1} \in F^{(k-1)}$ and $F^{(k-1)}$ is singular at o_{k-1}, then we let

- $\beta_k : S_k \to S_{k-1}$ be the t-chart of the blow-up of S_{k-1} at o_{k-1}, with $p_k := p_{k-1}/t$ and $v_k := v_{k-1}/t$,
- $P_k := (t = 0) \subset S_k$ be the exceptional divisor, and
- $F^{(k)} := (\beta_k)_*^{-1} F^{(k-1)}$.

Otherwise we stop, and we let $k_0 := k - 1$. Since F is singular at 0, $k_0 \geq 1$, and we show below that $k_0 < +\infty$.

The composite $S_k \to \widetilde{W}_*$ is given by $p = t^k p_k$ and $v = t^k v_k$, and maps P_k to 0. Thus $\operatorname{center}_{\widetilde{W}_*} P_k = \{0\}$, $a(P_k, \widetilde{W}_*, 0) = 2k+1$, and $\operatorname{mult}_{P_k} H = \operatorname{wt}_{(k,k,1)}(H)$ for any $H \in \Bbbk[[p, v, t]] \backslash \{0\}$, where $\operatorname{wt}_{(k,k,1)}(H)$ is the weighted order of H for the weights $(k, k, 1)$ on (p, v, t). Let

$$w_{jk} := \operatorname{wt}_{(k,k,1)}(H_j^*), \quad H_j^{(k)} := t^{-w_{jk}} H_j^* \quad \text{and} \quad \lambda_{*k} := \sum_j \kappa_j w_{jk} - 2k.$$

Then $H_j^{(k)}$ is a regular function on S_k not divisible by t, $a(P_k, \widetilde{W}_*, B^*) = 1 - \lambda_{*k}$, and since P_k is the only exceptional prime divisor of $S_k \to \widetilde{W}_*$ on S_k, the crepant pullback B_k^+ of B^* to S_k is

$$B_k^+ = \sum_j \kappa_j \operatorname{div}(H_j^{(k)}) + \lambda_{*k} P_k.$$

Let $\lambda_{*0} := 0$, so that $B_0^+ = B^*$.

We show by induction on k that for any $1 \leq k \leq k_0$,

$$\operatorname{mult}_{o_{k-1}} F^{(k-1)} = 2 \quad \text{and} \quad 0 \leq \lambda_{*k} \leq 1 - a_1.$$

By induction, $\lambda_{*,k-1} \geq 0$, so $B_{k-1}^+ \geq 0$. Since $F^{(k-1)} = (\beta_1 \circ \cdots \circ \beta_{k-1})_*^{-1} F$ and $\operatorname{mult}_F B^* = 1$, $F^{(k-1)}$ is a component of B_{k-1}^+ with coefficient one, so $\operatorname{mult}_{o_{k-1}} F^{(k-1)} \leq \operatorname{mult}_{o_{k-1}} B_{k-1}^+$. Since $S_k \to S_{k-1}$ is the blow-up of the closed point o_{k-1} of a regular threefold near o_{k-1}, and since (S_k, B_k^+) is crepant to $(\widetilde{W}_*, B^*)$, by (B.61),

$$1 - \lambda_{*k} = a(P_k, \widetilde{W}_*, B^*) = 3 - \operatorname{mult}_{o_{k-1}} B_{k-1}^+ \geq a_1.$$

Since $F^{(k-1)}$ is singular at o_{k-1},

$$2 \leq \operatorname{mult}_{o_{k-1}} F^{(k-1)} \leq \operatorname{mult}_{o_{k-1}} B_{k-1}^+ = 2 + \lambda_{*k} \leq 3 - a_1.$$

Thus $\operatorname{mult}_{o_{k-1}} F^{(k-1)} = 2$ and $0 \leq \lambda_{*k} \leq 1 - a_1$.

Since the pullback of F to S_{k-1} is $F^{(k-1)} + \operatorname{mult}_{P_{k-1}}(F) P_{k-1}$, $\operatorname{mult}_{o_{k-1}} F^{(k-1)} = 2$, and $\operatorname{mult}_{o_{k-1}} P_{k-1} = 1$ for $k \geq 2$, we obtain $\operatorname{mult}_{P_k} F = 2 + \operatorname{mult}_{P_{k-1}} F$, so

$$\operatorname{wt}_{(k,k,1)}(F) = \operatorname{mult}_{P_k} F = 2k \quad \text{and} \quad F^{(k)} = \operatorname{div}(t^{-2k} F) \quad \text{for any } 1 \leq k \leq k_0.$$

If $k_0 = +\infty$, then for any monomial $p^{j_1} v^{j_2} t^{j_3}$ with non-zero coefficient in F and any $k \geq 1$, we have $k(j_1 + j_2) + j_3 \geq 2k$, so $j_1 + j_2 \geq 2$. Then $F \in (p, v)^2$, so F is singular along the curve $(p = v = 0)$, which contradicts $\operatorname{Sing} F \subset \{0\}$ (**Step 2**). Thus $k_0 < +\infty$, and $F^{(k_0)}$ is smooth at o_{k_0} or $o_{k_0} \notin F^{(k_0)}$.

For $1 \leq k \leq k_0$, let $r_j(k) := \operatorname{wt}_{(k,k,1)}(R_j^*)$ and $R_j^{(k)} := t^{-r_j(k)} R_j^*$. Then $H_j^{(k)} = (t^{-2k} F)^{e_j} R_j^{(k)}$ and $w_{jk} = 2ke_j + r_j(k)$. Since $\sum_j e_j \kappa_j = 1$,

$$\lambda_{*k} = \sum_j \kappa_j r_j(k).$$

Since $R^*_{j_0}(0) = 0$ (**Step 2**), $r_{j_0}(k) \geq 1$, so $\lambda_{*k} \geq \kappa_{j_0} > 0$. Thus

$$0 < \lambda_{*k} < 1 \quad \text{for any } 1 \leq k \leq k_0. \tag{B.62}$$

Step 4. In this step we perform the $(2,2,2,1)$ tower on W_{N_i} and show that, for $i \gg 0$, a_i is computed on the last axial chart.

By Lemma 10.2(2) with $w = (k,k,1)$,

$$\mathrm{wt}_{(k,k,1)}(h(p^2, -v^2, pv, t)) = \mathrm{wt}_k(h)$$

for any h without monomials divisible by xy, where wt_k is the weighted order of Lemma B.10(4). Since $h_{*,j}$ has no monomials divisible by xy (Set-up 10.3), $\mathrm{wt}_k(h_{*,j}) = w_{jk}$. For d large, on Z_d the condition $\mathrm{wt}_k \geq w_{jk}$ is closed and the condition $\mathrm{wt}_k \leq w_{jk}$ is open, and both hold at the generic point. Hence, possibly discarding finitely many i, for any i,

$$\mathrm{wt}_k(h_{ij}) = w_{jk} \quad \text{for any } 1 \leq j \leq n \text{ and } 1 \leq k \leq k_0,$$

and $N_i \geq 4k_0 + 2$. For any i we use the objects $\varphi_1, \dots, \varphi_{k_0}$, E_k, (x_k, y_k, z_k), $f : Y \to W_{N_i}$, λ_{ik}, and h'_{ij} of Lemma B.10. Since $\mathrm{wt}_k(h_{ij}) = w_{jk}$,

$$\lambda_{ik} = \sum_j c_{ij} w_{jk} - 2k.$$

Since c_{ij} is increasing in i with limit κ_j, λ_{ik} is increasing in i with limit λ_{*k}. By (B.62), possibly discarding finitely many i, we may assume that $0 < \lambda_{ik} < 1$ for any i and any $1 \leq k \leq k_0$.

For $1 \leq k \leq k_0$ and $\star \in \{x, y\}$, let $\widetilde{X}_{k,\star} := \mathbb{A}^3_{s,Z,T}$ with the group $G_{k,\star}$ be the cover of the $\star$-chart of φ_k given by Lemma B.10(2), so that $G_{k,x} = G_k$ and

$$G_{k,y} = \{(g, \alpha) \mid g \in G,\ \alpha^2 = \chi_y(g)\chi_t(g)^{-2(k-1)}\},$$

acting on $\mathbb{A}^3_{s,Z,T}$ by the formula of Lemma B.10(2). Let $X_{k,\star} := \widetilde{X}_{k,\star}/G_{k,\star}$ with the quotient map $\pi_{k,\star} : \widetilde{X}_{k,\star} \to X_{k,\star}$. By Lemma B.10(2), $\widetilde{X}_{k,\star}$, $G_{k,\star}$, and $X_{k,\star}$ do not depend on i. Let $\widetilde{D}_{i,k,\star}$ be the boundary of Lemma B.10(4) on $\widetilde{X}_{k,\star}$, and let $B_{i,k,\star}$ be the $\mathbb{R}$-divisor on $X_{k,\star}$ with

$$K_{\widetilde{X}_{k,\star}} + \widetilde{D}_{i,k,\star} = \pi^*_{k,\star}(K_{X_{k,\star}} + B_{i,k,\star}).$$

Let

$$\Gamma_1 := (\Gamma \backslash \{0\}) \cup \{\lambda_{ik} \mid i \geq 1,\ 1 \leq k \leq k_0\},$$

a DCC set, and let $\Gamma_{1,+}$ be the DCC set of finite sums of elements of Γ_1, including 0. Since $|G_{k,\star}| = 2|G| \leq 2M$ and h'_{ij}, s, and T are regular functions, by the Hurwitz formula

$$B_{i,k,\star} \in \Gamma_2 := \left\{ \frac{e - 1 + b}{e} \;\middle|\; 1 \leq e \leq 2M,\ b \in \Gamma_{1,+} \right\} \backslash \{0\},$$

a DCC set.

By Lemma B.10(2) and (3), Y is covered by the x- and y-charts of φ_k for $1 \leq k \leq k_0$ and by the t-chart $W_{N_i - 4k_0}$ of φ_{k_0}, and $f^{-1}(0)$ is contained in the union of the x- and y-charts and the origin of $W_{N_i-4k_0}$. Since (W_{N_i}, D_i) is klt near 0, the infimum defining a_i is the minimal log discrepancy of the crepant quotient $(W_{N_i}, D_i)/G$ at the image of 0, hence a minimum, so a_i has a computing lift (Definition 2.4).

We claim that, possibly discarding finitely many i, for any i and any computing lift E of a_i, $\mathrm{center}_Y E$ is the origin of $W_{N_i-4k_0}$. Otherwise, since $\#(\{1, \dots, k_0\} \times \{x, y\}) = 2k_0$, possibly passing to a subsequence, there exist $(k, \star)$ and, for any i, a computing lift E_i of a_i such that the generic point of $\mathrm{center}_Y E_i$ belongs to the $\star$-chart of φ_k. Let E'_i be a prime divisor over $\widetilde{X}_{k,\star}$ above E_i (Lemma B.10(5)), let Q_i be the restriction of E'_i, a prime divisor over $X_{k,\star}$, and let η_i

be the generic point of $\mathrm{center}_{X_{k,\star}}(Q_i)$. Since $\pi_{k,\star}$ is crepant and the ramification index of $\pi_{k,\star}$ along E_i' is $e_{E_i'}$, by Lemma B.10(5),

$$a(Q_i, X_{k,\star}, B_{i,k,\star}) = \frac{a(E_i', \widetilde{X}_{k,\star}, \widetilde{D}_{i,k,\star})}{e_{E_i'}} = \frac{a(E_i, W_{N_i}, D_i)}{e_{E_i}} = a_i.$$

Conversely, let Q be a prime divisor over $X_{k,\star}$ with $\mathrm{center}_{X_{k,\star}}(Q) = \overline{\{\eta_i\}}$, and let E' be a prime divisor over $\widetilde{X}_{k,\star}$ whose restriction is Q. Since $\mathrm{center}_{W_{N_i}} E_i = \{0\}$, $\overline{\{\eta_i\}}$ is contained in the image of $f^{-1}(0)$, so $\mathrm{center}_{\widetilde{X}_{k,\star}}(E')$ is contained in the preimage of $f^{-1}(0)$, and by Lemma B.10(5), E' is above a prime divisor E over W_{N_i} with $\mathrm{center}_{W_{N_i}}(E) = \{0\}$. Since $\pi_{k,\star}$ is crepant, by Lemma B.10(5),

$$a(Q, X_{k,\star}, B_{i,k,\star}) = \frac{a(E', \widetilde{X}_{k,\star}, \widetilde{D}_{i,k,\star})}{e_{E'}} = \frac{a(E, W_{N_i}, D_i)}{e_E} \geq a_i.$$

Thus

$$a_i = \mathrm{mld}(X_{k,\star} \ni \eta_i, B_{i,k,\star}) \quad \text{for any } i.$$

Since $X_{k,\star}$ is a normal threefold which does not depend on i and $B_{i,k,\star} \in \Gamma_2$, by Theorem 2.22(1), the set $\{a_i\}_i$ satisfies the ACC, which contradicts the assumption that $(a_i)_{i\geq 1}$ is strictly increasing. This proves the claim.

Let $(x', y', z') := (x_{k_0}, y_{k_0}, z_{k_0})$ be the coordinates of Lemma B.10 on

$$W_{N_i-4k_0} = (x'y' + z'^2 + t^{N_i-4k_0} = 0),$$

with the action of G of Lemma B.10(1). Let

$$\begin{aligned} h_{ij}^{(k_0)} &:= t^{-w_{j,k_0}} h_{ij}(t^{2k_0}x', t^{2k_0}y', t^{2k_0}z', t) \quad \text{for } 1 \leq j \leq n, \\ h_{i,n+1}^{(k_0)} &:= t, \quad \text{and} \quad c_{i,n+1} := \lambda_{ik_0}. \end{aligned} \tag{B.63}$$

By Lemma B.10(4), the crepant pullback of D_i to $W_{N_i-4k_0}$ is

$$D_i' := \sum_{j=1}^{n+1} c_{ij} \,\mathrm{div}(h_{ij}^{(k_0)}).$$

By Lemma B.10(5), any prime divisor E' over $W_{N_i-4k_0}$ with center the origin is above a prime divisor E over W_{N_i} with $\mathrm{center}_{W_{N_i}}(E) = \{0\}$ and $a(E', W_{N_i-4k_0}, D_i')/e_{E'} = a(E, W_{N_i}, D_i)/e_E$. Hence $\mathrm{mld}((W_{N_i-4k_0} \ni 0, D_i')/G) \geq a_i$, and since by the claim any computing lift E of a_i is a prime divisor over $W_{N_i-4k_0}$ with center the origin, for any i,

$$a_i = \mathrm{mld}((W_{N_i-4k_0} \ni 0, D_i')/G). \tag{B.64}$$

Step 5. In this step we show that the data on $W_{N_i-4k_0}$ satisfy Set-up 10.3 with the limit $B_{k_0}^+$, and we derive a contradiction.

Let $\Gamma^\sharp := \Gamma \cup \{\lambda_{ik_0} \mid i \geq 1\}$, which is a DCC subset of $[0,1]$, and let $\kappa_{n+1} := \lambda_{*k_0}$. The data $(W_{N_i-4k_0} \ni 0)$, G, D_i' satisfy Set-up 10.3 for $\Gamma^\sharp$ and M: $c_{i,n+1} = \lambda_{ik_0}$ is increasing in i with limit $\kappa_{n+1} > 0$ by (B.62), and since h_{ij} has no monomials divisible by xy and $\mathrm{wt}_{k_0}(h_{ij}) = w_{j,k_0}$, $h_{ij}^{(k_0)}$ is a polynomial without monomials divisible by $x'y'$ with the character $\chi_{h_{ij}}\chi_t^{-w_{j,k_0}}$ (resp. χ_t for $j = n+1$), which does not depend on i. By (B.64), the values of the data $(W_{N_i-4k_0} \ni 0, G, D_i')$ are the values a_i, which are strictly increasing.

For any i, the tuple $(h_{ij}^{(k_0)})_j$ is given by (B.63), where $w_{j,k_0} = \mathrm{wt}_{k_0}(h_{ij})$ does not depend on i. Let $m := \max_j w_{j,k_0}$. Since for $\mathrm{wt}_{k_0}(h) \geq w_{j,k_0}$ the d-jet of $t^{-w_{j,k_0}} h(t^{2k_0}x', t^{2k_0}y', t^{2k_0}z', t)$ is a linear function of the $(d+m)$-jet of h, the d-jet of $(h_{ij}^{(k_0)})_j$ is the image of the $(d+m)$-jet of $(h_{ij})_j$ under a linear map μ_d. Let Z_d' be the closure of $\mu_d(Z_{d+m})$ for the set Z_{d+m} of Definition 2.5.

Since the image under μ_d of the generic point of Z_{d+m} is the generic point of Z'_d, and the preimage in Z_{d+m} of a non-empty open subset of Z'_d is non-empty and open,

$$h^{(k_0)}_{*,j} := t^{-w_{j,k_0}} h_{*,j}(t^{2k_0}x', t^{2k_0}y', t^{2k_0}z', t) \quad \text{for } 1 \le j \le n \quad \text{and} \quad h^{(k_0)}_{*,n+1} := t$$

form a compatible generic limit of $(h^{(k_0)}_{ij})_j$. Let p', v' be the coordinates of the double cover with $(x', y', z') = (p'^2, -v'^2, p'v')$. Then

$$h^{(k_0)}_{*,j}(p'^2, -v'^2, p'v', t) = t^{-w_{j,k_0}} H^*_j(t^{k_0}p', t^{k_0}v', t) \quad \text{for } 1 \le j \le n,$$

which is the expansion of $H^{(k_0)}_j$ at o_{k_0} under $(p', v') = (p_{k_0}, v_{k_0})$. Since after the substitution $(x', y', z') = (p'^2, -v'^2, p'v')$ the d-jet of the result is a linear function of the d-jet of the input, the closure of the image of Z_{d+m} under the composite of μ_d and the substitution $(x', y', z') = (p'^2, -v'^2, p'v')$ shows, by the argument for Z'_d, that the tuple $(h^{(k_0)}_{*,j}(p'^2, -v'^2, p'v', t))_j$ is a compatible generic limit of the tuples $(h^{(k_0)}_{ij}(p'^2, -v'^2, p'v', t))_j$, as Set-up 10.3 requires. Thus, for the data $(W_{N_i-4k_0} \ni 0, G, D'_i)$, the boundary B^* of Set-up 10.3, which we denote by B'^*, is the pullback of $B^+_{k_0}$ to the completion $S' := \operatorname{Spec} \Bbbk[[p', v', t]]$ of S_{k_0} at o_{k_0}. The group of Set-up 10.3 for the data $(W_{N_i-4k_0} \ni 0, G, D'_i)$ is

$$\widetilde{G}' := \{(g, \alpha') \mid g \in G,\ \alpha'^2 = \chi_x(g)\chi_t(g)^{-2k_0}\},$$

and $(g, \alpha) \to (g, \alpha\chi_t(g)^{-k_0})$ is an isomorphism $\widetilde{G} \to \widetilde{G}'$ compatible with the actions under $p' = t^{-k_0}p$ and $v' = t^{-k_0}v$. Since $(\widetilde{W}_*, B^*)$ is lc near 0, $(S_{k_0}, B^+_{k_0})$ is lc near o_{k_0}. The base change to S' of a log resolution of $(S_{k_0}, B^+_{k_0})$ near o_{k_0} is a log resolution of (S', B'^*), so (S', B'^*) is lc (resp. klt) near 0 if and only if $(S_{k_0}, B^+_{k_0})$ is lc (resp. klt) near o_{k_0}. By **Step 3**, $F^{(k_0)}$ is smooth at o_{k_0} or $o_{k_0} \notin F^{(k_0)}$.

Case (a): $F^{(k_0)}$ is smooth at o_{k_0}. Let F' be the completion of $F^{(k_0)}$ at o_{k_0}, which is $\widetilde{G}'$-stable since F is $\widetilde{G}$-stable. Thus F' is a smooth prime divisor on S' through 0 with $\operatorname{mult}_{F'} B'^* = 1$, that is, a smooth floor of B'^*. Let $e_{n+1} := 0$ and $R'_{n+1} := t$. Then $h^{(k_0)}_{*,j}(p'^2, -v'^2, p'v', t) = F'^{e_j} R'_j$ for $1 \le j \le n+1$, where R'_j is the expansion of $R^{(k_0)}_j$ at o_{k_0} for $j \le n$. Since F divides no R^*_j and $F^{(k_0)} \ne P_{k_0}$, F' divides no R'_j. Since $(S_{k_0}, B^+_{k_0})$ is lc near o_{k_0} and $F^{(k_0)}$ is a smooth component of $B^+_{k_0}$ with coefficient one, by [Kaw07, Theorem], the pair $(F', B_{F'})$ is lc, where

$$B_{F'} := \sum_{j=1}^{n+1} \kappa_j \operatorname{div}(R'_j|_{F'})$$

is the $\mathbb{R}$-divisor B_F of Set-up 10.3 for the data $(W_{N_i-4k_0} \ni 0, G, D'_i)$ and the floor F' (here $\kappa_{n+1} = \lambda_{*k_0}$). If $(F', B_{F'})$ is klt, then by Theorem B.5 or Theorem B.6, according as F' is deck-odd or deck-even, and otherwise by Theorem B.8, for the data $(W_{N_i-4k_0} \ni 0, G, D'_i)$, $(a_i)_{i\ge1}$ is not strictly increasing, a contradiction.

Case (b): $o_{k_0} \notin F^{(k_0)}$. We first show that any lc center of $(\widetilde{W}_*, B^*)$ containing 0 is contained in F.

By **Step 1**, $\{0\}$ is not an lc center. Let $P \ne F$ be a prime divisor on $\widetilde{W}_*$, and assume that $\operatorname{mult}_P B^* = 1$. Then $B^* \ge F + P$. Since F is singular at 0, $\operatorname{mult}_0 F \ge 2$, so

$$\operatorname{mult}_0 B^* \ge \operatorname{mult}_0 F + \operatorname{mult}_0 P \ge 3.$$

For the prime divisor P_1 of **Step 3**, we obtain $a(P_1, \widetilde{W}_*, B^*) = 3 - \operatorname{mult}_0 B^* \le 0$, which contradicts (B.61). Thus F is the only lc center of $(\widetilde{W}_*, B^*)$ of dimension 2 containing 0.

Let V be an lc center of $(\widetilde{W}_*, B^*)$ with $\dim V = 1$ and $0 \in V$, and assume that $V \not\subset F$. Since $\widetilde{W}_*$ is the spectrum of a local ring, $V \cap F = \{0\}$. Let $\mathfrak{a} := \prod_j (H^*_j \mathcal{O}_{\widetilde{W}_*})^{\kappa_j}$. Then $a(P, \widetilde{W}_*, \mathfrak{a}) = a(P, \widetilde{W}_*, B^*)$ for any prime divisor P over $\widetilde{W}_*$, so $(\widetilde{W}_*, \mathfrak{a})$ is lc and not klt at

0, and the lc centers of $(\widetilde{W}_*, \mathfrak{a})$ are the lc centers of $(\widetilde{W}_*, B^*)$. By [Kaw15, Theorem 1.2 and Definition 2.4], the smallest lc center V_0 of $(\widetilde{W}_*, \mathfrak{a})$ at 0 exists and is contained in any lc center of $(\widetilde{W}_*, \mathfrak{a})$ containing 0. Since F and V are lc centers of $(\widetilde{W}_*, \mathfrak{a})$ containing 0,

$$V_0 \subset V \cap F = \{0\},$$

so $\{0\}$ is an lc center of $(\widetilde{W}_*, B^*)$, which contradicts (B.61). Thus $V \subset F$.

We next show that $(S_{k_0}, B^+_{k_0})$ is klt near o_{k_0}. Let P be a prime divisor over S_{k_0} with $o_{k_0} \in \operatorname{center}_{S_{k_0}}(P)$, and assume that $a(P, S_{k_0}, B^+_{k_0}) = 0$. Then $a(P, \widetilde{W}_*, B^*) = 0$, so $\operatorname{center}_{\widetilde{W}_*} P$ is an lc center of $(\widetilde{W}_*, B^*)$ containing 0, and $\operatorname{center}_{\widetilde{W}_*} P \neq \{0\}$ by (B.61). By the previous two paragraphs, $\operatorname{center}_{\widetilde{W}_*} P \subset F$, so $\operatorname{mult}_P F > 0$. Since $F = t^{2k_0} \cdot t^{-2k_0} F$ on S_{k_0} (**Step 3**),

$$\operatorname{mult}_P F = 2k_0 \operatorname{mult}_P t + \operatorname{mult}_P F^{(k_0)}.$$

If $\operatorname{mult}_P t > 0$, then $\operatorname{center}_{S_{k_0}}(P) \subset P_{k_0}$, whose image in $\widetilde{W}_*$ is $\{0\}$, so $\operatorname{center}_{\widetilde{W}_*} P = \{0\}$, a contradiction. Thus $\operatorname{mult}_P F^{(k_0)} > 0$, so

$$\operatorname{center}_{S_{k_0}}(P) \subset F^{(k_0)},$$

which contradicts $o_{k_0} \in \operatorname{center}_{S_{k_0}}(P)$ and $o_{k_0} \notin F^{(k_0)}$. Since $(S_{k_0}, B^+_{k_0})$ is lc near o_{k_0}, $(S_{k_0}, B^+_{k_0})$ is klt near o_{k_0}.

Thus (S', B'^*) is klt near 0, and there exists a real number $\theta > 1$ such that $(S', \theta B'^*)$ is lc near 0. Let $\widetilde{W}'_0 := \mathbb{A}^3$ with coordinates p', v', t. By Theorem 2.18(1) for $\widetilde{W}'_0$, $H = \widetilde{G}'$, $\mathfrak{c} = \mathcal{O}$, the coefficients $\kappa_1, \dots, \kappa_{n+1}$, the compatible generic limit $(h^{(k_0)}_{*,j}(p'^2, -v'^2, p'v', t))_j$ above and θ, there exists an infinite set I of indices such that for any $i \in I$,

$$\Big(\widetilde{W}'_0,\ \theta \sum_{j=1}^{n+1} \kappa_j \operatorname{div}\big(h^{(k_0)}_{ij}(p'^2, -v'^2, p'v', t)\big)\Big)$$

is lc near 0. Since $c_{ij} \le \kappa_j$ for any j, for any $i \in I$,

$$\Big(\widetilde{W}'_0,\ \theta \sum_{j=1}^{n+1} c_{ij} \operatorname{div}\big(h^{(k_0)}_{ij}(p'^2, -v'^2, p'v', t)\big)\Big)$$

is lc near 0. Since $\widetilde{W}'_0 \to W'_0 := (x'y' + z'^2 = 0)$ is étale outside the curve $(p' = v' = 0)$, for any $i \in I$ the pair

$$\Big(W'_0,\ \theta \sum_{j=1}^{n+1} c_{ij} \operatorname{div}\big(h^{(k_0)}_{ij}|_{W'_0}\big)\Big)$$

is lc near 0. Since the data $(W_{N_i - 4k_0} \ni 0, G, D'_i)$ satisfy Set-up 10.3 with labels in the DCC set $\Gamma^\sharp$, by Proposition 3.14(3), the set $\{a_i \mid i \in I\}$ is contained in a set satisfying the ACC. This contradicts the assumption that $(a_i)_{i \ge 1}$ is strictly increasing.

The theorem follows. □

Lemma B.12 (cf. [LR12, Lemma 2.7(2) and Proposition 3.1]). *Let $\Bbbk \supset \mathbb{C}$ be an algebraically closed field, $X := \operatorname{Spec} \Bbbk[[x_1, x_2, x_3]]$ with closed point 0, and S a prime divisor on X. Assume that S is not normal and (X, S) is lc. Then there exist formal coordinates $\zeta_1, \zeta_2, \zeta_3$ of X, that is, a regular system of parameters of $\Bbbk[[x_1, x_2, x_3]]$, such that the ideal of S is generated by a power series f of one of the following forms.*

(1) f has multiplicity 3 at 0.
(2) $f = \zeta_1^2 + \zeta_2^2(\zeta_3^2 + \zeta_2^{q-2})$ for an integer $q \ge 3$.
(3) $f = \zeta_1^2 + \zeta_2^2 \zeta_3$.

Proof. Let $R := \Bbbk[[x_1, x_2, x_3]]$, let $\mathfrak{m}$ be the maximal ideal of R, and let $f \in R$ be a generator of the ideal of S. Then $f \in \mathfrak{m}$, and $\operatorname{mult}_0 f \geq 2$, since if $\operatorname{mult}_0 f = 1$, then $\mathcal{O}_S = R/(f)$ is a regular local ring and S is normal. Let E be the exceptional divisor of the blow-up of X at 0. Since $a(E, X, 0) = 3$, $\operatorname{mult}_E f = \operatorname{mult}_0 f$, and (X, S) is lc,

$$0 \leq a(E, X, S) = 3 - \operatorname{mult}_0 f.$$

Thus $\operatorname{mult}_0 f \in \{2, 3\}$, and if $\operatorname{mult}_0 f = 3$, then (1) holds. In the rest of the proof we assume that $\operatorname{mult}_0 f = 2$.

For formal coordinates u_1, u_2, u_3 of X and $w \in (\mathbb{N}^+)^3$ with $\gcd(w_1, w_2, w_3) = 1$, let V_w be the exceptional divisor of the weighted blow-up of X with weights w on (u_1, u_2, u_3). Then $a(V_w, X, 0) = w_1 + w_2 + w_3$, and $\operatorname{mult}_{V_w}$ is the monomial valuation with weights w on (u_1, u_2, u_3), so $\operatorname{mult}_{V_w} f$ is the least value of $w_1 i_1 + w_2 i_2 + w_3 i_3$ over the monomials $u_1^{i_1} u_2^{i_2} u_3^{i_3}$ of f. Since (X, S) is lc, $a(V_w, X, S) \geq 0$, that is,

$$\operatorname{mult}_{V_w} f \leq w_1 + w_2 + w_3. \tag{B.65}$$

As $\operatorname{mult}_0 f = 2$, the quadratic part f_2 of f is non-zero. Since $\Bbbk$ is infinite, f_2 does not vanish at some point of $\Bbbk^3$, so after a linear change of the coordinates we may assume that $f_2(1, 0, 0) \neq 0$. We denote the new coordinates by x, y, z, and let $R' := \Bbbk[[y, z]]$. Then $f(x, 0, 0)$ has order 2 in x, so by the Weierstrass preparation theorem, f is a unit times $x^2 + \varphi_1 x + \varphi_0$ with $\varphi_1, \varphi_0 \in (y, z)R'$. Since

$$x^2 + \varphi_1 x + \varphi_0 = \Big(x + \frac{\varphi_1}{2}\Big)^2 + \varphi_0 - \frac{\varphi_1^2}{4},$$

possibly replacing f with $x^2 + \varphi_1 x + \varphi_0$ and x with $x + \frac{\varphi_1}{2}$, we may assume that

$$f = x^2 + g \quad \text{with } g \in R'.$$

Setting $x = 0$ in $f \in \mathfrak{m}^2$, we obtain $g \in (y, z)^2$. If $\operatorname{mult}_0 g \geq 5$, then for $w = (5, 2, 2)$ on (x, y, z) the monomial x^2 of f has weight 10 and any monomial of g has weight at least $2 \operatorname{mult}_0 g \geq 10$, so $\operatorname{mult}_{V_w} f = 10 > 5 + 2 + 2$, which contradicts (B.65). Thus

$$\operatorname{mult}_0 g \leq 4.$$

We show that $\pi^2 \mid g$ for some prime element π of the factorial ring R'. Assume the contrary. By the Weierstrass division theorem, $\mathcal{O}_S = R/(f)$ is a free R'-module with basis $1, x$, and $x^2 = -g$ in $\mathcal{O}_S$. Let Q be the fraction field of R'. Then the fraction field of the domain $\mathcal{O}_S$ is $Q \oplus Qx$, and the Q-automorphism σ of $Q \oplus Qx$ with $\sigma(x) = -x$ preserves $\mathcal{O}_S$. Let $\xi = r_0 + r_1 x$, with $r_0, r_1 \in Q$, be integral over $\mathcal{O}_S$. Since $\mathcal{O}_S$ is finite over R' and $\sigma(\mathcal{O}_S) = \mathcal{O}_S$, ξ and $\sigma(\xi)$ are integral over R'. Thus $2r_0 = \xi + \sigma(\xi)$ and $r_0^2 + g r_1^2 = \xi\sigma(\xi)$, which are elements of Q integral over the normal ring R', belong to R', so $r_0 \in R'$ and $g r_1^2 \in R'$. Writing $r_1 = \gamma_1/\gamma_2$ with coprime $\gamma_1, \gamma_2 \in R'$, we obtain $\gamma_2^2 \mid g$, so γ_2 is a unit and $r_1 \in R'$. Hence $\xi \in \mathcal{O}_S$, and S is normal, a contradiction.

Since $\Bbbk \supset \mathbb{C}$ is algebraically closed, any unit of R' has a k-th root in R' for any $k \in \mathbb{N}^+$. Moreover, g is not a square in R': if $g = \gamma^2$ with $\gamma \in R'$, then $f = (x + \sqrt{-1}\gamma)(x - \sqrt{-1}\gamma)$ with $x \pm \sqrt{-1}\gamma \in \mathfrak{m}$, which contradicts the primality of f. Fix a prime element π of R' with $\pi^2 \mid g$, and write $g = \pi^2 h$ with $h \in R'$. If h is a unit, then h has a square root in R' and g is a square, which is not the case. Thus $h \in (y, z)R'$, so $\operatorname{mult}_0 h \geq 1$, and since

$$2 \operatorname{mult}_0 \pi + \operatorname{mult}_0 h = \operatorname{mult}_0 g \leq 4,$$

we have $\operatorname{mult}_0 \pi = 1$. Hence there exists a linear form λ in y, z with $R' = \Bbbk[[\pi, \lambda]]$. Possibly replacing (y, z) with (π, λ), we may assume that

$$f = x^2 + y^2 h \quad \text{with } h \in (y, z)R'.$$

Let $n \in \mathbb{N}^+ \cup \{+\infty\}$ be the order of $h(0, z) \in \Bbbk[[z]]$, with $n = +\infty$ if $h(0, z) = 0$, and write $h = h(0, z) + y h_1$ with $h_1 \in R'$. Assume that $n \geq 3$, and let $w := (10, 7, 2)$ on (x, y, z). The monomial x^2 of f has weight 20, any monomial of $y^2 h(0, z)$ has weight at least $14 + 2n \geq 20$,

and any monomial of y^3h_1 has weight at least 21. Thus $\mathrm{mult}_{V_w} f = 20 > 10+7+2$, which contradicts (B.65). Therefore $n \in \{1,2\}$.

If $n=1$, then the linear part of h is not a multiple of y, so $(\zeta_1,\zeta_2,\zeta_3) := (x,y,h)$ are formal coordinates of X, and $f = \zeta_1^2 + \zeta_2^2\zeta_3$, which is (3).

If $n=2$, then by the Weierstrass preparation theorem in z, $h = \upsilon\cdot(z^2+\psi_1 z+\psi_0)$ for a unit υ of R' and $\psi_1,\psi_0 \in y\Bbbk[[y]]$. Let $z' := z + \frac{\psi_1}{2}$ and $\psi := \psi_0 - \frac{\psi_1^2}{4} \in y\Bbbk[[y]]$, so that $h = \upsilon\cdot(z'^2+\psi)$. If $\psi = 0$, then $g = y^2h = \upsilon\cdot(yz')^2$ is a square in R', as υ has a square root in R', which is not the case. Thus $\psi = y^{q-2}\mu$ for an integer $q \geq 3$ and a unit μ of $\Bbbk[[y]]$. Let $\alpha,\beta \in R'$ be units with $\alpha^q = \upsilon\mu$ and $\beta^2 = \upsilon\alpha^{-2}$, and let $(\zeta_1,\zeta_2,\zeta_3) := (x,\alpha y,\beta z')$, which are formal coordinates of X. Then

$$\zeta_2^2(\zeta_3^2+\zeta_2^{q-2}) = \alpha^2\beta^2y^2z'^2 + \alpha^q y^q = \upsilon y^2(z'^2+\mu y^{q-2}) = y^2h,$$

so $f = \zeta_1^2 + \zeta_2^2(\zeta_3^2+\zeta_2^{q-2})$, which is (2). The lemma follows. □

Theorem B.13 is the case of non-normal floors in the proof of Theorem 10.4.

Theorem B.13. *In Set-up 10.3, assume that B^* is lc and has a non-normal floor F. Then $(a_i)_{i\geq 1}$ is not strictly increasing.*

Proof. Assume that there exists a sequence of data as in Set-up 10.3 such that B^* is lc, B^* has a non-normal floor F, and $(a_i)_{i\geq 1}$ is strictly increasing. We derive a contradiction. Since (W_{N_1}, D_1) is klt near 0,

$$a_i \geq a_1 > 0 \quad \text{for any } i.$$

Since $\sum_j e_j\kappa_j = 1$,

$$B^* = \mathrm{div}(F) + \sum_j \kappa_j \,\mathrm{div}(R_j^*). \tag{B.66}$$

Step 1. In this step we bound the log discrepancies of B^* at the prime divisors with center the closed point. By Lemma B.7 with $\epsilon = a_1$,

$$a(V, \widetilde{W}_*, B^*) \geq a_1 > 0 \tag{B.67}$$

for any prime divisor V over $\widetilde{W}_*$ with center the closed point.

Step 2. In this step we show that F is a pinch point. By (B.66), $B^* \geq \mathrm{div}(F)$, so $(\widetilde{W}_*, \mathrm{div}(F))$ is lc. Since F is not normal, by Lemma B.12 there exist formal coordinates ζ_1,ζ_2,ζ_3 of $\widetilde{W}_*$ such that the ideal (F) is generated by a power series f of one of the forms (1)–(3) of Lemma B.12. In case (1), let E be the exceptional divisor of the blow-up of the closed point. Then $a(E,\widetilde{W}_*,0) = 3 = \mathrm{mult}_E F$, so $a(E,\widetilde{W}_*,B^*) \leq 0$ by (B.66), which contradicts (B.67). In case (2), let V be the prime divisor over $\widetilde{W}_*$ with center the closed point given by the monomial valuation in ζ_1,ζ_2,ζ_3 whose weight vector is the primitive integral vector on the ray of $(q,2,q-2)$. Each of the monomials ζ_1^2, $\zeta_2^2\zeta_3^2$, and ζ_2^q of f has weight equal to the sum of the entries of the weight vector of V, so

$$a(V,\widetilde{W}_*,0) = \mathrm{mult}_V f,$$

and $a(V,\widetilde{W}_*,B^*) \leq 0$ by (B.66), which contradicts (B.67). Thus case (3) holds, that is,

$$(F) = (\zeta_1^2 + \zeta_2^2\zeta_3).$$

Let $\Sigma := (\zeta_1 = \zeta_2 = 0)$, let $\widetilde{F} := \mathrm{Spec}\,\Bbbk[[\widetilde{\zeta}_1,\widetilde{\zeta}_2]]$, and let $\nu : \widetilde{F} \to F$ be the morphism given by

$$(\zeta_1,\zeta_2,\zeta_3) = (\widetilde{\zeta}_1\widetilde{\zeta}_2, \widetilde{\zeta}_1, -\widetilde{\zeta}_2^2).$$

The morphism ν is finite and birational, since $\widetilde{\zeta}_2^2 = -\zeta_3$ and $\widetilde{\zeta}_2 = \zeta_1/\zeta_2$, and $\widetilde{F}$ is smooth, so ν is the normalization of F. The conductor of ν is the ideal (ζ_1,ζ_2) of F, so Σ is the non-normal locus of F. Moreover $\widetilde{\Sigma} := \nu^{-1}(\Sigma)_{\mathrm{red}} = (\widetilde{\zeta}_1 = 0)$, the ring $\mathcal{O}_\Sigma$ is $\Bbbk[[\zeta_3|_\Sigma]]$, and

$$\nu|_{\widetilde{\Sigma}}^*(\zeta_3|_\Sigma) = -\widetilde{\zeta}_2^2|_{\widetilde{\Sigma}}.$$

Step 3. In this step we show that

$$\Sigma = (p = v = 0). \tag{B.68}$$

The deck involution ι is the element $(1,-1)$ of $\widetilde{G}$, so ι preserves F, hence preserves the non-normal locus Σ of F. By the universal property of the normalization, $\iota|_F$ lifts to an automorphism $\widetilde{\iota}$ of $\widetilde{F}$ with $\widetilde{\iota}^2 = \mathrm{id}_{\widetilde{F}}$, and $\widetilde{\iota}$ preserves $\widetilde{\Sigma}$. Let $\beta \in \Bbbk$ be the number with $\widetilde{\iota}^*\widetilde{\zeta}_2 \equiv \beta\widetilde{\zeta}_2$ modulo $(\widetilde{\zeta}_1, \widetilde{\zeta}_2^2)$. Since $\widetilde{\iota}^2 = \mathrm{id}_{\widetilde{F}}$, $\beta^2 = 1$. On $\widetilde{\Sigma}$,

$$\nu^*\iota^*\zeta_3 = \widetilde{\iota}^*\nu^*\zeta_3 \equiv -\beta^2\widetilde{\zeta}_2^2 = -\widetilde{\zeta}_2^2 \pmod{\widetilde{\zeta}_2^3}.$$

Since $\nu|_{\widetilde{\Sigma}}^*$ is the injection $\Bbbk[[\zeta_3|_\Sigma]] \to \Bbbk[[\widetilde{\zeta}_2]]$ given by $\zeta_3 \to -\widetilde{\zeta}_2^2$, comparing the coefficients of $\widetilde{\zeta}_2^2$ we obtain $\iota^*\zeta_3 \equiv \zeta_3$ modulo ζ_3^2 in $\mathcal{O}_\Sigma$. Assume that ι acts non-trivially on Σ, and let $\gamma\zeta_3^m$, with $\gamma \in \Bbbk\backslash\{0\}$ and an integer $m \geq 2$, be the lowest term of $\iota^*\zeta_3 - \zeta_3$ in $\mathcal{O}_\Sigma = \Bbbk[[\zeta_3]]$. Then

$$(\iota^*)^2\zeta_3 \equiv \zeta_3 + 2\gamma\zeta_3^m \pmod{\zeta_3^{m+1}},$$

which contradicts $(\iota|_\Sigma)^2 = \mathrm{id}_\Sigma$. Thus ι acts trivially on Σ. In particular $2p = p - \iota^*p$ and $2v = v - \iota^*v$ vanish on Σ, so $\Sigma \subset (p = v = 0)$. Since Σ and $(p = v = 0)$ are integral curves, (B.68) holds.

Step 4. In this step we write F in a normal form. By **Step 2** and (B.68),

$$(F) = (\zeta_1^2 + \zeta_2^2\zeta_3) \subset (\zeta_1, \zeta_2)^2 = (p, v)^2.$$

Hence the homogeneous part F_2 of degree 2 of F belongs to $\Bbbk[p, v]$, and F_2 is a non-zero constant times the square of the linear part of ζ_1. Thus F_2 is a non-zero constant times u_1^2 for a linear form $u_1 \in \Bbbk p + \Bbbk v$. Since F_2 is semi-invariant, u_1 is a $\widetilde{G}$-eigenvector. Since $\widetilde{G}$ acts diagonally on $\Bbbk p + \Bbbk v$, there exists a $\widetilde{G}$-eigenvector $u_2 \in \Bbbk p + \Bbbk v$ such that the linear change $(p, v) \to (u_1, u_2)$ is a linear automorphism of determinant one commuting with $\widetilde{G}$, or the composite of the interchange of p and v with such an automorphism. Let χ_{u_1} and χ_{u_2} be the characters of u_1 and u_2, let $\rho := \chi_{u_1}/\chi_{u_2}$, and let n_ρ be the order of ρ in the group of characters of $\widetilde{G}$.

In the coordinates (u_1, u_2, t), $F(u_1, 0, 0)$ has order 2 in u_1. By the Weierstrass preparation theorem, F is a unit times $u_1^2 + g_1u_1 + g_0$ with $g_1, g_0 \in \Bbbk[[u_2, t]]$. Possibly replacing F with $u_1^2 + g_1u_1 + g_0$ and each R_j^* with a unit multiple, we may assume that $F = u_1^2 + g_1u_1 + g_0$. For any $\widetilde{g} \in \widetilde{G}$, the power series $\chi_{u_1}(\widetilde{g})^{-2}\widetilde{g}^*F$ is monic of degree 2 in u_1 with coefficients in $\Bbbk[[u_2, t]]$ and generates (F), so $\widetilde{g}^*F = \chi_{u_1}(\widetilde{g})^2F$ by the uniqueness in the Weierstrass preparation theorem. Thus g_1 and g_0 are semi-invariant with characters χ_{u_1} and $\chi_{u_1}^2$. Since $\chi_{u_1}(\iota) = -1$, F is ι-invariant, that is, F is deck-even. Since ι acts on (u_2, t) by $(-u_2, t)$, g_1 is odd in u_2 and g_0 is even in u_2. Since $F \in (u_1, u_2)^2$, $g_0 = F(0, u_2, t) \in (u_2^2)$. Hence there exist $\phi, \theta \in \Bbbk[[u_2^2, t]]$ such that

$$g_1 = 2u_2\phi(u_2^2, t), \quad g_0 = u_2^2\big(\phi(u_2^2, t)^2 - \theta(u_2^2, t)\big).$$

Since u_2 has the character χ_{u_2}, ϕ and θ are semi-invariant with the characters ρ and ρ^2. Let $u_1' := u_1 + u_2\phi(u_2^2, t)$. Then (u_1', u_2, t) are formal coordinates, and

$$F = u_1'^2 - u_2^2\theta(u_2^2, t). \tag{B.69}$$

Let $\mathcal{I}_\Sigma$ be the ideal of Σ. By (B.68), $\mathcal{I}_\Sigma = (\zeta_1, \zeta_2) = (u_1', u_2)$, and $\mathcal{I}_\Sigma/\mathcal{I}_\Sigma^2$ is a free $\mathcal{O}_\Sigma$-module of rank 2. For a power series $f \in \mathcal{I}_\Sigma^2$, the class of f in $\mathcal{I}_\Sigma^2/\mathcal{I}_\Sigma^3$ is a quadratic form $\mathfrak{q}_f$ over $\mathcal{O}_\Sigma$ on $\mathcal{I}_\Sigma/\mathcal{I}_\Sigma^2$. The ideal of $\mathcal{O}_\Sigma$ generated by the determinant of $\mathfrak{q}_f$ in a basis of $\mathcal{I}_\Sigma/\mathcal{I}_\Sigma^2$ depends only on the ideal (f): a change of basis multiplies the determinant by the square of a unit, and multiplying f by a unit multiplies the determinant by a unit. For $f = \zeta_1^2 + \zeta_2^2\zeta_3$ and the basis ζ_1, ζ_2, $\mathfrak{q}_f = \zeta_1^2 + \zeta_3|_\Sigma\zeta_2^2$, whose determinant $\zeta_3|_\Sigma$ generates the maximal ideal of $\mathcal{O}_\Sigma$. For the basis u_1', u_2, $\mathfrak{q}_F = u_1'^2 - \theta(0, t)u_2^2$ by (B.69), with determinant $-\theta(0, t) \in \mathcal{O}_\Sigma = \Bbbk[[t]]$. Since $(F) = (\zeta_1^2 + \zeta_2^2\zeta_3)$, $\theta(0, t)$ generates $t\Bbbk[[t]]$, that is, there exists $\xi \in \Bbbk\backslash\{0\}$ such that

$$\theta(u_2^2, t) - \xi t \in (u_2^2, t^2). \tag{B.70}$$

Since θ has character ρ^2 and t has character χ_t,

$$\chi_t = \rho^2. \tag{B.71}$$

Step 5. In this step we bound $\sum_j k_j\kappa_j$ from above and fix the weights. Let E_Σ be the exceptional divisor of the blow-up of Σ. Then $a(E_\Sigma, \widetilde{W}_*, 0) = 2$, and $\operatorname{mult}_{E_\Sigma} F = 2$ by (B.69). Since B^* is lc, (B.66) implies

$$0 \le a(E_\Sigma, \widetilde{W}_*, B^*) = -\sum_j \kappa_j \operatorname{mult}_{E_\Sigma} R_j^*.$$

Since $\kappa_j > 0$, $R_j^* \notin \mathcal{I}_\Sigma$ for any j. Thus for any j there exists an integer $k_j \ge 0$ such that $R_j^*(0,0,t)$ is t^{k_j} times a unit of $\Bbbk[[t]]$. Let

$$\sigma_i := \sum_j k_j c_{ij}, \quad \sigma_* := \sum_j k_j\kappa_j.$$

Let $\ell > \max_j k_j$ be an integer, let $w^{(\ell)} := (2\ell+1, 2\ell, 2)$, and let V_ℓ be the prime divisor over $\widetilde{W}_*$ with center the closed point given by the monomial valuation with weight vector $w^{(\ell)}$ on (u_1', u_2, t). By (B.70), the terms of $u_2^2\theta(u_2^2, t) - \xi u_2^2 t$ are multiples of u_2^4 or $u_2^2t^2$, so they have $w^{(\ell)}$-weight at least $4\ell+4$. Hence $\operatorname{in}_{w^{(\ell)}} F = u_1'^2 - \xi u_2^2 t$, of weight $4\ell+2$. The terms of $R_j^* - R_j^*(0,0,t)$ belong to (u_1', u_2), so they have weight at least $2\ell > 2k_j$, and $\operatorname{in}_{w^{(\ell)}} R_j^*$ is a non-zero multiple of t^{k_j}. Thus $\operatorname{mult}_{V_\ell} H_j^* = (4\ell+2)e_j + 2k_j$. Since $a(V_\ell, \widetilde{W}_*, 0) = 4\ell+3$ and $\sum_j e_j\kappa_j = 1$,

$$a(V_\ell, \widetilde{W}_*, B^*) = 4\ell + 3 - \sum_j \kappa_j \operatorname{mult}_{V_\ell} H_j^* = 1 - 2\sigma_*.$$

By (B.67),

$$0 \le \sigma_* < \frac{1}{2}. \tag{B.72}$$

Let $N_{\widetilde{G}}$ be the lattice of Definition 2.3 for the diagonal action of $\widetilde{G}$ in the coordinates (u_1, u_2, t). Let $\widetilde{g} \in \widetilde{G}$ be an element with $\rho(\widetilde{g}) = e^{2\pi\sqrt{-1}/n_\rho}$. Let w_2 be a real number with $e^{2\pi\sqrt{-1}w_2} = \chi_{u_2}(\widetilde{g})$ and $w_2 > \max\{1, 2k_1, \dots, 2k_n\}/n_\rho$, and let

$$w = (w_1, w_2, w_3) := \Big(w_2 + \frac{1}{n_\rho}, w_2, \frac{2}{n_\rho}\Big). \tag{B.73}$$

By (B.71),

$$\chi_{u_1}(\widetilde{g}) = \rho(\widetilde{g})\chi_{u_2}(\widetilde{g}) = e^{2\pi\sqrt{-1}w_1} \quad \text{and} \quad \chi_t(\widetilde{g}) = \rho(\widetilde{g})^2 = e^{2\pi\sqrt{-1}w_3},$$

so $w \in N_{\widetilde{G}} \cap \mathbb{R}^3_{>0}$. Moreover

$$2w_2 > w_3 \quad \text{and} \quad w_2 > k_j w_3 \text{ for any } j.$$

The vector w is primitive in $N_{\widetilde{G}}$: for any $\omega \in N_{\widetilde{G}}$, $e^{2\pi\sqrt{-1}(\omega_1-\omega_2)}$ is a value of ρ, so $\omega_1 - \omega_2 \in \frac{1}{n_\rho}\mathbb{Z}$, while $w = mw'$ with an integer $m \ge 2$ and $w' \in N_{\widetilde{G}}$ would imply $w_1' - w_2' = \frac{1}{mn_\rho}$.

Let $\mu := \max_j(2e_jw_1 + k_jw_3)$, and let d be a positive integer with $d\min\{w_2, w_3\} > \mu$. Since $w_1 > w_2$, any monomial in u_1, u_2, t of degree at least d has w-weight larger than μ, and $d > k_j$ for any j. Let ϕ_0 be the sum of the terms of $\phi(u_2^2, t)$ of degree less than d in u_2, t. It is a polynomial in u_2^2, t, semi-invariant with character ρ, and $u_1' - (u_1 + u_2\phi_0) \in (u_1, u_2, t)^{d+1}$. Thus, by (B.69), in the coordinates $(u_1 + u_2\phi_0, u_2, t)$,

$$H_j^* \equiv \big((u_1 + u_2\phi_0)^2 - u_2^2\theta(u_2^2, t)\big)^{e_j} R_j^* \pmod{(u_1, u_2, t)^{d+1}} \quad \text{for any } j. \tag{B.74}$$

Step 6. In this step we transport the normal form of **Step 4** to the members h_{ij} by automorphisms of $\mathbb{A}^4$ fixing t and $xy + z^2$. Let $\mathcal{J}_{d+1}$ and $Z_{d+1} \subset \mathcal{J}_{d+1}$ be as in Definition 2.5 on $\mathbb{A}^3$ for the tuples $(H_{ij})_j$ of Set-up 10.3 and the group $\widetilde{G}$.

Let $\mathcal{Z}$ be the set of tuples

$$\big((\overline{H}_j)_j,\ (u_1,u_2),\ \phi_0',\ \xi',\ \theta',\ (R_j')_j\big)$$

as follows.

(a) $(\overline{H}_j)_j$ is a point of $\mathcal{J}_{d+1}$.
(b) $(p,v)\to(u_1,u_2)$ is a linear change of the type in **Step 4** with u_1,u_2 semi-invariant with the characters χ_{u_1},χ_{u_2}.
(c) ϕ_0' is a polynomial in u_2^2,t of degree less than d with the character ρ.
(d) ξ' is a constant.
(e) θ' is a polynomial in u_2^2,t of degree at most d.
(f) R_j' is a polynomial in u_1,u_2,t of degree at most d for any j.

We require the following conditions.

(1) For any j,
$$\overline{H}_j\equiv\big((u_1+u_2\phi_0')^2-u_2^2\theta'\big)^{e_j}R_j'\pmod{(u_1,u_2,t)^{d+1}}.$$
(2) $\theta'-\xi' t\in(u_2^2,t^2)$ and $\xi'\neq 0$.
(3) For any j, the coefficients of $1,t,\dots,t^{k_j-1}$ in $R_j'(0,0,t)$ vanish, and the coefficient of t^{k_j} does not.

Conditions (1)–(3) are finitely many polynomial equations and inequations in the coefficients, so $\mathcal{Z}$ is constructible, and so is the image of $\mathcal{Z}$ in $\mathcal{J}_{d+1}$ by Chevalley's theorem. By (B.70) and (B.74), the $(d+1)$-jet of $(H_j^*)_j$, together with the linear change of **Step 4**, ϕ_0, ξ and the truncations in degree at most d of θ and of the R_j^*, is a point of $\mathcal{Z}$ with coordinates in $\Bbbk$. Since $(H_j^*)_j$ is a compatible generic limit of the tuples $(H_{ij})_j$ (Set-up 10.3), the image of $\mathcal{Z}$ contains the generic point of Z_{d+1}, hence contains a non-empty open subset of Z_{d+1}. Therefore, for all but finitely many i, the fiber of $\mathcal{Z}$ over the $(d+1)$-jet of $(H_{ij})_j$ is non-empty and constructible, hence has a $\mathbb{C}$-point. For any such i, let $(p,v)\to(u_1,u_2)$, $\phi_{0,i}$, $\xi_i\neq 0$, θ_i, and R_{ij} be the coordinates of such a $\mathbb{C}$-point.

By Lemma B.4(1)(b), the linear change $(p,v)\to(u_1,u_2)$ is induced through

$$(x,y,z)=(p^2,-v^2,pv)$$

by a linear automorphism τ_i of $\mathbb{A}^4$ fixing t and $xy+z^2$ which conjugates G to a diagonal group, with $\chi_x=\chi_{u_1}^2$, $\chi_y=\chi_{u_2}^2$ and $\chi_z=\chi_{u_1}\chi_{u_2}$ in the new coordinates. Since $\phi_{0,i}(-y,t)$ is semi-invariant with character $\rho=\chi_z/\chi_y$, Lemma B.4(1)(a), with x and y interchanged, implies that

$$\tau_i'\colon(x,y,z,t)\to\big(x+2z\phi_{0,i}(-y,t)-y\phi_{0,i}(-y,t)^2,\ y,\ z-y\phi_{0,i}(-y,t),\ t\big)$$

is an automorphism of $\mathbb{A}^4$ preserving $xy+z^2$ and t and commuting with G. It is induced by $(u_1,u_2,t)\to(u_1+u_2\phi_{0,i}(u_2^2,t),u_2,t)$, as

$$(u_1+u_2\phi_{0,i})^2=x+2z\phi_{0,i}-y\phi_{0,i}^2\quad\text{and}\quad(u_1+u_2\phi_{0,i})u_2=z-y\phi_{0,i}.$$

The automorphisms τ_i and τ_i' preserve W_N for any N. We replace the coordinates with their images under τ_i and τ_i', G with the conjugate diagonal group, and each h_{ij} with the polynomial without monomials divisible by xy whose difference with h_{ij} belongs to $(xy+z^2+t^{N_i})$ in the new coordinates, which is G-semi-invariant with the character of h_{ij} by its uniqueness, as G is diagonal and preserves $(xy+z^2+t^{N_i})$. By Lemma B.4(1)(c), the last replacement changes $h_{ij}(u_1^2,-u_2^2,u_1u_2,t)$ by an element of (t^{N_i}). Passing to the infinite set of such i with $N_i>d$, we may assume that the following hold for any i.

(1) G is diagonal, π_0 is $(x,y,z,t)=(u_1^2,-u_2^2,u_1u_2,t)$ on $\widetilde{W}_0=\mathbb{A}^3_{u_1,u_2,t}$, and $\widetilde{G}$ acts on (u_1,u_2,t) diagonally with characters $\chi_{u_1},\chi_{u_2},\chi_t$.
(2) (W_{N_i},D_i) satisfies the hypotheses of Proposition 3.14 with $N=N_i$, $D_i=\sum_j c_{ij}\operatorname{div}(h_{ij})$ with the labels c_{ij} of Set-up 10.3, and $a_i=\operatorname{mld}((W_{N_i}\ni 0,D_i)/G)$.

(3) For any j,

$$h_{ij}(u_1^2,-u_2^2,u_1u_2,t)\equiv\big(u_1^2-u_2^2\theta_i(u_2^2,t)\big)^{e_j}R_{ij}\pmod{(u_1,u_2,t)^{d+1}},$$

where $\theta_i(u_2^2,t)-\xi_i t\in(u_2^2,t^2)$ and $R_{ij}(0,0,t)\equiv\lambda_{ij}t^{k_j}$ modulo t^{k_j+1} with $\lambda_{ij}\neq 0$.

Since $2w_2>w_3$,

$$\mathrm{in}_w\big(u_1^2-u_2^2\theta_i(u_2^2,t)\big)=u_1^2-\xi_iu_2^2t,\quad\text{of weight } 2w_1=2w_2+w_3.$$

Since $w_1>w_2>k_jw_3$, $\mathrm{in}_w R_{ij}=\lambda_{ij}t^{k_j}$. Since monomials of degree at least $d+1$ have w-weight larger than $\mu\geq 2e_jw_1+k_jw_3$,

$$\mathrm{in}_w\big(h_{ij}(u_1^2,-u_2^2,u_1u_2,t)\big)=\lambda_{ij}(u_1^2-\xi_iu_2^2t)^{e_j}t^{k_j}\quad\text{for any } i,j. \tag{B.75}$$

Step 7. In this step we show that

$$a_i\geq\frac{1-2\sigma_i}{n_\rho}\quad\text{for any } i. \tag{B.76}$$

Since π_0 is quasi-étale and $\widetilde{G}\to G$ is surjective with kernel $\{\mathrm{id},\iota\}$, we have $\widetilde{W}_0/\widetilde{G}=W_0/G$, and for any boundary D on W_0 the quotient pair of (W_0,D) by G is the quotient pair of $(\widetilde{W}_0,D|_{\widetilde{W}_0})$ by $\widetilde{G}$, where $D|_{\widetilde{W}_0}$ is the pullback of D. In particular

$$\mathrm{mld}((W_0\ni 0,D)/G)=\mathrm{mld}((\widetilde{W}_0\ni 0,D|_{\widetilde{W}_0})/\widetilde{G}),$$

and (W_0,D) is lc near 0 if and only if $(\widetilde{W}_0,D|_{\widetilde{W}_0})$ is lc near 0.

Let $\alpha_i:=\sum_j e_jc_{ij}$. Since c_{ij} is increasing in i with limit κ_j, $\alpha_i\leq\sum_j e_j\kappa_j=1$ and $\sigma_i\leq\sigma_*<\frac{1}{2}$ by (B.72), and $\lim_i\alpha_i=1$, $\lim_i\sigma_i=\sigma_*$. Let

$$\Delta_i:=\alpha_i\,\mathrm{div}(u_1^2-\xi_iu_2^2t)+\sigma_i\,\mathrm{div}(t)\quad\text{on }\mathbb{A}^3_{u_1,u_2,t},\qquad a_i':=\mathrm{mld}((\mathbb{A}^3\ni 0,\Delta_i)/\widetilde{G}).$$

By (B.71), $u_1^2-\xi_iu_2^2t$ is semi-invariant. We use [Liu26a, Proposition 8.11(2)(3)] in the notation of [Liu26a, Set-up 8.4] on $\mathbb{A}^3$, with the group $\widetilde{G}$, the semi-invariant polynomials $f_1:=u_1^2-\xi_iu_2^2t$ and $f_2:=t$, which are reduced at 0 and coprime, and the coefficients $b_1:=\alpha_i\leq 1$ and $b_2:=\sigma_i<\frac{1}{2}$ (omitting f_2 if $\sigma_i=0$). For $\omega\in\mathbb{R}^3_{\geq 0}$ let

$$\psi_i(\omega):=\omega_1+\omega_2+\omega_3-\alpha_i\min\{2\omega_1,2\omega_2+\omega_3\}-\sigma_i\omega_3,$$

so that

$$\psi_i(\omega)=\omega_1+\omega_2+\omega_3-b_1\,\mathrm{wt}_\omega(f_1)-b_2\,\mathrm{wt}_\omega(f_2).$$

For any $\omega\in\mathbb{R}^3_{>0}$, the boundary

$$\alpha_i\,\mathrm{div}(\mathrm{in}_\omega f_1)+\sigma_i\,\mathrm{div}(t)$$

on the torus $\mathbb{G}_m^3$ is lc: $\mathrm{div}(t)$ and the divisors of monomials are empty on the torus, and if $\mathrm{in}_\omega f_1=f_1$, then f_1 is smooth on the torus since $\partial f_1/\partial t=-\xi_iu_2^2$, and $\alpha_i\leq 1$. Thus [Liu26a, Proposition 8.11(2)(3)] implies that $(\mathbb{A}^3,\Delta_i)$ is lc near 0 if and only if $\psi_i\geq 0$ on $\mathbb{R}^3_{\geq 0}$, and that in this case

$$a_i'=\min_{\omega\in N_{\widetilde{G}}\cap\mathbb{R}^3_{>0}}\psi_i(\omega). \tag{B.77}$$

Let $\psi_i^{(1)}$ be ψ_i with α_i replaced by 1. Then, for any $\omega\in\mathbb{R}^3_{\geq 0}$,

$$\begin{aligned}\psi_i^{(1)}(\omega)&=\Big(\frac{1}{2}-\sigma_i\Big)\omega_3+\Big|\omega_1-\omega_2-\frac{1}{2}\omega_3\Big|,\\ \psi_i(\omega)&=\psi_i^{(1)}(\omega)+(1-\alpha_i)\min\{2\omega_1,2\omega_2+\omega_3\}.\end{aligned} \tag{B.78}$$

Since $\sigma_i<\frac{1}{2}$ and $\alpha_i\leq 1$, $\psi_i\geq 0$ on $\mathbb{R}^3_{\geq 0}$, so $(\mathbb{A}^3,\Delta_i)$ is lc near 0 and (B.77) holds.

Let $\omega \in N_{\widetilde{G}} \cap \mathbb{R}^3_{>0}$. Then $\omega_1 - \omega_2 \in \frac{1}{n_\rho}\mathbb{Z}$, and $\omega_3 - 2(\omega_1 - \omega_2) \in \mathbb{Z}$ by (B.71). We show that $\psi_i^{(1)}(\omega) \geq \frac{1-2\sigma_i}{n_\rho}$.

- If $\omega_1 - \omega_2 > 0$, then $\omega_1 - \omega_2 \geq \frac{1}{n_\rho}$. If moreover $\omega_3 \leq 2(\omega_1 - \omega_2)$, then
$$\psi_i^{(1)}(\omega) = \omega_1 - \omega_2 - \sigma_i\omega_3 \geq (1-2\sigma_i)(\omega_1 - \omega_2).$$
If $\omega_3 \geq 2(\omega_1 - \omega_2)$, then
$$\psi_i^{(1)}(\omega) = (1-\sigma_i)\omega_3 - (\omega_1 - \omega_2) \geq (1-2\sigma_i)(\omega_1 - \omega_2).$$
In both cases $\psi_i^{(1)}(\omega) \geq \frac{1-2\sigma_i}{n_\rho}$.
- If $\omega_1 - \omega_2 < 0$, then $\omega_1 - \omega_2 \leq -\frac{1}{n_\rho}$, and
$$\psi_i^{(1)}(\omega) = (1-\sigma_i)\omega_3 - (\omega_1 - \omega_2) \geq \frac{1}{n_\rho}.$$
- If $\omega_1 = \omega_2$, then ω_3 is a positive integer, and $\psi_i^{(1)}(\omega) = (1-\sigma_i)\omega_3 \geq 1 - \sigma_i$.

Since $n_\rho \geq 1$ and $0 \leq \sigma_i < \frac{1}{2}$, by (B.78) we obtain

$$a_i' \geq \frac{1-2\sigma_i}{n_\rho}. \tag{B.79}$$

Let m_0 be a positive integer with $m_0 w \in \mathbb{Z}^3$, and let $w^\flat := m_0(2w_1, 2w_2, w_1+w_2, w_3) \in (\mathbb{N}^+)^4$, a weight vector on (x,y,z,t). Since $\mathrm{in}_{m_0 w} = \mathrm{in}_w$, Lemma 10.2(2) for the weight vector $m_0 w$ on (u_1, u_2, t) in place of (p, v, t) gives
$$(\mathrm{in}_{w^\flat} h)(u_1^2, -u_2^2, u_1u_2, t) = \mathrm{in}_w\left(h(u_1^2, -u_2^2, u_1u_2, t)\right)$$
for any non-zero polynomial h without monomials divisible by xy. Let $g_{ij} := \mathrm{in}_{w^\flat} h_{ij}$. By (B.75),
$$g_{ij}(u_1^2, -u_2^2, u_1u_2, t) = \lambda_{ij}(u_1^2 - \xi_i u_2^2 t)^{e_j} t^{k_j},$$
so the restriction of g_{ij} to W_0 is the restriction of $\lambda_{ij}(x + \xi_i y t)^{e_j} t^{k_j}$. For any $i \gg 0$,
$$N_i w_3 > 2(w_1 + w_2),$$
so $\mathrm{in}_{w^\flat}(xy + z^2 + t^{N_i}) = xy + z^2$. Let
$$B_i := \mathrm{div}(xy + z^2 + t^{N_i}) + \sum_j c_{ij}\,\mathrm{div}(h_{ij}), \quad B_i^{\mathrm{in}} := \mathrm{div}(xy + z^2) + \sum_j c_{ij}\,\mathrm{div}(g_{ij})$$
on $\mathbb{A}^4$, and let $D_i' := \sum_j c_{ij}\,\mathrm{div}(g_{ij}|_{W_0})$ on W_0. By Lemma 6.4(1) with $n = 4$ and the weight vector $w^\flat$,
$$\mathrm{mld}((\mathbb{A}^4 \ni 0, B_i)/G) \geq \mathrm{mld}((\mathbb{A}^4 \ni 0, B_i^{\mathrm{in}})/G).$$
For W_{N_i}, the pair (W_{N_i}, D_i) is klt near 0, so $a_i = \mathrm{mld}((\mathbb{A}^4 \ni 0, B_i)/G)$ by Theorem 2.17(1) with $n = 4$, $c = 1$, $H = G$, and $\mathfrak{a} = \prod_j (h_{ij})^{c_{ij}}$. The pullback of D_i' to $\widetilde{W}_0$ is Δ_i, which is lc near 0. Thus (W_0, D_i') is lc near 0, and by Theorem 2.17(1) with $n = 4$, $c = 1$, $H = G$, and $\mathfrak{a} = \prod_j (g_{ij})^{c_{ij}}$,
$$\mathrm{mld}((\mathbb{A}^4 \ni 0, B_i^{\mathrm{in}})/G) = \mathrm{mld}((W_0 \ni 0, D_i')/G) = a_i'.$$
Together with (B.79), $a_i \geq a_i' \geq \frac{1-2\sigma_i}{n_\rho}$, which is (B.76).

Step 8. In this step we show that
$$a_i \leq \frac{1-2\sigma_i}{n_\rho} + 2w_1(1-\alpha_i) \quad \text{for any } i \gg 0, \tag{B.80}$$
and conclude the proof. By Proposition 3.14(2)(b), for any i with $N_i \geq N_{\mathrm{ref}}(M, \Gamma)$,
$$a_i \leq \mathrm{mld}((W_0 \ni 0, D_{0,i})/G).$$

Let E_w be the toric prime divisor over $\mathbb{A}^3/\widetilde{G} = W_0/G$ given by the primitive vector w of $N_{\widetilde{G}}$. Since $w \in \mathbb{R}^3_{>0}$, the center of E_w is the image of 0. By **Step 7**, the quotient pair of $(W_0, D_{0,i})$ by G is the quotient pair of $(\widetilde{W}_0, D_{0,i}|_{\widetilde{W}_0})$ by $\widetilde{G}$, where

$$D_{0,i}|_{\widetilde{W}_0} = \sum_j c_{ij} \operatorname{div}\left(h_{ij}(u_1^2, -u_2^2, u_1u_2, t)\right) \quad \text{on } \widetilde{W}_0 = \mathbb{A}^3_{u_1,u_2,t}.$$

By [Liu26a, Lemma 8.5], in the notation of [Liu26a, Set-up 8.4] on $\mathbb{A}^3$ (omitting the j with $h_{ij}(0) \neq 0$, for which $e_j = k_j = 0$ and which do not affect log discrepancies of divisors with center 0), the log discrepancy of $(\widetilde{W}_0, D_{0,i}|_{\widetilde{W}_0})/\widetilde{G}$ at E_w is

$$w_1 + w_2 + w_3 - \sum_j c_{ij} \operatorname{wt}_w\left(h_{ij}(u_1^2, -u_2^2, u_1u_2, t)\right).$$

By (B.75),

$$\operatorname{wt}_w\left(h_{ij}(u_1^2, -u_2^2, u_1u_2, t)\right) = 2e_jw_1 + k_jw_3.$$

Hence

$$\begin{aligned}\operatorname{mld}((W_0 \ni 0, D_{0,i})/G) &\leq w_1 + w_2 + w_3 - \sum_j c_{ij}(2e_jw_1 + k_jw_3)\\ &= w_2 - w_1 + (1-\sigma_i)w_3 + 2w_1(1-\alpha_i).\end{aligned}$$

By (B.73), $w_2 - w_1 + (1-\sigma_i)w_3 = \frac{1-2\sigma_i}{n_\rho}$, which implies (B.80).

Since $\lim_i \alpha_i = 1$ and $\lim_i \sigma_i = \sigma_*$, (B.76) and (B.80) imply

$$\lim_i a_i = \frac{1-2\sigma_*}{n_\rho}.$$

Since $(a_i)_{i\geq 1}$ is strictly increasing, $a_i < \frac{1-2\sigma_*}{n_\rho}$ for any i. On the other hand, $\sigma_i \leq \sigma_*$, so (B.76) implies $a_i \geq \frac{1-2\sigma_*}{n_\rho}$ for any i, a contradiction. The theorem follows. □

Proof of Theorem 10.4. Assume that $(a_i)_{i\geq 1}$ is strictly increasing. We derive a contradiction. By Set-up 10.3, $(H_j^*)_j$ is a compatible generic limit of the tuples $(H_{ij})_j$. We use wt_w and in_w of Definition 2.3, also for power series over $\Bbbk$.

Step 1. In this step we prove that

$$a(P, \widetilde{W}_*, B^*) \geq a_1 \quad \text{for any prime divisor } P \text{ over } \widetilde{W}_* \text{ with center the closed point,} \tag{B.81}$$

and we reduce to the case when no component of $\lfloor B^* \rfloor$ is $\widetilde{G}$-stable.

Since $a_1 > 0$ by Set-up 10.3, $a_i \geq a_1 > 0$ for any i, and Lemma B.7 with $\epsilon = a_1$ implies (B.81).

Assume that some component F of $\lfloor B^* \rfloor$ is $\widetilde{G}$-stable. Since any prime divisor on $\widetilde{W}_*$ passes through the closed point, F is a floor of B^*. If F is normal, then Theorem B.11 contradicts the strict increase of a_i, and otherwise Theorem B.13 does. Thus we may assume that no component of $\lfloor B^* \rfloor$ is $\widetilde{G}$-stable.

Step 2. In this step we show that $\lfloor B^* \rfloor$ consists of two smooth prime divisors F_1, F_2 meeting transversely, we construct eigenparameters adapted to $F_1 + F_2$, and we prove (B.83) below.

Let P_{bl} be the exceptional divisor of the blow-up of the closed point of $\widetilde{W}_*$. By (B.81), $3 - \operatorname{mult}_0 B^* = a(P_{\mathrm{bl}}, \widetilde{W}_*, B^*) > 0$. Since B^* is $\widetilde{G}$-invariant, $\widetilde{G}$ permutes the components of $\lfloor B^* \rfloor$, and no orbit has length 1. Since each component has multiplicity at least 1 at 0 and $\operatorname{mult}_0 B^* < 3$, $\lfloor B^* \rfloor$ has exactly two components F_1, F_2, they are exchanged by $\widetilde{G}$, and $\operatorname{mult}_0 F_1 = \operatorname{mult}_0 F_2 = 1$. Thus F_1 and F_2 are smooth.

We show that F_1 and F_2 meet transversely. Otherwise there exists a regular system of parameters $(\zeta_1, \zeta_2, \zeta_3)$ of $\Bbbk[[p, v, t]]$ such that $F_1 = (\zeta_1 = 0)$ and $F_2 = (\zeta_1 + \phi'(\zeta_2, \zeta_3) = 0)$ for some $\phi' \in (\zeta_2, \zeta_3)^2$. The exceptional prime divisor P' of the weighted blow-up with weights $(2, 1, 1)$

in $(\zeta_1,\zeta_2,\zeta_3)$ satisfies $a(P',\widetilde{W}_*,0)=4$ and $\mathrm{mult}_{P'}F_1,\mathrm{mult}_{P'}F_2\geq 2$, so $a(P',\widetilde{W}_*,B^*)\leq 0$, which contradicts (B.81). Thus $\Sigma:=F_1\cap F_2$ is a smooth curve.

Let $\mathfrak{m}_*$ be the maximal ideal of $\Bbbk[[p,v,t]]$. Let $\widetilde{G}_1\subset\widetilde{G}$ be the stabilizer of F_1, a subgroup of index 2, and fix $g_0\in\widetilde{G}\backslash\widetilde{G}_1$. The ideal of F_1 is $\widetilde{G}_1$-stable, and the image of a generator in $\mathfrak{m}_*/\mathfrak{m}_*^2$ spans a $\widetilde{G}_1$-stable line. Hence the Reynolds projection of a generator to the character of the line is again a generator, whose linear part is the linear part of that generator. Let f_1 be the projected generator and χ_{f_1} the $\widetilde{G}_1$-character of f_1. Let $\lambda_0\in\Bbbk^*$ be a number with $\lambda_0^2=\chi_{f_1}(g_0^2)$ and let $f_2:=\lambda_0^{-1}g_0^*f_1$, a generator of the ideal of F_2. Since $\widetilde{G}$ is abelian, any $g\in\widetilde{G}_1$ acts on f_1 and on f_2 by $\chi_{f_1}(g)$, while $g_0^*f_1=\lambda_0 f_2$ and $g_0^*f_2=\lambda_0 f_1$. Thus

$$u_1:=f_1+f_2\quad\text{and}\quad u_2:=f_1-f_2$$

are $\widetilde{G}$-semi-invariant, $f_1f_2=(u_1^2-u_2^2)/4$, and $\rho:=\chi_{u_1}/\chi_{u_2}$ is the non-trivial character of $\widetilde{G}/\widetilde{G}_1$.

Since F_1 and F_2 meet transversely, the linear parts of u_1,u_2 are linearly independent. The deck involution $\iota=(1,-1)\in\widetilde{G}$ acts on $\mathfrak{m}_*/\mathfrak{m}_*^2$ with eigenvalue -1 on the span of p,v and eigenvalue 1 on the span of t. Exactly one of the following holds.

(a) $\iota\in\widetilde{G}_1$. Then $\rho(\iota)=1$, so $\chi_{u_1}(\iota)=\chi_{u_2}(\iota)$, which is -1 because the 1-eigenspace of ι on $\mathfrak{m}_*/\mathfrak{m}_*^2$ is of dimension 1. Thus ι acts trivially on the cotangent space $\mathfrak{m}_*/((u_1,u_2)+\mathfrak{m}_*^2)$ of Σ at 0, hence on Σ, so Σ is the fixed curve $(p=v=0)$ of ι. We let $s:=t$, whose restriction to Σ is a parameter.

(b) $\iota\notin\widetilde{G}_1$. Then $\rho(\iota)=-1$, so exactly one of u_1,u_2 is ι-invariant and the other is ι-anti-invariant. We let s be a $\widetilde{G}$-semi-invariant element of $\mathfrak{m}_*$ whose linear part is not in the span of those of u_1,u_2. Then $s|_\Sigma$ is a parameter of Σ, and s is ι-anti-invariant because the 1-eigenspace of ι on $\mathfrak{m}_*/\mathfrak{m}_*^2$ is of dimension 1.

We let (p',v',t') be (u_1,u_2,t) in case (a), and in case (b) the triple consisting of the ι-anti-invariant element of $\{u_1,u_2\}$, then s, then the ι-invariant element of $\{u_1,u_2\}$. Thus p',v' are ι-anti-invariant, t' is ι-invariant, and (p',v',t') is a regular system of parameters of $\Bbbk[[p,v,t]]$ consisting of $\widetilde{G}$-semi-invariants. The linear part of t' is a non-zero multiple of t. Rescaling t' in case (b), rescaling $s=v'$ in case (b), and rescaling u_1,u_2 by a common constant in case (a), we may assume that the linear part of t' is t and that the linear parts of p',v' in p,v form a matrix of determinant 1. After the rescalings, and after possibly interchanging u_1 and u_2 in case (b), which replaces ρ by $\rho^{-1}=\rho$, there exists $\gamma\in\Bbbk^*$ such that

$$f_1f_2\in\Bbbk^*\cdot(u_1^2-\gamma u_2^2),\qquad (u_1,u_2,s)=\begin{cases}(p',v',t') & \text{in case (a)},\\ (t',p',v') & \text{in case (b)}.\end{cases}\tag{B.82}$$

The characters $\chi_{p'},\chi_{v'}$ form the pair χ_p,χ_v, possibly in the other order, and $\chi_{t'}=\chi_t$. In case (a), after interchanging u_1 and u_2 and multiplying both by $\sqrt{-1}$ if necessary, $\chi_{p'}=\chi_p$, the linear parts of p',v' in p,v still form a matrix of determinant 1, and (B.82) holds after replacing γ with γ^{-1}. In case (b), if $\chi_{p'}\neq\chi_p$, we replace h_{ij} by $h_{ij}(y,x,-z,t)$ and G by the conjugate of G under $(x,y,z,t)\to(y,x,-z,t)$ for all i. The coordinate change $(x,y,z,t)\to(y,x,-z,t)$ preserves $xy+z^2+t^{N_i}$ and the condition that no monomial is divisible by xy, conjugates G to the diagonal group with χ_x and χ_y interchanged, and does not change a_i. On $\widetilde{W}_0$ the coordinate change $(x,y,z,t)\to(y,x,-z,t)$ is induced by $(p,v,t)\to(\sqrt{-1}v,\sqrt{-1}p,t)$, which interchanges χ_p and χ_v. We replace $h_{*,j}$, H_j^*, B^*, F_1, F_2, f_1, f_2, g_0, u_1, u_2, and s with their transforms. Thus we may assume that

$$\chi_{p'}=\chi_p,\quad \chi_{v'}=\chi_v\quad\text{and}\quad \chi_{t'}=\chi_t.$$

Let $e_j:=\mathrm{mult}_{F_1}H_j^*$, which equals $\mathrm{mult}_{F_2}H_j^*$ since H_j^* is $\widetilde{G}$-semi-invariant and g_0 exchanges F_1 and F_2, and let $R_j^*:=H_j^*/(f_1f_2)^{e_j}$, so that F_1,F_2 do not divide R_j^*. Here e_j and R_j^* are not those of Set-up 10.3, which are defined for a floor. Since $\mathrm{mult}_{F_1}B^*=1$, $\sum_j e_j\kappa_j=1$. The exceptional prime divisor P_Σ of the blow-up of Σ satisfies $a(P_\Sigma,\widetilde{W}_*,F_1+F_2)=0$. Since $(\widetilde{W}_*,B^*)$

is lc and $\kappa_j > 0$, $\mathrm{mult}_{P_\Sigma} R_j^* = 0$, that is, $R_j^*|_\Sigma \neq 0$. We let δ_j be the non-negative integer with

$$R_j^*|_\Sigma = s^{\delta_j} \cdot (\text{a unit}).$$

In case (b), H_j^* and $f_1 f_2$ are ι-invariant, so $R_j^*|_\Sigma$ is ι-invariant, and since ι preserves Σ and $\iota^* s = -s$, δ_j is even.

Let $m > \max_j \delta_j$ be an integer and let P'' be the exceptional prime divisor of the weighted blow-up with weights $(m, m, 1)$ in (u_1, u_2, s). Then $a(P'', \widetilde{W}_*, 0) = 2m + 1$, $\mathrm{mult}_{P''}(f_1 f_2) = 2m$ by (B.82), and $\mathrm{mult}_{P''} R_j^* = \delta_j$, because $R_j^* - R_j^*|_\Sigma(s) \in (u_1, u_2)$. Thus, with $\beta_* := \sum_j \delta_j \kappa_j$,

$$a(P'', \widetilde{W}_*, B^*) = 2m + 1 - \sum_j \kappa_j (2m e_j + \delta_j) = 1 - \beta_*,$$

and (B.81) implies

$$\beta_* < 1. \tag{B.83}$$

Step 3. In this step we compute the quotient values of the model pairs. Let ν be the order of $\chi_s|_{\widetilde{G}_1}$. Let $\widetilde{G}$ act on $\mathbb{A}^3$ with coordinates $\overline{u}_1, \overline{u}_2, \overline{s}$ by the characters $\chi_{u_1}, \chi_{u_2}, \chi_s$. The action is faithful, since $\widetilde{G}$ acts faithfully on $\widetilde{W}_0$ and (u_1, u_2, s) is a regular system of parameters. Let $N_{\widetilde{G}}$ be the lattice of Definition 2.3 for the action of $\widetilde{G}$ on $\mathbb{A}^3$ in the coordinates $(\overline{u}_1, \overline{u}_2, \overline{s})$. For $\gamma' \in \mathbb{C}^*$, $0 < \alpha \leq 1$ and $0 \leq \beta < 1$, let

$$\Delta'^{\gamma'}_{\alpha,\beta} := \alpha \,\mathrm{div}(\overline{u}_1^2 - \gamma' \overline{u}_2^2) + \beta \,\mathrm{div}(\overline{s}) \quad \text{on } \mathbb{A}^3.$$

For any $\omega = (\omega_1, \omega_2, \omega_3) \in \mathbb{R}^3_{>0}$, $\mathrm{in}_\omega(\overline{u}_1^2 - \gamma' \overline{u}_2^2)$ is a monomial if $\omega_1 \neq \omega_2$, and is $\overline{u}_1^2 - \gamma' \overline{u}_2^2$ itself if $\omega_1 = \omega_2$, whose zero set on the torus $(\mathbb{C}^*)^3$ has two disjoint smooth components, each of coefficient $\alpha \leq 1$. Hence all the weighted leading boundaries of $\Delta'^{\gamma'}_{\alpha,\beta}$ are lc on the torus, and by [Liu26a, Proposition 8.11(2)(3)], $(\mathbb{A}^3, \Delta'^{\gamma'}_{\alpha,\beta})$ is lc near 0 if and only if the function

$$\begin{aligned} \psi_{\alpha,\beta}(\omega) &:= \omega_1 + \omega_2 + \omega_3 - 2\alpha \min(\omega_1, \omega_2) - \beta \omega_3 \\ &= |\omega_1 - \omega_2| + 2(1 - \alpha) \min(\omega_1, \omega_2) + (1 - \beta)\omega_3 \end{aligned} \tag{B.84}$$

is non-negative on $\mathbb{R}^3_{\geq 0}$, which holds, and

$$\mathrm{mld}((\mathbb{A}^3 \ni 0, \Delta'^{\gamma'}_{\alpha,\beta})/\widetilde{G}) = \min_{\omega \in N_{\widetilde{G}} \cap \mathbb{R}^3_{>0}} \psi_{\alpha,\beta}(\omega).$$

Let $\omega \in N_{\widetilde{G}} \cap \mathbb{R}^3_{>0}$ and let $g \in \widetilde{G}$ be the image of ω under $\mathbb{R}^3 \to \mathbb{T}^3$. If $g \in \widetilde{G}_1$, then $\chi_s(g)^\nu = 1$, so $\omega_3 \in \frac{1}{\nu}\mathbb{N}^+$ and

$$\psi_{\alpha,\beta}(\omega) \geq (1 - \beta)\omega_3 \geq \frac{1 - \beta}{\nu}.$$

If $g \notin \widetilde{G}_1$, then $\rho(g) = -1$ and $g^2 \in \widetilde{G}_1$, so $\omega_1 - \omega_2 \in \frac{1}{2} + \mathbb{Z}$ and $\omega_3 \in \frac{1}{2\nu}\mathbb{N}^+$, and since $\nu \geq 1$ and $\beta \geq 0$,

$$\psi_{\alpha,\beta}(\omega) \geq \frac{1}{2} + \frac{1 - \beta}{2\nu} \geq \frac{1 - \beta}{\nu}.$$

Thus

$$\mathrm{mld}((\mathbb{A}^3 \ni 0, \Delta'^{\gamma'}_{\alpha,\beta})/\widetilde{G}) \geq \frac{1 - \beta}{\nu}. \tag{B.85}$$

Fix $g_1 \in \widetilde{G}_1$ with $\chi_s(g_1) = e^{2\pi\sqrt{-1}/\nu}$. Since $\rho(g_1) = 1$, $\chi_{u_1}(g_1) = \chi_{u_2}(g_1)$. Fix a real number $\ell \geq 1$ such that $\ell > \max_j \delta_j/\nu$ and $e^{2\pi\sqrt{-1}\ell} = \chi_{u_1}(g_1)$, and let

$$w^{(\ell)} := \left(\ell, \ell, \frac{1}{\nu}\right) \in N_{\widetilde{G}} \cap \mathbb{R}^3_{>0}. \tag{B.86}$$

The vector $w^{(\ell)}$ is primitive in $N_{\widetilde{G}}$: if $w^{(\ell)}/k' \in N_{\widetilde{G}}$ for an integer $k' \geq 2$, then the image of $w^{(\ell)}/k'$ in $\widetilde{G}$ has trivial ρ, so belongs to $\widetilde{G}_1$, while the third entry $1/(k'\nu)$ of $w^{(\ell)}/k'$ is not in $\frac{1}{\nu}\mathbb{Z}$. By (B.84),

$$\psi_{\alpha,\beta}(w^{(\ell)}) = \frac{1-\beta}{\nu} + 2\ell(1-\alpha). \tag{B.87}$$

Step 4. In this step we correct a coordinate change on the cover to an automorphism of the germ W_N. Let $\mathfrak{m}$ be the maximal ideal of $\mathbb{C}[[x,y,z,t]]$, $q := xy + z^2$, and $q_N := q + t^N$. For a power series $\phi \in \mathbb{C}[[x,y,z,t]]$ we let $[\phi]_{W_N}$, resp. $[\phi]_{W_0}$, be the unique power series without monomials divisible by xy such that $\phi - [\phi]_{W_N} \in (q_N)$, resp. $\phi - [\phi]_{W_0} \in (q)$. The reductions replace xy by $-z^2 - t^N$, resp. $-z^2$, so they do not lower the degree. Let $\pi^\sharp$ be the homomorphism

$$\pi^\sharp : \mathbb{C}[[x,y,z,t]] \to \mathbb{C}[[p,v,t]], \qquad (x,y,z,t) \to (p^2, -v^2, pv, t).$$

Then $\operatorname{Ker}\pi^\sharp = (q)$ and $\operatorname{Im}\pi^\sharp$ is the ring of ι-invariants.

Let $N \geq 3$ and let σ be a $\mathbb{C}$-automorphism of $\mathbb{C}[[p,v,t]]$ satisfying the following.

(1) $\sigma(p), \sigma(v), \sigma(t)$ are $\widetilde{G}$-semi-invariant with the characters χ_p, χ_v, χ_t.
(2) $\sigma(p), \sigma(v)$ are ι-anti-invariant and $\sigma(t)$ is ι-invariant.
(3) The linear parts of $\sigma(p), \sigma(v)$ in p, v form a matrix Λ_σ with $\det\Lambda_\sigma = 1$, and

$$\sigma(t) - t \in \mathfrak{m}_{p,v,t}^2,$$

where $\mathfrak{m}_{p,v,t}$ is the maximal ideal of $\mathbb{C}[[p,v,t]]$.

We let $\sigma_0^\flat$ be the endomorphism of $\mathbb{C}[[x,y,z,t]]$ sending x,y,z,t to the preimages without monomials divisible by xy of

$$\sigma(p)^2, \quad -\sigma(v)^2, \quad \sigma(p)\sigma(v), \quad \text{and} \quad \sigma(t)$$

under $\pi^\sharp$. The preimages exist by (2), and they are semi-invariant with the characters $\chi_x, \chi_y, \chi_z, \chi_t$ by (1), because $\chi_p^2 = \chi_x$, $\chi_v^2 = \chi_z^2/\chi_x = \chi_y$ and $\chi_p\chi_v = \chi_z$. Thus $\sigma_0^\flat$ commutes with G and

$$\pi^\sharp \circ \sigma_0^\flat = \sigma \circ \pi^\sharp. \tag{B.88}$$

Modulo $\mathfrak{m}^2$, $\sigma_0^\flat(x), \sigma_0^\flat(y), \sigma_0^\flat(z)$ are the linear forms in x,y,z induced by Λ_σ through

$$(x,y,z) = (p^2, -v^2, pv),$$

and $\sigma_0^\flat(t)$ is t plus a linear form in x,y,z. Hence the Jacobian matrix of $\sigma_0^\flat$ is invertible, and $\sigma_0^\flat$ is an automorphism. By (B.88), $\pi^\sharp(\sigma_0^\flat(q)) = \sigma(\pi^\sharp(q)) = 0$, so $\sigma_0^\flat(q) = \upsilon q$ for a power series υ. The quadratic part of $\sigma_0^\flat(q)$ is $(\det\Lambda_\sigma)^2 q = q$, by a direct computation for diagonal matrices, for shears $\left(\begin{smallmatrix} 1 & b \\ 0 & 1 \end{smallmatrix}\right)$, and for $\left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$. Thus υ is a unit with $\upsilon(0) = 1$, and υ is G-invariant because q and $\sigma_0^\flat(q)$ are semi-invariant with the character χ_q. Therefore

$$\upsilon^{-1}\sigma_0^\flat(q_N) = q + R_N, \qquad R_N := \upsilon^{-1}\sigma_0^\flat(t)^N \in \mathfrak{m}^N. \tag{B.89}$$

We construct a G-equivariant automorphism ϖ_N of $\mathbb{C}[[x,y,z,t]]$ with $\varpi_N(t) = t$ and

$$\varpi_N(x) - x, \ \varpi_N(y) - y, \ \varpi_N(z) - z \in \mathfrak{m}^{N-1},$$

and a power series $g_N(t)$, such that

$$\varpi_N(q + R_N) = q + g_N(t). \tag{B.90}$$

Assume that $k \geq N$ and that we have $q + \phi(t) + R$ with $R \in \mathfrak{m}^k$ semi-invariant with the character χ_q. Write the homogeneous part of degree k of R as $r_0 t^k + xr_1 + yr_2 + zr_3$ with $r_0 \in \mathbb{C}$ and r_1, r_2, r_3 homogeneous of degree $k-1$, assigning each monomial divisible by one of x,y,z to the first of x,y,z dividing it. Then r_1, r_2, r_3 are semi-invariant with the characters $\chi_q/\chi_x = \chi_y$, $\chi_q/\chi_y = \chi_x$, and $\chi_q/\chi_z = \chi_z$, since $\chi_q = \chi_x\chi_y = \chi_z^2$. The G-equivariant automorphism

$$(x,y,z,t) \to \left(x - r_2, \ y - r_1, \ z - \frac{r_3}{2}, \ t\right)$$

sends $q+\phi(t)+R$ to

$$q+\phi(t)+r_0t^k+\Big(r_1r_2+\frac{r_3^2}{4}+R\big(x-r_2,y-r_1,z-\tfrac{r_3}{2},t\big)-r_0t^k-xr_1-yr_2-zr_3\Big),$$

and the last summand belongs to $\mathfrak{m}^{2k-2}+\mathfrak{m}^{k+1}\subset\mathfrak{m}^{k+1}$ because $k\geq 3$. Starting from $k=N$, $\phi=0$, and $R=R_N$, which has the character $\chi_t^N=\chi_q$, and iterating, the composite of the automorphisms above converges in the $\mathfrak{m}$-adic topology. We let ϖ_N be the limit, which satisfies (B.90) and the congruences modulo $\mathfrak{m}^{N-1}$ above. The coefficient of t^N in R_N is 1, since $v(0)=1$ and $\sigma_0^\flat(t)\equiv t$ modulo $(x,y,z)+\mathfrak{m}^2$. Thus

$$g_N(t)=t^Nh_N(t),\qquad h_N(0)=1,$$

and h_N is invariant because g_N has the character χ_t^N.

Let $\eta_N(t)$ be the inverse, under composition, of the power series $th_N(t)^{1/N}$ with $h_N(0)^{1/N}=1$. Then $\eta_N(t)$ is t times an invariant unit with value 1 at 0, and $g_N(\eta_N(t))=t^N$. Let $\eta_N^\sharp$ be the automorphism of $\mathbb{C}[[x,y,z,t]]$ fixing x,y,z with $\eta_N^\sharp(t)=\eta_N(t)$, and let

$$\sigma_N^\flat:=\eta_N^\sharp\circ\varpi_N\circ\sigma_0^\flat.$$

By (B.89) and (B.90),

$$\sigma_N^\flat(q_N)=\eta_N^\sharp\varpi_N(v)\cdot\big(q+g_N(\eta_N(t))\big)=\eta_N^\sharp\varpi_N(v)\cdot q_N, \tag{B.91}$$

and $\eta_N^\sharp\varpi_N(v)$ is a G-invariant unit. Thus $\sigma_N^\flat$ induces a G-equivariant automorphism of the completion of W_N at 0. Moreover, for any $\phi\in\mathbb{C}[[x,y,z,t]]$,

$$\sigma_N^\flat(\phi)-\eta_N^\sharp(\sigma_0^\flat(\phi))\in\mathfrak{m}^{N-1}. \tag{B.92}$$

Step 5. In this step we show that for all but finitely many i there exist semi-invariant polynomials h'_{ij} without monomials divisible by xy, with the characters of the h_{ij}, such that $(W_{N_i},\sum_j c_{ij}\operatorname{div}(h'_{ij}))$ is klt near 0 with quotient value a_i, and such that the h'_{ij} satisfy (B.96) below.

Let w be the weight vector on (p,v,t) whose entries at the coordinates in the places of u_1,u_2,s in (B.82) are the entries $\ell,\ell,\frac{1}{\nu}$ of $w^{(\ell)}$ in (B.86), and let $w^\flat$ be the weight vector on (x,y,z,t) of Lemma 10.2(2). Explicitly,

$$(w,w^\flat)=\begin{cases}\big((\ell,\ell,\frac{1}{\nu}),\ (2\ell,2\ell,2\ell,\frac{1}{\nu})\big) & \text{in case (a)},\\ \big((\ell,\frac{1}{\nu},\ell),\ (2\ell,\frac{2}{\nu},\ell+\frac{1}{\nu},\ell)\big) & \text{in case (b)}.\end{cases} \tag{B.93}$$

By Lemma 10.2(2), for any non-zero power series ϕ without monomials divisible by xy,

$$\pi^\sharp(\operatorname{in}_{w^\flat}(\phi))=\operatorname{in}_w(\pi^\sharp(\phi)).$$

For $\gamma'\in\mathbb{C}^*$ let

$$Q_j^{\gamma'}:=\begin{cases}(p^2-\gamma'v^2)^{e_j}t^{\delta_j}=\pi^\sharp\big((x+\gamma'y)^{e_j}t^{\delta_j}\big) & \text{in case (a)},\\ (t^2-\gamma'p^2)^{e_j}v^{\delta_j}=\pi^\sharp\big((t^2-\gamma'x)^{e_j}(-y)^{\delta_j/2}\big) & \text{in case (b)},\end{cases} \tag{B.94}$$

where δ_j is even in case (b). Let σ_* be the $\Bbbk$-automorphism of $\Bbbk[[p,v,t]]$ with $\sigma_*(p')=p$, $\sigma_*(v')=v$, and $\sigma_*(t')=t$. By **Step 2**, σ_*^{-1} satisfies (1)–(3) of **Step 4** over $\Bbbk$, hence so does σ_*. By (B.82), $\sigma_*(f_1f_2)$ is a non-zero constant times $p^2-\gamma v^2$, resp. $t^2-\gamma p^2$, and since $\ell>\delta_j/\nu$ and $R_j^*-R_j^*|_\Sigma(s)\in(u_1,u_2)$,

$$\operatorname{in}_w(\sigma_*(H_j^*))=\xi_jQ_j^\gamma\quad\text{for some }\xi_j\in\Bbbk^*.$$

Fix an integer $J>\nu\max_j(2\ell e_j+\delta_j/\nu)$. Any monomial of degree at least J in p,v,t, resp. in x,y,z,t, has w-weight, resp. $w^\flat$-weight, greater than $\max_j\operatorname{wt}_w(Q_j^\gamma)$, because all weights in (B.93) are at least $1/\nu$. Let $\mathcal{I}_J$ be the set of tuples

$$\big(\operatorname{jet}_J((H_j)_j),\ \sigma(p),\sigma(v),\sigma(t),\ \gamma',\ (\xi'_j)_j\big),$$

where $(H_j)_j$ is a tuple as in Definition 2.5, satisfying the following.

(i) $\sigma(p), \sigma(v), \sigma(t)$ are polynomials of degree less than J satisfying (1)–(3) of **Step 4**.
(ii) $\gamma'\xi'_j \neq 0$ for any j.
(iii) For any j, the monomials of $H_j(\sigma(p), \sigma(v), \sigma(t))$ of degree less than J and of w-weight less than $\mathrm{wt}_w(Q_j^\gamma)$ vanish, and those of w-weight $\mathrm{wt}_w(Q_j^\gamma)$ add up to $\xi'_j Q_j^{\gamma'}$.

Conditions (i) and (iii) are polynomial identities in the coefficients, with $\det \Lambda_\sigma = 1$ among them, and (ii) is an open condition. Let $Z_J \subset \mathcal{J}_J$ be as in Definition 2.5 for the tuples $(H_{ij})_j$ and their limit $(H^*_j)_j$. The data σ_*, γ, and ξ_j truncated to degree less than J form a $\Bbbk$-point of $\mathcal{I}_J$ over the generic point of Z_J. Since the image of $\mathcal{I}_J$ in $\mathcal{J}_J$ is constructible and contains the generic point of Z_J, it contains a non-empty open subset of Z_J. Hence, by Definition 2.5, for all but finitely many i there exist polynomials defining an automorphism σ_i of $\mathbb{C}[[p, v, t]]$ satisfying (1)–(3) of **Step 4**, and $\gamma_i, \xi_{ij} \in \mathbb{C}^*$, such that by the choice of J,

$$\mathrm{in}_w(\sigma_i(H_{ij})) = \xi_{ij} Q_j^{\gamma_i} \quad \text{for any } j. \tag{B.95}$$

Possibly discarding finitely many i, we may assume that $N_i \geq J+2$ and $N_i w^\flat(t) > w^\flat(xy)$. Let $\sigma^\flat_{N_i}$ be the automorphism $\sigma^\flat_N$ of **Step 4** for $\sigma = \sigma_i$ and $N = N_i$, and let $\widehat{h}_{ij} := \sigma^\flat_{N_i}(h_{ij})$. By (B.91), the pair $(W_{N_i}, \sum_j c_{ij} \operatorname{div}(\widehat{h}_{ij}))$ on the completion is the pullback of the completion of (W_{N_i}, D_i) under a G-equivariant automorphism, so it is klt with quotient value a_i by [Liu26a, Lemma 3.3(2)(3)]. By Lemma B.4(2) with $J+1$ in place of J, there exist semi-invariant polynomials h'_{ij} without monomials divisible by xy, with the characters of the h_{ij}, such that

$$h'_{ij} - \widehat{h}_{ij} \in \mathfrak{m}^{J+1} + (q_{N_i})$$

and (W_{N_i}, D'_i), $D'_i := \sum_j c_{ij} \operatorname{div}(h'_{ij})$, is klt near 0 with quotient value a_i.

We compute $\mathrm{in}_{w^\flat}(h'_{ij})$. Let $g_{ij} := [\sigma^\flat_{0,i}(h_{ij})]_{W_0}$, where $\sigma^\flat_{0,i}$ is the automorphism $\sigma^\flat_0$ of **Step 4** for $\sigma = \sigma_i$. By (B.88), $\pi^\sharp(g_{ij}) = \sigma_i(H_{ij})$. By Lemma B.4(1)(c), for any polynomial ϕ, $[\phi]_{W_{N_i}} - [\phi]_{W_0}$ is a multiple of t^{N_i}. Applying the polynomial case to the truncations of a power series ϕ, since the reductions do not lower the degree, $[\phi]_{W_{N_i}} - [\phi]_{W_0}$ is a multiple of t^{N_i} for any power series ϕ. Writing $\phi = [\phi]_{W_{N_i}} + q_{N_i}\phi_1$, we have

$$\eta^\sharp_{N_i}(q_{N_i}) - q_{N_i} = \eta_{N_i}(t)^{N_i} - t^{N_i} \in \mathfrak{m}^{N_i}, \quad \text{so} \quad [\eta^\sharp_{N_i}(\phi)]_{W_{N_i}} - \eta^\sharp_{N_i}([\phi]_{W_{N_i}}) \in \mathfrak{m}^{N_i}.$$

Since h'_{ij} has no monomials divisible by xy and $[\cdot]_{W_{N_i}}$ kills (q_{N_i}) and does not lower degrees,

$$h'_{ij} = [h'_{ij}]_{W_{N_i}} \equiv [\widehat{h}_{ij}]_{W_{N_i}} \mod \mathfrak{m}^{J+1}.$$

Together with (B.92) and $N_i - 1 \geq J+1$, we obtain

$$h'_{ij} \equiv [\widehat{h}_{ij}]_{W_{N_i}} \equiv [\eta^\sharp_{N_i}(\sigma^\flat_{0,i}(h_{ij}))]_{W_{N_i}} \equiv \eta^\sharp_{N_i}([\sigma^\flat_{0,i}(h_{ij})]_{W_{N_i}}) \equiv \eta^\sharp_{N_i}(g_{ij}) \mod \mathfrak{m}^{J+1}.$$

Since $\eta^\sharp_{N_i}$ replaces t by t times a unit with value 1 at 0,

$$\mathrm{in}_{w^\flat}(\eta^\sharp_{N_i}(g_{ij})) = \mathrm{in}_{w^\flat}(g_{ij}).$$

By the choice of J and (B.95),

$$\pi^\sharp\big(\mathrm{in}_{w^\flat}(h'_{ij})\big) = \mathrm{in}_w(\sigma_i(H_{ij})) = \xi_{ij} Q_j^{\gamma_i} \quad \text{for any } j. \tag{B.96}$$

In particular

$$\mathrm{wt}_{w^\flat}(h'_{ij}) = \mathrm{wt}_w(Q_j^{\gamma_i}) = 2\ell e_j + \delta_j/\nu.$$

Step 6. In this step we prove that for all but finitely many i,

$$\mu_i \leq a_i \leq \mu_i + 2\ell(1-\alpha_i), \qquad \text{where } \alpha_i := \sum_j e_j c_{ij}, \quad \beta_i := \sum_j \delta_j c_{ij}, \quad \mu_i := \frac{1-\beta_i}{\nu}, \tag{B.97}$$

and we conclude the proof. Since $c_{ij} \in (0, \kappa_j]$ is increasing in i with limit κ_j, $\sum_j e_j \kappa_j = 1$, and (B.83) holds, we have

$$0 < \alpha_i \le 1, \qquad 0 \le \beta_i \le \beta_* < 1, \qquad \lim_i \alpha_i = 1, \qquad \text{and} \qquad \lim_i \beta_i = \beta_*.$$

Let i be as in **Step 5**, and let the coordinates $(\overline{u}_1, \overline{u}_2, \overline{s})$ of **Step 3** be (p, v, t) in case (a) and (t, p, v) in case (b). Then $\widetilde{G}$ acts on $\widetilde{W}_0$ by the characters of **Step 3**, because $\chi_{p'} = \chi_p$, $\chi_{v'} = \chi_v$ and $\chi_{t'} = \chi_t$, and $w = w^{(\ell)}$. By (B.94),

$$\sum_j c_{ij} \operatorname{div}(Q_j^{\gamma_i}) = \Delta^{\gamma_i}_{\alpha_i, \beta_i}.$$

Let

$$D_i^{\mathrm{in}} := \sum_j c_{ij} \operatorname{div}(\mathrm{in}_{w^\flat}(h'_{ij})|_{W_0}).$$

By (B.96), $\pi_0^* D_i^{\mathrm{in}} = \Delta^{\gamma_i}_{\alpha_i, \beta_i}$. Since π_0 is the quotient by $\langle \iota \rangle$, étale in codimension 1, and $\widetilde{G}/\langle \iota \rangle = G$, the pair (W_0, D_i^{in}) is lc near 0 by **Step 3** and

$$\operatorname{mld}((W_0 \ni 0, D_i^{\mathrm{in}})/G) = \operatorname{mld}((\widetilde{W}_0 \ni 0, \Delta^{\gamma_i}_{\alpha_i, \beta_i})/\widetilde{G}) \ge \mu_i \tag{B.98}$$

by (B.85).

We prove the lower bound. Let d_w be a positive integer with $d_w w^\flat \in \mathbb{Z}^4$, and for $\tau \in \mathbb{C}$ let

$$q_{N_i,\tau} := \tau^{-d_w \operatorname{wt}_{w^\flat}(q)} q_{N_i}(\tau^{d_w w^\flat} \cdot (x, y, z, t)), \qquad h'_{ij,\tau} := \tau^{-d_w \operatorname{wt}_{w^\flat}(h'_{ij})} h'_{ij}(\tau^{d_w w^\flat} \cdot (x, y, z, t)),$$

where $\tau^{d_w w^\flat} \cdot$ is the diagonal action of τ with the weights $d_w w^\flat$. Since $N_i w^\flat(t) > w^\flat(xy)$, $q_{N_i,\tau}$ and $h'_{ij,\tau}$ are polynomials in τ, x, y, z, t with $q_{N_i,0} = q$ and $h'_{ij,0} = \mathrm{in}_{w^\flat}(h'_{ij})$. Let B_{br} be the branch divisor of $\mathbb{A}^4 \to \mathbb{A}^4/G$, that is, $K_{\mathbb{A}^4}$ is the pullback of $K_{\mathbb{A}^4/G} + B_{\mathrm{br}}$. It is $\mathbb{Q}$-Cartier because $\mathbb{A}^4/G$ is $\mathbb{Q}$-factorial, and it is fixed by the diagonal actions above. For any τ, the G-invariant polynomials $q_{N_i,\tau}^{|G|}$ and $(h'_{ij,\tau})^{|G|}$ define the $\mathbb{R}$-ideal

$$\mathfrak{b}_\tau := (q_{N_i,\tau}^{|G|})^{1/|G|} \prod_j \left((h'_{ij,\tau})^{|G|}\right)^{c_{ij}/|G|}$$

on $\mathbb{A}^4/G$, and

$$\operatorname{mld}(\mathbb{A}^4/G \ni \overline{0}, B_{\mathrm{br}}, \mathfrak{b}_\tau) = \operatorname{mld}((\mathbb{A}^4 \ni 0, \operatorname{div}(q_{N_i,\tau}), \prod_j (h'_{ij,\tau})^{c_{ij}})/G),$$

where $\overline{0}$ is the image of 0. Let $G_{\mathrm{r}} \subset G$ be the subgroup generated by the elements acting non-trivially on exactly one coordinate. Then $\mathbb{A}^4/G = (\mathbb{A}^4/G_{\mathrm{r}})/(G/G_{\mathrm{r}})$, where $\mathbb{A}^4/G_{\mathrm{r}}$ is an affine space with coordinates the powers of x, y, z, t and G/G_{r} acts on it freely in codimension 1. By Theorem 2.16 for the family $(\mathfrak{b}_\tau)_{\tau \in \mathbb{A}^1}$ on $\mathbb{A}^4/G$ with the boundary B_{br} and the point $\overline{0}$,

$$\operatorname{mld}(\mathbb{A}^4/G \ni \overline{0}, B_{\mathrm{br}}, \mathfrak{b}_0) \le \operatorname{mld}(\mathbb{A}^4/G \ni \overline{0}, B_{\mathrm{br}}, \mathfrak{b}_\tau) \quad \text{for a general } \tau \in \mathbb{C}^*.$$

For $\tau \ne 0$, $q_{N_i,\tau}$ and $h'_{ij,\tau}$ are constant multiples of the pullbacks of q_{N_i} and h'_{ij} under a diagonal automorphism commuting with G, so the right-hand side is

$$\operatorname{mld}((\mathbb{A}^4 \ni 0, \operatorname{div}(q_{N_i}), \prod_j (h'_{ij})^{c_{ij}})/G).$$

By Theorem 2.17(1) with $n = 4$, $c = 1$, $X = W_{N_i}$, $H = G$, and the equation q_{N_i} of W_{N_i} as f_1 there, the right-hand side equals the quotient value a_i of $(W_{N_i} \ni 0, D'_i)$. Since (W_0, D_i^{in}) is lc near 0, Theorem 2.17(1) with q as f_1 there and $X = W_0$ implies that the left-hand side is $\operatorname{mld}((W_0 \ni 0, D_i^{\mathrm{in}})/G)$. By (B.98), $a_i \ge \mu_i$.

We prove the upper bound. Let $E^{(\ell)}$ be the toric prime divisor over $\widetilde{W}_0/\widetilde{G} = W_0/G$ corresponding to the primitive vector $w^{(\ell)} \in N_{\widetilde{G}}$ (see [Liu26a, Set-up 8.4 and Lemma 8.5]). Write

$(W_0, D'_{0,i})/G = (W_0/G, B'_{0,i})$ with $D'_{0,i} := \sum_j c_{ij} \operatorname{div}(h'_{ij}|_{W_0})$, and $(W_0, 0)/G = (W_0/G, B_0)$. Since the pullback of $h'_{ij}|_{W_0}$ to $\widetilde{W}_0$ is $\pi^\sharp(h'_{ij})$ and

$$\operatorname{wt}_w(\pi^\sharp(h'_{ij})) = 2\ell e_j + \frac{\delta_j}{\nu}$$

by (B.96), [Liu26a, Lemma 8.5], and (B.87) imply

$$a(E^{(\ell)}, W_0/G, B'_{0,i}) = 2\ell + \frac{1}{\nu} - \sum_j c_{ij}\left(2\ell e_j + \frac{\delta_j}{\nu}\right) = \psi_{\alpha_i,\beta_i}(w^{(\ell)}) = \mu_i + 2\ell(1-\alpha_i),$$

and $a(E^{(\ell)}, W_0/G, B_0) = 2\ell + 1/\nu$ by [Liu26a, Lemma 8.5]. Let $E_0^{(\ell)}$ be a prime divisor over W_0 whose image over W_0/G is $E^{(\ell)}$. By [Liu26a, Lemma 2.6(1)(2)],

$$\frac{a(E_0^{(\ell)}, W_0, 0)}{e_{E_0^{(\ell)}}} = 2\ell + \frac{1}{\nu} \qquad \text{and} \qquad \frac{a(E_0^{(\ell)}, W_0, D'_{0,i})}{e_{E_0^{(\ell)}}} = \mu_i + 2\ell(1-\alpha_i).$$

By Proposition 3.14(2)(a) with $\lambda = 2\ell + 1/\nu$, which does not depend on i, and with $D_N = D'_i$, for all but finitely many i the birational transform $E_i^{(\ell)}$ of $E_0^{(\ell)}$ under the inverse of $\varphi : W_{N_i} \dashrightarrow W_0$ is a prime divisor over W_{N_i} with center 0, $e_{E_i^{(\ell)}} = e_{E_0^{(\ell)}}$, and $a(E_i^{(\ell)}, W_{N_i}, D'_i) = a(E_0^{(\ell)}, W_0, D'_{0,i})$. Since a_i is the quotient value of $(W_{N_i} \ni 0, D'_i)$,

$$a_i \le \frac{a(E_i^{(\ell)}, W_{N_i}, D'_i)}{e_{E_i^{(\ell)}}} = \mu_i + 2\ell(1-\alpha_i).$$

This proves (B.97).

Let $\mu_* := (1-\beta_*)/\nu$. Since $\beta_i \le \beta_*$, $\mu_i \ge \mu_*$, and since $\lim_i \alpha_i = 1$ and $\lim_i \beta_i = \beta_*$, (B.97) implies

$$a_i \ge \mu_* \ \text{ for all but finitely many } i \qquad \text{and} \qquad \lim_i a_i = \mu_*.$$

A strictly increasing sequence with limit μ_* satisfies $a_i < \mu_*$ for any i, a contradiction. □

B.2. **Limits without coefficient-one components.** In this subsection we prove Proposition 10.5 of Section 10.

Proof of Proposition 10.5. We use the notation of Set-up 10.3 and write 0 also for the closed point of $\widetilde{W}_*$. We let $B_i := \sum_{j=1}^n c_{ij} \operatorname{div}(H_{ij})$ on $\widetilde{W}_0$. Then B_i is the pullback of $D_{0,i}$ under π_0, and π_0 is étale in codimension 1, since ι fixes only the line $(p = v = 0)$. The homomorphism $\widetilde{G} \to G$, $(g, \alpha) \to g$, is surjective with kernel $\{(1,1),(1,-1)\}$, which acts on $\widetilde{W}_0$ by $\{\mathrm{id}_{\widetilde{W}_0}, \iota\}$. Thus the composite $\widetilde{W}_0 \to W_0 \to W_0/G$ is the quotient by $\widetilde{G}$. For a prime divisor E over $\widetilde{W}_0$ we let $\widetilde{e}_E := |I_{\widetilde{G}}(E)|$, which is the ramification index of E over its restriction over W_0/G by Lemma 2.15(1).

By Definitions 2.2 and 2.4, for any i and any prime divisor F over W_{N_i} with center 0,

$$a_i \le \frac{a(F, W_{N_i}, D_i)}{e_F}, \tag{B.99}$$

with equality if F is a computing lift of a_i. Since the a_i are strictly increasing and $0 < a_i \le 2$ for any i by Set-up 10.3, $\lim_i a_i \in (0, 2]$ exists.

We divide the proof into several steps.

Step 1. In this step we prove that

$$a(E^*, \widetilde{W}_*, B^*) \ge \lim_i a_i \tag{B.100}$$

for any prime divisor E^* over $\widetilde{W}_*$ with center 0.

Let

$$\lambda := a(E^*, \widetilde{W}_*, 0) \quad \text{and} \quad s_i := \lambda - \sum_{j=1}^{n} c_{ij} \operatorname{mult}_{E^*} H_j^*.$$

By Set-up 10.3, $(H_j^*)_j$ is a compatible generic limit of the tuples $(H_{ij})_j$ of $\widetilde{G}$-semi-invariant polynomials on $\widetilde{W}_0 = \mathbb{A}^3$. By Lemma 3.15, for any $i \gg 0$ there exists a prime divisor E_i over $\widetilde{W}_0$ with center 0 such that

$$a(E_i, \widetilde{W}_0, 0) = \lambda \quad \text{and} \quad \operatorname{mult}_{E_i} H_{ij} = \operatorname{mult}_{E^*} H_j^* \text{ for any } j.$$

Thus

$$a(E_i, \widetilde{W}_0, B_i) = s_i.$$

By Lemma 2.15 for the group $\{\mathrm{id}_{\widetilde{W}_0}, \iota\}$, E_i has a restriction $E_{i,0}$ over W_0, with ramification index $r_i \in \{1, 2\}$ (Definition 2.2). The center of $E_{i,0}$ on W_0 is 0. Since π_0 is étale in codimension 1 and B_i is the pullback of $D_{0,i}$, $(\widetilde{W}_0, 0)/\{\mathrm{id}_{\widetilde{W}_0}, \iota\} = (W_0, 0)$ and $(\widetilde{W}_0, B_i)/\{\mathrm{id}_{\widetilde{W}_0}, \iota\} = (W_0, D_{0,i})$, and Lemma 2.15(2) implies that

$$a(E_i, \widetilde{W}_0, 0) = r_i a(E_{i,0}, W_0, 0) \quad \text{and} \quad a(E_i, \widetilde{W}_0, B_i) = r_i a(E_{i,0}, W_0, D_{0,i}).$$

Since ramification indices are multiplicative along $\widetilde{W}_0 \to W_0 \to W_0/G$, $\widetilde{e}_{E_i} = r_i e_{E_{i,0}}$, where $e_{E_{i,0}} = |I_G(E_{i,0})|$. Hence

$$\frac{a(E_{i,0}, W_0, 0)}{e_{E_{i,0}}} = \frac{\lambda}{\widetilde{e}_{E_i}} \le \lambda \quad \text{and} \quad \frac{a(E_{i,0}, W_0, D_{0,i})}{e_{E_{i,0}}} = \frac{s_i}{\widetilde{e}_{E_i}}. \tag{B.101}$$

Let $N_\lambda = N_\lambda(M, \Gamma, \lambda)$ be the integer of Proposition 3.14(2)(a). For any i with $N_i \ge N_\lambda$, Proposition 3.14(2)(a), with $N = N_i$, $D_N = D_i$, $D_0 = D_{0,i}$, and $F_0 = E_{i,0}$, and the first equality of (B.101) imply that the birational transform E_i' of $E_{i,0}$ under φ^{-1} is a prime divisor over W_{N_i} with center 0 such that

$$I_G(E_i') = I_G(E_{i,0}) \quad \text{and} \quad a(E_i', W_{N_i}, D_i) = a(E_{i,0}, W_0, D_{0,i}).$$

Since $\lim_i N_i = +\infty$, (B.99), the previous display and (B.101) imply that for any $i \gg 0$,

$$0 < a_i \le \frac{a(E_i', W_{N_i}, D_i)}{e_{E_i'}} = \frac{s_i}{\widetilde{e}_{E_i}} \le s_i,$$

where the last inequality holds since $s_i > 0$ and $\widetilde{e}_{E_i} \ge 1$. Since $\lim_i c_{ij} = \kappa_j$ for any j,

$$\lim_i a_i \le \lim_i s_i = \lambda - \sum_{j=1}^{n} \kappa_j \operatorname{mult}_{E^*} H_j^* = a(E^*, \widetilde{W}_*, B^*),$$

which is (B.100).

Step 2. In this step we prove that $(\widetilde{W}_*, B^*)$ is lc and

$$\operatorname{mld}(\widetilde{W}_* \ni 0, B^*) \ge \lim_i a_i. \tag{B.102}$$

Since any non-empty closed subset of $\widetilde{W}_*$ contains 0, (B.100) and $\lim_i a_i > 0$ imply that $(\widetilde{W}_*, B^*)$ is lc, and (B.102) follows from (B.100).

Step 3. In this step we prove that $(\widetilde{W}_*, B^*)$ is not klt and $\lfloor B^* \rfloor = 0$.

Assume that $(\widetilde{W}_*, B^*)$ is klt. By the last sentence of Theorem 2.18(1) with $\theta = 1$, there exists a real number $\theta_0 > 1$ such that

$$\Big(\widetilde{W}_0, \theta_0 \sum_{j=1}^{n} \kappa_j \operatorname{div}(H_{ij})\Big)$$

is lc near 0 for infinitely many i. For any such i, $c_{ij} \le \kappa_j$ for any j, so $(\widetilde{W}_0, \theta_0 B_i)$ is lc near 0. Since π_0 is étale in codimension 1 and B_i is the pullback of $D_{0,i}$, $(W_0, \theta_0 D_{0,i})$ is lc near 0, that is, θ_0 is a margin of $(W_0, D_{0,i})$ near 0 (Definition 2.9). By Proposition 3.14(3), with $D_N = D_i$, $D_0 = D_{0,i}$ and $\theta = \theta_0$, the numbers a_i for such i belong to a set satisfying the ACC. This contradicts the assumption that the a_i are strictly increasing. Thus $(\widetilde{W}_*, B^*)$ is not klt.

If $\lfloor B^* \rfloor \ne 0$, then, since $(\widetilde{W}_*, B^*)$ is lc by **Step 2**, Theorem 10.4 implies that $(a_i)_{i \ge 1}$ is not strictly increasing, a contradiction. Thus $\lfloor B^* \rfloor = 0$.

Step 4. In this step we find the curve C and prove (1).

Since [Kaw15] works over $\mathbb{K}[[x_1, x_2, x_3]]$ for any field $\mathbb{K}$ of characteristic zero, [Kaw15, Theorem 1.2] holds for the lc and not klt pair $(\widetilde{W}_*, B^*)$ on the regular scheme $\widetilde{W}_*$ of dimension 3: the smallest lc center C of $(\widetilde{W}_*, B^*)$ exists, that is, C is an lc center contained in any lc center, and C is normal. By (B.102) and $\lim_i a_i > 0$, $\{0\}$ is not an lc center. An lc center Z with $\dim Z = 2$ on the regular scheme $\widetilde{W}_*$ is a prime divisor with coefficient one in B^*, which does not exist by **Step 3**. Thus $\dim C = 1$. Since C is normal, C is regular, that is, C is a smooth curve. Any lc center Z satisfies $C \subset Z$ and $\dim Z \le 1$, so $Z = C$. Hence C is the unique lc center of $(\widetilde{W}_*, B^*)$, and $0 \in C$. Since any H_j^* is $\widetilde{G}$-semi-invariant, any element of $\widetilde{G}$ preserves B^* and maps lc centers to lc centers. Thus C is $\widetilde{G}$-stable.

Since the smallest lc center C satisfies $\dim C = 1$, [Kaw15, Proposition 6.1] implies that $\mathrm{mld}(\widetilde{W}_* \ni 0, B^*) \le 1$. Together with **Steps 2** and 3, we obtain (1).

Step 5. In this step we prove (2).

Let $\mathfrak{m}_C$ be the maximal ideal of $\mathcal{O}_C$. Since C is a regular curve with closed point 0 and residue field $\Bbbk$, $\mathcal{O}_C \cong \Bbbk[[u]]$ for a parameter u. Since $\iota = (1, -1) \in \widetilde{G}$, ι preserves C.

If ι acts trivially on C, then $p|_C = \iota^* p|_C = -p|_C$ and $v|_C = -v|_C$, so $p|_C = v|_C = 0$. Thus $C \subset (p = v = 0)$, and $C = (p = v = 0)$ since both are integral curves.

Assume that ι acts non-trivially on C. Since ι has order 2, acts non-trivially on $\mathcal{O}_C \cong \Bbbk[[u]]$ and $\mathrm{char}\, \Bbbk = 0$, ι acts on $\mathfrak{m}_C/\mathfrak{m}_C^2$ by -1. Let $\mathfrak{m}_0 := (p, v, t)$. The surjection

$$\mathfrak{m}_0/\mathfrak{m}_0^2 \to \mathfrak{m}_C/\mathfrak{m}_C^2$$

is $\widetilde{G}$-equivariant and $\iota^* t = t$, so it maps t to 0, and its restriction

$$\rho : \Bbbk p \oplus \Bbbk v \to \mathfrak{m}_C/\mathfrak{m}_C^2$$

is surjective. Thus $\mathrm{Ker}\, \rho$ is a $\widetilde{G}$-stable line. Since $\widetilde{G}$ acts on $\Bbbk p \oplus \Bbbk v$ diagonally, a $\widetilde{G}$-stable line in $\Bbbk p \oplus \Bbbk v$ is $\Bbbk p$ or $\Bbbk v$ if $\chi_p \ne \chi_v$, and any line consists of $\widetilde{G}$-semi-invariant forms if $\chi_p = \chi_v$. Thus there exist $\widetilde{G}$-semi-invariant linear forms P, Q in p, v such that the change of coordinates $(p, v) \to (P, Q)$ has determinant one and Q spans $\mathrm{Ker}\, \rho$. Then $P|_C$ is a parameter of $\mathcal{O}_C$, and $Q|_C, t|_C \in \mathfrak{m}_C^2$. Hence there exist power series $\psi, \eta \in \Bbbk[[P]]$ of order at least 2 with

$$Q|_C = \psi(P|_C) \quad \text{and} \quad t|_C = \eta(P|_C).$$

Since $P, Q - \psi(P), t - \eta(P)$ are regular parameters at 0, $(Q - \psi(P) = t - \eta(P) = 0)$ is an integral curve containing C, so

$$C = (Q - \psi(P) = t - \eta(P) = 0).$$

Since $\iota^* P = -P$, $\iota^* Q = -Q$, $\iota^* t = t$, and $\iota(C) = C$, applying ι^* to $Q|_C = \psi(P|_C)$ and $t|_C = \eta(P|_C)$, we obtain $-Q|_C = \psi(-P|_C)$ and $t|_C = \eta(-P|_C)$. As $P|_C$ is a parameter of $\mathcal{O}_C$, $\psi(-P) = -\psi(P)$ and $\eta(-P) = \eta(P)$. Hence $\psi = P f(P^2)$ and $\eta = g(P^2)$ for power series f and g, and $f(0) = g(0) = 0$ since ψ and η have order at least 2. This proves (2).

Step 6. In this step we prove the first sentence of (3). Let

$$B_{J_0}^* := \sum_{j \in J_0} \kappa_j \,\mathrm{div}(H_j^*) \quad \text{and} \quad B_{J_1}^* := \sum_{j \in J_1} \kappa_j \,\mathrm{div}(H_j^*),$$

so that $B^* = B_{J_0}^* + B_{J_1}^*$.

Since C is the unique lc center of $(\widetilde{W}_*, B^*)$ by **Step 4** and H_j^* does not vanish identically on C for any $j \in J_1$, there exists a real number $\theta > 1$ such that $(\widetilde{W}_*, B^*_{J_0} + \theta B^*_{J_1})$ is lc.

The double cover $\pi_0 : \widetilde{W}_* \to W_*$ is the quotient by ι and is étale in codimension 1, and H_j^* is the pullback of $h_{*,j}$ for any j. Since $H_j^* \neq 0$, $h_{*,j} \notin (xy + z^2)$. Thus

$$\Big(W_*,\ \sum_{j\in J_0} \kappa_j \operatorname{div}(h_{*,j}) + \theta \sum_{j\in J_1} \kappa_j \operatorname{div}(h_{*,j})\Big)$$

is lc. By [Liu26b, Lemma D.14],

$$\Big(\operatorname{Spec} \Bbbk[[x,y,z,t]],\ \operatorname{div}(xy+z^2) + \sum_{j\in J_0} \kappa_j \operatorname{div}(h_{*,j}) + \theta \sum_{j\in J_1} \kappa_j \operatorname{div}(h_{*,j})\Big)$$

is lc.

For any positive integer d and any i with $N_i \geq d$, the d-jet of $xy + z^2 + t^{N_i}$ is that of $xy + z^2$. Hence $(xy + z^2, (h_{*,j})_j)$ is a compatible generic limit of the tuples $(xy + z^2 + t^{N_i}, (h_{ij})_j)$ on $\mathbb{A}^4$, with $\{\operatorname{jet}_d(xy + z^2)\} \times Z_d$ in place of Z_d in Definition 2.5. By Theorem 2.18(1) with 1 in place of θ,

$$\Big(\mathbb{A}^4,\ \operatorname{div}(xy+z^2+t^{N_i}) + \sum_{j\in J_0} \kappa_j \operatorname{div}(h_{ij}) + \theta \sum_{j\in J_1} \kappa_j \operatorname{div}(h_{ij})\Big)$$

is lc near 0 for infinitely many i. For any such i, since $W_{N_i} = (xy + z^2 + t^{N_i} = 0)$ is normal, [Kaw07, Theorem] implies that

$$\Big(W_{N_i},\ \sum_{j\in J_0} \kappa_j \operatorname{div}(h_{ij}) + \theta \sum_{j\in J_1} \kappa_j \operatorname{div}(h_{ij})\Big)$$

is lc near 0. Since $c_{ij} \leq \kappa_j$ for any j,

$$\Big(W_{N_i},\ D_i^\theta := \sum_{j\in J_0} \kappa_j \operatorname{div}(h_{ij}) + \theta \sum_{j\in J_1} c_{ij} \operatorname{div}(h_{ij})\Big) \text{ is lc near } 0 \tag{B.103}$$

for infinitely many i. This is the first sentence of (3).

Step 7. In this step we prove the remaining part of (3).

Possibly passing to a subsequence, we may assume that (B.103) holds for any i. For any i let F_i be a computing lift of a_i, let w_{ij} and R_i be as in (3), and let

$$\mu_i := \sum_{j\in J_1} c_{ij} w_{ij} \quad \text{and} \quad \beta_i := a_i - R_i + \mu_i.$$

Since h_{ij} is G-semi-invariant and $|G| \leq M$, $h_{ij}^{M!}$ is G-invariant, hence a regular function on W_{N_i}/G. Let $\overline{F}_i$ be the prime divisor over W_{N_i}/G induced by F_i. Since

$$\operatorname{mult}_{F_i}(h_{ij}^{M!}) = e_{F_i} \operatorname{mult}_{\overline{F}_i}(h_{ij}^{M!}),$$

$$w_{ij} = \frac{\operatorname{mult}_{F_i} h_{ij}}{e_{F_i}} = \frac{\operatorname{mult}_{\overline{F}_i}(h_{ij}^{M!})}{M!} \in \frac{1}{M!}\mathbb{N}. \tag{B.104}$$

By the equality case of (B.99),

$$a_i = \frac{a(F_i, W_{N_i}, 0)}{e_{F_i}} - \sum_{j=1}^{n} c_{ij} w_{ij},$$

so (B.103) implies that

$$0 \leq \frac{a(F_i, W_{N_i}, D_i^\theta)}{e_{F_i}} = a_i - \sum_{j\in J_0} (\kappa_j - c_{ij}) w_{ij} - (\theta - 1) \sum_{j\in J_1} c_{ij} w_{ij} = a_i - R_i - (\theta - 1)\mu_i.$$

Since $c_{ij} \le \kappa_j$, $R_i \ge 0$. Thus

$$0 \le R_i + (\theta - 1)\mu_i \le a_i \le 2. \tag{B.105}$$

Let $\gamma := \min(\Gamma\backslash\{0\})$, which exists since Γ satisfies the DCC. For any $j \in J_1$, $c_{ij} \ge \gamma$, so (B.105) implies that $w_{ij} \le 2/((\theta-1)\gamma)$. By (B.104), w_{ij} belongs to a finite set independent of i. Possibly passing to a subsequence, we may assume that $w_{ij} = w_j$ does not depend on i for any $j \in J_1$.

We prove that the β_i belong to a finite set independent of i. Let n_z and n_t be the integers of Proposition 3.14(1). They do not depend on i since G does not. Let

$$\mathfrak{a}_i := (z^{M!})^{(n_z-1)/M!}(t^{M!})^{(n_t-1)/M!}\prod_{j\in J_0}(h_{ij}^{M!})^{\kappa_j/M!}$$

on W_{N_i}/G. The boundary $\sum_{j\in J_0} c_{ij}\operatorname{div}(h_{ij}) \le D_i$ is a klt labelled boundary with labels in Γ. Thus Proposition 3.14(1), with $D_N = \sum_{j\in J_0} c_{ij}\operatorname{div}(h_{ij})$, and Lemma 2.15(2) imply that for any prime divisor F over W_{N_i} and the prime divisor E over W_{N_i}/G induced by F,

$$a\Big(E, W_{N_i}/G, (z^{M!})^{(n_z-1)/M!}(t^{M!})^{(n_t-1)/M!}\prod_{j\in J_0}(h_{ij}^{M!})^{c_{ij}/M!}\Big) = \frac{1}{e_F}a\Big(F, W_{N_i}, \sum_{j\in J_0} c_{ij}\operatorname{div}(h_{ij})\Big).$$

Since $\operatorname{mult}_E(h_{ij}^{M!})/M! = \operatorname{mult}_F h_{ij}/e_F$ as in (B.104), both sides change by

$$-\sum_{j\in J_0}(\kappa_j - c_{ij})\operatorname{mult}_F h_{ij}/e_F$$

when c_{ij} is replaced by κ_j for $j \in J_0$, which gives

$$a(E, W_{N_i}/G, \mathfrak{a}_i) = \frac{1}{e_F}a\Big(F, W_{N_i}, \sum_{j\in J_0}\kappa_j\operatorname{div}(h_{ij})\Big). \tag{B.106}$$

Since

$$\sum_{j\in J_0}\kappa_j\operatorname{div}(h_{ij}) \le D_i^{\theta},$$

(B.103) and (B.106) imply that $(W_{N_i}/G, \mathfrak{a}_i)$ is lc near the image of 0. Possibly shrinking W_{N_i}/G to a neighborhood of the image of 0, we may assume that $(W_{N_i}/G, \mathfrak{a}_i)$ is lc. By (B.106) with $F = F_i$,

$$a(\overline{F}_i, W_{N_i}/G, \mathfrak{a}_i) = \frac{a(F_i, W_{N_i}, 0)}{e_{F_i}} - \sum_{j\in J_0}\kappa_j w_{ij} = a_i + \sum_{j\in J_1} c_{ij}w_{ij} - \sum_{j\in J_0}(\kappa_j - c_{ij})w_{ij} = \beta_i.$$

By Proposition 3.14(1), the Cartier index of $K_{W_{N_i}/G}$ at the image of 0 is the order of a character of G, which divides $|G|$ and hence $\operatorname{lcm}(1,\dots,M)$. Thus $\operatorname{lcm}(1,\dots,M)K_{W_{N_i}/G}$ is Cartier near the image of 0. Since the lc pairs $(W_{N_i}/G, \mathfrak{a}_i)$ have dimension 3 and the exponents of $\mathfrak{a}_i$ belong to the finite set

$$\Big\{\frac{n_z-1}{M!}, \frac{n_t-1}{M!}\Big\} \cup \Big\{\frac{\kappa_j}{M!}\ \Big|\ j \in J_0\Big\}$$

independent of i, [Nak16a, Theorem 1.2] implies that the β_i belong to a discrete subset of $\mathbb{R}$ independent of i. By (B.105),

$$0 \le \beta_i \le a_i + \mu_i \le 2 + \frac{2}{\theta - 1}.$$

Thus the β_i belong to a finite set independent of i. Possibly passing to a subsequence, we may assume that $\beta_i = \beta$ does not depend on i.

Hence $a_i = \beta + R_i - \sum_{j\in J_1} c_{ij}w_j$ for any i. For any $i < i'$,

$$R_{i'} - R_i = (a_{i'} - a_i) + \sum_{j\in J_1}(c_{i'j} - c_{ij})w_j > 0,$$

since the a_i are strictly increasing, the c_{ij} are increasing in i and $w_j \geq 0$. Thus R_i is strictly increasing. Since $0 \leq R_i \leq 2$ by (B.105), $\lim_i R_i$ exists and $\lim_i R_i > R_i \geq 0$ for any i. This proves (3). □

Proposition B.14. *Notation and conditions as in Set-up 10.3. Then the following hold.*

(1) Assume that the a_i are strictly increasing and, possibly passing to a subsequence, that the conclusions of Proposition 10.5 hold. Let I_1 and u_i, $i \in I_1$, be as follows.
  (i) If ι acts trivially on C, let I_1 be the set of all indices and $u_i := t$.
  (ii) If ι acts non-trivially on C, let I_1 be an infinite set of indices and, for $i \in I_1$, let X_i be a non-zero G-semi-invariant linear form in x, y, z such that the tuples $((h_{ij})_j, X_i)$, $i \in I_1$, have a compatible generic limit $((h_{*,j})_j, X_*)$ with $X_*(p^2, -v^2, pv) = P^2$ for the coordinate P of Proposition 10.5(2), and let $u_i := X_i$.

Then there exist a real number $\delta > 0$ and an infinite subset $I_0 \subset I_1$ such that for any $i \in I_0$, the pair

$$\Big(W_{N_i}, \sum_j \kappa_j \operatorname{div}(h_{ij}) + \delta \operatorname{div}(u_i)\Big)$$

is lc near 0 and any computing lift F_i of a_i satisfies

$$\sum_j (\kappa_j - c_{ij}) \frac{\operatorname{mult}_{F_i} h_{ij}}{e_{F_i}} + \delta \frac{\operatorname{mult}_{F_i} u_i}{e_{F_i}} \leq a_i \leq 2.$$

In particular, $\operatorname{mult}_{F_i} u_i / e_{F_i} \leq 2/\delta$.

(2) Let I be an infinite set of indices and $\rho > 0$ a real number. For any $i \in I$, let X_i be a non-zero G-semi-invariant linear form in x, y, z. Assume that $\operatorname{mult}_F X_i / e_F \leq \rho$ for any $i \in I$ and any computing lift F of a_i. Let $i \in I$ with $N_i \geq 2\lceil M\rho \rceil + 3$, and let F_i be a computing lift of a_i. Then there exist an integer n_i with $1 \leq n_i \leq \lceil M\rho \rceil$ and a G-equivariant projective birational morphism $f_i : Y_i \to W_{N_i}$, isomorphic over $W_{N_i} \backslash \{0\}$, which is the composite of n_i blow-ups, each of them the blow-up of the unique singular point of the variety obtained before it, satisfying the following.
  (a) The generic point η_i of $\operatorname{center}_{Y_i} F_i$ has a G-invariant open neighborhood V_i in Y_i which is G-equivariantly isomorphic to a G-invariant open subset of $\mathbb{A}^3$ or of $\mathbb{G}_m \times \mathbb{A}^2$, where G acts on the three coordinates by characters.
  (b) Any exceptional prime divisor of f_i is G-stable,

$$\#\{E \mid E \text{ is an exceptional prime divisor of } f_i \text{ and } E \cap V_i \neq \emptyset\} \leq 2,$$

and for any such E, $E \cap V_i$ is a coordinate divisor of V_i.

(3) In (2), for $1 \leq k \leq n_i$ let $Y_i^{(k)}$ be the variety obtained by the first k blow-ups, let $\tau_k : Y_i \to Y_i^{(k)}$ be the composite of the remaining $n_i - k$ blow-ups, let E_k° be the exceptional divisor of the k-th blow-up, and let $E_k := \tau_{k*}^{-1} E_k^\circ$. Then

$$\sum_{k=1}^{n_i} k \frac{\operatorname{mult}_{F_i} E_k}{e_{F_i}} \leq \frac{\operatorname{mult}_{F_i} X_i}{e_{F_i}} \leq \rho,$$

where $\operatorname{mult}_{F_i} E_k$ is the order along F_i of a local equation of E_k at η_i if $\eta_i \in E_k$, and $\operatorname{mult}_{F_i} E_k := 0$ if $\eta_i \notin E_k$.

Proof. We divide the proof into six steps.

Step 1. In this step we choose the limit u_* of u_i and show that the limit pair stays lc after adding $\delta \operatorname{div}(u_*)$. If ι acts trivially on C, let $u_* := t$. If ι acts non-trivially on C, let $u_* := X_*$ be the limit in (1)(ii), so that $X_*(p^2, -v^2, pv) = P^2$. Let $U^* := u_*(p^2, -v^2, pv, t)$, so $U^* = t$ or $U^* = P^2$.

We show that $U^*|_C \neq 0$. If ι acts trivially on C, then $C = (p = v = 0)$ by Proposition 10.5(2), so $t|_C$ is a regular parameter of C. If ι acts non-trivially on C, then $C = (Q - Pf(P^2) = t - g(P^2) = 0)$ for the coordinates P, Q of Proposition 10.5(2), so $P|_C$ is a regular parameter of C and $P^2|_C \neq 0$.

Since any non-empty closed subset of $\widetilde{W}_*$ contains 0, C is the unique lc center of the lc pair $(\widetilde{W}_*, B^*)$ by Proposition 10.5(1). Since moreover $U^*|_C \neq 0$, there exists a real number $\delta > 0$ such that $(\widetilde{W}_*, B^* + \delta \operatorname{div}(U^*))$ is lc. We fix δ.

Since the double cover $\pi_0 : \widetilde{W}_* \to W_*$ of Set-up 10.3 is étale outside the curve $(p = v = 0)$ and $\dim \widetilde{W}_* = 3$, $K_{\widetilde{W}_*} = \pi_0^* K_{W_*}$. Since $H_j^* = h_{*,j} \circ \pi_0$ and $U^* = u_* \circ \pi_0$,

$$K_{\widetilde{W}_*} + B^* + \delta \operatorname{div}(U^*) = \pi_0^*\big(K_{W_*} + D_* + \delta \operatorname{div}(u_*|_{W_*})\big),$$

so the pair

$$\big(W_*, D_* + \delta \operatorname{div}(u_*|_{W_*})\big)$$

is lc. Since H_j^* and U^* are non-zero, neither $h_{*,j}$ nor u_* belongs to $(xy + z^2)$. By inversion of adjunction over $\Bbbk[[x, y, z, t]]$ [Liu26b, Lemma D.14],

$$\Big(\operatorname{Spec} \Bbbk[[x, y, z, t]], \operatorname{div}(xy + z^2) + \sum_j \kappa_j \operatorname{div}(h_{*,j}) + \delta \operatorname{div}(u_*)\Big) \text{ is lc.} \tag{B.107}$$

Step 2. In this step we take the limit of the tuples augmented by u_i.

Whether or not ι acts trivially on C, for any positive integer d, the d-jet of $xy + z^2 + t^{N_i}$ is the d-jet of $xy + z^2$ for $N_i > d$. Since $\lim_i N_i = +\infty$, possibly passing to a subsequence of I_1, the tuples $(xy + z^2 + t^{N_i}, (h_{ij})_j, u_i)$ have the compatible generic limit $(xy + z^2, (h_{*,j})_j, u_*)$.

Step 3. In this step we prove (1). By (B.107) and Theorem 2.18(1) (in dimension 4, with $\mathfrak{c} = \mathcal{O}$, the exponents $1, \kappa_1, \dots, \kappa_n, \delta$ and $\theta = 1$), there exists an infinite subset $I_0 \subset I_1$ such that

$$\Big(\mathbb{A}^4, \operatorname{div}(xy + z^2 + t^{N_i}) + \sum_j \kappa_j \operatorname{div}(h_{ij}) + \delta \operatorname{div}(u_i)\Big)$$

is lc near 0 for any $i \in I_0$. Since W_{N_i} is normal and neither h_{ij} nor u_i vanishes identically on W_{N_i}, inversion of adjunction [Kaw07, Theorem] implies that

$$\Big(W_{N_i}, \sum_j \kappa_j \operatorname{div}(h_{ij}) + \delta \operatorname{div}(u_i)\Big)$$

is lc near 0 for any $i \in I_0$.

Let $i \in I_0$ and let F_i be a computing lift of a_i. Then, by Definition 2.4, $\operatorname{center}_{W_{N_i}} F_i = \{0\}$ and

$$a(F_i, W_{N_i}, D_i) = a_i e_{F_i}.$$

Thus

$$0 \leq a\Big(F_i, W_{N_i}, \sum_j \kappa_j \operatorname{div}(h_{ij}) + \delta \operatorname{div}(u_i)\Big) = a_i e_{F_i} - \sum_j (\kappa_j - c_{ij}) \operatorname{mult}_{F_i} h_{ij} - \delta \operatorname{mult}_{F_i} u_i.$$

By Set-up 10.3, $a_i \leq 2$. Dividing by e_{F_i}, we obtain the inequality of (1). Since c_{ij} is increasing in i with limit κ_j, $\kappa_j - c_{ij} \geq 0$, and since $\operatorname{mult}_{F_i} h_{ij} \geq 0$, $\delta \operatorname{mult}_{F_i} u_i / e_{F_i} \leq a_i \leq 2$. This proves (1).

Step 4. In this step we construct f_i and prove (2). Let $m \geq 3$ be an integer, let $\sigma_m : W_m' \to W_m$ be the blow-up of $W_m = (xy + z^2 + t^m = 0) \subset \mathbb{A}^4$ at 0, and let E be the exceptional divisor of σ_m. Then W_m' is covered by the following four charts.

- The x-chart: $x = x'$, $y = x'y'$, $z = x'z'$, $t = x't'$. On the x-chart, W_m' is $(y' + z'^2 + x'^{m-2}t'^m = 0)$, which is isomorphic to $\mathbb{A}^3$ with coordinates x', z', t', and $E = (x' = 0)$.
- The y-chart, given by interchanging x and y in the x-chart.
- The z-chart: $z = z'$, $x = z'x'$, $y = z'y'$, $t = z't'$. On the z-chart, W_m' is $(x'y' + 1 + z'^{m-2}t'^m = 0)$ and $E = (z' = 0)$. Since $x'y' = -1$ on E, E is contained in the open subset $(x' \neq 0)$, which is isomorphic to an open subset of $\mathbb{G}_m \times \mathbb{A}^2$ with coordinates x', z', t' via $y' = -(1 + z'^{m-2}t'^m)/x'$.

- The t-chart: $x = tx'$, $y = ty'$, $z = tz'$. On the t-chart, W'_m is $(x'y' + z'^2 + t^{m-2} = 0)$, that is, W_{m-2} with coordinates x', y', z', t, and $E = (t = 0) = (x'y' + z'^2 = 0) \subset \mathbb{A}^3$, which is reduced and irreducible.

On the t-chart, $x' \neq 0$ or $y' \neq 0$ at any point of E other than the origin o', as $x' = y' = 0$ and $x'y' + z'^2 = 0$ imply $z' = 0$. The open subset $(x' \neq 0)$ of the t-chart is isomorphic to an open subset of $\mathbb{G}_m \times \mathbb{A}^2$ with coordinates x', z', t via $y' = -(z'^2 + t^{m-2})/x'$, and similarly for $(y' \neq 0)$. Thus $W'_m \backslash \{o'\}$ is smooth, and o' is a singular point of W'_m if $m \geq 4$. Moreover, E is the quadric $(xy + z^2 = 0) \subset \mathbb{P}^3$, which is integral, and $\sigma_{m*}^{-1}(t = 0)$ is $(t' = 0)$ on the x-, y- and z-charts and does not intersect the t-chart, since $t = x't'$ on the x-chart, $t = y't'$ on the y-chart and $t = z't'$ on the z-chart. If G acts on W_m by characters on x, y, z, t, then G acts on W'_m, each chart is G-invariant, being the locus where the given semi-invariant coordinate generates the pullback of the maximal ideal of 0, and x', y', z', t' are G-semi-invariant.

Now fix i as in (2), and write $N := N_i$ and $F := F_i$. Let $Y^{(0)} := W_N$, $o_0 := 0$, and $\mathcal{U}_0 := W_N$ with coordinates $(x_0, y_0, z_0, t) := (x, y, z, t)$. Let $k \geq 0$, and assume that $Y^{(k)}$, a G-fixed point $o_k \in Y^{(k)}$ and a G-invariant open neighborhood $\mathcal{U}_k$ of o_k in $Y^{(k)}$ are constructed, where $\mathcal{U}_k$ is identified with an open neighborhood of 0 in W_{N-2k} with G-semi-invariant coordinates x_k, y_k, z_k, t satisfying

$$x = t^k x_k, \quad y = t^k y_k, \quad \text{and} \quad z = t^k z_k,$$

and o_k is the origin. Assume that $\operatorname{center}_{Y^{(k)}} F = \{o_k\}$. Then X_i/t^k is a linear form in x_k, y_k, z_k, which vanishes at o_k, and $\operatorname{mult}_F t \geq 1$ since $\operatorname{center}_{W_N} F = \{0\}$. Thus

$$k + 1 \leq k \operatorname{mult}_F t + \operatorname{mult}_F(X_i/t^k) = \operatorname{mult}_F X_i \leq \rho e_F \leq M\rho,$$

where $e_F \leq |G| \leq M$. In particular, $k \leq \lceil M\rho \rceil - 1$, so $N - 2k \geq 5$.

Let $\varphi_{k+1} : Y^{(k+1)} \to Y^{(k)}$ be the blow-up of o_k, let E°_{k+1} be the exceptional divisor, let $\mathcal{U}_{k+1}$ be the t-chart of the blow-up of $\mathcal{U}_k$ at o_k, with coordinates $(x_{k+1}, y_{k+1}, z_{k+1}, t) := (x', y', z', t)$, let o_{k+1} be the origin of $\mathcal{U}_{k+1}$, and for $0 \leq j < r \leq k + 1$ let $\varphi_{j,r} : Y^{(r)} \to Y^{(j)}$ be the composite of $\varphi_{j+1}, \ldots, \varphi_r$. Then $\mathcal{U}_{k+1}$ is an open neighborhood of 0 in W_{N-2k-2}, and $x = t^{k+1} x_{k+1}$, $y = t^{k+1} y_{k+1}$, $z = t^{k+1} z_{k+1}$. By the chart description with $m = N - 2k$, by induction on k the following hold:

- $Y^{(k+1)} \backslash \{o_{k+1}\}$ is smooth, and o_{k+1} is a singular point of $Y^{(k+1)}$ if $N - 2k - 2 \geq 2$.
- $E^\circ_{k+1} \cap \mathcal{U}_{k+1} = (t = 0)$, and $\mathcal{U}_{k+1}$ does not intersect $\varphi^{-1}_{j,k+1*} E^\circ_j$ for $1 \leq j \leq k$.
- E°_{k+1} is G-stable, as it is the exceptional divisor of the blow-up of the G-fixed point o_k.

For the second item, $E^\circ_k \cap \mathcal{U}_k = (t = 0)$ for $k \geq 1$ by induction, so $\varphi^{-1}_{k+1*} E^\circ_k$ does not intersect $\mathcal{U}_{k+1}$, and for $1 \leq j \leq k - 1$, $\mathcal{U}_k$ does not intersect $\varphi^{-1}_{j,k*} E^\circ_j$, so $\varphi^{-1}_{j,k+1*} E^\circ_j$, which is contained in $\varphi^{-1}_{k+1}(\varphi^{-1}_{j,k*} E^\circ_j)$, does not intersect $\varphi^{-1}_{k+1}(\mathcal{U}_k) \supset \mathcal{U}_{k+1}$. By the bound $k \leq \lceil M\rho \rceil - 1$ above, there exists a least integer $n_i \geq 1$ with $\operatorname{center}_{Y^{(n_i)}} F \neq \{o_{n_i}\}$, and $n_i \leq \lceil M\rho \rceil$. Let $Y_i := Y^{(n_i)}$ and $f_i : Y_i \to W_N$ the composite. For $k < n_i$, o_k is the unique singular point of $Y^{(k)}$, since $N - 2k \geq 5$. The morphism f_i is projective, birational, G-equivariant and isomorphic over $W_N \backslash \{0\}$.

Let η_i be the generic point of $\operatorname{center}_{Y_i} F$. Since $\operatorname{center}_{Y^{(n_i-1)}} F = \{o_{n_i-1}\}$, $\eta_i \in E_{n_i}$ and $\eta_i \neq o_{n_i}$, and η_i belongs to one of the four charts of the blow-up of $\mathcal{U}_{n_i-1}$ at o_{n_i-1}. We define V_i as follows.

- If η_i belongs to the x-chart, let V_i be the x-chart, and similarly for the y-chart.
- Otherwise, if η_i belongs to the z-chart, let V_i be the open subset $(x' \neq 0)$ of the z-chart.
- Otherwise, let V_i be the open subset $(x_{n_i} \neq 0)$ or $(y_{n_i} \neq 0)$ of $\mathcal{U}_{n_i}$ containing η_i.

In each case, V_i is a G-invariant open neighborhood of η_i, as the coordinates of each chart are G-semi-invariant, and V_i is G-equivariantly isomorphic to a G-invariant open subset of $\mathbb{A}^3$ or of $\mathbb{G}_m \times \mathbb{A}^2$ with G-semi-invariant coordinates. This proves (2)(a).

Since V_i is contained in the preimage of $\mathcal{U}_{n_i-1}$, the exceptional prime divisors of f_i intersecting V_i are among E_{n_i} and, if $n_i \geq 2$, E_{n_i-1}. On the x-chart, $E_{n_i} = (x' = 0)$ and $E_{n_i-1} = (t' = 0)$,

since $E^\circ_{n_i-1} \cap \mathcal{U}_{n_i-1} = (t = 0)$ and $t = x't'$. The y-chart is symmetric. On the z-chart, $E_{n_i} = (z' = 0)$ and $E_{n_i-1} = (t' = 0)$. On $\mathcal{U}_{n_i}$, we have $E_{n_i} = (t = 0)$, and $E_{n_i-1} \cap \mathcal{U}_{n_i} = \emptyset$. Since any E°_k is G-stable, so is any E_k, and this proves (2)(b).

Step 5. In this step we compute the order along E_k of a polynomial without monomials divisible by xy. Let $1 \le k \le n_i$. Let $\mathrm{wt}_{(k,k,k,1)}$ and $\mathrm{in}_{(k,k,k,1)}$ be the weight and the initial form for the weight $(k, k, k, 1)$ on (x, y, z, t) (Definition 2.3). On $\mathcal{U}_k$,

$$\mathrm{div}(t) = E^\circ_k \cap \mathcal{U}_k = (x_k y_k + z_k^2 = 0) \subset \mathbb{A}^3,$$

which is reduced and irreducible, so $\mathrm{mult}_{E_k} t = 1$. Let h be a polynomial in x, y, z, t without monomials divisible by xy, and let $w := \mathrm{wt}_{(k,k,k,1)}(h)$. Since

$$x^{m_1} y^{m_2} z^{m_3} t^{m_4} = t^{k(m_1+m_2+m_3)+m_4} x_k^{m_1} y_k^{m_2} z_k^{m_3},$$

we have $h = t^w(h_0(x_k, y_k, z_k) + t h_1)$ on $\mathcal{U}_k$, where h_1 is regular on $\mathcal{U}_k$ and

$$h_0 := \mathrm{in}_{(k,k,k,1)} h(x_k, y_k, z_k, 1)$$

is a non-zero polynomial without monomials divisible by $x_k y_k$. Since any non-zero multiple of $x_k y_k + z_k^2$ has a monomial divisible by $x_k y_k$, h_0 does not vanish identically on $(x_k y_k + z_k^2 = 0) \subset \mathbb{A}^3$. Thus

$$\mathrm{mult}_{E_k} h = \mathrm{wt}_{(k,k,k,1)}(h). \tag{B.108}$$

In particular, $\mathrm{mult}_{E_k} \ell = k$ for any non-zero linear form ℓ in x, y, z.

Step 6. In this step we prove (3). By (B.108), $\mathrm{mult}_{E_k} X_i = k$ for $1 \le k \le n_i$, so

$$f_i^* \,\mathrm{div}(X_i) = f_{i*}^{-1} \,\mathrm{div}(X_i) + \sum_{k=1}^{n_i} k E_k.$$

Since V_i is smooth, all divisors on V_i are Cartier, and taking orders along F of local equations at η_i,

$$\mathrm{mult}_F X_i = \mathrm{mult}_F\big(f_{i*}^{-1} \,\mathrm{div}(X_i)\big) + \sum_{k=1}^{n_i} k \,\mathrm{mult}_F E_k \ge \sum_{k=1}^{n_i} k \,\mathrm{mult}_F E_k,$$

where E_k with $\eta_i \notin E_k$ contributes 0. Dividing by e_F and using $\mathrm{mult}_F X_i / e_F \le \rho$, we obtain (3). □

Proposition B.15. *In Set-up 10.3, for any i let $Y_{i,0} := W_{N_i}$, for any integer k with $1 \le k < N_i/2$ let $Y_{i,k} \to Y_{i,k-1}$ be the blow-up of the unique singular point of $Y_{i,k-1}$, and let $E_{i,k}$ be the exceptional divisor of $Y_{i,k} \to Y_{i,k-1}$. For $k < k' < N_i/2$ we also write $E_{i,k}$ for $\tau_*^{-1} E_{i,k}$, where $\tau : Y_{i,k'} \to Y_{i,k}$ is the composite of the blow-ups. Assume that there exists a positive integer m_0 with the following property. For any i with $N_i > 2m_0$, there exist a computing lift F_i of a_i and an integer $m_i \in \{1, \dots, m_0\}$ such that $\mathrm{center}_{Y_{i,k}} F_i$ is the singular point of $Y_{i,k}$ for any $k < m_i$, and $\mathrm{center}_{Y_{i,m_i}} F_i$ is a curve or E_{i,m_i}. Then $(a_i)_{i \ge 1}$ is not strictly increasing.*

Proof. Suppose that data as in Set-up 10.3 satisfy the assumption of the proposition and that the a_i are strictly increasing. Possibly truncating the sequence, we may assume that $N_i \ge 2m_0 + 3$ for any i. Let $\gamma := \min(\Gamma \setminus \{0\})$, which exists since Γ satisfies the DCC and contains the labels $c_{ij} \ne 0$, and let

$$\Lambda := \{\sigma - k \mid \sigma \in \Gamma_+,\ k \in \{1, \dots, m_0\}\} \cap [-m_0, 1).$$

Since Γ_+ satisfies the DCC, so does Λ. We construct a set $\mathcal{A} \subset \mathbb{R}$ satisfying the ACC and depending only on Γ, M, and m_0 such that $a_i \in \mathcal{A}$ for any i, which contradicts the strict increase of the a_i.

In **Steps 1–10** we fix i. We write $Y_k := Y_{i,k}$ for $0 \le k \le m_0$, $E_k := E_{i,k}$ for $1 \le k \le m_0$, o_k for the singular point of Y_k, $g_k : Y_k \to Y_{k-1}$ for the blow-up, and $f_k : Y_k \to W_{N_i}$ for the composite morphism.

Step 1. In this step we describe Y_k for $0 \le k \le m_0$. We show by induction on k that Y_k is smooth outside o_k and that o_k has a G-invariant open neighborhood

$$\text{(B.109)}\quad U_k = (x'_k y'_k + z'^2_k + t^{N_i-2k} = 0) \subset \mathbb{A}^4_{x'_k,y'_k,z'_k,t}, \qquad (x,y,z,t) = (t^k x'_k, t^k y'_k, t^k z'_k, t) \text{ on } U_k,$$

in which o_k is the origin, $E_k \cap U_k = (t=0)$ if $k \ge 1$, and $E_{k'} \cap U_k = \emptyset$ for $1 \le k' < k$. For $k = 0$ let $(x'_0, y'_0, z'_0) := (x, y, z)$ and $U_0 := W_{N_i}$. Let $1 \le k \le m_0$, so that $N_i - 2k + 2 \ge 5$. The blow-up of the origin of U_{k-1} is covered by the x'-, y'-, z'- and t-charts, on which x'_{k-1}, y'_{k-1}, z'_{k-1}, resp. t generates the pullback of the maximal ideal of the origin. On the t-chart, $(x'_{k-1}, y'_{k-1}, z'_{k-1}) = t(x'_k, y'_k, z'_k)$, the equation is the one of U_k, and, if $k \ge 2$, the t-chart does not meet $g_{k*}^{-1}(E_{k-1} \cap U_{k-1})$, where $E_{k-1} \cap U_{k-1} = (t=0)$. The x'-chart is the graph

$$\frac{y'_{k-1}}{x'_{k-1}} = -\Big(\frac{z'_{k-1}}{x'_{k-1}}\Big)^2 - x'^{N_i-2k}_{k-1}\Big(\frac{t}{x'_{k-1}}\Big)^{N_i-2k+2},$$

so it is isomorphic to $\mathbb{A}^3$ with coordinates x'_{k-1}, z'_{k-1}/x'_{k-1}, t/x'_{k-1}. Exchanging x'_{k-1} and y'_{k-1}, the y'-chart is isomorphic to $\mathbb{A}^3$ with coordinates y'_{k-1}, z'_{k-1}/y'_{k-1}, t/y'_{k-1}. The z'-chart has the equation $\widetilde{x}\widetilde{y} + 1 + z'^{N_i-2k}_{k-1}\widetilde{t}^{N_i-2k+2} = 0$ in the coordinates $\widetilde{x} := x'_{k-1}/z'_{k-1}$, $\widetilde{y} := y'_{k-1}/z'_{k-1}$, z'_{k-1}, $\widetilde{t} := t/z'_{k-1}$. Where $\widetilde{x} = \widetilde{y} = 0$, the equation forces $z'_{k-1}\widetilde{t} \ne 0$, so the partial derivative in $\widetilde{t}$ does not vanish, and the z'-chart is smooth. This proves the claim. Since G acts diagonally on x, y, z, t, it acts on each Y_k, each chart above is G-invariant, and G acts diagonally on the coordinates of each chart.

Since $N_i - 2k + 2 \ge 3$, the tangent cone of U_{k-1} at the origin is $(x'_{k-1}y'_{k-1} + z'^2_{k-1} = 0)$. Hence, with $\overline{x}, \overline{y}, \overline{z}, \overline{t}$ the classes of $x'_{k-1}, y'_{k-1}, z'_{k-1}, t$,

$$E_k \cong Q := (\overline{x}\,\overline{y} + \overline{z}^2 = 0) \subset \mathbb{P}^3_{[\overline{x}:\overline{y}:\overline{z}:\overline{t}]} \quad \text{and} \quad \mathcal{O}_{E_k}(-E_k) \cong \mathcal{O}_Q(1),$$

and o_k is the vertex $[0:0:0:1]$ of the integral quadric cone Q. On Q, $\overline{x} = \overline{y} = 0$ forces $\overline{z} = 0$, so the x'-chart and the y'-chart cover $E_k \backslash \{o_k\}$. On the x'-chart, $E_k = (x'_{k-1} = 0)$ and, if $k \ge 2$, $E_{k-1} = (t/x'_{k-1} = 0)$. On the y'-chart, $E_k = (y'_{k-1} = 0)$ and, if $k \ge 2$, $E_{k-1} = (t/y'_{k-1} = 0)$. In particular, for $k \ge 2$, $E_k \cap E_{k-1}$ is the smooth conic $Q \cap (\overline{t} = 0)$, and E_k and E_{k-1} are transversal along it.

By (3.33), the form $dx \wedge dz \wedge dt/x$ generates $\omega_{W_{N_i}}$ near 0 and the form $dx'_k \wedge dz'_k \wedge dt/x'_k$ generates ω_{U_k}, and by (B.109) the first equals $t^k dx'_k \wedge dz'_k \wedge dt/x'_k$ on U_k. Thus

$$a(E_k, W_{N_i}, 0) = k+1 \quad \text{and} \quad (\operatorname{mult}_{E_k} x, \operatorname{mult}_{E_k} y, \operatorname{mult}_{E_k} z, \operatorname{mult}_{E_k} t) = (k,k,k,1).$$

For $1 \le k \le m_0$ let

$$\mu_k := \sum_{j=1}^n c_{ij} \operatorname{mult}_{E_k} h_{ij}, \qquad \vartheta_k := \mu_k - k, \quad \text{and} \quad D_k^+ := \sum_{j=1}^n c_{ij} f_{k*}^{-1} \operatorname{div}(h_{ij}),$$

and let $\mu_0 := 0$. Then $1 - a(E_k, W_{N_i}, D_i) = \vartheta_k$, so near $f_k^{-1}(0)$

$$K_{Y_k} + D_{Y_k} = f_k^*(K_{W_{N_i}} + D_i), \qquad D_{Y_k} := D_k^+ + \sum_{k'=1}^k \vartheta_{k'} E_{k'}.$$

Since (W_{N_i}, D_i) is klt near 0, $D_i \ge 0$, and $\mu_k \in \Gamma_+$, we have $\vartheta_k \in \Lambda$. The coefficient of a prime divisor P in D_k^+ is the coefficient of $f_k(P)$ in D_i, so it belongs to $\Gamma_+ \cap (0,1)$ if it is non-zero.

Step 2. In this step we prove (B.111) and (B.112) below. Let h be one of the marks h_{ij}, and let $\mathcal{E}(h) \subset \mathbb{N}^4$ be the set of exponents α of the monomials $x^{\alpha_1}y^{\alpha_2}z^{\alpha_3}t^{\alpha_4}$ occurring in the expansion of h in $\mathbb{C}[[x,y,z,t]]$. By Set-up 10.3, $\alpha_1\alpha_2 = 0$ for any $\alpha \in \mathcal{E}(h)$. Let $|\alpha| := \alpha_1 + \alpha_2 + \alpha_3$. For $r \in \mathbb{R}_{\ge 0}$, the weight of Definition 2.3 with respect to $(r,r,r,1)$ is

$$\operatorname{wt}_{(r,r,r,1)}(h) = \min\{r|\alpha| + \alpha_4 \mid \alpha \in \mathcal{E}(h)\}.$$

We claim that

$$\operatorname{mult}_{E_k} h = \operatorname{wt}_{(k,k,k,1)}(h) \quad \text{for } 1 \le k \le m_0. \tag{B.110}$$

Let $\epsilon := \operatorname{wt}_{(k,k,k,1)}(h)$, let $h^{\le}$ be the sum of the terms of the expansion of h of total degree at most ϵ, and let $h^{=}$ be the sum of the terms $x^{\alpha_1}y^{\alpha_2}z^{\alpha_3}t^{\alpha_4}$ of the expansion of h with

$$k|\alpha| + \alpha_4 = \epsilon.$$

Then

$$h - h^{\le} \in (x, y, z, t)^{\epsilon+1}\mathcal{O}_{\mathbb{A}^4,0},$$

so $\operatorname{mult}_{E_k}(h - h^{\le}) \ge \epsilon + 1$ by **Step 1**. By (B.109), $f_k^* h^{\le} = t^{\epsilon}(h_k^{\circ} + t h_k')$ for polynomials h_k° and h_k', where

$$h_k^{\circ}(x_k', y_k', z_k') = h^{=}(x_k', y_k', z_k', 1).$$

Distinct $\alpha, \alpha' \in \mathcal{E}(h)$ with $k|\alpha| + \alpha_4 = k|\alpha'| + \alpha_4'$ differ in one of the first three coordinates, and the map

$$(\alpha_1, \alpha_2, \alpha_3) \to (2\alpha_1 + \alpha_3, 2\alpha_2 + \alpha_3)$$

is injective on $\{\alpha_1\alpha_2 = 0\}$, so under the double cover π_0 of Set-up 10.3, given by

$$(x, y, z) = (p^2, -v^2, pv),$$

the monomials $x^{\alpha_1}y^{\alpha_2}z^{\alpha_3}$ with $\alpha_1\alpha_2 = 0$ map to pairwise distinct monomials up to sign. Hence they are linearly independent modulo $xy + z^2$, and h_k° does not vanish on $E_k \cap U_k = (x_k'y_k' + z_k'^2 = 0) \subset \mathbb{A}^3$. Thus $\operatorname{mult}_{E_k} h = \epsilon$.

The function

$$r \to \operatorname{wt}_{(r,r,r,1)}(h)$$

is the minimum of affine functions with non-negative slopes, so it is concave and increasing, and $\operatorname{wt}_{(0,0,0,1)}(h) \ge 0$. Let $\operatorname{mult}_{E_0} h := 0$ and $\Delta_k := \operatorname{mult}_{E_k} h - \operatorname{mult}_{E_{k-1}} h$ for $1 \le k \le m_0$. By (B.110),

$$\Delta_1 \ge \Delta_2 \ge \cdots \ge \Delta_{m_0} \ge 0,$$

where $\Delta_1 \ge \Delta_2$ follows from

$$\Delta_1 = \operatorname{wt}_{(1,1,1,1)}(h) \ge \operatorname{wt}_{(1,1,1,1)}(h) - \operatorname{wt}_{(0,0,0,1)}(h) \ge \operatorname{wt}_{(2,2,2,1)}(h) - \operatorname{wt}_{(1,1,1,1)}(h) = \Delta_2.$$

Since $\sum_{k'=1}^{k} \Delta_{k'} = \operatorname{mult}_{E_k} h$, we obtain $0 \le \Delta_k \le \frac{1}{k}\operatorname{mult}_{E_k} h$. Multiplying by c_{ij} and adding over j,

$$0 \le \mu_k - \mu_{k-1} \le \frac{\mu_k}{k} \quad \text{for } 1 \le k \le m_0. \tag{B.111}$$

If $\vartheta_k < 0$, then $\mu_k < k$. If $k \ge 2$ and $\vartheta_{k-1} < 0$, then $\mu_k \le \frac{k}{k-1}\mu_{k-1} < k$ by (B.111). Thus for $1 \le k \le m_0$,

$$\vartheta_k < 0 \;\Rightarrow\; \mu_k - \mu_{k-1} < 1, \qquad \text{and} \qquad \vartheta_{k-1} < 0 \;\Rightarrow\; \vartheta_k < 0 \;\text{ if } k \ge 2. \tag{B.112}$$

Step 3. In this step we bound the restriction of D_k^+ to E_k for $1 \le k \le m_0$. Let h be one of the marks h_{ij}. Near o_{k-1} we have

$$f_{k-1*}^{-1}\operatorname{div}(h) = f_{k-1}^{*}\operatorname{div}(h) - (\operatorname{mult}_{E_{k-1}} h)E_{k-1},$$

and $f_{k-1*}^{-1}\operatorname{div}(h)$ is Cartier on U_{k-1}, being $\operatorname{div}(t^{-\operatorname{mult}_{E_{k-1}} h} f_{k-1}^{*} h)$ (for $k = 1$ it is $\operatorname{div}(h)$). If $k \ge 2$, then $g_k^* E_{k-1} = E_{k-1} + E_k$ near E_k by **Step 1**. Hence, for any k,

$$f_{k*}^{-1}\operatorname{div}(h) = g_k^* f_{k-1*}^{-1}\operatorname{div}(h) - (\operatorname{mult}_{E_k} h - \operatorname{mult}_{E_{k-1}} h)E_k \quad \text{near } E_k.$$

The pullback $g_k^* f_{k-1*}^{-1}\operatorname{div}(h)$ restricts to the trivial line bundle on E_k and $\mathcal{O}_{E_k}(-E_k) \cong \mathcal{O}_Q(1)$, so $f_{k*}^{-1}\operatorname{div}(h)|_{E_k}$ is a Cartier divisor in $|\mathcal{O}_Q(\operatorname{mult}_{E_k} h - \operatorname{mult}_{E_{k-1}} h)|$. Since Q is an integral surface of degree 2 in $\mathbb{P}^3$, for any integral curve $C' \subset Q$ and any $A \in |\mathcal{O}_Q(N')|$ we have

$\mathrm{mult}_{C'} A \cdot \deg C' \le 2N'$. Multiplying by c_{ij}, adding over j and using (B.112), we obtain for any integral curve $C' \subset E_k$

$$\mathrm{mult}_{C'}(D_k^+|_{E_k}) \le \frac{2(\mu_k - \mu_{k-1})}{\deg C'}, \quad \text{which is } < \frac{2}{\deg C'} \text{ if } \vartheta_k < 0. \tag{B.113}$$

In particular, if $k \ge 2$, $\vartheta_k < 0$ and $C' = E_k \cap E_{k-1}$, then $\mathrm{mult}_{C'}(D_k^+|_{E_k}) \le \mu_k - \mu_{k-1} < 1$.

Step 4. In this step we treat the case when $\mathrm{center}_{Y_{m_i}} F_i = E_{m_i}$. Then $F_i = E_{m_i}$, $e_{F_i} \le |G| \le M$ and, by **Step 1**, $e_{F_i} a_i = a(E_{m_i}, W_{N_i}, D_i) = m_i + 1 - \mu_{m_i}$. Thus a_i belongs to

$$\mathcal{A}_1 := \Big\{ \frac{k+1-\sigma}{e} \;\Big|\; k \in \{1, \dots, m_0\},\ e \in \{1, \dots, M\},\ \sigma \in \Gamma_+ \Big\},$$

which satisfies the ACC since Γ_+ satisfies the DCC.

Step 5. By **Step 4**, it remains to treat the case when $C_i := \mathrm{center}_{Y_{m_i}} F_i$ is a curve. Until the end of the proof, we assume that C_i is a curve, and we let ξ_{C_i} be the generic point of C_i. In this step we fix coordinates near ξ_{C_i}, compute the quotient value at ξ_{C_i}, and divide into three cases. Since $\mathrm{center}_{Y_{m_i-1}} F_i = o_{m_i-1}$, we have $C_i \subset g_{m_i}^{-1}(o_{m_i-1}) = E_{m_i}$ and $\xi_{C_i} \ne o_{m_i}$. By **Step 1**, ξ_{C_i} belongs to the x'-chart or the y'-chart of g_{m_i}. If ξ_{C_i} belongs to the x'-chart, let $\zeta_1 := x'_{m_i-1}$, $\zeta_2 := t/x'_{m_i-1}$, and let V be the x'-chart. Otherwise, let $\zeta_1 := y'_{m_i-1}$, $\zeta_2 := t/y'_{m_i-1}$, and let V be the y'-chart. Then V is a G-invariant open subset of $\mathbb{A}^3$ on which G acts diagonally, ζ_1, ζ_2 are G-semi-invariant coordinates on V,

$$E_{m_i} \cap V = (\zeta_1 = 0), \qquad E_{m_i-1} \cap V = (\zeta_2 = 0) \ \text{ if } m_i \ge 2,$$

and $\xi_{C_i} \notin E_{m_i-1}$ unless $m_i \ge 2$ and $C_i = E_{m_i} \cap E_{m_i-1}$, in which case $C_i \cap V = (\zeta_1 = \zeta_2 = 0)$.

Let $G_F := I_G(F_i)$ and $e := e_{F_i} = |G_F|$. For a subgroup H of G containing G_F and an H-invariant open subset $U \subset Y_{m_i}$ containing ξ_{C_i}, let $D_{U/H}$ be the crepant quotient boundary of $D_{Y_{m_i}}|_U$ on U/H and $\overline{\xi}_{C_i} \in U/H$ the image of ξ_{C_i}. By Definition 2.2 and Lemma 2.15(2)(3), $\mathrm{mld}(U/H \ni \overline{\xi}_{C_i}, D_{U/H})$ is the infimum of $a(P, U, D_{Y_{m_i}}|_U)/|I_H(P)|$ over the prime divisors P over U with $\mathrm{center}_U P = gC_i$ for some $g \in H$. A prime divisor P over U whose center is gC_i for some $g \in H$ has center 0 on W_{N_i}, so $a(P, U, D_{Y_{m_i}}|_U) = a(P, W_{N_i}, D_i) > 0$, and since $I_H(P) = I_G(P) \cap H$,

$$\frac{a(P, U, D_{Y_{m_i}}|_U)}{|I_H(P)|} \ge \frac{a(P, W_{N_i}, D_i)}{e_P} \ge a_i,$$

with equalities for $P = F_i$, as $I_H(F_i) = G_F$. Hence

$$\mathrm{mld}(U/H \ni \overline{\xi}_{C_i}, D_{U/H}) = a_i, \tag{B.114}$$

and the infimum is attained at $P = F_i$.

By (B.112), exactly one of the following holds:

(i) $\vartheta_{m_i} \ge 0$, and $\vartheta_{m_i-1} \ge 0$ if $m_i \ge 2$ and $C_i = E_{m_i} \cap E_{m_i-1}$.
(ii) $\vartheta_{m_i} < 0$, and $\vartheta_{m_i-1} \ge 0$ if $m_i \ge 2$ and $C_i = E_{m_i} \cap E_{m_i-1}$.
(iii) $m_i \ge 2$, $C_i = E_{m_i} \cap E_{m_i-1}$, and $\vartheta_{m_i-1} < 0$, so that $\vartheta_{m_i} < 0$.

Step 6. In this step we treat case (i). Then $D_{Y_{m_i}} \ge 0$ near ξ_{C_i}. Possibly replacing V with a G-invariant open neighborhood of ξ_{C_i}, we may assume that $D_{Y_{m_i}}|_V \ge 0$. Let $G_V \subset \mathbb{T}^3$ be the image of G under the characters of the three coordinates of V. As G acts faithfully on V, $|G_V| = |G| \le M$, so $G_V \subset \mu_{M!}^3$, and V/G is an open subset of $X_V := \mathbb{A}^3/G_V$, one of finitely many threefolds depending only on M. Let $\overline{D}_V$ be the closure in X_V of $D_{V/G}$. By (B.114) with $H = G$ and $U = V$,

$$\mathrm{mld}(X_V \ni \overline{\xi}_{C_i}, \overline{D}_V) = \mathrm{mld}(V/G \ni \overline{\xi}_{C_i}, D_{V/G}) = a_i.$$

A prime divisor on V has coefficient $b \in (\Gamma_+ \cup \Lambda) \cap [0,1)$ in $D_{Y_{m_i}}$ by **Step 1**, and the image of the prime divisor on V/G has coefficient $1-(1-b)/e'$ in $D_{V/G}$, where $e' \in \{1,\dots,M\}$ is the ramification index (Lemma 2.15(4)). Hence the coefficients of $\overline{D}_V$ belong to the DCC set

$$\Gamma_{\mathrm{E}} := \left\{1-(1-b)/e' \;\middle|\; b \in (\Gamma_+ \cup \Lambda) \cap [0,1),\ e' \in \{1,\dots,M\}\right\} \cap (0,1).$$

By Theorem 2.22(1) for each of the finitely many X_V and for Γ_{E}, a_i belongs to a set $\mathcal{A}_2$ satisfying the ACC and depending only on Γ, M, and m_0.

Step 7. In this step we pass to a surface germ over an algebraically closed field in cases (ii) and (iii). Let $Y^\circ := Y_{m_i} \backslash \{o_{m_i}\}$, which is smooth by **Step 1**, and $D_{Y^\circ} := D_{Y_{m_i}}|_{Y^\circ}$. The group G_F fixes C_i pointwise: any $g \in G_F$ fixes F_i, so $g(C_i) = C_i$, and g acts trivially on $\kappa(F_i)$, which contains $\kappa(C_i)$. By (B.114) for $H = G_F$ and $U = Y^\circ$, we have $\mathrm{mld}(Y^\circ/G_F \ni \overline{\xi}_{C_i}, D_{Y^\circ/G_F}) = a_i$, and the infimum of **Step 5** is attained at $P = F_i$. We use [Liu26b, Lemma 3.22(1)–(3)] for the sub-pair (Y°, D_{Y°), the curve $C_i \cap Y^\circ$ and the group G_F. By [Liu26b, Lemma 3.22(1)] there exists a G_F-invariant rational function φ on Y°, regular at ξ_{C_i}, such that $\varphi|_{C_i}$ is not constant. Let Y_φ be the generic fiber of φ over $\operatorname{Spec}\mathbb{C}(\varphi)$, on which G_F acts over $\mathbb{C}(\varphi)$, and let $o_\varphi \in Y_\varphi$ be the closed point corresponding to ξ_{C_i}, so that $\mathcal{O}_{Y_\varphi, o_\varphi} = \mathcal{O}_{Y_{m_i}, \xi_{C_i}}$ [Liu26b, Lemma 3.22(2)]. Let Ω be an algebraic closure of $\mathbb{C}(\varphi)$, o_S a closed point of $Y_\varphi \times_{\mathbb{C}(\varphi)} \Omega$ over o_φ, S the connected component of $Y_\varphi \times_{\mathbb{C}(\varphi)} \Omega$ containing o_S, and B_S the pullback of D_{Y° to S. The group G_F acts on S over Ω and fixes o_S, since G_F acts trivially on $\kappa(o_\varphi) = \kappa(C_i)$. The action is faithful, since the function field of Y_{m_i} embeds into the function field of S. By [Liu26b, Lemma 3.22(2)(3)],

$$\mathrm{mld}((S \ni o_S, B_S)/G_F) = a_i, \tag{B.115}$$

and any prime divisor F' over S with center o_S such that $\mathrm{mult}_{F'}$ restricts to mult_{F_i} on

$$\mathcal{O}_{Y_{m_i}, \xi_{C_i}} \subset \mathcal{O}_{S, o_S}$$

satisfies $I_{G_F}(F') = G_F$ and $a(F', S, B_S) = e a_i$. Since a constant field extension in characteristic zero is unramified at the divisorial valuation mult_{F_i}, there exists such a prime divisor F'. We fix one.

The homomorphism $\mathcal{O}_{Y_\varphi, o_\varphi} \to \mathcal{O}_{S, o_S}$ is flat with

$$\mathfrak{m}_{o_\varphi} \mathcal{O}_{S, o_S} = \mathfrak{m}_{o_S},$$

so it preserves lengths of modules of finite length and regular systems of parameters. Let $R_1 \subset S$ be the pullback of E_{m_i}, so $R_1 = (\zeta_1 = 0)$ is a smooth curve through o_S. If $C_i = E_{m_i} \cap E_{m_i - 1}$, let $R_2 \subset S$ be the pullback of $E_{m_i - 1}$, so $R_2 = (\zeta_2 = 0)$ and (ζ_1, ζ_2) is a regular system of parameters of $\mathcal{O}_{S, o_S}$. Let D_S^+ be the pullback of $D_{m_i}^+$ to S. In characteristic zero, the prime components of D_S^+ are reduced, pairwise distinct and different from R_1 and R_2, with coefficients in $\Gamma_+ \cap (0,1)$. Near o_S,

$$B_S = \vartheta_{m_i} R_1 + D_S^+ \ \text{ if } \xi_{C_i} \notin E_{m_i - 1}, \qquad B_S = \vartheta_{m_i} R_1 + \vartheta_{m_i - 1} R_2 + D_S^+ \ \text{ if } C_i = E_{m_i} \cap E_{m_i - 1}.$$

As

$$\mathcal{O}_{E_{m_i}, \xi_{C_i}} = \mathcal{O}_{Y_{m_i}, \xi_{C_i}} / (\zeta_1),$$

we have

$$(D_S^+ \cdot R_1)_{o_S} = \mathrm{mult}_{C_i}(D_{m_i}^+|_{E_{m_i}}). \tag{B.116}$$

The coefficients of B_S are less than 1, and by Definition 2.2 and (B.115) any prime divisor P over S with center o_S has $a(P, S, B_S) \geq |I_{G_F}(P)| a_i > 0$. Hence (S, B_S) is sub-klt near o_S.

Let ϖ be a uniformizer of $\mathcal{O}_{F'}$. Since $I_{G_F}(F') = G_F$, for any $g \in G_F$ we have

$$g^* \varpi \equiv \chi_{F'}(g) \varpi \bmod \varpi^2 \mathcal{O}_{F'}$$

for a homomorphism $\chi_{F'} : G_F \to \kappa(F')^\times$, whose values are roots of unity and thus belong to Ω. Comparing the lowest terms in $\mathcal{O}_{F'}$, any G_F-semi-invariant $\psi \in \mathcal{O}_{S, o_S}$ with character χ_ψ satisfies

$$\chi_\psi = \chi_{F'}^{\mathrm{mult}_{F'} \psi}. \tag{B.117}$$

Step 8. In this step we treat case (ii). Let $B_+ := B_S - \vartheta_{m_i} R_1$ near o_S, so $B_+ = D_S^+$, or $B_+ = D_S^+ + \vartheta_{m_i-1} R_2$ with $\vartheta_{m_i-1} \in \Lambda \cap [0,1)$ if $C_i = E_{m_i} \cap E_{m_i-1}$. The coefficients of B_+ belong to the DCC set Γ_{E} of **Step 6**. If $\xi_{C_i} \notin E_{m_i-1}$, then by (B.116) and (B.113) with $C' = C_i$,

$$(B_+ \cdot R_1)_{o_S} = \mathrm{mult}_{C_i}(D_{m_i}^+|_{E_{m_i}}) < 2.$$

If $C_i = E_{m_i} \cap E_{m_i-1}$, then $(R_2 \cdot R_1)_{o_S} = 1$, and by (B.116), (B.113) with $C' = C_i$, and $\vartheta_{m_i-1} = \mu_{m_i-1} - m_i + 1$,

$$(B_+ \cdot R_1)_{o_S} \le (\mu_{m_i} - \mu_{m_i-1}) + \vartheta_{m_i-1} = 1 + \vartheta_{m_i} < 1.$$

By **Step 7**, $(S \ni o_S)$ is a smooth surface germ over Ω, R_1 is a smooth curve through o_S with coefficient $\vartheta_{m_i} \in \Lambda \cap (-\infty, 0)$ in B_S, (S, B_S) is sub-klt near o_S, and the group G_F acts faithfully on $(S \ni o_S)$ and preserves R_1 and B_S. By (B.115) and Proposition 3.13(B) for the DCC sets $\Lambda \cap (-\infty, 0)$ and Γ_{E}, a_i belongs to a set $\mathcal{A}_3$ satisfying the ACC and depending only on Γ, M, and m_0.

Step 9. In this step we assume case (iii) and show that F' is the exceptional prime divisor of a weighted blow-up in ζ_1, ζ_2. Near o_S write $D_S^+ = \sum_j \nu_j \operatorname{div}(\psi_j)$, where the $\operatorname{div}(\psi_j)$ are pairwise distinct prime divisors through o_S, $\psi_j \in \mathcal{O}_{S,o_S}$ is divisible by neither ζ_1 nor ζ_2, and $\nu_j \in \Gamma_+ \cap (0,1)$. Let

$$\lambda_j := (\operatorname{div}(\psi_j) \cdot R_1)_{o_S} \in \mathbb{N}^+.$$

By (B.116) and (B.113) with $C' = C_i$,

$$\sum_j \nu_j \lambda_j \le \mu_{m_i} - \mu_{m_i-1} < 1. \tag{B.118}$$

Let $q := \gcd(\mathrm{mult}_{F'} \zeta_1, \mathrm{mult}_{F'} \zeta_2)$, let

$$(\omega_1, \omega_2) := \frac{1}{q}(\mathrm{mult}_{F'} \zeta_1, \mathrm{mult}_{F'} \zeta_2),$$

let $\pi : S' \to S$ be the weighted blow-up of o_S with weights (ω_1, ω_2) in ζ_1, ζ_2, and let P be the exceptional prime divisor of π. Then mult_P is the monomial valuation with weights (ω_1, ω_2) on $\Omega[[\zeta_1, \zeta_2]]$ and $\kappa(P) = \Omega(\zeta_2^{\omega_1}/\zeta_1^{\omega_2})$. By (B.117),

$$\chi_{\zeta_1}^{\omega_2} = \chi_{F'}^{q\omega_1\omega_2} = \chi_{\zeta_2}^{\omega_1},$$

so G_F fixes P and acts trivially on $\kappa(P)$, that is, $I_{G_F}(P) = G_F$. By Definition 2.2 and (B.115),

$$0 < e a_i = a(F', S, B_S) \le a(P, S, B_S). \tag{B.119}$$

Suppose that $F' \ne P$. Then the center of F' on S' is a closed point $o' \in P$. Since

$$\pi^* \operatorname{div}(\zeta_k) = \omega_k P + \pi_*^{-1} R_k \quad \text{for } k \in \{1, 2\},$$

we have

$$q\omega_k = \mathrm{mult}_{F'} \zeta_k = \omega_k \,\mathrm{mult}_{F'} P + \mathrm{mult}_{F'} \pi_*^{-1} R_k \quad \text{for } k \in \{1, 2\}.$$

As $\pi_*^{-1} R_1 \cap \pi_*^{-1} R_2 = \emptyset$, one of $\mathrm{mult}_{F'} \pi_*^{-1} R_1$ and $\mathrm{mult}_{F'} \pi_*^{-1} R_2$ is zero. Then $\mathrm{mult}_{F'} P = q$, and both are zero, that is, $o' \notin \pi_*^{-1} R_1 \cup \pi_*^{-1} R_2$. Hence S' is smooth at o', P is a smooth curve near o', and

$$K_{S'} + (1 - a(P, S, B_S))P + \pi_*^{-1} D_S^+ = \pi^*(K_S + B_S) \quad \text{near } o'.$$

By (B.119) and $q \ge 1$,

$$a(F', S', P + \pi_*^{-1} D_S^+) = a(F', S, B_S) - q a(P, S, B_S) \le 0.$$

Since F' is exceptional over S' with center o', $(S', P + \pi_*^{-1} D_S^+)$ is not plt near o', and by inversion of adjunction,

$$(\pi_*^{-1} D_S^+ \cdot P)_{o'} \ge 1. \tag{B.120}$$

On the other hand, for any j,

$$\pi^* \operatorname{div}(\psi_j) = \pi_*^{-1} \operatorname{div}(\psi_j) + (\operatorname{mult}_P \psi_j)P \quad \text{and} \quad P^2 = -\frac{1}{\omega_1\omega_2},$$

so

$$(\pi_*^{-1} \operatorname{div}(\psi_j) \cdot P)_{o'} \le \pi_*^{-1} \operatorname{div}(\psi_j) \cdot P = \frac{\operatorname{mult}_P \psi_j}{\omega_1\omega_2}.$$

Since $\mathcal{O}_{S,o_S}/(\zeta_1)$ is a discrete valuation ring with uniformizer ζ_2 in which ψ_j has order λ_j, the monomial $\zeta_2^{\lambda_j}$ occurs in the expansion of ψ_j in $\Omega[[\zeta_1,\zeta_2]]$, so $\operatorname{mult}_P \psi_j \le \omega_2\lambda_j$. By (B.118),

$$(\pi_*^{-1} D_S^+ \cdot P)_{o'} \le \sum_j \nu_j \frac{\omega_2\lambda_j}{\omega_1\omega_2} = \frac{1}{\omega_1}\sum_j \nu_j\lambda_j < 1,$$

which contradicts (B.120). Thus $F' = P$, so $(\operatorname{mult}_{F'} \zeta_1, \operatorname{mult}_{F'} \zeta_2) = (\omega_1,\omega_2)$ and $q = 1$.

Step 10. In this step we prove the ACC in case (iii) and conclude the proof. Let $\mathfrak{J}_j \subset \Omega[\zeta_1,\zeta_2]$ be the monomial ideal generated by the monomials occurring in the expansion of ψ_j in $\widehat{\mathcal{O}}_{S,o_S} = \Omega[[\zeta_1,\zeta_2]]$. Let $\overline{G}_F \subset \mathbb{T}^2$ be the image of G_F under $(\chi_{\zeta_1},\chi_{\zeta_2})$. Then $\overline{G}_F \cong G_F$, as G_F acts faithfully on S. Let $N_{\overline{G}_F} \subset \mathbb{R}^2$ and wt_w be as in Definition 2.3. For $w = (w_1,w_2) \in \mathbb{R}^2_{>0}$ let

$$\Psi(w) := (1-\vartheta_{m_i})w_1 + (1-\vartheta_{m_i-1})w_2 - \sum_j \nu_j \operatorname{wt}_w(\mathfrak{J}_j).$$

For coprime positive integers ω_1',ω_2', let $P_{\omega'}$ be the exceptional prime divisor of the weighted blow-up of o_S with weights $\omega' = (\omega_1',\omega_2')$ in ζ_1,ζ_2. Since $\operatorname{mult}_{P_{\omega'}}$ is the monomial valuation with weights ω', we have

$$\operatorname{mult}_{P_{\omega'}} \psi_j = \operatorname{wt}_{\omega'}(\mathfrak{J}_j).$$

Since $a(P_{\omega'},S,0) = \omega_1' + \omega_2'$, we have $a(P_{\omega'},S,B_S) = \Psi(\omega')$. Since $g \in G_F$ fixes $P_{\omega'}$ and acts on

$$\zeta_2^{\omega_1'}/\zeta_1^{\omega_2'} \in \kappa(P_{\omega'}) = \Omega(\zeta_2^{\omega_1'}/\zeta_1^{\omega_2'})$$

by $\chi_{\zeta_2}(g)^{\omega_1'}\chi_{\zeta_1}(g)^{-\omega_2'}$, g belongs to $I_{G_F}(P_{\omega'})$ if and only if $\chi_{\zeta_1}(g)^{\omega_2'} = \chi_{\zeta_2}(g)^{\omega_1'}$, and the number of such g is the index of $\mathbb{Z}\omega'$ in $N_{\overline{G}_F} \cap \mathbb{R}\omega'$, since $\overline{G}_F \cong G_F$ and, ω_1',ω_2' being coprime, the quotient map $\mathbb{R}^2 \to \mathbb{T}^2$ of Definition 2.3 maps $\mathbb{R}\omega'$ onto

$$\{u \in \mathbb{T}^2 \mid \omega_2'u_1 - \omega_1'u_2 \in \mathbb{Z}\}$$

with kernel $\mathbb{Z}\omega'$. Hence the primitive vector of $N_{\overline{G}_F}$ on $\mathbb{R}_{>0}\omega'$ is $\omega'/|I_{G_F}(P_{\omega'})|$, and by Definition 2.2 and (B.115),

$$\Psi\big(\omega'/|I_{G_F}(P_{\omega'})|\big) = \frac{a(P_{\omega'},S,B_S)}{|I_{G_F}(P_{\omega'})|} \ge a_i.$$

Any element of $N_{\overline{G}_F} \cap \mathbb{R}^2_{>0}$ is a positive integral multiple of such a primitive vector, Ψ is positively homogeneous of degree 1, and $a_i > 0$, so $\Psi \ge a_i$ on $N_{\overline{G}_F} \cap \mathbb{R}^2_{>0}$. By **Step 9**, $P_\omega = F'$ for $\omega = (\omega_1,\omega_2)$, and $I_{G_F}(F') = G_F$, so $\Psi(\omega/e) = a(F',S,B_S)/e = a_i$. Therefore

$$a_i = \min\{\Psi(w) \mid w \in N_{\overline{G}_F} \cap \mathbb{R}^2_{>0}\}. \tag{B.121}$$

Since $\nu_j \ge \gamma$ and $\lambda_j \ge 1$, (B.118) implies $\gamma\cdot\#\{j\} \le \sum_j \nu_j\lambda_j < 1$. Since $\Lambda \subset [-m_0,1)$ contains ϑ_{m_i} and ϑ_{m_i-1}, dividing Ψ by m_0+1 we obtain

$$\frac{\Psi(w)}{m_0+1} = w_1 + w_2 - \frac{m_0+\vartheta_{m_i}}{m_0+1}\operatorname{wt}_w((\zeta_1)) - \frac{m_0+\vartheta_{m_i-1}}{m_0+1}\operatorname{wt}_w((\zeta_2)) - \sum_j \frac{\nu_j}{m_0+1}\operatorname{wt}_w(\mathfrak{J}_j),$$

whose coefficients belong to the DCC subset

$$\Lambda' := \{0\} \cup \left\{\frac{m_0+\vartheta}{m_0+1} \;\middle|\; \vartheta \in \Lambda\right\} \cup \left\{\frac{b}{m_0+1} \;\middle|\; b \in \Gamma_+ \cap (0,1)\right\}$$

of $[0,1)$. Adding copies of the unit ideal (1), for which $\mathrm{wt}_w((1))=0$, with coefficient 0, we obtain a tuple $\mathfrak{J}$ of exactly $2+\lfloor 1/\gamma\rfloor$ non-zero monomial ideals and a tuple $\mathbf{r}$ of coefficients in Λ'. For $\mathcal{M}\subset\mathbb{N}^2$, wt_w takes equal values on the monomial ideals

$$(\zeta_1^{\alpha_1}\zeta_2^{\alpha_2}\mid\alpha\in\mathcal{M})\subset\Omega[\zeta_1,\zeta_2]\quad\text{and}\quad(x_1^{\alpha_1}x_2^{\alpha_2}\mid\alpha\in\mathcal{M})\subset\mathbb{C}[x_1,x_2].$$

By (B.121), $a_i/(m_0+1)=\psi_{\min}(\overline{G}_F,\mathfrak{J},\mathbf{r})$ in the notation of Definition 2.3. By [Liu26a, Theorem 4.5] for $d=2$, the number $2+\lfloor 1/\gamma\rfloor$ of ideals and the DCC set Λ', a_i belongs to a set $\mathcal{A}_4$ satisfying the ACC and depending only on Γ and m_0.

By **Steps 4, 6, 8**, and 10, for any i

$$a_i\in\mathcal{A}:=\mathcal{A}_1\cup\mathcal{A}_2\cup\mathcal{A}_3\cup\mathcal{A}_4,$$

and $\mathcal{A}$ satisfies the ACC, which contradicts the strict increase of the a_i. □

B.3. **Curves on which the deck involution is non-trivial.** In this subsection we prove Theorem 10.7.

Set-up B.16. Notation and conditions as in Set-up 10.3. Assume that the a_i are strictly increasing, let C be the unique lc center of B^* through 0, a smooth $\widetilde{G}$-stable curve by Proposition 10.5(1), and assume that ι acts non-trivially on C. Let P,Q be $\widetilde{G}$-semi-invariant linear forms in p,v with $dP\wedge dQ=dp\wedge dv$, and let $f,g\in\Bbbk[[s]]$ with $f(0)=g(0)=0$ and

$$C=(Q-Pf(P^2)=t-g(P^2)=0),$$

which exist by Proposition 10.5(2).

We need the following proposition.

Proposition B.17. *Notation and conditions as in Set-up B.16. Let x'_*,y'_*,z'_* be the linear forms in x,y,z over $\Bbbk$ with*

$$x'_*(p^2,-v^2,pv)=P^2,\quad y'_*(p^2,-v^2,pv)=-Q^2,\quad\text{and}\quad z'_*(p^2,-v^2,pv)=PQ.$$

Then, after passing to a subsequence, there exist G-semi-invariant linear forms x'_i,y'_i,z'_i in x,y,z with $x'_iy'_i+z_i'^2=xy+z^2$ such that the tuples $((h_{ij})_j,x'_i,y'_i,z'_i)$ have compatible generic limit $((h_{*,j})_j,x'_*,y'_*,z'_*)$. Moreover, for any choice of a computing lift F_i of a_i for any i, after passing to a further subsequence and replacing $\Bbbk$ with an algebraically closed field containing $\Bbbk$, the following hold.

(1) The center of F_i on the blow-up of W_{N_i} at 0 is a G-fixed closed point q_i of the chart $(x'_i\neq 0)$. Let τ_i be the value of t/x'_i at q_i, and let $\xi_1:=x'_i$, $\xi_2:=z'_i/x'_i$, and $\xi_3:=t/x'_i-\tau_i$. Then the chart $(x'_i\neq 0)$ is $\mathbb{A}^3$ with coordinates ξ_1,ξ_2,ξ_3, the group G acts diagonally on ξ_1,ξ_2,ξ_3, $q_i=(\xi_1=\xi_2=\xi_3=0)$,

$$x'_i=\xi_1,\quad z'_i=\xi_1\xi_2,\quad t=\xi_1(\xi_3+\tau_i),\quad\text{and}\quad y'_i=-\xi_1\xi_2^2-\xi_1^{N_i-1}(\xi_3+\tau_i)^{N_i},$$

and, for any $\gamma\in\mathbb{C}$ with $\tau_i\neq-\gamma$ for any i, the tuples $((h_{ij})_j,x'_i,y'_i,z'_i,\tau_i+\gamma)$ have compatible generic limit $((h_{*,j})_j,x'_*,y'_*,z'_*,g'(0)+\gamma)$.

(2) The integers $m_j:=\mathrm{mult}_0 h_{ij}$ do not depend on i, and $\beta_i:=\sum_j c_{ij}m_j-1$ is increasing with limit

$$\beta_*:=\sum_j\kappa_jm_j-1\in(-1,0).$$

(3) For any j, $h'_{ij}:=\xi_1^{-m_j}h_{ij}$ is a polynomial in ξ_1,ξ_2,ξ_3 not divisible by ξ_1. Let

$$B_i:=\sum_j c_{ij}\,\mathrm{div}(h'_{ij})+\beta_i\,\mathrm{div}(\xi_1).$$

Then

$$\mathrm{mld}((\mathbb{A}^3\ni q_i,B_i)/G)=\frac{a(F_i,\mathbb{A}^3,B_i)}{e_{F_i}}=a_i.$$

Moreover, $\operatorname{mult}_{F_i}\xi_1$ *and* $a(F_i,\mathbb{A}^3,0)-\sum_j\kappa_j\operatorname{mult}_{F_i}h'_{ij}$ *are bounded from above independently of* i*, and* $\lim_i\max_j\operatorname{mult}_{F_i}h'_{ij}=+\infty$.

(4) *Let*

$$h'_{*,j}:=\xi_1^{-m_j}h_{*,j}\big(\xi_1,-\xi_1\xi_2^2,\xi_1\xi_2,\xi_1(\xi_3+g'(0))\big),$$

where $h_{*,j}$ *is written as a power series in* x'_*,y'_*,z'_*,t*, in the order of the four arguments. Then* $(h'_{*,j})_j$ *is a compatible generic limit of the tuples* $(h'_{ij})_j$ *in the coordinates* ξ_1,ξ_2,ξ_3 *(Definition 2.5), and the non-klt locus of*

$$\Big(\operatorname{Spec}\Bbbk[[\xi_1,\xi_2,\xi_3]][\xi_1^{-1}],\sum_j\kappa_j\operatorname{div}(h'_{*,j})\Big)$$

is $C_1\cap(\xi_1\neq0)$*, where*

$$C_1:=\Big(\xi_2-f(\xi_1)=\xi_3+g'(0)-\frac{g(\xi_1)}{\xi_1}=0\Big)\subset\operatorname{Spec}\Bbbk[[\xi_1,\xi_2,\xi_3]],$$

and C_1 *passes through* 0 *and is transverse to* $(\xi_1=0)$.

Proof. We divide the proof into five steps. Before **Step 1** we construct x'_i,y'_i,z'_i.

Since P and Q are $\widetilde{G}$-semi-invariant, x'_*,y'_*,z'_* are G-semi-invariant. Since the coordinate change from (p,v) to (P,Q) has determinant one, a direct computation for the elementary matrices, which generate $SL_2(\Bbbk)$, shows that

$$x'_*y'_*+z'^2_*=xy+z^2.$$

By Lemma 2.6(4) for the compatible generic limit $(h_{*,j})_j$ of Set-up 10.3 and the non-zero power series x'_*,y'_*,z'_*, after passing to a subsequence there exist G-semi-invariant polynomials x'_i,y'_i,z'_i with the characters of x'_*,y'_*,z'_* such that $((h_{*,j})_j,x'_*,y'_*,z'_*)$ is a compatible generic limit of the tuples $((h_{ij})_j,x'_i,y'_i,z'_i)$, with coefficient field $\Bbbk$. Replacing x'_i,y'_i,z'_i with their parts linear in x,y,z keeps $((h_{*,j})_j,x'_*,y'_*,z'_*)$ a compatible generic limit. For any d, let lin_d be the linear map of the jet space taking the parts linear in x,y,z of the last three members. Then lin_d fixes the d-jet of $((h_{*,j})_j,x'_*,y'_*,z'_*)$ and sends the d-jets of the tuples to those of the new tuples. Let Z_d be the closed subset of Definition 2.5 for the limit $((h_{*,j})_j,x'_*,y'_*,z'_*)$. Then $\overline{\operatorname{lin}_d(Z_d)}$ is irreducible with generic point the d-jet of $((h_{*,j})_j,x'_*,y'_*,z'_*)$, and for any non-empty open subset $U'\subset\overline{\operatorname{lin}_d(Z_d)}$, $\operatorname{lin}_d^{-1}(U')\cap Z_d$ is a non-empty open subset of Z_d. Vanishing of a member is a closed condition, so the members stay non-zero for all but finitely many i. The identity $x'y'+z'^2=xy+z^2$ holds for x'_*,y'_*,z'_* and is a polynomial identity among the coefficients of degree 1, which taking linear parts does not change, so by the last sentence of Lemma 2.6(4) it holds for x'_i,y'_i,z'_i for all but finitely many i, which we discard. In particular, x'_i satisfies the conditions on X_i in Proposition B.14(1), and we take $X_i=x'_i$ there, so that $u_i=x'_i$. By Lemma 2.6(2), Set-up 10.3 and the propositions used below stay valid after passing to subsequences.

Step 1. In this step we show that, after passing to a subsequence, the smallest k such that the center of F_i on $Y_{i,k}$ is not the singular point of $Y_{i,k}$, with $Y_{i,k}$ defined below, is a constant k_0 and the center of F_i on Y_{i,k_0} is a closed point, and we bound the orders along F_i at that point.

By Proposition B.14(1), after passing to a subsequence there exists $\delta>0$ such that for any i the pair

$$\Big(W_{N_i},\sum_j\kappa_j\operatorname{div}(h_{ij})+\delta\operatorname{div}(x'_i)\Big)$$

is lc near 0. In particular $(W_{N_i},\sum_j\kappa_j\operatorname{div}(h_{ij}))$ is lc near 0. By Proposition B.14(1), $a_i\le2$ and $\operatorname{mult}_{F_i}x'_i/e_{F_i}\le2/\delta$. Since $e_{F_i}\le|G|\le M$,

$$\operatorname{mult}_{F_i}x'_i\le\frac{2M}{\delta}\quad\text{and}\quad a(F_i,W_{N_i},D_i)\le2M. \tag{B.122}$$

By Proposition 10.5(3), after passing to a further subsequence, $(R_i)_i$ is strictly increasing with a positive limit, where J_0 and R_i are as there, and

$$R_i \le \Big(\sum_{j\in J_0}(\kappa_j - c_{ij})\Big)\max_{j'}\operatorname{mult}_{F_i} h_{ij'}$$

since $e_{F_i}\ge 1$. Since $\lim_i R_i > 0$ and $\lim_i c_{ij} = \kappa_j$,

$$\lim_i \max_j \operatorname{mult}_{F_i} h_{ij} = +\infty. \tag{B.123}$$

For any i let $Y_{i,0} := W_{N_i}$, for any integer k with $1\le k < N_i/2$ let $Y_{i,k}\to Y_{i,k-1}$ be the blow-up of the unique singular point of $Y_{i,k-1}$, and let $E_{i,k}$ be the exceptional divisor of $Y_{i,k}\to Y_{i,k-1}$. For $k<k'<N_i/2$ we also write $E_{i,k}$ for its strict transform on $Y_{i,k'}$. The $Y_{i,k}$ and $E_{i,k}$ are those of Proposition B.15. Let $k_i\ge 1$ be the smallest integer k such that the center of F_i on $Y_{i,k}$ is not the singular point. Since $\operatorname{mult}_F x_i'/e_F \le 2/\delta$ for every computing lift F of a_i by Proposition B.14(1), Proposition B.14(2) and (3) hold with $X_i = x_i'$ and $\rho = 2/\delta$. The morphism of Proposition B.14(2) is the composite $Y_{i,n_i}\to W_{N_i}$ with $n_i\le \lceil 2M/\delta\rceil$, and the center of F_i on Y_{i,n_i} is not the singular point. Thus, for i with $N_i\ge 2\lceil 2M/\delta\rceil+3$, we have $k_i\le n_i\le\lceil 2M/\delta\rceil$ and, with $\operatorname{mult}_{F_i}E_{i,k}$ the order along F_i of a local equation of $E_{i,k}$ at the generic point ϑ_i of $\operatorname{center}_{Y_{i,n_i}}F_i$ if $\vartheta_i\in E_{i,k}$, and $\operatorname{mult}_{F_i}E_{i,k}:=0$ otherwise,

$$\sum_{k=1}^{k_i}\operatorname{mult}_{F_i}E_{i,k} \le \sum_{k=1}^{n_i} k\operatorname{mult}_{F_i}E_{i,k} \le \operatorname{mult}_{F_i}x_i' \le \frac{2M}{\delta}. \tag{B.124}$$

If the set I' of the indices i such that the center of F_i on Y_{i,k_i} is a curve or E_{i,k_i} were infinite, then Proposition B.15 with $m_0=\lceil 2M/\delta\rceil$, for the data restricted to I', would show that $(a_i)_{i\in I'}$ is not strictly increasing, a contradiction. Thus, after passing to a subsequence, $k_i=k_0$ does not depend on i, and the center q_i of F_i on Y_{i,k_0} is a closed point other than the singular point.

For $0\le k<N_i/2$, the singular point of $Y_{i,k}$ has a neighborhood with coordinates $x_i'/t^k, y_i'/t^k, z_i'/t^k, t$ and equation $(x_i'/t^k)(y_i'/t^k)+(z_i'/t^k)^2+t^{N_i-2k}=0$, in which $E_{i,k}=(t=0)$ for $k\ge 1$ and no other exceptional divisor of $Y_{i,k}\to W_{N_i}$ appears. Blowing up the singular point of Y_{i,k_0-1}, we see that the points of E_{i,k_0} other than the singular point of Y_{i,k_0} belong to the images of two open immersions $\phi_i,\phi_i^\dagger:\mathbb{A}^3\to Y_{i,k_0}$, where ϕ_i is given in coordinates w_1,w_2,w_3 on $\mathbb{A}^3$ by

$$\begin{aligned} &x_i' = w_1^{k_0}w_3^{k_0-1},\quad z_i' = w_1^{k_0}w_3^{k_0-1}w_2,\quad t = w_1w_3,\\ &\text{and}\quad y_i' = -w_1^{k_0}w_3^{k_0-1}\big(w_2^2 + w_1^{N_i-2k_0}w_3^{N_i-2k_0+2}\big),\end{aligned} \tag{B.125}$$

and $\phi_i^\dagger$ is given by

$$\begin{aligned} &y_i' = w_1^{k_0}w_3^{k_0-1},\quad z_i' = w_1^{k_0}w_3^{k_0-1}w_2,\quad t = w_1w_3,\\ &\text{and}\quad x_i' = -w_1^{k_0}w_3^{k_0-1}\big(w_2^2 + w_1^{N_i-2k_0}w_3^{N_i-2k_0+2}\big).\end{aligned}$$

On $\phi_i(\mathbb{A}^3)$, $E_{i,k_0}=(w_1=0)$, $E_{i,k_0-1}=(w_3=0)$ if $k_0\ge 2$, and no other exceptional divisor of $Y_{i,k_0}\to W_{N_i}$ intersects $\phi_i(\mathbb{A}^3)$. Since $w_1 = x_i'/t^{k_0-1}$, $w_2=z_i'/x_i'$ and $w_3 = t^{k_0}/x_i'$, the group G acts diagonally on w_1,w_2,w_3. Possibly passing to a subsequence, either $q_i\in\phi_i(\mathbb{A}^3)$ for any i, or $q_i\in\phi_i^\dagger(\mathbb{A}^3)\backslash\phi_i(\mathbb{A}^3)$ for any i. In the second case, we use $\phi_i^\dagger$ in place of ϕ_i, with x_i' and y_i' exchanged in the formulas for w_1,w_2,w_3 above, until we show that this case does not occur. Let $(0,\sigma_i,\tau_i):=\phi_i^{-1}(q_i)$. Possibly passing to a subsequence, either $\tau_i=0$ for any i or $\tau_i\ne 0$ for any i. Let $r_i := w_1w_3$ if $k_0\ge 2$ and $\tau_i=0$, and $r_i:=w_1$ otherwise. Thus r_i is the product of the local equations at q_i of the exceptional divisors through q_i.

For $2\le k\le k_0$, near the singular point of $Y_{i,k-1}$ the only exceptional divisor of $Y_{i,k-1}\to W_{N_i}$ is $E_{i,k-1}$, which is Cartier there and cut out by t, and the blow-up of the singular point of

$(xy + z^2 + t^{N_i-2k+2} = 0)$ has exceptional divisor of log discrepancy 2 along which t vanishes to order 1. Thus

$$a(E_{i,k}, W_{N_i}, 0) = a(E_{i,k-1}, W_{N_i}, 0) + 1,$$

and since $a(E_{i,1}, W_{N_i}, 0) = 2$,

$$a(E_{i,k}, W_{N_i}, 0) = k + 1 \quad \text{for } 1 \le k \le k_0.$$

For $k \in \{k_0 - 1, k_0\}$ with $k \ge 1$ let $b_{jk} := \operatorname{mult}_{E_{i,k}} h_{ij}$, and let $b_{j0} := 0$. Since (W_{N_i}, D_i) is klt, $\sum_j c_{ij} b_{jk} < k + 1$, so $b_{jk} < (k+1)/c_{1j}$ as $c_{ij} \ge c_{1j} > 0$. Possibly passing to a subsequence, the b_{jk} do not depend on i. Let $\lambda_0^* := 0$ and

$$\lambda_k^* := \sum_j \kappa_j b_{jk} - k \quad \text{for } k \in \{k_0 - 1, k_0\},\ k \ge 1,$$

the coefficient of $E_{i,k}$ in the crepant pullback of $(W_{N_i}, \sum_j \kappa_j \operatorname{div}(h_{ij}))$. Since $(W_{N_i}, \sum_j \kappa_j \operatorname{div}(h_{ij}))$ is lc near 0, $-k \le \lambda_k^* \le 1$. Let

$$\widetilde{h}_{ij} := w_1^{-b_{jk_0}} w_3^{-b_{j,k_0-1}} h_{ij}(\phi_i(w)),$$

a polynomial in w_1, w_2, w_3. Then

$$K_{\mathbb{A}^3} + \sum_j \kappa_j \operatorname{div}(\widetilde{h}_{ij}) + \lambda_{k_0}^* \operatorname{div}(w_1) + \lambda_{k_0-1}^* \operatorname{div}(w_3) = \phi_i^*\Big(K_{W_{N_i}} + \sum_j \kappa_j \operatorname{div}(h_{ij})\Big),$$

so for any prime divisor F over $\mathbb{A}^3$ with center $\phi_i^{-1}(q_i)$,

$$a(F, \mathbb{A}^3, 0) - \sum_j \kappa_j \operatorname{mult}_F \widetilde{h}_{ij} = a\Big(F, W_{N_i}, \sum_j \kappa_j \operatorname{div}(h_{ij})\Big) + \lambda_{k_0}^* \operatorname{mult}_F w_1 + \lambda_{k_0-1}^* \operatorname{mult}_F w_3, \tag{B.126}$$

$$\operatorname{mult}_F h_{ij} = \operatorname{mult}_F \widetilde{h}_{ij} + b_{jk_0} \operatorname{mult}_F w_1 + b_{j,k_0-1} \operatorname{mult}_F w_3. \tag{B.127}$$

Since $\kappa_j \ge c_{ij}$,

$$a\Big(F_i, W_{N_i}, \sum_j \kappa_j \operatorname{div}(h_{ij})\Big) \le a(F_i, W_{N_i}, D_i).$$

Since $\lambda_k^* \le 1$, $\operatorname{mult}_{F_i} w_3 = 0$ if $k_0 \ge 2$ and $\tau_i \ne 0$, and $\operatorname{mult}_{F_i} r_i \le \sum_{k \le k_0} \operatorname{mult}_{F_i} E_{i,k}$, (B.122), (B.124) and (B.126) imply

$$\operatorname{mult}_{F_i} r_i \le \frac{2M}{\delta} \quad \text{and} \quad a(F_i, \mathbb{A}^3, 0) - \sum_j \kappa_j \operatorname{mult}_{F_i} \widetilde{h}_{ij} \le 2M + \frac{2M}{\delta}. \tag{B.128}$$

Since the b_{jk} are fixed, (B.123), (B.124) and (B.127) imply

$$\lim_i \max_j \operatorname{mult}_{F_i} \widetilde{h}_{ij} = +\infty. \tag{B.129}$$

Let E_{q_i} be the exceptional divisor of the blow-up of $\mathbb{A}^3$ at $\phi_i^{-1}(q_i)$. Then $a(E_{q_i}, \mathbb{A}^3, 0) = 3$, $\operatorname{mult}_{E_{q_i}} w_1 = 1$, and $\operatorname{mult}_{E_{q_i}} w_3 \le 1$, so (B.126) for $F = E_{q_i}$, lc near 0, and $\lambda_k^* \ge -k$ imply

$$\sum_j \kappa_j \operatorname{mult}_{\phi_i^{-1}(q_i)} \widetilde{h}_{ij} \le 2k_0 + 2. \tag{B.130}$$

Step 2. In this step we take a compatible generic limit that includes the coordinates of q_i, and we compute the limit of the translated tuples $(\widetilde{h}_{ij})_j$.

Fix $\gamma' \in \mathbb{C}$ with $\gamma' \ne -\sigma_i$ for any i, and let $\gamma \in \mathbb{C}$ with $\gamma \ne -\tau_i$ for any i. Let

$$\Theta_i := \big((h_{ij})_j, x_i', y_i', z_i', \sigma_i + \gamma', \tau_i + \gamma\big),$$

a tuple of G-semi-invariant functions on $\mathbb{A}^4$ with characters independent of i (constants are G-invariant). By Lemma 2.6(3) for the limit $((h_{*,j})_j, x_*', y_*', z_*')$ and the constant tuples $(\sigma_i+\gamma', \tau_i+\gamma)$,

after passing to a subsequence and replacing $\Bbbk$ with an algebraically closed field containing it, there exist $\sigma_*, \tau_* \in \Bbbk$ such that

$$\Theta_* := ((h_{*,j})_j, x'_*, y'_*, z'_*, \sigma_* + \gamma', \tau_* + \gamma)$$

is a compatible generic limit of the tuples Θ_i. The last two members of Θ_* are constants since being constant is a closed condition on jets. A log resolution of B^* stays a log resolution after extending the algebraically closed field $\Bbbk$, and the log discrepancy of each exceptional divisor does not change, so the base change of C is an lc center of the extended B^* through 0, and it is the unique one by Proposition 10.5(1) for the extended limit.

If a member of Θ_* is a constant $\alpha \in \mathbb{C}$, then the closure of the 1-jets of that member is the point α, so that member of Θ_i equals α for all but finitely many i. In particular, since either $\tau_i = 0$ for any i or $\tau_i \neq 0$ for any i, the member $\tau_* + \gamma$ equals γ, that is, $\tau_* = 0$, if and only if $\tau_i = 0$ for any i.

Let

$$\widetilde{h}^\circ_{ij}(\xi_1, \xi_2, \xi_3) := \widetilde{h}_{ij}(\xi_1, \xi_2 + \sigma_i, \xi_3 + \tau_i),$$

the polynomials $\widetilde{h}_{ij}$ in the coordinates translated to $\phi_i^{-1}(q_i)$. By (B.130), $\operatorname{mult}_0 \widetilde{h}^\circ_{ij} \leq (2k_0+2)/\kappa_j$. By (B.125), x'_i, y'_i, z'_i, t are divisible by w_1 (and by $w_1 w_3$ if $k_0 \geq 2$), and the translation does not move $w_1 = \xi_1$. Hence a monomial of degree ℓ in x, y, z, t contributes to $\widetilde{h}^\circ_{ij}$ only terms divisible by $\xi_1^{\ell - b_{jk_0}}$, and the part of h_{ij} of degree at least $d + b_{jk_0}$ does not change the d-jet of $\widetilde{h}^\circ_{ij}$. The term of (B.125) depending on N_i is divisible by $w_1^{N_i - 2k_0}$. The linear change from x, y, z to x'_i, y'_i, z'_i preserves $xy + z^2$, so the change has determinant ± 1 and the coefficients of the inverse change are polynomials in those of x'_i, y'_i, z'_i. Dividing a polynomial by a monomial and translating are polynomial in the coefficients and in σ_i, τ_i. Hence for any d there exist d' and a polynomial map Φ_d, independent of i, such that the d-jet of $(\widetilde{h}^\circ_{ij})_j$ is the image under Φ_d of the d'-jet of Θ_i for any i with $N_i \gg 0$.

Let ϕ_* be (B.125) with x'_i, y'_i, z'_i replaced by x'_*, y'_*, z'_* and the term $w_1^{N_i - 2k_0} w_3^{N_i - 2k_0 + 2}$ deleted, so that ϕ_* maps into W_0, and let $\phi_*^\dagger$ be the formula for $\phi_i^\dagger$ with x'_i, y'_i, z'_i replaced by x'_*, y'_*, z'_* and the term $w_1^{N_i - 2k_0} w_3^{N_i - 2k_0 + 2}$ deleted. In the second case of **Step 1**, $\phi_*^\dagger$ is used in place of ϕ_* in the rest of this step. Let

$$\widetilde{h}_{*,j} := w_1^{-b_{jk_0}} w_3^{-b_{j,k_0-1}} h_{*,j}(\phi_*(w)), \qquad \widetilde{h}^\circ_{*,j}(\xi_1, \xi_2, \xi_3) := \widetilde{h}_{*,j}(\xi_1, \xi_2 + \sigma_*, \xi_3 + \tau_*),$$

with $h_{*,j}$ written as a power series in x'_*, y'_*, z'_*, t. Since every component of ϕ_* is divisible by w_1, $h_{*,j}(\phi_*(w)) \in \Bbbk[w_2, w_3][[w_1]]$. It is divisible there by $w_1^{b_{jk_0}} w_3^{b_{j,k_0-1}}$: for $N_i > a + 2k_0$ the coefficient of a monomial $w_1^a w_2^c w_3^e$ in $h_{ij}(\phi_i(w))$ is the value of a polynomial Ψ_{ace}, independent of i, at the coefficients of degree at most a of the members of Θ_i, and the value of Ψ_{ace} at the corresponding coefficients of Θ_* is the coefficient of $w_1^a w_2^c w_3^e$ in $h_{*,j}(\phi_*(w))$. For $a < b_{jk_0}$ or $e < b_{j,k_0-1}$ the value of Ψ_{ace} at Θ_i vanishes for any i with $N_i > a + 2k_0$ since $\widetilde{h}_{ij}$ is a polynomial, so Ψ_{ace} vanishes at the generic point of the closure of the $(a+1)$-jets of the Θ_i with $N_i > a + 2k_0$, that is, for Θ_* (Definition 2.5). Translating w_2, w_3 by constants is an automorphism of $\Bbbk[w_2, w_3][[w_1]]$, so

$$\widetilde{h}^\circ_{*,j} \in \Bbbk[\xi_2, \xi_3][[\xi_1]] \subset \Bbbk[[\xi_1, \xi_2, \xi_3]].$$

The d-jet of $(\widetilde{h}^\circ_{*,j})_j$ is the image under Φ_d of the d'-jet of Θ_*. The closure Z'_d of the image under Φ_d of the closure of the d'-jets of the Θ_i is irreducible with generic point the image of the generic point, and the preimage of a non-empty open subset of Z'_d is non-empty and open. Thus $(\widetilde{h}^\circ_{*,j})_j$ is a compatible generic limit of the tuples $(\widetilde{h}^\circ_{ij})_j$ (Definition 2.5, with H trivial). Moreover $\widetilde{h}^\circ_{*,j} \neq 0$: for $d > (2k_0 + 2)/\kappa_j$ the d-jets of the $\widetilde{h}^\circ_{ij}$ are non-zero and belong to Z'_d for all but finitely many i, and vanishing of the j-th member is a closed condition.

Step 3. In this step we show that $k_0 = 1$, that $q_i \in \phi_i(\mathbb{A}^3)$, and that q_i converges to the point where the strict transform of the image of C intersects the exceptional divisor.

The double cover $\pi_0 : \widetilde{W}_* \to W_*$ is étale outside $(p = v = 0)$, and $\pi_0^*(K_{W_*} + D_*) = K_{\widetilde{W}_*} + B^*$ since $\pi_0^* h_{*,j} = H_j^*$. By the ramification formula and Proposition 10.5(1), (W_*, D_*) is lc with non-klt locus the image $\overline{C}$ of C. Since $C = (Q - Pf(P^2) = t - g(P^2) = 0)$,

$$\overline{C} = \big(z'_* - x'_* f(x'_*) = y'_* + x'_* f(x'_*)^2 = t - g(x'_*) = 0\big),$$

a smooth curve parametrized by x'_*.

Let $r_* := \xi_1\xi_3$ if $k_0 \geq 2$ and $\tau_* = 0$, and $r_* := \xi_1$ otherwise, so that $r_i(\xi_1, \xi_2 + \sigma_i, \xi_3 + \tau_i) = r_*$ by **Step 2**. On $U_* := \operatorname{Spec} \Bbbk[[\xi_1, \xi_2, \xi_3]][r_*^{-1}]$, the functions $w_1 = \xi_1$ and, if $k_0 \geq 2$, $w_3 = \xi_3 + \tau_*$ are units. The morphism $\phi_* : \mathbb{A}^3 \to W_0$ restricts to an isomorphism of $(w_1 w_3 \neq 0)$ (of $(w_1 \neq 0)$ if $k_0 = 1$) onto $W_0 \cap (x'_* t \neq 0)$ (onto $W_0 \cap (x'_* \neq 0)$ if $k_0 = 1$), with inverse

$$w_1 = \frac{x'_*}{t^{k_0 - 1}}, \quad w_2 = \frac{z'_*}{x'_*}, \quad \text{and} \quad w_3 = \frac{t^{k_0}}{x'_*}.$$

Let $V_* := \mathbb{A}^3 \times_{W_0} W_*$ be the base change of ϕ_* along $W_* \to W_0$. Then $V_* \to W_*$ is an isomorphism over $(x'_* t \neq 0)$ (over $(x'_* \neq 0)$ if $k_0 = 1$), and $\operatorname{Spec} \Bbbk[[\xi_1, \xi_2, \xi_3]]$ is the completion of V_* at its unique point over $(0, \sigma_*, \tau_*) \in \mathbb{A}^3$, in which the pullback of $h_{*,j}$ is $h_{*,j}(\phi_*(w))$ of **Step 2**. Since w_1 and, if $k_0 \geq 2$, w_3 are units on U_*, the pullback D_{U_*} of D_* to U_* is $\sum_j \kappa_j \operatorname{div}(\widetilde{h}^\circ_{*,j})$. Since log resolutions and log discrepancies are compatible with completion at $(0, \sigma_*, \tau_*)$ and with localization, the non-klt locus of (U_*, D_{U_*}) is the preimage of $\overline{C}$ under $U_* \to V_* \to W_*$. If $k_0 \geq 2$ and $g = 0$, then $\overline{C} \subset (t = 0)$ does not meet $W_* \cap (x'_* t \neq 0)$, so its preimage in U_* is empty. Otherwise, on $\overline{C} \backslash \{0\}$ we have

$$(w_1, w_2, w_3) = \Big(\frac{x'_*}{g(x'_*)^{k_0 - 1}}, f(x'_*), \frac{g(x'_*)^{k_0}}{x'_*}\Big),$$

and $\operatorname{mult}_0 g \geq 1$. If $k_0 \geq 2$, then w_1 has order at most 0 in x'_* on $\overline{C}$, so the closure of $\phi_*^{-1}(\overline{C})$ does not intersect $(w_1 = 0)$. If $k_0 = 1$, then the closure of $\phi_*^{-1}(\overline{C})$ is the curve $(w_2 = f(w_1),\ w_3 = g(w_1)/w_1)$, which intersects $(w_1 = 0)$ only at $q_* := (0, 0, g'(0))$. Therefore, in the first case of **Step 1**, unless $k_0 = 1$ and $(0, \sigma_*, \tau_*) = q_*$, the pair (U_*, D_{U_*}) is klt.

In the second case of **Step 1**, let $r_*^\dagger$ and $\widetilde{h}^{\dagger\circ}_{*,j}$ be r_* and $\widetilde{h}^\circ_{*,j}$ defined with $\phi_*^\dagger$ in place of ϕ_*, let

$$U_*^\dagger := \operatorname{Spec} \Bbbk[[\xi_1, \xi_2, \xi_3]][(r_*^\dagger)^{-1}], \quad \text{and let} \quad D_{U_*^\dagger} := \sum_j \kappa_j \operatorname{div}(\widetilde{h}^{\dagger\circ}_{*,j}) \quad \text{on } U_*^\dagger.$$

Possibly replacing ϕ_* with $\phi_*^\dagger$ and exchanging x'_* and y'_*, the argument of the previous paragraph shows that the non-klt locus of $(U_*^\dagger, D_{U_*^\dagger})$ is the preimage of $\overline{C}$. In the chart $\phi_*^\dagger$ the coordinate $w_2 = z'_*/y'_*$ equals $-1/f(x'_*)$ on $\overline{C}$, which has a pole at $x'_* = 0$ since $f(0) = 0$, and $\overline{C} \subset (y'_* = 0)$ if $f = 0$, so the closure of the preimage of $\overline{C}$ does not intersect the exceptional divisor $(w_1 = 0)$, and the preimage of $\overline{C}$ in $U_*^\dagger$ is empty. Thus $(U_*^\dagger, D_{U_*^\dagger})$ is klt.

Suppose that (U_*, D_{U_*}) is klt, or, in the second case of **Step 1**, that $(U_*^\dagger, D_{U_*^\dagger})$ is klt. In the second case we use $\phi_i^\dagger$, $r_*^\dagger$, and $\widetilde{h}^{\dagger\circ}_{*,j}$ in place of ϕ_i, r_*, and $\widetilde{h}^\circ_{*,j}$ until the end of the paragraph. Since ϕ_i is an open immersion and the translation by $(0, \sigma_i, \tau_i)$ is an automorphism of $\mathbb{A}^3$ sending 0 to $\phi_i^{-1}(q_i)$, through their composite F_i is a prime divisor over $\mathbb{A}^3$ with center 0, the translation preserves $a(F_i, \mathbb{A}^3, 0)$, and $\operatorname{mult}_{F_i} \widetilde{h}^\circ_{ij} = \operatorname{mult}_{F_i} \widetilde{h}_{ij}$. By **Step 2** and (B.128), the hypotheses of Proposition 6.2 hold for the polynomials $\widetilde{h}^\circ_{ij}$ in the coordinates ξ_1, ξ_2, ξ_3 (in the coordinates ξ_1, ξ_3, ξ_2 if $r_* = \xi_1\xi_3$), the prime divisors F_i, $r = r_*$, $R_0 = 2M/\delta$, and $C_0 = 2M + 2M/\delta$. Thus $\operatorname{mult}_{F_i} \widetilde{h}_{ij}$ is bounded for all but finitely many i, contradicting (B.129).

Thus the second case of **Step 1** does not occur, $k_0 = 1$, $q_i \in \phi_i(\mathbb{A}^3)$, $\sigma_* = 0$, and $\tau_* = g'(0)$. By **Step 2**, $\sigma_i = 0$ for all but finitely many i, and $\tau_i = 0$ for all but finitely many i if $g'(0) = 0$. Since $\overline{C}$ is G-stable and the closure of $\phi_*^{-1}(\overline{C})$ intersects $(w_1 = 0)$ only at q_*, the point q_* is G-fixed. If $g'(0) \neq 0$, then G acts trivially on w_3, since w_3 is G-semi-invariant and takes the

value $g'(0) \neq 0$ at the G-fixed point q_*. Thus, for $g'(0) = 0$ and for $g'(0) \neq 0$, $q_i = \phi_i(0,0,\tau_i)$ is G-fixed and $w_3 - \tau_i$ is G-semi-invariant. For $k_0 = 1$, $\phi_i(\mathbb{A}^3)$ is the chart $(x_i' \neq 0)$ of the blow-up of W_{N_i} at 0, and in the coordinates $\xi_1 = w_1$, $\xi_2 = w_2$, $\xi_3 = w_3 - \tau_i$, (B.125) is the formula in (1). The compatible generic limit in (1) for γ is a sub-tuple of Θ_*. For any $\gamma_1 \in \mathbb{C}$ with $\tau_i \neq -\gamma_1$ for any i, translating the last member by the constant $\gamma_1 - \gamma$ maps the closures Z_d of Definition 2.5 isomorphically onto closed subsets, so the tuples with $\tau_i + \gamma_1$ have the compatible generic limit with $g'(0) + \gamma_1$ along the subsequence of **Step 2**. This proves (1).

Step 4. In this step we prove (3) and (4), and the first part of (2).

Since no monomial of the part of h_{ij} of degree m_j is divisible by xy, the part of h_{ij} of degree m_j does not vanish identically on the cone $(xy + z^2 = 0)$. Since $N_i \geq 3$, $E_{i,1}$ is the cone $(xy + z^2 = 0)$ and $\operatorname{mult}_{E_{i,1}} h_{ij}$ is the least degree of a homogeneous part of h_{ij} not vanishing on it, so $b_{j1} = m_j$. Thus the m_j do not depend on i, and $h_{ij}' = \widetilde{h}_{ij}^{\circ}$ is a polynomial in ξ_1, ξ_2, ξ_3 not divisible by ξ_1. Since $a(E_{i,1}, W_{N_i}, 0) = 2$, the coefficient of $E_{i,1}$ in the crepant pullback of (W_{N_i}, D_i) is $\sum_j c_{ij} m_j - 1 = \beta_i$, so

$$K_{\mathbb{A}^3} + B_i = \phi_i^*(K_{W_{N_i}} + D_i).$$

Since the c_{ij} are increasing with limits κ_j, β_i is increasing with limit β_*.

Any prime divisor F over $\mathbb{A}^3$ with center q_i is a prime divisor over W_{N_i} with center 0, $a(F, \mathbb{A}^3, B_i) = a(F, W_{N_i}, D_i)$, and the groups $I_G(F)$ computed over $\mathbb{A}^3$ and over W_{N_i} are equal since ϕ_i is G-equivariant and birational. Hence, by Definition 2.2, $\operatorname{mld}((\mathbb{A}^3 \ni q_i, B_i)/G) \geq a_i$, and F_i, whose center is q_i by (1), satisfies $a(F_i, \mathbb{A}^3, B_i)/e_{F_i} = a_i$. With $r_i = \xi_1$, (B.128) and (B.129) are the bounds and the divergence in (3). This proves (3).

With $k_0 = 1$, $b_{j1} = m_j$, $\sigma_* = 0$, and $\tau_* = g'(0)$, the polynomials $\widetilde{h}_{ij}^{\circ}$ are the h_{ij}' and the $\widetilde{h}_{*,j}^{\circ}$ are the $h_{*,j}'$ of (4), so $(h_{*,j}')_j$ is a compatible generic limit of the tuples $(h_{ij}')_j$ by **Step 2**. By **Step 3** with $r_* = \xi_1$, the non-klt locus of $(U_*, \sum_j \kappa_j \operatorname{div}(h_{*,j}'))$ is the preimage of $\overline{C}$ under $U_* \to V_* \to W_*$. Since $(w_1, w_2, w_3) = (x_*', f(x_*'), g(x_*')/x_*')$ on $\overline{C} \backslash \{0\}$ and $(\xi_1, \xi_2, \xi_3) = (w_1, w_2, w_3 - g'(0))$, the preimage of $\overline{C}$ under $U_* \to V_* \to W_*$ is $C_1 \cap (\xi_1 \neq 0)$. Since $f(0) = 0$ and $g(\xi_1)/\xi_1$ takes the value $g'(0)$ at 0, C_1 passes through 0. The curve C_1 is parametrized by ξ_1, so it is transverse to $(\xi_1 = 0)$. This proves (4).

Step 5. In this step we show that $\beta_* \in (-1, 0)$, which completes the proof of (2).

Since $\lim_i R_i > 0$ (Proposition 10.5(3)), $J_0 \neq \emptyset$. For $j \in J_0$, H_j^* vanishes on $C \ni 0$, so the constant term of $h_{*,j}$ is 0, and by **Step 2** (applied to constant terms) $h_{ij}(0) = 0$ for all but finitely many i. Thus $m_j \geq 1$ and $\beta_* > -1$.

Suppose that $\beta_* \geq 0$. Let

$$B_i^* := \sum_j \kappa_j \operatorname{div}(h_{ij}') + \beta_* \operatorname{div}(\xi_1) \geq 0.$$

Since $\lambda_1^* = \beta_*$,

$$K_{\mathbb{A}^3} + B_i^* = \phi_i^*\Big(K_{W_{N_i}} + \sum_j \kappa_j \operatorname{div}(h_{ij})\Big),$$

so $(\mathbb{A}^3, B_i^*)$ is lc near q_i. Moreover, since $\beta_* - \beta_i = \sum_j (\kappa_j - c_{ij}) m_j$,

$$B_i^* - B_i = \sum_j (\kappa_j - c_{ij})\big(\operatorname{div}(h_{ij}') + m_j \operatorname{div}(\xi_1)\big) \geq 0. \tag{B.131}$$

Possibly passing to a subsequence, the characters of ξ_1, ξ_2, ξ_3 do not depend on i, so $\mathbb{A}^3/G$ does not depend on i. As a quotient of $\mathbb{A}^3$ by a finite group, $\mathbb{A}^3/G$ is $\mathbb{Q}$-factorial and klt. Let $\eta \in \mathbb{A}^3/G$ be the image of $q_i = 0$. Let B_i° be the part of B_i^* consisting of the prime components through q_i. Since q_i is G-fixed, B_i° is G-invariant and the images of the other components do not contain η.

Let $(\mathbb{A}^3/G, B_{G,i}) := (\mathbb{A}^3, B_i^\circ)/G$. The coefficient of $(\xi_1 = 0)$ in B_i° is β_*, since ξ_1 divides no h'_{ij}, and any other coefficient of B_i° belongs to the finite set

$$\Lambda := \Big\{ \sum_j \kappa_j n_j \;\Big|\; n_j \in \mathbb{N},\ \sum_j \kappa_j n_j \le 1 \Big\}$$

since $(\mathbb{A}^3, B_i^*)$ is lc near q_i and $\kappa_j > 0$. By Lemma 2.15(4), a prime divisor of $\mathbb{A}^3$ with coefficient b in B_i° and ramification index e over $\mathbb{A}^3/G$ has image with coefficient $1-(1-b)/e$ in $B_{G,i}$, and e divides $|G|$. Thus $B_{G,i} \ge 0$ is $\mathbb{R}$-Cartier, the coefficients of $B_{G,i}$ belong to the finite set

$$\Lambda_G := \Big\{ 1 - \frac{1-b}{e} \;\Big|\; b \in \Lambda \cup \{\beta_*\},\ e \text{ divides } |G| \Big\} \cap \mathbb{R}_{>0},$$

and $(\mathbb{A}^3/G, B_{G,i})$ is lc near η. By Theorem 2.22(3) ([Kaw24, Theorem 1.4]) for $\mathbb{A}^3/G$ and Λ_G, there exist a positive integer L independent of i and prime divisors E'_i over $\mathbb{A}^3/G$ computing $\mathrm{mld}(\mathbb{A}^3/G \ni \eta, B_{G,i})$ with $a(E'_i, \mathbb{A}^3/G, 0) \le L$.

Let $\widetilde{E}'_i$ be a prime divisor over $\mathbb{A}^3$ whose restriction to $\mathbb{A}^3/G$ is E'_i (Definition 2.2), which exists by Lemma 2.15(3). Then $\mathrm{center}_{\mathbb{A}^3} \widetilde{E}'_i = \{q_i\}$, the preimage of η, and $e_{\widetilde{E}'_i} \le M$. Let $(\mathbb{A}^3/G, \Delta_G) := (\mathbb{A}^3, 0)/G$. Then $\Delta_G \ge 0$, so

$$a(\widetilde{E}'_i, \mathbb{A}^3, 0) = e_{\widetilde{E}'_i} a(E'_i, \mathbb{A}^3/G, \Delta_G) \le M a(E'_i, \mathbb{A}^3/G, 0) \le ML.$$

Since $(\mathbb{A}^3, B_i^*)$ is lc near q_i and $\beta_* \ge 0$, for any j

$$\kappa_j \operatorname{mult}_{\widetilde{E}'_i} h'_{ij} \le a(\widetilde{E}'_i, \mathbb{A}^3, 0).$$

Since $(\mathbb{A}^3, \mathrm{div}(\xi_1\xi_2\xi_3))$ is lc, $\operatorname{mult}_{\widetilde{E}'_i} \xi_1 \le a(\widetilde{E}'_i, \mathbb{A}^3, 0)$. Hence the integers $a(\widetilde{E}'_i, \mathbb{A}^3, 0)$, $\operatorname{mult}_{\widetilde{E}'_i} h'_{ij}$, $\operatorname{mult}_{\widetilde{E}'_i} \xi_1$, and $e_{\widetilde{E}'_i}$ belong to a finite set independent of i. Since $\mathrm{center}_{\mathbb{A}^3} \widetilde{E}'_i = \{q_i\}$ and no component of $B_i^* - B_i^\circ$ passes through q_i,

$$a'_i := \mathrm{mld}(\mathbb{A}^3/G \ni \eta, B_{G,i}) = \frac{a(\widetilde{E}'_i, \mathbb{A}^3, B_i^*)}{e_{\widetilde{E}'_i}} = \frac{a(\widetilde{E}'_i, \mathbb{A}^3, 0) - \sum_j \kappa_j \operatorname{mult}_{\widetilde{E}'_i} h'_{ij} - \beta_* \operatorname{mult}_{\widetilde{E}'_i} \xi_1}{e_{\widetilde{E}'_i}},$$

so a'_i belongs to a finite set independent of i. Possibly passing to a subsequence, $a'_i = a'$ does not depend on i. By (3) and (B.131), and since F_i maps to a prime divisor over $\mathbb{A}^3/G$ with center η,

$$a' \le \frac{a(F_i, \mathbb{A}^3, B_i^*)}{e_{F_i}} \le \frac{a(F_i, \mathbb{A}^3, B_i)}{e_{F_i}} = a_i.$$

Conversely, since $\widetilde{E}'_i$ has center q_i, (3) and (B.131) imply

$$\begin{aligned} a_i &\le \frac{a(\widetilde{E}'_i, \mathbb{A}^3, B_i)}{e_{\widetilde{E}'_i}} = a' + \sum_j (\kappa_j - c_{ij}) \frac{\operatorname{mult}_{\widetilde{E}'_i} h'_{ij} + m_j \operatorname{mult}_{\widetilde{E}'_i} \xi_1}{e_{\widetilde{E}'_i}} \\ &\le a' + ML \sum_j (\kappa_j - c_{ij}) \Big(\frac{1}{\kappa_j} + m_j \Big). \end{aligned}$$

Thus $\lim_i a_i = a' \le a_i$ for any i, which is impossible for a strictly increasing sequence. Therefore $\beta_* \in (-1, 0)$. □

We need the following lemma.

Lemma B.18. *In Set-up 10.3, assume that the a_i are strictly increasing, let C be the curve of Proposition 10.5(1), and let ϖ, y_1, y_2 be $\widetilde{G}$-semi-invariant regular parameters of $\Bbbk[[p, v, t]]$ with $C = (y_1 = y_2 = 0)$. Then, possibly after exchanging y_1 and y_2, there exist coprime integers*

$w_1 \geq w_2 \geq 1$, Laurent series $\phi_r \in \Bbbk((\varpi))$ for $1 \leq r < w_1/w_2$, and integers ρ_1, ρ_2 satisfying the following, where

$$\widetilde{y}_1 := y_1 - \sum_{1 \leq r < w_1/w_2} \phi_r(\varpi) y_2^r.$$

For any sufficiently large integer L, let

$$\alpha_1 := w_1 L + \rho_1, \qquad \alpha_2 := w_2 L + \rho_2, \qquad V_1 := \varpi^{-\alpha_1}\widetilde{y}_1, \quad \text{and} \quad V_2 := \varpi^{-\alpha_2} y_2$$

in $\Bbbk[[p, v, t]][\varpi^{-1}]$. Then:

(1) There exists a $\widetilde{G}$-stable prime divisor ν_L over $\widetilde{W}_*$ with center 0, whose valuation we also denote by ν_L, such that $\nu_L(\varpi) = 1$, $\nu_L(\widetilde{y}_1) = \alpha_1$, $\nu_L(y_2) = \alpha_2$, and

$$0 < a(\nu_L, \widetilde{W}_*, B^*) \leq 1.$$

(2) One of the following holds, where the rings are subrings of $\Bbbk[[p, v, t]][\varpi^{-1}]$.

(a) $e := 0$, $Y := \mathbb{A}^3_\Bbbk$ with coordinates ϖ, V_1, V_2, and $\mathcal{Y} := \operatorname{Spec} \Bbbk[[p, v, t]][V_1, V_2]$.

(b) There exist an integer $e \geq 1$ and a polynomial $c \in \Bbbk[\varpi]$ with $c(0) \neq 0$, both independent of L, such that, for $V_3 := \varpi^{-e}(V_1^{w_2} - c(\varpi)V_2^{w_1})$, the hypersurface

$$Y := (\varpi^e V_3 = V_1^{w_2} - c(\varpi)V_2^{w_1}) \subset \mathbb{A}^4_\Bbbk$$

with coordinates ϖ, V_1, V_2, V_3 is normal, and $\mathcal{Y} := \operatorname{Spec} \Bbbk[[p, v, t]][V_1, V_2, V_3]$.

In cases (a) and (b), $\varpi\mathcal{O}(\mathcal{Y})$ is a prime ideal, $(\varpi = 0) \subset \mathcal{Y}$ is isomorphic to $(\varpi = 0) \subset Y$, and ν_L is the valuation of the prime divisor $(\varpi = 0)$ on $\mathcal{Y}$.

(3) The function $\widetilde{y}_1$ is $\widetilde{G}$-semi-invariant, and $\widetilde{G}$ acts diagonally on the coordinates of Y. If $e \geq 1$, then the equation of Y is $\widetilde{G}$-semi-invariant and $c(\chi_\varpi(\sigma)\varpi) = c(\varpi)$ for any $\sigma \in \widetilde{G}$. The ring $\mathcal{O}(\mathcal{Y})$ is $\widetilde{G}$-stable, and the morphism $\Psi : \mathcal{Y} \to \widetilde{W}_*$ induced by the inclusion is $\widetilde{G}$-equivariant, is an isomorphism over $(\varpi \neq 0)$, and maps $(\varpi = 0)$ to 0. The elements ϖ, V_1, V_2, and V_3 if $e \geq 1$, generate a maximal ideal $\mathfrak{n}$ of $\mathcal{O}(\mathcal{Y})$, and the homomorphism from $\Bbbk[[p, v, t]]$ to the completion $\widehat{\mathcal{O}}_{Y,0}$ of Y at the origin sending ϖ, $\widetilde{y}_1$, and y_2 to ϖ, $\varpi^{\alpha_1}V_1$, and $\varpi^{\alpha_2}V_2$ extends to a $\widetilde{G}$-equivariant homomorphism $\mathcal{O}(\mathcal{Y}) \to \widehat{\mathcal{O}}_{Y,0}$ which induces an isomorphism

$$\widehat{\mathcal{O}}_{\mathcal{Y},\mathfrak{n}} \cong \widehat{\mathcal{O}}_{Y,0}. \tag{B.132}$$

Near $\mathfrak{n}$, $\mathcal{Y}$ is normal, $(\varpi = 0)$ is the only exceptional prime divisor of Ψ, and

$$K_{\mathcal{Y}} = \Psi^* K_{\widetilde{W}_*} + (\alpha_1 + \alpha_2 + e)\operatorname{div}(\varpi).$$

(4) For any j, $H_j^* = \varpi^{n_j}\widetilde{H}_j$ with $n_j := \nu_L(H_j^*)$ and

$$\widetilde{H}_j \in \mathcal{O}_{\mathcal{Y},\mathfrak{n}} \backslash \varpi\mathcal{O}_{\mathcal{Y},\mathfrak{n}}.$$

The number $\beta_L := \sum_j \kappa_j n_j - \alpha_1 - \alpha_2 - e$ satisfies $0 \leq \beta_L < 1$, and near $\mathfrak{n}$,

$$\Psi^*(K_{\widetilde{W}_*} + B^*) = K_{\mathcal{Y}} + \sum_j \kappa_j \operatorname{div}(\widetilde{H}_j) + \beta_L \operatorname{div}(\varpi).$$

Proof. We have $\Bbbk[[p, v, t]] = \Bbbk[[\varpi, y_1, y_2]]$. For $\sigma \in \widetilde{G}$ let σ^* be the induced automorphism of $\Bbbk[[p, v, t]]$. The functions H_j^* are $\widetilde{G}$-semi-invariant. By Proposition 10.5(1), $(\widetilde{W}_*, B^*)$ is lc, C is an lc center of $(\widetilde{W}_*, B^*)$, and

$$\operatorname{mld}(\widetilde{W}_* \ni 0, B^*) > 0. \tag{B.133}$$

We divide the proof into six steps.

Step 1. In this step we construct w_1, w_2 and the ϕ_r, and show that $\widetilde{y}_1$ is $\widetilde{G}$-semi-invariant.

The field $\Bbbk((\varpi))$ is contained in the local ring $\mathcal{O}_{\widetilde{W}_*,C}$ and maps isomorphically onto the residue field of $\mathcal{O}_{\widetilde{W}_*,C}$, so the completion of $\mathcal{O}_{\widetilde{W}_*,C}$ is $\Bbbk((\varpi))[[y_1, y_2]]$. Let $S := \operatorname{Spec} \Bbbk((\varpi))[[y_1, y_2]]$

with closed point ξ, and let $\mathfrak{a} := \prod_j (H_j^*)^{\kappa_j}$ on S. The ring $\mathcal{O}_{\widetilde{W}_*,C}$ is an excellent regular local ring of dimension 2, so $S \to \widetilde{W}_*$ is a regular morphism, and log resolutions of $(\widetilde{W}_*, B^*)$ pull back to log resolutions of $(S, \mathfrak{a})$, with the relative canonical divisors pulled back [JM12, Lemma 1.2]. After localizing at the generic point of C, any prime divisor over $\widetilde{W}_*$ with center C is obtained by finitely many blow-ups of closed points, which commute with completion, so the prime divisors F over $\widetilde{W}_*$ with center C correspond to the prime divisors F_S over S with center ξ, with $a(F_S, S, \mathfrak{a}) = a(F, \widetilde{W}_*, B^*)$ (cf. [dFEM11, Lemma 2.21]). Thus $(S, \mathfrak{a})$ is lc and $\mathrm{mld}(S \ni \xi, \mathfrak{a}) = 0$. Any $\sigma \in \widetilde{G}$ preserves the ideals (y_1, y_2) and (H_j^*), so σ^* extends to an automorphism of $\Bbbk((\varpi))[[y_1, y_2]]$, not necessarily over $\Bbbk((\varpi))$, which fixes ξ and preserves $\mathfrak{a}$. Let m, $\xi_0 = \xi, \xi_1, \dots, \xi_{m-1}$, $S_0 = S, S_1, \dots, S_m$, and $E_1, \dots, E_m$ be the integer, the points, the surfaces, and the exceptional curves of Lemma 3.4 for $(S \ni \xi, \mathfrak{a})$ over the field $\Bbbk((\varpi))$. By Lemma 3.4(1)(3), $\xi_1, \dots, \xi_{m-1}$ are $\Bbbk((\varpi))$-points,

$$a(E_k, S, \mathfrak{a}) \in (0,1) \text{ for } 1 \le k < m, \qquad a(E_m, S, \mathfrak{a}) = 0, \tag{B.134}$$

and any $\sigma \in \widetilde{G}$ fixes $\xi_1, \dots, \xi_{m-1}$ and E_m.

By Lemma 3.4(2), there exist coprime integers $w_1 \ge w_2 \ge 1$ and regular parameters ζ_1, ζ_2 of $\Bbbk((\varpi))[[y_1, y_2]]$ such that E_m is the exceptional curve of the weighted blow-up of S with weights (w_1, w_2) in (ζ_1, ζ_2). For a non-zero $h \in \Bbbk((\varpi))[[y_1, y_2]]$, $\mathrm{mult}_{E_m} h$ is the smallest weighted degree $w_1 a + w_2 b$ of a monomial $\zeta_1^a \zeta_2^b$ of h. If $w_1 = w_2 = 1$, then the sum defining $\widetilde{y}_1$ is empty and E_m is the exceptional curve of the blow-up of ξ, which is the weighted blow-up with weights $(1,1)$ in $(\widetilde{y}_1, y_2) = (y_1, y_2)$.

Assume that $w_1 > w_2$. For a non-zero $h \in \Bbbk((\varpi))[[y_1, y_2]]$, let $\mathrm{in}_{(w_1,w_2)} h$ be the lowest weight part of h for the weights (w_1, w_2) on ζ_1, ζ_2 (Definition 2.3). Since $w_1 > w_2$, the only monomial of weighted degree $r w_2 < w_1$ is ζ_2^r. Let $\mathfrak{m}$ be the maximal ideal of $\Bbbk((\varpi))[[y_1, y_2]]$. The classes of y_1, y_2 form a basis of $\mathfrak{m}/\mathfrak{m}^2$, so possibly exchanging y_1 and y_2, we may assume that

$$y_2 \equiv \gamma_1 \zeta_1 + \gamma_2 \zeta_2 \mod \mathfrak{m}^2, \qquad \gamma_2 \ne 0.$$

Then $\mathrm{in}_{(w_1,w_2)} y_2 = \gamma_2 \zeta_2$ and $\mathrm{mult}_{E_m} y_2^r = r w_2$ for any r. By induction on r, there exist unique $\phi_r \in \Bbbk((\varpi))$ for $1 \le r < w_1/w_2$ with $\mathrm{mult}_{E_m} \widetilde{y}_1 \ge w_1$: if $\mathrm{mult}_{E_m}(y_1 - \sum_{r'<r} \phi_{r'} y_2^{r'}) \ge r w_2$, then the terms of weighted degree $r w_2$ of $y_1 - \sum_{r'<r} \phi_{r'} y_2^{r'}$ form a multiple of ζ_2^r, and ϕ_r is the unique element cancelling them. For the uniqueness, if $\phi_r' \in \Bbbk((\varpi))$ also satisfy

$$\mathrm{mult}_{E_m} \Big(y_1 - \sum_r \phi_r' y_2^r \Big) \ge w_1$$

and $\phi_r' \ne \phi_r$ for some r, then

$$\mathrm{mult}_{E_m} \Big(\sum_r (\phi_r - \phi_r') y_2^r \Big) = \min\{ r w_2 \mid \phi_r \ne \phi_r' \} < w_1,$$

a contradiction. Since $\mathrm{mult}_{E_m} \widetilde{y}_1 > w_2$, the coefficient of ζ_2 in $\widetilde{y}_1$ is zero. Since the classes of $\widetilde{y}_1, y_2$ in $\mathfrak{m}/\mathfrak{m}^2$ form a basis, $\widetilde{y}_1 \equiv \gamma_1' \zeta_1 \bmod \mathfrak{m}^2$ with $\gamma_1' \ne 0$. Thus

$$\mathrm{in}_{(w_1,w_2)} \widetilde{y}_1 = \gamma_1' \zeta_1 + \gamma_3 \zeta_2^{w_1}, \qquad \mathrm{in}_{(w_1,w_2)} y_2 = \gamma_2 \zeta_2,$$

with $\gamma_3 = 0$ unless $w_2 = 1$. Thus $\mathrm{in}_{(w_1,w_2)} \widetilde{y}_1$ and $\mathrm{in}_{(w_1,w_2)} y_2$ are the images of ζ_1, ζ_2 under a graded automorphism of $\Bbbk((\varpi))[\zeta_1, \zeta_2]$, so E_m is also the exceptional curve of the weighted blow-up with weights (w_1, w_2) in the regular parameters $\widetilde{y}_1, y_2$. In both cases $w_1 = w_2$ and $w_1 > w_2$,

$$\mathrm{mult}_{E_m} \widetilde{y}_1 = w_1, \qquad \mathrm{mult}_{E_m} y_2 = w_2, \tag{B.135}$$

and the residue of $\widetilde{y}_1^{w_2} y_2^{-w_1}$ along E_m is transcendental over $\Bbbk((\varpi))$, because E_m is the exceptional curve of the weighted blow-up with weights (w_1, w_2) in $(\widetilde{y}_1, y_2)$.

Let $\sigma \in \widetilde{G}$. Since σ fixes E_m, $\mathrm{mult}_{E_m} \sigma^* h = \mathrm{mult}_{E_m} h$ for any non-zero $h \in \Bbbk((\varpi))[[y_1, y_2]]$. Thus

$$\chi_{y_1}(\sigma)^{-1}\sigma^*\widetilde{y}_1 = y_1 - \sum_r \chi_{y_1}(\sigma)^{-1}\chi_{y_2}(\sigma)^r \phi_r(\chi_\varpi(\sigma)\varpi) y_2^r$$

satisfies $\mathrm{mult}_{E_m} \sigma^*\widetilde{y}_1 = \mathrm{mult}_{E_m} \widetilde{y}_1 \geq w_1$. By the uniqueness of the ϕ_r, $\sigma^*\widetilde{y}_1 = \chi_{y_1}(\sigma)\widetilde{y}_1$, so $\widetilde{y}_1$ is $\widetilde{G}$-semi-invariant.

Step 2. In this step we construct ν_L and prove the inequalities in (1).

Let $Z_0 := \widetilde{W}_*$ and $C_0 := C$. Inductively for $0 \leq k < m$, let $Z_{k+1} \to Z_k$ be the blow-up of C_k with exceptional divisor F_{k+1}, and, if $k+1 < m$, let C_{k+1} be the closure of the image of ξ_{k+1} under $S_{k+1} \to Z_{k+1}$. We show by induction on k that Z_k is regular, $Z_k \times_{\widetilde{W}_*} S = S_k$, $F_k \times_{\widetilde{W}_*} S = E_k$ if $k \geq 1$, and, if $k < m$, $C_k \cong C$ and $C_k \times_{Z_k} S_k = \xi_k$. For $k = 0$ the claims hold, as $C \times_{\widetilde{W}_*} S = \xi$. Let $0 \leq k < m$ and assume that Z_k is regular, $Z_k \times_{\widetilde{W}_*} S = S_k$, and $F_k \times_{\widetilde{W}_*} S = E_k$ if $k \geq 1$. Since the generic point of C_k has residue field $\kappa(\xi_k) = \Bbbk((\varpi))$, the function field of C, the morphism $C_k \to C$ is proper and birational, hence an isomorphism onto the regular curve C. Thus $C_k \times_{Z_k} S_k = C_k \times_C (C \times_{\widetilde{W}_*} S) = \xi_k$, Z_{k+1} is regular, and $F_{k+1} \to C_k \cong C$ is a $\mathbb{P}^1$-bundle. Since blowing up commutes with flat base change, $Z_{k+1} \times_{\widetilde{W}_*} S = S_{k+1}$ and $F_{k+1} \times_{\widetilde{W}_*} S = E_{k+1}$. Since any $\sigma \in \widetilde{G}$ fixes ξ_k, the curves C_k are $\widetilde{G}$-stable and $\widetilde{G}$ acts on any Z_k.

Let $\pi : Z_m \to \widetilde{W}_*$ be the composite morphism. The exceptional prime divisors of π are the strict transforms of $F_1, \dots, F_m$, still denoted $F_1, \dots, F_m$, and by the equality $F_k \times_{\widetilde{W}_*} S = E_k$, $\mathrm{mult}_{F_k} h = \mathrm{mult}_{E_k} h$ for any non-zero $h \in \Bbbk[[p, v, t]]$, and $a(F_k, \widetilde{W}_*, B^*) = a(E_k, S, \mathfrak{a})$. Let B_{Z_m} be the $\mathbb{R}$-divisor with $K_{Z_m} + B_{Z_m} = \pi^*(K_{\widetilde{W}_*} + B^*)$. By (B.134),

$$B_{Z_m} = \pi_*^{-1} B^* + \sum_{k=1}^{m} (1 - a(E_k, S, \mathfrak{a}))F_k \geq 0, \qquad \mathrm{mult}_{F_m} B_{Z_m} = 1. \tag{B.136}$$

Let $F_{m,0}$ be the fiber of $F_m \to C$ over 0, so that $F_{m,0} \cong \mathbb{P}^1_\Bbbk$ and $\pi(F_{m,0}) = 0$, let $\eta_{F_{m,0}}$ be the generic point of $F_{m,0}$, and let q be a local equation of F_m at $\eta_{F_{m,0}}$. Since F_m is a $\mathbb{P}^1$-bundle over C, ϖ restricts to a uniformizer of the discrete valuation ring $R := \mathcal{O}_{F_m, \eta_{F_{m,0}}}$, so ϖ, q are regular parameters of the regular local ring $\mathcal{O} := \mathcal{O}_{Z_m, \eta_{F_{m,0}}}$ of dimension 2. For an integer $L \geq 1$ let ν_L be the exceptional prime divisor of the weighted blow-up of $\operatorname{Spec} \mathcal{O}$ with weights $(1, L)$ in (ϖ, q). Then ν_L is a prime divisor over $\widetilde{W}_*$ with center $\pi(F_{m,0}) = 0$, and $\nu_L(\varpi) = 1$. The valuation ν_L is determined by the ideals (ϖ) and (q) of $\mathcal{O}$, which are preserved by any $\sigma \in \widetilde{G}$, since σ preserves F_m, $(\varpi = 0)$, and $F_{m,0}$. Thus ν_L is $\widetilde{G}$-stable. Since $a(\nu_L, Z_m, 0) = 1 + L$, (B.136) implies $\nu_L(B_{Z_m}) \geq \nu_L(q) = L$, so

$$a(\nu_L, \widetilde{W}_*, B^*) = a(\nu_L, Z_m, B_{Z_m}) = 1 + L - \nu_L(B_{Z_m}) \leq 1,$$

and $a(\nu_L, \widetilde{W}_*, B^*) > 0$ by (B.133). The residue field of ν_L is $\Bbbk(F_{m,0})(\tau)$, where τ is the residue of $q\varpi^{-L}$ and is transcendental over $\Bbbk(F_{m,0})$. For $\psi \in \mathcal{O}^*$, $\nu_L(\psi) = 0$ and the residue of ψ is the image of ψ in $\Bbbk(F_{m,0})$. For $\psi' \in \mathcal{O}$, $\nu_L(q\psi') \geq L$.

Step 3. In this step we define ρ_1, ρ_2 and the dichotomy of (2), with e and c.

The ring $\mathcal{O}[\varpi^{-1}]$ is factorial, q is a prime element of $\mathcal{O}[\varpi^{-1}]$, and $\mathcal{O}[\varpi^{-1}]/(q)$ is the fraction field of $R = \mathcal{O}/(q)$. Since $\mathrm{mult}_{F_m} h = \mathrm{mult}_{E_m} h$ for any non-zero $h \in \Bbbk[[p, v, t]][\varpi^{-1}]$, by (B.135) and $\widetilde{y}_1, y_2 \in \mathcal{O}[\varpi^{-1}]$ there exist integers ρ_1, ρ_2, units $\psi_1, \psi_2 \in \mathcal{O}^*$ and elements $\psi_1', \psi_2' \in \mathcal{O}[\varpi^{-1}]$ with

$$\widetilde{y}_1 = q^{w_1}(\varpi^{\rho_1}\psi_1 + q\psi_1'), \qquad y_2 = q^{w_2}(\varpi^{\rho_2}\psi_2 + q\psi_2'). \tag{B.137}$$

The integers ρ_1, ρ_2 and the images $\lambda_1, \lambda_2 \in \Bbbk(F_{m,0})^*$ of ψ_1, ψ_2 depend only on $\widetilde{y}_1$ and y_2. Let

$$\Lambda := \varpi^{w_1\rho_2 - w_2\rho_1} \widetilde{y}_1^{w_2} y_2^{-w_1}.$$

By (B.137), $\mathrm{mult}_{F_m}\Lambda = 0$ and the restriction $\Theta := \Lambda|_{F_m}$ is the image of $\psi_1^{w_2}\psi_2^{-w_1}$ in R, so $\Theta \in R^*$ has image $\lambda_1^{w_2}\lambda_2^{-w_1}$ in $\Bbbk(F_{m,0})$. By the end of **Step 1**, Θ is transcendental over $\Bbbk((\varpi))$.

If $\lambda_1^{w_2}\lambda_2^{-w_1} \notin \Bbbk$, we are in case (a) and let $e := 0$. Assume that $\lambda_1^{w_2}\lambda_2^{-w_1} \in \Bbbk$. Two polynomials in $\Bbbk[\varpi]$ of degree less than n which are congruent modulo $\varpi^n R$ are equal. If $\Theta \in \Bbbk[\varpi] + \varpi^n R$ for any n, the corresponding polynomials of degree less than n are thus compatible and define $c_\infty \in \Bbbk[[\varpi]]$ with

$$\Theta - c_\infty \in \bigcap_n \varpi^n R = 0,$$

where $\Bbbk[[\varpi]] = \mathcal{O}(C) \subset R$ through $F_m \to C$, contradicting the transcendence of Θ over $\Bbbk((\varpi))$. Thus there exists a largest integer e with $\Theta \in \Bbbk[\varpi] + \varpi^e R$, and $e \geq 1$. We write

$$\Theta = c(\varpi) + \varpi^e \Theta', \tag{B.138}$$

where $c \in \Bbbk[\varpi]$ is the unique polynomial of degree less than e with $\Theta - c(\varpi) \in \varpi^e R$, and $\Theta' \in R$. The image $\overline{\Theta'}$ of Θ' in $\Bbbk(F_{m,0})$ is not in $\Bbbk$ by the maximality of e, and $c(0) = \lambda_1^{w_2}\lambda_2^{-w_1} \neq 0$. This is case (b). Neither e nor c depends on L.

In case (b), let $\sigma \in \widetilde{G}$. The function Λ is $\widetilde{G}$-semi-invariant, σ preserves F_m and $F_{m,0}$, and σ acts on $\Bbbk(F_{m,0})$ over $\Bbbk$. Since $\sigma^*\Theta = \chi_\Lambda(\sigma)\Theta$ and the image of Θ in $\Bbbk(F_{m,0})$ is $c(0) \in \Bbbk^*$, $\chi_\Lambda(\sigma) = 1$. Thus

$$\Theta = \sigma^*\Theta \in c(\chi_\varpi(\sigma)\varpi) + \varpi^e R,$$

and by the uniqueness of c,

$$c(\chi_\varpi(\sigma)\varpi) = c(\varpi). \tag{B.139}$$

Fix an integer $\mu_0 \geq 0$ with $\varpi^{\mu_0}\psi_1', \varpi^{\mu_0}\psi_2' \in \mathcal{O}$ and, in case (b),

$$\varpi^{\mu_0}\Big((\psi_1 + \varpi^{-\rho_1}q\psi_1')^{w_2} - c(\varpi)(\psi_2 + \varpi^{-\rho_2}q\psi_2')^{w_1} - \psi_1^{w_2} + c(\varpi)\psi_2^{w_1}\Big) \in q\mathcal{O}.$$

From now on, let L be an integer with

$$\begin{gathered} L > \mu_0 + \max\{\rho_1, \rho_2, e\}, \qquad \alpha_1 \geq 1, \qquad \alpha_2 \geq 1, \\ \varpi^{r\alpha_2}\phi_r \in \varpi\Bbbk[[\varpi]] \text{ for } 1 \leq r < w_1/w_2. \end{gathered} \tag{B.140}$$

Any sufficiently large L satisfies (B.140).

Step 4. In this step we prove (1) and (2).

By (B.137),

$$V_1 = (q\varpi^{-L})^{w_1}(\psi_1 + \varpi^{-\rho_1}q\psi_1'), \qquad V_2 = (q\varpi^{-L})^{w_2}(\psi_2 + \varpi^{-\rho_2}q\psi_2'),$$

and $\nu_L(\varpi^{-\rho_i}q\psi_i') \geq L - \mu_0 - \rho_i > 0$ by (B.140). Thus $\nu_L(V_1) = \nu_L(V_2) = 0$, that is, $\nu_L(\widetilde{y}_1) = \alpha_1$ and $\nu_L(y_2) = \alpha_2$, which proves (1), and the residues of V_1, V_2 are

$$\overline{V_1} = \lambda_1\tau^{w_1}, \qquad \overline{V_2} = \lambda_2\tau^{w_2}.$$

In case (b), by (B.138), the function

$$\psi_1^{w_2} - c(\varpi)\psi_2^{w_1} = \psi_2^{w_1}(\psi_1^{w_2}\psi_2^{-w_1} - c(\varpi))$$

restricts to $\varpi^e\psi_2^{w_1}\Theta'$ on F_m, so

$$\psi_1^{w_2} - c(\varpi)\psi_2^{w_1} = \varpi^e\psi_2^{w_1}\widetilde{\Theta}' + q\psi_3$$

for a lift $\widetilde{\Theta}' \in \mathcal{O}$ of Θ' and some $\psi_3 \in \mathcal{O}$. Since $\overline{\Theta'} \neq 0$, $\widetilde{\Theta}' \in \mathcal{O}^*$. By the choice of μ_0,

$$V_3 = (q\varpi^{-L})^{w_1w_2}\big(\psi_2^{w_1}\widetilde{\Theta}' + \varpi^{-e}q\psi_3 + \varpi^{-e-\mu_0}q\psi_4\big)$$

for some $\psi_4 \in \mathcal{O}$. Since $L > e + \mu_0$, $\nu_L(V_3) = 0$ and $\overline{V_3} = \lambda_2^{w_1}\overline{\Theta'}\tau^{w_1w_2}$.

Since $\Bbbk$ is algebraically closed and $\Bbbk(F_{m,0})$ is the function field of $F_{m,0} \cong \mathbb{P}^1_{\Bbbk}$, any element of $\Bbbk(F_{m,0})\backslash\Bbbk$ is transcendental over $\Bbbk$. The field $\Bbbk(\overline{V_1}, \overline{V_2}, \overline{V_3})$, with $\overline{V_3}$ omitted in case (a), contains

$$\overline{V_1}^{w_2}\overline{V_2}^{-w_1} = \lambda_1^{w_2}\lambda_2^{-w_1} \in \Bbbk(F_{m,0})\backslash\Bbbk \qquad \text{in case (a)},$$
$$\overline{V_3V_2}^{-w_1} = \overline{\Theta'} \in \Bbbk(F_{m,0})\backslash\Bbbk \qquad \text{in case (b)},$$

and $\overline{V_2} = \lambda_2\tau^{w_2}$ is transcendental over $\Bbbk(F_{m,0})$. Thus

$$\operatorname{trdeg}_{\Bbbk} \Bbbk(\overline{V_1}, \overline{V_2}, \overline{V_3}) \geq 2. \tag{B.141}$$

In case (b), Y is normal: the derivatives of $\varpi^e V_3 - V_1^{w_2} + c(\varpi)V_2^{w_1}$ with respect to V_3, V_1, V_2 are ϖ^e, $-w_2V_1^{w_2-1}$, $w_1c(\varpi)V_2^{w_1-1}$, so Y is smooth if $w_2 = 1$, and otherwise the singular locus of Y is contained in the curve $(\varpi = V_1 = V_2 = 0)$, while $\dim Y = 3$. A hypersurface regular in codimension 1 is normal.

Let

$$D := \Bbbk[V_1, V_2] \text{ in case (a)}, \qquad D := \Bbbk[V_1, V_2, V_3]/(V_1^{w_2} - c(0)V_2^{w_1}) \text{ in case (b)},$$

the coordinate ring of $(\varpi = 0) \subset Y$. Since w_1, w_2 are coprime, D is a domain of dimension 2. By (B.140), $\alpha_1 \geq 1$ and $\varpi^{r\alpha_2}\phi_r \in \varpi\Bbbk[[\varpi]]$, so

$$y_2 = \varpi^{\alpha_2}V_2 \in \varpi\mathcal{O}(\mathcal{Y}), \qquad y_1 = \varpi^{\alpha_1}V_1 + \sum_r \varpi^{r\alpha_2}\phi_r(\varpi)V_2^r \in \varpi\mathcal{O}(\mathcal{Y}). \tag{B.142}$$

Any $h \in \Bbbk[[p, v, t]]$ satisfies

$$h - h(0) \in (\varpi, y_1, y_2) \subset \varpi\mathcal{O}(\mathcal{Y}).$$

Thus $\mathcal{O}(\mathcal{Y})/\varpi\mathcal{O}(\mathcal{Y})$ is generated over $\Bbbk$ by the images of V_1, V_2 (and V_3), and since

$$\varpi^e V_3 = V_1^{w_2} - c(\varpi)V_2^{w_1}$$

in case (b), there exists a surjection

$$D \to \mathcal{O}(\mathcal{Y})/\varpi\mathcal{O}(\mathcal{Y}).$$

The valuation ν_L is non-negative on $\Bbbk[[p, v, t]]$, as the center of ν_L is 0, and on V_1, V_2, V_3, so $\nu_L \geq 0$ on $\mathcal{O}(\mathcal{Y})$. Let $\mathfrak{p} := \{h \in \mathcal{O}(\mathcal{Y}) \mid \nu_L(h) > 0\}$, a prime ideal containing ϖ. The composite surjection

$$D \to \mathcal{O}(\mathcal{Y})/\varpi\mathcal{O}(\mathcal{Y}) \to \mathcal{O}(\mathcal{Y})/\mathfrak{p} \subset \kappa(\nu_L)$$

maps onto a domain of transcendence degree at least 2 over $\Bbbk$ by (B.141), and D is a domain of dimension 2, so the composite is injective. Thus

$$\mathfrak{p} = \varpi\mathcal{O}(\mathcal{Y}), \qquad \mathcal{O}(\mathcal{Y})/\varpi\mathcal{O}(\mathcal{Y}) \cong D.$$

The ring $\mathcal{O}(\mathcal{Y})$ is noetherian, so $\mathcal{O}(\mathcal{Y})_{\mathfrak{p}}$ is a noetherian local domain whose maximal ideal is generated by $\varpi \neq 0$, hence a discrete valuation ring. The valuation ring of ν_L dominates $\mathcal{O}(\mathcal{Y})_{\mathfrak{p}}$, and the fraction field of each of them is $\operatorname{Frac}\Bbbk[[p, v, t]]$, so ν_L is the valuation of $\mathcal{O}(\mathcal{Y})_{\mathfrak{p}}$, normalized by $\nu_L(\varpi) = 1$. This proves (2).

Step 5. In this step we prove (3).

The functions ϖ, $V_1 = \varpi^{-\alpha_1}\widetilde{y}_1$, and $V_2 = \varpi^{-\alpha_2}y_2$ are $\widetilde{G}$-semi-invariant by **Step 1**. In case (b), $\chi_{V_1}^{w_2}\chi_{V_2}^{-w_1} = \chi_\Lambda = 1$ by the identity $\alpha_2w_1 - \alpha_1w_2 = w_1\rho_2 - w_2\rho_1$ and **Step 3**, so by (B.139) the function $V_1^{w_2} - c(\varpi)V_2^{w_1}$ and the equation of Y are semi-invariant with character $\chi_{V_2}^{w_1}$, and V_3 is semi-invariant with character $\chi_{V_2}^{w_1}\chi_\varpi^{-e}$. Thus $\widetilde{G}$ acts diagonally on the coordinates of Y, preserves $\mathcal{O}(\mathcal{Y})$, and Ψ is $\widetilde{G}$-equivariant. Since

$$\mathcal{O}(\mathcal{Y})[\varpi^{-1}] = \Bbbk[[p, v, t]][\varpi^{-1}],$$

Ψ is an isomorphism over $(\varpi \neq 0)$, and by (B.142), Ψ maps $(\varpi = 0)$ to 0.

By (B.140), $\varpi^{r\alpha_2}\phi_r \in \Bbbk[[\varpi]]$, so the substitution sending ϖ to ϖ and y_1, y_2 to the expressions in (B.142) defines a $\widetilde{G}$-equivariant homomorphism $\Bbbk[[p,v,t]] \to \widehat{\mathcal{O}}_{Y,0}$, sending $\widetilde{y}_1$ to $\varpi^{\alpha_1}V_1$. The ring $\widehat{\mathcal{O}}_{Y,0}$ is a domain, since Y is normal, so the homomorphism extends to

$$\Bbbk[[p,v,t]][\varpi^{-1}] \to \widehat{\mathcal{O}}_{Y,0}[\varpi^{-1}],$$

which sends $\varpi^{-\alpha_1}\widetilde{y}_1$, $\varpi^{-\alpha_2}y_2$ and, in case (b), $\varpi^{-e}(V_1^{w_2} - c(\varpi)V_2^{w_1})$ to the coordinates V_1, V_2, V_3 of Y. Thus it restricts to a homomorphism $\mathcal{O}(\mathcal{Y}) \to \widehat{\mathcal{O}}_{Y,0}$. Let $\mathfrak{n} \subset \mathcal{O}(\mathcal{Y})$ be the ideal generated by the coordinates of Y, and let $\mathfrak{m}_0$ be the maximal ideal of $\widehat{\mathcal{O}}_{Y,0}$. For any $n \geq 1$, any $h \in \Bbbk[[p,v,t]]$ is congruent modulo $(\varpi, y_1, y_2)^n \subset \mathfrak{n}^n$ to a polynomial in ϖ, y_1, y_2, and y_1, y_2 are polynomials in V_1, V_2 with coefficients in $\Bbbk[[\varpi]]$. Thus $\mathcal{O}(\mathcal{Y})/\mathfrak{n}^n$ is generated over $\Bbbk$ by the images of the coordinates of Y. The composite

$$\mathcal{O}(Y)/(\text{coordinates})^n \to \mathcal{O}(\mathcal{Y})/\mathfrak{n}^n \to \widehat{\mathcal{O}}_{Y,0}/\mathfrak{m}_0^n$$

is the natural isomorphism and the first map is surjective, so both maps are isomorphisms. For $n = 1$, $\mathfrak{n}$ is a maximal ideal with residue field $\Bbbk$. Passing to the limit, we obtain (B.132). Since $\widehat{\mathcal{O}}_{Y,0}$ is normal, $\mathcal{O}_{\mathcal{Y},\mathfrak{n}}$ is normal. By (2), near $\mathfrak{n}$ the exceptional locus of Ψ, contained in $(\varpi = 0)$, is the prime divisor $(\varpi = 0)$.

The canonical module of $\widetilde{W}_*$ is generated by $d\varpi \wedge dy_1 \wedge dy_2$. Since $\phi_r \in \Bbbk((\varpi))$,

$$d\varpi \wedge dy_1 \wedge dy_2 = d\varpi \wedge d\widetilde{y}_1 \wedge dy_2 = \varpi^{\alpha_1+\alpha_2} d\varpi \wedge dV_1 \wedge dV_2.$$

By (B.132), near $\mathfrak{n}$ the canonical module of $\mathcal{Y}$ is generated by $d\varpi \wedge dV_1 \wedge dV_2$ in case (a), and in case (b) by the residue form of the hypersurface Y, which equals $\varpi^{-e} d\varpi \wedge dV_1 \wedge dV_2$ over $(\varpi \neq 0)$, as the derivative of the equation of Y with respect to V_3 is ϖ^e. Thus near $\mathfrak{n}$,

$$K_{\mathcal{Y}} = \Psi^* K_{\widetilde{W}_*} + (\alpha_1 + \alpha_2 + e)\operatorname{div}(\varpi).$$

This proves (3).

Step 6. In this step we prove (4).

By (2), $\varpi\mathcal{O}_{\mathcal{Y},\mathfrak{n}}$ is the only height-one prime ideal of $\mathcal{O}_{\mathcal{Y},\mathfrak{n}}$ containing ϖ, and the valuation of $\varpi\mathcal{O}_{\mathcal{Y},\mathfrak{n}}$ is ν_L. Let $\widetilde{H}_j := \varpi^{-n_j} H_j^*$. Then $\nu_L(\widetilde{H}_j) = 0$, and $\widetilde{H}_j$ is regular at any other height-one prime ideal, where ϖ is a unit. Since $\mathcal{O}_{\mathcal{Y},\mathfrak{n}}$ is normal,

$$\widetilde{H}_j \in \mathcal{O}_{\mathcal{Y},\mathfrak{n}} \backslash \varpi\mathcal{O}_{\mathcal{Y},\mathfrak{n}}.$$

By (3), near $\mathfrak{n}$,

$$\begin{aligned}\Psi^*(K_{\widetilde{W}_*} + B^*) &= K_{\mathcal{Y}} - (\alpha_1 + \alpha_2 + e)\operatorname{div}(\varpi) + \sum_j \kappa_j \big(\operatorname{div}(\widetilde{H}_j) + n_j \operatorname{div}(\varpi)\big) \\ &= K_{\mathcal{Y}} + \sum_j \kappa_j \operatorname{div}(\widetilde{H}_j) + \beta_L \operatorname{div}(\varpi).\end{aligned}$$

No $\operatorname{div}(\widetilde{H}_j)$ contains $(\varpi = 0)$, so by (2), $a(\nu_L, \widetilde{W}_*, B^*) = 1 - \beta_L$, and (1) implies $0 \leq \beta_L < 1$. This proves (4). □

We need the following lemma.

Lemma B.19. *Notation and conditions as in Set-up B.16. Let $\varpi := P$, let (y_1, y_2) be one of $(Q - Pf(P^2), t - g(P^2))$ and $(t - g(P^2), Q - Pf(P^2))$ for which the conclusion of Lemma B.18 for (ϖ, y_1, y_2) holds without exchanging y_1 and y_2, and let $w_1, w_2, e, c, Y, \mathcal{Y}, \mathfrak{n}, \widetilde{H}_j, \beta_L$ be the objects there for a sufficiently large integer L. Let $Y_0 := \mathbb{A}^3_{\mathbb{C}}$ with coordinates ϖ, V_1, V_2 if $e = 0$, and*

$$Y_0 := (\varpi^e V_3' = V_1'^{w_2} - V_2^{w_1}) \subset \mathbb{A}^4_{\mathbb{C}}$$

with coordinates ϖ, V_1', V_2, V_3' if $e \geq 1$. Then the following hold.

(1) *There exist a diagonal action of* $\widetilde{G}$ *on* Y_0 *and a* $\widetilde{G}$*-equivariant isomorphism from the completion of* Y *at the origin to the completion of* $Y_0 \times_{\mathbb{C}} \operatorname{Spec} \Bbbk$ *at* 0*, given by the identity if* $e = 0$*, and by* $V_1 = \gamma(\varpi)V_1'$ *and* $V_3 = c(\varpi)V_3'$ *for a* $\widetilde{G}$*-invariant unit* γ *of* $\Bbbk[[\varpi]]$ *with* $\gamma^{w_2} = c$ *if* $e \geq 1$*.*

(2) Y_0 *is klt near* 0*, and* $Y_0/\widetilde{G}$ *is* $\mathbb{Q}$*-factorial and klt near the image of* 0*.*

(3) *Let*

$$B_Y := \sum_j \kappa_j \operatorname{div}(\widetilde{H}_j) + \beta_L \operatorname{div}(\varpi)$$

on $\mathcal{Y}$ *near* $\mathfrak{n}$*. Then* $B_Y \geq 0$*, the pair* $(\mathcal{Y}, B_Y)$ *is lc near* $\mathfrak{n}$*, and*

$$\operatorname{mld}((\widehat{Y} \ni 0, B_Y)/\widetilde{G}) > 0,$$

where $\widehat{Y} := \operatorname{Spec} \widehat{\mathcal{O}}_{Y,0}$ *and* B_Y *is pulled back to* $\widehat{Y}$ *by the isomorphism* (B.132) *of Lemma B.18(3).*

Proof. The functions P, $Q - Pf(P^2)$, $t - g(P^2)$ are regular parameters of $\Bbbk[[p, v, t]]$ and

$$C = (Q - Pf(P^2) = t - g(P^2) = 0),$$

so $C = \operatorname{Spec} \Bbbk[[P|_C]]$. Let $\sigma \in \widetilde{G}$. Since C is $\widetilde{G}$-stable, restricting $\sigma^* t = \chi_t(\sigma)t$ and $\sigma^* Q = \chi_Q(\sigma)Q$ to C implies

$$g(\chi_P(\sigma)^2 P^2) = \chi_t(\sigma) g(P^2), \qquad \chi_P(\sigma) Pf(\chi_P(\sigma)^2 P^2) = \chi_Q(\sigma) Pf(P^2),$$

so $\sigma^*(t - g(P^2)) = \chi_t(\sigma)(t - g(P^2))$ and $\sigma^*(Q - Pf(P^2)) = \chi_Q(\sigma)(Q - Pf(P^2))$. Thus the sequences

$$(P, Q - Pf(P^2), t - g(P^2)) \quad \text{and} \quad (P, t - g(P^2), Q - Pf(P^2))$$

satisfy the hypotheses of Lemma B.18, and an order as in the statement exists. Let Ψ be the morphism of Lemma B.18(3).

(1) The characters of $\widetilde{G}$ take values in the group of roots of unity of $\mathbb{C} \subset \Bbbk$. If $e = 0$, $Y = Y_0 \times_{\mathbb{C}} \operatorname{Spec} \Bbbk$, and we let $\widetilde{G}$ act on Y_0 by the characters of ϖ, V_1, V_2. Let $e \geq 1$. Since $c(0) \neq 0$, the binomial series of $(c/c(0))^{1/w_2}$ and a w_2-th root of $c(0)$ define $\gamma \in \Bbbk[[\varpi]]^*$ with $\gamma^{w_2} = c$, and γ is a power series in $c(\varpi)$. By Lemma B.18(3), $c(\varpi)$ is $\widetilde{G}$-invariant, hence so is γ, and the equation $\varpi^e V_3 - V_1^{w_2} + c(\varpi)V_2^{w_1}$ is semi-invariant. As $c(0) \neq 0$, $\chi_\varpi^e \chi_{V_3} = \chi_{V_1}^{w_2} = \chi_{V_2}^{w_1}$. We let $\widetilde{G}$ act on ϖ, V_1', V_2, V_3' by the characters of ϖ, V_1, V_2, V_3, and the action preserves Y_0. The automorphism of $\Bbbk[[\varpi, V_1, V_2, V_3]]$ given by $V_1 = \gamma(\varpi)V_1'$, $V_3 = c(\varpi)V_3'$, fixing ϖ and V_2, is $\widetilde{G}$-equivariant and maps

$$\varpi^e V_3 - V_1^{w_2} + c(\varpi)V_2^{w_1} \quad \text{to} \quad c(\varpi)\big(\varpi^e V_3' - V_1'^{w_2} + V_2^{w_1}\big),$$

so the automorphism induces the isomorphism of (1).

(3) We prove (3) before (2). By Lemma B.18(4), $\widetilde{H}_j \in \mathcal{O}_{\mathcal{Y},\mathfrak{n}}$ and $\beta_L \geq 0$, so $B_Y \geq 0$, and $K_{\mathcal{Y}} + B_Y = \Psi^*(K_{\widetilde{W}_*} + B^*)$ near $\mathfrak{n}$. Since Ψ is birational, for any prime divisor F over $\mathcal{Y}$ with $\mathfrak{n} \in \overline{\operatorname{center}_{\mathcal{Y}} F}$,

$$a(F, \mathcal{Y}, B_Y) = a(F, \widetilde{W}_*, B^*) \geq 0, \tag{B.143}$$

as $(\widetilde{W}_*, B^*)$ is lc by Proposition 10.5(1). Thus $(\mathcal{Y}, B_Y)$ is lc near $\mathfrak{n}$. Near $\mathfrak{n}$, $\mathcal{Y}$ is normal by Lemma B.18(3) and Gorenstein, as $\widehat{\mathcal{O}}_{\mathcal{Y},\mathfrak{n}}$ is a hypersurface ring by (B.132). The scheme $\mathcal{Y}$ is of finite type over $\widetilde{W}_*$, hence excellent, so $\widehat{Y} \to \mathcal{Y}$ is a regular morphism. Thus, near $\mathfrak{n}$, log resolutions of $(\mathcal{Y}, B_Y)$ pull back to log resolutions of $(\widehat{Y}, B_Y)$ (cf. [dFEM11, Remark 2.10]), with the relative canonical divisors pulled back, as dualizing sheaves commute with flat base change, and the formal prime divisors over $\widehat{Y}$ with center the closed point correspond to the prime divisors over $\mathcal{Y}$ with center $\mathfrak{n}$ (cf. [dFEM11, Lemma 2.21]). We also write F for the formal prime

divisor corresponding to a prime divisor F over $\mathcal{Y}$. Then $a(F,\widehat{Y},B_Y)=a(F,\mathcal{Y},B_Y)$. For such a prime divisor F, $\operatorname{center}_{\widetilde{W}_*}F=\Psi(\mathfrak{n})=0$, so by (B.143) and $e_F\le|\widetilde{G}|$,

$$\frac{a(F,\widehat{Y},B_Y)}{e_F}\ge\frac{\operatorname{mld}(\widetilde{W}_*\ni 0,B^*)}{|\widetilde{G}|},$$

which is positive by Proposition 10.5(1). Thus $\operatorname{mld}((\widehat{Y}\ni 0,B_Y)/\widetilde{G})>0$.

(2) If $e=0$, Y_0 is smooth. Let $e\ge 1$. By (1) and Lemma B.18(3),

$$\widehat{\mathcal{O}}_{Y_0\times_{\mathbb{C}}\operatorname{Spec}\Bbbk,0}\cong\widehat{\mathcal{O}}_{Y,0}\cong\widehat{\mathcal{O}}_{\mathcal{Y},\mathfrak{n}}.$$

By the pull-back of log resolutions in (3) with boundary 0, $\widehat{Y}$ is klt if $\mathcal{Y}$ is klt near $\mathfrak{n}$. Since klt singularities are compatible with completion at a closed point of a variety [dFEM11, Proposition 2.11] and with the extension of the base field from $\mathbb{C}$ to $\Bbbk$ [dFEM11, Lemma 2.14], Y_0 is klt near 0 if $\mathcal{Y}$ is klt near $\mathfrak{n}$. Let F be an exceptional prime divisor over $\mathcal{Y}$ with $\mathfrak{n}\in\overline{\operatorname{center}_{\mathcal{Y}}F}$. If $\operatorname{center}_{\mathcal{Y}}F\not\subset(\varpi=0)$, then $\mathcal{Y}$ is regular at the generic point of $\operatorname{center}_{\mathcal{Y}}F$, as Ψ is an isomorphism over $(\varpi\ne 0)$, so $a(F,\mathcal{Y},0)\ge 2$. If $\operatorname{center}_{\mathcal{Y}}F\subset(\varpi=0)$, then $\operatorname{center}_{\widetilde{W}_*}F=0$ as Ψ maps $(\varpi=0)$ to 0, and since $B_Y\ge 0$ is a combination of Cartier divisors, (B.143) and Proposition 10.5(1) imply

$$a(F,\mathcal{Y},0)\ge a(F,\mathcal{Y},B_Y)=a(F,\widetilde{W}_*,B^*)\ge\operatorname{mld}(\widetilde{W}_*\ni 0,B^*)>0.$$

Thus $\mathcal{Y}$ is klt near $\mathfrak{n}$, and Y_0 is klt near 0.

We show that $\operatorname{Cl}(Y_0)=0$. If $e=0$, then $Y_0=\mathbb{A}^3_{\mathbb{C}}$. Let $e\ge 1$. Solving the equation of Y_0 for V_3' over $(\varpi\ne 0)$, the projection to the coordinates ϖ,V_1',V_2 maps $Y_0\cap(\varpi\ne 0)$ isomorphically onto $(\varpi\ne 0)\subset\mathbb{A}^3_{\mathbb{C}}$, so $\operatorname{Cl}(Y_0\cap(\varpi\ne 0))=0$. Since w_1,w_2 are coprime, $Y_0\cap(\varpi=0)=(V_1'^{w_2}=V_2^{w_1})\times\mathbb{A}^1_{V_3'}$ is a prime divisor and equals $\operatorname{div}(\varpi)$. Thus $\operatorname{Cl}(Y_0)=0$, and $Y_0/\widetilde{G}$ is $\mathbb{Q}$-factorial and klt near the image of 0 [KM98, Lemma 5.16 and Proposition 5.20]. □

Proof of Theorem 10.7. Assume that $(a_i)_{i\ge 1}$ is strictly increasing. We derive a contradiction.

Step 1. In this step we collect the data on W_{N_i} and on the limit.

By Proposition 10.5(1), possibly passing to a subsequence, C is the unique lc center of B^* through 0, so the conditions of Set-up B.16 hold. Let P,Q,f,g be as there. For any i let F_i be a computing lift of a_i. By Proposition B.17, possibly passing to a subsequence and replacing $\Bbbk$ with an algebraically closed field containing it (and C,P,Q,f,g with their base changes), we have the following.

(1) x_i',y_i',z_i' are the linear forms of Proposition B.17, q_i is the center of F_i on the blow-up of W_{N_i} at 0, a closed point of the chart $(x_i'\ne 0)$, τ_i is the value of t/x_i' at q_i, and $\xi_1=x_i'$, $\xi_2=z_i'/x_i'$, $\xi_3=t/x_i'-\tau_i$ are the coordinates of $(x_i'\ne 0)\cong\mathbb{A}^3$, on which G acts diagonally, and q_i is their origin (Proposition B.17(1)).

(2) $m_j=\operatorname{mult}_0 h_{ij}$ and $\beta_i=\sum_j c_{ij}m_j-1\ge -1$ (Proposition B.17(2)), and, by Proposition B.17(3), $h_{ij}'=\xi_1^{-m_j}h_{ij}$ and

$$B_i=\sum_j c_{ij}\operatorname{div}(h_{ij}')+\beta_i\operatorname{div}(\xi_1).$$

(3) $(h_{*,j}')_j$ is the compatible generic limit of Proposition B.17(4).

(4) $R_0>0$ is a real number with $\operatorname{mult}_{F_i}\xi_1\le R_0$ and $a(F_i,\mathbb{A}^3,0)-\sum_j\kappa_j\operatorname{mult}_{F_i}h_{ij}'\le R_0$ for any i, which exists by Proposition B.17(3).

(5) C_1 is the curve of Proposition B.17(4), so $C_1=(\zeta_2=\zeta_3=0)$ for

$$\zeta_2:=\xi_2-f(\xi_1)\quad\text{and}\quad\zeta_3:=\xi_3+g'(0)-\frac{g(\xi_1)}{\xi_1}.$$

By Proposition B.14(1) with $X_i := x_i'$, possibly passing to a subsequence, $(W_{N_i}, \sum_j \kappa_j \operatorname{div}(h_{ij}))$ is lc near 0 for any i.

Let $\varpi := P$, and let $\{y_1, y_2\} = \{Q - Pf(P^2), t - g(P^2)\}$, ordered so that Lemma B.18 for (ϖ, y_1, y_2) needs no exchange of y_1 and y_2. By Lemmas B.18 and B.19, for a sufficiently large integer L there exist coprime positive integers $w_1 \geq w_2$, Laurent series $\phi_r \in \Bbbk((\varpi))$ for $1 \leq r < w_1/w_2$, positive integers α_1, α_2, an integer $e \geq 0$, and, if $e \geq 1$, a $\widetilde{G}$-invariant polynomial $c \in \Bbbk[\varpi]$ with $c(0) \neq 0$, satisfying the following, where $\widetilde{y}_1 := y_1 - \sum_r \phi_r(\varpi) y_2^r$ and $k := \alpha_1 + \alpha_2 + e$.

(a) The functions ϖ, $\widetilde{y}_1$, y_2 are $\widetilde{G}$-semi-invariant. Let ν_L be the prime divisor over $\widetilde{W}_*$ of Lemma B.18(1), given by $(\varpi = 0)$ on Y, where $Y := \mathbb{A}^3_{\Bbbk}$ with coordinates ϖ, $V_1 := \widetilde{y}_1/\varpi^{\alpha_1}$, $V_2 := y_2/\varpi^{\alpha_2}$ if $e = 0$, and
$$Y := (\varpi^e V_3 = V_1^{w_2} - c(\varpi) V_2^{w_1}) \subset \mathbb{A}^4_{\Bbbk}$$
if $e \geq 1$ (Lemma B.18(2)). Then the center of ν_L is the closed point, and $n_j := \nu_L(H_j^*)$ satisfies
$$0 \leq \beta_L := \sum_j \kappa_j n_j - k < 1.$$

(b) Let $Y_0 := \mathbb{A}^3$ with coordinates ϖ, V_1', V_2 if $e = 0$, and $Y_0 := (\varpi^e V_3' = V_1'^{w_2} - V_2^{w_1}) \subset \mathbb{A}^4$ if $e \geq 1$, where the isomorphism of Lemma B.19(1) fixes ϖ and V_2, is the identity if $e = 0$ (so $V_1' = V_1$), and is given by $V_1 = \gamma_*(\varpi) V_1'$ and $V_3 = c(\varpi) V_3'$ for the $\widetilde{G}$-invariant unit γ_* of that lemma with $\gamma_*^{w_2} = c$ if $e \geq 1$. Then $\widetilde{G}$ acts diagonally on Y_0, by the characters of ϖ, V_1, V_2, V_3 on ϖ, V_1', V_2, V_3'. Moreover Y_0 is klt near 0, and $Y_0/\widetilde{G}$ is $\mathbb{Q}$-factorial and klt near the image x_0 of 0.

Let $m_0 \geq 0$ be an integer with $\varpi^{m_0} \phi_r \in \Bbbk[[\varpi]]$ for any r. Since $\alpha_1 = w_1 L + \rho_1$ and $\alpha_2 = w_2 L + \rho_2$ for integers ρ_1, ρ_2 independent of L (Lemma B.18), while ϕ_r and m_0 do not depend on L, possibly enlarging L we may assume that $\alpha_1 \geq 3$ and $\alpha_2 \geq m_0 + 3$.

Step 2. In this step we specialize the generators ζ_2, ζ_3 of the ideal of C_1 to functions of high order along F_i.

Since each of x_i', y_i', z_i', t is ξ_1 times a polynomial in ξ_1, ξ_2, ξ_3 and h_{ij} has order m_j at 0, each h_{ij}' is a polynomial in ξ_1, ξ_2, ξ_3. Since (W_{N_i}, D_i) is klt near 0, $a_i > 0$, and the exceptional divisor E_{q_i} of the blow-up of q_i satisfies

$$0 < e_{E_{q_i}} a_i \leq a(E_{q_i}, \mathbb{A}^3, B_i) = 3 - \beta_i - \sum_j c_{ij} \operatorname{mult}_{q_i} h_{ij}' \tag{B.144}$$

by Proposition B.17(3), so $\operatorname{mult}_{q_i} h_{ij}' < 4/c_{1j}$, as $\beta_i \geq -1$ and $c_{ij} \geq c_{1j} > 0$. Off $(\xi_1 = 0)$, the chart $(x_*' \neq 0)$ of the blow-up of W_* at 0, with $\xi_1 = P^2$, $\xi_2 = Q/P$ and $\xi_3 + g'(0) = t/P^2$ on it, is the quotient of $(P \neq 0)$ by ι, which acts freely there, and $\operatorname{Spec} \Bbbk[[\xi_1, \xi_2, \xi_3]]$ is the completion of $(x_*' \neq 0)$ at $\xi_1 = \xi_2 = \xi_3 = 0$. Since

$$H_j^* = P^{2m_j} h_{*,j}'(P^2, Q/P, t/P^2 - g'(0)), \tag{B.145}$$

the pullback of $\sum_j \kappa_j \operatorname{div}(h_{*,j}')$ to $(P \neq 0)$ is $B^*|_{(P \neq 0)}$. Let $\mathcal{X} := \operatorname{Spec} \Bbbk[[\xi_1, \xi_2, \xi_3]][\xi_1^{-1}]$ and $\mathfrak{a} := \prod_j (h_{*,j}')^{\kappa_j}$. Since B^* is lc, and since log resolutions and log discrepancies are compatible with completion at that point and with localization, the pair $(\mathcal{X}, \mathfrak{a})$ is lc. By Proposition B.17(4), the reduced closure of $\operatorname{Nklt}(\mathcal{X}, \mathfrak{a})$ is C_1, so its ideal is (ζ_2, ζ_3).

Write $f(\xi_1) = \sum_{n \geq 1} f_n \xi_1^n$ and $g(\xi_1) = \sum_{n \geq 1} g_n \xi_1^n$, so that $\zeta_3 = \xi_3 - \sum_{n \geq 1} g_{n+1} \xi_1^n$. Let T and d be integers with

$$T > \max\left((m_0 + \alpha_1 + \alpha_2 + e)R_0,\ \max_j n_j\right) \quad \text{and} \quad d > \max_j n_j.$$

Let λ be the finite tuple of elements of $\Bbbk$ consisting of

(i) f_n and g_{n+1} for $1 \leq n < T$,
(ii) the coefficients of ϖ^n in ϕ_r for $-m_0 \leq n < T$ and $1 \leq r < w_1/w_2$, and

(iii) if $e \geq 1$, the coefficients of c and the number $c(0)^{-1}$.

Since $\alpha_1 \geq 3$ and $\alpha_2 \geq m_0 + 3$, the pullbacks of $\xi_1 = P^2$, $\xi_2 = Q/P$, and $\xi_3 = t/P^2 - g'(0)$ to Y are ϖ times regular functions, so a monomial of degree d' in ξ_1, ξ_2, ξ_3 pulls back to a function of order at least d' along $(\varpi = 0)$. Write the pullback of H_j^* to Y as $\sum_n \Theta_{j,n}\varpi^n$, with $\Theta_{j,n} \in \Bbbk[V_1, V_2]$ if $e = 0$, and, reducing modulo the equation $\varpi^e V_3 - V_1^{w_2} + c(\varpi)V_2^{w_1}$ of Y, with $\Theta_{j,n} \in \Bbbk[V_1, V_2, V_3]$ of V_1-degree less than w_2 if $e \geq 1$. Since the coordinate ring of Y is free over $\Bbbk[\varpi, V_2, V_3]$ with basis $1, V_1, \dots, V_1^{w_2-1}$ if $e \geq 1$, and ϖ generates the prime ideal of $(\varpi = 0)$,

$$n_j = \nu_L(H_j^*) = \min\{n \mid \Theta_{j,n} \neq 0\}.$$

Let θ_j be a non-zero coefficient of Θ_{j,n_j}. By (B.145) and since $T, d > n_j$, the coefficients of $\Theta_{j,n}$ for $n \leq n_j$ are polynomials in the coefficients of $\mathrm{jet}_d((h'_{*,j})_j)$ and in λ. Add the numbers θ_j^{-1} to λ, let m_λ be the length of λ, and let Ψ consist of the polynomial functions on $\mathcal{J}_d \times \mathbb{A}^{m_\lambda}$, written in the jet coordinates and the coordinates of $\mathbb{A}^{m_\lambda}$, given by

(i) the coefficients of the monomials of $\Theta_{j,n}$ for $n < n_j$ and any j,
(ii) $\theta_j \cdot \theta_j^{-1} - 1$ for any j,
(iii) $c(0) \cdot c(0)^{-1} - 1$ if $e \geq 1$, and
(iv) the coordinate functions of $\mathbb{A}^{m_\lambda}$ at which the corresponding entry of λ is 0.

Then Ψ vanishes at $(\mathrm{jet}_d((h'_{*,j})_j), \lambda)$.

By Proposition 6.3(A), with the coordinates ξ_1, ξ_2, ξ_3 in place of z_1, z_2, z_3, ξ_1 in place of r, $C_0 := R_0$, the h'_{ij} in place of the $g_{i,k}$, the F_i as prime divisors over the chart $(x'_i \neq 0) \cong \mathbb{A}^3$ of the blow-up of W_{N_i} at 0, and the limit $(h'_{*,j})_j$ (their orders at q_i are less than $4/c_{1j}$ by (B.144), conditions (ii) and (iii) of Proposition 6.3 hold by Proposition B.17(3), and $(\mathcal{X}, \mathfrak{a})$ is lc), with T in place of both L and T, with m_λ, λ, λ_i in place of q, β, β_i, and with the truncations

$$\xi_2 - \sum_{1 \leq n < T} f_n \xi_1^n \quad \text{and} \quad \xi_3 - \sum_{1 \leq n < T} g_{n+1} \xi_1^n$$

of ζ_2, ζ_3 in degree less than T as $\Phi_1(\lambda, \xi), \Phi_2(\lambda, \xi)$, possibly passing to a subsequence, for any i there exists a complex tuple λ_i with $\Psi(\mathrm{jet}_d((h'_{ij})_j), \lambda_i) = 0$ such that the following holds. Let $f_i^\sharp$, $\phi_{r,i}^\sharp$, and $c_i^\sharp$ be the truncations of f, ϕ_r, and c in degree less than T with each coordinate of λ replaced by the corresponding coordinate of λ_i, and let

$$g_i^\sharp(\xi_1) := \tau_i \xi_1 + \sum_{1 \leq n < T} g_{n+1,i} \xi_1^{n+1},$$

where $g_{n+1,i}$ is the entry of λ_i replacing g_{n+1}. Then

$$\mathrm{mult}_{F_i}\left(\frac{z'_i}{x'_i} - f_i^\sharp(x'_i)\right) \geq T \quad \text{and} \quad \mathrm{mult}_{F_i}\left(\frac{t - g_i^\sharp(x'_i)}{x'_i}\right) \geq T. \tag{B.146}$$

Since $\Psi(\mathrm{jet}_d((h'_{ij})_j), \lambda_i) = 0$, $c_i^\sharp(0) \neq 0$ if $e \geq 1$, and $f_i^\sharp$, $\phi_{r,i}^\sharp$, $c_i^\sharp$ have no monomial absent from f, ϕ_r, c. Thus $Q - Pf_i^\sharp(P^2)$, $\phi_{r,i}^\sharp(P)$, and $c_i^\sharp(P)$ have the characters of $Q - Pf(P^2)$, $\phi_r(\varpi)$, and $c(\varpi)$. Since a compatible generic limit has the characters of the members (Definition 2.5) and the limits of x'_i and z'_i pull back to P^2 and PQ on $\widetilde{W}_*$ (Proposition B.17), the characters of x'_i and z'_i are those of P^2 and PQ. Since $t/x'_i - \tau_i = \xi_3$ and t/x'_i are both G-semi-invariant, $\tau_i \neq 0$ only if $\chi_t = \chi_{x'_i} = \chi_{P^2}$. As moreover $g_{n+1,i} = 0$ whenever $g_{n+1} = 0$, $t - g_i^\sharp(P^2)$ has the character of t.

Step 3. In this step, for any i, we construct a $\widetilde{G}$-equivariant morphism $\psi_i : Y_i \to W_{N_i}$ from a germ $(Y_i \ni 0)$ étale-locally isomorphic to $(Y_0 \ni 0)$, such that F_i lifts to a prime divisor with center 0 and $\mathrm{mld}((Y_i \ni 0, B_{Y,i})/\widetilde{G}) = a_i$ for the $\mathbb{R}$-divisor $B_{Y,i}$ with $K_{Y_i} + B_{Y,i} = \psi_i^*(K_{W_{N_i}} + D_i)$.

Let $\rho_i : \mathbb{A}^3_{P,Q,t} \dashrightarrow W_{N_i}$ be given by

$$x'_i = P^2, \quad z'_i = PQ, \quad \text{and} \quad y'_i = -Q^2 - \frac{t^{N_i}}{P^2}. \tag{B.147}$$

Since $x_i'y_i' + z_i'^2 + t^{N_i} = 0$ on W_{N_i}, ρ_i is a morphism on $(P \neq 0)$ onto $(x_i' \neq 0)$, and it is the quotient by $\iota : (P, Q, t) \to (-P, -Q, t)$ there. Since the characters of x_i' and z_i' are those of P^2 and PQ, with $\widetilde{G}$ acting diagonally on $\mathbb{A}^3_{P,Q,t}$ by the characters of P, Q, t, ρ_i is $\widetilde{G}$-equivariant, with $\widetilde{G}$ acting on W_{N_i} through $\widetilde{G} \to G$, whose kernel is $\{\mathrm{id}, \iota\}$. Thus ρ_i induces a birational map $\mathbb{A}^3_{P,Q,t}/\widetilde{G} \dashrightarrow W_{N_i}/G$. Then the following holds.

(∗) Let Y' be a normal variety with a $\widetilde{G}$-equivariant birational map $Y' \dashrightarrow \mathbb{A}^3_{P,Q,t}$ whose composite with ρ_i is a morphism $Y' \to W_{N_i}$, and let D be a G-invariant $\mathbb{R}$-divisor on W_{N_i} and B' the $\mathbb{R}$-divisor on Y' with $K_{Y'} + B'$ the pullback of $K_{W_{N_i}} + D$. Then for any prime divisor F' over Y' and the prime divisor F over W_{N_i} onto which F' maps,

$$\frac{a(F', Y', B')}{e_{F'}} = \frac{a(F, W_{N_i}, D)}{e_F},$$

with $e_{F'}$ and e_F the orders of the inertia groups in $\widetilde{G}$ and in G.

(∗) follows from the ramification formula for the quotients by $\widetilde{G}$ and G, since F' and F induce prime divisors over $\mathbb{A}^3_{P,Q,t}/\widetilde{G}$ and W_{N_i}/G which correspond under the birational map induced by ρ_i.

Let

$$(y_{1,i}, y_{2,i}) := \begin{cases} (Q - Pf_i^\sharp(P^2), t - g_i^\sharp(P^2)) & \text{if } y_1 = Q - Pf(P^2), \\ (t - g_i^\sharp(P^2), Q - Pf_i^\sharp(P^2)) & \text{otherwise,} \end{cases}$$

let $\widetilde{y}_{1,i} := y_{1,i} - \sum_r \phi_{r,i}^\sharp(P) y_{2,i}^r$, and let

$$Y_i := \mathbb{A}^3_{P,V_1,V_2} \text{ if } e = 0, \qquad Y_i := (P^e V_3 = V_1^{w_2} - c_i^\sharp(P) V_2^{w_1}) \subset \mathbb{A}^4 \text{ if } e \geq 1.$$

Let $\psi_i' : Y_i \dashrightarrow \mathbb{A}^3_{P,Q,t}$ be given by

$$\widetilde{y}_{1,i} = P^{\alpha_1} V_1 \quad \text{and} \quad y_{2,i} = P^{\alpha_2} V_2,$$

that is,

$$y_{2,i} = P^{\alpha_2} V_2 \quad \text{and} \quad y_{1,i} = P^{\alpha_1} V_1 + \sum_r \phi_{r,i}^\sharp(P) P^{r\alpha_2} V_2^r,$$

and $Q = y_{1,i} + Pf_i^\sharp(P^2)$, $t = y_{2,i} + g_i^\sharp(P^2)$ if $y_1 = Q - Pf(P^2)$ (and with $y_{1,i}, y_{2,i}$ exchanged otherwise), and let $\psi_i := \rho_i \circ \psi_i'$. Since $\alpha_2 \geq m_0 + 3$, $\phi_{r,i}^\sharp(P) P^{r\alpha_2}$ is a polynomial divisible by P^3, and since moreover $\alpha_1 \geq 3$ and $g_i^\sharp(0) = f_i^\sharp(0) = 0$, the functions Q and t are polynomials in P, V_1, V_2 with $t \in (P^2)$. By (B.147), x_i', y_i', z_i', t, and hence x, y, z, t, are polynomials on Y_i. Thus $\psi_i : Y_i \to W_{N_i}$ is a morphism, and it maps the origin 0 of Y_i to 0, as $N_i \geq 2$. By **Step 2**, P, V_1, V_2, V_3 are $\widetilde{G}$-semi-invariant with the characters of ϖ, V_1, V_2, V_3 in (b), so $\widetilde{G}$ acts diagonally on Y_i and ψ_i is $\widetilde{G}$-equivariant.

If $e = 0$, let $U_i := Y_i = Y_0$, $u_i := 0$, and $\sigma_i = \sigma_i^0 := \mathrm{id}_{U_i}$. If $e \geq 1$, let $\gamma_{i,0} \in \mathbb{C}$ with $\gamma_{i,0}^{w_2} = c_i^\sharp(0)$, and let U_i be a neighborhood of $u_i := (0, \gamma_{i,0})$ in

$$\{(y, \gamma) \in Y_i \times \mathbb{A}^1 \mid \gamma^{w_2} = c_i^\sharp(P)\},$$

with $\widetilde{G}$ acting trivially on γ, containing no other point $(0, \gamma)$ with $\gamma^{w_2} = c_i^\sharp(0)$. Since the points $(0, \gamma)$ with $\gamma^{w_2} = c_i^\sharp(0)$ are $\widetilde{G}$-fixed, U_i can be taken $\widetilde{G}$-stable. Consider

$$\sigma_i : U_i \to Y_i,\ (y, \gamma) \to y, \qquad \sigma_i^0 : U_i \to Y_0,\ (P, V_1, V_2, V_3, \gamma) \to \Big(P, \frac{V_1}{\gamma}, V_2, \frac{V_3}{c_i^\sharp(P)}\Big).$$

Since $c_i^\sharp$ is $\widetilde{G}$-invariant and $c_i^\sharp(0) \neq 0$, the morphisms σ_i and σ_i^0 are $\widetilde{G}$-equivariant and étale at u_i, with $\sigma_i^{-1}(0) = (\sigma_i^0)^{-1}(0) = \{u_i\}$. Since Y_0 is normal near 0 by (b), possibly shrinking Y_i and U_i to $\widetilde{G}$-stable open neighborhoods of 0 and u_i, U_i is normal, being étale over Y_0, and Y_i is normal. By Lemma 2.13, prime divisors with centers 0, u_i, and 0 over Y_i, U_i, and Y_0 correspond.

Corresponding prime divisors have equal log discrepancies with respect to boundaries related by pullback under σ_i and σ_i^0, equal multiplicities of functions related by pullback, and equal inertia groups in $\widetilde{G}$. We denote corresponding prime divisors by one letter. Since $\gcd(w_1,w_2)=1$ and $c_i^\sharp(0)\neq 0$, $S_i := \operatorname{div}(P)$ is a prime divisor on Y_i.

Let F_i' be a prime divisor over Y_i which maps onto F_i under ψ_i. Then $\operatorname{mult}_{F_i'}\circ\psi_i^* = e_i' \operatorname{mult}_{F_i}$ on the rational functions on W_{N_i}, with $e_i'\in\{1,2\}$, as ψ_i has degree 2. Since $\psi_i^*\xi_1 = \psi_i^* x_i' = P^2$, by Proposition B.17(3),

$$\operatorname{mult}_{F_i'} P = \frac{e_i'}{2}\operatorname{mult}_{F_i}\xi_1 \le R_0,$$

and

$$Q - P f_i^\sharp(P^2) = P\big(\xi_2 - f_i^\sharp(\xi_1)\big) \quad\text{and}\quad t - g_i^\sharp(P^2) = P^2\big(t - g_i^\sharp(x_i')\big)/x_i',$$

so $\operatorname{mult}_{F_i'} y_{1,i}\ge T$ and $\operatorname{mult}_{F_i'} y_{2,i}\ge T$ by (B.146). Since $P^{m_0}\phi_{r,i}^\sharp(P)$ is a polynomial,

$$\operatorname{mult}_{F_i'} V_1 \ge T-(m_0+\alpha_1)R_0,\quad \operatorname{mult}_{F_i'} V_2 \ge T-\alpha_2 R_0,\quad\text{and}\quad \operatorname{mult}_{F_i'} V_3 \ge T-(m_0+\alpha_1+\alpha_2+e)R_0,$$

where the last one is for $e\ge 1$. Since $T > (m_0+\alpha_1+\alpha_2+e)R_0$, the right-hand sides are positive, so the center of F_i' on Y_i is 0.

Since $h_{ij}\circ\psi_i = P^{2m_j}h_{ij}'(P^2, Q/P, t/P^2-\tau_i)$ and $\Psi(\operatorname{jet}_d((h_{ij}')_j),\lambda_i)=0$, the vanishing of the coefficients of $\Theta_{j,n}$ for $n<n_j$ and the non-vanishing of θ_j, which determine n_j in **Step 2**, hold for the pullbacks $h_{ij}\circ\psi_i$, with P in place of ϖ and with the equation of Y_i in place of that of Y, since the monomials of h_{ij}' of degree at least d and the terms of f,g,ϕ_r,c of degree at least T contribute only to P^n with $n\ge\min(d,T)>n_j$, and, possibly passing to a subsequence, the summand $t^{N_i}/P^2\in(P^{2N_i-2})$ of y_i' satisfies $2N_i-2>\max_j n_j$. Thus $\operatorname{mult}_{S_i}(h_{ij}\circ\psi_i)=n_j$, so

$$h_{ij}\circ\psi_i = P^{n_j}\widetilde{h}_{ij}$$

with $\widetilde{h}_{ij}$ a $\widetilde{G}$-semi-invariant rational function on Y_i whose only possible poles are along S_i and with $\operatorname{mult}_{S_i}\widetilde{h}_{ij}=0$. Since Y_i is normal, $\widetilde{h}_{ij}$ is regular.

The form $\omega_i := dP\wedge dV_1\wedge dV_2/P^e$ generates ω_{Y_i}, and $dx_i'\wedge dz_i'\wedge dt/x_i'$ generates $\omega_{W_{N_i}}$. Since

$$dP\wedge dy_{1,i}\wedge dy_{2,i} = \pm dP\wedge dQ\wedge dt \quad\text{and}\quad dP\wedge d\widetilde{y}_{1,i}\wedge dy_{2,i} = dP\wedge dy_{1,i}\wedge dy_{2,i},$$

by (B.147),

$$\psi_i^*\frac{dx_i'\wedge dz_i'\wedge dt}{x_i'} = 2dP\wedge dQ\wedge dt = \pm 2P^{\alpha_1+\alpha_2}dP\wedge dV_1\wedge dV_2 = \pm 2P^k\omega_i.$$

Thus, for

$$B_{Y,i} := \sum_j c_{ij}\operatorname{div}(\widetilde{h}_{ij}) + \Big(\sum_j c_{ij}n_j - k\Big)\operatorname{div}(P) \quad\text{and}\quad B_{Y,i}^* := \sum_j \kappa_j\operatorname{div}(\widetilde{h}_{ij}) + \beta_L\operatorname{div}(P),$$

$K_{Y_i}+B_{Y,i}$ and $K_{Y_i}+B_{Y,i}^*$ are the pullbacks of $K_{W_{N_i}}+D_i$ and of $K_{W_{N_i}}+\sum_j\kappa_j\operatorname{div}(h_{ij})$. By $(*)$, any prime divisor F' over Y_i with center 0 satisfies $a(F',Y_i,B_{Y,i})/e_{F'}\ge a_i$, since the corresponding F has center $\psi_i(0)=0$, and equality holds for $F'=F_i'$. Thus

$$\operatorname{mld}((Y_i\ni 0, B_{Y,i})/\widetilde{G}) = a_i. \tag{B.148}$$

By $(*)$ and **Step 1**, possibly shrinking Y_i, $(Y_i,B_{Y,i}^*)$ is sub-lc, and $B_{Y,i}^*\ge 0$ as $\beta_L\ge 0$. Thus $(Y_i,B_{Y,i}^*)$ is lc, and $b_i := \operatorname{mld}((Y_i\ni 0,B_{Y,i}^*)/\widetilde{G})\ge 0$.

Step 4. In this step we show that b_i belongs to a finite set and is computed by a prime divisor E_i' with bounded invariants.

Possibly replacing Y_0 with a $\widetilde{G}$-stable open neighborhood of 0, by (b), $X_0 := Y_0/\widetilde{G}$ is $\mathbb{Q}$-factorial and klt. Since Y_0 is a hypersurface, K_{Y_0} is Cartier. By (b), there exists $\epsilon>0$ such that $(Y_0,\epsilon\operatorname{div}(\varpi))$ is klt near 0. Let Γ_0 be the set of positive numbers $1-(1-\mu)/\eta$, where η divides

$|\widetilde{G}|$ and $\mu \in [0,1]$ belongs to the additive monoid generated by β_L and the κ_j. The set Γ_0 is finite. Let L_0 be the integer of Theorem 2.22(3) for X_0 and Γ_0, and let

$$\Lambda_1 := \frac{2ML_0}{\min_j \kappa_j}.$$

Fix i. Let $\overline{u}_i$ be the image of u_i in $U_i/\widetilde{G}$, and let L_i' be the integer of Theorem 2.22(2) for the normal threefold $U_i/\widetilde{G}$ and the finite set consisting of the numbers $1-1/\eta$ with $\eta > 1$ dividing $|\widetilde{G}|$, the numbers $\kappa_j/|\widetilde{G}|$, and $\beta_L/|\widetilde{G}|$ if $\beta_L > 0$. Since $\mathcal{O}_{U_i,u_i}$ is finite over $\mathcal{O}_{U_i/\widetilde{G},\overline{u}_i}$, there exists an integer δ_i with

$$\mathfrak{m}_{u_i}^{\delta_i} \subset \mathfrak{m}_{\overline{u}_i}\mathcal{O}_{U_i,u_i}.$$

Since the Reynolds operator is $\mathcal{O}_{U_i/\widetilde{G},\overline{u}_i}$-linear and maps $(\mathfrak{m}_{\overline{u}_i}\mathcal{O}_{U_i,u_i})^{L_i'}$ into $\mathfrak{m}_{\overline{u}_i}^{L_i'}$,

$$\mathfrak{m}_{u_i}^{\delta_i L_i'} \cap \mathcal{O}_{U_i/\widetilde{G},\overline{u}_i} \subset \mathfrak{m}_{\overline{u}_i}^{L_i'}.$$

Since σ_i^0 induces a $\widetilde{G}$-equivariant isomorphism $\widehat{\mathcal{O}}_{Y_0,0} \cong \widehat{\mathcal{O}}_{U_i,u_i}$, there exist $\widetilde{G}$-semi-invariant regular functions $\widetilde{h}_{ij}^\flat$ on Y_0, with the characters of $\widetilde{h}_{ij}$, namely polynomial truncations of the elements of $\widehat{\mathcal{O}}_{Y_0,0}$ corresponding to the $\sigma_i^*\widetilde{h}_{ij}$, such that

$$\sigma_i^*\widetilde{h}_{ij} - (\sigma_i^0)^*\widetilde{h}_{ij}^\flat \in \mathfrak{m}_{u_i}^{\max(\delta_i L_i', \lfloor \Lambda_1 \rfloor + 1)}$$

for any j. Then $(\sigma_i^*\widetilde{h}_{ij})^{|\widetilde{G}|}$ and $((\sigma_i^0)^*\widetilde{h}_{ij}^\flat)^{|\widetilde{G}|}$ are $\widetilde{G}$-invariant and congruent modulo $\mathfrak{m}_{\overline{u}_i}^{L_i'}$. Let

$$B_i^\flat := \sum_j \kappa_j \operatorname{div}(\widetilde{h}_{ij}^\flat) + \beta_L \operatorname{div}(\varpi)$$

on Y_0. Writing the quotient values at u_i of the pullbacks of $B_{Y,i}^*$ and of $B_i^\flat$ as minimal log discrepancies at $\overline{u}_i$ of the boundary of $(U_i,0)/\widetilde{G}$ with the $\mathbb{R}$-ideals

$$\prod_j \big((\sigma_i^*\widetilde{h}_{ij})^{|\widetilde{G}|}\big)^{\frac{\kappa_j}{|\widetilde{G}|}} \big((\sigma_i^*P)^{|\widetilde{G}|}\big)^{\frac{\beta_L}{|\widetilde{G}|}} \quad \text{and} \quad \prod_j \big(((\sigma_i^0)^*\widetilde{h}_{ij}^\flat)^{|\widetilde{G}|}\big)^{\frac{\kappa_j}{|\widetilde{G}|}} \big((\sigma_i^*P)^{|\widetilde{G}|}\big)^{\frac{\beta_L}{|\widetilde{G}|}},$$

where $\sigma_i^*P = (\sigma_i^0)^*\varpi$, Theorem 2.22(2) and Lemma 2.13 imply

$$b_i = \operatorname{mld}((Y_0 \ni 0, B_i^\flat)/\widetilde{G}) = \operatorname{mld}(X_0 \ni x_0, \Delta_i^\flat),$$

where $\Delta_i^\flat$ is the sum of the components through x_0 of the boundary of $(Y_0, B_i^\flat)/\widetilde{G}$. Since X_0 is $\mathbb{Q}$-factorial, $\Delta_i^\flat$ is $\mathbb{R}$-Cartier. Since $b_i \geq 0$, $(X_0, \Delta_i^\flat)$ is lc near x_0, so $\operatorname{mult}_\Pi B_i^\flat \leq 1$ for any prime divisor Π on Y_0 through 0, and $\Delta_i^\flat \in \Gamma_0$.

By Theorem 2.22(3), there exists a prime divisor E_i over X_0 which computes $\operatorname{mld}(X_0 \ni x_0, \Delta_i^\flat)$ with $a(E_i, X_0, 0) \leq L_0$. Let E_i' be a prime divisor over Y_0 whose valuation restricts to a multiple of $\operatorname{mult}_{E_i}$. Since the boundary Δ_0 of $(Y_0,0)/\widetilde{G}$ satisfies $\Delta_0 \geq 0$ and $e_{E_i'} \leq |\widetilde{G}| \leq 2M$,

$$a(E_i', Y_0, 0) \leq e_{E_i'} a(E_i, X_0, 0) \leq 2ML_0.$$

Since $(Y_0, B_i^\flat)$ is lc near 0 and $B_i^\flat \geq 0$,

$$\sum_j \kappa_j \operatorname{mult}_{E_i'} \widetilde{h}_{ij}^\flat \leq a(E_i', Y_0, 0),$$

so $\operatorname{mult}_{E_i'} \widetilde{h}_{ij}^\flat \leq \lfloor \Lambda_1 \rfloor$, as $\operatorname{mult}_{E_i'} \widetilde{h}_{ij}^\flat \in \mathbb{N}$. Since

$$\operatorname{mult}_{E_i'}\big(\sigma_i^*\widetilde{h}_{ij} - (\sigma_i^0)^*\widetilde{h}_{ij}^\flat\big) \geq \lfloor \Lambda_1 \rfloor + 1,$$

we have

$$\operatorname{mult}_{E_i'} \sigma_i^*\widetilde{h}_{ij} = \operatorname{mult}_{E_i'} \widetilde{h}_{ij}^\flat$$

for any j, so

$$b_i \le \frac{a(E_i', Y_i, B^*_{Y,i})}{e_{E_i'}} = \frac{a(E_i', Y_0, 0) - \sum_j \kappa_j \operatorname{mult}_{E_i'} \widetilde{h}^\flat_{ij} - \beta_L \operatorname{mult}_{E_i'} \varpi}{e_{E_i'}} = a(E_i, X_0, \Delta_i^\flat) = b_i. \tag{B.149}$$

Since K_{Y_0} is Cartier, $a(E_i', Y_0, 0)$ is an integer in $[1, 2ML_0]$. Moreover $\operatorname{mult}_{E_i'} \widetilde{h}^\flat_{ij} \in [0, \Lambda_1]$ are integers, $\epsilon \operatorname{mult}_{E_i'} \varpi < a(E_i', Y_0, 0)$, and $e_{E_i'} \le 2M$. Thus, by (B.149), the b_i belong to a finite set, and for $\Lambda_2 := \max(\Lambda_1, 2ML_0/\epsilon)$,

$$\operatorname{mult}_{E_i'} \widetilde{h}_{ij} \le \Lambda_2 \quad \text{and} \quad \operatorname{mult}_{E_i'} P \le \Lambda_2$$

for any i and j.

Step 5. In this step we compare a_i and b_i and conclude.

By the definitions of $B_{Y,i}$ and $B^*_{Y,i}$,

$$B^*_{Y,i} - B_{Y,i} = \sum_j (\kappa_j - c_{ij}) \big(\operatorname{div}(\widetilde{h}_{ij}) + n_j \operatorname{div}(P) \big) \ge 0.$$

Thus $b_i \le a_i$ by (B.148). By (B.148) and (B.149),

$$a_i \le \frac{a(E_i', Y_i, B_{Y,i})}{e_{E_i'}} = b_i + \sum_j (\kappa_j - c_{ij}) \frac{\operatorname{mult}_{E_i'} \widetilde{h}_{ij} + n_j \operatorname{mult}_{E_i'} P}{e_{E_i'}}$$
$$\le b_i + (1 + \max_j n_j) \Lambda_2 \sum_j (\kappa_j - c_{ij}).$$

Since the b_i belong to a finite set, possibly passing to a subsequence, $b_i = b$ for any i. Since $c_{ij} \to \kappa_j$,

$$b \le a_i \quad \text{for any } i \quad \text{and} \quad \lim_{i \to +\infty} a_i = b.$$

Since the a_i are strictly increasing, $a_1 < \lim_{i \to +\infty} a_i = b \le a_1$, a contradiction. □

B.4. **Axes fixed by the deck involution.**

Proposition B.20. *(1) Let $n \ge 1$ be an integer, $\kappa_1, \dots, \kappa_n$ positive real numbers, $B_0 \ge 0$ and $R' \ge 1$ real numbers, and $N_1, N_2, \dots$ integers with $N_i \ge 2$ for any i and $\lim_i N_i = +\infty$. For any i let $h_{i1}, \dots, h_{in}$ be polynomials in x, y, z, t with non-zero restrictions to W_{N_i}, and let $(h_{*,j})_j$, with coefficient field $\Bbbk$, be a compatible generic limit of the tuples $(h_{ij})_j$ along an infinite subset $\mathcal{I} \subset \mathbb{N}^+$ (Definition 2.5, with H trivial) with non-zero restrictions to W_0. Assume that*

$$\Big(\operatorname{Spec} \Bbbk[[x, y, z, t]]/(xy + z^2), \sum_j \kappa_j \operatorname{div}(h_{*,j}) \Big)$$

is klt outside the axis $(x = y = z = 0)$. Then there exists a real number T such that for all but finitely many $i \in \mathcal{I}$, any prime divisor F over W_{N_i} with center 0 satisfying

$$a(F, W_{N_i}, 0) - \sum_j \kappa_j \operatorname{mult}_F h_{ij} \le B_0 \quad \text{and} \quad \min(\operatorname{mult}_F x, \operatorname{mult}_F y, \operatorname{mult}_F z) \le R'$$

satisfies

$$\sum_j \operatorname{mult}_F h_{ij} \le T \quad \text{and} \quad a(F, W_{N_i}, 0) \le B_0 + T \max_j \kappa_j.$$

(2) *Notation and conditions as in Set-up 10.3. Assume that the a_i are strictly increasing and ι acts trivially on the curve C of Proposition 10.5(1), and let $\delta > 0$ be a real number such that*

$$\Big(W_{N_i}, \sum_j \kappa_j \operatorname{div}(h_{ij}) + \delta \operatorname{div}(t)\Big)$$

is lc near 0 for infinitely many i. Then, after passing to a subsequence of these i, any computing lifts F_i of a_i satisfy

$$\lim_{i\to+\infty} \min(\operatorname{mult}_{F_i} x, \operatorname{mult}_{F_i} y, \operatorname{mult}_{F_i} z) = +\infty \quad \textit{and} \quad \operatorname{mult}_{F_i} t \le \Big\lceil \frac{2M}{\delta} \Big\rceil.$$

Proof. Klt pairs and log discrepancies over $\operatorname{Spec} \Bbbk[[x,y,z,t]]$ and over $\operatorname{Spec} \Bbbk[[x,y,z,t]]/(xy+z^2)$ are understood in the framework of [dFEM11, Section 2].

We first prove (1). We let

$$f_0 := xy + z^2, \quad f_i := xy + z^2 + t^{N_i} \ (i \ge 1),$$

so that $W_0 = (f_0 = 0)$ and $W_{N_i} = (f_i = 0)$ in $\mathbb{A}^4$, we let $W_* := \operatorname{Spec} \Bbbk[[x,y,z,t]]/(f_0)$, and we let $\mathfrak{m}$ be the maximal ideal of $\mathcal{O}_{\mathbb{A}^4,0}$. Let $\kappa_{\max} := \max_j \kappa_j$ and $\rho := 4\kappa_{\max} + 1$, and for any i and any prime divisor F over W_{N_i} let $\sigma_i(F) := \sum_j \operatorname{mult}_F h_{ij}$.

Step 1. In this step we prove that for any i and any prime divisor F over W_{N_i} with center 0,

$$\operatorname{mult}_F x \le 2a(F, W_{N_i}, 0) \quad \text{and} \quad \operatorname{mult}_F y \le 2a(F, W_{N_i}, 0). \tag{B.150}$$

We show that the polynomial xf_i is non-degenerate with respect to the Newton polyhedron of xf_i. The Newton polyhedron of xf_i is the translate by $(1,0,0,0)$ of the Newton polyhedron of f_i, so the compact faces of the Newton polyhedra of xf_i and f_i correspond bijectively, and for a compact face σ the face polynomial $(xf_i)_\sigma$ equals $x(f_i)_\sigma$, where x is invertible on $(\mathbb{C}^*)^4$. If σ contains the exponent of xy, then $\partial(f_i)_\sigma/\partial x = y$ has no zero on $(\mathbb{C}^*)^4$. Otherwise $(f_i)_\sigma \in \{z^2, t^{N_i}, z^2 + t^{N_i}\}$, and $\partial(f_i)_\sigma/\partial z$ or $\partial(f_i)_\sigma/\partial t$ has no zero on $(\mathbb{C}^*)^4$. By [Amb03, Lemma 1.2], a regular simplicial subdivision of the Newton fan of xf_i defines a toric log resolution $\pi : Y \to \mathbb{A}^4$ of $(\mathbb{A}^4, \operatorname{div}(xf_i))$ near 0, and $\pi_*^{-1} \operatorname{div}(x)$ is a torus-invariant prime divisor on Y. Thus the prime divisors on Y with non-zero coefficient in the pullback of $\operatorname{div}(f_i) + \frac{1}{2}\operatorname{div}(x)$ are $\pi_*^{-1} W_{N_i}$, with coefficient 1, and torus-invariant prime divisors E_w for primitive vectors $w = (w_1, w_2, w_3, w_4) \in \mathbb{N}^4$. Since $\operatorname{mult}_{E_w} f_i = \min(w_1+w_2, 2w_3, N_i w_4)$, $\operatorname{mult}_{E_w} x = w_1$, and $\min(w_1+w_2, 2w_3) \le \frac{1}{2}(w_1+w_2+2w_3)$,

$$a\Big(E_w, \mathbb{A}^4, \operatorname{div}(f_i) + \frac{1}{2}\operatorname{div}(x)\Big) = \frac{w_1}{2} + w_2 + w_3 + w_4 - \min(w_1+w_2, 2w_3, N_i w_4) \ge \frac{w_2}{2} + w_4 \ge 0.$$

Thus the pairs

$$\Big(\mathbb{A}^4, \operatorname{div}(f_i) + \frac{1}{2}\operatorname{div}(x)\Big) \quad \text{and} \quad \Big(\mathbb{A}^4, \operatorname{div}(f_i) + \frac{1}{2}\operatorname{div}(y)\Big) \tag{B.151}$$

are lc near 0, the second one since the automorphism $(x,y,z,t) \to (y,x,z,t)$ of $\mathbb{A}^4$ preserves f_i and exchanges $\operatorname{div}(x)$ and $\operatorname{div}(y)$.

Let F be a prime divisor over W_{N_i} with center 0. By Lemma 5.4(1), with $V = \mathbb{A}^4$ and W_{N_i} in place of T, the trivial group, and the functions x and y, there exist an integer $k \ge 1$ and a prime divisor $\widehat{F}$ over $\mathbb{A}^4$ with center 0 such that

$$\operatorname{mult}_{\widehat{F}} f_i = k, \quad \operatorname{mult}_{\widehat{F}} x = \operatorname{mult}_F x, \quad \operatorname{mult}_{\widehat{F}} y = \operatorname{mult}_F y, \quad \text{and} \quad a(\widehat{F}, \mathbb{A}^4, 0) = a(F, W_{N_i}, 0) + k.$$

Since the first pair in (B.151) is lc near 0,

$$0 \le a\Big(\widehat{F}, \mathbb{A}^4, \operatorname{div}(f_i) + \frac{1}{2}\operatorname{div}(x)\Big) = a(F, W_{N_i}, 0) - \frac{1}{2}\operatorname{mult}_F x,$$

and with y in place of x we obtain the second inequality of (B.150).

If moreover $a(F,W_{N_i},0)-\sum_j \kappa_j \operatorname{mult}_F h_{ij} \le B_0$, then $a(F,W_{N_i},0)\le B_0+\kappa_{\max}\sigma_i(F)$, and (B.150) implies

$$\operatorname{mult}_F x+\operatorname{mult}_F y+\max_j \operatorname{mult}_F h_{ij} \le 4a(F,W_{N_i},0)+\sigma_i(F)\le 4B_0+\rho\sigma_i(F). \tag{B.152}$$

Step 2. In this step we fix exponents and an integer $m\ge 1$ on the limit side.

The condition that $(W_*,\sum_j \kappa_j \operatorname{div}(h_{*,j}))$ is klt outside the axis consists, on a log resolution of $(W_*,\sum_j \operatorname{div}(h_{*,j}))$ [dFEM11, Section 2], of finitely many strict inequalities between log discrepancies, each continuous in the coefficients κ_j. Thus there exist rational numbers $\mu_j>\kappa_j$ such that

$$\Big(W_*,\sum_j \mu_j \operatorname{div}(h_{*,j})\Big) \text{ is klt outside the axis } (x=y=z=0).$$

Since

$$\lim_{\lambda\to 1}\Big(\min_j(\lambda\mu_j-\kappa_j)-(1-\lambda)\rho\Big)=\min_j(\mu_j-\kappa_j)>0,$$

there exist rational numbers $0<\lambda_0<\lambda_1<1$ such that

$$\epsilon:=\min_j(\lambda_0\mu_j-\kappa_j)>(1-\lambda_0)\rho. \tag{B.153}$$

Let Q be a positive integer such that $q_j:=Q\mu_j\in\mathbb{Z}$ for any j, and let

$$\mathfrak{b}_*:=\Big(f_0^Q\prod_j h_{*,j}^{q_j}\Big)\subset \Bbbk[[x,y,z,t]],\qquad \mathfrak{b}_i:=\Big(f_i^Q\prod_j h_{ij}^{q_j}\Big)\subset \mathcal{O}_{\mathbb{A}^4,0},$$

$$\text{and}\quad \mathcal{J}_*:=\mathcal{J}(\mathfrak{b}_*^{\lambda_1/Q})=\mathcal{J}\Big(f_0^{\lambda_1}\prod_j h_{*,j}^{\lambda_1\mu_j}\Big).$$

Since

$$\Big(W_*,\sum_j \mu_j \operatorname{div}(h_{*,j})\Big)=\Big(W_*,\frac{1}{Q}\operatorname{div}(h_*|_{W_*})\Big),\quad \text{where } h_*:=\prod_j h_{*,j}^{q_j},$$

is klt outside the axis, Lemma 6.1(3), with $f_*=f_0$, this h_*, the exponent $1/Q$ in place of μ, $\theta_2=\lambda_1$, and the axis $(x=y=z=0)$ in place of V, shows that $\mathcal{J}_*$ is the unit ideal near any point of W_* outside the axis. Thus

$$\operatorname{Supp}\big(\Bbbk[[x,y,z,t]]/(\mathcal{J}_*+(f_0))\big)\subset(x=y=z=0),$$

so $x,y,z\in\sqrt{\mathcal{J}_*+(f_0)}$, and there exists an integer $m\ge 1$ such that

$$r^m\in\mathcal{J}_*+(f_0)\quad\text{for any } r\in\{x,y,z\}. \tag{B.154}$$

Step 3. In this step we spread (B.154) to the members. Fix an integer $L>mR'$, and let $\mathfrak{n}_*$ be the maximal ideal of $\Bbbk[[x,y,z,t]]$. By (B.154), for any $r\in\{x,y,z\}$ there exists $b_r^*\in\Bbbk[[x,y,z,t]]$ with $r^m-f_0b_r^*\in\mathcal{J}_*$. Possibly replacing b_r^* with its part of degree less than L, we may assume that b_r^* is a polynomial, and then, since $f_0\mathfrak{n}_*^L\subset\mathfrak{n}_*^L$,

$$g_r:=r^m-f_0b_r^*\in\mathcal{J}_*+\mathfrak{n}_*^L=\mathcal{J}\Big(f_0^{\lambda_1}\prod_j h_{*,j}^{\lambda_1\mu_j}\Big)+\mathfrak{n}_*^L.$$

Since $f_i\equiv f_0 \pmod{\mathfrak{m}^{N_i}}$ and $\lim_i N_i=+\infty$, $(f_0,h_{*,1},\dots,h_{*,n})$ is a compatible generic limit of the tuples $(f_i,h_{i1},\dots,h_{in})$ along $\mathcal{I}$, with coefficient field $\Bbbk$. Let

$$\ell:=3\cdot\#\{\alpha\in\mathbb{N}^4\mid \alpha_1+\cdots+\alpha_4<L\},$$

and let $u_{r,\alpha}$, for $r\in\{x,y,z\}$ and $\alpha\in\mathbb{N}^4$ with $\alpha_1+\cdots+\alpha_4<L$, be ℓ variables, let

$$\Phi_r:=r^m-f_0\sum_\alpha u_{r,\alpha}x^{\alpha_1}y^{\alpha_2}z^{\alpha_3}t^{\alpha_4},$$

a polynomial with complex coefficients, and let $\beta\in\Bbbk^\ell$ be the vector of the coefficients of b_x^*,b_y^*,b_z^*, so that $g_r=\Phi_r(\beta)$. By Lemma 6.1(1), with the exponents $\lambda_0,\lambda_0\mu_1,\dots,\lambda_0\mu_n$ in place

of $\gamma_1,\dots,\gamma_N$, the exponents $\lambda_1,\lambda_1\mu_1,\dots,\lambda_1\mu_n$ in place of $\gamma'_1,\dots,\gamma'_N$, the polynomials Φ_x,Φ_y,Φ_z, the parameter β, the integer L, and $(\mathbb{A}^4\ni 0)$ in place of $(Y_i\ni y_i)$, for all but finitely many $i\in\mathcal{I}$ there exists $\beta_i\in\mathbb{C}^\ell$ such that, for b_{ix},b_{iy},b_{iz} the polynomials whose vector of coefficients is β_i and $g_{ir}:=r^m-f_0b_{ir}=\Phi_r(\beta_i)$, for any $r\in\{x,y,z\}$,

$$\begin{aligned} &g_{ir}=g'_{ir}+g''_{ir}\quad\text{with}\quad g'_{ir}\in\mathcal{J}\Big(f_i^{\lambda_0}\prod_j h_{ij}^{\lambda_0\mu_j}\Big)=\mathcal{J}(\mathfrak{b}_i^{\lambda_0/Q})\\ &\text{and}\quad g''_{ir}\in\mathfrak{m}^L. \end{aligned}\tag{B.155}$$

Possibly removing finitely many indices from $\mathcal{I}$, we may assume that (B.155) holds and $N_i>mR'$ for any $i\in\mathcal{I}$.

Step 4. In this step we bound $\sigma_i(F)$. Let $i\in\mathcal{I}$, let F be a prime divisor over W_{N_i} with center 0 satisfying $a(F,W_{N_i},0)-\sum_j\kappa_j\operatorname{mult}_F h_{ij}\le B_0$ and $\min(\operatorname{mult}_F x,\operatorname{mult}_F y,\operatorname{mult}_F z)\le R'$, and let $r\in\{x,y,z\}$ with $\operatorname{mult}_F r\le R'$. Let $j_F:=\operatorname{mult}_F\operatorname{Jac}_{W_{N_i}}$, let $x_\bullet\in\{x,y,z,t\}$ with $\operatorname{mult}_F(\partial f_i/\partial x_\bullet|_{W_{N_i}})=j_F$, and let

$$k:=\lfloor 4B_0+\rho\sigma_i(F)+mR'\rfloor+1.$$

By (B.152),

$$\begin{aligned} &k>mR',\\ &\text{and}\quad k\le\rho\sigma_i(F)+4B_0+mR'+1. \end{aligned}\tag{B.156}$$

Since $\operatorname{Jac}_{W_{N_i}}$ contains $x|_{W_{N_i}}$ and $y|_{W_{N_i}}$, $j_F\le\min(\operatorname{mult}_F x,\operatorname{mult}_F y)$, so (B.152) and the definition of k give $k>2j_F$ and $k-j_F>\max(\operatorname{mult}_F r,\max_j\operatorname{mult}_F h_{ij})$, where $\operatorname{mult}_F z\le R'\le mR'$ is used if $r=z$. By Lemma 5.4(2), with $\mathcal{X}=\mathbb{A}^4$, $f=f_i$, which defines W_{N_i}, normal near 0 since singular only at 0, j_F in place of j, $x_\bullet$ in place of x, and the functions $r,h_{i1},\dots,h_{in}$, there exists a prime divisor $\widehat{F}$ over $\mathbb{A}^4$ with center 0 such that

$$\begin{aligned} &\operatorname{mult}_{\widehat{F}} f_i=k,\quad \operatorname{mult}_{\widehat{F}} r=\operatorname{mult}_F r,\quad \operatorname{mult}_{\widehat{F}} h_{ij}=\operatorname{mult}_F h_{ij}\text{ for any }j,\\ &\text{and}\quad a(\widehat{F},\mathbb{A}^4,0)=a(F,W_{N_i},0)+k. \end{aligned}\tag{B.157}$$

Since $\operatorname{center}_{\mathbb{A}^4}\widehat{F}=\{0\}$, $\operatorname{mult}_{\widehat{F}} t\ge 1$ and $\operatorname{mult}_{\widehat{F}} b_{ir}\ge 0$. Since $f_0=f_i-t^{N_i}$ and $N_i>mR'$, (B.156) and (B.157) imply

$$\operatorname{mult}_{\widehat{F}}(f_0b_{ir})\ge\min(k,N_i)>mR'\ge m\operatorname{mult}_F r=\operatorname{mult}_{\widehat{F}} r^m,$$

so $\operatorname{mult}_{\widehat{F}} g_{ir}=m\operatorname{mult}_F r\le mR'$. Since $\operatorname{mult}_{\widehat{F}} g''_{ir}\ge L>mR'$, (B.155) implies

$$\operatorname{mult}_{\widehat{F}} g'_{ir}=m\operatorname{mult}_F r.$$

Since $g'_{ir}\in\mathcal{J}(\mathfrak{b}_i^{\lambda_0/Q})$, using that the multiplier ideal in (B.155), taken on $\operatorname{Spec}\mathbb{C}[[x,y,z,t]]$, is generated by that of $\mathfrak{b}_i^{\lambda_0/Q}$ on $\mathbb{A}^4$ [dFEM11, Proposition A.14],

$$\operatorname{mult}_{\widehat{F}} g'_{ir}+a(\widehat{F},\mathbb{A}^4,0)>\frac{\lambda_0}{Q}\operatorname{mult}_{\widehat{F}}\mathfrak{b}_i=\lambda_0k+\sum_j\lambda_0\mu_j\operatorname{mult}_F h_{ij}.$$

By (B.157), (B.156) and the hypothesis $a(F,W_{N_i},0)-\sum_j\kappa_j\operatorname{mult}_F h_{ij}\le B_0$,

$$\begin{aligned} \sum_j(\lambda_0\mu_j-\kappa_j)\operatorname{mult}_F h_{ij}&<m\operatorname{mult}_F r+\Big(a(F,W_{N_i},0)-\sum_j\kappa_j\operatorname{mult}_F h_{ij}\Big)+(1-\lambda_0)k\\ &\le mR'+B_0+(1-\lambda_0)\big(\rho\sigma_i(F)+4B_0+mR'+1\big). \end{aligned}$$

Since $\operatorname{mult}_F h_{ij}\ge 0$ for any j, the left side is at least $\epsilon\sigma_i(F)$. By (B.153),

$$\sigma_i(F)<T:=\frac{B_0+mR'+(1-\lambda_0)(4B_0+mR'+1)}{\epsilon-(1-\lambda_0)\rho},$$

and T depends only on B_0, R', m, λ_0 and the κ_j, μ_j, not on i or F. By the hypothesis on F and $\sigma_i(F) < T$,

$$a(F, W_{N_i}, 0) \le B_0 + \kappa_{\max}\sigma_i(F) \le B_0 + T\kappa_{\max}.$$

This proves (1).

We prove (2). Let $\mathcal{I}_0$ be the set of the indices i such that

$$\Big(W_{N_i}, \sum_j \kappa_j \operatorname{div}(h_{ij}) + \delta \operatorname{div}(t)\Big)$$

is lc near 0, which is infinite by assumption. We pass to the subsequence indexed by $\mathcal{I}_0$. Possibly replacing $\mathcal{I}_0$ with an infinite subset, we may assume that (1)–(3) of Proposition 10.5 hold.

The hypotheses of (1), for the κ_j, N_i, h_{ij}, and $(h_{*,j})_j$, hold along $\mathcal{I}_0$: since $H_j^* \ne 0$, $h_{*,j}|_{W_0} \ne 0$. Since ι acts trivially on C, $C = (p = v = 0)$ by Proposition 10.5(2), and the image of C in W_* is the axis $(x = y = z = 0)$. The double cover $\pi_0 : \widetilde{W}_* \to W_*$ is the quotient by ι, which is ramified only along $C = (p = v = 0)$ with $\dim C = 1$, so $K_{\widetilde{W}_*} + B^* = \pi_0^*(K_{W_*} + D_*)$. Thus, since π_0 is finite, (W_*, D_*) is klt outside the axis if and only if $(\widetilde{W}_*, B^*)$ is klt outside C. By Proposition 10.5(1), B^* is lc and the unique lc center of B^* through 0 is C. Since any non-empty closed subset of $\widetilde{W}_*$ contains 0, C is the unique lc center of B^*, so $(\widetilde{W}_*, B^*)$ is klt outside C.

Next, let $i \in \mathcal{I}_0$ and let F_i be a computing lift of a_i. Then $\operatorname{center}_{W_{N_i}} F_i = \{0\}$ and, by the definition of computing lifts,

$$a(F_i, W_{N_i}, 0) - \sum_j c_{ij} \operatorname{mult}_{F_i} h_{ij} = a(F_i, W_{N_i}, D_i) = e_{F_i} a_i.$$

Since $i \in \mathcal{I}_0$, $c_{ij} \le \kappa_j$ for any j, and $a_i \le 2$ by Set-up 10.3,

$$\delta \operatorname{mult}_{F_i} t \le a\Big(F_i, W_{N_i}, \sum_j \kappa_j \operatorname{div}(h_{ij})\Big) \le e_{F_i} a_i \le 2e_{F_i},$$

so

$$\frac{\operatorname{mult}_{F_i} t}{e_{F_i}} \le \frac{2}{\delta}. \tag{B.158}$$

Since moreover $e_{F_i} \le |G| \le M$,

$$a(F_i, W_{N_i}, 0) - \sum_j \kappa_j \operatorname{mult}_{F_i} h_{ij} = e_{F_i} a_i - \sum_j (\kappa_j - c_{ij}) \operatorname{mult}_{F_i} h_{ij} \le 2M. \tag{B.159}$$

For a positive integer R', let $T_{R'}$ be the real number of (1) with $B_0 = 2M$, and let $\mathcal{S}_{R'}$ be the set of the indices $i \in \mathcal{I}_0$ such that some computing lift F of a_i satisfies $\min(\operatorname{mult}_F x, \operatorname{mult}_F y, \operatorname{mult}_F z) \le R'$. We show that $\mathcal{S}_{R'}$ is finite. Suppose that $\mathcal{S}_{R'}$ is infinite. Proposition 10.5 holds for the subsequence indexed by $\mathcal{S}_{R'}$, and C does not change. For any $i \in \mathcal{S}_{R'}$ fix a computing lift F_i of a_i with $\min(\operatorname{mult}_{F_i} x, \operatorname{mult}_{F_i} y, \operatorname{mult}_{F_i} z) \le R'$, and let

$$w_{ij} := \frac{\operatorname{mult}_{F_i} h_{ij}}{e_{F_i}} \text{ for any } j \quad \text{and} \quad R_i := \sum_{j \in J_0} (\kappa_j - c_{ij}) w_{ij},$$

where J_0 is the set of the indices j such that H_j^* vanishes identically on C. By Proposition 10.5(3), there exists an infinite subset $\mathcal{S}' \subset \mathcal{S}_{R'}$ along which $(R_i)_i$ is strictly increasing with $\lim_i R_i > 0$. On the other hand, by (B.159) and (1), $w_{ij} \le \operatorname{mult}_{F_i} h_{ij} \le T_{R'}$ for any j and all but finitely many $i \in \mathcal{S}'$, so

$$R_i \le T_{R'} \sum_j (\kappa_j - c_{ij}),$$

and the right side tends to 0 since $\lim_i c_{ij} = \kappa_j$ for any j. This contradicts $\lim_i R_i > 0$. Thus $\mathcal{S}_{R'}$ is finite.

Since $\mathcal{S}_1 \subset \mathcal{S}_2 \subset \cdots$ are finite, for any positive integer R' and any $i \in \mathcal{I}_0 \backslash \mathcal{S}_{R'}$, any computing lift F_i of a_i satisfies $\min(\mathrm{mult}_{F_i} x, \mathrm{mult}_{F_i} y, \mathrm{mult}_{F_i} z) > R'$, so

$$\lim_{i \in \mathcal{I}_0} \min(\mathrm{mult}_{F_i} x, \mathrm{mult}_{F_i} y, \mathrm{mult}_{F_i} z) = +\infty.$$

Finally, by (B.158) and $e_{F_i} \leq M$,

$$\mathrm{mult}_{F_i} t \leq \frac{2M}{\delta} \leq \left\lceil \frac{2M}{\delta} \right\rceil. \qquad \square$$

Lemma B.21. *Let $n \geq 2$ be an integer or $n = +\infty$, where $W_\infty := W_0$, let H be a finite diagonal abelian group preserving the equation of W_n up to a character and acting faithfully on W_n, and let $(\kappa_j)_j$ be a finite tuple of positive real numbers. Then the following hold.*

(1) Let $\Bbbk \supset \mathbb{C}$ be an algebraically closed field, and let $(g_j)_j$ be a tuple of H-semi-invariant elements of $\Bbbk[[x,y,z,t]]$ with non-zero restrictions to W_n such that $(W_n, \sum_j \kappa_j \operatorname{div}(g_j))$ is lc near 0, where, as in Definition 2.2, the pair is taken on the completion of $W_n \times_{\mathbb{C}} \operatorname{Spec} \Bbbk$ at 0. Then there exists an integer L, depending only on n, H, and $(\kappa_j)_j$, satisfying the following. Let $(g'_j)_j$ be a tuple of H-semi-invariant elements of $\Bbbk[[x,y,z,t]]$ with $g'_j - g_j \in (x,y,z,t)^L$ for any j. Then
   (a) $g'_j|_{W_n} \neq 0$ for any j,
   (b) the pair $(W_n, \sum_j \kappa_j \operatorname{div}(g'_j))$ is lc near 0, and
   (c)

$$\mathrm{mld}\Big(\Big(W_n \ni 0, \sum_j \kappa_j \operatorname{div}(g'_j)\Big)/H\Big) = \mathrm{mld}\Big(\Big(W_n \ni 0, \sum_j \kappa_j \operatorname{div}(g_j)\Big)/H\Big).$$

(2) For any i let $(g_{ij})_j$ be a tuple of H-semi-invariant polynomials whose characters do not depend on i, and let $(g_{*,j})_j$, with coefficient field $\Bbbk$, be a compatible generic limit of the tuples $(g_{ij})_j$ along an infinite subset $\mathcal{I} \subset \mathbb{N}^+$ (Definition 2.5) with non-zero restrictions to W_n. If $(W_n, \sum_j \kappa_j \operatorname{div}(g_{*,j}))$ is lc near 0, then for all but finitely many $i \in \mathcal{I}$, the pair $(W_n, \sum_j \kappa_j \operatorname{div}(g_{ij}))$ is lc near 0 and

$$\mathrm{mld}\Big(\Big(W_n \ni 0, \sum_j \kappa_j \operatorname{div}(g_{ij})\Big)/H\Big) = \mathrm{mld}\Big(\Big(W_n \ni 0, \sum_j \kappa_j \operatorname{div}(g_{*,j})\Big)/H\Big).$$

If $(W_n, \sum_j \kappa_j \operatorname{div}(g_{*,j}))$ is not lc near 0, then $(W_n, \sum_j \kappa_j \operatorname{div}(g_{ij}))$ is not lc near 0 for all but finitely many $i \in \mathcal{I}$.

(3) In Set-up 10.3, let $\mathcal{I} \subset \mathbb{N}^+$ be the infinite subset along which $(h_{*,j})_j$ is a compatible generic limit of the tuples $(h_{ij})_j$. If $(W_0, \sum_j \kappa_j \operatorname{div}(h_{*,j}))$ is lc near 0, then

$$\limsup_{i \in \mathcal{I}} a_i \leq \mathrm{mld}\Big(\Big(W_0 \ni 0, \sum_j \kappa_j \operatorname{div}(h_{*,j})\Big)/G\Big).$$

Proof. Let $f_n := xy + z^2 + t^n$ if $n \neq +\infty$ and $f_\infty := xy + z^2$, let $\mathfrak{m} := (x,y,z,t)$ and $q := |H|$, and for $g \in \Bbbk[[x,y,z,t]]$ and an integer m let $g^{<m}$ be the sum of the terms of g of total degree less than m. Let $\pi : W_n \to X := W_n/H$ be the quotient morphism, let $\overline{0} := \pi(0)$, and let Δ_H be the branch divisor of π, so that $K_{W_n} = \pi^*(K_X + \Delta_H)$. If $n \neq +\infty$, the threefold X is klt and Δ_H is $\mathbb{Q}$-Cartier by Proposition 3.14(1) with $M := q$ and $D_N := 0$. If $n = +\infty$, then $X = \widetilde{W}_0/\widetilde{H}$, where $\widetilde{H}$ is the group of lifts of H to $\widetilde{W}_0 \cong \mathbb{A}^3$, defined as $\widetilde{G}$ in Set-up 10.3, so X is klt and Δ_H is $\mathbb{Q}$-Cartier.

For a tuple $\phi = (\phi_j)_j$ of H-semi-invariant elements of $\Bbbk[[x,y,z,t]]$ with non-zero restrictions to W_n, let

$$\beta(\phi) := \mathrm{mld}\Big(\Big(W_n \ni 0, \sum_j \kappa_j \operatorname{div}(\phi_j)\Big)/H\Big),$$

and let $\mathfrak{a}(\phi) := \prod_j (\phi_j^q)^{\kappa_j/q}$, an $\mathbb{R}$-ideal on the completion of $X_{\Bbbk}$ at $\overline{0}$, since ϕ_j^q is H-invariant (Lemma 2.12(1)). For a prime divisor F over W_n with center 0 and the image $\overline{F}$ of F over X, we have

$$a\Big(F, W_n, \sum_j \kappa_j \operatorname{div}(\phi_j)\Big) = e_F \cdot a(\overline{F}, X, \Delta_H, \mathfrak{a}(\phi)),$$

and any prime divisor over X with center $\overline{0}$ is such an image $\overline{F}$, by Lemma 2.15(2) and (3), on the completions by Lemma 2.12(3). Hence

$$\beta(\phi) = \operatorname{mld}(X \ni \overline{0}, \Delta_H, \mathfrak{a}(\phi)). \tag{B.160}$$

Statements about pairs on W_n, W_0 and their quotients by H defined by finitely many polynomials over $\Bbbk$, such as Theorem 2.22(2) and (3) for $X_{\Bbbk}$ with the constants of X, reduce to $\Bbbk = \mathbb{C}$ as follows. Such data are defined over the algebraic closure $\Bbbk_0 \subset \Bbbk$ of the field generated over $\mathbb{C}$ by their finitely many coefficients. Since $\Bbbk_0$ is an algebraically closed field of characteristic 0 and cardinality $2^{\aleph_0}$, there exists an isomorphism $\tau : \Bbbk_0 \to \mathbb{C}$. There exists an integer r prime to q such that $\tau(\zeta) = \zeta^r$ for any root of unity ζ of order dividing q, so τ maps W_n, W_0, H, and X to themselves. Log discrepancies, inertia groups and minimal log discrepancies do not change under τ and under the extension $\Bbbk_0 \subset \Bbbk$, since they are computed on a log resolution defined over $\Bbbk_0$.

Let $\kappa_{\min} := \min_j \kappa_j$, and let $\Gamma_H \subset \mathbb{R}_{>0}$ be the set of finite sums of the numbers κ_j/q and the non-zero coefficients of Δ_H, which is a DCC set. Let L_2 and L_3 be the integers of Theorem 2.22(2) and (3) for X and Γ_H, and let

$$L := \max\Big\{q(L_2+4),\ \Big\lfloor \frac{qL_3}{\kappa_{\min}} \Big\rfloor + 1,\ \Big\lfloor \frac{2}{\kappa_{\min}} \Big\rfloor + 1\Big\},$$

which depends only on n, H, and $(\kappa_j)_j$.

(1) We first show the following: if $\phi = (\phi_j)_j$ and $\phi' = (\phi'_j)_j$ are tuples of H-semi-invariant polynomials over $\Bbbk$ with non-zero restrictions to W_n and $\phi_j - \phi'_j \in \mathfrak{m}^L$ for any j, then $\beta(\phi) = \beta(\phi')$. Let $(x_1, x_2, x_3, x_4) := (x, y, z, t)$. The polynomial

$$\phi_j^q - \phi_j'^q = (\phi_j - \phi'_j) \sum_{k=0}^{q-1} \phi_j^k \phi_j'^{q-1-k}$$

is H-invariant and belongs to $\mathfrak{m}^{q(L_2+4)}$, so it is a linear combination of H-invariant monomials $\prod_{k=1}^4 x_k^{\alpha_k}$ with $\sum_k \alpha_k \ge q(L_2+4)$, since H is diagonal. Such a monomial is the product of the H-invariant monomial $\prod_k x_k^{\alpha_k - q\lfloor \alpha_k/q \rfloor}$ and of $\sum_k \lfloor \alpha_k/q \rfloor \ge L_2$ factors of the form x_k^q, each of which vanishes at $\overline{0}$ on X. Hence

$$(\phi_j^q) + \mathfrak{m}_{\overline{0}}^{L_2} = (\phi_j'^q) + \mathfrak{m}_{\overline{0}}^{L_2} \quad \text{in } \mathcal{O}_{X_{\Bbbk},\overline{0}}$$

for any j. By Theorem 2.22(2) with $B = \Delta_H$, (B.160), and Lemma 2.12(2), $\beta(\phi) = \beta(\phi')$. Here β may equal $-\infty$.

Next we prove the non-vanishing. Let E_0 be the exceptional prime divisor of the blow-up of W_n at 0. The tangent cone of W_n at 0 is the quadric cone defined by the quadratic part of f_n, which is integral, so $\operatorname{mult}_{E_0}$ is the $\mathfrak{m}$-adic order on W_n. By adjunction on $\operatorname{Bl}_0 \mathbb{A}^4$, $a(E_0, W_n, 0) = 2$. Since $(W_n, \sum_j \kappa_j \operatorname{div}(g_j))$ is lc near 0, $\kappa_{\min} \operatorname{mult}_{E_0}(g_j|_{W_n}) \le 2$, so

$$\operatorname{mult}_{E_0}(g_j|_{W_n}) < L \le \operatorname{mult}_{E_0}\big((g'_j - g_j)|_{W_n}\big).$$

Hence $\operatorname{mult}_{E_0}(g'_j|_{W_n}) = \operatorname{mult}_{E_0}(g_j|_{W_n})$ and $g'_j|_{W_n} \ne 0$. For $m \ge L$, also $g_j'^{<m} - g_j \in \mathfrak{m}^L$, so $\operatorname{mult}_{E_0}(g_j'^{<m}|_{W_n}) = \operatorname{mult}_{E_0}(g_j|_{W_n})$ and $g_j'^{<m}|_{W_n} \ne 0$.

Next we show that some truncation of $(g_j)_j$ is lc near 0. For $m \ge 1$, the tuple $(g_j^{<m}|_{W_n})_j$ agrees with $(g_j)_j$ modulo $\mathfrak{m}^m$. Hence on the klt variety $W_n \times_{\mathbb{C}} \operatorname{Spec} \Bbbk$, the sequence $\{(g_j^{<m})_j\}_{m\ge 1}$ of tuples of ideals stabilizes modulo every power of $\mathfrak{m}$, so its generic limit in the sense of [Kaw21, Section 3], constructed over a point, is the tuple of ideals generated by the g_j on the completion.

By [Kaw21, Theorem 3.10], there exists an integer $m_0 \geq L$ such that $(W_n, \sum_j \kappa_j \operatorname{div}(g_j^{<m_0}))$ is lc near 0, that is, $\beta((g_j^{<m_0})_j) \geq 0$.

We now prove the lc property for any $(g'_j)_j$ as in (1). For $m \geq L$, the polynomials $g_j'^{<m}$ and $g_j^{<m_0}$ differ by an element of $\mathfrak{m}^L$, so $\beta((g_j'^{<m})_j) = \beta((g_j^{<m_0})_j) \geq 0$ by the first paragraph of (1). Let F be a prime divisor over W_n with center 0, and take $m \geq L$ with $m > \max_j \operatorname{mult}_F g'_j$. Then

$$a\Big(F, W_n, \sum_j \kappa_j \operatorname{div}(g'_j)\Big) = a\Big(F, W_n, \sum_j \kappa_j \operatorname{div}(g_j'^{<m})\Big) \geq e_F \beta\big((g_j'^{<m})_j\big) \geq 0,$$

so $(W_n, \sum_j \kappa_j \operatorname{div}(g'_j))$ is lc near 0.

Let $g^\circ := (g_j^{<L})_j$, so that $g_j'^{<L} = g_j^{<L}$ for any j. Let $\widetilde{g} = (\widetilde{g}_j)_j$ be a tuple of H-semi-invariant elements of $\Bbbk[[x, y, z, t]]$ with $\widetilde{g}_j - g_j \in \mathfrak{m}^L$ for any j. By the lc property above, $(W_n, \sum_j \kappa_j \operatorname{div}(\widetilde{g}_j))$ and $(W_n, \sum_j \kappa_j \operatorname{div}(g_j^{<L}))$ are lc near 0, with non-zero restrictions of the entries to W_n. We show $\beta(\widetilde{g}) = \beta(g^\circ)$. For $\widetilde{g} = (g_j)_j$ and $\widetilde{g} = (g'_j)_j$ this implies (1).

Let F be a prime divisor over W_n with center 0, and let $m \geq L$ be an integer with $m > \operatorname{mult}_F \widetilde{g}_j$ for any j. Since $\operatorname{mult}_F(\widetilde{g}_j - \widetilde{g}_j^{<m}) \geq m$, we have $\operatorname{mult}_F \widetilde{g}_j^{<m} = \operatorname{mult}_F \widetilde{g}_j$. The polynomials $\widetilde{g}_j^{<m}$ and $g_j^{<L}$ differ by an element of $\mathfrak{m}^L$, so

$$\beta(g^\circ) = \beta\big((\widetilde{g}_j^{<m})_j\big) \leq \frac{a(F, W_n, \sum_j \kappa_j \operatorname{div}(\widetilde{g}_j^{<m}))}{e_F} = \frac{a(F, W_n, \sum_j \kappa_j \operatorname{div}(\widetilde{g}_j))}{e_F}.$$

Taking the infimum over F, we obtain $\beta(g^\circ) \leq \beta(\widetilde{g})$.

Conversely, the crepant quotient of $(W_n, \sum_j \kappa_j \operatorname{div}(g_j^{<L}))$ is (X, B°) with

$$B^\circ := \Delta_H + \frac{1}{q} \sum_j \kappa_j \operatorname{div}\big((g_j^{<L})^q\big),$$

which is lc near $\overline{0}$. The $\mathbb{R}$-divisor $B^\circ \geq 0$ is $\mathbb{R}$-Cartier, since Δ_H is $\mathbb{Q}$-Cartier and each $(g_j^{<L})^q$ is an H-invariant polynomial, hence a regular function on $X_\Bbbk$, and $B^\circ \in \Gamma_H$. By Theorem 2.22(3) and (B.160), there exists a prime divisor $\overline{E}$ over X with center $\overline{0}$ such that $a(\overline{E}, X, B^\circ) = \beta(g^\circ)$ and $a(\overline{E}, X, 0) \leq L_3$. Let E be a prime divisor over W_n with image $\overline{E}$. Then

$$a(E, W_n, 0) = e_E \cdot a(\overline{E}, X, \Delta_H) \leq qL_3.$$

Since $a(E, W_n, \sum_j \kappa_j \operatorname{div}(g_j^{<L})) \geq 0$,

$$\kappa_{\min} \operatorname{mult}_E g_j^{<L} \leq a(E, W_n, 0) \leq qL_3 < \kappa_{\min} L$$

for any j. Since $\operatorname{mult}_E(\widetilde{g}_j - g_j^{<L}) \geq L$, we have $\operatorname{mult}_E \widetilde{g}_j = \operatorname{mult}_E g_j^{<L}$, so

$$\beta(\widetilde{g}) \leq \frac{a(E, W_n, \sum_j \kappa_j \operatorname{div}(\widetilde{g}_j))}{e_E} = \frac{a(E, W_n, \sum_j \kappa_j \operatorname{div}(g_j^{<L}))}{e_E} = \beta(g^\circ).$$

This proves (1).

(2) Let L be the integer of (1). Since f_n does not depend on i, by Definition 2.5, $(f_n, (g_{*,j})_j)$ is a compatible generic limit of the tuples $(f_n, (g_{ij})_j)$ along $\mathcal{I}$ with coefficient field $\Bbbk$. By Theorem 2.18(3), applied to the tuples $(f_n, (g_{ij})_j)$ on $\mathbb{A}^4$ with $\mathfrak{c} := \mathcal{O}_{\mathbb{A}^4}$, there exist an integer $d \geq L$, a non-empty open subset Z_d° of the closed subset Z_d of d-jets in Definition 2.5, and an H-equivariant log resolution near 0 of the family of pairs defined by the d-jets in Z_d°, with constant dual complex, discrepancies, multiplicities, decomposition groups and inertia groups. By Definition 2.5, Z_d° contains the d-jets $(f_n, (g_{ij}^{<d})_j)$ for all but finitely many $i \in \mathcal{I}$, and $(f_n, (g_{*,j}^{<d})_j)$ is a $\Bbbk$-point of Z_d° over the generic point of Z_d. Since $g_{*,j}|_{W_n} \neq 0$ and $\bigcap_{e \geq 1} \big((f_n) + \mathfrak{m}^e\big) = (f_n)$ in $\Bbbk[[x, y, z, t]]$ by the Krull intersection theorem, there exists an integer d' with $g_{*,j} \notin (f_n) + \mathfrak{m}^{d'}$ for any j. The d'-jets $(f_n, (\phi_j)_j)$ with $\phi_j \notin (f_n) + \mathfrak{m}^{d'}$ for any j form an open subset of $Z_{d'}$ containing its generic point, so by Definition 2.5, $g_{ij}|_{W_n} \neq 0$ for any j and all but finitely many $i \in \mathcal{I}$.

For a point $(f_n, (\phi_j)_j)$ of Z_d°, let

$$D_\phi := \operatorname{div}(f_n) + \sum_j \kappa_j \operatorname{div}(\phi_j) \quad \text{on } \mathbb{A}^4.$$

The lc property near 0 of $(\mathbb{A}^4, D_\phi)$ and the quotient value $\operatorname{mld}((\mathbb{A}^4 \ni 0, D_\phi)/H)$ are, by Lemma 2.15, the lc property near $\overline{0}$ and the minimal log discrepancy at $\overline{0}$ of the crepant quotient $(\mathbb{A}^4, D_\phi)/H$, and after shrinking Z_d° they are read off from a log resolution of the family of the quotient pairs $(\mathbb{A}^4, D_\phi)/H$, together with the fiber over $\overline{0}$, whose restrictions to the fibers are log resolutions with constant discrepancies and multiplicities. Hence they do not depend on the point. If $\phi_j|_{W_n} = 0$ for some j, then D_ϕ has coefficient greater than 1 along W_n, so $(\mathbb{A}^4, D_\phi)$ is not lc near 0. Otherwise, by [Kaw07, Theorem], $(\mathbb{A}^4, D_\phi)$ is lc near 0 if and only if $(W_n, \sum_j \kappa_j \operatorname{div}(\phi_j))$ is lc near 0, and if $(W_n, \sum_j \kappa_j \operatorname{div}(\phi_j))$ is lc near 0, $\operatorname{mld}((\mathbb{A}^4 \ni 0, D_\phi)/H) = \beta(\phi)$ by Theorem 2.17(1). For the $\Bbbk$-point, [Kaw07, Theorem] and Theorem 2.17(1) apply by the reduction to $\mathbb{C}$ in the third paragraph of this proof. Hence there exists a finite subset $\mathcal{I}_0 \subset \mathcal{I}$ such that for any $i \in \mathcal{I}\backslash\mathcal{I}_0$, $g_{ij}|_{W_n} \neq 0$ for any j, and the pair $(\mathbb{A}^4, D_{(g_{ij}^{<d})_j})$ is lc near 0 if and only if $(\mathbb{A}^4, D_{(g_{*,j}^{<d})_j})$ is lc near 0. For $i \in \mathcal{I}\backslash\mathcal{I}_0$ with $(\mathbb{A}^4, D_{(g_{ij}^{<d})_j})$ lc near 0, the entries of both d-jets have non-zero restrictions to W_n and

$$\beta\big((g_{ij}^{<d})_j\big) = \beta\big((g_{*,j}^{<d})_j\big).$$

Assume that $(W_n, \sum_j \kappa_j \operatorname{div}(g_{*,j}))$ is lc near 0. Since $g_{*,j} - g_{*,j}^{<d} \in \mathfrak{m}^L$ for any j, (1) shows that $g_{*,j}^{<d}|_{W_n} \neq 0$ for any j, that the pair $(W_n, \sum_j \kappa_j \operatorname{div}(g_{*,j}^{<d}))$ is lc near 0, and that

$$\beta\big((g_{*,j}^{<d})_j\big) = \beta\big((g_{*,j})_j\big).$$

Hence $(\mathbb{A}^4, D_{(g_{*,j}^{<d})_j})$ is lc near 0, and for $i \in \mathcal{I}\backslash\mathcal{I}_0$, so is $(\mathbb{A}^4, D_{(g_{ij}^{<d})_j})$. Thus $g_{ij}^{<d}|_{W_n} \neq 0$ for any j, and the pair $(W_n, \sum_j \kappa_j \operatorname{div}(g_{ij}^{<d}))$ is lc near 0 with quotient value $\beta((g_{*,j})_j)$. By (1) with $\Bbbk := \mathbb{C}$, applied to $(g_{ij}^{<d})_j$ and $g'_j := g_{ij}$, and by Lemma 2.12(2), $(W_n, \sum_j \kappa_j \operatorname{div}(g_{ij}))$ is lc near 0 and $\beta((g_{ij})_j) = \beta((g_{*,j})_j)$.

Assume that $(W_n, \sum_j \kappa_j \operatorname{div}(g_{*,j}))$ is not lc near 0. If $(\mathbb{A}^4, D_{(g_{*,j}^{<d})_j})$ were lc near 0, then $g_{*,j}^{<d}|_{W_n} \neq 0$ for any j and $(W_n, \sum_j \kappa_j \operatorname{div}(g_{*,j}^{<d}))$ would be lc near 0, so by (1), applied to $(g_{*,j}^{<d})_j$ and $g'_j := g_{*,j}$, the pair $(W_n, \sum_j \kappa_j \operatorname{div}(g_{*,j}))$ would be lc near 0. Hence $(\mathbb{A}^4, D_{(g_{*,j}^{<d})_j})$ is not lc near 0, and for $i \in \mathcal{I}\backslash\mathcal{I}_0$, $(\mathbb{A}^4, D_{(g_{ij}^{<d})_j})$ is not lc near 0. If $(W_n, \sum_j \kappa_j \operatorname{div}(g_{ij}))$ were lc near 0, then by (1) with $\Bbbk := \mathbb{C}$, applied to $(g_{ij})_j$ and $g'_j := g_{ij}^{<d}$, the restrictions $g_{ij}^{<d}|_{W_n}$ would be non-zero and $(W_n, \sum_j \kappa_j \operatorname{div}(g_{ij}^{<d}))$ lc near 0, so $(\mathbb{A}^4, D_{(g_{ij}^{<d})_j})$ would be lc near 0, a contradiction. This proves (2).

(3) Let

$$b_* := \operatorname{mld}\Big(\Big(W_0 \ni 0, \sum_j \kappa_j \operatorname{div}(h_{*,j})\Big)/G\Big).$$

Since G preserves $xy + z^2 + t^{N_i}$ up to a character, $\chi_x\chi_y = \chi_z^2$, so G preserves $xy + z^2$ up to a character, and G acts faithfully on W_0, since no coordinate vanishes on W_0. By (2) with $n = +\infty$ and $H = G$, there exists a finite subset $\mathcal{I}_1 \subset \mathcal{I}$ such that for any $i \in \mathcal{I}\backslash\mathcal{I}_1$, the pair $(W_0, \sum_j \kappa_j \operatorname{div}(h_{ij}))$ is lc near 0 and

$$\operatorname{mld}\Big(\Big(W_0 \ni 0, \sum_j \kappa_j \operatorname{div}(h_{ij})\Big)/G\Big) = b_*.$$

Let X, Γ_H, and L_3 be as at the beginning of the proof, for $n = +\infty$ and $H = G$. For $i \in \mathcal{I}\backslash\mathcal{I}_1$, the crepant quotient of $(W_0, \sum_j \kappa_j \operatorname{div}(h_{ij}))$ has coefficients in Γ_H, so by Theorem 2.22(3) and

(B.160) there exists a prime divisor E_i over W_0 with center 0 such that

$$a\Big(E_i, W_0, \sum_j \kappa_j \operatorname{div}(h_{ij})\Big) = e_{E_i} b_* \quad \text{and} \quad a(E_i, W_0, 0) \le e_{E_i} L_3.$$

Since $a(E_i, W_0, \sum_j \kappa_j \operatorname{div}(h_{ij})) \ge 0$, we have $\kappa_j \operatorname{mult}_{E_i} h_{ij} \le e_{E_i} L_3$ for any j. By Proposition 3.14(2)(b) with $N := N_i$ and $D_N := D_i$, for $N_i \ge N_{\mathrm{ref}}(M, \Gamma)$,

$$\begin{aligned} a_i &\le \operatorname{mld}\big((W_0 \ni 0, D_{0,i})/G\big) \le \frac{a(E_i, W_0, D_{0,i})}{e_{E_i}} \\ &= b_* + \sum_j (\kappa_j - c_{ij}) \frac{\operatorname{mult}_{E_i} h_{ij}}{e_{E_i}} \le b_* + L_3 \sum_j \frac{\kappa_j - c_{ij}}{\kappa_j}. \end{aligned}$$

Since $\lim_i c_{ij} = \kappa_j$ for any j and $\lim_i N_i = +\infty$, we obtain $\limsup_{i \in \mathcal{I}} a_i \le b_*$. □

Lemma B.22. *Let the double cover $\widetilde{W}_0 \to W_0$, given by $(x, y, z, t) = (p^2, -v^2, pv, t)$, and its deck involution ι be as in Set-up 10.3. Then the following hold.*

(1) Let $n_i \ge 2$ be integers with $\lim_i n_i = +\infty$, let G be a finite diagonal abelian group preserving the equations of the W_{n_i} up to characters and acting faithfully on the W_{n_i}, and let $(\kappa_j)_j$ be a finite tuple of real numbers in $(0, 1]$. For any i let $(g_{ij})_j$ be marks on $(W_{n_i} \ni 0)$ which are polynomials without monomials divisible by xy, hence G-semi-invariant polynomials (since G is diagonal and a polynomial without monomials divisible by xy is determined by its restriction to W_{n_i}), whose characters do not depend on i, such that (W_{n_i}, B_i) is lc near 0, where $B_i := \sum_j \kappa_j \operatorname{div}(g_{ij})$, and let $b_i := \operatorname{mld}((W_{n_i} \ni 0, B_i)/G)$. Let $(g_{*,j})_j$, with coefficient field $\Bbbk$, be a compatible generic limit of the tuples $(g_{ij})_j$ along an infinite subset $\mathcal{I} \subset \mathbb{N}^+$ such that $(g_{*,j}(p^2, -v^2, pv, t))_j$ is a compatible generic limit of the tuples $(g_{ij}(p^2, -v^2, pv, t))_j$ along $\mathcal{I}$, and let

$$B_* := \sum_j \kappa_j \operatorname{div}\big(g_{*,j}(p^2, -v^2, pv, t)\big) \quad \text{on } \operatorname{Spec} \Bbbk[[p, v, t]].$$

Assume that $(W_0, \sum_j \kappa_j \operatorname{div}(g_{*,j}))$ is lc near 0 with

$$b_* := \operatorname{mld}\Big(\Big(W_0 \ni 0, \sum_j \kappa_j \operatorname{div}(g_{*,j})\Big)/G\Big) > 0,$$

and that one of the following holds.

(i) B_* is klt.

(ii) $\lfloor B_* \rfloor \ne 0$.

(iii) $\lfloor B_* \rfloor = 0$, B_* has exactly one lc center C of dimension 1 through 0, C is smooth, and ι acts non-trivially on C.

Then $\liminf_{i \in \mathcal{I}} b_i \ge b_*$.

(2) Let $\Bbbk \supset \mathbb{C}$ be an algebraically closed field, let G be a finite diagonal abelian group preserving the equation of W_0 up to a character and acting faithfully on W_0, let $(\kappa_j)_j$ be a finite tuple of positive real numbers, and let $(g_j)_j$ be G-semi-invariant elements of $\Bbbk[[x, y, z, t]]$ without monomials divisible by xy such that $(W_0, \sum_j \kappa_j \operatorname{div}(g_j))$ is lc near 0. Let d_j be the least total degree in x, y, z of a non-zero term of g_j, let P_j be the sum of the terms of g_j of total degree d_j in x, y, z, for an integer $k_0 \ge 0$ let G' be the group G acting on

x, y, z, t by the characters $\chi_x\chi_t^{-k_0}$, $\chi_y\chi_t^{-k_0}$, $\chi_z\chi_t^{-k_0}$, χ_t, and let

$$b_g := \operatorname{mld}\Big(\Big(W_0 \ni 0, \sum_j \kappa_j \operatorname{div}(g_j)\Big)/G\Big),$$

$$b_{P,G} := \operatorname{mld}\Big(\Big(W_0 \ni 0, \sum_j \kappa_j \operatorname{div}(P_j)\Big)/G\Big),$$

$$\text{and} \quad b_{P,G'} := \operatorname{mld}\Big(\Big(W_0 \ni 0, \sum_j \kappa_j \operatorname{div}(P_j)\Big)/G'\Big).$$

Assume that

$$\sum_j \kappa_j d_j = 1.$$

Then $(W_0, \sum_j \kappa_j \operatorname{div}(P_j))$ is lc near 0 and $b_{P,G'} \geq b_{P,G} \geq b_g$.

(3) *Let $\Bbbk \supset \mathbb{C}$ be an algebraically closed field, let $w = (w_x, w_y, w_z)$ be positive integers with $w_x + w_y = 2w_z$, let $n \geq 2$ be an integer or $n = +\infty$, where $W_\infty := W_0$, let G be a finite diagonal abelian group preserving the equation of W_n up to a character and acting faithfully on W_n and on W_0, let $(\kappa_j)_j$ be a finite tuple of positive real numbers, $(d_j)_j$ non-negative integers and $\mu \geq 0$, and let $(Q_j)_j$ be G-semi-invariant elements of $\Bbbk[[t]][x, y, z]$ of w-degree at most d_j in x, y, z, with non-zero restrictions to W_n, whose w-homogeneous parts P_j of degree d_j have non-zero restrictions to W_0. Let*

$$B_Q := \sum_j \kappa_j \operatorname{div}(Q_j) + \mu \operatorname{div}(t) \text{ on } W_n \quad \text{and} \quad B_P := \sum_j \kappa_j \operatorname{div}(P_j) + \mu \operatorname{div}(t) \text{ on } W_0.$$

If (W_n, B_Q) is lc near 0, then

$$\operatorname{mld}\big((W_n \ni 0, B_Q)/G\big) \geq \operatorname{mld}\big((W_0 \ni 0, B_P)/G\big).$$

Proof. (1) Since G preserves $xy + z^2 + t^{n_i}$ up to a character, $\chi_x\chi_y = \chi_z^2$, so G preserves $xy + z^2$ up to a character, and G acts faithfully on W_0, since no coordinate vanishes on W_0. A non-zero polynomial without monomials divisible by xy is not divisible by $xy + z^2$, so $g_{ij}|_{W_0} \neq 0$. Let $B_*^0 := \sum_j \kappa_j \operatorname{div}(g_{*,j})$ on W_0. For $u \in [0, 1)$ let

$$\phi_i(u) := \operatorname{mld}\big((W_{n_i} \ni 0, (1-u)B_i)/G\big), \qquad \phi_*(u) := \operatorname{mld}\big((W_0 \ni 0, (1-u)B_*^0)/G\big),$$

so that $\phi_i(0) = b_i$ and $\phi_*(0) = b_*$. Since log discrepancies are affine in the boundary, for a klt variety W, an $\mathbb{R}$-divisor $B \geq 0$ on W such that (W, B) is lc near 0, and $c \in (0, 1]$, the pair $(W, (1-c)B)$ is klt near 0.

Step 1. In this step we show that for any $u \in (0, 1)$, $\phi_i(u) = \phi_*(u) \geq b_*$ for all but finitely many $i \in \mathcal{I}$. Since W_0 is klt and (W_0, B_*^0) is lc near 0, the pair $(W_0, (1 - \frac{u}{2})B_*^0)$ is klt near 0. By Lemma B.21(2) with $n = +\infty$ and $H = G$, applied with the labels $(1 - \frac{u}{2})\kappa_j$ and with the labels $(1-u)\kappa_j$, for all but finitely many $i \in \mathcal{I}$, the pair $(W_0, (1 - \frac{u}{2}) \sum_j \kappa_j \operatorname{div}(g_{ij}))$ is lc near 0 and

$$\operatorname{mld}\Big(\Big(W_0 \ni 0, (1-u)\sum_j \kappa_j \operatorname{div}(g_{ij})\Big)/G\Big) = \phi_*(u).$$

Also $(W_{n_i}, (1-u)B_i)$ is klt near 0. Let

$$M := |G|, \quad \Gamma := \{(1-u)\kappa_j\}_j \subset [0, 1], \quad N := n_i, \quad D_N := (1-u)B_i, \quad \text{and} \quad \theta := \frac{1 - u/2}{1-u} > 1.$$

For such i with $n_i \geq N'_\theta(M, \Gamma, \theta)$, by Proposition 3.14(3),

$$\phi_i(u) = \operatorname{mld}\Big(\Big(W_0 \ni 0, (1-u)\sum_j \kappa_j \operatorname{div}(g_{ij})\Big)/G\Big) = \phi_*(u).$$

Since $\lim_i n_i = +\infty$ and $\phi_*(u) \geq \phi_*(0) = b_*$ by $B_*^0 \geq 0$, **Step 1** follows.

Step 2. In this step we show that for any i, the function ϕ_i is increasing and continuous on $[0,1)$. Since $B_i \geq 0$, ϕ_i is increasing. Let $X_i := W_{n_i}/G$ with the image $\overline{0}$ of 0 and the branch divisor Δ_i of $W_{n_i} \to X_i$, and let $\overline{B}_i := \frac{1}{M}\sum_j \kappa_j \operatorname{div}(g_{ij}^M)$ on X_i. Then $(W_{n_i}, (1-u)B_i)/G = (X_i, \Delta_i + (1-u)\overline{B}_i)$, so by Lemma 2.15(0), (2) and (3),

$$\phi_i(u) = \operatorname{mld}\big(X_i \ni \overline{0}, \Delta_i + (1-u)\overline{B}_i\big).$$

Computing on a log resolution of $(X_i, \Delta_i + \overline{B}_i)$ near $\overline{0}$, ϕ_i is the minimum of finitely many affine functions of u on $[0,1)$, hence continuous.

Step 3. In this step we derive a contradiction from $\liminf_{i\in\mathcal{I}} b_i < b_*$. Since $\widetilde{W}_0 \to W_0$ is étale outside the curve $(p = v = 0)$, which has codimension 2, $K_{\widetilde{W}_0} + \widetilde{D}$ is the pullback of $K_{W_0} + D$ for any $\mathbb{R}$-divisor D on W_0 with pullback $\widetilde{D}$. In particular, $K_{\widetilde{W}_0} + B_*$ is the pullback of $K_{W_0} + B_*^0$, and $W_0/G = \widetilde{W}_0/\widetilde{G}$. Hence B_* is lc, and

$$\operatorname{mld}(\widetilde{W}_0 \ni 0, B_*) \geq b_* > 0,$$

since the log discrepancy with respect to B_* of a prime divisor F over $\widetilde{W}_0$ with center 0 is a positive integer multiple of the log discrepancy of the image of F over W_0/G with respect to the crepant quotient of (W_0, B_*^0) (Lemma 2.15(2) and (5)).

Suppose that $\liminf_{i\in\mathcal{I}} b_i < b_*$. Then there exist a real number $\epsilon \in (0, b_*)$ and an infinite subset $\mathcal{I}_\epsilon \subset \mathcal{I}$ such that $b_i \leq b_* - \epsilon$ for any $i \in \mathcal{I}_\epsilon$. By Lemma 2.6(2), $(g_{*,j})_j$ and $(g_{*,j}(p^2, -v^2, pv, t))_j$ are compatible generic limits of the tuples $(g_{ij})_j$ and $(g_{ij}(p^2, -v^2, pv, t))_j$ along any infinite subset of $\mathcal{I}_\epsilon$. In case (i), by Theorem 2.18(1), applied on $\mathbb{A}^3$ with the coordinates p, v, t to the tuples $(g_{ij}(p^2, -v^2, pv, t))_j$ along $\mathcal{I}_\epsilon$ with $H := \widetilde{G}$, $c_j := \kappa_j$, $\mathfrak{c} := \mathcal{O}_{\mathbb{A}^3}$, and $\theta := 1$, there exist an infinite subset $\mathcal{I}' \subset \mathcal{I}_\epsilon$ and a real number $\theta' > 1$ such that $(\mathbb{A}^3, \theta'\widetilde{B}_i)$ is lc near 0 for any $i \in \mathcal{I}'$, where

$$\widetilde{B}_i := \sum_j \kappa_j \operatorname{div}\big(g_{ij}(p^2, -v^2, pv, t)\big).$$

Since $K_{\widetilde{W}_0} + \theta'\widetilde{B}_i$ is the pullback of $K_{W_0} + \theta'\sum_j \kappa_j \operatorname{div}(g_{ij})$, the pair $(W_0, \theta'\sum_j \kappa_j \operatorname{div}(g_{ij}))$ is lc near 0 for any $i \in \mathcal{I}'$. In cases (ii) and (iii) let $\mathcal{I}' := \mathcal{I}_\epsilon$.

Let $(\alpha_m)_{m\geq 1}$ be a strictly increasing sequence in $(b_* - \epsilon, b_*)$, and let $\sigma_0 := 1$ and $i_0 := 0$. For $m \geq 1$, assume that σ_{m-1} and i_{m-1} are defined, and let $u_m := \min\{\sigma_{m-1}/2, 1/m\}$. By **Step 1**, there exists $i_m \in \mathcal{I}'$ with $i_m > i_{m-1}$ and $\phi_{i_m}(u_m) \geq b_* > \alpha_m$. Since $\phi_{i_m}(0) = b_{i_m} \leq b_* - \epsilon < \alpha_m$, by **Step 2** there exists $\sigma_m \in (0, u_m)$ such that

$$\phi_{i_m}(\sigma_m) = \operatorname{mld}\big((W_{n_{i_m}} \ni 0, (1-\sigma_m)B_{i_m})/G\big) = \alpha_m.$$

Then $\sigma_m < \sigma_{m-1}/2$ and $\sigma_m < 1/m$, so $(\sigma_m)_m$ is strictly decreasing with $\lim_m \sigma_m = 0$. For any m, $(W_{n_{i_m}}, (1-\sigma_m)B_{i_m})$ is klt near 0. The labels $(1-\sigma_m)\kappa_j$ are increasing in m with $\lim_m (1-\sigma_m)\kappa_j = \kappa_j$, and

$$\Gamma' := \{(1-\sigma_m)\kappa_j \mid m \geq 1 \text{ and } j\} \subset (0,1]$$

is a DCC set, as a finite union of increasing sequences.

In case (i), for any m the pair

$$\Big(W_0, \theta'(1-\sigma_m)\sum_j \kappa_j \operatorname{div}(g_{i_m j})\Big)$$

is lc near 0, since $i_m \in \mathcal{I}'$ and $1 - \sigma_m \leq 1$. By Proposition 3.14(3) with M, Γ', and θ', applied to $N := n_{i_m}$ and $D_N := (1-\sigma_m)B_{i_m}$ for $m \geq 1$, the set $\{\alpha_m\}_m$ satisfies the ACC, which contradicts that $(\alpha_m)_m$ is strictly increasing.

In cases (ii) and (iii), the data $N_m := n_{i_m}$, G, and $D_m := \sum_j (1-\sigma_m)\kappa_j \operatorname{div}(g_{i_m j})$ for $m \geq 1$, with M, the DCC set Γ', the marks $g_{i_m j}$ and the labels $(1-\sigma_m)\kappa_j$, satisfy Set-up 10.3, with the compatible generic limits $(g_{*,j})_j$ and $(g_{*,j}(p^2, -v^2, pv, t))_j$ along $\{i_m\}_{m\geq 1}$, whose limit boundary B^* is B_*. Their quotient values are the α_m. Since B_* is lc with $\operatorname{mld}(\widetilde{W}_0 \ni 0, B_*) > 0$, in case

(iii) the data also satisfy Set-up 10.6, and Theorem 10.4 in case (ii) and Theorem 10.7 in case (iii) imply that $(\alpha_m)_m$ is not strictly increasing, a contradiction. This proves (1).

(2) Let $q := |G|$. We first show that for any integer $k_0 \geq 0$, the pair $(W_0, \sum_j \kappa_j \operatorname{div}(P_j))$ is lc near 0 and $b_{P,G'} \geq b_g$. Since the monomials of g_j have one G-character and those of P_j have one total degree d_j in x, y, z, P_j is G'-semi-invariant, and $P_j|_{W_0} \neq 0$, since $P_j \neq 0$ has no monomials divisible by xy.

Let F be a prime divisor over $W_0 \times_{\mathbb{C}} \operatorname{Spec} \Bbbk$ with center 0, viewed as a divisorial valuation of $\Bbbk(W_0)$ and extended to $\Bbbk[[x, y, z, t]]$ by $\mathfrak{m}$-adic continuity (the formal prime divisors of Definition 2.2). Let k be an integer with $k \equiv k_0 \pmod q$ and $k \operatorname{mult}_F t > \max_j \operatorname{mult}_F P_j$. The morphism

$$\psi_k : \mathbb{A}^4 \to \mathbb{A}^4, \quad (x', y', z', t) \to (x, y, z, t) = (t^k x', t^k y', t^k z', t),$$

maps 0 to 0 and W_0 birationally onto W_0, since $xy + z^2 = t^{2k}(x'y' + z'^2)$. Since $\chi_t^{k-k_0} = 1$, ψ_k is equivariant, where G' acts on the copy with coordinates x', y', z', t and G on the copy with coordinates x, y, z, t. Let F_k be the prime divisor over W_0 with $\operatorname{mult}_{F_k} h = \operatorname{mult}_F(\psi_k^* h)$ for any $h \in \Bbbk(W_0)$. Then F_k has center 0, and the inertia group of F for G' equals the inertia group of F_k for G, so $e_F = e_{F_k}$. For any j, write $g_j = \sum_{m \geq d_j} g_{j,m}$, where $g_{j,m}$ is the sum of the terms of g_j of total degree m in x, y, z, so that $g_{j,d_j} = P_j$ and

$$g_j \circ \psi_k = t^{kd_j} \Big(P_j(x', y', z', t) + \sum_{m > d_j} t^{k(m-d_j)} g_{j,m}(x', y', z', t) \Big).$$

The multiplicity along F of the sum over $m > d_j$ is at least $k \operatorname{mult}_F t > \operatorname{mult}_F P_j$, so

$$\operatorname{mult}_{F_k} g_j = kd_j \operatorname{mult}_F t + \operatorname{mult}_F P_j.$$

Moreover $x^{-1} dx \wedge dz \wedge dt$ generates ω_{W_0} near 0 and

$$\psi_k^*(x^{-1} dx \wedge dz \wedge dt) = t^k x'^{-1} dx' \wedge dz' \wedge dt,$$

so $a(F_k, W_0, 0) = a(F, W_0, 0) + k \operatorname{mult}_F t$. Since $\sum_j \kappa_j d_j = 1$,

$$a\Big(F_k, W_0, \sum_j \kappa_j \operatorname{div}(g_j)\Big) = a\Big(F, W_0, \sum_j \kappa_j \operatorname{div}(P_j)\Big).$$

The left-hand side is non-negative, since $(W_0, \sum_j \kappa_j \operatorname{div}(g_j))$ is lc near 0. Since F is arbitrary, $(W_0, \sum_j \kappa_j \operatorname{div}(P_j))$ is lc near 0, and dividing by $e_F = e_{F_k}$ and taking the infimum over F, we obtain $b_{P,G'} \geq b_g$.

For $k_0 = 0$ we have $G' = G$, so $b_{P,G} \geq b_g$. Each P_j is G-semi-invariant without monomials divisible by xy and homogeneous of total degree d_j in x, y, z, and $(W_0, \sum_j \kappa_j \operatorname{div}(P_j))$ is lc near 0. Hence the inequality $b_{P,G'} \geq b_g$ for $(g_j)_j := (P_j)_j$, for which $b_g = b_{P,G}$, reads $b_{P,G'} \geq b_{P,G}$. This proves (2).

(3) If (W_0, B_P) is not lc near 0, then the right-hand side is $-\infty$. Thus we may assume that it is lc near 0. Let L be an integer at least the integers of Lemma B.21(1) for W_n, G and for W_0, G, with the labels $(\kappa_j)_j$ and, if $\mu > 0$, the additional label μ of t. Replacing each coefficient $\gamma \in \Bbbk[[t]]$ of each Q_j by the part of γ of t-degree less than L changes Q_j and P_j by elements of $(x, y, z, t)^L$, keeps the G-semi-invariance and the bounds on w-degrees, and replaces P_j by the w-homogeneous part of degree d_j of the new Q_j. By Lemma B.21(1) for (W_n, B_Q) and (W_0, B_P), we may assume that the Q_j are polynomials. Quotient values of pairs defined by polynomials over $\Bbbk$ do not change if $\Bbbk$ is replaced by the algebraic closure $\Bbbk_0$ in $\Bbbk$ of the field generated over $\mathbb{C}$ by the coefficients, and $\Bbbk_0$ is isomorphic to $\mathbb{C}$ by an isomorphism mapping W_n, W_0, and G to themselves. Thus we may assume that $\Bbbk = \mathbb{C}$.

For $\tau \in \mathbb{C}$ let $f_{n,\tau} := xy + z^2 + \tau^{2w_z} t^n$ if $n \neq +\infty$ and $f_{\infty,\tau} := xy + z^2$, and let

$$Q_{j,\tau} := \tau^{d_j} Q_j(\tau^{-w_x} x, \tau^{-w_y} y, \tau^{-w_z} z, t).$$

Since Q_j has w-degree at most d_j in x, y, z, $Q_{j,\tau}$ is a polynomial in x, y, z, t, τ, and $Q_{j,0} = P_j$ and $f_{n,0} = xy + z^2$. For $\tau \neq 0$ the diagonal automorphism

$$\lambda_\tau : \mathbb{A}^4 \to \mathbb{A}^4, \quad (x, y, z, t) \to (\tau^{-w_x} x, \tau^{-w_y} y, \tau^{-w_z} z, t),$$

commutes with G, fixes 0, and satisfies

$$f_{n,1} \circ \lambda_\tau = \tau^{-2w_z} f_{n,\tau} \quad \text{and} \quad Q_{j,1} \circ \lambda_\tau = \tau^{-d_j} Q_{j,\tau},$$

since $w_x + w_y = 2w_z$.

Let $q := |G|$, let $\rho : \mathbb{A}^4 \to X := \mathbb{A}^4/G$ be the quotient morphism with $\overline{0} := \rho(0)$, let $(x_1, x_2, x_3, x_4) := (x, y, z, t)$, and let r_k be the order of the subgroup of G acting only on x_k, so that $K_{\mathbb{A}^4} = \rho^* K_X + \sum_k (r_k - 1) \operatorname{div}(x_k)$. For $\tau \in \mathbb{C}$ let

$$\mathfrak{a}_\tau := \prod_j (Q_{j,\tau})^{\kappa_j} \cdot (t)^\mu \quad \text{on } \mathbb{A}^4$$

and

$$\mathfrak{A}_\tau := \prod_k (x_k^q)^{(r_k-1)/q} \cdot (f_{n,\tau}^q)^{1/q} \cdot \prod_j (Q_{j,\tau}^q)^{\kappa_j/q} \cdot (t^q)^{\mu/q}$$

on X, the restriction to $X \times \{\tau\}$ of an $\mathbb{R}$-ideal sheaf $\mathfrak{A}$ on $X \times \mathbb{A}^1$. By Lemma 2.15(2) and (3),

$$\operatorname{mld}\big((\mathbb{A}^4 \ni 0, \operatorname{div}(f_{n,\tau}), \mathfrak{a}_\tau)/G\big) = \operatorname{mld}(X \ni \overline{0}, \mathfrak{A}_\tau). \tag{B.161}$$

Let $G_{\mathrm{r}} \subset G$ be the subgroup generated by the elements acting only on one coordinate. Then $\mathbb{A}^4/G_{\mathrm{r}} \cong \mathbb{A}^4$ with the coordinates $x_k^{r_k}$, and G/G_{r} acts diagonally on it and freely in codimension 1: the subgroup of G acting only on x_k consists of the diagonal matrices with an r_k-th root of unity at position k and 1 elsewhere, so if $g \in G$ acts trivially on $x_k^{r_k}$ for any $k \neq k'$, then g is the product of elements of G acting only on one coordinate, and $g \in G_{\mathrm{r}}$. Thus $X = (\mathbb{A}^4/G_{\mathrm{r}})/(G/G_{\mathrm{r}})$, and by Theorem 2.16 with $\Delta = 0$ and $T = \mathbb{A}^1$, the function $\tau \to \operatorname{mld}(X \ni \overline{0}, \mathfrak{A}_\tau)$ is lower semi-continuous. For $\tau \neq 0$, λ_τ induces an automorphism of X fixing $\overline{0}$ which maps $\mathfrak{A}_\tau$ to $\mathfrak{A}_1$, so

$$\operatorname{mld}(X \ni \overline{0}, \mathfrak{A}_\tau) = \operatorname{mld}(X \ni \overline{0}, \mathfrak{A}_1).$$

Hence

$$\operatorname{mld}(X \ni \overline{0}, \mathfrak{A}_0) \leq \operatorname{mld}(X \ni \overline{0}, \mathfrak{A}_1).$$

The polynomials $f_{n,1}$ and $f_{n,0}$ define W_n and W_0, normal klt threefolds on which G acts faithfully, and $\mathfrak{a}_1|_{W_n}$ and $\mathfrak{a}_0|_{W_0}$ are non-zero, with lc pairs $(W_n, \mathfrak{a}_1|_{W_n})$ and $(W_0, \mathfrak{a}_0|_{W_0})$ near 0. By Theorem 2.17(1) with $H := G$ and (B.161), since B_Q is the divisor of $\mathfrak{a}_1|_{W_n}$ and B_P that of $\mathfrak{a}_0|_{W_0}$,

$$\operatorname{mld}\big((W_n \ni 0, B_Q)/G\big) = \operatorname{mld}(X \ni \overline{0}, \mathfrak{A}_1) \geq \operatorname{mld}(X \ni \overline{0}, \mathfrak{A}_0) = \operatorname{mld}\big((W_0 \ni 0, B_P)/G\big).$$

This proves (3). □

Proof of Theorem 10.8. Let $W_\infty := W_0$. Assume, to the contrary, that data as in Set-up 10.3 satisfy the hypotheses of Theorem 10.8 and that $(a_i)_{i \geq 1}$ is strictly increasing. By Lemma 2.6(2), $(h_{*,j})_j$ and $(H_j^*)_j$ remain compatible generic limits, with coefficient field $\Bbbk$, along any subsequence, and B^*, the d_j and $C = (p = v = 0)$ do not change, so the hypotheses of Theorem 10.8 hold after any passage to a subsequence below. Then $e_F \leq |G| \leq M$ for any prime divisor F over W_{N_i}.

Step 1. In this step we fix computing lifts F_i of a_i and bound $\lim_i a_i$ from above.

Since ι acts trivially on C, Proposition B.14(1) with $u_i = t$ yields a real number $\delta > 0$ such that, after passing to a subsequence,

$$\Big(W_{N_i}, \sum_j \kappa_j \operatorname{div}(h_{ij}) + \delta \operatorname{div}(t)\Big) \text{ is lc near 0 for any } i. \tag{B.162}$$

For any i let F_i be a computing lift of a_i, so that $a_i = a(F_i, W_{N_i}, D_i)/e_{F_i}$. By Proposition 10.5(3), after passing to a further subsequence, the numbers R_i defined in it are strictly increasing with a positive limit. Let

$$\beta_i := \frac{a(F_i, W_{N_i}, \sum_j \kappa_j \operatorname{div}(h_{ij}))}{e_{F_i}}.$$

Since $c_{ij} \le \kappa_j$ and $\operatorname{mult}_{F_i} h_{ij} \ge 0$ for any i and j,

$$a_i - \beta_i = \sum_j (\kappa_j - c_{ij}) \frac{\operatorname{mult}_{F_i} h_{ij}}{e_{F_i}} \ge R_i. \tag{B.163}$$

By (B.162), $e_{F_i}\beta_i \ge \delta \operatorname{mult}_{F_i} t > 0$. By Set-up 10.3, $a_i \le 2$. Thus, for any i,

$$0 < \beta_i \le a_i \le 2. \tag{B.164}$$

By Proposition B.20(2) and (B.162), after passing to a subsequence,

$$m_i := \min(\operatorname{mult}_{F_i} x, \operatorname{mult}_{F_i} y, \operatorname{mult}_{F_i} z) \to +\infty \quad \text{and} \quad \operatorname{mult}_{F_i} t \le \lceil 2M/\delta \rceil. \tag{B.165}$$

Since $xy + z^2 + t^{N_i} = 0$ on W_{N_i}, the minimum of $\operatorname{mult}_{F_i}(xy)$, $\operatorname{mult}_{F_i}(z^2)$, and $N_i \operatorname{mult}_{F_i} t$ is attained at least twice. Since $\operatorname{mult}_{F_i}(xy) \ge 2m_i$ and $\operatorname{mult}_{F_i}(z^2) \ge 2m_i$, we obtain

$$2m_i \le N_i \operatorname{mult}_{F_i} t. \tag{B.166}$$

Since $\pi_0 : \widetilde{W}_* \to W_*$ is étale outside $(p = v = 0)$ and B^* is lc, (W_*, D_*) is lc near 0. Let

$$b_* := \operatorname{mld}\big((W_* \ni 0, D_*)/G\big).$$

Since the a_i are increasing and $a_i \le 2$, the limit $a_* := \lim_i a_i$ exists, and by Lemma B.21(3),

$$a_i \le a_* \le b_* \quad \text{for any } i. \tag{B.167}$$

Step 2. In this step we define the number k_i of rescalings and show that $k_i \to +\infty$.

For any i and j, we write the monomials of h_{ij} as $x^\alpha t^e := x^{\alpha_1} y^{\alpha_2} z^{\alpha_3} t^e$, with $|\alpha| := \alpha_1 + \alpha_2 + \alpha_3$. Let q_i be the least positive integer q such that, for some j, some monomial $x^\alpha t^e$ with non-zero coefficient in h_{ij} satisfies

$$e + q(|\alpha| - d_j) < 0, \tag{B.168}$$

and let $q_i := +\infty$ if no such q exists. If (B.168) holds for q, then $|\alpha| < d_j$ and (B.168) holds for any integer $q' \ge q$. Let

$$k_i := \min\left\{ q_i - 1, \left\lfloor \frac{m_i}{\operatorname{mult}_{F_i} t} \right\rfloor - 1 \right\}. \tag{B.169}$$

We show that $q_i \to +\infty$. Fix a positive integer q, and let $d := (q+1)\max_j d_j + 1$. If $x^\alpha t^e$ satisfies (B.168) for q and j, then $|\alpha| < d_j$ and $e < qd_j$, so $|\alpha| + e < d$. By the definition of d_j, for any j the coefficients in $h_{*,j}$ of the monomials $x^\alpha t^e$ with $|\alpha| < d_j$ vanish. By Lemma 2.8, for all but finitely many i the coefficients in h_{ij} of the monomials $x^\alpha t^e$ with $|\alpha| < d_j$ and $|\alpha| + e < d$ vanish for any j, that is, $q_i > q$. Thus $q_i \to +\infty$. By (B.165), $m_i/\operatorname{mult}_{F_i} t \to +\infty$. Hence $k_i \to +\infty$, and after passing to a subsequence there exists an integer k_0 with $0 \le k_0 < |G|$ such that

$$k_i \ge 0 \quad \text{and} \quad k_i \equiv k_0 \pmod{|G|} \quad \text{for any } i.$$

Step 3. In this step we rescale x, y, z by t^{k_i} and show that the boundary $\sum_j \kappa_j \operatorname{div}(h_{ij})$ is crepant under the rescaling.

Let $n_i := N_i - 2k_i$. Since $k_i \le m_i/\operatorname{mult}_{F_i} t - 1$, (B.166) implies $2k_i \le N_i - 2$, that is, $n_i \ge 2$. On a second copy of $\mathbb{A}^4$ with coordinates X, Y, Z, t let $W_{n_i} := (XY + Z^2 + t^{n_i} = 0)$, and let

$$\psi_i : \mathbb{A}^4 \to \mathbb{A}^4, \quad (X, Y, Z, t) \to (t^{k_i}X, t^{k_i}Y, t^{k_i}Z, t).$$

Since $xy + z^2 + t^{N_i}$ pulls back to $t^{2k_i}(XY + Z^2 + t^{n_i})$, ψ_i restricts to a birational morphism $W_{n_i} \to W_{N_i}$ with $\psi_i(0) = 0$. Let

$$g_{ij} := t^{-k_i d_j} h_{ij}(t^{k_i} X, t^{k_i} Y, t^{k_i} Z, t). \tag{B.170}$$

A monomial $x^\alpha t^e$ of h_{ij} becomes $X^\alpha t^{e+k_i(|\alpha|-d_j)}$ with the coefficient of $x^\alpha t^e$ in h_{ij}, and distinct monomials of h_{ij} become distinct monomials. Since $k_i < q_i$, $e + k_i(|\alpha| - d_j) \geq 0$ for any monomial $x^\alpha t^e$ with non-zero coefficient in h_{ij}. Thus g_{ij} is a polynomial without monomials divisible by XY.

Let G' be the group G acting on X, Y, Z, t by the characters $\chi_x \chi_t^{-k_0}$, $\chi_y \chi_t^{-k_0}$, $\chi_z \chi_t^{-k_0}$, χ_t. Since $k_i \equiv k_0 \pmod{|G|}$, ψ_i is equivariant for the action of G' on the copy with coordinates X, Y, Z, t and of G on the copy with coordinates x, y, z, t. Hence G' preserves $XY + Z^2 + t^{n_i}$ up to a character, each g_{ij} is G'-semi-invariant with character $\chi_{h_{ij}} \chi_t^{-k_0 d_j}$, which does not depend on i, and G' acts on the common function field of W_{n_i} and W_{N_i} as G does. In particular G' acts faithfully on W_{n_i}, and also on W_0, as W_0 is not contained in any coordinate hyperplane. Moreover F_i is a prime divisor over W_{n_i} whose inertia group in G' is $I_G(F_i)$, of order e_{F_i}. By (B.169),

$$\operatorname{mult}_{F_i} X = \operatorname{mult}_{F_i} x - k_i \operatorname{mult}_{F_i} t \geq m_i - k_i \operatorname{mult}_{F_i} t \geq \operatorname{mult}_{F_i} t > 0,$$

and similarly for Y and Z. Thus the center of F_i on W_{n_i} is 0.

The generator $dx \wedge dz \wedge dt/x$ of $\omega_{W_{N_i}}$ pulls back under ψ_i to $t^{k_i} dX \wedge dZ \wedge dt/X$, and $dX \wedge dZ \wedge dt/X$ generates $\omega_{W_{n_i}}$. Hence, by (B.170) for the second equality, any prime divisor F over W_{n_i} satisfies

$$a(F, W_{N_i}, 0) = a(F, W_{n_i}, 0) + k_i \operatorname{mult}_F t \quad \text{and} \quad \operatorname{mult}_F h_{ij} = k_i d_j \operatorname{mult}_F t + \operatorname{mult}_F g_{ij}.$$

Since $\sum_j \kappa_j d_j = 1$, for any real number μ we obtain

$$a\Big(F, W_{N_i}, \sum_j \kappa_j \operatorname{div}(h_{ij}) + \mu \operatorname{div}(t)\Big) = a\Big(F, W_{n_i}, \sum_j \kappa_j \operatorname{div}(g_{ij}) + \mu \operatorname{div}(t)\Big). \tag{B.171}$$

Since ψ_i is continuous and $\psi_i(0) = 0$, (B.162) and (B.171) with $\mu = \delta$ imply

$$\Big(W_{n_i}, \sum_j \kappa_j \operatorname{div}(g_{ij}) + \delta \operatorname{div}(t)\Big) \text{ is lc near 0 for any } i. \tag{B.172}$$

Let

$$b_i' := \operatorname{mld}\Big(\Big(W_{n_i} \ni 0, \sum_j \kappa_j \operatorname{div}(g_{ij})\Big)/G'\Big).$$

By (B.171) with $\mu = 0$ and $F = F_i$,

$$b_i' \leq \frac{a(F_i, W_{n_i}, \sum_j \kappa_j \operatorname{div}(g_{ij}))}{e_{F_i}} = \beta_i. \tag{B.173}$$

Step 4. In this step we take a joint compatible generic limit of the h_{ij} and the g_{ij}, and bound the quotient value of the new limit from below by b_*.

After passing to a subsequence, either $n_i = n$ for a fixed integer $n \geq 2$ and any i, or $n_i \to +\infty$, in which case we let $n := +\infty$. After passing to a further subsequence, one of the following holds.

(a) $k_i = \lfloor m_i / \operatorname{mult}_{F_i} t \rfloor - 1$ for any i.
(b) $k_i = q_i - 1 < \lfloor m_i / \operatorname{mult}_{F_i} t \rfloor - 1$ for any i.

In case (a), $m_i < (k_i + 2) \operatorname{mult}_{F_i} t$, so by (B.165)

$$\min(\operatorname{mult}_{F_i} X, \operatorname{mult}_{F_i} Y, \operatorname{mult}_{F_i} Z) = m_i - k_i \operatorname{mult}_{F_i} t < 2 \operatorname{mult}_{F_i} t \leq 2\lceil 2M/\delta \rceil. \tag{B.174}$$

In case (b), $q_i = k_i + 1 < +\infty$, so for some j some monomial $x^\alpha t^e$ with non-zero coefficient in h_{ij} satisfies (B.168) with $q = k_i + 1$. Then $|\alpha| < d_j$, and the monomial $X^\alpha t^r$ of g_{ij} with $r := e + k_i(|\alpha| - d_j)$ satisfies $0 \leq r < d_j - |\alpha|$. The triples (j, α, r) with $|\alpha| < d_j$ and $0 \leq r < d_j$

are finitely many. Thus, after passing to a subsequence, there exists one triple (j_0, α_0, r_0) with $|\alpha_0| < d_{j_0}$ such that the coefficient of $X^{\alpha_0}t^{r_0}$ in g_{ij_0} is non-zero for any i.

Let P_j be the sum of the terms of $h_{*,j}$ of total degree d_j in x, y, z. By the definition of d_j, $P_j \neq 0$. Fix a monomial $x^\alpha t^e$ with $|\alpha| = d_j$ and non-zero coefficient in $h_{*,j}$. By Lemma 2.8, for all but finitely many i the coefficient of $x^\alpha t^e$ in h_{ij} is non-zero, so $X^\alpha t^e$ has non-zero coefficient in g_{ij} and $\operatorname{mult}_0 g_{ij} \leq d_j + e$. Thus the multiplicities $\operatorname{mult}_0 g_{ij}$ are bounded. Since the h_{ij} are G-semi-invariant and the g_{ij} are G'-semi-invariant, we regard $(h_{ij})_j$ and $(g_{ij})_j$ as tuples for the trivial group in Definition 2.5. The semi-invariance of the limits is recovered below by Lemma 2.8. By Lemma 2.6(3), after passing to a subsequence there exist a compatible generic limit $((h'_{*,j})_j, (Q_j)_j)$ of the tuples $((h_{ij})_j, (g_{ij})_j)$, with coefficient field $\Bbbk'$, and an embedding $\Bbbk \to \Bbbk'$ over $\mathbb{C}$ under which $h_{*,j}$ maps to $h'_{*,j}$ for any j. We regard $\Bbbk$ as a subfield of $\Bbbk'$ via the embedding, so that $h'_{*,j} = h_{*,j}$. Since lc and quotient values do not change under the extension $\Bbbk \subset \Bbbk'$ [Kaw21, Remark 2.6], $(W_0, \sum_j \kappa_j \operatorname{div}(h_{*,j}))$ over $\Bbbk'$ is lc near 0 with quotient value b_*.

Semi-invariance under G' with the character of g_{ij}, and the absence of monomials divisible by XY, are closed conditions on d-jets satisfied by g_{ij} for any i. Thus, by Lemma 2.8 for the joint limit, Q_j is G'-semi-invariant and has no monomials divisible by XY. For a positive integer d and any i with $k_i \geq d$, any monomial $X^\alpha t^{e'}$ of g_{ij} with $|\alpha| > d_j$ satisfies $e' \geq k_i(|\alpha| - d_j) \geq d$, so the d-jet of g_{ij} has no such monomial. Thus, by Lemma 2.8, Q_j has no monomial $X^\alpha t^{e'}$ with $|\alpha| > d_j$. For $|\alpha| = d_j$, the coefficient of $X^\alpha t^e$ in g_{ij} equals the coefficient of $x^\alpha t^e$ in h_{ij} for any i, so, by Lemma 2.8, the coefficient of $X^\alpha t^e$ in Q_j equals the coefficient of $x^\alpha t^e$ in $h_{*,j}$. Hence

$$Q_j \in \Bbbk'[[t]][X, Y, Z], \quad \deg_{X,Y,Z} Q_j \leq d_j, \quad \text{and} \quad \deg_{X,Y,Z} \big(Q_j - P_j(X, Y, Z, t)\big) < d_j. \tag{B.175}$$

Since $P_j \neq 0$ and $Q_j \neq 0$ have no monomials divisible by XY, P_j has non-zero restriction to W_0 and Q_j has non-zero restriction to W_n. In case (b), the coefficient of $X^{\alpha_0}t^{r_0}$ in g_{ij_0} is non-zero for any i, so by Lemma 2.8

$$\text{the coefficient of } X^{\alpha_0}t^{r_0} \text{ in } Q_{j_0} \text{ is non-zero, and } |\alpha_0| < d_{j_0}. \tag{B.176}$$

For a positive integer d, let $Z_d \subset \mathcal{J}_d$ be as in Definition 2.5 for the limit $(Q_j)_j$ of the tuples $(g_{ij})_j$, and let $\mathcal{J}'_d$ be the affine space of d-jets of the tuples $(g_{ij}(p^2, -v^2, pv, t))_j$. Since the substitution $(X, Y, Z) = (p^2, -v^2, pv)$ does not lower degrees, it induces a linear map $\mathcal{J}_d \to \mathcal{J}'_d$ sending the d-jet of $(g_{ij})_j$ to that of $(g_{ij}(p^2, -v^2, pv, t))_j$ and the d-jet of $(Q_j)_j$ to that of $(Q_j(p^2, -v^2, pv, t))_j$. The closure of the image of Z_d is irreducible, with generic point the image of the generic point of Z_d, and the preimage in Z_d of a non-empty open subset of the closure is a non-empty open subset of Z_d. Thus $(Q_j(p^2, -v^2, pv, t))_j$ is a compatible generic limit of the tuples $(g_{ij}(p^2, -v^2, pv, t))_j$. If $n = +\infty$, we let

$$B'_* := \sum_j \kappa_j \operatorname{div}(Q_j(p^2, -v^2, pv, t)) \quad \text{on } \operatorname{Spec} \Bbbk'[[p, v, t]].$$

We show that

$$\Big(W_n, \sum_j \kappa_j \operatorname{div}(Q_j) + \delta \operatorname{div}(t)\Big) \text{ is lc near } 0, \tag{B.177}$$

where, as in Definition 2.2, the pair is taken on the completion of $W_n \times_{\mathbb{C}} \operatorname{Spec} \Bbbk'$ at 0. Let $f_i := XY + Z^2 + t^{n_i}$, and let $f_* := XY + Z^2 + t^n$ if $n < +\infty$ and $f_* := XY + Z^2$ if $n = +\infty$. For a positive integer d and all but finitely many i, the d-jets of f_i and t are the $\mathbb{C}$-points $\operatorname{jet}_d(f_*)$ and $\operatorname{jet}_d(t)$, as $f_i = f_*$ if $n < +\infty$ and $n_i \to +\infty$ if $n = +\infty$, so the d-jet of $(f_*, (Q_j)_j, t)$ maps to the generic point of

$$Z'_d := \{\operatorname{jet}_d(f_*)\} \times Z_d \times \{\operatorname{jet}_d(t)\},$$

and $(f_*, (Q_j)_j, t)$ is a compatible generic limit of the tuples $(f_i, (g_{ij})_j, t)$. Let

$$\mathfrak{b}_i := (f_i) \prod_j (g_{ij})^{\kappa_j} (t)^\delta,$$

an $\mathbb{R}$-ideal on the germ $0 \in \mathbb{A}^4$. The closed subsets Z'_d of Definition 2.5 for the tuples $(f_i, (g_{ij})_j, t)$, shrunk to dense open subsets over which the ideals generated by the universal d-jets and $\mathfrak{m}^d$ are flat, form a family of approximations of $(\mathfrak{b}_i)_i$ [Kaw21, Definition 3.1]. The generic points of the Z'_d give an embedding $\mathbb{K} \subset \Bbbk'$ of the union $\mathbb{K}$ of their function fields. Under the embedding $\mathbb{K} \subset \Bbbk'$ the generic limit $\mathfrak{a}_{\mathbb{K}}$ of $(\mathfrak{b}_i)_i$ on $\operatorname{Spec} \mathbb{K}[[X, Y, Z, t]]$ [Kaw21, Definition 3.2] maps to $(f_*) \prod_j (Q_j)^{\kappa_j} (t)^{\delta}$. Assume that (B.177) fails. Then there exists a prime divisor F over W_n with center 0 and

$$a\Big(F, W_n, \sum_j \kappa_j \operatorname{div}(Q_j) + \delta \operatorname{div}(t)\Big) < 0.$$

Fix an integer $L > \max_j \operatorname{mult}_F Q_j$. With $\widehat{\mathfrak{m}} := (X, Y, Z, t)$ and $\mathfrak{m}$ the maximal ideal of $0 \in \mathbb{A}^4_{\Bbbk'}$, the $\widehat{\mathfrak{m}}$-primary ideal $(Q_j) + \widehat{\mathfrak{m}}^L$ is the completion of an $\mathfrak{m}$-primary ideal $\mathfrak{a}_j$ on $\mathbb{A}^4_{\Bbbk'}$ [Kaw21, Lemma 2.7(i)], and $\operatorname{mult}_F((Q_j) + \widehat{\mathfrak{m}}^L) = \operatorname{mult}_F Q_j$ as $\operatorname{mult}_F \widehat{\mathfrak{m}} \geq 1$. By [dFEM11, Lemma 2.21, Proposition A.14 and Remark A.15], the prime divisors with center 0 over $(f_* = 0) \subset \mathbb{A}^4_{\Bbbk'}$ and over the completion of $(f_* = 0)$ at 0, with their log discrepancies, correspond to each other, so the pair

$$\Big((f_* = 0), \prod_j (\mathfrak{a}_j \mathcal{O}_{(f_*=0)})^{\kappa_j} (t)^{\delta}\Big) \text{ is not lc at } 0.$$

The coefficients of f_* and of generators of the $\mathfrak{a}_j$ belong to the algebraic closure in $\Bbbk'$ of a subfield finitely generated over $\mathbb{Q}$, which embeds in $\mathbb{C}$, and the lc property does not change under extensions of algebraically closed coefficient fields [dFEM11, Lemma 2.14(ii)]. Thus, by [EMY03, Theorem 1.6], the pair

$$\Big(\mathbb{A}^4_{\Bbbk'}, \operatorname{div}(f_*) + \prod_j \mathfrak{a}_j^{\kappa_j} (t)^{\delta}\Big) \text{ is not lc at } 0,$$

and passing back to the completion and using $(Q_j) \subset (Q_j) + \widehat{\mathfrak{m}}^L$, the pair

$$\Big(\operatorname{Spec} \Bbbk'[[X, Y, Z, t]], \operatorname{div}(f_*) + \sum_j \kappa_j \operatorname{div}(Q_j) + \delta \operatorname{div}(t)\Big)$$

is not lc. Since base change along $\mathbb{K} \subset \Bbbk'$ takes a log resolution to a log resolution and preserves log discrepancies [Kaw21, Remark 2.6], the pair $(\operatorname{Spec} \mathbb{K}[[X, Y, Z, t]], \mathfrak{a}_{\mathbb{K}})$ is not lc either, so it has a prime divisor $\widehat{E}$ with center the closed point and negative log discrepancy. By [Kaw21, Remark 3.8(i)] for $\widehat{E}$, for infinitely many i there exists a prime divisor E_i over $\mathbb{A}^4$ with center 0 such that

$$a\Big(E_i, \mathbb{A}^4, \operatorname{div}(f_i) + \sum_j \kappa_j \operatorname{div}(g_{ij}) + \delta \operatorname{div}(t)\Big) = a\big(\widehat{E}, \operatorname{Spec} \mathbb{K}[[X, Y, Z, t]], \mathfrak{a}_{\mathbb{K}}\big) < 0.$$

On the other hand, $W_{n_i} = (f_i = 0)$ is a normal hypersurface in $\mathbb{A}^4$, so by [EMY03, Theorem 1.6] and (B.172),

$$\operatorname{mld}\Big(\mathbb{A}^4 \ni 0, W_{n_i} + \sum_j \kappa_j \operatorname{div}(g_{ij}) + \delta \operatorname{div}(t)\Big) = \operatorname{mld}\Big(W_{n_i} \ni 0, \sum_j \kappa_j \operatorname{div}(g_{ij}) + \delta \operatorname{div}(t)\Big) \geq 0,$$

a contradiction. Thus (B.177) holds. The double cover $\operatorname{Spec} \Bbbk'[[p, v, t]] \to W_0$ is étale outside $(p = v = 0)$, so if $n = +\infty$, then B'_* is lc by (B.177).

Let

$$b'_* := \operatorname{mld}\Big(\Big(W_n \ni 0, \sum_j \kappa_j \operatorname{div}(Q_j)\Big)/G'\Big).$$

By (B.177), any prime divisor F over W_n with center 0 satisfies

$$a\Big(F, W_n, \sum_j \kappa_j \operatorname{div}(Q_j)\Big) \geq \delta \operatorname{mult}_F t \geq \delta,$$

and the inertia group of F in G' has order at most M. Thus $b'_* \geq \delta/M > 0$. By Lemma B.22(2) for $g_j := h_{*,j}$ (which have no monomials divisible by xy by Set-up 10.3), the group G, the labels κ_j, the degrees d_j and the integer k_0 of **Step 2**, the pair $(W_0, \sum_j \kappa_j \operatorname{div}(P_j))$ is lc near 0 and

$$\operatorname{mld}\Big(\Big(W_0 \ni 0, \sum_j \kappa_j \operatorname{div}(P_j)\Big)/G'\Big) \geq b_*.$$

By (B.175), (B.177), and Lemma B.22(3) with $w = (1,1,1)$ and $\mu = 0$,

$$b'_* \geq \operatorname{mld}\Big(\Big(W_0 \ni 0, \sum_j \kappa_j \operatorname{div}(P_j)\Big)/G'\Big) \geq b_*. \tag{B.178}$$

It remains to show that, after passing to a subsequence,

$$\liminf_i \beta_i \geq b_*. \tag{B.179}$$

Step 5. In this step we show (B.179) when $n < +\infty$.

By Lemma B.21(2) for W_n, the group G', the tuples $(g_{ij})_j$ and the limit $(Q_j)_j$, and by (B.177), $b'_i = b'_*$ for all but finitely many i. Thus (B.173) and (B.178) imply (B.179).

Step 6. In this step we show (B.179) when $n = +\infty$ and B'_* satisfies one of the conditions (i)–(iii) of Lemma B.22(1).

Since $c_{ij} \in \Gamma \subset [0,1]$, $\kappa_j \in (0,1]$. By **Step 3**, (B.172), **Step 4** and $b'_* > 0$, Lemma B.22(1) holds for the integers n_i, the group G', the labels κ_j, the marks g_{ij} and the limit $(Q_j)_j$. Thus $\liminf_i b'_i \geq b'_*$, and (B.173) and (B.178) imply (B.179).

Step 7. In this step we show (B.179) in the remaining case: $n = +\infty$, B'_* is not klt, $\lfloor B'_* \rfloor = 0$, and B'_* does not satisfy Lemma B.22(1)(iii).

First we determine the lc centers of B'_*. As in **Step 4**, for the tuples $(g_{ij}(p^2, -v^2, pv, t))_j$ on the smooth germ $0 \in \mathbb{A}^3_{p,v,t}$ the union $\mathbb{K}'$ of the function fields of the corresponding Z_d embeds in $\Bbbk'$, and the generic limit $B'_{\mathbb{K}'}$ on $\operatorname{Spec} \mathbb{K}'[[p,v,t]]$ [Kaw21, Definition 3.2] of the $\mathbb{R}$-ideals $\prod_j (g_{ij}(p^2, -v^2, pv, t))^{\kappa_j}$ maps to $\prod_j (Q_j(p^2, -v^2, pv, t))^{\kappa_j}$, whose divisor is B'_*. Since $\operatorname{Spec} \Bbbk'[[p,v,t]] \to W_0$ is finite and étale in codimension 1, the log discrepancy with respect to B'_* of a prime divisor over $\operatorname{Spec} \Bbbk'[[p,v,t]]$ with center 0 is a positive multiple of the log discrepancy with respect to $(W_0, \sum_j \kappa_j \operatorname{div}(Q_j))$ of the image prime divisor F', which is at least $|I_{G'}(F')| b'_* \geq b'_* > 0$ by the definition of b'_*. Thus B'_* has positive minimal log discrepancy at 0. Since B'_* is lc and not klt and has no component with coefficient one, $B'_{\mathbb{K}'}$ is lc and not klt, has no component with coefficient one, and has positive minimal log discrepancy at the closed point [Kaw21, Remark 2.6], so by [Kaw21, Theorem 4.10] the smallest lc center of $B'_{\mathbb{K}'}$ is a regular curve $C'_{\mathbb{K}'}$. Any lc center of $B'_{\mathbb{K}'}$ contains $C'_{\mathbb{K}'}$, is not a component of $B'_{\mathbb{K}'}$ as $\lfloor B'_* \rfloor = 0$, and is not the closed point, as the minimal log discrepancy at the closed point is positive. Thus $C'_{\mathbb{K}'}$ is the unique lc center of $B'_{\mathbb{K}'}$, and the non-klt locus of $B'_{\mathbb{K}'}$ is $C'_{\mathbb{K}'} \cong \operatorname{Spec} \mathbb{K}'[[s]]$, whose closed point is a $\mathbb{K}'$-point. Since base change along $\mathbb{K}' \subset \Bbbk'$ takes a log resolution to a log resolution and preserves log discrepancies [Kaw21, Remark 2.6], the non-klt locus of B'_* is the base change $C' \cong \operatorname{Spec} \Bbbk'[[s]]$ of $C'_{\mathbb{K}'}$, which is regular and irreducible, so C' is the unique lc center of B'_*. Since B'_* is ι-invariant, $\iota(C') = C'$. If ι acted non-trivially on C', then B'_* would satisfy Lemma B.22(1)(iii). Hence ι acts trivially on C', so C' is contained in the fixed locus $(p = v = 0)$ of ι, and $C' = (p = v = 0)$. Since the double cover is étale outside $(p = v = 0)$ and B'_* is klt outside C',

$$\Big(W_0, \sum_j \kappa_j \operatorname{div}(Q_j)\Big) \text{ is klt outside the axis } (X = Y = Z = 0). \tag{B.180}$$

Next we show that case (b) of **Step 4** does not occur. Let $\Bbbk''$ be an algebraic closure of the fraction field of $\Bbbk'[[t]]$. By (B.175), $Q_j \in \Bbbk''[X,Y,Z]$, and we let

$$B'' := \sum_j \kappa_j \operatorname{div}\left(Q_j(p^2, -v^2, pv, t)\right) \quad \text{on } \mathbb{A}^2_{\Bbbk''} = \operatorname{Spec} \Bbbk''[p,v].$$

The homomorphism

$$\Bbbk'[[p,v,t]] = \Bbbk'[[t]][[p,v]] \to \Bbbk''[[p,v]]$$

induces a morphism $\operatorname{Spec} \Bbbk''[[p,v]] \to \operatorname{Spec} \Bbbk'[[p,v,t]]$ which maps the closed point to the generic point of C' and is flat with geometrically regular fibers, as we are in characteristic zero, so it is a regular morphism, and base change along it takes a log resolution $V \to S$, where $S := \operatorname{Spec} \Bbbk'[[p,v,t]]$, to a log resolution and preserves the coefficients of the boundary and of $K_{V/S}$, as $K_{V/S}$ is defined by the zeroth Fitting ideal of $\Omega_{V/S}$ [dFEM11, Remark A.12]. Thus a log resolution of B'_* pulls back, over the generic point of C', to a log resolution with the coefficients of B'_*. Since C' is an lc center of the lc pair B'_*, $\lfloor B'_* \rfloor = 0$, the $Q_j(p^2, -v^2, pv, t)$ are polynomials in p, v over $\Bbbk''$, and lc and klt at 0 can be checked on the completion, the pair $(\mathbb{A}^2_{\Bbbk''}, B'')$ is lc and not klt at 0, and any component of B'' through 0 has coefficient less than one. By (B.175), the image of Q_j in $\Bbbk''[X,Y,Z]/(XY + Z^2)$ has a representative without monomials divisible by XY of degree at most d_j, and $\sum_j \kappa_j d_j = 1$. By Lemma 7.5(2), any Q_j is homogeneous of degree d_j in X, Y, Z. In case (b), the coefficient of X^{α_0} in Q_{j_0} is a non-zero element of $\Bbbk'[[t]]$ by (B.176), and $|\alpha_0| < d_{j_0}$, a contradiction. Hence case (a) holds, so (B.174) holds.

Next we bound $a(F_i, W_{n_i}, 0)$. By (B.180), there exists a real number $\zeta \in (0,1)$ such that the pair

$$\Big(W_0, \sum_j \kappa_j \operatorname{div}(Q_j) + \zeta \operatorname{div}(Z) + \zeta \operatorname{div}(t)\Big)$$

is klt outside the axis ($X = Y = Z = 0$), as on a log resolution of

$$\Big(W_0, \sum_j \operatorname{div}(Q_j) + \operatorname{div}(Z) + \operatorname{div}(t)\Big)$$

the finitely many prime divisors whose centers on W_0 are not contained in the axis have positive log discrepancies with respect to $(W_0, \sum_j \kappa_j \operatorname{div}(Q_j))$. As for $(f_*, (Q_j)_j, t)$ in **Step 4**, $((Q_j)_j, Z, t)$ is a compatible generic limit of the tuples $((g_{ij})_j, Z, t)$, as the d-jets of Z and t do not depend on i. By (B.173), (B.164) and $e_{F_i} \le M$,

$$\begin{aligned} a(F_i, W_{n_i}, 0) - \sum_j \kappa_j \operatorname{mult}_{F_i} g_{ij} - \zeta \operatorname{mult}_{F_i} Z - \zeta \operatorname{mult}_{F_i} t &\le a\Big(F_i, W_{n_i}, \sum_j \kappa_j \operatorname{div}(g_{ij})\Big) \\ &= e_{F_i} \beta_i \le 2M, \end{aligned}$$

and by (B.174),

$$\min(\operatorname{mult}_{F_i} X, \operatorname{mult}_{F_i} Y, \operatorname{mult}_{F_i} Z) \le 2\lceil 2M/\delta \rceil.$$

By Proposition B.20(1) for the germs W_{n_i}, the marks $((g_{ij})_j, Z, t)$ with labels $((\kappa_j)_j, \zeta, \zeta)$, $B_0 := 2M$, and $R' := 2\lceil 2M/\delta \rceil$, there exists a constant T such that, for all but finitely many i,

$$a(F_i, W_{n_i}, 0) \le 2M + T \max(\{\kappa_j\}_j \cup \{\zeta\}) \le 2M + T, \tag{B.181}$$

as $\kappa_j \le 1$ and $\zeta < 1$.

Finally we move F_i to W_0. Let $\lambda := 2M + T$ and $\Gamma' := \{\kappa_j/2\}_j$. Since W_{n_i} is klt, (B.172) implies that $(W_{n_i}, \sum_j (\kappa_j/2) \operatorname{div}(g_{ij}))$ is klt near 0. By (B.181), $a(F_i, W_{n_i}, 0)/e_{F_i} \le \lambda$. By Proposition 3.14(2)(a) for M, Γ', λ, the group G' and the labelled boundaries $\sum_j (\kappa_j/2) \operatorname{div}(g_{ij})$, for any i with $n_i \ge N_\lambda(M, \Gamma', \lambda)$ the birational transform $F_{i,0}$ of F_i under

$$(X, Y, Z, t) \to (X, Y + t^{n_i}/X, Z, t)$$

is a prime divisor over W_0 with center 0 and inertia group $I_G(F_i)$ in G', with

$$a(F_{i,0}, W_0, 0) = a(F_i, W_{n_i}, 0) \quad \text{and} \quad \operatorname{mult}_{F_{i,0}}(g_{ij}|_{W_0}) = \operatorname{mult}_{F_i}(g_{ij}|_{W_{n_i}}) \quad \text{for any } j,$$

as the label of g_{ij} is $\kappa_j/2 > 0$. Hence, by Proposition 3.14(2)(a) and (B.173),

$$a\Big(F_{i,0}, W_0, \sum_j \kappa_j \operatorname{div}(g_{ij})\Big) = a\Big(F_i, W_{n_i}, \sum_j \kappa_j \operatorname{div}(g_{ij})\Big) = e_{F_i}\beta_i.$$

By Lemma B.21(2) for $W_\infty = W_0$, the group G', the tuples $(g_{ij})_j$ and the limit $(Q_j)_j$, and by (B.177), for all but finitely many i the pair $(W_0, \sum_j \kappa_j \operatorname{div}(g_{ij}))$ is lc near 0 and

$$\operatorname{mld}\Big(\Big(W_0 \ni 0, \sum_j \kappa_j \operatorname{div}(g_{ij})\Big)/G'\Big) = b'_*.$$

Thus, for all but finitely many i,

$$\beta_i = \frac{a(F_{i,0}, W_0, \sum_j \kappa_j \operatorname{div}(g_{ij}))}{e_{F_i}} \geq b'_*,$$

and (B.178) implies (B.179).

Step 8. In this step we conclude the proof.

By **Steps 5–7**, (B.179) holds after passing to a subsequence. By (B.167) and (B.164),

$$b_* \geq a_* \geq \limsup_i a_i \geq \limsup_i \beta_i \geq \liminf_i \beta_i \geq b_*.$$

Thus $\lim_i(a_i - \beta_i) = 0$, so $\lim_i R_i \leq 0$ by (B.163). Since R_i is strictly increasing with a positive limit along the subsequence of **Step 1**, and hence along any further subsequence, we obtain a contradiction. The theorem follows. □

*Proof of Theorem 10.9, **Steps 1–4**.* Assume, to the contrary, that $(a_i)_{i\geq 1}$ is strictly increasing.

Step 1. In this step we fix computing lifts F_i of a_i, bound $\operatorname{mult}_{F_i} t$, and fix the number b_* bounding $\lim_i a_i$.

After passing to a subsequence, the conclusions of Proposition 10.5 hold, and by hypothesis the curve C of Proposition 10.5 is $(p = v = 0)$, on which ι acts trivially. By Proposition B.14(1)(i), with $u_i = t$, there exists a real number $\epsilon > 0$ such that, after passing to a subsequence,

$$\Big(W_{N_i}, \sum_j \kappa_j \operatorname{div}(h_{ij}) + \epsilon \operatorname{div}(t)\Big) \text{ is lc near 0 for any } i,$$

and any computing lift F_i of a_i satisfies

$$\sum_j (\kappa_j - c_{ij}) \frac{\operatorname{mult}_{F_i} h_{ij}}{e_{F_i}} + \epsilon \frac{\operatorname{mult}_{F_i} t}{e_{F_i}} \leq a_i \leq 2. \tag{B.182}$$

Since $C \not\subset (t = 0)$ and C is the only lc center of B^* through 0, there exists a real number $\eta_0 > 0$ such that $(\widetilde{W}_*, B^* + \eta_0 \operatorname{div}(t))$ is lc.

We let $\eta := \min\{\epsilon, \eta_0\}$. Then

$$\begin{aligned} &\Big(W_{N_i}, \sum_j \kappa_j \operatorname{div}(h_{ij}) + \eta \operatorname{div}(t)\Big) \text{ is lc near 0 for any } i, \text{ and} \\ &\big(\widetilde{W}_*, B^* + \eta \operatorname{div}(t)\big) \text{ is lc.} \end{aligned} \tag{B.183}$$

For any i we fix a computing lift F_i of a_i, and we let

$$\beta_i := \frac{1}{e_{F_i}}\Big(a(F_i, W_{N_i}, 0) - \sum_j \kappa_j \operatorname{mult}_{F_i} h_{ij}\Big).$$

Since F_i computes a_i,

$$a_i = \frac{1}{e_{F_i}}\Big(a(F_i, W_{N_i}, 0) - \sum_j c_{ij} \operatorname{mult}_{F_i} h_{ij}\Big).$$

Since $c_{ij} \le \kappa_j$,

$$a_i - \beta_i = \sum_j (\kappa_j - c_{ij}) \frac{\operatorname{mult}_{F_i} h_{ij}}{e_{F_i}} \ge 0,$$

and (B.182) with $\eta \le \epsilon$ implies

$$\eta \frac{\operatorname{mult}_{F_i} t}{e_{F_i}} \le \beta_i \le a_i \le 2.$$

By Proposition B.20(2) with ϵ in place of δ, after passing to a subsequence,

$$\gamma_i := \min\{\operatorname{mult}_{F_i} x, \operatorname{mult}_{F_i} y, \operatorname{mult}_{F_i} z\} \to +\infty \quad \text{and} \quad \operatorname{mult}_{F_i} t \le \lceil 2M/\epsilon \rceil.$$

The pair $(\widetilde{W}_*, B^*)$ is the pullback of (W_*, D_*) under the double cover $\pi_0 : \widetilde{W}_* \to W_*$, which is étale outside the curve $(p = v = 0)$. Since B^* is lc, (W_*, D_*) is lc near 0. We let

$$b_* := \operatorname{mld}\big((W_* \ni 0, D_*)/G\big).$$

Since $(a_i)_{i \ge 1}$ is strictly increasing, Lemma B.21(3) implies

$$\lim_i a_i \le b_*. \tag{B.184}$$

We shall prove that, after passing to a subsequence, $\liminf_i \beta_i \ge b_*$.

By hypothesis, $\sum_j \kappa_j d_j < 1$. By Set-up 10.3, $\operatorname{mult}_C H_j^* = 2d_j$ for any j, so

$$\operatorname{mult}_C(B^*) = 2 \sum_j \kappa_j d_j.$$

Let E_C be the exceptional prime divisor of the blow-up of C in $\widetilde{W}_*$. Then

$$a(E_C, \widetilde{W}_*, B^*) = 2 - 2 \sum_j \kappa_j d_j > 0.$$

Step 2. In this step we construct the first prime divisor E of log discrepancy zero over the generic point of C, and the prime divisors ν_k over $\widetilde{W}_*$ with center the closed point.

The completion of the local ring of $\widetilde{W}_*$ at the generic point of C is $\Bbbk((t))[[p, v]]$. We let $\Sigma := \operatorname{Spec} \Bbbk((t))[[p, v]]$, with closed point o_0, and $\mathfrak{a} := \prod_j (H_j^*)^{\kappa_j}$, an $\mathbb{R}$-ideal on Σ. For any prime divisor E' over Σ with center o_0, the closure of E' is a prime divisor over $\widetilde{W}_*$ with center C, and $a(E', \Sigma, \mathfrak{a}) = a(E', \widetilde{W}_*, B^*)$, by the compatibility of log resolutions and log discrepancies with localization and completion [dFEM11, Lemma 2.21 and Proposition A.14]. Since B^* is lc and C is an lc center of B^*, $(\Sigma, \mathfrak{a})$ is lc and $\operatorname{mld}(\Sigma \ni o_0, \mathfrak{a}) = 0$. By Lemma 3.4(1) and (2) over the field $\Bbbk((t))$, there exist an integer $r \ge 1$ and closed points $o_1, \dots, o_{r-1}$ satisfying the following, where $\Sigma_0 := \Sigma$ and $\Sigma_{r'+1} \to \Sigma_{r'}$ is the blow-up of $o_{r'}$ with exceptional curve $E_{r'+1}$:

(1) For any $1 \le r' < r$, $a(E_{r'}, \Sigma, \mathfrak{a}) \in (0, 1)$, and $a(E_r, \Sigma, \mathfrak{a}) = 0$.
(2) For any $1 \le r' < r$, $o_{r'}$ is a $\Bbbk((t))$-point of $E_{r'}$.
(3) $E := E_r$ is the exceptional divisor of the weighted blow-up of Σ with coprime weights in some regular parameters of $\Bbbk((t))[[p, v]]$.

Any element of $\widetilde{G}$ induces an automorphism of $\Bbbk[[p, v, t]]$ preserving C and multiplying each H_j^* by a constant, hence an automorphism of Σ, not necessarily over $\Bbbk((t))$, fixing o_0 and preserving $\mathfrak{a}$. By Lemma 3.4(3), $\widetilde{G}$ fixes $o_1, \dots, o_{r-1}$ and E.

Next we extend the blow-ups over $\operatorname{Spec} \Bbbk[[t]]$. The scheme $\widetilde{W}_*$ is formally smooth over $\operatorname{Spec} \Bbbk[[t]]$ of relative dimension 2, and $T_0 := C$ is a section of $\widetilde{W}_* \to \operatorname{Spec} \Bbbk[[t]]$. Let X_1 be the blow-up of $\widetilde{W}_*$ along T_0, and for $1 \le r' < r$ let $T_{r'}$ be the closure in $X_{r'}$ of the image of $o_{r'}$ and $X_{r'+1}$ the blow-up of $X_{r'}$ along $T_{r'}$. The blow-up along a section of a scheme formally smooth over $\operatorname{Spec} \Bbbk[[t]]$ is formally smooth over $\operatorname{Spec} \Bbbk[[t]]$, since it is flat over $\operatorname{Spec} \Bbbk[[t]]$ and fiber-wise it is the blow-up of a regular surface at a closed point. Thus X_1 is formally smooth over $\operatorname{Spec} \Bbbk[[t]]$, and $\Sigma_1 = \Sigma \times_{\widetilde{W}_*} X_1$ since blow-ups commute with the flat morphism $\Sigma \to \widetilde{W}_*$. We show by induction

on r' that $X_{r'}$ is formally smooth over $\operatorname{Spec}\Bbbk[[t]]$ with $\Sigma_{r'} = \Sigma \times_{\widetilde{W}_*} X_{r'}$ for $1 \le r' \le r$, and that $T_{r'}$ is a section of $X_{r'} \to \operatorname{Spec}\Bbbk[[t]]$ for $1 \le r' < r$. Assume that this holds for $X_{r'}$ with $1 \le r' < r$. Then $T_{r'}$ is integral with function field $\Bbbk((t))$, the residue field of $o_{r'}$, and it is contained in the exceptional locus of $X_{r'} \to \widetilde{W}_*$, which is proper over $C \cong \operatorname{Spec}\Bbbk[[t]]$. Thus $T_{r'} \to \operatorname{Spec}\Bbbk[[t]]$ is finite and birational, hence an isomorphism since $\operatorname{Spec}\Bbbk[[t]]$ is normal, and $T_{r'}$ is a section of $X_{r'} \to \operatorname{Spec}\Bbbk[[t]]$. Hence $X_{r'+1}$ is formally smooth over $\operatorname{Spec}\Bbbk[[t]]$, and $\Sigma_{r'+1} = \Sigma \times_{\widetilde{W}_*} X_{r'+1}$ since blow-ups commute with the flat morphism $\Sigma \to \widetilde{W}_*$. Thus $X := X_r$ is formally smooth over $\operatorname{Spec}\Bbbk[[t]]$, the exceptional prime divisors of $X \to \widetilde{W}_*$ are the closures of $E_1, \dots, E_r$, and $\widetilde{G}$ acts on X. We denote the closure of E in X again by E. It is a $\mathbb{P}^1$-bundle over $T_{r-1} \cong \operatorname{Spec}\Bbbk[[t]]$. Let B_X be the $\mathbb{R}$-divisor with $K_X + B_X$ the pullback of $K_{\widetilde{W}_*} + B^*$. The coefficient of the closure of $E_{r'}$ in B_X is $1 - a(E_{r'}, \Sigma, \mathfrak{a}) \in (0,1]$, that of E is 1, and the other components of B_X are the birational transforms of components D of B^*. Thus $B_X \ge 0$.

We let $E^0 := E|_{t=0} \cong \mathbb{P}^1_{\Bbbk}$. Let ς be a local equation of E at the generic point of E^0, and ζ a rational function on X, regular at the generic point of E^0, whose restriction to E^0 is a coordinate of $\mathbb{P}^1_{\Bbbk}$. Since $E \to \operatorname{Spec}\Bbbk[[t]]$ is smooth, (t, ς) is a regular system of parameters of $\mathcal{O}_{X,E^0}$, and the completion of $\mathcal{O}_{X,E^0}$ is $\Bbbk(\zeta)[[t,\varsigma]]$. For any positive integer k we let ν_k be the monomial valuation in (t,ς) with weights $(1,k)$ at the generic point of E^0. Then ν_k is a prime divisor over $\widetilde{W}_*$ with center the closed point, since E^0 maps to $C \cap (t=0) = \{0\}$. Moreover, ν_k is $\widetilde{G}$-stable, since $\widetilde{G}$ preserves E and $(t=0)$. Since $a(\nu_k, X, 0) = 1 + k$ and $B_X \ge 0$ has coefficient 1 along E,

$$a(\nu_k, \widetilde{W}_*, B^*) = a(\nu_k, X, B_X) \le 1 + k - k = 1.$$

Since $\nu_k(t) = 1$, (B.183) implies $a(\nu_k, \widetilde{W}_*, B^*) \ge \eta$. Thus

$$0 < \eta \le a(\nu_k, \widetilde{W}_*, B^*) \le 1 \quad \text{for any positive integer } k. \tag{B.185}$$

Step 3. In this step we choose $\widetilde{G}$-semi-invariant coordinates p', v in which E is monomial, and integers ρ_p, ρ_v such that the monomial valuation in (t, p', v) with weights $(1, u_1 k + \rho_p, u_2 k + \rho_v)$ has log discrepancy in $[\eta, 1]$ with respect to B^* for any $k \gg 0$.

Let $\mathfrak{m}$ be the maximal ideal of $\Bbbk((t))[[p,v]]$. By **Step 2**, E is the monomial valuation with coprime weights (u_1, u_2), $u_1 \ge u_2$, in some regular parameters (y_1, y_2) of $\Bbbk((t))[[p,v]]$. Since E_C restricts to the blow-up of o_0, and

$$a(E_C, \Sigma, \mathfrak{a}) = 2 - 2\sum_j \kappa_j d_j > 0 = a(E, \Sigma, \mathfrak{a}),$$

$(u_1, u_2) \ne (1,1)$, so $u_1 > u_2$. Any element of $\mathfrak{m}$ with E-value greater than u_2 has class in $\mathfrak{m}/\mathfrak{m}^2$ in the line spanned by the class of y_1, since any monomial in y_1, y_2 other than y_2 has weight greater than u_2. Since the classes of p and v form a basis of $\mathfrak{m}/\mathfrak{m}^2$, $\min\{E(p), E(v)\} = u_2$. Possibly exchanging p and v, which corresponds to $(x,y,z) \to (-y,-x,z)$ on the W_{N_i} and preserves the hypotheses of the theorem, we may assume that $E(v) = u_2$. Then the part of v of weight u_2 is a non-zero multiple of y_2, so E is the monomial valuation in (y_1, v) with weights (u_1, u_2), and the class of p is not a multiple of the class of v. Expanding p in (y_1, v), the monomials of weight less than u_1 are the v^m with $mu_2 < u_1$. Thus there exists a unique polynomial

$$\phi = \sum_{1 \le m < u_1/u_2} \phi_m(t) v^m, \quad \phi_m \in \Bbbk((t)),$$

with $E(p - \phi) \ge u_1$: if ϕ' is another one, then $E(\phi - \phi') \ge u_1$, while $E(\phi - \phi') = \min\{mu_2 \mid \phi_m \ne \phi'_m\} < u_1$ unless $\phi = \phi'$. We let $p' := p - \phi$. The part of p' of weight u_1 is a non-zero multiple of y_1 plus, if $u_2 = 1$, a multiple of v^{u_1}, so E is the monomial valuation in (p', v) with weights (u_1, u_2).

For $\widetilde{g} = (g, \alpha) \in \widetilde{G}$ we let $\chi_p(\widetilde{g}) := \alpha$, $\chi_v(\widetilde{g}) := \chi_z(g)\alpha^{-1}$, and $\chi_t(\widetilde{g}) := \chi_t(g)$, so that $\widetilde{g}^* p = \chi_p(\widetilde{g}) p$, $\widetilde{g}^* v = \chi_v(\widetilde{g}) v$, $\widetilde{g}^* t = \chi_t(\widetilde{g}) t$, and $\widetilde{g}^* E = E$ (**Step 2**). Thus $\chi_p(\widetilde{g})^{-1} \widetilde{g}^* \phi$ is also a

polynomial of the form above with $E(p-\chi_p(\widetilde{g})^{-1}\widetilde{g}^*\phi)\geq u_1$, and the uniqueness of ϕ implies $\widetilde{g}^*\phi=\chi_p(\widetilde{g})\phi$. Thus p' is $\widetilde{G}$-semi-invariant with character χ_p, and the coefficient of $t^{e'}v^m$ in ϕ is zero unless $\chi_t^{e'}\chi_v^m=\chi_p$. Since $\iota\in\widetilde{G}$ acts by $(p,v,t)\to(-p,-v,t)$, $\phi_m=0$ for any even m. We fix an integer $c_0\geq 0$ such that $t^{c_0}\phi_m\in\Bbbk[[t]]$ for any m.

We let $\delta_j:=E(H_j^*)$ for any j. Since $a(E,\Sigma,0)=u_1+u_2$ and $a(E,\Sigma,\mathfrak{a})=0$,

$$\sum_j \kappa_j\delta_j=u_1+u_2. \tag{B.186}$$

Since $t^{c_0}p'$ and v are regular on X, $E(p')=u_1$ and $E(v)=u_2$, in $\Bbbk(\zeta)((t))[[\varsigma]]$ we may write

$$p'=\varsigma^{u_1}\big(\xi_1(t,\zeta)+\varsigma\xi_1'\big)\quad\text{and}\quad v=\varsigma^{u_2}\big(\xi_2(t,\zeta)+\varsigma\xi_2'\big),$$

with non-zero $\xi_1\in t^{-c_0}\Bbbk(\zeta)[[t]]$ and $\xi_2\in\Bbbk(\zeta)[[t]]$, $\xi_1'\in t^{-c_0}\Bbbk(\zeta)[[t,\varsigma]]$ and $\xi_2'\in\Bbbk(\zeta)[[t,\varsigma]]$. We let ρ_p,ρ_v be the integers and $\xi_1^\circ(\zeta),\xi_2^\circ(\zeta)$ the non-zero elements of $\Bbbk(\zeta)$ with

$$\xi_1\in\xi_1^\circ t^{\rho_p}+t^{\rho_p+1}\Bbbk(\zeta)[[t]]\quad\text{and}\quad\xi_2\in\xi_2^\circ t^{\rho_v}+t^{\rho_v+1}\Bbbk(\zeta)[[t]].$$

We fix a positive integer k_1, depending only on ϕ, $\xi_1,\xi_2,\xi_1',\xi_2'$ and the H_j^*, such that the finitely many inequalities between affine functions of k used in **Steps 3** and 4 hold for any integer $k\geq k_1$. Then for any $k\geq k_1$,

$$\nu_k(p')=u_1k+\rho_p\quad\text{and}\quad\nu_k(v)=u_2k+\rho_v.$$

For any integer $k\geq k_1$ we let

$$V_1:=\frac{p'}{t^{u_1k+\rho_p}}\quad\text{and}\quad V_2:=\frac{v}{t^{u_2k+\rho_v}},$$

and we let ν_k^0 be the monomial valuation in (t,p',v) with weights $(1,u_1k+\rho_p,u_2k+\rho_v)$, that is, the order along $(t=0)$ in the coordinates t,V_1,V_2, through

$$p=t^{u_1k+\rho_p}V_1+\phi(t,t^{u_2k+\rho_v}V_2)\quad\text{and}\quad v=t^{u_2k+\rho_v}V_2,$$

whose coefficients are in $\Bbbk[[t]]$ since $k\geq k_1$. Then ν_k^0 is a $\widetilde{G}$-stable prime divisor over $\widetilde{W}_*$ with center the closed point. Since

$$dt\wedge dp\wedge dv=t^{(u_1+u_2)k+\rho_p+\rho_v}dt\wedge dV_1\wedge dV_2,$$

we have

$$a(\nu_k^0,\widetilde{W}_*,0)=(u_1+u_2)k+\rho_p+\rho_v+1.$$

Since $\nu_k(t)=1$ and $\nu_k(V_1)=\nu_k(V_2)=0$, $\nu_k\geq\nu_k^0$ on $\Bbbk[[p,v,t]]$, with equality if the residues $\overline{V}_1,\overline{V}_2$ of V_1,V_2 in the residue field of ν_k are algebraically independent over $\Bbbk$. Let τ be the residue of ς/t^k, which is transcendental over $\Bbbk(\zeta)$. Then

$$\overline{V}_1=\tau^{u_1}\xi_1^\circ(\zeta)\quad\text{and}\quad\overline{V}_2=\tau^{u_2}\xi_2^\circ(\zeta).$$

We let

$$\lambda:=t^{u_1\rho_v-u_2\rho_p}\frac{\xi_1^{u_2}}{\xi_2^{u_1}}=\sum_{n\geq 0}\lambda_n(\zeta)t^n\in\Bbbk(\zeta)[[t]],\quad\lambda_0=\frac{(\xi_1^\circ)^{u_2}}{(\xi_2^\circ)^{u_1}}\neq 0,$$

so that $\overline{V}_1^{u_2}=\lambda_0(\zeta)\overline{V}_2^{u_1}$. The restriction of p'^{u_2}/v^{u_1} to E is not in $\Bbbk((t))$, since it restricts to a coordinate of the exceptional curve over Σ of the weighted blow-up in (p',v). The expansion of $p'^{u_2}/v^{u_1}|_E$ at the generic point of E^0 is $\xi_1^{u_2}/\xi_2^{u_1}$, and the expansion map from the function field of E to $\Bbbk(\zeta)((t))$ is injective and is the identity on $\Bbbk((t))$. Thus $\lambda\notin\Bbbk((t))$.

Case 1: $\lambda_0\notin\Bbbk$. Then $\Bbbk(\overline{V}_1,\overline{V}_2)$ contains $\lambda_0(\zeta)$, over which ζ is algebraic, and $\overline{V}_2$, which is transcendental over $\Bbbk(\zeta)$. Thus $\overline{V}_1,\overline{V}_2$ are algebraically independent over $\Bbbk$, and $\nu_k=\nu_k^0$ for any $k\geq k_1$.

Case 2: $\lambda_0\in\Bbbk$. Since $\lambda\notin\Bbbk((t))$, we may let $e:=\min\{n\mid\lambda_n\notin\Bbbk\}\geq 1$ and

$$c(t):=\sum_{n<e}\lambda_nt^n\in\Bbbk[t],$$

so that $c(0)=\lambda_0\neq 0$. Since ξ_2 is t^{ρ_v} times a unit of $\Bbbk(\zeta)[[t]]$,

$$\frac{V_1^{u_2}}{V_2^{u_1}}=t^{u_1\rho_v-u_2\rho_p}\frac{(\xi_1+\varsigma\xi_1')^{u_2}}{(\xi_2+\varsigma\xi_2')^{u_1}}=\lambda+\varsigma\theta$$

for some $\theta=\sum_{n\geq 0}\theta_n\varsigma^n$ with $\theta_n\in t^{-c_1(n+1)}\Bbbk(\zeta)[[t]]$ and c_1 independent of k, so that

$$\nu_k(\theta_n\varsigma^{n+1})\geq (n+1)(k-c_1).$$

Since $\nu_k(\varsigma\theta)\geq k-c_1>e$ for $k\geq k_1$, $\nu_k(V_1^{u_2}V_2^{-u_1}-c(t))=e$ with residue of the quotient by t^e equal to $\lambda_e(\zeta)$. Thus for any $k\geq k_1$,

$$\nu_k\big(V_1^{u_2}-c(t)V_2^{u_1}\big)=e,\quad \text{and the residue of } \frac{V_1^{u_2}-c(t)V_2^{u_1}}{t^e} \text{ is } \overline{V}_2^{u_1}\lambda_e(\zeta).$$

Case 2a: $\lambda_0\in\Bbbk$ and $u_2=1$. We let $\psi(t):=c(t)t^{\rho_p-u_1\rho_v}$. Then

$$\frac{p'-\psi(t)v^{u_1}}{t^{u_1k+\rho_p}}=V_1-c(t)V_2^{u_1},$$

so $\nu_k(p'-\psi v^{u_1})=u_1k+\rho_p+e$, and the residue of $(p'-\psi v^{u_1})/t^{u_1k+\rho_p+e}$ is $\overline{V}_2^{u_1}\lambda_e(\zeta)$, which is algebraically independent of $\overline{V}_2$ over $\Bbbk$ by the argument of Case 1. For a fixed $k\geq k_1$, ψ is the unique Laurent polynomial with exponents in $[\rho_p-u_1\rho_v,\rho_p-u_1\rho_v+e)$ satisfying

$$\nu_k(p'-\psi v^{u_1})\geq u_1k+\rho_p+e,$$

since for two of them the difference $\psi-\psi'\neq 0$ satisfies $\nu_k((\psi-\psi')v^{u_1})\leq u_1k+\rho_p+e-1$. Since ν_k is $\widetilde{G}$-stable, the argument of the uniqueness of ϕ implies that ψv^{u_1} is $\widetilde{G}$-semi-invariant with character χ_p. Since $\psi\neq 0$ and ι multiplies p' by -1 and ψv^{u_1} by $(-1)^{u_1}$, u_1 is odd. Since $E(\psi v^{u_1})=u_1$, E is the monomial valuation in $(p'-\psi v^{u_1},v)$ with weights $(u_1,1)$. We replace ϕ by $\phi+\psi v^{u_1}$, p' by $p'-\psi v^{u_1}$, ρ_p by ρ_p+e, and c_0 by $\max\{c_0,u_1\rho_v-\rho_p\}$, and we enlarge k_1 so that the inequalities fixed above hold for the new ϕ, ξ_1, ξ_1', ρ_p, and c_0. Then all the properties of ϕ, p', and c_0 above hold, except that ϕ has v-degree at most u_1/u_2, and $\nu_k=\nu_k^0$ for any $k\geq k_1$ by the argument of Case 1.

Case 2b: $\lambda_0\in\Bbbk$ and $u_2\geq 2$. We let $V_3:=(V_1^{u_2}-c(t)V_2^{u_1})/t^e$. Then $\nu_k(V_3)=0$ and $\nu_k(t)=1$. The field $\Bbbk(\overline{V}_1,\overline{V}_2,\overline{V}_3)$ contains $\overline{V}_2$ and $\lambda_e(\zeta)=\overline{V}_3/\overline{V}_2^{u_1}$, so it has transcendence degree 2 over $\Bbbk$. Since u_1,u_2 are coprime, $\Bbbk[V_1,V_2,V_3]/(V_1^{u_2}-\lambda_0V_2^{u_1})$ is an integral domain of dimension 2, so

$$\Bbbk[V_1,V_2,V_3]/(V_1^{u_2}-\lambda_0V_2^{u_1})\cong\Bbbk[\overline{V}_1,\overline{V}_2,\overline{V}_3]$$

via $(V_1,V_2,V_3)\to(\overline{V}_1,\overline{V}_2,\overline{V}_3)$. Thus ν_k is the valuation of the prime divisor $(t=0)$ on the normal hypersurface

$$Y:=\big(t^eV_3=V_1^{u_2}-c(t)V_2^{u_1}\big)\subset\mathbb{A}^4_{\Bbbk},\tag{B.187}$$

whose fiber over $t=0$ is reduced and irreducible, where the comparison of Y with $\operatorname{Spec}\Bbbk[[p,v,t]][V_1,V_2,V_3]$ near $(t=0)$, as for ν_L and $\mathcal{Y}$ in Lemma B.18(2)(3), is omitted. Since the partial derivative in V_3 of the equation of Y is t^e, the form $t^{-e}dt\wedge dV_1\wedge dV_2$ generates ω_Y at the generic point of $(t=0)$. Since

$$dt\wedge dp\wedge dv=t^{(u_1+u_2)k+\rho_p+\rho_v}dt\wedge dV_1\wedge dV_2,$$

we have

$$\begin{aligned}a(\nu_k,\widetilde{W}_*,0)&=(u_1+u_2)k+\rho_p+\rho_v+e+1\\&=a(\nu_k^0,\widetilde{W}_*,0)+e.\end{aligned}\tag{B.188}$$

In all cases, for any j we expand

$$H_j^*=\sum_{\alpha\in\mathbb{N}^2}f_{j,\alpha}(t)p'^{\alpha_1}v^{\alpha_2},\quad f_{j,\alpha}\in\Bbbk((t)),$$

where $t^{c_0\alpha_2} f_{j,\alpha} \in \Bbbk[[t]]$ since $p = p' + \phi$. Then

$$\delta_j = \min\{u_1\alpha_1 + u_2\alpha_2 \mid f_{j,\alpha} \neq 0\},$$

and we let

$$\rho_j := \min\{e' + \alpha_1\rho_p + \alpha_2\rho_v \mid u_1\alpha_1 + u_2\alpha_2 = \delta_j, \text{ the coefficient of } t^{e'} \text{ in } f_{j,\alpha} \text{ is non-zero}\}.$$

Any term $\Theta = f_{j,\alpha} p'^{\alpha_1} v^{\alpha_2}$ of the expansion of H_j^* with $u_1\alpha_1 + u_2\alpha_2 \geq \delta_j + 1$ satisfies, for $k \geq k_1$,

$$\min\{\nu_k(\Theta), \nu_k^0(\Theta)\} \geq (\delta_j + 1)\big(k - c_0 - |\rho_p| - |\rho_v|\big) \geq k\delta_j + \rho_j + e\delta_j + 1,$$

where we let $e := 0$ in Case 1. The first inequality holds since $\nu_k(\Theta) = \nu_k^0(\Theta)$ is the t-order of $f_{j,\alpha}$ plus $\alpha_1(u_1k + \rho_p) + \alpha_2(u_2k + \rho_v)$, the t-order of $f_{j,\alpha}$ is at least $-c_0\alpha_2$, and $\alpha_1 + \alpha_2 \leq u_1\alpha_1 + u_2\alpha_2$. Thus for any $k \geq k_1$,

$$\begin{aligned} &\nu_k^0(H_j^*) = k\delta_j + \rho_j \quad \text{and} \\ &H_j^* \equiv t^{k\delta_j + \rho_j} \Phi_j(t, V_1, V_2) \mod t^{k\delta_j + \rho_j + e\delta_j + 1} \Bbbk[[t, V_1, V_2]], \end{aligned}$$

where

$$\Phi_j := \sum_{u_1\alpha_1 + u_2\alpha_2 = \delta_j} f_{j,\alpha} t^{\alpha_1\rho_p + \alpha_2\rho_v - \rho_j} V_1^{\alpha_1} V_2^{\alpha_2} \in \Bbbk[[t]][V_1, V_2]$$

does not depend on k, and $\Phi_j^0 := \Phi_j(0, V_1, V_2) \neq 0$ is weighted homogeneous of degree δ_j for the weights (u_1, u_2).

In Case 2b we let m_j be the multiplicity of the factor $V_1^{u_2} - \lambda_0 V_2^{u_1}$ in Φ_j^0. Since $V_1^{u_2} - \lambda_0 V_2^{u_1}$ has weighted degree u_1u_2, $m_j u_1 u_2 \leq \delta_j$. Let $\Phi_j = \sum_{n \geq 0} (V_1^{u_2} - c(t) V_2^{u_1})^n R_n$ be the expansion with $R_n \in \Bbbk[[t]][V_1, V_2]$ of V_1-degree less than u_2. By the uniqueness of the expansion, the reduction modulo t of the expansion of Φ_j is the expansion of Φ_j^0 in powers of $V_1^{u_2} - \lambda_0 V_2^{u_1}$, so $R_n \in t\Bbbk[[t]][V_1, V_2]$ for $n < m_j$ and $R_{m_j} \notin t\Bbbk[[t]][V_1, V_2]$. On Y, $\Phi_j = \sum_n t^{en} V_3^n R_n$, and the residues of the monomials $V_1^{\alpha_1} V_2^{\alpha_2} V_3^{\alpha_3}$ with $\alpha_1 < u_2$ are linearly independent over $\Bbbk$, since $\Bbbk[V_1, V_2, V_3]/(V_1^{u_2} - \lambda_0 V_2^{u_1})$ is free over $\Bbbk[V_2, V_3]$ with basis $1, V_1, \ldots, V_1^{u_2 - 1}$. Thus

$$0 \leq \nu_k(\Phi_j) \leq e m_j,$$

the lower bound since $\nu_k(V_1) = \nu_k(V_2) = 0$. Since $m_j \leq \delta_j$ and $\nu_k \geq k\delta_j + \rho_j + e\delta_j + 1$ on $t^{k\delta_j + \rho_j + e\delta_j + 1} \Bbbk[[t, V_1, V_2]]$, $\nu_k(H_j^*) = k\delta_j + \rho_j + \nu_k(\Phi_j)$, so

$$0 \leq \nu_k(H_j^*) - \nu_k^0(H_j^*) \leq e m_j.$$

By $m_j u_1 u_2 \leq \delta_j$, (B.186) and $u_1 > u_2 \geq 2$,

$$\sum_j \kappa_j m_j \leq \frac{u_1 + u_2}{u_1 u_2} = \frac{1}{u_1} + \frac{1}{u_2} < 1. \tag{B.189}$$

By (B.188) and (B.189),

$$\begin{aligned} a(\nu_k^0, \widetilde{W}_*, B^*) &\leq a(\nu_k, \widetilde{W}_*, B^*) - e\Big(1 - \sum_j \kappa_j m_j\Big) \\ &\leq a(\nu_k, \widetilde{W}_*, B^*). \end{aligned}$$

By (B.185) and the three cases, $a(\nu_k^0, \widetilde{W}_*, B^*) \leq 1$ for any $k \geq k_1$. Since $\nu_k^0(t) = 1$ and ν_k^0 has center the closed point, (B.183) implies $a(\nu_k^0, \widetilde{W}_*, B^*) \geq \eta$. We let

$$\mu := \sum_j \kappa_j \rho_j - \rho_p - \rho_v.$$

By (B.186),

$$\begin{aligned} a(\nu_k^0, \widetilde{W}_*, B^*) &= (u_1 + u_2)k + \rho_p + \rho_v + 1 - \sum_j \kappa_j (k\delta_j + \rho_j) \\ &= 1 - \mu, \end{aligned}$$

so

$$0 \le \mu \le 1 - \eta. \tag{B.190}$$

Step 4. In this step we transport the W_{N_i} by cone shears and weighted scalings, take the largest scaling k_i for which the marks stay regular and F_i keeps center the origin, and express β_i on the new germs.

Since ϕ is odd in v, $\phi(t,v) = vf_\phi(-v^2,t)$ for a polynomial $f_\phi(y',t) \in t^{-c_0}\Bbbk[[t]][y']$. We let $x' := p'^2$, $y' := -v^2$, and $z' := p'v$. Since $p = p' + vf_\phi(y',t)$,

$$x = x' + 2z'f_\phi(y',t) - y'f_\phi(y',t)^2, \quad y = y', \quad \text{and} \quad z = z' - y'f_\phi(y',t),$$

and $xy + z^2 = x'y' + z'^2$. Since ϕ has character χ_p, f_ϕ has character $\chi_p/\chi_v = \chi_x/\chi_z$, and since $\chi_x\chi_y = \chi_z^2$, the three right-hand sides above are semi-invariant with characters χ_x, χ_y, and χ_z.

We let $\mathcal{P}$ be the set of polynomials $\vartheta = \sum_m \vartheta_m(t)v^m$ with $\vartheta_m \in \mathbb{C}[t,t^{-1}]$ satisfying the following:

(i) $\vartheta_m = 0$ unless m is odd and $1 \le m \le u_1/u_2$.
(ii) $t^{c_0}\vartheta_m \in \mathbb{C}[t]$ for any m.
(iii) The coefficient of $t^{e'}v^m$ in ϑ is zero unless $\chi_t^{e'}\chi_v^m = \chi_p$.

For $\vartheta \in \mathcal{P}$ we write $\vartheta(t,v) = vf_\vartheta(-v^2,t)$. For an integer $k \ge k_1$ we let $\Psi_{k,\vartheta} : \mathbb{A}^4 \to \mathbb{A}^4$, $(X',Y',Z',t) \to (x,y,z,t)$, be

$$\begin{aligned}
x &= t^{2(u_1k+\rho_p)}X' + 2t^{(u_1+u_2)k+\rho_p+\rho_v}Z'f_\vartheta\big(t^{2(u_2k+\rho_v)}Y',t\big) \\
&\quad - t^{2(u_2k+\rho_v)}Y'f_\vartheta\big(t^{2(u_2k+\rho_v)}Y',t\big)^2, \\
y &= t^{2(u_2k+\rho_v)}Y', \\
z &= t^{(u_1+u_2)k+\rho_p+\rho_v}Z' - t^{2(u_2k+\rho_v)}Y'f_\vartheta\big(t^{2(u_2k+\rho_v)}Y',t\big).
\end{aligned}$$

Since $k \ge k_1$ and ϑ satisfies (i) and (ii), each term above is a polynomial divisible by a positive power of t. Thus $\Psi_{k,\vartheta}$ is a morphism sending the origin to the origin, and it is birational with inverse defined over $\mathbb{C}[t,t^{-1}]$. For an integer $N \ge 2$ we let $n := N - 2((u_1+u_2)k + \rho_p + \rho_v)$. By the identity $xy + z^2 = x'y' + z'^2$,

$$xy + z^2 + t^N = t^{2((u_1+u_2)k+\rho_p+\rho_v)}\big(X'Y' + Z'^2 + t^n\big),$$

so for $n \ge 2$, $\Psi_{k,\vartheta}$ restricts to a birational morphism from

$$W_n = (X'Y' + Z'^2 + t^n = 0) \tag{B.191}$$

to W_N. The map $(x',y',z') \to (x,y,z)$ has Jacobian determinant 1 and fixes t, and the scaling multiplies $dx' \wedge dy' \wedge dz' \wedge dt$ by $t^{3((u_1+u_2)k+\rho_p+\rho_v)}$ and the equation by $t^{2((u_1+u_2)k+\rho_p+\rho_v)}$. Thus the residue generator of ω_{W_N} pulls back to $t^{(u_1+u_2)k+\rho_p+\rho_v}$ times the residue generator of ω_{W_n}, and for any prime divisor F over W_n,

$$a(F,W_N,0) = a(F,W_n,0) + \big((u_1+u_2)k + \rho_p + \rho_v\big)\operatorname{mult}_F t.$$

By (iii), $\Psi_{k,\vartheta}$ is G-equivariant when G acts on X',Y',Z',t by the characters $\chi_x\chi_t^{-2(u_1k+\rho_p)}$, $\chi_y\chi_t^{-2(u_2k+\rho_v)}$, $\chi_z\chi_t^{-(u_1+u_2)k-\rho_p-\rho_v}$, and χ_t. On the double covers, $\Psi_{k,\vartheta}$ is induced by

$$(P,V,t) \to (p,v,t) = \big(t^{u_1k+\rho_p}P + \vartheta(t,t^{u_2k+\rho_v}V), t^{u_2k+\rho_v}V, t\big),$$

where $(X',Y',Z') = (P^2, -V^2, PV)$. For $k \ge k_1$, the formulas for x,y,z above with f_ϕ in place of f_ϑ define a morphism

$$\Psi_{k,\phi} : \operatorname{Spec}\Bbbk[[X',Y',Z',t]] \to \operatorname{Spec}\Bbbk[[x,y,z,t]]$$

over $\operatorname{Spec}\Bbbk[[t]]$.

For an index i, $\vartheta \in \mathcal{P}$ and an integer $k \geq k_1$, we let $n := N_i - 2((u_1+u_2)k + \rho_p + \rho_v)$ and consider the conditions

$$\begin{aligned} &n \geq 2; \\ &\Psi^*_{k,\vartheta} h_{ij} \in t^{k\delta_j+\rho_j}\mathbb{C}[X',Y',Z',t]/(X'Y' + Z'^2 + t^n) \text{ for any } j; \\ &\text{the center of } F_i \text{ on } W_n \text{ is the origin.} \end{aligned} \tag{B.192}$$

We show that (B.192) holds for a fixed $k \geq k_1$ and all but finitely many i, with $\vartheta \in \mathcal{P}$ whose coefficients of $t^{e'}v^m$ with $e' < \ell$, for an integer ℓ fixed below, are specializations of those of ϕ. First, at the limit: by the expansion $H_j^* = \sum_\alpha f_{j,\alpha} p'^{\alpha_1} v^{\alpha_2}$ of **Step 3**,

$$\begin{aligned} &H_j^*\big(t^{u_1k+\rho_p}P + \phi(t, t^{u_2k+\rho_v}V), t^{u_2k+\rho_v}V, t\big) \\ &= \sum_\alpha f_{j,\alpha} t^{\alpha_1(u_1k+\rho_p)+\alpha_2(u_2k+\rho_v)} P^{\alpha_1} V^{\alpha_2}, \end{aligned}$$

and by the definition of ρ_j and the bound on the terms of weight greater than δ_j in **Step 3**, each coefficient on the right-hand side has t-order at least $k\delta_j + \rho_j$. Since the right-hand side is invariant under ι,

$$\Psi^*_{k,\phi} h_{*,j} \in t^{k\delta_j+\rho_j}\Bbbk[[X',Y',Z',t]]/(X'Y' + Z'^2).$$

Second, the condition is a finite jet condition: let $c^{(k)} := \max_j(k\delta_j + \rho_j)$. Since $\Psi^*_{k,\vartheta}x$, $\Psi^*_{k,\vartheta}y$ and $\Psi^*_{k,\vartheta}z$ are divisible by t, a monomial in x, y, z, t of degree at least $c^{(k)}$ pulls back into $(t^{c^{(k)}})$, and the coefficients of $t^{e'}$ in ϑ with $e' \geq c^{(k)}$ contribute only to $(t^{c^{(k)}})$. For $n \geq c^{(k)}$ we have

$$(t^{k\delta_j+\rho_j}, X'Y' + Z'^2 + t^n) = (t^{k\delta_j+\rho_j}, X'Y' + Z'^2).$$

Thus for $n \geq c^{(k)}$ the second condition of (B.192) is the vanishing of the elements of a finite set $\mathcal{Q}$ of polynomials with rational coefficients in the $c^{(k)}$-jets of the h_{ij} and the coefficients of $t^{e'}$, $e' < c^{(k)}$, of ϑ, and the elements of $\mathcal{Q}$ vanish at the $c^{(k)}$-jets of the $h_{*,j}$ and the coefficients of ϕ.

For any positive integer d, the image of $\mathrm{jet}_d((h_{*,j})_j)$ is the generic point of the irreducible closed subset $Z_d \subset \mathcal{J}_d$ of Definition 2.5, and by Lemma 2.8, any non-empty open subset of Z_d contains $\mathrm{jet}_d((h_{ij})_j)$ for all but finitely many i. Let $\ell' \geq c^{(k)}$ be an integer. The coefficients of $t^{e'}$, $e' < \ell'$, of ϕ are finitely many elements of $\Bbbk$, so they belong to a finitely generated extension $\mathbb{L}$ of the function field of $Z_{\ell'}$. Thus there exist a non-empty open subset $U \subset Z_{\ell'}$ and an integral variety V' with function field $\mathbb{L}$ and a surjective morphism $V' \to U$ of finite type (by Chevalley's theorem after shrinking U) such that the coefficients of $t^{e'}$, $e' < \ell'$, of ϕ are regular functions on V'. For any i with ℓ'-jet of $(h_{ij})_j$ in U and any $\mathbb{C}$-point of V' over the ℓ'-jet, the values of the coefficients of $t^{e'}$, $e' < \ell'$, of ϕ define, together with the monomials $t^{e'}v^m$ of ϕ with $e' < \ell'$, an element $\vartheta \in \mathcal{P}$. The elements of $\mathcal{Q}$ vanish at the generic point of V', hence on V', so ϑ satisfies the second condition of (B.192) whenever $n \geq c^{(k)}$. Since $N_i \to +\infty$, $n \geq \max\{2, c^{(k)}\}$ for $i \gg 0$. For the third condition, the inverse of the shear is $x' = x - 2zf_\vartheta(y,t) - yf_\vartheta(y,t)^2$, $y' = y$, and $z' = z + yf_\vartheta(y,t)$, and $\mathrm{mult}_{F_i} f_\vartheta(y,t) \geq -c_0 \,\mathrm{mult}_{F_i} t$ by (ii). Thus

$$\min\{\mathrm{mult}_{F_i} X', \mathrm{mult}_{F_i} Y', \mathrm{mult}_{F_i} Z'\} \geq \gamma_i - \big(2(u_1+u_2)k + 2|\rho_p| + 2|\rho_v| + 2c_0\big)\,\mathrm{mult}_{F_i} t,$$

which is positive for $i \gg 0$ since $\gamma_i \to +\infty$ and $\mathrm{mult}_{F_i} t \leq \lceil 2M/\epsilon \rceil$ (**Step 1**). Since also $\mathrm{mult}_{F_i} t \geq 1$, the center of F_i on W_n is the origin for $i \gg 0$.

We let

$$\Lambda := \max\left(\{k_1\} \cup \left\{ |\rho_p - m\rho_v| \;\middle|\; 1 \leq m \leq \frac{u_1}{u_2} \right\}\right).$$

Let $i_0 := 0$. For any positive integer ℓ, the last two paragraphs with

$$k = \ell + \Lambda \quad \text{and} \quad \ell' = \max\{\ell, c^{(\ell+\Lambda)}\}$$

give an index $i_\ell > i_{\ell-1}$ and an element $\phi^\circ_{i_\ell} \in \mathcal{P}$ which satisfies (B.192) for i_ℓ and $k = \ell + \Lambda$, and whose coefficients of $t^{e'}$, $e' < \max\{\ell, c^{(\ell+\Lambda)}\}$, are the values at a $\mathbb{C}$-point of the variety V' of the previous paragraph of the corresponding coefficients of ϕ. Passing to the subsequence $(i_\ell)_\ell$, we

let $\ell_{i_\ell} := \ell$. Thus $\ell_i \to +\infty$, and ϕ_i° satisfies (B.192) with $k = \ell_i + \Lambda$ for any i. We let $\mathcal{P}_i$ be the set of $\vartheta \in \mathcal{P}$ whose coefficients of $t^{e'}$, $e' < \ell_i$, are those of ϕ_i°. Since the first condition of (B.192) bounds k from above, we may let

$$k_i := \max\{k \geq k_1 \mid \text{(B.192) holds for } i,\ k \text{ and some } \vartheta \in \mathcal{P}_i\} \geq \ell_i + \Lambda,$$

and we fix $\phi_i \in \mathcal{P}_i$ such that (B.192) holds for i, k_i, and ϕ_i. Thus $k_i \to +\infty$.

Since the characters of G have orders dividing $|G| \leq M$, after passing to a subsequence the class of k_i modulo $|G|$ does not depend on i. We let G' be the group G acting on X', Y', Z', t by the characters above with $k = k_i$, which do not depend on i. We let

$$n_i := N_i - 2\big((u_1+u_2)k_i + \rho_p + \rho_v\big), \quad \Psi_i := \Psi_{k_i,\phi_i} : W_{n_i} \to W_{N_i},$$

and let g_{ij} be the representative without monomials divisible by $X'Y'$ of

$$t^{-k_i\delta_j-\rho_j}\Psi_i^* h_{ij} \in \mathbb{C}[X', Y', Z', t]/(X'Y' + Z'^2 + t^{n_i}).$$

Then g_{ij} is G'-semi-invariant, and

$$\operatorname{mult}_{F_i} h_{ij} = \operatorname{mult}_{F_i} g_{ij} + (k_i\delta_j + \rho_j)\operatorname{mult}_{F_i} t.$$

Since Ψ_i is birational and equivariant for the actions of G' on W_{n_i} and of G on W_{N_i}, the inertia group of F_i for the action of G on W_{N_i} equals the inertia group of F_i for the action of G' on W_{n_i}. By the formula for $a(F, W_N, 0)$ above and (B.186), with μ as in **Step 3**,

$$\beta_i = \frac{1}{e_{F_i}}\Big(a(F_i, W_{n_i}, 0) - \sum_j \kappa_j \operatorname{mult}_{F_i} g_{ij} - \mu \operatorname{mult}_{F_i} t\Big). \tag{B.193}$$

Since

$$\Psi_i^* \operatorname{div}(h_{ij}) = \operatorname{div}(g_{ij}) + (k_i\delta_j + \rho_j)\operatorname{div}(t),$$

the formula for $a(F, W_N, 0)$ above and (B.186) also imply

$$K_{W_{n_i}} + \sum_j \kappa_j \operatorname{div}(g_{ij}) + (\mu+\eta)\operatorname{div}(t) = \Psi_i^*\Big(K_{W_{N_i}} + \sum_j \kappa_j \operatorname{div}(h_{ij}) + \eta \operatorname{div}(t)\Big).$$

Since Ψ_i is a birational morphism sending the origin to 0, (B.183) implies

$$\Big(W_{n_i}, \sum_j \kappa_j \operatorname{div}(g_{ij}) + (\mu+\eta)\operatorname{div}(t)\Big) \text{ is lc near the origin.} \tag{B.194}$$

We let

$$\Delta_i' := \sum_j \kappa_j \operatorname{div}(g_{ij}) + \mu \operatorname{div}(t),$$

an $\mathbb{R}$-divisor on W_{n_i}. By (B.190), $\Delta_i' \geq 0$. We let $b_i' := \operatorname{mld}((W_{n_i} \ni 0, \Delta_i')/G')$. Since F_i has center the origin of W_{n_i}, (B.193) implies $b_i' \leq a(F_i, W_{n_i}, \Delta_i')/e_{F_i} = \beta_i$. For any prime divisor F over W_{n_i} with center the origin, (B.194) and $\operatorname{mult}_F t \geq 1$ imply $a(F, W_{n_i}, \Delta_i') \geq \eta \operatorname{mult}_F t \geq \eta$, and $e_F \leq |G| \leq M$. Thus

$$\frac{\eta}{M} \leq b_i' \leq \beta_i. \tag{B.195}$$

The proof is continued below.

Proof of Theorem 10.9, continued. We keep the integers k_i and n_i, the elements $\phi_i \in \mathcal{P}_i$, the group G' and the marks g_{ij} on W_{n_i} of (B.192), which satisfy (B.193)–(B.195). We write X', Y', Z', t for the coordinates of $W_{n_i} = (X'Y' + Z'^2 + t^{n_i} = 0)$, assign to X', Y', Z' the weights

$$w := (2u_1, 2u_2, u_1 + u_2),$$

and let $(X', Y', Z') = (P^2, -V^2, PV)$, so that P and V have weights u_1 and u_2. The weighted scaling at level k in $\Psi_{k,\phi}$ is

$$x' = t^{2(u_1k+\rho_p)}X', \quad y' = t^{2(u_2k+\rho_v)}Y', \quad \text{and} \quad z' = t^{(u_1+u_2)k+\rho_p+\rho_v}Z'.$$

Let $A_j \in \Bbbk[[t]][X', Y', Z']$ be the w-homogeneous polynomial of degree δ_j without monomials divisible by $X'Y'$ such that, on W_0, $\Psi_{k,\phi}^* h_{*,j} - t^{k\delta_j+\rho_j} A_j$ consists of terms of w-degree greater than δ_j. The scaling multiplies a monomial of w-degree D in x', y', z' by t^{kD} times a power of t independent of k, so A_j does not depend on k, and $A_j(P^2, -V^2, PV, t) \neq 0$ because δ_j is the (u_1, u_2)-weighted order of H_j^* in p', v. Each of x, y, z is replaced under the cone shears of ϕ and of the ϕ_i by a polynomial in x', y', z' without constant term, of bounded degree and with coefficients of bounded pole order in t. Hence there exists a constant c_2, independent of i, such that for any j and any D, any monomial of w-degree D in X', Y', Z' of the Laurent polynomial

$$t^{-k_i\delta_j-\rho_j} \Psi_i^* h_{ij} \in \mathbb{C}[X', Y', Z', t, t^{-1}],$$

whose reduction modulo $X'Y' + Z'^2 + t^{n_i}$ to a representative without monomials divisible by $X'Y'$ is g_{ij}, has a coefficient of t-order at least

$$k_i(D - \delta_j) - c_2(D + 1). \tag{B.196}$$

Step 5. In this step we take the limit of the pairs on W_{n_i} and prove that the quotient value of the limit is at least b_*.

Possibly passing to a subsequence, we may assume that either $n_i = n$ for any i and an integer $n \geq 2$, or $\lim_i n_i = +\infty$, in which case we let $n := +\infty$ and $W_\infty := W_0$. Possibly passing to a further subsequence, we may assume that the characters of the g_{ij} do not depend on i. Any automorphism of $\Bbbk$ fixing the coefficients of the $h_{*,j}$, acting on coefficients, fixes the H_j^*, hence fixes $o_1, \dots, o_{r-1}$, E, and ν_k by Lemma 3.4(3), and hence fixes ϕ by the uniqueness of ϕ and, in Case 2a, of the term added to ϕ in **Step 3**. Since $\Bbbk$ is algebraically closed of characteristic zero, any coefficient of ϕ thus belongs to the field generated over $\mathbb{C}$ by the coefficients of the $h_{*,j}$, so it is a rational function on Z_d for some d, whose value at any $\mathbb{C}$-point of V' over the jet of $(h_{ij})_j$ is, for $i \gg 0$, its value at that jet. Possibly enlarging $\Bbbk$, let $(g_{*,j})_j$ be a compatible generic limit of the tuples $(g_{ij})_j$, the orders at 0 of the g_{ij} being bounded since the coefficient in g_{ij} of a fixed monomial of A_j is non-zero for $i \gg 0$, taken jointly with the tuples $(h_{ij})_j$ and the coefficients of the ϕ_i, such that $(g_{*,j}(P^2, -V^2, PV, t))_j$ is a compatible generic limit of the tuples $(g_{ij}(P^2, -V^2, PV, t))_j$. Since $\phi_i \in \mathcal{P}_i$ and $\ell_i \to +\infty$, the limit of the ϕ_i is ϕ. With Δ_i' and b_i' as in (B.195), let

$$\Delta_*' := \sum_j \kappa_j \operatorname{div}(g_{*,j}) + \mu \operatorname{div}(t) \quad \text{on } W_n.$$

We show that $g_{*,j} \in \Bbbk[[t]][X', Y', Z']$ has w-degree at most δ_j and that the w-homogeneous part of $g_{*,j}$ of degree δ_j is A_j. For $D \geq \delta_j + 1$ and $k_i > c_2$, the bound (B.196) satisfies $k_i(D-\delta_j) - c_2(D+1) \geq k_i - c_2(\delta_j + 2)$, which tends to $+\infty$. The reduction of $X'Y'$ to $-Z'^2 - t^{n_i}$ does not raise the w-degree, since $w(X'Y') = 2(u_1 + u_2) = w(Z'^2)$, and it only adds non-negative powers of t. Hence any monomial of w-degree greater than δ_j has a coefficient in g_{ij} whose t-order tends to $+\infty$, and no such monomial occurs in $g_{*,j}$. The monomials of w-degree at most δ_j are finitely many, so $g_{*,j}$ is a polynomial in X', Y', Z' with coefficients in $\Bbbk[[t]]$. In w-degree δ_j the scaling at level k_i multiplies $x'^a y'^b z'^c$ by $t^{k_i\delta_j + (2a+c)\rho_p + (2b+c)\rho_v}$. Hence, up to terms which come from monomials of higher w-degree through t^{n_i} and whose t-orders tend to $+\infty$ by (B.196), the coefficients of g_{ij} in w-degree δ_j do not depend on k_i and are given by polynomials, independent of i, in the coefficients of h_{ij} and ϕ_i, which at the coefficients of $h_{*,j}$ and ϕ give the coefficients of A_j. Thus their limits are the coefficients of A_j.

We show that $(W_n, \Delta_*' + \eta \operatorname{div}(t))$ is lc near 0. If $n < +\infty$, then by Lemma B.21(2), for $H := G'$, the marks g_{ij} and t, and the labels κ_j and $\mu + \eta$, if $(W_n, \Delta_*' + \eta \operatorname{div}(t))$ were not lc near 0, then $(W_n, \Delta_i' + \eta \operatorname{div}(t))$ would not be lc near 0 for any $i \gg 0$, contrary to (B.194). Let $n = +\infty$, and assume that $(W_0, \Delta_*' + \eta \operatorname{div}(t))$ is not lc near 0. Let $\mathfrak{a}_*$ be the $\mathbb{R}$-ideal $\prod_j (g_{*,j})^{\kappa_j} \cdot (t)^{\mu+\eta}$ on $\operatorname{Spec} \Bbbk[[X', Y', Z', t]]$, and let $\mathfrak{a}_i$ be the $\mathbb{R}$-ideal $\prod_j (g_{ij})^{\kappa_j} \cdot (t)^{\mu+\eta}$ on $\mathbb{A}^4$. Since $(W_0, \mathfrak{a}_*|_{W_0})$ is not lc near 0 and every point of W_0 specializes to 0, $\operatorname{mld}(W_0 \ni 0, \mathfrak{a}_*|_{W_0}) = -\infty$. By [Liu26b,

Lemma D.14] for $R = \Bbbk[[X', Y', Z', t]]$ and $f = X'Y' + Z'^2$, there exists a prime divisor E over $\operatorname{Spec} \Bbbk[[X', Y', Z', t]]$ with center the closed point and

$$a(E, \operatorname{Spec} \Bbbk[[X', Y', Z', t]], \operatorname{div}(X'Y' + Z'^2) + \mathfrak{a}_*) < 0.$$

The tuple $(X'Y' + Z'^2, (g_{*,j})_j, t)$ is a compatible generic limit of the tuples

$$(X'Y' + Z'^2 + t^{n_i}, (g_{ij})_j, t).$$

By Lemma 3.15 for the trivial group, for any $i \gg 0$ there exists a prime divisor E_i over $\mathbb{A}^4$ with center 0 such that, for any j,

$$a(E_i, \mathbb{A}^4, 0) = a(E, \operatorname{Spec} \Bbbk[[X', Y', Z', t]], 0), \quad \operatorname{mult}_{E_i}(X'Y' + Z'^2 + t^{n_i}) = \operatorname{mult}_E(X'Y' + Z'^2),$$
$$\operatorname{mult}_{E_i} g_{ij} = \operatorname{mult}_E g_{*,j}, \quad \text{and} \quad \operatorname{mult}_{E_i} t = \operatorname{mult}_E t,$$

so

$$a(E_i, \mathbb{A}^4, \operatorname{div}(X'Y' + Z'^2 + t^{n_i}) + \mathfrak{a}_i) < 0.$$

By (B.194) and Theorem 2.17(1), for the trivial subgroup H,

$$\operatorname{mld}(\mathbb{A}^4 \ni 0, \operatorname{div}(X'Y' + Z'^2 + t^{n_i}), \mathfrak{a}_i) = \operatorname{mld}(W_{n_i} \ni 0, \Delta'_i + \eta \operatorname{div}(t)) \geq 0,$$

a contradiction.

In particular (W_n, Δ'_*) is lc near 0. For any prime divisor E over W_n with center 0,

$$a(E, W_n, \Delta'_*) = a(E, W_n, \Delta'_* + \eta \operatorname{div}(t)) + \eta \operatorname{mult}_E t \geq \eta, \tag{B.197}$$

so $b'_* := \operatorname{mld}((W_n \ni 0, \Delta'_*)/G') \geq \eta/M$.

We show that

$$b_A := \operatorname{mld}\Big(\Big(W_0 \ni 0, \sum_j \kappa_j \operatorname{div}(A_j) + \mu \operatorname{div}(t)\Big)/G'\Big) \geq b_*, \tag{B.198}$$

where $b_* = \operatorname{mld}((W_* \ni 0, D_*)/G)$ is the value in (B.184). Let E be a prime divisor over W_0, in the coordinates X', Y', Z', t, with center 0. Let k run over the integers in the residue class of the k_i modulo $|G|$, which does not depend on i. For any such k, $\Psi_{k,\phi}$ is equivariant for G' on the new germ and G on the old germ. Possibly replacing ϕ with a truncation of sufficiently high t-order, which changes $\Psi^*_{k,\phi} h_{*,j}$ only by terms of order along E greater than $(k\delta_j + \rho_j) \operatorname{mult}_E t + \operatorname{mult}_E A_j$, we may assume that ϕ is a Laurent polynomial in t. For $k \gg 0$, $\Psi_{k,\phi}$ is then a birational morphism of germs $(W_0 \ni 0) \to (W_0 \ni 0)$ which is an isomorphism over $(t \neq 0)$, and $\Psi^*_{k,\phi}$ multiplies a generator of ω_{W_0} by $t^{(u_1+u_2)k+\rho_p+\rho_v}$. As for (B.196), any monomial of w-degree $D \geq \delta_j + 1$ of $t^{-k\delta_j-\rho_j} \Psi^*_{k,\phi} h_{*,j}$ has a coefficient of t-order at least $k(D - \delta_j) - c_2(D + 1) \geq k - c_2(\delta_j + 2)$ for $k > c_2$, so the terms of $\Psi^*_{k,\phi} h_{*,j}$ of w-degree greater than δ_j have order along E at least $(k\delta_j + \rho_j + k - c_2(\delta_j + 2)) \operatorname{mult}_E t$, and for $k \gg 0$,

$$\operatorname{mult}_E \Psi^*_{k,\phi} h_{*,j} = (k\delta_j + \rho_j) \operatorname{mult}_E t + \operatorname{mult}_E A_j.$$

By (B.186) and $\mu = \sum_j \kappa_j \rho_j - \rho_p - \rho_v$,

$$a(E, W_*, D_*) = a\Big(E, W_0, \sum_j \kappa_j \operatorname{div}(A_j) + \mu \operatorname{div}(t)\Big) + k\Big(u_1 + u_2 - \sum_j \kappa_j \delta_j\Big) \operatorname{mult}_E t, \tag{B.199}$$

where the last term is zero. The center of E on the old germ is 0 because $\Psi_{k,\phi}(0) = 0$. Let $e_E := |I_{G'}(E)|$. By the equivariance of $\Psi_{k,\phi}$, $I_{G'}(E)$ maps isomorphically onto $I_G(E)$, so by (B.199) and the definition of b_*,

$$\frac{1}{e_E} a\Big(E, W_0, \sum_j \kappa_j \operatorname{div}(A_j) + \mu \operatorname{div}(t)\Big) \geq b_*,$$

and (B.198) follows.

By Lemma B.22(3), with $w = (2u_1, 2u_2, u_1 + u_2)$, which satisfies $w_x + w_y = 2w_z$, with $n = \lim_i n_i$, G' in place of G, δ_j in place of d_j, the polynomials $g_{*,j}$ and A_j in place of Q_j and P_j of the statement, and the label μ of t,

$$b'_* \geq b_A.$$

With (B.198),

$$b'_* \geq b_*. \tag{B.200}$$

Step 6. In this step we prove $\liminf_i \beta_i \geq b_*$ when $n < +\infty$, or when $n = +\infty$ and the limit on the double cover is not of the form treated in **Step 7**.

If $n < +\infty$, then by Lemma B.21(2), for $H := G'$, the marks g_{ij} and t, and the labels κ_j and μ (the mark t omitted if $\mu = 0$), $b'_i = b'_*$ for any $i \gg 0$. By (B.195) and (B.200), $\liminf_i \beta_i \geq b'_* \geq b_*$.

Let $n = +\infty$, let $\widetilde{W} := \operatorname{Spec} \Bbbk[[P, V, t]]$, and let

$$B'_* := \sum_j \kappa_j \operatorname{div}\left(g_{*,j}(P^2, -V^2, PV, t)\right) + \mu \operatorname{div}(t) \quad \text{on } \widetilde{W}.$$

The double cover $\widetilde{W} \to W_0$ is étale outside $(P = V = 0)$, and the pullbacks of $K_{W_0} + \Delta'_*$ and $K_{W_0} + \Delta'_* + \eta \operatorname{div}(t)$ are $K_{\widetilde{W}} + B'_*$ and $K_{\widetilde{W}} + B'_* + \eta \operatorname{div}(t)$. Hence $(\widetilde{W}, B'_* + \eta \operatorname{div}(t))$ is lc, and as in (B.197), any prime divisor over $\widetilde{W}$ with center 0 has log discrepancy at least η with respect to $(\widetilde{W}, B'_*)$. One of the following holds.

(1) $(\widetilde{W}, B'_*)$ is klt near 0.
(2) $\lfloor B'_* \rfloor \neq 0$ near 0.
(3) $(\widetilde{W}, B'_*)$ is not klt near 0 and $\lfloor B'_* \rfloor = 0$ near 0.

In case (3), 0 is not an lc center and no lc center is a divisor, so by [Kaw15, Theorem 1.2] the smallest lc center C' of $(\widetilde{W}, B'_*)$, which is contained in any lc center, exists and is normal. Thus C' is the unique lc curve through 0 and C' is smooth. Since ι preserves B'_*, $\iota(C') = C'$. If ι fixes C' pointwise, then $C' \subset (P = V = 0)$, so $C' = (P = V = 0)$.

Assume that (1) or (2) holds, or that (3) holds and ι acts non-trivially on C'. Then by Lemma B.22(1), for the integers n_i, the group G', the marks g_{ij} and t, and the labels κ_j and μ (the mark t omitted if $\mu = 0$), we have $\liminf_i b'_i \geq b'_*$. By (B.195) and (B.200), $\liminf_i \beta_i \geq b_*$.

Step 7. In this step we treat the case where $n = +\infty$, (3) holds and $C' = (P = V = 0)$.

Near 0, the non-klt locus of $(\widetilde{W}, B'_*)$ is C', so (W_0, Δ'_*) is klt outside the axis $(X' = Y' = Z' = 0)$. By (B.193), $\beta_i \leq a_i \leq 2$ (Proposition B.14(1)) and $e_{F_i} \leq M$,

$$a(F_i, W_{n_i}, 0) - \sum_j \kappa_j \operatorname{mult}_{F_i} g_{ij} - \mu \operatorname{mult}_{F_i} t = e_{F_i} \beta_i \leq 2M, \tag{B.201}$$

and by (B.194), $\eta \operatorname{mult}_{F_i} t \leq e_{F_i} \beta_i \leq 2M$.

Assume first that there exist a real number R' and an infinite set I of indices with $\min(\operatorname{mult}_{F_i} X', \operatorname{mult}_{F_i} Y', \operatorname{mult}_{F_i} Z') \leq R'$ for any $i \in I$. By Proposition B.20(1) for the integers n_i ($i \in I$), the marks g_{ij} and t, the labels κ_j and μ (the mark t omitted if $\mu = 0$), the bound $2M$ and R', there exists a real number c_3 such that

$$a(F_i, W_{n_i}, 0) \leq \lambda := 2M + c_3 \max\{\kappa_j, \mu\}$$

for any $i \in I$ with $i \gg 0$. The pair $(W_{n_i}, \frac{1}{2}\Delta'_i)$ is klt near 0, since (W_{n_i}, Δ'_i) is lc and $a(E, W_{n_i}, 0) \geq 1$ for any exceptional prime divisor E over the canonical threefold W_{n_i}. By Proposition 3.14(2)(a) for W_{n_i}, the group G', the labelled boundary $\frac{1}{2}\Delta'_i$ with labels in the finite set

$$\Gamma' := \begin{cases} \{\frac{1}{2}\kappa_j\}_j \cup \{\frac{1}{2}\mu\} & \text{if } \mu > 0, \\ \{\frac{1}{2}\kappa_j\}_j & \text{if } \mu = 0 \end{cases}$$

(if $\mu = 0$, then $\frac{1}{2}\Delta'_i$ does not contain $\operatorname{div}(t)$), and λ, for $i \in I$ with $n_i \geq N_\lambda(M, \Gamma', \lambda)$, the birational transform $F_{i,0}$ of F_i under $(X', Y', Z', t) \to (X', Y' + t^{n_i}/X', Z', t)$ is a prime divisor over W_0 with center 0 whose inertia group in G' equals the inertia group of F_i, and

$$a(F_{i,0}, W_0, 0) = a(F_i, W_{n_i}, 0), \quad \operatorname{mult}_{F_{i,0}}(g_{ij}|_{W_0}) = \operatorname{mult}_{F_i} g_{ij},$$
$$\text{and} \quad \operatorname{mult}_{F_{i,0}} t = \operatorname{mult}_{F_i} t \text{ if } \mu > 0,$$

since all labels of $\frac{1}{2}\Delta'_i$ are positive. With (B.201),

$$\beta_i = \frac{1}{e_{F_{i,0}}} a\Big(F_{i,0}, W_0, \sum_j \kappa_j \operatorname{div}(g_{ij}|_{W_0}) + \mu \operatorname{div}(t)\Big)$$
$$\geq \operatorname{mld}\Big(\Big(W_0 \ni 0, \sum_j \kappa_j \operatorname{div}(g_{ij}|_{W_0}) + \mu \operatorname{div}(t)\Big)/G'\Big).$$

By Lemma B.21(2), for $n = +\infty$, $H := G'$, the marks $g_{ij}|_{W_0}$ and t, and the labels κ_j and μ (the mark t omitted if $\mu = 0$), the right side equals b'_* for any $i \gg 0$, since (W_0, Δ'_*) is lc near 0. By (B.200), $\liminf_{i\in I} \beta_i \geq b_*$.

Assume next that

$$\lim_i \min(\operatorname{mult}_{F_i} X', \operatorname{mult}_{F_i} Y', \operatorname{mult}_{F_i} Z') = +\infty.$$

We derive a contradiction. Let

$$\widetilde{Q}_j := g_{*,j}(P^2, -V^2, PV, t) \quad \text{and} \quad \widetilde{A}_j := A_j(P^2, -V^2, PV, t) \quad \text{in } \Bbbk[[t]][P, V].$$

The completed germ of $\widetilde{W}$ at the generic point of C' is $\operatorname{Spec} \Bbbk((t))[[P, V]]$. Since t is a unit on it, the restriction of B'_* to it is $\sum_j \kappa_j \operatorname{div}(\widetilde{Q}_j)$. Let $\mathbb{K}$ be an algebraic closure of $\Bbbk((t))$. Since the curve $C' = (P = V = 0)$ is the unique lc center of $(\widetilde{W}, B'_*)$ through 0 and $\lfloor B'_* \rfloor = 0$, the pair $(\mathbb{A}^2_{\mathbb{K}}, \sum_j \kappa_j \operatorname{div}(\widetilde{Q}_j))$, which is the base change to $\mathbb{K}$ of the restriction of $(\widetilde{W}, B'_*)$ to $\operatorname{Spec} \Bbbk((t))[[P, V]]$, is lc near 0 and not klt at 0, and any component through 0 has coefficient less than one, as base change along the regular morphism $\operatorname{Spec} \mathbb{K}[[P, V]] \to \operatorname{Spec} \Bbbk((t))[[P, V]]$ preserves log resolutions and log discrepancies [Kaw21, Section 2]. Any monomial of $\widetilde{Q}_j$ has (u_1, u_2)-weighted degree at most δ_j, $\widetilde{A}_j$ is the part of degree δ_j, and $\sum_j \kappa_j \delta_j = u_1 + u_2$ by (B.186). By (B.198) and $b_* > 0$, any prime divisor over W_0 with center 0 has positive log discrepancy with respect to

$$\Big(W_0, \sum_j \kappa_j \operatorname{div}(A_j) + \mu \operatorname{div}(t)\Big).$$

Since $\widetilde{W} \to W_0$ is étale outside $(P = V = 0)$ and $\mu \geq 0$, the pair $(\widetilde{W}, \sum_j \kappa_j \operatorname{div}(\widetilde{A}_j))$ has positive minimal log discrepancy at the closed point. Let

$$\widetilde{G'} := \{(g, \alpha) \mid g \in G', \ \alpha^2 = \chi_{X'}(g)\}$$

act by

$$(P, V, t) \to (\alpha P, \chi_{Z'}(g)\alpha^{-1} V, \chi_t(g) t),$$

so that $\iota \in \widetilde{G'}$. By Proposition 7.4(2), with δ_j in place of d_j, the weights (u_1, u_2) in place of (w_1, w_2), where $u_1 > u_2$, the coordinates (P, V) in place of (p, v), the polynomials $\widetilde{Q}_j$ and $\widetilde{A}_j$ in place of Q_j and P_j of the statement, and the group $\widetilde{G'}$, there exists $\psi \in \Bbbk[[t]][Y']$ such that $\sigma(V, t) := V\psi(-V^2, t)$ has only exponents m of V with $0 < mu_2 < u_1$, and such that for $\widehat{P} := P + \sigma(V, t)$ and $\widehat{V} := V$,

$$\widetilde{Q}_j(P, V, t) = \widetilde{A}_j(\widehat{P}, \widehat{V}, t) \quad \text{for any } j.$$

The substitution

$$\widehat{X} := X' + 2Z'\psi(Y', t) - Y'\psi(Y', t)^2, \quad \widehat{Y} := Y', \quad \text{and} \quad \widehat{Z} := Z' - Y'\psi(Y', t)$$

satisfies $\widehat{X}\widehat{Y}+\widehat{Z}^2 = X'Y'+Z'^2$. Any element $(P,V,t)\to(\alpha P,\beta V,\gamma t)$ of $\widetilde{G'}$ satisfies

$$(\alpha\beta)^2 = \chi^2_{Z'} = \gamma^{n_i},$$

because G' preserves $X'Y'+Z'^2+t^{n_i}$ up to a character. Hence, by the last sentence of Proposition 7.4(2) for $N := n_i$ and the group $\widetilde{G'}$, with ψ in place of ϕ there, the substitution is a G'-equivariant automorphism of the completed germ of any W_{n_i}, and on W_0 any $g_{*,j}$, written in $\widehat{X},\widehat{Y},\widehat{Z},t$, has no monomial of w-degree less than δ_j.

For a polynomial $\psi'\in\mathbb{C}[t][Y']$ whose exponents m' of Y' satisfy $(2m'+1)u_2<u_1$ and such that $Y'\psi'(Y',t)$ has the character of Z' (as $Y'\psi(Y',t)$ does, by the G'-equivariance of the substitution of ψ), let $\widehat{X}_{\psi'},\widehat{Y}_{\psi'},\widehat{Z}_{\psi'}$ be given by the substitution above with ψ' in place of ψ, and consider the condition

$(*)_i$ for any j, after writing g_{ij} in $\widehat{X}_{\psi'},\widehat{Y}_{\psi'},\widehat{Z}_{\psi'},t$ and reducing $\widehat{X}_{\psi'}\widehat{Y}_{\psi'}$ to $-\widehat{Z}^2_{\psi'}-t^{n_i}$, any monomial

$$t^e\widehat{X}^{m_1}_{\psi'}\widehat{Y}^{m_2}_{\psi'}\widehat{Z}^{m_3}_{\psi'}\quad\text{with}\quad D := 2u_1m_1+2u_2m_2+(u_1+u_2)m_3<\delta_j,$$

where D is its w-degree, satisfies $e\ge\delta_j-D$.

Condition $(*)_i$ says that $t^{-\delta_j}g_{ij}$ is regular after the scaling $\widehat{X}_{\psi'}=t^{2u_1}X''$, $\widehat{Y}_{\psi'}=t^{2u_2}Y''$, and $\widehat{Z}_{\psi'}=t^{u_1+u_2}Z''$. Each of X',Z' is a polynomial in $\widehat{X}_{\psi'},\widehat{Y}_{\psi'},\widehat{Z}_{\psi'}$ without constant term, so a monomial of total degree m_0 in X',Y',Z' only contributes monomials of total degree at least m_0 in $\widehat{X}_{\psi'},\widehat{Y}_{\psi'},\widehat{Z}_{\psi'}$, hence of w-degree at least $2u_2m_0$. Thus only the monomials of g_{ij} of total degree less than $\delta_j/(2u_2)$ and t-order less than δ_j, and the coefficients of ψ' of t-order less than $\max_j\delta_j$, enter $(*)_i$, and for $n_i\ge\max_j\delta_j$ the term t^{n_i} does not enter. Hence $(*)_i$ is a finite system $S(\mathbf{c}_i,\psi')=0$ of polynomial equations in the finitely many coefficients $\mathbf{c}_i$ of $(g_{ij})_j$ entering $(*)_i$ and in finitely many coefficients of ψ'. For the corresponding coefficients $\mathbf{c}_*$ of $(g_{*,j})_j$, the truncation of ψ at t-order $\max_j\delta_j$ solves $S(\mathbf{c}_*,\cdot)=0$. As for $U\subset Z_{\ell'}$ and $V'\to U$ in **Step 4**, with the jets of the tuples $(g_{ij})_j$ in place of those of the $(h_{ij})_j$, the system S in place of $\mathcal{Q}$ and the coefficients of the truncation of ψ in place of those of ϕ, possibly passing to a subsequence, there exist polynomials $\psi_i\in\mathbb{C}[t][Y']$ as ψ' above satisfying $(*)_i$. Let

$$\sigma_i(V,t) := V\psi_i(-V^2,t) = \sum_{m\in2\mathbb{Z}+1,\ 0<m<u_1/u_2}\sigma_{i,m}(t)V^m.$$

We show that k_i+1 satisfies the conditions of (B.192) for any $i\gg0$. Let $\widehat{X}_i := \widehat{X}_{\psi_i}$, $\widehat{Y}_i := \widehat{Y}_{\psi_i}$, and $\widehat{Z}_i := \widehat{Z}_{\psi_i}$. Since $\widehat{X}_i,\widehat{Y}_i,\widehat{Z}_i$ are combinations of X',Y',Z' with regular coefficients,

$$\min(\operatorname{mult}_{F_i}\widehat{X}_i,\operatorname{mult}_{F_i}\widehat{Y}_i,\operatorname{mult}_{F_i}\widehat{Z}_i)\ge\min(\operatorname{mult}_{F_i}X',\operatorname{mult}_{F_i}Y',\operatorname{mult}_{F_i}Z')\to+\infty,$$

whereas $\operatorname{mult}_{F_i}t\le 2M/\eta$. Thus $\widehat{X}_i/t^{2u_1}$, $\widehat{Y}_i/t^{2u_2}$, $\widehat{Z}_i/t^{u_1+u_2}$, and t have positive order along F_i for $i\gg0$, so F_i has center the origin after one more scaling. Also $n_i-2(u_1+u_2)\ge2$ for $i\gg0$, which is the first condition of (B.192) for k_i+1, and the marks $t^{-\delta_j}g_{ij}$ are regular after one more scaling by $(*)_i$. In the cover coordinates, $p-\phi_i(t,v)=t^{u_1k_i+\rho_p}P$ and $v=t^{u_2k_i+\rho_v}V$, and we let $\widehat{P}_i := P+\sigma_i(V,t)$, so

$$\widehat{P}_i = t^{-(u_1k_i+\rho_p)}\Big(p-\phi_i(t,v)+\sum_m t^{(u_1-mu_2)k_i+\rho_p-m\rho_v}\sigma_{i,m}(t)v^m\Big).$$

Hence the composite of the cone shear of ϕ_i, the scaling at level k_i, the substitution of ψ_i and one more scaling is the composite of the cone shear of

$$\phi_i^+ := \phi_i-\sum_m t^{(u_1-mu_2)k_i+\rho_p-m\rho_v}\sigma_{i,m}(t)v^m$$

with the scaling at level k_i+1. Since $u_1-mu_2\ge1$, $\sigma_{i,m}\in\mathbb{C}[t]$, $k_i\ge\ell_i+\Lambda$, and $\Lambda\ge|\rho_p-m\rho_v|$ for $1\le m\le u_1/u_2$, the difference $\phi_i^+-\phi_i$ has no pole in t and vanishes to t-order at least ℓ_i. The exponents of v in $\phi_i^+-\phi_i$ are odd and less than u_1/u_2, and $\phi_i^+-\phi_i$ has the characters of ϕ_i

by the G'-equivariance of the substitution of ψ_i. Thus $\phi_i^+ \in \mathcal{P}_i$. Hence $k_i + 1$ satisfies (B.192) with ϕ_i^+, contrary to the maximality of k_i.

Step 8. In this step we conclude the proof.

By **Steps 5–7**, possibly passing to a subsequence, $\liminf_i \beta_i \geq b_*$. Since F_i computes a_i and $c_{ij} \leq \kappa_j$,

$$0 \leq a_i - \beta_i = \sum_j (\kappa_j - c_{ij}) w_{ij}.$$

By (B.184), $\limsup_i (a_i - \beta_i) \leq b_* - \liminf_i \beta_i \leq 0$, so $\lim_i (a_i - \beta_i) = 0$. By Proposition 10.5(3), possibly passing to a further subsequence, $(R_i)_i$ is strictly increasing with a positive limit. But $0 \leq R_i \leq a_i - \beta_i$, so $\lim_i R_i = 0$, a contradiction. Thus $(a_i)_{i \geq 1}$ is not strictly increasing. □

Appendix C. Proofs for unbounded groups

This section proves Theorems 12.2, 12.3 and 12.4.

C.1. **Quadratic rank two.** This subsection proves Theorem 12.2.

Proof of Theorem 12.2, Steps A–D. Let $(W \ni o)$ be an isolated Gorenstein terminal hypersurface threefold germ of quadratic rank 2, G a finite abelian group acting faithfully on W and fixing o, and h a mark on $(W \ni o)$ such that $(W, \frac{1}{m}\operatorname{div}(h))$ is lc near o. By [Liu26a, Lemma 2.6], the radial function q of $(W \ni o, G, h, m)$ satisfies

$$q(c) = \inf_F \frac{a(F, W, 0) - \frac{c}{m}\operatorname{mult}_F h}{e_F} \quad \text{for any } c \in [0, 1], \tag{C.1}$$

where F runs over the prime divisors over W with center o.

Assume that the theorem does not hold for m and ϵ. Then for any integer $i \geq 2$ there exist $(W_i \ni o_i)$, G_i, and h_i as in the theorem with $|G_i| \geq i$, and a real number $s_i \in (1 - \frac{1}{i}, 1]$, such that

$$q_i(s_i) - q_i(1) \geq \epsilon,$$

where q_i is the radial function of $(W_i \ni o_i, G_i, h_i, m)$. In particular $s_i < 1$.

Step A. In this step we choose coefficients $\lambda_i < 1$ at which the values $q_i(\lambda_i)$ strictly increase.

The blow-up of o_i in a smooth fourfold germ $(M_i \ni o_i)$ containing W_i induces prime divisors E_i^+ and E_i^- over W_i, the planes whose union is the projectivized tangent cone, with

$$a(E_i^+, W_i, 0) = a(E_i^-, W_i, 0) = 2. \tag{C.2}$$

By (C.1) with $c = 0$, $q_i(0) \leq 2$ for any i. By Lemma 4.5, after passing to a subsequence, there exist $\lambda_i \in [s_i, 1)$, strictly increasing with limit 1, such that

$$\alpha_i := q_i(\lambda_i) \text{ is strictly increasing with limit } \alpha_\infty > 0.$$

Discarding finitely many indices, we may assume that $\alpha_i > 0$ for any i. For any prime divisor F over a neighborhood of o_i, the number $a(F, W_i, \frac{c}{m}\operatorname{div}(h_i))$ is affine in c, positive at $c = 0$ since W_i is terminal, and non-negative at $c = 1$ since $(W_i, \frac{1}{m}\operatorname{div}(h_i))$ is lc near o_i. Thus $(W_i, \frac{\lambda_i}{m}\operatorname{div}(h_i))$ is klt near o_i. The set

$$\Gamma := \{\frac{\lambda_i}{m}\}_i \subset [0, 1]$$

is a DCC set.

Step B. In this step we replace $(W_i \ni o_i, h_i)$ with polynomial data in split form.

By [Liu26a, Lemma 5.4], we may take M_i with a faithful action of G_i such that $W_i = (f_i = 0)$ for a G_i-semi-invariant regular function f_i on M_i, there exist G_i-eigenparameters $x_{i,1}, \dots, x_{i,4}$ of M_i at o_i, and h_i is the restriction of a G_i-semi-invariant regular function on M_i.

By [Liu26b, Definition 8.1] with an auxiliary mark (one exists since W_i is Gorenstein terminal, and the conclusions do not involve it), there exist formal coordinates x, y, z, t of M_i at o_i, on which G_i acts diagonally, such that the completed local equation of W_i is the G_i-semi-invariant series

$$\Phi_i := xy + \widehat{g}_i(z, t) \quad \text{or} \quad \Phi_i := x^2 + y^2 + \widehat{g}_i(z, t).$$

In the second case we may assume that $\chi_x \neq \chi_y$, since otherwise $x + \sqrt{-1}y$ and $x - \sqrt{-1}y$ are G_i-eigenfunctions and the first case holds. Since the quadratic part of Φ_i has rank 2,

$$n_i := \operatorname{mult}_0 \widehat{g}_i \geq 3.$$

Both (x, y, z, t) and $(x_{i,1}, \dots, x_{i,4})$ induce G_i-eigenbases of the cotangent space of M_i at o_i. Reordering $x_{i,1}, \dots, x_{i,4}$, we may assume that $x_{i,1}, x_{i,2}, x_{i,3}, x_{i,4}$ have the characters of x, y, z, t. Let Υ_i be the G_i-equivariant automorphism of $\mathbb{C}[[x_{i,1}, \dots, x_{i,4}]]$ mapping x, y, z, t to $x_{i,1}, x_{i,2}, x_{i,3}, x_{i,4}$. The expansion $\widehat{f_i}$ of f_i is a unit times Φ_i, so there exists a G_i-semi-invariant, hence G_i-invariant, unit v_i with

$$v_i \Upsilon_i(\widehat{f_i}) = \Phi_i(x_{i,1}, x_{i,2}, x_{i,3}, x_{i,4}).$$

By Lemma 4.6 with $n=4$, λ_i in place of the coefficient t, $\mathcal{P}=\emptyset$, $\psi=\Upsilon_i$, $u=1$, $v=v_i$, and a sufficiently large integer $L>n_i$, there exist a germ $(W_i' \ni 0)=(f_i'=0)\subset(\mathbb{A}^4\ni 0)$, where f_i' is the truncation of $\Phi_i(x_{i,1},\dots,x_{i,4})$ in degrees less than L, with the diagonal action of G_i in which the coordinates have the characters of x,y,z,t, and a G_i-semi-invariant polynomial h_i', such that W_i' is isolated Gorenstein terminal at 0, $(W_i', \frac{1}{m}\operatorname{div}(h_i'))$ is lc near 0 by Lemma 4.6(3), and

$$\operatorname{mld}\Big((W_i'\ni 0, \tfrac{\lambda_i}{m}\operatorname{div}(h_i'))/G_i\Big)=\operatorname{mld}\Big((W_i\ni o_i, \tfrac{\lambda_i}{m}\operatorname{div}(h_i))/G_i\Big)=\alpha_i$$

by Lemma 4.6(1). Since the completions of W_i at o_i and of W_i' at 0 are G_i-equivariantly isomorphic, G_i acts faithfully on W_i'. We denote the coordinates of $\mathbb{A}^4$ again by x,y,z,t, and in the second case we replace x and y with $x+\sqrt{-1}y$ and $x-\sqrt{-1}y$. Then $f_i'=xy+g_i(z,t)$, where g_i is the truncation of $\widehat{g}_i$ in degrees less than L, so $\operatorname{mult}_0 g_i=n_i$. In the second case, an element $\rho\in G_i$ with $\chi_x(\rho)\neq\chi_y(\rho)$ before the replacement satisfies $\chi_y(\rho)=-\chi_x(\rho)$, since $\chi_x^2=\chi_y^2$, and hence exchanges the new coordinates x and y.

Possibly replacing $(W_i\ni o_i, h_i)$ with $(W_i'\ni 0, h_i')$, we may assume that the following hold for any i.

(1) $W_i=(f_i=0)\subset\mathbb{A}^4$ with $f_i=xy+g_i(z,t)$ is an isolated Gorenstein terminal germ at 0, where g_i is a polynomial with $\operatorname{mult}_0 g_i=n_i\geq 3$.
(2) G_i acts faithfully on W_i, linearly on $\mathbb{A}^4$, diagonally on z,t and monomially on $\{x,y\}$, and preserves f_i up to a character. Either G_i is diagonal, or some element of G_i exchanges x and y.
(3) h_i is a G_i-semi-invariant polynomial and $(W_i, \frac{1}{m}\operatorname{div}(h_i))$ is lc near 0.
(4) $\alpha_i=q_i(\lambda_i)$, where q_i is the radial function of $(W_i\ni 0, G_i, h_i, m)$.

As in Step A, $(W_i, \frac{\lambda_i}{m}\operatorname{div}(h_i))$ is klt near 0. From now on, prime divisors over W_i have center 0, and $e_F=|I_{G_i}(F)|$.

Let $G_i^{\mathrm{s}}\subset G_i$ be the subgroup preserving each of x and y. If $\tau\in G_i$ exchanges x and y, say $\tau(x)=c_\tau y$, then for any $\rho\in G_i^{\mathrm{s}}$ the equality $\tau(\rho(x))=\rho(\tau(x))$ reads $\chi_x(\rho)c_\tau y=c_\tau\chi_y(\rho)y$. Thus

$$\chi_x(\rho)=\chi_y(\rho)\quad\text{for any } \rho\in G_i^{\mathrm{s}} \text{ if } G_i\neq G_i^{\mathrm{s}}. \tag{C.3}$$

Any polynomial is congruent modulo f_i to a unique element of $\mathbb{C}[z,t]+x\mathbb{C}[x,z,t]+y\mathbb{C}[y,z,t]$. Replacing h_i with the unique such representative, which is again G_i-semi-invariant by uniqueness, we may write

$$h_i=h_{i,0}(z,t)+\sum_{j\geq 1}\big(x^j h_{i,j}^+(z,t)+y^j h_{i,j}^-(z,t)\big). \tag{C.4}$$

The blow-up of $0\in\mathbb{A}^4$ induces prime divisors $E_i^\pm$ over W_i satisfying (C.2), and

$$\operatorname{mult}_{E_i^+} h_i\geq\operatorname{mult}_0 h_i.$$

Since $(W_i, \frac{1}{m}\operatorname{div}(h_i))$ is lc near 0, $\frac{1}{m}\operatorname{mult}_{E_i^+} h_i\leq a(E_i^+, W_i, 0)=2$, so

$$\operatorname{mult}_0 h_i\leq 2m.$$

Step C. In this step we show that $(n_i)_i$ is bounded. Assume, after passing to a subsequence, that $n_i\to+\infty$ and that $n_i>4m+2$ for any i.

Let

$$\mathbf{b}_i:=\Big(\Big\lfloor\frac{n_i}{2}\Big\rfloor, \Big\lceil\frac{n_i}{2}\Big\rceil, 1, 1\Big).$$

The lowest $\mathbf{b}_i$-weight part of f_i is $xy+g_i^{(n_i)}$, where $g_i^{(n_i)}\neq 0$ is the homogeneous part of degree n_i of g_i. The polynomial $xy+g_i^{(n_i)}$ is irreducible and not divisible by any coordinate. By [Liu26b, Lemma 4.12], the $\mathbf{b}_i$-weighted blow-up induces a prime divisor E_i° over W_i with $a(E_i^\circ, W_i, 0)=n_i+2-n_i=2$ and

$$\operatorname{mult}_{E_i^\circ} h_i\geq\operatorname{wt}_{\mathbf{b}_i}(h_i).$$

Since $(W_i, \frac{1}{m}\operatorname{div}(h_i))$ is lc near 0, $\operatorname{wt}_{\mathbf{b}_i}(h_i) \le 2m$. In (C.4), any monomial divisible by x or y has $\mathbf{b}_i$-weight at least $\lfloor \frac{n_i}{2} \rfloor > 2m$, and a monomial of $h_{i,0}$ has $\mathbf{b}_i$-weight equal to the degree. Thus

$$h_{i,0} \neq 0 \quad \text{and} \quad \operatorname{mult}_0 h_{i,0} \le 2m. \tag{C.5}$$

Let $\mathbb{T}^2 := \mathbb{R}^2/\mathbb{Z}^2$, and let $\Pi_i \supset \Pi_i^{\mathrm{s}}$ be the images of $G_i \supset G_i^{\mathrm{s}}$ under the homomorphism to $\mathbb{T}^2$ mapping ρ to the class of (ξ_z, ξ_t) with $\chi_z(\rho) = e^{2\pi\sqrt{-1}\xi_z}$ and $\chi_t(\rho) = e^{2\pi\sqrt{-1}\xi_t}$. Then $[\Pi_i : \Pi_i^{\mathrm{s}}] \le 2$. After passing to a subsequence, Π_i and Π_i^{s} converge in the Hausdorff distance to closed subgroups $\Pi \supset \Pi^{\mathrm{s}}$ of $\mathbb{T}^2$ with $[\Pi : \Pi^{\mathrm{s}}] \le 2$, so the identity components satisfy $\Pi^0 = (\Pi^{\mathrm{s}})^0$. Moreover, $\Pi_i \subset \Pi$ and $\Pi_i^{\mathrm{s}} \subset \Pi^{\mathrm{s}}$ for $i \gg 0$ by the following fact, used again in Step D. If finite subgroups of a real torus converge in the Hausdorff distance to a closed subgroup, then they are contained in it from some index on, since the closed subgroup is the common kernel of finitely many characters $\kappa_1, \dots, \kappa_k$ of the torus trivial on it, and from some index on each κ_j maps the finite subgroups to subgroups of S^1 contained in a small neighborhood of 1, which are trivial.

For a prime divisor F over W_i, the group $I_{G_i}(F)$ acts faithfully on $\mathfrak{m}_F/\mathfrak{m}_F^2$, where $\mathfrak{m}_F$ is the maximal ideal of the valuation ring of F [Liu26a, Lemma 2.6(1)]. Let $\rho_F \in I_{G_i}(F)$ act on $\mathfrak{m}_F/\mathfrak{m}_F^2$ by $e^{2\pi\sqrt{-1}/e_F}$. Since ρ_F acts trivially on the residue field of F, comparing the leading terms of z and $\rho_F(z) = \chi_z(\rho_F)z$ in a uniformizer which is a ρ_F-eigenvector shows $\chi_z(\rho_F) = e^{2\pi\sqrt{-1}\operatorname{mult}_F z/e_F}$, and similarly for t. Thus the class of

$$\xi_F := \frac{1}{e_F}(\operatorname{mult}_F z, \operatorname{mult}_F t) \in \mathbb{R}^2_{>0}$$

belongs to Π_i, and to Π_i^{s} if $I_{G_i}(F) \subset G_i^{\mathrm{s}}$.

Assume that Π is finite. Then $|\Pi_i| \le |\Pi|$ for $i \gg 0$. Let $H_i \subset G_i$ be the kernel of $G_i \to \Pi_i$. Since $g_i \neq 0$ is fixed by H_i, $\chi_x\chi_y = 1$ on H_i. If G_i is diagonal, then $H_i = G_i \cap \mathbf{T}_{(1,-1,0,0)}$, $|H_i| \to +\infty$, and $[G_i : H_i] \le |\Pi|$. By [Liu26b, Theorem 8.3] with the labelled boundaries $\frac{\lambda_i}{m}\operatorname{div}(h_i)$ and labels in Γ, the α_i do not form a strictly increasing sequence of positive numbers, a contradiction. If $G_i \neq G_i^{\mathrm{s}}$, then by (C.3) any element of $H_i \cap G_i^{\mathrm{s}}$ acts on x and y by a scalar ζ with $\zeta^2 = 1$, so $|H_i| \le 4$ and $|G_i| \le 4|\Pi|$, contradicting $|G_i| \to +\infty$. Thus Π is infinite.

Let $\rho \in G_i^{\mathrm{s}}$ and $\xi = (\xi_x, \xi_z, \xi_t) \in \mathbb{Q}^3_{>0}$ satisfy

$$\big(\chi_x(\rho), \chi_z(\rho), \chi_t(\rho)\big) = \big(e^{2\pi\sqrt{-1}\xi_x}, e^{2\pi\sqrt{-1}\xi_z}, e^{2\pi\sqrt{-1}\xi_t}\big) \quad \text{and} \quad \xi_x < \operatorname{wt}_{(\xi_z,\xi_t)}(g_i).$$

We claim that

$$\alpha_i \le \xi_z + \xi_t - \frac{\lambda_i}{m}\min\big\{\operatorname{wt}_{(\xi_z,\xi_t)}(h_{i,0}),\ \xi_x,\ \operatorname{wt}_{(\xi_z,\xi_t)}(g_i) - \xi_x\big\}. \tag{C.6}$$

Let e_ξ be the positive rational number such that $e_\xi\xi \in \mathbb{Z}^3$ is primitive. Let e_ρ be the order of ρ. Since $\rho^{e_\rho} = 1$, $e_\rho\xi \in \mathbb{Z}^3$, so e_ρ/e_ξ is a positive integer. Let $p := \gcd(e_\xi\xi_z, e_\xi\xi_t)$, and let P_ξ be the monomial prime divisor over $\mathbb{A}^2_{z,t}$ with the coprime weights $\frac{e_\xi}{p}(\xi_z, \xi_t)$. Then p and $e_\xi\xi_x$ are coprime, and

$$e_\xi\xi_x < p\operatorname{mult}_{P_\xi} g_i = e_\xi \operatorname{wt}_{(\xi_z,\xi_t)}(g_i).$$

By [Liu26b, Proposition 3.13(1)] for G_i^{s}, with $u = x$, the integer p and the u-value $e_\xi\xi_x$, there exists a prime divisor F over W_i with

$$(\operatorname{mult}_F x, \operatorname{mult}_F y, \operatorname{mult}_F z, \operatorname{mult}_F t) = e_\xi\big(\xi_x, \operatorname{wt}_{(\xi_z,\xi_t)}(g_i) - \xi_x, \xi_z, \xi_t\big),$$
$$a(F, W_i, 0) = e_\xi(\xi_z + \xi_t),$$

and $I_{G_i^{\mathrm{s}}}(F)$ is the kernel of $\chi_x^p\chi_{P_\xi}^{-e_\xi\xi_x}$ on the inertia group J_{P_ξ} of P_ξ for the action on $\mathbb{A}^2_{z,t}$, where χ_{P_ξ} is the character on the cotangent space of P_ξ. The element ρ acts on z, t through the one-parameter subgroup of the weights of P_ξ, with parameter $e^{2\pi\sqrt{-1}p/e_\xi}$, so $\rho \in J_{P_\xi}$ and $\chi_{P_\xi}(\rho) = e^{2\pi\sqrt{-1}p/e_\xi}$. Moreover, $\chi_x(\rho) = e^{2\pi\sqrt{-1}\xi_x}$, so

$$\chi_x^p\chi_{P_\xi}^{-e_\xi\xi_x}(\rho) = e^{2\pi\sqrt{-1}p\xi_x}e^{-2\pi\sqrt{-1}p\xi_x} = 1.$$

Hence $\rho \in I_{G_i^{\mathrm{s}}}(F) \subset I_{G_i}(F)$: passing from G_i^{s} to G_i only enlarges the inertia group, and $e_F \geq e_\rho \geq e_\xi$. By (C.4),

$$\operatorname{mult}_F h_i \geq e_\xi \min\bigl\{\operatorname{wt}_{(\xi_z,\xi_t)}(h_{i,0}), \xi_x, \operatorname{wt}_{(\xi_z,\xi_t)}(g_i) - \xi_x\bigr\}.$$

Since $a(F, W_i, 0) - \frac{\lambda_i}{m}\operatorname{mult}_F h_i > 0$, (C.1) implies

$$\alpha_i \leq \frac{a(F, W_i, 0) - \frac{\lambda_i}{m}\operatorname{mult}_F h_i}{e_F} \leq \frac{a(F, W_i, 0) - \frac{\lambda_i}{m}\operatorname{mult}_F h_i}{e_\xi},$$

and (C.6) follows.

Let $\xi^\infty \in \mathbb{R}^2_{>0}$ have class in Π^{s}. Since $\Pi_i^{\mathrm{s}} \to \Pi^{\mathrm{s}}$, there exist $\rho_i \in G_i^{\mathrm{s}}$ and $(\xi_{i,z}, \xi_{i,t}) \in \mathbb{Q}^2_{>0}$ with

$$\bigl(\chi_z(\rho_i), \chi_t(\rho_i)\bigr) = \bigl(e^{2\pi\sqrt{-1}\xi_{i,z}}, e^{2\pi\sqrt{-1}\xi_{i,t}}\bigr) \quad \text{and} \quad (\xi_{i,z}, \xi_{i,t}) \to \xi^\infty. \tag{C.7}$$

Then

$$\operatorname{wt}_{(\xi_{i,z},\xi_{i,t})}(g_i) \geq n_i \min\{\xi_{i,z}, \xi_{i,t}\} \to +\infty,$$

and we take $\xi_{i,x}$ with

$$e^{2\pi\sqrt{-1}\xi_{i,x}} = \chi_x(\rho_i) \quad \text{and} \quad \Bigl|\xi_{i,x} - \frac{1}{2}\operatorname{wt}_{(\xi_{i,z},\xi_{i,t})}(g_i)\Bigr| \leq \frac{1}{2}. \tag{C.8}$$

For $i \gg 0$, $\rho = \rho_i$ and $\xi = (\xi_{i,x}, \xi_{i,z}, \xi_{i,t})$ satisfy the hypotheses of (C.6). If Π^{s} contains the classes of vectors in $\mathbb{R}^2_{>0}$ of arbitrarily small norm, which is the case if the identity component of Π^{s} is $\mathbb{T}^2$ or a circle with direction in $\mathbb{R}^2_{>0}$, then (C.6) implies $\alpha_\infty \leq \xi_z^\infty + \xi_t^\infty$ for any such ξ^∞. Letting $|\xi^\infty| \to 0$, we obtain $\alpha_\infty \leq 0$, a contradiction. Since Π is infinite, after possibly interchanging z and t, the identity component of Π and of Π^{s} is the image $\mathbf{T}_0$ of $\mathbb{R}(\omega_1, -\omega_2)$ in $\mathbb{T}^2$ for coprime integers $\omega_1 > 0$ and $\omega_2 \geq 0$. Since $\gcd(\omega_1, \omega_2) = 1$,

$$\text{the class of } \xi \in \mathbb{R}^2 \text{ belongs to } \mathbf{T}_0 \iff \omega_2\xi_z + \omega_1\xi_t \in \mathbb{Z}. \tag{C.9}$$

Since Π is infinite and $\Pi_i \subset \Pi$, the order $r_i' := |\Pi_i \cap \mathbf{T}_0|$ tends to $+\infty$, and $\Pi_i \cap \mathbf{T}_0$ is generated by the class of $\frac{1}{r_i'}(\omega_1, -\omega_2)$. All monomials $z^j t^k$ of the G_i-semi-invariant polynomial $h_{i,0}$ have the character of $h_{i,0}$ on G_i, so their $\mathbf{T}_0$-weights $\omega_1 j - \omega_2 k$ are congruent modulo r_i'. By (C.5), $h_{i,0}$ has a monomial $z^{j_0}t^{k_0}$ with $j_0 + k_0 \leq 2m$. Let $z^{k_z}t^{k_t}$ be the monomial of least degree among the monomials of $h_{i,0}$ of $\mathbf{T}_0$-weight $\omega_1 j_0 - \omega_2 k_0$. Since two monomials $z^j t^k$ and $z^{j'}t^{k'}$ satisfy $\omega_1 j - \omega_2 k = \omega_1 j' - \omega_2 k'$ if and only if $(j' - j, k' - k) \in \mathbb{Z}(\omega_2, \omega_1)$, the minimality of $k_z + k_t \leq j_0 + k_0$ implies the first equality below. After passing to a subsequence, k_z and k_t do not depend on i, and the sum $h_{i,0}^\flat$ of the terms of $h_{i,0}$ of the $\mathbf{T}_0$-weight $\omega_1 k_z - \omega_2 k_t$ satisfies

$$h_{i,0}^\flat = z^{k_z}t^{k_t}u_i(z^{\omega_2}t^{\omega_1}),\ u_i(0) \neq 0,\ k_z + k_t \leq 2m,\ \operatorname{mult}_0(h_{i,0} - h_{i,0}^\flat) \to +\infty, \tag{C.10}$$

where u_i is a polynomial in one variable. The last condition holds since any other monomial $z^j t^k$ of $h_{i,0}$ satisfies

$$|\omega_1 j - \omega_2 k| \geq r_i' - |\omega_1 k_z - \omega_2 k_t|, \quad \text{hence} \quad j + k \geq \frac{r_i' - 2m\max\{\omega_1, \omega_2\}}{\max\{\omega_1, \omega_2\}}.$$

Let $c_h, c_z, c_t \geq 0$ be fixed real numbers. We show that for $i \gg 0$,

$$\begin{aligned} &\bigl(W_i, c_h \operatorname{div}(h_i) + c_z \operatorname{div}(z) + c_t \operatorname{div}(t)\bigr) \text{ is lc near } 0 \\ \iff &\bigl(\mathbb{A}^2_{z,t}, c_h \operatorname{div}(h_{i,0}^\flat) + c_z \operatorname{div}(z) + c_t \operatorname{div}(t)\bigr) \text{ is lc near } 0. \end{aligned} \tag{C.11}$$

Let

$$\begin{aligned} B_i &:= c_h \operatorname{div}(h_i) + c_z \operatorname{div}(z) + c_t \operatorname{div}(t) \text{ on } \mathbb{A}^4, \\ B_i^0 &:= c_h \operatorname{div}(h_{i,0}) + c_z \operatorname{div}(z) + c_t \operatorname{div}(t) \text{ on } \mathbb{A}^2_{z,t}. \end{aligned}$$

By [Kaw07, Theorem], the left side is equivalent to the lc of $(\mathbb{A}^4, \operatorname{div}(f_i) + B_i)$ near 0. Let L' be an integer satisfying [Liu26b, Lemma 4.14] for $n = 4$ and for $n = 2$, at most four functions, and the finite set $\{1, c_h, c_z, c_t\}\backslash\{0\}$. Since $f_i - xy = g_i$ has order $n_i \geq L'$ for $i \gg 0$, [Liu26b,

Lemma 4.14], used in both directions, with lc near 0 compared with lc of the formal pairs by [dFEM11, Proposition 2.11], shows that the lc of $(\mathbb{A}^4, \operatorname{div}(f_i) + B_i)$ near 0 is equivalent to the lc of $(\mathbb{A}^4, \operatorname{div}(x) + \operatorname{div}(y) + B_i)$ near 0. By (C.4), $h_i|_{(x=y=0)} = h_{i,0} \neq 0$, so [Kaw07, Theorem] for the divisors $(x = 0) \subset \mathbb{A}^4$ and $(y = 0) \subset \mathbb{A}^3_{y,z,t}$ shows that the lc of $(\mathbb{A}^4, \operatorname{div}(x) + \operatorname{div}(y) + B_i)$ near 0 is equivalent to the lc of $(\mathbb{A}^2_{z,t}, B_i^0)$ near 0. By (C.10), $h_{i,0} - h_{i,0}^\flat \in (z,t)^{L'}$ for $i \gg 0$, and [Liu26b, Lemma 4.14], again with [dFEM11, Proposition 2.11], proves (C.11).

By (C.10), near 0,

$$\operatorname{div}(h_{i,0}^\flat) = k_z \operatorname{div}(z) + k_t \operatorname{div}(t).$$

Since $(W_i, \frac{1}{m}\operatorname{div}(h_i))$ is lc near 0, (C.11) with $(c_h, c_z, c_t) = (\frac{1}{m}, 0, 0)$ implies $k_z \leq m$ and $k_t \leq m$. Let

$$k_* := \max\{k_z, k_t\}, \quad \gamma_z := 1 - \frac{k_z}{m}, \quad \gamma_t := 1 - \frac{k_t}{m}, \quad \sigma := 1 - \frac{k_*}{m} = \min\{\gamma_z, \gamma_t\} \geq 0,$$

and for a subgroup $H \subset \mathbb{T}^2$ let

$$\mathcal{L}(H) := \inf\big\{\gamma_z\xi_z + \gamma_t\xi_t \;\big|\; \xi \in \mathbb{R}^2_{>0}, \text{ the class of } \xi \text{ belongs to } H\big\}.$$

For a prime divisor F over W_i let $\eta_F := a(F, W_i, 0) - \operatorname{mult}_F z - \operatorname{mult}_F t$. We show that for $i \gg 0$ and any prime divisor F over W_i,

$$a(F, W_i, 0) - \frac{1}{m}\operatorname{mult}_F h_i \geq \gamma_z \operatorname{mult}_F z + \gamma_t \operatorname{mult}_F t + \sigma\eta_F. \tag{C.12}$$

If $k_* = 0$, then $h_{i,0}(0) \neq 0$, so $h_i(0) \neq 0$, $\operatorname{mult}_F h_i = 0$, and $\gamma_z = \gamma_t = \sigma = 1$. Hence (C.12) is an equality. If $k_* > 0$, then near 0

$$\frac{1}{k_*}\operatorname{div}(h_{i,0}^\flat) + \Big(1 - \frac{k_z}{k_*}\Big)\operatorname{div}(z) + \Big(1 - \frac{k_t}{k_*}\Big)\operatorname{div}(t) = \operatorname{div}(z) + \operatorname{div}(t),$$

which is lc. By (C.11), for any F,

$$a(F, W_i, 0) \geq \frac{1}{k_*}\operatorname{mult}_F h_i + \Big(1 - \frac{k_z}{k_*}\Big)\operatorname{mult}_F z + \Big(1 - \frac{k_t}{k_*}\Big)\operatorname{mult}_F t.$$

Multiplying by $\frac{k_*}{m}$ and adding $(1 - \frac{k_*}{m})a(F, W_i, 0)$ to both sides, we obtain (C.12). By [Liu26b, Lemma B.16], for any prime divisor F over W_i with center 0,

$$\eta_F \geq 0, \quad \text{and} \quad e_F \text{ is even and } \eta_F \geq \frac{e_F}{2} \text{ if } I_{G_i}(F) \not\subset G_i^{\mathrm{s}}. \tag{C.13}$$

We show that $\alpha_\infty \leq \mathcal{L}(\Pi^{\mathrm{s}})$. Let $\xi^\infty \in \mathbb{R}^2_{>0}$ have class in Π^{s}, and let ρ_i, $\xi_{i,z}$, $\xi_{i,t}$, and $\xi_{i,x}$ satisfy (C.7) and (C.8). The monomials of $h_{i,0}^\flat$ have $(\xi_{i,z}, \xi_{i,t})$-weight at least $k_z\xi_{i,z} + k_t\xi_{i,t}$, and by (C.10) the other monomials of $h_{i,0}$ have weight at least

$$\operatorname{mult}_0(h_{i,0} - h_{i,0}^\flat)\min\{\xi_{i,z}, \xi_{i,t}\} \to +\infty.$$

Moreover, $\xi_{i,x}$ and $\operatorname{wt}_{(\xi_{i,z},\xi_{i,t})}(g_i) - \xi_{i,x}$ tend to $+\infty$. By (C.6), for $i \gg 0$,

$$\alpha_i \leq \Big(1 - \frac{\lambda_i k_z}{m}\Big)\xi_{i,z} + \Big(1 - \frac{\lambda_i k_t}{m}\Big)\xi_{i,t}.$$

Letting $i \to +\infty$, we obtain $\alpha_\infty \leq \gamma_z\xi_z^\infty + \gamma_t\xi_t^\infty$, so

$$\alpha_\infty \leq \mathcal{L}(\Pi^{\mathrm{s}}). \tag{C.14}$$

Let $i \gg 0$ and let F be a prime divisor over W_i. Dividing (C.12) by e_F, we obtain

$$\frac{a(F, W_i, 0) - \frac{1}{m}\operatorname{mult}_F h_i}{e_F} \geq \gamma_z\xi_{F,z} + \gamma_t\xi_{F,t} + \sigma\frac{\eta_F}{e_F}.$$

If the class of ξ_F belongs to Π^{s}, then the right side is at least $\mathcal{L}(\Pi^{\mathrm{s}})$ by $\sigma \geq 0$ and (C.13). The class of ξ_F belongs to Π^{s} if $I_{G_i}(F) \subset G_i^{\mathrm{s}}$, and for any F if $\Pi = \Pi^{\mathrm{s}}$. Otherwise $\rho_F \notin G_i^{\mathrm{s}}$ and $\rho_F^2 \in G_i^{\mathrm{s}}$, since $[G_i : G_i^{\mathrm{s}}] = 2$, so the class of $2\xi_F$ belongs to $\Pi_i^{\mathrm{s}} \subset \Pi^{\mathrm{s}}$, and by (C.13) the right side is at least $\frac{1}{2}\mathcal{L}(\Pi^{\mathrm{s}}) + \frac{1}{2}\sigma$. If $\omega_2 \geq 1$, then the open segment $\omega_2\xi_z + \omega_1\xi_t = 1$ between $(\frac{1}{\omega_2}, 0)$ and

$(0,\frac{1}{\omega_1})$ is contained in $\mathbb{R}^2_{>0}$, and since $\mathbf{T}_0 \subset \Pi^{\mathrm{s}}$, by (C.9), letting ξ tend to $(\frac{1}{\omega_2},0)$ and to $(0,\frac{1}{\omega_1})$ along it, we obtain

$$\mathcal{L}(\Pi^{\mathrm{s}}) \le \mathcal{L}(\mathbf{T}_0) \le \min\Big\{\frac{\gamma_z}{\omega_2},\frac{\gamma_t}{\omega_1}\Big\} \le \sigma,$$

so

$$\frac{1}{2}\mathcal{L}(\Pi^{\mathrm{s}}) + \frac{1}{2}\sigma \ge \mathcal{L}(\Pi^{\mathrm{s}}).$$

Hence, if $G_i = G_i^{\mathrm{s}}$, or $\Pi = \Pi^{\mathrm{s}}$, or $\omega_2 \ge 1$, then by (C.1), Lemma 4.2(1) and (C.14),

$$\alpha_i = q_i(\lambda_i) \ge q_i(1) \ge \mathcal{L}(\Pi^{\mathrm{s}}) \ge \alpha_\infty > \alpha_i,$$

a contradiction.

It remains that, discarding finitely many indices, $G_i \ne G_i^{\mathrm{s}}$ for any i, $\Pi \ne \Pi^{\mathrm{s}}$ and $\omega_2 = 0$, that is, $\mathbf{T}_0$ is the image of $\mathbb{R}(1,0)$. If $\Pi_i \subset \Pi^{\mathrm{s}}$ for infinitely many i, then $\Pi \subset \Pi^{\mathrm{s}}$. Thus for $i \gg 0$, $\Pi_i \not\subset \Pi^{\mathrm{s}}$, so $\Pi_i \ne \Pi_i^{\mathrm{s}}$, and $\Pi_i \cap \Pi^{\mathrm{s}} = \Pi_i^{\mathrm{s}}$ since $[\Pi_i : \Pi_i^{\mathrm{s}}] \le 2$. The kernel of $G_i \to \Pi_i$ is contained in G_i^{s}, since an element of the kernel exchanging x and y would imply $\Pi_i = \Pi_i^{\mathrm{s}}$. Let $R_i \subset G_i$ be the preimage of $\Pi_i \cap \mathbf{T}_0$. Then $R_i \subset G_i^{\mathrm{s}}$, $|R_i| \to +\infty$, and $[G_i : R_i] \le [\Pi : \mathbf{T}_0]$ for $i \gg 0$. By (C.3), any $\rho \in R_i$ acts on x and y by one scalar $\zeta(\rho)$, and $\chi_t(\rho) = 1$.

Assume that z does not divide g_i. Since $g_i(0,t) \ne 0$ is fixed by R_i, $\zeta^2 = 1$ on R_i. Let $x' := \frac{1}{2}(x+y)$ and $y' := \frac{1}{2\sqrt{-1}}(x-y)$, the coordinates of Step B before the replacement. Then $f_i = x'^2 + y'^2 + g_i(z,t)$ and G_i is diagonal on x', y', z, t. The kernel of ζ on R_i is $G_i \cap \mathbf{T}_{(0,0,1,0)}$ for the coordinates (x', y', z, t). It has index at most 2 in R_i, so the order of $G_i \cap \mathbf{T}_{(0,0,1,0)}$ tends to $+\infty$. The germ W_i is cDV, since for a general $c \in \mathbb{C}$ the section $W_i \cap (z = ct)$ is $(xy + t^{n_i} \cdot (\text{unit}) = 0)$, an A_{n_i-1} singularity. By the last sentence of [Liu26b, Theorem 8.5] in the coordinates (z, x', y', t), with the labelled boundaries $\frac{\lambda_i}{m}\operatorname{div}(h_i)$ and labels in Γ, the α_i do not form a strictly increasing sequence, a contradiction.

Assume that z divides g_i. If z^2 divided g_i, then W_i would be singular along $(x = y = z = 0)$. Since W_i has an isolated singularity, z^2 does not divide g_i. Thus $g_i = zg_i'$ with $g_i'(0,0) = 0$ and $g_i'(0,t) \ne 0$. Since $zg_i'(0,t)$ has the character of xy on R_i, $\chi_z = \zeta^2$ on R_i. Hence $R_i = G_i \cap \mathbf{T}_{(1,1,2,0)}$, with $|R_i| \to +\infty$ and $[G_i : R_i] \le [\Pi : \mathbf{T}_0]$. By [Liu26b, Theorem 8.7], where G_i may exchange x and y, with the labelled boundaries $\frac{\lambda_i}{m}\operatorname{div}(h_i)$ and labels in Γ, the α_i do not form a strictly increasing sequence of positive numbers, a contradiction. Thus $(n_i)_i$ is bounded.

Step D. In this step we reduce to three configurations of a fixed circle.

By Step C, after passing to a subsequence, $n_i = n$ for any i, and there exist integers $d_1, d_2 \ge 0$ with $d_1 + d_2 = n$ such that the coefficient of $z^{d_1}t^{d_2}$ in g_i is non-zero for any i. After passing to a subsequence, either G_i is diagonal for any i, or $G_i \ne G_i^{\mathrm{s}}$ for any i. In the first case let $(x', y') := (x, y)$. In the second case let x' and y' be as in Step C, so that $f_i = x'^2 + y'^2 + g_i$. In both cases G_i is diagonal on x', y', z, t, and we regard G_i as a subgroup of $\mathbb{T}^4 := \mathbb{R}^4/\mathbb{Z}^4$, the class of ξ acting on each $u \in \{x', y', z, t\}$ by $e^{2\pi\sqrt{-1}\xi_u}$. Since $z^{d_1}t^{d_2}$ has the character of f_i, after passing to a subsequence the G_i converge in the Hausdorff distance to a closed subgroup $\mathcal{H} \subset \mathbb{T}^4$ whose Lie algebra is contained in

$$\mathfrak{s} := \begin{cases} \{\xi \in \mathbb{R}^4 \mid \xi_{x'} + \xi_{y'} = d_1\xi_z + d_2\xi_t\} & \text{if } G_i \text{ is diagonal,} \\ \{\xi \in \mathbb{R}^4 \mid 2\xi_{x'} = 2\xi_{y'} = d_1\xi_z + d_2\xi_t\} & \text{otherwise.} \end{cases}$$

By the fact in Step C for $\mathbb{T}^4$, $G_i \subset \mathcal{H}$ for $i \gg 0$.

Assume that the Lie algebra of $\mathcal{H}$ contains a vector $\xi^\circ \in \mathbb{R}^4_{>0}$, and let $\epsilon' > 0$. Since $G_i \to \mathcal{H}$, there exist vectors $\xi^{(i)} \in \mathbb{R}^4$ with classes in G_i and $\xi^{(i)} \to \epsilon'\xi^\circ$, so $\xi^{(i)} \in \mathbb{R}^4_{>0}$ for $i \gg 0$. Let ξ' be the primitive vector on the ray $\mathbb{R}_{>0}\xi^{(i)}$ of the lattice of vectors with classes in G_i. Then $\sum_j \xi'_j \le \sum_j \xi^{(i)}_j$. By Lemma 2.15(1)(2) for the monomial prime divisor F' over $\mathbb{A}^4$ whose weight vector is the primitive vector of $\mathbb{Z}^4$ on $\mathbb{R}_{>0}\xi'$, whose valuation restricts to $e_{F'}$ times that of the

toric prime divisor E' of ξ' over $\mathbb{A}^4/G_i$ (cf. [Liu26a, Lemma 8.5]), the prime divisor E', which has center the image of 0, has log discrepancy

$$\frac{a(F',\mathbb{A}^4,\Delta_i)}{e_{F'}}=\sum_j \xi'_j-\mathrm{wt}_{\xi'}(f_i)-\frac{\lambda_i}{m}\mathrm{wt}_{\xi'}(h_i)\le\sum_j\xi'_j$$

with respect to $(\mathbb{A}^4,\Delta_i)/G_i$, where

$$\Delta_i:=\mathrm{div}(f_i)+\frac{\lambda_i}{m}\mathrm{div}(h_i).$$

By the last sentence of Theorem 2.10 with $H=G_i$, $\alpha_i\le\sum_j\xi_j^{(i)}$. Letting $i\to+\infty$ and then $\epsilon'\to 0$, we obtain $\alpha_\infty\le 0$, a contradiction.

By the previous paragraph, the Lie algebra of $\mathcal{H}$ contains no vector of $\mathbb{R}^4_{>0}$. Since

$$(1,n-1,1,1)\in\mathfrak{s}\text{ in the first case},\qquad \left(\frac{n}{2},\frac{n}{2},1,1\right)\in\mathfrak{s}\text{ in the second case},$$

the Lie algebra of $\mathcal{H}$ is a proper subspace of $\mathfrak{s}$. Thus $\dim\mathcal{H}^0\le 2$, and $\dim\mathcal{H}^0\le 1$ if $G_i\ne G_i^{\mathrm{s}}$. If $\dim\mathcal{H}^0=0$, then $|G_i|\le|\mathcal{H}|$ for $i\gg 0$, a contradiction. If $\dim\mathcal{H}^0=2$, then G_i is diagonal, $\mathrm{mult}_0\, g_i=n$, the coefficient of $z^{d_1}t^{d_2}$ in g_i is non-zero, and $\alpha_i\ge\alpha_1>0$, so the last sentence of [Liu26b, Proposition 8.10], with the labelled boundaries $\frac{\lambda_i}{m}\mathrm{div}(h_i)$ and labels in Γ, contradicts the strict increase of α_i. Hence $\mathcal{H}^0=\mathbf{T}:=\mathbf{T}_{(w)}$ for a primitive vector $w=(w_x,w_y,w_z,w_t)\in\mathbb{Z}^4$ of weights on x,y,z,t which is neither in $\mathbb{R}^4_{>0}$ nor in $\mathbb{R}^4_{<0}$, with $w_x+w_y=d_1w_z+d_2w_t$, and with $w_x=w_y$ if $G_i\ne G_i^{\mathrm{s}}$, since then $\mathbf{T}$ acts on the span of x',y' by a scalar. Let $e_0:=[\mathcal{H}:\mathcal{H}^0]$ and $r_i:=|G_i\cap\mathbf{T}|$. Then $[G_i:G_i\cap\mathbf{T}]\le e_0$ and $r_i\to+\infty$. Since all monomials of g_i have the character of xy on $G_i\cap\mathbf{T}\cong\mu_{r_i}$,

$$jw_z+kw_t\equiv w_x+w_y\pmod{r_i}\quad\text{for any monomial } z^jt^k \text{ of } g_i. \tag{C.15}$$

In each case below, the cited result of [Liu26b] is used with the labelled boundaries $\frac{\lambda_i}{m}\mathrm{div}(h_i)$, labels in Γ, the circle $\mathbf{T}$ and the bound e_0, and the conclusion contradicts the strict increase of the positive numbers α_i.

Case $w_z=w_t=0$. Then $w_x+w_y=0$, so $w_x\ne w_y$, G_i is diagonal, and $\mathbf{T}=\mathbf{T}_{(1,-1,0,0)}$ up to orientation. We use [Liu26b, Theorem 8.3] for $W_i=(xy-(-g_i)=0)$.

Case exactly one of w_z,w_t is zero. Interchanging z and t and orienting $\mathbf{T}$, we may assume that $w_t=0<w_z$. If $d_1\ge 1$, then by (C.15) no monomial of g_i is a power of t once $r_i>nw_z$, so z divides g_i, and as in Step C z^2 does not divide g_i. A monomial zt^k of g_i and (C.15) imply $(d_1-1)w_z\equiv 0\pmod{r_i}$, so $d_1=1$. Thus $d_1\in\{0,1\}$.

- If $d_1=0$ and $w_x=w_y=0$, then $\mathbf{T}=\mathbf{T}_{(0,0,1,0)}$, W_i is cDV as in Step C, and we use the last sentence of [Liu26b, Theorem 8.5] in the coordinates (z,x',y',t).
- If $d_1=1$, then $g_i=zg_i'$ with $g_i'(0,0)=0$ and $g_i'(0,t)\ne 0$, and $w_x+w_y=w_z$. If $G_i\ne G_i^{\mathrm{s}}$, then $\mathbf{T}=\mathbf{T}_{(1,1,2,0)}$ and we use [Liu26b, Theorem 8.7]. If G_i is diagonal and $w_x,w_y\ge 0$, then $\mathbf{T}=\mathbf{T}_{(w_x,w_y,w_x+w_y,0)}$ with $\gcd(w_x,w_y)=1$, and we use [Liu26b, Theorem 8.8].
- Otherwise G_i is diagonal and w_x,w_y are non-zero of opposite signs, which is configuration (2) below.

Case $w_zw_t>0$. Orienting $\mathbf{T}$, we may assume that $w_z,w_t>0$. Then $w_x+w_y=d_1w_z+d_2w_t>0$. If $w_x,w_y>0$, in particular if $G_i\ne G_i^{\mathrm{s}}$ (then $w_x=w_y$), then $w\in\mathbb{R}^4_{>0}$, which is impossible. If $w_x=0$ or $w_y=0$, then $\mathbf{T}=\mathbf{T}_{(0,w_y,w_z,w_t)}$ or $\mathbf{T}_{(w_x,0,w_z,w_t)}$ with $\gcd(w_x+w_y,w_z,w_t)=1$, the sum of the terms of g_i of $\mathbf{T}$-weight w_x+w_y contains the monomial $z^{d_1}t^{d_2}$ and is non-zero, and we use [Liu26b, Theorem 8.9]. Otherwise, interchanging x and y, we may assume that $w_x<0<w_y$. By (C.15), once $r_i>n\max\{w_z,w_t\}$, any monomial of g_i of degree n has $\mathbf{T}$-weight w_x+w_y, so the sum g_{i0} of the terms of g_i of $\mathbf{T}$-weight w_x+w_y satisfies $\mathrm{mult}_0\, g_{i0}=n$, which is configuration (3) below.

Case w_z, w_t are non-zero of opposite signs. Two distinct monomials of degree n in z, t have $\mathbf{T}$-weights with a non-zero difference of absolute value at most $n|w_z - w_t|$. By (C.15), once $r_i > n|w_z - w_t|$, the homogeneous part of degree n of g_i is a non-zero multiple of $z^{d_1}t^{d_2}$. Possibly rescaling z or t, we may assume that the homogeneous part of degree n of g_i is $z^{d_1}t^{d_2}$. If G_i is diagonal, then, orienting $\mathbf{T}$, we may assume that $w_z < 0 < w_t$, which is configuration (1) below. Assume that $G_i \neq G_i^{\mathrm{s}}$, so $w_x = w_y$. Interchanging z and t and orienting $\mathbf{T}$, we may write $\mathbf{T} = \mathbf{T}_{(-A,-A,-C,D)}$ with $A \geq 0$, $C, D > 0$, and $d_1C - d_2D = 2A$. If $d_1 = 0$, then $-d_2D = 2A \geq 0$ implies $d_2 = 0$, contradicting $d_1 + d_2 = n \geq 3$. Thus $d_1 \geq 1$.

- If $d_2 = 0$, then $2A = nC$. Let $d := \gcd(n, 2)$ and $c' := \frac{dC}{2}$. Then c' is a positive integer with $A = \frac{nc'}{d}$ and $C = \frac{2c'}{d}$, and $\gcd(c', D) = 1$, since a common prime factor of c' and D divides A, C and D. Since $\widehat{\mathcal{O}}_{W_i,0} = \mathbb{C}[[x, y, z, t]]/(xy + g_i)$ with $\operatorname{mult}_0 g_i(z, 0) = n$, we use the second part of [Liu26b, Theorem 8.6] with n, c' in place of c, and $v = D$.
- If $d_2 \geq 1$, then $\gcd(A, C, D) = 1$ since $(-A, -A, -C, D)$ is primitive, and we use [Liu26b, Theorem 8.13] with $(a, b) = (d_1, d_2)$.

Thus G_i is diagonal, and we have reached the following situation, where q_i is the radial function of $(W_i \ni 0, G_i, h_i, m)$, G_i acts faithfully on W_i, and h_i is a G_i-semi-invariant polynomial:

(C.16)
$$\begin{cases} W_i = (f_i = 0) \subset \mathbb{A}^4,\ f_i = xy + g_i(z,t),\ \operatorname{mult}_0 g_i = n, \\ G_i \text{ diagonal},\ \operatorname{mult}_0 h_i \leq 2m, \\ \left(W_i, \frac{1}{m}\operatorname{div}(h_i)\right) \text{ lc and } \left(W_i, \frac{\lambda_i}{m}\operatorname{div}(h_i)\right) \text{ klt near } 0, \\ \alpha_i = q_i(\lambda_i) \text{ strictly increasing with limit } \alpha_\infty > 0, \\ r_i = |G_i \cap \mathbf{T}| \to +\infty,\ [G_i : G_i \cap \mathbf{T}] \leq e_0, \\ \text{one of the following holds:} \\ (1)\ w_z < 0 < w_t, \\ \quad \text{the homogeneous part of degree } n \text{ of } g_i \text{ is } z^{d_1}t^{d_2}; \\ (2)\ \mathbf{T} = \mathbf{T}_{(w_x,w_y,w_z,0)},\ w_z > 0,\ w_x w_y < 0,\ d_1 \in \{0, 1\}; \\ (3)\ w_x < 0 < w_z, w_t, \text{ and the sum } g_{i0} \text{ of the terms of } g_i \text{ of } \mathbf{T}\text{-weight } w_x + w_y \\ \quad \text{satisfies } \operatorname{mult}_0 g_{i0} = n. \end{cases}$$

The proof is continued below.

Proof of Theorem 12.2, continued. We are in the situation of (C.16). Thus

$$W_i = (xy + g_i(z,t) = 0) \subset \mathbb{A}^4_{x,y,z,t},$$

the groups G_i act diagonally, $\mathbf{T} = \mathbf{T}_{(w)}$ is a fixed primitive circle with $w = (w_x, w_y, w_z, w_t)$, $r_i := |G_i \cap \mathbf{T}| \to +\infty$, $[G_i : G_i \cap \mathbf{T}] \leq e_0$, $\operatorname{mult}_0 h_i \leq 2m$, $(W_i, \frac{1}{m}\operatorname{div}(h_i))$ is lc near 0, and $\alpha_i = q_i(\lambda_i)$ is strictly increasing with limit $\alpha_\infty > 0$. Moreover one of the following holds.

(1) $w_z < 0 < w_t$, and the lowest-degree part of g_i is $z^{d_1}t^{d_2}$ with $d_1, d_2 \geq 0$.
(2) $w_t = 0 < w_z$, $w_x w_y < 0$, and the coefficient of $z^{d_1}t^{d_2}$ in g_i is non-zero, with $d_1 \in \{0, 1\}$.
(3) $w_x < 0 < w_z, w_t$, and the block g_{i0} of g_i of $\mathbf{T}$-weight $w_x + w_y$ satisfies $g_{i0} \neq 0$ and $\operatorname{mult}_0 g_{i0} = \operatorname{mult}_0 g_i \geq 3$.

We write

$$(z_1, z_2, z_3, z_4) := (x, y, z, t),$$

and w_j for the weight of z_j. Let $c := \frac{1}{m}$ and $w_{\max} := \max_j |w_j|$. In (1) and (2), $d_1 + d_2 = n \geq 3$. For a prime divisor F over W_i with center 0 and a function u on W_i we write $e_F := |I_{G_i}(F)|$ and $[u]_F := \operatorname{mult}_F u / e_F$, omitting the subscript F when F is fixed, and we let

$$\sigma_F := \frac{a(F, W_i, c\operatorname{div}(h_i))}{e_F}, \quad \text{so that} \quad q_i(1) = \inf_F \sigma_F.$$

Equations. In case (1), let $\varpi := z^{w_t/\gcd(w_z,w_t)}t^{|w_z|/\gcd(w_z,w_t)}$, the primitive $\mathbf{T}$-invariant monomial in z, t. By Hensel's lemma for the coprime factors z^{d_1} and t^{d_2} of the lowest-degree part of g_i and

the Weierstrass preparation theorem in $\mathbb{C}[[z,t]]$, $g_i = \upsilon_i\phi_i\psi_i$ with a unit υ_i and

$$\phi_i = z^{d_1} + \sum_{j<d_1} \phi_{i,j}(t)z^j, \qquad \psi_i = t^{d_2} + \sum_{j<d_2} \psi_{i,j}(z)t^j \tag{C.17}$$

($\phi_i = 1$ if $d_1 = 0$, $\psi_i = 1$ if $d_2 = 0$). By the uniqueness of the factors, ϕ_i and ψ_i are G_i-semi-invariant with the characters of z^{d_1} and t^{d_2}, and, after normalizing the coefficient of $z^{d_1}t^{d_2}$ in g_i, υ_i is G_i-invariant with $\upsilon_i(0) = 1$. Fix the integer L_i of the paragraph after the next, and replace ϕ_i and ψ_i by their truncations in degrees less than L_i. They are still monic of degree d_1 in z, resp. d_2 in t, with the characters above, and $g_i \equiv \upsilon_i\phi_i\psi_i$ modulo $(z,t)^{L_i}$. The $\mathbf{T}$-weights of the monomials of a G_i-semi-invariant series are congruent modulo r_i. Since $j < d_1$ in ϕ_i and $j < d_2$ in ψ_i,

$$\begin{aligned} &z^{j_1}t^{j_2} \text{ in } \phi_i - z^{d_1}: \quad w_zj_1 + w_tj_2 = w_zd_1 + kr_i,\ k \ge 1,\ j_2 \ge (kr_i - |w_z|d_1)/w_t,\\ &z^{j_1}t^{j_2} \text{ in } \psi_i - t^{d_2}: \quad w_zj_1 + w_tj_2 = w_td_2 - kr_i,\ k \ge 1,\ j_1 \ge (kr_i - w_td_2)/|w_z|. \end{aligned} \tag{C.18}$$

Since $\phi_i\psi_i$, which is congruent to $\upsilon_i^{-1}g_i$ modulo $(z,t)^{L_i}$, has multiplicity $d_1 + d_2$ at 0, and a monomial in z,t of weight $w_zd_1 + w_td_2$ and degree at least $d_1 + d_2$ is $z^{d_1}t^{d_2}\varpi^j$ with $j \ge 0$, the block of $\phi_i\psi_i$ of weight $w_zd_1 + w_td_2$ is $z^{d_1}t^{d_2}(1 + \mu_i)$ with a G_i-invariant $\mu_i \in \varpi^{\ell_i}\mathbb{C}[[\varpi]]$, and $\ell_i \to +\infty$ by (C.18). Let $\widetilde{\psi}_i := \psi_i(1 + \mu_i)^{-1}$. Since $1 + \mu_i$ has weight 0, the block of $\phi_i\widetilde{\psi}_i$ of weight $w_zd_1 + w_td_2$ is exactly $z^{d_1}t^{d_2}$, and the other blocks of $\phi_i\widetilde{\psi}_i$ have weights $w_zd_1 + w_td_2 + kr_i$ with $k \ne 0$.

In case (2), the monomials of g_i have z-exponents congruent to d_1 modulo $r_i/\gcd(w_z, r_i)$, and $d_1 \in \{0,1\}$. Hence $g_i = z^{d_1}\upsilon_i\psi_i$ with ψ_i as in (C.17), a G_i-invariant unit υ_i, and $\psi_{i,j}(z) \in z^{\ell_i}\mathbb{C}[[z]]$ for $j < d_2$, where $\ell_i \to +\infty$. We let $\phi_i := z^{d_1}$, $\mu_i := 0$, and $\widetilde{\psi}_i := \psi_i$.

In cases (1) and (2), the automorphism $x \to \upsilon_i(1+\mu_i)x$ fixing y,z,t and the unit $\upsilon_i^{-1}(1 + \mu_i)^{-1}$ transform $xy + g_i$ into $xy + \phi_i\widetilde{\psi}_i$ modulo $(z,t)^{L_i}$. By Lemma 4.6 with λ_i in place of the coefficient t and with the automorphism and the unit above, possibly replacing (W_i, h_i), we may assume that $W_i = (f_i = 0)$, where f_i is the truncation of $xy + \phi_i\widetilde{\psi}_i$ in degrees less than an integer $L_i \ge i|G_i|$, and L_i is so large that the completion of W_i at 0 is G_i-isomorphic to $(xy + \phi_i\widetilde{\psi}_i = 0)$ by an automorphism of $\mathbb{C}[[x,y,z,t]]$ congruent to the identity modulo $(x,y,z,t)^{L_i/2}$. Such an L_i exists since W_i has an isolated singularity: if $(x,y,z,t)^{L_i'} \subset (f) + (x,y,z,t)J(f)$ for the Jacobian ideal $J(f)$, then an error of order $L \ge 2L_i'$ is removed modulo order $2L - 2L_i'$ by a coordinate change and a unit, which can be made G_i-equivariant by projecting to the characters. By Lemma 4.6(1) and (3), $q_i(\lambda_i) = \alpha_i$ is kept, $(W_i, c\,\mathrm{div}(h_i))$ is still lc near 0, and $\mathrm{mult}_0 h_i \le 2m$ is kept since $L_i > 2m$. A monomial of degree at least $L_i/2$ has $[\cdot]_F \ge L_i/(2|G_i|) \ge i/2$ for any F with center 0. Hence for any prime divisor F over W_i with center 0 and $[h_i]_F \le i/4$,

$$\sigma_F = \frac{1}{e_F}a\big(F', (xy + \phi_i\widetilde{\psi}_i = 0), c\,\mathrm{div}(h_i')\big),$$

where F' and h_i' are the images of F and h_i under the isomorphism of completions above, and every monomial u of $h_i' - h_i$ has $[u]_{F'} \ge i/2$. Below, such terms u are treated together with the terms u with $[u]_F > \Lambda_1$, for the number Λ_1 of (C.22).

In case (3) we keep $f_i = xy + g_i$. Since $w_z, w_t > 0$, any monomial of $g_i - g_{i0}$ has weight $w_x + w_y + kr_i$ with $k \ge 1$, and

$$\deg g_{i0} \le D_0 := \left\lfloor \frac{w_x + w_y}{\min(w_z, w_t)} \right\rfloor.$$

Marks and references. We replace h_i by the polynomial congruent to h_i modulo f_i without monomials divisible by xy, using $xy = xy - f_i \in \mathbb{C}[z,t]$ on W_i. Fix a monomial of h_i of degree $\mathrm{mult}_0 h_i$, and let κ_i be its $\mathbf{T}$-weight. Since $\mathrm{mult}_0 h_i \le 2m$, the numbers κ_i are bounded. Possibly passing to a subsequence, $\kappa := \kappa_i$ does not depend on i. Let h_i^0 be the block of h_i of weight κ. Any other block of h_i has weight $\kappa + kr_i$ with $k \ne 0$, and is called *positive* or *negative* according

to the sign of k. A monomial $x^{a_1}y^{a_2}z^{a_3}t^{a_4}$ in a negative block satisfies $\sum_{w_j<0}|w_j|a_j \geq |k|r_i - |\kappa|$, and one in a positive block satisfies $\sum_{w_j>0} w_j a_j \geq kr_i - |\kappa|$. Let

$$f_i^0 := xy + z^{d_1}t^{d_2} \text{ in (1) and (2)}, \qquad f_i^0 := xy + g_{i0} \text{ in (3)}, \quad \text{and} \quad V_{0,i} := (f_i^0 = 0).$$

Then $(W_i \ni 0, G_i, h_i)$ is a marked germ over $(V_{0,i}, h_i^0)$ in the sense of Set-up 11.1: the blocks of f_i other than f_i^0 have weights $\mathrm{wt}_w(f_i^0) + kr_i$ with $k \neq 0$. The monomials not divisible by xy form a basis of $\mathbb{C}[V_{0,i}]$, hence $h_i^0|_{V_{0,i}} \neq 0$. The variety $V_{0,i}$ is normal. It is klt, since at 0 [Liu26b, Lemma B.16] implies $a(F, V_{0,i}, 0) \geq \mathrm{mult}_F z + \mathrm{mult}_F t > 0$, and at the other singular points $V_{0,i}$ is locally analytically the product of a curve and an A-type surface singularity. The group G_i acts faithfully on $V_{0,i}$ since no coordinate vanishes on $V_{0,i}$. Since $(W_i, c\,\mathrm{div}(h_i))$ is lc near 0, and the monomials of $f_i - f_i^0$ and $h_i - h_i^0$ have degree at least

$$\frac{r_i - \max\{|\kappa|, |\mathrm{wt}_w(f_i^0)|\}}{w_{\max}} \to +\infty$$

by the block weights above, $(V_{0,i}, c\,\mathrm{div}(h_i^0))$ is lc near 0 for $i \gg 0$ by Lemma 2.19, with lc near 0 compared with lc of the formal pairs by [dFEM11, Proposition 2.11], and [Kaw07, Theorem]. Let H, Z_0, o, $\overline{o}$, $Y^{\pm}$, $B^{\pm}$, $\delta_{\mathbf{T}}$, $Q_{\mathbf{T}}$, $\beta_i^{\pm}$, and $\beta_i := \beta_{\mathbf{T}}$ be as in Set-up 11.1 for $(V_{0,i}, G_i, h_i^0)$ and c. We have $\delta_{\mathbf{T}} = c\kappa - w_z - w_t$, which does not depend on i. By Proposition 11.2(5),

$$\begin{aligned} &\beta_i^- \geq \beta_i^+ + \frac{|\delta_{\mathbf{T}}|}{Q_{\mathbf{T}}} \text{ if } \delta_{\mathbf{T}} < 0, \qquad \beta_i^+ \geq \beta_i^- + \frac{\delta_{\mathbf{T}}}{Q_{\mathbf{T}}} \text{ if } \delta_{\mathbf{T}} > 0, \\ &\beta_i^+ = \beta_i^- \text{ if } \delta_{\mathbf{T}} = 0, \end{aligned} \tag{C.19}$$

and $\beta_i^{\pm}$ belong to a finite subset of $[0,2]$ independent of i.

The upper bound. The equations satisfy

$$\deg f_i^0 \leq n \text{ in cases (1) and (2)}, \qquad \deg f_i^0 \leq \max\{2, D_0\} \text{ in case (3)},$$

and in cases (1) and (2), resp. (3), they belong to class (a), resp. (b), of Proposition 11.7. In case (3), $w_y > 0$ since $w_x < 0 < w_x + w_y$. By Proposition 11.4 there exist real numbers N_0, N_1 such that

$$\alpha_i = q_i(\lambda_i) \leq \beta_i + \frac{N_0}{r_i} + \frac{N_1}{m}(1 - \lambda_i) \quad \text{for any } i \gg 0.$$

The numbers β_i belong to a finite set. Possibly passing to a subsequence, $\beta_i = \beta$ for any i, and since $r_i \to +\infty$ and $\lambda_i \to 1$,

$$\beta \geq \alpha_\infty > \alpha_i > 0 \quad \text{for any } i \gg 0. \tag{C.20}$$

The dichotomy. By (C.20), $\beta_i > 0$, so the hypotheses of Proposition 11.7 hold. After passing to a subsequence, Proposition 11.7 gives a pair $(\mathcal{A}_{\Bbbk}, C_*)$ satisfying (1) and (2) there. We show that

$$q_i(1) \geq \beta_i = \beta \quad \text{for any } i \gg 0. \tag{C.21}$$

This finishes the proof: by Lemma 4.2(1) and (C.20),

$$\alpha_i = q_i(\lambda_i) \geq q_i(1) \geq \beta \geq \alpha_\infty > \alpha_i,$$

a contradiction.

If $(\mathcal{A}_{\Bbbk}, C_*)$ is not klt, then by Proposition 11.7(2), $\beta_i = \beta$ and $\sigma_F \geq \beta$ for any prime divisor F over W_i with center 0, which is (C.21).

In the rest of the proof $(\mathcal{A}_{\Bbbk}, C_*)$ is klt, so by Proposition 11.7(1) there exists $\theta > 1$ such that $(W_i, \theta c\,\mathrm{div}(h_i))$ is lc near 0 for any $i \gg 0$. Let F be a prime divisor over W_i with center 0. If $\sigma_F > 2$, then $\sigma_F \geq \beta_i$ since $\beta_i \in [0,2]$. Hence we may assume $\sigma_F \leq 2$. Since $a(F, W_i, 0) \geq \theta c\,\mathrm{mult}_F h_i$ and $a(F, W_i, 0) - c\,\mathrm{mult}_F h_i \leq 2e$, and by [Liu26b, Lemma B.16],

$$[h_i] \leq \Lambda_1 := \frac{2}{c(\theta - 1)}, \qquad [z] + [t] \leq \frac{a(F, W_i, 0)}{e} \leq \Lambda_0 := \frac{2\theta}{\theta - 1}. \tag{C.22}$$

Let $z_j, z_{j'}$ be coordinates with $w_j < 0 < w_{j'}$. The monomial $z_j^{w_{j'}} z_{j'}^{|w_j|}$ is $\mathbf{T}$-invariant, so $G_i \cap \mathbf{T}$ acts trivially on it, and the character of G_i on it has order at most e_0. Hence a power of it with exponent at most e_0 is a G_i-invariant function vanishing at 0, so its multiplicity along F is a positive multiple of the ramification index e of F over W_i/G_i. If $w_j = 0$, then $G_i \cap \mathbf{T}$ acts trivially on z_j, the character of G_i on z_j has order at most e_0, and the multiplicity along F of a G_i-invariant power of z_j with exponent at most e_0 is a positive multiple of e. Thus

$$w_{j'}[z_j] + |w_j|[z_{j'}] \geq \frac{1}{e_0}, \qquad [z_j] \geq \frac{1}{e_0} \text{ if } w_j = 0. \tag{C.23}$$

Fix a real number ϵ_1 with $0 < \epsilon_1 < \min(1/(4e_0 w_{\max}), 1/(2Q_{\mathbf{T}}))$. By (C.23), if $[z_j] < \epsilon_1$ for one coordinate z_j of one sign, then $[z_{j'}] \geq 3/(4e_0 w_{\max}) > \epsilon_1$ for any coordinate $z_{j'}$ of the other sign. We use the following estimate.

Claim C.1. *Let $\pm \in \{+, -\}$. Let f° and h° be G_i-semi-invariant polynomials in x, y, z, t such that the $\mathbf{T}$-blocks of f° have weights $\mathrm{wt}_w(f_i^0) \pm k r_i$ with $k \geq 0$, the block of weight $\mathrm{wt}_w(f_i^0)$ is f_i^0, the blocks of h° have weights $\kappa \pm k r_i$ with $k \geq 0$, and the block of weight κ is h_i^0. Let $W^\circ := (f^\circ = 0)$, and let F be a prime divisor over W° with center 0, $e_F := |I_{G_i}(F)|$, $[u]_F := \mathrm{mult}_F u / e_F$, $\mathfrak{r}^+ := \min_{w_j > 0}[z_j]/w_j$, and $\mathfrak{r}^- := \min_{w_j < 0}[z_j]/|w_j|$. Then*

$$\frac{a(F, W^\circ, c\,\mathrm{div}(h^\circ))}{e} \geq \beta_i^+ - \delta_{\mathbf{T}} \mathfrak{r}^+ \text{ if } \pm = +, \qquad \frac{a(F, W^\circ, c\,\mathrm{div}(h^\circ))}{e} \geq \beta_i^- + \delta_{\mathbf{T}} \mathfrak{r}^- \text{ if } \pm = -.$$

In case (3), if $\pm = +$ and $\delta_{\mathbf{T}} \geq 0$, then moreover

$$\frac{a(F, W^\circ, c\,\mathrm{div}(h^\circ))}{e} \geq \beta_i^- + \frac{\delta_{\mathbf{T}}[x]}{|w_x|}.$$

Proof. We treat $\pm = +$. The case $\pm = -$ is symmetric. Let z_{j_0} be a coordinate with $w_{j_0} > 0$ and $[z_{j_0}] = w_{j_0} \mathfrak{r}^+$. Adjoin ξ with $\xi^{w_{j_0}} = z_{j_0}$, and let $\widetilde{z}_j := \xi^{-w_j} z_j$ and $\varrho := \xi^{r_i}$. The functions $\widetilde{z}_j$ and ϱ are invariant under $G_i \cap \mathbf{T}$. By the assumption on the blocks,

$$f^\circ = \xi^{\mathrm{wt}_w(f_i^0)} \Phi(\varrho, \widetilde{z}), \quad h^\circ = \xi^\kappa \Psi(\varrho, \widetilde{z}), \quad \Phi(0, \widetilde{z}) = f_i^0(\widetilde{z}), \quad \text{and} \quad \Psi(0, \widetilde{z}) = h_i^0(\widetilde{z}),$$

with $\widetilde{z}_{j_0} = 1$. Let $X := (\Phi = 0)$ in the coordinates ϱ and $\widetilde{z}_j$, $j \neq j_0$, and

$$S_0 := X \cap (\varrho = 0) = V_{0,i} \cap (z_{j_0} = 1).$$

Let Ω_{W° and Ω_X be the Poincaré residues of $dz_1 \wedge \cdots \wedge dz_4 / f^\circ$ and of $d\varrho \wedge \bigwedge_{j \neq j_0} d\widetilde{z}_j / \Phi$. Comparing the residue forms, Ω_{W° is a non-zero constant multiple of

$$\xi^{\sum_j w_j - \mathrm{wt}_w(f_i^0)} \varrho^{-1} \Omega_X,$$

hence

$$a(F, W^\circ, c\,\mathrm{div}(h^\circ)) = a(F, X, S_0 + c\,\mathrm{div}(\Psi)) - \delta_{\mathbf{T}} \mathrm{mult}_F \xi,$$

where F also denotes the extension of F to the root field. Under the extension to the root field and the quotient by $G_i \cap \mathbf{T}$, the ratio of the log discrepancy of F to its inertia order does not change, and $\mathrm{mult}_F \xi / e = \mathfrak{r}^+$. We have $[\widetilde{z}_j] \geq 0$ for $w_j > 0$ by the choice of j_0, and $[\widetilde{z}_j] > 0$ for $w_j \leq 0$, so the center of F on X is a point $o_F \in S_0$ whose image in Y^+ is a point of the fiber of $Y^+ \to Z_0$ over o. The surface S_0 is a normal Cartier divisor on X, so X is normal near S_0. Since S_0 is klt by Proposition 11.2(2), (X, S_0) is plt near S_0 by inversion of adjunction, hence X is klt near S_0. Let

$$\widetilde{G} := \{(g, \zeta) \in G_i \times \mathbb{C}^* \mid g \text{ multiplies } z_{j_0} \text{ by } \zeta^{w_{j_0}}\}$$

act by g on $z_1, \ldots, z_4$ and by $\xi \to \zeta \xi$, and let $\widetilde{R} := \{(s, s) \mid s \in G_i \cap \mathbf{T}\}$, where $\mathbf{T} = \mathbb{C}^*$. The group $\widetilde{G}/\widetilde{R}$ acts diagonally on X in the coordinates ϱ and $\widetilde{z}_j$, and its image in the automorphisms of S_0 is that of the stabilizer of $(z_{j_0} = 1)$ in the group generated by G_i and $\mathbf{T}$. By Proposition 11.2(2) and (3), the hypotheses of Lemma 11.5 with $\widetilde{G}/\widetilde{R}$ in place of H, $u = \varrho$, $c_S = 1$, and $\mathfrak{a} = \Psi^c$ hold at o_F, and the number β_0 of Lemma 11.5 at o_F satisfies $\beta_0 \geq \beta_i^+$ by the definition of β_i^+. This proves the first inequality.

In case (3), let $\widetilde{x} := \xi^{|w_x|}x$ on the chart. Let P be a prime divisor over Z_0/H whose center on Y^+/H is contained in the fiber over $\overline{o}$ and in the image of the root section at z_{j_0}, let ν be an extension of mult_P to a $\mathbb{Q}$-valued valuation on the rational functions on Z_0, and consider the numbers $\vartheta_\pm$ of Proposition 11.2(4) for ν. Then z_{j_0} attains the maximum defining ϑ_-, and x is the only coordinate of negative weight. Since $\widetilde{x} = \psi_x\psi_{j_0}^{|w_x|/w_{j_0}}$ for the functions ψ_x and ψ_{j_0} of Proposition 11.2(4), we have

$$\nu(\widetilde{x}) = \nu(\psi_x) + \frac{|w_x|\nu(\psi_{j_0})}{w_{j_0}} = |w_x|(\vartheta_+ - \vartheta_-).$$

Hence, by Proposition 11.2(4)(a), the log discrepancy of P for $(Y^+, B^+)/H$ with the boundary $(\delta_{\mathbf{T}}/|w_x|)\,\mathrm{div}(\widetilde{x})$ added is

$$a(P,(Y^+,B^+)/H) - \delta_{\mathbf{T}}(\vartheta_+ - \vartheta_-) = \varphi(\nu) - \delta_{\mathbf{T}}\vartheta_+ = a(P,(Y^-,B^-)/H) \geq \beta_i^- \geq 0. \tag{C.24}$$

Let $\mathfrak{a}' := \Psi^c\widetilde{x}^{\delta_{\mathbf{T}}/|w_x|}$. Then $(S_0, \mathfrak{a}'|_{S_0})$ is lc near o_F: for the prime divisors with center in the fiber over $\overline{o}$ by (C.24), since the quotient of $(S_0, \mathfrak{a}'|_{S_0})$ by the stabilizer of $(z_{j_0} = 1)$ is crepant onto a neighborhood in Y^+/H of the image of o_F, and, for the prime divisors whose center is not contained in the fiber, by Proposition 11.2(3), since x divides every non-constant $\mathbf{T}$-invariant monomial, so that $\widetilde{x}$ vanishes on S_0 only over the fiber. Lemma 11.5 with $\mathfrak{a} = \mathfrak{a}'$ and $[\widetilde{x}] = [x] + |w_x|\mathfrak{r}^+$, where the number β_0 of Lemma 11.5 at o_F, an infimum over the prime divisors with center the image of o_F, is at least β_i^- by (C.24), then implies

$$\frac{a(F,W^\circ,c\,\mathrm{div}(h^\circ))}{e} \geq \beta_i^- + \frac{\delta_{\mathbf{T}}}{|w_x|}([x] + |w_x|\mathfrak{r}^+) - \delta_{\mathbf{T}}\mathfrak{r}^+ = \beta_i^- + \frac{\delta_{\mathbf{T}}[x]}{|w_x|}.$$

□

Cases (1) and (2). Let

$$\Theta^+ := \frac{\widetilde{\psi}_i}{t^{d_2}} \quad\text{and}\quad \Theta^- := \frac{\phi_i}{z^{d_1}(1+\mu_i)} \text{ in case (1)}, \qquad \Theta^+ := 1 \quad\text{and}\quad \Theta^- := \frac{\psi_i}{t^{d_2}} \text{ in case (2)}.$$

For $x^\pm := x/\Theta^\pm$, the completed equation of W_i is

$$\begin{aligned} xy + \phi_i\widetilde{\psi}_i &= \Theta^+\cdot(x^+y + \phi_i t^{d_2}) = \Theta^-\cdot(x^-y + z^{d_1}\psi_i) && \text{in (1)},\\ xy + z^{d_1}\psi_i &= \Theta^-\cdot(x^-y + z^{d_1}t^{d_2}) && \text{in (2)}. \end{aligned} \tag{C.25}$$

Regarding $x^\pm$ as coordinates of weight w_x, by (C.18) in case (1) and by $\psi_{i,j}(z) \in z^{\ell_i}\mathbb{C}[[z]]$ for $j < d_2$ in case (2), the $\mathbf{T}$-blocks of $x^+y + \phi_i t^{d_2}$ and of $xy + z^{d_1}\psi_i$ in case (2) have weights $w_z d_1 + w_t d_2 + kr_i$ with $k \geq 0$, and the $\mathbf{T}$-blocks of $x^-y + z^{d_1}\psi_i$ have weights $w_z d_1 + w_t d_2 - kr_i$ with $k \geq 0$. The block of weight $w_z d_1 + w_t d_2$ of each of $x^+y + \phi_i t^{d_2}$, $xy + z^{d_1}\psi_i$, and $x^-y + z^{d_1}\psi_i$ is $x^\pm y + z^{d_1}t^{d_2}$.

Assume that $[z_j] \geq \epsilon_1$ for any z_j with $w_j < 0$. In case (1), z is such a coordinate. By (C.18), a monomial of $\psi_i/t^{d_2} - 1$ is $z^{j_1}t^{j_2-d_2}$ with $j_1 \geq (r_i - w_t d_2)/|w_z|$, and by (C.22) we have $[z^{j_1}t^{j_2-d_2}] \geq j_1\epsilon_1 - d_2\Lambda_0$. Moreover, $\mu_i \in \varpi^{\ell_i}\mathbb{C}[[\varpi]]$ with $[\varpi] \geq \epsilon_1$. Hence $[\Theta^+ - 1] > \Lambda_1$ for $i \gg 0$, and $\mathrm{mult}_F\,\Theta^+ = 0$.

Since Θ^+ is a function of z, t,

$$\frac{dx}{x}\wedge dz\wedge dt = \frac{dx^+}{x^+}\wedge dz\wedge dt.$$

By (C.25), for $W^\circ := (x^+y + \phi_i t^{d_2} = 0)$ (resp. $W^\circ := W_i$ in case (2)), the substitution

$$x^+ = x/\Theta^+$$

is a birational map $W_i \dashrightarrow W^\circ$ that pulls back the residue form of W° to the residue form of W_i, and under it F is a prime divisor over W° with center 0 and with inertia group $I_{G_i}(F)$. Let h° be the sum of h_i^0 and the positive blocks of h_i, evaluated at (x^+, y, z, t). The monomials of $h_i(x,y,z,t) - h^\circ(x^+,y,z,t)$ are multiples of $(\Theta^+)^{a_1} - 1$ with $a_1 \geq 1$ or monomials of negative

blocks. We have $[u] > \Lambda_1$ for a multiple u of $(\Theta^+)^{a_1} - 1$, and $[u] \geq \epsilon_1(r_i - |\kappa|)/w_{\max} > \Lambda_1$ for a monomial u of a negative block, for $i \gg 0$. Since $[h_i] \leq \Lambda_1$, $\operatorname{mult}_F h^\circ = \operatorname{mult}_F h_i$. Claim C.1 with sign + implies, with the number $\mathfrak{r}^+$ of Claim C.1 for W° and F,

$$\sigma_F \geq \beta_i^+ - \delta_{\mathbf{T}}\mathfrak{r}^+. \tag{C.26}$$

Assume that $[z_j] \geq \epsilon_1$ for any z_j with $w_j > 0$. In case (1), t is such a coordinate, the monomials of $\phi_i/z^{d_1} - 1$ are $z^{j_1-d_1}t^{j_2}$ with $j_2 \geq (r_i - |w_z|d_1)/w_t$, and $[\varpi] \geq \epsilon_1$. In case (2), z is such a coordinate and the monomials of $\psi_i/t^{d_2} - 1$ are divisible by z^{ℓ_i} with exponent of t at least $-d_2$. Replacing x^+, Θ^+, the positive blocks of h_i, and the sign + in Claim C.1 with x^-, Θ^-, the negative blocks of h_i, and the sign −, the argument for (C.26) gives

$$\sigma_F \geq \beta_i^- + \delta_{\mathbf{T}}\mathfrak{r}^-. \tag{C.27}$$

By (C.23), at least one of the assumptions of (C.26) and (C.27) holds, and if one of them fails, then the other holds with $\mathfrak{r}^- \leq \epsilon_1$, resp. $\mathfrak{r}^+ \leq \epsilon_1$. If $\delta_{\mathbf{T}} \leq 0$, then $\beta_i = \beta_i^+$. Then (C.26) implies $\sigma_F \geq \beta_i^+$, and (C.27) with $\mathfrak{r}^- \leq \epsilon_1 < 1/(2Q_{\mathbf{T}})$ and (C.19) implies

$$\sigma_F \geq \beta_i^+ + \frac{|\delta_{\mathbf{T}}|}{Q_{\mathbf{T}}} - |\delta_{\mathbf{T}}|\epsilon_1 \geq \beta_i^+.$$

If $\delta_{\mathbf{T}} \geq 0$, then $\beta_i = \beta_i^-$, by (C.19) if $\delta_{\mathbf{T}} = 0$. Then (C.27) implies $\sigma_F \geq \beta_i^-$, and (C.26) with $\mathfrak{r}^+ \leq \epsilon_1 < 1/(2Q_{\mathbf{T}})$ and (C.19) implies

$$\sigma_F \geq \beta_i^- + \frac{\delta_{\mathbf{T}}}{Q_{\mathbf{T}}} - \delta_{\mathbf{T}}\epsilon_1 \geq \beta_i^-.$$

Thus $\sigma_F \geq \beta_i$, which is (C.21) in cases (1) and (2).

Case (3). Fix a real number $\Lambda_2 > 2|w_x|\Lambda_1$. Let $\epsilon_2 := 1/(2e_0 w_{\max})$.

Assume that $[x] > \Lambda_2/r_i$. All $\mathbf{T}$-blocks of f_i have weights $w_x + w_y + kr_i$ with $k \geq 0$. A monomial $x^{a_1}y^{a_2}z^{a_3}t^{a_4}$ of a negative block of h_i has $|w_x|a_1 \geq r_i - |\kappa|$, hence

$$[x^{a_1}y^{a_2}z^{a_3}t^{a_4}] \geq \frac{(r_i - |\kappa|)\Lambda_2}{|w_x|r_i} > \Lambda_1 \quad \text{for } i \gg 0.$$

Since $[h_i] \leq \Lambda_1$, $\operatorname{mult}_F h^\circ = \operatorname{mult}_F h_i$ for the sum h° of h_i^0 and the positive blocks of h_i. Thus Claim C.1 with sign +, $f^\circ = f_i$, and this h° implies

$$\sigma_F \geq \beta_i^+ - \delta_{\mathbf{T}}\mathfrak{r}^+ \geq \beta_i^+ \text{ if } \delta_{\mathbf{T}} < 0, \qquad \sigma_F \geq \beta_i^- + \frac{\delta_{\mathbf{T}}[x]}{|w_x|} \geq \beta_i^- \text{ if } \delta_{\mathbf{T}} \geq 0.$$

Assume that $[x] \leq \Lambda_2/r_i$. By (C.23) for the pairs (x, y), (x, z), and (x, t), we have $[y], [z], [t] \geq \epsilon_2$ for $i \gg 0$. For instance,

$$|w_x|[z] \geq \frac{1}{e_0} - \frac{w_z\Lambda_2}{r_i}.$$

The monomials of $g_i - g_{i0}$ have degree at least $(w_x + w_y + r_i)/w_{\max}$, so $[g_i - g_{i0}] \to +\infty$ uniformly in F, while $[g_{i0}] \geq \epsilon_2 \operatorname{mult}_0 g_{i0} \geq 3\epsilon_2 > [x]$. Let $y^\circ := -g_{i0}/x$. Then $[y^\circ] > 0$, and $[y - y^\circ] = [g_i - g_{i0}] - [x] > \Lambda_1$ for $i \gg 0$. Both W_i and $V_{0,i}$ are birational to $\mathbb{A}^3_{x,z,t}$ by the projection, with inverses given by $y = -g_i/x$, resp. $y = y^\circ$, and under the projections both residue forms equal $\frac{dx}{x} \wedge dz \wedge dt$. Hence F is a prime divisor over $V_{0,i}$ with center 0, $a(F, W_i, 0) = a(F, V_{0,i}, 0)$, and the inertia groups coincide. A monomial $x^{a_1}y^{a_2}z^{a_3}t^{a_4}$ of a positive block of h_i has $\sum_{w_j>0} w_j a_j \geq r_i - |\kappa|$, hence

$$[x^{a_1}y^{a_2}z^{a_3}t^{a_4}] \geq \frac{\epsilon_2(r_i - |\kappa|)}{w_{\max}} > \Lambda_1.$$

Let h° be the sum of h_i^0 and the negative blocks of h_i. Then $h^\circ(x, y, z, t) - h^\circ(x, y^\circ, z, t)$ is $y - y^\circ$ times a polynomial in x, y, y°, z, t, all with $[\cdot] \geq 0$, so $\operatorname{mult}_F h_i = \operatorname{mult}_F(h^\circ|_{V_{0,i}})$. Claim C.1

with sign $-$, $f^\circ = xy + g_{i0}$, and h° the sum of h_i^0 and the negative blocks of h_i, together with $\mathfrak{r}^- = [x]/|w_x|$, implies

$$\sigma_F \geq \beta_i^- + \frac{\delta_{\mathbf{T}}[x]}{|w_x|}.$$

If $\delta_{\mathbf{T}} \geq 0$, then $\sigma_F \geq \beta_i^- = \beta_i$, by (C.19) if $\delta_{\mathbf{T}} = 0$. If $\delta_{\mathbf{T}} < 0$, then by (C.19), for $i \gg 0$,

$$\sigma_F \geq \beta_i^+ + \frac{|\delta_{\mathbf{T}}|}{Q_{\mathbf{T}}} - \frac{|\delta_{\mathbf{T}}|\Lambda_2}{|w_x|r_i} \geq \beta_i^+ = \beta_i.$$

Thus $\sigma_F \geq \beta_i$ for any prime divisor F over W_i with center 0 and any $i \gg 0$, which is (C.21) in case (3). This finishes the proof of Theorem 12.2. □

C.2. **Quadratic rank one.** In this subsection we prove Theorem 12.3 of Section 12. To show that the hypersurfaces defined by the first faces (C.50) in the proof of Theorem 12.3 are normal and klt, we need the following proposition.

Proposition C.2. *Let $W_0 \subset \mathbb{A}^4$ be one of the following hypersurfaces.*

(i) $W_0 = (x^2+y^2z+z^{n_1}t^{n_2} = 0)$, where n_1 and n_2 are non-negative integers with $n_1+n_2 \geq 3$.
(ii) $W_0 = (x^2 + y^2z + yt^{n_3} = 0)$, where $n_3 \geq 2$ is an integer.

Then W_0 is normal and klt.

Proof. Let (z_1, z_2, z_3, z_4) be the coordinates of $\mathbb{A}^4$, such as (x, y, z, t), (P, Q, s, t), or (X, s, z, t). For a hypersurface $V = (g = 0) \subset \mathbb{A}^4$ with $\partial g/\partial z_{j_4}|_V \neq 0$, we let

$$\Omega_V := dz_{j_1} \wedge dz_{j_2} \wedge dz_{j_3}/(\partial g/\partial z_{j_4})$$

be the residue form of V, where $\{j_1, j_2, j_3, j_4\} = \{1, 2, 3, 4\}$. Up to sign, Ω_V does not depend on j_4, and $\operatorname{div}_V(\Omega_V) = 0$ when V is normal. For example, $\Omega_{W_0} = dy \wedge dz \wedge dt/(2x)$.

Assume that W_0 is in (i) with $n_1 \geq 1$, and let $f := x^2 + y^2z + z^{n_1}t^{n_2}$. Let

$$V_1 := (PQ + s^{2n_1-2}t^{n_2} = 0) \subset \mathbb{A}^4_{P,Q,s,t},$$

$$\pi_1(P, Q, s, t) := \Big(\frac{s(P+Q)}{2}, \frac{P-Q}{2\sqrt{-1}}, s^2, t\Big), \quad \text{and} \quad \iota_1(P, Q, s, t) := (-Q, -P, -s, t).$$

Since $f \circ \pi_1 = s^2(PQ + s^{2n_1-2}t^{n_2})$, π_1 maps V_1 to W_0. The involution ι_1 multiplies $P+Q$, $P-Q$, s, t by $-1, 1, -1, 1$, so the ι_1-invariant polynomials are generated by $s(P+Q)$, $P-Q$, s^2, t, and $(P+Q)^2$, and $(P+Q)^2 = (P-Q)^2 - 4s^{2n_1-2}t^{n_2}$ on V_1. Thus π_1 induces $\mathbb{C}[W_0] \cong \mathbb{C}[V_1]^{\iota_1}$. The fixed locus of ι_1 on V_1 is contained in the curve $V_1 \cap (P+Q = s = 0)$, so π_1 is quasi-étale. On V_1 we have $d(P-Q) \equiv \frac{P+Q}{P} dP$ modulo ds and dt, so $\pi_1^*\Omega_{W_0} = -\sqrt{-1}\Omega_{V_1}$ with $\Omega_{V_1} = dP \wedge ds \wedge dt/P$.

Assume that W_0 is in (ii), and let $f := x^2+y^2z+yt^{n_3}$. Let $V_2 := (X^2+s^2z+t^{n_3} = 0) \subset \mathbb{A}^4_{X,s,z,t}$,

$$\pi_2(X, s, z, t) := (sX, s^2, z, t), \qquad \iota_2(X, s, z, t) := (-X, -s, z, t).$$

Here $f \circ \pi_2 = s^2(X^2 + s^2z + t^{n_3})$, the ι_2-invariant polynomials are generated by sX, s^2, z, t, and X^2, and $X^2 = -s^2z - t^{n_3}$ on V_2. Thus π_2 induces $\mathbb{C}[W_0] \cong \mathbb{C}[V_2]^{\iota_2}$. The fixed locus of ι_2 on V_2 is the curve $(X = s = t = 0)$, so π_2 is quasi-étale, and $\pi_2^*\Omega_{W_0} = ds \wedge dz \wedge dt/X = 2\Omega_{V_2}$. If $n_3 = 2$, then with $P := X + \sqrt{-1}t$ and $Q := X - \sqrt{-1}t$ we have $V_2 = (PQ + s^2z = 0)$.

Let n, n' be non-negative integers with $n + n' \geq 2$ and $V_{n,n'} := (PQ + s^n t^{n'} = 0) \subset \mathbb{A}^4$. The singular locus of $V_{n,n'}$ is contained in the union of lines $(P = Q = 0) \cap (st = 0)$. Thus the hypersurface $V_{n,n'}$ is non-singular in codimension 1, hence normal. Moreover

$$\Omega_{V_{n,n'}} = dP \wedge ds \wedge dt/P = st \cdot d\log P \wedge d\log s \wedge d\log t,$$

and the pullback of $d\log P \wedge d\log s \wedge d\log t$ to a log resolution of $(V_{n,n'}, \operatorname{div}(PQst))$ has at most simple poles. Hence any prime divisor E over $V_{n,n'}$ with center in the singular locus satisfies $a(E, V_{n,n'}, 0) \geq \operatorname{mult}_E(st) > 0$, and $V_{n,n'}$ is klt.

In (i) with $n_1 \geq 1$, $V_1 = V_{2n_1-2,n_2}$ and $(2n_1-2)+n_2 \geq 2$. Thus W_0 is the quotient of the normal klt variety V_1 by $\langle \iota_1 \rangle$, hence normal, and W_0 is klt by Lemma 2.15(2)(3), since $K_{V_1} = \pi_1^* K_{W_0}$.

If W_0 is in (i) with $n_1 = 0$, then $n_2 \geq 3$, the singular locus of W_0 is the z-axis, and near a point of the z-axis with $z \neq 0$, W_0 is analytically the product of an A_{n_2-1} surface singularity and a smooth curve. Hence W_0 is normal, and klt outside 0. The Cartier divisor $W_0 \cap (z=t)$ is the Du Val surface $(x^2+y^2t+t^{n_2}=0)$ of type D_{n_2+1}, hence klt, so W_0 is klt near 0 by inversion of adjunction.

Assume that W_0 is in (ii). If $n_3 \geq 3$, then V_2 is in (i) with $n_1 = 0$ and $n_2 = n_3$ in the coordinates X, s, z, t, so V_2 is normal klt by the previous paragraph. If $n_3 = 2$, then $V_2 \cong V_{2,1}$, which is normal klt. Since π_2 is quasi-étale and $W_0 = V_2/\langle \iota_2 \rangle$, W_0 is normal klt by Lemma 2.15(2)(3). □

Proof of Theorem 12.3. We divide the proof into steps.

Step 1. In this step we construct the sequence to be excluded and replace it by polynomial models.

Let $(W \ni o)$, G, and h be as in the theorem, and let q be the radial function of $(W \ni o, G, h, m)$. For any prime divisor P over W whose center contains o and any $t \in [0,1]$,

$$a\Big(P, W, \frac{t}{m}\operatorname{div}(h)\Big) = (1-t)a(P,W,0) + ta\Big(P, W, \frac{1}{m}\operatorname{div}(h)\Big).$$

Since W is terminal and $(W, \frac{1}{m}\operatorname{div}(h))$ is lc near o, the right-hand side is decreasing in t and positive for $t < 1$. Thus q is decreasing, so $q(t) - q(1) \geq 0$ for any $t \in [0,1]$, and $(W, \frac{t}{m}\operatorname{div}(h))$ is klt near o for any $t \in [0,1)$. If h is a unit, then q is constant.

By [Liu26a, Lemma 11.3(1)(3)], there exist a smooth fourfold germ $(Y \ni o)$ with a faithful action of G, a G-equivariant closed embedding $W = (f=0) \subset Y$ for a regular G-semi-invariant function f on Y, and G-semi-invariant elements x, y, z, t of the maximal ideal of $\widehat{\mathcal{O}}_{Y,o}$ with $\widehat{\mathcal{O}}_{Y,o} = \mathbb{C}[[x,y,z,t]]$, on which G acts diagonally, such that the expansion of f is $x^2 + g(y,z,t)$ with $\operatorname{mult}_0 g \geq 3$ and the cubic part g_3 of g is the residual cubic. By hypothesis, g_3 is not the cube of a linear form. Let $\mathcal{C} := (g_3 = 0) \subset \mathbb{P}^2$, where y, z, t are homogeneous coordinates of $\mathbb{P}^2$. Since $x^2 + g$ is G-semi-invariant with the character of x^2, g_3 is G-semi-invariant.

Assume that $\mathcal{C}$ is a triangle or a secant conic [Liu26b, Definition-Lemma 6.4]. If $\mathcal{C}$ is a triangle, let y', z', t' be linear forms in y, z, t with $g_3 = y'z't'$. If $\mathcal{C}$ is a secant conic, let t' be the linear factor of g_3 and let y', z' define the lines tangent to the conic at the points of intersection of the conic and the line $(t'=0)$. The conics tangent to $(y'=0)$ at $(y'=t'=0)$ and to $(z'=0)$ at $(z'=t'=0)$ form the pencil spanned by $y'z'$ and t'^2, so after scaling y' and t' we have $g_3 = (y'z' + t'^2)t' = y'z't' + t'^3$. By [Liu26a, Lemma 11.3(2)] in the first case, and by the uniqueness of the factorization of g_3 in the second case, G permutes the three lines $(y'=0)$, $(z'=0)$, and $(t'=0)$, and fixes $(t'=0)$ if $\mathcal{C}$ is a secant conic. Thus G acts on y', z', t' by monomial transformations. If G fixes each of the three lines, then y', z', t' are G-semi-invariant, and we replace y, z, t with y', z', t', so that

$$g_3 = yzt, \quad \text{resp.} \quad g_3 = yzt + t^3.$$

Otherwise we replace y, z, t with G-semi-invariant linear combinations of y', z', t'. If $\mathcal{C}$ is a cuspidal cubic, a tangent conic or a double line, then by [Liu26b, Proposition 3.17(1)(2)] and [Liu26b, Lemma C.3], where for the double line $g_3 = L^2M$ with L and M G-semi-invariant by [Liu26a, Lemma 11.3(2)], so that $g_3 = y^2z$ after a G-equivariant linear change, after a G-equivariant linear change of y, z, t and a G-equivariant automorphism of $\mathbb{C}[[x,y,z,t]]$, the expansion of f is a G-invariant unit times $x^2 + g(y,z,t)$, where g is as follows. For the cuspidal cubic, $g = y^2z + c_0yt^\ell + b(z,t)$, where $c_0 \in \mathbb{C}$, $\ell \geq 2$, and $b \in (z,t)^3$ has part independent of z equal to t^3 times a G-invariant unit. For the tangent conic, $g = y^2z + yt^2 + b(z,t)$ with $b \in (z,t)^4$. For the double line, $g = y^2z + c_0yt^\ell + b(z,t)$ with $c_0 \in \mathbb{C}$, $\ell \geq 3$, and $\operatorname{mult}_0 b \geq 4$.

Let $x_1, \dots, x_4$ be G-semi-invariant regular parameters of Y at o with the characters of x, y, z, t, let $\widehat{f}$ be the expansion of f in $\mathbb{C}[[x_1, \dots, x_4]]$, and let ψ be a G-equivariant automorphism of $\mathbb{C}[[x_1, \dots, x_4]]$ and v_f a G-invariant unit with

$$v_f \psi(\widehat{f}) = x_1^2 + g(x_2, x_3, x_4).$$

Possibly shrinking Y to a G-invariant neighborhood of o, we may assume that h is the restriction of a G-semi-invariant regular function on Y. Let $t_* \in [0, 1)$. By Lemma 4.6 with $n = 4$, $\mathcal{P} = \emptyset$, $u = 1$, ψ, v_f in place of v, t_* in place of t, and a sufficiently large integer $\ell_0 \geq 4$ in place of L, writing x, y, z, t for the coordinates of $\mathbb{A}^4$, we obtain $W' = (f' = 0) \subset \mathbb{A}^4$ with $f' = x^2 + g_{<\ell_0}(y, z, t)$, where $g_{<\ell_0}$ is the sum of the terms of g of degree less than ℓ_0, and a G-semi-invariant polynomial h' with the character of h, such that $(W' \ni 0)$ is isolated Gorenstein terminal and G acts diagonally on $\mathbb{A}^4$ and faithfully on W'. By Lemma 4.6(1)(3), $(W', \frac{1}{m}\operatorname{div}(h'))$ is lc near 0, and for any subgroup $G' \subset G$,

$$\operatorname{mld}\Big(\Big(W' \ni 0, \frac{t_*}{m}\operatorname{div}(h')\Big)/G'\Big) = \operatorname{mld}\Big(\Big(W \ni o, \frac{t_*}{m}\operatorname{div}(h)\Big)/G'\Big). \tag{C.28}$$

Since $\ell_0 \geq 4$, the cubic part of $g_{<\ell_0}$ is g_3. If $h(o) = 0$, then $h'(0) = 0$, since $h' \equiv \psi(\widehat{h}) \pmod{(x, y, z, t)^{\ell_0}}$.

Set-up 11.1(1)–(3) of [Liu26b] hold for $(W' \ni 0)$, G, $s = 1$, $\mu_1 = 1$, and $h_1 = h'$, since marks may be units [Liu26b, Definition 2.7]. Condition $(4')$ there with $2m$ in place of m, that is, $(W', \frac{1}{2m}\operatorname{div}(h'))$ is klt near 0 and $a(P, W', 0) - \frac{1}{2m}\operatorname{mult}_P h' > 0$ for any prime divisor P over W' with center 0, holds, since $(W', \frac{1}{2m}\operatorname{div}(h'))$ is klt near 0, W' being terminal and $(W', \frac{1}{m}\operatorname{div}(h'))$ lc near 0. Possibly replacing h' with its normal form [Liu26b, Definition-Lemma 11.5(1)], which is congruent to h' modulo f', we may assume that h' has degree at most 1 in x. We call $(W' \ni 0, G, h')$ a *polynomial model* of $(W \ni o, G, h)$ at t_*.

Let E_0 be the prime divisor over W' given by the weights $(3, 2, 2, 2)$ on x, y, z, t. It is well defined, since the part $x^2 + g_3$ of f' of least weight is irreducible, $-g_3$ not being a square. By [Liu26b, Lemma 4.12], $a(E_0, W', 0) = 3$ and

$$\operatorname{mult}_{E_0} h' \geq \operatorname{wt}_{(3,2,2,2)}(h') \geq 2 \operatorname{mult}_0 h',$$

where $\operatorname{mult}_0 h'$ is the least total degree of a monomial of h'. Since $(W', \frac{1}{m}\operatorname{div}(h'))$ is lc near 0, $3 - \frac{1}{m}\operatorname{mult}_{E_0} h' \geq 0$. Thus

$$2 \operatorname{mult}_0 h' \leq \operatorname{mult}_{E_0} h' \leq 3m, \qquad \text{so} \qquad \operatorname{mult}_0 h' \leq \Big\lfloor \frac{3m}{2} \Big\rfloor. \tag{C.29}$$

Moreover, for the polynomial model at $t_* = 0$, (C.28) implies

$$q(0) = \operatorname{mld}\big((W' \ni 0, 0)/G\big) \leq a(E_0, W', 0) = 3.$$

Assume that the theorem does not hold for m and ϵ. Then for any positive integer i there exist data $(W_i \ni o_i, G_i, h_i)$ as in the theorem and $s_i \in (1 - \frac{1}{i}, 1]$ with $|G_i| \geq i$ and $q_i(s_i) - q_i(1) \geq \epsilon$, where q_i is the radial function of $(W_i \ni o_i, G_i, h_i, m)$. Thus $s_i < 1$, and h_i is not a unit. Since $q_i(0) \leq 3$ for any i, by Lemma 4.5, possibly passing to a subsequence, there exist $t_i \in [s_i, 1)$, strictly increasing with limit 1, such that

$$\alpha_i := q_i(t_i) < \alpha_{i+1}, \qquad \alpha_\infty := \lim_{i \to +\infty} \alpha_i > 0, \quad \text{and} \quad b_i := q_i(1) \leq \alpha_\infty - \frac{\epsilon}{2}. \tag{C.30}$$

For any i, we replace $(W_i \ni o_i, G_i, h_i)$ with a polynomial model at t_i, and q_i with the radial function of the polynomial model. By (C.28), the value of the new q_i at t_i is α_i, while b_i in (C.30) keeps denoting the value of the data before the replacement. By the definition of polynomial models, by the first paragraph of this step, and by (C.29), we may assume that the following hold for any i.

(a) $W_i = (f_i = 0) \subset \mathbb{A}^4$ with $f_i = x^2 + g_i(y,z,t)$, where $g_i \in (y,z,t)^3$ is a polynomial whose cubic part $g_{i,3}$ defines the residual cubic $\mathcal{C}_i$, $(W_i \ni 0)$ is isolated Gorenstein terminal, and G_i acts diagonally on x,y,z,t and faithfully on W_i. If $\mathcal{C}_i$ is a triangle, resp. a secant conic, and G_i fixes each of the three lines, then $g_{i,3} = yzt$, resp. $g_{i,3} = yzt + t^3$.
(b) h_i is a G_i-semi-invariant polynomial of degree at most 1 in x with $h_i(0) = 0$, $\operatorname{mult}_{E_0} h_i \le 3m$, and $\operatorname{mult}_0 h_i \le \lfloor 3m/2 \rfloor$.
(c) $(W_i, \frac{1}{m}\operatorname{div}(h_i))$ is lc near 0, and $(W_i, \frac{t_i}{m}\operatorname{div}(h_i))$ is klt near 0.

Step 2. In this step we take the limit of the groups G_i and compute quotient values along weighted blow-ups.

By [Liu26b, Lemma 4.6], possibly passing to a subsequence, the G_i converge in the Hausdorff distance to a closed subgroup $H \subset \mathbb{T}^4$, and for $i \gg 0$, $G_i \subset H$ and G_i intersects any connected component of H. Since $|G_i| \to +\infty$, $\dim H \ge 1$. Let $\rho : \mathbb{R}^4 \to \mathbb{T}^4$ be the quotient map of [Liu26a, Definition 2.15], and $\mathfrak{h} \subset \mathbb{R}^4$ the linear subspace with $\rho(\mathfrak{h}) = H^0$ [Liu26a, Lemma 4.3(1)]. For $\mathbf{a} = (a_x, a_y, a_z, a_t) \in \mathbb{Z}^4$ we write $\mathbf{x}^{\mathbf{a}} := x^{a_x} y^{a_y} z^{a_z} t^{a_t}$. For a finite diagonal group G let $N_G := \rho^{-1}(G)$, so that $v \in N_G$ if and only if $\langle v, \mathbf{a} \rangle \in \mathbb{Z}$ for any G-invariant Laurent monomial $\mathbf{x}^{\mathbf{a}}$. By [Liu26b, Lemma 4.7], if $W = (f = 0) \subset \mathbb{A}^4$ is normal and klt near 0, G is a finite diagonal group acting faithfully on W, and $D = \sum_j c_j \operatorname{div}(g_j)$ is a boundary with $c_j \ge 0$ and G-semi-invariant polynomials g_j such that (W, D) is lc near 0, then

$$\operatorname{mld}((W \ni 0, D)/G) \le \sum_{k=1}^{4} v_k - \operatorname{wt}_v(f) - \sum_j c_j \operatorname{wt}_v(g_j) \quad \text{for any } v \in N_G \cap \mathbb{R}^4_{>0};$$

moreover, for a sequence of such data whose groups converge in the Hausdorff distance to a closed subgroup of $\mathbb{T}^4$ with identity component $\rho(V)$ for a linear subspace V containing a vector with all coordinates positive, $\operatorname{mld}((W \ni 0, D)/G)$ tends to 0 along the sequence. If $\mathfrak{h}$ contained a vector with all coordinates positive, then by the second part of [Liu26b, Lemma 4.7] for the pairs $(W_i, \frac{t_i}{m}\operatorname{div}(h_i))$ of **Step 1**(c), $\alpha_i \to 0$, contrary to $\alpha_\infty > 0$. Thus

$$\mathfrak{h} \cap \mathbb{R}^4_{>0} = \emptyset. \tag{C.31}$$

Let χ be a character of $\mathbb{T}^4$ with $\chi|_H \ne 1$. If $\chi(g) \ne 1$ for some $g \in H^0$, then G_i contains elements $g_i' \to g$, and $\chi(g_i') \ne 1$ for $i \gg 0$. Otherwise $\chi|_{H^0} = 1$, and χ is a constant different from 1 on a connected component of H, which G_i intersects. In both cases

$$\chi|_{G_i} \ne 1 \quad \text{for any } i \gg 0. \tag{C.32}$$

Assume that $\dim H = 1$. By [Liu26b, Lemma 4.6], $H^0 = \rho(\mathbb{R}w)$ for a primitive vector $w = (w_x, w_y, w_z, w_t) \in \mathbb{Z}^4$, and with $\mathbf{T} := \mathbf{T}_{(w)}$, $r_i := |G_i \cap \mathbf{T}| \to +\infty$, and $[G_i : G_i \cap \mathbf{T}] = |H/H^0| =: e_0$ for $i \gg 0$. By **Step 1**, [Liu26b, Definition-Lemma 11.5(2)] holds for $(W_i \ni 0)$, G_i, and h_i with $2m$ in place of m. Thus, if $r_i > 2(6m-1)\max_j |w_j|$, then there exists a unique integer κ_i with $|\kappa_i| < r_i/2$ in the residue class modulo r_i of the $\mathbf{T}$-weights of the monomials of h_i, and, with h_i^0 the sum of the terms of h_i of $\mathbf{T}$-weight κ_i, any monomial of $h_i - h_i^0$ has total degree at least $(r_i - |\kappa_i|)/\max_j |w_j|$. Let $\mathbf{x}^{\mathbf{a}_i}$ be a monomial of h_i of total degree $\operatorname{mult}_0 h_i \le \lfloor 3m/2 \rfloor$. Then

$$|\operatorname{wt}_w(\mathbf{x}^{\mathbf{a}_i})| \le \lfloor 3m/2 \rfloor \max_j |w_j| < r_i/2 \quad \text{for } i \gg 0,$$

so $\operatorname{wt}_w(\mathbf{x}^{\mathbf{a}_i}) = \kappa_i$. Possibly passing to a subsequence, $\kappa_i = \kappa$ for any i. Then $h_i^0 \ne 0$, and h_i^0 has degree at most 1 in x. The data $(W_i, \frac{t_i}{m}\operatorname{div}(h_i))$ and G_i satisfy Set-up 5.1 of [Liu26b] with e_0 and the DCC set $\{t_j/m\}_{j \ge 1}$, and each of f_i and h_i has a monomial of total degree at most $\max\{2, \lfloor 3m/2 \rfloor\}$. By [Liu26b, Lemma 5.3], for $r_i > 2\max\{2, \lfloor 3m/2 \rfloor\}\max_j |w_j|$, the principal block f_i^0 of f_i is the sum of the terms of f_i of $\mathbf{T}$-weight $2w_x$, the $\mathbf{T}$-weight of x^2. Thus for $i \gg 0$,

$$\deg \mathbf{x}^{\mathbf{a}} \ge \frac{r_i - 2|w_x|}{\max_j |w_j|}, \qquad \text{resp.} \qquad \deg \mathbf{x}^{\mathbf{a}} \ge \frac{r_i - |\kappa|}{\max_j |w_j|}, \tag{C.33}$$

for any monomial $\mathbf{x}^{\mathbf{a}}$ of $f_i - f_i^0$, resp. $h_i - h_i^0$.

We compute the quotient value α_i along a weighted blow-up. Write $(z_1, z_2, z_3, z_4) := (x, y, z, t)$. Fix i, and let $\omega \in \mathbb{Z}^4_{>0}$ be a primitive vector such that the part $\mathrm{in}_\omega(f_i)$ of f_i of least ω-weight is irreducible and does not divide $\mathrm{in}_\omega(h_i)$. The objects below depend on i through ω, G_i, f_i, h_i, and t_i. By [Liu26b, Lemma 4.12], the weights ω define a prime divisor E_ω over W_i with center 0 and

$$a(E_\omega, W_i, 0) = \sum_{j=1}^{4} \omega_j - \mathrm{wt}_\omega(f_i), \qquad \mathrm{mult}_{E_\omega} h_i = \mathrm{wt}_\omega(h_i).$$

Let $R_\omega := G_i \cap \mathbf{T}_{(\omega)}$ and $r_\omega := |R_\omega|$. Since $\mathrm{in}_\omega(f_i) \in (x, y, z, t)^2$ is irreducible, the divisor

$$E_\omega = (\mathrm{in}_\omega(f_i) = 0) \subset \mathbb{P}(\omega)$$

is not contained in a coordinate hyperplane, and an element of G_i fixes a point of $\mathbb{P}(\omega)$ with non-zero coordinates if and only if it belongs to $\mathbf{T}_{(\omega)}$. Thus $I_{G_i}(E_\omega) = R_\omega$, and the coefficient of the image of E_ω in the boundary of the quotient of the weighted blow-up is

$$\gamma_i := 1 - \frac{a(E_\omega, W_i, \frac{t_i}{m}\operatorname{div}(h_i))}{r_\omega} = 1 - \frac{1}{r_\omega}\Big(\sum_{j=1}^{4} \omega_j - \mathrm{wt}_\omega(f_i) - \frac{t_i}{m}\mathrm{wt}_\omega(h_i)\Big). \tag{C.34}$$

By **Step 1**(c), $\gamma_i < 1$.

For $k \in \{1, \dots, 4\}$, let $\mathbb{A}^4_k$ be the affine space with coordinates s and z'_j for $j \neq k$, and $\pi_k : \mathbb{A}^4_k \to \mathbb{A}^4$ the morphism given by $z_k = s^{\omega_k}$ and $z_j = s^{\omega_j} z'_j$ for $j \neq k$. The group

$$\widetilde{G}_k := \{(g, \xi) \in G_i \times \mathbb{C}^* \mid \xi^{\omega_k} = \chi_{z_k}(g)\}$$

acts diagonally on $\mathbb{A}^4_k$ by $s \to \xi s$ and $z'_j \to \chi_{z_j}(g)\xi^{-\omega_j} z'_j$, the morphism π_k is equivariant along the projection $\widetilde{G}_k \to G_i$, and the quotient of $\mathbb{A}^4_k$ by the kernel μ_{ω_k} of $\widetilde{G}_k \to G_i$ is the k-th chart of the weighted blow-up of $\mathbb{A}^4$ with weights ω. The subgroup

$$\widetilde{R}_k := \{(\xi^\omega, \xi) \mid \xi^\omega \in R_\omega\} \cong R_\omega$$

of $\widetilde{G}_k$ acts on s alone, so $\mathbb{A}^4_k/\widetilde{R}_k$ is the affine space with coordinates $\varsigma := s^{r_\omega}$ and z'_j. Write

$$\pi_k^* f_i = s^{\mathrm{wt}_\omega(f_i)} f_{i,k}, \qquad \pi_k^* h_i = s^{\mathrm{wt}_\omega(h_i)} h_{i,k}.$$

Since f_i and h_i are R_ω-semi-invariant, the exponents of s in $f_{i,k}$ and $h_{i,k}$ are divisible by r_ω, so $f_{i,k}$ and $h_{i,k}$ are polynomials in ς and the z'_j. Let $W_{i,k} := (f_{i,k} = 0)$ in the affine space with coordinates ς and z'_j, with the faithful diagonal action of $G_{i,k} := \widetilde{G}_k/\widetilde{R}_k$, and let

$$D_{i,k} := \frac{t_i}{m}\operatorname{div}(h_{i,k}|_{W_{i,k}}) + \gamma_i \operatorname{div}(\varsigma|_{W_{i,k}}).$$

The variety $W_{i,k}$ is normal near $(\varsigma = 0)$: it is a hypersurface, near $(\varsigma = 0)$ it is smooth outside $(\varsigma = 0)$ since $(W_i \ni 0)$ is an isolated singularity, and $W_{i,k} \cap (\varsigma = 0)$ is the reduced surface $(\mathrm{in}_\omega(f_i) = 0)$. The pullback of

$$K_{\mathbb{A}^4} + \operatorname{div}(f_i) + \frac{t_i}{m}\operatorname{div}(h_i)$$

to $\mathbb{A}^4_k$ is

$$K_{\mathbb{A}^4_k} + \operatorname{div}(f_{i,k}) + \frac{t_i}{m}\operatorname{div}(h_{i,k}) + \Big(1 - \sum_j \omega_j + \mathrm{wt}_\omega(f_i) + \frac{t_i}{m}\mathrm{wt}_\omega(h_i)\Big)\operatorname{div}(s),$$

and the quotient by $\widetilde{R}_k$ is ramified along $(s = 0)$ with index r_ω. By adjunction on $\mathbb{A}^4$ and on $\mathbb{A}^4_k$, the quotients $(W_{i,k}, D_{i,k})/G_{i,k}$, $k = 1, \dots, 4$, are open subsets covering the quotient pair (Q'_i, Δ'_i) of the strict transform of W_i in the weighted blow-up by G_i, and (Q'_i, Δ'_i) is the crepant pullback of the quotient pair (Q_i, Δ_i) of $(W_i, \frac{t_i}{m}\operatorname{div}(h_i))$ by G_i. We call (Q'_i, Δ'_i) the *return* along E_ω. The fiber of Q'_i over the image $\overline{o}_i$ of 0 is the image of E_ω.

Since the quotient value is the minimal log discrepancy of the quotient pair [Liu26b, Definition 2.7], $\alpha_i = \mathrm{mld}(Q_i \ni \bar{o}_i, \Delta_i)$. The pair (Q_i, Δ_i) is klt near $\bar{o}_i$, so α_i is attained by a prime divisor F_i over Q_i with center $\bar{o}_i$. Let $Z_i := \mathrm{center}_{Q'_i} F_i$, which is contained in the image of E_ω. Any prime divisor over Q'_i with center Z_i has center $\bar{o}_i$ on Q_i, so α_i is the minimal log discrepancy of (Q'_i, Δ'_i) at the generic point of Z_i. Let $\Gamma := \{t_j/m\}_{j\geq 1} \cup \{\gamma_j\}_{j\geq 1}$, and let $\mathcal{D}(\Gamma)$ be the set of numbers $1 - \frac{1-c}{n}$, where $n \in \mathbb{N}^+$ and $c \leq 1$ is a finite sum of elements of $\Gamma \cup \{0\}$.

If $\Gamma \subset [0,1]$ is a DCC set, then so is $\mathcal{D}(\Gamma)$, since finite sums of elements of a DCC set form a DCC set. If moreover $\gamma_i \geq 0$, then the coefficients of Δ'_i near the fiber over $\bar{o}_i$ belong to $\mathcal{D}(\Gamma)$, as Δ'_i is the quotient of the effective boundaries $D_{i,k}$, whose coefficients are finite sums of elements of Γ and less than 1 over a neighborhood of $\bar{o}_i$, the pairs $(W_{i,k}, D_{i,k})$ being klt there. We obtain the following.

(R) Assume that for any i a primitive vector $\omega_i \in \mathbb{Z}^4_{>0}$ with $\mathrm{in}_{\omega_i}(f_i)$ irreducible and not dividing $\mathrm{in}_{\omega_i}(h_i)$ is given, form E_ω, R_ω, $W_{i,k}$, $D_{i,k}$, ς, and γ_i with $\omega = \omega_i$, and assume that $(\gamma_i)_i$ is an increasing sequence in $[0,1)$ and, for any i, any k and any closed point $o' \in W_{i,k}$ with $\varsigma(o') = 0$, the germ $(W_{i,k} \ni o')$ is smooth or a hypersurface threefold germ of quadratic rank at least 3. Then $(\alpha_i)_i$ is not strictly increasing.

We prove (R). The set Γ is a DCC set contained in $[0,1)$, since $(t_j)_j$ and $(\gamma_j)_j$ are increasing sequences in $[0,1)$. Possibly passing to a subsequence, $\dim Z_i$ is constant. If $\dim Z_i = 2$, then Z_i is the image of E_ω and $\alpha_i = 1 - \gamma_i$, so $(\alpha_i)_i$ is decreasing.

If $\dim Z_i = 1$, let ϑ_i be a rational function on Q'_i which is regular at the generic point of Z_i and non-constant on Z_i, and let Ω be an algebraic closure of $\mathbb{C}(\vartheta_1)$, regarded as an algebraic closure of $\mathbb{C}(\vartheta_i)$ through the $\mathbb{C}$-isomorphism $\mathbb{C}(\vartheta_1) \to \mathbb{C}(\vartheta_i)$ sending ϑ_1 to ϑ_i. Let $(X_{i,\Omega}, B_{i,\Omega})$ and c'_i be the surface pair over Ω and its closed point given by Lemma 3.5 for $(X, B) = (Q'_i, \Delta'_i)$, $C = Z_i$, the trivial group in place of H, and ϑ_i in place of t. Since (Q'_i, Δ'_i) is klt near Z_i, as the crepant pullback of (Q_i, Δ_i), which is klt near $\bar{o}_i$, with $\Delta'_i \geq 0$ near the fiber over $\bar{o}_i$, which contains Z_i, $(X_{i,\Omega}, B_{i,\Omega})$ is klt near c'_i by Lemma 3.5(3), so by Lemma 3.5(0)(2), $\mathrm{mld}(X_{i,\Omega} \ni c'_i, B_{i,\Omega})$ is the minimal log discrepancy of (Q'_i, Δ'_i) at the generic point of Z_i, that is, α_i. The coefficients of $B_{i,\Omega}$ are coefficients of Δ'_i, hence belong to $\mathcal{D}(\Gamma)$. By Theorem 2.21(1) with $\Bbbk = \Omega$, the set $\{\alpha_i\}$ satisfies the ACC.

If $\dim Z_i = 0$, let o_i^+ be a closed point of the strict transform W_i^+ of W_i over Z_i, $G_{o_i^+} \subset G_i$ its stabilizer, and S the set of k such that the k-th chart contains o_i^+. The weighted blow-up is the quotient of $\{(s, z'_1, \dots, z'_4) \in \mathbb{A}^5 \mid (z'_1, \dots, z'_4) \neq 0\}$ by $\mathbb{C}^*$ acting by $\xi \cdot (s, z'_j) = (\xi s, \xi^{-\omega_j} z'_j)$, with $z_j = s^{\omega_j} z'_j$, the k-th chart being the image of $(z'_k = 1) \cong \mathbb{A}^4_k$. Let $n_S := \sum_{k\in S} \omega_k$ and $T := (\prod_{k\in S} z'_k = 1) \subset \mathbb{A}^5$. Each $\mathbb{C}^*$-orbit in $(\prod_{k\in S} z'_k \neq 0)$ intersects T in a μ_{n_S}-orbit, so T/μ_{n_S} is the intersection B_S of the charts with index in S. The subgroup $\widetilde{R} := \widetilde{R}_k$ of $G_i \times \mathbb{C}^*$ does not depend on k. For $k \in S$, adjoining an ω_k-th root of the unit z'_k on T and adjoining an n_S-th root of the unit $\prod_{l\in S} z'_l$ on $\mathbb{A}^4_k \cap (\prod_{l\in S} z'_l \neq 0)$ give varieties isomorphic over B_S by $(s, z') \to \xi \cdot (s, z')$ with $\xi^{n_S} = \prod_{l\in S} z'_l$, an isomorphism commuting with the action $s \to \xi s$ of $\widetilde{R}$. The group

$$\widetilde{G} := \left\{ (g, \xi) \in G_{o_i^+} \times \mathbb{C}^* \;\middle|\; \xi^{n_S} = \prod_{k\in S} \chi_{z_k}(g) \right\}$$

acts on T by $s \to \xi s$ and $z'_j \to \xi^{-\omega_j} \chi_{z_j}(g) z'_j$, with $T/\widetilde{G} = B_S/G_{o_i^+}$, and contains $\widetilde{R}$, which acts on s alone. Thus $T/\widetilde{R}$ is smooth with coordinates ς and the z'_j, and, for the morphism $\pi : T \to \mathbb{A}^4$, $\pi^* f_i = s^{\mathrm{wt}_\omega(f_i)} f_i^\sharp$ and $\pi^* h_i = s^{\mathrm{wt}_\omega(h_i)} h_i^\sharp$ for polynomials $f_i^\sharp$, $h_i^\sharp$ in ς and the z'_j. Let $W^\sharp := (f_i^\sharp = 0) \subset T/\widetilde{R}$ and

$$D^\sharp := \frac{t_i}{m} \operatorname{div}(h_i^\sharp|_{W^\sharp}) + \gamma_i \operatorname{div}(\varsigma|_{W^\sharp}).$$

After adjoining the roots of z'_k and of $\prod_{l\in S} z'_l$, the germs of $(W^\sharp, D^\sharp)$ and of $(W_{i,k}, D_{i,k})$ at closed points with a common image in W_i^+/R_ω have a common étale neighborhood. Hence the quotient of $(W^\sharp, D^\sharp)$ by $\widetilde{G}/\widetilde{R}$ is the open subset over B_S of the quotient pair of W_i^+ by $G_{o_i^+}$, which is étale over (Q'_i, Δ'_i) at the image of o_i^+, as $G_{o_i^+}$ is the stabilizer of o_i^+. Moreover, by the hypothesis of (R), for any closed point $o^\sharp \in W^\sharp$ over o_i^+, the germ $(W^\sharp \ni o^\sharp)$ is smooth or a hypersurface threefold germ of quadratic rank at least 3. Let $o^\sharp$ be a closed point of $W^\sharp$ over o_i^+ and $G^\sharp \subset \widetilde{G}/\widetilde{R}$ its stabilizer, acting on $W^\sharp$ through its image in the automorphism group of $W^\sharp$. Since $W^\sharp/G^\sharp \to W^\sharp/(\widetilde{G}/\widetilde{R})$ is étale at the image of $o^\sharp$,

$$\alpha_i = \mathrm{mld}((W^\sharp \ni o^\sharp, D^\sharp)/G^\sharp),$$

where $D^\sharp$ is a labelled boundary with labels t_i/m and γ_i in Γ, and $(W^\sharp, D^\sharp)$ is klt near $o^\sharp$. By Theorem 10.1, the set $\{\alpha_i\}$ satisfies the ACC. In each case $(\alpha_i)_i$ is not strictly increasing.

Step 3. In this step we list the cases.

By [Liu26b, Definition-Lemma 6.4] and the hypothesis on the residual cubic, possibly passing to a subsequence, the type of $\mathcal{C}_i$ does not depend on i and is one of the following: smooth cubic, nodal cubic, cuspidal cubic, secant conic, tangent conic, triangle, three concurrent lines, double line. If the $\mathcal{C}_i$ are triangles or secant conics, then, possibly passing to a subsequence, either G_i fixes each of the three lines for any i, or the image of G_i in the permutations of the three lines is a fixed non-trivial cyclic group: generated by a 3-cycle or a transposition for the triangle, and the exchange of $(y' = 0)$ and $(z' = 0)$ for the secant conic.

Assume that the $\mathcal{C}_i$ are triangles or secant conics and G_i fixes each of the three lines. By **Step 1**(a), the monomials x^2 and yzt of f_i, and t^3 for the secant conic, have the character of f_i. Thus $\chi_x^2 = \chi_y\chi_z\chi_t$, and $\chi_t^3 = \chi_x^2$ for the secant conic, on G_i for any i, hence on H by (C.32). Thus

$$\mathfrak{h} \subset \mathfrak{t} := \{2w_x = w_y + w_z + w_t\},$$

and, for the secant conic,

$$\mathfrak{h} \subset \mathfrak{t}' := \mathfrak{t} \cap \{3w_t = 2w_x\},$$

where $\dim \mathfrak{t} = 3$, $\dim \mathfrak{t}' = 2$, and $(3,2,2,2) \in \mathfrak{t}'$. Thus, if $\dim H = 3$, or $\dim H = 2$ and the $\mathcal{C}_i$ are secant conics, then $(3,2,2,2) \in \mathfrak{h}$. If $\dim H = 1$, then we write $w = (u, w_y, w_z, w_t)$ with $u := w_x$ and $2u = w_y + w_z + w_t$. Possibly replacing w with $-w$, we may assume that $u \geq 0$. If $u = 0$, then at most one of w_y, w_z, w_t is zero, and if one is zero, then w is $(0,1,-1,0)$ up to the order of y, z, t, since w is primitive. If $u > 0$, then at most two of w_y, w_z, w_t are zero, and if two are zero, then w is $(1,0,0,2)$ up to the order of y, z, t, since w is primitive. Thus the cases of Rows 3a–3f of Table 1 exhaust the possibilities.

In Table 1, H, $\mathfrak{h}$ and the circle $\mathbf{T} = \mathbf{T}_{(w)}$ are as in **Step 2**, and a circle is given by $w = (w_x, w_y, w_z, w_t)$. In Rows 4, 5 and 6, (c, v), (v, c_1), and (c_2, c_3) are pairs of coprime integers. In Rows 7b–7h, a circle satisfies $w_z = 2(w_x - w_y)$, and the *first face* of f_i along the circle, that is, its initial part in (C.50) (Row 7b), is one of

$$\mathrm{I}: \ x^2 + y^2 z + z^{n_1} t^{n_2}, \qquad \mathrm{II}: \ x^2 + y^2 z + y t^{n_3},$$

where $n_1, n_2, n_3 \in \mathbb{N}$, with $n_1 + n_2 \geq 4$ or $(n_1, n_2) = (0,3)$ in I, and $n_3 \geq 2$ in II. The cases $(n_1, n_2) = (0,3)$ and $n_3 = 2$ arise from the cuspidal cubic and the tangent conic, respectively.

Step 4. In this step we exclude Rows 1 and 2.

Assume that the $\mathcal{C}_i$ are smooth cubics or nodal cubics. Since q_i is decreasing,

$$q_i(0) \geq \alpha_i \geq \frac{\alpha_\infty}{2} > 0 \quad \text{for } i \gg 0.$$

By [Liu26a, Proposition 11.10],

$$|G_i| \leq \frac{162}{q_i(0)} \leq \frac{324}{\alpha_\infty} \quad \text{for } i \gg 0,$$

contrary to $|G_i| \to +\infty$.

Row	Residual cubic $\mathcal{C}_i$	Limit H of the G_i
1	smooth cubic	$\dim H \geq 1$
2	nodal cubic	$\dim H \geq 1$
3a	triangle, $g_{i,3} = yzt$, or secant conic, $g_{i,3} = yzt + t^3$; G_i fixes each line	$\mathfrak{h} \cap \mathbb{R}^4_{>0} \neq \emptyset$, in particular if $\dim H = 3$, or $\dim H = 2$ for the secant conic
3b	triangle, as in 3a	$\dim H = 2$, $\mathfrak{h} \cap \mathbb{R}^4_{>0} = \emptyset$
3c	as in 3a	circle (u, w_y, w_z, w_t), $u \geq 0$, $w_y w_z w_t \neq 0$, not all positive
3d	as in 3a	circle, $u > 0$, exactly one of w_y, w_z, w_t zero
3e	as in 3a	circle, $u > 0$, two of w_y, w_z, w_t zero, $(1, 0, 0, 2)$ up to order
3f	as in 3a	circle $(0, 1, -1, 0)$ up to the order of y, z, t
3g	triangle with a 3-cycle of the lines, or secant conic with $(y' = 0)$, $(z' = 0)$ exchanged	limit of the diagonal kernels in the circle of $(3, 2, 2, 2)$
3h	triangle with $(y' = 0)$, $(z' = 0)$ exchanged	circles (u, w_y, w_y, w_t) and (u, w_y, w_z, w_z), $w_y < 0$; $(1, 1, 1, 0)$ and $(1, 0, 0, 2)$
4	three concurrent lines	circles $(3c, 2c, 2c, v)$, $cv \leq 0$
5	cuspidal cubic, $x^2 + y^2 z + t^3 + \cdots$	circles $(3v, c_1, 6v - 2c_1, 2v)$
5′	cuspidal cubic	circle $(3, 0, 6, 2)$
6	tangent conic, $x^2 + y^2 z + yt^2 + \cdots$	circles $(2c_2 + c_3, 2c_2, 2c_3, c_2 + c_3)$
7a	double line, $f_i = x^2 + y^2 z + C_i(t)y + B_i(z, t)$	$\mathrm{mult}_0(C_i y + B_i)$ unbounded
7b	double line, and the circles $\mathbf{T}_{(0,1,-2,0)}$, $\mathbf{T}_{(3,4,-2,2)}$ of Row 5 and $\mathbf{T}_{(1,0,2,1)}$, $\mathbf{T}_{(1,2,-2,0)}$ of Row 6	$\dim H \neq 1$
7c	first face I or II	circle, $w_z w_t < 0$
7d	as in 7c	circle, $w_z w_t > 0$
7e	as in 7c	circle, $w_z = 2w_t$
7f	as in 7c	circle, $w_z = 0$
7g	as in 7c	circle, $w_t = 0 \neq w_z$
7h	as in 7c	circles of Row 7c, of Row 7d with $w_y < 0$, and $(0, -1, 2, 0)$, $(-1, -2, 2, 0)$

TABLE 1. Cases in the proof of Theorem 12.3.

Step 5. In this step we exclude Rows 3a and 3b.

Row 3a does not occur by (C.31). Assume that we are in Row 3b: the $\mathcal{C}_i$ are triangles, G_i fixes each of the three lines, $\dim H = 2$ and $\mathfrak{h} \cap \mathbb{R}^4_{>0} = \emptyset$.

Let $W_\triangle := (x^2 + yzt = 0) \subset \mathbb{A}^4$ and let $\pi_\triangle : \mathbb{A}^3 \to W_\triangle$ be the morphism

$$(p_1, p_2, p_3) \to (\sqrt{-1} p_1 p_2 p_3, p_1^2, p_2^2, p_3^2).$$

It is the quotient by the group of the sign changes of an even number of the coordinates p_1, p_2, p_3, whose non-trivial elements fix only coordinate axes, so $\pi_\triangle$ is quasi-étale. For $\mathbf{a} \in \mathbb{N}^4$, the pullback $\pi_\triangle^* \mathbf{x}^\mathbf{a}$ is a non-zero constant times $p^{\tau(\mathbf{x}^\mathbf{a})}$, where

$$\tau(\mathbf{x}^\mathbf{a}) := a_x(1, 1, 1) + 2(a_y, a_z, a_t) \in \mathbb{N}^3.$$

Let $|\tau| := \tau_1 + \tau_2 + \tau_3$ for $\tau \in \mathbb{R}^3$. The total degree of $\mathbf{x}^\mathbf{a}$ is

$$a_x + a_y + a_z + a_t = \frac{|\tau(\mathbf{x}^\mathbf{a})| - a_x}{2} \leq |\tau(\mathbf{x}^\mathbf{a})|,$$

and if $a_x \leq 1$, then $a_x \equiv \tau(\mathbf{x}^\mathbf{a})_1 \pmod 2$ is determined by $\tau(\mathbf{x}^\mathbf{a})$. The vectors $\tau(\mathbf{x}^\mathbf{a})$ of the monomials $\mathbf{x}^\mathbf{a}$ of degree at most 1 in x are the elements of

$$\mathcal{L}_\triangle \cap \mathbb{N}^3, \qquad \mathcal{L}_\triangle := \{\mu \in \mathbb{Z}^3 \mid \mu_1 \equiv \mu_2 \equiv \mu_3 \pmod 2\},$$

each attained by exactly one such monomial. Let $\mathbb{T}_\triangle \subset \mathbb{T}^4$ be the subgroup defined by $\chi_x^2 = \chi_y \chi_z \chi_t$. The map sending $\chi_x^{e_x} \chi_y^{e_y} \chi_z^{e_z} \chi_t^{e_t}$ to $e_x(1, 1, 1) + 2(e_y, e_z, e_t)$ is an isomorphism from

the character group of $\mathbb{T}_\triangle$ onto $\mathcal{L}_\triangle$, which extends τ. In particular, $\mathbb{T}_\triangle$ is connected. For $\zeta \in \mathbb{R}^3$ let

$$\varphi(\zeta) := (\zeta_1+\zeta_2+\zeta_3, 2\zeta_1, 2\zeta_2, 2\zeta_3) \in \mathbb{R}^4.$$

Then

$$\mathbb{T}_\triangle = \{\rho(\varphi(\zeta)) \mid \zeta \in \mathbb{R}^3\},$$

and for a monomial $\mathbf{x}^{\mathbf{a}}$,

$$\mathrm{wt}_{\varphi(\zeta)}(\mathbf{x}^{\mathbf{a}}) = \langle \tau(\mathbf{x}^{\mathbf{a}}), \zeta\rangle, \qquad \chi_{\mathbf{x}^{\mathbf{a}}}(\rho(\varphi(\zeta))) = \exp(2\pi\sqrt{-1}\langle \tau(\mathbf{x}^{\mathbf{a}}), \zeta\rangle).$$

By **Step 3**, $H \subset \mathbb{T}_\triangle$. Since $\dim H = 2$, there exists a primitive element μ of $\mathcal{L}_\triangle$ with $H^0 = \{\rho(\varphi(\zeta)) \mid \langle \mu, \zeta\rangle = 0\}$. Since $\varphi(\zeta) \in \mathbb{R}^4_{>0}$ if and only if $\zeta \in \mathbb{R}^3_{>0}$, by (C.31) the plane $\mu^\perp$ does not intersect $\mathbb{R}^3_{>0}$, so the entries of μ do not have opposite signs. Possibly replacing μ with $-\mu$, we may assume that $\mu \in \mathbb{N}^3\backslash\{0\}$. Write $\mu = \varpi(1,1,1) + 2\nu$ with $\varpi \in \{0,1\}$ and $\nu \in \mathbb{N}^3$, and let

$$U := x^\varpi y^{\nu_1} z^{\nu_2} t^{\nu_3},$$

so that $\tau(U) = \mu$. Then $H^0 = \ker(\chi_U|_{\mathbb{T}_\triangle})$, since μ is primitive. Let n_U be the order of $\chi_U|_{G_i}$. Since $G_i \subset H$ intersects any connected component of H and $\chi_U|_{H^0} = 1$, $\chi_U(G_i) = \chi_U(H)$, so n_U does not depend on $i \gg 0$, and

$$H = \{g \in \mathbb{T}_\triangle \mid \chi_U(g)^{n_U} = 1\}.$$

Let Ψ be the monomial in y,z,t with $\tau(\Psi) = (1+\varpi)\mu$. The vectors $\tau(\mathbf{x}^{\mathbf{a}})$ of the monomials $\mathbf{x}^{\mathbf{a}}$ in y,z,t belong to $2\mathbb{N}^3$, and $2\mathbb{Z}^3 \cap \mathbb{Z}\mu = \mathbb{Z}(1+\varpi)\mu$. Thus a monomial in y,z,t has the character of yzt on H^0 if and only if it is $yzt\Psi^k$ for an integer k. Let $\overline{f}_i$ be the sum of the terms of f_i with the character of x^2 on H^0. If $k<0$, then $\deg(yzt\Psi^k) \le 2 < 3 \le \mathrm{mult}_0\, g_i$. Since $g_{i,3} = yzt$, we obtain

$$\overline{f}_i = x^2 + yztV_i(\Psi)$$

for a polynomial V_i in one variable with $V_i(0) = 1$. Let $\mathbf{x}^{\mathbf{a}_i}$ be a monomial of h_i of total degree $\mathrm{mult}_0\, h_i$, and $\overline{h}_i$ the sum of the terms of h_i with the character of $\mathbf{x}^{\mathbf{a}_i}$ on H^0. Since $\mathrm{mult}_0\, h_i \le \lfloor 3m/2\rfloor$ by **Step 1**(b), possibly passing to a subsequence, $\mathbf{a}_i = \mathbf{a}_1$ for any i. Let $\mathbf{e} := \tau(\mathbf{x}^{\mathbf{a}_1}) \in \mathbb{N}^3$. Since f_i and h_i are G_i-semi-invariant, by (C.32) applied to the finitely many monomials of total degree at most d, for any integer $d \ge 1$ we have

$$f_i - \overline{f}_i \in (x,y,z,t)^d \quad \text{and} \quad h_i - \overline{h}_i \in (x,y,z,t)^d \quad \text{for } i \gg 0. \tag{C.35}$$

The vectors $\tau(\mathbf{x}^{\mathbf{a}})$ of the monomials $\mathbf{x}^{\mathbf{a}}$ of $\overline{h}_i$ are of the form $\mathbf{e} + k\mu$ with $k \in \mathbb{Z}$. The total degree of the monomial $\mathbf{x}^{\mathbf{a}}$ with $\tau(\mathbf{x}^{\mathbf{a}}) = \mathbf{e} + k\mu$ is strictly increasing in k, since $|\mu| \ge 2$, and it is at least $\mathrm{mult}_0\, h_i$ for the monomials of h_i. Thus $k \ge 0$ for the monomials of $\overline{h}_i$, and the term of $\overline{h}_i$ with $k = 0$ is a non-zero multiple of $\mathbf{x}^{\mathbf{a}_1}$.

Let $\overline{W}_i := (\overline{f}_i = 0) \subset \mathbb{A}^4$, and let Φ_i be the automorphism of $\mathbb{C}[[x,y,z,t]]$ fixing y,z,t with $\Phi_i(x) = xV_i(\Psi)^{1/2}$, where the square root has value 1 at 0. Then $\Phi_i(\overline{f}_i) = V_i(\Psi)(x^2+yzt)$, so Φ_i induces an isomorphism between the completions of $W_\triangle$ and of $\overline{W}_i$ at 0. Here and below, by [Liu26a, Lemma 3.3(2)(3)], the prime divisors with center 0 over a pair given by polynomials correspond bijectively, preserving log discrepancies, to the prime divisors with center the closed point over the corresponding pair on the spectrum of the completed local ring at 0. In particular, $\overline{W}_i$ is normal and klt near 0. For a monomial $\mathbf{x}^{\mathbf{a}}$, the pullback of $\Phi_i(\mathbf{x}^{\mathbf{a}})$ to $\mathbb{C}[[p_1,p_2,p_3]]$ is a unit times $p^{\tau(\mathbf{x}^{\mathbf{a}})}$. Hence the pullbacks of $\Phi_i(\overline{h}_i)$ and $\Phi_i(U)$ are units times $p^{\mathbf{e}}$ and p^μ. Since a pair is lc near 0 if and only if the log discrepancies of the prime divisors over it with center 0 are non-negative, lc near 0 is determined by the completed pair. Since $\pi_\triangle$ is quasi-étale, for any real number $c \ge 0$, the pair

$$\Big(\overline{W}_i, \frac{1}{m}\operatorname{div}(\overline{h}_i) + c\operatorname{div}(U)\Big)$$

is lc near 0 if and only if the pair

$$\Big(\mathbb{A}^3, \frac{1}{m}\operatorname{div}(p^{\mathbf{e}}) + c\operatorname{div}(p^\mu)\Big)$$

is lc near 0, that is,

$$\frac{\mathbf{e}_j}{m} + c\mu_j \le 1 \quad \text{for } j = 1,2,3. \tag{C.36}$$

We show that $\mathbf{e}_j \le m$ for any j. By **Step 1**(c) and Theorem 2.17(1) with the trivial group, the pair

$$\Big(\mathbb{A}^4, \operatorname{div}(f_i) + \frac{1}{m}\operatorname{div}(h_i)\Big)$$

is lc near 0. By (C.35) and Lemma 2.19 with $\Lambda = \{1, \frac{1}{m}\}$, for $i \gg 0$, the pair

$$\Big(\mathbb{A}^4, \operatorname{div}(\overline{f}_i) + \frac{1}{m}\operatorname{div}(\overline{h}_i)\Big)$$

is lc near 0. By adjunction, $(\overline{W}_i, \frac{1}{m}\operatorname{div}(\overline{h}_i))$ is lc near 0, and (C.36) with $c = 0$ implies $\mathbf{e}_j \le m$. Let

$$\lambda := \min_{j:\mu_j>0} \frac{1 - \mathbf{e}_j/m}{\mu_j} \ge 0.$$

We show the upper bound $\alpha_\infty \le \lambda/n_U$. Let $\zeta \in \mathbb{R}^3_{>0}$ with $\langle \mu, \zeta\rangle = 1/n_U$. Then

$$\chi_U(\rho(\varphi(\zeta))) = \exp(2\pi\sqrt{-1}/n_U),$$

so $\rho(\varphi(\zeta)) \in H$. Since the G_i converge to H, there exist $\omega_i \in N_{G_i}$ with $\omega_i \to \varphi(\zeta)$. For $i \gg 0$, $\omega_i \in \mathbb{R}^4_{>0}$. By [Liu26b, Lemma 4.7] applied to $(W_i, \frac{t_i}{m}\operatorname{div}(h_i))$ and ω_i,

$$\alpha_i \le \sum_{j=1}^4 \omega_{i,j} - \operatorname{wt}_{\omega_i}(f_i) - \frac{t_i}{m}\operatorname{wt}_{\omega_i}(h_i).$$

Let $d_i := \max_j |\omega_{i,j} - \varphi(\zeta)_j|$. For a monomial $\mathbf{x}^{\mathbf{a}}$,

$$\operatorname{wt}_{\omega_i}(\mathbf{x}^{\mathbf{a}}) \ge \langle \tau(\mathbf{x}^{\mathbf{a}}), \zeta\rangle - d_i|\tau(\mathbf{x}^{\mathbf{a}})|.$$

For the monomials $\mathbf{x}^{\mathbf{a}}$ of $\overline{f}_i$, $\tau(\mathbf{x}^{\mathbf{a}}) = (2,2,2) + k(1+\varpi)\mu$ with $k \ge 0$, and for the monomials $\mathbf{x}^{\mathbf{a}}$ of $\overline{h}_i$, $\tau(\mathbf{x}^{\mathbf{a}}) = \mathbf{e} + k\mu$ with $k \ge 0$. Since $\langle \mu, \zeta\rangle - d_i|\mu| > 0$ for $i \gg 0$, the ω_i-weights of the monomials of $\overline{f}_i$, resp. $\overline{h}_i$, are at least $2|\zeta| - 6d_i$, resp. $\langle \mathbf{e}, \zeta\rangle - d_i|\mathbf{e}|$. The monomials of $f_i - \overline{f}_i$ and $h_i - \overline{h}_i$ have ω_i-weights at least $\min_j \omega_{i,j}$ times their degrees, which tend to $+\infty$ by (C.35). Since $\sum_j \varphi(\zeta)_j = 3|\zeta|$ and $t_i \to 1$, we obtain

$$\alpha_\infty \le 3|\zeta| - 2|\zeta| - \frac{1}{m}\langle \mathbf{e}, \zeta\rangle = \sum_{j=1}^3 \Big(1 - \frac{\mathbf{e}_j}{m}\Big)\zeta_j.$$

For j with $\mu_j > 0$, letting $\zeta \in \mathbb{R}^3_{>0}$ with $\langle \mu, \zeta\rangle = 1/n_U$ tend to $\frac{1}{n_U\mu_j}$ times the j-th standard basis vector, we obtain $\alpha_\infty \le \frac{1-\mathbf{e}_j/m}{n_U\mu_j}$. Thus $\alpha_\infty \le \lambda/n_U$. If $\lambda = 0$, then $\alpha_\infty \le 0$, contrary to (C.30).

Assume that $\lambda > 0$. We show the lower bound $\alpha_i \ge \lambda/n_U$ for $i \gg 0$. By (C.36) with $c = \lambda$, the pair

$$\Big(\overline{W}_i, \frac{1}{m}\operatorname{div}(\overline{h}_i) + \lambda\operatorname{div}(U)\Big)$$

is lc near 0. By Theorem 2.17(1) with the trivial group, the pair

$$\Big(\mathbb{A}^4, \operatorname{div}(\overline{f}_i) + \frac{1}{m}\operatorname{div}(\overline{h}_i) + \lambda\operatorname{div}(U)\Big)$$

is lc near 0. By (C.35) and Lemma 2.19 with $\Lambda = \{1, \frac{1}{m}, \lambda\}$, for $i \gg 0$, the pair

$$\Big(\mathbb{A}^4, \operatorname{div}(f_i) + \frac{1}{m}\operatorname{div}(h_i) + \lambda\operatorname{div}(U)\Big)$$

is lc near 0, so by adjunction the pair

$$\Big(W_i, \frac{1}{m}\operatorname{div}(h_i) + \lambda\operatorname{div}(U)\Big)$$

is lc near 0. By [Liu26b, Lemma 5.8] with $Z = \{0\}$, $D = \frac{1}{m}\operatorname{div}(h_i)$ and the G_i-semi-invariant function U, which vanishes at 0 and whose character has order n_U on G_i, for any prime divisor P over W_i with center 0,

$$\frac{a(P, W_i, \frac{t_i}{m}\operatorname{div}(h_i))}{|I_{G_i}(P)|} \geq \frac{a(P, W_i, \frac{1}{m}\operatorname{div}(h_i))}{|I_{G_i}(P)|} \geq \frac{\lambda}{n_U}.$$

Thus $\alpha_i \geq \lambda/n_U \geq \alpha_\infty$ for $i \gg 0$, contrary to $\alpha_i < \alpha_\infty$, which holds since $(\alpha_i)_i$ is strictly increasing. Thus Row 3b does not occur.

Proof of Theorem 12.3, continued. We continue with rows 3c–3h of Table 1. In rows 3g and 3h, where G_i permutes the three lines, we replace y, z, t by the linear forms in y, z, t defining the three lines, so that the cubic part of g_i is yzt, resp. $yzt + t^3$, and G_i acts on y, z, t by monomial substitutions. The linear change of coordinates keeps $f_i = x^2 + g_i(y, z, t)$ with $g_i \in (y, z, t)^3$ and keeps h_i of degree at most 1 in x. Let $G_i' \subset G_i$ be the subgroup of the elements inducing the trivial permutation of the three lines, so $[G_i : G_i'] \leq 3$ by **Step 3** and G_i' acts diagonally on x, y, z, t. In rows 3c–3f, $G_i' = G_i$, the circle $\mathbf{T}$, $w = (u, w_y, w_z, w_t)$, and r_i are as in **Steps 2** and **3**, and $R_i := G_i \cap \mathbf{T}$. In rows 3g and 3h we determine the circle of the G_i' below, and there $R_i := G_i' \cap \mathbf{T} = G_i \cap \mathbf{T}$. Since f_i is R_i-semi-invariant, the $\mathbf{T}$-weights of the monomials of f_i are congruent modulo r_i. Applied to x^2 and yzt for $i \gg 0$, the congruence gives

$$w_y + w_z + w_t = 2u. \tag{C.37}$$

Possibly replacing w with $-w$, we may assume that $u \geq 0$. The $\mathbf{T}$-weight of $\mathbf{x}^{\mathbf{k}}$ is $\langle w, \mathbf{k}\rangle$.

Row 3c: $u \geq 0$ and $w_y w_z w_t \neq 0$. By row 3a, w is not a vector with all coordinates positive. Since $u \geq 0$, (C.37) then forces both signs among w_y, w_z, w_t. Possibly renaming y, z, t and replacing w with $-w$ when $u = 0$, we may assume that w has one of the forms

$$\begin{aligned} &\mathrm{A}: \quad w = (u, -\overline{n}_1, -\overline{n}_2, \overline{n}_3), \qquad \overline{n}_3 = 2u + \overline{n}_1 + \overline{n}_2, \\ &\mathrm{B}: \quad w = (u, -\overline{n}_1, \overline{n}_2, \overline{n}_3), \qquad \overline{n}_2 + \overline{n}_3 = 2u + \overline{n}_1, \quad \overline{n}_2 \leq \overline{n}_3, \quad \text{and} \quad u > 0, \end{aligned}$$

with $\overline{n}_1, \overline{n}_2, \overline{n}_3 \in \mathbb{N}^+$. For $u = 0$, if w has form B, then $-w$ has form A after renaming. For a secant conic, let $t' \in \{y, z, t\}$ be the coordinate whose cube occurs in g_i. Then $3w_{t'} = 2u$ by [Liu26b, Lemma 5.3]. In form A we have $3\overline{n}_3 > 2u$ and $-3\overline{n}_1, -3\overline{n}_2 < 0 \leq 2u$, so a secant conic has form B. There $\overline{n}_3 > u$, since $2\overline{n}_3 \geq \overline{n}_2 + \overline{n}_3 = 2u + \overline{n}_1$, so $w_{t'} = 2u/3 = \overline{n}_2$, $t' = z$, and the cubic part of g_i is $yzt + z^3$.

We use the base vectors $v_0^{\pm}$ of Table 2, with the signs $\pm$ shown there. In any row of Table 2, $v_0^{\pm}$ has x-coordinate 1, the $v_0^{\pm}$-weights of x^2 and yzt are 2, and the coordinates of $v_0^{\pm}$-weight 0 have $\mathbf{T}$-weights of one sign: t for A^+ and B^+, y and z for A^-, y for B^-, and z for the last row, which is used in row 3f.

base	w	$v_0^{\pm}$	monomials of $v_0^{\pm}$-weight 2 besides x^2, yzt	substitution
A^+	A	$(1,1,1,0)$	none	none
A^-	A	$(1,0,0,2)$	$c'y^n t$, $(n-1)\overline{n}_1 = \overline{n}_2$, $n \geq 3$	$z \to z - c'y^{n-1}$
			$c'z^n t$, $(n-1)\overline{n}_2 = \overline{n}_1$, $n \geq 3$	$y \to y - c'z^{n-1}$
B^+	B	$(1,1,1,0)$	$c'y^2t^n$, $n\overline{n}_3 = 2u + 2\overline{n}_1$, $n \geq 2$	$z \to z - c'yt^{n-1}$
B^-	B	$(1,0,1,1)$	$c'y^n t^2$, $n\overline{n}_1 = 2\overline{n}_3 - 2u$, $n \geq 2$	$z \to z - c'y^{n-1}t$
H^-	$(0,1,-1,0)$	$(1,1,0,1)$	$c'y^2z^2$	$t \to t - c'yz$

TABLE 2. Base vectors and the monomials of $v_0^{\pm}$-weight 2 of the principal block of f_i other than x^2 and yzt, with $c' \in \mathbb{C}^*$, and the substitutions removing them. In each row at most one such monomial occurs.

We first verify the rows $\mathrm{A}^{\pm}$ and $\mathrm{B}^{\pm}$ of Table 2. Let $\mathbf{x}^{\mathbf{k}} \neq x^2$ be a monomial of the principal block f_i^0 of f_i (see **Step 2**) of $v_0^{\pm}$-weight at most 2. Then $k_x = 0$, $k_y + k_z + k_t \geq 3$, $\langle w, \mathbf{k}\rangle = 2u$, and a cubic monomial other than yzt (and z^3 for a secant conic) does not occur.

- A^+: $k_y + k_z \le 2$, so $\overline{n}_3 k_t = 2u + \overline{n}_1 k_y + \overline{n}_2 k_z < 2\overline{n}_3$, and $k_t \ge 1$ since $\mathbf{k} \ne 0$. Thus $k_t = 1$ and $\overline{n}_1 k_y + \overline{n}_2 k_z = \overline{n}_1 + \overline{n}_2$, so $(k_y, k_z) \in \{(1,1), (2,0), (0,2)\}$. The pairs $(2,0)$ and $(0,2)$ correspond to y^2t and z^2t, which do not occur. Hence $\mathbf{x}^{\mathbf{k}} = yzt$.
- A^-: $2k_t \le 2$, and $k_t = 0$ would imply $\langle w, \mathbf{k}\rangle \le 0$, with equality only for $\mathbf{k} = 0$. Thus $k_t = 1$ and $\overline{n}_1 k_y + \overline{n}_2 k_z = \overline{n}_1 + \overline{n}_2$, so $\mathbf{x}^{\mathbf{k}} \in \{yzt, y^n t, z^n t\}$ with $(n-1)\overline{n}_1 = \overline{n}_2$, resp. $(n-1)\overline{n}_2 = \overline{n}_1$, and $n \ge 3$ since y^2t and z^2t do not occur. Both $\overline{n}_2 \ge 2\overline{n}_1$ and $\overline{n}_1 \ge 2\overline{n}_2$ cannot hold.
- B^+: $k_y + k_z \le 2$. Since $\overline{n}_3 > u$ and $\overline{n}_2 + \overline{n}_3 > 2u$, the cases $(k_y, k_z) = (0,0), (0,1), (0,2)$ are impossible. If $(k_y, k_z) = (1,0)$, then $\overline{n}_3 k_t = \overline{n}_2 + \overline{n}_3$, so $k_t = 2$ and $\mathbf{x}^{\mathbf{k}} = yt^2$, which does not occur. If $(k_y, k_z) = (1,1)$, then $k_t = 1$. If $(k_y, k_z) = (2,0)$, then $\mathbf{x}^{\mathbf{k}} = y^2 t^n$ with $n\overline{n}_3 = 2u + 2\overline{n}_1$, and $n \ge 2$ since y^2t does not occur.
- B^-: $k_z + k_t \le 2$ and $k_z + k_t \ge 1$ since $u > 0$. If $(k_z, k_t) = (1,0)$ or $(0,1)$, then $\overline{n}_1 k_y = \overline{n}_2 - 2u < \overline{n}_1$, resp. $\overline{n}_1 k_y = \overline{n}_3 - 2u < \overline{n}_1$, so $k_y = 0$ and $\mathbf{x}^{\mathbf{k}}$ is linear. If $(k_z, k_t) = (1,1)$, then $k_y = 1$. If $(k_z, k_t) = (2,0)$, then $\overline{n}_1 k_y = 2\overline{n}_2 - 2u \le \overline{n}_1$, so $\mathbf{x}^{\mathbf{k}} \in \{z^2, yz^2\}$, which do not occur. If $(k_z, k_t) = (0,2)$, then $\mathbf{x}^{\mathbf{k}} = y^n t^2$ with $n\overline{n}_1 = 2\overline{n}_3 - 2u$, and $n \ge 2$ since t^2 and yt^2 do not occur.

The monomial z^3 of a secant conic has $v_0^{\pm}$-weight 3 for $\mathrm{B}^{\pm}$. Since f_i is G_i-semi-invariant with character $\chi_x^2 = \chi_y\chi_z\chi_t$, a monomial of Table 2 with non-zero coefficient c' forces the equality of characters under which the substitution is G_i-equivariant. For example, y^2t^n forces $\chi_z = \chi_y\chi_t^{n-1}$. The substituted polynomial has the $\mathbf{T}$-weight and the $v_0^{\pm}$-weight of the replaced coordinate, and degree at least 2. Hence the substitution is a G_i-equivariant automorphism of $\mathbb{A}^4$ inducing the identity on the tangent space at 0 and preserving $\mathbf{T}$-weights and $v_0^{\pm}$-weights. It cancels the monomial against yzt, keeps f_i of the form $x^2 + g_i(y,z,t)$ with cubic part yzt, resp. $yzt + z^3$, keeps h_i linear in x, and preserves the quotient values and the lc property of all pairs below. We treat the signs $+$ and $-$ separately. For the sign $\pm$, possibly applying the substitution for $v_0^{\pm}$, we may therefore assume that x^2 and yzt are the only monomials of f_i^0 of $v_0^{\pm}$-weight at most 2. The substitution preserves $\mathbf{T}$-weights, so it maps h_i^0 to the principal block of the new h_i and does not change κ.

Let $\sigma_0 > 0$ be a number such that $v_0^{\pm} \pm \sigma w \in \mathbb{R}^4_{>0}$ for any $\sigma \in (0, \sigma_0)$ and the bases $\mathrm{A}^{\pm}$, resp. $\mathrm{B}^{\pm}$, of Table 2 for w of form A, resp. B, with $2u\sigma_0 \le 1$ for the sign $+$. For positive integers N and L let

$$v := Nv_0^{\pm} \pm Lw, \qquad \sigma := L/N,$$

let $\gcd(v)$ be the greatest common divisor of the coordinates of v, and let $E_v := E_\omega$ for $\omega := v/\gcd(v)$, as in **Step 2**. For any $\mathbf{k}$,

$$\mathrm{wt}_v(\mathbf{x}^{\mathbf{k}}) = N\langle v_0^{\pm}, \mathbf{k}\rangle \pm L\langle w, \mathbf{k}\rangle.$$

Fix real numbers $0 < \sigma' < \sigma'' < \sigma_0$, and let $c_* > 0$ be the minimum of the coordinates of $v_0^{\pm} \pm \sigma w$ over $\sigma \in [\sigma', \sigma'']$. Assume that $\sigma \in [\sigma', \sigma'']$. The monomials of f_i^0 have v-weight $N\,\mathrm{wt}_{v_0^{\pm}}(\mathbf{x}^{\mathbf{k}}) \pm 2uL$, so the least one, $2N \pm 2uL \le 3N$, is attained exactly by x^2 and yzt. Any monomial of $f_i - f_i^0$ has degree at least $(r_i - 2u)/\max_j |w_j|$ by (C.33), hence v-weight at least $c_* N(r_i - 2u)/\max_j |w_j| > 3N$ for $i \gg 0$. Thus $\mathrm{in}_v f_i = x^2 + yzt$, which is irreducible, and [Liu26b, Lemma 4.12] implies

$$a(E_v, W_i, 0) = \frac{1}{\gcd(v)}\Big(\sum_j v_j - \mathrm{wt}_v(f_i)\Big) = \frac{N \pm uL}{\gcd(v)},$$

because the coordinates of $v_0^{\pm}$ and of w sum to 3 and $3u$.

Recall h_i^0 and κ from **Step 2**, and let $\delta_{\mathbf{T}} := \kappa/m - u$. Let $mJ_{\pm} := \mathrm{wt}_{v_0^{\pm}}(h_i^0)$. By **Step 2**, h_i^0 contains a monomial of degree $\mathrm{mult}_0 h_i \le \lfloor 3m/2 \rfloor$, and $v_0^{\pm} \in \{0,1,2\}^4$. Hence $J_{\pm} \in \frac{1}{m}\mathbb{Z} \cap [0,3]$, and passing to a subsequence we may assume that J_+ and J_- do not depend on i. The part of h_i^0 of least v-weight is the part of $v_0^{\pm}$-weight $mJ_{\pm}$, of v-weight $NmJ_{\pm} \pm L\kappa$. Since h_i is linear in x,

x^2+yzt does not divide the part of h_i^0 of $v_0^\pm$-weight $mJ_\pm$. Any monomial of $h_i-h_i^0$ has degree at least $(r_i-|\kappa|)/\max_j|w_j|$ by (C.33), hence larger v-weight for $i\gg 0$. By [Liu26b, Lemma 4.12], $\operatorname{mult}_{E_v} h_i = (NmJ_\pm \pm L\kappa)/\gcd(v)$. Therefore, for $\sigma\in[\sigma',\sigma'']$, $t\in[0,1]$ and $i\gg 0$,

$$a\Big(E_v, W_i, \frac{t}{m}\operatorname{div}(h_i)\Big) = \frac{N}{\gcd(v)}\big(1-\Upsilon_\pm(\sigma,t)\big), \qquad \Upsilon_\pm(\sigma,t) := tJ_\pm \pm \sigma\Big(\frac{t\kappa}{m}-u\Big). \tag{C.38}$$

Since $h_i(0)=0$, we have $\operatorname{mult}_{E_v} h_i>0$, that is, $J_\pm \pm \sigma\kappa/m>0$. Since $(W_i, \frac{1}{m}\operatorname{div}(h_i))$ is lc near 0, (C.38) with $t=1$ implies

$$\Upsilon_\pm(\sigma,1) = J_\pm \pm \sigma\delta_{\mathbf{T}} \le 1 \qquad \text{for any } \sigma\in(0,\sigma_0).$$

For $i\gg 0$ there exists an integer L_i coprime to r_i with $L_i/r_i\in[\sigma',\sigma'']$ [Liu26b, Lemma 4.9]. Let $v_i := r_i v_0^\pm \pm L_i w$. A prime number dividing $\gcd(v_i)$ and L_i divides $(v_i)_x \mp L_i u = r_i$, and a prime number dividing $\gcd(v_i)$ and r_i divides $L_i w_j$ for any j, hence divides L_i since w is primitive. Thus $\gcd(v_i)$ is coprime to L_i and to r_i. For a coordinate j of $v_0^\pm$-weight 0 we have $(v_i)_j = \pm L_i w_j$, so $\gcd(v_i)$ divides the non-zero integer w_j. Moreover $(v_i)_j/\gcd(v_i) \equiv \pm L_i \gcd(v_i)^{-1} w_j \pmod{r_i}$ for any j, where $\gcd(v_i)^{-1}$ is the inverse modulo r_i, with L_i and $\gcd(v_i)$ invertible modulo r_i, so any element of R_i is the value of the one-parameter subgroup of weight $v_i/\gcd(v_i)$ at a root of unity. That one-parameter subgroup acts trivially on E_{v_i}. Hence $R_i\subset I_{G_i}(E_{v_i})$ and r_i divides $|I_{G_i}(E_{v_i})|$. Thus $e_i := |I_{G_i}(E_{v_i})|/r_i$ is a positive integer with $e_i\le[G_i:R_i]$, and $[G_i:R_i]$ is bounded independently of i.

We show that $(J_\pm,\delta_{\mathbf{T}})\ne(1,0)$ for each sign $\pm$. Otherwise $\Upsilon_\pm(\sigma,1)=1$ for any σ, and (C.38) with $N=r_i$, $L=L_i$ implies

$$\alpha_i = q_i(t_i) \le \frac{a(E_{v_i}, W_i, \frac{t_i}{m}\operatorname{div}(h_i))}{|I_{G_i}(E_{v_i})|} = \frac{(1-t_i)\operatorname{mult}_{E_{v_i}} h_i}{m|I_{G_i}(E_{v_i})|} \le (1-t_i)\Big(1+\frac{\sigma''|\kappa|}{m}\Big), \tag{C.39}$$

which tends to 0, contrary to $\alpha_i\to\alpha_\infty>0$ in (C.30).

We now choose the sign, with each $J_\pm$ computed after the substitution for $v_0^\pm$. If $J_+>0$, we use $+$. If $J_+=0$, then h_i^0 has a monomial of v_0^+-weight 0, for $v_0^+=(1,1,1,0)$, that is, a power t^{n_t}, with $n_t\ge 1$ since $h_i(0)=0$. Then $\kappa = n_t\overline{n}_3>0$. The coordinates of v_0^--weight 0 have negative $\mathbf{T}$-weights, so no monomial of $\mathbf{T}$-weight κ has v_0^--weight 0, that is, $J_->0$, and we use $-$. Let $J:=J_\pm>0$ for the sign used. Letting $\sigma\to 0$ in the lc inequality above, $J\le 1$. If $J<1$, then $0<\Upsilon_\pm(\sigma,1)<1$ for any small $\sigma>0$. If $J=1$, then $\pm\delta_{\mathbf{T}}\le 0$ by the lc inequality, and $\pm\delta_{\mathbf{T}}\ne 0$ by (C.39), so again $0<\Upsilon_\pm(\sigma,1)<1$ for any small $\sigma>0$. Hence we may choose $\sigma'<\sigma''$ in $(0,\sigma_0)$ such that

$$0<\Upsilon_\pm(\sigma,1)<1 \qquad \text{for any } \sigma\in[\sigma',\sigma''].$$

There exists $\underline{t}<1$ such that the function $t\to\pm(\frac{t\kappa}{m}-u)$ has a constant sign, possibly zero, on $[\underline{t},1)$, and $\Upsilon_\pm>0$ on $[\sigma',\sigma'']\times[\underline{t},1]$: the first holds near 1 if $\delta_{\mathbf{T}}\ne 0$, the function being $\pm(t-1)\kappa/m$ if $\delta_{\mathbf{T}}=0$, and the second holds by continuity. Possibly truncating the sequence, we may assume that $t_i\in[\underline{t},1)$ for any i. Passing to a subsequence with r_i growing fast enough and using [Liu26b, Lemma 4.9], we let L_i be integers coprime to r_i such that $\sigma_i := L_i/r_i\in[\sigma',\sigma'']$ is increasing if $\pm(\frac{t\kappa}{m}-u)\ge 0$ on $[\underline{t},1)$ and decreasing otherwise. With $v_i := r_i v_0^\pm \pm L_i w$, $\gcd(v_i)$ divides a fixed non-zero integer and $e_i\le[G_i:R_i]$ is bounded, so passing to a subsequence we may assume that $\gcd(v_i)$ does not depend on i and that $e_i=e$ does not depend on i. Then

$$\Upsilon_\pm(\sigma_{i+1},t_{i+1})-\Upsilon_\pm(\sigma_i,t_i) = \pm\Big(\frac{t_{i+1}\kappa}{m}-u\Big)(\sigma_{i+1}-\sigma_i) + (t_{i+1}-t_i)\Big(J\pm\frac{\sigma_i\kappa}{m}\Big)\ge 0,$$

since the first term is non-negative by the choice of σ_i, and the second is non-negative because $t_{i+1}\ge t_i$ and $J\pm\sigma_i\kappa/m>0$. By (C.38) and $|I_{G_i}(E_{v_i})| = er_i$, the exceptional coefficient (C.34) of the return along E_{v_i} is

$$\gamma_i = 1-\frac{a(E_{v_i}, W_i, \frac{t_i}{m}\operatorname{div}(h_i))}{er_i} = 1-\frac{1-\Upsilon_\pm(\sigma_i,t_i)}{e\gcd(v_i)}. \tag{C.40}$$

Since $(W_i, \frac{t_i}{m}\operatorname{div}(h_i))$ is klt near 0, $\gamma_i < 1$. Since $\Upsilon_\pm(\sigma_i, t_i) > 0$, $\gamma_i > 1 - \frac{1}{e\gcd(v_i)} \geq 0$. Thus $(\gamma_i)_i$ is an increasing sequence in $(0,1)$.

It remains to see the centers of the return. Let $\omega_i := v_i/\gcd(v_i)$. For $k = 1, \dots, 4$, the surface $W_{i,k} \cap (\varsigma = 0)$ of **Step 2**, with $\omega = \omega_i$, is defined by $\operatorname{in}_{\omega_i}(f_i) = x^2 + yzt$ with $z_k = 1$ in the coordinates z'_j, that is,

$$1 + z'_2 z'_3 z'_4 = 0, \qquad z_1'^2 + z'_3 z'_4 = 0, \qquad z_1'^2 + z'_2 z'_4 = 0, \quad \text{and} \quad z_1'^2 + z'_2 z'_3 = 0.$$

The first is smooth. Each of the others is smooth outside the origin and of quadratic rank 3 at the origin. Since $W_{i,k} \cap (\varsigma = 0)$ is a hyperplane section of $W_{i,k}$, any closed point $o' \in W_{i,k}$ with $\varsigma(o') = 0$ is a smooth point or a point of quadratic rank at least 3 of $W_{i,k}$. Since moreover $\operatorname{in}_{\omega_i}(f_i) = x^2 + yzt$ is irreducible and does not divide $\operatorname{in}_{\omega_i}(h_i)$, row 3c does not occur by (R).

Rows 3d and 3e: $u > 0$ and $w_y w_z w_t = 0$. Let $u > 0$ and $i \gg 0$. In rows 3d and 3e we show that, for r_i large, $\operatorname{Sing} W_i$ contains a curve through 0, contrary to the isolatedness of the singularity of W_i at 0.

In row 3d exactly one of w_y, w_z, w_t is zero, say $w_y = 0$ after renaming. Then $w_z, w_t \neq 0$ and $w_z + w_t = 2u$. The monomials y^n, xy^n, zy^n, and ty^n have $\mathbf{T}$-weights 0, u, w_z, and w_t. None of them is congruent to $2u$ modulo r_i for $r_i > 2u + |w_z| + |w_t|$, because $w_z \equiv 2u$ if and only if $w_t \equiv 0$, and $w_t \equiv 2u$ if and only if $w_z \equiv 0$. By [Liu26b, Lemma 5.3], none of them occurs in f_i, so $f_i \in (x,z,t)^2$ and the y-axis $(x = z = t = 0)$ is contained in $\operatorname{Sing} W_i$.

In row 3e two of w_y, w_z, w_t are zero, so $w = (1,0,0,2)$ after renaming by (C.37), since w is primitive. A monomial $\mathbf{x}^\mathbf{k}$ of f_i has $\mathbf{T}$-weight $k_x + 2k_t \equiv 2 \pmod{r_i}$. For $r_i > 2$, the congruence excludes $k_x = k_t = 0$ and $(k_x, k_t) = (1,0)$. Hence

$$f_i = t\varphi_i(y,z) + F_i, \qquad F_i \in (x,t)^2,$$

for a polynomial φ_i. On the plane $\Pi := (x = t = 0)$ the functions f_i, $\partial f_i/\partial x$, $\partial f_i/\partial y$, and $\partial f_i/\partial z$ vanish, and $\partial f_i/\partial t|_\Pi = \varphi_i$. Since x^2 is the only monomial of f_i of degree at most 2, $\varphi_i \in (y,z)^2$. Thus $\Pi \cap (\varphi_i = 0)$ is a curve through 0 or the plane Π, and it is contained in $\operatorname{Sing} W_i$.

Row 3f: $w = (0,1,-1,0)$. We use the last row H^- of Table 2, so that

$$v = N(1,1,0,1) - Lw = (N, N-L, L, N),$$

positive for $\sigma = L/N \in (0,1)$. A monomial $\mathbf{x}^\mathbf{k}$ of f_i^0 has $k_y = k_z$ and v_0^--weight $k_x + k_y + k_t$. The monomials of v_0^--weight at most 2 are therefore among x^2, xt, t^2, yz, xyz, yzt, and y^2z^2. Since the quadratic part of f_i is x^2 and the cubic part of g_i is yzt or $yzt + t^3$, only x^2, yzt, and $c'y^2z^2$ can occur. The monomial t^3 has v_0^--weight 3. The monomial $c'y^2z^2$ forces $\chi_t = \chi_y\chi_z$, and the G_i-equivariant substitution $t \to t - c'yz$ removes it as in row 3c. With $v_0^- = (1,1,0,1)$, the sign $-$, $u = 0$, and $\sigma_0 = 1$, the argument of row 3c from the choice of σ_0 to (C.39) proves (C.38), the lc inequality, and (C.39). Here $\delta_\mathbf{T} = \kappa/m$.

Let $\kappa^+ := \max\{\kappa, 0\}$ and $\kappa^- := \max\{-\kappa, 0\}$. Since h_i^0 is a sum of monomials of $\mathbf{T}$-weight κ, we have $h_i^0 = y^{\kappa^+} z^{\kappa^-} P_i(x, t, yz)$ for a polynomial P_i in three variables. Let n_P be the least degree of a monomial of P_i. The coordinates x, t and the product yz have v_0^--weight 1 and z has v_0^--weight 0, so $mJ_- = \kappa^+ + n_P$ and

$$\Upsilon_-(\sigma, t) = tJ_- - \frac{\sigma t \kappa}{m}.$$

The lc inequality of row 3c implies $J_- \leq 1$. If $J_- = 1$, then $\kappa \geq 0$, and $\kappa \neq 0$ by (C.39). Thus $0 < \Upsilon_-(\sigma, 1) < 1$ for small $\sigma > 0$. If $0 < J_- < 1$, then $0 < \Upsilon_-(\sigma, 1) < 1$ for small $\sigma > 0$ as well. If $J_- = 0$, then $\kappa^+ = n_P = 0$, so $h_i^0 = z^{\kappa^-} P_i$ with $P_i(0) \neq 0$. Since $h_i(0) = 0$, we get $\kappa^- \geq 1$, $\kappa < 0$, and $0 < \Upsilon_-(\sigma, 1) = -\sigma\kappa/m < 1$ for small $\sigma > 0$. Moreover $\gcd(v_i) = \gcd(r_i, r_i - L_i, L_i) = 1$ for L_i coprime to r_i. The choice of σ_i and the formula (C.40) of row 3c hold with $\gcd(v_i) = 1$, so $(\gamma_i)_i$ is increasing in $(0,1)$. The lowest v_i-weight part of f_i is $x^2 + yzt$, so the surfaces $W_{i,k} \cap (\varsigma = 0)$ are the four surfaces of row 3c, and the conclusion of row 3c excludes row 3f.

Row 3g: a permutation of order 3, or the exchange for a secant conic. Let $g \in G_i$ and $g' \in G_i'$. Since G_i is abelian, $gg'g^{-1} = g'$, so the characters of g' on two coordinates permuted by g are equal. If the permutation has order 3, then $\chi_y = \chi_z = \chi_t$ on G_i', and $\chi_x^2 = \chi_y\chi_z\chi_t$. Thus

$$G_i' \subset \mathcal{S} := \{(\lambda_x, \lambda, \lambda, \lambda) \mid \lambda_x^2 = \lambda^3\}.$$

For a secant conic with y and z exchanged, $\chi_y = \chi_z$ on G_i', $\chi_x^2 = \chi_y\chi_z\chi_t = \chi_t^3$, and thus

$$G_i' \subset \mathcal{S} := \{(\lambda_x, \lambda_y, \lambda_y, \lambda_t) \mid \lambda_y^2 = \lambda_t^2,\ \lambda_x^2 = \lambda_t^3\}.$$

For either permutation, $\mathcal{S}$ is a closed subgroup of $\mathbb{T}^4$ with identity component $\mathbf{T}_{(3,2,2,2)}$. In particular, G_i' acts by scalars on the span of the permuted coordinates, so G_i' is diagonal in the G_i-eigenparameters of the polynomial model as well as in x, y, z, t. Since $|G_i'| \geq |G_i|/3 \to +\infty$, by [Liu26b, Lemma 4.6], after passing to a subsequence the G_i' have a Hausdorff limit $H' \subset \mathcal{S}$ with $\dim H' \geq 1$, so $H'^0 = \mathbf{T}_{(3,2,2,2)}$ and the Lie algebra of H'^0 contains the positive vector $(3, 2, 2, 2)$. Since $(W_i, \frac{t_i}{m}\operatorname{div}(h_i))$ is klt near 0, [Liu26b, Lemma 4.7] implies

$$\operatorname{mld}\Big(\Big(W_i \ni 0, \frac{t_i}{m}\operatorname{div}(h_i)\Big)/G_i'\Big) \to 0.$$

For any prime divisor F over W_i, $I_{G_i'}(F) \subset I_{G_i}(F)$, so $\alpha_i = q_i(t_i)$ is at most the left-hand side, contrary to $\alpha_i \to \alpha_\infty > 0$. Hence row 3g does not occur.

Row 3h: a transposition for a triangle. Renaming y, z, t, we may assume that G_i exchanges y and z up to scalars. Then $[G_i : G_i'] = 2$, $\chi_y = \chi_z$ on G_i', and

$$G_i' \subset \mathcal{S}' := \{(\lambda_x, \lambda_y, \lambda_y, \lambda_t) \mid \lambda_x^2 = \lambda_y^2\lambda_t\},$$

a 2-dimensional torus containing $\mathbf{T}_{(3,2,2,2)}$. As in row 3g, G_i' is diagonal in the G_i-eigenparameters of the polynomial model. By [Liu26b, Lemma 4.6], after passing to a subsequence the G_i' have a Hausdorff limit $H' \subset \mathcal{S}'$ with $\dim H' \geq 1$. If $\dim H' = 2$, then $H'^0 = \mathcal{S}'$ contains $\mathbf{T}_{(3,2,2,2)}$, and the argument of row 3g leads to a contradiction. Hence $H'^0 = \mathbf{T}_{(w)}$ is a circle with $w_y = w_z$, $r_i = |G_i' \cap \mathbf{T}| \to +\infty$, and $2u = 2w_y + w_t$ by (C.37). We go through the signs of w_y and w_t.

- If $w_y = 0$, then $w = (1, 0, 0, 2)$, excluded as in row 3e.
- If $w_t = 0$, then $w = (1, 1, 1, 0)$, excluded as in row 3d: the monomials t^n, xt^n, yt^n, and zt^n have $\mathbf{T}$-weights 0, 1, 1, and 1, so $f_i \in (x, y, z)^2$ for $r_i > 2$, and the t-axis is contained in $\operatorname{Sing} W_i$.
- If $w_y, w_t > 0$, then w has all coordinates positive, excluded as in row 3g.
- If $w_y < 0$, then $w_t = 2u - 2w_y > 0$ and $w = (u, -\overline{n}_1, -\overline{n}_1, 2u + 2\overline{n}_1)$ with $\overline{n}_1 \in \mathbb{N}^+$. We call the vectors $w = (u, -\overline{n}_1, -\overline{n}_1, 2u + 2\overline{n}_1)$ the *first family*.
- If $w_y > 0 > w_t$ and $u = 0$, then $-w = (0, -w_y, -w_y, 2w_y)$ is in the first family.
- If $w_y > 0 > w_t$ and $u > 0$, then, renaming the coordinates so that the negative weight is on y and G_i exchanges z and t, $w = (u, -\overline{n}_1, \overline{n}_2, \overline{n}_2)$ with $\overline{n}_1 = 2(\overline{n}_2 - u) > 0$. We call the vectors $w = (u, -\overline{n}_1, \overline{n}_2, \overline{n}_2)$ the *second family*.

In each family G_i preserves the $\mathbf{T}$-weights and the $v_0^{\pm}$-weights below, because the exchanged coordinates have equal $\mathbf{T}$-weights and equal $v_0^{\pm}$-weights. Hence G_i acts on the weighted blow-ups E_v below, and G_i preserves the principal block h_i^0 of h_i, which is unique and is computed with G_i' [Liu26b, Definition-Lemma 11.5(1)].

The first family is the form A of row 3c with $\overline{n}_1 = \overline{n}_2$, and the bases $\mathrm{A}^{\pm}$ are symmetric in y and z. The monomials of Table 2 for A^- require $\overline{n}_2 \geq 2\overline{n}_1$ or $\overline{n}_1 \geq 2\overline{n}_2$, so no substitution is used. With G_i' in place of G_i for the blocks and with G_i acting on E_{v_i}, the argument of row 3c shows that $(\gamma_i)_i$ is an increasing sequence in $(0, 1)$. Here (R) is applied in the G_i-semi-invariant coordinates of the polynomial model (x, $y \pm cz$, t for the first family and x, y, $z \pm ct$ for the second, $c \in \mathbb{C}^*$), in which G_i is diagonal: the exchanged coordinates have equal v_i-weights, so E_{v_i} and γ_i do not change, and the surfaces $W_{i,k} \cap (\varsigma = 0)$ are the slices of $(x^2 + yzt = 0)$ on which one of x, $y \pm cz$, t, resp. x, y, $z \pm ct$, is 1, which are smooth or have only points of quadratic rank

3, for instance $x^2 + t(1 - y_-^2)/(4c) = 0$ for $y_+ = 1$, where $y_\pm := y \pm cz$. Hence the first family does not occur.

For the second family we use only the base B^-, $v_0^- = (1, 0, 1, 1)$, which is symmetric in z and t:

$$v = Nv_0^- - Lw = (N - uL, \overline{n}_1 L, N - \overline{n}_2 L, N - \overline{n}_2 L), \qquad 0 < \sigma = L/N < 1/\overline{n}_2.$$

Here $\overline{n}_2 = \overline{n}_3$, so $2\overline{n}_2 = 2u + \overline{n}_1$, and a monomial $y^n t^2$ of Table 2 would have $n\overline{n}_1 = 2\overline{n}_2 - 2u = \overline{n}_1$, that is, $n = 1 < 2$. Hence no such monomial occurs. Hence x^2 and yzt are the only monomials of f_i^0 of v_0^--weight at most 2 without substitution. With the sign $-$ and $\sigma_0 = 1/\overline{n}_2$, the argument of row 3c from the choice of σ_0 to (C.39) proves (C.38), the lc inequality, and (C.39). If $J_- > 0$, the choice of the interval $[\sigma', \sigma'']$ is the one of row 3c. If $J_- = 0$, then h_i^0 has a monomial of v_0^--weight 0, that is, a power y^{n_y}, with $n_y \geq 1$ since $h_i(0) = 0$. Then $\kappa = -\overline{n}_1 n_y$, $\delta_{\mathbf{T}} = -\overline{n}_1 n_y/m - u < 0$, and

$$0 < \Upsilon_-(\sigma, 1) = \sigma\Big(u + \frac{\overline{n}_1 n_y}{m}\Big) < 1 \qquad \text{for } 0 < \sigma < \min\Big\{\frac{1}{\overline{n}_2}, \frac{1}{u + \overline{n}_1 n_y/m}\Big\}.$$

For $J_- > 0$ and for $J_- = 0$, the choice of σ_i, the bound $\gcd(v_i) \mid \overline{n}_1$, and the formula (C.40) of row 3c hold, so $(\gamma_i)_i$ is increasing in $(0, 1)$. The lowest v_i-weight part of f_i is $x^2 + yzt$, and with (R) applied in the coordinates x, y, $z \pm ct$ as for the first family, the conclusion of row 3c excludes the second family, and hence row 3h.

Proof of Theorem 12.3, continued (rows 4–6). Row 4: a polynomial model. The residual cubic of $(W_i \ni o_i)$ consists of three concurrent lines. By [Liu26b, Lemma E.15(1)(2)] for $(W_i \ni o_i)$, G_i, and the single mark 1 with coefficient 1 in [Liu26b, Set-up 11.1] (allowed by [Liu26b, Remark 11.6]), possibly passing to a subsequence, there exist a faithful diagonal action of G_i on $\mathbb{A}^4_{x,y,z,t}$, a G_i-semi-invariant polynomial f_i', and a G_i-equivariant isomorphism

$$\Phi_i : \widehat{\mathcal{O}}_{W_i, o_i} \to \widehat{\mathcal{O}}_{W_i', 0}, \qquad W_i' := (f_i' = 0) \subset \mathbb{A}^4,$$

such that $(W_i' \ni 0)$ is an isolated Gorenstein terminal germ, G_i acts faithfully on W_i', and either f_i' is of the form (I) for any i or f_i' is of the form (II) for any i:

$$\text{(I)} \qquad f_i' = x^2 + y^2 z + z^3 + a_{i,1}(t)z^2 + a_{i,2}(t)z + a_{i,3}(t)y + a_{i,4}(t),$$

$$\text{(II)} \qquad f_i' = x^2 + y^3 + z^3 + a_{i,1}(t)yz + a_{i,2}(t)y + a_{i,3}(t)z + a_{i,4}(t).$$

Here the $a_{i,j}$ are polynomials with

$$\operatorname{mult}_0 a_{i,1} \geq 2, \qquad \operatorname{mult}_0 a_{i,2} \geq 3, \qquad \operatorname{mult}_0 a_{i,3} \geq 3, \quad \text{and} \quad \operatorname{mult}_0 a_{i,4} \geq 4.$$

We carry the mark to W_i'. Let $x_1, \dots, x_4$ be G_i-eigenparameters at $o_i = 0$ of the ambient space $\mathbb{A}^4 \supset W_i = (f_i = 0)$ of **Step 1**(a), ordered so that x_k has the character of the k-th coordinate of $\mathbb{A}^4_{x,y,z,t}$. Such an order exists because Φ_i induces a G_i-equivariant isomorphism between the Zariski cotangent spaces of $(W_i \ni o_i)$ and $(W_i' \ni 0)$, both of dimension 4. We identify $\mathbb{C}[[x, y, z, t]]$ with $\mathbb{C}[[x_1, \dots, x_4]]$ through $(x, y, z, t) = (x_1, \dots, x_4)$, and let ψ_i be the G_i-equivariant automorphism of $\mathbb{C}[[x_1, \dots, x_4]]$ sending x_k to a lift of $\Phi_i(x_k|_{W_i})$ with the character of x_k, so that ψ_i induces Φ_i. Let $\widehat{f_i}$ and $\widehat{h_i}$ be the expansions of the equation f_i of W_i and of h_i. Since ψ_i induces Φ_i, the series $\psi_i(\widehat{f_i})$ and f_i' generate one ideal, so $\psi_i(\widehat{f_i}) = v_i^{-1} f_i'$ for a G_i-semi-invariant unit v_i, and v_i is G_i-invariant as $v_i(0) \neq 0$. By Lemma 4.6 with $n = 4$, the label t_i, $\mathcal{P} = \emptyset$, $\psi = \psi_i$, $u = 1$, $v = v_i$, and a sufficiently large integer $L > \deg f_i'$ (so that $f' = f_i'$ there), there exists a G_i-semi-invariant polynomial $h_i' \equiv \psi_i(\widehat{h_i}) \pmod{(x, y, z, t)^L}$ such that $(W_i', \frac{1}{m}\operatorname{div}(h_i'))$ is lc near 0 and

$$\operatorname{mld}\Big(\Big(W_i' \ni 0, \frac{t_i}{m}\operatorname{div}(h_i')\Big)\Big/G_i\Big) = \alpha_i.$$

Since W_i' is terminal, $(W_i', \frac{1}{m}\operatorname{div}(h_i'))$ is lc near 0, and $t_i < 1$, the pair $(W_i', \frac{t_i}{m}\operatorname{div}(h_i'))$ is klt near 0. In the rest of row 4 we replace $(W_i \ni o_i)$, f_i, and h_i with $(W_i' \ni 0)$, f_i', and h_i'.

The circle in rows 4–6. In rows 5 and 6, by **Step 1**(a) and [Liu26b, Proposition 3.17(1)(2)], we use coordinates x, y, z, t of $\mathbb{A}^4$ in which G_i acts diagonally, $o_i = 0$, and the equation f_i of W_i has the form of the row in Table 1:

$$f_i = x^2 + g_i \text{ (row 5)}, \qquad f_i = x^2 + y^2 z + y t^2 + \phi_i \text{ (row 6)},$$

where $g_i \in (y, z, t)^3 \subset \mathbb{C}[y, z, t]$ has cubic part $y^2 z + t^3$ and $\phi_i \in (z, t)^4 \subset \mathbb{C}[z, t]$. In each of rows 4–6, f_i contains the three monomials of

$$x^2 + y^2 z + z^3 \text{ or } x^2 + y^3 + z^3 \text{ (row 4)}, \qquad x^2 + y^2 z + t^3 \text{ (row 5)}, \quad \text{and} \quad x^2 + y^2 z + y t^2 \text{ (row 6)},$$

on which G_i acts by one character. The identity component of the group of diagonal matrices acting on the three monomials of the row by one character is a 2-dimensional torus $\mathbf{S}$ whose primitive circles are the $\mathbf{T}_{(w)}$ with

$$w = (3c, 2c, 2c, v), \qquad w = (3v, c_1, 6v - 2c_1, 2v), \quad \text{and} \quad w = (2c_2 + c_3, 2c_2, 2c_3, c_2 + c_3),$$

respectively, for coprime integers c, v, resp. v, c_1, resp. c_2, c_3. The torus $\mathbf{S}$ contains the strictly positive vector $(3, 2, 2, 1)$, resp. $(3, 1, 4, 2)$, resp. $(3, 2, 2, 2)$. Since each G_i is contained in the closed group of diagonal matrices acting on the three monomials by one character, whose identity component is $\mathbf{S}$, the limit H of **Step 2** satisfies $H^0 \subset \mathbf{S}$. Since $\mathbf{S}$ contains a strictly positive vector, (C.31) gives $H^0 \neq \mathbf{S}$, and $\dim H \geq 1$ by **Step 2**, so $\dim H = 1$. Thus $H^0 = \mathbf{T} = \mathbf{T}_{(w)}$ with w, r_i, and e_0 as in **Step 2**, $\mathbf{T}$ is a primitive circle in $\mathbf{S}$, and $w \notin (\mathbb{N}^+)^4 \cup (-\mathbb{N}^+)^4$ by (C.31). Let $R_i := G_i \cap \mathbf{T}$, so $r_i = |R_i|$ and $[G_i : R_i] = e_0$ for $i \gg 0$. Since $R_i \subset G_i$ and the G_i-semi-invariant polynomial f_i contains x^2,

$$\mathrm{wt}_w(M) \in \mathrm{wt}_w(x^2) + r_i \mathbb{Z} \quad \text{for any monomial } M \text{ of } f_i. \tag{C.41}$$

Row 4: the circles. Here $w = (3c, 2c, 2c, v)$ with $\gcd(c, v) = 1$, and $w \notin (\mathbb{N}^+)^4 \cup (-\mathbb{N}^+)^4$ means $cv \leq 0$. Possibly replacing w with $-w$, we may assume that one of the following holds.

(1) $w = (3, 2, 2, 0)$. Let $\mathcal{M} := \{z^2, z, y, 1\}$ in (I) and $\mathcal{M} := \{yz, y, z, 1\}$ in (II). Then $\mathrm{wt}_w(M) \in \{0, 2, 4\}$ for any $M \in \mathcal{M}$ and $\mathrm{wt}_w(t) = 0$, so any monomial of $a_{i,j}(t)M$ has w-weight in $\{0, 2, 4\}$, while $\mathrm{wt}_w(x^2) = 6$. For $r_i > 6$, $\{0, 2, 4\} \cap (6 + r_i \mathbb{Z}) = \emptyset$, so (C.41) implies $a_{i,j} = 0$ for any j. Thus $f_i = x^2 + y^2 z + z^3$ or $f_i = x^2 + y^3 + z^3$, and W_i is singular along the t-axis, contrary to the isolatedness of $(W_i \ni 0)$.

(2) $w = (0, 0, 0, 1)$. Then $R_i = \mu_{r_i}(0, 0, 0, 1)$. Since $(W_i \ni 0)$ is a singular isolated Gorenstein terminal germ, it is an isolated singular cDV germ. We apply the last sentence of [Liu26b, Theorem 8.5] to W_i with (t, x, y, z) in place of (x, u_1, u_2, u_3), to G_i, e_0, the DCC set $\{\frac{t_i}{m}\}_i$ and the boundaries $\frac{t_i}{m}\,\mathrm{div}(h_i)$: the numbers

$$\alpha_i = \mathrm{mld}\Big(\Big(W_i \ni 0, \frac{t_i}{m}\,\mathrm{div}(h_i)\Big)\Big/G_i\Big)$$

do not form a strictly increasing sequence with $r_i \to +\infty$, a contradiction.

(3) $w = (-3c, -2c, -2c, v)$ for coprime $c, v \in \mathbb{N}^+$. We use kind (1) of [Liu26b, Theorem 10.4] for the type D_4 with the weights $(3, 2, 2)$ on x, y, z, the integers c, v, e_0, and the DCC set $\{\frac{t_i}{m}\}_i$. In (I) we take the polynomial $x^2 + y^2 z + z^3$ and the basis $z^2, z, y, 1$ of $\mathbb{C}[y, z]/(yz, y^2 + 3z^2)$, and in (II) the polynomial $x^2 + y^3 + z^3$ and the basis $yz, y, z, 1$ of $\mathbb{C}[y, z]/(y^2, z^2)$. Then $\mathbf{T}$ is the circle $\mathbf{T}_{(-3c,-2c,-2c,v)}$ of [Liu26b, Theorem 10.4], $a_{i,j} \in t\mathbb{C}[t]$, $(W_i \ni 0)$ is isolated Gorenstein terminal, G_i is diagonal and faithful on W_i and acts on any monomial of f_i by the character of f_i, $[G_i : R_i] \leq e_0$, and $(W_i, \frac{t_i}{m}\,\mathrm{div}(h_i))$ is klt near 0. Since $0 < \alpha_1 < \alpha_2 < \cdots$ and $r_i \to +\infty$, this contradicts [Liu26b, Theorem 10.4].

Row 5: the cuspidal cubic. Here $w = (3v, c_1, 6v - 2c_1, 2v)$ with $\gcd(v, c_1) = 1$, and $\mathrm{wt}_w(x^2) = 6v$. Possibly replacing w with $-w$, we may assume that $v \geq 0$, and $c_1 = 1$ if $v = 0$. Since $w \notin (\mathbb{N}^+)^4$, we have $c_1 \leq 0$ or $c_1 \geq 3v$, and one of the following holds.

(1) $w \in \{(0, 1, -2, 0), (3, 4, -2, 2)\}$. The circles $\mathbf{T}_{(0,1,-2,0)}$ and $\mathbf{T}_{(3,4,-2,2)}$ are treated in rows 7a–7h of Table 1, with the first face I with $(n_1, n_2) = (0, 3)$.

(2) $c_1 = 3v$. Then $(v, c_1) = (1, 3)$ and $w = (3, 3, 0, 2)$. The monomials z^n, xz^n, yz^n, and tz^n have w-weights $0, 3, 3, 2$, and $\{0, 2, 3\} \cap (6 + r_i\mathbb{Z}) = \emptyset$ for $r_i > 6$. By (C.41), $f_i \in (x, y, t)^2$, so W_i is singular along the z-axis, contrary to the isolatedness of $(W_i \ni 0)$.
(3) $v > 0$, $(v, c_1) \neq (1, 4)$, and either $c_1 \leq 0$ or $c_1 > 3v$. We use Lemma 3.11. Case (3) includes the circle $\mathbf{T}_{(3,0,6,2)}$ of row 5′ of Table 1.

In case (3), fix s_0, y_0, I given by Lemma 3.11(1), a compact interval $I' \subset I$ and a point σ_* in the interior of I'. For integers L_i with $\gcd(L_i, r_i) = 1$ and $\sigma_i := L_i/r_i \in I'$, to be fixed after (C.44), let

$$S_i := r_i s_0 + vL_i, \qquad Y_i := r_i y_0 + c_1 L_i, \qquad \omega_i := (3S_i, Y_i, 6S_i - 2Y_i, 2S_i), \quad \text{and} \quad d_i := 3S_i - Y_i.$$

Since h_i has x-degree at most 1 by **Step 1**(b) and the least degree of a monomial of h_i of w-weight κ is bounded by (C.29), Lemma 3.11(3) implies the following for $i \gg 0$: $\mathrm{in}_{\omega_i} f_i = x^2 + y^2 z + t^3$, the strict transform of W_i in the ω_i-weighted blow-up of $\mathbb{A}^4$ intersects the exceptional divisor in one prime divisor E_i, and $\mathrm{mult}_{E_i} h_i = r_i j_i + L_i \kappa$ for bounded integers j_i. Possibly passing to a subsequence, $j_i = j$ for any i. By Lemma 3.11(3),

$$a(E_i, W_i, 0) = 5S_i - Y_i = r_i\delta_0 + L_i\delta_1, \qquad (\delta_0, \delta_1) := (5s_0 - y_0, 5v - c_1). \tag{C.42}$$

Row 6: a conic and a tangent line. Here $w = (2c_2 + c_3, 2c_2, 2c_3, c_2 + c_3)$ with $\gcd(c_2, c_3) = 1$, and $\mathrm{wt}_w(x^2) = 4c_2 + 2c_3$. Since $w \notin (\mathbb{N}^+)^4 \cup (-\mathbb{N}^+)^4$, we have $c_2 c_3 \leq 0$. Possibly replacing w with $-w$, we may assume that one of the following holds.

(1) $(c_2, c_3) = (1, 0)$ and $w = (2, 2, 0, 1)$. The monomials z^n, xz^n, yz^n, and tz^n have w-weights $0, 2, 2, 1$, and $\{0, 1, 2\} \cap (4 + r_i\mathbb{Z}) = \emptyset$ for $r_i > 4$. By (C.41), $f_i \in (x, y, t)^2$, so W_i is singular along the z-axis, contrary to the isolatedness of $(W_i \ni 0)$.
(2) $(c_2, c_3) \in \{(0, 1), (1, -1)\}$, that is, $w \in \{(1, 0, 2, 1), (1, 2, -2, 0)\}$. The circles $\mathbf{T}_{(1,0,2,1)}$ and $\mathbf{T}_{(1,2,-2,0)}$ are treated in rows 7a–7h of Table 1, with the first face II with $n_3 = 2$.
(3) $c_2 > 0 > c_3$ and $(c_2, c_3) \neq (1, -1)$. Let $k := -c_3$. Then c_2, k are coprime positive integers with $(c_2, k) \neq (1, 1)$, and we use Lemma 3.12.

In case (3), fix u_0, v_0, I given by Lemma 3.12(1), a compact interval $I' \subset I$ and a point σ_* in the interior of I'. For integers L_i with $\gcd(L_i, r_i) = 1$ and $\sigma_i := L_i/r_i \in I'$, to be fixed after (C.44), let

$$N_{1,i} := r_i u_0 + c_2 L_i, \qquad N_{2,i} := r_i v_0 - kL_i, \quad \text{and} \quad \omega_i := (2N_{1,i} + N_{2,i}, 2N_{1,i}, 2N_{2,i}, N_{1,i} + N_{2,i}).$$

As in row 5, Lemma 3.12(3) implies the following for $i \gg 0$: the lowest ω_i-weighted part of f_i is $x^2 + y^2 z + yt^2$, the strict transform of W_i in the ω_i-weighted blow-up of $\mathbb{A}^4$ intersects the exceptional divisor in one prime divisor E_i, $\mathrm{mult}_{E_i} h_i = r_i j + L_i \kappa$ for an integer j independent of i after passing to a subsequence, and

$$a(E_i, W_i, 0) = N_{1,i} + 2N_{2,i} = r_i\delta_0 + L_i\delta_1, \qquad (\delta_0, \delta_1) := (u_0 + 2v_0, c_2 - 2k). \tag{C.43}$$

Rows 5 and 6: the coefficients. In case (3) of rows 5 and 6, (C.34) with R_i in the role of R_ω (so with r_i in place of r_ω), (C.42), and (C.43) give

$$\gamma_i := 1 - \frac{a(E_i, W_i, 0) - \frac{t_i}{m}\mathrm{mult}_{E_i} h_i}{r_i} = 1 - \delta_0 + \frac{t_i j}{m} + \Big(\frac{t_i\kappa}{m} - \delta_1\Big)\sigma_i. \tag{C.44}$$

Since $a(E_i, W_i, 0) \leq r_i$ by Lemma 3.11(2), resp. Lemma 3.12(2), and $\mathrm{mult}_{E_i} h_i \geq 0$, we have $\gamma_i \geq 0$. Since $(W_i, \frac{t_i}{m}\mathrm{div}(h_i))$ is klt near 0, we have $\gamma_i < 1$. As t_i is increasing, the sign of $\frac{t_i\kappa}{m} - \delta_1$ is constant for $i \gg 0$. Since $r_i \to +\infty$, [Liu26b, Lemma 4.9] implies that, possibly passing to a subsequence, we may choose the L_i such that $\sigma_i \in I'$ for any i, the sequence $(\sigma_i)_i$ is increasing if $\frac{t_i\kappa}{m} - \delta_1 \geq 0$ for $i \gg 0$, and $(\sigma_i)_i$ is decreasing if $\frac{t_i\kappa}{m} - \delta_1 < 0$ for $i \gg 0$. We take σ_{i+1} between σ_i and σ_*. By (C.44),

$$\gamma_{i+1} - \gamma_i = \frac{(t_{i+1} - t_i)(j + \kappa\sigma_{i+1})}{m} + \Big(\frac{t_i\kappa}{m} - \delta_1\Big)(\sigma_{i+1} - \sigma_i) \geq 0,$$

since $j + \kappa\sigma_{i+1} = \mathrm{mult}_{E_{i+1}} h_{i+1}/r_{i+1} \geq 0$. Thus

$$\Gamma' := \left\{\frac{t_i}{m}\right\}_i \cup \{\gamma_i\}_i \subset [0,1]$$

is a DCC set.

Rows 5 and 6: the returns. Since $\omega_i \equiv L_i w \pmod{r_i}$ and $\gcd(L_i, r_i) = 1$, the group R_i is the subgroup of order r_i of $\mathbf{T}_{(\omega_i)}$, and by Lemma 3.9 for G_i, w, and L_i, on each chart of the ω_i-weighted blow-up of $\mathbb{A}^4$ the group R_i acts through $\mu_{r_i}^{\mathrm{rad}}$, only on the chart parameter, and the quotient by G_i is the quotient chart.

Let F_i be a computing lift of α_i. We use the notation introduced after (C.34), with $\omega = \omega_i$ and with R_i in the role of R_ω. Choose a chart of the ω_i-weighted blow-up, with hypersurface $\widehat{W}_i \subset \mathbb{A}^4_{\mathrm{rad}}$ the quotient by $\mu_{r_i}^{\mathrm{rad}}$ of the strict transform of W_i in the cover of the chart, and a point $p_i \in \widehat{W}_i$ over the center of F_i on the quotient by R_i of the strict transform of W_i. Thus $\widehat{W}_i$ is the hypersurface of the quotient chart at p_i. Let ς be the r_i-th power of the chart parameter, $\widetilde{h}_i$ the strict transform of h_i, and

$$\widehat{D}_i := \gamma_i \,\mathrm{div}(\varsigma) + \frac{t_i}{m}\,\mathrm{div}(\widetilde{h}_i),$$

the corresponding boundary on $\widehat{W}_i$. Let $\widehat{G}_i$ be the stabilizer of p_i in the group of the quotient chart at p_i. If p_i is a closed point, then $(\widehat{W}_i, \widehat{D}_i)$ is klt near p_i and $\alpha_i = \mathrm{mld}((\widehat{W}_i \ni p_i, \widehat{D}_i)/\widehat{G}_i)$, since the quotient of $(\widehat{W}_i, \widehat{D}_i)$ by the group of the quotient chart at p_i is an open subset of the crepant pullback (Q_i', Δ_i') of the quotient pair (Q_i, Δ_i) in **Step 2**, which is klt near $\overline{o}_i$, and the induced morphism from $\widehat{W}_i/\widehat{G}_i$ to Q_i' is étale at the image of p_i. The labels of $\widehat{D}_i$ belong to Γ'. Possibly passing to a subsequence, one of the following holds for any i.

(a) p_i maps to the generic point of the image of E_i or of a curve. In the first case, the central surface $(\varsigma = 0)$ has coefficient γ_i in $\widehat{D}_i$, so $\alpha_i = (1-\gamma_i)/e_i$ for the order $e_i \leq e_0$ of the inertia of the image of E_i, which we may take independent of i after passing to a subsequence. In the second case, by Lemma 3.5(0)(2)(3) at the generic point of the curve (with the trivial group in place of H), α_i is the minimal log discrepancy at a closed point of an lc surface pair over an algebraically closed field whose coefficients belong to a DCC set given by Γ', and Theorem 2.21(1) gives the ACC. Since $(\gamma_i)_i$ is increasing, $(\alpha_i)_i$ is not strictly increasing.

(b) p_i is a closed point at which the central surface $(\varsigma = 0)$ is smooth. Then $\widehat{W}_i$ is smooth at p_i, as it contains the Cartier divisor $(\varsigma = 0)$, smooth at p_i. By Theorem 10.1 with the DCC set Γ', the α_i belong to an ACC set, contrary to their strict increase.

(c) p_i is the A_2 point of the central surface of the z-chart in row 5, resp. the A_3 point in row 6 (Lemma 3.11(4), resp. Lemma 3.12(4)). Since p_i is the only singular point of the central surface of the z-chart, $\widehat{G}_i$ is the group of the quotient chart at p_i.

It remains to exclude (c). Write $\omega_i = (\omega_{i,1}, \ldots, \omega_{i,4})$, and let x_1, y_1, z_1, t_1 be the coordinates of the cover $\mathbb{A}^4_{(3)}$ of the z-chart of the ω_i-weighted blow-up of $\mathbb{A}^4$, so that

$$(x,y,z,t) = (x_1 z_1^{\omega_{i,1}}, y_1 z_1^{\omega_{i,2}}, z_1^{\omega_{i,3}}, t_1 z_1^{\omega_{i,4}}), \qquad (X,Y,T,\varsigma) = (x_1, y_1, t_1, z_1^{r_i}),$$

where (X,Y,T,ς) are the coordinates of Lemma 3.11(4), resp. Lemma 3.12(4), in which $p_i = 0$, so $(\widehat{W}_i \ni p_i) \subset (\mathbb{A}^4_{X,Y,T,\varsigma} \ni 0)$. Near p_i the central surface $\widehat{W}_i \cap (\varsigma = 0)$ is smooth off p_i, so $\widehat{W}_i$ is smooth at the points of $(\varsigma = 0)$ other than p_i. Since the weighted blow-up is an isomorphism over $W_i \backslash \{0\}$ and R_i acts freely on $(z_1 \neq 0)$, $\widehat{W}_i$ is smooth at the points of $(\varsigma \neq 0)$ near p_i, as $(W_i \ni 0)$ is isolated. Hence the germ $(\widehat{W}_i \ni p_i)$ is isolated.

Since the hyperplane section $(\varsigma = 0)$ of $(\widehat{W}_i \ni p_i)$ is Du Val, $(\widehat{W}_i \ni p_i)$ is an isolated cDV point, hence Gorenstein and terminal. By Lemma 3.11(4), resp. Lemma 3.12(4), for $i \gg 0$ the equation of $\widehat{W}_i$ at p_i has no linear term in ς and no quadratic term involving ς, and the central

surface is $(X^2+Y^2+T^3=0)$, resp. $(X^2+Y^2+YT^2=0)$. Thus the quadratic part of the equation is X^2+Y^2, and $(\widehat{W}_i \ni p_i)$ has quadratic rank 2. By [Liu26b, Proposition 3.15(4)] with $(u_1,u_2,z,t):=(X,Y,T,\varsigma)$, there exists a $\widehat{G}_i$-equivariant isomorphism

$$\widehat{\mathcal{O}}_{\widehat{W}_i,p_i} \cong \mathbb{C}[[x',y',z',t']]/(Q(x',y')+\widehat{g}_i(z',t')), \qquad Q \in \{x'y', x'^2+y'^2\},$$

where $\widehat{G}_i$ acts diagonally on x',y',z',t' with the characters of X,Y,T,ς, and z',t' correspond to T,ς. Modulo (t') and (ς), the isomorphism induces an isomorphism between $(Q(x',y')+\widehat{g}_i(z',0)=0)$ and the completion of the A_2 surface germ $(X^2+Y^2+T^3=0)$, resp. of the A_3 surface germ

$$(X^2+Y^2+YT^2=0) = \Big(X^2+\Big(Y+\frac{1}{2}T^2\Big)^2-\frac{1}{4}T^4=0\Big).$$

Since $(Q(x',y')+\varphi(z')=0)$ with $\operatorname{mult}_0\varphi=n$ is an A_{n-1} surface germ, $\operatorname{mult}_0\widehat{g}_i(z',0)=3$, resp. 4.

By Lemma 3.11(4), resp. Lemma 3.12(4), the group $\widehat{G}_i$ contains

$$\mu_{d_i}(-3,-3,-2,|c_1-3v|), \quad \text{resp.} \quad \mu_{N_{2,i}}(-2,-2,-1,k),$$

in the coordinates (X,Y,T,ς), with index at most $2[G_i:R_i]\le 2e_0$. Here $k>0$. Let

$$\mathbf{T}' := \mathbf{T}_{(-3,-3,-2,|c_1-3v|)}, \quad \text{resp.} \quad \mathbf{T}' := \mathbf{T}_{(-2,-2,-1,k)},$$

acting on x',y',z',t'. Since $\omega_{i,3}=2d_i$, resp. $\omega_{i,3}=2N_{2,i}$, Lemma 3.11(2), resp. Lemma 3.12(2), with its constant Λ for the compact interval I', implies

$$|\widehat{G}_i\cap\mathbf{T}'| \ge \frac{\omega_{i,3}}{2} \ge \frac{r_i}{2\Lambda} \to +\infty, \qquad [\widehat{G}_i:\widehat{G}_i\cap\mathbf{T}'] \le 2e_0.$$

We apply the second part of [Liu26b, Theorem 8.6], with $n=3$, $c=1$, and $|c_1-3v|$ in place of v, resp. with $n=4$, $c=1$, and k in place of v, with $2e_0$ in place of e_0, Γ' in place of Γ, and the coordinates x,y,z,t there corresponding to x',y',z',t'. The circle of [Liu26b, Theorem 8.6] is $\mathbf{T}'$, and the data $(\widehat{W}_i\ni p_i,\widehat{G}_i,\widehat{D}_i)$ satisfy the hypotheses there: $(\widehat{W}_i\ni p_i)$ is an isolated Gorenstein terminal germ of quadratic rank 2, $\widehat{G}_i$ is a finite abelian group acting faithfully on $\widehat{W}_i$, linearly on the span of x',y' and diagonally on z',t', $\operatorname{mult}_0\widehat{g}_i(z',0)=n$, $[\widehat{G}_i:\widehat{G}_i\cap\mathbf{T}']\le 2e_0$, and $\widehat{D}_i$ is a labelled boundary with labels in Γ' such that $(\widehat{W}_i,\widehat{D}_i)$ is klt near p_i. Since the quotient values satisfy $0<\alpha_1<\alpha_2<\cdots$ and $|\widehat{G}_i\cap\mathbf{T}'|\to+\infty$, this contradicts [Liu26b, Theorem 8.6].

The proof continues below.

Proof of Theorem 12.3, continued (rows 7a–7h). In row 7a, and in rows 7b–7h except for the four circles below, the residual cubic of W_i is L^2M for linearly independent linear forms L and M. From (C.50) on we also treat the circles $\mathbf{T}_{(0,1,-2,0)}$ and $\mathbf{T}_{(3,4,-2,2)}$ of the cuspidal cubic y^2z+t^3 and the circles $\mathbf{T}_{(1,2,-2,0)}$ and $\mathbf{T}_{(1,0,2,1)}$ of the tangent conic y^2z+yt^2, which Lemmas 3.11 and 3.12 do not cover. For a prime divisor F over W_i with center o_i we let

$$\mathcal{A}_i(F) := \frac{a(F,W_i,\frac{1}{m}\operatorname{div}(h_i))}{e_F},$$

and for a regular function g near o_i we write $[g]_F := \operatorname{mult}_F g/e_F$.

Row 7a: the prepared equation and unbounded remainders. The linear forms L and M are G_i-semi-invariant by [Liu26a, Lemma 11.3(1)(2)]. Instead of the polynomial model of **Step 1**, we use the model given by [Liu26b, Lemma C.3] and Lemma 4.6 at the label t_i, applied to the data before the replacement in **Step 1** with $\mathcal{P}=\{P\}$ for a prime divisor P computing b_i. Thus we may assume the following: $o_i=0\in\mathbb{A}^4$, G_i acts diagonally on the coordinates x,y,z,t of $\mathbb{A}^4$, $W_i=(f_i=0)$ with

$$f_i = x^2+y^2z+C_i(t)y+B_i(z,t), \qquad \operatorname{mult}_0 C_i\ge 3, \quad \text{and} \quad \operatorname{mult}_0 B_i\ge 4, \tag{C.45}$$

h_i is a G_i-semi-invariant polynomial of degree at most 1 in x (reducing modulo f_i, as in the normal form of [Liu26b, Definition-Lemma 11.5(1)]), α_i does not change, and, by Lemma 4.6(2),

$\mathcal{A}_i(P_i) = b_i$ for a prime divisor P_i over W_i with center 0. Since (a)–(c) of **Step 1** and (C.29) use only Lemma 4.6(1)(3), they hold for f_i and h_i in (C.45).

Let $\chi_x, \chi_y, \chi_z, \chi_t$ be the characters of G_i on the coordinates. Since f_i is semi-invariant and contains x^2 and y^2z, $\chi_y^2\chi_z = \chi_x^2$. Let $\chi_{\mathbf{c}} := \chi_y$ and $\chi_p := \chi_x\chi_y^{-1}$. Then

$$(\chi_x, \chi_y, \chi_z, \chi_t) = (\chi_p\chi_{\mathbf{c}}, \chi_{\mathbf{c}}, \chi_p^2, \chi_t). \tag{C.46}$$

Let Σ_i be the kernel of $(\chi_p, \chi_t) : G_i \to \mathbb{T}^2$. We claim that $|\Sigma_i| \le 2$. An element $g \in \Sigma_i$ multiplies x and y by $\chi_{\mathbf{c}}(g)$ and fixes z and t, and G_i acts faithfully, so $\chi_{\mathbf{c}}(\Sigma_i) = \mu_{|\Sigma_i|}$. A monomial $y^jz^at^b$ of f_i has the character χ_x^2, so $\chi_{\mathbf{c}}^{j-2} = 1$ on Σ_i and $j \equiv 2 \pmod{|\Sigma_i|}$. If $|\Sigma_i| \ge 3$, then $C_i = B_i = 0$, and $W_i = (x^2 + y^2z = 0)$ is singular along the plane $(x = y = 0)$, which contradicts the isolatedness of $(W_i \ni 0)$.

Assume that, possibly passing to a subsequence, $\operatorname{mult}_0(C_iy + B_i) \to +\infty$. We derive a contradiction in the rest of row 7a. Let $w^\natural := (2m+2, 2m+1, 2, 1)$ and $W^\natural := (x^2+y^2z = 0) \subset \mathbb{A}^4$. For $i \gg 0$, $\operatorname{in}_{w^\natural} f_i = x^2 + y^2z$, which is irreducible, so by [Liu26b, Lemma 4.12] the $w^\natural$-weighted blow-up defines a prime divisor $E^\natural$ over W_i with center 0 and $a(E^\natural, W_i, 0) = 2$, and

$$\operatorname{mult}_{E^\natural} h_i \ge \operatorname{wt}_{w^\natural}(h_i).$$

Since $(W_i, \frac{1}{m}\operatorname{div}(h_i))$ is lc, $\operatorname{wt}_{w^\natural}(h_i) \le 2m$. Any monomial divisible by x or y has $w^\natural$-weight at least $2m+1$. Thus h_i has a monomial z^at^b with $2a + b \le 2m$.

Let $\overline{G}_i \subset \mathbb{T}^2$ be the image of (χ_p, χ_t). Since $|\Sigma_i| \le 2$, $|\overline{G}_i| \to +\infty$. By [Liu26b, Lemma 4.6], possibly passing to a subsequence, the $\overline{G}_i$ converge in the Hausdorff distance to a closed subgroup $\overline{H} \subset \mathbb{T}^2$ with $\dim \overline{H} \ge 1$ and $\overline{G}_i \subset \overline{H}$ for any i. Assume that the Lie algebra of $\overline{H}$ contains a vector (φ_1, φ_2) with $\varphi_1, \varphi_2 > 0$. Since $\overline{G}_i \to \overline{H}$, there exist $g_i \in G_i$ and $\varphi_{1,i}, \varphi_{2,i}$ with $\varphi_{j,i} \to \varphi_j$ for $j = 1, 2$ and

$$(\chi_p(g_i), \chi_t(g_i)) = (e^{2\pi\sqrt{-1}\varphi_{1,i}}, e^{2\pi\sqrt{-1}\varphi_{2,i}}).$$

Let N_0 be a positive integer, and let $N_{0,i} \in [N_0, N_0 + 1)$ with $\chi_{\mathbf{c}}(g_i) = e^{2\pi\sqrt{-1}N_{0,i}}$. By (C.46), $\rho(\mathbf{v}_i) = g_i$ for

$$\mathbf{v}_i := (\varphi_{1,i} + N_{0,i}, N_{0,i}, 2\varphi_{1,i}, \varphi_{2,i}) \in N_{G_i} \cap \mathbb{R}^4_{>0}.$$

Since $\operatorname{mult}_0(C_iy + B_i) \to +\infty$, for $i \gg 0$

$$\operatorname{wt}_{\mathbf{v}_i}(f_i) = 2\varphi_{1,i} + 2N_{0,i}.$$

By [Liu26b, Lemma 4.7] applied to the klt pairs $(W_i, \frac{t_i}{m}\operatorname{div}(h_i))$, $\alpha_i \le \varphi_{1,i} + \varphi_{2,i}$, so $\alpha_\infty \le \varphi_1 + \varphi_2$. Since the Lie algebra of $\overline{H}$ contains the positive multiples of (φ_1, φ_2), $\alpha_\infty \le 0$, a contradiction. Thus $\dim \overline{H} = 1$, and the Lie algebra of $\overline{H}$ contains no vector with both coordinates positive. Hence there exist integers $k_0 \ge 1$ and $k_1, k_2 \ge 0$ with $\gcd(k_1, k_2) = 1$ and

$$\overline{H} = \{(p, \tau) \in \mathbb{T}^2 \mid (p^{k_1}\tau^{k_2})^{k_0} = 1\}. \tag{C.47}$$

Since h_i is G_i-semi-invariant, the ratio $z^{a'-a}t^{b'-b}$ of any two monomials z^at^b and $z^{a'}t^{b'}$ of h_i is G_i-invariant, so the character $(p, \tau) \to p^{2(a'-a)}\tau^{b'-b}$ of $\mathbb{T}^2$ is trivial on $\overline{G}_i$. A character of $\mathbb{T}^2$ which is trivial on $\overline{G}_i$ for infinitely many i is trivial on $\overline{H}$, so it is an integral multiple of $k_0(k_1, k_2)$. Let $k_3, k_4 \ge 0$ be the integers such that $(2k_3, k_4)$ is the least positive multiple of $k_0(k_1, k_2)$ with even first entry, and let $\Pi := z^{k_3}t^{k_4}$. For any positive integer D, only finitely many characters $(p, \tau) \to p^{2(a'-a)}\tau^{b'-b}$ come from pairs of monomials of degree at most D, so for $i \gg 0$ any two monomials z^at^b and $z^{a'}t^{b'}$ of h_i of degree at most D satisfy $(a' - a, b' - b) \in \mathbb{Z}(k_3, k_4)$. Let $z^{n_z}t^{n_t}$ be the monomial z^at^b of h_i with $2a + b$ least. Then $2n_z + n_t \le 2m$, as h_i has a monomial z^at^b with $2a + b \le 2m$, and possibly passing to a subsequence and multiplying h_i by constants, n_z and n_t do not depend on i and the coefficient of $z^{n_z}t^{n_t}$ in h_i is 1. Since $2k_3 + k_4 > 0$, for any D and $i \gg 0$ the monomials of h_i of degree at most D which are not divisible by x or y are of the form $z^{n_z}t^{n_t}\Pi^j$ with $j \ge 0$.

We show that $2n_z \le m$ and $n_t \le m$. By [Kaw07, Theorem], the pair $(\mathbb{A}^4, B_{\mathbb{A}^4,i})$ is lc near 0, where

$$B_{\mathbb{A}^4,i} := \operatorname{div}(f_i) + \frac{1}{m}\operatorname{div}(h_i).$$

By Lemma 2.19 with the coefficients $\{1, \frac{1}{m}\}$, since $f_i - x^2 - y^2 z \in (x,y,z,t)^L$ for any L and $i \gg 0$, the pair $(\mathbb{A}^4, W^\natural + \frac{1}{m}\operatorname{div}(h_i))$ is lc near 0. The normalization of $W^\natural$ is $\nu : \mathbb{A}^3_{\eta,y,t} \to W^\natural$, $(\eta, y, t) \to (\eta y, y, -\eta^2, t)$, with conductor $\operatorname{div}(y)$, so that $\nu^* K_{W^\natural} = K_{\mathbb{A}^3} + \operatorname{div}(y)$ [Kol13, 5.7] and the different of $(\mathbb{A}^4, W^\natural + \frac{1}{m}\operatorname{div}(h_i))$ on $\mathbb{A}^3$ is $\operatorname{div}(y) + \frac{1}{m}\operatorname{div}(\nu^* h_i)$. By [Kaw07, Theorem], the pair

$$\Big(\mathbb{A}^3, \operatorname{div}(y) + \frac{1}{m}\operatorname{div}(\nu^* h_i)\Big)$$

is lc near 0, and by adjunction to $(y = 0)$, $(\mathbb{A}^2_{\eta,t}, \frac{1}{m}\operatorname{div}(\psi_i))$ is lc near 0 for $\psi_i := h_i(0, 0, -\eta^2, t)$. By the previous paragraph, $\psi_i = \pm\eta^{2n_z} t^{n_t} \upsilon_i + \psi_i'$, where υ_i is a polynomial with $\upsilon_i(0) = 1$ and $\psi_i' \in (\eta, t)^L$ for any L and $i \gg 0$. By Lemma 2.19 with the coefficient $\frac{1}{m}$, $(\mathbb{A}^2, \frac{1}{m}\operatorname{div}(\eta^{2n_z} t^{n_t}))$ is lc near 0, that is, $2n_z \le m$ and $n_t \le m$.

Let

$$\begin{aligned} &\Lambda_1 := 1 - \frac{2n_z}{m}, \qquad \Lambda_2 := 1 - \frac{n_t}{m}, \quad \text{and} \\ &\Lambda_0 := \min\Big(\Big\{\frac{\Lambda_1}{k_0 k_1} \;\Big|\; k_1 > 0\Big\} \cup \Big\{\frac{\Lambda_2}{k_0 k_2} \;\Big|\; k_2 > 0\Big\}\Big). \end{aligned} \tag{C.48}$$

Then $\Lambda_0, \Lambda_1, \Lambda_2 \in [0, 1]$. For $\mathbf{v} = (v_x, v_y, v_z, v_t) \in \mathbb{R}^4_{\ge 0}$ let

$$\Psi(\mathbf{v}) := \sum_j v_j - \operatorname{wt}_{\mathbf{v}}(x^2 + y^2 z) - \frac{1}{m}\operatorname{wt}_{\mathbf{v}}(z^{n_z} t^{n_t}) = \Lambda_1 \frac{v_z}{2} + \Lambda_2 v_t + \Big|v_x - v_y - \frac{v_z}{2}\Big|. \tag{C.49}$$

We claim that $\Psi(\mathbf{v}) \ge \Lambda_0$ for any $\mathbf{v} \in N_{G_i} \cap \mathbb{R}^4_{>0}$. Let $\delta := v_x - v_y$ and $\varrho := v_z/2 > 0$. Since $\rho(\mathbf{v}) \in G_i$, (C.46) and (C.47) imply

$$2\varrho - 2\delta \in \mathbb{Z} \qquad \text{and} \qquad k_0(k_1\delta + k_2 v_t) \in \mathbb{Z}.$$

If $\delta \ge 0$, then $\Lambda_1\varrho + |\delta - \varrho| \ge \Lambda_1\delta$, since $\Lambda_1 \le 1$. Recall that k_1 and k_2 are not both zero.

(1) If $k_1 = 0$, then $k_2 = 1$, $v_t \ge 1/k_0$, and $\Psi(\mathbf{v}) \ge \Lambda_2 v_t \ge \Lambda_2/k_0 = \Lambda_0$.

(2) If $k_1 > 0$, $k_2 = 0$ and $\delta > 0$, then $k_1 = 1$, $\delta \ge 1/k_0$, and

$$\Psi(\mathbf{v}) \ge \Lambda_1\delta \ge \Lambda_1/k_0 = \Lambda_0.$$

(3) If $k_1 > 0$, $k_2 = 0$ and $\delta = 0$, then $k_1 = 1$, $2\varrho \ge 1$, and

$$\Psi(\mathbf{v}) \ge (\Lambda_1 + 1)\varrho \ge \Lambda_1 \ge \Lambda_0.$$

(4) If $k_1, k_2 > 0$ and $\delta \ge 0$, then $k_1\delta + k_2 v_t \ge 1/k_0$, so

$$\Psi(\mathbf{v}) \ge \Lambda_1\delta + \Lambda_2 v_t \ge \min\Big(\frac{\Lambda_1}{k_1}, \frac{\Lambda_2}{k_2}\Big)(k_1\delta + k_2 v_t) \ge \Lambda_0.$$

(5) If $k_1 > 0$ and $\delta < 0$, then $2\varrho - 2\delta$ is a positive integer, so $\Psi(\mathbf{v}) \ge \varrho - \delta \ge \frac{1}{2}$. Assume $\Lambda_0 > \frac{1}{2}$. Since $\Lambda_0 \le \Lambda_1/(k_0k_1) \le 1/(k_0k_1)$ and $\Lambda_0 \le 1/(k_0k_2)$ if $k_2 > 0$, we have $k_0 = k_1 = 1$ and $k_2 \le 1$. If $k_2 = 0$, then $\delta \in \mathbb{Z}$, so $-\delta \ge 1$ and $\Psi(\mathbf{v}) > 1 \ge \Lambda_0$. If $k_2 = 1$, then $\Lambda_0 = \min(\Lambda_1, \Lambda_2)$ and $\delta + v_t \in \mathbb{Z}$. If $\delta + v_t \le -1$, then $-\delta > 1$ and $\Psi(\mathbf{v}) > 1 \ge \Lambda_0$. If $v_t \ge -\delta$, then

$$\Psi(\mathbf{v}) \ge \Lambda_0\varrho + \Lambda_0(-\delta) + (\varrho - \delta) = (1 + \Lambda_0)(\varrho - \delta) \ge \frac{1 + \Lambda_0}{2} \ge \Lambda_0.$$

Let

$$\Delta^\natural := W^\natural + \frac{1}{m}\operatorname{div}(z^{n_z} t^{n_t}).$$

We show that $a(E, \mathbb{A}^4, \Delta^\natural)/e_E \ge \Lambda_0$ for any prime divisor E over $\mathbb{A}^4$ with center 0, where $e_E := |I_{G_i}(E)|$. The restrictions of $x^2 + y^2 z$ to the faces of the Newton polyhedron of $x^2 + y^2 z$,

namely x^2, y^2z, and x^2+y^2z, have no singular points on $(\mathbb{C}^*)^4$. Hence a smooth toric modification $\pi : Y^\natural \to \mathbb{A}^4$ whose fan refines the cones $\{2v_x \le 2v_y + v_z\}$ and $\{2v_x \ge 2v_y + v_z\}$ of $\mathbb{R}^4_{\ge 0}$ is a log resolution of $(\mathbb{A}^4, W^\natural)$, and $S := \pi_*^{-1}W^\natural$ is smooth and intersects the toric strata transversally. For a ray $\mathfrak{r}$ of the fan with primitive vector $\mathbf{v}_\mathfrak{r}$ and toric divisor $D_\mathfrak{r}$,

$$a(D_\mathfrak{r}, \mathbb{A}^4, \Delta^\natural) = \Psi(\mathbf{v}_\mathfrak{r}),$$

and Ψ is linear on each cone of the fan. Let $w_E := (\operatorname{mult}_E x, \operatorname{mult}_E y, \operatorname{mult}_E z, \operatorname{mult}_E t)$. Since $(Y^\natural, S + \sum_\mathfrak{r} D_\mathfrak{r})$ is lc,

$$a(E, \mathbb{A}^4, \Delta^\natural) = a\Big(E, Y^\natural, S + \sum_\mathfrak{r} D_\mathfrak{r}\Big) + \sum_\mathfrak{r} \Psi(\mathbf{v}_\mathfrak{r}) \operatorname{mult}_E D_\mathfrak{r} \ge \Psi(w_E).$$

The order along E of a G_i-invariant Laurent monomial is a multiple of e_E [Liu26a, Lemma 2.6], so $w_E/e_E \in N_{G_i} \cap \mathbb{R}^4_{>0}$, and

$$\frac{a(E, \mathbb{A}^4, \Delta^\natural)}{e_E} \ge \Psi(w_E/e_E) \ge \Lambda_0.$$

For $i \gg 0$, $\operatorname{in}_{w^\natural}(h_i) = z^{n_z}t^{n_t}$: the monomials divisible by x or y have $w^\natural$-weight at least $2m+1 > 2n_z + n_t$, and the other monomials of degree at most $2m+1$ are $z^{n_z}t^{n_t}\Pi^j$ with $j \ge 0$. For $\lambda \in \mathbb{C}$ let

$$f_{i,\lambda} := \lambda^{-4m-4} f_i(\lambda^{2m+2}x, \lambda^{2m+1}y, \lambda^2 z, \lambda t), \qquad h_{i,\lambda} := \lambda^{-2n_z-n_t} h_i(\lambda^{2m+2}x, \lambda^{2m+1}y, \lambda^2 z, \lambda t).$$

They are polynomials in λ, x, y, z, t with $f_{i,0} = x^2 + y^2 z$ and $h_{i,0} = z^{n_z}t^{n_t}$, and they are G_i-semi-invariant. Let $\mathcal{X}_i := \mathbb{A}^4/G_i$, let $\Delta_{\mathcal{X}_i}$ be the $\mathbb{Q}$-divisor on $\mathcal{X}_i$ such that $K_{\mathbb{A}^4}$ is the pull-back of $K_{\mathcal{X}_i} + \Delta_{\mathcal{X}_i}$ under the quotient map, and let $\mathfrak{a}_i$ be the $\mathbb{R}$-ideal on $\mathcal{X}_i \times \mathbb{A}^1_\lambda$ given by

$$(f_{i,\lambda}^{|G_i|})^{1/|G_i|} (h_{i,\lambda}^{|G_i|})^{1/(m|G_i|)}.$$

For $\lambda \ne 0$ the automorphism

$$(x, y, z, t) \to (\lambda^{2m+2}x, \lambda^{2m+1}y, \lambda^2 z, \lambda t)$$

of $\mathbb{A}^4$ commutes with G_i, so

$$\operatorname{mld}(\mathcal{X}_i \ni \overline{0}, \Delta_{\mathcal{X}_i}, \mathfrak{a}_{i,\lambda})$$

does not depend on $\lambda \ne 0$, and for $\lambda = 1$ it is the quotient value of $(\mathbb{A}^4 \ni 0, B_{\mathbb{A}^4,i})$ by G_i. Let $G_i^{\mathrm{pr}} \subset G_i$ be the subgroup generated by pseudo-reflections. Then $\mathbb{A}^4/G_i^{\mathrm{pr}} \cong \mathbb{A}^4$ and G_i/G_i^{pr} acts on it freely in codimension 1, so $\mathcal{X}_i$ satisfies the hypothesis of Theorem 2.16. By the lower semicontinuity at $\lambda = 0$ and the previous paragraph,

$$\operatorname{mld}((\mathbb{A}^4 \ni 0, B_{\mathbb{A}^4,i})/G_i) \ge \operatorname{mld}((\mathbb{A}^4 \ni 0, \Delta^\natural)/G_i) \ge \Lambda_0.$$

By Theorem 2.17(1),

$$\operatorname{mld}((\mathbb{A}^4 \ni 0, B_{\mathbb{A}^4,i})/G_i) = \operatorname{mld}\Big(\Big(W_i \ni 0, \frac{1}{m}\operatorname{div}(h_i)\Big)/G_i\Big) \le \mathcal{A}_i(P_i) = b_i.$$

Thus $b_i \ge \Lambda_0$.

Conversely, let $\varphi_1, \varphi_2 > 0$ with $k_0(k_1\varphi_1 + k_2\varphi_2) = 1$. Then

$$(e^{2\pi\sqrt{-1}\varphi_1}, e^{2\pi\sqrt{-1}\varphi_2}) \in \overline{H},$$

and, as in the paragraph before (C.47) with $N_0 > m(\varphi_1 + \varphi_2)$, there exist

$$\mathbf{v}_i = (\varphi_{1,i} + N_{0,i}, N_{0,i}, 2\varphi_{1,i}, \varphi_{2,i}) \in N_{G_i} \cap \mathbb{R}^4_{>0}$$

with $\varphi_{j,i} \to \varphi_j$ for $j = 1, 2$ and $N_{0,i} \in [N_0, N_0 + 1)$. Since $2n_z, n_t \le m$, for $i \gg 0$ any monomial divisible by x or y has $\mathbf{v}_i$-weight at least $N_0 > 2n_z\varphi_{1,i} + n_t\varphi_{2,i}$. For $i \gg 0$,

$$\operatorname{wt}_{\mathbf{v}_i}(f_i) = 2\varphi_{1,i} + 2N_{0,i} \quad \text{and} \quad \operatorname{wt}_{\mathbf{v}_i}(h_i) = 2n_z\varphi_{1,i} + n_t\varphi_{2,i},$$

and [Liu26b, Lemma 4.7] implies

$$\alpha_i \le \varphi_{1,i} + \varphi_{2,i} - \frac{t_i}{m}(2n_z\varphi_{1,i} + n_t\varphi_{2,i}) \to \Lambda_1\varphi_1 + \Lambda_2\varphi_2.$$

The infimum of $\Lambda_1\varphi_1 + \Lambda_2\varphi_2$ over such (φ_1, φ_2) is Λ_0. Hence $\alpha_\infty \le \Lambda_0 \le b_i$, which contradicts $b_i < \alpha_\infty$.

Row 7b: compact limits and first faces. By row 7a, $\mathrm{mult}_0(C_iy + B_i)$ is bounded. By (C.46), G_i is contained in the 3-dimensional torus $\mathbb{T}^\flat := \{(pc, c, p^2, \tau)\} \subset \mathbb{T}^4$, and the monomial $x^{j_1}y^{j_2}z^{j_3}t^{j_4}$ restricts to the character of $\mathbb{T}^\flat$ with exponents $(j_1 + 2j_3, j_1 + j_2, j_4)$ in (p, c, τ). By [Liu26b, Lemma 4.6], possibly passing to a subsequence, the G_i converge to a closed subgroup $H \subset \mathbb{T}^\flat$ with $G_i \subset H$ for any i.

(1) If $\dim H = 3$, then the Lie algebra of H contains $(2, 1, 2, 1)$, and [Liu26b, Lemma 4.7] implies $\alpha_i \to 0$, a contradiction.

(2) If $\dim H = 2$, then H^0 is the kernel of a primitive character with exponents (k_1', k_2', k_3'). The Lie algebra of H is

$$\{(\varphi_1 + \varphi_3, \varphi_3, 2\varphi_1, \varphi_2) \mid k_1'\varphi_1 + k_2'\varphi_3 + k_3'\varphi_2 = 0\},$$

and it contains no vector with all coordinates positive by [Liu26b, Lemma 4.7], so we may assume $k_1', k_2', k_3' \ge 0$. A monomial of f_i has the character χ_x^2, with exponents $(2, 2, 0)$. Thus for a monomial z^at^b, resp. t^jy, of $C_iy + B_i$ of bounded degree and $i \gg 0$, the vector $(2a - 2, -2, b)$, resp. $(-2, -1, j)$, is an integral multiple of (k_1', k_2', k_3'). This is impossible if $k_2' = 0$. If $k_2' > 0$, the multiple is negative, so $b = 0$ and $a \le 1$, resp. $j \le 0$, which contradicts (C.45). Hence, for any D and $i \gg 0$, $C_iy + B_i$ has no monomial of degree at most D, contrary to the boundedness of $\mathrm{mult}_0(C_iy + B_i)$.

(3) If $\dim H = 0$, then $|G_i|$ is bounded, a contradiction.

Therefore $\dim H = 1$, and we use w, $\mathbf{T} = \mathbf{T}_{(w)}$, r_i, e_0, κ, f_i^0, and h_i^0 of (C.33) for f_i and h_i of (C.45). By (C.46), $w = (u + w_y, w_y, 2u, w_t)$ for integers u, w_y, w_t. For monomials μ and μ_0 such that μ/μ_0 is G_i-invariant, we call the integer $(\mathrm{wt}_w(\mu) - \mathrm{wt}_w(\mu_0))/r_i$ the *shift* of μ relative to μ_0. A term of f_i, resp. h_i, has positive, zero or negative shift if the shift of the monomial relative to x^2, resp. to a monomial of h_i^0, is positive, zero or negative.

Let ι_i be the minimum of $j - 2$ over the monomials t^jy of C_iy and of $a + \frac{b}{2} - 1$ over the monomials z^at^b of B_i. Then $\iota_i \ge 1$ by (C.45), and ι_i is bounded since $\mathrm{mult}_0(C_iy + B_i)$ is bounded, so possibly passing to a subsequence $\iota := \iota_i$ does not depend on i. The monomials attaining ι have $(\iota + 1, \iota, 2, 1)$-weight $2\iota + 2$ and bounded degree, so for $i \gg 0$ their $\mathbf{T}$-weight is $2(u + w_y)$:

$$jw_t + w_y = 2(u + w_y) \text{ for } t^jy, \qquad 2au + bw_t = 2(u + w_y) \text{ for } z^at^b.$$

The case $u = w_t$ is row 7e. If $u \ne w_t$, the linear equations $2au + bw_t = 2(u + w_y)$ and $a + \frac{b}{2} = \iota + 1$ have at most one common solution (a, b), and a monomial z^at^b and the monomial $t^{\iota+2}y$ cannot both attain ι, since together the equations imply $2(u - w_t)(a + 1) = 0$. Hence, after multiplying a coordinate by a constant, $\mathrm{in}_{(\iota+1,\iota,2,1)}(f_i)$ is one of

$$\begin{aligned} &\mathrm{I}: \ x^2 + y^2z + z^{n_1}t^{n_2}, \quad n_1 + n_2 \ge 4, \quad 2w_y = 2(n_1 - 1)u + n_2w_t, \\ &\mathrm{II}: \ x^2 + y^2z + yt^{n_3}, \quad n_3 \ge 3, \quad 2u + w_y = n_3w_t, \end{aligned} \tag{C.50}$$

with $\iota = n_1 + \frac{n_2}{2} - 1$, resp. $\iota = n_3 - 2$. For the four circles of the cuspidal cubic and the tangent conic named at the beginning, instead of the polynomial model of **Step 1**, we use the model given by [Liu26b, Proposition 3.17] and Lemma 4.6 at the label t_i, applied to the data before the replacement in **Step 1** with $\mathcal{P}$ consisting of one prime divisor over that data computing b_i, ψ the automorphism and v the inverse of the unit given there. Thus we may assume that $f_i = x^2 + y^2z + C_i(t)y + B_i(z, t)$, where $B_i(0, t)$ is t^3 times an invariant unit, resp. $C_i = t^2$ and $B_i \in (z, t)^4$, and we let P_i be the corresponding prime divisor over W_i, so that $\mathcal{A}_i(P_i) = b_i$ as in row 7a. The circles $(0, 1, -2, 0)$ and $(3, 4, -2, 2)$ satisfy the relation of I with $(n_1, n_2) = (0, 3)$,

and $(1,2,-2,0)$ and $(1,0,2,1)$ that of II with $n_3=2$. From now on we allow $(n_1,n_2)=(0,3)$ and $n_3=2$ in (C.50). Then $n_1+n_2\geq 3$, $n_2\geq 3$ if $n_1=0$, and $n_3\geq 2$, which are the conditions of Proposition C.2 and of the class (c) in (iv) of Proposition 11.7. In particular, by Proposition C.2, $V_0:=(f^0=0)\subset\mathbb{A}^4$ is normal and klt for f^0 the polynomial I or II.

Row 7c: weights u and w_t of opposite signs. Assume $uw_t<0$. Possibly replacing w with $-w$, we may assume $u>0>w_t$, so that $w_z>0>w_t$.

In face I, a monomial z^at^b of f_i^0 satisfies $2(a-n_1)u=(n_2-b)w_t$. If $a<n_1$, then $b<n_2$ and $a+\frac{b}{2}-1<\iota$, which is impossible. Hence the monomials of f_i^0 not divisible by x or y are of the form $z^{n_1}t^{n_2}\Pi_{\mathrm{I}}^j$ with $j\geq 0$, where $\Pi_{\mathrm{I}}:=z^{|w_t|/k_5}t^{2u/k_5}$ and $k_5:=\gcd(2u,w_t)$. A monomial t^jy of f_i^0 would satisfy $j=\frac{n_2}{2}+\frac{(n_1+1)u}{w_t}<\frac{n_2}{2}$, hence $j-2<\iota$, which is impossible. Thus $f_i^0=x^2+y^2z+z^{n_1}t^{n_2}V_i$ for a polynomial V_i in Π_{I} with $V_i(0)=1$, and V_i is G_i-invariant, since a monomial of V_i is the ratio of two monomials of f_i. By Lemma 4.6, taking tV_i^{1/n_2} as the new coordinate t if $n_2>0$, resp. zV_i^{1/n_1} and $yV_i^{-1/(2n_1)}$ as the new coordinates z and y if $n_2=0$, we may assume $f_i^0=x^2+y^2z+z^{n_1}t^{n_2}$. The coordinate changes commute with G_i and $\mathbf{T}$, so they preserve the principal blocks.

In face II, the only monomial t^jy of f_i^0 is $t^{n_3}y$, and a monomial z^at^b of f_i^0 satisfies

$$(b-2n_3)|w_t|=2u(a+1).$$

Thus $f_i^0=x^2+y^2z+yt^{n_3}+t^{2n_3}\Xi_i$ for a polynomial Ξ_i such that $z\Xi_i$ is invariant under G_i and $\mathbf{T}$. With $\varpi_i:=t^{n_3}(1-\sqrt{1-4z\Xi_i})/(2z)$, we take $\sqrt{1-4z\Xi_i}(y+\varpi_i)$ and $z/(1-4z\Xi_i)$ as the new coordinates y and z. The coordinate change

$$(y,z)\to\big(\sqrt{1-4z\Xi_i}(y+\varpi_i),z/(1-4z\Xi_i)\big)$$

commutes with G_i and $\mathbf{T}$ and transforms $y^2z+yt^{n_3}+t^{2n_3}\Xi_i$ into $y^2z+yt^{n_3}$. By Lemma 4.6, we may assume $f_i^0=x^2+y^2z+yt^{n_3}$.

Row 7d: weights u and w_t with $uw_t>0$. Assume $uw_t>0$. Possibly replacing w with $-w$, we may assume $u,w_t>0$. If $w_y>0$, then $w_x,w_y,w_z,w_t>0$, and [Liu26b, Lemma 4.7] implies $\alpha_i\to 0$, a contradiction. Hence $w_y\leq 0$.

In face I, $2w_y=2(n_1-1)u+n_2w_t\leq 0$ and $n_1+n_2\geq 3$ imply $n_1=0$ and $u\geq\frac{n_2}{2}w_t\geq\frac{3}{2}w_t$. A monomial z^at^b of f_i^0 with $a\geq 1$ would satisfy $2au=(n_2-b)w_t$, hence

$$a+\frac{b}{2}-1=\iota-a\Big(\frac{u}{w_t}-1\Big)<\iota,$$

which is impossible. A monomial t^jy of f_i^0 satisfies $j=\frac{n_2}{2}+\frac{u}{w_t}\geq n_2$. Thus

$$f_i^0=x^2+y^2z+t^{n_2}(1+\lambda_iyt^{j-n_2})$$

for some $\lambda_i\in\mathbb{C}$. The monomial yt^{j-n_2} is invariant under G_i and $\mathbf{T}$, so by Lemma 4.6, taking $t(1+\lambda_iyt^{j-n_2})^{1/n_2}$ as the new coordinate t, we may assume $f_i^0=x^2+y^2z+t^{n_2}$. In face II, a monomial z^at^b of f_i^0 satisfies $2(a+1)u=(2n_3-b)w_t$, so $2a+b=2n_3-2-2(a+1)(\frac{u}{w_t}-1)$. Since $u\geq\frac{n_3}{2}w_t\geq w_t$ by $w_y\leq 0$, either $u>w_t$ and $2a+b<2\iota+2$, which is impossible, or $u=w_t$, $n_3=2$, and $\mathbf{T}=\mathbf{T}_{(1,0,2,1)}$, the circle of row 7e, where $2a+b=2$ and $B_i\in(z,t)^4$ has no such monomial. Hence $f_i^0=x^2+y^2z+yt^{n_3}$. In faces I and II, y is the only coordinate of y,z,t which may have negative weight, and x occurs in f_i only in x^2. Hence any monomial of C_iy+B_i has $\mathbf{T}$-weight at least $w_y>2(u+w_y)-r_i$. Since the coordinate change of face I commutes with $\mathbf{T}$, the $\mathbf{T}$-weights of the terms of $f_i-f_i^0$ are among these, and any term of $f_i-f_i^0$ has positive shift.

If $w_y<0$, the circle is treated in row 7h. Assume $w_y=0$. Then $\mathbf{T}=\mathbf{T}_{(u,0,2u,w_t)}$ with $2u=n_2w_t$ in face I and $2u=n_3w_t$ in face II. Since w is primitive, $\gcd(u,w_t)=1$, so $w_t\in\{1,2\}$. We treat the circles $\mathbf{T}_{(u,0,2u,w_t)}$ together with the circle $\mathbf{T}_{(1,1,0,2)}$ of row 7f, for which $f_i^0=x^2+y^2z+z^{n_1}t$

with $n_1 \geq 3$. Write $(z_1, \dots, z_4) = (x, y, z, t)$, let $j_\circ$ be the index with $w_{j_\circ} = 0$ ($j_\circ = 2$, resp. $j_\circ = 3$ for $\mathbf{T}_{(1,1,0,2)}$), let $\mathcal{J}_+ := \{1, \dots, 4\} \backslash \{j_\circ\}$, and let

$$c_{\mathbf{T}} := \sum_j w_j - 2(u + w_y) = u + w_t.$$

All weights are non-negative, and r_i exceeds the weights of f_i^0 and h_i^0 for $i \gg 0$, so any term of $f_i - f_i^0$ and of $h_i - h_i^0$ has positive shift. The $z_{j_\circ}$-axis is contained in W_i. Over the generic point of the $z_{j_\circ}$-axis, the weighted blow-up of the coordinates z_j, $j \in \mathcal{J}_+$, with the weights w_j defines a prime divisor with log discrepancy $c_{\mathbf{T}}$ along which h_i has order κ by [Liu26b, Lemma 4.12] over the field $\mathbb{C}(z_{j_\circ})$, since $\mathrm{in}_w(f_i) = f^0$ is irreducible over $\mathbb{C}(z_{j_\circ})$ and $\mathrm{in}_w(h_i) = h_i^0$ is not divisible by f^0. Since $(W_i, \frac{1}{m}\operatorname{div}(h_i))$ is lc,

$$\frac{\kappa}{m} \leq c_{\mathbf{T}}, \qquad \text{so} \qquad \gamma_i' := 1 - \frac{c_{\mathbf{T}} - \kappa/m}{r_i} \leq 1,$$

where γ_i' is the coefficient (C.34) at the label 1.

For $k \in \mathcal{J}_+$ let $S_k := V_0 \cap (z_k = 1) \subset (z_k = 1) \cong \mathbb{A}^3$, for $V_0 = (f^0 = 0)$, whose quotient by the stabilizer of $(z_k = 1)$ in $\mathbf{T}$ is the root section at z_k as in Set-up 11.1, let $\Phi_k := S_k \cap (z_{j_\circ} = 0)$ be the fiber over the origin of the $z_{j_\circ}$-axis, and let $H_{i,k}$ be the group of elements of $G_i\mathbf{T}$ fixing z_k. It acts diagonally on $(z_k = 1)$, and $|H_{i,k}| = w_k[G_i : R_i] \leq w_k e_0$. We write X, Y, Z, T for the restrictions of x, y, z, t to $(z_k = 1)$. The root sections and their fibers are as follows.

(1) Face I, $w_y = 0$: at x, z, t the sections are $1 + Y^2Z + T^{n_2}$, the A_{n_2-1} surface $X^2 + Y^2 + T^{n_2}$, and $X^2 + Y^2Z + 1$. The sections at x and t are smooth along Φ_k, since $\partial_T = n_2 T^{n_2-1} \neq 0$, resp. $\partial_X = 2X \neq 0$, there.

(2) Face II, $w_y = 0$: at x, z, t the sections are $1 + Y^2Z + YT^{n_3}$, with $\Phi_x = \emptyset$, the A_{2n_3-1} surface $X^2 + Y^2 + YT^{n_3} = X^2 + (Y + \frac{1}{2}T^{n_3})^2 - \frac{1}{4}T^{2n_3}$, and $X^2 + Y^2Z + Y$, which is smooth along Φ_t since $\partial_Y = 1$ there.

(3) $\mathbf{T}_{(1,1,0,2)}$: at x, y, t the sections are $1 + Y^2Z + Z^{n_1}T$, with $\Phi_x = \emptyset$, $X^2 + Z + Z^{n_1}T$, which is smooth along Φ_y, and the D_{n_1+1} surface $X^2 + Y^2Z + Z^{n_1}$.

Since $f_i - f^0, h_i - h_i^0 \in (x, y, z, t)^L$ for any L and $i \gg 0$, [Kaw07, Theorem] and Lemma 2.19 with the coefficients $\{1, \frac{1}{m}\}$ show that $(V_0, \frac{1}{m}\operatorname{div}(h_i^0))$ is lc near 0. Hence the pair $(S_k, \frac{1}{m}\operatorname{div}(h_i^0|_{S_k}))$ is lc near Φ_k: for a prime divisor P over S_k with center in Φ_k and $a(P, S_k, \frac{1}{m}\operatorname{div}(h_i^0|_{S_k})) < 0$, the Gauss extension of $N \operatorname{mult}_P$, $N \gg 0$, with a fixed positive value on the radial coordinate is a multiple of $\operatorname{mult}_{P'}$ for a prime divisor P' over V_0 with center 0 and negative log discrepancy. Let $\beta_{S,i}$ be the minimum, over $k \in \mathcal{J}_+$ and over the closed points and the generic points of curves ξ of Φ_k, of the quotient value of $(S_k, \frac{1}{m}\operatorname{div}(h_i^0|_{S_k}))$ by $H_{i,k}$ at the image of ξ. Since $m(K_{S_k} + \frac{1}{m}\operatorname{div}(h_i^0|_{S_k}))$ is Cartier, S_k is Gorenstein and $|H_{i,k}| \leq e_0 \max_j w_j$, the number $\beta_{S,i}$ belongs to $\frac{1}{N'}\mathbb{Z} \cap [0, 2]$ for $N' := m \cdot (e_0 \max_j w_j)!$. Possibly passing to a subsequence, $\beta_S := \beta_{S,i}$ does not depend on i.

Let F be a prime divisor over W_i with center 0, and let $k \in \mathcal{J}_+$ attain $\mathcal{R}(F) := \min_{j \in \mathcal{J}_+}[z_j]_F / w_j$. In the chart $z_k = \vartheta^{w_k}$, $z_j = \vartheta^{w_j} Z_j$ $(j \neq k)$ of the w-weighted blow-up of $\mathbb{A}^4$, divided by R_i, with radial coordinate $\varsigma := \vartheta^{r_i}$, the strict transforms of f_i and h_i are

$$f^0(Z) + \sum_{n \geq 1} \varsigma^n f_i^{[n]}(Z) \qquad \text{and} \qquad h_i^0(Z) + \sum_{n \geq 1} \varsigma^n h_i^{[n]}(Z),$$

where $Z_k := 1$. Let $W_{i,k} \subset \mathbb{A}^4_{\varsigma, (Z_j)_{j \neq k}}$ be the hypersurface defined by the first, $h_{i,k}$ the second, and $H'_{i,k}$ the finite diagonal group acting on $\mathbb{A}^4_{\varsigma, (Z_j)_{j \neq k}}$ induced by G_i and the root of z_k. The images of $H'_{i,k}$ and of $H_{i,k}$ in the automorphism group of $(\varsigma = 0) = (z_k = 1)$ coincide. Then $W_{i,k} \cap (\varsigma = 0) = S_k$, $W_{i,k}$ is normal and klt near S_k by inversion of adjunction, and $H'_{i,k}$ acts faithfully on $W_{i,k}$ and on S_k, as S_k is contained in no coordinate hyperplane, and no non-identity element of $H'_{i,k}$ acts trivially on an irreducible component of S_k, since S_k is normal, each connected

component of S_k contains a point with non-zero coordinates (an irreducible component of the surface S_k contained in $(z_j = 0)$, $j \neq k$, would be $(z_j = 0) \cap (z_k = 1)$, and then, as f^0 is $\mathbf{T}$-homogeneous and $w_k > 0$, f^0 would vanish on $(z_j = 0) \cap (z_k \neq 0)$, so z_j would divide f^0, which contains the monomials x^2 and y^2z), and an element of the diagonal group $H'_{i,k}$ fixing such a point acts trivially on $(\varsigma = 0) \supset S_k$. The pair $(W_i, \frac{1}{m}\operatorname{div}(h_i))$ with G_i is crepant to $(W_{i,k}, \gamma'_i S_k + \frac{1}{m}\operatorname{div}(h_{i,k}))$ with $H'_{i,k}$, and F induces a prime divisor F_k over $W_{i,k}$ whose center is a closed point or the generic point of a curve ξ of Φ_k, since $[Z_j]_F \geq 0$ for $j \in \mathcal{J}_+$ and $[z_{j_\circ}]_F > 0$. Moreover

$$\mathcal{A}_i(F) = \frac{a(F_k, W_{i,k}, \gamma'_i S_k + \frac{1}{m}\operatorname{div}(h_{i,k}))}{|I_{H'_{i,k}}(F_k)|}, \qquad \frac{\operatorname{mult}_{F_k}\varsigma}{|I_{H'_{i,k}}(F_k)|} = r_i\mathcal{R}(F).$$

By Lemma 11.5 with $c_S = \gamma'_i \leq 1$, $\mathcal{A}_i(F) \geq \beta_0(\xi) \geq \beta_S$, where $\beta_0(\xi)$ is the quotient value at ξ of the root-section pair. Thus $b_i = \mathcal{A}_i(P_i) \geq \beta_S$.

Conversely, let ξ_i be a closed point or the generic point of a curve of Φ_k at which β_S is attained. A translation moving a closed point ξ_i to 0 commutes with the stabilizer of ξ_i, so by Theorem 2.17(1)(3), β_S is the quotient value at the image of ξ_i of the ambient pair $(\mathbb{A}^3, S_k + \frac{1}{m}\operatorname{div}(h_i^0))$ by $H_{i,k}$. The quotients $\mathbb{A}^3/H_{i,k}$ are finitely many fixed threefolds, and the coefficients of the quotient boundaries belong to a fixed finite set, since the components of $\operatorname{div}(h_i^0)$ through ξ_i have multiplicity at most m. By Theorem 2.22(3), β_S is computed by a prime divisor E'_i over $\mathbb{A}^3$ with $a(E'_i, \mathbb{A}^3, 0)$ bounded, so the orders along E'_i of the coordinates, of the equation of S_k and of h_i^0 are bounded by a constant C_2. Let $E_i'^\sharp$ be the Gauss extension of E'_i to $\mathbb{A}^4_{\varsigma,(Z_j)_{j\neq k}}$ with $\operatorname{mult}_{E_i'^\sharp}\varsigma = N''_i$, where $N''_i > C_2$ is a bounded integer in the residue class modulo $|H'_{i,k}|$ for which $I_{H'_{i,k}}(E_i'^\sharp) = I_{H_{i,k}}(E'_i)$, which exists by [Liu26b, Lemma 3.1], applied to the stabilizer of ξ_i with the coordinates translated to ξ_i (over the residue field of ξ_i if ξ_i is the generic point of a curve), and for which $a(E_i'^\sharp, \mathbb{A}^4, 0) = a(E'_i, \mathbb{A}^3, 0) + N''_i$. Then the terms with ς^n, $n \geq 1$, do not change the orders along $E_i'^\sharp$, all coordinates x, y, z, t have positive order along $E_i'^\sharp$, and by Theorem 2.17(1) at the label t_i,

$$\alpha_i \leq \beta_S + \frac{(1-t_i)\operatorname{mult}_{E'_i} h_i^0}{m|I_{H_{i,k}}(E'_i)|} + \frac{N''_i(c_{\mathbf{T}} - t_i\kappa/m)}{r_i|I_{H_{i,k}}(E'_i)|} \leq \beta_S + \frac{C_1}{m}(1-t_i) + \frac{C_0}{r_i}$$

for constants C_0, C_1 independent of i. Hence $\alpha_\infty \leq \beta_S \leq b_i$, which contradicts $b_i < \alpha_\infty$.

Row 7e: $u = w_t$. The equations for the monomials attaining ι imply $w_y = \iota u$, so

$$w = u(\iota + 1, \iota, 2, 1).$$

Here $\iota \geq 1$ for the double line, $\iota = \frac{1}{2}$ for I with $(n_1, n_2) = (0, 3)$ and $\iota = 0$ for II with $n_3 = 2$. If $\iota > 0$, then all coordinates of w have the sign of u, and [Liu26b, Lemma 4.7] implies $\alpha_i \to 0$, a contradiction. If $\iota = 0$, then $w = (1, 0, 2, 1)$ after orientation, a circle of row 7d with $w_y = 0$.

Row 7f: zero weights. By rows 7c and 7d, it remains to treat $u = 0$ or $w_t = 0$, and $w_t = 0 \neq u$ is row 7g.

(1) If $u = w_y = 0$, then $w = (0, 0, 0, 1)$ after orientation and $G_i \cap \mathbf{T} = \mu_{r_i}(0, 0, 0, 1)$. Here W_i is isolated Gorenstein terminal and singular at 0, as the quadratic rank of W_i is 1, hence isolated singular cDV, and h_i is a G_i-semi-invariant polynomial. As in row 4(2), the last sentence of [Liu26b, Theorem 8.5], applied to W_i with (t, x, y, z) in place of (x, u_1, u_2, u_3), to G_i, e_0, the DCC set $\{\frac{t_i}{m}\}_i$, and $\frac{t_i}{m}\operatorname{div}(h_i)$, contradicts the strict increase of α_i.

(2) If $u = 0$ and $w_y w_t \neq 0$, then f_i has $\mathbf{T}$-weight $2w_y \neq 0$ and the monomials z^a have weight 0, so $B_i(z, 0) = 0$ for $i \gg 0$. On the z-axis, $\partial_x f_i = \partial_y f_i = \partial_z f_i = 0$ and $\partial_t f_i = \partial_t B_i(z, 0)$. As $(W_i \ni 0)$ is isolated, B_i has a monomial $z^a t$, so $w_t = 2w_y$, and $w = (1, 1, 0, 2)$ after orientation. By (C.50), f_i is in face I with $n_2 = 1$ and $n_1 \geq 3$, the monomials of f_i^0 other than x^2 and y^2z are $z^j t$, and by Lemma 4.6, taking t times a G_i-invariant polynomial in

z as the new coordinate t, we may assume $f_i^0 = x^2 + y^2 z + z^{n_1} t$. The circle $\mathbf{T}_{(1,1,0,2)}$ is treated in row 7d.

(3) If $u = w_t = 0$, then $w = (1,1,0,0)$, f_i has $\mathbf{T}$-weight 2, and the monomials of $C_i y$ and B_i have weights 1 and 0. Thus $f_i = x^2 + y^2 z$ for $r_i \geq 3$, and W_i is not normal, a contradiction.

Row 7g: $w_t = 0 \neq u$**.** On the t-axis

$$\partial_x f_i = 0, \quad \partial_y f_i = C_i(t), \quad \partial_z f_i = \partial_z B_i(0,t), \quad \text{and} \quad \partial_t f_i = \partial_t B_i(0,t).$$

Since $(W_i \ni 0)$ is isolated, one of $B_i(0,t)$, C_i, and $\partial_z B_i(0,t)$ is non-zero, and comparing $\mathbf{T}$-weights with $2(u + w_y)$ for $i \gg 0$, $w_y = -u$, $w_y = -2u$, or $w_y = 0$, respectively.

(1) If $w_y = 0$, then $w_y \neq -u, -2u$, so $C_i = 0$ and $B_i = z\widetilde{B}_i$ for a polynomial $\widetilde{B}_i$. On $(x = z = 0)$, $\partial_x f_i = \partial_y f_i = \partial_t f_i = 0$ and $\partial_z f_i = y^2 + \widetilde{B}_i(0,t)$. Thus the curve $\{x = z = 0,\ y^2 + \widetilde{B}_i(0,t) = 0\}$ is contained in $\operatorname{Sing} W_i$, which contradicts the isolatedness of $(W_i \ni 0)$.

(2) If $w_y = -u$, then $w = (0,-1,2,0)$ after orientation, and f_i has $\mathbf{T}$-weight 0. Since $t^j y$ has weight -1, $C_i = 0$. Since $z^a t^b$ has weight $2a$, $B_i = B_i(0,t) + B_i'$ with B_i' of degree at least $\frac{r_i}{2}$ in z. Since $C_i = 0$ and $\partial_z B_i'(0,t) = 0$, the isolatedness of $(W_i \ni 0)$ implies $B_i(0,t) \neq 0$, and by (C.50) $B_i(0,t) = t^{n_2} V_i(t)$ with $n_1 = 0$, $n_2 \geq 4$ ($n_2 = 3$ for the cusp circle) and $V_i(0) = 1$. The polynomial V_i is invariant under G_i and $\mathbf{T}$, so by Lemma 4.6, taking $t V_i^{1/n_2}$ as the new coordinate t, we may assume $f_i^0 = x^2 + y^2 z + t^{n_2}$, and any term of $f_i - f_i^0$ has z-degree at least $\frac{r_i}{2}$ and positive shift.

(3) If $w_y = -2u$, then $w = (-1,-2,2,0)$ after orientation, and f_i has $\mathbf{T}$-weight -2. Any $t^j y$ has weight -2, so $C_i(t) y$ belongs to f_i^0. A monomial $z^a t^b$ has weight $2a \equiv -2 \pmod{r_i}$, so $a \geq \frac{r_i}{2} - 1$, and $z^a t^b$ has positive shift as $2a > -2$. Since $B_i(0,t) = \partial_z B_i(0,t) = 0$ for $r_i \geq 6$, the isolatedness implies $C_i \neq 0$, and $C_i = t^{n_3} V_i(t)$ with $n_3 \geq 3$ ($n_3 = 2$ for the tangent circle) and $V_i(0) = 1$. By Lemma 4.6, taking $t V_i^{1/n_3}$ as the new coordinate t, we may assume $f_i^0 = x^2 + y^2 z + y t^{n_3}$.

In (2) the root sections at z and y are the A_{n_2-1} surface $X^2 + Y^2 + T^{n_2}$ and the smooth surface $X^2 + Z + T^{n_2}$. In (3) the root section at z is the A_{2n_3-1} surface $X^2 + Y^2 + YT^{n_3}$, and those at x and y are the smooth surfaces $1 + Y^2 Z + YT^{n_3}$ and $X^2 + Z + T^{n_3}$. In both, $w_y < 0 < w_z$ and any term of $f_i - f_i^0$ has positive shift, and the circles are treated in row 7h.

Row 7h: circles with positive and negative weights. It remains to treat the circles of row 7c, the circles of row 7d with $w_y < 0$, and the circles $\mathbf{T}_{(0,-1,2,0)}$ and $\mathbf{T}_{(-1,-2,2,0)}$ of row 7g. For the circles of row 7d with $w_y < 0$ and of row 7g, any term of $f_i - f_i^0$ has positive shift. In row 7c, $w_z > 0 > w_t$, and in row 7d with $w_y < 0$ and in row 7g, $w_z > 0 > w_y$. Hence $\mathbf{T}$ has a positive weight and a negative weight, $f_i^0 = f^0$ is I or II, and $V_0 = (f^0 = 0)$ is normal and klt. The group G_i acts faithfully on V_0, as V_0 is contained in no coordinate hyperplane, and $h_i^0|_{V_0} \neq 0$, as $h_i^0 \neq 0$ has degree at most 1 in x. Thus (V_0, G_i, h_i^0) with $c := \frac{1}{m}$, $\mathbf{T}$, κ, and e_0 satisfy Set-up 11.1, with $\delta_{\mathbf{T}} = \frac{\kappa}{m} - u - w_t$, and by [Liu26b, Lemma 5.3], $(W_i \ni 0, G_i, h_i)$ is a marked germ over (V_0, h_i^0). As in row 7d, $(V_0, \frac{1}{m}\operatorname{div}(h_i^0))$ is lc near 0. Let $\beta_i := \beta_{\mathbf{T}}$ for (V_0, G_i, h_i^0).

By Proposition 11.4, there exist constants C_0, C_1 with $\alpha_i \leq \beta_i + C_0/r_i + \frac{C_1}{m}(1 - t_i)$ for $i \gg 0$. By Proposition 11.2(5), β_i belongs to a finite set, so possibly passing to a subsequence $\beta := \beta_i$ does not depend on i. Thus $\alpha_\infty \leq \beta$, and $\beta > 0$. By Proposition 11.7, whose hypothesis (iv) holds with the class (c), possibly passing to a subsequence, (1) or (2) of that proposition holds. If (2) holds, then any prime divisor E over W_i with center 0 satisfies $a(E, W_i, \frac{1}{m}\operatorname{div}(h_i))/e_E \geq \beta$, so $b_i = \mathcal{A}_i(P_i) \geq \beta \geq \alpha_\infty$, which contradicts $b_i < \alpha_\infty$.

If (1) holds, there exists $\theta > 1$ with $(W_i, \frac{\theta}{m}\operatorname{div}(h_i))$ lc near 0 for $i \gg 0$. Since V_0 is klt, there exists $\zeta > 0$ with $(V_0, \zeta \operatorname{div}(xyzt))$ lc near 0, and, by [Kaw07, Theorem] and Lemma 2.19 with the coefficients $\{1, \zeta\}$, since $f_i - f^0 \in (x,y,z,t)^L$ for any L and $i \gg 0$, $(W_i, \zeta \operatorname{div}(xyzt))$ is lc near 0

for $i \gg 0$. We show that $\mathcal{A}_i(F) \geq \beta$ for any prime divisor F over W_i with center 0. This implies $b_i \geq \beta \geq \alpha_\infty$, a contradiction. By Proposition 11.2(5), $\beta \leq 2$, so we may assume $\mathcal{A}_i(F) \leq 2$. Let $[\cdot] := [\cdot]_F$. Since $a(F, W_i, 0)/e_F - [h_i]/m \leq 2$ and $a(F, W_i, 0)/e_F - \theta[h_i]/m \geq 0$,

$$\frac{a(F, W_i, 0)}{e_F} \leq \frac{2\theta}{\theta - 1}, \qquad [h_i] \leq \frac{2m}{\theta - 1}, \quad \text{and} \quad [x], [y], [z], [t] \leq C_3 := \frac{2\theta}{(\theta - 1)\zeta}. \tag{C.51}$$

Let $\mathcal{J}_\pm$ be the set of j with $\pm w_j > 0$. For $j \in \mathcal{J}_-$ and $k \in \mathcal{J}_+$, the monomial $\Pi_{j,k} := z_j^{w_k} z_k^{|w_j|}$ is $\mathbf{T}$-invariant, so the character of $\Pi_{j,k}$ on G_i is trivial on R_i, and $\Pi_{j,k}^{e_0!}$ is G_i-invariant and vanishes at 0. The order along F of a G_i-invariant function vanishing at 0 is a positive multiple of e_F, so

$$w_k[z_j] + |w_j|[z_k] \geq \frac{1}{e_0!} \qquad \text{for } j \in \mathcal{J}_-,\ k \in \mathcal{J}_+, \tag{C.52}$$

and $[z_j] \geq 1/e_0!$ if $w_j = 0$. Let $\epsilon_0 > 0$ with $\epsilon_0 < 1/(2Q_{\mathbf{T}})$ and $\epsilon_0 \leq 1/(2e_0! \max_j |w_j|)$. By (C.52), if $[z_j] < \epsilon_0$ for some $j \in \mathcal{J}_\mp$, then $[z_k] \geq \epsilon_0$ for any $k \in \mathcal{J}_\pm$. Let

$$\mathcal{R}_+(F) := \min_{k \in \mathcal{J}_+} \frac{[z_k]}{w_k}, \qquad \mathcal{R}_-(F) := \min_{j \in \mathcal{J}_-} \frac{[z_j]}{|w_j|}.$$

We claim:

(a) If $[z_j] \geq \epsilon_0$ for any $j \in \mathcal{J}_-$, then

$$\mathcal{A}_i(F) \geq \beta^+ - \delta_{\mathbf{T}} \mathcal{R}_+(F).$$

(b) If $[z_k] \geq \epsilon_0$ for any $k \in \mathcal{J}_+$, then

$$\mathcal{A}_i(F) \geq \beta^- + \delta_{\mathbf{T}} \mathcal{R}_-(F).$$

We deduce $\mathcal{A}_i(F) \geq \beta$ from the claim. Assume $\delta_{\mathbf{T}} \leq 0$, so that $\beta = \beta^+$. If $[z_j] \geq \epsilon_0$ for any $j \in \mathcal{J}_-$, then (a) implies $\mathcal{A}_i(F) \geq \beta^+$. Otherwise $[z_j] < \epsilon_0$ for some $j \in \mathcal{J}_-$, so $\mathcal{R}_-(F) < \epsilon_0$ and $[z_k] \geq \epsilon_0$ for any $k \in \mathcal{J}_+$, and (b) and Proposition 11.2(5) imply

$$\mathcal{A}_i(F) \geq \beta^- - |\delta_{\mathbf{T}}|\epsilon_0 \geq \beta^+ + \frac{|\delta_{\mathbf{T}}|}{Q_{\mathbf{T}}} - \frac{|\delta_{\mathbf{T}}|}{2Q_{\mathbf{T}}} \geq \beta.$$

The case $\delta_{\mathbf{T}} > 0$ is symmetric.

For the circles of row 7c, we write f_i in other G_i-semi-invariant coordinates of $\mathbb{C}[[x, y, z, t]]$ with the characters of x, y, z, t, namely those of (C.53) and (C.54) below, and for the circles of row 7c we read z_j, (C.52), $\mathcal{R}_\pm(F)$ and the claim in the new coordinates. In the new coordinates (C.52) holds, as the proof of (C.52) uses only the characters on G_i of the coordinates. By [Liu26a, Lemma 3.3(2)(3)], log discrepancies, orders and inertia groups of prime divisors over W_i with center 0 can be computed in the completion. Let $\mathrm{disc}_i := 4zB_i - C_i^2$. In face I, by [Liu26b, Lemma 5.3], the terms of $C_i y$ and of $B_i - z^{n_1} t^{n_2}$ have degree at least $(r_i - 2|u + w_y|)/\max_j |w_j|$, so the term of least degree of disc_i is $4z^{n_1+1}t^{n_2}$. By Hensel's lemma and the Weierstrass preparation theorem in $\mathbb{C}[[z, t]]$, $\mathrm{disc}_i = 4\mathfrak{u}_i A_i B_{1,i}$ for a unit $\mathfrak{u}_i$ with $\mathfrak{u}_i(0) = 1$, a Weierstrass polynomial A_i in z of degree $n_1 + 1$ and a Weierstrass polynomial $B_{1,i}$ in t of degree n_2. By the uniqueness in Hensel's lemma and in the Weierstrass preparation theorem, A_i and $B_{1,i}$ are G_i-semi-invariant and $\mathfrak{u}_i$ is G_i-invariant. We divide x and y by $\sqrt{\mathfrak{u}_i}$ and f_i by $\mathfrak{u}_i$, and then translate y by a series in z, t so that the coefficient of y depends only on t. The discriminant in y becomes $4A_iB_{1,i}$, so in the new coordinates f_i is a unit times

$$\begin{aligned} &x^2 + zy^2 + C_i(t)y + \frac{A_iB_{1,i} + C_i(t)^2/4}{z}, \qquad A_i(0,t)t^{n_2} = -\frac{C_i(t)^2}{4}, \\ &A_i = z^{n_1+1} + \sum_{j \leq n_1} A_{i,j}(t) z^j, \qquad B_{1,i} = t^{n_2} + \sum_{j < n_2} B_{1,i,j}(z) t^j, \end{aligned} \tag{C.53}$$

with a new series $C_i(t)$ of order at least $\iota + 2$. Since $w_t < 0 < u$, any term of $A_i - z^{n_1+1}$ has negative shift relative to z^{n_1+1}, and any term of $B_{1,i} - t^{n_2}$ has positive shift relative to t^{n_2}. A term

$t^j y$ of $C_i(t)y$ of non-negative shift would satisfy $j|w_t| \le \frac{n_2}{2}|w_t| - (n_1+1)u$, hence $j < \frac{n_2}{2} < \iota + 2$. Thus any term of $C_i(t)y$ has negative shift.

In face II, $\mathrm{disc}_i(0,t) = -C_i(t)^2$ has order $2n_3$, so $\mathrm{disc}_i = -\mathfrak{u}_i B_{1,i}$ for a G_i-invariant unit $\mathfrak{u}_i$ with $\mathfrak{u}_i(0) = 1$ and a Weierstrass polynomial $B_{1,i}$ in t of degree $2n_3$. Dividing x and y by $\sqrt{\mathfrak{u}_i}$ and f_i by $\mathfrak{u}_i$, translating y by a series in z,t so that the coefficient of y depends only on t, and replacing y with $-y$ if necessary, we see that in the new coordinates f_i is a unit times

$$x^2 + zy^2 + t^{n_3}y + \frac{t^{2n_3} - B_{1,i}}{4z}, \qquad B_{1,i} = t^{2n_3} + \sum_{j<2n_3} B_{1,i,j}(z)t^j, \tag{C.54}$$

and any term of $B_{1,i} - t^{2n_3}$ has positive shift relative to t^{2n_3}.

Since the terms of $f_i - f^0$ have degree at least $(r_i - 2|u + w_y|)/\max_j |w_j| \to +\infty$ by [Liu26b, Lemma 5.3], and the factors in Hensel's lemma and in the Weierstrass preparation theorem depend continuously on the series in the (z,t)-adic topology, in faces I and II the coordinate changes are the identity modulo terms of degree at least D_i for integers $D_i \to +\infty$. Hence, in the new coordinates, $h_i - h_i^0$ is the sum of terms of non-zero shift and of terms of degree at least D_i.

We prove (a). The proof of (b) is symmetric, and we list the changes at the end. Let $C_4 := C_3 + 2m/(\theta - 1)$. We find elements $\widetilde{x}, \widetilde{y}$ of the completion of the function field of W_i along F and a series $\widetilde{f}_i(\widetilde{x}, \widetilde{y}, z, t)$ such that $[\widetilde{x} - x], [\widetilde{y} - y] > C_4$, f_i is a unit times $\widetilde{f}_i(\widetilde{x}, \widetilde{y}, z, t)$, and $\widetilde{f}_i - f^0$ is a sum of terms of positive shift.

(1) In face II of row 7c, resp. in row 7d with $w_y < 0$ and in row 7g, $\widetilde{x} := x$, $\widetilde{y} := y$, and $\widetilde{f}_i$ is the series of (C.54), resp. $\widetilde{f}_i := f_i$.

(2) In face I of row 7c, let $\mathfrak{v} := A_i/z^{n_1+1}$,

$$\widetilde{x} := x/\sqrt{\mathfrak{v}}, \qquad \widetilde{y} := (y + C_i/(2z))/\sqrt{\mathfrak{v}}, \quad \text{and} \quad \widetilde{f}_i := \widetilde{x}^2 + z\widetilde{y}^2 + z^{n_1}B_{1,i},$$

where $\sqrt{\mathfrak{v}}$ has residue 1. By (C.53), f_i is a unit times $\mathfrak{v}\widetilde{f}_i$. A term of $A_i - z^{n_1+1}$ or of $C_i(t)y$ has negative shift, so the degree in t of the term is at least $(r_i - c')/|w_t|$ for a constant c' independent of i. Since $[t] \ge \epsilon_0$ and $[z] \le C_3$, $[\mathfrak{v} - 1]$ and $[C_i/z]$ exceed any bound for $i \gg 0$, uniformly in F. Thus $\sqrt{\mathfrak{v}}$ exists, and $[\widetilde{x} - x], [\widetilde{y} - y] > C_4$.

In (2),

$$d\widetilde{y} \wedge dz \wedge dt/(2\widetilde{x}) = dy \wedge dz \wedge dt/(2x),$$

as $\mathfrak{v}$ and C_i/z depend only on z,t. Let $\widetilde{W}_i := (\widetilde{f}_i = 0)$. Since $\widetilde{x}, \widetilde{y}, z, t$ have positive order along F, substitution gives a G_i-equivariant homomorphism from $\mathbb{C}[[\widetilde{x}, \widetilde{y}, z, t]]/(\widetilde{f}_i)$ to the completion of the function field of W_i along F, whose image is dense, as $x = \sqrt{\mathfrak{v}}\widetilde{x}$ and $y = \sqrt{\mathfrak{v}}\widetilde{y} - C_i/(2z)$ (resp. the formulas of (b)) with $\sqrt{\mathfrak{v}}$ and C_i/z in the closure of $\mathbb{C}(z,t)$. The homomorphism is injective by Abhyankar's inequality, as the residue field has transcendence degree 2. Thus the function fields of W_i and of $\widetilde{W}_i$ are dense subfields of one complete discretely valued field, and the identification of completions given by $(x,y,z,t) \to (\widetilde{x}, \widetilde{y}, z, t)$ transforms F into a prime divisor $\widetilde{F}$ over $\widetilde{W}_i$ with $a(\widetilde{F}, \widetilde{W}_i, 0) = a(F, W_i, 0)$, $I_{G_i}(\widetilde{F}) = I_{G_i}(F)$, residue field equal to that of F, and $[g(\widetilde{x}, \widetilde{y}, z, t)]_{\widetilde{F}} = [g(x,y,z,t)]_F$ for any polynomial g with $[g]_F \le C_4$, since all coordinates have non-negative order.

Let $h_i^{\ge}$ be the sum of h_i^0 and of the terms of h_i of positive shift, in the coordinates of (C.53) and (C.54) for faces I and II of row 7c. By the choice of the coordinates of (C.53) and (C.54) and [Liu26b, Lemma 5.3], any term of $h_i - h_i^{\ge}$ has non-positive shift and degree at least $D_i' := \min(D_i, (r_i - |\kappa|)/\max_j |w_j|)$. Such a monomial has $\mathbf{T}$-weight at most κ, so the degree of the monomial in the coordinates z_j with $w_j \le 0$ is at least $(D_i' - |\kappa|)/(\max_j |w_j| + 1)$. As $[z_j] \ge \epsilon_0$ for $j \in \mathcal{J}_-$ and $[z_j] \ge 1/e_0!$ if $w_j = 0$, the order of $h_i - h_i^{\ge}$ exceeds $C_4 \ge [h_i]$ for $i \gg 0$, uniformly in F. Hence $[h_i] = [h_i^{\ge}(\widetilde{x}, \widetilde{y}, z, t)]_{\widetilde{F}}$.

For $N \gg 0$, possibly replacing $\widetilde{f}_i$ and $h_i^{\ge}$ with the sums of their terms of shift at most N and of degree at most N in the coordinates of non-positive weight, and $\widetilde{x}$ with the root near $\widetilde{x}$ of the

truncated equation given by Hensel's lemma ($\partial_{\widetilde{x}}\widetilde{f}_i = 2\widetilde{x}$ and $[\widetilde{x}] \le C_3$), we may assume that $\widetilde{f}_i$ and $h_i^{\ge}$ are polynomials, and symmetrically in (b). The replacement changes neither $\mathcal{A}_i(F)$ nor any order at most C_4 along F, since a term of shift n has order at least $(nr_i - c'')\mathcal{R}_+(F)$ for a constant c'' independent of i, with $r_i\mathcal{R}_+(F) \ge 1/(e_0 \max_j |w_j|)$, and a coordinate of non-positive weight has order at least $\min\{\epsilon_0, 1/e_0!\}$.

Let $k \in \mathcal{J}_+$ attain $\mathcal{R}_+(F)$. As in row 7d, in the root chart at z_k divided by R_i, with radial coordinate ς, $\widetilde{f}_i$ and $h_i^{\ge}$ become $f^0(Z)+\sum_{n\ge1}\varsigma^n f_i^{[n]}(Z)$ and $h_i^0(Z)+\sum_{n\ge1}\varsigma^n h_i^{[n]}(Z)$ with $Z_k = 1$, since they have only terms of non-negative shift. They define a hypersurface $W_{i,k}$, normal and klt near $S_k := W_{i,k} \cap (\varsigma = 0) = V_0 \cap (z_k = 1)$, whose quotient by $\mathbf{T}_k$ is the root section at z_k, and a polynomial $h_{i,k}$, with the action of a finite diagonal group $H'_{i,k}$, which is faithful on $W_{i,k}$ and on S_k, as S_k is contained in no coordinate hyperplane, and no non-identity element of $H'_{i,k}$ acts trivially on an irreducible component of S_k (each as in row 7d). By Proposition 11.2(2)(3), the root-section pair $(S_k, \frac{1}{m}\operatorname{div}(h_i^0|_{S_k}))$ is lc near Φ_k, the preimage in S_k of $o \in Z_0$ (Set-up 11.1). The prime divisor $\widetilde{F}$ induces a prime divisor F_k over $W_{i,k}$ whose center is a closed point or the generic point of a curve ξ of Φ_k, since $[Z_j] \ge 0$ for $j \in \mathcal{J}_+$ and $[Z_j] > 0$ otherwise. Let γ'_i be the coefficient (C.34) at the label 1, as in row 7d. Since $\delta_{\mathbf{T}} = \kappa/m - c_{\mathbf{T}}$,

$$\gamma'_i = 1 - \frac{c_{\mathbf{T}} - \kappa/m}{r_i} = 1 + \frac{\delta_{\mathbf{T}}}{r_i},$$

and

$$\mathcal{A}_i(F) = \frac{a(F_k, W_{i,k}, \gamma'_i S_k + \frac{1}{m}\operatorname{div}(h_{i,k}))}{|I_{H'_{i,k}}(F_k)|}, \qquad \frac{\operatorname{mult}_{F_k}\varsigma}{|I_{H'_{i,k}}(F_k)|} = r_i\mathcal{R}_+(F).$$

By Lemma 11.5 with $c_S = \gamma'_i$,

$$\mathcal{A}_i(F) \ge \beta_0(\xi) + (1-\gamma'_i)r_i\mathcal{R}_+(F) \ge \beta^+ - \delta_{\mathbf{T}}\mathcal{R}_+(F),$$

where $\beta_0(\xi) \ge \beta^+$ is the quotient value at ξ of the root-section pair. This proves (a).

In (b) the roles of $\mathcal{J}_+$ and $\mathcal{J}_-$ are exchanged, the sum $h_i^{\le}$ of h_i^0 and of the terms of h_i of negative shift replaces $h_i^{\ge}$, the coefficient is $1 - \delta_{\mathbf{T}}/r_i$, and $\widetilde{f}_i - f^0$ is a sum of terms of negative shift:

(1) In face I of row 7c, $\mathfrak{v} := B_{1,i}/t^{n_2}$, $\widetilde{x} := x/\sqrt{\mathfrak{v}}$, $\widetilde{y} := (y + C_i/(2z))/\sqrt{\mathfrak{v}} - C_i/(2z)$, and

$$\widetilde{f}_i := \widetilde{x}^2 + z\widetilde{y}^2 + C_i\widetilde{y} + (A_i t^{n_2} + C_i^2/4)/z,$$

which is a power series, without pole along $z = 0$ since $A_i(0,t)t^{n_2} = -C_i(t)^2/4$ by (C.53).

(2) In face II of row 7c, $\mathfrak{v} := B_{1,i}/t^{2n_3}$, $\widetilde{x} := x/\sqrt{\mathfrak{v}}$,

$$\widetilde{y} := (y + t^{n_3}/(2z))/\sqrt{\mathfrak{v}} - t^{n_3}/(2z), \qquad \widetilde{f}_i := f^0(\widetilde{x}, \widetilde{y}, z, t).$$

(3) In row 7d with $w_y < 0$ and in row 7g, let $\mathcal{E}_i := f_i - f^0$, which does not involve x, let $\widetilde{x} := x$, $\widetilde{y} := y$, and $\widetilde{f}_i := f^0$, and replace z with $\widetilde{z} := z + \mathcal{E}_i/y^2$.

In (1) and (2), $[\mathfrak{v} - 1]$ exceeds any bound since $[z] \ge \epsilon_0$ and $[t] \le C_3$, and $[C_i(\mathfrak{v}^{-1/2} - 1)/z]$ exceeds any bound since $[C_i] \ge 0$ and $[z] \le C_3$. In (3), the terms of $\mathcal{E}_i$ have positive shift, so their degrees in the coordinates z_k, $k \in \mathcal{J}_+$, tend to $+\infty$, and since $[y] \le C_3$ by (C.51), $[\widetilde{z} - z]$ exceeds any bound. Moreover $\partial\widetilde{z}/\partial z = 1 + \partial_z\mathcal{E}_i/y^2$ has residue 1 along F, and the residue forms $dx \wedge dy \wedge dt/(y^2 + \partial_z\mathcal{E}_i)$ and $dx \wedge dy \wedge dt/y^2$ differ by the unit $y^2/(y^2 + \partial_z\mathcal{E}_i)$ with residue 1. The rest of the proof of (a) holds without change. This proves (b), and finishes row 7h.

Rows 1–6 and 7a–7h cover Table 1, and each of them contradicts (C.30). This proves Theorem 12.3. □

C.3. **Type cE.**

Proof of Theorem 12.4. We write τ for the variable of the radial functions. If N and η do not exist, then there exist data $(W_i \ni o_i, G_i, h_i)$, $i \geq 2$, satisfying the conditions on $(W \ni o, G, h)$ except $|G| \geq N$, with radial functions q_i, and real numbers $\tau_i' \in (1 - 1/i, 1]$ with

$$|G_i| \geq i \quad \text{and} \quad q_i(\tau_i') - q_i(1) \geq \epsilon.$$

Then $\tau_i' < 1$. By Lemma 4.2(3), $q_i(0) \leq 3$ for any i. By Lemma 4.5 with $s_i := \tau_i'$, possibly passing to a subsequence, there exist $\tau_i \in [\tau_i', 1)$, strictly increasing with limit 1, and, with $\alpha_i := q_i(\tau_i)$, prime divisors E_i over W_i with center o_i satisfying

$$\alpha_i < \alpha_{i+1}, \qquad \lim_i \alpha_i = \alpha_\infty > 0, \quad \text{and} \quad q_i(1) \leq \frac{a(E_i, W_i, \frac{1}{m}\operatorname{div}(h_i))}{e_{E_i}} < \alpha_\infty - \frac{\epsilon}{4}, \tag{C.55}$$

where e_{E_i} is the order of the inertia group of E_i in G_i. Since W_i is terminal and $(W_i, \frac{1}{m}\operatorname{div}(h_i))$ is lc near o_i, the identity

$$a\Big(E, W_i, \frac{\tau_i}{m}\operatorname{div}(h_i)\Big) = (1-\tau_i)a(E, W_i, 0) + \tau_i a\Big(E, W_i, \frac{1}{m}\operatorname{div}(h_i)\Big)$$

shows that $(W_i, \frac{\tau_i}{m}\operatorname{div}(h_i))$ is klt near o_i. The set $\Gamma := \{\tau_i/m\}_i$ is a DCC set, because $(\tau_i)_i$ is increasing.

Polynomial data. For any i, let $(\mathcal{X}_i \ni o_i)$, x, y, z, t, and

$$\widehat{f}_i = x^2 + y^3 + y\theta_i(z,t) + \omega_i(z,t)$$

be given by Lemma 3.7 for $(W_i \ni o_i, G_i, h_i)$, and denote also by h_i a G_i-semi-invariant regular function on $\mathcal{X}_i$ whose restriction to W_i is h_i. Let f', h', and $W' = (f' = 0)$ be defined by Lemma 4.6 with a sufficiently large integer $L > 6$, $n = 4$, $Y = \mathcal{X}_i$, G_i-eigenparameters $\ell_1, \dots, \ell_4$ of $\mathcal{X}_i$ at o_i with the characters of x, y, z, t, the G_i-equivariant automorphism ψ of $\mathbb{C}[[\ell_1, \dots, \ell_4]]$, identified with $\mathbb{C}[[x, y, z, t]]$ through $(\ell_1, \dots, \ell_4) = (x, y, z, t)$, that sends the expansion in $\ell_1, \dots, \ell_4$ of any element of $\widehat{\mathcal{O}}_{\mathcal{X}_i, o_i}$ to its expansion in x, y, z, t, the G_i-invariant unit v with $v\psi(\widehat{f}) = \widehat{f}_i$ for the expansion $\widehat{f}$ of a G_i-semi-invariant equation of W_i in $\mathcal{X}_i$, $u = 1$, the label τ_i, and $\mathcal{P} = \{E_i\}$. Since f' equals the truncation of $\widehat{f}_i$ in degrees less than L, we have $f' \equiv \widehat{f}_i \pmod{(x, y, z, t)^L}$, and the truncation is again of the form $x^2 + y^3 + y\theta'(z,t) + \omega'(z,t)$ in the row of $\widehat{f}_i$. Dividing h' by f', which is monic of degree 2 in x, we may assume that h' is x-linear, that is, $h' = h_0'(y,z,t) + xh_1'(y,z,t)$ with h_0', h_1' zero or G_i-semi-invariant. By Lemma 4.6(1) with $G' = G_i$, $q_i(\tau_i)$ does not change. By Lemma 4.6(2), E_i corresponds to a prime divisor over W' whose log discrepancy with respect to $(W', 0)$, multiplicity of h', and inertia group equal those of E_i, so (C.55) still holds. By Lemma 4.6(3), $(W', \frac{1}{m}\operatorname{div}(h'))$ is lc near 0. Possibly replacing $(W_i \ni o_i, G_i, h_i)$ with $(W' \ni 0, G_i, h')$, we may assume the following.

(1) $W_i = (f_i = 0) \subset \mathbb{A}^4$ with $f_i = x^2 + y^3 + y\theta_i(z,t) + \omega_i(z,t)$ a polynomial in one of the rows (E_6), (E_7), (E_8), and G_i acts diagonally on x, y, z, t.
(2) h_i is an x-linear G_i-semi-invariant polynomial and $(W_i, \frac{1}{m}\operatorname{div}(h_i))$ is lc near 0.
(3) (C.55) holds.

Below we change the formal coordinates x, y, z, t by G_i-equivariant automorphisms ψ of $\mathbb{C}[[x, y, z, t]]$ a few times. After any such change ψ, Lemma 4.6(1)–(3) with ψ, $u = v = 1$, and $G' = G_i$, together with the division by f', show that (2), (3) above, and the part $f_i - x^2 \in \mathbb{C}[y, z, t]$ of (1) above continue to hold for the polynomial truncations, provided that x is not changed and the corrections of y, z, t do not involve x. After a diagonal scaling of z or of t, or a replacement of t by t times a G_i-invariant unit of $\mathbb{C}[[t]]$, (1) itself continues to hold.

The circle. Let $\theta_{i,3}$ and $\omega_{i,j}$ be the homogeneous parts of degree 3 of θ_i and of degree j of ω_i, and let Λ_i be $\omega_{i,4}$, $\theta_{i,3}$, $\omega_{i,5}$ in the rows (E_6), (E_7), (E_8), respectively. Possibly passing to a subsequence, the row and the set of monomials of Λ_i do not depend on i. Fix a monomial $z^{a_1}t^{a_2}$ of Λ_i, with $a_1 + a_2 = 4, 3, 5$ in the three rows. Any $g \in G_i$ acts on x, y, z, t by multiplication by

$(\lambda_x,\lambda_y,\lambda_z,\lambda_t)\in(S^1)^4$. Since x^2, y^3 and $z^{a_1}t^{a_2}$, resp. $yz^{a_1}t^{a_2}$, are monomials of the semi-invariant f_i, the group G_i is contained in the compact subgroup $\mathcal{S}\subset(S^1)^4$ defined by

$$\lambda_x^2=\lambda_y^3=\lambda_z^{a_1}\lambda_t^{a_2}\ \text{ in the rows (E}_6\text{), (E}_8\text{)},\qquad \lambda_x^2=\lambda_y^3,\ \lambda_y^2=\lambda_z^{a_1}\lambda_t^{a_2}\ \text{ in the row (E}_7\text{)}.$$

The identity component $\mathcal{S}^0$ is a 2-dimensional torus, and the Lie algebra of $\mathcal{S}^0$ contains the positive vector

$$\mathbf{p}=(6,4,3,3),\quad (9,6,4,4),\quad\text{and}\quad (15,10,6,6)$$

in the rows (E$_6$), (E$_7$), (E$_8$), respectively, where a vector $w\in\mathbb{R}^4$ stands for the one-parameter subgroup $\sigma\to(e^{2\pi\sqrt{-1}\sigma w_j})_j$. By [Liu26b, Lemma 4.6], possibly passing to a subsequence, the G_i converge in the Hausdorff topology to a closed subgroup $\mathcal{H}\subset\mathcal{S}$, and $G_i\subset\mathcal{H}$ for any $i\gg0$. Possibly truncating the sequence, $G_i\subset\mathcal{H}$ for any i.

If $\dim\mathcal{H}=0$, then $\mathcal{H}$ is finite and $|G_i|\le|\mathcal{H}|$, contradicting $|G_i|\ge i$.

If $\dim\mathcal{H}=2$, then $\mathcal{H}^0=\mathcal{S}^0$. Fix $\sigma\in(0,\frac{1}{2\max_j p_j})$ with $\sigma\sum_j p_j<\alpha_\infty/2$. By Hausdorff convergence there exist $g_i\in G_i$ acting on x,y,z,t by $(e^{2\pi\sqrt{-1}u_{i,j}})_j$ with $u_i:=(u_{i,j})_j\in(0,1)^4$ and $\lim_i u_i=\sigma\mathbf{p}$. Then u_i belongs to the lattice N_{G_i} (Definition 2.3), and $u_i=k_iv_i$ for a primitive vector $v_i\in N_{G_i}$ and a positive integer k_i. The toric prime divisor E_{v_i} of $\mathbb{A}^4/G_i$ given by v_i has center the image of 0, and by [Liu26a, Set-up 8.4 and Lemma 8.5] (if $h_i(0)\ne0$, we apply it without h_i, since then $\mathrm{wt}_{v_i}(h_i)=0$ and $\mathrm{div}(h_i)$ does not pass through 0) the log discrepancy of E_{v_i} with respect to the quotient of $(\mathbb{A}^4,W_i+\frac{\tau_i}{m}\mathrm{div}(h_i))$ is

$$\sum_j v_{i,j}-\mathrm{wt}_{v_i}(f_i)-\frac{\tau_i}{m}\mathrm{wt}_{v_i}(h_i)\le\sum_j u_{i,j}.$$

By Theorem 2.17(1) with $n=4$, $c=1$, $H=G_i$, and $\mathfrak{a}=(h_i)^{\tau_i/m}$, α_i equals the quotient value of $(\mathbb{A}^4\ni0,W_i+\frac{\tau_i}{m}\mathrm{div}(h_i))$ and G_i. Hence $\alpha_i\le\sum_j u_{i,j}$, whose limit is $\sigma\sum_j p_j<\alpha_\infty/2$, contradicting (C.55).

Since $\dim\mathcal{H}\le\dim\mathcal{S}=2$, $\mathcal{H}^0$ is a circle $\mathbf{T}=\mathbf{T}_{(w)}$ with a primitive weight vector

$$w=(w_x,w_y,w_z,w_t)\in\mathbb{Z}^4.$$

The relation $\lambda_x^2=\lambda_y^3$ implies $2w_x=3w_y$, so $w=(3u,2u,w_z,w_t)$ for an integer u, and the remaining relation reads

$$6u=a_1w_z+a_2w_t\ \text{ in the rows (E}_6\text{), (E}_8\text{)},\qquad 4u=a_1w_z+a_2w_t\ \text{ in the row (E}_7\text{)}.\tag{C.56}$$

Let $R_i:=G_i\cap\mathbf{T}$, $r_i:=|R_i|$, and $e_0:=|\mathcal{H}/\mathcal{H}^0|$. Then

$$[G_i:R_i]\le e_0\quad\text{and}\quad r_i\to+\infty.\tag{C.57}$$

The weights w_x,w_y,w_z,w_t are not all of one strict sign. Otherwise, possibly replacing w with $-w$, $w\in\mathbb{Z}^4_{>0}$. The argument in the case $\dim\mathcal{H}=2$ shows $\alpha_i\le\sum_j u_{i,j}$ for any $g_i\in G_i$ acting by $(e^{2\pi\sqrt{-1}u_{i,j}})_j$ with $u_i\in(0,1)^4$. Taking for g_i the generator of R_i acting by $(e^{2\pi\sqrt{-1}w_j/r_i})_j$, that is, $u_i=w/r_i$, where $r_i>\max_j w_j$, we obtain $\alpha_i\le\sum_j w_j/r_i\to0$, contradicting (C.55). In particular $w_z\ne w_t$: if $w_z=w_t$, then (C.56) implies $w=\frac{u}{2}(6,4,3,3)$, $\frac{u}{3}(9,6,4,4)$, $\frac{u}{5}(15,10,6,6)$ in the three rows, with $u\ne0$, so that all four weights have the sign of u. For any other monomial $z^{a_1'}t^{a_2'}$ of Λ_i, the character $\lambda_z^{a_1-a_1'}\lambda_t^{a_2-a_2'}$ is trivial on G_i for any i, hence on $\mathcal{H}\supset\mathbf{T}$, so $(a_1-a_1')w_z+(a_2-a_2')w_t=0$. Since $a_1+a_2=a_1'+a_2'$, we obtain $(a_1-a_1')(w_z-w_t)=0$, hence $a_1'=a_1$. Thus Λ_i is a non-zero multiple of $z^{a_1}t^{a_2}$. Since the diagonal scalings of z and of t commute with G_i, scaling a coordinate with positive exponent in Λ_i, we may assume that $\Lambda_i=z^{a_1}t^{a_2}$.

A pure power. Assume that $a_1=0$ or $a_2=0$. Possibly interchanging z and t, we may assume that $a_2=0$, so $z^{a_1}=z^4$, z^3, z^5 and the lowest weighted part of $f_i(x,y,z,0)$ for the weights $\mathbf{p}'$, the first three coordinates of $\mathbf{p}$, is

$$f_0:=x^2+y^3+z^4,\quad x^2+y^3+yz^3,\quad\text{and}\quad x^2+y^3+z^5$$

in the three rows, of weighted degree $d = 12, 18, 30$, respectively. Any other monomial of $f_i(x, y, z, 0)$ has larger weighted degree. By [Kol98, Construction 2.6], a G_i-equivariant automorphism of $\mathbb{C}[[x, y, z]]$, extended to $\mathbb{C}[[x, y, z, t]]$ by $t \to t$, transforms $f_i(x, y, z, 0)$ into f_0, since every monomial in x, y, z of weighted degree larger than d belongs to the ideal generated by the partial derivatives of f_0. The automorphism can be taken G_i-equivariant, since f_0 is a sum of monomials of f_i, so G_i acts on f_0 by the character of f_i, and each coordinate correction may be projected to the character of the coordinate it corrects. After the change of coordinates, the expansion $\widehat{f_i}$ of f_i satisfies $\widehat{f_i}(x, y, z, 0) = f_0$, and the quadratic part of $\widehat{f_i}$ is x^2. By (C.56) with $a_2 = 0$, $(3u, 2u, w_z)$ is proportional to $\mathbf{p}'$, so $w = (c_0\mathbf{p}', v_0)$ for coprime integers c_0, v_0. We distinguish four cases.

- If $c_0 = 0$, then $\mathbf{T} = \mathbf{T}_{(0,0,0,1)}$ and $R_i = \mu_{r_i}(0, 0, 0, 1)$. By [Liu26b, Theorem 8.5] with t in the role of x, the labels in Γ, the index bound e_0 of (C.57), and the klt pairs $(W_i, \frac{\tau_i}{m}\operatorname{div}(h_i))$, the numbers α_i do not form a strictly increasing sequence. This contradicts (C.55).
- If $c_0v_0 > 0$, then all four weights have one strict sign, which we have excluded.
- If $v_0 = 0$, then $w = \pm(\mathbf{p}', 0)$. By [AGZV85, Section 8.3], $\widehat{f_i}$ is right equivalent to

$$f_0 + \sum_j \varphi_j(t) M_j(y, z), \qquad \varphi_j \in t\mathbb{C}[[t]],$$

  where the M_j form a weighted homogeneous monomial basis of $\mathbb{C}[y, z]/(\partial_y(f_0 - x^2), \partial_z(f_0 - x^2))$, of weighted degrees $d_j \le d - 2$. The equivalence may be taken G_i-equivariant, because each correction of the coordinates may be projected to the character of the coordinate, and each correction term $\varphi_j(t)M_j$ to the character of f_i. Since t has weight 0, the generator of R_i acts on a non-zero term $\varphi_j(t)M_j$ and on f_0 by $e^{\pm 2\pi\sqrt{-1}d_j/r_i}$ and $e^{\pm 2\pi\sqrt{-1}d/r_i}$, respectively. Hence $r_i \mid d - d_j$, and $2 \le d - d_j \le d$. Thus for $r_i > d$ all φ_j vanish and $\widehat{f_i} = f_0$ does not depend on t, contradicting that W_i has an isolated singularity at 0.
- If $c_0v_0 < 0$, then possibly replacing w with $-w$, we may assume $c_0 < 0 < v_0$, so $\mathbf{T} = \mathbf{T}_{(c_0\mathbf{p}', v_0)}$. The data $(W_i \ni 0, G_i, \frac{\tau_i}{m}\operatorname{div}(h_i))$ are of kind (2) of [Liu26b, Theorem 10.4] with $c = -c_0$, $v = v_0$, the type E_6, E_7, E_8, the labels in Γ, the index bound e_0 of (C.57), and $r_i \to +\infty$. By [Liu26b, Theorem 10.4], the positive numbers α_i are not strictly increasing, contradicting (C.55).

Therefore $a_1, a_2 \ge 1$.

A zero weight on z or t. Assume that $w_zw_t = 0$. Possibly interchanging z and t, we may assume that $w_t = 0$. Then $w_z \ne 0$ and, by (C.56), $u \ne 0$. Any monomial of f_i has $\mathbf{T}$-weight congruent to $6u$ modulo r_i, because f_i is R_i-semi-invariant with the character of x^2. For $r_i > 6|u|$, no power of t has $\mathbf{T}$-weight congruent to $6u$ modulo r_i, so the t-axis is contained in W_i. Since W_i is smooth at the generic point of the t-axis, f_i contains a monomial xt^k, yt^k, or zt^k. Their weights $3u$, $2u$, w_z are congruent to $6u$ modulo $r_i \to +\infty$, so only the last is possible, and $w_z = 6u$. Since w is primitive, possibly replacing w with $-w$, we obtain

$$w = (3, 2, 6, 0).$$

In the row (E_7), (C.56) reads $4 = 6a_1$, which is impossible. In the rows (E_6), (E_8), it reads $6 = 6a_1$, so $z^{a_1}t^{a_2} = zt^{a_2}$ with $a_2 = 3, 4$, respectively.

The case $w = (3, 2, 6, 0)$. The monomials of f_i of $\mathbf{T}$-weight 6 are x^2, y^3 and the monomials zt^k of ω_i, and $k \ge a_2$ for these. Hence the weight-6 part of f_i is $x^2 + y^3 + zt^{a_2}\nu_i(t)$ for a power series ν_i with $\nu_i(0) = 1$. Since $zt^{a_2}\nu_i(t)$ is G_i-semi-invariant, any monomial of ν_i is G_i-invariant, so $t\nu_i(t)^{1/a_2}$ is a G_i-semi-invariant parameter with the character of t. Possibly replacing t with $t\nu_i(t)^{1/a_2}$, we may assume that $\nu_i = 1$. All weights are non-negative and all monomials of f_i have

weight congruent to 6 modulo r_i. Thus, for $r_i > 6$,

$$f_i = F_0 + \sum_{j\ge 1} f_{i,j}, \qquad F_0 := x^2 + y^3 + zt^{a_2}, \tag{C.58}$$

where $f_{i,j}$ is $\mathbf{T}$-homogeneous of weight $6 + jr_i$. Let κ_i be the least $\mathbf{T}$-weight of a monomial of h_i. The prime divisor E_w over W_i given by the monomial valuation with weights w has lowest part F_0 on f_i, which is irreducible, and has center the t-axis, which contains 0. By [Liu26b, Lemma 4.12] over the field $\mathbb{C}(t)$ with $n = 3$ and the weights $(3, 2, 6)$ on x, y, z, at the generic point of the t-axis,

$$a(E_w, W_i, 0) = 3 + 2 + 6 - 6 = 5, \qquad \operatorname{mult}_{E_w} h_i = \kappa_i,$$

the equality holding because the lowest part of the x-linear h_i is not divisible by F_0. Since $(W_i, \frac{1}{m}\operatorname{div}(h_i))$ is lc near 0, $0 \le \kappa_i \le 5m$. Possibly passing to a subsequence, $\kappa_i = \kappa$ for any i. For $r_i > 5m$, any monomial of h_i has weight $\kappa + jr_i$ with $j \ge 0$, so

$$h_i = h_i^0 + \sum_{j\ge 1} h_{i,j},$$

where $h_i^0 \ne 0$, and h_i^0 and $h_{i,j}$ are $\mathbf{T}$-homogeneous of weights κ and $\kappa + jr_i$, respectively.

Let $\pi_Y : Y \to \mathbb{A}^4$ be the weighted blow-up of the t-axis with weights $(3, 2, 6)$ on x, y, z, that is, the toric morphism given by the star subdivision at w, with exceptional prime divisor E_Y. The charts of x and of y are the quotients of $\mathbb{A}^4_{\varrho,y_1,z_1,t}$ by μ_3 and of $\mathbb{A}^4_{\varrho,x_1,z_1,t}$ by μ_2 given by $x = \varrho^3$, $y = \varrho^2 y_1$, $z = \varrho^6 z_1$, and by $x = \varrho^3 x_1$, $y = \varrho^2$, $z = \varrho^6 z_1$, respectively. We describe the chart of z. Let $\mathbb{A}^4_{\varrho,x_1,y_1,t} \to Y$ be the quotient by μ_6 given by

$$x = \varrho^3 x_1, \qquad y = \varrho^2 y_1, \quad \text{and} \quad z = \varrho^6,$$

where $\zeta \in \mu_6$ acts by

$$(\varrho, x_1, y_1, t) \to (\zeta\varrho, \zeta^{-3} x_1, \zeta^{-2} y_1, t).$$

The circle $\mathbf{T}$ acts on $\mathbb{A}^4_{\varrho,x_1,y_1,t}$ by weight 1 on ϱ and weight 0 on x_1, y_1, t, compatibly with the chart map. Let $\varpi := \varrho^{r_i}$, so that $\mathbb{A}^4_{\varpi,x_1,y_1,t}$ is the quotient of the chart by R_i, and let H_i be the group of lifts of G_i to the chart modulo R_i, so that $|H_i| \le 6[G_i : R_i] \le 6e_0$ and H_i acts diagonally on ϖ, x_1, y_1, t. By (C.58),

$$\begin{gathered} f_i = \varrho^6\Big(F_0' + \sum_{j\ge 1} \varpi^j f_{i,j}'\Big), \qquad h_i = \varrho^\kappa\Big(h_i^{0\prime} + \sum_{j\ge 1} \varpi^j h_{i,j}'\Big), \\ F_0' = x_1^2 + y_1^3 + t^{a_2}, \end{gathered} \tag{C.59}$$

where F_0', $f_{i,j}'$, $h_i^{0\prime}$, and $h_{i,j}'$ are polynomials in x_1, y_1, t. Let $X_i := (f_i' = 0) \subset \mathbb{A}^4_{\varpi,x_1,y_1,t}$ with $f_i' := F_0' + \sum_{j\ge 1} \varpi^j f_{i,j}'$, and let $S := X_i \cap (\varpi = 0) = (F_0' = 0)$, a surface of type D_4 or E_6 at the origin for $a_2 = 3$ or 4, and smooth elsewhere near $(t = 0)$. In the charts of x and of y, the corresponding surfaces $(1 + y_1^3 + z_1 t^{a_2} = 0)$ and $(x_1^2 + 1 + z_1 t^{a_2} = 0)$ are smooth near $(t = 0)$, since $y_1 \ne 0$, resp. $x_1 \ne 0$, there. Since $\operatorname{mult}_{E_Y}$ of x, y, z is $3, 2, 6$ and E_Y is $(\varrho = 0)$, for $\tau \in [0, 1]$, the crepant pull-back of $(\mathbb{A}^4, W_i + \frac{\tau}{m}\operatorname{div}(h_i))$ to the chart, divided by R_i, is $(\mathbb{A}^4_{\varpi,x_1,y_1,t}, X_i + \Delta_i^\tau)$ with the group H_i, where

$$\begin{aligned} \Delta_i^\tau &:= \frac{\tau}{m}\operatorname{div}(h_i') + \Big(1 - \frac{\gamma(\tau)}{r_i}\Big)\operatorname{div}(\varpi), \\ \gamma(\tau) &:= 5 - \frac{\tau\kappa}{m}, \quad \text{and} \quad h_i' := h_i^{0\prime} + \sum_{j\ge 1} \varpi^j h_{i,j}', \end{aligned} \tag{C.60}$$

and, restricting to X_i, the crepant pull-back of $(W_i, \frac{\tau}{m}\operatorname{div}(h_i))$ divided by R_i is $(X_i, \Delta_i^\tau|_{X_i})$, which we also write (X_i, Δ_i^τ). The coefficient of $\operatorname{div}(\varpi)$ in (C.60) follows from

$$a\Big(E_Y, \mathbb{A}^4, W_i + \frac{\tau}{m}\operatorname{div}(h_i)\Big) = 3 + 2 + 6 - 6 - \frac{\tau\kappa}{m} = \gamma(\tau),$$

since the μ_6-cover is not ramified along $(\varrho = 0)$, because μ_6 acts on x_1, y_1 by ζ^{-3}, ζ^{-2}, and the cover $\varrho \to \varpi = \varrho^{r_i}$ is ramified along $(\varrho = 0)$ with index r_i. An element of H_i acting trivially on S acts trivially on x_1, y_1, t, so it is the class of a lift of some $g \in G_i$ acting on x, y, z, t by an element of $\mathbf{T}$, hence $g \in R_i$ and the class is trivial. Thus H_i acts faithfully on S and on X_i. Since S is a normal Cartier divisor of X_i with klt singularities, X_i is normal near S and, by inversion of adjunction, klt near S.

The preimage of $0 \in \mathbb{A}^4$ in Y intersects $(\pi_Y)_*^{-1} W_i$ in the curve $\Sigma := (x^2 + y^3 = 0) \subset \mathbb{P}(3, 2, 6)$, which is $(\varpi = t = 0) \cap S$ in the charts. Since π_Y is proper, any prime divisor E over W_i with center 0 has center on the charts a (not necessarily closed) point of Σ. By [Liu26a, Lemma 2.6(1)(2)], for any such E,

$$\frac{a(E, W_i, \frac{\tau}{m} \operatorname{div}(h_i))}{e_E} = \frac{a(E, X_i, \Delta_i^\tau)}{|I_{H_i}(E)|}. \tag{C.61}$$

We show that the pairs $(S, \frac{1}{m} \operatorname{div}(\overline{h}_i))$, $\overline{h}_i := h_i^{0\prime}|_S$, are lc near Σ for any $i \gg 0$, and similarly in the charts of x and of y. The pair (X_i, Δ_i^1) is lc near Σ, since it is crepant to $(W_i, \frac{1}{m} \operatorname{div}(h_i))$ over a neighborhood of 0. By inversion of adjunction [Kaw07, Theorem], the pair

$$\Big(\mathbb{A}^4,\ B_i + \Big(1 - \frac{\gamma(1)}{r_i}\Big) \operatorname{div}(\varpi)\Big), \qquad B_i := X_i + \frac{1}{m} \operatorname{div}(h_i'),$$

is lc near Σ. The exponents $1, \frac{1}{m}, 1$ are fixed. If $(\mathbb{A}^4, B_i + \operatorname{div}(\varpi))$ were not lc at a point of Σ for infinitely many i, then the lc thresholds of ϖ with respect to $(\mathbb{A}^4, B_i)$ at the points of Σ would be less than 1 and at least $1 - \gamma(1)/r_i \to 1$, contradicting Theorem 2.23(1). Possibly truncating the sequence, $(\mathbb{A}^4, B_i + \operatorname{div}(\varpi))$ is lc near Σ, and by adjunction [Kaw21, Theorem 2.5(ii)] twice, $(S, \frac{1}{m} \operatorname{div}(\overline{h}_i))$ is lc near Σ.

Let β_i' be the infimum of the quotient values of $(S, \frac{1}{m} \operatorname{div}(\overline{h}_i))$ and H_i, in all three charts, at the images of the (not necessarily closed) points of Σ. The surfaces S are Gorenstein, so the log discrepancies of $(S, \frac{1}{m} \operatorname{div}(\overline{h}_i))$ belong to $\frac{1}{m}\mathbb{N}$, and by [Liu26a, Lemma 2.6(1)(2)] the quotient values belong to $\frac{1}{m \cdot (6e_0)!}\mathbb{N}$. At the generic point of Σ, the only prime divisor over S with center Σ is Σ, and the quotient value at the image of Σ equals $1 - c_\Sigma \le 1$, where $c_\Sigma \ge 0$ is the coefficient of the image of Σ in the quotient pair. Hence

$$\beta_i' \in \frac{1}{m \cdot (6e_0)!}\mathbb{Z} \cap [0, 1],$$

and possibly passing to a subsequence, $\beta_i' = \beta'$ for any i. At a closed point $\xi \in \Sigma$ other than the origin of a chart, we translate the coordinates to move ξ to the origin and replace H_i with the stabilizer of ξ. The stabilizer acts diagonally on the translated coordinates, since a coordinate not vanishing at ξ has a character trivial on the stabilizer. For a prime divisor E with center ξ, the group $\operatorname{Stab}_{H_i}(E)$ fixes the center ξ of E, so it equals the stabilizer of E in the stabilizer of ξ, and $I_{H_i}(E)$ equals the inertia group of E in the stabilizer of ξ. Hence by [Liu26a, Definition 2.5] the quotient values at ξ and the ratios $a(E, X_i, \Delta_i^\tau)/|I_{H_i}(E)|$ for prime divisors E with center ξ do not change. Below we make such a translation at any closed point of Σ without further mention.

Lower bound. Let E be a prime divisor over W_i with center 0, and let ξ be the center of E on one of the charts, a (not necessarily closed) point of Σ. By (C.61), by Lemma 11.5 with $X = X_i$, $H = H_i$, $u = \varpi$, $S_0 = S$, $\mathfrak{a} = (h_i')^{1/m}$, and $c_S = 1 - \gamma(1)/r_i$, by the definition of β_i', and since $\gamma(1) \ge 0$,

$$\frac{a(E, W_i, \frac{1}{m} \operatorname{div}(h_i))}{e_E} = \frac{a(E, X_i, \Delta_i^1)}{|I_{H_i}(E)|} \ge \beta_i' + \frac{\gamma(1)}{r_i} \cdot \frac{\operatorname{mult}_E \varpi}{|I_{H_i}(E)|} \ge \beta'.$$

Applied to $E = E_i$ of (C.55), this implies $\beta' < \alpha_\infty - \epsilon/4$.

Upper bound. Possibly passing to a subsequence, the actions of H_i on the coordinates of each chart do not depend on i, since $|H_i| \le 6e_0$. Write H for the group H_i in the chart of z, acting on $\mathbb{A}^3_{x_1, y_1, t} \supset S$, and similarly in the charts of x and of y. The quotient of the pair

$(\mathbb{A}^3, S+\frac{1}{m}\operatorname{div}(h_i^{0\prime}))$ by H is a pair on the fixed klt threefold $\mathbb{A}^3/H$ whose coefficients are of the form $1-(1-b)/e$ with $b\in\frac{1}{m}\mathbb{Z}\cap[0,1]$ by lc and $e\le 6e_0$, so they belong to a finite set Γ' independent of i. Since the quotient values belong to a discrete set, there exists a (not necessarily closed) point $\xi_i\in\Sigma$ in one of the charts at which the infimum β' is attained. If ξ_i is the generic point of Σ, we take the chart of z, where $\Sigma=(x_1^2+y_1^3=t=0)$ is an H-invariant integral curve through the origin. Theorem 2.17(1)(3) with $n=3$, $c=1$, and $X=S$ implies that β' equals the quotient value at ξ_i of $(\mathbb{A}^3, S+\frac{1}{m}\operatorname{div}(h_i^{0\prime}))$ and H. Let C_2 be the maximum of the integers L of Theorem 2.22(3) for the fixed klt threefolds $\mathbb{A}^3/H$ of the three charts and the set Γ'. By Theorem 2.22(3) on $\mathbb{A}^3/H$ with the coefficient set Γ', there exists a prime divisor over $\mathbb{A}^3/H$ computing the quotient value at the image of ξ_i with log discrepancy at most C_2 with respect to $(\mathbb{A}^3/H,0)$. Let $\widetilde{E}_i$ be a prime divisor over $\mathbb{A}^3$ above it. Then $a(\widetilde{E}_i,\mathbb{A}^3,0)\le 6e_0C_2$, since the branch divisor B_H of $\mathbb{A}^3\to\mathbb{A}^3/H$ satisfies $B_H\ge 0$. Since $(\mathbb{A}^3,\operatorname{div}(x_1))$, $(\mathbb{A}^3,\operatorname{div}(y_1))$, $(\mathbb{A}^3,\operatorname{div}(t))$, and $(\mathbb{A}^3, S+\frac{1}{m}\operatorname{div}(h_i^{0\prime}))$ are lc near ξ_i, we obtain

$$\operatorname{mult}_{\widetilde{E}_i} F_0',\ \operatorname{mult}_{\widetilde{E}_i} h_i^{0\prime},\ \operatorname{mult}_{\widetilde{E}_i} x_1,\ \operatorname{mult}_{\widetilde{E}_i} y_1,\ \operatorname{mult}_{\widetilde{E}_i} t\le C_3:=6me_0C_2.$$

By [Liu26b, Lemma 3.1], whose proof uses only the generic point of $\widetilde{E}_i$ and so covers also the case that ξ_i is the generic point of Σ, there exists a residue class modulo $|I_H(\widetilde{E}_i)|$ such that for any positive integer k in it, the Gauss extension E_i' of $\widetilde{E}_i$ to $\mathbb{A}^4_{\varpi,x_1,y_1,t}$ with value k on ϖ (Definition 2.3) is a prime divisor with $I_H(E_i')=I_H(\widetilde{E}_i)$. Since $|I_H(\widetilde{E}_i)|\le|H|\le 6e_0$, we take such an integer k in the interval $(C_3, C_3+6e_0]$. Then $\operatorname{mult}_{E_i'}\varpi=k$, and for any polynomial $P=\sum_{j\ge0}\varpi^j P_j$ with $P_j\in\mathbb{C}[x_1,y_1,t]$ and $\operatorname{mult}_{\widetilde{E}_i}P_0\le C_3$, we have $\operatorname{mult}_{E_i'}P=\operatorname{mult}_{\widetilde{E}_i}P_0$. Thus $\operatorname{mult}_{E_i'}f_i'=\operatorname{mult}_{\widetilde{E}_i}F_0'$ and $\operatorname{mult}_{E_i'}h_i'=\operatorname{mult}_{\widetilde{E}_i}h_i^{0\prime}$, and since $a(E_i',\mathbb{A}^4,0)=a(\widetilde{E}_i,\mathbb{A}^3,0)+k$,

$$\begin{aligned}a\big(E_i',\mathbb{A}^4,X_i+\Delta_i^{\tau_i}\big)&=a\Big(\widetilde{E}_i,\mathbb{A}^3,S+\frac{\tau_i}{m}\operatorname{div}(h_i^{0\prime})\Big)+\frac{k\gamma(\tau_i)}{r_i}\\&\le a\Big(\widetilde{E}_i,\mathbb{A}^3,S+\frac{1}{m}\operatorname{div}(h_i^{0\prime})\Big)+\frac{(1-\tau_i)C_3}{m}+\frac{5k}{r_i}.\end{aligned}$$

The center of E_i' on $\mathbb{A}^4$ is ξ_i, and any prime divisor over X_i with center ξ_i has center 0 over W_i. By Theorem 2.17(1)(3) with $n=4$, $c=1$, $X=X_i$, and the klt pair $(X_i,\Delta_i^{\tau_i})$, and by [Liu26a, Lemma 2.6(1)(2)(3)], by which the image on the common quotient X_i/H_i, a variety birational to W_i/G_i, of any prime divisor over X_i with center ξ_i is the image of a prime divisor over W_i with center 0, dividing by $|I_H(E_i')|=|I_H(\widetilde{E}_i)|\ge1$, and using

$$a\Big(\widetilde{E}_i,\mathbb{A}^3,S+\frac{1}{m}\operatorname{div}(h_i^{0\prime})\Big)=\beta'|I_H(\widetilde{E}_i)|,$$

we obtain

$$\alpha_i\le\beta'+\frac{(1-\tau_i)C_3}{m}+\frac{5(C_3+6e_0)}{r_i}.\tag{C.62}$$

Letting $i\to+\infty$, (C.62) implies $\alpha_\infty\le\beta'$, which contradicts $\beta'<\alpha_\infty-\epsilon/4$. Thus $w_z,w_t\ne0$.

Mixed circles. We have $a_1,a_2\ge1$ and $w_z,w_t\ne0$. If w_z and w_t had one strict sign, then by (C.56) u would have the sign of w_z, and all four weights would have one strict sign, which we have excluded. Possibly replacing w with $-w$ and interchanging z and t, we may assume

$$w=(3u,2u,w_z,w_t),\qquad u\ge0,\quad\text{and}\quad w_z<0<w_t,\tag{C.63}$$

where $u=0$ is allowed. Let

$$F_0:=x^2+y^3+z^{a_1}t^{a_2}\ \text{ in the rows (E$_6$), (E$_8$) (for $a_1=1$, this is the polynomial F_0 of (C.58))},$$
$$F_0:=x^2+y^3+y\Lambda_i\ \text{ in the row (E$_7$)},$$

let $d:=12,18,30$ be the $\mathbf{p}$-degree of F_0 in the three rows, and let $J(F_0)\subset\mathbb{C}[x,y,z,t]$ be the ideal generated by the partial derivatives of F_0. Then F_0 is $\mathbf{T}$-homogeneous of weight $6u$

by (C.56), and any other monomial of f_i of $\mathbf{T}$-weight $6u$ has $\mathbf{p}$-degree larger than d, because $\Lambda_i = z^{a_1}t^{a_2}$, $\theta_i \in (z,t)^3$, $\omega_i \in (z,t)^4$, $\omega_{i,4} = 0$ in the rows (E_7), (E_8), and $\theta_{i,3} = 0$ in the row (E_8) (Lemma 3.7(2)).

We claim that any monomial $\mathfrak{n}$ of $\mathbf{T}$-weight $6u$ and $\mathbf{p}$-degree larger than d belongs to $J(F_0)$. If $x \mid \mathfrak{n}$, use $\partial_x F_0 = 2x$. In the rows (E_6), (E_8), if $y^2 \mid \mathfrak{n}$, use $\partial_y F_0 = 3y^2$. If $\mathfrak{n} = z^{p_1}t^{p_2}$, then $\mathfrak{n}$ and $z^{a_1}t^{a_2}$ both have $\mathbf{T}$-weight $6u$, so $(p_1 - a_1)w_z + (p_2 - a_2)w_t = 0$, and $p_1 + p_2 > a_1 + a_2$. Hence $p_1 > a_1$ and $p_2 > a_2$, and $\mathfrak{n}$ is a multiple of $z^{a_1}t^{a_2} = a_1^{-1}z\partial_z F_0$. If $\mathfrak{n} = yz^{p_1}t^{p_2}$, the $\mathbf{T}$-weights imply $(3p_1 - 2a_1)w_z + (3p_2 - 2a_2)w_t = 0$, and the $\mathbf{p}$-degrees imply $3(p_1 + p_2) > 2(a_1 + a_2)$. Hence $3p_1 > 2a_1$ and $3p_2 > 2a_2$. For $(a_1, a_2) \neq (4,1), (1,4)$ this implies $p_1 \geq a_1$ and $p_2 \geq a_2$. For $(a_1, a_2) = (4,1)$, resp. $(1,4)$, it implies that $\mathfrak{n}$ is a multiple of $z^3t = \frac{1}{4}\partial_z F_0$, resp. $zt^3 = \frac{1}{4}\partial_t F_0$. In the row ($\mathrm{E}_7$),

$$z\partial_z F_0 + t\partial_t F_0 = 3y\Lambda_i, \qquad \Lambda_i^2 = \Lambda_i \partial_y F_0 - 3y \cdot y\Lambda_i,$$

so $y\Lambda_i, \Lambda_i^2 \in J(F_0)$. If $\mathfrak{n} = yz^{p_1}t^{p_2}$, then $(p_1 - a_1)w_z + (p_2 - a_2)w_t = 0$ and $p_1 + p_2 > 3$, so $\mathfrak{n}$ is a multiple of $y\Lambda_i$. If $\mathfrak{n} = z^{p_1}t^{p_2}$, then $(2p_1 - 3a_1)w_z + (2p_2 - 3a_2)w_t = 0$ and $2(p_1 + p_2) > 9$, so $2p_1 > 3a_1$, $2p_2 > 3a_2$, and $\mathfrak{n}$ is a multiple of Λ_i^2. If $y^2 \mid \mathfrak{n}$, then

$$\mathfrak{n} \equiv -\frac{1}{3}\Lambda_i \mathfrak{n}/y^2 \pmod{J(F_0)},$$

where the $\mathbf{T}$-weight and the $\mathbf{p}$-degree of $\Lambda_i\mathfrak{n}/y^2$ equal those of $\mathfrak{n}$, and it is reduced by the previous cases. This proves the claim.

Let $\mathfrak{n}$ be a monomial of the weight-$6u$ part of $\widehat{f}_i$ other than those of F_0, of least $\mathbf{p}$-degree. By the claim (with the representations given in the proof of the claim, in which $c_1 = 0$ and c_2, c_3, c_4 do not involve x, since $\widehat{f}_i - x^2$ does not involve x), $\mathfrak{n} = \sum_{j=1}^4 c_j \partial_j F_0$ for polynomials c_j, where ∂_j is the derivative by the j-th coordinate of (x,y,z,t), and we may take c_j $\mathbf{T}$-homogeneous of the weight of the j-th coordinate, of $\mathbf{p}$-degree larger than that of the coordinate, and G_i-semi-invariant with the character of the coordinate, by projecting to the character of the coordinate. The substitution $(x,y,z,t) \to (x - c_1, y - c_2, z - c_3, t - c_4)$ removes $\mathfrak{n}$ and changes the weight-$6u$ part only in larger $\mathbf{p}$-degrees. The composition of such substitutions converges formally, since $\mathbf{p} \in \mathbb{Z}^4_{>0}$, and by [Kol98, Construction 2.6] they define a $\mathbf{T}$-equivariant and G_i-equivariant automorphism ψ of $\mathbb{C}[[x,y,z,t]]$ after which the weight-$6u$ part of $\widehat{f}_i$ is exactly F_0. By Lemma 4.6 with this ψ, (2), (3), and $f_i - x^2 \in \mathbb{C}[y,z,t]$ continue to hold for the polynomial truncations, and the truncation in degrees less than $L > d$ keeps the weight-$6u$ part F_0.

Blocks. The prime divisor $E_{\mathbf{p}}$ over W_i given by the monomial valuation with weights $\mathbf{p}$ has lowest part F_0 on f_i, which is irreducible. By [Liu26b, Lemma 4.12],

$$a(E_{\mathbf{p}}, W_i, 0) = \sum_j p_j - d = a_{\mathbf{p}}, \qquad a_{\mathbf{p}} := 4, 5, 7 \text{ in the three rows},$$

and $n_i := \mathrm{mult}_{E_{\mathbf{p}}} h_i$ equals $\mathrm{wt}_{\mathbf{p}}(h_i)$, since the lowest $\mathbf{p}$-weight part of the x-linear polynomial h_i is not divisible by F_0, which is monic of degree 2 in x. Since $(W_i, \frac{1}{m}\mathrm{div}(h_i))$ is lc near 0, $n_i \leq m a_{\mathbf{p}}$. Possibly passing to a subsequence, a monomial of h_i of $\mathbf{p}$-degree n_i has a $\mathbf{T}$-weight κ independent of i. All monomials of h_i have $\mathbf{T}$-weights congruent to κ modulo r_i, and the monomials of $\mathbf{p}$-degree at most $m a_{\mathbf{p}}$ have $\mathbf{T}$-weights bounded independently of i. Thus for $i \gg 0$,

$$f_i = F_0 + \sum_{j \neq 0} f_{i,j}, \qquad h_i = h_i^0 + \sum_{j \neq 0} h_{i,j}, \tag{C.64}$$

where $f_{i,j}$ and h_i^0, $h_{i,j}$ are $\mathbf{T}$-homogeneous of weights $6u + jr_i$ and κ, $\kappa + jr_i$, respectively, and h_i^0 contains the lowest $\mathbf{p}$-weight part of h_i. In particular h_i^0 is a non-zero x-linear G_i-semi-invariant polynomial, and $h_i^0|_{V_0} \neq 0$ for

$$V_0 := (F_0 = 0) \subset \mathbb{A}^4.$$

By [Liu26b, Lemma 5.3], the least degree L_i of a monomial of $f_i - F_0$ or $h_i - h_i^0$ satisfies $L_i \to +\infty$.

Marked germs over V_0. The hypersurface V_0 is regular in codimension 1, since the singular locus of V_0 is contained in the z-axis and the t-axis, so V_0 is normal. Near a point of the z-axis or the t-axis other than 0, V_0 is the product of a line and a Du Val surface germ $x^2+y^3+t^{a_2}$, $x^2+y^3+z^{a_1}$, $x^2+y^3+yt^{a_2}$, or $x^2+y^3+yz^{a_1}$ with $a_1, a_2 \le 4$, resp. $a_1, a_2 \le 2$ in the row (E_7), after an analytic change of the coordinate t, resp. z. At 0, F_0 is Newton non-degenerate: a face polynomial containing x^2 or y^3 has a derivative not vanishing on the torus, and $y^3 + y\Lambda_i = y(y^2 + \Lambda_i)$ has $\partial_y = 2y^2 \ne 0$ on the zero locus of $y^2 + \Lambda_i$ in the torus. For $p \in \mathbb{R}^4_{\ge 0}\backslash\{0\}$, the minimum $\mathrm{wt}_p(F_0)$ is at most the convex combination of the weights of x^2, y^3, $z^{a_1}t^{a_2}$ with coefficients $\frac12, \frac13, \frac16$, resp. of x^2, y^3, $y\Lambda_i$ with coefficients $\frac12, \frac14, \frac14$. In [Liu26a, Set-up 8.4] in dimension 4, with the trivial group and $B = \mathrm{div}(F_0)$, the function ψ of [Liu26a, Set-up 8.4(3)] is $\psi(p) = \sum_j p_j - \mathrm{wt}_p(F_0)$, and

$$\psi(p) \ge \Big(1 - \frac{a_1}{6}\Big)p_z + \Big(1 - \frac{a_2}{6}\Big)p_t \ge 0,$$
$$\text{resp.}\quad \psi(p) \ge \Big(1 - \frac{a_1}{4}\Big)p_z + \Big(1 - \frac{a_2}{4}\Big)p_t \ge 0,$$

and $\psi(p) > 0$ for $p \in \mathbb{R}^4_{>0}$. By the Newton non-degeneracy, $\mathrm{div}(\mathrm{in}_p F_0)$ is reduced and smooth on $(\mathbb{C}^*)^4$ for any $p \in \mathbb{R}^4_{>0}$, so the toric model is lc near the fiber over 0 by [Liu26a, Lemma 8.6(1)]. Since $\psi \ge 0$ on $\mathbb{R}^4_{\ge 0}\backslash\{0\}$, the boundary of the crepant pull-back of $(\mathbb{A}^4, V_0)$ in [Liu26a, Lemma 8.5] is at most the boundary of the toric model, so $(\mathbb{A}^4, V_0)$ is lc near 0. By [Liu26a, Lemma 8.6(2)], any prime divisor over $\mathbb{A}^4$ with center 0 has log discrepancy at least $\psi(v) > 0$ with respect to $(\mathbb{A}^4, V_0)$ for some $v \in \mathbb{Z}^4_{>0}$. Since moreover V_0 is smooth or a line times a Du Val surface germ near any point other than 0, $(\mathbb{A}^4, V_0)$ is plt near 0, and by inversion of adjunction V_0 is klt near 0. Thus $\mathbf{T}$, V_0, G_i, e_0, h_i^0, and $c = \frac{1}{m}$ are as in Set-up 11.1, and by (C.64), $(W_i \ni 0, G_i, h_i)$ is a marked germ over (V_0, h_i^0).

Next, $(V_0, \frac1m \mathrm{div}(h_i^0))$ is lc near 0 for $i \gg 0$. By inversion of adjunction [Kaw07, Theorem], $(\mathbb{A}^4, W_i + \frac1m \mathrm{div}(h_i))$ is lc near 0. Since $f_i \equiv F_0$ and $h_i \equiv h_i^0$ modulo $(x,y,z,t)^{L_i}$ with $L_i \to +\infty$, Lemma 2.19 with the exponents $1, \frac1m$ implies that $(\mathbb{A}^4, V_0 + \frac1m \mathrm{div}(h_i^0))$ is lc near 0 for $i \gg 0$, and by adjunction [Kaw21, Theorem 2.5(ii)], so is $(V_0, \frac1m \mathrm{div}(h_i^0))$.

The upper bound. Let β_i be the number $\beta_{\mathbf{T}}$ of Set-up 11.1 for (V_0, G_i, h_i^0) and $c = \frac1m$. By Proposition 11.2(5), the β_i belong to a finite set depending only on m, e_0, and w. Possibly passing to a subsequence,

$$\beta_i = \beta \quad \text{for any } i. \tag{C.65}$$

By Proposition 11.4 with $\mathbf{T}$, e_0, m, κ, the data (V_0, G_i, h_i^0) and the marked germs $(W_i \ni 0, G_i, h_i)$, we obtain for $i \gg 0$

$$\alpha_i = q_i(\tau_i) \le \beta + \frac{N_0}{r_i} + \frac{N_1}{m}(1 - \tau_i), \tag{C.66}$$

with N_0, N_1 independent of i. Letting $i \to +\infty$, (C.55), (C.57), and (C.66) imply

$$\beta \ge \alpha_\infty > 0. \tag{C.67}$$

The proof continues below.

Proof of Theorem 12.4, continued. By Proposition 11.7 with $c = \frac1m$ (F_0 is of class (d) by (C.63), and $\beta_i = \beta > 0$ by (C.65) and (C.67)), after passing to a subsequence, one of the alternatives (1) and (2) of Proposition 11.7 holds for any i.

Assume that (1) holds. Then there exists a real number $\theta > 1$ such that $(W_i, \frac{\theta}{m}\mathrm{div}(h_i))$ is lc near 0 for any $i \gg 0$. Let E be a prime divisor over W_i with center 0. Then $a(E, W_i, \frac{\theta}{m}\mathrm{div}(h_i)) \ge 0$, that is,

$$\frac{\theta - 1}{m}\,\mathrm{mult}_E h_i \le a\Big(E, W_i, \frac1m \mathrm{div}(h_i)\Big).$$

Thus for any $\tau \in [0,1]$,

$$\begin{aligned} a\Big(E, W_i, \frac{\tau}{m}\operatorname{div}(h_i)\Big) &= a\Big(E, W_i, \frac{1}{m}\operatorname{div}(h_i)\Big) + \frac{1-\tau}{m}\operatorname{mult}_E h_i \\ &\le \frac{\theta-\tau}{\theta-1} a\Big(E, W_i, \frac{1}{m}\operatorname{div}(h_i)\Big). \end{aligned} \tag{C.68}$$

By (C.68) with $E = E_i$ and $\tau = \tau_i$, and by (C.55),

$$\alpha_i \le \frac{a(E_i, W_i, \frac{\tau_i}{m}\operatorname{div}(h_i))}{e_{E_i}} \le \frac{\theta-\tau_i}{\theta-1} \cdot \frac{a(E_i, W_i, \frac{1}{m}\operatorname{div}(h_i))}{e_{E_i}} < \frac{\theta-\tau_i}{\theta-1}\Big(\alpha_\infty - \frac{\epsilon}{4}\Big)$$

for any $i \gg 0$. Since $\lim_i \tau_i = 1$, the right-hand side tends to $\alpha_\infty - \frac{\epsilon}{4}$, which contradicts $\lim_i \alpha_i = \alpha_\infty$.

Assume that (2) holds. Then any prime divisor F' over W_i with center 0 satisfies $a(F', W_i, \frac{1}{m}\operatorname{div}(h_i))/e_{F'} \ge \beta$ for any $i \gg 0$. Taking $F' = E_i$, we obtain by (C.55) and (C.67)

$$\beta \le \frac{a(E_i, W_i, \frac{1}{m}\operatorname{div}(h_i))}{e_{E_i}} < \alpha_\infty - \frac{\epsilon}{4} \le \beta - \frac{\epsilon}{4},$$

a contradiction.

Thus the circle **T** of (C.63) also leads to a contradiction. Since the other circles were excluded above, the sequence of data assumed at the beginning of the proof does not exist, and Theorem 12.4 follows. □

Appendix D. Proofs for prepared limits

This section proves the results of Sections 13 and 14.

D.1. **Prepared limits.** This subsection proves Theorems 13.1, 13.2, 13.3 and 13.8, Construction-Theorems 13.6 and 13.7, and Lemma 13.5 of Section 13, and it states and proves Theorem D.6, which the proofs of Construction-Theorem 13.7 and Theorem 13.8 use. We need the following lemma.

Lemma D.1. *Let m be a positive integer, and let $\theta > 0$ and $\alpha \ge 0$ be real numbers. Then there exists a positive integer $N_0 = N_0(m, \theta, \alpha)$ satisfying the following. Let*

$$W := (xy + g(z,t) = 0) \subset \mathbb{A}^4,$$

where $g \ne 0$ is a polynomial with $\operatorname{mult}_0 g \ge N_0$, let h be a polynomial not divisible by $xy+g$, and let $h_0 := h(0,0,z,t)$.

(1) If $(W, \frac{1}{m}\operatorname{div}(h))$ is lc near 0, then $h_0 \ne 0$, $(\mathbb{A}^2, \frac{1}{m}\operatorname{div}(h_0))$ is lc near 0, and $\operatorname{mult}_0 h_0 \le 2m$.

(2) Let $\ell \in \mathbb{C}[z,t]$ be non-zero. If $h_0 \ne 0$ and

$$\Big(\mathbb{A}^2, \frac{\theta}{m}\operatorname{div}(h_0) + \alpha\operatorname{div}(\ell)\Big) \text{ is lc near } 0,$$

then

$$\Big(W, \frac{\theta}{m}\operatorname{div}(h) + \alpha\operatorname{div}(\ell)\Big) \text{ is lc near } 0.$$

Proof. All pairs in this proof are considered near 0. Let $B \ge 0$ be an $\mathbb{R}$-divisor on $\mathbb{A}^4$ and $\widehat{B}$ the pullback of B to $\widehat{\mathbb{A}} := \operatorname{Spec}\mathbb{C}[[x,y,z,t]]$. The pair $(\mathbb{A}^4, B)$ is lc near 0 if and only if $a(E, \mathbb{A}^4, B) \ge 0$ for any prime divisor E over $\mathbb{A}^4$ with center 0, and $(\widehat{\mathbb{A}}, \widehat{B})$ is lc if and only if $a(\widehat{E}, \widehat{\mathbb{A}}, \widehat{B}) \ge 0$ for any prime divisor $\widehat{E}$ over $\widehat{\mathbb{A}}$ with center the closed point. By Lemma 2.12(3) with the trivial group, any such $\widehat{E}$ restricts to a prime divisor E over $\mathbb{A}^4$ with center 0 and $a(E, \mathbb{A}^4, B) = a(\widehat{E}, \widehat{\mathbb{A}}, \widehat{B})$, and any such E arises in this way. Thus

$$(\mathbb{A}^4, B) \text{ is lc near } 0 \iff (\widehat{\mathbb{A}}, \widehat{B}) \text{ is lc.} \tag{D.1}$$

Let L_0 be the integer of Lemma 2.19 for 4 variables, 2 functions and $\Lambda = \{1, \frac{1}{m}\}$. Let L_0' be the integer of Lemma 2.19 for 4 variables, 3 functions and $\Lambda = \{1, \frac{\theta}{m}, \alpha\}$ if $\alpha > 0$, and for 4 variables, 2

functions and $\Lambda = \{1, \frac{\theta}{m}\}$ if $\alpha = 0$. Let $N_0 := \max\{L_0, L_0'\}$. Then $g \in (x,y,z,t)^{L_0} \cap (x,y,z,t)^{L_0'}$. Since $g \in \mathbb{C}[z,t]\backslash\{0\}$, the polynomial $xy+g$ is irreducible and $\operatorname{Sing} W \subset (x=y=g=0)$, so $\dim \operatorname{Sing} W \le 1$ and W is a normal hypersurface. Thus we may use [Kaw07, Theorem] along W, where the different of a boundary is the restriction of the boundary to W.

(1) Assume that $(W, \frac{1}{m}\operatorname{div}(h))$ is lc. Since $xy+g \nmid h$, by [Kaw07, Theorem],

$$\Big(\mathbb{A}^4, \operatorname{div}(xy+g) + \frac{1}{m}\operatorname{div}(h)\Big) \text{ is lc.}$$

By (D.1) and Lemma 2.19 with $\Bbbk = \mathbb{C}$, applied to the pairs of functions $(xy+g, h)$ and (xy, h), which are congruent modulo $(x,y,z,t)^{L_0}$, the pair $(\mathbb{A}^4, \operatorname{div}(xy) + \frac{1}{m}\operatorname{div}(h))$ is lc. If $x \mid h$, then the coefficient of $(x=0)$ in $\operatorname{div}(xy) + \frac{1}{m}\operatorname{div}(h)$ is at least $1 + \frac{1}{m}$, which contradicts lc. Thus $h(0,y,z,t) \neq 0$, and by [Kaw07, Theorem] along $(x=0) \cong \mathbb{A}^3_{y,z,t}$,

$$\Big(\mathbb{A}^3_{y,z,t}, \operatorname{div}(y) + \frac{1}{m}\operatorname{div}(h(0,y,z,t))\Big) \text{ is lc.}$$

Thus the coefficient of $(y=0)$ in $\operatorname{div}(y) + \frac{1}{m}\operatorname{div}(h(0,y,z,t))$ is at most 1, so $y \nmid h(0,y,z,t)$, that is, $h_0 \neq 0$, and by [Kaw07, Theorem] along $(y=0) \cong \mathbb{A}^2_{z,t}$, $(\mathbb{A}^2, \frac{1}{m}\operatorname{div}(h_0))$ is lc. Let E_1 be the exceptional prime divisor of the blow-up of $\mathbb{A}^2$ at 0. Then

$$0 \le a\Big(E_1, \mathbb{A}^2, \frac{1}{m}\operatorname{div}(h_0)\Big) = 2 - \frac{1}{m}\operatorname{mult}_0 h_0,$$

so $\operatorname{mult}_0 h_0 \le 2m$.

(2) Since the restriction of $h(0,y,z,t)$ to $(y=0)$ is $h_0 \neq 0$ and $y \nmid \ell$, by [Kaw07, Theorem] along $(y=0) \subset \mathbb{A}^3_{y,z,t}$, the pair

$$\Big(\mathbb{A}^3_{y,z,t}, \operatorname{div}(y) + \frac{\theta}{m}\operatorname{div}(h(0,y,z,t)) + \alpha\operatorname{div}(\ell)\Big)$$

is lc. Since $x \nmid h$ and $x \nmid y\ell$, by [Kaw07, Theorem] along $(x=0) \subset \mathbb{A}^4$,

$$\Big(\mathbb{A}^4, \operatorname{div}(xy) + \frac{\theta}{m}\operatorname{div}(h) + \alpha\operatorname{div}(\ell)\Big) \text{ is lc.}$$

By (D.1) and Lemma 2.19 with $\Bbbk = \mathbb{C}$, applied to the triples (xy, h, ℓ) and $(xy+g, h, \ell)$ (to the pairs (xy, h) and $(xy+g, h)$ if $\alpha = 0$), which are congruent modulo $(x,y,z,t)^{L_0'}$,

$$\Big(\mathbb{A}^4, \operatorname{div}(xy+g) + \frac{\theta}{m}\operatorname{div}(h) + \alpha\operatorname{div}(\ell)\Big) \text{ is lc.}$$

Since $xy+g \nmid h\ell$, by [Kaw07, Theorem] along W,

$$\Big(W, \frac{\theta}{m}\operatorname{div}(h) + \alpha\operatorname{div}(\ell)\Big) \text{ is lc.}$$

□

Proof of Theorem 13.1. Let $C_m > 0$ be the constant of Lemma 5.7, and let $L(m) := C_m + 2$. Let q be the radial function of $(W \ni o, G, h, m)$ (Definition 4.1) and $b := q(1)$.

In **Steps 1–5** we assume that $xy+g$ and h are polynomials and that no monomial of h is divisible by xy, and we regard $(W \ni o)$ as the germ at 0 of $(xy+g=0) \subset \mathbb{A}^4$ (Lemma 2.12(2)(3)), with $G \subset GL_4(\mathbb{C})$ diagonal. Let $\chi_x, \chi_y, \chi_z, \chi_t$ be the characters of the coordinates, $\overline{G} \subset GL_2(\mathbb{C})$ the image of G acting on the (z,t)-plane, G_0 the kernel of $G \to \overline{G}$, and $k := |G_0|$. In **Steps 1–6**, $xy+g$ is G-semi-invariant, and in **Steps 1–5** so is h: if

$$Q = \sum_{\nu_1,\nu_2 \ge 0} Q_{\nu_1,\nu_2}(z,t) x^{\nu_1} y^{\nu_2} \in \mathbb{C}[[x,y,z,t]]$$

and no monomial of $Q(xy+g)$ is divisible by xy, then $Q_{\nu_1-1,\nu_2-1} = -gQ_{\nu_1,\nu_2}$ for $\nu_1, \nu_2 \ge 1$, so $Q_{\nu_1,\nu_2} \in \bigcap_n (z,t)^n = 0$ as $g \in (z,t)$, and $Q(xy+g) = 0$. For $\gamma \in G$, the series

$$\gamma(xy+g) - \chi_x(\gamma)\chi_y(\gamma)(xy+g) \in \mathbb{C}[[z,t]]$$

and the difference of h and its component of character χ_h are multiples of $xy+g$ without monomials divisible by xy, hence 0. Let

$$h_0 := h(0,0,z,t), \quad h_x := h(x,0,z,t), \quad \text{and} \quad h_y := h(0,y,z,t).$$

Since $h|_W \neq 0$, $xy+g \nmid h$. When $h_0 \neq 0$ and $(\mathbb{A}^2, \frac{1}{m}\operatorname{div}(h_0))$ is lc near 0, let

$$\beta := \operatorname{mld}\Big(\Big(\mathbb{A}^2 \ni 0, \frac{1}{m}\operatorname{div}(h_0)\Big)/\overline{G}\Big).$$

The form

$$\operatorname{Res}\frac{dx\wedge dy\wedge dz\wedge dt}{xy+g} = \frac{dy\wedge dz\wedge dt}{y} = -\frac{dx\wedge dz\wedge dt}{x}$$

generates ω_W, and as a rational form on $\mathbb{A}^3_{y,z,t}$, resp. $\mathbb{A}^3_{x,z,t}$, it has divisor $-(y=0)$, resp. $-(x=0)$. The projections of W to $\mathbb{A}^3_{y,z,t}$ and $\mathbb{A}^3_{x,z,t}$ are birational and G-equivariant, where G acts on $\mathbb{A}^3_{y,z,t}$ by χ_y, χ_z, χ_t and on $\mathbb{A}^3_{x,z,t}$ by χ_x, χ_z, χ_t. Thus for any prime divisor F over W,

$$a(F,W,0) = a(F,\mathbb{A}^3_{y,z,t},0) - \operatorname{mult}_F y = a(F,\mathbb{A}^3_{x,z,t},0) - \operatorname{mult}_F x, \tag{D.2}$$

and the inertia groups in G of F as a prime divisor over W, over $\mathbb{A}^3_{y,z,t}$, and over $\mathbb{A}^3_{x,z,t}$ coincide.

Step 1. In this step we assume that $\operatorname{mult}_0 g \geq N_0(m,1,0)$, and show that $h_0 \neq 0$, $(\mathbb{A}^2, \frac{1}{m}\operatorname{div}(h_0))$ is lc near 0, $\operatorname{mult}_0 h_0 \leq 2m$, and

$$\beta \in \frac{1}{mM!}\mathbb{Z} \cap [0,2]. \tag{D.3}$$

The first three claims are Lemma D.1(1). By Definition 2.2, β is the infimum of

$$\frac{a(E,\mathbb{A}^2,0) - \frac{1}{m}\operatorname{mult}_E h_0}{e_E}$$

over the prime divisors E over $\mathbb{A}^2$ with center 0. Since $a(E,\mathbb{A}^2,0), \operatorname{mult}_E h_0 \in \mathbb{Z}$ and e_E divides $|\overline{G}|$, which divides $M!$, the quotients in the infimum belong to $\frac{1}{mM!}\mathbb{Z}$, and they are non-negative because $(\mathbb{A}^2, \frac{1}{m}\operatorname{div}(h_0))$ is lc. The exceptional prime divisor E_1 of the blow-up of $\mathbb{A}^2$ at 0 satisfies $a(E_1,\mathbb{A}^2,0) = 2$, so $\beta \leq 2$.

Step 2. In this step we keep the assumption of **Step 1** and prove

$$\begin{aligned}\operatorname{mld}\Big(\Big(\mathbb{A}^3_{y,z,t} \ni 0, (y=0) + \frac{1}{m}\operatorname{div}(h_y)\Big)/G\Big) &= \beta \\ &= \operatorname{mld}\Big(\Big(\mathbb{A}^3_{x,z,t} \ni 0, (x=0) + \frac{1}{m}\operatorname{div}(h_x)\Big)/G\Big).\end{aligned} \tag{D.4}$$

Let $\gamma \in G$ act trivially on z and t. Then $\gamma g = g$, and since $g \neq 0$ has the character of $xy+g$, $\chi_x(\gamma)\chi_y(\gamma) = 1$. Thus $\gamma = 1$ if $\chi_y(\gamma) = 1$ or $\chi_x(\gamma) = 1$. Hence G acts faithfully on $\mathbb{A}^3_{y,z,t}$ and on $\mathbb{A}^3_{x,z,t}$, $\chi_x\chi_y = 1$ on G_0, and $\chi_y|_{G_0}$ and $\chi_x|_{G_0}$ are faithful. Since $h_0 \neq 0$ is a sum of monomials of h in z,t, the character of h is trivial on G_0. As G_0 is cyclic of order k and acts on y by the faithful character χ_y and trivially on z,t, we have $h_y \in \mathbb{C}[\overline{y},z,t]$ with $\overline{y} := y^k$, and $\pi_0 : \mathbb{A}^3_{y,z,t} \to \mathbb{A}^3_{\overline{y},z,t}$ is the quotient by G_0. Since $K_{\mathbb{A}^3_{y,z,t}} = \pi_0^* K_{\mathbb{A}^3_{\overline{y},z,t}} + (k-1)(y=0)$ and $\pi_0^*(\overline{y}=0) = k(y=0)$,

$$K_{\mathbb{A}^3_{y,z,t}} + (y=0) + \frac{1}{m}\operatorname{div}(h_y) = \pi_0^*\Big(K_{\mathbb{A}^3_{\overline{y},z,t}} + (\overline{y}=0) + \frac{1}{m}\operatorname{div}(h_y)\Big). \tag{D.5}$$

The group $\overline{G} = G/G_0$ acts diagonally on $\mathbb{A}^3_{\overline{y},z,t}$, and $\mathbb{A}^3_{y,z,t}/G = \mathbb{A}^3_{\overline{y},z,t}/\overline{G}$. By (D.5), the pairs on both sides of (D.5) are pullbacks of one pair on $\mathbb{A}^3_{y,z,t}/G$, so by Lemma 2.15(2)(3)(5) the left-hand side of (D.4) is

$$\operatorname{mld}\Big(\Big(\mathbb{A}^3_{\overline{y},z,t} \ni 0, (\overline{y}=0) + \frac{1}{m}\operatorname{div}(h_y)\Big)/\overline{G}\Big).$$

By **Step 1** and Theorem 2.17(1) with $n=3$, $c=1$, the group $\overline{G}$ acting diagonally on $\mathbb{A}^3_{\overline{y},z,t}$, $H=\overline{G}$, $f_1=\overline{y}$, and $\mathfrak{a}=(h_y)^{1/m}$, whose restriction to $(\overline{y}=0)\cong\mathbb{A}^2_{z,t}$ is $(h_0)^{1/m}$,

$$\beta=\operatorname{mld}\Big(\Big(\mathbb{A}^2\ni 0,\frac{1}{m}\operatorname{div}(h_0)\Big)/\overline{G}\Big)=\operatorname{mld}\Big(\Big(\mathbb{A}^3_{\overline{y},z,t}\ni 0,(\overline{y}=0)+\frac{1}{m}\operatorname{div}(h_y)\Big)/\overline{G}\Big),$$

which proves the first equality of (D.4). Exchanging x and y, that is, replacing h with $h(y,x,z,t)$ and conjugating G by the exchange, preserves the assumptions of this step, h_0, and $\overline{G}$, and turns h_y into $h_x(y,z,t)$, on which y has the character χ_x. Thus the first equality of (D.4) for $h(y,x,z,t)$ and the conjugate of G, with y renamed x, is the second equality.

Step 3. In this step we assume in addition that $\operatorname{mult}_0 g>(2mC_m+1)M$, and construct a prime divisor F over W with center 0 such that

$$\frac{a(F,W,0)}{e_F}\le C_m,\qquad \frac{\operatorname{mult}_F h}{me_F}\le C_m,\quad\text{and}\quad \frac{a(F,W,0)-\frac{1}{m}\operatorname{mult}_F h}{e_F}\le\beta. \tag{D.6}$$

Since

$$\frac{a(F,W,0)-\frac{\tau}{m}\operatorname{mult}_F h}{e_F}=\tau\frac{a(F,W,0)-\frac{1}{m}\operatorname{mult}_F h}{e_F}+(1-\tau)\frac{a(F,W,0)}{e_F},$$

Definitions 4.1 and 2.2 and (D.6) imply $b\le\beta$ and

$$q(\tau)\le\tau\beta+(1-\tau)C_m\quad\text{for any }\tau\in[0,1]. \tag{D.7}$$

By the Chinese remainder theorem, for positive integers k_1,k_2 and integers c,c' with $\gcd(c,k_1)=\gcd(c',k_2)=1$, the set

$$\{n\in\mathbb{Z}\mid c'n\equiv c\pmod{k_2},\ \gcd(n,k_1)=1\} \tag{D.8}$$

is a non-empty union of residue classes modulo k_1k_2.

Since the character of h is trivial on G_0 (**Step 2**), the polynomial h_0 is a mark on $(\mathbb{A}^2\ni 0)$ for the action of $\overline{G}$, with $(\mathbb{A}^2,\frac{1}{m}\operatorname{div}(h_0))$ lc near 0. By Lemma 5.7, there exists a computing lift v_0 of β, that is, a prime divisor over $\mathbb{A}^2$ with center 0 and $(a(v_0,\mathbb{A}^2,0)-\frac{1}{m}\operatorname{mult}_{v_0}h_0)/e_{v_0}=\beta$, such that

$$\frac{a(v_0,\mathbb{A}^2,0)}{e_{v_0}}\le C_m\quad\text{and}\quad\frac{\operatorname{mult}_{v_0}h_0}{e_{v_0}}\le mC_m. \tag{D.9}$$

Let $J:=I_{\overline{G}}(v_0)$, so $e_{v_0}=|J|$, and let $I\subset G$ be the preimage of J, so $|I|=ke_{v_0}$ and I is cyclic. Let $\mathcal{O}_{v_0}$ be the valuation ring of v_0. Let $\mathfrak{m}_{v_0}$ be the maximal ideal of $\mathcal{O}_{v_0}$. By Lemma 2.15(1), J acts on $\mathfrak{m}_{v_0}/\mathfrak{m}_{v_0}^2$ by a faithful character λ, and projecting a uniformizer to the λ-isotypic part gives a rational function π on $\mathbb{A}^2_{z,t}$ with $\gamma\pi=\lambda(\gamma)\pi$ for any $\gamma\in J$ and $\operatorname{mult}_{v_0}\pi=1$. We regard λ as a character of I, with kernel G_0.

We choose an integer n with

$$k\operatorname{mult}_{v_0}h_0<n\le k\operatorname{mult}_{v_0}h_0+ke_{v_0},\qquad\gcd(n,k)=1,\quad\text{and}\quad\lambda^n=\chi_y^k\text{ on }I. \tag{D.10}$$

Let γ_1 be a generator of I and ζ a primitive (ke_{v_0})-th root of unity, and write $\chi_y(\gamma_1)=\zeta^c$ and $\lambda(\gamma_1)=\zeta^{kc'}$ with integers c,c'. Since λ is faithful on $I/G_0\cong J$, $\gcd(c',e_{v_0})=1$. Since $G_0=\langle\gamma_1^{e_{v_0}}\rangle$ and $\chi_y|_{G_0}$ is faithful, $\gcd(c,k)=1$. The condition $\lambda^n=\chi_y^k$ reads $c'n\equiv c$ modulo e_{v_0}, so (D.8) with $k_1=k$ and $k_2=e_{v_0}$ provides n.

Let $\sigma:S'\to\mathbb{A}^2_{z,t}$ be a proper birational morphism from a smooth surface on which v_0 is a prime divisor E_0, and let $X':=S'\times\mathbb{A}^1_y$. Near the generic point η of $E_0\times\{0\}$, π and y are local parameters of X'. Let F be the exceptional prime divisor over η of the weighted blow-up of X' with weights (k,n) for (π,y), a primitive weight vector since $\gcd(k,n)=1$. For rational functions ϕ_j on $\mathbb{A}^2_{z,t}$, $\phi_j\pi^{-\operatorname{mult}_{v_0}\phi_j}$ is a unit at η which does not involve y, so

$$\operatorname{mult}_F\Big(\sum_j\phi_jy^j\Big)=\min_j\big(k\operatorname{mult}_{v_0}\phi_j+jn\big). \tag{D.11}$$

Since, near η,

$$K_{X'} = (\sigma \times \mathrm{id}_{\mathbb{A}^1_y})^* K_{\mathbb{A}^3_{y,z,t}} + (a(v_0, \mathbb{A}^2, 0) - 1)(E_0 \times \mathbb{A}^1)$$

and the weighted blow-up has log discrepancy $k + n$,

$$a(F, \mathbb{A}^3_{y,z,t}, 0) = k + n + k(a(v_0, \mathbb{A}^2, 0) - 1) = ka(v_0, \mathbb{A}^2, 0) + n.$$

Since the projection $W \to \mathbb{A}^3_{y,z,t}$ is birational, F is a prime divisor over W, and by (D.2),

$$a(F, W, 0) = ka(v_0, \mathbb{A}^2, 0).$$

We show that $I \subset I_G(F)$. Let $\gamma \in I$. For a non-zero rational function ϕ on $\mathbb{A}^2_{z,t}$ with $n_\phi := \mathrm{mult}_{v_0} \phi$, write $\phi = \pi^{n_\phi}\phi'$ with $\phi' \in \mathcal{O}^*_{v_0}$. Since the image of γ in $\overline{G}$ belongs to J, $\gamma\phi' - \phi'$ has positive order along v_0, so

$$\mathrm{mult}_{v_0} \left(\gamma\phi - \lambda(\gamma)^{n_\phi}\phi\right) > n_\phi.$$

If $kp + jn = kp' + j'n$ for integers p, p', j, j', then $(p - p', j - j')$ is an integer multiple of $(n, -k)$ since $\gcd(k, n) = 1$, so by (D.10),

$$\lambda(\gamma)^p \chi_y(\gamma)^j = \lambda(\gamma)^{p'} \chi_y(\gamma)^{j'}.$$

For a non-zero polynomial $\Phi = \sum_j \phi_j y^j$ in y whose coefficients ϕ_j are rational functions on $\mathbb{A}^2_{z,t}$, with $\mathrm{mult}_F \Phi = N_\Phi$, let $\mu_\Phi(\gamma) := \lambda(\gamma)^p \chi_y(\gamma)^j$ for any (p, j) with $kp + jn = N_\Phi$. Since $\gamma\Phi = \sum_j (\gamma\phi_j)\chi_y(\gamma)^j y^j$, (D.11) implies

$$\mathrm{mult}_F \left(\gamma\Phi - \mu_\Phi(\gamma)\Phi\right) > N_\Phi,$$

and in particular $\mathrm{mult}_F \gamma\Phi = N_\Phi$. Thus γ preserves F. Since $W \to \mathbb{A}^3_{y,z,t}$ is birational, any rational function on W of order 0 along F is Φ_1/Φ_2 with Φ_1, Φ_2 polynomials in y whose coefficients are rational functions on $\mathbb{A}^2_{z,t}$, and $N_{\Phi_1} = N_{\Phi_2}$, so $\mu_{\Phi_1}(\gamma) = \mu_{\Phi_2}(\gamma) =: \mu$, and

$$\gamma\frac{\Phi_1}{\Phi_2} - \frac{\Phi_1}{\Phi_2} = \frac{\Phi_2(\gamma\Phi_1 - \mu\Phi_1) - \Phi_1(\gamma\Phi_2 - \mu\Phi_2)}{\Phi_2 \cdot \gamma\Phi_2}$$

has positive order along F. Hence γ acts trivially on the residue field of F, and $\gamma \in I_G(F)$. Thus $e_F \geq |I| = ke_{v_0}$.

By (D.11), $\mathrm{mult}_F y = n$, $\mathrm{mult}_F z, \mathrm{mult}_F t \geq k$, and $\mathrm{mult}_F x = \mathrm{mult}_F g - n = k\,\mathrm{mult}_{v_0} g - n$ since $x = -g/y$ on W. As $\mathrm{mult}_{v_0} z, \mathrm{mult}_{v_0} t \geq 1$, $\mathrm{mult}_{v_0} g \geq \mathrm{mult}_0 g$. By (D.9) and $e_{v_0} \leq M$, $\mathrm{mult}_{v_0} h_0 \leq mC_mM$, so $\mathrm{mult}_0 g > 2\,\mathrm{mult}_{v_0} h_0 + e_{v_0}$ and, by (D.10),

$$\mathrm{mult}_F x \geq k(\mathrm{mult}_0 g - \mathrm{mult}_{v_0} h_0 - e_{v_0}) > k\,\mathrm{mult}_{v_0} h_0 \quad \text{and} \quad \mathrm{mult}_F y = n > k\,\mathrm{mult}_{v_0} h_0.$$

In particular, F has center 0 on W. Each monomial of $h - h_0$ is divisible by x or by y, so $\mathrm{mult}_F(h - h_0) > k\,\mathrm{mult}_{v_0} h_0 = \mathrm{mult}_F h_0$, and $\mathrm{mult}_F h = k\,\mathrm{mult}_{v_0} h_0$. Since $e_F \geq ke_{v_0}$ and

$$a(v_0, \mathbb{A}^2, 0) - \frac{1}{m}\,\mathrm{mult}_{v_0} h_0 = e_{v_0}\beta \geq 0,$$

$$\frac{a(F, W, 0) - \frac{1}{m}\,\mathrm{mult}_F h}{e_F} \leq \frac{a(v_0, \mathbb{A}^2, 0) - \frac{1}{m}\,\mathrm{mult}_{v_0} h_0}{e_{v_0}} = \beta,$$

$$\frac{a(F, W, 0)}{e_F} \leq \frac{a(v_0, \mathbb{A}^2, 0)}{e_{v_0}},$$

$$\text{and} \quad \frac{\mathrm{mult}_F h}{me_F} \leq \frac{\mathrm{mult}_{v_0} h_0}{me_{v_0}},$$

and (D.6) follows from (D.9).

Step 4. In this step we keep the assumption of **Step 1**, and show that $b \geq \beta$ if there exists a rational number $\theta > 1$ such that $(\mathbb{A}^2, \frac{\theta}{m}\,\mathrm{div}(h_0))$ is lc near 0 and

$$\mathrm{mult}_0 g \geq N_0(m, \theta, 0) \quad \text{and} \quad \mathrm{mult}_0 g > \frac{4mM}{\theta - 1}.$$

By Lemma D.1(2) with $\alpha = 0$ and $\ell = 1$, $(W, \frac{\theta}{m}\operatorname{div}(h))$ is lc near 0. Let F be a prime divisor over W with center 0 and let $b_F := (a(F, W, 0) - \frac{1}{m}\operatorname{mult}_F h)/e_F$. By Definition 2.2, it suffices to show $b_F \geq \beta$. If $b_F > 2$, then $b_F > \beta$ by (D.3). Assume that $b_F \leq 2$. Since $a(F, W, 0) - \frac{\theta}{m}\operatorname{mult}_F h \geq 0$ and $e_F \leq M$,

$$\frac{\theta - 1}{m}\operatorname{mult}_F h \leq a(F, W, 0) - \frac{1}{m}\operatorname{mult}_F h = e_F b_F \leq 2M, \quad \text{so} \quad \operatorname{mult}_F h \leq \frac{2mM}{\theta - 1}.$$

Since $\operatorname{mult}_F z, \operatorname{mult}_F t \geq 1$,

$$\operatorname{mult}_F x + \operatorname{mult}_F y = \operatorname{mult}_F g \geq \operatorname{mult}_0 g > \frac{4mM}{\theta - 1},$$

so $\operatorname{mult}_F x > \operatorname{mult}_F h$ or $\operatorname{mult}_F y > \operatorname{mult}_F h$. Assume that $\operatorname{mult}_F x > \operatorname{mult}_F h$. Each monomial of $h - h_y$ is divisible by x, so $\operatorname{mult}_F(h - h_y) > \operatorname{mult}_F h$ and $\operatorname{mult}_F h_y = \operatorname{mult}_F h$. The prime divisor F has center 0 on $\mathbb{A}^3_{y,z,t}$, and by (D.2),

$$b_F = \frac{a(F, \mathbb{A}^3_{y,z,t}, 0) - \operatorname{mult}_F y - \frac{1}{m}\operatorname{mult}_F h_y}{e_F} = \frac{a(F, \mathbb{A}^3_{y,z,t}, (y = 0) + \frac{1}{m}\operatorname{div}(h_y))}{e_F} \geq \beta$$

by Definition 2.2 and (D.4). If $\operatorname{mult}_F y > \operatorname{mult}_F h$, then each monomial of $h - h_x$ is divisible by y, so $\operatorname{mult}_F h_x = \operatorname{mult}_F h$, the prime divisor F has center 0 on $\mathbb{A}^3_{x,z,t}$, and by (D.2), Definition 2.2, and (D.4),

$$b_F = \frac{a(F, \mathbb{A}^3_{x,z,t}, (x = 0) + \frac{1}{m}\operatorname{div}(h_x))}{e_F} \geq \beta.$$

Step 5. Let $N_2 := \max\{3, N_0(m, 1, 0), \lfloor (2mC_m + 1)M \rfloor + 1\}$. In this step we show that there exists an integer $N(m, M) \geq N_2$ such that $b = \beta$ whenever $\operatorname{mult}_0 g \geq N(m, M)$, and conclude the polynomial case.

Suppose that no such integer exists. Then for any $i \in \mathbb{N}^+$ there exist $(W_i = (xy + g_i = 0) \ni 0)$, G_i, and h_i as in **Steps 1–4** with $\operatorname{mult}_0 g_i \geq \max\{N_2, i\}$ and $b_i \neq \beta_i$, where b_i and β_i are b and β for $(W_i \ni 0, G_i, h_i)$. By **Step 3**, $b_i < \beta_i$. A diagonal subgroup of $GL_4(\mathbb{C})$ of order at most M is contained in $\mu^4_{M!}$, and β_i takes finitely many values by (D.3). Passing to a subsequence, we may assume that $G_i = G$, that the character of h_i does not depend on i, and that $\beta_i = \beta_*$ for any i. Then $\beta_* > b_i \geq 0$, and $\overline{G}$ does not depend on i. Let $h_{0,i} := h_i(0, 0, z, t)$. By **Step 1**, the tuples $(h_{0,i}, z, t)$ consist of non-zero $\overline{G}$-semi-invariant polynomials whose characters do not depend on i, with $\operatorname{mult}_0 h_{0,i} \leq 2m$. By Lemma 2.6(1), after passing to an infinite subset $\mathcal{I} \subset \mathbb{N}^+$, they have a compatible generic limit $(h_{0,*}, z_*, t_*)$ with coefficient field $\Bbbk$ (Definition 2.5). For any d, let $Z_d \subset \mathcal{J}_d$ be the irreducible closed subset of Definition 2.5 for the limit $(h_{0,*}, z_*, t_*)$, and let $V_d \subset \mathcal{J}_d$ be the closed subset on which the second entry is the d-jet of z. The d-jets of $(h_{0,i}, z, t)$ belong to V_d, and any non-empty open subset of Z_d contains the d-jets of $(h_{0,i}, z, t)$ for all but finitely many $i \in \mathcal{I}$. Thus $Z_d \subset V_d$, and $z_* = z$. Likewise $t_* = t$. By Lemma 2.6(2), $(h_{0,*})$, $(h_{0,*}, z)$, and $(h_{0,*}, t)$ are compatible generic limits of $(h_{0,i})$, $(h_{0,i}, z)$, and $(h_{0,i}, t)$ along $\mathcal{I}$, with coefficient field $\Bbbk$. Let $\mathcal{A}_\Bbbk := \operatorname{Spec} \Bbbk[[z, t]]$ and $D_* := \frac{1}{m}\operatorname{div}(h_{0,*})$.

By Theorem 2.18 with $n = 2$, $H = \overline{G}$, $\mathfrak{c} = \mathcal{O}$, the tuples $(h_{0,i})$ and $c_1 = \frac{1}{m}$, there exists an infinite subset $\mathcal{I}' \subset \mathcal{I}$ such that (1)–(3) of Theorem 2.18 hold for $i \in \mathcal{I}'$. By Theorem 2.18(1) with $\theta = 1$, since $(\mathbb{A}^2, \frac{1}{m}\operatorname{div}(h_{0,i}))$ is lc near 0 for any i, $(\mathcal{A}_\Bbbk, D_*)$ is lc near 0. By Theorem 2.18(2),

$$\operatorname{mld}((\mathcal{A}_\Bbbk \ni 0, D_*)/\overline{G}) = \beta_*. \tag{D.12}$$

Case 1: $(\mathcal{A}_\Bbbk, D_*)$ is klt near 0. By Theorem 2.18(1), there exists $\theta' > 1$ such that $(\mathbb{A}^2, \frac{\theta'}{m}\operatorname{div}(h_{0,i}))$ is lc near 0 for any $i \in \mathcal{I}'$, and we may assume that θ' is rational. By **Step 4**, $b_i \geq \beta_i$ for any $i \in \mathcal{I}'$ with $\operatorname{mult}_0 g_i \geq N_0(m, \theta', 0)$ and $\operatorname{mult}_0 g_i > 4mM/(\theta' - 1)$, which contradicts $b_i < \beta_i$.

Case 2: $(\mathcal{A}_\Bbbk, D_*)$ is not klt near 0. By Definition 2.2 and (D.12), $a(E, \mathcal{A}_\Bbbk, D_*) \geq e_E\beta_* > 0$ for any prime divisor E over $\mathcal{A}_\Bbbk$ with center the closed point. As $(\mathcal{A}_\Bbbk, D_*)$ is lc but not klt, some prime divisor E over $\mathcal{A}_\Bbbk$ has $a(E, \mathcal{A}_\Bbbk, D_*) = 0$, so the center of E is not the closed point. Since

the local ring of $\mathcal{A}_{\Bbbk}$ at the generic point of a curve is a discrete valuation ring, E is a prime divisor P on $\mathcal{A}_{\Bbbk}$, and P is a component of D_* with coefficient 1. If D_* had two components with coefficient 1, or P were singular at 0, then $a(E_1, \mathcal{A}_{\Bbbk}, D_*) = 2 - \operatorname{mult}_0 D_* \le 0$ for the exceptional prime divisor E_1 of the blow-up of the closed point. Thus P is the unique component of D_* with coefficient 1, P is regular, and $\overline{G}$ preserves P.

Let $\ell \in \{z, t\}$ be a coordinate whose differential does not vanish on the tangent line of P. Then $\ell|_P$ is a uniformizer of P. Let $\mathfrak{p}$ be the ideal of P. Since $\mathfrak{p}$ is principal and $\overline{G}$-invariant, it has a $\overline{G}$-semi-invariant generator u, and $\Bbbk[[z,t]] = \Bbbk[[u,\ell]]$ with $\overline{G}$ acting diagonally on u, ℓ with characters χ_u, χ_ℓ. Write $h_{0,*} = u^m h'$ with $u \nmid h'$, and let j be the integer with $h'(0,\ell) \in \ell^j \Bbbk[[\ell]]^*$. Let r be the order of χ_ℓ on $\overline{G}$, $e' := |\overline{G}|/r = |\operatorname{Ker} \chi_\ell|$, and $\alpha := 1 - \frac{j}{m}$.

We show that

$$\beta_* \le \frac{\alpha}{r}, \quad \text{in particular} \quad 0 \le j < m. \tag{D.13}$$

Let γ_1 be a generator of the cyclic group $\overline{G}$ and ζ a primitive (re')-th root of unity, and write $\chi_u(\gamma_1) = \zeta^c$ and $\chi_\ell(\gamma_1) = \zeta^{e'c'}$ with integers c, c'. Then $\gcd(c', r) = 1$. Since $\operatorname{Ker} \chi_\ell = \langle \gamma_1^r \rangle$ and $\overline{G}$ acts faithfully on $\mathcal{A}_{\Bbbk}$, χ_u is faithful on $\operatorname{Ker} \chi_\ell$, so $\gcd(c, e') = 1$. By (D.8) with $k_1 = e'$ and $k_2 = r$, there exists an integer $n' > je'$ with $\gcd(n', e') = 1$ and $c'n' \equiv c$ modulo r, that is, $\chi_u^{e'} = \chi_\ell^{n'}$. Let E be the exceptional prime divisor of the weighted blow-up of $\mathcal{A}_{\Bbbk}$ with weights (n', e') for (u, ℓ). Then $a(E, \mathcal{A}_{\Bbbk}, 0) = n' + e'$, and the residue field of E is generated over $\Bbbk$ by the residue of $u^{e'} \ell^{-n'}$, on which $\overline{G}$ acts by $\chi_u^{e'} \chi_\ell^{-n'} = 1$. Thus $e_E = |\overline{G}| = re'$. Since $n' > je'$, $\operatorname{mult}_E h' = je'$ and $\operatorname{mult}_E h_{0,*} = mn' + je'$. By Definition 2.2 and (D.12),

$$\beta_* \le \frac{a(E, \mathcal{A}_{\Bbbk}, D_*)}{e_E} = \frac{n' + e' - n' - \frac{j}{m} e'}{re'} = \frac{\alpha}{r}.$$

Since $\beta_* > 0$, $\alpha > 0$, which proves (D.13).

We show that

$$\left(\mathcal{A}_{\Bbbk}, D_* + \alpha \operatorname{div}(\ell)\right) \text{ is lc.} \tag{D.14}$$

Let

$$\Delta := \frac{1}{m} \operatorname{div}(h') + \alpha \operatorname{div}(\ell).$$

Then $D_* + \alpha \operatorname{div}(\ell) = P + \Delta$, $P \not\subset \operatorname{Supp} \Delta$, and the restriction of Δ to $P \cong \operatorname{Spec} \Bbbk[[\ell]]$ is $(\frac{j}{m} + \alpha)[0] = [0]$, so the local intersection number $(\Delta \cdot P)_0$ is 1. Since $(P, \Delta|_P) = (P, [0])$ is lc, (D.14) follows from inversion of adjunction on the regular surface $\mathcal{A}_{\Bbbk}$ [Tan18, Theorem 5.1(1)].

By Theorem 2.18 with $n = 2$, $H = \overline{G}$, $\mathfrak{c} = \mathcal{O}$, the tuples $(h_{0,i}, \ell)$ and $(c_1, c_2) = (\frac{1}{m}, \alpha)$, and by (D.14) and Theorem 2.18(1) with $\theta = 1$, there exists an infinite subset $\mathcal{I}'' \subset \mathcal{I}$ such that

$$\left(\mathbb{A}^2, \frac{1}{m} \operatorname{div}(h_{0,i}) + \alpha \operatorname{div}(\ell)\right) \text{ is lc near 0 for any } i \in \mathcal{I}''.$$

By Lemma D.1(2) with $\theta = 1$, for any $i \in \mathcal{I}''$ with $\operatorname{mult}_0 g_i \ge N_0(m, 1, \alpha)$,

$$\left(W_i, \frac{1}{m} \operatorname{div}(h_i) + \alpha \operatorname{div}(\ell)\right) \text{ is lc near 0.} \tag{D.15}$$

Let F be a prime divisor over W_i with center 0. By (D.15), $a(F, W_i, 0) - \frac{1}{m} \operatorname{mult}_F h_i \ge \alpha \operatorname{mult}_F \ell$. The group G acts on ℓ by χ_ℓ through $G \to \overline{G}$, so ℓ^r is G-invariant, and by Lemma 2.15 for W_i and G, e_F divides $\operatorname{mult}_F(\ell^r) = r \operatorname{mult}_F \ell > 0$. Thus $r \operatorname{mult}_F \ell \ge e_F$. By (D.13),

$$\frac{a(F, W_i, 0) - \frac{1}{m} \operatorname{mult}_F h_i}{e_F} \ge \frac{\alpha \operatorname{mult}_F \ell}{e_F} \ge \frac{\alpha}{r} \ge \beta_* = \beta_i.$$

By Definition 2.2, $b_i \ge \beta_i$, which contradicts $b_i < \beta_i$.

Thus $N(m,M)$ exists. Assume that $\operatorname{mult}_0 g \geq N(m,M)$. By **Step 1**, $h_0 \neq 0$ and $(\mathbb{A}^2, \frac{1}{m}\operatorname{div}(h_0))$ is lc near 0, and

$$b = \beta = \operatorname{mld}\Big(\Big(\mathbb{A}^2 \ni 0, \frac{1}{m}\operatorname{div}(h_0)\Big)/\overline{G}\Big).$$

By (D.6), the prime divisor F of **Step 3** satisfies $(a(F,W,0) - \frac{1}{m}\operatorname{mult}_F h)/e_F \leq b$, so F is a computing lift of b, with $a(F,W,0)/e_F \leq C_m \leq L(m)$ and $\operatorname{mult}_F h/(me_F) \leq C_m \leq L(m)$. By (D.7) and (D.3),

$$q(\tau) \leq \tau b + (1-\tau)C_m \quad \text{for any } \tau \in [0,1], \quad \text{and} \quad b \leq 2. \tag{D.16}$$

This proves the theorem when $xy+g$ and h are polynomials and no monomial of h is divisible by xy.

Step 6. In this step we prove the theorem in general. Let $(Y \ni o)$ be the smooth fourfold germ $(\mathcal{X} \ni o)$ of the completed equivariant presentation $xy+g(z,t)$ of (ii), on whose elements x,y,z,t the group G acts diagonally. Possibly replacing Y with a G-invariant open neighborhood of o, let f be a G-semi-invariant regular function on Y with $W = (f=0)$. Since the ideal of W in $\widehat{\mathcal{O}}_{Y,o}$ is $(xy+g)$, the expansion $\widehat{f}$ of f is $w(xy+g(z,t))$ for a unit w. Since $\widehat{f}$ and $xy+g$ are G-semi-invariant, w is a G-semi-invariant unit, hence G-invariant. The component of character χ_h of any lift of h to Y is a G-semi-invariant lift of h, which we still denote by h. Let $x_1,\dots,x_4$ be G-semi-invariant regular parameters of Y at o lifting the images of x,y,z,t in $\mathfrak{m}_{Y,o}/\mathfrak{m}_{Y,o}^2$, with the characters of x,y,z,t. Then $\widehat{\mathcal{O}}_{Y,o} = \mathbb{C}[[x_1,\dots,x_4]]$, and we let ψ be the G-equivariant automorphism of $\mathbb{C}[[x_1,\dots,x_4]]$ with

$$\psi(x) = x_1, \quad \psi(y) = x_2, \quad \psi(z) = x_3, \quad \text{and} \quad \psi(t) = x_4,$$

and $v := \psi(w)^{-1}$, a G-invariant unit. Then $v\psi(\widehat{f}) = x_1x_2 + g(x_3,x_4)$. By Lemma 4.2(1), b has a computing lift F_1.

Let $\tau \in [0,1)$. By Lemma 4.6 with $n=4$, the number τ of (v), $\mathcal{P} = \{F_1\}$, ψ, v, and $u=1$, for a sufficiently large integer $L' > \operatorname{mult}_0 g$ in place of L there exist

- the germ $(W'_\tau \ni 0)$ with $W'_\tau = (x_1x_2 + g_{<L'}(x_3,x_4) = 0) \subset \mathbb{A}^4$, where $g_{<L'}$ is the truncation of g in degrees less than L', with the diagonal action of G in which $x_1,\dots,x_4$ have the characters of x,y,z,t,
- a G-equivariant isomorphism between the completions of W at o and of W'_τ at 0, and
- a G-semi-invariant polynomial h'_τ with the character of h

satisfying (1)–(3) of Lemma 4.6. Since $L' > \operatorname{mult}_0 g$, $g_{<L'} \neq 0$ and $\operatorname{mult}_0 g_{<L'} = \operatorname{mult}_0 g$. Possibly replacing h'_τ with the polynomial obtained by substituting $-g_{<L'}(x_3,x_4)$ for x_1x_2 repeatedly, which is congruent to h'_τ modulo $x_1x_2 + g_{<L'}(x_3,x_4)$ and is G-semi-invariant with the character of h, we may assume that no monomial of h'_τ is divisible by x_1x_2. The group G is cyclic of order at most M, acts faithfully on W'_τ by the isomorphism of completions, preserves the factors x_1 and x_2, and $(W'_\tau, \frac{1}{m}\operatorname{div}(h'_\tau))$ is lc near 0 by Lemma 4.6(3). Let q'_τ be the radial function of $(W'_\tau \ni 0, G, h'_\tau, m)$ and $b'_\tau := q'_\tau(1)$. By Lemma 4.6(2), the prime divisor F'_1 over W'_τ corresponding to F_1 satisfies

$$\frac{a(F'_1,W'_\tau,0) - \frac{1}{m}\operatorname{mult}_{F'_1} h'_\tau}{e_{F'_1}} = \frac{a(F_1,W,0) - \frac{1}{m}\operatorname{mult}_{F_1} h}{e_{F_1}} = b,$$

so $b'_\tau \leq b$ by Definition 2.2. By Lemma 4.6(1) with $G' = G$ and by (D.16) for W'_τ,

$$q(\tau) = q'_\tau(\tau) \leq \tau b'_\tau + (1-\tau)C_m \leq \min\{b,2\} + C_m(1-\tau).$$

Since q is decreasing (Lemma 4.2(1)), $b \leq q(\tau) \leq 2 + C_m(1-\tau)$ for any $\tau \in [0,1)$, so $b \leq 2$. Moreover, $q(\tau) \leq b + C_m(1-\tau)$ for any $\tau \in [0,1]$. Since W is Gorenstein, K_W is Cartier near o,

and Lemma 4.4 with $\lambda = C_m$ and $t_0 = 0$ implies that b has a computing lift F with

$$\frac{\operatorname{mult}_F h}{m e_F} \le C_m \le L(m) \quad \text{and} \quad \frac{a(F,W,0)}{e_F} \le b + C_m \le C_m + 2 = L(m).$$

The theorem follows. □

Proof of Theorem 13.2. **Step 1.** In this step we treat the factor-preserving case and reduce the exchange case to polynomial data.

Let $N'(m,M) := \max\{N(m,M), N_1\}$ and $L(m,M) := \max\{L(m), 2L_1\}$, where $N(m,M)$ and $L(m)$ are the constants of Theorem 13.1 and N_1 and L_1 are those of Claim D.2 below. The change of coordinates in [Liu26b, Definition 8.1] uses only that W is klt of quadratic rank 2 and that G is abelian. Possibly changing the presentation in this way, we may assume that G acts diagonally on z,t and monomially on $\{x,y\}$. This does not change $\operatorname{mult}_0 g$, which is the split radical order of W [Liu26b, Definition 8.1], so that still $\operatorname{mult}_0 g \ge N'(m,M)$. If G preserves the factors x and y, then Theorem 13.1(1) yields a computing lift F of $q(1)$ with $a(F,W,0)/e_F \le L(m) \le L(m,M)$.

Assume that some element of G exchanges the factors. Then the second case of [Liu26b, Definition 8.1] occurs before the replacement of x,y by $x \pm \sqrt{-1}y$ there: there exist G-eigen formal coordinates X,Y,z,t in which the completed equation is $X^2+Y^2+g(z,t)$, and $\chi_X \ne \chi_Y$, since otherwise G would act diagonally on $X \pm \sqrt{-1}Y$ and preserve the factors. Let

$$x := X + \sqrt{-1}Y \quad \text{and} \quad y := X - \sqrt{-1}Y,$$

so that $xy = X^2+Y^2$. Since X^2+Y^2 is semi-invariant, $\chi_Y^2 = \chi_X^2$, so $\chi_Y(\sigma) = \pm\chi_X(\sigma)$ for any $\sigma \in G$. The elements with $\chi_Y(\sigma) = \chi_X(\sigma)$ multiply x and y by $\chi_X(\sigma)$, and those with $\chi_Y(\sigma) = -\chi_X(\sigma)$ map x and y to $\chi_X(\sigma)y$ and $\chi_X(\sigma)x$.

Claim D.2. *There exist $N_1 \ge 3$ and $L_1 > 0$, depending only on m and M, satisfying the following. Let $g \in \mathbb{C}[z,t]$ with $\operatorname{mult}_0 g \ge N_1$, and assume that $W := (xy + g(z,t) = 0) \subset \mathbb{A}^4$ has an isolated terminal singularity at 0. Let G be a finite cyclic group with $|G| \le M$, acting faithfully on W by linear automorphisms of $\mathbb{A}^4$, diagonally on z,t and monomially on $\{x,y\}$, such that some element of G exchanges the factors x and y. Let h be a G-semi-invariant polynomial, non-zero on W, such that no monomial of h is divisible by xy and $(W, \frac{1}{m}\operatorname{div}(h))$ is lc near 0. Let q be the radial function of $(W \ni 0, G, h, m)$. Then $q(1)$ has a computing lift F with*

$$\frac{a(F,W,0)}{e_F} \le L_1.$$

Proof of Claim D.2. **Step A.** In this step we set up a contradicting sequence and exclude the case of a klt plane limit.

Suppose to the contrary that no such N_1 and L_1 exist. Then for each $i \ge 1$ there exist $g_i \in \mathbb{C}[z,t]$, $W_i := (xy + g_i = 0) \subset \mathbb{A}^4$, a group G_i and a polynomial h_i satisfying the hypotheses of Claim D.2 for (g,W,G,h) with $N_1 = i$, such that

$$n_i := \operatorname{mult}_0 g_i \ge i \quad \text{and} \quad \frac{a(F,W_i,0)}{e_F} > i$$

for any computing lift F of $b_i := q_i(1)$, where q_i is the radial function of $(W_i \ni 0, G_i, h_i, m)$. Possibly replacing (x,y) with $(\varrho x, \varrho^{-1}y)$ for some $\varrho \in \mathbb{C}^*$, which preserves $xy + g_i$ and the condition on the monomials of h_i, we may assume that a generator of G_i, which exchanges the factors since the elements preserving them form a proper subgroup, maps x to y. Since $|G_i| \le M$, the pair consisting of the representation of G_i on $\mathbb{A}^4$ and the character of h_i then takes finitely many values. Possibly passing to a subsequence, we may assume that $G_i = G$, the representation and $\chi_{h_i} = \chi_h$ do not depend on i, and $n_i \ge \max\{N(m,M), N_0(m,1,0)\}$ for any i, where N_0 is the integer of Lemma D.1. Let $H \subset G$ be the subgroup of the elements preserving x and y, so that $[G:H] = 2$, and let $\overline{G} \supset \overline{H}$ be the images of $G \supset H$ in $GL_2(\mathbb{C})$ acting on the (z,t)-plane.

Let $h_{0,i} := h_i(0,0,z,t)$. Since h_i is non-zero on W_i, h_i is not divisible by $xy+g_i$. By Lemma D.1(1),

$$h_{0,i} \neq 0, \qquad (\mathbb{A}^2, \tfrac{1}{m}\operatorname{div}(h_{0,i})) \text{ is lc near } 0, \quad \text{and} \quad \operatorname{mult}_0 h_{0,i} \leq 2m. \tag{D.17}$$

The group H is cyclic, $|H| \leq M$, and H acts faithfully on W_i, diagonally on x,y,z,t, preserving the factors. By Theorem 13.1(2) applied to (W_i, H, h_i), where $xy+g_i$ and h_i are polynomials and no monomial of h_i is divisible by xy,

$$\beta_i := \operatorname{mld}((W_i \ni 0, \tfrac{1}{m}\operatorname{div}(h_i))/H) = \operatorname{mld}((\mathbb{A}^2 \ni 0, \tfrac{1}{m}\operatorname{div}(h_{0,i}))/\overline{H}),$$

and, by Theorem 13.1(1), β_i has a computing lift F_i with $a(F_i, W_i, 0)/|I_H(F_i)| \leq L(m)$. Since $I_H(F_i) = I_G(F_i) \cap H$ and $a(F_i, W_i, \frac{1}{m}\operatorname{div}(h_i)) \geq 0$,

$$b_i \leq \frac{a(F_i, W_i, \frac{1}{m}\operatorname{div}(h_i))}{e_{F_i}} \leq \frac{a(F_i, W_i, \frac{1}{m}\operatorname{div}(h_i))}{|I_H(F_i)|} = \beta_i \quad \text{and} \quad \frac{a(F_i, W_i, 0)}{e_{F_i}} \leq L(m). \tag{D.18}$$

If $b_i \geq \beta_i$, then F_i is a computing lift of b_i by (D.18), which contradicts the choice of the sequence for $i > L(m)$. It therefore suffices to prove that $b_i \geq \beta_i$ for infinitely many i. If $\beta_i = 0$, then $b_i \geq 0 = \beta_i$. Thus we may assume that $\beta_i > 0$ for any i.

By (D.17) and [Liu26b, Construction-Proposition 4.1(a)], possibly passing to an infinite subset of indices, the $\overline{G}$-semi-invariant polynomials $h_{0,i}$ have a compatible generic limit $h_* \in \Bbbk[[z,t]]$ with coefficient field $\Bbbk$ (Definition 2.5 with $n=2$, $N=1$ and the group $\overline{G}$). The limit h_* is also a compatible generic limit for the subgroup $\overline{H} \subset \overline{G}$. Let

$$C_i := \tfrac{1}{m}\operatorname{div}(h_{0,i}) \quad \text{and} \quad C_* := \tfrac{1}{m}\operatorname{div}(h_*).$$

By Theorem 2.18(1)(2) (with the group $\overline{H}$, $\mathfrak{c} = \mathcal{O}$, and $\theta = 1$ in (1)), possibly passing to an infinite subset of indices, $(\mathcal{A}_\Bbbk, C_*)$ is lc near 0 and $\beta_i = \beta := \operatorname{mld}((\mathcal{A}_\Bbbk \ni 0, C_*)/\overline{H})$ for any i. In particular $\beta > 0$.

Assume that $(\mathcal{A}_\Bbbk, C_*)$ is klt near 0. By Theorem 2.18(1), possibly decreasing the number given there, there exists a rational number $\theta' > 1$ such that $(\mathbb{A}^2, \theta' C_i)$ is lc near 0 for any i. By Lemma D.1(2) with θ' in place of θ and $\alpha = 0$, $(W_i, \frac{\theta'}{m}\operatorname{div}(h_i))$ is lc near 0 when $n_i \geq N_0(m, \theta', 0)$. For such i and a computing lift F of b_i (Lemma 4.2(1)), $a(F, W_i, 0) \geq \theta' \operatorname{mult}_F h_i/m$, so

$$b_i e_F = a(F, W_i, 0) - \frac{\operatorname{mult}_F h_i}{m} \geq \Big(1 - \frac{1}{\theta'}\Big) a(F, W_i, 0), \quad \text{hence} \quad \frac{a(F, W_i, 0)}{e_F} \leq \frac{\theta' b_i}{\theta' - 1} \leq \frac{\theta' \beta}{\theta' - 1}$$

by (D.18). This contradicts the choice of the sequence for $i \gg 0$. Thus $(\mathcal{A}_\Bbbk, C_*)$ is lc and not klt near 0.

Step B. In this step we bring the marks h_i to a weighted normal form.

By [Liu26b, Definition-Proposition 5.7] (case (a), $n=2$, $\mathfrak{c} = \mathcal{O}$), since $\beta > 0$ and $(\mathcal{A}_\Bbbk, C_*)$ is not klt near 0, C_* has exactly one floor Σ: Σ is a smooth curve through 0 with coefficient one in C_*, and

$$\beta = \frac{1 - \operatorname{mult}_0((C_* - \Sigma)|_\Sigma)}{n_\Sigma},$$

where n_Σ is the order of the image of $\overline{H}$ in the automorphism group of $T_0\Sigma$. Since C_* is $\overline{G}$-invariant and Σ is unique, Σ is $\overline{G}$-invariant. As $\overline{G}$ is diagonal, there exist linear $\overline{G}$-eigenforms z_1, z_2 in z,t with coefficients in $\Bbbk$ such that z_1 vanishes on $T_0\Sigma$ and z_2 does not. Then

$$\Sigma = (z_1 = \varphi(z_2)), \qquad \varphi \in \Bbbk[[z_2]], \ \operatorname{mult}_0 \varphi \geq 2,$$

and φ is semi-invariant with character χ_{z_1}, because Σ is invariant and $z_2|_\Sigma$ is a parameter of Σ. Let $\zeta := z_1 - \varphi(z_2)$. Since Σ has coefficient one in C_*, there exist $h' \in \Bbbk[[\zeta, z_2]]$ not divisible by ζ and an integer $k \geq 0$ with

$$h_* = \zeta^m h', \qquad h'(0, z_2) = z_2^k \cdot (\text{unit}).$$

Then $\mathrm{mult}_0((C_* - \Sigma)|_\Sigma) = k/m$, and n_Σ is the order n_2 of χ_{z_2} on H. Let $\alpha := 1 - k/m$. We obtain

$$\beta = \frac{\alpha}{n_2}, \qquad 0 \le k < m, \qquad 0 < \alpha \le 1. \tag{D.19}$$

Since $[G : H] = 2$, the order of χ_{z_2} on G is n_2 or $2n_2$, so

$$z_2^{n_2} \text{ is } H\text{-invariant} \quad \text{and} \quad z_2^{2n_2} \text{ is } G\text{-invariant}. \tag{D.20}$$

Let

$$r := k + 1 \quad \text{and} \quad d := mr + k,$$

and let ζ, z_2 have weights $r, 1$. Any monomial of h' divisible by ζ has weight at least $r > k$, and the ζ-free part of h' is z_2^k times a unit. Hence

$$h_* = c\zeta^m z_2^k + (\text{terms of weight} > d), \qquad c \in \Bbbk\backslash\{0\}. \tag{D.21}$$

Let $\varphi_{\le d}$ be the truncation of φ in degrees at most d. Since $\varphi - \varphi_{\le d}$ has weight at least $d+1 > r$, (D.21) holds with ζ replaced by $z_1 - \varphi_{\le d}(z_2)$. From now on $\zeta := z_1 - \varphi_{\le d}(z_2)$. A monomial of degree j in z, t is a sum of monomials of weight at least j in ζ, z_2, so the part of weight at most d of h_* in ζ, z_2 depends only on $\mathrm{jet}_{d+1}(h_*)$. Thus the following conditions are finitely many polynomial identities with complex coefficients in $\mathrm{jet}_{d+1}(h_*)$ and in the coefficients of z_1, z_2, of the inverse linear change of coordinates, of $\varphi_{\le d}$, and c, together with $c \neq 0$:

(1) z_1, z_2 are linear eigenforms with the characters χ_{z_1}, χ_{z_2}, and the inverse matrix is inverse to the matrix of (z_1, z_2).
(2) $\varphi_{\le d}$ only contains monomials z_2^j with $\chi_{z_2}^j = \chi_{z_1}$.
(3) In the coordinates $\zeta = z_1 - \varphi_{\le d}(z_2)$ and z_2, the terms of h_* of weight less than d vanish and the part of weight d is $c\zeta^m z_2^k$.

By Lemma 2.7, for all but finitely many i there exist complex values of the coefficients of z_1, z_2, of the inverse linear change of coordinates, and of $\varphi_{\le d}$, and $c_i \neq 0$, such that conditions (1)–(3) hold with $h_{0,i}$ in place of h_* and c_i in place of c. For such i, the change of coordinates $(z, t) \to (\zeta, z_2)$ is a G-equivariant polynomial automorphism of $\mathbb{A}^2$ with invertible linear part, and the characters of ζ and z_2 are χ_{z_1} and χ_{z_2}. We use the coordinates x, y, ζ, z_2 on $\mathbb{A}^4$ from now on, so that t no longer occurs. In the coordinates x, y, ζ, z_2, $g_i \in \mathbb{C}[\zeta, z_2]$, $\mathrm{mult}_0 g_i = n_i$, no monomial of h_i is divisible by xy, and $h_i - h_{0,i}$ consists of monomials divisible by x or y. Let x, y, ζ, z_2 have weights

$$(d+1,\ d+1,\ r,\ 1).$$

Since any monomial divisible by x or y has weight at least $d+1$, and any monomial of g_i has weight at least n_i, we obtain

$$h_i = c_i\zeta^m z_2^k + (\text{terms of weight} > d) \quad \text{and} \quad g_i = (\text{terms of weight} \ge n_i). \tag{D.22}$$

Since d does not depend on i and $n_i \ge i$, possibly passing to a subsequence, we may assume that $n_i > 2d + 2$ for any i.

Step C. In this step we introduce two charts of the weighted blow-up with weights $(d+1, d+1, r, 1)$ and compare log discrepancies on them.

Fix i and drop the index: $W = (xy + g = 0)$, h, $c \neq 0$, and $\mathrm{mult}_0 g > 2d + 2$ satisfy (D.22). The z_2-chart is the morphism $\mathbb{A}^4_{X,Y,v,\ell} \to \mathbb{A}^4$ given by

$$(x, y, \zeta, z_2) = (\ell^{d+1}X,\ \ell^{d+1}Y,\ \ell^r v,\ \ell).$$

By (D.22), $g(\ell^r v, \ell) = \ell^{2d+2}\psi_2(v, \ell)$, so that $xy + g = \ell^{2d+2}(XY + \psi_2)$ on $\mathbb{A}^4_{X,Y,v,\ell}$, and $h = \ell^d h_2$ with

$$\psi_2 \in \ell\mathbb{C}[v, \ell]\backslash\{0\}, \qquad h_2 \in \mathbb{C}[X, Y, v, \ell], \quad \text{and} \quad h_2(X, Y, v, 0) = cv^m.$$

Let $V_2 := (XY + \psi_2 = 0) \subset \mathbb{A}^4$. The variety V_2 is birational to W via the z_2-chart, and it contains $\Gamma := (X = Y = \ell = 0)$. The ζ-chart is the morphism $\mathbb{A}^4_{X,Y,w,\xi} \to \mathbb{A}^4$ given by

$$(x, y, \zeta, z_2) = (\xi^{d+1}X,\ \xi^{d+1}Y,\ \xi^r,\ \xi w).$$

Similarly $g(\xi^r, \xi w) = \xi^{2d+2}\psi_1(w, \xi)$, so that $xy + g = \xi^{2d+2}(XY + \psi_1)$ on $\mathbb{A}^4_{X,Y,w,\xi}$, and $h = \xi^d h_1$ with

$$\psi_1 \in \xi\mathbb{C}[w, \xi]\backslash\{0\}, \qquad h_1 \in \mathbb{C}[X, Y, w, \xi], \quad \text{and} \quad h_1(X, Y, w, 0) = cw^k.$$

Let $V_1 := (XY + \psi_1 = 0)$. The ζ-chart maps V_1 to W. The projection $W \to \mathbb{A}^3_{y,\zeta,z_2}$ is birational, as $x = -g/y$ on W, and the composite $V_1 \to W \to \mathbb{A}^3_{y,\zeta,z_2}$ is $(Y, w, \xi) \to (\xi^{d+1}Y, \xi^r, \xi w)$ on the open set $(Y \neq 0)$, where $X = -\psi_1/Y$. Hence $V_1 \to W$ is generically finite of degree r.

Claim D.3. *Let $n' \in \{3, 4\}$, and $\psi \in \mathbb{C}[u_1, \dots, u_{n'-2}]$ a non-zero polynomial. Then*

$$V := (XY + \psi = 0) \subset \mathbb{A}^{n'}$$

is a normal integral variety with klt singularities.

Proof. Since $XY + \psi$ is irreducible and $\dim \operatorname{Sing} V \leq n' - 3$, V is normal. Moreover, V is a surface with Du Val singularities of type A if $n' = 3$, and a threefold whose general hyperplane section through any singular point is such a surface if $n' = 4$, so V is klt by inversion of adjunction. □

By Claim D.3, V_1 and V_2 are normal and klt. Let $J \subset G$ be a subgroup. Since x, y, ζ, z_2 are permuted up to scalars by G and the weights of x and y are equal, J acts linearly on $\mathbb{A}^4_{X,Y,v,\ell}$, preserving V_2, diagonally on v, ℓ and monomially on $\{X, Y\}$, and the z_2-chart is J-equivariant. Let

$$J' := \{(\sigma, \upsilon) \in J \times \mathbb{C}^\times \mid \upsilon^r = \chi_\zeta(\sigma)\}.$$

It acts on $\mathbb{A}^4_{X,Y,w,\xi}$ by $\xi \to \upsilon\xi$, $w \to \chi_{z_2}(\sigma)\upsilon^{-1}w$, and on (X, Y) by the action of σ on (x, y) multiplied by υ^{-d-1}. The ζ-chart is equivariant for $J' \to J$, $(\sigma, \upsilon) \to \sigma$, whose kernel μ_r acts on ξ faithfully. Thus J' is a finite abelian group of order $r|J|$ acting faithfully on V_1, and $V_1 \to W$ induces a birational map $V_1/J' \dashrightarrow W/J$. An element τ of J or J' exchanges the factors X and Y if and only if the image of τ in G exchanges x and y.

The residue $\Omega := x^{-1}dx \wedge d\zeta \wedge dz_2$ generates ω_W. Up to sign, Ω pulls back to $\ell^r\Omega_2$ on V_2 and to $r\xi^r\Omega_1$ on V_1, where $\Omega_2 := X^{-1}dX \wedge dv \wedge d\ell$ and $\Omega_1 := X^{-1}dX \wedge d\xi \wedge dw$ generate ω_{V_2} and ω_{V_1}. Since $\frac{d}{m} - r = \frac{k}{m}$, the $\mathbb{Q}$-divisors

$$B_2 := \tfrac{1}{m}\operatorname{div}(h_2) + \tfrac{k}{m}\operatorname{div}(\ell) \text{ on } V_2 \quad \text{and} \quad B_1 := \tfrac{1}{m}\operatorname{div}(h_1) + \tfrac{k}{m}\operatorname{div}(\xi) \text{ on } V_1$$

satisfy $B_2 \geq 0$, $B_1 \geq 0$, and the following. For a prime divisor F over W, regarded as a prime divisor over V_2 through the birational z_2-chart,

$$a(F, W, \tfrac{1}{m}\operatorname{div}(h)) = a(F, V_2, B_2). \tag{D.23}$$

Let F' be a prime divisor over V_1 whose restriction is F, with ramification index ρ (Definition 2.2). Then for $J \subset G$,

$$a(F', V_1, B_1) = \rho a(F, W, \tfrac{1}{m}\operatorname{div}(h)) \quad \text{and} \quad |I_{J'}(F')| = \rho|I_J(F)|. \tag{D.24}$$

The first equality follows from the ramification formula $\rho\operatorname{mult}_F\Omega + \rho - 1 = \operatorname{mult}_{F'}(r\xi^r\Omega_1)$ and from $\rho\operatorname{mult}_F h = d\operatorname{mult}_{F'}\xi + \operatorname{mult}_{F'}h_1$. For the second one, the ramification index of F' over W/J equals $|I_{J'}(F')|$, as $V_1/J' \dashrightarrow W/J$ is birational, and also equals $\rho|I_J(F)|$, as ramification indices are multiplicative along $V_1 \to W \to W/J$. In particular (V_2, B_2) and (V_1, B_1) are lc near the inverse images $(\ell = 0)$ and $(\xi = 0)$ of $0 \in W$, and

$$\frac{a(F, W, \frac{1}{m}\operatorname{div}(h))}{|I_J(F)|} = \frac{a(F', V_1, B_1)}{|I_{J'}(F')|}. \tag{D.25}$$

Step D. In this step we locate the centers of the prime divisors whose inertia exchanges the factors.

Let F be a prime divisor over W with center 0 such that $J := I_G(F) \not\subset H$, and let $\sigma \in J$ exchange the factors. Since $\sigma^* x$ is a non-zero multiple of y and J preserves mult_F, $\mathrm{mult}_F x = \mathrm{mult}_F y$. Since $xy = -g$ on W and $\mathrm{mult}_F \zeta, \mathrm{mult}_F z_2 > 0$,

$$2\,\mathrm{mult}_F x = \mathrm{mult}_F g \geq \mathrm{mult}_0 g \cdot \min\{\mathrm{mult}_F \zeta, \mathrm{mult}_F z_2\} \geq \mathrm{mult}_0 g \cdot \min\{\mathrm{mult}_F \zeta/r, \mathrm{mult}_F z_2\}.$$

As $\mathrm{mult}_0 g > 2d+2$,

$$\frac{\mathrm{mult}_F x}{d+1} = \frac{\mathrm{mult}_F y}{d+1} > \min\{\mathrm{mult}_F \zeta/r,\ \mathrm{mult}_F z_2\}. \tag{D.26}$$

If $\mathrm{mult}_F z_2 \leq \mathrm{mult}_F \zeta/r$, then $\mathrm{mult}_F v \geq 0$, and $\mathrm{mult}_F X, \mathrm{mult}_F Y, \mathrm{mult}_F \ell > 0$ by (D.26). Hence the center of F on V_2 exists and is contained in Γ. If $\mathrm{mult}_F \zeta/r < \mathrm{mult}_F z_2$, then for any prime divisor F' over V_1 whose restriction is F, $\mathrm{mult}_{F'} \xi, \mathrm{mult}_{F'} w, \mathrm{mult}_{F'} X, \mathrm{mult}_{F'} Y > 0$ by (D.26), so the center of F' on V_1 is 0. Thus one of the following holds.

(A0) The center of F on V_2 is 0.
(A1) The center of F on V_2 is the point $p = (0,0,v_0,0)$ with $v_0 \neq 0$.
(A2) The center of F on V_2 is the generic point of Γ.
(B) The center of F' on V_1 is 0.

In each case the center is invariant under J (resp. J'), as J is contained in the decomposition group of F.

Step E. In this step we prove that $a(F, W, \frac{1}{m}\mathrm{div}(h))/e_F \geq \beta$ for F as in **Step D**. We need two claims.

Claim D.4. *Let $n' \in \{3,4\}$, and let J_0 be a finite abelian group acting faithfully on $\mathbb{A}^{n'}$ by linear automorphisms in coordinates $X, Y, u_1, \dots, u_{n'-2}$, such that any element of J_0 multiplies each u_j by a scalar and either multiplies X and Y by scalars or maps them to scalar multiples of Y and X. Let $\psi \in u_1\mathbb{C}[u_1, \dots, u_{n'-2}]\backslash\{0\}$, let $\varpi \in \mathbb{C}[X, Y, u_1, \dots, u_{n'-2}]$ with $XY + \psi$ and ϖ semi-invariant and*

$$\varpi_0 := \varpi|_{u_1=0} \in \mathbb{C}[u_2, \dots, u_{n'-2}]\backslash\{0\},$$

and let $0 \leq k < m$ be an integer. Let $V := (XY + \psi = 0)$ and

$$B := \tfrac{1}{m}\mathrm{div}(\varpi|_V) + \tfrac{k}{m}\mathrm{div}(u_1|_V),$$

and assume that J_0 acts faithfully on V and (V, B) is lc near 0. Then

$$\mathrm{mld}((V \ni 0, B)/J_0) \geq \mathrm{mld}\Big(\Big(\mathbb{A}^{n'} \ni 0, \mathrm{div}(X) + \mathrm{div}(Y) + \tfrac{1}{m}\mathrm{div}(\varpi_0) + \tfrac{k}{m}\mathrm{div}(u_1)\Big)/J_0\Big).$$

Proof. After a linear change of X, Y, the abelian group J_0 acts diagonally. By Claim D.3 and Theorem 2.17(1) (omitting the factor $(u_1)^{k/m}$ if $k = 0$),

$$\mathrm{mld}((V \ni 0, B)/J_0) = \mathrm{mld}\Big(\Big(\mathbb{A}^{n'} \ni 0, \mathrm{div}(XY + \psi) + \tfrac{1}{m}\mathrm{div}(\varpi) + \tfrac{k}{m}\mathrm{div}(u_1)\Big)/J_0\Big).$$

For $\gamma \in \mathbb{A}^1_\gamma$ let $\psi^\gamma := \psi(\gamma u_1, u_2, \dots, u_{n'-2})$ and $\varpi^\gamma := \varpi(X, Y, \gamma u_1, u_2, \dots, u_{n'-2})$, so that $\psi^0 = 0$ and $\varpi^0 = \varpi_0$. For $\gamma \neq 0$, the automorphism $u_1 \to \gamma u_1$ commutes with J_0, fixes 0 and $\mathrm{div}(u_1)$, and pulls $XY + \psi$ and ϖ back to $XY + \psi^\gamma$ and ϖ^γ. Hence the quotient value at 0 of the pair

$$D_\gamma := \mathrm{div}(XY + \psi^\gamma) + \tfrac{1}{m}\mathrm{div}(\varpi^\gamma) + \tfrac{k}{m}\mathrm{div}(u_1)$$

does not depend on $\gamma \neq 0$, and $(\mathbb{A}^{n'}, D_0)$ is the pair on the right-hand side of the claim. Let $J_1 \subset J_0$ be the subgroup generated by pseudoreflections. Since J_1 is diagonal, $\mathbb{A}^{n'}/J_1$ is an affine space with coordinates powers of the coordinates, and J_0/J_1 acts on it freely in codimension 1. Hence $Z := \mathbb{A}^{n'}/J_0$ satisfies the hypothesis of Theorem 2.16. Let $\pi : \mathbb{A}^{n'} \to Z$ be the quotient, let $\Delta_\pi \geq 0$ be the divisor with $K_{\mathbb{A}^{n'}} = \pi^*(K_Z + \Delta_\pi)$, and let $\kappa_1, \kappa_2, \kappa_3$ be the orders of the characters of $XY + \psi^\gamma$, ϖ^γ, and u_1, which do not depend on γ. Then $(XY + \psi^\gamma)^{\kappa_1}$ and $(\varpi^\gamma)^{\kappa_2}$

define regular functions Φ_1, Φ_2 on $Z \times \mathbb{A}^1_\gamma$, and $u_1^{\kappa_3}$ defines a regular function Φ_3 on Z. By the ramification formula for the finite morphism π, for any γ the quotient value at 0 of D_γ equals

$$\mathrm{mld}\Big(Z \ni \pi(0),\ \Delta_\pi + \tfrac{k}{m\kappa_3}\operatorname{div}(\Phi_3),\ \mathfrak{A}_\gamma\Big), \qquad \mathfrak{A} := (\Phi_1)^{1/\kappa_1}(\Phi_2)^{1/(m\kappa_2)}, \tag{D.27}$$

where

$$\Delta_\pi + \tfrac{k}{m\kappa_3}\operatorname{div}(\Phi_3) \geq 0$$

is $\mathbb{R}$-Cartier since Z is $\mathbb{Q}$-factorial. By Theorem 2.16, the value (D.27) is a lower semi-continuous function of γ, so the value at $\gamma = 0$ is at most the value at $\gamma \neq 0$. The claim follows. □

Claim D.5. *Let $n' \in \{3, 4\}$, and let J_0 be a finite abelian group acting faithfully on $\mathbb{A}^{n'}$ by linear automorphisms in coordinates $X, Y, u_1, \dots, u_{n'-2}$, such that any element of J_0 multiplies each u_j by a scalar and either multiplies X and Y by scalars or maps them to scalar multiples of Y and X. Let E be a prime divisor over $\mathbb{A}^{n'}$ with center 0, $I := I_{J_0}(E)$, and*

$$\delta_E := a(E, \mathbb{A}^{n'}, 0) - \operatorname{mult}_E X - \operatorname{mult}_E Y - \sum_{j=1}^{n'-2} \operatorname{mult}_E u_j.$$

(1) δ_E is a non-negative integer.
(2) If ϑ is an I-invariant regular function vanishing at 0, then $\operatorname{mult}_E \vartheta \in |I| \cdot \mathbb{N}^+$.
(3) If an element of I exchanges X and Y, then $|I|$ is even and $\delta_E \geq |I|/2$.

Proof. (1) holds as $(\mathbb{A}^{n'}, \operatorname{div}(XY\prod_j u_j))$ is lc. The group I is cyclic, acts trivially on the residue field of E, and acts faithfully on the conormal line of the valuation ring of E through a character χ_E, so that χ_E has order $|I|$ [Ser79, Chapter IV, Proposition 7 and Corollary 2], applied to the completion of the valuation ring of E. If ϑ is semi-invariant with character χ_ϑ, comparing the leading terms of $\sigma^*\vartheta$ and ϑ along E yields $\chi_\vartheta(\sigma) = \chi_E(\sigma)^{\operatorname{mult}_E \vartheta}$ for any $\sigma \in I$. This implies (2). For (3), let

$$\omega := d\log X \wedge d\log Y \wedge d\log u_1 \wedge \cdots \wedge d\log u_{n'-2}.$$

Then $\sigma^*\omega = \omega$ if σ preserves X and Y, and $\sigma^*\omega = -\omega$ if σ exchanges them. On the other hand, $\omega = e^{\delta_E}\omega'$ for a local equation e of E and a logarithmic form ω' along E with non-zero residue. Comparing residues yields $\sigma^*\omega = \chi_E(\sigma)^{\delta_E}\omega$ for $\sigma \in I$. Hence $\sigma \to \chi_E(\sigma)^{\delta_E}$ is the non-trivial homomorphism $I \to \{\pm 1\}$, so $|I|$ is even and $\delta_E \equiv |I|/2$ modulo $|I|$. Since $\delta_E \geq 0$, $\delta_E \geq |I|/2$. □

Now let F and $J = I_G(F)$ be as in **Step D**, so that $e_F = |J| = |I_J(F)|$. By (D.23) in cases (A0)–(A2), by (D.25) in case (B), by Claim D.4, and by the definition of the quotient value, it suffices to prove that $a(E, \mathbb{A}^{n'}, R)/|I| \geq \beta$ for the reference $\mathbb{Q}$-divisor R of each case below, where E runs through the prime divisors over the ambient affine space $\mathbb{A}^{n'}$ with center the given point, and $I := I_{J_0}(E)$, where J_0 is the acting group.

Case (A0). Apply Claim D.4 to J acting on $\mathbb{A}^4_{X,Y,v,\ell}$, with $u_1 = \ell$, $u_2 = v$, $\varpi = h_2$, $\varpi_0 = cv^m$: the quotient value $\mathrm{mld}((V_2 \ni 0, B_2)/J)$ is at least that of the reference $\mathbb{Q}$-divisor

$$\operatorname{div}(X) + \operatorname{div}(Y) + \operatorname{div}(v) + \tfrac{k}{m}\operatorname{div}(\ell),$$

whose log discrepancy at E is $\delta_E + \alpha \operatorname{mult}_E \ell$. If $I \subset H$, then $\ell^{n_2} = z_2^{n_2}$ is I-invariant by (D.20), so $\operatorname{mult}_E \ell \geq |I|/n_2$ by Claim D.5(2) and

$$\frac{\delta_E + \alpha \operatorname{mult}_E \ell}{|I|} \geq \frac{\alpha}{n_2} = \beta.$$

Otherwise ℓ^{2n_2} is I-invariant, and by Claim D.5(2)(3) and $\alpha \leq 1 \leq n_2$,

$$\frac{\delta_E + \alpha \operatorname{mult}_E \ell}{|I|} \geq \frac{1}{2} + \frac{\alpha}{2n_2} \geq \frac{\alpha}{n_2} = \beta. \tag{D.28}$$

Case (B). Apply Claim D.4 to J' acting on $\mathbb{A}^4_{X,Y,w,\xi}$, with $u_1=\xi$, $u_2=w$, $\varpi=h_1$, $\varpi_0=cw^k$. The reference $\mathbb{Q}$-divisor

$$\operatorname{div}(X)+\operatorname{div}(Y)+\tfrac{k}{m}\operatorname{div}(w)+\tfrac{k}{m}\operatorname{div}(\xi)$$

has log discrepancy $\delta_E+\alpha\operatorname{mult}_E(\xi w)$ at E. Since $\xi w=z_2$, the function $(\xi w)^{n_2}$ is invariant under the elements of J' over H, which are those preserving X and Y, and $(\xi w)^{2n_2}$ is J'-invariant. The argument of Case (A0) with ξw in place of ℓ yields the bound β.

Case (A1). Since J fixes p and acts on v by a scalar, $v-v_0$ is J-invariant. Apply Claim D.4 at p with $u_1=\ell$, $u_2=v-v_0$, $\varpi=h_2$. Here $\varpi_0=cv^m$ is a unit near p. The reference $\mathbb{Q}$-divisor

$$\operatorname{div}(X)+\operatorname{div}(Y)+\tfrac{k}{m}\operatorname{div}(\ell)$$

has log discrepancy $\delta_E+\operatorname{mult}_E(v-v_0)+\alpha\operatorname{mult}_E\ell$ at E, and

$$\frac{\delta_E+\operatorname{mult}_E(v-v_0)+\alpha\operatorname{mult}_E\ell}{|I|}\geq\frac{\operatorname{mult}_E(v-v_0)}{|I|}\geq 1\geq\beta$$

by Claim D.5(1)(2) and (D.19).

Case (A2). The group J acts trivially on the residue field of F, and $v|_F$ is the pullback of the non-constant function $v|_\Gamma$. Hence v is J-invariant and J fixes Γ pointwise. Let $\Bbbk_v$ be an algebraic closure of $\mathbb{C}(v)$, and let $\overline{\eta}$ be the image of the generic point of Γ in V_2/J. By the formula for crepant quotients [Liu26a, Lemma 2.6(2)],

$$\frac{a(F,V_2,B_2)}{|J|}=\frac{a(F,V_2,B_2)}{|I_J(F)|}\geq\operatorname{mld}(V_2/J\ni\overline{\eta},B_{2,V_2/J}),$$

where $K_{V_2}+B_2$ is the pullback of $K_{V_2/J}+B_{2,V_2/J}$. By [Liu26b, Lemma 3.22(2)(3)] with J, Γ, v, $\mathbb{C}(v)$, and $\Bbbk_v$ in place of H, C, t, k, and Ω, the right-hand side equals $\operatorname{mld}((V_v\ni 0,B_v)/J)$, where

$$V_v:=(XY+\psi_2(v,\ell)=0)\subset\mathbb{A}^3_{X,Y,\ell}\text{ over }\Bbbk_v\quad\text{and}\quad B_v:=\tfrac{1}{m}\operatorname{div}(h_2)+\tfrac{k}{m}\operatorname{div}(\ell).$$

Since (V_2,B_2) is lc near the inverse image $(\ell=0)$ of 0, which contains Γ, $\operatorname{mld}((V_v\ni 0,B_v)/J)\geq 0$, so (V_v,B_v) is lc near 0. It therefore suffices to prove $\operatorname{mld}((V_v\ni 0,B_v)/J)\geq\beta$. The field $\Bbbk_v$ is algebraically closed of characteristic zero with transcendence degree over $\mathbb{Q}$ equal to that of $\mathbb{C}$, hence isomorphic to $\mathbb{C}$. Transporting all data along such an isomorphism, Claims D.3–D.5, Theorem 2.16, and Theorem 2.17 hold over $\Bbbk_v$. Apply Claim D.4 with $n'=3$, $u_1=\ell$, $\varpi=h_2$. Here $\varpi_0=cv^m\in\Bbbk_v\backslash\{0\}$ is a unit. The reference $\mathbb{Q}$-divisor

$$\operatorname{div}(X)+\operatorname{div}(Y)+\tfrac{k}{m}\operatorname{div}(\ell)$$

has log discrepancy $\delta_E+\alpha\operatorname{mult}_E\ell$ at E, and the argument of Case (A0) yields the bound β.

Step F. In this step we conclude the proof of Claim D.2.

Let i be an index for which conditions (1)–(3) of **Step B** hold and $n_i>2d+2$. All but finitely many i are such. Let F be a prime divisor over W_i with center 0. If $I_G(F)\subset H$, then $I_G(F)=I_H(F)$ and

$$\frac{a(F,W_i,\frac{1}{m}\operatorname{div}(h_i))}{e_F}=\frac{a(F,W_i,\frac{1}{m}\operatorname{div}(h_i))}{|I_H(F)|}\geq\beta_i=\beta.$$

Otherwise $a(F,W_i,\frac{1}{m}\operatorname{div}(h_i))/e_F\geq\beta$ by **Step E**. Thus $b_i\geq\beta=\beta_i$, which contradicts the reduction in **Step A**. □

Step 2. In this step we prove the theorem in the exchange case.

By **Step 1** we may assume that some element of G exchanges the factors. Let f be a G-semi-invariant local equation of W in a smooth fourfold germ with G-eigenparameters X,Y,z,t whose classes in $\mathfrak{m}_o/\mathfrak{m}_o^2$ are those of the formal coordinates X,Y,z,t of **Step 1**, and replace h by the χ_h-component of a regular function on the fourfold germ restricting to h. Sending X,Y,z,t to G-semi-invariant lifts, with the characters of X,Y,z,t, of their images in $\mathbb{C}[[X,Y,z,t]]/(X^2+Y^2+g)$

under the completed presentation defines a G-equivariant automorphism Ψ of $\mathbb{C}[[X,Y,z,t]]$, as the linear part of Ψ is the identity, and Ψ maps the ideal $(\widehat{f})$, where $\widehat{f}$ is the expansion of f in X,Y,z,t, onto (X^2+Y^2+g). Thus $\nu\Psi(\widehat{f}) = X^2+Y^2+g(z,t)$ for a unit ν, which is G-invariant since it is semi-invariant. Let F be a computing lift of $q(1)$ (Lemma 4.2(1)) and $\lambda \in [0,1)$. By Lemma 4.6 with $n=4$, $\mathcal{P}=\{F\}$, the exponent λ in place of t, Ψ in place of ψ, ν in place of v, $u=1$, and a sufficiently large truncation degree $N_2 > \operatorname{mult}_0 g$ in place of L, there exist a polynomial G-semi-invariant mark h' and the germ $W' = (X^2+Y^2+g'(z,t)=0)$, where g' is the truncation of g in degrees less than N_2, with the following properties:

- W' is isolated Gorenstein terminal at 0, G acts faithfully and diagonally on X,Y,z,t, and $(W', \frac{1}{m}\operatorname{div}(h'))$ is lc near 0.
- The radial function q' of $(W' \ni 0, G, h', m)$ satisfies $q'(\lambda) = q(\lambda)$.
- F corresponds to a prime divisor F° over W' with $a(F^\circ, W', 0) = a(F, W, 0)$, $\operatorname{mult}_{F^\circ} h' = \operatorname{mult}_F h$, and $I_G(F^\circ) = I_G(F)$, so $q'(1) \le q(1)$.

In the coordinates $x = X + \sqrt{-1}Y$ and $y = X - \sqrt{-1}Y$, $W' = (xy + g' = 0)$ with $\operatorname{mult}_0 g' = \operatorname{mult}_0 g \ge N_1$, and G acts monomially on $\{x,y\}$ and exchanges the factors. Possibly replacing xy with $-g'$ in the monomials of h' divisible by xy, finitely many times, we may assume that no monomial of h' is divisible by xy. The restriction $h'|_{W'}$ and the character of h' do not change. By Claim D.2, $q'(1)$ has a computing lift F'' with $a(F'', W', 0)/e_{F''} \le L_1$. By Lemma 4.3(1) for q' and F'', with L_1 as the bound there, and by the definition of $q'(\lambda)$,

$$q(\lambda) = q'(\lambda) \le q'(1) + L_1(1-\lambda) \le q(1) + L_1(1-\lambda),$$

$$q(1) \le q(\lambda) = q'(\lambda) \le \frac{a(F'', W', 0)}{e_{F''}} \le L_1.$$

Thus $q(\lambda) \le q(1) + L_1(1-\lambda)$ for any $\lambda \in [0,1]$, and $q(1) \le L_1$. Since W is Gorenstein, K_W is Cartier near o, and Lemma 4.4, applied with the linear bound $q(\lambda) \le q(1) + L_1(1-\lambda)$ on $[0,1]$, yields a computing lift F_0 of $q(1)$ with

$$\frac{a(F_0, W, 0)}{e_{F_0}} \le q(1) + L_1 \le 2L_1 \le L(m, M).$$

The theorem follows. □

Theorem D.6. *Let m and M be positive integers. For any positive integer i, let $(W_i \ni o_i)$ be an isolated Gorenstein terminal hypersurface threefold germ, let G_i be a finite abelian group of order at most M acting faithfully on W_i and fixing o_i, and let h_i be a mark on $(W_i \ni o_i)$. Assume that for any i the following hold.*

(i) $(W_i, \frac{1}{m}\operatorname{div}(h_i))$ is lc near o_i.

(ii) There exists a G_i-equivariant isomorphism

$$\widehat{\mathcal{O}}_{W_i, o_i} \cong \mathbb{C}[[x,y,z,t]]/(x^2 + y^2 z + R_i)$$

with $R_i \in \mathbb{C}[[y,z,t]]$, for an action of G_i on $\mathbb{C}[[x,y,z,t]]$ in which x,y,z,t are semi-invariant.

Assume moreover that $\lim_{i\to+\infty} \operatorname{mult}_0 R_i = +\infty$. Let q_i be the radial function of $(W_i \ni o_i, G_i, h_i, m)$ (Definition 4.1). Then for any sequence $(\tau_i)_{i\ge1}$ in $[0,1]$ with $\lim_{i\to+\infty}\tau_i = 1$,

$$\lim_{i\to+\infty} \big(q_i(\tau_i) - q_i(1)\big) = 0.$$

Proof. **Step 1.** In this step we fix a contradicting sequence, polynomial models of W_i and a compatible generic limit, and we bound the limit ambient pair from below.

Suppose that the theorem does not hold. Possibly passing to a subsequence, there exist a real number $\delta > 0$ and a sequence $(\tau_i)_{i\ge1}$ in $[0,1]$ with $\lim_{i\to+\infty}\tau_i = 1$ such that, with $b_i = q_i(1)$,

$$q_i(\tau_i) - b_i \ge \delta \quad \text{for any } i. \tag{D.29}$$

Thus $\tau_i < 1$, and $q_i(\tau_i) \geq \delta$ since $b_i \geq 0$. Since W_i is terminal and $(W_i, \frac{1}{m}\operatorname{div}(h_i))$ is lc near o_i, $(W_i, \frac{\tau_i}{m}\operatorname{div}(h_i))$ is klt near o_i. Let F_i be a computing lift of b_i (Lemma 4.2(1)), so that

$$b_i = \frac{a(F_i, W_i, \frac{1}{m}\operatorname{div}(h_i))}{e_{F_i}}.$$

By [Liu26a, Lemma 5.4], we fix a G_i-equivariant closed embedding of W_i into a smooth fourfold germ $(Y_i \ni o_i)$ with $W_i = (f_i = 0)$ for a regular G_i-semi-invariant function f_i on Y_i. Let $\mathfrak{m}_{Y_i}$ and $\mathfrak{m}_{W_i}$ be the maximal ideals of $\mathcal{O}_{Y_i,o_i}$ and $\mathcal{O}_{W_i,o_i}$. Since W_i is singular at o_i, $f_i \in \mathfrak{m}_{Y_i}^2$, so

$$\mathfrak{m}_{Y_i}/\mathfrak{m}_{Y_i}^2 = \mathfrak{m}_{W_i}/\mathfrak{m}_{W_i}^2,$$

and the classes of x, y, z, t in $\mathfrak{m}_{W_i}/\mathfrak{m}_{W_i}^2$ lift to G_i-semi-invariant regular parameters of Y_i at o_i, which we denote again by x, y, z, t. We let h_i also denote a regular G_i-semi-invariant function on Y_i restricting to h_i, and $\widehat{f}_i, \widehat{h}_i \in \mathbb{C}[[x, y, z, t]]$ the expansions of f_i, h_i. Let $\mathfrak{n}$ be the maximal ideal of $\mathbb{C}[[x, y, z, t]]$. Since $\widehat{f}_i$ and $x^2 + y^2 z + R_i$ belong to $\mathfrak{n}^2$, the isomorphism in (ii) lifts to a G_i-equivariant automorphism ω_i of $\mathbb{C}[[x, y, z, t]]$ with $\omega_i((\widehat{f}_i)) = (x^2 + y^2 z + R_i)$. Thus

$$v_i \omega_i(\widehat{f}_i) = x^2 + y^2 z + R_i$$

for a unit v_i, which is G_i-invariant since both $\omega_i(\widehat{f}_i)$ and $x^2 + y^2 z + R_i$ are semi-invariant ($x^2 + y^2 z + R_i$ is semi-invariant by the uniqueness in the Weierstrass division theorem) and $v_i(0) \neq 0$.

By Theorem 2.23(1) with $n = 4$, $\Delta = 0$, and D with coefficients in $\frac{1}{m}\mathbb{N}^+$, there exists $\epsilon_0 \in (0, 1)$ depending only on m such that

$$\operatorname{lct}(Y' \ni o'; (\theta_1)(\theta_2)^{1/m}) \notin (1 - \epsilon_0, 1) \tag{D.30}$$

for any smooth fourfold germ $(Y' \ni o')$ and any non-zero regular functions θ_1, θ_2 on Y' with $\theta_1(o') = 0$.

We apply Construction-Theorem 2.11 to $W_i \subset Y_i$ with $n = 4$, $H = G_i$, $N = 1$, $h_1 = h_i$, $\beta_1 = \frac{\tau_i}{m}$, and $\mathcal{P} = \{F_i\}$, and in (3) with $\psi = \omega_i$ and $v = v_i$. By (3), there exist integers $N_i' \geq 8m/\epsilon_0$ and $L_i \geq 8m/\epsilon_0$ such that, with

$$R_i' := (R_i)_{<L_i}, \quad f_i' := x^2 + y^2 z + R_i', \quad \text{and} \quad h_i' := \omega_i(\widehat{h}_i)_{<N_i'},$$

the conclusions (1) and (2) hold for f_i', h_i', the germ $W_i' := (f_i' = 0) \subset \mathbb{A}^4$ at 0, and the diagonal action of G_i on $\mathbb{A}^4$ in which x, y, z, t have the characters above. Here $R_i' \in \mathbb{C}[y, z, t]$ and $\operatorname{mult}_0 R_i' \geq \operatorname{mult}_0 R_i$. By (3) and (1), $(W_i' \ni 0)$ is an isolated Gorenstein terminal germ, G_i acts faithfully on W_i', $(W_i', \frac{\tau_i}{m}\operatorname{div}(h_i'))$ is klt near 0, and

$$q_i(\tau_i) = \operatorname{mld}((W_i' \ni 0, \tfrac{\tau_i}{m}\operatorname{div}(h_i'))/G_i). \tag{D.31}$$

By (2), F_i corresponds to a prime divisor F_i' over W_i' with center 0 such that $e_{F_i'} = e_{F_i}$, $\operatorname{mult}_{F_i'} h_i' = \operatorname{mult}_{F_i} h_i$, and

$$a(F_i', W_i', 0) = a(F_i, W_i, 0).$$

Thus for any $c \geq 0$,

$$\frac{a(F_i', W_i', c\operatorname{div}(h_i'))}{e_{F_i'}} = \frac{a(F_i, W_i, c\operatorname{div}(h_i))}{e_{F_i}}, \quad \text{in particular} \quad b_i = \frac{a(F_i', W_i', \frac{1}{m}\operatorname{div}(h_i'))}{e_{F_i'}}. \tag{D.32}$$

We show that $(W_i', \frac{1}{m}\operatorname{div}(h_i'))$ is lc near 0. By [EMY03, Corollary 1.8] and [dFEM11, Proposition 2.11], (i) implies

$$\operatorname{lct}(\operatorname{Spec}\mathbb{C}[[x, y, z, t]]; (\widehat{f}_i)(\widehat{h}_i)^{1/m}) \geq 1,$$

so, applying ω_i,

$$\operatorname{lct}\left(\operatorname{Spec}\mathbb{C}[[x, y, z, t]]; (x^2 + y^2 z + R_i)(\omega_i(\widehat{h}_i))^{1/m}\right) \geq 1.$$

Since $f_i' - (x^2 + y^2 z + R_i) \in \mathfrak{n}^{L_i}$, $h_i' - \omega_i(\widehat{h}_i) \in \mathfrak{n}^{N_i'}$, and

$$L_i, N_i' \geq \frac{8m}{\epsilon_0} = \frac{4}{\frac{\epsilon_0}{2} \cdot \frac{1}{m}},$$

Theorem 2.23(2) with $\epsilon = \epsilon_0/2$ implies $\operatorname{lct}(\mathbb{A}^4 \ni 0; (f_i')(h_i')^{1/m}) \geq 1 - \epsilon_0/2$, hence ≥ 1 by (D.30). By [EMY03, Corollary 1.8], $(W_i', \frac{1}{m}\operatorname{div}(h_i'))$ is lc near 0.

Since f_i' is monic of degree 2 in x, there exists a unique polynomial

$$\widetilde{h}_i = \alpha_{i,0}(y,z,t) + x\alpha_{i,1}(y,z,t) \quad \text{with} \quad h_i' - \widetilde{h}_i \in (f_i').$$

Then $\widetilde{h}_i$ is G_i-semi-invariant with the character of h_i', and $\widetilde{h}_i|_{W_i'} = h_i'|_{W_i'}$. Let

$$u_i(z,t) := \alpha_{i,0}(0,z,t).$$

Possibly passing to a subsequence, we may assume that $\operatorname{mult}_0 R_i > 4m+4$ for any i. By Lemma 4.2(3) and (D.29),

$$b_i < q_i(\tau_i) \leq 3. \tag{D.33}$$

Let $w := (2m+2, 2m+1, 2, 1)$, a weight vector for (x,y,z,t). Since $2w_x = 2w_y + w_z = 4m+4$ and any monomial of R_i' has degree greater than $4m+4$, $\operatorname{in}_w(f_i') = x^2 + y^2 z$, which is irreducible and not divisible by any coordinate. Since $\operatorname{in}_w(\widetilde{h}_i)$ has degree at most 1 in x, it is not divisible by $x^2 + y^2 z$. By [Liu26b, Lemma 4.12], there exists a prime divisor $E_{i,w}$ over W_i' with center 0 such that

$$a(E_{i,w}, W_i', 0) = w_y + w_z + w_t - w_x \quad \text{and} \quad \operatorname{mult}_{E_{i,w}} h_i' = \operatorname{wt}_w(\widetilde{h}_i).$$

Thus $a(E_{i,w}, W_i', 0) = 2$, so $\operatorname{wt}_w(\widetilde{h}_i) \leq 2m$ since $(W_i', \frac{1}{m}\operatorname{div}(h_i'))$ is lc. Since any monomial divisible by x or y has w-weight at least $2m+1$ and the w-weight of a monomial is at least the degree of the monomial,

$$\operatorname{mult}_0 \widetilde{h}_i \leq 2m, \quad \text{and} \quad \operatorname{wt}_{(2,1)}(u_i) \leq 2m, \tag{D.34}$$

where $\operatorname{wt}_{(2,1)}$ is the weight with values 2 on z and 1 on t.

Any element of G_i is a diagonal matrix whose entries are roots of unity of order at most M, so the $G_i \subset GL_4(\mathbb{C})$ belong to a finite set. Possibly passing to a subsequence, we may assume that $G_i = G$ for a fixed group G, and that the characters $\chi_x, \chi_y, \chi_z, \chi_t$ of the coordinates, χ_f of f_i' and χ_h of $\widetilde{h}_i$ do not depend on i. Since x^2 and $y^2 z$ are monomials of f_i',

$$\chi_z = (\chi_x \chi_y^{-1})^2. \tag{D.35}$$

Let n_1 and n_2 be the orders of $\chi_x \chi_y^{-1}$ and χ_t.

By (D.34), the orders of f_i' and $\widetilde{h}_i$ at 0 are bounded. Possibly passing to an infinite subset $\mathcal{I}$ of indices, we let (f_*, h_*) be a compatible generic limit of the pairs $(f_i', \widetilde{h}_i)$ along $\mathcal{I}$ with coefficient field $\Bbbk$ (Definition 2.5, Lemma 2.6(1)). For a positive integer N_0, the N_0-jet of (f_*, h_*) is the generic point of an irreducible closed subset of $\mathcal{J}_{N_0}$ any non-empty open subset of which contains the N_0-jets of $(f_i', \widetilde{h}_i)$ for all but finitely many $i \in \mathcal{I}$. Thus a closed or an open condition on N_0-jets which holds for the N_0-jets of $(f_i', \widetilde{h}_i)$ for all but finitely many $i \in \mathcal{I}$ holds for the N_0-jet of (f_*, h_*). Since $\lim_{i \to +\infty} \operatorname{mult}_0 R_i' = +\infty$, by (D.34) we obtain

$$f_* = x^2 + y^2 z, \quad h_* = \alpha_{*,0}(y,z,t) + x\alpha_{*,1}(y,z,t) \quad \text{with } \alpha_{*,0}, \alpha_{*,1} \in \Bbbk[[y,z,t]],$$

and $u_*(z,t) := \alpha_{*,0}(0,z,t)$ satisfies $\operatorname{wt}_{(2,1)}(u_*) \leq 2m$, in particular $u_* \neq 0$. Moreover, f_*, h_*, $\alpha_{*,0}$, and u_* are semi-invariant, and u_* has the character χ_h. Let $\mathcal{X}_* := \operatorname{Spec} \Bbbk[[x,y,z,t]]$, which is the scheme $\mathcal{A}_{\Bbbk}$ of Lemma 3.15 and Theorem 2.18 for $n = 4$. Log discrepancies and lc pairs on schemes of finite type over $\mathcal{X}_*$ are as in [dFEM11, Section 2].

We show that for any prime divisor Q over $\mathcal{X}_*$ with center the closed point,

$$a(Q, \mathcal{X}_*, (f_*)(h_*)^{1/m}) \geq \delta. \tag{D.36}$$

By Lemma 3.15, for all but finitely many $i \in \mathcal{I}$ there exists a prime divisor Q_i over $\mathbb{A}^4$ with center 0 such that $a(Q_i, \mathbb{A}^4, 0) = a(Q, \mathcal{X}_*, 0)$, $\operatorname{mult}_{Q_i} f_i' = \operatorname{mult}_Q f_*$ and $\operatorname{mult}_{Q_i} \widetilde{h}_i = \operatorname{mult}_Q h_*$. By Theorem 2.17(1) for the single equation $f_1 = f_i'$ and $\mathfrak{a} = (\widetilde{h}_i)^{\tau_i/m}$, since $(W_i', \frac{\tau_i}{m}\operatorname{div}(h_i'))$ is klt near 0, and by (D.31) and Definition 2.2,

$$a(Q_i, \mathbb{A}^4, (f_i')(\widetilde{h}_i)^{\tau_i/m}) \geq e_{Q_i} \cdot \operatorname{mld}((\mathbb{A}^4 \ni 0, \operatorname{div}(f_i'), (\widetilde{h}_i)^{\tau_i/m})/G) = e_{Q_i} \cdot q_i(\tau_i) \geq \delta.$$

The left-hand side equals $a(Q, \mathcal{X}_*, 0) - \operatorname{mult}_Q f_* - \frac{\tau_i}{m}\operatorname{mult}_Q h_*$, and $\lim_{i\to+\infty} \tau_i = 1$. Thus (D.36) holds.

Step 2. In this step we reduce (D.36) to a plane pair, exclude the case when the plane pair is klt, and describe the non-klt case.

Let $\pi : \widetilde{\mathcal{X}}_* \to \mathcal{X}_*$ be the blow-up of the ideal (x, y), E the exceptional divisor of π, $S := \pi_*^{-1}\operatorname{div}(f_*)$, and $\operatorname{pr} : E \to Z := \operatorname{Spec} \Bbbk[[z, t]]$ the induced $\mathbb{P}^1$-bundle. Since $\operatorname{mult}_{(x,y)} f_* = 2$ and $h_* \notin (x, y)$ by $u_* \neq 0$,

$$K_{\widetilde{\mathcal{X}}_*} + E + S + \tfrac{1}{m}\operatorname{div}(\pi^* h_*) = \pi^*\big(K_{\mathcal{X}_*} + \operatorname{div}(f_*) + \tfrac{1}{m}\operatorname{div}(h_*)\big), \tag{D.37}$$

and E is not a component of $\operatorname{div}(\pi^* h_*)$. Writing $E = \operatorname{Proj}_Z \Bbbk[[z, t]][\overline{x}, \overline{y}]$, we have

$$S \cap E = (\overline{x}^2 + z\overline{y}^2 = 0),$$

which is contained in the chart $(\overline{y} \neq 0)$. With $p := \overline{x}/\overline{y}$, that is, $x = yp$ on $\widetilde{\mathcal{X}}_*$, we have

$$S \cap E = \operatorname{Spec} \Bbbk[[z, t]][p]/(p^2 + z) \cong T := \operatorname{Spec} \Bbbk[[p, t]],$$

and $\operatorname{pr}|_{S\cap E} : T \to Z$ is the double cover given by $z = -p^2$. Since $h_* - u_* \in (x, y)$, $(\pi^* h_*)|_E = \operatorname{pr}^* u_*$, whose restriction to T is $\overline{u}_*(p, t) := u_*(-p^2, t)$.

We show

$$\operatorname{mld}(T \ni 0, \tfrac{1}{m}\operatorname{div}(\overline{u}_*)) \geq \delta. \tag{D.38}$$

We use [EMY03, Corollary 1.8] for regular schemes of finite type over $\mathcal{X}_*$, passing from smooth varieties over $\Bbbk$ to such schemes along the regular morphism $\mathcal{X}_* \to \mathbb{A}^4_{\Bbbk}$, under whose base change regularity [dFEM11, Remark 2.10], proper birational morphisms and relative canonical divisors [JM12, Lemma 1.2] are preserved, and from $\mathbb{C}$ to $\Bbbk$ by [dFEM11, Lemma 2.14]. Let $y_0 \in S \cap E$ be the point $p = t = 0$. Then p, t, y, s with $s := z + p^2$ are regular parameters of $\widetilde{\mathcal{X}}_*$ at y_0, and $E = (y = 0)$, $S = (s = 0)$ near y_0. Neither E nor $S \cap E$ is contained in the support of $\operatorname{div}(\pi^* h_*)$, since $(\pi^* h_*)|_{S\cap E} = \overline{u}_* \neq 0$. By adjunction along an embedded log resolution [dFEM11, Theorem 2.4], first for $E \subset \widetilde{\mathcal{X}}_*$ and then for $S \cap E \subset E$,

$$\operatorname{mld}(T \ni 0, \tfrac{1}{m}\operatorname{div}(\overline{u}_*)) \geq \operatorname{mld}(\widetilde{\mathcal{X}}_* \ni y_0, E + S + \tfrac{1}{m}\operatorname{div}(\pi^* h_*)).$$

Any prime divisor over $\widetilde{\mathcal{X}}_*$ with center y_0 is a prime divisor over $\mathcal{X}_*$ with center the closed point, so the right-hand side is $\geq \delta$ by (D.37) and (D.36). Thus (D.38) holds.

Since $\operatorname{mld}(T \ni 0, \frac{1}{m}\operatorname{div}(\overline{u}_*)) \geq \delta > 0$ by (D.38), the pair $(T, \frac{1}{m}\operatorname{div}(\overline{u}_*))$ is lc near 0. Let ι be the involution of T with $\iota^* p = -p$ and $\iota^* t = t$, and for $g \in G$ let g act on T by $g^* p = \chi_x \chi_y^{-1}(g) p$ and $g^* t = \chi_t(g) t$. Since $\overline{u}_*(-p, t) = \overline{u}_*(p, t)$, u_* is semi-invariant and $\chi_z = (\chi_x \chi_y^{-1})^2$ by (D.35), $\operatorname{div}(\overline{u}_*)$ is invariant under ι and G.

Suppose that $(T, \frac{1}{m}\operatorname{div}(\overline{u}_*))$ is klt near 0, and fix $\epsilon_1 > 0$ such that $(T, c\operatorname{div}(\overline{u}_*))$ is lc near 0, where $c := \frac{1+\epsilon_1}{m}$. Since h_* and u_* are power series, we use [EMY03, Corollary 1.8] below for the truncations $h_{*,<L'}$ and $u_{*,<L'} = h_{*,<L'}(0, 0, z, t)$ in place of h_* and u_*, where $L' > 2m$ is an integer at least the integers L of Lemma 2.19 for $(n, N, \Lambda) = (2, 1, \{c\})$ and $(4, 2, \{1, c\})$. The pairs defined by f_*, $h_{*,<L'}$, and $u_{*,<L'}$ come from schemes of finite type over $\Bbbk$. Since $\Bbbk$ is algebraically closed, by Lemma 2.19, $(T, c\operatorname{div}(\overline{u}_*))$ lc implies $(T, c\operatorname{div}(u_{*,<L'}(-p^2, t)))$ lc, and

$$(\mathcal{X}_*, \operatorname{div}(f_*) + c\operatorname{div}(h_{*,<L'}))\ \text{lc} \quad \Longrightarrow \quad (\mathcal{X}_*, \operatorname{div}(f_*) + c\operatorname{div}(h_*))\ \text{lc}.$$

Since

$$K_T = \operatorname{pr}|_T^*\big(K_Z + \tfrac{1}{2}\operatorname{div}(z)\big),$$

the pair $(T, c\,\mathrm{div}(\overline{u}_*))$ is the crepant pullback of $(Z, \frac{1}{2}\,\mathrm{div}(z)+c\,\mathrm{div}(u_*))$ under the finite morphism $\mathrm{pr}\,|_T$, so $(Z, \frac{1}{2}\,\mathrm{div}(z)+c\,\mathrm{div}(u_*))$ is lc, and so is $(Z, c\,\mathrm{div}(u_*))$. Since pr is smooth, $(E, c\,\mathrm{div}(\mathrm{pr}^*\,u_*))$ is lc. Since $(T, c\,\mathrm{div}(\overline{u}_*))$ is lc, [EMY03, Corollary 1.8] for the smooth divisor $S \cap E \cong T$ on E implies that $(E, S \cap E + c\,\mathrm{div}(\mathrm{pr}^*\,u_*))$ is lc near $S \cap E$, hence lc. Since

$$(S + c\,\mathrm{div}(\pi^* h_*))|_E = S \cap E + c\,\mathrm{div}(\mathrm{pr}^*\,u_*),$$

[EMY03, Corollary 1.8] implies that $(\widetilde{\mathcal{X}}_*, E + S + c\,\mathrm{div}(\pi^* h_*))$ is lc near E. Any closed point of $\widetilde{\mathcal{X}}_*$ maps to the closed point of $\mathcal{X}_*$, so belongs to E. Thus $(\widetilde{\mathcal{X}}_*, E + S + c\,\mathrm{div}(\pi^* h_*))$ is lc, and by (D.37) with $\frac{1}{m}$ replaced by c, followed by the implication above,

$$(\mathcal{X}_*, \mathrm{div}(f_*) + c\,\mathrm{div}(h_*)) \text{ is lc.}$$

By Theorem 2.18(1) with $n = 4$, $H = G$, the tuples $(f_i', \widetilde{h}_i)$, the coefficients 1 and c, $\theta = 1$, and $\mathfrak{c} = \mathcal{O}$, there exists an infinite subset $\mathcal{I}' \subset \mathcal{I}$ such that for any $i \in \mathcal{I}'$,

$$(\mathbb{A}^4, \mathrm{div}(f_i') + c\,\mathrm{div}(\widetilde{h}_i)) \text{ is lc near } 0,$$

and for such i, $(W_i', c\,\mathrm{div}(h_i'))$ is lc near 0 by [EMY03, Corollary 1.8]. By (D.32),

$$a(F_i, W_i, c\,\mathrm{div}(h_i)) \geq 0,$$

that is,

$$\frac{\epsilon_1}{m}\,\mathrm{mult}_{F_i}\,h_i \leq a(F_i, W_i, \tfrac{1}{m}\,\mathrm{div}(h_i)) = e_{F_i} b_i.$$

Thus by Definition 2.2 and (D.33),

$$q_i(\tau_i) \leq \frac{a(F_i, W_i, \frac{\tau_i}{m}\,\mathrm{div}(h_i))}{e_{F_i}} = b_i + \frac{(1-\tau_i)\,\mathrm{mult}_{F_i}\,h_i}{m e_{F_i}} \leq b_i + \frac{(1-\tau_i) b_i}{\epsilon_1} \leq b_i + \frac{3(1-\tau_i)}{\epsilon_1},$$

which contradicts (D.29) for $i \in \mathcal{I}'$ with $i \gg 0$.

Thus $(T, \frac{1}{m}\,\mathrm{div}(\overline{u}_*))$ is lc but not klt near 0, and some prime divisor P over T has log discrepancy 0. By (D.38), the center of P on T is not the closed point, so it is a curve $\Gamma \ni 0$ with $\mathrm{mult}_\Gamma\,\overline{u}_* = m$. If there were two such curves, or if Γ were singular at 0, the blow-up of 0 would have log discrepancy at most 0, contrary to (D.38). Thus Γ is unique and smooth at 0, hence invariant under ι and G. The tangent line $T_0\Gamma$ of Γ at 0 is ι-invariant, so $T_0\Gamma = (p = 0)$ or $T_0\Gamma = (t = 0)$.

In the case $T_0\Gamma = (p = 0)$, $\Gamma = (p = \kappa(t))$ for some $\kappa \in t\Bbbk[[t]]$, and $\iota(\Gamma) = \Gamma$ implies $\kappa = -\kappa$, so $\Gamma = (p = 0)$. Since $\overline{u}_*(p, t) = u_*(-p^2, t)$, the order of $\overline{u}_*$ along $(p = 0)$ is twice the order of u_* along $(z = 0)$. Thus m is even, $u_* = z^{m/2}\rho(z, t)$ with $\rho(0, t) \neq 0$, and we let $j := \mathrm{mult}_0\,\rho(0, t)$. For an integer $N > j$, the monomial valuation of T with values N on p and 1 on t is a prime divisor over T with center 0, log discrepancy $N + 1$ with respect to T, and value $Nm + j$ on $\overline{u}_*$. By (D.38), $1 - j/m \geq \delta$, so

$$0 \leq j < m.$$

In the case $T_0\Gamma = (t = 0)$, $\Gamma = (t = \overline{\phi}(p))$ for some $\overline{\phi} \in p\Bbbk[[p]]$, and $\iota(\Gamma) = \Gamma$ implies $\overline{\phi}(-p) = \overline{\phi}(p)$, so $\overline{\phi}(p) = \phi(-p^2)$ for some $\phi \in z\Bbbk[[z]]$, that is, $\Gamma = (t = \phi(z))$ on T. For $g \in G$, $g(\Gamma) = \Gamma$ means $\phi(\chi_z(g)z) = \chi_t(g)\phi(z)$, so

(D.39) $$t' := t - \phi(z) \text{ is semi-invariant with the character } \chi_t.$$

Writing $u_*(z, t' + \phi(z)) = \sum_{e \geq 0} c_e(z) t'^e$, $\overline{u}_*$ has order m along $\Gamma = (t' = 0)$ if and only if $c_e = 0$ for $e < m$ and $c_m \neq 0$. Thus $u_*(z, t' + \phi(z)) = t'^m \rho(z, t')$ with $\rho(z, 0) \neq 0$, and we let $j := \mathrm{mult}_0\,\rho(z, 0)$. For an integer $N > 2j$, the monomial valuation of T with values 1 on p and N on t' has log discrepancy $N + 1$ with respect to T and value $Nm + 2j$ on $\overline{u}_*$. By (D.38), $1 - 2j/m \geq \delta$, so

$$0 \leq 2j < m.$$

Step 3. In this step we bound a toric function on the lattice of G from below, fix two weight vectors, and specialize the lowest weight parts of h_* to $\widetilde{h}_i$.

In the case $T_0\Gamma = (p=0)$ let $\phi := 0$ and $t' := t$. In both cases we use the coordinates x, y, z, t', in which G acts diagonally with the characters $\chi_x, \chi_y, \chi_z, \chi_t$ by (D.39), and we let

$$(k_1, k_2) := (\tfrac{m}{2}, j) \text{ in the case } T_0\Gamma = (p=0), \qquad (k_1, k_2) := (j, m) \text{ in the case } T_0\Gamma = (t=0).$$

For $w = (w_x, w_y, w_z, w_{t'}) \in \mathbb{R}^4_{\geq 0}$, let

$$\psi(w) := w_x + w_y + w_z + w_{t'} - \mathrm{wt}_w(x^2 + y^2 z) - \tfrac{1}{m}\,\mathrm{wt}_w(z^{k_1} t'^{k_2}).$$

Since $\mathrm{wt}_w(x^2 + y^2 z) = \min(2w_x, 2w_y + w_z)$,

$$\psi(w) = \Big|w_x - w_y - \frac{w_z}{2}\Big| + \Big(1 - \frac{j}{m}\Big) w_{t'} \quad \text{or} \quad \psi(w) = \Big|w_x - w_y - \frac{w_z}{2}\Big| + \Big(1 - \frac{2j}{m}\Big)\frac{w_z}{2} \tag{D.40}$$

in the case $T_0\Gamma = (p=0)$, respectively $T_0\Gamma = (t=0)$. In particular $\psi \geq 0$ on $\mathbb{R}^4_{\geq 0}$. Let

$$\gamma := \Big(1 - \frac{j}{m}\Big)\frac{1}{n_2} \quad \text{or} \quad \gamma := \Big(1 - \frac{2j}{m}\Big)\frac{1}{n_1}$$

in the case $T_0\Gamma = (p=0)$, respectively $T_0\Gamma = (t=0)$. Then $0 < \gamma \leq 1$ by **Step 2**. Let $N_G \subset \mathbb{R}^4$ be the lattice of G in the coordinates x, y, z, t' [Liu26a, Definition 2.15], that is, the set of w with

$$(e^{2\pi\sqrt{-1}w_x}, e^{2\pi\sqrt{-1}w_y}, e^{2\pi\sqrt{-1}w_z}, e^{2\pi\sqrt{-1}w_{t'}}) = (\chi_x(g), \chi_y(g), \chi_z(g), \chi_t(g)) \quad \text{for some } g \in G.$$

Then $\langle \mathbf{e}, w\rangle \in \mathbb{Z}$ for any $w \in N_G$ and any exponent vector $\mathbf{e}$ of a G-invariant Laurent monomial in x, y, z, t'.

We show

$$\psi(w) \geq \gamma \quad \text{for any } w \in N_G \cap \mathbb{R}^4_{>0}. \tag{D.41}$$

Since $(x/y)^{n_1}$, t'^{n_2} and $y^2 z/x^2$ are G-invariant by (D.35),

$$w_x - w_y \in \tfrac{1}{n_1}\mathbb{Z}, \quad w_{t'} \in \tfrac{1}{n_2}\mathbb{Z}, \quad \text{and} \quad w_z - 2(w_x - w_y) \in \mathbb{Z}.$$

In the case $T_0\Gamma = (p=0)$, $\psi(w) \geq (1 - j/m) w_{t'} \geq \gamma$ since $w_{t'} \geq 1/n_2$. In the case $T_0\Gamma = (t=0)$, $0 < 1 - 2j/m \leq 1$. If $w_x - w_y > 0$, then $w_x - w_y \geq 1/n_1$ and

$$\psi(w) \geq \Big(1 - \frac{2j}{m}\Big)\Big(\Big|w_x - w_y - \frac{w_z}{2}\Big| + \frac{w_z}{2}\Big) \geq \Big(1 - \frac{2j}{m}\Big)(w_x - w_y) \geq \gamma.$$

If $w_x - w_y < 0$, then $\psi(w) \geq w_z/2 - (w_x - w_y) \geq 1/n_1 \geq \gamma$. If $w_x = w_y$, then $w_z \in \mathbb{N}^+$, so $\psi(w) = (2 - 2j/m) w_z/2 \geq 1 - 2j/m \geq \gamma$. Thus (D.41) holds.

For a weight vector w for (x, y, z, t'), consider the conditions

(a) $2w_x = 2w_y + w_z$.
(b) $w_z > j w_{t'}$ in the case $T_0\Gamma = (p=0)$, and $w_{t'} > j w_z$ in the case $T_0\Gamma = (t=0)$.
(c) $w_y > \Theta_w := k_1 w_z + k_2 w_{t'}$.

Under (a)–(c), $\mathrm{in}_w(x^2 + y^2 z) = x^2 + y^2 z$, and we show

$$\mathrm{in}_w\big(h_*(x, y, z, t' + \phi(z))\big) \in \Bbbk^* \cdot z^{k_1} t'^{k_2}. \tag{D.42}$$

Since $w_x = w_y + \frac{w_z}{2} \geq w_y$ by (a), any monomial divisible by x or y has w-weight at least $w_y > \Theta_w$ by (c), and

$$\alpha_{*,0}(0, z, t' + \phi(z)) = u_*(z, t' + \phi(z))$$

is $z^{m/2}\rho(z, t')$ with $\mathrm{mult}_0\,\rho(0, t') = j$, respectively $t'^m \rho(z, t')$ with $\mathrm{mult}_0\,\rho(z, 0) = j$. The coefficient of $z^{k_1} t'^{k_2}$ in

$$u_*(z, t' + \phi(z))$$

is non-zero, and any other monomial of $u_*(z, t' + \phi(z))$ has w-weight greater than Θ_w by (b).

Let $w^{(1)} \in (\mathbb{N}^+)^4$ satisfy (a)–(c), for example

$$w^{(1)} = \big((m+2)(j+1), (m+1)(j+1), 2j+2, 1\big) \quad \text{or}$$
$$w^{(1)} = \big((m+1)(2j+1)+1, (m+1)(2j+1), 2, 2j+1\big)$$

in the case $T_0\Gamma = (p = 0)$, respectively $T_0\Gamma = (t = 0)$. Let $g \in G$ with $\chi_t(g) = e^{2\pi\sqrt{-1}/n_2}$, respectively $\chi_x\chi_y^{-1}(g) = e^{2\pi\sqrt{-1}/n_1}$, and let $w^g \in [0,1)^4$ be the vector with

$$(e^{2\pi\sqrt{-1}w^g_x}, e^{2\pi\sqrt{-1}w^g_y}, e^{2\pi\sqrt{-1}w^g_z}, e^{2\pi\sqrt{-1}w^g_{t'}}) = (\chi_x(g), \chi_y(g), \chi_z(g), \chi_t(g)),$$

so that $w^g \in N_G$, $w^g_z - 2(w^g_x - w^g_y) \in \mathbb{Z}$ by (D.35), and $w^g_{t'} - 1/n_2 \in \mathbb{Z}$, respectively $w^g_x - w^g_y - 1/n_1 \in \mathbb{Z}$. Let $w^{(2)} \in w^g + \mathbb{Z}^4$ be a vector such that, in the case $T_0\Gamma = (p = 0)$,

$$w^{(2)}_{t'} = \frac{1}{n_2}, \quad w^{(2)}_x - w^{(2)}_y > \frac{j}{2n_2}, \quad w^{(2)}_z = 2(w^{(2)}_x - w^{(2)}_y), \quad \text{and} \quad w^{(2)}_y > \Theta_{w^{(2)}},$$

and in the case $T_0\Gamma = (t = 0)$,

$$w^{(2)}_x - w^{(2)}_y = \frac{1}{n_1}, \quad w^{(2)}_z = \frac{2}{n_1}, \quad w^{(2)}_{t'} > \frac{2j}{n_1}, \quad \text{and} \quad w^{(2)}_y > \Theta_{w^{(2)}}.$$

Such a vector is obtained by fixing $w^{(2)}_{t'}$, $w^{(2)}_x - w^{(2)}_y$, and $w^{(2)}_z$ in the respective classes modulo $\mathbb{Z}$, and then $w^{(2)}_y \in w^g_y + \mathbb{Z}$. Then $w^{(2)} \in N_G \cap \mathbb{Q}^4_{>0}$ satisfies (a)–(c), and by (D.40),

$$\psi(w^{(2)}) = \gamma. \tag{D.43}$$

Moreover, $w^{(2)}$ is primitive in N_G: if $w^{(2)} = nw'$ with $w' \in N_G$ and an integer $n \geq 2$, then $\psi(w') = \gamma/n < \gamma$, contrary to (D.41).

We specialize (D.42) for $w^{(1)}$ and $w^{(2)}$ to $\widetilde{h}_i$. Fix an integer N_0 greater than $\Theta_w / \min(w_x, w_y, w_z, w_{t'})$ for $w \in \{w^{(1)}, w^{(2)}\}$, and write

$$\phi(z) = \sum_{k=1}^{N_0-1} \lambda_k z^k \bmod z^{N_0}, \qquad \lambda = (\lambda_k)_k \in \Bbbk^{N_0-1}.$$

For a pair (f, h) and $\lambda' \in \Bbbk^{N_0-1}$, consider the following conditions.

(A) $\lambda'_k = 0$ for any k with $\chi_z^k \neq \chi_t$.
(B) for $w \in \{w^{(1)}, w^{(2)}\}$, the coefficient in $h(x, y, z, t' + \sum_k \lambda'_k z^k)$ of any monomial in x, y, z, t' of w-weight at most Θ_w, other than $z^{k_1}t'^{k_2}$, is zero.
(C) the coefficient of $z^{k_1}t'^{k_2}$ in $h(x, y, z, t' + \sum_k \lambda'_k z^k)$ is non-zero.

Since the substitution of $t' + \sum_k \lambda'_k z^k$ for t does not decrease degrees and the monomials of w-weight at most Θ_w have degree less than N_0, (A)–(C) are polynomial conditions with complex coefficients in the coordinates of the N_0-jet of (f, h) and λ'. They hold for (f_*, h_*) and λ: (A) by (D.39), and (B), (C) by (D.42), since $\phi(z) - \sum_k \lambda_k z^k \in (z^{N_0})$ and any monomial divisible by z^{N_0} has w-weight at least $N_0 w_z > \Theta_w$. By Lemma 2.7, for all but finitely many $i \in \mathcal{I}$ there exists $\lambda_i \in \mathbb{C}^{N_0-1}$ such that (A)–(C) hold for $(f'_i, \widetilde{h}_i)$ and λ_i. Let $\phi_i(z) := \sum_k \lambda_{i,k} z^k$. By (A), $t' = t - \phi_i(z)$ defines a G-equivariant automorphism of $\mathbb{A}^4$ fixing 0, and we regard f'_i and $\widetilde{h}_i$ as polynomials in x, y, z, t'. Since $f'_i = x^2 + y^2 z + R'_i(y, z, t' + \phi_i(z))$ and $\operatorname{mult}_0 R'_i(y, z, t' + \phi_i(z)) = \operatorname{mult}_0 R'_i$, by (B) and (C) we obtain for $i \gg 0$ and $w \in \{w^{(1)}, w^{(2)}\}$:

$$\operatorname{in}_w(f'_i) = x^2 + y^2 z \quad \text{and} \quad \operatorname{in}_w(\widetilde{h}_i) \in \mathbb{C}^* \cdot z^{k_1}t'^{k_2}. \tag{D.44}$$

Step 4. In this step we show that

$$b_i \geq \gamma \quad \text{for } i \gg 0. \tag{D.45}$$

Let $X := \mathbb{A}^4/G$, $\overline{0}$ the image of 0, and $\Delta_X \geq 0$ the $\mathbb{Q}$-divisor on X such that $K_{\mathbb{A}^4}$ is the pullback of $K_X + \Delta_X$. Let $G_0 \subset G$ be the subgroup generated by the elements acting non-trivially on exactly one coordinate. Then $\mathbb{A}^4/G_0 \cong \mathbb{A}^4$, G/G_0 acts on it diagonally and freely in codimension 1, and $X = (\mathbb{A}^4/G_0)/(G/G_0)$. Thus X and Δ_X satisfy the hypotheses of Theorem 2.16.

Let

$$B := \operatorname{div}(x^2 + y^2 z) + \tfrac{1}{m} \operatorname{div}(z^{k_1}t'^{k_2})$$

on $\mathbb{A}^4$. Then ψ is the function of [Liu26a, Set-up 8.4] for $d=4$ and B. By [CLS11, Theorem 11.1.9], there exists a regular subdivision Σ as in [Liu26a, Set-up 8.4(4)]. For $w\in\mathbb{R}^4_{>0}$, let

$$B_w := \operatorname{div}(\operatorname{in}_w(x^2+y^2z)) + \tfrac{1}{m}\operatorname{div}(z^{k_1}t'^{k_2}).$$

Since $\operatorname{in}_w(x^2+y^2z)$ is x^2, y^2z or x^2+y^2z, the zero locus of $\operatorname{in}_w(x^2+y^2z)$ in $(\mathbb{C}^*)^4$ is empty or a smooth reduced divisor, and $z^{k_1}t'^{k_2}$ has no zero on $(\mathbb{C}^*)^4$, so $((\mathbb{C}^*)^4, B_w)$ is lc. By [Liu26a, Lemma 8.6(1)], the toric model $\mu_\Sigma : (V_\Sigma, \Delta_\Sigma + B'_\Sigma) \to X$ of $(\mathbb{A}^4, B)/G$ for Σ [Liu26a, Set-up 8.4(4)] is lc near $\mu_\Sigma^{-1}(\overline{0})$. By [Liu26a, Lemma 8.5], the quotient pair of $(\mathbb{A}^4, B)$ pulls back to $(V_\Sigma, \Delta_\Sigma + B'_\Sigma - \sum_v \psi(v)E_v)$, where v runs over the primitive generators of the rays of Σ and E_v is the toric divisor of v. Since $\psi\geq 0$ on $\mathbb{R}^4_{\geq 0}$,

$$\Delta_\Sigma + B'_\Sigma - \sum_v \psi(v)E_v \leq \Delta_\Sigma + B'_\Sigma,$$

so the quotient pair is lc near $\overline{0}$, and $(\mathbb{A}^4, B)$ is lc near 0 by Lemma 2.15(2). Thus by [Liu26a, Lemma 8.6(2)], any prime divisor over X with center $\overline{0}$ has log discrepancy, with respect to the quotient pair of $(\mathbb{A}^4, B)$, at least $\psi(w)$ for some $w\in N_G\cap\mathbb{R}^4_{>0}$, hence at least γ by (D.41), that is,

$$\operatorname{mld}((\mathbb{A}^4\ni 0, B)/G) \geq \gamma. \tag{D.46}$$

Let $w := w^{(1)}$, $r := |G|$, and for $i\gg 0$ let

$$f_{i,\xi} := \xi^{-2w_x} f'_i(\xi^{w_x}x, \xi^{w_y}y, \xi^{w_z}z, \xi^{w_{t'}}t'), \qquad h_{i,\xi} := \xi^{-\Theta_w}\widetilde{h}_i(\xi^{w_x}x, \xi^{w_y}y, \xi^{w_z}z, \xi^{w_{t'}}t').$$

By (D.44), $f_{i,\xi}$ and $h_{i,\xi}$ are polynomials in x,y,z,t',ξ, semi-invariant with the characters χ_f and χ_h, $f_{i,0}=x^2+y^2z$, and $h_{i,0}$ is a non-zero multiple of $z^{k_1}t'^{k_2}$. Since $f_{i,\xi}^r$ and $h_{i,\xi}^r$ are G-invariant, they define an $\mathbb{R}$-ideal sheaf $\mathfrak{A}_i := (f_{i,\xi}^r)^{1/r}(h_{i,\xi}^r)^{1/(mr)}$ on $X\times\mathbb{A}^1_\xi$. By Definition 2.2 and Lemma 2.15(1)(2), for any $\xi\in\mathbb{C}$,

$$\operatorname{mld}(X\ni\overline{0}, \Delta_X, \mathfrak{A}_{i,\xi}) = \operatorname{mld}((\mathbb{A}^4\ni 0, \operatorname{div}(f_{i,\xi}) + \tfrac{1}{m}\operatorname{div}(h_{i,\xi}))/G).$$

For $\xi\neq 0$, the diagonal automorphism of $\mathbb{A}^4$ with weights w and value ξ commutes with G, fixes 0 and maps the pair for ξ to the pair for $\xi=1$, so the value above does not depend on $\xi\neq 0$. For $\xi=0$, the value above is $\geq\gamma$ by (D.46). By Theorem 2.16, it is lower semi-continuous in ξ, so the value for $\xi=1$ is at least the value for $\xi=0$. Thus

$$\operatorname{mld}((\mathbb{A}^4\ni 0, \operatorname{div}(f'_i) + \tfrac{1}{m}\operatorname{div}(\widetilde{h}_i))/G) \geq \gamma.$$

Since $(W'_i, \frac{1}{m}\operatorname{div}(h'_i))$ is lc near 0 (**Step 1**), Theorem 2.17(1) for the single equation $f_1 = f'_i$ and $\mathfrak{a} = (\widetilde{h}_i)^{1/m}$, together with Definition 2.2 and (D.32), implies

$$\begin{aligned} b_i &= \frac{a(F'_i, W'_i, \frac{1}{m}\operatorname{div}(h'_i))}{e_{F'_i}} \geq \operatorname{mld}((W'_i\ni 0, \tfrac{1}{m}\operatorname{div}(h'_i))/G) \\ &= \operatorname{mld}((\mathbb{A}^4\ni 0, \operatorname{div}(f'_i), (\widetilde{h}_i)^{1/m})/G) \geq \gamma. \end{aligned}$$

Thus (D.45) holds.

Step 5. In this step we bound $q_i(\tau_i)$ from above and conclude.

Let $w := w^{(2)}$, which is primitive in N_G and positive by **Step 3**. By [Liu26a, Lemma 8.5] for $d=4$ and the boundary

$$\operatorname{div}(f'_i) + \tfrac{\tau_i}{m}\operatorname{div}(\widetilde{h}_i),$$

the toric prime divisor E_w over X has center $\overline{0}$ and log discrepancy

$$w_x+w_y+w_z+w_{t'} - \operatorname{wt}_w(f'_i) - \frac{\tau_i}{m}\operatorname{wt}_w(\widetilde{h}_i) = \psi(w) + (1-\tau_i)\frac{\Theta_w}{m} = \gamma + (1-\tau_i)\frac{\Theta_w}{m}$$

with respect to the quotient pair, by (D.44) and (D.43). Since $(W_i', \frac{\tau_i}{m}\operatorname{div}(h_i'))$ is klt near 0, by (D.31) and Theorem 2.17(1) for the single equation $f_1 = f_i'$ and $\mathfrak{a} = (\widetilde{h}_i)^{\tau_i/m}$,

$$q_i(\tau_i) = \operatorname{mld}((\mathbb{A}^4 \ni 0, \operatorname{div}(f_i'), (\widetilde{h}_i)^{\tau_i/m})/G) \leq \gamma + (1-\tau_i)\frac{\Theta_w}{m}.$$

By (D.45), $q_i(\tau_i) - b_i \leq (1-\tau_i)\Theta_w/m$ for $i \gg 0$, where Θ_w does not depend on i. Since $\lim_{i\to+\infty}\tau_i = 1$, we obtain a contradiction with (D.29). The theorem follows. □

Proof of Theorem 13.3. We divide the proof into several steps.

Step 1. In this step we formulate (D.47) and fix the operations on families used in the proof of (D.47).

Let T be a variety, $\mathcal{X}$ a scheme flat and of finite type over T with a section $\sigma : T \to \mathcal{X}$, $\mathcal{B} \geq 0$ a $\mathbb{Q}$-divisor on $\mathcal{X}$, and N a positive integer such that $N(K_{\mathcal{X}/T} + \mathcal{B})$ is Cartier. For a point $t \in T$ let X_t be the fiber of $\mathcal{X}$ over t, $x_t := \sigma(t)$ and $B_t := \mathcal{B}|_{X_t}$. For an algebraically closed field $\Bbbk \supset \mathbb{C}(T)$ let $(X_\Bbbk, B_\Bbbk)$ and $x_\Bbbk$ be the base changes of $(\mathcal{X}, \mathcal{B})$ and σ along $\operatorname{Spec}\Bbbk \to T$. Let $\mathcal{S} \subset \mathbb{R}_{\geq 0}$ be the set of non-negative integral combinations of finitely many positive real numbers, with $1 \in \mathcal{S}$, as in Construction-Theorem 2.14. Let $t_i \in T$, $i \in \mathbb{N}^+$, be closed points, let $(X_i, B_i, x_i) := (X_{t_i}, B_{t_i}, x_{t_i})$, and let b, κ be positive real numbers. Assume the following.

- (H1) For any closed point $t \in T$ the following hold.
  - (i) X_t is a normal threefold.
  - (ii) No component of $\mathcal{B}$ contains an irreducible component of X_t.
  - (iii) (X_t, B_t) is klt near x_t.
  - (iv) $N(K_{X_t} + B_t)$ is the restriction of $N(K_{\mathcal{X}/T} + \mathcal{B})$.
- (H2) $D_i \geq 0$ is a $\mathbb{Q}$-divisor on X_i such that ND_i is Cartier near x_i and $\operatorname{mld}(X_i \ni x_i, B_i + D_i) = b$.
- (H3) Any prime divisor E over X_i computing $\operatorname{mld}(X_i \ni x_i, B_i + D_i)$ satisfies $\operatorname{mult}_E \mathfrak{m}_{x_i} \leq \kappa$.
- (H4) For any algebraically closed field $\Bbbk \supset \mathbb{C}(T)$ and any prime divisor E over $X_\Bbbk$, we have $a(E, X_\Bbbk, B_\Bbbk) \in \mathcal{S}$.
- (H5) Any proper closed subset of T contains t_i for only finitely many i.

Let $\mathcal{E}_i$ be the set of prime divisors over X_i computing $\operatorname{mld}(X_i \ni x_i, B_i + D_i)$. We prove that, after passing to a subsequence,

$$\sup_i \inf_{E\in\mathcal{E}_i} a(E, X_i, B_i) < +\infty. \tag{D.47}$$

A sequence $(t_i)_i$ of closed points is *generic in* T if any proper closed subset of T contains t_i for only finitely many i, so $(t_i)_i$ is generic in T by (H5). By Chevalley's theorem, $(t_i)_i$ stays generic, after passing to a subsequence, when we replace T with a non-empty open subset (*shrinking* T) or with a variety T'' admitting a dominant generically finite morphism to T together with closed points t_i'' over t_i (*replacing* T *with a cover* T'').

We use the following spreading. Let $\Bbbk_0$ be an algebraic closure of $\mathbb{C}(T)$. Any finitely many varieties, projective morphisms, prime divisors, Cartier divisors and linear equivalences over $X_{\Bbbk_0}$ are defined over a finite extension of $\mathbb{C}(T)$, hence over a variety T'' with a dominant generically finite morphism $T'' \to T$. We call the corresponding objects over T'' the *spreads* of the objects over $X_{\Bbbk_0}$. Possibly replacing T with the cover T'' and shrinking T, by generic flatness and generic smoothness we may assume the following.

- The spread varieties are flat over T with normal fibers.
- A resolution of the generic fiber spreads to a family of resolutions of the fibers whose exceptional prime divisors are geometrically integral over T and form a relatively simple normal crossing divisor.
- The coefficients of the spread divisors and the log discrepancies of the spread exceptional divisors are constant over T.
- T is smooth.

In particular, a spread variety whose generic fiber is klt, resp. terminal, near the preimage of the section has klt, resp. terminal, fibers near the preimage of the section and is itself klt there, and a morphism which is an isomorphism outside the section on the generic fiber is an isomorphism outside the section on any fiber.

Step 2. In this step we construct terminal models of the (X_i, B_i).

The field $\Bbbk_0$ is an algebraically closed field of characteristic 0 whose cardinality and transcendence degree over $\mathbb{Q}$ equal those of $\mathbb{C}$, so $\Bbbk_0 \cong \mathbb{C}$ as fields. By (H1) and the spreading of a log resolution of $(X_{\Bbbk_0}, B_{\Bbbk_0})$, the pair $(X_{\Bbbk_0}, B_{\Bbbk_0})$ is klt near $x_{\Bbbk_0}$, and by (H4) all log discrepancies of $(X_{\Bbbk_0}, B_{\Bbbk_0})$ belong to $\mathcal{S}$. Through the isomorphism $\Bbbk_0 \cong \mathbb{C}$, Construction-Theorem 2.14(1)–(4) with the trivial group yields, after shrinking $X_{\Bbbk_0}$ to a neighborhood of $x_{\Bbbk_0}$, a projective birational morphism $\mu_{\Bbbk_0} : Y_{\Bbbk_0} \to X_{\Bbbk_0}$ and a $\mathbb{Q}$-divisor $B_{Y_{\Bbbk_0}} \geq 0$ such that

$$K_{Y_{\Bbbk_0}} + B_{Y_{\Bbbk_0}} = \mu_{\Bbbk_0}^*(K_{X_{\Bbbk_0}} + B_{\Bbbk_0}),$$

the coefficients of $B_{Y_{\Bbbk_0}}$ belong to the finite set $\Lambda_{\mathcal{S}}$, $Y_{\Bbbk_0}$ is $\mathbb{Q}$-factorial and terminal, and $M_{\mathcal{S}} K_{Y_{\Bbbk_0}}$ is Cartier.

We spread $\mu_{\Bbbk_0}$, $B_{Y_{\Bbbk_0}}$ and the Cartier divisor $M_{\mathcal{S}} K_{Y_{\Bbbk_0}}$. Possibly replacing T with a cover and shrinking T, we obtain a projective morphism $\mu : \mathcal{Y} \to \mathcal{X}$ over T whose fiber over t_i is a projective birational morphism $\mu_i : Y_i \to X_i$ over a neighborhood of x_i such that Y_i is terminal, $M_{\mathcal{S}} K_{Y_i}$ is Cartier, and $B_{Y_i} \geq 0$ has coefficients in $\Lambda_{\mathcal{S}}$ and satisfies $K_{Y_i} + B_{Y_i} = \mu_i^*(K_{X_i} + B_i)$. Let $\Gamma_i := B_{Y_i} + \mu_i^* D_i \geq 0$. Then

$$K_{Y_i} + \Gamma_i = \mu_i^*(K_{X_i} + B_i + D_i). \tag{D.48}$$

Since $\mu_i^*(N D_i)$ is Cartier and (Y_i, Γ_i) is lc near $\mu_i^{-1}(x_i)$ by (D.48) and

$$\operatorname{mld}(X_i \ni x_i, B_i + D_i) = b > 0,$$

we have $\mu_i^* D_i \in \frac{1}{N}\mathbb{Z} \cap [0,1]$ near $\mu_i^{-1}(x_i)$. Thus the coefficients of Γ_i near $\mu_i^{-1}(x_i)$ belong to the finite set

$$I_0 := \left\{ \beta + \frac{j}{N} \;\middle|\; \beta \in \Lambda_{\mathcal{S}} \cup \{0\},\ j \in \mathbb{N} \right\} \cap [0,1].$$

Let $N_0 := N M_{\mathcal{S}}$. Since $N(K_{Y_i} + \Gamma_i) = \mu_i^* N(K_{X_i} + B_i + D_i)$ is Cartier near $\mu_i^{-1}(x_i)$ by (H1), (H2), and (D.48), and $M_{\mathcal{S}} K_{Y_i}$ is Cartier, $N_0 \Gamma_i$ is Cartier near $\mu_i^{-1}(x_i)$. Since $B_{Y_i} = (K_{Y_i} + B_{Y_i}) - K_{Y_i}$ is $\mathbb{Q}$-Cartier and $B_{Y_i} \geq 0$, for any prime divisor E over Y_i we have

$$a(E, X_i, B_i) = a(E, Y_i, B_{Y_i}) = a(E, Y_i, 0) - \operatorname{mult}_E B_{Y_i} \leq a(E, Y_i, 0). \tag{D.49}$$

Any prime divisor E over Y_i with $\operatorname{center}_{Y_i} E \subset \mu_i^{-1}(x_i)$ has $\operatorname{center}_{X_i} E = \{x_i\}$, so by (D.48)

$$a(E, Y_i, \Gamma_i) = a(E, X_i, B_i + D_i) \geq b, \tag{D.50}$$

with equality if and only if E computes $\operatorname{mld}(X_i \ni x_i, B_i + D_i)$.

For any i fix a prime divisor E_i over X_i computing $\operatorname{mld}(X_i \ni x_i, B_i + D_i)$, and let

$$Z_i := \operatorname{center}_{Y_i} E_i \subset \mu_i^{-1}(x_i).$$

Possibly passing to a subsequence, we may assume that $\dim Z_i$ is independent of i. By (D.49), it suffices to find, for any i, a prime divisor E_i' over Y_i with $\operatorname{center}_{Y_i} E_i' \subset \mu_i^{-1}(x_i)$, $a(E_i', Y_i, \Gamma_i) = b$, and $a(E_i', Y_i, 0)$ bounded independently of i.

Step 3. In this step we treat the case $\dim Z_i \geq 1$.

If $\dim Z_i = 2$, then $E_i = Z_i$ is a prime divisor on Y_i, and $E_i' := E_i$ satisfies $a(E_i', Y_i, 0) = 1$.

Assume $\dim Z_i = 1$. Let η_i be the generic point of Z_i and $k(\eta_i)$ the residue field at η_i, a finitely generated extension of $\mathbb{C}$ of transcendence degree 1. Since Y_i is a terminal threefold, it is smooth in codimension 2, hence at η_i, so $\mathcal{O}_{Y_i, \eta_i}$ is a regular local ring of dimension 2, and

$$\widehat{\mathcal{O}}_{Y_i, \eta_i} \cong k(\eta_i)[[u, v]]$$

[Mat89, Theorem 28.3]. Let $\mathcal{P}_i := \operatorname{Spec} k(\eta_i)[[u, v]]$ with closed point 0, let $\Gamma_{i,j}$, $j \in J_i$, be the components of Γ_i containing Z_i, with coefficients $\gamma_{i,j} \in I_0$ and local equations $\varphi_{i,j} \in k(\eta_i)[[u, v]]$,

and let $\mathfrak{a}_i := \prod_j (\varphi_{i,j})^{\gamma_{i,j}}$, an $\mathbb{R}$-ideal on $\mathcal{P}_i$ with exponents in I_0. The prime divisors over Y_i with center Z_i correspond to the prime divisors over $\mathcal{P}_i$ with center 0, with equal log discrepancies with respect to Γ_i and $\mathfrak{a}_i$, and with respect to 0, by [Kaw21, Remark 2.6 and Lemma 2.7(ii)] applied to $\operatorname{Spec} \mathcal{O}_{Y_i,\eta_i}$, the germ at a closed point of the generic fiber of a function on Y_i not constant on Z_i. Thus (D.50) and $a(E_i, Y_i, \Gamma_i) = b$ imply $\operatorname{mld}(\mathcal{P}_i \ni 0, \mathfrak{a}_i) = b$.

Let $\overline{k(\eta_i)}$ be an algebraic closure of $k(\eta_i)$, $\overline{\mathcal{P}}_i := \operatorname{Spec} \overline{k(\eta_i)}[[u, v]]$, and

$$\overline{\mathfrak{a}}_i := \mathfrak{a}_i \mathcal{O}_{\overline{\mathcal{P}}_i}.$$

Then $\operatorname{mld}(\overline{\mathcal{P}}_i \ni 0, \overline{\mathfrak{a}}_i) = b$ [Kaw21, Remark 2.6]. Since $b > 0$, some prime divisor Q over $\overline{\mathcal{P}}_i$ with center 0 computes $\operatorname{mld}(\overline{\mathcal{P}}_i \ni 0, \overline{\mathfrak{a}}_i)$. Let n_i be a positive integer with $n_i > \operatorname{mult}_Q \varphi_{i,j}$ for any j, and let

$$\mathfrak{a}'_i := \prod_j (\varphi_{i,j}, \mathfrak{m}^{n_i})^{\gamma_{i,j}},$$

where $\mathfrak{m}$ is the maximal ideal of $\overline{k(\eta_i)}[[u, v]]$. For any prime divisor Q' over $\overline{\mathcal{P}}_i$ with center 0, we have

$$\operatorname{mult}_{Q'}(\varphi_{i,j}, \mathfrak{m}^{n_i}) = \min\{\operatorname{mult}_{Q'} \varphi_{i,j}, n_i \operatorname{mult}_{Q'} \mathfrak{m}\} \le \operatorname{mult}_{Q'} \varphi_{i,j},$$

so

$$a(Q', \overline{\mathcal{P}}_i, \mathfrak{a}'_i) \ge a(Q', \overline{\mathcal{P}}_i, \overline{\mathfrak{a}}_i) \ge b, \tag{D.51}$$

and both are equalities for $Q' = Q$ since $n_i \operatorname{mult}_Q \mathfrak{m} \ge n_i > \operatorname{mult}_Q \varphi_{i,j}$. Thus $\operatorname{mld}(\overline{\mathcal{P}}_i \ni 0, \mathfrak{a}'_i) = b$, and by (D.51) any prime divisor computing $\operatorname{mld}(\overline{\mathcal{P}}_i \ni 0, \mathfrak{a}'_i)$ computes $\operatorname{mld}(\overline{\mathcal{P}}_i \ni 0, \overline{\mathfrak{a}}_i)$.

As in **Step 2**, $\overline{k(\eta_i)} \cong \mathbb{C}$. Under an isomorphism $\overline{k(\eta_i)} \cong \mathbb{C}$, $\mathfrak{a}'_i$ is an $\mathfrak{m}$-primary $\mathbb{R}$-ideal on $\operatorname{Spec} \mathbb{C}[[u, v]]$ with exponents in I_0, and it corresponds to an $\mathfrak{m}_0$-primary $\mathbb{R}$-ideal on $\mathbb{A}^2_{\mathbb{C}}$ with exponents in I_0, with equal log discrepancies at corresponding prime divisors with center 0 [Kaw21, Remark 2.6 and Lemma 2.7]. By [MN18, Theorem 1.3] there exists a positive integer L_1 depending only on I_0 such that for any i some prime divisor Q_i over $\overline{\mathcal{P}}_i$ with center 0 computes $\operatorname{mld}(\overline{\mathcal{P}}_i \ni 0, \mathfrak{a}'_i)$, hence $\operatorname{mld}(\overline{\mathcal{P}}_i \ni 0, \overline{\mathfrak{a}}_i)$, and satisfies $a(Q_i, \overline{\mathcal{P}}_i, 0) \le L_1$.

We descend Q_i to $\mathcal{P}_i$. By [Kaw21, Lemma 2.7(ii)], Q_i is the base change of a prime divisor over $\mathbb{A}^2_{\overline{k(\eta_i)}}$ with center 0, which is reached by finitely many blow-ups at closed points since a prime divisor over a smooth surface with center a closed point is. Hence Q_i is a prime divisor on a regular surface mapping to $\overline{\mathcal{P}}_i$ by a composite of finitely many blow-ups at closed points, and the centers of the blow-ups are defined over a finite Galois extension k'_i of $k(\eta_i)$ in $\overline{k(\eta_i)}$. Blowing up at each stage the $\operatorname{Gal}(k'_i/k(\eta_i))$-orbit of the center instead, we obtain a birational morphism $\mathcal{P}''_i \to \mathcal{P}_i$ such that the base change $\mathcal{P}''_i \times_{k(\eta_i)} k'_i$ contains a prime divisor Q'_i inducing Q_i. Since $k'_i/k(\eta_i)$ is separable, the morphism

$$\mathcal{P}''_i \times_{k(\eta_i)} k'_i \to \mathcal{P}''_i$$

is finite étale, so the image Q''_i of Q'_i is a prime divisor on $\mathcal{P}''_i$ with ramification index 1, that is, $\operatorname{mult}_{Q''_i} g = \operatorname{mult}_{Q'_i} g$ for any $g \in k(\eta_i)[[u, v]]$ and

$$a(Q''_i, \mathcal{P}_i, 0) = a(Q'_i, \mathcal{P}_i \times_{k(\eta_i)} k'_i, 0).$$

Together with [Kaw21, Remark 2.6],

$$a(Q''_i, \mathcal{P}_i, \mathfrak{a}_i) = a(Q_i, \overline{\mathcal{P}}_i, \overline{\mathfrak{a}}_i) = b \quad \text{and} \quad a(Q''_i, \mathcal{P}_i, 0) = a(Q_i, \overline{\mathcal{P}}_i, 0) \le L_1.$$

Let E'_i be the prime divisor over Y_i with center Z_i corresponding to Q''_i. Then $a(E'_i, Y_i, \Gamma_i) = b$ and $a(E'_i, Y_i, 0) \le L_1$, which finishes the case $\dim Z_i = 1$.

Step 4. In this step we assume that $Z_i = \{z_i\}$ is a closed point for any i, take a generic limit of the $(Y_i \ni z_i, \Gamma_i)$, and finish the case when the limit has minimal log discrepancy b or is klt.

Let $T_1 \subset \mathcal{Y}$ be the reduced preimage of $\sigma(T)$ under μ, a variety projective over T after passing to the irreducible component containing infinitely many z_i, let $\mathcal{Y}_1 := \mathcal{Y} \times_T T_1$, and let $\sigma_1 : T_1 \to \mathcal{Y}_1$ be the diagonal section. The fiber of $(\mathcal{Y}_1, \sigma_1)$ over the closed point $z_i \in T_1$ is

$(Y_i \ni z_i)$. Possibly passing to a subsequence and replacing T_1 with the closure of $\{z_i\}_i$, we may assume that $(z_i)_i$ is generic in T_1. Then $T_1 \to T$ is surjective by (H5), since its image is closed and contains t_i for any i. If $z_i \notin \operatorname{Supp} \Gamma_i$ for infinitely many i, then $E_i' := E_i$ works for these i, since $a(E_i, Y_i, 0) = a(E_i, Y_i, \Gamma_i) = b$. So we may assume that $z_i \in \operatorname{Supp} \Gamma_i$ for any i. Let φ_i be a local equation of $N_0\Gamma_i$ at z_i.

By [dFEM11, Section 3, before Lemma 3.1] and [Kaw21, Definitions 3.1 and 3.2], possibly replacing T_1 with a cover, shrinking T_1, and passing to a subsequence, there exist for any positive integer k a variety $\mathcal{Z}_k$, dominant morphisms $\mathcal{Z}_{k+1} \to \mathcal{Z}_k \to T_1$, closed points $\zeta_{k,i} \in \mathcal{Z}_k$ over z_i such that $\zeta_{k+1,i}$ maps to $\zeta_{k,i}$, and a regular function Φ_k on $\mathcal{Y}_1 \times_{T_1} \mathcal{Z}_k$ near the section, such that the following hold, where for any k and i we let $\Phi_{k,i}$ be the restriction of Φ_k to the fiber $(Y_i \ni z_i)$ over $\zeta_{k,i}$.

(L1) Any non-empty open subset of $\mathcal{Z}_k$ contains $\zeta_{k,i}$ for all but finitely many i.

(L2) $\varphi_i - \Phi_{k,i} \in \mathfrak{m}_{z_i}^k$, and $\Phi_{k+1} - \Phi_k$ vanishes to order at least k along the section at the generic point of $\mathcal{Z}_{k+1}$.

Let $\Bbbk$ be an algebraic closure of $\bigcup_k \mathbb{C}(\mathcal{Z}_k)$, let $(Y_\Bbbk \ni z_\Bbbk)$ be the base change of $(\mathcal{Y}_1, \sigma_1)$ to $\operatorname{Spec} \Bbbk$, let $\Phi_{k,\Bbbk}$ be the base change of Φ_k to $Y_\Bbbk$, let $\widehat{Y} := \operatorname{Spec} \widehat{\mathcal{O}}_{Y_\Bbbk, z_\Bbbk}$ with closed point $\widehat{z}$ and maximal ideal $\mathfrak{m}$, and let

$$\widehat{\varphi} := \lim_k \Phi_{k,\Bbbk} \in \widehat{\mathcal{O}}_{Y_\Bbbk, z_\Bbbk}, \qquad \widehat{\Gamma} := \frac{1}{N_0} \operatorname{div}(\widehat{\varphi}), \qquad \text{and} \qquad b_* := \operatorname{mld}(\widehat{Y} \ni \widehat{z}, \widehat{\Gamma}),$$

where $\widehat{\varphi} \neq 0$ by [dFEM11, Remark 3.5] and the limit of lc thresholds below. The field $\Bbbk$ is algebraically closed of characteristic 0 and generated over $\mathbb{C}$ by countably many elements up to algebraic extension, so $\Bbbk \cong \mathbb{C}$ as fields, as in **Step 2**. Any finitely many varieties, morphisms and divisors over $Y_\Bbbk$ are defined over a finite extension of $\mathbb{C}(\mathcal{Z}_k)$ for some k, hence over a variety $\mathcal{Z}'$ with a dominant generically finite morphism $\mathcal{Z}' \to \mathcal{Z}_k$. By (L1) and the argument for covers in **Step 1**, possibly passing to a subsequence, there exist closed points $\zeta_i' \in \mathcal{Z}'$ over $\zeta_{k,i}$ such that any non-empty open subset of $\mathcal{Z}'$ contains ζ_i' for all but finitely many i, and the spreading of **Step 1** over $\mathcal{Z}'$ yields the corresponding data over $(Y_i \ni z_i)$ as the fibers over ζ_i'. For $k' \geq k$, the pullbacks of the spread data to $\mathcal{Z}' \times_{\mathcal{Z}_k} \mathcal{Z}_{k'}$ have, over the points over ζ_i' and $\zeta_{k',i}$, fibers equal to their fibers over ζ_i'. We call the procedure above *spreading over the limit*.

Let P be a prime divisor over $\widehat{Y}$ with center $\widehat{z}$. By Lemma 2.12(3) with the trivial group, P corresponds to a prime divisor $P_\Bbbk$ over $Y_\Bbbk$ with center $z_\Bbbk$ such that

$$a(P_\Bbbk, Y_\Bbbk, 0) = a(P, \widehat{Y}, 0) \quad \text{and} \quad \operatorname{mult}_{P_\Bbbk} g = \operatorname{mult}_P g \text{ for any } g \in \mathcal{O}_{Y_\Bbbk, z_\Bbbk}.$$

Moreover $P_\Bbbk$ is a prime divisor on a projective birational morphism to $Y_\Bbbk$ which is an isomorphism outside $z_\Bbbk$. Spreading $P_\Bbbk$ over the limit as in [Kaw21, Remark 3.8(i)], we obtain for all but finitely many i a prime divisor P_i over Y_i with center z_i such that $a(P_i, Y_i, 0) = a(P, \widehat{Y}, 0)$ and $\operatorname{mult}_{P_i} \mathfrak{m}_{z_i} = \operatorname{mult}_P \mathfrak{m}$. Let k be an integer with $k \operatorname{mult}_P \mathfrak{m} > \operatorname{mult}_P \widehat{\varphi}$. Since $\widehat{\varphi} - \Phi_{k,\Bbbk} \in \mathfrak{m}^k$ by (L2) and $k \operatorname{mult}_P \mathfrak{m} > \operatorname{mult}_P \widehat{\varphi}$, we have $\operatorname{mult}_P \Phi_{k,\Bbbk} = \operatorname{mult}_P \widehat{\varphi}$, so $\operatorname{mult}_{P_i} \Phi_{k,i} = \operatorname{mult}_P \widehat{\varphi}$ for all but finitely many i, and then $\operatorname{mult}_{P_i} \varphi_i = \operatorname{mult}_P \widehat{\varphi}$ by (L2) and $k \operatorname{mult}_{P_i} \mathfrak{m}_{z_i} > \operatorname{mult}_P \widehat{\varphi}$. Thus, for all but finitely many i,

$$a(P_i, Y_i, \Gamma_i) = a(P, \widehat{Y}, \widehat{\Gamma}), \quad a(P_i, Y_i, 0) = a(P, \widehat{Y}, 0), \quad \text{and} \quad \operatorname{mult}_{P_i} \varphi_i = \operatorname{mult}_P \widehat{\varphi}. \tag{D.52}$$

By (D.52) and (D.50), $a(P, \widehat{Y}, \widehat{\Gamma}) \geq b$ for any such P, so

$$b_* \geq b. \tag{D.53}$$

The pairs (Y_i, Γ_i) are lc near z_i and the Y_i are fibers of one family of klt varieties. By [dFEM11, Propositions 2.11 and 3.3 and Corollary 3.4], after passing to a subsequence,

$$\operatorname{lct}(\widehat{Y} \ni \widehat{z}; \widehat{\Gamma}) = \lim_{i \to +\infty} \operatorname{lct}(Y_i \ni z_i; \Gamma_i) \geq 1.$$

In particular $(\widehat{Y}, \widehat{\Gamma})$ is lc.

If $b_* = b$, then some prime divisor P over $\widehat{Y}$ with center $\widehat{z}$ computes b_*, since $(\widehat{Y}, \widehat{\Gamma})$ is lc and has a log resolution [dFEM11, Theorems 2.3 and 2.4]. By (D.52) and (D.50), $E_i' := P_i$ satisfies $a(E_i', Y_i, \Gamma_i) = b$ and $a(E_i', Y_i, 0) = a(P, \widehat{Y}, 0)$, which is independent of i.

If $(\widehat{Y}, \widehat{\Gamma})$ is klt near $\widehat{z}$, then $\operatorname{lct}(\widehat{Y} \ni \widehat{z}; \widehat{\Gamma}) > 1$, so there exists $\epsilon > 0$ such that

$$\operatorname{lct}(Y_i \ni z_i; \Gamma_i) \geq 1 + \epsilon$$

for all but finitely many i. Then

$$0 \leq a(E_i, Y_i, (1+\epsilon)\Gamma_i) = b - \epsilon \operatorname{mult}_{E_i} \Gamma_i, \quad \text{so} \quad a(E_i, Y_i, 0) = b + \operatorname{mult}_{E_i} \Gamma_i \leq b\Big(1 + \frac{1}{\epsilon}\Big),$$

and $E_i' := E_i$ works.

By (D.53), we may thus assume that $b_* > b$ and that $(\widehat{Y}, \widehat{\Gamma})$ is lc and not klt near $\widehat{z}$.

Step 5. In this step we construct, on the limit, Kawakita's higher model at the smallest lc center, and spread it.

The germ $(Y_{\Bbbk} \ni z_{\Bbbk})$ is the base change of the $\mathbb{Q}$-factorial terminal threefold $Y_{\Bbbk_0}$ of **Step 2** along an extension of algebraically closed fields followed by localization at a $\Bbbk$-point, so it is terminal, and, as in [Kaw24, Section 4, the first paragraph after Theorem 4.2], it is $\mathbb{Q}$-factorial: a small $\mathbb{Q}$-factorialization of $Y_{\Bbbk}$ near $z_{\Bbbk}$ spreads to a small projective birational morphism over a variety over $\Bbbk_0$, whose fiber over a general closed point is a small projective birational morphism to the $\mathbb{Q}$-factorial $Y_{\Bbbk_0}$, hence an isomorphism, so the small $\mathbb{Q}$-factorialization is an isomorphism. Since $b_* > 0$, $\widehat{z}$ is not an lc center of $(\widehat{Y}, \widehat{\Gamma})$. Through $\Bbbk \cong \mathbb{C}$, [Kaw24, Section 4, the second paragraph after Theorem 4.2], used for the completion $\widehat{Y}$ of $(Y_{\Bbbk} \ni z_{\Bbbk})$, implies that there exists a smallest lc center $\widehat{\Sigma}$ of $(\widehat{Y}, \widehat{\Gamma})$ through $\widehat{z}$, that is, an lc center through $\widehat{z}$ contained in any lc center through $\widehat{z}$, and $\widehat{\Sigma}$ is a normal surface or a regular curve.

If $\widehat{\Sigma}$ is a regular curve, then through $\Bbbk \cong \mathbb{C}$, [Kaw24, Lemma 4.3, Propositions 5.1 and 5.2, Theorem 5.3, and the paragraph after Definition 6.1] yield a projective birational morphism $\pi_{\Bbbk} : V_{\Bbbk} \to Y_{\Bbbk}$ over a neighborhood of $z_{\Bbbk}$ with the following properties, where $\widehat{V} := V_{\Bbbk} \times_{Y_{\Bbbk}} \widehat{Y}$ with the induced morphism $\widehat{\pi} : \widehat{V} \to \widehat{Y}$.

- $\pi_{\Bbbk}$ is an isomorphism outside $z_{\Bbbk}$, and $V_{\Bbbk}$ is $\mathbb{Q}$-factorial and terminal.
- The $\mathbb{Q}$-divisor $\widehat{\Delta}_V$ defined by $K_{\widehat{V}} + \widehat{\Delta}_V = \widehat{\pi}^*(K_{\widehat{Y}} + \widehat{\Gamma})$ satisfies $\widehat{\Delta}_V \geq 0$.
- $V_{\Bbbk}$ is smooth at the closed points over $\widehat{z}$ of the strict transform $\widehat{\Sigma}_V$ of $\widehat{\Sigma}$.

In the curve case, $\widehat{\Sigma}_V \to \widehat{\Sigma}$ is a proper birational morphism onto a regular curve, hence an isomorphism, so $\widehat{\Sigma}_V \cap \widehat{\pi}^{-1}(\widehat{z})$ is one closed point $\widehat{v}$. Let $v_{\Bbbk} \in V_{\Bbbk}$ be the image of $\widehat{v}$. If $\widehat{\Sigma}$ is a normal surface, let $V_{\Bbbk} := Y_{\Bbbk}$, $\pi_{\Bbbk} := \mathrm{id}_{Y_{\Bbbk}}$, and $\widehat{\Delta}_V := \widehat{\Gamma}$. In the surface case, we let $\widehat{\Sigma}_V$, $\{v_{\Bbbk}\}$, $\{\widehat{v}\}$, and $\{v_i\}$ be empty. Whether $\widehat{\Sigma}$ is a curve or a surface, let $P_1, \dots, P_n$ be the $\pi_{\Bbbk}$-exceptional prime divisors, all with center $z_{\Bbbk}$ on $Y_{\Bbbk}$, write

$$K_{V_{\Bbbk}} = \pi_{\Bbbk}^* K_{Y_{\Bbbk}} + \sum_{j=1}^{n} d_j P_j, \qquad d_j \in \mathbb{Q},$$

and let N_1 be a positive integer such that $N_1 P_j$ is Cartier for any j and $N_1 N_0 \widehat{\Delta}_V$ is Cartier near $\widehat{\pi}^{-1}(\widehat{z})$.

Spreading $\pi_{\Bbbk}$, $v_{\Bbbk}$, the P_j and the Cartier divisors $N_1 P_j$ over the limit, we obtain for all but finitely many i a projective birational morphism $\pi_i : V_i \to Y_i$ over a neighborhood of z_i and π_i-exceptional prime divisors $P_{1,i}, \dots, P_{n,i}$, all with center z_i, with the following properties. Let Δ_{V_i} be defined by

$$K_{V_i} + \Delta_{V_i} = \pi_i^*(K_{Y_i} + \Gamma_i).$$

(M1) π_i is an isomorphism outside z_i, V_i is terminal, and, in the curve case, V_i is smooth at the closed point $v_i \in \pi_i^{-1}(z_i)$ given by the spread of $v_{\Bbbk}$.

(M2) $K_{V_i} = \pi_i^* K_{Y_i} + \sum_j d_j P_{j,i}$, and $N_1 P_{j,i}$ is Cartier for any j.

(M3) $\Delta_{V_i} \geq 0$, and the coefficients of Δ_{V_i} near $\pi_i^{-1}(z_i)$ belong to the finite set

$$I_1 := I_0 \cup \{\mathrm{mult}_{P_j} \widehat{\Delta}_V \mid 1 \leq j \leq n\}.$$

After shrinking, (M1) and (M2) follow from the spreading of **Step 1**. For (M3), any component of Δ_{V_i} other than the $P_{j,i}$ is $(\pi_i)_*^{-1}\Gamma'$ for a component Γ' of Γ_i, and by (D.52) applied to the divisors over $\widehat{Y}$ induced by the P_j,

$$\mathrm{mult}_{P_{j,i}} \Delta_{V_i} = 1 - a(P_{j,i}, Y_i, \Gamma_i) = 1 - a(P_j, \widehat{Y}, \widehat{\Gamma}) = \mathrm{mult}_{P_j} \widehat{\Delta}_V \geq 0$$

for all but finitely many i. By (D.50), any prime divisor E over V_i with $\mathrm{center}_{V_i} E \subset \pi_i^{-1}(z_i)$ satisfies

$$a(E, V_i, \Delta_{V_i}) = a(E, Y_i, \Gamma_i) \geq b. \tag{D.54}$$

Step 6. In this step we state (D.55) below and construct the limit data for the proof of (D.55). We prove that, after passing to a subsequence, any prime divisor E over Y_i with $\mathrm{center}_{Y_i} E = \{z_i\}$ and, in the curve case, $\mathrm{center}_{V_i} E \neq \{v_i\}$ satisfies

$$a(E, Y_i, \Gamma_i) \geq b_*. \tag{D.55}$$

We adapt the arguments for [Kaw15, Lemma 5.5] and [Kaw26, Lemma 4.11] to the families of **Step 5**.

Let $\widehat{V}_{\widehat{z}} := \widehat{\pi}^{-1}(\widehat{z})$, so that $\widehat{\Sigma}_V \cap \widehat{V}_{\widehat{z}} = \{\widehat{v}\}$ in the curve case.

The pair $(\widehat{V}, \widehat{\Delta}_V)$ is plt near any point of $\widehat{V}_{\widehat{z}} \backslash \widehat{\Sigma}_V$. Let Q be a prime divisor over $\widehat{V}$ with $a(Q, \widehat{V}, \widehat{\Delta}_V) = 0$ whose center on $\widehat{V}$ intersects $\widehat{V}_{\widehat{z}} \backslash \widehat{\Sigma}_V$. Then $\mathrm{center}_{\widehat{Y}} Q$ is an lc center of $(\widehat{Y}, \widehat{\Gamma})$ containing $\widehat{z}$, and $\mathrm{center}_{\widehat{Y}} Q \neq \{\widehat{z}\}$ since $b_* > 0$. Hence $\mathrm{center}_{\widehat{Y}} Q \supset \widehat{\Sigma}$. If $\dim \mathrm{center}_{\widehat{Y}} Q = 1$, then $\mathrm{center}_{\widehat{Y}} Q = \widehat{\Sigma}$ is a curve and $\mathrm{center}_{\widehat{V}} Q$ is an irreducible closed subset of $\widehat{\pi}^{-1}(\widehat{\Sigma}) = \widehat{\Sigma}_V \cup \widehat{V}_{\widehat{z}}$ dominating $\widehat{\Sigma}$, that is, $\mathrm{center}_{\widehat{V}} Q = \widehat{\Sigma}_V$, which does not intersect $\widehat{V}_{\widehat{z}} \backslash \widehat{\Sigma}_V$. Thus $\mathrm{center}_{\widehat{Y}} Q$ is a prime divisor, hence a component of $\lfloor \widehat{\Gamma} \rfloor$, and $Q = \widehat{\pi}_*^{-1} \mathrm{center}_{\widehat{Y}} Q$ is a component of $\lfloor \widehat{\Delta}_V \rfloor$. Since $\widehat{\Delta}_V \geq 0$, $(\widehat{V}, \widehat{\Delta}_V)$ is plt there. Similarly, at any point $y \in \widehat{Y} \backslash (\{\widehat{z}\} \cup \widehat{\Sigma})$ the lc centers of $(\widehat{Y}, \widehat{\Gamma})$ through y are components of $\lfloor \widehat{\Gamma} \rfloor$, so $(\widehat{Y}, \widehat{\Gamma})$ is plt at y.

An auxiliary function. By prime avoidance, there exists $\widehat{\psi} \in \mathfrak{m}$ vanishing along any component of $\widehat{\Gamma} - \lfloor \widehat{\Gamma} \rfloor$ and along no component of $\lfloor \widehat{\Gamma} \rfloor$. Fix a log resolution $\widetilde{V} \to \widehat{V}$ of $(\widehat{V}, \widehat{\Delta}_V + \widehat{\pi}^* \operatorname{div}(\widehat{\psi}))$ [dFEM11, Theorems 2.3 and 2.4]. Only finitely many prime divisors Q on $\widetilde{V}$ are exceptional over $\widehat{V}$ or components of the strict transform of $\widehat{\Delta}_V + \widehat{\pi}^* \operatorname{div}(\widehat{\psi})$, and for any $\epsilon \geq 0$, whether the pair

$$\big(\widehat{V}, \widehat{\Delta}_V + \epsilon \widehat{\pi}^* \operatorname{div}(\widehat{\psi})\big)$$

is lc near a point is decided by the log discrepancies of those Q whose centers contain it. By the plt property above, those Q with $a(Q, \widehat{V}, \widehat{\Delta}_V) = 0$ and center intersecting $\widehat{V}_{\widehat{z}} \backslash \widehat{\Sigma}_V$ are components of $\lfloor \widehat{\Delta}_V \rfloor$, along which $\widehat{\psi}$ does not vanish. Hence there exists $\epsilon_1 > 0$ such that

$$\big(\widehat{V}, \widehat{\Delta}_V + \epsilon_1 \widehat{\pi}^* \operatorname{div}(\widehat{\psi})\big) \text{ is lc near any point of } \widehat{V}_{\widehat{z}} \backslash \widehat{\Sigma}_V. \tag{D.56}$$

A resolution. For a projective birational morphism $\rho_{\Bbbk} : R_{\Bbbk} \to V_{\Bbbk}$ which is an isomorphism outside $\pi_{\Bbbk}^{-1}(z_{\Bbbk})$, let $\widehat{R} := R_{\Bbbk} \times_{Y_{\Bbbk}} \widehat{Y}$ with induced morphism $\widehat{\rho} : \widehat{R} \to \widehat{V}$, let $S = \sum_{j \in J} S_j$ be the reduced preimage of $\widehat{z}$ in $\widehat{R}$, which we identify with the reduced preimage of $z_{\Bbbk}$ in $R_{\Bbbk}$, let

$$\widehat{\Gamma}^R := (\widehat{\pi} \circ \widehat{\rho})_*^{-1} \widehat{\Gamma}, \quad \widehat{\Gamma}^R_{=1} := (\widehat{\pi} \circ \widehat{\rho})_*^{-1} \lfloor \widehat{\Gamma} \rfloor, \quad \text{and} \quad \widehat{\Gamma}^R_{<1} := (\widehat{\pi} \circ \widehat{\rho})_*^{-1} (\widehat{\Gamma} - \lfloor \widehat{\Gamma} \rfloor),$$

let

$$\widehat{\Omega}_{<1} := S \cap \operatorname{Supp} \widehat{\Gamma}^R_{<1},$$

and let $\widehat{\Omega}_v := \widehat{\rho}^{-1}(\widehat{v})$. Then $\widehat{\Omega}_{<1}$ and $\widehat{\Omega}_v$ are closed subsets of S, and S is projective over $\Bbbk$. We choose $\rho_{\Bbbk}$ with the following properties.

(a) $R_{\Bbbk}$ is smooth, and $\mathfrak{m}_{z_{\Bbbk}} \mathcal{O}_{R_{\Bbbk}}$ is invertible.

(b) Any point of $\widehat{\Omega}_{<1}$ is contained in some S_j with $\epsilon_1 \operatorname{mult}_{S_j} \widehat{\psi} \geq b_*$.

(c) $S + \widehat{\Gamma}^R_{=1}$ is a simple normal crossing divisor near any point of $S \backslash (\widehat{\Omega}_{<1} \cup \widehat{\Omega}_v)$.

Since $(Y_{\Bbbk} \ni z_{\Bbbk})$ is terminal, $V_{\Bbbk}$ is smooth outside $\pi_{\Bbbk}^{-1}(z_{\Bbbk})$ after shrinking $Y_{\Bbbk}$ to a neighborhood of $z_{\Bbbk}$. First let $\rho_{\Bbbk}$ be the composite of the blow-up of $\mathfrak{m}_{z_{\Bbbk}} \mathcal{O}_{V_{\Bbbk}}$ and a resolution which is an isomorphism over the non-singular locus. Then (a) holds, and (a) is kept below since we only blow up smooth centers.

To obtain (c), we first make $S + \widehat{\Gamma}^R_{=1}$ a simple normal crossing divisor near any point of $S \backslash \widehat{\Omega}_v$. Let

$$U := \widehat{R} \backslash \big(\widehat{\Omega}_v \cup \overline{\widehat{\rho}_*^{-1} \widehat{\Sigma}_V} \big).$$

Since $\overline{\widehat{\rho}_*^{-1} \widehat{\Sigma}_V} \cap S \subset \widehat{\Omega}_v$, we have $S \cap U = S \backslash \widehat{\Omega}_v$. A point of $U \backslash S$ is a point $y \in \widehat{Y} \backslash \{\widehat{z}\}$, with $y \notin \widehat{\Sigma}$ in the curve case. At such y, $\widehat{Y}$ is regular since $Y_{\Bbbk}$ is smooth outside $z_{\Bbbk}$, and $(\widehat{Y}, \widehat{\Gamma})$ is plt, because any lc center of $(\widehat{Y}, \widehat{\Gamma})$ through y contains $\widehat{z}$, hence $\widehat{\Sigma}$, and so it is a component of $\lfloor \widehat{\Gamma} \rfloor$, equal to $\widehat{\Sigma}$ in the surface case. Thus the components of $\lfloor \widehat{\Gamma} \rfloor$ through y are normal and pairwise disjoint by connectedness for plt pairs, and any component L of $\lfloor \widehat{\Gamma} \rfloor$ through y satisfies $\dim \mathcal{O}_{L,y} \leq 1$, so $\lfloor \widehat{\Gamma} \rfloor$ is regular at y. Hence the locus $U_{\mathrm{bad}} \subset U$ where $S + \widehat{\Gamma}^R_{=1}$ is not a simple normal crossing divisor is a closed subset of U contained in S.

By [Tem18, Theorem 1.1.3] for U and the boundary formed by the components of $S + \widehat{\Gamma}^R_{=1}$, finitely many blow-ups along regular centers contained in the preimage of U_{bad} turn $S + \widehat{\Gamma}^R_{=1}$, where S is the reduced preimage of $\widehat{z}$ on the new model, into a simple normal crossing divisor on the preimage of U. Each center is contained in the preimage of $\widehat{z}$, which is the support of the current S, so it is a closed point, a curve, or a component of the Cartier divisor S on the regular $\widehat{R}$, whose blow-up is an isomorphism. We blow up the closure $\overline{\Pi}$ of each center Π in $\widehat{R}$, a closed subset of S with $\overline{\Pi} \backslash \Pi \subset \widehat{\Omega}_v$. Before blowing up a curve $\overline{\Pi}$, we blow up the singular points of $\overline{\Pi}$, which are contained in $\widehat{\Omega}_v$, and then those of the strict transforms of $\overline{\Pi}$, until the strict transform of $\overline{\Pi}$ is regular. The blow-ups of the singular points are isomorphisms over the preimage of U, so afterwards $S + \widehat{\Gamma}^R_{=1}$ is a simple normal crossing divisor near any point of $S \backslash \widehat{\Omega}_v$.

To obtain (b), for a closed point $p \in \widehat{\Omega}_{<1}$ let $o(p) := \max\{\operatorname{mult}_{S_j} \widehat{\psi} \mid p \in S_j\}$, a positive integer, and let $\Omega_{<b_*}$ be the closure in S of

$$\{p \in \widehat{\Omega}_{<1} \mid \epsilon_1 o(p) < b_*\},$$

a closed subset of $\widehat{\Omega}_{<1}$ with $\dim \Omega_{<b_*} \leq 1$ since no S_j is a component of $\operatorname{Supp} \widehat{\Gamma}^R_{<1}$. For any integer c, the points of $\widehat{\Omega}_{<1}$ with $o \geq c$ form the intersection of $\widehat{\Omega}_{<1}$ with the union of the S_j with $\operatorname{mult}_{S_j} \widehat{\psi} \geq c$, a closed subset. Since $\widehat{\psi}$ vanishes along $\operatorname{Supp}(\widehat{\Gamma} - \lfloor \widehat{\Gamma} \rfloor)$, $(\widehat{\pi} \circ \widehat{\rho})_*^{-1} \operatorname{div}(\widehat{\psi})$ contains $\widehat{\Omega}_{<1}$. Hence, for a closed point p, resp. an irreducible curve Π, contained in $\widehat{\Omega}_{<1}$, any prime divisor S' over $\widehat{R}$ with center p, resp. Π, satisfies

$$\operatorname{mult}_{S'} \widehat{\psi} \geq o(p) + 1, \quad \text{resp.} \quad \operatorname{mult}_{S'} \widehat{\psi} \geq \max\{\operatorname{mult}_{S_j} \widehat{\psi} \mid \Pi \subset S_j\} + 1, \tag{D.57}$$

and after a blow-up along a center in S, the new $\widehat{\Omega}_{<1}$ is the union of the strict transform of the old one and a closed subset of the exceptional locus.

Assume that $\Omega_{<b_*} \neq \emptyset$, and let $o_{\min}$ be the minimum of $o(p)$ over the closed points $p \in \widehat{\Omega}_{<1}$ with $\epsilon_1 o(p) < b_*$. Since the points with $o \geq o_{\min} + 1$ form a closed subset and the general points of any component of $\Omega_{<b_*}$ satisfy $\epsilon_1 o < b_*$, any closed point $p \in \widehat{\Omega}_{<1}$ with $o(p) = o_{\min}$ is contained in a component of $\Omega_{<b_*}$ at whose general points $o = o_{\min}$, and $o \geq o_{\min}$ at any point of such a component. We blow up, one by one, the components of $\Omega_{<b_*}$ at whose general points $o = o_{\min}$, or rather their strict transforms, on which $o \geq o_{\min}$ also after the blow-ups for the earlier components.

Let Π be such a component. If Π is a closed point, we blow up Π. If Π is a curve not contained in $\widehat{\Omega}_v$, we first use [Tem18, Theorem 1.1.9] for the open subset of $\widehat{R} \backslash \widehat{\Omega}_v$ where $S + \widehat{\Gamma}^R_{=1}$ is a

simple normal crossing divisor, the boundary formed by the components of $S+\widehat{\Gamma}^R_{=1}$, and the closed subscheme $\Pi\backslash\widehat{\Omega}_v$. The centers of the blow-ups in [Tem18, Theorem 1.1.9] are regular and contained in the preimages of finitely many closed points of $\Pi\backslash\widehat{\Omega}_v$. Afterwards $S+\widehat{\Gamma}^R_{=1}$ is a simple normal crossing divisor near any point of $S\backslash\widehat{\Omega}_v$, and the strict transform of Π is regular and has simple normal crossings with $S+\widehat{\Gamma}^R_{=1}$ at any point outside $\widehat{\Omega}_v$. If Π is a curve, we then blow up closed points over $\widehat{v}$ until the strict transform of Π is regular, and blow up the strict transform of Π.

The blow-up of a closed point near which a reduced divisor D is simple normal crossing, and the blow-up along a regular curve having simple normal crossings with a simple normal crossing reduced divisor D, turn D together with the exceptional divisor into a simple normal crossing divisor near the preimage of the center. Thus after the blow-ups for each Π, $S+\widehat{\Gamma}^R_{=1}$ is a simple normal crossing divisor near any point of $S\backslash\widehat{\Omega}_v$. Each exceptional prime divisor of the blow-ups for Π has, before the blow-ups for Π, center Π or a closed point of Π, so $\operatorname{mult}\widehat{\psi}\geq o_{\min}+1$ along it by (D.57), and o does not change at the points of $\widehat{\Omega}_{<1}$ outside the preimages of the centers. Hence, after the blow-ups for all components of $\Omega_{<b_*}$ at whose general points $o=o_{\min}$, we have $o(p)\geq o_{\min}+1$ for any closed point $p\in\widehat{\Omega}_{<1}$ with $\epsilon_1 o(p)<b_*$. Thus $o_{\min}$ increases by at least 1 in each round, and $\epsilon_1 o_{\min}<b_*$ as long as $\Omega_{<b_*}\neq\emptyset$, so after finitely many rounds $\Omega_{<b_*}=\emptyset$, which is (b).

All blow-ups above have centers in the preimage of $\widehat{z}$, so they are blow-ups of $R_{\Bbbk}$ along closed subsets of the preimage of $z_{\Bbbk}$. Thus (a) and (b) hold, and (c) holds since $S+\widehat{\Gamma}^R_{=1}$ is a simple normal crossing divisor near any point of $S\backslash\widehat{\Omega}_v$.

For $j\in J$ let

$$\alpha_j:=a(S_j,\widehat{Y},\widehat{\Gamma})\geq b_*,\qquad \nu_j:=\operatorname{mult}_{S_j}\mathfrak{m}\geq 1,\qquad\text{and}\qquad \beta_j:=\operatorname{mult}_{S_j}\widehat{\varphi},$$

where $\alpha_j\geq b_*$ since $\operatorname{center}_{\widehat{Y}}S_j=\{\widehat{z}\}$. Then locally

$$\widehat{\rho}^*\widehat{\pi}^*\widehat{\varphi}=\widehat{\varphi}^R\prod_j s_j^{\beta_j},$$

where s_j is a local equation of S_j and $\widehat{\varphi}^R$ is a local equation of $N_0\widehat{\Gamma}^R$. Let k_1 be a positive integer with $k_1\nu_j>\operatorname{mult}_{S_j}\widehat{\psi}$ for any j, and let $\mathfrak{d}:=(\widehat{\psi})+\mathfrak{m}^{k_1}$, an $\mathfrak{m}$-primary ideal with $\operatorname{mult}_{S_j}\mathfrak{d}=\operatorname{mult}_{S_j}\widehat{\psi}$ for any j. Since $\mathfrak{d}\supset(\widehat{\psi})$, (D.56) implies

$$\big(\widehat{V},\widehat{\Delta}_V,(\mathfrak{d}\mathcal{O}_{\widehat{V}})^{\epsilon_1}\big)\text{ is lc near any point of }\widehat{V}_{\widehat{z}}\backslash\widehat{\Sigma}_V.\tag{D.58}$$

By [Kaw21, Lemma 2.7], $\mathfrak{d}$ is the completion of an $\mathfrak{m}_{z_{\Bbbk}}$-primary ideal $\mathfrak{d}_{\Bbbk}$ of $\mathcal{O}_{Y_{\Bbbk},z_{\Bbbk}}$.

Step 7. In this step we spread the data of **Step 6** and prove (D.55).

Let N_2 be a positive integer divisible by N_0N_1 such that $N_2\sum_j d_jP_j$ is Cartier, so that $N_2\widehat{\Delta}_V$ is Cartier near $\widehat{V}_{\widehat{z}}$. We spread $\rho_{\Bbbk}$, the S_j, the closed subsets $\widehat{\Omega}_{<1}$ and $\widehat{\Omega}_v$ of S, the ideal $\mathfrak{d}_{\Bbbk}$, and the Cartier divisor $N_2\sum_j d_jP_j$ over the limit, together with the data of **Step 5**. We obtain a variety $\mathcal{Z}'$ with a dominant generically finite morphism to some $\mathcal{Z}_{k_0}$, families $\mathcal{R}\to\mathcal{V}\to\mathcal{Y}'$ over $\mathcal{Z}'$, where $\mathcal{Y}':=\mathcal{Y}_1\times_{T_1}\mathcal{Z}'$, and for all but finitely many i the fibers $\rho_i:R_i\to V_i$, prime divisors $S_{j,i}$ on R_i, closed subsets $\Omega_{<1,i}$ and $\Omega_{v,i}$ of $S_i:=\sum_j S_{j,i}$, and an $\mathfrak{m}_{z_i}$-primary ideal $\mathfrak{d}_i$, with the following properties.

- $\mathcal{R}$ and R_i are smooth, and $\mathcal{V}$ and V_i are klt.
- S_i is the reduced preimage of z_i, $\Omega_{v,i}=\rho_i^{-1}(v_i)$, and
$$\mathfrak{m}_{z_i}\mathcal{O}_{R_i}=\mathcal{O}_{R_i}\Big(-\sum_j\nu_jS_{j,i}\Big).$$
- $\operatorname{mult}_{S_{j,i}}\mathfrak{d}_i=\operatorname{mult}_{S_j}\mathfrak{d}$ for any j, and by (b) any point of $\Omega_{<1,i}$ is contained in some $S_{j,i}$ with $\epsilon_1\operatorname{mult}_{S_j}\mathfrak{d}\geq b_*$.

By (D.52) applied to the divisors induced by the S_j, $a(S_{j,i}, Y_i, \Gamma_i) = \alpha_j$ and $\operatorname{mult}_{S_{j,i}} \varphi_i = \beta_j$ for all but finitely many i.

Let $\Gamma_i^R := (\pi_i \circ \rho_i)_*^{-1}\Gamma_i$ and let φ_i^R be a local equation of $N_0\Gamma_i^R$, so that

$$(\pi_i \circ \rho_i)^*\varphi_i = \varphi_i^R \prod_j s_{j,i}^{\beta_j}$$

for local equations $s_{j,i}$ of $S_{j,i}$. Then

$$K_{R_i} + \sum_{j\in J}(1-\alpha_j)S_{j,i} + \Gamma_i^R = (\pi_i \circ \rho_i)^*(K_{Y_i} + \Gamma_i). \tag{D.59}$$

For $k \geq k_0$ with $k\nu_j > \beta_j$ for any j, we have $\operatorname{mult}_{S_j} \Phi_{k,\Bbbk} = \beta_j$ since $\widehat{\varphi} - \Phi_{k,\Bbbk} \in \mathfrak{m}^k$, so $\operatorname{mult}_{S_{j,i}} \Phi_{k,i} = \beta_j$ for all but finitely many i, and

$$(\pi_i \circ \rho_i)^*\Phi_{k,i} = \Phi_{k,i}^R \prod_j s_{j,i}^{\beta_j}$$

with $\Phi_{k,i}^R$ regular near S_i. Since

$$(\pi_i \circ \rho_i)^*(\varphi_i - \Phi_{k,i}) \in \mathfrak{m}_{z_i}^k \mathcal{O}_{R_i} = \mathcal{O}_{R_i}\Big(-\sum_j k\nu_j S_{j,i}\Big),$$

we obtain

$$\varphi_i^R - \Phi_{k,i}^R \in \mathcal{O}_{R_i}\Big(-\sum_{j\in J}(k\nu_j - \beta_j)S_{j,i}\Big). \tag{D.60}$$

Similarly, let ϖ be a local equation of the Cartier divisor $N_2\sum_j d_j P_j$, and let

$$\vartheta_i := (\pi_i^*\varphi_i)^{N_2/N_0}\varpi^{-1}, \quad \vartheta_{k,i} := (\pi_i^*\Phi_{k,i})^{N_2/N_0}\varpi^{-1}, \quad \text{and} \quad \widehat{\vartheta} := (\widehat{\pi}^*\widehat{\varphi})^{N_2/N_0}\varpi^{-1},$$

so that ϑ_i is a local equation of $N_2\Delta_{V_i}$. Since

$$\varphi_i^{N_2/N_0} - \Phi_{k,i}^{N_2/N_0} \in \mathfrak{m}_{z_i}^k,$$

$$\vartheta_i - \vartheta_{k,i} \in \mathfrak{m}_{z_i}^k \mathcal{O}_{V_i}\Big(N_2\sum_j d_j P_{j,i}\Big). \tag{D.61}$$

We use [Kaw21, Theorem 3.11], which says the following. Let $\mathcal{M}$ be a klt variety and $\mathcal{M} \to \mathcal{H}$ a morphism to a variety whose closed fibers $\mathcal{M}_s$ are klt, let $\mathfrak{a} = \prod_j \mathfrak{a}_j^{r_j}$ be an $\mathbb{R}$-ideal on $\mathcal{M}$, and let $\mathcal{C} \subset \mathcal{M}$ be an irreducible closed subset dominating $\mathcal{H}$ such that $\operatorname{mld}(\mathcal{M} \ni \eta_{\mathcal{C}}, \mathfrak{a}) = 0$, where $\eta_{\mathcal{C}}$ is the generic point of $\mathcal{C}$, and a prime divisor $\mathcal{Q}$ over $\mathcal{M}$ computes it. Then, after replacing $\mathcal{M}$ and $\mathcal{H}$ with dense open subsets, for any closed point $s \in \mathcal{H}$ any connected component $\mathcal{Q}_s$ of the fiber of $\mathcal{Q}$ over s is a prime divisor over a component of $\mathcal{M}_s$, and $(\mathcal{M}_s, \mathfrak{b})$ is lc near the center $\mathcal{C}_s$ of $\mathcal{Q}_s$ on $\mathcal{M}_s$ for any $\mathbb{R}$-ideal $\mathfrak{b} = \prod_j \mathfrak{b}_j^{r_j}$ on $\mathcal{M}_s$ with

$$\mathfrak{a}_j\mathcal{O}_{\mathcal{M}_s} + \mathfrak{p}_j = \mathfrak{b}_j + \mathfrak{p}_j \quad \text{for any } j, \qquad \text{where} \quad \mathfrak{p}_j := \{f \in \mathcal{O}_{\mathcal{M}_s} \mid \operatorname{mult}_{\mathcal{Q}_s} f > \operatorname{mult}_{\mathcal{Q}} \mathfrak{a}_j\}.$$

We apply it to $\mathcal{V}' \to \mathcal{Z}'$ equal to $\mathcal{R} \to \mathcal{Z}'$ or $\mathcal{V} \to \mathcal{Z}'$, whose total spaces and closed fibers are klt, in the following form. Let V_i', $V_{\Bbbk}'$, and $\widehat{V}'$ be the corresponding fibers, base change, and limit. Let $\Delta' \geq 0$ be a spread Cartier divisor on $\mathcal{V}'$, N' a positive integer, and, for $k \geq k_0$, θ_k a regular function near the preimage of the section on the pull-back of $\mathcal{V}'$ to $\mathcal{Z}' \times_{\mathcal{Z}_{k_0}} \mathcal{Z}_k$, with restriction $\theta_{k,i}$ to V_i' and base change $\theta_{k,\Bbbk}$ to $V_{\Bbbk}'$, such that $\widehat{\theta} := \lim_k \theta_{k,\Bbbk}$ exists on $\widehat{V}'$ and

$$\widehat{\Delta} := \Delta' + \frac{1}{N'}\operatorname{div}(\widehat{\theta}) \geq 0.$$

Let $\mathfrak{c}$ be a spread $\mathbb{R}$-ideal, and $\Pi \subset V'_{\Bbbk}$ an irreducible locally closed subset contained in the preimage of $z_{\Bbbk}$, with generic point η_Π and with ideal I_Π of $\overline{\Pi}$. *Claim:* assume that $s \geq 0$, that $(\widehat{V}', \widehat{\Delta}, \mathfrak{c} I_\Pi^s)$ is lc near η_Π, that P_0 is a prime divisor over $\widehat{V}'$ with center $\overline{\Pi}$ and

$$a(P_0, \widehat{V}', \widehat{\Delta}, \mathfrak{c} I_\Pi^s) = 0,$$

and that for some $k \geq k_0$ we have

$$\operatorname{mult}_{P_0}(\widehat{\theta} - \theta_{k,\Bbbk}) > \operatorname{mult}_{P_0} \widehat{\theta},$$

and functions θ_i on V'_i satisfy $\theta_i - \theta_{k,i} \in \mathfrak{q}_i$ for ideals $\mathfrak{q}_i$ with $\operatorname{mult}_{P_{0,i}} \mathfrak{q}_i > \operatorname{mult}_{P_0} \widehat{\theta}$, where $P_{0,i}$ is the spread of P_0. Then there exists a non-empty open subset $\Pi^\circ \subset \Pi$ such that, for all but finitely many i, the triple

$$\Big(V'_i, \Delta'_i + \frac{1}{N'} \operatorname{div}(\theta_i), \mathfrak{c}_i\Big)$$

is lc near the spread Π_i° of Π°.

We prove the claim. Since $\overline{\Pi}$ is contained in the preimage of $\widehat{z}$, P_0 is a prime divisor over $\widehat{Y}$ with center $\widehat{z}$, so by Lemma 2.12(3) with the trivial group it restricts to a prime divisor $P_{0,\Bbbk}$ over $Y_{\Bbbk}$ with center $z_{\Bbbk}$. Since $\widehat{V}' \to V'_{\Bbbk}$ is the base change of $\widehat{Y} \to Y_{\Bbbk}$, the valuation of $P_{0,\Bbbk}$ is the restriction of that of P_0, and $K_{\widehat{V}'} - (\text{pull-back of } K_{\widehat{Y}})$ is the pull-back of $K_{V'_{\Bbbk}} - (\text{pull-back of } K_{Y_{\Bbbk}})$, $P_{0,\Bbbk}$ has center $\overline{\Pi}$ on $V'_{\Bbbk}$, with $a(P_{0,\Bbbk}, V'_{\Bbbk}, \cdot) = a(P_0, \widehat{V}', \cdot)$ and $\operatorname{mult}_{P_{0,\Bbbk}} = \operatorname{mult}_{P_0}$ on spread divisors, functions, and ideals. Spreading Π and $P_{0,\Bbbk}$ over the limit at the level k, we obtain a variety $\mathcal{Z}_\Pi$ with a dominant morphism to $\mathcal{Z}' \times_{\mathcal{Z}_{k_0}} \mathcal{Z}_k$, the pull-back $\mathcal{V}'_\Pi \to \mathcal{Z}_\Pi$ of $\mathcal{V}'$, the closure $\Pi_{\mathcal{Z}}$ in $\mathcal{V}'_\Pi$ of the spread of Π, which dominates $\mathcal{Z}_\Pi$, and a prime divisor $P_{0,\mathcal{Z}}$ over $\mathcal{V}'_\Pi$ with center $\Pi_{\mathcal{Z}}$ inducing $P_{0,i}$ on the fibers V'_i. Shrinking $\mathcal{V}'_\Pi$ to a neighborhood of $\Pi_{\mathcal{Z}}$, let δ be a local equation of Δ' and

$$\mathfrak{a} := (\delta) \cdot (\theta_k)^{1/N'} \cdot \mathfrak{c} \cdot I^s_{\Pi_{\mathcal{Z}}},$$

where $I_{\Pi_{\mathcal{Z}}}$ is the ideal of $\Pi_{\mathcal{Z}}$ and $\eta_{\Pi_{\mathcal{Z}}}$ its generic point. We show that

$$\operatorname{mld}(\mathcal{V}'_\Pi \ni \eta_{\Pi_{\mathcal{Z}}}, \mathfrak{a}) = 0$$

and that $P_{0,\mathcal{Z}}$ computes it. The local ring of $\mathcal{V}'_\Pi$ at $\eta_{\Pi_{\mathcal{Z}}}$ is a localization of the generic fiber of $\mathcal{V}'_\Pi \to \mathcal{Z}_\Pi$, whose base change to $\Bbbk$ is $V'_{\Bbbk}$ with $\Pi_{\mathcal{Z}}$ inducing Π, so by [Kaw21, Remark 2.6] it suffices to show that $a(P_{0,\Bbbk}, V'_{\Bbbk}, \mathfrak{a}_{\Bbbk}) = 0$ and that $(V'_{\Bbbk}, \mathfrak{a}_{\Bbbk})$ is lc near η_Π, where

$$\mathfrak{a}_{\Bbbk} := (\delta) \cdot (\theta_{k,\Bbbk})^{1/N'} \cdot \mathfrak{c} \cdot I^s_\Pi \quad \text{and} \quad \widehat{\mathfrak{a}} := (\delta) \cdot (\widehat{\theta})^{1/N'} \cdot \mathfrak{c} \cdot I^s_\Pi.$$

By the inequality on k in the claim, $\operatorname{mult}_{P_0} \theta_{k,\Bbbk} = \operatorname{mult}_{P_0} \widehat{\theta}$, so

$$a(P_{0,\Bbbk}, V'_{\Bbbk}, \mathfrak{a}_{\Bbbk}) = a(P_0, \widehat{V}', \widehat{\mathfrak{a}}) = a(P_0, \widehat{V}', \widehat{\Delta}, \mathfrak{c} I^s_\Pi) = 0.$$

For the lc property, let w be a general closed point of Π, $N_\Pi := \dim \Pi$, and $\mathfrak{m}_w$ the maximal ideal of w. Then Π is smooth at w, and $(\widehat{V}', \widehat{\mathfrak{a}})$ is lc near w since its lc locus is open and contains η_Π. Since $\widehat{V}' = V'_{\Bbbk} \times_{Y_{\Bbbk}} \widehat{Y}$ and w maps to $z_{\Bbbk}$, the completions of the local rings of $\widehat{V}'$ and $V'_{\Bbbk}$ at w are isomorphic. Let $\widehat{V}'_w$ be the spectrum of $\widehat{\mathcal{O}}_{V'_{\Bbbk},w}$. Let $\widetilde{V}'_w \to \widehat{V}'_w$ be the base change of a log resolution of $(\widehat{V}', \widehat{\mathfrak{a}} \cdot I_\Pi)$ on which P_0 is a prime divisor [dFEM11, Theorems 2.3 and 2.4], let Ξ be the simple normal crossing divisor on $\widetilde{V}'_w$ formed by the exceptional locus and the cosupport of $\widehat{\mathfrak{a}} \cdot I_\Pi$, and let $P_{0,w}$ be a component of the base change of P_0 meeting the preimage of w. Since w is general, any intersection of components of Ξ mapping into $\overline{\Pi}$ is smooth over Π near w by generic smoothness. Let $g_1, \dots, g_{N_\Pi} \in \mathcal{O}_{\widehat{V}'_w}$ restrict to a regular system of parameters of $\overline{\Pi}$ at w, so that $\mathfrak{m}_w = I_\Pi + (g_1, \dots, g_{N_\Pi})$. Let y be a point over w and ξ_j local equations at y of the components Ξ_j, $j \in J_y$, of Ξ through y. Since

$$I_\Pi \mathcal{O}_{\widetilde{V}'_w} = \mathcal{O}_{\widetilde{V}'_w}\Big(-\sum_{j \in J_y} \ell_j \Xi_j\Big)$$

near y with $\ell_j := \operatorname{mult}_{\Xi_j} I_\Pi$, and y maps into $\overline{\Pi}$, some ℓ_j is positive, so $\bigcap_{j\in J_y}\Xi_j$ maps into $\overline{\Pi}$ and is smooth over Π at y. Hence $g_1,\dots,g_{N_\Pi}$ and the ξ_j, $j\in J_y$, are part of a regular system of parameters at y, and

$$\mathfrak{m}_w\mathcal{O}_{\widetilde{V}'_w,y} = \Big(g_1,\dots,g_{N_\Pi},\prod_{j\in J_y}\xi_j^{\ell_j}\Big) \supset (g_1,\dots,g_{N_\Pi}).$$

Writing the pull-back of $(\widehat{V}'_w,\widehat{\mathfrak{a}})$ near y as $(\widetilde{V}'_w,\sum_{j\in J_y}(1-a_j)\Xi_j)$ with $a_j\ge 0$, a monomial computation shows that $(\widetilde{V}'_w,\sum_{j\in J_y}(1-a_j)\Xi_j)$ with the $\mathbb{R}$-ideal $(g_1,\dots,g_{N_\Pi})^{N_\Pi}$ is lc near y. Thus $(\widehat{V}'_w,\widehat{\mathfrak{a}}\cdot\mathfrak{m}_w^{N_\Pi})$ is lc.

Let Θ_w be a component of $P_{0,w}$ over w and Q_w the exceptional divisor of the ordinary blow-up of $\widetilde{V}'_w$ along Θ_w. For a component $\Xi_j\ne P_{0,w}$ of Ξ, $P_{0,w}\cap\Xi_j$ is smooth over Π of relative dimension

$$\dim\widetilde{V}'_w - 2 - N_\Pi < \dim\Theta_w = \dim\widetilde{V}'_w - 1 - N_\Pi,$$

so $P_{0,w}$ is the only component of Ξ containing Θ_w. At the generic point of Θ_w, with ξ a local equation of $P_{0,w}$, we thus have

$$\mathfrak{m}_w\mathcal{O}_{\widetilde{V}'_w} = (g_1,\dots,g_{N_\Pi},\xi^{\operatorname{mult}_{P_0}I_\Pi}),\qquad \Theta_w=\{g_1=\dots=g_{N_\Pi}=\xi=0\},\qquad\text{and}$$
$$K_{\widetilde{V}'_w} - (\text{pull-back of } K_{\widehat{V}'_w}+\widehat{\mathfrak{a}}) = -P_{0,w},$$

the last since $a(P_0,\widehat{V}',\widehat{\mathfrak{a}})=0$. Hence $\operatorname{mult}_{Q_w}\mathfrak{m}_w=1$ and $a(Q_w,\widehat{V}'_w,\widehat{\mathfrak{a}})=(N_\Pi+1)-1=N_\Pi$, so

$$a(Q_w,\widehat{V}'_w,\widehat{\mathfrak{a}}\cdot\mathfrak{m}_w^{N_\Pi})=0,$$

and Q_w computes $\operatorname{lct}(\widehat{\mathfrak{a}}\cdot\mathfrak{m}_w^{N_\Pi})=1$. Moreover $\operatorname{mult}_{Q_w}\ge\operatorname{mult}_{P_{0,w}}=\operatorname{mult}_{P_0}$ on $\mathcal{O}_{\widehat{V}'_w}$, with equality at $\widehat{\theta}$ since $\operatorname{div}(\widehat{\theta})$ is supported in Ξ. Thus

$$\operatorname{mult}_{Q_w}(\widehat{\theta}-\theta_{k,\Bbbk})\ge\operatorname{mult}_{P_0}(\widehat{\theta}-\theta_{k,\Bbbk})>\operatorname{mult}_{P_0}\widehat{\theta}=\operatorname{mult}_{Q_w}\widehat{\theta},$$

so on $\widehat{V}'_w$, by [dFEM11, Proposition 2.20],

$$\operatorname{lct}(\mathfrak{a}_\Bbbk\cdot\mathfrak{m}_w^{N_\Pi})=\operatorname{lct}(\widehat{\mathfrak{a}}\cdot\mathfrak{m}_w^{N_\Pi})=1.$$

By [dFEM11, Proposition 2.11], $(V'_\Bbbk,\mathfrak{a}_\Bbbk\cdot\mathfrak{m}_w^{N_\Pi})$ is lc near w, so $(V'_\Bbbk,\mathfrak{a}_\Bbbk)$ is lc near w, hence near η_Π.

Consider [Kaw21, Theorem 3.11] for $\mathcal{V}'_\Pi\to\mathcal{Z}_\Pi$, $\mathfrak{a}$, $\Pi_\mathcal{Z}$, and $P_{0,\mathcal{Z}}$. The dense open subset of $\mathcal{V}'_\Pi$ obtained there restricts to a non-empty open subset $\Pi^\circ\subset\Pi$. By (L1) and the argument for covers in **Step 1**, for all but finitely many i the dense open subset of $\mathcal{Z}_\Pi$ obtained there contains the point over ζ'_i and $\zeta_{k,i}$, over which the fiber of $P_{0,\mathcal{Z}}$ is $P_{0,i}$ and $\operatorname{mult}_{P_{0,i}}\theta_{k,i}=\operatorname{mult}_{P_0}\widehat{\theta}$. Let

$$\mathfrak{p}:=\{f\in\mathcal{O}_{V'_i}\mid\operatorname{mult}_{P_{0,i}}f>\operatorname{mult}_{P_0}\widehat{\theta}\}\quad\text{and}\quad\mathfrak{b}:=(\delta_i)\cdot(\theta_i)^{1/N'}\cdot\mathfrak{c}_i\cdot(I_{\Pi_\mathcal{Z}}\mathcal{O}_{V'_i})^s.$$

Since $\operatorname{mult}_{P_{0,i}}\mathfrak{q}_i>\operatorname{mult}_{P_0}\widehat{\theta}$, $\theta_i-\theta_{k,i}\in\mathfrak{q}_i\subset\mathfrak{p}$, so $(\theta_{k,i})+\mathfrak{p}=(\theta_i)+\mathfrak{p}$, and the other factors of $\mathfrak{b}$ are the restrictions of the factors of $\mathfrak{a}$. Since after shrinking $\mathcal{Z}_\Pi$ the fiber of $\Pi_\mathcal{Z}$ is irreducible and is the center of $P_{0,i}$, $(V'_i,\mathfrak{b})$ is lc near Π_i°, and so is the triple in the claim, which has the factor $I_{\Pi_\mathcal{Z}}^s$ removed. This proves the claim.

We prove that the following hold for all but finitely many i.

(i) $(R_i,S_i+\Gamma_i^R)$ is lc near any point of $S_i\backslash(\Omega_{<1,i}\cup\Omega_{v,i})$.

(ii) $(V_i,\Delta_{V_i},\mathfrak{d}_i^{\epsilon_1})$ is lc near any point of $\pi_i^{-1}(z_i)\backslash\{v_i\}$.

Both follow from the claim by noetherian induction. Let $\mathcal{D}\subset R_\Bbbk$, resp. $\mathcal{D}\subset V_\Bbbk$, be the locally closed subset $S\backslash(\widehat{\Omega}_{<1}\cup\widehat{\Omega}_v)$, resp. $\pi_\Bbbk^{-1}(z_\Bbbk)\backslash\{v_\Bbbk\}$. For an irreducible component Π of $\mathcal{D}$ we find P_0, s and k as below, obtain Π°, and repeat with the irreducible components of $\mathcal{D}\backslash\bigcup\Pi^\circ$, which have smaller dimension. After finitely many stages $\mathcal{D}$ is the union of finitely many locally closed subsets Π°, and the spreads of the Π° cover the spread $\mathcal{D}_i$ of $\mathcal{D}$ for all but finitely many i. At each stage only finitely many indices are discarded.

For (i), let $\mathcal{V}' = \mathcal{R}$, let Δ' be the spread of S on $\mathcal{R}$, let $\rho_{\mathcal{R}} : \mathcal{R} \to \mathcal{V}$ and $\pi_{\mathcal{V}} : \mathcal{V} \to \mathcal{Y}'$ be the morphisms of the families, let s_j be local equations of the spreads of the S_j, and let $\theta_k := \Phi_k^R$ be the regular function near Δ' with

$$(\pi_{\mathcal{V}} \circ \rho_{\mathcal{R}})^* \Phi_k = \Phi_k^R \prod_j s_j^{\beta_j},$$

which exists by the argument for $\Phi_{k,i}$ above at the generic point of $\mathcal{Z}' \times_{\mathcal{Z}_{k_0}} \mathcal{Z}_k$. By the computation giving (D.60), on $\widehat{R}$ we have $\widehat{\theta} = \lim_k \Phi_{k,\Bbbk}^R = \widehat{\varphi}^R$. Moreover $\theta_{k,i} = \Phi_{k,i}^R$ and $\theta_i = \varphi_i^R$, and we let $N' = N_0$ and $\mathfrak{c} = \mathcal{O}_{\mathcal{R}}$. Let $\Pi \subset \mathcal{D}$ be irreducible, with generic point η_Π. Possibly shrinking Π, Π is smooth, and by (c), near η_Π, $S + \widehat{\Gamma}^R = S + \widehat{\Gamma}_{=1}^R$ is reduced and simple normal crossing. Let

$$e_\Pi := \#\{\text{components of } S + \widehat{\Gamma}^R \text{ containing } \Pi\},$$

so that $\operatorname{codim} \Pi \geq e_\Pi$. Let P_0 be the exceptional divisor of the ordinary blow-up of $\widehat{R}$ along Π if $\operatorname{codim} \Pi \geq 2$, and $P_0 := \Pi$ if $\operatorname{codim} \Pi = 1$, and let $s := \operatorname{codim} \Pi - e_\Pi$. A computation with monomial valuations in local coordinates shows that $(\widehat{R}, S + \widehat{\Gamma}^R, I_\Pi^s)$ is lc near η_Π, and $a(P_0, \widehat{R}, S + \widehat{\Gamma}^R, I_\Pi^s) = \operatorname{codim} \Pi - e_\Pi - s = 0$. Moreover

$$\operatorname{mult}_{P_0} \widehat{\varphi}^R = N_0 \cdot \#\{\text{components of } \widehat{\Gamma}_{=1}^R \text{ containing } \Pi\} \leq 3N_0.$$

Since $\Pi \subset S$, by (D.60) the ideal

$$\mathfrak{q}_i := \mathcal{O}_{R_i}\Big(- \sum_j (k\nu_j - \beta_j) S_{j,i} \Big)$$

satisfies $\operatorname{mult}_{P_{0,i}} \mathfrak{q}_i \geq \min_j (k\nu_j - \beta_j) > 3N_0$ once $k\nu_j - \beta_j > 3N_0$ for any j. By (L2), $\widehat{\varphi} - \Phi_{k,\Bbbk} \in \mathfrak{m}^k$, so the computation giving (D.60) gives on $\widehat{R}$

$$\operatorname{mult}_{P_0}(\widehat{\varphi}^R - \Phi_{k,\Bbbk}^R) \geq \min_j (k\nu_j - \beta_j) > 3N_0.$$

Thus the claim with such k implies (i).

For (ii), let $\mathcal{V}' = \mathcal{V}$, $\Delta' = 0$, $\theta_k = (\pi_{\mathcal{V}}^* \Phi_k)^{N_2/N_0} \varpi^{-1}$, so that $\widehat{\theta} = \widehat{\vartheta}$ and $\theta_{k,i} = \vartheta_{k,i}$, $\theta_i = \vartheta_i$, $N' = N_2$, and $\mathfrak{c} = \mathfrak{d}^{\epsilon_1}$. Let $\Pi \subset \mathcal{D}$ be irreducible, with generic point η_Π. By (D.58), $(\widehat{V}, \widehat{\Delta}_V, \mathfrak{d}^{\epsilon_1})$ is lc near η_Π. Let

$$s := \max\{s' \in \mathbb{R}_{\geq 0} \mid (\widehat{V}, \widehat{\Delta}_V, \mathfrak{d}^{\epsilon_1} I_\Pi^{s'}) \text{ is lc near } \eta_\Pi\},$$

and let P_0 be a prime divisor over $\widehat{V}$ with center $\overline{\Pi}$ and

$$a(P_0, \widehat{V}, \widehat{\Delta}_V, \mathfrak{d}^{\epsilon_1} I_\Pi^s) = 0,$$

which exists on a log resolution [dFEM11, Theorems 2.3 and 2.4] and restricts to a prime divisor over $V_\Bbbk$ with center $\overline{\Pi}$ by Lemma 2.12(3) for $\widehat{Y}$, since $\widehat{V} \to V_\Bbbk$ is the base change of $\widehat{Y} \to Y_\Bbbk$. Since $\overline{\Pi} \subset \widehat{V}_{\widehat{z}}$, $\operatorname{mult}_{P_0} \mathfrak{m} \geq 1$. By (D.61), the ideal

$$\mathfrak{q}_i := \mathfrak{m}_{z_i}^k \mathcal{O}_{V_i}\Big(N_2 \sum_j d_j P_{j,i} \Big)$$

satisfies

$$\operatorname{mult}_{P_{0,i}} \mathfrak{q}_i \geq k \operatorname{mult}_{P_0} \mathfrak{m} - N_2 \sum_j d_j \operatorname{mult}_{P_0} P_j > \operatorname{mult}_{P_0} \widehat{\vartheta}$$

for k large, depending on P_0, and since $\widehat{\varphi} - \Phi_{k,\Bbbk} \in \mathfrak{m}^k$ by (L2), the computation giving (D.61) gives, for this k,

$$\operatorname{mult}_{P_0}(\widehat{\vartheta} - \theta_{k,\Bbbk}) > \operatorname{mult}_{P_0} \widehat{\vartheta}.$$

Thus the claim with this k implies (ii). Since only finitely many Π occur, only finitely many values of k are used.

Now let E be a prime divisor over Y_i with $\operatorname{center}_{Y_i} E = \{z_i\}$ such that $\operatorname{center}_{V_i} E \neq \{v_i\}$ in the curve case. Since $\mathfrak{m}_{z_i} \mathcal{O}_{R_i}$ is invertible with support S_i, the generic point p of $\operatorname{center}_{R_i} E$ is

contained in S_i, and $p \notin \Omega_{v,i}$ since $\operatorname{center}_{V_i} E \neq \{v_i\}$. Let $j_0 \in J$ with $p \in S_{j_0,i}$. Since R_i is smooth, $\operatorname{mult}_E S_{j_0,i} \geq 1$.

If $p \notin \Omega_{<1,i}$, then by (i), $a(E, R_i, S_i + \Gamma_i^R) \geq 0$, so by (D.59) and $\alpha_j \geq b_*$,

$$a(E, Y_i, \Gamma_i) = a(E, R_i, S_i + \Gamma_i^R) + \sum_{j \in J} \alpha_j \operatorname{mult}_E S_{j,i} \geq b_* \operatorname{mult}_E S_{j_0,i} \geq b_*.$$

If $p \in \Omega_{<1,i}$, we choose j_0 with $p \in S_{j_0,i}$ and $\epsilon_1 \operatorname{mult}_{S_{j_0}} \mathfrak{d} \geq b_*$. The generic point of $\operatorname{center}_{V_i} E$ is

$$\rho_i(p) \in \pi_i^{-1}(z_i) \backslash \{v_i\},$$

so by (ii) and (D.54),

$$a(E, Y_i, \Gamma_i) = a(E, V_i, \Delta_{V_i}) \geq \epsilon_1 \operatorname{mult}_E \mathfrak{d}_i \geq \epsilon_1 \operatorname{mult}_{S_{j_0,i}}(\mathfrak{d}_i) \operatorname{mult}_E S_{j_0,i} \geq b_*,$$

where the middle inequality holds since on the smooth R_i

$$\mathfrak{d}_i \mathcal{O}_{R_i} \subset \mathcal{O}_{R_i}\big(- \operatorname{mult}_{S_{j_0,i}}(\mathfrak{d}_i) S_{j_0,i}\big).$$

This proves (D.55).

By (D.50), $a(E_i, Y_i, \Gamma_i) = b < b_*$, so (D.55) implies that $\widehat{\Sigma}$ is a curve and $\operatorname{center}_{V_i} E_i = \{v_i\}$. Any prime divisor over V_i with center v_i has center z_i on Y_i, so by (D.54),

$$b = \operatorname{mld}(V_i \ni v_i, \Delta_{V_i}), \tag{D.62}$$

and any prime divisor over V_i computing $\operatorname{mld}(V_i \ni v_i, \Delta_{V_i})$ computes $\operatorname{mld}(X_i \ni x_i, B_i + D_i)$.

The proof continues below.

Proof of Theorem 13.3, continued. **Step 8.** In this step we treat the remaining case and finish the proof of (D.47). Let $\pi_i : V_i \to Y_i$, v_i, and Δ_{V_i} be as in (M1)–(M3).

Since V_i is smooth at v_i, near v_i the $\mathbb{Q}$-divisor Δ_{V_i} is the $\mathbb{R}$-ideal $\prod_P \mathcal{O}_{V_i}(-P)^{\operatorname{mult}_P \Delta_{V_i}}$, where P runs over the components of Δ_{V_i} through v_i, with exponents in $I_1 \backslash \{0\}$ by (M3). By [Kaw26, Theorem 1.4], there exist a positive integer L_0 depending only on I_1 and, for any i, a prime divisor E_i' over V_i computing $\operatorname{mld}(V_i \ni v_i, \Delta_{V_i})$ with

$$a(E_i', V_i, 0) \leq L_0.$$

By **Step 7**, E_i' computes $\operatorname{mld}(X_i \ni x_i, B_i + D_i)$, so by (H3),

$$\operatorname{mult}_{E_i'} \mathfrak{m}_{x_i} \leq \kappa. \tag{D.63}$$

Let $P_{j,i}$, d_j, and N_1 be as in (M2), and let $d_0 := \max\{0, d_1, \dots, d_n\}$. Since $\pi_i(P_{j,i}) = z_i$ for any j and $N_1 \sum_j P_{j,i}$ is Cartier,

$$\mathfrak{m}_{z_i}^{N_1} \mathcal{O}_{V_i} \subset \mathcal{O}_{V_i}\Big(- N_1 \sum_j P_{j,i}\Big) \quad \text{near } \pi_i^{-1}(z_i),$$

so any prime divisor E over V_i satisfies

$$\sum_{j=1}^{n} \operatorname{mult}_E P_{j,i} \leq \operatorname{mult}_E \mathfrak{m}_{z_i}. \tag{D.64}$$

Since $\mu_i(z_i) = x_i$,

$$\operatorname{mult}_{E_i'} \mathfrak{m}_{z_i} \leq \operatorname{mult}_{E_i'} \mathfrak{m}_{x_i}.$$

By (D.49), (D.64), and (D.63),

$$a(E_i', X_i, B_i) \leq a(E_i', Y_i, 0) = a(E_i', V_i, 0) + \sum_{j=1}^{n} d_j \operatorname{mult}_{E_i'} P_{j,i} \leq L_0 + d_0 \operatorname{mult}_{E_i'} \mathfrak{m}_{z_i} \leq L_0 + d_0 \kappa.$$

Together with the cases $\dim Z_i \geq 1$, $b_* = b$, and $(\widehat{Y}, \widehat{\Gamma})$ klt near $\widehat{z}$, treated before (D.55), this proves (D.47).

Proof of (1). Fix m, M, d and suppose that no constant $L(m, M, d)$ as in (1) exists. Then for any positive integer i there exist an isolated Gorenstein terminal germ $(W_i \ni 0) = (f_i = 0) \subset \mathbb{A}^4$ with $\deg f_i \le d$, a finite diagonal abelian group G_i with $|G_i| \le M$ preserving f_i up to a character and acting faithfully on W_i, and a polynomial mark h_i such that $(W_i, \frac{1}{m}\operatorname{div}(h_i))$ is lc near 0, $b_i := q_i(1) > 0$, and

$$\frac{a(F, W_i, 0)}{e_F} > i \quad \text{for any computing lift } F \text{ of } b_i. \tag{D.65}$$

Since $GL_4(\mathbb{C})$ has only finitely many diagonal subgroups of order at most M, after passing to a subsequence, $G_i = G$ for any i and the characters $\chi_{f_i} = \chi_f$ and $\chi_{h_i} = \chi_h$ do not depend on i. Let $r := |G| \le M$.

Let $\mathcal{A}$ be the affine space of G-semi-invariant polynomials in the coordinates $y_1, \dots, y_4$ of $\mathbb{A}^4$ of degree at most d with character χ_f, so $f_i \in \mathcal{A}$ for any i. Possibly passing to a subsequence, we may assume that $(f_i)_i$ is generic in an irreducible closed subset $T \subset \mathcal{A}$ in the sense of **Step 1**. Let

$$\mathcal{W} := \{(x, f) \in \mathbb{A}^4 \times T \mid f(x) = 0\}, \qquad \pi_{\mathcal{W}} : \mathcal{W} \to \mathcal{X} := \mathcal{W}/G,$$

where G acts on the factor $\mathbb{A}^4$, and let $\mathcal{B}$ be the $\mathbb{Q}$-divisor on $\mathcal{X}$ with coefficient $1 - 1/e$ along the image of any prime divisor of $\mathcal{W}$ along which $\pi_{\mathcal{W}}$ has ramification index e. For a closed point $t \in T$ let W_t, X_t and B_t be the fibers over t. Thus $W_i = W_{f_i}$. Let $\pi_{W_i} : W_i \to X_i := W_i/G$ be the quotient morphism, let $x_i := \pi_{W_i}(0)$, and let $(X_i, B_i) := (W_i, 0)/G$ be the crepant quotient near x_i (Definition 2.2), so that

$$K_{W_i} = \pi_{W_i}^*(K_{X_i} + B_i).$$

We check (H1)–(H5) for the pairs $(X_i \ni x_i, B_i)$ and the $\mathbb{Q}$-divisors D_i of (d) below, with $N := mr$ and $\mathcal{S} := \frac{1}{r}\mathbb{N}$, where (H5) holds since $(f_i)_i$ is generic in T.

(a) Removing the zero polynomial from T, $\mathcal{W} \to T$ is flat. Since $|G|$ is invertible in $\mathbb{C}$, forming $\mathcal{W}/G$ commutes with base change along T, so $X_t = W_t/G$, and $\mathcal{X} \to T$ is flat. The image of $\{0\} \times T$ is a section of $\mathcal{X} \to T$, whose value at t we denote by x_t. Possibly shrinking T, after stratifying the fixed loci of the subgroups of G in $\mathcal{W}$ by generic flatness, the boundary of the crepant quotient $(W_t, 0)/G$ near x_t is the fiber B_t of $\mathcal{B}$ over t for any closed point $t \in T$. Since B_t is a divisor on X_t, no component of $\mathcal{B}$ contains an irreducible component of X_t. Thus the pairs $(X_i \ni x_i, B_i)$ are the fibers of a family of finite type with a section.

(b) The sheaf $\omega_{\mathcal{W}/T}$ is generated near $\{0\} \times T$ by the residue form

$$\operatorname{Res}_f := \operatorname{Res}\Big(\frac{dy_1 \wedge \cdots \wedge dy_4}{f}\Big),$$

and G acts on Res_f by the character $\chi_f^{-1} \prod_{j=1}^4 \chi_{y_j}$, whose order divides r. Thus $\operatorname{Res}_f^{\otimes r}$ is a G-invariant generator of $\omega_{\mathcal{W}/T}^{\otimes r}$, and since

$$K_{\mathcal{W}/T} = \pi_{\mathcal{W}}^*(K_{\mathcal{X}/T} + \mathcal{B}),$$

$\operatorname{Res}_f^{\otimes r}$ descends to a generator of $\mathcal{O}_{\mathcal{X}}(r(K_{\mathcal{X}/T} + \mathcal{B}))$ near the section. Hence $r(K_{\mathcal{X}/T} + \mathcal{B})$, and so $N(K_{\mathcal{X}/T} + \mathcal{B})$, is Cartier near the section.

(c) Let $\psi_{\mathcal{W}} : \mathcal{W}' \to \mathcal{W}$ be a log resolution. By generic flatness and generic smoothness, possibly shrinking T, $\psi_{\mathcal{W}}$ restricts to a log resolution of W_t for any closed point $t \in T$ and of the base change of $\mathcal{W}$ to any algebraically closed field containing $\mathbb{C}(T)$, and for each exceptional prime divisor E' of $\psi_{\mathcal{W}}$, the log discrepancy $a(E', \mathcal{W}, 0)$ equals the log discrepancies of the restrictions of E'. Since $(f_i)_i$ is generic in T and W_i is terminal near 0 for any i, the general fiber of $\mathcal{W} \to T$ is terminal near the section. The union $\Upsilon \subset \mathcal{W}$ of the centers on $\mathcal{W}$ of the exceptional prime divisors E' of $\psi_{\mathcal{W}}$ with $a(E', \mathcal{W}, 0) \le 1$ is a closed subset not containing the generic point of the section. Possibly shrinking T, we replace $\mathcal{X}$ by $\mathcal{X} \setminus \pi_{\mathcal{W}}(\Upsilon)$, an open neighborhood of the section. Thus W_t is Gorenstein and

terminal and X_t is a normal threefold for any closed point $t \in T$, by Lemma 2.15(0)(2)(3), (X_t, B_t) is klt, and by (a) and (b), $N(K_{X_t} + B_t)$ is the restriction of $N(K_{\mathcal{X}/T} + \mathcal{B})$. By (a)–(c), (H1) holds.

(d) Let $u_i := h_i^r$. Since the order of χ_h divides r, u_i is G-invariant and is a regular function on X_i near x_i. Let

$$D_i := \frac{1}{mr} \operatorname{div}_{X_i}(u_i).$$

Then $ND_i = mrD_i$ is Cartier, $\pi_{W_i}^* D_i = \frac{1}{m} \operatorname{div}_{W_i}(h_i)$, and

$$K_{W_i} + \frac{1}{m} \operatorname{div}(h_i) = \pi_{W_i}^*(K_{X_i} + B_i + D_i).$$

By Lemma 2.15, $(X_i, B_i + D_i)$ is lc near x_i and $\operatorname{mld}(X_i \ni x_i, B_i + D_i) = b_i$, and $b_i \in \frac{1}{mr}\mathbb{N}$ by Lemma 4.2(1). By Theorem 5.1(3), $0 < b_i \le 3$. Thus, after passing to a subsequence, $b_i = b$ for a positive rational number b independent of i. This is (H2).

(e) Let E be a prime divisor computing $\operatorname{mld}(X_i \ni x_i, B_i + D_i)$ and F a prime divisor over W_i over E. Then F is a computing lift of b_i. Since $a(F', W_i, \frac{1}{m} \operatorname{div}(h_i)) \ge e_{F'} b_i > 0$ for any prime divisor F' over W_i with center 0, $\operatorname{mld}(W_i \ni 0, \frac{1}{m} \operatorname{div}(h_i)) > 0$, and Theorem 5.1(3) implies $\operatorname{mult}_F \mathfrak{m}_0 / e_F \le 3/\eta_m$. Let $k \in \{1, \dots, 4\}$ be an index with $\operatorname{mult}_F y_k = \operatorname{mult}_F \mathfrak{m}_0$. Since y_k^r is G-invariant and vanishes at 0, it belongs to $\mathfrak{m}_{x_i}$, so

$$\operatorname{mult}_E \mathfrak{m}_{x_i} \le \operatorname{mult}_E(y_k^r) = \frac{r \operatorname{mult}_F y_k}{e_F} = \frac{r \operatorname{mult}_F \mathfrak{m}_0}{e_F} \le \frac{3M}{\eta_m} =: \kappa.$$

This is (H3).

(f) Let $\Bbbk \supset \mathbb{C}(T)$ be an algebraically closed field and $W_\Bbbk$, $X_\Bbbk$, $B_\Bbbk$ the base changes along $\operatorname{Spec} \Bbbk \to T$. By (a), $X_\Bbbk = W_\Bbbk / G$ and $(X_\Bbbk, B_\Bbbk) = (W_\Bbbk, 0)/G$ near the image of the section, and by (c), $W_\Bbbk$ is Gorenstein and terminal, so by Lemma 2.15(1)–(3), for any prime divisor E over $X_\Bbbk$ and a prime divisor F over $W_\Bbbk$ over E, $a(E, X_\Bbbk, B_\Bbbk) = a(F, W_\Bbbk, 0)/e_F$, where $a(F, W_\Bbbk, 0)$ is a positive integer and e_F divides r. Thus $(X_\Bbbk, B_\Bbbk)$ is klt and

$$a(E, X_\Bbbk, B_\Bbbk) \in \frac{1}{r}\mathbb{N}^+ \subset \mathcal{S}$$

for any prime divisor E over $X_\Bbbk$. This is (H4).

By (D.47) with $N = mr$ and $\mathcal{S} = \frac{1}{r}\mathbb{N}$, after passing to a subsequence there exist a positive real number L' and, for any i, a prime divisor E_i computing $\operatorname{mld}(X_i \ni x_i, B_i + D_i)$ with $a(E_i, X_i, B_i) \le L'$. Let F_i be a prime divisor over W_i over E_i. Then $\operatorname{center}_{W_i} F_i \subset \pi_{W_i}^{-1}(x_i) = \{0\}$, F_i is a computing lift of b_i, and by Lemma 2.15(1)(2),

$$\frac{a(F_i, W_i, 0)}{e_{F_i}} = a(E_i, X_i, B_i) \le L',$$

which contradicts (D.65) for $i > L'$. Thus $L(m, M, d)$ exists.

Proof of (2). Let $R := \mathbb{C}[[x_1, \dots, x_4]]$ with maximal ideal $\mathfrak{n}$, and let $f \in R$ and the G-equivariant isomorphism $\widehat{\mathcal{O}}_{W,0} \cong R/(f)$ be the presentation in (2), after replacing $x_1, \dots, x_4$ with eigenvectors of the abelian group G in their span and f with a G-semi-invariant generator of (f), which changes neither (f) nor $(f) + \mathfrak{n} \cdot (\partial_1 f, \dots, \partial_4 f)$. Thus G acts diagonally on R, f is G-semi-invariant, and

$$\mathfrak{n}^c \subset (f) + \mathfrak{n} \cdot (\partial_1 f, \dots, \partial_4 f).$$

Let

$$f' := f_{<2c+1}, \qquad W' := (f' = 0) \subset \mathbb{A}^4, \qquad \widehat{W} := \operatorname{Spec} R/(f), \quad \text{and} \quad \widehat{W'} := \operatorname{Spec} R/(f'),$$

where $f_{<2c+1}$ is the sum of the terms of f of degree less than $2c+1$ and G acts diagonally on $\mathbb{A}^4$ with the characters of the x_j. Then f' is a G-semi-invariant polynomial with character χ_f, $\deg f' \le 2c$, and $f' - f \in \mathfrak{n}^{2c+1}$.

We show that the germ $(W' \ni 0)$ is isolated Gorenstein terminal. Since $f - f' \in \mathfrak{n}^{2c+1}$ and $\partial_j f - \partial_j f' \in \mathfrak{n}^{2c}$,

$$\mathfrak{n}^c \subset (f) + \mathfrak{n} \cdot (\partial_1 f, \dots, \partial_4 f) \subset (f') + \mathfrak{n} \cdot (\partial_1 f', \dots, \partial_4 f') + \mathfrak{n}^{c+1},$$

so by Nakayama's lemma

$$\mathfrak{n}^c \subset (f') + \mathfrak{n} \cdot (\partial_1 f', \dots, \partial_4 f').$$

Hence $W' \backslash \{0\}$ is smooth near 0 by the Jacobian criterion, and W' is normal near 0. By Lemma 3.2(2) with $L = 2c + 1$, there exist a G-equivariant automorphism ω of R and a G-invariant unit v of R with $v\omega(f) = f'$, so ω induces a G-equivariant isomorphism

$$\overline{\omega} : R/(f) \xrightarrow{\cong} R/(f') = \widehat{\mathcal{O}}_{W',0}.$$

By Lemma 2.12(3) for W' and W and by $\overline{\omega}$, any prime divisor F'_0 over W' with center 0 corresponds to a prime divisor F_0 over W with center 0 with $a(F'_0, W', 0) = a(F_0, W, 0)$. Since W is terminal and $W' \backslash \{0\}$ is smooth near 0, W' is terminal near 0. Since G acts faithfully on W, it acts faithfully on $\widehat{W}$, hence on W'.

Let $\widehat{h} \in R$ be a G-semi-invariant lift with character χ_h of the image of h in $R/(f)$, and let $\widehat{h}' := \omega(\widehat{h})$, a G-semi-invariant element of R with character χ_h and non-zero restriction to $\widehat{\mathcal{O}}_{W',0}$. Since $(W, \frac{1}{m}\operatorname{div}(h))$ is klt near 0, so is $(\widehat{W}, \frac{1}{m}\operatorname{div}(\widehat{h}))$ [Kaw21, Remark 2.6], and hence so is $(\widehat{W}', \frac{1}{m}\operatorname{div}(\widehat{h}'))$. By Lemma 3.2(1) with $H = G$, $V_0 = W'$, one function $g_1 = \widehat{h}'$, $\beta_1 = \frac{1}{m}$, and $\gamma = 4$, which is larger than $\operatorname{mld}(\widehat{W}' \ni 0, \frac{1}{m}\operatorname{div}(\widehat{h}'))$ since $\dim \widehat{W}' = 3$, there exists an integer L_γ such that $h' := (\widehat{h}')_{<L_\gamma}$ satisfies the following.

(T1) h' is a G-semi-invariant polynomial with character χ_h, non-zero on W', and $(W', \frac{1}{m}\operatorname{div}(h'))$ is klt near 0.

(T2) The pairs $(\widehat{W}', \frac{1}{m}\operatorname{div}(\widehat{h}'))$ and $(\widehat{W}', \frac{1}{m}\operatorname{div}(h'))$ satisfy

$$\operatorname{mld}\Big(\Big(\widehat{W}' \ni 0, \frac{1}{m}\operatorname{div}(\widehat{h}')\Big)/G\Big) = \operatorname{mld}\Big(\Big(\widehat{W}' \ni 0, \frac{1}{m}\operatorname{div}(h')\Big)/G\Big).$$

(T3) Any computing lift $\widehat{F}'$ of the second pair in (T2) satisfies $\operatorname{mult}_{\widehat{F}'} h' = \operatorname{mult}_{\widehat{F}'} \widehat{h}'$ and is a computing lift of the first pair.

Let q' be the radial function of $(W' \ni 0, G, h', m)$ and $b' := q'(1)$. By Lemma 2.12(2) for W', by (T2), by $\overline{\omega}$ and by Lemma 2.12(2) for W,

$$\begin{aligned} b' &= \operatorname{mld}\Big(\Big(\widehat{W}' \ni 0, \frac{1}{m}\operatorname{div}(h')\Big)/G\Big) = \operatorname{mld}\Big(\Big(\widehat{W}' \ni 0, \frac{1}{m}\operatorname{div}(\widehat{h}')\Big)/G\Big) \\ &= \operatorname{mld}\Big(\Big(\widehat{W} \ni 0, \frac{1}{m}\operatorname{div}(\widehat{h})\Big)/G\Big) = \operatorname{mld}\Big(\Big(W \ni 0, \frac{1}{m}\operatorname{div}(h)\Big)/G\Big) = b > 0. \end{aligned}$$

Thus $(W' \ni 0)$, G, and h' satisfy the hypotheses of (1) with $\deg f' \le 2c$, and (1) implies that there exists a computing lift F' of b' with

$$\frac{a(F', W', 0)}{e_{F'}} \le L(m, M, 2c).$$

By Lemma 2.12(3), F' is the restriction of a formal prime divisor $\widehat{F}'$ over $\widehat{W}'$ with center the closed point, with inertia group equal to the inertia group of F', and with

$$a(\widehat{F}', \widehat{W}', 0) = a(F', W', 0) \quad \text{and} \quad a\Big(\widehat{F}', \widehat{W}', \frac{1}{m}\operatorname{div}(h')\Big) = a\Big(F', W', \frac{1}{m}\operatorname{div}(h')\Big).$$

Thus $\widehat{F}'$ is a computing lift of the second pair in (T2), and by (T3) it is a computing lift of the first pair. Since $\overline{\omega}$ is a G-equivariant isomorphism mapping $\widehat{h}$ to $\widehat{h}'$, $\widehat{F}'$ corresponds to a formal prime divisor $\widehat{F}$ over $\widehat{W}$ computing the quotient value of $(\widehat{W} \ni 0, \frac{1}{m}\operatorname{div}(\widehat{h}))$, with inertia group equal to the inertia group of $\widehat{F}'$, and with

$$a(\widehat{F}, \widehat{W}, 0) = a(\widehat{F}', \widehat{W}', 0).$$

By Lemma 2.12(3) for W, $\widehat{F}$ restricts to a computing lift F of b with

$$\frac{a(F,W,0)}{e_F} = \frac{a(F',W',0)}{e_{F'}} \le L(m,M,2c).$$

Thus (2) holds with $L := L(m,M,2c)$.

Proof of (3). Let $\mathcal{I}$ be the infinite set of indices and $Z_k \subset \mathcal{J}_k$ the irreducible closed subsets of Definition 2.5 for the pairs (f_i,h_i) and the compatible generic limit (f_*,h_*). Let $\mathfrak{n}_*$ be the maximal ideal of $\Bbbk[[x_1,\dots,x_4]]$. Suppose that $\operatorname{Sing}(W_*)$ is contained in the closed point. By the Jacobian criterion [Mat89, p. 239, Theorem 30.8 and the paragraph following it], $\operatorname{Sing}(W_*)$ is the zero locus of $(\partial_1 f_*,\dots,\partial_4 f_*)$ on W_*, so there exists a positive integer c_0 with

$$(f_*,\partial_1 f_*,\dots,\partial_4 f_*) \supset \mathfrak{n}_*^{c_0}.$$

Let $c := c_0+1$. Then

$$\mathfrak{n}_*^{c} = \mathfrak{n}_* \cdot \mathfrak{n}_*^{c_0} \subset (f_*) + \mathfrak{n}_* \cdot (\partial_1 f_*,\dots,\partial_4 f_*).$$

For a field $\mathbb{K} \supset \mathbb{C}$ and $g \in \mathbb{K}[[x_1,\dots,x_4]]$, let $\mathfrak{n}_{\mathbb{K}} := (x_1,\dots,x_4)$ and

$$J_g := (g) + \mathfrak{n}_{\mathbb{K}} \cdot (\partial_1 g,\dots,\partial_4 g).$$

By Nakayama's lemma,

$$\mathfrak{n}_{\mathbb{K}}^{c} \subset J_g \quad \text{if and only if} \quad \mathfrak{n}_{\mathbb{K}}^{c} \subset J_g + \mathfrak{n}_{\mathbb{K}}^{c+1}. \tag{D.66}$$

Modulo $\mathfrak{n}_{\mathbb{K}}^{c+1}$, the ideal J_g is spanned over $\mathbb{K}$ by the products of $g_{<c+1}$ with the monomials of degree at most c and of $x_k\partial_j(g_{<c+1})$ with the monomials of degree at most $c-1$, for $j,k \in \{1,\dots,4\}$. Let $U_{\mathbb{K}}$ be the $\mathbb{K}$-vector space with basis the symbols e_μ and $e_{j,k,\nu}$, for monomials μ of degree at most c, monomials ν of degree at most $c-1$, and $j,k \in \{1,\dots,4\}$, and let

$$\lambda_g : U_{\mathbb{K}} \to \mathbb{K}[[x_1,\dots,x_4]]/\mathfrak{n}_{\mathbb{K}}^{c+1}$$

be the $\mathbb{K}$-linear map sending e_μ to the class of $\mu g_{<c+1}$ and $e_{j,k,\nu}$ to the class of $\nu x_k\partial_j(g_{<c+1})$. Then

$$\operatorname{Im}\lambda_g = (J_g + \mathfrak{n}_{\mathbb{K}}^{c+1})/\mathfrak{n}_{\mathbb{K}}^{c+1},$$

and the matrix entries of λ_g in the monomial bases are polynomials with integer coefficients in the coefficients of $g_{<c+1}$. Let $\overline{\lambda}_g$ be the composite of λ_g with the projection to $\mathbb{K}[[x_1,\dots,x_4]]/\mathfrak{n}_{\mathbb{K}}^{c}$. Since

$$\dim_{\mathbb{K}}\left(\operatorname{Im}\lambda_g \cap \mathfrak{n}_{\mathbb{K}}^{c}/\mathfrak{n}_{\mathbb{K}}^{c+1}\right) = \operatorname{rank}\lambda_g - \operatorname{rank}\overline{\lambda}_g,$$

the right hand side of (D.66) holds if and only if

$$\operatorname{rank}\lambda_g - \operatorname{rank}\overline{\lambda}_g = \dim_{\mathbb{K}}\mathfrak{n}_{\mathbb{K}}^{c}/\mathfrak{n}_{\mathbb{K}}^{c+1} = \binom{c+3}{3}.$$

The rank condition depends only on the $(c+1)$-jet of g and does not change under extension of $\mathbb{K}$. There exists a dense open subset $Z^\circ \subset Z_{c+1}$ on which $\operatorname{rank}\lambda_g$ and $\operatorname{rank}\overline{\lambda}_g$, for g the first component of a jet, are equal to their values at the generic point of Z_{c+1}. Since the $(c+1)$-jet of (f_*,h_*) maps to the generic point of Z_{c+1} and $\mathfrak{n}_*^{c} \subset J_{f_*}$, the condition holds at the generic point of Z_{c+1}, hence on Z°. By Definition 2.5, Z° contains the $(c+1)$-jets of (f_i,h_i) for all but finitely many $i \in \mathcal{I}$, and by (D.66),

$$\mathfrak{n}^{c} \subset (f_i) + \mathfrak{n} \cdot (\partial_1 f_i,\dots,\partial_4 f_i)$$

for all but finitely many $i \in \mathcal{I}$. For any such i, $(W_i \ni 0)$, G, and h_i satisfy the hypotheses of (2) with the presentation f_i and the constant c, so some computing lift F of b_i satisfies $a(F,W_i,0)/e_F \le L(m,M,2c)$, where $L(m,M,2c)$ is the constant of (1). This contradicts $\min_F a(F,W_i,0)/e_F \to +\infty$. Thus $\operatorname{Sing}(W_*)$ is not contained in the closed point, that is, $\dim\operatorname{Sing}(W_*) \ge 1$. □

Lemma D.7. *Let m be a positive integer and $\epsilon > 0$ a rational number. For any i let f_i and h_i be non-zero elements of $\mathbb{C}[[x_1, \dots, x_4]]$ such that $\widehat{W}_i := \operatorname{Spec} \mathbb{C}[[x_1, \dots, x_4]]/(f_i)$ is normal and integral and $h_i|_{\widehat{W}_i} \neq 0$, and let (f_*, h_*), with coefficient field $\Bbbk$, be a compatible generic limit of the pairs (f_i, h_i) along an infinite subset $\mathcal{I} \subset \mathbb{N}^+$ (Definition 2.5, with H trivial). If*

$$\left(\operatorname{Spec} \Bbbk[[x_1, \dots, x_4]], (f_*)^1 (h_*)^{(1+\epsilon)/m}\right)$$

is lc, then there exists an infinite subset $\mathcal{I}' \subset \mathcal{I}$ such that $(\widehat{W}_i, \frac{1+\epsilon}{m} \operatorname{div}(h_i))$ is lc for any $i \in \mathcal{I}'$.

Proof. Let L_i be positive integers with $L_i > \max\{\operatorname{mult}_0 f_i, \operatorname{mult}_0 h_i\}$ and $\lim_{i \to +\infty} L_i = +\infty$, and let f_i' and h_i' be the parts of f_i and h_i of degree less than L_i. Then f_i' and h_i' are non-zero regular functions on $\mathbb{A}^4$ with

$$f_i - f_i', \ h_i - h_i' \in (x_1, \dots, x_4)^{L_i}.$$

For any positive integer d, the d-jets of (f_i', h_i') and (f_i, h_i) coincide when $L_i \geq d$, so (f_*, h_*) is also a compatible generic limit of the (f_i', h_i') along $\mathcal{I}$. By the hypothesis and Theorem 2.18(1), there exists an infinite subset $\mathcal{I}'' \subset \mathcal{I}$ such that

$$\left(\mathbb{A}^4, \operatorname{div}(f_i') + \frac{1+\epsilon}{m} \operatorname{div}(h_i')\right) \text{ is lc near } 0 \quad \text{for any } i \in \mathcal{I}'',$$

so, by [dFEM11, Proposition 2.11(i)],

$$(\operatorname{Spec} \mathbb{C}[[x_1, \dots, x_4]], (f_i')^1 (h_i')^{(1+\epsilon)/m}) \text{ is lc} \quad \text{for any } i \in \mathcal{I}''.$$

Let L_0 be the positive integer of Lemma 2.19 for $n = 4$, $N = 2$, and $\Lambda = \{1, \frac{1+\epsilon}{m}\}$, and let

$$\mathcal{I}' := \{i \in \mathcal{I}'' \mid L_i \geq L_0\},$$

an infinite set. Let $i \in \mathcal{I}'$. By Lemma 2.19,

$$(\operatorname{Spec} \mathbb{C}[[x_1, \dots, x_4]], (f_i)^1 (h_i)^{(1+\epsilon)/m}) \text{ is lc.}$$

Since ϵ is rational, Lemma D.9(1), applied with $S = \widehat{W}_i$ in $X = \operatorname{Spec} \mathbb{C}[[x_1, \dots, x_4]]$ and with $B = \frac{1+\epsilon}{m} \operatorname{div}(h_i)$, implies that

$$\left(\widehat{W}_i, \frac{1+\epsilon}{m} \operatorname{div}(h_i)\right) \text{ is lc.} \quad \square$$

We need the following proposition in the proof of Theorem 13.8.

Proposition D.8. *(1) Let $\Bbbk \supset \mathbb{C}$ be an algebraically closed field, $c > 0$ a rational number, and $f, h \in \Bbbk[[x_1, \dots, x_4]]$ with $W = V(f)$ normal and not klt, such that $(\operatorname{Spec} \Bbbk[[x_1, \dots, x_4]], (f)^1 (h)^c)$ is lc with positive minimal log discrepancy at the closed point. Then $(\operatorname{Spec} \Bbbk[[x_1, \dots, x_4]], (f)^1 (h)^{c(1+\epsilon)})$ is lc for some rational $\epsilon > 0$.*

(2) Let m and M be positive integers, and let $(W_i \ni o_i, G_i, h_i, \tau_i)$ be a radial failure sequence with constants (m, a, δ) such that, for any i, $|G_i| \leq M$, $b_i > 0$, $(W_i, \frac{1}{m} \operatorname{div}(h_i))$ is klt near o_i, and $(W_i \ni o_i)$ has quadratic rank 1 or 2. After passing to a subsequence and fixing the group $G = G_i$, its linear action on $x_1, \dots, x_4$, and the characters, let (f_*, h_*) be a compatible generic limit, with coefficient field $\Bbbk$, of completed equivariant presentations $f_i \in \mathbb{C}[[x_1, \dots, x_4]]$ of the $(W_i \ni o_i)$ and of G-semi-invariant lifts $h_i \in \mathbb{C}[[x_1, \dots, x_4]]$ of the marks, and let

$$W_* := (f_* = 0) \subset \mathcal{X}_* := \operatorname{Spec} \Bbbk[[x_1, \dots, x_4]].$$

Assume that W_* is normal and not klt. Then there exist a rational number $\epsilon > 0$ and an infinite set I of indices such that for any $i \in I$, any computing lift F of b_i satisfies

$$\frac{a(F, W_i, 0)}{e_F} \leq 3\left(1 + \frac{1}{\epsilon}\right).$$

Proof. We prove (1). Let $\mathcal{X}_* := \operatorname{Spec} \Bbbk[[x_1, \dots, x_4]]$, so that $W = (f = 0) \subset \mathcal{X}_*$. Log discrepancies, lc and klt pairs and log resolutions over $\mathcal{X}_*$ are as in [dFEM11, Section 2]. Let o_* be the closed point of $\mathcal{X}_*$, and let $\mathfrak{a} := (f)^1(h)^c$. For any prime divisor P over $\mathcal{X}_*$,

$$a(P, \mathcal{X}_*, W) = a(P, \mathcal{X}_*, \mathfrak{a}) + c \operatorname{mult}_P h \geq a(P, \mathcal{X}_*, \mathfrak{a}), \tag{D.67}$$

so $(\mathcal{X}_*, W)$ is lc. Lc places and lc centers of $(\mathcal{X}_*, \mathfrak{a})$ are as in Definition 2.1, with $a(P, \mathcal{X}_*, \mathfrak{a})$ in place of $a(E, X, B)$. Since $\operatorname{mld}(\mathcal{X}_* \ni o_*, \mathfrak{a}) > 0$, $\{o_*\}$ is not an lc center of $(\mathcal{X}_*, \mathfrak{a})$.

Step 1. In this step we find an lc center Z_1 of $(\mathcal{X}_*, \mathfrak{a})$ with $\dim Z_1 = 1$ and $h \notin I_{Z_1}$, where I_{Z_1} is the ideal of Z_1.

Since W is normal, W is integral. Let $\pi : Y \to \mathcal{X}_*$ be a log resolution of $(\mathcal{X}_*, (fh))$ [dFEM11, Theorem 2.4], and let $W_Y := \pi_*^{-1} W$. Since $W_Y + \operatorname{Exc}(\pi)$ is a simple normal crossing divisor, $\pi|_{W_Y} : W_Y \to W$ is a log resolution of W. Since W is not klt, by (D.69) with $(X, S, B) = (\mathcal{X}_*, W, 0)$ there exists a π-exceptional prime divisor E on Y with $a(E, \mathcal{X}_*, W) \leq 0$, and $a(E, \mathcal{X}_*, W) = 0$ since $(\mathcal{X}_*, W)$ is lc.

Since

$$\operatorname{mult}_E f = a(E, \mathcal{X}_*, 0) - a(E, \mathcal{X}_*, W) = a(E, \mathcal{X}_*, 0) > 0,$$

we have $\operatorname{center}_{\mathcal{X}_*} E \subset W$. Since $(\mathcal{X}_*, W)$ is plt near any regular point of W, $\operatorname{center}_{\mathcal{X}_*} E \subset \operatorname{Sing} W$. Since W is normal and $\dim W = 3$, $\dim \operatorname{Sing} W \leq 1$. For any prime divisor P over $\mathcal{X}_*$ with $\operatorname{center}_{\mathcal{X}_*} P = \{o_*\}$, (D.67) implies $a(P, \mathcal{X}_*, W) \geq a(P, \mathcal{X}_*, \mathfrak{a}) > 0$, so $\operatorname{center}_{\mathcal{X}_*} E \neq \{o_*\}$. Thus $Z_1 := \operatorname{center}_{\mathcal{X}_*} E$ satisfies $\dim Z_1 = 1$. By (D.67),

$$0 \leq a(E, \mathcal{X}_*, \mathfrak{a}) = a(E, \mathcal{X}_*, W) - c \operatorname{mult}_E h = -c \operatorname{mult}_E h,$$

so $a(E, \mathcal{X}_*, \mathfrak{a}) = \operatorname{mult}_E h = 0$. Thus Z_1 is an lc center of $(\mathcal{X}_*, \mathfrak{a})$, and $h \notin I_{Z_1}$, since otherwise $\operatorname{mult}_E h \geq \operatorname{mult}_E I_{Z_1} > 0$.

Step 2. In this step we show that any lc center of $(\mathcal{X}_*, \mathfrak{a})$ contains Z_1.

Since π is a log resolution of $(\mathcal{X}_*, (fh))$, the center on Y of any lc place of $(\mathcal{X}_*, \mathfrak{a})$ is an irreducible component of an intersection of prime divisors P on Y with $a(P, \mathcal{X}_*, \mathfrak{a}) = 0$. Thus $(\mathcal{X}_*, \mathfrak{a})$ has finitely many lc centers, and any lc center of $(\mathcal{X}_*, \mathfrak{a})$ contains a minimal one with respect to inclusion. Since $\mathcal{X}_*$ is local, any non-empty closed subset of $\mathcal{X}_*$ contains o_*. Thus any proper irreducible closed subset of Z_1 is $\{o_*\}$, which is not an lc center, so Z_1 is a minimal lc center. Let $Z_2, \dots, Z_n$ be the other minimal lc centers of $(\mathcal{X}_*, \mathfrak{a})$, with reduced structures, so that $Z_1, \dots, Z_n$ are the minimal lc centers. It suffices to show that $n = 1$.

Similar to [Kaw15, Proposition 3.8], we first replace the factors (f) and (h) of $\mathfrak{a}$ by larger ideals. Let $Z := \bigcup_{j=1}^n Z_j$ with ideal I_Z. For any $j \in \{1, \dots, n\}$, fix an lc place E_j of $(\mathcal{X}_*, \mathfrak{a})$ with $\operatorname{center}_{\mathcal{X}_*} E_j = Z_j$. Since $\operatorname{mult}_{E_j} I_Z > 0$ for any j, there exists a positive integer N_0 such that

$$N_0 \operatorname{mult}_{E_j} I_Z \geq \max\{\operatorname{mult}_{E_j} f, \operatorname{mult}_{E_j} h\} \quad \text{for any } j \in \{1, \dots, n\}.$$

Let

$$\mathfrak{a}' := \big((f) + I_Z^{N_0}\big)^1 \big((h) + I_Z^{N_0}\big)^c.$$

For any prime divisor P over $\mathcal{X}_*$, $\operatorname{mult}_P((f) + I_Z^{N_0}) = \min\{\operatorname{mult}_P f, N_0 \operatorname{mult}_P I_Z\}$, and similarly for h, so

$$a(P, \mathcal{X}_*, \mathfrak{a}') \geq a(P, \mathcal{X}_*, \mathfrak{a}) \geq 0,$$

with equality for $P = E_j$ by the choice of N_0. Thus $(\mathcal{X}_*, \mathfrak{a}')$ is lc and $E_1, \dots, E_n$ are lc places of $(\mathcal{X}_*, \mathfrak{a}')$. Let P be an lc place of $(\mathcal{X}_*, \mathfrak{a}')$. Since

$$0 \leq a(P, \mathcal{X}_*, \mathfrak{a}) \leq a(P, \mathcal{X}_*, \mathfrak{a}') = 0,$$

P is an lc place of $(\mathcal{X}_*, \mathfrak{a})$. Since $(f) + I_Z^{N_0}$ and $(h) + I_Z^{N_0}$ are the unit ideal on $\mathcal{X}_* \backslash Z$,

$$a(P', \mathcal{X}_*, \mathfrak{a}') = a(P', \mathcal{X}_*, 0) > 0 \quad \text{for any prime divisor } P' \text{ over } \mathcal{X}_* \text{ with } \operatorname{center}_{\mathcal{X}_*} P' \not\subset Z,$$

so $\operatorname{center}_{\mathcal{X}_*} P \subset Z$. Thus $\operatorname{center}_{\mathcal{X}_*} P$ is an lc center of $(\mathcal{X}_*, \mathfrak{a})$ contained in Z_j for some j, and $\operatorname{center}_{\mathcal{X}_*} P = Z_j$ by the minimality of Z_j. Therefore the lc centers of $(\mathcal{X}_*, \mathfrak{a}')$ are exactly $Z_1, \dots, Z_n$.

Suppose that $n \geq 2$, and let $Z_{\geq 2} := \bigcup_{j=2}^{n} Z_j$ with ideal $I_{Z_{\geq 2}}$. For any $j \geq 2$, $Z_1 \not\subset Z_j$ by the minimality of Z_j, so $Z_1 \cap Z_j$ is a proper closed subset of Z_1 containing o_*, and $Z_1 \cap Z_j = \{o_*\}$. Thus $Z_1 \cap Z_{\geq 2} = \{o_*\}$, and

$$J := I_{Z_1} + I_{Z_{\geq 2}}$$

is $\mathfrak{n}_*$-primary, where $\mathfrak{n}_*$ is the maximal ideal of $\Bbbk[[x_1, \ldots, x_4]]$. Fix generators $\alpha_1, \ldots, \alpha_{N_1}$ of I_{Z_1} and $\beta_1, \ldots, \beta_{N_2}$ of $I_{Z_{\geq 2}}$, and let $g_1 : \mathcal{X}' \to \mathcal{X}_*$ be the blow-up of J. Then g_1 is an isomorphism over $\mathcal{X}_* \backslash \{o_*\}$, and $\mathcal{X}'$ is the closure in $\mathcal{X}_* \times \mathbb{P}^{N_1+N_2-1}$ of the graph of

$$[\alpha_1 : \cdots : \alpha_{N_1} : \beta_1 : \cdots : \beta_{N_2}] : \mathcal{X}_* \backslash \{o_*\} \to \mathbb{P}^{N_1+N_2-1}.$$

Let $[y_1 : \cdots : y_{N_1} : y'_1 : \cdots : y'_{N_2}]$ be the homogeneous coordinates of $\mathbb{P}^{N_1+N_2-1}$. The closure of $g_1^{-1}(Z_1 \backslash \{o_*\})$ is contained in $\mathcal{X}_* \times (y_1 = \cdots = y_{N_1} = 0)$, the closure of $g_1^{-1}(Z_{\geq 2} \backslash \{o_*\})$ is contained in $\mathcal{X}_* \times (y'_1 = \cdots = y'_{N_2} = 0)$, and

$$(y_1 = \cdots = y_{N_1} = 0) \cap (y'_1 = \cdots = y'_{N_2} = 0) = \emptyset \quad \text{in } \mathbb{P}^{N_1+N_2-1}.$$

Since $\mathcal{X}'$ is regular over $\mathcal{X}_* \backslash \{o_*\}$, by [dFEM11, Theorem 2.3] there exists a proper birational morphism $g_2 : \mathcal{X}'' \to \mathcal{X}'$ from a regular scheme $\mathcal{X}''$ which is an isomorphism over $g_1^{-1}(\mathcal{X}_* \backslash \{o_*\})$. Let $g := g_1 \circ g_2 : \mathcal{X}'' \to \mathcal{X}_*$, which is a proper birational morphism and an isomorphism over $\mathcal{X}_* \backslash \{o_*\}$. For any j, let Z''_j be the closure of $g^{-1}(Z_j \backslash \{o_*\})$, and let $Z''_{\geq 2} := \bigcup_{j=2}^{n} Z''_j$. Since $g_2(Z''_1)$ and $g_2(Z''_{\geq 2})$ are contained in the closures of $g_1^{-1}(Z_1 \backslash \{o_*\})$ and $g_1^{-1}(Z_{\geq 2} \backslash \{o_*\})$ respectively,

$$Z''_1 \cap Z''_{\geq 2} = \emptyset.$$

Let Nklt be the non-klt locus on $\mathcal{X}''$ of the pullback of $(\mathcal{X}_*, \mathfrak{a}')$, that is, the union of the centers on $\mathcal{X}''$ of the prime divisors P over $\mathcal{X}_*$ with $a(P, \mathcal{X}_*, \mathfrak{a}') \leq 0$. Since $(\mathcal{X}_*, \mathfrak{a}')$ is lc, Nklt is the union of the centers on $\mathcal{X}''$ of the lc places of $(\mathcal{X}_*, \mathfrak{a}')$. Let P be an lc place of $(\mathcal{X}_*, \mathfrak{a}')$, and let $\operatorname{center}_{\mathcal{X}_*} P = Z_j$. Since $Z_j \neq \{o_*\}$, g is an isomorphism over the generic point of Z_j. Since $g(\operatorname{center}_{\mathcal{X}''} P) = Z_j$, the generic point of $\operatorname{center}_{\mathcal{X}''} P$ is the generic point of Z''_j, so $\operatorname{center}_{\mathcal{X}''} P = Z''_j$. Since E_j is an lc place of $(\mathcal{X}_*, \mathfrak{a}')$ with $\operatorname{center}_{\mathcal{X}_*} E_j = Z_j$ for any j,

$$\text{Nklt} = Z''_1 \cup Z''_{\geq 2}.$$

Since g is proper and $g(Z''_j) \supset Z_j \backslash \{o_*\}$, $g(Z''_j) = Z_j \ni o_*$ for any j. Thus $\text{Nklt} \cap g^{-1}(o_*)$ is the union of the non-empty closed subsets $Z''_1 \cap g^{-1}(o_*)$ and $Z''_{\geq 2} \cap g^{-1}(o_*)$, whose intersection is empty, so $\text{Nklt} \cap g^{-1}(o_*)$ is not connected. Since $\mathcal{X}''$ is regular and g is a proper birational morphism which is an isomorphism over $\mathcal{X}_* \backslash \{o_*\}$, $\text{Nklt} \cap g^{-1}(o_*)$ is connected by [Kaw15, Corollary 3.2], a contradiction. Thus $n = 1$.

Step 3. In this step we conclude the proof of (1).

Let P be an lc place of $(\mathcal{X}_*, \mathfrak{a})$. By **Step 2**, $Z_1 \subset \operatorname{center}_{\mathcal{X}_*} P$. If $\operatorname{mult}_P h > 0$, then h vanishes at the generic point of $\operatorname{center}_{\mathcal{X}_*} P$, so $h \in I_{Z_1}$, which contradicts **Step 1**. Thus $\operatorname{mult}_P h = 0$ for any lc place P of $(\mathcal{X}_*, \mathfrak{a})$.

Let $\mathcal{P}$ be the finite set of prime divisors P on Y with $\operatorname{mult}_P h > 0$. Since $(\mathcal{X}_*, \mathfrak{a})$ is lc and no $P \in \mathcal{P}$ is an lc place of $(\mathcal{X}_*, \mathfrak{a})$, $a(P, \mathcal{X}_*, \mathfrak{a}) > 0$ for any $P \in \mathcal{P}$. Thus there exists a rational number $\epsilon > 0$ such that

$$\epsilon c \operatorname{mult}_P h \leq a(P, \mathcal{X}_*, \mathfrak{a}) \quad \text{for any } P \in \mathcal{P}.$$

Then for any prime divisor P on Y,

$$a(P, \mathcal{X}_*, (f)^1 (h)^{c(1+\epsilon)}) = a(P, \mathcal{X}_*, \mathfrak{a}) - \epsilon c \operatorname{mult}_P h \geq 0.$$

Since π is a log resolution of $(\mathcal{X}_*, (fh))$, $(\mathcal{X}_*, (f)^1 (h)^{c(1+\epsilon)})$ is lc. This proves (1).

We prove (2).

Step 4. In this step we show that there exists a rational number $\epsilon > 0$ such that

$$(\mathcal{X}_*, (f_*)^1 (h_*)^{(1+\epsilon)/m}) \text{ is lc.}$$

By the hypotheses and Definition 4.1(1), for any i, $(W_i \ni o_i) = (f_i = 0)$ is an isolated Gorenstein terminal hypersurface threefold germ of quadratic rank 1 or 2, and $(W_i, \frac{1}{m}\operatorname{div}(h_i))$ is klt near o_i. For any positive integer d, the d-jets of pairs of G-semi-invariant power series with the characters of (f_i, h_i) form a linear subspace of the affine space of d-jets of all pairs of power series, so (f_*, h_*) is also a compatible generic limit of the (f_i, h_i) in the sense of Definition 2.5 with $H = \{1\}$. By Lemma 13.4(2) with $H = \{1\}$, $(\mathcal{X}_*, (f_*)^1(h_*)^{1/m})$ is lc with positive minimal log discrepancy at the closed point. Since $W_* = (f_* = 0) \subset \mathcal{X}_*$ is normal and not klt, by (1) with $c = 1/m$, there exists a rational number $\epsilon > 0$ such that

$$(\mathcal{X}_*, (f_*)^1(h_*)^{(1+\epsilon)/m}) \text{ is lc.}$$

Step 5. In this step we prove (2) with ϵ as in **Step 4**.

Let $\mathcal{I}$ be the infinite set of indices along which (f_*, h_*) is a compatible generic limit of the (f_i, h_i) with $H = \{1\}$ (**Step 4**). For any i, let $\widehat{W}_i := (f_i = 0) \subset \operatorname{Spec} \mathbb{C}[[x_1, \dots, x_4]]$ be the completion of W_i at o_i. Then $\widehat{W}_i$ is normal and integral since W_i is normal near o_i, and $h_i|_{\widehat{W}_i} \neq 0$ since $h_i|_{W_i} \neq 0$. By **Step 4** and Lemma D.7, there exists an infinite subset $I \subset \mathcal{I}$ such that $(\widehat{W}_i, \frac{1+\epsilon}{m}\operatorname{div}(h_i))$ is lc for any $i \in I$, so, by [dFEM11, Proposition 2.11(i)],

$$\Big(W_i, (1+\epsilon)\frac{1}{m}\operatorname{div}(h_i)\Big) \text{ is lc near } o_i \quad \text{for any } i \in I.$$

Let $i \in I$, and let F be a computing lift of b_i. By Definition 2.4, $\operatorname{center}_{W_i} F = \{o_i\}$ and

$$a\Big(F, W_i, \frac{1}{m}\operatorname{div}(h_i)\Big) = e_F b_i \quad \text{and} \quad \frac{a(F, W_i, 0)}{e_F} = b_i + \frac{\operatorname{mult}_F h_i}{m e_F}.$$

Thus

$$0 \le a\Big(F, W_i, (1+\epsilon)\frac{1}{m}\operatorname{div}(h_i)\Big) = e_F b_i - \epsilon\frac{\operatorname{mult}_F h_i}{m},$$

so

$$\frac{\operatorname{mult}_F h_i}{m e_F} \le \frac{b_i}{\epsilon}.$$

Since $(W_i \ni o_i)$ is an isolated Gorenstein terminal hypersurface threefold germ of quadratic rank 1 or 2, $b_i = q_i(1) \le 3$ by Lemma 4.2(3). Thus

$$\frac{a(F, W_i, 0)}{e_F} = b_i + \frac{\operatorname{mult}_F h_i}{m e_F} \le b_i\Big(1 + \frac{1}{\epsilon}\Big) \le 3\Big(1 + \frac{1}{\epsilon}\Big). \quad \square$$

Lemma D.9. *Let $\Bbbk$ be an algebraically closed field of characteristic zero, and R a complete local $\Bbbk$-algebra with residue field $\Bbbk$ which is normal, Gorenstein, and a quotient of $\Bbbk[[x_1, \dots, x_n]]$ for some n. Let $X := \operatorname{Spec} R$ with closed point o, let $\ell \in R$ be an element such that $S := (\ell = 0)$ is normal and integral, and let $B = \sum_j b_j \operatorname{div}(g_j)$ with rational numbers $b_j \ge 0$ and elements $g_j \in R$ not vanishing identically on S.*

(1) If $(X, S + B)$ is lc, then $(S, B|_S)$ is lc and $\operatorname{mld}(S \ni o, B|_S) \ge \operatorname{mld}(X \ni o, S + B)$.

(2) If $(S, B|_S)$ is klt, then $(X, S + B)$ is plt.

Proof. Since X is a closed subscheme of $\operatorname{Spec} \Bbbk[[x_1, \dots, x_n]]$, log resolutions of pairs on X and on schemes projective over X exist by [dFEM11, Theorems 2.3–2.4]. To principalize an ideal, we first blow it up. Moreover, R is excellent and has a dualizing complex, as a quotient of the complete regular local ring $\Bbbk[[x_1, \dots, x_n]]$. Since R is a Gorenstein local ring and S is a Cartier divisor, $\omega_X(S) \cong \mathcal{O}_X$, S is Gorenstein, and the residue map Res_S induces an isomorphism $\omega_X(S)|_S \cong \omega_S$. Let ϑ be a generator of $\omega_X(S)$. We take $K_X + S$ and K_S to be the divisors of ϑ and of $\operatorname{Res}_S \vartheta$, so that

$$K_X + S = 0 \quad \text{and} \quad K_S = 0.$$

Since B is a $\mathbb{Q}$-linear combination of principal divisors, $K_X + S + B \sim_{\mathbb{Q}} 0$.

Let $\pi : Y \to X$ be a log resolution of $(X, S + B)$ such that $\mathfrak{m}_o \mathcal{O}_Y$ is invertible, let $S_Y := \pi_*^{-1} S$, and let Σ_Y be the reduced divisor with support $\operatorname{Exc}(\pi) \cup \pi^{-1}(o) \cup \pi_*^{-1} \operatorname{Supp} B$. Then $S_Y + \Sigma_Y$

is a simple normal crossing divisor on the regular scheme Y since π is a log resolution. Let $K_Y + S_Y$ be the divisor of $\pi^*\vartheta$ as a rational section of $\omega_Y(S_Y)$, and define Δ_Y by

$$K_Y + S_Y + \Delta_Y = \pi^*(K_X + S + B), \quad \text{that is,} \quad \Delta_Y = \pi^*B - (K_Y + S_Y).$$

Then $\operatorname{Supp}\Delta_Y \subset \Sigma_Y$, and $a(E, X, S + B) = 1 - \operatorname{mult}_E \Delta_Y$ for any prime divisor $E \neq S_Y$ on Y. Let $\pi_S := \pi|_{S_Y} : S_Y \to S$, a projective birational morphism. Since π is an isomorphism at the generic point of S_Y, $\pi^*\vartheta$ generates $\omega_Y(S_Y)$ there, and

$$\operatorname{Res}_{S_Y}(\pi^*\vartheta) = \pi_S^*(\operatorname{Res}_S \vartheta),$$

a rational section of ω_{S_Y} with divisor $K_{S_Y} := (K_Y + S_Y)|_{S_Y}$. Thus

$$K_{S_Y} + \Delta_Y|_{S_Y} = \pi_S^*(K_S + B|_S). \tag{D.68}$$

Since $S_Y + \Sigma_Y$ is a simple normal crossing divisor and $S_Y \not\subset \Sigma_Y$, any prime divisor F on S_Y is contained in at most one component E of Σ_Y, and then $\operatorname{mult}_F(E|_{S_Y}) = 1$. By (D.68), for any prime divisor F on S_Y,

$$a(F, S, B|_S) = \begin{cases} a(E, X, S + B) & \text{if } F \subset E \text{ for a component } E \text{ of } \Sigma_Y, \\ 1 & \text{if no component of } \Sigma_Y \text{ contains } F. \end{cases} \tag{D.69}$$

Any prime divisor over S (resp. over X) is a prime divisor on S_Y (resp. on Y) for a suitable choice of π.

(1) Assume that $(X, S + B)$ is lc, and let F be a prime divisor over S. Choose π such that F is a prime divisor on S_Y. Since $(X, S + B)$ is lc, $a(E, X, S + B) \geq 0$ for any component E of Σ_Y, so $a(F, S, B|_S) \geq 0$ by (D.69). Thus $(S, B|_S)$ is lc. Assume moreover that $\operatorname{center}_S F = \{o\}$. Then $F \subset \pi^{-1}(o)$. Since $\mathfrak{m}_o\mathcal{O}_Y$ is invertible, $\operatorname{Supp}\pi^{-1}(o)$ is the union of the components of Σ_Y with center o on X, so $F \subset E$ for one of them. By (D.69),

$$a(F, S, B|_S) = a(E, X, S + B) \geq \operatorname{mld}(X \ni o, S + B).$$

Thus $\operatorname{mld}(S \ni o, B|_S) \geq \operatorname{mld}(X \ni o, S + B)$.

(2) We follow the argument for [KM98, Theorem 5.48], with [Mur24, Theorem 8.6.1] as the vanishing theorem. Assume that $(S, B|_S)$ is klt, and suppose that $(X, S + B)$ is not plt. Then there exists a prime divisor $E_1 \neq S$ over X with $a(E_1, X, S + B) \leq 0$. Choose π such that E_1 is a prime divisor on Y, and write $S_Y + \Delta_Y = \sum_E d_E E$, so that $d_{S_Y} = 1$ and $d_{E_1} \geq 1$. Since $B \geq 0$, $d_E < 0$ only if E is π-exceptional. Let

$$D := \sum_{E \neq S_Y,\ d_E \geq 1} \lfloor d_E \rfloor E, \quad \Xi := \sum_{d_E < 0} \lceil -d_E \rceil E, \quad \text{and} \quad \Delta' := \Delta_Y + \Xi - D.$$

Then $D \geq E_1$, $\Xi \geq 0$ is π-exceptional, $\lfloor S_Y + \Delta_Y \rfloor = S_Y + D - \Xi$, and Δ' has simple normal crossing support and coefficients in $[0, 1)$. Since $K_Y + S_Y + \Delta_Y = \pi^*(K_X + S + B) \sim_{\mathbb{Q}} 0$,

$$\Xi - S_Y - D \sim_{\mathbb{Q}} K_Y + \Delta'.$$

Since Y is regular and the coefficients of Δ' are smaller than one, [Mur24, Theorem 8.6.1] implies that the only possible associated point of $R^1\pi_*\mathcal{O}_Y(\Xi - S_Y - D)$ is the generic point of X. Since π is an isomorphism over a dense open subset of X,

$$R^1\pi_*\mathcal{O}_Y(\Xi - S_Y - D) = 0.$$

Since $\Xi \geq 0$ is π-exceptional and X is normal, $\pi_*\mathcal{O}_Y(\Xi) = \mathcal{O}_X$. Thus the exact sequence

$$0 \to \mathcal{O}_Y(\Xi - S_Y - D) \to \mathcal{O}_Y(\Xi) \to \mathcal{O}_{S_Y + D}(\Xi) \to 0$$

implies that $\pi_*\mathcal{O}_{S_Y+D}(\Xi)$ is a quotient of $\mathcal{O}_X$, that is, a cyclic R-module.

Suppose that $S_Y \cap \operatorname{Supp} D = \emptyset$. Then

$$\pi_*\mathcal{O}_{S_Y+D}(\Xi) = \pi_*\mathcal{O}_{S_Y}(\Xi) \oplus \pi_*\mathcal{O}_D(\Xi).$$

Since Ξ has no common component with $S_Y + D$, $\pi_*\mathcal{O}_{S_Y} \subset \pi_*\mathcal{O}_{S_Y}(\Xi)$ and $\pi_*\mathcal{O}_D \subset \pi_*\mathcal{O}_D(\Xi)$, and both $\pi_*\mathcal{O}_{S_Y}$ and $\pi_*\mathcal{O}_D$ are non-zero since $S_Y \neq 0$ and $D \neq 0$. By Nakayama's lemma, a cyclic

R-module is not the direct sum of two non-zero finitely generated R-modules, a contradiction. Thus some component E_2 of D intersects S_Y. Since $d_{E_2} \geq 1$, E_2 is a component of Σ_Y. Let F be a component of $E_2 \cap S_Y$. By (D.69),

$$a(F, S, B|_S) = a(E_2, X, S + B) = 1 - d_{E_2} \leq 0,$$

which contradicts the klt property of $(S, B|_S)$. Thus $(X, S + B)$ is plt. □

Proof of Construction-Theorem 13.6. We divide the proof into six steps. Let $\mathfrak{n} := (x, y, z, t)$ be the maximal ideal of $\mathbb{C}[[x, y, z, t]]$.

Step 1. In this step we fix the group, the characters, the row, and a monomial of f_i.

By Lemma 3.7, for any i there exist a smooth fourfold germ $(\mathcal{X}_i \ni o_i)$ with a faithful G_i-action, a G_i-equivariant closed embedding $W_i \subset \mathcal{X}_i$, and G_i-semi-invariant elements x, y, z, t with $\widehat{\mathcal{O}}_{\mathcal{X}_i, o_i} = \mathbb{C}[[x, y, z, t]]$, on which G_i acts diagonally, such that $\widehat{\mathcal{O}}_{W_i, o_i} = \mathbb{C}[[x, y, z, t]]/(f_i)$ with

$$f_i = x^2 + y^3 + y\alpha_i(z, t) + \beta_i(z, t), \quad \alpha_i \in (z, t)^3, \quad \text{and} \quad \beta_i \in (z, t)^4,$$

in one of the rows E_6, E_7, E_8 of Lemma 3.7. By Lemma 3.7(3), the expansion of h_i is congruent modulo f_i to a unique G_i-semi-invariant element $\varphi_i(y, z, t) + x\xi_i(y, z, t)$ with the character of h_i. Let $\widehat{W}_i := \operatorname{Spec} \mathbb{C}[[x, y, z, t]]/(f_i)$. Possibly replacing h_i with the element $\varphi_i + x\xi_i$ of $\mathbb{C}[[x, y, z, t]]$, whose restriction to $\widehat{W}_i$ is that of h_i, we may assume that $h_i = \varphi_i + x\xi_i$, as in the statement. Since G_i acts faithfully on $\mathcal{X}_i$ and diagonally on x, y, z, t, the diagonal representation of G_i is faithful. Since G_i is cyclic and $|G_i| \leq M$, there exist finitely many possibilities, up to isomorphism, for the group G_i, the diagonal representation of G_i, and the characters of f_i and h_i. Passing to a subsequence, we may assume the following.

- $G_i = G$ is a fixed cyclic group with a fixed faithful diagonal action on x, y, z, t, also regarded as an action on $\mathbb{A}^4 = \operatorname{Spec} \mathbb{C}[x, y, z, t]$.
- The character χ_f of f_i and the character χ_h of h_i do not depend on i.
- The row of f_i does not depend on i.
- $\tau_i > 0$, which is possible since $\tau_i \to 1$.

By Lemma 3.7(2), the homogeneous part $\beta_{i,4}$, $\alpha_{i,3}$, $\beta_{i,5}$ of degree 4 of β_i, of degree 3 of α_i, of degree 5 of β_i in row E_6, E_7, E_8 respectively is non-zero. Since there exist finitely many monomials of degree at most 5, passing to a subsequence, we may assume that there exists a monomial $\mathbf{m}_f$, of degree 4 in z, t in row E_6, equal to y times a monomial of degree 3 in z, t in row E_7, and of degree 5 in z, t in row E_8, whose coefficient in f_i is non-zero for any i.

Step 2. In this step we construct polynomials f_i' and h_i' satisfying (D.70) and (D.71), and we bound $\operatorname{mult}_0 h_i'$.

Since $(W_i \ni o_i)$ is an isolated singularity, the ideal

$$(f_i, \partial_x f_i, \partial_y f_i, \partial_z f_i, \partial_t f_i) \subset \mathbb{C}[[x, y, z, t]]$$

is $\mathfrak{n}$-primary, so there exists a positive integer c_i with

$$\mathfrak{n}^{c_i} \subset (f_i) + \mathfrak{n} \cdot (\partial_x f_i, \partial_y f_i, \partial_z f_i, \partial_t f_i).$$

The right-hand side is contained in $\mathfrak{n}^2$, so $c_i \geq 2$. Let L_i be an integer with $L_i > 2c_i + i$, and let $f_i' := \operatorname{jet}_{L_i}(f_i)$. Since $L_i > 2c_i + i \geq 5$,

$$f_i' = x^2 + y^3 + y\alpha_i' + \beta_i' \quad \text{with} \quad \alpha_i' := \operatorname{jet}_{L_i - 1}(\alpha_i), \ \beta_i' := \operatorname{jet}_{L_i}(\beta_i), \quad \text{and} \quad \operatorname{jet}_6(f_i') = \operatorname{jet}_6(f_i);$$

moreover f_i' is semi-invariant with character χ_f. By Lemma 3.2(2), there exist a G-equivariant automorphism σ_i of $\mathbb{C}[[x, y, z, t]]$ and a G-invariant unit v_i with

$$v_i \sigma_i(f_i) = f_i' \quad \text{and} \quad \sigma_i(u) - u \in \mathfrak{n}^{L_i - c_i + 1} \text{ for any } u \in \{x, y, z, t\}.$$

Let $W_i' := (f_i' = 0) \subset \mathbb{A}^4$, whose completion at 0 is $\widehat{W}_i' := \operatorname{Spec} \mathbb{C}[[x, y, z, t]]/(f_i')$. Since σ_i maps (f_i) onto (f_i'), it induces a G-equivariant isomorphism

$$\mathbb{C}[[x, y, z, t]]/(f_i) \to \mathbb{C}[[x, y, z, t]]/(f_i')$$

sending the class of h_i to the class of $\sigma_i(h_i)$. Thus $(W_i' \ni 0)$ is an isolated Gorenstein terminal threefold germ on which G acts faithfully (Lemma 2.12(2)(3)). For $\beta \in \{\frac{\tau_i}{m}, \frac{1}{m}\}$, the pair $(\widehat{W}_i', \beta \operatorname{div}(\sigma_i(h_i)))$ is isomorphic to $(\widehat{W}_i, \beta \operatorname{div}(h_i))$, which is klt since $(W_i, \frac{1}{m}\operatorname{div}(h_i))$ is klt near o_i and $\beta \le \frac{1}{m}$ [dFEM11, Proposition 2.11(i)]. By Lemma 3.2(1) with $H = G$, $V_0 = W_i'$, $N = 1$, and $g_1 = \sigma_i(h_i)$, applied with $\beta_1 = \frac{1}{m}$ and $\gamma = 0$, and with $\beta_1 = \frac{\tau_i}{m}$ and

$$\gamma := \operatorname{mld}\big(\widehat{W}_i' \ni 0, \tfrac{\tau_i}{m}\operatorname{div}(\sigma_i(h_i))\big) + 1,$$

there exists an integer $N_i \ge L_i$, at least the integer L_γ of each application, such that

$$g_i := \operatorname{jet}_{N_i}(\sigma_i(h_i)),$$

which is semi-invariant with character χ_h, satisfies the following, by (a) of the first application and by (c) with $H' = G$ of the second.

- $(W_i', \frac{1}{m}\operatorname{div}(g_i|_{W_i'}))$ is klt near 0.
- The quotient values at 0 satisfy

$$\operatorname{mld}\big((\widehat{W}_i' \ni 0, \tfrac{\tau_i}{m}\operatorname{div}(g_i))/G\big) = \operatorname{mld}\big((\widehat{W}_i' \ni 0, \tfrac{\tau_i}{m}\operatorname{div}(\sigma_i(h_i)))/G\big).$$

Since f_i' is monic of degree 2 in x, any element of $\mathbb{C}[[y,z,t]][x]$ has a unique remainder of x-degree at most 1 on division by f_i'. Let h_i' be the remainder of g_i. Since $g_i, f_i' \in \mathbb{C}[x,y,z,t]$, $h_i' = \varphi_i' + x\xi_i'$ with $\varphi_i', \xi_i' \in \mathbb{C}[y,z,t]$, h_i' is semi-invariant with character χ_h, and $h_i'|_{W_i'} = g_i|_{W_i'}$. Thus the first item above gives the klt property in (D.70) below, and the equality of quotient values in (D.70) is the chain

$$\begin{aligned}\operatorname{mld}\big((W_i' \ni 0, \tfrac{\tau_i}{m}\operatorname{div}(h_i'))/G\big) &= \operatorname{mld}\big((\widehat{W}_i' \ni 0, \tfrac{\tau_i}{m}\operatorname{div}(g_i))/G\big)\\ &= \operatorname{mld}\big((\widehat{W}_i' \ni 0, \tfrac{\tau_i}{m}\operatorname{div}(\sigma_i(h_i)))/G\big)\\ &= \operatorname{mld}\big((\widehat{W}_i \ni o_i, \tfrac{\tau_i}{m}\operatorname{div}(h_i))/G\big) = q_i(\tau_i),\end{aligned}$$

given by Lemma 2.12(2) with $h_i'|_{W_i'} = g_i|_{W_i'}$, the second item, the isomorphism induced by σ_i, and Lemma 2.12(2) with Definition 4.1:

$$(W_i', \tfrac{1}{m}\operatorname{div}(h_i')) \text{ is klt near } 0 \quad \text{and} \quad \operatorname{mld}\big((W_i' \ni 0, \tfrac{\tau_i}{m}\operatorname{div}(h_i'))/G\big) = q_i(\tau_i). \tag{D.70}$$

Moreover,

$$f_i - f_i' \in \mathfrak{n}^{L_i} \quad \text{and} \quad h_i - h_i' \in \mathfrak{n}^{L_i - c_i + 1}. \tag{D.71}$$

The first congruence holds by the definition of f_i'. For the second, $\sigma_i(h_i) - h_i$ belongs to the ideal generated by $\sigma_i(u) - u$ for $u \in \{x,y,z,t\}$, so $\sigma_i(h_i) - h_i \in \mathfrak{n}^{L_i - c_i + 1}$, and

$$\sigma_i(h_i) - g_i \in \mathfrak{n}^{N_i} \subset \mathfrak{n}^{L_i}.$$

For any integers $k \ge 0$ and $e_0 \in \{0,1\}$,

$$x^{2k+e_0} \equiv (-y^3 - y\alpha_i' - \beta_i')^k x^{e_0} \pmod{f_i'} \quad \text{and} \quad (-y^3 - y\alpha_i' - \beta_i')^k x^{e_0} \in \mathfrak{n}^{3k+e_0} \subset \mathfrak{n}^{2k+e_0},$$

so the remainder on division by f_i' maps $\mathfrak{n}^{N'} \cap \mathbb{C}[[y,z,t]][x]$ into $\mathfrak{n}^{N'}$ for any integer $N' \ge 0$. Since h_i has x-degree at most 1, $h_i' - h_i$ is the remainder of $g_i - h_i \in \mathfrak{n}^{L_i - c_i + 1}$, which proves (D.71). Since $L_i - c_i + 1 > c_i + i + 1 > i$, (D.71) implies

$$\operatorname{jet}_d(f_i') = \operatorname{jet}_d(f_i) \quad \text{and} \quad \operatorname{jet}_d(h_i') = \operatorname{jet}_d(h_i) \quad \text{for any positive integer } d \le i+1.$$

Let wt_w be the weighted order for the weight $w := (3,2,2,2)$ on (x,y,z,t), for a polynomial ψ let $\operatorname{in}_w \psi$ be the sum of the terms of ψ of weight $\operatorname{wt}_w(\psi)$, and let E_0 be the prime divisor over W_i' of the weighted blow-up of $\mathbb{A}^4$ with weight w. Every monomial of $f_i' - x^2 - y^3 = y\alpha_i' + \beta_i'$ has weight at least 8, so $\operatorname{in}_w f_i' = x^2 + y^3$, which is irreducible and whose zero locus is not contained in any coordinate hyperplane. Since h_i' has x-degree at most 1, $\operatorname{in}_w h_i'$ is not divisible by $x^2 + y^3$. By [Liu26b, Lemma 4.12],

$$a(E_0, W_i', 0) = 3 + 2 + 2 + 2 - 6 = 3 \quad \text{and} \quad \operatorname{mult}_{E_0} h_i' = \operatorname{wt}_w(h_i').$$

Since $(W_i', \frac{1}{m}\operatorname{div}(h_i'))$ is klt near 0 by (D.70), $3 - \frac{1}{m}\operatorname{mult}_{E_0} h_i' > 0$, that is, $\operatorname{mult}_{E_0} h_i' < 3m$. Hence h_i' has a monomial $x^{e_0}y^{k_1}z^{k_2}t^{k_3}$ with non-zero coefficient such that

$$e_0 \in \{0,1\} \quad \text{and} \quad 3e_0 + 2(k_1+k_2+k_3) < 3m; \quad \text{in particular} \quad e_0 + k_1 + k_2 + k_3 < \frac{3m}{2}.$$

There exist finitely many such monomials. Passing to a subsequence, we may assume that there exists such a monomial $\mathbf{m}_h$ whose coefficient in h_i' is non-zero for any i. Then

$$\operatorname{mult}_0 h_i' \le \deg \mathbf{m}_h < \frac{3m}{2}.$$

Since $\operatorname{jet}_6(f_i') = \operatorname{jet}_6(f_i)$, the coefficient of $\mathbf{m}_f$ in f_i' is non-zero.

Step 3. In this step we construct (f_*, h_*) and prove (1).

The polynomials f_i' and h_i' are non-zero and G-semi-invariant with

$$\operatorname{mult}_0 f_i' = 2 \quad \text{and} \quad \operatorname{mult}_0 h_i' < \frac{3m}{2}.$$

By Lemma 2.6(1), there exists an infinite subset $\mathcal{I} \subset \mathbb{N}^+$ along which the pairs (f_i', h_i') have a compatible generic limit (f_*, h_*), with coefficient field $\Bbbk$ and with irreducible closed subsets $Z_d \subset \mathcal{J}_d$ (Definition 2.5). Since

$$\operatorname{jet}_d(f_i, h_i) = \operatorname{jet}_d(f_i', h_i') \quad \text{for any } i \ge d$$

by **Step 2** (so $\operatorname{mult}_0 h_i = \operatorname{mult}_0 h_i' < \frac{3m}{2}$ for $i \ge \frac{3m}{2}$), (f_*, h_*) is also a compatible generic limit of the (f_i, h_i) along $\mathcal{I}$. Let $\mathcal{X}_* := \operatorname{Spec} \Bbbk[[x,y,z,t]]$ with maximal ideal $\mathfrak{n}_*$ and with the diagonal action of G, and let $W_* := (f_* = 0) \subset \mathcal{X}_*$.

By Definition 2.5, (f_*, h_*) satisfies the following closed conditions, since each (f_i', h_i') does.

- $f = x^2 + y^3 + y\alpha(z,t) + \beta(z,t)$ with $\alpha \in (z,t)^3$ and $\beta \in (z,t)^4$, together with $\beta_4 = 0$ in row E_7 and $\beta_4 = \alpha_3 = 0$ in row E_8, where α_3 and β_4 are the homogeneous parts of degree 3 of α and of degree 4 of β.
- f and h are semi-invariant with characters χ_f and χ_h.
- h has x-degree at most 1.

It also satisfies the open conditions that the coefficients of $\mathbf{m}_f$ in f and of $\mathbf{m}_h$ in h are non-zero, since for d large the open subset of $\mathcal{J}_d$ that they define has non-empty intersection with Z_d. Thus

$$f_* = x^2 + y^3 + y\alpha_*(z,t) + \beta_*(z,t) \quad \text{and} \quad h_* = \varphi_*(y,z,t) + x\xi_*(y,z,t),$$

where $\alpha_* \in (z,t)^3$, $\beta_* \in (z,t)^4$, f_* is in the row fixed in **Step 1**, f_* and h_* are semi-invariant with characters χ_f and χ_h, and the coefficients of $\mathbf{m}_f$ in f_* and of $\mathbf{m}_h$ in h_* are non-zero. In particular, $h_* \ne 0$ has x-degree at most 1, so $h_* \notin (f_*)$ by the uniqueness of the Weierstrass division by f_*, that is,

$$h_*|_{W_*} \ne 0.$$

Let $\Phi(z,t)$ be the homogeneous part of degree 4 of β_*, of degree 3 of α_*, and of degree 5 of β_*, in rows E_6, E_7, and E_8 respectively. Since the coefficient of $\mathbf{m}_f$ in f_* is non-zero, $\Phi \ne 0$, so $\Phi(1,\zeta)$ is a non-zero polynomial in ζ of degree at most 5. Thus there exists $c' \in \{0,1,2,3,4,5\}$ with

$$\lambda := \Phi(1,c') \ne 0.$$

Let Φ_i be the corresponding homogeneous part of f_i, and $\lambda_i := \Phi_i(1,c')$. Since $\operatorname{jet}_6(f_i') = \operatorname{jet}_6(f_i)$, $\lambda_i \ne 0$ is an open condition on the 6-jet of f_i', satisfied by f_*. Removing finitely many indices from $\mathcal{I}$, we may assume that $\lambda_i \ne 0$ for any $i \in \mathcal{I}$.

Let $\ell := t - c'z$, so that $\Bbbk[[x,y,z,t]] = \Bbbk[[x,y,z,\ell]]$, let

$$\overline{f}_* := f_*(x,y,z,c'z) = x^2 + y^3 + y\alpha_*(z,c'z) + \beta_*(z,c'z) \in \Bbbk[[x,y,z]],$$

and let $S_* := W_* \cap (\ell = 0) = \operatorname{Spec} \Bbbk[[x,y,z]]/(\overline{f}_*)$. Let the weights of (x,y,z) be $(6,4,3)$, $(9,6,4)$, and $(15,10,6)$ in rows E_6, E_7, and E_8 respectively.

- In row E_6,
$$\overline{f}_* = x^2 + y^3 + \lambda z^4 + (\text{terms of weight at least } 13),$$
since the terms of $y\alpha_*(z, c'z)$ have weight at least 13 and those of $\beta_*(z, c'z) - \lambda z^4$ have weight at least 15.
- In row E_7,
$$\overline{f}_* = x^2 + y^3 + \lambda y z^3 + (\text{terms of weight at least } 20),$$
since the terms of $y\alpha_*(z, c'z) - \lambda y z^3$ have weight at least 22 and, since $\beta_* \in (z, t)^5$, the terms of $\beta_*(z, c'z)$ have weight at least 20.
- In row E_8,
$$\overline{f}_* = x^2 + y^3 + \lambda z^5 + (\text{terms of weight at least } 34),$$
since $\alpha_* \in (z, t)^4$ implies that the terms of $y\alpha_*(z, c'z)$ have weight at least 34, and the terms of $\beta_*(z, c'z) - \lambda z^5$ have weight at least 36.

Replacing z with $\lambda^{-1/4}z$, $\lambda^{-1/3}z$, and $\lambda^{-1/5}z$ respectively, $\overline{f}_*$ is the sum of p_0 and terms of weight larger than the weighted degree 12, 18, 30 of p_0, where
$$p_0 := x^2 + y^3 + z^4, \quad x^2 + y^3 + yz^3, \quad \text{and} \quad x^2 + y^3 + z^5$$
in rows E_6, E_7, E_8 respectively. The Milnor algebra $\Bbbk[[x, y, z]]/(\partial_x p_0, \partial_y p_0, \partial_z p_0)$ has the monomial basis
$$\{y^{e_0}z^k \mid e_0 \le 1,\ k \le 2\}, \quad \{1, y, z, yz, z^2, z^3, z^4\}, \quad \text{and} \quad \{y^{e_0}z^k \mid e_0 \le 1,\ k \le 3\},$$
whose largest weights are 10, 16, and 28, all smaller than the weighted degree of p_0. By [AGZV85, Section 12.6, Theorem], whose proof is a formal elimination of terms of increasing weight and holds for power series over $\Bbbk$, $\overline{f}_*$ is equal to p_0 up to a formal coordinate change of $\Bbbk[[x, y, z]]$. Thus S_* is the completion of a Du Val surface singularity of type E_6, E_7, or E_8 respectively, so S_* is normal and klt [dFEM11, Remark 2.10 and Proposition 2.11(i)]. For any $i \in \mathcal{I}$, f_i is in the row of f_* and $\Phi_i(1, c') = \lambda_i \neq 0$, so the three expansions above hold for $\overline{f}_i := f_i(x, y, z, c'z) \in \mathbb{C}[[x, y, z]]$, with $\lambda_i z^4$, $\lambda_i y z^3$, and $\lambda_i z^5$ as the terms of weight 12, 18, and 30 other than $x^2 + y^3$. By [AGZV85, Section 12.6, Theorem], $(\ell = 0)$ cuts out on $\widehat{W}_i$ a Du Val surface singularity of type E_6, E_7, or E_8 respectively.

Since S_* is normal, hence integral, $\overline{f}_*$ is irreducible. Since the reduction modulo ℓ of a non-unit of $\Bbbk[[x, y, z, \ell]]$ is a non-unit, f_* is irreducible, so W_* is integral. As a hypersurface, W_* is Cohen–Macaulay. A point of S_* at which W_* is singular is a singular point of the Cartier divisor S_*, so $\operatorname{Sing}(W_*) \cap S_* \subset \{0\}$. Since $\dim(V \cap S_*) \ge \dim V - 1$ for any non-empty closed subset $V \subset W_*$,
$$\dim \operatorname{Sing}(W_*) \le 1.$$
By Serre's criterion, W_* is normal. By Lemma D.9(2) with $(X, S, B) = (W_*, S_*, 0)$, (W_*, S_*) is plt. Thus $a(E, W_*, 0) \ge a(E, W_*, S_*) > 0$ for any exceptional prime divisor E over W_*, and W_* is klt. Since W_* is Gorenstein, $a(E, W_*, 0) \in \mathbb{Z}$, so $a(E, W_*, 0) \ge 1$, and W_* is canonical. Since S_* is a Du Val hyperplane section, W_* is cDV [Kol98, Definition 2.3]. This proves (1).

Step 4. In this step we prove (2).

Let $\partial_x, \partial_y, \partial_z$ be the partial derivatives in the coordinates (x, y, z, ℓ), and let
$$\mathcal{T}_* := \Bbbk[[x, y, z, \ell]]/(f_*, \partial_x f_*, \partial_y f_*, \partial_z f_*).$$
Since setting $\ell = 0$ commutes with $\partial_x, \partial_y, \partial_z$,
$$\mathcal{T}_*/\ell\mathcal{T}_* = \Bbbk[[x, y, z]]/(\overline{f}_*, \partial_x \overline{f}_*, \partial_y \overline{f}_*, \partial_z \overline{f}_*).$$
For a power series $g \in \Bbbk[[x, y, z]]$, the ideal $(g, \partial_x g, \partial_y g, \partial_z g)$ does not change under formal coordinate changes and under replacing g with ug for a unit u, and $p_0 \in (\partial_x p_0, \partial_y p_0, \partial_z p_0)$ by the Euler relation. By **Step 3**, $\dim_{\Bbbk} \mathcal{T}_*/\ell\mathcal{T}_*$ is the dimension n_0 of the Milnor algebra of p_0, so $n_0 = 6, 7, 8$ in rows E_6, E_7, E_8 respectively. Thus $\ell\mathcal{T}_*$ is primary to the maximal ideal of the

complete local ring $\mathcal{T}_*$, so $\mathcal{T}_*$ is ℓ-adically complete. By Nakayama's lemma, lifts $\gamma_1, \dots, \gamma_{n_0} \in \mathcal{T}_*$ of a $\Bbbk$-basis of $\mathcal{T}_*/\ell\mathcal{T}_*$ satisfy

$$\mathcal{T}_* = \sum_{j=1}^{n_0} \Bbbk[[\ell]]\gamma_j.$$

By the Jacobian criterion, the ideal of $\operatorname{Sing}(W_*)$ in $\Bbbk[[x,y,z,\ell]]$ contains $f_*, \partial_x f_*, \partial_y f_*, \partial_z f_*$, so $\mathcal{O}_{\operatorname{Sing}(W_*)}$ is a quotient of $\mathcal{T}_*$ and $\operatorname{Sing}(W_*)$ is finite over $\operatorname{Spec}\Bbbk[[\ell]]$. Let Γ be the union of the 1-dimensional irreducible components of $\operatorname{Sing}(W_*)$ with the reduced structure. If $\Gamma = \emptyset$, the sum in (2) is 0, so assume that $\Gamma \neq \emptyset$. Then $\mathcal{O}_\Gamma$ is a quotient of $\mathcal{O}_{\operatorname{Sing}(W_*)}$, hence of $\mathcal{T}_*$, and Γ is finite over $\operatorname{Spec}\Bbbk[[\ell]]$. Let $\Gamma_1, \dots, \Gamma_k$ be the irreducible components of Γ. Since $\mathcal{O}_\Gamma$ is a reduced complete local ring of dimension 1 with residue field $\Bbbk$, the normalization of $\mathcal{O}_\Gamma$ is $\prod_{j=1}^k \Bbbk[[s_j]]$, where $\Bbbk[[s_j]]$ is the normalization of $\mathcal{O}_{\Gamma_j}$. Since Γ_j is finite over $\operatorname{Spec}\Bbbk[[\ell]]$ and $\dim \Gamma_j = 1$, $\ell \neq 0$ in $\Bbbk[[s_j]]$, so ℓ is a unit times $s_j^{r_j}$ for a positive integer r_j, so r_j is the degree of $\Bbbk[[s_j]]$ over $\Bbbk[[\ell]]$, as in (2). Since $\mathcal{O}_\Gamma \otimes_{\Bbbk[[\ell]]} \Bbbk((\ell))$ is reduced and finite over $\Bbbk((\ell))$, it is a product of fields, hence equal to the normalization $\prod_j \Bbbk((s_j))$ of $\mathcal{O}_\Gamma \otimes_{\Bbbk[[\ell]]} \Bbbk((\ell))$, of dimension $\sum_j r_j$ over $\Bbbk((\ell))$. Since $\mathcal{O}_\Gamma$ is a quotient of $\mathcal{T}_*$,

$$\sum_{j=1}^{k} r_j = \dim_{\Bbbk((\ell))} \mathcal{O}_\Gamma \otimes_{\Bbbk[[\ell]]} \Bbbk((\ell)) \le \dim_{\Bbbk((\ell))} \mathcal{T}_* \otimes_{\Bbbk[[\ell]]} \Bbbk((\ell)) \le n_0 \le 8.$$

This proves (2).

Step 5. In this step we prove the inequality of (3) for prime divisors over $\mathcal{X}_*$, and that $(W_*, \frac{1}{m}\operatorname{div}(h_*))$ is lc with $\operatorname{mld}(W_* \ni 0, \frac{1}{m}\operatorname{div}(h_*)) \ge a$.

Let $i \in \mathcal{I}$, and let

$$B_i := \operatorname{div}(f_i') + \tfrac{\tau_i}{m}\operatorname{div}(h_i') \quad \text{on } \mathbb{A}^4.$$

By **Step 2**, (D.70) and Theorem 2.10 with $H = G$, the image over $\mathbb{A}^4/G$ of any prime divisor P' over $\mathbb{A}^4$ with center 0 has log discrepancy at least $q_i(\tau_i)$ with respect to $(\mathbb{A}^4, B_i)/G$. Since the ramification index of P' over the image of P' is $e_{P'}$, that log discrepancy equals $a(P', \mathbb{A}^4, B_i)/e_{P'}$. Since $q_i(\tau_i) \ge a$ (Definition 4.1),

$$\frac{a(P', \mathbb{A}^4, 0) - \operatorname{mult}_{P'} f_i' - \frac{\tau_i}{m}\operatorname{mult}_{P'} h_i'}{e_{P'}} = \frac{a(P', \mathbb{A}^4, B_i)}{e_{P'}} \ge a. \tag{D.72}$$

Let P be a prime divisor over $\mathcal{X}_*$ with center the closed point, and let $G_P \subset G$ be the stabilizer of P and $I_P := I_G(P) \subset G_P$ its inertia group, so that $e_P = |I_P|$. By Lemma 2.12(3) with $X = \mathbb{A}^4_\Bbbk := \operatorname{Spec}\Bbbk[x,y,z,t]$, $H = G$, and $B = 0$, P restricts to a prime divisor $P^\flat$ over $\mathbb{A}^4_\Bbbk$ with center 0, that is, mult_P restricts to $\operatorname{mult}_{P^\flat}$ on $\Bbbk[x,y,z,t]$, $a(P^\flat, \mathbb{A}^4_\Bbbk, 0) = a(P, \mathcal{X}_*, 0)$, and the decomposition group, that is, the stabilizer, and the inertia group of $P^\flat$ are G_P and I_P. Fix an integer d with $(d+1)\operatorname{mult}_P \mathfrak{n}_* > \max\{\operatorname{mult}_P f_*, \operatorname{mult}_P h_*\}$, so that

$$\operatorname{mult}_P f_* = \operatorname{mult}_{P^\flat} \operatorname{jet}_{d+1}(f_*) \quad \text{and} \quad \operatorname{mult}_P h_* = \operatorname{mult}_{P^\flat} \operatorname{jet}_{d+1}(h_*).$$

Let $Y_\Bbbk \to \mathbb{A}^4_\Bbbk$ be a projective birational morphism from a normal variety on which $g(P^\flat)$ is a prime divisor for any $g \in G$. For any $g \in G_P \backslash I_P$, choose a rational function ϖ_g on $P^\flat$ with $g^*\varpi_g \neq \varpi_g$. Any $g \in I_P$ restricts to the identity on a dense open subset of $P^\flat$. As in [Kaw21, Remark 3.8(i)], possibly enlarging d, there exist the following, where base changes are taken along a morphism $\operatorname{Spec}\Bbbk \to T$ onto the generic point of T.

- A normal variety T and a dominant morphism $T \to Z_{d+1}$ which is finite over a non-empty open subset of Z_{d+1}.
- A projective birational morphism $Y_T \to \mathbb{A}^4_T$ whose base change is $Y_\Bbbk \to \mathbb{A}^4_\Bbbk$.
- Prime divisors on Y_T whose base changes are the prime divisors $g(P^\flat)$, birational self-maps of Y_T whose base changes are the birational actions of the elements of G, and rational functions whose base changes are the functions ϖ_g.

For a closed point $\mathbf{t} \in T$, let $P_{\mathbf{t}}$ be the fiber over $\mathbf{t}$ of the divisor on Y_T corresponding to $P^\flat$, and let $(f_{\mathbf{t}}, h_{\mathbf{t}})$ be the pair of polynomials of degree at most d whose $(d+1)$-jet is the image of $\mathbf{t}$ in Z_{d+1}. By [Kaw21, Remark 3.8(i)], possibly shrinking T, for any closed point $\mathbf{t} \in T$, $P_{\mathbf{t}}$ is a prime divisor over $\mathbb{A}^4$ with center 0 such that

$$\begin{aligned} a(P_{\mathbf{t}}, \mathbb{A}^4, 0) = a(P, \mathcal{X}_*, 0), \quad \operatorname{mult}_{P_{\mathbf{t}}} \mathfrak{n} = \operatorname{mult}_P \mathfrak{n}_*, \\ \operatorname{mult}_{P_{\mathbf{t}}} f_{\mathbf{t}} = \operatorname{mult}_P f_*, \quad \text{and} \quad \operatorname{mult}_{P_{\mathbf{t}}} h_{\mathbf{t}} = \operatorname{mult}_P h_*. \end{aligned} \tag{D.73}$$

Possibly shrinking T further, for any closed point $\mathbf{t} \in T$ the following hold.

- For any $g \in G$, $g(P_{\mathbf{t}})$ is the fiber over $\mathbf{t}$ of the divisor corresponding to $g(P^\flat)$, and the fibers for distinct divisors $g(P^\flat)$ are distinct. Thus the stabilizer of $P_{\mathbf{t}}$ in G is G_P.
- For any $g \in G_P \backslash I_P$, the restriction to $P_{\mathbf{t}}$ of the rational function corresponding to $g^* \varpi_g - \varpi_g$ is non-zero. Thus $g \notin I_G(P_{\mathbf{t}})$.
- Any $g \in I_P$ acts as the identity on a dense open subset U of the divisor on Y_T corresponding to $P^\flat$, since it does so on $P^\flat$, and $U \cap P_{\mathbf{t}}$ is dense in $P_{\mathbf{t}}$. Thus $g \in I_G(P_{\mathbf{t}})$.

Hence $I_G(P_{\mathbf{t}}) = I_P$ and $e_{P_{\mathbf{t}}} = e_P$. The image of T contains a non-empty open subset $Z^\circ_{d+1} \subset Z_{d+1}$ over which T is finite and surjective, and Z°_{d+1} contains $\operatorname{jet}_{d+1}(f_i', h_i')$ for all but finitely many $i \in \mathcal{I}$. For any such i, let $P_i := P_{\mathbf{t}}$ for a closed point $\mathbf{t} \in T$ over $\operatorname{jet}_{d+1}(f_i', h_i')$. Since

$$(f_{\mathbf{t}}, h_{\mathbf{t}}) = \operatorname{jet}_{d+1}(f_i', h_i'),$$

(D.73) and the choice of d give

$$(d+1) \operatorname{mult}_{P_i} \mathfrak{n} > \max\{\operatorname{mult}_{P_i} \operatorname{jet}_{d+1}(f_i'), \operatorname{mult}_{P_i} \operatorname{jet}_{d+1}(h_i')\},$$

so

$$\operatorname{mult}_{P_i} f_i' = \operatorname{mult}_P f_* \quad \text{and} \quad \operatorname{mult}_{P_i} h_i' = \operatorname{mult}_P h_*.$$

Thus (D.72) with $P' = P_i$ implies

$$\frac{a(P, \mathcal{X}_*, 0) - \operatorname{mult}_P f_* - \frac{\tau_i}{m} \operatorname{mult}_P h_*}{e_P} \geq a$$

for all but finitely many $i \in \mathcal{I}$. Since $\tau_i \to 1$,

$$\frac{a(P, \mathcal{X}_*, 0) - \operatorname{mult}_P f_* - \frac{1}{m} \operatorname{mult}_P h_*}{e_P} \geq a, \tag{D.74}$$

which is the inequality of (3).

Since W_i' is normal near 0 (**Step 2**), by (D.70) and [EMY03, Corollary 1.8], the pair

$$\big(\mathbb{A}^4, \operatorname{div}(f_i') + \tfrac{1}{m} \operatorname{div}(h_i')\big)$$

is lc near 0 for any $i \in \mathcal{I}$. By Theorem 2.18(1), the pair $(\mathcal{X}_*, W_* + \frac{1}{m} \operatorname{div}(h_*))$ is lc. By **Step 3**, W_* is normal and integral and $h_*|_{W_*} \neq 0$. By Lemma D.9(1) and by (D.74) with $e_P \geq 1$, $(W_*, \frac{1}{m} \operatorname{div}(h_*))$ is lc and

$$\operatorname{mld}\big(W_* \ni 0, \tfrac{1}{m} \operatorname{div}(h_*)\big) \geq \operatorname{mld}\big(\mathcal{X}_* \ni 0, W_* + \tfrac{1}{m} \operatorname{div}(h_*)\big) \geq a.$$

Step 6. In this step we prove that $(W_*, \frac{1}{m} \operatorname{div}(h_*))$ is not klt, which completes the proof of (3).

Let $D_* := \frac{1}{m} \operatorname{div}(h_*|_{W_*})$, and suppose that (W_*, D_*) is klt. Then $(W_*, (1+\epsilon)D_*)$ is klt for some rational number $\epsilon > 0$. By Lemma D.9(2), the pair

$$\big(\mathcal{X}_*, W_* + \tfrac{1+\epsilon}{m} \operatorname{div}(h_*)\big) = \big(\mathcal{X}_*, (f_*)^1 (h_*)^{(1+\epsilon)/m}\big)$$

is plt, hence lc. By **Step 3**, (f_*, h_*) is a compatible generic limit of the (f_i, h_i) along $\mathcal{I}$, also with H trivial in Definition 2.5, since the G-semi-invariant jets with the characters χ_f and χ_h form a linear subspace of the space of all jets. Thus, by Lemma D.7, there exists an infinite subset $\mathcal{I}' \subset \mathcal{I}$ such that $(\widehat{W_i}, \frac{1+\epsilon}{m} \operatorname{div}(h_i))$ is lc for any $i \in \mathcal{I}'$, and so is $(W_i, \frac{1+\epsilon}{m} \operatorname{div}(h_i))$ near o_i [dFEM11, Proposition 2.11(i)].

Since $\tau_i \to 1$, we may fix $i \in \mathcal{I}'$ with $(a-\delta)(1+\frac{1-\tau_i}{\epsilon}) < a$, and let F be a computing lift of b_i (Lemma 4.2(1)), so that

$$a\big(F, W_i, \tfrac{1}{m}\operatorname{div}(h_i)\big) = b_i e_F.$$

Since $(W_i, \frac{1+\epsilon}{m}\operatorname{div}(h_i))$ is lc near o_i,

$$0 \le a\big(F, W_i, \tfrac{1+\epsilon}{m}\operatorname{div}(h_i)\big) = b_i e_F - \frac{\epsilon}{m}\operatorname{mult}_F h_i, \quad \text{so} \quad \frac{1}{m}\operatorname{mult}_F h_i \le \frac{b_i e_F}{\epsilon}.$$

By Lemma 4.2(2) and $0 \le b_i \le a-\delta$,

$$q_i(\tau_i) \le b_i + (1-\tau_i)\frac{\operatorname{mult}_F h_i}{m e_F} \le b_i\Big(1+\frac{1-\tau_i}{\epsilon}\Big) \le (a-\delta)\Big(1+\frac{1-\tau_i}{\epsilon}\Big).$$

This contradicts $q_i(\tau_i) \ge a$. Thus $(W_*, \frac{1}{m}\operatorname{div}(h_*))$ is not klt. Together with **Step 5**, this proves (3). □

Proof of Lemma 13.5. The proof is similar to [Fuj11, Proof of Theorem 9.1], with [Mur24, Theorem 8.6.1] in place of the vanishing theorem used there. We divide the proof into five steps.

Since R is a quotient of $\Bbbk[[x_1,\dots,x_n]]$, X is a closed subscheme of $\operatorname{Spec}\Bbbk[[x_1,\dots,x_n]]$, so log resolutions of pairs on schemes of finite type over X exist [dFEM11, Theorems 2.3 and 2.4]. To make the inverse image of an ideal a divisor as well, we first blow up the ideal and then take a log resolution. Moreover, X is excellent and has a dualizing complex, both coming from the complete regular local ring $\Bbbk[[x_1,\dots,x_n]]$, and the canonical divisors of X and of the schemes proper and birational over X are taken with respect to the dualizing complexes induced by that of X.

Step 1. In this step we show that the lc centers of (X,B) are the images of the strata of the coefficient-one part of the boundary on a log resolution.

Let $\pi : Y \to X$ be a log resolution of (X,B), and let Δ_Y be the $\mathbb{Q}$-divisor on Y with

$$K_Y + \Delta_Y = \pi^*(K_X + B) \quad \text{and} \quad \pi_*\Delta_Y = B.$$

Here Y is regular, $\operatorname{Supp}\Delta_Y \cup \operatorname{Exc}(\pi)$ is a simple normal crossing divisor since π is a log resolution, and $\operatorname{mult}_P \Delta_Y \le 1$ for any prime divisor P on Y as (X,B) is lc. Some coefficients of Δ_Y may be negative. Let $\Delta_Y^{=1}$ be the sum of the components of Δ_Y with coefficient 1. A *stratum* of $\Delta_Y^{=1}$ is an irreducible component of the intersection of one or more distinct components of $\Delta_Y^{=1}$.

Let E be a prime divisor over X, let $V := \operatorname{center}_Y E$ with generic point η, and let $r := \operatorname{codim}_Y V$. Choose a regular system of parameters $y_1,\dots,y_r$ of $\mathcal{O}_{Y,\eta}$ and an integer $c \in [0,r]$ such that the components of $\operatorname{Supp}\Delta_Y$ through η are $(y_j = 0)$ for $1 \le j \le c$, with coefficients $d_1,\dots,d_c$ in Δ_Y. On $\operatorname{Spec}\mathcal{O}_{Y,\eta}$ we have $\Delta_Y = \sum_{j=1}^c d_j(y_j=0)$, and the pair $(\operatorname{Spec}\mathcal{O}_{Y,\eta}, \sum_{j=1}^r (y_j=0))$ is lc, as $\sum_{j=1}^r (y_j=0)$ is a reduced simple normal crossing divisor on a regular scheme. Thus

(D.75)

$$\begin{aligned} a(E,X,B) = a(E,Y,\Delta_Y) &= a\Big(E, Y, \sum_{j=1}^r (y_j=0)\Big) + \sum_{j=1}^c (1-d_j)\operatorname{mult}_E y_j + \sum_{j=c+1}^r \operatorname{mult}_E y_j \\ &\ge \sum_{j=1}^c (1-d_j)\operatorname{mult}_E y_j + \sum_{j=c+1}^r \operatorname{mult}_E y_j. \end{aligned}$$

Since y_j vanishes on $V = \operatorname{center}_Y E$, $\operatorname{mult}_E y_j > 0$ for any j, and $d_j \le 1$ for any j. Thus, if $a(E,X,B) = 0$, then $c = r$ and $d_1 = \cdots = d_r = 1$ by (D.75), so V is a stratum of $\Delta_Y^{=1}$. Conversely, let V be a stratum of $\Delta_Y^{=1}$ and $r := \operatorname{codim}_Y V$. Then exactly r components of $\operatorname{Supp}\Delta_Y$ contain V, all with coefficient 1, so the exceptional prime divisor E_V of the blow-up of Y along V satisfies

$$a(E_V, Y, \Delta_Y) = r - r = 0.$$

Since the lc centers of (X, B) are the centers on X of the prime divisors E over X with $a(E, X, B) = 0$, we obtain

$$\{\text{lc centers of } (X, B)\} = \{\pi(V) \mid V \text{ is a stratum of } \Delta_Y^{=1}\}. \tag{D.76}$$

In particular, (X, B) has finitely many lc centers. Any lc center of (X, B) is an irreducible proper closed subset of X, and it contains o as X is local.

For a stratum V of $\Delta_Y^{=1}$, the blow-up $\beta : Y_V \to Y$ of Y along V is again a log resolution of (X, B), on which E_V and $\beta_*^{-1}D$ for the components D of $\Delta_Y^{=1}$ have coefficient 1. Blowing up, one at a time, a stratum mapping onto each lc center which is not the image of a component of $\Delta_Y^{=1}$, we may assume that

$$\text{any lc center of } (X, B) \text{ is } \pi(D) \text{ for a component } D \text{ of } \Delta_Y^{=1}. \tag{D.77}$$

Then (D.77) holds on any log resolution of (X, B) factoring through Y.

Step 2. In this step, for a union Z of lc centers, we construct a log resolution and divisors S, Ξ, and Δ' on it such that no stratum of $(\Delta')^{=1}$ maps into Z.

Let Z be a non-empty union of lc centers of (X, B) with the reduced structure, and let $I_Z \subset \mathcal{O}_X$ be the ideal of Z. Blowing up Y along $I_Z\mathcal{O}_Y$ and then taking a log resolution, we may assume that $\pi^{-1}(Z)$ is a divisor and that

$$\operatorname{Supp} \Delta_Y \cup \operatorname{Exc}(\pi) \cup \pi^{-1}(Z)$$

is a simple normal crossing divisor. By **Step 1**, (D.76) and (D.77) hold on Y. Let

- S be the sum of the components D of $\Delta_Y^{=1}$ with $\pi(D) \subset Z$,
- $\Xi := \lceil -\Delta_Y^{<1} \rceil$, where $\Delta_Y^{<1}$ is the part of Δ_Y consisting of the components with coefficients <1, and
- $\Delta' := \Delta_Y + \Xi - S$.

By (D.77), $\pi(S) = Z$. A component P of Δ_Y which is not π-exceptional has coefficient $\operatorname{mult}_{\pi(P)} B \in [0, 1]$ in Δ_Y as $\pi_*\Delta_Y = B$, so it is not a component of Ξ. A component of Δ_Y with coefficient $d < 1$ has coefficient $\lceil -d \rceil \geq 0$ in Ξ. Thus $\Xi \geq 0$, Ξ is π-exceptional, no component of S is a component of Ξ, and, since X is normal,

$$\pi_*\mathcal{O}_Y(\Xi) = \mathcal{O}_X.$$

A component of Δ_Y with coefficient $d < 1$ has coefficient $d + \lceil -d \rceil \in [0, 1)$ in Δ', and a component of $\Delta_Y^{=1}$ has coefficient 0 in Δ' if it is a component of S, and 1 otherwise. Thus Δ' has simple normal crossing support and coefficients in $[0, 1]$, and

$$(\Delta')^{=1} = \Delta_Y^{=1} - S.$$

We show

$$\pi(V) \not\subset Z \text{ for } V = Y \text{ and for any stratum } V \text{ of } (\Delta')^{=1}. \tag{D.78}$$

Since Z is a proper closed subset of X, $\pi(Y) = X \not\subset Z$. Let V be an irreducible component of $D_1 \cap \cdots \cap D_c$ for distinct components $D_1, \dots, D_c$ of $(\Delta')^{=1}$, and suppose that $\pi(V)$ is contained in Z. Then a component Γ of $\pi^{-1}(Z)$ contains V. Since $\operatorname{codim}_Y V = c$ and

$$\Theta := \operatorname{Supp} \Delta_Y \cup \operatorname{Exc}(\pi) \cup \pi^{-1}(Z)$$

is a simple normal crossing divisor, at most c components of Θ contain V, so $\Gamma = D_k$ for some k. Then $\pi(D_k) \subset Z$, so D_k is a component of S, which contradicts $(\Delta')^{=1} = \Delta_Y^{=1} - S$.

Step 3. In this step we show that, in **Step 2**,

$$\pi_*\mathcal{O}_S = \mathcal{O}_Z, \tag{D.79}$$

and that S is connected.

Let $\mathcal{L} := \mathcal{O}_Y(\Xi - S)$ and $\mathcal{O}_S(\Xi) := \mathcal{O}_Y(\Xi) \otimes \mathcal{O}_S$. We have the exact sequence

$$0 \to \mathcal{L} \to \mathcal{O}_Y(\Xi) \to \mathcal{O}_S(\Xi) \to 0. \tag{D.80}$$

Since

$$(\Xi - S) - (K_Y + \Delta') = -(K_Y + \Delta_Y) = -\pi^*(K_X + B),$$

for a positive integer N such that $N(K_X + B)$ is Cartier and $N\Delta'$ has integer coefficients, we have

$$\mathcal{L}^{\otimes N} \cong \mathcal{O}_Y(N(K_Y + \Delta')) \otimes \mathcal{M}, \quad \text{where } \mathcal{M} := \pi^*\mathcal{O}_X(-N(K_X + B)).$$

Since X is local, $\mathcal{O}_X(-N(K_X + B)) \cong \mathcal{O}_X$, so $\mathcal{M} \cong \mathcal{O}_Y$ and $\pi^*\pi_*\mathcal{M} \to \mathcal{M}$ is surjective. By [Mur24, Theorem 8.6.1], any associated point of $R^1\pi_*\mathcal{L}$ is the generic point of $\pi(V)$ for $V = Y$ or for a stratum V of $(\Delta')^{=1}$. By (D.78), no associated point of $R^1\pi_*\mathcal{L}$ is contained in Z.

A non-zero coherent subsheaf of $R^1\pi_*\mathcal{L}$ supported in Z has an associated point in Z, which is an associated point of $R^1\pi_*\mathcal{L}$. Thus $R^1\pi_*\mathcal{L}$ has no non-zero coherent subsheaf supported in Z. Since $\pi_*\mathcal{O}_S(\Xi)$ is supported in $\pi(S) = Z$, the connecting map $\pi_*\mathcal{O}_S(\Xi) \to R^1\pi_*\mathcal{L}$ of the push-forward of (D.80) is zero, so

$$\mathcal{O}_X = \pi_*\mathcal{O}_Y(\Xi) \to \pi_*\mathcal{O}_S(\Xi)$$

is surjective with kernel $\pi_*\mathcal{L}$. For a regular function ψ on X, since no component of S is a component of Ξ, $\psi \in \pi_*\mathcal{L}$ if and only if $\operatorname{mult}_D \pi^*\psi \geq 1$ for any component D of S, that is, ψ vanishes on $\pi(D)$ for any component D of S. Since $\pi(S) = Z$, $\pi_*\mathcal{L} = I_Z$, and the induced map

$$\mathcal{O}_Z \to \pi_*\mathcal{O}_S(\Xi)$$

is an isomorphism. The map $\mathcal{O}_Z \to \pi_*\mathcal{O}_S(\Xi)$ is the composite of $\mathcal{O}_Z \to \pi_*\mathcal{O}_S$, which is injective because a regular function ψ on X with $(\pi^*\psi)|_S = 0$ vanishes on $\pi(S) = Z$, and of $\pi_*\mathcal{O}_S \to \pi_*\mathcal{O}_S(\Xi)$, which is injective because S is reduced and no component of S is a component of Ξ. Thus $\mathcal{O}_Z \to \pi_*\mathcal{O}_S$ and $\pi_*\mathcal{O}_S \to \pi_*\mathcal{O}_S(\Xi)$ are isomorphisms, which proves (D.79). Since X is affine, (D.79) implies $H^0(S, \mathcal{O}_S) = H^0(Z, \mathcal{O}_Z)$, which is a quotient of R, hence a local ring. Thus 0 and 1 are the only idempotents of $H^0(S, \mathcal{O}_S)$, and S is connected.

Step 4. In this step we prove the first part of the lemma: the existence, normality and invariance of the minimal lc center C of (X, B), and the description of C when $\dim X = 3$.

Let C_1 and C_2 be lc centers of (X, B). We show that $C_1 \cap C_2$ contains an lc center. We may assume that $C_1 \not\subset C_2$ and $C_2 \not\subset C_1$. Apply **Steps 2 and 3** to $Z = C_1 \cup C_2$. For any component D of S, $\pi(D)$ is an lc center by (D.76), so $\pi(D)$ is irreducible and $\pi(D) \subset C_1$ or $\pi(D) \subset C_2$. If $\pi(D) \subset C_1 \cap C_2$ for some D, we are done. Otherwise $S = S_1 + S_2$, where S_k is the sum of the components D of S with $\pi(D) \subset C_k$ for $k \in \{1, 2\}$. By (D.77), $C_1 = \pi(D)$ for a component D of $\Delta_Y^{=1}$, and D is a component of S_1 as $C_1 \not\subset C_2$, so $S_1 \neq 0$. Similarly $S_2 \neq 0$. Since S is connected by **Step 3**, there exist a component D_1 of S_1 and a component D_2 of S_2 with $D_1 \cap D_2 \neq \emptyset$. Let V be an irreducible component of $D_1 \cap D_2$. Then V is a stratum of $\Delta_Y^{=1}$, so $\pi(V)$ is an lc center by (D.76), and

$$\pi(V) \subset \pi(D_1) \cap \pi(D_2) \subset C_1 \cap C_2.$$

Since (X, B) has finitely many lc centers, there exists an lc center C which is minimal with respect to inclusion. For any lc center C', $C \cap C'$ contains an lc center, which is equal to C by minimality, so $C \subset C'$. Thus C is the unique minimal lc center and is contained in all lc centers. An automorphism σ of X with $\sigma^*B = B$ maps lc centers of (X, B) to lc centers of (X, B) and preserves inclusions, so $\sigma(C) = C$. In particular, C is invariant under any finite group of automorphisms of X preserving B.

We show that C is normal. Apply **Steps 2 and 3** to $Z = C$. For any component D of S and any irreducible component V of the intersection of two distinct components of S, $\pi(D)$ and $\pi(V)$ are lc centers by (D.76) contained in C, so

$$\pi(D) = \pi(V) = C$$

by minimality. Moreover, D and V are regular.

Since C is integral, $H^0(C, \mathcal{O}_C)$ is a domain. Let ψ be an element of the function field of C which is integral over $H^0(C, \mathcal{O}_C)$. For any component D of S, $\pi|_D : D \to C$ is dominant and D is normal, so $\psi_D := (\pi|_D)^*\psi$ is a regular function on D. For two distinct components D and D'

of S and an irreducible component V of $D \cap D'$, the restrictions $\psi_D|_V$ and $\psi_{D'}|_V$ are both equal to $(\pi|_V)^*\psi$ at the generic point of V, which maps to the generic point of C. Since $D \cap D'$ is reduced, $\psi_D|_{D\cap D'} = \psi_{D'}|_{D\cap D'}$. Near a point $q \in Y$ where $S = (y_1 \cdots y_c = 0)$ for part $y_1, \dots, y_c$ of a regular system of parameters of $\mathcal{O}_{Y,q}$, we have the exact sequence

$$0 \to \mathcal{O}_{Y,q}/(y_1 \cdots y_c) \to \mathcal{O}_{Y,q}/(y_1 \cdots y_{c-1}) \oplus \mathcal{O}_{Y,q}/(y_c) \to \mathcal{O}_{Y,q}/(y_1 \cdots y_{c-1}, y_c) \to 0,$$

because $(y_1 \cdots y_{c-1}) \cap (y_c) = (y_1 \cdots y_c)$, and $\mathcal{O}_{Y,q}/(y_c)$ is regular with regular parameters the images of $y_1, \dots, y_{c-1}$. Since $(y_1 \cdots y_{c-1}, y_c)/(y_c)$ defines a reduced simple normal crossing divisor on the regular scheme $\operatorname{Spec} \mathcal{O}_{Y,q}/(y_c)$, a function on it is determined by its restrictions to the components $(y_j = y_c = 0)$ for $j < c$. By induction on c, a family of regular functions on the components of S whose restrictions to the pairwise intersections coincide is the family of restrictions of a regular function on S. Thus the functions ψ_D are the restrictions of a regular function ψ_S on S. By (D.79), $\psi_S = \pi^*\psi'$ for some $\psi' \in H^0(C, \mathcal{O}_C)$. Since $\pi|_D$ is dominant for a component D of S, and $(\pi|_D)^*\psi' = \psi_D = (\pi|_D)^*\psi$, we have $\psi = \psi' \in H^0(C, \mathcal{O}_C)$. Thus $H^0(C, \mathcal{O}_C)$ is integrally closed, and C is normal.

Assume that $\dim X = 3$ and $\operatorname{mld}(X \ni o, B) > 0$. Then $\{o\}$ is not an lc center, and C is a proper closed subset of X, so $\dim C \in \{1, 2\}$. If $\dim C = 1$, then C is normal of dimension 1 and $H^0(C, \mathcal{O}_C)$ is a quotient of R, so $C \cong \operatorname{Spec} \Bbbk[[s]]$.

Step 5. In this step we prove the alternatives (1) and (2) for the minimal lc center C of (W_*, D).

Let $\mathfrak{n}_*$ be the maximal ideal of $\Bbbk[[x_1, \dots, x_4]]$. By **Steps 1–4** applied to (W_*, D), the minimal lc center C of (W_*, D) is normal and G-invariant, and $\dim C \in \{1, 2\}$. We show that $\dim C = 2$ implies (1) and $\dim C = 1$ implies (2).

Assume that $\dim C = 2$. Since W_* is normal, $\mathcal{O}_{W_*,C}$ is a discrete valuation ring, so C is the only prime divisor over W_* with center C. Since C is an lc center, $a(C, W_*, D) = 1 - \operatorname{mult}_C D = 0$, so C is a component of D with coefficient 1. Any lc center C' of (W_*, D) contains C and is a proper irreducible closed subset of W_*, so $C' = C$ as $\dim W_* = 3$. Thus C is the only lc center. Any exceptional prime divisor E over W_* satisfies $\dim \operatorname{center}_{W_*} E \le 1$, so $\operatorname{center}_{W_*} E$ is not an lc center and $a(E, W_*, D) > 0$. Thus (W_*, D) is plt, which proves (1).

Assume that $\dim C = 1$. By **Step 4**, $C \cong \operatorname{Spec} \Bbbk[[s]]$ is regular. Let $J \subset \Bbbk[[x_1, \dots, x_4]]$ be the ideal of C. Then $f_* \in J$, J is G-stable, and

$$\overline{J} := (J + \mathfrak{n}_*^2)/\mathfrak{n}_*^2 \subset \mathfrak{n}_*/\mathfrak{n}_*^2$$

is a G-stable subspace with $\dim \overline{J} = 4 - 1 = 3$, since $\mathfrak{n}_*/(J + \mathfrak{n}_*^2)$ is the cotangent space of $\Bbbk[[s]]$. Since G is finite and acts diagonally and $\Bbbk$ is algebraically closed of characteristic zero, there exist G-eigenvectors $\overline{u}_1, \overline{u}_2, \overline{u}_3$ forming a basis of $\overline{J}$ and a G-eigenvector $\overline{s} \in (\mathfrak{n}_*/\mathfrak{n}_*^2) \backslash \overline{J}$. For a character χ of G, the projector

$$p_\chi(\psi) := \frac{1}{|G|} \sum_{g \in G} \chi(g)^{-1} g \cdot \psi$$

maps $\Bbbk[[x_1, \dots, x_4]]$ to the χ-semi-invariant elements, preserves J and $\mathfrak{n}_*^2$, and induces the identity on the χ-eigenvectors of $\mathfrak{n}_*/\mathfrak{n}_*^2$. Applying to each of the lifts of $\overline{u}_1, \overline{u}_2, \overline{u}_3$ in J and of $\overline{s}$ in $\mathfrak{n}_*$ the projector p_χ, where χ is the character of the class of that lift in $\mathfrak{n}_*/\mathfrak{n}_*^2$, we obtain G-semi-invariant elements $u_1, u_2, u_3 \in J$ and $s \in \mathfrak{n}_*$ (whose image in $H^0(C, \mathcal{O}_C)$ is a uniformizer) with classes $\overline{u}_1, \overline{u}_2, \overline{u}_3, \overline{s}$ in $\mathfrak{n}_*/\mathfrak{n}_*^2$. Since $\overline{s}, \overline{u}_1, \overline{u}_2, \overline{u}_3$ form a basis of $\mathfrak{n}_*/\mathfrak{n}_*^2$, (s, u_1, u_2, u_3) are G-eigenparameters of $\Bbbk[[x_1, \dots, x_4]]$, that is,

$$\Bbbk[[x_1, \dots, x_4]] = \Bbbk[[s, u_1, u_2, u_3]]$$

with G acting diagonally on s, u_1, u_2, u_3. In particular, f_* and h_* are G-semi-invariant elements of $\Bbbk[[s, u_1, u_2, u_3]]$. Since $(u_1, u_2, u_3) \subset J$ and $\Bbbk[[x_1, \dots, x_4]]/(u_1, u_2, u_3) \cong \Bbbk[[s]]$ and $\Bbbk[[x_1, \dots, x_4]]/J$ are 1-dimensional domains, the surjection between them is an isomorphism, so

$$C = (u_1 = u_2 = u_3 = 0), \quad J = (u_1, u_2, u_3).$$

We show $h_* \in J$. Otherwise h_* is a unit at the generic point of C, so $\mathrm{mult}_E h_* = 0$ for any prime divisor E over W_* with center C. Then

$$a(E, W_*, D) = a(E, W_*, 0) > 0$$

as W_* is klt, which contradicts the fact that C is an lc center of (W_*, D).

Assume that $C \subset \mathrm{Sing}(W_*)$. Since $f_* \in J = (u_1, u_2, u_3)$, we may write

$$f_* = \sum_{k=1}^{3} \gamma_k(s) u_k + f_2 \quad \text{with} \quad \gamma_1, \gamma_2, \gamma_3 \in \Bbbk[[s]] \quad \text{and} \quad f_2 \in J^2.$$

The localization $\Bbbk[[x_1, \dots, x_4]]_J$ is a regular local ring with maximal ideal $J\Bbbk[[x_1, \dots, x_4]]_J$ and regular parameters u_1, u_2, u_3, and with residue field the fraction field of $\Bbbk[[s]]$. The local ring of W_* at the generic point of C is $\Bbbk[[x_1, \dots, x_4]]_J/(f_*)$, which is not regular as $C \subset \mathrm{Sing}(W_*)$, so $f_* \in J^2\Bbbk[[x_1, \dots, x_4]]_J$. Since the classes of u_1, u_2, u_3 form a basis of $J\Bbbk[[x_1, \dots, x_4]]_J/J^2\Bbbk[[x_1, \dots, x_4]]_J$, $\gamma_1 = \gamma_2 = \gamma_3 = 0$. Thus $f_* = f_2 \in (u_1, u_2, u_3)^2$, which proves (2). □

Lemma D.10. *Let $\Bbbk \supset \mathbb{C}$ be an algebraically closed field, let a finite abelian group G act diagonally on $\Bbbk[[x, y, z, t]]$ with semi-invariant x, y, z, t, let $\mathfrak{n} := (x, y, z, t)$, and let $f \in \mathfrak{n}^2$ be a semi-invariant element with quadratic part x^2. Then there exists a unique $\varphi \in (y, z, t)\Bbbk[[y, z, t]]$ with $(\partial_x f)(\varphi, y, z, t) = 0$. Moreover, φ is semi-invariant of character χ_x, and there exists a semi-invariant element x' of character χ_x with $x' - (x - \varphi) \in (x - \varphi)\mathfrak{n}$ and*

$$f = x'^2 + f(\varphi, y, z, t).$$

Proof. Since G acts diagonally and f is semi-invariant, the quadratic part x^2 of f is semi-invariant of character χ_f, so $\chi_x^2 = \chi_f$. Since $\partial_x f - 2x \in \mathfrak{n}^2$, the formal implicit function theorem gives a unique $\varphi \in (y, z, t)\Bbbk[[y, z, t]]$ with $(\partial_x f)(\varphi, y, z, t) = 0$, and $\varphi \in (y, z, t)^2$ since $2\varphi = -(\partial_x f - 2x)(\varphi, y, z, t)$. For $\sigma \in G$, applying σ to $(\partial_x f)(\varphi, y, z, t) = 0$ and using that $\partial_x f$ is semi-invariant, we see that $\chi_x(\sigma)^{-1}\sigma(\varphi)$ also satisfies it, so $\sigma(\varphi) = \chi_x(\sigma)\varphi$ by the uniqueness. The Taylor formula in x at φ gives $U \in \Bbbk[[x, y, z, t]]$ with

$$f = f(\varphi, y, z, t) + (x - \varphi)(\partial_x f)(\varphi, y, z, t) + (x - \varphi)^2 U = f(\varphi, y, z, t) + (x - \varphi)^2 U,$$

and $U(0) = 1$ by comparing quadratic parts, since $\varphi \in (y, z, t)^2$. Since $f - f(\varphi, y, z, t)$ and $(x - \varphi)^2$ are semi-invariant of character $\chi_f = \chi_x^2$, U is invariant. Let $U^{1/2}$ be the square root of U with constant term 1. For $\sigma \in G$, $\sigma(U^{1/2})$ is a square root of U with constant term 1, so $U^{1/2}$ is invariant. Then $x' := (x - \varphi)U^{1/2}$ is semi-invariant of character χ_x, $f = x'^2 + f(\varphi, y, z, t)$, and

$$x' - (x - \varphi) = (x - \varphi)(U^{1/2} - 1) \in (x - \varphi)\mathfrak{n}.$$

The lemma follows. □

Proof of Construction-Theorem 13.7. By Definition 4.1(3), $b_i = q_i(1) \le a - \delta$ and $q_i(\tau_i) \ge a$ for any i. Let F_i be the computing lift of b_i given in the statement. By Lemma 2.12(3), a prime divisor F over W_i with center o_i is regarded as a formal prime divisor over $\operatorname{Spec} \widehat{\mathcal{O}}_{W_i, o_i}$ with equal log discrepancies and equal inertia groups, so mult_F is defined on $\widehat{\mathcal{O}}_{W_i, o_i}$. We divide the proof into four steps.

Step 1. In this step we prepare the equations and the marks, replace them by polynomials, and take the limit.

Fix i. Since G_i is abelian, lifting a basis of G_i-eigenvectors of $\mathfrak{m}_{o_i}/\mathfrak{m}_{o_i}^2$ to semi-invariant elements of $\widehat{\mathcal{O}}_{W_i, o_i}$ yields a G_i-equivariant isomorphism $\widehat{\mathcal{O}}_{W_i, o_i} \cong \mathbb{C}[[x, y, z, t]]/(f_i)$, where G_i acts diagonally on $\mathbb{C}[[x, y, z, t]]$ and f_i is semi-invariant. Since $(W_i \ni o_i)$ has quadratic rank 1, the quadratic part of f_i is a non-zero multiple of x_0^2 for a linear form x_0 unique up to a constant,

so the line $\mathbb{C}x_0$ is G_i-stable and we may take $x = x_0$. By Lemma D.10 with $\Bbbk = \mathbb{C}$, possibly multiplying f_i by a non-zero constant and replacing x with x', we may assume that

$$f_i = x^2 + g_i(y,z,t), \qquad g_i \in (y,z,t)^3,$$

where $g_i \in (y,z,t)^3$ since $\varphi \in (y,z,t)^2$ by $\partial_x f_i - 2x \in \mathfrak{n}^2$, $\mathfrak{n} := (x,y,z,t)$.

Since $(W_i \ni o_i)$ is of type cD, the cubic part $g_{i,3}$ of g_i is not the cube of a linear form [Liu26b, Definition 9.1], and $g_{i,3} \neq 0$, since otherwise any hyperplane section of $(W_i \ni o_i)$ has multiplicity at least 3 or, by the splitting lemma [Kol98, 2.5(1)(2) and Proposition 2.7], an equation $u^2 + q(v,w)$ with $q \in (v,w)^4$, and neither is Du Val. Thus $g_{i,3}$ is either reduced, or the product of the square of a linear form and a linear form not proportional to it. Possibly passing to a subsequence, we may assume that one of the following holds for any i.

(R) $g_{i,3}$ is reduced.
(D) $g_{i,3}$ is the product of the square of a linear form and a linear form not proportional to it.

In case (D), the double line and the simple line of $(g_{i,3} = 0)$ have different multiplicities, so each is G_i-stable, and we may choose the eigenparameters y, z, t with $g_{i,3} = zy^2$. We then remove the higher terms divisible by y^2 or by yz, degree by degree. Let $k \geq 4$ and assume that $f_i \equiv x^2 + zy^2 + r_{<k} \pmod{\mathfrak{n}^k}$, where the polynomial $r_{<k} \in \mathbb{C}[y,z,t]$ has no monomial divisible by y^2 or by yz. Write the degree-k part of $f_i - x^2 - zy^2$ as $y^2u + yzv + r_k$, where r_k has no monomial divisible by y^2 or by yz, and u, v are homogeneous of degree $k-2$. Since f_i is semi-invariant with $\chi_{f_i} = \chi_y^2\chi_z$, u and v are semi-invariant with $\chi_u = \chi_z$ and $\chi_v = \chi_y$. The equivariant substitution $z \to z - u$, $y \to y - \frac{1}{2}v$ removes $y^2u + yzv$ and changes f_i by terms of degree at least $k+1$, since $k \geq 4$. The composite of the substitutions for $k = 4, 5, \dots$ converges in the $\mathfrak{n}$-adic topology, so in case (D)

$$f_i = x^2 + zy^2 + \gamma_i(t)y + \psi_i(z,t), \qquad \operatorname{mult}_0 \gamma_i \geq 3, \quad \text{and} \quad \operatorname{mult}_0 \psi_i \geq 4. \tag{D.81}$$

(D.81) is the form in Construction-Theorem 13.7(6). We let $R_i := \gamma_i(t)y + \psi_i(z,t)$. By [Liu26b, Definition-Lemma 11.5(1)],

$$h_i = \alpha_i(y,z,t) + x\xi_i(y,z,t) \quad \text{in } \widehat{\mathcal{O}}_{W_i,o_i}$$

for unique semi-invariant $\alpha_i, \xi_i \in \mathbb{C}[[y,z,t]]$.

Since G_i is cyclic and $|G_i| \leq M$, there exist finitely many possibilities for G_i together with the characters of x, y, z, t, f_i, h_i up to isomorphism. Possibly passing to a subsequence, we may assume that $G_i = G$ with fixed characters for any i. We let G act diagonally on $\mathbb{A}^4 = \operatorname{Spec}\mathbb{C}[x,y,z,t]$ through the fixed characters.

We now replace (f_i, h_i) by polynomials. Since W_i is isolated, there exists a positive integer κ_i with

$$\mathfrak{n}^{\kappa_i} \subset (f_i) + \mathfrak{n} \cdot (\partial_x f_i, \partial_y f_i, \partial_z f_i, \partial_t f_i).$$

Fix an integer $L_i > \max\{2\kappa_i, i\}$ and let f_i' be the sum of the terms of f_i of degree less than L_i. By Lemma 3.2(2), $W_i' := (f_i' = 0) \subset \mathbb{A}^4$ has a G-equivariant isomorphism $\widehat{\mathcal{O}}_{W_i',0} \cong \widehat{\mathcal{O}}_{W_i,o_i}$. Moreover, $f_i' = x^2 + g_i'$ with g_i' the sum of the terms of g_i of degree less than L_i, and in case (D) $f_i' = x^2 + zy^2 + \gamma_i'(t)y + \psi_i'(z,t)$, where $\gamma_i'(t)y$ and ψ_i' are the sums of the terms of $\gamma_i(t)y$ and ψ_i of degree less than L_i. Through the isomorphism, h_i is a semi-invariant element of $\widehat{\mathcal{O}}_{W_i',0}$. Since $f_i'(x,0,0,0) = x^2$, by the Weierstrass division theorem [GLS07, Theorem I.1.8] with respect to f_i', we write h_i as $\widehat{\alpha}_i + x\widehat{\xi}_i$ with unique $\widehat{\alpha}_i, \widehat{\xi}_i \in \mathbb{C}[[y,z,t]]$. Since G acts diagonally, the uniqueness of $(\widehat{\alpha}_i, \widehat{\xi}_i)$ shows that $\widehat{\alpha}_i + x\widehat{\xi}_i$ is semi-invariant with the character of h_i. With L_i and W_i' fixed, let $L_\gamma^{(\tau_i)}$ and $L_\gamma^{(1)}$ be the integers L_γ given by Lemma 3.2(1), with $H = G$, $V_0 = W_i'$, $\operatorname{mld}(\widehat{W}_i' \ni \widehat{0}) + 1$ in place of γ, where $\widehat{W}_i'$ is the completion of W_i' at 0 with closed point $\widehat{0}$, and the klt pairs with boundaries $\frac{\tau_i}{m}\operatorname{div}(h_i)$ and $\frac{1}{m}\operatorname{div}(h_i)$ on $\widehat{W}_i'$ respectively. As Lemma 3.2(1)(c) requires, $\operatorname{mld}(\widehat{W}_i' \ni \widehat{0}) + 1 > \operatorname{mld}(\widehat{W}_i' \ni \widehat{0}, B)$ for both boundaries B. Let $h_i' = \alpha_i' + x\xi_i'$, where,

for an integer $L_i' \geq \max(L_i, L_\gamma^{(\tau_i)}, L_\gamma^{(1)})$, α_i' and ξ_i' are the sums of the terms of $\widehat{\alpha}_i$ and $\widehat{\xi}_i$ of degree less than L_i'. By Lemma 3.2(1) and Lemma 2.12(2)(3), the following hold.

(1) The pair $(W_i', \frac{1}{m}\operatorname{div}(h_i'))$ is klt near 0, and the radial function of $(W_i' \ni 0, G, h_i', m)$ takes the values $q_i(\tau_i)$ and b_i at τ_i and at 1.
(2) The computing lifts F of b_i over W_i correspond to the computing lifts F' of b_i over W_i', with $a(F', W_i', 0) = a(F, W_i, 0)$, $e_{F'} = e_F$ and $\operatorname{mult}_{F'} h_i' = \operatorname{mult}_F h_i$. We let F_i' correspond to F_i.

Since f_i' and h_i' are polynomials, the members of the compatible generic limit below are regular functions on $\mathbb{A}^4$, as [Liu26b, Construction-Proposition 4.1(a)] below and Theorem 2.18 and [Kaw21, Definition 3.1] in **Step 3** require.

Let $w := (3,2,2,2)$ be the weights of x, y, z, t. The w-leading form of f_i' is $x^2 + g_{i,3}'$, which is irreducible since $g_{i,3}' = g_{i,3}$ is a non-zero cubic form. By [Liu26b, Lemma 4.12], the weighted blow-up with weights w yields a prime divisor E_i over W_i' with center 0 and

$$a(E_i, W_i', 0) = 3+2+2+2-6 = 3, \qquad \operatorname{mult}_{E_i} h_i' = \min(2\operatorname{mult}_0 \alpha_i', 3 + 2\operatorname{mult}_0 \xi_i'),$$

because $2\operatorname{mult}_0 \alpha_i'$ and $3 + 2\operatorname{mult}_0 \xi_i'$ have different parities and the w-leading form of h_i' has degree at most 1 in x, so it is not divisible by $x^2 + g_{i,3}'$. Since $(W_i', \frac{1}{m}\operatorname{div}(h_i'))$ is lc, $\operatorname{mult}_{E_i} h_i' \leq 3m$, so h_i' has a non-zero monomial of degree at most $\frac{3m}{2}$, so $\operatorname{mult}_0 h_i' \leq \frac{3m}{2}$ for any i, as Definition 2.5 requires.

By [Liu26b, Construction-Proposition 4.1(a)], possibly passing to a subsequence, there exists a compatible generic limit (f_*, h_*) of the pairs (f_i', h_i') (Definition 2.5), with coefficient field $\Bbbk$. Since the automorphism of Lemma 3.2(2) is the identity modulo $\mathfrak{n}^{L_i - \kappa_i + 1}$ and $L_i > 2\kappa_i$, by the uniqueness in the Weierstrass division theorem [GLS07, Theorem I.1.8] with respect to f_i and to $f_i' \equiv f_i \pmod{\mathfrak{n}^{L_i}}$, we have $f_i' \equiv f_i$ and $h_i' \equiv \alpha_i + x\xi_i$ modulo $\mathfrak{n}^{\lceil L_i/2 \rceil}$, so for any d the d-jets of (f_i', h_i') and of $(f_i, \alpha_i + x\xi_i)$ are equal for all but finitely many i, and (f_*, h_*) is also a compatible generic limit of the pairs $(f_i, \alpha_i + x\xi_i)$. For any integer $d \geq 1$ let $Z_d \subset \mathcal{J}_d$ be as in Definition 2.5 for (f_*, h_*).

The characters are fixed by Definition 2.5. The conditions that the coefficient of x^2 in f is 1, that the other coefficients of f at monomials divisible by x or of degree at most 2 vanish, that in case (D) moreover the coefficient of zy^2 in f is 1 and the other coefficients of f at monomials of degree 3 or divisible by y^2 or by yz vanish, and that the coefficients of h at monomials divisible by x^2 vanish, prescribe finitely many coefficients of each jet $(f, h) \in \mathcal{J}_d$, so they define closed subsets of $\mathcal{J}_d$, and by Lemma 2.8

$$f_* = x^2 + g_*(y,z,t), \quad g_* \in (y,z,t)^3, \quad \text{and} \quad h_* = \alpha_*(y,z,t) + x\xi_*(y,z,t),$$

with the characters fixed above, and in case (D)

$$f_* = x^2 + zy^2 + \gamma_*(t)y + \psi_*(z,t), \qquad \operatorname{mult}_0 \gamma_* \geq 3, \quad \text{and} \quad \operatorname{mult}_0 \psi_* \geq 4. \tag{D.82}$$

Since the non-zero cubic forms that are not cubes, and the reduced cubic forms, form constructible subsets, Lemma 2.8 implies that the cubic part $g_{*,3}$ of g_* is non-zero and not a cube, and reduced in case (R).

We show that $h_* \notin (f_*)$. By Definition 2.5, $h_* \neq 0$. Since h_* has degree at most 1 in x and f_* is monic of degree 2 in x, the uniqueness in the Weierstrass division by f_* implies $h_* \notin (f_*)$.

Step 2. In this step we construct the D_n sections and show that W_* is normal, canonical cDV and not isolated.

For $\lambda, \mu \in \Bbbk$ let $\ell_{\lambda,\mu} := y + \lambda z + \mu t$. We show that there exist an integer $n \geq 4$ and a non-zero polynomial $Q \in \Bbbk[\lambda, \mu]$ such that, for any $\lambda, \mu \in \mathbb{C}$ with $Q(\lambda, \mu) \neq 0$, the form $\ell_{\lambda,\mu}$ cuts out a D_n section of W_*, of W_i' and of W_i (in the coordinates of **Step 1**) for all but finitely many i.

Case (R). Let $n := 4$ and let $Q(\lambda, \mu)$ be the discriminant of the binary cubic form $g_{*,3}|_{(\ell_{\lambda,\mu}=0)}$. Since a general line intersects the reduced plane cubic $(g_{*,3} = 0)$ in three distinct points, $Q \neq 0$. If $Q(\lambda, \mu) \neq 0$, then in the coordinates x, z, t of $(\ell_{\lambda,\mu} = 0)$, with weights $(3,2,2)$, the weighted

leading part of $f_*|_{(\ell_{\lambda,\mu}=0)}$ is $x^2+c_3(z,t)$, where the binary cubic form $c_3 := g_{*,3}|_{(\ell_{\lambda,\mu}=0)}$ has three distinct linear factors, and it becomes $x^2+z^2t+t^3$ after a linear change of z,t. Terms of degree $e\geq 4$ have weight at least $2e\geq 8>6$ and are contained in $(x)+(z,t)^3$, which is contained in the Jacobian ideal $(x,\partial_z c_3,\partial_t c_3)$ of the leading part $x^2+c_3(z,t)$ since c_3 has three distinct linear factors, so they are removed by coordinate changes of higher weight [Kol98, Construction 2.6]. Thus $(\ell_{\lambda,\mu}=0)$ cuts out a D_4 section of W_*.

Case (D). We first show that $(\gamma_*,\psi_*)\neq(0,0)$. Assume the contrary. Then for any d, all coefficients of the d-jet of $\gamma_i'(t)y+\psi_i'(z,t)$ vanish for all but finitely many i by Lemma 2.8. Since $f_i\equiv f_i' \pmod{\mathfrak{n}^{L_i}}$ and $L_i\to+\infty$, $\lim_{i\to+\infty}\operatorname{mult}_0 R_i=+\infty$. By **Step 1**, (D.81) gives G-equivariant isomorphisms

$$\widehat{\mathcal{O}}_{W_i,o_i}\cong\mathbb{C}[[x,y,z,t]]/(x^2+y^2z+R_i),$$

so Theorem D.6 gives $\lim_{i\to+\infty}(q_i(\tau_i)-q_i(1))=0$, which contradicts $q_i(\tau_i)-q_i(1)\geq a-(a-\delta)=\delta$.

Let $N_\gamma := \operatorname{mult}_0\gamma_*\geq 3$ and $N_\psi := \operatorname{mult}_0\psi_*\geq 4$, allowing one of them to be $+\infty$, and let

$$n := \min(2N_\gamma, N_\psi+1)\geq 5.$$

Let $\gamma_{*,n/2}$ be the coefficient of $t^{n/2}$ in γ_* ($\gamma_{*,n/2}:=0$ if n is odd), let $\psi_{*,n-1}(z,t)$ be the degree-$(n-1)$ part of ψ_*, and let

$$c(\rho) := \psi_{*,n-1}(1,\rho)-\tfrac14\gamma_{*,n/2}^2\rho^n\in\Bbbk[\rho].$$

If $2N_\gamma=n$, then

$$\gamma_{*,n/2}\neq 0\quad\text{and}\quad\deg\psi_{*,n-1}(1,\rho)\leq n-1,$$

so $\deg c=n$. Otherwise $n=N_\psi+1<2N_\gamma$, so $\gamma_{*,n/2}=0$ and $c(\rho)=\psi_{*,N_\psi}(1,\rho)\neq 0$ since $\psi_{*,N_\psi}\neq 0$. Thus $c\neq 0$. Let $Q(\lambda,\mu):=\mu^{n+1}c(-\lambda/\mu)$, a non-zero polynomial.

Let $\lambda,\mu\in\Bbbk$ with $Q(\lambda,\mu)\neq 0$. Then $\mu\neq 0$ since μ divides Q, and we let $\rho:=-\lambda/\mu$. On $(\ell_{\lambda,\mu}=0)$ we use the coordinates x,y,z, so $t=\rho z-\mu^{-1}y$, and the weights $(n-1,n-2,2)$, for which x^2 and zy^2 have weight $2n-2$. A monomial y^kz^{e-k} of degree e has weight $2e+k(n-4)$. The terms of $\psi_*(z,t)$ have degree $e\geq n-1$, and weight $2n-2$ occurs only for $e=n-1$ and $k=0$, with sum $\psi_{*,n-1}(1,\rho)z^{n-1}$. The terms of $\gamma_*(t)y$ coming from t^e, $e\geq N_\gamma\geq\frac{n}{2}$, have weight at least $n-2+2e\geq 2n-2$, with equality only for $e=\frac{n}{2}$ and the term $\gamma_{*,n/2}\rho^{n/2}yz^{n/2}$. Thus the weighted leading part of $f_*|_{(\ell_{\lambda,\mu}=0)}$ is

$$x^2+zy^2+\gamma_{*,n/2}\rho^{n/2}yz^{n/2}+\psi_{*,n-1}(1,\rho)z^{n-1}=x^2+z\Big(y+\tfrac12\gamma_{*,n/2}\rho^{n/2}z^{n/2-1}\Big)^2+c(\rho)z^{n-1},$$

where the terms with $\gamma_{*,n/2}$ are absent if n is odd. Since $c(\rho)\neq 0$, after the change of y of weight $n-2$ and rescaling, the weighted leading part is the D_n polynomial $x^2+zy^2+z^{n-1}$. Terms of higher weight, which are contained in the Jacobian ideal $(x,yz,y^2+(n-1)z^{n-2})$ of $x^2+zy^2+z^{n-1}$, are removed by coordinate changes of higher weight [Kol98, Construction 2.6]. Thus $(\ell_{\lambda,\mu}=0)$ cuts out a D_n section of W_*.

Members. In case (D), the coefficients of γ_* in degrees less than $\frac{n}{2}$ and of ψ_* in degrees less than $n-1$ vanish, and for $\lambda,\mu\in\mathbb{C}$ with $Q(\lambda,\mu)\neq 0$, $Q(\lambda,\mu)$ is a polynomial in the coefficients of the n-jet of f_* which does not vanish there. By Lemma 2.8, for all but finitely many i, the coefficients of γ_i' in degrees less than $\frac{n}{2}$ and of ψ_i' in degrees less than $n-1$ vanish, and $Q(\lambda,\mu)$, as a polynomial in the coefficients of the n-jet, does not vanish at the n-jet of f_i'. The computation of the weighted leading part above, applied to f_i', then shows that $(\ell_{\lambda,\mu}=0)$ cuts out a D_n section of W_i'. In case (R) the argument is identical, with the discriminant Q. Terms of degree at least n have weight at least $2n>2n-2$, so the type of the section depends only on the n-jet of the equation. Since $f_i\equiv f_i' \pmod{\mathfrak{n}^{L_i}}$ and $L_i\geq n$ for $i\gg 0$, $(\ell_{\lambda,\mu}=0)$ also cuts out a D_n section of W_i.

Normality and singularities. Fix λ,μ with $Q(\lambda,\mu)\neq 0$ and let $S:=W_*\cap(\ell_{\lambda,\mu}=0)$. Since S has an isolated singularity and S is singular at any singular point of W_* on S, $\operatorname{Sing}(W_*)\cap S$

is contained in the closed point, so $\dim \operatorname{Sing}(W_*) \le 1$. Since the hypersurface W_* is Cohen–Macaulay of dimension 3 and regular in codimension 1, it is normal by Serre's criterion. In particular, W_* is integral. Since S is Du Val, it is klt, so (W_*, S) is plt by Lemma D.9(2) (with $X = W_*$ and $B = 0$), and W_* is klt. Since W_* is Gorenstein, it is canonical, and since S is a Du Val section, W_* is cDV.

Finally, by Lemma 4.3(2) and **Step 1**(2), $a(F', W_i', 0)/e_{F'} \ge \delta/(1-\tau_i) \to +\infty$ for any computing lift F' of b_i over W_i', so Theorem 13.3(3) for the (f_i', h_i') gives $\dim \operatorname{Sing}(W_*) \ge 1$.

Step 3. In this step we show Construction-Theorem 13.7(4) and the description of the minimal lc center in (5).

Let $D := \frac{1}{m}\operatorname{div}(h_*|_{W_*})$ and $\mathcal{X}_* := \operatorname{Spec} \Bbbk[[x,y,z,t]]$. We show that

$$\operatorname{mld}(\mathcal{X}_* \ni 0, \Delta_*) \ge a, \qquad \text{where} \qquad \Delta_* := \operatorname{div}(f_*) + \tfrac{1}{m}\operatorname{div}(h_*) \text{ on } \mathcal{X}_*. \tag{D.83}$$

Let $\Bbbk_0 \subset \Bbbk$ be the union of the function fields of the Z_d, embedded through the d-jets of (f_*, h_*), so $f_*, h_* \in \Bbbk_0[[x,y,z,t]]$, and let $\mathcal{X}_{\Bbbk_0} := \operatorname{Spec} \Bbbk_0[[x,y,z,t]]$. Let $Z_d^\circ \subset Z_d$ be a dense open subset over which the ideal generated by the universal d-jets and $\mathfrak{n}^d$ is flat. The Z_d° with the jet projections form a family of approximations of the sequences of ideals (f_i') and (h_i') on $\mathbb{A}^4$ [Kaw21, Definition 3.1], with generic limit $((f_*), (h_*))$ [Kaw21, Definition 3.2]. Let P be a prime divisor over $\mathcal{X}_{\Bbbk_0}$ with center the closed point. By [Kaw21, Remark 3.8(i)], for infinitely many i there exists a prime divisor P_i over $\mathbb{A}^4$ with center 0 such that

$$a(P_i, \mathbb{A}^4, 0) = a(P, \mathcal{X}_{\Bbbk_0}, 0), \qquad \operatorname{mult}_{P_i} f_i' = \operatorname{mult}_P f_*, \quad \text{and} \quad \operatorname{mult}_{P_i} h_i' = \operatorname{mult}_P h_*.$$

Fix such i. By **Step 1**(1) and Definition 2.2, for any prime divisor F over W_i' with center 0,

$$a\Big(F, W_i', \frac{\tau_i}{m}\operatorname{div}(h_i')\Big) \ge e_F q_i(\tau_i) \ge a,$$

so $\operatorname{mld}(W_i' \ni 0, \frac{\tau_i}{m}\operatorname{div}(h_i')) \ge a$. By Theorem 2.10 for the hypersurface $W_i' = (f_i' = 0) \subset \mathbb{A}^4$ and the trivial subgroup,

$$a(P_i, \mathbb{A}^4, \operatorname{div}(f_i') + \tfrac{\tau_i}{m}\operatorname{div}(h_i')) \ge \operatorname{mld}(W_i' \ni 0, \tfrac{\tau_i}{m}\operatorname{div}(h_i')) \ge a.$$

Thus

$$a(P, \mathcal{X}_{\Bbbk_0}, 0) - \operatorname{mult}_P f_* - \frac{\tau_i}{m}\operatorname{mult}_P h_* \ge a$$

for infinitely many i, and letting $\tau_i \to 1$,

$$a(P, \mathcal{X}_{\Bbbk_0}, 0) - \operatorname{mult}_P f_* - \frac{1}{m}\operatorname{mult}_P h_* \ge a.$$

Hence

$$\operatorname{mld}\big(\mathcal{X}_{\Bbbk_0} \ni 0, \operatorname{div}(f_*) + \tfrac{1}{m}\operatorname{div}(h_*)\big) \ge a,$$

and (D.83) follows since the minimal log discrepancy does not change under the extension $\Bbbk_0 \subset \Bbbk$ [Kaw21, Remark 2.6].

Lc. Let

$$\Delta_i := \operatorname{div}(f_i') + \tfrac{1}{m}\operatorname{div}(h_i') \quad \text{on } \mathbb{A}^4.$$

For any i, $(W_i', \frac{1}{m}\operatorname{div}(h_i'))$ is klt near 0, so $(\mathbb{A}^4, \Delta_i)$ is lc near 0 by [EMY03, Corollary 1.8]. By Theorem 2.18(1), with $\mathfrak{c} = \mathcal{O}$, $C_i = \Delta_i$ and $\theta = 1$, possibly passing to a subsequence, the pair $(\mathcal{X}_*, \Delta_*)$ is lc. By Lemma D.9(1), with $X = \mathcal{X}_*$, $S = W_*$ (normal by **Step 2**) and $B = \frac{1}{m}\operatorname{div}(h_*)$, and by (D.83),

$$(W_*, D) \text{ is lc} \qquad \text{and} \qquad \operatorname{mld}(W_* \ni 0, D) \ge \operatorname{mld}(\mathcal{X}_* \ni 0, \Delta_*) \ge a.$$

Not klt. Assume that (W_*, D) is klt. Computing on one log resolution, there exists a rational number $\epsilon > 0$ such that $(W_*, (1+\epsilon)D)$ is klt. By Lemma D.9(2), with $X = \mathcal{X}_*$, $S = W_*$ and $B = \frac{1+\epsilon}{m}\operatorname{div}(h_*)$, the pair

$$\Big(\mathcal{X}_*, \Delta_* + \frac{\epsilon}{m}\operatorname{div}(h_*)\Big)$$

is plt, in particular lc. By Theorem 2.18(1), with $\mathfrak{c} = \mathcal{O}$, $C_i = \Delta_i + \frac{\epsilon}{m}\operatorname{div}(h_i')$ and $\theta = 1$, and by Lemma 2.6(2), possibly passing to a subsequence, $(\mathbb{A}^4, \Delta_i + \frac{\epsilon}{m}\operatorname{div}(h_i'))$ is lc near 0 for any i. By [EMY03, Corollary 1.8], $(W_i', \frac{1+\epsilon}{m}\operatorname{div}(h_i'))$ is lc near 0, so $a(F_i', W_i', 0) - \frac{1+\epsilon}{m}\operatorname{mult}_{F_i'} h_i' \geq 0$. By **Step 1**(2),

$$0 \leq a(F_i, W_i, 0) - \frac{1+\epsilon}{m}\operatorname{mult}_{F_i} h_i = e_{F_i} b_i - \frac{\epsilon}{m}\operatorname{mult}_{F_i} h_i, \qquad \text{so} \qquad \frac{\epsilon \operatorname{mult}_{F_i} h_i}{m e_{F_i}} \leq b_i \leq a - \delta,$$

which contradicts Lemma 4.3(2) for $i \gg 0$. Thus (W_*, D) is not klt.

The minimal lc center. By Lemma 13.5, the minimal lc center C of (W_*, D) is either a component of D with coefficient one, in which case (W_*, D) is plt, or a regular curve with $C \cong \operatorname{Spec} \Bbbk[[s]]$.

If C is a curve, let $s \to (x(s), y(s), z(s), t(s))$ parametrize C, and let $v_C := \min(\operatorname{mult}_0 y(s), \operatorname{mult}_0 z(s), \operatorname{mult}_0 t(s)$ 1. Since $x(s)^2 = -g_*(y(s), z(s), t(s))$ and $g_* \in (y, z, t)^3$, $2\operatorname{mult}_0 x(s) \geq 3v_C$, so $\operatorname{mult}_0 x(s) > v_C$. Since C is regular, the minimum of the four orders is 1, so $v_C = 1$ and the tangent vector $\mathbf{v} := (y'(0), z'(0), t'(0))$ is non-zero. A linear form ℓ in y, z, t satisfies that $\ell|_C$ is a regular parameter of C if and only if $\ell(\mathbf{v}) \neq 0$. We let

$$Q'(\lambda, \mu) := Q(\lambda, \mu) \cdot (y'(0) + \lambda z'(0) + \mu t'(0)),$$

which is non-zero since $\mathbf{v} \neq 0$. If C is a component of D, we let $Q' := Q$.

Step 4. In this step we choose ℓ_1, ℓ_2, ℓ_3 and the form ℓ_j of bounded order.

For $\lambda_1, \mu_1, \lambda_2, \mu_2, \lambda_3, \mu_3 \in \mathbb{C}$, the forms ℓ_{λ_k, μ_k}, $k \in \{1, 2, 3\}$, are linearly independent if and only if

$$\det \begin{pmatrix} 1 & \lambda_1 & \mu_1 \\ 1 & \lambda_2 & \mu_2 \\ 1 & \lambda_3 & \mu_3 \end{pmatrix} \neq 0.$$

The product of the determinant and $\prod_{k=1}^3 Q'(\lambda_k, \mu_k)$ is a non-zero polynomial with coefficients in $\Bbbk$. Since $\mathbb{C}$ is infinite, a non-zero polynomial over $\Bbbk$ does not vanish identically on $\mathbb{C}^6$. We let $\ell_k := \ell_{\lambda_k, \mu_k}$ for a point of $\mathbb{C}^6$ where it does not vanish. By **Steps 2** and 3, possibly passing to a subsequence, which by Lemma 2.6(2) keeps (f_*, h_*), each ℓ_k cuts out a D_n section of W_*, of the W_i' and of the W_i, and if C is a curve, $\ell_k|_C$ is a regular parameter of C for any k.

Since $x^2 = -g_i(y, z, t)$ with $g_i \in (y, z, t)^3$ and ℓ_1, ℓ_2, ℓ_3 span the linear forms in y, z, t, $\min_{1 \leq k \leq 3} \operatorname{mult}_F \ell_k = \operatorname{mult}_F \mathfrak{m}_{o_i}$ for any prime divisor F over W_i with center o_i. For any i choose $k_i \in \{1, 2, 3\}$ with $\operatorname{mult}_{F_i} \ell_{k_i} = \operatorname{mult}_{F_i} \mathfrak{m}_{o_i}$. Some $j \in \{1, 2, 3\}$ equals k_i for infinitely many i. Passing to the subsequence of such i, which by Lemma 2.6(2) keeps (f_*, h_*), by the choice of k_i and Theorem 5.1(3) we have

$$\operatorname{mult}_{F_i} \ell_j = \operatorname{mult}_{F_i} \mathfrak{m}_{o_i} \leq \frac{3M}{\eta_m} \quad \text{for any } i.$$

Lifts. Let $N \geq n$ be an integer. For any i choose a regular function $\ell_{j,i}$ on a neighborhood of o_i in W_i whose image in

$$\widehat{\mathcal{O}}_{W_i, o_i} = \mathbb{C}[[x, y, z, t]]/(f_i)$$

is congruent to ℓ_j modulo $\mathfrak{n}^N$. Since

$$\operatorname{mult}_{F_i}(\ell_{j,i} - \ell_j) \geq N \operatorname{mult}_{F_i} \mathfrak{m}_{o_i} > \operatorname{mult}_{F_i} \ell_j,$$

we have $\operatorname{mult}_{F_i} \ell_{j,i} = \operatorname{mult}_{F_i} \ell_j$. Let $\widetilde{\ell}$ be a lift of $\ell_{j,i}$ to $\mathbb{C}[[x, y, z, t]]$ with $\widetilde{\ell} - \ell_j \in \mathfrak{n}^N$, and complete ℓ_j to linear coordinates on $\mathbb{C}^4$. The automorphism of $\mathbb{C}[[x, y, z, t]]$ sending ℓ_j to $\widetilde{\ell}$ and fixing the other coordinates is congruent to the identity modulo $\mathfrak{n}^N$, so the completion of $W_i \cap (\ell_{j,i} = 0)$ at o_i is isomorphic to $\mathbb{C}[[x, y, z, t]]/(f_i^\sharp, \ell_j)$ for some $f_i^\sharp \equiv f_i \pmod{\mathfrak{n}^N}$. Since the type of the section depends only on the n-jet of the equation (**Step 2**), $(\ell_{j,i} = 0)$ is a D_n section of W_i. □

Proof of Theorem 13.8. We divide the proof into several steps. We write $(x, y, z, t) = (x_1, \dots, x_4)$, and for an algebraically closed field $\Bbbk \supset \mathbb{C}$ we let $\mathfrak{n}_*$ be the maximal ideal of $\Bbbk[[x, y, z, t]]$. Let $\epsilon_1 := \frac{1}{m \cdot M!}$ and $\mathfrak{n} := (x, y, z, t) \subset \mathbb{C}[[x, y, z, t]]$.

Step 1. In this step we bound some computing lift on a germ $(x^2 + y^2 z + R = 0)$ with $\operatorname{mult}_0 R$ large.

Let $(W \ni o)$ be a normal threefold germ, G a finite group acting faithfully on W and fixing o, and h a mark on $(W \ni o)$ such that $(W, \frac{1}{m} \operatorname{div}(h))$ is lc near o. Let q be the radial function of $(W \ni o, G, h, m)$. If moreover W is Gorenstein and $|G| \le M$, then for any prime divisor F over W with center o, $a(F, W, 0) \in \mathbb{Z}$, $\operatorname{mult}_F h \in \mathbb{N}$, and e_F divides $|G|$, hence divides $M!$. Thus

$$\frac{a(F, W, 0) - \frac{1}{m} \operatorname{mult}_F h}{e_F} \in \epsilon_1 \mathbb{Z} \tag{D.84}$$

for any such F.

We show that there exist a positive integer N_0 and a real number $\epsilon_0 \in (0, 1)$, depending only on m and M, satisfying the following: for any $(W \ni o, G, h)$ satisfying the hypotheses of Theorem D.6 for one index, with $\operatorname{mult}_0 R \ge N_0$ for the element $R \in \mathbb{C}[[y, z, t]]$ of the presentation there, the radial function q of $(W \ni o, G, h, m)$ satisfies

$$q(s) - q(1) < \frac{\epsilon_1}{2} \quad \text{for any } s \in (1 - \epsilon_0, 1). \tag{D.85}$$

Otherwise, for any positive integer j, there exist $(W'_j \ni o'_j, G'_j, h'_j)$ satisfying the hypotheses of Theorem D.6 for one index, with $\operatorname{mult}_0 R'_j \ge j$ for the element $R'_j \in \mathbb{C}[[y, z, t]]$ of the presentation there, and a real number $\sigma_j \in (1 - 1/j, 1)$, such that the radial function q'_j of $(W'_j \ni o'_j, G'_j, h'_j, m)$ satisfies $q'_j(\sigma_j) - q'_j(1) \ge \epsilon_1/2$. Since $\lim_{j \to +\infty} \operatorname{mult}_0 R'_j = +\infty$ and $\lim_{j \to +\infty} \sigma_j = 1$, this contradicts Theorem D.6.

Fix $s_0 \in (1 - \epsilon_0, 1)$ and let

$$L_0 := 3 + \frac{\epsilon_1}{2(1 - s_0)}.$$

Let $(W \ni o, G, h)$ satisfy the hypotheses of Theorem D.6 for one index with $\operatorname{mult}_0 R \ge N_0$ and $b := q(1) > 0$. We show that some computing lift F of b satisfies

$$\frac{a(F, W, 0)}{e_F} < L_0. \tag{D.86}$$

By Definition 2.2 and (D.85), there exists a prime divisor F over W with center o such that $(a(F, W, 0) - \frac{s_0}{m} \operatorname{mult}_F h)/e_F < b + \frac{\epsilon_1}{2}$. Then

$$b \le \frac{a(F, W, 0) - \frac{1}{m} \operatorname{mult}_F h}{e_F} = \frac{a(F, W, 0) - \frac{s_0}{m} \operatorname{mult}_F h}{e_F} - (1 - s_0) \frac{\operatorname{mult}_F h}{m e_F} < b + \frac{\epsilon_1}{2}.$$

Since b and the middle term $(a(F, W, 0) - \frac{1}{m} \operatorname{mult}_F h)/e_F$ belong to $\epsilon_1 \mathbb{Z}$ by Lemma 4.2(1) and (D.84), and their difference is in $[0, \frac{\epsilon_1}{2})$, the middle term is equal to b. Thus F is a computing lift of b and $(1 - s_0) \operatorname{mult}_F h/(m e_F) < \epsilon_1/2$. Since $b \le 3$ by Lemma 4.2(3),

$$\frac{a(F, W, 0)}{e_F} = b + \frac{\operatorname{mult}_F h}{m e_F} < 3 + \frac{\epsilon_1}{2(1 - s_0)} = L_0.$$

Step 2. In this step we fix the presentations in which the limits of (1) and (2) are formed, and pass closed conditions from the members to the limits.

After passing to a subsequence, $G = G_i$ acts on $\mathbb{C}[[x, y, z, t]]$ through a linear representation independent of i, there exist G-semi-invariant $f_i \in \mathbb{C}[[x, y, z, t]]$ and G-equivariant isomorphisms

$$\widehat{\mathcal{O}}_{W_i, o_i} \cong \mathbb{C}[[x, y, z, t]]/(f_i),$$

and, for G-semi-invariant lifts $h_i \in \mathbb{C}[[x, y, z, t]]$ of the marks, which we fix, the characters χ_f of f_i and χ_h of h_i are independent of i. The equations are of the following form.

(i) In quadratic rank 1, x, y, z, t are semi-invariant, $f_i = x^2 + g_i(y, z, t)$, and the residual cubic of W_i (Definition 2.2) is the cubic part of g_i.
(ii) In quadratic rank 2, $f_i = xy + g_i(z, t)$, where z, t are semi-invariant, G preserves the span of x and y, and G may exchange the factors x and y [Liu26b, Definition 8.1], whose construction of x, y, z, t uses only that W_i is klt and G is abelian, not the mark.

Presentations as in (i) exist by the following equivariant splitting, which we use again in **Step 4**. Let G act diagonally on $\Bbbk[[x, y, z, t]]$ with semi-invariant x, y, z, t, and let $f \in \mathfrak{n}_*^2$ be semi-invariant with quadratic part x^2. Any equation of quadratic rank 1 takes the form above after a G-equivariant linear change of coordinates and multiplication by a non-zero constant, since the quadratic part is $c\ell^2$ with $c \in \Bbbk\backslash\{0\}$ and the line spanned by the linear form ℓ is G-stable. By Lemma D.10, for the φ and x' given there and $g := f(\varphi, y, z, t)$,

$$f = x'^2 + g(y, z, t). \tag{D.87}$$

For any positive integer d, the coefficients of the d-jets of φ, $x'/(x - \varphi)$, and g are polynomials in the coefficients of the $(d + 2)$-jet of f, and before the normalization of the quadratic part they are regular functions of the coefficients of a finite jet of f on each of the open subsets U_u, one for each coordinate u, of jets whose quadratic part f_2 has rank 1 and $\partial_u^2 f_2 \neq 0$ (take $\frac{1}{2}\partial_u f_2$ as the coordinate in place of u and multiply by $\frac{1}{2}\partial_u^2 f_2$). The U_u cover the jets whose quadratic part has rank 1. Hence forming the splitting before or after the compatible generic limit leads to limit hypersurfaces which differ by a G-equivariant coordinate change over $\Bbbk$ and a unit, and the normality and the irreducibility of the limit do not depend on the presentation.

Let $\mathcal{I}$ be an infinite set of indices and (f_*, h_*) a compatible generic limit of the (f_i, h_i) along $\mathcal{I}$ with coefficient field $\Bbbk$, formed in the presentations above, let $W_* := (f_* = 0) \subset \mathcal{X}_*$, and for any positive integer d let $\mathcal{J}_d$ be as in Definition 2.5. By Lemma 2.8, for any closed subset $\Sigma \subset \mathcal{J}_d$ containing the d-jets of (f_i, h_i) for all but finitely many $i \in \mathcal{I}$, the d-jet of (f_*, h_*) belongs to Σ. Hence

$$f_* = x^2 + g_*(y, z, t) \quad \text{with} \quad \operatorname{mult}_0 g_* \geq 3 \quad \text{in quadratic rank 1,}$$
$$f_* = xy + g_*(z, t) \quad \text{in quadratic rank 2.}$$

In quadratic rank 1, the set Σ_3 of cubic forms in y, z, t which are zero or the cube of a linear form is closed, being the affine cone over the image of the third Veronese embedding of $\mathbb{P}^2$. If the cubic part of g_* belonged to Σ_3, then, since the 4-jets (p_1, p_2) such that the cubic part of $p_1(0, y, z, t)$ belongs to Σ_3 form a closed subset of $\mathcal{J}_4$, Lemma 2.8 would imply that the cubic part of g_i belongs to Σ_3 for all but finitely many $i \in \mathcal{I}$, contradicting the hypothesis on the residual cubics. Thus, in quadratic rank 1,

$$\operatorname{mult}_0 g_* = 3, \quad \text{and the cubic part of } g_* \text{ is not the cube of a linear form.} \tag{D.88}$$

Step 3. In this step, in quadratic rank 1, we assume that W_* is not normal, and show that, possibly permuting y, z, t simultaneously in the presentations of the f_i and in f_*, there exists a G-equivariant automorphism α of $\Bbbk[[x, y, z, t]]$ such that $\alpha(x) = x$ and

$$f_* = \alpha(x^2 + y^2 z). \tag{D.89}$$

Suppose that g_* is squarefree in $\Bbbk[[y, z, t]]$. By the Jacobian criterion [Mat89, Theorem 30.4(ii)],

$$\operatorname{Sing}(W_*) \subset V(x, g_*, \partial_y g_*, \partial_z g_*, \partial_t g_*),$$

and the right-hand side is the singular locus of the reduced surface $(g_* = 0) \subset \operatorname{Spec} \Bbbk[[y, z, t]]$ [Mat89, Theorem 30.8 and the paragraph following it], so $\dim \operatorname{Sing}(W_*) \leq 1$. Thus the threefold W_* is R_1, and W_* is S_2 as a hypersurface. By Serre's criterion, W_* is normal, a contradiction.

Therefore g_* has a prime factor p such that $g_* = p^2 v$ with $v \in \Bbbk[[y, z, t]]$. By (D.88), $\operatorname{mult}_0 p = \operatorname{mult}_0 v = 1$. Let p_1 and v_1 be the linear parts of p and v. Then p_1 and v_1 are linearly independent, since otherwise the cubic part $p_1^2 v_1$ of g_* is the cube of a linear form. For any $\gamma \in G$, $\gamma(p)$ is a prime element and $\gamma(p)^2$ divides $\gamma(g_*) = \chi_f(\gamma) g_*$. If $\gamma(p) \notin \Bbbk[[y, z, t]]^* \cdot p$, then

$\gamma(p)^2$ divides v, contradicting $\mathrm{mult}_0 v = 1$. Thus $\gamma(p) = u_\gamma p$ for a unit u_γ, so $\gamma(p_1) = u_\gamma(0)p_1$, and p_1 is semi-invariant. Let χ_1 be the character of p_1 and

$$y' := \frac{1}{|G|}\sum_{\gamma\in G}\chi_1(\gamma)^{-1}\gamma(p).$$

Then y' is semi-invariant of character χ_1, $y' \in (p)$, and the linear part of y' is p_1. Thus $y' = pw$ for a unit w with $w(0) = 1$. Let

$$z' := \frac{g_*}{y'^2} = \frac{v}{w^2},$$

which is semi-invariant of character $\chi_f\chi_1^{-2}$ with linear part v_1. Let $t' \in \{y, z, t\}$ be a coordinate such that p_1, v_1, t' are linearly independent. Since p_1, v_1, t' are semi-invariant linear forms forming a basis of the span of y, z, t, on which G acts diagonally, the characters $\chi_1, \chi_f\chi_1^{-2}, \chi_{t'}$ are those of y, z, t up to order. Possibly permuting y, z, t simultaneously in the presentations of the f_i and in f_*, we may assume that

$$\chi_1 = \chi_y, \quad \chi_f\chi_1^{-2} = \chi_z, \quad \text{and} \quad \chi_{t'} = \chi_t.$$

Let α be the $\Bbbk$-algebra endomorphism of $\Bbbk[[x, y, z, t]]$ with $\alpha(x) = x$, $\alpha(y) = y'$, $\alpha(z) = z'$, and $\alpha(t) = t'$. Since the linear parts x, p_1, v_1, t' form a basis, α is an automorphism. Since $\alpha(x), \alpha(y), \alpha(z), \alpha(t)$ are semi-invariant of characters $\chi_x, \chi_y, \chi_z, \chi_t$, α is G-equivariant. Finally $\alpha(x^2 + y^2z) = x^2 + y'^2z' = x^2 + g_* = f_*$.

Step 4. In this step we prove (1) in quadratic rank 1. Assume that W_* is not normal, and let α be as in (D.89).

Fix an integer $d \geq \max\{3, N_0\}$. Let Λ be the set of pairs (u, μ) with $u \in \{x, y, z, t\}$ and μ a monomial in x, y, z, t with $1 \leq \deg\mu \leq d-1$ and $\chi_\mu = \chi_u$. For $\lambda = (\lambda_{u,\mu}) \in \Bbbk^\Lambda$, let α_λ be the endomorphism of $\Bbbk[[x, y, z, t]]$ with

$$\alpha_\lambda(u) = \sum_{(u,\mu)\in\Lambda}\lambda_{u,\mu}\mu \quad \text{for any } u \in \{x, y, z, t\},$$

and let $\Psi(\lambda)$ be the determinant of the linear part of α_λ. Then α_λ is G-equivariant, α_λ is an automorphism if $\Psi(\lambda) \neq 0$, and the coefficients of the d-jet of $\alpha_\lambda(x^2 + y^2z)$ are polynomials in λ with integer coefficients. Let $\lambda_* \in \Bbbk^\Lambda$ be the coefficients of the d-jets of $\alpha(x), \alpha(y), \alpha(z), \alpha(t)$. Since the d-jet of $\alpha(x^2 + y^2z)$ only depends on the d-jets of $\alpha(x), \alpha(y), \alpha(z), \alpha(t)$, (D.89) implies

$$\mathrm{jet}_d(\alpha_{\lambda_*}(x^2 + y^2z)) = \mathrm{jet}_d(f_*) \quad \text{and} \quad \Psi(\lambda_*) \neq 0.$$

Let $\Phi_1, \dots, \Phi_r$ be the coefficients of $\mathrm{jet}_d(\alpha_\lambda(x^2 + y^2z))$ minus the corresponding coordinates of the f-component of a point of $\mathcal{J}_d$. The Φ_k are polynomials with complex coefficients in λ and in the coordinates of $\mathcal{J}_d$. By Lemma 2.7 with these Φ_k, with Ψ, and with $\#\Lambda$ further variables, for all but finitely many $i \in \mathcal{I}$ there exists $\lambda_i \in \mathbb{C}^\Lambda$ such that

$$\alpha_{\lambda_i}(x^2 + y^2z) - f_i \in \mathfrak{n}^d \quad \text{and} \quad \Psi(\lambda_i) \neq 0.$$

For such i, α_{λ_i} is a G-equivariant automorphism of $\mathbb{C}[[x, y, z, t]]$ and $\alpha_{\lambda_i}^{-1}(\mathfrak{n}^d) = \mathfrak{n}^d$, so

$$\alpha_{\lambda_i}^{-1}(f_i) = x^2 + y^2z + S_i$$

with $S_i \in \mathfrak{n}^d$ semi-invariant of character χ_f. Since $d \geq 3$, the quadratic part of $\alpha_{\lambda_i}^{-1}(f_i)$ is x^2. Let φ_i be the element φ of Lemma D.10 for $\alpha_{\lambda_i}^{-1}(f_i)$. Then $2\varphi_i = -(\partial_x S_i)(\varphi_i, y, z, t)$ with $\partial_x S_i \in \mathfrak{n}^{d-1}$, so $\varphi_i \in (y, z, t)^{d-1}$. By Lemma D.10,

$$\alpha_{\lambda_i}^{-1}(f_i) = x'^2 + y^2z + R_i, \quad R_i := \varphi_i^2 + S_i(\varphi_i, y, z, t) \in (y, z, t)^d,$$

since $\varphi_i^2 \in (y, z, t)^{2d-2}$ and $2d - 2 \geq d$, where x' is semi-invariant of character χ_x and (x', y, z, t) is a regular system of parameters. Thus there exists a G-equivariant isomorphism

$$\widehat{\mathcal{O}}_{W_i,o_i} \cong \mathbb{C}[[x, y, z, t]]/(x^2 + y^2z + R_i) \quad \text{with} \quad R_i \in \mathbb{C}[[y, z, t]],\ \mathrm{mult}_0 R_i \geq d \geq N_0,$$

in which x, y, z, t are semi-invariant. Thus, by Definition 4.1(1), $|G| \le M$ and the klt hypothesis, $(W_i \ni o_i, G, h_i)$ satisfies the hypotheses of Theorem D.6 for one index, with R_i as the element of the presentation there. Since $b_i > 0$, (D.86) implies that some computing lift F_i of b_i satisfies $a(F_i, W_i, 0)/e_{F_i} < L_0$ for all but finitely many $i \in \mathcal{I}$. This proves (1) in quadratic rank 1.

Step 5. In this step we prove (1) in quadratic rank 2. Assume that W_* is not normal. If $g_* \neq 0$, then by the Jacobian criterion [Mat89, Theorem 30.4(ii)]

$$\operatorname{Sing}(W_*) \subset V(x, y, g_*) \quad \text{and} \quad \dim V(x, y, g_*) = \dim \Bbbk[[z, t]]/(g_*) \le 1,$$

so the threefold W_* is R_1. As W_* is S_2 as a hypersurface, W_* is normal by Serre's criterion, a contradiction. Thus $g_* = 0$. Let $N' := N'(m, M)$ and $L(m, M)$ be the constants of Theorem 13.2. The coefficients of f at the monomials $z^{j_1} t^{j_2}$ with $j_1 + j_2 < N'$ vanish at $\operatorname{jet}_{N'}(f_*, h_*)$, so by Lemma 2.7 with $\Psi = 1$ they vanish at $\operatorname{jet}_{N'}(f_i, h_i)$ for all but finitely many $i \in \mathcal{I}$, that is,

$$\operatorname{mult}_0 g_i \ge N'.$$

For such i, Theorem 13.2 implies that $b_i = q_i(1)$ has a computing lift F_i with

$$a(F_i, W_i, 0)/e_{F_i} \le L(m, M).$$

This proves (1) in quadratic rank 2.

Step 6. In this step we prove (2). Assume that $\min_F a(F, W_i, 0)/e_F \to +\infty$ over the computing lifts F of b_i.

Fix the presentations of **Step 2** after passing to a subsequence. In quadratic rank 1, let $\kappa := \mathrm{id}_{\mathbb{C}[[x,y,z,t]]}$. In quadratic rank 2, let κ be a $\mathbb{C}$-linear automorphism of $\mathbb{C}[[x, y, z, t]]$ fixing z and t and mapping x and y to a basis of semi-invariant linear forms of the span of x and y. Then G acts diagonally in the coordinates $\kappa(x), \kappa(y), z, t$. Since κ is linear and defined over $\mathbb{C}$, rewriting power series in the coordinates $\kappa(x), \kappa(y), z, t$ induces linear isomorphisms of the jet spaces $\mathcal{J}_d$ compatible with truncation. Thus rewriting carries compatible generic limits of the (f_i, h_i) to compatible generic limits of the rewritten tuples, and does not change the germs, the pairs or the singular loci. Below, we use Theorem 13.3(3), whose hypotheses ask for a diagonal action, and Lemma 13.4, in the coordinates $\kappa(x), \kappa(y), z, t$.

By Lemma 13.4(1) for the constant sequence with all terms equal to (f_i, h_i), the pair

$$(\operatorname{Spec} \mathbb{C}[[x, y, z, t]], (f_i)^1 (h_i)^{1/m} \mathfrak{n}^{\eta_m})$$

is lc, hence so is $(\operatorname{Spec} \mathbb{C}[[x, y, z, t]], (f_i)^1 (h_i)^{1/m})$. The log discrepancy of $(\operatorname{Spec} \mathbb{C}[[x, y, z, t]], (f_i)^1 (h_i)^{1/m})$ at the exceptional divisor of the blow-up of the closed point is $4 - \operatorname{mult}_0 f_i - \frac{1}{m} \operatorname{mult}_0 h_i$, and $\operatorname{mult}_0 f_i = 2$, so

$$\operatorname{mult}_0 h_i \le 2m.$$

Thus the orders at 0 of the f_i and h_i are bounded, and by Lemma 2.6(1), there exist an infinite set $\mathcal{I}$ of indices and a compatible generic limit (f_*, h_*) of the (f_i, h_i) along $\mathcal{I}$, with coefficient field $\Bbbk$, formed in the presentations of **Step 2**. Let $W_* := (f_* = 0) \subset \mathcal{X}_*$.

By (1) and the hypothesis of (2), W_* is normal, hence integral. If W_* were not klt, then Proposition D.8(2) would imply the existence of a rational number $\epsilon > 0$ and an infinite set $\mathcal{I}'$ of indices such that any computing lift F of b_i satisfies $a(F, W_i, 0)/e_F \le 3(1 + 1/\epsilon)$ for any $i \in \mathcal{I}'$, contradicting the hypothesis of (2). Thus W_* is klt, hence canonical since it is a hypersurface, which is Gorenstein.

By Theorem 13.3(3) for the sequence indexed by $\mathcal{I}$, $\dim \operatorname{Sing}(W_*) \ge 1$, so W_* is not isolated.

Since W_* is integral, (f_*) is a prime ideal and $\operatorname{div}(f_*) = W_*$. By Lemma 13.4(2), the pair

$$(\mathcal{X}_*, (f_*)^1 (h_*)^{1/m}) = \Big(\mathcal{X}_*, W_* + \frac{1}{m} \operatorname{div}(h_*)\Big)$$

is lc with positive minimal log discrepancy at the closed point 0. If $h_* \in (f_*)$, then

$$\operatorname{mult}_{W_*} \Big(W_* + \frac{1}{m} \operatorname{div}(h_*)\Big) \ge 1 + \frac{1}{m},$$

contradicting the lc property. Thus $h_*|_{W_*} \neq 0$, and Lemma D.9(1), with $R = \Bbbk[[x,y,z,t]]$, $\ell = f_*$ and $B = \frac{1}{m}\operatorname{div}(h_*)$, implies that $(W_*, \frac{1}{m}\operatorname{div}(h_*|_{W_*}))$ is lc and

$$\operatorname{mld}\Big(W_* \ni 0, \frac{1}{m}\operatorname{div}(h_*|_{W_*})\Big) \geq \operatorname{mld}\Big(\mathcal{X}_* \ni 0, W_* + \frac{1}{m}\operatorname{div}(h_*)\Big) > 0.$$

This proves (2). □

D.2. **Quadratic rank two along the singular locus.** In this subsection we prove Theorem 14.2 of Section 14, and we state and prove Lemmas D.11 and D.15.

Lemma D.11. *Let m be a positive integer, G a finite abelian group acting linearly on $\mathbb{A}^4$ with coordinates $x_1, \dots, x_4$, S a variety of finite type over $\mathbb{C}$, and $(f_s)_{s\in S}$ the family of polynomials defined by a morphism from S to a vector space $\mathcal{P}$ of polynomials in $x_1, \dots, x_4$ with $\dim \mathcal{P} < +\infty$. Assume that any f_s is G-semi-invariant with one fixed character, and that for any point $s \in S$, the germ*

$$(W_s \ni 0) := \big((f_s = 0) \ni 0\big)$$

is a normal klt hypersurface threefold germ of quadratic rank 2 with a completed presentation $xy + g_s(z,t)$ with $\operatorname{mult}_0 g_s \geq 3$, on which G acts faithfully. Then there exists a positive real number L_1 satisfying the following. Let $s \in S$ be a point, $\widehat{W}_s$ the completion of W_s at 0, $h \in \mathbb{C}[[x_1, \dots, x_4]]$ a G-semi-invariant element with $h|_{\widehat{W}_s} \neq 0$, and

$$b := \operatorname{mld}\Big(\big(\widehat{W}_s \ni 0, \tfrac{1}{m}\operatorname{div}(h)\big)/G\Big).$$

(1) If $b \geq 0$, then b has a computing lift P with $a(P, W_s, 0) \leq L_1$.
(2) If $b < 0$, then there exists a prime divisor P over W_s with center 0, $a(P, W_s, 0) \leq L_1$, and $a(P, W_s, \frac{1}{m}\operatorname{div}(h)) < 0$.

Proof. We let $r := |G|$. Possibly changing the coordinates linearly, we may assume that G acts diagonally on $\mathbb{A}^4$ with coordinates $x_1, \dots, x_4$. We let N_0 be a positive integer with $\deg f_s \leq N_0$ for any $s \in S$. For a field $\Bbbk$, an element $v \in \Bbbk[[x_1, \dots, x_n]]$, and a positive integer N, we let $v_{<N}$ be the sum of the terms of v of degree less than N. Let $\mathfrak{n} := (x_1, \dots, x_n)$, in $\mathbb{C}[[x_1, \dots, x_n]]$ or in $\mathbb{C}[x_1, \dots, x_n]$. If a finite abelian group acts diagonally and v is semi-invariant, then $v_{<N}$ is semi-invariant with the character of v. Since $v - v_{<N}$ belongs to the N-th power of the maximal ideal,

$$\operatorname{mult}_P(v - v_{<N}) \geq N \tag{D.90}$$

for any prime divisor P with center the closed point. A prime divisor over an algebraic germ $(V \ni o)$ with center o is regarded as a formal prime divisor over the completion by Lemma 2.12(3), so that $\operatorname{mult}_P u$ is defined for formal functions u. We use the following four facts.

(a) Let H be a finite abelian group acting faithfully on a normal Gorenstein germ $(V \ni o)$ and fixing o, $\pi : V \to Q := V/H$ the quotient, $x_Q := \pi(o)$, and $B_Q \geq 0$ the $\mathbb{Q}$-divisor with $K_V = \pi^*(K_Q + B_Q)$ (Lemma 2.15(0)). Then $|H|(K_Q + B_Q)$ is Cartier near x_Q, since H acts on a generator of ω_V near o by a character whose order divides $|H|$, and by Lemma 2.15(2),

$$a(P, V, \pi^*\Delta) = e_P a(E, Q, B_Q + \Delta) \tag{D.91}$$

for any prime divisor P over V with image E over Q and any $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor $\Delta \geq 0$ or $\mathbb{R}$-ideal Δ on Q near x_Q. Thus

$$\operatorname{mld}((V \ni o, \pi^*\Delta)/H) = \operatorname{mld}(Q \ni x_Q, B_Q + \Delta),$$

and P is a computing lift of $\operatorname{mld}((V \ni o, \pi^*\Delta)/H)$ if and only if E computes $\operatorname{mld}(Q \ni x_Q, B_Q + \Delta)$. For a semi-invariant function u and a positive integer m', the function $u^{|H|}$ is invariant and

$$\frac{1}{m'}\operatorname{div}(u) = \pi^*\Big(\frac{1}{m'|H|}\operatorname{div}\big(u^{|H|}\big)\Big).$$

(b) Let (V,Δ) be a pair, where Δ is either an $\mathbb{R}$-divisor with $\Delta\geq 0$ or an $\mathbb{R}$-ideal, over a variety or over the spectrum of a complete local ring (log resolutions exist by [dFEM11, Theorems 2.3–2.4]), and $o\in V$ a closed point. Then (V,Δ) is lc near o if and only if $\mathrm{mld}(V\ni o,\Delta)\geq 0$, in which case the minimal log discrepancy is attained by a prime divisor on a log resolution principalizing the maximal ideal of o, and otherwise $\mathrm{mld}(V\ni o,\Delta)=-\infty$.

(c) Let $s\in S$ be a point. Since W_s has multiplicity 2 at 0 and the tangent cone $(xy=0)$ of the completed presentation is reduced, the blow-up of W_s at 0 has an exceptional prime divisor E_s^+ with $a(E_s^+,W_s,0)=2$, and $\mathrm{mult}_{E_s^+}v\geq N$ for any $v\in(f_s)+\mathfrak{n}^N$. Thus if $h\in(f_s)+\mathfrak{n}^{2m+1}$, then

$$a\big(E_s^+,W_s,\tfrac{1}{m}\operatorname{div}(h)\big)\leq 2-\frac{2m+1}{m}<0. \tag{D.92}$$

(d) Let T_0 be an irreducible variety and $\tau_i\in T_0$ points such that any non-empty open subset of T_0 contains τ_i for all but finitely many i. Finitely many data over the geometric generic point of T_0 (varieties, morphisms, divisors, log resolutions, regular functions) are the base change of data over a variety T' with a generically finite dominant morphism $T'\to T_0$, and the data over T' extend to families over a dense open subset $T'^{\circ}\subset T'$. The image of T'° contains a dense open subset of T_0, hence τ_i for all but finitely many i. For such i, we fix a point $\tau_i'\in T'^{\circ}$ over τ_i and use the fibers over τ_i'. Spreading out a log resolution, we may assume that the fibers over T'° are log resolutions whose prime divisors have the log discrepancies of the corresponding prime divisors on the log resolution of the geometric generic fiber. In particular, normality and the klt, terminal, and lc properties of the geometric generic fiber hold for the fibers over T'°, and conversely.

We divide the proof into five steps.

Step 1. In this step we compare a formal power series with its polynomial truncations on one fixed germ. Let $n\in\{3,4\}$, H a finite abelian group acting diagonally on $\mathbb{A}^n$ with coordinates $x_1,\dots,x_n$, $V:=\mathbb{A}^3$ if $n=3$, and $V:=(\Phi=0)\subset\mathbb{A}^4$ for an H-semi-invariant polynomial Φ with $(V\ni 0)$ normal if $n=4$, and m' a positive integer. Assume that H acts faithfully on V.

Claim D.12. *There exists a positive integer N_V depending only on V, H, and m' satisfying the following. Let $u\in\mathbb{C}[[x_1,\dots,x_n]]$ be an H-semi-invariant element with $u|_{\widehat{V}}\neq 0$, and $N\geq N_V$ an integer with $u_{<N}|_V\neq 0$. Then the following hold.*

(1) We have

$$\begin{aligned}\beta&:=\mathrm{mld}\Big(\big(\widehat{V}\ni 0,\tfrac{1}{m'}\operatorname{div}(u)\big)/H\Big)\\&=\mathrm{mld}\Big(\big(V\ni 0,\tfrac{1}{m'}\operatorname{div}(u_{<N})\big)/H\Big).\end{aligned}$$

(2) Let P be a prime divisor over V with center 0 and $m'a(P,V,0)<N$. If $\beta\geq 0$ and P is a computing lift of the right side of (1), then P is a computing lift of β. If $a(P,V,\frac{1}{m'}\operatorname{div}(u_{<N}))<0$, then $a(P,V,\frac{1}{m'}\operatorname{div}(u))<0$.

Proof. Let $r_H:=|H|$, and let $\pi:V\to Q$, x_Q, and B_Q be as in (a). Let Γ be the set of coefficients of B_Q together with $\frac{1}{m'r_H}$, and L_Q the integer of Theorem 2.22(2) for Q and Γ. Since $\mathfrak{m}_{x_Q}\widehat{\mathcal{O}}_{V,0}$ is primary to the maximal ideal, there exists N_V with

$$\mathfrak{n}^{N_V}\widehat{\mathcal{O}}_{V,0}\subset\mathfrak{m}_{x_Q}^{L_Q}\widehat{\mathcal{O}}_{V,0}.$$

Since $\widehat{\mathcal{O}}_{Q,x_Q}=\widehat{\mathcal{O}}_{V,0}^H$ (Lemma 2.12(1)) and the Reynolds operator of H is $\widehat{\mathcal{O}}_{Q,x_Q}$-linear, any H-invariant element of $\mathfrak{n}^{N_V}\widehat{\mathcal{O}}_{V,0}$ belongs to $\mathfrak{m}_{x_Q}^{L_Q}\widehat{\mathcal{O}}_{Q,x_Q}$.

Let $w:=u^{r_H}$ and $w_N:=(u_{<N})^{r_H}$, both H-invariant. For $N\geq N_V$ we have $w-w_N\in\mathfrak{n}^N$, so $w-w_N\in\mathfrak{m}_{x_Q}^{L_Q}\widehat{\mathcal{O}}_{Q,x_Q}$. For two integers $N,N'\geq N_V$ with $u_{<N}|_V\neq 0$ and $u_{<N'}|_V\neq 0$, the regular function $w_N-w_{N'}$ belongs to

$$\mathfrak{m}_{x_Q}^{L_Q}\widehat{\mathcal{O}}_{Q,x_Q}\cap\mathcal{O}_{Q,x_Q}=\mathfrak{m}_{x_Q}^{L_Q}$$

by faithful flatness. Thus the ideal

$$\mathfrak{a} := (w_N) + \mathfrak{m}_{x_Q}^{L_Q} \subset \mathcal{O}_{Q,x_Q}$$

does not depend on N, and $w \in \mathfrak{a}\widehat{\mathcal{O}}_{Q,x_Q}$. By Theorem 2.22(2) and (D.91),

$$\begin{aligned}\operatorname{mld}\Big(\big(V \ni 0, \tfrac{1}{m'}\operatorname{div}(u_{<N})\big)/H\Big) &= \operatorname{mld}\big(Q \ni x_Q, B_Q, (w_N)^{\frac{1}{m'r_H}}\big)\\ &= \operatorname{mld}\big(Q \ni x_Q, B_Q, \mathfrak{a}^{\frac{1}{m'r_H}}\big)\end{aligned}$$

for any $N \geq N_V$ with $u_{<N}|_V \neq 0$. We let β' be the right side, which does not depend on N.

We show $\beta = \beta'$. By Definition 2.2 and Lemma 2.12(3), β is the infimum of $a(P, V, \frac{1}{m'}\operatorname{div}(u))/e_P$ over the prime divisors P over V with center 0. Let P be a prime divisor over V with center 0, and $N' > \max\{N_V, \operatorname{mult}_P u\}$ an integer. By (D.90), $\operatorname{mult}_P u_{<N'} = \operatorname{mult}_P u$. In particular, $u_{<N'}|_V \neq 0$. Thus if $\beta' \geq 0$, then

$$a\big(P, V, \tfrac{1}{m'}\operatorname{div}(u)\big) = a\big(P, V, \tfrac{1}{m'}\operatorname{div}(u_{<N'})\big) \geq e_P\beta',$$

so $\beta \geq \beta'$. Conversely, let E be a prime divisor over Q with center x_Q which computes β' if $\beta' \geq 0$, and has negative log discrepancy with respect to $(Q, B_Q, \mathfrak{a}^{1/(m'r_H)})$ otherwise, and let P be a prime divisor over V over E. Since $w \in \mathfrak{a}\widehat{\mathcal{O}}_{Q,x_Q}$, we have $\operatorname{mult}_E w \geq \operatorname{mult}_E \mathfrak{a}$, so by (D.91)

$$\frac{a(P, V, \frac{1}{m'}\operatorname{div}(u))}{e_P} = a(E, Q, B_Q) - \frac{\operatorname{mult}_E w}{m'r_H} \leq a\big(E, Q, B_Q, \mathfrak{a}^{\frac{1}{m'r_H}}\big).$$

Thus $\beta \leq \beta'$ if $\beta' \geq 0$, and $\beta < 0$ otherwise, in which case $\beta = \beta' = -\infty$ by (b) and $e_P \leq r_H$ for any P. This proves (1).

For (2), if $\beta \geq 0$ and P is a computing lift of β for $u_{<N}$, then

$$\operatorname{mult}_P u_{<N} = m'\big(a(P, V, 0) - e_P\beta\big) \leq m'a(P, V, 0) < N,$$

so $\operatorname{mult}_P u = \operatorname{mult}_P u_{<N}$ by (D.90). Thus $a(P, V, \frac{1}{m'}\operatorname{div}(u)) = e_P\beta$, and P is a computing lift of β by (1). If $a(P, V, \frac{1}{m'}\operatorname{div}(u_{<N})) < 0$, then either $\operatorname{mult}_P u_{<N} < N$ and $\operatorname{mult}_P u = \operatorname{mult}_P u_{<N}$, or $\operatorname{mult}_P u \geq N > m'a(P, V, 0)$ by (D.90). In both cases $a(P, V, \frac{1}{m'}\operatorname{div}(u)) < 0$. □

Step 2. In this step we reduce the lemma to polynomial marks and fix a sequence. Assume that there exists a positive real number L such that (1) and (2) of the lemma hold with L in place of L_1 for any point $s \in S$ and any G-semi-invariant polynomial h with $h|_{W_s} \neq 0$. For such h, by Lemma 2.12(2),

$$\operatorname{mld}\Big(\big(W_s \ni 0, \tfrac{1}{m}\operatorname{div}(h)\big)/G\Big) = \operatorname{mld}\Big(\big(\widehat{W_s} \ni 0, \tfrac{1}{m}\operatorname{div}(h)\big)/G\Big).$$

We show that $L_1 := L$ satisfies the lemma. Let s and h be as in the lemma, and N_{W_s} the integer of Claim D.12 for $V = W_s$, $H = G$, and $m' = m$. Since

$$h \notin (f_s)\mathbb{C}[[x_1, \dots, x_4]] = \bigcap_N \big((f_s) + \mathfrak{n}^N\big)$$

by the Krull intersection theorem, there exists an integer $N > \max\{N_{W_s}, mL\}$ with $h_{<N}|_{W_s} \neq 0$. By Claim D.12(1), b is the quotient value of $(W_s \ni 0, \frac{1}{m}\operatorname{div}(h_{<N}))/G$. By assumption there exists a prime divisor P over W_s with center 0 and $ma(P, W_s, 0) \leq mL < N$ which is a computing lift of b for $h_{<N}$ if $b \geq 0$, and has $a(P, W_s, \frac{1}{m}\operatorname{div}(h_{<N})) < 0$ otherwise. By Claim D.12(2), P satisfies (1), resp. (2), for h.

For a point s and a G-semi-invariant polynomial h with $h|_{W_s} \neq 0$, we let $b(s, h)$ be the quotient value of $(W_s \ni 0, \frac{1}{m}\operatorname{div}(h))/G$ and $a_{\min}(s, h)$ the minimum of $a(P, W_s, 0)$ over the computing lifts P of $b(s, h)$ if $b(s, h) \geq 0$, and over the prime divisors P over W_s with center 0 and $a(P, W_s, \frac{1}{m}\operatorname{div}(h)) < 0$ otherwise. The minimum exists since $a(P, W_s, 0) \in \mathbb{N}^+$ for the klt Gorenstein germ W_s, and the set is non-empty by (a) and (b). Suppose that $a_{\min}(s, h)$ is not bounded from above over the points $s \in S$ and the G-semi-invariant polynomials h with

$h|_{W_s} \neq 0$. Then for any $i \in \mathbb{N}^+$ there exist a point $s_i \in S$ and a G-semi-invariant polynomial h_i with $h_i|_{W_{s_i}} \neq 0$ and

$$a_{\min}(s_i, h_i) > i.$$

We let $b_i := b(s_i, h_i)$. Possibly passing to a subsequence, we may assume that the character of h_i does not depend on i, and that $b_i > 0$ for any i, or $b_i < 0$ for any i, or $b_i = 0$ for any i. Let T be a minimal closed subset of S containing infinitely many s_i. Then T is irreducible, and possibly passing to a subsequence, we may assume that $s_i \in T$ for any i. Thus any non-empty open subset of T contains s_i for all but finitely many i. Let $\overline{s}$ be the geometric generic point of T. Since the W_{s_i} are normal and klt near 0, (d) for T, the points s_i, and a log resolution of $W_{\overline{s}}$ near 0 shows that $W_{\overline{s}}$ is normal and klt near 0.

Step 3. In this step we assume that $b_i > 0$ for any i and derive a contradiction. For a point s of S or the geometric generic point $\overline{s}$ of T, we let $\pi_s : W_s \to X_s := W_s/G$, $o_s := \pi_s(0)$, and B_s be as in (a), and for a G-semi-invariant polynomial h we let $D_h := \frac{1}{mr}\operatorname{div}(h^r)$ on X_s, so that $\pi_s^* D_h = \frac{1}{m}\operatorname{div}(h|_{W_s})$ and $r(K_{X_s} + B_s)$ is Cartier near o_s. By (D.91), for any prime divisor P over W_s with image E over X_s,

$$\begin{aligned} a(P, W_s, 0) &= e_P a(E, X_s, B_s), \\ a\big(P, W_s, \tfrac{1}{m}\operatorname{div}(h)\big) &= e_P a(E, X_s, B_s + D_h), \end{aligned} \tag{D.93}$$

and $b(s, h) = \operatorname{mld}(X_s \ni o_s, B_s + D_h)$.

Since $W_{\overline{s}}$ is normal and klt near 0, the pair $(X_{\overline{s}}, B_{\overline{s}})$ is klt near $o_{\overline{s}}$ by (D.93). By [BCHM10, Corollary 1.4.3] (the characteristic zero statement remains true after extension of the algebraically closed base field, as one sees by descending the finitely many data to a countably generated field and embedding the algebraic closure of that field in $\mathbb{C}$), possibly shrinking $X_{\overline{s}}$ to a neighborhood of $o_{\overline{s}}$, there exists a projective birational morphism $\rho : Y \to X_{\overline{s}}$ from a $\mathbb{Q}$-factorial variety Y whose exceptional prime divisors are exactly the exceptional prime divisors E over $X_{\overline{s}}$ with $a(E, X_{\overline{s}}, B_{\overline{s}}) \leq 1$. Let $K_Y + B_Y = \rho^*(K_{X_{\overline{s}}} + B_{\overline{s}})$. Then $B_Y \geq 0$, and

$$a(E, Y, B_Y) = a(E, X_{\overline{s}}, B_{\overline{s}}) > 1$$

for any exceptional prime divisor E over Y, so Y is terminal. By (d) for T and the points s_i, with the points $s_i' \in T'^{\circ}$ over s_i, spreading out ρ gives, for all but finitely many i, a projective birational morphism $\rho_i : Y_i \to X_{s_i}$ over a neighborhood of o_{s_i} with Y_i terminal, $K_{Y_i} + B_{Y_i} = \rho_i^*(K_{X_{s_i}} + B_{s_i})$, and $B_{Y_i} \geq 0$. Let

$$m_1 := mr^2, \qquad \Delta_i := B_{Y_i} + \rho_i^* D_{h_i}.$$

Then $K_{Y_i} + \Delta_i = \rho_i^*(K_{X_{s_i}} + B_{s_i} + D_{h_i})$, and for any prime divisor E over Y_i,

$$\begin{aligned} a(E, X_{s_i}, B_{s_i} + D_{h_i}) &= a(E, Y_i, \Delta_i), \\ a(E, X_{s_i}, B_{s_i}) &= a(E, Y_i, B_{Y_i}) \leq a(E, Y_i, 0). \end{aligned} \tag{D.94}$$

Since $b_i > 0$, the pair $(X_{s_i}, B_{s_i} + D_{h_i})$ is lc near o_{s_i} by (b), so (Y_i, Δ_i) is lc near $\rho_i^{-1}(o_{s_i})$.

Fix a computing lift P_i of b_i, let E_i be the image of P_i over X_{s_i}, which computes

$$\operatorname{mld}(X_{s_i} \ni o_{s_i}, B_{s_i} + D_{h_i}) = b_i,$$

and let η_i be the generic point of $\operatorname{center}_{Y_i} E_i \subset \rho_i^{-1}(o_{s_i})$. Any prime divisor over Y_i with center η_i has center o_{s_i} on X_{s_i}. By (D.94), $b_i = \operatorname{mld}(Y_i \ni \eta_i, \Delta_i)$, and any prime divisor E computing $\operatorname{mld}(Y_i \ni \eta_i, \Delta_i)$ computes $\operatorname{mld}(X_{s_i} \ni o_{s_i}, B_{s_i} + D_{h_i})$. By (D.93) and (D.94), a prime divisor P over W_{s_i} over such E is a computing lift of b_i with

$$a(P, W_{s_i}, 0) = e_P a(E, X_{s_i}, B_{s_i}) \leq r a(E, Y_i, 0).$$

Thus it suffices to find such E with $a(E, Y_i, 0)$ bounded from above independently of i. Possibly passing to a subsequence, we may assume that $\dim \operatorname{center}_{Y_i} E_i$ does not depend on i.

If $\dim \operatorname{center}_{Y_i} E_i = 2$, then $E := E_i$ satisfies $a(E, Y_i, 0) = 1$.

Assume that $\dim \operatorname{center}_{Y_i} E_i = 1$. Since Y_i is terminal, Y_i is smooth at η_i. Since

$$r(K_{Y_i} + B_{Y_i}) = \rho_i^*\big(r(K_{X_{s_i}} + B_{s_i})\big)$$

and rK_{Y_i} are Cartier near η_i, so is rB_{Y_i}. Since $mr\rho_i^* D_{h_i} = \operatorname{div}(\rho_i^* h_i^r)$, $m_1\Delta_i$ is Cartier near η_i. Let u_i be a local equation of $m_1\Delta_i$ at η_i. Let $\kappa := \kappa(\eta_i)$, so that $\widehat{\mathcal{O}}_{Y_i,\eta_i} \cong \kappa[[w_1, w_2]]$ for a coefficient field κ, let $\overline{\kappa}$ be an algebraic closure of κ, $Y_i' := \operatorname{Spec} \overline{\kappa}[[w_1, w_2]]$ with closed point y_i', and Δ_i' the pullback of Δ_i to Y_i'. By Lemma 5.7 with m_1 in place of m for the smooth surface germ $(Y_i' \ni y_i')$, the trivial group, and the mark u_i, there exists a prime divisor E' over Y_i' computing $\operatorname{mld}(Y_i' \ni y_i', \Delta_i')$ with $a(E', Y_i', 0) \le C_{m_1}$. The finitely many blow-ups of closed points that give E' are defined over a finite extension κ' of κ, and $\kappa'[[w_1, w_2]]$ is finite étale over $\kappa[[w_1, w_2]]$. Hence E' maps to a prime divisor E over Y_i with center η_i and ramification index 1, and

$$a(E, Y_i, 0) = a(E', Y_i', 0), \qquad a(E, Y_i, \Delta_i) = a(E', Y_i', \Delta_i').$$

Conversely, the base change to Y_i' of a log resolution is a log resolution, so

$$\operatorname{mld}(Y_i \ni \eta_i, \Delta_i) \ge \operatorname{mld}(Y_i' \ni y_i', \Delta_i').$$

Thus E computes $\operatorname{mld}(Y_i \ni \eta_i, \Delta_i)$ and $a(E, Y_i, 0) \le C_{m_1}$.

Assume that $\xi_i := \operatorname{center}_{Y_i} E_i$ is a closed point. Let $(Y_{s'})_{s' \in T'^\circ}$, $(X_{s'})$, $(o_{s'})$, $(B_{s'})$, $(B_{Y_{s'}})$, and $(\rho_{s'} : Y_{s'} \to X_{s'})$ be the families over T'° obtained by spreading out Y, $X_{\overline{s}}$, $o_{\overline{s}}$, $B_{\overline{s}}$, B_Y, and ρ, with fibers Y_i, X_{s_i}, o_{s_i}, B_{s_i}, B_{Y_i}, and ρ_i over s_i'. For a geometric point $\overline{s}'$ of T'°, $Y_{\overline{s}'}$, $X_{\overline{s}'}$, $B_{\overline{s}'}$, $B_{Y_{\overline{s}'}}$, and $\rho_{\overline{s}'}$ are their base changes to $\overline{s}'$. The points (s_i', ξ_i) belong to the closed subset $\mathcal{F}$ of the total space of the family $(Y_{s'})_{s' \in T'^\circ}$ consisting of the fibers $\rho_{s'}^{-1}(o_{s'})$. Possibly passing to a subsequence, we fix an irreducible closed subset $T_1 \subset \mathcal{F}$ containing the (s_i', ξ_i) such that any non-empty open subset of T_1 contains (s_i', ξ_i) for all but finitely many i, and we let $(\overline{s}', \overline{\xi})$ be the geometric generic point of T_1, over an algebraically closed field $\Bbbk_1$.

The germ $(Y_{\overline{s}'} \ni \overline{\xi})$ is terminal. Let r' be the Cartier index of $K_{Y_{\overline{s}'}}$ at $\overline{\xi}$, $\pi' : V \to Y_{\overline{s}'}$ the index 1 cover near $\overline{\xi}$, and $\overline{\xi}' \in V$ the point over $\overline{\xi}$. Then π' is quasi-étale, the group $\mu_{r'}$ acts faithfully on V with quotient $Y_{\overline{s}'}$, and $(V \ni \overline{\xi}')$ is Gorenstein terminal, hence smooth or an isolated cDV hypersurface singularity. Let $n \in \{3, 4\}$ be the embedding dimension of V at $\overline{\xi}'$, and fix $\mu_{r'}$-eigenfunctions $x_1, \dots, x_n$ generating the maximal ideal of $\mathcal{O}_{V,\overline{\xi}'}$. If $n = 4$, then

$$\widehat{\mathcal{O}}_{V,\overline{\xi}'} \cong \Bbbk_1[[x_1, \dots, x_4]]/(\overline{\Phi})$$

for a semi-invariant $\overline{\Phi}$, and there exists a positive integer c with

$$\mathfrak{n}_1^c \subset (\overline{\Phi}) + \mathfrak{n}_1 \cdot (\partial_1\overline{\Phi}, \dots, \partial_4\overline{\Phi}), \tag{D.95}$$

where $\mathfrak{n}_1$ is the maximal ideal of $\Bbbk_1[[x_1, \dots, x_4]]$, since the singularity is isolated. Since K_V is Cartier and

$$K_V + \pi'^* B_{Y_{\overline{s}'}} = \pi'^* \rho_{\overline{s}'}^*(K_{X_{\overline{s}'}} + B_{\overline{s}'}),$$

$r\pi'^* B_{Y_{\overline{s}'}}$ is Cartier near $\overline{\xi}'$. Since $\mu_{r'}$ acts on the fiber at $\overline{\xi}'$ of the invertible sheaf $\mathcal{O}_V(-r\pi'^* B_{Y_{\overline{s}'}})$ by a character χ, the projection of a local equation to χ is again a local equation by Nakayama's lemma. Thus $r\pi'^* B_{Y_{\overline{s}'}} = \operatorname{div}(\overline{\lambda})$ for a $\mu_{r'}$-semi-invariant $\overline{\lambda}$. By Nakayama's lemma, inclusion (D.95) is equivalent to

$$\mathfrak{n}_1^c \subset (\overline{\Phi}) + \mathfrak{n}_1 \cdot (\partial_1\overline{\Phi}, \dots, \partial_4\overline{\Phi}) + \mathfrak{n}_1^{c+1},$$

a condition of maximal rank on the image of $\overline{\Phi}$ in $\Bbbk_1[x_1, \dots, x_4]/\mathfrak{n}_1^{c+2}$. By (d) for T_1 and the points (s_i', ξ_i), for all but finitely many i we obtain a quasi-étale $\mu_{r'}$-quotient $\pi_i' : V_i \to Y_i$ near ξ_i with a point ξ_i' over ξ_i, V_i Gorenstein terminal of embedding dimension n at ξ_i', eigenfunctions $x_1, \dots, x_n$ with the characters above, and a $\mu_{r'}$-semi-invariant λ_i with $r\pi_i'^* B_{Y_i} = \operatorname{div}(\lambda_i)$. If $n = 4$, then

$$\widehat{\mathcal{O}}_{V_i,\xi_i'} \cong \mathbb{C}[[x_1, \dots, x_4]]/(\Phi_i), \qquad \mathfrak{n}^c \subset (\Phi_i) + \mathfrak{n} \cdot (\partial_1\Phi_i, \dots, \partial_4\Phi_i),$$

and Φ_i has the character of $\overline{\Phi}$. Let

$$\Psi_i := \lambda_i^{mr} \cdot \left(\pi_i'^* \rho_i^* h_i^r\right)^r.$$

Then $\frac{1}{m_1}\operatorname{div}(\Psi_i) = \pi_i'^*\Delta_i$ and $K_{V_i} = \pi_i'^* K_{Y_i}$. By (D.91) for π_i', whose divisor B_Q is 0 since π_i' is quasi-étale, and $\Delta = \Delta_i$,

$$\operatorname{mld}\Big(\big(V_i \ni \xi_i', \tfrac{1}{m_1}\operatorname{div}(\Psi_i)\big)/\mu_{r'}\Big) = \operatorname{mld}(Y_i \ni \xi_i, \Delta_i) = b_i,$$

and for a computing lift F'' of b_i for Ψ_i with image E over Y_i, the prime divisor E computes $\operatorname{mld}(Y_i \ni \xi_i, \Delta_i)$ and $a(E, Y_i, 0) = a(F'', V_i, 0)/e_{F''}$. Thus it suffices to find such F'' with $a(F'', V_i, 0)/e_{F''}$ bounded from above independently of i.

If $n = 4$, then by Lemma 3.2(2) with $L = 2c+1$ and $\Phi_{0,i} := (\Phi_i)_{<2c+1}$, there exist a $\mu_{r'}$-equivariant automorphism φ_i of $\mathbb{C}[[x_1, \dots, x_4]]$ and an invariant unit v_i with $v_i\varphi_i(\Phi_i) = \Phi_{0,i}$. Let $V_i' := (\Phi_{0,i} = 0) \subset \mathbb{A}^4$ and $\Psi_i' := \varphi_i(\Psi_i)$, where Ψ_i is lifted to $\mathbb{C}[[x_1, \dots, x_4]]$. Then $\deg \Phi_{0,i} \le 2c$, and φ_i induces a $\mu_{r'}$-equivariant isomorphism $\widehat{\mathcal{O}}_{V_i',0} \cong \widehat{\mathcal{O}}_{V_i,\xi_i'}$. By Lemma 2.12, $(V_i' \ni 0)$ is an isolated Gorenstein terminal germ on which $\mu_{r'}$ acts faithfully,

$$\operatorname{mld}\Big(\big(\widehat{V}_i' \ni 0, \tfrac{1}{m_1}\operatorname{div}(\Psi_i')\big)/\mu_{r'}\Big) = b_i,$$

and the prime divisors P' over V_i' with center 0 correspond to the prime divisors P over V_i with center ξ_i' with $a(P', V_i', 0) = a(P, V_i, 0)$, $e_{P'} = e_P$, and $\operatorname{mult}_{P'} \Psi_i' = \operatorname{mult}_P \Psi_i$. Let $L_4 := L(m_1, r', 2c)$ be the constant of Theorem 13.3(1). Let N_i be an integer with $N_i \ge N_{V_i'}$ (Claim D.12 for V_i', $\mu_{r'}$, and $m' = m_1$), $N_i > m_1 r' L_4$, and $(\Psi_i')_{<N_i}|_{V_i'} \ne 0$. By Claim D.12(1),

$$\operatorname{mld}\Big(\big(V_i' \ni 0, \tfrac{1}{m_1}\operatorname{div}\left((\Psi_i')_{<N_i}\right)\big)/\mu_{r'}\Big) = b_i > 0,$$

so the pair $(V_i', \frac{1}{m_1}\operatorname{div}((\Psi_i')_{<N_i}))$ is lc near 0 by (a) and (b). By Theorem 13.3(1), b_i has a computing lift F' for the polynomial mark $(\Psi_i')_{<N_i}$ with $a(F', V_i', 0) \le e_{F'} L_4 \le r' L_4 < N_i/m_1$. By Claim D.12(2), F' is a computing lift of b_i for Ψ_i', and the corresponding prime divisor F'' over V_i is a computing lift of b_i for Ψ_i with $a(F'', V_i, 0)/e_{F''} \le L_4$.

If $n = 3$, then $\widehat{\mathcal{O}}_{V_i,\xi_i'} \cong \mathbb{C}[[x_1, x_2, x_3]]$ with the diagonal action of $\mu_{r'}$ given by the characters of x_1, x_2, x_3, which do not depend on i. Let $Q := \mathbb{A}^3/\mu_{r'}$ and $\pi : \mathbb{A}^3 \to Q$ the quotient morphism. Since π_i' is quasi-étale and $\mu_{r'}$ acts on $\mathbb{A}^3$ as on $\widehat{\mathcal{O}}_{V_i,\xi_i'}$, so is π, and $K_{\mathbb{A}^3} = \pi^* K_Q$. Let $\Gamma := \frac{1}{m_1 r'}\mathbb{N}^+$, and L_3 the integer of Theorem 2.22(3) for the $\mathbb{Q}$-factorial klt variety Q and Γ. Let N_i be an integer with $N_i \ge N_{\mathbb{A}^3}$ (Claim D.12 for $\mathbb{A}^3$, $\mu_{r'}$, and $m' = m_1$), $N_i > m_1 r' L_3$, and $(\Psi_i)_{<N_i} \ne 0$. By Claim D.12(1) and (a), (b), the pair $(Q, \frac{1}{m_1 r'}\operatorname{div}((\Psi_i)^{r'}_{<N_i}))$ is lc near the image x_Q of 0 with mld b_i at x_Q, and the coefficients of $\frac{1}{m_1 r'}\operatorname{div}((\Psi_i)^{r'}_{<N_i})$ belong to Γ. By Theorem 2.22(3), there exists a prime divisor E' over Q with $a(E', Q, 0) \le L_3$ computing

$$\operatorname{mld}\Big(Q \ni x_Q, \tfrac{1}{m_1 r'}\operatorname{div}\left((\Psi_i)^{r'}_{<N_i}\right)\Big).$$

A prime divisor F' over $\mathbb{A}^3$ over E' is a computing lift of b_i for $(\Psi_i)_{<N_i}$ with

$$a(F', \mathbb{A}^3, 0) = e_{F'} a(E', Q, 0) \le r' L_3 < N_i/m_1$$

by (D.91). By Claim D.12(2), F' is a computing lift of b_i for Ψ_i, and the corresponding prime divisor F'' over V_i satisfies $a(F'', V_i, 0)/e_{F''} \le L_3$.

In all cases, $a(E, Y_i, 0)$ is bounded from above independently of i (by 1, C_{m_1}, L_4, or L_3), so for all but finitely many i, $a_{\min}(s_i, h_i) \le r a(E, Y_i, 0)$ is bounded from above independently of i, contradicting $a_{\min}(s_i, h_i) > i$.

Step 4. In this step we assume that $b_i \le 0$ for any i and construct a generic limit. If $b_i < 0$, then $h_i \notin (f_{s_i}) + \mathfrak{n}^{2m+1}$ for $i \ge 2$, since otherwise $E_{s_i}^+$ satisfies $a(E_{s_i}^+, W_{s_i}, 0) = 2 < a_{\min}(s_i, h_i)$ and (D.92). If $b_i = 0$, then $(W_{s_i}, \frac{1}{m}\operatorname{div}(h_i))$ is lc near 0 by (a) and (b), so $h_i \notin (f_{s_i}) + \mathfrak{n}^{2m+1}$ by (D.92). In both cases $\operatorname{mult}_0 h_i \le 2m$. Since $\operatorname{mult}_0 f_{s_i} = 2$ and $\operatorname{mult}_0 h_i \le 2m$, Lemma 2.6(1)

gives, possibly after passing to a subsequence, a compatible generic limit (f_*, h_*) of the pairs (f_{s_i}, h_i) (Definition 2.5 with $H = G$ and $N = 2$), with coefficient field $\Bbbk$ and irreducible closed subsets $Z_d \subset \mathcal{J}_d$. By Lemma 2.6(2), (f_*, h_*) is a compatible generic limit with coefficient field $\Bbbk$ of the pairs (f_{s_i}, h_i) along any infinite set of indices i. By Lemma 2.8, a constructible subset of $\mathcal{J}_d$ that contains the d-jets of infinitely many pairs (f_{s_i}, h_i) contains the d-jet of (f_*, h_*).

For $d > N_0$, the d-jets of the f_{s_i} are the f_{s_i} themselves, so f_* is a polynomial of degree at most N_0. Let $W_*^{\mathrm{alg}} := (f_* = 0) \subset \mathbb{A}^4_{\Bbbk}$, $W_* := \operatorname{Spec} \Bbbk[[x_1, \dots, x_4]]/(f_*)$ the completion of W_*^{alg} at 0, and $\mathfrak{n}_* := (x_1, \dots, x_4)$, in $\Bbbk[[x_1, \dots, x_4]]$ or in $\Bbbk[x_1, \dots, x_4]$. The image of S in $\mathcal{P}$ is constructible and contains the dense subset $\{f_{s_i}\}$ of the closure Z^f of $\{f_{s_i}\}$, so it contains the generic point ζ of Z^f, and f_* is a geometric point over ζ. The set of (not necessarily closed) points of S at which the geometric fiber is normal and klt near 0 is constructible by spreading out log resolutions and contains all closed points, so it is S. Thus W_*^{alg} is normal and klt near 0, and so is W_* by [dFEM11, Remark 2.10 and Proposition 2.11(i)]. The $(2m+1)$-jets of the pairs (f_{s_i}, h_i) with $i \geq 2$ belong to the set of jets (f, h) with $f \in \mathfrak{n}^2$ of non-zero quadratic part and $h \notin (f) + \mathfrak{n}^{2m+1}$, which is constructible: for such f, multiplication by f is injective from polynomials of degree at most $2m-2$ to $\mathbb{C}[x_1, \dots, x_4]/\mathfrak{n}^{2m+1}$, so the condition on h is the non-vanishing of a maximal minor. Thus

$$h_* \notin (f_*) + \mathfrak{n}_*^{2m+1}; \quad \text{in particular, } h_*|_{W_*} \neq 0.$$

Claim D.13. *Let $k \in \mathbb{N}^+ \cup \{+\infty\}$, with $\frac{1}{+\infty} := 0$.*

(1) If

$$\left(W_{s_i}, \tfrac{1}{m} \operatorname{div}(h_i), \mathfrak{n}^{1/k}\right)$$

is lc near 0 for any i, then $(W_*, \frac{1}{m} \operatorname{div}(h_*), \mathfrak{n}_*^{1/k})$ is lc near 0.

(2) If $(W_*, \frac{1}{m} \operatorname{div}(h_*), \mathfrak{n}_*^{1/k})$ is lc near 0, then

$$\left(W_{s_i}, \tfrac{1}{m} \operatorname{div}(h_i), \mathfrak{n}^{1/k}\right)$$

is lc near 0 for infinitely many i.

Proof. If $k = +\infty$, let $\mathfrak{a}_i := (h_i)$, $\mathfrak{a}_* := (h_*)$, and $\gamma := \frac{1}{m}$. If $k < +\infty$, let $\mathfrak{a}_i := (h_i^k)\mathfrak{n}^m$, $\mathfrak{a}_* := (h_*^k)\mathfrak{n}_*^m$, and $\gamma := \frac{1}{mk}$. Then the pairs in (1) and (2) are $(W_{s_i}, \mathfrak{a}_i^\gamma)$ and $(W_*, \mathfrak{a}_*^\gamma)$, and they are lc near 0 if and only if $\operatorname{lct}(W_{s_i} \ni 0; \mathfrak{a}_i) \geq \gamma$, resp. $\operatorname{lct}(W_* \ni 0; \mathfrak{a}_*) \geq \gamma$. The germs $(W_{s_i} \ni 0)$ belong to the family $(W_s)_{s \in S}$ of klt germs of finite type, and the $\mathfrak{a}_i$ are proper ideals, since $h_i(0) = 0$ as W_{s_i} is klt and $b_i \leq 0$. After the base change to $\Bbbk$, which preserves lc thresholds [Kaw21, Remark 2.6], $((f_*), \mathfrak{a}_* + (f_*))$ is a generic limit of the $((f_{s_i}), \mathfrak{a}_i + (f_{s_i}))$ in the sense of [dFEM11, Section 3]: the closures of the images of the $Z_{\max\{d, N_0+1\}}$ in its Hilbert schemes under $(f, h) \to ((f), \mathfrak{a}_h + (f) + \mathfrak{n}^d)$, where $\mathfrak{a}_h$ is formed from h as $\mathfrak{a}_i$ from h_i, satisfy its conditions $(\star)$ and $(\star\star)$. By [dFEM11, Proposition 3.3 and Corollary 3.4], possibly passing to a subsequence,

$$\operatorname{lct}(W_* \ni 0; \mathfrak{a}_*) = \lim_{i \to +\infty} \operatorname{lct}(W_{s_i} \ni 0; \mathfrak{a}_i),$$

which implies (1). By [dFEM11, Theorem 1.1], the set $\{\operatorname{lct}(W_{s_i} \ni 0; \mathfrak{a}_i)\}_i$ satisfies the ACC. For (2), if

$$\operatorname{lct}(W_{s_i} \ni 0; \mathfrak{a}_i) < \gamma \leq \operatorname{lct}(W_* \ni 0; \mathfrak{a}_*)$$

for all but finitely many i, then the limit above contains a strictly increasing subsequence, contradicting the ACC. □

Claim D.14. *Let $N > N_0$ be an integer, and let*

$$\mathfrak{b}_i := \left((h_i)_{<N}\right) + \mathfrak{n}^N \subset \mathbb{C}[x_1, \dots, x_4],$$
$$\mathfrak{b}_* := \left((h_*)_{<N}\right) + \mathfrak{n}_*^N \subset \Bbbk[x_1, \dots, x_4].$$

Assume that $\mathrm{mld}(W_* \ni 0, \mathfrak{b}_*^{1/m}) < 0$, *resp.* ≤ 0. *Then there exists a positive integer* I *such that for all but finitely many* i, *there exists a prime divisor* P_i *over* W_{s_i} *with center* 0, $a(P_i, W_{s_i}, 0) = I$, *and* $a(P_i, W_{s_i}, \frac{1}{m}\operatorname{div}(h_i)) < 0$, *resp.* ≤ 0.

Proof. By [dFEM11, Lemma 2.21 and Proposition A.14], the prime divisors P' over W_* with center the closed point correspond to the prime divisors P over W_*^{alg} with center 0 with $a(P', W_*, 0) = a(P, W_*^{\mathrm{alg}}, 0)$ and $\mathrm{mult}_{P'} \mathfrak{c} = \mathrm{mult}_P \mathfrak{c}$ for any ideal $\mathfrak{c} \subset \Bbbk[x_1, \dots, x_4]$. Since $\mathfrak{b}_*$ is an ideal of the polynomial ring,

$$\mathrm{mld}\big(W_*^{\mathrm{alg}} \ni 0, \mathfrak{b}_*^{1/m}\big) = \mathrm{mld}\big(W_* \ni 0, \mathfrak{b}_*^{1/m}\big).$$

Since $N > N_0$, the N-jet of (f_*, h_*) is $(f_*, (h_*)_{<N})$ and maps to the generic point of Z_N, and the N-jet of (f_{s_i}, h_i) is $(f_{s_i}, (h_i)_{<N})$. Let $\overline{\zeta}_N$ be the geometric generic point of Z_N through which the N-jet of (f_*, h_*) factors, $(f_{\overline{\zeta}_N}, h_{\overline{\zeta}_N})$ the N-jet at $\overline{\zeta}_N$, $W_{\overline{\zeta}_N} := (f_{\overline{\zeta}_N} = 0) \subset \mathbb{A}^4$ over the residue field of $\overline{\zeta}_N$, and

$$\mathfrak{b}_{\overline{\zeta}_N} := (h_{\overline{\zeta}_N}) + (x_1, \dots, x_4)^N.$$

Then $(W_*^{\mathrm{alg}}, \mathfrak{b}_*)$ is the base change of $(W_{\overline{\zeta}_N}, \mathfrak{b}_{\overline{\zeta}_N})$ to $\Bbbk$. A log resolution of $(W_{\overline{\zeta}_N}, \mathfrak{b}_{\overline{\zeta}_N})$ that principalizes the maximal ideal of 0 base changes to a log resolution over $\Bbbk$, so

$$\mathrm{mld}\big(W_{\overline{\zeta}_N} \ni 0, \mathfrak{b}_{\overline{\zeta}_N}^{1/m}\big) = \mathrm{mld}\big(W_*^{\mathrm{alg}} \ni 0, \mathfrak{b}_*^{1/m}\big).$$

By (b), there exists a prime divisor P over $W_{\overline{\zeta}_N}$ with center 0 and $a(P, W_{\overline{\zeta}_N}, \mathfrak{b}_{\overline{\zeta}_N}^{1/m}) < 0$, resp. ≤ 0. We let $I := a(P, W_{\overline{\zeta}_N}, 0)$. By (d) for Z_N and the N-jets of the (f_{s_i}, h_i), for all but finitely many i we obtain a prime divisor P_i over W_{s_i} with center 0, $a(P_i, W_{s_i}, 0) = I$, and $a(P_i, W_{s_i}, \mathfrak{b}_i^{1/m}) < 0$, resp. ≤ 0. Since $h_i \in \mathfrak{b}_i \mathbb{C}[[x_1, \dots, x_4]]$, we have $\mathrm{mult}_{P_i} h_i \ge \mathrm{mult}_{P_i} \mathfrak{b}_i$, so

$$a\big(P_i, W_{s_i}, \tfrac{1}{m}\operatorname{div}(h_i)\big) \le a\big(P_i, W_{s_i}, \mathfrak{b}_i^{1/m}\big).$$

□

Step 5. In this step we derive a contradiction when $b_i < 0$ for any i and when $b_i = 0$ for any i. For a prime divisor P_* over W_* with center the closed point and an integer $N > \max\{N_0, \mathrm{mult}_{P_*} h_*\}$, we have $\mathrm{mult}_{P_*} \mathfrak{b}_* = \mathrm{mult}_{P_*} h_*$ by (D.90) and $\mathrm{mult}_{P_*} \mathfrak{n}_*^N \ge N$, so

$$a\big(P_*, W_*, \mathfrak{b}_*^{1/m}\big) = a\big(P_*, W_*, \tfrac{1}{m}\operatorname{div}(h_*)\big). \tag{D.96}$$

Assume that $b_i < 0$ for any i. Then $(W_{s_i}, \frac{1}{m}\operatorname{div}(h_i))$ is not lc near 0 for any i, so by Claim D.13(2) with $k = +\infty$, $(W_*, \frac{1}{m}\operatorname{div}(h_*))$ is not lc near 0. By (b), there exists a prime divisor P_* over W_* with center the closed point and $a(P_*, W_*, \frac{1}{m}\operatorname{div}(h_*)) < 0$. Let $N > \max\{N_0, \mathrm{mult}_{P_*} h_*\}$ be an integer. By (D.96), $\mathrm{mld}(W_* \ni 0, \mathfrak{b}_*^{1/m}) < 0$. By Claim D.14, $a_{\min}(s_i, h_i) \le I$ for all but finitely many i, a contradiction.

Assume that $b_i = 0$ for any i. Then $(W_{s_i}, \frac{1}{m}\operatorname{div}(h_i))$ is lc near 0 for any i, so by Claim D.13(1) with $k = +\infty$ and by (b),

$$\beta_* := \mathrm{mld}\big(W_* \ni 0, \tfrac{1}{m}\operatorname{div}(h_*)\big) \ge 0.$$

If $\beta_* = 0$, then by (b) there exists a prime divisor P_* over W_* with center the closed point and $a(P_*, W_*, \frac{1}{m}\operatorname{div}(h_*)) = 0$. Let $N > \max\{N_0, \mathrm{mult}_{P_*} h_*\}$ be an integer. By (D.96),

$$\mathrm{mld}(W_* \ni 0, \mathfrak{b}_*^{1/m}) \le 0.$$

By Claim D.14, for all but finitely many i there exists a prime divisor P_i over W_{s_i} with center 0, $a(P_i, W_{s_i}, 0) = I$, and $a(P_i, W_{s_i}, \frac{1}{m}\operatorname{div}(h_i)) \le 0$. Since $(W_{s_i}, \frac{1}{m}\operatorname{div}(h_i))$ is lc near 0, equality holds, so P_i is a computing lift of $b_i = 0$ and $a_{\min}(s_i, h_i) \le I$, a contradiction. If $\beta_* > 0$, let $W' \to W_*$ be a log resolution of $(W_*, \operatorname{div}(h_*), \mathfrak{n}_*)$ [dFEM11, Theorems 2.3–2.4] with $\mathfrak{n}_* \mathcal{O}_{W'} = \mathcal{O}_{W'}(-\sum_j c_j E_j)$

for prime divisors E_j and integers $c_j \geq 0$. Any E_j with $c_j > 0$ has center the closed point, so $a(E_j, W_*, \frac{1}{m}\operatorname{div}(h_*)) \geq \beta_* > 0$. Let k be a positive integer with

$$\frac{1}{k} < \min_{c_j>0} \frac{a(E_j, W_*, \frac{1}{m}\operatorname{div}(h_*))}{c_j}.$$

Then the log discrepancies of the E_j with respect to $(W_*, \frac{1}{m}\operatorname{div}(h_*), \mathfrak{n}_*^{1/k})$ are non-negative, so $(W_*, \frac{1}{m}\operatorname{div}(h_*), \mathfrak{n}_*^{1/k})$ is lc near 0. By Claim D.13(2), there exist infinitely many i such that $(W_{s_i}, \frac{1}{m}\operatorname{div}(h_i), \mathfrak{n}^{1/k})$ is lc near 0. For such i and any prime divisor P over W_{s_i} with center 0,

$$a\big(P, W_{s_i}, \tfrac{1}{m}\operatorname{div}(h_i)\big) \geq \frac{\operatorname{mult}_P \mathfrak{n}}{k} \geq \frac{1}{k},$$

so $b_i \geq \frac{1}{kr} > 0$, a contradiction. The lemma follows. □

We need the following lemma in the proof of Lemma D.16.

Lemma D.15. *Let $\Bbbk \supset \mathbb{C}$ be an algebraically closed field, G a finite abelian group acting diagonally on $\Bbbk[[z,t]]$, and $g \neq 0$ a G-semi-invariant element of $\Bbbk[[z,t]]$. Then there exist a G-equivariant automorphism ψ of $\Bbbk[[z,t]]$, a G-invariant unit υ of $\Bbbk[[z,t]]$, and a G-semi-invariant polynomial $q \in \Bbbk[z,t]$ with the character of g such that*

$$\psi(g) = \upsilon q.$$

Proof. Let $\mathcal{O} := \Bbbk[[z,t]]$, and let $\mathfrak{m}$ be the maximal ideal of $\mathcal{O}$. Let $\mathfrak{p}_1, \dots, \mathfrak{p}_n$ be the distinct prime ideals of height one of $\mathcal{O}$ containing g, and let d_j be the multiplicity of $\mathfrak{p}_j$ in g for any j. Since $\mathcal{O}$ is a regular local ring, it is factorial, so $\mathfrak{p}_j$ is principal for any j. Since $\sigma(g) = \chi_g(\sigma)g$ for any $\sigma \in G$, the group G permutes $\mathfrak{p}_1, \dots, \mathfrak{p}_n$, and $d_j = d_k$ whenever $\sigma(\mathfrak{p}_j) = \mathfrak{p}_k$ for some $\sigma \in G$.

Step 1. Fix j_0, let $G_{j_0} \subset G$ be the stabilizer of $\mathfrak{p}_{j_0}$, and let χ be the character by which G_{j_0} acts on the $\Bbbk$-vector space $\mathfrak{p}_{j_0}/\mathfrak{m}\mathfrak{p}_{j_0}$ of dimension 1. Let p' be a generator of $\mathfrak{p}_{j_0}$, and let

$$p_{j_0} := \frac{1}{|G_{j_0}|} \sum_{\tau \in G_{j_0}} \chi(\tau)^{-1} \tau(p').$$

Then $\tau(p_{j_0}) = \chi(\tau)p_{j_0}$ for any $\tau \in G_{j_0}$. Since $\tau(p') - \chi(\tau)p' \in \mathfrak{m}\mathfrak{p}_{j_0}$ for any $\tau \in G_{j_0}$, we have $p_{j_0} - p' \in \mathfrak{m}\mathfrak{p}_{j_0} = \mathfrak{m}p'$, so p_{j_0} is p' times a unit and generates $\mathfrak{p}_{j_0}$. Let $\sigma_1, \dots, \sigma_{n'} \in G$ be representatives of G/G_{j_0}, and take $\sigma_k(p_{j_0})$ as the generator of $\sigma_k(\mathfrak{p}_{j_0})$ for any k. For $\sigma \in G$ and $k \in \{1, \dots, n'\}$, write $\sigma\sigma_k = \sigma_{k'}\tau$ with $\tau \in G_{j_0}$. Then

$$\sigma\big(\sigma_k(p_{j_0})\big) = \chi(\tau)\sigma_{k'}(p_{j_0}). \tag{D.97}$$

Fixing p_{j_0} in this way for one index j_0 in each G-orbit of $\{\mathfrak{p}_1, \dots, \mathfrak{p}_n\}$, we obtain generators $p_1, \dots, p_n$ of $\mathfrak{p}_1, \dots, \mathfrak{p}_n$ such that for any $\sigma \in G$ and any j there exist k and $\xi \in \Bbbk^\times$ with $\sigma(p_j) = \xi p_k$. Since d_j is constant on the G-orbits, $\prod_j p_j^{d_j}$ is G-semi-invariant, hence so is the unit $\upsilon' := g/\prod_j p_j^{d_j}$. Since the action is diagonal, $\sigma(\upsilon')(0) = \upsilon'(0) \neq 0$ for any $\sigma \in G$, so υ' is G-invariant.

Step 2. Let $p := \prod_{j=1}^n p_j$, which is G-semi-invariant by **Step 1**. Since p is reduced and $\operatorname{char}\Bbbk = 0$, the curve $(p=0) \subset \operatorname{Spec}\Bbbk[[z,t]]$ is regular outside the closed point, so $(p, \partial_z p, \partial_t p)$ is $\mathfrak{m}$-primary, and there exists a positive integer c with

$$\mathfrak{m}^c \subset (p) + \mathfrak{m} \cdot (\partial_z p, \partial_t p).$$

For any $j \neq k$, we have $p_j \notin \mathfrak{p}_k$, and $\bigcap_{N\geq 1}(\mathfrak{p}_k + \mathfrak{m}^N) = \mathfrak{p}_k$ by the Krull intersection theorem for $\mathcal{O}/\mathfrak{p}_k$. Fix a positive integer N_1 with $p_j \notin \mathfrak{p}_k + \mathfrak{m}^{N_1}$ for any $j \neq k$, and an integer

$$N_2 \geq \max\{2c+1, N_1 + c\}.$$

For each index j_0 fixed in **Step 1**, let $\widetilde{p}_{j_0} := \mathrm{jet}_{N_2}(p_{j_0})$ (Definition 2.5), and let $\widetilde{p}_j := \sigma_k(\widetilde{p}_{j_0})$ if $p_j = \sigma_k(p_{j_0})$. Since the action is diagonal, $\tau(\widetilde{p}_{j_0}) = \chi(\tau)\widetilde{p}_{j_0}$ for any τ in the stabilizer G_{j_0} of $\mathfrak{p}_{j_0}$, so

$$\sigma\big(\sigma_k(\widetilde{p}_{j_0})\big) = \chi(\tau)\sigma_{k'}(\widetilde{p}_{j_0})$$

for σ, k, k', and τ as in (D.97). Thus $\sigma(\widetilde{p}_j) = \xi\widetilde{p}_k$ whenever $\sigma(p_j) = \xi p_k$. Moreover, $\widetilde{p}_j - p_j \in \mathfrak{m}^{N_2}$ for any j. Let $\widetilde{p} := \prod_{j=1}^n \widetilde{p}_j$. Then $\widetilde{p}$ is G-semi-invariant with the character of p, and $\widetilde{p} - p \in \mathfrak{m}^{N_2}$. By Lemma 3.2(2) in the variables z, t, with $F = p$, $F_0 = \widetilde{p}$, and $L = N_2$, there exist a G-equivariant automorphism ψ of $\mathcal{O}$ and a G-invariant unit v with $v\psi(p) = \widetilde{p}$ and

$$\psi(z) - z,\ \psi(t) - t \in \mathfrak{m}^{N_2-c+1}.$$

In particular,

$$\psi(a) - a \in \mathfrak{m}^{N_2-c+1} \quad \text{for any } a \in \mathcal{O}. \tag{D.98}$$

The elements $\psi(p_1), \dots, \psi(p_n)$ are irreducible and pairwise non-associate, and

$$\prod_{j=1}^n \psi(p_j) = v^{-1} \prod_{j=1}^n \widetilde{p}_j \quad \text{with} \quad \widetilde{p}_1, \dots, \widetilde{p}_n \in \mathfrak{m}.$$

Since the left-hand side has exactly n irreducible factors and $\widetilde{p}_1, \dots, \widetilde{p}_n$ are non-units, unique factorization in $\mathcal{O}$ implies that $\widetilde{p}_k$ is irreducible for any k and that there exists a permutation π of $\{1, \dots, n\}$ with $(\psi(p_j)) = (\widetilde{p}_{\pi(j)})$ for any j. Suppose that $k := \pi(j) \neq j$ for some j. Since $\widetilde{p}_k - p_k \in \mathfrak{m}^{N_2}$ and $N_2 - c + 1 > N_1$, (D.98) implies

$$p_j \in \psi(p_j) + \mathfrak{m}^{N_2-c+1} \subset (\widetilde{p}_k) + \mathfrak{m}^{N_2-c+1} \subset \mathfrak{p}_k + \mathfrak{m}^{N_2-c+1} \subset \mathfrak{p}_k + \mathfrak{m}^{N_1},$$

which contradicts the choice of N_1. Thus π is the identity, and $\psi(p_j) = v_j\widetilde{p}_j$ for a unit v_j, for any j.

Step 3. Let $q := \prod_{j=1}^n \widetilde{p}_j^{d_j}$, a polynomial. By **Step 2**, q is G-semi-invariant with the character of $\prod_j p_j^{d_j}$, which is the character of g by **Step 1**. Since ψ is G-equivariant, $\psi(g)$ is G-semi-invariant with the character of g. Thus

$$v := \frac{\psi(g)}{q} = \psi(v') \prod_{j=1}^n v_j^{d_j}$$

is a G-semi-invariant unit with trivial character, that is, a G-invariant unit, and $\psi(g) = vq$. □

Lemma D.16. *Let $k_1 \geq 3$, M, and m be positive integers, and let G be a finite cyclic group of order at most M acting on the coordinates x, y, z, t so that it permutes the lines $\mathbb{C}x$ and $\mathbb{C}y$ and is linear on the (z,t)-plane. For any i let $(W_i \ni o_i)$ be an isolated Gorenstein terminal hypersurface threefold germ of quadratic rank 2 with a faithful action of G fixing o_i, and let h_i be a mark with $(W_i, \frac{1}{m}\operatorname{div}(h_i))$ klt near o_i. Let (f_*, h_*), $f_* = xy + g_*(z,t)$, be a compatible generic limit (Definition 2.5), with coefficient field $\Bbbk$, of completed G-equivariant presentations $xy + g_i(z,t)$ of the $(W_i \ni o_i)$ with $\operatorname{mult}_0 g_i \in [3, k_1]$ and of the h_i, where the characters of the equations and of the marks do not depend on i, and let $W_* := (f_* = 0) \subset \operatorname{Spec} \Bbbk[[x,y,z,t]]$. Assume that W_* is normal and canonical, that $(W_*, \frac{1}{m}\operatorname{div}(h_*))$ is lc, and that $\operatorname{Nklt}(W_*, \frac{1}{m}\operatorname{div}(h_*))$ and $\operatorname{Sing}(W_*)$ have no common point other than 0. Let $\mathfrak{n}$ be the maximal ideal of $\mathbb{C}[[x,y,z,t]]$, and for a power series f let*

$$\operatorname{Jac}(f) := (\partial_x f, \partial_y f, \partial_z f, \partial_t f).$$

Then there exist a family $(f_{0,s})_{s\in S}$ of polynomials $f_{0,s} = xy + q_{0,s}(z,t)$ with $\operatorname{mult}_0 q_{0,s} \in [3, k_1]$ for any $s \in S$, satisfying the hypotheses of Lemma D.11, a rational number $\mu > \frac{1}{m}$, and a positive integer n with the following property. For any rational number $\theta \in (0,1)$, any positive integers Q and p' with $p' = Q\mu$, and any integer $R \geq 2$, there exists a positive integer c such that for any positive integer N and all but finitely many i, the following hold.

(1) There exist a G-semi-invariant polynomial f_i such that

$$(W_i' \ni 0) := ((f_i = 0) \ni 0) \subset (\mathbb{A}^4 \ni 0)$$

is an isolated Gorenstein terminal germ, a G-equivariant isomorphism between the completions of W_i at o_i and of W_i' at 0, and a G-semi-invariant polynomial H_i such that $(W_i', \frac{1}{m}\operatorname{div}(H_i))$ is klt near 0, with the following property: for any subgroup $G' \subset G$,

$$\operatorname{mld}\big((W_i \ni o_i, \tfrac{1}{m}\operatorname{div}(h_i))/G'\big) = \operatorname{mld}\big((W_i' \ni 0, \tfrac{1}{m}\operatorname{div}(H_i))/G'\big),$$

and any computing lift of one side corresponds, through the isomorphism, to a computing lift of the other side with equal log discrepancies, equal inertia groups, and equal orders of the mark.

(2) There exists a closed point $s_i \in S$ such that $f_i - f_{0,s_i} \in \mathfrak{n}^N$.

(3) There exist G-semi-invariant polynomials $\zeta_{k,i}$ and $\gamma_{k,i}$, $1 \le k \le n$, with $\zeta_{k,i}$ of the character of $f_i\gamma_{k,i}$, such that the following hold, where $\mathfrak{a}_i := (f_i^Q H_i^{p'})$ and

$$I_i := (\zeta_{1,i} + f_i\gamma_{1,i}, \dots, \zeta_{n,i} + f_i\gamma_{n,i}).$$

(a) $\zeta_{k,i} \in \mathcal{J}(\mathbb{A}^4, \mathfrak{a}_i^{\theta/Q})$ for any k.

(b) $(\zeta_{k,i} + f_i\gamma_{k,i})(0) = 0$ for any k.

(c)

$$\mathfrak{n}^c \subset (f_{0,s_i}) + \mathfrak{n} \cdot \operatorname{Jac}(f_{0,s_i}) + I_i^R.$$

Proof. Since G is cyclic, possibly replacing (z,t) with a linear eigenbasis of the (z,t)-plane, simultaneously in the g_i, the h_i, g_*, and h_*, we may assume that G acts diagonally on (z,t). Lemma 3.2 is used below in a linear eigenbasis of $\mathbb{A}^4$, which exists since G is cyclic. We denote the power series presenting h_i again by h_i, and we write $\mathcal{X}_* := \operatorname{Spec}\Bbbk[[x,y,z,t]]$, with maximal ideal $\mathfrak{n}_*$, and for any power series ϕ, G-semi-invariant or not, $\operatorname{jet}_e(\phi)$ denotes its part of degree less than e (Definition 2.5).

Step 1. In this step we construct the family $(f_{0,s})_{s\in S}$.

We first show $\operatorname{mult}_0 g_* \in [3, k_1]$. For any i, $\operatorname{jet}_{k_1+1}(xy + g_i, h_i)$ belongs to the constructible subset of $\mathcal{J}_{k_1+1}$ of the pairs (p_1, p_2) such that $p_1 - xy$ is non-zero and has no terms of degree less than 3. Thus $\operatorname{mult}_0 g_* \in [3, k_1]$ by Lemma 2.8.

By Lemma D.15, there exist a G-equivariant automorphism ψ of $\Bbbk[[z,t]]$, a G-invariant unit $V' \in \Bbbk[[z,t]]$, and a polynomial $q_0 \in \Bbbk[z,t]$ with the character of g_*, such that

$$g_*(\psi(z), \psi(t)) = V' q_0.$$

In particular $\operatorname{mult}_0 q_0 = \operatorname{mult}_0 g_* \in [3, k_1]$. Since $\Bbbk$ is algebraically closed of characteristic 0, there exists $w \in \Bbbk[[z,t]]$ with $w^2 = V'$. For any $\sigma \in G$ we have $\sigma(w)^2 = \sigma(V') = V'$ and $\sigma(w)(0) = w(0) \neq 0$, so $\sigma(w) = w$. Let Ψ be the automorphism of $\Bbbk[[x,y,z,t]]$ with

$$\Psi(x) = wx, \quad \Psi(y) = wy, \quad \Psi(z) = \psi(z), \quad \text{and} \quad \Psi(t) = \psi(t).$$

Since G permutes the lines $\mathbb{C}x$ and $\mathbb{C}y$ and w is G-invariant, Ψ is G-equivariant. Moreover,

$$\Psi(f_*) = w^2 xy + V' q_0 = V' f_0, \quad \text{where } f_0 := xy + q_0. \tag{D.99}$$

Let $H_* := \Psi(h_*)$ and $W_0 := (f_0 = 0) \subset \mathcal{X}_*$. By (D.99), Ψ induces a G-equivariant isomorphism from $(W_*, \operatorname{div}(h_*))$ to $(W_0, \operatorname{div}(H_*))$, so the hypotheses of the lemma on W_* and $(W_*, \frac{1}{m}\operatorname{div}(h_*))$ hold for W_0 and $(W_0, \frac{1}{m}\operatorname{div}(H_*))$.

Let $r := \#\{(a,b) \in \mathbb{N}^2 : a + b \le \deg q_0\}$, let $\lambda \in \Bbbk^r$ be the vector of the coefficients of q_0, and let $S_1 \subset \mathbb{A}^r$ be the smallest closed subvariety with $\lambda \in S_1(\Bbbk)$. Then S_1 is irreducible, and $\lambda \notin S'(\Bbbk)$ for any proper closed subset $S' \subsetneq S_1$. For a closed point $s \in S_1$, let $q_{0,s}$ be the polynomial with coefficient vector s, and let $f_{0,s} := xy + q_{0,s}$.

Since W_0 is normal and canonical, the germ at 0 of $(f_0 = 0) \subset \mathbb{A}^4_{\Bbbk}$ is normal and klt [dFEM11, Proposition 2.11(i)]. A log resolution of the generic fiber of $(f_{0,s} = 0)_{s\in S_1}$ near the zero section has discrepancies equal to those of the base change of the generic fiber to $\Bbbk$ along λ [dFEM11,

Lemma 2.14], and it restricts to log resolutions of the fibers over a dense open subset of S_1. Thus there exists a dense open subset $S \subset S_1$ such that for any closed point $s \in S$:

- The germ $((f_{0,s} = 0) \ni 0)$ is normal and klt.
- The polynomial $f_{0,s}$ is G-semi-invariant with the character of f_0, a closed condition which holds at λ.
- We have $\operatorname{mult}_0 q_{0,s} = \operatorname{mult}_0 q_0 \in [3, k_1]$, since $\operatorname{mult}_0 q_{0,s} \geq \operatorname{mult}_0 q_0$ is a closed condition and $\operatorname{mult}_0 q_{0,s} \leq \operatorname{mult}_0 q_0$ an open condition, both holding at λ.

Since $\operatorname{mult}_0 q_{0,s} \geq 3$, the germ $((f_{0,s} = 0) \ni 0)$ is of quadratic rank 2 with completed presentation $xy + q_{0,s}$. Since G acts faithfully on W_i, G acts faithfully on $\mathbb{A}^4$. Since $f_{0,s} \in \mathfrak{n}^2$, the Zariski tangent space of $(f_{0,s} = 0)$ at 0 is that of $\mathbb{A}^4$, on which G acts faithfully, so G acts faithfully on $(f_{0,s} = 0)$. Thus $(f_{0,s})_{s \in S}$ satisfies the hypotheses of Lemma D.11. Possibly shrinking S, we may assume that $S = \{\varphi_S \neq 0\}$ for a regular function φ_S on S_1.

Step 2. In this step we choose Z, $\xi_1, \dots, \xi_n$, μ, θ', N_0, and c.

Since $(W_0, \frac{1}{m}\operatorname{div}(H_*))$ is lc, $H_*|_{W_0} \neq 0$. Let

$$Z := \operatorname{Nklt}\left(W_0, \tfrac{1}{m}\operatorname{div}(H_*)\right) \cup \{0\}.$$

Since H_* is G-semi-invariant, Z is a G-stable closed subset of W_0. Let $I_Z \subset \Bbbk[[x, y, z, t]]$ be the ideal of Z. Since G is abelian and I_Z is G-stable, I_Z is generated by G-semi-invariant elements $\xi_1, \dots, \xi_n$, and $\xi_k(0) = 0$ for any k since $0 \in Z$.

Since $(W_0, \frac{1}{m}\operatorname{div}(H_*))$ is klt on $W_0 \backslash Z$ and $(W_0, \operatorname{div}(H_*))$ has a log resolution [dFEM11, Theorems 2.3–2.4], there exists a rational number $\mu > \frac{1}{m}$ such that $(W_0, \mu \operatorname{div}(H_*))$ is klt on $W_0 \backslash Z$.

Let $\theta \in (0, 1)$ be a rational number, and let Q and p' be positive integers with $p' = Q\mu$. Choose a rational number $\theta' \in (\theta, 1)$, and let

$$\mathfrak{a}_* := (f_0^Q H_*^{p'}), \qquad \mathcal{J}_* := \mathcal{J}(\mathcal{X}_*, \mathfrak{a}_*^{\theta'/Q}) = \mathcal{J}(f_0^{\theta'} H_*^{\theta'\mu}),$$

where the equality holds since $p'\theta'/Q = \theta'\mu$. By Lemma 6.1(3), with f_0, H_*, θ', and Z in place of f_*, h_*, θ_2, and V, the ideal $\mathcal{J}_*$ is the unit ideal near any point of $W_0 \backslash Z$. Thus the zero locus of $\mathcal{J}_* + (f_0)$ is contained in Z, the zero locus of $(\xi_1, \dots, \xi_n)$. Hence $\xi_k \in \sqrt{\mathcal{J}_* + (f_0)}$ for any k, so there exists a positive integer N_0 such that

$$\xi_k^{N_0} \in \mathcal{J}_* + (f_0) \quad \text{for any } 1 \leq k \leq n. \tag{D.100}$$

For any k, choose $\gamma_k \in \Bbbk[[x, y, z, t]]$ such that

$$\zeta_k := \xi_k^{N_0} - f_0\gamma_k \in \mathcal{J}_*.$$

Since $\mathcal{J}_*$ is G-stable, possibly replacing γ_k with the component of γ_k of character $\chi_{\xi_k}^{N_0}\chi_{f_0}^{-1}$, we may assume that γ_k and ζ_k are G-semi-invariant.

Let $R \geq 2$ be an integer. The zero locus of $(f_0) + \mathfrak{n}_* \cdot \operatorname{Jac}(f_0)$ is $\{0\} \cup \operatorname{Sing}(W_0)$, and the zero locus of $(\xi_1^{N_0}, \dots, \xi_n^{N_0})^R$ is Z. Since $Z \cap \operatorname{Sing}(W_0) \subset \{0\}$ by hypothesis, and since f_0 and the ξ_k vanish at 0, the ideal

$$(f_0) + \mathfrak{n}_* \cdot \operatorname{Jac}(f_0) + (\xi_1^{N_0}, \dots, \xi_n^{N_0})^R$$

is $\mathfrak{n}_*$-primary. Thus there exists a positive integer c such that

$$\mathfrak{n}_*^c \subset (f_0) + \mathfrak{n}_* \cdot \operatorname{Jac}(f_0) + (\xi_1^{N_0}, \dots, \xi_n^{N_0})^R. \tag{D.101}$$

Step 3. In this step we fix a positive integer N and construct, for all but finitely many i, the data of (1)–(3).

Let $\epsilon := (\theta' - \theta)/Q$, choose a positive integer ℓ with $\lfloor \ell\epsilon \rfloor - 3 \geq c + 1$, and let

$$N_1 := \max\{0, \lfloor \ell\theta'/Q \rfloor - 3\}.$$

Then $N_1 \geq \lfloor \ell\epsilon \rfloor - 3 \geq c+1$ since $\theta'/Q > \epsilon$. Since $\mathfrak{a}_*$ and $\mathfrak{n}_*^\ell$ are contained in $\mathfrak{a}_* + \mathfrak{n}_*^\ell$, and the blow-up of the closed point is a log resolution of $\mathfrak{n}_*^\ell$ whose exceptional divisor has discrepancy 3,

$$\mathcal{J}\big((\mathfrak{a}_* + \mathfrak{n}_*^\ell)^{\theta'/Q}\big) \supset \mathcal{J}_* + \mathcal{J}\big((\mathfrak{n}_*^\ell)^{\theta'/Q}\big) = \mathcal{J}_* + \mathfrak{n}_*^{N_1}.$$

Let $\rho_k := \mathrm{jet}_{N_1}(\zeta_k)$, a polynomial. As $\zeta_k \in \mathcal{J}_*$ and $\zeta_k - \rho_k \in \mathfrak{n}_*^{N_1}$, the display gives

$$\rho_k \in \mathcal{J}\big((\mathfrak{a}_* + \mathfrak{n}_*^\ell)^{\theta'/Q}\big) \quad \text{for any } k.$$

Let $d := \max\{N, c+1, \ell, N_1\}$. For any monomial ω of degree c in x, y, z, t, (D.101) implies that there exist polynomials A_ω, $B_{\omega,u,v}$, and $C_{\omega,\nu}$ of degree at most c with coefficients in $\Bbbk$ such that

$$\omega \equiv A_\omega f_0 + \sum_{u,v\in\{x,y,z,t\}} B_{\omega,u,v} u\partial_v f_0 + \sum_{\substack{\nu\in\mathbb{N}^n \\ \nu_1+\cdots+\nu_n=R}} C_{\omega,\nu} \prod_{k=1}^n \xi_k^{N_0\nu_k} \mod \mathfrak{n}_*^{c+1}. \tag{D.102}$$

Let $\beta \in \Bbbk^{n_\beta}$ be the vector of the following finitely many elements of $\Bbbk$:

- The coordinates of λ.
- The coefficients of $\mathrm{jet}_d(\Psi(x))$, $\mathrm{jet}_d(\Psi(y))$, $\mathrm{jet}_d(\Psi(z))$, $\mathrm{jet}_d(\Psi(t))$, and $\mathrm{jet}_d(V'^{-1})$.
- The coefficients of $\mathrm{jet}_d(\gamma_k)$ and of the terms of positive degree of $\mathrm{jet}_d(\xi_k^{N_0})$, for $1 \leq k \leq n$.
- The coefficients of the A_ω, $B_{\omega,u,v}$, and $C_{\omega,\nu}$.

Let $\eta := (\mathrm{jet}_d(f_*, h_*), \beta)$, and let T be the smallest closed subvariety of $\mathbb{A}^{\dim \mathcal{J}_d + n_\beta}$ with $\eta \in T(\Bbbk)$. Let

$$\mathfrak{b} := \big(\mathrm{jet}_\ell(f_0^Q H_*^{p'})\big) + (x,y,z,t)^\ell \subset \Bbbk[x,y,z,t].$$

Since $H_* = \Psi(h_*)$, $\Psi(u) \in \mathfrak{n}_*$ for any $u \in \{x,y,z,t\}$, $\rho_k = \mathrm{jet}_{N_1}(\xi_k^{N_0} - f_0\gamma_k)$, and $d \geq \ell, N_1$, the coefficients of $\mathrm{jet}_\ell(f_0^Q H_*^{p'})$ and of the ρ_k are values at η of polynomials with complex coefficients. The completion of $\mathfrak{b}$ at 0 is $\mathfrak{a}_* + \mathfrak{n}_*^\ell$, multiplier ideals commute with completion at 0 by [dFEM11, Proposition A.14] and flat base change, and the completion is faithfully flat, so $\rho_k \in \mathcal{J}(\mathfrak{b}^{\theta'/Q})$ near $0 \in \mathbb{A}^4_\Bbbk$ for any k. By Lemma 6.1(4), with η, $\rho_1, \ldots, \rho_n$, and θ'/Q in place of λ, $\phi_1, \ldots, \phi_r$, and γ, there exists a polynomial Θ with complex coefficients with $\Theta(\eta) \neq 0$ such that for any closed point $\eta' \in T$ with $\Theta(\eta') \neq 0$, the specializations $\mathfrak{b}_{\eta'}$ and $\rho_{k,\eta'}$ of $\mathfrak{b}$ and of the ρ_k at η' satisfy, near $0 \in \mathbb{A}^4$,

$$\rho_{k,\eta'} \in \mathcal{J}\big(\mathfrak{b}_{\eta'}^{\theta'/Q}\big) \quad \text{for any } k.$$

The following identities hold at the limit, and each of them is a finite set of polynomial identities with complex coefficients in $\mathrm{jet}_d(f_*, h_*)$ and β:

(i) By (D.99),
$$\mathrm{jet}_d\big(V'^{-1}\Psi(f_*)\big) = \mathrm{jet}_d(f_0).$$

(ii) For any $\sigma \in G$ and $u \in \{x,y,z,t\}$,
$$\sigma(\mathrm{jet}_d(\Psi(u))) = \mathrm{jet}_d(\Psi(\sigma(u))),$$
$\mathrm{jet}_d(V'^{-1})$ is G-invariant, $\mathrm{jet}_d(\gamma_k)$ and $\mathrm{jet}_d(\xi_k^{N_0})$ are G-semi-invariant of characters $\chi_{\xi_k}^{N_0}\chi_{f_0}^{-1}$ and $\chi_{\xi_k}^{N_0}$, and $\lambda \in S_1$.

(iii) The finitely many generators of the ideal of T vanish at η.

(iv) Identity (D.102) holds for any monomial ω of degree c.

Moreover,

$$\varphi_S(\lambda) \cdot \det\big(\partial(\Psi(x), \Psi(y), \Psi(z), \Psi(t))/\partial(x,y,z,t)\big)(0) \cdot V'^{-1}(0) \cdot \Theta(\eta) \neq 0.$$

By Lemma 2.7, applied once, for the tuples $(xy + g_i, h_i)$ with limit (f_*, h_*), for all but finitely many i there exists $\beta_i \in \mathbb{C}^{n_\beta}$ satisfying (i)–(iv) and the non-vanishing above with η_i in place of η, where

$$\eta_i := (\mathrm{jet}_d(xy + g_i, h_i), \beta_i).$$

By (iii), η_i is a closed point of T, and $\Theta(\eta_i) \neq 0$. We define the following data from β_i.

- Let $s_i \in S$ be the closed point given by the coordinates of β_i corresponding to λ.
- Let Ψ_i be the endomorphism of $\mathbb{C}[[x,y,z,t]]$ given by the polynomials corresponding to the $\mathrm{jet}_d(\Psi(u))$, $u \in \{x,y,z,t\}$, which is G-equivariant by (ii). Their constant terms vanish by (iii), and Ψ_i is an automorphism since the Jacobian determinant at 0 is non-zero.
- Let U_i be the G-invariant polynomial corresponding to $\mathrm{jet}_d(V'^{-1})$, a unit in $\mathbb{C}[[x,y,z,t]]$ since $U_i(0) \neq 0$.
- Let $\gamma_{k,i}$ and $\widetilde{\xi}_{k,i}$ be the G-semi-invariant polynomials corresponding to $\mathrm{jet}_d(\gamma_k)$ and $\mathrm{jet}_d(\xi_k^{N_0})$.

Let $F_i := U_i\Psi_i(xy+g_i)$. By (i), $F_i - f_{0,s_i} \in \mathfrak{n}^d$, and Ψ_i induces a G-equivariant isomorphism from the completion of W_i at o_i to $\operatorname{Spec}\mathbb{C}[[x,y,z,t]]/(F_i)$.

We now construct f_i and H_i. Since $(W_i \ni o_i)$ is isolated, $(F_i)+\mathfrak{n}\cdot\mathrm{Jac}(F_i)$ is $\mathfrak{n}$-primary, so there exists a positive integer e_i with

$$\mathfrak{n}^{e_i} \subset (F_i) + \mathfrak{n}\cdot\mathrm{Jac}(F_i).$$

Let $L_i > 2e_i + d$ be an integer, and let $f_i := \mathrm{jet}_{L_i}(F_i)$, a G-semi-invariant polynomial with the character of F_i. By Lemma 3.2(2), there exist a G-equivariant automorphism ϕ_i of $\mathbb{C}[[x,y,z,t]]$ and a G-invariant unit v_i such that

$$v_i\phi_i(F_i) = f_i, \qquad \phi_i(u) - u \in \mathfrak{n}^{L_i-e_i+1} \subset \mathfrak{n}^d \text{ for any } u \in \{x,y,z,t\}.$$

Let $W_i' := (f_i = 0) \subset \mathbb{A}^4$. Then $\phi_i \circ \Psi_i$ induces a G-equivariant isomorphism between the completions of W_i at o_i and of W_i' at 0. Thus $(W_i' \ni 0)$ is an isolated Gorenstein terminal germ [dFEM11, Lemma 2.21 and Proposition A.14]. Since $f_i - F_i \in \mathfrak{n}^{L_i} \subset \mathfrak{n}^d$ and $F_i - f_{0,s_i} \in \mathfrak{n}^d$,

$$f_i - f_{0,s_i} \in \mathfrak{n}^d \subset \mathfrak{n}^N,$$

which is (2). Let $\widehat{h}_i := \phi_i(\Psi_i(h_i))$. Since $(W_i, \frac{1}{m}\operatorname{div}(h_i))$ is klt near o_i, the completed pair $(\widehat{W}_i', \frac{1}{m}\operatorname{div}(\widehat{h}_i))$ is klt [dFEM11, Proposition 2.11(i)], where $\widehat{W}_i'$ is the completion of W_i' at 0. Let κ_i be a real number with

$$\kappa_i > \mathrm{mld}\big(\widehat{W}_i' \ni 0, \tfrac{1}{m}\operatorname{div}(\widehat{h}_i)\big).$$

By Lemma 3.2(1), with G, f_i, W_i', $\widehat{h}_i$, $\frac{1}{m}$, and κ_i in place of H, f_0, V_0, g_1, β_1, and γ, and with an integer $N_i' \geq \max\{d, L_{\kappa_i}\}$ in place of L, the polynomial $H_i := \mathrm{jet}_{N_i'}(\widehat{h}_i)$ satisfies the following: $(W_i', \frac{1}{m}\operatorname{div}(H_i))$ is klt near 0 by (1)(a), and by (1)(c),

$$\mathrm{mld}\big((\widehat{W}_i' \ni 0, \tfrac{1}{m}\operatorname{div}(\widehat{h}_i))/G'\big) = \mathrm{mld}\big((\widehat{W}_i' \ni 0, \tfrac{1}{m}\operatorname{div}(H_i))/G'\big)$$

for any subgroup $G' \subset G$, and any computing lift P of one side satisfies $\mathrm{mult}_P H_i = \mathrm{mult}_P \widehat{h}_i$ and is a computing lift of the other side. By Lemma 2.12(2) and (3), applied to $(W_i \ni o_i, \frac{1}{m}\operatorname{div}(h_i))$ and to $(W_i' \ni 0, \frac{1}{m}\operatorname{div}(H_i))$, and to $(W_i \ni o_i)$ and $(W_i' \ni 0)$ with zero boundary, with the action of any subgroup $G' \subset G$, and since $\phi_i \circ \Psi_i$ is G-equivariant, the quotient values of the germs and of their completions are equal, and computing lifts correspond with equal log discrepancies, inertia groups, and orders of the mark. Thus f_i and H_i satisfy (1).

We now construct $\zeta_{k,i}$. Since $\phi_i(u) - u \in \mathfrak{n}^d$ for any coordinate u and $N_i' \geq d$, we have $H_i \equiv \Psi_i(h_i) \mod \mathfrak{n}^d$. Since $f_i \equiv f_{0,s_i} \mod \mathfrak{n}^d$ and $d \geq \ell, N_1$, the specializations at η_i of $\mathfrak{b}$ and of ρ_k are

$$\mathfrak{b}_{\eta_i} = \big(\mathrm{jet}_\ell(f_i^Q H_i^{p'})\big) + (x,y,z,t)^\ell = \mathfrak{a}_i + (x,y,z,t)^\ell \quad \text{and} \quad \rho_{k,i} := \mathrm{jet}_{N_1}(\widetilde{\xi}_{k,i} - f_{0,s_i}\gamma_{k,i}).$$

Since $\eta_i \in T$ and $\Theta(\eta_i) \neq 0$, the choice of Θ and [Mus02, Proposition 2.1], in dimension 4, give near 0

$$\rho_{k,i} \in \mathcal{J}\big((\mathfrak{a}_i + (x,y,z,t)^\ell)^{\theta'/Q}\big) \subset \mathcal{J}(\mathbb{A}^4, \mathfrak{a}_i^{\theta/Q}) + (x,y,z,t)^{\lfloor \ell\epsilon \rfloor - 3} \subset \mathcal{J}(\mathbb{A}^4, \mathfrak{a}_i^{\theta/Q}) + (x,y,z,t)^{c+1}.$$

Thus near 0 we may write $\rho_{k,i} = z'/u' + v'$ with a polynomial $z' \in \mathcal{J}(\mathbb{A}^4, \mathfrak{a}_i^{\theta/Q})$, a polynomial u' with $u'(0) \neq 0$, and $v' \in (x,y,z,t)^{c+1}\mathcal{O}_{\mathbb{A}^4,0}$. Let w' be the sum of the terms of degree at most c

of $u'^{-1} \in \mathbb{C}[[x,y,z,t]]$. Then $\zeta_{k,i} := z'w'$ is a polynomial in $\mathcal{J}(\mathbb{A}^4, \mathfrak{a}_i^{\theta/Q})$ with $\rho_{k,i} - \zeta_{k,i} \in \mathfrak{n}^{c+1}$. By (ii) at β_i, $\rho_{k,i}$ is G-semi-invariant of character $\chi_{\xi_k}^{N_0}$. Since $\mathcal{J}(\mathbb{A}^4, \mathfrak{a}_i^{\theta/Q})$ and $\mathfrak{n}^{c+1}$ are G-stable, possibly replacing $\zeta_{k,i}$ with its component of character $\chi_{\xi_k}^{N_0}$, which is the character of $f_i\gamma_{k,i}$ (f_i has the character χ_{f_0} of f_{0,s_i} and $\gamma_{k,i}$ has the character $\chi_{\xi_k}^{N_0}\chi_{f_0}^{-1}$), we may assume that $\zeta_{k,i}$ is G-semi-invariant. Since $\rho_{k,i}$ is the N_1-jet of $\widetilde{\xi}_{k,i} - f_{0,s_i}\gamma_{k,i}$, $N_1 \geq c+1$, and $f_i - f_{0,s_i} \in \mathfrak{n}^d \subset \mathfrak{n}^{c+1}$,

$$\zeta_{k,i} + f_i\gamma_{k,i} \equiv \rho_{k,i} + f_{0,s_i}\gamma_{k,i} \equiv \widetilde{\xi}_{k,i} \mod \mathfrak{n}^{c+1}.$$

Since $\xi_k(0) = 0$ and β only contains the terms of positive degree of $\mathrm{jet}_d(\xi_k^{N_0})$, $\widetilde{\xi}_{k,i}(0) = 0$, so $(\zeta_{k,i} + f_i\gamma_{k,i})(0) = 0$.

Finally we prove the inclusion of (3). Let

$$\mathfrak{q}_i := (f_{0,s_i}) + \mathfrak{n} \cdot \mathrm{Jac}(f_{0,s_i}) + I_i^R.$$

By (iv) at β_i, for any monomial ω of degree c,

$$\omega \in (f_{0,s_i}) + \mathfrak{n} \cdot \mathrm{Jac}(f_{0,s_i}) + (\widetilde{\xi}_{1,i}, \dots, \widetilde{\xi}_{n,i})^R + \mathfrak{n}^{c+1}.$$

For any k, the congruence of $\zeta_{k,i} + f_i\gamma_{k,i}$ and $\widetilde{\xi}_{k,i}$ modulo $\mathfrak{n}^{c+1}$ implies

$$(\widetilde{\xi}_{1,i}, \dots, \widetilde{\xi}_{n,i})^R \subset I_i^R + \mathfrak{n}^{c+1}.$$

Thus

$$\mathfrak{n}^c \subset \mathfrak{q}_i + \mathfrak{n}^{c+1} = \mathfrak{q}_i + \mathfrak{n} \cdot \mathfrak{n}^c.$$

By Nakayama's lemma, $\mathfrak{n}^c \subset \mathfrak{q}_i$, which is the inclusion of (3). Since only the application of Lemma 2.7 above excludes finitely many i, the lemma follows. □

Proof of Theorem 14.2. We divide the proof into five steps.

Step 1. In this step we reduce the theorem to an upper bound for $a(P_i, W_i, 0)$.

Since $(W_i, \frac{1}{m}\mathrm{div}(h_i))$ is klt near o_i, we have $b_i > 0$, and since W_i is singular at o_i, Lemma 4.2(3) implies

$$0 < b_i \leq 2. \tag{D.103}$$

It suffices to find a real number L_2 and, on an infinite subfamily, a computing lift P_i of b_i with

$$a(P_i, W_i, 0) \leq L_2, \tag{D.104}$$

since then $a(P_i, W_i, 0)/e_{P_i} \leq L_2$ by $e_{P_i} \geq 1$.

Step 2. In this step we fix the constants and the polynomial data of Lemma D.16.

Since G is abelian, we fix linear coordinates $x_1, \dots, x_4$ of $\mathbb{A}^4$ on which G acts diagonally. The ideal $\mathfrak{n} \cdot \mathrm{Jac}(f)$ of Lemma D.16 does not depend on the choice of linear coordinates. We fix a family $(f_{0,s})_{s\in S}$ of polynomials $f_{0,s} = xy + q_{0,s}(z,t)$ with $\mathrm{mult}_0\, q_{0,s} \in [3, k_1]$ for any $s \in S$, satisfying the hypotheses of Lemma D.11, a rational number $\mu > \frac{1}{m}$, and a positive integer n with the property of Lemma D.16. For any closed point $s \in S$ let $W_{0,s} := (f_{0,s} = 0) \subset \mathbb{A}^4$. We fix the following.

(1) A real number L_1 with the property of Lemma D.11 for the family $(f_{0,s})_{s\in S}$, and $J := \frac{k_1 L_1}{2}$. Since $f_{0,s} = xy + q_{0,s}(z,t)$ with $\mathrm{mult}_0\, q_{0,s} \leq k_1$, Lemma 5.6 implies that for any closed point $s \in S$ and any prime divisor P over $W_{0,s}$ with center 0 and $a(P, W_{0,s}, 0) \leq L_1$,

$$\mathrm{mult}_P \mathrm{Jac}_{W_{0,s}} \leq \frac{k_1}{2}\big(a(P, W_{0,s}, 0) - 1\big) \leq J. \tag{D.105}$$

(2) A rational number $\theta \in (0,1)$ with

$$a_0 := \theta\mu - \frac{1}{m} - (1-\theta)\Big(1 + \frac{k_1}{m}\Big) > 0;$$

it exists since $\mu > \frac{1}{m}$.

(3) A positive integer Q with $p' := Q\mu \in \mathbb{Z}$.

(4) The real numbers

$$\alpha := \frac{a_0}{2-\theta}, \quad \beta := \frac{2M + (1-\theta)(2k_1M+1)}{2-\theta}, \quad \text{and} \quad T := \max\left\{1, \frac{2\beta}{\alpha}\right\} + 1.$$

(5) An integer $R \geq 2$ with

$$R > 2J + 2mL_1 + 3, \quad R > \frac{2}{\alpha}\Big(\frac{k_1}{m} + 2k_1M + 2\Big), \quad \text{and} \quad R \geq 2 + \frac{2}{\alpha}\Big(\frac{k_1}{m} + 2k_1M\Big). \tag{D.106}$$

(6) The positive integer c of Lemma D.16 for θ, Q, p', R, and $N := 2c+4$.

By Lemma D.16, possibly removing finitely many indices, we may assume that for any i there exist f_i, $W_i' = (f_i = 0)$, H_i, s_i, and $\zeta_{k,i}, \gamma_{k,i}$ for $1 \leq k \leq n$, satisfying Lemma D.16(1)–(3) with $N = 2c+4$. Write $\mathfrak{a}_i$ and I_i for the ideals defined in Lemma D.16(3), and let $\widehat{W}_{0,s_i}$ be the completion of W_{0,s_i} at 0. Since $\widehat{\mathcal{O}}_{W_i',0} \cong \widehat{\mathcal{O}}_{W_i,o_i}$, we have $\widehat{\mathcal{O}}_{W_i',0} \cong \mathbb{C}[[x,y,z,t]]/(xy+g_i)$, so Lemma 5.6 holds on $(W_i' \ni 0)$ with $\operatorname{mult}_0 g_i \leq k_1$. By Lemma D.16(1),

$$\operatorname{mld}\Big((W_i' \ni 0, \frac{1}{m}\operatorname{div}(H_i))/G\Big) = b_i, \tag{D.107}$$

and any computing lift of the left-hand side corresponds to a computing lift of b_i over W_i with equal log discrepancy.

Step 3. In this step we fix i and bound the order of I_i from below by the order of H_i.

Let F be a prime divisor over W_i' with center 0, and let

$$n_F := \operatorname{mult}_F H_i, \quad j_F := \operatorname{mult}_F \operatorname{Jac}_{W_i'}, \quad \text{and} \quad \lambda_F := \operatorname{mult}_F I_i.$$

Assume that $a(F, W_i', \frac{1}{m}\operatorname{div}(H_i)) \leq 2M$ and $\lambda_F < +\infty$. We show

$$\lambda_F > \alpha n_F - \beta. \tag{D.108}$$

Since $\lambda_F < +\infty$, there exists $k \in \{1, \dots, n\}$ with $\operatorname{mult}_F(\zeta_{k,i} + f_i\gamma_{k,i}) = \lambda_F$. Possibly reordering the coordinates, we may assume that $\operatorname{mult}_F(\partial f_i/\partial x_1) = j_F$. Let $k_F := n_F + 2j_F + \lambda_F + 1$. Since $H_i|_{W_i'} \neq 0$, $(\zeta_{k,i} + f_i\gamma_{k,i})|_{W_i'} \neq 0$, $k_F > 2j_F$, and $k_F - j_F > \max\{n_F, \lambda_F\}$, Lemma 5.4(2) for the integer k_F implies that there exists a prime divisor P over $\mathbb{A}^4$ with center 0 such that

$$\operatorname{mult}_P f_i = k_F, \quad \operatorname{mult}_P H_i = n_F, \quad \operatorname{mult}_P(\zeta_{k,i} + f_i\gamma_{k,i}) = \lambda_F, \quad \text{and} \quad a(P, \mathbb{A}^4, 0) = a(F, W_i', 0) + k_F.$$

Since $\operatorname{mult}_P(f_i\gamma_{k,i}) \geq k_F > \lambda_F$, we have $\operatorname{mult}_P \zeta_{k,i} = \lambda_F$. Computing $\mathcal{J}(\mathbb{A}^4, \mathfrak{a}_i^{\theta/Q})$ on a log resolution of $\mathfrak{a}_i$ on which P is a divisor, $\zeta_{k,i} \in \mathcal{J}(\mathbb{A}^4, \mathfrak{a}_i^{\theta/Q})$ implies

$$\operatorname{mult}_P \zeta_{k,i} \geq \Big\lfloor \frac{\theta}{Q} \operatorname{mult}_P \mathfrak{a}_i \Big\rfloor - a(P, \mathbb{A}^4, 0) + 1 > \frac{\theta}{Q} \operatorname{mult}_P \mathfrak{a}_i - a(P, \mathbb{A}^4, 0).$$

Since $\operatorname{mult}_P \mathfrak{a}_i = Qk_F + p'n_F$ and $p' = Q\mu$, we obtain

$$\lambda_F + a(F, W_i', 0) + k_F > \theta k_F + \theta\mu n_F.$$

Since

$$a(F, W_i', 0) = a\Big(F, W_i', \frac{1}{m}\operatorname{div}(H_i)\Big) + \frac{n_F}{m} \leq 2M + \frac{n_F}{m},$$

Lemma 5.6 implies $2j_F \leq k_1(a(F, W_i', 0) - 1) \leq k_1(\frac{n_F}{m} + 2M)$. Thus

$$\theta\mu n_F < \lambda_F + a(F, W_i', 0) + (1-\theta)k_F \leq \lambda_F + 2M + \frac{n_F}{m} + (1-\theta)\Big(n_F + \lambda_F + 1 + \frac{k_1 n_F}{m} + 2k_1M\Big),$$

that is,

$$a_0 n_F < (2-\theta)\lambda_F + 2M + (1-\theta)(2k_1M+1) = (2-\theta)(\lambda_F + \beta).$$

Dividing by $2-\theta$, we obtain (D.108).

Step 4. In this step we fix i and transport prime divisors between the germs of $(f_i = 0)$ and $(f_{0,s_i} = 0)$ at 0.

Since $f_i - f_{0,s_i} \in \mathfrak{n}^{2c+4} \subset \mathfrak{n}^3$, the quadratic parts of f_i and f_{0,s_i} are equal, and they are non-zero since $\operatorname{mult}_0 W_{0,s_i} = 2$. Since G acts linearly and the common quadratic part of the G-semi-invariant polynomials f_i and f_{0,s_i} is non-zero, $\chi_{f_i} = \chi_{f_{0,s_i}}$. The ideal $I_i^R\mathbb{C}[[x_1,\dots,x_4]]$ is G-stable, being generated by products of the G-semi-invariant elements $\zeta_{k,i} + f_i\gamma_{k,i}$, and by Lemma D.16(3)(c),

$$\mathfrak{n}^c \subset (f_{0,s_i}) + \mathfrak{n}\cdot \operatorname{Jac}(f_{0,s_i}) + I_i^R.$$

By Lemma 3.2(3) with $F_0 = f_{0,s_i}$, $F = f_i$, $I = I_i^R\mathbb{C}[[x_1,\dots,x_4]]$, and $L = 2c+4$, there exist a G-equivariant automorphism Φ of $\mathbb{C}[[x_1,\dots,x_4]]$ and a G-invariant unit v with $v\Phi(f_i) - f_{0,s_i} \in \Phi(I_i^R)$. Let

$$f_i' := v\Phi(f_i), \quad I_i' := \Phi(I_i)\mathbb{C}[[x_1,\dots,x_4]], \quad H_i' := \Phi(H_i), \quad \text{and}$$
$$W_i'' := (f_i' = 0) \subset \operatorname{Spec}\mathbb{C}[[x_1,\dots,x_4]].$$

Then

$$f_i' - f_{0,s_i} \in I_i'^R. \tag{D.109}$$

Since $(\Phi(f_i)) = (f_i')$, Φ induces a G-equivariant isomorphism from the completion of W_i' at 0 onto W_i'', and the Jacobian ideal of a hypersurface is intrinsic. Together with [Liu26a, Lemma 3.3(2)(3)], we obtain a bijection (injective by [dFEM11, Lemma 2.21 and Remark A.15]) $F \to F''$ from the set of prime divisors over W_i' with center 0 onto the set of prime divisors over W_i'' with center the closed point, such that

$$\begin{gathered} a(F'', W_i'', 0) = a(F, W_i', 0), \quad I_G(F'') = I_G(F), \quad \operatorname{mult}_{F''}\operatorname{Jac}_{W_i''} = \operatorname{mult}_F \operatorname{Jac}_{W_i'}, \\ \operatorname{mult}_{F''} H_i' = \operatorname{mult}_F H_i, \quad \text{and} \quad \operatorname{mult}_{F''} I_i' = \operatorname{mult}_F I_i. \end{gathered} \tag{D.110}$$

By (D.107), (D.110) and since $(W_i', \frac{1}{m}\operatorname{div}(H_i))$ is klt near 0, we have:

(i)
$$\operatorname{mld}((W_i'' \ni 0, \tfrac{1}{m}\operatorname{div}(H_i'))/G) = b_i,$$
and any computing lift of it corresponds to a computing lift of b_i over W_i with equal log discrepancy.

(ii) $a(F'', W_i'', \frac{1}{m}\operatorname{div}(H_i')) > 0$ for any prime divisor F'' over W_i'' with center the closed point.

(iii) (D.108) holds with F'' in place of F for any prime divisor F'' over W_i'' with center the closed point, $\operatorname{mult}_{F''} I_i' < +\infty$, and
$$a(F'', W_i'', \tfrac{1}{m}\operatorname{div}(H_i')) \le 2M,$$
where $n_{F''} := \operatorname{mult}_{F''} H_i'$ and $\lambda_{F''} := \operatorname{mult}_{F''} I_i'$.

By [Liu26a, Lemma 3.3(2)(3)], prime divisors over W_{0,s_i} with center 0 correspond (injectively by [dFEM11, Lemma 2.21 and Remark A.15]) to prime divisors over $\widehat{W}_{0,s_i}$ with center the closed point, and corresponding prime divisors have equal log discrepancies, equal inertia groups, and equal orders of polynomials. We denote a prime divisor over W_{0,s_i} and the corresponding prime divisor over $\widehat{W}_{0,s_i}$ by one letter.

Assume that $I_i' \subset (f_{0,s_i})$ or $I_i' \subset (f_i')$. If $I_i' \subset (f_{0,s_i})$, then by (D.109), $f_i' = f_{0,s_i}(1+u)$ for some $u \in (f_{0,s_i}^{R-1}) \subset \mathfrak{n}$. If $I_i' \subset (f_i')$, then by (D.109), $f_{0,s_i} = f_i'(1-u)$ for some $u \in (f_i'^{R-1}) \subset \mathfrak{n}$. In each case $(f_i') = (f_{0,s_i})$ since $R \ge 2$, so $W_i'' = \widehat{W}_{0,s_i}$, and $H_i'|_{\widehat{W}_{0,s_i}} \ne 0$ since $H_i|_{W_i'} \ne 0$. By (i),

$$\operatorname{mld}\Big((\widehat{W}_{0,s_i} \ni 0, \frac{1}{m}\operatorname{div}(H_i'))/G\Big) = b_i \ge 0,$$

so by Lemma D.11, it has a computing lift P with $a(P, W_{0,s_i}, 0) \le L_1$. By (i), (D.104) holds for i with $L_2 = L_1$.

It remains to treat the indices i with $I_i' \not\subset (f_{0,s_i})$ and $I_i' \not\subset (f_i')$. Since $\widehat{W}_{0,s_i}$ and W_i'' are integral and the generators $\Phi(\zeta_{k,i} + f_i\gamma_{k,i})$ of I_i' belong to $\mathfrak{n}$, for any prime divisor F over $\widehat{W}_{0,s_i}$ or over W_i'' with center the closed point,

$$1 \le \operatorname{mult}_F I_i' < +\infty. \tag{D.111}$$

Let F be a prime divisor over one of $\widehat{W}_{0,s_i}$ and W_i'' with center the closed point, j_F the Jacobian order of F and $\lambda_F := \operatorname{mult}_F I_i'$, and assume that $R\lambda_F > 2j_F + 2\lambda_F$. By (D.109), (D.111), and Lemma 8.1(1) with $f_0 = f_{0,s_i}$, $f' = f_i'$, $I = I_i'$, and $P' = R$, F corresponds to a prime divisor $\widetilde{F}$ over the other germ with center the closed point such that the following hold.

(a) The log discrepancies of $\widetilde{F}$ and of F with boundary 0 are equal, and $I_G(\widetilde{F}) = I_G(F)$.
(b) The Jacobian order of $\widetilde{F}$ is j_F, and $\operatorname{mult}_{\widetilde{F}} I_i' = \lambda_F$.
(c)

$$\begin{gathered}\operatorname{mult}_{\widetilde{F}} H_i' = \operatorname{mult}_F H_i' \text{ if } \operatorname{mult}_F H_i' < R\lambda_F - j_F, \quad \text{and} \\ \operatorname{mult}_{\widetilde{F}} H_i' \ge \min\{\operatorname{mult}_F H_i', R\lambda_F - j_F\}.\end{gathered} \tag{D.112}$$

For the second inequality, by Lemma 8.1(1)(v), $\widetilde{F}$ satisfies the assumptions of Lemma 8.1(1) with $\widehat{W}_{0,s_i}$ and W_i'' exchanged, and the prime divisor associated with it is F. Thus

$$\operatorname{mult}_F H_i' = \operatorname{mult}_{\widetilde{F}} H_i' \quad \text{if} \quad \operatorname{mult}_{\widetilde{F}} H_i' < R\lambda_F - j_F.$$

Step 5. In this step we show (D.104) for the indices i left at the end of **Step 4**.

Fix an index i as at the end of **Step 4**, and let $L_2 := \max\{L_1, \frac{T}{m} + 2M\}$. Let F be a computing lift of $\operatorname{mld}((W_i'' \ni 0, \frac{1}{m}\operatorname{div}(H_i'))/G) = b_i$, which exists since b_i has a computing lift over W_i and the correspondences of Lemma D.16(1) and (D.110) are bijective, and let

$$n_F := \operatorname{mult}_F H_i', \quad j_F := \operatorname{mult}_F \operatorname{Jac}_{W_i''}, \quad \text{and} \quad \lambda_F := \operatorname{mult}_F I_i'.$$

Since $e_F \le |G| \le M$, (D.103) implies

$$a(F, W_i'', 0) = \frac{n_F}{m} + e_F b_i \le \frac{n_F}{m} + 2M, \quad \text{so} \quad a\Big(F, W_i'', \frac{1}{m}\operatorname{div}(H_i')\Big) \le 2M.$$

If $n_F \le T$, then $a(F, W_i'', 0) \le \frac{T}{m} + 2M$, and by (i), (D.104) holds for i.

Assume that $n_F > T$. Since $n_F > T > \frac{2\beta}{\alpha}$, (iii) and (D.111) imply $\lambda_F > \alpha n_F - \beta > \frac{\alpha n_F}{2}$. Since $n_F \ge 1$, Lemma 5.6 implies

$$2j_F \le k_1\big(a(F, W_i'', 0) - 1\big) < k_1\Big(\frac{n_F}{m} + 2M\Big) \le \Big(\frac{k_1}{m} + 2k_1M\Big)n_F.$$

By (D.106),

$$\begin{gathered}2j_F + 2\lambda_F < \Big(\frac{k_1}{m} + 2k_1M\Big)n_F + 2\lambda_F \le \frac{(R-2)\alpha n_F}{2} + 2\lambda_F < R\lambda_F \quad \text{and} \\ 2j_F + 2n_F < \Big(\frac{k_1}{m} + 2k_1M + 2\Big)n_F < \frac{R\alpha n_F}{2} < R\lambda_F.\end{gathered}$$

In particular $n_F < R\lambda_F - j_F$. By (a) and (D.112), F corresponds to a prime divisor $\widetilde{F}$ over $\widehat{W}_{0,s_i}$ with $a(\widetilde{F}, W_{0,s_i}, 0) = a(F, W_i'', 0)$, $e_{\widetilde{F}} = e_F$, and $\operatorname{mult}_{\widetilde{F}} H_i' = n_F < +\infty$. Thus $H_i'|_{\widehat{W}_{0,s_i}} \ne 0$, and

$$\frac{a(\widetilde{F}, W_{0,s_i}, \frac{1}{m}\operatorname{div}(H_i'))}{e_{\widetilde{F}}} = \frac{a(F, W_i'', \frac{1}{m}\operatorname{div}(H_i'))}{e_F} = b_i.$$

Let

$$b_{0,i} := \operatorname{mld}\Big((\widehat{W}_{0,s_i} \ni 0, \frac{1}{m}\operatorname{div}(H_i'))/G\Big).$$

Then

$$b_{0,i} \le b_i. \tag{D.113}$$

We show $b_{0,i} \geq 0$. Otherwise, by Lemma D.11, there exists a prime divisor P over W_{0,s_i} with center 0, $a(P, W_{0,s_i}, 0) \leq L_1$, and $a(P, W_{0,s_i}, \frac{1}{m}\operatorname{div}(H_i')) < 0$, that is, $\operatorname{mult}_P H_i' > ma(P, W_{0,s_i}, 0)$. Let $j_P := \operatorname{mult}_P \operatorname{Jac}_{W_{0,s_i}}$ and $\lambda_P := \operatorname{mult}_P I_i'$. By (D.105), (D.111), and (D.106), $j_P \leq J$, $\lambda_P \geq 1$,

$$(R-2)\lambda_P \geq R-2 > 2J \geq 2j_P \quad \text{and} \quad R\lambda_P - j_P \geq R-J > mL_1 \geq ma(P, W_{0,s_i}, 0).$$

Thus $R\lambda_P > 2j_P + 2\lambda_P$, and by (a) and (D.112), P corresponds to a prime divisor $\widetilde{P}$ over W_i'' with center the closed point, $a(\widetilde{P}, W_i'', 0) = a(P, W_{0,s_i}, 0)$, and

$$\operatorname{mult}_{\widetilde{P}} H_i' \geq \min\{\operatorname{mult}_P H_i', R\lambda_P - j_P\} > ma(P, W_{0,s_i}, 0).$$

Then $a(\widetilde{P}, W_i'', \frac{1}{m}\operatorname{div}(H_i')) < 0$, which contradicts (ii). Thus $b_{0,i} \geq 0$.

By Lemma D.11, $b_{0,i}$ has a computing lift P with $a(P, W_{0,s_i}, 0) \leq L_1$. Let $j_P := \operatorname{mult}_P \operatorname{Jac}_{W_{0,s_i}}$ and $\lambda_P := \operatorname{mult}_P I_i'$. Since $b_{0,i} \geq 0$,

$$\operatorname{mult}_P H_i' = m\big(a(P, W_{0,s_i}, 0) - e_P b_{0,i}\big) \leq mL_1.$$

As in the previous paragraph, $R\lambda_P > 2j_P + 2\lambda_P$ and $R\lambda_P - j_P > mL_1 \geq \operatorname{mult}_P H_i'$. By (a) and (D.112), P corresponds to a prime divisor $\widetilde{P}$ over W_i'' with center the closed point, $a(\widetilde{P}, W_i'', 0) = a(P, W_{0,s_i}, 0)$, $e_{\widetilde{P}} = e_P$, and $\operatorname{mult}_{\widetilde{P}} H_i' = \operatorname{mult}_P H_i'$. Thus

$$b_i \leq \frac{a(\widetilde{P}, W_i'', \frac{1}{m}\operatorname{div}(H_i'))}{e_{\widetilde{P}}} = \frac{a(P, W_{0,s_i}, \frac{1}{m}\operatorname{div}(H_i'))}{e_P} = b_{0,i}.$$

By (D.113), $b_{0,i} = b_i$, so $\widetilde{P}$ is a computing lift of $\operatorname{mld}((W_i'' \ni 0, \frac{1}{m}\operatorname{div}(H_i'))/G) = b_i$ with $a(\widetilde{P}, W_i'', 0) \leq L_1$. By (i), (D.104) holds for i.

By **Steps 4** and 5, (D.104) holds with $L_2 = \max\{L_1, \frac{T}{m} + 2M\}$ for any i, and **Step 1** implies the theorem. □

D.3. **Limits of types cE and D_n.** In this subsection we prove Theorems 14.4 and 14.5, after the following lemma.

Lemma D.17. *(1) Let M and N be positive integers. Then there exists a positive integer $d(N, M)$ satisfying the following. Let G be a finite abelian group of order at most M acting faithfully and diagonally on $\mathbb{C}[[x_1, \dots, x_4]]$, let $\mathfrak{n}$ be the maximal ideal of $\mathbb{C}[[x_1, \dots, x_4]]$, and let $f \in \mathbb{C}[[x_1, \dots, x_4]]$ be a G-semi-invariant element such that the closed point is an isolated singular point of $(f = 0)$. Then there exists a G-semi-invariant polynomial g with the character of f such that 0 is an isolated singular point of $(g = 0) \subset \mathbb{A}^4$ and*

$$\deg g \leq d(N, M), \qquad g - f \in \mathfrak{n}^N.$$

Moreover, let $f' \in \mathbb{C}[[x_1, \dots, x_4]]$ with $f' - f \in \mathfrak{n}^N$, for instance $f' = g$. Then the following hold.

(a) If $N \geq 16$, $(f = 0)$ is of type cE, and E_j is the type of a general hyperplane section of $(f = 0)$ through the closed point, then a general hyperplane section of $(f' = 0)$ through the closed point is Du Val of type E_j.

(b) If $N \geq 2n + 6$ for an integer $n \geq 4$ and a linear form ℓ cuts out a Du Val section of type D_n of $(f = 0)$, then ℓ cuts out a Du Val section of type D_n of $(f' = 0)$.

(2) Let m, M, d be positive integers, and let $L(m, M, d)$ be the constant of Theorem 13.3(1). Let $(V \ni 0) = (g = 0) \subset \mathbb{A}^4$ be an isolated Gorenstein terminal germ with $\deg g \leq d$, let G be a finite diagonal abelian group of order at most M preserving g up to a character and acting faithfully on V, let $\widehat{V}$ be the completion of V at 0, and let $h \in \mathbb{C}[[x_1, \dots, x_4]]$ be a G-semi-invariant element whose restriction to $\widehat{V}$ is non-zero and not a unit. Assume that

$$b := \operatorname{mld}((\widehat{V} \ni 0, \tfrac{1}{m}\operatorname{div}(h))/G) > 0.$$

Then b has a computing lift F, a prime divisor over V with center 0 regarded as a formal prime divisor over $\widehat{V}$ (Lemma 2.12(3)), such that

$$\frac{a(F,V,0)}{e_F} \leq L(m,M,d).$$

Proof. We prove (1). We first fix G, the characters $\chi_1,\dots,\chi_4$ of G on $x_1,\dots,x_4$, and the character χ of f. Since G acts diagonally, a power series is χ-semi-invariant if and only if any monomial occurring in it has character χ. In particular, any truncation of f by degree is χ-semi-invariant. Since the closed point is a singular point of $(f=0)$, $f\in\mathfrak{n}^2$.

Let $\mathcal{T}_N$ be the affine space of χ-semi-invariant elements of $\mathfrak{n}^2/\mathfrak{n}^N$ (a point if $N\leq 2$), and for any integer $k\geq N$ let $\mathcal{P}_k$ be the affine space of χ-semi-invariant polynomials in $(x_1,\dots,x_4)^2$ of degree at most k. The truncation $\pi_k:\mathcal{P}_k\to\mathcal{T}_N$ is a surjective linear map, hence open. Let

$$\mathcal{Z}_k := \{(z,g)\in\mathbb{A}^4\times\mathcal{P}_k \mid g(z)=\partial_1 g(z)=\cdots=\partial_4 g(z)=0\}$$

with the projection $\mathcal{Z}_k\to\mathcal{P}_k$. Then $\{0\}\times\mathcal{P}_k\subset\mathcal{Z}_k$. Since the dimension at a point of the fiber through it is upper semicontinuous on $\mathcal{Z}_k$, the set

$$\mathcal{P}_k^\circ := \{g\in\mathcal{P}_k \mid 0 \text{ is an isolated singular point of } (g=0)\},$$

which consists of the g such that $(0,g)$ is an isolated point of the fiber over g, is open in $\mathcal{P}_k$. Thus $\mathcal{U}_k := \pi_k(\mathcal{P}_k^\circ)$ is open in $\mathcal{T}_N$. Since $\mathcal{P}_k\subset\mathcal{P}_{k+1}$, $\pi_{k+1}|_{\mathcal{P}_k}=\pi_k$ and $\mathcal{P}_k^\circ=\mathcal{P}_{k+1}^\circ\cap\mathcal{P}_k$, we have $\mathcal{U}_k\subset\mathcal{U}_{k+1}$. Since $\mathcal{T}_N$ is Noetherian, there exists an integer $k_0\geq N$ with $\mathcal{U}_k=\mathcal{U}_{k_0}$ for any $k\geq k_0$.

We show that the class $[f]$ of f in $\mathcal{T}_N$ is in $\mathcal{U}_{k_0}$. By the Jacobian criterion, $(f,\partial_1 f,\dots,\partial_4 f)$ is $\mathfrak{n}$-primary, so $\mathfrak{n}^{N_1}\subset(f,\partial_1 f,\dots,\partial_4 f)$ for a positive integer N_1. Let $k:=\max\{N_1+1,N\}$, and let g_1 be the sum of the homogeneous parts of f of degree at most k. Then $g_1\in\mathcal{P}_k$ and $\pi_k(g_1)=[f]$. Since $f-g_1\in\mathfrak{n}^{N_1+2}$, the ideal $I_1:=(g_1,\partial_1 g_1,\dots,\partial_4 g_1)$ satisfies

$$\mathfrak{n}^{N_1}\subset I_1+\mathfrak{n}^{N_1+1},$$

so $\mathfrak{n}^{N_1}\subset I_1$ by Nakayama's lemma. Since $\mathbb{C}[[x_1,\dots,x_4]]$ is faithfully flat over $\mathcal{O}_{\mathbb{A}^4,0}$, we have the inclusion

$$\mathfrak{m}_{\mathbb{A}^4,0}^{N_1}\subset(g_1,\partial_1 g_1,\dots,\partial_4 g_1)\subset\mathcal{O}_{\mathbb{A}^4,0}.$$

Thus $g_1\in\mathcal{P}_k^\circ$ and

$$[f]\in\mathcal{U}_k\subset\mathcal{U}_{\max\{k,k_0\}}=\mathcal{U}_{k_0}.$$

Therefore $\pi_{k_0}^{-1}([f])\cap\mathcal{P}_{k_0}^\circ$ is a non-empty open subset of the affine space $\pi_{k_0}^{-1}([f])$, so it contains a polynomial g. Then g is χ-semi-invariant, $\deg g\leq k_0$, $g-f\in\mathfrak{n}^N$, and 0 is an isolated singular point of $(g=0)$.

The integer k_0 depends on $(G,\chi_1,\dots,\chi_4,\chi)$. Since G acts faithfully and diagonally and $|G|\leq M$, G is a subgroup of the group of diagonal matrices whose entries are $M!$-th roots of unity, so there exist only finitely many possibilities for $(G,\chi_1,\dots,\chi_4,\chi)$. We let $d(N,M)$ be the maximum of the corresponding integers k_0.

Now let $f'\in\mathbb{C}[[x_1,\dots,x_4]]$ with $f'-f\in\mathfrak{n}^N$. Let H be a hyperplane through 0, let $R:=\mathbb{C}[[v_1,v_2,v_3]]$ with maximal ideal $\mathfrak{m}$ be the completed local ring of H at 0, and assume that $f|_H$ defines a Du Val germ of type E_6, E_7, E_8, or D_n. For $u\in R$ let

$$T(u) := (u)+\mathfrak{m}\cdot\Big(\frac{\partial u}{\partial v_1},\frac{\partial u}{\partial v_2},\frac{\partial u}{\partial v_3}\Big).$$

Let Φ be the normal form of the type of $f|_H$, namely

$$v_1^2+v_2^3+v_3^4\ (E_6),\quad v_1^2+v_2^3+v_2v_3^3\ (E_7),\quad v_1^2+v_2^3+v_3^5\ (E_8),\quad\text{and}\quad v_1^2+v_2^2v_3+v_3^{n-1}\ (D_n),$$

let $J_\Phi\subset R$ be the ideal generated by the partial derivatives of Φ with respect to v_1,v_2,v_3, and let $c:=6$ for the types E_6,E_7,E_8 and $c:=n$ for the type D_n. Then J_Φ is

$$(v_1,v_2^2,v_3^3)\ (E_6),\quad (v_1,3v_2^2+v_3^3,v_2v_3^2)\ (E_7),\quad (v_1,v_2^2,v_3^4)\ (E_8),\quad\text{and}\quad (v_1,v_2v_3,v_2^2+(n-1)v_3^{n-2})\ (D_n),$$

and $\mathfrak{m}^{c-1} \subset J_\Phi$, for E_7 since $v_2^3, v_3^5 \in (3v_2^2 + v_3^3, v_2v_3^2)$, and for D_n since $v_2^{n-1}, v_3^{n-1} \in (v_2v_3, v_2^2 + (n-1)v_3^{n-2})$ as $n \geq 4$. Thus

$$\mathfrak{m}^c \subset (\Phi) + \mathfrak{m}J_\Phi = T(\Phi).$$

Since $f|_H$ and Φ define singular germs, $f|_H, \Phi \in \mathfrak{m}^2$, so an isomorphism $R/(\Phi) \cong R/(f|_H)$ lifts to an automorphism ψ of R with $(\psi(\Phi)) = (f|_H)$. By the chain rule $T(\psi(\Phi)) = \psi(T(\Phi))$, and by the product rule $T(u)$ does not change when u is multiplied by a unit, so

$$\mathfrak{m}^c \subset T(f|_H).$$

Since $f'|_H - f|_H \in \mathfrak{m}^N$ and $N > 2c$ under the hypothesis of (1)(a), resp. (1)(b) ($16 > 12$ for type cE, $2n + 6 > 2n$ for type D_n), Lemma 3.2(2) with G trivial, $F = f|_H$, $F_0 = f'|_H$, and $L = N$ shows that $f'|_H$ is a unit times $\psi'(f|_H)$ for an automorphism ψ' of R, so $f'|_H$ defines a Du Val germ of the type of $f|_H$. Taking $H = (\ell = 0)$, we obtain (1)(b). If $(f = 0)$ is of type cE and E_j is the type of a general hyperplane section, then the hyperplanes H with $f|_H$ of type E_j contain a non-empty open subset of the space of hyperplanes through 0, and $f'|_H$ is of type E_j for any such H. Thus a general hyperplane section of $(f' = 0)$ is Du Val of type E_j, which is (1)(a). This proves (1).

We prove (2). Let $r := |G|$, let $\pi : V \to Q := V/G$ be the quotient, $y := \pi(0)$, and let B be the branch divisor of π, so that $K_V = \pi^*(K_Q + B)$ and $B \in \{1 - \frac{1}{e} \mid 2 \leq e \leq M\}$. Since r annihilates the character of h, $w := h^r$ is G-invariant, and the restriction of w to $\widehat{V}$, still denoted by w, belongs to $\widehat{\mathcal{O}}_{Q,y}$ by Lemma 2.12(1). Since V is normal at 0, $\widehat{V}$ is integral, so $w \neq 0$.

For a prime divisor P over Q with center y and a non-zero $u \in \widehat{\mathcal{O}}_{Q,y}$, we let $\operatorname{mult}_P u$ be the order of u along the formal prime divisor over the completion of Q at y corresponding to P by Lemma 2.12(3) (unique by [dFEM11, Lemma 2.21 and Remark A.15]), and

$$\lambda(P, u) := a(P, Q, B) - \frac{\operatorname{mult}_P u}{mr}.$$

For a prime divisor F over V with center 0 over P,

$$a(F, V, 0) = e_F a(P, Q, B) \quad \text{and} \quad \operatorname{mult}_F u = e_F \operatorname{mult}_P u.$$

Since $\operatorname{mult}_F w = r \operatorname{mult}_F h$, we obtain $a(F, V, \frac{1}{m}\operatorname{div}(h))/e_F = \lambda(P, w)$, so by Definition 2.2 and Lemma 2.12(3),

$$b = \inf_P \lambda(P, w), \tag{D.114}$$

where P runs through the prime divisors over Q with center y. For a G-semi-invariant polynomial h'' with the character of h and $h''|_V \neq 0$, the identity $a(F, V, \frac{1}{m}\operatorname{div}(h''))/e_F = \lambda(P, h''^r)$ implies

$$\operatorname{mld}((V \ni 0, \tfrac{1}{m}\operatorname{div}(h''))/G) = \inf_P \lambda(P, h''^r) = \operatorname{mld}(Q \ni y, B, (h''^r)^{\frac{1}{mr}}). \tag{D.115}$$

Let $\Gamma := \{1 - \frac{1}{e} \mid 2 \leq e \leq M\} \cup \{\frac{1}{mr}\}$. By Theorem 2.22(2) for Q and Γ, there exists a positive integer N' such that for any ideals $\mathfrak{a}_1, \mathfrak{a}_2 \subset \mathcal{O}_{Q,y}$,

$$\mathfrak{a}_1 + \mathfrak{m}_{Q,y}^{N'} = \mathfrak{a}_2 + \mathfrak{m}_{Q,y}^{N'} \implies \operatorname{mld}(Q \ni y, B, \mathfrak{a}_1^{1/(mr)}) = \operatorname{mld}(Q \ni y, B, \mathfrak{a}_2^{1/(mr)}).$$

Let $w_0 \in \mathcal{O}_{Q,y}$ with $w_0 - w \in \widehat{\mathfrak{m}}_{Q,y}^{N'}$, let $\mathfrak{a} := (w_0) + \mathfrak{m}_{Q,y}^{N'}$, and let $\beta := \operatorname{mld}(Q \ni y, B, \mathfrak{a}^{1/(mr)})$. For any $u \in \mathcal{O}_{Q,y}$ with $u - w \in \widehat{\mathfrak{m}}_{Q,y}^{N'}$, we have

$$u - w_0 \in \widehat{\mathfrak{m}}_{Q,y}^{N'} \cap \mathcal{O}_{Q,y} = \mathfrak{m}_{Q,y}^{N'},$$

so $(u) + \mathfrak{m}_{Q,y}^{N'} = \mathfrak{a}$ and

$$\operatorname{mld}(Q \ni y, B, (u)^{\frac{1}{mr}}) = \beta. \tag{D.116}$$

We show that $\beta = b$. Since $w - w_0 \in \widehat{\mathfrak{m}}_{Q,y}^{N'}$, $w \in \mathfrak{a}\widehat{\mathcal{O}}_{Q,y}$, so $\operatorname{mult}_P w \geq \operatorname{mult}_P \mathfrak{a}$ and

$$\lambda(P, w) \leq a(P, Q, B, \mathfrak{a}^{1/(mr)}) \quad \text{for any } P.$$

Thus $b \le \beta$ by (D.114). Conversely, fix P, let $k > \max\{N', \operatorname{mult}_P w\}$ be an integer, and let $u \in \mathcal{O}_{Q,y}$ with $u - w \in \widehat{\mathfrak{m}}^k_{Q,y}$. Then $\operatorname{mult}_P(u-w) \ge k > \operatorname{mult}_P w$, so $\operatorname{mult}_P u = \operatorname{mult}_P w$, and by (D.116),

$$\lambda(P,w) = \lambda(P,u) \ge \operatorname{mld}(Q \ni y, B, (u)^{\frac{1}{mr}}) = \beta.$$

Thus $b \ge \beta$ by (D.114).

Since $\pi^{-1}(y) = \{0\}$, there exists a positive integer N_2 with

$$\mathfrak{m}^{N_2}_{V,0} \subset \mathfrak{m}_{Q,y}\mathcal{O}_{V,0}.$$

Let $L := L(m,M,d)$, and let s be an integer with $s \ge N_2N'$, $s > mML$, and $h|_{\widehat{V}} \notin \widehat{\mathfrak{m}}^s_{V,0}$. Let h' be the sum of the homogeneous parts of h of degree less than s. Then h' is a G-semi-invariant polynomial with the character of h, $h - h' \in \mathfrak{n}^s$, and $h'|_V$ is non-zero and not a unit. The polynomial h'^r is G-invariant, so $u := h'^r|_V \in \mathcal{O}_{Q,y}$, and

$$u - w \in \widehat{\mathfrak{m}}^s_{V,0} \subset \widehat{\mathfrak{m}}^{N_2N'}_{V,0} \subset \mathfrak{m}^{N'}_{Q,y}\widehat{\mathcal{O}}_{V,0}.$$

Since $u - w$ is G-invariant and the Reynolds operator $\frac{1}{r}\sum_{\sigma\in G}\sigma$ is $\widehat{\mathcal{O}}_{Q,y}$-linear and the identity on G-invariant elements, $u - w \in \widehat{\mathfrak{m}}^{N'}_{Q,y}$. By (D.115), (D.116) and $\beta = b$,

$$\operatorname{mld}((V \ni 0, \tfrac{1}{m}\operatorname{div}(h'))/G) = b > 0.$$

In particular, $(V, \frac{1}{m}\operatorname{div}(h'))$ is lc near 0. By Theorem 13.3(1) for the polynomial mark h', b has a computing lift F for h' with $a(F,V,0)/e_F \le L$. Then $\operatorname{mult}_F h'/(me_F) = a(F,V,0)/e_F - b < L$ since $b > 0$, and since $e_F \le M$ and $\operatorname{mult}_F \mathfrak{n} \ge 1$,

$$\operatorname{mult}_F h' < mML < s \le \operatorname{mult}_F(h - h').$$

Hence $\operatorname{mult}_F h = \operatorname{mult}_F h'$, so $a(F,V,\frac{1}{m}\operatorname{div}(h))/e_F = b$. Thus F is a computing lift of b and $a(F,V,0)/e_F \le L$. The lemma follows. □

Proof of Theorems 14.4 and 14.5. We prove Theorems 14.4 and 14.5 simultaneously. We say that we are in case (E) under the hypotheses of Theorem 14.4, and in case (N) under the hypotheses of Theorem 14.5. The cases differ in the Jacobian slope λ fixed below, in the integer N and the property of W'_i in **Step 4**, in the proof of (D.120), and in the reason for $b_i > 0$ in **Step 6**.

Let $\widehat{\mathcal{O}} := \mathbb{C}[[x_1,\dots,x_4]]$, with $x_1,\dots,x_4$ the parameters of Set-up 14.3(3), and let $\mathfrak{n}$ be the maximal ideal of $\widehat{\mathcal{O}}$. Let $\mathcal{I} \subset \mathbb{N}^+$ be the infinite subset along which (f_*,h_*) is a compatible generic limit of the tuples (f_i,h_i), and $\Bbbk$ the coefficient field of (f_*,h_*) (Definition 2.5). Let $\mathcal{X}_* := \operatorname{Spec}\Bbbk[[x_1,\dots,x_4]]$, with maximal ideal $\mathfrak{n}_*$, so that $W_* = (f_* = 0) \subset \mathcal{X}_*$. For $g \in \widehat{\mathcal{O}}$, resp. $g \in \Bbbk[[x_1,\dots,x_4]]$, let

$$T(g) := (g) + \mathfrak{n}\cdot(\partial_1 g,\dots,\partial_4 g), \quad \text{resp.} \quad T(g) := (g) + \mathfrak{n}_*\cdot(\partial_1 g,\dots,\partial_4 g).$$

In case (E), let $\lambda := 15$. In case (N), fix a rational number $\kappa_n > 0$ such that $(S_n, \operatorname{Jac}_{S_n}^{\kappa_n})$ is lc near 0 for the surface S_n of Theorem 8.4, which exists since S_n is klt, and let $\lambda := 1/\kappa_n$. We divide the proof into six steps.

Step 1. In this step we fix the exponents $\mu,\theta_1,\theta_2,\theta_3$ and the integer P'.

By hypothesis, $(W_*, \frac{1}{m}\operatorname{div}(h_*))$ is lc, so $h_*|_{W_*} \ne 0$. For the generic point ξ of any of the finitely many 1-dimensional components of $\operatorname{Sing}(W_*)$, the pair $(W_*, \frac{1}{m}\operatorname{div}(h_*|_{W_*}))$ is klt at ξ, that is, $a(E,W_*,\frac{1}{m}\operatorname{div}(h_*|_{W_*})) > 0$ for the finitely many prime divisors E on a log resolution of $(W_*, \operatorname{div}(h_*|_{W_*}))$ near ξ whose centers contain ξ [dFEM11, Theorems 2.3 and 2.4]. Since

$$a\big(E, W_*, \nu\operatorname{div}(h_*|_{W_*})\big) = a(E,W_*,0) - \nu\operatorname{mult}_E h_*$$

is continuous in ν, there exists a rational number $\mu > \frac{1}{m}$ such that

$$\Big(W_*, \mu\operatorname{div}(h_*|_{W_*})\Big) \text{ is klt at } \xi \text{ for any such } \xi.$$

If $\mathrm{Sing}(W_*)$ has no 1-dimensional component, let $\mu > \frac{1}{m}$ be any rational number. Since $\theta\mu - \frac{1}{m} - (1-\theta)(1+\frac{2\lambda}{m})$ tends to $\mu - \frac{1}{m} > 0$ as θ tends to 1, there exist rational numbers $0 < \theta_1 < \theta_2 < 1$ with the inequality in the hypotheses of Theorem 8.2:

$$\theta_1\mu - \frac{1}{m} - (1-\theta_1)\Big(1+\frac{2\lambda}{m}\Big) > 0.$$

Let $P' \geq 2$ and $L_1 > 0$ be the integer and the real number of Theorem 8.2 for $m, M, \lambda, 3, \mu, \theta_1, \theta_2$ (denoted P' and C there, with 3 as the constant B_0 there). Fix a rational number $\theta_3 \in (\theta_2, 1)$, and let

$$\mathcal{J}_* := \mathcal{J}\big(\mathcal{X}_*, f_*^{\theta_3} h_*^{\theta_3\mu}\big), \qquad \mathcal{J}_i := \mathcal{J}\big(\operatorname{Spec}\widehat{\mathcal{O}}, f_i^{\theta_2} h_i^{\theta_2\mu}\big) \text{ for any } i \in \mathcal{I}.$$

Step 2. In this step we show that there exists a positive integer c with

$$\mathfrak{n}_*^c \subset T(f_*) + \mathcal{J}_*^{P'}. \tag{D.117}$$

Let $V_* \subset W_*$ be the union of the closed point and the non-klt locus of $(W_*, \mu\,\mathrm{div}(h_*|_{W_*}))$, a closed subset. By Lemma 6.1(3) with θ_3 and V_* in place of θ_2 and V, $\mathcal{J}_*$ is the unit ideal near any point of $W_*\backslash V_*$, in particular at the generic point ξ of any 1-dimensional component of $\mathrm{Sing}(W_*)$, since $\xi \notin V_*$ by **Step 1**. Since W_* is normal of dimension 3, $\dim \mathrm{Sing}(W_*) \leq 1$. At any point of $\mathcal{X}_*$ other than the closed point, $\mathfrak{n}_*$ is the unit ideal, so $T(f_*)$ coincides there with $(f_*, \partial_1 f_*, \dots, \partial_4 f_*)$, whose zero set is $\mathrm{Sing}(W_*)$ by the Jacobian criterion [Mat89, Theorem 30.8 and the paragraph following it]. Thus any irreducible component of the zero set $V(T(f_*))$ is either the closed point or the closure $\overline{\{\xi\}}$ of the generic point ξ of such a component. For any such ξ, since $\xi \notin V(\mathcal{J}_*)$, the set $V(\mathcal{J}_*) \cap \overline{\{\xi\}}$ is a proper closed subset of the 1-dimensional irreducible local scheme $\overline{\{\xi\}}$, so it is contained in the closed point. Hence

$$V\big(T(f_*) + \mathcal{J}_*^{P'}\big) = V(T(f_*)) \cap V(\mathcal{J}_*)$$

is contained in the closed point, and (D.117) holds for some $c \geq 1$.

Step 3. In this step we show that, possibly replacing $\mathcal{I}$ with a cofinite subset,

$$\mathfrak{n}^c \subset T(f_i) + \mathcal{J}_i^{P'} \quad \text{for any } i \in \mathcal{I}. \tag{D.118}$$

We use the substitution $\Phi(\mathbf{g}, \beta)$ of Lemma 6.1 for pairs $\mathbf{g} = (g, g')$. Let $z_1, \dots, z_r$ be generators of $\mathcal{J}_*$. Let $u_{j,\alpha}$, for $1 \leq j \leq r$ and $\alpha \in \mathbb{N}^4$ with $|\alpha| < c+1$, be the variables $u_1, \dots, u_q$, where $q = r \cdot \#\{\alpha \in \mathbb{N}^4 \mid |\alpha| \leq c\}$, let

$$\Phi_j := \sum_{|\alpha|<c+1} u_{j,\alpha} x^\alpha \quad \text{for } 1 \leq j \leq r,$$

and let $\beta \in \Bbbk^q$ have $u_{j,\alpha}$-entry the coefficient of x^α in z_j. Then $\Phi_j((f_*, h_*), \beta) - z_j \in \mathfrak{n}_*^{c+1}$, so

$$\Phi_j((f_*, h_*), \beta) \in \mathcal{J}_* + \mathfrak{n}_*^{c+1} \quad \text{for } 1 \leq j \leq r.$$

Let $\Psi_1, \dots, \Psi_{17}$ be the polynomials $y_{1,(0,0,0,0)}$ and $x_k y_{1,\alpha}$, for $1 \leq k \leq 4$ and $\alpha \in \mathbb{N}^4$ with $|\alpha| = 1$. Thus $\Psi_1((g,g'),\beta'), \dots, \Psi_{17}((g,g'),\beta')$ generate $T(g)$ for any pair of power series (g, g') and any vector β'. Since

$$(z_1, \dots, z_r)^{P'} \subset \big(\Phi_1((f_*, h_*), \beta), \dots, \Phi_r((f_*, h_*), \beta)\big)^{P'} + \mathfrak{n}_*^{c+1},$$

(D.117) gives

$$\mathfrak{n}_*^c \subset T(f_*) + \big(\Phi_1((f_*, h_*), \beta), \dots, \Phi_r((f_*, h_*), \beta)\big)^{P'} + \mathfrak{n}_* \cdot \mathfrak{n}_*^c,$$

and Nakayama's lemma implies

$$\mathfrak{n}_*^c \subset \big(\Psi_1((f_*, h_*), \beta), \dots, \Psi_{17}((f_*, h_*), \beta)\big) + \big(\Phi_1((f_*, h_*), \beta), \dots, \Phi_r((f_*, h_*), \beta)\big)^{P'}.$$

By Set-up 14.3(3), under the isomorphism $\widehat{\mathcal{O}}_{\mathcal{X}_i, o_i} \cong \widehat{\mathcal{O}}$ given by $x_1, \dots, x_4$, f_i is a unit times the image of a regular function on $\mathcal{X}_i$, and h_i is the image of a regular function on $\mathcal{X}_i$. The data of Lemma 6.1(1) are taken as follows, with $\mathbf{f}_i = (f_i, h_i)$ and $\mathbf{f}_* = (f_*, h_*)$ along $\mathcal{I}$.

(1) $(Y_i \ni y_i) = (\mathcal{X}_i \ni o_i)$ and $\mathfrak{m} = \mathfrak{n}$.
(2) $(\gamma_1, \gamma_2) = (\theta_2, \theta_2\mu)$ and $(\gamma_1', \gamma_2') = (\theta_3, \theta_3\mu)$.
(3) $\Phi_1, \dots, \Phi_r$, β, and $L = c + 1$.
(4) $\Psi_1, \dots, \Psi_{17}$, c, and $P = P'$.

By Lemma 6.1(1), for all but finitely many $i \in \mathcal{I}$ there exists $\beta_i \in \mathbb{C}^q$ such that

$$\mathfrak{n}^c \subset \big(\Psi_1((f_i, h_i), \beta_i), \dots, \Psi_{17}((f_i, h_i), \beta_i)\big) + \mathcal{J}_i^{P'}.$$

The first ideal on the right is $T(f_i)$. Thus (D.118) holds for all but finitely many $i \in \mathcal{I}$.

Step 4. In this step we fix $i \in \mathcal{I}$ and construct a polynomial g_i, of degree at most an integer d not depending on i, with

$$g_i - f_i \in \mathfrak{n}^{2c+4} \quad \text{and} \quad \mathfrak{n}^c \subset T(g_i) + \mathcal{J}_i^{P'}. \tag{D.119}$$

Let

$$N := \max\{2c+4, 16\} \text{ in case (E)}, \qquad N := \max\{2c+4, 2n+6\} \text{ in case (N)},$$

and $d := d(N, M)$ the integer of Lemma D.17(1). Since $(f_i = 0)$ has an isolated singularity at the closed point and G acts faithfully and diagonally, Lemma D.17(1) yields a G-semi-invariant polynomial g_i with the character of f_i such that

$$\deg g_i \le d, \qquad g_i - f_i \in \mathfrak{n}^N,$$

and $W_i' := (g_i = 0) \subset \mathbb{A}^4$ has an isolated singularity at 0. In case (E), since $N \ge 16$, Lemma D.17(1)(a) shows that a general hyperplane section of W_i' through 0 is Du Val of type E_6, E_7, or E_8, so W_i' is of type cE. In case (N), since $N \ge 2n+6$, ℓ_i cuts out a Du Val section of type D_n of W_i' by Lemma D.17(1)(b). In both cases $(W_i' \ni 0)$ is an isolated cDV germ, hence normal, Gorenstein and terminal. Since $N \ge 3$, $g_i - f_i \in \mathfrak{n}^3$, so the quadratic part of g_i equals the quadratic part of f_i. In particular $g_i \in \mathfrak{n}^2$, so G acts on the cotangent space of W_i' at 0 as on $\mathfrak{n}/\mathfrak{n}^2$. Since G acts faithfully on W_i, hence on $\mathfrak{n}/\mathfrak{n}^2$, G acts faithfully on W_i'.

Since $N \ge c+1$, the elements $g_i - f_i$ and $x_k\partial_{k'}(g_i - f_i)$, for $1 \le k, k' \le 4$, belong to $\mathfrak{n}^N \subset \mathfrak{n}^{c+1}$. Thus

$$T(g_i) + \mathfrak{n}^{c+1} = T(f_i) + \mathfrak{n}^{c+1}.$$

By (D.118),

$$\mathfrak{n}^c \subset T(g_i) + \mathcal{J}_i^{P'} + \mathfrak{n} \cdot \mathfrak{n}^c,$$

and Nakayama's lemma implies $\mathfrak{n}^c \subset T(g_i) + \mathcal{J}_i^{P'}$. Since $N \ge 2c+4$, (D.119) holds.

Step 5. In this step we show that, for $(V \ni v) \in \{(W_i \ni o_i), (W_i' \ni 0)\}$,

$$\operatorname{mult}_F \operatorname{Jac}_V \le \lambda a(F, V, 0) \quad \text{for any prime divisor } F \text{ over } V \text{ with center } v, \tag{D.120}$$

and that $a(E_0, W_i, 0) = 3$ for some prime divisor E_0 over W_i with center o_i.

In case (E), a general hyperplane section of V through v is a Du Val germ of type E_6, E_7, or E_8. For such a surface germ, the quadratic part of an equation has rank 1. The quadratic part of a general hyperplane section of V is the restriction of that of V to a general hyperplane, so the quadratic part of an equation of V has rank 1. Since V is moreover normal, Theorem 8.6 implies (D.120) with $\lambda = 15$.

In case (N), the linear form ℓ_i is a regular parameter of $(\mathcal{X}_i \ni o_i)$ cutting out a Du Val section of type D_n of W_i (by hypothesis), and a regular parameter of $(\mathbb{A}^4 \ni 0)$ cutting out one of W_i' (**Step 4**). By Theorem 8.4 with $\ell_V = \ell_i$,

$$\kappa_n \operatorname{mult}_F \operatorname{Jac}_V \le a(F, V, 0) - \operatorname{mult}_F \ell_i \le a(F, V, 0),$$

which is (D.120) with $\lambda = 1/\kappa_n$.

In both cases the quadratic part of f_i has rank 1, and $(W_i \ni o_i)$ is an isolated cDV germ. Lemma 3.1(2) yields a prime divisor E_0 over W_i with center o_i and $a(E_0, W_i, 0) = 3$.

Step 6. In this step we find the computing lifts F_i and the constant L.

The pair $(W_i, \frac{1}{m}\operatorname{div}(h_i))$ is lc near o_i, and $b_i > 0$: in case (E) by hypothesis, and in case (N) since the pair is klt near o_i, so that the quotient pair by G is klt. By **Steps 4** and 5, the hypotheses of Theorem 8.2 hold for $f = f_i$, $f_0 = g_i$, and $h = h_i$, with the constants of **Step 1**. By Set-up 14.3(3), the algebraicity hypothesis of Theorem 8.2 holds, with $(\mathcal{X}_i \ni o_i)$, via the G-semi-invariant regular parameters $x_1, \dots, x_4$, as the G-equivariant étale neighborhood of $(\mathbb{A}^4 \ni 0)$.

If $\mathcal{J}_i = \widehat{\mathcal{O}}$, Theorem 8.2(1) yields a computing lift F_i of b_i with

$$\frac{a(F_i, W_i, 0)}{e_{F_i}} \le L_1.$$

If $\mathcal{J}_i \ne \widehat{\mathcal{O}}$, then by (D.119) Theorem 8.2(2) yields a G-equivariant automorphism φ of $\widehat{\mathcal{O}}$ such that $h_0 := \varphi(h_i)|_{\widehat{W}_i'}$ is non-zero and a non-unit on the completion $\widehat{W}_i'$ of W_i' at 0. By Theorem 8.2(2)(a) with $t = 1$ and $H = G$,

$$\operatorname{mld}\Big(\Big(\widehat{W}_i' \ni 0, \frac{1}{m}\operatorname{div}(h_0)\Big)/G\Big) = b_i > 0.$$

Since $\deg g_i \le d$, Lemma D.17(2), with $V = W_i'$, $g = g_i$, and $h = \varphi(h_i)$, yields a computing lift E of $\operatorname{mld}((\widehat{W}_i' \ni 0, \frac{1}{m}\operatorname{div}(h_0))/G)$ with

$$\frac{a(E, W_i', 0)}{e_E} \le L(m, M, d),$$

where $L(m, M, d)$ is the constant of Theorem 13.3(1). By Theorem 8.2(2)(b), there exists a computing lift F_i of b_i with

$$\frac{a(F_i, W_i, 0)}{e_{F_i}} \le L(m, M, d).$$

Let $L := \max\{L_1, L(m, M, d)\}$, which does not depend on i. Whether or not $\mathcal{J}_i = \widehat{\mathcal{O}}$, $a(F_i, W_i, 0)/e_{F_i} \le L$ for any $i \in \mathcal{I}$, which proves Theorems 14.4 and 14.5. □

Appendix E. Proofs for bounded Jacobian orders

We now prove the results of Section 15.

E.1. **Signed smooth surfaces.** In this subsection we prove Theorem 15.2.

Construction E.1. Let M be a positive integer, and let x_1, x_2 be the coordinates of $\mathbb{A}^2$. The entries of a diagonal matrix of order at most M are roots of unity of order at most M, so there exist only finitely many diagonal subgroups $G_0 \subset GL_2(\mathbb{C})$ of order at most M. For any such G_0, let

$$\phi_0 : \mathbb{A}^2_{\mathbb{C}} \to Q_0 := \mathbb{A}^2_{\mathbb{C}}/G_0$$

be the quotient morphism. The stabilizer in G_0 of a point of $\mathbb{A}^2\backslash\{0\}$ acts trivially on x_1 or on x_2, so it is generated by pseudo-reflections, and Q_0 is smooth outside $\phi_0(0)$ by the Chevalley–Shephard–Todd theorem. We fix a log resolution $\pi_0 : Y_0 \to Q_0$ which is an isomorphism over $Q_0\backslash\{\phi_0(0)\}$.

Let $\Bbbk \supset \mathbb{C}$ be an algebraically closed field. Let $G \subset GL_2(\Bbbk)$ be a diagonal subgroup of order at most M, acting on $\mathbb{A}^2_{\Bbbk}$. Then $G = G_0$ for one of the groups G_0 above. Since G is diagonal, $\Bbbk[x_1, x_2]^G$ is spanned by the G-invariant monomials, so the quotient morphism $\phi' : \mathbb{A}^2_{\Bbbk} \to Q' := \mathbb{A}^2_{\Bbbk}/G$ is the base change of ϕ_0 to $\Bbbk$. Let $\overline{0} := \phi'(0)$, and let $\pi' : Y' \to Q'$ be the base change of π_0 to $\Bbbk$. Then π' is a log resolution of Q' which is an isomorphism over $Q'\backslash\{\overline{0}\}$, and

$$K_{Y'} = \pi'^* K_{Q'} + \sum_k \kappa_k E_k',$$

where E_k' runs over the exceptional curves of π' and the numbers κ_k are the corresponding coefficients for π_0. For $i \in \{1, 2\}$, let n_i be the order of the subgroup of G fixing the curve

$(x_i = 0)$ pointwise. A non-trivial diagonal matrix fixes pointwise no prime divisor of $\mathbb{A}^2$ other than $(x_1 = 0)$ and $(x_2 = 0)$, so the ramification divisor of ϕ' is

$$R(\phi') = \sum_{i=1}^{2} (n_i - 1) \operatorname{div}(x_i).$$

Since Q_0 has quotient singularities, Q_0 is klt, so $\kappa_k > -1$ for any k. Let $L_1 \geq 1$ be an integer with $|\kappa_k| \leq L_1$ for all the coefficients κ_k of all the resolutions π_0 above, and let $\mathcal{K}_M \subset (0,1)$ be the finite set of the positive numbers among the $-\kappa_k$. Both L_1 and $\mathcal{K}_M$ depend only on M.

Proposition E.2. *Let M, N, N_0 be positive integers, $c_0 \geq 0$ a real number, and $\Gamma \subset [0,1]$ a DCC set. Then there exists a positive real number L depending only on M, N, N_0, c_0, Γ satisfying the following. Let $(T \ni o)$ be a smooth surface germ over an algebraically closed field $\Bbbk \supset \mathbb{C}$, G a finite abelian group of order at most M acting faithfully on T and fixing o, and*

$$B_T = \sum_j c_j \operatorname{div}(f_j) + b' \operatorname{div}(u),$$

a finite sum, where the f_j and u are non-zero G-semi-invariant regular functions on T, $c_j \in \Gamma \backslash \{0\}$, and $b' \in [-c_0, 0]$. Assume that

(1) (T, B_T) is sub-lc near o,
(2) $\mu := \operatorname{mld}((T \ni o, B_T)/G) > 0$, and
(3) some computing lift F of μ satisfies $\operatorname{mult}_F u \leq N$ and $\operatorname{mult}_F \mathfrak{m}_o \leq N_0$.

Then some computing lift P of μ satisfies

$$a(P, T, 0) \leq L \quad \text{and} \quad \operatorname{mult}_P u \leq L.$$

Proof. We divide the proof into several steps.

Step 1. In this step we relate T/G to a diagonal quotient of $\mathbb{A}^2$. Possibly replacing T with a G-invariant affine open neighborhood of o, we may assume that T is affine. Since G is abelian, there exist G-semi-invariant regular functions $x_1, x_2 \in \mathfrak{m}_o$ whose classes form a basis of $\mathfrak{m}_o/\mathfrak{m}_o^2$. Since $\widehat{\mathcal{O}}_{T,o} = \Bbbk[[x_1, x_2]]$, an element of G fixing x_1 and x_2 acts trivially on $\Bbbk(T)$, hence is the identity. Thus G acts faithfully on $\mathbb{A}^2_\Bbbk$ through the characters of x_1, x_2, as a diagonal group of order at most M, and the morphism

$$\psi := (x_1, x_2) : T \to \mathbb{A}^2_\Bbbk$$

is G-equivariant and étale at o. Let ϕ', Q', $\overline{0}$, $\pi' : Y' \to Q'$, E'_k, κ_k, n_1, n_2, L_1, and $\mathcal{K}_M$ be as in Construction E.1 for the diagonal action of G on $\mathbb{A}^2_\Bbbk$. Let $\phi : T \to Q := T/G$ be the quotient morphism, $\overline{o} := \phi(o)$, and $\overline{\psi} : Q \to Q'$ the morphism with $\overline{\psi} \circ \phi = \phi' \circ \psi$. Since $\phi^{-1}(\overline{o}) = \{o\}$, $\phi'^{-1}(\overline{0}) = \{0\}$, and taking G-invariants commutes with completion in characteristic zero, ψ induces an isomorphism

$$\widehat{\mathcal{O}}_{Q',\overline{0}} = \Bbbk[[x_1, x_2]]^G \to \left(\widehat{\mathcal{O}}_{T,o}\right)^G = \widehat{\mathcal{O}}_{Q,\overline{o}}.$$

Thus $\overline{\psi}$ is étale at $\overline{o}$. Possibly replacing T with a smaller G-invariant affine open neighborhood of o, we may assume that ψ and $\overline{\psi}$ are étale and $\overline{\psi}^{-1}(\overline{0}) = \{\overline{o}\}$. Then $K_T = \psi^* K_{\mathbb{A}^2}$ and $K_Q = \overline{\psi}^* K_{Q'}$, so by Construction E.1,

$$R(\phi) = K_T - \phi^* K_Q = \psi^* \left(K_{\mathbb{A}^2} - \phi'^* K_{Q'}\right) = \sum_{i=1}^{2} (n_i - 1) \operatorname{div}(x_i). \tag{E.1}$$

Step 2. In this step we write the boundary of the quotient. For a G-semi-invariant regular function g on T, the function $g^{|G|}$ is G-invariant, hence a regular function on Q, and

$$\phi^* \operatorname{div}_Q(g^{|G|}) = |G| \operatorname{div}(g).$$

Let

$$B_Q := \sum_j \frac{c_j}{|G|} \operatorname{div}_Q(f_j^{|G|}) + \sum_{i=1}^{2} \frac{n_i - 1}{|G|} \operatorname{div}_Q(x_i^{|G|}) + \frac{b'}{|G|} \operatorname{div}_Q(u^{|G|}),$$

$$B_Q^0 := \sum_{i=1}^{2} \frac{n_i - 1}{|G|} \operatorname{div}_Q(x_i^{|G|}).$$

By (E.1), $\phi^* B_Q = B_T + R(\phi)$ and $\phi^* B_Q^0 = R(\phi)$, so

$$\phi^*(K_Q + B_Q) = K_T + B_T \quad \text{and} \quad \phi^*(K_Q + B_Q^0) = K_T.$$

For a prime divisor P over T, let P_Q be the restriction of P, whose ramification index is e_P (see Definition 2.2). Then

$$a(P, T, B_T) = e_P a(P_Q, Q, B_Q) \quad \text{and} \quad a(P, T, 0) = e_P a(P_Q, Q, B_Q^0)$$

for any prime divisor P over T, and any prime divisor over Q is P_Q for some prime divisor P over T [Liu26a, Lemma 2.6(1)–(3)]. Moreover $\mu = \operatorname{mld}(Q \ni \overline{o}, B_Q)$, and a prime divisor P over T with center o is a computing lift of μ if and only if $a(P_Q, Q, B_Q) = \mu$ (see Definition 2.2). Since (T, B_T) is sub-lc near o, (Q, B_Q) is sub-lc near $\overline{o}$. Since $B_Q^0 \geq 0$,

$$a(P_Q, Q, 0) \geq a(P_Q, Q, B_Q^0) = a(P, T, 0)/e_P > 0,$$

and since $e_P \leq |G| \leq M$,

$$a(P, T, 0) \leq M a(P_Q, Q, 0) \tag{E.2}$$

for any prime divisor P over T. Moreover, for any G-semi-invariant regular function g on T,

$$\operatorname{mult}_P g = \frac{1}{|G|} \operatorname{mult}_P(g^{|G|}) = \frac{e_P}{|G|} \operatorname{mult}_{P_Q}(g^{|G|}). \tag{E.3}$$

Step 3. In this step we pull back the fixed resolution. Let $Y := Y' \times_{Q'} Q$ and let $\pi : Y \to Q$ be the projection. Since $\overline{\psi}$ is étale and $\overline{\psi}^{-1}(\overline{0}) = \{\overline{o}\}$, Y is smooth, π is a projective birational morphism which is an isomorphism over $Q \backslash \{\overline{o}\}$, $\pi^{-1}(\overline{o}) \cong \pi'^{-1}(\overline{0})$, the exceptional curves of π are the preimages E_k of the E'_k, $\sum_k E_k$ is an SNC divisor, and

$$K_Y = \pi^* K_Q + \sum_k \kappa_k E_k.$$

Let P' be a prime divisor over Y whose center on Y is a closed point $y \in \pi^{-1}(\overline{o})$. Then $a(P', Q, 0) = a(P', Y, 0) + \sum_k \kappa_k \operatorname{mult}_{P'} E_k$. Since $\sum_k E_k$ is a reduced SNC divisor, $(Y, \sum_k E_k)$ is lc near y, so $\sum_k \operatorname{mult}_{P'} E_k \leq a(P', Y, 0)$. Since $|\kappa_k| \leq L_1$,

$$a(P', Q, 0) \leq (1 + L_1) a(P', Y, 0). \tag{E.4}$$

For a G-invariant regular function g on T, regarded as a regular function on Q, let

$$\operatorname{div}_Y(g) := \pi^* \operatorname{div}_Q(g)$$

be the divisor of $g \circ \pi$. Let $B_Y := \pi^* B_Q - \sum_k \kappa_k E_k$. Then $K_Y + B_Y = \pi^*(K_Q + B_Q)$ and $B_Y = B_Y^+ + B_Y^-$, where

$$B_Y^+ := \sum_j \frac{c_j}{|G|} \operatorname{div}_Y(f_j^{|G|}) + \sum_{i=1}^{2} \frac{n_i - 1}{|G|} \operatorname{div}_Y(x_i^{|G|}) + \sum_{\kappa_k < 0} (-\kappa_k) E_k,$$

$$B_Y^- := \frac{b'}{|G|} \operatorname{div}_Y(u^{|G|}) - \sum_{\kappa_k > 0} \kappa_k E_k.$$

Here $B_Y^+ \geq 0$ and $B_Y^- \leq 0$.

Step 4. In this step we locate F on Y. Since the center of F_Q on Q is $\bar{o}$, the center of F_Q on Y is either the generic point of some E_k or a closed point $y \in \pi^{-1}(\bar{o})$.

Assume that the center of F_Q on Y is the generic point of E_k. Then $F_Q = E_k$, and we let $P := F$. By (E.2),

$$a(P, T, 0) \le M a(E_k, Q, 0) = M(1 + \kappa_k) \le M(1 + L_1),$$

and $\operatorname{mult}_P u = \operatorname{mult}_F u \le N$.

Assume that the center of F_Q on Y is a closed point y. Any prime divisor P' over Y with center y has center $\bar{o}$ on Q, so $a(P', Y, B_Y) = a(P', Q, B_Q) \ge \mu$. Since $a(F_Q, Y, B_Y) = \mu$,

$$\operatorname{mld}(Y \ni y, B_Y) = \mu,$$

and F_Q computes $\operatorname{mld}(Y \ni y, B_Y)$. By (E.3) applied with $P = F$, and since for any G-semi-invariant g

$$\operatorname{mult}_{F_Q}(g^{|G|}) = \operatorname{mult}_{F_Q} \operatorname{div}_Y(g^{|G|}),$$

we obtain

$$\operatorname{mult}_{F_Q} \operatorname{div}_Y(u^{|G|}) = \frac{|G|}{e_F} \operatorname{mult}_F u \le |G| N. \tag{E.5}$$

In the rest of the proof we treat this case.

Step 5. In this step we select P_Q by Lemma 3.3. Since $\mathfrak{m}_o \mathcal{O}_{T,o} = (x_1, x_2)$,

$$\operatorname{mult}_F \mathfrak{m}_o = \min\{\operatorname{mult}_F x_1, \operatorname{mult}_F x_2\}.$$

Possibly reordering indices, we may assume that the minimum is attained by x_1, so $\operatorname{mult}_F x_1 \le N_0$. Let D_Y be the reduced divisor with

$$\operatorname{Supp} D_Y = \operatorname{Supp} \operatorname{div}_Y(u^{|G|} x_1^{|G|}).$$

Possibly shrinking Y to a neighborhood of y, we may assume that any irreducible component of D_Y contains y.

Let D_1 be an irreducible component of D_Y. Since $y \in D_1$ and F_Q has center y, $\operatorname{mult}_{F_Q} D_1 \ge 1$. Since

$$\operatorname{div}_Y(u^{|G|}) \ge \big(\operatorname{mult}_{D_1} \operatorname{div}_Y(u^{|G|})\big) D_1,$$

(E.5) implies

$$\operatorname{mult}_{D_1} \operatorname{div}_Y(u^{|G|}) \le |G| N.$$

Since $b' \in [-c_0, 0]$ and $|\kappa_k| \le L_1$, we obtain $B_Y^- \in [-c_1, 0]$, where $c_1 := c_0 N + L_1$.

Let

$$\Theta := \{1\} \cup \bigcup_{i=1}^{M} \frac{1}{i}(\Gamma \setminus \{0\}) \cup \left\{ \frac{i'}{i} \;\middle|\; 1 \le i' < i \le M \right\} \cup \mathcal{K}_M.$$

Then $\Theta \subset (0, 1]$ is a non-empty DCC set depending only on M and Γ. Near y, any E_k is the divisor of a local equation, and the terms of B_Y^+ with non-zero coefficients have coefficients $c_j/|G|$, $(n_i - 1)/|G|$ with $1 \le n_i - 1 < |G|$, and $-\kappa_k \in \mathcal{K}_M$, all in Θ. Thus near y, B_Y^+ is a finite sum $\sum d \operatorname{div}(g)$ with $d \in \Theta$ and non-zero regular functions g.

Let $P_1, \dots, P_m = F_Q$ be the exceptional curves of the sequence of point blow-ups extracting F_Q, where the center y_1 of the first blow-up is y and the center y_i of the i-th blow-up belongs to P_{i-1} for $i \ge 2$. Let P_Q be the first of $P_1, \dots, P_m$ with $a(P_Q, Y, B_Y) = \mu$, and let $\lambda := 2 + (1 + c_1)/\delta(\Theta)$. By Lemma 3.3 for $(Y \ni y, D_Y, B_Y^-, B_Y^+, F_Q)$ in place of $(T \ni p, D, B_-, B_+, E)$, with c_1 in place of c_0,

$$a(P_Q, Y, 0) \le \lambda \operatorname{mult}_{P_Q} D_Y \le \lambda \operatorname{mult}_{F_Q} D_Y. \tag{E.6}$$

Since $\operatorname{div}_Y(u^{|G|}x_1^{|G|}) \geq 0$ has integral coefficients and support $\operatorname{Supp} D_Y$, $\operatorname{div}_Y(u^{|G|}x_1^{|G|}) \geq D_Y$. By (E.3) for F and the choice of x_1,

$$\operatorname{mult}_{F_Q} D_Y \leq \operatorname{mult}_{F_Q} \operatorname{div}_Y(u^{|G|}x_1^{|G|}) = \frac{|G|}{e_F}\big(\operatorname{mult}_F u + \operatorname{mult}_F x_1\big) \leq M(N+N_0).$$

Thus (E.6) implies

$$a(P_Q, Y, 0) \leq \lambda M(N+N_0). \tag{E.7}$$

Step 6. In this step we lift P_Q and conclude. The center of P_Q on Y is y, so the center of P_Q on Q is $\overline{o}$ and

$$a(P_Q, Q, B_Q) = a(P_Q, Y, B_Y) = \mu.$$

By **Step 2**, there exists a prime divisor P over T whose restriction is P_Q. The center of P on T belongs to $\phi^{-1}(\overline{o}) = \{o\}$, so P is a computing lift of μ. By (E.2), (E.4) with $P' = P_Q$, and (E.7),

$$a(P, T, 0) \leq Ma(P_Q, Q, 0) \leq M(1+L_1)a(P_Q, Y, 0) \leq M^2(1+L_1)\lambda(N+N_0).$$

Since $y_i \in P_{i-1}$ for $2 \leq i \leq m$,

$$\operatorname{mult}_{P_Q} \operatorname{div}_Y(u^{|G|}) \leq \operatorname{mult}_{F_Q} \operatorname{div}_Y(u^{|G|}).$$

By (E.3) and (E.5),

$$\operatorname{mult}_P u = \frac{e_P}{|G|} \operatorname{mult}_{P_Q} \operatorname{div}_Y(u^{|G|}) \leq \frac{e_P}{|G|} \cdot \frac{|G|}{e_F} \operatorname{mult}_F u \leq MN.$$

Let

$$L := M^2(1+L_1)\lambda(N+N_0).$$

Since $\lambda \geq 2$, $L \geq M(1+L_1)$ and $L \geq MN \geq N$, so the bounds of **Step 4** and **Step 6** are at most L. The number L depends only on M, N, N_0, c_0, Γ, since L_1 depends only on M, $c_1 = c_0 N + L_1$, and Θ depends only on M and Γ. The proposition follows. □

Proof of Theorem 15.2. Possibly shrinking T to a G-invariant affine neighborhood of o, we may assume that

$$D_T = \operatorname{div}(u)$$

for a G-semi-invariant regular function u. Then $\operatorname{mult}_F u = \operatorname{mult}_F D_T \leq N$. By Proposition E.2 with $(M, N, N_0, c_0, \Gamma) = (M, N, N_0, 1, \{\frac{1}{m}\})$ and $B_T = \frac{1}{m}\operatorname{div}(h) + b' \operatorname{div}(u)$ for $b' = -1$, some computing lift P of μ satisfies $a(P, T, 0) \leq L$ and $\operatorname{mult}_P D_T = \operatorname{mult}_P u \leq L$, where L depends only on m, M, N, N_0. □

E.2. **Smooth returns.** In this subsection we prove Proposition 15.3 of Section 15.

Proof of Proposition 15.3. Let X_0 be the smooth fourfold germ containing W, and let

$$D := \frac{1}{m}\operatorname{div}(h).$$

Since F is a computing lift of b,

$$a(P, W, D) \geq be_P \quad \text{and} \quad a(F, W, D) = be_F \tag{E.8}$$

for any prime divisor P over W with center o. In particular,

$$\operatorname{center}_W F = \{o\} \quad \text{and} \quad \operatorname{mld}(W \ni o, D) \geq b > 0.$$

Since W is a hypersurface, ω_W is invertible. Possibly replacing X_0 with a G-invariant open neighborhood of o, we may assume that (W, D) is lc and that ω_W is generated by a global section ω. Then ω is a rational differential 3-form on W, and we let $K_W := \operatorname{div}_W(\omega) = 0$. We divide the proof into several steps.

Step 1. In this step we construct the blow-ups. Let $V_0 := W$. Whenever X_j is constructed, let $\pi_j : X_j \to X_0$ be the composite of the blow-ups, $V_j := (\pi_j)_*^{-1} W \subset X_j$,

$$Z_j := \operatorname{center}_{V_j} F,$$

which exists since $V_j \to W$ is proper, and ξ_j the generic point of Z_j. Let $j \geq 1$ be an integer, and assume that we have constructed G-equivariant projective birational morphisms $X_{j-1} \to \cdots \to X_0$ satisfying the following.

(1) $X_{j-1} \to X_0$ is an isomorphism over $X_0 \backslash \{o\}$.
(2) X_{j-1} is smooth at ξ_{j-1}.

For $j = 1$, (1) and (2) hold since X_0 is smooth and $Z_0 = \{o\}$. If V_{j-1} is smooth at ξ_{j-1}, we stop and let $n := j - 1$. Otherwise $Z_{j-1} \subset \operatorname{Sing}(V_{j-1})$. Since V_{j-1} is an integral threefold, $\dim Z_{j-1} \leq 2$, so

$$\kappa_j := \operatorname{codim}_{X_{j-1}} Z_{j-1} \in \{2, 3, 4\}.$$

Let $GZ_{j-1} := \bigcup_{\sigma \in G} \sigma(Z_{j-1})$ with the reduced structure. Let $X_{j-1}^\circ \subset X_{j-1}$ be the complement of the union of the following closed subsets:

- $\operatorname{Sing}(X_{j-1})$.
- $\sigma(\operatorname{Sing}(Z_{j-1}))$ for $\sigma \in G$.
- $\sigma(Z_{j-1}) \cap \sigma'(Z_{j-1})$ for $\sigma, \sigma' \in G$ with $\sigma(Z_{j-1}) \neq \sigma'(Z_{j-1})$.

By (2), X_{j-1}° is a smooth G-invariant open subset of X_{j-1} containing the generic point of $\sigma(Z_{j-1})$ for any $\sigma \in G$, and $GZ_{j-1} \cap X_{j-1}^\circ$ is the disjoint union of the smooth closed subvarieties $\sigma(Z_{j-1}) \cap X_{j-1}^\circ$, each contained in $\operatorname{Sing}(V_{j-1})$.

Let $\phi_j : X_j \to X_{j-1}$ be the blow-up of GZ_{j-1}. Since GZ_{j-1} is G-invariant, G acts on X_j and ϕ_j is G-equivariant. Over X_{j-1}°, ϕ_j is the blow-up of the smooth variety X_{j-1}° along the smooth closed subvariety $GZ_{j-1} \cap X_{j-1}^\circ$, so $\phi_j^{-1}(X_{j-1}^\circ)$ is smooth. Since ϕ_j is proper, $\phi_j(Z_j) = Z_{j-1}$. Thus ξ_j maps to ξ_{j-1}, which belongs to X_{j-1}°, and (2) holds for j. Moreover, Z_{j-1} maps onto $Z_0 = \{o\}$, and o is fixed by G, so GZ_{j-1} maps to o and (1) holds for j.

Step 2. In this step we show that the blow-ups ϕ_j of **Step 1** stop at some $n \leq N$. The computation below is [Hic05, Lemme 4.2] at the centers of F instead of along an arc (see also [Hic05, Remarque 4.3]).

Let $j \geq 1$ be an integer such that V_{j-1} is singular at ξ_{j-1}, and let $\kappa := \kappa_j$. Since X_{j-1}° and $Z_{j-1} \cap X_{j-1}^\circ$ are smooth, there exist an open subset $U' \subset X_{j-1}^\circ$ containing ξ_{j-1} and disjoint from $\sigma(Z_{j-1})$ for any $\sigma \in G$ with $\sigma(Z_{j-1}) \neq Z_{j-1}$, regular functions $x_1, \ldots, x_\kappa, w_1, \ldots, w_{4-\kappa}$ on U' defining an étale morphism $(x, w) : U' \to \mathbb{A}^4$, and a regular function f on U', satisfying

$$Z_{j-1} \cap U' = (x_1 = \cdots = x_\kappa = 0) \quad \text{and} \quad V_{j-1} \cap U' = (f = 0).$$

On $V_{j-1} \cap U'$, $\operatorname{Jac}_{V_{j-1}}$ is generated by the restrictions of $\partial f / \partial x_i$ for $1 \leq i \leq \kappa$ and $\partial f / \partial w_i$ for $1 \leq i \leq 4 - \kappa$ (see Definition 5.2).

Let r_j be the multiplicity of V_{j-1} at ξ_{j-1}, that is,

$$r_j = \max\{r \in \mathbb{N} \mid f \in (x_1, \ldots, x_\kappa)^r \mathcal{O}_{X_{j-1}, Z_{j-1}}\}.$$

Since V_{j-1} is singular at ξ_{j-1}, $r_j \geq 2$ by the Jacobian criterion. Possibly shrinking U' to a neighborhood of ξ_{j-1}, we may assume that $f \in (x_1, \ldots, x_\kappa)^{r_j}$ on U'. Since $f \in (x_1, \ldots, x_\kappa)^2$ and $x_i \notin (x_1, \ldots, x_\kappa)^2$ at ξ_{j-1}, $x_i|_{V_{j-1}} \neq 0$ for $1 \leq i \leq \kappa$. Possibly reordering indices, we may assume that

$$\operatorname{mult}_F x_1 \leq \operatorname{mult}_F x_i \quad \text{for } 2 \leq i \leq \kappa.$$

Then ξ_j belongs to the open subset $U_j' \subset \phi_j^{-1}(U')$ on which $\phi_j^* x_i / \phi_j^* x_1$ is regular for $2 \leq i \leq \kappa$. Let $y_j := \phi_j^* x_1$ and $x_i' := \phi_j^* x_i / y_j$ for $2 \leq i \leq \kappa$, and write w_i for $\phi_j^* w_i$. Since U_j' is the base

change by (x, w) of an affine chart of the blow-up of $\mathbb{A}^4$ along a coordinate linear subspace, the functions $y_j, x'_2, \dots, x'_\kappa, w_1, \dots, w_{4-\kappa}$ define an étale morphism $U'_j \to \mathbb{A}^4$, and

$$x_1 = y_j, \qquad x_i = y_j x'_i \quad (2 \le i \le \kappa), \quad \text{and} \quad (y_j = 0) = \phi_j^{-1}(Z_{j-1}) \cap U'_j. \tag{E.9}$$

The divisor $(y_j = 0)$ on U'_j is irreducible. The function $f' := y_j^{-r_j} \phi_j^* f$ is regular on U'_j, and the restriction of f' to $(y_j = 0)$ is the image of f in $(x_1, \dots, x_\kappa)^{r_j}/(x_1, \dots, x_\kappa)^{r_j+1}$, which is non-zero at ξ_{j-1}. Thus

$$\operatorname{div}(\phi_j^* f) = r_j(y_j = 0) + \operatorname{div}(f'),$$

where $\operatorname{div}(f')$ does not contain $(y_j = 0)$, and

$$V_j \cap U'_j = (f' = 0).$$

By the chain rule and (E.9), with partial derivatives on U'_j taken with respect to y_j, $x'_2, \dots, x'_\kappa$, $w_1, \dots, w_{4-\kappa}$,

$$\begin{aligned}
\phi_j^* \frac{\partial f}{\partial x_1} &= y_j^{r_j - 1} \Big(r_j f' + y_j \frac{\partial f'}{\partial y_j} - \sum_{i=2}^{\kappa} x'_i \frac{\partial f'}{\partial x'_i} \Big), \\
\phi_j^* \frac{\partial f}{\partial x_i} &= y_j^{r_j - 1} \frac{\partial f'}{\partial x'_i} \quad (2 \le i \le \kappa), \quad \text{and} \\
\phi_j^* \frac{\partial f}{\partial w_i} &= y_j^{r_j} \frac{\partial f'}{\partial w_i} \quad (1 \le i \le 4 - \kappa).
\end{aligned}$$

Since $f' = 0$ on $V_j \cap U'_j$,

$$\operatorname{Jac}_{V_{j-1}} \cdot \mathcal{O}_{V_j \cap U'_j} \subset y_j^{r_j - 1} \operatorname{Jac}_{V_j}.$$

By (E.9), $Z_j \cap U'_j \subset (y_j = 0)$, and y_j is regular at ξ_j. Thus $\operatorname{mult}_F y_j$ is a positive integer, and

$$\operatorname{mult}_F \operatorname{Jac}_{V_{j-1}} \ge (r_j - 1) \operatorname{mult}_F y_j + \operatorname{mult}_F \operatorname{Jac}_{V_j}. \tag{E.10}$$

We also compare the dualizing sheaves of V_{j-1} and V_j. Let

$$\alpha := dx_1 \wedge \cdots \wedge dx_\kappa \wedge dw_1 \wedge \cdots \wedge dw_{4-\kappa} \quad \text{and} \quad \alpha' := dy_j \wedge dx'_2 \wedge \cdots \wedge dx'_\kappa \wedge dw_1 \wedge \cdots \wedge dw_{4-\kappa}.$$

By (E.9), $\phi_j^* \alpha = y_j^{\kappa - 1} \alpha'$. By adjunction, the Poincaré residues

$$\omega_{j-1} := \operatorname{Res}_{V_{j-1}}(\alpha/f) \quad \text{and} \quad \omega'_j := \operatorname{Res}_{V_j}(\alpha'/f')$$

generate $\omega_{V_{j-1}}$ on $V_{j-1} \cap U'$ and ω_{V_j} on $V_j \cap U'_j$ respectively. Since ϕ_j is an isomorphism outside $(y_j = 0)$, as rational differential 3-forms on V_j,

$$\phi_j^* \omega_{j-1} = \operatorname{Res}_{V_j}\big(y_j^{\kappa_j - 1 - r_j} \alpha'/f'\big) = y_j^{\kappa_j - 1 - r_j} \omega'_j. \tag{E.11}$$

Now let $j \ge 1$ be an integer such that V_{i-1} is singular at ξ_{i-1} for $1 \le i \le j$. Since Jac_{V_j} is an ideal of $\mathcal{O}_{V_j}$ near Z_j, (E.10) for $i = 1, \dots, j$ implies

$$\operatorname{mult}_F \operatorname{Jac}_W \ge \sum_{i=1}^{j} (r_i - 1) \operatorname{mult}_F y_i + \operatorname{mult}_F \operatorname{Jac}_{V_j} \ge j.$$

Since $\operatorname{mult}_F \operatorname{Jac}_W \le N$, the blow-ups ϕ_j of **Step 1** stop at some $n \le N$, and

$$\sum_{j=1}^{n} (r_j - 1) \operatorname{mult}_F y_j \le \operatorname{mult}_F \operatorname{Jac}_W \le N. \tag{E.12}$$

Let $\nu : Y \to V_n$ be the normalization, and let $g : Y \to W$ be the composite of ν and $V_n \to W$. Then G acts on Y, and g is G-equivariant, proper and birational. By (1) and since W is normal, g is an isomorphism over $W \backslash \{o\}$. Let

$$Z := \operatorname{center}_Y F,$$

so $\nu(Z) = Z_n$, and let $U := \nu^{-1}((V_n)_{\mathrm{reg}})$, where $(V_n)_{\mathrm{reg}}$ is the non-singular locus of V_n. Then $\nu|_U : U \to (V_n)_{\mathrm{reg}}$ is an isomorphism, U is a smooth G-invariant open subset of Y, and since V_n is smooth at ξ_n, U contains the generic point of $\sigma(Z)$ for any $\sigma \in G$.

Step 3. In this step we prove Proposition 15.3(1) and (2), and bound $\operatorname{mult}_F \mathfrak{m}_{U,Z}$. Let

$$K_U := \operatorname{div}_U(g^*\omega), \quad \text{so} \quad K_{U/W} = K_U - g^*K_W = \operatorname{div}_U(g^*\omega)$$

is an integral Cartier divisor on the smooth variety U. Since g is an isomorphism over $W\backslash\{o\}$ and $\operatorname{div}_W(\omega) = 0$, $g(\operatorname{Supp} K_{U/W}) \subset \{o\}$. For any prime divisor P on U, $\operatorname{mult}_P K_{U/W} = a(P, W, 0) - 1$, which is non-negative if $g(P) = \{o\}$ since W is canonical, and 0 otherwise. Hence $K_{U/W} \geq 0$, and $K_{U/W}$ is G-invariant since $a(\sigma(P), W, 0) = a(P, W, 0)$ for any $\sigma \in G$. Let

$$B_U := \frac{1}{m} \operatorname{div}_U(g^*h) - K_{U/W}.$$

Since $K_U = g^*K_W + K_{U/W}$ and $\operatorname{div}_U(g^*h) = g^* \operatorname{div}_W(h)$, we have $K_U + B_U = g^*(K_W + D)$ on U, so

$$a(P, W, 0) = a(P, U, 0) + \operatorname{mult}_P K_{U/W} \quad \text{and} \quad a(P, W, D) = a(P, U, B_U) \tag{E.13}$$

for any prime divisor P over U, in particular for any prime divisor P over Y with center Z. Since (W, D) is lc, (U, B_U) is sub-lc.

We show that $\operatorname{mult}_F K_{U/W} \leq N$. If $n = 0$, then $Y = W$ and $K_{U/W} = 0$. Assume that $n \geq 1$, and let $\omega_n := \omega'_n$. Then

$$\frac{g^*\omega}{\omega_n} = \frac{\omega}{\omega_0} \cdot \prod_{j=1}^{n} \frac{\omega_{j-1}}{\omega'_j} \cdot \prod_{j=1}^{n-1} \frac{\omega'_j}{\omega_j}.$$

Here ω/ω_0 is a unit of $\mathcal{O}_{W,o}$, and ω'_j/ω_j is a unit of $\mathcal{O}_{V_j,Z_j}$ for $1 \leq j \leq n-1$, since the numerator and the denominator generate the dualizing sheaf there. Since $\operatorname{center}_W F = \{o\}$ and $\operatorname{center}_{V_j} F = Z_j$, we have $\operatorname{mult}_F(\omega/\omega_0) = \operatorname{mult}_F(\omega'_j/\omega_j) = 0$. By (E.11), $\omega_{j-1}/\omega'_j = y_j^{\kappa_j - 1 - r_j}$. Since ω_n generates ω_U at the generic point of Z, $g^*\omega/\omega_n$ is a local equation of $K_{U/W}$ at the generic point of Z. Thus

$$\operatorname{mult}_F K_{U/W} = \sum_{j=1}^{n} (\kappa_j - 1 - r_j) \operatorname{mult}_F y_j.$$

Since $\kappa_j \in \{2, 3, 4\}$ and $r_j \geq 2$, we have $|\kappa_j - 1 - r_j| \leq r_j - 1$, and (E.12) implies

$$\operatorname{mult}_F K_{U/W} \leq \sum_{j=1}^{n} (r_j - 1) \operatorname{mult}_F y_j \leq N.$$

Since $\operatorname{mld}(W \ni o, D) > 0$, $b > 0$ and $|G| \leq M$, Theorem 5.1(3) implies $\operatorname{mult}_F \mathfrak{m}_{W,o} \leq 3M/\eta_m$. Since $g(Z) = \{o\}$, $g^*\mathfrak{m}_{W,o} \cdot \mathcal{O}_{U,Z} \subset \mathfrak{m}_{U,Z}$, so

$$1 \leq \operatorname{mult}_F \mathfrak{m}_{U,Z} \leq \operatorname{mult}_F \mathfrak{m}_{W,o} \leq \frac{3M}{\eta_m}. \tag{E.14}$$

Step 4. In this step we fix the constant L and prove Proposition 15.3(3) when $\dim Z = 2$. Let

$$N_0 := \left\lfloor \frac{3M}{\eta_m} \right\rfloor,$$

which depends only on m and M and is positive by (E.14). Let L_1 be the constant L of Theorem 15.2 for m, M, N, N_0, and let

$$L := \max\{N + 1, 2L_1\}.$$

Since (W, D) is lc, $\frac{1}{m} \operatorname{mult}_P h \leq a(P, W, 0)$ for any prime divisor P over W. Since $e_P \geq 1$, any computing lift P of b with $a(P, W, 0) \leq L$ satisfies the bounds of Proposition 15.3(3). Since $\operatorname{center}_W F = \{o\}$, $\dim Z \leq 2$.

Assume that $\dim Z = 2$. Then Z is a prime divisor on the normal variety Y, so F is the valuation of Z, and $a(F, U, 0) = 1$ since the generic point of Z belongs to U. By (E.13) and **Step 3**,

$$a(F, W, 0) = 1 + \operatorname{mult}_F K_{U/W} \le N + 1.$$

Since $N + 1 \le L$, $P := F$ satisfies the bounds of Proposition 15.3(3).

Step 5. In this step we prove Proposition 15.3(3) when $\dim Z = 1$. Let $G' := I_G(F)$, so $|G'| = e_F \le M$. By Lemma 2.15(1), G' is cyclic. Let $\sigma \in G'$. Since σ preserves the valuation of F, $\sigma(Z) = Z$. By the definition of the inertia group [Liu26a, Definition 2.5], σ acts trivially on the residue field of the valuation ring of F, which contains the residue field of the generic point of Z. Since Z is integral, $\sigma|_Z = \mathrm{id}_Z$. Since G' fixes Z pointwise, there exists a G'-invariant function t, regular at the generic point of Z, with $t|_Z$ non-constant.

Since G acts faithfully on W, h is a mark and $K_{U/W}$ is G-invariant, G' acts faithfully on U and preserves B_U. Let $k := \mathbb{C}(t)$, U_k the generic fiber of t on the open subset of U on which t is regular, B_k the restriction of B_U, Ω an algebraic closure of k, c' a closed point of $U_k \otimes_k \Omega$ over the point given by the generic point of Z, U_Ω the connected component of $U_k \otimes_k \Omega$ containing c', B_Ω the restriction of $B_k \otimes_k \Omega$ to U_Ω, and $\varpi : U_\Omega \to U$ the induced morphism. By Lemma 3.5(0) for the sub-pair (U, B_U), the curve $Z \cap U$, the group G', and the function t, G' acts faithfully on U_Ω, fixes c', and ϖ is G'-equivariant. Since U is smooth and k has characteristic zero, $(U_\Omega \ni c')$ is a smooth surface germ over Ω. Since (U, B_U) is sub-lc, so is (U_k, B_k). Since Ω/k is separable, the base change of a log resolution of (U_k, B_k) is a log resolution of $(U_k \otimes_k \Omega, B_k \otimes_k \Omega)$, on which each prime divisor E has the log discrepancy of the restriction of E, so (U_Ω, B_Ω) is sub-lc.

Let ψ be a local equation of $K_{U/W}$ at the generic point of Z. Since $\varpi(c')$ is the generic point of Z, near c' we have

$$B_\Omega = \frac{1}{m} \operatorname{div}(\varpi^* g^* h) - \varpi^* K_{U/W} \quad \text{and} \quad \varpi^* K_{U/W} = \operatorname{div}(\varpi^* \psi),$$

where $\varpi^* g^* h$ is a regular G'-semi-invariant function and $\varpi^* K_{U/W} \ge 0$ is a G'-invariant integral Cartier divisor.

Let P'' be a prime divisor over U_Ω with center c'. By Lemma 3.5(0) and (2), there exists a prime divisor P over U with center $Z \cap U$ such that

$$a(P'', U_\Omega, B_\Omega) = a(P, U, B_U), \quad \operatorname{mult}_{P''}(\varpi^* \zeta) = \operatorname{mult}_P \zeta, \quad \text{and} \quad I_{G'}(P'') = I_{G'}(P) \tag{E.15}$$

for any non-zero rational function ζ on U. Then P is a prime divisor over W with center o, and $I_{G'}(P) = G' \cap I_G(P)$, so $|I_{G'}(P'')| \le e_P$. Since $a(P, W, D) \ge 0$, (E.13) and (E.8) imply

$$\frac{a(P'', U_\Omega, B_\Omega)}{|I_{G'}(P'')|} = \frac{a(P, W, D)}{|I_{G'}(P)|} \ge \frac{a(P, W, D)}{e_P} \ge b. \tag{E.16}$$

By Lemma 3.5(0) and (1), there exists a prime divisor F' over U_Ω with center c' such that $\operatorname{mult}_{F'}(\varpi^* \zeta) = \operatorname{mult}_F \zeta$ for any non-zero rational function ζ on U. For $P'' := F'$, the prime divisor P in (E.15) satisfies $\operatorname{mult}_P \zeta = \operatorname{mult}_{F'}(\varpi^* \zeta) = \operatorname{mult}_F \zeta$ for any non-zero rational function ζ on U, so $P = F$. Hence $a(F', U_\Omega, B_\Omega) = a(F, W, D) = b e_F$ and $I_{G'}(F') = I_{G'}(F) = G'$, so $|I_{G'}(F')| = e_F$. By (E.16),

$$\operatorname{mld}((U_\Omega \ni c', B_\Omega)/G') = b > 0,$$

and F' is a computing lift of $\operatorname{mld}((U_\Omega \ni c', B_\Omega)/G')$. Moreover, by **Step 3** and (E.14),

$$\operatorname{mult}_{F'}(\varpi^* K_{U/W}) = \operatorname{mult}_{F'}(\varpi^* \psi) = \operatorname{mult}_F K_{U/W} \le N,$$

$$\operatorname{mult}_{F'} \mathfrak{m}_{U_\Omega, c'} \le \operatorname{mult}_{F'}(\varpi^* \mathfrak{m}_{U,Z}) = \operatorname{mult}_F \mathfrak{m}_{U,Z} \le \frac{3M}{\eta_m},$$

and since $\operatorname{mult}_{F'}$ takes integer values, $\operatorname{mult}_{F'} \mathfrak{m}_{U_\Omega, c'} \le N_0$.

By Theorem 15.2 for the germ $(U_\Omega \ni c')$ over Ω, the cyclic group G', the function $\varpi^* g^* h$ and the divisor $\varpi^* K_{U/W}$ as D_T, there exists a computing lift P'' of $\operatorname{mld}((U_\Omega \ni c', B_\Omega)/G') = b$ with

$$a(P'', U_\Omega, 0) \le L_1 \quad \text{and} \quad \operatorname{mult}_{P''}(\varpi^* \psi) \le L_1.$$

Let P be the prime divisor over U in (E.15) for P''. Since P'' computes b, equality holds in (E.16), so $a(P,W,D) = be_P$ and P is a computing lift of b. By (E.15),

$$\operatorname{mult}_P B_U = \frac{1}{m}\operatorname{mult}_P(g^*h) - \operatorname{mult}_P \psi = \frac{1}{m}\operatorname{mult}_{P''}(\varpi^* g^* h) - \operatorname{mult}_{P''}(\varpi^*\psi) = \operatorname{mult}_{P''} B_\Omega,$$

so

$$a(P,U,0) = a(P,U,B_U) + \operatorname{mult}_P B_U = a(P'',U_\Omega,B_\Omega) + \operatorname{mult}_{P''} B_\Omega = a(P'',U_\Omega,0) \le L_1.$$

Since the center of P on U is $Z \cap U$,

$$\operatorname{mult}_P K_{U/W} = \operatorname{mult}_P \psi = \operatorname{mult}_{P''}(\varpi^*\psi) \le L_1.$$

By (E.13),

$$a(P,W,0) = a(P,U,0) + \operatorname{mult}_P K_{U/W} \le 2L_1 \le L,$$

so P satisfies the bounds of Proposition 15.3(3).

Step 6. In this step we prove Proposition 15.3(4), the case when Z is a closed point. Then $Z \in U$, and (E.14) gives $\operatorname{mult}_F \mathfrak{m}_{U,Z} \le 3M/\eta_m$. Let P be a prime divisor on U with $Z \in P$. Since U is smooth and $Z = \operatorname{center}_U F$, $\operatorname{mult}_F P \ge 1$. Since $K_{U/W} \ge 0$,

$$0 \le \operatorname{mult}_P K_{U/W} \le \operatorname{mult}_F K_{U/W} \le N.$$

Since g^*h is regular, $\operatorname{mult}_P \operatorname{div}_U(g^*h) \in \mathbb{N}$. Thus

$$\operatorname{mult}_P B_U = \frac{1}{m}\operatorname{mult}_P \operatorname{div}_U(g^*h) - \operatorname{mult}_P K_{U/W} \in \frac{1}{m}\mathbb{Z} \cap [-N, +\infty).$$

By (E.13), $\operatorname{mult}_P B_U = 1 - a(P,U,B_U) = 1 - a(P,W,D) \le 1$ since (W,D) is lc. Hence the coefficients of $B_U = \frac{1}{m}\operatorname{div}(g^*h) - K_{U/W}$ near Z belong to $\frac{1}{m}\mathbb{Z} \cap [-N, 1]$.

The proposition follows. □

E.3. **Jacobian lifts.** In this subsection we prove Proposition 15.5 of Section 15.

Proof of Proposition 15.5. For a power series g in variables $y_1, \dots, y_n$ over a field we let

$$\partial_j := \partial/\partial y_j \quad \text{and} \quad J(g) := (g, \partial_1 g, \dots, \partial_n g).$$

Let $\mathbb{K}$ be a field and $g \in \mathbb{K}[[y_1, \dots, y_n]]$. For any $\mathbb{K}$-automorphism φ of $\mathbb{K}[[y_1, \dots, y_n]]$ and any unit u, the Jacobian matrix of φ is invertible, so the chain rule and $\partial_j(ug) = u\partial_j g + g\partial_j u$ imply

$$J(u\varphi(g)) = \varphi(J(g)). \tag{E.17}$$

We let $D_i := \frac{1}{m}\operatorname{div}(h_i)$, $D_* := \frac{1}{m}\operatorname{div}(h_*)$, $J_i := J(f_i)$, and $J_* := J(f_*)$ in the variables $x_1, \dots, x_4$. Let $\mathfrak{n}$ and $\mathfrak{n}_*$ be the maximal ideals of $\mathbb{C}[[x_1, \dots, x_4]]$ and $\Bbbk[[x_1, \dots, x_4]]$, and o_* the closed point of $\mathcal{X}_*$. Then $\operatorname{Jac}_{W_i} = J_i \mathcal{O}_{W_i}$, and, by the Jacobian criterion,

$$V(J_*) = \operatorname{Sing}(W_*) \quad \text{and} \quad V(J_i) = \operatorname{Sing}(W_i) \subset \{o_i\}. \tag{E.18}$$

Since W_* is normal, $\dim \operatorname{Sing}(W_*) \le 1$. We divide the proof into four steps. In **Steps 2** and 3 we assume that no lc center of (W_*, D_*) is contained in $\operatorname{Sing}(W_*)$.

Step 1. In this step we show that a lower bound of an lc threshold on a completion passes to algebraic truncations, and that it passes from (f_*, h_*) to (f_i, h_i) for all but finitely many i.

Let $\mathbb{K} \supset \mathbb{C}$ be an algebraically closed field, $(X \ni x)$ a klt germ of a variety of dimension d over $\mathbb{K}$, $\widehat{X} := \operatorname{Spec} \widehat{\mathcal{O}}_{X,x}$, $\mathfrak{m}_{\widehat{X}}$ the maximal ideal of $\widehat{\mathcal{O}}_{X,x}$, and c and γ positive real numbers such that

$$\operatorname{lct}(X \ni x; \mathfrak{a}') \notin (c - \gamma, c) \quad \text{for any non-zero ideal } \mathfrak{a}' \text{ on a neighborhood of } x. \tag{E.19}$$

Let N be a positive integer with $d/N < \gamma$, and let $\mathfrak{a} \subset \widehat{\mathcal{O}}_{X,x}$ be a non-zero proper ideal with $\operatorname{lct}(\widehat{X} \ni x; \mathfrak{a} + \mathfrak{m}_{\widehat{X}}^N) \ge c$. We show the following.

(i) For any integer $N' \ge N$ and any non-zero ideal $\mathfrak{a}'$ on a neighborhood of x with

$$\mathfrak{a}'\widehat{\mathcal{O}}_{X,x} + \mathfrak{m}_{\widehat{X}}^{N'} = \mathfrak{a} + \mathfrak{m}_{\widehat{X}}^{N'},$$

we have $\operatorname{lct}(X \ni x; \mathfrak{a}') \ge c$.

(ii) $\mathrm{lct}(\widehat{X} \ni x; \mathfrak{a}) \geq c$.

For (i), since $N' \geq N$, we have

$$\mathfrak{a}'\widehat{\mathcal{O}}_{X,x} + \mathfrak{m}_{\widehat{X}}^N = \mathfrak{a} + \mathfrak{m}_{\widehat{X}}^N,$$

which is a proper ideal. By [dFEM11, Proposition 2.11] and by Theorem 2.23(2) on the completion,

$$\mathrm{lct}(X \ni x; \mathfrak{a}') = \mathrm{lct}(\widehat{X} \ni x; \mathfrak{a}'\widehat{\mathcal{O}}_{X,x}) \geq \mathrm{lct}(\widehat{X} \ni x; \mathfrak{a} + \mathfrak{m}_{\widehat{X}}^N) - \frac{d}{N} > c - \gamma.$$

Thus $\mathrm{lct}(X \ni x; \mathfrak{a}') \geq c$ by (E.19). For (ii), let $N' \geq N$ be an integer. Since

$$\mathcal{O}_{X,x}/\mathfrak{m}_x^{N'} = \widehat{\mathcal{O}}_{X,x}/\mathfrak{m}_{\widehat{X}}^{N'},$$

where $\mathfrak{m}_x$ is the maximal ideal of $\mathcal{O}_{X,x}$, there exists a non-zero ideal $\mathfrak{a}^{(N')}$ on a neighborhood of x with

$$\mathfrak{a}^{(N')}\widehat{\mathcal{O}}_{X,x} + \mathfrak{m}_{\widehat{X}}^{N'} = \mathfrak{a} + \mathfrak{m}_{\widehat{X}}^{N'}.$$

By Theorem 2.23(2) on the completion, by [dFEM11, Proposition 2.11] and by (i),

$$\mathrm{lct}(\widehat{X} \ni x; \mathfrak{a}) \geq \mathrm{lct}(X \ni x; \mathfrak{a}^{(N')}) - \frac{d}{N'} \geq c - \frac{d}{N'}.$$

Letting $N' \to +\infty$ we obtain (ii).

Next let $\epsilon > 0$ be a rational number, N_0 a positive integer with $N_0/m \in \mathbb{Z}$ and $N_0\epsilon \in \mathbb{Z}$, and

$$\mathfrak{a}_i := (f_i)^{N_0}(h_i)^{N_0/m}J_i^{N_0\epsilon} \subset \mathbb{C}[[x_1, \dots, x_4]] \quad \text{and} \quad \mathfrak{a}_* := (f_*)^{N_0}(h_*)^{N_0/m}J_*^{N_0\epsilon} \subset \Bbbk[[x_1, \dots, x_4]].$$

We show that, if $(\mathcal{X}_*, (f_*)^1(h_*)^{1/m}J_*^{\epsilon})$ is lc, that is, $\mathrm{lct}(\mathcal{X}_* \ni o_*; \mathfrak{a}_*) \geq 1/N_0$, then

$$\mathrm{lct}(\mathcal{X} \ni o_i; \mathfrak{a}_i) \geq \frac{1}{N_0} \quad \text{for all but finitely many } i. \tag{E.20}$$

By [EM08, Theorem 1.1 and Remark 3.3], the lc thresholds at 0 of non-zero ideals on $\mathbb{A}^4_{\mathbb{C}}$ satisfy the ACC. Thus there exists $\gamma > 0$ satisfying (E.19) for $(X \ni x) = (\mathbb{A}^4_{\mathbb{C}} \ni 0)$ and $c = 1/N_0$. We fix an integer N with $4/N < \gamma$. Since $f_i \in \mathfrak{n}$ and $f_ih_i \neq 0$, $\mathfrak{a}_i$ is a non-zero proper ideal, so by (ii) we only need to prove

$$\mathrm{lct}(\mathcal{X} \ni o_i; \mathfrak{a}_i + \mathfrak{n}^N) \geq \frac{1}{N_0} \quad \text{for all but finitely many } i.$$

The ideal $\mathfrak{a}_i + \mathfrak{n}^N$ depends only on the $(N+1)$-jet of (f_i, h_i). Recall from Definition 2.5 that the $(N+1)$-jet of (f_*, h_*) is a $\Bbbk$-point of the jet space $\mathcal{J}_{N+1}$ over the generic point ζ of an irreducible closed subset $Z_{N+1} \subset \mathcal{J}_{N+1}$, and that any non-empty open subset of Z_{N+1} contains the $(N+1)$-jet z_i of (f_i, h_i) for all but finitely many i. For a point $z \in \mathcal{J}_{N+1}$ with residue field $\kappa(z)$, let (f_z, h_z) be the pair of polynomials over $\kappa(z)$ of degree at most N represented by z, and

$$\mathfrak{b}_z := (f_z)^{N_0}(h_z)^{N_0/m}J(f_z)^{N_0\epsilon} + (x_1, \dots, x_4)^N \subset \kappa(z)[x_1, \dots, x_4].$$

Then

$$\mathfrak{a}_i + \mathfrak{n}^N = \mathfrak{b}_{z_i}\mathbb{C}[[x_1, \dots, x_4]] \quad \text{and} \quad \mathfrak{a}_* + \mathfrak{n}_*^N = \mathfrak{b}_\zeta\Bbbk[[x_1, \dots, x_4]].$$

By [dFEM11, Proposition 2.11 and Lemma 2.14], lc thresholds are invariant under completion and under extension of the coefficient field, so

$$\mathrm{lct}(\mathcal{X} \ni o_i; \mathfrak{a}_i + \mathfrak{n}^N) = \mathrm{lct}(\mathbb{A}^4_{\mathbb{C}} \ni 0; \mathfrak{b}_{z_i}),$$

$$\mathrm{lct}(\mathbb{A}^4_{\kappa(\zeta)} \ni 0; \mathfrak{b}_\zeta) = \mathrm{lct}(\mathcal{X}_* \ni o_*; \mathfrak{a}_* + \mathfrak{n}_*^N) \geq \mathrm{lct}(\mathcal{X}_* \ni o_*; \mathfrak{a}_*) \geq \frac{1}{N_0},$$

where we also write $\mathfrak{b}_\zeta$ for the extension of $\mathfrak{b}_\zeta$ to $\overline{\kappa(\zeta)}[x_1, \dots, x_4]$.

A log resolution of $(\mathbb{A}^4_{\kappa(\zeta)}, \mathfrak{b}_\zeta)$ is defined over a finite extension of $\kappa(\zeta)$, that is, over the function field of a variety Z' with a finite dominant morphism $Z' \to Z_{N+1}$. Spreading it out, we obtain a dense open subset $U' \subset Z'$, the ideal $\mathfrak{b} \subset \mathcal{O}_{\mathbb{A}^4_{U'}}$ given by the formula of $\mathfrak{b}_z$ with the coefficients

of the universal $(N+1)$-jet, whose fiber over a closed point $z' \in U'$ with image z is $\mathfrak{b}_z$, and a projective birational morphism $Y' \to \mathbb{A}^4_{U'}$ satisfying the following.

(1) For any closed point $z' \in U'$ with image $z \in Z_{N+1}$, the fiber of Y' over z' is a log resolution of $(\mathbb{A}^4_{\mathbb{C}}, \mathfrak{b}_z)$.
(2) The prime components of the exceptional divisor and of the divisor of $\mathfrak{b}\mathcal{O}_{Y'}$ are smooth over U' with irreducible fibers, and the log discrepancy of $\mathbb{A}^4$ and the order of $\mathfrak{b}$ along any of them are constant on U'.
(3) For any such prime component E', the set of points of U' over which E' intersects the preimage of the zero section is either U' or empty.

Since the lc threshold at 0 of an ideal on $\mathbb{A}^4$ is the minimum, over the prime components E of a log resolution whose images contain 0, of $a(E, \mathbb{A}^4)$ divided by the order of the ideal along E, we obtain

$$\operatorname{lct}(\mathbb{A}^4_{\mathbb{C}} \ni 0; \mathfrak{b}_z) = \operatorname{lct}(\mathbb{A}^4_{\kappa(\zeta)} \ni 0; \mathfrak{b}_\zeta) \geq \frac{1}{N_0}$$

for any closed point z in the image of U'. The image of U' contains a dense open subset of Z_{N+1}, which contains z_i for all but finitely many i. This proves (E.20).

Step 2. In this step we show that, for the generic point ξ of any curve component of $\operatorname{Sing}(W_*)$, there exists a rational number $\rho_\xi > 0$ such that

$$a(E, \mathcal{X}_*, (f_*)^1(h_*)^{1/m}) \geq \rho_\xi \operatorname{mult}_E J_* \tag{E.21}$$

for any prime divisor E over $\mathcal{X}_*$ with $\operatorname{center}_{\mathcal{X}_*} E = \overline{\{\xi\}}$.

Now we fix ξ. Let $\kappa(\xi)$ be the residue field of $\mathcal{X}_*$ at ξ and $\mathbb{K}_\xi$ an algebraic closure of $\kappa(\xi)$. Since $\widehat{\mathcal{O}}_{\mathcal{X}_*,\xi}$ is a complete regular local ring of dimension 3 containing $\mathbb{Q}$, $\widehat{\mathcal{O}}_{\mathcal{X}_*,\xi} \cong \kappa(\xi)[[y_1, y_2, y_3]]$. We let

$$R_\xi := \mathbb{K}_\xi[[y_1, y_2, y_3]], \quad T := \operatorname{Spec} R_\xi/(g), \quad \text{and} \quad D_T := \frac{1}{m} \operatorname{div}(\widetilde{h}|_T),$$

where $g, \widetilde{h} \in R_\xi$ are the images of f_*, h_*, and we let o_T be the closed point of T and $\mathfrak{m}_\xi$ the maximal ideal of R_ξ. The morphism $T \to W_*$ is the composite of the localization at ξ, the completion and the extension of the coefficient field from $\kappa(\xi)$ to $\mathbb{K}_\xi$. It is the base change of $\operatorname{Spec} R_\xi \to \mathcal{X}_*$, which is a regular morphism as $\Bbbk[[x_1, \dots, x_4]]$ is excellent of characteristic zero (for the extension of the coefficient field by [Mat89, Theorem 28.10] and André's theorem [Mat89, p. 260]), so log resolutions pull back to log resolutions along it (cf. [dFEM11, Remark 2.10]) and log discrepancies are preserved by it: on the regular scheme $\mathcal{X}_*$ relative canonical divisors pull back [JM12, Lemma 1.2], and on the Cartier divisor $W_* \subset \mathcal{X}_*$, which is Gorenstein, they are given by the dualizing sheaves, which are compatible with the base change along $\operatorname{Spec} R_\xi \to \mathcal{X}_*$. Since W_* is normal and canonical, T is a normal canonical surface germ, hence klt. Moreover, (T, D_T) is lc, $\widetilde{h}|_T \neq 0$, and, since ξ is a generic point of $\operatorname{Sing}(W_*) = V(J_*)$, $\operatorname{Sing}(T) \subset \{o_T\}$ and $V(J_*\mathcal{O}_T) \subset \{o_T\}$. We show that

$$a(E, T, D_T) > 0 \quad \text{for any prime divisor } E \text{ over } T \text{ with } \operatorname{center}_T E = \{o_T\}. \tag{E.22}$$

Let $Y \to W_*$ be a log resolution of (W_*, D_*). Then the base change $Y_T \to T$ of $Y \to W_*$ is a log resolution of (T, D_T). If $a(E, T, D_T) = 0$ and $\operatorname{center}_T E = \{o_T\}$, then $\operatorname{center}_{Y_T} E$ is a stratum of the coefficient-one part of the boundary on Y_T with image o_T, which is a component of the preimage of a stratum of the coefficient-one part of the boundary on Y with image $\overline{\{\xi\}}$. Then $\overline{\{\xi\}} \subset \operatorname{Sing}(W_*)$ is an lc center of (W_*, D_*), contrary to the assumption of **Steps 2** and 3 that no lc center of (W_*, D_*) is contained in $\operatorname{Sing}(W_*)$.

Let $Y'_T \to T$ be a log resolution of $(T, D_T, J_*\mathcal{O}_T)$ [dFEM11, Theorems 2.3 and 2.4]. Since $V(J_*\mathcal{O}_T) \subset \{o_T\}$, any prime divisor E on Y'_T with $\operatorname{mult}_E J_* > 0$ has center o_T, and by (E.22) it satisfies $a(E, T, D_T) > 0$. As there exist finitely many such E, we may choose a rational number $\rho_\xi > 0$ with $\rho_\xi \operatorname{mult}_E J_* \leq a(E, T, D_T)$ for any prime divisor E on Y'_T. If $\operatorname{mult}_E J_* = 0$, the

inequality follows from the lc property of (T, D_T). Since the boundary on Y'_T has simple normal crossing support,

$$(T, D_T, (J_*\mathcal{O}_T)^{\rho_\xi}) \text{ is lc.} \tag{E.23}$$

To apply **Step 1**(i) and [EMY03, Corollary 1.8], which are results on varieties, we algebraize T. By the Jacobian criterion, $V(J(g)) = \mathrm{Sing}(T) \subset \{o_T\}$, so there exists a positive integer k with $\mathfrak{m}_\xi^{k-1} \subset J(g)$, so

$$\mathfrak{m}_\xi^k \subset (g) + \mathfrak{m}_\xi(\partial_1 g, \partial_2 g, \partial_3 g).$$

Let $p \in \mathbb{K}_\xi[y_1, y_2, y_3]$ be the sum of the terms of g of degree at most $2k$. By Lemma 3.2(2) with the trivial group, $c = k$ and $L = 2k + 1$, $p = u\varphi(g)$ for an automorphism φ of R_ξ and a unit u, so φ induces an isomorphism from T to the completion $\widehat{T}_0 = \mathrm{Spec}\, R_\xi/(p)$ at 0 of $T_0 := (p = 0) \subset \mathbb{A}^3_{\mathbb{K}_\xi}$. Possibly shrinking $\mathbb{A}^3_{\mathbb{K}_\xi}$ to a neighborhood of 0, we may assume that T_0 is normal and klt [dFEM11, Proposition 2.11]. We let $h_\xi := \varphi(\widetilde{h})$ and $J_\xi := \varphi(J_*R_\xi)$, and I_ξ be a positive integer with $I_\xi/m \in \mathbb{Z}$ and $I_\xi\rho_\xi \in \mathbb{Z}$. By (E.23),

$$\mathrm{lct}(\widehat{T}_0 \ni 0; \mathfrak{a}_\xi) \geq \frac{1}{I_\xi}, \qquad \mathfrak{a}_\xi := (h_\xi)^{I_\xi/m} J_\xi^{I_\xi\rho_\xi} \mathcal{O}_{\widehat{T}_0},$$

where $\mathfrak{a}_\xi$ is a non-zero proper ideal since $h_\xi|_{\widehat{T}_0} \neq 0$ and $J_\xi\mathcal{O}_{\widehat{T}_0}$ is primary to the maximal ideal.

By [dFEM11, Theorem 1.1] on the single klt germ $(T_0 \ni 0)$, the lc thresholds at 0 of non-zero ideals on T_0 satisfy the ACC, so there exists $\gamma_\xi > 0$ satisfying (E.19) with γ_ξ in place of γ for $(X \ni x) = (T_0 \ni 0)$ and $c = 1/I_\xi$. We fix an integer N_ξ with $2/N_\xi < \gamma_\xi$. For an integer $N' \geq N_\xi$, let $h_\xi^{(N')} \in \mathbb{K}_\xi[y_1, y_2, y_3]$ with $h_\xi - h_\xi^{(N')} \in \mathfrak{m}_\xi^{N'}$, and let $J_\xi^{(N')} \subset \mathbb{K}_\xi[y_1, y_2, y_3]$ be generated by polynomials congruent modulo $\mathfrak{m}_\xi^{N'}$ to a fixed finite set of generators of J_ξ. Since $h_\xi|_{\widehat{T}_0} \neq 0$ and $J_\xi\mathcal{O}_{\widehat{T}_0}$ is primary to the maximal ideal, by Nakayama's lemma, for $N' \gg 0$, $h_\xi^{(N')}$ does not vanish on T_0 and $J_\xi^{(N')}\mathcal{O}_{T_0}$ is primary to the maximal ideal of 0. Since

$$\mathrm{lct}(\widehat{T}_0 \ni 0; \mathfrak{a}_\xi + \mathfrak{m}_\xi^{N_\xi}) \geq \mathrm{lct}(\widehat{T}_0 \ni 0; \mathfrak{a}_\xi)$$

and the products of ideals generated by generators congruent modulo $\mathfrak{m}_\xi^{N'}$ are congruent modulo $\mathfrak{m}_\xi^{N'}$, **Step 1**(i) implies

$$(T_0, \tfrac{1}{m}\,\mathrm{div}(h_\xi^{(N')}|_{T_0}), (J_\xi^{(N')}\mathcal{O}_{T_0})^{\rho_\xi}) \text{ is lc near } 0 \quad \text{for any } N' \gg 0.$$

The polynomials p, $h_\xi^{(N')}$ and the generators of $J_\xi^{(N')}$ have coefficients in a subfield of $\mathbb{K}_\xi$ finitely generated over $\mathbb{Q}$, which embeds into $\mathbb{C}$, and the lc property is invariant under extension of algebraically closed coefficient fields [dFEM11, Lemma 2.14(ii)]. Thus [EMY03, Corollary 1.8] holds over $\mathbb{K}_\xi$, and, since T_0 is normal near 0,

$$(\mathbb{A}^3_{\mathbb{K}_\xi}, (p)^1(h_\xi^{(N')})^{1/m}(J_\xi^{(N')})^{\rho_\xi}) \text{ is lc near } 0 \quad \text{for any } N' \gg 0.$$

Let

$$\mathfrak{c}_\xi := (p)^{I_\xi}(h_\xi)^{I_\xi/m} J_\xi^{I_\xi\rho_\xi} \subset R_\xi \quad \text{and} \quad \mathfrak{c}_\xi^{(N')} := (p)^{I_\xi}(h_\xi^{(N')})^{I_\xi/m}(J_\xi^{(N')})^{I_\xi\rho_\xi}.$$

Then

$$\mathfrak{c}_\xi^{(N')} R_\xi + \mathfrak{m}_\xi^{N'} = \mathfrak{c}_\xi + \mathfrak{m}_\xi^{N'},$$

and by Theorem 2.23(2) on the completion and [dFEM11, Proposition 2.11],

$$\mathrm{lct}(\mathrm{Spec}\, R_\xi \ni 0; \mathfrak{c}_\xi) \geq \mathrm{lct}(\mathbb{A}^3_{\mathbb{K}_\xi} \ni 0; \mathfrak{c}_\xi^{(N')}) - \frac{3}{N'} \geq \frac{1}{I_\xi} - \frac{3}{N'}.$$

Letting $N' \to +\infty$, $(\mathrm{Spec}\, R_\xi, (p)^1(h_\xi)^{1/m} J_\xi^{\rho_\xi})$ is lc. Since $(p) = \varphi((g))$, applying φ^{-1} we see that $(\mathrm{Spec}\, R_\xi, (g)^1(\widetilde{h})^{1/m}(J_*R_\xi)^{\rho_\xi})$ is lc. The pair $(\mathrm{Spec}\, R_\xi, (g)^1(\widetilde{h})^{1/m}(J_*R_\xi)^{\rho_\xi})$ is the pullback of

$(\mathcal{X}_*, (f_*)^1(h_*)^{1/m} J_*^{\rho_\xi})$ along $\operatorname{Spec} R_\xi \to \mathcal{X}_*$, so, by the regularity of $\operatorname{Spec} R_\xi \to \mathcal{X}_*$ and by [JM12, Lemma 1.2], as for $T \to W_*$ above, the pair $(\mathcal{X}_*, (f_*)^1(h_*)^{1/m} J_*^{\rho_\xi})$ is lc at ξ, which is (E.21).

Step 3. In this step we prove (1).

By Definition 2.5, the orders of f_i and h_i at o_i are bounded. Thus Lemma 13.4 with the diagonal group $G \subset GL_4(\mathbb{C})$ implies that $(\mathcal{X}_*, (f_*)^1(h_*)^{1/m})$ is lc and $a(E, \mathcal{X}_*, (f_*)^1(h_*)^{1/m}) > 0$ for any prime divisor E over $\mathcal{X}_*$ with center o_*.

Let $Y_* \to \mathcal{X}_*$ be a log resolution of $(\mathcal{X}_*, (f_*)(h_*)J_*)$ [dFEM11, Theorems 2.3 and 2.4], and let E be a prime divisor on Y_* with $\operatorname{mult}_E J_* > 0$. By (E.18), $\operatorname{center}_{\mathcal{X}_*} E \subset \operatorname{Sing}(W_*)$, and since $\dim \operatorname{Sing}(W_*) \le 1$, either $\operatorname{center}_{\mathcal{X}_*} E = \{o_*\}$ or $\operatorname{center}_{\mathcal{X}_*} E = \overline{\{\xi\}}$ for the generic point ξ of a curve component of $\operatorname{Sing}(W_*)$. In the first case $a(E, \mathcal{X}_*, (f_*)^1(h_*)^{1/m}) > 0$ by the positivity of the minimal log discrepancy at o_*, and in the second case, by (E.21),

$$a(E, \mathcal{X}_*, (f_*)^1(h_*)^{1/m}) \ge \rho_\xi \operatorname{mult}_E J_* > 0.$$

Since $a(E, \mathcal{X}_*, (f_*)^1(h_*)^{1/m}) \in \frac{1}{m}\mathbb{Z}$, we may let ϵ be the minimum of the finite set of rational numbers

$$\{1\} \cup \left\{ \frac{a(E, \mathcal{X}_*, (f_*)^1(h_*)^{1/m})}{\operatorname{mult}_E J_*} \;\middle|\; E \text{ a prime divisor on } Y_*,\ \operatorname{mult}_E J_* > 0 \right\}.$$

Then $\epsilon > 0$, and $a(E, \mathcal{X}_*, (f_*)^1(h_*)^{1/m} J_*^{\epsilon}) \ge 0$ for any prime divisor E on Y_* (for $\operatorname{mult}_E J_* = 0$ by the lc property of $(\mathcal{X}_*, (f_*)^1(h_*)^{1/m})$), so

$$(\mathcal{X}_*, (f_*)^1(h_*)^{1/m} J_*^{\epsilon}) \text{ is lc.}$$

Let N_0 be a positive integer with $N_0/m \in \mathbb{Z}$ and $N_0\epsilon \in \mathbb{Z}$. By (E.20), for all but finitely many i,

$$(\mathcal{X}, (f_i)^1(h_i)^{1/m} J_i^{\epsilon}) \text{ is lc.} \tag{E.24}$$

We fix such an i and show that $(W_i, D_i, \operatorname{Jac}_{W_i}^{\epsilon})$ is lc near o_i. By (E.18), J_i is $\mathfrak{n}$-primary or the unit ideal, so there exists a positive integer k with $\mathfrak{n}^{k-1} \subset J_i$, hence

$$\mathfrak{n}^k \subset (f_i) + \mathfrak{n}(\partial_1 f_i, \dots, \partial_4 f_i).$$

Let f_i' be the sum of the terms of f_i of degree at most $2k$. By Lemma 3.2(2) with the trivial group, $c = k$ and $L = 2k+1$, and by (E.17), there exist an automorphism φ_i of $\mathbb{C}[[x_1, \dots, x_4]]$ and a unit u_i with

$$f_i' = u_i \varphi_i(f_i) \quad \text{and} \quad J(f_i') = \varphi_i(J_i).$$

We let $h_i' := \varphi_i(h_i)$ and

$$\mathfrak{a}_i' := (f_i')^{N_0} (h_i')^{N_0/m} J(f_i')^{N_0\epsilon}.$$

Let γ and N be as in **Step 1** for $(\mathbb{A}^4_{\mathbb{C}} \ni 0)$ and $c = 1/N_0$. By (E.24), $\operatorname{lct}(\mathcal{X} \ni o_i; \mathfrak{a}_i') \ge 1/N_0$, so

$$\operatorname{lct}(\mathcal{X} \ni o_i; \mathfrak{a}_i' + \mathfrak{n}^N) \ge \frac{1}{N_0}.$$

For an integer $N' \ge N$, let $h_i'^{(N')} \in \mathbb{C}[x_1, \dots, x_4]$ with $h_i' - h_i'^{(N')} \in \mathfrak{n}^{N'}$. We let

$$\mathfrak{a}_i'^{(N')} := (f_i')^{N_0} (h_i'^{(N')})^{N_0/m} J(f_i')^{N_0\epsilon}.$$

Since

$$\mathfrak{a}_i'^{(N')} + \mathfrak{n}^{N'} = \mathfrak{a}_i' + \mathfrak{n}^{N'},$$

Step 1(i) implies

$$(\mathbb{A}^4_{\mathbb{C}}, (f_i')^1 (h_i'^{(N')})^{1/m} J(f_i')^{\epsilon}) \text{ is lc near } 0 \quad \text{for any } N' \ge N \text{ with } N' > \operatorname{mult}_0 h_i'.$$

The hypersurface $W_i' := (f_i' = 0) \subset \mathbb{A}^4_{\mathbb{C}}$ is normal near 0, since φ_i induces an isomorphism $W_i \cong \widehat{W_i'}$, where $\widehat{W_i'}$ is the completion of W_i' at 0. Since $h_i|_{W_i} \ne 0$, $h_i'^{(N')}$ does not vanish on W_i'

for $N' \gg 0$, and $J(f_i')\mathcal{O}_{W_i'}$ is primary to the maximal ideal of 0 or the unit ideal near 0. Thus, by [EMY03, Corollary 1.8],

$$(W_i', \tfrac{1}{m}\operatorname{div}(h_i'^{(N')}|_{W_i'}), (J(f_i')\mathcal{O}_{W_i'})^{\epsilon}) \text{ is lc near } 0 \quad \text{for any } N' \gg 0.$$

Passing to the completion [dFEM11, Proposition 2.11] and applying φ_i^{-1}, we see that

$$(W_i, \tfrac{1}{m}\operatorname{div}(h_i^{(N')}), \operatorname{Jac}_{W_i}^{\epsilon}) \text{ is lc for any } N' \gg 0, \qquad h_i^{(N')} := \varphi_i^{-1}(h_i'^{(N')}),$$

where $h_i - h_i^{(N')} \in \mathfrak{n}^{N'}$. Let P be a prime divisor over W_i. If $\operatorname{center}_{W_i} P = \{o_i\}$, then, choosing $N' \gg 0$ with $N' > \operatorname{mult}_P h_i$, we have

$$\operatorname{mult}_P(h_i - h_i^{(N')}) \ge N' > \operatorname{mult}_P h_i,$$

so $\operatorname{mult}_P h_i^{(N')} = \operatorname{mult}_P h_i$ and

$$a(P, W_i, D_i, \operatorname{Jac}_{W_i}^{\epsilon}) = a(P, W_i, \tfrac{1}{m}\operatorname{div}(h_i^{(N')}), \operatorname{Jac}_{W_i}^{\epsilon}) \ge 0.$$

If $\operatorname{center}_{W_i} P \ne \{o_i\}$, then $\operatorname{mult}_P \operatorname{Jac}_{W_i} = 0$ by (E.18), and

$$a(P, W_i, D_i, \operatorname{Jac}_{W_i}^{\epsilon}) = a(P, W_i, D_i) > 0$$

since (W_i, D_i) is klt near o_i. Thus $(W_i, D_i, \operatorname{Jac}_{W_i}^{\epsilon})$ is lc near o_i for all but finitely many i.

Let i be an index with $(W_i, D_i, \operatorname{Jac}_{W_i}^{\epsilon})$ lc near o_i, and F a computing lift of b_i (Definition 2.4). Then $\operatorname{center}_{W_i} F = \{o_i\}$ and $a(F, W_i, D_i) = e_F b_i$, and $e_F = |I_G(F)| \le |G| \le M$. Thus

$$\epsilon \operatorname{mult}_F \operatorname{Jac}_{W_i} \le a(F, W_i, D_i) = e_F b_i \le MB_0,$$

so

$$\operatorname{mult}_F \operatorname{Jac}_{W_i} \le \frac{MB_0}{\epsilon} \le \left\lceil \frac{MB_0}{\epsilon} \right\rceil.$$

This proves (1).

Step 4. In this step we prove (2). We assume that D_* has a coefficient-one component through o_*, and we let S be one of them.

By Definition 2.5, f_* and h_* are G-semi-invariant with the characters of f_i and h_i, so G acts on W_* and preserves D_*. By Lemma 13.5, (W_*, D_*) has a unique minimal lc center C, which is contained in any lc center of (W_*, D_*), is normal and G-invariant, and satisfies $\dim C \in \{1, 2\}$. If $C \not\subset \operatorname{Sing}(W_*)$, then no lc center V of (W_*, D_*) is contained in $\operatorname{Sing}(W_*)$, as $C \subset V$. We are then in the first alternative of (2).

Assume that $C \subset \operatorname{Sing}(W_*)$. Since $\dim \operatorname{Sing}(W_*) \le 1$, $\dim C = 1$, so the normal curve C is regular. Any coefficient-one component of D_* through o_* is an lc center, hence contains C. Since C is an lc center of (W_*, D_*) with $\dim C = \dim W_* - 2$, there exists an exceptional prime divisor E over W_* with $\operatorname{center}_{W_*} E = C$ and $a(E, W_*, D_*) = 0$, so (W_*, D_*) is not plt at the generic point of C.

It remains to determine the generic transverse type. By Lemma 13.5(2) with the trivial group, there exist regular parameters (s, u_1, u_2, u_3) of $\Bbbk[[x_1, \dots, x_4]]$ with $C = (u_1 = u_2 = u_3 = 0)$. Let $\mathbb{K}_C$ be an algebraic closure of $\Bbbk((s))$. Since $\Bbbk[[s]]\backslash\{0\}$ consists of units at the generic point of C, the generic transverse surface of W_* along C (Definition 15.1 and the paragraph after it) is

$$T_C := \operatorname{Spec} \mathbb{K}_C[[u_1, u_2, u_3]]/(f_*), \qquad f_* \in \Bbbk[[s]][[u_1, u_2, u_3]] \subset \mathbb{K}_C[[u_1, u_2, u_3]].$$

Assume that T_C is Du Val. We let D_{T_C} and S_C be the pullbacks of D_* and S to T_C. Since $C \subset S$, S_C is a non-zero reduced curve through the closed point of T_C, $S_C \le D_{T_C}$, and (T_C, D_{T_C}) is lc (as for $T \to W_*$ in **Step 2**, $T_C \to W_*$ being the composite of the localization at the generic point of C, the completion and the extension of the coefficient field from $\Bbbk((s))$ to $\mathbb{K}_C$), so (T_C, S_C) is lc. Since T_C is Du Val, $T_C \cong \operatorname{Spec} \mathbb{K}_C[[y_1, y_2, y_3]]/(q)$ for the ADE polynomial q of the type of T_C, which has rational coefficients. Over $\mathbb{C}$, the surface $(q = 0)$ is the quotient of $\mathbb{A}^2$ by a finite

subgroup of SL_2 whose invariants are polynomials with algebraic coefficients, so $(q=0)\subset \mathbb{A}^3_{\mathbb{K}_C}$ is the quotient of $\mathbb{A}^2_{\mathbb{K}_C}$ by a finite subgroup of $SL_2(\mathbb{K}_C)$, as $\mathbb{K}_C\supset\overline{\mathbb{Q}}$, and

$$T_C\cong \operatorname{Spec}\mathbb{K}_C[[z_1,z_2]]^Q$$

for a finite subgroup $Q\subset SL_2(\mathbb{K}_C)$ acting linearly, which is cyclic, binary dihedral, binary tetrahedral, binary octahedral, or binary icosahedral according as T_C is of type A_n, D_n, E_6, E_7, or E_8. We let $\widetilde{T}_C:=\operatorname{Spec}\mathbb{K}_C[[z_1,z_2]]$. A non-identity element of Q has no eigenvalue 1, so the quotient morphism $\pi:\widetilde{T}_C\to T_C$ is étale outside the closed point. Thus $S':=\pi^{-1}(S_C)$ is a reduced Q-invariant curve through 0, $K_{\widetilde{T}_C}+S'=\pi^*(K_{T_C}+S_C)$, and $(\widetilde{T}_C,S')$ is lc.

Since $(\widetilde{T}_C,S')$ is lc and S' is reduced, S' is smooth or has an ordinary node at 0, so it has one smooth branch or two smooth branches with distinct tangent lines.

If S' has one smooth branch with tangent line $\ell\subset\mathbb{K}_C^2$, then Q preserves ℓ and, by complete reducibility, a complementary line. Thus Q is diagonal in $SL_2(\mathbb{K}_C)$, hence cyclic, and T_C is of type A_n. If S' has two branches with tangent lines $\ell_1\neq\ell_2$, then Q permutes $\{\ell_1,\ell_2\}$, and the kernel Q_0 of $Q\to\operatorname{Sym}\{\ell_1,\ell_2\}$ is diagonal in a basis of $\ell_1\oplus\ell_2$, hence cyclic, with $[Q:Q_0]\le 2$. The binary tetrahedral, octahedral and icosahedral groups have orders 24, 48, 120 and maximal element orders 6, 8, 10, so they have no cyclic subgroup of index at most 2. Thus Q is cyclic or binary dihedral, and T_C is of type A_n or D_n. This proves (2). □

E.4. **One signed axis.** In this subsection we prove Proposition 15.6 of Section 15.

Proof of Proposition 15.6. We divide the proof into five steps. Elements of G are denoted by γ. For a G-equivariant proper birational morphism $g':Y'\to W$ and a smooth open subset $U'\subset Y'$, we let $K_{U'/W}:=K_{U'}-g'^*K_W$, a divisor on U' since W is Gorenstein. Then

$$a(P,W,0)=a(P,U',0)+\operatorname{mult}_P K_{U'/W} \tag{E.25}$$

for any prime divisor P over W whose center on Y' intersects U'.

Step 1. In this step we describe the blow-ups used below and their effect on $\operatorname{mult}_F K_{U'/W}$.

Let $g':Y'\to W$ be a G-equivariant proper birational morphism, $Z':=\operatorname{center}_{Y'}F$ with generic point ξ', and $U'\subset Y'$ a smooth G-invariant open subset satisfying the following.

(C1) $\xi'\in U'$.

(C2) $K_{U'/W}\ge 0$ and $g'(\operatorname{Supp}K_{U'/W})\subset\{o\}$.

By hypothesis, (Y_0,U_0) satisfies (C1) and (C2). For such (Y',U') we also write Z' for the closed subset $Z'\cap U'$ of U', whose generic point is ξ', and likewise for Z_j and Z in **Steps 2–4**.

Let $\Sigma\subset U'$ be an irreducible closed subset containing Z' and smooth at ξ', of codimension 2 or 3 in U', with $g'(\Sigma)=\{o\}$, such that $\gamma\Sigma=\Sigma$ for any $\gamma\in G$ with $\xi'\in\gamma\Sigma$. Let $U'^\circ\subset U'$ be the complement of the union of the closed subsets $\operatorname{Sing}(\gamma\Sigma)$ and $\gamma\Sigma\cap\gamma'\Sigma$, where $\gamma,\gamma'\in G$ and $\gamma\Sigma\neq\gamma'\Sigma$. By the condition on Σ, U'° is a G-invariant open subset containing $G\cdot\xi'$, and $(G\cdot\Sigma)\cap U'^\circ$ is a disjoint union of smooth closed subsets of codimension $\operatorname{codim}_{U'}\Sigma$. Let $\pi:Y''\to Y'$ be the normalization of the blow-up of Y' along the closure of $(G\cdot\Sigma)\cap U'^\circ$ with reduced structure, $g'':=g'\circ\pi$, $U'':=\pi^{-1}(U'^\circ)$, and E the exceptional divisor of $\pi|_{U''}$. Since the blown-up subscheme is G-invariant, G acts on Y'' and g'' is G-equivariant. Over U'°, π is the blow-up of a smooth center, so U'' is smooth and

$$K_{U''/W}=\pi^*K_{U'^\circ/W}+(\operatorname{codim}_{U'}\Sigma-1)E. \tag{E.26}$$

The generic point of $\operatorname{center}_{Y''}F$ is contained in $\pi^{-1}(U'^\circ)=U''$, since π maps it to $\xi'\in U'^\circ$, and

$$\dim\operatorname{center}_{Y''}F\ge\dim Z'.$$

Since $g''(E)=g'(\Sigma)=\{o\}$, (Y'',U'') satisfies (C1) and (C2). We call (Y'',U'') the *blow-up of* Σ *near* Z'.

Let u_i, $1\le i\le\operatorname{codim}_{U'}\Sigma$, be part of a regular system of parameters of $\mathcal{O}_{U',\xi'}$ generating the ideal $\mathcal{I}_\Sigma$ of Σ, ordered so that $\operatorname{mult}_F u_1=\min_i\operatorname{mult}_F u_i$. On the chart of the blow-up where

u_1 generates the ideal of E, the functions u_i/u_1 are regular and $\operatorname{mult}_F(u_i/u_1) \geq 0$. Thus the valuation ring of F contains

$$\mathcal{O}_{U',\xi'}[u_i/u_1 \mid 2 \leq i \leq \operatorname{codim}_{U'} \Sigma],$$

so $\operatorname{center}_{Y''} F$ intersects the chart and

$$\operatorname{mult}_F E = \operatorname{mult}_F u_1 = \operatorname{mult}_F \mathcal{I}_\Sigma \geq 1, \tag{E.27}$$

where $\operatorname{mult}_F \mathcal{I}_\Sigma \geq 1$ because $\Sigma \supset Z'$. Since $g'(\Sigma) = \{o\}$, we have

$$g'^{*}\mathfrak{m}_o \cdot \mathcal{O}_{U',\xi'} \subset \mathcal{I}_\Sigma,$$

so

$$\operatorname{mult}_F \mathcal{I}_\Sigma \leq \operatorname{mult}_F \mathfrak{m}_o \leq N_0. \tag{E.28}$$

By (E.26) and (E.27),

$$\operatorname{mult}_F K_{U''/W} = \operatorname{mult}_F K_{U'/W} + (\operatorname{codim}_{U'} \Sigma - 1) \operatorname{mult}_F \mathcal{I}_\Sigma. \tag{E.29}$$

Finally, if Z' is a prime divisor, then $\mathcal{O}_{U',\xi'}$ is a discrete valuation ring dominated by the valuation ring of F, so F is the prime divisor Z', $a(F, U', 0) = 1$, and by (E.25)

$$a(F, W, 0) = 1 + \operatorname{mult}_F K_{U'/W}. \tag{E.30}$$

Step 2. In this step we move the center of F off the strict transform of $\operatorname{Supp} K_{U_0/W}$ by at most N blow-ups.

We construct (Y_j, U_j) satisfying (C1) and (C2) for $j = 0, 1, \ldots$, each (Y_{j+1}, U_{j+1}) being a blow-up of some Σ_j near Z_j as in **Step 1**, where

$$Z_j := \operatorname{center}_{Y_j} F, \qquad \xi_j := \text{the generic point of } Z_j, \qquad g_j : Y_j \to W, \quad \text{and} \quad \pi_j : Y_{j+1} \to Y_j.$$

For $i \leq j$ let

$$\rho_{i,j} := \pi_i \circ \cdots \circ \pi_{j-1} : Y_j \to Y_i,$$

the identity if $i = j$. Then $\rho_{i,j}(\xi_j) = \xi_i$, and $\dim Z_j$ is increasing in j. Let $\mathcal{E}_j$ be the sum of the prime divisors on U_j that are exceptional over Y_0. For each j, let $\{E_i\}_{i \in J}$ be the set of components of $\mathcal{E}_j$ containing ξ_j.

Let D_0 be the sum of the components of $\operatorname{Supp} K_{U_0/W}$ containing z_0. Since $K_{U_0/W} \geq 0$ is an integral divisor, $D_0 \leq K_{U_0/W}$, so $\operatorname{mult}_F D_0 \leq \operatorname{mult}_F K_{U_0/W} \leq N$.

If $D_0 = 0$, then $\operatorname{mult}_F K_{U_0/W} = 0$. We let (Y_1, U_1) be the blow-up of $\Sigma_0 := \{z_0\}$ near z_0. Here Σ_0 satisfies the condition of **Step 1** since $g_0(z_0) = \operatorname{center}_W F = \{o\}$ and $\xi_0 = z_0 \in \gamma\Sigma_0$ implies $\gamma\Sigma_0 = \Sigma_0$. By (E.29) and (E.28), $\operatorname{mult}_F K_{U_1/W} \leq 2N_0$. We let $j_0 := 1$.

If $D_0 \neq 0$, let $D_j := (\rho_{0,j})_*^{-1} D_0$ on U_j. As long as $Z_j \subset D_j$ and $\dim Z_j \leq 1$, we let (Y_{j+1}, U_{j+1}) be the blow-up of $\Sigma_j := Z_j$ near Z_j. Here Σ_j satisfies the condition of **Step 1** since $g_j(Z_j) = \operatorname{center}_W F = \{o\}$ and $\xi_j \in \gamma Z_j$ implies $\gamma Z_j = Z_j$. Let E be the exceptional divisor of $\pi_j|_{U_{j+1}}$. Near ξ_{j+1}, we have $\pi_j^* D_j = D_{j+1} + (\operatorname{mult}_{Z_j} D_j)E$, where $\operatorname{mult}_{Z_j} D_j \geq 1$ is the multiplicity of D_j at ξ_j. By (E.27),

$$\operatorname{mult}_F D_{j+1} = \operatorname{mult}_F D_j - (\operatorname{mult}_{Z_j} D_j) \operatorname{mult}_F \mathcal{I}_{Z_j} \leq \operatorname{mult}_F D_j - 1.$$

Since $\operatorname{mult}_F D_0 \leq N$, and $\operatorname{mult}_F D_j$ is a non-negative integer which is positive if and only if $Z_j \subset D_j$, there exists $j_0 \leq N$ such that $Z_{j_0} \not\subset D_{j_0}$ or $\dim Z_{j_0} = 2$. Moreover $j_0 \geq 1$ since $z_0 \in D_0$. By (E.29) and (E.28), each blow-up raises $\operatorname{mult}_F K_{U_j/W}$ by at most $2N_0$.

Hence, whether $D_0 = 0$ or $D_0 \neq 0$,

$$\operatorname{mult}_F K_{U_{j_0}/W} \leq N + 2NN_0 = \frac{N_1}{2}. \tag{E.31}$$

If $\dim Z_j = 2$ for some $j \leq j_0$, then $a(F, W, 0) \leq 1 + \frac{N_1}{2}$ by (E.30), and the first alternative of Proposition 15.6 holds with the computing lift F. Otherwise $\dim Z_{j_0} \leq 1$, and the following hold for $j = j_0$.

(a) $\#J \geq 1$, and near ξ_j we have $K_{U_j/W} = \sum_{i\in J} c_i E_i$ for positive integers c_i.
(b) $\mathcal{E}_j$ is SNC near ξ_j.
(c) For any component E' of $\mathcal{E}_j$ and any $\gamma \in G$ with $\xi_j \in E' \cap \gamma E'$, we have $\gamma E' = E'$.

For (a), the components of $K_{U_{j_0}/W}$ are components of $\mathcal{E}_{j_0}$ or of the form $(\rho_{0,j_0})_*^{-1}P_0$ for a component P_0 of $K_{U_0/W}$. Since $\rho_{0,j_0}(\xi_{j_0}) = z_0$, we have $\xi_{j_0} \notin (\rho_{0,j_0})_*^{-1}P_0$ for any component P_0 of $K_{U_0/W}$ with $z_0 \notin P_0$, and $\xi_{j_0} \notin D_{j_0}$. By (E.26), each exceptional divisor of $\pi_{j'}$ has coefficient at least $\operatorname{codim}_{U_{j'}} \Sigma_{j'} - 1 \geq 1$ in $K_{U_{j'+1}/W}$, and for a prime divisor P_0 on $U_{j'+1}$ and $j'' > j'$, the coefficient of $(\rho_{j'+1,j''})_*^{-1}P_0$ in $K_{U_{j''}/W}$ equals the coefficient of P_0 in $K_{U_{j'+1}/W}$. The exceptional divisor of π_{j_0-1} contains Z_{j_0}, so $\#J \geq 1$.

We show (b) and (c) for $j = j_0$. More generally, let $j \geq 0$ and, for each $i < j$, let (Y_{i+1}, U_{i+1}) be the blow-up of some Σ_i near Z_i as in **Step 1**, where π_i satisfies the following.

(∗) For some regular system of parameters $\mathbf{w}$ of $\mathcal{O}_{U_i,\xi_i}$, the ideal $\mathcal{I}_{\Sigma_i} \cdot \mathcal{O}_{U_i,\xi_i}$ is generated by members of $\mathbf{w}$, and each component of $\mathcal{E}_i$ through ξ_i is the zero locus near ξ_i of a member of $\mathbf{w}$.

We show that (b) and (c) hold for j. For (b), we use induction on j, the case $j = 0$ being trivial since $\mathcal{E}_0 = 0$. Assume that $\mathcal{E}_j$ is SNC near ξ_j. By (∗) for π_j, on each chart of the blow-up, the components of $\mathcal{E}_{j+1}$ through ξ_{j+1} are coordinate hyperplanes of a regular system of parameters, so $\mathcal{E}_{j+1}$ is SNC near ξ_{j+1}. For the blow-ups π_j, $j < j_0$, of this step we show by induction on j that (b) holds for j and π_j satisfies (∗): (b) for j gives (∗) for π_j, since

$$\mathcal{I}_{\Sigma_j} \cdot \mathcal{O}_{U_j,\xi_j} = \mathfrak{m}_{\xi_j}$$

is generated by a regular system of parameters containing the local equations of the components of $\mathcal{E}_j$ through ξ_j, and (b) for j with (∗) for π_j gives (b) for $j+1$. For (c), let $E' := (\rho_{i+1,j})_*^{-1}E'_{i+1}$ for a component E'_{i+1} of the exceptional divisor of $\pi_i|_{U_{i+1}}$, $i < j$. The components of the exceptional divisor of $\pi_i|_{U_{i+1}}$ are the preimages of the pairwise disjoint translates $\gamma'\Sigma_i \cap U_i^\circ$, $\gamma' \in G$, and γ permutes them. If $\xi_j \in E' \cap \gamma E'$, then $\xi_{i+1} \in E'_{i+1} \cap \gamma E'_{i+1}$, so $\pi_i(E'_{i+1})$ and $\pi_i(\gamma E'_{i+1})$ are translates of $\Sigma_i \cap U_i^\circ$ containing ξ_i. Since distinct translates are disjoint in U_i°, $\pi_i(E'_{i+1}) = \pi_i(\gamma E'_{i+1})$, so $\gamma E'_{i+1} = E'_{i+1}$ and $\gamma E' = E'$.

Step 3. In this step we blow up intersections of two components of $\mathcal{E}_j$ until only one component contains the center of F, keeping $\operatorname{mult}_F K_{U_j/W} \leq N_1$.

Let $j \geq j_0$ satisfy (a)–(c) with $\dim Z_j \leq 1$. Let $v_i := \operatorname{mult}_F E_i$ for $i \in J$, positive integers, and

$$Q_j := \sum_{i\in J} v_i.$$

Assume $\#J \geq 2$. Possibly reordering indices, we may assume $1, 2 \in J$ and $v_1 \leq v_2$. By (b), there exist $y_1, y_2 \in \mathcal{O}_{U_j,\xi_j}$, part of a regular system of parameters, with $E_1 = (y_1 = 0)$ and $E_2 = (y_2 = 0)$ near ξ_j. Let Σ_j be the irreducible component of $E_1 \cap E_2$ containing Z_j. Then Σ_j is smooth at ξ_j, $\operatorname{codim}_{U_j} \Sigma_j = 2$, $\mathcal{I}_{\Sigma_j} \cdot \mathcal{O}_{U_j,\xi_j} = (y_1, y_2)$, and $g_j(\Sigma_j) \subset g_j(E_1) = \{o\}$ by (a) and (C2). If $\xi_j \in \gamma\Sigma_j$ for some $\gamma \in G$, then $\xi_j \in \gamma E_1 \cap \gamma E_2$, so $\gamma E_1 = E_1$ and $\gamma E_2 = E_2$ by (c), and $\gamma\Sigma_j = \Sigma_j$ as the unique component of $E_1 \cap E_2$ through ξ_j. Let (Y_{j+1}, U_{j+1}) be the blow-up of Σ_j near Z_j and E the exceptional divisor of $\pi_j|_{U_{j+1}}$.

By (E.27), $\operatorname{mult}_F E = v_1$, and ξ_{j+1} is contained in the chart with regular parameters $y_1, y_2/y_1$, on which $E = (y_1 = 0)$, $(\pi_j)_*^{-1}E_2 = (y_2/y_1 = 0)$, and $(\pi_j)_*^{-1}E_1$ does not appear. Here $\operatorname{mult}_F(y_2/y_1) = v_2 - v_1$. If $v_1 = v_2$, then y_2/y_1 is a unit at ξ_{j+1}, since the valuation ring of F dominates $\mathcal{O}_{U_{j+1},\xi_{j+1}}$. For $i \in J\backslash\{1,2\}$, $E_i \not\supset \Sigma_j$ by (b), so $\pi_j^* E_i = (\pi_j)_*^{-1}E_i$, which contains ξ_{j+1}, and $\operatorname{mult}_F \pi_j^* E_i = \operatorname{mult}_F E_i = v_i$. Thus the components of $\mathcal{E}_{j+1}$ through ξ_{j+1} are E, $(\pi_j)_*^{-1}E_i$ for $i \in J\backslash\{1,2\}$, and, if $v_1 < v_2$, $(\pi_j)_*^{-1}E_2$. Hence $Q_{j+1} = Q_j - v_1$. By (E.29) with $\operatorname{codim}_{U_j} \Sigma_j = 2$, $\operatorname{mult}_F K_{U_{j+1}/W} = \operatorname{mult}_F K_{U_j/W} + v_1$. Therefore

$$\operatorname{mult}_F K_{U_{j+1}/W} + Q_{j+1} = \operatorname{mult}_F K_{U_j/W} + Q_j. \tag{E.32}$$

By (E.26), the coefficient of E in $K_{U_{j+1}/W}$ is c_1+c_2+1, so (a) holds for $j+1$. By (b) for j, y_1, y_2 extend to a regular system of parameters of $\mathcal{O}_{U_j,\xi_j}$ containing the local equations of the components of $\mathcal{E}_j$ through ξ_j, and $\mathcal{I}_{\Sigma_j}\cdot\mathcal{O}_{U_j,\xi_j}=(y_1,y_2)$, so π_j satisfies (∗) of **Step 2**. Hence (b) and (c) hold for $j+1$ by **Step 2**.

Since Q_j is a positive integer which decreases strictly, after at most Q_{j_0} blow-ups we reach j with $\dim Z_j=2$ or $\#J=1$. By (a), $Q_{j_0}\le\sum_{i\in J}c_iv_i=\operatorname{mult}_F K_{U_{j_0}/W}$. By (E.32) and (E.31), for any j reached in **Step 3**,

$$\operatorname{mult}_F K_{U_j/W}\le\operatorname{mult}_F K_{U_j/W}+Q_j=\operatorname{mult}_F K_{U_{j_0}/W}+Q_{j_0}\le 2\operatorname{mult}_F K_{U_{j_0}/W}\le N_1.$$

If $\dim Z_j=2$, then $a(F,W,0)\le 1+N_1$ by (E.30), and the first alternative holds with the computing lift F.

Otherwise let $(Y,U):=(Y_j,U_j)$, $g:=g_j$, $Z:=Z_j$, $\xi:=\xi_j$, $S:=E_1$ the unique component in (a), and $\alpha:=c_1$. Then S is a smooth prime divisor near ξ, $g(S)=\{o\}$, $\dim Z\le 1$, and near ξ,

$$K_{U/W}=\alpha S,\qquad \alpha\ge 1,\quad\text{and}\quad \alpha\operatorname{mult}_F S=\operatorname{mult}_F K_{U/W}\le N_1. \tag{E.33}$$

Since $\operatorname{mult}_F S\ge 1$, we have $\alpha\le N_1$ and $\operatorname{mult}_F S\le N_1$. Let $B_U:=\frac{1}{m}\operatorname{div}(g^*h)-K_{U/W}$, so that $B_U=\frac{1}{m}\operatorname{div}(g^*h)-\alpha S$ near ξ by (E.33). Since $K_U+B_U=g^*(K_W+\frac{1}{m}\operatorname{div}(h))$,

$$a(P,U,B_U)=a\Big(P,W,\frac{1}{m}\operatorname{div}(h)\Big) \tag{E.34}$$

for any prime divisor P over U.

Step 4. In this step we assume $\dim Z=1$ and find a computing lift P of b with $a(P,W,0)\le 2L_0$, where L_0 depends only on m,M,N,N_0.

Let $G':=I_G(F)$ and $r:=|G'|\le M$. By Lemma 2.15(1), G' is cyclic. Since G' fixes the valuation of F and acts trivially on its residue field $\kappa(F)$, it preserves Z and acts trivially on the residue field of the generic point ξ of Z, which is a subfield of $\kappa(F)$, so G' fixes Z pointwise. For any prime divisor P over W with center o, we have $|I_{G'}(P)|\le|I_G(P)|$ and $a(P,W,\frac{1}{m}\operatorname{div}(h))>0$, so

$$\frac{a(P,W,\frac{1}{m}\operatorname{div}(h))}{|I_{G'}(P)|}\ge\frac{a(P,W,\frac{1}{m}\operatorname{div}(h))}{|I_G(P)|}\ge b, \tag{E.35}$$

with equalities for $P=F$ since $I_{G'}(F)=G'=I_G(F)$.

Possibly replacing U with a G'-invariant open neighborhood of ξ, we may assume that $K_{U/W}=\alpha S$ on U, that $S=(s=0)$ for a regular function s on U, and that (U,B_U) is sub-klt, since $(W,\frac{1}{m}\operatorname{div}(h))$ is klt near o and (E.34) holds. Since $K_{U/W}$ is G-invariant, S is G'-invariant, and since h is G-semi-invariant, B_U is G'-invariant. Let t_0 be a rational function on U which is regular at ξ and non-constant on Z, and let $t:=\frac{1}{r}\sum_{\gamma\in G'}\gamma^*t_0$. Then t is G'-invariant and regular at ξ, and $t|_Z=t_0|_Z$ is non-constant since G' fixes Z pointwise.

We use Lemma 3.5 with $(X,B):=(U,B_U)$, $C:=Z$, $H:=G'$, and t, and with the notation k, U_k, c, Ω, c', U_Ω, B_Ω, and $\varpi:U_\Omega\to U$ of that lemma. The local rings of U_k are local rings of U and $\operatorname{char}k=0$, so U_Ω is a smooth surface over Ω. Since t is G'-invariant and G' fixes Z pointwise, G' acts faithfully on U_Ω over Ω and fixes c' by Lemma 3.5(0). Moreover, near c',

$$B_\Omega=\frac{1}{m}\operatorname{div}(\varpi^*g^*h)-\alpha\varpi^*S,$$

where ϖ^*g^*h is a non-zero regular G'-semi-invariant function and $\alpha\varpi^*S=\alpha\operatorname{div}(\varpi^*s)$ is a G'-invariant Cartier divisor with $\alpha\varpi^*S\ge 0$. By Lemma 3.5(3), (U_Ω,B_Ω) is sub-klt near c'. Let

$$\mu:=\operatorname{mld}((U_\Omega\ni c',B_\Omega)/G').$$

We show $\mu=b$. By Lemma 3.5(0) and (1), there exists a prime divisor F' over U_Ω with center c' such that $\operatorname{mult}_{F'}(\varpi^*\zeta)=\operatorname{mult}_F\zeta$ for any non-zero rational function ζ on U. Thus F is the

prime divisor which Lemma 3.5(2) gives for F', and by Lemma 3.5(0), (2) and (E.34),

$$\frac{a(F', U_\Omega, B_\Omega)}{|I_{G'}(F')|} = \frac{a(F, U, B_U)}{|I_{G'}(F)|} = \frac{a(F, W, \frac{1}{m}\operatorname{div}(h))}{|I_G(F)|} = b.$$

Conversely, by Lemma 3.5(0) and (2), any prime divisor P' over U_Ω with center c' restricts to a prime divisor P over U with center Z with $a(P, U, B_U) = a(P', U_\Omega, B_\Omega)$ and $I_{G'}(P) = I_{G'}(P')$. Since $g(Z) = \{o\}$, P is a prime divisor over W with center o, so by (E.34) and (E.35),

$$\frac{a(P', U_\Omega, B_\Omega)}{|I_{G'}(P')|} = \frac{a(P, U, B_U)}{|I_{G'}(P)|} \geq b. \tag{E.36}$$

Hence $\mu = b > 0$ and F' is a computing lift of μ.

Since

$$(g \circ \varpi)^* \mathfrak{m}_o \cdot \mathcal{O}_{U_\Omega, c'} \subset \mathfrak{m}_{c'},$$

we have

$$\operatorname{mult}_{F'} \mathfrak{m}_{c'} \leq \operatorname{mult}_{F'} \big((g \circ \varpi)^* \mathfrak{m}_o\big) = \operatorname{mult}_F \mathfrak{m}_o \leq N_0, \qquad \operatorname{mult}_{F'}(\alpha \varpi^* S) = \alpha \operatorname{mult}_F S \leq N_1$$

by (E.33). By Theorem 15.2 for the smooth surface germ $(U_\Omega \ni c')$ over Ω with the faithful action of the cyclic group G' of order at most M, the function $\varpi^* g^* h$, the divisor $\alpha \varpi^* S$, and the bounds N_1 on $\operatorname{mult}_{F'}(\alpha \varpi^* S)$ and N_0 on $\operatorname{mult}_{F'} \mathfrak{m}_{c'}$, there exists a computing lift P' of μ with

$$a(P', U_\Omega, 0) \leq L_0 \quad \text{and} \quad \operatorname{mult}_{P'}(\alpha \varpi^* S) \leq L_0,$$

where L_0 depends only on m, M, N_1, N_0, hence only on m, M, N, N_0.

Let P be the restriction of P' as in (E.36). Since $\operatorname{mult}_P \zeta = \operatorname{mult}_{P'}(\varpi^* \zeta)$ for any non-zero rational function ζ on U, we have $\operatorname{mult}_P B_U = \operatorname{mult}_{P'} B_\Omega$ and $\alpha \operatorname{mult}_P S = \operatorname{mult}_{P'}(\alpha \varpi^* S) \leq L_0$. Hence

$$a(P, U, 0) = a(P, U, B_U) + \operatorname{mult}_P B_U = a(P', U_\Omega, B_\Omega) + \operatorname{mult}_{P'} B_\Omega = a(P', U_\Omega, 0) \leq L_0.$$

By (E.25) and $K_{U/W} = \alpha S$,

$$a(P, W, 0) = a(P, U, 0) + \alpha \operatorname{mult}_P S \leq 2L_0.$$

Since P' computes $\mu = b$, by (E.36), (E.34) and $|I_{G'}(P)| \leq |I_G(P)|$,

$$b \leq \frac{a(P, W, \frac{1}{m}\operatorname{div}(h))}{|I_G(P)|} \leq \frac{a(P, U, B_U)}{|I_{G'}(P)|} = \frac{a(P', U_\Omega, B_\Omega)}{|I_{G'}(P')|} = b.$$

Thus P is a computing lift of b, and the first alternative holds with

$$L(m, M, N, N_0) := \max\{1 + N_1, 2L_0\},$$

which also covers the divisorial cases of **Steps 2 and 3**.

Step 5. In this step we assume $\dim Z = 0$, let $z := Z$, and prove (1)–(5) of Proposition 15.6 for $g : Y \to W$ and U.

By (C1), U is a smooth G-invariant open neighborhood of $G \cdot z$, and $z = \operatorname{center}_Y F$ is a closed point. By **Step 3**, the first sentence of (1) and $\alpha \operatorname{mult}_F S \leq N_1$ hold. Since

$$g^* \mathfrak{m}_o \cdot \mathcal{O}_{U,z} \subset \mathfrak{m}_{U,z},$$

we have $\operatorname{mult}_F \mathfrak{m}_{U,z} \leq \operatorname{mult}_F \mathfrak{m}_o \leq N_0$. This proves (1).

The stabilizer G_z is abelian, and it acts faithfully on $(U \ni z)$, since an element of G acting trivially near z acts trivially on a dense open subset of U, hence on W, as g is G-equivariant and birational. For $\gamma \in G_z$, γS is a component of the G-invariant divisor $K_{U/W}$ through z, so $\gamma S = S$. Since G_z is finite abelian and preserves the smooth prime divisor S through z, there exist regular G_z-eigenparameters s, x_1, x_2 at z with $S = (s = 0)$. Let $n := \operatorname{mult}_S(g^* h)$ and write $g^* h = s^n h_U$, where h_U is regular at z, not divisible by s, and G_z-semi-invariant since $g^* h$ and s are. Near z,

$$B_U = \frac{1}{m}\operatorname{div}(g^* h) - \alpha S = \Big(\frac{n}{m} - \alpha\Big) S + \frac{1}{m}\operatorname{div}(h_U).$$

Since $K_U + B_U = g^*(K_W + \frac{1}{m}\operatorname{div}(h))$ and $(W, \frac{1}{m}\operatorname{div}(h))$ is klt near o, (U, B_U) is sub-klt near z, so $\frac{n}{m} - \alpha < 1$. Thus $0 \le n \le m(\alpha+1) - 1$, and the coefficient of S in B_U satisfies

$$\frac{n}{m} - \alpha \in \frac{1}{m}\mathbb{Z} \cap [-N_1, 1).$$

This proves (2).

Let P be a prime divisor over U with center z. Any element of $I_G(P)$ fixes the valuation of P, hence fixes $z = \operatorname{center}_U P$, so $I_G(P) \subset G_z$ and $I_{G_z}(P) = I_G(P)$. By (E.25),

$$a(P, W, 0) = a(P, U, 0) + \alpha \operatorname{mult}_P s.$$

Since S is smooth, (U, S) is lc near z, so $a(P, U, 0) \ge \operatorname{mult}_P s$, and

$$a(P, W, 0) \le (1+\alpha)a(P, U, 0) \le (1+N_1)a(P, U, 0).$$

From $g^*h = s^n h_U$, $\operatorname{mult}_P(g^*h) = n \operatorname{mult}_P s + \operatorname{mult}_P h_U$, and $a(P, W, \frac{1}{m}\operatorname{div}(h)) = a(P, U, B_U)$ by (E.34). This proves (3).

Any prime divisor P over U with center z is a prime divisor over W with center $g(z) = o$, so by (3),

$$\frac{a(P, U, B_U)}{|I_{G_z}(P)|} = \frac{a(P, W, \frac{1}{m}\operatorname{div}(h))}{|I_G(P)|} \ge b,$$

with equality for $P = F$. Thus

$$\operatorname{mld}((U \ni z, B_U)/G_z) = b,$$

and any computing lift of the left-hand side is a computing lift of b. This proves (4).

Assume that $\eta > 0$ and

$$a\Big(P, W, \frac{1}{m}\operatorname{div}(h)\Big) \ge \eta \operatorname{mult}_P \mathfrak{m}_o$$

for any prime divisor P over W with center o. Since $g(S) = \{o\}$ and $S = (s = 0)$ is prime, s divides any element of $g^*\mathfrak{m}_o$ in $\mathcal{O}_{U,z}$. Hence $\operatorname{mult}_P \mathfrak{m}_o \ge \operatorname{mult}_P s$ for any prime divisor P over U with center z, and by (3),

$$a(P, U, B_U) = a\Big(P, W, \frac{1}{m}\operatorname{div}(h)\Big) \ge \eta \operatorname{mult}_P \mathfrak{m}_o \ge \eta \operatorname{mult}_P s.$$

This proves (5). The proposition follows. □

E.5. **Signed smooth points.** In this subsection we prove Theorem 15.7 of Section 15.

Proof of Theorem 15.7, first part. If $(V \ni o)$ is a smooth threefold germ and s, p, q are regular parameters at o, then $(V, \operatorname{div}(spq))$ is lc near o, so

$$a(P, V, 0) \ge \operatorname{mult}_P s + \operatorname{mult}_P p + \operatorname{mult}_P q \tag{E.37}$$

for any prime divisor P over V with center o. We divide the proof into steps.

Step 1. In this step we pass to a root of s and to the inertia group of a computing lift.

Assume that no such L exists. Then there exist, for any $i \ge 1$, data $(V_i \ni o_i)$, G_i, s_i, p_i, q_i, h_i, and β_i satisfying the hypotheses of Theorem 15.7, with $S_i := (s_i = 0)$, $B_i := \frac{1}{m}\operatorname{div}(h_i) + \beta_i S_i$, $b_i := \operatorname{mld}((V_i \ni o_i, B_i)/G_i)$ and computing lifts F_i of b_i with $\operatorname{mult}_{F_i} s_i \le R'$, such that

$$\min\{a(P, V_i, 0) \mid P \text{ is a computing lift of } b_i\} \to +\infty. \tag{E.38}$$

For any prime divisor P over V_i with center o_i, $a(P, V_i, B_i) \in \frac{1}{m}\mathbb{Z}$ and $|I_{G_i}(P)| \le M$, so $b_i \in \frac{1}{m \cdot M!}\mathbb{Z} \cap (0, B_0]$, a finite set. Since $\beta_i \in \frac{1}{m}\mathbb{Z} \cap [-A_0, 1)$ and $|G_i| \le M$, possibly passing to a subsequence, we may assume that $b_i = b$, $\beta_i = \beta$, and $k := \operatorname{mult}_{F_i} s_i \in \{1, \dots, R'\}$ do not depend on i, and that $G_i = G$ with the characters $\chi_s, \chi_p, \chi_q, \chi_h$ of s_i, p_i, q_i, h_i independent of i. Here $g^* s_i = \chi_s(g) s_i$ for $g \in G$.

Let V_i' be the germ at $o_i' := (o_i, 0)$ of the subscheme $(t^k = s_i)$ of $V_i \times \mathbb{A}^1_t$, and $\pi_i : V_i' \to V_i$ the projection. Since $(s_i, p_i, q_i) : V_i \to \mathbb{A}^3$ is étale at o_i, the morphism $(t, p_i, q_i) : V_i' \to \mathbb{A}^3$ is étale at o_i'. Moreover, V_i' is irreducible because $\operatorname{mult}_{S_i} s_i = 1$. The abelian group

$$G' := \{(g, \zeta) \in G \times \mathbb{C}^* \mid \zeta^k = \chi_s(g)\}, \qquad |G'| = k|G|,$$

acts faithfully on V_i' by g on V_i and by $(g, \zeta)^* t = \zeta t$, diagonally in (t, p_i, q_i), so that t is a G'-eigenfunction with character $\chi_t(g, \zeta) := \zeta$, and π_i induces an isomorphism $V_i'/G' \cong V_i/G$. Since $K_{V_i'} = \pi_i^* K_{V_i} + (k-1)(t=0)$, we have $K_{V_i'} + B_i' = \pi_i^*(K_{V_i} + B_i)$ for

$$B_i' := \frac{1}{m} \operatorname{div}(h_i \circ \pi_i) + (k\beta - k + 1)(t = 0).$$

Let P' be a prime divisor over V_i' with center o_i', and P the prime divisor over V_i with center o_i such that $\operatorname{mult}_{P'}(\zeta \circ \pi_i) = n_1 \operatorname{mult}_P \zeta$ for any non-zero rational function ζ on V_i, where n_1 is the ramification index. Any P arises in this way. Comparing coefficients along P' in $K_{V_i'} = \pi_i^* K_{V_i} + (k-1)(t=0)$ and in $K_{V_i'} + B_i' = \pi_i^*(K_{V_i} + B_i)$, and since ramification indices multiply along $V_i' \to V_i \to V_i/G$ and the ramification index of a Galois quotient along a prime divisor is the order of the inertia group of that prime divisor, we obtain

$$\begin{aligned} n_1 a(P, V_i, 0) = a(P', V_i', 0) + (k-1) \operatorname{mult}_{P'} t, \qquad a(P', V_i', B_i') = n_1 a(P, V_i, B_i), \\ |I_{G'}(P')| = n_1 |I_G(P)|. \end{aligned} \tag{E.39}$$

Thus (V_i', B_i') is sub-klt, $\operatorname{mld}((V_i' \ni o_i', B_i')/G') = b$, P' is a computing lift of b if and only if P is one, and, since $\operatorname{mult}_{P'} s_i = k \operatorname{mult}_{P'} t$,

$$a(P', V_i', B_i') = n_1 a(P, V_i, B_i) \geq n_1 \eta \operatorname{mult}_P s_i = k\eta \operatorname{mult}_{P'} t \geq \eta \operatorname{mult}_{P'} t. \tag{E.40}$$

Let τ be a uniformizer of the valuation ring R_i of F_i. Then $s_i = \tau^k u_1$ with $u_1 \in R_i^*$, and over R_i the equation $t^k = s_i$ becomes $(t/\tau)^k = u_1$, which is étale over R_i. Hence any prime divisor F_i' over V_i' over F_i has $n_1 = 1$ and $\operatorname{mult}_{F_i'} t = 1$, and F_i' is a computing lift of b by (E.39). Fix such F_i' and let $\Gamma_i := I_{G'}(F_i')$. The image of t spans the cotangent line of the valuation ring of F_i', on which Γ_i acts faithfully. Thus Γ_i is cyclic and $\chi_t|_{\Gamma_i}$ is faithful. For any prime divisor P' over V_i' with center o_i', $I_{\Gamma_i}(P') = \Gamma_i \cap I_{G'}(P')$, so

$$\frac{a(P', V_i', B_i')}{|I_{\Gamma_i}(P')|} \geq \frac{a(P', V_i', B_i')}{|I_{G'}(P')|} \geq b,$$

with equalities for $P' = F_i'$. Hence $\operatorname{mld}((V_i' \ni o_i', B_i')/\Gamma_i) = b$, and any computing lift of b for Γ_i is a computing lift of b for G'. Since G' has finitely many subgroups, possibly passing to a subsequence, we may assume that $\Gamma := \Gamma_i$ and the characters of $t, p_i, q_i, h_i \circ \pi_i$ on Γ do not depend on i. Let $e := |\Gamma| \leq R'M$.

If P' is a computing lift of b for $(V_i' \ni o_i', B_i')$ and Γ, then P is a computing lift of b_i, and by (E.39) and $\operatorname{mult}_{P'} t \leq a(P', V_i', 0)$ (by (E.37)), $a(P, V_i, 0) \leq k a(P', V_i', 0)$. By (E.38),

$$\min\{a(P', V_i', 0) \mid P' \text{ is a computing lift of } b \text{ for } (V_i' \ni o_i', B_i') \text{ and } \Gamma\} \to +\infty. \tag{E.41}$$

From now on we write p, q for p_i, q_i, h_i for $h_i \circ \pi_i$, β for $k\beta - k + 1 \in \frac{1}{m}\mathbb{Z} \cap (-\infty, 1)$, B_i for $B_i' = \frac{1}{m} \operatorname{div}(h_i) + \beta(t = 0)$ and F_i for F_i'. Since $\operatorname{mult}_{F_i} t = 1$ and $|I_\Gamma(F_i)| = e$,

$$a(F_i, V_i', 0) - \frac{1}{m} \operatorname{mult}_{F_i} h_i = be + \beta. \tag{E.42}$$

Step 2. In this step we replace p, q by Γ-eigenparameters along which F_i has unbounded orders.

Let $Y_{i,0} := V_i'$ and $c_{i,0} := o_i'$. Assume that $c_{i,0}, \dots, c_{i,j}$ are closed points, and that $c_{i,j}$ is the center of F_i on a smooth threefold $Y_{i,j}$ with regular parameters $t, p_{i,j}, q_{i,j}$ at $c_{i,j}$, where $p_{i,0} := p$ and $q_{i,0} := q$. Let $Y_{i,j+1}$ be the open subset of the blow-up of $c_{i,j}$ on which $p_{i,j}/t$ and $q_{i,j}/t$ are

regular. The morphism $(t, p_{i,j}/t, q_{i,j}/t)$ from $Y_{i,j+1}$ to $\mathbb{A}^3$ is étale, and $(t=0) \subset Y_{i,j+1}$ is the exceptional divisor $E_{i,j+1}$. Since

$$\operatorname{mult}_{F_i} t = 1 \le \min(\operatorname{mult}_{F_i} p_{i,j}, \operatorname{mult}_{F_i} q_{i,j}),$$

the center $c_{i,j+1}$ of F_i on the blow-up is contained in $E_{i,j+1} \subset Y_{i,j+1}$. If $c_{i,j+1}$ is a closed point, let $p_{i,j+1} := p_{i,j}/t - \lambda$ and $q_{i,j+1} := q_{i,j}/t - \lambda'$, where $\lambda, \lambda' \in \mathbb{C}$ are the values of $p_{i,j}/t$, $q_{i,j}/t$ at $c_{i,j+1}$. By induction, for $n \ge 1$ such that $c_{i,1}, \dots, c_{i,n-1}$ are closed points,

$$dt \wedge dp \wedge dq = t^{2n} dt \wedge d\Big(\frac{p_{i,n-1}}{t}\Big) \wedge d\Big(\frac{q_{i,n-1}}{t}\Big) \text{ on } Y_{i,n},$$

so, writing $\operatorname{div}(h_i)$ also for the divisor of the pullback of h_i,

$$\begin{gathered} K_{Y_{i,n}} + \frac{1}{m}\operatorname{div}(h_i) + (\beta - 2n)E_{i,n} \text{ is the pullback of } K_{V_i'} + B_i, \\ a(P, V_i', 0) = a(P, Y_{i,n}, 0) + 2n \operatorname{mult}_P t \end{gathered} \tag{E.43}$$

for any prime divisor P over $Y_{i,n}$. If moreover $c_{i,n}$ is a closed point, then

$$p = \alpha_{i,n}(t) + t^n p_{i,n}, \qquad q = \alpha'_{i,n}(t) + t^n q_{i,n}$$

for polynomials $\alpha_{i,n}, \alpha'_{i,n} \in \mathbb{C}[t]$ of degree at most n without constant term.

The group $\Gamma = I_\Gamma(F_i)$ fixes each $c_{i,j}$ and acts trivially on the residue field of $c_{i,j}$, so Γ acts on each $Y_{i,j}$. If $p_{i,j}$ has character $\chi_p \chi_t^{-j}$, then $p_{i,j}/t$ has character $\chi_p \chi_t^{-j-1}$. Since $g \in \Gamma$ fixes $c_{i,j+1}$, $\chi_p \chi_t^{-j-1}(g)\lambda = \lambda$, so $\lambda \ne 0$ only if $\chi_p = \chi_t^{j+1}$ on Γ. By induction, the coefficient of t^j in $\alpha_{i,n}$ is non-zero only if $\chi_p = \chi_t^j$ on Γ, and similarly for $\alpha'_{i,n}$. Thus $p - \alpha_{i,n}(t)$ and $q - \alpha'_{i,n}(t)$ are Γ-semi-invariant with characters χ_p and χ_q.

We claim that for any $n \ge 1$, the points $c_{i,1}, \dots, c_{i,n}$ are closed for all but finitely many i. Otherwise, possibly passing to a subsequence, there exists $n \ge 1$ such that $c_{i,1}, \dots, c_{i,n-1}$ are closed points and $c_{i,n}$ is not, for any i. If $c_{i,n}$ is the generic point of $E_{i,n}$, then $F_i = E_{i,n}$ and $a(F_i, V_i', 0) = 1 + 2n$ by (E.43), which contradicts (E.41).

Otherwise $c_{i,n}$ is the generic point ξ of a curve $\Sigma_i \subset E_{i,n}$. Since Γ acts trivially on the residue field of F_i, it fixes Σ_i pointwise. One of $p_{i,n-1}/t$, $q_{i,n-1}/t$, say ψ, is non-constant on Σ_i. The function ψ is Γ-semi-invariant and $\psi|_{\Sigma_i}$ is Γ-invariant and non-zero, so ψ is Γ-invariant. Let $X_i \subset Y_{i,n}$ be a Γ-invariant open subset containing ξ, let $B_{X_i} := \frac{1}{m}\operatorname{div}(h_i) + (\beta - 2n)E_{i,n}$ on X_i, and let X_Ω, c', B_Ω, and ϖ denote the objects that Lemma 3.5 constructs from the sub-pair (X_i, B_{X_i}), the curve Σ_i with generic point ξ, the group Γ, and the function ψ. By (E.43), $a(P, X_i, B_{X_i}) = a(P, V_i', B_i)$ for any prime divisor P over X_i. The surface germ $(X_\Omega \ni c')$ is smooth, Γ acts faithfully on X_Ω fixing c' by Lemma 3.5(0), and (X_Ω, B_Ω) is sub-klt near c' by Lemma 3.5(3). By Lemma 3.5(1), there exists a prime divisor F_i'' over X_Ω with center c' and $\operatorname{mult}_{F_i''}(\varpi^*\zeta) = \operatorname{mult}_{F_i} \zeta$ for any non-zero rational function ζ on X_i. By Lemma 3.5(2), the restriction of F_i'' is F_i, $a(F_i'', X_\Omega, B_\Omega) = a(F_i, V_i', B_i)$, and $I_\Gamma(F_i'') = I_\Gamma(F_i)$. Conversely, by Lemma 3.5(2), any prime divisor P'' over X_Ω with center c' restricts to a prime divisor P over V_i' with center o_i' such that $a(P, V_i', B_i) = a(P'', X_\Omega, B_\Omega)$ and $I_\Gamma(P) = I_\Gamma(P'')$. Hence

$$\operatorname{mld}((X_\Omega \ni c', B_\Omega)/\Gamma) = b, \qquad \operatorname{mult}_{F_i''} t = \operatorname{mult}_{F_i} t = 1,$$

and F_i'' is a computing lift of b. Since $n \ge 1$ and $\beta < 1$, $2n - \beta > 0$. Let $D_\Omega := \lceil 2n - \beta \rceil (t = 0)$ on X_Ω and $j_1 := m(\lceil 2n - \beta \rceil - (2n - \beta)) \in \mathbb{N}$, so that

$$B_\Omega = \frac{1}{m}\operatorname{div}(h_i t^{j_1}) - D_\Omega.$$

We have $\operatorname{mult}_{F_i''} D_\Omega = \lceil 2n - \beta \rceil$, and $\operatorname{mult}_{F_i''} \mathfrak{m}_{c'} \le \operatorname{mult}_{F_i''} t = 1$. By Theorem 15.2 with $(X_\Omega \ni c')$, Γ, $h_i t^{j_1}$, D_Ω, and $R'M$ in place of $(T \ni o)$, G, h, D_T, and M, there exists a computing lift P'' of b with $a(P'', X_\Omega, 0)$ and $\operatorname{mult}_{P''} t \le \operatorname{mult}_{P''} D_\Omega$ bounded by a constant depending only on m, $R'M$, n, and β. By Lemma 3.5(2), applied also to the sub-pair $(X_i, 0)$, P'' restricts to a computing

lift P of b for $(V_i' \ni o_i', B_i)$ and Γ with $a(P, Y_{i,n}, 0) = a(P'', X_\Omega, 0)$ and $\mathrm{mult}_P t = \mathrm{mult}_{P''} t$. By (E.43), $a(P, V_i', 0)$ is bounded independently of i, which contradicts (E.41). The claim follows.

By the claim, possibly passing to a subsequence, there exist integers $n_i \to +\infty$ such that $c_{i,1}, \dots, c_{i,n_i}$ are closed points. We replace p and q by $p - \alpha_{i,n_i}(t)$ and $q - \alpha'_{i,n_i}(t)$, which are regular Γ-eigenparameters with the characters χ_p, χ_q. Since p_{i,n_i} and q_{i,n_i} vanish at c_{i,n_i},

$$\mathrm{mult}_{F_i} p > n_i \quad \text{and} \quad \mathrm{mult}_{F_i} q > n_i. \tag{E.44}$$

Step 3. In this step we take a compatible generic limit h_* of the marks and show that $(\operatorname{Spec} \Bbbk((t))[[p,q]], \frac{1}{m}\operatorname{div}(h_*))$ is lc and not klt.

Since (t,p,q) is étale at o_i', $\widehat{\mathcal{O}}_{V_i', o_i'} = \mathbb{C}[[t,p,q]]$, and we regard h_i as an element of $\mathbb{C}[[t,p,q]]$, semi-invariant with character χ_h under Γ acting diagonally on $\mathbb{A}^3$ with the characters χ_t, χ_p, χ_q. The exceptional divisor $E_{i,1}$ of the blow-up of o_i' satisfies $0 < a(E_{i,1}, V_i', B_i) = 3 - \frac{1}{m}\mathrm{mult}_{o_i'} h_i - \beta$, so $\mathrm{mult}_{o_i'} h_i < m(3-\beta)$ and the orders of the h_i at o_i' are bounded. By Lemma 2.6(1), there exist an infinite subset $\mathcal{I} \subset \mathbb{N}^+$ and a compatible generic limit $h_* \in \Bbbk[[t,p,q]]$ of the h_i along $\mathcal{I}$, with coefficient field $\Bbbk$ (Definition 2.5, with $H = \Gamma$). By Lemma 2.6(2), we may replace $\mathcal{I}$ by any infinite subset. Let $\widehat{V}_* := \operatorname{Spec} \Bbbk[[t,p,q]]$ with closed point 0, $C := (p = q = 0) \subset \widehat{V}_*$, and $\widehat{B}_* := \frac{1}{m}\operatorname{div}(h_*) + \beta(t=0)$.

Let T be a prime divisor over $\widehat{V}_*$ with center 0. We show that for infinitely many $i \in \mathcal{I}$ there exists a prime divisor T_i over V_i' with center o_i' such that

$$a(T_i, V_i', 0) = a(T, \widehat{V}_*, 0), \quad \mathrm{mult}_{T_i} h_i = \mathrm{mult}_T h_*, \quad \text{and} \quad \mathrm{mult}_{T_i} t = \mathrm{mult}_T t. \tag{E.45}$$

Let $\nu_T := \mathrm{mult}_T(t,p,q) \geq 1$ and fix an integer d with $d\nu_T > \mathrm{mult}_T h_*$, so that

$$\mathrm{mult}_T h_* = \mathrm{mult}_T h_*^{<d},$$

where $h_*^{<d}$ is the d-jet of h_*. The tuple $(t,p,q,h_*^{<d})$ is a compatible generic limit of the tuples of polynomials $(t,p,q,h_i^{<d})$ along $\mathcal{I}$, since taking d-jets is linear. By Lemma 3.15, for all but finitely many $i \in \mathcal{I}$ there exists a prime divisor T_i^0 over $\mathbb{A}^3$ with center 0 such that $a(T_i^0, \mathbb{A}^3, 0) = a(T, \widehat{V}_*, 0)$, $\mathrm{mult}_{T_i^0} t = \mathrm{mult}_T t$, $\mathrm{mult}_{T_i^0}(t,p,q) = \nu_T$, and $\mathrm{mult}_{T_i^0} h_i^{<d} = \mathrm{mult}_T h_*$. The terms of h_i of degree at least d have T_i^0-order at least $d\nu_T$, so $\mathrm{mult}_{T_i^0} h_i = \mathrm{mult}_T h_*$. By Lemma 2.12(3), applied to $(\mathbb{A}^3 \ni 0)$ and to $(V_i' \ni o_i')$, whose completions are both $\mathbb{C}[[t,p,q]]$, T_i^0 corresponds to a prime divisor T_i over V_i' with center o_i' satisfying (E.45).

Since (V_i', B_i) is sub-klt, (E.45) implies that

$$a(T, \widehat{V}_*, \widehat{B}_*) = a(T_i, V_i', B_i) > 0 \quad \text{for any prime divisor } T \text{ over } \widehat{V}_* \text{ with center } 0. \tag{E.46}$$

We show that $(\widehat{V}_*, \widehat{B}_*)$ is lc on $(t \neq 0)$. By [Tem18, Theorem 1.1.3], there exists a log resolution $Y \to \widehat{V}_*$ of $(\widehat{V}_*, \operatorname{div}(th_*))$. Let $K_Y + B_Y$ be the pullback of $K_{\widehat{V}_*} + \widehat{B}_*$. Since $\operatorname{Supp} B_Y$ is snc, if the components of B_Y not contained in the inverse image of $(t=0)$ have coefficients at most one, then $(\widehat{V}_*, \widehat{B}_*)$ is lc on $(t \neq 0)$. Otherwise let P_Y be a component of B_Y not contained in the inverse image of $(t=0)$ with $\mathrm{mult}_{P_Y} B_Y > 1$. Let $y \in P_Y$ be a closed point over 0, and y_1, y_2, y_3 regular parameters at y with $P_Y = (y_1 = 0)$ and $\operatorname{Supp} B_Y \subset (y_1 y_2 y_3 = 0)$ near y. For an integer $w \geq 1$, the prime divisor T_w of the weighted blow-up of y with weights $(w,1,1)$ in (y_1, y_2, y_3) has center 0 on $\widehat{V}_*$ and

$$a(T_w, \widehat{V}_*, \widehat{B}_*) = w(1 - \mathrm{mult}_{P_Y} B_Y) + (1 - \mathrm{mult}_{(y_2=0)} B_Y) + (1 - \mathrm{mult}_{(y_3=0)} B_Y) < 0$$

for $w \gg 0$, which contradicts (E.46).

By Lemma 2.12(3), F_i corresponds to a prime divisor F_i^0 over $\mathbb{A}^3$ with center 0 such that $a(F_i^0, \mathbb{A}^3, 0) = a(F_i, V_i', 0)$ and $\mathrm{mult}_{F_i^0}$ equals mult_{F_i} on polynomials in t,p,q. By (E.42) and $be > 0$,

$$\mathrm{mult}_{F_i} h_i = m(a(F_i, V_i', 0) - be - \beta) < m(a(F_i, V_i', 0) + |\beta|).$$

Let $\Lambda_i := \lceil m(a(F_i, V_i', 0) + |\beta|)\rceil$ and $h_i^{\le\Lambda_i}$ the sum of the terms of h_i of degree at most Λ_i in t, p, q. Since t, p, q have F_i-order at least 1, $\operatorname{mult}_{F_i} h_i^{\le\Lambda_i} = \operatorname{mult}_{F_i} h_i$, so

$$a(F_i, V_i', 0) - \frac{1}{m}\operatorname{mult}_{F_i} h_i^{\le\Lambda_i} = be + \beta.$$

By (E.37) and (E.44), $a(F_i, V_i', 0) > 2n_i$, so $\Lambda_i \to +\infty$. Hence for any d, the d-jets of $h_i^{\le\Lambda_i}$ and h_i coincide for $i \gg 0$, and h_* is a compatible generic limit of the polynomials $h_i^{\le\Lambda_i}$ along $\mathcal{I}$, with coefficient field $\Bbbk$ and H trivial. Thus Propositions 6.2 and 6.3 are available for

$$(z_1, z_2, z_3) = (t, p, q), \quad r = t, \quad N = 1, \quad g_{i,1} = h_i^{\le\Lambda_i}, \quad \kappa_1 = \frac{1}{m},$$
$$R_0 = 1, \quad C_0 = \max\{be + \beta, 0\}, \quad \text{and} \quad F_i^0 \text{ in place of } F_i,$$

and

$$\operatorname{mult}_{F_i^0} h_i^{\le\Lambda_i} = \operatorname{mult}_{F_i} h_i \to +\infty$$

because $a(F_i, V_i', 0) \to +\infty$. If $(\widehat{V}_*, \widehat{B}_*)$ were klt on $(t \ne 0)$, Proposition 6.2 would bound $\operatorname{mult}_{F_i^0} h_i^{\le\Lambda_i}$ for all but finitely many $i \in \mathcal{I}$, contrary to the previous display. Hence $(\widehat{V}_*, \widehat{B}_*)$ is not klt on $(t \ne 0)$. Let Z be the reduced closure in $\widehat{V}_*$ of $\operatorname{Nklt}(\widehat{V}_*, \widehat{B}_*) \cap (t \ne 0)$, and I_Z the ideal of Z.

We show that $C \subset Z$. Otherwise there exists $f \in I_Z$ with $f(t, 0, 0) \ne 0$. Let n_f be the order of $f(t, 0, 0)$ in t, and let $f = f_1, \dots, f_{n'}$ generate I_Z. We take the following data in the notation of Proposition 6.3(A).

(1) We let $L = T = d = n_f + 1$.
(2) The polynomials $\Phi_1, \dots, \Phi_{n'}$ are the truncations of $f_1, \dots, f_{n'}$ in degree at most n_f, with their coefficients as variables.
(3) The Ψ_j are the coefficients of $t^0, \dots, t^{n_f-1}$ in the truncation of $f(t, 0, 0)$, and $u_0 c_{n_f} - 1$, where c_{n_f} is the coefficient of t^{n_f} and u_0 an extra variable with value $1/c_{n_f}$.

By Proposition 6.3(A), for all but finitely many $i \in \mathcal{I}$ there exists a polynomial $\widetilde{f}_i$ in t, p, q of degree at most n_f with $\operatorname{mult}_{F_i^0} \widetilde{f}_i \ge n_f + 1$ such that $\widetilde{f}_i(t, 0, 0)$ is a non-zero multiple of t^{n_f}. The terms of $\widetilde{f}_i$ divisible by p or q have F_i-order greater than n_i by (E.44), and $\operatorname{mult}_{F_i} t^{n_f} = n_f$. Thus $\operatorname{mult}_{F_i^0} \widetilde{f}_i = n_f$ when $n_i \ge n_f$, a contradiction.

Let ξ_C be the generic point of C. Since $\xi_C \in (t \ne 0)$ and the non-klt locus is closed there, ξ_C is a non-klt point. The local ring $\mathcal{O}_{\widehat{V}_*, \xi_C}$ contains $\Bbbk((t))$, which maps isomorphically onto the residue field of $\mathcal{O}_{\widehat{V}_*, \xi_C}$, so the completion of $\mathcal{O}_{\widehat{V}_*, \xi_C}$ is $\Bbbk((t))[[p, q]]$. Let

$$\widehat{S} := \operatorname{Spec} \Bbbk((t))[[p, q]]$$

with closed point 0, and $U_C := \operatorname{Spec} \mathcal{O}_{\widehat{V}_*, \xi_C}$. The completion morphism $\widehat{S} \to U_C$ is flat and regular, so a log resolution of $(U_C, \operatorname{div}(h_*))$ [Tem18, Theorem 1.1.3] pulls back to a log resolution of $(\widehat{S}, \operatorname{div}(h_*))$ on which each prime divisor Q has the coefficient of the restriction of Q, and any prime divisor T over $\widehat{S}$ with center 0 restricts to a prime divisor T' over U_C with center ξ_C such that

$$a\Big(T, \widehat{S}, \frac{1}{m}\operatorname{div}(h_*)\Big) = a\Big(T', U_C, \frac{1}{m}\operatorname{div}(h_*)\Big);$$

this is the proof of Lemma 2.12(3), with the excellent local scheme U_C in place of the germ $(X \ni x)$. Hence

$$\Big(\widehat{S}, \frac{1}{m}\operatorname{div}(h_*)\Big) \text{ is lc and not klt near } 0. \tag{E.47}$$

Step 4. In this step we show that $(\widehat{S}, \frac{1}{m}\operatorname{div}(h_*))$ is not plt near 0.

Assume that it is plt near 0. Since the pair is plt and, by (E.47), not klt near 0, some prime divisor $(u=0)$ through 0, with $u\in\Bbbk((t))[[p,q]]$ irreducible, has coefficient one, that is, $h_*=u^m v$ with $u\nmid v$. If $\operatorname{mult}_0 u\geq 2$, the exceptional divisor of the blow-up of 0 has log discrepancy at most $2-\operatorname{mult}_0 u\leq 0$, a contradiction. Thus u is a regular parameter, and possibly exchanging p and q, u is a unit times $p-\sigma(q)$ for some $\sigma\in\Bbbk((t))[[q]]$ with $\sigma(0)=0$. Let $j_0<+\infty$ be the order of $v(\sigma(q),q)$ in q. For an integer $w>j_0$, the monomial valuation with weights $(w,1)$ in $(p-\sigma,q)$ takes the values w on u and j_0 on v, so the exceptional divisor of the corresponding weighted blow-up has log discrepancy $w+1-\frac{1}{m}(mw+j_0)=1-\frac{j_0}{m}$. Since the pair is plt, $j_0<m$.

Fix an integer $w_1>j_0$, let $\sigma_0\in\Bbbk((t))[q]$ be the truncation of σ in q-degree at most w_1, and let $N_0\geq 0$ be an integer with $t^{N_0}\sigma_0\in\Bbbk[[t]][q]$. We assign to p,q the weights $w_1,1$. Since $\sigma-\sigma_0$ has q-order at least w_1+1 and the terms of $v(p+\sigma_0,q)$ divisible by p have weight at least $w_1>j_0$, the lowest-weight part of $h_*(t,p+\sigma_0,q)$ is $\mu(t)p^m q^{j_0}$ for some $\mu(t)\in\Bbbk((t))^*$, of weight $\nu:=mw_1+j_0$. Let ν_0 be the order of $\mu(t)$ in t. For $h\in\mathbb{C}[[t,p,q]]$ and $\sigma'\in\mathbb{C}((t))[q]$ with $t^{N_0}\sigma'\in\mathbb{C}[[t]][q]$ (resp. $h\in\Bbbk[[t,p,q]]$ and $\sigma'\in\Bbbk((t))[q]$ with $t^{N_0}\sigma'\in\Bbbk[[t]][q]$) and $\sigma'(0)=0$, the coefficient of $p^j q^{j'}$ in $h(t,p+\sigma',q)$ has t-order at least $-N_0 j'\geq -N_0(jw_1+j')$, since each factor σ' in a term contributes q-degree at least 1. Fix an integer $N_1\geq N_0(\nu+1)+\nu_0+1$ with $N_1(1-\frac{j_0}{m})>be+\beta+\frac{\nu_0}{m}-1$, which is possible since $j_0<m$. After the substitution $q=t^{N_1}q'$, $p=t^{N_1 w_1}p'$ and division by $t^{N_1\nu+\nu_0}$, a term of weight $\omega\geq\nu+1$ obtains t-order at least

$$N_1\omega-N_0\omega-N_1\nu-\nu_0\geq N_1-N_0(\nu+1)-\nu_0\geq 1.$$

Thus

$$t^{-N_1\nu-\nu_0}h\big(t,t^{N_1w_1}p'+\sigma'(t,t^{N_1}q'),t^{N_1}q'\big)$$

belongs to $\mathbb{C}[[t,p',q']]$ (resp. $\Bbbk[[t,p',q']]$) and is congruent to $\gamma p'^m q'^{j_0}$ modulo t for some $\gamma\neq 0$ if and only if the finitely many coefficients of $h(t,p+\sigma',q)$ of weight at most ν satisfy the following conditions on their coefficients of t^a, $a\leq N_1\nu+\nu_0$: (1) the coefficients of weight $\omega<\nu$ vanish up to t-degree $N_1(\nu-\omega)+\nu_0$, (2) the coefficients of weight ν other than that of $p^m q^{j_0}$ vanish up to t-degree ν_0, and (3) the coefficient of $p^m q^{j_0}$ has t-order exactly ν_0. They are polynomial equations and one inequation in the coefficients of a d-jet of h, for some d, and in finitely many coefficients $t^a q^j$, $a\geq -N_0$, $1\leq j\leq w_1$, of σ'. They hold for h_* and σ_0, hence for h_* and the truncation of σ_0 in a sufficiently high t-degree. By Chevalley's theorem, the d-jets for which some σ' with complex coefficients satisfies them form a constructible subset of the jet space $\mathcal{J}_d$ containing the generic point of Z_d, the irreducible closed subset of Definition 2.5 for the compatible generic limit h_* of the h_i along $\mathcal{I}$, hence containing a non-empty open subset of Z_d. Possibly replacing $\mathcal{I}$ with an infinite subset, we may assume that for any $i\in\mathcal{I}$ there exists $\sigma_{0,i}\in\mathbb{C}[t,t^{-1}][q]$ with $t^{N_0}\sigma_{0,i}\in\mathbb{C}[t][q]$ and $\sigma_{0,i}(0)=0$ such that

$$\begin{gathered} h_i':=t^{-N_1\nu-\nu_0}h_i\big(t,t^{N_1w_1}p'+\sigma_{0,i}(t,t^{N_1}q'),t^{N_1}q'\big)\in\mathbb{C}[[t,p',q']],\\ h_i'\equiv\gamma_i p'^m q'^{j_0}\pmod t,\qquad \gamma_i\neq 0.\end{gathered}\tag{E.48}$$

Since $N_1>N_0$, the formulas $q=t^{N_1}q'$ and $p=t^{N_1w_1}p'+\sigma_{0,i}(t,t^{N_1}q')$ define a birational morphism $\rho_i:\mathbb{A}^3\to\mathbb{A}^3$ with $\rho_i(0)=0$. Let V_i'' be the germ at $o_i'':=(o_i',0)$ of the fiber product of $(t,p,q):V_i'\to\mathbb{A}^3$ and ρ_i. Then $(t,p',q'):V_i''\to\mathbb{A}^3$ is étale at o_i'', $V_i''\to V_i'$ is birational, and $h_i\circ(V_i''\to V_i')=t^{N_1\nu+\nu_0}h_i'$ with h_i' regular at o_i'', because $(t^{N_1\nu+\nu_0})\widehat{R}\cap R=(t^{N_1\nu+\nu_0})R$ for the local ring R of V_i'' at o_i'' and the completion $\widehat{R}=\mathbb{C}[[t,p',q']]$ of R. By (E.44), for $i\gg 0$,

$$\begin{aligned}\operatorname{mult}_{F_i}q'&=\operatorname{mult}_{F_i}q-N_1>n_i-N_1>0,\\ \operatorname{mult}_{F_i}p'&\geq\min\big(\operatorname{mult}_{F_i}p,\operatorname{mult}_{F_i}q-N_0\big)-N_1w_1>0;\end{aligned}$$

since F_i has center o_i' on V_i' and t,p',q' have positive F_i-orders, F_i has center o_i'' on V_i''. Since $dt\wedge dp\wedge dq=t^{N_1(w_1+1)}dt\wedge dp'\wedge dq'$, the pullback of $K_{V_i'}+B_i$ to V_i'' is $K_{V_i''}+\frac{1}{m}\operatorname{div}(h_i')+\beta''(t=0)$ with

$$\beta'':=\beta+\frac{N_1\nu+\nu_0}{m}-N_1(w_1+1)=\beta+\frac{\nu_0}{m}-N_1\Big(1-\frac{j_0}{m}\Big).$$

By (E.48), h_i' is not divisible by t, and for any rational number $\delta \in (0,1)$ the pair

$$\Big((t=0), \frac{\delta}{m}\operatorname{div}(\gamma_i p'^m q'^{j_0})\Big)$$

is klt because $\frac{\delta j_0}{m} < \delta < 1$. By inversion of adjunction, the pair

$$\Big(V_i'', (t=0) + \frac{\delta}{m}\operatorname{div}(h_i')\Big)$$

is plt near o_i''. Letting $\delta \to 1$, we see that $(V_i'', (t=0) + \frac{1}{m}\operatorname{div}(h_i'))$ is lc near o_i''. Since $\operatorname{mult}_{F_i} t = 1$,

$$\begin{aligned} be = a(F_i, V_i', B_i) &= a\Big(F_i, V_i'', \frac{1}{m}\operatorname{div}(h_i') + \beta''(t=0)\Big) \\ &\geq 1 - \beta'' = 1 - \beta - \frac{\nu_0}{m} + N_1\Big(1 - \frac{j_0}{m}\Big). \end{aligned}$$

This contradicts the choice of N_1.

Step 5. In this step we find the first prime divisor of log discrepancy zero over $\widehat{S}$ and treat the case without component of coefficient one.

By **Step 4** and (E.47), there exists a prime divisor E_0 over $\widehat{S}$ with center 0 and $a(E_0, \widehat{S}, \frac{1}{m}\operatorname{div}(h_*)) = 0$. For an integer $N \geq 1$, let

$$\mathfrak{a}_N := \big((h_*) + (p,q)^N\big)^{1/m},$$

an $\mathbb{R}$-ideal on $\widehat{S}$, given by the truncation of h_* modulo $(p,q)^N$. For any prime divisor T over $\widehat{S}$, $\operatorname{mult}_T \mathfrak{a}_N \leq \frac{1}{m}\operatorname{mult}_T h_*$, with equality if $N \operatorname{mult}_T(p,q) \geq \operatorname{mult}_T h_*$. Thus $(\widehat{S}, \mathfrak{a}_N)$ is lc near 0, any prime divisor with $a(\cdot, \widehat{S}, \mathfrak{a}_N) = 0$ satisfies $a(\cdot, \widehat{S}, \frac{1}{m}\operatorname{div}(h_*)) = 0$, and $\operatorname{mld}(\widehat{S} \ni 0, \mathfrak{a}_N) = 0$ for $N \geq \operatorname{mult}_{E_0} h_*$. For such N, Lemma 3.4 over the base field $\Bbbk((t))$ yields a tower of blow-ups at points $y_{N,0} = 0, y_{N,1}, \dots$ with exceptional curves $E_{N,1}, \dots, E_{N,j_N}$, where $a(E_{N,j_N}, \widehat{S}, \mathfrak{a}_N) = 0$. For $N' \geq N$, $\mathfrak{a}_{N'}$ has larger orders than $\mathfrak{a}_N$, so any prime divisor with $a(\cdot, \widehat{S}, \mathfrak{a}_N) = 0$ satisfies $a(\cdot, \widehat{S}, \mathfrak{a}_{N'}) = 0$. If $E_{N,j} = E_{N',j}$ for $j \leq j'$ with $j' < \min(j_N, j_{N'})$, then E_{N,j_N} has center $y_{N,j'}$ and $a(E_{N,j_N}, \widehat{S}, \mathfrak{a}_{N'}) = 0$, so $y_{N',j'} = y_{N,j'}$ by the uniqueness in Lemma 3.4(1) for $\mathfrak{a}_{N'}$, and $E_{N,j'+1} = E_{N',j'+1}$. If $j_{N'} > j_N$, then $E_{N',j_N} = E_{N,j_N}$ has $a(\cdot, \widehat{S}, \mathfrak{a}_{N'}) = 0$, contradicting Lemma 3.4(1) for $\mathfrak{a}_{N'}$. Hence the tower for $\mathfrak{a}_{N'}$ is the beginning of the tower for $\mathfrak{a}_N$, and j_N is decreasing in N. Thus the tower does not depend on $N \gg 0$. Let E be the last divisor of the tower. Taking N large with respect to the finitely many divisors of the tower, their log discrepancies with respect to $\mathfrak{a}_N$ and $\frac{1}{m}\operatorname{div}(h_*)$ coincide: they belong to $(0,1)$ except $a(E, \widehat{S}, \frac{1}{m}\operatorname{div}(h_*)) = 0$. If an exceptional curve $E_{N,j}$ with $j < j_N$ contains a closed point other than $y_{N,j}$ which is the center of a prime divisor T with $a(T, \widehat{S}, \frac{1}{m}\operatorname{div}(h_*)) = 0$, then increasing N so that $a(T, \widehat{S}, \mathfrak{a}_N) = 0$ contradicts Lemma 3.4(1). Thus E is the unique prime divisor over $\widehat{S}$ of log discrepancy zero with respect to $(\widehat{S}, \frac{1}{m}\operatorname{div}(h_*))$ such that the exceptional divisors created before E in the tower of point blow-ups producing E have positive log discrepancy. The group Γ acts on $\Bbbk[[t,p,q]]$ preserving (p,q) and (h_*), hence acts on $\widehat{S}$, not necessarily over $\Bbbk((t))$, fixing 0 and preserving $\mathfrak{a}_N$, so Γ fixes E by Lemma 3.4(3).

By Lemma 3.4(2), E is the exceptional divisor of the weighted blow-up with coprime weights (w_1, w_2), $w_1 \geq w_2 \geq 1$, in some regular parameters x_1, x_2 of $\Bbbk((t))[[p,q]]$. Let $v_E := \operatorname{mult}_E$, so that $v_E(x_1) = w_1$ and $v_E(x_2) = w_2$. Possibly exchanging p and q, $v_E(q) = w_2 = \min(v_E(p), v_E(q))$. If $w_1 = w_2$, then $w_1 = w_2 = 1$ and E is the exceptional divisor of the blow-up of 0. In this case let $\sigma := 0$. If $w_1 > w_2$, the part of degree w_2 of the graded ring

$$\bigoplus_{\varrho} \{v_E \geq \varrho\}/\{v_E > \varrho\} = \Bbbk((t))[X_1, X_2],$$

where X_j is the class of x_j, is spanned by X_2, so the class of q is a non-zero multiple of X_2. Subtracting from p successively the multiples of powers q^j with $jw_2 < w_1$ which cancel the lowest-weight part of p, we obtain $\sigma \in \Bbbk((t))[q]$ with $\sigma(0) = 0$, $\deg_q \sigma < w_1/w_2$, and $v_E(p-\sigma) \geq w_1$. Since $p-\sigma, q$ are regular parameters and $p-\sigma$ has no part of weight w_2, as $v_E(p-\sigma) \geq w_1 > w_2$, the linear part of $p-\sigma$ in x_1, x_2 is a non-zero multiple of x_1, so $v_E(p-\sigma) = w_1$. For $w_1 = w_2$ and for $w_1 > w_2$, the classes of $p-\sigma$ and q generate the graded ring, so v_E is the monomial valuation with weights (w_1, w_2) in $(p-\sigma, q)$. Since the log discrepancy of E with respect to $\widehat{S}$ is w_1+w_2, $v_E(h_*) = m(w_1+w_2)$. Let Φ be the part of weight $m(w_1+w_2)$, for the weights (w_1, w_2) of (p, q), in

$$h_*(t, p+\sigma, q) = \Phi + (\text{terms of weight greater than } m(w_1+w_2)). \tag{E.49}$$

It is the class of h_* in the graded ring of v_E, written in the generators given by $p-\sigma$ and q. Another such choice of σ, or of the exchange of p and q, changes Φ by a graded automorphism of $\Bbbk((t))[p, q]$, which maps the irreducible factors over an algebraic closure of $\Bbbk((t))$ bijectively onto those of the new Φ, preserving multiplicities. In particular, when the hypotheses of Definition-Lemma 7.13 hold for the data listed at the end of this step, E is the divisor of Definition-Lemma 7.13(1) by the uniqueness of E above, and with the polynomial σ of Definition-Lemma 7.13(1), Φ is the polynomial of Definition-Lemma 7.13(1).

Let L' be a finite extension of $\Bbbk((t))$ over which Φ splits into absolutely irreducible factors. Then any irreducible factor ϕ of Φ in $L'[p, q]$, of multiplicity c, satisfies

$$c \leq m. \tag{E.50}$$

The pair $(\operatorname{Spec} L'[[p, q]], \frac{1}{m}\operatorname{div}(h_*))$ is lc near 0 by (E.47), as $L'[[p, q]]$ is finite étale over $\Bbbk((t))[[p, q]]$. If $\phi = p$, then for a rational number $\epsilon \in (0, 1/c)$, any monomial $p^j q^{j'}$ of $h_*(t, p+\sigma, q)$ has $(w_1+\epsilon)j + w_2 j' \geq m(w_1+w_2) + \epsilon c$, so the monomial valuation with weights $(w_1+\epsilon, w_2)$ in $(p-\sigma, q)$, a positive rational multiple of the valuation of a prime divisor, has log discrepancy at most

$$w_1 + \epsilon + w_2 - (w_1+w_2) - \frac{\epsilon c}{m} = \epsilon\Big(1 - \frac{c}{m}\Big).$$

Since the pair is lc, $c \leq m$. If $\phi = q$, then by the computation for $\phi = p$ with $p-\sigma$ and q exchanged, the monomial valuation with weights $(w_1, w_2+\epsilon)$ in $(p-\sigma, q)$ has log discrepancy at most $\epsilon(1-\frac{c}{m})$, so $c \leq m$. Otherwise, since over an algebraic closure of $\Bbbk((t))$ the irreducible polynomials in p, q that are weighted homogeneous for the weights (w_1, w_2) are, up to constants, p, q, and $p^{w_2} - \lambda q^{w_1}$ with $\lambda \neq 0$, of weighted degrees w_1, w_2, and $w_1 w_2$, we have $\phi = p^{w_2} - \lambda q^{w_1}$ up to a constant, with $\lambda \in L'^*$, and ϕ corresponds to the point y of E given by $\theta = \lambda$, where $\theta := (p-\sigma)^{w_2}/q^{w_1}$ (over L'). The weighted blow-up Y producing E is regular at y, with regular parameters $u, \theta - \lambda$ at y, where u is a local equation of E, and the pullback of h_* is $u^{m(w_1+w_2)}\widetilde{h}$ with $\widetilde{h}|_E$ vanishing to order c at y. For an integer $k' \geq c$, the monomial valuation with weights $(k', 1)$ in $(u, \theta-\lambda)$ takes a value at least c on $\widetilde{h}$, and since $a(E, \widehat{S}, \frac{1}{m}\operatorname{div}(h_*)) = 0$, the corresponding prime divisor has log discrepancy at most $1 - \frac{c}{m}$. Thus $c \leq m$.

Assume that Φ has no geometric component of coefficient one, that is, $c < m$ in (E.50) for any ϕ. Then $\widehat{B}_*$ has no component of coefficient one through ξ_C. Otherwise h_* is divisible by f_C^m for an irreducible $f_C \in \mathcal{O}_{\widehat{V}_*, \xi_C}$ vanishing at ξ_C, so u'^m divides h_* in $\Bbbk((t))[[p, q]]$ for an irreducible factor u' of f_C there. The class of u' in the graded ring of v_E is non-constant because $v_E(u') > 0$, and the m-th power of the class of u' divides Φ, which contradicts $c < m$.

The hypotheses of Definition-Lemma 7.13 hold for the group Γ, m, β, $\bar{b} = b$, η, R', the germs $(V_i' \ni o_i')$ for $i \in \mathcal{I}$ with the regular Γ-eigenparameters t, p, q in place of s_i, p_i, q_i, the marks h_i, the $\mathbb{Q}$-divisors $B_i = \frac{1}{m}\operatorname{div}(h_i) + \beta(t=0)$, the prime divisors F_i, and the limit h_*:

(i) The pair (V_i', B_i) is sub-klt and $\operatorname{mld}((V_i' \ni o_i', B_i)/\Gamma) = b$ by **Step 1**.
(ii) Hypothesis (ii) holds by (E.40).
(iii) We have $a(F_i, V_i', B_i) = be$ with $e = |I_\Gamma(F_i)|$ by (E.42), $\operatorname{mult}_{F_i} t = 1 \leq R'$, and $\min(\operatorname{mult}_{F_i} p, \operatorname{mult}_{F_i} q) \to +\infty$ by (E.44).

(iv) Hypothesis (iv) holds by **Step 3** and the previous paragraph.

Hence Φ is the polynomial of Definition-Lemma 7.13(1), and $\frac{1}{m}\operatorname{div}(\Phi)$ has no component of coefficient one on $\mathbb{A}^2_{\Bbbk((t))}$. By Definition-Lemma 7.13(2), on an infinite subset of $\mathcal{I}$ there exist computing lifts of b for $(V_i' \ni o_i', B_i)$ and Γ whose log discrepancies with respect to $(V_i', 0)$ are bounded independently of i. This contradicts (E.41).

It remains to treat the case in which $(\widehat{S}, \frac{1}{m}\operatorname{div}(h_*))$ is not plt near 0 and Φ has a geometric component of coefficient one, that is, $c = m$ in (E.50) for some ϕ.

Lemma E.3. *Let $e, m \geq 1$ and $N > 2me + 2m + 2$ be integers. Let $(V \ni o)$ be a smooth threefold germ with regular parameters t, x, y at o, and let G be a cyclic group of order e acting faithfully on V and fixing o. Let $h = U(t)(xy)^m + t^N R$ be a mark on $(V \ni o)$, where U is a polynomial with $U(0) \neq 0$ and R is regular, let β be a real number, and let*

$$B := \frac{1}{m}\operatorname{div}(h) + \beta \operatorname{div}(t), \qquad b := \operatorname{mld}((V \ni o, B)/G).$$

Assume that the following hold.

(1) The function t is G-semi-invariant, and $\zeta := \chi_t$ is faithful.
(2) Either (a) x and y are G-semi-invariant, or (b) a generator σ of G satisfies $\sigma^* x = cy$ and $\sigma^* y = cx$ for some $c \in \mathbb{C}^*$.
(3) The value b has a computing lift F with $\operatorname{mult}_F t = 1$ and $I_G(F) = G$.

In case (a) let $r_x, r_y \in \{1, \dots, e\}$ be the integers with $\chi_x = \zeta^{r_x}$ and $\chi_y = \zeta^{r_y}$. In case (b) let $z := x + y$ and $w := x - y$, and let $r_z, r_w \in \{1, \dots, e\}$ be the integers with $\chi_z = \zeta^{r_z}$ and $\chi_w = \zeta^{r_w}$. Let T be the prime divisor over V given by the weights $(1, r_x, r_y)$ on (t, x, y) in case (a), resp. $(1, r_z, r_w)$ on (t, z, w) in case (b). Then T is a computing lift of b with $a(T, V, 0) \leq 1 + 2e$, and

$$b = \frac{1-\beta}{e} \quad \text{in case (a)}, \qquad b = \frac{1-\beta}{e} + \frac{1}{2} \quad \text{in case (b)}.$$

Proof. In case (b), $\sigma^* z = cz$ and $\sigma^* w = -cw$, so z and w are G-semi-invariant with

$$\chi_w(\sigma) = -\chi_z(\sigma).$$

Since $\zeta(\sigma)$ is a primitive e-th root of unity and $\zeta(\sigma)^{r_w - r_z} = -1$, the integer e is even and $r_w - r_z \equiv \frac{e}{2} \pmod{e}$. As $r_z, r_w \in \{1, \dots, e\}$,

$$|r_z - r_w| = \frac{e}{2}. \tag{E.51}$$

In case (a) let $(v_1, v_2) := (x, y)$, $(r_1, r_2) := (r_x, r_y)$ and $\kappa := 0$. In case (b) let $(v_1, v_2) := (z, w)$, $(r_1, r_2) := (r_z, r_w)$ and $\kappa := \frac{1}{2}$. In both cases t, v_1, v_2 are G-semi-invariant regular parameters at o with $\chi_{v_j} = \zeta^{r_j}$, and $xy = \frac{1}{4}(z^2 - w^2)$ in case (b).

For a prime divisor E over V with center o, by the definition of the quotient value and of computing lifts,

$$b \leq \frac{a(E, V, B)}{e_E} \quad \text{for any such } E, \qquad b = \frac{a(F, V, B)}{e}. \tag{E.52}$$

We compute the right-hand side of (E.52) for $E = T$. The valuation mult_T is monomial in t, v_1, v_2 with weights $1, r_1, r_2$, so T has center o,

$$\operatorname{mult}_T t = 1, \qquad a(T, V, 0) = 1 + r_1 + r_2 \leq 1 + 2e.$$

On the chart of the weighted blow-up where t generates the ideal of the exceptional divisor, the function field of T is generated by the restrictions of v_1/t^{r_1} and v_2/t^{r_2}. Since $\chi_{v_j} = \zeta^{r_j} = \chi_{t^{r_j}}$, the functions v_1/t^{r_1} and v_2/t^{r_2} are G-invariant, so $I_G(T) = G$ and $e_T = e$. Since $U(0) \neq 0$, $\operatorname{mult}_T(U(t)(xy)^m) = m(r_x + r_y)$ in case (a). In case (b), $\operatorname{mult}_T(z^2 - w^2) = 2\min(r_z, r_w)$ because $r_z \neq r_w$, so $\operatorname{mult}_T(U(t)(xy)^m) = 2m \min(r_z, r_w)$. In both cases

$$\operatorname{mult}_T(U(t)(xy)^m) \leq 2me < N \leq \operatorname{mult}_T(t^N R),$$

hence

$$\operatorname{mult}_T h = \begin{cases} m(r_x + r_y) & \text{in case (a),} \\ 2m \min(r_z, r_w) & \text{in case (b).} \end{cases}$$

Thus $a(T,V,B) = 1-\beta$ in case (a), and $a(T,V,B) = 1 + |r_z - r_w| - \beta = 1 - \beta + \frac{e}{2}$ in case (b) by (E.51). By (E.52), $b \le \frac{1-\beta}{e} + \kappa$.

Since $\operatorname{mult}_F t = 1$, $a(F,V,B) = a(F,V,0) - \frac{1}{m}\operatorname{mult}_F h - \beta$. By (E.52), Lemma E.3 follows once we prove

$$a(F,V,0) - \frac{1}{m}\operatorname{mult}_F h \ge 1 + \kappa e : \tag{E.53}$$

indeed, (E.53) implies

$$b = \frac{a(F,V,B)}{e} \ge \frac{1-\beta}{e} + \kappa = \frac{a(T,V,B)}{e_T} \ge b,$$

so equality holds throughout and T is a computing lift of b.

Case (a). Let $H := (t = 0)$. Since $h|_H = U(0)(xy)^m$,

$$\frac{1}{m}\operatorname{div}(h|_H) = \operatorname{div}(x|_H) + \operatorname{div}(y|_H),$$

which is reduced snc on the smooth surface H near o, so $(H, \frac{1}{m}\operatorname{div}(h|_H))$ is lc. By [Kaw07, Theorem], $(V, \operatorname{div}(t) + \frac{1}{m}\operatorname{div}(h))$ is lc near o. Thus any prime divisor E over V with center o satisfies

$$a(E,V,0) - \frac{1}{m}\operatorname{mult}_E h \ge \operatorname{mult}_E t. \tag{E.54}$$

For $E = F$, (E.54) is (E.53) with $\kappa = 0$.

Case (b). We prove (E.53) with $\kappa e = \frac{e}{2}$. Possibly shrinking V to a G-stable neighborhood of o, we may assume that $(t,z,w) : V \to \mathbb{A}^3$ is étale and G-equivariant for the diagonal action with characters $\zeta, \zeta^{r_z}, \zeta^{r_w}$. Let $r := \min(r_z, r_w)$. By (E.51), $r \le \frac{e}{2}$.

Orders along F. Let $\mathcal{O}_F$ be the valuation ring of mult_F and $\mathfrak{m}_F \subset \mathcal{O}_F$ the maximal ideal. Since $I_G(F) = G$, the group G acts trivially on the residue field $k(F)$, so G acts $k(F)$-linearly on the $k(F)$-vector spaces

$$\mathfrak{m}_F^j / \mathfrak{m}_F^{j+1} = (\mathfrak{m}_F / \mathfrak{m}_F^2)^{\otimes j}$$

of dimension 1, through χ_F^j for a character χ_F of G. A non-zero G-semi-invariant regular function f has non-zero image in $\mathfrak{m}_F^{\operatorname{mult}_F f} / \mathfrak{m}_F^{\operatorname{mult}_F f + 1}$, so $\chi_f = \chi_F^{\operatorname{mult}_F f}$. For $f = t$ we obtain $\chi_F = \zeta$. Hence

$$\chi_f = \zeta^{\operatorname{mult}_F f} \quad \text{for any non-zero } G\text{-semi-invariant } f, \\ \operatorname{mult}_F z \ge r_z \quad \text{and} \quad \operatorname{mult}_F w \ge r_w, \tag{E.55}$$

where the inequalities hold because $\operatorname{mult}_F z \equiv r_z$ and $\operatorname{mult}_F w \equiv r_w \pmod e$, both $\operatorname{mult}_F z$ and $\operatorname{mult}_F w$ are positive, and $r_z, r_w \le e$.

A chart and a quotient. Let $\psi_T : V^\circ \to V$ be the base change of

$$\mathbb{A}^3_{(t,z',w')} \to \mathbb{A}^3_{(t,z,w)}, \qquad (t, z', w') \to (t, t^{r_z} z', t^{r_w} w'),$$

along (t,z,w). Then V° is the chart of the weighted blow-up defining T on which T is the zero locus E° of t. The variety V° is smooth with regular parameters t, z', w' along E°, the group G acts on V° with $\chi_t = \zeta$ and z', w' invariant, and

$$\psi_T^*(dt \wedge dz \wedge dw) = t^{r_z + r_w} dt \wedge dz' \wedge dw', \qquad \psi_T^* h = t^{2mr} h^\circ,$$

where

$$h^\circ = 4^{-m} U(t) \big(t^{2r_z - 2r} z'^2 - t^{2r_w - 2r} w'^2\big)^m + t^{N - 2mr} \psi_T^* R,$$

which is regular since $N > 2mr$. The monomial z^{2m} has coefficient $4^{-m}U(0) \neq 0$ in the expansion of h in t, z, w, so $\chi_h = \zeta^{2mr_z} = \zeta^{2mr_w} = \zeta^{2mr}$ by (E.51). Thus h° is G-invariant. By (E.55), $z' = z/t^{r_z}$ and $w' = w/t^{r_w}$ belong to $\mathcal{O}_F$, so F has a center on V°, contained in E°. Since G acts on V° through ζ on t only, $W := V^\circ/G$ is smooth with regular parameters $\tau := t^e$, z', w' along the image of E°. Let $\rho : V^\circ \to W$ be the quotient morphism and h_W the regular function with $\rho^* h_W = h^\circ$. By the Hurwitz formula, $K_{V^\circ} + E^\circ = \rho^*(K_W + \operatorname{div}(\tau))$.

Let F_W be the restriction of F for ρ, and r_F the ramification index of F over F_W, that is, the positive integer with

$$\operatorname{mult}_F(g \circ \rho) = r_F \operatorname{mult}_{F_W} g$$

for any non-zero rational function g on W. By (E.55), $e \mid \operatorname{mult}_F(g \circ \rho)$ for any such g, so $e \mid r_F$. As $\operatorname{mult}_F \tau = e$, also $r_F \mid e$. Thus $r_F = e$ and $\operatorname{mult}_{F_W} \tau = 1$. Since $\operatorname{mult}_F t = 1$ and $r_F = e$, the formula for $\psi_T^*(dt \wedge dz \wedge dw)$ and the Hurwitz formula for ρ imply

$$a(F, V, 0) = a(F, V^\circ, E^\circ) + 1 + r_z + r_w = e a(F_W, W, \operatorname{div}(\tau)) + 1 + r_z + r_w,$$

hence

$$a(F, V, 0) = e\big(a(F_W, W, 0) - 1\big) + 1 + r_z + r_w, \qquad \operatorname{mult}_F h = 2mr + e \operatorname{mult}_{F_W} h_W. \tag{E.56}$$

Equivariant arcs. Let $\mathcal{S} := \mathbb{C}[[u]]^3$ be the arc space of $\mathbb{A}^3$, with points $\phi = (\phi_0, \phi_1, \phi_2)$, and $\mathcal{S}^\circ \subset \mathcal{S}$ the open cylinder $(\phi_0(0) \neq 0)$. For $\phi \in \mathcal{S}$ let γ_ϕ be the arc of V through o lifting

$$\vartheta \to \big(\vartheta \phi_0(\vartheta^e),\ \vartheta^{r_z} \phi_1(\vartheta^e),\ \vartheta^{r_w} \phi_2(\vartheta^e)\big) \quad \text{in the coordinates } (t, z, w).$$

For $\phi \in \mathcal{S}^\circ$, γ_ϕ lifts to V° with $z' = \phi_1 \phi_0^{-r_z}$ and $w' = \phi_2 \phi_0^{-r_w}$ evaluated at ϑ^e, and

$$\rho \circ \gamma_\phi(\vartheta) = \mu(\phi)(\vartheta^e)$$

for the arc $\mu(\phi)$ of W with coordinates $(u\phi_0^e, \phi_1 \phi_0^{-r_z}, \phi_2 \phi_0^{-r_w})$. For an arc γ of V or W and a function f on V or W, let $\operatorname{mult}_\gamma f$ be the order at 0 of the power series $f \circ \gamma$. Thus

$$\operatorname{mult}_{\gamma_\phi} h = 2mr + e \operatorname{mult}_{\mu(\phi)} h_W \quad \text{for any } \phi \in \mathcal{S}^\circ. \tag{E.57}$$

On jets of any order, $\phi_0 \to \phi_0^e$ is étale on units and for fixed ϕ_0 the map

$$(\phi_1, \phi_2) \to (\phi_1 \phi_0^{-r_z}, \phi_2 \phi_0^{-r_w})$$

is an isomorphism, so μ is étale and surjective onto the cylinder $\mathcal{T}$ of the arcs δ of W based on the image of E° with $\operatorname{mult}_\delta \tau = 1$, and $\operatorname{codim} \mathcal{T} = 1$. Hence $\operatorname{codim}_{\mathcal{S}} \mu^{-1}(\mathcal{Z}) = \operatorname{codim} \mathcal{Z} - 1$ for any cylinder $\mathcal{Z} \subset \mathcal{T}$.

Let $Y \to W$ be a log resolution on which F_W is a divisor, and let $\operatorname{Cont}(F_W)$ be the closure of the image in the arc space of W of the arcs of Y which meet F_W at a general point with contact order 1. By [ELM04, Theorem 2.1], $\operatorname{codim} \operatorname{Cont}(F_W) = a(F_W, W, 0)$, and the arcs $\delta \in \operatorname{Cont}(F_W)$ with $\operatorname{mult}_\delta \tau = \operatorname{mult}_{F_W} \tau = 1$ and $\operatorname{mult}_\delta h_W = \operatorname{mult}_{F_W} h_W$ form a dense open subset $\operatorname{Cont}^\circ(F_W)$ of $\operatorname{Cont}(F_W)$. Let $k := \operatorname{mult}_F h$ and

$$\mathcal{C} := \{\phi \in \mathcal{S} \mid \operatorname{mult}_{\gamma_\phi} h \geq k\}.$$

Since the center of F_W is contained in the image of E°, $\operatorname{Cont}^\circ(F_W) \subset \mathcal{T}$. By (E.56) and (E.57), $\mu^{-1}(\operatorname{Cont}^\circ(F_W)) \subset \mathcal{C}$, so $\operatorname{codim}_{\mathcal{S}} \mathcal{C} \leq a(F_W, W, 0) - 1$, and (E.56) implies

$$a(F, V, 0) \geq e \operatorname{codim}_{\mathcal{S}} \mathcal{C} + 1 + r_z + r_w. \tag{E.58}$$

Degeneration to the initial form. Write $\phi_j = \sum_{j' \geq 0} \phi_{jj'} u^{j'}$, and assign to $\phi_{0j'}, \phi_{1j'}, \phi_{2j'}$ the arc degrees $1 + ej'$, $r_z + ej'$, $r_w + ej'$ and the weight 1. Let $\mathcal{P}$ be the polynomial ring in the $\phi_{jj'}$ of arc degree $< k$. Since t, z, w have positive ϑ-order along any γ_ϕ, for any $f \in \widehat{\mathcal{O}}_{V,o} = \mathbb{C}[[t, z, w]]$ and $j < k$ the coefficient $c_j(f)$ of ϑ^j in $f \circ \gamma_\phi$ belongs to $\mathcal{P}$ and is homogeneous of arc degree j, and a monomial $t^{n_0} z^{n_1} w^{n_2}$ contributes to $c_j(f)$ a polynomial homogeneous of weight $n_0 + n_1 + n_2$.

Let $\mathfrak{b} := (c_j(h) \mid j < k)$, so that $\mathcal{C}$ is the preimage of $V(\mathfrak{b}) \subset \operatorname{Spec} \mathcal{P}$ and $\operatorname{codim}_{\mathcal{S}} \mathcal{C} = \operatorname{ht} \mathfrak{b}$. Since $N > 2m$ and $U(t) - U(0)$ is divisible by t,

$$h = h_0 + (\text{terms of degree} > 2m), \qquad h_0 := 4^{-m} U(0)(z^2 - w^2)^m.$$

Thus $c_j(h)$ has no part of weight $< 2m$, and the weight-$2m$ part of $c_j(h)$ is $c_j(h_0)$. Let $\operatorname{in}(\mathfrak{b})$ be the ideal spanned by the lowest-weight parts of the elements of $\mathfrak{b}$. Then

$$\mathfrak{b}^0 := (c_j(h_0) \mid j < k) \subset \operatorname{in}(\mathfrak{b}).$$

Since $1 + ej', r_z + ej', r_w + ej' \geq 1$ and $\mathfrak{b}$ is homogeneous for the arc degree, for any integer $\nu \geq 0$ the space $\operatorname{in}(\mathfrak{b})_\nu$ is the graded space of $\mathfrak{b}_\nu$ for the weight filtration, and

$$\dim_{\mathbb{C}} \operatorname{in}(\mathfrak{b})_\nu = \dim_{\mathbb{C}} \mathfrak{b}_\nu < +\infty.$$

Hence the Hilbert functions of $\mathcal{P}/\mathfrak{b}$ and $\mathcal{P}/\operatorname{in}(\mathfrak{b})$ are equal, and

$$\operatorname{codim}_{\mathcal{S}} \mathcal{C} = \operatorname{ht} \mathfrak{b} = \operatorname{ht} \operatorname{in}(\mathfrak{b}) \geq \operatorname{ht} \mathfrak{b}^0. \tag{E.59}$$

For any $\phi \in \mathcal{S}$, $\operatorname{mult}_{\gamma_\phi} z \in r_z + e\mathbb{N}$ unless $z \circ \gamma_\phi = 0$, $\operatorname{mult}_{\gamma_\phi} w \in r_w + e\mathbb{N}$ unless $w \circ \gamma_\phi = 0$, and $r_z \not\equiv r_w \pmod e$. So

$$\operatorname{mult}_{\gamma_\phi}(z^2 - w^2) = 2 \min(\operatorname{mult}_{\gamma_\phi} z, \operatorname{mult}_{\gamma_\phi} w),$$

and

$$V(\mathfrak{b}^0) = \left\{ \phi \;\middle|\; \operatorname{mult}_{\gamma_\phi} z,\ \operatorname{mult}_{\gamma_\phi} w \geq \frac{k}{2m} \right\} = \{\phi_{1j'} = 0 \text{ for } j' < i_1,\ \phi_{2j'} = 0 \text{ for } j' < i_2\},$$

where i_1, i_2 are the least elements of $\mathbb{N}$ with $\alpha_1 := r_z + ei_1 \geq \frac{k}{2m}$ and $\alpha_2 := r_w + ei_2 \geq \frac{k}{2m}$. Thus $\operatorname{ht} \mathfrak{b}^0 = i_1 + i_2$. Since $\alpha_1 - \alpha_2 \equiv r_z - r_w \equiv \frac{e}{2} \pmod e$, $|\alpha_1 - \alpha_2| \geq \frac{e}{2}$. Since $\alpha_1, \alpha_2 \geq \frac{k}{2m}$, (E.58) and (E.59) give

$$a(F, V, 0) - \frac{k}{m} \geq e(i_1 + i_2) + 1 + r_z + r_w - 2 \min(\alpha_1, \alpha_2) = 1 + |\alpha_1 - \alpha_2| \geq 1 + \frac{e}{2},$$

which is (E.53) in case (b). The lemma follows. □

Proof of Theorem 15.7, continued. Types of components of coefficient one. A component of coefficient one is a factor ϕ with $c = m$ in (E.50). Over an algebraic closure of $\Bbbk((t))$, the irreducible polynomials in p, q that are weighted homogeneous for the weights (w_1, w_2) are p, q, and $p^{w_2} - \lambda q^{w_1}$ with $\lambda \neq 0$, of weighted degrees w_1, w_2, and $w_1 w_2$. If $(p^{w_2} - \lambda q^{w_1})^m$ divides Φ, which has weighted degree $m(w_1 + w_2)$ by (E.49), then $w_1 w_2 \leq w_1 + w_2$, so $(w_1 - 1)(w_2 - 1) \leq 1$, and $w_2 = 1$ as w_1, w_2 are coprime. If moreover $w_1 \geq 2$, then $2mw_1 > m(w_1 + 1)$, so $p - \lambda q^{w_1}$ is the only component of coefficient one of the form $p - \lambda q^{w_1}$. If $w_1 = w_2 = 1$, the components of coefficient one are lines, and there exist two of them only if $\Phi = \kappa(\ell_1 \ell_2)^m$ for linear forms ℓ_1, ℓ_2 and $\kappa \in \Bbbk((t))^*$. The coordinate $p - \sigma$ is Γ-semi-invariant: for $g \in \Gamma$, since Γ fixes E, the polynomial $g^*(p - \sigma) - \chi_p(g)(p - \sigma) \in \Bbbk((t))[q]$ has q-degree less than w_1/w_2 and v_E-value at least w_1, hence vanishes. Since h_* and the homogeneous coordinates q and $p - \sigma$ of Φ are Γ-semi-invariant, the Galois group of $\overline{\Bbbk((t))}/\Bbbk((t))$ and the semilinear action of Γ permute the components of coefficient one. Hence the components p and q, and a component which is the unique non-axis component of coefficient one, are defined over $\Bbbk((t))$ and Γ-stable. The remaining components of coefficient one are two non-axis lines $\ell_1 \neq \ell_2$ with $\Phi = \kappa(\ell_1 \ell_2)^m$. They are defined over the splitting field of $\ell_1 \ell_2$, of degree at most 2 over $\Bbbk((t))$. Since $\Bbbk$ is algebraically closed of characteristic zero, the splitting field is contained in $\Bbbk((t^{1/2}))$.

A second root. Since $\Gamma = I_\Gamma(F_i)$ and $\operatorname{mult}_{F_i} t = 1$, the group Γ is cyclic and $\chi_t|_\Gamma$ is faithful. Let $n := 2$ if the lines ℓ_1, ℓ_2 occur and are not defined over $\Bbbk((t))$, and $n := 1$ otherwise. If $n = 1$, let $(V_i^\sharp \ni o_i^\sharp, B_i^\sharp) := (V_i' \ni o_i', B_i)$, $\Gamma^\sharp := \Gamma$, $e^\sharp := e$, $\beta^\sharp := \beta$, and $F_i^\sharp := F_i$. If $n = 2$, let $(V_i^\sharp \ni o_i^\sharp)$ be the smooth germ with regular parameters t_1, p, q over $(V_i' \ni o_i')$ given by $t_1^2 = t$, and let

$$\Gamma^\sharp := \{(g, \xi) \in \Gamma \times \mathbb{C}^* \mid \xi^2 = \chi_t(g)\}.$$

Since Γ is cyclic and $\chi_t|_\Gamma$ is faithful, $(g,\xi)\to\xi$ is an isomorphism $\Gamma^\sharp\to\mu_{2e}$, so $\Gamma^\sharp$ is cyclic of order $e^\sharp := 2e$, acts on p,q through g, and acts on t_1 through the faithful character $(g,\xi)\to\xi$. Let $\beta^\sharp := 2\beta-1$. By the Hurwitz formula, the crepant boundary on $V_i^\sharp$ over (V_i', B_i) is

$$B_i^\sharp := \frac{1}{m}\operatorname{div}(h_i) + \beta^\sharp \operatorname{div}(t_1).$$

Since the quotient morphisms identify $V_i^\sharp/\Gamma^\sharp$ with V_i'/Γ and the ramification index over V_i'/Γ is the order of the inertia group, for any prime divisor $P^\sharp$ over $V_i^\sharp$ and its restriction P to V_i', both sides of

$$\frac{a(P^\sharp, V_i^\sharp, B_i^\sharp)}{|I_{\Gamma^\sharp}(P^\sharp)|} = \frac{a(P, V_i', B_i)}{|I_\Gamma(P)|}$$

are the log discrepancy of the restriction of $P^\sharp$ to V_i'/Γ with respect to the crepant quotient boundary, so the equality holds. Thus $\operatorname{mld}((V_i^\sharp \ni o_i^\sharp, B_i^\sharp)/\Gamma^\sharp) = b$. As $\operatorname{mult}_{F_i} t = 1$, F_i has a unique extension $F_i^\sharp$, with $\operatorname{mult}_{F_i^\sharp} t_1 = 1$ and $k(F_i^\sharp) = k(F_i)$. Hence $I_{\Gamma^\sharp}(F_i^\sharp) = \Gamma^\sharp$, and $F_i^\sharp$ is a computing lift of b. If $n=1$, let $t_1 := t$. In both cases $V_i^\sharp$ has regular parameters t_1, p, q with $t = t_1^n$, $\operatorname{mult}_{F_i^\sharp} t_1 = 1$, and the lines ℓ_1, ℓ_2, if they occur, are defined over $\Bbbk((t_1))$. Each of them is $\Gamma^\sharp$-stable, or a generator of $\Gamma^\sharp$ interchanges them.

Stable components. Assume that some component ϕ of coefficient one is $\Gamma^\sharp$-stable and defined over $\Bbbk((t_1))$. Let p_1 be a $\Gamma^\sharp$-semi-invariant equation of ϕ, namely $p-\sigma$, q, $p-\sigma-\lambda q^{w_1}$, or a linear form in p,q (in which case $\sigma=0$), and let p_2 be the element of $\{p-\sigma, q\}$ for which p_1, p_2 are weighted homogeneous coordinates. Let ω_1^0, ω_2^0 be their weights. Then $\Phi = p_1^m\Phi'$, and Φ' contains the monomial p_2^m with a non-zero coefficient, since otherwise p_1^{m+1} divides Φ, contrary to (E.50). Let

$$\omega_1 := (m+1)\omega_1^0 + 1, \qquad \omega_2 := (m+1)\omega_2^0.$$

A monomial $p_1^{j_1}p_2^{j_2}$ of the expansion of h_* in $\Bbbk((t_1))[[p_1,p_2]]$ has $j_1\omega_1^0 + j_2\omega_2^0 \ge m(\omega_1^0+\omega_2^0)$, and

$$j_1\omega_1 + j_2\omega_2 = (m+1)(j_1\omega_1^0 + j_2\omega_2^0) + j_1.$$

If equality holds in the first inequality, the monomial occurs in $p_1^m\Phi'$, so $j_1 \ge m$, with $j_1 = m$ only for $j_1 = j_2 = m$. If not, then

$$j_1\omega_1 + j_2\omega_2 - (m+1)m(\omega_1^0+\omega_2^0) - m \ge (m+1) + j_1 - m \ge 1.$$

Hence h_* has the unique weighted initial monomial $\kappa(p_1p_2)^m$ for the weights (ω_1,ω_2), with $\kappa\in\Bbbk((t_1))^*$.

Exchanged components. Otherwise $w_1 = w_2 = 1$ and $\Phi = \kappa(\ell_1\ell_2)^m$, where a generator g_0 of $\Gamma^\sharp$ interchanges the non-axis lines ℓ_1, ℓ_2. Normalize $\ell_j = p + a_j q$ with $a_j\in\Bbbk((t_1))$. Then $g_0^*\ell_1$ is a multiple of ℓ_2 with p-coefficient $\chi_p(g_0)$, so $g_0^*\ell_1 = c_{g_0}\ell_2$ and $g_0^*\ell_2 = c_{g_0}\ell_1$ for $c_{g_0} := \chi_p(g_0)$. Let $(p_1,p_2) := (\ell_1,\ell_2)$ and $\omega_1 = \omega_2 := 1$. The weighted initial form of h_* is $\kappa(p_1p_2)^m$.

Scaling. For stable and for exchanged components, let $c_0 := \operatorname{mult}_0\kappa$. Fix an integer N with

$$N > 2me^\sharp + 2m + 2.$$

Let $\Xi : (p,q)\to(p_1,p_2)$ be the change of coordinates, the composite of $p\to p-\sigma$ and of $(p-\sigma,q)\to(p_1,p_2)$. The components of Ξ and of Ξ^{-1} are polynomials of bounded degree with coefficients in $t_1^{-c_1}\Bbbk[[t_1]]$ for a constant c_1, and, as Ξ is triangular or linear in p,q, the Jacobian determinant $\Delta := \det\partial(p,q)/\partial(p_1,p_2)$ of Ξ^{-1} belongs to $\Bbbk((t_1))^*$. Hence there exists a constant c_2 such that the coefficient of $p_1^{j_1}p_2^{j_2}$ in h_* belongs to $t_1^{-c_2(j_1+j_2+1)}\Bbbk[[t_1]]$. For an integer $k_0\ge 1$ substitute

$$p_1 = t_1^{k_0\omega_1}x, \qquad p_2 = t_1^{k_0\omega_2}y,$$

and divide by $t_1^{k_0m(\omega_1+\omega_2)+c_0}$. A monomial of weighted excess $\delta\ge 1$ over $m(\omega_1+\omega_2)$ acquires $t_1^{k_0\delta-c_0}$, and $\delta \ge j_1+j_2-m(\omega_1+\omega_2)$. Fix k_0 so large that $k_0\delta - c_0 - c_2(j_1+j_2+1)\ge N$ for all

such monomials (first for the finitely many with $j_1 + j_2 \le 2m(\omega_1 + \omega_2)$, then for the others by the linear bound) and that the old parameters p, q become polynomials in x, y with coefficients in $\Bbbk[[t_1]]$ and without constant term. Monomials of excess $\delta < 0$, and monomials other than $(p_1p_2)^m$ of excess $\delta = 0$, have coefficient 0 in h_*.

The conditions used above involve finitely many t_1-jets of finitely many coefficients: the vanishing of those of the monomials with $\delta < 0$, and of the monomials other than $(p_1p_2)^m$ with $\delta = 0$, up to t_1-order $N + c_0 + k_0 m(\omega_1 + \omega_2)$, the t_1-jet of the coefficient of $(p_1p_2)^m$ up to order $c_0 + N$, the pole bounds, and the order $\operatorname{mult}_0 \Delta$. They hold for h_* and Ξ, and still hold after we replace the coefficients of Ξ by their truncations in a sufficiently high t_1-degree, which we do. If $n = 2$, we write the h_i and h_* in t_1, p, q, so that $h_*(t_1^2, p, q)$ is the compatible generic limit. By Lemma 2.6(4) for the components of $t_1^{c_1}\Xi$, in the exchanged case for $t_1^{c_1}(p_1 + p_2)$ and $t_1^{c_1}(p_1 - p_2)$ instead, which are $\Gamma^\sharp$-semi-invariant since g_0^* maps $p_1 + p_2$ and $p_1 - p_2$ to $c_{g_0}(p_1 + p_2)$ and $-c_{g_0}(p_1 - p_2)$ and g_0 generates $\Gamma^\sharp$, after passing to an infinite subset there exist changes Ξ_i of the coordinates p, q of $V_i^\sharp$, with $t_1^{c_1}\Xi_i$ of total degree in t_1, p, q at most that of $t_1^{c_1}\Xi$ (truncating the $t_1^{c_1}\Xi_i$ above the total degree of $t_1^{c_1}\Xi$ keeps the compatible generic limit, as the coefficients of $t_1^{c_1}\Xi$ in higher degree vanish), with coefficients in $t_1^{-c_1}\mathbb{C}[t_1]$ and with $\Gamma^\sharp$-semi-invariant components (in the exchanged case, with $\Gamma^\sharp$-semi-invariant sum and difference of the components), such that $t_1^{c_1}\Xi$ and h_* are a compatible generic limit of the $t_1^{c_1}\Xi_i$ and the h_i. The vanishing conditions, and the vanishing or constancy of the coefficients that fix the form of Ξ, are polynomial identities among finitely many coefficients, so they hold for h_i and Ξ_i by Lemma 2.6(4). The conditions of the form $\neq 0$, namely that the coefficient of $(p_1p_2)^m$ and Δ have the orders c_0 and $\operatorname{mult}_0 \Delta$, hold for them by the definition of compatible generic limits. On the germs $V_i^\sharp$ we apply Ξ_i and the substitution $p_1 = t_1^{k_0\omega_1}x$, $p_2 = t_1^{k_0\omega_2}y$. Let $N_3 := k_0 m(\omega_1 + \omega_2) + c_0$, and let $\widetilde{h}_i$ be h_i written in t_1, x, y and divided by $t_1^{N_3}$. Let $U_i(t_1)$ be the truncation in degree $< N$ of the coefficient of $(xy)^m$ in $\widetilde{h}_i$. Since $(t_1^N)\widehat{\mathcal{O}} \cap \mathcal{O} = (t_1^N)\mathcal{O}$ for the local ring $\mathcal{O}$ at the new origin, we obtain

$$\widetilde{h}_i = U_i(t_1)(xy)^m + t_1^N R_i, \qquad U_i(0) \neq 0,\ R_i \text{ regular}. \tag{E.60}$$

The substitution defines a $\Gamma^\sharp$-equivariant birational morphism from a smooth germ $(\widetilde{V}_i \ni \widetilde{o}_i)$ with regular parameters t_1, x, y to $V_i^\sharp$, mapping $\widetilde{o}_i$ to $o_i^\sharp$. Since the coefficients of Ξ_i have bounded poles and, by (E.44),

$$\min(\operatorname{mult}_{F_i^\sharp} p, \operatorname{mult}_{F_i^\sharp} q) \ge \min(\operatorname{mult}_{F_i} p, \operatorname{mult}_{F_i} q) \to +\infty,$$

for $i \gg 0$ the prime divisor $F_i^\sharp$ has center $\widetilde{o}_i$ on $\widetilde{V}_i$. The parameters x, y are $\Gamma^\sharp$-semi-invariant in the stable case, and $g_0^* x = c_{g_0}\chi_{t_1}(g_0)^{-k_0}y$ and $g_0^* y = c_{g_0}\chi_{t_1}(g_0)^{-k_0}x$ in the exchanged case.

Applying the lemma. Let $\psi_i : \widetilde{V}_i \to V_i'$ be the composite of the scaling, Ξ_i and, if $n = 2$, the morphism $V_i^\sharp \to V_i'$. Let $N_2 := k_0(\omega_1 + \omega_2) + \operatorname{mult}_0 \Delta$. Since the Jacobian determinant of Ξ_i^{-1} has order $\operatorname{mult}_0 \Delta$,

$$\psi_i^* t = t_1^n, \qquad \psi_i^* h_i = t_1^{N_3}\widetilde{h}_i, \quad \text{and} \quad \psi_i^*(dt \wedge dp \wedge dq) = t_1^{n-1+N_2} u_i dt_1 \wedge dx \wedge dy \tag{E.61}$$

for units u_i. Let $\beta' := n\beta + \frac{N_3}{m} - (n - 1 + N_2)$ and

$$\widetilde{B}_i := \frac{1}{m}\operatorname{div}(\widetilde{h}_i) + \beta' \operatorname{div}(t_1).$$

By (E.61),

$$K_{\widetilde{V}_i} + \widetilde{B}_i = \psi_i^*\big(K_{V_i'} + B_i\big).$$

As Ξ_i and the scaling are $\Gamma^\sharp$-equivariant and birational, log discrepancies and inertia groups of prime divisors over $\widetilde{V}_i$ with center $\widetilde{o}_i$ are those over $(V_i^\sharp, B_i^\sharp)$. Hence

$$\operatorname{mld}((\widetilde{V}_i \ni \widetilde{o}_i, \widetilde{B}_i)/\Gamma^\sharp) \ge b,$$

and equality holds since $F_i^\sharp$ has center $\widetilde{o}_i$ on $\widetilde{V}_i$ and is a computing lift of b for $(V_i^\sharp \ni o_i^\sharp, B_i^\sharp)$ and $\Gamma^\sharp$.

In Lemma E.3 we take $G := \Gamma^\sharp$, $e := e^\sharp$, $(V \ni o) := (\widetilde{V}_i \ni \widetilde{o}_i)$, t_1 in place of t, $h := \widetilde{h}_i$, $U := U_i$, $R := R_i$, $\beta := \beta'$, and $F := F_i^\sharp$, so that $B = \widetilde{B}_i$. The group $\Gamma^\sharp$ is cyclic of order $e^\sharp$ and χ_{t_1} is faithful on it by *A second root*, so $\Gamma^\sharp$ acts faithfully on $\widetilde{V}_i$ and (1) holds. The mark $\widetilde{h}_i$ has the form $U_i(t_1)(xy)^m + t_1^N R_i$ of the lemma with $N > 2me^\sharp + 2m + 2$ by (E.60) and *Scaling*. Condition (2) holds in case (a) for stable components and in case (b) for exchanged components, with the generator g_0 and $c := c_{g_0}\chi_{t_1}(g_0)^{-k_0}$. Condition (3) holds by the previous paragraph, as $\operatorname{mult}_{F_i^\sharp} t_1 = 1$ and $I_{\Gamma^\sharp}(F_i^\sharp) = \Gamma^\sharp$ by *A second root*. By the lemma, there exists a computing lift T_i over $\widetilde{V}_i$ given by weights with weight 1 on t_1, so that

$$\operatorname{mult}_{T_i} t_1 = 1, \qquad a(T_i, \widetilde{V}_i, 0) \le 1 + 2e^\sharp.$$

Return. Let P_i be the restriction of T_i for ψ_i, and $r_i \ge 1$ the ramification index of T_i over P_i, that is, the integer with

$$\operatorname{mult}_{T_i}(g \circ \psi_i) = r_i \operatorname{mult}_{P_i} g$$

for any non-zero rational function g on V_i'. Since $\psi_i(\widetilde{o}_i) = o_i'$, P_i has center o_i'. By (E.61),

$$r_i a(P_i, V_i', 0) = a(T_i, \widetilde{V}_i, 0) + (n - 1 + N_2)\operatorname{mult}_{T_i} t_1 \le 2e^\sharp + n + N_2.$$

Since $\widetilde{V}_i \to V_i^\sharp$ is $\Gamma^\sharp$-equivariant and birational and $V_i^\sharp/\Gamma^\sharp = V_i'/\Gamma$, the induced morphism $\widetilde{V}_i/\Gamma^\sharp \to V_i'/\Gamma$ is birational, so the restrictions of T_i and P_i to V_i'/Γ coincide, and by the Hurwitz formula

$$\frac{a(P_i, V_i', B_i)}{|I_\Gamma(P_i)|} = \frac{a(T_i, \widetilde{V}_i, \widetilde{B}_i)}{|I_{\Gamma^\sharp}(T_i)|} = b.$$

Thus P_i is a computing lift of b for $(V_i' \ni o_i', B_i)$ and Γ with $a(P_i, V_i', 0) \le 2e^\sharp + n + N_2 \le 4e + 2 + N_2$, a bound independent of i. This contradicts (E.41). □

Appendix F. Proofs for coefficient-one components and transverse types D and E

We prove the results of Section 16 and Section 17.

F.1. **Coefficient-one components.** In this subsection we prove Proposition 16.4 and Theorems 16.5 and 16.6, and we state and prove Corollary F.1.

Proof of Proposition 16.4. Write $(x_1, x_2, x_3, x_4) := (x, y, z, s)$ and $W_0 := (f_0 = 0) \subset \mathbb{A}^4$. Let

$$B := \operatorname{div}(f) + \frac{1}{m}\operatorname{div}(h) + \operatorname{div}(s) \quad \text{and} \quad B^0 := \operatorname{div}(f_0) + \frac{1}{m}\operatorname{div}(h_0) + \operatorname{div}(s)$$

be the $\mathbb{R}$-divisors on $\mathcal{X}$ near 0 and on $\mathbb{A}^4$ respectively. We divide the proof into five steps.

Step 1. In this step we show that $(\mathbb{A}^4, B^0)$ is lc near 0.

For any $N \in \mathbb{N}^+$, let $V_N := (xy - z^N = 0) \subset \mathbb{A}^4$. Since $V_N = \mathbb{A}^3/\mu_N$ for the action $\zeta \cdot (p, q, s) = (\zeta p, \zeta^{-1} q, s)$, via $(p, q, s) \to (p^N, q^N, pq, s)$, which is étale in codimension 1, and $\operatorname{div}_{V_N}(z) + \operatorname{div}_{V_N}(s)$ pulls back to $\operatorname{div}(pqs)$, the pair

$$\big(V_N, \operatorname{div}_{V_N}(z) + \operatorname{div}_{V_N}(s)\big) \text{ is lc.} \tag{F.1}$$

Let

$$\Delta_0 := \frac{1}{m}\operatorname{div}_{W_0}(h_0) + \operatorname{div}_{W_0}(s).$$

In row A, $W_0 = V_n$ and

$$\frac{1}{m}\operatorname{div}_{W_0}(h_0) = \operatorname{div}_{W_0}(z),$$

so (W_0, Δ_0) is lc by (F.1).

In row D, let $U := (v^2 + y^2 + u^{2n-4} = 0) \subset \mathbb{A}^4$, where u, v, y, s are the coordinates of $\mathbb{A}^4$, let ι be the involution $(u, v, y, s) \to (-u, -v, y, s)$ of U, and let

$$\pi_U : U \to \mathbb{A}^4, \qquad (u, v, y, s) \to (uv, y, u^2, s).$$

Since $f_0(uv, y, u^2, s) = u^2(v^2 + y^2 + u^{2n-4})$, $\pi_U(U) \subset W_0$. Since $v^2 = -y^2 - u^{2n-4}$ on U, the ring $\mathbb{C}[U]^\iota$ is generated by y, s, u^2, uv, so π_U induces a surjection $\mathbb{C}[x, y, z, s]/(f_0) \to \mathbb{C}[U]^\iota$, which is an isomorphism because $\mathbb{C}[x, y, z, s]/(f_0)$ and $\mathbb{C}[U]^\iota$ are integral domains of dimension 3. Thus π_U is the quotient map by ι. The fixed locus of ι is $(u = v = y = 0)$, and

$$\dim(u = v = y = 0) = \dim U - 2,$$

so $K_U = \pi_U^* K_{W_0}$. Since

$$\frac{1}{m} \operatorname{div}_{W_0}(h_0) = \frac{1}{2} \operatorname{div}_{W_0}(z),$$

$\pi_U^* \Delta_0 = \operatorname{div}_U(us)$. Let $P := v + \sqrt{-1}y$ and $Q := -v + \sqrt{-1}y$. Then $PQ = -v^2 - y^2 = u^{2n-4}$ on U, so the coordinates (P, Q, u, s) define an isomorphism $U \cong V_{2n-4}$ under which $\pi_U^* \Delta_0$ corresponds to the boundary of (F.1) with $N = 2n - 4$. By (F.1), (W_0, Δ_0) is lc.

In rows A and D, W_0 is normal, being the quotient of the normal variety $\mathbb{A}^3$, resp. U, and f_0 divides neither h_0 nor s. By [Kaw07, Theorem], $(\mathbb{A}^4, B^0)$ is lc near W_0, in particular near 0.

Step 2. In this step we show that $(W, \frac{1}{m} \operatorname{div}_W(h) + \operatorname{div}_W(s))$ is lc near 0.

By Lemma 2.12(2)(3), $\operatorname{mld}(\mathcal{X} \ni 0, B)$ and $\operatorname{mld}(\mathbb{A}^4 \ni 0, B^0)$ do not change when $\mathcal{X}$ and $\mathbb{A}^4$ are replaced by their common completion $\operatorname{Spec} \mathbb{C}[[x, y, z, s]]$ at 0. We regard f and h as elements of $\mathbb{C}[[x, y, z, s]]$. Let

$$\omega := (n, n, 2, 1) \text{ in row A}, \qquad \omega := (n - 1, n - 2, 2, 1) \text{ in row D},$$

as weights of (x, y, z, s). Then f_0 and h_0 are ω-homogeneous, of ω-weights $2n$ and $2m$ in row A, and $2n - 2$ and m in row D, all less than J. Since any monomial of degree at least J has ω-weight at least J, and $f - f_0, h - h_0 \in (x, y, z, s)^J$,

$$\operatorname{in}_\omega f = f_0, \qquad \operatorname{in}_\omega h = h_0, \quad \text{and} \quad \operatorname{in}_\omega s = s.$$

Since f_0 divides neither h_0 nor s, f divides neither h nor s, so $\operatorname{div}_W(h)$ and $\operatorname{div}_W(s)$ are defined. By Lemma 6.4(3) with the trivial group and the weight ω, and by **Step 1**,

$$\operatorname{mld}(\mathcal{X} \ni 0, B) \geq \operatorname{mld}(\mathbb{A}^4 \ni 0, B^0) \geq 0.$$

Thus $(\mathcal{X}, B)$ is lc near 0. Since W is normal and $B = W + \frac{1}{m} \operatorname{div}(h) + \operatorname{div}(s)$, the pair

$$\Big(W, \frac{1}{m} \operatorname{div}_W(h) + \operatorname{div}_W(s)\Big)$$

is lc near 0 by [Kaw07, Theorem].

Step 3. In this step we show that G acts faithfully on W and that

$$\operatorname{mld}\Big(\Big(W \ni 0, \frac{1}{m} \operatorname{div}(h)\Big)/G\Big) \geq \frac{1}{R}. \tag{F.2}$$

Let $\sigma \in G \backslash \{1\}$. Since G acts faithfully on $(\mathcal{X} \ni 0)$ and $x_1, \dots, x_4$ are G-semi-invariant regular parameters, there exists j such that σ multiplies x_j by a root of unity $\zeta \neq 1$, so any point of $\mathcal{X}$ fixed by σ is contained in $(x_j = 0)$. Since $f_0 \in (x, y, z, s)^2$ and $f - f_0 \in (x, y, z, s)^J$ with $J \geq 2$, $f \in (x, y, z, s)^2$, so $x_j \notin (f)$, so $W \not\subset (x_j = 0)$ and σ acts non-trivially on W.

Let E be a prime divisor over W with center 0. Since χ_s has order R, s^R is G-invariant, so $s^R = \pi_G^* \psi$ for the quotient map $\pi_G : W \to W/G$ and a regular function ψ on W/G vanishing at $\pi_G(0)$. Let E' be the restriction of E, a prime divisor over W/G. Since the ramification index of E over E' is e_E (Lemma 2.15(1)),

$$R \operatorname{mult}_E s = \operatorname{mult}_E(\pi_G^* \psi) = e_E \operatorname{mult}_{E'} \psi \in e_E \mathbb{N}^+.$$

By **Step 2**,

$$a\Big(E, W, \frac{1}{m}\operatorname{div}(h)\Big) = a\Big(E, W, \frac{1}{m}\operatorname{div}(h) + \operatorname{div}(s)\Big) + \operatorname{mult}_E s \geq \operatorname{mult}_E s \geq \frac{e_E}{R}.$$

Since $\operatorname{mld}((W \ni 0, \frac{1}{m}\operatorname{div}(h))/G)$ is the infimum of $a(E, W, \frac{1}{m}\operatorname{div}(h))/e_E$ over the prime divisors E over W with center 0 (Definition 2.2), (F.2) holds.

Step 4. In this step we construct a prime divisor F over W with center 0, an element $g \in I_G(F)$ of order k, and an integer $c \in [k+1, 2k]$ such that

$$a(F, W, 0) = c + \frac{k}{R} \qquad \text{and} \qquad \operatorname{mult}_F h = mc. \tag{F.3}$$

Let $g \in G$ be an element such that $\chi_s(g)$ has order R, and let k be the order of g. Then $R \mid k$ and $k \leq |G| \leq M$. Fix a primitive k-th root of unity ξ_0. Then $\chi_s(g) = \xi_0^{n_0 k/R}$ for an integer n_0 coprime to R. By the Chinese remainder theorem, there exists an integer n_0' coprime to k such that $n_0' \equiv n_0 \bmod R$, so $\xi := \xi_0^{n_0'}$ is a primitive k-th root of unity with $\chi_s(g) = \xi^{k/R}$. For any j, let α_j be an integer such that $\chi_{x_j}(g) = \xi^{\alpha_j}$, with $\alpha_4 = k/R$. Since G acts faithfully on $\mathcal{X}$, so does $\langle g \rangle$. If a positive integer k' divides $k, \alpha_1, \dots, \alpha_4$, then $g^{k/k'}$ fixes $x_1, \dots, x_4$, so $k' = 1$. Thus

$$\gcd(k, \alpha_1, \alpha_2, \alpha_3, \alpha_4) = 1. \tag{F.4}$$

Since $f - f_0 \in (x, y, z, s)^J$ with $J > n$, the monomials of f_0 occur in the G-semi-invariant function f, so they all have the character χ_f. Thus $\chi_x \chi_y = \chi_z^n$ in row A and $\chi_x^2 = \chi_y^2 \chi_z$ in row D.

- In row A, let $c \in [k+1, 2k]$ and $w_1 \in [1, k]$ be the integers with $c \equiv \alpha_3$ and $w_1 \equiv \alpha_1 \bmod k$, and let
$$w = (w_1, w_2, w_3, w_4) := \Big(w_1, nc - w_1, c, \frac{k}{R}\Big).$$
Then $w_2 \equiv n\alpha_3 - \alpha_1 \equiv \alpha_2 \bmod k$.
- In row D, let $c \in [k+1, 2k]$ and $w_2 \in [1, k]$ be the integers with $c \equiv \alpha_1 - \alpha_2$ and $w_2 \equiv \alpha_2 \bmod k$, and let
$$w = (w_1, w_2, w_3, w_4) := \Big(w_2 + c, w_2, 2c, \frac{k}{R}\Big).$$
Then $w_1 \equiv \alpha_1$ and $w_3 \equiv 2\alpha_1 - 2\alpha_2 \equiv \alpha_3 \bmod k$.

In both rows $w \in (\mathbb{N}^+)^4$ and $w_j \equiv \alpha_j \bmod k$ for any j, so g multiplies x_j by ξ^{w_j}. Since $\gcd(w_1, \dots, w_4)$ divides $w_4 = k/R$, hence k, it divides $\alpha_1, \dots, \alpha_4$, so w is primitive by (F.4).

In row A, xy and z^n have w-weight nc. In row D, x^2 and $y^2 z$ have w-weight $2(w_2 + c)$, and since $n \geq 4$ and $c > k \geq w_2$,

$$\operatorname{wt}_w(z^{n-1}) = 2(n-1)c > 2(w_2 + c).$$

In rows A and D, h_0 has w-weight mc. Since $c \leq 2k \leq 2M$, and $w_2 \leq k \leq M$ in row D,

$$nc,\ mc \leq 2M(n+m) < J \quad \text{and, in row D,} \quad 2(w_2 + c) \leq 6M < J,$$

and any monomial of degree at least J has w-weight at least J. Thus

$$\operatorname{in}_w f = xy - z^n \text{ in row A}, \qquad \operatorname{in}_w f = x^2 + y^2 z \text{ in row D}, \qquad \operatorname{in}_w h = h_0, \quad \text{and} \quad \operatorname{wt}_w(h) = mc.$$

Let $\phi : \mathcal{X}' \to \mathcal{X}$ be the weighted blow-up with weights w on $x_1, \dots, x_4$. Since $\operatorname{in}_w f$ is irreducible, the intersection of $\phi_*^{-1} W$ with the exceptional divisor of ϕ is the prime divisor

$$F := (\operatorname{in}_w f = 0) \subset \operatorname{Proj} \mathbb{C}[x_1, \dots, x_4],$$

and F has center 0 on W. By [Liu26b, Lemma 4.12],

$$a(F, W, 0) = \sum_{j=1}^{4} w_j - \operatorname{wt}_w(f) = c + \frac{k}{R} \qquad \text{and} \qquad \operatorname{mult}_F h = \operatorname{wt}_w(h) = mc,$$

which is (F.3).

Since g multiplies x_j by ξ^{w_j} for any j, ϕ is g-equivariant, and g acts on the exceptional divisor $\operatorname{Proj}\mathbb{C}[x_1,\dots,x_4]$ of ϕ, graded by w, by multiplying any homogeneous element of degree l by ξ^l. Thus g fixes any quotient of two homogeneous elements of equal degree, so g acts as the identity on $\operatorname{Proj}\mathbb{C}[x_1,\dots,x_4]$. The residue field of mult_F is the function field of

$$(\operatorname{in}_w f = 0) \subset \operatorname{Proj}\mathbb{C}[x_1,\dots,x_4].$$

Thus $g(F) = F$ and g acts trivially on the residue field of mult_F, so $g \in I_G(F)$ and $k \mid e_F$.

Step 5. In this step we conclude the proof.

By (F.3) and $k \mid e_F$,

$$\frac{a(F, W, \frac{1}{m}\operatorname{div}(h))}{e_F} = \frac{c + k/R - c}{e_F} = \frac{k}{Re_F} \leq \frac{1}{R}.$$

By (F.2), equality holds, $e_F = k$, and $\operatorname{mld}((W \ni 0, \frac{1}{m}\operatorname{div}(h))/G) = 1/R$ is computed by F. Since $c \leq 2k$, (F.3) implies

$$\frac{a(F, W, 0)}{e_F} = \frac{c}{k} + \frac{1}{R} \leq 2 + \frac{1}{R} \qquad \text{and} \qquad \frac{\operatorname{mult}_F h}{me_F} = \frac{c}{k} \leq 2.$$

Together with **Step 2**, the proposition follows. □

Corollary F.1. *In Set-up 16.2, after replacing $x_1,\dots,x_4$ with G-eigen-coordinates that are linear combinations of $x_1,\dots,x_4$ independent of i, the conditions of Set-up 15.4 hold with $B_0 = a$. Assume that the minimal lc center of $(W_*, \frac{1}{m}\operatorname{div}(h_*))$ is a regular curve $C \subset \operatorname{Sing}(W_*)$, and let $R'' := \lceil Ma/\epsilon \rceil$ for the number ϵ of Theorem 16.1 with $B_0 = a$. Then the following hold for the computing lifts F of b_i.*

(1) $\min_F \operatorname{mult}_F \operatorname{Jac}_{W_i} \to +\infty$ *as* $i \to +\infty$.

(2) After passing to an infinite subsequence, there exist G-eigenparameters $(s_i, u_{i1}, u_{i2}, u_{i3})$ of the ambient smooth germs of the W_i at o_i and G-eigenparameters (s, u_1, u_2, u_3) of $\Bbbk[[x_1,\dots,x_4]]$ with $C = (u_1 = u_2 = u_3 = 0)$ satisfying the following.

(a) $s_i, u_{i1}, u_{i2}, u_{i3}$ have the characters of s, u_1, u_2, u_3, respectively.

(b) (f_*, h_*) written in (s, u_1, u_2, u_3) is a compatible generic limit of the tuples (f_i, h_i) written in $(s_i, u_{i1}, u_{i2}, u_{i3})$.

(c) $\operatorname{mult}_F s_i \leq R''$ *for any* F.

(d) $\min_F \min_{1\leq j\leq 3} \operatorname{mult}_F u_{ij} \to +\infty$ *as* $i \to +\infty$.

Proof. By Set-up 16.2(1), $(W_i \ni o_i, G_i, h_i, \tau_i)$ is a radial failure sequence. By Definition 4.1(1), for any i, $(W_i \ni o_i)$ is an isolated Gorenstein terminal, hence canonical, hypersurface threefold germ, and G_i is a cyclic, hence abelian, group acting faithfully on W_i. By Definition 4.1(3), $\tau_i \to 1$, $b_i = q_i(1) \leq a - \delta$, and $q_i(\tau_i) \geq a$.

The prepared limit (f_*, h_*) of Set-up 16.2(2) is given by Construction-Theorem 13.9, which, after passing to a subsequence, fixed $G = G_i$ and the linear action of G on $x_1,\dots,x_4$ by (ii) there, and the characters of the f_i and the h_i by (iii) there. Since G is cyclic, possibly replacing $x_1,\dots,x_4$ with G-eigen-coordinates that are linear combinations of $x_1,\dots,x_4$ independent of i, we may assume that G acts diagonally, and we write (f_*, h_*) in the new coordinates as well. Since moreover $0 < b_i \leq a - \delta < a$, Set-up 16.2(1)–(4) imply the conditions of Set-up 15.4 with $B_0 = a$. This is the first sentence of the corollary.

(1) By Lemma 4.2(1), b_i has a computing lift. Since $(W_i \ni o_i)$ is an isolated singularity, Jac_{W_i} is non-zero, so $\operatorname{mult}_F \operatorname{Jac}_{W_i} \in \mathbb{N}$ for any F, and

$$\kappa_i := \min_F \operatorname{mult}_F \operatorname{Jac}_{W_i},$$

the minimum over the computing lifts F of b_i, is defined. Suppose that (1) does not hold. Then there exist $N \in \mathbb{N}^+$ and an infinite subset $\mathcal{I} \subset \mathbb{N}^+$ such that $\kappa_i \leq N$ for any $i \in \mathcal{I}$. Since

$|G_i| \le M$, $b_i > 0$, and $(W_i, \frac{1}{m}\operatorname{div}(h_i))$ is klt near o_i by Set-up 16.2(1), Theorem 15.8(1) implies that, for $L := L(m, M, N)$ and any $i \in \mathcal{I}$,

$$a \le q_i(\tau_i) \le b_i + L(1-\tau_i) \le a - \delta + L(1-\tau_i).$$

Thus $\delta \le L(1-\tau_i)$ for any $i \in \mathcal{I}$, which contradicts $\tau_i \to 1$.

(2) By the first sentence of the corollary, the conditions of Set-up 15.4 hold with $B_0 = a$. Since C is regular and, as the minimal lc center of $(W_*, \frac{1}{m}\operatorname{div}(h_*))$, G-invariant (Lemma 13.5), there exist G-eigenparameters (s, u_1, u_2, u_3) of $\Bbbk[[x_1, \dots, x_4]]$ with $C = (u_1 = u_2 = u_3 = 0)$. By (1), $\min_F \operatorname{mult}_F \operatorname{Jac}_{W_i} \to +\infty$. Thus, by Theorem 16.1(2) with $B_0 = a$, after passing to a subsequence, the data of Theorem 16.1(1) with $k = k_i \to +\infty$ consist of G-eigenparameters $(s_i, u_{i1}, u_{i2}, u_{i3})$ with the characters of s, u_1, u_2, u_3, respectively, such that the tuples (f_i, h_i) written in $(s_i, u_{i1}, u_{i2}, u_{i3})$ have a compatible generic limit, which is (f_*, h_*) written in (s, u_1, u_2, u_3), and satisfy

$$\operatorname{mult}_F s_i \le \left\lceil \frac{Ma}{\epsilon} \right\rceil = R'' \quad \text{for any computing lift } F \text{ of } b_i, \qquad \min_F \min_{1\le j\le 3} \operatorname{mult}_F u_{ij} \to +\infty.$$

This is (2). □

We need the following lemmas for the proof of Theorem 16.5.

Lemma F.2. *Let $\mathbb{L} \supset \mathbb{C}$ be an algebraically closed field and $\overline{R}$ the complete local ring of a singular Du Val surface germ over $\mathbb{L}$. Then there exists a non-trivial finite subgroup $\Pi \subset SL_2(\mathbb{C})$ acting linearly on $\mathbb{L}[[p,q]]$ such that $\overline{R} \cong \mathbb{L}[[p,q]]^\Pi$, the quotient morphism is étale outside the closed point, and any finite morphism $\operatorname{Spec}\mathbb{L}[[p,q]] \to \operatorname{Spec}\overline{R}$ étale in codimension 1 is isomorphic over $\operatorname{Spec}\overline{R}$ to the quotient morphism.*

Proof. By [Kol13, Theorem 2.29(2) and Example 3.26] there exists a polynomial $f_\Pi \in \mathbb{Z}[u_1, u_2, u_3]$, one of the normal forms of the Du Val singularities listed there, such that $\overline{R} \cong \mathbb{L}[[u_1, u_2, u_3]]/(f_\Pi)$. The germ at 0 of $(f_\Pi = 0) \subset \mathbb{C}^3$ is a Du Val singularity, so by [KM98, Theorem 4.20 and Remark 4.21(1)] (see also [Bri68, Satz 2.10 and 2.11]) it is analytically isomorphic to $(\mathbb{C}^2/\Pi \ni 0)$ for a non-trivial finite subgroup $\Pi \subset SL_2(\mathbb{C})$, which acts on $\mathbb{C}[[p,q]]$ by linear substitutions. Passing to completions, there exist $\phi_1, \phi_2, \phi_3 \in \mathbb{C}[[p,q]]^\Pi$ such that the homomorphism sending u_j to ϕ_j induces $\mathbb{C}[[u_1, u_2, u_3]]/(f_\Pi) \cong \mathbb{C}[[p,q]]^\Pi$.

Let $\mathfrak{n}_\mathbb{L}$ and $\mathfrak{n}_\mathbb{C}$ be the maximal ideals of $\mathbb{L}[[p,q]]^\Pi$ and $\mathbb{C}[[p,q]]^\Pi$. Since the Reynolds operator $\frac{1}{|\Pi|}\sum_{\gamma\in\Pi}\gamma$ preserves the degree in p, q, the homogeneous parts of degree n of the elements of $\mathbb{L}[[p,q]]^\Pi$ form $\mathbb{L} \otimes_\mathbb{C} \mathbb{C}[p,q]^\Pi_n$ for any n, so

$$\mathfrak{n}_\mathbb{L}/\mathfrak{n}_\mathbb{L}^2 = \mathbb{L} \otimes_\mathbb{C} \mathfrak{n}_\mathbb{C}/\mathfrak{n}_\mathbb{C}^2.$$

The classes of ϕ_1, ϕ_2, ϕ_3 span $\mathfrak{n}_\mathbb{C}/\mathfrak{n}_\mathbb{C}^2$, so the homomorphism $\mathbb{L}[[u_1, u_2, u_3]] \to \mathbb{L}[[p,q]]^\Pi$ sending u_j to ϕ_j is surjective, with kernel containing f_Π. Since f_Π has an isolated singularity at 0, $\mathbb{L}[[u_1, u_2, u_3]]/(f_\Pi)$ is a 2-dimensional normal local domain, and $\mathbb{L}[[p,q]]^\Pi$ is also 2-dimensional. Hence

$$\mathbb{L}[[p,q]]^\Pi \cong \mathbb{L}[[u_1, u_2, u_3]]/(f_\Pi) \cong \overline{R}.$$

Any $\gamma \in \Pi\backslash\{1\}$ has eigenvectors $p_\gamma, q_\gamma \in \mathbb{C}p + \mathbb{C}q$ with eigenvalues $\lambda, \lambda^{-1} \ne 1$. If a prime ideal $\mathfrak{q}$ of $\mathbb{L}[[p,q]]$ contains $\gamma(b) - b$ for any $b \in \mathbb{L}[[p,q]]$, then $(\lambda-1)p_\gamma, (\lambda^{-1}-1)q_\gamma \in \mathfrak{q}$, so $\mathfrak{q}$ is the maximal ideal. Thus the inertia groups of Π at the non-closed points of $\operatorname{Spec}\mathbb{L}[[p,q]]$ are trivial, and the quotient morphism $\operatorname{Spec}\mathbb{L}[[p,q]] \to \operatorname{Spec}\overline{R}$ is étale outside the closed point [Gro71, Exposé V, Corollaire 2.3].

Let U be the punctured spectrum of $\overline{R}$, which is regular since the quotient morphism is étale over U. By [Gro71, Exposé X, Théorème 3.1], a finite morphism $\pi_1 : \operatorname{Spec}\mathbb{L}[[p,q]] \to \operatorname{Spec}\overline{R}$ étale in codimension 1 is étale over U. The punctured spectrum of $\mathbb{L}[[p,q]]$ has no non-trivial connected finite étale cover: by [Gro71, Exposé X, Théorème 3.1], the normalization of $\mathbb{L}[[p,q]]$ in such a cover is finite étale over $\mathbb{L}[[p,q]]$, hence trivial as $\mathbb{L}$ is algebraically closed. Thus π_1 and the quotient morphism restrict to universal covers of U, which are isomorphic over U, and taking normalizations, π_1 is isomorphic over $\operatorname{Spec}\overline{R}$ to the quotient morphism. □

For a group G acting on a ring $\mathcal{R} \supset \Bbbk$ by ring automorphisms fixing $\Bbbk$, or on a $\Bbbk$-vector space $\mathcal{R}$ by $\Bbbk$-linear maps, and a character $\chi : G \to \Bbbk^*$, an element $y \in \mathcal{R}$ is *G-semi-invariant with character χ* if $g(y) = \chi(g)y$ for any $g \in G$.

Lemma F.3. *Let $\Bbbk \supset \mathbb{C}$ be an algebraically closed field and Γ a finite abelian group acting on $\Bbbk((s))$ by $\Bbbk$-automorphisms with $g(s) \in \Bbbk^* s$ for $g \in \Gamma$. Then the following hold.*

(1) If Γ acts semilinearly on a $\Bbbk((s))$-vector space $\mathcal{V}$ of dimension 1, then $\mathcal{V} = \Bbbk((s))v$ for a vector v with $g(v) = \chi(g)v$ for any $g \in \Gamma$, for a character $\chi : \Gamma \to \Bbbk^*$.

(2) If Γ acts semilinearly on a $\Bbbk((s))$-vector space $\mathcal{V}$ of dimension 2, then any Γ-stable line of $\mathcal{V}$ has a Γ-stable complement.

(3) Let Γ act on a ring $\mathcal{R} \supset \Bbbk$ by ring automorphisms fixing $\Bbbk$, let $\mathfrak{a} \subset \mathcal{R}$ be a Γ-stable $\Bbbk$-subspace, and let $l : \mathfrak{a} \to \mathcal{V}'$ be a Γ-equivariant $\Bbbk$-linear map to a $\Bbbk$-vector space $\mathcal{V}'$ on which Γ acts by $\Bbbk$-linear maps. If $l(y_0)$ is Γ-semi-invariant with character χ for some $y_0 \in \mathfrak{a}$, then

$$y := \frac{1}{|\Gamma|} \sum_{g \in \Gamma} \chi(g)^{-1} g(y_0) \in \mathfrak{a}$$

is Γ-semi-invariant with character χ and $l(y) = l(y_0)$.

Proof. (1) Let Γ_0 be the kernel of the action on $\Bbbk((s))$. Then Γ_0 acts $\Bbbk((s))$-linearly on $\mathcal{V}$ by a character whose values are roots of unity of $\Bbbk((s))$, hence belong to $\Bbbk^*$. Since $\Bbbk^*$ is divisible, the character of Γ_0 extends to a character $\chi : \Gamma \to \Bbbk^*$. The twisted action $g \star v' := \chi(g)^{-1} g(v')$ is semilinear and trivial on Γ_0, so it is an action of the group Γ/Γ_0 of automorphisms of $\Bbbk((s))$. Let v_1 be a basis and $\lambda_g \in \Bbbk((s))^*$ with $g \star v_1 = \lambda_g v_1$. By Dedekind's lemma on distinct field automorphisms, the map $c \to \sum_{g \in \Gamma/\Gamma_0} \lambda_g g(c)$ is not identically zero on $\Bbbk((s))$. For c with non-zero value, $v := \sum_{g \in \Gamma/\Gamma_0} g \star (cv_1)$ is non-zero and $g \star v = v$, that is, $g(v) = \chi(g)v$.

(2) Let ℓ be a Γ-stable line and ϱ a $\Bbbk((s))$-linear projection of $\mathcal{V}$ onto ℓ. For $g \in \Gamma$, the map $g \circ \varrho \circ g^{-1}$ is $\Bbbk((s))$-linear, maps $\mathcal{V}$ into ℓ and is the identity on ℓ. Thus

$$\varrho' := \frac{1}{|\Gamma|} \sum_{g \in \Gamma} g \circ \varrho \circ g^{-1}$$

is a Γ-equivariant projection onto ℓ, and $\operatorname{Ker} \varrho'$ is a Γ-stable complement.

(3) The element y belongs to $\mathfrak{a}$ since $\mathfrak{a}$ is a Γ-stable $\Bbbk$-subspace, $g(y) = \chi(g)y$ since χ is a character and Γ acts trivially on $\Bbbk$, and, l being Γ-equivariant and $\Bbbk$-linear,

$$l(y) = \frac{1}{|\Gamma|} \sum_{g \in \Gamma} \chi(g)^{-1} g(l(y_0)) = \frac{1}{|\Gamma|} \sum_{g \in \Gamma} \chi(g)^{-1} \chi(g) l(y_0) = l(y_0).$$

□

Lemma F.4. *Let G be a finite abelian group, m a positive integer, and $\Bbbk \supset \mathbb{C}$ an algebraically closed field. For any i let $(\mathcal{X}_i \ni o_i)$ be a smooth fourfold germ with an action of G fixing o_i and with G-eigenparameters $(t_i, u_{i1}, u_{i2}, u_{i3})$ whose characters do not depend on i, let f_i, h_i be G-semi-invariant regular functions on $\mathcal{X}_i$ whose characters do not depend on i, let $V_i := (f_i = 0)$, and let $\mathcal{F}_i$ be a set of prime divisors over V_i with center o_i. Let $f_*, h_* \in \Bbbk[[t, u_1, u_2, u_3]]$, and let G act on $\Bbbk((t))[[u_1, u_2, u_3]]$ through the characters of $t_i, u_{i1}, u_{i2}, u_{i3}$. Let Ψ be an automorphism of $\Bbbk((t))[[u_1, u_2, u_3]]$ over $\Bbbk((t))$ such that $x := \Psi(u_1)$, $y := \Psi(u_2)$, and $z := \Psi(u_3)$ are G-semi-invariant. Let c_Ψ be the t-order of the determinant of the linear part of Ψ. Let $w \in (\mathbb{N}^+)^3$, let $f_0, h_0 \in \mathbb{C}[x, y, z]$ be w-homogeneous polynomials, and let $\gamma := w_1 + w_2 + w_3 - \operatorname{wt}_w(f_0)$. Let $Q \in \Bbbk((t))[u_1, u_2, u_3]$ be a G-semi-invariant polynomial, let $\sigma_f, \sigma_h \in \Bbbk((t))^*$, and let c_f, c_h be the integers with $\sigma_f \in t^{c_f}\Bbbk[[t]]^*$ and $\sigma_h \in t^{c_h}\Bbbk[[t]]^*$. Assume that the following hold.*

(i) For any i, V_i is a normal threefold germ and $h_i|_{V_i} \neq 0$.

(ii) (f_*, h_*) is a compatible generic limit, with coefficient field $\Bbbk$, of the tuples (f_i, h_i) written in $(t_i, u_{i1}, u_{i2}, u_{i3})$.

(iii) $\operatorname{mult}_F t_i$ *is bounded independently of* i *and* $F \in \mathcal{F}_i$, *and*

$$\min_{F\in\mathcal{F}_i} \min_{1\le j\le 3} \operatorname{mult}_F u_{ij} \to +\infty.$$

(iv) $(f_0 = 0) \subset \mathbb{A}^3$ *is a normal surface,* $\gamma > 0$, *and* $\operatorname{wt}_w(h_0) = m\gamma$.

(v) Qf_* *has the character of* h_*, *and, in* $\Bbbk((t))[[x, y, z]]$ *with the weights* w *on* x, y, z,

$$\operatorname{in}_w f_* = \sigma_f f_0, \qquad \operatorname{in}_w(h_* - Qf_*) = \sigma_h h_0.$$

Then for any positive integer J *there exists a positive integer* N *such that* $\alpha := N\gamma - c_\Psi - c_f \ge 0$ *and* $\mu := Nm\gamma + c_h \ge 0$, *so that* $\frac{\mu}{m} - \alpha = \frac{c_h}{m} + c_\Psi + c_f$, *and, after passing to an infinite subsequence, there exist* G*-equivariant birational morphisms*

$$\pi_i : (\mathcal{X}_i' \ni o_i') \to (\mathcal{X}_i \ni o_i)$$

from charts of proper birational models of $\mathcal{X}_i$, *smooth at* o_i' *with* G*-eigenparameters* (t_i, x_i, y_i, z_i), *with the following properties, where* $V_i' := (\pi_i)_*^{-1} V_i$, *the induced morphism* $V_i' \to V_i$ *is again written* π_i, *and* $S_i := (t_i = 0) \subset V_i'$.

(1) $V_i' = (f_i' = 0)$ *near* o_i' *for the* G*-semi-invariant function* $f_i' := f_0(x_i, y_i, z_i) + t_i^J \xi_i$, V_i' *is normal near* o_i', *and*

$$\pi_i^* \operatorname{div}(h_i|_{V_i}) = \operatorname{div}(h_i'|_{V_i'}) + \mu S_i$$

for the G*-semi-invariant function* $h_i' := h_0(x_i, y_i, z_i) + t_i^J \eta_i$, *where* ξ_i, η_i *are regular functions on* $\mathcal{X}_i'$.

(2) $K_{V_i'} = \pi_i^* K_{V_i} + \alpha S_i$.

(3) Any $F \in \mathcal{F}_i$ *has center* o_i' *on* V_i', *and any prime divisor over* V_i' *with center* o_i' *has center* o_i *on* V_i.

Proof. We write u, v, and v' for the columns with entries u_1, u_2, u_3, resp. x, y, z, resp. v_1', v_2', v_3', so that $v = \Psi(u)$, and we let $d := \max\{\operatorname{wt}_w(f_0), m\gamma\}$. Let $\mathfrak{n} := (u_1, u_2, u_3) = (x, y, z)$ be the maximal ideal of $\Bbbk((t))[[u_1, u_2, u_3]]$.

Step 1. In this step we fix the constants. Let k_0 be a positive integer with $k_0 \min_j w_j > d$, and let Ψ^0 be the automorphism of $\Bbbk((t))[[u_1, u_2, u_3]]$ over $\Bbbk((t))$ such that $\Psi^0(u_j)$ is the sum of the terms of degree at most k_0 in u of $\Psi(u_j)$. The action of G is diagonal in u, so $\Psi^0(u_j)$ is G-semi-invariant with the character of $\Psi(u_j)$. Since $\Psi^0(u_j) - \Psi(u_j) \in \mathfrak{n}^{k_0+1}$ and any monomial of degree $k_0 + 1$ in x, y, z has w-weight larger than d, the lowest w-weight parts $\operatorname{in}_w f_*$ and $\operatorname{in}_w(h_* - Qf_*)$ and the linear part do not change when x, y, z are replaced with $\Psi^0(u_1), \Psi^0(u_2), \Psi^0(u_3)$. Possibly replacing Ψ with Ψ^0, we may assume that the entries of $\Psi(u)$ are polynomials of degree at most k_0 in u.

Let $L \in GL_3(\Bbbk((t)))$ be the linear part of Ψ, and write $\Psi(u) = Lu + \Psi_{\ge 2}(u)$, where $\Psi_{\ge 2}(u)$ has only terms of degree at least 2 in u. Let $c_0 \ge 0$ be an integer with $t^{c_0} Q \in \Bbbk[[t]][u_1, u_2, u_3]$, and $c_1 \ge 0$ an integer such that the coefficients of $\Psi(u)$ and of $L^{-1}\Psi_{\ge 2}(u)$ and the entries of L^{-1} have t-order at least $-c_1/2$. The inverse automorphism satisfies

$$\Psi^{-1}(v) = L^{-1}v - L^{-1}\Psi_{\ge 2}\big(\Psi^{-1}(v)\big).$$

Solving it recursively, the part of degree k in v of $\Psi^{-1}(v)$ is a sum of products indexed by rooted trees with k leaves in which any vertex that is not a leaf has at least two children. A leaf contributes an entry of L^{-1} and any other vertex a coefficient of $L^{-1}\Psi_{\ge 2}$. Such a tree has at most $k - 1$ vertices that are not leaves, so the coefficients of degree k of $\Psi^{-1}(v)$ have t-order at least $-c_1 k$. Since $f_*, h_* \in \Bbbk[[t, u_1, u_2, u_3]]$, $t^{c_0} Q \in \Bbbk[[t]][u_1, u_2, u_3]$, and a monomial in x, y, z of w-weight j has degree at most j, substituting $u = \Psi^{-1}(v)$ implies that, for $c_2 := c_0 + c_1$ and any $j \ge 1$, any coefficient a_j of a monomial of w-weight j in the expansion of f_* or of $h_* - Qf_*$ in $\Bbbk((t))[[x, y, z]]$ satisfies

$$a_j \in t^{-c_2 j}\Bbbk[[t]]. \tag{F.5}$$

Let N_1 be a positive integer divisible by the exponent $\exp(G)$ of G with $N_1 \geq c_1 + 1$, and N a positive integer divisible by $\exp(G)$ with

$$N \geq J + N_1 + c_1 + c_2(d+1) + |c_f| + |c_h| + |c_\Psi|. \tag{F.6}$$

Let

$$t^{Nw} := \operatorname{diag}(t^{Nw_1}, t^{Nw_2}, t^{Nw_3}).$$

Since $\gamma \geq 1$, we have $\alpha \geq N - |c_\Psi| - |c_f| \geq 0$ and $\mu \geq N - |c_h| \geq 0$. Let

$$\Phi(t,U) := t^{-N_1} L^{-1} \Psi_{\geq 2}(t^{N_1} U),$$
$$\Lambda(t) := L^{-1} \operatorname{diag}(t^{Nw_1 - N_1}, t^{Nw_2 - N_1}, t^{Nw_3 - N_1}),$$

where U is the column with entries U_1, U_2, U_3. A term of degree $k \geq 2$ in U of Φ has a coefficient of t-order at least $N_1(k-1) - c_1/2 \geq 1$, and the entries in the j-th column of Λ have t-order at least $Nw_j - N_1 - c_1/2 \geq 1$. Thus $\Phi \in t\Bbbk[[t]][U_1, U_2, U_3]$ has no terms of degree less than 2 in U, and $\Lambda \in t \operatorname{Mat}_3(\Bbbk[[t]])$.

Comparing the lowest w-weight parts of $g(f_*)$ and f_* for $g \in G$ shows that σ_f is G-semi-invariant. Thus the t-exponents of σ_f are congruent to c_f modulo the order of the character of t, and $\sigma_f t^{-c_f} \in \Bbbk[[t]]^*$ is G-invariant. Since $h_* - Qf_*$ is G-semi-invariant, $\sigma_h t^{-c_h} \in \Bbbk[[t]]^*$ is G-invariant as well.

Step 2. In this step we compute on a formal chart. Since $\Phi \equiv 0$ and $\Lambda \equiv 0$ modulo t and Φ has no terms of degree less than 2 in U, the equation

$$U + \Phi(t,U) = \Lambda(t) v' \tag{F.7}$$

has a unique solution $U = U_*(t,v') \in t\Bbbk[[t, v_1', v_2', v_3']]^3$, obtained by recursion on the degree in t. Multiplying (F.7) by $t^{N_1} L$, we obtain $\Psi(t^{N_1} U_*) = t^{Nw} v'$, that is, $t^{N_1} U_* = \Psi^{-1}(t^{Nw} v')$. Thus, for $\phi \in \Bbbk((t))[[u_1, u_2, u_3]]$, the series $\phi(t, t^{N_1} U_*)$ is the result of substituting $v = t^{Nw} v'$ into the expansion of ϕ in $\Bbbk((t))[[x,y,z]]$, and a monomial of w-weight j acquires the factor t^{Nj}. Let

$$\widehat{h}_* := t^{c_0}(h_* - Qf_*), \qquad \kappa_f := N \operatorname{wt}_w(f_0) + c_f, \quad \text{and} \quad \kappa_h := \mu + c_0.$$

Let (d_0, c) be $(\operatorname{wt}_w(f_0), c_f)$ or $(m\gamma, c_h)$. By hypothesis the monomials of w-weight less than d_0 do not occur, and by (F.5) and (F.6), for $j \geq d_0 + 1$ the coefficients of w-weight j acquire, after division by $t^{Nd_0 + c}$, the t-order at least

$$Nj - c_2 j - Nd_0 - c \geq (N - c_2)(j - d_0) - c_2 d_0 - c \geq N - c_2(d_0 + 1) - c \geq J.$$

For $\widehat{h}_*$ the factor t^{c_0} cancels against t^{c_0} in κ_h. Therefore

$$\begin{aligned} f_*(t, t^{N_1} U_*) &\in t^{\kappa_f}\big(\sigma_f t^{-c_f} f_0(v') + t^J \Bbbk[[t, v']]\big), \\ \widehat{h}_*(t, t^{N_1} U_*) &\in t^{\kappa_h}\big(\sigma_h t^{-c_h} h_0(v') + t^J \Bbbk[[t, v']]\big). \end{aligned} \tag{F.8}$$

Step 3. In this step we specialize the coordinate change to the (f_i, h_i). Let $N_2 := J + \max\{\kappa_f, \kappa_h\}$ and $N_3 := N_2 + c_1$. Let $\Psi^{<N_3}(u)$, $Q^{<N_3}$, ϵ_f, and ϵ_h be the truncations of $\Psi(u)$, $t^{c_0} Q$, $\sigma_f t^{-c_f}$, and $\sigma_h t^{-c_h}$, that is, the series given by deleting from any coefficient in $\Bbbk((t))$ the terms of t-order at least N_3. The entries of $\Psi^{<N_3}(u)$ and $Q^{<N_3}$ are G-semi-invariant with the characters of the entries of $\Psi(u)$ and of $t^{c_0} Q$, and $\epsilon_f, \epsilon_h \in \Bbbk[t]$ are G-invariant with non-zero constant terms. Since $\operatorname{wt}_w(f_0) \geq 1$, (F.6) gives $N_3 \geq \kappa_f + c_1 \geq N - |c_f| + c_1 > c_\Psi + c_1$, so the linear part $L^{<N_3}$ of $\Psi^{<N_3}$ satisfies $\det L^{<N_3} = t^{c_\Psi} \epsilon_\Psi$ for a polynomial $\epsilon_\Psi \in \Bbbk[t]$ with $\epsilon_\Psi(0) \neq 0$. Since $L^{-1}(L - L^{<N_3})$ has entries of t-order at least $N_3 - c_1/2 \geq 1$, $(L^{<N_3})^{-1} = (I - L^{-1}(L - L^{<N_3}))^{-1} L^{-1}$ has entries of t-order at least $-c_1/2$, so the entries of $(L^{<N_3})^{-1} - L^{-1} = (L^{<N_3})^{-1}(L - L^{<N_3})L^{-1}$ have t-order at least $N_3 - c_1$. Let $\Phi^{<N_3}$, $\Lambda^{<N_3}$, and $U_*^{<N_3}$ be defined from $\Psi^{<N_3}$ as Φ, Λ, and U_* are defined from Ψ. The coefficients of $\Phi^{<N_3} - \Phi$ and $\Lambda^{<N_3} - \Lambda$ have t-order at least $N_3 - c_1$. Subtracting the equations (F.7) for $\Psi^{<N_3}$ and for Ψ,

$$(U_*^{<N_3} - U_*) + \big(\Phi^{<N_3}(t, U_*^{<N_3}) - \Phi^{<N_3}(t, U_*)\big) = (\Lambda^{<N_3} - \Lambda)v' - (\Phi^{<N_3} - \Phi)(t, U_*),$$

and the bracket on the left is $(U_*^{<N_3} - U_*)$ multiplied by a matrix congruent to 0 modulo t. Thus $U_*^{<N_3} - U_* \in t^{N_3-c_1}\Bbbk[[t, v']]^3$. Since $f_*, t^{c_0}h_* \in \Bbbk[[t, u_1, u_2, u_3]]$ and $N_3 - c_1 = N_2$, (F.8) implies

$$\begin{aligned} f_*(t, t^{N_1}U_*^{<N_3}) &\equiv t^{\kappa_f}\epsilon_f f_0(v') \pmod{t^{\kappa_f+J}}, \\ \bigl(t^{c_0}h_* - Q^{<N_3}f_*\bigr)(t, t^{N_1}U_*^{<N_3}) &\equiv t^{\kappa_h}\epsilon_h h_0(v') \pmod{t^{\kappa_h+J}}. \end{aligned} \tag{F.9}$$

On the formal chart t and $u = t^{N_1}U_*^{<N_3}$ belong to (t), so a monomial of degree k in t, u_1, u_2, u_3 belongs to (t^k). Hence both sides of (F.9) modulo t^{N_2} depend only on the parts of degree less than N_2 of f_* and h_*, and on the finitely many coefficients in $\Bbbk$ of $\Psi^{<N_3}$, $Q^{<N_3}$, ϵ_f, and ϵ_h. The solution $U_*^{<N_3}$ modulo t^{N_2} is computed by N_2 substitutions in (F.7), and the inverse of $L^{<N_3}$ involves $\epsilon_\Psi(0)^{-1}$. Thus (F.9) is equivalent to finitely many polynomial identities in the coefficients of $\Psi^{<N_3}$, $Q^{<N_3}$, ϵ_f, ϵ_h, the coefficients of the N_2-jet of (f_*, h_*), and $\epsilon_\Psi(0)^{-1}$, together with the conditions $\epsilon_\Psi(0) \neq 0$, $\epsilon_f(0) \neq 0$, and $\epsilon_h(0) \neq 0$, and the finitely many identities saying that $\det L^{<N_3}$ has no terms of t-order less than c_Ψ and that $\Phi^{<N_3}$ and $\Lambda^{<N_3}$ have no terms of t-order at most 0.

By Lemma 2.6(4) for the non-zero coefficients in $\Bbbk$ of $\Psi^{<N_3}$, $Q^{<N_3}$, ϵ_f, ϵ_h and for $\epsilon_f(0)^{-1}$, $\epsilon_h(0)^{-1}$, $\epsilon_\Psi(0)^{-1}$, regarded as constant power series, after passing to an infinite subsequence the numbers above have specializations in $\mathbb{C}$, namely the constant terms of the corresponding members, for which the finitely many identities above hold. The conditions $\epsilon_\Psi(0) \neq 0$, $\epsilon_f(0) \neq 0$, $\epsilon_h(0) \neq 0$ become the identities

$$\epsilon_\Psi(0)\epsilon_\Psi(0)^{-1} = \epsilon_f(0)\epsilon_f(0)^{-1} = \epsilon_h(0)\epsilon_h(0)^{-1} = 1.$$

Let Ψ_i, L_i, Q_i, $\epsilon_{f,i}$, $\epsilon_{h,i}$, Φ_i, Λ_i, and $\epsilon_{\Psi,i}$ be the specializations, written in $t_i, u_{i1}, u_{i2}, u_{i3}$. A specialization creates no new monomial, so $\Psi_i, Q_i, \Phi_i, \Lambda_i$ are G-semi-invariant with the characters of $\Psi, t^{c_0}Q, \Phi, \Lambda$, and $\epsilon_{f,i}, \epsilon_{h,i} \in \mathbb{C}[t_i]$ are G-invariant with non-zero constant terms.

Step 4. In this step we construct the charts and prove (1)–(3). Let u_i be the column with entries u_{i1}, u_{i2}, u_{i3}, and let $\mathcal{X}_i'$ be a neighborhood of $o_i' := (o_i, 0, 0)$ in the closed subscheme of $\mathcal{X}_i \times \mathbb{A}^3_U \times \mathbb{A}^3_{v'}$ defined by

$$u_i = t_i^{N_1}U, \qquad U + \Phi_i(t_i, U) = \Lambda_i(t_i)v'. \tag{F.10}$$

The coefficients of Φ_i and Λ_i are regular near $t_i = 0$ since $\epsilon_{\Psi,i}(0) \neq 0$. Since $\Phi_i \equiv 0$ and $\Lambda_i \equiv 0$ modulo t_i and Φ_i has no terms of degree less than 2 in U, the Jacobian matrix of the six equations (F.10) with respect to (u_i, U) is the identity at o_i'. Thus $\mathcal{X}_i'$ is smooth of dimension 4 at o_i' with regular parameters

$$(t_i, x_i, y_i, z_i) := (t_i, v_1', v_2', v_3'),$$

and the completion of the local ring $\mathcal{O}_i := \mathcal{O}_{\mathcal{X}_i', o_i'}$ is $\mathbb{C}[[t_i, v']]$, in which U is the unique solution in $t_i\mathbb{C}[[t_i, v']]^3$ of the second equation of (F.10). Let $\pi_i : \mathcal{X}_i' \to \mathcal{X}_i$ be the projection. Multiplying the second equation by $t_i^{N_1}L_i$, we see that over $t_i \neq 0$ the equations (F.10) are equivalent to

$$U = t_i^{-N_1}u_i, \qquad v' = t_i^{-Nw}\Psi_i(t_i, u_i).$$

Hence π_i is birational, and near o_i', where $\mathcal{X}_i'$ is irreducible, $\mathcal{X}_i'$ is an open subset of the closure of the graph of the rational map $(U, v') : \mathcal{X}_i \dashrightarrow (\mathbb{P}^1)^6$, which is proper over $\mathcal{X}_i$. Let G act on U by the characters of u_i and on v' by the characters of x, y, z. Since N_1 and N are divisible by $\exp(G)$, the equations (F.10) are G-equivariant. Thus π_i is G-equivariant and (t_i, x_i, y_i, z_i) are G-eigenparameters. Moreover

$$t_i,\ u_{i1},\ u_{i2},\ u_{i3} \in t_i\mathcal{O}_i. \tag{F.11}$$

We prove (1). By (F.11), π_i^* maps the ideal $(t_i, u_{i1}, u_{i2}, u_{i3})^{N_2}$ into $t_i^{N_2}\mathbb{C}[[t_i, v']]$, so $\pi_i^*f_i$ and $\pi_i^*h_i$ modulo $t_i^{N_2}$ depend only on the N_2-jets of f_i and h_i. By the identities of **Step 3** for the specializations, (F.9) specializes to

$$\pi_i^*f_i \in t_i^{\kappa_f}\bigl(\epsilon_{f,i}f_0(v') + t_i^J\mathbb{C}[[t_i, v']]\bigr), \qquad \pi_i^*\widehat{h}_i \in t_i^{\kappa_h}\bigl(\epsilon_{h,i}h_0(v') + t_i^J\mathbb{C}[[t_i, v']]\bigr),$$

where $\widehat{h}_i := t_i^{c_0} h_i - Q_i f_i$. Since $\mathcal{O}_i \to \mathbb{C}[[t_i, v']]$ is faithfully flat, $t_i^n \mathbb{C}[[t_i, v']] \cap \mathcal{O}_i = t_i^n \mathcal{O}_i$ for any $n \geq 0$. Thus

$$f_i' := \frac{\pi_i^* f_i}{t_i^{\kappa_f} \epsilon_{f,i}}, \qquad h_i' := \frac{\pi_i^* \widehat{h}_i}{t_i^{\kappa_h} \epsilon_{h,i}}$$

are G-semi-invariant regular functions near o_i', and

$$f_i' = f_0(x_i, y_i, z_i) + t_i^J \xi_i \quad \text{and} \quad h_i' = h_0(x_i, y_i, z_i) + t_i^J \eta_i$$

for some $\xi_i, \eta_i \in \mathcal{O}_i$.

Since $f_i' \equiv f_0(x_i, y_i, z_i)$ modulo t_i, the Cartier divisor $(t_i = 0)$ of the hypersurface $(f_i' = 0)$ is the surface $(f_0(x_i, y_i, z_i) = 0)$, which is normal near o_i' because its completion at o_i' is $\mathbb{C}[[x, y, z]]/(f_0)$ and $(f_0 = 0) \subset \mathbb{A}^3$ is normal. Hence $(f_i' = 0)$ is normal, in particular integral, near o_i' by Serre's criterion. It is not contained in $(t_i = 0)$ and $\pi_i^* f_i = t_i^{\kappa_f} \epsilon_{f,i} f_i'$, so $V_i' = (f_i' = 0)$ near o_i', and $\operatorname{div}(t_i|_{V_i'}) = S_i$ is the reduced surface $(f_0(x_i, y_i, z_i) = 0)$. Since $f_i|_{V_i} = 0$, on V_i' we have

$$t_i^{c_0} \pi_i^* h_i = \pi_i^* \widehat{h}_i = t_i^{\mu + c_0} \epsilon_{h,i} h_i',$$

hence $\pi_i^* h_i = t_i^{\mu} \epsilon_{h,i} h_i'$ and

$$\pi_i^* \operatorname{div}(h_i|_{V_i}) = \operatorname{div}(h_i'|_{V_i'}) + \mu S_i.$$

We prove (2). On $\mathcal{X}_i'$, the second equation of (F.10) multiplied by $t_i^{N_1} L_i$ is $\Psi_i(t_i, u_i) = t_i^{Nw} v'$. Differentiating it with t_i fixed, and using $\partial_{u_i} \Psi_i = L_i(I + \partial_U \Phi_i(t_i, U))$, we obtain

$$\pi_i^*(dt_i \wedge du_{i1} \wedge du_{i2} \wedge du_{i3}) = \frac{t_i^{N(w_1+w_2+w_3)}}{\det L_i \cdot \det(I + \partial_U \Phi_i)} dt_i \wedge dv_1' \wedge dv_2' \wedge dv_3'.$$

Since $\det L_i = t_i^{c_\Psi} \epsilon_{\Psi,i}$ and $\partial_U \Phi_i \equiv 0$ modulo t_i, the denominator is $t_i^{c_\Psi}$ times a unit near o_i'. Thus, near o_i',

$$K_{\mathcal{X}_i'} = \pi_i^* K_{\mathcal{X}_i} + (N(w_1 + w_2 + w_3) - c_\Psi) \operatorname{div}(t_i).$$

By adjunction, $K_{V_i} = (K_{\mathcal{X}_i} + V_i)|_{V_i}$ and $K_{V_i'} = (K_{\mathcal{X}_i'} + V_i')|_{V_i'}$, and $\pi_i^* V_i = V_i' + \kappa_f \operatorname{div}(t_i)$ near o_i'. Since $N(w_1 + w_2 + w_3) - c_\Psi - \kappa_f = N\gamma - c_\Psi - c_f = \alpha$,

$$K_{V_i'} = \pi_i^* K_{V_i} + \alpha S_i.$$

We prove (3). Let R_0 be an upper bound of $\operatorname{mult}_F t_i$ for $F \in \mathcal{F}_i$. The coefficients of Ψ_i are Laurent polynomials in t_i with exponents at least $-c_1/2$, and the entries of Ψ_i have no constant terms in u_i. Thus for $F \in \mathcal{F}_i$ and $j \in \{1, 2, 3\}$,

$$\operatorname{mult}_F v_j' \geq \min_{1 \leq j' \leq 3} \operatorname{mult}_F u_{ij'} - \left(\frac{c_1}{2} + N w_j\right) R_0, \qquad \operatorname{mult}_F U_j \geq \operatorname{mult}_F u_{ij} - N_1 R_0,$$

where v_j' and U_j are regarded as rational functions on V_i. By hypothesis the right-hand sides tend to $+\infty$ uniformly in $F \in \mathcal{F}_i$. Possibly passing to a further subsequence, $t_i, U_1, U_2, U_3, v_1', v_2', v_3'$ have positive values at any $F \in \mathcal{F}_i$. Then the center of F on the closure of the graph of (U, v') is o_i', so F has center o_i' on V_i'. Finally $\pi_i(o_i') = o_i$, so any prime divisor over V_i' with center o_i' has center o_i on V_i. □

Proof of Theorem 16.5. Suppose that a radial failure sequence $(W_i \ni o_i, G_i, h_i, \tau_i)$ with constants (m, a, δ) satisfies Set-up 16.2, that the minimal lc center of (W_*, B_*), where $B_* := \frac{1}{m} \operatorname{div}(h_*)$, is a regular curve $C \subset \operatorname{Sing}(W_*)$, and that (W_*, B_*) has exactly one coefficient-one branch along C (Definition 16.3). Let $B_i := \frac{1}{m} \operatorname{div}(h_i)$. For any i, let $(\mathcal{X}_i \ni o_i)$ be the smooth fourfold germ with $W_i = (f_i = 0) \subset \mathcal{X}_i$ (Definition 4.1(1) and Construction-Theorem 13.9).

Step 1. In this step we pass to the transverse germ of W_* along C. By Corollary F.1(2), after passing to an infinite subsequence, there exist G-eigenparameters $(s_i, u_{i1}, u_{i2}, u_{i3})$ of $(\mathcal{X}_i \ni o_i)$ and G-eigenparameters (s, u_1, u_2, u_3) of $\Bbbk[[x_1, \dots, x_4]]$ with $C = (u_1 = u_2 = u_3 = 0)$ such that

the tuples (f_i, h_i) written in $(s_i, u_{i1}, u_{i2}, u_{i3})$ have the compatible generic limit (f_*, h_*) written in (s, u_1, u_2, u_3), and such that the set $\mathcal{F}_i$ of computing lifts of b_i (Definition 2.4) satisfies

$$\text{(F.12)} \qquad \operatorname{mult}_F s_i \le R'' \ \text{ for any } F \in \mathcal{F}_i, \qquad \min_{F\in\mathcal{F}_i} \min_{1\le j\le 3} \operatorname{mult}_F u_{ij} \to +\infty.$$

By Corollary F.1(2)(a), the characters of $s_i, u_{i1}, u_{i2}, u_{i3}$ are those of s, u_1, u_2, u_3. Let χ_s be the character of s.

The group G acts on $\Bbbk[[s, u_1, u_2, u_3]]$ fixing $\Bbbk$, hence on $\Bbbk((s))$ by $g(s) = \chi_s(g)s$, and semilinearly on the ring

$$\mathcal{O}_\eta := \Bbbk((s))[[u_1, u_2, u_3]]/(f_*),$$

where η is the generic point of C, which is the completion of the local ring $\mathcal{O}_{W_*,C}$ of W_* at η, since $\Bbbk((s)) \subset \mathcal{O}_{W_*,C}$ maps isomorphically onto the residue field and u_1, u_2, u_3 generate the maximal ideal of $\mathcal{O}_{W_*,C}$. Let $\mathfrak{m}_\eta$ be the maximal ideal of $\mathcal{O}_\eta$ and $B_\eta := \frac{1}{m}\operatorname{div}(h_*)$ on $\operatorname{Spec}\mathcal{O}_\eta$. Since W_* is the spectrum of a quotient of a complete local ring, it is excellent. Thus $\mathcal{O}_{W_*,C} \to \mathcal{O}_\eta$ is a regular morphism, and $\mathcal{O}_\eta$ is a normal local domain of dimension 2. Since C is an lc center of the lc pair (W_*, B_*), some log resolution of (W_*, B_*) carries a prime divisor E_C with center C and $a(E_C, W_*, B_*) = 0$. The pullback of the log resolution along the regular morphism $\operatorname{Spec}\mathcal{O}_\eta \to \operatorname{Spec}\mathcal{O}_{W_*,C}$ is a log resolution of $(\operatorname{Spec}\mathcal{O}_\eta, B_\eta)$, any component E' of the pullback of a prime divisor E satisfies $a(E', \operatorname{Spec}\mathcal{O}_\eta, B_\eta) = a(E, W_*, B_*)$ (compatibility of log resolutions and discrepancies with completion [Kol13, 2.16]), and the components of the pullback of E_C have center the closed point. Thus $(\operatorname{Spec}\mathcal{O}_\eta, B_\eta)$ is lc, and some prime divisor E_η over $\operatorname{Spec}\mathcal{O}_\eta$ with center the closed point satisfies

$$\text{(F.13)} \qquad a(E_\eta, \operatorname{Spec}\mathcal{O}_\eta, B_\eta) = 0.$$

Step 2. In this step we show that the generic transverse surface of W_* along C is of type A_{r-1} and that the coefficient-one branch is smooth. Let $\overline{\Bbbk((s))}$ be an algebraic closure of $\Bbbk((s))$, and let T, D_T, and $\pi : T' \to T$ be as in Definition 16.3, so that $T = \operatorname{Spec}\overline{\Bbbk((s))}[[u_1, u_2, u_3]]/(f_*)$, a singular Du Val germ, and $D_T = \frac{1}{m}\operatorname{div}(h_*|_T)$. As in **Step 1**, (T, D_T) is lc (by the regular base change $T \to \operatorname{Spec}\mathcal{O}_\eta$ [Kaw21, Remark 2.6]). By Lemma F.2, $T' = \operatorname{Spec}\overline{\Bbbk((s))}[[p', q']]$ and π is the quotient by a finite subgroup of $SL_2(\overline{\Bbbk((s))})$ acting linearly on p', q', which we call the *deck group*. It is étale in codimension 1, so (T', π^*D_T) is lc. By hypothesis the coefficient-one part of π^*D_T is one prime divisor $\widetilde{\Sigma}'$.

Since $\pi^*D_T \ge \widetilde{\Sigma}'$, the pair $(T', \widetilde{\Sigma}')$ is lc. The exceptional divisor of the blow-up of the closed point has log discrepancy $2 - \operatorname{mult}_0 \widetilde{\Sigma}'$ with respect to $(T', \widetilde{\Sigma}')$, so $\operatorname{mult}_0 \widetilde{\Sigma}' \le 2$. Suppose that $\operatorname{mult}_0 \widetilde{\Sigma}' = 2$. Then $\widetilde{\Sigma}'$ is an irreducible plane curve germ of multiplicity 2, so $\widetilde{\Sigma}' = (q''^2 = p''^{2n_1+1})$ for regular parameters p'', q'' of T' and an integer $n_1 \ge 1$. The prime divisor E' of the weighted blow-up with weights $(2, 2n_1+1)$ on (p'', q'') satisfies

$$a(E', T', \widetilde{\Sigma}') = 2 + (2n_1+1) - 2(2n_1+1) = 1 - 2n_1 < 0,$$

a contradiction. Thus $\widetilde{\Sigma}'$ is smooth.

Since π^*D_T is invariant under the deck group and $\widetilde{\Sigma}'$ is the only component of π^*D_T with coefficient one, the deck group preserves $\widetilde{\Sigma}'$, hence the tangent line of $\widetilde{\Sigma}'$ at the closed point, and by complete reducibility a complementary line. Thus, for linear coordinates in a basis of the tangent line and of the complementary line, again written p', q', the deck group consists of the matrices $\operatorname{diag}(\zeta^j, \zeta^{-j})$, $0 \le j < r$, for a primitive r-th root of unity ζ, where $r \ge 2$ since T is singular. We identify the deck group with the group $\boldsymbol{\mu}_r \subset \Bbbk^*$ of r-th roots of unity, $\zeta^j \in \boldsymbol{\mu}_r$ corresponding to $\operatorname{diag}(\zeta^j, \zeta^{-j})$, that is, acting by $(p', q') \to (\zeta^j p', \zeta^{-j} q')$. Then $T = \operatorname{Spec}\overline{\Bbbk((s))}[[p'^r, q'^r, p'q']]$ is a germ of type A_{r-1}.

Step 3. In this step we descend the coefficient-one branch to a prime divisor Σ of $\operatorname{Spec}\mathcal{O}_\eta$ and prove that $r\Sigma$ is Cartier. Let $\mathfrak{n} := (u_1, u_2, u_3) \subset \Bbbk((s))[[u_1, u_2, u_3]]$. Since $C \subset \operatorname{Sing}(W_*)$, we have $f_* \in \mathfrak{n}^2$. By [GP19, Theorem 4.8], over any field $\mathbb{K} \supset \Bbbk((s))$, $u_1u_2 - u_3^r \in \Bbbk((s))[[u_1, u_2, u_3]]$

is contact k_1-determined for $k_1 := 2\tau - \mathrm{mult}_0(u_1u_2 - u_3^r) + 2 = 2r - 2$, where $\tau = r - 1$ is the Tjurina number of $u_1u_2 - u_3^r$. By **Step 2**,

$$\overline{\Bbbk((s))}[[u_1, u_2, u_3]]/(f_*) \cong \overline{\Bbbk((s))}[[u_1, u_2, u_3]]/(u_1u_2 - u_3^r).$$

Since $f_* \in \mathfrak{n}^2$, the isomorphism lifts [GLS07, Remark I.2.9.1(3)] to an automorphism Φ of $\overline{\Bbbk((s))}[[u_1, u_2, u_3]]$ over $\overline{\Bbbk((s))}$ with $\upsilon\Phi(f_*) = u_1u_2 - u_3^r$ for a unit υ. Let Φ_{k_1} be the automorphism of $\overline{\Bbbk((s))}[[u_1, u_2, u_3]]$ sending u_j to the part of $\Phi(u_j)$ of degree at most k_1, υ_{k_1} the part of υ of degree at most k_1, and $F' \subset \overline{\Bbbk((s))}$ a finite Galois extension of $\Bbbk((s))$ containing their coefficients. Then

$$\upsilon_{k_1}\Phi_{k_1}(f_*) - (u_1u_2 - u_3^r) \in \mathfrak{n}^{k_1+1}F'[[u_1, u_2, u_3]], \quad \text{so} \quad F'[[u_1, u_2, u_3]]/(f_*) \cong F'[[u_1, u_2, u_3]]/(u_1u_2 - u_3^r).$$

Let

$$\mathcal{O}_{\eta,F'} := F'[[u_1, u_2, u_3]]/(f_*) = \mathcal{O}_\eta \otimes_{\Bbbk((s))} F'.$$

Since $F'[[u_1, u_2, u_3]]/(u_1u_2 - u_3^r) \cong F'[[p'^r, q'^r, p'q']]$, and a finite morphism to T from $\operatorname{Spec}\overline{\Bbbk((s))}[[p, q]]$, étale in codimension 1, is unique up to isomorphism over T by Lemma F.2, we may choose the coordinates p', q' of the cover of **Step 2** so that

$$\mathcal{O}_{\eta,F'} = F'[[p', q']]^{\boldsymbol{\mu}_r} = F'[[p'^r, q'^r, p'q']].$$

The group $\mathrm{Gal}(\overline{\Bbbk((s))}/F')$ acts on $\overline{\Bbbk((s))}[[p', q']]$ through the coefficients, compatibly with π, and fixes $h_* \in \mathcal{O}_{\eta,F'}$. Thus it fixes $\widetilde{\Sigma}'$ and the tangent line of $\widetilde{\Sigma}'$. If $r \geq 3$, the tangent line of $\widetilde{\Sigma}'$ is one of the eigenlines $(p' = 0)$ and $(q' = 0)$ of $\boldsymbol{\mu}_r$. If $r = 2$, then $\boldsymbol{\mu}_r = \{\pm 1\}$ and the tangent line is defined over F'. For any $r \geq 2$, possibly exchanging p', q' (and ζ, ζ^{-1}) or changing p', q' by an F'-linear transformation, we may assume that $\widetilde{\Sigma}'$ is tangent to $(p' = 0)$. Then $\widetilde{\Sigma}' = (\varpi = 0)$ for $\varpi := p' - \vartheta(q')$ with a unique $\vartheta \in q'\overline{\Bbbk((s))}[[q']]$. By uniqueness ϑ is fixed by the Galois group, so $\vartheta \in F'[[q']]$. Since $\zeta(\varpi)$ vanishes on $\widetilde{\Sigma}'$, the element

$$\zeta(\varpi) - \zeta\varpi = \zeta\vartheta(q') - \vartheta(\zeta^{-1}q')$$

belongs to $(\varpi) \cap F'[[q']] = 0$. Thus ϖ is $\boldsymbol{\mu}_r$-semi-invariant with the character $\zeta \to \zeta$, $\varpi^r \in \mathcal{O}_{\eta,F'}$, and h_* has order m along $(\varpi = 0)$. Let $\Sigma_{F'} \subset \operatorname{Spec}\mathcal{O}_{\eta,F'}$ be the image of $(\varpi = 0)$. The cover is étale in codimension 1 and $(\varpi = 0)$ is $\boldsymbol{\mu}_r$-stable, so $\mathrm{div}(\varpi^r) = r\Sigma_{F'}$ on $\operatorname{Spec}\mathcal{O}_{\eta,F'}$.

The prime divisor $\Sigma_{F'}$ is the only prime divisor of $\operatorname{Spec}\mathcal{O}_{\eta,F'}$ along which h_* has order at least m. Suppose that $P_{F'} \neq \Sigma_{F'}$ is another such prime divisor. Then a component $(\psi_1 = 0) \neq (\varpi = 0)$ of the preimage of $P_{F'}$ satisfies $\psi_1^m \mid h_*$ in $F'[[p', q']]$. Since $F'[[p', q']]/(\psi_1, \varpi)$ has finite length, some prime factor of ψ_1 in $\overline{\Bbbk((s))}[[p', q']]$ defines a prime divisor other than $\widetilde{\Sigma}'$ along which h_* has order at least m, which contradicts the lc property of π^*D_T and the uniqueness of $\widetilde{\Sigma}'$. The group $\mathrm{Gal}(F'/\Bbbk((s)))$ acts on $\mathcal{O}_{\eta,F'} = \mathcal{O}_\eta \otimes_{\Bbbk((s))} F'$ through F' and fixes h_*, hence fixes $\Sigma_{F'}$. Since $\mathcal{O}_\eta \to \mathcal{O}_{\eta,F'}$ is finite étale with the Galois orbits as fibers, $\Sigma_{F'}$ is the preimage of a prime divisor Σ of $\operatorname{Spec}\mathcal{O}_\eta$. Thus Σ is the only prime divisor of $\operatorname{Spec}\mathcal{O}_\eta$ along which h_* has order at least m, the order of h_* along Σ is m, and Σ is G-stable since $g(h_*) \in \Bbbk^* h_*$ for any $g \in G$.

Let

$$\mathcal{M} := \mathcal{O}_{\operatorname{Spec}\mathcal{O}_\eta}(-r\Sigma) \subset \mathcal{O}_\eta$$

be the divisorial ideal of $r\Sigma$. Since $\mathcal{O}_\eta \to \mathcal{O}_{\eta,F'}$ is finite étale, $\mathcal{M} \otimes_{\mathcal{O}_\eta} \mathcal{O}_{\eta,F'}$ is the divisorial ideal of $r\Sigma_{F'}$, that is, $\varpi^r\mathcal{O}_{\eta,F'}$, which is free of rank 1 over $\mathcal{O}_{\eta,F'}$. As $\mathcal{O}_\eta$-modules,

$$\mathcal{M} \otimes_{\mathcal{O}_\eta} \mathcal{O}_{\eta,F'} = \mathcal{M} \otimes_{\Bbbk((s))} F' \cong \mathcal{M}^{\oplus[F':\Bbbk((s))]},$$

and $\mathcal{O}_{\eta,F'}$ is free of rank $[F' : \Bbbk((s))]$ over $\mathcal{O}_\eta$. Thus $\mathcal{M}$ is a direct summand of a free $\mathcal{O}_\eta$-module, hence flat, hence free, being finitely generated over the Noetherian local ring $\mathcal{O}_\eta$. The ideal $\mathcal{M}$ is G-stable and $\mathcal{M}/\mathfrak{m}_\eta\mathcal{M}$ is a semilinear G-line. By Lemma F.3(1), Lemma F.3(3) for $\mathfrak{a} = \mathcal{M}$ and the quotient map $\mathcal{M} \to \mathcal{M}/\mathfrak{m}_\eta\mathcal{M}$, and Nakayama's lemma, $\mathcal{M}$ has a G-semi-invariant generator x_0. Let χ_0 be the character of x_0. Thus

$$\mathrm{div}(x_0) = r\Sigma \ \text{ on } \ \operatorname{Spec}\mathcal{O}_\eta.$$

Step 4. In this step we construct a smooth cover $\operatorname{Spec}\widetilde{\mathcal{O}}_\eta \to \operatorname{Spec}\mathcal{O}_\eta$ defined over $\Bbbk((s))$ and the coordinates x, y, z of $\mathcal{O}_\eta$. Since $x_0\mathcal{O}_{\eta,F'} = \varpi^r\mathcal{O}_{\eta,F'}$, we have $x_0 = \epsilon\varpi^r$ for a unit ϵ of $\mathcal{O}_{\eta,F'}$. Possibly replacing F' with a larger finite Galois extension, we may assume that $\epsilon = \epsilon_1^r$ for a unit ϵ_1 of $\mathcal{O}_{\eta,F'}$, since the residue field of the complete local ring $\mathcal{O}_{\eta,F'}$ is F'. The field of fractions of $F'[[p',q']]$ has degree r over that of $\mathcal{O}_{\eta,F'}$, and the $\boldsymbol{\mu}_r$-conjugates $\zeta^j\epsilon_1\varpi$, $0 \le j < r$, of $\epsilon_1\varpi$ are distinct. Thus $\omega^r - x_0$, where ω is a variable, is the minimal polynomial of $\epsilon_1\varpi$. Let

$$\mathcal{O}'_\eta := \mathcal{O}_\eta[\omega]/(\omega^r - x_0) \quad \text{and} \quad \mathcal{O}'_{\eta,F'} := \mathcal{O}'_\eta \otimes_{\Bbbk((s))} F' = \mathcal{O}_{\eta,F'}[\omega]/(\omega^r - x_0).$$

Then $\omega \to \epsilon_1\varpi$ maps $\mathcal{O}'_{\eta,F'}$ isomorphically onto a subring of $F'[[p',q']]$ whose field of fractions is that of $F'[[p',q']]$. As $F'[[p',q']]$ is normal and finite over $\mathcal{O}_{\eta,F'}$, it is the normalization of $\mathcal{O}'_{\eta,F'}$. In particular $\omega^r - x_0$ is irreducible over the field of fractions of $\mathcal{O}_\eta$, so $\mathcal{O}'_\eta$ is a domain.

Let $\widetilde{\mathcal{O}}_\eta$ be the normalization of the domain $\mathcal{O}'_\eta$, and

$$\widetilde{G} := \{(g,\xi) \mid g \in G,\ \xi \in \Bbbk,\ \xi^r = \chi_0(g)\},$$

an abelian group of order $r|G|$ acting on $\mathcal{O}'_\eta$ by g on $\mathcal{O}_\eta$ and by $\omega \to \xi\omega$. The action is well defined since $g(x_0) = \xi^r x_0$, and it extends to $\widetilde{\mathcal{O}}_\eta$. The kernel of $\widetilde{G} \to G$ is $\{(1,\xi) \mid \xi \in \boldsymbol{\mu}_r\}$, which we write $\boldsymbol{\mu}_r$. Under the isomorphism $\omega \to \epsilon_1\varpi$, the element $(1,\zeta) \in \boldsymbol{\mu}_r$ acts on $F'[[p',q']]$ as the deck transformation ζ.

The ring $\widetilde{\mathcal{O}}_\eta \otimes_{\Bbbk((s))} F'$ is integral over $\mathcal{O}'_{\eta,F'}$ and contained in the field of fractions of $\mathcal{O}'_{\eta,F'}$, hence contained in the normalization $F'[[p',q']]$ of $\mathcal{O}'_{\eta,F'}$, a finite $\mathcal{O}_\eta$-module. As $\mathcal{O}_\eta$ is Noetherian, the submodule $\widetilde{\mathcal{O}}_\eta \otimes_{\Bbbk((s))} F'$ of $F'[[p',q']]$ is finite, and so is its direct summand $\widetilde{\mathcal{O}}_\eta$, as

$$\widetilde{\mathcal{O}}_\eta \otimes_{\Bbbk((s))} F' \cong \widetilde{\mathcal{O}}_\eta^{\oplus[F':\Bbbk((s))]}.$$

The homomorphisms

$$\widetilde{\mathcal{O}}_\eta \to \widetilde{\mathcal{O}}_\eta \otimes_{\Bbbk((s))} F' \to F'[[p',q']]$$

are injective and integral, the first one faithfully flat, and $F'[[p',q']]$ is local, so $\widetilde{\mathcal{O}}_\eta$ and $\widetilde{\mathcal{O}}_\eta \otimes_{\Bbbk((s))} F'$ are local. The first homomorphism is a flat local homomorphism of Noetherian local rings whose fibers are finite products of finite separable field extensions, and $\widetilde{\mathcal{O}}_\eta$ is normal, so $\widetilde{\mathcal{O}}_\eta \otimes_{\Bbbk((s))} F'$ satisfies (R_1) and (S_2) by [Mat89, Theorem 23.9], that is, it is normal. Thus

$$\widetilde{\mathcal{O}}_\eta \otimes_{\Bbbk((s))} F' = F'[[p',q']].$$

Let $\widetilde{\mathfrak{m}}$ and κ be the maximal ideal and the residue field of $\widetilde{\mathcal{O}}_\eta$. The ring

$$\kappa \otimes_{\Bbbk((s))} F' = F'[[p',q']]/\widetilde{\mathfrak{m}}F'[[p',q']]$$

is local, reduced, and Artinian, hence a field mapping onto the residue field F', so $\kappa = \Bbbk((s))$ and $\widetilde{\mathfrak{m}}F'[[p',q']] = (p',q')$. Thus

$$\dim_{\Bbbk((s))} \widetilde{\mathfrak{m}}/\widetilde{\mathfrak{m}}^2 = 2 = \dim \widetilde{\mathcal{O}}_\eta,$$

and $\widetilde{\mathcal{O}}_\eta$ is regular. Being finite over the complete local ring $\mathcal{O}_\eta$, the ring $\widetilde{\mathcal{O}}_\eta$ is complete, and $\Bbbk((s)) \subset \widetilde{\mathcal{O}}_\eta$ maps isomorphically onto κ. Thus $\widetilde{\mathcal{O}}_\eta$ is the power series ring over $\Bbbk((s))$ in any regular system of parameters. Since

$$\widetilde{\mathcal{O}}_\eta/(\omega) \otimes_{\Bbbk((s))} F' = F'[[p',q']]/(\epsilon_1\varpi)$$

is regular of dimension 1, $\omega \notin \widetilde{\mathfrak{m}}^2$. The field of fractions of $\widetilde{\mathcal{O}}_\eta$ is a cyclic extension of degree r of that of $\mathcal{O}_\eta$ with group $\boldsymbol{\mu}_r$, and $\mathcal{O}_\eta$ is normal. Hence $\widetilde{\mathcal{O}}_\eta^{\boldsymbol{\mu}_r} = \mathcal{O}_\eta$.

Let $p := \omega$. The element p is $\widetilde{G}$-semi-invariant with character $(g,\xi) \to \xi$. The cotangent space $\widetilde{\mathfrak{m}}/\widetilde{\mathfrak{m}}^2$ is a 2-dimensional semilinear $\widetilde{G}$-space in which the class of p spans a $\widetilde{G}$-stable line. By Lemma F.3(2)(1)(3), the last for $\mathfrak{a} = \widetilde{\mathfrak{m}}$ and the quotient map onto $\widetilde{\mathfrak{m}}/\widetilde{\mathfrak{m}}^2$, there exists a $\widetilde{G}$-semi-invariant $q \in \widetilde{\mathfrak{m}}$ whose class spans a $\widetilde{G}$-stable complement. Thus $\widetilde{\mathcal{O}}_\eta = \Bbbk((s))[[p,q]]$ with $\widetilde{G}$-semi-invariant p, q. After base change to F', the cotangent representation of $\boldsymbol{\mu}_r$ has

the characters $\zeta \to \zeta$ and $\zeta \to \zeta^{-1}$, and p has the character $\zeta \to \zeta$. Hence q has the character $\zeta \to \zeta^{-1}$. Therefore, as the $\boldsymbol{\mu}_r$-invariant monomials $p^{j_1}q^{j_2}$ are those with $j_1 \equiv j_2 \pmod r$,

$$\mathcal{O}_\eta = \Bbbk((s))[[p,q]]^{\boldsymbol{\mu}_r} = \Bbbk((s))[[p^r, q^r, pq]].$$

Let $x := p^r = x_0$, $y := q^r$, and $z := pq$, so that $xy = z^r$. They are G-semi-invariant, being $\widetilde{G}$-semi-invariant and $\boldsymbol{\mu}_r$-invariant. Let $\widetilde{\Sigma} := \operatorname{div}(p)$ on $\operatorname{Spec}\widetilde{\mathcal{O}}_\eta$. Since $\operatorname{div}(x) = r\Sigma$, $\widetilde{\Sigma}$ is the preimage of Σ with ramification index 1, and x is a unit outside Σ. Thus $\operatorname{Spec}\widetilde{\mathcal{O}}_\eta \to \operatorname{Spec}\mathcal{O}_\eta$ is étale in codimension 1.

Step 5. In this step we determine the cofactor of the mark along $\widetilde{\Sigma}$. Let h be the image of h_* in $\widetilde{\mathcal{O}}_\eta$ and $\widetilde{B}_\eta := \frac{1}{m}\operatorname{div}(h)$ on $\operatorname{Spec}\widetilde{\mathcal{O}}_\eta$. As the cover is étale in codimension 1 and $(\operatorname{Spec}\mathcal{O}_\eta, B_\eta)$ is lc, (F.13) implies that $(\operatorname{Spec}\widetilde{\mathcal{O}}_\eta, \widetilde{B}_\eta)$ is lc and that

$$a(\widetilde{E}_\eta, \operatorname{Spec}\widetilde{\mathcal{O}}_\eta, \widetilde{B}_\eta) = 0$$

for a prime divisor $\widetilde{E}_\eta$ over $\operatorname{Spec}\widetilde{\mathcal{O}}_\eta$ with center the closed point. The order of h along $\widetilde{\Sigma}$ is the order m of h_* along Σ. Thus

$$h = p^m\varphi \quad \text{and} \quad p \nmid \varphi.$$

Let ν be the integer with $\varphi(0,q) \in q^\nu\Bbbk((s))[[q]]^*$, which exists since $p \nmid \varphi$. Let E_ν be the prime divisor of the weighted blow-up with weights $(\nu+1, 1)$ on (p,q). Then $a(E_\nu, \operatorname{Spec}\widetilde{\mathcal{O}}_\eta, 0) = \nu + 2$ and $\operatorname{mult}_{E_\nu} h = m(\nu+1) + \nu$, so

$$a(E_\nu, \operatorname{Spec}\widetilde{\mathcal{O}}_\eta, \widetilde{B}_\eta) = 1 - \nu/m \geq 0,$$

that is, $\nu \leq m$.

Suppose that $\nu < m$, and let ρ be a rational number with $0 < \rho < 1 - \nu/m$. For an integer $k \geq \nu$ let $\varphi_{\leq k} \in \Bbbk((s))[p,q]$ be the sum of the terms of φ of degree at most k. Then $q^{-\nu}\varphi_{\leq k}(0,q)$ is a polynomial with non-zero constant term. Let

$$X_k := \operatorname{Spec}\Bbbk((s))[p,q]\backslash\{p = q^{-\nu}\varphi_{\leq k}(0,q) = 0\},$$

let $D := (p = 0) \subset X_k$, and let

$$\Delta_k := \frac{1}{m}\operatorname{div}(\varphi_{\leq k}) + \rho\operatorname{div}(q) \quad \text{on } X_k.$$

After base change to $\overline{\Bbbk((s))}$, D is smooth and not contained in $\operatorname{Supp}\Delta_k$, and

$$\Delta_k|_D = (\nu/m + \rho)\cdot(\text{origin}) \quad \text{with } \nu/m + \rho < 1.$$

The pair $(X_k, D + \Delta_k)$ is the base change of a pair $(X_{k,0}, D_0 + \Delta_{k,0})$ defined over a subfield $K_0 \subset \Bbbk((s))$ finitely generated over $\mathbb{Q}$ and containing the coefficients of $\varphi_{\leq k}$, and K_0 embeds into $\mathbb{C}$. The conditions above hold for the base change $(X_{k,\mathbb{C}}, D_\mathbb{C} + \Delta_{k,\mathbb{C}})$ of $(X_{k,0}, D_0 + \Delta_{k,0})$ to $\mathbb{C}$, so $(X_{k,\mathbb{C}}, D_\mathbb{C} + \Delta_{k,\mathbb{C}})$ is lc near $D_\mathbb{C}$ by [EMY03, Corollary 1.8]. By [dFEM11, Lemma 2.14 and Proposition 2.11(i)],

$$\Big(\operatorname{Spec}\Bbbk((s))[[p,q]],\ \operatorname{div}(p) + \frac{1}{m}\operatorname{div}(\varphi_{\leq k}) + \rho\operatorname{div}(q)\Big) \text{ is lc.}$$

Let E be a prime divisor over $\operatorname{Spec}\widetilde{\mathcal{O}}_\eta$ with center the closed point. Since $\varphi - \varphi_{\leq k} \in \widetilde{\mathfrak{m}}^{k+1}$ and $\operatorname{mult}_E\widetilde{\mathfrak{m}} \geq 1$, we have $\operatorname{mult}_E\varphi_{\leq k} = \operatorname{mult}_E\varphi$ for $k \geq \operatorname{mult}_E\varphi$. As

$$\widetilde{B}_\eta = \operatorname{div}(p) + \frac{1}{m}\operatorname{div}(\varphi),$$

taking such k we obtain

$$a\big(E, \operatorname{Spec}\widetilde{\mathcal{O}}_\eta, \widetilde{B}_\eta + \rho\operatorname{div}(q)\big) \geq 0,$$

so

$$a(E, \operatorname{Spec}\widetilde{\mathcal{O}}_\eta, \widetilde{B}_\eta) \geq \rho\operatorname{mult}_E q > 0.$$

This contradicts the choice of $\widetilde{E}_\eta$. Thus $\nu = m$.

Step 6. In this step we derive a contradiction from Lemma F.4 and Proposition 16.4. By Lemma F.3(3) for $\mathfrak{a} = \mathfrak{n}$ and the quotient map $\mathfrak{n} \to \mathcal{O}_\eta$, applied to lifts in $\mathfrak{n}$ of x, y, z and to the characters of x, y, z respectively, there exist G-semi-invariant lifts of x, y, z in $\mathfrak{n}$, again written x, y, z. Since $f_* \in \mathfrak{n}^2$, we have

$$\mathfrak{m}_\eta/\mathfrak{m}_\eta^2 = \mathfrak{n}/\mathfrak{n}^2,$$

and x, y, z span it. Hence the lifts are regular parameters of $\Bbbk((s))[[u_1, u_2, u_3]]$. Let Ψ be the automorphism of $\Bbbk((s))[[u_1, u_2, u_3]]$ over $\Bbbk((s))$ with $\Psi(u_1) = x$, $\Psi(u_2) = y$, and $\Psi(u_3) = z$. The kernel (f_*) of $\Bbbk((s))[[x, y, z]] \to \mathcal{O}_\eta$ contains the prime element $xy - z^r$, and (f_*) and $(xy - z^r)$ are prime ideals of height one. Thus $f_* = \sigma' \cdot (xy - z^r)$ for a unit σ' of $\Bbbk((s))[[x, y, z]]$. Let

$$w := (r(m+1), r, m+2), \qquad f_0 := xy - z^r, \quad \text{and} \quad h_0 := z^m.$$

Then f_0 and h_0 are w-homogeneous, $(f_0 = 0) \subset \mathbb{A}^3$ is a normal surface, and

$$\begin{aligned} \mathrm{wt}_w(f_0) &= r(m+1) + r = r(m+2), \\ \gamma &= r(m+1) + r + (m+2) - r(m+2) = m+2, \\ \mathrm{wt}_w(h_0) &= m(m+2) = m\gamma. \end{aligned} \tag{F.14}$$

Moreover $\mathrm{in}_w f_* = \sigma_f f_0$ for $\sigma_f := \sigma'(0) \in \Bbbk((s))^*$.

With the weights $m+1$ and 1 on p and q, the weight of $x^{j_1} y^{j_2} z^{j_3}$ is the w-weight, so the inclusion

$$\mathcal{O}_\eta = \Bbbk((s))[[x, y, z]]/(f_0) \subset \Bbbk((s))[[p, q]]$$

respects the weights, and for any weight j the induced map from the part of weight j of $\Bbbk((s))[x, y, z]/(f_0)$ to that of $\Bbbk((s))[p, q]$ is injective. Since $\nu = m$, $\varphi(0, q) = \sigma_h q^m$ modulo q^{m+1} for some $\sigma_h \in \Bbbk((s))^*$, and $\varphi - \varphi(0, q) \in p\Bbbk((s))[[p, q]]$ has weight at least $m+1$. Thus

$$h = \sigma_h z^m + (\text{terms of weight larger than } m(m+2)).$$

We construct Q by induction on the weight $j \in \{0, 1, \dots, m(m+2)\}$, starting from $Q = 0$ and $j = 0$. Assume that $h_* - Qf_*$ has no terms of w-weight less than j, and let H_j be the part of w-weight j of $h_* - Qf_*$. The image of $h_* - Qf_*$ in $\mathcal{O}_\eta$ is h, so by the injectivity above $H_j \equiv 0$ if $j < m(m+2)$, and $H_j \equiv \sigma_h z^m$ if $j = m(m+2)$, modulo $f_0\Bbbk((s))[x, y, z]$. Write $H_j = f_0 Q'_j$, resp. $H_j - \sigma_h z^m = f_0 Q'_j$, with $Q'_j \in \Bbbk((s))[x, y, z]$. Here $\sigma_h z^m$, the part of weight $m(m+2)$ of h, is G-semi-invariant with the character χ_{H_j} of H_j. Since f_0 is G-semi-invariant with a character χ_{f_0}, as xy and z^r have equal images in $\mathcal{O}_\eta$, Lemma F.3(3) for $\mathfrak{a} = \mathcal{R} = \Bbbk((s))[x, y, z]$, the space $\mathcal{V}' = \Bbbk((s))[x, y, z]$ with $g \in G$ acting by $\chi_{f_0}(g)^{-1} g$, and the map $Q' \to f_0 Q'$ gives a G-semi-invariant Q''_j with character $\chi_{H_j}\chi_{f_0}^{-1}$ and $f_0 Q''_j = H_j$ (resp. $= H_j - \sigma_h z^m$). Replacing Q with $Q + \sigma_f^{-1} Q''_j$, the part of w-weight j of $h_* - Qf_*$ becomes 0, resp. $\sigma_h z^m$, only higher weights change, and Qf_* has the character of h_*. Finally, replacing Q with the sum of the terms of degree at most k_2 in u of $Q(\Psi(u))$ for an integer k_2 with $k_2 \min\{r, m+2\} > m(m+2)$ changes Qf_* only in w-weights larger than $m(m+2)$. Thus $Q \in \Bbbk((s))[u_1, u_2, u_3]$ and

$$\mathrm{in}_w(h_* - Qf_*) = \sigma_h h_0.$$

By Lemma F.4 with $t := s$, $t_i := s_i$, the ambient germs $(\mathcal{X}_i \ni o_i)$, $V_i := W_i$, the sets $\mathcal{F}_i$ of computing lifts of b_i, the data Ψ, w, f_0, h_0, Q, σ_f, σ_h above, and $J := 1 + 4M(r + m + 2)$, the number J of Proposition 16.4 for row A with $n = r$, there exist N, α, and μ and, after passing to an infinite subsequence, π_i, $(\mathcal{X}'_i \ni o'_i)$, V'_i, S_i, f'_i, and h'_i satisfying (1)–(3) of Lemma F.4. Let $\beta := \mu/m - \alpha$, which does not depend on i.

By Lemma F.4(1), $f'_i - f_0(x_i, y_i, z_i)$ and $h'_i - h_0(x_i, y_i, z_i)$ belong to $(s_i^J) \subset (x_i, y_i, z_i, s_i)^J$, and $V'_i = (f'_i = 0)$ is normal, hence integral, near o'_i, and $S_i = \mathrm{div}(s_i|_{V'_i})$ near o'_i, as $f'_i \equiv f_0(x_i, y_i, z_i)$ modulo s_i and $(f_0 = 0) \subset \mathbb{A}^3$ is a normal surface. The group G has order at most M and acts faithfully on W_i (Definition 4.1 and Set-up 16.2), hence on $(\mathcal{X}'_i \ni o'_i)$, since π_i is G-equivariant and birational. The parameters x_i, y_i, z_i, s_i are G-semi-invariant, and the character of s_i is χ_s.

Let R be the order of χ_s. By Proposition 16.4 for row A with $n = r$, $(V_i', \frac{1}{m}\operatorname{div}(h_i') + S_i)$ is lc near o_i', and some prime divisor E_i over V_i' with center o_i' satisfies

$$\frac{a(E_i, V_i', \frac{1}{m}\operatorname{div}(h_i'))}{e_{E_i}} = \frac{1}{R}, \qquad \frac{a(E_i, V_i', 0)}{e_{E_i}} \le 2 + \frac{1}{R}. \tag{F.15}$$

Let P be a prime divisor over V_i' with center o_i'. By Lemma F.4(3), P has center o_i on W_i, and since π_i is G-equivariant and birational, the inertia groups of P over V_i' and over W_i are equal. By Lemma F.4(1)(2),

$$\pi_i^*(K_{W_i} + B_i) = K_{V_i'} + \frac{1}{m}\operatorname{div}(h_i') + \beta S_i \quad \text{and} \quad \pi_i^* K_{W_i} = K_{V_i'} - \alpha S_i,$$

so

$$\begin{aligned} a(P, W_i, B_i) &= a\Big(P, V_i', \frac{1}{m}\operatorname{div}(h_i')\Big) - \beta \operatorname{mult}_P s_i, \\ a(P, W_i, 0) &= a(P, V_i', 0) + \alpha \operatorname{mult}_P s_i. \end{aligned} \tag{F.16}$$

Since $(V_i', \frac{1}{m}\operatorname{div}(h_i') + S_i)$ is lc near o_i', and since s_i^R is G-invariant, so that $R \operatorname{mult}_P s_i$ is a positive multiple of e_P,

$$a\Big(P, V_i', \frac{1}{m}\operatorname{div}(h_i')\Big) \ge \operatorname{mult}_P s_i \ge \frac{e_P}{R}. \tag{F.17}$$

By (F.15) and (F.17) for $P = E_i$, we have $\operatorname{mult}_{E_i} s_i / e_{E_i} = 1/R$, so (F.16) implies

$$\frac{a(E_i, W_i, B_i)}{e_{E_i}} = \frac{1-\beta}{R}.$$

Since E_i has center o_i on W_i and (W_i, B_i) is klt near o_i (Set-up 16.2(1)), we have $1 - \beta > 0$. By (F.16) and (F.17), any $F \in \mathcal{F}_i$, which has center o_i' on V_i' by Lemma F.4(3), satisfies

$$b_i = \frac{a(F, W_i, B_i)}{e_F} \ge \frac{(1-\beta)\operatorname{mult}_F s_i}{e_F} \ge \frac{1-\beta}{R}.$$

Thus $b_i = (1-\beta)/R$, the prime divisor E_i is a computing lift of b_i, and by (F.16) and (F.15),

$$\frac{a(E_i, W_i, 0)}{e_{E_i}} = \frac{a(E_i, V_i', 0)}{e_{E_i}} + \frac{\alpha}{R} \le 2 + \frac{1+\alpha}{R}.$$

The right-hand side does not depend on i. The infinite subsequence is a radial failure sequence with constants (m, a, δ), so this contradicts Lemma 4.3(2). □

We prove the normal form of the transverse pair used in the proof of Theorem 16.6.

Proposition F.5. *Let $\Bbbk \supset \mathbb{C}$ be an algebraically closed field, $E := \Bbbk((s))$ with an algebraic closure $\overline{E}$, m a positive integer, and $f, h \in E[[u_1, u_2, u_3]]$ with $h \notin (f)$. Let G be a finite abelian group acting on $E[[u_1, u_2, u_3]]$, trivially on $\Bbbk$, by $g(s) = \chi_s(g)s$ and $g(u_j) = \chi_j(g)u_j$ with $\chi_s(g), \chi_j(g) \in \Bbbk^*$, such that $g(f) \in \Bbbk^* f$ and $g(h) \in \Bbbk^* h$ for any $g \in G$ and G acts faithfully on $R := E[[u_1, u_2, u_3]]/(f)$. Assume that*

(i) R is normal,
(ii) $\overline{R} := \overline{E}[[u_1, u_2, u_3]]/(f)$ is a singular Du Val germ with $(\operatorname{Spec}\overline{R}, \frac{1}{m}\operatorname{div}(h))$ lc, and
(iii) there exists a finite morphism $\pi : \operatorname{Spec}\overline{E}[[p, q]] \to \operatorname{Spec}\overline{R}$ étale in codimension 1 such that the $\mathbb{Q}$-divisor $\pi^*\frac{1}{m}\operatorname{div}(h)$ has exactly two prime components with coefficient one.

Then there exists an integer $k \ge 1$ with the following property. Let t be a k-th root of s, so that $E \subset \Bbbk((t))$, and let

$$\widetilde{G} := \{(g, \zeta) \in G \times \Bbbk^* \mid \zeta^k = \chi_s(g)\}$$

act on $\Bbbk((t))[[u_1, u_2, u_3]]$, trivially on $\Bbbk$, by $(g, \zeta)(t) = \zeta t$ and $(g, \zeta)(u_j) = \chi_j(g)u_j$. There exist $\sigma_f, \sigma_h \in \Bbbk((t))^*$, a polynomial $Q \in \Bbbk((t))[u_1, u_2, u_3]$ and an automorphism Ψ of $\Bbbk((t))[[u_1, u_2, u_3]]$ over $\Bbbk((t))$ such that the following hold, where

$$x := \Psi(u_1), \qquad y := \Psi(u_2), \quad \text{and} \quad z := \Psi(u_3).$$

(1) x, y, z, and Q are $\widetilde{G}$-semi-invariant, and Qf has the character of h.
(2) There exist $w \in (\mathbb{N}^+)^3$ and $f_0, h_0 \in \mathbb{C}[x,y,z]$ satisfying one of the following.
  (a) There exists an integer $n \geq 2$ with $f_0 = xy - z^n$ or $f_0 = x^2 + y^2 - z^n$, $h_0 = z^m$, and $w = (n,n,2)$.
  (b) The integer m is even, and there exists an integer $r \geq 1$ with $f_0 = x^2 - zy^2 + 4z^{r+1}$, $h_0 = z^{m/2}$, and $w = (r+1, r, 2)$.
(3) The surface pair $\left(S_0, \frac{1}{m}\operatorname{div}(h_0|_{S_0})\right)$, where $S_0 := (f_0 = 0) \subset \mathbb{A}^3_{\Bbbk}$, is lc near 0.
(4) In $\Bbbk((t))[[x,y,z]] = \Bbbk((t))[[u_1,u_2,u_3]]$, with the weights w on x,y,z,

$$\operatorname{in}_w f = \sigma_f f_0, \qquad \operatorname{in}_w(h - Qf) = \sigma_h h_0.$$

Proof. We divide the proof into six steps. We write $u := (u_1,u_2,u_3)$. All fields below have characteristic zero.

Step 1. In this step we construct a finite extension E' of E and a power series ring $\mathcal{O}_0 = E'[[p,q]]$ finite over $R \otimes_E E'$, and we show that h is the m-th power of a product of two regular parameters of $\mathcal{O}_0$ up to a unit.

Since $\overline{E} \supset \mathbb{C}$ is algebraically closed, by [Kol13, Theorem 2.29(2) and Example 3.26] there exists a polynomial $f_\Pi \in \mathbb{Z}[u]$, one of the normal forms of the Du Val singularities listed there, such that $\overline{R} \cong \overline{E}[[u]]/(f_\Pi)$. The germ at 0 of $(f_\Pi = 0) \subset \mathbb{C}^3$ is a Du Val singularity, so by Lemma F.2 with $\mathbb{L} = \mathbb{C}$ there exist a non-trivial finite subgroup $\Pi \subset SL_2(\mathbb{C})$, acting on $\mathbb{C}[[p,q]]$ by linear substitutions, and $\phi_1, \phi_2, \phi_3 \in \mathbb{C}[[p,q]]^\Pi$ such that the homomorphism sending u_j to ϕ_j induces $\mathbb{C}[[u]]/(f_\Pi) \cong \mathbb{C}[[p,q]]^\Pi$.

Let $\mathbb{K} \supset \mathbb{C}$ be a field, and let $\mathfrak{n}_\mathbb{K}$ and $\mathfrak{n}_\mathbb{C}$ be the maximal ideals of $\mathbb{K}[[p,q]]^\Pi$ and $\mathbb{C}[[p,q]]^\Pi$. Since the Reynolds operator $\frac{1}{|\Pi|}\sum_{\gamma\in\Pi}\gamma$ preserves the degree in p,q, the homogeneous parts of degree i of the elements of $\mathbb{K}[[p,q]]^\Pi$ form $\mathbb{K} \otimes_\mathbb{C} \mathbb{C}[p,q]_i^\Pi$ for any i, so

$$\mathfrak{n}_\mathbb{K}/\mathfrak{n}_\mathbb{K}^2 = \mathbb{K} \otimes_\mathbb{C} \mathfrak{n}_\mathbb{C}/\mathfrak{n}_\mathbb{C}^2.$$

The classes of ϕ_1, ϕ_2, ϕ_3 span $\mathfrak{n}_\mathbb{C}/\mathfrak{n}_\mathbb{C}^2$, so the homomorphism $\mathbb{K}[[u]] \to \mathbb{K}[[p,q]]^\Pi$ sending u_j to ϕ_j is surjective, with kernel containing f_Π. Since f_Π has an isolated singularity at 0, $\mathbb{K}[[u]]/(f_\Pi)$ is a 2-dimensional normal local domain, and $\mathbb{K}[[p,q]]^\Pi$ is also 2-dimensional. Hence

$$\mathbb{K}[[p,q]]^\Pi = \mathbb{K}[[\phi_1,\phi_2,\phi_3]] \cong \mathbb{K}[[u]]/(f_\Pi). \tag{F.18}$$

Since $\overline{R}$ is singular, $f, f_\Pi \in (u)^2$. Thus the isomorphism $\overline{E}[[u]]/(f) \cong \overline{E}[[u]]/(f_\Pi)$ lifts to an $\overline{E}$-automorphism η of $\overline{E}[[u]]$, and $v \cdot \eta(f) = f_\Pi$ for a unit v of $\overline{E}[[u]]$. The Tjurina number $\tau(f_\Pi)$ is finite, so by [GP19, Theorem 4.8], over any field $\mathbb{K}$ of characteristic zero, f_Π is contact k_0-determined for $k_0 := 2\tau(f_\Pi) - \operatorname{mult}_0 f_\Pi + 2$: for any $g_1 \in \mathbb{K}[[u]]$ with $g_1 - f_\Pi \in (u)^{k_0+1}$, there exist a $\mathbb{K}$-automorphism η' of $\mathbb{K}[[u]]$ and a unit v' of $\mathbb{K}[[u]]$ with $v' \cdot \eta'(g_1) = f_\Pi$. Let η_{k_0} be the $\overline{E}$-automorphism of $\overline{E}[[u]]$ sending u_j to the part of $\eta(u_j)$ of degree at most k_0 for $j \in \{1,2,3\}$, let v_{k_0} be the part of v of degree at most k_0, and let $E' \subset \overline{E}$ be the field generated over E by the coefficients of $\eta_{k_0}(u_1)$, $\eta_{k_0}(u_2)$, $\eta_{k_0}(u_3)$, and v_{k_0}. Then $[E' : E] < +\infty$, $v_{k_0} \cdot \eta_{k_0}(f) \in E'[[u]]$, and $v_{k_0} \cdot \eta_{k_0}(f) - f_\Pi \in (u)^{k_0+1}$. Thus by (F.18) for $\mathbb{K} = E'$,

$$R' := R \otimes_E E' = E'[[u]]/(f) \cong E'[[u]]/(f_\Pi) \cong \mathcal{O}_0^\Pi, \qquad \text{where } \mathcal{O}_0 := E'[[p,q]]. \tag{F.19}$$

Via (F.19), $R \subset R' = \mathcal{O}_0^\Pi \subset \mathcal{O}_0$.

Any $\gamma \in \Pi\backslash\{1\}$ has eigenvectors $p', q' \in \mathbb{C}p + \mathbb{C}q$ with eigenvalues $\lambda, \lambda^{-1} \neq 1$. If a prime ideal $\mathfrak{q}$ of $\mathcal{O}_0$ contains $\gamma(a) - a$ for any $a \in \mathcal{O}_0$, then $(\lambda - 1)p', (\lambda^{-1} - 1)q' \in \mathfrak{q}$, so $\mathfrak{q}$ is the maximal ideal. Thus the inertia groups of Π at the non-closed points of $\operatorname{Spec}\mathcal{O}_0$ are trivial, and $\operatorname{Spec}\mathcal{O}_0 \to \operatorname{Spec}R'$ is étale outside the closed point [Gro71, Exposé V, Corollaire 2.3]. Let $\overline{\mathcal{O}}_0 := \overline{E}[[p,q]]$. We identify $\overline{R} = \overline{\mathcal{O}}_0^\Pi$ by the $\overline{E}$-linear extension of (F.19) (its second isomorphism becomes (F.18) for $\mathbb{K} = \overline{E}$), which restricts to the inclusion $R \subset \mathcal{O}_0$ of (F.19). Since the computation of inertia groups above uses only the linear action of Π on p,q, the quotient morphism $\operatorname{Spec}\overline{\mathcal{O}}_0 \to \operatorname{Spec}\overline{R}$ is also étale outside the closed point. By Lemma F.2 with $\mathbb{L} = \overline{E}$,

π and the quotient morphism by Π are both isomorphic over $\operatorname{Spec}\overline{R}$ to the quotient morphism of that lemma, so the quotient morphism by Π is isomorphic over $\operatorname{Spec}\overline{R}$ to π.

Let $D := \pi^*\frac{1}{m}\operatorname{div}(h)$, which is $\frac{1}{m}\operatorname{div}(h)$ on $\operatorname{Spec}\overline{\mathcal{O}}_0$, let D_1, D_2 be the prime components of D with coefficient one, and let $\pi_Y : Y \to \operatorname{Spec}\overline{\mathcal{O}}_0$ be the blow-up of the closed point 0, with exceptional curve F_1. Since π is finite and étale in codimension 1 and $(\operatorname{Spec}\overline{R}, \frac{1}{m}\operatorname{div}(h))$ is lc, $(\operatorname{Spec}\overline{\mathcal{O}}_0, D)$ is lc, so

$$0 \le a(F_1, \operatorname{Spec}\overline{\mathcal{O}}_0, D) = 2 - \operatorname{mult}_0 D \le 2 - \operatorname{mult}_0 D_1 - \operatorname{mult}_0 D_2 \le 0.$$

Thus $D = D_1 + D_2$ and $\operatorname{mult}_0 D_1 = \operatorname{mult}_0 D_2 = 1$. If D_1 and D_2 were tangent at 0, then $a(F_1, \operatorname{Spec}\overline{\mathcal{O}}_0, D) = 0$, the curves $(\pi_Y)_*^{-1}D_1$, $(\pi_Y)_*^{-1}D_2$, and F_1 would pass through one point y_1, and the exceptional curve F_2 of the blow-up of y_1 would satisfy $a(F_2, \operatorname{Spec}\overline{\mathcal{O}}_0, D) = 2 - 3 < 0$, a contradiction. Thus $D_i = (\ell_i = 0)$ for $i \in \{1, 2\}$, where ℓ_1, ℓ_2 form a regular system of parameters of $\overline{\mathcal{O}}_0$, and $h = (\ell_1\ell_2)^m v_h$ for a unit v_h of $\overline{\mathcal{O}}_0$.

The homogeneous part of h of lowest degree in p, q has coefficients in E' and equals $v_h(0)(L_1L_2)^m$, where L_i is the linear part of ℓ_i. Possibly replacing E' with a finite extension, we may assume that the lines $(L_1 = 0)$ and $(L_2 = 0)$ are defined over E'. By the Weierstrass preparation theorem, the ideal (ℓ_i) has a unique generator in $p + q\overline{E}[[q]]$ if the coefficient of p in L_i is non-zero, and in $q + p\overline{E}[[p]]$ otherwise. We replace ℓ_i with the generator. Any $\omega \in \operatorname{Gal}(\overline{E}/E')$ acts on $\overline{\mathcal{O}}_0$ through the coefficients and fixes h, so ω permutes the ideals $(\ell_1), (\ell_2)$. Since ω fixes the line $(L_i = 0)$ and $(L_1 = 0) \ne (L_2 = 0)$, we have $(\omega(\ell_i)) = (\ell_i)$, so $\omega(\ell_i) = \ell_i$ by the uniqueness of the generator. Thus $\ell_1, \ell_2 \in \mathcal{O}_0$, and $v_h = h/(\ell_1\ell_2)^m$ is fixed by $\operatorname{Gal}(\overline{E}/E')$, so

$$h = (\ell_1\ell_2)^m v_h, \qquad \ell_1, \ell_2 \in \mathcal{O}_0 \text{ a regular system of parameters}, \quad \text{and} \quad v_h \in \mathcal{O}_0^*. \tag{F.20}$$

Step 2. In this step we take a normal closure of $\operatorname{Frac}\mathcal{O}_0$ and show that the integral closure of R in it is a power series ring $E_1[[p, q]]$.

Let $\mathcal{L} := \operatorname{Frac} R$. Since G acts faithfully on R, $\mathcal{L}/\mathcal{L}^G$ is Galois with group G. Since $\mathcal{O}_0$ is finite over $R' = \mathcal{O}_0^\Pi$ and R' is finite over R, $[\operatorname{Frac}\mathcal{O}_0 : \mathcal{L}] < +\infty$. Let $\mathcal{N}$ be the normal closure of $\operatorname{Frac}\mathcal{O}_0$ over $\mathcal{L}^G$, $\mathcal{G} := \operatorname{Gal}(\mathcal{N}/\mathcal{L}^G)$, and $\mathcal{O}_1$ the integral closure of R in $\mathcal{N}$, which is also the integral closure of $\mathcal{O}_0$ in $\mathcal{N}$. Since R is a complete normal local domain, $\mathcal{O}_1$ is a finite R-algebra and a complete local domain with $\operatorname{Frac}\mathcal{O}_1 = \mathcal{N}$. Any $\gamma \in \mathcal{G}$ restricts to an element $g_\gamma \in G$ on $\mathcal{L}$, so $\gamma(R) = R$ and $\gamma(\mathcal{O}_1) = \mathcal{O}_1$.

Let $\mathfrak{p}$ be a height-one prime ideal of R. Since $R \to R'$ is finite étale and $R' \to \mathcal{O}_0$ is étale outside the closed point (**Step 1**), $R_\mathfrak{p} \otimes_R \mathcal{O}_0$ is finite étale over $R_\mathfrak{p}$. For $\gamma \in \mathcal{G}$, the ideal $\gamma^{-1}(\mathfrak{p})$ also has height one, so the integral closure $R_\mathfrak{p} \otimes_R \gamma(\mathcal{O}_0)$ of $R_\mathfrak{p}$ in $\gamma(\operatorname{Frac}\mathcal{O}_0)$ is finite étale over $R_\mathfrak{p}$. The field $\mathcal{N}$ is the compositum of the fields $\gamma(\operatorname{Frac}\mathcal{O}_0)$, $\gamma \in \mathcal{G}$. Let $\mathcal{T}_\mathfrak{p}$ be the tensor product over $R_\mathfrak{p}$ of the rings $R_\mathfrak{p} \otimes_R \gamma(\mathcal{O}_0)$, $\gamma \in \mathcal{G}$, a finite étale $R_\mathfrak{p}$-algebra, and $\mathcal{T}'_\mathfrak{p}$ the image of $\mathcal{T}_\mathfrak{p}$ in $\mathcal{N}$. Then $\mathcal{T}'_\mathfrak{p}$ is a direct factor of $\mathcal{T}_\mathfrak{p}$, as the kernel is a minimal prime ideal, and $\mathcal{T}'_\mathfrak{p}$ is normal with fraction field $\mathcal{N}$. Thus $R_\mathfrak{p} \otimes_R \mathcal{O}_1 = \mathcal{T}'_\mathfrak{p}$ is finite étale over $R_\mathfrak{p}$. Let $\mathfrak{q}$ be a height-one prime ideal of $\mathcal{O}_0$ and $\mathfrak{p} := \mathfrak{q} \cap R$, of height one. Since $R_\mathfrak{p} \otimes_R \mathcal{O}_0$ and $R_\mathfrak{p} \otimes_R \mathcal{O}_1$ are étale over $R_\mathfrak{p}$, $\mathcal{O}_1$ is étale over $\mathcal{O}_0$ at any prime ideal over $\mathfrak{q}$. The ring $\mathcal{O}_0$ is regular and $\mathcal{O}_1$ is normal, so by the purity of the branch locus [Gro71, Exposé X, Théorème 3.1], the set of points of $\operatorname{Spec}\mathcal{O}_1$ at which $\operatorname{Spec}\mathcal{O}_1 \to \operatorname{Spec}\mathcal{O}_0$ is not étale is empty or of pure codimension 1. Since it contains no point of codimension at most 1, it is empty, and $\mathcal{O}_1$ is finite étale over $\mathcal{O}_0$.

Let E_1 be the residue field of $\mathcal{O}_1$, a finite extension of E'. Since $\mathcal{O}_1$ is a complete local ring finite étale over $\mathcal{O}_0 = E'[[p, q]]$, it contains a coefficient field $E_1 \supset E'$, and

$$\mathcal{O}_1 = E_1 \otimes_{E'} \mathcal{O}_0 = E_1[[p, q]]. \tag{F.21}$$

Any element $a \in \mathcal{N}$ algebraic over E is integral over $E \subset R$, so $a \in \mathcal{O}_1$. Its constant term $a(0) \in E_1$ in $E_1[[p, q]]$ and a are roots in $\mathcal{O}_1$ of the minimal polynomial of a over E, which is separable, and their residues are equal, so $a = a(0)$ by the uniqueness in Hensel's lemma. Thus E_1 is the set of elements of $\mathcal{N}$ algebraic over E. Since $\mathcal{G}$ preserves E, $\mathcal{G}$ preserves E_1.

Step 3. In this step we show that $E_1 = \Bbbk((t))$ with $s = t^k$, that $\mathcal{G}_0 := \mathrm{Gal}(\mathcal{N}/\mathcal{L}E_1)$ equals Π, and that $\mathcal{G}/\mathcal{G}_0 \cong \widetilde{G}$.

Let $k := [E_1 : E]$. Since $\Bbbk$ is algebraically closed of characteristic zero, $E_1 = \Bbbk((t))$ for a k-th root t of s. For $\gamma \in \mathcal{G}$, $\gamma(t)^k = \gamma(s) = \chi_s(g_\gamma)t^k$, so $\gamma(t)/t \in E_1$ is a root of $X^k - \chi_s(g_\gamma)$, whose k roots belong to $\Bbbk$. Thus

$$\gamma(t) = \zeta_\gamma t \quad \text{with} \quad \zeta_\gamma \in \Bbbk^*,\ \zeta_\gamma^k = \chi_s(g_\gamma), \tag{F.22}$$

and γ is trivial on $\Bbbk \subset \mathcal{L}$, since $\gamma|_{\mathcal{L}} = g_\gamma$ and G acts trivially on $\Bbbk$.

Let $R_1 := R \otimes_E E_1 = E_1[[u]]/(f)$. By (F.19) and (F.21), $R_1 = E_1 \otimes_{E'} R' \subset E_1 \otimes_{E'} \mathcal{O}_0 = \mathcal{O}_1$, and Π acts on $\mathcal{O}_1$ by E_1-linear substitutions of p, q with

$$\mathcal{O}_1^\Pi = E_1 \otimes_{E'} \mathcal{O}_0^\Pi = R_1.$$

In particular R_1 is a domain with $\mathrm{Frac}\, R_1 = \mathcal{L}E_1$, and $\mathcal{N}^\Pi = \mathrm{Frac}(\mathcal{O}_1^\Pi) = \mathcal{L}E_1$. Since Π acts faithfully on $\mathcal{N}$, Artin's theorem shows $\Pi = \mathrm{Gal}(\mathcal{N}/\mathcal{L}E_1)$, that is,

$$\mathcal{G}_0 = \Pi, \qquad \mathcal{O}_1^{\mathcal{G}_0} = R_1 = E_1[[u]]/(f). \tag{F.23}$$

Let N_0 be the order of the group $\chi_s(G) \subset \Bbbk^*$. Since $s^{N_0} \in E^G \subset \mathcal{L}^G$ and $\Bbbk$ contains the kN_0-th roots of unity, $\mathcal{L}^G E_1 = \mathcal{L}^G(t)$ (as $E = E^G(s)$ and $s = t^k$) is the splitting field over $\mathcal{L}^G$ of $X^{kN_0} - s^{N_0}$, hence Galois over $\mathcal{L}^G$. Thus $\mathcal{L}E_1$, the compositum of $\mathcal{L}$ and $\mathcal{L}^G E_1$, is Galois over $\mathcal{L}^G$, and $\mathcal{G}_0 \lhd \mathcal{G}$. By (F.22), the map sending $\gamma\mathcal{G}_0$ to (g_γ, ζ_γ) is an injective homomorphism $\mathcal{G}/\mathcal{G}_0 = \mathrm{Gal}(\mathcal{L}E_1/\mathcal{L}^G) \to \widetilde{G}$. Since $\mathcal{L} \otimes_E E_1$ is a localization of the domain R_1 and finite over the field $\mathcal{L}$, $\mathcal{L} \otimes_E E_1 = \mathcal{L}E_1$. The homomorphism is surjective: for $(g, \zeta) \in \widetilde{G}$, let τ_ζ be the automorphism of E_1 trivial on $\Bbbk$ with $\tau_\zeta(t) = \zeta t$. Since $\tau_\zeta(s) = \zeta^k s = g(s)$, $g \otimes \tau_\zeta$ is an automorphism of $\mathcal{L}E_1$ fixing $\mathcal{L}^G$. Thus

$$\mathcal{G}/\mathcal{G}_0 \cong \widetilde{G},$$

the action of $\mathcal{G}$ on $R_1 = \mathcal{O}_1^{\mathcal{G}_0}$ factors through $\widetilde{G}$, and it is induced by the action of $\widetilde{G}$ on $E_1[[u]]$ of the proposition.

Step 4. In this step we choose regular parameters p, q of $\mathcal{O}_1$ on which $\mathcal{G}$ acts by constant monomial matrices.

Let $\mathfrak{m}_1$ be the maximal ideal of $\mathcal{O}_1$, with $\dim_{E_1} \mathfrak{m}_1/\mathfrak{m}_1^2 = 2$. Any $\gamma \in \mathcal{G}$ acts on $\mathfrak{m}_1/\mathfrak{m}_1^2$ with $\gamma(a\xi) = \gamma(a)\gamma(\xi)$ for $a \in E_1$ and $\xi \in \mathfrak{m}_1/\mathfrak{m}_1^2$. Since $\gamma(h) = g_\gamma(h) \in \Bbbk^* h$ and $\mathcal{O}_1$ is factorial, γ permutes the prime ideals $(\ell_1), (\ell_2)$ of (F.20). Thus γ permutes the lines $E_1 \cdot [\ell_1], E_1 \cdot [\ell_2] \subset \mathfrak{m}_1/\mathfrak{m}_1^2$, where $[\ell_i]$ is the class of ℓ_i, and $\mathfrak{m}_1/\mathfrak{m}_1^2 = E_1 \cdot [\ell_1] \oplus E_1 \cdot [\ell_2]$.

Let $\Lambda \subset \mathfrak{m}_1/\mathfrak{m}_1^2$ be the $\Bbbk[[t]]$-submodule generated by $\gamma([\ell_1]), \gamma([\ell_2])$ for $\gamma \in \mathcal{G}$. Since any $\gamma \in \mathcal{G}$ preserves $\Bbbk[[t]]$ by (F.22) and permutes the lines $E_1 \cdot [\ell_1], E_1 \cdot [\ell_2]$, Λ is $\mathcal{G}$-stable and $\Lambda = \Lambda_1 \oplus \Lambda_2$, where $\Lambda_i := \Lambda \cap E_1 \cdot [\ell_i]$ is free of rank 1 over $\Bbbk[[t]]$. The group $\mathcal{G}$ acts $\Bbbk$-linearly on $\Lambda/t\Lambda$, which equals $\Lambda_1^\circ \oplus \Lambda_2^\circ$, where Λ_i° is the image of Λ_i, and permutes $\Lambda_1^\circ, \Lambda_2^\circ$. Let $\iota : \Lambda/t\Lambda \to \Lambda$ be a $\Bbbk$-linear section of the reduction with $\iota(\Lambda_i^\circ) \subset \Lambda_i$, and

$$\iota' := \frac{1}{|\mathcal{G}|} \sum_{\gamma \in \mathcal{G}} \gamma \circ \iota \circ \gamma^{-1}.$$

Then ι' is a $\mathcal{G}$-equivariant $\Bbbk$-linear section with $\iota'(\Lambda_i^\circ) \subset \Lambda_i$. Choose $e_i^\circ \in \Lambda_i^\circ \backslash \{0\}$ and let $e_i := \iota'(e_i^\circ)$. By Nakayama's lemma, e_1, e_2 form a $\Bbbk[[t]]$-basis of Λ, hence an E_1-basis of $\mathfrak{m}_1/\mathfrak{m}_1^2$, and for any $\gamma \in \mathcal{G}$,

$$(\gamma(e_1), \gamma(e_2)) = (e_1, e_2)\rho(\gamma),$$

where $\rho(\gamma) \in GL_2(\Bbbk)$ is the monomial matrix of γ on $\Lambda/t\Lambda$ in the basis e_1°, e_2°.

Write $e_i = c_i[\ell_i]$ with $c_i \in E_1^*$, let $\varsigma : \mathfrak{m}_1/\mathfrak{m}_1^2 \to \mathfrak{m}_1$ be the E_1-linear section with $\varsigma(e_i) = c_i\ell_i$, and

$$\varsigma' := \frac{1}{|\mathcal{G}|} \sum_{\gamma \in \mathcal{G}} \gamma \circ \varsigma \circ \gamma^{-1}.$$

For $\gamma \in \mathcal{G}$, the map $\gamma \circ \varsigma \circ \gamma^{-1}$ is E_1-linear, is a section of the quotient map from $\mathfrak{m}_1$ to $\mathfrak{m}_1/\mathfrak{m}_1^2$, and sends e_i into the ideal (ℓ_i), since γ permutes the ideals $(\ell_1), (\ell_2)$ as it permutes the lines $E_1 \cdot [\ell_1], E_1 \cdot [\ell_2]$. Thus ς' is a $\mathcal{G}$-equivariant E_1-linear section, and $\varsigma'(e_1)$, $\varsigma'(e_2)$ form a regular system of parameters of $\mathcal{O}_1$ with $(\varsigma'(e_i)) = (\ell_i)$. Replacing p, q with $\varsigma'(e_1), \varsigma'(e_2)$, we have $\mathcal{O}_1 = E_1[[p, q]]$ and, by (F.20) and (F.22),

$$\begin{aligned} &h = (pq)^m v_h \text{ with } v_h \in \mathcal{O}_1^*, \qquad \gamma(t) = \zeta_\gamma t, \\ &\text{and} \quad (\gamma(p), \gamma(q)) = (p, q)\rho(\gamma) \quad \text{for any } \gamma \in \mathcal{G}. \end{aligned} \tag{F.24}$$

Since the entries of $\rho(\gamma)$ belong to $\Bbbk$, which $\mathcal{G}$ fixes, $\rho : \mathcal{G} \to GL_2(\Bbbk)$ is a homomorphism. Since $\mathcal{G}_0 = \Pi$ acts on $\mathcal{O}_1$ by E_1-linear substitutions of the coordinates p, q of **Step 1**, it acts faithfully on $\mathfrak{m}_1/\mathfrak{m}_1^2$, so $\rho|_{\mathcal{G}_0}$ is injective. For $\gamma \in \mathcal{G}_0$, the determinant of the E_1-linear map γ on $\mathfrak{m}_1/\mathfrak{m}_1^2$ does not depend on the basis, and it equals 1 in the basis given by the coordinates of **Step 1** by (F.23). Thus $\rho(\mathcal{G}_0) \subset SL_2(\Bbbk)$. Moreover $\mathcal{G}_0 = \Pi \neq \{1\}$.

Step 5. In this step we compute $R_1 = \mathcal{O}_1^{\mathcal{G}_0}$ and the action of $\widetilde{G}$ on it, and we prove (2) and (3).

Since $\rho(\mathcal{G}_0)$ is a finite monomial subgroup of $SL_2(\Bbbk)$, the subgroup $\rho(\mathcal{G}_0)_{\mathrm{diag}}$ of diagonal matrices in $\rho(\mathcal{G}_0)$ satisfies

$$\rho(\mathcal{G}_0)_{\mathrm{diag}} = \{\mathrm{diag}(\lambda, \lambda^{-1}) \mid \lambda \in \boldsymbol{\mu}_N\}$$

for an integer $N \geq 1$, where $\boldsymbol{\mu}_N \subset \Bbbk^*$ is the group of N-th roots of unity, and either $\rho(\mathcal{G}_0)$ equals $\rho(\mathcal{G}_0)_{\mathrm{diag}}$ or $\rho(\mathcal{G}_0)$ contains a matrix

$$J_\alpha := \begin{pmatrix} 0 & \alpha_2 \\ \alpha_1 & 0 \end{pmatrix} \quad \text{with} \quad \alpha_1\alpha_2 = -1.$$

Case (A): $\rho(\mathcal{G}_0)$ is diagonal. Then $n := N = |\mathcal{G}_0| \geq 2$, and

$$R_1 = E_1[[p, q]]^{\mathcal{G}_0} = E_1[[\overline{x}, \overline{y}, \overline{z}]], \qquad \overline{x} := p^n,\ \overline{y} := q^n,\ \overline{z} := pq.$$

The homomorphism $E_1[[x, y, z]]/(xy - z^n) \to R_1$ sending x, y, z to $\overline{x}, \overline{y}, \overline{z}$ is surjective, and it is an isomorphism since both sides are 2-dimensional domains. By (F.24), $v_h = h/\overline{z}^m$ is $\mathcal{G}_0$-invariant, so $v_h \in R_1^*$. Let $V_A := \Bbbk\overline{x} \oplus \Bbbk\overline{y} \subset R_1$. By (F.24), any $\gamma \in \mathcal{G}$ maps V_A to itself by a monomial matrix with entries in $\Bbbk$, $\gamma(\overline{z}) \in \Bbbk^*\overline{z}$, and $\gamma(\overline{xy}) \in \Bbbk^*\overline{xy}$ in $\mathrm{Sym}^2 V_A$. Since the action of $\mathcal{G}$ on R_1 factors through the abelian group $\widetilde{G}$, V_A has a basis x', y' of $\widetilde{G}$-semi-invariant vectors, with characters $\chi_{x'}, \chi_{y'}$. If $\chi_{x'} = \chi_{y'}$, then $\widetilde{G}$ acts on V_A by scalars, so $\overline{x}, \overline{y}$ are semi-invariant, and we let $f_0 := xy - z^n$. Otherwise write

$$\overline{xy} = c_{11}x'^2 + c_{12}x'y' + c_{22}y'^2 \quad \text{in } \mathrm{Sym}^2 V_A, \qquad c_{11}, c_{12}, c_{22} \in \Bbbk.$$

If $c_{12} \neq 0$, the character of $\overline{xy}$ is $\chi_{x'}\chi_{y'}$, which differs from $\chi_{x'}^2$ and $\chi_{y'}^2$, so $c_{11} = c_{22} = 0$. Replacing $(\overline{x}, \overline{y})$ with $(c_{12}x', y')$, we let $f_0 := xy - z^n$. If $c_{12} = 0$, then $c_{11}c_{22} \neq 0$, since the quadratic form $\overline{xy}$ has rank 2. Replacing $(\overline{x}, \overline{y})$ with $(c_{11}^{1/2}x', c_{22}^{1/2}y')$, we let $f_0 := x^2 + y^2 - z^n$. In any case let $h_0 := z^m$ and $w := (n, n, 2)$. Then $\overline{x}, \overline{y}, \overline{z}$ are $\widetilde{G}$-semi-invariant, the homomorphism sending x, y, z to $\overline{x}, \overline{y}, \overline{z}$ induces $E_1[[x, y, z]]/(f_0) \cong R_1$, and

$$h = h_0(\overline{x}, \overline{y}, \overline{z})v_h, \qquad h_0(\overline{x}, \overline{y}, \overline{z}) = (pq)^m, \quad \text{and} \quad v_h \in R_1^*.$$

Case (D): $\rho(\mathcal{G}_0)$ is not diagonal. Then $J_\alpha^2 = -I$ belongs to $\rho(\mathcal{G}_0)_{\mathrm{diag}}$, so $N = 2r$ for an integer $r \geq 1$. Possibly replacing q with $\alpha_1 q$, which changes v_h and conjugates ρ by a diagonal matrix in (F.24), we may assume that $\rho(\mathcal{G}_0)$ is generated by $\mathrm{diag}(\lambda, \lambda^{-1})$, $\lambda \in \boldsymbol{\mu}_{2r}$, and

$$J := \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad (J(p), J(q)) = (q, -p).$$

Let

$$\overline{z} := (pq)^2, \qquad \overline{y} := p^{2r} + q^{2r}, \quad \text{and} \quad \overline{x} := pq\big(p^{2r} - q^{2r}\big).$$

Then $\overline{x}, \overline{y}, \overline{z}$ are $\mathcal{G}_0$-invariant, and $\overline{x}^2 - \overline{z}\overline{y}^2 + 4\overline{z}^{r+1} = 0$ since $(p^{2r} - q^{2r})^2 = \overline{y}^2 - 4\overline{z}^r$. The invariant ring of $\rho(\mathcal{G}_0)_{\mathrm{diag}}$ is $E_1[[a_1, a_2, c]]$ with $a_1 := p^{2r}$, $a_2 := q^{2r}$, $c := pq$, and $J(a_1, a_2, c) = (a_2, a_1, -c)$.

Let $\psi := \psi_0(a_1,a_2,c^2) + c\psi_1(a_1,a_2,c^2)$ be J-invariant, where ψ_0,ψ_1 are power series. Then $\psi = \frac{1}{2}(\psi + J\psi) = \psi_0^+ + c\psi_1^-$ with

$$\psi_0^+ := \tfrac{1}{2}\big(\psi_0(a_1,a_2,c^2) + \psi_0(a_2,a_1,c^2)\big), \qquad \psi_1^- := \tfrac{1}{2}\big(\psi_1(a_1,a_2,c^2) - \psi_1(a_2,a_1,c^2)\big).$$

Here ψ_0^+ is symmetric in a_1,a_2, and $\psi_1^- = (a_1-a_2)\psi_2$ with ψ_2 symmetric in a_1,a_2. Since $a_1+a_2=\overline{y}$, $a_1a_2=c^{2r}$, $c^2=\overline{z}$, and $c(a_1-a_2)=\overline{x}$, we have $\psi \in E_1[[\overline{x},\overline{y},\overline{z}]]$, so $\overline{x},\overline{y},\overline{z}$ generate R_1. As in Case (A), the homomorphism sending x,y,z to $\overline{x},\overline{y},\overline{z}$ induces

$$E_1[[x,y,z]]/(f_0) \cong R_1, \qquad f_0 := x^2 - zy^2 + 4z^{r+1}.$$

Since $J(h)=h$ and $J((pq)^m) = (-1)^m(pq)^m$, (F.24) implies $J(v_h) = (-1)^m v_h$. As J fixes $v_h(0)\neq 0$, m is even. Let $h_0 := z^{m/2}$ and $w := (r+1,r,2)$. Then $v_h = h/\overline{z}^{m/2}$ is $\mathcal{G}_0$-invariant, and

$$h = h_0(\overline{x},\overline{y},\overline{z})v_h, \qquad h_0(\overline{x},\overline{y},\overline{z}) = (pq)^m, \quad \text{and} \quad v_h \in R_1^*.$$

Let $\gamma\in\mathcal{G}$ and write $\rho(\gamma) = \operatorname{diag}(\beta_1,\beta_2)J^j$ with $j\in\{0,1\}$. Since $\mathcal{G}_0 \lhd \mathcal{G}$ (**Step 3**) and $J\in\rho(\mathcal{G}_0)$, $\operatorname{diag}(\beta_1,\beta_2)$ normalizes $\rho(\mathcal{G}_0)$, so

$$\operatorname{diag}(\beta_1,\beta_2)J\operatorname{diag}(\beta_1,\beta_2)^{-1} = J\operatorname{diag}(\beta_2/\beta_1,\beta_1/\beta_2) \in \rho(\mathcal{G}_0),$$

that is, $\beta_2/\beta_1 \in \boldsymbol{\mu}_{2r}$ and $\beta_1^{2r} = \beta_2^{2r}$. Since $\overline{x},\overline{y},\overline{z}$ are J-invariant polynomials in p,q with coefficients in $\Bbbk$, γ acts on them as the substitution of $\beta_1 p, \beta_2 q$ for p,q:

$$\gamma(\overline{x}) = \beta_1^{2r+1}\beta_2\overline{x}, \qquad \gamma(\overline{y}) = \beta_1^{2r}\overline{y}, \quad \text{and} \quad \gamma(\overline{z}) = \beta_1^2\beta_2^2\overline{z}.$$

Thus $\overline{x},\overline{y},\overline{z}$ are $\widetilde{G}$-semi-invariant.

In Case (A) and Case (D), $f_0,h_0\in\mathbb{C}[x,y,z]$ satisfy (2). For (3), the computations above with $\Bbbk[p,q]$ in place of $E_1[[p,q]]$ show that

$$\Bbbk[p,q]^{\rho(\mathcal{G}_0)} = \Bbbk[\overline{x},\overline{y},\overline{z}] \cong \Bbbk[x,y,z]/(f_0),$$

so the quotient morphism $\pi_0 : \mathbb{A}^2_{\Bbbk} = \operatorname{Spec}\Bbbk[p,q] \to S_0$ is finite and, as in **Step 1**, étale outside the origin. Since $h_0(\overline{x},\overline{y},\overline{z}) = (pq)^m$, π_0 pulls $\frac{1}{m}\operatorname{div}(h_0|_{S_0})$ back to $\operatorname{div}(pq)$. Since π_0 is finite and étale in codimension 1 and $(\mathbb{A}^2_{\Bbbk}, \operatorname{div}(pq))$ is lc, (3) holds.

Step 6. In this step we lift $\overline{x},\overline{y},\overline{z}$ to coordinates of $E_1[[u]]$ and prove (1) and (4).

By **Step 3**, $\widetilde{G}$ acts on $E_1[[u]]$ as in the proposition, and the quotient homomorphism from $E_1[[u]]$ to R_1 is $\widetilde{G}$-equivariant. Since $(g,\zeta)\in\widetilde{G}$ acts on $E[[u]]$ as g, the elements f and h are $\widetilde{G}$-semi-invariant. By Lemma F.3(3), with $\Gamma = \widetilde{G}$ acting on $E_1 = \Bbbk((t))$, $\mathfrak{a} = (u)$, and l the restriction of the quotient homomorphism to (u), there exist $\widetilde{G}$-semi-invariant lifts $x,y,z\in(u)$ of $\overline{x},\overline{y},\overline{z}$. Since $f\in(u)^2$ (**Step 1**), the classes of $\overline{x},\overline{y},\overline{z}$ form a basis of $(u)/((u)^2+(f))$, so x,y,z form a regular system of parameters of $E_1[[u]]$. Let Ψ be the E_1-automorphism of $E_1[[u]]$ with $\Psi(u_1)=x$, $\Psi(u_2)=y$, $\Psi(u_3)=z$. Then $E_1[[x,y,z]] = E_1[[u]]$, and x,y,z are $\widetilde{G}$-semi-invariant.

By **Step 5**, the kernel (f) of $E_1[[x,y,z]] \to R_1$ is generated by $f_0 = f_0(x,y,z)$. Thus $f = v_f f_0$ for a unit v_f of $E_1[[x,y,z]]$, and since f_0 is w-homogeneous, $\operatorname{in}_w f = \sigma_f f_0$ with $\sigma_f := v_f(0)\in E_1^*$.

By **Step 5**, $v_h = h/h_0(\overline{x},\overline{y},\overline{z}) \in R_1^*$ is $\widetilde{G}$-semi-invariant. By Lemma F.3(3), with $\mathfrak{a} = E_1[[u]]$ and l the quotient homomorphism, there exists a lift $\widetilde{v}_h \in E_1[[u]]$ of v_h semi-invariant with the character of v_h. Then $h - \widetilde{v}_h h_0$ is semi-invariant and maps to 0 in R_1, so $h - \widetilde{v}_h h_0 = f\widehat{Q}$ for a semi-invariant $\widehat{Q}\in E_1[[u]]$. Let $Q\in E_1[u]$ be the sum of the terms of $\widehat{Q}$ of degree at most $\operatorname{wt}_w(h_0)$ in u. Since $\widetilde{G}$ multiplies each u_j by an element of $\Bbbk^*$, Q is semi-invariant with the character of $\widehat{Q}$, and since $f\widehat{Q} = h - \widetilde{v}_h h_0$ has the character of h, Qf has the character of h. Since $(u) = (x,y,z)$ and $w\in(\mathbb{N}^+)^3$, any element of $(u)^{\operatorname{wt}_w(h_0)+1}$ has only terms of w-weight greater than $\operatorname{wt}_w(h_0)$. Thus $f(\widehat{Q}-Q)$ has only terms of w-weight greater than $\operatorname{wt}_w(h_0)$, and

$$\operatorname{in}_w(h - Qf) = \operatorname{in}_w\big(\widetilde{v}_h h_0 + f(\widehat{Q}-Q)\big) = \sigma_h h_0, \qquad \sigma_h := \widetilde{v}_h(0)\in E_1^*.$$

This proves (4), and with the semi-invariance of x,y,z and the character of Qf shown above, it proves (1). Since $E_1 = \Bbbk((t))$ and $s = t^k$ (**Step 3**), the proposition follows. □

Proof of Theorem 16.6. Assume, to the contrary, that $(W_i \ni o_i, G_i, h_i, \tau_i)$ is a radial failure sequence with constants (m, a, δ) satisfying Set-up 16.2, such that the minimal lc center of $(W_*, \frac{1}{m}\operatorname{div}(h_*))$ is a regular curve $C \subset \operatorname{Sing}(W_*)$ and $(W_*, \frac{1}{m}\operatorname{div}(h_*))$ has exactly two coefficient-one branches along C (Definition 16.3). For any i, let $(\mathcal{X}_i \ni o_i)$ be the smooth fourfold germ with $W_i = (f_i = 0) \subset \mathcal{X}_i$ (Definition 4.1(1)). By Construction-Theorem 13.9(iii), the characters of f_i and h_i do not depend on i. We divide the proof into five steps.

Step 1. In this step we fix parameters $(s_i, u_{i1}, u_{i2}, u_{i3})$ along C and show that $\mathcal{O}_{W_i,o_i}/(s_i)$ is reduced.

By Corollary F.1(2)(b)–(d), possibly passing to a subsequence, there exist G-eigenparameters $(s_i, u_{i1}, u_{i2}, u_{i3})$ of $\mathcal{X}_i$ at o_i such that the tuples (f_i, h_i) written in $(s_i, u_{i1}, u_{i2}, u_{i3})$ have a compatible generic limit, equal to (f_*, h_*) written in G-eigenparameters (s, u_1, u_2, u_3) of $\Bbbk[[x_1, \dots, x_4]]$ with $C = (u_1 = u_2 = u_3 = 0)$, and such that for any i and any computing lift F of b_i,

$$\operatorname{mult}_F s_i \le R'', \qquad \min_F \min_{1\le j\le 3} \operatorname{mult}_F u_{ij} \to +\infty \ \text{ as } i \to +\infty, \tag{F.25}$$

where R'' is the number of Corollary F.1. We write $u_i := (u_{i1}, u_{i2}, u_{i3})$ and $u := (u_1, u_2, u_3)$. By Corollary F.1(2)(a), there exist characters $\chi_s, \chi_1, \chi_2, \chi_3$ of G such that for any $g \in G$, any i and any $j \in \{1, 2, 3\}$,

$$g(s_i) = \chi_s(g) s_i, \quad g(u_{ij}) = \chi_j(g) u_{ij}, \quad g(s) = \chi_s(g) s, \quad \text{and} \quad g(u_j) = \chi_j(g) u_j.$$

By Lemma 4.2(1), b_i has a computing lift for any i. By Corollary F.1(1), possibly passing to a subsequence, $\operatorname{mult}_F \operatorname{Jac}_{W_i} > 0$ for any i and any computing lift F of b_i. Since Jac_{W_i} is the unit ideal near a smooth point of W_i, W_i is singular at o_i for any i.

Fix i. We have $\widehat{\mathcal{O}}_{\mathcal{X}_i,o_i} = \mathbb{C}[[s_i, u_i]]$ and $\widehat{\mathcal{O}}_{W_i,o_i} = \mathbb{C}[[s_i, u_i]]/(f_i)$. Let

$$\overline{f}_i(u_i) := f_i(0, u_i) \in \mathbb{C}[[u_i]].$$

If $\overline{f}_i = 0$, then $f_i = s_i f_i'$ for some $f_i' \in \mathbb{C}[[s_i, u_i]]$. Since W_i is normal, $\widehat{\mathcal{O}}_{W_i,o_i}$ is a normal domain, so f_i' is a unit and W_i is smooth at o_i, a contradiction. Thus $\overline{f}_i \ne 0$. Assume that $\overline{f}_i = \varphi^2\varphi'$ for some $\varphi, \varphi' \in \mathbb{C}[[u_i]]$ with $\varphi(0) = 0$. Let $Z \subset \operatorname{Spec} \mathbb{C}[[s_i, u_i]]$ be the closed subset defined by

$$s_i = \varphi(u_i) = \frac{\partial f_i}{\partial s_i}(0, u_i) = 0.$$

Since W_i is singular at o_i, $\frac{\partial f_i}{\partial s_i}(0, 0) = 0$. Thus by Krull's height theorem in $\mathbb{C}[[u_i]]$, $\dim Z \ge 1$. Modulo s_i we have

$$f_i \equiv \varphi^2\varphi', \qquad \frac{\partial f_i}{\partial u_{ij}} \equiv \frac{\partial(\varphi^2\varphi')}{\partial u_{ij}} \in (\varphi), \qquad \text{and} \qquad \frac{\partial f_i}{\partial s_i} \equiv \frac{\partial f_i}{\partial s_i}(0, u_i),$$

so f_i, $\frac{\partial f_i}{\partial s_i}$, and $\frac{\partial f_i}{\partial u_{ij}}$ for $j \in \{1, 2, 3\}$ vanish on Z. Since o_i is an isolated singular point of W_i (Definition 4.1(1)), the ideal $I_i \subset \mathcal{O}_{\mathcal{X}_i,o_i}$ generated by f_i, $\frac{\partial f_i}{\partial s_i}$, and $\frac{\partial f_i}{\partial u_{ij}}$ for $j \in \{1, 2, 3\}$ is primary to the maximal ideal, hence so is $I_i\mathbb{C}[[s_i, u_i]]$. Thus Z is the closed point, which contradicts $\dim Z \ge 1$. Therefore $\overline{f}_i$ is square-free, and

$$\mathcal{O}_{W_i,o_i}/(s_i) \subset \widehat{\mathcal{O}}_{W_i,o_i}/(s_i) \cong \mathbb{C}[[u_i]]/(\overline{f}_i)$$

is reduced. Since $\overline{f}_i \ne 0$, $s_i|_{W_i} \ne 0$. Any height-one prime $\mathfrak{p}$ of $\mathcal{O}_{W_i,o_i}$ with $s_i \in \mathfrak{p}$ is a minimal prime of the reduced ring $\mathcal{O}_{W_i,o_i}/(s_i)$, so $\mathcal{O}_{W_i,\mathfrak{p}}/(s_i)$ is a field. Since W_i is normal, we obtain:

$$s_i \text{ is a uniformizer of the DVR } \mathcal{O}_{W_i,\mathfrak{p}} \text{ for any height-one prime } \mathfrak{p} \ni s_i \text{ of } \mathcal{O}_{W_i,o_i}. \tag{F.26}$$

Step 2. In this step we bring the limit at the generic point of C into normal form.

Let $\overline{\Bbbk((s))}$ be an algebraic closure of $\Bbbk((s))$. The (u)-adic completion of $\Bbbk[[s, u]]_{(u)}$ is $\Bbbk((s))[[u]]$, so $\Bbbk((s))[[u]]/(f_*)$ is the completion of the local ring of W_* at the generic point of C. We check

the hypotheses of Proposition F.5 for $f := f_*$, $h := h_*$ and the action of G on $\Bbbk((s))[[u]]$ by $g(s) = \chi_s(g)s$ and $g(u_j) = \chi_j(g)u_j$, trivial on $\Bbbk$.

(1) Since $(W_*, \frac{1}{m}\operatorname{div}(h_*))$ is lc, $h_*|_{W_*} \neq 0$. Since $\Bbbk[[s,u]]/(f_*)$ is a domain, the homomorphism
$$\Bbbk[[s,u]]/(f_*) \to \Bbbk((s))[[u]]/(f_*)$$
is injective. Thus $h_* \notin (f_*)$ in $\Bbbk((s))[[u]]$.

(2) The group G is cyclic (Definition 4.1(1)), hence finite abelian, and the power series f_* and h_* are G-semi-invariant, so $g(f_*) \in \Bbbk^* f_*$ and $g(h_*) \in \Bbbk^* h_*$ for any $g \in G$.

(3) Let $g \in G$ act trivially on $\Bbbk((s))[[u]]/(f_*)$. Since the map $\Bbbk[[s,u]]/(f_*) \to \Bbbk((s))[[u]]/(f_*)$ is injective by (1), g acts trivially on $\Bbbk[[s,u]]/(f_*)$. Since $C \subset \operatorname{Sing}(W_*)$ contains the closed point of W_*, $f_* \in (s,u_1,u_2,u_3)^2$, so the classes of s,u_1,u_2,u_3 form a basis of the cotangent space of W_* at the closed point. Thus $\chi_s(g) = \chi_1(g) = \chi_2(g) = \chi_3(g) = 1$, so g acts trivially on $\mathbb{C}[[s_i,u_i]]$, hence on W_i. Since G acts faithfully on W_i, $g = 1$.

(4) Since W_* is normal (Set-up 16.2(3)) and localizations and completions of excellent normal local rings are normal, $\Bbbk((s))[[u]]/(f_*)$ is normal.

(5) Let $T := \operatorname{Spec} \overline{\Bbbk((s))}[[u]]/(f_*)$, the generic transverse surface of W_* along C (Definition 15.1). The morphism $T \to W_*$ is regular (completion at the generic point of C followed by the extension of the coefficient field to $\overline{\Bbbk((s))}$ [Kaw21, Remark 2.6]). Thus the base change of a log resolution $Y \to W_*$ of $(W_*, \frac{1}{m}\operatorname{div}(h_*))$ is a log resolution of $(T, \frac{1}{m}\operatorname{div}(h_*))$, since the base change $Y \times_{W_*} T \to Y$ of $T \to W_*$ is regular and hence pulls back Y and the strata of a simple normal crossing divisor on Y to regular schemes [dFEM11, Remark 2.10]. Moreover, since relative canonical divisors pull back along the regular base change $T \to W_*$, any prime divisor of the base change lying over a prime divisor D of Y has log discrepancy $a(D, W_*, \frac{1}{m}\operatorname{div}(h_*))$. Since $(W_*, \frac{1}{m}\operatorname{div}(h_*))$ is lc, so is $(T, \frac{1}{m}\operatorname{div}(h_*))$, and T is a singular Du Val germ by Definition 16.3.

(6) By the hypothesis of the theorem and Definition 16.3, for the smooth uniformizing cover $\pi : T' = \operatorname{Spec} \overline{\Bbbk((s))}[[p,q]] \to T$, the pullback $\pi^* \frac{1}{m}\operatorname{div}(h_*)$ has exactly two prime components with coefficient one.

By Proposition F.5, there exists an integer $k \geq 1$ such that, for a k-th root t of s and the group
$$\widetilde{G} := \{(g,\zeta) \in G \times \Bbbk^* \mid \zeta^k = \chi_s(g)\}$$
acting on $\Bbbk((t))[[u]]$ by $(g,\zeta)(t) = \zeta t$ and $(g,\zeta)(u_j) = \chi_j(g)u_j$, there exist non-zero elements σ_f and σ_h of $\Bbbk((t))$, a polynomial $Q \in \Bbbk((t))[u_1,u_2,u_3]$, an automorphism Ψ of $\Bbbk((t))[[u]]$ over $\Bbbk((t))$, a vector $w \in (\mathbb{N}^+)^3$ and polynomials $f_0, h_0 \in \mathbb{C}[x,y,z]$ satisfying (1)–(4) of Proposition F.5 for $x := \Psi(u_1)$, $y := \Psi(u_2)$, and $z := \Psi(u_3)$. Since $\chi_s(g)$ is a root of unity, $\widetilde{G} \subset G \times \mathbb{C}^*$, and $\widetilde{G}$ is an abelian group with
$$|\widetilde{G}| = k|G| \leq kM.$$
By Proposition F.5(2), either $f_0 = xy - z^n$ or $f_0 = x^2 + y^2 - z^n$ with $n \geq 2$, $h_0 = z^m$, and $w = (n,n,2)$, or $f_0 = x^2 - zy^2 + 4z^{r+1}$ with $r \geq 1$, $h_0 = z^{m/2}$, and $w = (r+1,r,2)$. In each case, $f_0 \in (x,y,z)^2$, $(f_0 = 0) \subset \mathbb{A}^3$ is a normal surface whose only singular point is the Du Val point 0 (of type A_{n-1}, resp. D_{r+2}), $z \nmid f_0$, $\deg h_0 \leq m$, the polynomials f_0 and h_0 are w-homogeneous, and
$$w_1 + w_2 + w_3 - \operatorname{wt}_w(f_0) = \begin{cases} 2, & w = (n,n,2), \\ 1, & w = (r+1,r,2), \end{cases} \qquad \operatorname{wt}_w(h_0) = m\big(w_1 + w_2 + w_3 - \operatorname{wt}_w(f_0)\big).$$

Step 3. In this step we take the k-th root of s_i on W_i.

For any i, let
$$\widetilde{\mathcal{X}}_i := \operatorname{Spec} \mathcal{O}_{\mathcal{X}_i,o_i}[t]/(t^k - s_i), \qquad V_i := \operatorname{Spec} \mathcal{O}_{W_i,o_i}[t]/(t^k - s_i),$$
let t_i be the class of t, and let $\phi_i : \widetilde{\mathcal{X}}_i \to \mathcal{X}_i$ be the projection, whose restriction $V_i \to W_i$ is again written ϕ_i. Then $V_i = (f_i = 0) \subset \widetilde{\mathcal{X}}_i$. The fiber of ϕ_i over o_i is $\operatorname{Spec} \mathbb{C}[t]/(t^k)$, so $\widetilde{\mathcal{X}}_i$ and V_i are

local with one closed point $\widetilde{o}_i$ over o_i. The maximal ideal of $\mathcal{O}_{\widetilde{\mathcal{X}}_i,\widetilde{o}_i}$ is $(s_i,u_i,t_i)=(t_i,u_i)$, so $(\widetilde{\mathcal{X}}_i\ni\widetilde{o}_i)$ is a smooth fourfold germ with regular parameters $(t_i,u_{i1},u_{i2},u_{i3})$.

By (F.26), V_i is regular in codimension 1, and as a hypersurface it is S2, so V_i is normal and $\phi_i:V_i\to W_i$ is a finite morphism of degree k, ramified in codimension 1 exactly along the reduced divisor $\operatorname{div}(t_i|_{V_i})$, with index k. Thus

$$K_{V_i}=\phi_i^*K_{W_i}+(k-1)\operatorname{div}(t_i|_{V_i}). \tag{F.27}$$

The group $\widetilde{G}$ acts on $\mathcal{O}_{\mathcal{X}_i,o_i}[t]$ by g on $\mathcal{O}_{\mathcal{X}_i,o_i}$ and by $t\to\zeta t$ for $(g,\zeta)\in\widetilde{G}$. Since

$$(g,\zeta)(t^k-s_i)=\chi_s(g)(t^k-s_i),$$

and f_i,h_i are G-semi-invariant, $\widetilde{G}$ acts on $\widetilde{\mathcal{X}}_i$ and on V_i, fixes $\widetilde{o}_i$, and ϕ_i is equivariant with respect to the projection $\widetilde{G}\to G$. The parameters t_i,u_{i1},u_{i2},u_{i3} are $\widetilde{G}$-eigen with the characters $(g,\zeta)\to\zeta$, $\chi_1(g)$, $\chi_2(g)$, $\chi_3(g)$, and f_i,h_i are $\widetilde{G}$-semi-invariant with characters independent of i. The projection $\widetilde{G}\to G$ is surjective with kernel $\{1\}\times\mu_k$. If $(g,\zeta)\in\widetilde{G}$ acts trivially on V_i, then g acts trivially on $\mathcal{O}_{W_i,o_i}\subset\mathcal{O}_{V_i,\widetilde{o}_i}$, so $g=1$ since G acts faithfully on W_i, and $\zeta t_i=t_i$ with $t_i\neq0$ gives $\zeta=1$. Thus $\widetilde{G}$ acts faithfully on V_i, and

$$V_i/\widetilde{G}=W_i/G.$$

Let E be a prime divisor over V_i, with restriction F over W_i and ramification index ρ over F (Definition 2.2). Since ϕ_i is finite of degree k, $\rho\le k$. The center of F on W_i is the image of the center of E on V_i. Since $V_i/\widetilde{G}=W_i/G$ and in characteristic zero the order of an inertia group is the ramification index over the quotient,

$$e_E=\rho e_F, \tag{F.28}$$

where $e_E:=|I_{\widetilde{G}}(E)|$ and $e_F:=|I_G(F)|$. Any prime divisor over W_i is the restriction of a prime divisor over V_i.

Step 4. In this step we construct charts over V_i by Lemma F.4.

For any i, let $\mathcal{F}_i$ be the set of prime divisors over V_i whose restrictions are computing lifts of b_i. Let $E\in\mathcal{F}_i$, with restriction F and ramification index ρ. Since F has center o_i, E has center $\widetilde{o}_i$. Since $s_i=t_i^k$ and $1\le\rho\le k$, by (F.25),

$$\operatorname{mult}_E t_i=\frac{\rho\operatorname{mult}_F s_i}{k}\le R'',\qquad \operatorname{mult}_E u_{ij}=\rho\operatorname{mult}_F u_{ij}\ge\operatorname{mult}_F u_{ij}\ \text{ for } j\in\{1,2,3\}.$$

Thus $\operatorname{mult}_E t_i$ is bounded independently of i and $E\in\mathcal{F}_i$, and

$$\min_{E\in\mathcal{F}_i}\min_{1\le j\le3}\operatorname{mult}_E u_{ij}\to+\infty\ \text{ as } i\to+\infty.$$

Written in $(t_i,u_{i1},u_{i2},u_{i3})$, the tuple (f_i,h_i) is $(f_i(t_i^k,u_i),h_i(t_i^k,u_i))$, where $f_i(s_i,u_i)$ and $h_i(s_i,u_i)$ are the expressions in $(s_i,u_{i1},u_{i2},u_{i3})$. For any positive integer d, the substitution $s=t^k$ does not decrease degrees, so it induces a linear map λ_d from the space of d-jets in (s,u) to the space of d-jets in (t,u). Let Z_d be the irreducible closed subset of Definition 2.5 for the tuples written in $(s_i,u_{i1},u_{i2},u_{i3})$, and Z_d' the closure of $\lambda_d(Z_d)$. The d-jet of $(f_*(t^k,u),h_*(t^k,u))$ is the image under λ_d of the d-jet of (f_*,h_*), so it maps to the generic point of Z_d'. For any non-empty open subset $U\subset Z_d'$, $Z_d\cap\lambda_d^{-1}(U)$ is a non-empty open subset of Z_d, so U contains the d-jets of the tuples written in $(t_i,u_{i1},u_{i2},u_{i3})$ for all but finitely many i. Since $\operatorname{mult}_0\varphi(t^k,u)\le k\operatorname{mult}_0\varphi$ for any non-zero $\varphi\in\mathbb{C}[[s,u]]$, the orders at 0 of the tuples written in $(t_i,u_{i1},u_{i2},u_{i3})$ are bounded. Thus by Definition 2.5, the tuples (f_i,h_i) written in $(t_i,u_{i1},u_{i2},u_{i3})$ have the compatible generic limit $(f_*(t^k,u),h_*(t^k,u))$ with coefficient field $\Bbbk$. In $\Bbbk((t))[[u]]$ we have $f_*(t^k,u)=f_*$ and $h_*(t^k,u)=h_*$ through $s=t^k$, so by Proposition F.5(4),

$$\operatorname{in}_w f_*(t^k,u)=\sigma_f f_0,\qquad \operatorname{in}_w\big(h_*(t^k,u)-Qf_*(t^k,u)\big)=\sigma_h h_0.$$

Let N_1 be the integer N of Theorem 5.10 for f_0 and m, with kM and m in place of M and L. The hypotheses of Lemma F.4 with N_1, $\widetilde{G}$, $(\widetilde{\mathcal{X}}_i\ni\widetilde{o}_i)$, and $(f_*(t^k,u),h_*(t^k,u))$ in place of

J, G, $(\mathcal{X}_i \ni o_i)$, and (f_*, h_*) hold by **Steps 2 and 3**, the first two paragraphs of **Step 4**, and Proposition F.5(1). By Lemma F.4, there exists a positive integer N such that

$$\alpha := N\big(w_1+w_2+w_3-\mathrm{wt}_w(f_0)\big)-c_\Psi-c_f \ge 0, \qquad \mu := Nm\big(w_1+w_2+w_3-\mathrm{wt}_w(f_0)\big)+c_h \ge 0,$$

where c_Ψ, c_f, c_h are the integers of Lemma F.4, and, possibly passing to a subsequence, there exist $\widetilde{G}$-equivariant birational morphisms

$$\pi_i : (\mathcal{X}_i' \ni o_i') \to (\widetilde{\mathcal{X}}_i \ni \widetilde{o}_i),$$

smooth at o_i' with $\widetilde{G}$-eigenparameters (t_i, x_i, y_i, z_i), such that the following hold, where V_i' denotes $(\pi_i)_*^{-1}V_i$, the morphism $V_i' \to V_i$ is again written π_i, and $S_i := (t_i = 0) \subset V_i'$.

(1) $V_i' = (f_i' = 0)$ near o_i', V_i' is normal near o_i', and
$$\pi_i^* \operatorname{div}(h_i|_{V_i}) = \operatorname{div}(h_i'|_{V_i'}) + \mu S_i,$$
where $f_i' := f_0(x_i, y_i, z_i) + t_i^{N_1}\xi_i$ and $h_i' := h_0(x_i, y_i, z_i) + t_i^{N_1}\eta_i$ are $\widetilde{G}$-semi-invariant and ξ_i, η_i are regular functions on $\mathcal{X}_i'$.

(2) $K_{V_i'} = \pi_i^* K_{V_i} + \alpha S_i$.

(3) Any $E \in \mathcal{F}_i$ has center o_i' on V_i', and any prime divisor over V_i' with center o_i' has center $\widetilde{o}_i$ on V_i.

Step 5. In this step we compare the charts with W_i and conclude by Theorem 5.10.

Since $f_i'|_{(t_i=0)} = f_0(x_i, y_i, z_i)$ and x_i, y_i, z_i restrict to regular parameters of the smooth threefold germ $(t_i = 0) \subset \mathcal{X}_i'$ at o_i', the germ $(S_i \ni o_i')$ is the Du Val germ $(f_0 = 0)$. In particular S_i is a prime divisor and $\operatorname{div}(t_i|_{V_i'}) = S_i$ near o_i'. Let

$$\beta := \frac{\mu}{m} - \alpha - (k-1), \qquad B_i' := \frac{1}{m}\operatorname{div}(h_i'|_{V_i'}) + \beta S_i.$$

Since $\pi_i^* \operatorname{div}(t_i|_{V_i}) = S_i$ near o_i', (F.27) and (1), (2) of **Step 4** imply that near o_i',

$$K_{V_i'} - (\alpha+k-1)S_i = (\phi_i \circ \pi_i)^* K_{W_i}, \qquad K_{V_i'} + B_i' = (\phi_i \circ \pi_i)^*\Big(K_{W_i} + \frac{1}{m}\operatorname{div}(h_i)\Big).$$

Thus for any prime divisor E over V_i' whose center contains o_i', with restriction F and ramification index ρ,

$$\text{(F.29)}\quad a(E, V_i', 0) + (\alpha+k-1)\operatorname{mult}_E t_i = \rho a(F, W_i, 0), \qquad a(E, V_i', B_i') = \rho a\Big(F, W_i, \frac{1}{m}\operatorname{div}(h_i)\Big).$$

We take $E = S_i$ in (F.29). Since $h_i'|_{S_i} = h_0|_{S_i} \ne 0$ by $z \nmid f_0$, $a(S_i, V_i', B_i') = 1 - \beta$. Since the center of F contains o_i and $(W_i, \frac{1}{m}\operatorname{div}(h_i))$ is klt near o_i (Set-up 16.2(1)), we obtain $\beta < 1$.

Let E be a prime divisor over V_i' with center o_i', with restriction F. By **Step 4**(3), E has center $\widetilde{o}_i$ on V_i, so F has center o_i. By (F.28) and (F.29),

$$\frac{a(E, V_i', B_i')}{e_E} = \frac{a(F, W_i, \frac{1}{m}\operatorname{div}(h_i))}{e_F} \ge b_i.$$

A computing lift of b_i is the restriction of some $E \in \mathcal{F}_i$, which has center o_i' by **Step 4**(3), and then equality holds. Thus

$$\operatorname{mld}((V_i' \ni o_i', B_i')/\widetilde{G}) = b_i,$$

and the restriction of any prime divisor E over V_i' with center o_i' and $a(E, V_i', B_i')/e_E = b_i$ is a computing lift of b_i.

We check the hypotheses of Theorem 5.10 for f_0 and m, with kM and m in place of M and L, the germ $(\mathcal{X}_i' \ni o_i')$ with the regular parameters x_i, y_i, z_i, t_i in place of x_1, x_2, x_3, s, the group $\widetilde{G}$, the functions $f := f_i'$ and $h := h_i'$, and the number β. Let χ' be the character of f_i' and let pr be the $\widetilde{G}$-equivariant projection onto the sum of the eigenspaces of the characters other than χ'. Each monomial of $f_0(x_i, y_i, z_i)$ is $\widetilde{G}$-semi-invariant, and $\operatorname{pr}(f_i') = 0$, so the sum of the monomials of $f_0(x_i, y_i, z_i)$ whose characters differ from χ' equals $-\operatorname{pr}(t_i^{N_1}\xi_i) \in (t_i)$. Since

$(t_i) \cap \mathbb{C}[x_i, y_i, z_i] = 0$, $f_0(x_i, y_i, z_i)$ is $\widetilde{G}$-semi-invariant with the character of f_i'. The monomial $h_0(x_i, y_i, z_i)$ is $\widetilde{G}$-semi-invariant. The group $\widetilde{G}$ acts faithfully on V_i, hence on V_i' and on $\mathcal{X}_i'$. Let $\mathfrak{n}_i$ be the maximal ideal of $\mathcal{O}_{\mathcal{X}_i', o_i'}$ and $W_{i,0} := (f_0(x_i, y_i, z_i) = 0) \subset \mathcal{X}_i'$. If $g \in \widetilde{G}$ acts trivially on $W_{i,0}$, then, since $f_0(x_i, y_i, z_i) \in \mathfrak{n}_i^2$, g fixes x_i, y_i, z_i, t_i, so g acts trivially on $\mathcal{X}_i'$ and $g = 1$. Moreover $V_i' = (f_i' = 0)$ is normal near o_i', $f_i' - f_0(x_i, y_i, z_i) \in (t_i)$, $h_i' - h_0(x_i, y_i, z_i) \in (t_i)$, $\deg h_0 \le m$, h_0 is non-zero on $(f_0 = 0)$, and $((f_0 = 0), \frac{1}{m}\operatorname{div}(h_0))$ is lc near 0 by Proposition F.5(3), since $f_0, h_0 \in \mathbb{C}[x, y, z]$. Finally $\beta < 1$, and

$$f_i' - f_0(x_i, y_i, z_i),\ h_i' - h_0(x_i, y_i, z_i) \in (t_i^{N_1}) \subset \mathfrak{n}_i^{N_1}.$$

Since $\operatorname{div}(t_i|_{V_i'}) = S_i$, the boundary B of Theorem 5.10 is B_i'.

By Theorem 5.10(2), there exists a prime divisor E_i over V_i' with center o_i' such that

$$\frac{a(E_i, V_i', B_i')}{e_{E_i}} = \operatorname{mld}((V_i' \ni o_i', B_i')/\widetilde{G}) = b_i, \qquad a(E_i, V_i', 0) \le N_1, \qquad \text{and} \qquad \operatorname{mult}_{E_i} t_i \le N_1.$$

Let F_i be the restriction of E_i and ρ_i the ramification index of E_i. Then F_i is a computing lift of b_i. Let $c_2 := (\alpha + k)N_1$, which does not depend on i. Since $\rho_i \ge 1$, $a(F_i, W_i, 0) > 0$ and $\alpha + k - 1 \ge 0$, (F.29) implies

$$a(F_i, W_i, 0) \le \rho_i a(F_i, W_i, 0) = a(E_i, V_i', 0) + (\alpha + k - 1)\operatorname{mult}_{E_i} t_i \le (\alpha + k)N_1 = c_2.$$

By Lemma 4.2(2) for the computing lift F_i of $b_i = q_i(1)$, and since $e_{F_i} \ge 1$ and $b_i \ge 0$,

$$q_i(\tau_i) \le b_i + (1 - \tau_i)\Big(\frac{a(F_i, W_i, 0)}{e_{F_i}} - b_i\Big) \le b_i + c_2(1 - \tau_i).$$

By Definition 4.1(3),

$$a \le q_i(\tau_i) \le b_i + c_2(1 - \tau_i) \le a - \delta + c_2(1 - \tau_i),$$

so $\delta \le c_2(1 - \tau_i)$ for any i. Since $\delta > 0$ and $\lim_{i \to +\infty} \tau_i = 1$, we obtain a contradiction. The theorem follows. □

F.2. **Transverse types E.** In this subsection we prove Theorem 17.4 and Proposition 17.5 of Section 17.

Set-up F.6. Let m, (f_0, w, d), $(\mathcal{X} \ni o)$, G, $(W \ni o)$, x, y, z, t, R, J, and h be as follows.

(1) m is a positive integer.
(2) (f_0, w, d), with $w = (w_1, w_2, w_3)$, is one of the triples
$$(x^2 + y^3 + yz^3, (9, 6, 4), 18) \quad \text{and} \quad (x^2 + y^3 + z^5, (15, 10, 6), 30),$$
called type E_7 and E_8, respectively.
(3) $(\mathcal{X} \ni o)$ is a smooth fourfold germ with an action of a finite group G fixing o, and $(W \ni o) \subset (\mathcal{X} \ni o)$ is a G-invariant isolated terminal hypersurface threefold germ on which G acts faithfully.
(4) $\widehat{\mathcal{O}}_{\mathcal{X},o} = \mathbb{C}[[x, y, z, t]]$ for G-semi-invariant x, y, z, t with characters $\chi_x, \chi_y, \chi_z, \chi_t$, and W is defined by the G-semi-invariant series $f := f_0(x, y, z) + R(y, z, t)$ with $R \in \mathbb{C}[[y, z, t]]$.
(5) J is a positive integer such that any monomial of R has weight greater than Jd for the weights $(Jw_1, Jw_2, Jw_3, 1)$ on (x, y, z, t).
(6) h is a mark on $(W \ni o)$ such that $(W, \frac{1}{m}\operatorname{div}(h))$ is lc near o.

By (2), f_0 is w-homogeneous of weight d, and

$$w_1 + w_2 + w_3 = d + 1.$$

Let $D := \frac{1}{m}\operatorname{div}(h)$, and let q be the radial function of $(W \ni o, G, h, m)$. Let $\widehat{W} := \operatorname{Spec} \widehat{\mathcal{O}}_{W,o}$ and

$$\widehat{\mathcal{X}} := \operatorname{Spec} \widehat{\mathcal{O}}_{\mathcal{X},o} = \operatorname{Spec} \mathbb{C}[[x, y, z, t]].$$

Let Y_J be the t-chart of the weighted blow-up of $\widehat{W}$ with the weights $(Jw_1, Jw_2, Jw_3, 1)$ on (x, y, z, t), with coordinates

$$x' := \frac{x}{t^{Jw_1}}, \qquad y' := \frac{y}{t^{Jw_2}}, \qquad z' := \frac{z}{t^{Jw_3}}, \quad \text{and} \quad t.$$

Let $S := (t = 0) \subset Y_J$, which is the surface $(f_0(x', y', z') = 0)$. For a prime divisor F over W with center o, let $\widehat{F}$ be the formal prime divisor over $\widehat{W}$ corresponding to F (Lemma 2.12(3)). Let $\operatorname{center}_{Y_J} F$ be the center of $\widehat{F}$ on Y_J if it exists. For any $\phi \in \mathbb{C}[[x, y, z, t]]$ we write

$$\operatorname{mult}_F \phi := \operatorname{mult}_{\widehat{F}}(\phi|_{\widehat{W}}).$$

Lemma F.7. *Notation and conditions as in Set-up F.6. Write the image of h in $\widehat{\mathcal{O}}_{W,o}$ as $u + xv$ with $u, v \in \mathbb{C}[[y, z, t]]$, and for any real number $r > 0$ let $\nu(r)$ be the minimum of*

$$r(\alpha_1 w_1 + \alpha_2 w_2 + \alpha_3 w_3) + \alpha_4$$

over the monomials $x^{\alpha_1} y^{\alpha_2} z^{\alpha_3} t^{\alpha_4}$ of $u + xv$, and let $\varphi(r) := 1 + r - \frac{\nu(r)}{m}$. Then the following hold.

(1) For any coprime positive integers N, n with $N \le Jn$, the exceptional divisor of the weighted blow-up of $\widehat{W}$ with the weights (Nw_1, Nw_2, Nw_3, n) on (x, y, z, t) defines a prime divisor $E_{N,n}$ over W with center o, and

$$a(E_{N,n}, W, 0) = N + n \quad \text{and} \quad \operatorname{mult}_{E_{N,n}} h = n\nu(N/n).$$

Moreover, $\varphi(r) \ge 0$ for any real number $r \in (0, J]$.

(2) If $J > m$, then any prime divisor F over W with center o and $\operatorname{center}_{Y_J} F \subset S$ satisfies

$$a(F, W, D) \ge \varphi(J) \operatorname{mult}_F t.$$

Proof. Let $g := u + xv \in \mathbb{C}[[x, y, z, t]]$. Since $f = x^2 + (f_0 - x^2) + R$ is a monic polynomial of degree 2 in x over $\mathbb{C}[[y, z, t]]$ and $f(x, 0, 0, 0) = x^2$, the Weierstrass division theorem implies that $\widehat{\mathcal{O}}_{W,o}$ is a free $\mathbb{C}[[y, z, t]]$-module with basis $1, x$. Thus u and v exist and are unique, and ν is well defined. Since $h \ne 0$ in $\mathcal{O}_{W,o} \subset \widehat{\mathcal{O}}_{W,o}$, we have $g \notin (f)$. Write

$$g = \sum_{\alpha \in \mathbb{N}^4} c_\alpha x^{\alpha_1} y^{\alpha_2} z^{\alpha_3} t^{\alpha_4}, \qquad U_\alpha := \alpha_1 w_1 + \alpha_2 w_2 + \alpha_3 w_3.$$

Then $c_\alpha = 0$ if $\alpha_1 \ge 2$, and $\nu(r) = \min_{c_\alpha \ne 0}(rU_\alpha + \alpha_4)$. For $r > 0$, the function $rU_\alpha + \alpha_4$ is increasing in (U_α, α_4) for the product order on $\mathbb{N}^2$, and the set of minimal elements of $\{(U_\alpha, \alpha_4) \mid c_\alpha \ne 0\}$ is finite. Thus ν is the minimum of finitely many affine functions of r. In particular, ν is continuous on $(0, +\infty)$.

(1) Let $r := N/n$ and $w' := (Nw_1, Nw_2, Nw_3, n)$. For any monomial $y^{\alpha_2} z^{\alpha_3} t^{\alpha_4}$ of R (so $\alpha_1 = 0$), we have $J(U_\alpha - d) + \alpha_4 > 0$ and $\alpha_4 \ge 0$, so since $0 < r \le J$,

$$NU_\alpha + n\alpha_4 - Nd = n\Big(\frac{r}{J}\big(J(U_\alpha - d) + \alpha_4\big) + \Big(1 - \frac{r}{J}\Big)\alpha_4\Big) > 0.$$

Thus $\operatorname{wt}_{w'}(R) > Nd = \operatorname{wt}_{w'}(f_0)$, and

$$\operatorname{in}_{w'} f = f_0.$$

Since $f_0 - x^2 \in \mathbb{C}[y, z]$ is not a square, f_0 is irreducible, so $\mathbb{C}[x, y, z, t]/(f_0)$ is an integral domain. Since $\gcd(w_1, w_2, w_3) = 1$, we have $\gcd(Nw_1, Nw_2, Nw_3, n) = \gcd(N, n) = 1$. By [Liu26b, Lemma 4.12], the exceptional divisor of the w'-weighted blow-up of $\widehat{W}$ is a formal prime divisor $\widehat{E}$ over $\widehat{W}$ with center the closed point,

$$a(\widehat{E}, \widehat{W}, 0) = N(w_1 + w_2 + w_3) + n - \operatorname{wt}_{w'}(f) = N(d + 1) + n - Nd = N + n,$$

and $\operatorname{mult}_{\widehat{E}}(\phi|_{\widehat{W}}) = \operatorname{wt}_{w'}(\phi)$ for any $\phi \in \mathbb{C}[[x, y, z, t]]$ with $f_0 \nmid \operatorname{in}_{w'} \phi$. Since the weights w' are positive and u, v do not involve x, $\operatorname{in}_{w'} g$ is a non-zero polynomial of degree at most 1 in x, which is not divisible by f_0. Thus

$$\operatorname{mult}_{\widehat{E}} h = \operatorname{wt}_{w'}(g) = \min_{c_\alpha \ne 0}(NU_\alpha + n\alpha_4) = n\nu(N/n).$$

By Lemma 2.12(3), $\widehat{E}$ restricts to a prime divisor $E_{N,n}$ over W with center o, with

$$a(E_{N,n}, W, 0) = a(\widehat{E}, \widehat{W}, 0) \quad \text{and} \quad \operatorname{mult}_{E_{N,n}} \phi = \operatorname{mult}_{\widehat{E}} \phi$$

for any $\phi \in \mathcal{O}_{W,o}$, which proves the first sentence of (1).

Let $r = N/n \in (0, J]$ be a rational number with $\gcd(N, n) = 1$. Then $N \le Jn$, and since (W, D) is lc near o,

$$0 \le a(E_{N,n}, W, D) = N + n - \frac{n}{m}\nu(N/n) = n\varphi(r).$$

Since ν is continuous and $\mathbb{Q} \cap (0, J]$ is dense in $(0, J]$, (1) follows.

(2) We divide the proof of (2) into four steps. In **Steps 1–3**, S also denotes the surface $(f_0 = 0)$ in $\mathbb{A}^3 := \operatorname{Spec} \mathbb{C}[x', y', z']$.

Step 1. In this step we show that for any $P \in \mathbb{C}[x', y', z']$ with $P|_S \neq 0$ such that any monomial of P has w-weight at most m, the pair $(S, \frac{1}{m}\operatorname{div}(P|_S))$ is lc.

Let $\Pi \subset \mathrm{SL}_2(\mathbb{C})$ be the binary octahedral or icosahedral group if f_0 is of type E_7 or E_8, respectively. By [DBW18, Sections 5.1-5.2], after rescaling, there exist Π-invariant homogeneous polynomials of degrees $2w_1, 2w_2, 2w_3$ on $\mathbb{A}^2$ which generate $\mathbb{C}[\mathbb{A}^2]^\Pi$ and define the quotient morphism $\pi : \mathbb{A}^2 \to S = \mathbb{A}^2/\Pi$. Thus π^* maps a w-homogeneous polynomial of weight U to a homogeneous polynomial of degree $2U$. Since an element of $\mathrm{SL}_2(\mathbb{C})$ of finite order with eigenvalue 1 is the identity, Π acts freely on $\mathbb{A}^2 \backslash \{0\}$, so π is étale outside 0. Let $p := \pi^* P$. Then $\pi^* \operatorname{div}(P|_S) = \operatorname{div}(p)$, and since π is finite and étale outside 0, we only need to show that $(\mathbb{A}^2, \frac{1}{m}\operatorname{div}(p))$ is lc.

The polynomial p is non-zero, Π-invariant, and $\deg p \le 2m$. If $\deg p = 0$, then $\operatorname{div}(p) = 0$. Assume that $\deg p \ge 1$.

Suppose that an irreducible polynomial ℓ_1 divides p with multiplicity at least m. Let Φ be the product of irreducible polynomials defining the distinct curves in the Π-orbit of $(\ell_1 = 0)$. Since p is Π-invariant, any irreducible factor of Φ divides p with multiplicity at least m, so $m \deg \Phi \le \deg p \le 2m$, that is,

$$1 \le \deg \Phi \le 2.$$

For any $\gamma \in \Pi$, $\gamma^*\Phi \in \mathbb{C}^*\Phi$, so the homogeneous part Φ' of Φ of degree $\deg \Phi$ satisfies $\gamma^*\Phi' \in \mathbb{C}^*\Phi'$. Thus the zero set of Φ' in $\mathbb{P}^1$ is a non-empty set of at most two points, invariant under the image $\overline{\Pi} \subset \mathrm{PGL}_2(\mathbb{C})$ of Π. A finite subgroup of $\mathrm{PGL}_2(\mathbb{C})$ preserving a point (resp. a set of two points) is cyclic (resp. cyclic or dihedral), while $\overline{\Pi} \cong S_4$ or A_5, a contradiction. Thus any coefficient of $\frac{1}{m}\operatorname{div}(p)$ is < 1, so $(\mathbb{A}^2, \frac{1}{m}\operatorname{div}(p))$ is klt at the generic point of any curve, and the non-klt locus Σ of $(\mathbb{A}^2, \frac{1}{m}\operatorname{div}(p))$ is a finite set.

If $\Sigma = \emptyset$, then $(\mathbb{A}^2, \frac{1}{m}\operatorname{div}(p))$ is klt. Assume that $\Sigma \neq \emptyset$. Let $H_\infty := \mathbb{P}^2 \backslash \mathbb{A}^2$ be the line at infinity, let $\overline{\operatorname{div}(p)}$ be the closure of $\operatorname{div}(p)$ in $\mathbb{P}^2$, a curve of degree $\deg p$, let Q be a general conic in $\mathbb{P}^2$, and let

$$\beta := 1 - \frac{\deg p}{2m} \in [0, 1), \qquad \Delta := H_\infty + \frac{1}{m}\overline{\operatorname{div}(p)} + \beta Q.$$

Then $K_{\mathbb{P}^2} + \Delta \sim_{\mathbb{Q}} 0$, since $-3 + 1 + \frac{\deg p}{m} + 2\beta = 0$. Let $\psi : Z \to \mathbb{P}^2$ be a log resolution of $(\mathbb{P}^2, H_\infty + \overline{\operatorname{div}(p)})$. Since Q is a general conic, by Bertini's theorem Q avoids the centers of the blow-ups composing ψ, and $\psi_*^{-1}Q$ is a smooth curve intersecting the other components of the boundary transversally. Since moreover $\beta < 1$, we have

$$\operatorname{Nklt}(\mathbb{P}^2, \Delta) = \operatorname{Nklt}\left(\mathbb{P}^2, H_\infty + \frac{1}{m}\overline{\operatorname{div}(p)}\right) = H_\infty \sqcup \Sigma.$$

Since $\Sigma \neq \emptyset$, $\operatorname{Nklt}(\mathbb{P}^2, \Delta)$ is disconnected. By Theorem 7.2 for the morphism $\mathbb{P}^2 \to \operatorname{Spec} \mathbb{C}$, $(\mathbb{P}^2, \Delta)$ is lc. Since $\frac{1}{m}\operatorname{div}(p) \le \Delta|_{\mathbb{A}^2}$, $(\mathbb{A}^2, \frac{1}{m}\operatorname{div}(p))$ is lc.

Step 2. In this step we show that $(\mathbb{A}^3, S + \frac{1}{m}\operatorname{div}(P_J))$ is lc on an open neighborhood of S in $\mathbb{A}^3$, where

$$P_J := \sum_{JU_\alpha + \alpha_4 = \nu(J)} c_\alpha x'^{\alpha_1} y'^{\alpha_2} z'^{\alpha_3}.$$

Since $Jw_i \geq 1$ for any i, any α with $JU_\alpha + \alpha_4 = \nu(J)$ satisfies $\alpha_1 + \alpha_2 + \alpha_3 + \alpha_4 \leq \nu(J)$, so P_J is a polynomial, and $P_J \neq 0$ as the minimum $\nu(J)$ is attained. Since $c_\alpha = 0$ if $\alpha_1 \geq 2$, P_J has degree at most 1 in x', so $f_0 \nmid P_J$, that is, $P_J|_S \neq 0$. By (1) with $r = J$, $\nu(J) \leq m(J+1)$, so since $J > m$, any α with $c_\alpha \neq 0$ and $JU_\alpha + \alpha_4 = \nu(J)$ satisfies

$$U_\alpha \leq \frac{\nu(J)}{J} \leq m + \frac{m}{J} < m + 1.$$

Since U_α is an integer, $U_\alpha \leq m$. By **Step 1**, $(S, \frac{1}{m}\operatorname{div}(P_J|_S))$ is lc. Since S is a normal divisor on $\mathbb{A}^3$ which is not contained in $(P_J = 0)$, [EMY03, Corollary 1.8] implies that $(\mathbb{A}^3, S + \frac{1}{m}\operatorname{div}(P_J))$ is lc on an open neighborhood U_S of S in $\mathbb{A}^3$.

Step 3. In this step, for any integer $L > \max\{5, \nu(J)\}$, let $f^{(L)}$ and $g^{(L)}$ be the sums of the terms of f and g of degree less than L, and let

$$B_L := (f^{(L)} = 0) + \frac{1}{m}\operatorname{div}(g^{(L)}) + \varphi(J)\operatorname{div}(t)$$

on $\mathbb{A}^4 := \operatorname{Spec}\mathbb{C}[x, y, z, t]$. We show that any prime divisor $F_\mathcal{X}$ over $\mathbb{A}^4$ with

$$\begin{aligned} &\operatorname{mult}_{F_\mathcal{X}} x \geq Jw_1 \operatorname{mult}_{F_\mathcal{X}} t, \quad \operatorname{mult}_{F_\mathcal{X}} y \geq Jw_2 \operatorname{mult}_{F_\mathcal{X}} t, \quad \operatorname{mult}_{F_\mathcal{X}} z \geq Jw_3 \operatorname{mult}_{F_\mathcal{X}} t, \\ &\text{and} \quad \operatorname{mult}_{F_\mathcal{X}} f^{(L)} > Jd \operatorname{mult}_{F_\mathcal{X}} t > 0 \end{aligned} \tag{F.30}$$

satisfies

$$a(F_\mathcal{X}, \mathbb{A}^4, B_L) \geq 0. \tag{F.31}$$

Let $\rho : \mathbb{A}^4_{x',y',z',t} \to \mathbb{A}^4$ be the morphism given by $(x', y', z', t) \to (t^{Jw_1}x', t^{Jw_2}y', t^{Jw_3}z', t)$, and let $T := (t = 0) \subset \mathbb{A}^4_{x',y',z',t}$, which we identify with $\mathbb{A}^3$. Since f_0 is w-homogeneous of weight d, any monomial of R has weight at least $Jd+1$ for the weights $(Jw_1, Jw_2, Jw_3, 1)$, $\deg f_0 \leq 5 < L$, and any α with $JU_\alpha + \alpha_4 = \nu(J)$ satisfies $\alpha_1 + \cdots + \alpha_4 \leq \nu(J) < L$, we have

$$f^{(L)} \circ \rho = t^{Jd} f_L \quad \text{and} \quad g^{(L)} \circ \rho = t^{\nu(J)} g_L$$

for polynomials f_L, g_L with $f_L|_T = f_0(x', y', z')$ and $g_L|_T = P_J$. The Jacobian determinant of ρ is $t^{J(w_1+w_2+w_3)} = t^{J(d+1)}$, so

$$K_{\mathbb{A}^4_{x',y',z',t}} = \rho^* K_{\mathbb{A}^4} + J(d+1)T,$$

and since $-J(d+1) + Jd + \frac{\nu(J)}{m} + \varphi(J) = 1$,

$$\rho^*(K_{\mathbb{A}^4} + B_L) = K_{\mathbb{A}^4_{x',y',z',t}} + T + (f_L = 0) + \frac{1}{m}\operatorname{div}(g_L).$$

Let $V := U_S \times \mathbb{A}^1_t \subset \mathbb{A}^4_{x',y',z',t}$. Then $T \cap V$ is identified with U_S, $T \cap V \not\subset (f_L g_L = 0)$, and the restriction of $(f_L = 0) + \frac{1}{m}\operatorname{div}(g_L)$ to $T \cap V$ is $(S + \frac{1}{m}\operatorname{div}(P_J))|_{U_S}$. By **Step 2**, $(U_S, (S + \frac{1}{m}\operatorname{div}(P_J))|_{U_S})$ is lc. By [EMY03, Corollary 1.8], $(V, T + (f_L = 0) + \frac{1}{m}\operatorname{div}(g_L))$ is lc around $T \cap V \supset S$.

Let $F_\mathcal{X}$ be a prime divisor over $\mathbb{A}^4$ satisfying (F.30). Then $\operatorname{mult}_{F_\mathcal{X}} x'$, $\operatorname{mult}_{F_\mathcal{X}} y'$, $\operatorname{mult}_{F_\mathcal{X}} z' \geq 0$, so $F_\mathcal{X}$ has a center on $\mathbb{A}^4_{x',y',z',t}$. Since $\operatorname{mult}_{F_\mathcal{X}} t > 0$ and

$$\operatorname{mult}_{F_\mathcal{X}} f_L = \operatorname{mult}_{F_\mathcal{X}} f^{(L)} - Jd \operatorname{mult}_{F_\mathcal{X}} t > 0,$$

the center of $F_\mathcal{X}$ on $\mathbb{A}^4_{x',y',z',t}$ is contained in $T \cap (f_L = 0) = S$. Thus

$$a(F_\mathcal{X}, \mathbb{A}^4, B_L) = a\Big(F_\mathcal{X}, \mathbb{A}^4_{x',y',z',t}, T + (f_L = 0) + \frac{1}{m}\operatorname{div}(g_L)\Big) \geq 0.$$

Step 4. In this step we prove (2). Let F be a prime divisor over W with center o and $\operatorname{center}_{Y_J} F \subset S$. Since the coordinate ring of Y_J is generated over $\widehat{\mathcal{O}}_{W,o}$ by x', y', z', and t vanishes at o,

$$\begin{aligned} &\operatorname{mult}_F x \geq Jw_1 \operatorname{mult}_F t, \quad \operatorname{mult}_F y \geq Jw_2 \operatorname{mult}_F t, \quad \operatorname{mult}_F z \geq Jw_3 \operatorname{mult}_F t, \\ &\text{and} \quad \operatorname{mult}_F t > 0. \end{aligned} \tag{F.32}$$

Suppose that $a(F, W, D) < \varphi(J) \operatorname{mult}_F t$. Let $\mu := \operatorname{mult}_{\widehat{F}} \operatorname{Jac}_{\widehat{W}}$, and let $\xi \in \{x, y, z, t\}$ be a coordinate with $\operatorname{mult}_F(\partial f/\partial \xi) = \mu$. Since $f \in (x, y, z, t)^2$ and $g \notin (f)$, the restrictions of x, y, z, t, g to $\widehat{W}$ are non-zero, and $\widehat{W}$ is normal as W is normal. Let k be an integer with

$$k > \max\{2\mu,\ Jd \operatorname{mult}_F t\} \quad \text{and} \quad k - \mu > \max\{\operatorname{mult}_F x, \operatorname{mult}_F y, \operatorname{mult}_F z, \operatorname{mult}_F t, \operatorname{mult}_F h\}.$$

By Lemma 5.4(2) with $\widehat{\mathcal{X}}$, x, y, z, t, $\widehat{W}$, $\widehat{F}$, μ, ξ, and x, y, z, t, g in place of $\mathcal{X}$, $x_1, \dots, x_4$, W, F, j, x, and $H_1, \dots, H_5$, there exists a formal prime divisor $\widehat{F}_{\mathcal{X}}$ over $\widehat{\mathcal{X}}$ with center the closed point such that $\operatorname{mult}_{\widehat{F}_{\mathcal{X}}} f = k$, the orders of x, y, z, t, g along $\widehat{F}_{\mathcal{X}}$ are $\operatorname{mult}_F x, \operatorname{mult}_F y, \operatorname{mult}_F z, \operatorname{mult}_F t, \operatorname{mult}_F h$, respectively, and

$$a(\widehat{F}_{\mathcal{X}}, \widehat{\mathcal{X}}, 0) = a(\widehat{F}, \widehat{W}, 0) + k = a(F, W, 0) + k.$$

Since $\widehat{\mathcal{X}}$ is the completion of $\mathbb{A}^4$ at 0, by Lemma 2.12(3), $\widehat{F}_{\mathcal{X}}$ restricts to a prime divisor $F_{\mathcal{X}}$ over $\mathbb{A}^4$ with center 0, with $\operatorname{mult}_{F_{\mathcal{X}}} \phi = \operatorname{mult}_{\widehat{F}_{\mathcal{X}}} \phi$ for any $\phi \in \mathbb{C}[x, y, z, t]$ and

$$a(F_{\mathcal{X}}, \mathbb{A}^4, 0) = a(\widehat{F}_{\mathcal{X}}, \widehat{\mathcal{X}}, 0).$$

Let L be an integer with $L > \max\{5, \nu(J), k, \operatorname{mult}_F h\}$. Since $\widehat{F}_{\mathcal{X}}$ has center the closed point and $f - f^{(L)}, g - g^{(L)} \in (x, y, z, t)^L$,

$$\operatorname{mult}_{F_{\mathcal{X}}} f^{(L)} = k \quad \text{and} \quad \operatorname{mult}_{F_{\mathcal{X}}} g^{(L)} = \operatorname{mult}_F h.$$

Thus $F_{\mathcal{X}}$ satisfies (F.30) by (F.32), and

$$a(F_{\mathcal{X}}, \mathbb{A}^4, B_L) = a(F, W, 0) + k - k - \frac{1}{m} \operatorname{mult}_F h - \varphi(J) \operatorname{mult}_F t = a(F, W, D) - \varphi(J) \operatorname{mult}_F t < 0,$$

which contradicts (F.31). Thus

$$a(F, W, D) \geq \varphi(J) \operatorname{mult}_F t.$$

The lemma follows. □

Lemma F.8. *Notation and conditions as in Set-up F.6. Let F be a prime divisor over W with center o and $\operatorname{center}_{Y_J} F \subset S$, and let n_t be the order of the kernel of $\chi_t|_{I_G(F)}$. Assume that*

$$J \geq m + \frac{e_F}{n_t}.$$

Then there exists an integer N coprime to n_t with

$$mn_t < N \leq mn_t + e_F$$

such that the prime divisor $E := E_{N,n_t}$ over W given by the weighted blow-up of $\widehat{W}$ with the weights (Nw_1, Nw_2, Nw_3, n_t) on (x, y, z, t) (Lemma F.7(1)) satisfies

$$\frac{a(E, W, 0)}{e_E} \leq m + 2 \quad \textit{and} \quad \frac{a(E, W, D)}{e_E} \leq \frac{a(F, W, D)}{e_F}.$$

In particular, if F is a computing lift of $q(1)$, then so is E.

Proof of Lemma F.8. We denote $\widehat{F}$ (Set-up F.6) again by F, and similarly for E below. Passing from W to $\widehat{W}$ changes neither the log discrepancies nor the inertia groups of F and E (Lemma 2.12(3)). Let $\mathcal{O}_F$ be the valuation ring of F in the fraction field of $\widehat{\mathcal{O}}_{W,o}$, $\mathfrak{m}_F$ the maximal ideal of $\mathcal{O}_F$, and $\kappa(F) := \mathcal{O}_F/\mathfrak{m}_F$. For $g \in G$ we have $g^* x = \chi_x(g) x$, and similarly for y, z, t. We divide the proof into four steps.

Step 1. In this step we choose a faithful character ψ of $I_G(F)$ with $\chi_t|_{I_G(F)} = \psi^{n_t}$.

The group $I_G(F)$ acts trivially on $\kappa(F)$, and on the $\kappa(F)$-line $\mathfrak{m}_F/\mathfrak{m}_F^2$ it acts through a faithful character $\psi_F : I_G(F) \to \mathbb{C}^*$ [Ser79, Chapter IV, §1, Remark 2 and Chapter IV, §2, Proposition 7 and Corollary 2]. The values of ψ_F are roots of unity in $\kappa(F)$, so $\psi_F(I_G(F)) \subset \mathbb{C}^*$. Thus $I_G(F)$ is cyclic of order e_F, any character of $I_G(F)$ is a power of ψ_F, and for any integer a the kernel of ψ_F^a has order $\gcd(e_F, a)$. Let ϖ be a uniformizer of $\mathcal{O}_F$. Since $\mathrm{center}_W F = \{o\}$, $\mathrm{mult}_F t \geq 1$ and $t\varpi^{-\mathrm{mult}_F t} \in \mathcal{O}_F^\times$. For $g \in I_G(F)$, the residues in $\kappa(F)$ of $g^*\varpi/\varpi$ and of $g^*(t\varpi^{-\mathrm{mult}_F t})/(t\varpi^{-\mathrm{mult}_F t})$ are $\psi_F(g)$ and 1, while $g^*t/t = \chi_t(g)$. Hence

$$\chi_t|_{I_G(F)} = \psi_F^{\mathrm{mult}_F t} \quad \text{and} \quad n_t = \gcd(e_F, \mathrm{mult}_F t). \tag{F.33}$$

In particular n_t divides $\mathrm{mult}_F t$ and e_F, so $n_t \leq \mathrm{mult}_F t$. Since $\gcd(\mathrm{mult}_F t/n_t, e_F/n_t) = 1$ and the reduction map

$$(\mathbb{Z}/e_F\mathbb{Z})^\times \to (\mathbb{Z}/(e_F/n_t)\mathbb{Z})^\times$$

is surjective, there exists an integer i coprime to e_F with $i \equiv \mathrm{mult}_F t/n_t$ modulo e_F/n_t. Then $\psi := \psi_F^i$ is faithful and $in_t \equiv \mathrm{mult}_F t$ modulo e_F, so by (F.33),

$$\chi_t|_{I_G(F)} = \psi^{n_t}. \tag{F.34}$$

Step 2. In this step we find an integer c with $\gcd(c, n_t) = 1$ and

$$(\chi_x, \chi_y, \chi_z)|_{I_G(F)} = (\psi^{cw_1}, \psi^{cw_2}, \psi^{cw_3}). \tag{F.35}$$

We first find a character θ of $I_G(F)$ with

$$(\chi_x, \chi_y, \chi_z)|_{I_G(F)} = (\theta^{w_1}, \theta^{w_2}, \theta^{w_3}).$$

Any monomial of f_0 has weight Jd for the weights $(Jw_1, Jw_2, Jw_3, 1)$, while any monomial of R has weight greater than Jd, so any monomial of f_0 is a monomial of $f_0 + R$. Since G acts diagonally on x, y, z, t and $f_0 + R$ is G-semi-invariant, any two monomials of f_0 have equal characters. Let $(\alpha, \beta, \gamma) := (\chi_x, \chi_y, \chi_z)|_{I_G(F)}$.

- In type E_7, $\alpha^2 = \beta^3 = \beta\gamma^3$, so $\beta^2 = \gamma^3$. Let $\rho := \beta\gamma^{-1}$ and $\theta := \alpha\rho^{-4}$. Then $\rho^3 = \beta$, $\rho^2 = \gamma$, and $\theta^2 = \beta^3\rho^{-8} = \rho$, so
$$(\theta^9, \theta^6, \theta^4) = (\theta\rho^4, \rho^3, \rho^2) = (\alpha, \beta, \gamma).$$
- In type E_8, $\alpha^2 = \beta^3 = \gamma^5$. Let $\theta := \alpha\beta\gamma^{-4}$. Then
$$(\theta^{15}, \theta^{10}, \theta^6) = \big(\alpha(\alpha^2)^7(\beta^3)^5\gamma^{-60},\ (\alpha^2)^5\beta(\beta^3)^3\gamma^{-40},\ (\alpha^2)^3(\beta^3)^2\gamma^{-24}\big) = (\alpha, \beta, \gamma).$$

Since ψ is faithful, $\theta = \psi^c$ for an integer c, and (F.35) holds. Since $\gcd(c, n_t)$ divides e_F and $\psi(I_G(F))$ is the group of e_F-th roots of unity, there exists

$$g \in I_G(F) \quad \text{with} \quad \psi(g) = \exp\big(2\pi\sqrt{-1}/\gcd(c, n_t)\big).$$

By (F.34) and (F.35), g^* fixes x, y, z, t, hence acts trivially on $\widehat{\mathcal{O}}_{W,o}$, which is a quotient of $\mathbb{C}[[x, y, z, t]]$, hence on $\mathcal{O}_{W,o} \subset \widehat{\mathcal{O}}_{W,o}$ and on W. Since G acts faithfully on W, $g = 1$, so $\gcd(c, n_t) = 1$.

Step 3. In this step we construct N and E and show $I_G(F) \subset I_G(E)$.

Let N be the integer in $\{mn_t + 1, \ldots, mn_t + e_F\}$ with $N \equiv c$ modulo e_F. Since n_t divides e_F, $\gcd(N, n_t) = \gcd(c, n_t) = 1$. Since $J \geq m + e_F/n_t$,

$$m < \frac{N}{n_t} \leq m + \frac{e_F}{n_t} \leq J. \tag{F.36}$$

By Lemma F.7(1) with $(N, n) = (N, n_t)$, $E := E_{N,n_t}$ is a prime divisor over W with center o, and

$$a(E, W, 0) = N + n_t \quad \text{and} \quad \mathrm{mult}_E h = n_t\nu(N/n_t). \tag{F.37}$$

Since $\psi^N = \psi^c$, (F.34) and (F.35) imply

$$(\chi_x, \chi_y, \chi_z, \chi_t)|_{I_G(F)} = (\psi^{Nw_1}, \psi^{Nw_2}, \psi^{Nw_3}, \psi^{n_t}).$$

Let $g \in I_G(F)$ and $\zeta := \psi(g)$. For the weights (Nw_1, Nw_2, Nw_3, n_t) on (x, y, z, t) and any integer $k \geq 0$, let $\mathfrak{b}_k \subset \mathbb{C}[[x, y, z, t]]$ be the ideal generated by the monomials of weight at least k. Then g^* multiplies any monomial of weight k by ζ^k. Thus g^* preserves $\mathfrak{b}_k$ and the image $\overline{\mathfrak{b}}_k$ of $\mathfrak{b}_k$ in $\widehat{\mathcal{O}}_{W,o}$, so g acts on the weighted blow-up $\operatorname{Proj} \bigoplus_{k\geq 0} \overline{\mathfrak{b}}_k$ of $\widehat{W}$, and g^* acts on the degree-k part of the graded ring

$$\mathcal{G} := \bigoplus_{k\geq 0} \overline{\mathfrak{b}}_k / \overline{\mathfrak{b}}_{k+1}$$

by multiplication by ζ^k. Such a graded automorphism induces the identity on $\operatorname{Proj} \mathcal{G}$, which contains E. Hence g maps E to E and acts trivially on the function field of E, that is, $g \in I_G(E)$. Therefore

$$I_G(F) \subset I_G(E) \quad \text{and} \quad e_F \leq e_E. \tag{F.38}$$

Step 4. In this step we prove the inequalities and the last sentence of the lemma.

By Lemma F.7(1), $\varphi(r) \geq 0$ for any $r \in (0, J]$. We show that φ is increasing on $(m, J]$. Let $m < r < r' \leq J$, and let $x^{\alpha_1} y^{\alpha_2} z^{\alpha_3} t^{\alpha_4}$ be a monomial of $u + xv$ with $\nu(r) = rU + \alpha_4$, where $U := \alpha_1 w_1 + \alpha_2 w_2 + \alpha_3 w_3$. If $U \geq m + 1$, then $\nu(r) \geq r(m + 1)$ and

$$\varphi(r) \leq 1 + r - \frac{r(m+1)}{m} = 1 - \frac{r}{m} < 0,$$

a contradiction. Thus $U \leq m$, so $\nu(r') \leq r'U + \alpha_4 = \nu(r) + (r' - r)U \leq \nu(r) + (r' - r)m$, and

$$\varphi(r') \geq \varphi(r) + (r' - r) - \frac{(r' - r)m}{m} = \varphi(r).$$

By (F.36),

$$0 \leq \varphi(N/n_t) \leq \varphi(J). \tag{F.39}$$

Since $J > m$ and $\operatorname{center}_{Y_J} F \subset S$, Lemma F.7(2) implies $a(F, W, D) \geq \varphi(J) \operatorname{mult}_F t$. Since $D = \frac{1}{m} \operatorname{div}(h)$, (F.37) implies $a(E, W, D) = N + n_t - \frac{n_t}{m} \nu(N/n_t) = n_t \varphi(N/n_t)$. Since $n_t \leq \operatorname{mult}_F t$, (F.39) implies

$$0 \leq a(E, W, D) = n_t \varphi(N/n_t) \leq \varphi(J) \operatorname{mult}_F t \leq a(F, W, D).$$

Since $e_F \leq e_E$ by (F.38) and $a(E, W, D) \geq 0$,

$$\frac{a(E, W, D)}{e_E} \leq \frac{a(F, W, D)}{e_F}. \tag{F.40}$$

By (F.37), $N \leq m n_t + e_F$, (F.38), and $n_t \leq e_F$,

$$\frac{a(E, W, 0)}{e_E} \leq \frac{N + n_t}{e_F} \leq 1 + \frac{(m + 1)n_t}{e_F} \leq m + 2.$$

Finally, $q(1) = \operatorname{mld}((W \ni o, D)/G)$. Assume that F is a computing lift of $q(1)$. Since $\operatorname{center}_W E = \{o\}$, the definition of the quotient value and (F.40) imply

$$q(1) \leq \frac{a(E, W, D)}{e_E} \leq \frac{a(F, W, D)}{e_F} = q(1).$$

Thus E is a computing lift of $q(1)$. The lemma follows. □

Proposition F.9. *In Set-up F.6, let $L_{\text{type}} := 14$ and 24 in types E_7 and E_8, respectively, and let $\mathfrak{m}_o$ be the maximal ideal of W at o. Then every prime divisor F over W with center o satisfies $\operatorname{center}_{Y_J} F \subset S$ or*

$$\operatorname{mult}_F \operatorname{Jac}_W \leq L_{\text{type}} J \operatorname{mult}_F \mathfrak{m}_o.$$

Proof. Recall that $\widehat{W} = (f = 0) \subset \widehat{\mathcal{X}}$ for $f = f_0 + R$ (Set-up F.6). Let F be a prime divisor over W with center o, with $\widehat{F}$ and mult_F as in Set-up F.6, and let $\mathrm{mult}_F 0 := +\infty$. On $\mathcal{O}_{W,o}$, mult_F is the multiplicity along F, and for any ideal $\mathfrak{a}$ of $\mathcal{O}_{W,o}$,

$$\mathrm{mult}_F \mathfrak{a} = \min\{\mathrm{mult}_F g \mid g \in \mathfrak{a}\widehat{\mathcal{O}}_{W,o}\}.$$

Let

$$v_x := \mathrm{mult}_F x, \quad v_y := \mathrm{mult}_F y, \quad v_z := \mathrm{mult}_F z, \quad \text{and} \quad v_t := \mathrm{mult}_F t,$$

and let

$$\rho := \min\left\{\frac{v_x}{w_1}, \frac{v_y}{w_2}, \frac{v_z}{w_3}\right\}.$$

Since $f \in (x, y, z, t)^2$, none of x, y, z, t vanishes on $\widehat{W}$. Since the center of $\widehat{F}$ on $\widehat{W}$ is the closed point, v_x, v_y, v_z, v_t are positive integers, so $\rho > 0$. Since $\mathfrak{m}_o\widehat{\mathcal{O}}_{W,o} = (x, y, z, t)$,

$$\mathrm{mult}_F \mathfrak{m}_o = \min\{v_x, v_y, v_z, v_t\}. \tag{F.41}$$

Step 1. In this step we prove that $\mathrm{center}_{Y_J} F \subset S$ if $\rho \geq Jv_t$.

Assume that $\rho \geq Jv_t$. Then $v_x \geq w_1\rho \geq Jw_1v_t$, and similarly for y and z, so

$$\mathrm{mult}_F x' = v_x - Jw_1v_t \geq 0, \quad \mathrm{mult}_F y' = v_y - Jw_2v_t \geq 0, \quad \text{and} \quad \mathrm{mult}_F z' = v_z - Jw_3v_t \geq 0.$$

Since t has weight 1, the ring of the affine scheme Y_J is generated over $\widehat{\mathcal{O}}_{W,o}$ by x', y', z'. Thus the valuation ring of $\widehat{F}$ contains the ring of Y_J, and $\widehat{F}$ has a center on Y_J, namely the prime ideal $\mathfrak{p}_F$ of the elements g with $\mathrm{mult}_F g > 0$. Since $v_t > 0$, $t \in \mathfrak{p}_F$. Hence $\mathrm{center}_{Y_J} F \subset (t = 0) = S$.

Step 2. In this step we prove that

$$\mathrm{mult}_F \mathrm{Jac}_W \leq L_{\mathrm{type}}\rho \tag{F.42}$$

if $\rho < Jv_t$, and we deduce the proposition.

Assume that $\rho < Jv_t$. Then

$$v_x \geq w_1\rho, \quad v_y \geq w_2\rho, \quad v_z \geq w_3\rho, \quad \text{and} \quad v_t > \frac{\rho}{J}.$$

Let $f_x := \partial f/\partial x$, and similarly f_y, f_z, f_t, R_y, R_z. Since $\widehat{W}$ is the hypersurface $(f = 0)$ in $\widehat{\mathcal{X}}$, the ideal $\mathrm{Jac}_W \widehat{\mathcal{O}}_{W,o}$ is generated by the restrictions of f_x, f_y, f_z, f_t to $\widehat{W}$. Thus

$$\mathrm{mult}_F \mathrm{Jac}_W \leq \min\{\mathrm{mult}_F f_x, \mathrm{mult}_F f_y, \mathrm{mult}_F f_z\}. \tag{F.43}$$

Since $R \in \mathbb{C}[[y, z, t]]$,

$$f_x = 2x, \qquad f_y = \frac{\partial f_0}{\partial y} + R_y, \qquad \text{and} \qquad f_z = \frac{\partial f_0}{\partial z} + R_z.$$

We bound R_y and R_z. Write

$$R = \sum_{(\alpha,\beta,\gamma)\in\Lambda} c_{\alpha\beta\gamma} y^\alpha z^\beta t^\gamma,$$

where $\Lambda \subset \mathbb{N}^3$ is the set of exponents with $c_{\alpha\beta\gamma} \neq 0$. For any $(\alpha, \beta, \gamma) \in \Lambda$, the weight $Jw_2\alpha + Jw_3\beta + \gamma$ is an integer greater than Jd, so

$$w_2\alpha + w_3\beta + \frac{\gamma}{J} \geq d + \frac{1}{J}.$$

Thus, for any $(\alpha, \beta, \gamma) \in \Lambda$ with $\alpha \geq 1$,

$$\mathrm{mult}_F\left(y^{\alpha-1}z^\beta t^\gamma\right) \geq \rho\Big(w_2(\alpha - 1) + w_3\beta + \frac{\gamma}{J}\Big) \geq \Big(d - w_2 + \frac{1}{J}\Big)\rho, \tag{F.44}$$

and for any $(\alpha, \beta, \gamma) \in \Lambda$ with $\beta \geq 1$,

$$\mathrm{mult}_F\left(y^\alpha z^{\beta-1} t^\gamma\right) \geq \Big(d - w_3 + \frac{1}{J}\Big)\rho. \tag{F.45}$$

Since mult_F is a valuation and $v_y, v_z, v_t \geq 1$, (F.44) and (F.45) imply $\operatorname{mult}_F R_y \geq (d - w_2 + \frac{1}{J})\rho$ and $\operatorname{mult}_F R_z \geq (d - w_3 + \frac{1}{J})\rho$, so

$$\operatorname{mult}_F R_y > (d - w_2)\rho, \qquad \operatorname{mult}_F R_z > (d - w_3)\rho. \tag{F.46}$$

By the definition of ρ, at least one of $v_x = w_1\rho$, $v_y = w_2\rho$, and $v_z = w_3\rho$ holds.

Assume that (f_0, w, d) is of type E_8. Then $f_0 = x^2 + y^3 + z^5$ and $2w_1 = 3w_2 = 5w_3 = d$. Thus $f_y = 3y^2 + R_y$ and $f_z = 5z^4 + R_z$. If $v_x = w_1\rho$, then $\operatorname{mult}_F f_x = w_1\rho = (d - w_1)\rho$. If $v_y = w_2\rho$, then $\operatorname{mult}_F(3y^2) = 2w_2\rho = (d - w_2)\rho$, so $\operatorname{mult}_F f_y = (d - w_2)\rho$ by (F.46). If $v_z = w_3\rho$, then $\operatorname{mult}_F(5z^4) = 4w_3\rho = (d - w_3)\rho$, so $\operatorname{mult}_F f_z = (d - w_3)\rho$ by (F.46). Since $w_1 > w_2 > w_3$ and $d - w_3 = 24$, (F.43) implies (F.42).

Assume that (f_0, w, d) is of type E_7, that is, $f_0 = x^2 + y^3 + yz^3$, $w = (9, 6, 4)$, and $d = 18$. Then

$$f_y = 3y^2 + z^3 + R_y, \qquad f_z = 3yz^2 + R_z, \qquad \operatorname{mult}_F R_y > 12\rho, \qquad \text{and} \qquad \operatorname{mult}_F R_z > 14\rho.$$

- If $v_x = 9\rho$, then $\operatorname{mult}_F f_x = 9\rho$.
- If $v_y = 6\rho$ and $v_z > 4\rho$, then $\operatorname{mult}_F(3y^2) = 12\rho < \operatorname{mult}_F(z^3)$, so $\operatorname{mult}_F f_y = 12\rho$.
- If $v_z = 4\rho$ and $v_y > 6\rho$, then $\operatorname{mult}_F(z^3) = 12\rho < \operatorname{mult}_F(3y^2)$, so $\operatorname{mult}_F f_y = 12\rho$.
- If $v_y = 6\rho$ and $v_z = 4\rho$, then $\operatorname{mult}_F(3yz^2) = 14\rho < \operatorname{mult}_F R_z$, so $\operatorname{mult}_F f_z = 14\rho$.

Since $v_y \geq 6\rho$, $v_z \geq 4\rho$, and at least one of $v_x = 9\rho$, $v_y = 6\rho$, and $v_z = 4\rho$ holds, the hypothesis of one of the four items above holds. By (F.43), (F.42) holds with $L_{\text{type}} = 14$.

We deduce the proposition. By **Step 1**, we may assume that $\rho < Jv_t$. If $\operatorname{mult}_F \mathfrak{m}_o = v_t$, then

$$\rho < Jv_t = J \operatorname{mult}_F \mathfrak{m}_o.$$

If $\operatorname{mult}_F \mathfrak{m}_o = v_x$, then $\rho \leq v_x/w_1 \leq J \operatorname{mult}_F \mathfrak{m}_o$, and similarly for v_y and v_z. Thus, by (F.41) and (F.42),

$$\operatorname{mult}_F \operatorname{Jac}_W \leq L_{\text{type}}\rho \leq L_{\text{type}} J \operatorname{mult}_F \mathfrak{m}_o.$$

The proposition follows. □

Proof of Theorem 17.4. Assume, to the contrary, that there exists a radial failure sequence $(W_i \ni o_i, G_i, h_i, \tau_i)$ with constants (m, a, δ) satisfying Set-up 17.1 whose prepared limit (f_*, h_*) is of type cE and has generic transverse type E_7 or E_8 along C (Definition 15.1). We divide the proof into seven steps.

Step 1. In this step we fix coordinates and bound the marked orders of computing lifts from below.

By Construction-Theorem 13.6, possibly passing to a subsequence, we may assume that $G := G_i$ and the representation of G on $\widehat{\mathcal{X}} := \operatorname{Spec} \mathbb{C}[[x, y, z, t]]$ do not depend on i, that G acts diagonally on the coordinates (x, y, z, t) of Lemma 3.7, that $\widehat{\mathcal{O}}_{W_i, o_i} = \mathbb{C}[[x, y, z, t]]/(f_i)$ for a G-semi-invariant f_i, that the characters of f_i and h_i do not depend on i, and that

$$f_i = x^2 + y^3 + y\alpha_i(z, t) + \beta_i(z, t) \quad \text{and} \quad h_i = \varphi_i(y, z, t) + x\xi_i(y, z, t).$$

Vanishing of a coefficient passes to the compatible generic limit, so

$$f_* = x^2 + y^3 + y\alpha_*(z, t) + \beta_*(z, t) \quad \text{and} \quad h_* = \varphi_*(y, z, t) + x\xi_*(y, z, t) \tag{F.47}$$

for some $\alpha_*, \beta_* \in \Bbbk[[z, t]]$ and $\varphi_*, \xi_* \in \Bbbk[[y, z, t]]$. Since G is cyclic, Set-up 15.4 holds with $B_0 := a$: items (2)–(5) there follow from the choices above, Definition 4.1, and Set-up 17.1(2), and items (6) and (7) from Construction-Theorem 13.6(1)(3) and Set-up 17.1(4)(5), since $\operatorname{Nklt}(W_*, \frac{1}{m} \operatorname{div}(h_*)) = C$ is non-empty.

We now replace f_i and h_i with polynomials, in two steps. Let $\mathfrak{n}$ be the maximal ideal of $\mathbb{C}[[x, y, z, t]]$. First, since o_i is an isolated singular point of W_i, there exists a positive integer c_i with

$$\mathfrak{n}^{c_i} \subset (f_i) + \mathfrak{n} \cdot (\partial f_i/\partial x, \partial f_i/\partial y, \partial f_i/\partial z, \partial f_i/\partial t).$$

Fix integers $L_i > 2c_i$ with $\lim_{i\to+\infty} L_i = +\infty$, and let $f_i^{<L_i}$ be the part of f_i of degree $< L_i$. By Lemma 3.2(2) with $\mathbb{C}$, f_i, c_i, L_i, and $f_i^{<L_i}$ in place of $\Bbbk$, F, c, L, and F_0, there exist a G-equivariant automorphism θ_i of $\mathbb{C}[[x,y,z,t]]$ and a G-invariant unit η_i with

$$\eta_i\theta_i(f_i) = f_i^{<L_i} \quad \text{and} \quad \theta_i(x_j) - x_j \in \mathfrak{n}^{L_i-c_i+1} \ \text{ for } \ (x_1,x_2,x_3,x_4) = (x,y,z,t).$$

Thus θ_i induces a G-equivariant isomorphism

$$\widehat{\mathcal{O}}_{W_i,o_i} \to \mathbb{C}[[x,y,z,t]]/(f_i^{<L_i})$$

sending the class of h_i to that of $\theta_i(h_i)$. We replace f_i with $f_i^{<L_i}$ and h_i with $\theta_i(h_i)$, written again as $\varphi_i(y,z,t)+x\xi_i(y,z,t)$ by means of $x^2 = -(y^3+y\alpha_i+\beta_i)$ for the new α_i,β_i. Since $y^3+y\alpha_i+\beta_i$ belongs to $\mathfrak{n}^3$, the new h_i is congruent to the old one modulo $\mathfrak{n}^{L_i-c_i+1}$. By Lemma 2.12(2)(3), replacing (f_i,h_i) in this way changes neither the quotient values nor the computing lifts, nor the numbers $a(F,W_i,0)$, $\operatorname{mult}_F h_i$, and $\operatorname{mult}_F \operatorname{Jac}_{W_i}$ of prime divisors F over W_i with center o_i.

Second, let $W_i^\circ := (f_i = 0) \subset \mathbb{A}^4$ for the polynomial f_i above. The completion $\widehat{W_i^\circ}$ of W_i° at 0 is $\operatorname{Spec}\widehat{\mathcal{O}}_{W_i,o_i}$, so by Definition 4.1(1), Set-up 17.1(2), and Lemma 2.12(2), $(W_i^\circ \ni 0)$ is isolated Gorenstein terminal, G acts faithfully on W_i°, and $(\widehat{W_i^\circ}, \frac{1}{m}\operatorname{div}(h_i))$ is klt. Let $L_{i,Ma}$ be the integer L_γ of Lemma 3.2(1) with G, W_i°, h_i, $\frac{1}{m}$, and Ma in place of H, V_0, g_1, β_1, and γ, fix an integer $N_i' \ge \max\{L_{i,Ma}, L_i\}$, and replace h_i with its part of degree $< N_i'$. By Lemma 3.2(1) and Lemma 2.12(2)(3), $(W_i, \frac{1}{m}\operatorname{div}(h_i))$ is still klt near o_i, b_i and its computing lifts do not change, and for any prime divisor F over W_i with center o_i and

$$a(F,W_i,\tfrac{1}{m}\operatorname{div}(h_i))/e_F \le Ma,$$

the numbers $a(F,W_i,0)$, $\operatorname{mult}_F h_i$, and $\operatorname{mult}_F \operatorname{Jac}_{W_i}$ do not change. Moreover $q_i(\tau_i) \ge a$ still holds, since any F for which $\operatorname{mult}_F h_i$ changes satisfies

$$\frac{a(F,W_i,\frac{\tau_i}{m}\operatorname{div}(h_i))}{e_F} \ge \frac{a(F,W_i,\frac{1}{m}\operatorname{div}(h_i))}{e_F} > Ma \ge a$$

for the old and for the new h_i. Since $L_i - c_i > L_i/2 \to +\infty$ and $N_i' \ge L_i$, any fixed jet of (f_i,h_i) is eventually not changed, so the compatible generic limit stays (f_*,h_*).

Let F be a computing lift of b_i (Definition 2.4). Then $e_F \le |G| \le M$ and

$$a(F,W_i,0) - \tfrac{1}{m}\operatorname{mult}_F h_i = e_F b_i \le Ma. \tag{F.48}$$

By Definition 4.1(3) and Lemma 4.2(2),

$$a \le q_i(\tau_i) \le b_i + (1-\tau_i)\frac{\operatorname{mult}_F h_i}{me_F}.$$

Since $b_i \le a - \delta$, we obtain

$$\frac{\operatorname{mult}_F h_i}{me_F} \ge \frac{\delta}{1-\tau_i} \quad \text{for any computing lift } F \text{ of } b_i. \tag{F.49}$$

Step 2. In this step we bring f_* to a normal form along C.

Write $I_C \subset \Bbbk[[x,y,z,t]]$ for the ideal of C. Since $C \subset \operatorname{Sing}(W_*)$, $2x = \partial f_*/\partial x \in I_C$. If $y \notin I_C$, then at the generic point of C the Hessian of f_* has the entries $\partial^2 f_*/\partial x^2 = 2$, $\partial^2 f_*/\partial x\partial y = 0$, and $\partial^2 f_*/\partial y^2 = 6y \ne 0$, so the quadratic part of the generic transverse surface of W_* along C (Definition 15.1), whose rank is the rank of the Hessian of f_* at the generic point of C, has rank at least 2, which is impossible because a Du Val singularity of type E_7 or E_8 has quadratic rank 1. Thus $x,y \in I_C$, so C is a G-invariant regular curve in the plane $\operatorname{Spec}\Bbbk[[z,t]] = (x=y=0)$. Possibly exchanging z and t, we may assume that $t|_C$ is a parameter of C. We let $s := t$ (this s is not the parameter s of Set-up 17.1(6), which we denote by s_c in **Step 5**). There exists a G-semi-invariant $\psi \in s\Bbbk[[s]]$ with the character χ_z such that $C = (x = y = z - \psi(s) = 0)$. Replacing z with $z - \psi(s)$, which is an automorphism of $\Bbbk[[z,s]]$ with invertible linear part, the forms in (F.47) are kept, with $\alpha_*,\beta_*,\varphi_*,\xi_*$ now series in (z,s), resp. (y,z,s), and $C = (x=y=z=0)$. In particular s restricts to a parameter of C, and s plays the role of t in Set-up F.6 below.

Write $\alpha_* = \sum_{j\geq 0}\alpha_{*,j}(s)z^j$ and $\beta_* = \sum_{j\geq 0}\beta_{*,j}(s)z^j$. Since $C \subset W_*$ and $C \subset \mathrm{Sing}(W_*)$, the restrictions of f_*, $\partial f_*/\partial y$, and $\partial f_*/\partial z$ to C vanish, so $\beta_{*,0} = \alpha_{*,0} = \beta_{*,1} = 0$. At the generic point of C, the quadratic part of f_* in (x, y, z) is $x^2+\alpha_{*,1}yz+\beta_{*,2}z^2$, of rank 1, so $\alpha_{*,1} = \beta_{*,2} = 0$. The cubic part of $f_*|_{x=0}$ is $y^3 + \alpha_{*,2}yz^2 + \beta_{*,3}z^3$, and the generic transverse type is E, so $y^3 + \alpha_{*,2}yz^2 + \beta_{*,3}z^3 = (c_1y + c_2z)^3$ for some c_1, c_2 in an algebraic closure of $\Bbbk((s))$. Comparing the coefficients of y^3 and y^2z, we obtain $c_1^3 = 1$ and $3c_1^2c_2 = 0$, so $c_2 = 0$ and $\alpha_{*,2} = \beta_{*,3} = 0$. Thus

$$\alpha_* \in z^3\Bbbk[[z, s]] \quad \text{and} \quad \beta_* \in z^4\Bbbk[[z, s]].$$

Over an algebraic closure of $\Bbbk((s))$, the surface $x^2 + y^3 + \alpha_{*,3}yz^3 + \beta_{*,4}z^4 + \cdots = 0$ is of type E_6 if $\beta_{*,4} \neq 0$, of type E_7 if $\beta_{*,4} = 0 \neq \alpha_{*,3}$, and of type E_8 if $\beta_{*,4} = \alpha_{*,3} = 0 \neq \beta_{*,5}$, by [AGZV85, Section 12.6, Theorem and Lemma], since in the three cases the principal parts $x^2 + y^3 + \beta_{*,4}z^4$, $x^2 + y^3 + \alpha_{*,3}yz^3$, and $x^2 + y^3 + \beta_{*,5}z^5$ are quasi-homogeneous for the weights $(6, 4, 3)$, $(9, 6, 4)$, and $(15, 10, 6)$ on (x, y, z), the other terms have higher weights, and their Milnor algebras have no basis monomial of weight higher than the degree.

In type E_7, $\beta_{*,4} = 0$ and $\alpha_{*,3} = s^{n_0}v(s)$ for an integer $n_0 \geq 0$ and $v \in \Bbbk[[s]]$ with $v(0) \neq 0$. Since v is G-semi-invariant and $v(0) \neq 0$, v is G-invariant, and so is the cube root $v^{1/3} \in \Bbbk[[s]]$ with $v^{1/3}(0)^3 = v(0)$. Replacing z with $v^{1/3}z$, we obtain

$$f_* = x^2 + y^3 + s^{n_0}yz^3 + yz^4\alpha'_*(z, s) + z^5\beta'_*(z, s) \tag{F.50}$$

for some $\alpha'_*, \beta'_* \in \Bbbk[[z, s]]$.

In type E_8, $\alpha_* \in z^4\Bbbk[[z, s]]$ and $\beta_* = z^5v(z, s)$ for some $v \in \Bbbk[[z, s]]$, and we show that $v(0, 0) \neq 0$. For any j, let $\beta_{(j)}$ and $\alpha_{(j)}$ be the homogeneous parts of degree j of β_* and α_* in the coordinates (z, t) of Construction-Theorem 13.6, before the changes of this step. Since the composite of the changes of this step is an automorphism of $\Bbbk[[z, t]]$ and $\alpha_* \in (z, s)^4$, $\beta_* \in (z, s)^5$, we have $\alpha_{(3)} = \beta_{(4)} = 0$. By Construction-Theorem 13.6(1) the surface $\Sigma' := (t - c'z = 0) \subset W_*$ is Du Val. Since $\alpha_{(3)} = \beta_{(4)} = 0$, Σ' is defined by $x^2 + y^3 + y\alpha_*(z, c'z) + \beta_*(z, c'z)$ with α_*, β_* as in (F.47) and $\alpha_*(z, c'z) \in z^4\Bbbk[[z]]$, so $\beta_{(5)}(1, c') \neq 0$, as otherwise any monomial of $x^2 + y^3 + y\alpha_*(z, c'z) + \beta_*(z, c'z)$ has weight at least 6 for the weights $(3, 2, 1)$ on (x, y, z), and Σ' is not Du Val. The degree-5 homogeneous part of an element of $(z, t)^5$ is transformed by the linear part of the automorphism, which is invertible, so the degree-5 homogeneous part $v(0, 0)z^5$ of $\beta_* = z^5v(z, s)$ is non-zero. Thus $v(0, 0) \neq 0$. Since v is G-semi-invariant, v is G-invariant, and replacing z with $v^{1/5}z$, we obtain

$$f_* = x^2 + y^3 + z^5 + yz^4\alpha'_*(z, s) \tag{F.51}$$

for some $\alpha'_* \in \Bbbk[[z, s]]$. In type E_8 we let $n_0 := 0$.

Let (f_0, w, d) be $(x^2 + y^3 + yz^3, (9, 6, 4), 18)$ in type E_7 and $(x^2 + y^3 + z^5, (15, 10, 6), 30)$ in type E_8, as in Set-up F.6.

Step 3. In this step we show that the mark h_* has w-weighted order exactly m along C.

Give x, y, z the weights w and s the weight 0, so that the monomial $x^{\nu_1}y^{\nu_2}z^{\nu_3}s^n$ has w-weight $w_1\nu_1 + w_2\nu_2 + w_3\nu_3$. We show that

$$\min\{w_1\nu_1 + w_2\nu_2 + w_3\nu_3 \mid x^{\nu_1}y^{\nu_2}z^{\nu_3}s^n \text{ has a non-zero coefficient in } h_*\} = m. \tag{F.52}$$

Let $\mathbb{K}$ be an algebraic closure of $\Bbbk((s))$ and $S_{\mathbb{K}} := \operatorname{Spec}\mathbb{K}[[x, y, z]]/(f_*)$, where s is viewed as an element of $\mathbb{K}$. The surface germ $S_{\mathbb{K}}$ is the generic transverse surface of W_* along C (Definition 15.1, with (s, x, y, z) in place of (s, u_1, u_2, u_3)). In type E_7, over $\mathbb{K}$ and only in this step, we replace z by $s^{n_0/3}z$, which does not change the w-weight of any monomial. By (F.50) and (F.51), the part of f_* of w-weight $\leq d$ is f_0, since $w_2 + 4w_3 > d$ and, in type E_7, $5w_3 > d$. The Milnor algebra

$$\mathbb{K}[x, y, z]/(\partial f_0/\partial x, \partial f_0/\partial y, \partial f_0/\partial z)$$

has the basis $1, z, z^2, z^3, z^4, y, yz$ in type E_7 and $y^{\nu_2}z^{\nu_3}$, $\nu_2 \leq 1$, $\nu_3 \leq 3$, in type E_8, of w-weights at most $16 < 18$, resp. $28 < 30$, so by [AGZV85, Section 12.6, Theorem and Lemma], whose

proof is formal, a formal coordinate change $x_j \to x_j - \rho_j$, where $(x_1, x_2, x_3) = (x, y, z)$ and ρ_j has w-weight $> w_j$, gives an isomorphism $S_{\mathbb{K}} \cong \operatorname{Spec} \mathbb{K}[[x, y, z]]/(f_0)$ preserving the w-filtration and in_w.

For an integer r, let $\mathfrak{a}_r \subset \mathbb{K}[[x, y, z]]$ be the ideal generated by the monomials of w-weight $\geq r$. For a non-zero $g \in \mathbb{K}[[x, y, z]]/(f_*)$, let

$$\operatorname{wt}_w(g) := \max\{r \in \mathbb{Z} \mid g \in (\mathfrak{a}_r + (f_*))/(f_*)\}.$$

Since $\operatorname{in}_w(f_*) = f_0$, the graded ring of $\mathbb{K}[[x, y, z]]/(f_*)$ for the w-filtration is $\mathbb{K}[x, y, z]/(f_0)$. If $g = \varphi(y, z) + x\xi(y, z)$, then the part of $\varphi + x\xi$ of lowest w-weight has x-degree at most 1, so f_0, which is monic of degree 2 in x, does not divide it. Thus $\operatorname{wt}_w(g)$ is the lowest w-weight of a monomial of $\varphi + x\xi$, and $\operatorname{in}_w(g) \neq 0$ in $\mathbb{K}[x, y, z]/(f_0)$.

We show that $\operatorname{lct}(S_{\mathbb{K}}; \operatorname{div}(g)) = 1/\operatorname{wt}_w(g)$ for any non-zero g in the maximal ideal of $\mathbb{K}[[x, y, z]]/(f_*)$. Let $r := \operatorname{wt}_w(g)$. By [Roa04, Table 1], after rescaling x, y, z by elements of $\mathbb{K}^\times$, $\mathbb{K}[x, y, z]/(f_0) = \mathbb{K}[u, v]^{\widetilde{\Gamma}}$, where $\widetilde{\Gamma} \subset \mathrm{SL}_2(\mathbb{K})$ is the binary octahedral group in type E_7 and the binary icosahedral group in type E_8, and x, y, z are $\widetilde{\Gamma}$-invariant forms of degrees $2w_1, 2w_2, 2w_3$. Let $\pi : \operatorname{Spec} \mathbb{K}[[u, v]] \to S_{\mathbb{K}}$ be the induced finite morphism. Since $\widetilde{\Gamma}$ acts freely outside the closed point, π is étale in codimension 1, so for any $c \geq 0$, $(S_{\mathbb{K}}, c \operatorname{div}(g))$ is lc, resp. klt, if and only if $(\operatorname{Spec} \mathbb{K}[[u, v]], c \operatorname{div}(\pi^* g))$ is lc, resp. klt. Since π^* doubles degrees, $\pi^* g$ has order $2r$ and the homogeneous part $g_{2r} := \pi^* \operatorname{in}_w(g)$ of degree $2r$ is a non-zero $\widetilde{\Gamma}$-invariant binary form. The image of $\widetilde{\Gamma}$ in $\mathrm{PGL}_2(\mathbb{K})$ is the octahedral, resp. icosahedral, group, which fixes no point of $\mathbb{P}^1_{\mathbb{K}}$, so the orbit of any root of g_{2r} consists of at least 2 roots, all of multiplicity μ. Thus $2\mu \leq 2r$, that is, $\mu \leq r$. Let $\gamma : X \to \operatorname{Spec} \mathbb{K}[[u, v]]$ be the blow-up of the closed point, E_0 the exceptional curve, and $g' := \gamma_*^{-1} \operatorname{div}(\pi^* g)$. Since $a(E_0, \operatorname{Spec} \mathbb{K}[[u, v]], 0) = 2$ and $\operatorname{mult}_{E_0} \pi^* g = 2r$,

$$K_X + E_0 + \tfrac{1}{r} g' = \gamma^* \big(K_{\operatorname{Spec} \mathbb{K}[[u,v]]} + \tfrac{1}{r} \operatorname{div}(\pi^* g) \big),$$

and $\frac{1}{r} g'|_{E_0}$ is $\frac{1}{r}$ times the divisor of roots of g_{2r} on $E_0 \cong \mathbb{P}^1_{\mathbb{K}}$, whose coefficients are ≤ 1. By [EMY03, Corollary 1.8], which extends to the regular surface X over $\mathbb{K}$ because completion and extension of an algebraically closed coefficient field preserve log discrepancies [dFEM11, Proposition A.14 and Lemma 2.14], $(X, E_0 + \frac{1}{r} g')$ is lc near E_0. Thus $(\operatorname{Spec} \mathbb{K}[[u, v]], \frac{1}{r} \operatorname{div}(\pi^* g))$ is lc and, because of E_0, not klt. Therefore $\operatorname{lct}(S_{\mathbb{K}}; \operatorname{div}(g)) = 1/r$.

By Set-up 17.1(5) and Construction-Theorem 13.6(3), $(W_*, \frac{1}{m} \operatorname{div}(h_*))$ is lc and $\operatorname{Nklt}(W_*, \frac{1}{m} \operatorname{div}(h_*)) = C$, so the pair is not klt at the generic point of C. The morphism $S_{\mathbb{K}} \to W_*$ is the base change of the regular morphism $\operatorname{Spec} \mathbb{K}[[x, y, z]] \to \mathcal{X}_*$, so the base change of a log resolution of $(W_*, \frac{1}{m} \operatorname{div}(h_*))$ is a log resolution of $(S_{\mathbb{K}}, \frac{1}{m} \operatorname{div}(h_*))$, cf. [dFEM11, Remark 2.10]. Over the regular scheme $\mathcal{X}_*$, relative canonical divisors pull back to relative canonical divisors under the base change to $\operatorname{Spec} \mathbb{K}[[x, y, z]]$ [JM12, Lemma 1.2], and since W_* is a Cartier divisor in $\mathcal{X}_*$, W_* is Gorenstein and the relative canonical divisors over W_* are determined by the dualizing sheaves, which commute with flat base change. Thus for any prime divisor P on a log resolution of $(W_*, \frac{1}{m} \operatorname{div}(h_*))$, any component of the base change of P has log discrepancy $a(P, W_*, \frac{1}{m} \operatorname{div}(h_*))$ with respect to $(S_{\mathbb{K}}, \frac{1}{m} \operatorname{div}(h_*))$, and $(S_{\mathbb{K}}, \frac{1}{m} \operatorname{div}(h_*))$ is lc and not klt, so h_* is not a unit and $\operatorname{lct}(S_{\mathbb{K}}; \operatorname{div}(h_*)) = \frac{1}{m}$. Since the coordinate changes over $\mathbb{K}$ above preserve the w-filtration, the value $\operatorname{wt}_w(h_*)$ computed in the coordinates (x, y, z) of **Step 2** equals the value computed in the coordinates above. Thus $\operatorname{wt}_w(h_*) = m$. Since

$$h_* = \varphi_*(y, z, s) + x\xi_*(y, z, s)$$

and the coefficient in $\mathbb{K}$ of $x^{\nu_1} y^{\nu_2} z^{\nu_3}$ is a power series in s, which is non-zero if and only if some $x^{\nu_1} y^{\nu_2} z^{\nu_3} s^n$ has a non-zero coefficient in h_*, (F.52) follows.

Step 4. In this step we construct a chart of a weighted blow-up of W_* on which h_* becomes divisible by s^{km}.

For a monomial $x^{\nu_1} y^{\nu_2} z^{\nu_3} s^n$ with $\nu_1 \leq 1$, let

$$\omega := w_1\nu_1 + w_2\nu_2 + w_3\nu_3 \quad \text{and} \quad n' := n_0(3\nu_1 + 2\nu_2 + \nu_3) + n \geq 0.$$

By (F.52), the monomials of h_* with $\omega = m$ form a non-empty set. Let n_1 be the minimum of n' over them. Fix an integer $k > \max\{n_0, n_1\}$ and let

$$\lambda = (\lambda_1, \lambda_2, \lambda_3) := kw + n_0(3, 2, 1) \quad \text{and} \quad d_\lambda := kd + 6n_0.$$

Thus

$$\lambda_1\nu_1 + \lambda_2\nu_2 + \lambda_3\nu_3 = k\omega + n_0(3\nu_1 + 2\nu_2 + \nu_3) \quad \text{and} \quad \lambda_1 + \lambda_2 + \lambda_3 = k(d+1) + 6n_0 = d_\lambda + k,$$

and x^2, y^3, and $s^{n_0}yz^3$ in type E_7, resp. x^2, y^3, and z^5 in type E_8, have weight d_λ for the weights $(\lambda_1, \lambda_2, \lambda_3, 1)$ on (x, y, z, s). Let ϖ be the morphism

$$(x, y, z, s) = (s^{\lambda_1}x', s^{\lambda_2}y', s^{\lambda_3}z', s),$$

the s-chart of the weighted blow-up with the weights $(\lambda_1, \lambda_2, \lambda_3, 1)$ on (x, y, z, s). It sends $x^{\nu_1}y^{\nu_2}z^{\nu_3}s^n$ to $s^{k\omega+n'}x'^{\nu_1}y'^{\nu_2}z'^{\nu_3}$. By (F.50) and (F.51), $f_* \circ \varpi = s^{d_\lambda}f'_*$ with

$$f'_* = f_0(x', y', z') + R_*(y', z', s), \tag{F.53}$$

where

$$R_* = \begin{cases} s^{4k}y'z'^4\alpha'_*(s^{\lambda_3}z', s) + s^{2k-n_0}z'^5\beta'_*(s^{\lambda_3}z', s) & \text{in type } E_7, \\ s^{4k}y'z'^4\alpha'_*(s^{\lambda_3}z', s) & \text{in type } E_8. \end{cases}$$

Since $k > n_0$, $R_* \in s\Bbbk[[y', z', s]]$. Any monomial of R_* is divisible by $y'z'^4$, or, in type E_7, by z'^5. Since $w_2 + 4w_3 > d$ in both types and $5w_3 = 20 > 18 = d$ in type E_7, for any positive integer j the weight of any monomial of R_* for the weights $(jw_1, jw_2, jw_3, 1)$ on (x', y', z', s) is greater than jd.

Let $Y_* := (f'_* = 0) \subset \operatorname{Spec}\Bbbk[[x', y', z', s]]$ and $\Sigma_* := (s = 0) \subset Y_*$. Since $R_* \in (s)$,

$$\Sigma_* = (f_0(x', y', z') = 0)$$

is a Du Val surface of type E_7, resp. E_8, hence a normal Cartier divisor, so Y_* is normal. Since

$$\varpi^*(dx \wedge dy \wedge dz \wedge ds) = s^{d_\lambda + k}dx' \wedge dy' \wedge dz' \wedge ds,$$

adjunction implies $K_{Y_*} = \varpi^*K_{W_*} + k\operatorname{div}(s)$.

A monomial of h_* with $\omega = m$ has $n' \geq n_1$, and a monomial with $\omega \geq m + 1$ is sent to a multiple of $s^{k\omega+n'}$ with $k\omega + n' \geq km + k > km + n_1$. Thus $h_* \circ \varpi = s^{km}h'_*$ with

$$h'_* = s^{n_1}\big(P_*(x', y', z') + sg_*\big), \quad g_* \in \Bbbk[[x', y', z', s]],$$

where $P_* := s^{-km-n_1}(h_{*,m} \circ \varpi)$ and $h_{*,m}$ is the sum of the terms of h_* with $\omega = m$ and $n' = n_1$. Thus P_* is a polynomial in (x', y', z'). Since P_* is a non-zero polynomial of x'-degree at most 1, it is non-zero on Σ_*, so $h'_*|_{Y_*} \neq 0$, and

$$K_{Y_*} + \tfrac{1}{m}\operatorname{div}(h'_*) = \varpi^*\big(K_{W_*} + \tfrac{1}{m}\operatorname{div}(h_*)\big). \tag{F.54}$$

Step 5. In this step we show that, on a subsequence, any computing lift of b_i has bounded order along the axis s and large orders along x, y, z.

By Set-up 17.1(4)(5), $C \subset \operatorname{Sing}(W_*)$ is the only lc center of $(W_*, \frac{1}{m}\operatorname{div}(h_*))$, hence the minimal lc center, and W_* is cDV by Construction-Theorem 13.6(1). Since Set-up 15.4 holds with $B_0 = a$ (**Step 1**), the hypotheses of Theorem 16.1 hold. Fix G-eigenparameters (s_c, u_1, u_2, u_3) of $\mathcal{X}_*$ with $C = (u_1 = u_2 = u_3 = 0)$ (Set-up 17.1(6)), let ϵ and n_c be the number ϵ and the integer n of Theorem 16.1, and let

$$N_1 := \lceil Ma/\epsilon \rceil \quad \text{and} \quad N_u := \lambda_1 N_1 + 1.$$

By Theorem 16.1(1) with N_u in place of k, possibly passing to a subsequence, there exist G-eigenparameters $(s_{c,i}, u_{i1}, u_{i2}, u_{i3})$ of $\widehat{\mathcal{X}}$ such that the tuples formed by f_i, h_i, $s_{c,i}$ and the u_{ij} have a compatible generic limit congruent to $(f_*, h_*, s_c, u_1, u_2, u_3)$ modulo $\mathfrak{n}_*^{N_u}$, where $\mathfrak{n}_*$ is the maximal ideal of $\Bbbk[[x, y, z, t]]$, and such that for any i and any computing lift F of b_i, which satisfies $a(F, W_i, \frac{1}{m}\operatorname{div}(h_i))/e_F = b_i \leq a$,

$$\operatorname{mult}_F \mathfrak{m}_{o_i} \leq N_1, \quad \text{and} \quad \operatorname{mult}_F \operatorname{Jac}_{W_i} \leq n_c(N_u + N_1) \quad \text{if} \quad \min_j \operatorname{mult}_F u_{ij} \leq N_u. \tag{F.55}$$

The hypotheses of Theorem 6.5 for (f_i, h_i) and their compatible generic limit along the subsequence above, which is (f_*, h_*) before the coordinate changes of **Step 2**, with Ma in place of B_0 and $n_c(N_u + N_1)$ in place of R, hold:

(a) f_i and h_i are polynomials (**Step 1**).
(b) W_i° is normal near 0.
(c) $h_i|_{W_i^\circ} \neq 0$.
(d) W_* is normal.
(e) $h_*|_{W_*} \neq 0$.
(f) $(W_*, \frac{1}{m}\operatorname{div}(h_*))$ is klt on $W_* \setminus \operatorname{Sing}(W_*)$, because

$$\operatorname{Nklt}(W_*, \tfrac{1}{m}\operatorname{div}(h_*)) = C \subset \operatorname{Sing}(W_*).$$

The coordinate changes of **Step 2** do not affect the normality of W_*, the non-vanishing of $h_*|_{W_*}$, and the klt property of $(W_*, \frac{1}{m}\operatorname{div}(h_*))$ outside $\operatorname{Sing}(W_*)$. Let T_W be the real number T of Theorem 6.5. If a computing lift F of b_i satisfies $\min_j \operatorname{mult}_F u_{ij} \leq N_u$, then by (F.48) and (F.55), $\operatorname{mult}_F h_i \leq T_W$ for all but finitely many i, while $\operatorname{mult}_F h_i \geq m\delta/(1-\tau_i)$ by (F.49) and $e_F \geq 1$. Since $\lim_{i\to+\infty} \tau_i = 1$, possibly truncating the sequence, we may assume that any computing lift F of b_i satisfies $\min_j \operatorname{mult}_F u_{ij} > N_u$. Since $\mathfrak{m}_{o_i}$ is generated by $s_{c,i}$ and the u_{ij}, and $\operatorname{mult}_F \mathfrak{m}_{o_i} \leq N_1 < N_u$, we obtain $\operatorname{mult}_F s_{c,i} = \operatorname{mult}_F \mathfrak{m}_{o_i} \leq N_1$.

We compare (s_c, u_1, u_2, u_3) with the coordinates (x, y, z, s) of **Step 2**. In $\Bbbk[[x, y, z, s]]$, both (x, y, z) and (u_1, u_2, u_3) generate the ideal of C, and both $s|_C$ and $s_c|_C$ are parameters of $C \cong \operatorname{Spec} \Bbbk[[s]]$. Thus $x, y, z \in (u_1, u_2, u_3)$ and $s - \upsilon s_c \in (u_1, u_2, u_3)$ for a unit υ. Let $J := m + M$ and fix an integer

$$N_3 > d_\lambda + Jd + km + N_u.$$

We realize coordinate changes on (f_*, h_*) by coordinate changes on (f_i, h_i) as follows. From here up to the list (1)–(3) below, (f_*, h_*) is written in the coordinates of **Step 1**, with z and t exchanged if they were exchanged in **Step 2**, and in that list in the coordinates of **Step 2**. Let $K_0 \subset \Bbbk$ be the subfield generated over $\mathbb{C}$ by the coefficients of f_* and h_*. Let σ be an automorphism of $\Bbbk[[x, y, z, t]]$ sending each coordinate to a G-eigenfunction with the character of that coordinate, whose coefficients are algebraic over K_0, and let N be a positive integer. The coefficients of the monomials of degree $< N$ in $\sigma(x), \sigma(y), \sigma(z), \sigma(t)$ are finitely many elements of $\Bbbk$, algebraic over the function field of $Z_{d'}$ for some $d' \geq N$, whose generic point is the d'-jet of (f_*, h_*) (Definition 2.5). After a finite dominant base change of $Z_{d'}$ and shrinking it to a dense open subset, the coefficients of σ of degree $< N$ are regular functions on it, and the linear part of σ stays invertible. Specializing them to the jets of (f_i, h_i) defines, for infinitely many i, maps σ_i sending x, y, z, t to polynomials without constant terms and with invertible linear part, hence automorphisms σ_i of $\mathbb{C}[[x, y, z, t]]$.

A condition of the form

$$\text{a fixed coefficient of degree } < N \text{ of } \sigma,\ f_* \circ \sigma, \text{ or } h_* \circ \sigma \text{ equals a fixed element of } \mathbb{C}$$

is a closed condition on the base change of $Z_{d'}$. If it holds at the generic point, then it holds for σ_i, $f_i \circ \sigma_i$, and $h_i \circ \sigma_i$. In particular, the σ_i send each coordinate to an eigenfunction with the character of that coordinate. Since f_i, h_i and σ_i are polynomials, so are $f_i \circ \sigma_i$ and $h_i \circ \sigma_i$. Possibly passing to a subsequence, the tuples $(f_i \circ \sigma_i, h_i \circ \sigma_i)$ have a compatible generic limit congruent to $(f_* \circ \sigma, h_* \circ \sigma)$ modulo $\mathfrak{n}_*^N$. Here the limit exists by Lemma 2.6(1). As the base change of $Z_{d'}$ is finite and dominant, any non-empty open subset of the base change contains the points defining σ_i for all but finitely many i, so the N-jets of the limit and of $(f_* \circ \sigma, h_* \circ \sigma)$ map to one point of the jet space, and the congruence is up to an embedding of coefficient fields (Lemma 2.6(3)). If finitely many further functions are added to the tuples (f_i, h_i) and (f_*, h_*) and transformed by σ_i, resp. σ, then the transformed tuples again have, on a subsequence, a compatible generic limit congruent modulo $\mathfrak{n}_*^N$ to the transformed tuple of the limit.

The coefficients of ψ are algebraic over K_0: the automorphisms of $\Bbbk$ over K_0, acting on coefficients, fix f_*, so they permute the finitely many irreducible components of $\operatorname{Sing}(W_*)$, of

which C is one, and send ψ to one of finitely many series, while an element of $\Bbbk$ with finitely many conjugates over K_0 is algebraic over K_0. The coefficients of $v^{1/3}$ and $v^{1/5}$ are then algebraic over K_0, since they are solved coefficient by coefficient.

If z and t were exchanged in **Step 2**, we exchange them in the f_i, h_i, $s_{\mathrm{c},i}$, and u_{ij} as well, and we apply the realization above to the composite σ of the other coordinate changes of **Step 2**, with N_3 in place of N, adding the functions $s_{\mathrm{c},i}$ and u_{ij} to the tuples. Possibly passing to a subsequence, we obtain polynomial G-eigenparameters of $\widehat{\mathcal{X}}$, which we again denote by (x, y, z, s), and we replace f_i, h_i, $s_{\mathrm{c},i}$, and u_{ij} with $f_i \circ \sigma_i$, $h_i \circ \sigma_i$, $s_{\mathrm{c},i} \circ \sigma_i$, and $u_{ij} \circ \sigma_i$, so that the following hold for any i.

(1) $f_i = x^2 + y^3 + y\alpha_i(z, s) + \beta_i(z, s)$, and $h_i = \varphi_i(y, z, s) + x\xi_i(y, z, s)$, since σ changes only (z, t).
(2) Any monomial of degree $< N_3$ whose coefficient in f_*, resp. h_*, is 0 has coefficient 0 in f_i, resp. h_i, and the coefficient of $s^{n_0}yz^3$ in type E_7, resp. of z^5 in type E_8, in f_i is 1.
(3) The elements x, y, z, and $s - v_i s_{\mathrm{c},i}$ belong to $(u_{i1}, u_{i2}, u_{i3}) + \mathfrak{n}^{N_{\mathrm{u}}}$ for a unit v_i, where $\mathfrak{n}$ is the maximal ideal of $\mathbb{C}[[x, y, z, s]]$.

Here (3) holds for all but finitely many i, and we discard the other i: we have

$$x, y, z, s - vs_{\mathrm{c}} \in (u_1, u_2, u_3),$$

the limits of $s_{\mathrm{c},i}$ and u_{ij} are congruent to s_{c} and u_j modulo $\mathfrak{n}_*^{N_{\mathrm{u}}}$, and in the expansions of x, y, z, s in the parameters $(s_{\mathrm{c},i}, u_{i1}, u_{i2}, u_{i3})$, (3) is the vanishing of the coefficients of $s_{\mathrm{c},i}^e$ with $e < N_{\mathrm{u}}$ for x, y, z and the non-vanishing of the coefficient of $s_{\mathrm{c},i}$ for s, finitely many conditions which hold for the limit.

Let F be a computing lift of b_i. Since F has center o_i, $\operatorname{mult}_F \mathfrak{n}^{N_{\mathrm{u}}} \geq N_{\mathrm{u}}$, so by (3),

$$\min\{\operatorname{mult}_F x, \operatorname{mult}_F y, \operatorname{mult}_F z\} \geq \min\{\min_j \operatorname{mult}_F u_{ij}, N_{\mathrm{u}}\} = N_{\mathrm{u}} \quad \text{and} \quad \operatorname{mult}_F s = \operatorname{mult}_F s_{\mathrm{c},i}.$$

Thus, for any computing lift F of b_i,

$$\operatorname{mult}_F s \leq N_1 \quad \text{and} \quad \min\{\operatorname{mult}_F x, \operatorname{mult}_F y, \operatorname{mult}_F z\} > \lambda_1 N_1. \tag{F.56}$$

Step 6. In this step we pass to the chart ϖ of **Step 4** on the sequence and show that it keeps b_i and the computing lifts.

Let ϖ be the morphism of **Step 4** in the coordinates (x, y, z, s) of **Step 5**. A monomial $y^{\nu_2}z^{\nu_3}s^n$ of degree $\nu_2 + \nu_3 + n \geq N_3$ is sent to $s^{\lambda_2\nu_2+\lambda_3\nu_3+n}y'^{\nu_2}z'^{\nu_3}$ with $\lambda_2\nu_2 + \lambda_3\nu_3 + n \geq N_3 > d_\lambda + Jd$. By **Step 5**(1)(2) and the computation of **Step 4**, $f_i \circ \varpi = s^{d_\lambda} f_i'$ and $h_i \circ \varpi = s^{km} h_i'$ with polynomials

$$f_i' = f_0(x', y', z') + R_i(y', z', s) \quad \text{and} \quad h_i' \in \mathbb{C}[x', y', z', s],$$

where $R_i \in s\mathbb{C}[y', z', s]$ and any monomial of R_i has weight $> Jd$ for the weights $(Jw_1, Jw_2, Jw_3, 1)$ on (x', y', z', s).

Let $Y_i := (f_i' = 0)$, p_i the origin, and $\Sigma_i := (s = 0) \subset Y_i$, which is the Du Val surface $(f_0(x', y', z') = 0)$. The morphism ϖ is G-equivariant for the characters

$$\chi_{x'} = \chi_x \chi_s^{-\lambda_1}, \quad \chi_{y'} = \chi_y \chi_s^{-\lambda_2}, \quad \text{and} \quad \chi_{z'} = \chi_z \chi_s^{-\lambda_3},$$

birational, an isomorphism over $(s \neq 0)$, and sends p_i to o_i. Thus G acts faithfully on Y_i. Since Σ_i is a normal Cartier divisor, Y_i is normal near p_i and smooth along $\Sigma_i \backslash \{p_i\}$, and $Y_i \backslash \Sigma_i$ is smooth near p_i because $\operatorname{Sing}(W_i) = \{o_i\} \subset (s = 0)$ near o_i. Thus $(Y_i \ni p_i)$ is an isolated Gorenstein cDV threefold germ, hence terminal. The function h_i' is G-semi-invariant, and as in **Step 4**,

$$K_{Y_i} + \tfrac{1}{m}\operatorname{div}(h_i') = \varpi^*\big(K_{W_i} + \tfrac{1}{m}\operatorname{div}(h_i)\big).$$

Thus for any prime divisor P over Y_i with center p_i, P is a prime divisor over W_i with center o_i, $I_G(P)$ does not depend on whether P is viewed over Y_i or over W_i, and

$$a(P, W_i, 0) = a(P, Y_i, 0) + k \operatorname{mult}_P s \quad \text{and} \quad a(P, W_i, \tfrac{1}{m}\operatorname{div}(h_i)) = a(P, Y_i, \tfrac{1}{m}\operatorname{div}(h_i')). \tag{F.57}$$

In particular, $(Y_i, \frac{1}{m}\operatorname{div}(h_i'))$ is klt near p_i.

Let $b_i' := \operatorname{mld}((Y_i \ni p_i, \frac{1}{m}\operatorname{div}(h_i'))/G)$. By (F.57), $b_i' \geq b_i$. Let F be a computing lift of b_i. By (F.56), $\operatorname{mult}_F x' = \operatorname{mult}_F x - \lambda_1 \operatorname{mult}_F s > 0$, similarly $\operatorname{mult}_F y', \operatorname{mult}_F z' > 0$, and $\operatorname{mult}_F s > 0$, so $\operatorname{center}_{Y_i} F = \{p_i\}$. Thus $b_i' = b_i$ by (F.57), and the computing lifts of b_i' are exactly the computing lifts of b_i.

Step 7. In this step we bound the log discrepancy of one computing lift of b_i and conclude.

By **Step 6**, $(Y_i \ni p_i)$, G, the coordinates (x', y', z', s), where s plays the role of t, the equation $f_i' = f_0 + R_i$, the integer J, and the mark h_i' satisfy Set-up F.6, and the value at 1 of the radial function of $(Y_i \ni p_i, G, h_i', m)$ is $b_i' = b_i > 0$. Let V_i and S_i be the chart Y_J and the surface S of Set-up F.6 for Y_i, and let L_{type} be the integer L_{type} of Proposition F.9.

Let T_W' be the real number T of Theorem 6.5 for the polynomials f_i, h_i in the coordinates of **Step 1** and their compatible generic limit, with Ma in place of B_0 and $(L_{\text{type}}J + d_\lambda)N_1$ in place of R. The hypotheses of Theorem 6.5 do not involve B_0 and R, and for f_i, h_i in the coordinates of **Step 1** they hold by **Step 5** and Lemma 2.6(2). For a prime divisor F over W_i with center o_i, the numbers $a(F, W_i, 0)$, $\operatorname{mult}_F h_i$, and $\operatorname{mult}_F \operatorname{Jac}_{W_i}$ do not depend on the coordinates.

Let F be a computing lift of b_i. By Proposition F.9,

$$\operatorname{center}_{V_i} F \subset S_i \quad \text{or} \quad \operatorname{mult}_F \operatorname{Jac}_{Y_i} \leq L_{\text{type}} J \operatorname{mult}_F \mathfrak{m}_{p_i}.$$

Suppose that $\operatorname{center}_{V_i} F \subset S_i$. Let n_t be the order of the kernel of $\chi_s|_{I_G(F)}$. Since $e_F \leq M$ and $n_t \geq 1$, $J = m + M \geq m + e_F/n_t$, so by Lemma F.8 there exists a computing lift E of $b_i' = b_i$ with $a(E, Y_i, 0) \leq (m+2)e_E$. Since E is a computing lift of b_i by **Step 6**, (F.57) and (F.56) imply

$$a(E, W_i, 0) = a(E, Y_i, 0) + k \operatorname{mult}_E s \leq (m+2)e_E + kN_1.$$

Suppose that the second alternative holds. Since $s \in \mathfrak{m}_{p_i}$, (F.56) implies

$$\operatorname{mult}_F \operatorname{Jac}_{Y_i} \leq L_{\text{type}} J \operatorname{mult}_F \mathfrak{m}_{p_i} \leq L_{\text{type}} J \operatorname{mult}_F s \leq L_{\text{type}} J N_1,$$

where the first inequality is the second alternative. Differentiating $f_i \circ \varpi = s^{d_\lambda} f_i'$ for the chart ϖ of **Step 6**, we obtain

$$s^{d_\lambda} \frac{\partial f_i'}{\partial x'} = s^{\lambda_1} \frac{\partial f_i}{\partial x} \circ \varpi,$$

similarly for y' and z' with λ_2 and λ_3, and, on Y_i,

$$s^{d_\lambda} \frac{\partial f_i'}{\partial s} = \frac{\partial f_i}{\partial s} \circ \varpi + \lambda_1 s^{\lambda_1 - 1} x' \frac{\partial f_i}{\partial x} \circ \varpi + \lambda_2 s^{\lambda_2 - 1} y' \frac{\partial f_i}{\partial y} \circ \varpi + \lambda_3 s^{\lambda_3 - 1} z' \frac{\partial f_i}{\partial z} \circ \varpi.$$

Since $\lambda_1, \lambda_2, \lambda_3 \geq 1$, we have

$$s^{d_\lambda} \operatorname{Jac}_{Y_i} \subset \operatorname{Jac}_{W_i} \mathcal{O}_{Y_i}.$$

Since $\operatorname{center}_{Y_i} F = \{p_i\}$ by **Step 6**,

$$\operatorname{mult}_F \operatorname{Jac}_{W_i} \leq d_\lambda \operatorname{mult}_F s + \operatorname{mult}_F \operatorname{Jac}_{Y_i} \leq (L_{\text{type}} J + d_\lambda) N_1.$$

By (F.48),

$$a(F, W_i, 0) - \frac{1}{m} \operatorname{mult}_F h_i = e_F b_i \leq Ma.$$

Thus, by the choice of T_W', $a(F, W_i, 0) \leq Ma + T_W'/m$ for all but finitely many i.

Let $N_4 := m + 2 + Ma + T_W'/m + kN_1$. In both cases, for all but finitely many i there exists a computing lift F_i of b_i with $a(F_i, W_i, 0) \leq N_4 e_{F_i}$. Since $a(F_i, W_i, \frac{1}{m}\operatorname{div}(h_i)) = e_{F_i} b_i > 0$,

$$\frac{\operatorname{mult}_{F_i} h_i}{m e_{F_i}} < \frac{a(F_i, W_i, 0)}{e_{F_i}} \leq N_4,$$

which contradicts (F.49) for any i with $\delta/(1 - \tau_i) \geq N_4$. The theorem follows. □

Proof of Proposition 17.5. We divide the proof into five steps. By Construction-Theorem 13.6, after passing to a subsequence, $G = G_i$, the diagonal action of G on the ambient eigenparameters (x, y, z_0, t), and the characters of the f_i and of the h_i are fixed, $f_i - x^2$ is a series in (y, z_0, t), h_i and h_* have x-degree at most 1, and

$$f_* = x^2 + y^3 + y\widetilde{\phi}(z_0, t) + \widetilde{\psi}(z_0, t)$$

for some $\widetilde{\phi}, \widetilde{\psi} \in \Bbbk[[z_0, t]]$. Let η be the generic point of C, and let $w := (6, 4, 3)$ be weights on (x, y, z).

Step 1. In this step we construct (x, y, z, s) and compute the lowest w-weight of h_*.

Since $C \subset \mathrm{Sing}(W_*)$ and $\partial f_*/\partial x = 2x$, x vanishes on C. Since the transverse type of W_* at η is E_6, the Hessian matrix of f_* in (x, y, z_0, t) has rank 1 at η. The minor of the Hessian matrix of f_* in the rows and columns of x and y is $2 \cdot 6y$, so y vanishes on C. Thus C is a G-invariant regular curve in the surface $(x = y = 0) = \operatorname{Spec} \Bbbk[[z_0, t]]$. The tangent line of C is G-stable, so after a G-equivariant linear change of (z_0, t) the tangent line is $(z_0 = 0)$ and $C = (x = y = z_0 - \lambda(t) = 0)$ for some $\lambda \in \Bbbk[[t]]$. Since C is G-invariant, $z_0 - \lambda(t)$ is a G-eigenfunction with the character of z_0. Letting $z := z_0 - \lambda(t)$ and $s := t$, we obtain G-eigenparameters (x, y, z, s) with $C = (x = y = z = 0)$. The changes of coordinates above involve only z_0 and t, so $f_* = x^2 + y^3 + y\widetilde{\phi}(z, s) + \widetilde{\psi}(z, s)$ for new series $\widetilde{\phi}, \widetilde{\psi} \in \Bbbk[[z, s]]$, and h_* has x-degree at most 1.

Expand f_* in $\Bbbk((s))[[x, y, z]]$. Since W_* is singular along C, $f_* \in (x, y, z)^2$. The quadratic part $x^2 + q_2(y, z)$ of f_* has rank 1, so $q_2 = 0$. The cubic part $y^3 + q_3(y, z)$ is the cube of a linear form, since the transverse type at η is E_6. Since $y\widetilde{\phi} + \widetilde{\psi}$ is linear in y, q_3 has no monomial $y^2 z$, so the cube is y^3 and $q_3 = 0$. Thus $\widetilde{\phi} \in (z^3)$ and $\widetilde{\psi} \in (z^4)$. Since the type at η is E_6 and not E_7 or E_8, the coefficient of z^4 in $\widetilde{\psi}$ is non-zero. We write it as $s^\nu \vartheta(s)$ with $\nu \in \mathbb{N}$ and $\vartheta(0) \neq 0$. Since G acts diagonally and f_* is G-semi-invariant, every monomial of f_* in $z^4\Bbbk[[s]]$ has the character of $s^\nu z^4$, so ϑ is G-invariant, and so is the fourth root $\vartheta^{1/4} \in \Bbbk[[s]]$ with a fixed value at 0. Substituting $\vartheta(s)^{-1/4} z$ for z, we obtain

$$f_* = x^2 + y^3 + s^\nu z^4 + yz^3\phi(z, s) + z^5\psi(z, s)$$

with $\phi, \psi \in \Bbbk[[z, s]]$.

Since h_* has x-degree at most 1, the set $\mathcal{E}$ of the statement gives

$$h_* = \sum_{\alpha \in \mathcal{E}} \zeta_\alpha x^{\alpha_1} y^{\alpha_2} z^{\alpha_3} s^{\alpha_4}$$

with $\zeta_\alpha \in \Bbbk \backslash \{0\}$. We show that $\min_{\alpha \in \mathcal{E}} w(\alpha) = m$. We need the following threshold computation.

Claim F.10. *Let $\mathbb{K} \supset \mathbb{C}$ be an algebraically closed field, $S := \operatorname{Spec} \mathbb{K}[[x, y, z]]/(x^2 + y^3 + z^4)$, and $g \in \mathbb{K}[[x, y, z]]$ such that the lowest w-weight part $\mathrm{in}_w(g)$ of g has weight $I \geq 1$ and is not divisible by $x^2 + y^3 + z^4$. Then for any real number $\theta \geq 0$, $(S, \theta \operatorname{div}(g|_S))$ is lc if and only if $\theta \leq \frac{1}{I}$, and klt if and only if $\theta < \frac{1}{I}$. If moreover $g = \mathrm{in}_w(g)$, then any component of $\operatorname{div}(g|_S)$ has coefficient at most $\frac{I}{2}$.*

Proof. By [Roa04, Table 1], after scaling the generators, $\mathbb{K}[x, y, z]/(x^2 + y^3 + z^4)$ is the ring of invariants of the binary tetrahedral group $\Xi \subset SL_2(\mathbb{K})$ acting on $\mathbb{K}[\sigma_1, \sigma_2]$, with x, y, z homogeneous of degrees $12, 8, 6$, that is, twice their w-weights. Let $\Pi := \operatorname{Spec} \mathbb{K}[[\sigma_1, \sigma_2]]$ and $\pi : \Pi \to S$ the quotient map. Since $\Xi \subset SL_2(\mathbb{K})$ acts freely on $\Pi \backslash \{0\}$, π is étale in codimension 1 and $K_\Pi = \pi^* K_S$. Since $\mathrm{in}_w(g)$ is non-zero in the graded ring $\mathbb{K}[x, y, z]/(x^2 + y^3 + z^4)$, $\pi^* g$ has multiplicity $2I$ at 0, and the lowest degree part of $\pi^* g$ is the non-zero Ξ-invariant binary form $\Phi := \pi^* \mathrm{in}_w(g)$ of degree $2I$.

The image of Ξ in $PGL_2(\mathbb{K})$ is the tetrahedral group A_4. The stabilizer of a point of $\mathbb{P}^1$ in a finite subgroup of $PGL_2(\mathbb{K})$ is cyclic, and the cyclic subgroups of A_4 have order at most 3, so any

orbit of A_4 on $\mathbb{P}^1$ has at least $12/3 = 4$ points. Let $q \in \mathbb{P}^1$ be a root of Φ. Since Φ is Ξ-invariant, Φ has multiplicity $\mu := \operatorname{mult}_q \Phi$ at any point of the orbit $A_4 q$, so $4\mu \le \#(A_4 q) \cdot \mu \le 2I$, that is,

$$\mu \le \frac{I}{2}.$$

Let $\pi' : \Pi' \to \Pi$ be the blow-up of 0, E_Π the exceptional curve, and $D' := (\pi')_*^{-1} \operatorname{div}(\pi^* g)$. Then $a(E_\Pi, \Pi, \theta \operatorname{div}(\pi^* g)) = 2 - 2I\theta$, and

$$K_{\Pi'} + (2I\theta - 1)E_\Pi + \theta D' = (\pi')^*(K_\Pi + \theta \operatorname{div}(\pi^* g)).$$

The restriction $D'|_{E_\Pi}$ is the divisor of Φ on $E_\Pi \cong \mathbb{P}^1$, so $\frac{1}{I} D'|_{E_\Pi}$ has coefficients at most $\frac{1}{2}$. By inversion of adjunction, $(\Pi', E_\Pi + \frac{1}{I} D')$ is lc near E_Π. Thus $(\Pi, \frac{1}{I} \operatorname{div}(\pi^* g))$ is lc near 0 and not klt, that is, $\operatorname{lct}(\Pi \ni 0; \operatorname{div}(\pi^* g)) = \frac{1}{I}$. Since

$$K_\Pi + \theta \operatorname{div}(\pi^* g) = \pi^*(K_S + \theta \operatorname{div}(g|_S)),$$

the first statement follows.

If $g = \operatorname{in}_w(g)$, then $\pi^* g = \Phi$, whose components are lines of multiplicity at most $\frac{I}{2}$. Since π is étale in codimension 1, the coefficient of a component Σ of $\operatorname{div}(g|_S)$ equals $\operatorname{mult}_\ell \Phi \le \frac{I}{2}$ for a line $\ell \subset \Pi$ over Σ. □

Let $\mathbb{K}$ be an algebraic closure of $\Bbbk((s))$ and $S_\eta := \operatorname{Spec} \mathbb{K}[[x, y, z]]/(f_*)$. Since the pair $(W_*, \frac{1}{m} \operatorname{div}(h_*))$ is lc with non-klt locus C, it is lc and not klt at η, and since localization at η, completion and the extension of the coefficient field from $\Bbbk((s))$ to $\mathbb{K}$ compose to a regular morphism $\operatorname{Spec} \mathbb{K}[[x, y, z]] \to \mathcal{X}_*$, along which log resolutions pull back (cf. [dFEM11, Remark 2.10]) and relative canonical divisors pull back (over the regular $\mathcal{X}_*$ by [JM12, Lemma 1.2], and over its Cartier divisor W_*, which is Gorenstein, since dualizing sheaves commute with flat base change), hence preserve log discrepancies and so change neither the lc nor the klt property, $(S_\eta, \frac{1}{m} \operatorname{div}(h_*))$ is lc and not klt. Substituting $s^{-\nu/4} z$ for z for a fourth root $s^{\nu/4} \in \mathbb{K}$, we have $f_* = x^2 + y^3 + z^4 + Q$ with all monomials of Q of w-weight greater than 12.

The monomials outside the Jacobian ideal (x, y^2, z^3) of $x^2 + y^3 + z^4$ are $1, y, z, yz, z^2, yz^2$, of w-weight at most 10. Thus if $Q \ne 0$, the lowest w-weight part of Q equals $2xa_1 + 3y^2 a_2 + 4z^3 a_3$ with w-homogeneous a_1, a_2, a_3 of w-weights $\operatorname{wt}_w(Q) - 6$, $\operatorname{wt}_w(Q) - 8$, $\operatorname{wt}_w(Q) - 9$, and substituting $x - a_1, y - a_2, z - a_3$ for x, y, z turns f_* into $x^2 + y^3 + z^4$ plus monomials of w-weight greater than $\operatorname{wt}_w(Q)$. The composition of the substitutions converges, yields an isomorphism of S_η with S of Claim F.10, and changes any series g only by monomials of w-weight greater than the weight of $\operatorname{in}_w(g)$.

Let $I := \min_{\alpha \in \mathcal{E}} w(\alpha)$. Since $(S_\eta, \frac{1}{m} \operatorname{div}(h_*))$ is not klt, h_* is not a unit, so $I \ge 1$. For $\alpha \in \mathcal{E}$ with $w(\alpha) = I$, the coefficient of $x^{\alpha_1} y^{\alpha_2} z^{\alpha_3}$ in h_* is a non-zero element of $\Bbbk[[s]]$, so the w-weight I part of h_* is non-zero. After the substitutions, the lowest w-weight part of h_* is the w-weight I part above with z rescaled, which has x-degree at most 1 and is not divisible by $x^2 + y^3 + z^4$. By Claim F.10, the pair $(S_\eta, \frac{1}{m} \operatorname{div}(h_*))$ is lc and not klt if and only if $\frac{1}{m} = \frac{1}{I}$. Thus

$$w(\alpha) \ge m \text{ for any } \alpha \in \mathcal{E}, \text{ with equality for some } \alpha \in \mathcal{E}.$$

Step 2. In this step we bound ν and c and fix r.

For an integer $r' > \max\{\nu', c\}$, let $\mathbf{w}_{r'} := (6r' + 3\nu', 4r' + 2\nu', 3r' + \nu', d_\nu)$ be weights on (x, y, z, s). Then

$$\operatorname{wt}_{\mathbf{w}_{r'}}(x^{\alpha_1} y^{\alpha_2} z^{\alpha_3} s^{\alpha_4}) = r' w(\alpha) + \nu'(3\alpha_1 + 2\alpha_2 + \alpha_3) + d_\nu \alpha_4. \tag{F.58}$$

The monomials x^2, y^3, $s^\nu z^4$ have $\mathbf{w}_{r'}$-weight $12r' + 6\nu'$, while $yz^3\phi$ and $z^5\psi$ have $\mathbf{w}_{r'}$-weights at least $13r'+5\nu'$ and $15r'+5\nu'$, both greater than $12r'+6\nu'$ since $r' > \nu'$. Thus $\operatorname{wt}_{\mathbf{w}_{r'}}(f_*) = 12r'+6\nu'$ and $\operatorname{in}_{\mathbf{w}_{r'}}(f_*) = x^2 + y^3 + s^\nu z^4$, which is irreducible: as a polynomial in x over $\Bbbk[y, z, s]$, the constant term $y^3 + s^\nu z^4$ has odd degree in y and is not a square. The weights $\mathbf{w}_{r'}$ are primitive:

if $d_\nu = 2$, then ν is odd, $\nu' = \nu$, and $6r' + 3\nu'$ is odd. By [Liu26b, Lemma 4.12], the $\mathbf{w}_{r'}$-weighted blow-up induces a prime divisor $E_{r'}$ over W_* with center the closed point, and

$$a(E_{r'}, W_*, 0) = (13r' + 6\nu' + d_\nu) - (12r' + 6\nu') = r' + d_\nu.$$

Since $\mathrm{in}_{\mathbf{w}_{r'}}(h_*)$ has x-degree at most 1, it is not divisible by $\mathrm{in}_{\mathbf{w}_{r'}}(f_*)$, so $\mathrm{mult}_{E_{r'}} h_* = \mathrm{wt}_{\mathbf{w}_{r'}}(h_*)$ by [Liu26b, Lemma 4.12]. By (F.58) and **Step 1**, the monomials of h_* with $w(\alpha) = m$ have $\mathbf{w}_{r'}$-weight at least $r'm + c$, with equality for some of them, and the monomials with $w(\alpha) \geq m+1$ have $\mathbf{w}_{r'}$-weight at least $r'(m+1) > r'm + c$. Thus $\mathrm{mult}_{E_{r'}} h_* = r'm + c$. Since the minimal log discrepancy of $(W_*, \frac{1}{m}\operatorname{div}(h_*))$ at the closed point is positive (Set-up 17.1),

$$0 < a(E_{r'}, W_*, \tfrac{1}{m}\operatorname{div}(h_*)) = r' + d_\nu - \frac{r'm + c}{m} = d_\nu - \frac{c}{m}.$$

For $\alpha \in \mathcal{E}$ with $w(\alpha) = m$, $3(3\alpha_1 + 2\alpha_2 + \alpha_3) \geq w(\alpha) = m$, so $c \geq \frac{\nu' m}{3} = \frac{d_\nu \nu m}{6}$. Since c is an integer,

$$\left\lceil \frac{d_\nu \nu m}{6} \right\rceil \leq c < d_\nu m.$$

In particular $\nu < 6$, so $0 \leq \nu \leq 5$, $\nu' \leq 5$, and $c < 2m$.

Let $r := 2m + 5 > \max\{\nu', c\}$, and let $\mathbf{w} := \mathbf{w}_r$. By the computation above,

$$\mathrm{in}_{\mathbf{w}}(f_*) = x^2 + y^3 + s^\nu z^4, \quad \mathrm{wt}_{\mathbf{w}}(f_*) = 12r + 6\nu', \quad \text{and} \quad \mathrm{wt}_{\mathbf{w}}(h_*) = mr + c. \tag{F.59}$$

Step 3. In this step we realize (x, y, z, s) on the W_i and show that the computing lifts of b_i concentrate along C.

By Lemma 4.3(2), any computing lift F of b_i satisfies

$$\frac{\mathrm{mult}_F h_i}{m e_F} \geq \frac{\delta}{1 - \tau_i}. \tag{F.60}$$

The conditions of Set-up 15.4 hold with $B_0 := a$. Moreover W_* is cDV by Construction-Theorem 13.6(1). By Lemma 13.5, Construction-Theorem 13.6(1) and (3), and Set-up 17.1(4), as $\frac{1}{m}\operatorname{div}(h_*)$ has no coefficient-one component, the minimal lc center C' of $(W_*, \frac{1}{m}\operatorname{div}(h_*))$ is a regular curve. As $C' \subset \mathrm{Nklt}(W_*, \frac{1}{m}\operatorname{div}(h_*)) = C$ by Set-up 17.1(5), we have $C' = C$, so C is the minimal lc center of $(W_*, \frac{1}{m}\operatorname{div}(h_*))$. By Theorem 16.1, there exist a rational number $\epsilon > 0$, a positive integer n, and G-eigenparameters $(\widehat{s}, \widehat{x}, \widehat{y}, \widehat{z})$ of $\mathcal{X}_*$ with $C = (\widehat{x} = \widehat{y} = \widehat{z} = 0)$ satisfying Theorem 16.1(1) and (2) with $(\widehat{s}, \widehat{x}, \widehat{y}, \widehat{z})$ in place of (s, u_1, u_2, u_3). Let

$$\Theta := \left\lceil \frac{Ma}{\epsilon} \right\rceil.$$

Fix a positive integer k, and pass to the infinite subfamily of Theorem 16.1(1) for this k, with G-eigenparameters $(\widehat{s}_i, \widehat{x}_i, \widehat{y}_i, \widehat{z}_i)$ at o_i. By Theorem 16.1(1)(b) with $B_0 = a$, any computing lift F of b_i satisfies

$$\mathrm{mult}_F \mathfrak{m}_{o_i} \leq \Theta. \tag{F.61}$$

We realize z and s on the W_i. Let $N_0 := (m+12)r + c + 6\nu' + 1$ and let $N \geq \max\{N_0, k\}$ be an integer. The series z and s of **Step 1** are G-eigenseries in (z_0, t) with coefficients in $\Bbbk$. The coefficients of the linear change of (z_0, t), of λ and of $\vartheta^{1/4}$ are algebraic over the subfield of $\Bbbk$ generated over $\mathbb{C}$ by the coefficients of f_* and h_*, since C is a component of $\mathrm{Sing}(W_*)$ and $\vartheta^{1/4}$ is solved coefficient by coefficient. The coefficients of the N-jets of z and s are the values at the generic point of regular functions on an irreducible variety V_N mapping dominantly to the irreducible jet stratum of the tuples of Theorem 16.1(1) (Definition 2.5). The vanishing of a polynomial in the coefficients is a closed condition, which holds for all but finitely many i once it holds for the limits. The non-vanishing of such a polynomial is an open condition, which holds on a dense open subset of V_N once it holds for the limits, and the image of that dense open subset contains a dense open subset of the jet stratum, which contains the jets of all but finitely many i (Definition 2.5). Evaluating the functions at a point of V_N over the jet of the i-th tuple, in that

dense open subset when the fiber over that jet intersects it, we obtain G-eigenpolynomials z_i, s_i in (z_0, t) of degree less than N.

The realized coordinate change from (z_0, t) to (z_i, s_i) is the specialization of the limit coordinate change from (z_0, t) to (z, s) on an irreducible jet stratum. Hence the coefficients of degree less than N of f_i, h_i, $\widehat{s}_i$, $\widehat{x}_i$, $\widehat{y}_i$, $\widehat{z}_i$ expressed in (x, y, z_i, s_i) are the specializations of the corresponding coefficients of f_*, h_* and of the limits in Theorem 16.1(1), expressed in (x, y, z, s). Thus finitely many such conditions hold together for all but finitely many i.

Since $\mathbf{w} \in (\mathbb{N}^+)^4$, the monomials of $\mathbf{w}$-weight at most $\max\{12r + 6\nu', mr + c\}$ have degree less than N_0. Thus, by (F.59), possibly passing to an infinite subfamily, for any i the following hold, where $\mathfrak{n}$ is the maximal ideal of the ambient complete local ring at o_i.

(R1) (x, y, z_i, s_i) are G-eigenparameters at o_i.
(R2) Expressed in (x, y, z_i, s_i), f_i has $\mathbf{w}$-weighted order $12r + 6\nu'$, and $\mathrm{in}_{\mathbf{w}}(f_i) = x^2 + y^3 + s_i^{\nu} z_i^4$.
(R3) Expressed in (x, y, z_i, s_i), h_i has $\mathbf{w}$-weighted order $mr + c$, that is, the $\mathbf{w}$-weight $mr + c$ part of h_i is non-zero and all other monomials have larger $\mathbf{w}$-weight.
(R4) $x, y, z_i \in (\widehat{x}_i, \widehat{y}_i, \widehat{z}_i) + \mathfrak{n}^k$.

For (R4), $x, y, z \in (\widehat{x}, \widehat{y}, \widehat{z})$ since the ideal of C is $(x, y, z) = (\widehat{x}, \widehat{y}, \widehat{z})$, and the limits of $\widehat{x}_i, \widehat{y}_i, \widehat{z}_i$ are congruent to $\widehat{x}, \widehat{y}, \widehat{z}$ modulo $\mathfrak{n}_*^k$ (Theorem 16.1(1)). Thus x, y, z belong to the ideal generated by the limits of $\widehat{x}_i, \widehat{y}_i, \widehat{z}_i$ plus $\mathfrak{n}_*^k$. In the expansions of x, y, z in the limits of $(\widehat{s}_i, \widehat{x}_i, \widehat{y}_i, \widehat{z}_i)$, the membership is the vanishing of the coefficients of the powers of the limit of $\widehat{s}_i$ with exponent less than k, a closed condition.

Assume that $k > \Theta$. We show that for all but finitely many i of the subfamily, any computing lift F of b_i satisfies

$$\min\{\mathrm{mult}_F x, \mathrm{mult}_F y, \mathrm{mult}_F z_i\} \ge k \quad \text{and} \quad \mathrm{mult}_F s_i \le \Theta. \tag{F.62}$$

Let $R := n(k + \Theta)$. The orders at o_i of f_i and h_i equal those of f_* and h_* for all but finitely many i, W_* is normal, $h_*|_{W_*} \ne 0$, and $(W_*, \frac{1}{m}\mathrm{div}(h_*))$ is klt on $W_* \setminus \mathrm{Sing}(W_*)$ since the non-klt locus C is contained in $\mathrm{Sing}(W_*)$. Theorem 6.5 requires polynomials. By Lemma 3.2(1)(2) and Lemma 2.12(2)(3), there exist polynomials $f_i^\flat, h_i^\flat$ with compatible generic limit (f_*, h_*) for which b_i, its computing lifts F, and $a(F, W_i, 0)$, $\mathrm{mult}_F h_i$, and $\mathrm{mult}_F \mathrm{Jac}_{W_i}$ for any prime divisor F over W_i with center o_i and $a(F, W_i, \frac{1}{m}\mathrm{div}(h_i)) \le Ma$ do not change. Thus the conclusion of Theorem 6.5 for the $(f_i^\flat, h_i^\flat)$ holds for the (f_i, h_i). Let T_k be the real number of Theorem 6.5 for R, with Ma in place of B_0. For a computing lift F of b_i, $a(F, W_i, 0) - \frac{1}{m}\mathrm{mult}_F h_i = e_F b_i \le Ma$, and since $\tau_i \to 1$, $\mathrm{mult}_F h_i \ge \frac{m\delta}{1-\tau_i} > T_k$ for $i \gg 0$ by (F.60). Thus by Theorem 6.5, for all but finitely many i, any computing lift F of b_i satisfies $\mathrm{mult}_F \mathrm{Jac}_{W_i} > R$. Since $a(F, W_i, \frac{1}{m}\mathrm{div}(h_i))/e_F = b_i \le a$, Theorem 16.1(1)(b) implies that

$$\mathrm{mult}_F((\widehat{x}_i, \widehat{y}_i, \widehat{z}_i)\mathcal{O}_{W_i}) \ge k + 1.$$

By (R4),

$$\mathrm{mult}_F x \ge \min\{\mathrm{mult}_F((\widehat{x}_i, \widehat{y}_i, \widehat{z}_i)\mathcal{O}_{W_i}), k\,\mathrm{mult}_F \mathfrak{m}_{o_i}\} \ge k,$$

and similarly $\mathrm{mult}_F y \ge k$ and $\mathrm{mult}_F z_i \ge k$. By (R1), $\mathfrak{m}_{o_i}$ is generated by x, y, z_i, s_i, so by (F.61) and $k > \Theta$, $\mathrm{mult}_F s_i = \mathrm{mult}_F \mathfrak{m}_{o_i} \le \Theta$. This proves (F.62).

Let $k_1 < k_2 < \cdots$ be integers with $k_1 > (6r + 3\nu')\Theta$. For each $j \ge 1$, we pass, by Theorem 16.1(1) with $k = k_j$ and with the ϵ, n, and $(\widehat{s}, \widehat{x}, \widehat{y}, \widehat{z})$ fixed above, from the family fixed at the beginning of **Step 3** to the infinite subfamily on which (R1)–(R4) and (F.62) hold with $k = k_j$ and $N = N_j := \max\{N_0, k_j\}$, and choose an index $i_j > i_{j-1}$ in it, where $i_0 := 0$. We replace the family by $\{i_j\}_{j \ge 1}$. For $i = i_j$ let $k_i := k_j$ and $N_i := N_j$. Then (R1)–(R3) hold for any i, $k_i \to +\infty$, $k_i > (6r + 3\nu')\Theta$, and any computing lift F of b_i satisfies (F.62) with $k = k_i$. Since the local ring at o_i of the ambient smooth fourfold is dense in its completion and projection to a character preserves regular functions, we replace x, y, z_i, s_i by regular G-eigenfunctions near o_i congruent to them modulo $\mathfrak{n}^{N_i}$. Then (R1)–(R4) and (F.62) still hold, and the coefficients of degree less than N_i of f_i and h_i expressed in them do not change.

Next we replace f_i and h_i by regular functions. Let $\mathcal{O}_i$ be the local ring at o_i of the ambient smooth fourfold $\mathcal{X}_i$ of W_i, so that $\mathfrak{n}$ is the maximal ideal of $\widehat{\mathcal{O}}_i = \mathbb{C}[[x, y, z_i, s_i]]$, let $I_i \subset \mathcal{O}_i$ be the ideal of W_i, and let χ_f and χ_h be the characters of the f_i and of the h_i. Since I_i is principal and f_i generates $I_i\widehat{\mathcal{O}}_i$, there exists a unit $v \in \widehat{\mathcal{O}}_i$ such that vf_i generates I_i. Let $v' \in \mathcal{O}_i$ be congruent to v^{-1} modulo $\mathfrak{n}^{N_i}$, and let $f_i^\sharp$ be the projection of $v'vf_i$ to χ_f. Then $v'vf_i - f_i = v(v' - v^{-1})f_i \in \mathfrak{n}^{N_i+1}$, and since I_i and $\mathfrak{n}^{N_i+1}$ are G-stable and f_i is semi-invariant with character χ_f, we have $f_i^\sharp \in I_i$ and $f_i^\sharp - f_i \in \mathfrak{n}^{N_i+1}$. Since $f_i^\sharp - f_i \in f_i\widehat{\mathcal{O}}_i$ and $f_i \notin \mathfrak{n}^{N_i+1}$, the coefficient of x^2 in f_i being 1 by (R2), we have $f_i^\sharp - f_i \in f_i\mathfrak{n}$, so $f_i^\sharp$ is a unit times f_i and generates $I_i\widehat{\mathcal{O}}_i$. As $J\widehat{\mathcal{O}}_i \cap \mathcal{O}_i = J$ for any ideal $J \subset \mathcal{O}_i$ by the faithful flatness of $\widehat{\mathcal{O}}_i$ over $\mathcal{O}_i$, $f_i^\sharp$ generates I_i.

Let $\widetilde{h}_i \in \mathcal{O}_i$ restrict to the mark on W_i. Since h_i and $\widetilde{h}_i$ have equal images in $\widehat{\mathcal{O}}_i/(f_i)$, namely the image of the mark, $\widetilde{h}_i - h_i \in f_i^\sharp\widehat{\mathcal{O}}_i$, and since $f_i^\sharp\mathcal{O}_i$ is dense in $f_i^\sharp\widehat{\mathcal{O}}_i$, some element of $f_i^\sharp\mathcal{O}_i$ is congruent to $\widetilde{h}_i - h_i$ modulo $\mathfrak{n}^{N_i}$. Subtracting it from $\widetilde{h}_i$ and projecting to χ_h, we obtain $h_i^\sharp \in \mathcal{O}_i$ with $h_i^\sharp - h_i \in \mathfrak{n}^{N_i}$, since h_i is semi-invariant with character χ_h, and $h_i^\sharp$ restricts to the mark on W_i, which is semi-invariant with character χ_h. We replace f_i and h_i by $f_i^\sharp$ and $h_i^\sharp$. The replacement keeps the coefficients of degree less than N_i of f_i and h_i in x, y, z_i, s_i, so (R1)–(R4) and (F.62) hold. Now f_i is a regular G-semi-invariant generator of the ideal of W_i near o_i, and h_i is a regular G-semi-invariant function near o_i which restricts to the mark on W_i.

Step 4. In this step we construct the charts Y_i and the groups Γ_i, and prove (1), (5), (6), and the first part of (2) of Proposition 17.5.

Since x, y, z_i, s_i are G-eigenfunctions and regular parameters at o_i by (R1), possibly shrinking $\mathcal{X}_i$ to a G-invariant open neighborhood of o_i, we may assume that x, y, z_i, s_i, f_i, h_i are regular on $\mathcal{X}_i$ and that the morphism $(x, y, z_i, s_i) : \mathcal{X}_i \to \mathbb{A}^4$ is étale. Let $\mathcal{Y}_i \subset \mathcal{X}_i \times \mathbb{A}^4$ be the closed subscheme defined by

$$x = u^{6r+3\nu'}x', \quad y = u^{4r+2\nu'}y', \quad z_i = u^{3r+\nu'}z', \quad \text{and} \quad s_i = u^{d_\nu}, \tag{F.63}$$

where x', y', z', u are the coordinates of $\mathbb{A}^4$, let $\widetilde{\varpi}_i : \mathcal{Y}_i \to \mathcal{X}_i$ be the first projection, and let $p_i := (o_i, 0)$. The second projection $\mathcal{Y}_i \to \mathbb{A}^4$ is étale, as the base change of (x, y, z_i, s_i) by the morphism $\mathbb{A}^4 \to \mathbb{A}^4$ given by (F.63). Thus $\mathcal{Y}_i$ is a smooth fourfold, x', y', z', u are regular parameters at p_i, and $\widetilde{\varpi}_i$ induces the homomorphism

$$\widehat{\mathcal{O}}_{\mathcal{X}_i,o_i} = \mathbb{C}[[x, y, z_i, s_i]] \to \widehat{\mathcal{O}}_{\mathcal{Y}_i,p_i} = \mathbb{C}[[x', y', z', u]]$$

which substitutes $u^{6r+3\nu'}x'$, $u^{4r+2\nu'}y'$, $u^{3r+\nu'}z'$, and u^{d_ν} for x, y, z_i, and s_i.

Let $f_i' := u^{-(12r+6\nu')}(f_i \circ \widetilde{\varpi}_i)$ and $l_i' := u^{-(mr+c)}(h_i \circ \widetilde{\varpi}_i)$. In $\widehat{\mathcal{O}}_{\mathcal{Y}_i,p_i}[u^{-1}]$,

$$f_i' = u^{-(12r+6\nu')}f_i(u^{6r+3\nu'}x', u^{4r+2\nu'}y', u^{3r+\nu'}z', u^{d_\nu}),$$
$$l_i' = u^{-(mr+c)}h_i(u^{6r+3\nu'}x', u^{4r+2\nu'}y', u^{3r+\nu'}z', u^{d_\nu}),$$

with f_i and h_i expressed in (x, y, z_i, s_i). The substitution turns $x^{\alpha_1}y^{\alpha_2}z_i^{\alpha_3}s_i^{\alpha_4}$ into

$$u^{\mathrm{wt}_{\mathbf{w}}(x^{\alpha_1}y^{\alpha_2}z^{\alpha_3}s^{\alpha_4})}x'^{\alpha_1}y'^{\alpha_2}z'^{\alpha_3},$$

so by (R2) and (R3), f_i' and l_i' are power series in (x', y', z', u),

$$f_i'(x', y', z', 0) = x'^2 + y'^3 + z'^4,$$

and $l_i'(x', y', z', 0)$ is the non-zero $\mathbf{w}$-weight $mr + c$ part of h_i with $x', y', z', 1$ substituted for x, y, z_i, s_i, a substitution which is injective on the monomials of a fixed $\mathbf{w}$-weight. By the faithful flatness of $\widehat{\mathcal{O}}_{\mathcal{Y}_i,p_i}$ over $\mathcal{O}_{\mathcal{Y}_i,p_i}$, an element of $\mathcal{O}_{\mathcal{Y}_i,p_i}$ divisible by a power of u in $\widehat{\mathcal{O}}_{\mathcal{Y}_i,p_i}$ is divisible by it in $\mathcal{O}_{\mathcal{Y}_i,p_i}$, so f_i' and l_i' are regular near p_i. Modulo $\mathfrak{n}^{N_i}$, $f_i - x^2$ is a series in (y, z_i, s_i) and h_i has x-degree at most 1, so modulo $u^{N_i-N_0+1}$, $f_i' - x'^2$ is a series in (y', z', u) and l_i' has x'-degree at most 1.

Since f_i generates the ideal of W_i and $f_i \circ \widetilde{\varpi}_i = u^{12r+6\nu'} f_i'$, near p_i we have $\widetilde{\varpi}_i^{-1}(W_i) = (u=0) \cup (f_i'=0)$, and Y_i is the germ at p_i of the algebraic hypersurface $(f_i'=0) \subset \mathcal{Y}_i$, with completion $\operatorname{Spec} \mathbb{C}[[x',y',z',u]]/(f_i')$. Since $h_i \circ \widetilde{\varpi}_i = u^{mr+c} l_i'$, the function $l_i = h_i/u^{mr+c}$ on Y_i equals $l_i'|_{Y_i}$ and is regular. If $f_i' = g_1 g_2$ near p_i, then $g_1(x',y',z',0)g_2(x',y',z',0) = x'^2 + y'^3 + z'^4$ is irreducible, so $g_1(p_i) \neq 0$ or $g_2(p_i) \neq 0$. Thus Y_i is an integral hypersurface germ. Since the expansion of $f_i'|_{(u=0)}$ at p_i is $x'^2 + y'^3 + z'^4$, near p_i the surface $S_i = Y_i \cap (u=0)$ is the preimage of the Du Val surface $(x'^2 + y'^3 + z'^4 = 0) \subset \mathbb{A}^3$ of type E_6 under the morphism $(x',y',z') : (u=0) \to \mathbb{A}^3$, which is étale as a base change of $\mathcal{Y}_i \to \mathbb{A}^4$, and p_i is isolated in the fiber over 0. Thus S_i is a Du Val surface of type E_6 at p_i, smooth elsewhere near p_i, and a Cartier divisor on Y_i with $\operatorname{div}(u) = S_i$. Since $l_i'(x',y',z',0)$ is non-zero of x'-degree at most 1, it is not divisible by $x'^2 + y'^3 + z'^4$, so l_i does not vanish on S_i.

Since $S_i \backslash \{p_i\}$ is smooth and S_i is Cartier, Y_i is smooth along $S_i \backslash \{p_i\}$. Over $(u \neq 0)$, (F.63) expresses x', y', z' in terms of x, y, z_i, u, so $\widetilde{\varpi}_i$ restricts to the étale cover $(u^{d_\nu} = s_i)$ of $\mathcal{X}_i \backslash (s_i = 0)$, and it maps $Y_i \backslash S_i = \widetilde{\varpi}_i^{-1}(W_i) \backslash (u=0)$ étale to $W_i \backslash (s_i = 0) \subset W_i \backslash \{o_i\}$, which is smooth by Definition 4.1(1). Thus Y_i has an isolated singularity at p_i, so Y_i is normal, and since the hyperplane section S_i is Du Val, $(Y_i \ni p_i)$ is an isolated cDV germ. Hence $(Y_i \ni p_i)$ is an isolated Gorenstein terminal hypersurface germ.

Let $\chi_x, \chi_y, \chi_z, \chi_s$ be the characters of x, y, z_i, s_i, and let

$$\Gamma_i := \{(\gamma, \xi) \in G_i \times \mathbb{C}^* \mid \xi^{d_\nu} = \chi_s(\gamma)\},$$

where (γ, ξ) multiplies x', y', z', u by

$$\chi_x(\gamma)\xi^{-(6r+3\nu')}, \quad \chi_y(\gamma)\xi^{-(4r+2\nu')}, \quad \chi_z(\gamma)\xi^{-(3r+\nu')}, \quad \text{and} \quad \xi,$$

respectively. Then Γ_i is abelian of order $d_\nu |G_i| \leq 2M$, contains the root cover group $\{1\} \times \mu_{d_\nu}$, and maps onto G_i. Since $\xi^{d_\nu} = \chi_s(\gamma)$, the element (γ, ξ), acting by γ on $\mathcal{X}_i$ and on $\mathbb{A}^4$ as on (x', y', z', u), multiplies each equation in (F.63) by a constant. Thus Γ_i acts on $\mathcal{Y}_i$ fixing p_i, $\widetilde{\varpi}_i$ is equivariant, x', y', z', u are Γ_i-eigenfunctions, f_i' and l_i' are Γ_i-semi-invariant, and Y_i is Γ_i-stable. Let $\varpi_i : Y_i \to W_i$ be the restriction of $\widetilde{\varpi}_i$, which is dominant since $\varpi_i|_{Y_i \backslash S_i}$ is étale. If (γ, ξ) acts trivially on Y_i, then γ acts trivially on W_i since ϖ_i is equivariant and dominant, so $\gamma = 1$ since G_i acts faithfully on W_i (Definition 4.1(1)), and $\xi = 1$ since $u|_{Y_i} \neq 0$. Thus Γ_i acts faithfully and diagonally on (x', y', z', u) and faithfully on Y_i. Since Γ_i acts on W_i through G_i, ϖ_i induces a dominant morphism $\overline{\varpi}_i : Y_i/\Gamma_i \to W_i/G_i$. A point $q \in Y_i \backslash S_i$ is determined by $\varpi_i(q)$ and $u(q)$, and $u(q)^{d_\nu} = s_i(\varpi_i(q))$, so $\deg \varpi_i \leq d_\nu$. Computing the degree of $Y_i \to W_i/G_i$ through Y_i/Γ_i and through W_i, we obtain

$$d_\nu |G_i| = |\Gamma_i| \leq |\Gamma_i| \deg \overline{\varpi}_i = |G_i| \deg \varpi_i \leq d_\nu |G_i|,$$

so

$$\overline{\varpi}_i \text{ is birational} \quad \text{and} \quad \deg \varpi_i = d_\nu. \tag{F.64}$$

Let $\widetilde{f}_*$ and $\widetilde{l}_*$ be defined from f_* and h_* in (x, y, z, s) as f_i' and l_i' are defined from f_i and h_i. Since $s^\nu z^4 = u^{12r+6\nu'} z'^4$ and $d_\nu \nu = 2\nu'$,

$$\widetilde{f}_* = x'^2 + y'^3 + z'^4 + u^{r-\nu'} y' z'^3 \phi(u^{3r+\nu'} z', u^{d_\nu}) + u^{3r-\nu'} z'^5 \psi(u^{3r+\nu'} z', u^{d_\nu}),$$

whose terms other than $x'^2 + y'^3 + z'^4$ have transverse w-weight at least 13, and by (F.58),

$$\widetilde{l}_* = \sum_{\alpha \in \mathcal{E}} \zeta_\alpha u^{r(w(\alpha)-m)+\nu'(3\alpha_1+2\alpha_2+\alpha_3)+d_\nu\alpha_4-c} x'^{\alpha_1} y'^{\alpha_2} z'^{\alpha_3}.$$

By **Step 1** and $r > c$,

$$r(w(\alpha) - m) + \nu'(3\alpha_1 + 2\alpha_2 + \alpha_3) + d_\nu \alpha_4 - c \geq 0$$

for any $\alpha \in \mathcal{E}$, with equality only if $w(\alpha) = m$, so $\widetilde{l}_*$ is a power series of x'-degree at most 1 with only terms of transverse w-weight at least m, and its u-free part has transverse w-weight

m. Since l_i' has x'-degree at most 1 modulo $u^{N_i-N_0+1}$, Weierstrass division by f_i' shows that the form l_i° of x'-degree at most 1 of l_i on Y_i is congruent to l_i' modulo $u^{N_i-N_0+1}$.

The d-jet of (f_i', l_i') is determined by the jet of (f_i, h_i) in (x, y, z_i, s_i) of degree bounded in terms of d, r, ν', m, and c, so for any d, once N_i is large, the d-jet of (f_i', l_i') belongs to the closure Z_d' of the d-jet of $(\widetilde{f}_*, \widetilde{l}_*)$. By Lemma 2.6(1), after passing to an infinite subfamily, the f_i' have a compatible generic limit f_*', with coefficient field $\Bbbk'$. Possibly replacing $\Bbbk'$ with an algebraic closure of the fraction field of the domain $\Bbbk \otimes_{\mathbb{C}} \Bbbk'$, we may assume that $\Bbbk' \supset \Bbbk$. By Lemma 2.6(3), after passing to a further infinite subfamily, the pairs (f_i', l_i') have a compatible generic limit (f_*', l_*), with coefficient field $\Bbbk''$, where f_*' now denotes the image of f_*' under an embedding $\Bbbk' \to \Bbbk''$. By Lemma 2.6(2), f_*' is still a compatible generic limit of the f_i' along the further subfamily. Since $\mathrm{jet}_d(l_i^\circ) = \mathrm{jet}_d(l_i')$ once $N_i \geq N_0 + d$, (f_*', l_*) is also a compatible generic limit of the pairs (f_i', l_i°).

The d-jet of (f_*', l_*) is the generic point of an irreducible closed subset Z_d'' whose non-empty open subsets contain the d-jets of all but finitely many i, so $Z_d'' \subset Z_d'$. Thus any coefficient of $(\widetilde{f}_*, \widetilde{l}_*)$ which vanishes, or equals 1, does so for (f_*', l_*): f_*' is $x'^2 + y'^3 + z'^4$ plus terms of transverse w-weight at least 13, and l_* has x'-degree at most 1 and only terms of transverse w-weight at least m. Each $l_i'(x', y', z', 0)$ is non-zero, and of transverse w-weight m since the d-jet of l_i' belongs to Z_d'. The vanishing of the u-free part of transverse w-weight m is a closed condition, so it fails at the generic point of Z_d'' for $d > m$. Thus $\mathrm{in}_w(l_*)$ has weight m and is non-zero. This proves (1) and the first sentence of (5).

Since $\mathrm{in}_w(l_*)$ is non-zero of x'-degree at most 1, it is not divisible by $x'^2 + y'^3 + z'^4$. By Claim F.10 with (x', y', z') in place of (x, y, z), $g = \mathrm{in}_w(l_*)$, $I = m$, and an algebraic closure of $\Bbbk''((u))$ in place of $\mathbb{K}$, the pair

$$(x'^2 + y'^3 + z'^4 = 0, \tfrac{1}{m}\operatorname{div}(\mathrm{in}_w(l_*)))$$

is lc, and the components of $\frac{1}{m}\operatorname{div}(\mathrm{in}_w(l_*))$ have coefficients at most $\frac{1}{2}$, so none has coefficient one. This proves (5).

We prove (6). From the end of **Step 3** on, x, y, z_i, s_i are regular functions near o_i, f_i is a regular G-semi-invariant generator of the ideal of W_i, and h_i is a regular G-semi-invariant function restricting to the mark on W_i, and **Steps 4** and **5** prove (1)–(5) for them. This step shows that $\mathcal{Y}_i$ is a smooth fourfold with an action of Γ_i fixing p_i, that x', y', z', u are regular Γ_i-eigenparameters at p_i, that $\widetilde{\varpi}_i$ is equivariant for the projection $\Gamma_i \to G_i$ and satisfies (F.63), that f_i' and l_i' are regular Γ_i-semi-invariant functions near p_i, that Y_i is the germ at p_i of the normal hypersurface $(f_i' = 0) \subset \mathcal{Y}_i$ with completion $\operatorname{Spec} \mathbb{C}[[x', y', z', u]]/(f_i')$, that $l_i = l_i'|_{Y_i}$, and that the expansions of the pairs (f_i', l_i') have the compatible generic limit (f_*', l_*) of (5). This proves (6).

Step 5. In this step we prove (3), the rest of (2), and (4) of Proposition 17.5.

Let P be a prime divisor over Y_i with center p_i. Since ϖ_i is dominant and generically finite, P restricts to a prime divisor F over W_i, with ramification index ρ (Definition 2.2). Since

$$\operatorname{mult}_F x = \frac{1}{\rho}\operatorname{mult}_P(u^{6r+3\nu'}x') > 0,$$

and similarly $\operatorname{mult}_F y, \operatorname{mult}_F z_i, \operatorname{mult}_F s_i > 0$, F has center o_i.

Let σ_i be the residue of $dx \wedge dy \wedge dz_i \wedge ds_i/f_i$ and σ_i' the residue of $dx' \wedge dy' \wedge dz' \wedge du/f_i'$, generators of ω_{W_i} and ω_{Y_i} near o_i and p_i. Since $ds_i = d_\nu u^{d_\nu - 1} du$,

$$dx \wedge dy \wedge dz_i \wedge ds_i = d_\nu u^{13r+6\nu'+d_\nu-1} dx' \wedge dy' \wedge dz' \wedge du,$$

and since $f_i \circ \widetilde{\varpi}_i = u^{12r+6\nu'} f_i'$ and $\widetilde{\varpi}_i$ is étale over $(u \neq 0)$, the pullback of σ_i is $d_\nu u^\kappa \sigma_i'$, where $\kappa = r + d_\nu - 1$. The order of σ_i along F is $a(F, W_i, 0) - 1$, the order of σ_i' along P is $a(P, Y_i, 0) - 1$, and P is tamely ramified of index ρ over F, so the order of the pullback of σ_i along P is

$\rho(a(F,W_i,0)-1)+\rho-1$. Thus

$$\rho a(F,W_i,0)=a(P,Y_i,0)+\kappa\,\mathrm{mult}_P u,$$

and since $h_i=u^{mr+c}l_i$ on Y_i,

$$\rho\,\mathrm{mult}_F h_i=\mathrm{mult}_P l_i+(mr+c)\,\mathrm{mult}_P u.$$

By (F.64), $Y_i\to Y_i/\Gamma_i$ and the composition $Y_i\to W_i\to W_i/G_i$ differ by the birational map $\overline{\varpi}_i$, so P and F induce one prime divisor $\overline{P}$ over W_i/G_i. In characteristic zero e_P is the ramification index of P over $\overline{P}$ and e_F is that of F over $\overline{P}$, and ramification indices are multiplicative along $Y_i\to W_i\to W_i/G_i$, so $e_P=\rho e_F$. This proves (3). Moreover $\rho\le\deg\varpi_i=d_\nu$ by (F.64).

Since $\mathrm{div}(u)=S_i$ and $\kappa+\beta=r+\frac{c}{m}$, (3) implies

$$a(P,Y_i,B_i)=a(P,Y_i,0)-\tfrac{1}{m}\,\mathrm{mult}_P l_i-\beta\,\mathrm{mult}_P u=\rho a(F,W_i,\tfrac{1}{m}\,\mathrm{div}(h_i)),$$

and dividing by $e_P=\rho e_F$,

$$\frac{a(P,Y_i,B_i)}{e_P}=\frac{a(F,W_i,\frac{1}{m}\,\mathrm{div}(h_i))}{e_F}\ge b_i. \tag{F.65}$$

Thus $\mathrm{mld}((Y_i\ni p_i,B_i)/\Gamma_i)\ge b_i$ (Definition 2.2). Conversely, let F be a computing lift of b_i and P a prime divisor over Y_i whose valuation extends that of F, with ramification index $\rho\in[1,d_\nu]$. By (F.62), $\mathrm{mult}_P u=\frac{\rho}{d_\nu}\,\mathrm{mult}_F s_i\in(0,\Theta]$ and

$$\mathrm{mult}_P x'=\rho\,\mathrm{mult}_F x-(6r+3\nu')\,\mathrm{mult}_P u\ge k_i-(6r+3\nu')\Theta>0,$$

and similarly $\mathrm{mult}_P y',\mathrm{mult}_P z'\ge k_i-(6r+3\nu')\Theta$. Thus P has center p_i on Y_i, and $a(P,Y_i,B_i)/e_P=b_i$ by (F.65). Hence $\mathrm{mld}((Y_i\ni p_i,B_i)/\Gamma_i)=b_i$, any prime divisor over Y_i whose valuation extends that of a computing lift of b_i has center p_i, and by (F.65) any computing lift of $\mathrm{mld}((Y_i\ni p_i,B_i)/\Gamma_i)$ restricts to a computing lift of b_i. This proves (2).

Since any computing lift P of $\mathrm{mld}((Y_i\ni p_i,B_i)/\Gamma_i)$ restricts to a computing lift of b_i by (2), the bounds $\mathrm{mult}_P u\le\Theta$ and $\mathrm{mult}_P x',\mathrm{mult}_P y',\mathrm{mult}_P z'\ge k_i-(6r+3\nu')\Theta$ above, with $k_i\to+\infty$, prove (4) with $R':=\Theta$. □

F.3. **Bounded computing lifts for types D and E.** In this subsection we prove Theorem 17.6 of Section 17.

Proof of Theorem 17.6, ***Steps 1–6****.* We divide the proof into several steps. We write

$$(x_1,x_2,x_3):=(x,y,z)\quad\text{and}\quad(w_1,w_2,w_3):=(w_x,w_y,w_z),$$

and for $\alpha\in\mathbb{N}^3$ we let $x^\alpha:=x_1^{\alpha_1}x_2^{\alpha_2}x_3^{\alpha_3}$, of w-weight $w\cdot\alpha:=\sum_{j=1}^3 w_j\alpha_j$. The coordinate u has weight 0.

Step 1. In this step we fix the Jacobian slope λ. Let

$$N_0:=\dim_{\Bbbk}\Bbbk[x,y,z]/(\partial_x f_0,\partial_y f_0,\partial_z f_0),$$

which is finite since 0 is an isolated singular point of $(f_0=0)\subset\mathbb{A}^3_{\Bbbk}$. Let $\Bbbk_1\supset\mathbb{C}$ be an algebraically closed field, $\Xi\in\Bbbk_1[x,y,z]$ a polynomial whose monomials have w-weight less than d, and $S':=(f_0+\Xi=0)\subset\mathbb{A}^3_{\Bbbk_1}$. We show that S' is normal and that any singular point of S' is a Du Val singularity of type A_n, D_n, or E_n with $n\le N_0$.

If f_0 is one of the polynomials of Proposition 7.7, then S' is normal with at worst Du Val singularities by the first paragraph of Proposition 7.7 with Ξ in place of F'. If $f_0=x^2+y^3+z^3$, then, since the binary cubics y^3+z^3 and zy^2+z^3 both have three distinct linear factors, there exist linear forms y',z' in y,z with $y^3+z^3=z'y'^2+z'^3$. Since $w_2=w_3=2$, the coordinate change $(x,y,z)\to(x,y',z')$ preserves w-weights and turns f_0 into the polynomial $x^2+z'y'^2+z'^3$ of Proposition 7.7 with $k=4$, so S' is normal with at worst Du Val singularities by the first paragraph of Proposition 7.7 with Ξ in place of F'.

Let $I\subset\Bbbk_1[x,y,z]$ be the ideal generated by the three partial derivatives of $f_0+\Xi$. For a non-zero polynomial φ, let $\mathrm{top}(\varphi)$ be the sum of the terms of φ of maximal w-weight. Since

$\partial_{x_j} f_0 \neq 0$ is weighted homogeneous of weight $d - w_j$ and the monomials of $\partial_{x_j}\Xi$ have weight less than $d - w_j$, we have

$$\operatorname{top}(\partial_{x_j}(f_0 + \Xi)) = \partial_{x_j} f_0 \quad \text{for } j = 1, 2, 3.$$

The graded ring associated with the filtration of $\Bbbk_1[x, y, z]/I$ by w-weight is $\Bbbk_1[x, y, z]/\operatorname{top}(I)$, where $\operatorname{top}(I)$ is the ideal generated by $\{\operatorname{top}(\varphi) \mid 0 \neq \varphi \in I\}$. Since

$$\operatorname{top}(I) \supset (\partial_x f_0, \partial_y f_0, \partial_z f_0),$$

we have

$$\dim_{\Bbbk_1} \Bbbk_1[x, y, z]/I \leq N_0.$$

The Milnor number of S' at a singular point p is the length of the localization of $\Bbbk_1[x, y, z]/I$ at p, hence at most N_0, and a Du Val singularity of type A_n, D_n, or E_n has Milnor number n.

For a Du Val surface germ $(S \ni p)$ of type A_n, D_n, or E_n with $n \leq N_0$, we have $\operatorname{lct}(S \ni p; \operatorname{Jac}_S) \in \mathbb{Q}_{>0}$, and $\operatorname{lct}(S \ni p; \operatorname{Jac}_S)$ depends only on the type, since a Du Val singularity is determined by the type up to analytic isomorphism, Jac_S does not change under analytic changes of coordinates and multiplication of the local equation by units, and the log canonical threshold does not change under extension of the algebraically closed field [dFEM11, Lemma 2.14(iii)]. Let γ be the minimum of 1 and the numbers $\operatorname{lct}(S \ni p; \operatorname{Jac}_S)$ for the finitely many types, and let $\lambda := 1/\gamma$. Then $(S', \operatorname{Jac}_{S'}^{\gamma})$ is lc for any $\Bbbk_1$ and Ξ.

Let $V = (f = 0)$ be a hypersurface threefold germ at 0 in a smooth fourfold germ with regular parameters (x, y, z, u) at 0 such that the restriction of f to $(u = 0)$ is $f_0(x, y, z) + \Xi(x, y, z)$ for a polynomial Ξ whose monomials have w-weight less than d, and let $S := V \cap (u = 0)$. Near 0, S is the preimage of $S' = (f_0 + \Xi = 0) \subset \mathbb{A}^3$, with $\Bbbk_1 = \mathbb{C}$, under the morphism $(x, y, z) : (u = 0) \to \mathbb{A}^3$, which is étale at 0. After shrinking, 0 is the only point of S over $0 \in S'$. Thus, near 0, S is normal, any singular point of S is a Du Val singularity of type A_n, D_n, or E_n with $n \leq N_0$, and Jac_S is the pullback of $\operatorname{Jac}_{S'}$. By Lemma 2.13, whose correspondence of prime divisors has ramification index 1, and since $(S', \operatorname{Jac}_{S'}^{\gamma})$ is lc,

$$\operatorname{mld}(S \ni 0, \operatorname{Jac}_S^{\gamma}) = \operatorname{mld}(S' \ni 0, \operatorname{Jac}_{S'}^{\gamma}) \geq 0.$$

Assume moreover that V is normal, and let $\mathcal{J}_{\mathrm{rel}} := (\partial_x f, \partial_y f, \partial_z f)$. Then $\mathcal{J}_{\mathrm{rel}}\mathcal{O}_S = \operatorname{Jac}_S$ and $\mathcal{J}_{\mathrm{rel}}\mathcal{O}_V \subset \operatorname{Jac}_V$. Since V is a normal local complete intersection and $S = \operatorname{div}(u|_V)$ is a normal Cartier divisor on V, [EM04, Theorem 1.1] implies

$$\operatorname{mld}(V \ni 0, S, \mathcal{J}_{\mathrm{rel}}^{\gamma}) = \operatorname{mld}(S \ni 0, \operatorname{Jac}_S^{\gamma}) \geq 0.$$

Thus $a(F, V, 0) - \operatorname{mult}_F u - \gamma \operatorname{mult}_F \mathcal{J}_{\mathrm{rel}} \geq 0$ for any prime divisor F over V with center 0, and hence

$$\gamma \operatorname{mult}_F \operatorname{Jac}_V \leq a(F, V, 0) - \operatorname{mult}_F u \quad \text{and} \quad \operatorname{mult}_F \operatorname{Jac}_V \leq \lambda a(F, V, 0). \tag{F.66}$$

By [Liu26a, Lemma 3.3(3)], (F.66) also holds for any prime divisor F over the completion of V at 0 with center the closed point.

Step 2. In this step we bound marked orders by multiplier ideals. For rational numbers $\mu > \frac{1}{m}$ and $\theta \in (0, 1)$ let

$$\epsilon := \theta\mu - \frac{1}{m} - (1 - \theta)\Big(1 + \frac{2\lambda}{m}\Big),$$

let L_1 be a real number, and let $L_2 := (1 + 2\lambda(1 - \theta))L_1 + 1 - \theta$.

Let $V = (f = 0)$ be a normal hypersurface germ at 0 in a smooth fourfold germ $(\mathcal{X} \ni 0)$ with regular parameters (x, y, z, u) at 0, let h and ζ be regular functions near 0 on $\mathcal{X}$ with non-zero restrictions to V, and let F be a prime divisor over V with center 0. Assume that any prime divisor P over $\mathcal{X}$ with center 0 and $\operatorname{mult}_P \zeta = \operatorname{mult}_F(\zeta|_V)$ satisfies

$$\operatorname{mult}_P \zeta + a(P, \mathcal{X}, 0) > \theta \operatorname{mult}_P f + \theta\mu \operatorname{mult}_P h. \tag{F.67}$$

By the definition of multiplier ideals, (F.67) holds for any P if $\zeta \in \mathcal{J}(\mathcal{X}, f^\theta h^{\theta\mu})$. We show

$$\begin{aligned} a(F,V,0) - \frac{1}{m}\operatorname{mult}_F h \le L_1 \quad &\text{and} \quad \operatorname{mult}_F \operatorname{Jac}_V \le \lambda a(F,V,0) \\ \Longrightarrow \quad & \epsilon \operatorname{mult}_F h < L_2 + (2-\theta)\operatorname{mult}_F(\zeta|_V). \end{aligned} \tag{F.68}$$

The ideal Jac_V is generated by the restrictions of the four partial derivatives of f, so there exists a coordinate $x_\bullet \in \{x,y,z,u\}$ with

$$\operatorname{mult}_F(\partial f/\partial x_\bullet) = \operatorname{mult}_F \operatorname{Jac}_V .$$

Let

$$k := \operatorname{mult}_F h + \operatorname{mult}_F(\zeta|_V) + 2\operatorname{mult}_F \operatorname{Jac}_V + 1.$$

Then $k > 2\operatorname{mult}_F \operatorname{Jac}_V$ and

$$k - \operatorname{mult}_F \operatorname{Jac}_V > \max\{\operatorname{mult}_F h, \operatorname{mult}_F(\zeta|_V)\},$$

so Lemma 5.4(2) on the completion $\operatorname{Spec} \mathbb{C}[[x,y,z,u]]$ of $\mathcal{X}$ at 0, with $H_1 = h$, $H_2 = \zeta$, and a formal prime divisor restricting to F, together with Lemma 2.12(3) for V and for $\mathcal{X}$, yields a prime divisor P over $\mathcal{X}$ with center 0 such that $\operatorname{mult}_P f = k$, $\operatorname{mult}_P h = \operatorname{mult}_F h$, $\operatorname{mult}_P \zeta = \operatorname{mult}_F(\zeta|_V)$, and $a(P,\mathcal{X},0) = a(F,V,0) + k$. By (F.67), $\operatorname{mult}_F(\zeta|_V) + a(F,V,0) + (1-\theta)k > \theta\mu \operatorname{mult}_F h$. Substituting k and using

$$\operatorname{mult}_F \operatorname{Jac}_V \le \lambda a(F,V,0) \quad \text{and} \quad a(F,V,0) \le L_1 + \frac{1}{m}\operatorname{mult}_F h,$$

we obtain

$$\begin{aligned} (\theta\mu - 1 + \theta)\operatorname{mult}_F h &< a(F,V,0) + 2(1-\theta)\operatorname{mult}_F \operatorname{Jac}_V + (2-\theta)\operatorname{mult}_F(\zeta|_V) + 1 - \theta \\ &\le (1 + 2\lambda(1-\theta))a(F,V,0) + (2-\theta)\operatorname{mult}_F(\zeta|_V) + 1 - \theta \\ &\le (1 + 2\lambda(1-\theta))\Big(L_1 + \frac{1}{m}\operatorname{mult}_F h\Big) + (2-\theta)\operatorname{mult}_F(\zeta|_V) + 1 - \theta. \end{aligned}$$

Since $\theta\mu - 1 + \theta - \frac{1}{m}(1 + 2\lambda(1-\theta)) = \epsilon$, (F.68) follows.

We apply (F.68) to limits. Let $\mathcal{I}'$ be an infinite set of indices, and for $i \in \mathcal{I}'$ let $\widetilde{f}_i$ and $\widetilde{h}_i$ be regular functions near 0 on a smooth fourfold germ with regular parameters (x,y,z,u) at 0 such that the orders of $\widetilde{f}_i$ and $\widetilde{h}_i$ at 0 are bounded, $\widetilde{W}_i := (\widetilde{f}_i = 0)$ is normal near 0, and $\widetilde{h}_i|_{\widetilde{W}_i} \ne 0$. Assume that $(\widetilde{f}_i, \widetilde{h}_i)$ have a compatible generic limit $(\widetilde{f}_*, \widetilde{h}_*)$ along $\mathcal{I}'$ with some coefficient field $\Bbbk_2$ (Definition 2.5, with H trivial) such that

$$\widetilde{W}_* := (\widetilde{f}_* = 0) \subset \operatorname{Spec} \Bbbk_2[[x,y,z,u]]$$

is normal and $\widetilde{h}_*|_{\widetilde{W}_*} \ne 0$. Let

$$\widetilde{\Delta}_* := \operatorname{div}(\widetilde{h}_*|_{\widetilde{W}_*}),$$

and let $V_* \subset \widetilde{W}_*$ be a closed subset containing $\widetilde{W}_* \cap (u=0)$ such that $(\widetilde{W}_* \backslash V_*, \frac{1}{m}\widetilde{\Delta}_*)$ is klt. Since the klt property of $(\widetilde{W}_* \backslash V_*, \frac{1}{m}\widetilde{\Delta}_*)$ is tested on the finitely many prime divisors of a log resolution, there exists a rational number $\mu > \frac{1}{m}$ such that $(\widetilde{W}_* \backslash V_*, \mu\widetilde{\Delta}_*)$ is lc. Since $\epsilon \to \mu - \frac{1}{m} > 0$ as $\theta \to 1$, there exist rational numbers $\theta < \theta_1 < 1$ with $\epsilon > 0$. Let

$$\mathcal{J}_* := \mathcal{J}(\widetilde{f}_*^{\theta_1} \widetilde{h}_*^{\theta_1\mu}).$$

By Lemma 6.1(3) with $\theta_2 = \theta_1$, $\mathcal{J}_*$ is the unit ideal near any point of $\widetilde{W}_* \backslash V_*$. Thus the zero locus of the ideal $(\widetilde{f}_*) + \mathcal{J}_*$ of $\Bbbk_2[[x,y,z,u]]$ is contained in V_*.

First let $V_* := \widetilde{W}_* \cap (u=0)$. Then u vanishes on the zero locus of $(\widetilde{f}_*) + \mathcal{J}_*$, so $u^\sigma - \phi_* \widetilde{f}_* \in \mathcal{J}_*$ for a positive integer σ and some $\phi_* \in \Bbbk_2[[x,y,z,u]]$. Let $N_1 := \sigma R' + 1$, let $\mathfrak{n}_*$ be the maximal ideal of $\Bbbk_2[[x,y,z,u]]$, and let $\phi_*^{<N_1}$ be the sum of the terms of ϕ_* of degree less than N_1, so that

$$u^\sigma - \phi_*^{<N_1} \widetilde{f}_* \in \mathcal{J}_* + \mathfrak{n}_*^{N_1}.$$

Writing the coefficients of $\phi_*^{<N_1}$ as the entries of a vector $\mathbf{b}$, Lemma 6.1(1) with $\mathbf{b}$ in place of β, $L = N_1$, $(\gamma_1, \gamma_2) = (\theta, \theta\mu)$, and $(\gamma_1', \gamma_2') = (\theta_1, \theta_1\mu)$ yields, for all but finitely many $i \in \mathcal{I}'$, the polynomial ϕ_i given by substituting $\mathbf{b}_i$ for $\mathbf{b}$ in $\phi_*^{<N_1}$, such that

$$\zeta_i := u^\sigma - \phi_i \widetilde{f}_i \in \mathcal{J}(\widetilde{f}_i^\theta \widetilde{h}_i^{\theta\mu}) + \mathfrak{m}^{N_1},$$

where, as the $\widetilde{f}_i$ and $\widetilde{h}_i$ are regular functions on smooth algebraic germs, the multiplier ideal is taken on $\operatorname{Spec} \mathbb{C}[[x, y, z, u]]$ and $\mathfrak{m}$ is the maximal ideal of $\mathbb{C}[[x, y, z, u]]$. Let F be a prime divisor over $\widetilde{W}_i$ with center 0 and $\operatorname{mult}_F u \le R'$. Since $\zeta_i|_{\widetilde{W}_i} = u^\sigma|_{\widetilde{W}_i}$, we have $\operatorname{mult}_F(\zeta_i|_{\widetilde{W}_i}) \le \sigma R' < N_1$, so any P as in (F.67) satisfies $\operatorname{mult}_P(\mathfrak{m}^{N_1}) \ge N_1 > \operatorname{mult}_P \zeta_i$, and (F.67) holds by Lemma 6.1(2) for a formal prime divisor restricting to P (Lemma 2.12(3)). By (F.68), if $(\widetilde{W}_* \backslash (u = 0), \frac{1}{m}\widetilde{\Delta}_*)$ is klt, then for all but finitely many $i \in \mathcal{I}'$ any prime divisor F over $\widetilde{W}_i$ with center 0 and

$$\operatorname{mult}_F u \le R', \quad a(F, \widetilde{W}_i, 0) - \frac{1}{m} \operatorname{mult}_F \widetilde{h}_i \le L_1, \quad \text{and} \quad \operatorname{mult}_F \operatorname{Jac}_{\widetilde{W}_i} \le \lambda a(F, \widetilde{W}_i, 0)$$

satisfies

$$\epsilon \operatorname{mult}_F \widetilde{h}_i < L_2 + (2 - \theta)\sigma R'. \tag{F.69}$$

In particular, $\operatorname{mult}_F \widetilde{h}_i$, $a(F, \widetilde{W}_i, 0)$, and $\operatorname{mult}_F \operatorname{Jac}_{\widetilde{W}_i}$ are bounded independently of i and F, since

$$a(F, \widetilde{W}_i, 0) \le L_1 + \frac{1}{m} \operatorname{mult}_F \widetilde{h}_i \quad \text{and} \quad \operatorname{mult}_F \operatorname{Jac}_{\widetilde{W}_i} \le \lambda a(F, \widetilde{W}_i, 0).$$

Next let $U_1, U_2, U_3 \in \Bbbk_2[[x, y, z, u]]$, and let $V_* := \widetilde{W}_* \cap \big((u = 0) \cup (U_1 = U_2 = U_3 = 0)\big)$, and define μ, θ, θ_1, ϵ, L_2, and $\mathcal{J}_*$ for the second V_* as for the first. For $j = 1, 2, 3$, uU_j vanishes on V_*, so there exist positive integers σ_j, σ_j' and $\phi_{*,j} \in \Bbbk_2[[x, y, z, u]]$ with

$$u^{\sigma_j} U_j^{\sigma_j'} - \phi_{*,j} \widetilde{f}_* \in \mathcal{J}_*.$$

Let Λ be a positive integer, $N_1' := \max_j(\sigma_j R' + \sigma_j' \Lambda) + 1$, and $U_j^{<N_1'}$ and $\phi_{*,j}^{<N_1'}$ the sums of the terms of U_j and $\phi_{*,j}$ of degree less than N_1', so that

$$u^{\sigma_j} (U_j^{<N_1'})^{\sigma_j'} - \phi_{*,j}^{<N_1'} \widetilde{f}_* \in \mathcal{J}_* + \mathfrak{n}_*^{N_1'} \quad \text{for } j = 1, 2, 3.$$

Writing the coefficients of $U_j^{<N_1'}$ and $\phi_{*,j}^{<N_1'}$, $j = 1, 2, 3$, as the entries of $\mathbf{b}$, Lemma 6.1(1) with $\mathbf{b}$ in place of β, $L = N_1'$, and the exponents above yields, for all but finitely many $i \in \mathcal{I}'$, the polynomials $U_{i,j}$ and $\phi_{i,j}$ given by substituting $\mathbf{b}_i$ for $\mathbf{b}$ in $U_j^{<N_1'}$ and $\phi_{*,j}^{<N_1'}$, such that

$$\zeta_{i,j} := u^{\sigma_j} U_{i,j}^{\sigma_j'} - \phi_{i,j} \widetilde{f}_i \in \mathcal{J}(\widetilde{f}_i^\theta \widetilde{h}_i^{\theta\mu}) + \mathfrak{m}^{N_1'}.$$

If $\operatorname{mult}_F U_{i,j} \le \Lambda$ and $\operatorname{mult}_F u \le R'$, then

$$\operatorname{mult}_F(\zeta_{i,j}|_{\widetilde{W}_i}) \le \sigma_j R' + \sigma_j' \Lambda < N_1'.$$

Thus, if $(\widetilde{W}_* \backslash V_*, \frac{1}{m}\widetilde{\Delta}_*)$ is klt, then for any Λ and all but finitely many $i \in \mathcal{I}'$, any prime divisor F over $\widetilde{W}_i$ with center 0 and

$$\operatorname{mult}_F u \le R', \quad a(F, \widetilde{W}_i, 0) - \frac{1}{m} \operatorname{mult}_F \widetilde{h}_i \le L_1, \quad \text{and} \quad \operatorname{mult}_F \operatorname{Jac}_{\widetilde{W}_i} \le \lambda a(F, \widetilde{W}_i, 0)$$

satisfies

$$\epsilon \operatorname{mult}_F \widetilde{h}_i \ge L_2 + (2 - \theta)(\sigma_j R' + \sigma_j' \Lambda) \implies \operatorname{mult}_F U_{i,j} > \Lambda \quad \text{for } j = 1, 2, 3. \tag{F.70}$$

Step 3. In this step we fix the quotient value b and the depths n_i^0. Since W_i is Gorenstein and klt, $a(E, W_i, 0)$ is a positive integer for any prime divisor E over W_i, and the computing lifts F_i

exist by (2) of the theorem. For any prime divisor F over W_i with center 0,

$$a(F, W_i, B_i) = a(F, W_i, 0) - \frac{1}{m}\operatorname{mult}_F h_i - \beta \operatorname{mult}_F u \in \frac{1}{m}\mathbb{Z}$$

since $\beta \in \frac{1}{m}\mathbb{Z}$, and e_F divides $|G|$. Thus $\operatorname{mld}((W_i \ni 0, B_i)/G) \in \frac{1}{m|G|}\mathbb{Z} \cap (0, L_0]$, and, possibly passing to an infinite subfamily, along which (f_*, h_*) is still a compatible generic limit by Lemma 2.6(2),

$$b := \operatorname{mld}((W_i \ni 0, B_i)/G) \quad \text{for any } i, \qquad 0 < b \le L_0.$$

By (3) of the theorem, the coefficients in $\Bbbk[[u]]$ of the monomials x^α of f_* are 0 for $w \cdot \alpha < d$ and are the coefficients of f_0 for $w \cdot \alpha = d$, and those of h_* are 0 for $w \cdot \alpha < m$. Only finitely many α satisfy $w \cdot \alpha \le \max\{d, m\}$. For any positive integer N, each of the following conditions is a closed condition on finitely many coordinates of a d'-jet, for d' large, which holds at the generic point of the set $Z_{d'}$ of Definition 2.5 for the limit (f_*, h_*), hence on $Z_{d'}$. Thus, for any positive integer N and all but finitely many i, the coefficient in $\mathbb{C}[[u]]$ of x^α in f_i is divisible by u^N if $w \cdot \alpha < d$ and is congruent to the coefficient of x^α in f_0 modulo u^N if $w \cdot \alpha = d$, and the coefficient of x^α in h_i is divisible by u^N if $w \cdot \alpha < m$. Let n_i^0 be the largest positive integer n divisible by $|G|$ satisfying the following.

(a) The coefficient of x^α in f_i is divisible by $u^{n(d - w\cdot\alpha)}$ for any α with $w \cdot \alpha < d$.
(b) The coefficient of x^α in h_i is divisible by $u^{n(m - w\cdot\alpha)}$ for any α with $w \cdot \alpha < m$.
(c) $nw_jR' < \operatorname{mult}_{F_i} x_j$ for $j = 1, 2, 3$.

For any fixed n divisible by $|G|$, (a) and (b) hold for $i \gg 0$ by the previous paragraph, and (c) holds for $i \gg 0$ since $\min_j \operatorname{mult}_{F_i} x_j \to +\infty$ by (2) of the theorem. For each i, (c) holds for only finitely many n. Thus, possibly removing finitely many indices, n_i^0 is defined and $n_i^0 \to +\infty$. Possibly removing finitely many more indices, the terms of w-weight d of $f_i(x, y, z, 0)$ form f_0 for any i. Let n_i' be positive integers with $n_i' \to +\infty$ and $n_i'/n_i^0 \to 0$.

Step 4. In this step we choose maximal preparations. Let $(\mathcal{X}_i \ni 0)$ be the smooth fourfold germ of the theorem containing W_i. Let χ_f and χ_h be the characters of the equations and of the marks, and χ_{x_j} and χ_u the characters of the coordinates. A *preparation* of index i is a pair (g, ξ) of the following data.

(a) The automorphism g of $\mathbb{A}^4$ over $\mathbb{A}^1_u$ given by new coordinates $x_j' := x_j + p_j$, $j = 1, 2, 3$, where $p_j \in u^{n_i'}\mathbb{C}[u][x, y, z]$ is a G-semi-invariant polynomial with character χ_{x_j} whose monomials have w-weight less than w_j.
(b) A G-semi-invariant polynomial $\xi \in u^{n_i'}\mathbb{C}[u][x, y, z]$ with character $\chi_h\chi_f^{-1}$ whose monomials have w-weight less than $m - d$. Thus $\xi = 0$ if $m \le d$.

A monomial of w-weight less than w_j involves only coordinates of weight less than w_j, so g is triangular with Jacobian determinant 1, and $g \equiv \mathrm{id}_{\mathbb{A}^4}$ modulo $u^{n_i'}$. Hence the functions $x_j' = x_j + p_j(x, y, z, u)$ on $\mathcal{X}_i$, $j = 1, 2, 3$, and u are regular G-eigenparameters at 0 with the characters of x_j and u, and g induces the automorphism $\varphi \to \varphi \circ g$ of $\widehat{\mathcal{O}}_{\mathcal{X}_i,0} = \mathbb{C}[[x, y, z, u]]$ substituting x_j' for x_j. Write

$$f_i = \sum_\alpha c_\alpha^g(u)x'^\alpha \quad \text{and} \quad h_i + \xi f_i = \sum_\alpha c_\alpha'^{g,\xi}(u)x'^\alpha$$

with $c_\alpha^g, c_\alpha'^{g,\xi} \in \mathbb{C}[[u]]$. Since $g \equiv \mathrm{id}_{\mathbb{A}^4}$ and $\xi \equiv 0$ modulo $u^{n_i'}$, c_α^g and $c_\alpha'^{g,\xi}$ are congruent modulo $u^{n_i'}$ to the coefficients of x^α in f_i and h_i, respectively. For a positive integer n divisible by $|G|$, let

$$f_i^{g,n} := \sum_\alpha c_\alpha^g(u)u^{n(w\cdot\alpha - d)}X^\alpha \quad \text{and} \quad h_i^{g,\xi,n} := \sum_\alpha c_\alpha'^{g,\xi}(u)u^{n(w\cdot\alpha - m)}X^\alpha,$$

power series in X_1, X_2, X_3 with coefficients Laurent series in u. We call (g, ξ, n) *admissible* if $f_i^{g,n}, h_i^{g,\xi,n} \in \mathbb{C}[[X_1, X_2, X_3, u]]$ and $\operatorname{mult}_{F_i} x_j' > nw_j \operatorname{mult}_{F_i} u$ for $j = 1, 2, 3$. By **Step 3** and $\operatorname{mult}_{F_i} u \le R'$, $(\mathrm{id}_{\mathbb{A}^4}, 0, n_i^0)$ is admissible.

Let (g,ξ,n) be admissible. After shrinking $\mathcal{X}_i$ to a G-stable open neighborhood of 0 on which the morphism $(x_1',x_2',x_3',u):\mathcal{X}_i\to\mathbb{A}^4$ is étale, let $\varpi:\mathcal{X}_i^{g,n}\to\mathcal{X}_i$ be the base change along (x_1',x_2',x_3',u) of the morphism

$$\mathbb{A}^4\to\mathbb{A}^4,\quad (X_1,X_2,X_3,u)\to(u^{nw_1}X_1,u^{nw_2}X_2,u^{nw_3}X_3,u),$$

so that $x_j'\circ\varpi=u^{nw_j}X_j$ and $u\circ\varpi=u$, and let $0\in\mathcal{X}_i^{g,n}$ be the point over $0\in\mathcal{X}_i$ and over the origin of $\mathbb{A}^4$ with coordinates (X_1,X_2,X_3,u). The projection $(X_1,X_2,X_3,u):\mathcal{X}_i^{g,n}\to\mathbb{A}^4$ is étale as a base change of (x_1',x_2',x_3',u), so X_1,X_2,X_3,u are regular parameters of $\mathcal{X}_i^{g,n}$ at 0, and ϖ induces on completions the substitution of $u^{nw_j}X_j$ for x_j'. Then $f_i\circ\varpi=u^{nd}f_i^{g,n}$ and $(h_i+\xi f_i)\circ\varpi=u^{nm}h_i^{g,\xi,n}$ in the completion $\mathbb{C}[[X_1,X_2,X_3,u]]$ of $\mathcal{X}_i^{g,n}$ at 0, so $f_i^{g,n}$ and $h_i^{g,\xi,n}$ are regular functions near 0, since

$$u^k\widehat{\mathcal{O}}_{\mathcal{X}_i^{g,n},0}\cap\mathcal{O}_{\mathcal{X}_i^{g,n},0}=u^k\mathcal{O}_{\mathcal{X}_i^{g,n},0}\quad\text{for any } k\ge 0.$$

Let $W_i^{g,n}:=(f_i^{g,n}=0)$ and

$$B_i^{g,\xi,n}:=\frac{1}{m}\operatorname{div}(h_i^{g,\xi,n}|_{W_i^{g,n}})+\beta\operatorname{div}(u|_{W_i^{g,n}}).$$

Since $|G|$ divides n, the morphism $\mathbb{A}^4\to\mathbb{A}^4$ above is equivariant for the diagonal actions of G on the coordinates (X_1,X_2,X_3,u) and (x_1',x_2',x_3',u) with characters $\chi_{x_1},\chi_{x_2},\chi_{x_3},\chi_u$, so G acts on $\mathcal{X}_i^{g,n}$, fixing 0, with eigenparameters X_1,X_2,X_3,u of characters $\chi_{x_1},\chi_{x_2},\chi_{x_3},\chi_u$. Moreover, the functions $f_i^{g,n}$ and $h_i^{g,\xi,n}$ have characters χ_f and χ_h, and ϖ restricts to a G-equivariant birational morphism $W_i^{g,n}\to W_i$ near 0, isomorphic over $(u\ne 0)$, with $\varpi(0)=0$. Since

$$\operatorname{mult}_{F_i}X_j=\operatorname{mult}_{F_i}x_j'-nw_j\operatorname{mult}_{F_i}u>0,$$

F_i is a prime divisor over $W_i^{g,n}$ with center 0.

The terms of $f_i^{g,n}$ with $w\cdot\alpha>d$ vanish on $(u=0)$, and $c_\alpha^g(0)$ is the coefficient of x^α in $f_i(x,y,z,0)$. By **Step 3**,

$$f_i^{g,n}(X_1,X_2,X_3,0)=f_0(X_1,X_2,X_3)+\Xi$$

for a polynomial Ξ whose monomials have w-weight less than d. By **Step 1**, the surface $W_i^{g,n}\cap(u=0)$ is normal with Du Val singularities of the types of **Step 1**. Thus $W_i^{g,n}$ is normal near 0, $(W_i^{g,n},(u=0))$ is plt near 0 by [KM98, Theorem 5.50(1)], so $W_i^{g,n}$ is klt near 0, and (F.66) holds for $V=W_i^{g,n}$.

Since g has Jacobian determinant 1,

$$\varpi^*(dx_1\wedge dx_2\wedge dx_3\wedge du)=u^{n(w_1+w_2+w_3)}dX_1\wedge dX_2\wedge dX_3\wedge du,$$

and $f_i\circ\varpi=u^{nd}f_i^{g,n}$. Since $w_1+w_2+w_3=d+1$ for any f_0 in the theorem, the pull-back of the residue of $dx_1\wedge dx_2\wedge dx_3\wedge du/f_i$, a generator of ω_{W_i}, is u^n times the residue of $dX_1\wedge dX_2\wedge dX_3\wedge du/f_i^{g,n}$, a generator of $\omega_{W_i^{g,n}}$. Moreover h_i and $h_i+\xi f_i$ coincide on W_i, and

$$\varpi^*\operatorname{div}((h_i+\xi f_i)|_{W_i})=nm\operatorname{div}(u)+\operatorname{div}(h_i^{g,\xi,n})\quad\text{on } W_i^{g,n}.$$

Thus any prime divisor F over $W_i^{g,n}$ satisfies

$$a(F,W_i,0)=a(F,W_i^{g,n},0)+n\operatorname{mult}_F u\quad\text{and}\quad a(F,W_i,B_i)=a(F,W_i^{g,n},B_i^{g,\xi,n}).\tag{F.71}$$

Since ϖ is G-equivariant, birational and $\varpi(0)=0$, any prime divisor over $W_i^{g,n}$ with center 0 is a prime divisor over W_i with center 0, and the decomposition and inertia groups of such a divisor for $W_i^{g,n}$ and for W_i coincide. By (F.71),

$$\operatorname{mld}((W_i^{g,n}\ni 0,B_i^{g,\xi,n})/G)\ge b,$$

and F_i attains b. Thus $\operatorname{mld}((W_i^{g,n}\ni 0,B_i^{g,\xi,n})/G)=b$, and F_i computes it.

By (F.71) and $a(F_i,W_i^{g,n},0)>0$, $n<a(F_i,W_i,0)/\operatorname{mult}_{F_i}u\le a(F_i,W_i,0)$ for any admissible (g,ξ,n). Thus the set of n with (g,ξ,n) admissible for some (g,ξ) is finite and non-empty. Let

n_i be the maximum of the set, and fix a preparation (g_i, ξ_i) with (g_i, ξ_i, n_i) admissible. Then $n_i \geq n_i^0 \to +\infty$. We let

$$c_{i,\alpha} := c_\alpha^{g_i}, \quad c'_{i,\alpha} := c_\alpha'^{g_i,\xi_i}, \quad f_i^{(n)} := f_i^{g_i,n},$$
$$h_i^{(n)} := h_i^{g_i,\xi_i,n}, \quad W_i^{(n)} := W_i^{g_i,n}, \quad \text{and} \quad B_i^{(n)} := B_i^{g_i,\xi_i,n},$$

and $f_i^{\mathrm{res}} := f_i^{(n_i)}$ and $h_i^{\mathrm{res}} := h_i^{(n_i)}$.

Step 5. In this step we take the common tops and the charts of fixed depth. The coefficients of x^α in $f_i \circ g_i^{-1}$ and $(h_i + \xi_i f_i) \circ g_i^{-1}$ are $c_{i,\alpha}$ and $c'_{i,\alpha}$. By Lemma 2.6(3), possibly passing to an infinite subfamily and replacing $\Bbbk$ with the field $\Bbbk'$ there, into which $\Bbbk$ embeds over $\mathbb{C}$ with f_*, h_* mapping to the first two entries of the new limit (from now on $\Bbbk$ denotes $\Bbbk'$, over which hypothesis (3) of the theorem still holds), the tuples

$$(f_i, h_i, f_i^{\mathrm{res}}, h_i^{\mathrm{res}}, f_i \circ g_i^{-1}, (h_i + \xi_i f_i) \circ g_i^{-1}),$$

whose entries are non-zero and G-semi-invariant with characters independent of i, have a compatible generic limit (Definition 2.5) whose first two entries are f_*, h_*. Let Q_f and Q_h be the limits of f_i^{res} and h_i^{res}. Let $h_0 := \mathrm{in}_w(h_*) \in \Bbbk[[u]][x, y, z]$. In the coordinates (X_1, X_2, X_3, u) we write f_0 and h_0 for $f_0(X_1, X_2, X_3)$ and $h_0(X_1, X_2, X_3)$.

The coefficient of X^α in f_i^{res} is $c_{i,\alpha} u^{n_i(w\cdot\alpha - d)}$. If $w \cdot \alpha > d$, then the coefficient of X^α in f_i^{res} is an element of $u^{n_i}\mathbb{C}[[u]]$, so the limit of the coefficient is 0 since $n_i \to +\infty$. If $w \cdot \alpha = d$, then $c_{i,\alpha}$ is congruent modulo $u^{n'_i}$ to the coefficient of x^α in f_i, whose limit is the coefficient of x^α in f_0. Since $n'_i \to +\infty$, the limit of $c_{i,\alpha}$ is the coefficient of x^α in f_0. If $w \cdot \alpha < d$, then $c_{i,\alpha} u^{-n_i(d - w\cdot\alpha)} \in \mathbb{C}[[u]]$ by admissibility. The coefficient of X^α in h_i^{res} is $c'_{i,\alpha} u^{n_i(w\cdot\alpha - m)}$, which is an element of $u^{n_i}\mathbb{C}[[u]]$ if $w \cdot \alpha > m$ and of $\mathbb{C}[[u]]$ if $w \cdot \alpha < m$ by admissibility, and the limit of $c'_{i,\alpha}$ for $w \cdot \alpha = m$ is the coefficient of x^α in h_*, that is, in h_0, as a series in u. Thus $Q_f, Q_h \in \Bbbk[[u]][X_1, X_2, X_3]$ and

(F.72) $Q_f - f_0$ and $Q_h - h_0$ are sums of terms of w-weight less than d and m, respectively.

In particular Q_f has maximal w-weight d with top part f_0, and Q_h has maximal w-weight m with top part h_0.

Let $\overline{n}$ be a positive integer divisible by $|G|$. If $n_i \geq \overline{n}$, then $(g_i, \xi_i, \overline{n})$ is admissible, since

$$c_{i,\alpha} u^{-\overline{n}(d - w\cdot\alpha)} = c_{i,\alpha} u^{-n_i(d-w\cdot\alpha)} \cdot u^{(n_i - \overline{n})(d - w\cdot\alpha)} \in \mathbb{C}[[u]] \quad \text{for } w \cdot \alpha < d,$$
$$c'_{i,\alpha} u^{-\overline{n}(m - w\cdot\alpha)} = c'_{i,\alpha} u^{-n_i(m-w\cdot\alpha)} \cdot u^{(n_i - \overline{n})(m - w\cdot\alpha)} \in \mathbb{C}[[u]] \quad \text{for } w \cdot \alpha < m, \quad \text{and}$$
$$\mathrm{mult}_{F_i} x'_j - \overline{n} w_j \,\mathrm{mult}_{F_i} u \geq \mathrm{mult}_{F_i} x'_j - n_i w_j \,\mathrm{mult}_{F_i} u > 0.$$

By **Step 4**, $\mathrm{mld}((W_i^{(\overline{n})} \ni 0, B_i^{(\overline{n})})/G) = b$. Let $h_{0,i} := \sum_{w\cdot\alpha = m} c'_{i,\alpha} X^\alpha$, the part of w-weight m of $h_i^{(\overline{n})}$, which does not depend on $\overline{n}$. The terms of $f_i^{(\overline{n})}$ with $w \cdot \alpha > d$ are elements of $(u^{\overline{n}})$, those with $w \cdot \alpha < d$ are elements of $(u^{n_i - \overline{n}})$, and, by **Step 3** and $n'_i \to +\infty$, the coefficients of those with $w \cdot \alpha = d$ differ from the coefficients of f_0 by elements of $u^{\overline{n}}\mathbb{C}[[u]]$ for $i \gg 0$. The terms of $h_i^{(\overline{n})}$ with $w \cdot \alpha > m$ are elements of $(u^{\overline{n}})$, those with $w \cdot \alpha < m$ are elements of $(u^{n_i - \overline{n}})$, and those with $w \cdot \alpha = m$ form $h_{0,i}$. Let $c_{*,\alpha}$ and $c'_{*,\alpha}$ be the coefficients in $\Bbbk[[u]]$ of x^α in the limits of $f_i \circ g_i^{-1}$ and $(h_i + \xi_i f_i) \circ g_i^{-1}$ in the compatible generic limit above, and let

$$f_*^{(\overline{n})} := \sum_\alpha c_{*,\alpha} u^{\overline{n}(w\cdot\alpha - d)} X^\alpha \quad \text{and} \quad h_*^{(\overline{n})} := \sum_\alpha c'_{*,\alpha} u^{\overline{n}(w\cdot\alpha - m)} X^\alpha$$

be the limits of $f_i^{(\overline{n})}$ and $h_i^{(\overline{n})}$. Thus, for all but finitely many i,

(F.73)
$$f_i^{(\overline{n})} - f_0,\ h_i^{(\overline{n})} - h_{0,i} \in u^{\overline{n}}\mathbb{C}[[X_1, X_2, X_3, u]] \quad \text{and}$$
$$f_*^{(\overline{n})} - f_0,\ h_*^{(\overline{n})} - h_0 \in u^{\overline{n}}\Bbbk[[X_1, X_2, X_3, u]],$$

and the limit of $h_{0,i}$ is h_0.

Step 6. In this step we show $b \le b_0$. Let

$$W_0 := (f_0 = 0) \subset \mathbb{A}^4_{X_1,X_2,X_3,u} \quad \text{and} \quad \widehat{W}_0 := \operatorname{Spec} \Bbbk[[X_1, X_2, X_3, u]]/(f_0),$$

a germ over $\mathbb{C}$ and a completion over $\Bbbk$, and let

$$\Delta_0 := \frac{1}{m} \operatorname{div}(h_0|_{\widehat{W}_0}) + \beta \operatorname{div}(u|_{\widehat{W}_0}) \quad \text{and} \quad b_0 := \operatorname{mld}((\widehat{W}_0 \ni 0, \Delta_0)/G),$$

where b_0 is the quotient value of Definition 2.2 on the completion. The group G acts faithfully on W_0 since no coordinate vanishes on W_0, and $h_0|_{\widehat{W}_0} \neq 0$ since $h_0 \neq 0$ has degree at most 1 in X_1 while f_0 is X_1-regular of order 2. We show

$$0 < b \le b_0. \tag{F.74}$$

Let E be a prime divisor over $\widehat{W}_0$ with center the closed point, and let $\overline{n}$ be a positive integer divisible by $|G|$ with

$$\overline{n} > 2 \operatorname{mult}_E \operatorname{Jac}_{W_0} + \operatorname{mult}_E(h_0|_{\widehat{W}_0}) + \operatorname{mult}_E u + 1.$$

The germ $(W_0 \ni 0)$ is defined over $\mathbb{C}$ and does not depend on i, and h_0 is the limit of $h_{0,i}$ by **Step 5**. We show that for infinitely many i there exists a prime divisor E' over the completion $\widehat{W}_{0,\mathbb{C}}$ of W_0 at 0 over $\mathbb{C}$, with center the closed point, such that

$$a(E', W_0, 0) = a(E, \widehat{W}_0, 0), \quad \operatorname{mult}_{E'}(h_{0,i}|_{W_0}) = \operatorname{mult}_E(h_0|_{\widehat{W}_0}),$$
$$\operatorname{mult}_{E'} u = \operatorname{mult}_E u, \quad \text{and} \quad \operatorname{mult}_{E'} \operatorname{Jac}_{W_0} = \operatorname{mult}_E \operatorname{Jac}_{W_0}.$$

Here, as Lemma 3.15 specializes formal prime divisors over $\mathbb{A}^n$ (cf. [Kaw21, Remark 3.8(i)]), E descends to a prime divisor E_T over $W_0 \times T$, where T is a variety whose function field, in $\Bbbk$, contains a field of definition of E, with a dominant morphism to the set $Z_{d'}$ of Definition 2.5 for the compatible generic limit of **Step 5** and some $d' > \operatorname{mult}_E(h_0|_{\widehat{W}_0})$, and E' is the fiber of E_T over a point of T over the d'-jet of the i-th member, and since G acts on $W_0 \times T$ through W_0, commuting with the projection to T, ϱE_T is the descent of ϱE for any $\varrho \in G$. After shrinking T to a non-empty open subset, the fibers of E_T over T are irreducible, since $\Bbbk$ is algebraically closed and hence the generic fiber is geometrically irreducible, and for any $\varrho \in G$ each of the conditions $\varrho E_T = E_T$ and "ϱ acts trivially on E_T" holds at the generic point of T if and only if it holds on the fiber over each point of T, so the decomposition and inertia groups of E' coincide with those of E. Fix such an i with $n_i \ge \overline{n}$ for which (F.73) holds.

Both f_0 and $f_i^{(\overline{n})}$ have character χ_f, and W_0 and $W_i^{(\overline{n})}$ are normal by **Steps 1 and 4**. Let $x_\bullet \in \{X_1, X_2, X_3, u\}$ be an eigenparameter with

$$\operatorname{mult}_{E'}(\partial f_0/\partial x_\bullet) = \operatorname{mult}_{E'} \operatorname{Jac}_{W_0}.$$

Since $h_i^{(\overline{n})} - h_{0,i} \in (u^{\overline{n}})$ by (F.73) and $\overline{n} > \operatorname{mult}_E(h_0|_{\widehat{W}_0})$, we have

$$\operatorname{mult}_{E'}(h_i^{(\overline{n})}|_{W_0}) = \operatorname{mult}_{E'}(h_{0,i}|_{W_0}) = \operatorname{mult}_E(h_0|_{\widehat{W}_0}).$$

By (F.73),

$$f_i^{(\overline{n})} - f_0 \in (u^{\overline{n}}) \quad \text{and} \quad \partial(f_i^{(\overline{n})} - f_0)/\partial x_\bullet \in (u^{\overline{n}-1}),$$

so

$$\operatorname{mult}_{E'}((f_i^{(\overline{n})} - f_0)|_{W_0}) \ge \overline{n} > 2 \operatorname{mult}_{E'} \operatorname{Jac}_{W_0} + \max\{\operatorname{mult}_{E'}(h_i^{(\overline{n})}|_{W_0}), \operatorname{mult}_{E'} u\}$$

and

$$\operatorname{mult}_{E'}(\partial(f_i^{(\overline{n})} - f_0)/\partial x_\bullet|_{W_0}) \ge \overline{n} - 1 > \operatorname{mult}_{E'} \operatorname{Jac}_{W_0}.$$

Thus, by Lemma 5.4(3)(b)(d) with $f := f_0$, $f' := f_i^{(\overline{n})}$, $F := E'$, $H_1 := h_i^{(\overline{n})}$, and $H_2 := u$, there exists a prime divisor E'' over the completion of $W_i^{(\overline{n})}$ at 0 with center the closed point, whose decomposition and inertia groups coincide with those of E', such that

$$a(E'', W_i^{(\overline{n})}, 0) = a(E', W_0, 0), \quad \operatorname{mult}_{E''} h_i^{(\overline{n})} = \operatorname{mult}_E(h_0|_{\widehat{W}_0}), \quad \text{and} \quad \operatorname{mult}_{E''} u = \operatorname{mult}_E u.$$

By **Step 5** and [Liu26a, Lemma 3.3(2)(3)], the quotient value of $(W_i^{(\overline{n})} \ni 0, B_i^{(\overline{n})})$ and G on the completion is b. Therefore

$$\frac{a(E,\widehat{W}_0,\Delta_0)}{e_E} = \frac{a(E,\widehat{W}_0,0) - \frac{1}{m}\operatorname{mult}_E(h_0|_{\widehat{W}_0}) - \beta \operatorname{mult}_E u}{e_E} = \frac{a(E'',W_i^{(\overline{n})},B_i^{(\overline{n})})}{e_{E''}} \geq b.$$

Since E is arbitrary and $b > 0$ by **Step 3**, (F.74) holds.

The proof continues below.

Proof of Theorem 17.6, continued. We keep the choices made before (F.74), and we let $e_i := e_{F_i}$ and $W_i^{\mathrm{res}} := W_i^{(n_i)} = (f_i^{\mathrm{res}} = 0)$. Since F_i computes b, the second identity of (F.71) with $(g,\xi,n) = (g_i,\xi_i,n_i)$ gives

$$a(F_i,W_i^{\mathrm{res}},0) - \frac{1}{m}\operatorname{mult}_{F_i} h_i^{\mathrm{res}} - \beta \operatorname{mult}_{F_i} u = be_i. \tag{F.75}$$

Step 7. In this step we prove that, with $\widehat{\mathcal{X}} := \operatorname{Spec} \Bbbk[[X_1,X_2,X_3,u]]$, any prime divisor E over $\widehat{\mathcal{X}}$ with center the closed point satisfies

$$\frac{a(E,\widehat{\mathcal{X}},0) - \operatorname{mult}_E Q_f - \frac{1}{m}\operatorname{mult}_E Q_h - \beta \operatorname{mult}_E u}{e_E} \geq b_0. \tag{F.76}$$

By (F.72), $Q_f, Q_h \in \Bbbk[[u]][X_1,X_2,X_3]$ are G-semi-invariant with maximal-weight parts f_0 and h_0. By Lemma 7.11 with the weights w_x, w_y, w_z of X_1, X_2, X_3, $(Q_1,Q_2) = (Q_f,Q_h)$, and $(c_1,c_2) = (1,\frac{1}{m})$, (F.76) follows from

$$\operatorname{mld}\Big(\Big(\widehat{\mathcal{X}} \ni 0, \operatorname{div}(f_0) + \frac{1}{m}\operatorname{div}(h_0) + \beta \operatorname{div}(u)\Big)/G\Big) \geq b_0. \tag{F.77}$$

We first remove pseudoreflections. Let G act on $\mathbb{A}^4_v$, with coordinates $v_1,\dots,v_4$, by

$$\varrho^* v_1 = \chi_x(\varrho)v_1, \quad \varrho^* v_2 = \chi_y(\varrho)v_2, \quad \varrho^* v_3 = \chi_z(\varrho)v_3, \quad \text{and} \quad \varrho^* v_4 = \chi_u(\varrho)v_4$$

for $\varrho \in G$. Let $\widetilde{\mathcal{X}}$ be the completion of $\widehat{\mathcal{X}} \times \mathbb{A}^4_v$ at the closed point, $\widetilde{W}_0 := (f_0 = 0) \subset \widetilde{\mathcal{X}}$, and

$$\widetilde{\Delta}_0 := \Delta_0 + \sum_{j=1}^4 \operatorname{div}(v_j) \ \text{ on } \widetilde{W}_0,$$

$$\widetilde{\Delta} := \operatorname{div}(f_0) + \frac{1}{m}\operatorname{div}(h_0) + \beta \operatorname{div}(u) + \sum_{j=1}^4 \operatorname{div}(v_j) \ \text{ on } \widetilde{\mathcal{X}}.$$

Since $b_0 \geq b > 0$ by (F.74), Lemma 3.6(3) and (5) with $X = W_0 \times_{\mathbb{C}} \Bbbk$ and $x = 0$, whose completion at 0 is $\widehat{W}_0$, the boundary $B = \Delta_0$ on $\widehat{W}_0$, and the characters $\chi_x,\chi_y,\chi_z,\chi_u$ imply $\operatorname{mld}((\widetilde{W}_0 \ni 0, \widetilde{\Delta}_0)/G) = b_0$. By Lemma 3.6(2) and (5) with $X = \mathbb{A}^4_{\Bbbk}$ and $x = 0$, whose completion at 0 is $\widehat{\mathcal{X}}$, for any prime divisor E over $\widehat{\mathcal{X}}$ with center the closed point there exists a prime divisor $\widetilde{E}$ over $\widetilde{\mathcal{X}}$ with center the closed point such that $I_G(\widetilde{E}) = I_G(E)$ and

$$a(\widetilde{E},\widetilde{\mathcal{X}},\widetilde{\Delta}) = a\Big(E,\widehat{\mathcal{X}},\operatorname{div}(f_0) + \frac{1}{m}\operatorname{div}(h_0) + \beta\operatorname{div}(u)\Big).$$

Thus (F.77) follows from

$$\operatorname{mld}((\widetilde{\mathcal{X}} \ni 0, \widetilde{\Delta})/G) \geq \operatorname{mld}((\widetilde{W}_0 \ni 0, \widetilde{\Delta}_0)/G) = b_0. \tag{F.78}$$

By Lemma 3.6(4) with $X = W_0 \subset \mathbb{A}^4$, G acts on $\widetilde{\mathcal{X}}$ without pseudoreflections.

Let I be a positive integer divisible by $|G|$. For $\varrho \in G$ let $\vartheta_x,\vartheta_y,\vartheta_z,\vartheta_u \in [0,1)$ be the numbers with $\varrho^* X_1 = \exp(2\pi\sqrt{-1}\vartheta_x)X_1$, $\varrho^* X_2 = \exp(2\pi\sqrt{-1}\vartheta_y)X_2$, $\varrho^* X_3 = \exp(2\pi\sqrt{-1}\vartheta_z)X_3$, and $\varrho^* u = \exp(2\pi\sqrt{-1}\vartheta_u)u$, and let

$$(\vartheta_1,\dots,\vartheta_4) := (\vartheta_x,\vartheta_y,\vartheta_z,\vartheta_u),$$

so that $\varrho^* v_j = \exp(2\pi\sqrt{-1}\vartheta_j)v_j$ for $1 \le j \le 4$. For a tuple $\ell = (\ell_0, \ell_f, \ell_h, \ell_1, \dots, \ell_4)$ of non-negative integers let $\mathcal{C}(\varrho, \ell)$ be the cylinder $C(g, k)$ of Lemma 7.10(1) on $\widetilde{\mathcal{X}}$ for $g = \varrho$, $k = \ell$, and $(Q_1, \dots, Q_6) = (f_0, h_0, v_1, \dots, v_4)$, that is, the cylinder of ϱ-twisted arcs at the image of 0 with u-order exactly $\vartheta_u + \ell_0$ and with orders of f_0^I, h_0^I, $v_1^I, \dots, v_4^I$ at least $\ell_f, \ell_h, \ell_1, \dots, \ell_4$. By that lemma with $(c_1, \dots, c_6) = (1, \frac{1}{m}, 1, 1, 1, 1)$, (F.78) follows from

$$\operatorname{codim} \mathcal{C}(\varrho, \ell) + \operatorname{age}(\varrho) - \beta(\vartheta_u + \ell_0) - \frac{\ell_f}{I} - \frac{\ell_h}{mI} - \sum_{j=1}^{4} \frac{\ell_j}{I} \ge b_0 \tag{F.79}$$

for any ϱ and ℓ with $\mathcal{C}(\varrho, \ell) \ne \emptyset$.

Fix ϱ and ℓ with $\mathcal{C} := \mathcal{C}(\varrho, \ell) \ne \emptyset$. Since the arcs of $\mathcal{C}$ are centered at the image of 0, $\vartheta_u + \ell_0 > 0$, and we fix an integer N_2 with $N_2(\vartheta_u + \ell_0) > \ell_h / I$ and let $h_0^{<N_2}$ be the truncation of h_0 modulo u^{N_2}, a polynomial in X_1, X_2, X_3, u. Along any arc of $\mathcal{C}$ the orders of h_0 and $h_0^{<N_2}$ are equal or both at least $N_2(\vartheta_u + \ell_0)$, so $\mathcal{C}$ does not change if h_0 is replaced by $h_0^{<N_2}$, and below [NS26, Lemma 2.17] and [NS26, Proposition 3.10 and Lemma 5.2] are applied to the polynomial data f_0, $h_0^{<N_2}$, u, $v_1, \dots, v_4$. Let N_3 be a jet level defining $\mathcal{C}$, let π_{N_3} be the truncation to N_3-jets, and let $\mathcal{C}_1 \subset \mathcal{C}$ be an irreducible component with $\operatorname{codim} \mathcal{C}_1 = \operatorname{codim} \mathcal{C}$. The set $\mathcal{S}$ of N_3-jets with u-order exactly $\vartheta_u + \ell_0$ is closed in the open set of jets whose coefficient of $\tau^{\vartheta_u + \ell_0}$ in u is non-zero, where we write τ for the variable t of the twisted arcs defined before Lemma 7.9. Since $\mathcal{C}$ is defined by closed conditions on N_3-jets in $\mathcal{S}$, $\pi_{N_3}(\mathcal{C})$ is closed in $\mathcal{S}$, hence so is its irreducible component $\pi_{N_3}(\mathcal{C}_1)$.

We show that, for any $\gamma \in \mathcal{C}_1$, the N_3-jet γ_0 with the u-coordinate of $\pi_{N_3}(\gamma)$ and vanishing $X_1, X_2, X_3, v_1, \dots, v_4$-coordinates belongs to $\pi_{N_3}(\mathcal{C}_1)$. Let $\mathbb{G}_m$ act on $\widetilde{\mathcal{X}}$ by

$$t \cdot (X_1, X_2, X_3, u, v_1, \dots, v_4) = (t^{w_x} X_1, t^{w_y} X_2, t^{w_z} X_3, u, t v_1, \dots, t v_4).$$

The action of $\mathbb{G}_m$ commutes with the diagonal action of G, hence acts on ϱ-twisted arcs. Since f_0 and h_0 are w-homogeneous of weights d and m with coefficients in $\Bbbk[[u]]$, $f_0(t \cdot \gamma) = t^d f_0(\gamma)$ and $h_0(t \cdot \gamma) = t^m h_0(\gamma)$. Thus the orders of u, f_0, h_0, $v_1, \dots, v_4$ along arcs are $\mathbb{G}_m$-invariant, $\mathcal{C}$ is $\mathbb{G}_m$-stable, and so is the irreducible component $\mathcal{C}_1$ of $\mathcal{C}$. As $t \to 0$, $t \cdot \pi_{N_3}(\gamma)$ tends to γ_0. Since $\gamma_0 \in \mathcal{S}$ and $\pi_{N_3}(\mathcal{C}_1)$ is closed in $\mathcal{S}$, $\gamma_0 \in \pi_{N_3}(\mathcal{C}_1)$.

We next construct $\zeta_1, \zeta_2, \zeta_3 \in \Bbbk[[\tau]]$ with $\zeta_1 \ne 0$, with orders larger than N_3, and with

$$f_0(\tau^{\vartheta_x}\zeta_1, \tau^{\vartheta_y}\zeta_2, \tau^{\vartheta_z}\zeta_3) = 0. \tag{F.80}$$

Since ϱ acts diagonally and f_0 is G-semi-invariant, any monomial of f_0 has the ϱ-eigenvalue of f_0.

- Let $f_0(x, y, z) = x^2 + y^3 + \psi(y, z)$ with $\psi \in \{z^4, yz^3, z^5, z^3\}$. Then $\iota := 3\vartheta_y - 2\vartheta_x \in \mathbb{Z}$, and
$$\psi(\tau^{\vartheta_y}, \tau^{\vartheta_z})/\tau^{3\vartheta_y}$$
is an integral power of τ. Choose an integer $n_y > N_3$ with $n_y \equiv \iota \pmod 2$, let $n_x := (3n_y + \iota)/2$, and choose an integer $n_z > N_3$ such that
$$\eta := \frac{\psi(\tau^{\vartheta_y + n_y}, \tau^{\vartheta_z + n_z})}{\tau^{3(\vartheta_y + n_y)}} \in \tau\Bbbk[[\tau]].$$
- Let $f_0(x, y, z) = x^2 + zy^2 + z^{k-1}$ with $k \ge 4$. Then $\iota := \vartheta_z + 2\vartheta_y - 2\vartheta_x \in \mathbb{Z}$ and $(k-2)\vartheta_z - 2\vartheta_y \in \mathbb{Z}$. Choose an integer $n_y > N_3$, then an integer $n_z > N_3$ with $n_z \equiv \iota \pmod 2$ and $(k-2)(\vartheta_z + n_z) > 2(\vartheta_y + n_y)$, and let $n_x := (n_z + 2n_y + \iota)/2$ and
$$\eta := \tau^{(k-2)(\vartheta_z + n_z) - 2(\vartheta_y + n_y)} \in \tau\Bbbk[[\tau]].$$

In both cases let $\zeta_2 := \tau^{n_y}$, $\zeta_3 := \tau^{n_z}$, and $\zeta_1 := \sqrt{-1}\tau^{n_x}(1 + \eta)^{1/2}$, where $(1 + \eta)^{1/2} \in \Bbbk[[\tau]]$ is the binomial series. Then

$$\tau^{2\vartheta_x}\zeta_1^2 = -\tau^{2(\vartheta_x + n_x)}(1 + \eta)$$

equals $-(y^3 + \psi(y, z))$, respectively $-(zy^2 + z^{k-1})$, at $(y, z) = (\tau^{\vartheta_y}\zeta_2, \tau^{\vartheta_z}\zeta_3)$, which is (F.80). For $n_y \gg 0$, $n_x > N_3$.

Let γ_1 be the ϱ-twisted arc with $X_1 = \tau^{\vartheta_x}\zeta_1$, $X_2 = \tau^{\vartheta_y}\zeta_2$, $X_3 = \tau^{\vartheta_z}\zeta_3$, $v_j = \tau^{\vartheta_j+N_3+1}$ for $1 \le j \le 4$, and with the u-coordinate of γ. Then $\pi_{N_3}(\gamma_1) = \gamma_0 \in \pi_{N_3}(\mathcal{C}_1)$, so $\gamma_1 \in \mathcal{C}_1$, since $\mathcal{C}_1$, an irreducible component of the cylinder $\mathcal{C}$ of level N_3, is the preimage of $\pi_{N_3}(\mathcal{C}_1)$. By (F.80), γ_1 is an arc of $\widetilde{W}_0$. Let $w_\varrho(f_0) \in [0,1)$ be the weight of f_0 in the sector of ϱ in [NS26, Theorem 5.3], and let $\mathrm{Jac}^{(\varrho)}$ be the ideal generated by the partial derivatives of the twisted equation

$$f_0^{(\varrho)} := \tau^{-w_\varrho(f_0)} f_0(\tau^{\vartheta_x}X_1, \tau^{\vartheta_y}X_2, \tau^{\vartheta_z}X_3),$$

which is the ideal $\mathrm{Jac}^{(\varrho)}$ of [NS26, Theorem 5.3]. Since x^2 is a monomial of $f_0(x,y,z)$,

$$2\vartheta_x - w_\varrho(f_0) \in \{0,1\},$$

and along γ_1

$$\frac{\partial f_0^{(\varrho)}}{\partial X_1} = 2\tau^{2\vartheta_x - w_\varrho(f_0)}X_1 = 2\tau^{2\vartheta_x - w_\varrho(f_0)}\zeta_1 \ne 0.$$

Therefore the least order n_J of $\mathrm{Jac}^{(\varrho)}$ along the ϱ-twisted arcs of $\widetilde{W}_0$ in $\mathcal{C}_1$ is finite.

Let $L_\varrho(\widetilde{W}_0)$ be the space of ϱ-twisted arcs of $\widetilde{W}_0$ at the image of 0, and let $\mathcal{C}_2 \subset \mathcal{C}_1$ be the open subset of arcs along which $\mathrm{Jac}^{(\varrho)}$ has order at most n_J. For a cylinder in $L_\varrho(\widetilde{W}_0)$, its codimension $\mathrm{codim}(\cdot, L_\varrho(\widetilde{W}_0))$ is taken in the arc space of the $\Bbbk[\tau]$-scheme defined by $f_0^{(\varrho)}$ in $\mathbb{A}^8_{\Bbbk[\tau]}$, whose arcs at the image of 0 are the ϱ-twisted arcs of $\widetilde{W}_0$, in the sense of [NS26, Definition 2.9] with $n = 7$, as in [NS26, Lemmas 2.17 and 5.2]. Then $\mathcal{C}_2$ contains an arc of $\widetilde{W}_0$ in $\mathcal{C}_1$ along which $\mathrm{Jac}^{(\varrho)}$ has order n_J, so $\mathcal{C}_2 \cap L_\varrho(\widetilde{W}_0) \ne \emptyset$, and $\mathrm{codim}\,\mathcal{C}_2 = \mathrm{codim}\,\mathcal{C}_1$, since $\mathcal{C}_2$ is a non-empty open subset of the irreducible cylinder $\mathcal{C}_1$. Let $\ell'_f := \lceil \ell_f/I - w_\varrho(f_0)\rceil$, so that $w_\varrho(f_0) + \ell'_f \ge \ell_f/I$ and the twisted equation of f_0 has order at least ℓ'_f along $\mathcal{C}_1$. By [NS26, Lemma 2.17],

$$\mathrm{codim}\left(\mathcal{C}_2 \cap L_\varrho(\widetilde{W}_0), L_\varrho(\widetilde{W}_0)\right) \le \mathrm{codim}\,\mathcal{C}_1 + n_J - \ell'_f. \tag{F.81}$$

Along any arc of $\mathcal{C}_2 \cap L_\varrho(\widetilde{W}_0)$, $\mathrm{Jac}^{(\varrho)}$ has order exactly n_J, u has order exactly $\vartheta_u + \ell_0$, and $h_0^I, v_1^I, \dots, v_4^I$ have orders at least $\ell_h, \ell_1, \dots, \ell_4$. Let $\widetilde{W}_0^{\mathrm{alg}} \subset \mathbb{A}^8_\Bbbk$ be the hypersurface defined by f_0 in the coordinates of $\widetilde{\mathcal{X}}$, so that $\widetilde{W}_0$ is the completion of $\widetilde{W}_0^{\mathrm{alg}}$ at 0, let $\mathcal{Y} := \widetilde{W}_0^{\mathrm{alg}}/G$ with quotient morphism ϖ_0, and let $o_\mathcal{Y} := \varpi_0(0)$. By Lemma 3.6(4) for $W_0 \subset \mathbb{A}^4$, near $o_\mathcal{Y}$ the morphism ϖ_0 is étale in codimension 1, and the image of the locus where $\mathbb{A}^8_\Bbbk \to \mathbb{A}^8_\Bbbk/G$ is not étale intersects $\mathcal{Y}$ in codimension at least 2. The locus where ϖ_0 is not étale and the intersection with $\mathcal{Y}$ of the image of the locus where $\mathbb{A}^8_\Bbbk \to \mathbb{A}^8_\Bbbk/G$ is not étale are stable under the $\mathbb{G}_m$-action on $\mathbb{A}^8_\Bbbk$ with weights $(w_x, w_y, w_z, 1, \dots, 1)$ on $(X_1, X_2, X_3, u, v_1, \dots, v_4)$, which commutes with G, preserves $\widetilde{W}_0^{\mathrm{alg}}$ as f_0 is w-homogeneous, and contracts $\mathbb{A}^8_\Bbbk$ to 0, so both codimension bounds hold on all of $\widetilde{W}_0^{\mathrm{alg}}$ and $\mathcal{Y}$, as [NS26, Section 3] requires. Since $\widetilde{W}_0^{\mathrm{alg}}$ is a normal hypersurface and ϖ_0 is étale in codimension 1, $\mathcal{Y}$ is normal and $rK_\mathcal{Y}$ is Cartier for some positive integer r. Let $\mathfrak{n}_{r,\mathcal{Y}}$ be the ideal of the r-th Nash subscheme of $\mathcal{Y}$ [EM09, Section 7]. Let $\mathcal{L}_1$ be an irreducible component of $\mathcal{C}_2 \cap L_\varrho(\widetilde{W}_0)$ with

$$\mathrm{codim}\left(\mathcal{L}_1, L_\varrho(\widetilde{W}_0)\right) = \mathrm{codim}\left(\mathcal{C}_2 \cap L_\varrho(\widetilde{W}_0), L_\varrho(\widetilde{W}_0)\right).$$

By [NS26, Lemma 5.1(2) and Proposition 2.12], the other contact orders of [NS26, Lemma 5.2] are constant on a dense open subset $\mathcal{L}_1^\circ \subset \mathcal{L}_1$. By [NS26, Proposition 3.10] and [NS26, Propositions 2.15 and 2.16], from which [NS26, Lemma 5.2] follows, applied to the G-saturation $G \cdot \mathcal{L}_1^\circ$, a G-invariant subcylinder of the cylinder of [NS26, Lemma 5.2] given by n_J and the contact orders constant on $\mathcal{L}_1^\circ$, with the codimension of $\mathcal{L}_1^\circ$ and, G being abelian, with the image of $\mathcal{L}_1^\circ$, the image $\mathcal{T}$ of $\mathcal{L}_1^\circ$ in the space of arcs of $\mathcal{Y}$ through $o_\mathcal{Y}$ is an irreducible cylinder along whose arcs $\mathfrak{n}_{r,\mathcal{Y}}$ has a constant order $n_\mathcal{Y}$, the order of any $\varphi \in \mathcal{O}_{\mathcal{Y},o_\mathcal{Y}}$ along the image of a twisted arc γ is $\mathrm{mult}_\gamma(\varpi_0^*\varphi)$, and

$$\mathrm{codim}\,\mathcal{T} - \frac{n_\mathcal{Y}}{r} = \mathrm{codim}\left(\mathcal{C}_2 \cap L_\varrho(\widetilde{W}_0), L_\varrho(\widetilde{W}_0)\right) + \mathrm{age}(\varrho) - w_\varrho(f_0) - n_J. \tag{F.82}$$

The cylinder $\mathcal{T}$ is fat by [dFEI08, Corollary 3.7], since the Jacobian ideal $\mathrm{Jac}_{\mathcal{Y}}$ [NS26, Definition 2.1(1)], whose pullback is one of the ideals of [NS26, Lemma 5.2], has finite order along the arcs of $\mathcal{T}$. By [dFEI08, Proposition 3.10, Proposition 2.12, Remark 2.11, Theorem 3.9], there exist a prime divisor $E_{\mathcal{Y}}$ over $\mathcal{Y}$ with center $o_{\mathcal{Y}}$ and a positive integer q such that the order of any $\varphi \in \mathcal{O}_{\mathcal{Y},o_{\mathcal{Y}}}$ along the generic arc of $\mathcal{T}$ is $q \operatorname{mult}_{E_{\mathcal{Y}}} \varphi$ and

$$\operatorname{codim} \mathcal{T} \geq q\big(\operatorname{mult}_{E_{\mathcal{Y}}} \widehat{K}_{\mathcal{Y}'/\mathcal{Y}} + 1\big),$$

where $\widehat{K}_{\mathcal{Y}'/\mathcal{Y}}$ is the relative Mather canonical divisor [dFEI08, Definition 1.6] of a resolution $\mathcal{Y}' \to \mathcal{Y}$ factoring through the Nash blow-up of $\mathcal{Y}$ on which $E_{\mathcal{Y}}$ is a divisor. Since

$$\mathfrak{n}_{r,\mathcal{Y}}\mathcal{O}_{\mathcal{Y}'} = \mathcal{O}_{\mathcal{Y}'}\big(-r\widehat{K}_{\mathcal{Y}'/\mathcal{Y}} + rK_{\mathcal{Y}'/\mathcal{Y}}\big) \quad \text{and} \quad n_{\mathcal{Y}} = q \operatorname{mult}_{E_{\mathcal{Y}}} \mathfrak{n}_{r,\mathcal{Y}}$$

[dFEI08, Proposition 1.7], [EM09, Section 7], we obtain

$$\operatorname{codim} \mathcal{T} - \frac{n_{\mathcal{Y}}}{r} \geq q a(E_{\mathcal{Y}}, \mathcal{Y}, 0). \tag{F.83}$$

By Lemma 2.15(3), there exists a prime divisor E over $\widetilde{W}_0^{\mathrm{alg}}$ whose valuation restricts to a positive integral multiple of the valuation of $E_{\mathcal{Y}}$, and the center of E is 0. Since ϖ_0 is étale in codimension 1, Lemma 2.15(1)(2)(4) with the boundary 0 imply that

$$a(E, \widetilde{W}_0^{\mathrm{alg}}, 0) = e_E a(E_{\mathcal{Y}}, \mathcal{Y}, 0) \quad \text{and} \quad \operatorname{mult}_E(\varpi_0^*\varphi) = e_E \operatorname{mult}_{E_{\mathcal{Y}}} \varphi \ \text{ for any } \varphi \in \mathcal{O}_{\mathcal{Y},o_{\mathcal{Y}}}.$$

For a non-zero G-semi-invariant polynomial g, $g^I = \varpi_0^*\varphi$ for some $\varphi \in \mathcal{O}_{\mathcal{Y},o_{\mathcal{Y}}}$, so

$$\operatorname{mult}_E g = \frac{e_E}{q} \operatorname{mult}_\gamma g$$

for the generic arc γ of $\mathcal{L}_1^\circ$. For $g = u, v_j, h_0^{<N_2}$, where the order of $h_0^{<N_2}$ along the arcs of $\mathcal{C}$ is at least $\min\{\ell_h/I, N_2(\vartheta_u + \ell_0)\} = \ell_h/I$, we obtain

$$\operatorname{mult}_E u = \frac{e_E}{q}(\vartheta_u + \ell_0), \qquad \operatorname{mult}_E v_j \geq \frac{e_E}{q} \cdot \frac{\ell_j}{I}, \quad \text{and} \quad \operatorname{mult}_E h_0^{<N_2} \geq \frac{e_E}{q} \cdot \frac{\ell_h}{I}.$$

By Lemma 2.12(3), we regard E as a prime divisor over $\widetilde{W}_0$ with center the closed point. Since $h_0 - h_0^{<N_2}$ is divisible by u^{N_2},

$$\operatorname{mult}_E\big(h_0 - h_0^{<N_2}\big) \geq N_2 \operatorname{mult}_E u = N_2 \frac{e_E}{q}(\vartheta_u + \ell_0) > \frac{e_E}{q} \cdot \frac{\ell_h}{I},$$

so $\operatorname{mult}_E h_0 \geq \frac{e_E}{q} \cdot \frac{\ell_h}{I}$. Since $\operatorname{mult}_E u$ is given by an equality, for β of either sign

$$\frac{q}{e_E} a(E, \widetilde{W}_0, \widetilde{\Delta}_0) = q a(E_{\mathcal{Y}}, \mathcal{Y}, 0) - \frac{q}{e_E}\Big(\frac{1}{m} \operatorname{mult}_E h_0 + \beta \operatorname{mult}_E u + \sum_{j=1}^{4} \operatorname{mult}_E v_j\Big)$$

$$\leq q a(E_{\mathcal{Y}}, \mathcal{Y}, 0) - \beta(\vartheta_u + \ell_0) - \frac{\ell_h}{mI} - \sum_{j=1}^{4} \frac{\ell_j}{I}.$$

Since $0 < b_0 \leq a(E, \widetilde{W}_0, \widetilde{\Delta}_0)/e_E$ and $q \geq 1$, we have $b_0 \leq q a(E, \widetilde{W}_0, \widetilde{\Delta}_0)/e_E$, so (F.82) and (F.83) imply

$$b_0 \leq \operatorname{codim}\big(\mathcal{C}_2 \cap L_\varrho(\widetilde{W}_0), L_\varrho(\widetilde{W}_0)\big) + \operatorname{age}(\varrho) - w_\varrho(f_0) - n_J$$

$$- \beta(\vartheta_u + \ell_0) - \frac{\ell_h}{mI} - \sum_{j=1}^{4} \frac{\ell_j}{I}.$$

By (F.81) and $w_\varrho(f_0) + \ell'_f \geq \ell_f/I$, the right side is at most the left side of (F.79). This proves (F.79), hence (F.77) and (F.76).

Step 8. Let $\mathbb{K}$ be an algebraic closure of $\Bbbk((u))$, $V_Q := (Q_f = 0) \subset \mathbb{A}^3_{\mathbb{K}}$, and $D_Q := \frac{1}{m} \operatorname{div}(Q_h|_{V_Q})$. In this step we prove that (V_Q, D_Q) has at most one non-klt point, and that, if it has one, (F.84)

holds for a triangular automorphism U commuting with G. By (F.72), $Q_f = f_0 + \Xi$, where any monomial of Ξ has w-weight less than d.

If $f_0(x,y,z) = x^2+y^3+z^3$, a constant linear change of (X_2, X_3), followed by a scaling of X_1 and of the equation, transforms f_0 into $X_1^2 + X_3X_2^2 + X_3^3$, which is the case $k = 4$ of Proposition 7.7, with the weights $(3,2,2)$. The change preserves the w-weights and the class of automorphisms $X_1 \to X_1 + a_1X_2 + a_2X_3 + a_3$, $X_2 \to X_2 + a_4$, $X_3 \to X_3 + a_5$. In this case Proposition 7.7 is used below for $X_1^2 + X_3X_2^2 + X_3^3$, and the automorphism of Proposition 7.7 is carried back by the inverse change.

By the first paragraph of Proposition 7.7, with X_1, X_2, X_3 in place of x, y, z, V_Q is normal with at worst Du Val singularities. We show that any prime divisor Γ on V_Q satisfies $\operatorname{mult}_\Gamma Q_h < m$. Let

$$Q_f^t := t^d Q_f(t^{-w_x}X_1, t^{-w_y}X_2, t^{-w_z}X_3), \qquad Q_h^t := t^m Q_h(t^{-w_x}X_1, t^{-w_y}X_2, t^{-w_z}X_3),$$

which belong to $\mathbb{K}[X_1, X_2, X_3, t]$ since Q_f and Q_h have maximal weights d and m, with $Q_f^0 = f_0$ and $Q_h^0 = h_0$. Let $\mathcal{V} := (Q_f^t = 0) \subset \mathbb{A}^3_{\mathbb{K}} \times \mathbb{A}^1_t$ with fibers $\mathcal{V}_t$ over $t \in \mathbb{A}^1$. For $t \neq 0$ let $t^w : \mathbb{A}^3_{\mathbb{K}} \to \mathbb{A}^3_{\mathbb{K}}$ be the automorphism $(X_1, X_2, X_3) \to (t^{w_x}X_1, t^{w_y}X_2, t^{w_z}X_3)$. Then $\mathcal{V}_t = t^w(V_Q)$ for $t \neq 0$, $\mathcal{V}_0 = (f_0 = 0)$, all fibers are normal, and $\mathcal{V}$ is normal. Since $h_0|_{\mathcal{V}_0} \neq 0$, the Cartier divisor $\mathcal{D} := \operatorname{div}(Q_h^t|_{\mathcal{V}})$ contains no component of $\mathcal{V}_0$, and the intersection cycle of $\mathcal{D}$ with $\mathcal{V}_0$ is $\operatorname{div}(h_0|_{\mathcal{V}_0})$. Assume $\operatorname{mult}_\Gamma Q_h \geq m$, and let $\widetilde{\Gamma}$ be the closure of $\{(t^w(p), t) \mid p \in \Gamma,\ t \neq 0\}$. Since $Q_h^t(t^w(p)) = t^m Q_h(p)$, $\mathcal{D} - m\widetilde{\Gamma} \geq 0$. Since $t^w(p) \to 0$ as $t \to 0$, the irreducible surface $\widetilde{\Gamma}$ intersects $\mathcal{V}_0$ and is not contained in $\mathcal{V}_0$, so the intersection cycle Γ_0 of $\widetilde{\Gamma}$ with $\mathcal{V}_0$ is a non-zero effective 1-cycle. Thus $\operatorname{div}(h_0|_{\mathcal{V}_0}) \geq m\Gamma_0$, and some component of $\frac{1}{m}\operatorname{div}(h_0|_{\mathcal{V}_0})$ has coefficient at least 1, contradicting hypothesis (3). Hence any component of D_Q has coefficient less than 1, and the non-klt locus of (V_Q, D_Q) is a finite set of points.

Let $\overline{V}_Q$ be the closure of V_Q in $\mathbb{P}_{\mathbb{K}}(w_x, w_y, w_z, 1)$, where the last coordinate x_0 has weight 1, let $\Delta_\infty := (x_0 = 0)|_{\overline{V}_Q}$, and let

$$\overline{Q_h} := x_0^m Q_h(x_0^{-w_x}X_1, x_0^{-w_y}X_2, x_0^{-w_z}X_3).$$

Then

$$\Delta_\infty = (f_0 = 0) \subset \mathbb{P}_{\mathbb{K}}(w_x, w_y, w_z)$$

is a reduced irreducible curve, and $\overline{Q_h}|_{\Delta_\infty} = h_0|_{\Delta_\infty} \neq 0$. The surface $\overline{V}_Q$ is normal, since V_Q is normal and Δ_∞ is reduced, and well formed, since the singular locus of $\mathbb{P}_{\mathbb{K}}(w_x, w_y, w_z, 1)$ is contained in the union of the lines $(x_0 = X_j = 0)$ for $1 \leq j \leq 3$, and f_0 vanishes identically on none of them. Since $\overline{V}_Q$ is a normal well-formed hypersurface of degree d in $\mathbb{P}_{\mathbb{K}}(w_x, w_y, w_z, 1)$ and $w_x + w_y + w_z + 1 = d + 2$, $K_{\overline{V}_Q} \sim_{\mathbb{Q}} \mathcal{O}(-2)|_{\overline{V}_Q}$. Since

$$\Delta_\infty \in |\mathcal{O}(1)|_{\overline{V}_Q}| \quad \text{and} \quad \operatorname{div}(\overline{Q_h}) \in |\mathcal{O}(m)|_{\overline{V}_Q}|,$$

the boundary

$$\overline{D}_Q := \Delta_\infty + \frac{1}{m}\operatorname{div}(\overline{Q_h}) \quad \text{satisfies} \quad K_{\overline{V}_Q} + \overline{D}_Q \sim_{\mathbb{Q}} 0.$$

Assume that (V_Q, D_Q) is not klt. Then the non-klt locus of $(\overline{V}_Q, \overline{D}_Q)$ is the disjoint union of Δ_∞ and the non-klt points of (V_Q, D_Q), and is disconnected. By Theorem 7.2 over the base $\operatorname{Spec}\mathbb{K}$, the non-klt locus has exactly two connected components, and $(\overline{V}_Q, \overline{D}_Q)$ is lc. Thus (V_Q, D_Q) is lc and has exactly one non-klt point o.

Since Q_f is monic quadratic in X_1 and the monomials of Ξ have weight less than d, division by Q_f in X_1 does not increase weights. Since h_0 has X_1-degree at most 1, the division yields

$$Q_h = Q_f\Phi + R$$

with $\Phi, R \in \Bbbk[[u]][X_1, X_2, X_3]$ G-semi-invariant, R of X_1-degree at most 1, and $R - h_0$ of weight less than m. Then $Q_f\Phi = (Q_h - h_0) - (R - h_0)$ has weight less than m, so Φ has maximal weight less than $m - d$, and $\Phi = 0$ if $m \leq d$. By Proposition 7.7(1) and (5) at o, with X_1, X_2, X_3 in

place of x, y, z, $F = Q_f$, $N = 1$, $Q_1 = R$, $\kappa_1 = \frac{1}{m}$, $d_1 = m$, and $P_1 = h_0$, there exists a unique triangular automorphism $U = (U_x, U_y, U_z)$,

$$U_x = X_1 + \phi_x(X_2, X_3), \quad U_y = X_2 + \phi_y(X_3), \quad \text{and} \quad U_z = X_3 + \phi_z,$$

with ϕ_x, ϕ_y, ϕ_z over $\Bbbk[[u]]$ of w-weights less than w_x, w_y, w_z, respectively, such that

$$Q_f = f_0(U), \qquad Q_h = h_0(U) + Q_f\Phi \tag{F.84}$$

in $\Bbbk[[u]][X_1, X_2, X_3]$. The inverse of U has coefficients in $\Bbbk[[u]]$, and $o = (U_x = U_y = U_z = 0)$ is the non-klt locus of (V_Q, D_Q) by Proposition 7.7(1) and (4). If f_0 is one of the polynomials of Proposition 7.7, then, since the characters of Q_f and f_0 are equal, U commutes with G by Proposition 7.7(3), applied to the group generated by extensions to $\mathbb{K}[X_1, X_2, X_3]$ of the elements of G. If $f_0 = x^2 + y^3 + z^3$, for $\varrho \in G$ the triple

$$U^\varrho := \big(\chi_x(\varrho)^{-1}\varrho^* U_x, \chi_y(\varrho)^{-1}\varrho^* U_y, \chi_z(\varrho)^{-1}\varrho^* U_z\big)$$

belongs to the class of automorphisms above and satisfies the first identity of (F.84), as Q_f and f_0 have the character χ_f, so $U^\varrho = U$ by the uniqueness in Proposition 7.7(1), that is, U commutes with G.

Step 9. In this step we prove that, possibly passing to an infinite subfamily, $\operatorname{mult}_{F_i} h_i^{\mathrm{res}}$ is bounded. Let

$$L_1 := ML_0 + \max\{\beta, 0\}R'.$$

Since $e_i \le |G| \le M$, $b \le L_0$ by hypothesis (1), and $\operatorname{mult}_{F_i} u \le R'$, (F.75) implies

$$a(F_i, W_i^{\mathrm{res}}, 0) - \frac{1}{m}\operatorname{mult}_{F_i} h_i^{\mathrm{res}} = be_i + \beta \operatorname{mult}_{F_i} u \le L_1.$$

Thus F_i satisfies the hypotheses of (F.69) with $\widetilde{f}_i := f_i^{\mathrm{res}}$, $\widetilde{h}_i := h_i^{\mathrm{res}}$, $(\widetilde{f}_*, \widetilde{h}_*) := (Q_f, Q_h)$, $\Bbbk_2 := \Bbbk$, $F := F_i$, and L_1: $\operatorname{mult}_{F_i} u \le R'$, the last display, and, by (F.66),

$$\operatorname{mult}_{F_i} \operatorname{Jac}_{W_i^{\mathrm{res}}} \le \lambda a(F_i, W_i^{\mathrm{res}}, 0).$$

Assume that $\operatorname{mult}_{F_i} h_i^{\mathrm{res}}$ is unbounded on any infinite subfamily. Possibly passing to a subsequence, $\operatorname{mult}_{F_i} h_i^{\mathrm{res}} \to +\infty$. Let $W_Q := \operatorname{Spec} \Bbbk[[X_1, X_2, X_3, u]]/(Q_f)$. Then W_Q is normal, since $(Q_f(X_1, X_2, X_3, 0) = 0)$ is normal by the first paragraph of Proposition 7.7. Since $\operatorname{mult}_{F_i} h_i^{\mathrm{res}} \to +\infty$, the bound (F.69) fails for $i \gg 0$, so $(W_Q, \frac{1}{m}\operatorname{div}(Q_h))$ is not klt on $D(u)$. Since $Q_f, Q_h \in \Bbbk[[u]][X_1, X_2, X_3]$, the points of $D(u)$ map to the generic point of $\operatorname{Spec} \Bbbk[[u]]$, and the klt property is preserved and reflected by the completion of $\operatorname{Spec} \Bbbk[[u]][X_1, X_2, X_3]/(Q_f)$ at the closed point and by the extension of the coefficient field from $\Bbbk((u))$ to $\mathbb{K}$, by [Kaw21, Remark 2.6]. Thus (V_Q, D_Q) is not klt, and (F.84) holds by **Step 8**. Let $\varsigma \in \Bbbk[[u]]^3$ be the point with $U(\varsigma) = 0$. Then the non-klt locus of $(W_Q, \frac{1}{m}\operatorname{div}(Q_h))$ on $D(u)$ is contained in the closure of ς. If $\varsigma(0) \ne 0$, some U_{x_j} is a unit of $\Bbbk[[X_1, X_2, X_3, u]]$, so $(U_x = U_y = U_z = 0)$ is empty and $(W_Q, \frac{1}{m}\operatorname{div}(Q_h))$ is klt on $D(u)$, a contradiction. Thus $\varsigma(0) = 0$, that is, U_x, U_y, U_z vanish at the closed point of $\operatorname{Spec} \Bbbk[[X_1, X_2, X_3, u]]$, and U induces an automorphism of $\Bbbk[[X_1, X_2, X_3, u]]$ fixing u with Jacobian determinant 1.

Fix a positive integer r divisible by $|G|$, a positive integer $\Lambda > r\max\{w_x, w_y, w_z\}R'$, and an integer $N_4 > \max\{r\max\{d, m\}, \Lambda\}$. Since U commutes with G and Φ is G-semi-invariant, any monomial of $U_{x_j} - X_j$ has the character of X_j, and any monomial of Φ has the character of Φ. We use (F.70) with Λ, with (U_x, U_y, U_z) in place of (U_1, U_2, U_3), and with the following choices in Lemma 6.1(1). The parameter vector of Lemma 6.1(1), here denoted $\mathbf{b}$, contains the coefficients of the monomials $u^k X^\alpha$ with $k < N_4$ of $U_{x_j} - X_j$ for $j \in \{1, 2, 3\}$ and of Φ, where X^α runs over the monomials allowed in $U_{x_j} - X_j$ by the form of U in (F.84), respectively over the monomials of w-weight less than $m - d$, with the character of X_j, respectively of Φ. The polynomials $U_{i,j} - X_j$ and Φ_i in X_1, X_2, X_3, u have the monomials $u^k X^\alpha$ above with the corresponding entries of $\mathbf{b}_i$ as coefficients, and all other coefficients zero, so that the $U_{i,j}$ differ from the polynomials of **Step 2** built from $\mathbf{b}_i$ by monomials of order $> \Lambda$ along F_i. Thus $U_i := (U_{i,1}, U_{i,2}, U_{i,3})$ is a

triangular polynomial automorphism over $\mathbb{C}[u]$ of the form of U commuting with G, and Φ_i is G-semi-invariant. The finite set $\mathcal{Q}$ in the last sentence of Lemma 6.1(1) contains the constant terms at $u=0$ of $U_{i,1}, U_{i,2}, U_{i,3}$, which vanish for the limit since $\varsigma(0)=0$, and the polynomials of the conditions (†) below on the (N_4-1)-jets of $(f_i^{\mathrm{res}}, h_i^{\mathrm{res}})$ and on the coefficients of U_i and Φ_i. Thus each condition in $\mathcal{Q}$ holds at the indices i if it holds for the limit.

Since $\operatorname{mult}_{F_i} h_i^{\mathrm{res}} \to +\infty$, the premise of (F.70) holds for $i \gg 0$. Since $(U_x = U_y = U_z = 0)$ contains the non-klt locus on $D(u)$, (F.70) implies, possibly passing to an infinite subfamily,

$$\operatorname{mult}_{F_i} U_{i,j} > \Lambda \quad \text{for } j \in \{1,2,3\}. \tag{F.85}$$

For an integer n and a triple $T=(T_1,T_2,T_3)$ we write

$$u^{nw}T := (u^{nw_1}T_1, u^{nw_2}T_2, u^{nw_3}T_3).$$

Let U_i^{-1} be the inverse triangular polynomial automorphism, and let

$$f_i^+ := u^{-rd} f_i^{\mathrm{res}}\big(U_i^{-1}(u^{rw}X^+), u\big),$$

$$h_i^+ := u^{-rm}\big(h_i^{\mathrm{res}} - f_i^{\mathrm{res}}\Phi_i\big)\big(U_i^{-1}(u^{rw}X^+), u\big),$$

with coordinates $X^+=(X_1^+,X_2^+,X_3^+)$. We show that $f_i^+, h_i^+ \in \mathbb{C}[[X_1^+,X_2^+,X_3^+,u]]$. Since the constant terms of $U_{i,1}, U_{i,2}, U_{i,3}$ vanish at $u=0$, any coordinate of $U_i^{-1}(u^{rw}X^+)$ is divisible by u. Thus a monomial of degree at least N_4 in X_1, X_2, X_3, u contributes only terms divisible by u^{N_4}, and the divisibility of $f_i^{\mathrm{res}}(U_i^{-1}(u^{rw}X^+),u)$ by u^{rd} and of $(h_i^{\mathrm{res}} - f_i^{\mathrm{res}}\Phi_i)(U_i^{-1}(u^{rw}X^+),u)$ by u^{rm} are finitely many polynomial conditions (†) on the coefficients of degree less than N_4. For the limit, the truncation of U^{-1} modulo u^{N_4} is the inverse of the truncation of U modulo u^{N_4}, and (F.84) implies

$$Q_f\big(U^{-1}(u^{rw}X^+)\big) = u^{rd} f_0(X^+), \qquad (Q_h - Q_f\Phi)\big(U^{-1}(u^{rw}X^+)\big) = u^{rm} h_0(X^+).$$

Thus the conditions (†) hold for the limit, since $N_4 > r\max\{d,m\}$, hence at the indices i. By (F.85), for $j \in \{1,2,3\}$, the rational function $X_j^+ = u^{-rw_j}U_{i,j}$ on W_i^{res} satisfies

$$\operatorname{mult}_{F_i} X_j^+ = \operatorname{mult}_{F_i} U_{i,j} - rw_j \operatorname{mult}_{F_i} u > \Lambda - rw_j R' > 0.$$

Let x' be the coordinates of g_i, so that $f_i^{\mathrm{res}} = u^{-n_i d} f_i$ and $h_i^{\mathrm{res}} = u^{-n_i m}(h_i + \xi_i f_i)$ under $x' = u^{n_i w}X$. Let g_i^+ be the triangular automorphism with coordinates

$$x_j^+ := u^{n_i w_j} U_{i,j}(u^{-n_i w}x', u) \quad \text{for } j \in \{1,2,3\}, \qquad \text{and let} \qquad \xi_i^+ := \xi_i - u^{n_i(m-d)}\Phi_i(u^{-n_i w}x', u).$$

Under $x^+ = u^{(n_i+r)w}X^+$ we have $X = U_i^{-1}(u^{rw}X^+)$ and $h_i^{\mathrm{res}} - f_i^{\mathrm{res}}\Phi_i = u^{-n_i m}(h_i + \xi_i^+ f_i)$, so $f_i^+ = u^{-(n_i+r)d} f_i$ and $h_i^+ = u^{-(n_i+r)m}(h_i + \xi_i^+ f_i)$, that is, $f_i^+ = f_i^{g_i^+, n_i+r}$ and $h_i^+ = h_i^{g_i^+, \xi_i^+, n_i+r}$. A monomial X^α of $U_{i,j} - X_j$, of weight $\omega < w_j$, gives the term $u^{n_i(w_j-\omega)}x'^\alpha$ of $x_j^+ - x_j'$ with the coefficient of X^α, an element of $\mathbb{C}[u]$, and a monomial X^α of Φ_i, of weight $\omega < m-d$, gives the term $u^{n_i(m-d-\omega)}x'^\alpha$ of $\xi_i - \xi_i^+$ with the coefficient of X^α. Thus the corrections added to g_i and ξ_i are polynomials in x', hence in x, with coefficients in $u^{n_i}\mathbb{C}[u]$, which is contained in $u^{n_i'}\mathbb{C}[u]$ for $i \gg 0$ since $n_i \geq n_i^0$ and $n_i'/n_i^0 \to 0$. A composite of triangular automorphisms with corrections of lower weights is again a triangular automorphism with corrections of lower weights, and g_i^+, ξ_i^+ have the characters of g_i, ξ_i, since U_i commutes with G, Φ_i is G-semi-invariant, and n_i and r are divisible by $|G|$. Since $f_i^+, h_i^+ \in \mathbb{C}[[X_1^+,X_2^+,X_3^+,u]]$ and, for $j \in \{1,2,3\}$,

$$\operatorname{mult}_{F_i} x_j^+ = (n_i+r)w_j \operatorname{mult}_{F_i} u + \operatorname{mult}_{F_i} X_j^+ > (n_i+r)w_j \operatorname{mult}_{F_i} u,$$

the triple (g_i^+, ξ_i^+, n_i+r) is admissible. This contradicts the maximality of n_i.

Step 10. In this step we prove $b = b_0$. By **Step 9**, possibly passing to an infinite subfamily, $\operatorname{mult}_{F_i} h_i^{\mathrm{res}} \leq \Lambda_0$ for a real number Λ_0. By (F.75),

$$a(F_i, W_i^{\mathrm{res}}, 0) = be_i + \frac{1}{m}\operatorname{mult}_{F_i} h_i^{\mathrm{res}} + \beta \operatorname{mult}_{F_i} u \leq L_3 := ML_0 + \frac{\Lambda_0}{m} + |\beta|R',$$

and, by (F.66),

$$\operatorname{mult}_{F_i} \operatorname{Jac}_{W_i^{\mathrm{res}}} \le \lambda L_3.$$

Let N_5 be an integer with $N_5 > 2\lambda L_3 + \Lambda_0 + R'$. The completions at 0 of the ambient germ $\mathcal{X}_i^{g_i,n_i}$ of W_i^{res} and of $\mathbb{A}^4$ with coordinates X_1, X_2, X_3, u are $\operatorname{Spec} \mathbb{C}[[X_1, X_2, X_3, u]]$, with the diagonal action of G. Let k_i be the integer in $[N_5, N_5 + e_i)$ in the residue class modulo e_i of Lemma 5.4(2) on $\operatorname{Spec} \mathbb{C}[[X_1, X_2, X_3, u]]$ for a formal prime divisor restricting to F_i (Lemma 2.12(3)). By Lemma 5.4(2) there, with $H_1 = h_i^{\mathrm{res}}$ and $H_2 = u$, and by Lemma 2.12(3) for $\mathbb{A}^4$, there exists a prime divisor $\widehat{F}_i$ over $\mathbb{A}^4$ with center 0 such that

$$\operatorname{mult}_{\widehat{F}_i} f_i^{\mathrm{res}} = k_i, \quad \operatorname{mult}_{\widehat{F}_i} h_i^{\mathrm{res}} = \operatorname{mult}_{F_i} h_i^{\mathrm{res}}, \quad \operatorname{mult}_{\widehat{F}_i} u = \operatorname{mult}_{F_i} u,$$
$$a(\widehat{F}_i, \mathbb{A}^4, 0) = a(F_i, W_i^{\mathrm{res}}, 0) + k_i, \quad \text{and} \quad I_G(\widehat{F}_i) = I_G(F_i).$$

Since $I_G(\widehat{F}_i) = I_G(F_i)$, we have $e_{\widehat{F}_i} = e_i$, so (F.75) implies

$$\frac{a(\widehat{F}_i, \mathbb{A}^4, 0) - \operatorname{mult}_{\widehat{F}_i} f_i^{\mathrm{res}} - \frac{1}{m} \operatorname{mult}_{\widehat{F}_i} h_i^{\mathrm{res}} - \beta \operatorname{mult}_{\widehat{F}_i} u}{e_{\widehat{F}_i}} = b,$$

and $\operatorname{mult}_{\widehat{F}_i} u \le R'$, $\operatorname{mult}_{\widehat{F}_i} f_i^{\mathrm{res}} < N_5 + M$, $\operatorname{mult}_{\widehat{F}_i} h_i^{\mathrm{res}} \le \Lambda_0$, and $a(\widehat{F}_i, \mathbb{A}^4, 0) < L_3 + N_5 + M$. By Lemma 7.10(2) with the compatible generic limit (Q_f, Q_h) of $(f_i^{\mathrm{res}}, h_i^{\mathrm{res}})$,

$$\operatorname{mld}\Big(\Big(\widehat{\mathcal{X}} \ni 0, \beta \operatorname{div}(u) + \operatorname{div}(Q_f) + \frac{1}{m} \operatorname{div}(Q_h)\Big)/G\Big) \le b.$$

By (F.76), the left side is at least b_0. Thus $b_0 \le b$, and, by (F.74),

$$b = b_0. \tag{F.86}$$

Step 11. In this step we find computing lifts of bounded log discrepancy through a chart of fixed depth. Since W_0 is a hypersurface, $a(E, \widehat{W}_0, 0) \in \mathbb{Z}$ for any prime divisor E over $\widehat{W}_0$. Since $\beta \in \frac{1}{m}\mathbb{Z}$ and $e_E \le |G|$, the numbers

$$\frac{a(E, \widehat{W}_0, 0) - \frac{1}{m} \operatorname{mult}_E h_0 - \beta \operatorname{mult}_E u}{e_E}$$

belong to the discrete set $\bigcup_{e=1}^{|G|} \frac{1}{me}\mathbb{Z}$. Thus their infimum $b_0 > 0$ over the prime divisors E over $\widehat{W}_0$ with center the closed point is attained by such a prime divisor E_0. Let $j_0 := \operatorname{mult}_{E_0} \operatorname{Jac}_{W_0}$, and fix a positive integer $\overline{n}$ divisible by $|G|$ with

$$\overline{n} > 2j_0 + \operatorname{mult}_{E_0} h_0 + \operatorname{mult}_{E_0} u + 1.$$

As in **Step 6** with E_0 in place of E, for the polynomials $h_{0,i}$ of (F.73), which have the compatible generic limit h_0, possibly passing to an infinite subfamily, there exist prime divisors E_i' over $\widehat{W}_{0,\mathbb{C}}$ with center the closed point such that

$$a(E_i', \widehat{W}_{0,\mathbb{C}}, 0) = a(E_0, \widehat{W}_0, 0), \quad \operatorname{mult}_{E_i'} h_{0,i} = \operatorname{mult}_{E_0} h_0,$$
$$\operatorname{mult}_{E_i'} u = \operatorname{mult}_{E_0} u, \quad \text{and} \quad \operatorname{mult}_{E_i'} \operatorname{Jac}_{W_0} = j_0.$$

The decomposition and inertia groups of E_i' coincide with those of E_0, as in **Step 6** (with T dominating $Z_{d'}$, its function field containing a field of definition of E_0, and G acting on $W_0 \times T$ through W_0).

Take i with $n_i \ge \overline{n}$ for which (F.73) holds, so that $f_i^{(\overline{n})} - f_0$ and $h_i^{(\overline{n})} - h_{0,i}$ belong to $(u^{\overline{n}})$. Since $\operatorname{mult}_{E_i'} u = \operatorname{mult}_{E_0} u \ge 1$, the order along E_i' of $(f_i^{(\overline{n})} - f_0)|_{W_0}$ is at least

$$\overline{n} \operatorname{mult}_{E_0} u > \max\{2j_0,\ j_0 + \operatorname{mult}_{E_0} h_0,\ j_0 + \operatorname{mult}_{E_0} u\},$$

the four partial derivatives of $f_i^{(\overline{n})} - f_0$ belong to $(u^{\overline{n}-1})$, so they have orders at least $\overline{n} - 1 > j_0$ along E_i', and $h_i^{(\overline{n})} - h_{0,i}$ has order at least $\overline{n} > \operatorname{mult}_{E_0} h_0$ along E_i', so $\operatorname{mult}_{E_i'} h_i^{(\overline{n})} = \operatorname{mult}_{E_0} h_0$.

By Lemma 5.4(3) with $f := f_0$, $f' := f_i^{(\overline{n})}$, $F := E_i'$, $H_1 := h_i^{(\overline{n})}$, and $H_2 := u$, there exists a prime divisor E_i over $W_i^{(\overline{n})}$ with center the closed point (over the completion, and by [Liu26a, Lemma 3.3(3)] it restricts to a prime divisor over $W_i^{(\overline{n})}$ with center 0 with equal invariants) such that

$$a(E_i, W_i^{(\overline{n})}, 0) = a(E_0, \widehat{W}_0, 0), \quad I_G(E_i) = I_G(E_0),$$
$$\operatorname{mult}_{E_i} h_i^{(\overline{n})} = \operatorname{mult}_{E_0} h_0, \quad \text{and} \quad \operatorname{mult}_{E_i} u = \operatorname{mult}_{E_0} u.$$

In particular $e_{E_i} = e_{E_0}$, and E_i has center 0 on W_i, since the chart of depth $\overline{n}$ maps its closed point to 0. By (F.71) for the preparation (g_i, ξ_i) and $n = \overline{n}$,

$$a(E_i, W_i, B_i) = a(E_i, W_i^{(\overline{n})}, B_i^{(\overline{n})}) \quad \text{and} \quad a(E_i, W_i, 0) = a(E_i, W_i^{(\overline{n})}, 0) + \overline{n} \operatorname{mult}_{E_i} u.$$

By (F.86),

$$\frac{a(E_i, W_i, B_i)}{e_{E_i}} = \frac{a(E_0, \widehat{W}_0, 0) - \frac{1}{m}\operatorname{mult}_{E_0} h_0 - \beta \operatorname{mult}_{E_0} u}{e_{E_0}} = b_0 = b = \operatorname{mld}((W_i \ni 0, B_i)/G),$$

so E_i is a computing lift of $\operatorname{mld}((W_i \ni 0, B_i)/G)$ with

$$a(E_i, W_i, 0) = a(E_0, \widehat{W}_0, 0) + \overline{n} \operatorname{mult}_{E_0} u$$

independent of i. This proves the bound $a(E_i, W_i, 0) \le L$ of Theorem 17.6 with

$$L := a(E_0, \widehat{W}_0, 0) + \overline{n} \operatorname{mult}_{E_0} u.$$

Step 12. In this step we prove that no radial failure sequence as in the last sentence of Theorem 17.6 exists. Assume that a radial failure sequence $(W_i \ni o_i, G_i, h_i, \tau_i)$ with constants (m, a, δ) satisfies Set-up 17.1, its prepared limit is of type cE, and its generic transverse type along C is E_6. By Proposition 17.5, there exist integers $\nu \in [0, 5]$ and c and an infinite subfamily with germs $(Y_i \ni p_i)$ in coordinates (x', y', z', u), groups Γ_i and regular functions l_i satisfying Proposition 17.5(1)–(6), where $d_\nu \in \{1, 2\}$, $\beta = c/m - (d_\nu - 1)$, $S_i = (u = 0)$, $B_i = \frac{1}{m}\operatorname{div}(l_i) + \beta S_i$, and κ is the positive integer of Proposition 17.5(3). Since there exist only finitely many diagonal groups of order at most $2M$ and finitely many characters of each, possibly passing to a subsequence, $\Gamma_i = \Gamma$ and the characters of the equations and of the marks are fixed. The germs $(Y_i \ni p_i)$ with B_i and Γ satisfy the hypotheses of Theorem 17.6 with $2M$ in place of M, $f_0 = x'^2 + y'^3 + z'^4$, $w = (6, 4, 3)$, $d = 12$, and $L_0 := a$.

- By Proposition 17.5(1), (2), and (6), $(Y_i \ni p_i)$ is an isolated Gorenstein terminal, hence normal klt Gorenstein, hypersurface germ in a smooth fourfold germ with regular Γ-eigenparameters (x', y', z', u), defined by the regular function f_i', and the mark l_i is the restriction of the regular Γ-semi-invariant function l_i'. By Proposition 17.5(5) and (6), the compatible generic limit (f_*', l_*) of the pairs (f_i', l_i') is also that of the pairs formed by the f_i' and the representatives of the l_i modulo the f_i' of x'-degree at most 1. Moreover, $|\Gamma| = d_\nu |G_i| \le 2M$.
- $\beta \in \frac{1}{m}\mathbb{Z}$, and $\beta < 1$ since $c < d_\nu m$.
- By Proposition 17.5(2) and Set-up 17.1, $\operatorname{mld}((Y_i \ni p_i, B_i)/\Gamma) = b_i \in (0, a - \delta]$, which is hypothesis (1).
- By Proposition 17.5(4), there exists a positive integer R' such that any computing lift P of $\operatorname{mld}((Y_i \ni p_i, B_i)/\Gamma)$ satisfies $\operatorname{mult}_P u \le R'$, and
$$\min\{\operatorname{mult}_P x', \operatorname{mult}_P y', \operatorname{mult}_P z'\} \to +\infty$$
for such P, which is hypothesis (2).
- By Proposition 17.5(1) and (5) and the first item, the equations of the Y_i and the l_i have the compatible generic limit (f_*', l_*) with coefficients in $\Bbbk''[[u]]$, where $f_* - f_0$ consists of terms of transverse w-weight greater than 12, $\operatorname{in}_w(l_*)$ has weight m, is non-zero and has x'-degree at most 1, and $(f_0 = 0, \frac{1}{m}\operatorname{div}(\operatorname{in}_w(l_*)))$ over an algebraic closure of $\Bbbk''((u))$ is lc

and has no coefficient-one component. This is hypothesis (3). As the weight-12 part of the limit of Γ-semi-invariant equations with a fixed character, f_0 has the character of the equations, which is the condition on characters in Theorem 17.6.

By the bound $a(E_i, W_i, 0) \le L$ proved above, applied to the $(Y_i \ni p_i, B_i)$ and Γ, possibly passing to an infinite subfamily, there exist a real number L and computing lifts P_i of $\mathrm{mld}((Y_i \ni p_i, B_i)/\Gamma)$ with $a(P_i, Y_i, 0) \le L$. By Proposition 17.5(2) and (3), P_i restricts to a computing lift F_i' of b_i, a prime divisor over W_i with center o_i, and, with ρ_i the ramification index of P_i over F_i',

$$e_{P_i} = \rho_i e_{F_i'}, \qquad \rho_i a(F_i', W_i, 0) = a(P_i, Y_i, 0) + \kappa \operatorname{mult}_{P_i} u.$$

Thus

$$\frac{a(F_i', W_i, 0)}{e_{F_i'}} = \frac{a(P_i, Y_i, 0) + \kappa \operatorname{mult}_{P_i} u}{e_{P_i}} \le L + \kappa R'.$$

This contradicts Lemma 4.3(2). □

F.4. **Transverse types D.** In this subsection we prove Propositions 17.7 and 17.8, Theorems 17.9, 17.10, 17.11, and 17.12, Proposition 17.13, and Theorem 17.14 of Section 17.

Proof of Proposition 17.7. By Construction-Theorem 13.6, we may assume that $G_i = G$ for any i, where G is a finite cyclic group with $|G| \le M$ acting diagonally on $\mathcal{X} = \operatorname{Spec} \mathbb{C}[[t_1, \dots, t_4]]$ and on $\mathcal{X}_* = \operatorname{Spec} \Bbbk[[x_1, \dots, x_4]]$, that $W_i = (f_i = 0) \subset \mathcal{X}$, and that $W_* = (f_* = 0) \subset \mathcal{X}_*$, with t_j corresponding to x_j in the compatible generic limit. Here $(x_1, \dots, x_4)$ are the coordinates (x, y, z, t) of that theorem. Let $\mathfrak{m}$ be the maximal ideal of $\mathbb{C}[[t_1, \dots, t_4]]$. For a G-semi-invariant element u we write χ_u for the character of u. We divide the proof into five steps.

Step 1. In this step we prove the first sentence of the proposition.

By Lemma 17.3(3), there exist G-eigenparameters (x, y, z, s) of $\mathcal{X}_*$ with $C = (x = y = z = 0)$, an invariant unit v and some $\phi \in (y, z)^3 \Bbbk[[y, z, s]]$ such that $f_* = v \cdot (x^2 + \phi)$ and $\chi_x^2 = \chi_{f_*}$. Write

$$\phi = c_0(s) y^3 + c_1(s) y^2 z + c_2(s) y z^2 + c_3(s) z^3 + R$$

with $c_0, c_1, c_2, c_3 \in \Bbbk[[s]]$ and $R \in (y, z)^4 \Bbbk[[y, z, s]]$. By Construction-Theorem 13.6, W_* is cDV and the f_i have the form of Lemma 3.7(1) in $t_1, \dots, t_4$. Since this is a closed condition on the jets (Definition 2.5),

$$f_* \in x_1^2 + x_2^3 + x_2 (x_3, x_4)^3 \Bbbk[[x_3, x_4]] + (x_3, x_4)^4 \Bbbk[[x_3, x_4]].$$

Thus the cubic part of f_* modulo x_1 is x_2^3. For an equation whose quadratic part is a non-zero multiple of x^2, coordinate changes and multiplication by units change the cubic part modulo x only by a linear change of the other coordinates and a non-zero factor. Hence for general elements $\kappa_1, \kappa_2, \kappa_3$ of $\Bbbk$ the hyperplane section $(s = \kappa_1 x + \kappa_2 y + \kappa_3 z)$ of W_* is a Du Val surface germ whose cubic part modulo x is a cube, of type E_6, E_7, or E_8. Since $\phi \in (y, z)^3$, in the coordinates (x, y, z) of the section, the equation $v \cdot (x^2 + \phi(y, z, \kappa_1 x + \kappa_2 y + \kappa_3 z))$ has quadratic part $v(0) x^2$ and, modulo x, cubic part

$$v(0) \big(c_0(0) y^3 + c_1(0) y^2 z + c_2(0) y z^2 + c_3(0) z^3 \big).$$

The normal forms $x^2 + y^3 + z^4$, $x^2 + y^3 + y z^3$, and $x^2 + y^3 + z^5$ of E_6, E_7, and E_8 have cubic part y^3. Thus

$$c_0(0) y^3 + c_1(0) y^2 z + c_2(0) y z^2 + c_3(0) z^3 = \ell^3$$

for a non-zero linear form ℓ in y, z. Since ℓ^3 is G-semi-invariant, ℓ is a G-eigenvector. Possibly replacing y with ℓ and z with a G-eigenvector in $\Bbbk y + \Bbbk z$ not proportional to ℓ, and renaming $c_0, \dots, c_3, R$, we may assume that $c_0(0) \ne 0$ and $c_1(0) = c_2(0) = c_3(0) = 0$.

Since $c_0(0) \ne 0$ and any monomial of ϕ has character χ_x^2, we have $\chi_y^3 = \chi_x^2$ and c_0 is invariant. Possibly replacing x with $c_0^{1/2} x$ and v with $c_0 v$, and dividing c_1, c_2, c_3, R by c_0, we may assume that $c_0 = 1$. Any monomial of $c_1(s) y^2 z$ has character χ_y^3, so $c_1(s) z$ is a G-eigenvector of character

χ_y. Possibly replacing y with $y-\frac{1}{3}c_1(s)z$ and renaming, we may assume that $c_1=0$. Let $\alpha:=c_2$ and $\gamma:=c_3$, so that

$$\phi=y^3+\alpha(s)yz^2+\gamma(s)z^3+R,\qquad \alpha(0)=\gamma(0)=0.$$

Let Ω be an algebraic closure of $\Bbbk((s))$. The generic transverse type of W_* along C is the type of the surface germ $(x^2+y^3+\alpha yz^2+\gamma z^3+R=0)$ over Ω, which is D_k with $k\geq 5$. The normal form $x^2+y^2z+z^{k-1}$ of D_k has cubic part y^2z modulo x, so by the invariance above, $y^3+\alpha yz^2+\gamma z^3=\ell_1^2\ell_2$ for non-proportional linear forms ℓ_1,ℓ_2 in y,z over Ω. Since the coefficient of y^2z is 0, ℓ_1 and ℓ_2 are proportional to $y-\lambda z$ and $y+2\lambda z$ for some $\lambda\in\Omega$ with $\lambda\neq 0$. Thus

$$y^3+\alpha yz^2+\gamma z^3=(y-\lambda z)^2(y+2\lambda z)=y^3-3\lambda^2yz^2+2\lambda^3z^3,$$

so $\alpha=-3\lambda^2\neq 0$, $\gamma=2\lambda^3$, and $\lambda=-3\gamma/(2\alpha)\in\Bbbk((s))$. Since $\lambda^2=-\alpha/3\in s\Bbbk[[s]]$, we have $\lambda=s^\nu w(s)$ for an integer $\nu\geq 1$ and some $w\in\Bbbk[[s]]$ with $w(0)\neq 0$. The group G acts on $\Bbbk((s))$ through χ_s, and since αyz^2 and γz^3 have character χ_y^3, the series α and γ are eigenvectors of characters $\chi_y^2\chi_z^{-2}$ and $\chi_y^3\chi_z^{-3}$. Hence λ has character $\chi_y\chi_z^{-1}$. Since $w(0)\neq 0$, w is invariant and

$$\chi_y=\chi_s^\nu\chi_z. \tag{F.87}$$

Possibly replacing y with wy, x with $w^{3/2}x$, and v with w^3v, where $w^{3/2}\in\Bbbk[[s]]$ is an invariant unit, and renaming R, we may assume that

$$f_*=v\cdot\big(x^2+y^3-3s^{2\nu}yz^2+2s^{3\nu}z^3+R\big),\qquad R\in(y,z)^4\Bbbk[[y,z,s]]. \tag{F.88}$$

The coordinate changes in this step are G-equivariant, fix s and preserve the ideal (x,y,z) of C. Thus the first sentence of the proposition holds.

Step 2. In this step we study the chart of (1) on W_*.

Let $r:=2\nu+1$, and let $\varrho:\Bbbk[[x,y,z,s]]\to\Bbbk[[X,V,Z,s]]$ be the homomorphism of $\Bbbk[[s]]$-algebras with

$$\varrho(x)=s^{3r+2\nu}X,\qquad \varrho(y)=s^{2r+\nu}\Big(\frac{2}{3}V+s^\nu Z\Big),\quad\text{and}\quad \varrho(z)=s^{2r}\Big(-\frac{1}{3}V+s^\nu Z\Big). \tag{F.89}$$

In $\Bbbk[[X,V,Z,s]][s^{-1}]$ we have

$$X=\frac{\varrho(x)}{s^{3r+2\nu}},\qquad V=\frac{\varrho(y)}{s^{2r+\nu}}-\frac{\varrho(z)}{s^{2r}},\quad\text{and}\quad Z=\frac{1}{3s^\nu}\Big(\frac{\varrho(y)}{s^{2r+\nu}}+\frac{2\varrho(z)}{s^{2r}}\Big). \tag{F.90}$$

By (F.87), $\chi_y\chi_s^{-2r-\nu}=\chi_z\chi_s^{-2r}$. Hence, letting G act diagonally on X,V,Z,s with

$$\chi_X=\chi_x\chi_s^{-3r-2\nu},\qquad \chi_V=\chi_z\chi_s^{-2r},\quad\text{and}\quad \chi_Z=\chi_z\chi_s^{-2r-\nu},$$

ϱ is G-equivariant. Since (F.89) is a linear substitution of determinant $s^{7r+4\nu}\neq 0$, ϱ is injective.

Since $\varrho(y-s^\nu z)=s^{2r+\nu}V$ and $\varrho(y+2s^\nu z)=3s^{2r+2\nu}Z$, we have

$$\varrho\big(y^3-3s^{2\nu}yz^2+2s^{3\nu}z^3\big)=\varrho\big((y-s^\nu z)^2(y+2s^\nu z)\big)=3s^{6r+4\nu}V^2Z.$$

Moreover $\varrho(x^2)=s^{6r+4\nu}X^2$, and $\varrho(R)\in s^{8r}(V,Z)^4\Bbbk[[V,Z,s]]$ since $R\in(y,z)^4$. Since $r=2\nu+1$, we have $8r-6r-4\nu=2$, and (F.88) implies

$$\begin{aligned}&\varrho(v^{-1}f_*)=s^{6r+4\nu}g_*,\qquad g_*:=X^2+3V^2Z+s^2\widetilde{R},\\ &\text{and}\quad \widetilde{R}:=s^{-8r}\varrho(R)\in(V,Z)^4\Bbbk[[V,Z,s]].\end{aligned} \tag{F.91}$$

We show that there exist G-eigenvectors $V',Z'\in\Bbbk[[V,Z,s]]$ of characters χ_V,χ_Z with

$$V'-V,\ Z'-Z\in(V,Z)^2,$$

and some $\Psi\in Z^4\Bbbk[[Z,s]]$, such that

$$g_*=X^2+3V'^2Z'+\Psi(Z',s). \tag{F.92}$$

Let $j \geq 4$ be an integer, and assume that $V_j, Z_j \in \Bbbk[[V, Z, s]]$ are G-eigenvectors of characters χ_V, χ_Z with $V_j - V, Z_j - Z \in (V, Z)^2$, and $\Psi_j \in Z^4\Bbbk[[s]][Z]$ is a polynomial of degree less than j in Z, such that

$$3V^2Z + s^2\widetilde{R} \equiv 3V_j^2Z_j + \Psi_j(Z_j, s) + P_j \mod (V, Z)^{j+1}$$

for a form P_j of degree j in V_j, Z_j with coefficients in $\Bbbk[[s]]$. For $j = 4$ we take $V_4 = V$, $Z_4 = Z$, and $\Psi_4 = 0$. Any monomial of degree j in V_j, Z_j other than Z_j^j is divisible by V_j^2 or equals $V_jZ_j^{j-1}$. Thus

$$P_j = V_j^2Q_j + V_jZ_jQ_j' + P_j(0, Z_j)$$

for forms Q_j, Q_j' of degree $j-2$, where $V_j^2Q_j$ consists of the monomials of P_j divisible by V_j^2. Since P_j has character $\chi_V^2\chi_Z$, Q_j and Q_j' have characters χ_Z and χ_V. Let $V_{j+1} := V_j + \frac{1}{6}Q_j'$ and $Z_{j+1} := Z_j + \frac{1}{3}Q_j$. Since $j - 2 \geq 2$ and $\Psi_j \in Z^4\Bbbk[[s]][Z]$, modulo $(V, Z)^{j+1}$ we have

$$3V_{j+1}^2Z_{j+1} \equiv 3V_j^2Z_j + V_j^2Q_j + V_jZ_jQ_j', \qquad \Psi_j(Z_{j+1}, s) \equiv \Psi_j(Z_j, s), \quad \text{and} \quad Z_{j+1}^j \equiv Z_j^j.$$

Hence the assumption holds for $j + 1$, with $\Psi_{j+1}(Z, s) := \Psi_j(Z, s) + P_j(0, Z)$. Since

$$V_{j+1} - V_j,\ Z_{j+1} - Z_j \in (V, Z)^{j-2},$$

the V_j, Z_j, and Ψ_j converge (V, Z)-adically to some V', Z', and Ψ, and (F.92) holds.

Over Ω, (F.89) is an invertible linear change of (x, y, z), and (X, V', Z') are coordinates over $\Bbbk[[s]] \subset \Omega$. Thus the generic transverse type of W_* along C is the type of

$$(X^2 + 3V'^2Z' + \Psi(Z', s) = 0)$$

over Ω. For a series $\Psi_\Omega \in Z^4\Omega[[Z]]$, the germ $(X^2+3V^2Z+\Psi_\Omega(Z) = 0)$ is not isolated if $\Psi_\Omega = 0$, and for $\Psi_\Omega \neq 0$ it is of type D_k if and only if $\Psi_\Omega \in Z^{k-1}\Omega[[Z]]\backslash Z^k\Omega[[Z]]$. Since the generic transverse type is D_k, the series $\Psi \in \Bbbk[[s]][[Z]] \subset \Omega[[Z]]$ has order $k-1$ in Z over $\Bbbk((s))$, that is,

$$\Psi \in Z^{k-1}\Bbbk[[Z, s]] \quad \text{and the coefficient of } Z^{k-1} \text{ in } \Psi \text{ is a non-zero element of } \Bbbk[[s]]. \tag{F.93}$$

Let Ψ_0 be the non-zero homogeneous part of least degree of Ψ in (Z, s), and let $n-1$ be the degree of Ψ_0, so $n \geq k$. Since $\Psi_0(1, \cdot)$ is a non-zero polynomial, there exists $c \in \mathbb{C}$ with $\Psi_0(1, c) \neq 0$. Then

$$\Psi(Z', cZ') = \Psi_0(1, c)Z'^{n-1} + (\text{terms of degree at least } n \text{ in } Z'),$$

so by (F.92), the section $(s = cZ')$ of $(g_* = 0)$ is a Du Val surface germ of type D_n. By finite determinacy [GLS07, Theorem I.2.23 and Corollary I.2.24], there exists an integer $N_3 \geq n$ such that for any polynomial $\psi \in Z^{n-1}\mathbb{C}[Z]\backslash Z^n\mathbb{C}[Z]$, any series in $\mathbb{C}[[X, V, Z]]$ congruent to $X^2 + 3V^2Z + \psi(Z)$ modulo $(X, V, Z)^{N_3+1}$ defines a Du Val surface germ of type D_n.

Since $x^2 + y^3 - 3s^{2\nu}yz^2 + 2s^{3\nu}z^3 + R$ is a Weierstrass polynomial of degree 2 in x, there exist unique $h_*^{(0)}, h_*^{(1)} \in \Bbbk[[y, z, s]]$ with

$$h_* \equiv h_*^{(0)} + xh_*^{(1)} \mod f_*,$$

and by uniqueness $h_*^{(0)}$ and $xh_*^{(1)}$ have character χ_{h_*}. Since $h_*|_{W_*} \neq 0$, we have $h_*^{(0)} + xh_*^{(1)} \neq 0$, so $\varrho(h_*^{(0)} + xh_*^{(1)}) \neq 0$. Let

$$N := \max\{j \in \mathbb{N} \mid s^j \text{ divides } \varrho(h_*^{(0)} + xh_*^{(1)})\}.$$

Since $\varrho(h_*^{(0)})$ does not involve X and $\varrho(xh_*^{(1)}) = s^{3r+2\nu}X\varrho(h_*^{(1)})$, the series

$$l_* := s^{-N}\varrho(h_*^{(0)} + xh_*^{(1)}) \in \Bbbk[[X, V, Z, s]]$$

has $l_*|_{s=0}$ non-zero of degree at most 1 in X. In particular,

$$l_*|_{s=0} \notin (X^2 + 3V^2Z)\Bbbk[[X, V, Z]]. \tag{F.94}$$

Step 3. In this step we realize the data of **Steps 1** and 2 on the members and show that the computing lifts of b_i have large orders along x, y, z.

By Set-up 17.1 and Construction-Theorem 13.6, the conditions of Set-up 15.4 hold with $B_0 := a$, since $b_i \le a - \delta$. Moreover C is the only lc center of $(W_*, \frac{1}{m}\operatorname{div}(h_*))$, hence its minimal lc center, since any lc center is contained in the non-klt locus C and is not the closed point, where the minimal log discrepancy is positive (Set-up 17.1(4) and (5)). We show

$$\lim_{i\to+\infty} \min\{\operatorname{mult}_F \operatorname{Jac}_{W_i} \mid F \text{ is a computing lift of } b_i\} = +\infty. \tag{F.95}$$

Otherwise there exist an integer J and infinitely many i such that some computing lift F of b_i satisfies $\operatorname{mult}_F \operatorname{Jac}_{W_i} \le J$. By Theorem 15.8(1) with J in place of N, there exists L independent of i with $q_i(t) \le b_i + L(1-t)$ for any $t \in [0,1]$ and any such i. Thus

$$a \le q_i(\tau_i) \le b_i + L(1-\tau_i) \le a - \delta + L(1-\tau_i)$$

for infinitely many i, which contradicts $\lim_{i\to+\infty} \tau_i = 1$.

Since C is a regular curve contained in $\operatorname{Sing}(W_*)$ (Set-up 17.1(5)), Theorem 16.1(2) with the G-eigenparameters (s,x,y,z) of $\mathcal{X}_*$ in place of (s,u_1,u_2,u_3) and with (F.95) gives, after passing to a subsequence, G-eigenparameters s_i, x_i, y_i, z_i of $\mathcal{X}$ (the $s_i, u_{i1}, u_{i2}, u_{i3}$ of that theorem) with the characters of s,x,y,z such that the following hold.

- The integers
$$\mu_i := \min\{\operatorname{mult}_F x_i, \operatorname{mult}_F y_i, \operatorname{mult}_F z_i \mid F \text{ is a computing lift of } b_i\}$$
satisfy $\lim_{i\to+\infty} \mu_i = +\infty$.
- Regard the elements of $\mathbb{C}[[t_1,\dots,t_4]] = \mathbb{C}[[x_i,y_i,z_i,s_i]]$ and of $\Bbbk[[x_1,\dots,x_4]] = \Bbbk[[x,y,z,s]]$ as power series in variables x,y,z,s via $x_i,y_i,z_i,s_i \to x,y,z,s$, with G acting on the variables by the characters of x,y,z,s. Then the tuples (f_i,h_i) have the compatible generic limit (f_*,h_*) with coefficient field $\Bbbk$.

The local ring at o_i of the ambient smooth fourfold of W_i is dense in $\mathbb{C}[[t_1,\dots,t_4]]$, and projection to a character preserves regular functions. Thus, replacing x_i,y_i,z_i,s_i with G-eigenfunctions congruent to them modulo $\mathfrak{m}^{\mu_i+i}$, we may assume that x_i,y_i,z_i,s_i are regular on a G-invariant affine neighborhood U_i of o_i in that fourfold: this does not decrease μ_i, and it changes the power series f_i, h_i only in degrees at least $\mu_i + i$, so for any integer $d \ge 1$ it keeps their d-jets for all but finitely many i, and hence it keeps the compatible generic limit. Let

$$\varrho_i : \mathbb{C}[[t_1,\dots,t_4]] = \mathbb{C}[[x_i,y_i,z_i,s_i]] \to \mathbb{C}[[X,V,Z,s]]$$

be the homomorphism given by (F.89) with x_i,y_i,z_i,s_i in place of x,y,z,s.

By Lemma 2.6(4) for v and for the non-zero coefficients ζ of the truncations in degrees at most N_3 of V', Z', and $\Psi(Z,cZ)$, regarded as constant power series, after passing to an infinite subset there exist, for any i, a G-semi-invariant polynomial v_i in x,y,z,s with the character of v and, for each such ζ, an invariant polynomial, the polynomial attached to ζ, such that (f_*,h_*,v) together with the ζ is a compatible generic limit with coefficient field $\Bbbk$ of the tuples formed by (f_i,h_i,v_i) and the polynomials attached to the ζ. Let ζ_i be the constant term of the polynomial attached to ζ. We write v_i also for the invariant element $v_i(x_i,y_i,z_i,s_i)$, which is regular on U_i. Replacing each non-zero coefficient ζ of the truncations of V', Z', and $\Psi(Z,cZ)$ with ζ_i, we obtain the *realized truncations* of V', Z', and $\Psi(Z,cZ)$. Let $N_1 := 6r + 4\nu + N + N_3$, and consider the following conditions.

(a) $v_i(0) \ne 0$, $\varrho_i(v_i^{-1} f_i) \in s^{6r+4\nu}\mathbb{C}[[X,V,Z,s]]$, and $g_i := s^{-6r-4\nu}\varrho_i(v_i^{-1}f_i)$ satisfies
$$g_i \equiv X^2 + 3V^2 Z \mod s^2.$$

(b) The realized truncations $V_i', Z_i' \in \mathbb{C}[X,V,Z,s]$ of V', Z' and $\psi_i \in \mathbb{C}[Z]$ of $\Psi(Z,cZ)$ satisfy $V_i' - V, Z_i' - Z \in (V,Z)^2$, $\psi_i \in Z^{n-1}\mathbb{C}[Z]\backslash Z^n\mathbb{C}[Z]$, and, for $\sigma_i := s - cZ_i'$,
$$g_i \equiv X^2 + 3V_i'^2 Z_i' + \psi_i(Z_i') \mod (\sigma_i) + (X, V_i', Z_i', \sigma_i)^{N_3+1}.$$

(c) The Weierstrass division of h_i by f_i with respect to x_i gives $h_i \equiv h_i^{(0)} + x_i h_i^{(1)} \mod f_i$ with $h_i^{(0)}, h_i^{(1)} \in \mathbb{C}[[y_i,z_i,s_i]]$, s^N divides $\varrho_i(h_i^{(0)} + x_i h_i^{(1)})$, and $s^{-N}\varrho_i(h_i^{(0)} + x_i h_i^{(1)})|_{s=0} \ne 0$.

Since ϱ_i maps a monomial of degree e' in x_i, y_i, z_i, s_i to a multiple of $s^{e'}$, (a), (b), and (c) are conditions on the coefficients in degrees at most N_1 of f_i, h_i, v_i and on the ζ_i, given by polynomial equations and the non-vanishing of finitely many polynomials, among them the coefficient of x_i^2 in f_i, since in each degree the coefficients of the quotient and the remainder of a Weierstrass division by a series of order 2 in x_i are polynomials in those of the dividend and the divisor and in the inverse of the coefficient of x_i^2. Conditions (a), (b), and (c) also hold at the limit, that is, for $f_*, h_*, v, x, y, z, s, \varrho$ and the coefficients ζ, by (F.91), (F.92), $\Psi_0(1,c) \neq 0$, the choice of N and (F.94). Hence the polynomial equations hold for all but finitely many i by Lemma 2.6(4), and so does the non-vanishing by Definition 2.5, since the product of the polynomials in question does not vanish at the (N_1+1)-jet of the limit and therefore on a non-empty open subset of the closed set Z_{N_1+1} of that definition. Removing finitely many i, (a), (b), and (c) hold for any i, and $\mu_i > (3r+2\nu)\cdot 3M/\eta_m$ for any i.

Let F be a computing lift of b_i. By Theorem 5.1(3),

$$\operatorname{mult}_F \mathfrak{m}_{o_i}/e_F \leq 3/\eta_m \quad \text{and} \quad \operatorname{mult}_F \mathfrak{m}_{o_i} \leq 3M/\eta_m < \mu_i.$$

Since $\mathfrak{m}_{o_i} = (x_i, y_i, z_i, s_i)\mathcal{O}_{W_i}$ and $\min\{\operatorname{mult}_F x_i, \operatorname{mult}_F y_i, \operatorname{mult}_F z_i\} \geq \mu_i$,

$$\operatorname{mult}_F s_i = \operatorname{mult}_F \mathfrak{m}_{o_i} \leq \frac{3e_F}{\eta_m} \leq \frac{3M}{\eta_m}. \tag{F.96}$$

Step 4. In this step we construct Y_i and prove the identities of (1).

Possibly shrinking U_i, the morphism $U_i \to \mathbb{A}^4$ given by (x_i, y_i, z_i, s_i) is étale. Let

$$\Pi_i := U_i \times_{\mathbb{A}^4} \mathbb{A}^4,$$

where $\mathbb{A}^4 \to \mathbb{A}^4$ is the morphism $(X, V, Z, s) \to (x, y, z, s)$ given by (F.89), let $\pi_i : \Pi_i \to U_i$ be the projection, and let $0 \in \Pi_i$ be the point over o_i and the origin. Then X, V, Z, s are regular parameters of Π_i at 0, G acts on Π_i fixing 0, $\pi_i^* s_i = s$, π_i is G-equivariant, and π_i is an isomorphism over $(s_i \neq 0)$ by (F.90). The completion of $\mathcal{O}_{\Pi_i,0}$ is $\mathbb{C}[[X, V, Z, s]]$, and π_i induces ϱ_i on completions. Let $(Y_i \ni 0)$ be the germ at 0 of the strict transform of W_i. By (a), $Y_i = (g_i = 0)$ near 0, and $S_i := (s = 0) \subset Y_i$ has completed local ring $\mathbb{C}[[X, V, Z]]/(X^2 + 3V^2 Z)$, which is a domain since $-3Z$ is not a square in the fraction field of $\mathbb{C}[[V, Z]]$. By (b) and the choice of N_3, the Cartier divisor $(\sigma_i = 0)$ of Y_i through 0 is a Du Val surface germ of type D_n, with n independent of i. Thus $(Y_i \ni 0)$ is cDV, hence normal, Gorenstein and canonical, and S_i is a reduced integral Cartier divisor with $\operatorname{div}(s|_{Y_i}) = S_i$.

Let $\omega := dx_i \wedge dy_i \wedge dz_i \wedge ds_i$. By (F.89) and (a),

$$\pi_i^* \omega = s^{7r+4\nu} dX \wedge dV \wedge dZ \wedge ds \quad \text{and} \quad \pi_i^* f_i = s^{6r+4\nu} \varrho_i(v_i) g_i.$$

Thus the residue of ω/f_i along W_i, which generates ω_{W_i} near o_i, pulls back to $s^r \varrho_i(v_i)^{-1}$ times the residue of $dX \wedge dV \wedge dZ \wedge ds/g_i$ along Y_i, which generates ω_{Y_i} near 0. Since $\varrho_i(v_i)$ is a unit,

$$K_{Y_i} = \pi_i^* K_{W_i} + r S_i. \tag{F.97}$$

By (c), $h_i|_{Y_i} = \varrho_i(h_i^{(0)} + x_i h_i^{(1)})|_{Y_i} = s^N l_i$ with $l_i := s^{-N} \varrho_i(h_i^{(0)} + x_i h_i^{(1)})|_{Y_i}$. The function $l_i = h_i/s^N$ of Y_i is regular at 0 since it is regular on the completion. By (c) and the argument for (F.94), l_i does not vanish on S_i. Thus $\pi_i^* \operatorname{div}(h_i) = \operatorname{div}(l_i) + N S_i$ on Y_i.

Let $\beta := N/m - r$ and $B_i := \frac{1}{m} \operatorname{div}(l_i) + \beta S_i$. Then $\beta \in \frac{1}{m}\mathbb{Z}$, and $\beta \geq -r$ since $N \geq 0$. By (F.97), for any $t \in [0, 1]$,

$$K_{Y_i} + tB_i + (t-1) r S_i = \pi_i^* \Big(K_{W_i} + \frac{t}{m} \operatorname{div}(h_i) \Big). \tag{F.98}$$

The morphism π_i is G-equivariant and birational, so any prime divisor F over Y_i is a prime divisor over W_i, the inertia groups in G of F over Y_i and over W_i are equal, and if F has center

0 on Y_i, then F has center o_i on W_i. By (F.97), (F.98) and $h_i|_{Y_i} = s^N l_i$, any such F satisfies

$$a(F, W_i, 0) = a(F, Y_i, 0) + r \operatorname{mult}_F s, \qquad \operatorname{mult}_F h_i = \operatorname{mult}_F l_i + N \operatorname{mult}_F s,$$

$$\text{and} \qquad a\Big(F, W_i, \frac{t}{m}\operatorname{div}(h_i)\Big) = a(F, Y_i, tB_i + (t-1)rS_i)$$

for $0 \le t \le 1$. By (F.89), $\pi_i^*\mathfrak{m}_{o_i} \subset (s)$, so S_i is a prime divisor over W_i with center o_i, and (F.98) with $t = 1$ implies

$$a\Big(S_i, W_i, \frac{1}{m}\operatorname{div}(h_i)\Big) = a(S_i, Y_i, B_i) = 1 - \beta.$$

Since $(W_i, \frac{1}{m}\operatorname{div}(h_i))$ is klt near o_i, $\beta < 1$, and by (F.98) with $t = 1$, (Y_i, B_i) is sub-klt.

Step 5. In this step we prove the remaining statements of (1) and (2).

Let F be a computing lift of b_i. By (F.90) for x_i, y_i, z_i, s_i, the definition of μ_i, and (F.96), we have

$$\operatorname{mult}_F X \ge \mu_i - (3r+2\nu)\operatorname{mult}_F s_i, \quad \operatorname{mult}_F V \ge \mu_i - (2r+\nu)\operatorname{mult}_F s_i, \quad \text{and} \quad \operatorname{mult}_F Z \ge \mu_i - (2r+2\nu)\operatorname{mult}_F s_i,$$

so

$$\min\{\operatorname{mult}_F X, \operatorname{mult}_F V, \operatorname{mult}_F Z\} \ge \mu_i - (3r+2\nu)\cdot\frac{3M}{\eta_m} > 0. \tag{F.99}$$

Since F has center o_i on U_i, $\operatorname{mult}_F s = \operatorname{mult}_F s_i > 0$, and $\mathcal{O}_{\Pi_i}$ is generated over $\mathcal{O}_{U_i}$ by X, V, Z, F has center 0 on Π_i, hence on Y_i. Together with **Step 4**, Definitions 2.2 and 2.4 give $\operatorname{mld}((Y_i \ni 0, B_i)/G) = b_i$, and any computing lift of $\operatorname{mld}((Y_i \ni 0, B_i)/G)$ is a computing lift of b_i. By (F.96) and $\pi_i^* s_i = s$, any computing lift F of b_i satisfies $\operatorname{mult}_F s/e_F \le 3/\eta_m$. Together with **Steps 1** and 4, this proves (1).

In the variables x, y, z, s of **Step 3**, ϱ_i becomes ϱ. By (a), the coefficient of x^2 in f_i is non-zero, since the coefficient of X^2 in $g_i|_{s=0}$ is the coefficient of x^2 in f_i divided by $v_i(0)$. Each coefficient of $g_i = s^{-6r-4\nu}\varrho(v_i^{-1}f_i)$ and of the series $\widehat{l_i} := s^{-N}\varrho(h_i^{(0)} + xh_i^{(1)})$, whose restriction to Y_i is l_i (**Step 4**), is a polynomial with complex coefficients independent of i in finitely many coefficients of f_i, h_i, v_i and in the inverses of $v_i(0)$ and of the coefficient of x^2 in f_i (**Step 3**). Evaluated at the corresponding coefficients of f_*, h_*, v and the inverses of $v(0)$ and of the coefficient of x^2 in f_*, each such polynomial gives the corresponding coefficient of g_* or of l_*, by (F.91) and the definition of l_* in **Step 2**. Hence, for any integer $d \ge 1$, there exist an integer $d' \ge d$ and a morphism Θ_d on the open subset $Z_{d'}^\circ$ of the closed set $Z_{d'}$ of Definition 2.5 for (f_i, h_i, v_i) (**Step 3** and Lemma 2.6(2)) where the coordinates corresponding to $v_i(0)$ and to the coefficient of x^2 in f_i do not vanish, sending the d'-jet of (f_i, h_i, v_i) to the d-jet of $(g_i, \widehat{l_i})$ and the d'-jet of (f_*, h_*, v) to the d-jet of (g_*, l_*). The d-jet of (g_*, l_*) therefore maps to the generic point of $\overline{\Theta_d(Z_{d'}^\circ)}$, and the preimage under Θ_d of a non-empty open subset of $\overline{\Theta_d(Z_{d'}^\circ)}$ is a non-empty open subset of $Z_{d'}$. Thus the pairs $(g_i, \widehat{l_i})$ formed by the equations of the Y_i and the series $\widehat{l_i}$ have the compatible generic limit (g_*, l_*) with coefficient field $\Bbbk$.

By (F.92) and (F.93), g_* has the form of (2) in the G-eigenparameters (X, V', Z', s), which preserve s.

We regard s as a non-zero element of Ω, so that

$$\Bbbk[[x, y, z, s]] \subset \Omega[[x, y, z]] \quad \text{and} \quad \Bbbk[[X, V, Z, s]] \subset \Omega[[X, V, Z]].$$

By **Step 2**, ϱ extends to an isomorphism $\Omega[[x, y, z]] \to \Omega[[X, V, Z]]$. By (F.91), it maps f_* to $s^{6r+4\nu}\varrho(v)g_*$, a unit multiple of g_*, and by the definitions of $h_*^{(0)}, h_*^{(1)}$ and l_* in **Step 2**, it maps h_* to $s^N l_*$ modulo $\varrho(f_*)$, where $s^N \in \Omega\backslash\{0\}$. By Definition 15.1, $T := (f_* = 0) \subset \operatorname{Spec}\Omega[[x, y, z]]$ is the generic transverse surface of W_* along C. Hence, over Ω,

$$\Big((g_* = 0), \frac{1}{m}\operatorname{div}(l_*)\Big) \cong \Big(T, \frac{1}{m}\operatorname{div}(h_*|_T)\Big).$$

Finally, (F.99) and $\lim_{i\to+\infty}\mu_i = +\infty$ imply that any computing lift F of b_i satisfies $\min\{\operatorname{mult}_F X, \operatorname{mult}_F V, \operatorname{mult}_F Z\} \to +\infty$. This proves (2). □

Lemma F.11. *Let $\mathbb{L} \supset \mathbb{C}$ be an algebraically closed field, and let $f_0 := x^2 + zy^2 + z^{k-1}$ for an integer $k \geq 4$, with $w := (k-1, k-2, 2)$, or $f_0 := x^2 + y^3 + z^3$, with $w := (3,2,2)$. Let $S_0 := (f_0 = 0) \subset \operatorname{Spec} \mathbb{L}[[x,y,z]]$, let $g \in \mathbb{L}[[x,y,z]]$ be non-zero of x-degree at most 1, and let $t \geq 0$ be a real number. Then $(S_0, t \operatorname{div}(g|_{S_0}))$ is lc near 0 if and only if $t \operatorname{wt}_w(g) \leq 1$, and klt near 0 if and only if $t \operatorname{wt}_w(g) < 1$.*

Proof. If $\operatorname{wt}_w(g) = 0$, then g is a unit, and both equivalences hold since S_0 is klt. Assume that $\operatorname{wt}_w(g) > 0$. Since y^3+z^3 and zy^2+z^3 are products of three pairwise non-proportional linear forms, a linear change of (y,z), which preserves wt_w and the x-degree, reduces the case $f_0 = x^2+y^3+z^3$ to the case $f_0 = x^2 + zy^2 + z^{k-1}$ with $k = 4$. By [Roa04, Table 1], there exist a binary dihedral group $\Gamma \subset SL_2(\mathbb{L})$ of order $4(k-2)$ and the quotient morphism $\psi : \operatorname{Spec} \mathbb{L}[[p_1,p_2]] \to S_0$ by Γ such that $\psi^* x$, $\psi^* y$, $\psi^* z$ are homogeneous of degrees $2(k-1)$, $2(k-2)$, 4, and such that ψ^* is an isomorphism

$$\psi^* : \mathbb{L}[x,y,z]/(f_0) \to \mathbb{L}[p_1,p_2]^\Gamma.$$

Since ψ^* maps a w-homogeneous polynomial of weight j to a form of degree $2j$ and is injective modulo f_0, and $\operatorname{in}_w(g)$, of x-degree at most 1, is not divisible by f_0, which is monic of degree 2 in x, the lowest form of $\psi^* g$ is $\psi^* \operatorname{in}_w(g) \neq 0$, of degree $2 \operatorname{wt}_w(g)$. Since $\Gamma \subset SL_2(\mathbb{L})$ contains no pseudo-reflection, ψ is étale in codimension 1, so for any real number $t' \geq 0$, $(S_0, t' \operatorname{div}(g|_{S_0}))$ is lc near 0 if and only if $(\operatorname{Spec} \mathbb{L}[[p_1,p_2]], t' \operatorname{div}(\psi^* g))$ is lc near 0. Since Γ is non-abelian and $\Delta := t' \operatorname{div}(\psi^* g)$ is Γ-invariant, by [CCJ20, Appendix A, Lemma A.6] (the lemma and its proof, blowing up the origin and using that the non-abelian group has no fixed point on the exceptional curve, hold over any algebraically closed field of characteristic 0 and for formal germs), $(\operatorname{Spec} \mathbb{L}[[p_1,p_2]], \Delta)$ is lc near 0 if and only if

$$\operatorname{mult}_0 \Delta = 2t' \operatorname{wt}_w(g) \leq 2.$$

This proves the first equivalence. Let

$$B_0 := \operatorname{wt}_w(g)^{-1} \operatorname{div}(g|_{S_0}).$$

If $t \operatorname{wt}_w(g) < 1$, then $(S_0, t \operatorname{div}(g|_{S_0}))$ is klt near 0, being a convex combination of the klt pair $(S_0, 0)$ and the lc pair (S_0, B_0). If $t \operatorname{wt}_w(g) \geq 1$, then $(S_0, t' \operatorname{div}(g|_{S_0}))$ is not lc for any $t' > t$, so $(S_0, t \operatorname{div}(g|_{S_0}))$ is not klt near 0. The lemma follows. □

Proof of Proposition 17.8. Let $n \geq 4$ be the integer of Construction-Theorem 13.7, let $\ell_{j,i}$ be the regular function on W_i given by Construction-Theorem 13.7(3), so that $(\ell_{j,i} = 0) \ni o_i$ is a Du Val surface germ of type D_n, let $\kappa_n, \kappa_{k+\nu_0} > 0$ be rational numbers such that the pairs

$$(S_n, \operatorname{Jac}_{S_n}^{\kappa_n}) \quad \text{and} \quad (S_{k+\nu_0}, \operatorname{Jac}_{S_{k+\nu_0}}^{\kappa_{k+\nu_0}})$$

are lc near 0, for the Du Val surfaces S_n and $S_{k+\nu_0}$ of Theorem 8.4, and let $\lambda := 1/\kappa_n$. Let $\mathcal{I}$ be the infinite set of indices along which (f_*, h_*) is a compatible generic limit of (f_i, h_i), and let $\mathfrak{m}$ be the maximal ideal of the local ring at o_i of the smooth G-invariant fourfold germ in which $W_i = (f_i = 0)$. By Construction-Theorem 13.7, we regard f_i and h_i as elements of $\mathbb{C}[[x,y,z,s]]$, the $\mathfrak{m}$-adic completion, through the G-eigencoordinates (x,y,z,t) of Construction-Theorem 13.7(6), with t written s, in which f_* has the form assumed in the proposition. The hypothesis of Construction-Theorem 13.7(6) holds, since the residual cubic zy^2 of W_* is a double line and a different line, hence by Definition 2.5 so is the residual cubic of W_i for all but finitely many $i \in \mathcal{I}$. We use Definition 2.5 through Lemma 2.7 with $m = 0$ there. Each time it is used, we remove the finitely many exceptional indices from $\mathcal{I}$. The subfamily of the proposition is the final $\mathcal{I}$ with finitely many indices removed. We divide the proof into five steps.

Step 1. In this step we prove that C is the s-axis, $\gamma = 0$ and, after the substitution of the proposition, $\phi = z^{k-1}(s^\nu + z\phi_1(z,s))$.

Let η be the generic point of C. Since $C \subset \operatorname{Sing}(W_*)$, f_* is contained in the square of the maximal ideal of $\mathcal{O}_{\mathcal{X}_*,\eta}$. Since moreover the generic transverse surface of W_* along C is of type D_k, the Hessian matrix of f_* in (x,y,z,s) at η has rank 1, the rank of the quadratic part of a

D_k equation. Since $\partial^2 f_*/\partial x^2 = 2$ and $\partial^2 f_*/\partial x\partial v = 0$ for $v \in \{y, z, s\}$, all second derivatives of f_* in y, z, s vanish at η. In particular $\partial^2 f_*/\partial y^2 = 2z$ and $\partial^2 f_*/\partial y\partial z = 2y$ vanish on C, and so does $\partial f_*/\partial x = 2x$ since $C \subset \mathrm{Sing}(W_*)$. Thus $C \subset (x = y = z = 0)$. Since C is a curve, C is the s-axis, and we take $(u_1, u_2, u_3) = (x, y, z)$ in Set-up 17.1. Restricting $\partial f_*/\partial y$, f_*, $\partial f_*/\partial z$, and $\partial^2 f_*/\partial z^2$ to C, we obtain

$$\gamma(s) = \phi(0, s) = \frac{\partial \phi}{\partial z}(0, s) = \frac{\partial^2 \phi}{\partial z^2}(0, s) = 0.$$

Thus $z^3 \mid \phi$. Moreover $\phi \neq 0$, since $(x^2 + zy^2 = 0)$ is singular along the surface $(x = y = 0)$ while W_* is normal. Write $\phi = z^{n_0}(a_0(s) + z\phi'(z, s))$ with $n_0 \geq 3$ and $0 \neq a_0 \in \Bbbk[[s]]$. Since $1 + z\phi'/a_0$ is a unit of $\Bbbk((s))[[z]]$, the coordinates $z(1 + z\phi'/a_0)^{1/n_0}$ and $y(1 + z\phi'/a_0)^{-1/(2n_0)}$ of $\Bbbk((s))[[x, y, z]]$, in place of z and y, turn f_* into $x^2 + zy^2 + a_0 z^{n_0}$. Over $\mathbb{K} := \overline{\Bbbk((s))}$, rescaling y and z turns $x^2 + zy^2 + a_0 z^{n_0}$ into $x^2 + zy^2 + z^{n_0}$, which is of type D_{n_0+1}. Thus $n_0 = k - 1$. Write $a_0 = s^\nu \phi_0(s)$ with $\phi_0(0) \neq 0$. Since f_* is G-semi-invariant, any monomial $z^{k-1} s^{\nu+j}$ with non-zero coefficient in f_* has the character of $z^{k-1}s^\nu$, so ϕ_0 is a G-invariant unit of $\Bbbk[[s]]$, and so are $\phi_0^{1/(2k-2)}$ and $\phi_0^{-1/(k-1)}$, given by a root of $\phi_0(0)$ and the binomial series. Substituting $\phi_0^{1/(2k-2)} y$ for y and $\phi_0^{-1/(k-1)} z$ for z, which is the substitution of the proposition with $\mu = \phi_0^{-1/(k-1)}$, preserves $x^2 + zy^2$ and $\gamma = 0$, and gives

$$\phi = z^{k-1}(s^\nu + z\phi_1(z, s)), \quad \phi_1(z, s) := \phi_0^{-k/(k-1)} \phi'(\phi_0^{-1/(k-1)} z, s).$$

In the new coordinates $a_0 = s^\nu$, and we let $\zeta := 1 + z\phi_1/s^\nu$, $\widetilde{z} := z\zeta^{1/(k-1)}$, and $\widetilde{y} := y\zeta^{-1/(2k-2)}$, so that $f_* = x^2 + \widetilde{z}\widetilde{y}^2 + s^\nu \widetilde{z}^{k-1}$.

By Lemma 2.6(4), applied to the G-invariant power series $\phi_0^{1/(2k-2)}$ and $\phi_0^{-1/(k-1)}$, after replacing $\mathcal{I}$ with an infinite subset there exist G-invariant polynomials α_i, α_i', $i \in \mathcal{I}$, such that $(f_*, h_*, \phi_0^{1/(2k-2)}, \phi_0^{-1/(k-1)})$ is a compatible generic limit of the $(f_i, h_i, \alpha_i, \alpha_i')$. By Definition 2.5, $\alpha_i(0)\alpha_i'(0) \neq 0$ for all but finitely many i, and we remove the other indices. For $i \in \mathcal{I}$ we substitute $\alpha_i(0, 0, 0, s)y$ for y and $\alpha_i'(0, 0, 0, s)z$ for z. For any positive integer D, one polynomial map Θ_D, independent of i, sends the D-jet of $(f_i, h_i, \alpha_i, \alpha_i')$ to the D-jet of (f_i, h_i) in the new coordinates, and the D-jet of $(f_*, h_*, \phi_0^{1/(2k-2)}, \phi_0^{-1/(k-1)})$ to that of (f_*, h_*) in the new coordinates. As the preimage under Θ_D of a non-empty open subset of the closure of the image of Z_D, where $Z_D \subset \mathcal{J}_D$ is the closed subset of Definition 2.5 for the tuples $(f_i, h_i, \alpha_i, \alpha_i')$, is a non-empty open subset of Z_D, (f_*, h_*) is a compatible generic limit of the (f_i, h_i) in the new coordinates.

Let $w := (k - 1, k - 2, 2)$ be weights on (x, y, z), with s in the coefficient field $\Bbbk((s))$, and let wt_w and in_w be the w-weight and the lowest w-weight part (Definition 2.3) on $\Bbbk((s))[[x, y, z]]$. Since $\widetilde{y} - y \in (yz)$ and $\widetilde{z} - z \in (z^2)$, for any non-zero $H \in \Bbbk((s))[[x, y, z]]$, the lowest w-weight part of H in $(x, \widetilde{y}, \widetilde{z})$ is $\mathrm{in}_w(H)$ with $\widetilde{y}, \widetilde{z}$ substituted for y, z.

Step 2. In this step we prove that the mark has lowest w-weight m along C, and we define c.

Since f_* is monic of degree 2 in x, Weierstrass division writes

$$h_* = f_*\xi_* + H_*, \qquad H_* \in \Bbbk[[y, z, s]] + x\Bbbk[[y, z, s]],$$

and $H_*|_{W_*} = h_*|_{W_*}$. Let $T := \operatorname{Spec} \mathbb{K}[[x, y, z]]/(f_*)$. Since $(W_*, \frac{1}{m}\operatorname{div}(h_*))$ is lc with $\mathrm{Nklt}(W_*, \frac{1}{m}\operatorname{div}(h_*)) = C \ni \eta$, the pair $(T, \frac{1}{m}\operatorname{div}(H_*))$ is lc and not klt (cf. [dFEM11, Proposition 2.11]): $T \to W_*$ is the base change of $W_* \subset \mathcal{X}_*$ by the regular morphism $\operatorname{Spec} \mathbb{K}[[x, y, z]] \to \mathcal{X}_*$ given by localizing at η, completing, and extending the coefficient field $\Bbbk((s))$ to $\mathbb{K}$, a log resolution pulls back along a regular morphism to a log resolution (cf. [dFEM11, Remark 2.10], and [JM12, Lemma 1.2] for the ambient $\mathcal{X}_*$), and relative canonical divisors are preserved since W_*, a Cartier divisor in the regular $\mathcal{X}_*$, is Gorenstein and its dualizing sheaf commutes with flat base change. By **Step 1**, after rescaling $\widetilde{y}, \widetilde{z}$,

$$T \cong \operatorname{Spec} \mathbb{K}[[x, \widetilde{y}, \widetilde{z}]]/(x^2 + \widetilde{z}\widetilde{y}^2 + \widetilde{z}^{k-1}).$$

In the coordinates $(x, \widetilde{y}, \widetilde{z})$, H_* has x-degree at most 1 and, by the last sentence of **Step 1**, w-weight $\mathrm{wt}_w(H_*)$. Thus, by Lemma F.11 with $\mathbb{L} = \mathbb{K}$, for any real number $t \geq 0$, $(T, t\,\mathrm{div}(H_*))$ is lc if and only if $t\,\mathrm{wt}_w(H_*) \leq 1$, and klt if and only if $t\,\mathrm{wt}_w(H_*) < 1$. Since the pair is lc and not klt for $t = 1/m$, $\mathrm{wt}_w(H_*) = m$. Thus

$$\mathrm{wt}_w(x^{n_1}y^{n_2}z^{n_3}s^{n_4}) \geq m$$

for any monomial of H_*, with equality for at least one monomial. Let

$$c := \min\{d(n_1+n_2)+n_4 \mid x^{n_1}y^{n_2}z^{n_3}s^{n_4} \text{ is a monomial of } H_* \text{ with } \mathrm{wt}_w(x^{n_1}y^{n_2}z^{n_3}s^{n_4}) = m\},$$

a non-negative integer, and let $Q_* \in \Bbbk[x,y,z]$ be the result of setting $s = 1$ in the sum of the terms of H_* with $\mathrm{wt}_w = m$ and $d(n_1+n_2)+n_4 = c$. Then $Q_* \neq 0$, Q_* is w-homogeneous of weight m, and Q_* has x-degree at most 1.

Step 3. In this step we prove that along the computing lifts of b_i, the orders of x, y, z tend to $+\infty$ while the order of s stays bounded.

Let F be a computing lift of b_i. Since $q_i(\tau_i) \geq b_i + \delta$ (Definition 4.1(3)) and F computes b_i,

$$\frac{a(F, W_i, 0) - \tau_i \,\mathrm{mult}_F h_i/m}{e_F} \geq b_i + \delta \quad \text{and} \quad \frac{a(F, W_i, 0) - \mathrm{mult}_F h_i/m}{e_F} = b_i.$$

Subtracting the second relation from the first, we obtain

$$\frac{\mathrm{mult}_F h_i}{m e_F} \geq \frac{\delta}{1-\tau_i}. \tag{F.100}$$

Since $b_i \leq 3$ (Lemma 4.2(3)) and $e_F \leq M$, $a(F, W_i, 0) \leq \frac{1}{m}\,\mathrm{mult}_F h_i + 3M$. By Theorem 8.4 with ℓ_V a lift of $\ell_{j,i}$ to the ambient fourfold (a regular parameter, since the Du Val surface ($\ell_{j,i} = 0$) has embedding dimension 3), $j_i := \mathrm{mult}_F \mathrm{Jac}_{W_i} \leq \lambda a(F, W_i, 0)$.

Since $\mathrm{Nklt}(W_*, \frac{1}{m}\,\mathrm{div}(h_*)) = C$, and a log resolution of $(W_*, \mathrm{div}(h_*))$ [dFEM11, Theorems 2.3–2.4] has finitely many prime divisors whose images are not contained in C, there exists a rational number $\mu_0 > \frac{1}{m}$ such that $(W_*, \mu_0\,\mathrm{div}(h_*))$ is klt on $W_* \backslash C$. Choose rational numbers $0 < \theta_1 < \theta_2 < 1$ such that

$$\varpi := \theta_1\mu_0 - \frac{1}{m} - (1-\theta_1)\left(1 + \frac{2\lambda}{m}\right) > 0,$$

which is possible since $\varpi \to \mu_0 - \frac{1}{m} > 0$ as $\theta_1 \to 1$. Let $\mathcal{J}_* := \mathcal{J}(f_*^{\theta_2} h_*^{\theta_2\mu_0})$. By Lemma 6.1(3) with $V = C$, $\mathcal{J}_*$ is the unit ideal near any point of $W_* \backslash C$. Since the ideal of C is (x, y, z) and the zero locus of $(f_*) + \mathcal{J}_*$ is contained in C, there exists a positive integer P such that

$$(x, y, z)^P \subset (f_*) + \mathcal{J}_*.$$

Let ρ be a positive integer and $\mathfrak{m}_*$ the maximal ideal of $\Bbbk[[x, y, z, s]]$, which is the ideal $\mathfrak{n}_*$ of Lemma 6.1 in the coordinates (x, y, z, s). For $\sigma \in \{x^P, y^P, z^P\}$ choose $a_\sigma \in \Bbbk[[x, y, z, s]]$ with $\sigma - a_\sigma f_* \in \mathcal{J}_*$, let A_σ be the sum of $u_{\sigma,\alpha}\alpha$ over the monomials α of degree less than ρ in x, y, z, s, with distinct variables $u_{\sigma,\alpha}$, and let $\Phi_\sigma := \sigma - A_\sigma y_{1,0}$ in the notation of Lemma 6.1 for the pairs (f, h). Let β be the point whose coordinate $u_{\sigma,\alpha}$ is the coefficient of α in a_σ, so that $a_\sigma - A_\sigma(\beta) \in \mathfrak{m}_*^\rho$ and

$$\Phi_\sigma((f_*, h_*), \beta) = (\sigma - a_\sigma f_*) + (a_\sigma - A_\sigma(\beta)) f_* \in \mathcal{J}_* + \mathfrak{m}_*^\rho.$$

By Lemma 6.1(1) and (2), applied to $(f_i, h_i)_{i\in\mathcal{I}}$ with the exponents $(\theta_1, \theta_1\mu_0) < (\theta_2, \theta_2\mu_0)$ and $L = \rho$, for all but finitely many $i \in \mathcal{I}$ there exists β_i such that the elements

$$z_{\sigma,i} := \Phi_\sigma((f_i, h_i), \beta_i) = \sigma - A_\sigma(\beta_i) f_i$$

satisfy $z_{\sigma,i}|_{W_i} = \sigma|_{W_i}$, and any prime divisor E over $\mathrm{Spec}\,\mathbb{C}[[x, y, z, s]]$ with center the closed point and $\mathrm{mult}_E z_{\sigma,i} < \rho$ satisfies

$$\mathrm{mult}_E z_{\sigma,i} + a(E, \mathrm{Spec}\,\mathbb{C}[[x, y, z, s]], 0) > \theta_1\,\mathrm{mult}_E f_i + \theta_1\mu_0\,\mathrm{mult}_E h_i.$$

Let F be a computing lift of b_i and $\sigma \in \{x^P, y^P, z^P\}$. Since W_i is normal, f_i is irreducible, and $f_i \in \mathfrak{m}^2$ does not divide σ, so $\sigma|_{W_i} \neq 0$. If $\operatorname{mult}_F(\sigma|_{W_i}) < \rho$, then Lemma 5.5(1), with $(f, h, g, \theta, \mu) = (f_i, h_i, z_{\sigma,i}, \theta_1, \mu_0)$ and the threshold ρ, gives

$$(\theta_1\mu_0 - 1 + \theta_1)\operatorname{mult}_F h_i < (2-\theta_1)\operatorname{mult}_F(\sigma|_{W_i}) + a(F, W_i, 0) + 2(1-\theta_1)j_i + 1 - \theta_1.$$

Since $j_i \le \lambda a(F, W_i, 0)$ and $a(F, W_i, 0) \le \frac{1}{m}\operatorname{mult}_F h_i + 3M$, the right side is at most

$$(2-\theta_1)\operatorname{mult}_F(\sigma|_{W_i}) + \big(1 + 2(1-\theta_1)\lambda\big)\left(\frac{1}{m}\operatorname{mult}_F h_i + 3M\right) + 1 - \theta_1.$$

Moving the terms with $\operatorname{mult}_F h_i$ to the left, we obtain

$$\varpi \operatorname{mult}_F h_i < (2-\theta_1)\operatorname{mult}_F(\sigma|_{W_i}) + 3M + (1-\theta_1)(1 + 6\lambda M). \tag{F.101}$$

As $\operatorname{mult}_F(x^P|_{W_i}) = P \operatorname{mult}_F x$, and similarly for y and z,

$$P \min\{\operatorname{mult}_F x, \operatorname{mult}_F y, \operatorname{mult}_F z\} \ge \min\left\{\rho, \frac{\varpi \operatorname{mult}_F h_i - 3M - (1-\theta_1)(1+6\lambda M)}{2-\theta_1}\right\}.$$

By (F.100), $\operatorname{mult}_F h_i \ge m\delta/(1-\tau_i) \to +\infty$. Since ρ is arbitrary,

$$\omega_i := \min_F \min\{\operatorname{mult}_F x, \operatorname{mult}_F y, \operatorname{mult}_F z\} \to +\infty,$$

the minimum over the computing lifts F of b_i. By Theorem 5.1(3), $\operatorname{mult}_F \mathfrak{m}/e_F \le 3/\eta_m$ and $\operatorname{mult}_F \mathfrak{m} \le 3M/\eta_m$. Thus for any i with $\omega_i > 3M/\eta_m$, $\operatorname{mult}_F \mathfrak{m} = \operatorname{mult}_F s$ and

$$\frac{\operatorname{mult}_F s}{e_F} \le \frac{3}{\eta_m} \quad \text{and} \quad \operatorname{mult}_F s \le \frac{3M}{\eta_m}. \tag{F.102}$$

Step 4. In this step we construct the charts and prove (1), (2), (4), and the inequality $\operatorname{mult}_E s/e_E \le 3/\eta_m$ of (3).

Let r be an integer with $r > c$, $r \ge N$, and $2(r-d) \ge N$, and let $N_1 := N + 2(k+m)(r+d+1)$.

For $i \in \mathcal{I}$, since f_i is monic of degree 2 in x (Construction-Theorem 13.7), Weierstrass division writes $h_i = f_i\xi_i + H_i$ with ξ_i semi-invariant and H_i of x-degree at most 1. Each coefficient of H_i is $\Psi(f_i, h_i)$ for a polynomial Ψ in finitely many coefficients of f and h, and the corresponding coefficient of H_* is $\Psi(f_*, h_*)$. Possibly replacing the ambient lift h_i of the mark with $\widetilde{h}_i - \widetilde{f}_i\xi_i'$, where $\widetilde{h}_i$ is a G-semi-invariant regular lift of the mark, $\widetilde{f}_i$ is a regular equation of W_i near o_i such that $f_i/\widetilde{f}_i$ is an invariant unit, and ξ_i' is a semi-invariant regular function with $(f_i/\widetilde{f}_i)\widetilde{\xi}_i - \xi_i' \in \mathfrak{m}^{N_1+i}$ for the quotient $\widetilde{\xi}_i$ of the Weierstrass division of $\widetilde{h}_i$ by f_i, whose remainder is again H_i, and each $v \in \{x, y, z, s\}$, written as a power series in a fixed system of regular parameters of the local ring at o_i, with the χ_v-eigencomponent of the part of v of degree less than $N_1 + i$, which are regular parameters of the local ring at o_i, we may assume that x, y, z, s are regular parameters of that local ring, and that the coefficients of degree less than $N_1 + i$ of f_i and h_i are those of f_i and H_i in the coordinates (x, y, z, s) before the replacement. The lift of the mark in the proposition is this h_i. Since the replacement changes each of x, y, z, s by an element of $\mathfrak{m}^{N_1+i}$, **Step 3** holds for the new parameters with ω_i replaced by $\min\{\omega_i, N_1 + i\}$, which still tends to $+\infty$.

Under the substitution

$$x = s^{(k-1)r+d}x', \quad y = s^{(k-2)r+d}y', \quad \text{and} \quad z = s^{2r}z', \tag{F.103}$$

a monomial becomes

$$x^{n_1}y^{n_2}z^{n_3}s^{n_4} = s^{r\operatorname{wt}_w(x^{n_1}y^{n_2}z^{n_3}s^{n_4}) + d(n_1+n_2) + n_4}x'^{n_1}y'^{n_2}z'^{n_3}. \tag{F.104}$$

Since

$$\operatorname{wt}_w(x^{n_1}y^{n_2}z^{n_3}s^{n_4}) \ge 2(n_1 + n_2 + n_3),$$

the exponent of s in (F.104) is at least $n_1 + n_2 + n_3 + n_4$. Since $r > c$,

$$N_1 \ge N + \max\{2(k-1)r + 2d, mr + c\},$$

so any monomial of degree at least N_1 becomes, after division by $s^{2(k-1)r+2d}$ or by s^{mr+c}, an element of $\mathfrak{n}^N$, where $\mathfrak{n}$ is the maximal ideal of $\mathbb{C}[[x', y', z', s]]$. We write $\mathfrak{n}_*$ for the maximal ideal of $\Bbbk[[x', y', z', s]]$.

The equations. By **Step 1** and (F.104),

$$g_* := s^{-2(k-1)r-2d} f_* = x'^2 + z'y'^2 + s^{\nu_0} z'^{k-1} + z'^k \cdot s^{2r-2d}\phi_1(s^{2r}z', s),$$

which is of the form in (4), and the last term is contained in $\mathfrak{n}_*^N$ since $2(r-d) \geq N$. Let $g_i := s^{-2(k-1)r-2d} f_i$ on the chart (F.103). By Lemma 2.7, the coefficients of f_i of degree less than N_1 vanish whenever those of f_* do, and those of x^2, zy^2, and $z^{k-1}s^\nu$ are 1. Thus by (F.104), g_i is regular near the origin and

$$g_i = x'^2 + z'y'^2 + s^{\nu_0} z'^{k-1} + R_i, \quad R_i \in \mathfrak{n}^N. \tag{F.105}$$

The marks. Let $l_i := s^{-mr} h_i$ and $l_* := s^{-mr} H_*$ on the chart (F.103). By Lemma 2.7, the coefficients of h_i of degree less than N_1 vanish whenever those of H_* do: for the monomials with $n_1 \geq 2$, with $\mathrm{wt}_w < m$, and with $\mathrm{wt}_w = m$ and $d(n_1+n_2)+n_4 < c$ (**Step 2**). By (F.104), a monomial with $\mathrm{wt}_w \geq m+1$ becomes a multiple of s^{mr+r}, and $r > c$. Thus

$$l_i = s^c(Q_i + sl_i') \quad \text{and} \quad l_* = s^c(Q_*(x', y', z') + sl_*'),$$

where l_i' and l_*' are regular near the origin (h_i being regular, divisibility by a power of s in the completion of the local ring of the chart at the origin implies it in that local ring) and Q_i is the polynomial in (x', y', z') whose coefficient at $x'^{n_1}y'^{n_2}z'^{n_3}$, for each monomial $x^{n_1}y^{n_2}z^{n_3}s^{n_4}$ of H_* with $\mathrm{wt}_w = m$ and $d(n_1+n_2)+n_4 = c$, is the coefficient of $x^{n_1}y^{n_2}z^{n_3}s^{n_4}$ in h_i, and whose other coefficients are 0. By Lemma 2.7 and $Q_* \neq 0$, we may assume $Q_i \neq 0$. Then Q_i is w-homogeneous of weight m in (x', y', z') and has x'-degree at most 1. As g_* is monic of degree 2 in x' and l_*, like H_*, has x'-degree at most 1, l_* is its own representative modulo g_* of x'-degree at most 1. By (F.104), (F.103) sends a monomial $x^{n_1}y^{n_2}z^{n_3}s^{n_4}$ of w-weight m to $s^{mr}s^{d(n_1+n_2)+n_4}x'^{n_1}y'^{n_2}z'^{n_3}$, of w-weight m in (x', y', z'), and a monomial of larger w-weight to a monomial of larger w-weight in (x', y', z'). Hence the part of l_* of lowest w-weight in (x', y', z') has weight m and is $\mathrm{in}_w(h_*)(s^d x', s^d y', z')$, where $\mathrm{in}_w(h_*) = \mathrm{in}_w(H_*)$ is the part of H_* of weight m (**Step 2**), as in (4). For any integer J and any i with $N_1 + i > J + 2(k+m)(r+d+1)$, the coefficients of g_i and l_i of degree less than J are, by (F.104), coefficients of f_i and H_i under a rule independent of i, and those of H_i are fixed polynomials in coefficients of f_i and h_i. Thus, by the argument with the maps Θ_D in **Step 1**, (g_*, l_*) is a compatible generic limit of the (g_i, l_i). This proves the first three sentences of (4).

Proof of (1). The map defined by (F.103) is G-equivariant, with x', y', z' of characters $\chi_x\chi_s^{-(k-1)r-d}$, $\chi_y\chi_s^{-(k-2)r-d}$, $\chi_z\chi_s^{-2r}$, and it is birational and an isomorphism over $(s \neq 0)$. Let $Y_i := (g_i = 0)$ near the origin, $\pi_i : Y_i \to W_i$ the induced morphism, and $S_i := (s = 0) \subset Y_i$. Since π_i is G-equivariant and birational, G acts faithfully on Y_i. Since $(W_i \ni o_i)$ is isolated and π_i is an isomorphism over $(s \neq 0)$, Y_i is smooth near the origin outside S_i. By (F.105) and $R_i \in \mathfrak{n}^N$ with $N \geq 2k+8$, finite determinacy [GLS07, Theorem I.2.23 and Corollary I.2.24] implies the following.

- If $\nu_0 = 0$, then the germ
$$(Y_i \cap (s = 0) \ni 0) = (x'^2 + z'y'^2 + z'^{k-1} + R_i|_{s=0} = 0)$$
is of type D_k.
- If $\nu_0 = 1$, then the germ
$$(Y_i \cap (s - z' = 0) \ni 0) = (x'^2 + z'y'^2 + z'^k + R_i|_{s=z'} = 0)$$
is of type D_{k+1}.

If $\nu_0 = 0$, a point of S_i at which Y_i is singular is a singular point of the Cartier divisor S_i, so $\mathrm{Sing}(Y_i) \cap S_i = \{0\}$ near the origin. If $\nu_0 = 1$, let $g^0 := x'^2 + z'y'^2 + sz'^{k-1}$. The ideal generated

by s, g^0, and the four partial derivatives of g^0 contains

$$(s, x', z'y', y'^2, z'^{k-1}) \supset \mathfrak{n}^{k-1}.$$

Replacing g^0 with $g^0 + R_i$ changes g^0 and the four partial derivatives of g^0 by elements of $\mathfrak{n}^{N-1}$, and $N - 1 \geq k$. Thus, by Nakayama's lemma, the ideal generated by s, g_i, and the partial derivatives of g_i contains $\mathfrak{n}^{k-1}$, so $\operatorname{Sing}(Y_i) \cap S_i = \{0\}$ near the origin. Thus $(Y_i \ni 0)$ is a hypersurface germ with an isolated singularity and a Du Val hyperplane section, hence normal and cDV, hence Gorenstein terminal.

With the weights w on (x', y', z'), by (F.105) the restriction $g_i|_{s=0} \in \mathbb{C}[[x', y', z']]$ has $\operatorname{in}_w(g_i|_{s=0}) = x'^2 + z'y'^2 + z'^{k-1}$ if $\nu_0 = 0$ and $\operatorname{in}_w(g_i|_{s=0}) = x'^2 + z'y'^2$ if $\nu_0 = 1$, since R_i has w-weight at least $2N > 2k - 2$. Both are irreducible polynomials, so $g_i|_{s=0}$ is irreducible and S_i is reduced and irreducible near the origin, so S_i is a prime divisor, as claimed in (1), and $\operatorname{div}(s|_{Y_i}) = S_i$. The Jacobian determinant of (F.103) is $s^{(2k-1)r+2d}$ and $f_i = s^{2(k-1)r+2d} g_i$, so by adjunction $K_{Y_i} = \pi_i^* K_{W_i} + rS_i$. Since $h_i = s^{mr} l_i$,

$$\pi_i^* \operatorname{div}(h_i) = \operatorname{div}(l_i) + mrS_i.$$

Thus

$$K_{Y_i} + \frac{1}{m} \operatorname{div}(l_i) = \pi_i^* \left(K_{W_i} + \frac{1}{m} \operatorname{div}(h_i) \right), \tag{F.106}$$

and $\frac{1}{m} \operatorname{div}(l_i) \geq 0$ since l_i is regular. Since Q_i is w-homogeneous with x'-degree at most 1 and $\operatorname{in}_w(g_i|_{s=0})$ is monic of degree 2 in x', Q_i is not divisible by $g_i|_{s=0}$. Thus $l_i/s^c = Q_i + sl_i'$ does not vanish on S_i, so $\operatorname{mult}_{S_i} l_i = c$. Since $\pi_i(S_i) = o_i$, by (F.106) $a(S_i, W_i, \frac{1}{m} \operatorname{div}(h_i)) = 1 - \frac{c}{m}$, which is positive because $(W_i, \frac{1}{m} \operatorname{div}(h_i))$ is klt near o_i. Thus $0 \leq c < m$. This proves (1).

Proof of (2) and of the fourth sentence of (4). Let E be a prime divisor over Y_i with center the origin. Since π_i maps the origin to o_i and is G-equivariant and birational, E is a prime divisor over W_i with center o_i, E defines one valuation of the common function field of Y_i and W_i, hence the inertia groups of E in G over Y_i and over W_i are equal, and (F.106) together with $s = \pi_i^* s$ implies

$$a(E, W_i, 0) = a(E, Y_i, 0) + r \operatorname{mult}_E s \quad \text{and} \quad \operatorname{mult}_E h_i = \operatorname{mult}_E l_i + mr \operatorname{mult}_E s.$$

Let F be a computing lift of b_i. Removing finitely many indices, we may assume

$$\min\{\omega_i, N_1 + i\} > ((k-1)r + d + 1) \cdot 3M/\eta_m$$

for any $i \in \mathcal{I}$. By (F.102),

$$\operatorname{mult}_F x' = \operatorname{mult}_F x - ((k-1)r + d) \operatorname{mult}_F s \geq \min\{\omega_i, N_1 + i\} - ((k-1)r + d) \frac{3M}{\eta_m} > 0,$$

and similarly

$$\operatorname{mult}_F y' = \operatorname{mult}_F y - ((k-2)r + d) \operatorname{mult}_F s > 0 \quad \text{and} \quad \operatorname{mult}_F z' = \operatorname{mult}_F z - 2r \operatorname{mult}_F s > 0.$$

Thus $\min\{\operatorname{mult}_F x', \operatorname{mult}_F y', \operatorname{mult}_F z'\} \to +\infty$, and F has center the origin of Y_i. By (F.106),

$$a\left(E, Y_i, \frac{1}{m} \operatorname{div}(l_i)\right) = a\left(E, W_i, \frac{1}{m} \operatorname{div}(h_i)\right)$$

for any prime divisor E over Y_i with center the origin. Thus, by the first paragraph of this proof of (2),

$$\operatorname{mld}\left(\left(Y_i \ni 0, \frac{1}{m} \operatorname{div}(l_i)\right)/G\right) = b_i,$$

the minimum being attained exactly by the computing lifts of b_i. This proves (2) and the fourth sentence of (4). The inequality $\operatorname{mult}_E s/e_E \leq 3/\eta_m$ of (3) is the first inequality of (F.102).

Step 5. In this step we prove the limit $\min_E \operatorname{mult}_E \operatorname{Jac}_{Y_i} \to +\infty$ of (3).

Let $Y_* := \operatorname{Spec} \Bbbk[[x', y', z', s]]/(g_*)$ and $S_* := (s = 0) \subset Y_*$. Since

$$g_* \equiv x'^2 + z'y'^2 + s^{\nu_0} z'^{k-1} \mod \mathfrak{n}_*^N,$$

the argument for (1) shows that Y_* has a Du Val section through the closed point, so $\dim \operatorname{Sing}(Y_*) \le 1$ and Y_* is normal. Let

$$B := \Bbbk[[x, y, z, s]][x', y', z']/(x - s^{(k-1)r+d}x', y - s^{(k-2)r+d}y', z - s^{2r}z')$$

be the ring of the chart (F.103). Then $g_* \in B$, $B[s^{-1}] = \Bbbk[[x, y, z, s]][s^{-1}]$, so

$$\operatorname{Spec} B/(g_*) \backslash (s = 0) \cong W_* \backslash (s = 0),$$

and the completion of B at (x', y', z', s) is $\Bbbk[[x', y', z', s]]$, so Y_* is the completion of $\operatorname{Spec} B/(g_*)$ at the origin. Since $Y_* \to \operatorname{Spec} B/(g_*)$, the completion of the excellent ring $B/(g_*)$, is a regular morphism, along which log resolutions pull back to log resolutions and relative canonical divisors are preserved, $B/(g_*)$ being a hypersurface ring and hence Gorenstein, and along which multiplier ideals are preserved (cf. [dFEM11, Proposition 2.11 and Remarks 2.12–2.13], and [JM12, Section 1.3] for multiplier ideals), the non-klt points of $(Y_*, \frac{1}{m}\operatorname{div}(l_*))$ outside S_* are contained in the preimage of C, that is, the s-axis of the chart, which is contained in $\operatorname{Sing}(Y_*)$ since W_* is singular along C. Thus

$$\operatorname{Nklt}\left(Y_*, \frac{1}{m}\operatorname{div}(l_*)\right) \subset V_* := \operatorname{Sing}(Y_*) \cup S_*.$$

Let $\lambda' := 1/\kappa_{k+\nu_0}$. As in **Step 3**, choose a rational number $\mu_0' > \frac{1}{m}$ such that $(Y_*, \mu_0' \operatorname{div}(l_*))$ is klt on $Y_* \backslash V_*$, and rational numbers $0 < \theta_1' < \theta_2' < 1$ such that

$$\varpi' := \theta_1'\mu_0' - \frac{1}{m} - (1 - \theta_1')\left(1 + \frac{2\lambda'}{m}\right) > 0.$$

Let

$$\mathcal{J}_*' := \mathcal{J}(g_*^{\theta_2'} l_*^{\theta_2'\mu_0'}).$$

Let F be a computing lift of b_i. By (2), $a(F, Y_i, 0) \le \frac{1}{m}\operatorname{mult}_F l_i + 3M$. By Theorem 8.4 and the section of Y_i of type $D_{k+\nu_0}$ given by $(s = 0)$ if $\nu_0 = 0$ and by $(s - z' = 0)$ if $\nu_0 = 1$, shown in **Step 4**, $j_i' := \operatorname{mult}_F \operatorname{Jac}_{Y_i} \le \lambda' a(F, Y_i, 0)$. By (2), (F.100) and (F.102),

$$\operatorname{mult}_F l_i = \operatorname{mult}_F h_i - mr \operatorname{mult}_F s \ge \frac{m\delta}{1 - \tau_i} - \frac{3mrM}{\eta_m} \to +\infty.$$

By Lemma 6.1(3) with $V = V_*$, $\mathcal{J}_*'$ is the unit ideal near any point of $Y_* \backslash V_*$. Since V_* is the zero locus of g_* and $s\partial g_*/\partial v$, $v \in \{x', y', z', s\}$, there exists a positive integer P' such that

$$\left(s\frac{\partial g_*}{\partial v}\right)^{P'} \in (g_*) + \mathcal{J}_*' \quad \text{for any } v \in \{x', y', z', s\}.$$

Let ρ be a positive integer. As in **Step 3**, with $(g_*, l_*, \mathcal{J}_*')$ in place of $(f_*, h_*, \mathcal{J}_*)$ and $(s\partial g_*/\partial v)^{P'}$ in place of σ, let A_v be defined as A_σ. Let $\Phi_v := (sy_{1,e_v})^{P'} - A_v y_{1,0}$, where y_{1,e_v} is the variable substituted by $\partial g/\partial v$. By Lemma 6.1(1) and (2), applied to $(g_i, l_i)_{i \in \mathcal{I}}$ with the exponents $(\theta_1', \theta_1'\mu_0') < (\theta_2', \theta_2'\mu_0')$ and $L = \rho$, for all but finitely many $i \in \mathcal{I}$ there exist elements $z_{v,i}$ with

$$z_{v,i}|_{Y_i} = \left(s\frac{\partial g_i}{\partial v}\right)^{P'}\Big|_{Y_i}$$

such that any prime divisor E over $\operatorname{Spec} \mathbb{C}[[x', y', z', s]]$ with center the closed point and $\operatorname{mult}_E z_{v,i} < \rho$ satisfies

$$\operatorname{mult}_E z_{v,i} + a(E, \operatorname{Spec} \mathbb{C}[[x', y', z', s]], 0) > \theta_1' \operatorname{mult}_E g_i + \theta_1'\mu_0' \operatorname{mult}_E l_i.$$

Choose v with $\operatorname{mult}_F(\partial g_i/\partial v|_{Y_i}) = j_i'$, so that

$$\operatorname{mult}_F(z_{v,i}|_{Y_i}) = P'(\operatorname{mult}_F s + j_i').$$

If $\operatorname{mult}_F(z_{v,i}|_{Y_i}) < \rho$, then Lemma 5.5(1), with $(f,h,g,\theta,\mu) = (g_i, l_i, z_{v,i}, \theta_1', \mu_0')$ and the threshold ρ, and the computation of **Step 3** with $(W_i, h_i, j_i, \lambda, \mu_0, \theta_1)$ replaced by $(Y_i, l_i, j_i', \lambda', \mu_0', \theta_1')$ give

$$\varpi' \operatorname{mult}_F l_i < (2-\theta_1')P'(\operatorname{mult}_F s + j_i') + 3M + (1-\theta_1')(1+6\lambda' M).$$

Thus

$$P'(\operatorname{mult}_F s + j_i') \geq \min\left\{\rho, \frac{\varpi' \operatorname{mult}_F l_i - 3M - (1-\theta_1')(1+6\lambda' M)}{2-\theta_1'}\right\}.$$

Since $\operatorname{mult}_F s \leq 3M/\eta_m$ by (F.102), $\operatorname{mult}_F l_i \to +\infty$ and ρ is arbitrary, $\min_F j_i' \to +\infty$, the minimum over the computing lifts F of b_i. This proves (3). □

Proof of Theorem 17.9. We divide the proof into five steps. Let

$$J := 1 + \max\{6M, 2mM, m, 4\}.$$

Let (x,y,z,s) be the G-eigenparameters of Proposition 17.8, in which

$$f_* = x^2 + zy^2 + z^{k-1}\big(s^\nu + z\phi_1(z,s)\big).$$

In $f_*(x,0,0,0) = x^2$ the coefficients of 1 and x vanish and the coefficient of x^2 does not. By Lemma 2.7 without further variables, possibly removing finitely many indices, $f_i(x,0,0,0)$ has order 2 in x for any i. As in Proposition 17.8(4), we take h_* and the h_i of x-degree at most 1, replacing them by their remainders of the Weierstrass division by f_* and f_i with respect to x. The remainders differ from h_* and h_i by multiples of f_* and f_i, and (f_*, h_*) is still a compatible generic limit of the (f_i, h_i), since any jet of a remainder is a regular function of bounded jets of the dividend and of the divisor.

Step 1. In this step we prove that m is even and that

$$\text{(F.107)} \qquad \operatorname{in}_w(h_*) = \lambda(s) z^{m/2} \quad \text{or} \quad \operatorname{in}_w(h_*) = \lambda(s)\big(y + \sigma\sqrt{-1}s^{\nu/2}z\big)^{m/2},$$

where $\lambda(s) \in \Bbbk[[s]]\backslash\{0\}$, $\sigma \in \{1,-1\}$, and the second form occurs only if $k=4$ and ν is even.

Let $\omega \in \mathbb{K}$ with $\omega^{2(k-1)} = s^\nu$, where $\mathbb{K}$ is an algebraic closure of $\Bbbk((s))$. The substitution $(y,z) \to (\omega y, \omega^{-2} z)$ transforms

$$x^2 + zy^2 + s^\nu z^{k-1} \quad \text{into} \quad x^2 + zy^2 + z^{k-1}$$

and preserves w-weights, so S^0 is a Du Val surface of type D_k. By [Roa04, Table 1 and (3)], there exist coordinates (v_1, v_2) on $\mathbb{A}^2_{\mathbb{K}}$, the binary dihedral group $\Gamma \subset SL_2$ of order $4(k-2)$ generated by

$$\rho(v_1,v_2) = (\zeta v_1, \zeta^{-1} v_2) \quad \text{and} \quad \iota(v_1,v_2) = (v_2, -v_1),$$

where ζ is a primitive $2(k-2)$-th root of unity, and a quotient morphism $\pi : \mathbb{A}^2 \to S^0$ by Γ, such that π^*x, π^*y, π^*z are Γ-invariant forms of degrees $2(k-1)$, $2(k-2)$, 4 generating the ring of Γ-invariants. Thus π^* sends a w-homogeneous polynomial of weight j to a form of degree $2j$. In particular, since $\operatorname{in}_w(h_*)$ has weight m by Proposition 17.8(4), $\pi^* \operatorname{in}_w(h_*)$ is a Γ-invariant form of degree $2m$. Since no element of $\Gamma\backslash\{1\} \subset SL_2$ has eigenvalue 1, Γ acts freely on $\mathbb{A}^2\backslash\{0\}$ and π is étale over $S^0\backslash\{0\}$.

Let Q be a coefficient-one component of $\frac{1}{m}\operatorname{div}(\operatorname{in}_w(h_*))$ on S^0, so that $\operatorname{mult}_Q \operatorname{in}_w(h_*) = m$. Since π is étale over the generic point of Q, the prime divisors in $\pi^{-1}(Q)$ have coefficient m in $\operatorname{div}(\pi^* \operatorname{in}_w(h_*))$, and they form one Γ-orbit of lines through 0. No line is Γ-stable: otherwise, by Maschke's theorem, Γ would also stabilize a complementary line and would be abelian, whereas $\rho\iota \neq \iota\rho$. Thus $\pi^{-1}(Q)$ consists of at least two lines, each with coefficient m in $\operatorname{div}(\pi^* \operatorname{in}_w(h_*))$, whose degree is $2m$. Hence $\pi^{-1}(Q)$ consists of exactly two lines, and

$$\pi^* \operatorname{in}_w(h_*) = \mu q^m$$

for some $\mu \in \mathbb{K}\backslash\{0\}$ and a quadratic form q whose zero set is a Γ-stable pair of distinct lines. Let $\overline{\rho}, \overline{\iota}$ be the images of ρ, ι in PGL_2.

- If $k \geq 5$, then $\overline{\rho}$ has order $k-2 \geq 3$, and $\overline{\rho}^j \neq 1$ fixes exactly the lines $(v_1 = 0)$ and $(v_2 = 0)$ for any $0 < j < k-2$. Since the pair is Γ-stable, $\overline{\rho}$ permutes its two lines, so $\overline{\rho}^2$ fixes each of them, and $0 < 2 < k-2$. Thus $q = v_1 v_2$ up to a constant.
- If $k = 4$, then $\langle \overline{\rho}, \overline{\iota} \rangle \cong (\mathbb{Z}/2)^2$. The stabilizer of one line of the pair has index 2, so it is generated by an involution, which fixes that line and, preserving the pair, also the other one. As an involution of $\mathbb{P}^1$ has exactly two fixed points, the pair is the fixed locus of $\overline{\rho}$, $\overline{\iota}$, or $\overline{\rho\iota}$, and q is a constant multiple of $v_1 v_2$, $v_1^2 + v_2^2$, or $v_1^2 - v_2^2$.

In each case $\rho^* q = \pm q$ and $\iota^* q = \pm q$, with at least one minus sign: $\iota^*(v_1 v_2) = -v_1 v_2$ and $\rho^*(v_1^2 \pm v_2^2) = -(v_1^2 \pm v_2^2)$. Since μq^m is Γ-invariant, $(-1)^m = 1$, so m is even, and q^2 is a Γ-invariant form of degree 4. Hence $q^2 = \pi^* \alpha$ for a w-homogeneous polynomial α of weight 2, that is, $\alpha \in \mathbb{K}z$ if $k \geq 5$ and $\alpha \in \mathbb{K}y + \mathbb{K}z$ if $k = 4$. Then $\pi^*(\mathrm{in}_w(h_*) - \mu \alpha^{m/2}) = 0$, so $\mathrm{in}_w(h_*) - \mu \alpha^{m/2}$, a polynomial of x-degree at most 1, is divisible by $x^2 + zy^2 + s^\nu z^{k-1}$ and hence vanishes:

$$\mathrm{in}_w(h_*) = \mu \alpha^{m/2}.$$

Assume that $k = 4$. Since

$$\pi^* \operatorname{div}(\alpha|_{S^0}) = \operatorname{div}(q^2)$$

is twice a reduced divisor and π is étale in codimension 1, $\operatorname{div}(\alpha|_{S^0})$ is not reduced. Let $(y_0, z_0) \neq (0,0)$ with $\alpha(y_0, z_0) = 0$. On the plane $(\alpha = 0)$, with coordinates (x, ϖ) given by $(y, z) = \varpi(y_0, z_0)$, the equation of S^0 restricts to

$$x^2 + \varpi^3 z_0 \big(y_0^2 + s^\nu z_0^2\big),$$

which is reduced unless $z_0(y_0^2 + s^\nu z_0^2) = 0$. Hence, for a fixed square root $s^{\nu/2}$ of s^ν in $\mathbb{K}$, α is a constant multiple of z or of $y + \sigma\sqrt{-1} s^{\nu/2} z$ with $\sigma \in \{1, -1\}$. If ν is odd, then s^ν is not a square in $\Bbbk((s))$, and an automorphism of $\mathbb{K}$ over $\Bbbk((s))$ sends $s^{\nu/2}$ to $-s^{\nu/2}$. It fixes $\mathrm{in}_w(h_*) \in \Bbbk[[s]][y, z]$, so by unique factorization in $\mathbb{K}[y, z]$, if α were proportional to $y + \sigma\sqrt{-1} s^{\nu/2} z$, then $y + \sigma\sqrt{-1} s^{\nu/2} z$ would be proportional to $y - \sigma\sqrt{-1} s^{\nu/2} z$, which is not the case. Thus α is proportional to z unless $k = 4$ and ν is even, in which case $s^{\nu/2} \in \Bbbk[[s]]$ is the monomial.

Let $\lambda(s)$ be the coefficient of $z^{m/2}$, respectively of $y^{m/2}$, in $\mathrm{in}_w(h_*)$. Then $\lambda(s) \in \Bbbk[[s]] \backslash \{0\}$ and (F.107) holds. We write $\lambda(s) = s^{n_0} u(s)$ with $u(0) \neq 0$.

Step 2. In this step we write the marks on the charts of Proposition 17.8 as $s^{c_0} l_i^\circ$ with $l_i^\circ - z'^{m/2} \in \mathfrak{n}_i^J$. Let $c_0 := n_0$ in the first case of (F.107) and $c_0 := n_0 + \nu m/4$ in the second. We use Proposition 17.8 with $N := \max\{c_0 + J, 2k + 8\}$, which is the integer N of Theorem 17.9(2), let $r \geq N$ be the integer given there, and pass to the infinite subfamily given there. Here h_i is still the remainder fixed at the beginning of the proof, and l_i is the mark of Proposition 17.8(1), formed with the lift given there and the chart

$$x = s^{(k-1)r+d} x', \quad y = s^{(k-2)r+d} y', \quad \text{and} \quad z = s^{2r} z'.$$

Let $\mathcal{X}_i := \operatorname{Spec} \mathbb{C}[[x', y', z', s]]$, with maximal ideal $\mathfrak{n}_i$. Then G_i acts on $\mathcal{X}_i$ diagonally in x', y', z', s, and $Y_i = (g_i = 0) \subset \mathcal{X}_i$ for the equation g_i of Proposition 17.8(1), which satisfies

$$g_i - \big(x'^2 + z' y'^2 + s^{\nu_0} z'^{k-1}\big) \in \mathfrak{n}_i^N,$$

with the integers $d = \lfloor \nu/2 \rfloor$ and $\nu_0 = \nu - 2d$ of Proposition 17.8.

For $j = (j_1, j_2, j_3) \in \mathbb{N}^3$ let $\mathrm{wt}_w(j) := (k-1)j_1 + (k-2)j_2 + 2j_3$, the w-weight of the monomial $x^{j_1} y^{j_2} z^{j_3} s^n$. Then

$$s^{-mr} x^{j_1} y^{j_2} z^{j_3} s^n = s^{r(\mathrm{wt}_w(j) - m) + d(j_1 + j_2) + n} x'^{j_1} y'^{j_2} z'^{j_3}. \tag{F.108}$$

Let $z'' := z'$ in the first case of (F.107) and $z'' := y' + \sigma\sqrt{-1} z'$ in the second. In the second case $y + \sigma\sqrt{-1} s^{\nu/2} z = s^{2r + \nu/2} z''$. Thus, by (F.108), $s^{-mr} \mathrm{in}_w(h_*) = s^{c_0} u(s) z''^{m/2}$.

Consider the following conditions on h_i.

(1) The coefficient of $x^{j_1} y^{j_2} z^{j_3} s^n$ in h_i vanishes whenever $\mathrm{wt}_w(j) < m$ and $n < rm + c_0 + J$.

(2) The sum of the terms of h_i with $\mathrm{wt}_w(j) = m$ and $n < c_0 + J$, after the substitution (F.108), is congruent modulo s^{c_0+J} to $s^{c_0}u_i(s)z''^{m/2}$ for a polynomial $u_i(s)$ with $u_i(0) \neq 0$.

They are conditions on finitely many coefficients of h_i. Condition (1) and the congruence in (2) are closed, and $u_i(0) \neq 0$ is open. By Proposition 17.8(4), h_* has no monomial of w-weight less than m, and its part of w-weight m is $\mathrm{in}_w(h_*)$, so h_* satisfies (1) and (2), with u_i the truncation of u in degree less than J. The monomials in (1) and (2) have degree less than $rm + c_0 + J + m$, since each entry of w is at least 2. By Lemma 2.7 without further variables, for the jets of order $rm + c_0 + J + m$, possibly removing finitely many indices, (1) and (2) hold for any i. Let θ be the exponent of s in (F.108). Then $\theta \geq r \geq c_0 + J$ if $\mathrm{wt}_w(j) \geq m + 1$, $\theta \geq n - rm \geq c_0 + J$ if $\mathrm{wt}_w(j) < m$ and $n \geq rm + c_0 + J$, and $\theta \geq c_0 + J$ if $\mathrm{wt}_w(j) = m$ and $n \geq c_0 + J$. Hence the following function is regular on $\mathcal{X}_i$, and its restriction to Y_i is that of l_i, since the restrictions to W_i of h_i and of the lift of Proposition 17.8 are equal:

$$s^{-mr}h_i\big(s^{(k-1)r+d}x', s^{(k-2)r+d}y', s^{2r}z', s\big) = s^{c_0}\big(u_i(s)z''^{m/2} + T_i\big), \tag{F.109}$$

where $T_i \in s^J\mathbb{C}[[x', y', z', s]]$. The left hand side of (F.109) is G_i-semi-invariant. For $n < J$, the coefficient of the monomial $s^{c_0+n}z'^{m/2}$ (first case), respectively $s^{c_0+n}y'^{m/2}$ (second case), in the left hand side of (F.109) equals the coefficient of s^n in $u_i(s)$. Comparing with $n = 0$, $\chi_s^n = 1$ whenever s^n occurs in $u_i(s)$, so $u_i(s)$ is a G_i-invariant unit. Let

$$l_i^\circ := z''^{m/2} + u_i(s)^{-1}T_i, \qquad l_i^\circ - z''^{m/2} \in s^J\mathbb{C}[[x', y', z', s]] \subset \mathfrak{n}_i^J.$$

Replacing l_i with $s^{c_0}l_i^\circ$ changes neither the divisor of the mark nor any mult_E, so Proposition 17.8(2) holds with $s^{c_0}l_i^\circ$ in place of l_i, and l_i° is G_i-semi-invariant.

In the second case, $k = 4$, ν is even and $\nu_0 = 0$. The monomials $y'^{m/2}$ and $y'^{m/2-1}z'$ of l_i° have non-zero coefficients, so $\chi_{y'} = \chi_{z'}$ and G_i acts on $\mathbb{C}y' + \mathbb{C}z'$ by scalars. In particular, any linear change of (y', z') is G_i-equivariant. We have

$$z'y'^2 + z'^3 = z'' \cdot z'\big(y' - \sigma\sqrt{-1}z'\big),$$

and no two of the linear forms z', $y' - \sigma\sqrt{-1}z'$, and z'' are proportional. Let y'' be a linear form with $\mathbb{C}y'' + \mathbb{C}z'' = \mathbb{C}y' + \mathbb{C}z'$. Then

$$z'(y' - \sigma\sqrt{-1}z') = \beta_1 y''^2 + \beta_2 y''z'' + \beta_3 z''^2$$

with $\beta_1 \neq 0$, and completing the square,

$$z'(y' - \sigma\sqrt{-1}z') = \beta_1\Big(y'' + \frac{\beta_2}{2\beta_1}z''\Big)^2 + \Big(\beta_3 - \frac{\beta_2^2}{4\beta_1}\Big)z''^2, \qquad \beta_3 - \frac{\beta_2^2}{4\beta_1} \neq 0.$$

After replacing (y', z') with constant multiples of $y'' + \frac{\beta_2}{2\beta_1}z''$ and z'', and l_i° with a constant multiple, the equation g_i satisfies $g_i - (x'^2 + z'y'^2 + z'^3) \in \mathfrak{n}_i^N$, and $l_i^\circ - z'^{m/2} \in s^J\mathbb{C}[[x', y', z', s]]$. Hence in either case of (F.107)

$$\begin{aligned} &g_i - \big(x'^2 + z'y'^2 + s^{\nu_0}z'^{k-1}\big) \in \mathfrak{n}_i^J, \\ &\qquad\qquad \text{and} \qquad l_i^\circ - z'^{m/2} \in s^J\mathbb{C}[[x', y', z', s]] \subset \mathfrak{n}_i^J, \end{aligned} \tag{F.110}$$

where G_i acts on $\mathcal{X}_i$ diagonally in x', y', z', s, and the mark on Y_i is $s^{c_0}l_i^\circ$.

Since $l_i^\circ - z'^{m/2} \in s^J\mathbb{C}[[x', y', z', s]]$, the restriction of l_i° to $S_i = (s = 0)$ is $z'^{m/2}$. It does not vanish identically on S_i, since the restriction of g_i to $(s = z' = 0)$ is congruent to x'^2 modulo $(x', y')^J$, hence non-zero. By Proposition 17.8(1), S_i is a prime divisor, so $\mathrm{mult}_{S_i} s = 1$. Thus $\mathrm{mult}_{S_i}(s^{c_0}l_i^\circ) = c_0$, so $c_0 = c$, the integer of Proposition 17.8(1). In particular, $0 \leq c < m$.

Step 3. In this step we prove that

$$\Big(Y_i, \frac{1}{m}\,\mathrm{div}(l_i^\circ) + \mathrm{div}(s)\Big) \text{ is lc near } 0. \tag{F.111}$$

First we prove that, on $\mathbb{A}^4$ with coordinates (x', y', z', s),

$$\Big(\mathbb{A}^4, \operatorname{div}(x'^2 + z'y'^2) + \frac{1}{2}\operatorname{div}(z') + \operatorname{div}(s)\Big) \text{ is lc near } 0. \tag{F.112}$$

By [EMY03, Corollary 1.8] with the prime divisor $(s = 0)$ it suffices that

$$\Big(\mathbb{A}^3, \operatorname{div}(x'^2 + z'y'^2) + \frac{1}{2}\operatorname{div}(z')\Big)$$

is lc near 0, which one checks on the blow-up of the ideal (x', y'), where the log pullback is $\widetilde{S} + E_1 + \frac{1}{2}\widetilde{Z}$ for the exceptional divisor E_1 and the strict transforms $\widetilde{S}$ and $\widetilde{Z}$ of $(x'^2 + z'y'^2 = 0)$ and $(z' = 0)$.

Consider the weights $(2, 1, 2, 1)$ on (x', y', z', s). The monomials x'^2 and $z'y'^2$ have weight 4, $s^{\nu_0} z'^{k-1}$ has weight $2k - 2 + \nu_0 \geq 6$, and any element of $\mathfrak{n}_i^J$ has weight at least $J \geq 5$. The monomial $z'^{m/2}$ has weight $m < J$. Let

$$B_i := \operatorname{div}(g_i) + \frac{1}{m}\operatorname{div}(l_i^\circ) + \operatorname{div}(s) \quad \text{on } \mathcal{X}_i.$$

By (F.110), the lowest-weight parts of g_i, l_i°, and s are $x'^2 + z'y'^2$, $z'^{m/2}$, and s, and

$$\operatorname{div}(x'^2 + z'y'^2) + \frac{1}{m}\operatorname{div}(z'^{m/2}) + \operatorname{div}(s) = \operatorname{div}(x'^2 + z'y'^2) + \frac{1}{2}\operatorname{div}(z') + \operatorname{div}(s)$$

is the boundary of (F.112). By Lemma 6.4(3) with the trivial group, the weights $(2, 1, 2, 1)$, the functions (g_i, l_i°, s), and the coefficients $(1, \frac{1}{m}, 1)$, and by (F.112),

$$\operatorname{mld}(\mathcal{X}_i \ni 0, B_i) \geq 0.$$

Hence $(\mathcal{X}_i, B_i)$ is lc near 0.

By Proposition 17.8(1), the germ $(Y_i \ni 0)$, $Y_i = (g_i = 0)$, is an isolated Gorenstein terminal germ, in particular normal, and neither l_i° nor s vanishes identically on Y_i. Hence (F.111) follows from the lc of $(\mathcal{X}_i, B_i)$ near 0 by adjunction on a log resolution of $(\mathcal{X}_i, \operatorname{div}(g_i l_i^\circ s))$ [dFEM11, Theorems 2.3 and 2.4], on which the strict transform of Y_i is a log resolution of the pair in (F.111).

Step 4. In this step we prove that

$$b_i = \frac{1 - c/m}{R},$$

where R is the order of the character χ_s of G_i, and we find a prime divisor E over Y_i with center 0, computing b_i, with $a(E, Y_i, 0)/e_E \leq 3$.

Let F be a prime divisor over Y_i with center 0. By (F.111),

$$\begin{aligned} a\Big(F, Y_i, \frac{1}{m}\operatorname{div}(s^c l_i^\circ)\Big) &= a\Big(F, Y_i, \frac{1}{m}\operatorname{div}(l_i^\circ) + \operatorname{div}(s)\Big) + \Big(1 - \frac{c}{m}\Big)\operatorname{mult}_F s \\ &\geq \Big(1 - \frac{c}{m}\Big)\operatorname{mult}_F s. \end{aligned} \tag{F.113}$$

The function s^R is G_i-invariant, so it is the pullback of a regular function on Y_i/G_i vanishing at the image of 0. On G_i-invariant functions, mult_F is e_F times the valuation of the image of F over Y_i/G_i, which takes positive integer values on functions vanishing at the image of 0. Thus $R \operatorname{mult}_F s \geq e_F$. By (F.113) and Proposition 17.8(2) (and Lemma 2.12(2)(3)),

$$b_i = \operatorname{mld}\Big(\big(Y_i \ni 0, \tfrac{1}{m}\operatorname{div}(s^c l_i^\circ)\big)/G_i\Big) = \inf_F \frac{a(F, Y_i, \frac{1}{m}\operatorname{div}(s^c l_i^\circ))}{e_F} \geq \frac{1 - c/m}{R},$$

the infimum over the prime divisors F over Y_i with center 0.

Let $\kappa \in G_i$ with $\chi_s(\kappa)$ of order R, and let e be the order of κ. Then $R \mid e$ and $e \leq |G_i| \leq M$. Since $\chi_s(\kappa)$ has order R and $R \mid e$, there exists a primitive e-th root of unity ξ with $\chi_s(\kappa) = \xi^{e/R}$. Let $w_{y'} \in \{1, \dots, e\}$ and $w' \in \{e + 1, \dots, 2e\}$ be the integers with

$$\chi_{y'}(\kappa) = \xi^{w_{y'}}, \qquad \chi_{x'}(\kappa)\chi_{y'}(\kappa)^{-1} = \xi^{w'}.$$

Since g_i is G_i-semi-invariant and both x'^2 and $z'y'^2$ occur in g_i, $\chi_{z'} = \chi_{x'}^2\chi_{y'}^{-2}$. We use the weights

$$(w_{x'}, w_{y'}, w_{z'}, w_s) := \Big(w_{y'} + w', w_{y'}, 2w', \frac{e}{R}\Big) \quad \text{on } (x', y', z', s), \tag{F.114}$$

so that $\chi_V(\kappa) = \xi^{w_V}$ for any $V \in \{x', y', z', s\}$. The monomials x'^2 and $z'y'^2$ have weight $2w_{x'}$. Since $k \geq 4$ and $w' > e \geq w_{y'}$, the monomial $s^{\nu_0}z'^{k-1}$ has weight at least $6w' > 2w_{x'}$. Since $2w_{x'} \leq 6e \leq 6M < J$, $mw' \leq 2me \leq 2mM < J$ and all weights are positive integers, (F.110) implies that the lowest-weight parts of g_i and l_i° are $x'^2 + z'y'^2$ and $z'^{m/2}$.

We prove that the vector (F.114) is primitive. Let p be a positive common divisor of $w_{x'}, w_{y'}, w_{z'}, w_s$. Then $p \mid w_s \mid e$, and $\chi_V(\kappa^{e/p}) = \xi^{w_V e/p} = 1$ for any $V \in \{x', y', z', s\}$. So $\kappa^{e/p}$ acts trivially on $\mathcal{X}_i$, hence on Y_i. Since G_i acts faithfully on Y_i by Proposition 17.8(1), $\kappa^{e/p} = 1$ and $p = 1$.

Let $\Psi_i : \mathcal{X}_i' \to \mathcal{X}_i$ be the weighted blow-up with weights (F.114), with exceptional divisor $\mathbb{P}(w_{x'}, w_{y'}, w_{z'}, w_s)$, and let $Y_i' := \Psi_{i*}^{-1}Y_i$. Since $x'^2 + z'y'^2$ is irreducible, the intersection

$$Y_i' \cap \mathbb{P}(w_{x'}, w_{y'}, w_{z'}, w_s) = (x'^2 + z'y'^2 = 0)$$

is irreducible. At a general point of $Y_i' \cap \mathbb{P}(w_{x'}, w_{y'}, w_{z'}, w_s)$, where $x'y'z's \neq 0$, $\mathcal{X}_i'$ is smooth because (F.114) is primitive, and Y_i' is smooth and intersects $\mathbb{P}(w_{x'}, w_{y'}, w_{z'}, w_s)$ transversally because

$$\partial(x'^2 + z'y'^2)/\partial x' = 2x' \neq 0.$$

Let E be the prime divisor over Y_i given by $Y_i' \cap \mathbb{P}(w_{x'}, w_{y'}, w_{z'}, w_s)$. Then E has center 0 and

$$\operatorname{mult}_E V = w_V \quad \text{for any } V \in \{x', y', z', s\}, \qquad \operatorname{mult}_E l_i^\circ = mw',$$

where the second equality holds since $z'^{m/2}$ is not divisible by $x'^2 + z'y'^2$. By adjunction

$$a(E, Y_i, 0) = w_{x'} + w_{y'} + w_{z'} + w_s - 2w_{x'} = w' + \frac{e}{R}, \qquad \operatorname{mult}_E(s^c l_i^\circ) = mw' + \frac{ce}{R}. \tag{F.115}$$

The residue field of E is the field of fractions of degree 0 of the graded ring $\mathbb{C}[x', y', z', s]/(x'^2+z'y'^2)$ with the weights (F.114). Since κ multiplies any homogeneous element of weight j by ξ^j, κ preserves E and acts trivially on the residue field of E. Thus $\langle\kappa\rangle \subset I_{G_i}(E)$ and $e_E \geq e$. By (F.115),

$$\frac{a(E, Y_i, \frac{1}{m}\operatorname{div}(s^c l_i^\circ))}{e_E} = \frac{(1 - c/m)e/R}{e_E} \leq \frac{1 - c/m}{R}.$$

Since $1 - c/m > 0$, together with the lower bound above, $b_i = (1 - c/m)/R$ and E computes b_i on Y_i. Moreover, since $w' \leq 2e$,

$$\frac{a(E, Y_i, 0)}{e_E} \leq \frac{w' + e/R}{e} \leq 2 + \frac{1}{R} \leq 3.$$

Step 5. In this step we compare E with W_i and conclude. By Proposition 17.8(2) (and Lemma 2.12(2)(3)), E is a prime divisor over W_i with center o_i whose inertia group over W_i is $I_{G_i}(E)$, and

$$a(E, W_i, 0) = a(E, Y_i, 0) + r \operatorname{mult}_E s, \qquad \operatorname{mult}_E h_i = \operatorname{mult}_E(s^c l_i^\circ) + mr \operatorname{mult}_E s.$$

Hence

$$a\Big(E, W_i, \frac{1}{m}\operatorname{div}(h_i)\Big) = a\Big(E, Y_i, \frac{1}{m}\operatorname{div}(s^c l_i^\circ)\Big) = b_i e_E,$$

so E is a computing lift of b_i. By Proposition 17.8(3), $\operatorname{mult}_E s/e_E \leq 3/\eta_m$. By **Step 4**,

$$\frac{a(E, W_i, 0)}{e_E} = \frac{a(E, Y_i, 0)}{e_E} + r\frac{\operatorname{mult}_E s}{e_E} \leq 3 + \frac{3r}{\eta_m}.$$

Thus $F_i := E$ is a computing lift of b_i with $a(F_i, W_i, 0)/e_{F_i} \leq 3 + 3r/\eta_m$. Together with **Steps 1 and 4**, this proves (1) and (2) of the theorem, and the last sentence follows from (2) by Lemma 4.3(2). □

Proof of Theorem 17.10, ***Steps 1–5****.* Suppose that a radial failure sequence $(W_i \ni o_i, G_i, h_i, \tau_i)$ satisfying the assumptions of Theorem 17.10 exists, in the notation of Set-up 17.1, with radial functions q_i. By Definition 4.1(3), $\lim_{i\to+\infty}\tau_i = 1$ and

$$q_i(\tau_i) \geq a \geq q_i(1) + \delta = b_i + \delta. \tag{F.116}$$

We derive a contradiction. For a finite group G acting on a germ $(X \ni x)$, a prime divisor E over X with $\operatorname{center}_X E = \{x\}$, and an $\mathbb{R}$-divisor Δ with coefficients of any sign, let

$$\varsigma_\Delta(E) := \frac{a(E, X, 0) - \operatorname{mult}_E \Delta}{e_E};$$

the quotient value $\operatorname{mld}((X \ni x, \Delta)/G)$ of Definition 2.2 is the infimum of $\varsigma_\Delta(E)$ over these E.

Step 1. In this step we fix the numerical data, express b_i on a smooth ambient germ, and construct ambient lifts of prime divisors.

By Lemma 4.2(1) and $|G_i| \leq M$, $b_i \in \frac{1}{m\cdot M!}\mathbb{Z} \cap (0, a-\delta]$, a finite set, so possibly passing to an infinite subsequence we may assume that $b_i = b$ for a fixed $b > 0$. For any i let E_i be a computing lift of b on $(W_i \ni o_i, \frac{1}{m}\operatorname{div}(h_i))$.

Let $w := (k-1, k-2, 2)$ be the weights on (x, y, z), with s of weight 0, and let

$$F_0 := x^2 + zy^2 + z^{k-1},$$

of w-weight $2k-2$. By Proposition 17.8, C is the s-axis and, after the substitution of (y, z) in Proposition 17.8, which we make,

$$f_* = x^2 + zy^2 + z^{k-1}\big(s^\nu + z\phi_1(z, s)\big)$$

for an integer $\nu \geq 0$ and $\phi_1 \in \Bbbk[[z, s]]$. Let $\mathbb{K}$ be an algebraic closure of $\Bbbk((s))$, let $d = \lfloor \nu/2 \rfloor$ and $\nu_0 = \nu - 2d$ be the integers of Proposition 17.8, write $\epsilon := \nu_0 \in \{0, 1\}$, let S^0 be the surface of Theorem 17.9,

$$S^0 = (x^2 + zy^2 + s^\nu z^{k-1} = 0) \subset \mathbb{A}^3_{\mathbb{K}},$$

and let $\operatorname{in}_w(h_*)$ be the lowest-weight part of Proposition 17.8(4). If $(S^0, \frac{1}{m}\operatorname{div}(\operatorname{in}_w(h_*)))$ has a coefficient-one component, then Theorem 17.9 shows that no radial failure sequence satisfies our assumptions. Thus we may assume that

$$\Big(S^0, \frac{1}{m}\operatorname{div}(\operatorname{in}_w(h_*))\Big) \text{ has no coefficient-one component.} \tag{F.117}$$

Let $(V_i \ni o_i)$ be the smooth G-invariant fourfold germ in which $W_i = (f_i = 0)$. We claim that for $\tau \in \{\tau_i, 1\}$ and any $\varpi \in \mathcal{O}_{V_i, o_i}$ with the character of h_i/f_i,

$$q_i(\tau) = \operatorname{mld}\Big(\Big(V_i \ni o_i, \operatorname{div}(f_i) + \frac{\tau}{m}\operatorname{div}(h_i + f_i\varpi)\Big)/G\Big). \tag{F.118}$$

Since $(h_i + f_i\varpi)|_{W_i} = h_i|_{W_i}$, we may assume that $\varpi = 0$. If V_i is an open subset of $\mathbb{A}^4$ with G diagonal, (F.118) is Theorem 2.17(1) with $c = 1$, $f_1 = f_i$, $H = G$, and $\mathfrak{a}$ the $\frac{\tau}{m}$-th power of the ideal of a semi-invariant polynomial lift of h_i. Otherwise we take finitely many eigenfunctions generating the algebraic local ring of V_i and obtain a G-equivariant embedding of V_i into a diagonal representation $\mathbb{A}^n$. Since V_i is smooth, the ideal of V_i near 0 is generated by a regular sequence $f_1, \dots, f_c$ of semi-invariant polynomials, and we let f_{c+1} be a semi-invariant polynomial lift of f_i. By Theorem 2.17(2) with $X = (f_1 = \cdots = f_c = 0)$, $S = W_i$, $H = G$, and $\mathfrak{a}$ the $\frac{\tau}{m}$-th power of the ideal of a semi-invariant polynomial lift of h_i (the mark is first lifted to a semi-invariant polynomial on $\mathbb{A}^n$), the claim follows, since the middle term of Theorem 2.17(2) is the right side of (F.118). In particular, any prime divisor $\widehat{E}$ over V_i with $\operatorname{center}_{V_i}\widehat{E} = \{o_i\}$ satisfies

$$\frac{a(\widehat{E}, V_i, 0) - \operatorname{mult}_{\widehat{E}} f_i - \frac{1}{m}\operatorname{mult}_{\widehat{E}}(h_i + f_i\varpi)}{e_{\widehat{E}}} \geq b.$$

Passing to the completion $\widehat{V}_i$ of V_i at o_i preserves the log discrepancies, orders, and inertia groups of prime divisors with center the closed point, so (F.118) holds also with $\widehat{V}_i$ in place of V_i, and the

inequality holds for any prime divisor $\widehat{E}$ over $\widehat{V}_i$ with center the closed point. By approximation, both hold for any $\varpi \in \widehat{\mathcal{O}}_{V_i,o_i}$ with the character of h_i/f_i.

We also need ambient lifts. Let $\mathcal{X}$ be $\mathbb{A}^4$ or $\operatorname{Spec}\mathbb{C}[[x,y,z,s]]$ with G diagonal on (x,y,z,s), let $W=(g=0)\subset\mathcal{X}$ be normal near 0 and G-invariant with G faithful on W, and let $h\in\mathcal{O}_{\mathcal{X},0}$ be semi-invariant with $h|_W\neq 0$ and $s|_W\neq 0$. Let E be a prime divisor over W with $\operatorname{center}_W E=\{0\}$. By Lemma 5.4(2) with $H_1=h$, $H_2=s$, and a coordinate $x_1\in\{x,y,z,s\}$ with

$$\operatorname{mult}_E(\partial g/\partial x_1)=\operatorname{mult}_E \operatorname{Jac}_W,$$

there exists a prime divisor $\widehat{E}$ over $\mathcal{X}$ with $\operatorname{center}_{\mathcal{X}}\widehat{E}=\{0\}$ such that

$$\begin{gathered}\operatorname{mult}_{\widehat{E}} g\le \operatorname{mult}_E h+\operatorname{mult}_E s+2\operatorname{mult}_E \operatorname{Jac}_W+e_E+2,\\ a(\widehat{E},\mathcal{X},0)=a(E,W,0)+\operatorname{mult}_{\widehat{E}} g,\\ \operatorname{mult}_{\widehat{E}} h=\operatorname{mult}_E h,\quad \operatorname{mult}_{\widehat{E}} s=\operatorname{mult}_E s,\quad I_G(\widehat{E})=I_G(E).\end{gathered}\tag{F.119}$$

By Lemma 5.4(2), such $\widehat{E}$ exists for any integer $\operatorname{mult}_{\widehat{E}} g$ in a fixed residue class modulo e_E with

$$\operatorname{mult}_{\widehat{E}} g>\operatorname{mult}_E h+\operatorname{mult}_E s+2\operatorname{mult}_E \operatorname{Jac}_W,$$

and any e_E consecutive integers contain such an integer, so some such $\widehat{E}$ satisfies the first inequality of (F.119). In particular, for any $\beta\in\mathbb{R}$ and the $\mathbb{R}$-divisors

$$\Delta_1:=\operatorname{div}(g)+\frac{1}{m}\operatorname{div}(h)+\beta\operatorname{div}(s)\quad\text{and}\quad \Delta_2:=\frac{1}{m}\operatorname{div}(h|_W)+\beta\operatorname{div}(s|_W)$$

on $\mathcal{X}$ and W, we have $\varsigma_{\Delta_1}(\widehat{E})=\varsigma_{\Delta_2}(E)$.

Step 2. In this step we pass, for $\epsilon\in\{0,1\}$, to a chart whose equation restricts to F_0 on $(s=0)$ and on which the boundary acquires a fixed signed axis coefficient β.

Fix integers $r\ge 2m+3$ and $N\ge 2(k-1)r+2k$, and write r_Y for the integer r of Proposition 17.8. By Proposition 17.8 applied with N, we obtain, for i in an infinite subfamily, the germs $(Y_i\ni 0)$ with coordinates (x',y',z',s), their equations $g_i=x'^2+z'y'^2+s^\epsilon z'^{k-1}+R_i$ with $R_i\in(x',y',z',s)^N$ of Proposition 17.8(1), and the marks l_i. By Proposition 17.8(4), the g_i and the l_i have a compatible generic limit (g_*,l_*) with $g_*=x'^2+z'y'^2+s^\epsilon z'^{k-1}+z'^k\psi(z',s)$, so g_* has no monomial of w-weight less than $2k-2$ and has part $x'^2+z'y'^2+s^\epsilon z'^{k-1}$ of w-weight $2k-2$, and the representative of l_* modulo g_* of x'-degree at most 1 has lowest w-weight part $\operatorname{in}_w(h_*)(s^dx',s^dy',z')$, of weight m. Since g_i has x'-order 2, the remainder l_i^{rem} of l_i under Weierstrass division by g_i satisfies $l_i^{\mathrm{rem}}|_{Y_i}=l_i|_{Y_i}$, and at each jet order it is a polynomial in the coefficients. Since the pull-back of f_i by the chart of Proposition 17.8 is a power of s times g_i, by uniqueness of the division $s^{mr_Y}l_i^{\mathrm{rem}}$ is the pull-back of the remainder h_i^{rem} of h_i under Weierstrass division by f_i with respect to x in $\widehat{\mathcal{O}}_{V_i,o_i}$ (for $i\gg 0$, f_i has x-order 2, as f_* does, by Lemma 2.7), and $h_i^{\mathrm{rem}}=h_i+f_i\varpi$ for some $\varpi\in\widehat{\mathcal{O}}_{V_i,o_i}$ with the character of h_i/f_i. Possibly replacing h_i with h_i^{rem} and l_i with l_i^{rem}, we may assume that l_* has x'-degree at most 1. Thus l_* has no monomial of w-weight less than m.

The case $\epsilon=0$. For a positive integer r', the *dilation of depth* r' of coordinates (x',y',z',s) is the substitution $x'=s^{(k-1)r'}\overline{x}$, $y'=s^{(k-2)r'}\overline{y}$, $z'=s^{2r'}\overline{z}$ with new coordinates $\overline{x},\overline{y},\overline{z}$. We apply the dilation of depth r:

$$x'=s^{(k-1)r}\overline{x},\quad y'=s^{(k-2)r}\overline{y},\quad\text{and}\quad z'=s^{2r}\overline{z}.$$

For a monomial $x'^{j_1}y'^{j_2}z'^{j_3}s^{j_4}$ of w-weight v, the exponent of s after substitution is $rv+j_4$. Let ϑ be the minimum of $r(v-m)+j_4$ over the monomials of l_*. Since l_* has w-lowest weight m, $\vartheta\ge 0$, and we let

$$\overline{f}_i:=s^{-(2k-2)r}g_i(s^{(k-1)r}\overline{x},s^{(k-2)r}\overline{y},s^{2r}\overline{z},s),\quad \overline{h}_i:=s^{-(mr+\vartheta)}l_i(s^{(k-1)r}\overline{x},s^{(k-2)r}\overline{y},s^{2r}\overline{z},s),$$

and $\beta := \vartheta/m$. The divisions by $s^{(2k-2)r}$ and $s^{mr+\vartheta}$ are regular for the limits, and their regularity is a finite set of vanishing conditions on jets, so, by Lemma 2.7, it holds for all but finitely many members.

The case $\epsilon = 1$. Let $s = u^2$, and let

$$G' := \{(\iota, \alpha) \mid \iota \in G,\ \alpha^2 = \chi_s(\iota)\}$$

act by α on u. The group G' is abelian and diagonal of order $2|G|$. By Proposition 17.8(1), $S_i = (s = 0)$ is a prime divisor on Y_i, so s is not a square in the function field of Y_i and the cover $u \to u^2$ of Y_i is connected of degree 2. We let

$$x' = u^{(k-1)r+1}\overline{x}, \quad y' = u^{(k-2)r+1}\overline{y}, \quad \text{and} \quad z' = u^{2r}\overline{z}.$$

For a monomial $x'^{j_1}y'^{j_2}z'^{j_3}s^{j_4}$ of w-weight v, the exponent of u after substitution is

$$rv + j_1 + j_2 + 2j_4.$$

Let ϑ be the minimum of $r(v-m) + j_1 + j_2 + 2j_4$ over the monomials of l_*. Thus $\vartheta \geq 0$. We let $\overline{f}_i$ be the substituted g_i divided by $u^{(2k-2)r+2}$, $\overline{h}_i$ the substituted l_i divided by $u^{mr+\vartheta}$, and $\beta := \vartheta/m - 1$. The divisions by $u^{(2k-2)r+2}$ and $u^{mr+\vartheta}$ are regular for the limits, and their regularity is a finite set of vanishing conditions on jets, so, by Lemma 2.7, it holds for all but finitely many members. From now on we write s for u and G for G', so $|G| \leq 2M$.

For $\epsilon \in \{0, 1\}$, let

$$\overline{\mathcal{X}}_i := \operatorname{Spec} \mathbb{C}[[\overline{x}, \overline{y}, \overline{z}, s]]$$

and $\overline{W}_i := (\overline{f}_i = 0) \subset \overline{\mathcal{X}}_i$. Since $r \geq 2m + 3$, any monomial of w-weight $v > 2k - 2$ acquires a positive power of s, and since $R_i \in (x', y', z', s)^N$ with $N \geq 2(k-1)r + 2k$, so does any monomial of R_i of w-weight at most $2k - 2$. Thus $\overline{f}_i(\overline{x}, \overline{y}, \overline{z}, 0) = F_0(\overline{x}, \overline{y}, \overline{z})$. Since F_0 is irreducible, $\overline{D}_i := \overline{W}_i \cap (s = 0)$ is a prime divisor on $\overline{W}_i$ with $\operatorname{mult}_{\overline{D}_i} s = 1$. Since $\overline{W}_i \cap (s = 0) = (F_0 = 0)$, the functions $\overline{x}$, $\overline{y}$, $\overline{z}$, s do not vanish identically on $\overline{W}_i$, so an element of G acting trivially on $\overline{W}_i$ fixes $\overline{x}, \overline{y}, \overline{z}, s$. For $\epsilon = 1$ such an element (ι, α) has $\alpha = 1$. Thus, for $\epsilon \in \{0, 1\}$, it fixes the coordinates x', y', z' of Y_i and the fourth coordinate of Y_i, which is s if $\epsilon = 0$ and s^2 if $\epsilon = 1$, so it is the identity, because the group of Proposition 17.8(1) acts faithfully on Y_i (and $\alpha = 1$ if $\epsilon = 1$). Thus G acts faithfully on $\overline{W}_i$.

Let E be a prime divisor over $\overline{W}_i$ with center in $(s = 0)$, and let E' be its restriction, a prime divisor over W_i, with ramification index ρ ($\rho = 1$ if $\epsilon = 0$). We claim that

$$\begin{aligned} e_E = \rho e_{E'}, \qquad & \rho a(E', W_i, 0) = a(E, \overline{W}_i, 0) + \kappa_0 \operatorname{mult}_E s, \\ & \frac{\rho}{m} \operatorname{mult}_{E'} h_i = \frac{1}{m} \operatorname{mult}_E \overline{h}_i + (\kappa_0 + \beta) \operatorname{mult}_E s, \end{aligned} \tag{F.120}$$

where $\kappa_0 := r_Y + r$ if $\epsilon = 0$ and $\kappa_0 := 2r_Y + r + 1$ if $\epsilon = 1$. By Proposition 17.8(2),

$$a(E', W_i, 0) = a(E', Y_i, 0) + r_Y \operatorname{mult}_{E'} s \quad \text{and} \quad \operatorname{mult}_{E'} h_i = \operatorname{mult}_{E'} l_i + mr_Y \operatorname{mult}_{E'} s,$$

and $I_G(E')$ is also the inertia group of E' for the action of G on Y_i. If $\epsilon = 0$, the dilation has Jacobian determinant $s^{(2k-1)r}$ and divides the equation by $s^{(2k-2)r}$, so

$$a(E', Y_i, 0) = a(E, \overline{W}_i, 0) + r \operatorname{mult}_E s \quad \text{and} \quad \operatorname{mult}_E l_i = \operatorname{mult}_E \overline{h}_i + (mr + \vartheta) \operatorname{mult}_E s.$$

If $\epsilon = 1$, the map $(\overline{x}, \overline{y}, \overline{z}, u) \to (x', y', z', s)$ has Jacobian determinant of u-order $(2k-1)r + 3$, the equation is divided by $u^{(2k-2)r+2}$, and comparing the residue forms yields $\rho a(E', Y_i, 0) = a(E, \overline{W}_i, 0) + (r+1) \operatorname{mult}_E u$. Moreover, $\rho \operatorname{mult}_{E'} s = 2 \operatorname{mult}_E u$ and $\rho \operatorname{mult}_{E'} l_i = \operatorname{mult}_E \overline{h}_i + (mr + \vartheta) \operatorname{mult}_E u$. The map $I_{G'}(E) \to I_G(E')$ is surjective with kernel $I_{G'}(E) \cap \{(1, \pm 1)\}$ [Ser79, Chapter I, §7, Proposition 22], of order ρ [Liu26a, Lemma 2.6(1)], so $e_E = \rho e_{E'}$. Combining the identities for $a(E', W_i, 0)$, $a(E', Y_i, 0)$, $\operatorname{mult}_{E'} h_i$, $\operatorname{mult}_{E'} l_i$, and $\rho \operatorname{mult}_{E'} s$ with $\beta = \vartheta/m$, resp. $\beta = \vartheta/m - 1$, proves (F.120).

Let

$$\overline{\Delta}_i := \frac{1}{m} \operatorname{div}(\overline{h}_i) + \beta \operatorname{div}(s) \quad \text{on } \overline{W}_i.$$

By (F.120),

$$\varsigma_{\overline{\Delta}_i}(E) = \varsigma_{\frac{1}{m}\operatorname{div}(h_i)}(E').$$

For $E = \overline{D}_i$, with $a(\overline{D}_i, \overline{W}_i, 0) = 1$ and $\operatorname{mult}_{\overline{D}_i} s = 1$, the klt property of $(W_i, \frac{1}{m}\operatorname{div}(h_i))$ implies

$$0 < \rho a\Big(E', W_i, \frac{1}{m}\operatorname{div}(h_i)\Big) = 1 - \frac{1}{m}\operatorname{mult}_{\overline{D}_i} \overline{h}_i - \beta \le 1 - \beta.$$

Thus $\beta < 1$, so $0 \le \vartheta < m$ if $\epsilon = 0$ and $0 \le \vartheta < 2m$ if $\epsilon = 1$. Since $r \ge 2m+3 > \vartheta$, the minimum defining ϑ is attained only at monomials of w-weight m, and the monomials of l_* of w-weight larger than m acquire positive powers of s. Thus $\beta \in \frac{1}{m}\mathbb{Z} \cap [-1, 1)$, and the limits $\overline{f}_*$, $\overline{h}_*$ of $\overline{f}_i$, $\overline{h}_i$ satisfy

$$\overline{f}_* = F_0(\overline{x}, \overline{y}, \overline{z}) + (\text{terms of } w\text{-weight} > 2k-2), \qquad \overline{h}_* = P(\overline{x}, \overline{y}, \overline{z}, s) + (\text{terms of } w\text{-weight} > m),$$

where $P \in \Bbbk[[s]][x, y, z]$ is w-homogeneous of weight m with $P|_{s=0} \ne 0$, and the higher terms have coefficients in $s\Bbbk[[s]]$. Since l_* has x'-degree at most 1, P has x-degree at most 1. By Proposition 17.8(4), the w-weight m part of l_* is $\operatorname{in}_w(h_*)(s^d x', s^d y', z')$, so over $\mathbb{K}$ the composite of $(x, y, z) \to (s^d x', s^d y', z')$, the substitutions of this step, and, if $\epsilon = 1$, the cover $s = u^2$, is a diagonal rescaling from (x, y, z) to $(\overline{x}, \overline{y}, \overline{z})$ carrying S^0 to $(F_0 = 0)$ and $\operatorname{in}_w(h_*)$ to a unit multiple of P. Thus, by (F.117), over $\mathbb{K}$ the pair

$$\Big(F_0 = 0, \frac{1}{m}\operatorname{div}(P)\Big)$$

has no coefficient-one component.

By (F.120) any prime divisor E over $\overline{W}_i$ with center 0 satisfies $\varsigma_{\overline{\Delta}_i}(E) \ge b$, and by Proposition 17.8(3)(4), $\operatorname{mult}_{E_i} s \le 3M/\eta_m$ and

$$\min\{\operatorname{mult}_{E_i} x', \operatorname{mult}_{E_i} y', \operatorname{mult}_{E_i} z'\} \to +\infty.$$

Thus, for $i \gg 0$, any prime divisor E over $\overline{W}_i$ whose restriction is E_i has center 0 and $\varsigma_{\overline{\Delta}_i}(E) = b$ by (F.120). Hence

$$\operatorname{mld}\big((\overline{W}_i \ni 0, \overline{\Delta}_i)/G\big) = b.$$

We keep the notation E_i for such a prime divisor over $\overline{W}_i$. Then $\operatorname{mult}_{E_i} s$ is bounded and

$$\min\{\operatorname{mult}_{E_i} \overline{x}, \operatorname{mult}_{E_i} \overline{y}, \operatorname{mult}_{E_i} \overline{z}\} \to +\infty.$$

Let

$$\widehat{\Delta}_i := \operatorname{div}(\overline{f}_i) + \frac{1}{m}\operatorname{div}(\overline{h}_i) + \beta \operatorname{div}(s) \quad \text{on } \overline{\mathcal{X}}_i.$$

We claim that the ambient pair also has quotient value b:

$$\operatorname{mld}\big((\overline{\mathcal{X}}_i \ni 0, \widehat{\Delta}_i)/G\big) = b. \tag{F.121}$$

Let $\Psi_i : \overline{\mathcal{X}}_i \to \widehat{V}_i$ be the composite of the chart of Proposition 17.8 and the substitutions of this step, let $\widehat{E}$ be a prime divisor over $\overline{\mathcal{X}}_i$ with center 0, let $\widehat{E}'$ be its restriction, a prime divisor over $\widehat{V}_i$, with ramification index ρ, and let κ^{amb} be the s-order of the Jacobian determinant of Ψ_i (with the ramification of $s = u^2$ counted) and σ the integer with $\Psi_i^* f_i = s^\sigma \overline{f}_i$, so that

$$e_{\widehat{E}} = \rho e_{\widehat{E}'}, \quad \rho a(\widehat{E}', \widehat{V}_i, 0) = a(\widehat{E}, \overline{\mathcal{X}}_i, 0) + \kappa^{\mathrm{amb}} \operatorname{mult}_{\widehat{E}} s,$$
$$\text{and} \quad \rho \operatorname{mult}_{\widehat{E}'} f_i = \operatorname{mult}_{\widehat{E}} \overline{f}_i + \sigma \operatorname{mult}_{\widehat{E}} s.$$

The residue forms of W_i and $\overline{W}_i$ differ by $s^{\kappa^{\mathrm{amb}} - \sigma}$, so comparing with (F.120) for $E = \overline{D}_i$, where $\operatorname{mult}_{\overline{D}_i} s = 1$, gives $\kappa^{\mathrm{amb}} - \sigma = \kappa_0$. By the definitions of l_i and $\overline{h}_i$, $\Psi_i^* h_i = s^{m(\kappa_0+\beta)} \overline{h}_i$, so

$$\rho \operatorname{mult}_{\widehat{E}'} h_i = \operatorname{mult}_{\widehat{E}} \overline{h}_i + m(\kappa_0 + \beta) \operatorname{mult}_{\widehat{E}} s.$$

Thus

$$\varsigma_{\widehat{\Delta}_i}(\widehat{E}) = \varsigma_{\operatorname{div}(f_i) + \frac{1}{m}\operatorname{div}(h_i)}(\widehat{E}') \ge b$$

by (F.118). Conversely, by (F.119) for E_i on $\overline{W}_i$, with $g = \overline{f}_i$ and $h = \overline{h}_i$, there exists a prime divisor $\widehat{E}$ over $\overline{\mathcal{X}}_i$ with center 0 and $\varsigma_{\widehat{\Delta}_i}(\widehat{E}) = b$, which proves (F.121).

Step 3. In this step we classify the sections ($s = 0$) of the charts below and bound Jacobian orders on the charts.

Let $\Bbbk_1 \supset \mathbb{C}$ be an algebraically closed field and $\varphi \in \Bbbk_1[x, y, z]$ a polynomial of maximal w-weight $2k-2$ with top part F_0 (Definition 7.1). The x-linear part of φ has w-weight less than $2k-2$, so completing the square in x by a polynomial of w-weight less than $k-1$ transforms φ into

$$x^2 + (z + a')y^2 + p_1(z)y + p_0(z), \tag{F.122}$$

where $a' \in \Bbbk_1$, p_0 has degree $k-1$ and leading coefficient 1, and $\deg p_1 < k/2$: a monomial y^3 has w-weight $3k-6 \le 2k-2$ only for $k = 4$, and then it would be a monomial of the top part F_0. Let $\Lambda := (z + a')p_0 - p_1^2/4$, of degree k with leading coefficient 1. At a point with $z + a' \neq 0$, completing the square in y and taking a square root of the unit $z + a'$ transforms the germ of $(\varphi = 0)$ into $(uv + \Lambda(z)/(z + a') = 0)$, which is smooth or of type A_{j-1} at a root of Λ of multiplicity $j \le k$. At a point with $z + a' = 0$ the surface is smooth if $p_1(-a') \neq 0$. If $p_1(-a') = 0$, we write $p_1 = (z + a')p_2$ and complete the square in y. With $Z := z + a'$ the equation becomes $x^2 + ZY^2 + \widetilde{p}_0(Z)$, where $\widetilde{p}_0$ has degree $k-1$ and leading coefficient 1 since $\deg((z + a')p_2^2) < k-1$. A singular point with $Z = 0$ and $Y \neq 0$ is of type A_1. A singular point with $Z = Y = 0$ satisfies $\widetilde{p}_0(0) = \widetilde{p}_0'(0) = 0$. If $\widetilde{p}_0$ has order 2 at 0, completing the square in Z shows type A_3, and if it has order $j \ge 3$, the type is D_{j+1} with $j \le k-1$. Thus $(\varphi = 0)$ is normal and any point of $(\varphi = 0)$ is smooth or of one of the types

$$A_j \ (1 \le j \le k-1), \qquad D_j \ (4 \le j \le k). \tag{F.123}$$

Since the Jacobian ideal of a Du Val surface germ is non-zero on a klt germ, the log canonical threshold of that ideal is positive. We fix a real number $c_{\mathrm{J}} > 0$ smaller than the log canonical thresholds of the Jacobian ideals of the germs in (F.123).

Let $\mathcal{X}$ be $\mathbb{A}^4$ or $\operatorname{Spec}\mathbb{C}[[x, y, z, s]]$, and let $W = (g = 0) \subset \mathcal{X}$ be a hypersurface germ at 0, in the formal case the completion at 0 of a normal algebraic hypersurface germ, such that $g(x, y, z, 0)$ is a polynomial in x, y, z of maximal w-weight $2k-2$ with top part F_0, and let $D := W \cap (s = 0) = (g(x, y, z, 0) = 0)$. Since D is a normal Cartier divisor on W, W is normal near 0. Since the ideal

$$\Big(\frac{\partial g}{\partial x}, \frac{\partial g}{\partial y}, \frac{\partial g}{\partial z}\Big)\mathcal{O}_D$$

is the Jacobian ideal Jac_D of D, we have $\operatorname{Jac}_D \subset \operatorname{Jac}_W \mathcal{O}_D$, so $(D, c_{\mathrm{J}} \operatorname{Jac}_W \mathcal{O}_D)$ is lc at 0 by the choice of c_{J}. Since W is a normal local complete intersection and $D \not\subset V(\operatorname{Jac}_W)$, [EM04, Theorem 1.1] with $X = W$, $Y = c_{\mathrm{J}} V(\operatorname{Jac}_W)$ and the closed subset $\{0\}$ implies

$$\operatorname{mld}\big(W \ni 0, D + c_{\mathrm{J}} V(\operatorname{Jac}_W)\big) = \operatorname{mld}\big(D \ni 0, c_{\mathrm{J}} \operatorname{Jac}_W \mathcal{O}_D\big) \ge 0.$$

Thus any prime divisor E over W with $\operatorname{center}_W E = \{0\}$ satisfies

$$c_{\mathrm{J}} \operatorname{mult}_E \operatorname{Jac}_W \le a(E, W, 0) - \operatorname{mult}_E s. \tag{F.124}$$

In the formal case, with W the completion of an algebraic germ W^{alg}, D and Jac_W are the completions of their algebraic counterparts, and (F.124) follows from the algebraic case by Lemma 2.12(3): a formal prime divisor E over W with center 0 restricts to a prime divisor E^{alg} over the algebraic germ W^{alg} with center 0 and $a(E^{\mathrm{alg}}, W^{\mathrm{alg}}, 0) = a(E, W, 0)$, and $\operatorname{mult}_E s$, $\operatorname{mult}_E \operatorname{Jac}_W$ are values of the valuation of E^{alg}. The charts used below are such completions, since they come from the algebraic chart Y_i by the cover $s = u^2$, dilations, and polynomial preparations. By **Step 2**, (F.124) holds for $W = \overline{W}_i$.

Step 4. In this step we dilate the charts $\overline{W}_i$ as deeply as possible along the critical weights.

Let $\exp(G)$ be the exponent of G. Below, r' is a positive integer divisible by $\exp(G)$, and we apply the dilation of depth r' of **Step 2** to the coordinates $(\overline{x},\overline{y},\overline{z},s)$ of $\overline{W}_i$:

$$\overline{x}=s^{(k-1)r'}\widetilde{x},\quad \overline{y}=s^{(k-2)r'}\widetilde{y},\quad \text{and}\quad \overline{z}=s^{2r'}\widetilde{z}.$$

Since $\exp(G)\mid r'$, the coordinates $\widetilde{x},\widetilde{y},\widetilde{z}$ have the characters of $\overline{x},\overline{y},\overline{z}$. The dilation has Jacobian determinant $s^{(2k-1)r'}$, so dividing the equation by $s^{(2k-2)r'}$ and the mark by $s^{mr'}$ keeps the value ς_Δ of any prime divisor with center in $(s=0)$, for Δ the sum of $\frac{1}{m}$ times the divisor of the mark and $\beta\,\mathrm{div}(s)$, by the computation of **Step 2**.

Since $\overline{f}_*$ and $\overline{h}_*$ have no monomials of w-weight less than $2k-2$ and m, respectively, and the regularity is a set of finitely many vanishing conditions on jets, by Lemma 2.7, for any fixed r' divisible by $\exp(G)$ the divisions by $s^{(2k-2)r'}$ and $s^{mr'}$ after the dilation of depth r' are regular for all but finitely many i. Since $\mathrm{mult}_{E_i}s$ is bounded and

$$\min\{\mathrm{mult}_{E_i}\overline{x},\mathrm{mult}_{E_i}\overline{y},\mathrm{mult}_{E_i}\overline{z}\}\to+\infty,$$

E_i has center the origin of the dilated chart for $i\gg 0$. Possibly passing to an infinite subsequence, there exist integers r_i^0 divisible by $\exp(G)$ with $\lim_{i\to+\infty}r_i^0=+\infty$ such that after the dilation of depth r_i^0 the divisions by $s^{(2k-2)r_i^0}$ and $s^{mr_i^0}$ are regular and E_i has center the origin. We fix integers $n_i\geq 1$ with $\lim_{i\to+\infty}n_i=+\infty$ and $\lim_{i\to+\infty}n_i/r_i^0=0$.

A *preparation* of $\overline{W}_i$ is a coordinate change

$$\overline{x}\to\overline{x}+T_i(\overline{y},\overline{z},s),\quad \overline{y}\to\overline{y}+U_i(\overline{z},s),\quad\text{and}\quad \overline{z}\to\overline{z}+a_i'(s),\tag{F.125}$$

where T_i and U_i are polynomials of w-weight less than $k-1$ and $k-2$, respectively, T_i, U_i, a_i' have coefficients in $s^{n_i}\mathbb{C}[s]$, and each correction has the character of the coordinate it modifies, together with a replacement of the ambient lift $\overline{h}_i$ by $\overline{h}_i+\overline{f}_i\gamma_i$, where γ_i is a polynomial in $\overline{x},\overline{y},\overline{z}$ of w-weight less than $m-(2k-2)$ with coefficients in $s^{n_i}\mathbb{C}[s]$ and the character of $\overline{h}_i/\overline{f}_i$ (so $\gamma_i=0$ if $m\leq 2k-2$). For $i\gg 0$ there exists $\varpi_i\in\widehat{\mathcal{O}}_{V_i,o_i}$ with

$$\Psi_i^*\varpi_i=s^{m(\kappa_0+\beta)-\sigma}\gamma_i,\quad\text{so that}\quad \Psi_i^*(h_i+f_i\varpi_i)=s^{m(\kappa_0+\beta)}\big(\overline{h}_i+\overline{f}_i\gamma_i\big),$$

and ϖ_i is a polynomial in the coordinates x,y,z,s of $\widehat{V}_i$, since a monomial of γ_i has w-weight less than $m-(2k-2)$ and coefficient in $s^{n_i}\mathbb{C}[s]$. If $\epsilon=0$, then

$$m(\kappa_0+\beta)-\sigma=(m-2k+2)(r_Y+r)+\vartheta-2d,$$

and the corresponding monomial of ϖ_i, written in (x,y,z,s), has exponent of s at least $n_i+r_Y+r-(m+2)d$. If $\epsilon=1$, the exponent of u of that monomial is at least n_i minus a constant independent of i, and it is even, since $\Psi_i^*\varpi_i$ is invariant under $(1,-1)\in G'$, as $\Psi_i^*h_i$ and $\Psi_i^*f_i$ are and γ_i has the character of $\overline{h}_i/\overline{f}_i$. The change (F.125) has Jacobian determinant 1, fixes the origin, and does not change $\overline{h}_i|_{\overline{W}_i}$. A positive integer r' is *admissible* if $\exp(G)\mid r'$, $r'\geq r_i^0$, and for some preparation, after the dilation of depth r', the equation divided by $s^{(2k-2)r'}$ and the ambient lift divided by $s^{mr'}$ are regular in $\mathbb{C}[[\widetilde{x},\widetilde{y},\widetilde{z},s]]$ and E_i has center the origin. The integer r_i^0 is admissible.

We show that the admissible integers are bounded. The form

$$\omega:=d\overline{y}\wedge d\overline{z}\wedge ds/(\partial\overline{f}_i/\partial\overline{x})$$

generates the canonical module of $\overline{W}_i$ near 0, and for the coordinates (X,Y,Z,s) after (F.125) we have $dY\wedge dZ\wedge ds=d\overline{y}\wedge d\overline{z}\wedge ds$ and $\partial\overline{f}_i/\partial X=\partial\overline{f}_i/\partial\overline{x}$. Let π be a uniformizer at the generic point of E_i on a higher model of $\overline{W}_i$. The differential of a function of order j along E_i belongs to $\pi^{j-1}(\mathcal{O}d\pi+\pi\Omega)$, and $d\pi\wedge d\pi=0$. Thus $dY\wedge dZ\wedge ds$ vanishes to order at least $\mathrm{mult}_{E_i}Y+\mathrm{mult}_{E_i}Z+\mathrm{mult}_{E_i}s-1$ along E_i, while it equals $\partial\overline{f}_i/\partial\overline{x}\cdot\omega$, which vanishes to order

$$\mathrm{mult}_{E_i}(\partial\overline{f}_i/\partial\overline{x})+a(E_i,\overline{W}_i,0)-1.$$

Hence

$$\operatorname{mult}_{E_i} Y + \operatorname{mult}_{E_i} Z + \operatorname{mult}_{E_i} s \le a(E_i, \overline{W}_i, 0) + \operatorname{mult}_{E_i}\Big(\frac{\partial \overline{f}_i}{\partial \overline{x}}\Big), \tag{F.126}$$

whose right side does not depend on the preparation. If r' is admissible, then

$$\operatorname{mult}_{E_i} Y > (k-2)r' \operatorname{mult}_{E_i} s \quad \text{and} \quad \operatorname{mult}_{E_i} Z > 2r' \operatorname{mult}_{E_i} s,$$

so $\operatorname{mult}_{E_i} Y + \operatorname{mult}_{E_i} Z + \operatorname{mult}_{E_i} s > kr'$. Thus the admissible integers are bounded, and we let r_i be the largest one, realized by a preparation Φ_i with lift correction γ_i. We let f_i' and h_i' be the equation and the ambient lift after Φ_i, the dilation of depth r_i, and the divisions by $s^{(2k-2)r_i}$ and s^{mr_i}, and we write $W_i' := (f_i' = 0)$ in

$$\mathcal{X}_i' := \operatorname{Spec} \mathbb{C}[[\widetilde{x}, \widetilde{y}, \widetilde{z}, s]].$$

Then $r_i \ge r_i^0 \to +\infty$.

Since (F.125) has Jacobian determinant 1, any prime divisor E over W_i' with center in $(s=0)$ and its restriction E', a prime divisor over W_i, with ramification index ρ, satisfy

$$e_E = \rho e_{E'}, \quad \rho a(E', W_i, 0) = a(E, W_i', 0) + (\kappa_0 + r_i) \operatorname{mult}_E s,$$
$$\text{and} \quad \frac{\rho}{m} \operatorname{mult}_{E'} h_i = \frac{1}{m} \operatorname{mult}_E h_i' + (\kappa_0 + r_i + \beta) \operatorname{mult}_E s,$$

and the composite map $\mathcal{X}_i' \to \widehat{V}_i$ satisfies the ambient identities of **Step 2** with $\kappa^{\mathrm{amb}} - \sigma = \kappa_0 + r_i$, for the ambient lift $h_i + f_i \varpi_i$. Since E_i has center the origin of W_i', the argument of **Step 2** proves

$$\operatorname{mld}\Big(\Big(W_i' \ni 0, \frac{1}{m}\operatorname{div}(h_i') + \beta \operatorname{div}(s)\Big)/G\Big) = \operatorname{mld}\Big(\Big(\mathcal{X}_i' \ni 0, \operatorname{div}(f_i') + \frac{1}{m}\operatorname{div}(h_i') + \beta \operatorname{div}(s)\Big)/G\Big) = b.$$

The monomials of w-weight v of the prepared equation are multiplied by $s^{r_i(v-2k+2)}$. Since Φ_i is the identity modulo s^{n_i} and $\overline{f}_i(\overline{x}, \overline{y}, \overline{z}, 0) = F_0(\overline{x}, \overline{y}, \overline{z})$, the polynomial $f_i'(\widetilde{x}, \widetilde{y}, \widetilde{z}, 0)$ has maximal w-weight $2k-2$ with top part F_0. Thus (F.124) holds for $W = W_i'$, and for the charts of any depth $r' \le r_i$ divisible by $\exp(G)$ obtained by Φ_i.

Step 5. In this step we compare the limits of the charts of maximal depth with the limit of the charts $\overline{W}_i$, and reduce the contradiction to an upper bound on the residual limit.

By Lemma 2.6(3), possibly passing to an infinite subset and replacing $\Bbbk$ with an extension field and $\mathbb{K}$ with an algebraic closure of $\Bbbk((s))$ for the new $\Bbbk$, the prepared equations and lifts and the functions f_i', h_i' have a compatible generic limit. Since Φ_i and γ_i are trivial modulo s^{n_i} and $n_i \to +\infty$, the limits of the prepared equations and lifts are $\overline{f}_*$ and $\overline{h}_*$. In f_i' and h_i' the coefficient of a monomial of w-weight v is multiplied by $s^{r_i(v-2k+2)}$ and $s^{r_i(v-m)}$, respectively. Since f_i' and h_i' are regular and $r_i \to +\infty$, the limits F_Q of f_i' and H_Q of h_i' (written in the coordinates x, y, z, s) satisfy

$$\begin{aligned} F_Q \in \Bbbk[[s]][x,y,z], \quad \max \operatorname{wt}_w(F_Q) = 2k-2, \quad \text{with top part } F_0, \\ H_Q \in \Bbbk[[s]][x,y,z], \quad \max \operatorname{wt}_w(H_Q) = m, \quad \text{with top part } P. \end{aligned} \tag{F.127}$$

Let $\overline{r}$ be a positive integer divisible by $\exp(G)$. For $i \gg 0$ we have $\overline{r} < r_i$, and Φ_i followed by the dilation of depth $\overline{r}$ and the divisions by $s^{(2k-2)\overline{r}}$ and $s^{m\overline{r}}$ is regular: a monomial of w-weight $v < 2k-2$ (resp. $v < m$) has coefficient divisible by $s^{r_i(2k-2-v)}$ (resp. $s^{r_i(m-v)}$) by the regularity at depth r_i, and the other monomials acquire non-negative powers of s. Let $f_i^{(\overline{r})}$ and $h_i^{(\overline{r})}$ be the equation and the ambient lift after the divisions by $s^{(2k-2)\overline{r}}$ and $s^{m\overline{r}}$. The orders of $\widetilde{x}, \widetilde{y}, \widetilde{z}$ along E_i at depth $\overline{r}$ exceed those at depth r_i, so E_i has center the origin, and, by the argument proving (F.121), the quotient values of the chart of depth $\overline{r}$ and of the ambient space $\operatorname{Spec} \mathbb{C}[[\widetilde{x}, \widetilde{y}, \widetilde{z}, s]]$ equal b. Their limits $f_*^{(\overline{r})}$ and $h_*^{(\overline{r})}$ satisfy

$$f_*^{(\overline{r})} - F_0 \in (s^{\overline{r}}), \qquad h_*^{(\overline{r})} - P \in (s^{\overline{r}}), \tag{F.128}$$

since the coefficients of w-weight $v < 2k-2$ (resp. $v < m$) are divisible by $s^{(r_i-\overline{r})(2k-2-v)}$ (resp. $s^{(r_i-\overline{r})(m-v)}$) with $r_i - \overline{r} \to +\infty$, and those of larger weight are multiplied by $s^{\overline{r}(v-2k+2)}$ (resp. $s^{\overline{r}(v-m)}$).

Let $\mathcal{X}_* := \operatorname{Spec} \Bbbk[[x,y,z,s]]$, and let a_P and a_Q be the quotient values with G at the closed point of $\mathcal{X}_*$ of

$$\Delta_P := \operatorname{div}(F_0) + \frac{1}{m}\operatorname{div}(P) + \beta \operatorname{div}(s) \quad \text{and} \quad \Delta_Q := \operatorname{div}(F_Q) + \frac{1}{m}\operatorname{div}(H_Q) + \beta \operatorname{div}(s),$$

respectively (in the notation of Lemma 7.11, with $Q_1 = F_Q$, $Q_2 = H_Q$, $P_1 = F_0$, $P_2 = P$). We claim that $a_Q \geq b$ and $a_P \geq b$. Let $\widehat{E}_0$ be a prime divisor over $\mathcal{X}_*$ with center the closed point. It is the exceptional divisor of a finite sequence of (weighted) blow-ups with centers defined by finitely many coefficients, and $a(\widehat{E}_0, \mathcal{X}_*, 0)$, $I_G(\widehat{E}_0)$, and the orders along $\widehat{E}_0$ of finitely many functions depend only on finite jets of the functions. Prescribing $a(\widehat{E}_0, \mathcal{X}_*, 0)$, $I_G(\widehat{E}_0)$, and the orders of F_Q, H_Q, s along $\widehat{E}_0$ is a finite set of polynomial equations and one non-vanishing condition, with complex coefficients, in the coefficients of a finite jet of (F_Q, H_Q) and the tuple $\mathbf{t} \in \Bbbk^{n'}$ of coefficients of the centers, so by Lemma 2.7 they hold, with some $\mathbf{t}_i \in \mathbb{C}^{n'}$ in place of $\mathbf{t}$, for the jets of (f_i', h_i') for all but finitely many i. Thus, for infinitely many i, $\widehat{E}_0$ specializes to prime divisors over $\mathcal{X}_i'$ with center 0 whose log discrepancies and inertia groups are $a(\widehat{E}_0, \mathcal{X}_*, 0)$ and $I_G(\widehat{E}_0)$ and whose orders of f_i', h_i', and s are the orders of F_Q, H_Q, and s along $\widehat{E}_0$. For

$$\Delta_i' := \operatorname{div}(f_i') + \frac{1}{m}\operatorname{div}(h_i') + \beta \operatorname{div}(s) \quad \text{on } \mathcal{X}_i',$$

their values of $\varsigma_{\Delta_i'}$ are at least b, so $\varsigma_{\Delta_Q}(\widehat{E}_0) \geq b$, and $a_Q \geq b$. For Δ_P we take $\overline{r}$ with

$$\overline{r} \operatorname{mult}_{\widehat{E}_0} s > \max\{\operatorname{mult}_{\widehat{E}_0} F_0, \operatorname{mult}_{\widehat{E}_0} P\}.$$

By (F.128), $\operatorname{mult}_{\widehat{E}_0} f_*^{(\overline{r})} = \operatorname{mult}_{\widehat{E}_0} F_0$ and $\operatorname{mult}_{\widehat{E}_0} h_*^{(\overline{r})} = \operatorname{mult}_{\widehat{E}_0} P$, and specializing $\widehat{E}_0$ to the intermediate charts of depth $\overline{r}$ shows $a_P \geq b$.

Since $a_P \geq b > 0$, by (F.127) and Lemma 7.11 with $n = 4$, $u = s$, $(x_1, x_2, x_3) = (x, y, z)$ with weights w, $c_1 = 1$, $c_2 = \frac{1}{m}$, and the given β,

$$a_P \leq a_Q. \tag{F.129}$$

The values $\varsigma_{\Delta_P}(\widehat{E})$ for the prime divisors $\widehat{E}$ over $\mathcal{X}_*$ with center the closed point belong to $\frac{1}{m \cdot |G|!}\mathbb{Z}$ and are at least b, so a_P is attained by a prime divisor $\widehat{E}_P$ over $\mathcal{X}_*$ with center the closed point. We fix $\overline{r}$ divisible by $\exp(G)$ with

$$\overline{r} \operatorname{mult}_{\widehat{E}_P} s > \max\{\operatorname{mult}_{\widehat{E}_P} F_0, \operatorname{mult}_{\widehat{E}_P} P\},$$

and let

$$\ell_P := \frac{1}{e_{\widehat{E}_P}}\Big(\frac{1}{m}\operatorname{mult}_{\widehat{E}_P} P + (\kappa_0 + \overline{r} + \beta)\operatorname{mult}_{\widehat{E}_P} s\Big).$$

By Lemma 2.7 for the limit $(f_*^{(\overline{r})}, h_*^{(\overline{r})})$, for infinitely many i, $\widehat{E}_P$ specializes to a prime divisor $\widehat{E}_{P,i}$ over the ambient space of the intermediate chart of depth $\overline{r}$, with center 0 and with the orders of $\widehat{E}_P$, so $\varsigma_{\Delta_i^{(\overline{r})}}(\widehat{E}_{P,i}) = a_P$ for

$$\Delta_i^{(\overline{r})} := \operatorname{div}(f_i^{(\overline{r})}) + \frac{1}{m}\operatorname{div}(h_i^{(\overline{r})}) + \beta \operatorname{div}(s).$$

Let $\widehat{E}'_{P,i}$ be the restriction of $\widehat{E}_{P,i}$, a prime divisor over $\widehat{V}_i$. The composite of Ψ_i and of Φ_i followed by the dilation of depth $\overline{r}$ satisfies the ambient identities of **Step 2** with $\kappa^{\mathrm{amb}} - \sigma = \kappa_0 + \overline{r}$. Thus, for the lift $h_i + f_i\varpi_i$ of **Step 4**,

$$\varsigma_{\operatorname{div}(f_i) + \frac{1}{m}\operatorname{div}(h_i + f_i\varpi_i)}(\widehat{E}'_{P,i}) = a_P,$$

and

$$\frac{\operatorname{mult}_{\widehat{E}'_{P,i}}(h_i + f_i\varpi_i)}{me_{\widehat{E}'_{P,i}}} = \ell_P.$$

By (F.118) with $\tau = \tau_i$,

$$q_i(\tau_i) \le a_P + (1-\tau_i)\ell_P \tag{F.130}$$

for infinitely many i, where ℓ_P does not depend on i. If, for an infinite subset of the indices along which (f'_i, h'_i) has a compatible generic limit (F_Q, H_Q) as in (F.127), the corresponding quotient value satisfies

$$a_Q \le b, \tag{F.131}$$

then $b \le a_P \le a_Q \le b$ by (F.129), so (F.130) and $\tau_i \to 1$ imply $q_i(\tau_i) < b + \delta$ for infinitely many i, contradicting (F.116). The proof continues below.

Proof of Theorem 17.10, continued. We keep the notation of **Steps 1–5**. As in (F.127), we write (x, y, z, s) for the coordinates $(\widetilde{x}, \widetilde{y}, \widetilde{z}, s)$ of $\mathcal{X}'_i$. The computing lift E_i is a prime divisor over W'_i with center the origin, and

$$a(E_i, W'_i, 0) = be_{E_i} + \frac{1}{m}\operatorname{mult}_{E_i} h'_i + \beta \operatorname{mult}_{E_i} s. \tag{F.132}$$

By **Step 2**, $\operatorname{mult}_{E_i} s$ is bounded independently of i, since (F.125) and the dilations do not change s. Let C_0 be the sum of $b|G|$ and an upper bound of the numbers $\operatorname{mult}_{E_i} s$. Since $e_{E_i} \le |G|$ and $\beta < 1$, we have $\operatorname{mult}_{E_i} s \le C_0$ and

$$a(E_i, W'_i, 0) \le \frac{1}{m}\operatorname{mult}_{E_i} h'_i + C_0 \quad \text{for any } i. \tag{F.133}$$

Let $\lambda := 1/c_{\mathrm{J}}$. Since $\operatorname{mult}_{E_i} s \ge 0$, (F.124) implies

$$\operatorname{mult}_{E_i} \operatorname{Jac}_{W'_i} \le \lambda a(E_i, W'_i, 0) \quad \text{for any } i. \tag{F.134}$$

Step 6. In this step we show that $\operatorname{mult}_{E_i} h'_i$ is not bounded on any infinite set of indices.

Assume the contrary. Possibly passing to a compatible infinite subfamily $(f'_i, h'_i)_{i\in I}$, we may assume that $\operatorname{mult}_{E_i} h'_i$ is bounded independently of i, and that (F_Q, H_Q) in (F.127) and a_Q are the compatible generic limit of $(f'_i, h'_i)_{i\in I}$ and the corresponding quotient value. By (F.133) and (F.134), $a(E_i, W'_i, 0)$ and $\operatorname{mult}_{E_i} \operatorname{Jac}_{W'_i}$ are bounded independently of i. For any i, let $\widehat{E}_i$ be the prime divisor over $\mathcal{X}'_i$ with center 0 given by (F.119) for E_i, the mark h'_i and the axis s. Thus $e_{\widehat{E}_i} = e_{E_i}$, $\operatorname{mult}_{\widehat{E}_i} h'_i = \operatorname{mult}_{E_i} h'_i$, $\operatorname{mult}_{\widehat{E}_i} s = \operatorname{mult}_{E_i} s$,

$$\operatorname{mult}_{\widehat{E}_i} f'_i \le \operatorname{mult}_{E_i} h'_i + \operatorname{mult}_{E_i} s + 2\operatorname{mult}_{E_i} \operatorname{Jac}_{W'_i} + e_{E_i} + 2,$$

and

$$a(\widehat{E}_i, \mathcal{X}'_i, 0) = a(E_i, W'_i, 0) + \operatorname{mult}_{\widehat{E}_i} f'_i.$$

Since $e_{E_i} \le |G|$, the numbers $\operatorname{mult}_{\widehat{E}_i} s$, $\operatorname{mult}_{\widehat{E}_i} f'_i$, $\operatorname{mult}_{\widehat{E}_i} h'_i$, and $a(\widehat{E}_i, \mathcal{X}'_i, 0)$ are bounded independently of i, and by (F.132),

$$\frac{a(\widehat{E}_i, \mathcal{X}'_i, 0) - \beta \operatorname{mult}_{\widehat{E}_i} s - \operatorname{mult}_{\widehat{E}_i} f'_i - \frac{1}{m}\operatorname{mult}_{\widehat{E}_i} h'_i}{e_{\widehat{E}_i}}$$
$$= \frac{a(E_i, W'_i, 0) - \beta \operatorname{mult}_{E_i} s - \frac{1}{m}\operatorname{mult}_{E_i} h'_i}{e_{E_i}} = b.$$

By Lemma 7.10(2), with $n = 4$, the axis $u = s$, the coordinates (x, y, z), the group G, the coefficient β of the axis, $N = 2$, $(Q_{i,1}, Q_{i,2}) = (f'_i, h'_i)$, $(c_1, c_2) = (1, \frac{1}{m})$ (the indices in I are dense

in any finite-jet stratum by Definition 2.5), and F_i the prime divisor over $\mathbb{A}^4$ to which $\widehat{E}_i$ restricts (Lemma 2.12(3), which keeps the log discrepancy, the orders, and the inertia group), we have

$$a_Q = \mathrm{mld}\Big(\big(\mathcal{X}_* \ni 0, \beta\,\mathrm{div}(s) + \mathrm{div}(F_Q) + \frac{1}{m}\,\mathrm{div}(H_Q)\big)/G\Big) \le b.$$

By (F.131), this contradicts the radial failure.

Step 7. In this step we assume that $\mathrm{mult}_{E_i} h_i' \to +\infty$, which holds by **Step 6**, and prove (F.138) below.

Let $L_0 := \Bbbk((s))$ and let

$$S_{L_0} := (F_Q = 0) \subset \mathbb{A}^3_{L_0}, \quad S := (F_Q = 0) \subset \mathbb{A}^3_{\mathbb{K}}, \quad \text{and} \quad S_{F_0} := (F_0 = 0) \subset \mathbb{A}^3_{\mathbb{K}},$$

and let $B := \frac{1}{m}\,\mathrm{div}(H_Q|_S)$. Let $\overline{S} \subset \mathbb{P}(1, k-1, k-2, 2)$ be the closure of S, the first homogeneous coordinate having weight 1, let $\Delta_\infty := \overline{S}\backslash S$, and let $\overline{B}$ be the closure of B. By (F.127), F_Q and H_Q have maximal w-weights $2k-2$ and m and top parts F_0 and P, and P has x-degree at most 1 (**Step 2**), so $P \notin (F_0)$ and $P|_{S_{F_0}} \neq 0$. Since

$$\frac{1}{m} \cdot m = 1 = (k-1) + (k-2) + 2 - (2k-2),$$

the pair $(S_{F_0}, \frac{1}{m}\,\mathrm{div}(P))$ is a critical pair for $w = (k-1, k-2, 2)$. It has no coefficient-one component by **Step 2**. Let E_w be the exceptional curve of the w-weighted blow-up $S_w \to S_{F_0}$, P_w the strict transform of $\mathrm{div}(P)$, and Diff the different of $\frac{1}{m}P_w$ on E_w. By criticality $a(E_w, S_{F_0}, \frac{1}{m}\,\mathrm{div}(P)) = 0$, so $K_{S_w} + E_w + \frac{1}{m}P_w$ is the pull-back of $K_{S_{F_0}} + \frac{1}{m}\,\mathrm{div}(P)$ and $\deg(K_{E_w} + \mathrm{Diff}) = 0$, where the points $p \in E_w$ have indices $r_p \in \{1, 2, k-2\}$ and Diff has coefficient $1 - (1-c_p)/r_p$ at p (computed on the index-1 cover at p, see [Kol13, (3.36.1) and Corollary 3.45(3)]), for the coefficient c_p in $\frac{1}{m}\,\mathrm{div}(P)$ of the ruling over p. The points of E_w with $r_p > 1$ are two points of index 2 and one point of index $k-2$, so

$$0 = \deg(K_{E_w} + \mathrm{Diff}) = -2 + \frac{1}{2} + \frac{1}{2} + \Big(1 - \frac{1}{k-2}\Big) + \sum_p \frac{c_p}{r_p},$$

and $\sum_p c_p/r_p = \frac{1}{k-2}$, so $c_p \le r_p/(k-2) \le 1$ as $k \ge 4$, with $c_p < 1$ by **Step 2**. Thus (E_w, Diff) is klt, so $(S_w, E_w + \frac{1}{m}P_w)$ is plt near E_w [Kol13, Theorem 4.9(1)], and $(S_{F_0}, \frac{1}{m}\,\mathrm{div}(P))$ is lc near the vertex, hence lc as F_0 and P are w-homogeneous. By Lemma 7.3, case (b), with the field $\mathbb{K}$, $F = F_Q$, $N = 1$, $Q_1 = H_Q$, and $\kappa_1 = \frac{1}{m}$, we have the following.

(1) S is normal with at worst Du Val singularities.
(2) $\overline{S}$ is normal and klt.
(3) $(\overline{S}, \overline{B} + \Delta_\infty)$ is plt near Δ_∞.

Moreover,

$$K_{\overline{S}} + \overline{B} + \Delta_\infty \sim_{\mathbb{Q}} 0,$$

since

$$K_{\overline{S}} \sim_{\mathbb{Q}} \big((2k-2) - (1 + (k-1) + (k-2) + 2)\big)\Delta_\infty = -2\Delta_\infty$$

for the hypersurface $\overline{S}$ of weighted degree $2k-2$, and $\overline{B} \sim_{\mathbb{Q}} \Delta_\infty$ as H_Q has maximal weight m.

Let $C' \neq \Delta_\infty$ be an irreducible component of $\mathrm{Nklt}(\overline{S}, \overline{B} + \Delta_\infty)$. Since Δ_∞ is the support of an ample $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor, $\dim C' = 1$ would imply $C' \cap \Delta_\infty \neq \emptyset$, contradicting the plt property near Δ_∞. Thus $\mathrm{Nklt}(S, B)$ is a finite set of closed points. In particular, no component of B has coefficient at least one. If $\mathrm{Nklt}(S, B) \neq \emptyset$, then

$$\mathrm{Nklt}(\overline{S}, \overline{B} + \Delta_\infty) = \Delta_\infty \sqcup \mathrm{Nklt}(S, B)$$

is disconnected, and Theorem 7.2, which holds over the algebraically closed field $\mathbb{K}$ of characteristic zero [FS23, Section 2.1], with $X = \overline{S}$, $Z = \mathrm{Spec}\,\mathbb{K}$ and f the structure morphism, implies that $(\overline{S}, \overline{B} + \Delta_\infty)$ is lc and that $\mathrm{Nklt}(\overline{S}, \overline{B} + \Delta_\infty)$ has exactly two connected components. Thus one of the following holds.

(a) (S, B) is klt.
(b) (S, B) is lc, and $\mathrm{Nklt}(S, B) = \{o\}$ for a closed point $o \in S$.

Assume (b). Since F_Q has maximal weight $2k-2$ and top part F_0, $F_Q - x^2$ has x-degree at most 1. Let $\widetilde{H}_Q \in \Bbbk[[s]][x, y, z]$ be the remainder of H_Q on division by F_Q as polynomials in x. Then $\widetilde{H}_Q$ has x-degree at most 1 and $\widetilde{H}_Q|_S = H_Q|_S$, and since P has x-degree at most 1, $\widetilde{H}_Q$ has maximal weight m and top part P. By Proposition 7.7(1) and (5), with $f_0 = F_0$, $F = F_Q$, $N = 1$, $Q_1 = \widetilde{H}_Q$, $\kappa_1 = \frac{1}{m}$, $d_1 = m$, $u = s$, and ξ_1, ξ_2, ξ_3 in place of the coordinates x', y', z' there, there exist unique polynomials

$$\xi_1 = x + g_1(y, z), \quad \xi_2 = y + g_2(z), \quad \text{and} \quad \xi_3 = z + g_3,$$

with coefficients in $\Bbbk[[s]]$ (in particular $g_3 \in \Bbbk[[s]]$), $\mathrm{wt}_w(g_1) < k-1$ and $\mathrm{wt}_w(g_2) < k-2$, such that

$$F_Q = F_0(\xi_1, \xi_2, \xi_3), \qquad \widetilde{H}_Q = P(\xi_1, \xi_2, \xi_3) \qquad \text{in } \Bbbk[[s]][x, y, z], \tag{F.135}$$

and $o = (\xi_1 = \xi_2 = \xi_3 = 0)$. Since ξ_1, ξ_2, ξ_3 have coefficients in $\Bbbk[[s]]$, o is an L_0-point of S_{L_0}, and the closure of o in $\operatorname{Spec} \Bbbk[[s]][x, y, z]$ is the section $\Gamma := (\xi_1 = \xi_2 = \xi_3 = 0)$ over $\operatorname{Spec} \Bbbk[[s]]$. Since G acts on L_0 through the character of s and diagonally on x, y, z, with F_Q and F_0 of one character and $\widetilde{H}_Q$ and P of one character, Proposition 7.7(3) implies that ξ_1, ξ_2, ξ_3 are G-semi-invariant with the characters of x, y, z. Let $\Upsilon_0 := (H_Q - \widetilde{H}_Q)/F_Q \in \Bbbk[[s]][x, y, z]$, so that

$$H_Q = \Upsilon_0 F_Q + P(\xi_1, \xi_2, \xi_3).$$

Since H_Q and $\widetilde{H}_Q = P(\xi_1, \xi_2, \xi_3)$ both have maximal weight m and top part P, $H_Q - P(\xi_1, \xi_2, \xi_3)$ has maximal weight less than m, and Υ_0 has maximal weight less than $m-(2k-2)$. In particular, $\Upsilon_0 = 0$ if $m \le 2k-2$.

Since a log resolution of $(S, \operatorname{div}(H_Q|_S))$ has finitely many prime divisors, there exists a rational number $\mu > \frac{1}{m}$ such that $(S, \mu \operatorname{div}(H_Q|_S))$ is klt in case (a), and klt on $S \backslash \{o\}$ in case (b). Since $S = S_{L_0} \times_{L_0} \mathbb{K}$, the pair $(S_{L_0}, \mu \operatorname{div}(H_Q|_{S_{L_0}}))$ is klt in case (a), and klt outside the image of o in case (b). Fix rational numbers $0 < \theta_1 < \theta_2 < 1$ with

$$\Xi := \theta_1 \mu - \frac{1}{m} - (1-\theta_1)\Big(1 + \frac{2\lambda}{m}\Big) > 0,$$

which exist since the left side tends to $\mu - \frac{1}{m} > 0$ as $\theta_1 \to 1$.

Let $W_Q := (F_Q = 0) \subset \mathcal{X}_*$ and $\mathcal{J}_* := \mathcal{J}(F_Q^{\theta_2} H_Q^{\theta_2 \mu})$ on $\mathcal{X}_*$. The surface $W_Q \cap (s = 0)$ is defined by the polynomial $F_Q(x, y, z, 0)$ of maximal weight $2k-2$ with top part F_0, so it is normal by (F.123). Thus W_Q is normal. Since $P \notin (F_0)$, $H_Q|_{W_Q} \neq 0$. Let $I_\Gamma \subset \Bbbk[[x, y, z, s]]$ be the unit ideal in case (a) and the ideal generated by ξ_1, ξ_2, ξ_3 in case (b), and let $V \subset W_Q$ be the closed subset defined by sI_Γ, which contains the closed point. Since $\Bbbk[[s]][x, y, z]$ is excellent, its completion $\Bbbk[[x, y, z, s]]$ at the origin is regular over it, and since $\Bbbk[[s]][x, y, z][s^{-1}] = L_0[x, y, z]$, the morphism $W_Q^\circ := W_Q \backslash (s = 0) \to S_{L_0}$ is regular. Thus for a log resolution $Y \to S_{L_0}$ of $(S_{L_0}, \operatorname{div}(H_Q|_{S_{L_0}}))$, the base change $Y \times_{S_{L_0}} W_Q^\circ \to W_Q^\circ$ is a log resolution of $(W_Q^\circ, \operatorname{div}(H_Q|_{W_Q^\circ}))$, and any prime divisor E' on $Y \times_{S_{L_0}} W_Q^\circ$ with image a prime divisor E on Y satisfies

$$a\big(E', W_Q^\circ, \mu \operatorname{div}(H_Q|_{W_Q^\circ})\big) = a\big(E, S_{L_0}, \mu \operatorname{div}(H_Q|_{S_{L_0}})\big).$$

Hence $(W_Q, \mu \operatorname{div}(H_Q|_{W_Q}))$ is klt on $W_Q \backslash V$. By Lemma 6.1(3), with $(f_*, h_*) = (F_Q, H_Q)$, $W_* = W_Q$, and the closed subset V, $\mathcal{J}_*$ is the unit ideal near any point of $W_Q \backslash V$. Thus sI_Γ is contained in the radical of $(F_Q) + \mathcal{J}_*$, and there exists a positive integer N_0 with

$$(sI_\Gamma)^{N_0} \subset (F_Q) + \mathcal{J}_*. \tag{F.136}$$

Let D' be a positive integer. Let $\varphi := 1$ if I_Γ is the unit ideal, and $\varphi \in \{\xi_1, \xi_2, \xi_3\}$ otherwise. By (F.136), there exists $g_\varphi \in \Bbbk[[x, y, z, s]]$ with $s^{N_0}\varphi^{N_0} - g_\varphi F_Q \in \mathcal{J}_*$. Let $\mathbf{p}_* \in \Bbbk^q$ be the vector

of the coefficients of the monomials of degree less than D' in g_φ for the values of φ above and, in case (b), in g_1, g_2, g_3, and Υ_0. For $\mathbf{p} \in \Bbbk^q$ or $\mathbf{p} \in \mathbb{C}^q$, let

(i) $g_\varphi(\mathbf{p})$, $g_1(\mathbf{p})$, $g_2(\mathbf{p})$, $g_3(\mathbf{p})$, and $\Upsilon_0(\mathbf{p})$ be the polynomials of degree less than D' with the coefficients listed in $\mathbf{p}_*$ replaced by the entries of $\mathbf{p}$,
(ii) $\xi_1(\mathbf{p}) := x + g_1(\mathbf{p})$, $\xi_2(\mathbf{p}) := y + g_2(\mathbf{p})$, and $\xi_3(\mathbf{p}) := z + g_3(\mathbf{p})$,
(iii) $\varphi(\mathbf{p})$ be 1, resp. $\xi_j(\mathbf{p})$ for $\varphi = \xi_j$, and
(iv) Φ_φ be the polynomial with complex coefficients in $y_{1,0}$ (the variable of Lemma 6.1 substituted by the first entry of a pair), in q further variables, and in x, y, z, s such that

$$\Phi_\varphi((f,h),\mathbf{p}) = s^{N_0}\varphi(\mathbf{p})^{N_0} - g_\varphi(\mathbf{p})f.$$

Since $\varphi - \varphi(\mathbf{p}_*) \in \mathfrak{n}_*^{D'}$ and $g_\varphi - g_\varphi(\mathbf{p}_*) \in \mathfrak{n}_*^{D'}$, where $\mathfrak{n}_*$ is the maximal ideal of $\Bbbk[[x,y,z,s]]$, we have

$$\Phi_\varphi((F_Q,H_Q),\mathbf{p}_*) \in \mathcal{J}_* + \mathfrak{n}_*^{D'}.$$

By Lemma 6.1(1) and (2), with $n = 4$, $N = 2$, $(f_{i,1}, f_{i,2}) = (f_i', h_i')$, $(\gamma_1,\gamma_2) = (\theta_1, \theta_1\mu)$, $(\gamma_1',\gamma_2') = (\theta_2, \theta_2\mu)$, $L = D'$, the polynomials Φ_φ, and $\mathbf{p}_*$ and $\mathbf{p}_i$ in place of β and β_i (the orders of f_i' and h_i' at 0 are bounded since $F_Q, H_Q \neq 0$), for all but finitely many i there exists $\mathbf{p}_i \in \mathbb{C}^q$ such that, for $\varphi_i := \varphi(\mathbf{p}_i)$, $g_{\varphi,i} := g_\varphi(\mathbf{p}_i)$, and

$$z_{\varphi,i} := \Phi_\varphi((f_i',h_i'),\mathbf{p}_i) = s^{N_0}\varphi_i^{N_0} - g_{\varphi,i}f_i',$$

any prime divisor E over $\mathcal{X}_i'$ with center the origin and $\operatorname{mult}_E z_{\varphi,i} < D'$ satisfies

$$\operatorname{mult}_E z_{\varphi,i} + a(E,\mathcal{X}_i',0) > \theta_1 \operatorname{mult}_E f_i' + \theta_1\mu \operatorname{mult}_E h_i',$$

as $\operatorname{mult}_E(\mathfrak{m}^{D'}) \geq D'$, where $\mathfrak{m}$ is the maximal ideal of $\mathbb{C}[[x,y,z,s]]$. Let $\Upsilon_{0,i} := \Upsilon_0(\mathbf{p}_i)$ and $\xi_{j,i} := \xi_j(\mathbf{p}_i)$.

Since $z_{\varphi,i}|_{W_i'} = s^{N_0}\varphi_i^{N_0}|_{W_i'}$, Lemma 5.5(1), with $\mathcal{X} = \mathcal{X}_i'$, $f = f_i'$, $h = h_i'$, $F = E_i$, $\theta = \theta_1$, the number μ above, $g = z_{\varphi,i}$, and $\rho = D'$, shows that if $N_0(\operatorname{mult}_{E_i} s + \operatorname{mult}_{E_i}\varphi_i) < D'$, then

$$\begin{aligned}(\theta_1\mu - 1 + \theta_1)\operatorname{mult}_{E_i} h_i' < (2-\theta_1)N_0(&\operatorname{mult}_{E_i} s + \operatorname{mult}_{E_i}\varphi_i)\\ &+ a(E_i,W_i',0) + 2(1-\theta_1)\operatorname{mult}_{E_i}\operatorname{Jac}_{W_i'} + 1 - \theta_1,\end{aligned}$$

and hence, by (F.134) and (F.133), with $C_1 := (1+2\lambda)C_0 + 1$,

$$\Xi \operatorname{mult}_{E_i} h_i' < C_1 + (2-\theta_1)D'. \tag{F.137}$$

Since $\Xi > 0$ and $\operatorname{mult}_{E_i} h_i' \to +\infty$, (F.137) fails for all but finitely many i, so

$$N_0(\operatorname{mult}_{E_i} s + \operatorname{mult}_{E_i}\varphi_i) \geq D'$$

for all but finitely many i. If I_Γ is the unit ideal, then $\varphi_i = 1$, and for $D' > N_0C_0$ this contradicts $\operatorname{mult}_{E_i} s \leq C_0$. Thus, for $D' > N_0C_0$, I_Γ is not the unit ideal, so we are in case (b) and $\xi_1, \xi_2, \xi_3 \in \mathfrak{n}_*$, that is, Γ passes through the closed point of $\operatorname{Spec}\Bbbk[[x,y,z,s]]$. Since $\operatorname{mult}_{E_i} s \leq C_0$, taking $\varphi = \xi_j$ for $j = 1,2,3$ gives

$$\min\{\operatorname{mult}_{E_i}\xi_{1,i}, \operatorname{mult}_{E_i}\xi_{2,i}, \operatorname{mult}_{E_i}\xi_{3,i}\} \geq \frac{D'}{N_0} - C_0 \quad \text{for all but finitely many } i. \tag{F.138}$$

Step 8. In this step we show that (F.138) contradicts the maximality of r_i.

Take the integer D' of **Step 7** with

$$D' \geq (2k-2+m)\exp(G) + 2k + m \quad \text{and} \quad \frac{D'}{N_0} > \big(1 + (k-1)\exp(G)\big)C_0.$$

Since $\operatorname{mult}_{E_i} s \leq C_0$, (F.138) implies

$$\begin{gathered}\operatorname{mult}_{E_i}\xi_{1,i} > (k-1)\exp(G)\operatorname{mult}_{E_i} s, \quad \operatorname{mult}_{E_i}\xi_{2,i} > (k-2)\exp(G)\operatorname{mult}_{E_i} s,\\ \text{and} \quad \operatorname{mult}_{E_i}\xi_{3,i} > 2\exp(G)\operatorname{mult}_{E_i} s\end{gathered} \tag{F.139}$$

for all but finitely many i. By the last sentence of Lemma 6.1(1), with $d' = D'$ and the finite set consisting of the coordinate functions $u_1, \dots, u_q$ and of the set $\mathcal{Q}$ defined below, the vectors $\mathbf{p}_i$ of **Step 7** can be taken such that, for all but finitely many i, an entry of $\mathbf{p}_i$ vanishes exactly when the corresponding entry of $\mathbf{p}_*$ does, and $\Psi(\mathrm{jet}_{D'}(f_i', h_i'), \mathbf{p}_i) = 0$ if and only if $\Psi(\mathrm{jet}_{D'}(F_Q, H_Q), \mathbf{p}_*) = 0$, for any $\Psi \in \mathcal{Q}$. Since **Step 7** uses only the first conclusion of Lemma 6.1(1) and (2), (F.138) holds for these $\mathbf{p}_i$. We take them so. Then the polynomials $\xi_{1,i} - x$, $\xi_{2,i} - y$, $\xi_{3,i} - z$, and $\Upsilon_{0,i}$ have coefficients in $\mathbb{C}[s]$, satisfy $\mathrm{wt}_w(\xi_{1,i} - x) < k - 1$, $\mathrm{wt}_w(\xi_{2,i} - y) < k - 2$, $\xi_{3,i} - z \in \mathbb{C}[s]$, and $\mathrm{wt}_w(\Upsilon_{0,i}) < m - (2k - 2)$, and have the characters of g_1, g_2, g_3, and Υ_0, respectively. Moreover, $\xi_{1,i}, \xi_{2,i}, \xi_{3,i} \in \mathfrak{m}$, so $(\xi_{1,i}, \xi_{2,i}, \xi_{3,i}, s)$ is a triangular change of the coordinates (x, y, z, s) of determinant one, and $\mathfrak{m}^{D'}$ is generated by the monomials of degree D' in (x, y, z, s) and also by those in $(\xi_{1,i}, \xi_{2,i}, \xi_{3,i}, s)$.

Consider the chart

$$\xi_{1,i} = s^{(k-1)\exp(G)} x'', \quad \xi_{2,i} = s^{(k-2)\exp(G)} y'', \quad \text{and} \quad \xi_{3,i} = s^{2\exp(G)} z''.$$

The monomial with exponents $(\alpha_1, \alpha_2, \alpha_3, \alpha_4)$ in $(\xi_{1,i}, \xi_{2,i}, \xi_{3,i}, s)$, of weight

$$v := (k-1)\alpha_1 + (k-2)\alpha_2 + 2\alpha_3,$$

becomes

$$s^{\exp(G)v + \alpha_4} x''^{\alpha_1} y''^{\alpha_2} z''^{\alpha_3}.$$

Thus $s^{-(2k-2)\exp(G)} f_i'$ and $s^{-m\exp(G)}(h_i' - \Upsilon_{0,i} f_i')$ are regular on the chart if, in the expansions of f_i' and of $h_i' - \Upsilon_{0,i} f_i'$ in $(\xi_{1,i}, \xi_{2,i}, \xi_{3,i}, s)$, the coefficients vanish of the monomials with

$$\exp(G)v + \alpha_4 < (2k-2)\exp(G), \quad \text{resp.} \quad \exp(G)v + \alpha_4 < m\exp(G).$$

Since $v \geq 2(\alpha_1 + \alpha_2 + \alpha_3)$, any such monomial satisfies, for $t = 2k - 2$, resp. $t = m$,

$$\alpha_1 + \alpha_2 + \alpha_3 + \alpha_4 \leq \frac{v}{2} + \alpha_4 < \frac{v}{2} + (t - v)\exp(G) \leq t\exp(G) \leq (2k - 2 + m)\exp(G) \leq D',$$

so its coefficient is $\Psi(\mathrm{jet}_{D'}(f_i', h_i'), \mathbf{p}_i)$ for a polynomial Ψ with complex coefficients, one for each such monomial, and $\Psi(\mathrm{jet}_{D'}(F_Q, H_Q), \mathbf{p}_*)$ is the coefficient of the monomial with exponents $(\alpha_1, \alpha_2, \alpha_3, \alpha_4)$ in the expansion of F_Q, resp. $H_Q - \Upsilon_0 F_Q$, in (ξ_1, ξ_2, ξ_3, s), as $\mathbf{p}_*$ gives $\xi_1, \xi_2, \xi_3, \Upsilon_0$ modulo $\mathfrak{n}_*^{D'}$. Each $\Psi(\mathrm{jet}_{D'}(F_Q, H_Q), \mathbf{p}_*)$ vanishes, since by (F.135),

$$F_Q = F_0(\xi_1, \xi_2, \xi_3), \qquad H_Q - \Upsilon_0 F_Q = P(\xi_1, \xi_2, \xi_3)$$

are w-homogeneous of weights $2k - 2$ and m in (ξ_1, ξ_2, ξ_3). Let $\mathcal{Q}$ be the finite set of the polynomials Ψ. By the choice of the $\mathbf{p}_i$, $\Psi(\mathrm{jet}_{D'}(f_i', h_i'), \mathbf{p}_i) = 0$ for any $\Psi \in \mathcal{Q}$ and all but finitely many i. Since $n_i / r_i^0 \to 0$ and $r_i \geq r_i^0$, we may fix such an index i satisfying (F.139) and $n_i \leq r_i$. By (F.139), E_i has center the origin of the chart.

Let (X, Y, Z, s) be the coordinates after the preparation (F.125) realizing r_i, so that

$$X = s^{(k-1)r_i} x, \quad Y = s^{(k-2)r_i} y, \quad \text{and} \quad Z = s^{2r_i} z,$$

and $s^{(2k-2)r_i} f_i'$ and $s^{mr_i} h_i'$ are the prepared equation and the prepared lift. For $\zeta \in \mathbb{C}$, a term $\zeta s^{\alpha_4} y^{\alpha_2} z^{\alpha_3}$ of $\xi_{1,i} - x$, of weight $v < k - 1$, equals

$$s^{-(k-1)r_i} \cdot \zeta s^{\alpha_4 + (k-1-v)r_i} Y^{\alpha_2} Z^{\alpha_3}.$$

Similarly, a term of weight $v < k - 2$ of $\xi_{2,i} - y$ acquires the factor $s^{(k-2-v)r_i}$, the polynomial $\xi_{3,i} - z \in \mathbb{C}[s]$ acquires s^{2r_i}, and a term $\zeta s^{\alpha_4} x^{\alpha_1} y^{\alpha_2} z^{\alpha_3}$ of $\Upsilon_{0,i}$, of weight $v < m - (2k - 2)$, contributes the term

$$\zeta s^{\alpha_4 + (m - (2k-2) - v)r_i} X^{\alpha_1} Y^{\alpha_2} Z^{\alpha_3}$$

to a polynomial $\widetilde{\gamma}$ with

$$s^{mr_i}(h_i' - \Upsilon_{0,i} f_i') = s^{mr_i} h_i' - \widetilde{\gamma} \cdot s^{(2k-2)r_i} f_i'.$$

All weight gaps are positive integers, so the coefficients of $\xi_{1,i} - x$, $\xi_{2,i} - y$, $\xi_{3,i} - z$, and $\widetilde{\gamma}$ in (X, Y, Z, s) are divisible by s^{r_i}, hence by s^{n_i}. Composing with (F.125) and with the correction of the lift there, we obtain a preparation with the following properties.

(1) The change of coordinates is triangular of the form (F.125), satisfies its weight bounds and character conditions, has determinant one, and has coefficients in $s^{n_i}\mathbb{C}[s]$.
(2) The lift $\overline{h}_i + \overline{f}_i\gamma_i^+$, where $\gamma_i^+ := \gamma_i - \widetilde{\gamma}$ with $\widetilde{\gamma}$ written in the coordinates of $\overline{\mathcal{X}}_i$ through (F.125), satisfies the weight bounds and the character conditions of (F.125), and γ_i^+ has coefficients in $s^{n_i}\mathbb{C}[s]$.
(3) The dilation of depth $r_i + \exp(G)$ divides the equation and the lift regularly by $s^{(2k-2)(r_i+\exp(G))}$ and $s^{m(r_i+\exp(G))}$.
(4) E_i has center the origin of the chart of the dilation of depth $r_i + \exp(G)$.

Since $\exp(G)$ divides r_i, the depth $r_i + \exp(G)$ is divisible by $\exp(G)$, so it is an admissible depth larger than r_i, which contradicts the maximality of r_i.

By **Step 6**, $\mathrm{mult}_{E_i} h_i' \to +\infty$, and **Steps 7** and **8** show that this is impossible. This completes the proof of Theorem 17.10. □

Lemma F.12. *Let $\Bbbk \supset \mathbb{C}$ be an algebraically closed field, G a finite abelian group acting diagonally on $\Bbbk[[x,y,z,s]]$, and $f = x^2 + g(y,z,s)$ a G-semi-invariant element with $g \in (y,z)^3\Bbbk[[y,z,s]]$. Assume that the cubic part $g_0(y,z)$ of $g(y,z,0)$ is a reduced binary cubic form. Then there exists a G-equivariant automorphism Φ of $\Bbbk[[x,y,z,s]]$ with $\Phi(x) = x$, $\Phi(s) = s$ and*

$$(\Phi(y), \Phi(z)) \equiv (y,z)A(s) \mod (y,z)^2$$

for a matrix $A(s) \in GL_2(\Bbbk[[s]])$ with $A(0)$ the identity, such that

$$\Phi(f) = x^2 + g_0(y,z).$$

Moreover, after a G-equivariant $\Bbbk$-linear change of (y,z), we have $g_0 = zy^2 + z^3$ if the image of G in the permutation group of the three linear factors of g_0 has order at most 2, and $g_0 = y^3 + z^3$ if the image has order 3. If moreover g_0 has a G-semi-invariant linear factor ℓ, then the image of G in the permutation group of the three linear factors of g_0 has order at most 2, and the linear change can be taken so that z is a non-zero constant multiple of ℓ.

Proof. For $\sigma \in G$ write $\sigma^* x = \chi_x(\sigma)x$, and similarly for y, z, s, and f. Then $\chi_x^2 = \chi_f$ and $\sigma^* g = \chi_f(\sigma)g$. Write $g = \sum_{d\geq 3} g_d(s;y,z)$, where $g_d \in \Bbbk[[s]][y,z]$ is homogeneous of degree d in y,z. Then $g_3(0;y,z) = g_0(y,z)$, and with

$$D_\sigma := \mathrm{diag}(\chi_y(\sigma), \chi_z(\sigma)),$$

we have

$$g_d(\chi_s(\sigma)s; (y,z)D_\sigma) = \chi_f(\sigma) g_d(s;y,z) \quad \text{for any } d \geq 3 \text{ and } \sigma \in G. \tag{F.140}$$

An automorphism fixing x and s and sending (y,z) to $(y,z)B(s)$, with $B(s) \in GL_2(\Bbbk[[s]])$, is G-equivariant if and only if

$$B(s) = D_\sigma B(\chi_s(\sigma)s) D_\sigma^{-1} \quad \text{for any } \sigma \in G. \tag{F.141}$$

We first make the cubic part constant. Let V_3 be the space of binary cubic forms over $\Bbbk$, so $\dim V_3 = 4$. The differential at the identity matrix I of the morphism $GL_2 \to V_3$, $B \to g_0((y,z)B)$, sends a matrix (b_{jk}) to

$$(b_{11}y + b_{21}z)\partial_y g_0 + (b_{12}y + b_{22}z)\partial_z g_0.$$

Since g_0 has three distinct roots in $\mathbb{P}^1$, its stabilizer in GL_2 is finite, so the differential above, a linear map between 4-dimensional spaces, is bijective. By the formal inverse function theorem, there exists a unique $B(s) \in GL_2(\Bbbk[[s]])$ with $B(0)$ the identity and

$$g_0((y,z)B(s)) = g_3(s;y,z).$$

For $\sigma \in G$, the matrix $B_\sigma := D_\sigma B(\chi_s(\sigma)s) D_\sigma^{-1}$ also equals the identity at $s = 0$, and by (F.140) for $d = 3$ it satisfies $g_0((y,z)B_\sigma) = g_3(s;y,z)$:

$$\begin{aligned} g_0((y,z)D_\sigma B(\chi_s(\sigma)s)D_\sigma^{-1}) &= \chi_f(\sigma)^{-1} g_0((y,z)D_\sigma B(\chi_s(\sigma)s)) \\ &= \chi_f(\sigma)^{-1} g_3(\chi_s(\sigma)s; (y,z)D_\sigma) = g_3(s;y,z). \end{aligned}$$

By uniqueness, $B(s)$ satisfies (F.141), and so does $A(s) := B(s)^{-1}$. Let Φ_1 be the automorphism fixing x and s with $(\Phi_1(y), \Phi_1(z)) = (y, z)A(s)$. Then Φ_1 is G-equivariant and, since $g_3(s; (y, z)A(s)) = g_0((y, z)A(s)B(s)) = g_0(y, z)$,

$$\Phi_1(f) \in x^2 + g_0(y, z) + (y, z)^4\Bbbk[[y, z, s]].$$

Next we remove the terms of degree at least 4 in y, z. Since g_0 has no multiple root, $\partial_y g_0$ and $\partial_z g_0$ form a regular sequence of quadrics in $\Bbbk[y, z]$, so the ideal they generate contains any form of degree $d \geq 3$. Let $d \geq 4$ and let Φ' be a G-equivariant automorphism fixing x and s with

$$\Phi'(f) \in x^2 + g_0 + e_d + (y, z)^{d+1}\Bbbk[[y, z, s]],$$

where $e_d \in \Bbbk[[s]][y, z]$ is homogeneous of degree d in y, z. Then $\sigma^* e_d = \chi_f(\sigma)e_d$. Applying a $\Bbbk$-linear section of the surjection $(a, b) \to a\partial_y g_0 + b\partial_z g_0$ from pairs of forms of degree $d-2$ to forms of degree d coefficientwise in s, we obtain $a, b \in \Bbbk[[s]][y, z]$ homogeneous of degree $d-2$ with $e_d = a\partial_y g_0 + b\partial_z g_0$. Since $\partial_y g_0$ and $\partial_z g_0$ have characters $\chi_f\chi_y^{-1}$ and $\chi_f\chi_z^{-1}$, projecting the identity $e_d = a\partial_y g_0 + b\partial_z g_0$ to the χ_f-isotypic component, we may assume that $\sigma^* a = \chi_y(\sigma)a$ and $\sigma^* b = \chi_z(\sigma)b$ for any $\sigma \in G$. The automorphism Φ'' fixing x and s with $\Phi''(y) = y - a$ and $\Phi''(z) = z - b$ is G-equivariant. Since

$$g_0(y - a, z - b) - g_0 + a\partial_y g_0 + b\partial_z g_0 \in (y, z)^{2(d-2)+1}$$

and, for a form $e \in \Bbbk[[s]][y, z]$ of degree $d' \geq d$ in y, z, $e(y - a, z - b) - e \in (y, z)^{d'-1+d-2}$, where $2(d-2)+1 \geq d+1$ and $d'-1+d-2 \geq d+1$, we have

$$\Phi''(\Phi'(f)) \in x^2 + g_0 + (y, z)^{d+1}\Bbbk[[y, z, s]].$$

The d-th correction has degree $d-2$, so the composite of Φ_1 and of the automorphisms Φ'' for $d = 4, 5, \ldots$ converges in the (y, z)-adic topology. The composite Φ fixes x and s, is G-equivariant, satisfies $\Phi(f) = x^2 + g_0$, and is congruent to Φ_1 modulo $(y, z)^2$.

Finally we normalize g_0. Let Π be the image of G in the permutation group of the three linear factors of g_0. The group Π is abelian, so $\#\Pi \in \{1, 2, 3\}$.

Assume that $\#\Pi \leq 2$, which holds if g_0 has a G-semi-invariant linear factor, since Π fixes it. Then Π fixes a linear factor ℓ of g_0, which we take to be the given one if one is given, so ℓ is G-semi-invariant. If $\chi_y \neq \chi_z$, then ℓ is a multiple of y or of z. If $\chi_y = \chi_z$, then any $\Bbbk$-linear change of (y, z) is G-equivariant. Thus after a G-equivariant linear change $z = \ell$, and

$$g_0 = z(a_1 y^2 + a_2 yz + a_3 z^2), \qquad a_1 \neq 0, \quad \text{and} \quad a_2^2 \neq 4a_1 a_3,$$

because g_0 is reduced. If $a_2 \neq 0$, then $y^2 z$ and yz^2 have the character χ_f, so $\chi_y = \chi_z$ and the change $y \to y - \frac{a_2}{2a_1} z$ is G-equivariant. After the change, $a_2 = 0$, and then $a_3 \neq 0$. Rescaling y and z, we obtain $g_0 = zy^2 + z^3$.

Assume that $\#\Pi = 3$. A root of g_0 at $[1 : 0]$ or $[0 : 1]$ would be fixed by the diagonal group G, so the roots of g_0 are three points ζ of $\Bbbk^*$ (in the coordinate z/y), and the coefficients of y^3 and z^3 in g_0 are non-zero. Let $\rho := \chi_z/\chi_y$. An element $\sigma \in G$ acts on the roots ζ by multiplication by $\rho(\sigma)^{\pm 1}$. For σ inducing a 3-cycle, $\rho(\sigma) \neq 1$ and $\rho(\sigma)^3 = 1$. Since $y^{3-j} z^j$ has character $\chi_y^3 \rho^j$, the monomials $y^2 z$ and yz^2 have characters different from that of y^3, so $g_0 = a_1 y^3 + a_2 z^3$ with $a_1 a_2 \neq 0$. Rescaling y and z, we obtain $g_0 = y^3 + z^3$. □

Proof of Theorem 17.11. Suppose that there exists a radial failure sequence $(W_i \ni o_i, G_i, h_i, \tau_i)$ with constants (m, a, δ) satisfying Set-up 17.1, whose prepared limit (f_*, h_*) is of type cD, with generic transverse type D_k $(k \geq 4)$ along C, and such that the residual cubic of W_* is reduced or a double line and a distinct line. By Definition 4.1(3), $\lim_{i \to +\infty} \tau_i = 1$ and $q_i(\tau_i) \geq a \geq b_i + \delta$ for any i. By Construction-Theorem 13.7(3), there exist an integer $n \geq 4$ and a linear form ℓ which lifts to D_n sections $(\ell_i = 0)$ of the W_i. By Set-up 17.1, $(W_*, \frac{1}{m}\operatorname{div}(h_*))$ has no coefficient-one component, and $\operatorname{Nklt}(W_*, \frac{1}{m}\operatorname{div}(h_*))$ is a regular G-invariant curve $C \subset \operatorname{Sing}(W_*)$. We divide the proof into steps.

Step 1. In this step we fix numerical data and treat the double-line case.

By Construction-Theorem 13.7, after passing to a subsequence, $G_i = G$ with $|G| \le M$, the representation of G and the characters are fixed, $f_i = x^2 + g_i(y,z,t)$ and $f_* = x^2 + g_*(y,z,t)$ in the G-eigenparameters (x,y,z,t) of Construction-Theorem 13.7. For any prime divisor F over W_i with center o_i, we have $a(F, W_i, \frac{1}{m}\operatorname{div}(h_i)) \in \frac{1}{m}\mathbb{Z}$ because W_i is Gorenstein, and e_F divides $|G|$. Hence $b_i \in \frac{1}{m|G|}\mathbb{Z}$, and $0 < b_i \le 3$ by Theorem 5.1(3). Possibly passing to a subsequence, we may assume that $b_i = b$ for any i. For any i fix a computing lift F_i of b, so that $a(F_i, W_i, \frac{1}{m}\operatorname{div}(h_i)) = be_{F_i}$. By the definition of q_i,

$$b + \delta \le q_i(\tau_i) \le \frac{a(F_i, W_i, \frac{\tau_i}{m}\operatorname{div}(h_i))}{e_{F_i}} = b + (1-\tau_i)\frac{\operatorname{mult}_{F_i} h_i}{me_{F_i}},$$

hence

$$\text{(F.142)}\quad \frac{\operatorname{mult}_{F_i} h_i}{me_{F_i}} \ge \frac{\delta}{1-\tau_i} \to +\infty \quad \text{and} \quad a(F_i, W_i, 0) = be_{F_i} + \frac{\operatorname{mult}_{F_i} h_i}{m} \le \frac{\operatorname{mult}_{F_i} h_i}{m} + 3M.$$

Let

$$S_n := (x^2 + y^2 z + z^{n-1} = 0) \subset \mathbb{A}^3,$$

and let $\kappa_n > 0$ be a rational number such that $(S_n, \operatorname{Jac}_{S_n}^{\kappa_n})$ is lc near 0. The number κ_n exists because S_n is klt and Jac_{S_n} is primary to the maximal ideal. Since $(\ell_i = 0)$ is a D_n section of W_i, Theorem 8.4 implies that

$$\text{(F.143)}\quad \kappa_n \operatorname{mult}_F \operatorname{Jac}_{W_i} \le a(F, W_i, 0) \quad \text{for any prime divisor } F \text{ over } W_i \text{ with center } o_i.$$

Assume that the residual cubic of W_* is a double line and a distinct line, that is, the product of the square of a linear form and a distinct linear form. As the linear factors have different multiplicities, each of them is G-invariant. Such ternary cubic forms form a constructible set containing the cubic part of g_*, so by Definition 2.5 the residual cubic of W_i is a double line and a distinct line for $i \gg 0$. By Construction-Theorem 13.7(6), f_* then has the presentation in the hypothesis of Theorem 17.10, with t in place of s, which contradicts Theorem 17.10. Thus from now on the residual cubic of W_* is reduced.

Step 2. In this step we show that W_* is a product of the axis and a D_4 surface.

Since $C \subset \operatorname{Sing}(W_*)$ and $\partial f_*/\partial x = 2x$, we have $C \subset (x = 0) \cong \operatorname{Spec} \Bbbk[[y,z,t]]$. The cotangent line of C at the closed point is spanned by the images of the eigenparameters y, z, t. Thus, possibly permuting y, z, t, we may assume that t restricts to a uniformizer of $\widehat{\mathcal{O}}_C$, and we write $s := t$. Then

$$C = (x = y - \psi_1(s) = z - \psi_2(s) = 0), \qquad \psi_1, \psi_2 \in s\Bbbk[[s]].$$

For $\sigma \in G$, the function

$$\sigma^*(y - \psi_1(s)) = \chi_y(\sigma) y - \psi_1(\chi_s(\sigma) s)$$

vanishes on the G-invariant curve C, and $\Bbbk[[s]] \to \widehat{\mathcal{O}}_C$ is an isomorphism, so

$$\psi_1(\chi_s(\sigma)s) = \chi_y(\sigma)\psi_1(s) \quad \text{and similarly} \quad \psi_2(\chi_s(\sigma)s) = \chi_z(\sigma)\psi_2(s).$$

Hence the translation

$$y \to y + \psi_1(s), \qquad z \to z + \psi_2(s)$$

is G-equivariant. After it, C is the s-axis $(x = y = z = 0)$ and $f_* = x^2 + g(y,z,s)$ for some $g \in \Bbbk[[y,z,s]]$.

Let $\mathbb{K}$ be an algebraic closure of $\Bbbk((s))$. Base changing the completion of W_* at the generic point of C to $\mathbb{K}$, we obtain the generic transverse surface $(x^2 + g(y,z,s) = 0) \subset \operatorname{Spec} \mathbb{K}[[x,y,z]]$, which is Du Val of type D_k. Write $g = \sum_{d \ge 0} g_d(s; y, z)$ with $g_d \in \Bbbk[[s]][y,z]$ homogeneous of degree d in y, z. Since g, $\partial_y g$ and $\partial_z g$ vanish on $C \subset \operatorname{Sing}(W_*)$, we have $g_0 = g_1 = 0$. The quadratic part of an equation of a D_k surface singularity has rank 1, and the rank does not depend on the coordinates or on a unit factor. Since x^2 already has rank 1, $g_2 = 0$ in $\mathbb{K}[y,z]$, hence in $\Bbbk[[s]][y,z]$. Thus

$$\text{(F.144)}\quad g \in (y,z)^3 \Bbbk[[y,z,s]].$$

Since $\psi_1(0)=\psi_2(0)=0$, the residual cubic of W_* is, up to a linear change of coordinates, the cubic part of g at the closed point, which is $g_3(0;y,z)$ by (F.144), so $g_3(0;y,z)$ is reduced, and in particular $g_3(s;y,z)$ is reduced over $\mathbb{K}$, so the generic transverse type is D_4.

By Lemma F.12, and after a G-equivariant linear change of (y,z), there exists a G-equivariant automorphism of $\Bbbk[[x,y,z,s]]$ fixing x and s which sends f_* to

$$f_0 := x^2 + P_0(y,z), \qquad P_0 \in \{zy^2+z^3,\ y^3+z^3\}. \tag{F.145}$$

We replace (f_*,h_*) by their images under the automorphism above, so that $f_*=f_0$. Since f_0 is monic of degree 2 in x, there exists a unique Q_* with $h_*-Q_*f_0$ of x-degree at most 1. Since $(h_*-Q_*f_0)|_{W_*}=h_*|_{W_*}$, possibly replacing h_* with $h_*-Q_*f_0$, we may assume that h_* has x-degree at most 1. Let x,y,z have weights $w=(3,2,2)$ and s weight 0. Then f_0 is weighted homogeneous of weight 6.

We pass the changes of coordinates above to the members as follows. Let $(f_*^{\mathrm{p}},h_*^{\mathrm{p}})$ be the compatible generic limit of the (f_i,h_i) of Construction-Theorem 13.7, before the changes above, and let ϑ be the composite of the translation by (ψ_1,ψ_2), the automorphism of Lemma F.12, and the linear change of (y,z). Then ϑ is a G-equivariant automorphism of $\Bbbk[[x,y,z,s]]$ fixing x and s, and

$$f_* = \vartheta(f_*^{\mathrm{p}}) \quad \text{and} \quad h_* = \vartheta(h_*^{\mathrm{p}}) - Q_*f_*.$$

The power series Q_* is G-semi-invariant by the uniqueness of the division by f_0. By Lemma 2.6(4) for the power series $\vartheta(y)$, $\vartheta(z)$, and Q_*, where, if $Q_*=0$, Q_* is omitted, $Q_i:=0$, and the last entries of the tuples below are dropped, possibly passing to a subsequence, there exist G-semi-invariant polynomials $\vartheta_i(y)$, $\vartheta_i(z)$, and Q_i with the characters of y, z, and Q_* such that $(f_*^{\mathrm{p}},h_*^{\mathrm{p}},\vartheta(y),\vartheta(z),Q_*)$ is a compatible generic limit of the tuples $(f_i,h_i,\vartheta_i(y),\vartheta_i(z),Q_i)$. By the last sentence of Lemma 2.6(4) and Definition 2.5, for $i\gg 0$ the polynomials $\vartheta_i(y)$ and $\vartheta_i(z)$ have no constant terms, and the coefficients of y and z in them form an invertible matrix, as for $\vartheta(y)$ and $\vartheta(z)$. Thus $x\to x$, $y\to\vartheta_i(y)$, $z\to\vartheta_i(z)$, $s\to s$ defines a G-equivariant automorphism ϑ_i of $\mathbb{C}[[x,y,z,s]]$. We replace (f_i,h_i) with $(\vartheta_i(f_i),\vartheta_i(h_i)-Q_i\vartheta_i(f_i))$ and keep the names x,y,z,s. For $i\gg 0$ the new f_i and h_i are non-zero with bounded orders at 0, since non-zero coefficients of f_* and h_* stay non-zero. For any positive integer e, on the closed subset of the jet space where the constant terms of the third and the fourth entries vanish, one morphism defined over $\mathbb{Q}$ sends the e-jets of $(f_i,h_i,\vartheta_i(y),\vartheta_i(z),Q_i)$, for the old (f_i,h_i), and of $(f_*^{\mathrm{p}},h_*^{\mathrm{p}},\vartheta(y),\vartheta(z),Q_*)$ to the e-jets of the new (f_i,h_i) and of (f_*,h_*). The closure of the image of the irreducible closed subset of Definition 2.5 for the tuples $(f_i,h_i,\vartheta_i(y),\vartheta_i(z),Q_i)$ is irreducible, the e-jet of (f_*,h_*) maps to the generic point of the closure, and the preimage of any non-empty open subset of the closure is a non-empty open subset. Hence (f_*,h_*) is a compatible generic limit of the new (f_i,h_i), and by Definition 2.5 the following holds:

(∗) Any polynomial identity with complex coefficients among finitely many coefficients of the presentations which holds for (f_*,h_*) holds for (f_i,h_i) for $i\gg 0$, and so does the non-vanishing of any one coefficient.

Step 3. In this step we show that $\mathrm{mult}_{F_i}s$ is bounded, compute the critical weight, and pass to a chart in which the F_i are centered at the origin and have quotient value b.

Since $\mathrm{Nklt}(W_*,\frac{1}{m}\operatorname{div}(h_*))=C$, considering a log resolution of $(W_*,\operatorname{div}(h_*))$, we may choose a rational number $\mu>\frac{1}{m}$ such that $(W_*,\mu\operatorname{div}(h_*))$ is klt on $W_*\backslash C$. Choose rational numbers $0<\theta_1<\theta_2<1$ with

$$\gamma := \theta_1\mu - \frac{1}{m} - (1-\theta_1)\Big(1+\frac{2}{m\kappa_n}\Big) > 0,$$

which is possible since $\gamma\to\mu-\frac{1}{m}>0$ as $\theta_1\to 1$. Let $\mathcal{J}_*:=\mathcal{J}(f_*^{\theta_2}h_*^{\theta_2\mu})$. By Lemma 6.1(3) with $V=C$, $\mathcal{J}_*$ is the unit ideal near any point of $W_*\backslash C$, so the zero locus of $(f_*)+\mathcal{J}_*$ is contained in $C=(x=y=z=0)$. Hence there exists a positive integer N_1 with

$$(x,y,z)^{N_1}\subset(f_*)+\mathcal{J}_*.$$

Fix $\omega \in \{x^{N_1}, y^{N_1}, z^{N_1}\}$ and a positive integer ρ, and write $\omega = a_\omega f_* + j_\omega$ with $a_\omega \in \Bbbk[[x,y,z,s]]$ and $j_\omega \in \mathcal{J}_*$. Let a_u be the polynomial of degree less than ρ in x,y,z,s whose coefficients are independent variables, let $\Theta_\omega := \omega - a_u y_{1,0}$, a polynomial in the variables $y_{k,\alpha}$, the coefficients of a_u, and x,y,z,s, with the substitution of Lemma 6.1 for pairs (f,h), so that $y_{1,0}$ is replaced by f, and let ς_ω be the vector of coefficients of the part of degree less than ρ of a_ω. Then

$$\Theta_\omega\big((f_*,h_*),\varsigma_\omega\big) = \omega - a_u(\varsigma_\omega) f_* \in \mathcal{J}_* + \mathfrak{n}_*^\rho,$$

where $\mathfrak{n}_*$ is the maximal ideal of $\Bbbk[[x,y,z,s]]$. Let $\mathfrak{m}$ be the maximal ideal of $\mathbb{C}[[x,y,z,s]]$. By Lemma 6.1(1)(2) with $L=\rho$, the pairs (f_i,h_i), and the exponents $(\theta_1,\theta_1\mu)$ for the members and $(\theta_2,\theta_2\mu)$ for the limit, for all but finitely many i there exists a complex vector $\varsigma_{\omega,i}$ such that $z_{\omega,i} := \omega - a_u(\varsigma_{\omega,i}) f_i$ satisfies the following: any prime divisor $\widehat{F}$ over $\operatorname{Spec}\mathbb{C}[[x,y,z,s]]$ with center the closed point and $\operatorname{mult}_{\widehat{F}} z_{\omega,i} < \rho$, hence $\operatorname{mult}_{\widehat{F}}(\mathfrak{m}^\rho) > \operatorname{mult}_{\widehat{F}} z_{\omega,i}$, satisfies

$$\operatorname{mult}_{\widehat{F}} z_{\omega,i} + a(\widehat{F}, \operatorname{Spec}\mathbb{C}[[x,y,z,s]],0) > \theta_1 \operatorname{mult}_{\widehat{F}} f_i + \theta_1\mu \operatorname{mult}_{\widehat{F}} h_i.$$

Moreover $z_{\omega,i}|_{W_i} = \omega|_{W_i}$, which is non-zero since $f_i \in \mathfrak{m}^2$ is irreducible, W_i being normal and singular at o_i. Let F be a prime divisor over W_i with center o_i and $\operatorname{mult}_F(\omega|_{W_i}) < \rho$. By Lemma 5.5(1) with $g = z_{\omega,i}$ and threshold ρ, and by (F.143), the identity

$$\Big(1 + 2(1-\theta_1)\kappa_n^{-1}\Big)\frac{\operatorname{mult}_F h_i}{m} + \gamma \operatorname{mult}_F h_i = (\theta_1\mu - 1 + \theta_1)\operatorname{mult}_F h_i$$

and $a(F, W_i, \frac{1}{m}\operatorname{div}(h_i)) \ge 0$ imply that

$$\gamma \operatorname{mult}_F h_i < (2-\theta_1)\operatorname{mult}_F(\omega|_{W_i}) + (1+2\kappa_n^{-1}) a\Big(F, W_i, \frac{1}{m}\operatorname{div}(h_i)\Big) + 1. \tag{F.146}$$

For $F = F_i$ we have $a(F_i, W_i, \frac{1}{m}\operatorname{div}(h_i)) = b e_{F_i} \le 3M$, so with $\gamma_0 := 3M(1+2\kappa_n^{-1})+1$, for all but finitely many i,

$$\operatorname{mult}_{F_i}(\omega|_{W_i}) \ge \min\Big\{\rho, \frac{\gamma \operatorname{mult}_{F_i} h_i - \gamma_0}{2-\theta_1}\Big\}.$$

By (F.142), $\operatorname{mult}_{F_i} h_i$ tends to $+\infty$. As ρ is arbitrary, $\operatorname{mult}_{F_i}(\omega|_{W_i})$ tends to $+\infty$ for $\omega = x^{N_1}, y^{N_1}, z^{N_1}$, that is, $\operatorname{mult}_{F_i} x$, $\operatorname{mult}_{F_i} y$, and $\operatorname{mult}_{F_i} z$ tend to $+\infty$. By Theorem 5.1(3), $\operatorname{mult}_{F_i} \mathfrak{m}_{o_i} \le 3M/\eta_m$, and $\mathfrak{m}_{o_i}$ is generated by the restrictions of x,y,z,s, so for $i \gg 0$

$$0 < \operatorname{mult}_{F_i} s = \operatorname{mult}_{F_i} \mathfrak{m}_{o_i} \le T_0 := \frac{3M}{\eta_m}. \tag{F.147}$$

Write $h_* = \sum_v h_{*,v}$, where $h_{*,v} \in \Bbbk[[s]][x,y,z]$ is the part of weight v, and let $v_{\min} := \min\{v \mid h_{*,v} \ne 0\}$. Let $S_0 := (f_0 = 0) \subset \mathbb{A}^3$ and $S_{\mathbb{K}} := S_0 \times_{\Bbbk} \mathbb{K}$. By (F.145), the generic transverse pair of $(W_*, \frac{1}{m}\operatorname{div}(h_*))$ along C is the germ at 0 of $(S_{\mathbb{K}}, \frac{1}{m}\operatorname{div}(h_*))$, and it is lc and not klt because C is a non-klt center and $(W_*, \frac{1}{m}\operatorname{div}(h_*))$ is lc. The surface $S_{\mathbb{K}}$ is the quotient $\mathbb{A}^2_{\mathbb{K}}/Q_8$ by the quaternion group $Q_8 \subset SL_2$, which acts freely outside 0, so the quotient morphism $\pi : \mathbb{A}^2_{\mathbb{K}} \to S_{\mathbb{K}}$ is étale in codimension 1. The invariants of Q_8 have degrees $4,4,6$, matching the weights $(2,2,3)$ of y,z,x, so a function of weight v on $S_{\mathbb{K}}$ pulls back to a binary form of degree $2v$ [CCJ20, Appendix A, (A.4.4)]. By Lemma F.11 over $\mathbb{K}$ with $g = h_*$ and $t = \frac{1}{m}$, where $\operatorname{wt}_w(h_*) = v_{\min}$, the pair $(S_{\mathbb{K}}, \frac{1}{m}\operatorname{div}(h_*))$ is lc near 0 if and only if $v_{\min} \le m$, and klt near 0 if and only if $v_{\min} < m$. Hence $v_{\min} = m$. Let

$$P := h_{*,m} \in \Bbbk[[s]][x,y,z],$$

which is weighted homogeneous of weight m and of x-degree at most 1, and let $c \ge 0$ be the minimum of the s-orders of the coefficients of P.

Let $J := 1 + 4M(m+6)$, the integer of Proposition 16.4 for row D with $n=4$. Fix an integer $r > c + J$ divisible by $|G|$, let $\widehat{\mathcal{X}}_i := \operatorname{Spec}\mathbb{C}[[x,y,z,s]]$ be the ambient germ of $W_i = (f_i = 0)$ and $\widehat{\mathcal{X}} := \operatorname{Spec}\mathbb{C}[[x',y',z',s]]$, and let $\delta_r : \widehat{\mathcal{X}} \to \widehat{\mathcal{X}}_i$ be the chart of the weighted blow-up with weights $(3r,2r,2r,1)$ on (x,y,z,s) given by

$$x = s^{3r}x', \quad y = s^{2r}y', \quad z = s^{2r}z', \quad \text{and} \quad s = s.$$

Since f_0 is weighted homogeneous of weight 6 and h_* has no part of weight less than m,

$$f_*\circ\delta_r = s^{6r}f_0(x',y',z'),\qquad h_*\circ\delta_r = s^{mr+c}h'_*,\qquad \text{and}\qquad h'_* := \sum_{v\geq m} s^{r(v-m)-c}h_{*,v}(x',y',z'),$$

with h'_* regular and $h'_*|_{s=0} = (s^{-c}P)|_{s=0}(x',y',z') \neq 0$. For a monomial ϖ in x,y,z of weight v, the monomial $s^j\varpi(x,y,z)$ becomes $s^{j+rv}\varpi(x',y',z')$, so the divisibility of $f_i\circ\delta_r$ by s^{6r} and of $h_i\circ\delta_r$ by s^{mr+c}, as well as the restrictions of the quotients to $(s=0)$, are determined by the coefficients of the finitely many monomials with $j+rv\leq\max\{6r,mr+c\}$. By $(*)$, for $i\gg 0$ the regular functions

$$f'_i := s^{-6r}f_i\circ\delta_r \quad\text{and}\quad h'_i := s^{-mr-c}h_i\circ\delta_r$$

are defined, $f'_i|_{s=0} = f_0(x',y',z')$, and $h'_i|_{s=0}$ is a non-zero polynomial of x'-degree at most 1. Let $W'_i := (f'_i=0)$. Since r is divisible by $|G|$, the coordinates x',y',z' have the characters of x,y,z, and G acts faithfully on W'_i, since δ_r is G-equivariant and birational onto W_i and G acts faithfully on W_i. The Cartier divisor $W'_i\cap(s=0) = S_0$ is a normal surface with the unique singular point 0, of type D_4, so W'_i is normal near 0. Outside $(s=0)$, δ_r (here and below, the charts, taken in the formal coordinates x,y,z,s, are the completions at the origin of the charts of the corresponding weighted blow-ups of algebraic germs, written in algebraic G-eigenfunctions congruent to x,y,z,s modulo a high power of the maximal ideal, and isomorphisms, birationality and prime divisors with center the origin are read on the algebraic charts by Lemma 2.12(3)) is an isomorphism onto an open subset of $W_i\backslash\{o_i\}$, which is smooth as $(W_i\ni o_i)$ is an isolated singularity (Definition 4.1(1)), so $(W'_i\ni 0)$ is an isolated cDV point, hence Gorenstein terminal.

Since

$$\delta_r^*(dx\wedge dy\wedge dz\wedge ds)/f_i = s^{7r-6r}(dx'\wedge dy'\wedge dz'\wedge ds)/f'_i,$$

any prime divisor F over W'_i satisfies

$$a(F,W_i,0) = a(F,W'_i,0) + r\operatorname{mult}_F s,\qquad \operatorname{mult}_F h_i = \operatorname{mult}_F h'_i + (mr+c)\operatorname{mult}_F s. \tag{F.148}$$

For the prime divisor $E := W'_i\cap(s=0)$ over W_i, whose center is o_i, (F.148) implies that

$$a\Big(E,W_i,\frac{1}{m}\operatorname{div}(h_i)\Big) = 1+r-\frac{mr+c}{m} = 1-\frac{c}{m},$$

which is positive since $(W_i,\frac{1}{m}\operatorname{div}(h_i))$ is klt near o_i. Thus

$$\beta := \frac{c}{m}\in[0,1),$$

and by (F.148), any prime divisor F over W'_i and any $t\in[0,1]$ satisfy

$$\begin{aligned} a\Big(F,W_i,\frac{t}{m}\operatorname{div}(h_i)\Big) &= a\Big(F,W'_i,\frac{1}{m}\operatorname{div}(h'_i)+\beta\operatorname{div}(s)\Big)\\ &\quad+(1-t)\Big(\frac{\operatorname{mult}_F h'_i}{m}+(r+\beta)\operatorname{mult}_F s\Big). \end{aligned} \tag{F.149}$$

By (F.142) and (F.147), $\operatorname{mult}_{F_i}x' = \operatorname{mult}_{F_i}x - 3r\operatorname{mult}_{F_i}s\to+\infty$, and similarly for y' and z', so F_i has center 0 on W'_i for $i\gg 0$. A prime divisor over W'_i with center 0 has center o_i on W_i, and the inertia groups of such a divisor over W_i and over W'_i are equal since δ_r is G-equivariant and birational. Hence (F.149) for $t=1$ implies that for $i\gg 0$

$$\operatorname{mld}\Big(\Big(W'_i\ni 0,\frac{1}{m}\operatorname{div}(h'_i)+\beta\operatorname{div}(s)\Big)/G\Big) = b, \tag{F.150}$$

computed by F_i.

For any e, the e-jets of (f'_i,h'_i) and of (f_0,h'_*) are the images of jets of bounded degree of (f_i,h_i) and of (f_*,h_*), respectively, under one linear map of jet spaces defined over $\mathbb{Q}$, so, as for the change of coordinates in **Step 2**, (f_0,h'_*) is a compatible generic limit of the (f'_i,h'_i): the part of weight m of h'_* is $s^{-c}P$, and

$$h'_* - s^{-c}P\in(s^{r-c}).$$

We rename x', y', z' as x, y, z and $s^{-c}P$ as P. Thus from now on $P \in \Bbbk[[s]][x, y, z]$ is weighted homogeneous of weight m, of x-degree at most 1, the s-orders of the coefficients of P have minimum 0, and

$$h'_* - P \in (s^{r-c}) \subset (x, y, z, s)^J. \tag{F.151}$$

Step 4. In this step we exclude the case where the $\mathbb{Q}$-divisor $\frac{1}{m}\operatorname{div}(P|_{S_\mathbb{K}})$ on $S_\mathbb{K}$ has a component with coefficient 1.

Assume that $\Gamma \subset S_\mathbb{K}$ is a prime divisor with $\operatorname{mult}_\Gamma(P|_{S_\mathbb{K}}) = m$. Since P is weighted homogeneous, Γ is stable under the weighted $\mathbb{G}_m$-action, so $\pi^{-1}(\Gamma)$ is a union of lines through 0 forming one Q_8-orbit, and $\psi := \pi^*(P|_{S_\mathbb{K}})$ is a binary form of degree $2m$. Since π is étale in codimension 1, each line in $\pi^{-1}(\Gamma)$ has multiplicity m in ψ. Since any Q_8-orbit in $\mathbb{P}^1$ has 2 or 4 points and $\deg\psi = 2m$, the orbit has 2 points, and

$$\psi = \xi(l_1 l_2)^m, \qquad \xi \in \mathbb{K}^*,$$

for linear forms l_1, l_2. The quadratic form $l_1 l_2$ is Q_8-semi-invariant, since the zero set of $l_1 l_2$ is Q_8-stable, and not Q_8-invariant, since there does not exist a Q_8-invariant binary form of degree 2. As the characters of Q_8 have order at most 2, the character of $l_1 l_2$ has order 2. Since ψ is Q_8-invariant, m is even, and $(l_1 l_2)^2 = \pi^*(L|_{S_\mathbb{K}})$ for a linear form L in y, z, the functions of weight 2 on $S_\mathbb{K}$ being the restrictions of linear forms in y, z. The curve $(L = 0) \cap S_\mathbb{K} = \Gamma$ is then non-reduced in $S_\mathbb{K}$, so $x^2 + P_0|_{L=0}$ is a square, that is, L divides P_0. Since P and $\xi L^{m/2}$ have x-degree at most 1, satisfy $P|_{S_\mathbb{K}} = \xi L^{m/2}|_{S_\mathbb{K}}$, and f_0 is monic of degree 2 in x, we obtain $P = \xi L^{m/2}$. Since $P \in \Bbbk[[s]][x, y, z]$, the minimum of the s-orders of the coefficients of P is 0, and $L \in \Bbbk[y, z]$, we have

$$P = \upsilon(s) L^{m/2}, \qquad \upsilon(s) \in \Bbbk[[s]], \quad \text{and} \quad \upsilon(0) \neq 0.$$

As P is G-semi-invariant, so is L, and G fixes the linear factor L of P_0. By the last sentence of Lemma F.12 for the field $\mathbb{C}$, for $g_0 = P_0$, and for ℓ a linear factor of P_0 with complex coefficients proportional to L, after a G-equivariant linear change of (y, z) with complex coefficients, which we also apply to the (f'_i, h'_i) and which multiplies $\upsilon(s)$ by a non-zero constant, we have

$$f_0 = x^2 + y^2 z + z^3, \qquad P = \upsilon(s) z^{m/2}.$$

By (F.151), $h'_* - \upsilon(s) z^{m/2} \in (x, y, z, s)^J$. Let $\upsilon_i \in \mathbb{C}[s]$ be the specialization at (f_i, h_i) of the part of degree less than J of υ. Then $\upsilon_i(0) \neq 0$ for $i \gg 0$. By $(*)$ applied to the J-jets, for $i \gg 0$

$$f'_i - f_0 \in (x, y, z, s)^J \quad \text{and} \quad \upsilon_i^{-1} h'_i - z^{m/2} \in (x, y, z, s)^J.$$

By Proposition 16.4 for row D with $n = 4$, applied to W'_i with the semi-invariant regular parameters x, y, z, s and the functions f'_i and $\upsilon_i^{-1} h'_i$, where υ_i is a unit near 0, the following hold for the order R of χ_s:

- The pair
$$\Big(W'_i, \frac{1}{m}\operatorname{div}(h'_i) + \operatorname{div}(s)\Big)$$
is lc near 0.
- $\operatorname{mld}((W'_i \ni 0, \frac{1}{m}\operatorname{div}(h'_i))/G) = \frac{1}{R}$, computed by a prime divisor F over W'_i with center 0 such that $a(F, W'_i, 0) \le (2 + \frac{1}{R}) e_F$ and $\operatorname{mult}_F h'_i \le 2m e_F$.

For any prime divisor F' over W'_i with center 0, the function s^R is G-invariant, so $e_{F'}$ divides $R \operatorname{mult}_{F'} s$ by [Liu26a, Lemma 9.2(3)]. Since $\operatorname{mult}_{F'} s > 0$, we have $\operatorname{mult}_{F'} s \ge e_{F'}/R$. For the prime divisor F above,

$$\frac{e_F}{R} = a\Big(F, W'_i, \frac{1}{m}\operatorname{div}(h'_i)\Big) = a\Big(F, W'_i, \frac{1}{m}\operatorname{div}(h'_i) + \operatorname{div}(s)\Big) + \operatorname{mult}_F s \ge \operatorname{mult}_F s \ge \frac{e_F}{R},$$

so $\operatorname{mult}_F s = e_F/R$ and

$$a\Big(F, W_i', \frac{1}{m}\operatorname{div}(h_i') + \beta \operatorname{div}(s)\Big) = \frac{e_F}{R} - \beta \operatorname{mult}_F s = (1-\beta)\frac{e_F}{R}.$$

On the other hand, any prime divisor F' over W_i' with center 0 satisfies

$$a\Big(F', W_i', \frac{1}{m}\operatorname{div}(h_i') + \beta \operatorname{div}(s)\Big) = a\Big(F', W_i', \frac{1}{m}\operatorname{div}(h_i') + \operatorname{div}(s)\Big) + (1-\beta)\operatorname{mult}_{F'} s \geq (1-\beta)\frac{e_{F'}}{R}.$$

By (F.150), $b = (1-\beta)/R$, computed by F. Since F has center o_i on W_i, and the inertia groups of F over W_i and over W_i' are equal, (F.149) for $t = \tau_i$ implies that

$$\begin{aligned} b + \delta \leq q_i(\tau_i) \leq \frac{a(F, W_i, \frac{\tau_i}{m}\operatorname{div}(h_i))}{e_F} &= b + (1-\tau_i)\Big(\frac{\operatorname{mult}_F h_i'}{m e_F} + (r+\beta)\frac{\operatorname{mult}_F s}{e_F}\Big) \\ &\leq b + (1-\tau_i)\Big(2 + \frac{r+1}{R}\Big). \end{aligned}$$

Since r and R do not depend on i and $\lim_{i\to+\infty}\tau_i = 1$, the last inequality fails for $i \gg 0$.

Thus the $\mathbb{Q}$-divisor $\frac{1}{m}\operatorname{div}(P|_{S_{\mathbb{K}}})$ on $S_{\mathbb{K}}$ has no component with coefficient 1. The proof continues below.

Proof of Theorem 17.11, continued. It remains to treat the case where the residual cubic of W_* is reduced and $\Delta_P := \frac{1}{m}\operatorname{div}(P|_{S_{\mathbb{K}}})$ has no coefficient-one component. Since $f_0 = x^2 + P_0$ with $P_0 \in \{zy^2 + z^3, y^3 + z^3\}$ by (F.145), and P has x-degree at most 1 and weight m, Lemma F.11 with $\mathbb{L} = \mathbb{K}$ ($k = 4$ if $P_0 = zy^2 + z^3$), $g = P$, and $t = \frac{1}{m}$, together with Lemma 2.12(2), shows that $(S_{\mathbb{K}}, \Delta_P)$ is lc near 0, hence lc, as its non-lc locus is closed and stable under the weighted $\mathbb{G}_m$-action.

Let $t \in \{1, \tau_i\}$, and let $\widetilde{h}$ be a G-semi-invariant lift of $h_i|_{W_i}$ to $\widehat{\mathcal{X}}_i$. We show that any prime divisor E over $\widehat{\mathcal{X}}_i$ with center 0 satisfies

$$a\Big(E, \widehat{\mathcal{X}}_i, \operatorname{div}(f_i) + \frac{t}{m}\operatorname{div}(\widetilde{h})\Big) \geq q_i(t) e_E. \tag{F.152}$$

The algebraic germ $(W_i \ni o_i)$ of Definition 4.1 embeds G-equivariantly, as a hypersurface $(f = 0)$, into a smooth germ Y of dimension 4, and its mark is the restriction of a G-semi-invariant regular function h on Y. Thus $f_i = v\psi(\widehat{f})$ and $\widetilde{h} = \psi(\widehat{h}) + f_i g$, where $\widehat{f}, \widehat{h}$ are the expansions of Construction-Theorem 2.11 in G-semi-invariant regular parameters of Y with the characters of x, y, z, s, ψ is a G-equivariant automorphism of $\mathbb{C}[[x, y, z, s]]$, v is a G-invariant unit, and g is a G-semi-invariant power series. By Set-up 17.1(2) and $t \leq 1$, the pair $(W_i, \frac{t}{m}\operatorname{div}(h_i))$ is klt near o_i, and Construction-Theorem 2.11(3) gives integers N_4 and then N_5, both larger than $\operatorname{mult}_E f_i + \operatorname{mult}_E \widetilde{h}$, such that $f_{(N_5)} := (f_i)_{<N_5}$ and $h_{(N_4)} := \psi(\widehat{h})_{<N_4}$ define a polynomial model $(X_0 \ni 0, \frac{t}{m}\operatorname{div}(h_{(N_4)}|_{X_0}))$, $X_0 := (f_{(N_5)} = 0) \subset \mathbb{A}^4$, with quotient value $q_i(t)$. By Lemma 2.12(3), E restricts to a prime divisor E' over $\mathbb{A}^4$ with center 0, $a(E', \mathbb{A}^4, 0) = a(E, \widehat{\mathcal{X}}_i, 0)$, and $e_{E'} = e_E$. Theorem 2.10 for the polynomial lift $\widetilde{h}_{(N_5)} := h_{(N_4)} + f_{(N_5)} g_{<N_5}$ of $h_{(N_4)}|_{X_0}$ gives

$$a\Big(E', \mathbb{A}^4, \operatorname{div}(f_{(N_5)}) + \frac{t}{m}\operatorname{div}(\widetilde{h}_{(N_5)})\Big) \geq q_i(t) e_{E'}.$$

As

$$f_i - f_{(N_5)},\ \widetilde{h} - \widetilde{h}_{(N_5)} \in (x, y, z, s)^{\min\{N_4, N_5\}},$$

we have $\operatorname{mult}_{E'} f_{(N_5)} = \operatorname{mult}_E f_i$ and $\operatorname{mult}_{E'} \widetilde{h}_{(N_5)} = \operatorname{mult}_E \widetilde{h}$, and (F.152) follows.

Step 5. In this step we choose, for any i, a chart of maximal depth.

For $\nu \in \mathbb{N}$ let σ_ν be the $\mathbb{C}[[s]]$-algebra endomorphism of $\mathbb{C}[[x, y, z, s]]$ with

$$\sigma_\nu(x) = s^{3\nu}x, \quad \sigma_\nu(y) = s^{2\nu}y, \quad \text{and} \quad \sigma_\nu(z) = s^{2\nu}z.$$

It is the pull-back by a birational morphism $\epsilon_\nu : \widehat{\mathcal{X}} \to \widehat{\mathcal{X}}$ with Jacobian determinant $s^{7\nu}$, and it is G-equivariant if $|G|$ divides ν. For a power series g and an integer ω,

$$\text{(F.153)}\quad s^{-\omega\nu}\sigma_\nu(g) \in \mathbb{C}[[x,y,z,s]] \iff \\ j \geq \nu(\omega - v) \text{ for any term of } g \text{ of } s\text{-degree } j \text{ and weight } v < \omega.$$

Fix $\nu \in |G|\mathbb{N}$. By (F.153), $s^{-6\nu}\sigma_\nu(f_i')$ and $s^{-m\nu}\sigma_\nu(h_i')$ are power series if and only if finitely many coefficients of f_i' and h_i' vanish, namely those of the terms of s-degree j and weight v with $v < 6$ and $j < \nu(6-v)$, resp. $v < m$ and $j < \nu(m-v)$. The corresponding coefficients of f_0 and h_*' vanish. A polynomial function on the jet space $\mathcal{J}_d$ vanishing at the generic point of Z_d vanishes on Z_d, and Z_d contains the d-jets of all but finitely many members (Definition 2.5). The sum of the terms of weight 6 of $f_i'(x,y,z,0)$ is $c_i x^2 + P_{0,i}(y,z)$ with $c_i \in \mathbb{C}$ and $P_{0,i}$ a binary cubic form. The conditions $c_i \neq 0$ and $P_{0,i}$ reduced are open conditions on a finite jet and hold for the limit, where $c_i x^2 + P_{0,i} = f_0$, hence for all but finitely many i. Moreover $\operatorname{mult}_{F_i} x - 3\nu \operatorname{mult}_{F_i} s$, $\operatorname{mult}_{F_i} y - 2\nu \operatorname{mult}_{F_i} s$, and $\operatorname{mult}_{F_i} z - 2\nu \operatorname{mult}_{F_i} s$ tend to $+\infty$. Possibly passing to a subsequence, we may therefore fix $\nu_i^0 \in |G|\mathbb{N}^+$ with $\lim_{i\to+\infty} \nu_i^0 = +\infty$, such that for any i, $c_i \neq 0$, $P_{0,i}$ is reduced, both $s^{-6\nu_i^0}\sigma_{\nu_i^0}(f_i')$ and $s^{-m\nu_i^0}\sigma_{\nu_i^0}(h_i')$ are power series and the center of F_i on $(\epsilon_{\nu_i^0})_*^{-1}W_i'$ is the origin.

A *chart* for i is a triple $\mathcal{C} = (\Xi, \zeta, \nu)$ satisfying the following.

(a) Ξ is the G-equivariant $\mathbb{C}[[s]]$-automorphism of $\mathbb{C}[[x,y,z,s]]$ with

$$\Xi(x) = x + \lambda_0 + \lambda_1 y + \lambda_2 z, \quad \Xi(y) = y + \lambda_3, \quad \text{and} \quad \Xi(z) = z + \lambda_4,$$

where $\lambda_0, \dots, \lambda_4 \in s^{\nu_i^0}\mathbb{C}[s]$.

(b) $\zeta \in s^{\nu_i^0}\mathbb{C}[s][x,y,z]$ is G-semi-invariant with $\chi_\zeta \chi_{f_i'} = \chi_{h_i'}$, and any term of ζ has weight less than $m-6$. In particular, $\zeta = 0$ if $m \leq 6$.

(c) $\nu \in |G|\mathbb{N}$, both

$$f_{\mathcal{C}} := s^{-6\nu}\sigma_\nu(\Xi(f_i')) \quad \text{and} \quad h_{\mathcal{C}} := s^{-m\nu}\sigma_\nu(\Xi(h_i' + f_i'\zeta))$$

are power series, and the center of F_i on $W_{\mathcal{C}} := (f_{\mathcal{C}} = 0)$ is the origin.

The chart is *admissible* if moreover $\nu \geq \nu_i^0$. We let $\widehat{\mathcal{X}}_{\mathcal{C}} := \operatorname{Spec}\mathbb{C}[[x,y,z,s]]$, and let $\epsilon_{\mathcal{C}}$ be the morphism from $\widehat{\mathcal{X}}_{\mathcal{C}}$ to $\widehat{\mathcal{X}}$ with $\epsilon_{\mathcal{C}}^* = \sigma_\nu \circ \Xi$. Since Ξ has Jacobian determinant 1, $\epsilon_{\mathcal{C}}^* f_i' = s^{6\nu} f_{\mathcal{C}}$ and $(h_i' + f_i'\zeta)|_{W_i'} = h_i'|_{W_i'}$, any prime divisor F over $W_{\mathcal{C}}$ with center 0 satisfies

$$\text{(F.154)}\qquad a(F, W_i', 0) = a(F, W_{\mathcal{C}}, 0) + \nu \operatorname{mult}_F s, \qquad \operatorname{mult}_F h_i' = \operatorname{mult}_F h_{\mathcal{C}} + m\nu \operatorname{mult}_F s.$$

Thus

$$a\Big(F, W_i', \frac{1}{m}\operatorname{div}(h_i') + \beta \operatorname{div}(s)\Big) = a\Big(F, W_{\mathcal{C}}, \frac{1}{m}\operatorname{div}(h_{\mathcal{C}}) + \beta \operatorname{div}(s)\Big).$$

As $\epsilon_{\mathcal{C}}$ sends the origin to the origin, F_i computes

$$b = \operatorname{mld}\Big(\Big(W_{\mathcal{C}} \ni 0, \frac{1}{m}\operatorname{div}(h_{\mathcal{C}}) + \beta \operatorname{div}(s)\Big)/G\Big).$$

In the coordinates $\overline{x}, \overline{y}, \overline{z}, s$ of $\widehat{\mathcal{X}}_i$ with $\delta_r^*(\overline{x}, \overline{y}, \overline{z}) = (s^{3r}x, s^{2r}y, s^{2r}z)$, the automorphism Ξ becomes

$$\overline{x} \to \overline{x} + s^{3r}\lambda_0 + s^r(\lambda_1\overline{y} + \lambda_2\overline{z}), \quad \overline{y} \to \overline{y} + s^{2r}\lambda_3, \quad \text{and} \quad \overline{z} \to \overline{z} + s^{2r}\lambda_4,$$

and $s^{mr+c} f_i'\zeta = (f_i\widetilde{\zeta}) \circ \delta_r$, where a term $s^j x^{k_1} y^{k_2} z^{k_3}$ of ζ of weight v contributes the term $s^{j+c+r(m-6-v)}\overline{x}^{k_1}\overline{y}^{k_2}\overline{z}^{k_3}$, with the coefficient of $s^j x^{k_1} y^{k_2} z^{k_3}$, to $\widetilde{\zeta}$. Since $v < m-6$, $\widetilde{\zeta}$ is a power series, and $h_i + f_i\widetilde{\zeta}$ is a G-semi-invariant lift of $h_i|_{W_i}$ to $\widehat{\mathcal{X}}_i$. Composing δ_r with $\epsilon_{\mathcal{C}}$, the Jacobian

determinants s^{7r} and $s^{7\nu}$ and the divisions by $s^{6r+6\nu}$ and $s^{(r+\beta+\nu)m}$ imply that any prime divisor E over $\widehat{\mathcal{X}}_{\mathcal{C}}$ with center 0 satisfies, for any $t \in [0,1]$,

$$\begin{aligned}(\text{F.155})\quad & a\Big(E, \widehat{\mathcal{X}}_i, \operatorname{div}(f_i) + \frac{t}{m}\operatorname{div}(h_i + f_i\widetilde{\zeta})\Big) \\ & = a\Big(E, \widehat{\mathcal{X}}_{\mathcal{C}}, \operatorname{div}(f_{\mathcal{C}}) + \frac{1}{m}\operatorname{div}(h_{\mathcal{C}}) + \beta\operatorname{div}(s)\Big) + (1-t)\Big(\frac{\operatorname{mult}_E h_{\mathcal{C}}}{m} + (\nu + r + \beta)\operatorname{mult}_E s\Big).\end{aligned}$$

For $t = 1$ the second term of the right-hand side of (F.155) vanishes, and (F.152) for $t = 1$ and the lift $h_i + f_i\widetilde{\zeta}$, with $q_i(1) = b$, implies

$$(\text{F.156})\qquad a\Big(E, \widehat{\mathcal{X}}_{\mathcal{C}}, \operatorname{div}(f_{\mathcal{C}}) + \frac{1}{m}\operatorname{div}(h_{\mathcal{C}}) + \beta\operatorname{div}(s)\Big) \geq be_E.$$

Assume that $\nu \geq 1$ and let $D := W_{\mathcal{C}} \cap (s = 0)$. The terms of $\Xi(f_i')$ of weight $v > 6$ acquire the factor $s^{\nu(v-6)}$ in $f_{\mathcal{C}}$, and Ξ does not change the terms of f_i' that are not divisible by s. Thus $D = (c_i x^2 + P_{0,i} + g_{<6} = 0) \subset \mathbb{A}^3$, where $g_{<6} \in \mathbb{C}[x,y,z]$ has weight less than 6. Replacing x with x plus an affine linear function of y, z, we obtain $D \cong (x^2 + g = 0)$ with $g = c_i^{-1}P_{0,i} + g_{\leq 2}$ and $\deg g_{\leq 2} \leq 2$. At any point the plane curve $(g = 0)$ has multiplicity at most 3. If the multiplicity is 3, then g is $c_i^{-1}P_{0,i}$ in translated coordinates, three distinct lines. If the multiplicity is 2, then $(g = 0)$ is a node, a cusp or a tacnode, since a cubic intersects a line with multiplicity at most 3 unless the line is a component of the cubic, and g has no double line as $P_{0,i}$ is reduced. Hence the singular points of D are of types A_1, A_2, A_3, and D_4. Since D is a normal Cartier divisor on the hypersurface $W_{\mathcal{C}}$, $W_{\mathcal{C}}$ is normal near D. For S the surface germ of type A_1, A_2, A_3, or D_4, the ideal Jac_S is primary to the maximal ideal and S is klt, so there exists a rational number $\kappa_S > 0$ with $(S, \operatorname{Jac}_S^{\kappa_S})$ lc. Let κ be the minimum of the four numbers κ_S. Let $\mathfrak{a} \subset \mathcal{O}_D$ be the image of $\operatorname{Jac}_{W_{\mathcal{C}}}$. Since $\mathfrak{a} \supset \operatorname{Jac}_D$, the pair $(D, \mathfrak{a}^{\kappa})$ is lc. Let F be a prime divisor over $W_{\mathcal{C}}$ with center 0, let $\kappa' \in (0, \kappa)$ be a rational number and $n > \kappa'$ an integer, and let $\varphi_1, \dots, \varphi_n$ be general elements of $\operatorname{Jac}_{W_{\mathcal{C}}}$, so that $\operatorname{mult}_F \varphi_j = \operatorname{mult}_F \operatorname{Jac}_{W_{\mathcal{C}}}$ for any j and $(D, \frac{\kappa'}{n}\sum_j \operatorname{div}(\varphi_j|_D))$ is klt, as $(D, \mathfrak{a}^{\kappa'})$ is klt (D being klt). By Lemma D.9(2) for $\mathbb{C}[[x,y,z,s]]/(f_{\mathcal{C}})$ and the element s, the pair

$$\Big(W_{\mathcal{C}}, D + \frac{\kappa'}{n}\sum_{j=1}^{n}\operatorname{div}(\varphi_j)\Big)$$

is plt near 0. Since $\operatorname{div}(s|_{W_{\mathcal{C}}}) = D$, we obtain

$$\kappa' \operatorname{mult}_F \operatorname{Jac}_{W_{\mathcal{C}}} < a(F, W_{\mathcal{C}}, 0) - \operatorname{mult}_F s.$$

Letting κ' tend to κ, any prime divisor F over $W_{\mathcal{C}}$ with center 0 satisfies

$$(\text{F.157})\qquad \kappa \operatorname{mult}_F \operatorname{Jac}_{W_{\mathcal{C}}} \leq a(F, W_{\mathcal{C}}, 0) - \operatorname{mult}_F s.$$

By (F.154) and (F.157), any admissible chart (Ξ, ζ, ν) for i satisfies

$$\nu \operatorname{mult}_{F_i} s = a(F_i, W_i', 0) - a(F_i, W_{\mathcal{C}}, 0) < a(F_i, W_i', 0),$$

and $\operatorname{mult}_{F_i} s > 0$. Thus ν is bounded independently of Ξ and ζ. The triple $(\mathrm{id}, 0, \nu_i^0)$ is an admissible chart for i. Hence there exists an admissible chart $\mathcal{C}_i = (\Xi_i, \zeta_i, \nu_i)$ for i with ν_i maximal. We let

$$f_i'' := f_{\mathcal{C}_i}, \quad h_i'' := h_{\mathcal{C}_i} \quad \text{and} \quad W_i'' := W_{\mathcal{C}_i}.$$

Then F_i computes

$$b = \operatorname{mld}\Big(\Big(W_i'' \ni 0, \frac{1}{m}\operatorname{div}(h_i'') + \beta\operatorname{div}(s)\Big)/G\Big).$$

Step 6. In this step we reduce to the inequality $\alpha_Q \leq b$ below.

For $N' \in |G|\mathbb{N}$ with $N' \leq \nu_i$, the triple $\mathcal{C}_i^{N'} := (\Xi_i, \zeta_i, N')$ is a chart for i: the inequalities of (F.153) at depth ν_i imply those at depth N', and $\epsilon_{\mathcal{C}_i}$ factors through $\epsilon_{\nu_i - N'}$, viewed as a morphism from $\widehat{\mathcal{X}}_{\mathcal{C}_i}$ to $\widehat{\mathcal{X}}_{\mathcal{C}_i^{N'}}$, which sends the origin to the origin. We let $f_i^{N'} := f_{\mathcal{C}_i^{N'}}$ and $h_i^{N'} := h_{\mathcal{C}_i^{N'}}$. Possibly passing to a subsequence and enlarging $\Bbbk$, the pairs (f_i'', h_i'') have a compatible generic

limit (F_Q, H_Q), and for any $N' \in |G|\mathbb{N}$ the pairs $(f_i^{N'}, h_i^{N'})$ have a compatible generic limit $(f_*^{N'}, h_*^{N'})$, all with coefficient field $\Bbbk$.

A term of $\Xi_i(f_i')$, resp. of $\Xi_i(h_i' + f_i'\zeta_i)$, of weight v is multiplied by $s^{\nu_i(v-6)}$ in f_i'', resp. by $s^{\nu_i(v-m)}$ in h_i''. Since $\lim_{i\to+\infty}\nu_i = +\infty$, the terms of weight $v > 6$, resp. $v > m$, do not contribute to the limit. Since $\Xi_i \equiv \mathrm{id}$ and $\zeta_i \equiv 0$ modulo $s^{\nu_i^0}$ and $\lim_{i\to+\infty}\nu_i^0 = +\infty$, the terms of weight 6, resp. m, contribute f_0, resp. P. Thus $F_Q, H_Q \in \Bbbk[[s]][x,y,z]$ and

$$\text{the top part of } F_Q \text{ is } f_0, \qquad \text{the top part of } H_Q \text{ is } P. \tag{F.158}$$

At depth N', a term of weight $v > 6$ acquires the factor $s^{N'(v-6)}$, and a term of $\Xi_i(f_i')$ of s-degree j and weight $v < 6$ satisfies $j \ge \nu_i(6-v)$ and becomes a term of s-degree $j - N'(6-v) \ge \nu_i - N'$. Similarly, a term of weight $v > m$ acquires the factor $s^{N'(v-m)}$, and a term of $\Xi_i(h_i' + f_i'\zeta_i)$ of s-degree j and weight $v < m$ satisfies $j \ge \nu_i(m-v)$ and becomes a term of s-degree $j - N'(m-v) \ge \nu_i - N'$. Thus

$$f_*^{N'} - f_0 \in s^{N'}\Bbbk[[x,y,z,s]] \quad \text{and} \quad h_*^{N'} - P \in s^{N'}\Bbbk[[x,y,z,s]]. \tag{F.159}$$

Let $\mathcal{X}_* := \operatorname{Spec}\Bbbk[[x,y,z,s]]$ and let

$$\alpha_P := \operatorname{mld}\Big(\Big(\mathcal{X}_* \ni 0, \operatorname{div}(f_0) + \frac{1}{m}\operatorname{div}(P) + \beta\operatorname{div}(s)\Big)/G\Big),$$
$$\alpha_Q := \operatorname{mld}\Big(\Big(\mathcal{X}_* \ni 0, \operatorname{div}(F_Q) + \frac{1}{m}\operatorname{div}(H_Q) + \beta\operatorname{div}(s)\Big)/G\Big)$$

be the quotient values (Definition 2.2).

We use the following specialization of prime divisors. Let $(g_{i,1}, \dots, g_{i,n})$ be tuples of power series in x, y, z, s with a compatible generic limit $(g_{*,1}, \dots, g_{*,n})$ with coefficient field $\Bbbk$, and let E be a prime divisor over $\mathcal{X}_*$ with center 0 and $\operatorname{mult}_E g_{*,k} < +\infty$ for any k. Then for all but finitely many i there exists a prime divisor E_i over $\widehat{\mathcal{X}}$ with center 0 such that $a(E_i, \widehat{\mathcal{X}}, 0) = a(E, \mathcal{X}_*, 0)$, $\operatorname{mult}_{E_i} s = \operatorname{mult}_E s$, $\operatorname{mult}_{E_i} g_{i,k} = \operatorname{mult}_E g_{*,k}$ for any k, and $I_G(E_i) = I_G(E)$ (cf. Lemma 3.15, the inertia condition being constructible in the G-equivariant family of blow-ups). The prime divisor E is obtained by finitely many blow-ups whose centres are defined by finitely many elements of $\Bbbk$, and $a(E, \mathcal{X}_*, 0)$, $\operatorname{mult}_E s$, $\operatorname{mult}_E g_{*,k}$, and $I_G(E)$ are determined by finite jets of the $g_{*,k}$ and of the equations of the centres. The d-jets of tuples $(g_1, \dots, g_n)$ for which there exists a prime divisor E' over $\widehat{\mathcal{X}}$ with center 0, $a(E', \widehat{\mathcal{X}}, 0) = a(E, \mathcal{X}_*, 0)$, $\operatorname{mult}_{E'} s = \operatorname{mult}_E s$, $\operatorname{mult}_{E'} g_k = \operatorname{mult}_E g_{*,k}$ for any k, and $I_G(E') = I_G(E)$ form a constructible subset of $\mathcal{J}_d$ which contains the generic point of Z_d, hence a dense open subset U_d of Z_d, and U_d contains the d-jets of all but finitely many members (Definition 2.5).

Suppose that a prime divisor E over $\mathcal{X}_*$ with center 0 satisfies

$$a\Big(E, \mathcal{X}_*, \operatorname{div}(f_0) + \frac{1}{m}\operatorname{div}(P) + \beta\operatorname{div}(s)\Big) < be_E.$$

Fix $N' \in |G|\mathbb{N}$ with $N'\operatorname{mult}_E s > \max\{\operatorname{mult}_E f_0, \operatorname{mult}_E P\}$. By (F.159), $\operatorname{mult}_E f_*^{N'} = \operatorname{mult}_E f_0$ and $\operatorname{mult}_E h_*^{N'} = \operatorname{mult}_E P$. The specialization of E to the pairs $(f_i^{N'}, h_i^{N'})$ contradicts (F.156) for $\mathcal{C} = \mathcal{C}_i^{N'}$. Thus any prime divisor E over $\mathcal{X}_*$ with center 0 satisfies

$$\frac{1}{e_E}a\Big(E, \mathcal{X}_*, \operatorname{div}(f_0) + \frac{1}{m}\operatorname{div}(P) + \beta\operatorname{div}(s)\Big) \ge b > 0.$$

As the left-hand side is in $\frac{1}{m\cdot|G|!}\mathbb{Z}$, there exists a prime divisor E over $\mathcal{X}_*$ with center 0 computing α_P, and $\alpha_P \ge b$. Fix $N' \in |G|\mathbb{N}$ with $N'\operatorname{mult}_E s > \max\{\operatorname{mult}_E f_0, \operatorname{mult}_E P\}$, and let E_i be the specialization of E to $(f_i^{N'}, h_i^{N'})$. By (F.152) and (F.155) with $t = \tau_i$,

$$a \le q_i(\tau_i) \le \alpha_P + (1-\tau_i)\gamma_E, \qquad \gamma_E := \frac{1}{e_E}\Big(\frac{\operatorname{mult}_E P}{m} + (N' + r + \beta)\operatorname{mult}_E s\Big), \tag{F.160}$$

for all but finitely many i. Since γ_E does not depend on i and $\lim_{i\to+\infty}\tau_i = 1$, we obtain $\alpha_P \ge a \ge b + \delta$.

The ring $\Bbbk[[x,y,z,s]]$ with the weights $3,2,2,0$, the polynomials $F_Q, H_Q \in \Bbbk[[s]][x,y,z]$ with top parts f_0, P by (F.158), and the diagonal action of G satisfy the conditions of Lemma 7.11, whose a_P and a_Q are α_P and α_Q here. Lemma 7.11 implies $\alpha_Q \ge \alpha_P \ge b+\delta$. It remains to show $\alpha_Q \le b$. Since F_i computes b on W_i'' and $e_{F_i} \le M$, for any i

$$a(F_i, W_i'', 0) = be_{F_i} + \frac{1}{m}\operatorname{mult}_{F_i} h_i'' + \beta \operatorname{mult}_{F_i} s \le bM + \frac{1}{m}\operatorname{mult}_{F_i} h_i'' + T_0. \tag{F.161}$$

Possibly passing to a subsequence, either $\operatorname{mult}_{F_i} h_i''$ is bounded or $\lim_{i\to+\infty} \operatorname{mult}_{F_i} h_i'' = +\infty$.

Step 7. In this step we show $\alpha_Q \le b$ if $\operatorname{mult}_{F_i} h_i''$ is bounded.

Let $j_i := \operatorname{mult}_{F_i} \operatorname{Jac}_{W_i''}$. By (F.161), $a(F_i, W_i'', 0)$ is bounded, and so is j_i by (F.157). Let k_i be the least integer with

$$k_i \ge \operatorname{mult}_{F_i} h_i'' + \operatorname{mult}_{F_i} s + 2j_i + 1$$

in the residue class modulo e_{F_i} of Lemma 5.4(2) for f_i'', F_i, h_i'', and s. Then

$$k_i > 2j_i, \qquad k_i - j_i > \max\{\operatorname{mult}_{F_i} h_i'', \operatorname{mult}_{F_i} s\}, \quad \text{and} \quad k_i \le \operatorname{mult}_{F_i} h_i'' + \operatorname{mult}_{F_i} s + 2j_i + |G|.$$

By Lemma 5.4(2), there exists a prime divisor $\widehat{F}_i$ over $\widehat{\mathcal{X}}$ with center 0, $I_G(\widehat{F}_i) = I_G(F_i)$, $\operatorname{mult}_{\widehat{F}_i} f_i'' = k_i$, $\operatorname{mult}_{\widehat{F}_i} h_i'' = \operatorname{mult}_{F_i} h_i''$, $\operatorname{mult}_{\widehat{F}_i} s = \operatorname{mult}_{F_i} s$, and $a(\widehat{F}_i, \widehat{\mathcal{X}}, 0) = a(F_i, W_i'', 0) + k_i$. Thus

$$a\Big(\widehat{F}_i, \widehat{\mathcal{X}}, \operatorname{div}(f_i'') + \frac{1}{m}\operatorname{div}(h_i'') + \beta \operatorname{div}(s)\Big) = a\Big(F_i, W_i'', \frac{1}{m}\operatorname{div}(h_i'') + \beta \operatorname{div}(s)\Big) = be_{\widehat{F}_i},$$

and $a(\widehat{F}_i, \widehat{\mathcal{X}}, 0)$, $\operatorname{mult}_{\widehat{F}_i} s$, $\operatorname{mult}_{\widehat{F}_i} f_i''$, and $\operatorname{mult}_{\widehat{F}_i} h_i''$ are bounded. By Lemma 7.10(2), with $n=4$, the axis $u=s$, the coordinates $(x_1,x_2,x_3)=(x,y,z)$, the group G, the coefficient β of the axis, $N=2$, $(Q_{i,1}, Q_{i,2}) = (f_i'', h_i'')$ with the compatible generic limit (F_Q, H_Q) along the subsequence fixed at the end of **Step 6** (Definition 2.5 passes to infinite subsets), $(c_1,c_2) = (1, \frac{1}{m})$, and $F_i = \widehat{F}_i$, we obtain $\alpha_Q \le b$, a contradiction.

Step 8. In this step we show that $\lim_{i\to+\infty} \operatorname{mult}_{F_i} h_i'' = +\infty$ is impossible.

Let $V := (F_Q = 0) \subset \mathbb{A}^3_{\mathbb{K}}$ and $\Delta_V := \frac{1}{m}\operatorname{div}(H_Q|_V)$. A constant linear change of (y,z) with complex coefficients preserves the weights and the form of Ξ in (a), and after it f_0 is the polynomial $x^2 + zy^2 + z^3$ of Proposition 7.7 with $k=4$, weights $(3,2,2)$ and degree 6. We use the changed coordinates only for Proposition 7.7 and Lemma 7.3. Elsewhere x,y,z,s are the original coordinates, in which G acts diagonally, and (i) and the G-equivariance of Φ below refer to them. By (F.158) and the first paragraph of Proposition 7.7, V is normal with at worst Du Val singularities. Let $X \subset \mathbb{P}(1,3,2,2)$ be the closure of V, $H_\infty := X \backslash V$, and B the closure of Δ_V. Then X is a hypersurface of degree 6, $K_X \sim_{\mathbb{Q}} -2H_\infty$, and $B \sim_{\mathbb{Q}} H_\infty$ since H_Q has top part P of weight m. Thus $K_X + B + H_\infty \sim_{\mathbb{Q}} 0$. By Lemma 7.3 (case (b) with $w=(3,2,2)$, $f_0 = x^2 + zy^2 + z^3$, $N=1$, $Q_1 = H_Q$, and $\kappa_1 = \frac{1}{m}$), X is normal and klt and, since $(S_{\mathbb{K}}, \Delta_P)$ is lc without coefficient-one component, $(X, B + H_\infty)$ is plt near H_∞.

Let $\Gamma \subset V$ be a curve contained in $\operatorname{Nklt}(V, \Delta_V)$. The closure of Γ intersects H_∞, as H_∞ is ample, and at an intersection point $(X, B + H_\infty)$ is not plt. Thus $\operatorname{Nklt}(V, \Delta_V)$ is finite. In particular, Δ_V has no coefficient-one component. Theorem 7.2 holds over $\mathbb{K}$: X and B are defined over a subfield of $\mathbb{K}$ finitely generated over $\mathbb{Q}$, whose algebraic closure in $\mathbb{K}$ embeds into $\mathbb{C}$, and log discrepancies and connected components of non-klt loci do not change when the base field is extended from the algebraic closure of the subfield in $\mathbb{K}$ to $\mathbb{K}$ or to $\mathbb{C}$. If $\operatorname{Nklt}(V, \Delta_V) \ne \emptyset$, then $\operatorname{Nklt}(X, B + H_\infty) = H_\infty \sqcup \operatorname{Nklt}(V, \Delta_V)$ is disconnected, and Theorem 7.2 for $X \to \operatorname{Spec} \mathbb{K}$ implies that $(X, B + H_\infty)$ is lc and $\operatorname{Nklt}(V, \Delta_V)$ is one point o. Thus either (V, Δ_V) is klt, or (V, Δ_V) is lc with $\operatorname{Nklt}(V, \Delta_V) = \{o\}$. A log resolution of $(V, \operatorname{div}(H_Q|_V))$ implies that there exists a rational number $\mu' > \frac{1}{m}$ with $(V, \mu' \operatorname{div}(H_Q|_V))$ klt outside $\operatorname{Nklt}(V, \Delta_V)$.

Fix such μ', and fix rational numbers $0 < \theta_1' < \theta_2' < 1$ with

$$\gamma' := \theta_1'\mu' - (1 - \theta_1') - \frac{1}{m}\Big(1 + \frac{2(1-\theta_1')}{\kappa}\Big) > 0,$$

which is possible since $\mu' > \frac{1}{m}$. Let $\gamma'_0 := (1 + \frac{2(1-\theta'_1)}{\kappa})(bM + T_0) + 1$ and

$$\mathcal{J}'_* := \mathcal{J}\big(\mathcal{X}_*, F_Q^{\theta'_2} H_Q^{\theta'_2\mu'}\big).$$

Let $I \subset \Bbbk[[s]][x,y,z]$ be an ideal whose zero locus in $\mathbb{A}^3_{\mathbb{K}}$ contains $\mathrm{Nklt}(V, \Delta_V)$. Let

$$V' := (F_Q = 0) \subset \mathbb{A}^3_{\Bbbk((s))}.$$

Since $V = V' \times_{\Bbbk((s))} \mathbb{K}$, the pair $(V', \mu' \operatorname{div}(H_Q|_{V'}))$ is klt outside the zero locus of I. Let $W_Q := (F_Q = 0) \subset \mathcal{X}_*$. Then W_Q is normal near 0, since $W_Q \cap (s = 0)$ is a normal surface by the argument for D in **Step 5**. The morphism $\mathcal{X}_* \to \operatorname{Spec} \Bbbk[[s]][x,y,z]$ is regular, as a completion of an excellent ring at a maximal ideal, and klt pairs and log resolutions pull back to klt pairs and log resolutions under regular morphisms. As

$$\operatorname{Spec} \Bbbk[[s]][x,y,z] \backslash (s = 0) = \operatorname{Spec} \Bbbk((s))[x,y,z],$$

the pair $(W_Q, \mu' \operatorname{div}(H_Q|_{W_Q}))$ is klt outside the zero locus of sI. By Lemma 6.1(3), $\mathcal{J}'_*$ is the unit ideal near any point of W_Q outside the zero locus of sI. Hence the zero locus of $(F_Q) + \mathcal{J}'_*$ is contained in that of sI, and there exists a positive integer N_0 with

$$(sI)^{N_0} \Bbbk[[x,y,z,s]] \subset (F_Q) + \mathcal{J}'_*. \tag{F.162}$$

Let $\phi \in sI$ and fix a positive integer N_2. For $g \in \Bbbk[[x,y,z,s]]$ let $g^{<N_2}$ be the sum of the terms of g of total degree less than N_2 in x,y,z,s, and let $\mathfrak{n}_*$ be the maximal ideal of $\Bbbk[[x,y,z,s]]$. Writing $\phi^{N_0} = a_\phi F_Q + j_\phi$ with $j_\phi \in \mathcal{J}'_*$ by (F.162), we obtain

$$(\phi^{<N_2})^{N_0} - a_\phi^{<N_2} F_Q \in \mathcal{J}'_* + \mathfrak{n}_*^{N_2}.$$

Let $\Theta_\phi := \phi_u^{N_0} - a_{u'} y_{1,0}$, a polynomial in the variables of Lemma 6.1(1), where ϕ_u is $\phi^{<N_2}$ with its coefficients not in $\mathbb{C}$ replaced with variables, $a_{u'}$ is the polynomial of total degree less than N_2 in x,y,z,s whose coefficients are further variables, and $y_{1,0}$ is the variable of Lemma 6.1 substituted by the first entry of a tuple. Let ς consist of the coefficients of $\phi^{<N_2}$ not in $\mathbb{C}$ and of $a_\phi^{<N_2}$, so that

$$\Theta_\phi((F_Q, H_Q), \varsigma) = (\phi^{<N_2})^{N_0} - a_\phi^{<N_2} F_Q.$$

By Lemma 6.1(1) with $L = N_2$, for the tuples (f_i'', h_i'') with the limit (F_Q, H_Q), the exponents $(\theta'_1, \theta'_1\mu') < (\theta'_2, \theta'_2\mu')$, and Θ_ϕ, for all but finitely many i there exists ς_i (the β_i of Lemma 6.1(1) for $\beta = \varsigma$). We let ϕ_i and a_i be $\phi^{<N_2}$ and $a_\phi^{<N_2}$ with the entries of ς replaced with those of ς_i, and $\xi_i := \phi_i^{N_0} - a_i f_i''$. Let $\mathfrak{m}$ be the maximal ideal of $\mathbb{C}[[x,y,z,s]]$. As $\operatorname{mult}_{P'}(\mathfrak{m}^{N_2}) \geq N_2$, Lemma 6.1(2) implies that any prime divisor P' over $\widehat{\mathcal{X}}$ with center the closed point and $\operatorname{mult}_{P'} \xi_i < N_2$ satisfies

$$\operatorname{mult}_{P'} \xi_i + a(P', \widehat{\mathcal{X}}, 0) > \theta'_1 \operatorname{mult}_{P'} f_i'' + \theta'_1\mu' \operatorname{mult}_{P'} h_i''.$$

As $\xi_i|_{W_i''} = \phi_i^{N_0}|_{W_i''}$, if $N_0 \operatorname{mult}_{F_i} \phi_i < N_2$, then Lemma 5.5(1) for f_i'', h_i'', F_i, θ'_1, μ', $g = \xi_i$, and $\rho = N_2$, together with $a(F_i, W_i'', 0) \leq \frac{1}{m} \operatorname{mult}_{F_i} h_i'' + bM + T_0$ by (F.161) and $j_i \leq a(F_i, W_i'', 0)/\kappa$ by (F.157), implies

$$\gamma' \operatorname{mult}_{F_i} h_i'' < (2 - \theta'_1) N_0 \operatorname{mult}_{F_i} \phi_i + \gamma'_0. \tag{F.163}$$

Thus

$$N_0 \operatorname{mult}_{F_i} \phi_i \geq \min\Big\{N_2, \frac{\gamma' \operatorname{mult}_{F_i} h_i'' - \gamma'_0}{2 - \theta'_1}\Big\}, \tag{F.164}$$

and, as $\lim_{i \to +\infty} \operatorname{mult}_{F_i} h_i'' = +\infty$,

$$N_0 \operatorname{mult}_{F_i} \phi_i \geq N_2 \quad \text{for all but finitely many } i. \tag{F.165}$$

Suppose that (V, Δ_V) is klt. Take $I = (1)$, $\phi = s$, and $N_2 > N_0 T_0$. Then $\phi^{<N_2} = s$ and $\phi_i = s$, and (F.165) gives $N_0 \operatorname{mult}_{F_i} s \geq N_2 > N_0 T_0$ for all but finitely many i, which contradicts $\operatorname{mult}_{F_i} s \leq T_0$.

Thus (V,Δ_V) is lc with $\mathrm{Nklt}(V,\Delta_V)=\{o\}$. The automorphisms of $\mathbb{K}$ over $\Bbbk((s))$ and the elements of G preserve (V,Δ_V), hence fix o. Thus $o\in\Bbbk((s))^3$ in the coordinates x,y,z, and o is G-fixed. Let $I\subset\Bbbk[[s]][x,y,z]$ be the ideal of the closure of o. If $I\Bbbk[[x,y,z,s]]$ is the unit ideal, which includes the case where a coordinate of o has a pole, then (F.162) reads $s^{N_0}\in(F_Q)+\mathcal{J}'_*$, and the argument of the klt case with $\phi=s$ leads to a contradiction. Thus $o=(o_1,o_2,o_3)\in(s\Bbbk[[s]])^3$.

Let t_o be the G-equivariant translation $x\to x+o_1$, $y\to y+o_2$, $z\to z+o_3$. It lowers weights, so $F_Q\circ t_o$ and $H_Q\circ t_o$ have top parts f_0 and P, and $(0,0,0)$ is the non-klt point. As $F_Q\circ t_o$ is monic of degree 2 in x, let $H_Q\circ t_o=R+(F_Q\circ t_o)\cdot\zeta_o$ with R of x-degree at most 1. Then R has top part P. Proposition 7.7(1)(5) over $\mathbb{K}$ with $F=F_Q\circ t_o$, $N=1$, $Q_1=R$, $d_1=m$, and $\kappa_1=\frac{1}{m}$ give, in the standard coordinates of Proposition 7.7, an automorphism $x\to x+g_1(y,z)$, $y\to y+g_2(z)$, $z\to z+c'$ with coefficients in $\Bbbk[[s]]$, g_1 of weight less than 3, and g_2 of weight less than 2, hence constant. Since $(0,0,0)$ is the non-klt point, Proposition 7.7(1), by which the automorphism sends the non-klt point $(0,0,0)$ to $(0,0,0)$, gives $g_1(0,0)=g_2=c'=0$. As a constant linear change of (y,z) preserves the form $x\to x+g_1(y,z)$, $y\to y$, $z\to z$ with g_1 a linear form, we obtain an automorphism Φ_0 of $\Bbbk[[s]][x,y,z]$ of the form $x\to x+\lambda_1y+\lambda_2z$, $y\to y$, $z\to z$ with $\lambda_1,\lambda_2\in\Bbbk[[s]]$, such that $\Phi:=\Phi_0\circ t_o^{-1}$ satisfies

$$F_Q=f_0\circ\Phi\quad\text{and}\quad H_Q=P\circ\Phi+F_Q\cdot\zeta_Q,\tag{F.166}$$

where $\zeta_Q\in\Bbbk[[s]][x,y,z]$ and any term of ζ_Q has weight less than $m-6$, since $H_Q-P\circ\Phi$ has weight less than m (the top parts coincide) and F_Q is monic of weight 6 in x with top part f_0. For any element of G, the conjugate of Φ by it is of the form $\Phi'_0\circ t_o^{-1}$ with Φ'_0 of the form $x\to x+\lambda'_1y+\lambda'_2z$, $y\to y$, $z\to z$, and satisfies (F.166), since $\chi_{F_Q}=\chi_{f_0}$ and $\chi_{H_Q}=\chi_P$. By the uniqueness in Proposition 7.7(1), Φ is G-equivariant. The ideal I of the closure of o is generated by $\Phi(x)$, $\Phi(y)$, and $\Phi(z)$.

Fix $\nu'\in|G|\mathbb{N}$ with $\nu'>0$, an integer $N_2>N_0(3\nu'+1)T_0$, and an integer N_3 with

$$N_3>\max\{\nu'\max\{6,m\},N_2\}.$$

The automorphism Φ and the polynomial ζ_Q are determined by finitely many power series in s over $\Bbbk$. We use their jets of order N_3. Consider the following conditions on a finite jet of (f''_i,h''_i) and on complex coefficients of Φ_i and $\zeta_{Q,i}$, where Φ_i and $\zeta_{Q,i}$ are the jets of Φ and ζ_Q with their coefficients in $\Bbbk$ replaced with complex numbers.

(i) Φ_i and $\zeta_{Q,i}$ are G-equivariant, resp. G-semi-invariant with $\chi_{\zeta_{Q,i}}\chi_{f''_i}=\chi_{h''_i}$, and the constant terms of $\Phi_i(x)$, $\Phi_i(y)$, $\Phi_i(z)$ are divisible by s.

(ii) The terms of $f''_i\circ\Phi_i^{-1}$, resp. of $(h''_i-f''_i\zeta_{Q,i})\circ\Phi_i^{-1}$, of total degree less than N_3 in x,y,z,s, of s-degree j and weight v, with $v<6$ and $j<\nu'(6-v)$, resp. $v<m$ and $j<\nu'(m-v)$, vanish.

(iii) $\mathrm{mult}_{F_i}\Phi_i(x)>3\nu'T_0$ and $\mathrm{mult}_{F_i}\Phi_i(y),\mathrm{mult}_{F_i}\Phi_i(z)>2\nu'T_0$.

By (F.166), $F_Q\circ\Phi^{-1}=f_0$ and $(H_Q-F_Q\zeta_Q)\circ\Phi^{-1}=P$ are weighted homogeneous of weights 6 and m, and replacing Φ, ζ_Q with their jets of order N_3 changes $F_Q\circ\Phi^{-1}$ and $(H_Q-F_Q\zeta_Q)\circ\Phi^{-1}$ only in s-degree greater than N_3. Thus the jets of Φ and ζ_Q satisfy (i) and (ii) for the limit (F_Q,H_Q). For $\phi\in\{s\Phi(x),s\Phi(y),s\Phi(z)\}\subset sI$, the coefficients of $\phi^{<N_2}$ are among those of the jet of $s\Phi$ of order N_3, as $N_3>N_2$.

By Lemma 6.1(1) with $L=N_2$, used once for the three polynomials Θ_ϕ, with ς consisting of the coefficients not in $\mathbb{C}$ of the jets of Φ and ζ_Q of order N_3 and of the three polynomials $a_\phi^{<N_2}$, and with (i) and (ii) as the additional polynomial conditions of the last sentence of Lemma 6.1(1), for all but finitely many i the entries of ς_i give Φ_i and $\zeta_{Q,i}$ satisfying (i) and (ii), and the polynomials ϕ_i are the sums of the terms of total degree less than N_2 of $s\Phi_i(x)$, $s\Phi_i(y)$, and $s\Phi_i(z)$, which satisfy (F.165) by the argument above. Since the other terms have order at least N_2 along F_i, we obtain $\mathrm{mult}_{F_i}s\Phi_i(x)\geq N_2/N_0>(3\nu'+1)T_0$, and similarly for y,z. As $\mathrm{mult}_{F_i}s\leq T_0$, (iii) follows.

Fix such i, and let $\Phi_i, \zeta_{Q,i}$ satisfy (i)–(iii). By (i), $\sigma_{\nu'}(\Phi_i^{-1}(x))$, $\sigma_{\nu'}(\Phi_i^{-1}(y))$, and $\sigma_{\nu'}(\Phi_i^{-1}(z))$ are divisible by s, so any term of f_i'' or h_i'' of total degree at least N_3 becomes divisible by s^{N_3}, and $N_3 > \nu' \max\{6, m\}$. With (ii) and (F.153), both

$$s^{-6\nu'}\sigma_{\nu'}(f_i'' \circ \Phi_i^{-1}) \quad \text{and} \quad s^{-m\nu'}\sigma_{\nu'}\big((h_i'' - f_i''\zeta_{Q,i}) \circ \Phi_i^{-1}\big)$$

are power series, and by (iii) and $\operatorname{mult}_{F_i} s \le T_0$ the center of F_i on the new chart is the origin. We express Φ_i^{-1}, $\zeta_{Q,i}$ and the dilation $\sigma_{\nu'}$ in the coordinates of $\widehat{\mathcal{X}}$. A correction of weight v to a coordinate of weight $\omega > v$ in Φ_i^{-1}, made at depth ν_i, is the correction multiplied by $s^{\nu_i(\omega - v)}$ before the dilation σ_{ν_i}. A term of weight $v < m - 6$ of $\zeta_{Q,i}$ is multiplied by $s^{\nu_i(m-6-v)}$ before the dilation σ_{ν_i}. The exponents $\omega - v$ and $m - 6 - v$ are positive integers and $\nu_i \ge \nu_i^0$ since $\mathcal{C}_i$ is admissible, so the coefficients are in $s^{\nu_i^0}\mathbb{C}[s]$. The composition of Ξ_i with the correction of Φ_i^{-1} is again of the form (a), and ζ_i plus the correction of $\zeta_{Q,i}$, composed with Ξ_i^{-1}, is again of the form (b). Since $\sigma_{\nu'} \circ \sigma_{\nu_i} = \sigma_{\nu_i + \nu'}$ and $\nu_i + \nu' \in |G|\mathbb{N}$, we obtain an admissible chart for i of depth $\nu_i + \nu'$, which contradicts the maximality of ν_i.

Thus neither case of **Step 6** occurs. This finishes the case of a reduced residual cubic, and the proof. □

Proof of Theorem 17.12. Suppose that $(W_i \ni o_i, G_i, h_i, \tau_i)$ is a radial failure sequence with constants (m, a, δ) which satisfies Set-up 17.1, whose prepared limit (f_*, h_*) is of type cE, and whose generic transverse type along C is D_k for an integer $k \ge 5$. We derive a contradiction.

Step 1. In this step we fix the chart of Proposition 17.7(1) and bound $a(F, Y_i, B_i)$ from below on it.

By Proposition 17.7(1), possibly passing to an infinite subfamily, let r, N, $(Y_i \ni 0)$, l_i, S_i, β, and B_i be as there.

The chart of Proposition 17.7(1) writes each of the parameters of the ambient germ of W_i as a positive power of s times a regular function on Y_i. Thus $\mathfrak{m}_{o_i}\mathcal{O}_{Y_i,0} = s\mathcal{O}_{Y_i,0}$, so any prime divisor F over Y_i with center 0 is a prime divisor over W_i with center o_i and $\operatorname{mult}_F \mathfrak{m}_{o_i} = \operatorname{mult}_F s$. Since $(W_i \ni o_i)$ is a hypersurface germ and $(W_i, \frac{1}{m}\operatorname{div}(h_i))$ is klt near o_i (Set-up 17.1(2)), Theorem 5.1(2) and Proposition 17.7(1) (with $t = 1$) imply

$$a(F, Y_i, B_i) \ge \eta_m \operatorname{mult}_F s \quad \text{for any prime divisor } F \text{ over } Y_i \text{ with center } 0. \tag{F.167}$$

Step 2. In this step we write the limit of the Y_i in a double-line form and show that the limit mark has lowest w-weight m, where $w := (k-1, k-2, 2)$.

By Proposition 17.7(2), possibly passing to an infinite subfamily, the equations g_i of the Y_i and the functions l_i, written in G-eigenparameters (X, V, Z, s) of the ambient germs of the Y_i, have a compatible generic limit (g_*, l_*) with coefficient field $\Bbbk$, where $g_* = X^2 + 3V^2Z + \Psi(Z, s)$ and $\Psi \in Z^{k-1}\Bbbk[[Z, s]]$ has order $k-1$ in Z over $\Bbbk((s))$. From now on (x, y, z, s) denote the parameters $(X, \sqrt{3}V, Z, s)$ on the ambient germs of the Y_i. We write $\Psi = z^{k-1}\phi(z, s)$ and $\alpha(s) := \phi(0, s)$, so $\alpha \ne 0$. We use the following for (g_i, l_i) and for the compatible generic limits constructed from them below.

(†) Any polynomial relation over $\mathbb{C}$ among finitely many coefficients which holds for the limit holds for all but finitely many i, and finitely many coefficients which are non-zero for the limit are non-zero for all but finitely many i (Lemma 2.8).

Fix an algebraic closure $\overline{\Bbbk((s))}$ of $\Bbbk((s))$ and let $\Sigma := (g_* = 0) \subset \operatorname{Spec}\overline{\Bbbk((s))}[[x, y, z]]$. By Proposition 17.7(2), over $\overline{\Bbbk((s))}$ the pair $(\Sigma, \frac{1}{m}\operatorname{div}(l_*|_\Sigma))$ is isomorphic to a generic transverse surface of W_* along C together with the restriction of $\frac{1}{m}\operatorname{div}(h_*)$. Since $(W_*, \frac{1}{m}\operatorname{div}(h_*))$ is lc, has no coefficient-one component, and has non-klt locus C (Set-up 17.1), we obtain:

$$\begin{gathered}\Sigma \text{ is Du Val of type } D_k, \text{ and } (\Sigma, \tfrac{1}{m}\operatorname{div}(l_*|_\Sigma)) \text{ is lc and not klt at } 0 \\ \text{and has no coefficient-one component.}\end{gathered} \tag{F.168}$$

We write

$$\alpha(s) = s^\nu \alpha_0(s), \qquad \nu \in \mathbb{N}, \quad \text{and} \quad \alpha_0(0) \neq 0.$$

Any monomial $s^e z^{k-1}$ with non-zero coefficient in g_* has the character χ_{g_*} of g_*. Since the coefficient of $s^\nu z^{k-1}$ is non-zero, $\chi_s^{e-\nu} = 1$ for any such e, so α_0 is G-invariant. Fix a square root ς_* of $\alpha_0(0)$ and let $\alpha_0^{1/2} \in \Bbbk[[s]]$ be the square root of α_0 with constant term ς_*. In the parameters $\alpha_0^{-1/2}x$, $\alpha_0^{-1/2}y$, z, s, renamed x, y, z, s, the series $\alpha_0^{-1}g_*$, renamed g_*, is

$$g_* = x^2 + zy^2 + s^\nu z^{k-1} + z^k \phi_1(z,s), \qquad \phi_1 \in \Bbbk[[z,s]]. \tag{F.169}$$

Since g_* is monic of degree 2 in x, the Weierstrass division theorem writes $l_* = v_* g_* + \widetilde{l}_*$ with $\widetilde{l}_*$ of x-degree at most 1. Since $\widetilde{l}_* - l_* \in (g_*)$, the restrictions of $\widetilde{l}_*$ and l_* to $(g_* = 0)$ coincide, and we replace l_* with $\widetilde{l}_*$.

We now normalize g_i and l_i as g_* and l_* above. By (†), for all but finitely many i the coefficients of $s^e z^{k-1}$ in g_i vanish for $e < \nu$, and the coefficients of $s^\nu z^{k-1}$ and of x^2 in g_i are non-zero. Thus the coefficient of z^{k-1} in $g_i(0,0,z,s)$ is $s^\nu \alpha_{0,i}(s)$ with $\alpha_{0,i}(0) \neq 0$, and $\alpha_{0,i}$ is G-invariant by the character argument above. For a square root ς_i of $\alpha_{0,i}(0)$ fixed in the next paragraph, we rename the parameters $\alpha_{0,i}^{-1/2}x$, $\alpha_{0,i}^{-1/2}y$ as x, y and the series $\alpha_{0,i}^{-1}g_i$ as g_i, where $\alpha_{0,i}^{1/2}$ has constant term ς_i, and then replace l_i with the remainder of the Weierstrass division of l_i by g_i with respect to x. The new l_i is semi-invariant with the old character and differs from the old l_i by an element of (g_i), so the function $l_i|_{Y_i}$ and the boundary B_i do not change. The parameters x, y, z differ from X, V, Z by units, so Proposition 17.7(2) implies

$$\lim_{i \to +\infty} \min\{\operatorname{mult}_F x, \operatorname{mult}_F y, \operatorname{mult}_F z \mid F \text{ is a computing lift of } b_i\} = +\infty. \tag{F.170}$$

By Lemma 2.6(4) for the constant ς_*, possibly passing to an infinite subfamily, there exist $\varsigma_i \in \mathbb{C}$ (the constant terms of the polynomials given there) such that the old pairs together with ς_i have the compatible generic limit formed by the old limit and ς_*. Then $\varsigma_i^2 = \alpha_{0,i}(0)$ for all but finitely many i, and since the coefficients of the new pairs are polynomials over $\mathbb{C}$ in the coefficients of the old pairs, in ς_i, and in the inverses of the coefficients of $s^\nu z^{k-1}$ and x^2, the new pairs (g_i, l_i) have the compatible generic limit (g_*, l_*) by the argument of Lemma 2.6(2).

Let $\mathbb{L} \supset \mathbb{C}$ be an algebraically closed field, and let

$$f_0 := x^2 + zy^2 + z^{k-1} \quad \text{and} \quad S_0 := (f_0 = 0) \subset \operatorname{Spec} \mathbb{L}[[x,y,z]].$$

For $j = (j_1, j_2, j_3) \in \mathbb{N}^3$ let $\operatorname{wt}_w(j) := (k-1)j_1 + (k-2)j_2 + 2j_3$.

Let $\psi := s^\nu + z\phi_1$. The parameters $z' := z\psi^{\frac{1}{k-1}}$ and $y' := y\psi^{-\frac{1}{2k-2}}$ satisfy

$$x^2 + zy^2 + z^{k-1}\psi = x^2 + z'y'^2 + z'^{k-1},$$

and z', y' are non-zero multiples of z, y plus terms of higher w-weight. Thus there exists an isomorphism $\Sigma \cong S_0$ over $\overline{\Bbbk((s))}$ which fixes x and changes the lowest w-weight part of any $g \in \overline{\Bbbk((s))}[[x,y,z]]$ by non-zero constant factors on the monomials of $\operatorname{in}_w(g)$. Write

$$l_* = \sum_j \gamma_j x^{j_1} y^{j_2} z^{j_3} s^{j_4}, \qquad j = (j_1, j_2, j_3, j_4) \in \{0,1\} \times \mathbb{N}^3, \ \gamma_j \in \Bbbk,$$

and $\operatorname{wt}_w(j) := \operatorname{wt}_w(j_1, j_2, j_3)$. Over $\overline{\Bbbk((s))}$ the coefficient of $x^{j_1} y^{j_2} z^{j_3}$ is non-zero if and only if $\gamma_j \neq 0$ for some j_4. By (F.168) and Lemma F.11 with $\mathbb{L} = \overline{\Bbbk((s))}$ and $t = \frac{1}{m}$, lc implies $\operatorname{wt}_w(j) \leq m$ for some j with $\gamma_j \neq 0$, and not klt implies $\operatorname{wt}_w(j) \geq m$ for any j with $\gamma_j \neq 0$. Let

$$Q := \sum_{\operatorname{wt}_w(j) = m} \gamma_j x^{j_1} y^{j_2} z^{j_3} s^{j_4} \in \Bbbk[[s]][x,y,z]. \tag{F.171}$$

Then $Q \neq 0$, Q has x-degree at most 1, and $l_* - Q$ consists of monomials with $\operatorname{wt}_w(j) \geq m+1$.

Step 3. In this step we take a root u of s and a weighted chart, and construct the germs W_i' with the group G'.

Let $\sigma := 1$ if ν is even and $\sigma := 2$ if ν is odd, and let $\nu' := \sigma\nu/2 \in \mathbb{N}$. Let

$$c := \min\{\nu'(j_1 + j_2) + \sigma j_4 \mid \mathrm{wt}_w(j) = m,\ \gamma_j \neq 0\} \in \mathbb{N}. \tag{F.172}$$

We write f_0 and S_0 also for $X^2 + ZY^2 + Z^{k-1}$ and its zero locus. If m is even, let N_1 be the integer N of Theorem 5.10 for f_0 and the integers m, $2M$, and $m/2$. If m is odd, let $N_1 := 1$. Let $r_1 := c + \nu' + N_1$ and

$$\omega = (\omega_1, \omega_2, \omega_3, \omega_4) := \big((k-1)r_1 + \nu', (k-2)r_1 + \nu', 2r_1, \sigma\big).$$

For $R = \mathbb{C}$ or $\Bbbk$, let $\iota : R[[x, y, z, s]] \to R[[X, Y, Z, u]]$ be the ring homomorphism with $\iota(x) = u^{\omega_1}X$, $\iota(y) = u^{\omega_2}Y$, $\iota(z) = u^{\omega_3}Z$, and $\iota(s) = u^{\sigma}$. It maps $x^{j_1}y^{j_2}z^{j_3}s^{j_4}$ to $u^{\omega\cdot j}X^{j_1}Y^{j_2}Z^{j_3}$, where

$$\omega \cdot j = r_1\,\mathrm{wt}_w(j) + \nu'(j_1 + j_2) + \sigma j_4 \geq j_1 + j_2 + j_3 + j_4. \tag{F.173}$$

Thus ι maps distinct monomials to distinct monomials, and for any integer n only finitely many monomials satisfy $\omega \cdot j \leq n$.

Let $\theta := (2k-2)r_1 + 2\nu'$. By (F.173), the monomials x^2, zy^2, $s^\nu z^{k-1}$ satisfy $\omega \cdot j = \theta$, and $z^{k+e}s^{j_4}$ satisfies

$$\omega \cdot j = \theta + 2(e+1)r_1 - 2\nu' + \sigma j_4 \geq \theta + 2r_1 - 2\nu'.$$

By **Step 2**, (F.172), and $r_1 \geq c + N_1$, any j with $\gamma_j \neq 0$ satisfies $\omega \cdot j \geq mr_1 + c$, and $\omega \cdot j \geq (m+1)r_1 \geq mr_1 + c + N_1$ if $\mathrm{wt}_w(j) \geq m+1$. Thus (F.169) and (F.171) imply

$$\begin{gathered} \iota(g_*) = u^\theta f'_*, \qquad f'_* - f_0 \in Z^k u^{2r_1 - 2\nu'}\Bbbk[[Z, u]], \\ \iota(l_*) = u^{mr_1+c}l'_*, \qquad \mathrm{in}_w(l'_*) = u^{-mr_1-c}\iota(Q), \qquad l'_* - \mathrm{in}_w(l'_*) \in u^{N_1}\Bbbk[[X, Y, Z, u]], \end{gathered} \tag{F.174}$$

where $\mathrm{in}_w(l'_*)$ is the lowest transverse w-weight part of l'_*, of weight m, and $l'_* - \mathrm{in}_w(l'_*)$ consists of terms of transverse w-weight at least $m+1$. Moreover $\mathrm{in}_w(l'_*) \in \Bbbk[[u]][X, Y, Z]$ has X-degree at most 1, and $\mathrm{in}_w(l'_*)|_{u=0} \neq 0$ by (F.172). Since $2r_1 - 2\nu' \geq N_1$, also $f'_* - f_0 \in (u^{N_1})$.

Only finitely many monomials satisfy $\omega \cdot j < \theta + N_1$ or $\omega \cdot j < mr_1 + c + N_1$. By (†), discarding finitely many i, the coefficients of g_i at the monomials with $\omega \cdot j < \theta + N_1$ are those of g_*, which are 0 or 1, and the coefficients of l_i at the monomials with $\omega \cdot j < mr_1 + c + N_1$ vanish where those of l_* vanish and are non-zero where those of l_* are non-zero. Thus $\iota(g_i) = u^\theta f'_i$ and $\iota(l_i) = u^{mr_1+c}l'_i$, where for $e < N_1$ the coefficient of l'_i at $X^{j_1}Y^{j_2}Z^{j_3}u^e$ is non-zero if and only if that of l'_* is, and

$$f'_i - f'_* \in (u^{N_1}), \qquad l'_i - \mathrm{in}_w(l'_i) \in (u^{N_1}), \tag{F.175}$$

where $\mathrm{in}_w(l'_i)$ is the part of l'_i of transverse w-weight m. In particular $f'_i|_{u=0} = f_0$, and $l'_i|_{u=0}$ is non-zero of X-degree at most 1, since $\mathrm{in}_w(l'_*)|_{u=0} \neq 0$. By (F.173), each coefficient of the d-jet of (f'_i, l'_i) is the coefficient of (g_i, l_i) at a monomial of degree less than $d + \theta + mr_1 + c$ independent of i, and likewise for the limits, so (f'_i, l'_i) has the compatible generic limit (f'_*, l'_*), with coefficients in $\Bbbk[[u]]$, by the argument of Lemma 2.6(2).

Let $W'_i := (f'_i = 0) \subset \mathrm{Spec}\,\mathbb{C}[[X, Y, Z, u]]$. Since $f'_i|_{u=0} = f_0$ and S_0 is a normal Du Val surface, W'_i is normal near 0 by Serre's criterion, and $(W'_i, (u = 0))$ is plt near 0. Thus W'_i is a klt Gorenstein hypersurface threefold germ, and $E_0 := (u = 0) \subset W'_i$ is a prime divisor. Since $l'_i|_{u=0}$ is non-zero of X-degree at most 1, it does not vanish on S_0. Thus l'_i is non-zero on W'_i and $\mathrm{mult}_{E_0} l'_i = 0$.

The parameters x, y, z, the new g_i and l_i of **Step 2**, and W'_i are formal. We now pass to algebraic germs. Let $(\mathcal{X}_i \ni 0)$ be the ambient germ of Y_i and $\mathfrak{m}$ the maximal ideal of $\widehat{\mathcal{O}}_{\mathcal{X}_i,0}$. The chart of Proposition 17.7(1) is algebraic, and G acts diagonally on algebraic regular parameters of $(\mathcal{X}_i \ni 0)$, one of which is s. Let $n_i > \theta + \omega_1 + mr_1 + c$ be integers with $\lim_{i\to+\infty} n_i = +\infty$. The new g_i is $\alpha_{0,i}^{-1}$ times an equation of Y_i, and the new l_i is the old l_i minus $v_i g_i$ for a G-semi-invariant series v_i. Truncating expansions in those parameters (for $\widetilde{g}_i$ and $\widetilde{l}_i$, truncating $\alpha_{0,i}^{-1}$ and v_i), we

obtain G-semi-invariant regular functions $\widetilde{x}_i, \widetilde{y}_i, \widetilde{z}_i, \widetilde{g}_i, \widetilde{l}_i$ on $\mathcal{X}_i$ with the characters of x, y, z, g_i, l_i such that

$$\widetilde{x}_i - x,\ \widetilde{y}_i - y,\ \widetilde{z}_i - z \in \mathfrak{m}^{n_i}, \qquad \widetilde{g}_i - g_i,\ \widetilde{l}_i - l_i \in \mathfrak{m}^{n_i} g_i.$$

By (F.173), $\iota(\mathfrak{m}^n) \subset (u^n)$ for any n.

Let $(\mathcal{X}'_i \ni 0)$ be the germ of

$$\operatorname{Spec} \mathcal{O}_{\mathcal{X}_i,0}[u, \widetilde{X}_i, \widetilde{Y}_i, \widetilde{Z}_i]/\big(u^\sigma - s,\ \widetilde{x}_i - u^{\omega_1}\widetilde{X}_i,\ \widetilde{y}_i - u^{\omega_2}\widetilde{Y}_i,\ \widetilde{z}_i - u^{\omega_3}\widetilde{Z}_i\big)$$

at the point where $u, \widetilde{X}_i, \widetilde{Y}_i, \widetilde{Z}_i$ vanish. It is smooth with regular parameters $\widetilde{X}_i, \widetilde{Y}_i, \widetilde{Z}_i, u$, its completion at 0 is $\mathbb{C}[[X, Y, Z, u]]$ with the induced map ι, and $\widetilde{X}_i - X, \widetilde{Y}_i - Y, \widetilde{Z}_i - Z \in (u^{n_i - \omega_1})$, since $\omega_1 \geq \omega_2, \omega_3$. By $\iota(g_i) = u^\theta f'_i$ and $\iota(l_i) = u^{mr_1+c} l'_i$, the functions $\iota(\widetilde{g}_i)$ and $\iota(\widetilde{l}_i)$ are divisible by u^θ and u^{mr_1+c} in the completion, hence in $\mathcal{O}_{\mathcal{X}'_i,0}$ by faithful flatness. Let

$$\widetilde{f}_i := u^{-\theta}\iota(\widetilde{g}_i), \qquad \widetilde{l}'_i := u^{-mr_1-c}\iota(\widetilde{l}_i), \quad \text{and} \quad \widetilde{W}'_i := (\widetilde{f}_i = 0) \subset \mathcal{X}'_i.$$

Then $\widetilde{f}_i - f'_i \in (u^{n_i}) f'_i$ and $\widetilde{l}'_i - l'_i \in (u^{n_i - mr_1 - c}) f'_i$.

Thus W'_i is the completion of $\widetilde{W}'_i$ at 0 and $\widetilde{l}'_i|_{W'_i} = l'_i|_{W'_i}$. Expanded in $(\widetilde{X}_i, \widetilde{Y}_i, \widetilde{Z}_i, u)$, the series $\widetilde{f}_i$ and $\widetilde{l}'_i$ differ from the expansions of f'_i and l'_i in (X, Y, Z, u) by elements of $(u^{n_i - \omega_1 - mr_1 - c})$. So for any d and all but finitely many i, the d-jets of $(\widetilde{f}_i, \widetilde{l}'_i)$ and (f'_i, l'_i) coincide. Hence (f'_*, l'_*) is a compatible generic limit of $(\widetilde{f}_i, \widetilde{l}'_i)$, and (F.175) holds for $(\widetilde{f}_i, \widetilde{l}'_i)$ after discarding finitely many i.

Since $\iota(\widetilde{g}_i) = u^\theta \widetilde{f}_i$, the morphism $\mathcal{X}'_i \to \mathcal{X}_i$ restricts to a dominant morphism $\widetilde{W}'_i \to Y_i$, again denoted ι. Let $\varpi_i : Y'_i \to Y_i$ be the normalization of $(u^\sigma = s) \subset Y_i \times \mathbb{A}^1_u$, a finite cyclic cover of degree σ. It is irreducible, since if $\sigma = 2$, then s is not the square of a rational function on Y_i, as $\operatorname{mult}_{S_i} s = 1$ by Proposition 17.7(1). Since $\widetilde{W}'_i$ is normal, $\iota^* s = u^\sigma$, and $\widetilde{X}_i = u^{-\omega_1}\iota^*\widetilde{x}_i$, $\widetilde{Y}_i = u^{-\omega_2}\iota^*\widetilde{y}_i$, $\widetilde{Z}_i = u^{-\omega_3}\iota^*\widetilde{z}_i$, the morphism ι is the composite of a birational morphism $\widetilde{W}'_i \to Y'_i$ and ϖ_i. The form $d\widetilde{x}_i \wedge d\widetilde{y}_i \wedge d\widetilde{z}_i \wedge ds$ pulls back to

$$\sigma u^{\omega_1+\omega_2+\omega_3+\sigma-1} d\widetilde{X}_i \wedge d\widetilde{Y}_i \wedge d\widetilde{Z}_i \wedge du,$$

and $\omega_1 + \omega_2 + \omega_3 + \sigma - 1 - \theta = r_1 + \sigma - 1$. Taking residues along $\widetilde{g}_i$ and $\widetilde{f}_i$, we obtain

$$K_{\widetilde{W}'_i} = \iota^* K_{Y_i} + (r_1 + \sigma - 1)\operatorname{div}(u). \tag{F.176}$$

Let

$$G' := \{(g, \xi) \in G \times \mathbb{C}^* \mid \xi^\sigma = \chi_s(g)\},$$

acting on (X, Y, Z, u) by the characters $\chi_x(g)\xi^{-\omega_1}$, $\chi_y(g)\xi^{-\omega_2}$, $\chi_z(g)\xi^{-\omega_3}$, ξ. It is a finite abelian group with $|G'| = \sigma|G| \leq 2M$, and ι is equivariant along the projection $G' \to G$. Thus f'_i and l'_i are G'-semi-invariant with characters independent of i, and f_0 has the character of f'_i, since any monomial of f_0 occurs in f'_i. Moreover G' acts on $(\widetilde{X}_i, \widetilde{Y}_i, \widetilde{Z}_i, u)$ by the characters of (X, Y, Z, u), inducing its action on W'_i, and $\widetilde{f}_i$, $\widetilde{l}'_i$ have the characters of f'_i, l'_i. If (g, ξ) acts trivially on $\widetilde{W}'_i$, then it acts trivially on Y'_i, hence on Y_i, which is birational to W_i, so $g = 1$, and $\xi = 1$ since (g, ξ) maps u to ξu. Let $W_0 := (f_0 = 0) \subset \operatorname{Spec} \mathbb{C}[[X, Y, Z, u]]$. Since none of X, Y, Z, u vanishes on W_0, an element acting trivially on W_0 acts trivially on X, Y, Z, u. Thus G' acts faithfully on $\widetilde{W}'_i$, hence on its completion W'_i, and on W_0. Since $G' \to G$ is surjective with kernel $\{(1, \xi) \mid \xi^\sigma = 1\}$, the covering group of ϖ_i,

$$\text{the rational map } \widetilde{W}'_i/G' \dashrightarrow W_i/G \text{ induced by } \iota \text{ is birational.} \tag{F.177}$$

For any prime divisor E over W'_i with center 0, $\operatorname{mult}_E \widetilde{X}_i \geq \min\{\operatorname{mult}_E X, n_i - \omega_1\}$, and similarly for Y and Z. By Lemma 2.12(2)(3), the prime divisors with center 0 over W'_i and over $\widetilde{W}'_i$ correspond, with equal inertia groups, equal log discrepancies, and equal orders of functions on $\mathcal{X}'_i$, and the quotient mlds of corresponding boundaries are equal. Below, prime divisors over W'_i are read on $\widetilde{W}'_i$ through the correspondence of Lemma 2.12(3).

Step 4. In this step we compare log discrepancies on W_i' and on W_i, show that the new axis coefficient is less than 1, and show that W_i' captures the computing lifts of b_i.

Let E be a prime divisor over W_i' with $\operatorname{mult}_E u > 0$, for instance with center 0 or $E = E_0$. Let F be the prime divisor over Y_i under E through ι, and ρ its ramification index over Y_i, so that $\operatorname{mult}_E \circ \iota^* = \rho \operatorname{mult}_F$ on rational functions on Y_i. Since $\iota(x), \iota(y), \iota(z), \iota(s) \in (u)$ and $\operatorname{mult}_E u > 0$, F has center 0 on Y_i, and $\rho \operatorname{mult}_F s = \sigma \operatorname{mult}_E u$. Factoring ι through the finite morphism ϖ_i, (F.176) implies $a(E, W_i', 0) + (r_1 + \sigma - 1) \operatorname{mult}_E u = \rho a(F, Y_i, 0)$. By Proposition 17.7(1) (with $t = 1$), $\iota(l_i) = u^{mr_1+c} l_i'$, and $h_i = s^N l_i$, we obtain

$$\begin{gathered} \rho a(F, W_i, 0) = a(E, W_i', 0) + \kappa \operatorname{mult}_E u, \qquad \kappa := \sigma r + r_1 + \sigma - 1, \\ \rho \operatorname{mult}_F h_i = \operatorname{mult}_E l_i' + (\sigma N + m r_1 + c) \operatorname{mult}_E u. \end{gathered} \tag{F.178}$$

Let

$$\beta_1 := \sigma\beta + \frac{c}{m} - \sigma + 1 \in \frac{1}{m}\mathbb{Z}, \qquad B_i' := \frac{1}{m} \operatorname{div}(l_i') + \beta_1 \operatorname{div}(u).$$

Since $\sigma N/m = \sigma\beta + \sigma r$, we have $\kappa = (\sigma N + m r_1 + c)/m - \beta_1$, so (F.178) implies

$$a(E, W_i', B_i') = \rho a\Big(F, W_i, \frac{1}{m} \operatorname{div}(h_i)\Big). \tag{F.179}$$

Since the ramification index of a finite quotient morphism along a prime divisor over its source equals the order of the inertia group of that divisor [Liu26a, Lemma 2.6(1)], comparing the quotient morphisms $\widetilde{W}_i' \to \widetilde{W}_i'/G'$ and $W_i \to W_i/G$ through the birational map of (F.177), and using the multiplicativity of ramification indices, we obtain

$$e_E = \rho e_F. \tag{F.180}$$

For $E = E_0$, we have $\operatorname{mult}_{E_0} l_i' = 0$ and $\operatorname{mult}_{E_0} u = 1$, so $a(E_0, W_i', B_i') = 1 - \beta_1$, and the prime divisor F_0 over Y_i under E_0 satisfies $\rho \operatorname{mult}_{F_0} s = \sigma$. By (F.179), Proposition 17.7(1) (with $t = 1$), and (F.167),

$$1 - \beta_1 = \rho a(F_0, Y_i, B_i) \geq \rho \eta_m \operatorname{mult}_{F_0} s = \sigma \eta_m > 0.$$

Thus

$$\beta_1 < 1. \tag{F.181}$$

Let F be a computing lift of b_i. By Proposition 17.7(1), F has center 0 on Y_i and satisfies

$$\operatorname{mult}_F s \leq 3e_F/\eta_m \leq 3M/\eta_m.$$

By (F.170), discarding finitely many i, we may assume that, for any such F,

$$\min\{\operatorname{mult}_F x, \operatorname{mult}_F y, \operatorname{mult}_F z\} > 3\omega_1 M/\eta_m.$$

Let E be a prime divisor over W_i' lying over F, and ρ its ramification index over Y_i. Then

$$\operatorname{mult}_E X = \rho\Big(\operatorname{mult}_F x - \frac{\omega_1}{\sigma} \operatorname{mult}_F s\Big) > 0,$$

and similarly $\operatorname{mult}_E Y > 0$, $\operatorname{mult}_E Z > 0$, $\operatorname{mult}_E u > 0$. Thus E has center 0 on W_i', and $a(E, W_i', B_i')/e_E = b_i$ by (F.179) and (F.180). Conversely, for any prime divisor E over W_i' with center 0, the prime divisor F over Y_i under E is a prime divisor over W_i with center o_i by **Step 1**, so

$$\frac{a(E, W_i', B_i')}{e_E} = \frac{a(F, W_i, \frac{1}{m} \operatorname{div}(h_i))}{e_F} \geq b_i.$$

Hence

$$\operatorname{mld}((W_i' \ni 0, B_i')/G') = b_i, \tag{F.182}$$

and the prime divisor F under any computing lift E of $\mathrm{mld}((W_i' \ni 0, B_i')/G')$ is a computing lift of b_i. Since $\rho \le \sigma$, for any such E we obtain

$$\begin{aligned} \mathrm{mult}_E u = \frac{\rho}{\sigma}\,\mathrm{mult}_F s \le R' := \Big\lfloor \frac{3M}{\eta_m} \Big\rfloor, \\ \min\{\mathrm{mult}_E X, \mathrm{mult}_E Y, \mathrm{mult}_E Z\} \ge \min\{\mathrm{mult}_F x, \mathrm{mult}_F y, \mathrm{mult}_F z\} - \omega_1 R', \end{aligned} \tag{F.183}$$

and the right-hand side of the second inequality tends to $+\infty$ by (F.170).

Step 5. In this step we treat the case where the pair $(S_0, \frac{1}{m}\,\mathrm{div}(\mathrm{in}_w(l_*')))$ over an algebraic closure $\mathbb{L}_u$ of $\Bbbk((u))$ has no coefficient-one component.

In Theorem 17.6 we take $\widetilde{W}_i'$, G', $2M$ in place of W_i, G, M, with f_0, w, $d = 2k-2$, with the parameters $(\widetilde{X}_i, \widetilde{Y}_i, \widetilde{Z}_i, u)$ of **Step 3** in place of (x, y, z, u), with the equations $\widetilde{f}_i$ and the marks $\widetilde{l}_i'$, and with R' of (F.183), $L_0 := a$, and $\beta := \beta_1$, so that the boundary B_i of Theorem 17.6 is

$$\widetilde{B}_i' := \frac{1}{m}\,\mathrm{div}(\widetilde{l}_i') + \beta_1\,\mathrm{div}(u),$$

whose pullback to W_i' is B_i' (**Step 3**). The group G' is abelian of order at most $2M$ and acts faithfully and diagonally on $(\widetilde{X}_i, \widetilde{Y}_i, \widetilde{Z}_i, u)$ with fixed characters of the equations and of the marks, and f_0 has the character of the equations (**Step 3**). Each $\widetilde{W}_i'$ is a G'-invariant normal klt Gorenstein hypersurface threefold germ at 0, since its completion W_i' is (**Step 3**), and $\beta_1 \in \frac{1}{m}\mathbb{Z}$ with $\beta_1 < 1$ by (F.181). The hypotheses (1)–(3) of Theorem 17.6 hold for the following reasons.

(1) By (F.182), **Step 3**, Set-up 17.1(2), and Definition 4.1(3),

$$\mathrm{mld}((\widetilde{W}_i' \ni 0, \widetilde{B}_i')/G') = b_i \quad \text{and} \quad 0 < b_i \le a - \delta < a.$$

(2) Computing lifts of $\mathrm{mld}((\widetilde{W}_i' \ni 0, \widetilde{B}_i')/G')$ exist by **Step 4**, and any such computing lift E corresponds to a computing lift of $\mathrm{mld}((W_i' \ni 0, B_i')/G')$ (**Step 3**), so it satisfies (F.183), and

$$\min\{\mathrm{mult}_E \widetilde{X}_i, \mathrm{mult}_E \widetilde{Y}_i, \mathrm{mult}_E \widetilde{Z}_i\} \ge \min\{\mathrm{mult}_E X, \mathrm{mult}_E Y, \mathrm{mult}_E Z, n_i - \omega_1\} \to +\infty.$$

(3) By **Step 3**, (f_*', l_*') is a compatible generic limit of $(\widetilde{f}_i, \widetilde{l}_i')$ with coefficients in $\Bbbk[[u]]$. By (F.174), $f_*' - f_0$ consists of terms of transverse w-weight at least $2k > 2k-2$, and $\mathrm{in}_w(l_*')$ has weight m, is non-zero, and has X-degree at most 1. By Lemma F.11 with $\mathbb{L} = \mathbb{L}_u$, $g = \mathrm{in}_w(l_*')$, whose w-weight is m, and $t = \frac{1}{m}$, the pair $(S_0, \frac{1}{m}\,\mathrm{div}(\mathrm{in}_w(l_*')))$ over $\mathbb{L}_u$ is lc, and it has no coefficient-one component by the assumption of this step.

By Theorem 17.6, possibly passing to an infinite subfamily, there exist a real number C_1 and computing lifts E_i of $\mathrm{mld}((\widetilde{W}_i' \ni 0, \widetilde{B}_i')/G')$ with $a(E_i, \widetilde{W}_i', 0) \le C_1$ for any i.

Step 6. In this step we treat the case where $(S_0, \frac{1}{m}\,\mathrm{div}(\mathrm{in}_w(l_*')))$ over $\mathbb{L}_u$ has a coefficient-one component P_1.

By [Roa04, Table 1 and (3)], there exists a quotient morphism $\pi : \mathbb{A}^2_{\mathbb{L}_u} \to S_0$ by a binary dihedral group $\Gamma \subset SL_2(\mathbb{L}_u)$ of order $4(k-2)$, which pulls X, Y, Z back to invariant forms of degrees $2(k-1)$, $2(k-2)$, 4. Let $\widehat{h} := \pi^* \mathrm{in}_w(l_*')$, a Γ-invariant form of degree $2m$. Since $\widehat{h}$ is a form, $\pi^{-1}(P_1)$ is a union of lines ℓ through 0, and since π is étale in codimension 1, $\mathrm{mult}_\ell \widehat{h} = m$. We choose coordinates p, q on $\mathbb{A}^2$ such that Γ is generated by $\mathrm{diag}(\zeta, \zeta^{-1})$, where ζ is a primitive $2(k-2)$-th root of unity, and by $(p, q) \to (q, -p)$. The first generator acts on $\mathbb{P}^1$ by $[p : q] \to [\zeta^2 p : q]$, of order $k - 2 \ge 3$. Thus the Γ-orbit of any point of $\mathbb{P}^1$ other than $[1 : 0]$ and $[0 : 1]$ has at least 3 points. Since $\widehat{h}$ is Γ-invariant of degree $2m$, any line in the Γ-orbit of ℓ has multiplicity m in $\widehat{h}$, so the Γ-orbit of ℓ is $\{(p = 0), (q = 0)\}$ and $\widehat{h} = \lambda'(pq)^m$ with $\lambda' \in \mathbb{L}_u^*$. The second generator maps $(pq)^m$ to $(-1)^m (pq)^m$, so m is even. The invariant $(pq)^2$ has degree 4, and by [Roa04, Table 1] the invariants of degree 4 are the multiples of $\pi^* Z$, since the other two generators have degrees $2(k-2) > 4$ and $2(k-1) > 4$. Thus $\widehat{h}$ is a multiple of $\pi^* Z^{m/2}$, so

$\mathrm{in}_w(l'_*)$ minus a multiple of $Z^{m/2}$ vanishes on S_0 and has X-degree at most 1. Such a polynomial is 0. Since $\mathrm{in}_w(l'_*) \in \Bbbk[[u]][X,Y,Z]$ and $\mathrm{in}_w(l'_*)|_{u=0} \neq 0$, we obtain

$$m \in 2\mathbb{N}^+, \qquad \mathrm{in}_w(l'_*) = \lambda(u) Z^{m/2}, \qquad \lambda \in \Bbbk[[u]], \quad \text{and} \quad \lambda(0) \neq 0. \tag{F.184}$$

Let $\lambda_i(u)$ be the coefficient of $Z^{m/2}$ in l'_i, viewed as a series in u. By (F.174) and (F.184), $l'_* - \lambda Z^{m/2} \in (u^{N_1})$, so by **Step 3** and (†), $l'_i - \lambda_i Z^{m/2} \in (u^{N_1})$ and $\lambda_i(0) \neq 0$. Since $\lambda_i(0) \neq 0$, the character of l'_i is $\chi_Z^{m/2}$. Thus $\chi_u^e = 1$ for any e such that $Z^{m/2}u^e$ has non-zero coefficient in l'_i, and λ_i is a G'-invariant unit. Since λ_i is a unit, we obtain, with (F.175) and $f'_* - f_0 \in (u^{N_1})$,

$$f'_i - f_0 \in (u^{N_1}), \qquad l'_i/\lambda_i - Z^{m/2} \in (u^{N_1}). \tag{F.185}$$

Let $\widetilde{\lambda}_i \in \mathbb{C}[u]$ be the truncation of λ_i below degree N_1, a G'-invariant unit with

$$1/\widetilde{\lambda}_i - 1/\lambda_i \in (u^{N_1}).$$

In Theorem 5.10 we take f_0, m, $2M$, $L := m/2$, for which $N = N_1$ since m is even, the germ $(\mathcal{X}'_i \ni 0)$ with the parameters $(\widetilde{X}_i, \widetilde{Y}_i, \widetilde{Z}_i, u)$ of **Step 3**, the group G', $f := \widetilde{f}_i$, $h := \widetilde{l}'_i/\widetilde{\lambda}_i$, $h_0 := \widetilde{Z}_i^{m/2}$, and $\beta := \beta_1$, so that $W = \widetilde{W}'_i$ and the boundary B of Theorem 5.10 is $\widetilde{B}'_i$ of **Step 5**. Let $\mathfrak{n}$ be the maximal ideal of $\mathcal{O}_{\mathcal{X}'_i,0}$. The hypotheses of Theorem 5.10 hold for the following reasons.

- $(f_0 = 0) \subset \mathbb{A}^3$ is a normal surface whose only singular point is the Du Val point 0, since f_0 is quasi-homogeneous with an isolated singularity at 0.
- $\widetilde{X}_i, \widetilde{Y}_i, \widetilde{Z}_i, u$, $f_0(\widetilde{X}_i, \widetilde{Y}_i, \widetilde{Z}_i)$, $\widetilde{f}_i$, and $\widetilde{l}'_i/\widetilde{\lambda}_i$ are G'-semi-invariant, and $\widetilde{f}_i$ has the character of f_0 (**Step 3**).
- $\widetilde{W}'_i$ is normal, and G' acts faithfully on $\widetilde{W}'_i$ and on $(f_0(\widetilde{X}_i, \widetilde{Y}_i, \widetilde{Z}_i) = 0)$, by the argument of **Step 3**.
- h_0 has degree $m/2$ and is non-zero on S_0, and $(S_0, \frac{1}{m}\operatorname{div}(Z^{m/2}|_{S_0}))$ is lc by Lemma F.11 with $\mathbb{L} = \mathbb{C}$, $g = Z^{m/2}$, and $t = \frac{1}{m}$.
- $\beta_1 < 1$ by (F.181).
- $\widetilde{f}_i - f_0(\widetilde{X}_i, \widetilde{Y}_i, \widetilde{Z}_i)$ and $\widetilde{l}'_i/\widetilde{\lambda}_i - \widetilde{Z}_i^{m/2}$ belong to $(u) \cap \mathfrak{n}^{N_1}$ for all but finitely many i: by **Step 3** and $1/\widetilde{\lambda}_i - 1/\lambda_i \in (u^{N_1})$, in the completion they are congruent modulo (u^{N_1}) to $f'_i - f_0$ and $l'_i/\lambda_i - Z^{m/2}$, so (F.185) and faithful flatness give this.

By Theorem 5.10(2), there exist computing lifts E_i of $\mathrm{mld}((\widetilde{W}'_i \ni 0, \widetilde{B}'_i)/G')$ with

$$a(E_i, \widetilde{W}'_i, 0) \leq C_1 := N_1.$$

Step 7. In this step we return the computing lifts E_i of **Steps 5** and 6 to W_i and contradict Definition 4.1(3).

By Lemma 2.12(2)(3), as in **Step 3**, the E_i correspond to computing lifts of the quotient value in (F.182) with $a(E_i, W'_i, 0) \leq C_1$, so the statements of **Step 4** hold for them. Let F_i be the prime divisor over W_i under E_i. By **Step 4**, F_i is a computing lift of b_i and $\mathrm{mult}_{E_i} u \leq R'$. By (F.178) and (F.180),

$$\frac{a(F_i, W_i, 0)}{e_{F_i}} = \frac{a(E_i, W'_i, 0) + \kappa \,\mathrm{mult}_{E_i} u}{e_{E_i}} \leq C_2 := C_1 + \kappa R'.$$

By Lemma 4.2(2) for F_i, for any $t \in [0,1]$,

$$q_i(t) \leq b_i + (1-t)\Big(\frac{a(F_i, W_i, 0)}{e_{F_i}} - b_i\Big) \leq b_i + (1-t)C_2.$$

For $t = \tau_i$, Definition 4.1(3) implies

$$a \leq q_i(\tau_i) \leq a - \delta + (1-\tau_i)C_2,$$

so $\delta \leq (1-\tau_i)C_2$ for any i of the infinite subfamily. This is impossible since $\lim_{i\to+\infty} \tau_i = 1$. □

Proof of Proposition 17.13. We write (x, y, z) for the eigenparameters (u_1, u_2, u_3) of Set-up 17.1, so that $C = (x = y = z = 0)$. From now on, u only denotes a root $s = u^{12}$. Let $\mathbb{K}$ be an algebraic closure of $\Bbbk((s))$, and let $\mathrm{wt}_{(3,2,2)}$ denote the weight with weights $3, 2, 2, 0$ on x, y, z, s. We divide the proof into seven steps.

Step 1. In this step we bring f_* to the form of the proposition.

By Set-up 17.1(6), $f_* \in (x, y, z)^2$. By Construction-Theorem 13.6 and Lemma 3.7(1), whose conditions on the coefficients of the f_i are closed and hence hold for f_*, the quadratic part of f_* is the square of a non-zero linear form and its residual cubic is the cube of a non-zero linear form. Since $f_* \in (x, y, z)^2$ is semi-invariant, the quadratic part of f_* is ℓ_0^2 for a linear eigenform $\ell_0 \neq 0$ in x, y, z. Possibly reordering x, y, z and replacing x with ℓ_0, we may assume that the quadratic part of f_* is x^2. By Lemma D.10 for f_* and the parameters x, y, z, s, there exist $\theta \in (y, z, s)\Bbbk[[y, z, s]]$ with $(\partial_x f_*)(\theta, y, z, s) = 0$ and an eigenparameter $\widetilde{x}$ with the character of x such that $\widetilde{x} - (x - \theta) \in (x - \theta)(x, y, z, s)$ and $f_* = \widetilde{x}^2 + R_1$ with $R_1 := f_*(\theta, y, z, s)$. Since $\partial_x f_* \in (x, y, z)$, both 0 and $\theta(0, 0, s)$ are solutions $X \in s\Bbbk[[s]]$ of $(\partial_x f_*)(X, 0, 0, s) = 0$, which has only one solution since $\partial_x^2 f_*$ is a unit. Thus $\theta \in (y, z)$. Replacing x with $\widetilde{x}$, we obtain

$$f_* = x^2 + R_1, \qquad R_1 \in (y, z)^2\Bbbk[[y, z, s]],$$

and still $C = (x = y = z = 0)$, since $f_* \in (x, y, z)^2$ and $\theta \in (y, z)$.

The germ of W_* at the generic point of C has type D_4, of quadratic rank 1, so the coefficients of y^2, yz, z^2 in R_1 vanish in $\Bbbk((s))$, that is, $R_1 \in (y, z)^3$. Let $\sum_{j=0}^{3} \lambda_j y^{3-j} z^j$, with $\lambda_j \in \Bbbk[[s]]$, be the part of R_1 of degree 3 in y, z. At $s = 0$ it is the cubic part of R_1 at the closed point, which is ℓ^3 for a non-zero linear eigenform ℓ in y, z, because the residual cubic of f_* is the cube of a non-zero linear form (the residual cubic being well defined up to a linear change of variables and a non-zero scalar). Possibly reordering y, z and replacing y with ℓ, we may assume that $\lambda_0(0) = 1$ and $\lambda_1(0) = \lambda_2(0) = \lambda_3(0) = 0$. Then λ_0 is G-invariant, and so is the cube root $\lambda_0^{1/3}$ with constant term 1. Replacing y with the eigenform $\lambda_0^{1/3}(y + \frac{\lambda_1}{3\lambda_0} z)$, which is y at $s = 0$, we obtain

$$f_* = x^2 + y^3 + \alpha(s) y z^2 + \gamma(s) z^3 + R, \qquad R \in (y, z)^4\Bbbk[[y, z, s]], \quad \text{and} \quad \alpha(0) = \gamma(0) = 0.$$

Since the generic transverse type along C is D_4, the cubic $y^3 + \alpha y z^2 + \gamma z^3$ has three distinct linear factors over $\mathbb{K}$, that is, $4\alpha^3 + 27\gamma^2 \neq 0$. Thus, with $\mathrm{mult}_0 0 := +\infty$,

$$v = 12 \min\left\{ \frac{\mathrm{mult}_0 \alpha}{2}, \frac{\mathrm{mult}_0 \gamma}{3} \right\} = \min\{6 \,\mathrm{mult}_0 \alpha, 4 \,\mathrm{mult}_0 \gamma\}$$

is finite, $v \in 2\mathbb{Z}$, and $v \geq 4$.

Step 2. In this step we prove that the representative of h_* of x-degree at most 1 has weight m.

Let $\psi_* + x\xi_*$, with $\psi_*, \xi_* \in \Bbbk[[y, z, s]]$, be the representative of h_* modulo f_* of x-degree at most 1, given by Weierstrass division by f_*. We prove

$$\mathrm{wt}_{(3,2,2)}(\psi_* + x\xi_*) = m. \tag{F.186}$$

Let $S := \operatorname{Spec} \mathbb{K}[[x, y, z]]/(f_*)$, the germ of W_* at the generic point of C after completion and base change to $\mathbb{K}$. Since C is the non-klt locus of the lc pair $(W_*, \frac{1}{m} \operatorname{div}(h_*))$ and a log resolution of $(W_*, \frac{1}{m} \operatorname{div}(h_*))$ induces one of $(S, \frac{1}{m} \operatorname{div}(h_*))$, the pair $(S, \frac{1}{m} \operatorname{div}(h_*))$ is lc and not klt.

Let $f_0 := x^2 + y^3 + \alpha y z^2 + \gamma z^3$. Over $\mathbb{K}$, a linear change of y, z and a scaling of x transform f_0 into $f_1 := x^2 + y^3 + z^3$. Any monomial in x, y, z of weight at least 5 belongs to the Jacobian ideal (x, y^2, z^2) of f_1, and any monomial of R has weight at least 8. Thus the part of weight $n_0 > 6$ of the equation is $\sum_j a_j \partial f_1 / \partial t_j$, where $(t_1, t_2, t_3) = (x, y, z)$ and

$$\mathrm{wt}\, a_j = n_0 - 6 + \mathrm{wt}\, t_j > \mathrm{wt}\, t_j,$$

and the substitution of $t_j - a_j$ for t_j removes it modulo weights greater than n_0. The composite of the substitutions converges and raises weights. It transforms S into

$$S_0 := (f_1 = 0) \subset \operatorname{Spec} \mathbb{K}[[x, y, z]],$$

and preserves the exceptional prime divisor E of the $(3,2,2)$-weighted blow-up. The lowest weight part of $\psi_* + x\xi_*$ has x-degree at most 1, so it is not divisible by f_0, and [Liu26b, Lemma 4.12] implies

$$n_E := \operatorname{mult}_E(h_*|_S) = \operatorname{wt}_{(3,2,2)}(\psi_* + x\xi_*).$$

By [Liu26b, Lemma 4.12] in the new coordinates, the representative h^0 of $h_*|_{S_0}$ of x-degree at most 1 has weight n_E. By Lemma F.11 for $\mathbb{K}$, f_1, and h^0, for any real number $t \geq 0$ the pair $(S_0, t\operatorname{div}(h^0|_{S_0}))$ is lc near 0 if and only if $tn_E \leq 1$, and klt near 0 if and only if $tn_E < 1$. The composite of the substitutions transforms $(S, t\operatorname{div}(h_*))$ into $(S_0, t\operatorname{div}(h^0|_{S_0}))$. As $(S, \frac{1}{m}\operatorname{div}(h_*))$ is lc and not klt, $n_E = m$, which proves (F.186).

By (F.186) and since v is even, the integer

$$c := \min\left\{ v\left(\tfrac{3}{2}k_1 + k_2\right) + 12k_4 \;\middle|\; 3k_1 + 2k_2 + 2k_3 = m,\ x^{k_1}y^{k_2}z^{k_3}s^{k_4} \text{ occurs in } \psi_* + x\xi_* \right\} \geq 0$$

is well defined. Here $k_1 \in \{0,1\}$ and $k_2, k_3, k_4 \in \mathbb{N}$.

Step 3. In this step we fix coordinates on $\mathcal{X}$ and prove that the computing lifts of b_i have large orders along x_i, y_i, z_i.

By Construction-Theorem 13.6, after passing to a subsequence, the group $G_i = G$, the diagonal action of G on the eigenparameters of $\mathcal{X} = \operatorname{Spec}\mathbb{C}[[x_1, \dots, x_4]]$ and the characters of f_i and h_i are fixed. Together with Set-up 17.1, the data satisfy Set-up 15.4 with $B_0 := a$. Moreover, W_* is cDV, and the minimal lc center of $(W_*, \frac{1}{m}\operatorname{div}(h_*))$ is $C \subset \operatorname{Sing}(W_*)$, since the non-klt locus is C and the minimal log discrepancy at the closed point is positive (Set-up 17.1(4) and (5)).

By Lemma 4.2(2) and Definition 4.1(3), any computing lift F of b_i satisfies

$$\frac{\operatorname{mult}_F h_i}{me_F} \geq \frac{\delta}{1-\tau_i}. \tag{F.187}$$

For any integer $N_1 > 0$, only finitely many i have a computing lift F of b_i with $\operatorname{mult}_F \operatorname{Jac}_{W_i} \leq N_1$: otherwise, by Theorem 15.8(1), for infinitely many i,

$$q_i(\tau_i) \leq b_i + L_1(1-\tau_i) \leq a - \delta + L_1(1-\tau_i),$$

with L_1 independent of i, and $a - \delta + L_1(1-\tau_i) < a$ for $i \gg 0$, which contradicts $q_i(\tau_i) \geq a$. Thus

$$\lim_{i\to+\infty} \min\{\operatorname{mult}_F \operatorname{Jac}_{W_i} \mid F \text{ is a computing lift of } b_i\} = +\infty,$$

which is the hypothesis of the second part of Theorem 16.1(2). By Theorem 16.1(2) for the G-eigenparameters (s,x,y,z) of $\mathcal{X}_*$, after passing to a subsequence, there exist, for any i, G-eigenparameters (s_i, x_i, y_i, z_i) of $\mathcal{X}$ with the characters of s, x, y, z, respectively, such that the tuples (f_i, h_i) written in (s_i, x_i, y_i, z_i) have a compatible generic limit with coefficient field $\Bbbk$, which is (f_*, h_*) written in (s,x,y,z), and

$$\lim_{i\to+\infty} \min_F \min\{\operatorname{mult}_F x_i, \operatorname{mult}_F y_i, \operatorname{mult}_F z_i\} = +\infty, \tag{F.188}$$

the minimum over the computing lifts F of b_i. By Theorem 5.1(3), $\operatorname{mult}_F \mathfrak{m}_{o_i} \leq 3M/\eta_m$ for any computing lift F of b_i. Since x_i, y_i, z_i, s_i generate $\mathfrak{m}_{o_i}$, (F.188) implies that for $i \gg 0$ and any computing lift F of b_i,

$$\operatorname{mult}_F s_i = \operatorname{mult}_F \mathfrak{m}_{o_i} \leq \frac{3M}{\eta_m}. \tag{F.189}$$

Let $\mathcal{O}_{W_i,o_i}$ be the local ring of the algebraic germ $(W_i \ni o_i)$ of Definition 4.1(1), whose completion is $\mathbb{C}[[x_1, \dots, x_4]]/(f_i)$, and $\mathbb{C}(W_i)$ its function field. Fix integers $n_i \geq 2$ with $\lim_{i\to+\infty} n_i = +\infty$. Since the completion map of $\mathcal{O}_{W_i,o_i}$ induces isomorphisms modulo the n-th powers of the maximal ideals for any n and G is finite abelian, we may replace s_i, x_i, y_i, z_i by G-semi-invariant elements of $\mathbb{C}[[x_1, \dots, x_4]]$ with the characters of s, x, y, z, congruent to s_i, x_i, y_i, z_i modulo $(x_1, \dots, x_4)^{n_i}$, whose restrictions to W_i belong to $\mathcal{O}_{W_i,o_i}$. The replacement changes no part of degree less than n_i of f_i and h_i written in (s_i, x_i, y_i, z_i), so the limit statement

above holds for the new parameters. The order along F of each new parameter is at least the minimum of n_i and the order of the old one, so (F.188), and hence (F.189), hold for the new parameters.

Closed and open conditions over $\mathbb{C}$ on finitely many coefficients of f_i and of the representative $\psi_i + x_i\xi_i$ of h_i modulo f_i of x_i-degree at most 1 which hold for f_* and $\psi_* + x\xi_*$ in x, y, z, s hold for all but finitely many i (Lemma 2.8). Weierstrass division by f_i, which is x_i-regular of order 2 for $i \gg 0$, is given on jets by polynomial maps. The coefficients of the chart equations and marks below are coefficients of f_i and $\psi_i + x_i\xi_i$, so closed and open conditions on them which hold for the limit hold for $i \gg 0$.

Step 4. In this step we construct Γ_i and the first chart, prove the first sentence of (6), and treat the case of three distinct roots.

Let χ_s be the character of s and of s_i. Since $f_*(x, y, z, 0) \equiv x^2 + y^3$ modulo $(x, y, z)^4$, **Step 3** implies $f_i(x_i, y_i, z_i, 0) \equiv x_i^2 + y_i^3$ modulo $(x_i, y_i, z_i)^4$ for $i \gg 0$. By Weierstrass preparation and completing the square, $f_i(x_i, y_i, z_i, 0)$ is a unit times $(x_i + \theta_i)^2 + \sigma_i$ with $\theta_i, \sigma_i \in \mathbb{C}[[y_i, z_i]]$, where σ_i has lowest form y_i^3, which is not a square. Thus $\mathbb{C}[[x_1, \dots, x_4]]/(f_i, s_i)$ is a domain, and so is its subring $\mathcal{O}_{W_i,o_i}/(s_i)$. As W_i is normal, s_i is a uniformizer at the prime divisor $(s_i = 0)$, so $u^{12} - s_i$ is an Eisenstein polynomial and

$$[\mathbb{C}(W_i)(u) : \mathbb{C}(W_i)] = 12.$$

Let

$$\Gamma_i := \{(g, \zeta) \in G \times \mu_{12|G|} \mid \zeta^{12} = \chi_s(g)\},$$

where (g, ζ) acts on $\mathbb{C}(W_i)$ as g and sends u to ζu. The action is well defined since $(\zeta u)^{12} = \chi_s(g)s_i$. Since $\chi_s(g)^{|G|} = 1$, each $g \in G$ has 12 elements ζ, so $|\Gamma_i| = 12|G|$. The group Γ_i is abelian, and it acts faithfully on $\mathbb{C}(W_i)(u)$ since G acts faithfully on W_i. Thus $\mathbb{C}(W_i)(u)$ is a Galois extension of $\mathbb{C}(W_i)^G$ with group Γ_i, which does not depend on i. We write $\Gamma := \Gamma_i$.

Fix an integer $r > \max\{c, \frac{3}{2}v, 6\}$, and let $\Phi : \operatorname{Spec}\mathbb{C}[[x', y', z', u]] \to \mathcal{X}$ be given by

$$x_i = u^{3r+\frac{3}{2}v}x', \qquad y_i = u^{2r+v}y', \qquad z_i = u^{2r}z', \quad \text{and} \quad s_i = u^{12}. \tag{F.190}$$

The formulas $x = u^{3r+\frac{3}{2}v}x'$, $y = u^{2r+v}y'$, $z = u^{2r}z'$, and $s = u^{12}$ define a morphism $\Phi_* : \operatorname{Spec}\Bbbk[[x', y', z', u]] \to \mathcal{X}_*$ over $\Bbbk$. Since $3r + \frac{3}{2}v > 2r + v > 2r > 12$, a monomial of degree d in x_i, y_i, z_i, s_i has u-order at least $12d$ after Φ, and so does a monomial of degree d in x, y, z, s after Φ_*. The group Γ acts diagonally on x', y', z', u, the element (g, ζ) multiplying x' by $\chi_x(g)\zeta^{-3r-3v/2}$, y' by $\chi_y(g)\zeta^{-2r-v}$, z' by $\chi_z(g)\zeta^{-2r}$ and u by ζ. Under Φ_*,

$$x^{k_1}y^{k_2}z^{k_3}s^{k_4} = u^{r(3k_1+2k_2+2k_3)+v(\frac{3}{2}k_1+k_2)+12k_4}x'^{k_1}y'^{k_2}z'^{k_3}. \tag{F.191}$$

By **Step 1**, the definition of v and (F.191), $f_* \circ \Phi_* = u^{6r+3v}f'_*$ with

$$f'_* = x'^2 + y'^3 + \alpha'(u)y'z'^2 + \gamma'(u)z'^3 + u^L R', \tag{F.192}$$

where

$$\alpha'(u) := u^{-2v}\alpha(u^{12}) \in \Bbbk[[u]], \qquad \gamma'(u) := u^{-3v}\gamma(u^{12}) \in \Bbbk[[u]], \qquad \text{and} \qquad L := 2r - 3v > 0,$$

$(\alpha'(0), \gamma'(0)) \neq (0, 0)$, and $R' \in (y', z')^4\Bbbk[[y', z', u]]$, because any monomial of R has u-order at least $8r$ after Φ_*, and $8r = 6r + 3v + L$. By (F.186) and (F.191), the monomials of $\psi_* + x\xi_*$ with $3k_1 + 2k_2 + 2k_3 = m$ have u-order at least $mr + c$, with equality for some of them, and the others have u-order at least $(m + 1)r > mr + c$. Thus $(\psi_* + x\xi_*) \circ \Phi_* = u^{mr+c}l'_*$, where $l'_*|_{u=0}$ is a non-zero polynomial in x', y', z' of x'-degree at most 1. In $\Bbbk((u))$,

$$4\alpha'^3 + 27\gamma'^2 = u^{-6v}(4\alpha^3 + 27\gamma^2)(u^{12}) \neq 0.$$

The binary cubic $y'^3 + \alpha'(0)y'z'^2 + \gamma'(0)z'^3$ is not a cube, since a cube with y'^3-coefficient 1 and no y'^2z'-term is y'^3. Thus it has either three distinct roots or one double root and one simple root. We let $j_0 := 0$ in the first case and $j_0 := 1$ in the second.

Let $f_i' := u^{-6r-3v}(f_i \circ \Phi)$ and $l_i' := u^{-mr-c}((\psi_i + x_i\xi_i) \circ \Phi)$. By **Step 3** for the finitely many monomials of u-order at most $6r+3v$, resp. $mr+c$, after Φ, for $i \gg 0$ the following hold.

(1) f_i' and l_i' are regular.
(2) $f_i'|_{u=0} = x'^2 + y'^3 + \alpha_i' y' z'^2 + \gamma_i' z'^3$ with constants $(\alpha_i', \gamma_i') \neq (0,0)$, and $y'^3 + \alpha_i' y' z'^2 + \gamma_i' z'^3$ has three distinct roots if $j_0 = 0$ and one double root if $j_0 = 1$.
(3) $l_i'|_{u=0}$ is a non-zero polynomial of x'-degree at most 1.

Moreover, for any positive integer d, each coefficient of the d-jet of (f_i', l_i') is the coefficient of $(f_i, \psi_i + x_i\xi_i)$ at a monomial independent of i, of degree at most $d'' := d + 6r + 3v + mr + c$. Thus the d-jets of (f_i', l_i') and of (f_*', l_*') are the values at the d''-jets of (f_i, h_i) and of (f_*, h_*) of one regular map ϖ over $\mathbb{Q}$ given by Weierstrass division (**Step 3**), defined near the generic point of the closed subset $Z_{d''}$ of Definition 2.5 for the tuples (f_i, h_i), and ϖ sends that generic point to the generic point of the closure of $\varpi(Z_{d''})$. So the tuples formed by f_i' and l_i' have the compatible generic limit (f_*', l_*'). With (F.192), this proves the first sentence of (6) in the Γ-eigenparameters x', y', z', u. Let

$$Z_i' := (f_i' = 0) \subset \operatorname{Spec} \mathbb{C}[[x', y', z', u]].$$

Let Y_i be the germ at the origin of the quotient by its u-power torsion of

$$\operatorname{Spec} \mathcal{O}_{W_i, o_i}[u, x', y', z']/(u^{12} - s_i,\ x_i - u^{3r+\frac{3}{2}v}x',\ y_i - u^{2r+v}y',\ z_i - u^{2r}z'),$$

with the induced action of Γ. Since completion is flat and $f_i \circ \Phi = u^{6r+3v} f_i'$ with $f_i'|_{u=0} \neq 0$, the completion of Y_i at the origin is Z_i'. Since $x' = u^{-3r-3v/2}x_i$, $y' = u^{-2r-v}y_i$ and $z' = u^{-2r}z_i$, the function field of Y_i is $\mathbb{C}(W_i)(u)$. The morphism $Y_i \to W_i$ is étale over $u \neq 0$, $Y_i \backslash (u=0)$ maps to $W_i \backslash \{o_i\}$, and $(u=0) \cap Y_i$ maps to o_i.

Let F be a computing lift of b_i, and let P be a divisorial valuation of $\mathbb{C}(W_i)(u)$ restricting to F, with ramification index $\rho \leq 12$. Then $\operatorname{mult}_P u = \frac{\rho}{12} \operatorname{mult}_F s_i \leq \frac{3M}{\eta_m}$ by (F.189), and

$$\begin{aligned} \operatorname{mult}_P x' &= \rho \operatorname{mult}_F x_i - \left(3r + \tfrac{3}{2}v\right) \operatorname{mult}_P u, \\ \operatorname{mult}_P y' &= \rho \operatorname{mult}_F y_i - (2r+v) \operatorname{mult}_P u, \quad \text{and} \\ \operatorname{mult}_P z' &= \rho \operatorname{mult}_F z_i - 2r \operatorname{mult}_P u. \end{aligned}$$

By (F.188), for any integer J and $i \gg 0$, any such F and P satisfy

$$\min\{\operatorname{mult}_P x', \operatorname{mult}_P y', \operatorname{mult}_P z'\} > J. \tag{F.193}$$

In particular, P has center the origin of Y_i.

Assume that $j_0 = 0$, and let $Z_i := Z_i'$, $g_i := f_i'$, and $S_i := (u=0) \subset Z_i$. By (2), S_i is a Du Val surface germ of type D_4 with singular locus the origin. Since $Y_i \backslash (u=0)$ is étale over the non-singular locus $W_i \backslash \{o_i\}$ (Definition 4.1(1)), Z_i is smooth outside S_i, and a singular point of Z_i on S_i is a singular point of S_i. Thus $(Z_i \ni 0)$ is an isolated cDV point, hence Gorenstein terminal.

Step 5. In this step we treat the case $j_0 = 1$ by a coordinate change which fixes x' and u, and prove the second sentence of (6).

Assume that $j_0 = 1$. The group Γ acts diagonally on y', z' and preserves $y'^3 + \alpha'(0)y'z'^2 + \gamma'(0)z'^3$ up to a character, so it preserves the double line and the simple line. Thus the double line and the simple line are defined by linear eigenforms. Possibly replacing y' with a multiple of an eigenform defining the double line and z' with a multiple of an eigenform defining the simple line, we may assume that $f_*' = x'^2 + R_2$ with $R_2 \in (y', z')^3 \Bbbk[[y', z', u]]$ and $R_2(y', z', 0) = z'y'^2$. We remove the terms of R_2 divisible by y' other than $z'y'^2$, by increasing total degree in y', z', u. Such a term is $\vartheta y'^{k_2} z'^{k_3} u^{k_4}$ with $\vartheta \in \Bbbk \backslash \{0\}$, $k_2 + k_3 \geq 3$, and $k_4 \geq 1$. If $k_2 \geq 2$, it is removed by subtracting $\vartheta y'^{k_2-2} z'^{k_3} u^{k_4}$ from z'. If $k_2 = 1$, then $k_3 \geq 2$, and it is removed by subtracting $\frac{1}{2}\vartheta z'^{k_3-1} u^{k_4}$ from y'. The corrections belong to $u(y', z')$, have degree at least 2, are eigen-series for the characters of z', resp. y', and change the other terms only in higher total degree. The

composite of the substitutions converges to a coordinate change over $\Bbbk[[u]]$ preserving x', u and the ideal (y',z'): the new coordinates are eigen-series $\eta_1,\eta_2\in(y',z')\Bbbk[[y',z',u]]$ with

$$\eta_1\equiv y' \quad\text{and}\quad \eta_2\equiv z' \quad \text{modulo } (y',z',u)^2,$$

and

$$f'_*=x'^2+\eta_2\eta_1^2+\phi(\eta_2,u),\qquad \phi\in z'^3\Bbbk[[z',u]],\quad\text{and}\quad \phi(z',0)=0.$$

The cubic part $\eta_2\eta_1^2+\phi_3(u)\eta_2^3$ in η_1,η_2, with ϕ_3 the coefficient of z'^3 in ϕ, is a non-zero constant times the image of $y'^3+\alpha' y'z'^2+\gamma' z'^3$ under an invertible linear change of y',z' over $\Bbbk[[u]]$. Since $4\alpha'^3+27\gamma'^2\neq 0$, the cubic part has no multiple factor over $\Bbbk((u))$, so $\phi_3\neq 0$.

Write η_1,η_2 in the coordinates x',y',z',u of **Step 4**. Then $\eta_1,\eta_2\in(y',z')\Bbbk[[y',z',u]]$ are Γ-semi-invariant, and the coefficients of y',z' in η_1,η_2 form an invertible 2×2 matrix. By Lemma 2.6(4) for the tuples (f'_i,l'_i) of **Step 4** and the series η_1,η_2, after passing to a subsequence there exist Γ-semi-invariant polynomials $\eta_{i,1},\eta_{i,2}\in\mathbb{C}[x',y',z',u]$ with the characters of η_1,η_2 such that $(f'_*,l'_*,\eta_1,\eta_2)$ is a compatible generic limit of the tuples $(f'_i,l'_i,\eta_{i,1},\eta_{i,2})$ with coefficient field $\Bbbk$. Possibly replacing $\eta_{i,j}$ with the sum of the terms of $\eta_{i,j}$ divisible by y' or z' and not by x', a coordinate projection on jets which fixes η_j, we may assume that $\eta_{i,j}\in(y',z')\mathbb{C}[y',z',u]$. Since the coefficients of y',z' in η_1,η_2 form an invertible matrix, Definition 2.5 shows that for $i\gg 0$ so do those in $\eta_{i,1},\eta_{i,2}$, so $x',\eta_{i,1},\eta_{i,2},u$ are Γ-eigenparameters of $\mathbb{C}[[x',y',z',u]]$ and the ideals $(\eta_{i,1},\eta_{i,2})$ and (y',z') coincide. Let g_i be f'_i written in $x',\eta_{i,1},\eta_{i,2},u$, and g_* be f'_* written in x',η_1,η_2,u. For any integer d', the d'-jets of g_i and l'_i written in $x',\eta_{i,1},\eta_{i,2},u$, and of g_* and l'_* written in x',η_1,η_2,u, are the values of one regular map over $\mathbb{Q}$, defined on the open subset where that matrix is invertible, at the d'-jets of $(f'_i,l'_i,\eta_{i,1},\eta_{i,2})$ and of $(f'_*,l'_*,\eta_1,\eta_2)$, respectively, so the g_i and the l'_i written in $x',\eta_{i,1},\eta_{i,2},u$ have the compatible generic limit formed by g_* and l'_* written in x',η_1,η_2,u. By the formula for f'_* above, $g_*=x'^2+z'y'^2+\phi(z',u)$ in the variables renamed x',y',z',u, with $\phi_3\neq 0$. This proves the second sentence of (6).

Let $Z_i:=Z'_i$ and $S_i:=(u=0)\subset Z_i$, so that $Z_i=(g_i=0)$ in the coordinates $x',\eta_{i,1},\eta_{i,2},u$. The germ Z_i is a hypersurface, hence Gorenstein, and $Y_i\backslash(u=0)$ is étale over the non-singular locus $W_i\backslash\{o_i\}$ as in **Step 4**, so Z_i is smooth near 0 outside S_i. By (2) of **Step 4**,

$$S_i=(x'^2+y'^3+\alpha'_i y'z'^2+\gamma'_i z'^3=0)$$

with a binary cubic without triple root, so $\dim\mathrm{Sing}(S_i)\le 1$. A singular point of Z_i on the Cartier divisor S_i is a singular point of S_i, so $\dim\mathrm{Sing}(Z_i)\le 1$, and Z_i is normal by Serre's criterion.

Let F and P be as in **Step 4**. Since $\eta_{i,1},\eta_{i,2}\in(y',z')$, (F.193) holds with $\eta_{i,1},\eta_{i,2}$ in place of y',z'.

Step 6. In this step we prove (1)–(4).

Let w_1,w_2,w_3 be x',y',z' if $j_0=0$ and $x',\eta_{i,1},\eta_{i,2}$ if $j_0=1$. Then $f_i\circ\Phi=u^{6r+3v}g_i$ with g_i written in w_1,w_2,w_3,u. Since the coordinate change of **Step 5** has a unit Jacobian determinant, the Jacobian determinant of Φ with respect to x_i,y_i,z_i,s_i and w_1,w_2,w_3,u is $12u^{7r+\frac{5}{2}v+11}$ times a unit. Since $7r+\frac{5}{2}v+11=(6r+3v)+\kappa$ with $\kappa:=r-\frac{v}{2}+11$, and $\kappa>0$ since $r>\frac{3}{2}v$,

$$\Phi^*\Big(\frac{dx_i\wedge dy_i\wedge dz_i\wedge ds_i}{f_i}\Big)=12u^{\kappa}\cdot(\text{unit})\cdot\frac{dw_1\wedge dw_2\wedge dw_3\wedge du}{g_i}.$$

The residues of $\frac{dx_i\wedge dy_i\wedge dz_i\wedge ds_i}{f_i}$ and $\frac{dw_1\wedge dw_2\wedge dw_3\wedge du}{g_i}$ generate ω_{W_i} and ω_{Z_i}, respectively. Since $S_i=(x'^2+y'^3+\alpha'_i y'z'^2+\gamma'_i z'^3=0)$ by (2) of **Step 4** is integral, the binary cubic not being a square, $\mathrm{div}(u)=S_i$ on Z_i, and

$$K_{Z_i}=\Phi^*K_{W_i}+\kappa S_i,\qquad \kappa>0. \tag{F.194}$$

On Z_i, $h_i\circ\Phi=(\psi_i+x_i\xi_i)\circ\Phi=u^{mr+c}l'_i$ since $f_i\circ\Phi=u^{6r+3v}g_i$. The equation of S_i is monic of degree 2 in x' and S_i is integral, so by (3) of **Step 4**, $l'_i|_{u=0}$ is non-zero on S_i, and $\mathrm{mult}_{S_i}(h_i\circ\Phi)=N$ with

$$N:=mr+c\ge 0.$$

Thus $l_i := l_i'|_{Z_i} = u^{-N}(h_i\circ\Phi)$ is regular on Z_i and does not contain S_i. Since $\psi_i+x_i\xi_i$ has x_i-degree at most 1 and the coordinate change of **Step 5** fixes x' and u with $\eta_{i,1},\eta_{i,2}\in(y',z')\mathbb{C}[y',z',u]$, l_i' written in w_1,w_2,w_3,u is a representative of l_i of x'-degree at most 1. Over an algebraic closure of $\Bbbk((u))$, Φ_*, followed if $j_0=1$ by the coordinate change $(y',z')\to(\eta_1,\eta_2)$ of **Step 5**, induces an automorphism of the power series ring in three variables fixing 0 which, by **Steps 4** and **5**, carries f_* to u^{6r+3v} times the limit of the g_i and $h_*\equiv\psi_*+x\xi_*$ modulo f_* to u^N times the limit of the l_i' written in w_1,w_2,w_3,u, which proves the last sentence of (6). Since v is even, $\kappa\in\mathbb{Z}$, and

$$\beta=\frac{N}{m}-\kappa\in\frac{1}{m}\mathbb{Z}\qquad\text{and}\qquad\beta=\frac{c}{m}+\frac{v}{2}-11\ge-9,$$

since $c\ge0$ and $v\ge4$. Let $B_i':=\frac{1}{m}\operatorname{div}(l_i)+\beta S_i$. By (F.194) and $h_i\circ\Phi=u^Nl_i$,

$$K_{Z_i}+B_i'=\Phi^*\Big(K_{W_i}+\frac{1}{m}\operatorname{div}(h_i)\Big).$$

Since Z_i is normal, so is Y_i. By Lemma 2.12, (F.194) and B_i' are pullbacks of the corresponding divisors on Y_i, and the prime divisors over Z_i with center the origin, with their log discrepancies, inertia groups in Γ and orders of functions, are those over Y_i with center the origin. Below we read prime divisors over Z_i on Y_i, whose function field is $\mathbb{C}(W_i)(u)$ by **Step 4**. Let P be S_i, the completion of the prime divisor $(u=0)$ of Y_i, or a prime divisor over Z_i with center the origin, F the prime divisor over W_i given by the restriction of P to $\mathbb{C}(W_i)$, and ρ the ramification index. Then (F.194) and $h_i\circ\Phi=u^Nl_i$ imply

$$\rho a(F,W_i,0)=a(P,Z_i,0)+\kappa\operatorname{mult}_P u,\qquad\rho\operatorname{mult}_F h_i=\operatorname{mult}_P l_i+N\operatorname{mult}_P u,$$

$$a(P,Z_i,B_i')=\rho a\Big(F,W_i,\frac{1}{m}\operatorname{div}(h_i)\Big).\tag{F.195}$$

For $P=S_i$, F has center o_i, and (F.195) implies $1-\beta=a(S_i,Z_i,B_i')>0$ since $(W_i,\frac{1}{m}\operatorname{div}(h_i))$ is klt. Thus $\beta\in\frac{1}{m}\mathbb{Z}\cap[-9,1)\subset[-10,1)$.

Now let P have center the origin of Z_i. Since x_i,y_i,z_i,s_i vanish at the origin, F has center o_i. By (F.190) and $r\ge6$, each of $x_i\circ\Phi$, $y_i\circ\Phi$, $z_i\circ\Phi$ is u^{12} times a regular function, and $s_i\circ\Phi=u^{12}$. Since x_i,y_i,z_i,s_i generate $\mathfrak{m}_{o_i}$,

$$\rho\operatorname{mult}_F\mathfrak{m}_{o_i}=12\operatorname{mult}_P u.$$

In the tower

$$\mathbb{C}(W_i)^G\subset\mathbb{C}(W_i)\subset\mathbb{C}(W_i)(u),$$

ramification indices multiply, and in characteristic zero e_P and e_F are the ramification indices of P and F over $\mathbb{C}(W_i)^G$. Thus $e_P=\rho e_F$. This proves (2).

Since B_i' need not be effective, we use [Liu26b, Definition 2.4]: the quotient value in (1) is the infimum of $a(P,Z_i,B_i')/e_P$ over the prime divisors P over Z_i with center the origin. By (F.195) and $e_P=\rho e_F$,

$$\frac{a(P,Z_i,B_i')}{e_P}=\frac{a(F,W_i,\frac{1}{m}\operatorname{div}(h_i))}{e_F}\ge b_i,$$

with equality if and only if F is a computing lift of b_i. By **Step 4**, for any computing lift F of b_i, any P restricting to F has center the origin of Z_i. Thus

$$\operatorname{mld}\big((Z_i\ni0,B_i')/\Gamma\big)=b_i,$$

any computing lift of b_i has center the origin of Z_i, and any computing lift of the quotient value in (1) restricts to a computing lift of b_i. This proves (1).

Let P be a computing lift of the quotient value in (1), and F the restriction of P. By Theorem 5.1(3), $\operatorname{mult}_F\mathfrak{m}_{o_i}/e_F\le b_i/\eta_m\le3/\eta_m$, so (2) implies

$$\frac{\operatorname{mult}_P u}{e_P}=\frac{\rho\operatorname{mult}_F\mathfrak{m}_{o_i}}{12\rho e_F}\le\frac{3}{12\eta_m}.$$

This proves (3). Finally, (F.193) and the last paragraph of **Step 5** prove (4) for the coordinates x', y', z' and, if $j_0 = 1$, for the coordinates $\eta_{i,1}, \eta_{i,2}$ after the coordinate change.

Step 7. In this step we prove (5).

Let P be a computing lift of the quotient value in (1), and F the restriction of P, a computing lift of b_i by (1). Since $a(F, W_i, 0) = e_F b_i + \frac{1}{m} \operatorname{mult}_F h_i$, (2), (3) and (F.187) imply

$$\frac{a(P, Z_i, 0)}{e_P} = \frac{a(F, W_i, 0)}{e_F} - \kappa \frac{\operatorname{mult}_P u}{e_P} \geq b_i + \frac{\operatorname{mult}_F h_i}{m e_F} - \frac{\kappa}{4\eta_m} \geq \frac{\delta}{1 - \tau_i} - \frac{\kappa}{4\eta_m},$$

which tends to $+\infty$.

For $j = 1, \dots, 4$, with $w_4 := u$, differentiating $f_i \circ \Phi = u^{6r+3v} g_i$ implies

$$u^{6r+3v} \frac{\partial g_i}{\partial w_j} \equiv \sum_{t \in \{x_i, y_i, z_i, s_i\}} \Big(\frac{\partial f_i}{\partial t} \circ \Phi \Big) \frac{\partial (t \circ \Phi)}{\partial w_j} \quad \text{modulo } (g_i),$$

and the $\partial(t \circ \Phi)/\partial w_j$ are regular. Thus

$$u^{6r+3v} \operatorname{Jac}_{Z_i} \subset (\Phi^* \operatorname{Jac}_{W_i}) \mathcal{O}_{Z_i},$$

and for any prime divisor P over Z_i with restriction F and ramification index ρ,

$$\rho \operatorname{mult}_F \operatorname{Jac}_{W_i} \leq (6r + 3v) \operatorname{mult}_P u + \operatorname{mult}_P \operatorname{Jac}_{Z_i}. \tag{F.196}$$

Suppose that there exist an integer N_2 and infinitely many i with a computing lift P of the quotient value in (1) satisfying $\operatorname{mult}_P \operatorname{Jac}_{Z_i} \leq N_2$. By (3) and $e_P \leq |\Gamma| \leq 12M$, $\operatorname{mult}_P u \leq 3M/\eta_m$. The restriction F of P is a computing lift of b_i with

$$\operatorname{mult}_F \operatorname{Jac}_{W_i} \leq \frac{3M(6r + 3v)}{\eta_m} + N_2$$

by (F.196), which happens for only finitely many i by **Step 3**, a contradiction. Thus $\min_P \operatorname{mult}_P \operatorname{Jac}_{Z_i} \to +\infty$, which proves (5). □

Proof of Theorem 17.14. Suppose that $(W_i \ni o_i, G_i, h_i, \tau_i)$ is a radial failure sequence with constants (m, a, δ) satisfying Set-up 17.1, whose prepared limit is of type cE and has generic transverse type D_4 along C. We divide the proof into eight steps.

Step 1. In this step we fix the charts Z_i and list the properties of them used below. By Proposition 17.13, after passing to an infinite subfamily, there exist $j_0 \in \{0, 1\}$, real numbers $\kappa > 0$ and $\beta \in \frac{1}{m}\mathbb{Z} \cap [-10, 1)$, groups Γ_i of order $12|G_i| \leq 12M$, charts $(Z_i \ni 0)$ with eigenparameters (x, y, z, u) and equations f_{Z_i}, and marks l_i with $h_i = u^N l_i$. Let Ω be an algebraic closure of $\Bbbk((u))$. Possibly passing to a further infinite subfamily, $\Gamma_i = \Gamma$ acts on (x, y, z, u) by fixed characters $\chi_x, \chi_y, \chi_z, \chi_u$, and f_{Z_i}, l_i have fixed characters. Let $S_i := (u = 0) \subset Z_i$ and

$$B_{Z_i} := \frac{1}{m} \operatorname{div}(l_i) + \beta S_i.$$

Let P be a prime divisor over Z_i with center the origin. By Proposition 17.13(2), the restriction F of P to W_i is a prime divisor over W_i with center o_i. Let ρ be the ramification index of P over F. The identities of Proposition 17.13(2) and $\beta = N/m - \kappa$ imply

$$a(P, Z_i, B_{Z_i}) = a(P, Z_i, 0) - \frac{1}{m} \operatorname{mult}_P l_i - \beta \operatorname{mult}_P u = \rho a\Big(F, W_i, \frac{1}{m} \operatorname{div}(h_i)\Big), \qquad e_P = \rho e_F,$$

so

$$\frac{a(P, Z_i, B_{Z_i})}{e_P} = \frac{a(F, W_i, \frac{1}{m} \operatorname{div}(h_i))}{e_F}.$$

With $c_u := 3/(12\eta_m) = 1/(4\eta_m)$, where η_m is the constant of Theorem 5.1(2), Proposition 17.13 and the last equality imply (R0)–(R4) below, and we prove (R5) after the list.

(R0) $Z_i = (f_{Z_i} = 0) \subset \operatorname{Spec} \mathbb{C}[[x, y, z, u]]$ is normal and Gorenstein, and smooth outside S_i near 0. If $j_0 = 0$, then Z_i is moreover klt with an isolated singularity at 0. The group Γ is abelian and acts faithfully and diagonally on (x, y, z, u) (diagonally since (x, y, z, u) are eigenparameters, and faithfully since an element acting trivially fixes u, hence is in G_i and acts trivially on the coordinates of W_i, which are functions of (x, y, z, u)), and f_{Z_i}, l_i are Γ-semi-invariant. Moreover $\beta \in \frac{1}{m}\mathbb{Z}$ and $\beta < 1$.

(R1) $\operatorname{mld}((Z_i \ni 0, B_{Z_i})/\Gamma) = b_i$. A prime divisor P over Z_i with center the origin is a computing lift of $\operatorname{mld}((Z_i \ni 0, B_{Z_i})/\Gamma)$ if and only if the restriction F of P to W_i is a computing lift of b_i, and any computing lift of b_i is such a restriction.

(R2) For P and F as in (R1),

$$e_P = \rho e_F \quad \text{and} \quad \rho a(F, W_i, 0) = a(P, Z_i, 0) + \kappa \operatorname{mult}_P u.$$

(R3) Any computing lift P of $\operatorname{mld}((Z_i \ni 0, B_{Z_i})/\Gamma)$ satisfies $\operatorname{mult}_P u \le c_u e_P$.

(R4) For any integer J, after passing to an infinite subfamily,

$$\min\{\operatorname{mult}_P x, \operatorname{mult}_P y, \operatorname{mult}_P z\} > J \quad \text{for any computing lift } P \text{ of } \operatorname{mld}((Z_i \ni 0, B_{Z_i})/\Gamma).$$

(R5) The (f_{Z_i}, l_i) have a compatible generic limit $(f_{Z,*}, l_*)$ with

$$f_{Z,*} = x^2 + g(y, z, u), \qquad g \in (y, z)^3 \Bbbk[[y, z, u]],$$

and l_* of x-degree at most 1. Over Ω, the pair

$$\Big(\operatorname{Spec} \Omega[[x, y, z]]/(f_{Z,*}), \frac{1}{m} \operatorname{div}(l_*) \Big)$$

is lc and not klt at 0.

The first sentence of (R5) is Proposition 17.13(6). Since C is the non-klt locus of the lc pair $(W_*, \frac{1}{m}\operatorname{div}(h_*))$, localizing at the generic point of C, completing, and extending the residue field to an algebraic closure of $\Bbbk((s))$, we obtain a pair which is lc and not klt at the closed point (pull back a log resolution along the regular morphism from the completion of $\mathcal{O}_{W_*,C}$, W_* being excellent, and then extend the coefficient field). Over $\Omega \supset \Bbbk((s))$, $s = u^{12}$, the automorphism of Proposition 17.13(6) carries f_* to a unit times a power of u times $f_{Z,*}$ and h_* to $u^N l_*$ modulo f_*. This proves the last sentence of (R5). Let $g_0(y, z)$ be the cubic part of $g(y, z, 0)$. If $j_0 = 0$, then g_0 is reduced by Proposition 17.13(6).

Step 2. In this step we show that point charts keep (R0)–(R4). Let c_x, c_y, c_z be positive integers, $r \in \{1, 2\}$, d_1, d_2 non-negative integers, and ψ the substitution

$$x = u'^{c_x} x', \quad y = u'^{c_y} y', \quad z = u'^{c_z} z', \quad \text{and} \quad u = u'^{r}.$$

Assume the following.

(a) $\psi^* f_{Z,*} = u'^{d_1} f'_*$ and $\psi^* l_* = u'^{d_2} l'_*$ with $f'_*, l'_* \in \Bbbk[[x', y', z', u']]$.

(b) The restriction of f'_* to $(u' = 0)$ satisfies

$$f'_*(x', y', z', 0) = c_1 x'^2 + c_2 z' y'^2 + c_3 z'^3, \qquad c_1 c_2 c_3 \neq 0.$$

Let

$$\beta' := r\beta + \frac{d_2}{m} - c_K, \qquad c_K := c_x + c_y + c_z + r - 1 - d_1,$$

and let

$$\Gamma' := \{(\gamma, \mu) \in \Gamma \times \mathbb{C}^* \mid \mu^r = \chi_u(\gamma)\}$$

act by $(\gamma, \mu) \cdot u' = \mu u'$ and $(\gamma, \mu) \cdot x' = \chi_x(\gamma)\mu^{-c_x} x'$, and similarly on y', z'. We show that, after passing to an infinite subfamily, the following hold.

(1) $\psi^* f_{Z_i} = u'^{d_1} f_{Z'_i}$ and $\psi^* l_i = u'^{d_2} l'_i$ with $f_{Z'_i}, l'_i \in \mathbb{C}[[x', y', z', u']]$.

(2) (f'_*, l'_*) is the compatible generic limit of the $(f_{Z'_i}, l'_i)$.

(3) (R0)–(R4) hold for $Z_i' := (f_{Z_i'} = 0)$, Γ', l_i', $S_i' := (u' = 0) \subset Z_i'$, and (x', y', z', u'), with β' in place of β and

$$\kappa' := c_K + r\kappa, \qquad c_u' := \frac{c_u}{r}.$$

(4) Z_i' is klt with an isolated singularity at 0.

The substitution ψ sends $x^{k_1}y^{k_2}z^{k_3}u^{k_4}$ to $u'^{c_xk_1+c_yk_2+c_zk_3+rk_4}x'^{k_1}y'^{k_2}z'^{k_3}$. Since $c_x, c_y, c_z, r \geq 1$, only finitely many monomials have exponent at most $\max\{d_1, d_2\}$. The divisibility of $\psi^* f_{Z_i}$ by u'^{d_1} (resp. of $\psi^* l_i$ by u'^{d_2}) is the vanishing of the coefficients of those with exponent less than d_1 (resp. d_2), and $f_{Z_i'}(x', y', z', 0)$ is determined by the coefficients of those with exponent d_1. By [Liu26b, Construction-Proposition 4.1], a coefficient that vanishes for $(f_{Z,*}, l_*)$ vanishes for the members of an infinite subfamily, and a coefficient that does not vanish is non-zero for all but finitely many of them. Thus, after passing to an infinite subfamily, the divisibility holds and

$$f_{Z_i'}(x', y', z', 0) = c_{i,1}x'^2 + c_{i,2}z'y'^2 + c_{i,3}z'^3, \qquad c_{i,1}c_{i,2}c_{i,3} \neq 0.$$

Since ψ is fixed, (f_*', l_*') is the compatible generic limit of the $(f_{Z_i'}, l_i')$. Here Z_i is the completion at the origin of the normal algebraic germ Z_i^{alg} of Proposition 17.13, on which x, y, z, u are regular functions, which stays true below since the charts are monomial and the coordinate changes polynomial. Thus Z_i' is the completion at the origin of ${Z_i'}^{\mathrm{alg}}$, the germ at the origin of Spec of the quotient by its u'-power torsion of

$$\mathcal{O}_{Z_i^{\mathrm{alg}},0}[u', x', y', z']/(u'^r - u,\ x - u'^{c_x}x',\ y - u'^{c_y}y',\ z - u'^{c_z}z'),$$

whose function field is that of Z_i^{alg} with u' adjoined, of degree r with Galois group the kernel $\{(1, \mu) \mid \mu^r = 1\}$ of the surjection $\Gamma' \to \Gamma$. Below, the étaleness of ψ off S_i', restrictions of prime divisors, ramification indices, and inertia groups are read on Z_i^{alg} and ${Z_i'}^{\mathrm{alg}}$, prime divisors with center the origin being read there by Lemma 2.12(3).

(R0). If (γ, μ) acts trivially, then $\mu = 1$, so each of $\chi_x, \chi_y, \chi_z, \chi_u$ takes the value 1 at γ, and $\gamma = 1$. For each $\gamma \in \Gamma$ there exist exactly r values of μ with $\mu^r = \chi_u(\gamma)$, so $|\Gamma'| = r|\Gamma|$. Thus Γ' is abelian and acts faithfully and diagonally. Since ψ is equivariant for $\Gamma' \to \Gamma$, $f_{Z_i'}$ and l_i' are semi-invariant. The surface S_i' is normal with a D_4 singularity at 0 and smooth elsewhere, because $z'(c_{i,2}y'^2 + c_{i,3}z'^2)$ is a reduced cubic. The threefold Z_i' is smooth along $S_i' \backslash \{0\}$, as S_i' is a Cartier divisor on Z_i'. Near 0 the threefold Z_i' is smooth off S_i', as ψ is étale there onto points of Z_i with $u \neq 0$, which are smooth by (R0). Thus Z_i' is a hypersurface with an isolated singularity at 0, hence normal and Gorenstein, and (Z_i', S_i') is plt near 0 by inversion of adjunction. In particular Z_i' is klt. Moreover $\beta' \in \frac{1}{m}\mathbb{Z}$, and $\beta' < 1$ by (R1) below.

Identities. We have $\psi^* f_{Z_i} = u'^{d_1} f_{Z_i'}$ and

$$\psi^*(dx \wedge dy \wedge dz \wedge du) = ru'^{c_x+c_y+c_z+r-1}dx' \wedge dy' \wedge dz' \wedge du',$$

so the residue forms satisfy $K_{Z_i'} = \psi^* K_{Z_i} + c_K S_i'$. Since $\psi^* S_i = rS_i'$ and

$$\psi^* \operatorname{div}(l_i) = \operatorname{div}(l_i') + d_2 S_i',$$

we have

$$K_{Z_i'} + B_{Z_i'} = \psi^*(K_{Z_i} + B_{Z_i}), \qquad B_{Z_i'} := \frac{1}{m}\operatorname{div}(l_i') + \beta' S_i'.$$

Let P' be a prime divisor over Z_i' with center the origin, P its restriction to the function field of Z_i, which has center the origin since $\operatorname{mult}_{P'} x = c_x \operatorname{mult}_{P'} u' + \operatorname{mult}_{P'} x' > 0$ and similarly for y, z, u, and ρ' the ramification index of P' over P. For $\mathbb{R}$-divisors Δ, Δ' with $K_{Z_i'} + \Delta' = \psi^*(K_{Z_i} + \Delta)$, log discrepancies multiply by the ramification index:

$$a(P', Z_i', \Delta') = \rho' a(P, Z_i, \Delta). \tag{F.197}$$

Applying (F.197) to $(\Delta,\Delta')=(B_{Z_i},B_{Z_i'})$ and to $(\Delta,\Delta')=(0,-c_KS_i')$, and using $\psi^*u=u'^r$, we obtain

$$\rho' a(P,Z_i,B_{Z_i})=a(P',Z_i',B_{Z_i'}),\quad \rho' a(P,Z_i,0)=a(P',Z_i',0)+c_K\operatorname{mult}_{P'}u',$$
$$\text{and}\quad \rho'\operatorname{mult}_P u=r\operatorname{mult}_{P'}u'.$$

Since $Z_i'/\Gamma' \dashrightarrow Z_i/\Gamma$ is birational and $e_P, e_{P'}$ are the ramification indices over Z_i/Γ and Z_i'/Γ', ramification indices multiply and $e_{P'}=\rho' e_P$, so

$$\frac{a(P',Z_i',B_{Z_i'})}{e_{P'}}=\frac{a(P,Z_i,B_{Z_i})}{e_P}. \tag{F.198}$$

(R2). The restriction of P' to W_i is F, with ramification index $\rho\rho'$. Multiplying (R2) for P by ρ', we obtain

$$e_{P'}=\rho\rho' e_F\quad\text{and}\quad \rho\rho' a(F,W_i,0)=a(P',Z_i',0)+\kappa'\operatorname{mult}_{P'}u'.$$

(R1). By (R1) for Z_i and (F.198), any prime divisor P' over Z_i' with center the origin satisfies $a(P',Z_i',B_{Z_i'})/e_{P'}\ge b_i$. Let F be a computing lift of b_i, and P a prime divisor over Z_i with center the origin restricting to F. By (R1), P is a computing lift, so $\operatorname{mult}_P u\le c_u e_P\le c_u|\Gamma|$ by (R3) and $e_P\le|\Gamma|$. Let $J_0:=c_u|\Gamma|\max\{c_x,c_y,c_z\}$. By (R4), after passing to an infinite subfamily, $\operatorname{mult}_P x>J_0$, and similarly for y,z. Such prime divisors P' over Z_i' restricting to P exist, since the function field of Z_i' is a finite extension of that of Z_i, and for any of them

$$\operatorname{mult}_{P'}x'=\rho'\Big(\operatorname{mult}_P x-\frac{c_x}{r}\operatorname{mult}_P u\Big)>0,\qquad \operatorname{mult}_{P'}u'=\frac{\rho'}{r}\operatorname{mult}_P u>0,$$

and similarly for y',z'. Thus P' has center the origin and $a(P',Z_i',B_{Z_i'})/e_{P'}=b_i$. This proves (R1) for Z_i'. (F.197) also holds for $P'=S_i'$, whose restriction to the function field of Z_i has center the origin as $c_x,c_y,c_z\ge1$, so (R1) for Z_i gives $1-\beta'\ge a(S_i',Z_i',B_{Z_i'})>0$, that is, $\beta'<1$.

(R3) and (R4). For a computing lift P', the restriction P is a computing lift, so

$$\frac{\operatorname{mult}_{P'}u'}{e_{P'}}=\frac{\operatorname{mult}_P u}{re_P}\le\frac{c_u}{r}.$$

Moreover $\operatorname{mult}_{P'}x'\ge\operatorname{mult}_P x-c_x\operatorname{mult}_P u\ge\operatorname{mult}_P x-J_0$, and similarly for y',z', so (R4) for Z_i with $J+J_0$ in place of J implies (R4) for Z_i'.

Step 3. In this step we show the following. Let $w:=(3,2,2)$ be the weights of (x,y,z). Let $f=x^2+g(y,z)\in\Omega[[x,y,z]]$ with $g\in(y,z)^3$ whose cubic part g_0 is reduced, and let $l\in\Omega[[x,y,z]]$ be non-zero of x-degree at most 1. Assume that $(S,\frac{1}{m}\operatorname{div}(l|_S))$ is lc and not klt at 0, where $S:=\operatorname{Spec}\Omega[[x,y,z]]/(f)$. Let $S^0:=(x^2+g_0=0)\subset\mathbb{A}^3_\Omega$. After a linear change of (y,z) we have $g_0=y^2z+z^3$. Let $\Pi\subset\mathrm{SL}_2(\Omega)$ be the binary dihedral group of order 8 and $\pi=(F_x,F_y,F_z):\mathbb{A}^2_\Omega\to S^0$ the quotient morphism by Π [Liu26b, Lemmas D.3 and D.4]. Then:

(1) $\operatorname{wt}_w(\operatorname{in}_w l)=m$.
(2) $\pi^*\operatorname{in}_w l$ is a binary form of degree $2m$, any line through 0 has multiplicity at most m in $\pi^*\operatorname{in}_w l$, and (S^0,B_{S^0}) is lc, where

$$B_{S^0}:=\frac{1}{m}\operatorname{div}(\operatorname{in}_w l|_{S^0}).$$

By Lemma F.12 over Ω, with the trivial group and f regarded as an element of $\Omega[[x,y,z,s]]$ independent of s, the automorphism Φ there fixes s. Since $A(0)$ is the identity, its reduction modulo s is an automorphism Φ_0 of $\Omega[[x,y,z]]$ with $\Phi_0(x)=x$, $\Phi_0(y)-y\in(y,z)^2$, $\Phi_0(z)-z\in(y,z)^2$, and $\Phi_0(f)=x^2+g_0$. Elements of $(y,z)^2$ have w-weight at least $4>2$, so Φ_0 preserves the w-filtration and $\operatorname{in}_w\Phi_0(l)=\operatorname{in}_w l$. Replacing x^2 with $-g_0$ in $\Phi_0(l)$ changes $\Phi_0(l)$ by a multiple of $\Phi_0(f)=x^2+g_0$ and keeps w-weights, as $\operatorname{wt}_w(x^2)=\operatorname{wt}_w(g_0)=6$. It keeps $\operatorname{in}_w\Phi_0(l)=\operatorname{in}_w l$, which has x-degree at most 1, and makes $\Phi_0(l)$ of x-degree at most 1. Possibly replacing (f,l)

with $(x^2+g_0,\Phi_0(l))$, we may assume that $f=x^2+g_0$, so that S is the completion of S^0 at 0. Let $k_0:=\mathrm{wt}_w(\mathrm{in}_w\, l)$.

By [Liu26b, Lemmas D.3 and D.4] and flat base change, $F_x,F_y,F_z\in\Omega[\xi_1,\xi_2]$ are Π-invariant forms of degrees $6,4,4$, π is étale over $S^0\backslash\{0\}$, and no line of Ω^2 is Π-invariant. Since π is étale in codimension 1, $\pi^*K_{S^0}=K_{\mathbb{A}^2}$. Hence, for a $\mathbb{Q}$-divisor $\Delta\ge 0$ on S^0, (S^0,Δ) is lc (resp. klt) near 0 if and only if $(\mathbb{A}^2,\pi^*\Delta)$ is lc (resp. klt) near 0. Let $\widehat{\mathbb{A}}^2:=\operatorname{Spec}\Omega[[\xi_1,\xi_2]]$, and let $\widehat{\pi}:\widehat{\mathbb{A}}^2\to S$ be the morphism induced by π, which is the quotient by Π and étale in codimension 1, so, for a $\mathbb{Q}$-divisor $\Delta\ge 0$ on S, (S,Δ) is lc (resp. klt) if and only if $(\widehat{\mathbb{A}}^2,\widehat{\pi}^*\Delta)$ is lc (resp. klt). In particular, with $\widetilde{l}:=\widehat{\pi}^*l$, the pair $(\widehat{\mathbb{A}}^2,\frac{1}{m}\operatorname{div}(\widetilde{l}))$ is lc and not klt near 0.

The ring $\Omega[x,y,z]/(x^2+g_0)$ is free over $\Omega[y,z]$ with basis $1,x$, so the non-zero form $\mathrm{in}_w\, l$, of x-degree at most 1, does not vanish on S^0. Since $(\deg F_x,\deg F_y,\deg F_z)=(6,4,4)=2w$, π^* doubles w-weights, so $\widetilde{l}$ has lowest-degree part $\pi^*\,\mathrm{in}_w\, l\neq 0$, of degree $2k_0$. For a line $\Lambda\subset\Omega^2$ through 0, let q_Λ be the multiplicity of Λ in $\pi^*\,\mathrm{in}_w\, l$, so $\sum_\Lambda q_\Lambda=2k_0$. The form $\pi^*\,\mathrm{in}_w\, l$ is Π-invariant, so $q_{\gamma\Lambda}=q_\Lambda$ for any $\gamma\in\Pi$. As the Π-orbit of Λ has at least two lines,

$$q_\Lambda\le k_0\quad\text{for any }\Lambda.$$

Let $\alpha:Y\to\mathbb{A}^2$ be the blow-up of 0, $E_0\subset Y$ the exceptional curve, and

$$\widehat{\alpha}:\widehat{Y}:=Y\times_{\mathbb{A}^2}\widehat{\mathbb{A}}^2\to\widehat{\mathbb{A}}^2$$

the blow-up of 0, with exceptional curve E_0. Then

$$a\Big(E_0,\widehat{\mathbb{A}}^2,\frac{1}{m}\operatorname{div}(\widetilde{l})\Big)=2-\frac{2k_0}{m}\ge 0,$$

so $k_0\le m$. Suppose that $k_0<m$. The crepant pullback of $(\widehat{\mathbb{A}}^2,\frac{1}{m}\operatorname{div}(\widetilde{l}))$ is

$$\Big(\widehat{Y},\Big(\frac{2k_0}{m}-1\Big)E_0+\frac{1}{m}L_{\widehat{Y}}\Big),\qquad L_{\widehat{Y}}:=\widehat{\alpha}_*^{-1}\operatorname{div}(\widetilde{l}).$$

The divisor $L_{\widehat{Y}}$ intersects $E_0\cong\mathbb{P}^1$ at the points $[\Lambda]$ with local intersection numbers $q_\Lambda\le k_0<m$. By inversion of adjunction, $(\widehat{Y},E_0+\frac{1}{m}L_{\widehat{Y}})$ is plt near E_0. Since $2k_0/m-1<1$, the crepant pullback is sub-klt near E_0, so $(\widehat{\mathbb{A}}^2,\frac{1}{m}\operatorname{div}(\widetilde{l}))$ is klt near 0, a contradiction. Hence $k_0=m$, which proves (1) and the first two parts of (2). On Y, the crepant pullback of $(\mathbb{A}^2,\pi^*B_{S^0})$ is $(Y,E_0+\sum_\Lambda\frac{q_\Lambda}{m}\Lambda_Y)$ with $\Lambda_Y:=\alpha_*^{-1}\Lambda$, the divisor $E_0+\sum_\Lambda\Lambda_Y$ has simple normal crossings, and all coefficients are ≤ 1. Thus $(\mathbb{A}^2,\pi^*B_{S^0})$ is lc, and so is (S^0,B_{S^0}).

Step 4. In this step we treat the case $j_0=1$. By Proposition 17.13(6), $f_{Z,*}=x^2+zy^2+\phi(z,u)$ with $\phi=\sum_{j\ge3}\phi_j(u)z^j$, $\phi(z,0)=0$, and $\phi_3\neq 0$. Write $\phi_3=u^\nu v_3(u)$ with $v_3(0)\neq 0$. Over Ω, the cubic part $zy^2+\phi_3(u)z^3$ of $f_{Z,*}$ is reduced, so by **Step 3** for the pair in (R5), any monomial $x^{k_1}y^{k_2}z^{k_3}u^{k_4}$ of l_* satisfies $3k_1+2k_2+2k_3\ge m$. Let $N_0>\nu$ be an integer, and apply **Step 2** with

$$(c_x,c_y,c_z)=(3N_0+\nu,2N_0+\nu,2N_0),\quad r=2,\quad d_1=6N_0+2\nu,\quad\text{and}\quad d_2=mN_0.$$

The exponent of $x^{k_1}y^{k_2}z^{k_3}u^{k_4}$ is $N_0(3k_1+2k_2+2k_3)+\nu(k_1+k_2)+2k_4$. Thus ψ^*l_* is divisible by u'^{mN_0}, and

$$u'^{-d_1}\psi^*f_{Z,*}=x'^2+z'y'^2+v_3(u'^2)z'^3+\sum_{j\ge4}\phi_j(u'^2)u'^{2(j-3)N_0-2\nu}z'^j,\tag{F.199}$$

whose restriction to $(u'=0)$ is $x'^2+z'y'^2+v_3(0)z'^3$. Moreover

$$c_K=N_0+1\quad\text{and}\quad\beta'=2\beta+N_0-(N_0+1)=2\beta-1<1.$$

Possibly replacing $(Z_i,\Gamma,l_i,\beta,\kappa,c_u)$ with $(Z_i',\Gamma',l_i',2\beta-1,N_0+1+2\kappa,c_u/2)$ and renaming u' as u, we may assume that (R0)–(R4) hold, that Z_i is klt with an isolated singularity at 0, and, by (F.199), that $f_{Z,*}=x^2+g(y,z,u)$ with $g\in(y,z)^3\Bbbk[[y,z,u]]$ and $g_0=zy^2+v_3(0)z^3$ reduced.

Over $\Omega \ni u'$, ψ induces an automorphism of $\Omega[[x,y,z]]$ fixing 0, so (R5) still holds, and l_* still has x-degree at most 1.

Step 5. In this step we bring the limit equation to a constant D_4 polynomial. By **Steps 1 and 4**, Z_i is klt with an isolated singularity at 0, and $f_{Z,*} = x^2 + g(y,z,u)$ with $g \in (y,z)^3\Bbbk[[y,z,u]]$ and g_0 reduced. By Lemma F.12 with $s = u$, followed by the Γ-equivariant $\Bbbk$-linear change of (y,z) given there, there exists a Γ-equivariant automorphism Φ of $\Bbbk[[x,y,z,u]]$ with $\Phi(x) = x$, $\Phi(u) = u$, and $(\Phi(y),\Phi(z)) \equiv (y,z)A$ modulo $(y,z)^2$ for a matrix $A \in GL_2(\Bbbk[[u]])$, such that $\Phi(f_{Z,*}) = f_0$ with

$$f_0 \in \{f_I := x^2 + zy^2 + z^3,\ f_{II} := x^2 + y^3 + z^3\}.$$

By Lemma 2.6(4) with $(\Phi^{-1}(y),\Phi^{-1}(z))$ as the adjoined power series, which are in (y,z) and have independent linear parts, as $(\Phi^{-1}(y),\Phi^{-1}(z)) \equiv (y,z)A^{-1}$ modulo $(y,z)^2$, on an infinite subfamily there exist Γ-semi-invariant polynomials y_i, z_i with the characters of y,z such that $(f_{Z,*}, l_*, \Phi^{-1}(y), \Phi^{-1}(z))$ is the compatible generic limit of the (f_{Z_i}, l_i, y_i, z_i). Possibly replacing y_i, z_i with their parts in (y,z), which keeps the compatible generic limit since, for each d, the coefficients of the monomials of degree less than d outside (y,z) vanish for $\Phi^{-1}(y), \Phi^{-1}(z)$, hence for y_i, z_i for all but finitely many i by Lemma 2.6(4), so the d-jets do not change for those i, we may assume that $y_i, z_i \in (y,z)$. As the linear parts of $\Phi^{-1}(y), \Phi^{-1}(z)$ in (y,z) are independent, so are those of y_i, z_i for all but finitely many i, and (x, y_i, z_i, u) are new eigenparameters of Z_i. The d-jets of the new equations and marks are polynomials in the d-jets of f_{Z_i}, l_i, y_i, z_i and in $\det_i^{-1}$, where $\det_i$ is the determinant of the linear parts of y_i, z_i in y,z, so, as $f_{Z,*} = \Phi(f_{Z,*})(x, \Phi^{-1}(y), \Phi^{-1}(z), u)$ and similarly for l_*, their compatible generic limit is exactly $(\Phi(f_{Z,*}), \Phi(l_*)) = (f_0, \Phi(l_*))$. Possibly replacing the coordinates (x,y,z,u) of Z_i with (x, y_i, z_i, u), we may assume that $(f_{Z,*}, l_*) = (f_0, \Phi(l_*))$ is the compatible generic limit of the (f_{Z_i}, l_i). In the new coordinates (R0)–(R3) still hold, so does (R4), since the new y,z are in the ideal (y,z), and so does the last sentence of (R5), since Φ induces an automorphism of $\Omega[[x,y,z]]$ fixing 0. Replacing x^2 with $x^2 - f_0$ in l_*, we write $l_* = q_* f_0 + r_*$ with r_* of x-degree at most 1. If $q_* \neq 0$, then by Lemma 2.6(4) with q_* adjoined there exist Γ-semi-invariant polynomials q_i with the character of q_*, and replacing l_i with $l_i - q_i f_{Z_i}$, which keeps $\operatorname{div}(l_i)$ on Z_i and hence B_{Z_i}, replaces l_* with r_*. Thus we may assume that l_* has x-degree at most 1. By (R5) and **Step 3** over Ω, we obtain

$$f_{Z,*} = f_0, \qquad l_* = \operatorname{in}_w l_* + (\text{terms of } w\text{-weight} > m), \tag{F.200}$$

where $\operatorname{in}_w l_* \in \Bbbk[[u]][x,y,z]$ is non-zero, of x-degree at most 1, of w-weight m, and (S^0, B_{S^0}) is lc over Ω, with S^0 and B_{S^0} as in **Step 3** for f_0 and l_*.

Step 6. In this step we treat the case where B_{S^0} has a component D with coefficient one. Let π and Π be as in **Step 3** for S^0. By **Step 3**(2), $\pi^* \operatorname{in}_w l_*$ is a binary form, so $\pi^{-1}(D)$ is a union of lines through 0. Since π is étale over $S^0 \backslash \{0\}$ and D has coefficient one in B_{S^0}, the lines in $\pi^{-1}(D)$ have multiplicity m in $\pi^* \operatorname{in}_w l_*$. By **Step 3**(2), $\pi^* \operatorname{in}_w l_*$ has degree $2m$, and the Π-orbit of such a line has at least two lines. Thus

$$\pi^* \operatorname{in}_w l_* = c_0 Q^m, \qquad c_0 \in \Omega^*,$$

where Q is the quadratic form vanishing on the lines Λ_1, Λ_2 forming one Π-orbit. Hence Q is Π-semi-invariant. The Π-invariant forms are generated by F_x, F_y, F_z, of degrees $6,4,4$, so Q is not invariant. As $\Pi/[\Pi,\Pi] \cong (\mathbb{Z}/2)^2$, the character of Q has order 2. As $c_0 Q^m$ is invariant, m is even. As Q^2 is invariant of degree 4, $Q^2 = \pi^* \ell$ for a linear form ℓ in y,z. Thus $\operatorname{in}_w l_* - c_0 \ell^{m/2}$ vanishes on S^0 and has x-degree at most 1, so $\operatorname{in}_w l_* = c_0 \ell^{m/2}$. Since $\pi^* \operatorname{div}(\ell|_{S^0}) = 2(\Lambda_1 + \Lambda_2)$, $\ell|_{S^0}$ vanishes to order 2 along the curve $\pi(\Lambda_1)$. If $\ell \nmid g_0$, then $g_0|_{(\ell=0)} = c' t^3$ with $c' \neq 0$ for a linear coordinate t on the line $(\ell = 0)$ of the (y,z)-plane, so $S^0 \cap (\ell = 0)$ is the reduced curve $(x^2 + c' t^3 = 0)$, along which $\ell|_{S^0}$ vanishes to order 1. Hence ℓ is a linear factor of g_0. Normalizing $\ell \in \Bbbk[y,z]$, $\lambda := c_0$ is a coefficient of $\operatorname{in}_w l_*$, so

$$\operatorname{in}_w l_* = \lambda \ell^{m/2}, \qquad \lambda \in \Bbbk[[u]] \backslash \{0\}.$$

For $\gamma \in \Gamma$, $\gamma \cdot \mathrm{in}_w l_*$ is a multiple of $\mathrm{in}_w l_*$, so $\gamma \cdot \ell$ is a multiple of ℓ. Since ℓ is a Γ-semi-invariant linear factor of g_0, the last sentence of Lemma F.12 gives, after a Γ-equivariant linear change of (y, z), which has complex coefficients, as the stabilizer of the reduced cubic $zy^2 + z^3$ in $GL_2(\Bbbk)$ is finite, which we also make on the coordinates of the Z_i, and which multiplies λ by a non-zero constant, $f_0 = f_I$ in **Step 5** and $\ell = z$. Write $\lambda = u^{n_\lambda}\lambda_1$ with an integer $n_\lambda \geq 0$ and $\lambda_1(0) \neq 0$. Comparing the characters of $\mathrm{in}_w l_*$, z, and u, the unit λ_1 is Γ-invariant. By Lemma 2.6(4) with λ_1 as the adjoined power series, on an infinite subfamily there exist Γ-invariant polynomials $\lambda_{1,i}$ such that $(f_{Z,*}, l_*, \lambda_1)$ is the compatible generic limit of the $(f_{Z_i}, l_i, \lambda_{1,i})$, and $\lambda_{1,i}(0) \neq 0$ for all but finitely many i since $\lambda_1(0) \neq 0$. Replacing l_i with $\lambda_{1,i}^{-1} l_i$, which keeps $\mathrm{div}(l_i)$, and l_* with $\lambda_1^{-1} l_*$, which keeps the compatible generic limit as the d-jets of $\lambda_{1,i}^{-1} l_i$ are polynomials in the d-jets of $\lambda_{1,i}, l_i$ and in $\lambda_{1,i}(0)^{-1}$, we may assume that $\lambda = u^{n_\lambda}$.

Let $M' := |\Gamma|$, let $J > \max\{6M', 2mM', m, 4\}$ be an integer, and let $N_1 \geq J + n_\lambda$ be an integer. We apply **Step 2** with

$$(c_x, c_y, c_z) = (3N_1, 2N_1, 2N_1), \quad r = 1, \quad d_1 = 6N_1, \quad \text{and} \quad d_2 = mN_1 + n_\lambda.$$

Then $c_K = N_1$ and $\beta' = \beta + n_\lambda/m$. By (F.200),

$$u'^{-d_1}\psi^* f_{Z,*} = x'^2 + z'y'^2 + z'^3 \quad \text{and} \quad u'^{-d_2}\psi^* l_* \in z'^{m/2} + u'^{N_1 - n_\lambda}\Bbbk[[x', y', z', u']].$$

A monomial of $u'^{-d_1}\psi^* f_{Z_i}$ (resp. $u'^{-d_2}\psi^* l_i$) of total degree less than J is u'^{-d_1} (resp. u'^{-d_2}) times the image under ψ^* of one of the finitely many monomials of f_{Z_i} (resp. l_i) of w-weight at most 6 (resp. m) and u-degree less than $J + 6N_1$ (resp. $J + n_\lambda + mN_1$). By [Liu26b, Construction-Proposition 4.1], after passing to an infinite subfamily, the coefficients of those monomials equal those of $f_{Z,*}$ (resp. l_*), which are 0 or 1. Thus

$$f_{Z_i'} - (x'^2 + z'y'^2 + z'^3) \in \mathfrak{n}'^J, \qquad l_i' - z'^{m/2} \in \mathfrak{n}'^J, \tag{F.201}$$

with $\mathfrak{n}'$ the maximal ideal of $\mathbb{C}[[x', y', z', u']]$. In particular $l_i'|_{S_i'} \neq 0$. Possibly replacing $(Z_i, l_i, \beta, \kappa)$ with $(Z_i', l_i', \beta + n_\lambda/m, N_1 + \kappa)$ and (x', y', z', u') with (x, y, z, u), we may assume that (R0)–(R4) and (F.201) hold. Let $\vartheta := 1 - \beta > 0$.

We show that $(Z_i, \frac{1}{m}\mathrm{div}(l_i) + S_i)$ is lc near 0.

(1) The pair

$$\Big(\mathbb{A}^4, \mathrm{div}(x^2 + y^2 z) + \frac{1}{2}\mathrm{div}(z) + \mathrm{div}(u)\Big) \tag{F.202}$$

is lc. The normalization of $V := (x^2 + y^2 z = 0) \subset \mathbb{A}^3$ is $\mathbb{A}^2 \to V$, $(y, v) \to (yv, y, -v^2)$, with conductor $(y = 0)$ of coefficient one, and $\frac{1}{2}\mathrm{div}(z)$ pulls back to $\mathrm{div}(v)$. Since $(\mathbb{A}^2, (y = 0) + (v = 0))$ is lc, [Kaw07, Theorem] implies that $(\mathbb{A}^3, V + \frac{1}{2}\mathrm{div}(z))$ is lc near V, hence lc. By [Kaw07, Theorem] with $(u = 0)$ in place of V, the pair in (F.202) is lc near $(u = 0)$. Away from $(u = 0)$ it is the product of $(\mathbb{A}^3, V + \frac{1}{2}\mathrm{div}(z))$ with $\mathbb{A}^1\backslash\{0\}$.

(2) The pair

$$\Big(\mathrm{Spec}\,\mathbb{C}[[x, y, z, u]], \mathrm{div}(f_{Z_i}) + \frac{1}{m}\mathrm{div}(l_i) + \mathrm{div}(u)\Big) \tag{F.203}$$

is lc near 0. Let $w' = (2, 1, 2, 1)$ be weights of $\mathbf{x} := (x, y, z, u)$. Let $J' \geq J$ be an integer with which Lemma 2.19 holds for $n = 4$, $N = 3$, and $\Lambda = \{1, \frac{1}{m}\}$. Possibly replacing f_{Z_i} and l_i with their truncations below degree J', which changes neither the w'-initial parts below nor, by Lemma 2.19, whether the pair in (F.203) is lc, in the proof of (2) we may assume that f_{Z_i} and l_i are polynomials. By (F.201) and $J > \max\{m, 4\}$,

$$\mathrm{in}_{w'} f_{Z_i} = x^2 + y^2 z, \qquad \mathrm{in}_{w'} l_i = z^{m/2}, \qquad \text{and} \qquad \mathrm{in}_{w'} u = u.$$

Let

$$f_{Z_i}^\sigma := \sigma^{-4} f_{Z_i}(\sigma^{w'}\mathbf{x}), \qquad l_i^\sigma := \sigma^{-m} l_i(\sigma^{w'}\mathbf{x}), \qquad \text{and} \qquad u^\sigma := \sigma^{-1} u(\sigma^{w'}\mathbf{x}) = u.$$

Every monomial of f_{Z_i} (resp. l_i) has w'-weight at least 4 (resp. m), so $f^\sigma_{Z_i}, l^\sigma_i \in \mathbb{C}[\mathbf{x}, \sigma]$, $f^\sigma_{Z_i}(\mathbf{x}, 0) = x^2 + y^2 z$, and $l^\sigma_i(\mathbf{x}, 0) = z^{m/2}$. By [Kaw07, Theorem] with $(\sigma = 0)$ in place of V and (F.202), the pair on $\mathbb{A}^4 \times \mathbb{A}^1$ with boundary

$$\operatorname{div}(f^\sigma_{Z_i}) + \frac{1}{m}\operatorname{div}(l^\sigma_i) + \operatorname{div}(u)$$

is lc near $(\sigma = 0)$. The non-lc locus of the pair on $\mathbb{A}^4 \times \mathbb{A}^1$ is closed and does not meet $(\sigma = 0)$, so it misses $(0, \sigma_0)$ for all but finitely many σ_0. For a general $\sigma_0 \in \mathbb{C}$, adding $(\sigma = \sigma_0)$ keeps it lc near $(0, \sigma_0)$, as $(\sigma = \sigma_0)$ is a general member of a free linear system, so by [Kaw07, Theorem] the restriction to $(\sigma = \sigma_0)$ is lc near 0. The automorphism $\mathbf{x} \to \sigma_0^{w'}\mathbf{x}$ carries the restriction of the pair on $\mathbb{A}^4 \times \mathbb{A}^1$ to $(\sigma = \sigma_0)$ to the pair on $\mathbb{A}^4$ with the boundary of (F.203), which is thus lc near 0. As a pair on $\mathbb{A}^4$ is lc near 0 if and only if its pullback to $\operatorname{Spec}\mathbb{C}[[x, y, z, u]]$ is lc, (2) follows.

(3) $(Z_i, \frac{1}{m}\operatorname{div}(l_i) + S_i)$ is lc near 0. Since $Z_i = (f_{Z_i} = 0)$ is normal, (2) and Lemma D.9(1), whose S is here $Z_i = (f_{Z_i} = 0) \subset \operatorname{Spec}\mathbb{C}[[x, y, z, u]]$, show that $(Z_i, \frac{1}{m}\operatorname{div}(l_i) + S_i)$ is lc.

Thus, for any prime divisor P over Z_i with center the origin,

$$a(P, Z_i, B_{Z_i}) = a\Big(P, Z_i, \frac{1}{m}\operatorname{div}(l_i) + S_i\Big) + \vartheta \operatorname{mult}_P u \geq \vartheta \operatorname{mult}_P u. \tag{F.204}$$

Let R be the order of χ_u. Since u^R is Γ-invariant, it is the pullback of a function on Z_i/Γ. As e_P is the ramification index of P over the image of P on Z_i/Γ, $R \operatorname{mult}_P u/e_P$ is a positive integer, so $\operatorname{mult}_P u/e_P \geq 1/R$. By (F.204),

$$\frac{a(P, Z_i, B_{Z_i})}{e_P} \geq \frac{\vartheta}{R} \quad \text{for any such } P.$$

Let $\gamma \in \Gamma$ be an element such that $\chi_u(\gamma)$ has order R, let n_γ be the order of γ, and let $n_3 := n_\gamma/R$. Let ζ be a primitive n_γ-th root of unity with $\chi_u(\gamma) = \zeta^{n_3}$, and let $n_1 \in [1, n_\gamma]$ and $n_2 \in [n_\gamma + 1, 2n_\gamma]$ be the integers with $\chi_y(\gamma) = \zeta^{n_1}$ and $\chi_x(\gamma)\chi_y(\gamma)^{-1} = \zeta^{n_2}$. Since f_I is semi-invariant, $\chi_x^2 = \chi_z\chi_y^2$, so $\chi_z(\gamma) = \zeta^{2n_2}$. Thus γ acts on (x, y, z, u) by ζ raised to the weights

$$w_E := (n_1 + n_2, n_1, 2n_2, n_3).$$

The integers $n_1 + n_2, n_1, 2n_2, n_3$ are coprime: a common divisor n divides n_1, n_2, n_3 and $n_\gamma = Rn_3$, so $\gamma^{n_\gamma/n}$ acts trivially and $n = 1$ by faithfulness. Since $n_2 > n_1$, $\operatorname{wt}_{w_E}(z^3) = 6n_2 > 2(n_1 + n_2)$. Let $\mathfrak{n}$ be the maximal ideal of $\mathbb{C}[[x, y, z, u]]$. Elements of $\mathfrak{n}^J$ have w_E-weight at least J, and $J > 6M' \geq 2(n_1 + n_2)$ and $J > 2mM' \geq mn_2$. By (F.201),

$$\operatorname{in}_{w_E} f_{Z_i} = x^2 + y^2 z, \qquad \operatorname{in}_{w_E} l_i = z^{m/2}.$$

Let E be the prime divisor over Z_i given by the weights w_E [Liu26b, Lemma 4.12]. Since $x^2 + y^2 z$ is irreducible and does not divide $z^{m/2}$, [Liu26b, Lemma 4.12] gives

$$a(E, Z_i, 0) = n_2 + n_3, \qquad \operatorname{mult}_E l_i = mn_2, \qquad \text{and} \qquad \operatorname{mult}_E u = n_3,$$

so $a(E, Z_i, B_{Z_i}) = \vartheta n_3$. The element γ acts on the w_E-graded ring by ζ^{degree}, hence trivially on E, so $n_\gamma \mid e_E$. Thus $a(E, Z_i, B_{Z_i})/e_E \leq \vartheta n_3/n_\gamma = \vartheta/R$. By the lower bound above, equality holds, E is a computing lift of $\operatorname{mld}((Z_i \ni 0, B_{Z_i})/\Gamma)$, $e_E = n_\gamma$, and

$$\frac{a(E, Z_i, 0)}{e_E} = \frac{n_2 + n_3}{n_\gamma} \leq 3. \tag{F.205}$$

Step 7. In this step we treat the case where B_{S^0} has no component with coefficient one. Let $i_0 := 0$. For $J = 1, 2, \ldots$, by (R4) there exists an infinite subfamily $\mathcal{I}_J$ of the present family such that for any $i \in \mathcal{I}_J$, any computing lift P of $\operatorname{mld}((Z_i \ni 0, B_{Z_i})/\Gamma)$ satisfies $\min\{\operatorname{mult}_P x, \operatorname{mult}_P y, \operatorname{mult}_P z\} > J$, and we choose $i_J \in \mathcal{I}_J$ with $i_J > i_{J-1}$. We replace the family by $\{i_J\}_{J \geq 1}$, which keeps the compatible generic limit in (F.200) by Lemma 2.6(2). Let $\mathfrak{n}$ be the maximal ideal of $\mathbb{C}[[x, y, z, u]]$. As f_0 has no terms of degree less than 2, Lemma 2.8

with $d = 2$ shows that $f_{Z_i} \in \mathfrak{n}^2$ for all but finitely many i, and we discard the other i, which again keeps the compatible generic limit.

Fix i, and let $\mathcal{O}_i$ be the local ring of Z_i^{alg} at 0, with maximal ideal $\mathfrak{m}_i$. By **Step 2**,

$$\widehat{\mathcal{O}}_i = \mathbb{C}[[x, y, z, u]]/(f_{Z_i}).$$

As $f_{Z_i} \in \mathfrak{n}^2$, the classes of x, y, z, u form a basis of $\mathfrak{m}_i/\mathfrak{m}_i^2 \cong \mathfrak{n}/\mathfrak{n}^2$. Possibly shrinking Z_i^{alg} to a Γ-invariant affine open neighborhood of 0, we may assume that Z_i^{alg} is affine. As Γ is finite abelian, the functions x, y, z, u and finitely many Γ-semi-invariant regular functions vanishing at 0 generate the coordinate ring of Z_i^{alg} and define a Γ-equivariant closed embedding of Z_i^{alg} into an affine space with a diagonal linear action of Γ, mapping 0 to the origin, in which x, y, z, u are the restrictions of coordinates $\widetilde{x}, \widetilde{y}, \widetilde{z}, \widetilde{u}$. Let $\mathfrak{a}$ be the ideal of Z_i^{alg} in the coordinate ring of the affine space.

The kernel of the restriction from the cotangent space of the affine space at the origin to $\mathfrak{m}_i/\mathfrak{m}_i^2$ consists of the differentials at the origin of the elements of $\mathfrak{a}$, and it is complementary to the span of the differentials of $\widetilde{x}, \widetilde{y}, \widetilde{z}, \widetilde{u}$. As projecting to characters keeps $\mathfrak{a}$, the kernel has a basis formed by the differentials at the origin of Γ-semi-invariant generators of an ideal $\mathfrak{b} \subset \mathfrak{a}$. By [Mat89, Theorem 14.2], the local ring at the origin of the zero locus of $\mathfrak{b}$ is regular of dimension 4, with regular parameters $\widetilde{x}, \widetilde{y}, \widetilde{z}, \widetilde{u}$. Let $\mathcal{Z}_i$ be a Γ-invariant open neighborhood of the origin in the zero locus of $\mathfrak{b}$ which is a smooth fourfold, with regular Γ-eigenparameters $\widetilde{x}, \widetilde{y}, \widetilde{z}, \widetilde{u}$ at 0, again denoted x, y, z, u. Then $\mathcal{Z}_i$ contains Z_i^{alg} near 0, Γ acts faithfully on $\mathcal{Z}_i$ by (R0), and $\widehat{\mathcal{O}}_{\mathcal{Z}_i,0} = \mathbb{C}[[x, y, z, u]]$.

The ideal $\mathfrak{q}_i$ of Z_i^{alg} in the regular local ring $\mathcal{O}_{\mathcal{Z}_i,0}$ is a prime ideal of height 1, hence principal [Mat89, Theorems 20.1 and 20.3]. Restriction to Z_i^{alg} induces on completions the surjection $\mathbb{C}[[x, y, z, u]] \to \widehat{\mathcal{O}}_i$ with kernel (f_{Z_i}), so $\mathfrak{q}_i\mathbb{C}[[x, y, z, u]] = (f_{Z_i})$. By Nakayama's lemma, projecting a generator of $\mathfrak{q}_i$ to the character of Γ on the line $\mathfrak{q}_i/\mathfrak{m}_{\mathcal{Z}_i,0}\mathfrak{q}_i$, where $\mathfrak{m}_{\mathcal{Z}_i,0}$ is the maximal ideal of $\mathcal{O}_{\mathcal{Z}_i,0}$, we obtain a Γ-semi-invariant generator, which is $\epsilon_i f_{Z_i}$ for a unit ϵ_i of $\mathbb{C}[[x, y, z, u]]$. As a quotient of Γ-semi-invariant elements with $\epsilon_i(0) \neq 0$, ϵ_i is Γ-invariant. Let $f_{Z_i}^\sharp$ be the product of $\epsilon_i f_{Z_i}$ and the polynomial formed by the terms of ϵ_i^{-1} of degree less than i, which is Γ-invariant with non-zero constant term. Then $f_{Z_i}^\sharp$ is a regular Γ-semi-invariant function on $\mathcal{Z}_i$ near 0 with the character of f_{Z_i}, $Z_i^{\mathrm{alg}} = (f_{Z_i}^\sharp = 0)$ near 0, and $f_{Z_i}^\sharp - f_{Z_i} \in \mathfrak{n}^i$.

The pullback of the mark h_i of W_i under the dominant rational map $Z_i^{\mathrm{alg}} \dashrightarrow W_i$, again denoted h_i, is a rational function on Z_i^{alg} with $h_i = u^{N'} l_i$ on Z_i for an integer $N' \geq 0$, by Proposition 17.13 and **Steps 2, 4, and 5**. Since $b\widehat{\mathcal{O}}_i \cap \mathcal{O}_i = b\mathcal{O}_i$ for any $b \in \mathcal{O}_i$ by faithful flatness [Mat89, Theorems 7.5(ii) and 8.14], a rational function on Z_i^{alg} whose image in the fraction field of $\widehat{\mathcal{O}}_i$ belongs to $\widehat{\mathcal{O}}_i$ belongs to $\mathcal{O}_i$. Thus $u^{-N'}h_i$ is a regular Γ-semi-invariant function on Z_i^{alg} near 0 whose image in $\widehat{\mathcal{O}}_i$ is the restriction of l_i to Z_i. Lifting $u^{-N'}h_i$ to a regular function on $\mathcal{Z}_i$ near 0 and projecting the lift to the character of l_i, we obtain a regular Γ-semi-invariant function $l_i^\flat$ on $\mathcal{Z}_i$ near 0 restricting to $u^{-N'}h_i$, so $l_i^\flat - l_i \in f_{Z_i}^\sharp \mathbb{C}[[x, y, z, u]]$. Let $l_i^\sharp$ be $l_i^\flat$ minus the product of $f_{Z_i}^\sharp$ and the polynomial formed by the terms of degree less than i of the Γ-semi-invariant series $(l_i^\flat - l_i)/f_{Z_i}^\sharp$. Then $l_i^\sharp$ is a regular Γ-semi-invariant function on $\mathcal{Z}_i$ near 0 with the character of l_i, restricting to $u^{-N'}h_i$ on Z_i^{alg}, and $l_i^\sharp - l_i \in \mathfrak{n}^i$.

For any d, the d-jets of $(f_{Z_i}^\sharp, l_i^\sharp)$ and of (f_{Z_i}, l_i) are equal for $i \geq d$. Hence, by Definition 2.5, the compatible generic limit (f_0, l_*) of the (f_{Z_i}, l_i) in (F.200) is also that of the $(f_{Z_i}^\sharp, l_i^\sharp)$. Let

$$B_i := \frac{1}{m}\operatorname{div}(l_i^\sharp) + \beta \operatorname{div}(u) \quad \text{on } Z_i^{\mathrm{alg}}.$$

As $\operatorname{div}(u) = S_i$ on Z_i, the pullback of B_i to Z_i is B_{Z_i}. By Lemma 2.12(2),

$$\operatorname{mld}((Z_i^{\mathrm{alg}} \ni 0, B_i)/\Gamma) = \operatorname{mld}((Z_i \ni 0, B_{Z_i})/\Gamma),$$

and by Lemma 2.12(3) with B_i and with 0, the restriction to Z_i^{alg} of any computing lift of the right-hand side is a computing lift of the left-hand side, any computing lift of the left-hand side is such a restriction, and the restriction keeps the orders of x, y, z, u and the log discrepancy with respect to 0.

We use Theorem 17.6 with $Z_i^{\mathrm{alg}} \subset \mathcal{Z}_i$, $f_{Z_i}^\sharp$, $l_i^\sharp$, Γ, $|\Gamma|$, a, and an integer $N_3 \geq c_u|\Gamma|$ in place of W_i in the smooth fourfold germ, f_i, h_i, G, M, L_0, and R', and with f_0, $w = (3,2,2)$, $d = 6$, and β, where f_I is the case (D_4) of Proposition 7.7 and $f_{II} = x^2 + y^3 + z^3$. The assumptions of Theorem 17.6 hold. The germ Z_i^{alg} is a Γ-invariant normal Gorenstein hypersurface germ in $\mathcal{Z}_i$, and it is klt near 0 since Z_i is (**Step 5**), as a log resolution of Z_i^{alg} pulls back to one of Z_i along the regular morphism $Z_i \to Z_i^{\mathrm{alg}}$, $\mathcal{O}_i$ being excellent. By (R0), Γ is abelian and acts faithfully and diagonally on (x, y, z, u), and $\beta \in \frac{1}{m}\mathbb{Z}$. The $f_{Z_i}^\sharp$ and their limit f_0 have the character of the f_{Z_i}, and the $l_i^\sharp$ have that of the l_i. Hypothesis (1) holds, as $\operatorname{mld}((Z_i^{\mathrm{alg}} \ni 0, B_i)/\Gamma) = b_i$ by (R1) and $0 < b_i \leq a - \delta$ by Set-up 17.1(2) and Definition 4.1(3). Computing lifts of $\operatorname{mld}((Z_i \ni 0, B_{Z_i})/\Gamma)$ exist by Lemma 4.2(1) and (R1), so Hypothesis (2) holds by (R3), $e_P \leq |\Gamma|$, and the choice of the i_J. Hypothesis (3) holds for $(f_*, h_*) = (f_0, l_*)$ by (F.200), **Step 5**, and the assumption of this step. By Theorem 17.6, on an infinite subfamily some computing lift of $\operatorname{mld}((Z_i^{\mathrm{alg}} \ni 0, B_i)/\Gamma)$ has log discrepancy at most L_1 with respect to $(Z_i^{\mathrm{alg}}, 0)$, for a constant L_1 independent of i. It is the restriction of a computing lift E_i of $\operatorname{mld}((Z_i \ni 0, B_{Z_i})/\Gamma)$, and $a(E_i, Z_i, 0) \leq L_1$.

Step 8. In this step we conclude. By **Steps 6 and 7**, after passing to an infinite subfamily, there exist computing lifts E_i of $\operatorname{mld}((Z_i \ni 0, B_{Z_i})/\Gamma)$ with

$$\frac{a(E_i, Z_i, 0)}{e_{E_i}} \leq L_2 := \max\{3, L_1\}.$$

By (R1), the restriction F_i of E_i to W_i is a computing lift of b_i. By (R2) and (R3),

$$\frac{a(F_i, W_i, 0)}{e_{F_i}} = \frac{a(E_i, Z_i, 0) + \kappa \operatorname{mult}_{E_i} u}{e_{E_i}} \leq L := L_2 + \kappa c_u.$$

This contradicts Lemma 4.3(2). □

Appendix G. Proofs for transverse type A

G.1. **Admission.**

Proof of Theorem 18.1. We divide the proof into five steps. By Construction-Theorems 13.7 and 13.6, possibly passing to a subsequence, we may assume the following.

- $G_i = G$ for a fixed cyclic group G with $|G| \leq M$ acting diagonally on $\mathcal{X} := \operatorname{Spec} \mathbb{C}[[x_1, \dots, x_4]]$.
- $W_i = (f_i = 0) \subset \mathcal{X}$.
- f_i and h_i are G-semi-invariant with characters independent of i.
- (f_*, h_*) is a compatible generic limit of the (f_i, h_i) with coefficient field $\Bbbk$ (Definition 2.5).

Let $\mathfrak{n}$ be the maximal ideal of $\mathbb{C}[[x_1, \dots, x_4]]$, let $\mathcal{O}_{W_i} := \mathbb{C}[[x_1, \dots, x_4]]/(f_i)$ with maximal ideal $\mathfrak{m}_{o_i}$, and let $\mathcal{K}_i$ be the fraction field of $\mathcal{O}_{W_i}$.

By Definitions 4.1, 2.2 and 2.4, for any prime divisor F over W_i with center o_i and any $t \in [0, 1]$,

$$q_i(t) \leq \frac{a(F, W_i, \frac{t}{m}\operatorname{div}(h_i))}{e_F} = \frac{a(F, W_i, 0)}{e_F} - \frac{t \operatorname{mult}_F h_i}{m e_F}, \tag{G.1}$$

and for $t = 1$ equality holds if and only if F is a computing lift of b_i. By Definition 4.1, $b_i \le a - \delta$ and $q_i(\tau_i) \ge a$, so

$$q_i(\tau_i) \ge b_i + \delta \quad \text{for any } i. \tag{G.2}$$

By Lemma 4.2(3), $b_i \le 3$. Thus, by Set-up 17.1, the W_i, the h_i and (f_*, h_*) satisfy Set-up 15.4 with $B_0 := 3$. By Set-up 17.1(4) and (5) and Lemma 13.5, the minimal lc center of $(W_*, \frac{1}{m}\operatorname{div}(h_*))$ is a curve contained in $\operatorname{Nklt}(W_*, \frac{1}{m}\operatorname{div}(h_*)) = C$, so it is C.

Step 1. In this step we construct the charts on the limit.

Let (x, y, z, s) be the G-eigenparameters of Lemma 17.3. Thus $C = (x = y = z = 0)$ and $f_* = v \cdot (x^2 + g(y, z, s))$, where v is an invariant unit, $g \in (y, z)^2$, and $g - Q \in (y, z)^3$ for the (y, z)-quadratic part Q of g. Since the generic transverse type along C is A_n, exactly one of the following holds, for some positive integers ν, ν_1, ν_2.

(i) $n \ge 2$ and $Q = s^\nu y^2$.
(ii) $n = 1$ and $Q = s^{\nu_1} y^2 + s^{\nu_2} z^2$.
(iii) $n = 1$ and $Q = s^\nu yz$.

Let χ_s be the character of s and

$$\Gamma_* := \{(\sigma, \alpha) \in G \times \mathbb{C}^* \mid \alpha^2 = \chi_s(\sigma)\}.$$

For any $\sigma \in G$ there exist exactly two $\alpha \in \mathbb{C}^*$ with $(\sigma, \alpha) \in \Gamma_*$. Thus Γ_* is an abelian group with $|\Gamma_*| = 2|G|$.

Fix an integer r with $r > 2\nu + 2$ in cases (i) and (iii), and with $r > 2(\nu_1 + \nu_2) + 2$ in case (ii). Let $\psi_1 : \operatorname{Spec} \Bbbk[[x', y', z', u]] \to \mathcal{X}_*$ be the morphism given by

$$\begin{aligned} &x = u^{r+\nu}x', \quad y = u^r y', \quad z = u^r z', \quad s = u^2 \quad \text{in cases (i) and (iii)},\\ &x = u^{r+\nu_1+\nu_2}x', \quad y = u^{r+\nu_2}y', \quad z = u^{r+\nu_1}z', \quad s = u^2 \quad \text{in case (ii)}. \end{aligned} \tag{G.3}$$

We let $(\sigma, \alpha) \in \Gamma_*$ act on $\Bbbk[[x', y', z', u]]$ by multiplying u by α and each of x', y', z' by the scalar for which ψ_1 is equivariant with respect to the homomorphism $\Gamma_* \to G$, $(\sigma, \alpha) \to \sigma$. For instance, in cases (i) and (iii), x' is multiplied by $\chi_x(\sigma)\alpha^{-r-\nu}$.

A term $\xi(s)y^j z^{j'}$ of $g - Q$, with $j + j' \ge 3$, pulls back into $u^{3r}(y', z')^3$ in cases (i) and (iii), and into $u^{3r+3\min\{\nu_1,\nu_2\}}(y', z')^3$ in case (ii), where $3r + 3\min\{\nu_1, \nu_2\} > 2r + 2\nu_1 + 2\nu_2$. Hence there exists $g_1 \in (y', z')^3\Bbbk[[y', z', u]]$ with

$$\psi_1^* f_* = \begin{cases} u^{2r+2\nu}\psi_1^* v \cdot (x'^2 + y'^2 + u^{r-2\nu}g_1) & \text{in case (i)},\\ u^{2r+2\nu_1+2\nu_2}\psi_1^* v \cdot (x'^2 + y'^2 + z'^2 + ug_1) & \text{in case (ii)},\\ u^{2r+2\nu}\psi_1^* v \cdot (x'^2 + y'z' + u^{r-2\nu}g_1) & \text{in case (iii)}. \end{cases} \tag{G.4}$$

Since $ds = 2udu$,

$$\psi_1^*(dx \wedge dy \wedge dz \wedge ds) = \begin{cases} \pm 2u^{3r+\nu+1} dx' \wedge dy' \wedge dz' \wedge du & \text{in cases (i) and (iii)},\\ \pm 2u^{3r+2\nu_1+2\nu_2+1} dx' \wedge dy' \wedge dz' \wedge du & \text{in case (ii)}. \end{cases}$$

In cases (ii) and (iii) we let $(\widetilde{x}, \widetilde{y}, \widetilde{z}) := (x', y', z')$ and $\pi_* := \psi_1$.

In case (i) we prepare the chart further. Let $g_2 := y'^2 + u^{r-2\nu}g_1$. Since $\partial g_2/\partial y' \equiv 2y'$ modulo $u^{r-2\nu}(y', z')^2$, the implicit function theorem yields a unique $\phi \in \Bbbk[[z', u]]$ with

$$\frac{\partial g_2}{\partial y'}(\phi, z', u) = 0,$$

and $\phi \in u^{r-2\nu}(z')^2$. Expanding g_2 in $y' - \phi$, we obtain

$$g_2 = g_3(z', u) + \lambda^2 \cdot (y' - \phi)^2, \qquad g_3(z', u) := g_2(\phi, z', u),$$

for a unit $\lambda \in \Bbbk[[y', z', u]]$ with $\lambda(0) = 1$. The series g_2 is Γ_*-semi-invariant with the character of y'^2. By the uniqueness of ϕ, ϕ is an eigenfunction with the character of y'. Thus λ^2 is invariant,

and so is λ, the unique square root of λ^2 with $\lambda(0)=1$. Let $y_1' := \lambda\cdot(y'-\phi)$, an eigenparameter. Then, in the coordinates (x', y_1', z', u),

$$\psi_1^* f_* = u^{2r+2\nu}\psi_1^* v\cdot(x'^2 + y_1'^2 + g_3(z',u)), \qquad g_3 = \phi^2 + u^{r-2\nu}g_1(\phi,z',u) \in u^{r-2\nu}\Bbbk[[z',u]].$$

Over the field $\Bbbk((u)) \supset \Bbbk((s))$, the substitution (G.3) and the replacement of y' by y_1' are invertible changes of coordinates of $\Bbbk((u))[[x',y',z']]$, and $u^{2r+2\nu}\psi_1^* v$ is a unit there. Since the generic transverse type of W_* along C is A_n, the series $x'^2 + y_1'^2 + g_3(z',u)$ defines a singularity of type A_n over an algebraic closure of $\Bbbk((u))$, so the order of g_3 in z' over $\Bbbk((u))$ is $n+1$. Thus

$$0 \neq g_3 \in z'^{n+1}\Bbbk[[z',u]] \cap u\Bbbk[[z',u]],$$

that is, any monomial $z'^j u^{j'}$ of g_3 satisfies $j \geq n+1$ and $j' \geq 1$. Let $w_0 := \mathrm{wt}_{(2,1)}(g_3)$ for the weights $(2,1)$ on (z',u), and let

$$n' := \left\lfloor \frac{w_0}{2} \right\rfloor, \qquad \varsigma := w_0 - 2n' \in \{0,1\}, \quad \text{and} \quad \Theta_0(z'') := u^{-w_0}\,\mathrm{in}_{(2,1)}(g_3)(u^2 z'', u).$$

Then $n' \geq n+1$, and Θ_0 is a non-zero polynomial with $n_0 := \mathrm{mult}_0\,\Theta_0 \geq n+1$. Let ψ_2 be the morphism given by

$$x' = u^{n'}x'', \qquad y_1' = u^{n'}y'', \quad \text{and} \quad z' = u^2 z'',$$

with the diagonal action of Γ_* defined as for ψ_1. Then

$$\psi_2^*(dx'\wedge dy_1'\wedge dz'\wedge du) = u^{2n'+2}dx''\wedge dy''\wedge dz''\wedge du$$

and

$$u^{-2n'}\psi_2^*(x'^2 + y_1'^2 + g_3) = x''^2 + y''^2 + g_5(z'',u), \qquad g_5 := u^{\varsigma}\Theta_0(z'') + u^{\varsigma+1}g_4(z'',u),$$

for some $g_4 \in z''^{n+1}\Bbbk[[z'',u]]$. In case (i) we let $(\widetilde{x},\widetilde{y},\widetilde{z}) := (x'',y'',z'')$, and we let π_* be the composite of ψ_1, the coordinate change $(x',y',z',u) \to (x',y_1',z',u)$ and ψ_2. If $\varsigma = 1$, any monomial of g_5 has z''-degree at least 3 and u-degree at least 1. In that case let $d \geq 4$ be the degree of the lowest-degree homogeneous part g_6 of g_5, and fix $\varpi \in \mathbb{C}$ with $g_6(1,\varpi) \neq 0$. Such ϖ exists since $g_6(1,\cdot)$ is a non-zero polynomial. Then

$$g_5(z'', \varpi z'') \in z''^d\cdot\Bbbk[[z'']]^\times.$$

In all cases, π_* pulls x,y,z,s back into $u^2\Bbbk[[\widetilde{x},\widetilde{y},\widetilde{z},u]]$, $\pi_*^* x$ is $\widetilde{x}$ times a power of u, and $\pi_*^* y, \pi_*^* z, \pi_*^* s \in \Bbbk[[\widetilde{y},\widetilde{z},u]]$. Moreover,

$$\begin{gathered}\pi_*^* f_* = u^{N_1}\widetilde{f}_*, \qquad \widetilde{f}_* \equiv v(0)\Xi_0 \pmod u,\\ \pi_*^*(dx\wedge dy\wedge dz\wedge ds) \in u^{N_1+\kappa}\cdot\Bbbk[[\widetilde{x},\widetilde{y},\widetilde{z},u]]^\times\cdot d\widetilde{x}\wedge d\widetilde{y}\wedge d\widetilde{z}\wedge du,\end{gathered} \tag{G.5}$$

where the integers N_1, κ and the polynomial Ξ_0 are the following.

(i) $N_1 = 2r+2\nu+2n'$ and $\kappa = r-\nu+3$. If $\varsigma = 0$, then $\Xi_0 = \widetilde{x}^2 + \widetilde{y}^2 + \Theta_0(\widetilde{z})$. If $\varsigma = 1$, then $\Xi_0 = \widetilde{x}^2 + \widetilde{y}^2$ and moreover

$$\widetilde{f}_* \equiv v(0)\cdot(\widetilde{x}^2 + \widetilde{y}^2 + u\Theta_0(\widetilde{z})) \pmod{u^2}.$$

(ii) $N_1 = 2r + 2\nu_1 + 2\nu_2$, $\kappa = r+1$, and $\Xi_0 = \widetilde{x}^2 + \widetilde{y}^2 + \widetilde{z}^2$.

(iii) $N_1 = 2r+2\nu$, $\kappa = r-\nu+1$, and $\Xi_0 = \widetilde{x}^2 + \widetilde{y}\widetilde{z}$.

Here we use $\pi_*^* v \equiv v(0)$ modulo u^2. In cases (i) and (iii), $\kappa \geq 1$ since $r > 2\nu+2$. In case (ii), $\kappa = r+1 \geq 1$. Let

$$N_3 := r+\nu+n'+2 \text{ in case (i)}, \qquad N_3 := r+\nu_1+\nu_2 \text{ in case (ii)}, \quad \text{and} \quad N_3 := r+\nu \text{ in case (iii)}.$$

Since x^2+g is monic of degree 2 in x and $g \in \Bbbk[[y,z,s]]$, there exist $h_{*,0}, h_{*,1} \in \Bbbk[[y,z,s]]$ with $h_* \equiv h_{*,0} + xh_{*,1}$ modulo f_*, not both zero since $h_*|_{W_*} \neq 0$. Then

$$\pi_*^*(h_{*,0} + xh_{*,1}) = \widetilde{h}_{*,0} + \widetilde{x}\widetilde{h}_{*,1} \quad \text{for some } \widetilde{h}_{*,0}, \widetilde{h}_{*,1} \in \Bbbk[[\widetilde{y},\widetilde{z},u]],$$

not both zero since π_*^* is injective. Let

$$N' := \max\{j \in \mathbb{N} \mid \widetilde{h}_{*,0}, \widetilde{h}_{*,1} \in u^j\Bbbk[[\widetilde{y},\widetilde{z},u]]\}.$$

Since $\widetilde{f}_*(\widetilde{x},0,0,0) = v(0)\widetilde{x}^2$, Weierstrass division by $\widetilde{f}_*$ with respect to $\widetilde{x}$ exists. Since

$$\pi_*^*(h_* - h_{*,0} - xh_{*,1}) \in (\widetilde{f}_*),$$

the Weierstrass remainder of $\pi_*^* h_*$ is

$$\widetilde{h}_{*,0} + \widetilde{x}\widetilde{h}_{*,1} \in u^{N'}\Bbbk[[\widetilde{x},\widetilde{y},\widetilde{z},u]] \backslash u^{N'+1}\Bbbk[[\widetilde{x},\widetilde{y},\widetilde{z},u]]. \tag{G.6}$$

Step 2. In this step we choose coordinates on the members and show that the computing lifts of b_i concentrate along C.

Let $T := \lceil 3MN_3/\eta_m \rceil$, and let $\epsilon > 0$ and $n_1 \geq 1$ be the rational number and the integer n of Theorem 16.1. By Theorem 16.1(1) with (s,x,y,z) for (s,u_1,u_2,u_3) and $T+1$ for k, possibly passing to a subsequence, there exist G-eigenparameters s_i^{c}, x_i^{c}, y_i^{c}, z_i^{c} of $\mathcal{X}$ (the $(s_i,u_{i1},u_{i2},u_{i3})$ there) and elements $g_{i,1},\dots,g_{i,N_2}$ as there. Let $\mathfrak{c}_i$ be the ideal generated by x_i^{c}, y_i^{c}, z_i^{c}, and let $\mathbf{t}_*$ be the compatible generic limit there of the tuples

$$\mathbf{t}_i := (f_i, h_i, s_i^{\mathrm{c}}, x_i^{\mathrm{c}}, y_i^{\mathrm{c}}, z_i^{\mathrm{c}}, g_{i,1},\dots,g_{i,N_2}).$$

Then the limits of $x_i^{\mathrm{c}}, y_i^{\mathrm{c}}, z_i^{\mathrm{c}}$ are congruent to x,y,z modulo $\mathfrak{n}_*^{T+1}$, where $\mathfrak{n}_*$ is the maximal ideal of $\mathcal{O}_{\mathcal{X}_*}$, and any prime divisor F over W_i with center o_i, $a(F,W_i,\frac{1}{m}\operatorname{div}(h_i))/e_F \leq 3$, and $\operatorname{mult}_F(\mathfrak{c}_i\mathcal{O}_{W_i}) \leq T+1$ satisfies

$$\operatorname{mult}_F \operatorname{Jac}_{W_i} \leq n_1(T+1+\lceil 3M/\epsilon \rceil). \tag{G.7}$$

Let $N_4 := T + N_1 + N' + d + 2$, where d is the integer of **Step 1** in case (i) with $\varsigma = 1$, and $d := 0$ otherwise. In case (i), let $\phi^{<N_4}$ and $\lambda^{<N_4}$ be the truncations of ϕ and λ in degrees less than N_4, which are Γ_*-eigenpolynomials over $\Bbbk$. Let π'_* be defined by the formulas of **Step 1**, with $\phi^{<N_4}$ and $\lambda^{<N_4}$ in place of ϕ and λ in case (i), and $\pi'_* := \pi_*$ in cases (ii) and (iii). In case (i), let the γ_j be the non-zero coefficients of $\phi^{<N_4}$ and $\lambda^{<N_4}$, regarded as constant series. By Lemma 2.6(4) for $\mathbf{t}_*$, the series x,y,z,s and, in case (i), the γ_j, possibly passing to a subsequence, there exist G-semi-invariant polynomials x_i,y_i,z_i,s_i with the characters of x,y,z,s and, in case (i), polynomials $\gamma_{i,j}$, such that the tuples formed by $\mathbf{t}_i$, x_i,y_i,z_i,s_i and the $\gamma_{i,j}$ have the compatible generic limit formed by $\mathbf{t}_*$, x,y,z,s and the γ_j. In case (i), let $\phi_i \in \mathbb{C}[z',u]$ and $\lambda_i \in \mathbb{C}[y',z',u]$ be the Γ_*-eigenpolynomials with the monomials of $\phi^{<N_4}$ and $\lambda^{<N_4}$ in which the constant term of $\gamma_{i,j}$ is the coefficient in place of γ_j. By Lemma 2.6(4), $\lambda_i(0) = 1$ for all but finitely many i, and we discard the other i. Below, the *tuples of* ***Step 2*** are the tuples formed by $\mathbf{t}_i$, x_i,y_i,z_i,s_i and the $\gamma_{i,j}$, and the *limit of* ***Step 2*** is their compatible generic limit. By Definition 2.5, a condition on finitely many coefficients of the tuples given by the vanishing and non-vanishing of polynomials with complex coefficients which holds for the limit holds at the generic point of the corresponding irreducible closed subset of the jet space, hence on a non-empty open subset of it, and hence for all but finitely many i. For instance, since the linear parts of x,y,z,s are linearly independent, (x_i,y_i,z_i,s_i) are regular parameters of $\mathbb{C}[[x_1,\dots,x_4]]$ for all but finitely many i, and we discard the other i. Since $\mathfrak{n}_*^{T+1}$ contains the limits of $x_i - x_i^{\mathrm{c}}$, $y_i - y_i^{\mathrm{c}}$, $z_i - z_i^{\mathrm{c}}$, and membership in $\mathfrak{n}^{T+1}$ is the vanishing of finitely many coefficients,

$$x_i - x_i^{\mathrm{c}},\ y_i - y_i^{\mathrm{c}},\ z_i - z_i^{\mathrm{c}} \in \mathfrak{n}^{T+1}$$

for all but finitely many i, and we discard the other i. Thus

$$(x_i,y_i,z_i) + \mathfrak{n}^{T+1} = \mathfrak{c}_i + \mathfrak{n}^{T+1},$$

so

$$\min\{\operatorname{mult}_F((x_i,y_i,z_i)\mathcal{O}_{W_i}), T+1\} = \min\{\operatorname{mult}_F(\mathfrak{c}_i\mathcal{O}_{W_i}), T+1\}$$

for any prime divisor F over W_i with center o_i.

We show that, after discarding finitely many i, any computing lift F of b_i satisfies

$$\min\{\operatorname{mult}_F x_i, \operatorname{mult}_F y_i, \operatorname{mult}_F z_i\} > T, \qquad \frac{\operatorname{mult}_F s_i}{e_F} \leq \frac{3}{\eta_m}, \quad \text{and} \quad \operatorname{mult}_F s_i \leq \frac{3M}{\eta_m}. \tag{G.8}$$

Let F be a computing lift of b_i with $\min\{\mathrm{mult}_F x_i, \mathrm{mult}_F y_i, \mathrm{mult}_F z_i\} \le T$. Then

$$\frac{a(F, W_i, \frac{1}{m}\operatorname{div}(h_i))}{e_F} = b_i \le 3$$

and $\mathrm{mult}_F(\mathfrak{c}_i \mathcal{O}_{W_i}) \le T$, so (G.7) holds. By Theorem 15.8(1) with $n_1(T + 1 + \lceil 3M/\epsilon \rceil)$ in place of N, and with L' the constant L there, $q_i(\tau_i) \le b_i + L'(1 - \tau_i)$. Since $\tau_i \to 1$, the last inequality contradicts (G.2) for any $i \gg 0$. Discarding finitely many i, the first inequality of (G.8) holds for any computing lift F of b_i. By Theorem 5.1(3), $\mathrm{mult}_F \mathfrak{m}_{o_i}/e_F \le 3/\eta_m$ and $\mathrm{mult}_F \mathfrak{m}_{o_i} \le 3M/\eta_m \le T$. Since

$$\mathrm{mult}_F \mathfrak{m}_{o_i} = \min\{\mathrm{mult}_F x_i, \mathrm{mult}_F y_i, \mathrm{mult}_F z_i, \mathrm{mult}_F s_i\},$$

we have $\mathrm{mult}_F \mathfrak{m}_{o_i} = \mathrm{mult}_F s_i$, and (G.8) follows.

Step 3. In this step we construct the degree 2 extensions, the actions of the Γ_i and the germs Z_i, and prove (4).

Since $(W_i \ni o_i)$ is terminal, $\mathcal{O}_{W_i}$ is a normal domain. Since $(W_i \ni o_i)$ has quadratic rank 1, $f_i \in \mathfrak{n}^2$, so

$$\mathfrak{m}_{o_i}/\mathfrak{m}_{o_i}^2 = \mathfrak{n}/\mathfrak{n}^2$$

and $s_i \notin \mathfrak{m}_{o_i}^2$. If some $u \in \mathcal{K}_i$ satisfied $u^2 = s_i$, then u would be integral over $\mathcal{O}_{W_i}$, hence $u \in \mathcal{O}_{W_i}$. Since $u^2 \in \mathfrak{m}_{o_i}$, we would then have $u \in \mathfrak{m}_{o_i}$ and $s_i \in \mathfrak{m}_{o_i}^2$, a contradiction. Thus

$$\mathcal{K}_i(u) := \mathcal{K}_i[u]/(u^2 - s_i)$$

is a field extension of $\mathcal{K}_i$ of degree 2. Let $\Gamma_i := \Gamma_*$. The group Γ_i acts on $\mathcal{K}_i(u)$, with (σ, α) acting on $\mathcal{K}_i$ through σ and multiplying u by α. The action is well defined since $\alpha^2 s_i = \chi_s(\sigma) s_i$. The subgroup $\{(1, \pm 1)\}$ has fixed field $\mathcal{K}_i$, and $\Gamma_i/\{(1, \pm 1)\} \cong G$ acts faithfully on $\mathcal{K}_i$. Hence Γ_i acts faithfully on $\mathcal{K}_i(u)$ and

$$\mathcal{K}_i(u)^{\Gamma_i} = \mathcal{K}_i^G. \tag{G.9}$$

Let

$$\widetilde{\mathcal{X}} := \operatorname{Spec} \mathbb{C}[[\widetilde{x}, \widetilde{y}, \widetilde{z}, u]],$$

and let $\pi_i : \widetilde{\mathcal{X}} \to \mathcal{X}$ be defined by the formulas of **Step 1** with (x_i, y_i, z_i, s_i), ϕ_i, λ_i in place of (x, y, z, s), ϕ, λ. It is equivariant for the diagonal action of Γ_i on $\widetilde{\mathcal{X}}$ defined as in **Step 1**.

The morphism π_i pulls x_i, y_i, z_i, s_i back into $u^2\mathbb{C}[[\widetilde{x}, \widetilde{y}, \widetilde{z}, u]]$, and the coefficient of u^j (the *u^j-layer*) of each pull-back is a polynomial in $\widetilde{x}, \widetilde{y}, \widetilde{z}$, since the substitutions of **Step 1** multiply the variables by positive powers of u and ϕ_i, λ_i are polynomials. Hence, for any $\theta \in \mathbb{C}[[x_1, \dots, x_4]]$ and any integer $j \ge 0$, the u^j-layer of $\pi_i^*\theta$ is a polynomial depending only on θ and x_i, y_i, z_i, s_i modulo $\mathfrak{n}^{j+1}$ and on ϕ_i and λ_i. Since x, y, z, s and, in case (i), the coefficients of $\phi^{<N_4}, \lambda^{<N_4}$ are the limits of x_i, y_i, z_i, s_i and of the coefficients of ϕ_i, λ_i, the coefficients of the u^j-layers of $\pi_i^* f_i$ and $\pi_i^* h_i$ have as compatible generic limit those of $\pi_*'^* f_*$ and $\pi_*'^* h_*$. For $j < N_4$, the u^j-layers of $\pi_*'^* f_*$ and $\pi_*'^* h_*$ equal those of $\pi_*^* f_*$ and $\pi_*^* h_*$.

Thus, as in **Step 2**, any condition on the coefficients of finitely many such layers given by the vanishing and non-vanishing of polynomials with complex coefficients which holds for $\pi_*^* f_*$ and $\pi_*^* h_*$ holds for all but finitely many i. By **Step 1**, the following conditions hold in the limit, so we may assume that they hold for any i.

(a) $\pi_i^* f_i = u^{N_1} \widetilde{f}_i$ for some $\widetilde{f}_i \in \mathbb{C}[[\widetilde{x}, \widetilde{y}, \widetilde{z}, u]]$.

(b) $\widetilde{f}_i \equiv \Xi_{0,i} \pmod{u}$, where $\Xi_{0,i}$ is a polynomial whose monomials with non-zero coefficient are the monomials of $v(0)\Xi_0$. Let c_i be the coefficient of $\widetilde{x}^2$ in $\Xi_{0,i}$, which is non-zero. In case (i) with $\varsigma = 0$ we write

$$\Xi_{0,i} = c_i \widetilde{x}^2 + c_i' \widetilde{y}^2 + \Theta_{0,i}(\widetilde{z})$$

with $\mathrm{mult}_0 \Theta_{0,i} = n_0$. In case (i) with $\varsigma = 1$,

$$\widetilde{f}_i \equiv c_i \widetilde{x}^2 + c_i' \widetilde{y}^2 + u\Theta_{0,i}(\widetilde{z}) \pmod{u^2}$$

with $c_i c_i' \neq 0$ and $\operatorname{mult}_0 \Theta_{0,i} = n_0$.

Moreover, since $d < N_4 - N_1$, in case (i) with $\varsigma = 1$ the coefficients of degree at most d of $\widetilde{f}_i(\widetilde{x}, \widetilde{y}, \widetilde{z}, \varpi\widetilde{z})$ have as compatible generic limit those of

$$\widetilde{f}_*(\widetilde{x}, \widetilde{y}, \widetilde{z}, \varpi\widetilde{z}) = \pi_*^* v \cdot (\widetilde{x}^2 + \widetilde{y}^2 + g_5(\widetilde{z}, \varpi\widetilde{z})). \tag{G.10}$$

Let $Z_i := (\widetilde{f}_i = 0) \subset \widetilde{\mathcal{X}}$, let $U_i := \operatorname{div}(u)$ on Z_i, and let

$$\mathcal{O}_{Z_i} := \mathbb{C}[[\widetilde{x}, \widetilde{y}, \widetilde{z}, u]]/(\widetilde{f}_i).$$

By (b),

$$U_i \cong (\Xi_{0,i} = 0) \subset \operatorname{Spec} \mathbb{C}[[\widetilde{x}, \widetilde{y}, \widetilde{z}]].$$

In cases (ii) and (iii), U_i is a surface singularity of type A_1. In case (i) with $\varsigma = 0$, U_i is of type A_{n_0-1}. In case (i) with $\varsigma = 1$, U_i is the union of two distinct planes $(c_i\widetilde{x}^2 + c_i'\widetilde{y}^2 = 0)$. By the splitting lemma, a series ζ in $\mathbb{C}[[\widetilde{x}, \widetilde{y}, \widetilde{z}]]$ or in $\Bbbk[[\widetilde{x}, \widetilde{y}, \widetilde{z}]]$ with 2-jet $c\widetilde{x}^2 + c'\widetilde{y}^2$, $cc' \neq 0$, is right equivalent to $\widetilde{x}^2 + \widetilde{y}^2 + \zeta_0(\widetilde{z})$, where ζ_0 is the restriction of ζ to the smooth curve

$$\Big(\frac{\partial \zeta}{\partial \widetilde{x}} = \frac{\partial \zeta}{\partial \widetilde{y}} = 0\Big).$$

In particular, the coefficients of $\widetilde{z}^j$, $j \leq d$, of ζ_0 are polynomials in $(cc')^{-1}$ and in the coefficients of degree at most d of ζ. By the choice of ϖ in **Step 1**, $g_5(\widetilde{z}, \varpi\widetilde{z})$ has order d, so the order of ζ_0 for $\zeta = \widetilde{f}_*(\widetilde{x}, \widetilde{y}, \widetilde{z}, \varpi\widetilde{z})$ is d, so by (G.10), possibly discarding finitely many i, in case (i) with $\varsigma = 1$ the section $(u = \varpi\widetilde{z})$ of Z_i is of type A_{d-1}. In any case Z_i has a Du Val hyperplane section through 0, so $(Z_i \ni 0)$ is a cDV germ. In particular, $\mathcal{O}_{Z_i}$ is a normal domain of dimension 3.

Solving the formulas defining π_i for $\widetilde{x}, \widetilde{y}, \widetilde{z}$, we regard $\widetilde{x}, \widetilde{y}, \widetilde{z}$ as elements of $\mathcal{K}_i(u)$. In case (i) we likewise regard the coordinates $x' = x_i/u^{r+\nu}$, $y' = y_i/u^r$, $z' = z_i/u^r$ of the first chart as elements of $\mathcal{K}_i(u)$, and in cases (ii) and (iii) we let $(x', y', z') := (\widetilde{x}, \widetilde{y}, \widetilde{z})$. Let

$$R_i := \mathcal{O}_{W_i}[u, x', y', z', \widetilde{x}, \widetilde{y}, \widetilde{z}] \subset \mathcal{K}_i(u).$$

Here $\mathcal{O}_{W_i}[u] \subset \mathcal{K}_i(u)$ is the subring generated by u over $\mathcal{O}_{W_i}$, a free $\mathcal{O}_{W_i}$-module with basis $1, u$. Since the formulas of **Step 1** multiply the variables by powers of u and ϕ_i, λ_i are polynomials,

$$u^j x',\ u^j y',\ u^j z',\ u^j \widetilde{x},\ u^j \widetilde{y},\ u^j \widetilde{z} \in \mathcal{O}_{W_i}[u]$$

for any sufficiently large integer j, so $\operatorname{Spec} R_i$ is the affine chart $(u^j \neq 0)$ of the blow-up of $\operatorname{Spec} \mathcal{O}_{W_i}[u]$ along $(u^j, u^j x', u^j y', u^j z', u^j \widetilde{x}, u^j \widetilde{y}, u^j \widetilde{z})$, and Γ_i acts on R_i. Let $\widehat{R}_i$ be the completion of R_i at the maximal ideal $(u, \widetilde{x}, \widetilde{y}, \widetilde{z}, y')$, which is generated by $u, \widetilde{x}, \widetilde{y}, \widetilde{z}$ since $\lambda_i(0) = 1$ in case (i). Then $\widehat{R}_i$ is a quotient of $\mathbb{C}[[\widetilde{x}, \widetilde{y}, \widetilde{z}, u]]$, into which the regular parameters x_i, y_i, z_i, s_i of $\mathbb{C}[[x_1, \dots, x_4]]$ pull back, and $u^{N_1} \widetilde{f}_i = \pi_i^* f_i = 0$ in $\widehat{R}_i$. Since completion is flat over the domain R_i, u is a non-zerodivisor in $\widehat{R}_i$, so $\widetilde{f}_i = 0$ in $\widehat{R}_i$. Moreover $\dim \widehat{R}_i = 3$, since R_i is an affine chart of a birational model of the 3-dimensional ring $\mathcal{O}_{W_i}[u]$ and the origin has residue field $\mathbb{C}$. Hence $\widehat{R}_i$ is a 3-dimensional quotient of the 3-dimensional domain $\mathcal{O}_{Z_i}$, so $\widehat{R}_i = \mathcal{O}_{Z_i}$. Moreover $R_i[u^{-1}] = \mathcal{O}_{W_i}[s_i^{-1}][u]$, that is, $\operatorname{Spec} R_i[u^{-1}] \to W_i \backslash (s_i = 0)$ is the finite étale morphism given by $u^2 = s_i$ followed by invertible changes of coordinates. Since W_i is regular outside o_i, $R_i[u^{-1}]$ is regular. Since $\mathcal{O}_{Z_i}$ is the completion of a local ring of the excellent ring R_i, Z_i is regular outside U_i. At a point of U_i where U_i is regular, Z_i is regular. In cases (ii), (iii), and (i) with $\varsigma = 0$, U_i is regular outside 0. In case (i) with $\varsigma = 1$, $\operatorname{Sing}(U_i)$ is the line $(\widetilde{x} = \widetilde{y} = u = 0)$, and by (b), along $(\widetilde{x} = \widetilde{y} = u = 0)$,

$$\frac{\partial \widetilde{f}_i}{\partial u} = \Theta_{0,i}(\widetilde{z}) \in \widetilde{z}^{n_0} \mathbb{C}[[\widetilde{z}]]^\times,$$

so Z_i is regular along the line outside 0. Hence $(Z_i \ni 0)$ is an isolated cDV germ, thus an isolated Gorenstein terminal hypersurface threefold germ. By (b), the quadratic part of $\widetilde{f}_i$ restricted to $(u = 0)$ is the quadratic part of $\Xi_{0,i}$, which has rank at least 2. Hence $(Z_i \ni 0)$ has quadratic rank at least 2. Since Γ_i acts faithfully on $\mathcal{K}_i(u) \subset \operatorname{Frac} \mathcal{O}_{Z_i}$, it acts faithfully on Z_i.

By (b), $\widetilde{f}_i(\widetilde{x},0,0,0)$ is $\widetilde{x}^2$ times a unit, so Weierstrass division by $\widetilde{f}_i$ with respect to $\widetilde{x}$ exists. It can be carried out layer by layer in u: the u^j-layer of the remainder of $\pi_i^*h_i$ depends only on the $u^{j'}$-layers, $j' \le j$, of $\pi_i^*h_i$ and $\widetilde{f}_i$, and it is a polynomial whose coefficients are polynomials in c_i^{-1} and in the coefficients of the $u^{j'}$-layers, $j' \le j$, of $\pi_i^*h_i$ and $\widetilde{f}_i$, since the u^0-layer $\Xi_{0,i}$ of $\widetilde{f}_i$ has degree 2 in $\widetilde{x}$ with leading coefficient c_i, so that each step is a division of polynomials by $\Xi_{0,i}$. Since $N' + N_1 < N_4$, by (G.6) and the principle above, possibly passing to a subsequence, the layers of index less than N' of the remainder of $\pi_i^*h_i$ vanish and the $u^{N'}$-layer is non-zero. Let l_i be $u^{-N'}$ times the remainder. Then l_i is a Γ_i-semi-invariant series of degree at most 1 in $\widetilde{x}$ whose u^0-layer is non-zero, and $\pi_i^*h_i \equiv u^{N'}l_i$ modulo $\widetilde{f}_i$. If $l_i \in (u, \widetilde{f}_i)$, then $l_i \in (u)$ by the uniqueness of the remainder, which is not the case. Hence, in $\mathcal{O}_{Z_i}$,

$$\pi_i^*h_i = u^{N'}l_i, \qquad u \nmid l_i, \tag{G.11}$$

and l_i is a mark on $(Z_i \ni 0)$. Any coefficient of $\widetilde{f}_i$ and of l_i is a polynomial, independent of i, in finitely many coefficients of the tuples of **Step 2**, in c_i^{-1} and in the inverse of the determinant of the linear parts of x_i, y_i, z_i, s_i. Hence, for any positive integer ℓ, the ℓ-jet of $(\widetilde{f}_i, l_i)$ is the image of a jet of the tuples of **Step 2** under a morphism, independent of i, defined where c_i and that determinant do not vanish, and by Definition 2.5 the $(\widetilde{f}_i, l_i)$ have the compatible generic limit (f'_*, l_*) whose ℓ-jet is the image under that morphism of the corresponding jet of the limit of **Step 2**. Then $\pi_*'^*f_* = u^{N_1}f'_*$ and $\pi_*'^*h_* \equiv u^{N'}l_*$ modulo f'_*, since $\phi - \phi^{<N_4}$ and $\lambda - \lambda^{<N_4}$ consist of terms of degree at least N_4, which pull back into $u^{N_4}\Bbbk[[\widetilde{x},\widetilde{y},\widetilde{z},u]]$, so that $\pi_*'^*\theta \equiv \pi_*^*\theta$ modulo u^{N_4} for any $\theta \in \Bbbk[[x_1,\dots,x_4]]$ (in cases (ii) and (iii), $\pi'_* = \pi_*$), and $N_4 > N_1 + N'$. As for $R_i[u^{-1}]$ above, over $(u \ne 0)$ the restriction of π'_* to $(f'_* = 0) \to W_*$ is the finite étale morphism given by $u^2 = s$ followed by invertible changes of coordinates. Since $\phi^{<N_4} \in (z')^2$ in case (i), over $(u \ne 0)$ the inverse image of $C = (x = y = z = 0)$ under π'_* is the curve $(\widetilde{x} = \widetilde{y} = \widetilde{z} = 0)$, which is regular with parameter u. This proves (4).

Step 4. In this step we prove (2) and all of (1) except the equality of quotient values.

Let $R_{i,0}$ be the local ring of R_i at the origin, and let P be a prime divisor over Z_i with center 0, or a component of U_i or of $\operatorname{div}(l_i)$. Then mult_P restricts to a divisorial valuation of $\mathcal{K}_i(u)$ with value group $\mathbb{Z}$, as follows. If P has center 0, each ideal $\{\operatorname{mult}_P \ge j\}$ of $\mathcal{O}_{Z_i}$ contains a power of the maximal ideal, so any element of $\mathcal{O}_{Z_i}$ is congruent modulo it to an element of $R_{i,0}$. Hence the restriction of mult_P to $\mathcal{K}_i(u)$ has value group $\mathbb{Z}$, its residue field equals that of mult_P, and its decomposition and inertia groups in Γ_i equal those of P. If P is a component of U_i or of $\operatorname{div}(l_i)$, the ideal of P contains the non-zero element u or $u^{N'}l_i = \pi_i^*h_i$ of R_i, so it contracts to a height-one prime $\mathfrak{p}$ of $R_{i,0}$, and since the formal fibers of the excellent ring $R_{i,0}$ are regular, a uniformizer of $(R_{i,0})_{\mathfrak{p}}$ is a uniformizer of P, so mult_P restricts to the valuation of $\mathfrak{p}$. The restriction of mult_P to $\mathcal{K}_i$ equals $\rho \operatorname{mult}_F$ for a prime divisor F over W_i and an integer $\rho \in \{1,2\}$, the ramification index of P over F. If P has center 0 on Z_i, then $\operatorname{mult}_P x_i, \operatorname{mult}_P y_i, \operatorname{mult}_P z_i, \operatorname{mult}_P s_i > 0$ since $\pi_i^*x_i, \dots, \pi_i^*s_i \in (u)$, so F has center o_i.

By (G.9), ramification indices multiply in the tower $\mathcal{K}_i^G \subset \mathcal{K}_i \subset \mathcal{K}_i(u)$, so $e_P = \rho e_F$ if P has center 0.

By (G.5) and (a), since the replacement of y' by $\lambda_i \cdot (y' - \phi_i)$ has unit Jacobian, the generators of ω_{W_i} and ω_{Z_i} given by residues satisfy

$$\pi_i^* \operatorname{Res}\Big(\frac{dx_i \wedge dy_i \wedge dz_i \wedge ds_i}{f_i}\Big) \in u^\kappa \cdot \mathcal{O}_{Z_i}^\times \cdot \operatorname{Res}\Big(\frac{d\widetilde{x} \wedge d\widetilde{y} \wedge d\widetilde{z} \wedge du}{\widetilde{f}_i}\Big).$$

Thus $K_{Z_i} = \pi_i^*K_{W_i} + \kappa U_i$. Let

$$\beta := \frac{N'}{m} - \kappa, \qquad D'_i := \frac{1}{m}\operatorname{div}(l_i) + \beta U_i.$$

Since l_i and u are eigenfunctions, D_i' is Γ_i-invariant. By (G.11),

$$K_{Z_i} + D_i' = \pi_i^*\Big(K_{W_i} + \frac{1}{m}\operatorname{div}(h_i)\Big).$$

Since $Z_i \to W_i$ is the composite

$$Z_i \to \operatorname{Spec} R_i \to \operatorname{Spec} \mathcal{O}_{W_i}[u] \to W_i$$

of the completion, the birational chart, and the finite morphism, log discrepancies multiply by the ramification index. Together with (G.11) we obtain, if P has center 0,

$$\begin{aligned} \rho a(F, W_i, 0) &= a(P, Z_i, 0) + \kappa \operatorname{mult}_P u, \\ \text{and} \quad \rho \operatorname{mult}_F h_i &= \operatorname{mult}_P l_i + N' \operatorname{mult}_P u, \end{aligned} \tag{G.12}$$

and, for both kinds of P,

$$a(P, Z_i, D_i') = \rho a\Big(F, W_i, \frac{1}{m}\operatorname{div}(h_i)\Big). \tag{G.13}$$

This proves (2).

Since $N' \geq 0$, $\beta \in \frac{1}{m}\mathbb{Z}$ and $\beta \geq -\kappa$. By (G.11), $D_i' = \frac{1}{m}\operatorname{div}(\pi_i^* h_i) - \kappa U_i$ is the pull-back of the $\mathbb{Q}$-divisor

$$D_{R_i} := \frac{1}{m}\operatorname{div}(\pi_i^* h_i) - \kappa \operatorname{div}(u) \quad \text{on } \operatorname{Spec} R_{i,0}.$$

Let $Y \to \operatorname{Spec} R_{i,0}$ be a log resolution of $(\operatorname{Spec} R_{i,0}, D_{R_i})$ and $Y' := Y \times_{\operatorname{Spec} R_{i,0}} Z_i$. Since $R_{i,0}$ is excellent, $R_{i,0} \to \mathcal{O}_{Z_i}$ is flat with regular fibers, so $Y' \to Z_i$ is a log resolution of (Z_i, D_i'), and, as for the components of U_i above, any prime divisor P on Y' satisfies

$$a(P, Z_i, D_i') = a(E, \operatorname{Spec} R_{i,0}, D_{R_i})$$

for the restriction E of P, a prime divisor on Y. Since $(W_i, \frac{1}{m}\operatorname{div}(h_i))$ is klt near o_i, as for (G.13),

$$a(E, \operatorname{Spec} R_{i,0}, D_{R_i}) = \rho a\Big(F, W_i, \frac{1}{m}\operatorname{div}(h_i)\Big) > 0,$$

where $\rho \operatorname{mult}_F$ is the restriction of mult_E to $\mathcal{K}_i$. Hence every prime divisor P on the log resolution Y' satisfies $a(P, Z_i, D_i') > 0$, so (Z_i, D_i') is sub-klt near 0. In particular any component of D_i' has coefficient less than 1. By **Step 3**, U_i is reduced with one or two components, so $(u) \subset \mathcal{O}_{Z_i}$ is a radical ideal. Since $u \nmid l_i$, some component P_1 of U_i satisfies $\operatorname{mult}_{P_1} l_i = 0$, and the coefficient of D_i' along P_1 is β. Thus $\beta < 1$, and

$$\beta \in \frac{1}{m}\mathbb{Z} \cap [-\kappa, 1).$$

Step 5. In this step we prove (3) and the equality of quotient values in (1).

Let F be a computing lift of b_i, and let P be a divisorial valuation of $\mathcal{K}_i(u)$ extending mult_F, with ramification index $\rho \in \{1, 2\}$. Then

$$2 \operatorname{mult}_P u = \operatorname{mult}_P s_i = \rho \operatorname{mult}_F s_i, \qquad \text{so} \quad \operatorname{mult}_P u \leq \operatorname{mult}_F s_i.$$

In cases (ii) and (iii), by (G.3), each of $\operatorname{mult}_P \widetilde{x}, \operatorname{mult}_P \widetilde{y}, \operatorname{mult}_P \widetilde{z}$ equals one of $\rho \operatorname{mult}_F x_i - j \operatorname{mult}_P u$, $\rho \operatorname{mult}_F y_i - j \operatorname{mult}_P u$, $\rho \operatorname{mult}_F z_i - j \operatorname{mult}_P u$ for some integer $j \leq N_3$. In case (i), each of $\operatorname{mult}_P x', \operatorname{mult}_P y', \operatorname{mult}_P z'$ equals one of

$$\rho \operatorname{mult}_F x_i - j \operatorname{mult}_P u, \qquad \rho \operatorname{mult}_F y_i - j \operatorname{mult}_P u, \quad \text{and} \quad \rho \operatorname{mult}_F z_i - j \operatorname{mult}_P u$$

for some integer $j \leq r + \nu$. Since $\phi_i \in (z')^2$ and λ_i is a polynomial in y', z', u, the new coordinate $y_1' = \lambda_i \cdot (y' - \phi_i)$ has value at least $\min\{\operatorname{mult}_P y', \operatorname{mult}_P z'\}$. Finally, ψ_2 subtracts at most $(n' + 2) \operatorname{mult}_P u$. In any case, by (G.8),

$$\begin{aligned} \min\{\operatorname{mult}_P \widetilde{x}, \operatorname{mult}_P \widetilde{y}, \operatorname{mult}_P \widetilde{z}\} &\geq \rho \min\{\operatorname{mult}_F x_i, \operatorname{mult}_F y_i, \operatorname{mult}_F z_i\} - N_3 \operatorname{mult}_P u \\ &> \rho T - \frac{3MN_3}{\eta_m} \geq 0. \end{aligned}$$

Thus mult_P is non-negative on R_i and positive on $u, \widetilde{x}, \widetilde{y}, \widetilde{z}$, so P is a prime divisor over Z_i with center 0. Hence any computing lift of b_i has center 0 on Z_i.

Let P be a prime divisor over Z_i with center 0, and F the restriction of P. By (G.13), $e_P = \rho e_F$ and (G.1),

$$\frac{a(P, Z_i, D_i')}{e_P} = \frac{a(F, W_i, \frac{1}{m}\operatorname{div}(h_i))}{e_F} \geq b_i,$$

with equality if and only if F is a computing lift of b_i. Since any computing lift of b_i extends to a prime divisor over Z_i with center 0,

$$\operatorname{mld}((Z_i \ni 0, D_i')/\Gamma_i) = b_i,$$

and any computing lift P of $\operatorname{mld}((Z_i \ni 0, D_i')/\Gamma_i)$ restricts to a computing lift F of b_i. For such P, $\operatorname{mult}_P u = \rho \operatorname{mult}_F s_i/2$ and $e_P = \rho e_F$, so by (G.8),

$$\frac{\operatorname{mult}_P u}{e_P} = \frac{\operatorname{mult}_F s_i}{2e_F} \leq \frac{3}{2\eta_m}, \qquad \operatorname{mult}_P u \leq \operatorname{mult}_F s_i \leq \frac{3M}{\eta_m}.$$

This proves (3) and completes the proof of (1) and of the theorem. □

G.2. **The signed A selection.** We prove Theorem 18.2 of Section 18.

Proof of Theorem 18.2. Let $f_i := xy + g_i(z, u)$, let $g_* := z^d \vartheta(z, u)$ and $f_* := xy + g_*$, so that (f_*, h_*) is the compatible generic limit of hypothesis (4) of Theorem 18.2, and let $\mathfrak{n}$ and $\mathfrak{n}_*$ be the maximal ideals of $\mathbb{C}[[x, y, z, u]]$ and $\Bbbk[[x, y, z, u]]$. Passing to an infinite subfamily keeps (f_*, h_*) a compatible generic limit (Lemma 2.6(2)). By Lemma 2.7, for finitely many coefficients of (f_*, h_*) that vanish and one coefficient of (f_*, h_*) that is non-zero, the corresponding coefficients of (f_i, h_i) vanish and the corresponding one is non-zero for all but finitely many i.

Step 1. In this step we reduce to a constant quotient value and show that the orders of x, y, and z along the computing lifts F_i of the theorem tend to $+\infty$.

Let N be a positive integer. Any computing lift F of b_i with $a(F, W_i, 0) \leq N$ and $\operatorname{mult}_F u \leq N$ satisfies

$$\operatorname{mult}_F h_i = m\big(a(F, W_i, 0) - \beta \operatorname{mult}_F u - b_i e_F\big) \leq m(1 + |\beta|)N,$$

so if infinitely many i had such a lift, the theorem would hold with $C_0 := m(1 + |\beta|)N$. Suppose that the theorem fails. Then the following holds.

(∗) For any positive integer N, there exist only finitely many i such that b_i has a computing lift F with $a(F, W_i, 0) \leq N$ and $\operatorname{mult}_F u \leq N$.

Since W_i is Gorenstein and $\beta \in \frac{1}{m}\mathbb{Z}$, $a(F, W_i, D_i) \in \frac{1}{m}\mathbb{Z}$ for any prime divisor F over W_i. Since $e_F \leq |G| \leq M$,

$$b_i \in \frac{1}{m \cdot M!}\mathbb{Z} \cap (0, B_0],$$

which is a finite set. Passing to an infinite subfamily, we may assume that $b_i = b_0$ for any i, where $b_0 > 0$. Since $\operatorname{mult}_{F_i} u \leq R'$, applying property (∗) with any $N \geq R'$, we obtain $a(F_i, W_i, 0) \to +\infty$. Since $e_{F_i} \leq M$ and $b_0 \leq B_0$,

$$a(F_i, W_i, 0) = \frac{1}{m}\operatorname{mult}_{F_i} h_i + \beta \operatorname{mult}_{F_i} u + b_0 e_{F_i} \leq \frac{1}{m}\operatorname{mult}_{F_i} h_i + |\beta| R' + B_0 M, \tag{G.14}$$

so $\operatorname{mult}_{F_i} h_i \to +\infty$.

Let $k := \operatorname{mult}_0 g_*$. Then $k \geq d \geq 2$, and we may assume that $\operatorname{mult}_0 g_i = k$ for any i. Let $j_i := \operatorname{mult}_{F_i} \operatorname{Jac}_{W_i}$. By Lemma 5.6,

$$j_i \leq \frac{k}{2}\big(a(F_i, W_i, 0) - 1\big) \leq \frac{k}{2} a(F_i, W_i, 0).$$

Let $Y \to W_*$ be a log resolution of $(W_*, \operatorname{div}(h_*|_{W_*}))$. Only finitely many prime divisors E on Y are exceptional over W_* or components of the strict transform of $\operatorname{div}(h_*|_{W_*})$, by hypothesis (4), $a(E, W_*, \frac{1}{m}\operatorname{div}(h_*|_{W_*})) > 0$ for those whose centers on W_* meet $(u \neq 0)\backslash C$, and any other prime divisor on Y has log discrepancy 1 with respect to $(W_*, \mu \operatorname{div}(h_*|_{W_*}))$ for any μ. Thus there exists

a rational number $\mu > \frac{1}{m}$ such that $(W_*, \mu \operatorname{div}(h_*|_{W_*}))$ is klt on $(u \neq 0)\backslash C$. Let $\theta_1 < \theta_2 < 1$ be positive rational numbers with

$$\epsilon := \theta_1\mu - \frac{1}{m} - (1-\theta_1)\Big(1 + \frac{k}{m}\Big) > 0;$$

they exist since the left side tends to $\mu - \frac{1}{m} > 0$ as θ_1 tends to 1.

Let $\Sigma_* := W_* \cap (C \cup (u=0))$, a closed subset of W_* containing the closed point. The germ W_* is normal, since $\operatorname{Sing}(W_*) \subset (x = y = 0,\ z\vartheta(z,u) = 0)$, and $(W_*, \mu \operatorname{div}(h_*|_{W_*}))$ is klt on $W_*\backslash\Sigma_*$. By Lemma 6.1(3), $\mathcal{J}_* := \mathcal{J}(f_*^{\theta_2} h_*^{\theta_2\mu})$ is the unit ideal near any point of $W_*\backslash\Sigma_*$. Thus the zero locus of $(f_*) + \mathcal{J}_*$ is contained in Σ_*, which is the zero locus of (f_*, ux, uy, uz), and there exists a positive integer N_1 with

$$(ux, uy, uz)^{N_1} \subset (f_*) + \mathcal{J}_*.$$

Let $\varphi \in \{x, y, z\}$, let L be a positive integer, and let $\mathcal{M}_L$ be the set of monomials of degree less than L in x, y, z, u. For a vector $\nu = (\nu_{\mathbf{x}})_{\mathbf{x}\in\mathcal{M}_L}$ let

$$\nu \cdot \mathcal{M}_L := \sum_{\mathbf{x}\in\mathcal{M}_L} \nu_{\mathbf{x}}\mathbf{x},$$

and let Θ_φ be the polynomial, in the sense of Lemma 6.1 with parameters indexed by $\mathcal{M}_L$, with

$$\Theta_\varphi((f,h),\nu) = (u\varphi)^{N_1} - (\nu \cdot \mathcal{M}_L)f$$

for any pair (f, h) of power series. Taking for $\nu_\varphi \cdot \mathcal{M}_L$ the part of degree less than L of an element $\varkappa \in \Bbbk[[x, y, z, u]]$ with $(u\varphi)^{N_1} - \varkappa f_* \in \mathcal{J}_*$, we obtain $\nu_\varphi \in \Bbbk^{\mathcal{M}_L}$ with

$$\Theta_\varphi((f_*, h_*), \nu_\varphi) \in \mathcal{J}_* + \mathfrak{n}_*^L.$$

By Lemma 6.1(1)(2), with the exponents $(\theta_1, \theta_1\mu)$ on the members and $(\theta_2, \theta_2\mu)$ on the limit, for any $i \gg 0$ there exists $\nu_{\varphi,i} \in \mathbb{C}^{\mathcal{M}_L}$ such that

$$\Theta_{\varphi,i} := (u\varphi)^{N_1} - (\nu_{\varphi,i} \cdot \mathcal{M}_L)f_i$$

satisfies the following: any prime divisor E over $\operatorname{Spec}\mathbb{C}[[x, y, z, u]]$ with center the closed point and $\operatorname{mult}_E \Theta_{\varphi,i} < L$ satisfies

$$\operatorname{mult}_E \Theta_{\varphi,i} + a(E, \operatorname{Spec}\mathbb{C}[[x, y, z, u]], 0) > \theta_1 \operatorname{mult}_E f_i + \theta_1\mu \operatorname{mult}_E h_i,$$

as $\operatorname{mult}_E(\mathfrak{n}^L) \geq L$. The restriction of $\Theta_{\varphi,i}$ to W_i is that of $(u\varphi)^{N_1}$, which is non-zero. We regard F_i as a prime divisor over

$$(f_i = 0) \subset \operatorname{Spec}\mathbb{C}[[x, y, z, u]]$$

by Lemma 2.12(3). By Lemma 5.5(1), with $g = \Theta_{\varphi,i}$ and $\rho = L$, if $N_1 \operatorname{mult}_{F_i}(u\varphi) < L$, then

$$(\theta_1\mu - 1 + \theta_1) \operatorname{mult}_{F_i} h_i < (2-\theta_1)N_1 \operatorname{mult}_{F_i}(u\varphi) + a(F_i, W_i, 0) + 2(1-\theta_1)j_i + 1 - \theta_1. \tag{G.15}$$

Since $j_i \leq \frac{k}{2}a(F_i, W_i, 0)$, (G.15) and (G.14) imply

$$\epsilon \operatorname{mult}_{F_i} h_i < (2-\theta_1)N_1 \operatorname{mult}_{F_i}(u\varphi) + L_1, \tag{G.16}$$

where $L_1 := (1 + (1-\theta_1)k)(|\beta|R' + B_0M) + 1$. If $N_1 \operatorname{mult}_{F_i}(u\varphi) < L$ for infinitely many i, then (G.16) bounds $\operatorname{mult}_{F_i} h_i$ on them, as $\epsilon > 0$, which contradicts $\operatorname{mult}_{F_i} h_i \to +\infty$. Since L is arbitrary,

$$\operatorname{mult}_{F_i}(u\varphi) = \operatorname{mult}_{F_i} u + \operatorname{mult}_{F_i} \varphi \to +\infty.$$

Since $\operatorname{mult}_{F_i} u \leq R'$, we obtain $\operatorname{mult}_{F_i} x, \operatorname{mult}_{F_i} y, \operatorname{mult}_{F_i} z \to +\infty$.

Step 2. In this step we pass to a chart of a fixed weighted dilation of the axis on which the limit equation becomes $xy + z^d$.

All coordinate changes below are given by algebraic power series, hence are defined on G-stable pointed étale neighborhoods of 0, to which we pull back W_i and h_i, and by Lemma 2.12(2)(3) passing to such a neighborhood changes neither the quotient values nor the prime divisors with center the closed point, their log discrepancies, the orders of functions along them and their inertia groups.

Modulo f_i, any element of $\mathbb{C}[[x,y,z,u]]$ is congruent to a unique element without monomials divisible by xy, obtained by replacing $x^a y^b$ with $(-g_i)^{\min\{a,b\}} x^{a-\min\{a,b\}} y^{b-\min\{a,b\}}$. The coefficients of the representative in each degree are polynomials in finitely many coefficients of the given element and of g_i. Possibly replacing h_i with the representative of h_i modulo f_i, which does not change $h_i|_{W_i}$, and h_* with the representative of h_* modulo f_*, we may assume that no monomial of h_i or of h_* is divisible by xy. After each coordinate change below we again replace h_i and h_* with their representatives modulo f_i and f_*.

Write $\vartheta(0,u) = u^{n_\vartheta}\vartheta_0(u)$ with $n_\vartheta \in \mathbb{N}$ and $\vartheta_0(0) \neq 0$. Let $N_z > n_\vartheta$ be an integer, with $dN_z + n_\vartheta$ even if d is odd, fixed but large for the finitely many conditions below. We define integers n, k_z, N_x, N_y as follows.

(1) If no element of G exchanges x and y, let $n := 1$, $k_z := N_z$, and let N_x, N_y be positive integers with $N_x + N_y = dN_z + n_\vartheta$.
(2) If some element of G exchanges x and y and $dN_z + n_\vartheta$ is even, let $n := 1$, $k_z := N_z$, and $N_x = N_y := (dN_z + n_\vartheta)/2$.
(3) Otherwise d is even, n_ϑ is odd and some element of G exchanges x and y. Let $n := 2$, $k_z := 2N_z$, and $N_x = N_y := dN_z + n_\vartheta$.

In each case $N_x + N_y = dk_z + nn_\vartheta$ and $k_z > nn_\vartheta$. Let

$$\psi : (X,Y,Z,s) \to (x,y,z,u) = (s^{N_x}X,\ s^{N_y}Y,\ s^{k_z}Z,\ s^n).$$

Then $f_i \circ \psi = s^{dk_z+nn_\vartheta} f_i^{(1)}$, where

$$f_i^{(1)} := XY + g_i^{(1)}(Z,s) \qquad \text{and} \qquad g_i^{(1)} := s^{-dk_z-nn_\vartheta} g_i(s^{k_z}Z, s^n).$$

A monomial $z^j u^e$ of g_i becomes $s^{jk_z+ne}Z^j$, and $jk_z + ne \leq dk_z + nn_\vartheta$ holds only for finitely many (j,e), all with $j < d$ or with $j = d$ and $e \leq n_\vartheta$, as $k_z > nn_\vartheta$. Since $g_* = z^d\vartheta(z,u)$, the coefficients in g_* of the monomials $z^j u^e$ with such (j,e) vanish, except the coefficient $\vartheta_0(0)$ of $z^d u^{n_\vartheta}$. Thus, for any $i \gg 0$, $g_i^{(1)} \in \mathbb{C}[[Z,s]]$ and $g_i^{(1)}(Z,0) = a_i Z^d$, where a_i is the coefficient of $z^d u^{n_\vartheta}$ in g_i, and we may assume $a_i \neq 0$. Moreover let $f_*^{(1)} := XY + g_*^{(1)}$, where

$$g_*^{(1)} := s^{-dk_z-nn_\vartheta} g_*(s^{k_z}Z, s^n) = s^{-nn_\vartheta} Z^d \vartheta(s^{k_z}Z, s^n).$$

Let N_0 be the largest integer such that s^{N_0} divides $h_* \circ \psi$. Only finitely many monomials of h_i become monomials of s-order less than N_0, and their coefficients in h_* vanish since ψ maps distinct monomials to distinct monomials, so s^{N_0} divides $h_i \circ \psi$ for any $i \gg 0$. Let

$$h_i^{(1)} := s^{-N_0} h_i \circ \psi \qquad \text{and} \qquad h_*^{(1)} := s^{-N_0} h_* \circ \psi.$$

Since $h_*^{(1)}(X,Y,Z,0) \neq 0$, we may assume $h_i^{(1)}(X,Y,Z,0) \neq 0$. The coefficients of $f_i^{(1)}$ and $h_i^{(1)}$ are coefficients of f_i and h_i, so $(f_*^{(1)}, h_*^{(1)})$ is a compatible generic limit of the pairs $(f_i^{(1)}, h_i^{(1)})$.

We rename s as u. Let $W_i^{(1)} := (f_i^{(1)} = 0) \ni 0$. Then $W_i^{(1)}$ has at most an isolated singularity at 0, hence is normal, if $g_i^{(1)}$ is reduced, and $g_i^{(1)}$ is reduced since g_i is reduced, ψ is étale over $(u \neq 0)$, and $g_i^{(1)}(Z,0) \neq 0$. The map ψ sends 0 to o_i and restricts to a dominant morphism $W_i^{(1)} \to W_i$ of degree n. If $n = 1$, G acts on X,Y,Z,u through ψ, monomially on $\{X,Y\}$ as $N_x = N_y$ when G exchanges x and y. Let $G^{(1)} := G$. If $n = 2$, let

$$G^{(1)} := \{(g,\alpha) \in G \times \mathbb{C}^* \mid \alpha^2 = \chi_u(g)\},$$

acting by $s \to \alpha s$ and on X,Y,Z through ψ. It is abelian of order $2|G| \leq 2M$, and it acts faithfully on $W_i^{(1)}$, since $\operatorname{div}(u|_{W_i}) = (xy + g_i(z,0) = 0)$ is reduced, so $u|_{W_i} \neq \phi^2$ for any rational function ϕ on W_i. In both cases $f_i^{(1)}$ and $h_i^{(1)}$ are $G^{(1)}$-semi-invariant with characters independent of i.

Let ω_i be the residue of $dx \wedge dy \wedge dz \wedge du/f_i$, a generator of ω_{W_i}, and $\omega_i^{(1)}$ that of $dX \wedge dY \wedge dZ \wedge du/f_i^{(1)}$. Since

$$\psi^*(dx \wedge dy \wedge dz \wedge du) = ns^{N_x+N_y+k_z+n-1} dX \wedge dY \wedge dZ \wedge ds,$$

we obtain

$$\psi^*\omega_i = nu^{\kappa_0}\omega_i^{(1)}, \qquad \kappa_0 := k_z + n - 1 \geq 0,$$

so the crepant pull-back of K_{W_i} is $K_{W_i^{(1)}} - \kappa_0 \operatorname{div}(u)$.

Let F' be a prime divisor over $W_i^{(1)}$, let F be the restriction of F', and let $\rho \in \{1, \dots, n\}$ be the ramification index of F' over F, so that $\operatorname{mult}_{F'}(\phi \circ \psi) = \rho \operatorname{mult}_F \phi$ for any rational function ϕ on W_i. Since ψ is equivariant for the projection $G^{(1)} \to G$ and induces a birational map $W_i^{(1)}/G^{(1)} \dashrightarrow W_i/G$, the inertia orders $e_{F'}$ and e_F are the ramification indices of F' and F over the prime divisor over W_i/G to which F restricts. Since $\psi^* h_i = u^{N_0} h_i^{(1)}$ and $\psi^* u = u^n$, we obtain

$$\begin{gathered} e_{F'} = \rho e_F, \qquad \rho a(F, W_i, 0) = a(F', W_i^{(1)}, 0) + \kappa_0 \operatorname{mult}_{F'} u, \\ \rho \operatorname{mult}_F h_i = \operatorname{mult}_{F'} h_i^{(1)} + N_0 \operatorname{mult}_{F'} u, \qquad \rho \operatorname{mult}_F u = n \operatorname{mult}_{F'} u, \end{gathered} \tag{G.17}$$

where on the left u is the coordinate of W_i. Let $\beta_1 := n\beta + \frac{N_0}{m} - \kappa_0 \in \frac{1}{m}\mathbb{Z}$ and

$$D_i^{(1)} := \frac{1}{m} \operatorname{div}(h_i^{(1)}) + \beta_1 \operatorname{div}(u).$$

By (G.17),

$$\frac{a(F', W_i^{(1)}, D_i^{(1)})}{e_{F'}} = \frac{a(F, W_i, D_i)}{e_F}. \tag{G.18}$$

Relations (G.17) and (G.18) only use that $\psi^*\omega_i$, $\psi^* h_i$, and $\psi^* u$ are the products of u^{κ_0}, u^{N_0}, and u^n with units. Thus they hold for the composite of ψ with the coordinate changes below, with other non-negative integers κ_0, N_0, n independent of i and with $\beta_1 := n\beta + \frac{N_0}{m} - \kappa_0$. For such a composite, assume that $f_i^{(1)}(X, Y, Z, 0) = XY + a_i Z^d$ with $a_i \neq 0$, and that $h_i^{(1)}(X, Y, Z, 0) \neq 0$ has no monomial divisible by XY. Then the following hold.

(a) $D_i^{(1)}$ is sub-klt and $\beta_1 < 1$.

(b) $\operatorname{mld}((W_i^{(1)} \ni 0, D_i^{(1)})/G^{(1)}) = b_0$, computed by a prime divisor F_i' over $W_i^{(1)}$ whose restriction is F_i, with $\operatorname{mult}_{F_i'} u \leq R'$ and with $a(F_i', W_i^{(1)}, 0)$, $\operatorname{mult}_{F_i'} X$, $\operatorname{mult}_{F_i'} Y$, and $\operatorname{mult}_{F_i'} Z$ tending to $+\infty$.

(c) Property (∗) holds for $(W_i^{(1)}, D_i^{(1)}, G^{(1)})$.

Proof of (a)–(c). Since any prime divisor over $W_i^{(1)}$ restricts to a prime divisor over W_i, (G.18) and the sub-klt property of D_i imply that $D_i^{(1)}$ is sub-klt. The subscheme $T_i := (u = 0) \subset W_i^{(1)}$, which is $(XY + a_i Z^d = 0) \subset (u = 0)$, is a prime divisor, $h_i^{(1)}|_{T_i} \neq 0$ since no non-zero series without monomials divisible by XY vanishes on T_i, and $a(T_i, W_i^{(1)}, D_i^{(1)}) = 1 - \beta_1$, so $\beta_1 < 1$. Since any prime divisor over $W_i^{(1)}$ with center 0 restricts to a prime divisor over W_i with center o_i, $\operatorname{mld}((W_i^{(1)} \ni 0, D_i^{(1)})/G^{(1)}) \geq b_0$ by (G.18). A prime divisor F_i' over $W_i^{(1)}$ whose restriction is F_i exists since $W_i^{(1)} \to W_i$ is dominant and generically finite. By (G.17), $\operatorname{mult}_{F_i'} u = \rho \operatorname{mult}_{F_i} u/n \leq R'$,

$$\operatorname{mult}_{F_i'} X = \rho \operatorname{mult}_{F_i} x - N_x \operatorname{mult}_{F_i'} u \geq \operatorname{mult}_{F_i} x - N_x R',$$

likewise for Y and Z, and $a(F_i', W_i^{(1)}, 0) \geq a(F_i, W_i, 0) - \kappa_0 R'$. Thus F_i' has center 0 for $i \gg 0$, and it computes b_0 by (G.18). Finally, if F' is a computing lift of b_0 over $W_i^{(1)}$ with $a(F', W_i^{(1)}, 0) \leq N$

and $\mathrm{mult}_{F'} u \le N$, then, by (G.17) and (G.18), the restriction F of F' is a computing lift of b_0 over W_i with $a(F, W_i, 0) \le (1+\kappa_0)N$ and $\mathrm{mult}_F u \le nN$, so property (∗) for W_i implies (c).

The assumption holds for ψ. We replace W_i, f_i, g_i, h_i, G, F_i, and D_i by $W_i^{(1)}$, $f_i^{(1)}$, $g_i^{(1)}$, $h_i^{(1)}$, $G^{(1)}$, F_i', and $D_i^{(1)}$, write again x, y, z for X, Y, Z, and replace M by $2M$. By (b), the conclusions of **Step 1** on $\mathrm{mult}_{F_i} u$, $a(F_i, W_i, 0)$ and the orders of x, y, z along F_i persist.

Write

$$g_i = \sum_{j<d} a_{i,j}(u) z^j + z^d \Psi_i(z, u)$$

with $a_{i,j} \in \mathbb{C}[[u]]$ and $\Psi_i \in \mathbb{C}[[z, u]]$, and likewise $g_* = z^d \Psi_*$. Then $\Psi_*(0,0) = \vartheta_0(0) \ne 0$ and $\Psi_i(0,0) = a_i \ne 0$. Since z^d and any monomial of $z^d \Psi_i$ have the character of f_i, Ψ_i is G-invariant.

- If $d \ge 3$, then, as $g_i(z,0) = a_i z^d$ with $a_i \ne 0$, the Weierstrass preparation theorem gives $g_i = \upsilon_i g_i'$ with $\upsilon_i \in \mathbb{C}[[z,u]]$ a unit and $g_i' \in \mathbb{C}[[u]][z]$ monic of degree d in z with $g_i'(z,0) = z^d$. By uniqueness, g_i' has the character of z^d and υ_i is G-invariant. We replace x by $\upsilon_i x$, or x and y by $\upsilon_i^{1/2} x$ and $\upsilon_i^{1/2} y$ if some element of G exchanges x and y, where $\upsilon_i^{1/2}$ has constant term a square root $\sqrt{a_i}$ of a_i, so that $\upsilon_i^{1/2}$ is G-invariant, divide f_i by υ_i, and write g_i for g_i'. Then $f_i = xy + g_i(z,u)$. Likewise $g_* = z^d \Psi_*$ gives $g_*' = z^d$, and the limit becomes $f_* = xy + z^d$.
- If $d = 2$, then, as $g_i(z,0) = a_i z^2$, $\partial_z g_i(0,0) = 0$ and $\partial_z^2 g_i(0,0) = 2a_i \ne 0$. By the implicit function theorem there exists a unique $\varsigma_i \in u\mathbb{C}[[u]]$ with $(\partial_z g_i)(\varsigma_i(u), u) = 0$, and by uniqueness ς_i has the character of z. Replacing z with $z + \varsigma_i(u)$, we obtain $g_i = g_i(\varsigma_i(u), u) + z^2 \Psi_i'(z,u)$ with $\Psi_i'(0,0) = a_i$. We then take $z(\Psi_i'/a_i)^{1/2}$ as the new z, where the square root has constant term 1, divide f_i by a_i, and take x/a_i, or $x/\sqrt{a_i}$ and $y/\sqrt{a_i}$ if some element of G exchanges x and y, as the new x, y. Thus $f_i = xy + z^2 + r_i(u)$ with $r_i(u) := a_i^{-1} g_i(\varsigma_i(u), u)$, and $f_* = xy + z^2$.

For $d \ge 3$ and for $d = 2$, the coefficients of the new f_i, h_i are polynomials in finitely many coefficients of the old ones and in a_i^{-1} (and $\sqrt{a_i}^{\pm 1}$, where, after passing to an infinite subfamily, the signs of the $\sqrt{a_i}$ are fixed so that $(f_i, h_i, \sqrt{a_i})$ has the compatible generic limit $(f_*, h_*, \sqrt{\vartheta_0(0)})$ by Lemma 2.6(4)), so the transforms of (f_*, h_*) form a compatible generic limit of the transforms of (f_i, h_i). Moreover, for $d = 2$, the series ς_* defined from $g_* = z^2 \Psi_*$ as ς_i from g_i is 0, and any coefficient of ς_i is a polynomial in finitely many coefficients of g_i and in a_i^{-1}, as is the corresponding coefficient of ς_*, so the order of ς_i tends to $+\infty$. The coordinate changes fix 0 and u and multiply the residue form by units, so (G.17)–(G.18) and (a)–(c) persist, and since the order of ς_i tends to $+\infty$ if $d = 2$ and $\mathrm{mult}_{F_i} u \ge 1$, the orders of x, y, z along F_i still tend to $+\infty$. Thus $f_* = xy + z^d$, $f_i = xy + g_i(z,u)$, where $g_i - z^d$ is a polynomial in z of degree less than d whose coefficient of $z^j u^e$ vanishes for $i \gg 0$ for any j and e, as $g_* = z^d$, and $g_i = z^2 + r_i(u)$ if $d = 2$.

Step 3. In this step we write the first prime divisor of log discrepancy zero of the limit pair at the generic point of C as a weighted blow-up in parameters adapted to x, y, z, and we lift the corresponding shear to W_i.

Let $V_u := \mathrm{Spec}\, \Bbbk((u))[[x,y,z]]/(xy + z^d)$. The completion at the generic point η of $C = (x = y = z = 0)$ of the transformed limit germ $(xy + z^d = 0)$ is $\mathcal{O}_{V_u}$, and $(V_u, \frac{1}{m}\mathrm{div}(h_*))$ differs from the completion at the generic point of C of the pair $(W_*, \frac{1}{m}\mathrm{div}(h_*))$ in hypothesis (4) of Theorem 18.2 by a change of coordinates over $\Bbbk((u))$, the multiplication of h_* by a unit of $\Bbbk((u))$, and, if $n = 2$, a base change by a quadratic extension of fields. The completion and the quadratic base change are regular morphisms, so they preserve log resolutions and log discrepancies (cf. [dFEM11, Proposition 2.11 and Lemma 2.14]): each of them is the base change of a regular morphism of the regular ambient schemes, along which log resolutions pull back to log resolutions (cf. [dFEM11, Remark 2.10]) and relative canonical divisors of the ambient schemes pull back [JM12, Lemma 1.2], and the germs are Cartier divisors in the regular ambient schemes,

hence Gorenstein, so that their relative canonical divisors are determined by the dualizing sheaves, which commute with flat base change. By the hypothesis on $(W_*, \frac{1}{m}\operatorname{div}(h_*))$ near $C \cap (u \neq 0)$, the pair $(V_u, \frac{1}{m}\operatorname{div}(h_*))$ is lc and not klt at the closed point, and no component of $\frac{1}{m}\operatorname{div}(h_*)$ has coefficient one. Let

$$\pi : S := \operatorname{Spec} \Bbbk((u))[[p,q]] \to V_u, \qquad (x,y,z) = (p^d, -q^d, pq),$$

the quotient by the group μ_d acting by $(p,q) \to (\zeta p, \zeta^{-1} q)$, which is étale in codimension 1. Let $\mathfrak{a} := (h_* \circ \pi)^{1/m}$. Then $(S, \mathfrak{a})$ is lc and $\operatorname{mld}(S \ni 0, \mathfrak{a}) = 0$. By Lemma 3.4, there exist the first prime divisor E_0 over S with $a(E_0, S, \mathfrak{a}) = 0$, regular parameters s_1, s_2 of $\Bbbk((u))[[p,q]]$ and coprime integers $w_1 \geq w_2 \geq 1$ such that E_0 is the exceptional divisor of the weighted blow-up with weights (w_1, w_2) in (s_1, s_2). Let $v := \operatorname{mult}_{E_0}$, so that v is the monomial valuation with $v(s_1) = w_1$ and $v(s_2) = w_2$.

Let $\widehat{G}$ be the group of automorphisms $\widehat{g}$ of S, not necessarily over $\Bbbk((u))$, that lift the action on V_u of an element of G (which acts through x, y, z, u), that is, $\pi \circ \widehat{g}$ equals that action composed with π. Then $\mu_d \subset \widehat{G}$, and any element of G lifts to $\widehat{G}$: if it multiplies x, y, z by χ_x, χ_y, χ_z, then $(p,q) \to (\chi_x^{1/d} p, \chi_z \chi_x^{-1/d} q)$ is a lift for a d-th root $\chi_x^{1/d}$ of χ_x, as $\chi_x \chi_y = \chi_z^d$. If it sends (x,y,z) to $(\chi_1 y, \chi_2 x, \chi_z z)$, then $(p,q) \to ((-\chi_1)^{1/d} q, \chi_z (-\chi_1)^{-1/d} p)$ is a lift for a d-th root $(-\chi_1)^{1/d}$ of $-\chi_1$. Thus $\widehat{G}$ is an extension of G by μ_d, of order $d|G|$. Any $\widehat{g} \in \widehat{G}$ fixes 0 and preserves $\mathfrak{a}$, so $v \circ \widehat{g} = v$ by Lemma 3.4(3).

Assume that $w_1 > w_2$. Then

$$T_+ := \{ap + bq \mid a, b \in \Bbbk((u)),\ v(ap+bq) > w_2\}$$

is the $\Bbbk((u))$-line spanned by the linear part of s_1, and $\widehat{g}(T_+) = T_+$ for any $\widehat{g} \in \widehat{G}$.

- If $d \geq 3$, then $\zeta \neq \zeta^{-1}$ for a generator ζ of μ_d, so $T_+ \in \{\Bbbk((u))p, \Bbbk((u))q\}$. Possibly exchanging the roles of x and y (and of p and q), we may assume $T_+ = \Bbbk((u))p$. A lift of an element of G exchanging x and y would interchange $\Bbbk((u))p$ and $\Bbbk((u))q$, so no element of G exchanges x and y.
- If $d = 2$, let $\Lambda := \Bbbk[[u]]p \oplus \Bbbk[[u]]q$ and $\Lambda_1' := T_+ \cap \Lambda$, a direct summand of rank 1. By Maschke's theorem and Nakayama's lemma there exist $\widehat{G}$-eigenvectors p', q' with $\Lambda_1' = \Bbbk[[u]]p'$ and $\Lambda = \Bbbk[[u]]p' \oplus \Bbbk[[u]]q'$. Let $\mathcal{N} \in GL_2(\Bbbk[[u]])$ be the matrix with $(p,q) = (p', q')\mathcal{N}$. Then $x' := p'^2$, $y' := -q'^2$, and $z' := p'q'$ are $\Bbbk[[u]]$-linear combinations of x, y, z, G acts diagonally on x', y', z', and $xy + z^2 = \det(\mathcal{N})^2 (x'y' + z'^2)$. We pass to the coordinates x', y', z', divide f_* by $\det(\mathcal{N})^2$, a unit of $\Bbbk[[u]]$, and rename (x', y', z', p', q') as (x, y, z, p, q). Then $T_+ = \Bbbk((u))p$.

Hence, if $d \geq 3$ and some element of G exchanges x and y, then $w_1 = w_2$, so $w_1 = w_2 = 1$ as w_1, w_2 are coprime.

Still assuming $w_1 > w_2$, we have $v(q) = w_2$ as $q \notin T_+$. By the Weierstrass preparation theorem $s_1 = \upsilon \cdot (p - \phi(q))$ with υ a unit and $\phi \in q^2 \Bbbk((u))[[q]]$, as the linear part of s_1 belongs to $\Bbbk((u))p$. Let $\sigma(q) := \sum_{j w_2 < w_1} \sigma_j q^j$ be the sum of the terms of ϕ with $j w_2 < w_1$. Then $v(\phi(q) - \sigma(q)) \geq w_1$, so the parts of lowest v-value of $p - \sigma(q)$ and q are $v(0)^{-1} \overline{s}_1$ plus a multiple of $\overline{s}_2^{w_1}$, and a non-zero multiple of $\overline{s}_2$, where $\overline{s}_1, \overline{s}_2$ are the classes of s_1, s_2 in the graded ring $\Bbbk((u))[\overline{s}_1, \overline{s}_2]$ of v. As they generate the graded ring, v is the monomial valuation with weights (w_1, w_2) in the regular parameters $p - \sigma(q)$ and q. As p is a $\widehat{G}$-eigenvector, write $\widehat{g}^* p = \chi_p(\widehat{g}) p$ for $\widehat{g} \in \widehat{G}$. Then the invariance $v \circ \widehat{g} = v$ implies

$$v\big(\sigma(q) - \chi_p(\widehat{g})^{-1} \widehat{g}^* \sigma(q)\big) = v\big(\chi_p(\widehat{g})^{-1} \widehat{g}^* (p - \sigma(q)) - (p - \sigma(q))\big) \geq w_1,$$

and the left side is a sum of terms q^j with $j w_2 < w_1$, so $\widehat{g}^*(p - \sigma(q)) = \chi_p(\widehat{g})(p - \sigma(q))$. For $\widehat{g} = \zeta \in \mu_d$ we obtain $\sigma_j(1 - \zeta^{-j-1}) = 0$, so $\sigma_j = 0$ unless d divides $j+1$. Let

$$U(y) := - \sum_{k' \geq 1,\ k' d w_2 < w_1 + w_2} \sigma_{k'd-1} (-y)^{k'} \in y \Bbbk((u))[y].$$

Since $q^{k'd} = (-y)^{k'}$, we obtain $(p-\sigma(q))q = z + U(y)$ and $(p-\sigma(q))^d = -(z+U(y))^d/y$ on V_u. Thus, under the automorphism of $\Bbbk((u))[[x,y,z]]$ given by the formula of Proposition 7.6(3) with z^d in place of g, the new coordinates pull back by π to $(p-\sigma(q))^d$, $-q^d$, and $(p-\sigma(q))q$, and U has the character of z since $p-\sigma(q)$ is a $\widehat{G}$-eigenvector. If $w_1 = w_2 = 1$, let $U := 0$. Then $v = \mathrm{mult}_0$, and no change of coordinates is made.

Let $c := w_1 + w_2$, and let $w := (dw_1, dw_2, c)$ be the weights of x, y, z, with u of weight 0. For series over $\Bbbk((u))$, wt_w and in_w are as in Definition 2.3. In the new coordinates, the monomials $x^a z^b$ and $y^a z^b$ pull back to $(p-\sigma(q))^{da+b}q^b$ and $\pm(p-\sigma(q))^b q^{da+b}$, which are distinct monomials in the parameters of v. Hence, for any series φ over $\Bbbk((u))$ without monomials divisible by xy, $v(\varphi\circ\pi)$ is the smallest w-weight of a monomial of φ. Since $a(E_0, S, \mathfrak{a}) = 0$ and $a(E_0, S, 0) = w_1 + w_2$, $v(h_*\circ\pi) = mc$. Thus $\mathrm{in}_w(h_*)$ has w-weight mc, $\mathrm{in}_w(h_*) \in \Bbbk((u))[x,y,z]$, and $\mathrm{in}_w(h_*)|_{V_u} \neq 0$.

The polynomial U and, for $d = 2$, the matrix $\mathcal{N}$ are constructed from (f_*, h_*), and their coefficients in each degree in u are algebraic over the subfield of $\Bbbk$ generated over $\mathbb{C}$ by finitely many coefficients of (f_*, h_*), since the blow-ups of Lemma 3.4 are centered at $\Bbbk((u))$-points. Let N_U be a non-negative integer with $u^{N_U}U \in y\Bbbk[[u]][y]$. By Lemma 2.6(4) for the non-zero coefficients of $u^{N_U}U$ and the non-zero entries of $\mathcal{N}$, after passing to an infinite subfamily there exist $U_i^\circ \in y\mathbb{C}((u))[y]$ (the coefficients of $u^{N_U}U_i^\circ$ being the parts in $\mathbb{C}[u]$ of the polynomials given by Lemma 2.6(4)) with $u^{N_U}U_i^\circ \in y\mathbb{C}[[u]][y]$ and, for $d = 2$, matrices $\mathcal{N}_i$ with entries in $\mathbb{C}[[u]]$, with the characters of the corresponding coefficients and entries (if an element g of G exchanges x and y, the entries of $\mathcal{N}$ are semi-invariant only for the subgroup G_0 of elements of G preserving x, and we lift the entries of the first column of $\mathcal{N}$ for G_0 and define the entries of the second column of $\mathcal{N}_i$ by the relations obtained by applying a lift $\widehat{g} \in \widehat{G}$ of g to $(p,q) = (p',q')\mathcal{N}$, which express the second column of $\mathcal{N}$ through the first, as $\widehat{g}$ multiplies p' and q' by constants and sends p to a constant multiple of q), such that the tuples formed by f_i, h_i, the coefficients of $u^{N_U}U_i^\circ$ and the entries of $\mathcal{N}_i$ have the compatible generic limit formed by f_*, h_*, the coefficients of $u^{N_U}U$ and the entries of $\mathcal{N}$. Thus U_i° has the character of z and $\mathcal{N}_i$ is equivariant as $\mathcal{N}$. Since $\det\mathcal{N}(0) \neq 0$, after passing to an infinite subfamily $\det\mathcal{N}_i(0) \neq 0$, so $\mathcal{N}_i \in GL_2(\mathbb{C}[[u]])$.

For $d = 2$, we pass to the coordinates x', y', z' of W_i given by the $\mathbb{C}[[u]]$-linear substitution of (x,y,z) that $\mathcal{N}_i$ induces on the quadratic forms $p^2, -q^2, pq$ in (p,q), as $\mathcal{N}$ does above, and rename them x, y, z. Since $f_i = xy + z^2 + r_i(u)$, after dividing by $\det(\mathcal{N}_i)^2$ we still have $f_i = xy + z^2 + r_i(u)$ with a new r_i tending to 0, and G acts diagonally on x, y, z.

Let Φ_i be the automorphism of $\mathbb{C}((u))[x,y,z]$ given by the formula of Proposition 7.6(3) with U_i° and g_i:

$$x \to x - \frac{g_i(z + U_i^\circ(y), u) - g_i(z,u)}{y}, \qquad y \to y, \qquad z \to z + U_i^\circ(y), \quad \text{and} \quad u \to u. \tag{G.19}$$

Since y divides U_i°, $(g_i(z+U_i^\circ(y),u) - g_i(z,u))/y \in \mathbb{C}((u))[y,z]$. The inverse of Φ_i is

$$x \to x - \frac{g_i(z - U_i^\circ(y), u) - g_i(z,u)}{y}, \qquad y \to y, \qquad z \to z - U_i^\circ(y), \quad \text{and} \quad u \to u,$$

the Jacobian determinant of Φ_i is 1, and Φ_i preserves $xy + g_i$. Let N_w be a positive integer with $dw_2N_w \geq 2N_U$ and $cN_w \geq N_U$, and so large that the largest power of u dividing the transform $h_*\circ\Phi_*^{-1}\circ\delta$ of h_*, for the automorphism Φ_* below, divides the part of w-weight mc of $h_*\circ\Phi_*^{-1}\circ\delta$ exactly (replacing N_w with $N_w + k_0$ multiplies the part of w-weight t by $u^{k_0 t}$), and let

$$\delta : (X,Y,Z,u) \to (x,y,z,u) = (u^{dw_1N_w}X,\ u^{dw_2N_w}Y,\ u^{cN_w}Z,\ u).$$

We take (X,Y,Z,u) as new coordinates, through $\Phi_i^{-1}\circ\delta$. The pull-backs of x, y, z are regular for $i \gg 0$: z pulls back to $u^{cN_w}Z - U_i^\circ(u^{dw_2N_w}Y)$. In the expansion of x by powers of U_i°, the term with $(U_i^\circ)^{n'}$, $n' \geq 2$, contains $(U_i^\circ)^{n'}/y$, whose pole of order at most $n'N_U$ is compensated by $u^{(n'-1)dw_2N_w}$. The term with U_i°/y contains $\partial_z g_i(u^{cN_w}Z, u)$, and $\partial_z g_i(u^{cN_w}Z,u) \in u^{N_U}\mathbb{C}[[Z,u]]$

and $\partial_z g_i(0,u) \in u^{N_U+1}\mathbb{C}[[u]]$ for $i \gg 0$ since $\partial_z g_* = dz^{d-1}$, so the pull-backs of x, y, z vanish at the origin. As Φ_i preserves $xy + g_i$,

$$f_i \circ \Phi_i^{-1} \circ \delta = u^{dcN_w}\big(XY + u^{-dcN_w} g_i(u^{cN_w} Z, u)\big),$$

and $u^{-dcN_w} g_i(u^{cN_w} Z, u)$ is regular for $i \gg 0$, with value Z^d at $u = 0$, since $g_* = z^d$. For the limit, we take (X, Y, Z, u) as coordinates through $\Phi_*^{-1} \circ \delta$, where Φ_* is the automorphism

$$x \to x - \frac{(z + U(y))^d - z^d}{y}, \qquad y \to y, \qquad z \to z + U(y), \quad \text{and} \quad u \to u$$

of $\Bbbk((u))[[x, y, z]]$. Then the equation of the limit becomes $XY + Z^d$. The residue form is multiplied by u^{cN_w}. Let N_h be the largest integer with u^{N_h} dividing the transform of h_*. As $\varphi \circ \Phi_i^{-1} \circ \delta \in (u)$ for $\varphi \in \{x, y, z\}$ and $i \gg 0$ (for $d = 2$, as $U_i^\circ \in y^2\mathbb{C}((u))[y]$), only finitely many monomials of h_i matter modulo u^{N_h}, so, as in **Step 2**, u^{N_h} divides the transform of h_i for $i \gg 0$, and we divide both transforms by u^{N_h}. The limit of the transforms is the transform of the limit. We rename (X, Y, Z) as (x, y, z).

The composite of the coordinate changes of **Steps 2 and 3** satisfies the assumption of (a)–(c), so (G.17)–(G.18) and (a)–(c) hold for it, with fixed integers $\kappa_0, N_0 \geq 0$ and $n \in \{1, 2\}$, and with $\beta_1 = n\beta + \frac{N_0}{m} - \kappa_0 < 1$. The orders of $z + U_i^\circ(y)$ and of $x - (g_i(z + U_i^\circ(y), u) - g_i(z, u))/y$ along F_i tend to $+\infty$, as $\operatorname{mult}_{F_i} u \leq R'$, the coefficients of U_i° have poles of order at most N_U, and $\partial_z g_i - dz^{d-1}$ tends to 0. Thus, for the lifts F_i of (b), $\operatorname{mult}_{F_i} u \leq R'$, and $a(F_i, W_i, 0)$, $\operatorname{mult}_{F_i} x$, $\operatorname{mult}_{F_i} y$, and $\operatorname{mult}_{F_i} z$ tend to $+\infty$.

The dilation δ preserves w-weights, so after it $\operatorname{in}_w(h_*)$ is, up to a unit of $\Bbbk((u))$, the lowest-weight part computed before δ. Since the pull-backs of x, y, z under $\Phi_*^{-1} \circ \delta$ are regular, as for the members with $g_* = z^d$, so are $h_* \circ \Phi_*^{-1} \circ \delta$ and $\operatorname{in}_w(h_* \circ \Phi_*^{-1} \circ \delta)$, and by the choice of N_w the u-order of $\operatorname{in}_w(h_* \circ \Phi_*^{-1} \circ \delta)$ equals that of $h_* \circ \Phi_*^{-1} \circ \delta$. Thus $\operatorname{in}_w(h_*) \in \Bbbk[[u]][x, y, z]$ has w-weight mc, and $\operatorname{in}_w(h_*)(x, y, z, 0) \neq 0$. We show that $(V_u, \frac{1}{m}\operatorname{div}(\operatorname{in}_w(h_*)))$ is lc. Let $\tau : S' \to S$ be the weighted blow-up with exceptional divisor E_0. Since $v(\operatorname{in}_w(h_*) \circ \pi) = v(h_* \circ \pi) = mc$ and $a(E_0, S, 0) = c$,

$$\tau^*\Big(K_S + \frac{1}{m}\operatorname{div}(h_* \circ \pi)\Big) = K_{S'} + E_0 + \frac{1}{m}\tau_*^{-1}\operatorname{div}(h_* \circ \pi),$$

and likewise with $\operatorname{in}_w(h_*)$ in place of h_*, and

$$\tau_*^{-1}\operatorname{div}(h_* \circ \pi)|_{E_0} = \tau_*^{-1}\operatorname{div}(\operatorname{in}_w(h_*) \circ \pi)|_{E_0},$$

since $\operatorname{in}_w(h_*) \circ \pi$ is the lowest-weight part of $h_* \circ \pi$ with respect to v. As $(S, \mathfrak{a})$ is lc, the pair

$$\Big(E_0, \operatorname{Diff}_{E_0}\Big(\frac{1}{m}\tau_*^{-1}\operatorname{div}(h_* \circ \pi)\Big)\Big)$$

is lc by adjunction. Let $\mathbb{K}$ be an algebraic closure of $\Bbbk((u))$. After base change to $\mathbb{K}$, inversion of adjunction [Kaw07, Theorem] on the weighted blow-up of $\mathbb{A}^2_{\mathbb{K}}$ in the polynomial parameters $p - \sigma(q), q$ of v implies, as $\operatorname{in}_w(h_*) \circ \pi$ is a polynomial and lc is preserved by completion and field extension [dFEM11, Proposition 2.11 and Lemma 2.14], that the pair

$$\Big(S', E_0 + \frac{1}{m}\tau_*^{-1}\operatorname{div}(\operatorname{in}_w(h_*) \circ \pi)\Big)$$

is lc near E_0, so the pair

$$\Big(S, \frac{1}{m}\operatorname{div}(\operatorname{in}_w(h_*) \circ \pi)\Big)$$

is lc near 0, and $(V_u, \frac{1}{m}\operatorname{div}(\operatorname{in}_w(h_*)))$ is lc since π is étale in codimension 1. It is not klt at the closed point, since E_0 has log discrepancy $c - \frac{1}{m} \cdot mc = 0$ with respect to

$$\Big(S, \frac{1}{m}\operatorname{div}(\operatorname{in}_w(h_*) \circ \pi)\Big).$$

As $\mathrm{in}_w(h_*)$ is a polynomial, the pair on $(xy+z^d=0)\subset \mathbb{A}^3_{\Bbbk}$ with boundary $\frac{1}{m}\operatorname{div}(\mathrm{in}_w(h_*))$ is lc and not klt at the origin, since the base change to $\mathbb{K}$ and the completion at the origin are regular morphisms (cf. [dFEM11, Proposition 2.11 and Lemma 2.14]).

We continue the proof below.

Proof of Theorem 18.2, continued. We keep the chart on which (G.17) holds and in which

$$W_i=(f_i=0),\quad f_i=xy+g_i(z,u),$$

and we keep the group G, the mark h_i, the coefficient β_1 of $\operatorname{div}(u)$, the computing lifts F_i of b_0, the coprime weights $w_1\ge w_2$, $c=w_1+w_2$, and $w=(dw_1,dw_2,c)$. Here $|G|\le M$, $g_i(z,u)-z^d$ is a polynomial in z of degree less than d, the limit equation is $xy+z^d$, and $\mathrm{in}_w(h_*)\in\Bbbk[[u]][x,y,z]$ is the non-zero lowest-weight part of h_*, of w-weight mc. The pair $(V,\frac{1}{m}\operatorname{div}(\mathrm{in}_w(h_*)))$, where $\mathbb{K}$ is an algebraic closure of $\Bbbk((u))$ and $V:=(xy+z^d=0)\subset\mathbb{A}^3_{\Bbbk}$, is lc and not klt at the origin. For any i,

$$\operatorname{mld}\Big(\Big(W_i\ni o_i,\frac{1}{m}\operatorname{div}(h_i)+\beta_1\operatorname{div}(u)\Big)/G\Big)=b_0,$$

the center of F_i on the chart is the origin, and

$$\operatorname{mult}_{F_i}u\le R',\qquad \min\{\operatorname{mult}_{F_i}x,\operatorname{mult}_{F_i}y,\operatorname{mult}_{F_i}z\}\to+\infty.$$

We let $\pi_{\mathbb{K}}:\mathbb{A}^2_{\mathbb{K}}\to V$, $(p,q)\to(p^d,-q^d,pq)$, be the cyclic cover with the formula of π, with deck group μ_d acting by $(p,q)\to(\zeta p,\zeta^{-1}q)$, and we let

$$\widetilde{h}:=\mathrm{in}_w(h_*)(p^d,-q^d,pq)\in\Bbbk[[u]][p,q],$$

which is homogeneous of degree mc for the weights (w_1,w_2) on (p,q). Since $\pi_{\mathbb{K}}$ is étale in codimension 1, $(\mathbb{A}^2_{\mathbb{K}},\frac{1}{m}\operatorname{div}(\widetilde{h}))$ is lc, and the components of coefficient one of $\frac{1}{m}\operatorname{div}(\mathrm{in}_w(h_*))$ are the images of the irreducible factors of $\widetilde{h}$ of multiplicity m.

Step 4. In this step we show that the following case does not occur.

(E) The $\mathbb{Q}$-divisor $\frac{1}{m}\operatorname{div}(\mathrm{in}_w(h_*))$ on V has a component of coefficient one.

Suppose that (E) holds. The irreducible (w_1,w_2)-homogeneous polynomials in $\mathbb{K}[p,q]$ are p, q, and $p^{w_2}-\lambda q^{w_1}$ with $\lambda\in\mathbb{K}^\times$, the last one of weight w_1w_2.

Assume first that $w_1>w_2$ and that a factor $p^{w_2}-\lambda q^{w_1}$ has multiplicity m. Since $mw_1w_2\le mc$, we have $(w_1-1)(w_2-1)\le 1$, so $w_2=1$ because w_1,w_2 are coprime. A generator ζ of μ_d maps $p-\lambda q^{w_1}$ to $\zeta(p-\lambda\zeta^{-w_1-1}q^{w_1})$, so the μ_d-orbit Ω of the factor satisfies $|\Omega|=d/\gcd(d,w_1+1)$. Since $\widetilde{h}$ is μ_d-invariant, any element of Ω divides $\widetilde{h}$ with multiplicity m, and thus

$$m|\Omega|w_1\le mc=m(w_1+1).$$

As $w_1\ge 2$, $|\Omega|=1$, that is, $d\mid w_1+1$. Then $p-\lambda q^{w_1}$ is the unique factor of $\widetilde{h}$ of multiplicity m other than p,q, so it is stable under $\mathrm{Gal}(\mathbb{K}/\Bbbk((u)))$ and under G. Thus $\lambda\in\Bbbk((u))$, and λq^{w_1} has the character of p. Let $j_0:=(w_1+1)/d$. The shear $p'=p-\lambda q^{w_1}$, $q'=q$ commutes with μ_d and descends to

$$z'=z-\lambda(-y)^{j_0},$$

because $pq-\lambda q^{w_1+1}=z-\lambda(-y)^{j_0}$. We apply the automorphism Φ_{g_i} of Proposition 7.6(3) with $U=-\lambda_i(-y)^{j_0}$, where, by Lemma 2.6(4), after passing to an infinite subfamily, $\lambda_i\in\mathbb{C}((u))$ are of bounded pole order, semi-invariant with the character of λ, and have compatible generic limit λ. By Proposition 7.6(3) for $R=\mathbb{C}((u))$, as $g_i\in\mathbb{C}((u))[z]$, Φ_{g_i} is a G-equivariant automorphism over $\mathbb{C}((u))$ with Jacobian determinant 1 fixing f_i. After it, the image of $(p-\lambda q^{w_1}=0)$ is $(p'=0)$, and we rename p',z' and the transformed x as p,z,x.

Assume next that $w_1=w_2=1$. A factor $p-\lambda q$ has the μ_d-orbit $\{p-\lambda\zeta^{-2j}q\}_j$ of $d/\gcd(d,2)$ elements, and at most two factors of degree 1 have multiplicity m, as $\deg\widetilde{h}=2m$. Hence a factor of multiplicity m other than p,q exists only if $d\in\{2,4\}$. If $d=4$, the orbit is $\{p-\lambda q,p+\lambda q\}$,

and we obtain $\widetilde{h} = \gamma(u)(p^2 - \lambda' q^2)^m$ with $\gamma(u) \in \mathbb{K}^\times$ and $\lambda' := \lambda^2$. The deck generator $\zeta = \sqrt{-1}$ maps $p^2 - \lambda' q^2$ to $-(p^2 - \lambda' q^2)$, so m is even by invariance, and

$$\mathrm{in}_w(h_*) = \gamma(u)(x - 2\lambda' z^2 - \lambda'^2 y)^{m/2} \quad \text{modulo } xy + z^4.$$

Comparing the coefficients of $x^{m/2}$ and of $x^{m/2-1}z^2$, we have $\gamma(u), \lambda' \in \Bbbk((u))$. The substitution $x = \lambda' X$, $y = \lambda'^{-1} Y$ preserves $xy + g_i$, has determinant 1, and maps $x - 2\lambda' z^2 - \lambda'^2 y$ to $\lambda'(X - Y - 2z^2)$. If G preserves x and y, then λ' is semi-invariant, and if an element $g \in G$ with $g^* x = \eta_1 y$ and $g^* z = \eta_3 z$ exchanges x and y, then $g^*(x - 2\lambda' z^2 - \lambda'^2 y)$ is a multiple of $x - 2\lambda' z^2 - \lambda'^2 y$, and comparing the coefficients of y and of z^2 gives $\lambda' \cdot g^* \lambda' = -\eta_1 \eta_3^{-2} \in \Bbbk^\times$. In both cases G acts on X, Y, z, u monomially with constant coefficients.

If $d = 2$, a unique line of multiplicity m in $\mathrm{div}(\widetilde{h})$ is defined over $\Bbbk((u))$ and is G-stable. Two such lines form a G-stable and $\mathrm{Gal}(\mathbb{K}/\Bbbk((u)))$-stable pair, which splits over $\Bbbk((u))$ or over $\Bbbk((u^{1/2}))$ because $\Bbbk$ is algebraically closed. In the second case we pass to the axis cover $u = s^2$ with the group $\{(g, \xi) \mid g \in G,\ \xi^2 = \chi_u(g)\}$, of order $2|G| \le 2M$, and we rename s as u. Let $\Lambda = \Bbbk[[u]]p \oplus \Bbbk[[u]]q$ and let $\Lambda_1, \Lambda_2 \subset \Lambda$ be the intersections of Λ with the lines of multiplicity m in $\mathrm{div}(\widetilde{h})$ if there exist two such lines, or with the line of multiplicity m and a G-stable complement if there exists only one. Then $\Lambda_1 \oplus \Lambda_2$ is a free G-stable lattice on which G permutes $\{\Lambda_1, \Lambda_2\}$. Projecting a generator of Λ_1 (and of Λ_2 if G preserves it) to an isotypic component and applying Nakayama's lemma, we obtain a basis $e_1 \in \Lambda_1$, $e_2 \in \Lambda_2$ on which G acts by constant monomial matrices. Let $\det_\Lambda(u) \in \Bbbk((u))^\times$ be the determinant of the matrix expressing p, q in terms of e_1, e_2. The induced linear change of (x, y, z) multiplies $xy + z^2$ by $\det_\Lambda^2$ and the residue form by $\det_\Lambda$, and the equation becomes $xy + z^2 + \det_\Lambda^{-2}(g_i(z, u) - z^2)$ in the new coordinates.

As for U_i° and $\mathcal{N}_i$ before (G.19), by Definition 2.5 and Lemma 2.6(4), possibly passing to an infinite subfamily, we apply the shear $z' = z - \lambda(-y)^{j_0}$, the substitution $x = \lambda' X$, $y = \lambda'^{-1} Y$, and the base change of Λ above to f_i and h_i with the coefficients λ_i of the shear above and, for the substitution and the base change, a coefficient $\lambda_i' \in \mathbb{C}((u))$ and a matrix with entries in $\mathbb{C}((u))$, of bounded pole order, whose compatible generic limits are λ' and the matrix expressing e_1, e_2 in terms of p, q, where λ_i' is semi-invariant as λ' if G preserves x and y, and otherwise λ_i' is $\lambda'(0)$, where $\lambda'(0)^2 = -\eta_1 \eta_3^{-2}$, times the exponential of a lift of $\log(\lambda'/\lambda'(0))$, which is invariant under the elements of G preserving x and anti-invariant under those exchanging x and y, so that $\lambda_i' \cdot g^* \lambda_i' = \lambda' \cdot g^* \lambda'$, and where one entry of the matrix in each G-orbit of entries is lifted for the subgroup of elements preserving x, y, Λ_1, and Λ_2 and the other entries are defined by the relations given by the constant monomial actions of G on (p, q) and on (e_1, e_2). We let $\det_{\Lambda,i}$ be the inverse of the determinant of the lifted matrix. All shears, scalings and base changes above have coefficients in $\Bbbk((u))$ with poles of bounded order, independent of i. We compose them with the dilation $x = u^{n_2 d w_1} x'$, $y = u^{n_2 d w_2} y'$, $z = u^{n_2 c} z'$ for a fixed sufficiently large $n_2 \in \exp(G)\mathbb{N}^+$, where $\exp(G)$ is the exponent of G, after which x, y, z are regular functions of x', y', z', u, since the dilation multiplies $x^{k_1} y^{k_2} z^{k_3}$ by $u^{n_2(dw_1 k_1 + dw_2 k_2 + c k_3)}$ and the shear term $\lambda_i(-y)^{j_0}$ has weight c. After it, all forward maps, the division of f_i by $\det_{\Lambda,i}^2$ (when $d = 2$), and the division of f_i by $u^{n_2 dc}$ are regular for i in an infinite subfamily, since only finitely many coefficients of f_i of bounded weight and bounded u-order have to vanish to a bounded order and they vanish on the limit, hence, by Definition 2.5, for all but finitely many i. Thus, possibly passing to an infinite subfamily (Lemma 2.6(2)) and renaming the coordinates, one of the following holds over $\Bbbk[[u]]$, with $\gamma(u) \neq 0$:

(i) $\widetilde{h} = p^m R(p, q)$ with $R(0, q) = \gamma(u) q^m$, and $(p = 0)$ is G-stable, or $\widetilde{h} = q^m R(p, q)$ with $R(p, 0) = \gamma(u) p^m$, and $(q = 0)$ is G-stable.
(ii) $\widetilde{h} = \gamma(u)(pq)^m$, that is, $\mathrm{in}_w(h_*) = \gamma(u) z^m$ up to a constant.
(iii) $d = 4$, m is even, and $\mathrm{in}_w(h_*) = \gamma(u)(x - y - 2z^2)^{m/2}$ modulo $xy + z^4$.

In case (i), $R(0,q)\neq 0$ because p has multiplicity exactly m in $\widetilde{h}$. For an integer $n_1\geq 1$ we make the dilation $p=u^{n_1}p'$, $q=q'$, that is,

$$x=u^{dn_1}x',\qquad y=y',\qquad \text{and}\qquad z=u^{n_1}z',$$

and we divide f_i by u^{dn_1} and h_i by u^{mn_1}. The residue form $dy\wedge dz\wedge du/y$ acquires u^{n_1} and the mark acquires u^{mn_1}, so the coefficient of $\operatorname{div}(u)$ changes by $n_1-\frac{mn_1}{m}=0$. Then $\widetilde{h}$ becomes

$$u^{-mn_1}(u^{n_1}p')^m R(u^{n_1}p',q')=\gamma(u)(p'q')^m+u^{n_1}p'^{m+1}R_1(u,p',q')$$

for a polynomial R_1, so after dividing by $u^{\operatorname{mult}_0\gamma}$, where $\operatorname{mult}_0\gamma$ is the u-order of γ, the coefficients other than that of $(p'q')^m$ vanish to order at least $n_1-\operatorname{mult}_0\gamma$. If $\widetilde{h}=q^mR(p,q)$, we make the dilation $x=x'$, $y=u^{dn_1}y'$, $z=u^{n_1}z'$, and the computation above holds with p and q exchanged. Thus case (i) reduces to case (ii) up to an arbitrarily high fixed order in u. In each case we divide the mark by $u^{\operatorname{mult}_0\gamma}$.

Let β_2 be the coefficient of $\operatorname{div}(u)$ after the operations of this step. Then $\beta_2=n'\beta_1+\frac{N_0'}{m}-\kappa_0'$, as for β_1 after (G.17), where $n'\in\{1,2\}$ is the degree of the axis cover and N_0', κ_0' are the exponents of u in the factors acquired by the mark and by the residue form under the maps of this step. For i in an infinite subfamily they do not depend on i, and β_2 does not depend on n_1, since the dilation by u^{n_1} adds mn_1 to N_0' and n_1 to κ_0'. Let $f_0:=xy+z^d$, where $d=4$ in case (iii), and let $h_{0,i}:=\gamma_i(0)z^m$, respectively $h_{0,i}:=\gamma_i(0)(x-y-2z^2)^{m/2}$, where $\gamma_i(u)$ is the coefficient of z^m in the transformed h_i divided by $u^{\operatorname{mult}_0\gamma}$, respectively divided by $u^{\operatorname{mult}_0\gamma}$ and by the coefficient $(-1)^{m/2}\binom{m}{m/2}$ of z^m in the representative of $(x-y-2z^2)^{m/2}$ modulo f_0 without monomials divisible by xy. For i in an infinite subfamily, $\gamma_i(0)\neq 0$ since $\gamma(u)u^{-\operatorname{mult}_0\gamma}$ is a unit. The surface $(f_0=0)\subset\mathbb{A}^3$ with the $\mathbb{Q}$-divisor $\frac{1}{m}\operatorname{div}(h_{0,i}|_{(f_0=0)})$ is lc near 0: on the cover $(p,q)\to(p^d,-q^d,pq)$ it is $(\mathbb{A}^2,\operatorname{div}(pq))$, respectively $(\mathbb{A}^2,\operatorname{div}(p^2-q^2))$. The section $(u=0)$ of the transformed W_i is the integral Du Val surface $(xy+z^d=0)$, the generic point of $(u=0)$ has center o_i on W_i, and the transformed mark has order 0 along it. Since $\frac{1}{m}\operatorname{div}(h_i)+\beta_1\operatorname{div}(u)$ is sub-klt, we have

$$\beta_2<1.$$

Let L_2 be the integer N of Theorem 5.10 for f_0 and for the integers m, $2M$, and m in the roles of m, M, and L ($\deg h_{0,i}=m$), where in case of an exchange we first replace x,y by the eigenparameters $x\pm ty$ with $t^2=\eta_1/\eta_2$ for an element $g\in G$ with $g^*x=\eta_1 y$ and $g^*y=\eta_2 x$, which changes only the polynomial f_0. We fix $n_1\geq L_2+\operatorname{mult}_0\gamma$. Since $\gamma_i(u)$ is semi-invariant, as z is, and $\gamma_i(0)\neq 0$, it is G-invariant. We divide the mark further by $\gamma_i(u)/\gamma_i(0)$, and in case (iii) we add to it $\gamma_i(0)f_i$ times the w-homogeneous polynomial of weight $mc-dc$ whose product with f_0 is $(x-y-2z^2)^{m/2}$ minus its representative modulo f_0 without monomials divisible by xy, so that, by the choice of n_1 in case (i), its part of w-weight mc is $h_{0,i}$ modulo (u^{L_2}) for all but finitely many i. Neither operation changes $\operatorname{div}(h_i|_{W_i})$, and after the dilation below we keep the lift h_i so obtained instead of replacing h_i with a representative modulo f_i. We fix a depth $r_0\geq mn_1+\operatorname{mult}_0\gamma+L_2$ divisible by $\exp(G)$, and we make the dilation

$$x=u^{r_0dw_1}x',\qquad y=u^{r_0dw_2}y',\qquad \text{and}\qquad z=u^{r_0c}z',$$

dividing f_i by u^{r_0dc} and h_i by u^{r_0mc}. The residue form acquires u^{r_0c} and the mark acquires u^{r_0mc}, and $r_0c=\frac{r_0mc}{m}$, so β_2 does not change. A monomial of w-weight v acquires $u^{r_0(v-dc)}$ in f_i and $u^{r_0(v-mc)}$ in h_i. Since the coefficients of f_i-f_0, a polynomial in x,y,z with monomials of weight less than dc, have u-order at least r_0dc+L_2 for all but finitely many i, as they vanish on the limit (Definition 2.5), since, as h_* has no monomial of weight less than mc, the coefficients in h_i of the finitely many monomials in x,y,z of weight less than mc have u-order at least r_0mc+L_2 for all but finitely many i, and since a monomial of weight greater than mc acquires at least u^{r_0}, which exceeds by at least L_2 the pole order, at most $mn_1+\operatorname{mult}_0\gamma$, created by the divisions of

h_i by u^{mn_1} and by $u^{\mathrm{mult}_0\,\gamma}$, for i in an infinite subfamily the transformed f_i and h_i are regular, the center of F_i is still the origin, and

$$f_i - f_0 \in (u) \cap \mathfrak{n}^{L_2}, \qquad h_i - h_{0,i} \in (u) \cap \mathfrak{n}^{L_2}.$$

By Theorem 5.10(1)(2), applied with $s = u$, $\beta = \beta_2$, and $h_0 := h_{0,i}$, there exists a prime divisor P over the transformed W_i with center the origin such that

$$\max\{a(P, W_i, 0), \mathrm{mult}_P\, h_i, \mathrm{mult}_P\, u\} \le L_2$$

and

$$\frac{1}{e_P} a\Big(P, W_i, \frac{1}{m}\,\mathrm{div}(h_i) + \beta_2\,\mathrm{div}(u)\Big) = \mathrm{mld}\Big(\Big(W_i \ni 0, \frac{1}{m}\,\mathrm{div}(h_i) + \beta_2\,\mathrm{div}(u)\Big)/G\Big).$$

Any prime divisor over the transformed W_i with center the origin restricts to a prime divisor over W_i with center o_i whose quotient-normalized log discrepancy does not change under the maps of this step, and the transform of F_i is centred at the origin with value b_0. Hence the mld on the right is b_0, and the restriction F of P is a computing lift of b_0. The exponents of u in the composite of the chart on which (G.17) holds with the maps of this step do not depend on i, so it satisfies the four relations of (G.17) with fixed integers in place of κ_0, N_0, and n, and with a ramification index $\rho \ge 1$. Thus $a(F, W_i, 0)$ and $\mathrm{mult}_F\, u$ are bounded independently of i. This contradicts property (∗). Hence (E) does not occur, that is, $\frac{1}{m}\,\mathrm{div}(\mathrm{in}_w(h_*))$ has no component of coefficient one.

Step 5. In this step we choose, for any i, a preparation, of maximal depth, of the chart on which (G.17) holds.

For a positive integer r divisible by $\exp(G)$, the substitution

$$x = u^{rdw_1} X, \qquad y = u^{rdw_2} Y, \qquad \text{and} \qquad z = u^{rc} Z \tag{G.20}$$

is G-equivariant, with X, Y, Z of the characters of x, y, z. We divide f_i by u^{rdc} and h_i by u^{rmc}. A monomial of w-weight v acquires $u^{r(v-dc)}$ in f_i and $u^{r(v-mc)}$ in h_i. For a fixed r, the divisions are regular for all but finitely many i, because the finitely many coefficients of weight less than dc, respectively mc, and of bounded u-order vanish on the limit, and the center of F_i on the new chart is the origin for $i \gg 0$, because $\mathrm{mult}_{F_i}\, u \le R'$ and $\mathrm{mult}_{F_i}\, x, \mathrm{mult}_{F_i}\, y, \mathrm{mult}_{F_i}\, z \to +\infty$. Since, for each fixed r, the regularity of the divisions and the position of the center hold for all but finitely many i, after discarding finitely many i there exist $r_i^0 \in \exp(G)\mathbb{N}^+$ with $r_i^0 \to +\infty$ such that for any i the divisions of f_i and h_i at depth r_i^0 are regular and the center of F_i at depth r_i^0 is the origin. We fix positive integers ℓ_i with

$$\ell_i \to +\infty \qquad \text{and} \qquad \ell_i / r_i^0 \to 0.$$

Let $\mathcal{P}_i$ be the set of finite compositions of the following maps, each applied to the current equation $xy + g_i(z, u)$ and the current lift h_i before the substitution (G.20).

(T) A translation $z \to z - \alpha_i(u)$, where $\alpha_i \in u^{\ell_i}\mathbb{C}[u]$ is semi-invariant with the character of z.

(H) The automorphism Φ_{g_i} of Proposition 7.6(3) for the current g_i and

$$U = U_i := \sum_{j \ge 1,\ jdw_2 < c} \sigma_{i,j}(u) y^j \quad \text{with} \quad \sigma_{i,j} \in u^{\ell_i}\mathbb{C}[u],$$

such that U_i is semi-invariant with the character of z.

(C) The change of lift $h_i \to h_i + f_i\varphi_i$, where φ_i is a regular function near 0 in the ideal (u^{ℓ_i}), semi-invariant of the character $\chi_{h_i}\chi_{f_i}^{-1}$, without any bound on the degree in x, y, z.

In (H), by Proposition 7.6(3) for $R = \mathbb{C}((u))$, as $g_i \in \mathbb{C}((u))[z]$, Φ_{g_i} is an automorphism fixing $xy + g_i$. If an element of G exchanges x and y, then $w_1 = w_2 = 1$, as **Step 3** excludes such elements when $w_1 > w_2$. Thus no $j \ge 1$ satisfies $jdw_2 < c$ and $U_i = 0$. The maps (T) and (H) are G-equivariant, fix y and u, have Jacobian determinant 1, and keep the equation in the form $xy + g(z, u)$ with $g(z, u) - z^d$ a polynomial in z of degree less than d. Hence they preserve the

residue form $dy \wedge dz \wedge du/y$. The map (C) changes no restriction $h_i|_{W_i}$. Any element of $\mathcal{P}_i$ is the identity modulo u^{ℓ_i} on coefficients.

We call a pair (ϖ, r) with $\varpi \in \mathcal{P}_i$ and $r \in \exp(G)\mathbb{Z}$, $r \geq r_i^0$, *admissible* if, after ϖ and the substitution (G.20), the divisions of the equation by u^{rdc} and of the lift by u^{rmc} are regular near the origin and the center of F_i on the new chart is the origin. The pair (id, r_i^0) is admissible. Since any $\varpi \in \mathcal{P}_i$ fixes y and u, and $\operatorname{mult}_{F_i} u \geq 1$ because u vanishes at o_i, any admissible pair (ϖ, r) satisfies $\operatorname{mult}_{F_i} Y > 0$, that is,

$$rdw_2 \operatorname{mult}_{F_i} u < \operatorname{mult}_{F_i} y. \tag{G.21}$$

Thus the set of r occurring in admissible pairs is finite and non-empty, and we let r_i be the largest one and $\varpi_i \in \mathcal{P}_i$ a map with (ϖ_i, r_i) admissible. We let

$$f_i' = xy + g_i'(z,u) \qquad \text{and} \qquad h_i'$$

be the equation and the lift after ϖ_i and the substitution (G.20) with $r = r_i$, divided by $u^{r_i dc}$ and $u^{r_i mc}$, and $W_i' := (f_i' = 0)$, with origin o_i'. Then $r_i \geq r_i^0 \to +\infty$.

The composite map $W_i' \dashrightarrow W_i$ is a G-equivariant morphism near o_i' mapping o_i' to o_i, and it is birational. Thus any prime divisor F over W_i' with center o_i' is a prime divisor over W_i with center o_i, and $I_G(F)$ does not change. The residue form acquires $u^{r_i c}$ and the mark acquires $u^{r_i mc}$, so

$$a(F, W_i, 0) = a(F, W_i', 0) + r_i c \operatorname{mult}_F u, \qquad \operatorname{mult}_F h_i = \operatorname{mult}_F h_i' + r_i mc \operatorname{mult}_F u,$$

and since $\frac{r_i mc}{m} = r_i c$, the coefficient β_1 of $\operatorname{div}(u)$ does not change:

$$a\Big(F, W_i, \frac{1}{m}\operatorname{div}(h_i) + \beta_1 \operatorname{div}(u)\Big) = a\Big(F, W_i', \frac{1}{m}\operatorname{div}(h_i') + \beta_1 \operatorname{div}(u)\Big).$$

Since F_i is centred at o_i',

$$\operatorname{mld}\Big(\Big(W_i' \ni o_i', \frac{1}{m}\operatorname{div}(h_i') + \beta_1 \operatorname{div}(u)\Big)/G\Big) = b_0,$$

and F_i is a computing lift of b_0.

Write $g_i(z,u) = z^d + \sum_{j<d} a_{i,j}(u) z^j$ after ϖ_i. Then

$$g_i'(z,u) = z^d + \sum_{j<d} u^{-r_i(d-j)c} a_{i,j}(u) z^j,$$

and $g_i'(z,0)$ is a monic polynomial of degree d vanishing at 0, since $o_i' \in W_i'$. The section $(u=0)$ of W_i' is the surface $(xy + g_i'(z,0) = 0) \subset \mathbb{A}^3$, which is smooth or of type $A_{\nu_0 - 1}$ at the origin, where $\nu_0 \leq d$ is the multiplicity of the root 0 of $g_i'(z,0)$. Thus W_i' is a normal Gorenstein germ with a Du Val Cartier section, hence cDV. Off $(u=0)$, W_i' is isomorphic to an open subset of $W_i \backslash (u=0)$, or of an étale cover of it if the axis cover was used in (G.17), which is smooth since $o_i \in (u=0)$. On $(u=0)$ off the origin, W_i' is smooth near o_i' because the section is. Hence $(W_i' \ni o_i')$ is an isolated cDV point, so it is Gorenstein terminal or smooth. If it is singular, then $2 \leq \operatorname{mult}_0 g_i' \leq \nu_0 \leq d$, and Lemma 5.6 implies that any prime divisor F over W_i' with center o_i' satisfies

$$\operatorname{mult}_F \operatorname{Jac}_{W_i'} \leq \frac{d}{2} a(F, W_i', 0). \tag{G.22}$$

If W_i' is smooth at o_i', then $\operatorname{Jac}_{W_i'}$ is the unit ideal near o_i' and (G.22) holds as well.

The proof is continued below.

Proof of Theorem 18.2, continued. We continue with the chart on which (G.17) holds, with coordinates x, y, z, u, group G, coefficient β_1 and weights $w = (dw_1, dw_2, c)$, with the computing lifts F_i of $b_i = b_0$, and, for any i, with the maximal admissible depth $r_i \geq r_i^0$ (so that (G.21) holds for $r = r_i$) realized by preparations (T)–(C) with coefficients in (u^{ℓ_i}), where $r_i^0 \to +\infty$ and

$\ell_i \to +\infty$. Let $\widetilde{f}_i$ and $\widetilde{h}_i$ be the equation and the mark after the preparations. The equation $\widetilde{f}_i$ has the form $xy + g(z,u)$. For any $r \in [0, r_i] \cap \exp(G)\mathbb{Z}$ let

$$f_{i,r} := \frac{\widetilde{f}_i(u^{rdw_1}x, u^{rdw_2}y, u^{rc}z, u)}{u^{rdc}} \quad \text{and} \quad h_{i,r} := \frac{\widetilde{h}_i(u^{rdw_1}x, u^{rdw_2}y, u^{rc}z, u)}{u^{rmc}}, \tag{G.23}$$

let $W_{i,r} := (f_{i,r} = 0)$, and let

$$D_{i,r} := \frac{1}{m}\operatorname{div}(h_{i,r}|_{W_{i,r}}) + \beta_1 \operatorname{div}(u|_{W_{i,r}}).$$

Then $f_{i,r_i} = f_i' = xy + g_i'(z,u)$, $h_{i,r_i} = h_i'$, and $W_{i,r_i} = W_i'$ are the residual equation, mark, and germ fixed above, and we let $D_i' := D_{i,r_i}$.

Let $\varphi \in \mathbb{C}[[u]]$ be the coefficient in $\widetilde{h}_i$ of a monomial $x^{\mathbf{e}} := x^{e_1}y^{e_2}z^{e_3}$, $\mathbf{e} \in \mathbb{N}^3$, of w-weight

$$\operatorname{wt}_w(x^{\mathbf{e}}) := e_1dw_1 + e_2dw_2 + e_3c.$$

The coefficient of $x^{\mathbf{e}}$ in $h_{i,r}$ is $u^{r(\operatorname{wt}_w(x^{\mathbf{e}})-mc)}\varphi$. Let $\operatorname{mult}_0\varphi$ be the u-order of φ. Since h_{i,r_i} is regular,

$$\operatorname{mult}_0\varphi \geq r_i(mc - \operatorname{wt}_w(x^{\mathbf{e}})) \quad \text{if } \operatorname{wt}_w(x^{\mathbf{e}}) < mc.$$

Thus

$$\operatorname{mult}_0\left(u^{r(\operatorname{wt}_w(x^{\mathbf{e}})-mc)}\varphi\right) \geq \begin{cases} (r_i - r)(mc - \operatorname{wt}_w(x^{\mathbf{e}})) & \text{if } \operatorname{wt}_w(x^{\mathbf{e}}) < mc, \\ r(\operatorname{wt}_w(x^{\mathbf{e}}) - mc) & \text{if } \operatorname{wt}_w(x^{\mathbf{e}}) \geq mc, \end{cases} \tag{G.24}$$

and (G.24) also holds for the coefficients of $\widetilde{f}_i$ and $f_{i,r}$, with dc in place of mc. In particular $f_{i,r}$ and $h_{i,r}$ are regular.

The substitution in (G.23) multiplies $dx \wedge dy \wedge dz \wedge du$ by $u^{r(dc+c)}$ and the equation by u^{rdc}, hence multiplies the residue form $\frac{dy}{y} \wedge dz \wedge du$ by u^{rc}, and it divides the mark by $(u^{rc})^m$. The preparations have Jacobian determinant 1, preserve the residue form, and change the mark by multiples of the equation. Thus, by (G.17), for any prime divisor P over $W_{i,r}$ with center the origin, the restriction of P is a prime divisor F with center o_i over the germ W_i of the theorem such that, with D_i, G, e_F, and b_i those of the theorem,

$$\frac{a(F, W_i, D_i)}{e_F} = \frac{a(P, W_{i,r}, D_{i,r})}{e_P}, \tag{G.25}$$

and $a(F, W_i, 0)$, $\operatorname{mult}_F u$, and $\operatorname{mult}_F h_i$ are bounded from above by a function of r, $a(P, W_{i,r}, 0)$, $\operatorname{mult}_P u$, and $\operatorname{mult}_P h_{i,r}$ which is independent of i. The coordinates x, y, z of $W_{i,r}$ are $u^{(r_i-r)dw_1}x$, $u^{(r_i-r)dw_2}y$, $u^{(r_i-r)c}z$ in the coordinates of W_i', and F_i has center the origin of W_i', so F_i has center the origin of $W_{i,r}$. By (G.25), any such P satisfies $a(P, W_{i,r}, D_{i,r})/e_P \geq b_i = b_0$, with equality for $P = F_i$. Thus

$$\operatorname{mld}((W_{i,r} \ni 0, D_{i,r})/G) = b_0 \quad \text{for any } r \in [0, r_i] \cap \exp(G)\mathbb{Z}. \tag{G.26}$$

Assume that some element $\iota \in G$ exchanges x and y. Then $w_1 = w_2 = 1$. Any $g \in G$ with $g(x) = \chi_x(g)x$ and $g(y) = \chi_y(g)y$ commutes with ι, so $\chi_x(g) = \chi_y(g)$. Since G is abelian, G acts diagonally in a basis x', y' of the plane spanned by x and y. The linear substitution from x, y to x', y' preserves the w-weights, the Jacobian ideals and the compatibility of the limits below, and we use the coordinates x', y', z, u in Lemmas 8.1(1), 7.11, 5.4(2), and 7.10(2) below, which assume a diagonal action.

Step 6. In this step we take the limit (f_Q, h_Q) of the residual equations and marks (f_i', h_i'), and we prove that $a_Q \geq a_0 > b_0$ for the quotient values a_0 and a_Q defined below.

By Lemma 2.6(3), possibly passing to an infinite subset of indices and enlarging $\Bbbk$, the tuples formed by the equation and the mark of the chart, by $\widetilde{f}_i, \widetilde{h}_i$ and by f_i', h_i' have a compatible generic limit extending (f_*, h_*). Here $f_* = xy + z^d$, and any monomial of h_* has w-weight at

least mc, the part of weight mc being $\mathrm{in}_w(h_*)$. Let (f_Q, h_Q) be the limit of (f_i', h_i'). We claim that

$$f_Q = xy + g_Q(z,u) \qquad \text{and} \qquad h_Q \in \Bbbk[[u]][x,y,z], \tag{G.27}$$

where $g_Q \in \Bbbk[[u]][z]$ is monic of degree d in z, and the part of h_Q of maximal w-weight is $\mathrm{in}_w(h_*)$, of weight mc. By (G.24) with $r = r_i$, the coefficient in h_i' of a monomial $\mathbf{x}$ with $\mathrm{wt}_w(\mathbf{x}) > mc$ has u-order at least r_i, and $r_i \to +\infty$. Hence the coefficient in h_Q is 0, and h_Q is a polynomial in x, y, z. Since the coefficient of a monomial $\mathbf{x}$ in $h_{i,r}$ is $u^{r(\mathrm{wt}_w(\mathbf{x})-mc)}$ times the coefficient in $\widetilde{h}_i$, a monomial $\mathbf{x}$ of weight mc has equal coefficients in h_i' and in $\widetilde{h}_i$, and the coefficient of $\mathbf{x}$ in $\widetilde{h}_i$ is congruent modulo u^{ℓ_i} to the coefficient of $\mathbf{x}$ in the mark of the chart. Since $\ell_i \to +\infty$, the coefficient in h_Q of a monomial of weight mc is the coefficient in h_*, that is, in $\mathrm{in}_w(h_*)$.

For f_Q, let $\mathbf{x}$ be a monomial in x, y, z. If $\mathrm{wt}_w(\mathbf{x}) > dc$, then, by (G.24) for the coefficients of $f_{i,r}$ with $r = r_i$, the coefficient of $\mathbf{x}$ in f_i' has u-order at least r_i, so the coefficient of $\mathbf{x}$ in f_Q is 0. If $\mathrm{wt}_w(\mathbf{x}) = dc$, then the coefficient of $\mathbf{x}$ in f_i' equals the coefficient of $\mathbf{x}$ in $\widetilde{f}_i$, which is congruent modulo u^{ℓ_i} to the coefficient of $\mathbf{x}$ in the equation of the chart, so the coefficient of $\mathbf{x}$ in f_Q is the coefficient of $\mathbf{x}$ in $f_* = xy + z^d$. Since the preparations preserve the shape $xy + g(z,u)$, z^n has weight nc, and $xy + z^d$ has coefficient 1 at z^d, the claim for f_Q follows.

Let $f_0 := xy + z^d$. Let $r' \in \exp(G)\mathbb{Z}$ be a positive integer. Then $r' \le r_i$ for any $i \gg 0$. By Lemma 2.6(3), possibly passing to an infinite subset of indices and enlarging $\Bbbk$, the tuples $(f_{i,r'}, h_{i,r'})$ have a compatible generic limit $(f_{*,r'}, h_{*,r'})$ jointly with the previous tuples, and

$$f_{*,r'} - f_0 \in (u^{r'}) \qquad \text{and} \qquad h_{*,r'} - \mathrm{in}_w(h_*) \in (u^{r'}). \tag{G.28}$$

By (G.24) with $r = r'$, the coefficient in $h_{i,r'}$ of a monomial $\mathbf{x}$ with $\mathrm{wt}_w(\mathbf{x}) < mc$ has u-order at least $r_i - r' \to +\infty$, and the coefficient of a monomial $\mathbf{x}$ with $\mathrm{wt}_w(\mathbf{x}) > mc$ belongs to $(u^{r'})$. A monomial $\mathbf{x}$ of weight mc has equal coefficients in $h_{i,r'}$ and in $\widetilde{h}_i$, and the coefficient of $\mathbf{x}$ in $\widetilde{h}_i$ is congruent modulo u^{ℓ_i} to the coefficient of $\mathbf{x}$ in the mark of the chart. Since $\ell_i \to +\infty$, the limit coefficient of $\mathbf{x}$ is the coefficient of $\mathbf{x}$ in $\mathrm{in}_w(h_*)$. This proves the second inclusion. For the first inclusion, by (G.24) for the coefficients of $f_{i,r}$ with $r = r'$, the coefficient in $f_{i,r'}$ of a monomial $\mathbf{x}$ with $\mathrm{wt}_w(\mathbf{x}) < dc$ has u-order at least $r_i - r'$, and the coefficient of a monomial $\mathbf{x}$ with $\mathrm{wt}_w(\mathbf{x}) > dc$ belongs to $(u^{r'})$. Moreover a monomial $\mathbf{x}$ of weight dc has equal coefficients in $f_{i,r'}$ and in $\widetilde{f}_i$, with limit the coefficient of $\mathbf{x}$ in $f_* = xy + z^d$.

Let $\widehat{\mathcal{X}} := \operatorname{Spec} \Bbbk[[x,y,z,u]]$, $W_0 := (f_0 = 0) \subset \widehat{\mathcal{X}}$,

$$D_0 := \frac{1}{m} \operatorname{div}(\mathrm{in}_w(h_*)|_{W_0}) + \beta_1 \operatorname{div}(u|_{W_0}), \qquad \text{and} \qquad a_0 := \mathrm{mld}((W_0 \ni 0, D_0)/G).$$

We prove that any prime divisor E over W_0 with center 0 satisfies $a(E, W_0, D_0)/e_E \ge b_0$, so that

$$a_0 \ge b_0. \tag{G.29}$$

Let $j_E := \mathrm{mult}_E \operatorname{Jac}_{W_0}$, let N be an integer with

$$N > 2j_E + \mathrm{mult}_E \mathrm{in}_w(h_*) + \mathrm{mult}_E u + 3,$$

and let $r' \in \exp(G)\mathbb{Z}$ with $r' \ge N$, so that (G.28) holds. Let $\psi \in \Bbbk[x,y,z,u]$ be the polynomial of degree less than N with $\mathrm{in}_w(h_*) - \psi \in \mathfrak{n}_*^N$. Since

$$\mathrm{mult}_E(\mathfrak{n}_*^N) \ge N > \mathrm{mult}_E \mathrm{in}_w(h_*),$$

we have $\mathrm{mult}_E \psi = \mathrm{mult}_E \mathrm{in}_w(h_*)$. Since $r' \ge N$, (G.28) implies $f_{*,r'} - f_0 \in \mathfrak{n}_*^N$ and $h_{*,r'} - \psi \in \mathfrak{n}_*^N$.

By Lemma 2.12(3), E restricts to a prime divisor E^{alg} over the germ $((f_0 = 0) \ni 0)$ in $\mathbb{A}^4_{\Bbbk}$ with center 0, with isomorphic residue field, equal log discrepancy, and equal decomposition and inertia groups. Moreover $\mathrm{mult}_{E^{\mathrm{alg}}} u = \mathrm{mult}_E u$, $\mathrm{mult}_{E^{\mathrm{alg}}} \psi = \mathrm{mult}_E \psi$, and $\mathrm{mult}_{E^{\mathrm{alg}}} \operatorname{Jac}_{W_0} = j_E$, since the valuation of E^{alg} is the restriction of the valuation of E. Let $R_0 \subset \Bbbk$ be a finitely generated $\mathbb{C}$-subalgebra containing the coefficients of ψ and of a projective birational morphism onto $(f_0 = 0) \subset \mathbb{A}^4_{\Bbbk}$ on which E^{alg} is a divisor. For a complex point ξ of $\operatorname{Spec} R_0$ let $\psi_\xi \in \mathbb{C}[x,y,z,u]$

be the specialization of ψ. Possibly replacing $\operatorname{Spec} R_0$ with a non-empty open subset, for any complex point ξ the specialization of E^{alg} is a prime divisor E_ξ over $W_{0,\mathbb{C}} := (f_0 = 0) \subset \mathbb{A}^4_{\mathbb{C}}$ with center 0 such that

$$a(E_\xi, W_{0,\mathbb{C}}, 0) = a(E, W_0, 0), \quad \operatorname{mult}_{E_\xi} u = \operatorname{mult}_E u, \quad \operatorname{mult}_{E_\xi} \psi_\xi = \operatorname{mult}_E \psi,$$
$$\operatorname{mult}_{E_\xi} \operatorname{Jac}_{W_{0,\mathbb{C}}} = j_E, \quad \text{and} \quad I_G(E_\xi) = I_G(E),$$

because the log discrepancies and orders of the specializations, and the stabilizer and inertia subgroups in the finite group G, are constant on a non-empty open subset of $\operatorname{Spec} R_0$. By Definition 2.5, possibly passing to an infinite subset of indices, for any i there exists a complex point ξ_i of $\operatorname{Spec} R_0$ such that

$$f_{i,r'} - f_0 \in \mathfrak{n}^N \qquad \text{and} \qquad h_{i,r'} - \psi_{\xi_i} \in \mathfrak{n}^N. \tag{G.30}$$

Fix i, and let $\widehat{\mathcal{X}}_{\mathbb{C}} := \operatorname{Spec} \mathbb{C}[[x, y, z, u]]$. We use Lemma 8.1(1) with the complex germ $(f_0 = 0) \subset \widehat{\mathcal{X}}_{\mathbb{C}}$, the completion of the normal germ $(W_{0,\mathbb{C}} \ni 0)$, in place of $\widehat{W}_0$, with $f' := f_{i,r'}$, whose zero locus is the completion of the algebraic germ $(W_{i,r'} \ni 0)$, normal since $f_{i,r'}$ has the form $xy + g(z, u)$ with $g \neq 0$, with $I := \mathfrak{n}$, N in place of P', and with F the divisor over the completion corresponding to E_{ξ_i} (Lemma 2.12(3)). Since $(N-2) \operatorname{mult}_{E_{\xi_i}} \mathfrak{n} \geq N - 2 > 2j_E$, there exists a prime divisor E'_i over $W_{i,r'}$ with center the origin such that

$$a(E'_i, W_{i,r'}, 0) = a(E, W_0, 0), \qquad e_{E'_i} = e_E,$$
$$\operatorname{mult}_{E'_i} u = \operatorname{mult}_E u, \qquad \text{and} \qquad \operatorname{mult}_{E'_i} \psi_{\xi_i} = \operatorname{mult}_E \psi,$$

where the equalities for u and ψ_{ξ_i} hold because

$$\max\{\operatorname{mult}_E u, \operatorname{mult}_E \psi\} < N - j_E \leq N \operatorname{mult}_{E_{\xi_i}} \mathfrak{n} - j_E.$$

By (G.30),

$$\operatorname{mult}_{E'_i}(h_{i,r'} - \psi_{\xi_i}) \geq N > \operatorname{mult}_{E'_i} \psi_{\xi_i},$$

so $\operatorname{mult}_{E'_i} h_{i,r'} = \operatorname{mult}_E \operatorname{in}_w(h_*)$. Thus

$$\frac{a(E, W_0, D_0)}{e_E} = \frac{a(E'_i, W_{i,r'}, D_{i,r'})}{e_{E'_i}} \geq b_0$$

by (G.26). This proves (G.29).

Since $a(E, W_0, 0)$, $\operatorname{mult}_E u$, and $\operatorname{mult}_E \operatorname{in}_w(h_*)$ are integers, $e_E \leq |G|$, and $\beta_1 \in \mathbb{Q}$, the values $a(E, W_0, D_0)/e_E$ belong to $\frac{1}{l_0}\mathbb{Z}$ for a positive integer l_0 independent of E. By (G.29), a_0 is finite, and we may fix a prime divisor E over W_0 with center 0 and $a_0 = a(E, W_0, D_0)/e_E$. Assume that $a_0 = b_0$, and let N, r', and E'_i be the integers and prime divisors constructed for E after (G.29). By (G.25), the restriction of E'_i to a prime divisor over W_i is a computing lift of $b_i = b_0$, and the log discrepancy of the restriction with respect to $(W_i, 0)$ and the orders of u and h_i along the restriction are bounded independently of i, because r', $a(E'_i, W_{i,r'}, 0)$, $\operatorname{mult}_{E'_i} u$, and $\operatorname{mult}_{E'_i} h_{i,r'}$ are independent of i. This contradicts the assumption at the beginning of the proof. Thus

$$a_0 > b_0. \tag{G.31}$$

By (G.29), a_0 is finite and non-negative, so Proposition 7.12, with weights $(dw_1, dw_2, c, 0)$ on (x, y, z, u) and the single mark $P_1 = \operatorname{in}_w(h_*)$ with coefficient $\frac{1}{m}$, implies

$$a_0 = \operatorname{mld}\Big(\Big(\widehat{\mathcal{X}} \ni 0, \operatorname{div}(f_0) + \beta_1 \operatorname{div}(u) + \frac{1}{m} \operatorname{div}(\operatorname{in}_w(h_*))\Big)/G\Big).$$

Let

$$a_Q := \operatorname{mld}\Big(\Big(\widehat{\mathcal{X}} \ni 0, \operatorname{div}(f_Q) + \beta_1 \operatorname{div}(u) + \frac{1}{m} \operatorname{div}(h_Q)\Big)/G\Big).$$

By (G.27), with u of weight 0, the parts of maximal weight of f_Q and h_Q are f_0 and $\mathrm{in}_w(h_*)$. By Lemma 7.11 with $(Q_1, Q_2) := (f_Q, h_Q)$, $(c_1, c_2) := (1, \frac{1}{m})$, and β_1 in place of β, any prime divisor E over $\widehat{\mathcal{X}}$ with center the closed point satisfies

$$\frac{a(E, \widehat{\mathcal{X}}, 0) - \beta_1 \operatorname{mult}_E u - \operatorname{mult}_E f_Q - \frac{1}{m} \operatorname{mult}_E h_Q}{e_E} \geq a_0.$$

Taking the infimum over E, we obtain $a_Q \geq a_0$, and by (G.31),

$$a_Q \geq a_0 > b_0. \tag{G.32}$$

Step 7. In this step we prove that $\operatorname{mult}_{F_i} h'_i \to +\infty$.

Assume that $\operatorname{mult}_{F_i} h'_i$ is bounded on an infinite subset of indices. We pass to that subset. Since $\operatorname{mult}_{F_i} u \leq R'$, $\operatorname{mult}_{F_i} u$ is bounded, and $e_{F_i} \leq |G|$. Since F_i is a computing lift of b_0 on W'_i by (G.25) and (G.26),

$$a(F_i, W'_i, 0) = b_0 e_{F_i} + \frac{1}{m} \operatorname{mult}_{F_i} h'_i + \beta_1 \operatorname{mult}_{F_i} u$$

is bounded. By (G.22), $j'_i := \operatorname{mult}_{F_i} \operatorname{Jac}_{W'_i}$ is bounded. Let $\varrho_i \in \{x, y, z, u\}$, resp. $\varrho_i \in \{x', y', z, u\}$ if some element of G exchanges x and y, be a coordinate with $\operatorname{mult}_{F_i}(\partial f'_i / \partial \varrho_i) = j'_i$, let k_0 be the residue class modulo e_{F_i} in Lemma 5.4(2) for f'_i, F_i, ϱ_i, and $(H_1, H_2) := (h'_i, u)$, and let n_i be the least integer with $n_i \equiv k_0 \pmod{e_{F_i}}$ and

$$n_i \geq 2j'_i + \max\{\operatorname{mult}_{F_i} h'_i, \operatorname{mult}_{F_i} u\} + 1.$$

Then $n_i > 2j'_i$, $n_i - j'_i > \max\{\operatorname{mult}_{F_i} h'_i, \operatorname{mult}_{F_i} u\}$, and $n_i \leq 2j'_i + \max\{\operatorname{mult}_{F_i} h'_i, \operatorname{mult}_{F_i} u\} + e_{F_i}$ is bounded. By Lemma 5.4(2) with n_i in place of k, there exists a prime divisor $\widehat{F}_i$ over $\widehat{\mathcal{X}}_{\mathbb{C}}$ with center the closed point 0 such that

$$\operatorname{mult}_{\widehat{F}_i} f'_i = n_i, \quad \operatorname{mult}_{\widehat{F}_i} h'_i = \operatorname{mult}_{F_i} h'_i, \quad \operatorname{mult}_{\widehat{F}_i} u = \operatorname{mult}_{F_i} u,$$
$$a(\widehat{F}_i, \widehat{\mathcal{X}}_{\mathbb{C}}, 0) = a(F_i, W'_i, 0) + n_i, \quad \text{and} \quad I_G(\widehat{F}_i) = I_G(F_i).$$

Thus $\operatorname{mult}_{\widehat{F}_i} u$, $\operatorname{mult}_{\widehat{F}_i} f'_i$, $\operatorname{mult}_{\widehat{F}_i} h'_i$, and $a(\widehat{F}_i, \widehat{\mathcal{X}}_{\mathbb{C}}, 0)$ are bounded, and

$$\frac{a(\widehat{F}_i, \widehat{\mathcal{X}}_{\mathbb{C}}, 0) - \beta_1 \operatorname{mult}_{\widehat{F}_i} u - \operatorname{mult}_{\widehat{F}_i} f'_i - \frac{1}{m} \operatorname{mult}_{\widehat{F}_i} h'_i}{e_{\widehat{F}_i}} = \frac{a(F_i, W'_i, D'_i)}{e_{F_i}} = b_0.$$

By Lemma 2.12(3), $\widehat{F}_i$ restricts to a prime divisor $\widehat{F}_i^{\mathrm{alg}}$ over $\mathbb{A}^4_{\mathbb{C}}$ with center 0, with equal orders of u, f'_i, and h'_i and with

$$a(\widehat{F}_i^{\mathrm{alg}}, \mathbb{A}^4_{\mathbb{C}}, 0) = a(\widehat{F}_i, \widehat{\mathcal{X}}_{\mathbb{C}}, 0) \qquad \text{and} \qquad I_G(\widehat{F}_i^{\mathrm{alg}}) = I_G(\widehat{F}_i).$$

Then Lemma 7.10(2) with $\widehat{F}_i^{\mathrm{alg}}$ in place of F_i, $(Q_{i,1}, Q_{i,2}) := (f'_i, h'_i)$, $(c_1, c_2) := (1, \frac{1}{m})$, β_1 in place of β, and the limit (f_Q, h_Q) implies $a_Q \leq b_0$, which contradicts (G.32). Thus $\operatorname{mult}_{F_i} h'_i \to +\infty$.

Step 8. In this step we prove that, over an algebraic closure of $\Bbbk((u))$, the residual limit pair is either klt, or lc with a single non-klt point, and in the second case we normalize the equation and the mark by Proposition 7.6.

Let $V_Q := (xy + g_Q(z, u) = 0) \subset \mathbb{A}^3_{\mathbb{K}}$, and let $B_Q := \frac{1}{m} \operatorname{div}(h_Q|_{V_Q})$. Here $h_Q|_{V_Q} \neq 0$, since the part $\mathrm{in}_w(h_*)$ of maximal weight of h_Q is non-zero on $(f_0 = 0)$. The surface V_Q is normal, and its singular points are the A_{j-1} points $(0, 0, z_0)$, where $z_0 \in \mathbb{K}$ is a root of g_Q of multiplicity $j \geq 2$.

Let $c' := c / \gcd(d, c)$, let X be the closure of V_Q in

$$\mathbb{P}_{\mathbb{K}}\Big(1, \frac{dw_1}{\gcd(d, c)}, \frac{dw_2}{\gcd(d, c)}, c'\Big)$$

with homogeneous coordinates t, x, y, z, in the order of the weights, where $xy+g_Q$ is homogenized to degree dc', let $\Delta_\infty := X \cap (t=0) = X \backslash V_Q$, and let $\overline{B}$ be the closure of B_Q in X. The pair

$$\Big((f_0=0), \frac{1}{m}\operatorname{div}(\operatorname{in}_w(h_*))\Big)$$

over $\mathbb{K}$ is lc and has no component of coefficient 1, the case of such a component being excluded in (E). By Lemma 7.3 (the first paragraph of the statement, which has no hypothesis at the vertex), X is a normal projective klt surface, $K_X + \overline{B} + \Delta_\infty \sim_{\mathbb{R}} 0$, hence $K_X + \Delta_\infty + \overline{B} \sim_{\mathbb{Q}} 0$ since the coefficients are rational, and $(X, \Delta_\infty + \overline{B})$ is plt near Δ_∞. Here Δ_∞ is the curve $(f_0 = 0)$ in $\mathbb{P}(dw_1/\gcd(d,c), dw_2/\gcd(d,c), c')$, irreducible since f_0 is, and the divisor of t, hence ample and $\mathbb{Q}$-Cartier.

Let $\Gamma \subset V_Q$ be a curve with $\operatorname{mult}_\Gamma B_Q \geq 1$. Since Δ_∞ is ample, the closure of Γ intersects Δ_∞, which is impossible because $(X, \Delta_\infty + \overline{B})$ is plt near Δ_∞. Thus any component of B_Q has coefficient less than 1. Since V_Q is klt, the non-klt locus Σ of (V_Q, B_Q) is finite, and $\Sigma \cap \Delta_\infty = \emptyset$ in X. If $\Sigma \neq \emptyset$, then the non-klt locus $\Delta_\infty \sqcup \Sigma$ of $(X, \Delta_\infty + \overline{B})$ is disconnected, and Theorem 7.2 with $Z := \operatorname{Spec}\mathbb{K}$ implies that Σ is a single point o. By Theorem 7.2, the pair $(X, \Delta_\infty + \overline{B})$ is lc. Thus one of the following holds.

(a) (V_Q, B_Q) is klt.
(b) (V_Q, B_Q) is lc, and the non-klt locus of (V_Q, B_Q) is a closed point o.

In case (b) we prove that

$$g_Q(z,u) = (z-\alpha(u))^d \qquad \text{and} \qquad o = (0,0,\alpha(u)) \tag{G.33}$$

for some $\alpha(u) \in \Bbbk[[u]]$ with the character of z. By Lemma 7.3(3) (in the case (b) of that lemma, over $\mathbb{K}$, with $xy+g_Q$, $xy+z^d$, 1, h_Q, and $\frac{1}{m}$ in place of F, f_0, N, Q_1, and κ_1), there exists a polynomial automorphism of $\mathbb{A}^3_{\mathbb{K}}$ whose coordinate functions vanish at o and which pulls f_0 back to $xy+g_Q$. Thus $(V_Q \ni o) \cong ((f_0=0) \ni 0)$.

Since $(V_Q \ni o) \cong ((f_0=0) \ni 0)$ is an A_{d-1} point, $o = (0,0,\alpha)$ for a root $\alpha \in \mathbb{K}$ of g_Q of multiplicity d, so $g_Q = (z-\alpha)^d$ as g_Q is monic of degree d. The coefficient $-d\alpha$ of z^{d-1} in g_Q belongs to $\Bbbk[[u]]$, so $\alpha = \alpha(u) \in \Bbbk[[u]]$. Since $g_Q = (z-\alpha(u))^d$ is G-semi-invariant and $\alpha(u)$ is the unique root of g_Q, $\alpha(u)$ has the character of z. This proves (G.33).

In case (b), let $h^\alpha_Q(x,y,z,u)$ be the representative of $h_Q(x,y,z+\alpha(u),u)$ modulo f_0 without monomials divisible by xy. Since $\alpha(u)$ has w-weight 0 and $\operatorname{in}_w(h_*)$ has no monomial divisible by xy, $h^\alpha_Q \in \Bbbk[[u]][x,y,z]$ has maximal w-weight mc, with $\operatorname{in}_w(h_*)$ as part of maximal weight. By (G.33), the substitution $z \to z+\alpha(u)$ maps $(V_Q \ni o, B_Q)$ to $((f_0=0) \ni 0, \frac{1}{m}\operatorname{div}(h^\alpha_Q))$, which is lc, not klt at 0, and without components of coefficient at least 1. Since $\frac{1}{m}\cdot mc = c = w_1 + w_2$, Proposition 7.6(1) over $\mathbb{K}$ implies that there exists a unique $U \in \mathbb{K}[y]$, divisible by y, whose monomials y^j satisfy $jdw_2 < c$, such that

$$h_Q(x,y,z+\alpha(u),u) \equiv \operatorname{in}_w(h_*)\Big(x - \frac{(z+U(y))^d - z^d}{y},\ y,\ z+U(y),\ u\Big) \pmod{f_0}, \tag{G.34}$$

and $U = 0$ if $w_1 = w_2 = 1$. By the equivariance in Proposition 7.6(1), with G acting on $\mathbb{K}$ through u, $U(y)$ has the character of z. By Proposition 7.6(2), $U \in \Bbbk((u))[y]$. Since U has finitely many coefficients, there exists $\nu \in \mathbb{N}$ with $u^\nu U \in \Bbbk[[u]][y]$, and ν is independent of i.

We continue the proof below.

Proof of Theorem 18.2, continued. Let $\mathcal{I} \subset \mathbb{N}^+$ be the infinite set of indices to which we have passed, along which (f_Q, h_Q) of (G.27) is the compatible generic limit of the (f'_i, h'_i). **Steps 9 and 10** use the following properties.

(1) Since the preparations and (G.23) fix u, $\operatorname{mult}_{F_i} u \leq R'$. Since F_i computes b_0,

$$a(F_i, W'_i, 0) = b_0 e_{F_i} + \frac{1}{m}\operatorname{mult}_{F_i} h'_i + \beta_1 \operatorname{mult}_{F_i} u. \tag{G.35}$$

(2) $j_i' := \mathrm{mult}_{F_i} \mathrm{Jac}_{W_i'}$ satisfies $j_i' \le \frac{d}{2} a(F_i, W_i', 0)$ by (G.22).

Step 9. In this step we show that case (a) does not occur, that $\alpha(0) = 0$ in case (b), and that the lifts F_i concentrate along the curve $(x = y = z - \alpha(u) = 0)$, see (G.40).

We first fix the exponents. A log resolution of $(V_Q, \mathrm{div}(h_Q|_{V_Q}))$ has finitely many prime divisors E, and for any such E the function $\mu' \to a(E, V_Q, \mu' \mathrm{div}(h_Q|_{V_Q}))$ is affine. For any such E other than the prime divisors with center o in case (b), $a(E, V_Q, B_Q) > 0$. Thus there exists a rational number $\mu > \frac{1}{m}$ such that $(V_Q, \mu \mathrm{div}(h_Q|_{V_Q}))$ is klt in case (a), and klt on $V_Q \backslash \{o\}$ in case (b). Fix rational numbers $0 < \theta_1 < \theta_2 < 1$ with

$$\epsilon := \theta_1 \mu - \frac{1}{m} - (1 - \theta_1)\Big(1 + \frac{d}{m}\Big) > 0, \tag{G.36}$$

which is possible since the left-hand side tends to $\mu - \frac{1}{m} > 0$ as $\theta_1 \to 1$.

Next we pass to the limit. Let $W_Q := (f_Q = 0) \subset \widehat{\mathcal{X}}$ and

$$\mathcal{J}_Q := \mathcal{J}(\widehat{\mathcal{X}}, f_Q^{\theta_2} h_Q^{\theta_2 \mu}).$$

The hypersurface W_Q is normal, since $\mathrm{Sing}(W_Q) \subset (x = y = g_Q = 0)$ and

$$\dim(x = y = g_Q = 0) \le 1.$$

Moreover $h_Q|_{W_Q} \ne 0$. Let $S_Q := W_Q \cap (u = 0)$ in case (a), and $S_Q := W_Q \cap ((u = 0) \cup C_\alpha)$ with $C_\alpha := (x = y = z - \alpha(u) = 0)$ in case (b). Let $V_{Q,\Bbbk((u))} := \mathrm{Spec}\, \Bbbk((u))[x, y, z]/(f_Q)$. The points of $W_Q \backslash S_Q$ map into $V_{Q,\Bbbk((u))}$, and in case (b) into the complement in $V_{Q,\Bbbk((u))}$ of the closed point $(x = y = z - \alpha(u) = 0)$, the image of o, under the morphism

$$W_Q \to \mathrm{Spec}\, \Bbbk[[u]][x, y, z]/(f_Q),$$

which is regular, being a localization followed by the completion of an excellent local ring. Hence a log resolution of

$$(V_{Q,\Bbbk((u))}, \mathrm{div}(h_Q|_{V_{Q,\Bbbk((u))}}))$$

pulls back to a log resolution over $W_Q \backslash S_Q$, with log discrepancies equal to those of the corresponding divisors over $V_{Q,\Bbbk((u))}$. Log discrepancies do not change under the base change from $\Bbbk((u))$ to an algebraic closure of $\Bbbk((u))$, so the choice of μ implies that $(W_Q, \mu \mathrm{div}(h_Q|_{W_Q}))$ is klt on $W_Q \backslash S_Q$. By Lemma 6.1(3), $\mathcal{J}_Q$ is the unit ideal near any point of $W_Q \backslash S_Q$. Thus the zero locus of $(f_Q) + \mathcal{J}_Q$ is contained in S_Q. The ideal of S_Q contains u in case (a) and $u \cdot (x, y, z - \alpha)$ in case (b), so $u \in \sqrt{(f_Q) + \mathcal{J}_Q}$, resp. $u \cdot (x, y, z - \alpha) \subset \sqrt{(f_Q) + \mathcal{J}_Q}$. Since $\Bbbk[[x, y, z, u]]$ is Noetherian, there exists a positive integer N with

$$u^N \in (f_Q) + \mathcal{J}_Q \text{ in case (a)}, \qquad \big(u \cdot (x, y, z - \alpha)\big)^N \subset (f_Q) + \mathcal{J}_Q \text{ in case (b)}. \tag{G.37}$$

If $\alpha(0) \ne 0$ in case (b), then $z - \alpha$ is a unit, and $u^N \in (f_Q) + \mathcal{J}_Q$ as well.

We spread (G.37) to the members. Let

$$\varphi_0 := u^N, \quad \varphi_1 := u^N x^N, \quad \varphi_2 := u^N y^N, \quad \text{and} \quad \varphi_3 := u^N \Big(\frac{1}{d!} \partial_z^{d-1} f_Q\Big)^N.$$

In case (b), $\frac{1}{d!} \partial_z^{d-1} f_Q = z - \alpha(u)$. In Lemma 6.1 we take $n = 4$, $(x_1, \dots, x_4) = (x, y, z, u)$, and the pairs (f_i', h_i') as the tuples. For $k \in \{0, 1, 2, 3\}$, φ_k is the value at f_Q of a polynomial Π_k^0 in $x_1, \dots, x_4$ and the variables of that lemma standing for the partial derivatives of the first entry, without parameter variables. Let $\mathcal{K} := \{0\}$ in case (a) and in case (b) with $\alpha(0) \ne 0$, and $\mathcal{K} := \{1, 2, 3\}$ in case (b) with $\alpha(0) = 0$. By (G.37), for any $k \in \mathcal{K}$ we may write

$$\varphi_k = q_k f_Q + \psi_k, \qquad q_k \in \Bbbk[[x, y, z, u]], \quad \text{and} \quad \psi_k \in \mathcal{J}_Q.$$

Let L be a positive integer, let $q_{k,L}$ be the sum of the terms of q_k of degree less than L, and let $\Pi_k := \Pi_k^0 - Q_k y_{1,0}$, where Q_k is a polynomial in $x_1, \dots, x_4$ and the parameter variables of

Lemma 6.1 whose value at a vector τ_k with entries in $\Bbbk$ is $q_{k,L}$. Then

$$\Pi_k(f_Q, h_Q, \tau_k) = \psi_k + (q_k - q_{k,L}) f_Q \in \mathcal{J}_Q + \mathfrak{n}_*^L f_Q \subset \mathcal{J}_Q + \mathfrak{n}_*^L.$$

By Lemma 6.1(1) and (2), with the exponents $(\theta_1, \theta_1\mu) < (\theta_2, \theta_2\mu)$, for all but finitely many $i \in \mathcal{I}$ there exists a complex vector $\tau_{k,i}$ such that $z_{k,i} := \Pi_k(f_i', h_i', \tau_{k,i})$ satisfies the following: any prime divisor P over $\widehat{\mathcal{X}}_{\mathbb{C}}$ with center the closed point and $\operatorname{mult}_P z_{k,i} < L$ satisfies

$$\operatorname{mult}_P z_{k,i} + a(P, \widehat{\mathcal{X}}_{\mathbb{C}}, 0) > \theta_1 \operatorname{mult}_P f_i' + \theta_1\mu \operatorname{mult}_P h_i'.$$

Since $z_{k,i} - \Pi_k^0(f_i') \in (f_i')$, the restrictions of $z_{k,i}$ and $\Pi_k^0(f_i')$ to W_i' coincide. Let

$$\omega_{k,i} := \operatorname{mult}_{F_i} \big(\Pi_k^0(f_i')|_{W_i'}\big).$$

If $\omega_{k,i} < L$, then $z_{k,i}|_{W_i'} \neq 0$, and Lemma 5.5(1) with $\mathcal{X} = \widehat{\mathcal{X}}_{\mathbb{C}}$, $\theta = \theta_1$, $f = f_i'$, $h = h_i'$, $g = z_{k,i}$, $\rho = L$, and $F = F_i$ gives

$$(\theta_1\mu - 1 + \theta_1) \operatorname{mult}_{F_i} h_i' < (2 - \theta_1)\omega_{k,i} + a(F_i, W_i', 0) + 2(1 - \theta_1) j_i' + 1 - \theta_1.$$

We have $j_i' \le \frac{d}{2} a(F_i, W_i', 0)$, and by (G.35), $a(F_i, W_i', 0) \le \frac{1}{m} \operatorname{mult}_{F_i} h_i' + B_1$ with

$$B_1 := B_0|G| + |\beta_1|R',$$

since $b_0 \le B_0$ and $e_{F_i} \le |G|$. Hence $\omega_{k,i} \ge L$, or

$$\epsilon \operatorname{mult}_{F_i} h_i' < (2 - \theta_1)\omega_{k,i} + \big(1 + (1 - \theta_1)d\big)B_1 + 1. \tag{G.38}$$

If $\omega_{k,i} < L$, the right-hand side of (G.38) is less than $(2 - \theta_1)L + (1 + (1 - \theta_1)d)B_1 + 1$, which does not depend on i. Since $\operatorname{mult}_{F_i} h_i' \to +\infty$ and $\epsilon > 0$ by (G.36), $\omega_{k,i} < L$ and (G.38) hold together for only finitely many i, so

(G.39)
$$\operatorname{mult}_{F_i} \big(\Pi_k^0(f_i')|_{W_i'}\big) \ge L \quad \text{for any } k \in \mathcal{K} \text{ and any } i \ge i_L, \text{ for an integer } i_L \text{ depending on } L.$$

If $\mathcal{K} = \{0\}$, take $L := NR' + 1$. Then (G.39) reads $N \operatorname{mult}_{F_i} u \ge NR' + 1$ for $i \gg 0$, which contradicts $\operatorname{mult}_{F_i} u \le R'$. Thus case (b) holds and $\alpha(0) = 0$.

For any i let

$$\alpha_i^\natural(u) := -\frac{1}{d!}\partial_z^{d-1} g_i'(0, u) \in \mathbb{C}[[u]],$$

and for a positive integer N' let $\widetilde{\alpha}_i \in \mathbb{C}[u]$ be the sum of the terms of $\alpha_i^\natural$ of degree less than N'. Since $\frac{1}{d!}\partial_z^{d-1} g_Q$ has the character of z, so do $\alpha_i^\natural$ and $\widetilde{\alpha}_i$. We show that for any $i \gg 0$,

$$\min\{\operatorname{mult}_{F_i} x,\ \operatorname{mult}_{F_i} y,\ \operatorname{mult}_{F_i}(z - \widetilde{\alpha}_i(u))\} \ge N'. \tag{G.40}$$

Take $L := N(N' + R')$. For $i \gg 0$, (G.39) with $k = 1$ reads

$$N(\operatorname{mult}_{F_i} u + \operatorname{mult}_{F_i} x) \ge L,$$

so $\operatorname{mult}_{F_i} x \ge N'$. Likewise, (G.39) with $k = 2$ and $k = 3$ implies $\operatorname{mult}_{F_i} y \ge N'$ and $\operatorname{mult}_{F_i}(\frac{1}{d!}\partial_z^{d-1} g_i') \ge N'$. Since g_i' is monic of degree d in z, $\frac{1}{d!}\partial_z^{d-1} g_i' = z - \alpha_i^\natural(u)$, so

$$z - \widetilde{\alpha}_i = \frac{1}{d!}\partial_z^{d-1} g_i' + (\alpha_i^\natural - \widetilde{\alpha}_i),$$

and both terms have F_i-order at least N'. Thus (G.40) holds. By Lemma 2.7 and $\alpha(0) = 0$, moreover $\widetilde{\alpha}_i(0) = 0$.

Step 10. In this step we construct, for $i \gg 0$, an admissible preparation of depth $r_i + k'$ for a fixed positive integer k', which contradicts the maximality of r_i.

Let $T_\alpha(x, y, z) := (x, y, z - \alpha(u))$. Since α has weight $0 < c$ and any monomial y^j of U has weight $jdw_2 < c$, the automorphism $\Phi_{z^d} \circ T_\alpha$ preserves the w-weight filtration of $\Bbbk((u))[x, y, z]$ and induces the identity on the associated graded ring. By (G.34), where Φ_{z^d} is the automorphism of

Proposition 7.6(3) for $R = \Bbbk((u))$, $g = z^d$, and U, and since Φ_{z^d} preserves $xy + z^d$, there exists $H \in \Bbbk((u))[x, y, z]$ with

$$h_Q\big((\Phi_{z^d} \circ T_\alpha)^{-1}(x, y, z)\big) = \mathrm{in}_w(h_*)(x, y, z) + H(x, y, z)(xy + z^d). \tag{G.41}$$

The left-hand side of (G.41) has top part $\mathrm{in}_w(h_*)$ of weight mc, and $xy + z^d$ is homogeneous of weight dc, so every monomial of H has w-weight less than $mc - dc$, and $H = 0$ if $mc \le dc$. Let ν' be the maximum of the u-pole orders of the coefficients of H, and $\nu' = 0$ if $H = 0$.

Fix $k' \in \exp(G)\mathbb{N}^+$ with

$$k'dw_2 \ge 2\nu + 1 \qquad \text{and} \qquad k'c \ge \nu + 1.$$

Fix a positive integer $N' \ge k'c + \nu + 1$ with $N' > R'(2\nu + k'(dw_1 + c))$ and $N' \ge k'mc + 2\nu + \nu'$. Let $U_{N'}$, resp. $H_{N'}$, be U, resp. H, with the terms of u-degree at least N' deleted from all coefficients, and let $\alpha_{N'}$ be the sum of the terms of α of degree less than N'. Let $\widetilde{\alpha}_i$ be as in **Step 9** for this N', so that $\widetilde{\alpha}_i$ has limit $\alpha_{N'}$. By Lemma 2.6(4), applied to the non-zero ones among $u^\nu U_{N'}$ and $u^{\nu'} H_{N'}$, possibly passing to an infinite subset of $\mathcal{I}$, there exist G-semi-invariant polynomials $U_i^\sharp$ and $H_i^\sharp$ with the characters of $u^\nu U_{N'}$ and $u^{\nu'} H_{N'}$, and with $U_i^\sharp = 0$ if $U_{N'} = 0$ and $H_i^\sharp = 0$ if $H_{N'} = 0$, such that $(f_Q, h_Q, u^\nu U_{N'}, u^{\nu'} H_{N'})$ is a compatible generic limit of the tuples $(f_i', h_i', U_i^\sharp, H_i^\sharp)$ along $\mathcal{I}$, zero entries omitted. Let

$$\widetilde{U}_i = \sum_{j \ge 1,\ jdw_2 < c} \widetilde{\rho}_{i,j}(u) y^j, \quad u^\nu \widetilde{\rho}_{i,j} \in \mathbb{C}[u], \qquad \text{and} \qquad \widetilde{H}_i \in u^{-\nu'}\mathbb{C}[u][x, y, z]$$

be $u^{-\nu} U_i^\sharp$ and $u^{-\nu'} H_i^\sharp$ with the terms whose monomials do not occur in $u^\nu U_{N'}$, resp. $u^{\nu'} H_{N'}$, deleted. Deleting the terms is a linear projection of each jet space fixing the jets of the limit tuple, so the limit tuple stays a compatible generic limit of the modified tuples. In particular $\widetilde{U}_i = 0$ if $w_1 = w_2 = 1$, and every monomial of $\widetilde{H}_i$, being a monomial of H, has w-weight less than $mc - dc$. Let $\widetilde{g}_i(z, u) := g_i'(z + \widetilde{\alpha}_i(u), u)$.

The new chart has coordinates X, Y, Z, u, related to the residual coordinates by

$$\begin{aligned} y &= u^{k'dw_2} Y, \\ z &= \widetilde{\alpha}_i(u) + u^{k'c} Z - \widetilde{U}_i(u^{k'dw_2} Y, u), \\ x &= u^{k'dw_1} X + \frac{\widetilde{g}_i(u^{k'c} Z, u) - \widetilde{g}_i\big(u^{k'c} Z - \widetilde{U}_i(u^{k'dw_2} Y, u), u\big)}{u^{k'dw_2} Y}. \end{aligned} \tag{G.42}$$

Equivalently, (X, Y, Z) is the image of (x, y, z) under $T_{\widetilde{\alpha}_i}$, then the automorphism $\Phi_{\widetilde{g}_i}$ of Proposition 7.6(3) for $g = \widetilde{g}_i$ and $U = \widetilde{U}_i$, then the dilation of depth k'. Since $\widetilde{\alpha}_i$ and $\widetilde{U}_i$ have the character of z and $k' \in \exp(G)\mathbb{Z}$, the map (G.42) is G-equivariant.

Regularity of (G.42). Let

$$a := u^{k'c} Z, \qquad b := \widetilde{U}_i(u^{k'dw_2} Y, u) = u^{k'dw_2} Y \cdot \widetilde{V}_i, \qquad \text{and} \qquad \widetilde{V}_i := \sum_j \widetilde{\rho}_{i,j}(u) u^{(j-1)k'dw_2} Y^{j-1}.$$

Then $u^\nu \widetilde{V}_i \in \mathbb{C}[u, Y]$, $a \in (u^{k'c}) \subset (u^{\nu+1})$, and $b \in (u^{k'dw_2 - \nu}) \subset (u^{\nu+1})$. By Taylor expansion of $\widetilde{g}_i(a - b, u)$ at a,

$$x = u^{k'dw_1} X + \widetilde{V}_i \delta_i, \qquad \delta_i := \sum_{n \ge 1} \frac{(-1)^{n+1}}{n!} \partial_z^n \widetilde{g}_i(a, u) b^{n-1}.$$

Here $\delta_i \equiv \partial_z \widetilde{g}_i(0, u) \mod (a, b)$. For the limit,

$$\partial_z g_Q(\alpha_{N'}(u), u) = d(\alpha_{N'} - \alpha)^{d-1} \in (u^{N'}),$$

so by Lemma 2.7, for $i \gg 0$,

$$\partial_z \widetilde{g}_i(0, u) \in (u^{N'}) \subset (u^{\nu+1}).$$

Hence $\delta_i \in (u^{\nu+1})$ and $\widetilde{V}_i\delta_i \in (u)$. Together with $\widetilde{\alpha}_i(0) = 0$, the right-hand sides of (G.42) are regular and belong to $(u)\cdot\mathbb{C}[[X,Y,Z,u]]$. Thus (G.42) is a regular map sending the origin to the origin.

Regularity of the equations. Since $f_i' = xy + \widetilde{g}_i(z - \widetilde{\alpha}_i(u), u)$ and $z - \widetilde{\alpha}_i = a - b$, (G.42) implies

$$f_i' = u^{k'dc}XY + \widetilde{g}_i(u^{k'c}Z, u).$$

Write $\widetilde{g}_i = \sum_{n\ge 0} g_{i,n}(u)z^n$. Then $f_i'' := u^{-k'dc}f_i'$ is regular if and only if $g_{i,n} \in (u^{k'c(d-n)})$ for any $n < d$. For the limit, $g_Q(z + \alpha_{N'}, u) = (z + \alpha_{N'} - \alpha)^d$, whose coefficient of z^n belongs to $(u^{N'(d-n)}) \subset (u^{k'c(d-n)})$. By Lemma 2.7 for the coefficients of $1, u, \dots, u^{k'c(d-n)-1}$ in $g_{i,n}$, $n < d$, the series f_i'' is regular for $i \gg 0$. For the mark, let

$$h_i'' := u^{-k'mc}\Big(h_i' - \widetilde{H}_i\big(u^{k'dw_1}X, u^{k'dw_2}Y, u^{k'c}Z, u\big)f_i'\Big),$$

where h_i' and f_i' are composed with (G.42). Then $u^{k'mc+\nu'}h_i'' \in \mathbb{C}[[X,Y,Z,u]]$, and h_i'' is regular if and only if $u^{k'mc+\nu'}h_i'' \in (u^{k'mc+\nu'})$. Since $x, y, z, u, a, b \in (u)$, the class of $u^{k'mc+\nu'}h_i''$ modulo $u^{k'mc+\nu'}$ is a polynomial in X, Y, Z, u whose coefficients are the members of a finite family Θ of polynomials in finitely many coefficients of h_i', g_i', $\widetilde{\alpha}_i$, $\widetilde{U}_i$, and $\widetilde{H}_i$. Since $x, y, z, u, a, b \in (u)$ and the poles of $\widetilde{V}_i$ and $\widetilde{H}_i$ are bounded by ν and ν', the values of the members of Θ at the coefficients of h_Q, g_Q, $\alpha_{N'}$, $U_{N'}$, and $H_{N'}$ do not involve the coefficients of $u^{k'mc+2\nu+\nu'}$ and of higher powers of u in α, $u^\nu U$, and $u^{\nu'}H$. As $N' \ge k'mc + 2\nu + \nu'$, the values at the truncations $\alpha_{N'}$, $U_{N'}$, $H_{N'}$ coincide with the values of the members of Θ at the coefficients of α, U, and H. With α, U, H, the map (G.42) for the limit is $(\Phi_{z^d}\circ T_\alpha)^{-1}$ composed with the dilation, since $g_Q(z+\alpha, u) = z^d$. By (G.41), the expression h_i'' formed with h_Q, g_Q, α, U, H in place of h_i', g_i', $\widetilde{\alpha}_i$, $\widetilde{U}_i$, $\widetilde{H}_i$ is

$$u^{-k'mc}\operatorname{in}_w(h_*)\big(u^{k'dw_1}X, u^{k'dw_2}Y, u^{k'c}Z\big) = \operatorname{in}_w(h_*)(X,Y,Z) \in \Bbbk[[u]][X,Y,Z],$$

since $\operatorname{in}_w(h_*)$ is homogeneous of weight mc. Thus the values of the members of Θ at the coefficients of h_Q, g_Q, $\alpha_{N'}$, $U_{N'}$, $H_{N'}$ vanish, and by Lemma 2.7, h_i'' is regular for $i \gg 0$.

Centredness of F_i. Let $\widehat{R}_i$ be the completion of the valuation ring of F_i, and $v := \operatorname{mult}_{F_i}$. By (G.40), $v(x), v(y), v(z - \widetilde{\alpha}_i) \ge N'$, and $1 \le v(u) \le R'$. In $\widehat{R}_i$ let

$$b^\flat := \widetilde{U}_i(y,u), \qquad a^\flat := z - \widetilde{\alpha}_i(u) + b^\flat,$$

and

$$Y^\flat := u^{-k'dw_2}y, \qquad Z^\flat := u^{-k'c}a^\flat, \qquad \text{and} \qquad X^\flat := u^{-k'dw_1}\Big(x - \frac{b^\flat}{y}\delta_i(a^\flat, b^\flat, u)\Big),$$

where

$$\delta_i(a^\flat, b^\flat, u) := \sum_{n\ge 1}\frac{(-1)^{n+1}}{n!}\partial_z^n\widetilde{g}_i(a^\flat, u)(b^\flat)^{n-1}.$$

Since $v(\widetilde{\rho}_{i,j}(u)y^j) \ge jN' - \nu R'$ for any $j \ge 1$,

$$v(b^\flat) \ge N' - \nu R', \qquad v(a^\flat) \ge N' - \nu R', \qquad \text{and} \qquad v\Big(\frac{b^\flat}{y}\Big) \ge -\nu R'. \tag{G.43}$$

In particular $\delta_i(a^\flat, b^\flat, u)$, a finite sum since $\deg_z \widetilde{g}_i = d$, is defined in $\widehat{R}_i$. Since $\partial_z\widetilde{g}_i(0,u) \in (u^{N'})$,

$$v(\delta_i(a^\flat, b^\flat, u)) \ge \min\{N', v(a^\flat), v(b^\flat)\} \ge N' - \nu R'.$$

Therefore

$$v(Y^\flat) \ge N' - k'dw_2R', \qquad v(Z^\flat) \ge N' - (\nu + k'c)R', \qquad \text{and} \qquad v(X^\flat) \ge N' - (2\nu + k'dw_1)R',$$

and all three are positive by $N' > R'(2\nu + k'(dw_1 + c))$, since $w_2 \le w_1$. Since $\widetilde{V}_i(Y^\flat) = b^\flat/y$, the substitution of $(X^\flat, Y^\flat, Z^\flat, u)$ into the right-hand sides of (G.42) returns (x, y, z). Thus

$$(X,Y,Z,u) \to (X^\flat, Y^\flat, Z^\flat, u)$$

defines a local homomorphism from the completed local ring of the new chart at the origin to $\widehat{R}_i$, which kills f_i'' and is compatible with (G.42). Hence F_i is centred at the origin of the new chart.

Admissibility. We undo the depth r_i. Conjugation by the dilation of depth r_i turns $T_{\widetilde{\alpha}_i}$ into the translation by $u^{r_i c}\widetilde{\alpha}_i(u)$, turns $\Phi_{\widetilde{g}_i}$ with $\widetilde{U}_i$ into the map (H) for the equation at depth 0 with $U = U_i'$, where

$$U_i' := \sum_j u^{r_i(c-jdw_2)}\widetilde{\rho}_{i,j}(u)y^j,$$

and turns the multiple $\widetilde{H}_i f_i'$ into a multiple of the equation at depth 0 in which any monomial $\mathbf{x}$ of $\widetilde{H}_i$ acquires the factor $u^{r_i(mc-dc-\mathrm{wt}_w(\mathbf{x}))}$. The exponents satisfy

$$c - jdw_2 \geq 1 \quad \text{and} \quad mc - dc - \mathrm{wt}_w(\mathbf{x}) \geq 1,$$

the poles are bounded by ν and ν', $r_i \geq r_i^0 \to +\infty$, and $\ell_i/r_i^0 \to 0$. Hence for $i \gg 0$ the coefficients of $u^{r_i c}\widetilde{\alpha}_i$, of U_i' and of the transformed multiple belong to $u^{\ell_i}\mathbb{C}[u]$. The new multiple of the equation is the transform of a polynomial with coefficients in $u^{\ell_i}\mathbb{C}[u]$ under the composite regular algebraic preparation, hence a regular algebraic power series in (u^{ℓ_i}), as allowed in (C). The characters are those of (T)–(C), and y and u are fixed. Hence ϖ_i, followed by the translation by $u^{r_i c}\widetilde{\alpha}_i$, the map (H) for the current equation with U_i', and the change of lift by the new multiple, is a finite composition of maps (T)–(C), so it belongs to $\mathcal{P}_i$.

Thus, for $i \gg 0$, there exists $\varpi \in \mathcal{P}_i$ such that $(\varpi, r_i + k')$ is admissible, with $r_i + k' \in \exp(G)\mathbb{Z}$. Since $r_i + k' > r_i$, this contradicts the maximality of r_i.

All cases of the proof have led to a contradiction. Hence (∗) fails, so the theorem holds. □

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